Step 1 of 2

8.001E

Refer to Example 5 in the text book.

The change in output current is 1% of the reference current.

$$\Delta I_o = \left(\frac{1}{100}\right)(100\ \mu A)$$

$$\Delta I_o = 1\ \mu A$$

The output resistance is,

$$r_{o2} = \frac{V_A' L}{I_D}$$

$$\frac{V_A' L}{I_D} = \frac{\Delta V_o}{\Delta I_o} \qquad \left(r_{o2} = \frac{\Delta V_o}{\Delta I_o}\right)$$

$$\frac{(20\times 10^6)L}{(100\times 10^{-6})} = \frac{1}{1\times 10^{-6}} \qquad (\text{since, } I_D = I_{REF})$$

$$L = \frac{100}{20\times 10^6}$$

Thus, the value of L is $\boxed{5\ \mu m}$.

Step 2 of 2

Choose the value of the aspect ratio as 10.

$$\frac{W}{L} = 10$$

$$W = 10L$$

$$W = 10(5\ \mu m)$$

Thus, the value of W is $\boxed{50\ \mu m}$.

Step 1 of 6

Refer to Figure 8.1 from the text book.

Consider the expression for the output current.

$$I_O = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(V_{GS} - V_t)^2$$

Substitute $100\ \mu\text{A}$ for I_O, $500\ \mu\text{A/V}^2$ for k_n', $0.5\ \mu\text{m}$ for L, $5\ \mu\text{m}$ for W and $0.4\ \text{V}$ for V_t.

$$100\ \mu = \frac{1}{2}(500\ \mu)\left(\frac{5\ \mu}{0.5\ \mu}\right)(V_{GS} - 0.4)^2$$

$$(V_{GS} - 0.4)^2 = \frac{100\ \mu}{\frac{1}{2}(500\ \mu)\left(\frac{5\ \mu}{0.5\ \mu}\right)}$$

$$(V_{GS} - 0.4)^2 = \frac{2}{50}$$

$$(V_{GS} - 0.4)^2 = 0.04$$

$$V_{GS} - 0.4 = \sqrt{0.04}$$

$$V_{GS} - 0.4 = \pm 0.2$$

Step 2 of 6

Solve for the value of V_{GS}.

$$V_{GS} - 0.4 = -0.2$$

$$V_{GS} = -0.2 + 0.4$$

$$V_{GS} = 0.2\ \text{V}$$

Or

$$V_{GS} - 0.4 = 0.2$$

$$V_{GS} = 0.2 + 0.4$$

$$V_{GS} = 0.6\ \text{V}$$

The value of V_{GS} should be less greater than the value of V_t for the transistor to operate in saturation region. Hence, the required value of V_{GS} is $0.6\ \text{V}$.

Step 3 of 6

Consider the expression for the drain current I_D.

$$I_D = \frac{V_{DD} - V_{GS}}{R}$$

Substitute $100\ \mu\text{A}$ for I_D, $1.3\ \text{V}$ for V_{DD} and $0.6\ \text{V}$ for V_{GS}.

$$100\ \mu = \frac{1.3 - 0.6}{R}$$

$$100\ \mu = \frac{0.7}{R}$$

Solve for the value of R.

$$R = \frac{0.7}{100\ \mu}$$

$$= 7\ \text{k}\Omega$$

Therefore, the value of the resistor R is $\boxed{7\ \text{k}\Omega}$.

Step 4 of 6

The value of the output voltage will be minimum when $V_O = V_{GS} - V_t$. Hence, calculate the value of the output voltage.

$$V_{O\text{min}} = V_{GS} - V_t$$

Substitute $0.6\ \text{V}$ for V_{GS} and $0.4\ \text{V}$ for V_t.

$$V_{O\text{min}} = 0.6 - 0.4$$

$$= 0.2\ \text{V}$$

Therefore, the minimum value of the output voltage $V_{O\text{min}}$ is $\boxed{0.2\ \text{V}}$.

Step 5 of 6

Consider the expression for the output resistance of the current source r_{o2}.

$$r_{o2} = \frac{V_A' L}{I_D}$$

Substitute $100\ \mu\text{A}$ for I_D, $0.5\ \mu\text{m}$ for L and $5\ \text{V/}\mu\text{A}$ for V_A'.

$$r_{o2} = \frac{(5/1\ \mu)(0.5\ \mu)}{100\ \mu}$$

$$= \frac{2.5}{100\ \mu}$$

$$= 25\ \text{k}\Omega$$

Therefore, the value of the output resistance of the current source r_{o2} is $\boxed{25\ \text{k}\Omega}$.

Step 6 of 6

Consider the expression for the output resistance of the current source r_{o2} in terms of change in output voltage to change in output current.

$$r_{o2} = \frac{\Delta V_O}{\Delta I_O}$$

Substitute $0.5\ \text{V}$ for ΔV_O and $25\ \text{k}\Omega$ for r_{o2}.

$$25\ \text{k} = \frac{0.5}{\Delta I_O}$$

Solve for the value of ΔI_O.

$$\Delta I_O = \frac{0.5}{25\ \text{k}}$$

$$= 20\ \mu\text{A}$$

Therefore, the value of the change in output current ΔI_O is $\boxed{20\ \mu\text{A}}$.

Step 1 of 5

Write the expression for the drain current of the transistor Q_1.

$$I_{D1}=\frac{1}{2}k_n'\left(\frac{W}{L}\right)_n V_{OV}^2$$

Since $I_{D1}=I_{REF}$, the drain current of the transistor Q_1 is,

$$I_{D1}=10\ \mu A$$

Substitute $I_{D1}=10\ \mu A$, $k_n'=200\ \mu A/V^2$, $L_1=1\ \mu m$ and $V_{OV}=0.2\ V$.

$$10\times10^{-6}=\frac{1}{2}(200\times10^{-6})\left(\frac{W_1}{1\times10^{-6}}\right)(0.2)^2$$

$$4W_1=10\times10^{-6}$$

$$W_1=\frac{10\times10^{-6}}{4}$$

$$=2.5\ \mu m$$

Thus, the value of W_1 is $\boxed{2.5\ \mu m}$.

Step 2 of 5

Since transistors Q_1 and Q_2 are matched, the expression for the current I_2 is,

$$I_2=I_{REF}\left(\frac{W_2/L_2}{W_1/L_1}\right)$$

Substitute $I_2=60\ \mu A$, $I_{REF}=10\ \mu A$ and $W_1=2.5\ \mu m$.

$$60\times10^{-6}=(10\times10^{-6})\left(\frac{W_2}{2.5\times10^{-6}}\right)\qquad(\text{since, } L_2=L_1)$$

$$W_2=\left(\frac{2.5}{10}\right)(60\times10^{-6})$$

$$=15\times10^{-6}$$

$$=15\ \mu m$$

Thus, the value of W_2 is $\boxed{15\ \mu m}$.

Step 3 of 5

Since transistors Q_1 and Q_3 are matched, the expression for the current I_3 is,

$$I_3=I_{REF}\left(\frac{W_3/L_3}{W_1/L_1}\right)$$

Substitute $I_3=20\ \mu A$, $I_{REF}=10\ \mu A$ and $W_1=2.5\ \mu m$.

$$20\times10^{-6}=(10\times10^{-6})\left(\frac{W_3}{2.5\times10^{-6}}\right)\qquad(\text{since, } L_2=L_1)$$

$$W_3=\left(\frac{2.5}{10}\right)(20\times10^{-6})$$

$$W_3=5\times10^{-6}\ m$$

$$=5\ \mu m$$

Thus, the value of W_3 is $\boxed{W_3=5\ \mu m}$.

Step 4 of 5

Equate the expressions of the currents I_3 and I_4.

$$\frac{1}{2}k_n\left(\frac{W_3}{L_3}\right)V_{OV}^2=\frac{1}{2}k_p\left(\frac{W_4}{L_4}\right)V_{OV}^2$$

$$W_4=\left(\frac{k_n}{k_p}\right)W_3$$

$$=\left(\frac{200}{80}\right)(5\ \mu m)$$

$$=12.5\ \mu m$$

Thus, the value of W_4 is $\boxed{W_4=12.5\ \mu m}$.

Step 5 of 5

The transistors Q_4 and Q_5 are matched current mirrors.

$$\frac{I_5}{I_4}=\frac{(W_5/L_5)}{(W_4/L_4)}$$

$$W_5=\left(\frac{I_5}{I_4}\right)W_4$$

$$W_5=\left(\frac{80}{20}\right)(12.5\ \mu m)$$

$$=50\ \mu m$$

Thus, the value of W_5 is $\boxed{50\ \mu m}$.

Step 1 of 7

Refer to Figure 8.1 from the text book.

Consider the expression for the value of V_{GS} .

$$V_{GS} = V_{Omin} + V_t$$

Substitute $0.3\ \text{V}$ for V_{Dmin} and $0.5\ \text{V}$ for V_t .

$$V_{GS} = 0.3 + 0.5$$
$$= 0.8\ \text{V}$$

Therefore, the value of the voltage V_{GS} is, $0.8\ \text{V}$.

Step 2 of 7

Consider the expression for the drain current, I_D .

$$I_D = \frac{V_{DD} - V_{GS}}{R}$$

Substitute $150\ \mu\text{A}$ for I_D , $1.8\ \text{V}$ for V_{DD} and $0.8\ \text{V}$ for V_{GS} .

$$150\ \mu = \frac{1.8 - 0.8}{R}$$

$$150\ \mu = \frac{1}{R}$$

Solve for the value of R .

$$R = \frac{1}{160\ \mu}$$
$$= 6.67\ \text{k}\Omega$$

Therefore, the value of the resistor R is, $\boxed{6.67\ \text{k}\Omega}$.

Step 3 of 7

Calculate the value of the change in the output current, ΔI_D .

$$\Delta I_D = (10\%) I_O$$
$$= 0.1 I_O$$

Substitute $150\ \mu\text{A}$ for I_D .

$$\Delta I_D = 0.1(150\ \mu)$$
$$= 15\ \mu\text{A}$$

Therefore, the value of the change in output current ΔI_D is, $15\ \mu\text{A}$.

Step 4 of 7

Calculate the value of the change in output voltage, ΔV_D .

$$\Delta V_O = V_O - V_{Dmin}$$

Substitute $1.8\ \text{V}$ for V_D and $0.3\ \text{V}$ for V_{Dmin} .

$$\Delta V_O = 1.8 - 0.3$$
$$= 1.5\ \text{V}$$

Therefore, the value of the output voltage ΔV_D is, $1.5\ \text{V}$.

Step 5 of 7

Consider the expression for the output resistance of the current source, r_{o2} in terms of change in output voltage to change in output current.

$$r_{o2} = \frac{\Delta V_O}{\Delta I_D}$$

Substitute $1.5\ \text{V}$ for ΔV_D and $15\ \mu\text{A}$ for ΔI_D .

$$r_{o2} = \frac{1.5}{15\ \mu}$$
$$= 100\ \text{k}\Omega$$

Therefore, the value of the output resistance r_{o2} is, $100\ \text{k}\Omega$.

Step 6 of 7

Consider the expression for the output resistance of the current source r_{o2} in terms of early effect and drain current.

$$r_{o2} = \frac{V_A' L}{I_D}$$

Substitute $150\ \mu\text{A}$ for I_D , $100\ \text{k}\Omega$ for r_{o2} and $10\ \text{V}/\mu\text{A}$ for V_A' .

$$100\ \text{k} = \frac{(10/1\ \mu)(L)}{150\ \mu}$$

Solve for the value of L .

$$L = \frac{(150\ \mu)(100\ \text{k})}{(10/1\ \mu)}$$
$$= 1.5\ \mu\text{m}$$

Therefore, the value of the length L of the device is, $\boxed{1.5\ \mu\text{m}}$.

Step 7 of 7

Consider the expression for the output current, I_D .

$$I_D = \frac{1}{2}(\mu_n C_{ox})\left(\frac{W}{L}\right)(V_{GS} - V_t)^2$$

Substitute $150\ \mu\text{A}$ for I_D , $400\ \mu\text{A}/\text{V}^2$ for $\mu_n C_{ox}$, $1.5\ \mu\text{m}$ for L , $0.8\ \text{V}$ for V_{GS} and $0.5\ \text{V}$ for V_t .

$$150\ \mu = \frac{1}{2}(400\ \mu)\left(\frac{W}{1.5\ \mu}\right)(0.8 - 0.5)^2$$

Solve for the value of W .

$$150\ \mu = \frac{1}{2}(400\ \mu)\left(\frac{W}{1.5\ \mu}\right)(0.8 - 0.5)^2$$

$$W = \frac{(2)(150\ \mu)(1.5\ \mu)}{(400\ \mu)(0.8 - 0.5)^2}$$

$$W = \frac{450\ \text{p}}{(400\ \mu)(0.3)^2}$$

$$W = \frac{450\ \text{p}}{(400\ \mu)(0.09)}$$

$$W = \frac{450\ \text{p}}{36\ \mu}$$
$$= 12.5\ \mu\text{m}$$

Therefore, the value of width of the device W is, $\boxed{12.5\ \mu\text{m}}$.

Step 1 of 2

8.003E

Write the expression of the output current of a BJT mirror with a nominal current transfer ratio.

$$I_O = I_{REF}\left(\frac{m}{1+\frac{m+1}{\beta}}\right)\left(1+\frac{V_O - V_{BE}}{V_{A2}}\right)$$

Substitute the nominal current transfer ratio $m = 1$.

$$I_O = I_{REF}\left(\frac{1}{1+\frac{1+1}{\beta}}\right)\left(1+\frac{V_O - V_{BE}}{V_{A2}}\right)$$

Substitute 1 mA for I_{REF}, 100 for β, 5 V for V_O and 0.7 V for V_{BE}.

$$I_O = \left(1\times10^{-3}\right)\left(\frac{1}{1+\frac{2}{100}}\right)\left(1+\frac{5-0.7}{100}\right) \qquad (\text{since, } V_{BE} = 0.7\text{ V})$$

$$= \left(1\times10^{-3}\right)\left(\frac{1}{1.02}\right)(1.043)$$

$$= 1.02\times10^{-3}$$

$$= 1.02\text{ mA}$$

Thus, the value of I_O is $\boxed{1.02\text{ mA}}$.

Step 2 of 2

Calculate the output resistance.

$$r_{o2} = \frac{V_A}{I_O}$$

$$= \frac{100}{1\times10^{-3}} \qquad \left(\begin{array}{l}\text{since, desired current mirror}\\ I_O = I_{REF}\end{array}\right)$$

$$= 100\text{ k}\Omega$$

Thus, the value of the resistance r_{o2} is $\boxed{100\text{ k}\Omega}$.

Step 1 of 6

Consider the circuit diagram shown in Figure 1.

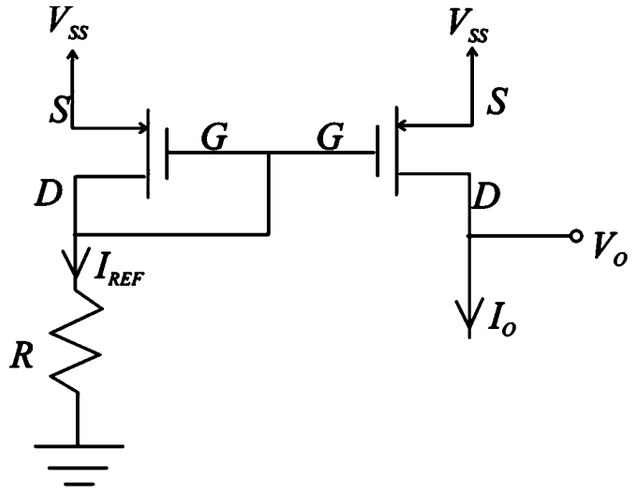


Figure 1

Step 2 of 6

Notice that the value of V_{DD} in NMOS circuit represents the values of V_{SS} in PMOS circuit.

Calculate the value for the maximum value of the drain to source voltage, $V_{DS\max}$.

$$V_{DS\max} = V_{O\max} - V_{SS}$$

Substitute $1.3\ \text{V}$ for V_{SS} and $1.1\ \text{V}$ for $V_{O\max}$.

$$\begin{aligned} V_{DS\max} &= 1.1 - 1.3 \\ &= -0.2\ \text{V} \end{aligned}$$

Therefore, the value of the voltage $V_{DS\max}$ is, $-0.2\ \text{V}$.

Step 3 of 6

The value of the voltage $V_{DS\max}$ should be equal to $V_{GS} - V_t$.

$$V_{DS\max} = V_{GS} - V_t$$

Substitute $-0.2\ \text{V}$ for $V_{DS\max}$ and $-0.4\ \text{V}$ for V_t.

$$\begin{aligned} -0.2 &= V_{GS} - (-0.4) \\ V_{GS} &= -0.2 - 0.4 \\ V_{GS} &= -0.6\ \text{V} \end{aligned}$$

Therefore, the value of the gate to source voltage V_{GS} is, $-0.6\ \text{V}$.

Step 4 of 6

Calculate the value of the voltage at the gate terminal, V_G.

$$\begin{aligned} V_G &= V_{SS} - V_{SG} \\ &= V_{SS} + V_{GS} \end{aligned}$$

Substitute $-0.6\ \text{V}$ for V_{GS} and $1.3\ \text{V}$ for V_{SS}.

$$\begin{aligned} V_G &= 1.3 - 0.6 \\ &= 0.7\ \text{V} \end{aligned}$$

Therefore, the value of the voltage V_G is, $0.7\ \text{V}$.

Step 5 of 6

Calculate the value of the resistance, R.

$$R = \frac{V_G}{I_{REF}}$$

Substitute $0.7\ \text{V}$ for V_G and $80\ \mu\text{A}$ for I_{REF}.

$$\begin{aligned} R &= \frac{0.7}{80\ \mu} \\ &= 8.75\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistor R, is $\boxed{8.75\ \text{k}\Omega}$.

Step 6 of 6

Consider the expression for the drain current, I_D.

$$I_D = \frac{1}{2}\left(\mu_p C_{ox}\right)\left(\frac{W}{L}\right)\left(V_{GS} - V_t\right)^2$$

Substitute $80\ \mu\text{A}$ for I_D, $80\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$, $-0.6\ \text{V}$ for V_{GS} and $-0.4\ \text{V}$ for V_t.

$$80\ \mu = \frac{1}{2}(80\ \mu)\left(\frac{W}{L}\right)\left(-0.6 - (-0.4)\right)^2$$

$$1 = \frac{1}{2}\left(\frac{W}{L}\right)(-0.6 + 0.4)^2$$

$$2 = \left(\frac{W}{L}\right)(-0.2)^2$$

$$2 = \left(\frac{W}{L}\right)(0.04)$$

Solve for the value of $\frac{W}{L}$.

$$\begin{aligned} \frac{W}{L} &= \frac{2}{0.04} \\ &= \frac{200}{4} \\ &= \frac{50}{1} \end{aligned}$$

Therefore, the ratio of the width to length $\frac{W}{L}$ should be $\boxed{\frac{50}{1}}$.

Step 1 of 3

Refer to Figure 7.30 in the text book.

Write the expression for the output current of a BJT mirror with a nominal current transfer ratio.

$$I_O = I_{REF}\left(\frac{m}{1+\frac{m+1}{\beta}}\right)\left(1+\frac{V_O - V_{BE}}{V_{A2}}\right)$$

Substitute the nominal current transfer ratio, $m = 1$ $\left(\text{since, } I_{S2} = I_{S1} = I_S\right)$.

$$I_O = I_{REF}\left(\frac{1}{1+\frac{1+1}{\beta}}\right)\left(1+\frac{V_O - V_{BE}}{V_{A2}}\right)$$

Determine the value of I_{REF}.

$$0.5\times10^{-3} = I_{REF}\left(\frac{1}{1+\frac{2}{100}}\right)\left(1+\frac{2-0.7}{50}\right) \qquad (\text{since, } V_{BE} = 0.7\text{ V})$$

$$0.5\times10^{-3} = I_{REF}\left(\frac{1}{1.02}\right)(1.026)$$

$$I_{REF} = \left(\frac{1.02}{1.026}\right)\left(0.5\times10^{-3}\right)$$

$$= 0.497\times10^{-3}$$

Thus, the value of the current I_{REF} is $\boxed{0.497\text{ mA}}$.

Step 2 of 3

Find the value for current limiting resistor.

$$R = \frac{V_{CC} - V_{BE}}{I_{REF}}$$

$$= \frac{5-0.7}{0.497\times10^{-3}}$$

$$= 8.65\text{ k}\Omega$$

Thus, the value of the resistance R is $\boxed{8.65\text{ k}\Omega}$.

Step 3 of 3

The transistor Q_1 and Q_2 must be in saturation to act as current mirror network.

$$V_{O\min} = V_{CEsat2}$$

$$= 0.3\text{ V}$$

Thus, the value of $V_{O\min}$ is $\boxed{0.3\text{ V}}$.

Determine the output current for $V_O = 5\text{ V}$.

$$I_O = I_{REF}\left(\frac{1}{1+\frac{1+1}{\beta}}\right)\left(1+\frac{V_O - V_{BE}}{V_{A2}}\right)$$

$$= \left(0.497\times10^{-3}\right)\left(\frac{1}{1+\frac{2}{100}}\right)\left(1+\frac{5-0.7}{50}\right) \qquad (\text{since, } V_{BE} = 0.7\text{ V})$$

$$= \left(0.497\times10^{-3}\right)\left(\frac{1}{1.02}\right)(1.086)$$

$$= 0.529\times10^{-3}\text{ A}$$

$$\approx 0.53\text{ mA}$$

Thus, the value of the current I_O is $\boxed{0.53\text{ mA}}$.

Step 1 of 5

Refer to Figure 7.23 in the textbook for the circuit.

Channel width of transistor Q_2 is five times the channel width of transistor Q_1.

$$W_2 = 5W_1$$

The transistors have equal channel length.

$$L_1 = L_2$$

The output current of the current mirror circuit is,

$$I_O = I_{REF}\left(\frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1}\right)$$
$$= I_{REF}\left(\frac{\frac{W_2}{L_2}}{\frac{W_1}{L_1}}\right)$$

Substitute L_1 for L_2, $5W_1$ for W_2, and $20\ \mu A$ for I_{REF} in the equation.

$$I_O = \left(20\times10^{-6}\right)\left(\frac{\frac{5W_1}{L_1}}{\frac{W_1}{L_1}}\right)$$
$$= \left(20\times10^{-6}\right)(5)$$
$$= 100\times10^{-6}$$

Hence, the output current is, $\boxed{I_O = 100\ \mu A}$.

Step 2 of 5

Minimum allowable value of V_o is equal to over drive voltage.

$$V_o(\min) = V_{DS2} = V_{OV}$$
$$= 0.2\ V$$

The over drive voltage is,

$$V_{OV} = V_{GS} - V_t$$
$$V_{GS} = V_{OV} + V_t$$
$$V_{GS} = 0.2 + 0.5$$
$$= 0.7\ V$$

Step 3 of 5

The drain current in linear region is,

$$I = I_{REF}\left(\frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1}\right)\left(1 + \frac{V_o - V_{GS}}{V_{A2}}\right)$$
$$1 + \frac{V_o - V_{GS}}{V_{A2}} = \frac{I}{I_{REF}\left(\frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1}\right)}$$

The condition for nominal value of current is $\frac{I}{I_{REF}\left(\frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1}\right)} = 1$. Substitute this in the equation.

$$1 + \frac{V_o - V_{GS}}{20} = 1$$
$$\frac{V_o - V_{GS}}{20} = 1 - 1$$
$$\frac{V_o - V_{GS}}{20} = 0$$
$$V_o = V_{GS}$$
$$= 0.7\ V$$

Thus, the output voltage is, $\boxed{V_O = 0.7\ V}$

Step 4 of 5

The output resistance is,

$$R_o = r_{o2} = \frac{V_A}{I_{D2}}$$
$$= \frac{20}{100\times10^{-6}}$$
$$= 200\times10^3$$

Step 5 of 5

Write the equation for output resistance.

$$R_o = r_{O2} = \frac{\Delta V_O}{\Delta I_O}$$

Rearrange the equation.

$$\Delta I_O = \frac{\Delta V_O}{R_o}$$

Substitute $200\ k\Omega$ for R_o and $1\ V$ for ΔV_O in the equation

$$\Delta I_D = \frac{1}{200\times10^3}$$
$$= 5\ \mu A$$

Thus, the change in drain current is, $\boxed{\Delta I_D = 5\ \mu A}$.

Step 1 of 5

Refer to Figure E7.19 in the text book.

Modify the circuit diagram by marking currents as shown in Figure 1.

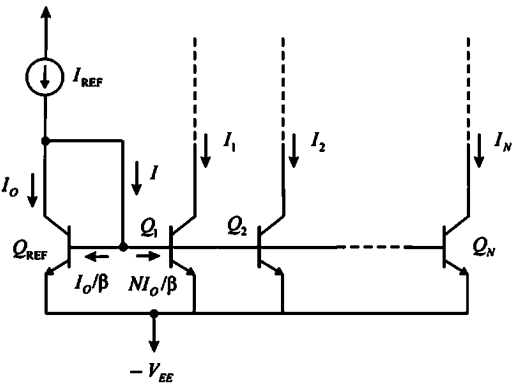


Figure 1

Step 2 of 5

The transistors pairs, (Q_{REF}, Q_2) , (Q_{REF}, Q_2) , ... , (Q_{REF}, Q_n) act as current mirror networks whose collector currents are equal to I_O .

Apply Kirchhoff's current law to the base node of the transistor, Q_1 .

$$I = \frac{I_O}{\beta} + \frac{NI_O}{\beta}$$
$$I = \frac{(1+N)I_O}{\beta}$$

Step 3 of 5

Apply Kirchhoff's current law to the collector node of the transistor Q_{REF} .

$$I_{REF} = I_O + I$$
$$I_{REF} = I_O + \frac{(1+N)I_O}{\beta}$$
$$I_{REF} = I_O\left[1 + \frac{(1+N)}{\beta}\right]$$
$$I_O = \frac{I_{REF}}{1 + \frac{(1+N)}{\beta}} \quad \text{..... (1)}$$

Since the entire transistor pairs are matched, the currents are,

$$\boxed{I_1 = I_2 = ... = I_N = \frac{I_{REF}}{1 + \frac{(1+N)}{\beta}}}$$

Step 4 of 5

For a maximum deviation of 10% from the unity, choose the value of I_{REF} as 1 A.

Calculate the value of I_o for $I_{REF} = 1\ \text{A}$.

$$I_O = 1 - \left(\frac{10}{100}\right)(1)$$
$$I_O = 1 - 0.1$$
$$I_O = 0.9\ \text{A}$$

Substitute $I_O = 0.9\ \text{A}$ in equation (1).

$$\frac{I_O}{I_{REF}} = \frac{1}{1 + \frac{(1+N)}{\beta}}$$
$$\frac{0.9}{1} = \frac{1}{1 + \frac{(1+N)}{100}}$$
$$0.9\left(1 + \frac{(1+N)}{100}\right) = 1$$
$$0.9 + \frac{0.9(1+N)}{100} = 1$$

Simplify further.

$$\frac{0.9(1+N)}{100} = 1 - 0.9$$
$$\frac{0.9(1+N)}{100} = 0.1$$
$$0.9(1+N) = 0.1 \times 100$$
$$0.9 + 0.9N = 10$$

Step 5 of 5

Simplify further.

$$0.9N = 10 - 0.9$$
$$0.9N = 9.1$$
$$N = \frac{9.1}{0.9}$$
$$N = 10.1$$

The deviation is less than 10%.

Choose an integer value such that, $N < 10$.

Therefore, the maximum possible number of outputs for an error not exceeding 10% is $\boxed{9}$.

Step 1 of 3

Refer to Figure P7.50 in the textbook for the circuit.

Redraw the circuit by representing currents I_2 and I_3 as shown in Figure 1.

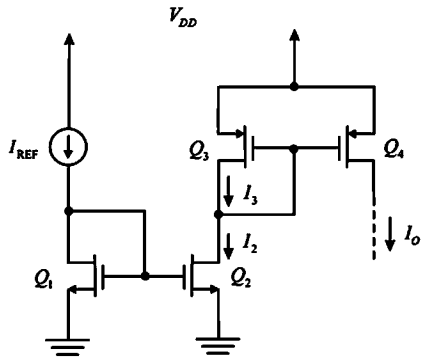


Figure 1

Step 2 of 3

The current reference I_{REF} is applied at the drain terminal of transistor Q_1.

From the circuit, the drain and gate terminals of Q_1 are connected, which pushes the transistor Q_1 into saturation region.

The gate voltage of transistor Q_1 is same as gate voltage of transistor Q_2

Hence transistors Q_1 and Q_2 acts as current mirror network

The current flowing in the drain terminal of transistor Q_2 is I_2.

The relation between I_{REF} and I_2 is,

$$\frac{I_2}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1}$$

Here, $(W/L)_x$ indicates x^{th} transistor aspect ratio.

$$I_2 = \left(\frac{(W/L)_2}{(W/L)_1}\right) I_{REF}$$

Step 3 of 3

The current flowing in the transistor Q_3 is I_3.

From the circuit diagram, the drain voltage of transistor Q_2 is equal to gate voltage of transistors Q_3 and Q_4 which in turn drives the transistor into saturation region.

Since the current flowing into the gate of the MOSFET is zero, the current I_3 is,

$$I_3 = I_2$$

Hence transistors Q_3 and Q_4 acts as current mirror network

Relation between I_3 and I_0 is,

$$\frac{I_0}{I_3} = \frac{(W/L)_4}{(W/L)_3}$$

$$I_0 = \left(\frac{(W/L)_4}{(W/L)_3}\right) I_3$$

Substitute I_2 for I_3 in the equation.

$$I_0 = \left(\frac{(W/L)_4}{(W/L)_3}\right) I_2$$

Substitute $\left(\frac{(W/L)_2}{(W/L)_1}\right) I_{REF}$ for I_2 in the equation.

$$I_0 = \left(\frac{(W/L)_4}{(W/L)_3}\right)\left(\frac{(W/L)_2}{(W/L)_1}\right) I_{REF}$$

Thus, the output current of the circuit in terms of reference current I_{REF} is,

$$\boxed{I_0 = \left(\frac{(W/L)_4}{(W/L)_3}\right)\left(\frac{(W/L)_2}{(W/L)_1}\right) I_{REF}}.$$

Step 1 of 6

Refer to Figure 8.12(a) from the text book.

Calculate the value of the output current I_O.

$$I_O = A_{is} I_{DI}$$

Substitute 5 for A_{is} and $100\ \mu A$ for I_{DI}.

$$I_O = (5)(100\ \mu)$$
$$= 500\ \mu A$$

Therefore, the value of the output current I_O is $500\ \mu A$.

Step 2 of 6

Calculate the value of the output resistance R_{out}.

$$R_{out} = \frac{V_A' L_1}{I_O}$$

Substitute $20\ V/\mu m$ for V_A', $40\ k\Omega$ for R_{out} and $500\ \mu A$ for I_O.

$$40\ k = \frac{\left(\frac{20}{1\mu}\right)(L_1)}{500\ \mu}$$
$$L_1 = \frac{(40\ k)(500\ \mu)(1\ \mu)}{20}$$
$$L_1 = 1\ \mu m$$

Therefore, the value of the length L_1 is $\boxed{1\ \mu m}$.

The value of the length L_2 is equal to the value of the length L_1, therefore, the value of the L_2 is $\boxed{1\ \mu m}$.

Step 3 of 6

Consider the expression for the input resistance.

$$R_{in} = \frac{1}{g_{m1}}$$

Substitute $1\ k\Omega$ for R_{in}.

$$1\ k = \frac{1}{g_{m1}}$$
$$g_{m1} = \frac{1}{1\ k}$$
$$g_{m1} = 1\ mS$$

Therefore, the value of the transconductance g_{m1} is $1\ mS$.

Step 4 of 6

Consider the expression for transconductance g_{m1} in terms of the over drive voltage and the bias current.

$$g_{m1} = \frac{2I_D}{V_{ov}}$$

Substitute $100\ \mu A$ for I_D and $1\ mA$ for g_{m1}.

$$1\ m = \frac{2(100\ \mu)}{V_{ov}}$$
$$V_{ov} = \frac{200\ \mu}{1\ m}$$
$$V_{ov} = 0.2\ V$$

Therefore, the value of the over drive voltage V_{ov} is $0.2\ V$.

Step 5 of 6

Consider the expression for the transconductance g_{m1}.

$$g_{m1} = \mu_n C_{ox} \left(\frac{W_1}{L_1}\right) V_{ov}$$

Substitute $0.2\ V$ for V_{ov}, $1\ mA$ for g_{m1}, $1\ \mu m$ for L_1 and $400\ \mu A/V^2$ for $\mu_n C_{ox}$.

$$1\ m = (400\ \mu)\left(\frac{W_1}{1\ \mu}\right)(0.2)$$
$$W_1 = \frac{(1\ m)(1\ \mu)}{(400\ \mu)(0.2)}$$
$$W_1 = 12.5\ \mu m$$

Therefore, the value of the width W_1 is $\boxed{12.5\ \mu m}$.

Step 6 of 6

Calculate the value of the current gain A_{is}.

$$A_{is} = \frac{(W_2/L_2)}{(W_1/L_1)}$$

Substitute $12.5\ \mu m$ for W_1, 5 for A_{is} and L_1 for L_2.

$$5 = \frac{(W_2/L_2)}{(20\ \mu/L_2)}$$
$$5 = \frac{W_2}{(12.5\ \mu)}$$
$$W_2 = (5)(12.5\ \mu)$$
$$W_2 = 62.5\ \mu m$$

Therefore, the value of the width W_2 is $\boxed{62.5\ \mu m}$.

Step 1 of 15

Refer to Figure P8.6 from the text book.

Identify the voltages in the circuit diagram shown in Figure 1.

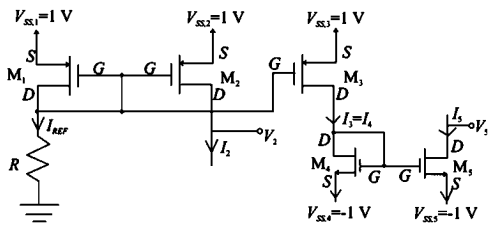


Figure 1

Step 2 of 15

Consider the expression for the voltage, $V_{DS\,max,2}$.

$$V_{DS\,max,2} = V_2 - V_{SS,2}$$

Substitute $1\ \text{V}$ for $V_{SS,2}$ and $0.8\ \text{V}$ for $V_{DS\,max,2}$.

$$\begin{aligned} V_{DS\,max,2} &= 0.8 - 1 \\ &= -0.2\ \text{V} \end{aligned}$$

Therefore, the value of the voltage $V_{DS\,max,2}$ is, $-0.2\ \text{V}$.

Step 3 of 15

The value of the voltage $V_{DS\,max,2}$ should be equal to $V_{GS,2} - V_{tp}$. Hence,

$$V_{DS\,max,2} = V_{GS,2} - V_{tp}$$

Substitute $-0.2\ \text{V}$ for $V_{DS\,max,2}$ and $-0.5\ \text{V}$ for V_{tp}.

$$\begin{aligned} -0.2 &= V_{GS,2} - (-0.5) \\ V_{GS,2} &= -0.2 - 0.5 \\ V_{GS,2} &= -0.7\ \text{V} \end{aligned}$$

Therefore, the value of the gate to source voltage $V_{GS,2}$ is, $-0.7\ \text{V}$.

Step 4 of 15

Calculate the value of the voltage at the gate terminal, $V_{G,2}$.

$$\begin{aligned} V_{G,2} &= V_{SS,2} - V_{SG,2} \\ &= V_{SS} + V_{GS,2} \end{aligned}$$

Substitute $-0.7\ \text{V}$ for $V_{GS,2}$ and $1\ \text{V}$ for $V_{SS,2}$.

$$\begin{aligned} V_{G,2} &= 1 - 0.7 \\ &= 0.3\ \text{V} \end{aligned}$$

Therefore, the value of the voltage $V_{G,2}$ is, $0.3\ \text{V}$.

Step 5 of 15

Calculate the value of the resistance, R.

$$R = \frac{V_{G,2}}{I_{REF}}$$

Substitute $0.3\ \text{V}$ for $V_{G,2}$ and $20\ \mu\text{A}$ for I_{REF}.

$$\begin{aligned} R &= \frac{0.3}{20\ \mu} \\ &= 15\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistor R, is $\boxed{15\ \text{k}\Omega}$.

Step 6 of 15

Consider the expression for the drain current, I_{REF}.

$$I_{REF} = \frac{1}{2}\left(\mu_p C_{ox}\right)\left(\frac{W_1}{L}\right)\left(V_{GS} - V_{tp}\right)^2$$

Substitute $20\ \mu\text{A}$ for I_{REF}, $100\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$, $-0.7\ \text{V}$ for V_{GS}, $0.5\ \mu\text{m}$ for L and $-0.5\ \text{V}$ for V_{tp}.

$$\begin{aligned} 20\ \mu &= \frac{1}{2}(100\ \mu)\left(\frac{W_1}{0.5\ \mu}\right)\left(-0.7 - (-0.5)\right)^2 \\ 1 &= \frac{1}{2}(5)\left(\frac{W_1}{0.5\ \mu}\right)(-0.7 + 0.5)^2 \\ 2 &= \left(\frac{W_1}{0.1\ \mu}\right)(-0.2)^2 \\ 2 &= \left(\frac{W_1}{0.1\ \mu}\right)(0.04) \end{aligned}$$

Solve for the value of W_1.

$$\begin{aligned} W_1 &= \left(\frac{2}{0.04}\right)(0.1\ \mu) \\ &= 5\ \mu\text{m} \end{aligned}$$

Therefore, the value of the width W_1 is, $\boxed{5\ \mu\text{m}}$.

Step 7 of 15

Consider the expression for the current, I_2.

$$I_2 = \frac{1}{2}\left(\mu_p C_{ox}\right)\left(\frac{W_2}{L}\right)\left(V_{GS} - V_{tp}\right)^2$$

Substitute $100\ \mu\text{A}$ for I_2, $100\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$, $-0.7\ \text{V}$ for V_{GS}, $0.5\ \mu\text{m}$ for L and $-0.5\ \text{V}$ for V_{tp}.

$$\begin{aligned} 100\ \mu &= \frac{1}{2}(100\ \mu)\left(\frac{W_2}{0.5\ \mu}\right)\left(-0.7 - (-0.5)\right)^2 \\ 1 &= \frac{1}{2}\left(\frac{W_2}{0.5\ \mu}\right)(-0.7 + 0.5)^2 \\ 2 &= \left(\frac{W_2}{0.5\ \mu}\right)(-0.2)^2 \\ 2 &= \left(\frac{W_2}{0.5\ \mu}\right)(0.04) \end{aligned}$$

Solve for the value of W_2.

$$\begin{aligned} W_2 &= \left(\frac{2}{0.04}\right)(0.5\ \mu) \\ &= 25\ \mu\text{m} \end{aligned}$$

Therefore, the value of the width W_2 is, $\boxed{25\ \mu\text{m}}$.

Step 8 of 15

Consider the expression for the drain current, I_3.

$$I_3 = \frac{1}{2}\left(\mu_p C_{ox}\right)\left(\frac{W_3}{L}\right)\left(V_{GS} - V_{tp}\right)^2$$

Substitute $40\ \mu\text{A}$ for I_3, $100\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$, $-0.7\ \text{V}$ for V_{GS}, $0.5\ \mu\text{m}$ for L and $-0.5\ \text{V}$ for V_{tp}.

$$\begin{aligned} 40\ \mu &= \frac{1}{2}(100\ \mu)\left(\frac{W_3}{0.5\ \mu}\right)\left(-0.7 - (-0.5)\right)^2 \\ 0.4 &= \frac{1}{2}\left(\frac{W_1}{0.5\ \mu}\right)(-0.7 + 0.5)^2 \\ 0.8 &= \left(\frac{W_1}{0.5\ \mu}\right)(-0.2)^2 \\ 0.8 &= \left(\frac{W_3}{0.5\ \mu}\right)(0.04) \end{aligned}$$

Solve for the value of W_2.

$$\begin{aligned} W_3 &= \left(\frac{0.8}{0.04}\right)(0.5\ \mu) \\ &= 10\ \mu\text{m} \end{aligned}$$

Therefore, the value of the width W_2 is, $\boxed{10\ \mu\text{m}}$.

Step 9 of 15

Consider the expression for the voltage, $V_{DS\,max,5}$.

$$V_{DS\,max,5} = V_3 - V_{SS,5}$$

Substitute $-1\ \text{V}$ for $V_{SS,5}$ and $-0.8\ \text{V}$ for $V_{DS\,max,5}$.

$$\begin{aligned} V_{DS\,max,5} &= -0.8 - (-1) \\ &= -0.8 + 1 \\ &= 0.2\ \text{V} \end{aligned}$$

Therefore, the value of the voltage $V_{DS\,max,5}$ is, $0.2\ \text{V}$.

Step 10 of 15

The value of the voltage $V_{DS\,max,5}$ should be equal to $V_{GS,5} - V_{tn}$. Hence,

$$V_{DS\,max,5} = V_{GS,5} - V_{tn}$$

Substitute $0.2\ \text{V}$ for $V_{DS\,max,5}$ and $0.5\ \text{V}$ for V_{tn}.

$$\begin{aligned} 0.2 &= V_{GS,5} - 0.5 \\ V_{GS,5} &= 0.2 + 0.5 \\ V_{GS,5} &= 0.7\ \text{V} \end{aligned}$$

Therefore, the value of the gate to source voltage $V_{GS,5}$ is, $0.7\ \text{V}$.

Step 11 of 15

Consider the expression for the drain current, I_4.

$$I_4 = \frac{1}{2}\left(\mu_n C_{ox}\right)\left(\frac{W_4}{L}\right)\left(V_{GS} - V_{tn}\right)^2$$

Substitute $40\ \mu\text{A}$ for I_4, $400\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$, $0.7\ \text{V}$ for V_{GS}, $0.5\ \mu\text{m}$ for L and $0.5\ \text{V}$ for V_{tn}.

$$\begin{aligned} 40\ \mu &= \frac{1}{2}(400\ \mu)\left(\frac{W_4}{0.5\ \mu}\right)(0.7 - 0.5)^2 \\ 1 &= \frac{1}{2}(10)\left(\frac{W_4}{0.5\ \mu}\right)(0.2)^2 \\ 0.2 &= \left(\frac{W_4}{0.5\ \mu}\right)(0.2)^2 \\ 0.2 &= \left(\frac{W_4}{0.5\ \mu}\right)(0.04) \end{aligned}$$

Step 12 of 15

Solve for the value of W_4.

$$\begin{aligned} W_4 &= \left(\frac{0.2}{0.04}\right)(0.5\ \mu) \\ &= 6.25\ \mu\text{m} \end{aligned}$$

Therefore, the value of the width W_4 is, $\boxed{6.25\ \mu\text{m}}$.

Step 13 of 15

Consider the expression for the drain current, I_5.

$$I_5 = \frac{1}{2}\left(\mu_n C_{ox}\right)\left(\frac{W_5}{L}\right)\left(V_{GS} - V_{tn}\right)^2$$

Substitute $80\ \mu\text{A}$ for I_4, $400\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$, $0.7\ \text{V}$ for V_{GS}, $0.5\ \mu\text{m}$ for L and $0.5\ \text{V}$ for V_{tn}.

$$\begin{aligned} 80\ \mu &= \frac{1}{2}(400\ \mu)\left(\frac{W_5}{0.5\ \mu}\right)(0.7 - 0.5)^2 \\ 1 &= \frac{1}{2}(5)\left(\frac{W_5}{0.5\ \mu}\right)(0.2)^2 \\ 2 &= \left(\frac{W_5}{0.1\ \mu}\right)(0.2)^2 \\ 2 &= \left(\frac{W_5}{0.1\ \mu}\right)(0.04) \end{aligned}$$

Solve for the value of W_5.

$$\begin{aligned} W_5 &= \left(\frac{2}{0.04}\right)(0.1\ \mu) \\ &= 5\ \mu\text{m} \end{aligned}$$

Therefore, the value of the width W_5 is, $\boxed{5\ \mu\text{m}}$.

Step 14 of 15

Calculate the value of the output resistance r_{o2} of the current source Q_2.

$$r_{o2} = \frac{V'_{Ap} L}{I_2}$$

Substitute $5\ \text{V/}\mu\text{m}$ for V'_{Ap}, $0.5\ \mu\text{m}$ for L and $100\ \mu\text{A}$ for I_2.

$$\begin{aligned} r_{o2} &= \frac{(5/\mu)(0.5\ \mu)}{100\ \mu} \\ &= 25\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the output resistance r_{o2} of the current source Q_2 is, $\boxed{25\ \text{k}\Omega}$.

Step 15 of 15

Calculate the value of the output resistance r_{o5} of the current sink Q_5.

$$r_{o5} = \frac{V'_{An} L}{I_5}$$

Substitute $5\ \text{V/}\mu\text{m}$ for V'_{An}, $0.5\ \mu\text{m}$ for L and $80\ \mu\text{A}$ for I_5.

$$\begin{aligned} r_{o5} &= \frac{(5/\mu)(0.5\ \mu)}{80\ \mu} \\ &= 31.25\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the output resistance r_{o5} of the current sink Q_5 is, $\boxed{31.25\ \text{k}\Omega}$.

Step 1 of 5

Write the formula for transconductance, g_m.

$$g_m = \sqrt{2\mu_n C_{ox}\left(\frac{W}{L}\right)I_D}$$

Here,

μ_n is the electron mobility

C_{ox} is the oxide capacitance

$\frac{W}{L}$ is the width-to-length ratio

I_D is the drain current

Substitute $387\times10^{-6}\ \text{A/V}^2$ for $\mu_n C_{ox}$, 10 for $\frac{W}{L}$ and $10\times10^{-6}\ \text{A}$ I_D

$$\begin{aligned} g_m &= \sqrt{2\left(387\times10^{-6}\right)(10)\left(10\times10^{-6}\right)} \\ &= \sqrt{77400\times10^{-12}} \\ &= 278.2085\times10^{-6}\ \text{A/V} \\ &\approx 0.28\ \text{mA/V} \end{aligned}$$

Thus, the transconductance g_m is $\boxed{0.28\ \text{mA/V}}$.

Step 2 of 5

Write the formula for intrinsic gain A_0.

$$A_0 = \frac{V'_A\sqrt{2\left(\mu_n C_{ox}\right)\left(\frac{W}{L}\right)L^2}}{\sqrt{I_D}}$$

Substitute $5\times10^6\ \text{V/m}$ for V'_A, $387\times10^{-6}\ \text{A/V}^2$ for $\mu_n C_{ox}$, 10 for $\frac{W}{L}$, $0.36\times10^{-6}\ \text{m}$ for L and $10\times10^{-6}\ \text{A}$ for I_D

$$\begin{aligned} A_0 &= \frac{\left(5\times10^6\right)\sqrt{2\left(387\times10^{-6}\right)(10)\left(0.36\times10^{-6}\right)^2}}{\sqrt{10\times10^{-6}}} \\ &= \frac{\left(5\times10^6\right)\sqrt{1003.104\times10^{-18}}}{3.16227\times10^{-3}} \\ &= \frac{\left(5\times10^6\right)\left(31.6718\times10^{-9}\right)}{3.16227\times10^{-3}} \\ &= \frac{0.15835}{3.16227\times10^{-3}} \\ &= 50\ \text{V/V} \end{aligned}$$

Thus, the intrinsic gain A_0 of the CS amplifier is $\boxed{50\ \text{V/V}}$.

Step 3 of 5

Determine the transconductance, g_m at $I_D = 100\ \mu\text{A}$.

Substitute $387\times10^{-6}\ \text{A/V}^2$ for $\mu_n C_{ox}$, 10 for $\frac{W}{L}$ and $100\times10^{-6}\ \text{A}$ I_D

$$\begin{aligned} g_m &= \sqrt{2\left(387\times10^{-6}\right)(10)\left(100\times10^{-6}\right)} \\ &= \sqrt{774000\times10^{-12}} \\ &= 879.77\times10^{-6}\ \text{A/V} \\ &\approx 0.88\ \text{mA/V} \end{aligned}$$

Thus, the transconductance g_m is $\boxed{0.88\ \text{mA/V}}$.

Determine the intrinsic gain A_0 at $I_D = 100\ \mu\text{A}$.

Substitute $5\times10^6\ \text{V/m}$ for V'_A, $387\times10^{-6}\ \text{A/V}^2$ for $\mu_n C_{ox}$, 10 for $\frac{W}{L}$, $0.36\times10^{-6}\ \text{m}$ for L and $100\times10^{-6}\ \text{A}$ for I_D

$$\begin{aligned} A_0 &= \frac{\left(5\times10^6\right)\sqrt{2\left(387\times10^{-6}\right)(10)\left(0.36\times10^{-6}\right)^2}}{\sqrt{100\times10^{-6}}} \\ &= \frac{\left(5\times10^6\right)\sqrt{1003.104\times10^{-18}}}{10\times10^{-3}} \\ &= \frac{\left(5\times10^6\right)\left(31.6718\times10^{-9}\right)}{10\times10^{-3}} \\ &= \frac{0.15835}{10\times10^{-3}} \\ &= 15.8\ \text{V/V} \end{aligned}$$

Thus, the intrinsic gain A_0 of the CS amplifier is $\boxed{15.8\ \text{V/V}}$.

Step 4 of 5

Determine the transconductance, g_m at $I_D = 1\ \text{mA}$.

Substitute $387\times10^{-6}\ \text{A/V}^2$ for $\mu_n C_{ox}$, 10 for $\frac{W}{L}$ and $10^{-3}\ \text{A}$ I_D

$$\begin{aligned} g_m &= \sqrt{2\left(387\times10^{-6}\right)(10)\left(10^{-3}\right)} \\ &= \sqrt{774\times10^{-8}} \\ &= 27.8\times10^{-4}\ \text{A/V} \\ &= 2.78\ \text{mA/V} \end{aligned}$$

Thus, the transconductance g_m is $\boxed{2.78\ \text{mA/V}}$.

Step 5 of 5

Determine the intrinsic gain A_0 at $I_D = 1\ \text{mA}$.

Substitute $5\times10^6\ \text{V/m}$ for V'_A, $387\times10^{-6}\ \text{A/V}^2$ for $\mu_n C_{ox}$, 10 for $\frac{W}{L}$, $0.36\times10^{-6}\ \text{m}$ for L and $10^{-3}\ \text{A}$ for I_D

$$\begin{aligned} A_0 &= \frac{\left(5\times10^6\right)\sqrt{2\left(387\times10^{-6}\right)(10)\left(0.36\times10^{-6}\right)^2}}{\sqrt{10^{-3}}} \\ &= \frac{\left(5\times10^6\right)\sqrt{1003.104\times10^{-18}}}{\sqrt{10^{-3}}} \\ &= \left(5\times10^6\right)\left(10.015\times10^{-7}\right) \\ &= 5\ \text{V/V} \end{aligned}$$

Thus, the intrinsic gain A_0 of the CS amplifier is $\boxed{5\ \text{V/V}}$.

Step 1 of 15

The PMOS current mirror which consists of three PMOS transistors is shown in Figure 1.

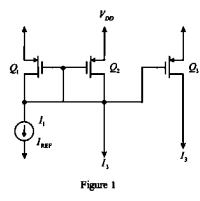


Figure 1

Step 2 of 15

Case1:

If effective channel width of diode connected transistor is $W_1 = 10\ \mu m$:

Output current relation of a current mirror network is,

$$\frac{I_2}{I_1} = \frac{\left(\frac{W_2}{L_2}\right)}{\left(\frac{W_1}{L_1}\right)}$$

For output current relationship for first and second transistors is,

$$\frac{I_1}{I_1} = \frac{\left(\frac{W_2}{L_2}\right)}{\left(\frac{W_1}{L_1}\right)}$$

$$\frac{I_1}{I_{REF}} = \frac{\left(\frac{W_2}{L}\right)}{\left(\frac{W_1}{L}\right)} \qquad \begin{pmatrix}\text{since, } L_2 = L_1 = L \text{ and} \\ I_1 = I_{REF}\end{pmatrix}$$

$$\frac{I_1}{I_{REF}} = \frac{W_2}{W_1}$$

$$\frac{I_1}{100\times10^{-6}} = \frac{20\times10^{-6}}{10\times10^{-6}}$$

$$I_1 = \left(100\times10^{-6}\right)\left(\frac{20\times10^{-6}}{10\times10^{-6}}\right)$$

$$= 200\times10^{-6}$$

$$= 200\ \mu A$$

Thus, the output current of second PMOS transister is $\boxed{200\ \mu A}$.

Step 3 of 15

The output current relationship for first and third transistors is,

$$\frac{I_3}{I_1} = \frac{\left(\frac{W_3}{L_3}\right)}{\left(\frac{W_1}{L_1}\right)}$$

$$\frac{I_3}{I_{REF}} = \frac{\left(\frac{W_3}{L}\right)}{\left(\frac{W_1}{L}\right)} \qquad (\text{since, } L_3 = L_1 = L)$$

$$\frac{I_3}{I_{REF}} = \frac{W_3}{W_1}$$

$$\frac{I_3}{100\times10^{-6}} = \frac{40\times10^{-6}}{10\times10^{-6}}$$

$$I_3 = \left(100\times10^{-6}\right)\left(\frac{40\times10^{-6}}{10\times10^{-6}}\right)$$

$$= 400\times10^{-6}$$

$$= 400\ \mu A$$

Thus, the output current of third PMOS transister is $\boxed{400\ \mu A}$.

Step 4 of 15

Case2:

If effective channel width of diode connected transistor is $W_1 = 20\ \mu m$:

The output current relation of a current mirror network is,

$$\frac{I_2}{I_1} = \frac{\left(\frac{W_2}{L_2}\right)}{\left(\frac{W_1}{L_1}\right)}$$

The output current relationship for first and second transistors is,

$$\frac{I_2}{I_1} = \frac{\left(\frac{W_2}{L_2}\right)}{\left(\frac{W_1}{L_1}\right)}$$

$$\frac{I_1}{I_{REF}} = \frac{\left(\frac{W_2}{L}\right)}{\left(\frac{W_1}{L}\right)} \qquad (\text{since, } L_2 = L_1 = L)$$

$$\frac{I_1}{I_{REF}} = \frac{W_2}{W_1}$$

$$\frac{I_2}{100\times10^{-6}} = \frac{10\times10^{-6}}{20\times10^{-6}} \qquad (\text{choose, } W_2 = 10\ \mu m)$$

$$I_2 = \left(100\times10^{-6}\right)\left(\frac{10\times10^{-6}}{20\times10^{-6}}\right)$$

$$= 50\times10^{-6}$$

$$= 50\ \mu A$$

Thus, the output current of second PMOS transistor is $\boxed{50\ \mu A}$

Step 5 of 15

The output current relationship for first and third transistors is,

$$\frac{I_3}{I_1} = \frac{\left(\frac{W_3}{L_3}\right)}{\left(\frac{W_1}{L_1}\right)}$$

$$\frac{I_3}{I_{REF}} = \frac{\left(\frac{W_3}{L}\right)}{\left(\frac{W_1}{L}\right)} \qquad (\text{since, } L_3 = L_1 = L)$$

$$\frac{I_3}{I_{REF}} = \frac{W_3}{W_1}$$

$$\frac{I_3}{100\times10^{-6}} = \frac{20\times10^{-6}}{20\times10^{-6}} \qquad (\text{choose, } W_3 = 20\ \mu m)$$

$$I_3 = \left(100\times10^{-6}\right)\left(\frac{20\times10^{-6}}{20\times10^{-6}}\right)$$

$$= 50\times10^{-6}$$

$$= 50\ \mu A$$

Thus, the output current of third PMOS transister is $\boxed{50\ \mu A}$.

So, for the three possible cases, four different output currents are obtained. Thus, the output currents are $\boxed{200\ \mu A, 400\ \mu A, 25\ \mu A, \text{and } 50\ \mu A}$.

Step 6 of 15

If two transistors are diode connected, then acts as equivalent transister whose width is equivalent to sum of individual widths. The other third transistor acts as current mirror network.

Case 1:

Effective channel width for two diode connected transistors bearing widths $10\ \mu m$ and $20\ \mu m$ is,

$$W_{eff} = 10\ \mu m + 20\ \mu m$$

$$= 30\ \mu m$$

The channel width of other transister is, $W_3 = 40\ \mu m$

The output current is,

$$\frac{I_3}{I_1} = \frac{\left(\frac{W_3}{L_3}\right)}{\left(\frac{W_{eff}}{L_1}\right)}$$

$$\frac{I_3}{I_{REF}} = \frac{\left(\frac{W_3}{L}\right)}{\left(\frac{W_{eff}}{L}\right)} \qquad (\text{since, } L_3 = L_1 = L)$$

$$\frac{I_3}{I_{REF}} = \frac{W_3}{W_{eff}}$$

$$\frac{I_3}{100\times10^{-6}} = \frac{40\times10^{-6}}{30\times10^{-6}}$$

$$I_3 = \left(100\times10^{-6}\right)\left(\frac{40\times10^{-6}}{30\times10^{-6}}\right)$$

$$= 133.33\times10^{-6}$$

$$= 133.33\ \mu A$$

Thus, the output current of third PMOS transistor is $\boxed{133.33\ \mu A}$.

Step 7 of 15

Case 2:

Effective channel width for two diode connected transistors bearing widths $10\ \mu m$ and $40\ \mu m$ is,

$$W_{eff} = 10\ \mu m + 40\ \mu m$$

$$= 50\ \mu m$$

The channel width of other transistor is $W_3 = 20\ \mu m$

The output current is,

$$\frac{I_3}{I_1} = \frac{\left(\frac{W_3}{L_3}\right)}{\left(\frac{W_{eff}}{L_1}\right)}$$

$$\frac{I_3}{I_{REF}} = \frac{\left(\frac{W_3}{L}\right)}{\left(\frac{W_{eff}}{L}\right)} \qquad (\text{since, } L_3 = L_1 = L)$$

$$\frac{I_3}{I_{REF}} = \frac{W_3}{W_{eff}}$$

$$\frac{I_3}{100\times10^{-6}} = \frac{20\times10^{-6}}{50\times10^{-6}}$$

Step 8 of 15

Simpally further.

$$I_3 = \left(100\times10^{-6}\right)\left(\frac{20\times10^{-6}}{50\times10^{-6}}\right)$$

$$= 40\times10^{-6}$$

$$= 40\ \mu A$$

Thus, the output current of third PMOS transistor is $\boxed{40\ \mu A}$.

Step 9 of 15

Case 3:

Effective channel width for two diode connected transistors bearing widths $20\ \mu m$ and $40\ \mu m$ is,

$$W_{eff} = 20\ \mu m + 40\ \mu m$$

$$= 60\ \mu m$$

The channel width of other transistor is $W_3 = 10\ \mu m$

The output current is,

$$\frac{I_3}{I_1} = \frac{\left(\frac{W_3}{L_3}\right)}{\left(\frac{W_{eff}}{L_1}\right)}$$

$$\frac{I_3}{I_{REF}} = \frac{\left(\frac{W_3}{L}\right)}{\left(\frac{W_{eff}}{L}\right)} \qquad (\text{since, } L_3 = L_1 = L)$$

$$\frac{I_3}{I_{REF}} = \frac{W_3}{W_{eff}}$$

$$\frac{I_3}{100\times10^{-6}} = \frac{10\times10^{-6}}{60\times10^{-6}}$$

$$I_3 = \left(100\times10^{-6}\right)\left(\frac{10\times10^{-6}}{60\times10^{-6}}\right)$$

$$= 16.67\times10^{-6}$$

$$= 16.67\ \mu A$$

Thus, the output current of third PMOS transister is $\boxed{16.67\ \mu A}$.

So, for the above three possible cases, three different output current are obtained. Thus, the output currents are $\boxed{40\ \mu A, 133.33\ \mu A, \text{and } 16.67\ \mu A}$.

Step 10 of 15

The drain current of a transister is,

$$I_D = \frac{1}{2}K_p'\left(\frac{W}{L}\right)\left(V_{SG} - V_t\right)^2$$

For $W_{eff} = 30\ \mu m$:

$$I_{REF} = \frac{1}{2}K_p'\left(\frac{W_{eff}}{L}\right)\left(V_{SG} - V_t\right)^2$$

Substitute $100\ \mu A$ for I_{REF} , $100\ \mu A/V^2$ for K_p' , $30\ \mu m$ for W_{eff} , $1\ \mu m$ for L , and $0.6\ V$ for V_t in the equation.

$$100\times10^{-6} = \frac{1}{2}\left(100\times10^{-6}\right)\left(\frac{30\times10^{-6}}{1\times10^{-6}}\right)\left(V_{SG} - 0.6\right)^2$$

$$100 = (50)(30)\left(V_{SG} - 0.6\right)^2$$

$$\left(V_{SG} - 0.6\right)^2 = \frac{100}{1500}$$

$$\left(V_{SG} - 0.6\right)^2 = 0.0667$$

$$V_{SG} - 0.6 = \sqrt{0.0667}$$

$$V_{SG} - 0.6 = 0.25826$$

$$V_{SG} = 0.6 + 0.25826$$

$$V_{SG} = 0.85826\ V$$

Thus, the source to gate voltage $\left(V_{SG}\right)$ is, $\boxed{V_{SG} = 0.85826\ V}$.

Step 11 of 15

For $W_{eff} = 50\ \mu m$:

$$I_{REF} = \frac{1}{2}K_p'\left(\frac{W_{eff}}{L}\right)\left(V_{SG} - V_t\right)^2$$

Substitute $100\ \mu A$ for I_{REF} , $100\ \mu A/V^2$ for K_p' , $50\ \mu m$ for W_{eff} , $1\ \mu m$ for L , and $0.6\ V$ for V_t in the equation.

$$100\times10^{-6} = \frac{1}{2}\left(100\times10^{-6}\right)\left(\frac{50\times10^{-6}}{1\times10^{-6}}\right)\left(V_{SG} - 0.6\right)^2$$

$$100 = (50)(50)\left(V_{SG} - 0.6\right)^2$$

$$\left(V_{SG} - 0.6\right)^2 = \frac{100}{2500}$$

$$\left(V_{SG} - 0.6\right)^2 = 0.04$$

Step 12 of 15

Simplify further.

$$V_{SG} - 0.6 = \sqrt{0.04}$$

$$V_{SG} - 0.6 = 0.2$$

$$V_{SG} = 0.6 + 0.2$$

$$V_{SG} = 0.8\ V$$

Thus, the source to gate voltage $\left(V_{SG}\right)$ is, $\boxed{V_{SG} = 0.8\ V}$

Step 13 of 15

For $W_{eff} = 60\ \mu m$:

$$I_{REF} = \frac{1}{2}K_p'\left(\frac{W_{eff}}{L}\right)\left(V_{SG} - V_t\right)^2$$

Substitute the parameters

$$100\times10^{-6} = \frac{1}{2}\left(100\times10^{-6}\right)\left(\frac{60\times10^{-6}}{1\times10^{-6}}\right)\left(V_{SG} - 0.6\right)^2$$

$$100 = (50)(60)\left(V_{SG} - 0.6\right)^2$$

$$\left(V_{SG} - 0.6\right)^2 = \frac{100}{3000}$$

$$\left(V_{SG} - 0.6\right)^2 = 0.0333$$

$$V_{SG} - 0.6 = \sqrt{0.0333}$$

$$V_{SG} - 0.6 = 0.18248$$

$$V_{SG} = 0.6 + 0.18248$$

$$V_{SG} = 0.78248\ V$$

Thus, the source to gate voltage $\left(V_{SG}\right)$ is, $\boxed{V_{SG} = 0.78248\ V}$

Step 1 of 4

Refer to Figure 7.4 (a) in the text book for the common-source amplifier circuit.

Write the formula for transconductance g_{m1} of transistor Q_1.

$$g_{m1}=\sqrt{2k_n'\left(\frac{W}{L}\right)_1 I_{REF}}$$

Substitute $387\times10^{-6}\ \text{A/V}^2$ for k_n', $\frac{7.2\mu\text{m}}{0.36\mu\text{m}}$ for $\frac{W}{L}$ and $100\times10^{-6}\ \text{A}$ for I_{REF}.

$$\begin{aligned}g_{m1}&=\sqrt{2\left(387\times10^{-6}\right)\left(\frac{7.2\times10^{-6}}{0.36\times10^{-6}}\right)\left(100\times10^{-6}\right)}\\&=\sqrt{2\left(387\times10^{-6}\right)(20)\left(100\times10^{-6}\right)}\\&=\sqrt{1.548\times10^{-6}}\\&=1.25\times10^{-3}\ \text{A/V}\\&=1.25\ \text{mA/V}\end{aligned}$$

Therefore, the transconductance g_{m1} of Q_1 is $\boxed{1.25\,\text{mA/V}}$.

Step 2 of 4

The resistance, r_{o1} is,

$$r_{o1}=\frac{V_{An}'L}{I_{D1}}$$
$$r_{o1}=\frac{V_{An}'L}{I_{REF}}$$

Substitute $5\times10^6\ \text{V/m}$ for V_{An}', $0.36\mu\text{m}$ for L and $100\mu\text{A}$ for I_{REF}

$$\begin{aligned}r_{o1}&=\frac{\left(5\times10^6\right)(0.36)}{100}\\&=18\times10^3\ \Omega\\&=18\ \text{k}\Omega\end{aligned}$$

Thus, the resistance r_{o1} is $\boxed{18\,\text{k}\Omega}$.

Step 3 of 4

The resistance, r_{o2} is,

$$r_{o2}=\frac{V_{Ap}'L}{I_{D1}}$$
$$r_{o2}=\frac{V_{Ap}'L}{I_{REF}}$$

Substitute $6\times10^6\ \text{V/m}$ for V_{An}', $0.36\mu\text{m}$ for L and $100\mu\text{A}$ for I_{REF}

$$\begin{aligned}r_{o2}&=\frac{\left(6\times10^6\right)(0.36)}{100}\\&=21.6\times10^3\ \Omega\\&=21.6\ \text{k}\Omega\end{aligned}$$

Thus, the resistance r_{o2} is $\boxed{21.6\,\text{k}\Omega}$.

Step 4 of 4

The voltage gain is,

$$A_v=-g_{m1}\left(r_{o1}\|r_{o2}\right)$$

Substitute $1.25\times10^{-3}\ \text{A/V}$ for g_{m1}, $18\times10^3\ \Omega$ for r_{o1} and $21.6\times10^3\ \Omega$ for r_{o2}

$$\begin{aligned}A_v&=-\left(1.25\times10^{-3}\right)\left[\left(18\times10^3\right)\|\left(21.6\times10^3\right)\right]\\&=-\left(1.25\times10^{-3}\right)\left[\frac{\left(18\times10^3\right)\left(21.6\times10^3\right)}{\left(18\times10^3\right)+\left(21.6\times10^3\right)}\right]\\&=-\left(1.25\times10^{-3}\right)\left(9.8182\times10^3\right)\\&=-12.3\ \text{V/V}\end{aligned}$$

Thus, the voltage gain A_v is $\boxed{-12.3\,\text{V/V}}$.

Step 1 of 13

Refer to Figure 8.7 from the text book.

(a)

Consider the expression for the current I_{REF} in terms of the current I_S.

$$I_{REF} = I_S\, e^{V_{BE}/V_T}$$

Apply natural logarithm.

$$\ln(I_{REF}) = \ln\left(I_S\, e^{V_{BE}/V_T}\right)$$
$$\ln(I_{REF}) = \ln(I_S) + \ln\left(e^{V_{BE}/V_T}\right)$$
$$\ln(I_{REF}) - \ln(I_S) = +\frac{V_{BE}}{V_T}\ln(e)$$
$$\ln\left(\frac{I_{REF}}{I_S}\right) = \frac{V_{BE}}{V_T}$$

Solve for the value of V_{BE}.

$$V_{BE} = V_T \ln\left(\frac{I_{REF}}{I_S}\right)$$

Substitute $25\ \text{mV}$ for V_T and $10^{-17}\ \text{A}$ for I_S.

$$V_{BE} = (25\text{m})\ln\left(\frac{I_{REF}}{10^{-17}}\right) \quad \text{..... (1)}$$

Therefore, the expression for the value of V_{BE} is, $(25\text{m})\ln\left(\frac{I_{REF}}{10^{-17}}\right)$.

Step 2 of 13

Recall equation (1).

$$V_{BE} = (25\text{m})\ln\left(\frac{I_{REF}}{10^{-17}}\right)$$

Substitute $10\ \mu\text{A}$ for I_{REF}.

$$V_{BE} = (25\ \text{m})\ln\left(\frac{10\ \mu}{10^{-17}}\right)$$
$$= (25\ \text{m})(27.63)$$
$$= 0.691\ \text{V}$$

Therefore, the value of V_{BE} for the value of the current $I_{REF} = 10\ \mu\text{A}$ is $0.691\ \text{V}$.

Step 3 of 13

Recall equation (1).

$$V_{BE} = (25\text{m})\ln\left(\frac{I_{REF}}{10^{-17}}\right)$$

Substitute $10\ \text{mA}$ for I_{REF}.

$$V_{BE} = (25\ \text{m})\ln\left(\frac{10\ \text{m}}{10^{-17}}\right)$$
$$= (25\ \text{m})(34.54)$$
$$= 0.864\ \text{V}$$

Therefore, the value of V_{BE} for the value of the current $I_{REF} = 10\ \text{mA}$ is $0.864\ \text{V}$.

Step 4 of 13

Write the range of voltage V_{BE}.

The range of the voltage V_{BE} is, $\boxed{0.693\ \text{V} \le V_{BE} \le 0.864\ \text{V}}$ for the range of the current I_{REF}, $10\ \mu\text{A} \le I_{REF} \le 10\ \text{mA}$.

Step 5 of 13

Consider the expression for the output current, I_O.

$$I_O = \frac{I_{REF}}{1 + \frac{2}{\beta}} \quad \text{..... (1)}$$

Substitute ∞ for β, since the value of β is very high.

$$I_O = \frac{I_{REF}}{1 + \frac{2}{\infty}}$$
$$= \frac{I_{REF}}{1 + 0}$$
$$= I_{REF}$$

The range of the output current is, $\boxed{10\ \mu\text{A} \le I_O \le 10\ \text{mA}}$ since, $I_O = I_{REF}$.

Step 6 of 13

(b)
Recall equation (1).

$$I_O = \frac{I_{REF}}{1 + \frac{2}{\beta}}$$

Substitute 100 for β.

$$I_O = \frac{I_{REF}}{1 + \frac{2}{100}}$$
$$= \frac{I_{REF}}{1 + 0.02}$$

$$I_O = \frac{I_{REF}}{1.02} \quad \text{..... (2)}$$

Therefore, the expression for the current I_O is $\frac{I_{REF}}{1.02}$.

Step 7 of 13

Recall equation (2).

$$I_O = \frac{I_{REF}}{1.02}$$

Substitute $10\ \mu\text{A}$ for I_{REF}.

$$I_O = \frac{10\ \mu}{1.02}$$
$$= 9.8\ \mu\text{A}$$

Therefore, the value of the output current for the value of current $I_{REF} = 10\ \mu\text{A}$ is, $9.8\ \mu\text{A}$.

Step 8 of 13

Recall equation (2).

$$I_O = \frac{I_{REF}}{1.02}$$

Substitute $10\ \text{mA}$ for I_{REF}.

$$I_O = \frac{10\ \text{m}}{1.02}$$
$$= 9.8\ \text{mA}$$

Therefore, the value of the output current for the value of current $I_{REF} = 10\ \text{mA}$ is, $9.8\ \text{mA}$.

Step 9 of 13

Write the range of output current I_O.

The range of the output current is, $\boxed{9.8\ \mu\text{A} \le I_O \le 9.8\ \text{mA}}$ for the range of the reference current, $10\ \mu\text{A} \le I_{REF} \le 10\ \text{mA}$.

Step 10 of 13

Recall equation (1).

$$I_O = \frac{I_{REF}}{1 + \frac{2}{\beta}}$$

Substitute 50 for β and $10\ \mu\text{A}$ for I_{REF}.

$$I_O = \frac{10\ \mu}{1 + \frac{2}{50}}$$
$$= \frac{10\ \mu}{1 + 0.04}$$
$$= \frac{10\ \mu}{1.04}$$
$$= 9.62\ \mu\text{A}$$

The value of the output current is, $\boxed{9.62\ \mu\text{A}}$.

Step 11 of 13

Recall equation (1).

$$I_O = \frac{I_{REF}}{1 + \frac{2}{\beta}}$$

Substitute 100 for β and $0.1\ \text{mA}$ for I_{REF}.

$$I_O = \frac{0.1\ \text{m}}{1 + \frac{2}{100}}$$
$$= \frac{0.1\ \text{m}}{1 + 0.02}$$
$$= \frac{0.1\ \text{m}}{1.02}$$
$$= 98.04\ \mu\text{A}$$

The value of the output current is, $\boxed{98.04\ \mu\text{A}}$.

Step 12 of 13

Recall equation (1).

$$I_O = \frac{I_{REF}}{1 + \frac{2}{\beta}}$$

Substitute 100 for β and $1\ \text{mA}$ for I_{REF}.

$$I_O = \frac{1\ \text{m}}{1 + \frac{2}{100}}$$
$$= \frac{1\ \text{m}}{1 + 0.02}$$
$$= \frac{1\ \text{m}}{1.02}$$
$$= 0.98\ \text{mA}$$

The value of the output current is, $\boxed{0.98\ \text{mA}}$.

Step 13 of 13

Recall equation (1).

$$I_O = \frac{I_{REF}}{1 + \frac{2}{\beta}}$$

Substitute 50 for β and $10\ \text{mA}$ for I_{REF}.

$$I_O = \frac{10\ \text{m}}{1 + \frac{2}{50}}$$
$$= \frac{10\ \text{m}}{1 + 0.04}$$
$$= \frac{10\ \text{m}}{1.04}$$
$$= 9.62\ \text{mA}$$

The value of the output current is, $\boxed{9.62\ \text{mA}}$.

Step 1 of 5

Determine the resistance r_π.

$$r_\pi = \frac{\beta}{g_m}$$
$$= \frac{100}{4\times 10^{-3}}$$
$$= 25\ \text{k}\Omega$$

The input resistance of the CE amplifier is,

$$R_{in} = r_\pi$$

Therefore, the input resistance R_{in} is $\boxed{25\ \text{k}\Omega}$.

Step 2 of 5

Determine the value of r_o.

$$r_o = \frac{|V_A|}{I_C}$$
$$= \frac{50}{0.1\times 10^{-3}}$$
$$= 0.5\ \text{M}\Omega$$

Thus, the value of r_o is $\boxed{0.5\ \text{M}\Omega}$.

Step 3 of 5

Determine the value of g_m.

$$g_m = \frac{I_C}{V_T}$$
$$= \frac{0.1\times 10^{-3}}{25\times 10^{-3}}$$
$$= 4\ \text{mA/V}$$

Thus, the value of g_m is $\boxed{4\ \text{mA/V}}$.

Step 4 of 5

Determine the value of A_o.

$$A_o = g_m r_o$$
$$= 4\times 10^{-3}\times 0.5\times 10^{6}$$
$$= 2000\ \text{V/V}$$

Thus, the value of A_o is $\boxed{2000\ \text{V/V}}$.

Step 5 of 5

Determine the amplifier voltage gain.

$$A_v = -\frac{1}{2} g_m r_o$$
$$= \frac{4\times 10^{-3}\times 0.5\times 10^{6}}{2}$$
$$= -1000\ \text{V/V}$$

Therefore, amplifier voltage gain A_v is $\boxed{-1000\ \text{V/V}}$.

Step 1 of 5

8.009P

Refer to Figure 8.7 from the text book.

Consider the circuit diagram shown in Figure 1.

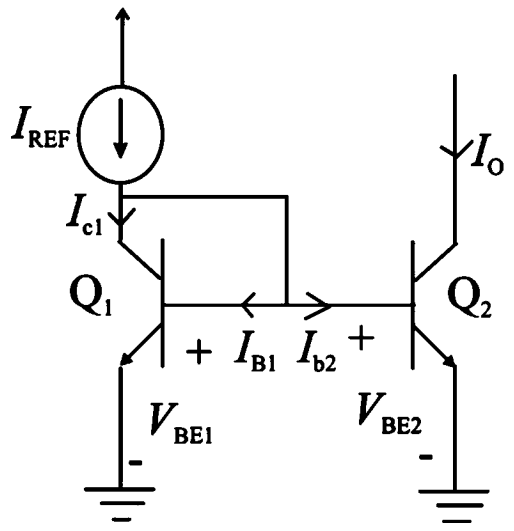


Figure 1

Step 2 of 5

Consider the relation for the base to emitter voltages for both the transistors.

$V_{BE1} = V_{BE2}$

Substitute $V_T \ln\left(\frac{I_{C1}}{I_{S1}}\right)$ for V_{BE1} and $V_T \ln\left(\frac{I_O}{I_{S2}}\right)$ for V_{BE2}.

$$V_T \ln\left(\frac{I_{C1}}{I_{S1}}\right) = V_T \ln\left(\frac{I_O}{I_{S2}}\right)$$

$$\frac{I_{C1}}{I_{S1}} = \frac{I_O}{I_{S2}}$$

$$\frac{I_O}{I_{C1}} = \frac{I_{S2}}{I_{S1}}$$

But $\frac{I_O}{I_{C1}} = \frac{I_{S2}}{I_{S1}}$ is equal to m. Hence,

$$I_{C1} = \frac{I_O}{m}$$

Therefore, the value of the current I_{C1} is $\frac{I_O}{m}$.

Step 3 of 5

Consider the expression for the reference current.

$I_{REF} = I_{C1} + I_{B1} + I_{B2}$

Substitute $\frac{I_{C1}}{\beta}$ for I_{B1} and $\frac{I_O}{\beta}$ for I_{B2}.

$$I_{REF} = I_{C1} + \frac{I_{C1}}{\beta} + \frac{I_O}{\beta}$$

$$= I_{C1}\left(1 + \frac{1}{\beta}\right) + \frac{I_O}{\beta}$$

Substitute $\frac{I_O}{m}$ for I_{C1}.

$$I_{REF} = \left(\frac{I_O}{m}\right)\left(1 + \frac{1}{\beta}\right) + \frac{I_O}{\beta}$$

$$= \left(\frac{I_O}{m}\right)\left(1 + \frac{1}{\beta} + \frac{m}{\beta}\right)$$

$$= \left(\frac{I_O}{m}\right)\left(1 + \frac{1+m}{\beta}\right)$$

$$\frac{I_O}{I_{REF}} = \frac{m}{1 + \frac{1+m}{\beta}} \quad \text{..... (1)}$$

Therefore, the expression for the current transfer ratio is $\boxed{\frac{m}{1 + \frac{1+m}{\beta}}}$.

Step 4 of 5

Recall equation (1).

$$\frac{I_O}{I_{REF}} = \frac{m}{1 + \frac{1+m}{\beta}}$$

Substitute ∞ for β.

$$\frac{I_O}{I_{REF}} = \frac{m}{1 + \frac{1+m}{\infty}}$$

$$= \frac{m}{1+0}$$

$$= m$$

$I_O = I_{REF} m$ .....(2)

Step 5 of 5

Recall equation (1).

$$\frac{I_O}{I_{REF}} = \frac{m}{1 + \frac{1+m}{\beta}}$$

Here, I_O is 10% lower to the value of equation (2).

$$\frac{(100\% - 10\%) I_{REF} m}{I_{REF}} = \frac{m}{1 + \frac{1+m}{\beta}}$$

Substitute 80 for β.

$$0.9 = \frac{1}{1 + \frac{1+m}{80}}$$

$$1 + \frac{1+m}{80} = \frac{1}{0.9}$$

$$1 + \frac{1+m}{80} = \frac{10}{9}$$

$$\frac{1+m}{80} = \frac{10}{9} - 1$$

$$\frac{1+m}{80} = \frac{10-9}{9}$$

$$\frac{1+m}{80} = \frac{1}{9}$$

$$1 + m = \frac{80}{9}$$

$$m = \frac{80}{9} - 1$$

$$m = \frac{80-9}{9}$$

$$= \frac{71}{9}$$

$$= 7.89$$

Therefore, the value of the current transfer ratio m is $\boxed{7.89}$.

Step 1 of 3

8.010E

Refer to Figure 8.18(a) from the text book.

Consider the small signal equivalent circuit diagram shown in Figure 1.

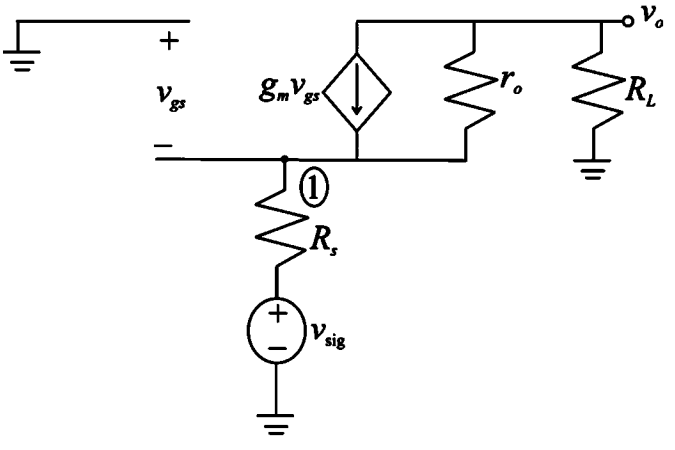


Figure 1

Step 2 of 3

Consider the nodal equation at output terminal.

$$\frac{v_o}{R_L}+\frac{v_o+v_{gs}}{r_o}+g_m v_{gs}=0$$

$$v_o\left(\frac{1}{R_L}+\frac{1}{r_o}\right)+v_{gs}\left(\frac{1}{r_o}+g_m\right)=0$$

$$v_o\left(\frac{r_o+R_L}{R_L r_o}\right)+v_{gs}\left(\frac{1+g_m r_o}{r_o}\right)=0$$

$$v_o\left(\frac{r_o+R_L}{R_L r_o}\right)+v_{gs}\left(\frac{(1+g_m r_o)R_L}{r_o}\right)=0$$

$$v_o(r_o+R_L)+v_{gs}(1+g_m r_o)R_L=0$$

$$v_{gs}(1+g_m r_o)R_L=-v_o(r_o+R_L)$$

$$v_{gs}=-v_o\frac{(r_o+R_L)}{(1+g_m r_o)R_L}$$

Therefore, the value of the voltage v_{gs} is $-v_o\dfrac{(r_o+R_L)}{(1+g_m r_o)R_L}$.

Step 3 of 3

Apply nodal analysis at node 1.

$$\frac{-v_{gs}-v_o}{r_o}-g_m v_{gs}+\frac{-v_{gs}-v_{sig}}{R_s}=0$$

$$-v_{gs}\left(\frac{1}{r_o}+g_m+\frac{1}{R_s}\right)-\frac{v_o}{r_o}-\frac{v_{sig}}{R_s}=0$$

$$v_{gs}\left(\frac{R_s+g_m r_o R_s+r_o}{r_o R_s}\right)+\frac{v_o}{r_o}+\frac{v_{sig}}{R_s}=0$$

Substitute $-v_o\dfrac{(r_o+R_L)}{(1+g_m r_o)R_L}$ for v_{gs}.

$$-v_o\frac{(r_o+R_L)}{(1+g_m r_o)R_L}\left(\frac{R_s(1+g_m r_o)}{r_o R_s}+\frac{r_o}{r_o R_s}\right)+\frac{v_o}{r_o}+\frac{v_{sig}}{R_s}=0$$

$$-v_o\frac{(r_o+R_L)}{(1+g_m r_o)R_L}\left(\frac{1+g_m r_o}{r_o}+\frac{1}{R_s}\right)+\frac{v_o}{r_o}+\frac{v_{sig}}{R_s}=0$$

$$-v_o\left(\frac{(r_o+R_L)(1+g_m r_o)}{r_o((1+g_m r_o)R_L)}+\frac{r_o+R_L}{R_s(1+g_m r_o)R_L}\right)+\frac{v_o}{r_o}+\frac{v_{sig}}{R_s}=0$$

$$-v_o\left(\frac{(r_o+R_L)}{r_o R_L}+\frac{r_o+R_L}{R_s(1+g_m r_o)R_L}\right)+\frac{v_o}{r_o}+\frac{v_{sig}}{R_s}=0$$

$$v_o\left(-\frac{1}{r_o}-\frac{1}{R_L}-\frac{r_o+R_L}{R_s(1+g_m r_o)R_L}+\frac{1}{r_o}\right)+\frac{v_{sig}}{R_s}=0$$

$$v_o\left(-\frac{1}{R_L}-\frac{r_o+R_L}{R_s(1+g_m r_o)R_L}\right)=-\frac{v_{sig}}{R_s}$$

$$v_o\left(\frac{1}{R_L R_s}\right)\left(R_s+\frac{r_o+R_L}{1+g_m r_o}\right)=\frac{v_{sig}}{R_s}$$

$$v_o\left(\frac{1}{R_L}\right)\left(R_s+\frac{r_o+R_L}{1+g_m r_o}\right)=v_{sig}$$

$$\frac{v_o}{v_{sig}}=\frac{R_L}{R_s+\dfrac{r_o+R_L}{1+g_m r_o}}$$

The term $\dfrac{r_o+R_L}{1+g_m r_o}$ represents the expression for the input resistance R_{in} of the circuit.

$$\frac{v_o}{v_{sig}}=\frac{R_L}{R_s+R_{in}}$$

Therefore, the expression for the voltage gain is $\boxed{\dfrac{R_L}{R_s+R_{in}}}$.

Step 1 of 2

8.010P

Consider the circuit diagram shown in Figure 1 for the pnp version of the current mirror circuit.

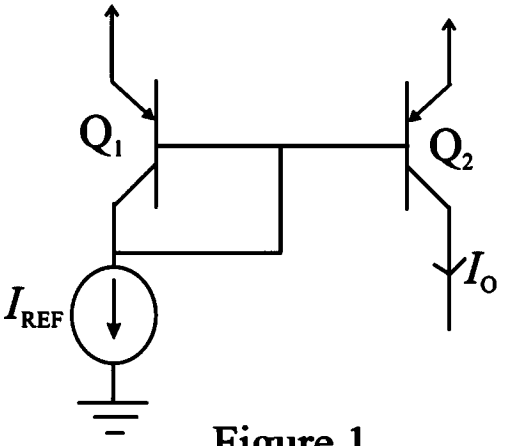


Figure 1

Step 2 of 2

Consider the expression for the current transfer ratio for the identical transistors.

$$\frac{I_O}{I_{REF}} = \frac{1}{1+\frac{2}{\beta}}$$

Substitute 50 for β.

$$\begin{aligned}\frac{I_O}{I_{REF}} &= \frac{1}{1+\frac{2}{50}} \\ &= \frac{1}{1+0.04} \\ &= \frac{1}{1.04} \\ &= 0.962\end{aligned}$$

Therefore, the value of the current transfer ratio is $\boxed{0.962}$.

Step 1 of 4

Consider the expression for the input resistance of the CG amplifier at $g_m r_o >> 1$.

$$R_{in} = \frac{1}{g_m} + \frac{R_L}{g_m r_o} \quad \text{..... (1)}$$

Substitute 0 for R_L.

$$R_{in} = \frac{1}{g_m} + \frac{0}{g_m r_o}$$
$$= \frac{1}{g_m}$$

Therefore, the value of the input resistance at $R_L = 0$ is $\boxed{\frac{1}{g_m}}$.

Step 2 of 4

Recall equation (1).

$$R_{in} = \frac{1}{g_m} + \frac{R_L}{g_m r_o}$$

Substitute r_o for R_L.

$$R_{in} = \frac{1}{g_m} + \frac{r_o}{g_m r_o}$$
$$= \frac{1}{g_m} + \frac{1}{g_m}$$
$$= \frac{2}{g_m}$$

Therefore, the value of the input resistance at $R_L = r_o$ is $\boxed{\frac{2}{g_m}}$.

Step 3 of 4

Recall equation (1).

$$R_{in} = \frac{1}{g_m} + \frac{R_L}{g_m r_o}$$

Substitute $(g_m r_o) r_o$ for R_L.

$$R_{in} = \frac{1}{g_m} + \frac{(g_m r_o) r_o}{g_m r_o}$$
$$= \frac{1}{g_m} + r_o$$

As $g_m r_o >> 1$, it implies $r_o >> \frac{1}{g_m}$.

$$R_{in} = \frac{1}{g_m} + \frac{(g_m r_o) r_o}{g_m r_o}$$
$$= \frac{1}{g_m} + r_o$$
$$= r_o$$

Therefore, the value of the input resistance at $R_L = (g_m r_o) r_o$ is $\boxed{r_o}$.

Step 4 of 4

Recall equation (1).

$$R_{in} = \frac{1}{g_m} + \frac{R_L}{g_m r_o}$$

Substitute ∞ for R_L.

$$R_{in} = \frac{1}{g_m} + \frac{\infty}{g_m r_o}$$
$$= \frac{1}{g_m} + \infty$$
$$= \infty$$

Therefore, the value of the input resistance at $R_L = \infty$ is $\boxed{\infty}$.

Step 1 of 4

Refer to Figure 8.7 from the text book.

Consider the expression for the current transfer ratio for the identical transistors.

$$\frac{I_O}{I_{REF}} = \frac{1}{1+\frac{2}{\beta}}$$

Substitute ∞ for β and $1\ \text{mA}$ for I_{REF}

$$\frac{I_O}{1\ \text{m}} = \frac{1}{1+\frac{2}{\infty}}$$

$$\frac{I_O}{1\ \text{m}} = \frac{1}{1+0}$$

$$\frac{I_O}{1\ \text{m}} = 1$$

$$I_O = 1\ \text{mA}$$

Therefore, the value of the current I_O is, $1\ \text{mA}$.

Step 2 of 4

Calculate the value of the output resistance r_{o2} .

$$r_{o2} = \frac{V_{A2}}{I_O}$$

Substitute $1\ \text{mA}$ for I_O and $90\ \text{V}$ for V_{A2} .

$$r_{o2} = \frac{90}{1\ \text{m}}$$
$$= 90\ \text{k}\Omega$$

Therefore, the value of the output resistance r_{o2} is, $90\ \text{k}\Omega$.

Step 3 of 4

Consider the expression for the output resistance r_{o2} in terms of change in output voltage and change in output current.

$$r_{o2} = \frac{\Delta V_O}{\Delta I_O}$$

Substitute $90\ \text{k}\Omega$ and $10\ \text{V} - 1\ \text{V}$ for V_{A2} .

$$90\ \text{k} = \frac{10-1}{\Delta I_O}$$

$$90\ \text{k} = \frac{9}{\Delta I_O}$$

$$\Delta I_O = \frac{9}{90\ \text{k}}$$

$$\Delta I_O = 0.1\ \text{mA}$$

Therefore, the value of the change in current ΔI_O is, $\boxed{0.1\ \text{mA}}$.

Step 4 of 4

Calculate the percentage change in the value of output current.

$$\Delta I_O\% = \frac{\Delta I_O}{I_O} \times 100$$

Substitute $0.1\ \text{mA}$ for ΔI_O and $1\ \text{mA}$ for I_O .

$$\Delta I_O\% = \frac{0.1\ \text{m}}{1\ \text{m}} \times 100$$
$$= 0.1 \times 100$$
$$= 10\%$$

Therefore, he value of the percentage change in the output current is, $\boxed{10\%}$.

Step 1 of 4

Consider the expression for the output resistance of the CG amplifier at $g_m r_o >> 1$.

$$R_{out} = r_o + (g_m r_o) R_s \quad \text{..... (1)}$$

Substitute 0 for R_s.

$$\begin{aligned} R_{out} &= r_o + (g_m r_o)(0) \\ &= r_o \end{aligned}$$

Therefore, the value of the input resistance at $R_s = 0$ is $\boxed{r_o}$.

Step 2 of 4

Recall equation (1).

$$R_{out} = r_o + (g_m r_o) R_s$$

Substitute r_o for R_s.

$$\begin{aligned} R_{out} &= r_o + (g_m r_o)(r_o) \\ &= r_o (1 + g_m r_o) \end{aligned}$$

As $g_m r_o >> 1$, it implies $1 + g_m r_o = g_m r_o$.

$$R_{out} = (g_m r_o) r_o$$

Therefore, the value of the input resistance at $R_s = r_o$ is $\boxed{(g_m r_o) r_o}$.

Step 3 of 4

Recall equation (1).

$$R_{out} = r_o + (g_m r_o) R_s$$

Substitute $(g_m r_o) r_o$ for R_s.

$$\begin{aligned} R_{out} &= r_o + (g_m r_o)(g_m r_o) r_o \\ &= r_o \left(1 + (g_m r_o)^2\right) \end{aligned}$$

As $g_m r_o >> 1$, it implies $1 + (g_m r_o)^2 = (g_m r_o)^2$.

$$R_{out} = (g_m r_o)^2 r_o$$

Therefore, the value of the input resistance at $R_s = g_m r_o$ is $\boxed{(g_m r_o)^2 r_o}$.

Step 4 of 4

Recall equation (1).

$$R_{out} = r_o + (g_m r_o) R_s$$

Substitute ∞ for R_s.

$$\begin{aligned} R_{out} &= r_o + (g_m r_o) \infty \\ &= r_o + \infty \\ &= \infty \end{aligned}$$

As $g_m r_o >> 1$, it implies $1 + (g_m r_o)^2 = (g_m r_o)^2$.

$$R_{out} = (g_m r_o)^2 r_o$$

Therefore, the value of the input resistance at $R_s = \infty$ is $\boxed{\infty}$.

Step 1 of 7

Refer to Figure P8.12 from the text book.

Consider the expression for the output current.

$$I_O = I_{REF}\left(\frac{1+\frac{3-V_O-V_{EB}}{|V_A|}}{1+\frac{2}{\beta}}\right)$$

Assume $0.7\ \text{V}$ for V_{EB}. Substitute $50\ \text{V}$ for $|V_A|$, $1\ \text{V}$ for V_O and 50 for β.

$$\begin{aligned}1\ \text{m} &= I_{REF}\left(\frac{1+\frac{3-1-0.7}{50}}{1+\frac{2}{50}}\right)\\ &= I_{REF}\left(\frac{1+\frac{1.3}{50}}{1+\frac{2}{50}}\right)\\ &= I_{REF}\left(\frac{\frac{50+1.3}{50}}{\frac{50+2}{50}}\right)\\ &= I_{REF}\left(\frac{51.3}{52}\right)\end{aligned}$$

$$1\ \text{m} = I_{REF}(0.987)$$

$$I_{REF} = \frac{1\ \text{m}}{0.987}$$

$$I_{REF} = 1.014\ \text{mA}$$

Therefore, the value of the reference current I_{REF} is $\boxed{1.014\ \text{mA}}$.

Step 2 of 7

Calculate the value of the collector voltage V_C.

$$V_C = V_{EE} - V_{EB}$$

Assume $0.7\ \text{V}$ for V_{EB}. Substitute $3\ \text{V}$ for V_{EE}.

$$\begin{aligned}V_C &= 3 - 0.7\\ &= 2.3\ \text{V}\end{aligned}$$

Therefore, the value of the collector voltage V_C is $2.3\ \text{V}$.

Step 3 of 7

Calculate the value of the resistance R.

$$R = \frac{V_C}{I_{REF}}$$

Substitute $2.3\ \text{V}$ for V_C and $1.014\ \text{mA}$ for I_{REF}.

$$\begin{aligned}R &= \frac{2.3}{1.014\ \text{m}}\\ &= 2.27\ \text{k}\Omega\end{aligned}$$

Therefore, the value of the resistance R is, $\boxed{2.27\ \text{k}\Omega}$.

Step 4 of 7

Calculate the value of the output resistance r_o.

$$r_o = \frac{V_A}{I_O}$$

Substitute $50\ \text{V}$ for V_A and $1\ \text{mA}$ for I_O.

$$\begin{aligned}r_o &= \frac{50}{1\ \text{m}}\\ &= 50\ \text{k}\Omega\end{aligned}$$

Therefore, the value of the output resistance r_o is, $50\ \text{k}\Omega$.

Step 5 of 7

The maximum value of the output voltage occurs for the minimum value of the emitter to collector voltage, that is for the value of $V_{EC} = 0.3\ \text{V}$. Hence, calculate the value of the maximum value of the output voltage.

$$V_{O\max} = V_{EE} - V_{EC}$$

Substitute $3\ \text{V}$ for V_{EE} and $0.3\ \text{V}$ for V_{EC}.

$$\begin{aligned}V_{O\max} &= 3 - 0.3\\ &= 2.7\ \text{V}\end{aligned}$$

Therefore, the maximum value of the output voltage is $\boxed{2.7\ \text{V}}$.

Step 6 of 7

Consider the expression for the value of the output resistance r_o in terms of the change in output voltage and the change in the output current.

$$r_{o2} = \frac{\Delta V_O}{\Delta I_O}$$

Substitute $50\ \text{k}\Omega$ and $2.7\ \text{V} - (-5)\ \text{V}$ for V_A.

$$50\ \text{k} = \frac{2.7-(-5)}{\Delta I_O}$$

$$50\ \text{k} = \frac{2.7+5}{\Delta I_O}$$

$$\Delta I_O = \frac{7.7}{50\ \text{k}}$$

$$\Delta I_O = 0.154\ \text{mA}$$

Therefore, the value of the change in current ΔI_O is, $\boxed{0.154\ \text{mA}}$.

Step 7 of 7

Calculate the percentage change in the value of output current.

$$\Delta I_O\% = \frac{\Delta I_O}{I_O} \times 100$$

Substitute $0.154\ \text{mA}$ for ΔI_O and $1\ \text{mA}$ for I_O.

$$\begin{aligned}\Delta I_O\% &= \frac{0.154\ \text{m}}{1\ \text{m}} \times 100\\ &= 0.154 \times 100\\ &= 15.4\%\end{aligned}$$

Therefore, he value of the percentage change in the output current is, $\boxed{15.4\%}$.

Step 1 of 6

Consider the small signal equivalent circuit diagram for the Common Source amplifier without load resistance. Neglect the resistance r_s in the circuit diagram.

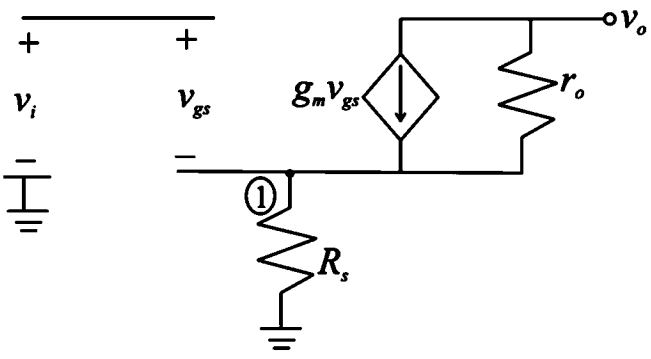


Figure 1

Step 2 of 6

Apply nodal analysis at the output terminal.

$$\frac{v_o-\left(-v_{gs}+v_i\right)}{r_o}+g_m v_{gs}=0$$

$$\frac{v_o+v_{gs}-v_i}{r_o}+g_m v_{gs}=0$$

$$v_o+v_{gs}-v_i+g_m v_{gs} r_o=0$$

$$v_{gs}\left(1+g_m r_o\right)=v_i-v_o$$

$$v_{gs}=\frac{v_i-v_o}{1+g_m r_o}$$

Therefore, the expression for the voltage v_{gs} is $\frac{v_i-v_o}{1+g_m r_o}$.

Step 3 of 6

Apply nodal analysis at node-1.

$$\frac{\left(-v_{gs}+v_i\right)-v_o}{r_o}-g_m v_{gs}+\frac{\left(-v_{gs}+v_i\right)}{R_s}=0$$

$$-v_{gs}\left(\frac{1}{r_o}+g_m+\frac{1}{R_s}\right)+v_i\left(\frac{1}{r_o}+\frac{1}{R_s}\right)-\frac{v_o}{r_o}=0$$

$$-v_{gs}\left(\frac{R_s+g_m R_s r_o+r_o}{r_o R_s}\right)+v_i\left(\frac{R_s+r_o}{r_o R_s}\right)-\frac{v_o R_s}{r_o R_s}=0$$

$$-v_{gs}\left(R_s\left(1+g_m r_o\right)+r_o\right)+v_i\left(R_s+r_o\right)-v_o R_s=0$$

Substitute $\frac{v_i-v_o}{1+g_m r_o}$ for v_{gs}.

$$-\left(\frac{v_i-v_o}{1+g_m r_o}\right)\left(R_s\left(1+g_m r_o\right)+r_o\right)+v_i\left(R_s+r_o\right)-v_o R_s=0$$

$$\left(v_o-v_i\right)\left(R_s+\frac{r_o}{1+g_m r_o}\right)+v_i\left(R_s+r_o\right)-v_o R_s=0$$

$$v_o\left(R_s+\frac{r_o}{1+g_m r_o}-R_s\right)+v_i\left(R_s+r_o-R_s-\frac{r_o}{1+g_m r_o}\right)=0$$

$$v_o\left(\frac{r_o}{1+g_m r_o}\right)+v_i\left(r_o-\frac{r_o}{1+g_m r_o}\right)=0$$

$$v_o\left(\frac{1}{1+g_m r_o}\right)+v_i\left(1-\frac{1}{1+g_m r_o}\right)=0$$

$$v_o\left(\frac{1}{1+g_m r_o}\right)+v_i\left(\frac{1+g_m r_o-1}{1+g_m r_o}\right)=0$$

$$v_o\left(\frac{1}{1+g_m r_o}\right)+v_i\left(\frac{g_m r_o}{1+g_m r_o}\right)=0$$

$$v_o+v_i\left(g_m r_o\right)=0$$

$$v_o=-v_i\left(g_m r_o\right)$$

$$\frac{v_o}{v_i}=-g_m r_o$$

Here, $\frac{v_o}{v_i}$ represents the gain open loop gain A_{vo}.

$$A_{vo}=-g_m r_o$$

Therefore, the expression for the gain A_{vo} is $\boxed{-g_m r_o}$.

Step 4 of 6

Consider the small signal equivalent circuit diagram for the Common Source amplifier with load resistance. Neglect the resistance r_s in the circuit diagram.

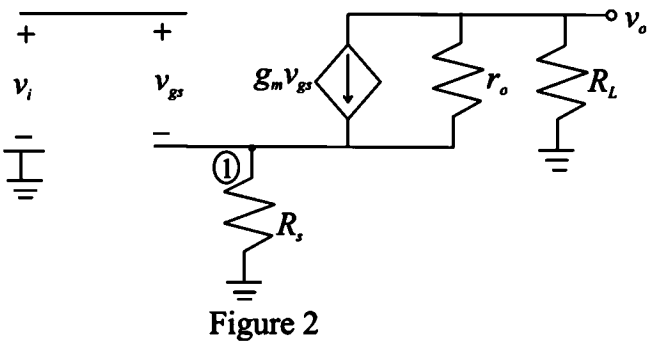


Figure 2

Step 5 of 6

Apply nodal analysis at the output terminal.

$$\frac{v_o-\left(-v_{gs}+v_i\right)}{r_o}+g_m v_{gs}+\frac{v_o}{R_L}=0$$

$$\frac{v_o+v_{gs}-v_i}{r_o}+g_m v_{gs}+\frac{v_o}{R_L}=0$$

$$v_{gs}\left(\frac{1}{r_o}+g_m\right)-\frac{v_i}{r_o}+v_o\left(\frac{1}{r_o}+\frac{1}{R_L}\right)=0$$

$$v_{gs}\left(\frac{1+g_m r_o}{r_o}\right)-\frac{v_i}{r_o}+v_o\left(\frac{R_L+r_o}{r_o R_L}\right)=0$$

$$v_{gs}\left(1+g_m r_o\right)-v_i+v_o\left(\frac{R_L+r_o}{R_L}\right)=0$$

$$v_{gs}\left(1+g_m r_o\right)=v_i-v_o\left(\frac{R_L+r_o}{R_L}\right)$$

$$v_{gs}=\frac{v_i}{\left(1+g_m r_o\right)}-v_o\left(\frac{R_L+r_o}{\left(1+g_m r_o\right)R_L}\right)$$

Therefore, the expression for the voltage v_{gs} is $\frac{v_i}{\left(1+g_m r_o\right)}-v_o\left(\frac{R_L+r_o}{\left(1+g_m r_o\right)R_L}\right)$.

Step 6 of 6

Apply nodal analysis at node-1.

$$\frac{\left(-v_{gs}+v_i\right)-v_o}{r_o}-g_m v_{gs}+\frac{\left(-v_{gs}+v_i\right)}{R_s}=0$$

$$-v_{gs}\left(\frac{1}{r_o}+g_m+\frac{1}{R_s}\right)+v_i\left(\frac{1}{r_o}+\frac{1}{R_s}\right)-\frac{v_o}{r_o}=0$$

$$-v_{gs}\left(\frac{R_s+g_m R_s r_o+r_o}{r_o R_s}\right)+v_i\left(\frac{R_s+r_o}{r_o R_s}\right)-\frac{v_o R_s}{r_o R_s}=0$$

$$-v_{gs}\left(R_s\left(1+g_m r_o\right)+r_o\right)+v_i\left(R_s+r_o\right)-v_o R_s=0$$

Substitute $\frac{v_i}{\left(1+g_m r_o\right)}-v_o\left(\frac{R_L+r_o}{\left(1+g_m r_o\right)R_L}\right)$ for v_{gs}.

$$-\left(\frac{v_i}{\left(1+g_m r_o\right)}-v_o\left(\frac{R_L+r_o}{\left(1+g_m r_o\right)R_L}\right)\right)\left(R_s\left(1+g_m r_o\right)+r_o\right)+v_i\left(R_s+r_o\right)-v_o R_s=0$$

$$v_o\left(\frac{\left(R_L+r_o\right)\left(R_s\left(1+g_m r_o\right)+r_o\right)}{\left(1+g_m r_o\right)R_L}\right)-\frac{v_i\left(R_s\left(1+g_m r_o\right)+r_o\right)}{\left(1+g_m r_o\right)}+v_i\left(R_s+r_o\right)-v_o R_s=0$$

$$v_o\left(\frac{R_L R_s\left(1+g_m r_o\right)+r_o R_L+r_o R_s\left(1+g_m r_o\right)+r_o^2}{\left(1+g_m r_o\right)R_L}-R_s\right)-v_i\left(R_s+\frac{r_o}{1+g_m r_o}-R_s-r_o\right)=0$$

$$v_o\left(R_s+\frac{r_o}{1+g_m r_o}+\frac{r_o R_s\left(1+g_m r_o\right)}{\left(1+g_m r_o\right)R_L}+\frac{r_o^2}{\left(1+g_m r_o\right)R_L}-R_s\right)-v_i\left(R_s+\frac{r_o}{1+g_m r_o}-R_s-r_o\right)=0$$

$$v_o\left(\frac{r_o}{1+g_m r_o}+\frac{r_o R_s\left(1+g_m r_o\right)}{\left(1+g_m r_o\right)R_L}+\frac{r_o^2}{\left(1+g_m r_o\right)R_L}\right)-v_i\left(\frac{r_o}{1+g_m r_o}-r_o\right)=0$$

$$v_o\left(\frac{1}{1+g_m r_o}+\frac{R_s\left(1+g_m r_o\right)}{\left(1+g_m r_o\right)R_L}+\frac{r_o}{\left(1+g_m r_o\right)R_L}\right)-v_i\left(\frac{1}{1+g_m r_o}-1\right)=0$$

$$v_o\left(\frac{1}{1+g_m r_o}+\frac{\left(1+g_m r_o\right)R_s}{\left(1+g_m r_o\right)R_L}+\frac{r_o}{\left(1+g_m r_o\right)R_L}\right)-v_i\left(\frac{1-1-g_m r_o}{1+g_m r_o}\right)=0$$

$$v_o\left(R_L+\left(1+g_m r_o\right)R_s+r_o\right)-v_i\left(-g_m r_o R_L\right)=0$$

$$v_o\left(R_L+\left(1+g_m r_o\right)R_s+r_o\right)=v_i\left(-g_m r_o R_L\right)$$

$$v_o\left(R_L+R_s+g_m r_o R_s+r_o\right)=v_i\left(-g_m r_o\right)R_L$$

$$v_o\left(R_L+R_s+\left(g_m R_s+1\right)r_o\right)=v_i\left(-g_m r_o\right)R_L$$

Here, the term $R_s+\left(g_m R_s+1\right)r_o$ is approximately equal to $\left(g_m R_s+1\right)r_o$ since $g_m r_o >> 1$.

$$v_o\left(R_L+\left(g_m R_s+1\right)r_o\right)=v_i\left(-g_m r_o\right)R_L$$

$$\frac{v_o}{v_i}=\left(-g_m r_o\right)\frac{R_L}{R_L+\left(g_m R_s+1\right)r_o}$$

Here, $\frac{v_o}{v_i}$ represents the gain open loop gain A_v.

$$A_v=\left(-g_m r_o\right)\frac{R_L}{R_L+\left(g_m R_s+1\right)r_o}$$

Therefore, the expression for the gain A_v is $\boxed{\left(-g_m r_o\right)\frac{R_L}{R_L+\left(g_m R_s+1\right)r_o}}$.

Refer to Figure P8.13 from the text book.

Identify the voltages and currents in the circuit diagram shown in Figure 1.

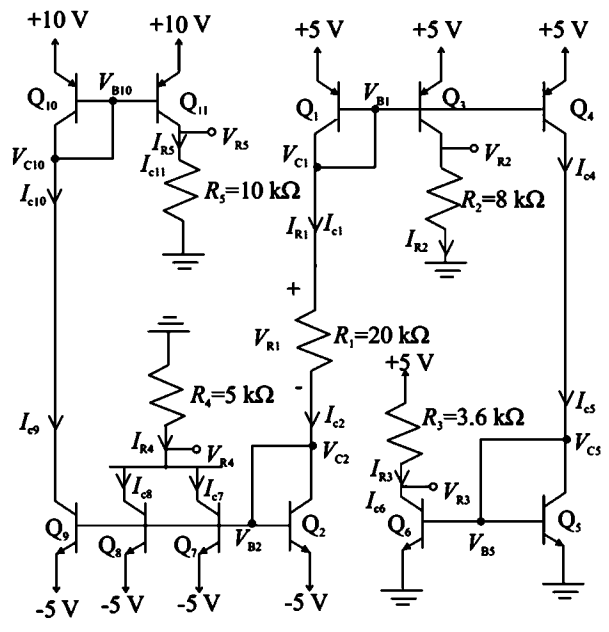


Figure 1

Step 2 of 14

Calculate the value of the voltage V_{B1}.

$$V_{B1} = V_{EE} - V_{BE}$$

Substitute 5 V for V_{EE} and 0.7 V for V_{BE}.

$$V_{B1} = 5 - 0.7$$
$$= 4.3 \text{ V}$$

Therefore, the value of the voltage V_{B1} is, 4.3 V.

The node voltage V_{C1} represent the nodal voltage of V_{B1}.

Step 3 of 14

Calculate the value of the voltage V_{B2}.

$$V_{B2} = V_{EE} + V_{BE}$$

Substitute -5 V for V_{EE} and 0.7 V for V_{BE}.

$$V_{B2} = -5 + 0.7$$
$$= -4.3 \text{ V}$$

Therefore, the value of the voltage V_{B2} is, -4.3 V.

The node voltage V_{C2} represent the nodal voltage of V_{B2}.

Step 4 of 14

Calculate the value of the current I_{R1}.

$$I_{R1} = \frac{V_{B1} - V_{B2}}{R_1}$$

Substitute -4.3 V for V_{B2}, 4.3 V for V_{B1} and 20 kΩ for R_1.

$$I_{R1} = \frac{4.3 - (-4.3)}{20 \text{ k}}$$
$$= \frac{4.3 + 4.3}{20 \text{ k}}$$
$$= \frac{8.6}{20 \text{ k}}$$
$$= 0.43 \text{ mA}$$

Therefore, the value of the current I_{R1} is, 0.43 mA.

Step 5 of 14

The currents I_{C1}, I_{R1}, I_{C2}, I_{C3}, I_{R2}, I_{C4}, I_{C5}, I_{C6}, I_{R3}, I_{C7}, I_{C8}, I_{C9}, I_{C10}, I_{R5} and I_{C11} is, $\boxed{0.43 \text{ mA}}$ and the value of the current I_{R4} is, $\boxed{0.86 \text{ mA}}$.

Step 6 of 14

Calculate the value of the current I_{R4}.

$$I_{R4} = I_{C8} + I_{C7}$$

Substitute 0.43 mA for I_{C7} and I_{C8}.

$$I_{R4} = 0.43 \text{ m} + 0.43 \text{ m}$$
$$= 0.86 \text{ mA}$$

Therefore, the value of the current is, 0.86 mA.

Step 7 of 14

Calculate the value of the voltage V_{R1}.

$$V_{R1} = V_{B1} - V_{B2}$$

Substitute -4.3 V for V_{B2} and 4.3 V for V_{B1}.

$$V_{R1} = 0.43 - (-0.43)$$
$$= 0.43 + 0.43$$
$$= 0.86 \text{ V}$$

Therefore, the value of the voltage V_{R1} is, 0.86 V.

Step 8 of 14

Calculate the value of the voltage V_{R2}.

$$V_{R2} = I_{C2} R_2$$

Substitute 0.43 mA for I_{C2} and 8 kΩ for R_2.

$$V_{R2} = (0.43 \text{ m})(8 \text{ k})$$
$$= 3.44 \text{ V}$$

Therefore, the value of the voltage V_{R2} is, 3.44 V.

Step 9 of 14

Calculate the value of the voltage V_{R3}.

$$V_{R3} = I_{C6} R_3$$

Substitute 0.43 mA for I_{C6} and 3.6 kΩ for R_3.

$$V_{R3} = (0.43 \text{ m})(3.6 \text{ k})$$
$$= 1.548 \text{ V}$$

Therefore, the value of the voltage V_{R3} is, 1.548 V.

Step 10 of 14

Calculate the value of the voltage V_{R4}.

$$V_{R4} = -I_{R4} R_4$$

Substitute 0.86 mA for I_{R4} and 5 kΩ for R_4.

$$V_{R4} = -(0.86 \text{ m})(5 \text{ k})$$
$$= -4.3 \text{ V}$$

Therefore, the value of the voltage V_{R4} is, -4.3 V.

Step 11 of 14

Calculate the value of the voltage V_{R5}.

$$V_{R5} = I_{C11} R_5$$

Substitute 0.43 mA for I_{C6} and 10 kΩ for R_5.

$$V_{R5} = (0.43 \text{ m})(10 \text{ k})$$
$$= 4.3 \text{ V}$$

Therefore, the value of the voltage V_{R5} is, 4.3 V.

Step 12 of 14

Calculate the value of the voltage V_{B5}.

$$V_{B5} = V_{BE}$$

Substitute 0.7 V for V_{BE}.

$$V_{B5} = 0.7 \text{ V}$$

Therefore, the value of the voltage V_{B5} is, 0.7 V.

The node voltage V_{C5} represent the nodal voltage of V_{B5}.

Step 13 of 14

Calculate the value of the voltage V_{B10}.

$$V_{B10} = V_{EE} - V_{BE}$$

Substitute 10 V for V_{EE} and 0.7 V for V_{BE}.

$$V_{B10} = 10 - 0.7$$
$$= 9.3 \text{ V}$$

Therefore, the value of the voltage V_{B10} is, 9.3 V.

The nodal voltages V_{C10}, V_{C9} represent the nodal voltage of V_{C10}.

Step 14 of 14

The value of the currents I_{C1}, I_{R1}, I_{C2}, I_{C3}, I_{R2}, I_{C4}, I_{C5}, I_{C6}, I_{R3}, I_{C7}, I_{C8}, I_{C9}, I_{C10}, I_{R5} and I_{C11} is, $\boxed{0.43 \text{ mA}}$ and the value of the current I_{R4} is, $\boxed{0.86 \text{ mA}}$.

The values of the voltages across the resistors are $\boxed{V_{R1} = 0.86 \text{ V}}$, $\boxed{V_{R2} = 3.44 \text{ V}}$, $\boxed{V_{R3} = 1.548 \text{ V}}$, $\boxed{V_{R4} = -4.3 \text{ V}}$ and $\boxed{V_{R5} = 4.3 \text{ V}}$.

The values of the nodal voltages are $\boxed{V_{B1} = 4.3 \text{ V}}$, $\boxed{V_{C1} = 4.3 \text{ V}}$, $\boxed{V_{B2} = -4.3 \text{ V}}$, $\boxed{V_{C2} = -4.3 \text{ V}}$, $\boxed{V_{B5} = 0.7 \text{ V}}$, $\boxed{V_{C5} = 0.7 \text{ V}}$, $\boxed{V_{B10} = 9.3 \text{ V}}$ and $\boxed{V_{C10} = 9.3 \text{ V}}$.

Step 1 of 4

Consider the expression for the input resistance of the Common Base amplifier.

$$R_{in} = r_e\left(\frac{r_o + R_L}{r_o + \frac{R_L}{\beta+1}}\right) \quad \text{..... (1)}$$

Substitute 0 for R_L.

$$\begin{aligned} R_{in} &= r_e\left(\frac{r_o + 0}{r_o + \frac{0}{\beta+1}}\right) \\ &= r_e\left(\frac{r_o}{r_o}\right) \\ &= r_e(1) \\ &= r_e \end{aligned}$$

Therefore, the expression for the input resistance at $R_L = 0$ is $\boxed{r_e}$.

Step 2 of 4

Recall equation (1).

$$R_{in} = r_e\left(\frac{r_o + R_L}{r_o + \frac{R_L}{\beta+1}}\right)$$

Substitute r_o for R_L.

$$R_{in} = r_e\left(\frac{r_o + r_o}{r_o + \frac{r_o}{\beta+1}}\right)$$

Here $\beta+1 >> r_o$ resulting $\frac{r_o}{\beta+1} \approx 0$.

$$\begin{aligned} R_{in} &= r_e\left(\frac{2r_o}{r_o + 0}\right) \\ &= r_e\left(\frac{2r_o}{r_o}\right) \\ &= 2r_e \end{aligned}$$

Therefore, the expression for the input resistance at $R_L = r_o$ is $\boxed{2r_e}$.

Step 3 of 4

Recall equation (1).

$$R_{in} = r_e\left(\frac{r_o + R_L}{r_o + \frac{R_L}{\beta+1}}\right)$$

Substitute $(\beta+1)r_o$ for R_L.

$$\begin{aligned} R_{in} &= r_e\left(\frac{r_o + (\beta+1)r_o}{r_o + \frac{(\beta+1)r_o}{\beta+1}}\right) \\ &= r_e\left(\frac{r_o + (\beta+1)r_o}{r_o + r_o}\right) \\ &= r_e\left(\frac{r_o + (\beta+1)r_o}{2r_o}\right) \\ &= r_e\left(\frac{1 + (\beta+1)}{2}\right) \end{aligned}$$

$$R_{in} = \frac{r_e + (\beta+1)r_e}{2}$$

Here $(\beta+1)r_e >> r_e$ resulting to $(\beta+1)r_e$.

$$R_{in} = \frac{(\beta+1)r_e}{2}$$

Here, the expression $(\beta+1)r_e$ is equal to r_π.

$$R_{in} = \frac{r_\pi}{2}$$

Therefore, the expression for the input resistance at $R_L = (\beta+1)r_o$ is $\boxed{\frac{r_\pi}{2}}$.

Step 4 of 4

Recall equation (1).

$$\begin{aligned} R_{in} &= r_e\left(\frac{r_o + R_L}{r_o + \frac{R_L}{\beta+1}}\right) \\ &= r_e\left(\frac{\frac{r_o}{R_L} + 1}{\frac{r_o}{R_L} + \frac{1}{\beta+1}}\right) \end{aligned}$$

As $R_L \to \infty$, it implies $\frac{1}{R_L} \to 0$.

$$\begin{aligned} R_{in} &= r_e\left(\frac{0+1}{0 + \frac{1}{\beta+1}}\right) \\ &= r_e\left(\frac{1}{\frac{1}{\beta+1}}\right) \\ &= r_e(\beta+1) \end{aligned}$$

Here, the expression $(\beta+1)r_e$ is equal to r_π.

$$R_{in} = r_\pi$$

Therefore, the expression for the input resistance at $R_L = \infty$ is $\boxed{r_\pi}$.

Step 1 of 14

Refer to Figure P8.14 from the text book.

(a)

Calculate the value of the voltage V_1.

$$V_1 = -V_{BE}$$

Substitute $0.7\ \text{V}$ for V_{BE}.

$$V_1 = -0.7\ \text{V}$$

Therefore, the value of the voltage V_1 is, $\boxed{-0.7\ \text{V}}$.

Step 2 of 14

Calculate the value of the voltage is I_{C1}.

$$I_{C1} = \frac{V_1 - (-2.7)}{R}$$

Substitute $-0.7\ \text{V}$ for V_1 and $10\ \text{k}\Omega$ for R.

$$\begin{aligned} I_{C1} &= \frac{-0.7 - (-2.7)}{10\ \text{k}} \\ &= \frac{-0.7 + 2.7}{10\ \text{k}} \\ &= \frac{2}{10\ \text{k}} \\ &= 0.2\ \text{mA} \end{aligned}$$

Therefore, the value of the current I_{C1} is, $0.2\ \text{mA}$.

The values of the currents I_{C2}, I_{C3}, I_{C4}, I_{C5} and I_{C6} represent the same value of the current I_{C1} due to the characteristics of the current mirror circuit.

Step 3 of 14

Calculate the value of the current I.

$$I = I_{C4} + I_{C3}$$

Substitute $0.2\ \text{mA}$ for I_{C3} and I_{C4}.

$$\begin{aligned} I &= 0.2\ \text{m} + 0.2\ \text{m} \\ &= 0.4\ \text{mA} \end{aligned}$$

Therefore, the value of the current I is, $\boxed{0.4\ \text{mA}}$.

Step 4 of 14

Calculate the value of the voltage V_2.

$$V_2 = 2.7 - V_{BE}$$

Substitute $0.7\ \text{V}$ for V_{BE}.

$$\begin{aligned} V_2 &= 2.7 - 0.7 \\ &= 2\ \text{V} \end{aligned}$$

Therefore, the value of the voltage V_2 is, $\boxed{2\ \text{V}}$.

Step 5 of 14

Calculate the value of the voltage V_3.

$$V_3 = V_{BE}$$

Substitute $0.7\ \text{V}$ for V_{BE}.

$$V_3 = 0.7\ \text{V}$$

Therefore, the value of the voltage V_3 is, $\boxed{0.7\ \text{V}}$.

Step 6 of 14

Calculate the value of the voltage V_4.

$$V_4 = -2.7 + I_{C5}R$$

Substitute $0.2\ \text{mA}$ for I_{C5} and $10\ \text{k}\Omega$ for R.

$$\begin{aligned} V_4 &= -2.7 + (0.2\ \text{m})(10\ \text{k}) \\ &= -2.7 + 2 \\ &= -0.7\ \text{V} \end{aligned}$$

Therefore, the value of the voltage V_4 is, $\boxed{-0.7\ \text{V}}$.

Step 7 of 14

Calculate the value of the voltage V_5.

$$V_5 = -2.7 + I_{C6}\left(\frac{R}{2}\right)$$

Substitute $0.2\ \text{mA}$ for I_{C5} and $10\ \text{k}\Omega$ for R.

$$\begin{aligned} V_5 &= -2.7 + (0.2\ \text{m})\left(\frac{10\ \text{k}}{2}\right) \\ &= -2.7 + 1 \\ &= -1.7\ \text{V} \end{aligned}$$

Therefore, the value of the voltage V_5 is, $\boxed{-1.7\ \text{V}}$.

Step 8 of 14

(b)

Calculate the value of the voltage V_1.

$$V_1 = -V_{BE}$$

Substitute $0.7\ \text{V}$ for V_{BE}.

$$V_1 = -0.7\ \text{V}$$

Therefore, the value of the voltage V_1 is, $\boxed{-0.7\ \text{V}}$.

Step 9 of 14

Calculate the value of the voltage is I_{C1}.

$$I_{C1} = \frac{V_1 - (-2.7)}{R}$$

Substitute $-0.7\ \text{V}$ for V_1 and $100\ \text{k}\Omega$ for R.

$$\begin{aligned} I_{C1} &= \frac{-0.7 - (-2.7)}{100\ \text{k}} \\ &= \frac{-0.7 + 2.7}{100\ \text{k}} \\ &= \frac{2}{100\ \text{k}} \\ &= 0.02\ \text{mA} \end{aligned}$$

Therefore, the value of the current I_{C1} is, $0.02\ \text{mA}$.

The values of the currents I_{C2}, I_{C3}, I_{C4}, I_{C5} and I_{C6} represent the same value of the current I_{C1} due to the characteristics of the current mirror circuit.

Step 10 of 14

Calculate the value of the current I.

$$I = I_{C4} + I_{C3}$$

Substitute $0.02\ \text{mA}$ for I_{C3} and I_{C4}.

$$\begin{aligned} I &= 0.02\ \text{m} + 0.02\ \text{m} \\ &= 0.04\ \text{mA} \end{aligned}$$

Therefore, the value of the current I is, $\boxed{0.04\ \text{mA}}$.

Step 11 of 14

Calculate the value of the voltage V_2.

$$V_2 = 2.7 - V_{BE}$$

Substitute $0.7\ \text{V}$ for V_{BE}.

$$\begin{aligned} V_2 &= 2.7 - 0.7 \\ &= 2\ \text{V} \end{aligned}$$

Therefore, the value of the voltage V_2 is, $\boxed{2\ \text{V}}$.

Step 12 of 14

Calculate the value of the voltage V_3.

$$V_3 = V_{BE}$$

Substitute $0.7\ \text{V}$ for V_{BE}.

$$V_3 = 0.7\ \text{V}$$

Therefore, the value of the voltage V_3 is, $\boxed{0.7\ \text{V}}$.

Step 13 of 14

Calculate the value of the voltage V_4.

$$V_4 = -2.7 + I_{C5}R$$

Substitute $0.02\ \text{mA}$ for I_{C5} and $100\ \text{k}\Omega$ for R.

$$\begin{aligned} V_4 &= -2.7 + (0.02\ \text{m})(100\ \text{k}) \\ &= -2.7 + 2 \\ &= -0.7\ \text{V} \end{aligned}$$

Therefore, the value of the voltage V_4 is, $\boxed{-0.7\ \text{V}}$.

Step 14 of 14

Calculate the value of the voltage V_5.

$$V_5 = -2.7 + I_{C6}\left(\frac{R}{2}\right)$$

Substitute $0.02\ \text{mA}$ for I_{C5} and $100\ \text{k}\Omega$ for R.

$$\begin{aligned} V_5 &= -2.7 + (0.02\ \text{m})\left(\frac{100\ \text{k}}{2}\right) \\ &= -2.7 + 1 \\ &= -1.7\ \text{V} \end{aligned}$$

Therefore, the value of the voltage V_5 is, $\boxed{-1.7\ \text{V}}$.

Step 1 of 5

8.015E

Consider the expression for the output resistance of the Common Base amplifier.

$$R_{out} = r_o + (R_e \| r_\pi) g_m r_o$$

$$R_{out} = r_o + \left(\frac{R_e r_\pi}{R_e + r_\pi} \right) g_m r_o \quad \text{..... (1)}$$

Substitute 0 for R_e.

$$\begin{aligned} R_{out} &= r_o + \left(\frac{(0) r_\pi}{0 + r_\pi} \right) g_m r_o \\ &= r_o + 0 \\ &= r_o \end{aligned}$$

Therefore, the expression for the output resistance at $R_e = 0$ is $\boxed{r_o}$.

Step 2 of 5

Recall equation (1).

$$R_{out} = r_o + \left(\frac{R_e r_\pi}{R_e + r_\pi} \right) g_m r_o$$

Substitute r_e for R_e and $\frac{1}{r_e} - \frac{1}{r_\pi}$ for g_m.

$$\begin{aligned} R_{out} &= r_o + \left(\frac{r_e r_\pi}{r_e + r_\pi} \right) \left(\frac{1}{r_e} - \frac{1}{r_\pi} \right) r_o \\ &= r_o + \left(\frac{r_e r_\pi}{r_e + r_\pi} \right) \left(\frac{r_\pi - r_e}{r_e r_\pi} \right) r_o \\ &= r_o + \left(\frac{r_\pi - r_e}{r_e + r_\pi} \right) r_o \end{aligned}$$

Here, $r_\pi >> r_e$. Hence, both terms $r_\pi + r_e$ and $r_\pi - r_e$ are equal to r_π.

$$\begin{aligned} R_{out} &= r_o + \left(\frac{r_\pi}{r_\pi} \right) r_o \\ &= r_o + r_o \\ &= 2r_o \end{aligned}$$

Therefore, the expression for the output resistance at $R_e = r_e$ is $\boxed{2r_o}$.

Step 3 of 5

Recall equation (1).

$$R_{out} = r_o + \left(\frac{R_e r_\pi}{R_e + r_\pi} \right) g_m r_o$$

Substitute r_π for R_e.

$$\begin{aligned} R_{out} &= r_o + \left(\frac{r_\pi \cdot r_\pi}{r_\pi + r_\pi} \right) g_m r_o \\ &= r_o + \left(\frac{r_\pi \cdot r_\pi}{2 r_\pi} \right) g_m r_o \\ &= r_o + \left(\frac{g_m r_\pi}{2} \right) r_o \\ &= r_o \left(1 + \left(\frac{g_m r_\pi}{2} \right) \right) \end{aligned}$$

Here, the term $g_m r_\pi$ is equal to β.

$$R_{out} = r_o \left(1 + \frac{\beta}{2} \right)$$

Therefore, the expression for the output resistance at $R_e = r_\pi$ is $\boxed{r_o \left(1 + \frac{\beta}{2} \right)}$.

Step 4 of 5

Recall equation (1).

$$R_{out} = r_o + \left(\frac{R_e r_\pi}{R_e + r_\pi} \right) g_m r_o$$

Substitute r_o for R_e.

$$R_{out} = r_o + \left(\frac{r_o \cdot r_\pi}{r_\pi + r_o} \right) g_m r_o$$

Here, $r_o >> r_\pi$.

$$\begin{aligned} R_{out} &= r_o + \left(\frac{r_o \cdot r_\pi}{r_o} \right) g_m r_o \\ &= r_o + g_m r_o r_\pi \\ &= r_o (1 + g_m r_\pi) \end{aligned}$$

Here, the term $g_m r_\pi$ is equal to β.

$$R_{out} = r_o (1 + \beta)$$

Therefore, the expression for the output resistance at $R_e = r_o$ is $\boxed{r_o (1 + \beta)}$.

Step 5 of 5

Recall equation (1).

$$\begin{aligned} R_{out} &= r_o + \left(\frac{R_e r_\pi}{R_e + r_\pi} \right) g_m r_o \\ &= r_o + \left(\frac{R_e r_\pi}{R_e \left(1 + \frac{r_\pi}{R_e} \right)} \right) g_m r_o \\ &= r_o + \left(\frac{r_\pi}{\left(1 + \frac{r_\pi}{R_e} \right)} \right) g_m r_o \end{aligned}$$

As $R_e \to \infty$, it implies $\frac{1}{R_e} \to 0$.

$$\begin{aligned} R_{out} &= r_o + \left(\frac{r_\pi}{1 + 0} \right) g_m r_o \\ &= r_o + g_m r_o r_\pi \\ &= r_o (1 + g_m r_\pi) \end{aligned}$$

Here, the term $g_m r_\pi$ is equal to β.

$$R_{out} = r_o (1 + \beta)$$

Therefore, the expression for the output resistance at $R_e = \infty$ is $\boxed{r_o (1 + \beta)}$.

8.015P

We don't have the solution to this problem yet.

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Step 1 of 4

Refer to Figure 8.28 in the text book.

Calculate the value of the output resistance, r_o.

$$r_o = \frac{V_A}{I_C}$$

Substitute $10\ \text{V}$ for V_A and $1\ \text{mA}$ for I_C.

$$r_o = \frac{10}{1\ \text{m}}$$
$$= 1\ \text{k}\Omega$$

Therefore, the value of the resistance r_o is $1\ \text{k}\Omega$.

Step 2 of 4

Calculate the value of the transconductance, g_m.

$$g_m = \frac{I_C}{V_T}$$

Substitute $25\ \text{mV}$ for V_T and $1\ \text{mA}$ for I_C.

$$g_m = \frac{1\ \text{m}}{25\ \text{m}}$$
$$= 0.04\ \text{S}$$

Therefore, the value of the transconductance, g_m is, $0.04\ \text{S}$.

Calculate the value of the resistance, r_π.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute $0.04\ \text{S}$ for g_m and 100 for β.

$$r_\pi = \frac{100}{0.04}$$
$$= 2.5\ \text{k}\Omega$$

Therefore, the value of the resistance r_π is, $2.5\ \text{k}\Omega$.

Step 3 of 4

Calculate the value of the output resistance R_{out} when the source terminal is degenerated.

$$R_{out} = \left[1 + g_m\left(R_e \| r_\pi\right)\right] r_o$$

$$R_{out} = \left[1 + g_m\left(\frac{R_e \cdot r_\pi}{R_e + r_\pi}\right)\right] r_o \quad \text{..... (1)}$$

Substitute $2.5\ \text{k}\Omega$ for r_π, $0.04\ \text{S}$ for g_m, $10\ \text{k}\Omega$ for r_o and $500\ \Omega$ for R_e.

$$R_{out} = \left[1 + (0.04)\left(\frac{(500)\cdot(2.5\ \text{k})}{500 + 2.5\ \text{k}}\right)\right](10\ \text{k})$$
$$= \left[1 + (0.04)\left(\frac{(500)\cdot(2.5\ \text{k})}{3000}\right)\right](10\ \text{k})$$
$$= \left[1 + (0.04)(416.7)\right](10\ \text{k})$$
$$= \left[1 + 16.7\right](10\ \text{k})$$

$$R_{out} = (17.7)(10\ \text{k})$$
$$= 177\ \text{k}\Omega$$

Therefore, the value of the output resistance R_{out} when the source terminal is degenerated is $\boxed{177\ \text{k}\Omega}$.

Step 4 of 4

Calculate the value of the output resistance, R_{out} without source degeneration.

Recall equation (1).

$$R_{out} = \left[1 + g_m\left(\frac{R_e \cdot r_\pi}{R_e + r_\pi}\right)\right] r_o$$

Substitute 0 for R_e

$$R_{out} = \left[1 + g_m\left(\frac{(0)\cdot r_\pi}{0 + r_\pi}\right)\right] r_o$$
$$= \left[1 + 0\right] r_o$$
$$= r_o$$

Substitute $10\ \text{k}\Omega$ for r_o.

$$R_{out} = 10\ \text{k}\Omega$$

Therefore, the value of the output resistance R_{out} without source degeneration is $\boxed{10\ \text{k}\Omega}$.

8.016P

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Step 1 of 3

Refer to Figure 8.29 in the text book.

Calculate the expression for the output voltage v_o.

$$v_o = -(g_{m1}v_i)(Kr_{o1}) \quad \text{..... (1)}$$

Therefore, the expression for the output voltage v_o is $-(g_{m1}v_i)(Kr_{o1})$.

Step 2 of 3

Consider the expression for the voltage v_1.

$$v_1 = -(g_{m1}v_{gs1})r_{o1}$$

Substitute v_i for v_{gs1}.

$$v_1 = -(g_{m1}v_i)r_{o1}$$

Therefore, the expression for the voltage v_1 is $-(g_{m1}v_i)r_{o1}$.

Step 3 of 3

Apply nodal analysis at output terminal.

$$\frac{v_o - v_1}{r_{o2}} + g_{m2}v_{gs2} = 0$$

Substitute $-v_1$ for v_{gs2}.

$$\frac{v_o - v_1}{r_{o2}} - v_1 g_{m2} = 0$$
$$v_o - v_1 - v_1 g_{m2} r_{o2} = 0$$
$$v_o = v_1 + v_1 g_{m2} r_{o2}$$
$$v_o = v_1(1 + g_{m2}r_{o2})$$

Here, $g_{m2}r_{o2} >> 1$ and hence $1 + g_{m2}r_{o2} \approx g_{m2}r_{o2}$.

$$v_o = v_1(g_{m2}r_{o2})$$

Substitute $-(g_{m1}v_i)r_{o1}$ for v_1.

$$v_o = -(g_{m1}v_i)(g_{m2}r_{o2})r_{o1} \quad \text{..... (2)}$$

Compare the equations (1) and (2).

$$K = g_{m2}r_{o2}$$

Therefore, the approximate value for the factor K is $\boxed{g_{m2}r_{o2}}$.

Step 1 of 6

Refer to Figure 8.12(a) from the text book.

Consider the expression for transconductance g_{m1} in terms of the over drive voltage and the bias current.

$$g_{m1} = \frac{2I_D}{V_{OV}}$$

Substitute $100\ \mu\text{A}$ for I_D.

$$g_{m1} = \frac{2(100\ \mu)}{V_{OV}}$$

$$V_{OV} = \frac{200\ \mu}{g_{m1}}$$

Therefore, the expression for the over drive voltage V_{OV} is $\frac{200\ \mu}{g_{m1}}$.

Step 2 of 6

Consider the expression for the transconductance g_{m1}.

$$g_{m1} = \mu_n C_{ox}\left(\frac{W_1}{L_1}\right)V_{OV}$$

Substitute $\frac{200\ \mu}{g_m}$ for V_{OV}, $10\ \mu\text{m}$ for W_1, $0.5\ \mu\text{m}$ for L_1 and $500\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$.

$$g_{m1} = (500\ \mu)\left(\frac{10\ \mu}{0.5\ \mu}\right)\left(\frac{200\ \mu}{g_m}\right)$$

$$g_{m1}^2 = 2\ \mu$$

$$g_{m1} = \sqrt{2\ \mu}$$

$$g_{m1} = 1.414\ \text{mS}$$

Therefore, the value of g_{m1} is $1.414\ \text{mS}$.

Step 3 of 6

Calculate the value of the input resistance R_{in}.

$$R_{in} = \frac{1}{g_{m1}}$$

Substitute $1.414\ \text{mS}$ for g_{m1}.

$$R_{in} = \frac{1}{1.414\ \text{m}}$$
$$= 707.11\ \Omega$$

Therefore, the value of the input resistance R_{in} is $\boxed{707.11\ \Omega}$.

Step 4 of 6

Calculate the value of the current gain A_{is}.

$$A_{is} = \frac{(W_2/L_2)}{(W_1/L_1)}$$

Substitute $10\ \mu\text{m}$ for W_1, $50\ \mu\text{m}$ for W_2 and $0.5\ \mu\text{m}$ for L_1 and L_2.

$$A_{is} = \frac{(50\ \mu/0.5\ \mu)}{(10\ \mu/0.5\ \mu)}$$
$$= 5$$

Therefore, the value of the current gain A_{is} is $\boxed{5}$.

Step 5 of 6

Calculate the value of the output current I_O.

$$I_O = A_{is} I_{D1}$$

Substitute 5 for A_{is} and $100\ \mu\text{A}$ for I_D.

$$I_O = (5)(100\ \mu)$$
$$= 500\ \mu\text{A}$$

Therefore, the value of the output current I_O is $500\ \mu\text{A}$.

Step 6 of 6

Calculate the value of the output resistance R_{out}.

$$R_{out} = \frac{V_A' L_2}{I_O}$$

Substitute $10\ V/\mu\text{m}$ for V_A', $0.5\ \mu\text{m}$ for L_2 and $500\ \mu\text{A}$ for I_O.

$$R_{out} = \frac{(10)(0.5)}{500\ \mu}$$
$$= 10\ \text{k}\Omega$$

Therefore, the value of the output resistance R_{out} is $\boxed{10\ \text{k}\Omega}$.

Step 1 of 3

Refer to Figure 7.10 and Example 7.4 in the text book.

The changed value of L is,

$$L = \frac{1}{2} \times 0.55\ \mu m$$

Find the changed value of V_A.

$$\begin{aligned} V_A &= L\left|V_A'\right| \\ &= \frac{1}{2} \times 0.55 \times 10^{-6} \times 5 \times 10^{6} \\ &= 1.375\ V \end{aligned}$$

Step 2 of 3

The new value of output resistance R_o is,

$$R_o = \left(g_{m3} r_{o3}\right) r_{o4}$$

$$\Rightarrow R_o = \frac{\frac{\left|V_A\right|}{\left|V_{OV}\right|} \times \left|V_A\right|}{\frac{I_D}{2}}$$

Substitute 1.375 V for V_A and 0.3 V for $\left|V_{OV}\right|$.

$$\begin{aligned} R_o &= \frac{\frac{1.375}{0.3}}{\frac{2}{100 \times 10^{-6}}} \times \frac{1.375}{} \\ &= 126\ k\Omega \end{aligned}$$

Therefore, the output resistance R_o is $\boxed{126\ k\Omega}$.

Step 3 of 3

Write the expression of the drain current I_D.

$$I_D = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} \left|V_{OV}^2\right| \left(1 + \frac{V_{SD}}{\left|V_A\right|}\right)$$

Substitute $100\ \mu A$ for I_D, $90\ \mu A/V^2$ for $\mu_p C_{ox}$, 0.3 V for V_{SD}, 1.375 V for V_A and 0.3 V for $\left|V_{OV}\right|$.

Find the value of $\frac{W}{L}$.

$$\begin{aligned} \frac{W}{L} &= \frac{I_D}{\frac{1}{2} \times \mu_p C_{ox} \times \left|V_{OV}^2\right| \left(1 + \frac{V_{SD}}{\left|V_A\right|}\right)} \\ &= \frac{100 \times 10^{-6}}{\frac{1}{2} \times 90 \times 10^{-6} \times (0.3)^2 \times \left(1 + \frac{0.3}{1.375}\right)} \\ &= 20.26 \end{aligned}$$

Thus, the value of $\frac{W}{L}$ is $\boxed{20.3}$.

Step 1 of 6

Refer to Figure 8.12(a) from the text book.

Calculate the value of the output current I_O.

$$I_O = A_{is} I_{D1}$$

Substitute 4 for A_{is} and $200\ \mu A$ for I_{D1}.

$$I_O = (4)(200\ \mu)$$
$$= 800\ \mu A$$

Therefore, the value of the output current I_O is $800\ \mu A$.

Step 2 of 6

Calculate the value of the output resistance R_{out}.

$$R_{out} = \frac{V_A' L}{I_O}$$

Substitute $20\ V/\mu m$ for V_A', $20\ k\Omega$ for R_{out} and $800\ \mu A$ for I_O.

$$20\ k = \frac{\left(\frac{20}{1\mu}\right)(L)}{800\ \mu}$$
$$L = \frac{(20\ k)(800\ \mu)(1\ \mu)}{20}$$
$$L = \frac{(20\ k)(800\ \mu)(1\ \mu)}{20}$$
$$L = 0.8\ \mu m$$

Therefore, the value of the length L is $\boxed{0.8\ \mu m}$.

Step 3 of 6

Consider the expression for the input resistance.

$$R_{in} = \frac{1}{g_{m1}}$$

Substitute $500\ \Omega$ for R_{in}.

$$500 = \frac{1}{g_{m1}}$$
$$g_{m1} = \frac{1}{500}$$
$$g_{m1} = 2\ mS$$

Therefore, the value of the transconductance g_{m1} is $2\ mS$.

Step 4 of 6

Consider the expression for transconductance g_{m1} in terms of the over drive voltage and the bias current.

$$g_{m1} = \frac{2I_D}{V_{OV}}$$

Substitute $100\ \mu A$ for I_D and $2\ mA$ for g_{m1}.

$$2\ m = \frac{2(200\ \mu)}{V_{OV}}$$
$$V_{OV} = \frac{400\ \mu}{2\ m}$$
$$V_{OV} = 0.2\ V$$

Therefore, the value of the over drive voltage V_{OV} is $0.2\ V$.

Step 5 of 6

Consider the expression for the transconductance g_{m1}.

$$g_{m1} = \mu_n C_{ox} \left(\frac{W_1}{L_1}\right) V_{OV}$$

Substitute $0.2\ V$ for V_{OV}, $2\ mA$ for g_{m1}, $0.8\ \mu m$ for L_1 and $400\ \mu A/V^2$ for $\mu_n C_{ox}$.

$$2\ m = (400\ \mu)\left(\frac{W_1}{0.8\ \mu}\right)(0.2)$$
$$W_1 = \frac{(2\ m)(0.8\ \mu)}{(400\ \mu)(0.2)}$$
$$W_1 = 20\ \mu m$$

Therefore, the value of the width W_1 is $\boxed{20\ \mu m}$.

Step 6 of 6

Calculate the value of the current gain A_{is}.

$$A_{is} = \frac{(W_2/L_2)}{(W_1/L_1)}$$

Substitute $20\ \mu m$ for W_1, 4 for A_{is} and L_1 for L_2.

$$4 = \frac{(W_2/L_2)}{(20\ \mu/L_2)}$$
$$4 = \frac{W_2}{(20\ \mu)}$$
$$W_2 = (4)(20\ \mu)$$
$$W_2 = 80\ \mu m$$

Therefore, the value of the width W_2 is $\boxed{80\ \mu m}$.

Step 1 of 2

8.019E

Refer to Figure 7.11 in the text book.

Write the expression for the voltage V_{OV} .

$$V_{OV} = V_{DD} - V_{G4} - V_t$$

Substitute 1.8 V for V_{DD} , 1.1 V for V_{G4} and 0.5 V for V_t .

$$\begin{aligned} V_{OV} &= 1.8 - 1.1 - 0.5 \\ &= 1.8 - 1.6 \\ &= 0.2\text{ V} \end{aligned}$$

Thus, the overdrive voltage at which the four transistors are operating is $\boxed{0.2\text{ V}}$.

Step 2 of 2

Find the allowable range of voltage at the output.

The minimum voltage at the output is,

$$\begin{aligned} v_{o\min} &= V_t \\ &= 0.5\text{ V} \end{aligned}$$

The maximum voltage at the output is,

$$\begin{aligned} v_{o\max} &= V_{DD} - V_t \\ &= 1.8 - 0.5 \\ &= 1.3\text{ V} \end{aligned}$$

Thus, the allowable range of voltage at the output is $\boxed{0.5\text{ V to }1.3\text{ V}}$.

Step 1 of 8

Refer to Figure P8.19 in the text book.

Draw the small signal equivalent circuit diagram.

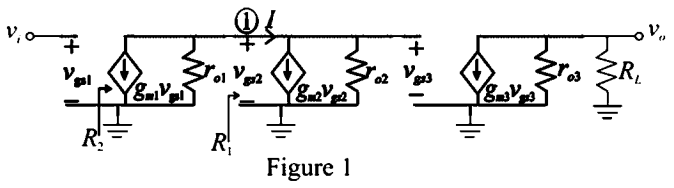


Figure 1

Step 2 of 8

Apply nodal analysis at node-1.

$$\frac{v_{gs3}}{r_{o2}}+\frac{v_{gs3}}{r_{o1}}+g_{m1}v_{gs1}+g_{m2}v_{gs3}=0 \quad \text{..... (1)}$$

$$v_{gs3}\left(\frac{1}{r_{o2}}+\frac{1}{r_{o1}}+g_{m2}\right)+g_{m1}v_{gs1}=0$$

$$v_{gs3}\left(\frac{1}{r_{o2}\,\|\,r_{o1}\,\|\,\frac{1}{g_{m2}}}\right)=-g_{m1}v_{gs1}$$

$$v_{gs3}=-g_{m1}\left(r_{o2}\,\|\,r_{o1}\,\|\,\frac{1}{g_{m2}}\right)v_{gs1}$$

Therefore, the value of the voltage v_{gs3} is $-g_{m1}\left(r_{o2}\,\|\,r_{o1}\,\|\,\frac{1}{g_{m2}}\right)v_{gs1}$.

Step 3 of 8

Consider the expression for the output voltage, v_o.

$$v_o=-g_{m3}v_{gs3}\left(R_L\,\|\,r_{o3}\right)$$

Substitute $-g_{m1}\left(r_{o2}\,\|\,r_{o1}\,\|\,\frac{1}{g_{m2}}\right)v_{gs1}$ for v_{gs3}.

$$v_o=-g_{m3}\left(R_L\,\|\,r_{o3}\right)\left(-g_{m1}v_{gs1}\right)$$

$$=g_{m1}g_{m3}\left(R_L\,\|\,r_{o3}\right)\left(r_{o2}\,\|\,r_{o1}\,\|\,\frac{1}{g_{m2}}\right)v_{gs1}$$

Substitute v_i for v_{gs1}.

$$v_o=g_{m1}g_{m3}\left(R_L\,\|\,r_{o3}\right)\left(r_{o2}\,\|\,r_{o1}\,\|\,\frac{1}{g_{m2}}\right)v_i \quad \text{..... (2)}$$

Therefore, the expression for the output voltage v_o is,

$$\boxed{g_{m1}g_{m3}\left(R_L\,\|\,r_{o3}\right)\left(r_{o2}\,\|\,r_{o1}\,\|\,\frac{1}{g_{m2}}\right)v_i}.$$

Step 4 of 8

Recall equation (2).

$$v_o=g_{m1}g_{m3}\left(R_L\,\|\,r_{o3}\right)\left(r_{o2}\,\|\,r_{o1}\,\|\,\frac{1}{g_{m2}}\right)v_i$$

$$\frac{v_o}{v_i}=g_{m1}g_{m3}\left(R_L\,\|\,r_{o3}\right)\left(r_{o2}\,\|\,r_{o1}\,\|\,\frac{1}{g_{m2}}\right)$$

Therefore, the expression for the voltage gain is $\boxed{g_{m1}g_{m3}\left(R_L\,\|\,r_{o3}\right)\left(r_{o2}\,\|\,r_{o1}\,\|\,\frac{1}{g_{m2}}\right)}$.

Step 5 of 8

Assume that a current of I flows through the drain of the MOSFET Q_2.

Calculate the current I.

$$I=\frac{v_{gs2}}{r_{o2}}+g_{m2}v_{gs2}$$

$$=v_{gs2}\left(\frac{1}{r_{o2}}+g_{m2}\right)$$

$$=\frac{v_{gs2}}{\frac{1}{g_{m2}}\,\|\,r_{o2}}$$

Therefore, the expression for the current I is $\frac{v_{gs2}}{\frac{1}{g_{m2}}\,\|\,r_{o2}}$.

Step 6 of 8

Calculate the value of the resistance R_1.

$$R_1=\frac{v_{gs2}}{I}$$

Substitute $\frac{v_{gs2}}{\frac{1}{g_{m2}}\,\|\,r_{o2}}$ for I.

$$R_1=\frac{v_{gs2}}{\frac{v_{gs2}}{\frac{1}{g_{m2}}\,\|\,r_{o2}}}$$

$$=\frac{1}{g_{m2}}\,\|\,r_{o2}$$

The value $\frac{1}{g_{m2}}\,\|\,r_{o2}$ is equal to $\frac{1}{g_{m2}}$ since $g_{m2}r_{o2}>>1$.

$$R_1=\frac{1}{g_{m2}}$$

Therefore, the value of the resistance R_1 is $\boxed{\frac{1}{g_{m2}}}$.

Step 7 of 8

Calculate the value of the resistance R_2.

$$R_2=r_{o1}\,\|\,R_1$$

Substitute $\frac{1}{g_{m2}}$ for R_1.

$$R_2=r_{o1}\,\|\,\frac{1}{g_{m2}}$$

Therefore, the expression for the resistance R_2 is $\boxed{r_{o1}\,\|\,\frac{1}{g_{m2}}}$.

Step 8 of 8

Apply nodal analysis at node 1.

$$\frac{v_{gs3}}{r_{o2}}+\frac{v_{gs3}}{r_{o1}}+g_{m1}v_{gs1}+g_{m2}v_{gs3}=0$$

Substitute ∞ for r_{o1} and r_{o2}.

$$\frac{v_{gs3}}{\infty}+\frac{v_{gs3}}{\infty}+g_{m1}v_{gs1}+g_{m2}v_{gs2}=0$$

$$0+0+g_{m1}v_{gs1}+g_{m2}v_{gs2}=0$$

$$g_{m1}v_{gs1}=-g_{m2}v_{gs2}=0$$

$$\frac{v_{gs2}}{v_{gs2}}=-\frac{g_{m1}}{g_{m2}}$$

Therefore, the voltage gain for the transistor Q_1 is $\boxed{-\frac{g_{m1}}{g_{m2}}}$.

Step 1 of 5

Refer to Figure 7.11 in the text book.

Write the expression for the trans-conductance g_{m1}.

$$g_{m1} = \frac{I_D}{V_{OV}/2}$$

Substitute $0.2\ \text{V}$ for V_{OV} and 0.2 mA for I_D

$$\begin{aligned} g_{m1} &= \frac{0.2\times10^{-3}}{\frac{0.2}{2}} \\ &= 2\times10^{-3} \\ &= 2\ \text{mA/V} \end{aligned}$$

Thus, the value of g_{m1} is $\boxed{2\ \text{mA/V}}$.

Step 2 of 5

Determine the value for the resistance r_o.

$$\begin{aligned} r_o &= \frac{|V_A|}{I_D} \\ &= \frac{2\ \text{V}}{0.2\ \text{mA}} \\ &= 10\ \text{k}\Omega \end{aligned}$$

Determine the output resistance of the current source R_{op}.

$$\begin{aligned} R_{op} &= (g_{m3}r_{o3})r_{o4} \\ &= (2\ \text{mA/V})(10\ \text{k}\Omega)(10\ \text{k}\Omega) \\ &= 200\ \text{k}\Omega \end{aligned}$$

Therefore, the output resistance of the current source is $\boxed{200\ \text{k}\Omega}$.

Step 3 of 5

Determine the output resistance of the amplifier R_{on}.

$$\begin{aligned} R_{on} &= (g_{m1}r_{o1})r_{o2} \\ &= \left(2\frac{\text{mA}}{\text{V}}\right)(10\ \text{k}\Omega)(10\ \text{k}\Omega) \\ &= 200\ \text{k}\Omega \end{aligned}$$

Therefore, the output resistance of the amplifier is $\boxed{200\ \text{k}\Omega}$.

Step 4 of 5

Determine the overall output resistance R_o.

$$\begin{aligned} R_o &= R_{on} \parallel R_{op} \\ &= \frac{(200\ \text{k}\Omega)(200\ \text{k}\Omega)}{200\ \text{k}\Omega + 200\ \text{k}\Omega} \\ &= \frac{200}{2}\ \text{k}\Omega \\ &= 100\ \text{k}\Omega \end{aligned}$$

Therefore, the overall output resistance is $\boxed{100\ \text{k}\Omega}$.

Step 5 of 5

Write the expression for the voltage gain of the amplifier.

$$\begin{aligned} A_v &= \frac{v_o}{v_i} \\ &= -g_{m1}\left[R_{on} \parallel R_{op}\right] \end{aligned}$$

Substitute $2\ \text{mA/V}$ for g_{m1}, $200\ \text{k}\Omega$ for R_{on} and $200\ \text{k}\Omega$ for R_{op}.

$$\begin{aligned} A_v &= -\left(2\times10^{-3}\right)\left[\frac{\left(200\times10^3\right)\left(200\times10^3\right)}{\left(200\times10^3\right)+\left(200\times10^3\right)}\right] \\ &= -\left(2\times10^{-3}\right)\left(100\times10^3\right) \\ &= -200\ \text{V/V} \end{aligned}$$

Therefore, the voltage gain of the amplifier is $\boxed{-200\ \text{V/V}}$.

Step 1 of 6

Refer to Figure P8.20 in the text book.

Draw the small signal equivalent circuit diagram.

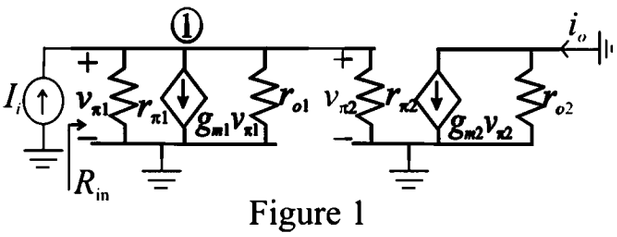


Figure 1

Step 2 of 6

Apply nodal analysis at node-1.

$$\frac{v_{\pi 1}}{r_{\pi 2}}+\frac{v_{\pi 1}}{r_{\pi 1}}+\frac{v_{\pi 1}}{r_{o1}}+g_{m1}v_{\pi 1}=i_i$$

$$v_{\pi 1}\left(\frac{1}{r_{\pi 2}}+\frac{1}{r_{\pi 1}}+\frac{1}{r_{o1}}+g_{m1}\right)=i_i$$

$$v_{\pi 1}\left(\frac{1}{r_{\pi 2}\parallel r_{\pi 1}\parallel r_{o1}\parallel \frac{1}{g_{m1}}}\right)=i_i$$

$$\frac{v_{\pi 1}}{i_i}=r_{\pi 2}\parallel r_{\pi 1}\parallel r_{o1}\parallel \frac{1}{g_{m1}}$$

As $r_o >> r_\pi$, the value of the r_{o1} can be neglected.

$$\frac{v_{\pi 1}}{i_i}=r_{\pi 2}\parallel r_{\pi 1}\parallel \frac{1}{g_{m1}} \quad \text{..... (1)}$$

Here, $\frac{v_{\pi 1}}{i_i}$ represents the input resistance R_{in} .

$$R_{in}=r_{\pi 2}\parallel r_{\pi 1}\parallel \frac{1}{g_{m1}}$$

Therefore, the expression fro the input resistance R_{in} is $\boxed{r_{\pi 2}\parallel r_{\pi 1}\parallel \frac{1}{g_{m1}}}$.

Step 3 of 6

Observe that the resistance r_{o2} is shorted. Therefore, the value of the output current i_i is equal to the value $g_{m2}v_{\pi 2}$ that is $g_{m2}v_{\pi 1}$ since, $v_{\pi 2}=v_{\pi 1}$.

Recall equation (1).

$$\frac{v_{\pi 1}}{i_i}=r_{\pi 2}\parallel r_{\pi 1}\parallel \frac{1}{g_{m1}}$$

Substitute $\frac{i_o}{g_{m2}}$ for $v_{\pi 1}$.

$$\frac{\frac{i_o}{g_{m2}\cdot i_i}=r_{\pi 2}\parallel r_{\pi 1}\parallel \frac{1}{g_{m1}}}{\frac{i_o}{i_i}=g_{m2}\left(r_{\pi 2}\parallel r_{\pi 1}\parallel \frac{1}{g_{m1}}\right)}$$

Therefore, the expression fro the current gain $\frac{i_o}{i_i}$ is, $\boxed{g_{m2}\left(r_{\pi 2}\parallel r_{\pi 1}\parallel \frac{1}{g_{m1}}\right)}$.

Step 4 of 6

Disconnect the input source and apply a voltage of V volts at the output terminals supplying a current of I amps into the circuit.

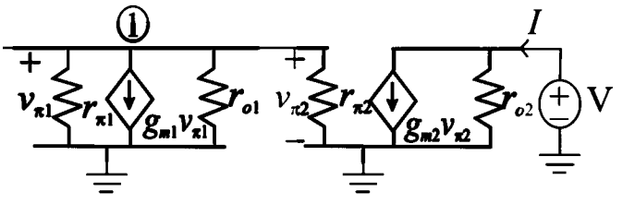


Figure 2

Step 5 of 6

Apply nodal analysis at node-1.

$$\frac{v_{\pi 2}}{r_{\pi 2}}+\frac{v_{\pi 2}}{r_{\pi 1}}+\frac{v_{\pi 2}}{r_{o1}}+g_{m1}v_{\pi 2}=0$$

$$v_{\pi 2}\left(\frac{1}{r_{\pi 2}}+\frac{1}{r_{\pi 1}}+\frac{1}{r_{o1}}+g_{m1}\right)=0$$

$$v_{\pi 2}=0\ \text{V}$$

Therefore, the value of the voltage $v_{\pi 2}$ is 0 V .

Step 6 of 6

Calculate the value of the current I .

$$I=\frac{V}{r_{o2}}+g_{m2}v_{\pi 2}$$

Substitute 0 V for $v_{\pi 2}$.

$$I=\frac{V}{r_{o2}}+g_{m2}(0)$$

$$I=\frac{V}{r_{o2}}+0$$

$$I=\frac{V}{r_{o2}}$$

$$\frac{V}{I}=r_{o2}$$

Here, $\frac{V}{I}$ represents the output resistance R_{out} .

$$R_{out}=r_{o2}$$

Therefore, the expression for the output resistance R_{out} is $\boxed{r_{o2}}$.

Step 1 of 6

Determine the transconductance of the CS and CG amplifiers.

$$\begin{aligned} g_m &= \frac{I_D}{\frac{|V_{OV}|}{2}} \\ &= \frac{0.1\text{ mA}}{\frac{0.2\text{ V}}{2}} \\ &= 1\text{ mA/V} \end{aligned}$$

Step 2 of 6

Determine the output resistance r_o.

$$\begin{aligned} r_o &= \frac{|V_A|}{I_D} \\ &= \frac{2\text{ V}}{0.1\text{ mA}} \\ &= 20\text{ k}\Omega \end{aligned}$$

The output resistances of the CS and CD amplifiers are,

$$\begin{aligned} r_{o1} &= r_o \\ &= 20\text{ k}\Omega \\ r_{o2} &= r_o \\ &= 20\text{ k}\Omega \end{aligned}$$

Step 3 of 6

(a)

The load resistance R_L is,

$$R_L = r_o$$

Refer to Table 7.1 in the text book.

The gain of the amplifier A_{v1} for $R_L = r_o$ is,

$$A_{v1} = -2$$

Thus, the gain of the amplifier A_{v1} is $\boxed{-2\text{ V/V}}$.

Write expression for the gain A_{v2} for $R_L = r_o$.

$$A_{v2} = \frac{1}{2}(g_m r_o)$$

Substitute 1 mA/V for g_m and $20\text{ k}\Omega$ for r_o.

$$\begin{aligned} A_{v2} &= \frac{1}{2}\left(1\times10^{-3}\right)\left(20\times10^{3}\right) \\ &= 10\text{ V/V} \end{aligned}$$

Thus, the gain A_{v2} is $\boxed{10\text{ V/V}}$.

Step 4 of 6

Determine the overall gain A_v.

$$\begin{aligned} A_v &= A_{v1}A_{v2} \\ &= (-2)(10) \\ &= -20\text{ V/V} \end{aligned}$$

Therefore, the overall gain A_v of the cascode amplifier is $\boxed{-20\text{ V/V}}$.

Step 5 of 6

(b)

Determine the value of $(g_m r_o) r_o$.

$$\begin{aligned} (g_m r_o) r_o &= \left(\left(1\times10^{-3}\right)\left(20\times10^{3}\right)\right)\left(20\times10^{3}\right) \\ &= 400\times10^{3} \\ &= 400\text{ k}\Omega \end{aligned}$$

The load resistance R_L is,

$$R_L = (g_m r_o) r_o$$

Refer to Table 7.1 in the text book.

Write the expression for the gain of the amplifier A_{v1} for $R_L = (g_m r_o) r_o$.

$$A_{v1} = -\frac{1}{2}(g_m r_o)$$

Substitute 1 mA/V for g_m and $20\text{ k}\Omega$ for r_o.

$$\begin{aligned} A_{v1} &= -\frac{1}{2}\left(1\times10^{-3}\right)\left(20\times10^{3}\right) \\ &= -10\text{ V/V} \end{aligned}$$

Thus, the gain of the amplifier A_{v1} is $\boxed{-10\text{ V/V}}$.

Write expression for the gain A_{v2} for $R_L = (g_m r_o) r_o$.

$$A_{v2} = g_m r_o$$

Substitute 1 mA/V for g_m and $20\text{ k}\Omega$ for r_o.

$$\begin{aligned} A_{v2} &= \left(1\times10^{-3}\right)\left(20\times10^{3}\right) \\ &= 20\text{ V/V} \end{aligned}$$

Thus, the gain A_{v2} is $\boxed{20\text{ V/V}}$.

Step 6 of 6

Determine the overall gain A_v.

$$\begin{aligned} A_v &= A_{v1}A_{v2} \\ &= (-10)(20) \\ &= -200\text{ V/V} \end{aligned}$$

Therefore, the overall gain A_v of the cascode amplifier is $\boxed{-200\text{ V/V}}$.

Step 1 of 8

(a)

Refer to Figure P8.21 (a) in the text book.

Draw the small signal equivalent circuit diagram.

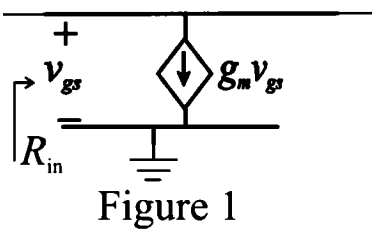


Figure 1

Step 2 of 8

Consider the expression for the input resistance R_{in} .

$$R_{in} = \frac{v_{gs}}{i}$$

Substitute $g_m v_{gs}$ for i .

$$R_{in} = \frac{v_{gs}}{g_m v_{gs}}$$

$$R_{in} = \frac{1}{g_m} \quad \text{..... (1)}$$

Therefore, the expression for the input resistance is $\frac{1}{g_m}$.

Step 3 of 8

Calculate the value of the transconductance, g_m .

$$g_m = u_n C_{ox}\left(\frac{W}{L}\right)V_{ov}$$

Substitute $\frac{2I_D}{g_m}$ for V_{ov} .

$$g_m = u_n C_{ox}\left(\frac{W}{L}\right)\left(\frac{2I_D}{g_m}\right)$$

$$g_m^2 = 2u_n C_{ox}\left(\frac{W}{L}\right)(I_D)$$

$$g_m = \sqrt{2u_n C_{ox}\left(\frac{W}{L}\right)(I_D)}$$

Substitute $200\ \mu\text{A/V}^2$ for $u_n C_{ox}$, 10 for $\frac{W}{L}$ and $0.1\ \text{mA}$ for I_D .

$$\begin{aligned} g_m &= \sqrt{2(200\ \mu)(10)(0.1\,\text{m})} \\ &= \sqrt{0.4\ \mu} \\ &= 0.633\ \text{mS} \end{aligned}$$

Therefore, the value of the transconductance g_m is $0.633\ \text{mS}$.

Step 4 of 8

Recall equation (1).

$$R_{in} = \frac{1}{g_m}$$

Substitute $0.633\ \text{mS}$ for g_m .

$$\begin{aligned} R_{in} &= \frac{1}{0.633\ \text{m}} \\ &= 1.58\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the input resistance R_{in} is $\boxed{1.58\ \text{k}\Omega}$.

Step 5 of 8

(b)

Refer to Figure P8.21 (b) in the text book.

Draw the small signal equivalent circuit diagram.

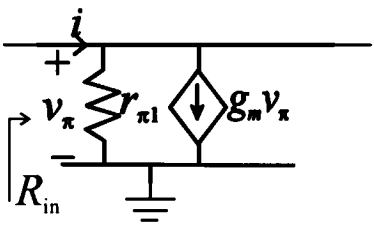


Figure 2

Step 6 of 8

Consider the expression for the input resistance, R_{in} .

$$R_{in} = \frac{v_\pi}{i}$$

Substitute $g_m v_\pi + \frac{v_\pi}{r_\pi}$ for i .

$$\begin{aligned} R_{in} &= \frac{v_\pi}{g_m v_\pi + \frac{v_\pi}{r_\pi}} \\ &= \frac{v_\pi}{v_\pi\left(g_m + \frac{1}{r_\pi}\right)} \\ &= \frac{1}{g_m + \frac{1}{r_\pi}} \end{aligned}$$

Substitute $\frac{\beta}{g_m}$ for r_π .

$$\begin{aligned} R_{in} &= \frac{1}{g_m + \frac{1}{\frac{\beta}{g_m}}} \\ &= \frac{1}{g_m + \frac{g_m}{\beta}} \\ &= \frac{\beta}{g_m\beta + g_m} \\ &= \frac{\beta}{g_m(\beta+1)} \end{aligned}$$

Here, $\beta + 1 << \beta$

$$R_{in} = \frac{1}{g_m} \quad \text{..... (2)}$$

Therefore, the expression for the input resistance is $\frac{1}{g_m}$.

Step 7 of 8

Calculate the value of the transconductance g_m .

$$g_m = \frac{I_C}{V_T}$$

Substitute $25\ \text{mV}$ for V_T and $0.1\ \text{mA}$ for I_D .

$$\begin{aligned} g_m &= \frac{0.1\ \text{m}}{25\ \text{m}} \\ &= 4\ \text{mS} \end{aligned}$$

Therefore, the value of the transconductance g_m is $4\ \text{mS}$.

Step 8 of 8

Recall equation (2).

$$R_{in} = \frac{1}{g_m}$$

Substitute $4\ \text{mS}$ for g_m .

$$\begin{aligned} R_{in} &= \frac{1}{4\ \text{m}} \\ &= 250\ \Omega \end{aligned}$$

Therefore, the value of the input resistance R_{in} is $\boxed{250\ \Omega}$.

Step 1 of 6

Refer to Figure 7.16 in the text book.

(a)

Write the expression of the drain current.

$$I_D = \frac{1}{2}k_p\left(\frac{W}{L}\right)_p V_{ov}^2$$
$$= \frac{1}{2}k_n\left(\frac{W}{L}\right)_n V_{ov}^2$$

Determine the drain current of transistor Q_1.

$$I_{D1} = I_1 - I_2$$
$$I_{D1} = 2I - I$$
$$I_{D1} = I$$

Step 2 of 6

Equate drain currents of PMOS and NMOS transistors.

$$I_{D1} = I_{D2} \qquad (\text{since, } I_{D1} = I \text{ and } I_{D2} = I)$$
$$\frac{1}{2}k_n\left(\frac{W}{L}\right)_1 V_{ov}^2 = \frac{1}{2}k_p\left(\frac{W}{L}\right)_2 V_{ov}^2$$
$$k_n\left(\frac{W}{L}\right)_1 = k_p\left(\frac{W}{L}\right)_2$$
$$\frac{(W/L)_2}{(W/L)_1} = \frac{k_n}{k_p}$$

Simplify further.

$$\frac{(W/L)_2}{(W/L)_1} = \frac{k_n}{\frac{1}{4}k_n}$$
$$\frac{(W/L)_2}{(W/L)_1} = 4$$

Thus, the relationship of $(W/L)_2$ to $(W/L)_1$ is $\boxed{(W/L)_2 = 4(W/L)_1}$

Step 3 of 6

(b)

To allow for the largest possible signal swing at the output terminal, the minimum required voltage across Q_1, namely, $|V_{OV}|$ or 0.2 V.

Thus, voltage across current source is $\boxed{0.2\text{ V}}$.

If a peak to peak signal swing of 0.1 V is allowed at the drain of Q_1, the highest dc bias voltage that can be used at the node is,

$$V = 1.8 - 0.2 - \frac{0.1}{2}$$
$$= 1.8 - 0.2 - 0.05$$
$$= 1.55\text{ V}$$

Thus, the minimum dc voltage required across current source I_1 for proper operation is $\boxed{1.55\text{ V}}$.

Step 4 of 6

(c)

Determine the source to gate voltage of transistor Q_2.

$$V_{SG2} = V_{DV} - V_{tp}$$
$$= 0.2 - (-0.5)$$
$$= 0.7\text{ V}$$

Thus, the source to gate voltage of transistor Q_2 is $\boxed{0.7\text{ V}}$.

Determine the largest value to which gate of transistor Q_2 used.

$$V_{G2max} = V_{D1max} - V_{SG2}$$
$$V_{G2max} = 1.55 - 0.7$$
$$= 0.85\text{ V}$$

Thus, the largest value of V_{G2} is $\boxed{0.85\text{ V}}$.

Step 5 of 6

(d)

Determine the minimum required voltage across current source for proper operation.

$$V = 2V_{OV}$$
$$V = 2(0.2)$$
$$= 0.4\text{ V}$$

Thus, the minimum dc voltage required across current-source I_2 for proper operation is $\boxed{0.4\text{ V}}$.

Step 6 of 6

(e)

The minimum output voltage is equal to the minimum voltage across current source I_2. $V_{min} = 0.4\text{ V}$

Determine the maximum output voltage.

$$V_{max} = V_{G2max} - V_{tp}$$
$$= 0.85 - (-0.5)$$
$$V_{max} = 1.35\text{ V}$$

Thus, the allowable range of signal swing at the output is $\boxed{0.4\text{ V to }1.35\text{ V}}$.

8.022P

We don't have the solution to this problem yet.

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Refer to Figure 7.19 in the text book.

From the above circuit, we have

The expression for the voltage gain A_v is,

$$A_v = -g_{m1}\left(R_{on} \parallel R_{op}\right)$$

Substitute $(g_{m2}r_{o2})(r_{o1} \parallel r_{\pi 2})$ for R_{on} and $(g_{m3}r_{o3})(r_{o4} \parallel r_{\pi 3})$ for R_{op}.

$$\begin{aligned}
A_v &= -g_{m1}\left(\left[(g_{m3}r_{o3})(r_{o4} \parallel r_{\pi 3})\right] \parallel \left[(g_{m2}r_{o2})(r_{o1} \parallel r_{\pi 2})\right]\right) \\
&= -g_{m1}\left(\left[(g_{m3}r_{o3})(r_{\pi 3})\right] \parallel \left[(g_{m2}r_{o2})(r_{\pi 2})\right]\right) \qquad (\text{Since}, r_o \gg r_\pi) \\
&= -g_{m1}\left(\left[(g_{m3}r_{\pi 3})(r_{o3})\right] \parallel \left[(g_{m2}r_{\pi 2})(r_{o1})\right]\right) \\
&= -g_{m1}\left(\left[\beta_3 (r_{o3})\right] \parallel \left[\beta_2 (r_{o2})\right]\right) \qquad (\text{Since}, \beta = g_m r_\pi \text{ and } r_{o1} = r_{o2})
\end{aligned}$$

Therefore, maximum voltage gain $\left|A_v\right|_{max}$ is $\boxed{g_{m1}\left(\beta_2 r_{o2} \parallel \beta_3 r_{o3}\right)}$.

8.023P

We don't have the solution to this problem yet.

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Step 1 of 8

Refer to Figure 6.19 in the text book.

Find the trans-conductance, g_{m2} of the *npn* transistor.

$$g_{m2} = \frac{I_C}{V_T}$$
$$= \frac{0.2\times10^{-3}}{25\times10^{-3}}$$
$$= 8\ \text{mA/V}$$

Find the value of $r_{\pi 2}$.

$$r_{\pi 2} = \frac{\beta}{g_{m_2}}$$
$$= \frac{100}{8\times10^{-3}}$$
$$= 12.5\,\text{k}\Omega$$

Step 2 of 8

Find the value of r_{o2}.

$$r_{o2} = \frac{V_A}{I_C}$$
$$= \frac{5}{0.2\times10^{-3}}$$
$$= 25\ \text{k}\Omega$$

Find the value of r_{o1}.

$$r_{o1} = \frac{V_A}{I_C}$$
$$= \frac{5}{0.2\times10^{-3}}$$
$$= 25\ \text{k}\Omega$$

Step 3 of 8

The output resistance for *npn* transistor is,

$$R_{on} = (g_{m2}r_{o2})(r_{o1} \| r_{\pi 2})$$

Substitute $25\ \text{k}\Omega$ for r_{o1}, $25\ \text{k}\Omega$ for r_{o2}, $8\ \text{mA/V}$ for g_{m2} and $12.5\ \text{k}\Omega$ for $r_{\pi 2}$.

$$R_{on} = (g_{m2}r_{o2})(r_{o1} \| r_{\pi 2})$$
$$= \left(8\times10^{-3}\times25\times10^{3}\right)\left(\frac{25\times10^{3}\times12.5\times10^{3}}{25\times10^{3}\times12.5\times10^{3}}\right)$$
$$= (200)\left(\frac{312.5\times10^{6}}{37.5\times10^{3}}\right)$$
$$= 1.67\ \text{M}\Omega$$

Therefore, the output resistance R_{on} for *npn* transistor is $\boxed{1.67\ \text{M}\Omega}$.

Step 4 of 8

Find the trans-conductance g_{m3} of the *pnp* transistor.

$$g_{m3} = \frac{I_C}{V_T}$$
$$= \frac{0.2\times10^{-3}}{25\times10^{-3}}$$
$$= 8\ \text{mA/V}$$

Find the value of $r_{\pi 3}$.

$$r_{\pi 3} = \frac{\beta}{g_{m3}}$$
$$= \frac{50}{8\times10^{-3}}$$
$$r_{\pi 3} = 6.25\,\text{k}\Omega$$

Step 5 of 8

Find the value of r_{o3}.

$$r_{o3} = \frac{V_A}{I_C}$$
$$= \frac{4}{0.2\times10^{-3}}$$
$$= 20\ \text{k}\Omega$$

Find the value of r_{o4}.

$$r_{o4} = \frac{V_A}{I_C}$$
$$= \frac{4}{0.2\times10^{-3}}$$
$$= 20\ \text{k}\Omega$$

Step 6 of 8

The output resistance for *pnp* transistor is,

$$R_{op} = (g_{m3}r_{o3})(r_{o4} \| r_{\pi 3})$$

Substitute $20\ \text{k}\Omega$ for r_{o1}, $20\ \text{k}\Omega$ for r_{o2}, $8\ \text{mA/V}$ for g_{m2} and $6.25\ \text{k}\Omega$ for $r_{\pi 2}$.

$$R_{op} = (g_{m3}r_{o3})(r_{o4} \| r_{\pi 3})$$
$$= \left(8\times10^{-3}\times20\times10^{3}\right)\left(\frac{20\times10^{3}\times6.25\times10^{3}}{20\times10^{3}+6.25\times10^{3}}\right)$$
$$= 0.762\times10^{6}$$
$$= 0.762\ \text{M}\Omega$$

Therefore, the output resistance R_{op} of the *pnp* transistor is $\boxed{0.762\ \text{M}\Omega}$.

Step 7 of 8

Write the expression for the voltage gain, A_v.

$$A_v = -g_{m1}\left(R_{on} \| R_{op}\right)$$

Substitute $g_{m1} = 8\ \text{mA/V}$, $R_{op} = 0.762\ \text{M}\Omega$ and $R_{on} = 1.67\ \text{M}\Omega$.

$$A_v = -8\times10^{-3}\times\left(\frac{1.67\times10^{6}\times0.762\times10^{6}}{1.67\times10^{6}+0.762\times10^{6}}\right)$$
$$= -8\times10^{-3}\times\frac{1.27254\times10^{12}}{2.432\times10^{6}}$$
$$= -8\times10^{-3}\times0.523\times10^{6}$$
$$= -4186$$

Hence, the voltage gain A_v of the BJT cascode amplifier is $\boxed{-4186\ \text{V/V}}$.

Step 8 of 8

Write the expression for the maximum available gain A_{vmax}.

$$A_{vmax} = -g_m\left(\beta_2 r_{o2} \| \beta_3 r_{o3}\right)$$

Substitute $g_m = 8\ \text{mA/V}$, $\beta_2 = 100$, $\beta_3 = 50$, $r_{o2} = 25\ \text{k}\Omega$ and $r_{o3} = 20\ \text{k}\Omega$.

$$A_{vmax} = -8\times10^{-3}\times\left(100\times25\times10^{3} \| 50\times20\times10^{3}\right)$$
$$= -8\times10^{-3}\times\left(\frac{100\times25\times10^{3}\times50\times20\times10^{3}}{100\times25\times10^{3}+50\times20\times10^{3}}\right)$$
$$= -5714.28\ \text{V/V}$$

Therefore, the value of the maximum achievable gain A_{vmax} is $\boxed{-5714.28\ \text{V/V}}$.

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Step 1 of 3

Refer to Figure 7.32 in the text book for the cascade MOS current mirror circuit.

The drain current I_D is,

$$\begin{aligned} I_D &= I_{REF} \\ &= 100\ \mu A \end{aligned}$$

Write the expression for the drain current, I_D.

$$I_D = \frac{1}{2}(\mu_n C_{ox})\left(\frac{W}{L}\right)V_{OV}^2$$

Substitute $100\ \mu A$ for I_D, $387\ \mu A/V^2$ for $\mu_n C_{ox}$ and $\frac{3.6\ \mu m}{0.36\ \mu m}$ for $\frac{W}{L}$.

$$100\times10^{-6} = \frac{1}{2}(387\times10^{-6})\left(\frac{3.6}{0.36}\right)V_{OV}^2 \qquad (\text{since, } I_D = I_{REF})$$

$$\begin{aligned} 100 &= 1935V_{OV}^2 \\ V_{OV}^2 &= 0.0517 \\ V_{OV} &= 0.2273\ V \end{aligned}$$

The minimum output voltage of cascade MOS mirror network is,

$$\begin{aligned} V_O &= V_t + 2V_{OV} \\ &= 0.5 + 2(0.2273) \\ &= 0.5 + 0.4546 \\ &\approx 0.95\ V \end{aligned}$$

Thus, the value of V_O is $\boxed{0.95\ V}$.

Step 2 of 3

Determine the output resistance of the NMOS transistor.

$$\begin{aligned} r_o &= \frac{V_A' L}{I_O} \\ &= \frac{(5)(0.36)}{100\times10^{-6}} \\ &= 1.8\times10^4 \\ &= 18\ k\Omega \end{aligned}$$

Step 3 of 3

Determine the output resistance of the circuit.

$$\begin{aligned} R_o &= g_{m3}r_{o3}r_{o2} \\ &= \left(\frac{2I_D}{V_{OV}}\right)r_o^2 \qquad (\text{since, } r_{o3} = r_{o2} = r_o) \\ &= \left(\frac{2(100\times10^{-6})}{0.2273}\right)(18\times10^3)^2 \\ &= 285\ k\Omega \end{aligned}$$

Thus, the value of the output resistance is $\boxed{285\ k\Omega}$.

Step 1 of 14

Refer to circuit diagram in Fig 7.1(b) in the textbook.

Draw the small-signal equivalent circuit for the figure.

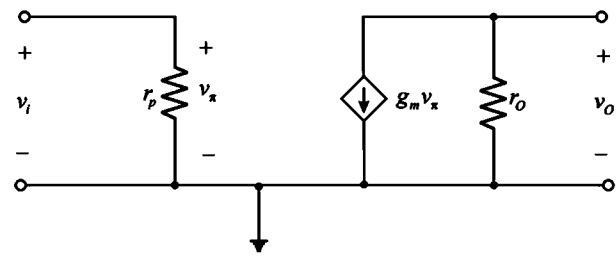


Figure 1

Step 2 of 14

At current $I = 10\mu A$:

Write the formula for trans-conductance.

$$g_m = \frac{I_C}{V_T}$$

Substitute 10×10^{-6} for I_C and 25×10^{-3} for V_T .

$$g_m = \frac{10 \times 10^{-6}}{25 \times 10^{-3}}$$
$$= 0.4 \text{ mA/V}$$

Step 3 of 14

Write the formula for input resistance.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute 100 for β and 0.4×10^{-3} for g_m .

$$r_\pi = \frac{100}{0.4 \times 10^{-3}}$$
$$= 250 \text{ k}\Omega$$

Step 4 of 14

Write the formula for output resistance.

$$r_o = \frac{V_A}{I_C}$$

Substitute 10 for V_A and 10×10^{-6} for I_C .

$$r_o = \frac{10}{10 \times 10^{-6}}$$
$$= 1 \text{ M}\Omega$$

Step 5 of 14

Write the formula for gain.

$$A_o = g_m r_o$$

Substitute 0.4×10^{-3} for g_m and 1×10^6 for r_o .

$$A_o = 0.4 \times 10^{-3} \times 1 \times 10^6$$
$$= 400 \text{ V/V}$$

Step 6 of 14

At current $I = 100\mu A$:

Write the formula for trans-conductance.

$$g_m = \frac{I_C}{V_T}$$

Substitute 100×10^{-6} for I_C and 25×10^{-3} for V_T .

$$g_m = \frac{100 \times 10^{-6}}{25 \times 10^{-3}}$$
$$= 4 \text{ mA/V}$$

Step 7 of 14

Write the formula for input resistance.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute 100 for β and 4×10^{-3} for g_m .

$$r_\pi = \frac{100}{4 \times 10^{-3}}$$
$$= 25 \text{ k}\Omega$$

Step 8 of 14

Write the formula for output resistance.

$$r_o = \frac{V_A}{I_C}$$

Substitute 10 for V_A and 100×10^{-6} for I_C .

$$r_o = \frac{10}{100 \times 10^{-6}}$$
$$= 0.1 \text{ M}\Omega$$

Step 9 of 14

Write the formula for gain.

$$A_o = g_m r_o$$

Substitute 4 mA/V for g_m and 0.1×10^6 for r_o

$$A_o = 4 \times 10^{-3} \times 0.1 \times 10^6$$
$$= 400 \text{ V/V}$$

Step 10 of 14

At current $I = 1\ \mu A$:

$$g_m = \frac{I_C}{V_T}$$

Substitute 1000×10^{-6} for I_C and 25×10^{-3} for V_T .

$$g_m = \frac{1000 \times 10^{-6}}{25 \times 10^{-3}}$$
$$= 40 \text{ mA/V}$$

Step 11 of 14

Write the formula for input resistance.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute 100 for β and 40×10^{-3} for g_m .

$$r_\pi = \frac{100}{40 \times 10^{-3}}$$
$$= 2.5 \text{ k}\Omega$$

Step 12 of 14

Write the formula for output resistance.

$$r_o = \frac{V_A}{I_C}$$

Substitute 10 for V_A and 1000×10^{-6} for I_C .

$$r_o = \frac{10}{1000 \times 10^{-6}}$$
$$= 0.01 \text{M}\Omega$$

Step 13 of 14

Write the formula for gain.

$$A_o = g_m r_o$$

Substitute 40×10^{-3} for g_m and 0.01×10^6 for r_o .

$$A_o = 40 \times 10^{-3} \times 0.01 \times 10^6$$
$$= 400 \text{ V/V}$$

Step 14 of 14

Thus, the values of small signal parameters g_m, r_π, r_o **and** A_o at collector currents $10\ \mu A$, $100\ \mu A$ **and** $1000 \mu A$ are listed in table 1.

Table 1

I_0 (in μA)	g_m (in A/V)	r_π (in kΩ)	r_o (in MΩ)	A_o (in V/V)
10	0.4	250	1	400
100	4	25	0.1	400
1000	40	2.5	0.01	400

Step 1 of 6

Refer to Figure 7.34 (a) in the text book.

Transfer ratio of the Wilson bipolar current mirror circuit is,

$$\begin{aligned}\frac{I_o}{I_{REF}} &= \frac{1}{1+\frac{2}{\beta(\beta+1)}} \\ &= \frac{1}{1+\frac{2}{100(100+1)}} \\ &= \frac{1}{1.000198} \\ &= 0.9998\end{aligned}$$

Step 2 of 6

Determine the percentage error in the transfer ratio.

$$\begin{aligned}\left(1-\frac{I_o}{I_{REF}}\right)\times 100 &= \frac{1-0.9998}{0.9998}\times 100 \\ &= \frac{0.0002}{0.9998}\times 100\end{aligned}$$

$$\left(1-\frac{I_o}{I_{REF}}\right)\times 100 = 0.02\% \quad \text{..... (1)}$$

Thus, the transfer ratio of the Wilson circuit is $\boxed{0.02\%}$.

Step 3 of 6

Determine the transfer ratio of a simple current mirror circuit.

$$\begin{aligned}\frac{I_o}{I_{REF}} &= \frac{1}{1+\frac{2}{\beta}} \\ &= \frac{1}{1+\frac{2}{100}} \\ &= \frac{1}{1+0.02} \\ &= 0.98039\end{aligned}$$

Step 4 of 6

Determine the percentage error in the transfer ratio.

$$\begin{aligned}\left(1-\frac{I_o}{I_{REF}}\right)\times 100 &= \frac{1-0.98039}{0.98039}\times 100 \\ &= \frac{0.01961}{0.98039}\times 100\end{aligned}$$

$$\left(1-\frac{I_o}{I_{REF}}\right)\times 100 = 2\% \quad \text{..... (2)}$$

Thus, the percentage error in the transfer ratio is $\boxed{2\%}$.

Hence, from equations (1) and (2), the transfer ratio error is 0.02% for Wilson as opposed to 2% for the simple circuit.

Step 5 of 6

Calculate the output resistance of a Wilson bipolar current mirror circuit.

$$\begin{aligned}R_o &= \frac{\beta_3 r_{o3}}{2} \\ &= \frac{100\times 100\times 10^3}{2} \qquad (\text{since, } \beta_3 = \beta \text{ and } r_{o3} = r_o) \\ &= 5\times 10^6\ \Omega \\ &= 5\ \text{M}\Omega\end{aligned}$$

Thus, the output resistance of a Wilson bipolar current mirror circuit is $\boxed{5\ \text{M}\Omega}$.

Step 6 of 6

Calculate the output resistance of a simple current mirror circuit.

$$\begin{aligned}R_o &= r_o \\ &= 100\times 10^3\ \Omega \\ &= 100\ \text{k}\Omega\end{aligned}$$

Thus, the output resistance of a simple current mirror circuit is $\boxed{100\ \text{k}\Omega}$.

Step 1 of 9

Refer to Figure 8.13 (b) in the text book.

Draw the small signal equivalent circuit diagram.

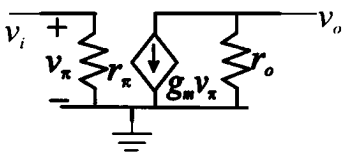


Figure 1

Step 2 of 9

Calculate the value of the transconductance g_m.

$$g_m = \frac{I_C}{V_T}$$

Substitute $0.5\ \text{mA}$ for I_C and $25\ \text{mA}$ for V_T.

$$g_m = \frac{0.5\ \text{m}}{25\ \text{m}}$$
$$= 0.02\ \text{S}$$

Therefore, the value of the transconductance g_m is $0.02\ \text{S}$.

Step 3 of 9

Calculate the value of the input resistance, R_{in}.

$$R_{in} = r_\pi$$

Substitute $\frac{\beta}{g_m}$ for r_π.

$$R_{in} = \frac{\beta}{g_m} \quad \ldots\ldots (1)$$

Substitute $0.02\ \text{S}$ for g_m and 100 for β.

$$R_{in} = \frac{100}{0.02}$$
$$= 5\ \text{k}\Omega$$

Therefore, the value of the input resistance R_{in} is $\boxed{5\ \text{k}\Omega}$.

Step 4 of 9

Calculate the value of the voltage gain.

$$A_{vo} = -g_m r_o$$

Substitute $\frac{V_A}{I_C}$ for r_o.

$$A_{vo} = -g_m\left(\frac{V_A}{I_C}\right)$$

Substitute $100\ \text{V}$ for V_A, $0.5\ \text{mA}$ for I_C and $0.02\ \text{S}$ for g_m.

$$A_{vo} = -(0.02)\left(\frac{100}{0.5\ \text{m}}\right)$$
$$= -4000$$

Therefore, the value of the voltage gain A_{vo} is $\boxed{-4000}$.

Step 5 of 9

Calculate the value of the output resistance, R_{out}.

$$R_{out} = r_o$$

Substitute $\frac{V_A}{I_C}$ for r_o.

$$R_{out} = \frac{V_A}{I_C}$$

Substitute $100\ \text{V}$ for V_A and $0.5\ \text{mA}$ for I_C.

$$R_{out} = \frac{100}{0.5\ \text{m}}$$
$$= 0.2\ \text{M}\Omega$$

Therefore, the value of the output resistance is $\boxed{0.2\ \text{M}\Omega}$.

Step 6 of 9

Recall equation (1).

$$R_{in} = \frac{\beta}{g_m}$$

Substitute $\frac{I_C}{V_T}$ for g_m.

$$R_{in} = \frac{\beta}{\left(\frac{I_C}{V_T}\right)}$$
$$= \frac{\beta V_T}{I_C}$$

The value of the input resistance R_{in} is inversely proportional to current I_C, that is,

$$R_{in} \alpha \frac{1}{I_C} \quad \ldots\ldots (2)$$

Step 7 of 9

Recall expression (2).

$$R_{in} \alpha \frac{1}{I_C}$$

Rewrite the expression for the two different cases.

$$\frac{R_{in2}}{R_{in1}} = \frac{I_{C1}}{I_{C2}}$$

Substitute $5R_{in1}$ for R_{in2}.

$$\frac{5R_{in1}}{R_{in1}} = \frac{I_{C1}}{I_{C2}}$$
$$5 = \frac{I_{C1}}{I_{C2}}$$
$$I_{C2} = \frac{I_{C1}}{5}$$
$$I_{C2} = 0.2 I_{C1}$$

Therefore, the value of the current I_C has to be changed to $\boxed{0.2}$ times the current I_C to attain the value of the input resistance 5 times its initial value.

Draw the small signal equivalent circuit diagram with resistance R_{sig} at the input terminal and the resistance R_L at the output terminal.

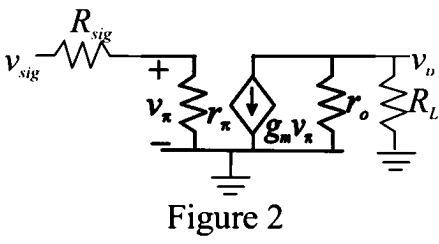


Figure 2

Step 8 of 9

Consider the expression for the voltage v_π.

$$v_\pi = v_{sig}\left(\frac{r_\pi}{r_\pi + R_{sig}}\right)$$

Therefore, the value of the voltage v_π is, $v_{sig}\left(\frac{r_\pi}{r_\pi + R_{sig}}\right)$.

Step 9 of 9

Consider the expression for the output voltage v_o.

$$v_o = -g_m v_\pi (r_o \| R_L)$$
$$= -g_m v_\pi \left(\frac{r_o \cdot R_L}{r_o + R_L}\right)$$

Substitute $v_{sig}\left(\frac{r_\pi}{r_\pi + R_{sig}}\right)$ for v_π.

$$v_o = -g_m\left(\frac{r_o \cdot R_L}{r_o + R_L}\right)\left(\frac{r_\pi}{r_\pi + R_{sig}}\right)(v_{sig})$$
$$\frac{v_o}{v_{sig}} = -g_m\left(\frac{r_o \cdot R_L}{r_o + R_L}\right)\left(\frac{r_\pi}{r_\pi + R_{sig}}\right)$$

Substitute $0.2\ \text{M}\Omega$ for r_o, $100\ \text{k}\Omega$ for R_L, $15\text{k}\Omega$ for R_{sig}, $5\ \text{k}\Omega$ for r_π and $0.02\ \text{S}$ for g_m.

$$\frac{v_o}{v_{sig}} = -(0.02)\left(\frac{(0.2\ \text{M})\cdot(100\ \text{k})}{0.2\ \text{M} + 100\ \text{k}}\right)\left(\frac{5\ \text{k}}{5\ \text{k} + 5\ \text{k}}\right)$$
$$= -(0.02)\left(\frac{(0.2\ \text{M})\cdot(100\ \text{k})}{210\ \text{k}}\right)\left(\frac{5\ \text{k}}{10\ \text{k}}\right)$$
$$= -(0.02)\left(\frac{(0.2\ \text{M})\cdot(100)}{210}\right)\left(\frac{1}{2}\right)$$
$$= -(0.01)(95.238\ \text{k})$$

$$\frac{v_o}{v_{sig}} = -952.38$$

Therefore, the value of the overall gain is $\boxed{-952.38}$.

Step 1 of 3

Refer to Example 7.6 and Figure 7.37 in the text book.

Determine the reference current for the matched transistors.

$$I_{REF} = I_O$$
$$= 10\ \mu A$$

Calculate the output resistance r_o.

$$r_o = \frac{V_A}{I_O}$$
$$= \frac{100}{10\times10^{-6}}$$
$$= 10\times10^{6}\ \Omega$$
$$= 10\ M\Omega$$

Thus, the output resistance of transistor Q_1 is $\boxed{10\ M\Omega}$.

Step 2 of 3

Calculate the trans-conductance g_m.

$$g_m = \frac{I_C}{V_T}$$
$$= \frac{10\times10^{-6}}{25\times10^{-3}}$$
$$= 0.4\times10^{-3}\ A/V$$
$$= 0.4\ mA/V$$

Calculate the resistance r_π.

$$r_\pi = \frac{\beta}{g_m}$$
$$= \frac{100}{0.4\times10^{-3}}$$
$$= 250\ k\Omega$$

The emitter resistance is,

$$R_E = R_3$$
$$= 11.5\ k\Omega$$

Step 3 of 3

Write the expression of the output resistance for the Widlar circuit as shown in Figure 7.37 (b) in the text book.

$$R_{out} \approx \left[1 + g_m\left(R_E \| r_\pi\right)\right] r_o$$

Substitute $R_E = 11.5\ k\Omega$, $r_\pi = 25\ k\Omega$, $r_o = 10\ M\Omega$ and $g_m = 0.4\ mA/V$.

$$R_{out} = \left[1 + \left(\frac{0.4}{1000}\right)\left(\frac{(11.5\times10^3)(250\times10^3)}{(11.5\times10^3)+(250\times10^3)}\right)\right](10\times10^6)$$
$$= \left[1 + \left(\frac{0.4}{1000}\right)\left(\frac{2.875\times10^6}{261.5\times10^3}\right)\right](10\times10^6)$$
$$= \left[1 + 4.396\right](10\times10^6)$$
$$\approx 54\ M\Omega$$

Thus, the value of R_{out} is $\boxed{54\ M\Omega}$.

Step 1 of 3

Calculate the value of the intrinsic gain A_o.

$$A_o = \frac{2V_A' L}{V_{OV}}$$

Substitute $0.2\ \text{V}$ for V_{OV}, $0.5\ \mu\text{m}$ for L and $10\ \text{V/}\mu\text{m}$ for V_A'.

$$A_o = \frac{2\left(\frac{10}{1\ \mu}\right)(0.5\ \mu)}{0.2}$$
$$= 50$$

Therefore, the value of the intrinsic gain is $\boxed{50}$.

Step 2 of 3

Calculate the value of the drain current I_D.

$$I_D = \frac{g_m V_{OV}}{2}$$

Substitute $2\ \text{mA/V}$ for g_m and $0.2\ \text{V}$ for V_{OV}.

$$I_D = \frac{(2\ \text{m})(0.2)}{2}$$
$$= 200\ \mu\text{A}$$

Therefore, the value of the drain current I_D is $\boxed{200\ \mu\text{A}}$.

Step 3 of 3

Consider the expression for the transconductance g_m.

$$g_m = k_n'\left(\frac{W}{L}\right)V_{OV}$$

Substitute $2\ \text{mA/V}$ for g_m, $400\ \mu\text{A/V}^2$ for k_n', $0.5\ \mu\text{m}$ for L and $0.2\ \text{V}$ for V_{OV}.

$$2\ \text{m} = (400\ \mu)\left(\frac{W}{0.5\ \mu}\right)(0.2)$$
$$W = \frac{(2\ \text{m})(0.5\ \mu)}{(400\ \mu)(0.2)}$$
$$W = 12.5\ \mu\text{m}$$

Therefore, the value of the width W is $\boxed{12.5\ \mu\text{m}}$.

Step 1 of 4

Refer to Figure 8.45(a) in the text book.

Calculate the value of the resistance r_{o1}.

$$r_{o1} = \frac{V_A}{I_D}$$

Substitute $5\ \text{V}$ for V_A and $200\ \mu\text{A}$ for I_D.

$$\begin{aligned} r_{o1} &= \frac{5}{200\ \mu} \\ &= 25\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance r_{o1} is $25\ \text{k}\Omega$.

Assume the value of the resistance r_{o3} is equal to resistance r_{o1}. Hence the value of the resistance r_{o3} is $25\ \text{k}\Omega$.

Step 2 of 4

Calculate the value of the transconductance g_m.

$$g_m = \frac{2I_D}{V_{OV}}$$

Substitute $0.2\ \text{V}$ for V_{DV} and $200\ \mu\text{A}$ for I_D.

$$\begin{aligned} g_m &= \frac{2(200\ \mu)}{0.2} \\ &= 2\ \text{mS} \end{aligned}$$

Therefore, the value of the transconductance g_m is $2\ \text{mS}$.

Step 3 of 4

Calculate the value of the resistance R_L.

$$R_L = r_{o1} \| r_{o3} \| \frac{1}{g_{mb}}$$

Substitute χg_m for g_{mb}.

$$R_L = r_{o1} \| r_{o3} \| \frac{1}{\chi g_m}$$

Substitute 0.2 for χ, $2\ \text{mS}$ for g_m and $25\ \text{k}\Omega$ for r_{o1} and r_{o3}.

$$\begin{aligned} R_L &= 25\ \text{k} \| 25\ \text{k} \| \frac{1}{(0.2)(2\ \text{m})} \\ &= 25\ \text{k} \| 25\ \text{k} \| \frac{1}{0.4\ \text{m}} \\ &= 25\ \text{k} \| 25\ \text{k} \| 2.5\ \text{k} \\ &= \frac{(25\ \text{k})(25\ \text{k})(2.5\ \text{k})}{(25\ \text{k})(25\ \text{k}) + (25\ \text{k})(2.5\ \text{k}) + (25\ \text{k})(2.5\ \text{k})} \end{aligned}$$

$$\begin{aligned} R_L &= \frac{1562.5\ \text{k}}{625 + 62.5 + 62.5} \\ &= \frac{1562.5\ \text{k}}{750} \\ &= 2.083\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance R_L is $2.083\ \text{k}\Omega$.

Step 4 of 4

Calculate the value of the voltage gain of the circuit.

$$\frac{v_o}{v_i} = \frac{R_L}{R_L + \frac{1}{g_m}}$$

Substitute $2.083\ \text{k}\Omega$ for R_L and $2\ \text{mS}$ for g_m.

$$\begin{aligned} \frac{v_o}{v_i} &= \frac{2.083\ \text{k}}{2.083\ \text{k} + \frac{1}{2\ \text{m}}} \\ &= \frac{2.083\ \text{k}}{2.083\ \text{k} + 500} \\ &= \frac{2.083\ \text{k}}{2.583\ \text{k}} \\ &= 0.81\ \text{V/V} \end{aligned}$$

Therefore, the value of the voltage gain is $\boxed{0.81\ \text{V/V}}$.

Step 1 of 7

Consider the expression for the drain current, I_D .

$$I_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)V_{OV}^2$$

$$V_{OV}^2 = \frac{I_D}{\frac{1}{2}k_n'\left(\frac{W}{L}\right)}$$

$$V_{OV} = \sqrt{\frac{I_D}{\frac{1}{2}k_n'\left(\frac{W}{L}\right)}}$$

$$V_{OV} = \sqrt{\frac{2I_DL}{k_n'W}}$$

Therefore, the expression for the overdrive voltage V_{OV} is $\sqrt{\frac{2I_DL}{k_n'W}}$.

Step 2 of 7

Consider the expression fro the intrinsic gain A_o .

$$A_o = \frac{2V_A'L}{V_{OV}}$$

Substitute $\sqrt{\frac{2I_DL}{k_n'W}}$ for V_{OV} .

$$A_o = \frac{2V_A'L}{\sqrt{\frac{2I_DL}{k_n'W}}}$$

$$= \frac{V_A'\sqrt{2k_n'WL}}{\sqrt{I_D}}$$

The value of the intrinsic gain A_o is inversely proportional to $\sqrt{I_D}$. That is,

$$A_o \alpha \frac{1}{\sqrt{I_D}} \quad \text{..... (1)}$$

Step 3 of 7

Recall expression (1).

$$A_o \alpha \frac{1}{\sqrt{I_D}}$$

Rewrite the equation for two different cases.

$$\frac{(A_o)_2}{(A_o)_1} = \frac{\left(\sqrt{I_D}\right)_1}{\left(\sqrt{I_D}\right)_2}$$

$$(A_o)_2 = (A_o)_1 \times \frac{\left(\sqrt{I_D}\right)_1}{\left(\sqrt{I_D}\right)_2} \quad \text{..... (2)}$$

Substitute $50\ \text{V/V}$ for $(A_o)_1$, $100\ \mu\text{A}$ for $\left(\sqrt{I_D}\right)_1$ and $25\ \mu\text{A}$ and $\left(\sqrt{I_D}\right)_2$.

$$(A_o)_2 = 50 \times \sqrt{\frac{100\ \mu}{25\ \mu}}$$

$$= 50 \times \sqrt{4}$$

$$= 50 \times 2$$

$$= 100$$

Therefore, the value of the intrinsic gain at $I_D = 25\ \mu\text{A}$ is $\boxed{100}$.

Step 4 of 7

Recall equation (2).

$$(A_o)_2 = (A_o)_1 \times \frac{\left(\sqrt{I_D}\right)_1}{\left(\sqrt{I_D}\right)_2}$$

Substitute $50\ \text{V/V}$ for $(A_o)_1$, $100\ \mu\text{A}$ for $\left(\sqrt{I_D}\right)_1$ and $400\ \mu\text{A}$ and $\left(\sqrt{I_D}\right)_2$.

$$(A_o)_2 = 50 \times \sqrt{\frac{100\ \mu}{400\ \mu}}$$

$$= 50 \times \sqrt{\frac{1}{4}}$$

$$= 50 \times \frac{1}{2}$$

$$= 25$$

Therefore, the value of the intrinsic gain at $I_D = 400\ \mu\text{A}$ is $\boxed{25}$.

Step 5 of 7

Consider the expression for the transconductance, g_m .

$$g_m = k_n'\left(\frac{W}{L}\right)V_{OV}$$

Substitute $\frac{2I_D}{g_m}$ for V_{OV} .

$$g_m^2 = k_n'\left(\frac{W}{L}\right)(2I_D)$$

$$g_m = \sqrt{k_n'\left(\frac{W}{L}\right)(2I_D)}$$

The value of g_m is directly proportional to $\sqrt{I_D}$.

$$g_m \alpha \sqrt{I_D} \quad \text{..... (3)}$$

Step 6 of 7

Recall expression (3).

$$g_m \alpha \sqrt{I_D}$$

Rewrite the equation for two different cases.

$$\frac{g_{m2}}{g_{m1}} = \frac{\sqrt{I_{D2}}}{\sqrt{I_{D1}}}$$

$$g_{m2} = g_{m1}\frac{\sqrt{I_{D2}}}{\sqrt{I_{D1}}} \quad \text{..... (4)}$$

Substitute $100\ \mu\text{A}$ for $\left(\sqrt{I_D}\right)_1$ and $25\ \mu\text{A}$ and $\left(\sqrt{I_D}\right)_2$.

$$g_{m2} = g_{m1}\sqrt{\frac{25\ \mu}{100\ \mu}}$$

$$= g_{m1}\sqrt{\frac{1}{4}}$$

$$= \frac{g_{m1}}{2}$$

$$= 0.5g_{m1}$$

Therefore, the value of the transconductance changes by the factor of $\boxed{0.5}$.

Step 7 of 7

Recall equation (4).

$$g_{m2} = g_{m1}\frac{\sqrt{I_{D2}}}{\sqrt{I_{D1}}} \quad \text{..... (4)}$$

Substitute $100\ \mu\text{A}$ for $\left(\sqrt{I_D}\right)_1$ and $400\ \mu\text{A}$ and $\left(\sqrt{I_D}\right)_2$.

$$g_{m2} = g_{m1}\sqrt{\frac{400\ \mu}{100\ \mu}}$$

$$= g_{m1}\sqrt{4}$$

$$= 2g_{m1}$$

Therefore, the value of the transconductance changes by the factor of $\boxed{2}$.

We don't have the solution to this problem yet.

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Step 1 of 3

Consider the expression for the intrinsic gain.

$$A_o = \frac{2V_A' L}{V_{OV}}$$

Substitute $0.2\ \text{V}$ for V_{OV}, $20\ \text{V/V}$ for A_o and $5\ \text{V/}\mu\text{m}$ for V_A'.

$$20 = \frac{2(5/1\ \mu)L}{0.2}$$

$$20 = \frac{(5/1\ \mu)L}{0.1}$$

$$L = \frac{2}{5}\ \mu$$

$$L = 0.4\ \mu\text{m}$$

Therefore, the value of the length L is $\boxed{0.4\ \mu\text{m}}$.

Step 2 of 3

Consider the expression for the transconductance, g_m.

$$g_m = k_n'\left(\frac{W}{L}\right)V_{OV}$$

Substitute $0.2\ \text{V}$ for V_{OV}, $2\ \text{mS}$ for g_m, $400\ \mu\text{A/V}^2$ for k_n' and $0.4\ \mu\text{m}$ for L.

$$2\ \text{m} = (400\ \mu)\left(\frac{W}{L}\right)(0.2)$$

$$2\ \text{m} = (80\ \mu)\left(\frac{W}{L}\right)$$

$$\frac{W}{L} = \frac{2\ \text{m}}{80\ \mu}$$

$$\frac{W}{L} = 25$$

Therefore, the value of the ratio $\frac{W}{L}$ is $\boxed{25}$.

Step 3 of 3

Consider the expression for the transconductance, g_m.

$$g_m = \frac{2I_D}{V_{OV}}$$

Substitute $2\ \text{mS}$ for g_m and $0.2\ \text{V}$ for V_{OV}.

$$2\ \text{m} = \frac{2I_D}{0.2}$$

$$2\ \text{m} = \frac{I_D}{0.1}$$

$$I_D = 0.2\ \text{mA}$$

Therefore, the value of the drain current I_D is $\boxed{0.2\ \text{mA}}$.

Step 1 of 5

8.030E

Refer to Figure 7.40 (b) in the text book for the voltage follower circuit using Darlington configuration.

The input resistance of a transistor is,

$$R_{in} = (\beta + 1)(r_e + R_E)$$

Here,

r_e is the small signal emitter resistance

R_E is the emitter resistance

The input resistance R_{in} of the Darlington circuit is,

$$\begin{aligned} R_{in} &= (\beta_1 + 1)(r_{e1} + R_{inQ_2}) \\ &= (\beta_1 + 1)\left[r_{e1} + (\beta_2 + 1)(r_{e2} + R_E)\right] \end{aligned}$$

Thus, the input resistance R_{in} of the circuit is $\boxed{(\beta_1 + 1)\left[r_{e1} + (\beta_2 + 1)(r_{e2} + R_E)\right]}$

Determine the output resistance, R_{out} .

$$\begin{aligned} R_{out} &= R_E \| \left[r_{e2} + R_{BQ_2}\right] \\ &= R_E \| \left[r_{e2} + \frac{r_{e1} + \left(\frac{R_{sig}}{\beta_1 + 1}\right)}{\beta_2 + 1}\right] \end{aligned}$$

Thus, the output resistance R_{out} of the circuit is $\boxed{R_E \| \left[r_{e2} + \frac{r_{e1} + \left(\frac{R_{sig}}{\beta_1 + 1}\right)}{\beta_2 + 1}\right]}$.

Step 2 of 5

Determine the voltage gain $\frac{v_o}{v_{sig}}$.

$$\begin{aligned} \frac{v_o}{v_{sig}} &= \frac{\text{total resistance at the output}}{r_{e2} + \text{total resistance at the input of } Q_2} \\ &= \frac{R_{out}}{r_{e2} + \frac{r_{e1} + \left(\frac{R_{sig}}{\beta_1 + 1}\right)}{\beta_2 + 1}} \\ &= \frac{R_E \| \left[r_{e2} + \frac{r_{e1} + \left(\frac{R_{sig}}{\beta_1 + 1}\right)}{\beta_2 + 1}\right]}{r_{e2} + \frac{r_{e1} + \left(\frac{R_{sig}}{\beta_1 + 1}\right)}{\beta_2 + 1}} \end{aligned}$$

Simplify further

$$\begin{aligned} \frac{v_o}{v_{sig}} &= \frac{R_E \| \left[r_{e2} + \frac{r_{e1} + \left(\frac{R_{sig}}{\beta_1 + 1}\right)}{\beta_2 + 1}\right]}{r_{e2} + \frac{r_{e1} + \left(\frac{R_{sig}}{\beta_1 + 1}\right)}{\beta_2 + 1}} \\ &= \frac{R_E \left[r_{e2} + \frac{r_{e1} + \left(\frac{R_{sig}}{\beta_1 + 1}\right)}{\beta_2 + 1}\right]}{\left[R_E + r_{e2} + \frac{r_{e1} + \left(\frac{R_{sig}}{\beta_1 + 1}\right)}{\beta_2 + 1}\right]\left[r_{e2} + \frac{r_{e1} + \left(\frac{R_{sig}}{\beta_1 + 1}\right)}{\beta_2 + 1}\right]} \end{aligned}$$

$$\frac{v_o}{v_{sig}} = \frac{R_E}{R_E + r_{e2} + \left[\frac{r_{e1} + \left(\frac{R_{sig}}{\beta_1 + 1}\right)}{\beta_2 + 1}\right]}$$

Thus, the voltage gain $\frac{v_o}{v_{sig}}$ is $\boxed{\frac{R_E}{R_E + r_{e2} + \left[\frac{r_{e1} + \left(\frac{R_{sig}}{\beta_1 + 1}\right)}{\beta_2 + 1}\right]}}$.

Step 3 of 5

Determine the resistance, r_{e2} .

$$\begin{aligned} r_{e2} &= \frac{V_T}{I_{E2}} \\ &= \frac{50 \times 10^{-3}}{5 \times 10^{-3}} \\ &= 10\ \Omega \end{aligned}$$

$$r_{e1} = r_{e2} = 10\ \Omega$$

Evaluate the input voltage R_{in} .

$$R_{in} = (\beta_1 + 1)\left[r_{e1} + (\beta_2 + 1)(r_{e2} + R_E)\right]$$

Substitute 100 for β_1 and β_2 , $10^3\ \Omega$ for R_E , $10\ \Omega$ for r_{e1} and r_{e2} .

$$\begin{aligned} R_{in} &= (100 + 1)\left[10 + (100 + 1)(10 + 10^3)\right] \\ &= 10.3 \times 10^6\ \Omega \\ &= 10.3\ \text{M}\Omega \end{aligned}$$

Thus, the input resistance R_{in} is $\boxed{10.3\ \text{M}\Omega}$.

Step 4 of 5

Determine the output resistance R_{out} .

$$\begin{aligned} R_{out} &= R_E \| \left[r_{e2} + \frac{r_{e1} + \left(\frac{R_{sig}}{\beta_1 + 1}\right)}{\beta_2 + 1}\right] \\ &= (10^3) \| \left[10 + \frac{10 + \left(\frac{100 \times 10^3}{100 + 1}\right)}{100 + 1}\right] \qquad R_{out} = 19.5\ \Omega \\ &= 10^3 \| 19.902 \qquad\qquad \approx 20\ \Omega \\ &= \frac{(10^3)(19.9)}{10^3 + 19.9} \end{aligned}$$

Thus, the output resistance R_{out} is $\boxed{20\ \Omega}$.

Step 5 of 5

Determine the voltage gain, $\frac{v_o}{v_{sig}}$.

$$\begin{aligned} \frac{v_0}{v_{sig}} &= \frac{R_E}{R_E + r_{e2} + \left[\frac{r_{e1} + \left(\frac{R_{sig}}{\beta_1 + 1}\right)}{\beta_2 + 1}\right]} \\ &= \frac{1000}{1000 + 10 + \left[\frac{10 + \left(\frac{100 \times 10^3}{100 + 1}\right)}{100 + 1}\right]} \\ &= \frac{1000}{1000 + 10 + 9.902} \\ &= 0.98\ \text{V/V} \end{aligned}$$

Thus, the voltage gain $\frac{v_0}{v_{sig}}$ is $\boxed{0.98\ \text{V/V}}$.

8.030P

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Step 1 of 3

8.031E

Refer to Figure 7.41 in the circuit for the amplifier circuits. Analysis of both the circuits is illustrated in Figure 7.42 in the text book. Both the circuits have the same input resistance and the same voltage gain.

Determine the resistance, r_e.

$$r_e = \frac{V_T}{2I}$$
$$= \frac{50\times10^{-3}}{2\times10^{-3}}$$
$$= 25\,\Omega$$

Determine the input resistance, R_{in}.

$$R_{in} = (\beta + 1)(2r_e)$$

Substitute 100 for β and $25\,\Omega$ for r_e.

$$R_{in} = (100 + 1)(2 \times 25)$$
$$= 5050\,\Omega$$
$$= 5.05\,k\Omega$$

Step 2 of 3

Determine α_2.

$$\alpha_2 = \frac{\beta}{\beta+1}$$
$$= \frac{100}{100+1}$$
$$= 0.99$$
$$\approx 1$$

Determine the ratio, $\frac{v_o}{v_i}$.

$$\frac{v_o}{v_i} = \frac{\alpha_2 R_L}{2r_e}$$

Substitute 1 for α_2, $5\times10^3\,\Omega$ for R_L and $25\,\Omega$ for r_e.

$$\frac{v_o}{v_i} = \frac{(1)(5\times10^3)}{2\times25}$$
$$= 100\ V/V$$

Thus, the voltage gain $\frac{v_o}{v_i}$ is $\boxed{100\ V/V}$.

Step 3 of 3

Determine the ratio, $\frac{v_o}{v_{sig}}$.

$$\frac{v_o}{v_{sig}} = \left(\frac{R_{in}}{R_{in} + R_{sig}}\right)\left(\frac{v_o}{v_i}\right)$$

Substitute $5050\,\Omega$ for R_{in}, $5\times10^3\,\Omega$ for R_{sig} and 100 V/V for $\frac{v_o}{v_i}$.

$$\frac{v_o}{v_{sig}} = \left(\frac{5050}{5050+5\times10^3}\right)(100)$$
$$\approx 50\ V/V$$

Thus, the gain $\frac{v_o}{v_{sig}}$ is $\boxed{50\ V/V}$.

Step 1 of 2

Calculate the value of the transconductance g_m of the *npn* transistor.

$$g_m = \frac{I_C}{V_T}$$

Substitute $25\ \text{mV}$ for V_T and $0.1\ \text{mA}$ for I_C .

$$g_m = \frac{0.1\ \text{m}}{25\ \text{m}}$$
$$= 4\ \text{mS}$$

Therefore, the value of the transconductance g_m of the *npn* transistor is $\boxed{4\ \text{mS}}$.

Step 2 of 2

Calculate the value of the transconductance of the NMOS transistor.

$$g_m = \frac{2I_D}{V_{OV}}$$

Substitute $4\ \text{mS}$ for g_m and $0.25\ \text{V}$ for V_{OV} .

$$4\ \text{m} = \frac{2I_D}{0.25}$$

$$I_D = \frac{(4\ \text{m})(0.25)}{2}$$

$$I_D = 0.5\ \text{mA}$$

Therefore, the value of the drain current I_D is $\boxed{0.5\ \text{mA}}$.

Step 1 of 2

Refer to Figure 7.41 (c) for the CD-CG amplifier.

(a)

Determine voltage gain $\frac{v_o}{v_i}$ by neglecting r_{o1}

$$v_o = \left(\frac{v_i}{2r_e}\right)R_L$$

$$v_o = \frac{R_L v_i}{2r_e}$$

$$\frac{v_o}{v_i} = \frac{R_L}{2r_e}$$

$$\frac{v_o}{v_i} = \frac{R_L}{2\left(\frac{V_{OV}}{2I}\right)}$$

$$\frac{v_o}{v_i} = \frac{IR_L}{V_{OV}}$$

Here,

R_L is the load resistance

V_{OV} is the overdrive voltage

Thus, the voltage gain is,

$$\boxed{\frac{v_o}{v_i} = \frac{IR_L}{V_{OV}}}.$$

Step 2 of 2

(b)

The voltage gain of the amplifier is,

$$\frac{v_o}{v_i} = \frac{IR_L}{V_{OV}}$$

Substitute 0.1×10^{-3} A for I, $20\times10^{3}\ \Omega$ for R_L and 10 V/V for $\frac{v_o}{v_i}$.

$$10 = \frac{(0.1\times10^{-3})(20\times10^{3})}{2V_{OV}}$$

$$V_{OV} = \frac{(0.1\times10^{-3})(20\times10^{3})}{(10)}$$

$$= 0.2\ \text{V}$$

Write the formula for the drain current.

$$I = \frac{1}{2}k_n'\left(\frac{W}{L}\right)V_{OV}^2$$

Substitute 200×10^{-6} A/V^2 for k_n', 0.1×10^{-3} A for I and 0.2 V for V_{OV}.

$$0.1\times10^{-3} = \frac{1}{2}(200\times10^{-6})\left(\frac{W}{L}\right)(0.2)^2$$

$$0.1\times10^{-3} = (100\times10^{-6})\left(\frac{W}{L}\right)(0.04)$$

$$0.1\times10^{-3} = (4\times10^{-6})\left(\frac{W}{L}\right)$$

$$\frac{W}{L} = 25$$

Thus, the $\frac{W}{L}$ ratio of the two transistors is $\boxed{25}$.

Step 1 of 4

Consider the expression for the transconductance, g_m.

$$g_m = \frac{2I_D}{V_{OV}}$$

Substitute $100\ \mu A$ for I_D and $0.5\ V$ for V_{OV}.

$$g_m = \frac{2(100\ \mu)}{0.5}$$
$$g_m = \frac{200\ \mu}{0.5}$$
$$g_m = 0.4\ mA/V$$

Therefore, the value of the transconductance g_m is $\boxed{0.4\ mA/V}$.

Step 2 of 4

Consider the expression for the resistance, r_o.

$$r_o = \frac{V_A' L}{I_D}$$

Substitute $100\ \mu A$ for I_D, $20\ V/\mu m$ for V_A' and $1\ \mu m$ for L.

$$r_o = \frac{(20/1\ \mu)(1\ \mu)}{100\ \mu}$$
$$= \frac{20}{100\ \mu}$$
$$= 200\ k\Omega$$

Therefore, the value of the resistance r_o is $\boxed{200\ k\Omega}$.

Step 3 of 4

Calculate the value of the intrinsic gain.

$$A_o = \frac{2V_A' L}{V_{OV}}$$

Substitute $0.5\ V$ for V_{OV}, $1\ \mu m$ for L and $20\ V/\mu m$ for V_A'.

$$A_o = \frac{2(20/1\ \mu)(1\ \mu)}{0.5}$$
$$A_o = \frac{40}{0.5}$$
$$A_o = 80$$

Therefore, the value of the intrinsic gain is $\boxed{80}$.

Step 4 of 4

Consider the expression for the transconductance, g_m.

$$g_m = k_n'\left(\frac{W}{L}\right)V_{OV}$$

Substitute $0.5\ V$ for V_{OV}, $1\ \mu m$ for L, $0.4\ mA/V$ for g_m and $190\ \mu A/V^2$ for k_n'.

$$0.4\ m = (190\ \mu)\left(\frac{W}{1\ \mu}\right)(0.5)$$
$$0.4\ m = (95)(W)$$
$$W = 4.21\ \mu m$$

Therefore, the value of the width W is $\boxed{4.21\ \mu m}$.

Step 1 of 4

Consider the expression for the transconductance, g_m.

$$g_m = \frac{2I_D}{V_{OV}}$$

Substitute $100\ \mu A$ for I_D and $0.2\ V$ for V_{OV}.

$$g_m = \frac{2(100\ \mu)}{0.2}$$
$$g_m = \frac{200\ \mu}{0.2}$$
$$g_m = 1\ mA/V$$

Therefore, the value of the transconductance g_m is $\boxed{1\ mA/V}$.

Step 2 of 4

Consider the expression for the resistance, r_o.

$$r_o = \frac{V_A' L}{I_D}$$

Substitute $100\ \mu A$ for I_D, $5\ V/\mu m$ for V_A' and $0.3\ \mu m$ for L.

$$r_o = \frac{(5/1\ \mu)(0.3\ \mu)}{100\ \mu}$$
$$= \frac{15}{100\ \mu}$$
$$= 150\ k\Omega$$

Therefore, the value of the resistance r_o is $\boxed{150\ k\Omega}$.

Step 3 of 4

Calculate the value of the intrinsic gain.

$$A_o = \frac{2V_A' L}{V_{OV}}$$

Substitute $0.2\ V$ for V_{OV}, $0.3\ \mu m$ for L and $5\ V/\mu m$ for V_A'.

$$A_o = \frac{2(5/1\ \mu)(0.3\ \mu)}{0.2}$$
$$A_o = \frac{3}{0.2}$$
$$A_o = 15$$

Therefore, the value of the intrinsic gain is, $\boxed{15}$.

Step 4 of 4

Consider the expression for the transconductance, g_m.

$$g_m = k_n'\left(\frac{W}{L}\right)V_{OV}$$

Substitute $0.2\ V$ for V_{OV}, $0.3\ \mu m$ for L, $1\ mA/V$ for g_m and $387\ \mu A/V^2$ for k_n'.

$$1\ m = (387\ \mu)\left(\frac{W}{0.3\ \mu}\right)(0.2)$$
$$1\ m = (258)(W)$$
$$W = 3.88\ \mu m$$

Therefore, the value of the width W is $\boxed{3.88\ \mu m}$.

Step 1 of 17

BJT cell calculations for $I_C = 0.1\ \text{mA}$.

Consider the expression for the transconductance of the npn transistor.

$$g_m = \frac{I_C}{V_T}$$

Substitute $0.1\ \text{mA}$ for I_C and $25\ \text{mA}$ for V_T .

$$g_m = \frac{0.1\ \text{m}}{25\ \text{m}}$$
$$= 4\ \text{mA/V}$$

Therefore, the value of the transconductance g_m is $4\ \text{mA/V}$.

Step 2 of 17

Calculate the value of the resistance r_o .

$$r_o = \frac{V_A}{I_C}$$

Substitute $0.1\ \text{mA}$ for I_C and $100\ \text{V}$ for V_A .

$$r_o = \frac{100}{0.1\ \text{m}}$$
$$= 1\ \text{M}\Omega$$

Therefore, the value of the resistance r_o is $1\ \text{M}\Omega$.

Step 3 of 17

Calculate the value of the intrinsic gain A_o .

$$A_o = g_m r_o$$

Substitute $1\ \text{M}\Omega$ for r_o and $4\ \text{mA/V}$ for g_m .

$$A_o = (4\ \text{m})(1\ \text{M})$$
$$= 4000\ \text{V/V}$$

Therefore, the value of the intrinsic gain A_o is $4000\ \text{V/V}$.

Step 4 of 17

Calculate the value of the input resistance R_{in} .

$$R_{in} = r_\pi$$

Substitute $\frac{\beta}{g_m}$ for r_π .

$$R_{in} = \frac{\beta}{g_m}$$

Substitute $4\ \text{mA/V}$ for g_m and 100 for β .

$$R_{in} = \frac{100}{4\ \text{m}}$$
$$= 25\ \text{k}\Omega$$

Therefore, the value of the input resistance R_{in} is $\boxed{25\ \text{k}\Omega}$.

Step 5 of 17

BJT cell calculations for $I_C = 1\ \text{mA}$.

Consider the expression for the transconductance of the npn transistor.

$$g_m = \frac{I_C}{V_T}$$

Substitute $1\ \text{mA}$ for I_C and $25\ \text{mA}$ for V_T .

$$g_m = \frac{1\ \text{m}}{25\ \text{m}}$$
$$= 40\ \text{mA/V}$$

Therefore, the value of the transconductance g_m is $40\ \text{mA/V}$.

Step 6 of 17

Calculate the value of the resistance r_o .

$$r_o = \frac{V_A}{I_C}$$

Substitute $1\ \text{mA}$ for I_C and $100\ \text{V}$ for V_A .

$$r_o = \frac{100}{1\ \text{m}}$$
$$= 100\ \text{k}\Omega$$

Therefore, the value of the resistance r_o is $100\ \text{k}\Omega$.

Step 7 of 17

Calculate the value of the intrinsic gain A_o .

$$A_o = g_m r_o$$

Substitute $100\ \text{k}\Omega$ for r_o and $4\ \text{mA/V}$ for g_m .

$$A_o = (4\ \text{m})(100\ \text{k})$$
$$= 400\ \text{V/V}$$

Therefore, the value of the intrinsic gain A_o is $400\ \text{V/V}$.

Step 8 of 17

Calculate the value of the input resistance R_{in} .

$$R_{in} = r_\pi$$

Substitute $\frac{\beta}{g_m}$ for r_π .

$$R_{in} = \frac{\beta}{g_m}$$

Substitute $40\ \text{mA/V}$ for g_m and 100 for β .

$$R_{in} = \frac{100}{40\ \text{m}}$$
$$= 2.5\ \text{k}\Omega$$

Therefore, the value of the input resistance R_{in} is $\boxed{2.5\ \text{k}\Omega}$.

Step 9 of 17

MOSFET cell calculations for $I_D = 0.1\ \text{mA}$.

Consider the expression for the drain current I_D .

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)V_{ov}^2$$

Substitute $0.1\ \text{mA}$ for I_D , $200\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$ and 40 for $\frac{W}{L}$.

$$0.1\ \text{m} = \frac{1}{2}(200\ \mu)(40)V_{ov}^2$$
$$0.1\ \text{m} = (4\ \text{m})V_{ov}^2$$
$$V_{ov}^2 = \frac{0.1\ \text{m}}{4\ \text{m}}$$
$$V_{ov} = \sqrt{25\ \text{m}}$$
$$V_{ov} = 0.158\ \text{V}$$

Therefore, the value of the overdrive voltage V_{ov} is $0.158\ \text{V}$.

Step 10 of 17

Calculate the value of the intrinsic gain g_m .

$$g_m = \frac{2I_D}{V_{ov}}$$

Substitute $0.158\ \text{V}$ for V_{ov} and $0.1\ \text{mA}$ for I_D .

$$g_m = \frac{2(0.1\ \text{m})}{0.158}$$
$$= 1.26\ \text{mA/V}$$

Therefore, the value of the transconductance is $1.26\ \text{mA/V}$.

Step 11 of 17

Calculate the value of the resistance r_o .

$$r_o = \frac{V_A}{I_D}$$

Substitute $10\ \text{V}$ for V_A and $0.1\ \text{mA}$ for I_D .

$$r_o = \frac{10}{0.1\ \text{m}}$$
$$= 100\ \text{k}\Omega$$

Therefore, the value of the resistance r_o is $100\ \text{k}\Omega$.

Step 12 of 17

Calculate the value of the intrinsic gain A_o .

$$A_o = \frac{2V_A}{V_{ov}}$$

Substitute $10\ \text{V}$ for V_A and $0.158\ \text{V}$ for V_{ov} .

$$A_o = \frac{2(10)}{0.158}$$
$$= 126.58$$

Therefore, the value of the intrinsic gain is 126.58 .

The values of the input resistance R_{in} for any value of drain current is ∞ .

Step 13 of 17

MOSFET cell calculations for $I_D = 1\ \text{mA}$.

Consider the expression for the drain current I_D .

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)V_{ov}^2$$

Substitute $1\ \text{mA}$ for I_D , $200\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$ and 40 for $\frac{W}{L}$.

$$1\ \text{m} = \frac{1}{2}(200\ \mu)(40)V_{ov}^2$$
$$1\ \text{m} = (4\ \text{m})V_{ov}^2$$
$$V_{ov}^2 = \frac{1\ \text{m}}{4\ \text{m}}$$
$$V_{ov} = \sqrt{0.25}$$
$$V_{ov} = 0.5\ \text{V}$$

Therefore, the value of the overdrive voltage V_{ov} is $0.5\ \text{V}$.

Step 14 of 17

Calculate the value of the intrinsic gain g_m .

$$g_m = \frac{2I_D}{V_{ov}}$$

Substitute $0.5\ \text{V}$ for V_{DV} and $1\ \text{mA}$ for I_D .

$$g_m = \frac{2(1\ \text{m})}{0.5}$$
$$= 4\ \text{mA/V}$$

Therefore, the value of the transconductance is $4\ \text{mA/V}$.

Step 15 of 17

Calculate the value of the resistance r_o .

$$r_o = \frac{V_A}{I_D}$$

Substitute $10\ \text{V}$ for V_A and $1\ \text{mA}$ for I_D .

$$r_o = \frac{10}{1\ \text{m}}$$
$$= 10\ \text{k}\Omega$$

Therefore, the value of the resistance r_o is $10\ \text{k}\Omega$.

Step 16 of 17

Calculate the value of the intrinsic gain A_o .

$$A_o = \frac{2V_A}{V_{ov}}$$

Substitute $10\ \text{V}$ for V_A and $0.5\ \text{V}$ for V_{ov} .

$$A_o = \frac{2(10)}{0.5}$$
$$= 40$$

Therefore, the value of the intrinsic gain is 40 .

The values of the input resistance R_{in} for any value of drain current is ∞ .

Step 17 of 17

Consider Table-1.

Table-1

Bias current	BJT Cell	MOSFET Cell		
	$I_C = 0.1\ \text{mA}$	$I_D = 1\ \text{mA}$	$I_D = 0.1\ \text{mA}$	$I_D = 1\ \text{mA}$
$g_m\ (\text{mA/V}^2)$	4	40	1.26	**4**
$r_o\ (\text{k}\Omega)$	1000	100	100	**10**
$A_O\ (\text{V/V})$	4000	400	**126.58**	**40**
$R_{in}\ (\text{k}\Omega)$	25	2.5	∞	∞

Therefore, the required values are calculated and values are displayed in Table-1.

Step 1 of 2

8.035P

Calculate the value of the intrinsic gain A_o .

$$A_o = \frac{2V_A' L}{V_{OV}}$$

Substitute $5\ \text{V/}\mu\text{m}$ for V_A' , $0.54\ \mu\text{m}$ for L and $18\ \text{V/V}$ for A_o .

$$18 = \frac{2(5/1\ \mu)(0.54\ \mu)}{V_{OV}}$$

$$V_{OV} = \frac{5.4}{18}$$

$$V_{OV} = 0.3\ \text{V}$$

Therefore, the value of the overdrive voltage V_{OV} is $0.3\ \text{V}$.

Step 2 of 2

Calculate the value of the bias current I_D .

$$I_D = \frac{1}{2}\mu_n C_{OX}\left(\frac{W}{L}\right)V_{OV}^2$$

Substitute $0.3\ \text{V}$ for V_{OV} , $400\ \mu\text{A/V}^2$ for $\mu_n C_{OX}$ and 8 for $\frac{W}{L}$.

$$I_D = \frac{1}{2}(400\ \mu)(8)(0.3)^2$$
$$= 144\ \mu\text{A}$$

Therefore, the required value of the bias current is $\boxed{144\ \mu\text{A}}$.

Step 1 of 9

Consider the expression for the bias current I_D.

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)V_{ov}^2$$

Substitute $25\ \mu A$ for I_D, $400\ \mu A/V^2$ for $\mu_n C_{ox}$ and 8 for $\frac{W}{L}$.

$$25\ \mu = \frac{1}{2}(400\ \mu)(8)V_{ov}^2$$
$$25\ \mu = (1600\ \mu)V_{ov}^2$$
$$V_{ov}^2 = \frac{25\ \mu}{1600\ \mu}$$
$$V_{ov} = \sqrt{15.625\ m}$$
$$V_{ov} = 0.125\ V$$

Therefore, the value of the overdrive voltage V_{ov} is $0.125\ V$.

Step 2 of 9

Calculate the value of the transconductance, g_m.

$$g_m = \frac{2I_D}{V_{ov}}$$

Substitute $0.125\ V$ for V_{ov} and $25\ \mu A$ for I_D.

$$g_m = \frac{2(25\ \mu)}{0.125}$$
$$= 0.4\ mA/V^2$$

Therefore, the value of the transconductance, g_m is $\boxed{0.4\ mA/V^2}$.

Step 3 of 9

Calculate the value of the intrinsic gain A_o.

$$A_o = \frac{2V_A' L}{V_{ov}}$$

Substitute $5\ V/\mu m$ for V_A', $0.36\ \mu m$ for L and $0.125\ V$ for V_{ov}.

$$A_o = \frac{2(5/1\ \mu)(0.36\ \mu)}{0.125}$$
$$= \frac{3.6}{0.125}$$
$$= 28.8$$

Therefore, the value of the intrinsic gain A_o is $\boxed{28.8}$.

Step 4 of 9

Calculations for the drain current $I_D = 250\ \mu A$.

Consider the expression for the bias current I_D.

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)V_{ov}^2$$

Substitute $250\ \mu A$ for I_D, $400\ \mu A/V^2$ for $\mu_n C_{ox}$ and 8 for $\frac{W}{L}$.

$$250\ \mu = \frac{1}{2}(400\ \mu)(8)V_{ov}^2$$
$$250\ \mu = (1600\ \mu)V_{ov}^2$$
$$V_{ov}^2 = \frac{250\ \mu}{1600\ \mu}$$
$$V_{ov} = \sqrt{156.25\ m}$$
$$V_{ov} = 0.395\ V$$

Therefore, the value of the overdrive voltage V_{ov} is $0.395\ V$.

Step 5 of 9

Calculate the value of the transconductance g_m.

$$g_m = \frac{2I_D}{V_{ov}}$$

Substitute $0.395\ V$ for V_{ov} and $250\ \mu A$ for I_D.

$$g_m = \frac{2(250\ \mu)}{0.395}$$
$$= 1.27\ mA/V^2$$

Therefore, the value of the transcondcutance g_m is $\boxed{1.27\ mA/V^2}$.

Step 6 of 9

Calculate the value of the intrinsic gain A_o.

$$A_o = \frac{2V_A' L}{V_{ov}}$$

Substitute $5\ V/\mu m$ for V_A', $0.36\ \mu m$ for L and $0.395\ V$ for V_{ov}.

$$A_o = \frac{2(5/1\ \mu)(0.36\ \mu)}{0.395}$$
$$= \frac{3.6}{0.395}$$
$$= 9.11$$

Therefore, the value of the intrinsic gain A_o is $\boxed{9.11}$.

Step 7 of 9

Consider the expression for the bias current I_D.

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)V_{ov}^2$$

Substitute $2.5\ mA$ for I_D, $400\ \mu A/V^2$ for $\mu_n C_{ox}$ and 8 for $\frac{W}{L}$.

$$2.5\ m = \frac{1}{2}(400\ \mu)(8)V_{ov}^2$$
$$2.5\ m = (1600\ \mu)V_{ov}^2$$
$$V_{ov}^2 = \frac{2.5\ m}{1600\ \mu}$$
$$V_{ov} = \sqrt{1.5625}$$
$$V_{ov} = 1.25\ V$$

Therefore, the value of the overdrive voltage V_{ov} is $1.25\ V$.

Step 8 of 9

Calculate the value of the transconductance g_m.

$$g_m = \frac{2I_D}{V_{ov}}$$

Substitute $1.25\ V$ for V_{ov} and $2.5\ mA$ for I_D.

$$g_m = \frac{2(2.5\ m)}{1.25}$$
$$= 4\ mA/V^2$$

Therefore, the value of the transconductance, g_m is $\boxed{4\ mA/V^2}$.

Step 9 of 9

Calculate the value of the intrinsic gain A_o.

$$A_o = \frac{2V_A' L}{V_{ov}}$$

Substitute $5\ V/\mu m$ for V_A', $0.36\ \mu m$ for L and $1.25\ V$ for V_{ov}.

$$A_o = \frac{2(5/1\ \mu)(0.36\ \mu)}{1.25}$$
$$= \frac{3.6}{1.25}$$
$$= 2.88$$

Therefore, the value of the intrinsic gain A_o is $\boxed{2.88}$.

Step 1 of 23

(a)

Calculate the value of the transconductance g_m.

$$g_m = \frac{2I_D}{V_{ov}}$$

Substitute 0.25 V for V_{ov} and $10\,\mu\text{A}$ for I_D.

$$g_m = \frac{2(10\,\mu)}{0.25}$$

$$= 80\,\text{mA/V}$$

Therefore, the value of transconductance g_m is $\boxed{80\ \text{mA/V}}$.

Step 2 of 23

Calculate the length L.

$$L = 2(0.18\,\mu)$$

$$= 0.36\,\mu\text{m}$$

Therefore, the value of length is $0.36\,\mu\text{m}$.

Step 3 of 23

Calculate the value of the resistance r_o.

$$r_o = \frac{V_A' L}{I_D}$$

Substitute $10\,\mu\text{A}$ for I_D, $0.36\,\mu\text{m}$ for L, and $5\text{ V/}\mu\text{m}$ for V_A'.

$$r_o = \frac{(5\text{ V}/\mu)(0.36\,\mu)}{10\,\mu}$$

$$= \frac{1.8}{10\,\mu}$$

$$= 180\,\text{k}\Omega$$

Therefore, the value of the resistance r_o is $\boxed{180\ \text{k}\Omega}$.

Step 4 of 23

Calculate the value of the intrinsic gain A_v.

$$A_v = \frac{2V_A' L}{V_{ov}}$$

Substitute 0.25 V for V_{ov}, $0.36\,\mu\text{m}$ for L, and $5\text{ V/}\mu\text{m}$ for V_A'.

$$A_v = \frac{2(5\text{ V}/\mu)(0.36\,\mu)}{0.25}$$

$$= \frac{3.6}{0.25}$$

$$= 14.4$$

Therefore, the value of the intrinsic gain is $\boxed{14.4}$.

Step 5 of 23

(b)

Consider the expression for the transconductance g_m.

$$g_m = \mu_n C_{ox}\left(\frac{W}{L}\right) V_{ov}$$

Substitute 0.25 V for V_{ov}, $0.36\,\mu\text{m}$ for L, $387\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$ and $80\ \text{mA/V}$ for g_m.

$$80\text{ m} = (387\text{ m})\left(\frac{W}{0.36\,\mu}\right)(0.25)$$

$$80\text{ m} = (268.75\text{ k})(W)$$

$$W = \frac{80\text{ m}}{268.75\text{ k}}$$

$$W = 0.3\,\mu\text{m}$$

Therefore, the value of the width W is $\boxed{0.3\ \mu\text{m}}$.

Step 6 of 23

Consider the expression for the drain current I_D.

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right) V_{ov}^2$$

Substitute 0.25 V for V_{ov}, $0.3\,\mu\text{m}$ for W, $0.36\,\mu\text{m}$ for L, $387\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$ and $100\ \mu\text{A}$ for I_D.

$$100\ \mu = \frac{1}{2}(387\,\mu)\left(\frac{0.3\,\mu}{0.36\,\mu}\right) V_{ov}^2$$

$$100\ \mu = (161.25\,\mu) V_{ov}^2$$

$$V_{ov}^2 = \frac{100\ \mu}{161.25\,\mu}$$

$$V_{ov}^2 = 0.62$$

$$V_{ov} = \sqrt{0.62}$$

$$V_{ov} = 0.79\text{ V}$$

Therefore, the value of the overdrive voltage V_{ov} is $\boxed{0.79\text{ V}}$.

Step 7 of 23

Calculate the value of the transconductance g_m.

$$g_m = \frac{2I_D}{V_{ov}}$$

Substitute 0.79 V for V_{ov} and $100\ \mu\text{A}$ for I_D.

$$g_m = \frac{2(100\ \mu)}{0.79}$$

$$= 0.25\ \text{mA/V}$$

Therefore, the value of the transconductance g_m is $\boxed{0.25\ \text{mA/V}}$.

Step 8 of 23

Calculate the value of the resistance r_o.

$$r_o = \frac{V_A' L}{I_D}$$

Substitute $100\ \mu\text{A}$ for I_D, $0.36\,\mu\text{m}$ for L, and $5\text{ V/}\mu\text{m}$ for V_A'.

$$r_o = \frac{(5\text{ V}/\mu)(0.36\,\mu)}{100\ \mu}$$

$$= \frac{1.8}{100\ \mu}$$

$$= 18\,\text{k}\Omega$$

Therefore, the value of the resistance r_o is $\boxed{18\ \text{k}\Omega}$.

Step 9 of 23

(c)

Consider the expression for the transconductance g_m.

$$g_m = \frac{2I_D}{V_{ov}}$$

Substitute 0.25 V for V_{ov} and $10\,\mu\text{A}$ for I_D.

$$g_m = \frac{2(10\,\mu)}{0.25}$$

$$= 80\,\text{mA/V}$$

Therefore, the value of transconductance g_m is $\boxed{80\ \text{mA/V}}$.

Step 10 of 23

Calculate the value of the length L.

$$L = 2(0.18\,\mu)$$

$$= 0.36\,\mu\text{m}$$

Therefore, the value of length is $0.36\,\mu\text{m}$.

Step 11 of 23

Calculate the value of the resistance r_o.

$$r_o = \frac{V_A' L}{I_D}$$

Substitute $10\,\mu\text{A}$ for I_D, $0.36\,\mu\text{m}$ for L, and $5\text{ V/}\mu\text{m}$ for V_A'.

$$r_o = \frac{(5\text{ V}/\mu)(0.36\,\mu)}{10\,\mu}$$

$$= \frac{1.8}{10\,\mu}$$

$$= 180\,\text{k}\Omega$$

Therefore, the value of the resistance r_o is $\boxed{180\ \text{k}\Omega}$.

Step 12 of 23

Calculate the value of the intrinsic gain A_v.

$$A_v = \frac{2V_A' L}{V_{ov}}$$

Substitute 0.25 V for V_{ov}, $0.36\,\mu\text{m}$ for L, and $5\text{ V/}\mu\text{m}$ for V_A'.

$$A_v = \frac{2(5\text{ V}/\mu)(0.36\,\mu)}{0.25}$$

$$= \frac{3.6}{0.25}$$

$$= 14.4$$

Therefore, the value of the intrinsic gain is $\boxed{14.4}$.

Step 13 of 23

(d)

Consider the expression for the transconductance g_m.

$$g_m = \mu_n C_{ox}\left(\frac{W}{L}\right) V_{ov}$$

Substitute 0.25 V for V_{ov}, $0.36\,\mu\text{m}$ for L, $387\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$ and $80\ \text{mA/V}$ for g_m.

$$80\text{ m} = (387\text{ m})\left(\frac{W}{0.36\,\mu}\right)(0.25)$$

$$80\text{ m} = (268.75\text{ k})(W)$$

$$W = \frac{80\text{ m}}{268.75\text{ k}}$$

$$W = 0.3\,\mu\text{m}$$

Therefore, the value of the width W is $\boxed{0.3\ \mu\text{m}}$.

Step 14 of 23

Consider the expression for the drain current I_D.

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right) V_{ov}^2$$

Substitute 0.25 V for V_{ov}, 3 mm for W, $0.36\,\mu\text{m}$ for L, $387\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$ and $10\ \mu\text{A}$ for I_D.

$$10\ \mu = \frac{1}{2}(387\,\mu)\left(\frac{3\text{ m}}{0.36\,\mu}\right) V_{ov}^2$$

$$10\ \mu = (1.6125\,\text{m}) V_{ov}^2$$

$$V_{ov}^2 = \frac{10\ \mu}{1.6125\,\mu}$$

$$V_{ov}^2 = 6.2$$

$$V_{ov} = \sqrt{6.2}$$

$$V_{ov} = 2.49\text{ V}$$

Therefore, the value of the overdrive voltage V_{ov} is $\boxed{2.49\text{ V}}$.

Step 15 of 23

Calculate the value of the transconductance g_m.

$$g_m = \frac{2I_D}{V_{ov}}$$

Substitute 2.49 V for V_{ov} and $10\ \mu\text{A}$ for I_D.

$$g_m = \frac{2(10\ \mu)}{2.49}$$

$$= 8.03\ \mu\text{A/V}$$

Therefore, the value of the transconductance g_m is $\boxed{8.03\ \mu\text{A/V}}$.

Step 16 of 23

Calculate the value of the resistance r_o.

$$r_o = \frac{V_A' L}{I_D}$$

Substitute $10\ \mu\text{A}$ for I_D, $0.36\,\mu\text{m}$ for L, and $5\text{ V/}\mu\text{m}$ for V_A'.

$$r_o = \frac{(5\text{ V}/\mu)(0.36\,\mu)}{10\,\mu}$$

$$= \frac{1.8}{10\,\mu}$$

$$= 180\,\text{k}\Omega$$

Therefore, the value of the resistance r_o is $\boxed{180\ \text{k}\Omega}$.

Step 17 of 23

Calculate the value of the intrinsic gain A_v.

$$A_v = \frac{2V_A' L}{V_{ov}}$$

Substitute 2.49 V for V_{ov}, $0.36\,\mu\text{m}$ for L, and $5\text{ V/}\mu\text{m}$ for V_A'.

$$A_v = \frac{2(5\text{ V}/\mu)(0.36\,\mu)}{2.49}$$

$$= \frac{3.6}{2.49}$$

$$= 1.45$$

Therefore, the value of the intrinsic gain is $\boxed{1.45}$.

Step 18 of 23

Calculate the value of the resistance r_o.

$$r_o = \frac{V_A' L}{I_D}$$

Substitute $10\ \mu\text{A}$ for I_D, $0.36\,\mu\text{m}$ for L, and $5\text{ V/}\mu\text{m}$ for V_A'.

$$r_o = \frac{(5\text{ V}/\mu)(0.36\,\mu)}{10\,\mu}$$

$$= \frac{1.8}{10\,\mu}$$

$$= 180\,\text{k}\Omega$$

Therefore, the value of the resistance r_o is $\boxed{180\ \text{k}\Omega}$.

Step 1 of 3

Calculate the value of the transconductance, g_m.

$$g_m = \frac{2I_D}{V_{OV}}$$

Substitute $0.15\ \text{V}$ for V_{OV} and $100\ \mu\text{A}$ for I_D.

$$g_m = \frac{2(100\ \mu)}{0.15}$$
$$= 1.33\ \text{mA/V}^2$$

Therefore, the value of the transconductance, g_m is $\boxed{1.33\ \text{mA/V}^2}$.

Step 2 of 3

Consider the expression for the bias current, I_D.

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)V_{OV}^2$$

Substitute $0.15\ \text{V}$ for V_{OV}, $100\ \mu\text{A}$ for I_D, $400\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$ and $0.5\ \mu\text{m}$ for L.

$$100\ \mu = \frac{1}{2}(400\ \mu)\left(\frac{W}{0.5\ \mu}\right)(0.15)^2$$
$$100\ \mu = (400)(W)(22.5\ \text{m})$$
$$100\ \mu = 9W$$
$$W = \frac{100\ \mu}{9}$$
$$W = 11.11\ \mu\text{m}$$

Therefore, the value of the width W is $\boxed{11.11\ \mu\text{m}}$.

Step 3 of 3

Calculate the value of the resistance, r_o.

$$r_o = \frac{V_A' L}{I_D}$$

Substitute $100\ \mu\text{A}$ for I_D, $0.5\ \mu\text{m}$ for L and $6\ \text{V/}\mu\text{m}$ for V_A'.

$$r_o = \frac{(6/1\ \mu)(0.5\ \mu)}{100\ \mu}$$
$$= \frac{3}{100\ \mu}$$
$$= 30\ \text{k}\Omega$$

Therefore, the value of the resistance r_o is $\boxed{30\ \text{k}\Omega}$.

Step 1 of 3

Required current-source-loaded CS amplifier is,

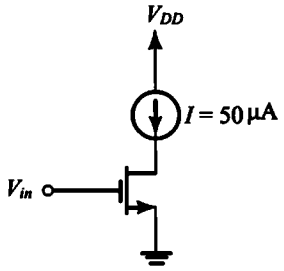


Figure 1

Step 2 of 3

Find channel width-length ratio, $\frac{W}{L}$.

$$I = \frac{1}{2}k_n'\left(\frac{W}{L}\right)V_{OV}^2$$

Trans-conductance parameter $k_n' = 200\ \mu A/V^2$

Early voltage $V_A' = 20\ V/\mu m$

Drain current $I = 50\ \mu A$

Over drive voltage $V_{OV} = 0.2\ V$ and

Open circuit voltage gain $A_{vo} = -100\ V/V$

Rewrite the equation.

$$\frac{W}{L} = \frac{2I}{k_n'V_{OV}^2}$$

Substitute corresponding values.

$$\frac{W}{L} = \frac{2 \times 50 \times 10^{-6}}{200 \times 10^{-6} \times 0.2^2}$$
$$= 12.5$$

Channel width-length ratio is $\boxed{12.5}$.

Find trans-conductance, g_m.

$$g_m = \frac{2I_D}{V_{OV}}$$
$$= \frac{2 \times 50 \times 10^{-6}}{0.2}$$
$$= 0.5\ mA/V$$

Thus, trans-conductance, g_m is $\boxed{0.5\ mA/V}$

Step 3 of 3

Input resistance of an active loaded or current-source-loaded CS amplifier is $\boxed{R_{in} = \infty}$.

Write the formula for voltage gain.

$$A_{vo} = -g_m r_o$$

Find output resistance, r_o

$$r_o = \frac{A_{vo}}{-g_m}$$

Substitute corresponding values.

$$r_o = \frac{-100}{\left(-0.5 \times 10^{-3}\right)}$$
$$= 200\ k\Omega$$

Find the length L

$$L = \frac{r_o I}{V_A'}$$

Substitute corresponding values.

$$L = \frac{200 \times 10^3 \times 50 \times 10^{-6} \times 10^{-6}}{20}$$
$$= 0.5\ \mu m$$

Thus, length of the channel, L is $\boxed{0.5\ \mu m}$.

Step 1 of 6

Refer to the circuit diagram of current –source load of the common source (CS) amplifier in Figure 7.3(a) in the textbook.

From the figure, it is clear that the PMOS device acts as the load to NMOS CS amplifier.

Write the expression for overdrive voltage V_{OV} .

$$|V_{OV}| = V_{SG} - |V_{tp}|$$

Here V_{tp} is threshold voltage.

V_{SG} is the source to gate voltage.

Rewrite the expression.

$$\begin{aligned}|V_{OV}| &= V_{SG} - |V_{tp}| \\ &= V_S - V_G - |V_{tp}|\end{aligned}$$

Rewrite the expression.

$$V_G = V_S - |V_{tp}| - |V_{OV}|$$

It is clear that, source voltage V_S at PMOS device is equal to V_{DD} .

Supply voltage V_{DD} is 2.5 V.

Substitute 2.5 for V_S , 0.5 for V_{tp} and 0.3 for V_{OV} .

$$\begin{aligned}V_G &= 2.5 - 0.5 - 0.3 \\ &= 2.5 - 0.8 \\ &= 1.7\ \text{V}\end{aligned}$$

Thus, gate voltage V_G is $\boxed{1.7\ \text{V}}$.

Step 2 of 6

(b)

There is a single current (drain current) flows through the entire circuit.

Drain current is,

$$\begin{aligned}I_D &= I \\ &= 100\ \mu\text{A}\end{aligned}$$

Write the expression for current.

$$I = \frac{1}{2}(\mu_p C_{ox})\left(\frac{W}{L}\right)_2 \left[V_{DD} - V_G - |V_{tp}|\right]^2$$

Substitute $\mu_p C_{ox}$ for 100, 2.5 for V_{DD} , 0.5 for V_{tp} and $100\ \mu\text{A}$ for I .

$$100\times10^{-6} = \frac{1}{2}(100)\left(\frac{W}{L}\right)_2 [2.5 - 1.7 - 0.5]^2$$

Solve for $\left(\frac{W}{L}\right)_2$.

$$\begin{aligned}\left(\frac{W}{L}\right)_2 &= \frac{100\times10^{-6}}{50(0.3)^2} \\ &= \frac{100\times10^{-6}}{50\left(0.8 - 100\times10^{-6}\right)^2} \\ &= 22.22\times10^{6}\end{aligned}$$

Thus, width to length ratio of PMOS transistor is $\boxed{22.22\times10^{-6}}$.

Step 3 of 6

Recall the current equation.

$$I = \frac{1}{2}(\mu_p C_{ox})\left(\frac{W}{L}\right)_2 \left[V_{DD} - V_G - |V_{tp}|\right]^2$$

Substitute $\mu_p C_{ox}$ for 200, 2.5 for V_{DD} , 0.5 for V_{tp} and $100\ \mu\text{A}$ for I .

$$100\times10^{-6} = \frac{1}{2}(200)\left(\frac{W}{L}\right)_2 [2.5 - 1.7 - 0.5]^2$$

Solve for $\left(\frac{W}{L}\right)_2$.

$$\begin{aligned}\left(\frac{W}{L}\right)_2 &= \frac{100\times10^{-6}}{100(0.3)^2} \\ &= \frac{100\times10^{-6}}{50\left(0.8 - 100\times10^{-6}\right)^2} \\ &= 11.11\times10^{6}\end{aligned}$$

Thus, width to length ratio of NMOS transistor is $\boxed{11.11\times10^{-6}}$.

Step 4 of 6

Refer to the circuit diagram of small-signal equivalent circuit of the amplifier in Figure 7.3(c) in the textbook.

Write the expression for gain of the circuit.

$$\begin{aligned}A_v &= \frac{v_o}{v_i} \\ &= -g_{m1}(r_{o1} \| r_{o2})\end{aligned}$$

Here g_{m1} is the trans-conductance of the NMOS transistor.

r_{o1} is the ac output resistance of the NMOS transistor.

r_{o2} is the ac output resistance of the PMOS transistor.

Step 5 of 6

Write the expression for early voltage.

$$V_{A1} = V'_{A1} L$$

L is the channel length.

V'_{A1} is the early voltage per length.

In the fabricated process, for CS amplifier the channel length is $0.4\ \mu\text{m}$.

Substitute $0.4\ \mu\text{m}$ for L and V'_{A1} for $20\ \text{V}/\mu\text{m}$ in the equation.

$$\begin{aligned}V_{A1} &= V'_{A1} L \\ &= (20\ \text{V}/\mu\text{m})(0.4\ \mu\text{m}) \\ &= 8\ \text{V}\end{aligned}$$

Write the expression for r_{o1} .

$$r_{o1} = \frac{|V_{A1}|}{I}$$

Here V_{A1} is the early voltage.

Substitute $20\ \text{V}/\mu\text{m}$ for V_{A1} and $100\ \mu\text{A}$ for I in the expression.

$$\begin{aligned}r_{o1} &= \frac{8\ \text{V}}{100\ \mu\text{A}} \\ &= 80\ \text{k}\Omega\end{aligned}$$

Thus, the ac output resistance r_{o1} of the NMOS transistor is $80\ \text{k}\Omega$.

Similarly the ac output resistance r_{o2} of the PMOS transistor is $80\ \text{k}\Omega$.

Step 6 of 6

Write the expression for trans-conductance of NMOS transistor.

$$\begin{aligned}g_{m1} &= \frac{2I}{|V_{OV}|} \\ &= \frac{2(100\ \mu\text{A})}{(0.3)} \\ &= 6.66\times10^{-04}\ \text{℧}\end{aligned}$$

Recall the gain of the circuit expression.

$$A_v = -g_{m1}(r_{o1} \| r_{o2})$$

Substitute 6.66×10^{-04} for g_{m1} , $80\ \text{k}\Omega$ for r_{o1} and $80\ \text{k}\Omega$ for r_{o2} .

$$\begin{aligned}A_v &= (6.66\times10^{-04}\ \text{℧})(8\ \text{k}\Omega \| 8\ \text{k}\Omega) \\ &= (6.66\times10^{-04}\ \text{℧})(4\ \text{k}\Omega) \\ &= 2.66\ \text{V/V}\end{aligned}$$

Therefore the voltage gain of the circuit is $\boxed{2.66\ \text{V/V}}$.

Step 1 of 7

Refer to Figure 8.15 (a) from the text book.

Calculate the value of the voltage v_{GS1}.

$$v_{GS2} = V_{tp} + V_{OV}$$

Substitute $-0.5\ \text{V}$ for v_{tp} and $-0.25\ \text{V}$ for v_{OV}.

$$\begin{aligned} v_{GS2} &= -0.5 - 0.25 \\ &= -0.75\ \text{V} \end{aligned}$$

Therefore, the value of the voltage v_{GS2} is $-0.75\ \text{V}$.

Step 2 of 7

Calculate the value of the voltage v_G.

$$\begin{aligned} v_G &= v_{DD} - V_{SG} \\ &= v_{DD} + V_{GS} \end{aligned}$$

Substitute $-0.75\ \text{V}$ for v_{GS2} and $1.8\ \text{V}$ for v_{DD}.

$$\begin{aligned} v_G &= 1.8 - 0.75 \\ &= 1.05\ \text{V} \end{aligned}$$

Therefore, the value of the voltage v_G is $\boxed{1.05\ \text{V}}$.

Step 3 of 7

Calculate the value of the transconductance g_{m1}.

$$g_{m1} = \frac{2I}{V_{OV}}$$

Substitute $100\ \mu\text{A}$ for I and $0.25\ \text{V}$ for V_{OV}.

$$\begin{aligned} g_{m1} &= \frac{2(100\ \mu)}{0.25} \\ &= 0.8\ \text{mA/V} \end{aligned}$$

Therefore, the value of the transconductance g_{m1} is $0.8\ \text{mA/V}$.

Step 4 of 7

Consider the expression for the voltage gain A_v.

$$A_v = -g_{m1}\left(r_{o1} \parallel r_{o2}\right)$$

Here, $r_{o1} = r_{o2}$.

$$A_v = -g_{m1}\left(\frac{r_{o1}}{2}\right)$$

Substitute $0.8\ \text{mA/V}$ for g_{m1} and $-40\ \text{V/V}$ for A_v.

$$\begin{aligned} -40 &= -(0.8\ \text{m})\left(\frac{r_{o1}}{2}\right) \\ 40 &= (0.4\ \text{m})(r_{o1}) \\ r_{o1} &= \frac{40}{0.4\ \text{m}} \\ r_{o1} &= 100\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance r_{o1} is $100\ \text{k}\Omega$.

Step 5 of 7

Consider the expression for the resistance r_o.

$$r_o = \frac{V_A' L}{I}$$

Substitute $100\ \text{k}\Omega$ for r_o, $100\ \mu\text{A}$ for I and $5\ \text{V/}\mu\text{m}$ for V_{OV}.

$$\begin{aligned} 100\ \text{k} &= \frac{(5/1\ \mu)L}{100\ \mu} \\ L &= \frac{(100\ \text{k})(100\ \mu)}{(5/1\ \mu)} \\ L &= \frac{10}{(5/1\ \mu)} \\ L &= 2\ \mu\text{m} \end{aligned}$$

Therefore, the value of the length L is $\boxed{2\ \mu\text{m}}$.

Step 6 of 7

Consider the expression for the current I through the transistor Q_1.

$$I = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_1 (V_{OV})^2$$

Substitute $400\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$, $100\ \mu\text{A}$ for I and $5\ \text{V/}\mu\text{m}$ for V_{OV}.

$$\begin{aligned} 100\ \mu &= \frac{1}{2}(400\ \mu)\left(\frac{W}{L}\right)_1 (0.25)^2 \\ 1 &= \frac{4}{2}\left(\frac{W}{L}\right)_1 (0.25)^2 \\ \left(\frac{W}{L}\right)_1 &= \frac{2}{4(0.25)^2} \\ \left(\frac{W}{L}\right)_1 &= \frac{2}{4(0.0625)} \\ \left(\frac{W}{L}\right)_1 &= \frac{2}{0.25} \\ &= 8 \end{aligned}$$

Therefore, the value of the ratio $\left(\frac{W}{L}\right)_1$ is $\boxed{8}$.

Step 7 of 7

Consider the expression for the current I through the transistor Q_2.

$$I = \frac{1}{2}\mu_p C_{ox}\left(\frac{W}{L}\right)_2 (V_{OV})^2$$

Substitute $100\ \mu\text{A/V}^2$ for $\mu_p C_{ox}$, $100\ \mu\text{A}$ for I and $5\ \text{V/}\mu\text{m}$ for V_{OV}.

$$\begin{aligned} 100\ \mu &= \frac{1}{2}(100\ \mu)\left(\frac{W}{L}\right)_2 (0.25)^2 \\ 1 &= \frac{1}{2}\left(\frac{W}{L}\right)_2 (0.25)^2 \\ \left(\frac{W}{L}\right)_2 &= \frac{2}{(0.25)^2} \\ \left(\frac{W}{L}\right)_2 &= \frac{2}{0.0625} \\ \left(\frac{W}{L}\right)_2 &= 32 \end{aligned}$$

Therefore, the value of the ratio $\left(\frac{W}{L}\right)_2$ is $\boxed{32}$.

Step 1 of 4

Consider the Figure P8.42 from the text book.

Draw the small signal equivalent circuit diagram.

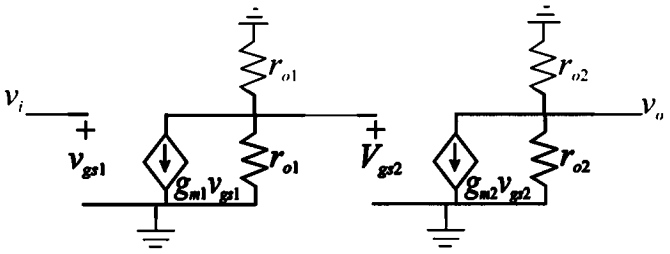


Figure 1

Step 2 of 4

Consider the expression for the voltage v_{gs2}.

$$\begin{aligned} v_{gs2} &= -g_{m1}v_{gs1}\left(r_{o1} \| r_{o1}\right) \\ &= -g_{m1}v_{gs1}\left(\frac{r_{o1}\cdot r_{o1}}{r_{o1}+r_{o1}}\right) \\ &= -g_{m1}v_{gs1}\left(\frac{r_{o1}\cdot r_{o1}}{2r_{o1}}\right) \\ &= -g_{m1}v_{gs1}\left(\frac{r_{o1}}{2}\right) \end{aligned}$$

Substitute v_i for v_{gs1}.

$$v_{gs2} = -g_{m1}v_i\left(\frac{r_{o1}}{2}\right)$$

Therefore, the expression for the voltage v_{gs2} is $-g_{m1}v_i\left(\frac{r_{o1}}{2}\right)$.

Step 3 of 4

Consider the expression for the output voltage.

$$\begin{aligned} v_o &= -g_{m2}v_{gs2}\left(r_{o2} \| r_{o2}\right) \\ &= -g_{m2}v_{gs2}\left(\frac{r_{o2}\cdot r_{o2}}{r_{o2}+r_{o2}}\right) \\ &= -g_{m2}v_{gs2}\left(\frac{r_{o2}\cdot r_{o2}}{2r_{o2}}\right) \\ &= -g_{m2}v_{gs2}\left(\frac{r_{o2}}{2}\right) \end{aligned}$$

Substitute $-g_{m1}v_i\left(\frac{r_{o1}}{2}\right)$ for v_{gs2}.

$$v_o = -g_{m2}\left(-g_{m1}v_i\left(\frac{r_{o1}}{2}\right)\right)\left(\frac{r_{o2}}{2}\right)$$

$$\frac{v_o}{v_i} = \frac{g_{m1}g_{m2}r_{o1}r_{o2}}{4}$$

Here, $r_{o1} = r_{o2}$ since $V_{An} = \left|V_{Ap}\right|$. Hence, substitute r_o for r_{o2} and r_{o1}.

$$\frac{v_o}{v_i} = \frac{g_{m1}g_{m2}r_o^2}{4} \quad \text{..... (1)}$$

Therefore, the expression for the overall voltage gain is, $\boxed{\frac{g_{m1}g_{m2}r_o^2}{4}}$.

Step 4 of 4

Recall equation (1).

$$\frac{v_o}{v_i} = \frac{g_{m1}g_{m2}r_o^2}{4}$$

Assume $g_{m1} = g_{m2}$.

$$\frac{v_o}{v_i} = \frac{g_m^2 r_o^2}{4}$$

Substitute $\frac{2I_D}{V_{OV}}$ for g_m and $\frac{V_A}{I_D}$ for r_o.

$$\begin{aligned} \frac{v_o}{v_i} &= \frac{1}{4}\left(\frac{2I_D}{V_{OV}}\right)^2\left(\frac{V_A}{I_D}\right)^2 \\ &= \frac{1}{4}\left(\frac{4I_D^2}{V_{OV}^2}\right)\left(\frac{V_A^2}{I_D^2}\right) \\ &= \frac{V_A^2}{V_{OV}^2} \end{aligned}$$

Substitute $5\ \text{V}$ for V_A and $400\ \text{V/V}$ for $\frac{v_o}{v_i}$.

$$400 = \frac{5^2}{V_{OV}^2}$$

$$V_{OV}^2 = \frac{25}{400}$$

$$V_{OV} = \sqrt{\frac{25}{400}}$$

$$V_{OV} = 0.25\ \text{V}$$

Therefore, the value of the overdrive voltage V_{OV} is, $\boxed{0.25\ \text{V}}$.

Step 1 of 12

(a)

Refer to Figure P7.13 in the textbook for the common source amplifier circuit.

Write the expression for drain current of an NMOS transistor.

$$I_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)V_{ov}^2$$

Here, $\frac{k_n'W}{L}$ is the conduction parameter.

Neglect the feedback current in the network.

Substitute $2\times10^{-3}\ \text{A/V}^2$ for $\frac{k_n'W}{L}$ and $200\times10^{-6}\ \text{A}$ for I_D.

$$200\times10^{-6} = \frac{1}{2}\left(2\times10^{-3}\right)V_{ov}^2$$

$$V_{ov}^2 = \left(\frac{200\times10^{-6}}{2\times10^{-3}}\right)(2)$$

$$V_{ov}^2 = 0.2$$

$$V_{ov} = 0.4472\ \text{V}$$

Calculate the gate-source voltage.

$$V_{ov} = V_{GS} - V_t$$

$$V_{GS} = V_{ov} + V_t$$

Substitute $0.5\ \text{V}$ for V_t and $0.4472\ \text{V}$ for V_{ov}.

$$V_{GS} = 0.4472 + 0.5$$
$$= 0.9472\ \text{V}$$
$$\approx 0.95\ \text{V}$$

Thus, the gate-to-source voltage of the network, V_{GS} is $\boxed{0.95\ \text{V}}$.

Step 2 of 12

Calculate the gate voltage of the network.

$$V_{GS} = 0.95$$

$$V_G - V_S = 0.95$$

Substitute 0 for V_S.

$$V_G - 0 = 0.95$$

$$V_G = 0.95\ \text{V}$$

Calculate the dc current in the feedback network.

$$V_G = I_F R$$

$$I_F = \frac{V_G}{R}$$

Substitute $0.95\ \text{V}$ for V_G and $2\ \text{M}\Omega$ for R.

$$I_F = \frac{0.95}{2\times10^6}$$
$$= 0.475\times10^{-6}\ \text{A}$$
$$= 0.475\ \mu\text{A}$$

Thus, the dc current in the feedback network, I_F is $\boxed{0.475\ \mu\text{A}}$.

The dc current in the feedback network is very small compared to $200\ \mu\text{A}$. So, neglecting the feedback current in the network is correct.

Step 3 of 12

Calculate the drain source voltage of the network.

$$V_{DS} = V_D - V_S$$

$$V_{DS} = I_F\left(R + R_F\right) - V_S$$

Substitute $0.475\ \mu\text{A}$ for I_F, $2\ \text{M}\Omega$ for R, $3\ \text{M}\Omega$ for R_F and $0\ \text{V}$ for V_S.

$$V_{DS} = \left(0.475\times10^{-6}\right)\left(2\times10^6 + 3\times10^6\right) - 0$$
$$= \left(0.475\times10^{-6}\right)\left(5\times10^6\right)$$
$$= 2.375\ \text{V}$$
$$\approx 2.4\ \text{V}$$

Thus, the drain source voltage of the network, V_{DS} is $\boxed{2.4\ \text{V}}$.

Step 4 of 12

Calculate the output resistance of the network.

$$R_o = \frac{V_{DS}}{I_D}$$

Substitute $2.4\ \text{V}$ for V_{DS} and $200\ \mu\text{A}$ for I_D.

$$R_o = \frac{2.4}{200\times10^{-6}}$$
$$= 12\times10^3\ \Omega$$
$$= 12\ \text{k}\Omega$$

Calculate the output resistance of transistor.

$$r_o = \frac{V_A}{I_D}$$

Substitute $20\ \text{V}$ for V_A and $200\ \mu\text{A}$ for.

$$r_o = \frac{20}{200\times10^{-6}}$$
$$= 0.1\times10^6$$
$$= 100\ \text{k}\Omega$$

Thus, the output resistance of transistor r_o is very large compared to the output resistance of the network R_o. Hence, the effect of r_o can be negligible.

Step 5 of 12

(b)

Draw the small signal equivalent circuit diagram.

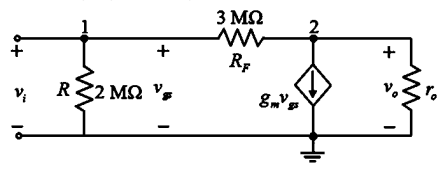


Figure 1

Step 6 of 12

Apply nodal analysis to node 2.

$$\frac{v_o}{r_o} + g_m v_{gs} + \frac{v_o - v_i}{R_F} = 0$$

$$\frac{v_o}{r_o} + g_m v_i + \frac{v_o}{R_F} - \frac{v_i}{R_F} = 0 \qquad \left(\text{Since } v_i = v_{gs}\right)$$

$$v_o\left(\frac{1}{r_o} + \frac{1}{R_F}\right) = \frac{v_i}{R_F} - g_m v_i$$

$$v_o\left(\frac{1}{r_o} + \frac{1}{R_F}\right) = v_i\left(\frac{1}{R_F} - g_m\right)$$

$$\frac{v_o}{v_i} = \frac{\frac{1}{R_F} - g_m}{\frac{1}{r_o} + \frac{1}{R_F}} \quad \ldots\ldots (1)$$

Calculate the transconductance of the network.

$$g_m = \frac{2I_D}{V_{OV}}$$
$$= \frac{2\left(200\times10^{-6}\right)}{0.4472}$$
$$= \frac{400\times10^{-6}}{0.4472}$$
$$= 894.45\times10^{-6}\ \text{A/V}$$
$$= 894.45\ \mu\text{A/V}$$

Step 7 of 12

Calculate the small signal voltage gain of the network from equation (1).

$$\frac{v_o}{v_i} = \frac{\frac{1}{R_F} - g_m}{\frac{1}{r_o} + \frac{1}{R_F}}$$

Substitute $3\times10^6\ \Omega$ for R_F, $100\times10^3\ \Omega$ for r_o and $894.45\ \mu\text{A/V}$ for g_m.

$$\frac{v_o}{v_i} = \frac{\frac{1}{3\times10^6} - 894.45\times10^{-6}}{\frac{1}{100\times10^3} + \frac{1}{3\times10^6}}$$
$$= \frac{-894.1167\times10^{-6}}{1.0333\times10^{-5}}$$
$$= -86.53\ \text{V/V}$$

Thus, the small signal voltage gain of the network, v_o/v_i is $\boxed{-86.53\ \text{V/V}}$.

Step 8 of 12

When the NMOS transistor remains in saturation, the largest output voltage is obtained at minimum drain to source voltage of the transistor.

$$v_{DS}(\text{min}) = V_{OV}$$
$$= 0.4472\ \text{V}$$

Largest signal voltage lies in the range of $v_{DS}(\text{min})$ to $v_{DS}(\text{max})$. Hence, the largest output voltage is,

$$v_o(\text{max}) = v_{DS}(\text{max}) - v_{DS}(\text{min})$$

Substitute $2.4\ \text{V}$ for $v_{DS}(\text{max})$ and $0.4472\ \text{V}$ for $v_{DS}(\text{min})$ in the equation.

$$v_o(\text{max}) = 2.4 - 0.4472$$
$$= 1.9528$$
$$\approx 1.95\ \text{V}$$

Thus, the peak of the largest output sine wave signal, $v_o(\text{max})$ is $\boxed{1.95\ \text{V}}$.

Calculate the input voltage at maximum output voltage.

$$\frac{v_o(\text{max})}{v_i} = -86.53$$

$$v_i = \frac{v_o(\text{max})}{-86.53}$$

Substitute $1.95\ \text{V}$ for $v_o(\text{max})$

$$v_i = -\frac{1.95}{86.53}$$

$$v_i = -0.0225\ \text{V}$$

$$v_i = -22.5\ \text{mV}$$

$$|v_i| = 22.5\ \text{mV}$$

Thus, the input voltage at maximum output voltage, $|v_i|$ is $\boxed{22.5\ \text{mV}}$.

Step 9 of 12

(c)

Draw the equivalent circuit diagram with dummy voltage source at the input.

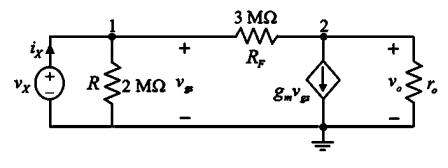


Figure 2

Step 10 of 12

Apply nodal analysis to node 2.

$$\frac{v_o}{r_o} + g_m v_{gs} + \frac{v_o - v_x}{R_F} = 0$$

$$\frac{v_o}{r_o} + g_m v_x + \frac{v_o}{R_F} - \frac{v_x}{R_F} = 0 \qquad \left(\text{Since } v_{gs} = v_x\right)$$

$$\frac{v_o}{r_o} + \frac{v_o}{R_F} = \frac{v_x}{R_F} - g_m v_x$$

$$v_o\left(\frac{1}{r_o} + \frac{1}{R_F}\right) = v_x\left(\frac{1}{R_F} - g_m\right)$$

$$v_o = \frac{v_x\left(\frac{1}{R_F} - g_m\right)}{\frac{1}{r_o} + \frac{1}{R_F}}$$

Step 11 of 12

Apply nodal analysis to node 1.

$$-i_x + \frac{v_x}{R} + \frac{v_x - v_o}{R_F} = 0$$

$$-i_x + \frac{v_x}{R} + \frac{v_x}{R_F} - \frac{v_o}{R_F} = 0$$

$$v_x\left(\frac{1}{R} + \frac{1}{R_F}\right) = \frac{v_o}{R_F} + i_x$$

Substitute $\frac{v_x\left(\frac{1}{R_F} - g_m\right)}{\frac{1}{r_o} + \frac{1}{R_F}}$ for v_o.

$$v_x\left(\frac{1}{R} + \frac{1}{R_F}\right) = \left(\frac{1}{R_F}\right)\left(\frac{v_x\left(\frac{1}{R_F} - g_m\right)}{\frac{1}{r_o} + \frac{1}{R_F}}\right) + i_x$$

$$v_x\left(\frac{1}{R} + \frac{1}{R_F}\right) - v_x\left(\frac{\left(\frac{1}{R_F}\right)\left(\frac{1}{R_F} - g_m\right)}{\frac{1}{r_o} + \frac{1}{R_F}}\right) = i_x$$

$$v_x\left(\frac{1}{R} + \frac{1}{R_F} - \left(\frac{\left(\frac{1}{R_F}\right)\left(\frac{1}{R_F} - g_m\right)}{\left(\frac{1}{R_F}\right)\left(\frac{R_F}{r_o} + 1\right)}\right)\right) = i_x$$

$$\frac{v_x}{i_x} = \frac{1}{\frac{1}{R} + \frac{1}{R_F} - \left(\frac{\frac{1}{R_F} - g_m}{\frac{R_F}{r_o} + 1}\right)}$$

$$R_{in} = \frac{1}{\frac{1}{R} + \frac{1}{R_F} - \left(\frac{\frac{1}{R_F} - g_m}{\frac{R_F}{r_o} + 1}\right)} \qquad \left(\text{Since } R_{in} = \frac{v_x}{i_x}\right)$$

Step 12 of 12

Calculate the small signal input resistance of the network.

$$R_{in} = \frac{1}{\frac{1}{R} + \frac{1}{R_F} - \left(\frac{\frac{1}{R_F} - g_m}{\frac{R_F}{r_o} + 1}\right)}$$

Substitute $3\times10^6\ \Omega$ for R_F, $2\times10^6\ \Omega$ for R, $100\times10^3\ \Omega$ for r_o and $894.45\ \mu\text{A/V}$ for g_m.

$$R_{in} = \frac{1}{\left(\frac{1}{2\times10^6}\right) + \left(\frac{1}{3\times10^6}\right) - \left(\frac{\frac{1}{3\times10^6} - 894.45\times10^{-6}}{\left(\frac{3\times10^6}{100\times10^3}\right) + 1}\right)}$$

$$= \frac{1}{8.333\times10^{-7} - \left(\frac{-894.4467\times10^{-6}}{31}\right)}$$
$$= 33.685\ \text{k}\Omega$$
$$\approx 33.7\ \text{k}\Omega$$

Thus, the small signal input resistance of the network, R_{in} is $\boxed{33.7\ \text{k}\Omega}$.

Step 1 of 5

Consider the Figure 8.16(a) from the text book.

Consider the expression for the output resistance.

$$R_{out} = r_{o1} \| r_{o2}$$

Assume $r_{o1} = r_{o2}$ and $I_{D1} = I_{D2}$ Substitute $\frac{V_A}{I_D}$ for r_{o1} and r_{o2}.

$$\begin{aligned} R_{out} &= \frac{\frac{V_A}{I_D} \cdot \frac{V_A}{I_D}}{\frac{V_A}{I_D} + \frac{V_A}{I_D}} \\ &= \frac{\frac{V_A}{I_D} \cdot \frac{V_A}{I_D}}{2\frac{V_A}{I_D}} \\ &= \frac{V_A}{2I_D} \end{aligned}$$

Substitute $5\ \text{V}$ for V_A and $100\ \text{k}\Omega$ for R_{out}.

$$\begin{aligned} 100\ \text{k} &= \frac{5}{2I_D} \\ I_D &= \frac{5}{2(100\ \text{k})} \\ I_D &= 25\ \mu\text{A} \end{aligned}$$

Therefore, the value of the drain current I_D is $25\ \mu\text{A}$.

Step 2 of 5

Consider the expression for the voltage gain A_v.

$$A_v = -g_{m1} R_{out}$$

Substitute $100\ \text{k}\Omega$ for R_{out} and $-40\ \text{V/V}$ for A_v.

$$\begin{aligned} -40\ \text{V/V} &= -g_{m1}(100\ \text{k}) \\ g_{m1} &= \frac{40}{100\ \text{k}} \\ g_{m1} &= 0.4\ \text{mA/V} \end{aligned}$$

Therefore, the value of the transconductance is $0.4\ \text{mA/V}$.

Step 3 of 5

Consider the expression for the transconductance g_{m1}.

$$g_{m1} = k_n' \left(\frac{W}{L}\right)_1 V_{OV1}$$

Substitute $\frac{2I_D}{g_{m1}}$ for V_{OV1}.

$$\begin{aligned} g_{m1} &= k_n' \left(\frac{W}{L}\right)_1 \frac{2I_D}{g_{m1}} \\ g_{m1}^2 &= 2I_D k_n' \left(\frac{W}{L}\right)_1 \\ g_{m1}^2 &= 2I_D k_n' \left(\frac{W}{L}\right)_1 \\ \left(\frac{W}{L}\right)_1 &= \frac{g_{m1}^2}{2I_D k_n'} \end{aligned}$$

Substitute $0.4\ \text{mA/V}$ for g_{m1}, $25\ \mu\text{A}$ for I_D and $400\ \mu\text{A/V}^2$ for k_n'.

$$\begin{aligned} \left(\frac{W}{L}\right)_1 &= \frac{(0.4\ \text{m})^2}{2(25\ \mu)(400\ \mu)} \\ &= \frac{0.16\ \mu}{2(25\ \mu)(400\ \mu)} \\ &= \frac{0.16\ \mu}{0.02\ \mu} \\ &= 8 \end{aligned}$$

Therefore, the ratio of $\left(\frac{W}{L}\right)_1$ is $\boxed{8}$.

Step 4 of 5

Consider the expression for the overdrive voltage V_{OV1}.

$$V_{OV1} = \frac{2I_D}{g_m}$$

Substitute $0.4\ \text{mA/V}$ for g_{m1} and $25\ \mu\text{A}$ for I_D.

$$\begin{aligned} V_{OV1} &= \frac{2(25\ \mu)}{0.4\ \text{m}} \\ &= 0.125\ \text{V} \end{aligned}$$

Therefore, the value of the voltage V_{OV1} is $0.125\ \text{V}$.

The values of the volages V_{OV2} and V_{OV3} are equal to V_{OV1} since transistors Q_2 and Q_3 are operated at V_{OV1}.

Step 5 of 5

Consider the expression for the drain current I_{D2}.

$$I_{D2} = \frac{1}{2} k_p' \left(\frac{W}{L}\right)_2 V_{OV2}^2$$

Substitute $25\ \mu\text{A}$ for I_{D2}, $0.125\ \text{V}$ for V_{OV2} and $100\ \mu\text{A/V}^2$ for k_p'.

$$\begin{aligned} 25\ \mu &= \frac{1}{2}(100\ \mu)\left(\frac{W}{L}\right)_2 (0.125)^2 \\ 1 &= \frac{1}{2}(4)\left(\frac{W}{L}\right)_2 (15.625\ \text{m}) \\ 1 &= \left(\frac{W}{L}\right)_2 (31.25\ \text{m}) \\ \left(\frac{W}{L}\right)_2 &= \frac{1}{31.25\ \text{m}} \\ \left(\frac{W}{L}\right)_2 &= 32 \end{aligned}$$

Therefore, value of the ratio $\left(\frac{W}{L}\right)_2$ is $\boxed{32}$.

Step 1 of 2

8.045P

The input dc bias for the maximum possible output voltage will be the average values of the voltages V_{IA} and V_{IB} since the characteristics are linear in segment III that is at after $v_{SG} = V_{SG}$.

Calculate the value of the input dc bias voltage.

$$V_I = \frac{V_{IA} + V_{IB}}{2}$$

Substitute 0.89 for V_{IA} and 0.935 for V_{IB}.

$$\begin{aligned} V_I &= \frac{0.89 + 0.935}{2} \\ &= \frac{1.825}{2} \\ &= 0.9125 \text{ V} \end{aligned}$$

Therefore, the required value of the input voltage V_I is $\boxed{0.9125 \text{ V}}$.

Step 2 of 2

Calculate the maximum value of the output voltage.

$$V_O = \frac{V_{OA} + V_{OB}}{2}$$

Substitute 2.47 for V_{IA} and 0.335 for V_{IB}.

$$\begin{aligned} V_O &= \frac{2.47 + 0.335}{2} \\ &= \frac{2.805}{2} \\ &= 1.4025 \text{ V} \end{aligned}$$

Therefore, the required value of the input voltage V_O is $\boxed{1.4025 \text{ V}}$.

Step 1 of 5

Calculate the value of the voltage V_{OA}.

$$V_{OA} = V_{DD} - |V_{OV3}|$$

Substitute $0.526\ \text{V}$ for $|V_{OV3}|$ and $5\ \text{V}$ for V_{DD}.

$$\begin{aligned} V_{OA} &= 5 - 0.526 \\ &= 4.474\ \text{V} \end{aligned}$$

Therefore, the value of the voltage V_{OA} is, $4.474\ \text{V}$.

Step 2 of 5

Calculate the value of the voltage V_{SG}.

$$V_{SG} = |V_{tp}| + |V_{OV3}|$$

Substitute $0.6\ \text{V}$ for $|V_{tp}|$ and $0.526\ \text{V}$ for $|V_{OV3}|$.

$$\begin{aligned} V_{SG} &= 0.6 + 0.526 \\ &= 1.126\ \text{V} \end{aligned}$$

Therefore, the value of the voltage V_{SG} is $1.126\ \text{V}$.

Step 3 of 5

The drain currents of the transistors Q_1 and Q_2 will be equal for the linear region of the characteristics in Figure 8.16(b) from the text book. That is,

$$I_{D1} = I_{D2}$$

Substitute $\frac{1}{2}\mu_n C_{OX}\left(\frac{W}{L}\right)_1 (V_I - V_{tn})^2\left(1 + \frac{V_o}{V_{An}}\right)$ for I_{D1} and

$\frac{1}{2}\mu_p C_{OX}\left(\frac{W}{L}\right)_2 \left(V_{SG} - |V_{tp}|\right)^2\left(1 + \frac{V_{DD} - V_o}{V_{Ap}}\right)$ for I_{D2}.

$$\frac{1}{2}\mu_n C_{OX}\left(\frac{W}{L}\right)_1 (V_I - V_{tn})^2\left(1 + \frac{V_o}{V_{An}}\right) = \frac{1}{2}\mu_p C_{OX}\left(\frac{W}{L}\right)_2 \left(V_{SG} - |V_{tp}|\right)^2\left(1 + \frac{V_{DD} - V_o}{V_{Ap}}\right)$$

The values $\left(\frac{W}{L}\right)_2$ and $\left(\frac{W}{L}\right)_1$ are equal. Hence,

$$\mu_n C_{OX}(V_I - V_{tn})^2\left(1 + \frac{V_o}{V_{An}}\right) = \mu_p C_{OX}\left(V_{SG} - |V_{tp}|\right)^2\left(1 + \frac{V_{DD} - V_o}{V_{Ap}}\right)$$

Substitute $200\ \mu\text{A/V}^2$ for $\mu_n C_{OX}$, $65\ \mu\text{A/V}^2$ for $\mu_p C_{OX}$, $0.6\ \text{V}$ for V_{tn} and $|V_{tp}|$,

$1.126\ \text{V}$ for V_{SG}, $5\ \text{V}$ for V_{DD}, $20\ \text{V}$ for V_{An} and $10\ \text{V}$ for V_{Ap}.

$$(200\ \mu)(V_I - 0.6)^2\left(1 + \frac{V_o}{20}\right) = (65\ \mu)(1.126 - 0.6)^2\left(1 + \frac{5 - V_o}{10}\right)$$

$$(200)(V_I - 0.6)^2\left(\frac{20 + V_o}{20}\right) = (65)(0.526)^2\left(\frac{10 + 5 - V_o}{10}\right)$$

$$(200)(V_I - 0.6)^2\left(\frac{20 + V_o}{2}\right) = (65)(0.276676)(15 - V_o)$$

$$(200)(V_I - 0.6)^2(20 + V_o) = (130)(0.276676)(15 - V_o)$$

$$(200)(V_I - 0.6)^2(20 + V_o) = (35.96788)(15 - V_o) \quad \ldots\ldots (1)$$

Let $V_O = V_{OA}$ and and $V_I = V_{IA}$ Substitute $4.474\ \text{V}$ for V_{OA}.

$$(200)(V_{IA} - 0.6)^2(20 + 4.474) = (35.96788)(15 - 4.474)$$

$$(200)(V_{IA} - 0.6)^2(24.474) = (35.96788)(10.526)$$

$$(V_{IA} - 0.6)^2(4894.8) = 378.588$$

$$(V_{IA} - 0.6)^2 = \frac{378.588}{4894.8}$$

$$(V_{IA} - 0.6)^2 = 0.0774$$

$$V_{IA} - 0.6 = \sqrt{0.0774}$$

$$V_{IA} - 0.6 = 0.278$$

$$V_{IA} = 0.278 + 0.6$$

$$V_{IA} = 0.878\ \text{V}$$

Therefore, the value of the voltage V_{IA} for $V_{OA} = 4.474\ \text{V}$ is $0.878\ \text{V}$.

Step 4 of 5

Recall equation (1).

$$(200)(V_I - 0.6)^2(20 + V_o) = (35.96788)(15 - V_o)$$

Let $V_O = V_{OB}$ and $V_I = V_{IB}$.

$$(200)(V_{IB} - 0.6)^2(20 + V_{OB}) = (35.96788)(15 - V_{OB})$$

Substitute $V_{IB} - V_{tn}$ for V_{OB}.

$$(200)(V_{IB} - 0.6)^2(20 + V_{IB} - V_{tn}) = (35.96788)(15 - V_{IB} - V_{tn})$$

Substitute $0.6\ \text{V}$ for $|V_{tp}|$.

$$(200)(V_{IB} - 0.6)^2(20 + V_{IB} - 0.6) = (35.96788)(15 - V_{IB} - 0.6)$$

$$(200)(V_{IB} - 0.6)^2(19.4 + V_{IB}) = (35.96788)(14.4 - V_{IB})$$

$$(V_{IB}^2 - 1.2V_{IB} + 0.36)(3880 + 200V_{IB}) = 517.94 - 35.97V_{IB}$$

$$3880V_{IB}^2 + 200V_{IB}^3 - 4656V_{IB} - 240V_{IB}^2 + 1396.8 + 72V_{IB} = 517.94 - 35.97V_{IB}$$

$$200V_{IB}^3 + 3640V_{IB}^2 - 4584V_{IB} + 1396.8 = 517.94 - 35.97V_{IB}$$

$$200V_{IB}^3 + 3640V_{IB}^2 - 4584V_{IB} + 1396.8 - 517.94 + 35.97V_{IB} = 0$$

$$200V_{IB}^3 + 3640V_{IB}^2 - 4548.03V_{IB} + 878.86 = 0$$

Solve for the value of V_{IB}, we get

$V_{IB} = 0.24\ \text{V}$, $V_{IB} = 0.95\ \text{V}$ and $V_{IB} = -19.39\ \text{V}$.

Chose the value of V_{IB} to be $V_{IB} = 0.95\ \text{V}$ so that $V_{IB} > V_{tn}$. Hence, the value of V_{IB} is $0.95\ \text{V}$

Step 5 of 5

Calculate the value of the voltage V_{OB}.

$$V_{OB} = V_{IB} - V_{tn}$$

Substitute $0.95\ \text{V}$ for V_{IB} and $0.6\ \text{V}$ for V_{tn}.

$$\begin{aligned} V_{OB} &= 0.95 - 0.6 \\ &= 0.35\ \text{V} \end{aligned}$$

Therefore, the value of the voltage V_{OB} is $0.35\ \text{V}$.

Therefore, the linear region at the output extends in the range $\boxed{0.35\ \text{V} \le V_O \le 4.474\ \text{V}}$ occurs for the input range $0.878\ \text{V} \le V_I \le 0.95\ \text{V}$.

8.047P

We don't have the solution to this problem yet.

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Step 1 of 16

Refer to Figure P8.48 from the text book.

(a)

No current flows through the gate's of the transistors Q_1 and Q_2 when the gate and the drain terminals are opened and hence the currents I_{d1} and I_{D2} are equal. That is,

$$I_{D1} = I_{D2}$$

Substitute $\frac{1}{2}k_n'\left(\frac{W}{L}\right)_1\left(V_G-(-1)-V_m\right)^2$ for I_{D1} and $\frac{1}{2}k_p'\left(\frac{W}{L}\right)_2\left(1-V_G-V_p\right)^2$ for I_{D2}.

$$\frac{1}{2}k_n'\left(\frac{W}{L}\right)_1\left(V_G-(-1)-V_m\right)^2=\frac{1}{2}k_p'\left(\frac{W}{L}\right)_2\left(1-V_G-V_p\right)^2$$

Substitute 0.5 V for V_m and V_p and $k_n'\left(\frac{W}{L}\right)_2$ for $k_p'\left(\frac{W}{L}\right)_1$.

$$\frac{1}{2}k_n'\left(\frac{W}{L}\right)_2\left(V_G-(-1)-0.5\right)^2=\frac{1}{2}k_n'\left(\frac{W}{L}\right)_2\left(1-V_G-0.5\right)^2$$

$$\left(V_G+1-0.5\right)^2=\left(1-V_G-0.5\right)^2$$

$$\left(V_G+0.5\right)^2=\left(0.5-V_G\right)^2$$

$$V_G+0.5=0.5-V_G$$

$$2V_G=0$$

$$V_G=0\text{ V}$$

Therefore, the value of the gate voltage V_G is **0 V**.

Step 2 of 16

Consider the expression for the current I_{m}.

$$I_{m}=\frac{1}{2}k_n'\left(\frac{W}{L}\right)_1\left(V_G-(-1)-V_m\right)^2$$

Substitute 1 mA/V^2 for $k_n'\left(\frac{W}{L}\right)_1$, 0.5 V for V_m and 0 V for V_G.

$$I_{m}=\frac{1}{2}(1\text{ m})\left(0-(-1)-0.5\right)^2$$

$$=\frac{1}{2}(1\text{ m})(1-0.5)^2$$

$$=(0.5\text{ m})(0.5)^2$$

$$=(0.5\text{ m})(0.25)$$

$$I_{m}=0.125\text{ mA}$$

Therefore, the value of the drain current I_{m} is, **0.125 mA**.

The value of the current I_{m} is equal to the value of the current I_{o1}. Therefore, the value of the current I_{o1} is, **0.125 mA**.

Step 3 of 16

(b)

Draw the small signal equivalent circuit diagram for infinite r_o.

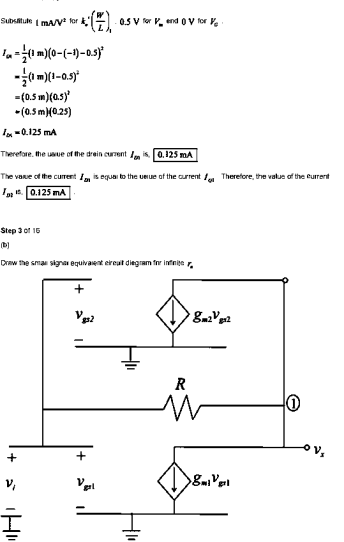


Figure 1

Step 4 of 16

Apply nodal analysis at node-1.

$$\frac{v_x-v_i}{R}+g_{m2}v_{gs2}+g_{m1}v_{gs1}=0$$

Substitute v_i for v_{gs1} and v_{gs2} and g_m for g_{m1} and g_{m2} since design specifications are similar for both the transistors.

$$\frac{v_x-v_i}{R}+g_mv_i+g_mv_i=0$$

$$\frac{v_x-v_i}{R}+2g_mv_i=0$$

$$v_x\left(\frac{1}{R}+\frac{2}{r_o}\right)-v_i\left(\frac{1}{R}-2g_m\right)=0$$

$$v_x\left(\frac{1}{R}+\frac{2}{r_o}\right)=v_i\left(\frac{1}{R}-2g_m\right)$$

$$\frac{v_x}{v_i}=\frac{\frac{1}{R}-2g_m}{\frac{1}{R}+\frac{2}{r_o}}$$

$$\frac{v_x}{v_i}=\frac{1-2g_mR}{1+\frac{2R}{r_o}}\quad\ldots\ldots(2)$$

Therefore, the expression for the voltage gain $\frac{v_x}{v_i}$ is, $\frac{1-2g_mR}{1+\frac{2R}{r_o}}$.

Step 5 of 16

Recall equation (2).

$$\frac{v_x}{v_i}=\frac{1-2g_mR}{1+\frac{2R}{r_o}}$$

Substitute $\frac{2I_D}{V_{ov}-V_t}$ for g_m.

$$\frac{v_x}{v_i}=\frac{1-2\left(\frac{2I_D}{V_{ov}-V_t}\right)R}{1+\frac{2R}{r_o}}$$

$$=\frac{1-\frac{4I_DR}{V_{ov}-V_t}}{1+\frac{2R}{r_o}}$$

Substitute 0.125 mA for I_D, 1 MΩ for R, 160 kΩ for r_o, 0.5 V for V_{ov} and 1 V for V_{tp}.

$$\frac{v_x}{v_i}=\frac{1-\frac{4(0.125\text{ m})(1\text{ M})}{1-0.5}}{1+\frac{2(1\text{ M})}{160\text{ k}}}$$

$$=\frac{1-\frac{500}{0.5}}{1+\frac{2\text{ M}}{160\text{ k}}}$$

$$=\frac{1-1000}{1+12.5}$$

$$=\frac{-999}{13.5}$$

$$\frac{v_x}{v_i}=-74\text{ V/V}$$

Therefore, the value of the voltage gain $\frac{v_x}{v_i}$ for finite r_o is **–74 V/V**.

Step 6 of 16

Refer to Figure 2.

Consider the expression for the voltage v_x.

$$v_x=v_i-i_xR$$

Therefore, the expression for the voltage v_x is, v_i-i_xR.

Step 7 of 16

Apply nodal analysis at node-1.

$$\frac{v_x-v_i}{R}+\frac{v_x}{r_{o1}}+\frac{v_x}{r_{o2}}+g_{m1}v_{gs1}+g_{m2}v_{gs2}=0$$

Substitute v_i for v_{gs1} and $-v_{gs2}$ - g_m for g_{m1} and g_{m2} and r_o for r_{o1} and r_{o2} since design specifications are similar for both the transistors.

$$\frac{v_x-v_i}{R}+\frac{v_x}{r_o}+\frac{v_x}{r_o}+g_mv_i+g_mv_i=0$$

$$\frac{v_x-v_i}{R}+\frac{2v_x}{r_o}+2g_mv_i=0$$

$$v_x\left(\frac{1}{R}+\frac{2}{r_o}\right)-v_i\left(\frac{1}{R}-2g_m\right)=0$$

$$v_x\left(\frac{1}{R}+\frac{2}{r_o}\right)=v_i\left(\frac{1}{R}-2g_m\right)$$

$$\frac{v_x}{v_i}=\frac{\frac{1}{R}-2g_m}{\frac{1}{R}+\frac{2}{r_o}}$$

$$\frac{v_x}{v_i}=\frac{1-2g_mR}{1+\frac{2R}{r_o}}\quad\ldots\ldots(2)$$

Therefore, the expression for the voltage gain $\frac{v_x}{v_i}$ is, $\frac{1-2g_mR}{1+\frac{2R}{r_o}}$.

Step 8 of 16

Recall equation (2).

$$\frac{v_x}{v_i}=\frac{1-2g_mR}{1+\frac{2R}{r_o}}$$

Substitute $\frac{2I_D}{V_{ov}-V_t}$ for g_m.

$$\frac{v_x}{v_i}=\frac{1-2\left(\frac{2I_D}{V_{ov}-V_t}\right)R}{1+\frac{2R}{r_o}}$$

$$=\frac{1-\frac{4I_DR}{V_{ov}-V_t}}{1+\frac{2R}{r_o}}$$

Substitute 0.125 mA for I_D, 1 MΩ for R, 160 kΩ for r_o, 0.5 V for V_{ov} and 1 V for V_{tp}.

$$\frac{v_x}{v_i}=\frac{1-\frac{4(0.125\text{ m})(1\text{ M})}{1-0.5}}{1+\frac{2(1\text{ M})}{160\text{ k}}}$$

$$=\frac{1-\frac{500}{0.5}}{1+\frac{2\text{ M}}{160\text{ k}}}$$

$$=\frac{1-1000}{1+12.5}$$

$$=\frac{-999}{13.5}$$

$$\frac{v_x}{v_i}=-74\text{ V/V}$$

Therefore, the value of the voltage gain $\frac{v_x}{v_i}$ for finite r_o is **–74 V/V**.

Step 9 of 16

Recall equation (2).

$$\frac{v_x}{v_i}=\frac{1-2g_mR}{1+\frac{2R}{r_o}}$$

Substitute $\frac{2I_D}{V_{ov}-V_t}$ for g_m.

$$\frac{v_x}{v_i}=\frac{1-2\left(\frac{2I_D}{V_{ov}-V_t}\right)R}{1+\frac{2R}{r_o}}$$

$$=\frac{1-\frac{4I_DR}{V_{ov}-V_t}}{1+\frac{2R}{r_o}}$$

Substitute 0.125 mA for I_D, 1 MΩ for R, 160 kΩ for r_o, 0.5 V for V_{ov} and 1 V for V_{tp}.

$$\frac{v_x}{v_i}=\frac{1-\frac{4(0.125\text{ m})(1\text{ M})}{1-0.5}}{1+\frac{2(1\text{ M})}{160\text{ k}}}$$

$$=\frac{1-\frac{500}{0.5}}{1+\frac{2\text{ M}}{160\text{ k}}}$$

$$=\frac{1-1000}{1+12.5}$$

$$=\frac{-999}{13.5}$$

$$\frac{v_x}{v_i}=-74\text{ V/V}$$

Therefore, the value of the voltage gain $\frac{v_x}{v_i}$ for finite r_o is **–74 V/V**.

Step 10 of 16

Refer to Figure 2.

Consider the expression for the voltage v_x.

$$v_x=v_i-i_xR$$

Therefore, the expression for the voltage v_x is, v_i-i_xR.

Step 11 of 16

Apply nodal analysis at node 1.

$$\frac{v_x-v_i}{R}+\frac{v_x}{r_{o1}}+\frac{v_x}{r_{o2}}+g_{m1}v_{gs1}+g_{m2}v_{gs2}=0$$

$$\frac{v_x-v_i}{R}+\frac{v_x}{r_o}+\frac{v_x}{r_o}+g_{m1}v_{gs}+g_{m2}v_{gs2}=0$$

$$\frac{v_x}{R}-\frac{v_i}{R}+\frac{v_x}{r_o}+\frac{v_x}{r_o}+g_{m1}v_{gs}+g_{m2}v_{gs2}=0$$

$$v_x\left(\frac{1}{R}+\frac{1}{r_o}+\frac{1}{r_o}\right)-\frac{v_i}{R}+g_mv_{gs}+g_mv_{gs2}=0$$

Substitute v_i-i_xR for v_x, v_i for v_{gs1} and v_{gs2}, g_m for g_{m1} and g_{m2} and r_o for r_{o1} and r_{o2} since design specifications are similar for both the transistors.

$$(v_i-i_xR)\left(\frac{1}{R}+\frac{1}{r_o}+\frac{1}{r_o}\right)-\frac{v_i}{R}+g_mv_i+g_mv_i=0$$

$$v_i\left(\frac{1}{R}+\frac{2}{r_o}\right)-i_xR\left(\frac{1}{R}+\frac{2}{r_o}\right)-\frac{v_i}{R}+2g_mv_i=0$$

$$v_i\left(\frac{1}{R}+\frac{2}{r_o}-\frac{1}{R}+2g_m\right)-i_x\left(1+\frac{2R}{r_o}\right)=0$$

$$v_i\left(\frac{2}{r_o}+2g_m\right)=i_x\left(1+\frac{2R}{r_o}\right)$$

$$\frac{v_i}{i_x}=\frac{1+\frac{2R}{r_o}}{\frac{2}{r_o}+2g_m}$$

$$=\frac{r_o+2R}{2(1+g_mr_o)}$$

Here, $\frac{v_i}{i_x}$ represents the input resistance R_{in}.

$$R_{in}=\frac{r_o+2R}{2(1+g_mr_o)}$$

Substitute $\frac{2I_D}{V_{ov}-V_t}$ for g_m.

$$R_{in}=\frac{r_o+2R}{2\left(1+\frac{2I_D}{V_{ov}-V_t}\right)r_o}$$

$$=\frac{r_o+2R}{2\left(1+\frac{2I_Dr_o}{V_{ov}-V_t}\right)}$$

Step 12 of 16

Substitute 1 MΩ for R, 160 kΩ for r_o and 0.5 V for V_{ov} and 1 V for V_{tp}.

$$R_{in}=\frac{160\text{ k}+2(1\text{ M})}{2\left(1+\frac{2(0.125\text{ m})(160\text{ k})}{1-0.5}\right)}$$

$$=\frac{160\text{ k}+2\text{ M}}{2\left(1+\frac{40}{0.5}\right)}$$

$$=\frac{2.16\text{ M}}{2(1+80)}$$

$$=\frac{2.16\text{ M}}{2(01)}$$

$$R_{in}=\frac{2.16\text{ M}}{162}$$

$$=13.33\text{ k}\Omega$$

Therefore, the value of the input resistance is **13.33 kΩ**.

Step 13 of 16

(d)

Consider the circuit diagram shown in Figure 3.

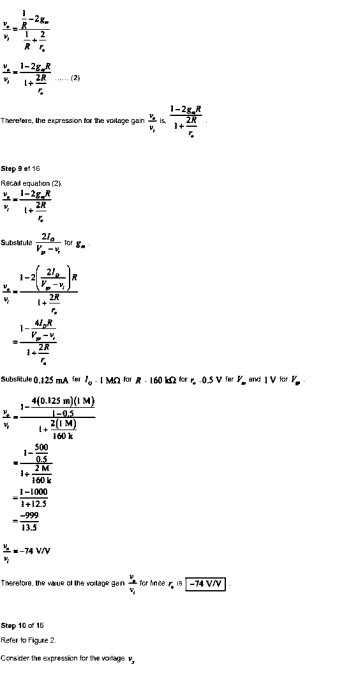


Figure 3

Step 14 of 16

Consider the expression for the voltage gain $\frac{v_x}{v_{sig}}$.

$$\frac{v_x}{v_{sig}}=\left(\frac{v_x}{v_i}\right)\left(\frac{R_{in}}{R_{in}+R_{sig}}\right)$$

Substitute 13.33 kΩ for R_{in}, 20 kΩ for R_{sig} and –74 V/V for $\frac{v_x}{v_i}$.

$$\frac{v_x}{v_{sig}}=(-74)\left(\frac{13.33\text{ k}}{13.33\text{ k}+20\text{ k}}\right)$$

$$=(-74)\left(\frac{13.33\text{ k}}{33.33\text{ k}}\right)$$

$$=(-74)(0.4)$$

$$=-29.6\text{ V/V}$$

Therefore, the value of the voltage $\frac{v_x}{v_{sig}}$ is, **–29.6 V/V**.

Step 15 of 16

(e)

The transistor Q_1 to be in saturation, it should satisfy the following condition.

$$V_{GS}\geq V_{tn}-V_t$$

Substitute V_G+1 for V_{GS} and V_D+1 for V_{DS}.

$$V_D+1\geq V_G+1-V_t$$

Substitute 0 V for V_G and 0.5 V for V_t.

$$V_D+1\geq 0+1-0.5$$

$$V_D\geq -0.5\text{ V}$$

Therefore, the minimum value of the drain voltage is **–0.5 V**.

Step 16 of 16

The transistor Q_2 to be in saturation, it should satisfy the following condition.

$$V_{SD}\geq V_{SG}-V_t$$

Substitute $1-V_D$ for V_{SD} and $1-V_G$ for V_{SG}.

$$1-V_D\geq 1-V_G-V_t$$

Substitute 0 V for V_G and 0.5 V for V_t.

$$1-V_D\geq 1-0-0.5$$

$$-V_D\geq -0.5$$

$$V_D\leq 0.5\text{ V}$$

Therefore, the maximum value of the drain voltage is **0.5 V**.

Therefore, the range of the output voltage is **–0.5 V ≤ V_D ≤ 0.5 V**.

Step 1 of 12

Refer to Figure P8.49 from the text book.

(a)

Calculate the value of the collector current I_{C3} of transistor Q_3.

$$I_{C3} = \frac{3 - V_{BE3}}{46\ \text{k}}$$

Substitute $0.7\ \text{V}$ for V_{BE3}.

$$I_{C3} = \frac{3 - 0.7}{46\ \text{k}}$$
$$= \frac{2.3}{46\ \text{k}}$$
$$= 0.05\ \text{mA}$$

Therefore, the value of the current I_{C3} is $0.05\ \text{mA}$.

Step 2 of 12

Consider the ratio of the currents I_{C3} and I_{C2} in terms of their respective transistor areas.

$$\frac{I_{C3}}{I_{C2}} = \frac{Ar_3}{Ar_2}$$

Here, Ar_3 represents the area of transistor Q_3

Ar_2 represents the area of transistor Q_2

Substitute $5Ar_3$ for Ar_2 and $0.05\ \text{mA}$ for I_{C3}.

$$\frac{0.05\ \text{m}}{I_{C2}} = \frac{Ar_3}{5Ar_3}$$
$$\frac{0.05\ \text{m}}{I_{C2}} = \frac{1}{5}$$
$$I_{C2} = (0.05\ \text{m})(5)$$
$$I_{C2} = 0.25\ \text{mA}$$

Therefore, the value of the current I_{C2} is, $0.25\ \text{mA}$.

Therefore, the value of the current I is, $\boxed{0.25\ \text{mA}}$.

Step 3 of 12

(b)

Calculate the value of the resistance r_{o1}.

$$r_{o1} = \frac{V_A}{I}$$

Substitute $0.25\ \text{mA}$ for I and $30\ \text{V}$ for V_A.

$$r_{o1} = \frac{30}{0.25\ \text{m}}$$
$$= 120\ \text{k}\Omega$$

Therefore, the value of the resistance r_{o1} is $\boxed{120\ \text{k}\Omega}$.

The value of the resistance r_{o2} is equal to the resistance r_{o1} as the value of the current is same for transistors Q_1 and Q_2. Hence, the value of the resistance r_{o2} is $\boxed{120\ \text{k}\Omega}$.

Step 4 of 12

Draw the small signal equivalent circuit diagram.

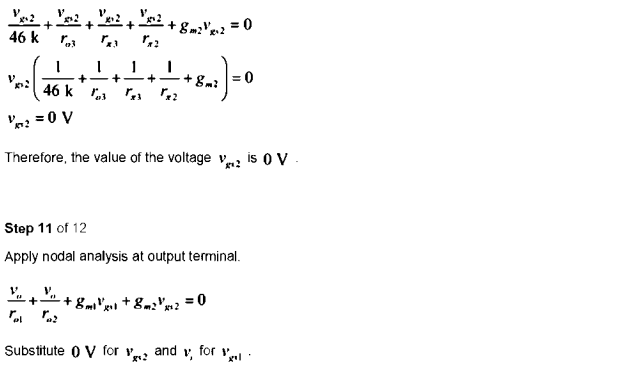


Figure 1

Step 5 of 12

Calculate the value of the total resistance at collector of transistor Q_1.

$$r_o = r_{o1} \| r_{o2}$$

Substitute $120\ \text{k}\Omega$ for r_{o1} and r_{o2}.

$$r_o = 120\ \text{k} \| 120\ \text{k}$$
$$= \frac{(120\ \text{k})(120\ \text{k})}{120\ \text{k} + 120\ \text{k}}$$
$$= \frac{(120\ \text{k})(120\ \text{k})}{2(120\ \text{k})}$$
$$= \frac{120\ \text{k}}{2}$$

$$r_o = 60\ \text{k}\Omega$$

Therefore, the value of the resistance r_o is $\boxed{60\ \text{k}\Omega}$.

Step 6 of 12

(c)

Calculate the value of the transconductance g_{m1}.

$$g_{m1} = \frac{I_{C1}}{V_T}$$

Substitute $0.25\ \text{mA}$ for I_{C1} and $25\ \text{mV}$ for V_T.

$$g_{m1} = \frac{0.25\ \text{m}}{25\ \text{m}}$$
$$= 10\ \text{mA/V}$$

Therefore, the value of the transconductance g_{m1} is, $\boxed{10\ \text{mA/V}}$.

Step 7 of 12

Calculate the value of the resistance $r_{\pi 1}$.

$$r_{\pi 1} = \frac{\beta_1}{g_{m1}}$$

Substitute $10\ \text{mA/V}$ for g_{m1} and 50 for β_1.

$$r_{\pi 1} = \frac{50}{10\ \text{m}}$$
$$= 5\ \text{k}\Omega$$

Therefore, the value of the resistance $r_{\pi 1}$ is $\boxed{5\ \text{k}\Omega}$.

Step 8 of 12

(d)

Consider the expression for the input resistance R_{in} as seen from the input terminals.

$$R_{in} = r_{\pi 1}$$

Substitute $5\ \text{k}\Omega$ for $r_{\pi 1}$.

$$R_{in} = 5\ \text{k}\Omega$$

Therefore, the value of the input resistance is, $\boxed{5\ \text{k}\Omega}$.

Step 9 of 12

Consider the expression for the output resistance R_{out} as seen from the output terminals.

$$R_{out} = r_{o1} \| r_{o2}$$

Substitute $120\ \text{k}\Omega$ for r_{o1} and r_{o2}.

$$R_{out} = 120\ \text{k} \| 120\ \text{k}$$
$$= \frac{(120\ \text{k})(120\ \text{k})}{120\ \text{k} + 120\ \text{k}}$$
$$= \frac{(120\ \text{k})(120\ \text{k})}{2(120\ \text{k})}$$
$$= \frac{120\ \text{k}}{2}$$

$$R_{out} = 60\ \text{k}\Omega$$

Therefore, the value of the output resistance R_{out} is $\boxed{60\ \text{k}\Omega}$.

Step 10 of 12

Apply nodal analysis at node-1.

$$\frac{v_{gs2}}{46\ \text{k}} + \frac{v_{gs2}}{r_{o3}} + \frac{v_{gs2}}{r_{\pi 3}} + \frac{v_{gs2}}{r_{\pi 2}} + g_{m2}v_{gs2} = 0$$
$$v_{gs2}\left(\frac{1}{46\ \text{k}} + \frac{1}{r_{o3}} + \frac{1}{r_{\pi 3}} + \frac{1}{r_{\pi 2}} + g_{m2}\right) = 0$$
$$v_{gs2} = 0\ \text{V}$$

Therefore, the value of the voltage v_{gs2} is $0\ \text{V}$.

Step 11 of 12

Apply nodal analysis at output terminal.

$$\frac{v_o}{r_{o1}} + \frac{v_o}{r_{o2}} + g_{m1}v_{gs1} + g_{m2}v_{gs2} = 0$$

Substitute $0\ \text{V}$ for v_{gs2} and v_i for v_{gs1}.

$$\frac{v_o}{r_{o1}} + \frac{v_o}{r_{o2}} + g_{m1}v_i + g_{m2}(0) = 0$$
$$v_o\left(\frac{1}{r_{o1}} + \frac{1}{r_{o2}}\right) + g_{m1}v_i = 0$$
$$v_o\left(\frac{r_{o2} + r_{o1}}{r_{o1}r_{o2}}\right) = -g_{m1}v_i$$

$$\frac{v_o}{v_i} = -\frac{g_{m1}r_{o1}r_{o2}}{r_{o2} + r_{o1}} \quad \ldots\ldots (1)$$

Step 12 of 12

Substitute $120\ \text{k}\Omega$ for r_{o1} and r_{o2} and $10\ \text{mA/V}$ for g_{m1} in equation (1).

$$\frac{v_o}{v_i} = -\frac{(10\ \text{m})(120\ \text{k})(120\ \text{k})}{120\ \text{k} + 120\ \text{k}}$$
$$= -\frac{(10\ \text{m})(120\ \text{k})(120\ \text{k})}{2(120\ \text{k})}$$
$$= -\frac{(10\ \text{m})(120\ \text{k})}{2}$$

$$A_v = -600\ \text{V/V}$$

Therefore, the value of the voltage gain A_v is, $\boxed{-600\ \text{V/V}}$.

Step 1 of 6

Refer to Figure 7.4(a) in the text book for the common source amplifier circuit.

Determine the overdrive voltage V_{OV1} for transistor Q_1.

$$V_{OV1} = \frac{2V'_{An}L}{A_v}$$

Here,

V'_{An} is the early voltage of the transistor Q_1

L is the length of the channel

A_v is the voltage gain of the amplifier

Substitute $5\times10^{6}\ \text{V/m}$ for V'_{An}, $0.18\times10^{-6}\ \text{m}$ for L and 10 V/V for A_v.

$$V_{OV1} = \frac{(2)(5\times10^{6})(0.18\times10^{-6})}{10}$$
$$= 0.18\ \text{V}$$

Step 2 of 6

Determine the overdrive voltage V_{OV2} of the transistor, Q_2.

$$V_{OV2} = \frac{2|V'_{Ap}|L}{A_v}$$

Here,

V'_{Ap} is the early voltage of the transistor Q_2

L is the length of the channel

A_v is the voltage gain of the amplifier

Substitute $6\times10^{6}\ \text{V/m}$ for V'_{An}, $0.18\times10^{-6}\ \text{m}$ for L and 10 V/V for A_v.

$$V_{OV2} = \frac{(2)(6\times10^{6})(0.18\times10^{-6})}{10}$$
$$= 0.216\ \text{V}$$

Thus, the overdrive voltages of the transistors in the design, V_{OV1} **and** V_{OV2} are, $\boxed{0.18\ \text{V}}$ and $\boxed{0.216\ \text{V}}$.

Step 3 of 6

Determine the transconductances, g_{m1} and g_{m2}, when $I = 50\ \mu\text{A}$.

$$g_{m1} = \frac{2I}{V_{OV1}}$$
$$= \frac{2\times50\times10^{-6}}{0.18}$$
$$= 0.56\ \text{mA/V}$$

$$g_{m2} = \frac{2I}{V_{OV2}}$$
$$= \frac{2\times50\times10^{-6}}{0.216}$$
$$= 0.463\ \text{mA/V}$$

Thus, the transconductances, g_{m1} **and** g_{m2} are, $\boxed{0.56\ \text{mA/V}}$ and $\boxed{0.463\ \text{mA/V}}$.

Step 4 of 6

Write the formula for transconductance, g_m for transistor Q_1.

$$g_m = \sqrt{2k'_n\left(\frac{W}{L}\right)_1 I_{REF}}$$
$$\left(\frac{W}{L}\right)_1 = \frac{g_m^2}{2k'_n I_{REF}}$$

Substitute $0.56\times10^{-3}\ \text{A/V}$ for g_m, $387\times10^{-6}\ \text{A/V}^2$ for k'_n and $50\times10^{-6}\ \text{A}$ for I_{REF}.

$$\left(\frac{W}{L}\right)_1 = \frac{(0.56\times10^{-3})^2}{2\times387\times10^{-6}\times50\times10^{-6}}$$
$$= 8.1$$

Thus, the $\frac{W}{L}$ ratio of the transistor Q_1 is $\boxed{8.1}$.

Step 5 of 6

Write the formula for transconductance, g_m for transistor Q_2.

$$g_m = \sqrt{2k'_p\left(\frac{W}{L}\right)_2 I_{REF}}$$
$$\left(\frac{W}{L}\right)_2 = \frac{g_m^2}{2k'_p I_{REF}}$$

Substitute $0.463\times10^{-3}\ \text{A/V}$ for g_m, $86\times10^{-6}\ \text{A/V}^2$ for k'_p and $50\times10^{-6}\ \text{A}$ for I_{REF}.

$$\left(\frac{W}{L}\right)_1 = \frac{(0.463\times10^{-3})^2}{2\times86\times10^{-6}\times50\times10^{-6}}$$
$$\approx 25$$

Thus, the $\frac{W}{L}$ ratio of the transistor Q_2 is $\boxed{25}$.

Step 6 of 6

The expression for the gain is,

$$A_v = \frac{2V'_{An}L}{V_{OV1}}$$

As gain is proportional to length, if the gain is increased by a factor of 2, L should also be increased by a factor of 2.

Determine the length of the channel.

$$L = 2(0.18\ \mu\text{m})$$
$$= 0.36\ \mu\text{m}$$

Thus, the required channel length for gain is increased by a factor of 2 is $\boxed{0.36\ \mu\text{m}}$.

As the width remains constant and the length is increased by a factor of 2 the total gate area is also increased by a factor of 2.

Step 1 of 6

Refer to Figure 8.18(a) from the text book.

Consider the expression for the input resistance R_{in} .

$$R_{in} = \frac{r_o + R_L}{1 + g_m r_o}$$

Substitute $2\ \text{mA/V}$ for g_m , $20\ \text{k}\Omega$ for R_L and $20\ \text{k}\Omega$ for r_o .

$$\begin{aligned} R_{in} &= \frac{20\ \text{k} + 20\ \text{k}}{1 + (2\ \text{m})(20\ \text{k})} \\ &= \frac{40\ \text{k}}{1 + 40} \\ &= \frac{40\ \text{k}}{41} \\ &= 975.61\ \Omega \end{aligned}$$

Therefore, the value of the input resistance R_{in} is, $\boxed{975.61\ \Omega}$.

Step 2 of 6

Consider the expression for the input resistance R_{out}.

$$R_{out} = r_o + R_s + g_m r_o R_s$$

Substitute $2\ \text{mA/V}$ for g_m , $1\ \text{k}\Omega$ for R_s and $20\ \text{k}\Omega$ for r_o .

$$\begin{aligned} R_{out} &= 20\ \text{k} + 1\ \text{k} + (2\ \text{m})(20\ \text{k})(1\ \text{k}) \\ &= 21\ \text{k} + 40\ \text{k} \\ &= 61\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the output resistance R_{out} is, $\boxed{61\ \text{k}\Omega}$.

Step 3 of 6

Consider the small signal equivalent circuit diagram shown in Figure 1.

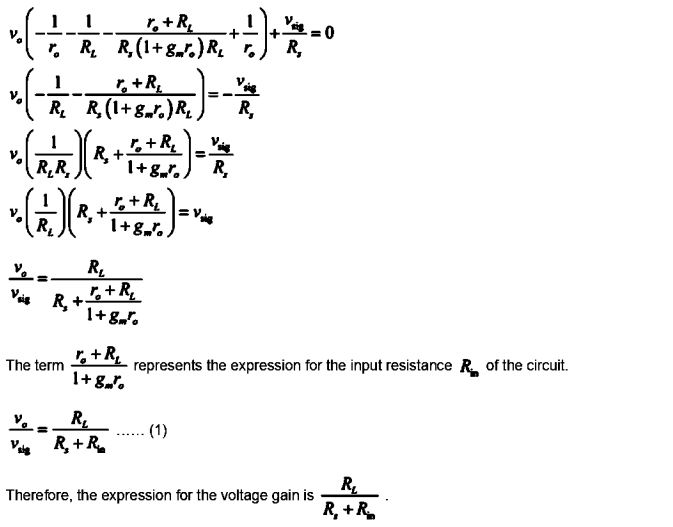


Figure 1

Step 4 of 6

Consider the nodal equation at output terminal.

$$\frac{v_o}{R_L} + \frac{v_o + v_{gs}}{r_o} + g_m v_{gs} = 0$$

$$v_o\left(\frac{1}{R_L} + \frac{1}{r_o}\right) + v_{gs}\left(\frac{1}{r_o} + g_m\right) = 0$$

$$v_o\left(\frac{r_o + R_L}{R_L r_o}\right) + v_{gs}\left(\frac{1 + g_m r_o}{r_o}\right) = 0$$

$$v_o\left(\frac{r_o + R_L}{R_L r_o}\right) + v_{gs}\left(\frac{(1 + g_m r_o) R_L}{r_o}\right) = 0$$

$$v_o(r_o + R_L) + v_{gs}(1 + g_m r_o) R_L = 0$$

$$v_{gs}(1 + g_m r_o) R_L = -v_o(r_o + R_L)$$

$$v_{gs} = -v_o \frac{(r_o + R_L)}{(1 + g_m r_o) R_L}$$

Therefore, the value of the voltage v_{gs} is $-v_o \dfrac{(r_o + R_L)}{(1 + g_m r_o) R_L}$.

Step 5 of 6

Apply nodal analysis at node 1.

$$\frac{-v_{gs} - v_o}{r_o} - g_m v_{gs} + \frac{-v_{gs} - v_{sig}}{R_s} = 0$$

$$-v_{gs}\left(\frac{1}{r_o} + g_m + \frac{1}{R_s}\right) - \frac{v_o}{r_o} - \frac{v_{sig}}{R_s} = 0$$

$$v_{gs}\left(\frac{R_s + g_m r_o R_s + r_o}{r_o R_s}\right) + \frac{v_o}{r_o} + \frac{v_{sig}}{R_s} = 0$$

Substitute $-v_o \dfrac{(r_o + R_L)}{(1 + g_m r_o) R_L}$ for v_{gs} .

$$-v_o \frac{(r_o + R_L)}{(1 + g_m r_o) R_L}\left(\frac{R_s(1 + g_m r_o)}{r_o R_s} + \frac{r_o}{r_o R_s}\right) + \frac{v_o}{r_o} + \frac{v_{sig}}{R_s} = 0$$

$$-v_o \frac{(r_o + R_L)}{(1 + g_m r_o) R_L}\left(\frac{1 + g_m r_o}{r_o} + \frac{1}{R_s}\right) + \frac{v_o}{r_o} + \frac{v_{sig}}{R_s} = 0$$

$$-v_o\left(\frac{(r_o + R_L)(1 + g_m r_o)}{r_o((1 + g_m r_o) R_L)} + \frac{r_o + R_L}{R_s(1 + g_m r_o) R_L}\right) + \frac{v_o}{r_o} + \frac{v_{sig}}{R_s} = 0$$

$$-v_o\left(\frac{(r_o + R_L)}{r_o R_L} + \frac{r_o + R_L}{R_s(1 + g_m r_o) R_L}\right) + \frac{v_o}{r_o} + \frac{v_{sig}}{R_s} = 0$$

$$v_o\left(-\frac{1}{r_o} - \frac{1}{R_L} - \frac{r_o + R_L}{R_s(1 + g_m r_o) R_L} + \frac{1}{r_o}\right) + \frac{v_{sig}}{R_s} = 0$$

$$v_o\left(-\frac{1}{R_L} - \frac{r_o + R_L}{R_s(1 + g_m r_o) R_L}\right) = -\frac{v_{sig}}{R_s}$$

$$v_o\left(\frac{1}{R_L R_s}\right)\left(R_s + \frac{r_o + R_L}{1 + g_m r_o}\right) = \frac{v_{sig}}{R_s}$$

$$v_o\left(\frac{1}{R_L}\right)\left(R_s + \frac{r_o + R_L}{1 + g_m r_o}\right) = v_{sig}$$

$$\frac{v_o}{v_{sig}} = \frac{R_L}{R_s + \dfrac{r_o + R_L}{1 + g_m r_o}}$$

The term $\dfrac{r_o + R_L}{1 + g_m r_o}$ represents the expression for the input resistance R_{in} of the circuit.

$$\frac{v_o}{v_{sig}} = \frac{R_L}{R_s + R_{in}} \quad \text{..... (1)}$$

Therefore, the expression for the voltage gain is $\dfrac{R_L}{R_s + R_{in}}$.

Step 6 of 6

Recall equation (1).

$$\frac{v_o}{v_{sig}} = \frac{R_L}{R_s + R_{in}}$$

Substitute $20\ \text{k}\Omega$ for R_L , $1\ \text{k}\Omega$ for R_s and $975.61\ \Omega$ for R_{in} .

$$\begin{aligned} \frac{v_o}{v_{sig}} &= \frac{20\ \text{k}}{1\ \text{k} + 975.61} \\ &= \frac{20\ \text{k}}{1.97561\ \text{k}} \\ &= 10.12\ \text{V/V} \end{aligned}$$

Therefore, the value of the voltage gain is $\boxed{10.12\ \text{V/V}}$.

Step 1 of 8

Consider the expression for the input resistance R_{in}.

$$R_{in} = \frac{r_o + R_L}{1 + g_m r_o}$$

Substitute $2\ \text{mA/V}$ for g_m, $20\ \text{k}\Omega$ for R_L and $20\ \text{k}\Omega$ for r_o.

$$\begin{aligned} R_{in} &= \frac{20\ \text{k} + 20\ \text{k}}{1 + (2\ \text{m})(20\ \text{k})} \\ &= \frac{40\ \text{k}}{1 + 40} \\ &= \frac{40\ \text{k}}{41} \\ &= 975.61\ \Omega \end{aligned}$$

Therefore, the value of the input resistance R_{in} is, $\boxed{975.61\ \Omega}$.

Step 2 of 8

Draw the small signal equivalent circuit diagram.

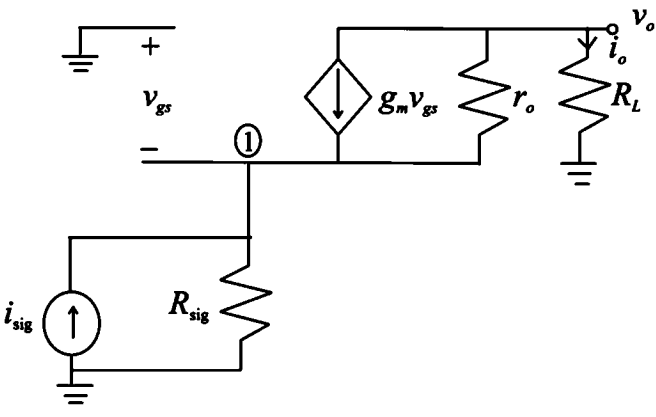


Figure 1

Step 3 of 8

Apply nodal analysis at output terminal.

$$\frac{v_o}{R_L} + \frac{v_o - (-v_{gs})}{r_o} + g_m v_{gs} = 0$$

$$\frac{v_o}{R_L} + \frac{v_o + v_{gs}}{r_o} + g_m v_{gs} = 0$$

$$v_o\left(\frac{1}{R_L} + \frac{1}{r_o}\right) + v_{gs}\left(\frac{1}{r_o} + g_m\right) = 0$$

$$v_o\left(\frac{r_o + R_L}{R_L r_o}\right) + v_{gs}\left(\frac{1 + g_m r_o}{r_o}\right) = 0$$

$$v_o\left(\frac{r_o + R_L}{R_L r_o}\right) + v_{gs}\left(\frac{1 + g_m r_o}{r_o R_L}\right) R_L = 0$$

$$v_o(r_o + R_L) + v_{gs}(1 + g_m r_o) R_L = 0$$

$$v_{gs}(1 + g_m r_o) R_L = -v_o(r_o + R_L)$$

$$v_{gs} = -v_o\left(\frac{r_o + R_L}{(1 + g_m r_o) R_L}\right)$$

Substitute $i_o R_L$ for v_o.

$$\begin{aligned} v_{gs} &= -i_o R_L\left(\frac{r_o + R_L}{(1 + g_m r_o) R_L}\right) \\ &= -i_o\left(\frac{r_o + R_L}{1 + g_m r_o}\right) \end{aligned}$$

Therefore, the expression for the voltage v_{gs} is $-i_o\left(\frac{r_o + R_L}{1 + g_m r_o}\right)$.

Step 4 of 8

Apply nodal analysis at node-1.

$$\frac{-v_{gs}}{R_s} + \frac{-v_{gs} - v_o}{r_o} - g_m v_{gs} - i_{sig} = 0$$

$$-v_{gs}\left(\frac{1}{R_s} + \frac{1}{r_o} + g_m\right) - \frac{v_o}{r_o} - i_{sig} = 0$$

Substitute $-i_o\left(\frac{r_o + R_L}{1 + g_m r_o}\right)$ for v_{gs} and $i_o R_L$ for v_o.

$$-\left(-i_o\left(\frac{r_o + R_L}{1 + g_m r_o}\right)\right)\left(\frac{1}{R_s} + \frac{1}{r_o} + g_m\right) - \frac{i_o R_L}{r_o} - i_{sig} = 0$$

$$i_o\left(\frac{r_o + R_L}{1 + g_m r_o}\right)\left(\frac{1}{R_s} + \frac{1}{r_o} + g_m\right) - \frac{i_o R_L}{r_o} - i_{sig} = 0$$

$$i_o\left(\frac{r_o + R_L}{1 + g_m r_o}\right)\left(\frac{r_o + R_s + g_m r_o R_s}{r_o R_s}\right) - \frac{i_o R_L}{r_o} - i_{sig} = 0$$

$$i_o\left(\frac{r_o + R_L}{1 + g_m r_o}\right)\left(\frac{r_o + R_s(1 + g_m r_o)}{r_o R_s}\right) - \frac{i_o R_L}{r_o} - i_{sig} = 0$$

$$i_o\left(\frac{r_o + R_L}{1 + g_m r_o}\right)\left(\frac{1}{R_s} + \frac{(1 + g_m r_o)}{r_o}\right) - \frac{i_o R_L}{r_o} - i_{sig} = 0$$

$$i_o\left(\frac{1}{R_s}\left(\frac{r_o + R_L}{1 + g_m r_o}\right) + \left(\frac{r_o + R_L}{1 + g_m r_o}\right)\left(\frac{1 + g_m r_o}{r_o}\right)\right) - \frac{i_o R_L}{r_o} - i_{sig} = 0$$

$$i_o\left(\frac{1}{R_s}\left(\frac{r_o + R_L}{1 + g_m r_o}\right) + \frac{r_o + R_L}{r_o}\right) - \frac{i_o R_L}{r_o} - i_{sig} = 0$$

$$i_o\left(\frac{1}{R_s}\left(\frac{r_o + R_L}{1 + g_m r_o}\right) + 1 + \frac{R_L}{r_o} - \frac{R_L}{r_o}\right) - i_{sig} = 0$$

$$i_o\left(\frac{1}{R_s}\left(\frac{r_o + R_L}{1 + g_m r_o}\right) + 1\right) - i_{sig} = 0$$

$$i_o\left(\frac{1}{R_s}\left(\frac{r_o + R_L}{1 + g_m r_o}\right) + 1\right) = i_{sig}$$

$$\frac{i_o}{i_{sig}} = \frac{1}{\frac{1}{R_s}\left(\frac{r_o + R_L}{1 + g_m r_o}\right) + 1}$$

$$\frac{i_o}{i_{sig}} = \frac{R_s}{\left(\frac{r_o + R_L}{1 + g_m r_o}\right) + R_s}$$

Here, the term $\frac{r_o + R_L}{1 + g_m r_o}$ represents the input resistance R_{in}.

$$\frac{i_o}{i_{sig}} = \frac{R_s}{R_{in} + R_s} \quad \text{..... (1)}$$

Therefore, the expression for the current gain is $\frac{R_s}{R_{in} + R_s}$.

Step 5 of 8

Recall equation (1).

$$\frac{i_o}{i_{sig}} = \frac{R_s}{R_{in} + R_s}$$

Substitute $20\ \text{k}\Omega$ for R_s and $975.61\ \Omega$ for R_{in}.

$$\begin{aligned} \frac{i_o}{i_{sig}} &= \frac{20\ \text{k}}{975.61 + 20\ \text{k}} \\ &= \frac{20\ \text{k}}{20.97561\ \text{k}} \\ &= 0.954\ \text{A/A} \end{aligned}$$

Therefore, the value of the current gain is $\boxed{0.954\ \text{A/A}}$.

Step 6 of 8

Recall equation (1).

$$\frac{i_o}{i_{sig}} = \frac{R_s}{R_{in} + R_s}$$

Substitute $\frac{r_o + R_L}{1 + g_m r_o}$ for R_{in}.

$$\frac{i_o}{i_{sig}} = \frac{R_s}{\frac{r_o + R_L}{1 + g_m r_o} + R_s}$$

Substitute $20\ \text{k}\Omega$ for R_s, $2\ \text{mA/V}$ for g_m, $200\ \text{k}\Omega$ for R_L and $20\ \text{k}\Omega$ for r_o.

$$\begin{aligned} \frac{i_o}{i_{sig}} &= \frac{20\ \text{k}}{\frac{20\ \text{k} + 200\ \text{k}}{1 + (2\ \text{m})(20\ \text{k})} + 20\ \text{k}} \\ &= \frac{20\ \text{k}}{\frac{220\ \text{k}}{1 + 40} + 20\ \text{k}} \\ &= \frac{20\ \text{k}}{5.366\ \text{k} + 20\ \text{k}} \\ &= \frac{20\ \text{k}}{25.366\ \text{k}} \end{aligned}$$

$$\begin{aligned} \frac{i_o}{i_{sig}} &= \frac{20\ \text{k}}{25.366\ \text{k}} \\ &= 0.789\ \text{A/A} \end{aligned}$$

Therefore, the value of the current gain when the load resistance is increased by 10 times is $0.789\ \text{A/A}$.

Step 7 of 8

Calculate the percentage change in the value of current gain.

$$\begin{aligned} \frac{i_o}{i_{sig}}\% &= \frac{0.954 - 0.789}{0.954} \times 100 \\ &= \frac{0.165}{0.954} \times 100 \\ &= 0.173 \times 100 \\ &= 17.3\% \end{aligned}$$

Therefore, the percentage change in the value of the current gain is $\boxed{17.3\%}$.

Step 8 of 8

Current buffer:

The circuit which transforms the current source with medium source resistance to an equal current with high source resistance is called a current buffer. Here, the Common Gate amplifier well behaved as a current buffer having a gain of $0.954\ \text{A/A}$ which is nearly equal to gain 1.

Step 1 of 4

Refer to Figure 8.20 from the text book.

Calculate the value of the transconductance g_m.

$$g_m = \frac{2I_D}{V_{OV}}$$

Substitute $0.2\ \text{mA}$ for I_D and $0.2\ \text{V}$ for V_{OV}.

$$g_m = \frac{2(0.2\ \text{m})}{0.2}$$
$$= 2\ \text{mA/V}$$

Therefore, the value of the transconductance is $2\ \text{mA/V}$.

Step 2 of 4

Calculate the value of the resistance r_o.

$$r_o = \frac{V_A}{I_D}$$

Substitute $0.2\ \text{mA}$ for I_D and $20\ \text{V}$ for V_A.

$$r_o = \frac{20}{0.2\ \text{m}}$$
$$= 100\ \text{k}\Omega$$

Therefore, the value of the resistance r_o $100\ \text{k}\Omega$.

Step 3 of 4

Consider the expression fro the output resistance R_{out}.

$$R_{out} = r_o + (1 + g_m r_o) R_s$$

Substitute $100\ \text{k}\Omega$ for r_o, $2\ \text{mA/V}$ for g_m and $500\ \text{k}\Omega$ for R_{out}.

$$500\ \text{k} = 100\ \text{k} + (1 + (2\ \text{m})(100\ \text{k}))R_s$$
$$500\ \text{k} - 100\ \text{k} = (1 + 200)R_s$$
$$400\ \text{k} = (201)R_s$$
$$R_s = \frac{400\ \text{k}}{201}$$
$$R_s = 1.99\ \text{k}\Omega$$

Therefore, the value of the resistance R_s is $\boxed{1.99\ \text{k}\Omega}$.

Step 4 of 4

Calculate the value of the of the voltage V_{BIAS}.

$$V_{BIAS} = I_D R_s + V_{OV} + V_t$$

Substitute $1.99\ \text{k}\Omega$ for R_s, $0.2\ \text{mA}$ for I_D, $0.5\ \text{V}$ for V_t and $0.2\ \text{V}$ for V_{OV}.

$$V_{BIAS} = (0.2\ \text{m})(1.99\ \text{k}) + 0.2 + 0.5$$
$$= 0.398 + 0.7$$
$$= 1.098\ \text{V}$$

Therefore, the value of the voltage V_{BIAS} is $\boxed{1.098\ \text{V}}$.

Step 1 of 7

Calculate the value of the resistance R_s.

$$R_s = \frac{0.3}{I_D}$$

Substitute $100\ \mu\text{A}$ for I_D.

$$R_s = \frac{0.3}{100\ \mu}$$
$$= 3\ \text{k}\Omega$$

Therefore, the value of the resistance R_s is $3\ \text{k}\Omega$.

Step 2 of 7

Calculate the value of the resistane r_o.

$$r_o = \frac{V_A}{I_D}$$

Substitute $100\ \mu\text{A}$ for I_D and $10\ \text{V}$ for V_A.

$$r_o = \frac{10}{100\ \mu}$$
$$= 100\ \text{k}\Omega$$

Therefore, the value of the resistance r_o is $100\ \text{k}\Omega$.

Step 3 of 7

Draw the small signal equivalent circuit diagram to find out the output resistance of the circuit. Apply a reference voltage at the output terminal.

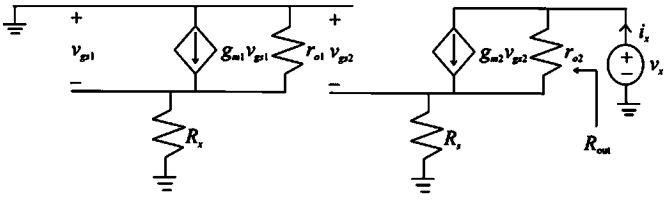


Figure 1

Step 4 of 7

Consider the expression for the voltage v_{gs2}.

$$v_{gs2} = v_{gs1} - v_{gs1}$$
$$= 0\ \text{V}$$

Therefore, the value of the voltage v_{gs2} is $0\ \text{V}$.

Step 5 of 7

Consider the current expression for the current through the resistance r_{o2}.

$$i_{r_{o2}} = i_x - g_{m2}v_{gs2}$$

Therefore, the expression for the current $i_{r_{o2}}$ is $i_x - g_{m2}v_{gs2}$.

Step 6 of 7

Calculate the value of the current through the resistance R_s of resistance Q_2.

$$i_{r_{o2}} = i_x - g_{m2}v_{gs2} + g_{m2}v_{gs2}$$
$$= i_x$$

Therefore, the expression for the current through R_s is i_x.

Step 7 of 7

Consider the expression for the output voltage v_x.

$$v_x = (i_x)(r_{o2}) + (i_x - g_{m2}v_{gs2})(R_s)$$

Substitute $0\ \text{V}$ for v_{gs2}.

$$v_x = (i_x)(r_{o2}) + (i_x - g_{m2}(0))(R_s)$$
$$= (i_x)(r_{o2}) + (i_x)(R_s)$$
$$= i_x(r_{o2} + R_s)$$
$$\frac{v_x}{i_x} = r_{o2} + R_s$$

Substitute $3\ \text{k}\Omega$ for R_s and $100\ \text{k}\Omega$ for r_{o2}.

$$\frac{v_x}{i_x} = 100\ \text{k} + 3\ \text{k}$$
$$= 103\ \text{k}\Omega$$

Here, $\frac{v_x}{i_x}$ represents the output resistance R_{out}.

$$R_{out} = 103\ \text{k}\Omega$$

Therefore, the value of the output resistance R_{out} is, $\boxed{103\ \text{k}\Omega}$.

Step 1 of 26

(a)

Refer to Figure P5.55 from the text book

The drain terminal v_{DS1} and v_{DS2} are equal and hence the drain currents I_1 and I_2 are equal. The drain terminal I_{D1} and I_{D2} is equal to current I_{SS} passes through drain of the transistor Q_1. Hence currents I_{D1} and I_{D2} are equal. That is,

$$I_{D1} = I_{D2}$$

Substitute the value of I_{D1} as I_{D2}.

$$I_{SS} = 100\ \mu A$$

Substitute the value of I_{SS} as $100\ \mu A$.

Step 2 of 26

Consider the expression for the current I_{D2}.

$$I_{D2} = \frac{1}{2} k_n \left(\frac{W}{L}\right)_2 (v_{GS} - V_T)^2$$

Substitute $100\ \mu A$ for I_{D2}, $4\ mA/V^2$ for $k_n\left(\frac{W}{L}\right)_2$ and $0.8\ V$ for V_T.

$$100 = \frac{1}{2}(4)(v_{GS} - 0.8)^2$$

$$50 = (v_{GS} - 0.8)^2$$

$$v_{GS} - 0.8 = \pm\sqrt{50}$$

$$v_{GS} = \pm 7.07 + 0.8$$

$$v_{GS} = -6.27\ V \text{ and } 8.27\ V$$

Consider the negative value of v_{GS} is not possible.

Hence, the value of v_{GS} is $8.27\ V$.

Step 3 of 26

Consider the expression for v_{GS} in terms of v_{out}.

$$v_{GS} = 5 - v_{out} + V_T$$

Substitute $8.27\ V$ for v_{GS} and $0.8\ V$ for V_T.

$$8.27 = 5 - v_{out} + 0.8$$

$$v_{out} = 5 + 0.8 - 8.27$$

$$v_{out} = -2.47\ V$$

Therefore, the value of output voltage v_{out} is $-2.47\ V$.

Step 4 of 26

Consider the expression for the current I_{D2}.

$$I_{D2} = \frac{1}{2} k_n \left(\frac{W}{L}\right)_2 (v_{GS} - V_T)^2$$

Substitute $100\ \mu A$ for I_{D2}, $4\ mA/V^2$ for $k_n\left(\frac{W}{L}\right)_2$ and $0.8\ V$ for V_T.

$$100 = \frac{1}{2}(4)(v_{GS} - 0.8)^2$$

$$50 = (v_{GS} - 0.8)^2$$

$$v_{GS} - 0.8 = \pm\sqrt{50}$$

$$v_{GS} = \pm 7.07 + 0.8$$

$$v_{GS} = -6.27\ V \text{ and } 8.27\ V$$

Consider the value of v_{GS} is not possible.

Hence, the value of v_{GS} is $8.27\ V$.

Step 5 of 26

Consider the expression for the current I_{D1}.

$$I_{D1} = \frac{1}{2} k_n \left(\frac{W}{L}\right)_1 (v_{GS} - V_T)^2$$

Substitute $200\ \mu A$ for I_{D1}, $4\ mA/V^2$ for $k_n\left(\frac{W}{L}\right)_1$ and $0.8\ V$ for V_T.

$$200 = \frac{1}{2}(4)(v_{GS} - 0.8)^2$$

$$100 = (v_{GS} - 0.8)^2$$

$$v_{GS} - 0.8 = \pm\sqrt{100}$$

$$v_{GS} = \pm 10 + 0.8$$

$$v_{GS} = -9.2\ V \text{ and } 10.8\ V$$

Consider the value of v_{GS} is not possible.

Hence, the value of v_{GS} is $10.8\ V$.

Step 6 of 26

Consider the expression for the output resistance R_{out}.

$$R_{out} = r_{o2} \left(1 + g_{m2} R_{SS}\right) \quad \ldots\ldots (1)$$

Therefore, the transconductance g_{m2} is $g_{m2} = \frac{2I_{D2}}{v_{GS2} - V_T}$.

Substitute $100\ \mu A$ for I_{D2}, $8.27\ V$ for v_{GS2} and $0.8\ V$ for V_T.

$$g_{m2} = \frac{2(100)}{8.27 - 0.8}$$

$$= \frac{200}{7.47}$$

$$= 26.77\ mA/V$$

Therefore, the value of g_{m2} is $26.77\ mA/V$.

Step 7 of 26

Consider the expression for the output resistance r_{o2}.

$$r_{o2} = \frac{1}{\lambda_2 I_{D2}}$$

Substitute $0.02\ V^{-1}$ for λ_2 and $100\ \mu A$ for I_{D2}.

$$r_{o2} = \frac{1}{(0.02)(100)}$$

$$= \frac{1}{2}$$

$$= 0.5\ M\Omega$$

Therefore, the value of r_{o2} is $0.5\ M\Omega$.

Step 8 of 26

Consider the expression for the output resistance R_{out}.

$$R_{out} = r_{o2}\left(1 + g_{m2} R_{SS}\right)$$

Substitute $0.5\ M\Omega$ for r_{o2}, $26.77\ mA/V$ for g_{m2} and $200\ k\Omega$ for R_{SS}.

$$R_{out} = 0.5\left(1 + (26.77)(200)\right)$$

$$= 0.5\left(1 + 5354\right)$$

$$= 2677.5\ M\Omega$$

Therefore, the value of the output resistance R_{out} is $2677.5\ M\Omega$.

Step 9 of 26

(b)

Refer to Figure P5.55 from the text book

Therefore, the value of v_{out} is $-2.47\ V$.

Step 10 of 26

Apply node analysis at the output terminal.

$$\frac{v_o - v_{in}}{r_{o2}} + \frac{v_o}{R_{out}} + g_{m2} v_{gs2} = 0$$

Substitute $0\ V$ for v_{in}.

$$\frac{v_o}{r_{o2}} + \frac{v_o}{R_{out}} + g_{m2} v_{gs2} = 0$$

$$v_o \left(\frac{1}{r_{o2}} + \frac{1}{R_{out}}\right) + g_{m2} v_{gs2} = 0$$

$$v_o \left(\frac{1}{r_{o2}} + \frac{1}{r_{o2}\left(1 + g_{m2} R_{SS}\right)}\right) + g_{m2} v_{gs2} = 0$$

$$v_o \left(\frac{2 + g_{m2} R_{SS}}{r_{o2}\left(1 + g_{m2} R_{SS}\right)}\right) + g_{m2} v_{gs2} = 0$$

$$v_o \left(2 + g_{m2} R_{SS}\right) + g_{m2} v_{gs2} r_{o2}\left(1 + g_{m2} R_{SS}\right) = 0$$

$$v_o \left(2 + g_{m2} R_{SS}\right) + v_{gs2}\left(1 + g_{m2} R_{SS}\right) = 0$$

$$v_{gs2} = -\frac{v_o \left(2 + g_{m2} R_{SS}\right)}{\left(1 + g_{m2} R_{SS}\right)} \quad \ldots\ldots (1)$$

Therefore, the expression for the output voltage v_o is: $-v_o\left(\frac{2 + g_{m2} R_{SS}}{1 + g_{m2} R_{SS}}\right)$

Step 11 of 26

Apply node analysis at node 2.

$$-i_x + \frac{-v_{gs2}}{r_{o2}} - g_{m2} v_{gs2} = 0$$

Substitute $-\frac{v_o\left(2 + g_{m2} R_{SS}\right)}{\left(1 + g_{m2} R_{SS}\right)}$ for v_{gs2}.

$$-i_x + \frac{-\left(-\frac{v_o\left(2 + g_{m2} R_{SS}\right)}{\left(1 + g_{m2} R_{SS}\right)}\right)}{r_{o2}} - g_{m2}\left(-\frac{v_o\left(2 + g_{m2} R_{SS}\right)}{\left(1 + g_{m2} R_{SS}\right)}\right) = 0$$

$$-i_x + \frac{\left(\frac{v_o\left(2 + g_{m2} R_{SS}\right)}{\left(1 + g_{m2} R_{SS}\right)}\right)}{r_{o2}} + g_{m2}\left(\frac{v_o\left(2 + g_{m2} R_{SS}\right)}{\left(1 + g_{m2} R_{SS}\right)}\right) = i_x$$

$$-i_x + \left(\frac{v_o\left(2 + g_{m2} R_{SS}\right)}{r_{o2}\left(1 + g_{m2} R_{SS}\right)}\right)

Step 1 of 11

Consider the expression for the input resistance of the Common-Base amplifier.

$$R_{in} = r_e\left(\frac{r_o + R_L}{r_o + \frac{R_L}{\beta+1}}\right) \quad \text{..... (1)}$$

Substitute 0 for R_L.

$$\begin{aligned} R_{in} &= r_e\left(\frac{r_o + 0}{r_o + \frac{0}{\beta+1}}\right) \\ &= r_e\left(\frac{r_o}{r_o}\right) \\ &= r_e(1) \\ &= r_e \end{aligned}$$

Therefore, the value of the input resistance R_{in} for $R_L = 0$ is r_e.

Step 2 of 11

Recall equation (1).

$$\begin{aligned} R_{in} &= r_e\left(\frac{r_o + R_L}{r_o + \frac{R_L}{\beta+1}}\right) \\ &= r_e\left(\frac{R_L}{R_L}\right)\left(\frac{\frac{r_o}{R_L} + 1}{\frac{r_o}{R_L} + \frac{1}{\beta+1}}\right) \\ &= r_e(1)\left(\frac{\frac{r_o}{R_L} + 1}{\frac{r_o}{R_L} + \frac{1}{\beta+1}}\right) \\ &= r_e\left(\frac{\frac{r_o}{R_L} + 1}{\frac{r_o}{R_L} + \frac{1}{\beta+1}}\right) \end{aligned}$$

Substitute ∞ for R_L.

$$\begin{aligned} R_{in} &= r_e\left(\frac{\frac{r_o}{\infty} + 1}{\frac{r_o}{\infty} + \frac{1}{\beta+1}}\right) \\ &= r_e\left(\frac{0+1}{0+\frac{1}{\beta+1}}\right) \\ &= r_e\left(\frac{1}{\frac{1}{\beta+1}}\right) \\ &= r_e(\beta+1) \end{aligned}$$

Therefore, the value of the input resistance R_{in} for $R_L = \infty$ is $r_e(\beta+1)$.

Therefore, as the value of the load resistance R_L increases the value of the input resistance R_{in} increases.

Step 3 of 11

Recall equation (1).

$$R_{in} = r_e\left(\frac{r_o + R_L}{r_o + \frac{R_L}{\beta+1}}\right)$$

$$\frac{R_{in}}{r_e} = \left(\frac{r_o}{r_o}\right)\left(\frac{1 + \frac{R_L}{r_o}}{1 + \left(\frac{R_L}{r_o}\right)\left(\frac{1}{\beta+1}\right)}\right)$$

$$\frac{R_{in}}{r_e} = (1)\left(\frac{1 + \frac{R_L}{r_o}}{1 + \left(\frac{R_L}{r_o}\right)\left(\frac{1}{\beta+1}\right)}\right)$$

$$\frac{R_{in}}{r_e} = \frac{1 + \frac{R_L}{r_o}}{1 + \left(\frac{R_L}{r_o}\right)\left(\frac{1}{\beta+1}\right)} \quad \text{..... (2)}$$

Therefore, the expression for $\frac{R_{in}}{r_e}$ is $\frac{1 + \frac{R_L}{r_o}}{1 + \left(\frac{R_L}{r_o}\right)\left(\frac{1}{\beta+1}\right)}$.

Step 4 of 11

Recall equation (2).

$$\frac{R_{in}}{r_e} = \frac{1 + \frac{R_L}{r_o}}{1 + \left(\frac{R_L}{r_o}\right)\left(\frac{1}{\beta+1}\right)}$$

Substitute 0 for $\frac{R_L}{r_o}$ and 100 for β.

$$\begin{aligned} \frac{R_{in}}{r_e} &= \frac{1+0}{1+(0)\left(\frac{1}{100+1}\right)} \\ &= \frac{1}{1} \\ &= 1 \end{aligned}$$

Therefore, the value of $\frac{R_{in}}{r_e}$ for $\frac{R_L}{r_o} = 0$ is 1.

Step 5 of 11

Recall equation (2).

$$\frac{R_{in}}{r_e} = \frac{1 + \frac{R_L}{r_o}}{1 + \left(\frac{R_L}{r_o}\right)\left(\frac{1}{\beta+1}\right)}$$

Substitute 1 for $\frac{R_L}{r_o}$ and 100 for β.

$$\begin{aligned} \frac{R_{in}}{r_e} &= \frac{1+1}{1+(1)\left(\frac{1}{100+1}\right)} \\ &= \frac{2}{1+\left(\frac{1}{101}\right)} \\ &= \frac{2}{1+9.9\,\mathrm{m}} \\ &= \frac{2}{1.01} \end{aligned}$$

$$\frac{R_{in}}{r_e} = 1.98$$

Therefore, the value of $\frac{R_{in}}{r_e}$ for $\frac{R_L}{r_o} = 1$ is 1.98.

Step 6 of 11

Recall equation (2).

$$\frac{R_{in}}{r_e} = \frac{1 + \frac{R_L}{r_o}}{1 + \left(\frac{R_L}{r_o}\right)\left(\frac{1}{\beta+1}\right)}$$

Substitute 10 for $\frac{R_L}{r_o}$ and 100 for β.

$$\begin{aligned} \frac{R_{in}}{r_e} &= \frac{1+10}{1+(10)\left(\frac{1}{100+1}\right)} \\ &= \frac{11}{1+\frac{10}{101}} \\ &= \frac{2}{1+0.099} \\ &= \frac{2}{1.099} \end{aligned}$$

$$\frac{R_{in}}{r_e} = 10$$

Therefore, the value of $\frac{R_{in}}{r_e}$ for $\frac{R_L}{r_o} = 10$ is 10.

Step 7 of 11

Recall equation (2).

$$\frac{R_{in}}{r_e} = \frac{1 + \frac{R_L}{r_o}}{1 + \left(\frac{R_L}{r_o}\right)\left(\frac{1}{\beta+1}\right)}$$

Substitute 100 for $\frac{R_L}{r_o}$ and 100 for β.

$$\begin{aligned} \frac{R_{in}}{r_e} &= \frac{1+100}{1+(100)\left(\frac{1}{100+1}\right)} \\ &= \frac{101}{1+\frac{100}{101}} \\ &= \frac{101}{1+0.99} \\ &= \frac{101}{1.99} \end{aligned}$$

$$\frac{R_{in}}{r_e} = 50.75$$

Step 8 of 11

Therefore, the value of $\frac{R_{in}}{r_e}$ for $\frac{R_L}{r_o} = 100$ is 50.75.

Step 9 of 11

Recall equation (2).

$$\frac{R_{in}}{r_e} = \frac{1 + \frac{R_L}{r_o}}{1 + \left(\frac{R_L}{r_o}\right)\left(\frac{1}{\beta+1}\right)}$$

Substitute 1000 for $\frac{R_L}{r_o}$ and 100 for β.

$$\begin{aligned} \frac{R_{in}}{r_e} &= \frac{1+1000}{1+(1000)\left(\frac{1}{100+1}\right)} \\ &= \frac{1001}{1+\frac{1000}{101}} \\ &= \frac{1001}{1+9.9} \\ &= \frac{1001}{10.9} \end{aligned}$$

$$\frac{R_{in}}{r_e} = 91.84$$

Therefore, the value of $\frac{R_{in}}{r_e}$ for $\frac{R_L}{r_o} = 1000$ is 91.84.

Step 10 of 11

Recall equation (2).

$$\begin{aligned} \frac{R_{in}}{r_e} &= \frac{1 + \frac{R_L}{r_o}}{1 + \left(\frac{R_L}{r_o}\right)\left(\frac{1}{\beta+1}\right)} \\ &= \frac{\left(\frac{R_L}{r_o}\right)\frac{1}{\left(\frac{R_L}{r_o}\right)} + 1}{\frac{1}{\left(\frac{R_L}{r_o}\right)} + \left(\frac{1}{\beta+1}\right)} \\ &= \frac{\frac{1}{\left(\frac{R_L}{r_o}\right)} + 1}{\frac{1}{\left(\frac{R_L}{r_o}\right)} + \left(\frac{1}{\beta+1}\right)} \end{aligned}$$

Substitute ∞ for $\frac{R_L}{r_o}$ and 100 for β.

$$\begin{aligned} \frac{R_{in}}{r_e} &= \frac{\frac{1}{\infty} + 1}{\frac{1}{\infty} + \left(\frac{1}{100+1}\right)} \\ &= \frac{0+1}{0+\left(\frac{1}{101}\right)} \\ &= 101 \end{aligned}$$

Therefore, the value of $\frac{R_{in}}{r_e}$ for $\frac{R_L}{r_o} = \infty$ is 101.

Step 11 of 11

Tabulate the results.

Table 1

$\frac{R_L}{r_o}$	0	1	10	100	1000	∞
$\frac{R_{in}}{r_e}$	1	1.98	10	50.75	91.84	101

Therefore, the required results are tabulated.

8.057P

Consider the expression for the input resistance of the Common-Base amplifier.

$$R_{in} = r_e\left(\frac{r_o + R_L}{r_o + \frac{R_L}{\beta+1}}\right)$$

Substitute $2r_e$ for R_{in} .

$$2r_e = r_e\left(\frac{r_o + R_L}{r_o + \frac{R_L}{\beta+1}}\right)$$

$$2 = \frac{r_o + R_L}{r_o + \frac{R_L}{\beta+1}}$$

$$2\left(r_o + \frac{R_L}{\beta+1}\right) = r_o + R_L$$

$$2r_o + \frac{2R_L}{\beta+1} = r_o + R_L$$

$$R_L - \frac{2R_L}{\beta+1} = 2r_o - r_o$$

$$R_L\left(1 - \frac{2}{\beta+1}\right) = r_o$$

$$R_L\left(\frac{\beta+1-2}{\beta+1}\right) = r_o$$

$$R_L\left(\frac{\beta-1}{\beta+1}\right) = r_o$$

$$R_L = r_o\left(\frac{\beta+1}{\beta-1}\right)$$

Therefore, the required value of load resistance R_L is $\boxed{r_o\left(\frac{\beta+1}{\beta-1}\right)}$.

Step 1 of 9

Consider the expression for the output resistance of the Common-Base amplifier.

$$R_{out} = r_o + g_m r_o (R_s \| r_\pi)$$
$$R_{out} = r_o (1 + g_m (R_s \| r_\pi))$$
$$\frac{R_{out}}{r_o} = 1 + g_m \left(\frac{R_s r_\pi}{R_s + r_\pi} \right)$$

Substitute $\frac{\beta}{r_\pi}$ for g_m.

$$\frac{R_{out}}{r_o} = 1 + \left(\frac{\beta}{r_\pi} \right) \left(\frac{R_s r_\pi}{R_s + r_\pi} \right)$$
$$= 1 + \frac{\beta R_s}{R_s + r_\pi}$$

Substitute $(\beta + 1) r_e$ for r_π.

$$\frac{R_{out}}{r_o} = 1 + \frac{\beta R_s}{R_s + (\beta + 1) r_e}$$

$$\frac{R_{out}}{r_o} = 1 + \frac{\beta \left(\frac{R_s}{r_e} \right)}{\frac{R_s}{r_e} + \beta + 1}$$

Therefore, the expression for $\frac{R_{out}}{r_o}$ is $\boxed{1 + \frac{\beta \left(\frac{R_s}{r_e} \right)}{\frac{R_s}{r_e} + \beta + 1}}$.

Step 2 of 9

Determine the value of $\frac{R_{out}}{r_o}$ when $R_s = 0$:

$$\frac{R_{out}}{r_o} = 1 + \frac{\beta \left(\frac{R_s}{r_e} \right)}{\frac{R_s}{r_e} + \beta + 1}$$

Substitute 0 for R_s and 100 for β.

$$\frac{R_{out}}{r_o} = 1 + \frac{(100) \left(\frac{0}{r_e} \right)}{\frac{0}{r_e} + 100 + 1}$$
$$= 1 + \frac{0}{0 + 101}$$
$$= 1$$

Therefore, the value of $\frac{R_{out}}{r_o}$ for $R_s = 0$ is 1.

Step 3 of 9

Determine the value of $\frac{R_{out}}{r_o}$ when $R_s = r_e$:

$$\frac{R_{out}}{r_o} = 1 + \frac{\beta \left(\frac{R_s}{r_e} \right)}{\frac{R_s}{r_e} + \beta + 1}$$

Substitute r_e for R_s and 100 for β.

$$\frac{R_{out}}{r_o} = 1 + \frac{(100) \left(\frac{r_e}{r_e} \right)}{\frac{r_e}{r_e} + 100 + 1}$$
$$= 1 + \frac{(100)(1)}{1 + 100 + 1}$$
$$= 1 + \frac{100}{102}$$
$$= 1 + 0.98$$

$$\frac{R_{out}}{r_o} = 1.98$$

Therefore, the value of $\frac{R_{out}}{r_o}$ for $R_s = r_e$ is 1.98.

Step 4 of 9

Determine the value of $\frac{R_{out}}{r_o}$ when $R_s = 2r_e$:

$$\frac{R_{out}}{r_o} = 1 + \frac{\beta \left(\frac{R_s}{r_e} \right)}{\frac{R_s}{r_e} + \beta + 1}$$

Substitute $2r_e$ for R_s and 100 for β.

$$\frac{R_{out}}{r_o} = 1 + \frac{(100) \left(\frac{2r_e}{r_e} \right)}{\frac{2r_e}{r_e} + 100 + 1}$$
$$= 1 + \frac{(100)(2)}{2 + 100 + 1}$$
$$= 1 + \frac{200}{103}$$
$$= 1 + 1.94$$

$$\frac{R_{out}}{r_o} = 2.94$$

Therefore, the value of $\frac{R_{out}}{r_o}$ for $R_s = 2r_e$ is 2.94.

Step 5 of 9

Determine the value of $\frac{R_{out}}{r_o}$ when $R_s = 10r_e$:

$$\frac{R_{out}}{r_o} = 1 + \frac{\beta \left(\frac{R_s}{r_e} \right)}{\frac{R_s}{r_e} + \beta + 1}$$

Substitute $10r_e$ for R_s and 100 for β.

$$\frac{R_{out}}{r_o} = 1 + \frac{(100) \left(\frac{10r_e}{r_e} \right)}{\frac{r_e}{r_e} + 100 + 1}$$
$$= 1 + \frac{(100)(10)}{10 + 100 + 1}$$
$$= 1 + \frac{1000}{111}$$
$$= 1 + 9.01$$

$$\frac{R_{out}}{r_o} = 10.01$$

Therefore, the value of $\frac{R_{out}}{r_o}$ for $R_s = 10r_e$ is 10.01.

Step 6 of 9

Determine the value of $\frac{R_{out}}{r_o}$ when $R_s = \frac{\beta}{2} r_e$:

$$\frac{R_{out}}{r_o} = 1 + \frac{\beta \left(\frac{R_s}{r_e} \right)}{\frac{R_s}{r_e} + \beta + 1}$$

Substitute $\left(\frac{\beta}{2} \right) r_e$ for R_s and 100 for β.

$$\frac{R_{out}}{r_o} = 1 + \frac{(100) \left(\frac{\left(\frac{100}{2} \right) r_e}{r_e} \right)}{\frac{\left(\frac{100}{2} \right) r_e}{r_e} + 100 + 1}$$
$$= 1 + \frac{(100)(50)}{50 + 100 + 1}$$
$$= 1 + \frac{5000}{151}$$
$$= 1 + 33.11$$

$$\frac{R_{out}}{r_o} = 34.11$$

Therefore, the value of $\frac{R_{out}}{r_o}$ for $R_s = \left(\frac{\beta}{2} \right) r_e$ is 34.11.

Step 7 of 9

Determine the value of $\frac{R_{out}}{r_o}$ when $R_s = \beta r_e$:

$$\frac{R_{out}}{r_o} = 1 + \frac{\beta \left(\frac{R_s}{r_e} \right)}{\frac{R_s}{r_e} + \beta + 1}$$

Substitute βr_e for R_s and 100 for β.

$$\frac{R_{out}}{r_o} = 1 + \frac{(100) \left(\frac{100 r_e}{r_e} \right)}{\frac{100 r_e}{r_e} + 100 + 1}$$
$$= 1 + \frac{(100)(100)}{100 + 100 + 1}$$
$$= 1 + \frac{10000}{201}$$
$$= 1 + 49.75$$

$$\frac{R_{out}}{r_o} = 50.75$$

Therefore, the value of $\frac{R_{out}}{r_o}$ for $R_s = \beta r_e$ is 50.75.

Step 8 of 9

Determine the value of $\frac{R_{out}}{r_o}$ when $R_s = 1000 r_e$:

$$\frac{R_{out}}{r_o} = 1 + \frac{\beta \left(\frac{R_s}{r_e} \right)}{\frac{R_s}{r_e} + \beta + 1}$$

Substitute $1000 r_e$ for R_s and 100 for β.

$$\frac{R_{out}}{r_o} = 1 + \frac{(100) \left(\frac{1000 r_e}{r_e} \right)}{\frac{1000 r_e}{r_e} + 100 + 1}$$
$$= 1 + \frac{(100)(1000)}{1000 + 100 + 1}$$
$$= 1 + \frac{100000}{1101}$$
$$= 1 + 90.823$$

$$\frac{R_{out}}{r_o} = 91.83$$

Therefore, the value of $\frac{R_{out}}{r_o}$ for $R_s = 1000 r_e$ is 91.83.

Step 9 of 9

Tabulate the results are as follows:

Table-1

R_s	0	r_e	$2r_e$	$10r_e$	$\left(\frac{\beta}{2} \right) r_e$	βr_e	$1000 r_e$
$\frac{R_{out}}{r_o}$	1	1.98	2.94	10.01	34.11	50.75	91.83

Therefore, the required values are tabulated.

Step 1 of 13

Refer to Figure P8.59 from the text book.

Calculate the value of the output voltage v_o.

$v_o = ki_{sig}R_{out}$ ..... (1)

Therefore, the expression fro the output voltage v_o is $ki_{sig}R_{out}$.

Step 2 of 13

Calculate the value of the resistance r_o.

$$r_o = \frac{V_A}{I_C}$$

Substitute $50\ \text{V}$ for V_A and $0.1\ \text{mA}$ for I_C.

$$\begin{aligned} r_o &= \frac{50}{0.1\ \text{m}} \\ &= \frac{50}{0.1\ \text{m}} \\ &= 500\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance r_o is $500\ \text{k}\Omega$.

Step 3 of 13

Calculate the value of the tranconductance g_m.

$$g_m = \frac{I_C}{V_T}$$

Substitute $0.1\ \text{mA}$ for I_C and $25\ \text{mV}$ for V_T.

$$\begin{aligned} g_m &= \frac{0.1\ \text{m}}{25\ \text{m}} \\ &= 4\ \text{mA/V} \end{aligned}$$

Therefore, the value of the transconductance g_m is, $4\ \text{mA/V}$.

Step 4 of 13

Calculate the value of the resistance r_π.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute $4\ \text{mA/V}$ for g_m and 100 for β.

$$\begin{aligned} r_\pi &= \frac{100}{4\ \text{m}} \\ &= 25\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance r_π is $25\ \text{k}\Omega$.

Step 5 of 13

Draw the small signal equivalent circuit diagram.

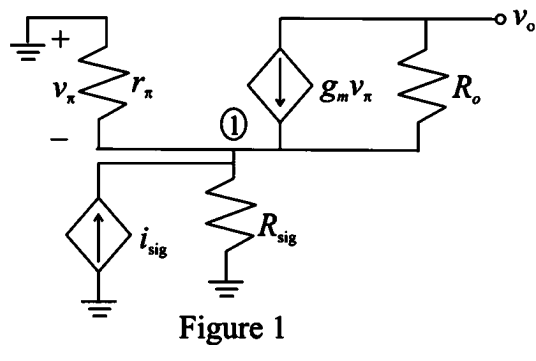


Figure 1

Step 6 of 13

Apply nodal analysis at output terminal.

$$\frac{v_o + v_\pi}{r_o} + g_m v_\pi = 0$$

$$\frac{v_o}{r_o} + \frac{v_\pi}{r_o} + g_m v_\pi = 0$$

$$\frac{v_o}{r_o} + \frac{v_\pi}{r_o} + \frac{g_m r_o v_\pi}{r_o} = 0$$

$$v_o + v_\pi + g_m r_o v_\pi = 0$$

$$v_o + v_\pi\left(1 + g_m r_o\right) = 0$$

$$v_\pi\left(1 + g_m r_o\right) = -v_o$$

$$v_\pi = -\frac{v_o}{1 + g_m r_o}$$

Therefore, the expression for the voltage v_π is $-\frac{v_o}{1 + g_m r_o}$.

Step 7 of 13

Apply nodal analysis at node-1.

$$\frac{-v_\pi}{r_\pi} - \frac{v_\pi}{R_{sig}} - i_{sig} - g_m v_\pi + \frac{-v_\pi - v_o}{r_o} = 0$$

$$-v_\pi\left(\frac{1}{r_\pi} + \frac{1}{R_{sig}} + g_m + \frac{1}{r_o}\right) - i_{sig} - \frac{v_o}{r_o} = 0$$

$$-v_\pi\left(\frac{R_{sig} + r_\pi}{r_\pi R_{sig}} + \frac{g_m r_o + 1}{r_o}\right) - \frac{v_o}{r_o} = i_{sig}$$

Substitute $-\frac{v_o}{1 + g_m r_o}$ for v_π.

$$\frac{v_o}{1 + g_m r_o}\left(\frac{R_{sig} + r_\pi}{r_\pi R_{sig}} + \frac{g_m r_o + 1}{r_o}\right) - \frac{v_o}{r_o} = i_{sig}$$

$$v_o\left(\frac{R_{sig} + r_\pi}{\left(1 + g_m r_o\right) r_\pi R_{sig}} + \frac{1}{r_o}\right) - \frac{v_o}{r_o} = i_{sig}$$

$$v_o\left(\frac{R_{sig} + r_\pi}{\left(1 + g_m r_o\right) r_\pi R_{sig}} + \frac{1}{r_o} - \frac{1}{r_o}\right) = i_{sig}$$

$$v_o\left(\frac{R_{sig} + r_\pi}{\left(1 + g_m r_o\right) r_\pi R_{sig}}\right) = i_{sig}$$

$$v_o = \left(\frac{\left(1 + g_m r_o\right) r_\pi R_{sig}}{R_{sig} + r_\pi}\right) i_{sig}$$ ..... (2)

Therefore, the expression for the output voltage is $\left(\frac{\left(1 + g_m r_o\right) r_\pi R_{sig}}{R_{sig} + r_\pi}\right) i_{sig}$.

Step 8 of 13

Recall equation (2).

$$v_o = \left(\frac{\left(1 + g_m r_o\right) r_\pi R_{sig}}{R_{sig} + r_\pi}\right) i_{sig}$$

Substitute $500\ \text{k}\Omega$ for r_o, $4\ \text{mA/V}$ for g_m, $10\ \text{k}\Omega$ for R_{sig} and $25\ \text{k}\Omega$ for r_π.

$$\begin{aligned} v_o &= \left(\frac{\left(1 + (4\ \text{m})(500\ \text{k})\right)(25\ \text{k})(10\ \text{k})}{10\ \text{k} + 25\ \text{k}}\right) i_{sig} \\ &= \left(\frac{\left(1 + 2000\right)(250\ \text{M})}{35\ \text{k}}\right) i_{sig} \\ &= \left(\frac{500250\ \text{M}}{35\ \text{k}}\right) i_{sig} \\ &= (14.29\ \text{M})\, i_{sig} \end{aligned}$$

Therefore, the expression for the output voltage v_o is $(14.29\ \text{M})\, i_{sig}$.

Step 9 of 13

Draw the small signal equivalent circuit to find the output resistance R_{out}.

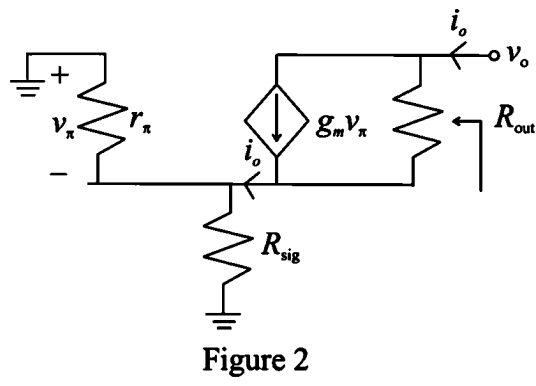


Figure 2

Step 10 of 13

Consider the expression for the voltage v_π.

$$v_\pi = -i_o\left(R_{sig} \| r_\pi\right)$$

Therefore, the expression for the voltage v_π is $-i_o\left(R_{sig} \| r_\pi\right)$.

Step 11 of 13

Consider the expression for the output voltage v_o.

$$v_o = \left(i_o - g_m v_\pi\right) r_o + i_o\left(R_{sig} \| r_\pi\right)$$

Substitute $-i_o\left(R_{sig} \| r_\pi\right)$ for v_π.

$$\begin{aligned} v_o &= \left(i_o + i_o g_m\left(R_{sig} \| r_\pi\right)\right) r_o + i_o\left(R_{sig} \| r_\pi\right) \\ &= i_o\left(r_o + g_m r_o\left(R_{sig} \| r_\pi\right) + \left(R_{sig} \| r_\pi\right)\right) \\ &= i_o\left(r_o + g_m r_o\left(R_{sig} \| r_\pi\right) + \left(R_{sig} \| r_\pi\right)\right) \\ &= i_o\left(r_o + \left(1 + g_m r_o\right)\left(R_{sig} \| r_\pi\right)\right) \end{aligned}$$

$$\frac{v_o}{i_o} = r_o + \left(1 + g_m r_o\right)\left(R_{sig} \| r_\pi\right)$$

Here, $\frac{v_o}{i_o}$ represents the output resistance R_o.

$$R_o = r_o + \left(1 + g_m r_o\right)\left(R_{sig} \| r_\pi\right)$$ ..... (3)

Therefore, the expression for the output resistance is $r_o + \left(1 + g_m r_o\right)\left(R_{sig} + r_\pi\right)$.

Step 12 of 13

Recall equation (3).

$$R_o = r_o + \left(1 + g_m r_o\right)\left(R_{sig} \| r_\pi\right)$$

Substitute $500\ \text{k}\Omega$ for r_o, $4\ \text{mA/V}$ for g_m, $10\ \text{k}\Omega$ for R_{sig} and $25\ \text{k}\Omega$ for r_π.

$$\begin{aligned} R_o &= 500\ \text{k} + \left(1 + (4\ \text{m})(500\ \text{k})\right)(10\ \text{k} \| 25\ \text{k}) \\ &= 500\ \text{k} + \left(1 + (4\ \text{m})(500\ \text{k})\right)\left(\frac{10\ \text{k} \cdot 25\ \text{k}}{10\ \text{k} + 25\ \text{k}}\right) \\ &= 500\ \text{k} + \left(1 + (4\ \text{m})(500\ \text{k})\right)(7.143\ \text{k}) \\ &= 500\ \text{k} + \left(1 + 2000\right)(7.143\ \text{k}) \\ &= 500\ \text{k} + (2001)(7.143\ \text{k}) \\ &= 500\ \text{k} + 14.29\ \text{M} \end{aligned}$$

$$R_o = 14.79\ \text{M}\Omega$$

Therefore, the value of the output resistance R_o is $\boxed{14.79\ \text{M}\Omega}$.

Step 13 of 13

Compare equations (1) and (2).

$$ki_{sig}R_{out} = (14.29\ \text{M})\, i_{sig}$$

$$kR_{out} = 14.29\ \text{M}$$

Substitute $14.79\ \text{M}\Omega$ for R_o.

$$ki_{sig}R_{out} = (14.29\ \text{M})\, i_{sig}$$

$$k(14.79\ \text{M}) = 14.29\ \text{M}$$

$$k = \frac{14.29\ \text{M}}{14.79\ \text{M}}$$

$$k = 0.966$$

Therefore, eh value of the current gain k is $\boxed{0.966}$.

Step 1 of 9

8.060P

Refer to Figure P8.60 from the text book.

Calculate the value of the emitter current I_E.

$$I_E = \frac{5 - V_{BE}}{4.3\ \text{k}}$$

Substitute $0.7\ \text{V}$ for V_{BE}.

$$\begin{aligned} I_E &= \frac{5 - 0.7}{4.3\ \text{k}} \\ &= \frac{4.3}{4.3\ \text{k}} \\ &= 1\ \text{mA} \end{aligned}$$

Therefore, the value of the emitter current is $1\ \text{mA}$.

The value of the collector current I_C is approximately equal to the emitter current I_E. Therefore, the value of the collector current is $\boxed{1\ \text{mA}}$.

Step 2 of 9

Calculate the value of the resistance r_o.

$$r_o = \frac{V_A}{I_C}$$

Substitute $100\ \text{V}$ for V_A and $1\ \text{mA}$ for I_C.

$$\begin{aligned} r_o &= \frac{100}{1\ \text{m}} \\ &= 100\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance r_o is $100\ \text{k}\Omega$.

Step 3 of 9

Calculate the value of the tranconductance g_m.

$$g_m = \frac{I_C}{V_T}$$

Substitute $1\ \text{mA}$ for I_C and $25\ \text{mV}$ for V_T.

$$\begin{aligned} g_m &= \frac{1\ \text{m}}{25\ \text{m}} \\ &= 40\ \text{mA/V} \end{aligned}$$

Therefore, the value of the transconductance g_m is, $40\ \text{mA/V}$.

Step 4 of 9

Calculate the value of the resistance r_π.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute $40\ \text{mA/V}$ for g_m and 100 for β.

$$\begin{aligned} r_\pi &= \frac{100}{40\ \text{m}} \\ &= 2.5\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance r_π is $2.5\ \text{k}\Omega$.

Step 5 of 9

Draw the small signal equivalent circuit to find the output resistance R_{out}.

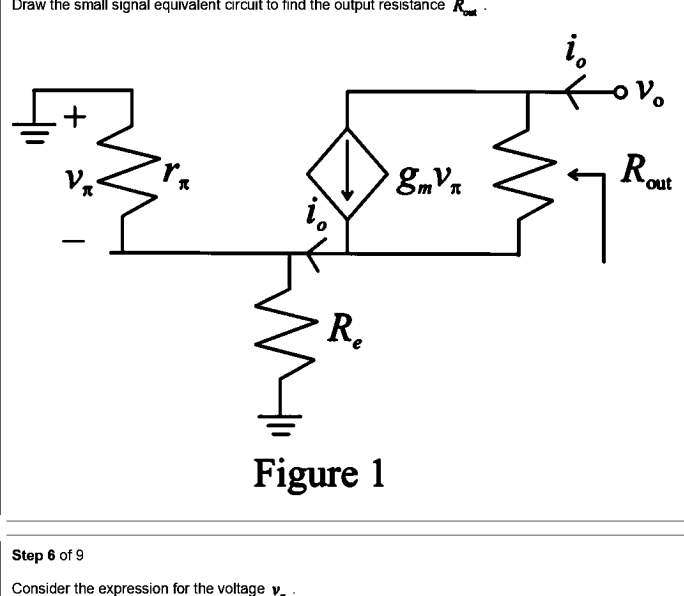


Figure 1

Step 6 of 9

Consider the expression for the voltage v_π.

$$v_\pi = -i_o (R_e \| r_\pi)$$

Therefore, the expression for the voltage v_π is $-i_o (R_e \| r_\pi)$.

Step 7 of 9

Consider the expression for the output voltage v_o.

$$v_o = (i_o - g_m v_\pi) r_o + i_o (R_e \| r_\pi)$$

Substitute $-i_o (R_e \| r_\pi)$ for v_π.

$$\begin{aligned} v_o &= (i_o + i_o g_m (R_e \| r_\pi)) r_o + i_o (R_e \| r_\pi) \\ &= i_o (r_o + g_m r_o (R_e \| r_\pi) + (R_e \| r_\pi)) \\ &= i_o (r_o + g_m r_o (R_e \| r_\pi) + (R_e \| r_\pi)) \\ &= i_o (r_o + (1 + g_m r_o)(R_e \| r_\pi)) \end{aligned}$$

$$\frac{v_o}{i_o} = r_o + (1 + g_m r_o)(R_e \| r_\pi)$$

Here, $\frac{v_o}{i_o}$ represents the output resistance R_o.

$$R_o = r_o + (1 + g_m r_o)(R_e \| r_\pi) \quad \text{..... (1)}$$

Therefore, the expression for the output resistance is $r_o + (1 + g_m r_o)(R_e \| r_\pi)$.

Step 8 of 9

Recall equation (1).

$$R_o = r_o + (1 + g_m r_o)(R_e \| r_\pi)$$

Substitute $100\ \text{k}\Omega$ for r_o, $40\ \text{mA/V}$ for g_m, $4.3\ \text{k}\Omega$ for R_e and $2.5\ \text{k}\Omega$ for r_π.

$$\begin{aligned} R_o &= 100\ \text{k} + (1 + (40\ \text{m})(100\ \text{k}))(4.3\ \text{k} \| 2.5\ \text{k}) \\ &= 100\ \text{k} + (1 + (40\ \text{m})(100\ \text{k}))\left(\frac{4.3\ \text{k} \times 2.5\ \text{k}}{4.3\ \text{k} + 2.5\ \text{k}}\right) \\ &= 100\ \text{k} + (1 + (40\ \text{m})(100\ \text{k}))\left(\frac{10.75\ \text{M}}{6.8\ \text{k}}\right) \\ &= 100\ \text{k} + (1 + 4000)(1.58\ \text{k}) \\ &= 100\ \text{k} + (4001)(1.58\ \text{k}) \\ &= 100\ \text{k} + 6.32\ \text{M} \end{aligned}$$

$$R_o = 6.42\ \text{M}\Omega$$

Therefore, the value of the output resistance R_o is $\boxed{6.42\ \text{M}\Omega}$.

Step 9 of 9

Calculate the change in the value of the collector current ΔI_C.

$$\Delta I_C = \frac{\Delta V}{R_{out}}$$

Substitute $10\ \text{V}$ for ΔV and $6.42\ \text{M}\Omega$ for R_{out}.

$$\begin{aligned} \Delta I_C &= \frac{10}{6.42\ \text{M}} \\ &= 1.57\ \mu\text{A} \end{aligned}$$

Therefore, the value of the change in the collector current ΔI_C is $\boxed{1.57\ \mu\text{A}}$.

Step 1 of 8

Consider the expression for the output resistance of the Common-Base amplifier.

$$R_{out} = r_o + g_m r_o (R_e \| r_x)$$
$$R_{out} = r_o (1 + g_m (R_e \| r_x))$$
$$\frac{R_{out}}{r_o} = 1 + g_m \left(\frac{R_e r_x}{R_e + r_x} \right)$$
$$g_m \left(\frac{R_e r_x}{R_e + r_x} \right) = \frac{R_{out}}{r_o} - 1$$
$$\frac{R_e r_x}{R_e + r_x} = \frac{1}{g_m} \left(\frac{R_{out}}{r_o} - 1 \right)$$
$$R_e r_x = \left(\frac{R_e + r_x}{g_m} \right) \left(\frac{R_{out}}{r_o} - 1 \right)$$
$$R_e r_x = \left(\frac{R_e}{g_m} \right) \left(\frac{R_{out}}{r_o} - 1 \right) + \left(\frac{r_x}{g_m} \right) \left(\frac{R_{out}}{r_o} - 1 \right)$$
$$R_e r_x - \left(\frac{R_e}{g_m} \right) \left(\frac{R_{out}}{r_o} - 1 \right) = \left(\frac{r_x}{g_m} \right) \left(\frac{R_{out}}{r_o} - 1 \right)$$
$$R_e \left(r_x - \left(\frac{1}{g_m} \right) \left(\frac{R_{out}}{r_o} - 1 \right) \right) = \left(\frac{r_x}{g_m} \right) \left(\frac{R_{out}}{r_o} - 1 \right)$$
$$R_e = \frac{\left(\frac{r_x}{g_m} \right) \left(\frac{R_{out}}{r_o} - 1 \right)}{r_x - \left(\frac{1}{g_m} \right) \left(\frac{R_{out}}{r_o} - 1 \right)}$$
$$R_e = \frac{(r_x) \left(\frac{R_{out}}{r_o} - 1 \right)}{r_x g_m - \left(\frac{R_{out}}{r_o} - 1 \right)} \quad \text{..... (1)}$$

Therefore, the expression for the resistance R_e is $\frac{(r_x) \left(\frac{R_{out}}{r_o} - 1 \right)}{r_x g_m - \left(\frac{R_{out}}{r_o} - 1 \right)}$.

Step 2 of 8

Calculate the value of the transconductance g_m.

$$g_m = \frac{I_C}{V_T}$$

Substitute $0.5\ \text{mA}$ for I_C and $25\ \text{mA}$ for V_T.

$$g_m = \frac{0.5\ \text{m}}{25\ \text{m}}$$
$$= 0.02\ \text{A/V}$$

Therefore, the value of the transconductance g_m is $0.02\ \text{A/V}$.

Step 3 of 8

Calculate the value of the resistance r_x.

$$r_x = \frac{\beta}{g_m}$$

Substitute $0.02\ \text{A/V}$ for g_m.

$$r_x = \frac{100}{0.02}$$
$$= 5\ \text{k}\Omega$$

Therefore, the value of r_x is $5\ \text{k}\Omega$.

Step 4 of 8

Recall equation (1).

$$R_e = \frac{(r_x) \left(\frac{R_{out}}{r_o} - 1 \right)}{r_x g_m - \left(\frac{R_{out}}{r_o} - 1 \right)}$$

Substitute 5 for $\frac{R_{out}}{r_o}$, $5\ \text{k}\Omega$ for r_x and $0.02\ \text{A/V}$ for g_m.

$$R_e = \frac{(5\ \text{k})(5-1)}{(5\ \text{k})(0.02) - (5-1)}$$
$$= \frac{(5\ \text{k})(4)}{100-4}$$
$$= \frac{20\ \text{k}}{96}$$
$$= 208.33\ \Omega$$

Therefore, the value of the resistance R_e for $\frac{R_{out}}{r_o} = 5$ is $\boxed{208.33\ \Omega}$.

Step 5 of 8

Recall equation (1).

$$R_e = \frac{(r_x) \left(\frac{R_{out}}{r_o} - 1 \right)}{r_x g_m - \left(\frac{R_{out}}{r_o} - 1 \right)}$$

Substitute 10 for $\frac{R_{out}}{r_o}$, $5\ \text{k}\Omega$ for r_x and $0.02\ \text{A/V}$ for g_m.

$$R_e = \frac{(5\ \text{k})(10-1)}{(5\ \text{k})(0.02) - (10-1)}$$
$$= \frac{(5\ \text{k})(9)}{100-9}$$
$$= \frac{45\ \text{k}}{91}$$
$$= 494.51\ \Omega$$

Therefore, the value of the resistance R_e for $\frac{R_{out}}{r_o} = 10$ is $\boxed{494.51\ \Omega}$.

Step 6 of 8

Recall equation (1).

$$R_e = \frac{(r_x) \left(\frac{R_{out}}{r_o} - 1 \right)}{r_x g_m - \left(\frac{R_{out}}{r_o} - 1 \right)}$$

Substitute 50 for $\frac{R_{out}}{r_o}$, $5\ \text{k}\Omega$ for r_x and $0.02\ \text{A/V}$ for g_m.

$$R_e = \frac{(5\ \text{k})(50-1)}{(5\ \text{k})(0.02) - (50-1)}$$
$$= \frac{(5\ \text{k})(49)}{100-49}$$
$$= \frac{245\ \text{k}}{51}$$
$$= 4.8\ \text{k}\Omega$$

Therefore, the value of the resistance R_e for $\frac{R_{out}}{r_o} = 50$ is $\boxed{4.8\ \text{k}\Omega}$.

Step 7 of 8

Consider the expression for the $\frac{R_{out}}{r_o}$.

$$\frac{R_{out}}{r_o} = 1 + g_m \left(\frac{R_e r_x}{R_e + r_x} \right)$$

Let the maximum possible factor of $\frac{R_{out}}{r_o}$ be a.

$$a = 1 + g_m \left(\frac{R_e r_x}{R_e + r_x} \right)$$

Substitute $5\ \text{k}\Omega$ for r_x and $0.02\ \text{A/V}$ for g_m.

$$a = 1 + (0.02) \left(\frac{R_e (5\ \text{k})}{R_e + 5\ \text{k}} \right)$$
$$a - 1 = (0.02) \left(\frac{R_e (5\ \text{k})}{R_e + 5\ \text{k}} \right)$$
$$a - 1 = \frac{R_e (100)}{R_e + 5\ \text{k}}$$
$$(a-1)(R_e + 5\ \text{k}) = R_e (100)$$
$$(a-1)(R_e) + (a-1)5\ \text{k} = R_e (100) \quad \text{..... (2)}$$

To calculate the value of resistance, it is required that the factor should be less than. That is $a - 1 < 100$ implies, $a < 101$. Therefore, the maximum value of the factor is $\boxed{100}$.

Step 8 of 8

Recall equation (2).

$$(a-1)(R_e) + (a-1)5\ \text{k} = R_e (100)$$

Substitute 100 for a.

$$(100-1)R_e + (100-1)(5\ \text{k}) = R_e (100)$$
$$99R_e + 495\ \text{k} = 100R_e$$
$$100R_e - 99R_e = 495\ \text{k}$$
$$R_e = 495\ \text{k}\Omega$$

Therefore, the value of the resistance R_e for the maximum value of the factor a is, $\boxed{495\ \text{k}\Omega}$.

Step 1 of 7

Refer to Figure 8.29 from the text book.

Calculate the expression for the output voltage v_o.

$$v_o = -(g_{m1}v_i)(Kr_{o1}) \quad \text{..... (1)}$$

Therefore, the expression for the output voltage v_o is $-(g_{m1}v_i)(Kr_{o1})$.

Step 2 of 7

Draw the small signal equivalent circuit diagram.

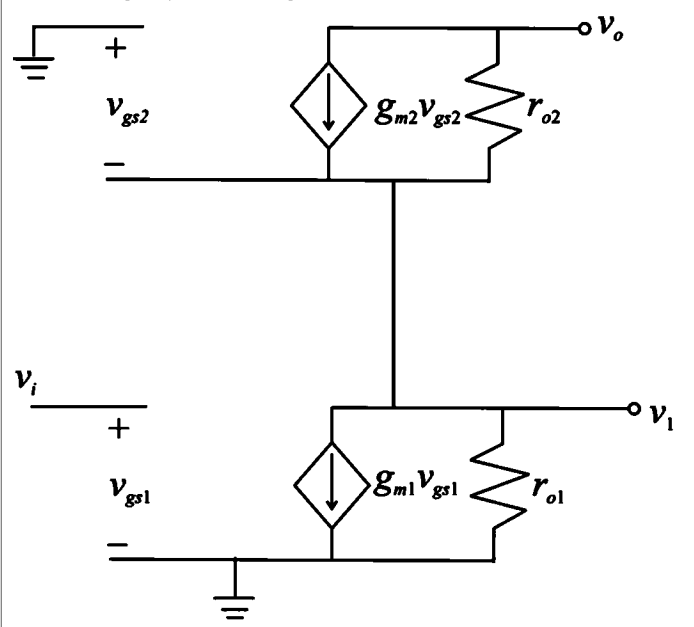


Figure 1

Step 3 of 7

Consider the expression for the voltage v_1.

$$v_1 = -(g_{m1}v_{gs1})r_{o1}$$

Substitute v_i for v_{gs1}.

$$v_1 = -(g_{m1}v_i)r_{o1}$$

Therefore, the expression for the voltage v_1 is $-(g_{m1}v_i)r_{o1}$.

Step 4 of 7

Apply nodal analysis at output terminal.

$$\frac{v_o - v_1}{r_{o2}} + g_{m2}v_{gs2} = 0$$

Substitute $-v_1$ for v_{gs2}.

$$\frac{v_o - v_1}{r_{o2}} - v_1 g_{m2} = 0$$
$$v_o - v_1 - v_1 g_{m2} r_{o2} = 0$$
$$v_o = v_1 + v_1 g_{m2} r_{o2}$$
$$v_o = v_1(1 + g_{m2}r_{o2})$$

Here, $g_{m2}r_{o2} >> 1$ and hence $1 + g_{m2}r_{o2} \approx g_{m2}r_{o2}$.

$$v_o = v_1(g_{m2}r_{o2})$$

Substitute $-(g_{m1}v_i)r_{o1}$ for v_1.

$$v_o = -(g_{m1}v_i)(g_{m2}r_{o2})r_{o1} \quad \text{..... (2)}$$

Step 5 of 7

Compare the equations (1) and (2).

$$K = g_{m2}r_{o2} \quad \text{..... (3)}$$

Therefore, the approximate value for the factor K is $g_{m2}r_{o2}$.

Step 6 of 7

Recall equation (3).

$$K = g_{m2}r_{o2}$$

Substitute $\frac{2I_D}{V_A}$ for g_{m2} and $\frac{V_A}{I_D}$ for r_{o2}.

$$K = \left(\frac{2I_D}{V_{ov}}\right)\left(\frac{V_A}{I_D}\right)$$
$$= \frac{2V_A}{V_{ov}}$$

Substitute 50 for K and $0.2\ \text{V}$ for V_{ov}.

$$50 = \frac{2V_A}{0.2}$$
$$50 = \frac{V_A}{0.1}$$
$$V_A = 5\ \text{V}$$

Therefore, the value of the voltage V_A is $\boxed{5\ \text{V}}$.

Step 7 of 7

Consider the expression for the voltage V_A.

$$V_A = V_A' L$$
$$L = \frac{V_A}{V_A'}$$

Substitute $5\ \text{V}$ for V_A and $5\ \text{V/}\mu\text{m}$ for V_A'.

$$L = \frac{5}{5/1\,\mu}$$
$$= 1\ \mu\text{m}$$

Therefore, the value of the length L is $\boxed{1\ \mu\text{m}}$.

Step 1 of 3

Refer to Figure 7.10 in the text book for the cascade current source amplifier.

The output resistance of cascade current source is,

$$R_o = (g_{m3} r_{o3}) r_{o4}$$

As the two transistors are identical, the output resistance is,

$$R_o = \left(\frac{2|V_A|}{|V_{OV}|}\right)\left(\frac{|V_A|}{I_D}\right)$$

$$R_o = \left(\frac{2|V_A|}{|V_{OV}|}\right)\left(\frac{|V_A|}{I}\right)$$

$$IR_o = \frac{2|V_A|^2}{|V_{OV}|}$$

Here, V_A is the early voltage, and V_{OV} is the overdrive voltage.

Hence, $\boxed{IR_o = \frac{2|V_A|^2}{|V_{OV}|}}$.

Step 2 of 3

(a)

Write the formula for the current supplied by the current source *I*.

$$I = \frac{1}{2}\mu_p C_{ox}\left(\frac{W}{L}\right)|V_{OV}|^2\left(1 + \frac{V_{SD}}{|V_A|}\right)$$

Substitute $100\times10^{-6}\ \text{A/V}^2$ for $\mu_p C_{ox}$, 0.2 V for V_{OV}, $0.1\times10^{-3}\ \text{A}$ for I, 4 V for $|V_A|$ and 0.2 V for V_{SD}.

$$0.1\times10^{-3} = \left(\frac{1}{2}\right)\left(100\times10^{-6}\right)\left(\frac{W}{L}\right)|0.2|^2\left(1 + \frac{0.2}{4}\right)$$

$$\frac{W}{L} = \frac{2\times0.1\times10^{-3}}{\left(100\times10^{-6}\right)|0.2|^2\left(1 + \frac{0.2}{4}\right)}$$

$$= 47.62$$

Thus, the $\frac{W}{L}$ ratio of the transistors is $\boxed{47.62}$.

Step 3 of 3

(b)

Write the formula for the current supplied by the current source *I*.

$$I = \frac{1}{2}\mu_p C_{ox}\left(\frac{W}{L}\right)|V_{OV}|^2\left(1 + \frac{V_{SD}}{|V_A|}\right)$$

Substitute $100\times10^{-6}\ \text{A/V}^2$ for $\mu_p C_{ox}$, 0.2 V for V_{OV}, $0.5\times10^{-3}\ \text{A}$ for I, 4 V for $|V_A|$ and 0.2 V for V_{SD}.

$$0.5\times10^{-3} = \left(\frac{1}{2}\right)\left(100\times10^{-6}\right)\left(\frac{W}{L}\right)|0.2|^2\left(1 + \frac{0.2}{4}\right)$$

$$\frac{W}{L} = \frac{2\times0.5\times10^{-3}}{\left(100\times10^{-6}\right)|0.2|^2\left(1 + \frac{0.2}{4}\right)}$$

$$= 238.1$$

Thus, the $\frac{W}{L}$ ratio of the transistors is $\boxed{238.1}$.

8.064P

We don't have the solution to this problem yet.

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Step 1 of 5

Refer to Figure 8.30(a) from the text book.

Consider the expression for the transconductance g_m.

$$g_m = \frac{2I_D}{V_{OV}}$$

Substitute $2\ \text{mA/V}$ for g_m and $0.25\ \text{V}$ for V_{OV}.

$$2\ \text{m} = \frac{2I_D}{0.25}$$
$$1\ \text{m} = \frac{I_D}{0.25}$$
$$I_D = 0.25\ \text{mA}$$

Therefore, the value of the drain current I_D is, $\boxed{0.25\ \text{mA}}$.

Step 2 of 5

Consider the expression for the output resistance R_o.

$$R_o = g_{m1} r_{o1} r_{o2}$$

Substitute $\frac{V_A' L}{I_D}$ and r_{o1} and r_{o2}.

$$R_o = g_{m1}\left(\frac{V_A' L}{I_D}\right)\left(\frac{V_A' L}{I_D}\right)$$
$$R_o = \frac{g_{m1}\left(V_A'\right)^2 L^2}{I_D^2}$$

Substitute $2\ \text{mA/V}$ for g_{m1}, $0.25\ \text{mA}$ for I_D, $5\ \text{V/}\mu\text{m}$ for V_A' and $200\ \text{k}\Omega$ for R_o.

$$200\ \text{k} = \frac{(2\ \text{m})(5/1\ \mu)^2 L^2}{(0.25\ \text{m})^2}$$
$$100\ \text{M} = \frac{(5/1\ \mu)^2 L^2}{(0.25\ \text{m})^2}$$
$$L^2 = \frac{(100\ \text{M})(0.25\ \text{m})^2}{(5/1\ \mu)^2}$$
$$L = \frac{(10\ \text{k})(0.25\ \text{m})}{(5/1\ \mu)}$$
$$L = \frac{2.5}{(5/1\ \mu)}$$
$$= 0.5\ \mu\text{m}$$

Therefore, the value of the length L is $\boxed{0.5\ \mu\text{m}}$.

Step 3 of 5

Consider the expression for the drain current I_D.

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)V_{OV}^2$$

Substitute $400\ \mu\text{A/V}^2$ for g_{m1}, $0.25\ \text{mA}$ for I_D and $0.25\ \text{V}$ for V_{OV}.

$$0.25\ \text{m} = \frac{1}{2}(400\ \mu)\left(\frac{W}{L}\right)(0.25)^2$$
$$\frac{W}{L} = \frac{2(0.25\ \text{m})}{(400\ \mu)(0.25)^2}$$
$$\frac{W}{L} = \frac{0.5\ \text{m}}{(400\ \mu)(0.0625)}$$
$$\frac{W}{L} = \frac{0.5\ \text{m}}{25\ \mu}$$
$$\frac{W}{L} = 20$$

Therefore, the value of the ratio $\frac{W}{L}$ is $\boxed{20}$.

Step 4 of 5

For the maximum negative signal swing at the output terminal, the value of the drain to source voltage V_{DS} of each transistor should be equal to V_{OV}. That is $V_{DS} = 0.25\ \text{V}$.

Calculate the value of the voltage V_{G2}.

$$V_{G2} = V_{DS} + V_{GS}$$

Substitute $V_{OV} + V_m$ for V_{GS}.

$$V_{G2} = V_{DS} + V_{OV} + V_m$$

Substitute $0.25\ \text{V}$ for V_{DS} and V_{OV} and $0.5\ \text{V}$ for V_m.

$$V_{G2} = 0.25 + 0.25 + 0.5$$
$$= 1\ \text{V}$$

Therefore, the value of the voltage V_{G2} is $\boxed{1\ \text{V}}$.

Step 5 of 5

Calculate the minimum possible output voltage V_{out}.

$$V_{out} = V_{DS} + V_{DS}$$
$$= 2V_{DS}$$

Substitute $0.25\ \text{V}$ for V_{DS}.

$$V_{out} = 2(0.25)$$
$$= 0.5\ \text{V}$$

Therefore, the value of the output voltage V_{out} is, $\boxed{0.5\ \text{V}}$.

Step 1 of 6

Refer to Figure 8.33 from the text book.

Calculate the value of the transconductance g_m.

$$g_{m1} = \frac{2I}{V_{OV}}$$

Substitute $0.2\ \text{mA}$ for I and $0.2\ \text{V}$ for V_{OV}.

$$\begin{aligned} g_{m1} &= \frac{2(0.2\ \text{m})}{0.2} \\ &= 2(1\ \text{m}) \\ &= 2\ \text{mA/V} \end{aligned}$$

Therefore, the value of the transconductance g_{m1} is $\boxed{2\ \text{mA/V}}$.

Step 2 of 6

Calculate the value of the resistance r_o.

$$r_o = \frac{|V_A|}{I}$$

Substitute $0.2\ \text{mA}$ for I and $4\ \text{V}$ for V_A.

$$\begin{aligned} r_o &= \frac{4}{0.2\ \text{m}} \\ &= 20\text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance r_o is $20\text{k}\Omega$.

Step 3 of 6

Calculate the value of the output resistance of the amplifier R_{on}.

$$R_{on} = (g_{m2}r_{o2})r_{o1}$$

Substitute $20\text{k}\Omega$ for r_{o1} and r_{o2} and $2\ \text{mA/V}$ for g_{m2}.

$$\begin{aligned} R_{on} &= (2\ \text{m})(20\text{k})(20\text{k}) \\ &= 800\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance R_{on} is $\boxed{800\ \text{k}\Omega}$.

Step 4 of 6

Calculate the value of the output resistance of the current source R_{op}.

$$R_{op} = (g_{m3}r_{o3})r_{o4}$$

Substitute $20\text{k}\Omega$ for r_{o3} and r_{o4} and $2\ \text{mA/V}$ for g_{m2}.

$$\begin{aligned} R_{op} &= (2\ \text{m})(20\text{k})(20\text{k}) \\ &= 800\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance R_{op} is $\boxed{800\ \text{k}\Omega}$.

Step 5 of 6

Calculate the value of the overall output resistance R_o.

$$R_o = R_{on} \| R_{op}$$

Substitute $800\ \text{k}\Omega$ for R_{op} and R_{on}.

$$\begin{aligned} R_o &= 800\ \text{k} \| 800\ \text{k} \\ &= \frac{(800\ \text{k})(800\ \text{k})}{800\ \text{k} + 800\ \text{k}} \\ &= \frac{(800\ \text{k})(800\ \text{k})}{1600\ \text{k}} \\ &= \frac{800\ \text{k}}{2} \end{aligned}$$

$$R_o = 400\ \text{k}\Omega$$

Therefore, the value of the overall output resistance R_o is $\boxed{400\ \text{k}\Omega}$.

Step 6 of 6

Calculate the value of the voltage gain A_v.

$$A_v = -g_{m1}R_o$$

Substitute $400\ \text{k}\Omega$ for R_o and $2\ \text{mA/V}$ for g_{m1}.

$$\begin{aligned} A_v &= -(2\ \text{m})(400\ \text{k}) \\ &= -800\ \text{V/V} \end{aligned}$$

Therefore, the value of the voltage gain A_v is $\boxed{-800\ \text{V/V}}$.

Step 1 of 7

Refer to Figure 8.33 from the text book.

Consider the expression for the gain A_v.

$$A_v = -g_m R_o$$

Substitute $-280\ \text{V/V}$ for A_v and $1\ \text{mA/V}$ for g_m.

$$-280 = -(1\ \text{m})R_o$$
$$R_o = \frac{280}{1\ \text{m}}$$
$$R_o = 280\ \text{k}\Omega$$

Therefore, the value of the output resistance R_o is $280\ \text{k}\Omega$.

Step 2 of 7

Consider the expression for the output resistance.

$$R_o = \left[(g_{o2}r_{o2})r_{o1}\right] \| \left[(g_{o3}r_{o3})r_{o4}\right]$$

Substitute $1\ \text{mA/V}$ for g_{o3} and g_{o2}, r_o for r_{o1}, r_{o2}, r_{o3} and r_{o4} and $280\ \text{k}\Omega$ for R_o.

$$280\ \text{k} = \left[(1\ \text{m})r_o^2\right] \| \left[(1\ \text{m})r_o^2\right]$$
$$280\ \text{k} = (1\ \text{m})r_o^2 (1 \| 1)$$
$$280\ \text{k} = (1\ \text{m})r_o^2 \left(\frac{1 \cdot 1}{1+1}\right)$$
$$280\ \text{k} = (1\ \text{m})r_o^2 \left(\frac{1}{2}\right)$$

$$280\ \text{M} = r_o^2 \left(\frac{1}{2}\right)$$
$$r_o^2 = 2(280\ \text{M})$$
$$r_o^2 = 560\ \text{M}$$
$$r_o = \sqrt{560\ \text{M}}$$

$$r_o = 23.66\ \text{k}\Omega$$

Therefore, the value of the resistance r_o is $\boxed{23.66\ \text{k}\Omega}$.

Step 3 of 7

Consider the expression for the current I.

$$I = \frac{g_m |V_{OV}|}{2}$$

Substitute $1\ \text{mA/V}$ for g_m and $0.25\ \text{V}$ for V_{OV}.

$$I = \frac{(1\ \text{m})(0.25)}{2}$$
$$= 0.125\ \text{mA}$$

Therefore, the value of the current I is $\boxed{0.125\ \text{mA}}$.

Step 4 of 7

Consider the expression for the resistance r_o.

$$r_o = \frac{V_A' L}{I}$$

Substitute $23.66\ \text{k}\Omega$ for r_o, $0.125\ \text{mA}$ for I and $5\ \text{V/1}\ \mu$ for V_A'.

$$23.66\ \text{k} = \frac{(5/1\ \mu)L}{0.125\ \text{m}}$$
$$L = \frac{(23.66\ \text{k})(0.125\ \text{m})}{(5/1\ \mu)}$$
$$L = \frac{2.9575}{(5/1\ \mu)}$$
$$L = 0.5915\ \mu\text{m}$$

Therefore, the value of the length L is $\boxed{0.5915\ \mu\text{m}}$.

Step 5 of 7

Consider the expression for the bias current I of transistor Q_1.

$$I = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_1 V_{ov}^2$$

Substitute $0.125\ \text{mA}$ for I, $400\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$ and $0.25\ \text{V}$ for V_{ov}.

$$0.125\ \text{m} = \frac{1}{2}(400\ \mu)\left(\frac{W}{L}\right)_1 (0.25)^2$$
$$0.125\ \text{m} = (200\ \mu)\left(\frac{W}{L}\right)_1 (0.25)^2$$
$$\left(\frac{W}{L}\right)_1 = \frac{0.125\ \text{m}}{(200\ \mu)(0.25)^2}$$
$$\left(\frac{W}{L}\right)_1 = \frac{0.125\ \text{m}}{12.5\ \mu}$$

$$\left(\frac{W}{L}\right)_1 = 10$$

Therefore, the value of the ratio $\left(\frac{W}{L}\right)_1$ is $\boxed{10}$.

Step 6 of 7

The value of the ratio $\left(\frac{W}{L}\right)_2$ is equal to the ratio $\left(\frac{W}{L}\right)_1$. Hence, the value of the ratio is $\left(\frac{W}{L}\right)_2$ is $\boxed{10}$.

Step 7 of 7

Consider the expression for the bias current I of transistor Q_3.

$$I = \frac{1}{2}\mu_p C_{ox}\left(\frac{W}{L}\right)_3 V_{ov}^2$$

Substitute $0.125\ \text{mA}$ for I, $100\ \mu\text{A/V}^2$ for $\mu_p C_{ox}$ and $0.25\ \text{V}$ for V_{ov}.

$$0.125\ \text{m} = \frac{1}{2}(100\ \mu)\left(\frac{W}{L}\right)_3 (0.25)^2$$
$$0.125\ \text{m} = (50\ \mu)\left(\frac{W}{L}\right)_3 (0.25)^2$$
$$\left(\frac{W}{L}\right)_3 = \frac{0.125\ \text{m}}{(50\ \mu)(0.25)^2}$$
$$\left(\frac{W}{L}\right)_3 = \frac{0.125\ \text{m}}{3.125\ \mu}$$

$$\left(\frac{W}{L}\right)_3 = 40$$

Therefore, the value of the ratio $\left(\frac{W}{L}\right)_3$ is $\boxed{40}$.

The value of the ratio $\left(\frac{W}{L}\right)_4$ is equal to the ratio $\left(\frac{W}{L}\right)_3$. Hence, the value of the ratio is $\left(\frac{W}{L}\right)_4$ is $\boxed{40}$.

Step 1 of 4

Refer to Figure 7.10 in the textbook for the circuit.

The output resistance of the cascade current source is,

$$\begin{aligned} R_o &= (g_m r_o) r_o \\ &= \left(\frac{|V_A|}{\frac{|V_{OV}|}{2}} \right) \left(\frac{|V_A|}{I} \right) \\ &= \frac{2|V_A|^2}{|V_{OV}| I} \\ &= \frac{2(5^2)}{0.2(100 \times 10^{-6})} \\ &= 2.5 \text{ M}\Omega \end{aligned}$$

Thus, the output resistance is, $\boxed{R_o = 2.5 \text{ M}\Omega}$.

Calculate the voltage V_{SG4}.

$$\begin{aligned} V_{SG4} &= |V_{tp}| + |V_{OV}| \\ &= 0.8 + 0.2 \\ &= 1 \text{ V} \end{aligned}$$

Calculate the voltage V_{G4}.

$$\begin{aligned} V_{G4} &= V_{DD} - V_{SG4} \\ &= 3.3 - 1 \\ &= 2.3 \text{ V} \end{aligned}$$

Thus, the voltage V_{G4} is, $\boxed{V_{G4} = 2.3 \text{ V}}$

Step 2 of 4

To allow largest possible signal swing at the output terminal, use minimum required voltage V_{OV} across Q_4.

Calculate the voltage V_{D4}.

$$\begin{aligned} V_{D4} &= V_{DD} - |V_{OV}| \\ &= 3.3 - 0.2 \\ &= 3.1 \text{ V} \end{aligned}$$

Since the two transistors are identical and are carrying equal currents, the voltage V_{SG3} is,

$$\begin{aligned} V_{SG3} &= V_{SG4} \\ &= 1 \text{ V} \end{aligned}$$

Calculate the voltage V_{G3}.

$$\begin{aligned} V_{G3} &= V_{D4} - V_{SG3} \\ &= 3.1 - 1 \\ &= 2.1 \text{ V} \end{aligned}$$

Thus, the voltage V_{G3} is, $\boxed{V_{G3} = 2.1 \text{ V}}$.

Step 3 of 4

The maximum voltage allowed at the output terminal of the current source is strained by the need to allow a minimum voltage of V_{OV} across Q_3.

The highest allowable voltage at the output is,

$$\begin{aligned} v_{D3\max} &= V_{D4} - |V_{OV}| \\ &= 3.1 - 0.2 \\ &= 2.9 \text{ V} \end{aligned}$$

Thus, the highest allowable voltage at the output is $\boxed{2.9 \text{ V}}$.

Step 4 of 4

The bias current in the circuit is,

$$\begin{aligned} I &= \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right) |V_{OV}|^2 \left[1 + \frac{|V_{SD}|}{|V_A|} \right] \\ &= \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right) |V_{OV}|^2 \left[1 + \frac{|V_{OV}|}{|V_A|} \right] \qquad \left(\text{Since, } V_{SD} = V_{OV} \right) \end{aligned}$$

Substitute $100 \ \mu\text{A}$ for I, $60 \ \mu\text{A/V}^2$ for $\mu_p C_{ox}$, 0.2 V for V_{OV}, and 5 V for $|V_A|$ in the equation.

$$100 \times 10^{-6} = \frac{1}{2} (60 \times 10^{-6}) \left(\frac{W}{L} \right) (0.2)^2 \left[1 + \frac{0.2}{5} \right]$$

$$\begin{aligned} \left(\frac{W}{L} \right) &= \frac{100 \times 10^{-6}}{\frac{1}{2} (60 \times 10^{-6}) (0.2)^2 \left[1 + \frac{0.2}{5} \right]} \\ &= 80.13 \end{aligned}$$

Thus, the $\frac{W}{L}$ ratio is, $\boxed{\frac{W}{L} = 80.13}$.

Step 1 of 3

8.069P

Refer to Figure P7.28 in the textbook for the circuit.

From the circuit diagram, the gate voltage of Q_1 is,

$$v_{gs1} = 0\ \text{V}$$

The small signal equivalent circuit is shown in Figure 1.

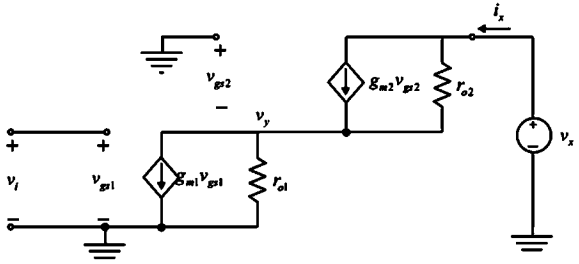


Figure 1

Step 2 of 3

From the circuit diagram, write following voltage relations:

$$v_i = v_{gs1} = 0\ \text{V}$$

and

$$v_y = -v_{gs2}$$

Since v_{gs1} is zero, the dependent voltage source $g_{m1}v_{gs1}$ is open. Then, the modified circuit diagram is shown in Figure 2.

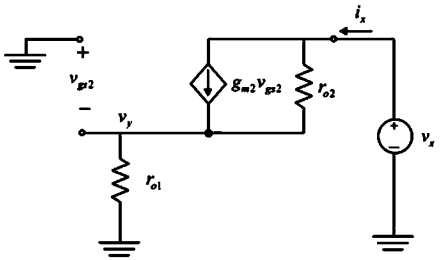


Figure 2

Step 3 of 3

Apply nodal analysis at node v_y

$$\frac{v_y}{r_{o1}} - g_{m2}v_{gs2} + \frac{v_y - v_x}{r_{o2}} = 0$$

$$\frac{v_y}{r_{o1}} + g_{m2}v_y + \frac{v_y}{r_{o2}} = \frac{v_x}{r_{o2}} \qquad (\text{since } v_{gs2} = -v_y)$$

$$v_y\left[\frac{1}{r_{o1}} + g_{m2} + \frac{1}{r_{o2}}\right] = \frac{v_x}{r_{o2}} \qquad \left(\text{since } g_{m2} \gg \left(\frac{1}{r_{o1}}\right), \left(\frac{1}{r_{o2}}\right)\right)$$

$$g_{m2}v_y = \frac{v_y}{r_{o2}}$$

$$v_y = \left(\frac{1}{g_{m2}r_{o2}}\right)v_x$$

Therefore the drain voltage of transistor Q_1 is smaller than applied voltage v_x by a factor of $\boxed{\frac{v_y}{v_x} = \frac{1}{g_{m2}r_{o2}}}$

Step 1 of 9

8.070P

Refer to Figure P7.29 in the textbook for the circuits (a), (b), and (c).

(a)

Given drain current is kept constant (I)

$$I = \frac{1}{2}(\mu_n C_{ox})\left(\frac{W}{L}\right)V_{OV}^2$$
$$= \frac{1}{2}k_n\left(\frac{W}{L}\right)V_{OV}^2$$

From the \circuit (a), the drain current is,

$$I = \frac{1}{2}k_n\left(\frac{W}{L}\right)V_{OV1}^2 \quad \text{..... (1)}$$

From the given circuit (b), the drain current is,

$$I = \frac{1}{2}k_n\left(\frac{W}{4L}\right)V_{OV2}^2 \quad \text{..... (2)}$$

Step 2 of 9

Equate equations (1) and (2).

$$\frac{1}{2}k_n\left(\frac{W}{L}\right)V_{OV1}^2 = \frac{1}{2}k_n\left(\frac{W}{4L}\right)V_{OV2}^2$$
$$V_{OV1}^2 = \left(\frac{1}{4}\right)V_{OV2}^2$$
$$\frac{V_{OV2}^2}{V_{OV1}^2} = 4$$
$$\sqrt{\frac{V_{OV2}^2}{V_{OV1}^2}} = \sqrt{4}$$
$$\frac{V_{OV2}}{V_{OV1}} = 2$$
$$V_{OV2} = 2V_{OV1} \quad \text{..... (3)}$$

Therefore, over drive voltage of circuit (b) is **double** that of the original circuit (a) is proved.

Step 3 of 9

The transconductance is,

$$g_m = \frac{2I_D}{V_{OV}}$$

Since drain current is kept constant, Transconductance varies inversely with the over drive voltage.

$$g_m \alpha \frac{1}{V_{OV}}$$
$$\frac{g_{m2}}{g_{m1}} = \frac{V_{OV1}}{V_{OV2}}$$
$$\frac{g_{m2}}{g_{m1}} = \frac{V_{OV1}}{2V_{OV1}} \qquad \left(\text{Since from equation (3), } V_{OV2} = 2V_{OV1}\right)$$
$$\frac{g_{m2}}{g_{m1}} = \frac{1}{2}$$
$$g_{m2} = \left(\frac{1}{2}\right)g_{m1}$$

Therefore, the transconductance of circuit (b) is **halved** that of the original circuit (a) is proved.

Step 4 of 9

The voltage gain is,

$$A_0 = g_m r_o$$
$$= \left(\frac{2I_D}{V_{OV}}\right)\left(\frac{V_A' L}{I_D}\right)$$
$$= \frac{2V_A' L}{V_{OV}}$$

The voltage gain is proportional to gate length and inversely proportional to overdrive voltage.

$$A_0 \alpha \frac{L}{V_{OV}}$$
$$\frac{A_{01}}{A_{02}} = \frac{L_1 V_{OV2}}{L_2 V_{OV1}}$$
$$\frac{A_{01}}{A_{02}} = \frac{L_1(2V_{OV1})}{(4L_1)V_{OV1}} \qquad \left(\text{Since, } V_{OV2} = 2V_{OV1} \text{ and } L_2 = 4L_1\right)$$
$$\frac{A_{01}}{A_{02}} = \frac{1}{2}$$
$$A_{02} = 2A_{01}$$

Therefore, the voltage gain of circuit (b) is **double** that of the original circuit (a) is proved.

Step 5 of 9

(b)

Since there is no change in gate width to length ratio (W/L), each transistor in circuit (c) have same over drive voltage as in circuit (a).

Therefore, over drive voltage of circuit (c) is **same as** that of the original circuit (a) is proved.

Step 6 of 9

The transconductance of circuit (a) is,

$$i_o = g_{m1}v_i$$
$$\frac{i_o}{v_i} = g_{m1}$$
$$G_m = g_{m1}$$

Step 7 of 9

For the circuit (c), the small signal model is shown in Figure 1.

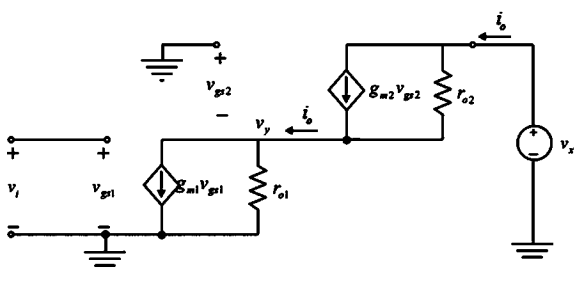


Figure 1

Step 8 of 9

From Figure 1, the output current is,

$$i_o \simeq g_{m2}v_{gs2} \quad \text{..... (4)}$$

Write the nodal equation for the circuit.

$$g_{m2}v_{gs2} + \frac{v_{gs2}}{r_{o1}} + \frac{v_{gs2}}{r_{o2}} = g_{m1}v_i$$
$$v_{gs2}\left(g_{m2} + \frac{1}{r_{o1}} + \frac{1}{r_{o2}}\right) = g_{m1}v_i$$
$$v_{gs2}g_{m2} = g_{m1}v_i \qquad \left(\text{Since } g_{m2} >> \left(\frac{1}{r_o}\right), \left(\frac{1}{r_o}\right)\right)$$

Substitute $g_{m1}v_i$ for $v_{gs2}g_{m2}$ in equation (4).

$$i_o \simeq g_{m1}v_i$$
$$\frac{i_o}{v_i} \simeq g_{m1}$$
$$G_m \simeq g_{m1}$$

Hence, equivalent transconductance of circuit (c) is equal to Transconductance of transistor for which input voltage is applied. That is g_{m1} .

Hence, transconductance of circuit (c) is **same** that of the original circuit (a).

Step 9 of 9

Voltage gain for cascaded transistor is,

$$A_{V0} = -A_0^2$$
$$= -\left(\frac{2V_A' L}{V_{OV}}\right)^2$$
$$= -(2V_A')^2\left(\frac{L}{V_{OV}}\right)^2$$

The overall voltage gain is proportional to gate length and inversely proportional to overdrive voltage.

In circuit (c), overdrive voltage and channel length are same as that of original circuit.

$$\frac{A_{V0}}{A_{01}} = \frac{-(2V_A')^2\left(\frac{L_3}{V_{OV3}}\right)^2}{\frac{2V_A' L_1}{V_{OV1}}} \qquad (\text{since } L_1 = L_3 \text{ and } V_{OV3} = V_{OV1})$$
$$\frac{A_{V0}}{A_0} = -\frac{2V_A' L}{V_{OV}}$$
$$A_{V0} = \left(-\frac{2V_A' L}{V_{OV}}\right)A_0$$
$$A_{V0} = (-A_0)A_0$$

Hence voltage gain of circuit (c) is increased by a factor of A_0 to the gain of original circuit (a).

Step 1 of 5

8.071P

Refer to Figure 8.33 from the text book.

The overdrive voltage for all the transformers will be same since all the transformers are matched.

Calculate the value of the overdrive voltage V_{OV}.

$$V_{OV} = V_t - V_m$$

Substitute $0.6\ \text{V}$ for V_t and $0.4\ \text{V}$ for V_m.

$$\begin{aligned} V_{OV} &= 0.6 - 0.4 \\ &= 0.2\ \text{V} \end{aligned}$$

Therefore, the value of the overdrive voltage V_{OV} is $\boxed{0.2\ \text{V}}$.

Step 2 of 5

Calculate the value of the voltage V_{S2}.

$$V_{S2} = V_{G2} - V_{GS2}$$

Substitute $V_{OV2} + V_m$ for V_{GS2}.

$$\begin{aligned} V_{S2} &= V_{G2} - (V_{OV2} + V_m) \\ &= V_{G2} - V_{OV2} - V_m \end{aligned}$$

Substitute $0.9\ \text{V}$ for V_{G2}, $0.2\ \text{V}$ for V_{OV2} and $0.4\ \text{V}$ for V_m.

$$\begin{aligned} V_{S2} &= 0.9 - 0.2 - 0.4 \\ &= 0.3\ \text{V} \end{aligned}$$

Therefore, the value of the voltage V_{S2} is $0.3\ \text{V}$.

Step 3 of 5

Calculate the minimum value of the output voltage $V_{o,\min}$.

$$V_{o,\min} = V_{S2} + V_{OV2}$$

Substitute $0.3\ \text{V}$ for V_{S2} and $0.2\ \text{V}$ for V_{OV2}.

$$\begin{aligned} V_{o,\min} &= 0.3 + 0.2 \\ &= 0.5\ \text{V} \end{aligned}$$

Therefore, the minimum value of the output voltage $V_{o,\min}$ is $0.5\ \text{V}$.

Step 4 of 5

Calculate the value of the voltage V_{S3}.

$$V_{S3} = V_{G3} + V_{GS3}$$

Substitute $V_{OV3} + V_m$ for V_{GS3}.

$$\begin{aligned} V_{S3} &= V_{G3} - (V_{OV3} + V_m) \\ &= V_{G3} - V_{OV3} - V_m \end{aligned}$$

Substitute $0.4\ \text{V}$ for V_{G3}, $0.2\ \text{V}$ for V_{OV} and $0.4\ \text{V}$ for V_m.

$$\begin{aligned} V_{S3} &= 0.4 + 0.2 + 0.4 \\ &= 1\ \text{V} \end{aligned}$$

Therefore, the value of the voltage V_{S3} is $1\ \text{V}$.

Step 5 of 5

Calculate the minimum value of the output voltage $V_{o,\max}$.

$$V_{o,\max} = V_{S3} - V_{OV3}$$

Substitute $1\ \text{V}$ for V_{S3} and $0.2\ \text{V}$ for V_{OV3}.

$$\begin{aligned} V_{o,\max} &= 1 - 0.2 \\ &= 0.8\ \text{V} \end{aligned}$$

Therefore, the maximum value of the output voltage $V_{o,\min}$ is $0.8\ \text{V}$.

Therefore, the range of the output voltage is $\boxed{0.5\ \text{V} \le V_o \le 0.8\ \text{V}}$.

Step 1 of 5

The transconductance, g_{m2} is,

$$g_{m2} = \sqrt{2k\frac{W}{L}I_D}$$

Substitute $400\ \mu A/V^2$ for k_p, 15 for $\frac{W}{L}$ and $0.2\ mA$ for I_D.

$$\begin{aligned} g_{m2} &= \sqrt{2\left(400\times10^{-6}\right)(15)\left(0.2\times10^{-3}\right)} \\ &= \sqrt{2400\times10^{-9}} \\ &= 1.55\ mA/V \end{aligned}$$

Step 2 of 5

The voltage, V_A is,

$V_A = V_A' L$ Substitute $5\ V/\mu m$ for V_A' and $0.36\ \mu m$ for L in the expression, $V_A = V_A' \times L$.

$$\begin{aligned} V_A &= V_A' L \\ &= 5\times10^{6}\left(0.36\times10^{-6}\right) \\ &= 1.8\ V \end{aligned}$$

The resistance, r_o is,

$$r_o = \frac{V_A}{I_D}$$

Substitute $1.8\ V$ for V_A and $0.2\ mA$ for I_D in the expression.

$$\begin{aligned} r_o &= \frac{V_A}{I_D} \\ &= \frac{1.8}{0.2\times10^{-3}} \\ &= 9\ k\Omega \end{aligned}$$

Step 3 of 5

The output resistance is,

$$R_o = (g_{m2}r_{o2})r_{o1}$$

Substitute $1.55\ mA/V$ for g_{m2}, $9\ k\Omega$ for r_{o1} and $9\ k\Omega$ for r_{o2} in the expression.

$$\begin{aligned} R_o &= (g_{m2}r_{o2})r_{o1} \\ &= \left(1.55\times10^{-3}\right)\left(9\times10^{3}\right)\left(9\times10^{3}\right) \\ &= 125.55\ k\Omega \end{aligned}$$

Step 4 of 5

The expression for the voltage gain is,

$$A_v = -g_{m1}\left(R_o \| R_L\right)$$

Substitute $1.55\ mA/V$ for g_{m2} and $100\ V/V$ for A_v in the equation.

$$\begin{aligned} A_v &= -g_{m1}\left(R_o \| R_L\right) \\ -100 &= -1.55\times10^{-3}\left(R_o \| R_L\right) \\ \left(R_o \| R_L\right) &= \frac{100}{1.55\times10^{-3}} \\ &= 64.516\ k\Omega \end{aligned}$$

Simplify further.

$$\begin{aligned} \frac{R_oR_L}{R_o+R_L} &= 64.516\times10^{3} \\ R_oR_L &= \left(R_o+R_L\right)\left(64.516\times10^{3}\right) \\ R_oR_L &= \left(64.516\times10^{3}\right)R_o + \left(64.516\times10^{3}\right)R_L \\ R_L\left(R_o-64.516\times10^{3}\right) &= \left(64.516\times10^{3}\right)R_o \\ R_L &= \frac{\left(64.516\times10^{3}\right)R_o}{R_o-64.516\times10^{3}} \end{aligned}$$

Substitute $125.55\ k\Omega$ for R_o in the expression.

$$\begin{aligned} R_L &= \frac{\left(64.51\times10^{3}\right)\left(125.55\times10^{3}\right)}{125.55\times10^{3}-64.51\times10^{3}} \\ &= 132.7\ k\Omega \end{aligned}$$

Hence, the value of R_L at which the voltage gain becomes $-100\ V/V$ is $\boxed{132.7\ k\Omega}$.

Step 5 of 5

The voltage gain of the CS amplifier is,

$$A_v = g_m r_o$$

Substitute $1.55\ mA/V$ for g_{m2} and $9\ k\Omega$ for r_{o1} in the equation.

$$\begin{aligned} A_v &= g_m r_o \\ &= \left(1.55\times10^{-3}\right)\left(9\times10^{3}\right) \\ &= 13.95\ V/V \end{aligned}$$

Therefore, the voltage gain of the CS amplifier is, $\boxed{A_v = 13.95\ V/V}$.

Step 1 of 9

Refer to Figure P7.34 in the textbook for the circuit.

(a)

All the transistors have same g_m and r_o.

$$g_{m1} = g_{m2} = g_{m3} = g_{m4} = g_m$$

From the circuit, the resistance r_o is,

$$R_1 = r_o$$

Thus, the resistance R_1 is, $\boxed{R_1 = r_o}$.

The cascade NMOS transistor Q_3 multiplies the resistance R_1 by $(g_{m3}r_o)$. Hence, the resistance R_2 is,

$$\begin{aligned} R_2 &= (g_{m3}r_o)R_1 \\ &= (g_{m3}r_o)r_o \\ &= g_m r_o^2 \qquad (\text{Since, } g_{m2} = g_m) \end{aligned}$$

Thus, the resistance R_2 is, $\boxed{R_2 = g_m r_o^2}$.

The cascade PMOS transistor Q_2 divides the resistance R_2 by $(g_{m2}r_o)$. Hence, the resistance R_3 is,

$$\begin{aligned} R_3 &= \frac{R_2}{g_{m2}r_o} \\ &= \frac{g_m r_o^2}{g_{m2}r_o} \\ &= \frac{g_m r_o^2}{g_m r_o} \qquad (\text{Since, } g_{m2} = g_m) \\ &= r_o \end{aligned}$$

Thus, the resistance R_3 is, $\boxed{R_3 = r_o}$.

Step 2 of 9

(b)

From the circuit, the current i_1 is,

$$i_1 = g_m v_i$$

Thus, the current i_1 is, $\boxed{i_1 = g_m v_i}$

Step 3 of 9

From the circuit, the current i_2 is,

$$i_2 = \frac{v_{gs2}}{r_o}$$

From the circuit, write the following current relation.

$i_3 + i_2 = i_1$ ..... (1)

The current i_3 is,

$$i_3 = g_{m2}v_{gs2}$$

Substitute $g_{m2}v_{gs2}$ for i_3, $\frac{v_{gs2}}{r_0}$ for i_2, and $g_m v_i$ for i_1 in equation (1).

$$g_m v_{gs2} + \frac{v_{gs1}}{r_o} = g_m v_i$$

$$v_{gs2}\left(g_m + \frac{1}{r_o}\right) = g_m v_i$$

$$v_{gs2} = \frac{g_m}{g_m + \frac{1}{r_o}} v_i$$

The current i_2 is,

$$i_2 = \frac{v_{gs2}}{r_o}$$

Substitute $\frac{g_m}{g_m + \frac{1}{r_o}} v_i$ for v_{gs2} in the equation.

$$\begin{aligned} i_2 &= \frac{v_i \frac{g_m}{r_o}}{\left(g_m + \frac{1}{r_o}\right)} \\ &= \frac{v_i g_m}{g_m r_o + 1} \end{aligned}$$

Thus, the current i_2 is, $\boxed{i_2 = \frac{v_i g_m}{g_m r_o + 1}}$

Step 4 of 9

Consider equation (1).

$i_1 = i_2 + i_3$

Substitute $\frac{v_i g_m}{g_m r_o + 1}$ for i_2 and $g_m v_i$ for i_1 in the equation

$$g_m v_i = v_i \frac{g_m}{(g_m r_o + 1)} + i_3$$

$$\begin{aligned} i_3 &= g_m v_i - v_i \frac{g_m}{(g_m r_o + 1)} \\ &= \frac{(g_m v_i)(g_m r_o + 1) - g_m v_i}{(g_m r_o + 1)} \\ &= \frac{g_m^2 r_o v_i + g_m v_i - g_m v_i}{(g_m r_o + 1)} \\ &= \left(\frac{g_m^2 r_o}{g_m r_o + 1}\right) v_i \end{aligned}$$

Thus, the current i_3 is, $\boxed{i_3 = \left(\frac{g_m^2 r_o}{g_m r_o + 1}\right) v_i}$

From the circuit, the current i_4 is,

$$\begin{aligned} i_4 &= g_m v_{gs2} + \frac{v_{gs2}}{r_o} \\ &= v_{gs2}\left(g_m + \frac{1}{r_o}\right) \\ &= v_i \frac{g_m}{g_m + \frac{1}{r_o}}\left(g_m + \frac{1}{r_o}\right) \qquad \left(\text{Since, } v_{gs2} = v_i \frac{g_m}{g_m + \frac{1}{r_o}}\right) \\ &= v_i g_m \end{aligned}$$

Thus, the current i_4 is, $\boxed{i_4 = g_m v_i}$.

From the circuit, the current i_5 is,

$$\begin{aligned} i_5 &= i_4 \\ &= v_i g_m \end{aligned}$$

Thus, the current i_5 is, $\boxed{i_5 = v_i g_m}$.

From the circuit, the current i_6 is zero.

Thus, the current i_6 is, $\boxed{i_6 = 0}$

From the circuit, the current i_7 is,

$$\begin{aligned} i_5 &= i_6 + i_7 \\ i_5 &= 0 + i_7 \qquad (\text{Since, } i_6 = 0) \\ i_7 &= i_5 \\ &= v_i g_m \qquad (\text{Since, } i_5 = v_i g_m) \end{aligned}$$

Thus, the current i_7 is, $\boxed{i_7 = v_i g_m}$

Step 5 of 9

(c)

From, the circuit, the voltage v_1 is,

$$\begin{aligned} v_1 &= A_m v_i \\ &= -g_m(r_o \| R_3)v_i \\ &= -g_m(r_o \| r_o)v_i \\ &= -\frac{g_m r_o}{2} v_i \end{aligned}$$

Thus, the voltage v_1 is, $\boxed{v_1 = -\frac{g_m r_o}{2} v_i}$.

Step 6 of 9

From, the circuit, the voltage v_2 is,

$$\begin{aligned} v_2 &= A_v v_i \\ &= -g_m(g_m r_o^2 \| R_2)v_i \\ &= -g_m(g_m r_o^2 \| g_m r_o^2)v_i \qquad (\text{Since, } R_2 = g_m r_o^2) \\ &= -\frac{g_m^2 r_o^2}{2} v_i \end{aligned}$$

Thus, the voltage v_2 is, $\boxed{v_2 = -\frac{g_m^2 r_o^2}{2} v_i}$.

Step 7 of 9

From, the circuit, the voltage v_3 is,

$$\begin{aligned} v_3 &= \frac{v_2}{-g_m r_o} \\ &= \frac{-\frac{g_m^2 r_o^2}{2} v_i}{-g_m r_o} \\ &= \frac{g_m r_o}{2} v_i \end{aligned}$$

Thus, the voltage v_3 is, $\boxed{v_1 = \frac{g_m r_o}{2} v_i}$.

Step 8 of 9

(d)

Consider the equation of voltage v_1.

$$v_1 = -\frac{g_m r_o}{2} v_i$$

Substitute 20 for $g_m r_o$ and $5\ \text{mV}$ for v_i.

$$\begin{aligned} v_1 &= -\frac{20}{2}(5\times10^{-3}) \\ &= -50\ \text{mV} \end{aligned}$$

Consider the equation of voltage v_2.

$$\begin{aligned} v_2 &= -\frac{g_m^2 r_o^2}{2} v_i \\ &= -\frac{(g_m r_o)^2}{2} v_i \end{aligned}$$

Substitute 20 for $g_m r_o$ and $5\ \text{mV}$ for v_i.

$$\begin{aligned} v_2 &= -\frac{20^2}{2}(5\times10^{-3}) \\ &= -1\ \text{V} \end{aligned}$$

Consider the equation of voltage v_3.

$$v_3 = \frac{g_m r_o}{2} v_i$$

Substitute 20 for $g_m r_o$ and $5\ \text{mV}$ for v_i.

$$\begin{aligned} v_3 &= \frac{20}{2}(5\times10^{-3}) \\ &= 50\ \text{mV} \end{aligned}$$

Step 9 of 9

The sketch of the waveforms is shown in Figure 1.

Figure 1

Step 1 of 8

8.074P

Refer to Figure P8.74 from the text book.

Consider the expression for the bias current I .

$$I=\frac{1}{2}\mu_p C_{ox}\left(\frac{W}{L}\right)V_{OV}^{\ 2}$$

Substitute $0.2\ \text{mA}$ for I , $100\ \mu\text{A/V}^2$ for $\mu_p C_{ox}$ and $0.2\ \text{V}$ for V_{OV} .

$$0.2\ \text{m}=\frac{1}{2}(100\ \mu)\left(\frac{W}{L}\right)(0.2)^2$$

$$0.2\ \text{m}=(50\ \mu)\left(\frac{W}{L}\right)(0.04)$$

$$0.2\ \text{m}=(2\ \mu)\left(\frac{W}{L}\right)$$

$$\frac{W}{L}=\frac{0.2\ \text{m}}{2\ \mu}$$

$$\frac{W}{L}=100$$

Therefore, the value of the ration $\frac{W}{L}$ is, $\boxed{100}$.

Step 2 of 8

For the largest possible output swing, the value of the drain to source voltage V_{DS} of each transistor should be equal to the overdrive voltage V_{OV} .

Calculate the value of the voltage V_{G1} .

$$V_{G1}=V_{DD}-|V_{tp}|-|V_{OV}|$$

Substitute $1.8\ \text{V}$ for V_{DD} , $0.4\ \text{V}$ for $|V_{tp}|$ and $0.2\ \text{V}$ for $|V_{OV}|$.

$$\begin{aligned}V_{G1}&=1.8-0.4-0.2\\&=1.2\ \text{V}\end{aligned}$$

Therefore, the value of the voltage V_{G1} is $\boxed{1.2\ \text{V}}$.

Step 3 of 8

Calculate the value of the voltage V_{G2} .

$$V_{G2}=V_{DD}-V_{DS1}-|V_{tp}|-|V_{OV}|$$

Substitute $1.8\ \text{V}$ for V_{DD} , $0.2\ \text{V}$ for V_{DS1} , $0.4\ \text{V}$ for $|V_{tp}|$ and $0.2\ \text{V}$ for $|V_{OV}|$.

$$\begin{aligned}V_{G2}&=1.8-0.2-0.4-0.2\\&=1\ \text{V}\end{aligned}$$

Therefore, the value of the voltage V_{G2} is, $\boxed{1\ \text{V}}$.

Step 4 of 8

Calculate the value of the voltage V_{G3} .

$$V_{G3}=V_{DD}-V_{DS1}-V_{DS2}-|V_{tp}|-|V_{OV}|$$

Substitute $1.8\ \text{V}$ for V_{DD} , $0.2\ \text{V}$ for V_{DS1} , $0.2\ \text{V}$ for V_{DS2} , $0.4\ \text{V}$ for $|V_{tp}|$ and $0.2\ \text{V}$ for $|V_{OV}|$.

$$\begin{aligned}V_{G3}&=1.8-0.2-0.2-0.4-0.2\\&=0.8\ \text{V}\end{aligned}$$

Therefore, the value of the voltage V_{G3} is, $\boxed{0.8\ \text{V}}$.

Step 5 of 8

Consider the expression for the output resistance.

$$R_o=(g_{m3}r_{o3})(g_{m2}r_{o2})r_{o1}$$

Here, $g_{m3}=g_{m2}$ and $r_{o1}=r_{o2}=r_{o3}$ since, the transistors are matched.

$$R_o=g_m^{\ 2}r_o^{\ 3}\ \ldots\ldots\ (1)$$

Therefore, the expression for the output resistance R_o is, $g_m^{\ 2}r_o^{\ 3}$.

Step 6 of 8

Calculate the value of the transconductance g_m .

$$g_m=\frac{2I}{V_{OV}}$$

Substitute $0.2\ \text{mA}$ for I and $0.2\ \text{V}$ for V_{OV} .

$$\begin{aligned}g_m&=\frac{2(0.2\ \text{m})}{0.2}\\&=2\ \text{mA/V}\end{aligned}$$

Therefore, the value of the transconductance g_m is, $2\ \text{mA/V}$.

Step 7 of 8

Calculate the value of the resistance r_o .

$$r_o=\frac{|V_A'|L}{I}$$

Substitute $0.2\ \text{mA}$ for I , $6\ \text{V}$ for V_{OV} and $0.13\ \mu\text{m}$ for L .

$$\begin{aligned}r_o&=\frac{(6/1\ \mu)(0.13\ \mu)}{0.2\ \text{m}}\\&=\frac{0.78}{0.2\ \text{m}}\\&=3.9\ \text{k}\Omega\end{aligned}$$

Therefore, the value of the resistance r_o is, $3.9\ \text{k}\Omega$.

Step 8 of 8

Recall equation (1).

$$R_o=g_m^{\ 2}r_o^{\ 3}$$

Substitute $3.9\ \text{k}\Omega$ for r_o and $2\ \text{mA/V}$ for g_m .

$$\begin{aligned}R_o&=(2\ \text{m})^2(3.9\ \text{k})^3\\&=\left(4\times10^{-6}\right)\left(5.9313\times10^{10}\right)\\&=237.252\ \text{k}\Omega\end{aligned}$$

Therefore, the value of the output resistance is, $\boxed{237.252\ \text{k}\Omega}$.

8.075P

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Step 1 of 8

Consider the circuit diagram shown in Figure 1.

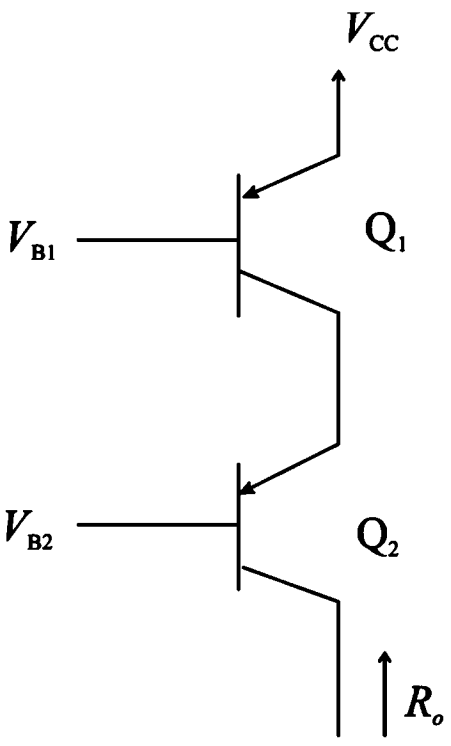


Figure 1

Step 2 of 8

Draw the small signal equivalent circuit diagram to calculate the value of output resistance R_o.

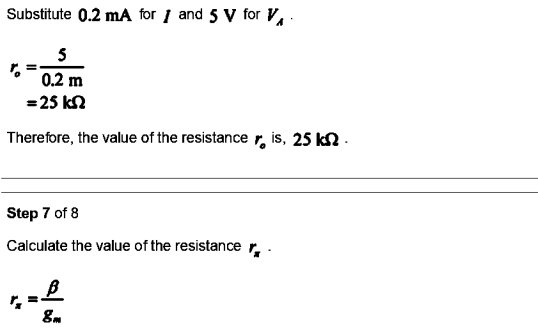


Figure 2

Step 3 of 8

Calculate the expression for the voltage $v_{\pi 2}$.

$$v_{\pi 2} = -i_x(r_{o1} \| r_{\pi 2})$$

Therefore, the expression for the voltage $v_{\pi 2}$ is $-i_x(r_{o1} \| r_{\pi 2})$.

Step 4 of 8

Consider the expression for the voltage v_x.

$$v_x = (i_x - g_{m2}v_{\pi 2})r_{o2} + (i_x - g_{m1}v_{\pi 1})(r_{o1} \| r_{\pi 2})$$

Substitute $0\ \text{V}$ for $v_{\pi 1}$.

$$\begin{aligned} v_x &= (i_x - g_{m2}v_{\pi 2})r_{o2} + (i_x - g_{m1}(0))(r_{o1} \| r_{\pi 2}) \\ &= (i_x - g_{m2}v_{\pi 2})r_{o2} + i_x(r_{o1} \| r_{\pi 2}) \\ &= i_x r_{o2} - g_{m2}v_{\pi 2}r_{o2} + i_x(r_{o1} \| r_{\pi 2}) \end{aligned}$$

Substitute $i_x(r_{o1} \| r_{\pi 2})$ for $v_{\pi 2}$.

$$\begin{aligned} v_x &= i_x r_{o2} - g_{m2}r_{o2}(i_x(r_{o1} \| r_{\pi 2})) + i_x(r_{o1} \| r_{\pi 2}) \\ &= i_x r_{o2} + g_{m2}r_{o2}i_x(r_{o1} \| r_{\pi 2}) + i_x(r_{o1} \| r_{\pi 2}) \\ &= i_x r_{o2} + (g_{m2}r_{o2} + 1)i_x(r_{o1} \| r_{\pi 2}) \end{aligned}$$

Here, $g_{m2}r_{o2} + 1$ is approximately equal to $g_{m2}r_{o2}$.

$$\begin{aligned} v_x &= i_x r_{o2} + (g_{m2}r_{o2})i_x(r_{o1} \| r_{\pi 2}) \\ &= i_x r_{o2}(1 + g_{m2}(r_{o1} \| r_{\pi 2})) \end{aligned}$$

Here, $1 + g_{m2}(r_{o1} \| r_{\pi 2})$ is approximately equal to $g_{m2}(r_{o1} \| r_{\pi 2})$.

$$\frac{v_x}{i_x} = g_{m2}r_{o2}(r_{o1} \| r_{\pi 2})$$

Here, $\frac{v_x}{i_x}$ represents the output resistance R_o.

$$R_o = r_{o2}(1 + g_{m2}(r_{o1} \| r_{\pi 2})) \quad \text{..... (1)}$$

Therefore, the expression for the output resistance R_o is $g_{m2}r_{o2}(r_{o1} \| r_{\pi 2})$.

Step 5 of 8

Calculate the value of the transconductance g_m.

$$g_m = \frac{I}{V_T}$$

Substitute $0.2\ \text{mA}$ for I and $25\ \text{mV}$ for V_T.

$$\begin{aligned} g_m &= \frac{0.2\ \text{m}}{25\ \text{m}} \\ &= 8\ \text{mA/V} \end{aligned}$$

Therefore, the value of the transconductance g_m is, $8\ \text{mA/V}$.

Step 6 of 8

Calculate the value of the resistance r_o.

$$r_o = \frac{V_A}{I}$$

Substitute $0.2\ \text{mA}$ for I and $5\ \text{V}$ for V_A.

$$\begin{aligned} r_o &= \frac{5}{0.2\ \text{m}} \\ &= 25\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance r_o is, $25\ \text{k}\Omega$.

Step 7 of 8

Calculate the value of the resistance r_π.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute $8\ \text{mA/V}$ for g_m and 50 for β.

$$\begin{aligned} r_\pi &= \frac{50}{8\ \text{m}} \\ &= 6.25\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance r_π is, $6.25\ \text{k}\Omega$.

Step 8 of 8

Recall equation (1), to determine the value of output resistance.

$$R_o = g_{m2}r_{o2}(r_{o1} \| r_{\pi 2})$$

Substitute $6.25\ \text{k}\Omega$ for r_π, $25\ \text{k}\Omega$ for r_{o1} and r_{o2} and $8\ \text{mA/V}$ for g_m.

$$\begin{aligned} R_o &= (25\ \text{k})(8\ \text{m})(25\ \text{k} \| 6.25\ \text{k}) \\ &= (200)\left(\frac{(25\ \text{k})(6.25\ \text{k})}{25\ \text{k} + 6.25\ \text{k}}\right) \\ &= (200)\left(\frac{156.25\ \text{M}}{31.25\ \text{k}}\right) \\ &= (200)(5000) \\ R_o &= 1\ \text{M}\Omega \end{aligned}$$

Therefore, the value of the output resistance R_o is $\boxed{1\ \text{M}\Omega}$.

Step 1 of 6

Write the expression for output resistance of the BJT cascode amplifier.

$$R_o = g_m r_o \left(r_o \| r_\pi\right)$$
$$= g_m r_o \left(\frac{r_o \times r_\pi}{r_o + r_\pi}\right)$$

Substitute $\frac{I_C}{V_T}$ for g_m, $\frac{|V_A|}{I_C}$ for r_o and $\frac{\beta V_T}{I_C}$ for r_π in the equation.

$$R_o = \left(\frac{I_C}{V_T}\right)\left(\frac{|V_A|}{I_C}\right)\left(\frac{\frac{|V_A|}{I_C} \times \frac{\beta V_T}{I_C}}{\frac{|V_A|}{I_C} + \frac{\beta V_T}{I_C}}\right)$$
$$= \left(\frac{|V_A|}{V_T}\right)\left(\frac{|V_A|\beta V_T}{I_C\left(|V_A| + \beta V_T\right)}\right)$$
$$= \frac{\beta |V_A^2|}{I_C\left(|V_A| + \beta V_T\right)}$$

Step 2 of 6

The expression for figure-of-merit of the BJT cascode amplifier is,

$$IR_o = I_C R_o \qquad (\text{Since } I = I_C)$$

Substitute $\frac{\beta |V_A^2|}{I_C\left(|V_A| + \beta V_T\right)}$ for R_o in the equation.

$$IR_o = I_C\left(\frac{\beta |V_A^2|}{I_C\left(|V_A| + \beta V_T\right)}\right)$$
$$= \frac{\beta |V_A^2|}{|V_A| + \beta V_T}$$
$$= \frac{\beta |V_A^2|}{\beta |V_A|\left(\frac{1}{\beta} + \frac{V_T}{|V_A|}\right)}$$
$$= \frac{|V_A|}{\left(\frac{V_T}{|V_A|} + \frac{1}{\beta}\right)}$$

Thus, the expression for figure-of-merit of the BJT cascode amplifier, IR_o is $\boxed{\frac{|V_A|}{\frac{V_T}{|V_A|} + \frac{1}{\beta}}}$.

Step 3 of 6

Evaluate the value of figure-of-merit.

$$IR_o = \frac{|V_A|}{\left(\frac{V_T}{|V_A|} + \frac{1}{\beta}\right)}$$

Substitute $5\ \text{V}$ for $|V_A|$, $25\ \text{mV}$ for V_T and 50 for β in the equation.

$$IR_o = \frac{5}{\left(\frac{25\times10^{-3}}{5}\right) + \frac{1}{50}}$$
$$= \frac{5}{\left(5\times10^{-3}\right) + \left(20\times10^{-3}\right)}$$
$$= \frac{5}{25\times10^{-3}}$$
$$= 200\ \text{V}$$

Thus, the value of figure-of-merit, IR_o is $\boxed{200\ \text{V}}$.

Step 4 of 6

Find the value of output resistance for $I = 0.1\ \text{mA}$.

$$IR_o = 200$$
$$R_o = \frac{200}{I}$$

Substitute $0.1\ \text{mA}$ for I in the equation.

$$R_o = \frac{200}{0.1\times10^{-3}}$$
$$= 2000\times10^3$$
$$= 2\ \text{M}\Omega$$

Thus, the value of output resistance R_o for $I = 0.1\ \text{mA}$ is $\boxed{2\ \text{M}\Omega}$.

Step 5 of 6

Find the value of output resistance for $I = 0.5\ \text{mA}$.

$$IR_o = 200$$
$$R_o = \frac{200}{I}$$

Substitute $0.5\ \text{mA}$ for I in the equation.

$$R_o = \frac{0.2\times10^3}{0.5\times10^{-3}}$$
$$= 0.4\times10^6$$
$$= 0.4\ \text{M}\Omega$$

Thus, the value of output resistance R_o for $I = 0.5\ \text{mA}$ is $\boxed{0.4\ \text{M}\Omega}$.

Step 6 of 6

Find the value of output resistance for $I = 1\ \text{mA}$.

$$IR_o = 200$$
$$R_o = \frac{200}{I}$$

Substitute $1\ \text{mA}$ for I in the equation.

$$R_o = \frac{0.2\times10^3}{1\times10^{-3}}$$
$$= 0.2\times10^6$$
$$= 0.2\ \text{M}\Omega$$

Thus, the value of output resistance R_o for $I = 1\ \text{mA}$ is $\boxed{0.2\ \text{M}\Omega}$.

Step 1 of 5

8.078P

Refer to Figure 7.19 in the textbook for the BJT cascode amplifier.

Write the expressions for R_{op} and R_{on}.

$$R_{op} = (g_{m3}r_{o3})(r_{o4} \| r_{o3})$$

And

$$R_{on} = (g_{m2}r_{o2})(r_{o1} \| r_{o2})$$

Step 2 of 5

Since all the transistors have equal β and r_o, the expression for R_{op} and R_{on} is,

$$\begin{aligned} R_{op} = R_{on} &= g_m r_o (r_o \| r_\pi) \\ &= g_m r_o \left(\frac{r_o \times r_\pi}{r_o + r_\pi} \right) \end{aligned}$$

Substitute $\frac{I_C}{V_T}$ for g_m, $\frac{|V_A|}{I_C}$ for r_o and $\frac{\beta V_T}{I_C}$ for r_π in the equation.

$$\begin{aligned} R_{op} = R_{on} &= \left(\frac{I_C}{V_T} \right) \left(\frac{|V_A|}{I_C} \right) \left(\frac{\frac{|V_A|}{I_C} \times \frac{\beta V_T}{I_C}}{\frac{|V_A|}{I_C} + \frac{\beta V_T}{I_C}} \right) \\ &= \left(\frac{|V_A|}{V_T} \right) \left(\frac{|V_A| \beta V_T}{I_C (|V_A| + \beta V_T)} \right) \\ &= \frac{\beta |V_A^2|}{I_C (|V_A| + \beta V_T)} \end{aligned}$$

Simplify further.

$$\begin{aligned} R_{op} = R_{on} &= \frac{\beta |V_A^2|}{I_C (|V_A| + \beta V_T)} \qquad (\text{Since, } I = I_C) \\ &= \frac{\beta |V_A^2|}{I_C \beta |V_A| \left(\frac{1}{\beta} + \frac{V_T}{|V_A|} \right)} \\ &= \frac{|V_A|}{I_C \left(\frac{V_T}{|V_A|} + \frac{1}{\beta} \right)} \end{aligned}$$

Step 3 of 5

The voltage gain is,

$$A_v = -g_{m1} (R_{on} \| R_{op})$$

Since all the transistors have equal β and r_o, the expression for voltage gain is,

$$A_v = -g_m (R_{on} \| R_{op})$$

Substitute $\frac{I_C}{V_T}$ for g_m, $\frac{|V_A|}{I_C \left(\frac{V_T}{|V_A|} + \frac{1}{\beta} \right)}$ for R_{op}, and $\frac{|V_A|}{I_C \left(\frac{V_T}{|V_A|} + \frac{1}{\beta} \right)}$ for R_{on} in the equation.

$$\begin{aligned} A_v &= -\frac{I_C}{V_T} \left[\left(\frac{|V_A|}{I_C \left(\frac{V_T}{|V_A|} + \frac{1}{\beta} \right)} \right) \| \left(\frac{|V_A|}{I_C \left(\frac{V_T}{|V_A|} + \frac{1}{\beta} \right)} \right) \right] \\ &= -\frac{I_C}{V_T} \left(\frac{|V_A|}{2 I_C \left(\frac{V_T}{|V_A|} + \frac{1}{\beta} \right)} \right) \\ &= -\frac{1}{2} \left(\frac{\frac{|V_A|}{V_T}}{\frac{V_T}{|V_A|} + \frac{1}{\beta}} \right) \end{aligned}$$

Therefore, it is proved that the voltage gain can be expressed as, $\boxed{A_v = -\frac{1}{2} \left(\frac{\frac{|V_A|}{V_T}}{\frac{V_T}{|V_A|} + \frac{1}{\beta}} \right)}$

Step 4 of 5

Consider the expression for voltage gain.

$$A_v = -\frac{1}{2} \left(\frac{\frac{|V_A|}{V_T}}{\frac{V_T}{|V_A|} + \frac{1}{\beta}} \right)$$

Step 5 of 5

Substitute $5\ \text{V}$ for V_A, $25\ \text{mV}$ for V_T and 50 for β in the equation.

$$\begin{aligned} A_v &= -\frac{1}{2} \left(\frac{\frac{5}{0.025}}{\frac{5}{0.025} + \frac{1}{50}} \right) \\ &= -\frac{1}{2} \left(\frac{200}{200 + 0.02} \right) \\ &= -\frac{100}{200.02} \\ &= -0.5 \end{aligned}$$

Thus, the voltage gain A_v is, $\boxed{-0.5\ \text{V/V}}$.

Step 1 of 6

Calculate the value of the transconductance g_m.

$$g_m = \frac{I}{V_T}$$

Substitute $0.2\ \text{mA}$ for I and $25\ \text{mV}$ for V_T.

$$\begin{aligned} g_m &= \frac{0.2\ \text{m}}{25\ \text{m}} \\ &= 8\ \text{mA/V} \end{aligned}$$

Therefore, the value of the transconductance g_m is, $8\ \text{mA/V}$.

Step 2 of 6

Calculate the value of the resistance r_o.

$$r_o = \frac{V_A}{I}$$

Substitute $0.2\ \text{mA}$ for I and $100\ \text{V}$ for V_A.

$$\begin{aligned} r_o &= \frac{100}{0.2\ \text{m}} \\ &= 500\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance r_o is, $500\ \text{k}\Omega$.

Step 3 of 6

Calculate the value of the resistance r_π.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute $8\ \text{mA/V}$ for g_m and 50 for β.

$$\begin{aligned} r_\pi &= \frac{50}{8\ \text{m}} \\ &= 6.25\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance r_π is, $6.25\ \text{k}\Omega$.

Step 4 of 6

Consider the expression for the resistance R_{op}.

$$R_{op} = \beta r_o$$

Substitute $500\ \text{k}\Omega$ for r_o and 50 for β.

$$\begin{aligned} R_{op} &= (50)(500\ \text{k}) \\ &= 25\ \text{M}\Omega \end{aligned}$$

Therefore, the value of the resistance R_{op} is $25\ \text{M}\Omega$.

Step 5 of 6

Consider the expression for the resistance R_{on}.

$$R_{on} = g_{m2} r_{o2} \left(r_{o1} \| r_{\pi 2} \right)$$

Substitute $6.25\ \text{k}\Omega$ for r_π, $500\ \text{k}\Omega$ for r_{o1} and r_{o2} and $8\ \text{mA/V}$ for g_m.

$$\begin{aligned} R_{on} &= (500\ \text{k})(8\ \text{m})(500\ \text{k} \| 6.25\ \text{k}) \\ &= (4000)\left(\frac{(500\ \text{k})(6.25\ \text{k})}{500\ \text{k} + 6.25\ \text{k}} \right) \\ &= (4000)\left(\frac{3125\ \text{M}}{506.25\ \text{k}} \right) \\ &= (4000)(6172.84) \end{aligned}$$

$$R_{on} = 24.69\ \text{M}\Omega$$

Therefore, the value of the output resistance R_{on} is $24.69\ \text{M}\Omega$.

Step 6 of 6

Consider the expression for the voltage gain A_v.

$$A_v = -g_m \left(R_{op} \| R_{on} \right)$$

Substitute $8\ \text{mA/V}$ for g_m, $24.69\ \text{M}\Omega$ for R_{on} and $25\ \text{M}\Omega$ for R_{op}.

$$\begin{aligned} A_v &= -(8\ \text{m})(25\ \text{M} \| 24.69\ \text{M}) \\ &= -(8\ \text{m})\left(\frac{25\ \text{M} \times 24.69\ \text{M}}{25\ \text{M} + 24.69\ \text{M}} \right) \\ &= -(8\ \text{m})\left(\frac{25\ \text{M} \times 24.69\ \text{M}}{49.69\ \text{M}} \right) \\ &= -(8\ \text{m})\left(\frac{617.25 \times 10^{12}}{49.69\ \text{M}} \right) \end{aligned}$$

$$\begin{aligned} A_v &= -(8\ \text{m})(12.42\ \text{M}) \\ &= -99.36\ \text{kV/V} \end{aligned}$$

Therefore, the value of the voltage gain A_v is $\boxed{-99.36\ \text{kV/V}}$.

Step 1 of 9

(a)

Refer to Figure P7.45 (a) in the textbook for the circuit.

The input resistance is,

$$\begin{aligned}R_{in} &= r_{\pi 1}\\ &= \frac{\beta}{g_m}\\ &= \frac{100}{4\times 10^{-2}}\\ &= 25\ \text{k}\Omega\end{aligned}$$

Thus, the input resistance is, $\boxed{R_{in} = 25\ \text{k}\Omega}$.

Finding trans-conductance g_m.

$$\begin{aligned}g_m &= \frac{I_C}{V_T}\\ &= \frac{100\times 10^{-6}}{25\times 10^{-3}}\\ &= 4\ \text{mA/V}\end{aligned}$$

Step 2 of 9

Calculate the value of r_o.

$$\begin{aligned}r_o &= \frac{V_A}{I_C}\\ &= \frac{5}{100\times 10^{-6}}\\ &= 50\ \text{k}\Omega\end{aligned}$$

The output resistance is,

$$\begin{aligned}R_o &= \frac{\beta r_o}{2}\\ &= \frac{(100)(50\times 10^3)}{2}\\ &= 2.5\ \text{M}\Omega\end{aligned}$$

Thus, the output resistance is, $\boxed{R_o = 2.5\ \text{M}\Omega}$

The open-circuit voltage gain is,

$$A_m = g_m(\beta r_\pi \parallel R_L)$$

From the circuit, $R_L = \beta r_o$. Substitute this in the equation.

$$\begin{aligned}A_m &= g_m(\beta r_\pi \parallel \beta r_o)\\ &= g_m\left(\frac{\beta r_o}{2}\right)\\ &= \left(4\times 10^{-3}\right)\left(\frac{100(50\times 10^3)}{2}\right)\\ &= 20000\ \text{V/V}\end{aligned}$$

Thus, the open-circuit voltage gain is, $\boxed{A_m = 20000\ \text{V/V}}$.

Step 3 of 9

(b)

Refer to Figure P7.45 (b) in the textbook for the circuit.

The input resistance is,

$$\begin{aligned}R_{in} &= r_{\pi 1}\\ &= \frac{\beta}{g_m}\\ &= \frac{100}{4\times 10^{-2}}\\ &= 25\ \text{k}\Omega\end{aligned}$$

Thus, the input resistance is, $\boxed{R_{in} = 25\ \text{k}\Omega}$

Step 4 of 9

Calculate the value of r_{o1}.

$$\begin{aligned}r_{o1} &= \frac{V_A}{I_C}\\ &= \frac{5}{100\times 10^{-6}}\\ &= 50\ \text{k}\Omega\end{aligned}$$

Calculate the value of r_{o2}.

$$\begin{aligned}r_{o2} &= \frac{V_A}{I_C}\\ &= \frac{5}{100\times 10^{-6}}\\ &= 50\ \text{k}\Omega\end{aligned}$$

Step 5 of 9

Calculate the value of g_{m1}.

$$\begin{aligned}g_{m1} &= \frac{I}{V_T}\\ &= \frac{100\times 10^{-6}}{25\times 10^{-3}}\\ &= 4\ \text{mA/V}\end{aligned}$$

Calculate the value of g_{m2}.

$$\begin{aligned}g_{m2} &= \frac{2I}{V_{OV}}\\ &= \frac{2(100\times 10^{-6})}{0.2}\\ &= 1\ \text{mA/V}\end{aligned}$$

The output resistance is,

$$\begin{aligned}R_o &= r_{o2} + (1+g_{m2}r_{o2})r_{o1}\\ &= (50\times 10^3) + (1+(1\times 10^{-2})(50\times 10^3))(50\times 10^3)\\ &= (50\times 10^3) + 51(50\times 10^3)\\ &= 2.6\ \text{M}\Omega\end{aligned}$$

Thus, the output voltage is, $\boxed{R_o = 2.6\ \text{M}\Omega}$

Calculate the output resistance of current source.

$$\begin{aligned}R_{L2} &= g_{m2}r_{o2}r_{o1}\\ &= (1\times 10^{-2})(50\times 10^3)(50\times 10^3)\\ R_o &= 2.5\ \text{M}\Omega\end{aligned}$$

The open-circuit voltage gain is,

$$\begin{aligned}A_m &= -g_{m1}\left(r_\pi \parallel \left(r_{o1} + \frac{1}{g_{m2}}\right)\right)(1+g_{m1}r_{o2})\frac{R_{L2}}{R_{L2}+R_o}\\ &= -g_{m1}\left(\frac{r_{o1}}{2}\right)(1+g_{m1}r_{o2})\frac{R_{L2}}{R_{L2}+R_o}\\ &= -\left(1\times 10^{-3}\right)\left(\frac{50\times 10^3}{2}\right)\left(1+(1\times 10^{-3})(50\times 10^3)\right)\frac{2.5\times 10^6}{2.5\times 10^6+2.6\times 10^6}\\ &= -2500\ \text{V/V}\end{aligned}$$

Thus, the open-circuit voltage gain is, $\boxed{A_m = -2500\ \text{V/V}}$.

Step 6 of 9

(c)

Refer to Figure P7.45 (c) in the textbook for the circuit.

Since the input resistance of MOSFET is infinity, the input resistance of the circuit is,

$\boxed{R_{in} = \infty}$

Calculate the value of g_{m1}.

$$\begin{aligned}g_{m1} &= \frac{2I}{V_{OV}}\\ &= \frac{2(100\times 10^{-6})}{0.2}\\ &= 1\ \text{mA/V}\end{aligned}$$

Calculate the value of g_{m2}.

$$\begin{aligned}g_{m2} &= \frac{2I}{V_{OV}}\\ &= \frac{2(100\times 10^{-6})}{0.2}\\ &= 1\ \text{mA/V}\end{aligned}$$

Calculate the value of r_{o1}.

$$\begin{aligned}r_{o1} &= \frac{V_A}{I_C}\\ &= \frac{5}{100\times 10^{-6}}\\ &= 50\ \text{k}\Omega\end{aligned}$$

Calculate the value of r_{o2}.

$$\begin{aligned}r_{o2} &= \frac{V_A}{I_C}\\ &= \frac{5}{100\times 10^{-6}}\\ &= 50\ \text{k}\Omega\end{aligned}$$

The output resistance is,

$$\begin{aligned}R_o &= r_{o2} + (1+g_{m2}r_{o2})r_{o1}\\ &= (50\times 10^3) + (1+(1\times 10^{-3})(50\times 10^3))(50\times 10^3)\\ &= (50\times 10^3) + 51(50\times 10^3)\\ &= 2.6\ \text{M}\Omega\end{aligned}$$

Thus, the output voltage is, $\boxed{R_o = 2.6\ \text{M}\Omega}$

Step 7 of 9

Calculate the output resistance of current source.

$$\begin{aligned}R_{L2} &= g_{m2}r_{o2}r_{o1}\\ &= (1\times 10^{-3})(50\times 10^3)(50\times 10^3)\\ R_o &= 2.5\ \text{M}\Omega\end{aligned}$$

The open-circuit voltage gain is,

$$\begin{aligned}A_m &= -g_{m1}\left(r_\pi \parallel \left(r_{o1} + \frac{1}{g_{m2}}\right)\right)(1+g_{m1}r_{o2})\frac{R_{L2}}{R_{L2}+R_o}\\ &= -g_{m1}\left(\frac{r_{o1}}{2}\right)(1+g_{m1}r_{o2})\frac{R_{L2}}{R_{L2}+R_o}\\ &= -\left(1\times 10^{-3}\right)\left(\frac{50\times 10^3}{2}\right)\left(1+(1\times 10^{-3})(50\times 10^3)\right)\frac{2.5\times 10^6}{2.5\times 10^6+2.6\times 10^6}\\ &= -625\ \text{V/V}\end{aligned}$$

Thus, the open-circuit voltage gain is, $\boxed{A_m = -625\ \text{V/V}}$.

Step 8 of 9

(d)

Refer to Figure P7.45 (d) in the textbook for the circuit.

Since the input resistance of MOSFET is infinity, the input resistance of the circuit is,

$\boxed{R_{in} = \infty}$.

Calculate the value of g_{m1}.

$$\begin{aligned}g_{m1} &= \frac{2I}{V_{OV}}\\ &= \frac{2(100\times 10^{-6})}{0.2}\\ &= 1\ \text{mA/V}\end{aligned}$$

Calculate the value of g_{m2}.

$$\begin{aligned}g_{m2} &= \frac{I_C}{V_T}\\ &= \frac{100\times 10^{-6}}{25\times 10^{-3}}\\ &= 4\ \text{mA/V}\end{aligned}$$

Calculate the value of r_{o1}.

$$\begin{aligned}r_{o1} &= \frac{V_A}{I_C}\\ &= \frac{5}{100\times 10^{-6}}\\ &= 50\ \text{k}\Omega\end{aligned}$$

Calculate the value of r_{o2}.

$$\begin{aligned}r_{o2} &= \frac{V_A}{I_C}\\ &= \frac{5}{100\times 10^{-6}}\\ &= 50\ \text{k}\Omega\end{aligned}$$

Calculate the value of r_π.

$$\begin{aligned}r_\pi &= \frac{\beta}{g_{m2}}\\ &= \frac{100}{4\times 10^{-2}}\\ &= 25\ \text{k}\Omega\end{aligned}$$

The output resistance is,

$$\begin{aligned}R_o &= r_{o2}\left(1+g_{m2}\left(r_{o1} \parallel r_\pi\right)\right)\\ &= (50\times 10^3)\left(1+(4\times 10^{-2})\left[(50\times 10^3) \parallel (25\times 10^3)\right]\right)\\ &= (50\times 10^3)\left(1+(4\times 10^{-2})(16.667\times 10^3)\right)\\ &= 3.383\ \text{M}\Omega\end{aligned}$$

Thus, the output voltage is, $\boxed{R_o = 3.383\ \text{M}\Omega}$.

Calculate the output resistance of current source.

$$\begin{aligned}R_{L2} &= R_o\\ &= 3.383\ \text{M}\Omega\end{aligned}$$

The emitter resistance of transistor Q_2 is,

$$\begin{aligned}r_{e2} &= \frac{\alpha}{g_{m2}}\\ &= \frac{\beta}{(\beta+1)g_{m2}}\\ &= \frac{100}{(100+1)(4\times 10^{-3})}\\ &= 247.525\ \Omega\end{aligned}$$

Step 9 of 9

The open-circuit voltage gain is,

$$\begin{aligned}A_m &= -g_{m1}(r_{o1} \parallel R_{in2})(1+g_{m2}r_{e2})\frac{R_{L2}}{R_{L2}+R_o}\\ &= -g_{m1}\left(r_o \parallel \frac{r_{e2}+R_{L2}}{\frac{R_{L2}}{\beta+1}+r_{e2}}\right)\frac{(1+g_{m2}r_{e2})}{2} \quad (\text{Since } R_{L2} = R_o)\\ &= -\left(1\times 10^{-3}\right)\left((50\times 10^3) \parallel \frac{(50\times 10^3)+(3.383\times 10^6)}{(50\times 10^3)+\left(\frac{3.383\times 10^6}{100+1}\right)}\right)\frac{\left(1+\left(4\times 10^{-3}\right)(50\times 10^3)\right)}{2}\\ &= -\left(1\times 10^{-3}\right)\left((50\times 10^3) \parallel (41.446\times 10^3)\right)\frac{(201)}{2}\\ &= -2277.5\ \text{V/V}\end{aligned}$$

Thus, the open-circuit voltage gain is, $\boxed{A_m = -2277.5\ \text{V/V}}$

8.081P

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Step 1 of 8

Refer to Figure 7.32 in the textbook for a cascode MOS current mirror circuit.

Write the expression for the current.

$$I_1 = \frac{1}{2}k_n'\left(\frac{W_1}{L_1}\right)V_{OV}^2$$

Substitute $20\ \mu A$ for I_1, $160\ \mu A/V^2$ for k_n', $4\ \mu m$ for W_1, and $1\ \mu m$ for L_1 in order to find the overdrive voltage, V_{OV}.

$$20\times10^{-6} = \frac{1}{2}\left(160\times10^{-6}\right)\left(\frac{4}{1}\right)V_{OV}^2$$

$$40 = (160)(4)V_{OV}^2$$

$$V_{OV} = \sqrt{\frac{40}{640}}$$

$$= 0.25\ V$$

Step 2 of 8

The overdrive voltages, $V_{OV1} = V_{OV2}$. Hence the overdrive voltage, $V_{OV2} = 0.25\ V$.

Find the ratio, $\frac{I_2}{I_1}$ by substituting $I_1 = \frac{1}{2}k_n'\frac{W_1}{L_1}V_{OV1}^2$ and $I_2 = \frac{1}{2}k_n'\frac{W_2}{L_2}V_{OV1}^2$ in the ratio $\frac{I_2}{I_1}$.

$$\frac{I_2}{I_1} = \frac{\frac{1}{2}k_n'\left(\frac{W_1}{L_1}\right)V_{OV}^2}{\frac{1}{2}k_n'\left(\frac{W_2}{L_2}\right)V_{OV}^2} \quad \{\text{since } V_{OV1} = V_{OV2}\}$$

$$= \frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1}$$

Substitute $20\ \mu A$ for I_1, $40\ \mu A$ for W_2, $4\ \mu m$ for W_1, and $1\ \mu m$ for L to calculate the I_2.

$$I_2 = I_1\frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1}$$

$$= (20\ \mu A)\left(\frac{40}{4}\right)$$

$$= 200\ \mu A$$

Therefore, the output current is, $\boxed{200\ \mu A}$.

Step 3 of 8

It is known that the current $I_2 = I_3$. Since, the output current, I_o as indicated in the circuit diagram for a cascade MOS current mirror is same as the current, I_3, the output current is also obtained as $200\ \mu A$.

The expression for the gate source voltage is,

$$V_{GS1} = V_{OV1} + V_t$$

Substitute $0.6\ V$ for V_t and $0.25\ V$ for V_{OV1} (since, $V_{OV1} = V_{OV2}$, $V_{OV1} = V_{OV2} = 0.25\ V$) in order to find the value of V_{GS1}.

$$V_{GS1} = 0.25 + 0.6$$

$$= 0.85\ V$$

Hence, the value of V_{G1} is obtained as 0.85 V.

Step 4 of 8

It is known that $V_{G4} = V_{GS1} + V_{GS4}$. Since the currents I_1 and I_4 are equal and the widths W_1 and W_4 are equal, the gate source voltage, V_{GS1} of the first component and the gate source voltage of the fourth component, V_{GS4} are equal, $V_{GS1} = V_{GS4}$. Substitute the relation between the gate source voltages, $V_{GS1} = V_{GS4}$ in the expression, $V_{G4} = V_{GS1} + V_{GS4}$.

$$V_{G4} = V_{GS1} + V_{GS4}$$

$$= V_{GS4} + V_{GS4}$$

$$= 2V_{GS4}$$

Since $V_{GS1} = V_{GS4}$, rearrange the expression.

$$V_{G4} = 2V_{GS4}$$

$$= 2V_{GS1}$$

Substitute $0.85\ V$ for V_{G1} to find the value of V_{G4}.

$$V_{G4} = 2V_{GS1}$$

$$= 2(0.85)$$

$$= 1.7\ V$$

Therefore, the voltages V_{G4} and V_{G3} are, $\boxed{1.7\ V \text{ and } 1.7\ V}$ respectively.

Step 5 of 8

The lowest possible voltage at the output is when Q_3 satisfies the condition, $V_{DS3} = V_{OV3}$. The lowest possible output voltage is, $V_{omin} = V_{G3} - V_{GS3} + V_{OV3}$. Since the gate source voltages, $V_{GS1} = V_{GS2}$ and the currents, $I_2 = I_3$ assume equal magnitude in their relation, the gate source voltage of the third component becomes same as the gate source voltage of the first one, $V_{GS3} = V_{GS1}$.
The lowest possible output voltage is,

$$V_{omin} = V_{G3} - V_{GS3} + V_{OV3}$$

Substitute $1.7\ V$ for V_{G3}, $0.85\ V$ for V_{GS3}, and $0.25\ V$ for V_{OV3} in order to obtain the value of V_{omin}.

$$V_{omin} = 1.7 - 0.85 + 0.25$$

$$= 1.1\ V$$

Hence, the current-source operation is possible at the lowest voltage is, $\boxed{1.1\ V}$.

Step 6 of 8

Write the expression for trans conductance, g_m of the components Q_1 and Q_3.

$$g_m = \frac{2I_D}{V_{OV}}$$

The trans conductance of the first and the third components are equal, $g_{m1} = g_{m3}$.

Substitute $200\ \mu A$ for I_D and $0.25\ V$ for V_{OV} to get the value of g_m.

$$g_m = \frac{2I_D}{V_{OV}}$$

$$= \frac{2(0.2)}{0.25}$$

$$= 1.6\ mA/V$$

Step 7 of 8

Write the expression for the output resistance, r_o for the first and the third components, Q_1 and Q_3.

$$r_o = \frac{V_A}{I_D}$$

Substitute $10\ V$ for V_A, and $200\ \mu A$ for I_D

$$r_o = \frac{10}{200\times10^{-6}}$$

$$= 50\ k\Omega$$

Hence, the values of g_m and r_o of Q_2 and Q_3 are, $\boxed{1.6\ mA/V}$ and $\boxed{50\ k\Omega}$ respectively.

Step 8 of 8

Write the expression for the output resistance.

$$R_o \approx g_{m3}r_{o3}r_{o2}$$

Substitute $1.6\ mA/V$ for g_{m3}, $20\ k\Omega$ for r_{o2}, and $20\ k\Omega$ for r_{o3}

$$R_o = \left(1.6\times10^{-3}\right)\left(20\times10^3\right)\left(20\times10^3\right)$$

$$= 0.64\ M\Omega$$

Hence, the output resistance, R_o of the mirror is, $\boxed{0.64\ M\Omega}$.

8.083P

Refer to Figure P7.66 in the textbook for the double-cascode current mirror circuit.

If the transistors output resistance be r_{oi} where subscript *i* represent the ith transistor. Let us apply a voltage source v_X at the output node to calculate the output resistance, R_o is shown in Figure 1.

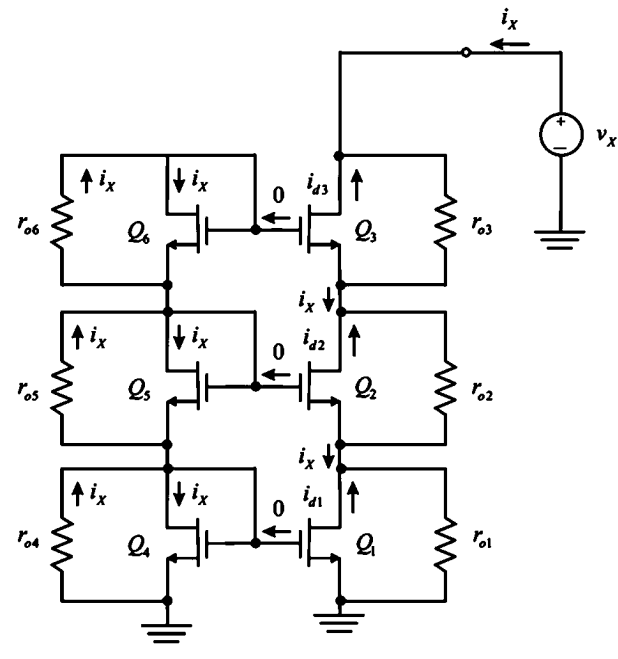


Figure 1

From Figure 1, the output resistance is,

$$R_o = \frac{v_X}{i_X} \quad \text{..... (1)}$$

Calculate the drain current of the transistor, Q_3.

$$\begin{aligned} i_{d3} &= g_{m3} v_{gs3} \\ &= g_{m3}\left(v_{g3} - v_{s3}\right) \\ &= g_{m3}\left(-\frac{i_X}{g_{m6}} - i_{d2} r_{o2}\right) \qquad \left(\text{since } v_{s3} = v_{d2} = i_{d2} r_{o2} \text{ and } v_{g3} = v_{d6} = -\frac{i_X}{g_{m6}}\right) \\ &= g_{m3}\left(-\frac{i_X}{g_{m6}} - \left(g_{m2} v_{gs2}\right) r_{o2}\right) \end{aligned}$$

Simplify further.

$$\begin{aligned} i_{d3} &= g_{m3}\left(-\frac{i_X}{g_{m6}} - \left[g_{m2}\left(v_{g2} - v_{s2}\right)\right] r_{o2}\right) \\ &= g_{m3}\left(-\frac{i_X}{g_{m6}} - \left[g_{m2}\left(-\frac{i_X}{g_{m5}} - i_{d1} r_{o1}\right)\right] r_{o2}\right) \qquad \left(\text{since } v_{s2} = v_{d1} = i_{d1} r_{o1} \text{ and } v_{g2} = v_{d5} = -\frac{i_X}{g_{m5}}\right) \\ &= g_{m3}\left(-\frac{i_X}{g_{m6}} - \left[g_{m2}\left(-\frac{i_X}{g_{m5}} - \left(g_{m1} v_{gs1}\right) r_{o1}\right)\right] r_{o2}\right) \\ &= g_{m3}\left(-\frac{i_X}{g_{m6}} - \left[g_{m2}\left(-\frac{i_X}{g_{m5}} - \left[g_{m1}\left(v_{g1} - v_{s1}\right)\right] r_{o1}\right)\right] r_{o2}\right) \end{aligned}$$

Substitute 0 for v_{s1} and $-\frac{i_X}{g_{m4}}$ for v_{g1} or v_{d4} in the expression of i_{d3}.

$$\begin{aligned} i_{d3} &= g_{m3}\left(-\frac{i_X}{g_{m6}} - \left[g_{m2}\left[-\frac{i_X}{g_{m5}} - \left(-\frac{i_X g_{m1}}{g_{m4}}\right) r_{o1}\right]\right] r_{o2}\right) \\ &= g_{m3}\left(-\frac{i_X}{g_{m6}} - \left(g_{m2}\left[-\frac{i_X}{g_{m5}} + \frac{i_X g_{m1} r_{o1}}{g_{m4}}\right]\right) r_{o2}\right) \\ &= g_{m3}\left(-\frac{i_X}{g_{m6}} - \left(-\frac{i_X g_{m2}}{g_{m5}} + \frac{i_X g_{m1} g_{m2} r_{o1}}{g_{m4}}\right) r_{o2}\right) \\ &= g_{m3}\left(-\frac{i_X}{g_{m6}} + \frac{i_X g_{m2} r_{o2}}{g_{m5}} - \frac{i_X g_{m1} g_{m2} r_{o1} r_{o2}}{g_{m4}}\right) \end{aligned}$$

$$= -\frac{i_X g_{m3}}{g_{m6}} + \frac{i_X g_{m2} g_{m3} r_{o2}}{g_{m5}} - \frac{i_X g_{m1} g_{m2} g_{m3} r_{o1} r_{o2}}{g_{m4}}$$

Therefore,

$$i_{d3} = \left(-\frac{g_{m3}}{g_{m6}} + \frac{g_{m2} g_{m3} r_{o2}}{g_{m5}} - \frac{g_{m1} g_{m2} g_{m3} r_{o1} r_{o2}}{g_{m4}}\right) i_X \quad \text{..... (2)}$$

Similarly, calculate the drain current of the transistor, Q_2.

$$\begin{aligned} i_{d2} &= g_{m2} v_{gs2} \\ &= g_{m2}\left[v_{g2} - v_{s2}\right] \\ &= g_{m2}\left[-\frac{i_X}{g_{m5}} - i_{d1} r_{o1}\right] \qquad \left(\text{since } v_{s2} = v_{d1} = i_{d1} r_{o1} \text{ and } v_{g2} = v_{d5} = -\frac{i_X}{g_{m5}}\right) \\ &= g_{m2}\left[-\frac{i_X}{g_{m5}} - \left(g_{m1} v_{gs1}\right) r_{o1}\right] \end{aligned}$$

Simplify further.

$$\begin{aligned} i_{d2} &= g_{m2}\left[-\frac{i_X}{g_{m5}} - \left(g_{m1}\left[v_{g1} - v_{s1}\right]\right) r_{o1}\right] \qquad \left(\text{since } v_{s1} = 0 \text{ and } v_{g1} = v_{d4} = -\frac{i_X}{g_{m4}}\right) \\ &= g_{m2}\left[-\frac{i_X}{g_{m5}} - \left(-\frac{i_X g_{m1}}{g_{m4}}\right) r_{o1}\right] \\ &= g_{m2}\left[-\frac{i_X}{g_{m5}} + \frac{i_X g_{m1} r_{o1}}{g_{m4}}\right] \\ &= -\frac{i_X g_{m2}}{g_{m5}} + \frac{i_X g_{m1} g_{m2} r_{o1}}{g_{m4}} \end{aligned}$$

Therefore,

$$i_{d2} = \left(-\frac{g_{m2}}{g_{m5}} + \frac{g_{m1} g_{m2} r_{o1}}{g_{m4}}\right) i_X \quad \text{..... (3)}$$

Similarly, calculate the drain current of the transistor, Q_1.

$$\begin{aligned} i_{d1} &= g_{m1} v_{gs1} \\ &= g_{m1}\left[v_{g1} - v_{s1}\right] \\ &= g_{m1}\left[-\frac{i_X}{g_{m4}} - 0\right] \qquad \left(\text{since } v_{s1} = 0 \text{ and } v_{g1} = v_{d4} = -\frac{i_X}{g_{m4}}\right) \end{aligned}$$

$$i_{d1} = -\left(\frac{g_{m1}}{g_{m4}}\right) i_X \quad \text{..... (4)}$$

Express the voltage v_X as the sum of voltage drop across r_{o3} and voltage across transistor Q_2.

$$\begin{aligned} v_X &= \left(i_X + i_{d3}\right) r_{o3} + v_{d2} \\ &= \left(i_X + i_{d3}\right) r_{o3} + \left(i_X + i_{d2}\right) r_{o2} + v_{d1} \\ &= \left(i_X + i_{d3}\right) r_{o3} + \left(i_X + i_{d2}\right) r_{o2} + \left(i_X + i_{d1}\right) r_{o1} \end{aligned}$$

Substitute equations (2), (3), and (4) in the expression of v_X.

$$\begin{aligned} v_X &= \left\{\begin{array}{l} \left[i_X + \left(-\frac{g_{m3}}{g_{m6}} + \frac{g_{m2} g_{m3} r_{o2}}{g_{m5}} - \frac{g_{m1} g_{m2} g_{m3} r_{o1} r_{o2}}{g_{m4}}\right) i_X\right] r_{o3} + \\ \left[i_X + \left(-\frac{g_{m2}}{g_{m5}} + \frac{g_{m1} g_{m2} r_{o1}}{g_{m4}}\right) i_X\right] r_{o2} + \left[i_X + \left[-\left(\frac{g_{m1}}{g_{m4}}\right) i_X\right]\right] r_{o1} \end{array}\right\} \\ &= \left\{\begin{array}{l} \left(i_X - \frac{g_{m3} i_X}{g_{m6}} + \frac{g_{m2} g_{m3} r_{o2} i_X}{g_{m5}} - \frac{g_{m1} g_{m2} g_{m3} r_{o1} r_{o2} i_X}{g_{m4}}\right) r_{o3} + \\ \left(i_X - \frac{g_{m2} i_X}{g_{m5}} + \frac{g_{m1} g_{m2} r_{o1} i_X}{g_{m4}}\right) r_{o2} + \left(i_X - \left(\frac{g_{m1}}{g_{m4}}\right) i_X\right) r_{o1} \end{array}\right\} \\ &= \left\{\begin{array}{l} \left(r_{o3} i_X - \frac{g_{m3} r_{o3} i_X}{g_{m6}} + \frac{g_{m2} g_{m3} r_{o2} r_{o3} i_X}{g_{m5}} - \frac{g_{m1} g_{m2} g_{m3} r_{o1} r_{o2} r_{o3} i_X}{g_{m4}}\right) + \\ \left(r_{o2} i_X - \frac{g_{m2} r_{o2} i_X}{g_{m5}} + \frac{g_{m1} g_{m2} r_{o1} r_{o2} i_X}{g_{m4}}\right) + \left(r_{o1} i_X - \left(\frac{g_{m1} r_{o1}}{g_{m4}}\right) i_X\right) \end{array}\right\} \end{aligned}$$

Simplify further.

$$\begin{aligned} v_X &= i_X \left[\begin{array}{l} r_{o3} - \frac{g_{m3} r_{o3}}{g_{m6}} + \frac{g_{m2} g_{m3} r_{o2} r_{o3}}{g_{m5}} - \frac{g_{m1} g_{m2} g_{m3} r_{o1} r_{o2} r_{o3}}{g_{m4}} + \\ r_{o2} - \frac{g_{m2} r_{o2}}{g_{m5}} + \frac{g_{m1} g_{m2} r_{o1} r_{o2}}{g_{m4}} + r_{o1} - \frac{g_{m1} r_{o1}}{g_{m4}} \end{array}\right] \\ \frac{v_X}{i_X} &= \left[\begin{array}{l} r_{o3} - \frac{g_{m3} r_{o3}}{g_{m6}} + \frac{g_{m2} g_{m3} r_{o2} r_{o3}}{g_{m5}} - \frac{g_{m1} g_{m2} g_{m3} r_{o1} r_{o2} r_{o3}}{g_{m4}} + \\ r_{o2} - \frac{g_{m2} r_{o2}}{g_{m5}} + \frac{g_{m1} g_{m2} r_{o1} r_{o2}}{g_{m4}} + r_{o1} - \frac{g_{m1} r_{o1}}{g_{m4}} \end{array}\right] \qquad \left(\text{since } R_o = \frac{v_X}{i_X}\right) \\ R_o &= \left[\begin{array}{l} r_{o3} - \frac{g_{m3} r_{o3}}{g_{m6}} + \frac{g_{m2} g_{m3} r_{o2} r_{o3}}{g_{m5}} - \frac{g_{m1} g_{m2} g_{m3} r_{o1} r_{o2} r_{o3}}{g_{m4}} + \\ r_{o2} - \frac{g_{m2} r_{o2}}{g_{m5}} + \frac{g_{m1} g_{m2} r_{o1} r_{o2}}{g_{m4}} + r_{o1} - \frac{g_{m1} r_{o1}}{g_{m4}} \end{array}\right] \end{aligned}$$

Therefore, the expression of the output resistance is,

$$\boxed{\begin{array}{l} r_{o3} - \frac{g_{m3} r_{o3}}{g_{m6}} + \frac{g_{m2} g_{m3} r_{o2} r_{o3}}{g_{m5}} - \frac{g_{m1} g_{m2} g_{m3} r_{o1} r_{o2} r_{o3}}{g_{m4}} + \\ r_{o2} - \frac{g_{m2} r_{o2}}{g_{m5}} + \frac{g_{m1} g_{m2} r_{o1} r_{o2}}{g_{m4}} + r_{o1} - \frac{g_{m1} r_{o1}}{g_{m4}} \end{array}}.$$

Step 1 of 3

Refer to Figure 7.34 (a) in the textbook for the Wilson's current mirror circuit diagram which is used to increase the output resistance of the circuit.

Refer to Figure 7.34 (b) in the textbook for the Wilson's current mirror circuit to determine the output resistance.

First, set reference current to zero to determine the output resistance.

Write the output resistance expression of the Wilson mirror.

$$R_o = \beta\left(\frac{r_o}{2}\right)$$

Write the expression for the output resistance, r_o

$$r_o = \frac{V_A}{I_C}$$

Substitute the expression for r_o in the expression for the output resistance, R_o to get the new expression for R_o.

$$\begin{aligned} R_o &= \beta\left(\frac{r_o}{2}\right) \\ &= \frac{\beta V_A}{2 I_C} \end{aligned}$$

Step 2 of 3

Substitute $100\ \text{V}$ for V_A and 1 mA for I_C, and 100 for β to find the output resistance.

$$\begin{aligned} R_o &= \frac{\beta V_A}{2 I_C} \\ &= \left(\frac{100}{2}\right)\left(\frac{100}{1\times10^{-3}}\right) \\ &= 5\ \text{M}\Omega \end{aligned}$$

Find the change in the current, I_o corresponding to a change of $+10\ \text{V}$ in the voltage at the collector of Q_3.

Write the expression for the change in current.

$$\Delta I_O = \frac{\Delta V_O}{R_O}$$

Substitute 10 V for ΔV_O and $5\ \text{M}\Omega$ for R_O to yield value of change in current.

$$\begin{aligned} \Delta I_O &= \frac{\Delta V_O}{R_O} \\ &= \frac{10}{5\times10^{6}} \\ &= 2\ \mu\text{A} \end{aligned}$$

Therefore, the change in the current, ΔI_O corresponding to a change of $+10\ \text{V}$ in the voltage at the collector of Q_3 is, $\boxed{2\ \mu\text{A}}$.

Step 3 of 3

Write the expression for the percentage change of the current.

$$\%\ \text{change of current} = \frac{\Delta I_O}{I_O}$$

Substitute $2\ \mu\text{A}$ for ΔI_O and 1 mA for I_O.

$$\begin{aligned} \%\ \text{change of current} &= \frac{2\times10^{-6}}{1\times10^{-3}} \\ &= 0.2\% \end{aligned}$$

Therefore, the percentage change in the current is, $\boxed{0.2\%}$.

Step 1 of 8

8.085P

Refer to Figure P7.71 in the text book.

Redraw the circuit diagram as shown in Figure 1.

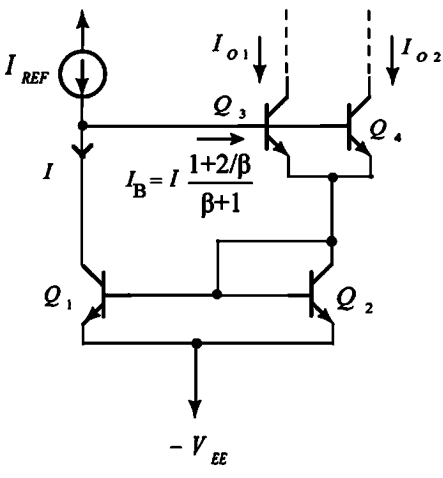


Figure 1

Step 2 of 8

From Figure 1, the reference current I_{ref} is,

$$\begin{aligned} I_{ref} &= I + \frac{\left(1+\frac{2}{\beta}\right)I}{(\beta+1)} \\ &= I + \frac{(\beta+2)I}{\beta(\beta+1)} \\ &= \left[\frac{\beta(\beta+1)+(\beta+2)}{\beta(\beta+1)}\right]I \\ &= \left[\frac{\beta^2+2\beta+2}{\beta(\beta+1)}\right]I \end{aligned}$$

Step 3 of 8

Calculate the current I_{01}.

$$\begin{aligned} I_{01} &= \frac{1}{2}I_B \\ &= \frac{1}{2}\frac{(\beta+2)}{\beta+1}I \end{aligned}$$

Calculate the current I_{02}.

$$\begin{aligned} I_{02} &= \frac{1}{2}I_B \\ &= \frac{1}{2}\frac{(\beta+2)}{\beta+1}I \end{aligned}$$

Step 4 of 8

Calculate the value of $\frac{I_{01}}{I_{ref}}$.

$$\begin{aligned} \frac{I_{01}}{I_{ref}} &= \frac{\frac{1}{2}\frac{(\beta+2)}{\beta+1}I}{\left[\frac{\beta^2+2\beta+2}{\beta(\beta+1)}\right]I} \\ &= \frac{1}{2}\frac{(\beta+2)\beta}{\beta^2+2\beta+2} \\ &= \frac{1}{2}\frac{\beta^2+2\beta}{\beta^2+2\beta\left(1+\frac{2}{\beta^2+2\beta}\right)} \\ &= \frac{1}{2}\frac{1}{\left(1+\frac{2}{\beta^2+2\beta}\right)} \qquad \left(\beta^2 >> 2\beta\right) \end{aligned}$$

$$I_{01} = \frac{1}{2}\frac{1}{\left(1+\frac{2}{\beta^2}\right)}I_{ref}$$

Step 5 of 8

Calculate the value of $\frac{I_{02}}{I_{ref}}$.

$$\begin{aligned} \frac{I_{02}}{I_{ref}} &= \frac{\frac{1}{2}\frac{(\beta+2)}{\beta+1}I}{\left[\frac{\beta^2+2\beta+2}{\beta(\beta+1)}\right]I} \\ &= \frac{1}{2}\frac{(\beta+2)\beta}{\beta^2+2\beta+2} \\ &= \frac{1}{2}\frac{\beta^2+2\beta}{\beta^2+2\beta\left(1+\frac{2}{\beta^2+2\beta}\right)} \\ &= \frac{1}{2}\frac{1}{\left(1+\frac{2}{\beta^2+2\beta}\right)} \qquad \left(\beta^2 >> 2\beta\right) \end{aligned}$$

$$I_{02} = \frac{1}{2}\frac{1}{\left(1+\frac{2}{\beta^2}\right)}I_{ref}$$

Step 6 of 8

(b)

The 0.7 mA reference current can be used to generate three output currents of

0.1 mA, 0.2 mA and 0.44 mA, by using three transistors in parallel having relative area ratios of 1, 2, 4.

When $I_{ref} = 0.7\ \text{mA}$, Calculate the current I_{01}.

$$\begin{aligned} I_{01} &= I_{ref} \times \frac{1}{2}\left(\frac{1}{1+\frac{2}{\beta^2}}\right) \\ &= 0.7\times10^{-3}\times\frac{1}{2}\left(\frac{1}{1+\frac{2}{50^2}}\right) \\ &= 0.7\times10^{-3}\times\frac{1}{2}\left(\frac{1}{1+\frac{2}{50^2}}\right) \\ &= 0.35\,\text{mA} \end{aligned}$$

Therefore, the actual value of I_{01} is $0.35\,\text{mA}$.

So, the current I_{02} is $0.35\,\text{mA}$.

Step 7 of 8

When $I_{ref} = 0.7\ \text{mA}$, Calculate the current I_{01}.

$$\begin{aligned} I_{01} &= I_{ref} \times \frac{2}{2}\left(\frac{1}{1+\frac{2}{\beta^2}}\right) \\ &= 0.7\times10^{-3}\times\frac{2}{2}\left(\frac{1}{1+\frac{2}{50^2}}\right) \end{aligned}$$

$$I_{01} = 0.7\,\text{mA}$$

Therefore, the actual value of I_{01}, I_{02} is $\boxed{0.7\text{mA}}$.

Step 8 of 8

Find the current, I_{01}.

$$\begin{aligned} I_{01} &= I_{ref} \times \frac{4}{2}\left(\frac{1}{1+\frac{2}{\beta^2}}\right) \\ &= 0.7\times10^{-3}\times\frac{4}{2}\left(\frac{1}{1+\frac{2}{50^2}}\right) \\ &= 1.4\ \text{mA} \end{aligned}$$

Therefore, the actual value of I_{01} is $\boxed{1.4\ \text{mA}}$.

Step 1 of 3

8.086P

Consider the circuit diagram shown in Figure 1.

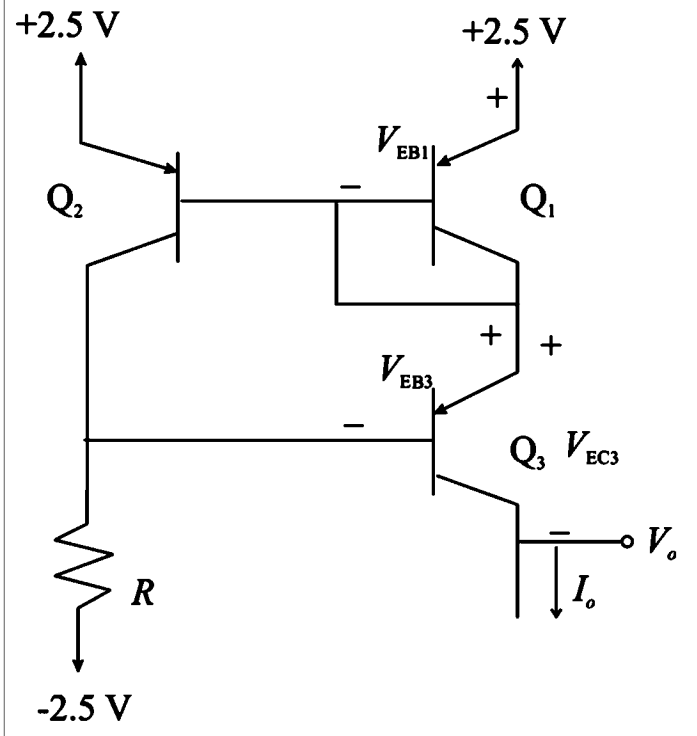


Figure 1

Step 2 of 3

Consider the expression for the current I_{REF} .

$$I_{REF} = \frac{2.5 + V_{EB1} + V_{EB3} - (-2.5)}{R}$$

Substitute -0.7 V for V_{EB1} and V_{EB3} and 0.1 mA for I_{REF} .

$$0.1\text{ m} = \frac{2.5 - 0.7 - 0.7 + 2.5}{R}$$

$$0.1\text{ m} = \frac{3.6}{R}$$

$$R = \frac{3.6}{0.1\text{ m}}$$

$$R = 36\text{ k}\Omega$$

Therefore, the value of the resistance R is, $\boxed{36\text{ k}\Omega}$.

Step 3 of 3

Maximum value of the output voltage is obtained when the value transistor Q_3 operates in saturation.

$$V_{o,max} = 2.5 + V_{EB1} + V_{EC3}$$

Substitute -0.7 V for V_{EB1} and assume -0.2 V for V_{EC3} .

$$V_{o,max} = 2.5 - 0.7 - 0.2$$
$$= 1.6\text{ V}$$

Therefore, the minimum value of the output voltage $V_{o,max}$ is, $\boxed{1.6\text{ V}}$.

Step 1 of 6

Consider the circuit diagram shown in Figure 1.

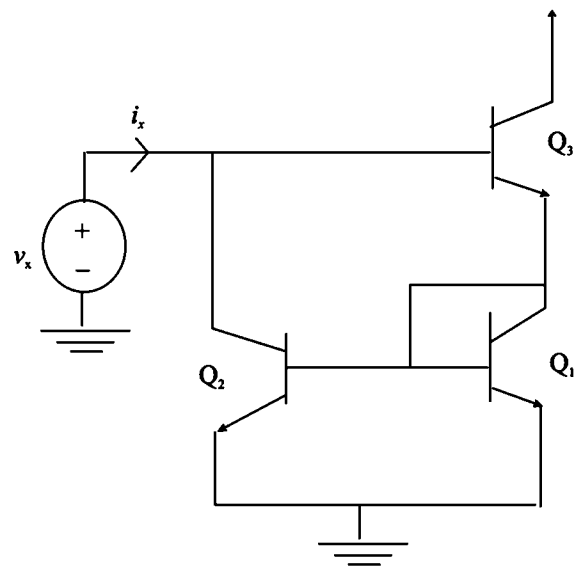


Figure 1

Step 2 of 6

Draw the small signal equivalent circuit diagram neglect the resistance r_o for all transistors.

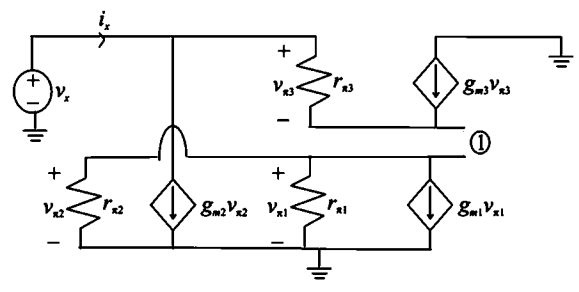


Figure 2

Step 3 of 6

Consider the expression of the voltage v_x.

$v_x = v_{\pi 3} + v_{\pi 2}$

Therefore, the expression for the voltage v_x is $v_{\pi 3} + v_{\pi 2}$.

Step 4 of 6

Apply nodal analysis at node-1.

$$-g_{m3}v_{\pi 3} + \frac{v_{\pi 3}}{r_{\pi 3}} + g_{m1}v_{\pi 1} + \frac{v_{\pi 1}}{r_{\pi 1}} + \frac{v_{\pi 1}}{r_{\pi 2}} = 0$$

$$v_{\pi 3}\left(\frac{1}{r_{\pi 3}} - g_{m3}\right) + v_{\pi 1}\left(g_{m1} + \frac{1}{r_{\pi 1}} + \frac{1}{r_{\pi 2}}\right) = 0$$

Substitute $v_x - v_{\pi 3}$ for $v_{\pi 1}$ since $v_{\pi 1} = v_{\pi 3}$.

$$v_{\pi 3}\left(\frac{1}{r_{\pi 3}} - g_{m3}\right) + (v_x - v_{\pi 3})\left(g_{m1} + \frac{1}{r_{\pi 1}} + \frac{1}{r_{\pi 2}}\right) = 0$$

$$v_{\pi 3}\left(\frac{1}{r_{\pi 3}} - g_{m3}\right) + v_x\left(g_{m1} + \frac{1}{r_{\pi 1}} + \frac{1}{r_{\pi 2}}\right) - v_{\pi 3}\left(g_{m1} + \frac{1}{r_{\pi 1}} + \frac{1}{r_{\pi 2}}\right) = 0$$

$$v_{\pi 3}\left(\frac{1}{r_{\pi 3}} - g_{m3} - g_{m1} - \frac{1}{r_{\pi 1}} - \frac{1}{r_{\pi 2}}\right) + v_x\left(g_{m1} + \frac{1}{r_{\pi 1}} + \frac{1}{r_{\pi 2}}\right) = 0$$

Substitute r_π for $r_{\pi 1}$, $r_{\pi 2}$ and $r_{\pi 3}$ and g_m for g_{m3} and g_{m1}.

$$v_{\pi 3}\left(\frac{1}{r_\pi} - g_m - g_m - \frac{1}{r_\pi} - \frac{1}{r_\pi}\right) + v_x\left(g_m + \frac{1}{r_\pi} + \frac{1}{r_\pi}\right) = 0$$

$$v_{\pi 3}\left(-\frac{1}{r_\pi} - 2g_m\right) + v_x\left(g_m + \frac{2}{r_\pi}\right) = 0$$

$$-v_{\pi 3}\left(\frac{1}{r_\pi} + 2g_m\right) + v_x\left(g_m + \frac{2}{r_\pi}\right) = 0$$

$$v_{\pi 3}\left(\frac{1}{r_\pi} + 2g_m\right) = v_x\left(g_m + \frac{2}{r_\pi}\right)$$

$$v_{\pi 3} = v_x \frac{\left(g_m + \frac{2}{r_\pi}\right)}{\left(\frac{1}{r_\pi} + 2g_m\right)}$$

Therefore, the expression for the voltage $v_{\pi 3}$ is $v_x \dfrac{\left(g_m + \frac{2}{r_\pi}\right)}{\left(\frac{1}{r_\pi} + 2g_m\right)}$.

Step 5 of 6

Consider the expression for the current i_x.

$$i_x = \frac{v_{\pi 3}}{r_{\pi 3}} + g_{m2}v_{\pi 2}$$

Substitute $v_x - v_{\pi 3}$ for $v_{\pi 2}$.

$$i_x = \frac{v_{\pi 3}}{r_{\pi 3}} + g_{m2}(v_x - v_{\pi 3})$$

$$= \frac{v_{\pi 3}}{r_{\pi 3}} + g_{m2}v_x - g_{m2}v_{\pi 3}$$

$$= v_{\pi 3}\left(\frac{1}{r_{\pi 3}} - g_{m2}\right) + g_{m2}v_x$$

Substitute $v_x \dfrac{\left(g_m + \frac{2}{r_\pi}\right)}{\left(\frac{1}{r_\pi} + 2g_m\right)}$ for $v_{\pi 3}$.

$$i_x = v_x \frac{\left(g_m + \frac{2}{r_\pi}\right)}{\left(\frac{1}{r_\pi} + 2g_m\right)}\left(\frac{1}{r_{\pi 3}} - g_{m2}\right) + g_{m2}v_x$$

$$= v_x \frac{\left(\frac{2 + g_m r_\pi}{r_\pi}\right)}{\left(\frac{1 + 2g_m r_\pi}{r_\pi}\right)}\left(\frac{1}{r_{\pi 3}} - g_{m2}\right) + g_{m2}v_x$$

$$= v_x \frac{(2 + g_m r_\pi)}{(1 + 2g_m r_\pi)}\left(\frac{1 - g_{m2}r_{\pi 3}}{r_{\pi 3}}\right) + g_{m2}v_x$$

$$i_x = v_x \frac{\left(2 + g_m r_\pi - 2g_{m2}r_{\pi 3} - g_m g_{m2} r_\pi r_{\pi 3}{}^2\right)}{(1 + 2g_m r_\pi) r_{\pi 3}} + g_{m2}v_x$$

Substitute r_π for $r_{\pi 3}$ and g_m for g_{m2}.

$$i_x = v_x \frac{\left(2 + g_m r_\pi - 2g_m r_\pi - g_m{}^2 r_\pi{}^2\right)}{(1 + 2g_m r_\pi) r_\pi} + g_m v_x$$

$$= v_x \frac{\left(2 - g_m r_\pi - g_m{}^2 r_\pi{}^2\right)}{(1 + 2g_m r_\pi) r_\pi} + g_m v_x$$

$$= v_x \left(\frac{\left(2 - g_m r_\pi - g_m{}^2 r_\pi{}^2\right)}{(1 + 2g_m r_\pi) r_\pi} + g_m\right)$$

$$= v_x \left(\frac{2 - g_m r_\pi - g_m{}^2 r_\pi{}^2 + g_m(1 + 2g_m r_\pi) r_\pi}{(1 + 2g_m r_\pi) r_\pi}\right)$$

$$i_x = v_x \left(\frac{2 - g_m r_\pi - g_m{}^2 r_\pi{}^2 + g_m r_\pi + 2g_m{}^2 r_\pi{}^2}{(1 + 2g_m r_\pi) r_\pi}\right)$$

$$= v_x \left(\frac{2 - g_m{}^2 r_\pi{}^2 + 2g_m{}^2 r_\pi{}^2}{(1 + 2g_m r_\pi) r_\pi}\right)$$

$$= v_x \left(\frac{2 + g_m{}^2 r_\pi{}^2}{(1 + 2g_m r_\pi) r_\pi}\right)$$

$$\frac{v_x}{i_x} = \frac{(1 + 2g_m r_\pi) r_\pi}{2 + g_m{}^2 r_\pi{}^2}$$

Here, $1 + 2g_m r_\pi \approx 2g_m r_\pi$ since $2g_m r_\pi >> 1$ and $2 + g_m{}^2 r_\pi{}^2 \approx g_m{}^2 r_\pi{}^2$ since $g_m{}^2 r_\pi{}^2 >> 2$.

$$\frac{v_x}{i_x} = \frac{(2g_m r_\pi) r_\pi}{g_m{}^2 r_\pi{}^2}$$

$$= \frac{2}{g_m}$$

Here, $\dfrac{v_x}{i_x}$ represents the incremental input resistance R_{in}.

$$R_{in} = \frac{2}{g_m}$$

Substitute $\dfrac{I_{REF}}{V_T}$ for g_m.

$$R_{in} = \frac{2}{\left(\frac{I_{REF}}{V_T}\right)}$$

$$R_{in} = \frac{2V_T}{I_{REF}} \quad \text{..... (1)}$$

Therefore, the expression for the incremental input resistance R_{in} is $\boxed{\dfrac{2V_T}{I_{REF}}}$.

Step 6 of 6

Recall equation (1).

$$R_{in} = \frac{2V_T}{I_{REF}}$$

Substitute $25\ \text{mV}$ for V_T and $0.2\ \text{mA}$ for I_{REF}.

$$R_{in} = \frac{2(25\ \text{m})}{0.2\ \text{m}}$$

$$= \frac{50\ \text{m}}{0.2\ \text{m}}$$

$$= 250\ \Omega$$

Therefore, the value of the input resistance R_{in} is $\boxed{250\ \Omega}$.

Step 1 of 16

Refer to Figure 8.41 from the text book.

(a)

Consider the expression for the current I_{REF}.

$$I_{REF} = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)V_{OV}^2$$

$$V_{OV}^2 = \frac{2I_{REF}}{\mu_n C_{ox}\left(\frac{W}{L}\right)}$$

$$V_{OV} = \sqrt{\frac{2I_{REF}}{\mu_n C_{ox}\left(\frac{W}{L}\right)}}$$

Substitute 10 for $\frac{W}{L}$, $180\ \mu A$ for I_{REF} and $400\ \mu A/V^2$ for $\mu_n C_{ox}$.

$$V_{OV} = \sqrt{\frac{2(180\ \mu)}{(400\ \mu)(10)}}$$
$$= \sqrt{\frac{36}{400}}$$
$$= \frac{6}{20}$$
$$= 0.3\ V$$

Therefore, the value of the overdrive voltage V_{OV} is $\boxed{0.3\ V}$

Step 2 of 16

Consider the expression for the voltage V_{GS}.

$$V_{GS} = V_{OV} + V_{tn}$$

Substitute $0.3\ V$ for V_{OV} and $0.5\ V$ for V_{tn}.

$$V_{GS} = 0.3 + 0.5$$
$$= 0.8\ V$$

Therefore, the value of the voltage V_{GS} is $\boxed{0.8\ V}$.

Step 3 of 16

(b)

Calculate the value of the voltage V_{DS3}.

$$V_{DS2} = V_{GS1} + V_{GS3}$$

Substitute $0.8\ V$ for V_{GS1} and V_{GS3}.

$$V_{DS2} = 0.8 + 0.8$$
$$= 1.6\ V$$

Therefore, the value of the voltage V_{DS3} is $1.6\ V$.

Step 4 of 16

Consider the value of the voltage V_{DS1}.

$$V_{DS1} = V_{GS1}$$

Substitute $0.8\ V$ for V_{GS1}.

$$V_{DS1} = 0.8\ V$$

Therefore, the value of the voltage V_{DS1} is $0.8\ V$.

Step 5 of 16

Calculate the value of the difference in the values of the drain to source voltages of the transistors Q_2 and Q_1.

$$\Delta V_{DS} = V_{DS2} - V_{DS1}$$

Substitute $0.8\ V$ for V_{DS1} and $1.6\ V$ for V_{DS3}.

$$\Delta V_{DS} = 1.6 - 0.8$$
$$= 0.8\ V$$

Therefore, the drain to source voltage difference ΔV_{DS} is $0.8\ V$.

Step 6 of 16

Calculate the value of the resistance r_o.

$$r_o = \frac{V_A}{I}$$

Substitute $18\ V$ for V_A and $180\ \mu A$ for I.

$$r_o = \frac{18}{180\ \mu}$$
$$= 100\ k\Omega$$

Therefore, the value of the resistance r_o is $100\ k\Omega$.

Step 7 of 16

Calculate the difference in the value of the current.

$$\Delta I = \frac{\Delta V}{r_o}$$

Substitute $100\ k\Omega$ for r_o and $0.8\ V$ for ΔV_{DS}.

$$\Delta I = \frac{0.8}{100\ k}$$
$$= 8\ \mu A$$

Therefore, the difference in the value of the current ΔI is $\boxed{8\ \mu A}$.

Step 8 of 16

Calculate the value of the current I_o.

$$I_o = I_{REF} - \Delta I$$

Substitute $180\ \mu A$ for I_o and $8\ \mu A$ for ΔI.

$$I_o = 180\ \mu - 8\ \mu$$
$$= 172\ \mu A$$

Therefore, the value of the current I_o is $\boxed{172\ \mu A}$.

Step 9 of 16

(c)

Refer to Figure P8.41 (c) from the text book.

Transistor Q_4 is added to the circuit to remove the difference in the values of the currents I_o and I_{REF}. Hence, the value of I_o will be approximately equal to I_{REF}. That is,

$\boxed{I_o \approx 180\ \mu A}$.

Step 10 of 16

(d)

Minimum value of the output voltage is obtained when the value of the drain to source voltage is equal to the value of the over drive voltage. That is, $V_{DS} = V_{OV}$.

Calculate the minimum value of the output voltage $V_{o,min}$.

$$V_{o,min} = V_{DS3} + V_{GS1}$$

Substitute $0.3\ V$ for V_{DS3} and $0.8\ V$ for V_{GS1}.

$$V_{o,min} = 0.3 + 0.8$$
$$= 1.1\ V$$

Therefore, the minimum value of the output voltage $V_{o,min}$ is $\boxed{1.1\ V}$.

Step 11 of 16

(e)

Draw the small signal equivalent circuit diagram to calculate the output resistance.

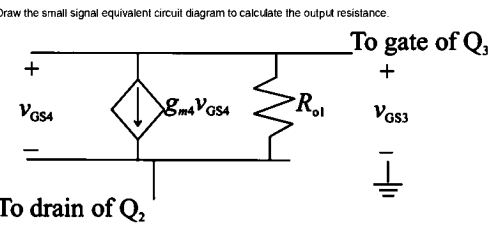


Figure 1

Step 12 of 16

Calculate the value of the voltage V_{GS4}.

$$V_{GS4} = -g_{m4}V_{GS4}r_{o4}$$
$$V_{GS4} + g_{m4}V_{GS4}r_{o4} = 0$$
$$V_{GS4}(1 + g_{m4}r_{o4}) = 0$$
$$V_{GS4} = 0\ V$$

The resistance r_{o4} will be shorted since the value of the voltage V_{GS4} is $0\ V$ implies that the transistor will not effect the output resistance.

Step 13 of 16

Calculate the value of the transconductance g_m.

$$g_m = \frac{2I_o}{V_{OV}}$$

Substitute $0.3\ V$ for V_{OV} and $180\ \mu A$ for I_o.

$$g_m = \frac{2(180\ \mu)}{0.3}$$
$$= 12\ \mu A/V$$

Therefore, the value of the transconductance g_m is $12\ \mu A/V$.

Step 14 of 16

Consider the expression for the output resistance R_o.

$$R_o = (g_{m3}r_{o3})r_{o2}$$

Substitute $12\ \mu A/V$ for g_{m3} and $100\ k\Omega$ for r_{o2} and r_{o3}.

$$R_o = (12\ \mu)(100\ k)(100\ k)$$
$$= 120\ k\Omega$$

Therefore, the value of the output resistance R_o is $\boxed{120\ k\Omega}$.

Step 15 of 16

(f)

Calculate the change in the value of the output current ΔI_o.

$$\Delta I_o = \frac{\Delta V_o}{R_o}$$

Substitute $1\ V$ for ΔV_o and $120\ k\Omega$ for R_o.

$$\Delta I_o = \frac{1}{120\ k}$$
$$= 4.55\ \mu A$$

Therefore, the change in the value of the output current is $\boxed{4.55\ \mu A}$.

Step 16 of 16

Calculate the percentage change in the value of the output current $\Delta I_o\%$.

$$\Delta I_o\% = \frac{\Delta I_o}{I_o} \times 100$$

Substitute $4.55\ \mu A$ for ΔI_o and $180\ \mu A$ for I_o.

$$\Delta I_o\% = \frac{4.55\ \mu}{180\ \mu} \times 100$$
$$= 25.28 \times 100$$
$$= 2.528\%$$

Therefore, the value of the change in the output current $\Delta I_o\%$ is $\boxed{2.528\%}$.

Step 1 of 3

8.089P

Refer to Figure 8.41 (a) from the text book.

Draw the small signal equivalent circuit diagram to calculate the input resistance. Neglect the resistances r_o for all the transistors.

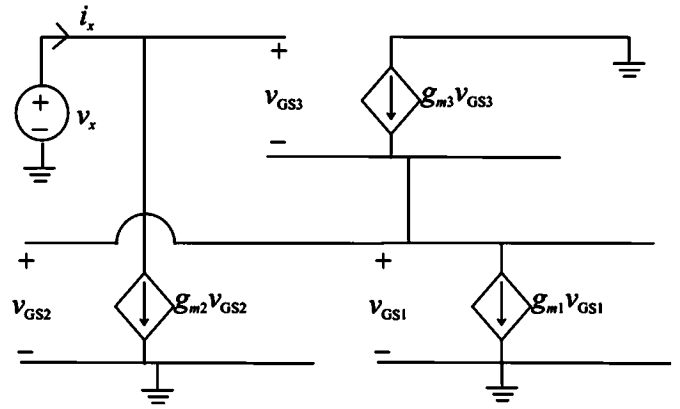


Figure 1

Step 2 of 3

Consider the expression for the voltage v_x.

$$v_x = v_{GS3} + v_{GS2}$$

Here, $v_{GS3} = v_{GS2}$ since the all the transistors have same current and parameters.

$$v_x = v_{GS2} + v_{GS2}$$
$$v_x = 2v_{GS2}$$
$$v_{GS2} = \frac{v_x}{2}$$

Therefore, the value of the voltage v_{GS2} is $\frac{v_x}{2}$.

Step 3 of 3

Consider the expression for the current i_x.

$$i_x = g_{m2}v_{GS2}$$

Substitute $\frac{v_x}{2}$ for v_{GS2}.

$$i_x = g_{m2}\left(\frac{v_x}{2}\right)$$
$$\frac{v_x}{i_x} = \frac{2}{g_{m2}}$$
$$= \frac{2}{g_m}$$

Here, $\frac{v_x}{i_x}$ represents the value of the input resistance R_{in}.

$$R_{in} = \frac{2}{g_m}$$

Therefore, the value of the incremental input resistance R_{in} is $\boxed{\frac{2}{g_m}}$.

Step 1 of 7

Refer to Figure 8.42 from the text book.

(a)

Calculate the value of the voltage V_{BE1}.

$$V_{BE1} = 0.7 + V_T \ln\left(\frac{I_{REF}}{I_S}\right)$$

Substitute $25\ \text{mA}$ for V_T, $200\ \mu\text{A}$ for I_{REF} and $1\ \text{mA}$ for I_S.

$$\begin{aligned} V_{BE1} &= 0.7 + (25\ \text{m}) \ln\left(\frac{200\ \mu}{1\ \text{m}}\right) \\ &= 0.7 + (25\ \text{m}) \ln(200\ \text{m}) \\ &= 0.7 + (25\ \text{m})(-1.61) \\ &= 0.7 - 40.25\ \text{m} \end{aligned}$$

$$V_{BE1} = 0.66\ \text{V}$$

Therefore, the value of the voltage is V_{BE1} is $0.66\ \text{V}$.

Step 2 of 7

Calculate the value of the voltage V_{BE2}.

$$V_{BE2} = 0.7 + V_T \ln\left(\frac{I_o}{I_S}\right)$$

Substitute $25\ \text{mA}$ for V_T, $20\ \mu\text{A}$ for I_o and $1\ \text{mA}$ for I_S.

$$\begin{aligned} V_{BE2} &= 0.7 + (25\ \text{m}) \ln\left(\frac{20\ \mu}{1\ \text{m}}\right) \\ &= 0.7 + (25\ \text{m}) \ln(20\ \text{m}) \\ &= 0.7 + (25\ \text{m})(-3.91) \\ &= 0.7 - 97.75\ \text{m} \end{aligned}$$

$$V_{BE2} = 0.60\ \text{V}$$

Therefore, the value of the voltage is V_{BE2} is $0.60\ \text{V}$.

Step 3 of 7

Consider the expression for the current I_o.

$$I_o = \frac{V_{BE1} - V_{BE2}}{R_E}$$

Substitute $20\ \mu\text{A}$ for I_o, $0.60\ \text{V}$ for V_{BE2} and $0.66\ \text{V}$ for V_{BE1}.

$$\begin{aligned} 20\ \mu &= \frac{0.66 - 0.6}{R_E} \\ R_E &= \frac{0.06}{20\ \mu} \\ R_E &= 3\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance R_E is $\boxed{3\ \text{k}\Omega}$.

Step 4 of 7

(b)

Calculate the value of the transconductance g_m.

$$g_m = \frac{I_o}{V_T}$$

Substitute $20\ \mu\text{A}$ for I_o and $25\ \text{mA}$ for V_T.

$$\begin{aligned} g_m &= \frac{20\ \mu}{25\ \text{m}} \\ &= 800\ \mu\text{A/V} \end{aligned}$$

Therefore, the value of the transconductance g_m is $800\ \mu\text{A/V}$.

Step 5 of 7

Calculate the value of the r_π.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute $800\ \mu\text{A/V}$ for g_m and 200 for β.

$$\begin{aligned} r_\pi &= \frac{200}{800\ \mu} \\ &= 250\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance r_π is $250\ \text{k}\Omega$.

Step 6 of 7

Calculate the value of the resistance r_o.

$$r_o = \frac{V_A}{I_o}$$

Substitute $20\ \mu\text{A}$ for I_o and $50\ \text{V}$ for V_A.

$$\begin{aligned} r_o &= \frac{50}{20\ \mu} \\ &= 2.5\ \text{M}\Omega \end{aligned}$$

Therefore, the value of the resistance r_o is $2.5\ \text{M}\Omega$.

Consider the expression for the output resistance.

$$R_o = \left[1 + g_m (R_E \| r_\pi)\right] r_o$$

Substitute $2.5\ \text{M}\Omega$ for r_o, $250\ \text{k}\Omega$ for r_π, $800\ \mu\text{A/V}$ for g_m and $3\ \text{k}\Omega$ for R_E.

$$\begin{aligned} R_o &= \left[1 + (800\ \mu)(3\ \text{k} \| 250\ \text{k})\right](2.5\ \text{M}) \\ &= \left[1 + (800\ \mu)\left(\frac{3\ \text{k} \times 250\ \text{k}}{3\ \text{k} + 250\ \text{k}}\right)\right](2.5\ \text{M}) \\ &= \left[1 + (800\ \mu)\left(\frac{3\ \text{k} \times 250\ \text{k}}{253\ \text{k}}\right)\right](2.5\ \text{M}) \\ &= \left[1 + (800\ \mu)(2.96\ \text{k})\right](2.5\ \text{M}) \end{aligned}$$

$$\begin{aligned} R_o &= [1 + 2.37](2.5\ \text{M}) \\ &= (3.37)(2.5\ \text{M}) \\ &= 8.425\ \text{M}\Omega \end{aligned}$$

Therefore, the value of the output resistance R_o is $\boxed{8.425\ \text{M}\Omega}$.

Step 7 of 7

Calculate the change in output current corresponding to a 5-V change in output voltage.

$$\begin{aligned} \Delta I_o &= \frac{\Delta V_o}{R_o} \\ &= \frac{5}{8.425 \times 10^6} \\ &= 0.59\ \mu\text{A} \end{aligned}$$

Therefore, the change in current is, $\boxed{0.59\ \mu\text{A}}$.

Step 1 of 15

Refer to Figure 8.42 from the text book.

Calculations for the ratio $\frac{I_O}{I_{REF}} = 0.8$.

Consider the current transfer ratio.

$$\frac{I_O}{I_{REF}} = 0.8$$

Substitute $100\ \mu A$ for I_{REF} .

$$\frac{I_O}{100\ \mu} = 0.8$$
$$I_O = 0.8(100\ \mu)$$
$$I_O = 80\ \mu A$$

Therefore, the value of the output current I_O is $80\ \mu A$.

Step 2 of 15

Consider the expression for the voltage across the emitter resistance.

$$I_O R_E = V_T \ln\left(\frac{I_{REF}}{I_O}\right)$$

Substitute 0.8 for $\frac{I_O}{I_{REF}}$, $80\ \mu A$ for I_O and $25\ mV$ for V_T .

$$(80\ \mu) R_E = (25\ m)\ln\left(\frac{1}{0.8}\right)$$
$$(80\ \mu) R_E = (25\ m)\ln(1.25)$$
$$(80\ \mu) R_E = (25\ m)(223.14)$$
$$R_E = \frac{(25\ m)(223.14)}{80\ \mu}$$
$$R_E = 69.73\ \Omega$$

Therefore, the value of the emitter resistance R_E is $69.73\ \Omega$.

Step 3 of 15

Calculate the value of the resistance r_o .

$$r_o = \frac{V_A}{I_O}$$

Substitute $50\ V$ for V_A and $80\ \mu A$ for I_O .

$$r_o = \frac{50}{80\ \mu}$$
$$= 625\ k\Omega$$

Therefore, the value of the resistance r_o is $625\ k\Omega$.

Step 4 of 15

Calculate the value of transconductance g_m .

$$g_m = \frac{I_O}{V_T}$$

Substitute $25\ mV$ for V_T and $80\ \mu A$ for I_O .

$$g_m = \frac{80\ \mu}{25\ m}$$
$$= 3.2\ mA/V$$

Therefore, the value of the transconductance g_m is $3.2\ mA/V$.

Step 5 of 15

Consider the expression for the output resistance.

$$R_o = \left[1 + g_m (R_E \| r_o)\right] r_o$$

As the value of β is very high, the resistance r_π can be neglected.

$$R_o = (1 + g_m R_E) r_o$$

Substitute $625\ k\Omega$ for r_o , $3.2\ mA/V$ for g_m and $69.73\ \Omega$ for R_E .

$$R_o = \left[1 + (3.2\ m)(69.73)\right](625\ k)$$
$$= \left[1 + 223.136\ m\right](625\ k)$$
$$= [1.22](625\ k)$$
$$= 764.46\ k\Omega$$

Therefore, the value of the output resistance R_o is $\boxed{764.46\ k\Omega}$.

Compare the value of R_o with r_o , the value of the output resistance increases with the introduction of the emitter resistance to the circuit.

Step 6 of 15

Calculations for the ratio $\frac{I_O}{I_{REF}} = 0.1$.

Consider the current transfer ratio.

$$\frac{I_O}{I_{REF}} = 0.1$$

Substitute $100\ \mu A$ for I_{REF} .

$$\frac{I_O}{100\ \mu} = 0.1$$
$$I_O = 0.1(100\ \mu)$$
$$I_O = 10\ \mu A$$

Therefore, the value of the output current I_O is $10\ \mu A$.

Step 7 of 15

Consider the expression for the voltage across the emitter resistance.

$$I_O R_E = V_T \ln\left(\frac{I_{REF}}{I_O}\right)$$

Substitute 0.1 for $\frac{I_O}{I_{REF}}$, $10\ \mu A$ for I_O and $25\ mV$ for V_T .

$$(10\ \mu) R_E = (25\ m)\ln\left(\frac{1}{0.1}\right)$$
$$(10\ \mu) R_E = (25\ m)\ln(1)$$
$$(10\ \mu) R_E = (25\ m)(2.3)$$
$$R_E = \frac{(25\ m)(2.3)}{10\ \mu}$$
$$R_E = 5.75\ k\Omega$$

Therefore, the value of the emitter resistance R_E is $5.75\ k\Omega$.

Step 8 of 15

Calculate the value of the resistance r_o .

$$r_o = \frac{V_A}{I_O}$$

Substitute $50\ V$ for V_A and $10\ \mu A$ for I_O .

$$r_o = \frac{50}{10\ \mu}$$
$$= 5\ M\Omega$$

Therefore, the value of the resistance r_o is $5\ M\Omega$.

Step 9 of 15

Calculate the value of transconductance g_m .

$$g_m = \frac{I_O}{V_T}$$

Substitute $25\ mV$ for V_T and $10\ \mu A$ for I_O .

$$g_m = \frac{10\ \mu}{25\ m}$$
$$= 0.4\ mA/V$$

Therefore, the value of the transconductance g_m is $0.4\ mA/V$.

Step 10 of 15

Consider the expression for the output resistance.

$$R_o = \left[1 + g_m (R_E \| r_o)\right] r_o$$

As the value of β is very high, the resistance r_π can be neglected.

$$R_o = (1 + g_m R_E) r_o$$

Substitute $5\ M\Omega$ for r_o , $0.4\ mA/V$ for g_m and $5.75\ k\Omega$ for R_E .

$$R_o = \left[1 + (0.4\ m)(5.75\ k)\right](5\ M)$$
$$= [1 + 2.3](5\ M)$$
$$= [3.3](5\ M)$$
$$= 16.5\ M\Omega$$

Therefore, the value of the output resistance R_o is $\boxed{16.5\ M\Omega}$.

Compare the value of R_o with r_o , the value of the output resistance increases with the introduction of the emitter resistance to the circuit.

Step 11 of 15

Calculations for the ratio $\frac{I_O}{I_{REF}} = 0.01$.

Consider the current transfer ratio.

$$\frac{I_O}{I_{REF}} = 0.01$$

Substitute $100\ \mu A$ for I_{REF} .

$$\frac{I_O}{100\ \mu} = 0.01$$
$$I_O = 0.01(100\ \mu)$$
$$I_O = 1\ \mu A$$

Therefore, the value of the output current I_O is $1\ \mu A$.

Step 12 of 15

Consider the expression for the voltage across the emitter resistance.

$$I_O R_E = V_T \ln\left(\frac{I_{REF}}{I_O}\right)$$

Substitute 0.01 for $\frac{I_O}{I_{REF}}$, $1\ \mu A$ for I_O and $25\ mV$ for V_T .

$$(1\ \mu) R_E = (25\ m)\ln\left(\frac{1}{0.01}\right)$$
$$(1\ \mu) R_E = (25\ m)\ln(100)$$
$$(1\ \mu) R_E = (25\ m)(4.61)$$
$$R_E = \frac{(25\ m)(4.61)}{1\ \mu}$$
$$R_E = 115.25\ k\Omega$$

Therefore, the value of the emitter resistance R_E is $115.25\ k\Omega$.

Step 13 of 15

Calculate the value of the resistance r_o .

$$r_o = \frac{V_A}{I_O}$$

Substitute $50\ V$ for V_A and $1\ \mu A$ for I_O .

$$r_o = \frac{50}{1\mu}$$
$$= 50\ M\Omega$$

Therefore, the value of the resistance r_o is $50\ M\Omega$.

Step 14 of 15

Calculate the value of transconductance g_m .

$$g_m = \frac{I_O}{V_T}$$

Substitute $25\ mV$ for V_T and $1\ \mu A$ for I_O .

$$g_m = \frac{1\ \mu}{25\ m}$$
$$= 0.04\ mA/V$$

Therefore, the value of the transconductance g_m is $0.04\ mA/V$.

Step 15 of 15

Consider the expression for the output resistance.

$$R_o = \left[1 + g_m (R_E \| r_o)\right] r_o$$

As the value of β is very high, the resistance r_π can be neglected.

$$R_o = (1 + g_m R_E) r_o$$

Substitute $50\ M\Omega$ for r_o , $0.04\ mA/V$ for g_m and $115.25\ k\Omega$ for R_E .

$$R_o = \left[1 + (0.04\ m)115.25\ k\right](50\ M)$$
$$= [1 + 4.61](50\ M)$$
$$= [5.61](50\ M)$$
$$= 280.5\ M\Omega$$

Therefore, the value of the output resistance R_o is $\boxed{280.5\ M\Omega}$.

Compare the value of R_o with r_o , the value of the output resistance increases with the introduction of the emitter resistance to the circuit.

Step 1 of 7

Refer to Figure P8.92 from the text book.

(a)

Calculate the value of the current I_O.

$$I_O = \frac{V_{BE1} + V_{BE2} - V_{BE3}}{R}$$

Here, $V_{BE1} = V_{BE2} = V_{BE3}$ since $I_O = I_{REF}$.

$$I_O = \frac{V_{BE} + V_{BE} - V_{BE}}{R}$$

$$I_O = \frac{V_{BE}}{R} \quad \text{..... (1)}$$

Therefore, the expression for the current I_O is $\frac{V_{BE}}{R}$.

Step 2 of 7

Consider the expression for the difference in the values of the base to emitter voltages of transistors Q_1 and Q_3.

$$V_{BE1} - V_{BE3} = V_T \ln\left(\frac{I_{REF}}{I_O}\right)$$

Substitute $0.7\ \text{V}$ for V_{BE1}, $25\ \text{mV}$ for V_T, $1\ \text{mA}$ for I_{REF} and $10\ \mu\text{A}$ for I_O.

$$0.7 - V_{BE3} = (25\ \text{m}) \ln\left(\frac{1\ \text{m}}{10\ \mu}\right)$$
$$0.7 - V_{BE3} = (25\ \text{m}) \ln(100)$$
$$0.7 - V_{BE3} = (25\ \text{m})(4.61)$$
$$0.7 - V_{BE3} = 115.25\ \text{m}$$

$$V_{BE3} = 0.7 - 115.25\ \text{m}$$
$$V_{BE3} = 0.585\ \text{V}$$

Therefore, the value of the voltage V_{BE3} is $0.585\ \text{V}$.

Step 3 of 7

Recall equation (1).

$$I_O = \frac{V_{BE}}{R}$$

Substitute $0.585\ \text{V}$ for V_{BE} and $10\ \mu\text{A}$ for I_O.

$$10\ \mu = \frac{0.585}{R}$$
$$R = \frac{0.585}{10\ \mu}$$
$$R = 58.5\ \text{k}\Omega$$

Therefore, the value of the resistance R is $\boxed{58.5\ \text{k}\Omega}$.

Step 4 of 7

(b)

Calculate the value of the transcondutance g_m.

$$g_m = \frac{I_O}{V_T}$$

Substitute $25\ \text{mV}$ for V_T and $10\ \mu\text{A}$ for I_O.

$$g_m = \frac{10\ \mu}{25\ \text{m}}$$
$$= 0.4\ \text{mA/V}$$

Therefore, the value of the transconductance g_m is $0.4\ \text{mA/V}$.

Step 5 of 7

Calculate the value of the resistance r_π.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute $0.4\ \text{mA/V}$ for g_m and 100 for β.

$$r_\pi = \frac{100}{0.4\ \text{m}}$$
$$= 250\ \text{k}\Omega$$

Therefore, he value of the resistance r_π is $250\ \text{k}\Omega$.

Step 6 of 7

Calculate the value of the resistance r_o.

$$r_o = \frac{V_A}{I}$$

Substitute $40\ \text{V}$ for V_A and $10\ \mu\text{A}$ for I_O.

$$r_o = \frac{40}{10\mu}$$
$$= 4\ \text{M}\Omega$$

Therefore, he value of the resistance r_o is $4\ \text{M}\Omega$.

Step 7 of 7

Consider the expression for the output resistance.

$$R_o = \left[1 + g_m (R \| r_\pi)\right] r_o$$

Substitute $4\ \text{M}\Omega$ for r_o, $250\ \text{k}\Omega$ for r_π, $0.4\ \text{mA/V}$ for g_m and $58.5\ \text{k}\Omega$ for R.

$$R_o = \left[1 + (0.4\ \text{m})(58.5\ \text{k} \| 250\ \text{k})\right](4\ \text{M})$$
$$= \left[1 + (0.4\ \text{m})\left(\frac{58.5\ \text{k} \times 250\ \text{k}}{58.5\ \text{k} + 250\ \text{k}}\right)\right](4\ \text{M})$$
$$= \left[1 + (0.4\ \text{m})\left(\frac{58.5\ \text{k} \times 250\ \text{k}}{308.5\ \text{k}}\right)\right](4\ \text{M})$$
$$= \left[1 + (0.4\ \text{m})(47.4\ \text{k})\right](4\ \text{M})$$

$$R_o = [1 + 18.964](4\ \text{M})$$
$$= (19.964)(4\ \text{M})$$
$$= 79.856\ \text{M}\Omega$$

Therefore, the value of the output resistance R_o is $\boxed{79.856\ \text{M}\Omega}$.

Step 1 of 3

8.093P

Refer to Figure P8.93 from the text book.

The values of the currents I_{D1} and I_{D2} are equal. Thus, the voltages V_{GS1} and V_{GS2} are equal since the transistors Q_1 and Q_2 are matched. Also, values of the voltages at the sources of Q_1 and Q_2 are equal. That is, $V_{S1} = V_{S2}$.

Hence, the base to emitter voltage of the transistor Q_6 is equal to the voltage drop across the resistor R. That is,

$$IR = V_{BE6}$$

Substitute $V_T \ln\left(\frac{I}{I_S}\right)$ for V_{BE6}.

$$IR = V_T \ln\left(\frac{I}{I_S}\right) \quad \text{..... (1)}$$

Therefore, the values of the voltage drop across the resistor R is $\boxed{V_T \ln\left(\frac{I}{I_S}\right)}$.

Step 2 of 3

Consider the expression for the emitter current I_E.

$$I_E = I_S e^{V_{EB}/V_T}$$

Substitute $0.7\ \text{V}$ for V_{EB}, $25\ \text{mV}$ for V_T and $1\ \text{mA}$ for I_E.

$$1\ \text{m} = I_S e^{0.7/25\ \text{m}}$$
$$1\ \text{m} = I_S e^{28}$$
$$I_S = \frac{1\ \text{m}}{1.45\times10^{12}}$$
$$I_S = 6.91\times10^{-16}\ \text{A}$$

Therefore, the value of the current I_S is $6.91\times10^{-16}\ \text{A}$.

Step 3 of 3

Recall equation (1), to determine the value of resistance.

$$IR = V_T \ln\left(\frac{I}{I_S}\right)$$

Substitute $6.91\times10^{-16}\ \text{A}$ for I_S, $200\ \mu\text{A}$ for I and $25\ \text{mV}$ for V_T.

$$(200\ \mu)R = (25\ \text{m})\ln\left(\frac{200\ \mu}{6.91\times10^{-16}}\right)$$
$$(200\ \mu)R = (25\ \text{m})\ln(2.89\times10^{11})$$
$$(200\ \mu)R = (25\ \text{m})(26.39)$$
$$(200\ \mu)R = 0.66$$

$$R = \frac{0.66}{200\ \mu}$$
$$R = 3.3\ \text{k}\Omega$$

Therefore, the value of the resistance R is $\boxed{3.3\ \text{k}\Omega}$.

Step 1 of 2

Refer to Figure 8.45 (b) from the text book.

Consider the circuit diagram shown in Figure 1 to calculate the output resistance of the circuit.

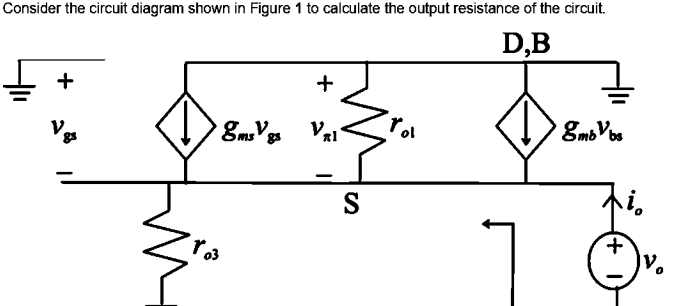


Figure 1

Step 2 of 2

Apply nodal analysis at output node.

$$-g_m v_{gs} - g_{mb} v_{bs} + \frac{v_o}{r_{o3} \| r_{o1}} - i_o = 0$$

Here, the voltages v_{bs} and v_{gs} are equal to the voltages $-v_o$.

$$g_m v_o + g_{mb} v_o + \frac{v_o}{r_{o3} \| r_{o1}} - i_o = 0$$

$$v_o \left(g_m + g_{mb} + \frac{1}{r_{o3} \| r_{o1}} \right) = i_o$$

$$v_o \left(\frac{1}{r_{o3} \| r_{o1} \| \frac{1}{g_m + g_{mb}}} \right) = i_o$$

$$\frac{v_o}{i_o} = r_{o3} \| r_{o1} \| \frac{1}{g_m + g_{mb}}$$

Here, $\frac{v_o}{i_o}$ represents the output resistance R_o.

$$R_o = r_{o3} \| r_{o1} \| \frac{1}{g_m + g_{mb}}$$

Therefore, the expression for the output resistance R_o is, $\boxed{r_{o3} \| r_{o1} \| \frac{1}{g_m + g_{mb}}}$.

Step 1 of 10

Refer to Figure 8.45 (a) from the text book.

At transistor Q_1, the difference in the output and the input voltage is equal to the gate to source voltage. That is, $V_{GS1} = 0.9\ \text{V}$.

Consider the expression for the value of the bias current.

$$I = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(V_{GS} - V_t)^2$$

Substitute $0.9\ \text{V}$ for V_{GS}, $0.6\ \text{V}$ for V_t, $20\ \mu\text{m}$ for W, $0.5\ \mu\text{m}$ for L and $200\ \mu\text{A/V}^2$ for k_n'.

$$\begin{aligned} I &= \frac{1}{2}(200\ \mu)\left(\frac{20\ \mu}{0.5\ \mu}\right)(0.9 - 0.6)^2 \\ &= (100\ \mu)(40)(0.3)^2 \\ &= (4\ \text{m})(0.09) \\ &= 360\ \mu\text{A} \end{aligned}$$

Therefore, the value of the bias current I is $\boxed{360\ \mu\text{A}}$.

Step 2 of 10

Calculate the value of the transconductance g_m.

$$g_m = \frac{2I}{V_{OV}}$$

Substitute $V_{GS} - V_t$ for V_{OV}.

$$g_m = \frac{2I}{V_{GS} - V_t}$$

Substitute $360\ \mu\text{A}$ for I, $0.9\ \text{V}$ for V_{GS} and $0.6\ \text{V}$ for V_t.

$$\begin{aligned} g_m &= \frac{2(360\ \mu)}{0.3} \\ &= 2.4\ \text{mA/V} \end{aligned}$$

Therefore, the value of the transconductance g_m is $\boxed{2.4\ \text{mA/V}}$.

Step 3 of 10

Calculate the value of the transconductance g_{mb}.

$$g_{mb} = \chi g_m$$

Substitute 0.2 for χ and $2.4\ \text{mA/V}$ for g_m.

$$\begin{aligned} g_{mb} &= (0.2)(2.4\ \text{m}) \\ &= 0.48\ \text{mA/V} \end{aligned}$$

Therefore, the value of the transconductance g_{mb} is $\boxed{0.48\ \text{mA/V}}$.

Step 4 of 10

Calculate the value of the resistance r_o.

$$r_o = \frac{V_A' L}{I}$$

Substitute $360\ \mu\text{A}$ for I, $0.5\ \mu\text{m}$ for L and $360\ \mu\text{A}$ for I.

$$\begin{aligned} r_o &= \frac{(20)(0.5)}{360\ \mu} \\ &= 27.78\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance r_o is $\boxed{27.78\ \text{k}\Omega}$.

Step 5 of 10

Calculate the value of the resistance R_L.

$$R_L = r_{o2} \| r_{o1} \| \frac{1}{g_{mb}}$$

Substitute $27.78\ \text{k}\Omega$ for r_{o1} and r_{o3} and $0.48\ \text{mA/V}$ for g_{mb}.

$$\begin{aligned} R_L &= 27.78\ \text{k} \| 27.78\ \text{k} \| \frac{1}{0.48\ \text{m}} \\ &= 27.78\ \text{k} \| 27.78\ \text{k} \| 2.08\ \text{k} \\ &= \frac{(27.78\ \text{k})(27.78\ \text{k})(2.08\ \text{k})}{(27.78\ \text{k})(27.78\ \text{k}) + (27.78\ \text{k})(2.08\ \text{k}) + (27.78\ \text{k})(2.08\ \text{k})} \\ &= \frac{1.605 \times 10^{12}}{771.8284\ \text{M} + 57.7824\ \text{M} + 57.7824\ \text{M}} \\ &= \frac{1.605 \times 10^{12}}{887.3932\ \text{M}} \end{aligned}$$

$$R_L = 1.81\ \text{k}\Omega$$

Therefore, the value of the resistance R_L is $1.81\ \text{k}\Omega$.

Step 6 of 10

Calculate the value of the voltage gain A_{vo}.

$$A_{vo} = \frac{R_L}{R_L + \frac{1}{g_m}}$$

Substitute $1.81\ \text{k}\Omega$ for R_L and $2.4\ \text{mA/V}$ for g_m.

$$\begin{aligned} A_{vo} &= \frac{1.81\ \text{k}}{1.81\ \text{k} + \frac{1}{2.4\ \text{m}}} \\ &= \frac{1.81\ \text{k}}{1.81\ \text{k} + 416.67} \\ &= \frac{1.81\ \text{k}}{2.227\ \text{k}} \\ &= 0.81\ \text{V/V} \end{aligned}$$

$$\begin{aligned} A_{vo} &= \frac{1.81\ \text{M}}{1.81\ \text{M} + \frac{1}{2.4\ \text{m}}} \\ &= \frac{1.81\ \text{M}}{1.81\ \text{M} + 416.67} \end{aligned}$$

Therefore, the value of the voltage gain is $\boxed{0.81\ \text{V/V}}$.

Step 7 of 10

Consider the circuit diagram shown in Figure 1 to calculate the output resistance of the circuit.

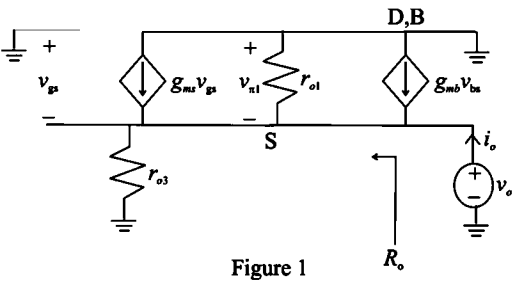


Figure 1

Step 8 of 10

Apply nodal analysis at output node.

$$-g_m v_{gs} - g_{mb} v_{bs} + \frac{v_o}{r_{o3} \| r_{o1}} - i_o = 0$$

Here, the voltages v_{bs} and v_{gs} are equal to the voltages $-v_o$.

$$\begin{aligned} g_m v_o + g_{mb} v_o + \frac{v_o}{r_{o3} \| r_{o1}} - i_o &= 0 \\ v_o\left(g_m + g_{mb} + \frac{1}{r_{o3} \| r_{o1}}\right) &= i_o \\ v_o\left(\frac{1}{r_{o3} \| r_{o1} \| \frac{1}{g_m + g_{mb}}}\right) &= i_o \\ \frac{v_o}{i_o} &= r_{o3} \| r_{o1} \| \frac{1}{g_m + g_{mb}} \end{aligned}$$

Here, $\frac{v_o}{i_o}$ represents the output resistance R_o.

$$R_o = r_{o3} \| r_{o1} \| \frac{1}{g_m + g_{mb}} \quad \text{..... (1)}$$

Therefore, the expression for the output resistance R_o is, $r_{o3} \| r_{o1} \| \frac{1}{g_m + g_{mb}}$.

Step 9 of 10

Recall equation (1).

$$R_o = r_{o3} \| r_{o1} \| \frac{1}{g_m + g_{mb}}$$

Substitute $27.78\ \text{k}\Omega$ for r_{o1} and r_{o3}, $2.4\ \text{mA/V}$ for g_m and $0.48\ \text{mA/V}$ for g_{mb}.

$$\begin{aligned} R_o &= 27.78\ \text{k} \| 27.78\ \text{k} \| \frac{1}{2.4\ \text{m} + 0.48\ \text{m}} \\ &= 27.78\ \text{k} \| 27.78\ \text{k} \| \frac{1}{2.88\ \text{m}} \\ &= 27.78\ \text{k} \| 27.78\ \text{k} \| 347.22 \\ &= \frac{(27.78\ \text{k})(27.78\ \text{k})(347.22)}{(27.78\ \text{k})(27.78\ \text{k}) + (27.78\ \text{k})(347.22) + (27.78\ \text{k})(347.22)} \\ R_o &= \frac{2.68 \times 10^{11}}{771.7284\ \text{M} + 9.6457716\ \text{M} + 9.6457716\ \text{M}} \\ &= \frac{2.68 \times 10^{11}}{791.0199432\ \text{M}} \\ &= 338.8\ \Omega \end{aligned}$$

Therefore, the value of the resistance R_o is $\boxed{338.8\ \Omega}$.

Step 10 of 10

Consider the expression for the voltage gain A_v when load resistor R_L is connected to the output.

$$A_v = \frac{R_L'}{R_L' + \frac{1}{g_m}}$$

Here, R_L' represents $R_L \| 2\ \text{k}\Omega$.

$$A_v = \frac{R_L \| 2\ \text{k}\Omega}{R_L \| 2\ \text{k}\Omega + \frac{1}{g_m}}$$

Substitute $1.81\ \text{k}\Omega$ for R_L and $2.4\ \text{mA/V}$ for g_m.

$$\begin{aligned} A_v &= \frac{1.81\ \text{k} \| 2\ \text{k}}{1.81\ \text{k} \| 2\ \text{k}\Omega + \frac{1}{2.4\ \text{m}}} \\ &= \frac{\frac{(1.81\ \text{k}) \cdot (2\ \text{k})}{1.81\ \text{k} + 2\ \text{k}}}{\frac{(1.81\ \text{k}) \cdot (2\ \text{k})}{1.81\ \text{k} + 2\ \text{k}} + 416.67} \\ &= \frac{\frac{(1.81\ \text{k}) \cdot (2\ \text{k})}{3.81\ \text{k}}}{\frac{(1.81\ \text{k}) \cdot (2\ \text{k})}{3.81\ \text{k}} + 416.67} \\ &= \frac{950.13}{950.13 + 416.67} \\ A_v &= \frac{950.13}{1366.8} \\ &= 0.7\ \text{V/V} \end{aligned}$$

Therefore, the value of the voltage gain A_v is $\boxed{0.7\ \text{V/V}}$.

Refer to Figure P8.96 from the text book.

(a)

Draw the small signal equivalent circuit diagram.

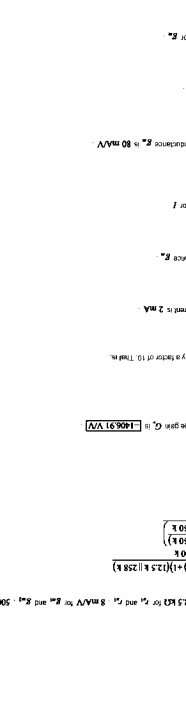


Figure 1

Step 2 of 22

Consider the expression for the voltage v_{o1}

$$v_{o1} = (g_{m1}R_1 + i_1)R_1$$

Substitute $i_1(R_1)$ for v_{o1}

$$v_{o1} = (g_{m1}R_1 + i_1)(R_1 \| R_2)$$

Therefore, the expression for the voltage v_{o1} is $i_1(g_{m1}R_1 + 1)(R_1 \| R_2)$ ..... (1)

Step 3 of 22

Consider the expression for the voltage v_{o2}.

$$v_{o2} = (g_{m2}R_2 + i_2)(R_2 \| R_3)$$

Substitute $i_1(R_1)$ for v_{o1}

$$v_{o2} = (g_{m2}R_2 + i_2)(R_2 \| R_3)$$

Therefore, the expression for the voltage v_{o2} is $i_1(g_{m1}R_1 + 1)(R_2 \| R_3)$

Step 4 of 22

Consider the expression for the voltage v_o.

$$v_o = v_{o1} + v_{o2} \quad \ldots\ldots (2)$$

Substitute $i_1(R_1)$ for v_{o1} and $i_1(g_{m1}R_1 + 1)(R_2 \| R_3)$ for v_{o2}

$$\begin{aligned} v_o &= i_1 R_1 + i_1(g_{m1}R_1 + 1)(R_2 \| R_3) \\ v_o &= i_1\left[R_1 + (g_{m1}R_1 + 1)(R_2 \| R_3)\right] \\ \frac{v_o}{i_1} &= R_1 + (g_{m1}R_1 + 1)(R_2 \| R_3) \end{aligned}$$

Here, the term $\frac{v_o}{i_1}$ represents the expression for the input resistance R_{in}.

Therefore, the expression for the input resistance is $R_1 + (g_{m1}R_1 + 1)(R_2 \| R_3)$ ..... (3)

Step 5 of 22

Calculate the value of the transconductance g_{m1}.

$$g_{m1} = \frac{I}{V_T}$$

Substitute 25 mV for V_T and 200 μA for I.

$$\begin{aligned} g_{m1} &= \frac{200\ \mu}{25\ m} \\ &= 8\ mA/V \end{aligned}$$

Therefore, the value of the transconductance g_{m1} is 8 mA/V.

Step 6 of 22

Calculate the value of the resistance r_o.

$$r_o = \frac{\beta}{g_m}$$

Substitute 100 for β and 8 mA/V for g_m.

$$\begin{aligned} r_o &= \frac{100}{8\ m} \\ &= 12.5\ k\Omega \end{aligned}$$

Therefore, the value of the resistance r_o is 12.5 kΩ.

Step 7 of 22

Calculate the value of the resistance r_s.

$$r_s = \frac{v_A}{I}$$

Substitute 50 V for V_A and 200 μA for I.

$$\begin{aligned} r_s &= \frac{50}{200\ \mu} \\ &= 250\ k\Omega \end{aligned}$$

Therefore, the new value of the resistance r_s is 25 kΩ.

Step 8 of 22

Recall equation (3).

$$R_{in} = r_{o1} + (g_{m1}r_{o1} + 1)(r_{o2} \| r_s)$$

Substitute 25 kΩ for r_s, 12.5 kΩ for r_{o1} and r_{o2} and 8 mA/V for g_{m1}.

$$\begin{aligned} R_{in} &= 12.5\ k + ((8\ m)(12.5\ k) + 1)(12.5\ k \| 25\ k) \\ &= 12.5\ k + (100 + 1)\left(\frac{(12.5\ k)(25\ k)}{12.5\ k + 25\ k}\right) \\ &= 12.5\ k + (101)\left(\frac{312.5\ k}{37.5}\right) \\ &= 12.5\ k + (101)(12.91\ k) \\ &= 1.31\ M\Omega \end{aligned}$$

Therefore, the value of the input resistance R_{in} is 1.31 MΩ.

Step 9 of 22

Consider the expression for the output voltage v_o.

$$v_o = -g_{m1}v_{o1}r_o$$

Solve for v_{o1}.

$$v_{o1} = -\frac{v_o}{g_{m1}r_o}$$

Therefore, the expression for the voltage v_{o1} is $-\frac{v_o}{g_{m1}r_o}$.

Step 10 of 22

Recall equation (1).

$$v_{o1} = i_1(g_{m1}R_1 + 1)(r_o \| r_s)$$

Substitute $\frac{v_{in}}{R_1}$ for i_1.

$$v_{o1} = \left(\frac{v_{in}}{R_1}\right)(g_{m1}R_1 + 1)(r_o \| r_s)$$

$$v_{o1} = v_{in}\left(\frac{g_{m1}R_1 + 1}{R_1}\right)(r_o \| r_s)$$

Substitute $-\frac{v_o}{g_{m1}r_o}$ for v_{o1}

$$v_{in}\left(\frac{g_{m1}R_1 + 1}{R_1}\right)(r_o \| r_s) = -\frac{v_o}{g_{m1}r_o}$$

Therefore, the expression for the voltage v_{in} is $-\frac{v_o}{g_{m1}r_o}\left(\frac{R_1}{(g_{m1}R_1 + 1)(r_o \| r_s)}\right)$.

Step 11 of 22

Recall equation (2).

$$v_o = v_{o1} + v_{o2}$$

Substitute $-v_{in}\left(\frac{g_{m1}R_1}{(g_{m1}R_1 + 1)(r_o \| r_s)}\right)$ for v_{o1} and $\frac{v_o}{g_{m2}r_o}$ for v_{o2}.

$$v_o = -v_{in}\left(\frac{g_{m1}R_1}{(g_{m1}R_1 + 1)(r_o \| r_s)}\right) - \frac{v_o}{g_{m2}r_o}$$

Therefore, the expression for the overall voltage gain G_v is $-\frac{g_{m1}R_1}{(g_{m1}R_1 + 1)(r_o \| r_s)} - \frac{1}{g_{m2}r_o}$.

Step 12 of 22

Consider the expression for the overall voltage gain G_v.

$$G_v = \frac{v_o}{v_{sig}}$$

$$= \frac{v_o}{v_i} \cdot \frac{v_i}{v_{sig}}$$

Therefore, the expression for the gain $\frac{v_o}{v_{sig}}$ is $\frac{R_M}{R_M + R_{in}}$.

Consider the expression $\frac{v_o}{v_{sig}}$ for the overall voltage gain G_v.

$$G_v = \frac{v_o}{v_{sig}}$$

Substitute $\frac{R_M}{R_M + R_{in}}$ for $\frac{v_i}{v_{sig}}$ and $-\frac{g_{m1}R_1}{(g_{m1}R_1 + 1)(r_o \| r_s)} - \frac{1}{g_{m2}r_o}$ for $\frac{v_o}{v_i}$.

$$G_v = \left(-\frac{g_{m1}R_1}{(g_{m1}R_1 + 1)(r_o \| r_s)} - \frac{1}{g_{m2}r_o}\right)\left(\frac{R_M}{R_M + R_{in}}\right)$$

Substitute $R_1 = r_o$ and $g_{m1}r_o + 1$ for R_{in}

$$G_v = -\left(\frac{g_{m1}r_o}{g_{m1}r_o(r_o \| r_s)} + \frac{1}{g_{m2}r_o}\right)\left(\frac{R_M}{R_M + R_{in}}\right)$$

$$G_v = -\frac{(g_{m2}r_o)(g_{m1}r_o(r_o \| r_s)) + (r_o \| r_s)}{(g_{m2}r_o)(g_{m1}r_o(r_o \| r_s))}\left(\frac{R_M}{R_M + R_{in}}\right)$$

$$G_v = -\frac{(g_{m2}r_o)(g_{m1}r_o + 1)(r_o \| r_s)}{R_M + R_{in}} \quad \ldots\ldots (4)$$

Therefore, the expression for the overall voltage gain G_v is $-\frac{(g_{m2}r_o)(g_{m1}r_o + 1)(r_o \| r_s)}{R_M + R_{in}}$.

Step 14 of 22

Recall equation (4).

$$G_v = -\frac{(g_{m2}r_o)(g_{m1}r_o + 1)(r_o \| r_s)}{R_M + R_{in}}$$

Substitute 20 kΩ for R_M, 12.5 kΩ for r_o and r_s, 8 mA/V for g_{m1} and g_{m2}, 500 kΩ for R_{in} and 1.31 MΩ for R_{in}.

$$\begin{aligned} G_v &= -\frac{((8\ m)(250\ k) + 1)((12.5\ k) \| (25\ k))}{1.31\ M + 500\ k} \\ &= -\frac{(2000)(100 + 1)\left(\frac{(12.5\ k)(25\ k)}{12.5\ k + 25\ k}\right)}{1.71\ M} \\ &= -\frac{(2000)(101)(11.91\ k)}{1.71\ M} \\ &= -\frac{2405.82\ M}{1.71\ M} \\ G_v &= -1406.91\ V/V \end{aligned}$$

Therefore, the value of the overall voltage gain G_v is $\boxed{-1406.91\ V/V}$

Step 15 of 22

Increase the value of the bias currents by a factor of 10. That is,

$$\begin{aligned} I &= 200\ \mu A \times 10 \\ &= 2\ mA \end{aligned}$$

Therefore, the new value of the bias current is 2 mA.

Step 16 of 22

Calculate the value of the transconductance g_m.

$$g_m = \frac{I}{V_T}$$

Substitute 25 mV for V_T and 2 mA for I

$$\begin{aligned} g_m &= \frac{2\ m}{25\ m} \\ &= 80\ mA/V \end{aligned}$$

Therefore, the new value of the transconductance g_m is 80 mA/V.

Step 17 of 22

Calculate the value of the resistance r_o.

$$r_o = \frac{\beta}{g_m}$$

Substitute 100 for β and 80 mA/V for g_m.

$$\begin{aligned} r_o &= \frac{100}{80\ m} \\ &= 1.25\ k\Omega \end{aligned}$$

Therefore, the new value of the resistance r_o is 1.25 kΩ.

Step 18 of 22

Calculate the value of the resistance r_s.

$$r_s = \frac{v_A}{I}$$

Substitute 50 V for V_A and 2 mA for I.

$$\begin{aligned} r_s &= \frac{50}{2\ m} \\ &= 25\ k\Omega \end{aligned}$$

Therefore, the new value of the resistance r_s is 25 kΩ.

Step 19 of 22

Recall equation (3).

$$R_{in} = r_{o1} + (g_{m1}r_{o1} + 1)(r_{o2} \| r_s)$$

Substitute 25 kΩ for r_s, 1.25 kΩ for r_{o1} and r_{o2} and 80 mA/V for g_{m1}.

$$\begin{aligned} R_{in} &= 1.25\ k + ((80\ m)(1.25\ k) + 1)(1.25\ k \| 1.25\ k) \\ &= 1.25\ k + (100 + 1)\left(\frac{(1.25\ k)(1.25\ k)}{1.25\ k + 1.25\ k}\right) \\ &= 1.25\ k + (101)\left(\frac{1.25\ k}{2.5}\right) \\ &= 1.25\ k + (101)(0.5\ k) \\ &= 51.25\ k\Omega \end{aligned}$$

Therefore, the value of the input resistance R_{in} is 51.25 kΩ.

Step 20 of 22

Recall equation (4).

$$G_v = -\frac{(g_{m2}r_o)(g_{m1}r_o + 1)(r_o \| r_s)}{R_M + R_{in}}$$

Substitute 20 kΩ for R_M, 1.25 kΩ for r_o and r_s, 8 mA/V for g_{m1} and g_{m2}, 51.25 kΩ for R_{in}.

$$\begin{aligned} G_v &= -\frac{((8\ m)(1.25\ k) + 1)((1.25\ k) \| (1.25\ k))}{20\ k + 51.25\ k} \\ &= -\frac{(11)\left(\frac{(1.25\ k)(1.25\ k)}{1.25\ k + 1.25\ k}\right)}{71.25\ k} \\ &= -\frac{(11)(0.625\ k)}{71.25\ k} \\ &= -\frac{6.875\ k}{71.25\ k} \\ G_v &= -0.096\ V/V \end{aligned}$$

Therefore, the value of the overall voltage gain G_v is $\boxed{-0.096\ V/V}$

Step 21 of 22

Calculate the power dissipated in the two transistors before the bias current is increased.

$$\begin{aligned} P &= 2IV_{CC} \\ &= 2(200\ \mu)(5) \\ &= 2\ mW \end{aligned}$$

Therefore, the value of the power dissipated P in the transistors before the bias current is increased is 2 mW.

Step 22 of 22

Calculate the power dissipated in the two transistors after the bias current is increased by a factor of 10.

$$\begin{aligned} P &= 2IV_{CC} \\ &= 2(2\ m)(5) \\ &= 20\ mW \end{aligned}$$

Therefore, the value of the power dissipated P in the transistors after the bias current is increased is 20 mW.

Therefore, the value of the power dissipation in the transistors **increases** as the value of the bias current is increased.

Step 1 of 19

8.097P

Refer to Figure P8.97 from the text book.

(a)

Consider the expression for the current through the source resistance of the transistor Q_1.

$$I_S = \frac{V_{BE2}}{6.8\,\text{k}\Omega}$$

Substitute **0.7 V** for V_{BE2}.

$$I_S = \frac{0.7}{6.8\,\text{k}}$$
$$= 0.1\,\text{mA}$$

Therefore, the value of the current through the source resistance is **0.1 mA**.

The value of the drain current is approximately equal to the source current. Hence, the value of the drain current is **0.1 mA**.

Step 2 of 19

Consider the expression for drain current I_D.

$$I_D = \frac{1}{2}k_n\left(\frac{W}{L}\right)(V_{GS}-V_t)^2$$

Substitute **0.1 mA** for I_D, **2 mA/V²** for k_n, **1 V** for V_t and neglect the value of the ratio $\frac{W}{L}$.

$$0.1\,\text{m} = \frac{1}{2}(2\,\text{m})(V_{GS}-1)^2$$
$$0.1 = (V_{GS}-1)^2$$
$$V_{GS}-1 = \sqrt{0.1}$$
$$V_{GS}-1 = 0.316$$
$$V_{GS} = 0.316+1$$
$$V_{GS} = 1.316\,\text{V}$$

Therefore, the value of the voltage V_{GS} is **1.316 V**.

Step 3 of 19

Calculate the value of the voltage at the gate terminal V_G.

$$V_G = V_{GS} + V_{RE}$$

Substitute **1.316 V** for V_{GS} and **0.7 V** for v_{RE}.

$$V_G = 1.316+0.7$$
$$= 2.016\,\text{V}$$

Therefore, the value of the voltage at the gate V_G is **2.016 V**.

Step 4 of 19

Assume the current through the resistor R_G be negligible. Hence, the value of the gate value is equal to the drain voltage of the transistor Q_1.

Calculate the value of the collector current I_C.

$$I_C = \frac{5-2.016}{\frac{3\,\text{k}}{2}}$$
$$= \frac{2.984}{3\,\text{k}}$$
$$= 1\,\text{mA}$$

Therefore, the value of the collector current I_C is **1 mA**.

Step 5 of 19

(b)

Calculate the value of the transconductance g_{m1}.

$$g_{m1} = \frac{2I_D}{V_{OV}}$$

Substitute $V_{GS}-V_t$ for V_{OV}.

$$g_{m1} = \frac{2I_D}{V_{GS}-V_t}$$

Substitute **0.1 mA** for I_D, **1.316 V** for V_{GS} and **1 V** for V_t.

$$g_{m1} = \frac{2(0.1\,\text{m})}{1.316-1}$$
$$= \frac{0.2\,\text{m}}{0.316}$$
$$= 0.63\,\text{mA/V}$$

Therefore, the value of the transconductance g_{m1} is **0.63 mA/V**.

Step 6 of 19

Calculate the value of the transconductance g_{m2}.

$$g_{m2} = \frac{I_C}{V_T}$$

Substitute **1 mA** for I_C and **25 mV** for V_T.

$$g_{m2} = \frac{1\,\text{m}}{25\,\text{m}}$$
$$= 40\,\text{mA/V}$$

Therefore, the value of the transconductance g_{m2} is **40 mA/V**.

Step 7 of 19

Calculate the value of the resistance r_π.

$$r_\pi = \frac{\beta}{g_{m2}}$$

Substitute **40 mA/V** for g_{m2} and **100** for β.

$$r_\pi = \frac{200}{40\,\text{m}}$$
$$= 5\,\text{k}\Omega$$

Therefore, the value of the resistance r_π is **5 kΩ**.

Step 8 of 19

(c)

Draw the small signal equivalent circuit diagram. Neglect the value of the resistances, r_π and R_G.

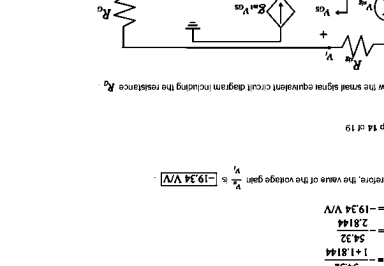


Figure 1

Step 9 of 19

Consider the expression for the output voltage v_o.

$$v_o = -g_{m2}v_x(R_C \parallel R_L)$$

Therefore, the expression for the voltage v_x is $\frac{-v_o}{g_{m2}(R_C \parallel R_L)}$.

Step 10 of 19

Consider the expression for the voltage v_x.

$$v_x = g_{m1}v_{gs}(r_\pi \parallel r_\pi)$$
$$v_{gs} = \frac{v_x}{g_{m1}(r_\pi \parallel r_\pi)}$$

Substitute $\frac{-v_o}{g_{m2}(R_C \parallel R_L)}$ for v_{gs}.

$$v_{gs} = \frac{-v_o}{g_{m2}(R_C \parallel R_L)g_{m1}(r_\pi \parallel r_\pi)}$$

Therefore, the expression for the voltage v_{gs} is $\frac{-v_o}{g_{m1}g_{m2}(R_C \parallel R_L)(R_S \parallel r_\pi)}$.

Step 11 of 19

Consider the expression to the input voltage v_i.

$$v_i = v_{gs} + v_x$$

Substitute $\frac{-v_o}{g_{m1}g_{m2}(R_C \parallel R_L)(R_S \parallel r_\pi)}$ for v_{gs} and $\frac{-v_o}{g_{m2}(R_C \parallel R_L)}$ for v_x.

$$v_i = \frac{-v_o}{g_{m1}g_{m2}(R_C \parallel R_L)(R_S \parallel r_\pi)} - \frac{v_o}{g_{m2}(R_C \parallel R_L)}$$
$$v_i = -\frac{v_o}{g_{m2}(R_C \parallel R_L)}\left[\frac{1}{g_{m1}(R_S \parallel r_\pi)}+1\right]$$
$$v_i = -\frac{v_o}{g_{m2}(R_C \parallel R_L)}\left[\frac{1+g_{m1}(R_S \parallel r_\pi)}{g_{m1}(R_S \parallel r_\pi)}\right]$$

$$\frac{v_o}{v_i} = -\frac{g_{m1}g_{m2}(R_C \parallel R_L)(R_S \parallel r_\pi)}{1+g_{m1}(R_S \parallel r_\pi)} \quad \ldots\ldots (1)$$

Therefore, the expression for the voltage gain $\frac{v_o}{v_i}$ is $-\frac{g_{m1}g_{m2}(R_C \parallel R_L)(R_S \parallel r_\pi)}{1+g_{m1}(R_S \parallel r_\pi)}$.

Step 12 of 19

Recall equation (1).

$$\frac{v_o}{v_i} = -\frac{g_{m1}g_{m2}(R_C \parallel R_L)(R_S \parallel r_\pi)}{1+g_{m1}(R_S \parallel r_\pi)}$$

Substitute **0.63 mA/V** for g_{m1}, **40 mA/V** for g_{m2}, **5 kΩ** for r_π, **3 kΩ** for R_C, **1 kΩ** for R_L and **6.8 kΩ** for R_S.

$$\frac{v_o}{v_i} = -\frac{(0.63\,\text{m})(40\,\text{m})(3\,\text{k}\parallel 1\,\text{k})(6.8\,\text{k}\parallel 5\,\text{k})}{1+(0.63\,\text{m})(6.8\,\text{k}\parallel 5\,\text{k})}$$
$$= -\frac{(0.63\,\text{m})(40\,\text{m})\left(\frac{(3\,\text{k})(1\,\text{k})}{3\,\text{k}+1\,\text{k}}\right)\left(\frac{(6.8\,\text{k})(5\,\text{k})}{6.8\,\text{k}+5\,\text{k}}\right)}{1+(0.63\,\text{m})\left(\frac{(6.8\,\text{k})(5\,\text{k})}{6.8\,\text{k}+5\,\text{k}}\right)}$$
$$= -\frac{(25.2\,\mu)\left(\frac{(3\,\text{k})(1\,\text{k})}{4\,\text{k}}\right)\left(\frac{(6.8\,\text{k})(5\,\text{k})}{11.8\,\text{k}}\right)}{1+(0.63\,\text{m})\left(\frac{(6.8\,\text{k})(5\,\text{k})}{11.8\,\text{k}}\right)}$$

Further simplification as follows:

$$\frac{v_o}{v_i} = -\frac{(25.2\,\mu)(0.75\,\text{k})(2.88\,\text{k})}{1+(0.63\,\text{m})(2.88\,\text{k})}$$
$$= -\frac{54.32}{1+1.8144}$$
$$= -\frac{54.32}{2.8144}$$
$$= -19.34\,\text{V/V}$$

Therefore, the value of the voltage gain $\frac{v_o}{v_i}$ is **−19.34 V/V**.

(d)

Draw the small signal equivalent circuit diagram including the resistance R_o.

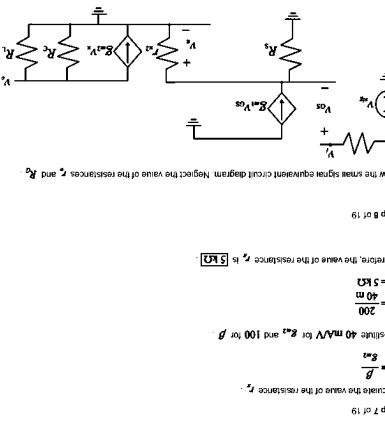


Figure 2

Step 15 of 19

Consider the expression for the voltage gain $\frac{v_o}{v_{sig}}$.

$$\frac{v_o}{v_{sig}} = \frac{R_o + R_C}{R_o}$$
$$\frac{v_o}{v_1} = 1 + \frac{R_C}{R_o}$$
$$\frac{R_C}{R_o} = \frac{v_o}{v_1} - 1$$
$$R_o = \frac{R_C}{\frac{v_o}{v_1} - 1} \quad \ldots\ldots (2)$$

Therefore, the expression for the input resistance R_o is $\frac{R_C}{1-\frac{v_o}{v_1}}$.

Step 16 of 19

Recall equation (2).

$$R_o = \frac{R_C}{1-\frac{v_o}{v_1}}$$

Substitute **10 MΩ** for R_C and -19.34 **V/V** for $\frac{v_o}{v_1}$.

$$R_o = \frac{10\,\text{M}}{1-(-19.34)}$$
$$= \frac{10\,\text{M}}{20.34}$$
$$= 491.64\,\text{k}\Omega$$

Therefore, the value of the input resistance R_o is **491.64 kΩ**.

Step 17 of 19

Consider the expression for the overall voltage gain $\frac{v_o}{v_{sig}}$.

$$\frac{v_o}{v_{sig}} = \frac{v_o}{v_1} \times \frac{v_1}{v_{sig}}$$

Substitute $\frac{R_o}{R_o + R_{sig}}$ for $\frac{v_1}{v_{sig}}$.

$$\frac{v_o}{v_{sig}} = \frac{v_o}{v_1} \times \frac{R_o}{R_o + R_{sig}} \quad \ldots\ldots (3)$$

Substitute **491.64 kΩ** for R_o, **500 kΩ** for R_{sig} and **−19.34 V/V** for $\frac{v_o}{v_1}$.

$$\frac{v_o}{v_{sig}} = (-19.34) \times \frac{491.64\,\text{k}}{491.64\,\text{k} + 500\,\text{k}}$$
$$= (-19.34) \times \frac{491.64\,\text{k}}{991.64\,\text{k}}$$
$$= (-19.34) \times 0.5$$
$$= -9.67\,\text{V/V}$$

Therefore, the value of the overall voltage gain $\frac{v_o}{v_{sig}}$ is **−9.67 V/V**.

Step 18 of 19

The value of the input resistance R_o as seen from the input terminals is equal to the value of the resistance R_G. Therefore, the value of the input resistance R_o is **10 MΩ**.

Recall equation (3).

$$\frac{v_o}{v_{sig}} = \frac{v_o}{v_1} \times \frac{R_o}{R_o + R_{sig}}$$

Substitute **10 MΩ** for R_o, **500 kΩ** for R_{sig} and **−19.34 V/V** for $\frac{v_o}{v_1}$.

$$\frac{v_o}{v_{sig}} = (-19.34) \times \frac{10\,\text{M}}{10\,\text{M} + 500\,\text{k}}$$
$$= (-19.34) \times \frac{10\,\text{M}}{10.5\,\text{M}}$$
$$= (-19.34) \times 0.95$$
$$= -18.38\,\text{V/V}$$

Therefore, the value of the overall voltage gain $\frac{v_o}{v_{sig}}$ is **−18.38 V/V**.

Step 1 of 7

Refer to the Figure P7.83 in the text book.

Determine the emitter resistance of second transistor.

$$r_{e2} = \frac{V_T}{I_E}$$

Substitute 25 mV for V_T and 10 mA for I_E.

$$r_{e2} = \frac{25 \times 10^{-3}}{10 \times 10^{-3}}$$
$$= 2.5\ \Omega$$

Determine the emitter resistance of first transistor.

$$r_{e1} = \beta r_{e2}$$

Substitute 100 for β and 2.5Ω for r_{e2}.

$$r_{e1} = (2.5\Omega)(100)$$
$$= 250\ \Omega$$

Step 2 of 7

Draw the following small signal equivalent circuit diagram.

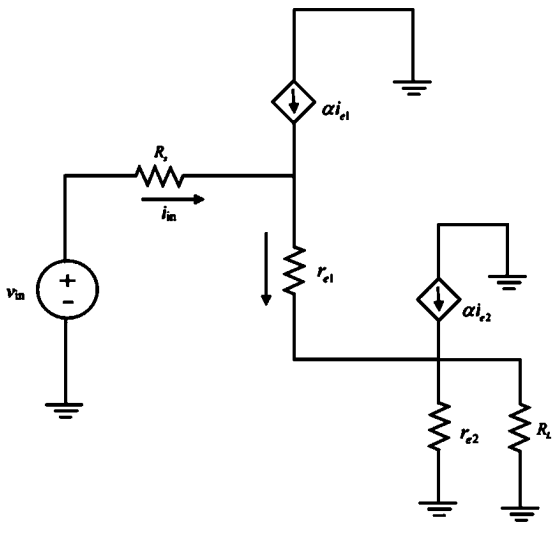


Figure 1

Step 3 of 7

Apply Kirchhoff's voltage law in Figure 1.

$$-v_{in} + R_s i_{in} + (\alpha i_{e1} + i_{in}) r_{e1} + (\alpha i_{e1} + i_{in} + \alpha i_{e2})(r_{e2} \| R_L) = 0$$

Substitute αi_{e1} for i_{c1}.

$$-v_{in} + R_s i_{in} + (i_{c1} + i_{in}) r_{e1} + (i_{c1} + i_{in} + i_{c2})(r_{e2} \| R_L) = 0$$

Substitute βi_{in} for i_{c1}

$$v_{in} = R_s i_{in} + (\beta i_{in} + i_{in}) r_{e1} + (\beta i_{in} + i_{in} + \beta(\beta i_{in} + i_{in}))(r_{e2} \| R_L)$$
$$v_{in} = i_{in}\left[R_s + (\beta+1) r_{e1} + (\beta+1)^2 (r_{e2} \| R_L)\right]$$
$$\frac{v_{in}}{i_{in}} = R_s + (\beta+1) r_{e1} + (\beta+1)^2 (r_{e2} \| R_L)$$
$$R_{in} = R_s + (\beta+1) r_{e1} + (\beta+1)^2 (r_{e2} \| R_L)$$

Substitute 100 for β, $250\ \Omega$ for r_{e1}, $2.5\ \Omega$ for r_{e2}, $1\ \text{k}\Omega$ for R_S and $100\ \text{k}\Omega$ for R_L

$$R_{in} = 10^3 + (100+1)(250\ \Omega) + (100+1)^2 (2.5 \| 10^5)$$
$$= 10^3 + (101)(250) + (101)^2 \left(\frac{(2.5)(10^5)}{2.5 + 10^5}\right)$$
$$= 51.75 \times 10^3$$
$$= 51.75\text{k}\Omega$$

Therefore, the value of input impedance is $\boxed{51.75\text{k}\Omega}$.

Step 4 of 7

Draw the following circuit diagram to determine the output impedance.

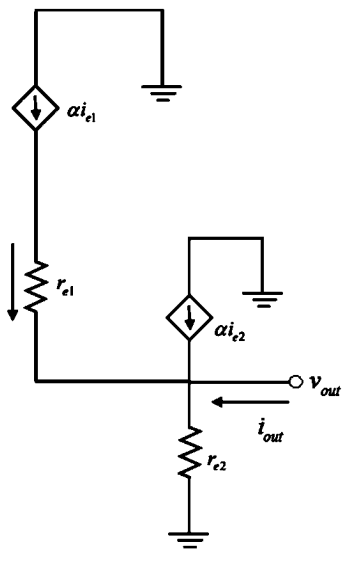


Figure 2

Step 5 of 7

From Figure 2, the output impedance of the circuit is,

$$Z_{out} = r_{e2}$$
$$= 2.5\Omega$$

Therefore, the value of output impedance is $\boxed{2.5\Omega}$.

Step 6 of 7

The expression of input voltage is,

$$v_{in} = i_{in}\left[R_s + (\beta+1) r_{e1} + (\beta+1)^2 (r_{e2} \| R_L)\right] \quad \ldots\ldots (1)$$

The expression of output voltage is,

$$v_{out} = i_{e2} r_{e2}$$

$$v_{out} = (1+\beta) i_{in} r_{e2} \quad \ldots\ldots (2)$$

Divide equation (2) with equation (1).

$$\frac{v_{out}}{v_{in}} = \frac{(1+\beta) i_{in} r_{e2}}{i_{in}\left[R_s + (\beta+1) r_{e1} + (\beta+1)^2 r_{e2}\right]}$$
$$= \frac{(1+\beta) r_{e2}}{R_s + (\beta+1) r_{e1} + (\beta+1)^2 (r_{e2})}$$

Substitute 100 for β, $250\ \Omega$ for r_{e1}, $2.5\ \Omega$ for r_{e2}, $100\ \text{k}\Omega$ for R_S and $1\ \text{k}\Omega$ for R_L

$$\frac{v_{out}}{v_{in}} = \frac{(1+100)(2.5)}{10^5 + (100+1)(250) + (100+1)^2 (2.5)}$$
$$= \frac{(101)(2.5)}{1.5 \times 10^5}$$
$$= 0.0016$$

Therefore, the open circuit voltage gain is $\boxed{0.0016\frac{\text{V}}{\text{V}}}$.

Step 7 of 7

Determine the voltage gain with load resistor.

$$A_v = A_{vo}\left[\frac{R_L}{R_L + Z_{out}}\right]$$
$$= (0.0016)\left[\frac{10^5}{10^5 + 2.5}\right]$$
$$= 0.0015\frac{\text{V}}{\text{V}}$$

Therefore, the voltage gain with load resistor is $\boxed{0.0015\frac{\text{V}}{\text{V}}}$.

Step 1 of 10

Refer to Figure 8.48 (a) from the text book.

Draw the small signal equivalent circuit diagram.

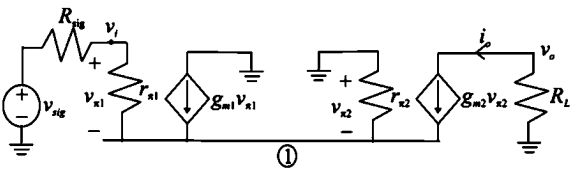


Figure 1

Step 2 of 10

Consider the expression for the voltage $v_{\pi 2}$.

$$v_{\pi 2} = v_{\pi 1} - v_i$$

Therefore, the expression for the voltage $v_{\pi 2}$ is $v_{\pi 1} - v_i$.

Step 3 of 10

Consider the expression for the voltage v_i.

$$v_i = i_i r_{\pi 1} + \left(i_i + g_{m1} v_{\pi 1} + g_{m2} v_{\pi 2}\right) r_{\pi 2}$$

Substitute $v_{\pi 1} - v_i$ for $v_{\pi 2}$.

$$v_i = i_i r_{\pi 1} + \left(i_i + g_{m1} v_{\pi 1} + g_{m2}\left(v_{\pi 1} - v_i\right)\right) r_{\pi 2}$$
$$v_i = i_i r_{\pi 1} + i_i r_{\pi 2} + g_{m1} v_{\pi 1} r_{\pi 2} + v_{\pi 1} g_{m2} r_{\pi 2} - v_i g_{m2} r_{\pi 2}$$

Substitute $i_i r_{\pi 1}$ for $v_{\pi 1}$.

$$v_i = i_i r_{\pi 1} + i_i r_{\pi 2} + i_i r_{\pi 1} g_{m1} r_{\pi 2} + i_i r_{\pi 1} g_{m2} r_{\pi 2} - v_i g_{m2} r_{\pi 2}$$
$$v_i + v_i g_{m2} r_{\pi 2} = i_i r_{\pi 1} + i_i r_{\pi 2} + i_i r_{\pi 1} g_{m2} r_{\pi 2} + i_i r_{\pi 1} g_{m2} r_{\pi 2}$$
$$v_i\left(1 + g_{m2} r_{\pi 2}\right) = i_i\left(r_{\pi 1} + r_{\pi 2} + r_{\pi 1} g_{m1} r_{\pi 2} + r_{\pi 1} g_{m2} r_{\pi 2}\right)$$

Let all the transistors have the same parameters. So that $g_{m1} = g_{m2}$ and $r_{\pi 1} = r_{\pi 2}$

$$v_i\left(1 + g_m r_\pi\right) = i_i\left(r_\pi + r_\pi + r_\pi g_m r_\pi + r_\pi g_m r_\pi\right)$$
$$v_i\left(1 + g_m r_\pi\right) = 2 i_i\left(r_\pi + r_\pi g_m r_\pi\right)$$
$$v_i\left(1 + g_m r_\pi\right) = 2 r_\pi i_i\left(1 + g_m r_\pi\right)$$
$$v_i = 2 r_\pi i_i$$

$$\frac{v_i}{i_i} = 2 r_\pi$$

Here, the term $\frac{v_i}{i_i}$ represents the input resistance R_{in} of the circuit.

$$R_{in} = 2 r_\pi$$

Therefore, the value of the input resistance R_{in} is $2 r_\pi$.

Step 4 of 10

Consider the expression for the output voltage v_o.

$$v_o = -g_{m2} v_{\pi 2} R_L$$
$$v_{\pi 2} = -\frac{v_o}{g_{m2} R_L}$$

Therefore, the expression for the voltage $v_{\pi 2}$ is $-\frac{v_o}{g_{m2} R_L}$.

Step 5 of 10

Apply nodal analysis at node-1.

$$-\frac{v_{\pi 1}}{r_{\pi 1}} - g_{m1} v_{\pi 1} - \frac{v_{\pi 2}}{r_{\pi 2}} - g_{m2} v_{\pi 2} = 0$$
$$-v_{\pi 1}\left(\frac{1}{r_{\pi 1}} + g_{m1}\right) - v_{\pi 2}\left(\frac{1}{r_{\pi 2}} + g_{m2}\right) = 0$$

Substitute $v_{\pi 2} + v_i$ for $v_{\pi 1}$.

$$-v_{\pi 2}\left(\frac{1}{r_{\pi 1}} + g_{m1}\right) - v_i\left(\frac{1}{r_{\pi 1}} + g_{m1}\right) - v_{\pi 2}\left(\frac{1}{r_{\pi 2}} + g_{m2}\right) = 0$$

Here, $g_{m1} = g_{m2}$ and $r_{\pi 1} = r_{\pi 2}$.

$$-v_{\pi 2}\left(\frac{1}{r_\pi} + g_m\right) - v_i\left(\frac{1}{r_\pi} + g_m\right) - v_{\pi 2}\left(\frac{1}{r_\pi} + g_m\right) = 0$$
$$-v_{\pi 2} - v_i - v_{\pi 2} = 0$$
$$-2 v_{\pi 2} - v_i = 0$$

Substitute $-\frac{v_o}{g_m R_L}$ for $v_{\pi 2}$.

$$-2\left(-\frac{v_o}{g_m R_L}\right) = v_i$$
$$\frac{2 v_o}{g_m R_L} = v_i$$

$$\frac{v_o}{v_i} = \frac{g_m R_L}{2} \quad \text{..... (1)}$$

Therefore, the expression for the voltage gain $\frac{v_o}{v_i}$ is $\frac{g_m R_L}{2}$.

Step 6 of 10

Consider the expression for the voltage v_i in terms of the voltage v_{sig}.

$$v_i = v_{sig}\left(\frac{R_{in}}{R_{in} + R_{sig}}\right)$$

Therefore, the expression for the voltage v_i is $v_{sig}\left(\frac{R_{in}}{R_{in} + R_{sig}}\right)$.

Step 7 of 10

Recall equation (1).

$$\frac{v_o}{v_i} = \frac{g_m R_L}{2}$$

Substitute $v_{sig}\left(\frac{R_{in}}{R_{in} + R_{sig}}\right)$ for v_i.

$$\frac{v_o}{v_{sig}\left(\frac{R_{in}}{R_{in} + R_{sig}}\right)} = \frac{g_m R_L}{2}$$
$$\frac{v_o}{v_{sig}} = \left(\frac{R_{in}}{R_{in} + R_{sig}}\right)\frac{g_m R_L}{2}$$

Substitute $2 r_\pi$ for R_{in}.

$$\frac{v_o}{v_{sig}} = \left(\frac{2 r_\pi}{2 r_\pi + R_{sig}}\right)\frac{g_m R_L}{2}$$
$$= \frac{g_m R_L r_\pi}{2 r_\pi + R_{sig}}$$

Here, the term $\frac{v_o}{v_{sig}}$ represents the overall voltage gain G_v.

$$G_v = \frac{g_m R_L r_\pi}{2 r_\pi + R_{sig}} \quad \text{..... (2)}$$

Therefore, the expression for the overall voltage gain G_v is $\frac{g_m R_L r_\pi}{2 r_\pi + R_{sig}}$.

Step 8 of 10

Calculate the value of the transconductance g_m.

$$g_m = \frac{I}{V_T}$$

Substitute $25\ \text{mV}$ for V_T and $0.5\ \text{mA}$ for I.

$$g_m = \frac{0.5\ \text{m}}{25\ \text{m}}$$
$$= 0.02\ \text{A/V}$$

Therefore, the value of the transconductance g_m is $0.02\ \text{A/V}$.

Step 9 of 10

Calculate the value of the resistance r_π.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute 0.02 for g_m and 100 for β.

$$r_\pi = \frac{100}{0.02}$$
$$= 5\ \text{k}\Omega$$

Therefore, the value of the resistance r_π is $5\ \text{k}\Omega$.

Step 10 of 10

Recall equation (2).

$$G_v = \frac{g_m R_L r_\pi}{2 r_\pi + R_{sig}}$$

Substitute $5\ \text{k}\Omega$ for r_π, $0.02\ \text{A/V}$ for g_m and $10\ \text{k}\Omega$ for R_L and R_{sig}.

$$G_v = \frac{(0.02)(10\ \text{k})(5\ \text{k})}{2(5\ \text{k}) + 10\ \text{k}}$$
$$= \frac{1\ \text{M}}{10\ \text{k} + 10\ \text{k}}$$
$$= \frac{1\ \text{M}}{20\ \text{k}}$$
$$= 50\ \text{V/V}$$

Therefore, the value of the overall voltage gain G_v is $\boxed{50\ \text{V/V}}$.

Step 1 of 4

8.100P

Refer to Figure 8.48 (c) from the text book.

Draw the small signal equivalent circuit diagram.

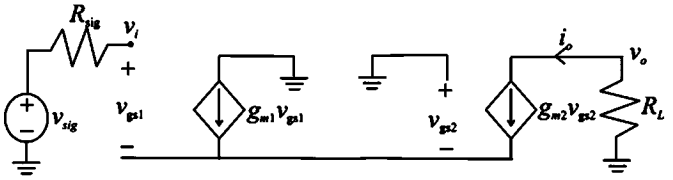


Figure 1

Step 2 of 4

Consider the expressions.

$g_{m1}v_{gs1} = -i_o$ and $g_{m2}v_{gs2} = i_o$

So that,

$v_{gs1} = -\dfrac{i_o}{g_{m1}}$ and $v_{gs2} = \dfrac{i_o}{g_{m2}}$

Step 3 of 4

Consider the expression for the voltage v_i.

$v_i = v_{gs1} - v_{gs2}$

Substitute $\dfrac{i_o}{g_{m2}}$ for v_{gs2} and $-\dfrac{i_o}{g_{m1}}$ for v_{gs1}.

$$v_i = -\frac{i_o}{g_{m1}} - \frac{i_o}{g_{m2}}$$
$$= -i_o\left(\frac{1}{g_{m1}} + \frac{1}{g_{m2}}\right)$$

Substitute $\dfrac{v_o}{R_L}$ for i_o and v_{sig} for v_i.

$$v_{sig} = \frac{v_o}{R_L}\left(\frac{1}{g_{m1}} + \frac{1}{g_{m2}}\right)$$

Let $g_{m1} = g_{m2}$.

$$v_{sig} = \frac{v_o}{R_L}\left(\frac{2}{g_m}\right)$$
$$\frac{v_o}{v_{sig}} = \frac{R_L g_m}{2}$$

The term $\dfrac{v_o}{v_{sig}}$ represents the overall voltage gain G_v.

$$G_v = \frac{R_L g_m}{2} \quad \text{..... (1)}$$

Therefore, the expression for overall voltage gain is $\dfrac{R_L g_m}{2}$.

Step 4 of 4

Recall equation (1).

$$G_v = \frac{R_L g_m}{2}$$

Substitute $5\ \text{mA/V}$ for g_m and $10\ \text{k}\Omega$ for R_L.

$$G_v = \frac{(10\ \text{k})(5\ \text{m})}{2}$$
$$= \frac{50}{2}$$
$$= 25\ \text{V/V}$$

Therefore, the value of the overall voltage gain G_v is $\boxed{25\ \text{V/V}}$.

Step 1 of 7

Refer to the circuits in Figure P7.86 in the textbook.

Calculate the trans-conductance (g_m).

$$g_m = \frac{I_C}{V_T}$$

Here,

The collector bias current is I_C

The thermal equivalent voltage is $V_T = 25\ \text{mV}$.

Substitute 0.5 mA for I_C and 25 mV for V_T.

$$\begin{aligned} g_m &= \frac{0.5\ \text{mA}}{25\ \text{mV}} \\ &= \frac{0.5\times10^{-3}}{25\times10^{-3}} \\ &= 20\times10^{-3} \\ &= 20\ \text{mA/V} \end{aligned}$$

Calculate the small signal equivalent base to emitter resistance.

$$r_\pi = \frac{\beta}{g_m}$$

Here,

Consider that the current gain β is, 100.

Substitute 100 for β and 20 mA/V for g_m .

$$\begin{aligned} r_\pi &= \frac{100}{20\times10^{-3}} \\ &= 5\times10^{3} \\ &= 5\ \text{k}\Omega \end{aligned}$$

Step 2 of 7

(a)

Refer to Figure P7.86 (a) in the textbook for a circuit diagram of a common-emitter amplifier.

Refer to Figure 7.1 (d) in the textbook for the small-signal equivalent circuit of current-source or active-loaded common-emitter amplifier. Neglect the resistance, r_o .

Calculate the output of the circuit.

$$\begin{aligned} v_o &= -g_m v_\pi R_C \\ &= -g_m R_C \left(\frac{r_\pi}{r_\pi + R_{sig}}\right) v_{sig} \end{aligned}$$

Calculate the overall gain, A_M of the common-emitter amplifier.

$$\begin{aligned} A_M &= \frac{v_o}{v_{sig}} \\ &= -g_m R_C \left(\frac{r_\pi}{r_\pi + R_{sig}}\right) \end{aligned}$$

Substitute 20 mA/V for g_m , $10\ \text{k}\Omega$ for R_C , $10\ \text{k}\Omega$ for R_{sig} , and $5\ \text{k}\Omega$ for r_π .

$$\begin{aligned} A_M &= -\left(20\times10^{-3}\right)\left(10\times10^{3}\right)\left[\frac{5\times10^{3}}{\left(5\times10^{3}\right)+\left(10\times10^{3}\right)}\right] \\ &= -200\left(\frac{5\times10^{3}}{15\times10^{3}}\right) \\ &= -66.67 \end{aligned}$$

Therefore, the overall gain, A_M is, $\boxed{-66.67\ \text{V/V}}$.

Step 3 of 7

(b)

Refer to Figure P7.86 (b) in the textbook for a circuit diagram of a cascode amplifier.

Refer to Figure 7.17 (b) in the textbook for the small-signal equivalent circuit representation of the output of the cascode amplifier. Neglect the resistance, r_o .

Calculate the overall gain, A_M of the cascode amplifier.

$$A_M = -\beta_1\alpha_2\left(\frac{R_C}{r_\pi + R_{sig}}\right)$$

Find the gain α_2 .

$$\begin{aligned} \alpha_2 &= \frac{\beta_2}{1+\beta_2} \\ &= \frac{100}{1+100} \\ &= 0.99 \end{aligned}$$

Substitute 100 for β_1 , 0.99 for α_2 , $10\ \text{k}\Omega$ for R_C , $10\ \text{k}\Omega$ for R_{sig} , and $5\ \text{k}\Omega$ for r_π .

$$\begin{aligned} A_M &= -(100)(0.99)\left[\frac{10\times10^{3}}{\left(5\times10^{3}\right)+\left(10\times10^{3}\right)}\right] \\ &= -99\left(\frac{10\times10^{3}}{15\times10^{3}}\right) \\ &= -66\ \text{V/V} \end{aligned}$$

Therefore, the overall gain, A_M is, $\boxed{-66\ \text{V/V}}$.

Step 4 of 7

(c)
Refer to Figure P7.86 (c) in the textbook for a circuit diagram of a common collector-common base cascade amplifier.

Calculate the overall gain, A_M of the common collector-common base cascade amplifier.

$$A_M = \frac{\beta R_C}{2r_\pi + R_{sig}}$$

Substitute 100 for β, $10\ \text{k}\Omega$ for R_C , $10\ \text{k}\Omega$ for R_{sig} , and $5\ \text{k}\Omega$ for r_π .

$$\begin{aligned} A_M &= \frac{(100)\left(10\times10^{3}\right)}{2\left(5\times10^{3}\right)+\left(10\times10^{3}\right)} \\ &= \frac{10^{6}}{2\left(10\times10^{3}\right)} \\ &= 50\ \text{V/V} \end{aligned}$$

Therefore, the overall gain, A_M is, $\boxed{50\ \text{V/V}}$.

Step 5 of 7

(d)

Refer to Figure P7.86 (d) in the textbook for a circuit diagram of a common collector-common emitter cascade amplifier.

Calculate the overall gain, A_M of the common collector-common emitter cascade amplifier.

$$A_M = -\frac{(\beta_1+1)\beta_2 R_C}{r_{\pi 1} + R_{sig} + (\beta_1+1)r_{\pi 2}}$$

Substitute 100 for β_1 **and** β_2 , $10\ \text{k}\Omega$ for R_C , $10\ \text{k}\Omega$ for R_{sig} , and $5\ \text{k}\Omega$ for $r_{\pi 1}$ **and** $r_{\pi 2}$.

$$\begin{aligned} A_M &= -\frac{(100+1)(100)\left(10\times10^{3}\right)}{\left(5\times10^{3}\right)+\left(10\times10^{3}\right)+(101)\left(5\times10^{3}\right)} \\ &= -\frac{101\times10^{6}}{520\times10^{3}} \\ &= -194\ \text{V/V} \end{aligned}$$

Therefore, the overall gain, A_M is, $\boxed{-194\ \text{V/V}}$.

Step 6 of 7

(e)

Refer to Figure P7.86 (e) in the textbook for a circuit diagram of the folded cascode amplifier.

Calculate the overall gain, A_M of the folded cascode amplifier.

$$A_M = -\beta_1\alpha_2\left(\frac{R_C}{r_\pi + R_{sig}}\right)$$

Substitute 100 for β_1 , 0.99 for α_2 , $10\ \text{k}\Omega$ for R_C , $10\ \text{k}\Omega$ for R_{sig} , and $5\ \text{k}\Omega$ for r_π .

$$\begin{aligned} A_M &= -(100)(0.99)\left[\frac{10\times10^{3}}{\left(5\times10^{3}\right)+\left(10\times10^{3}\right)}\right] \\ &= -99\left(\frac{10\times10^{3}}{15\times10^{3}}\right) \\ &= -66\ \text{V/V} \end{aligned}$$

Therefore, the overall gain, A_M is, $\boxed{-66\ \text{V/V}}$.

Step 7 of 7

(f)

Refer to Figure P7.86 (f) in the textbook for a circuit diagram of a common collector-common base cascade amplifier.

Calculate the overall gain, A_M of the common collector-common base cascade amplifier.

$$A_M = -\frac{(\beta_1+1)\alpha_2 R_C}{R_{sig} + (\beta_1+1)2r_e}$$

Find the small-signal emitter resistance, r_e .

$$\begin{aligned} r_e &= \frac{V_T}{I_e} \\ &= \frac{25\ \text{mV}}{0.5\ \text{mA}} \\ &= 50\ \Omega \end{aligned}$$

Substitute 100 for β_1 , 0.99 for α_2 , $10\ \text{k}\Omega$ for R_C , $10\ \text{k}\Omega$ for R_{sig} , and $50\ \Omega$ for r_e .

$$\begin{aligned} A_M &= \frac{(100+1)(0.99)\left(10\times10^{3}\right)}{\left(10\times10^{3}\right)+(100+1)2(50)} \\ &= \frac{10^{6}}{20.1\times10^{3}} \\ &= 50\ \text{V/V} \end{aligned}$$

Therefore, the overall gain, A_M is, $\boxed{50\ \text{V/V}}$.

Step 1 of 4

(a)

According to ohms law relation between V, I and R is $V = RI$

Resistance R is $1\ \text{k}\Omega$

Voltage V is $5\ \text{V}$

$$V = RI$$
$$5\ \text{V} = (1000\ \Omega)I$$
$$I = \frac{5\ \text{V}}{1000\ \Omega}$$
$$I = 5\ \text{mA}$$

Current I is $\boxed{5\ \text{mA}}$

Step 2 of 4

(b)

Voltage V is $5\ \text{V}$

Current I is $1\ \text{mA}$

$$V = RI$$
$$5\ \text{V} = R(1\times10^{-3}\ \text{A})$$
$$R = \frac{5\ \text{V}}{10^{-3}\ \text{A}}$$
$$R = 5\ \text{k}\Omega$$

Resistance R is $\boxed{5\ \text{k}\Omega}$

Step 3 of 4

(c)

Resistance R is $10\ \text{k}\Omega$

Current I is $0.1\ \text{mA}$

$$V = RI$$
$$V = (10\times10^{3}\ \Omega)(0.1\times10^{-3}\ \text{A})$$
$$V = 1\ \text{V}$$

Voltage V is $\boxed{1\ \text{V}}$

Step 4 of 4

(d)

Resistance R is $100\ \Omega$

Voltage V is $5\ \text{V}$

$$V = RI$$
$$5\ \text{V} = (100\ \Omega)I$$
$$I = \frac{5\ \text{V}}{100\ \Omega}$$
$$I = 50\ \text{mA}$$

Current I is $\boxed{50\ \text{mA}}$

1.002E

Calculate the source resistance.

$$R_s = \frac{v_{oc}}{i_{sc}}$$

Here,

v_{oc}, is the open circuit voltage.

i_{sc}, is the short circuit voltage.

Substitute $10\ \text{mV}$ for v_{oc} and $10\ \mu\text{A}$ for i_{sc}.

$$\begin{aligned} R_s &= \frac{10\times10^{-3}}{10\times10^{-6}} \\ &= \frac{1}{1\times10^{-3}} \\ &= 1\times10^{3} \\ &= 1\ \text{k}\Omega \end{aligned}$$

Therefore, the source resistance, R_s is $\boxed{1\ \text{k}\Omega}$.

Step 1 of 7

(a)

Value of resistance R is $1\ \text{k}\Omega$

Current flowing through $1\ \text{k}\Omega$ resistor is $20\ \text{mA}$

Power (P) dissipated in the resistor is I^2R

$$P = I^2R$$
$$= \left(20\times10^{-3}\right)^2(1000\ \Omega)$$
$$= 0.4\ \text{W}$$

Power (P) dissipated in $1\ \text{k}\Omega$ resistor is $0.4\ \text{W}$

For safe operation standard component having power rating greater than $0.4\ \text{W}$ should be selected and hence $1/2\ \text{W}$ or $1\ \text{W}$ or $2\ \text{W}$ power rating components are used.

Step 2 of 7

(b)

Value of resistance R is $1\ \text{k}\Omega$

Current flowing through $1\ \text{k}\Omega$ resistor is $40\ \text{mA}$

Power (P) dissipated in the resistor is I^2R

$$P = I^2R$$
$$= \left(40\times10^{-3}\right)^2(1000\ \Omega)$$
$$= 1.6\ \text{W}$$

Power (P) dissipated in $1\ \text{k}\Omega$ resistor is $1.6\ \text{W}$

For safe operation standard component having power rating greater than $1.6\ \text{W}$ should be selected and hence $2\ \text{W}$ power rating components are used

---

Step 3 of 7

(c)

Value of resistance R is $100\ \text{k}\Omega$

Current flowing through $100\ \text{k}\Omega$ resistor is $1\ \text{mA}$

Power (P) dissipated in the resistor is I^2R

$$P = I^2R$$
$$= \left(1\times10^{-3}\right)^2(100000\ \Omega)$$
$$= 0.1\ \text{W}$$

Power (P) dissipated in $100\ \text{k}\Omega$ resistor is $0.1\ \text{W}$

For safe operation standard component having power rating greater than $0.1\ \text{W}$ should be selected and hence $1/8\ \text{W}$ or $1/4\ \text{W}$ or $1/2\ \text{W}$ or $1\ \text{W}$ or $2\ \text{W}$ power rating components are used

---

Step 4 of 7

(d)

Value of resistance R is $10\ \text{k}\Omega$

Current flowing through $10\ \text{k}\Omega$ resistor is $4\ \text{mA}$

Power (P) dissipated in the resistor is I^2R

$$P = I^2R$$
$$= \left(4\times10^{-3}\right)^2(10000\ \Omega)$$
$$= 0.16\ \text{W}$$

Power (P) dissipated in $10\ \text{k}\Omega$ resistor is $0.16\ \text{W}$

For safe operation standard component having power rating greater than $0.16\ \text{W}$ should be selected and hence $1/4\ \text{W}$ or $1/2\ \text{W}$ or $1\ \text{W}$ or $2\ \text{W}$ power rating components are used

---

Step 5 of 7

(e)

Value of resistance R is $1\ \text{k}\Omega$

Voltage drop across the resistance is $20\ \text{V}$

Power (P) dissipated in the resistor is $\dfrac{V^2}{R}$

$$P = \frac{V^2}{R}$$
$$= \frac{(20)^2}{1\ \text{k}\Omega}$$
$$= 0.4\ \text{W}$$

Power (P) dissipated in $1\ \text{k}\Omega$ resistor is $0.4\ \text{W}$

For safe operation standard component having power rating greater than $0.4\ \text{W}$ should be selected and hence $1/2\ \text{W}$ or $1\ \text{W}$ or $2\ \text{W}$ power rating components are used

---

Step 6 of 7

(f)

Value of resistance R is $1\ \text{k}\Omega$

Voltage drop across the resistance is $11\ \text{V}$

Power (P) dissipated in the resistor is $\dfrac{V^2}{R}$

$$P = \frac{V^2}{R}$$
$$= \frac{(11)^2}{1\ \text{k}\Omega}$$
$$= 0.121\ \text{W}$$

Step 7 of 7

Power (P) dissipated in $1\ \text{k}\Omega$ resistor is $0.121\ \text{W}$

For safe operation standard component having power rating greater than $0.121\ \text{W}$ should be selected and hence $1/8\ \text{W}$ or $1/4\ \text{W}$ or $1/2\ \text{W}$ or $1\ \text{W}$ or $2\ \text{W}$ power rating components are used

Step 1 of 6

Draw the circuit diagram.

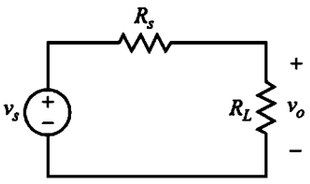


Figure 1

Step 2 of 6

The output voltage that appears across the load resistance is,

$$v_o = v_s\left(\frac{R_L}{R_L + R_s}\right)$$

Calculate the output voltage for $R_L = 100\ \text{k}\Omega$.

Substitute 100×10^3 for R_L, 1×10^3 for R_s and 10 mV for v_s.

$$\begin{aligned} v_o &= v_s\left(\frac{R_L}{R_L + R_s}\right) \\ &= \left(10\times10^{-3}\right)\left(\frac{100\times10^3}{100\times10^3 + 1\times10^3}\right) \\ &= \left(10\times10^{-3}\right)\left(\frac{100}{101}\right) \\ &= 9.9\ \text{mV} \end{aligned}$$

Therefore, the output voltage, v_o for $R_L = 100\ \text{k}\Omega$ is $\boxed{9.9\ \text{mV}}$.

Step 3 of 6

Calculate the output voltage for $R_L = 10\ \text{k}\Omega$.

$$v_o = v_s\left(\frac{R_L}{R_L + R_s}\right)$$

Substitute 10×10^3 for R_L, 1×10^3 for R_s and 10 mV for v_s.

$$\begin{aligned} v_o &= v_s\left(\frac{R_L}{R_L + R_s}\right) \\ &= \left(10\times10^{-3}\right)\left(\frac{10\times10^3}{10\times10^3 + 1\times10^3}\right) \\ &= \left(10\times10^{-3}\right)\left(\frac{10}{11}\right) \\ &= 9.1\ \text{mV} \end{aligned}$$

Therefore, the output voltage, v_o for $R_L = 10\ \text{k}\Omega$ is $\boxed{9.1\ \text{mV}}$.

Step 4 of 6

Calculate the output voltage for $R_L = 1\ \text{k}\Omega$.

$$v_o = v_s\left(\frac{R_L}{R_L + R_s}\right)$$

Substitute 1×10^3 for R_L, 1×10^3 for R_s and 10 mV for v_s.

$$\begin{aligned} v_o &= v_s\left(\frac{R_L}{R_L + R_s}\right) \\ &= \left(10\times10^{-3}\right)\left(\frac{1\times10^3}{1\times10^3 + 1\times10^3}\right) \\ &= \left(10\times10^{-3}\right)\left(\frac{1}{2}\right) \\ &= 5\ \text{mV} \end{aligned}$$

Therefore, the output voltage, v_o for $R_L = 1\ \text{k}\Omega$ is $\boxed{5\ \text{mV}}$.

Step 5 of 6

Calculate the output voltage for $R_L = 100\ \Omega$.

$$v_o = v_s\left(\frac{R_L}{R_L + R_s}\right)$$

Substitute 100 for R_L, 1×10^3 for R_s and 10 mV for v_s.

$$\begin{aligned} v_o &= v_s\left(\frac{R_L}{R_L + R_s}\right) \\ &= \left(10\times10^{-3}\right)\left(\frac{100}{100 + 1\times10^3}\right) \\ &= \left(10\times10^{-3}\right)\left(\frac{100}{1100}\right) \\ &= 0.9\ \text{mV} \end{aligned}$$

Therefore, the output voltage, v_o for $R_L = 100\ \Omega$ is $\boxed{0.9\ \text{mV}}$.

Step 6 of 6

The output voltage is 80 % of the source voltage.
That is, $v_o = 0.8\ v_s$.

Write the formula for output voltage.

$$v_o = v_s\left(\frac{R_L}{R_L + R_s}\right)$$

Substitute $1\ \text{k}\Omega$ for R_s and $0.8v_s$ for v_o.

$$\begin{aligned} 0.8v_s &= v_s\left(\frac{R_L}{R_L + 1\times10^3}\right) \\ 0.8 &= \left(\frac{R_L}{R_L + 1\times10^3}\right) \\ 0.8R_L + 800 &= R_L \\ 0.2R_L &= 800 \\ R_L &= 4000\ \Omega \\ &= 4\ \text{k}\Omega \end{aligned}$$

Therefore, the load resistance, R_L is $\boxed{4\ \text{k}\Omega}$.

Step 1 of 5

(a)

Resistance R is $1\ \text{k}\Omega$

Current I is $5\ \text{mA}$

According to ohms law $V = IR$

$$V = IR$$
$$= (5\times10^{-3}\ \text{A})(1000\ \Omega)$$
$$= 5\ \text{V}$$

Voltage V is $\boxed{5\ \text{V}}$

According to power law $P = VI$

$$P = VI$$
$$= (5\ \text{V})(5\times10^{-3}\ \text{A})$$
$$= 0.025\ \text{W}$$

Power P is $\boxed{0.025\ \text{W}}$

Step 2 of 5

(b)

Voltage V is $5\ \text{V}$

Current I is $1\ \text{mA}$

According to ohms law $V = IR$

$$V = IR$$
$$5\ \text{V} = (1\times10^{-3}\ \text{A})R$$
$$R = \frac{5\ \text{V}}{(1\times10^{-3}\ \text{A})}$$
$$R = 5\ \text{k}\Omega$$

Resistance R is $\boxed{5\ \text{k}\Omega}$

According to power law $P = VI$

$$P = VI$$
$$= (5\ \text{V})(1\times10^{-3}\ \text{A})$$
$$= 0.005\ \text{W}$$

Power P is $\boxed{0.005\ \text{W}}$

Step 3 of 5

(c)

Voltage V is $10\ \text{V}$

Power P is $100\ \text{mW}$

According to power law $P = VI$

$$P = VI$$
$$(100\times10^{-3}\ \text{W}) = (10\ \text{V})I$$
$$I = \frac{(100\times10^{-3}\ \text{W})}{(10\ \text{V})}$$
$$I = 10\ \text{mA}$$

Current I is $\boxed{10\ \text{mA}}$

According to ohms law $V = IR$

$$V = IR$$
$$10\ \text{V} = (10\times10^{-3}\ \text{A})(R)$$
$$R = \frac{10\ \text{V}}{(10\times10^{-3}\ \text{A})}$$
$$R = 1\ \text{k}\Omega$$

Resistance R is $\boxed{1\ \text{k}\Omega}$

Step 4 of 5

(d)

Current I is $1\ \text{mA}$

Power P is $1\ \text{mW}$

According to power law $P = VI$

$$P = VI$$
$$(1\times10^{-3}\ \text{W}) = V(1\times10^{-3}\ \text{A})$$
$$V = 1\ \text{V}$$

Voltage V is $\boxed{1\ \text{V}}$

According to ohms law $V = IR$

$$V = IR$$
$$1\ \text{V} = (1\times10^{-3}\ \text{A})(R)$$
$$R = \frac{1\ \text{V}}{(1\times10^{-3}\ \text{A})}$$
$$R = 1\ \text{k}\Omega$$

Resistance R is $\boxed{1\ \text{k}\Omega}$

Step 5 of 5

(e)

Resistance R is $1\ \text{k}\Omega$

Power P is $1\ \text{W}$

Power P is I^2R

$$P = I^2R$$
$$1\ \text{W} = I^2(1000\ \Omega)$$
$$I = \sqrt{\frac{1}{1000}}\ \text{A}$$
$$I = 0.032\ \text{A}$$

Current I is $\boxed{32\ \text{mA}}$

According to ohms law $V = IR$

$$V = IR$$
$$= (32\times10^{-3}\ \text{A})(1000\ \Omega)$$
$$= 32\ \text{V}$$

Voltage V is $\boxed{32\ \text{V}}$

Step 1 of 6

Draw the circuit diagram.

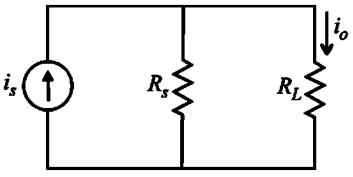


Figure 1

Step 2 of 6

Write the expression for output current using current division rule.

$$i_o = i_s\left(\frac{R_s}{R_s + R_L}\right)$$

Calculate the output voltage for $R_L = 1\ \text{k}\Omega$ by substituting 1×10^3 for R_L , 100×10^3 for R_s and 10×10^{-6} for i_s .

$$\begin{aligned} i_o &= \left(10\times10^{-6}\right)\left(\frac{100\times10^3}{100\times10^3 + 1\times10^3}\right) \\ &= \left(10\times10^{-6}\right)\left(\frac{100}{101}\right) \\ &= 9.9\ \mu\text{A} \end{aligned}$$

Therefore, the output current, i_o for $R_L = 1\ \text{k}\Omega$ is $\boxed{9.9\ \mu\text{A}}$.

Step 3 of 6

Calculate the output voltage for $R_L = 10\ \text{k}\Omega$ by substituting 10×10^3 for R_L , 100×10^3 for R_s and 10×10^{-6} for i_s in the output current expression.

$$\begin{aligned} i_o &= \left(10\times10^{-6}\right)\left(\frac{100\times10^3}{100\times10^3 + 10\times10^3}\right) \\ &= \left(10\times10^{-6}\right)\left(\frac{100}{110}\right) \\ &= 9.1\ \mu\text{A} \end{aligned}$$

Therefore, the output current, i_o for $R_L = 10\ \text{k}\Omega$ is $\boxed{9.1\ \mu\text{A}}$.

Step 4 of 6

Calculate the output voltage for $R_L = 100\ \text{k}\Omega$ by substituting 100×10^3 for R_L , 100×10^3 for R_s and 10×10^{-6} for i_s in the output current expression.

$$\begin{aligned} i_o &= \left(10\times10^{-6}\right)\left(\frac{100\times10^3}{100\times10^3 + 100\times10^3}\right) \\ &= \left(10\times10^{-6}\right)\left(\frac{100}{200}\right) \\ &= 5\ \mu\text{A} \end{aligned}$$

Therefore, the output current, i_o for $R_L = 100\ \text{k}\Omega$ is $\boxed{5\ \mu\text{A}}$.

Step 5 of 6

Calculate the output voltage for $R_L = 1\ \text{M}\Omega$ by substituting 1×10^6 for R_L , 100×10^3 for R_s and 10×10^{-6} for i_s in the output current expression.

$$\begin{aligned} i_o &= \left(10\times10^{-6}\right)\left(\frac{100\times10^3}{100\times10^3 + 1\times10^6}\right) \\ &= \left(10\times10^{-6}\right)\left(\frac{100}{1100}\right) \\ &= 0.9\ \mu\text{A} \end{aligned}$$

Therefore, the output current, i_o for $R_L = 100\ \text{k}\Omega$ is $\boxed{0.9\ \mu\text{A}}$.

Step 6 of 6

The output current is 80 % of the source current.

That is, $i_o = 0.8i_s$.

Write the formula for output current.

$$i_o = i_s\left(\frac{R_s}{R_L + R_s}\right)$$

Substitute $100\ \text{k}\Omega$ for R_s and $0.8i_s$ for i_o .

$$0.8i_s = i_s\left(\frac{100\times10^3}{R_L + 100\times10^3}\right)$$

$$0.8 = \left(\frac{100\times10^3}{R_L + 100\times10^3}\right)$$

$$0.8R_L + 80\times10^3 = 100\times10^3$$

$$0.8R_L = 20\times10^3$$

$$\begin{aligned} R_L &= 25\times10^3\ \Omega \\ &= 25\ \text{k}\Omega \end{aligned}$$

Therefore, the load resistance, R_L is $\boxed{25\ \text{k}\Omega}$.

Step 1 of 19

1.004P

Determine different resistances with different possible combinations like parallel, series, and series-parallel.

As possible parallel combinations:

Connect all the three resistors in parallel as shown in Figure 1.

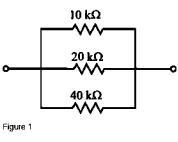


Figure 1

Step 2 of 19

Calculate the equivalent resistance.

$$\begin{aligned}R_{eq1} &= (10\times10^3)\,\|\,(20\times10^3)\,\|\,(40\times10^3)\\ &= (10\times10^3)\,\|\,(20\times10^3\,\|\,40\times10^3)\\ &= 10\times10^3\,\|\left(\frac{(20\times10^3)(40\times10^3)}{20\times10^3+40\times10^3}\right)\\ &= 10\times10^3\,\|\,(13.333\times10^3)\\ &= \frac{(10\times10^3)(13.333\times10^3)}{10\times10^3+13.333\times10^3}\\ &= \frac{133.33\times10^3}{23.333}\\ &= 5.714\times10^3\ \Omega\end{aligned}$$

Therefore, the equivalent resistance value is, $R_{eq1} = 5.714\,k\Omega$.

Step 3 of 19

Connect the two resistors **10 kΩ and 20 kΩ** in parallel as shown in Figure 2.

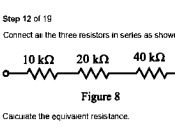


Figure 2

Step 4 of 19

Calculate the equivalent resistance.

$$\begin{aligned}R_{eq2} &= (10\times10^3)\,\|\,(20\times10^3)\\ &= \frac{(10\times10^3)(20\times10^3)}{10\times10^3+20\times10^3}\\ &= \frac{200\times10^3}{30}\\ &= 6.67\times10^3\ \Omega\end{aligned}$$

Therefore, the equivalent resistance value is, $R_{eq2} = 6.67\,k\Omega$.

Step 5 of 19

Connect the two resistors **20 kΩ and 40 kΩ** in parallel as shown in Figure 3.

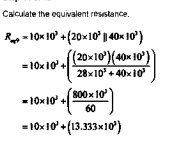


Figure 3

Step 6 of 19

Calculate the equivalent resistance

$$\begin{aligned}R_{eq3} &= 20\times10^3\,\|\,40\times10^3\\ &= \frac{(20\times10^3)(40\times10^3)}{20\times10^3+40\times10^3}\\ &= \frac{800\times10^3}{60}\\ &= 13.333\times10^3\ \Omega\end{aligned}$$

Therefore, the equivalent resistance value is, $R_{eq3} = 13.333\,k\Omega$.

Step 7 of 19

Connect the two resistors **10 kΩ and 40 kΩ** in parallel as shown in Figure 4.

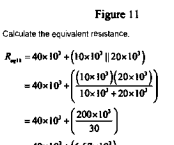


Figure 4

Step 8 of 19

Calculate the equivalent resistance.

$$\begin{aligned}R_{eq4} &= (10\times10^3)\,\|\,(40\times10^3)\\ &= \frac{(10\times10^3)(40\times10^3)}{10\times10^3+40\times10^3}\\ &= \frac{400\times10^3}{50}\\ &= 8\,k\Omega\end{aligned}$$

Therefore, the equivalent resistance value is, $R_{eq4} = 8\,k\Omega$.

Step 9 of 19

As possible series combinations:

Connect the two resistors **10 kΩ and 20 kΩ** are in series as shown in Figure 5.

10 kΩ **20 kΩ**

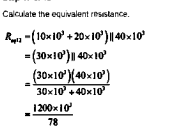


Figure 5

Calculate the equivalent resistance.

$$\begin{aligned}R_{eq5} &= 10\times10^3+20\times10^3\\ &= 30\times10^3\end{aligned}$$

Therefore, the equivalent resistance value is, $R_{eq5} = 30\,k\Omega$.

Step 10 of 19

Connect the two resistors **10 kΩ and 40 kΩ** are in series as shown in Figure 6.

10 kΩ **40 kΩ**

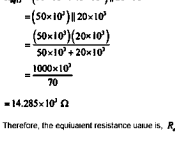


Figure 6

Calculate the equivalent resistance.

$$\begin{aligned}R_{eq6} &= 10\times10^3+40\times10^3\\ &= 50\times10^3\end{aligned}$$

Therefore, the equivalent resistance value is, $R_{eq6} = 50\,k\Omega$.

Step 11 of 19

Connect the two resistors **20 kΩ and 40 kΩ** are in series as shown in Figure 7.

20 kΩ **40 kΩ**

Figure 7

Calculate the equivalent resistance.

$$\begin{aligned}R_{eq7} &= 20\times10^3+40\times10^3\\ &= 60\times10^3\end{aligned}$$

Therefore, the equivalent resistance value is, $R_{eq7} = 60\,k\Omega$.

Step 12 of 19

Connect all the three resistors in series as shown in Figure 8.

10 kΩ **20 kΩ** **40 kΩ**

Figure 8

Calculate the equivalent resistance.

$$\begin{aligned}R_{eq8} &= 10\times10^3+20\times10^3+40\times10^3\\ &= 70\times10^3\end{aligned}$$

Therefore, the equivalent resistance value is, $R_{eq8} = 70\,k\Omega$.

Step 13 of 19

All possible series-parallel combinations

Connect the parallel combination of **20 kΩ and 40 kΩ** resistor in series with **10 kΩ** resistor as shown in Figure 9

Figure 9

Step 14 of 19

Calculate the equivalent resistance.

$$\begin{aligned}R_{eq9} &= 10\times10^3+(20\times10^3\,\|\,40\times10^3)\\ &= 10\times10^3+\left(\frac{(20\times10^3)(40\times10^3)}{20\times10^3+40\times10^3}\right)\\ &= 10\times10^3+\left(\frac{800\times10^3}{60}\right)\\ &= 10\times10^3+(13.333\times10^3)\\ &= 23.333\times10^3\end{aligned}$$

Therefore, the equivalent resistance value is, $R_{eq9} = 23.333\,k\Omega$.

Step 15 of 19

Connect the parallel combination of **10 kΩ and 40 kΩ** resistor in series with **20 kΩ** resistor as shown in Figure 10.

Figure 10

Calculate the equivalent resistance.

$$\begin{aligned}R_{eq10} &= 20\times10^3+(10\times10^3\,\|\,40\times10^3)\\ &= 20\times10^3+\left(\frac{(10\times10^3)(40\times10^3)}{10\times10^3+40\times10^3}\right)\\ &= 20\times10^3+\left(\frac{400\times10^3}{50}\right)\\ &= 20\times10^3+(8\times10^3)\\ &= 28\times10^3\end{aligned}$$

Therefore, the equivalent resistance value is, $R_{eq10} = 28\ k\Omega$.

Step 16 of 19

Connect the parallel combination of **10 kΩ and 20 kΩ** resistor in series with **40 kΩ** resistor as shown in Figure 11.

Figure 11

Calculate the equivalent resistance.

$$\begin{aligned}R_{eq11} &= 40\times10^3+(10\times10^3\,\|\,20\times10^3)\\ &= 40\times10^3+\left(\frac{(10\times10^3)(20\times10^3)}{10\times10^3+20\times10^3}\right)\\ &= 40\times10^3+\left(\frac{200\times10^3}{30}\right)\\ &= 40\times10^3+(6.67\times10^3)\\ &= 46.67\times10^3\ \Omega\\ &= 46.67\ k\Omega\end{aligned}$$

Therefore, the equivalent resistance value is, $R_{eq11} = 46.67\ k\Omega$.

Connect the series combination of **10 kΩ and 20 kΩ** resistor in parallel with **40 kΩ** resistor as shown in Figure 12.

Figure 12

Step 17 of 19

Calculate the equivalent resistance.

$$\begin{aligned}R_{eq12} &= ((10\times10^3+20\times10^3)\,\|\,40\times10^3\\ &= (30\times10^3)\,\|\,40\times10^3\\ &= \frac{(30\times10^3)(40\times10^3)}{30\times10^3+40\times10^3}\\ &= \frac{1200\times10^2}{78}\\ &= 17.143\times10^3\end{aligned}$$

Therefore, the equivalent resistance value is, $R_{eq12} = 17.143\ k\Omega$.

Step 18 of 19

Connect the series combination of **10 kΩ and 40 kΩ** resistor in parallel with **20 kΩ** resistor as shown in Figure 13.

Figure 13

Calculate the equivalent resistance.

$$\begin{aligned}R_{eq13} &= ((10\times10^3+40\times10^3)\,\|\,20\times10^3\\ &= (50\times10^2)\,\|\,20\times10^3\\ &= \frac{(50\times10^3)(20\times10^3)}{50\times10^3+20\times10^3}\\ &= \frac{1000\times10^3}{70}\\ &= 14.285\times10^3\ \Omega\end{aligned}$$

Therefore, the equivalent resistance value is, $R_{eq13} = 14.285\,k\Omega$.

Step 19 of 19

Connect the series combination of **20 kΩ and 40 kΩ** resistor in parallel with **10 kΩ** resistor as shown in Figure 14.

Figure 14

Calculate the equivalent resistance.

$$\begin{aligned}R_{eq14} &= (20\times10^3+40\times10^3)\,\|\,10\times10^3\\ &= (60\times10^3)\,\|\,10\times10^3\\ &= \frac{(60\times10^3)(10\times10^3)}{60\times10^3+18\times10^3}\\ &= \frac{600\times10^3}{70}\\ &= 8.57\times10^3\ \Omega\end{aligned}$$

Therefore, the equivalent resistance value is, $R_{eq14} = 8.57\,k\Omega$.

As possible individual resistors are,

$R_{eq11} = 10\,k\Omega$

$R_{eq1b} = 20\,k\Omega$

$R_{eq12} = 40\,k\Omega$

Therefore, the total possible resistances are **17**.

The values from lowest values to highest values are

$R_{eq} = 5.714\,k\Omega$, $R_{eq\,2} = 6.67\,k\Omega$, $R_{eq\,4} = 8\,k\Omega$, $R_{eq\,14} = 8.57\,k\Omega$, $R_{eq\,13} = 10\,k\Omega$, $R_{eq\,3} = 13.333\,k\Omega$, $R_{eq\,13} = 14.285\,k\Omega$, $R_{eq\,12} = 17.143\ k\Omega$, $R_{eq\,5} = 20\,k\Omega$, $R_{eq\,6} = 23.333\,k\Omega$, $R_{eq\,10} = 28\ k\Omega$, $R_{eq\,9} = 30\,k\Omega$, $R_{eq\,12} = 40\,k\Omega$, $R_{eq\,11} = 46.67\ k\Omega$ $R_{eq\,6} = 50\,k\Omega$, $R_{eq\,7} = 60\,k\Omega$, $R_{eq\,8} = 70\,k\Omega$

Step 1 of 2

1.005E

The time period of the sine-wave signal is, $T = 1\,\text{ms}$.

Calculate the frequency.

$$f = \frac{1}{T}$$

Substitute $1\,\text{ms}$ for T .

$$\begin{aligned} f &= \frac{1}{1\times10^{-3}} \\ &= 1\times10^{3} \\ &= 1000\ \text{Hz} \end{aligned}$$

Therefore, the frequency, f is $\boxed{1000\ \text{Hz}}$.

Step 2 of 2

Calculate the angular frequency.

$$\omega = 2\pi f$$

Substitute $1000\,\text{Hz}$ for f .

$$\begin{aligned} \omega &= 2\pi(1000) \\ &= 2\pi(1000) \\ &= 2\pi\times10^{3}\ \text{rad/s} \end{aligned}$$

Therefore, the angular frequency, ω is $\boxed{2\pi\times10^{3}\ \text{rad/s}}$.

Step 1 of 8

The original resistor value is, $R = 10\,\text{k}\Omega$

Calculate the equivalent resistance of two parallel connected resistors.

$$R_{eq} = R \parallel R_{Shunt}$$
$$= \frac{RR_{Shunt}}{R + R_{Shunt}}$$

The equivalent resistance value is reduced by 1%.

Calculate the shunt resistance value.

$$0.99\left(10\times10^3\right) = 10\times10^3 \parallel R_{Shunt}$$
$$\left(9.9\times10^3\right) = \frac{\left(10\times10^3\right)R_{Shunt}}{10\times10^3 + R_{Shunt}}$$
$$\left(9.9\times10^3\right)\left(10\times10^3 + R_{Shunt}\right) = \left(10\times10^3\right)R_{Shunt}$$
$$\left(9.9\right)\left(10\times10^3 + R_{Shunt}\right) = 10R_{Shunt}$$

$$99\times10^3 + 9.9R_{Shunt} = 10R_{Shunt}$$
$$99\times10^3 = 10R_{Shunt} - 9.9R_{Shunt}$$
$$99\times10^3 = 0.1R_{Shunt}$$
$$R_{Shunt} = \frac{99\times10^3}{0.1}$$

$$R_{Shunt} = 990\ \text{k}\Omega$$

Therefore, the value of shunt resistor, R_{Shunt} when the combined resistance is reduced by 1%, is

$\boxed{990\ \text{k}\Omega}$.

Step 2 of 8

The equivalent resistance value is reduced by 5%.

Calculate the shunt resistance value.

$$0.95\left(10\times10^3\right) = 10\times10^3 \parallel R_{Shunt}$$
$$\left(9.5\times10^3\right) = \frac{\left(10\times10^3\right)R_{Shunt}}{10\times10^3 + R_{Shunt}}$$
$$\left(9.5\times10^3\right)\left(10\times10^3 + R_{Shunt}\right) = \left(10\times10^3\right)R_{Shunt}$$
$$\left(9.5\right)\left(10\times10^3 + R_{Shunt}\right) = 10R_{Shunt}$$

$$95\times10^3 + 9.5R_{Shunt} = 10R_{Shunt}$$
$$95\times10^3 = 10R_{Shunt} - 9.5R_{Shunt}$$
$$95\times10^3 = 0.5R_{Shunt}$$
$$R_{Shunt} = \frac{95\times10^3}{0.5}\ \Omega$$

$$R_{Shunt} = 190\ \text{k}\Omega$$

Therefore, the value of shunt resistor, R_{Shunt} when the combined resistance is reduced by 5%, is

$\boxed{190\ \text{k}\Omega}$.

Step 3 of 8

The equivalent resistance value is reduced by 10%.

Calculate the shunt resistance value.

$$0.90\left(10\times10^3\right) = 10\times10^3 \parallel R_{Shunt}$$
$$\left(9\times10^3\right) = \frac{\left(10\times10^3\right)R_{Shunt}}{10\times10^3 + R_{Shunt}}$$
$$\left(9\times10^3\right)\left(10\times10^3 + R_{Shunt}\right) = \left(10\times10^3\right)R_{Shunt}$$
$$\left(9\right)\left(10\times10^3 + R_{Shunt}\right) = 10R_{Shunt}$$

$$90\times10^3 + 9R_{Shunt} = 10R_{Shunt}$$
$$90\times10^3 = 10R_{Shunt} - 9R_{Shunt}$$
$$90\times10^3 = 1R_{Shunt}$$
$$R_{Shunt} = \frac{90\times10^3}{1}\ \Omega$$

$$R_{Shunt} = 90\ \text{k}\Omega$$

Therefore, the value of shunt resistor, R_{Shunt} when the combined resistance is reduced by 10%, is

$\boxed{90\ \text{k}\Omega}$.

Step 4 of 8

The equivalent resistance value is reduced by 50%.

Calculate the shunt resistance value.

$$0.5\left(10\times10^3\right) = 10\times10^3 \parallel R_{Shunt}$$
$$\left(5\times10^3\right) = \frac{\left(10\times10^3\right)R_{Shunt}}{10\times10^3 + R_{Shunt}}$$
$$\left(5\times10^3\right)\left(10\times10^3 + R_{Shunt}\right) = \left(10\times10^3\right)R_{Shunt}$$
$$\left(5\right)\left(10\times10^3 + R_{Shunt}\right) = 10R_{Shunt}$$

Step 5 of 8

Simplify further.

$$50\times10^3 + 5R_{Shunt} = 10R_{Shunt}$$
$$50\times10^3 = 10R_{Shunt} - 5R_{Shunt}$$
$$50\times10^3 = 5R_{Shunt}$$
$$R_{Shunt} = \frac{50\times10^3}{5}\ \Omega$$

$$R_{Shunt} = 10\ \text{k}\Omega$$

Therefore, the value of shunt resistor, R_{Shunt} when the combined resistance is reduced by 10%, is

$\boxed{10\ \text{k}\Omega}$.

Step 6 of 8

Calculate the resultant resistance if $R_{Shunt} = 1\text{M}\Omega$.

$$R_{eq} = \frac{\left(10\times10^3\right)\left(1\times10^6\right)}{\left(10\times10^3\right) + \left(1\times10^6\right)}$$
$$= \frac{\left(10\times10^3\right)\left(1\times10^6\right)}{1.01\times10^6}$$
$$= \frac{10\times10^3}{1.01}$$
$$= 9.9\ \text{k}\Omega$$

Therefore, the resultant resistance, R_{eq} if $R_{Shunt} = 1\text{M}\Omega$ is $\boxed{9.9\ \text{k}\Omega}$.

Step 7 of 8

Calculate the resultant resistance if $R_{Shunt} = 100\,\text{k}\Omega$.

$$R_{eq} = \frac{\left(10\times10^3\right)\left(100\times10^3\right)}{\left(10\times10^3\right) + \left(100\times10^3\right)}$$
$$= \frac{\left(10\times10^3\right)\left(100\times10^3\right)}{110\times10^3}$$
$$= \frac{100\times10^3}{11}$$
$$= 9.091\ \text{k}\Omega$$

Therefore, the resultant resistance, R_{eq} if $R_{Shunt} = 100\,\text{k}\Omega$ is $\boxed{9.091\ \text{k}\Omega}$.

Step 8 of 8

If Calculate the resultant resistance if $R_{Shunt} = 10\,\text{k}\Omega$.

$$R_{eq} = \frac{\left(10\times10^3\right)\left(10\times10^3\right)}{\left(10\times10^3\right) + \left(10\times10^3\right)}$$
$$= \frac{\left(10\times10^3\right)\left(10\times10^3\right)}{20\times10^3}$$
$$= \frac{100\times10^3}{20}$$
$$= 5\ \text{k}\Omega$$

Therefore, the resultant resistance, R_{eq} if $R_{Shunt} = 10\,\text{k}\Omega$ is $\boxed{5\ \text{k}\Omega}$.

Step 1 of 3

(a)

The frequency of the sine wave is, $f = 60\,\text{Hz}$.

The time period is,

$$T = \frac{1}{f}$$

Substitute $60\ \text{Hz}$ for f .

$$\begin{aligned} T &= \frac{1}{60} \\ &= 16.7 \times 10^{-3}\ \text{s} \\ &= 16.7\ \text{ms} \end{aligned}$$

Therefore, the time period of the sine waveform, T is $\boxed{16.7\ \text{ms}}$.

Step 2 of 3

(b)

The frequency of the sine wave is, $f = 10^{-3}\,\text{Hz}$.

The time period is,

$$T = \frac{1}{f}$$

Substitute $10^{-3}\ \text{Hz}$ for f .

$$\begin{aligned} T &= \frac{1}{10^{-3}} \\ &= 10^{3} \\ &= 1000\ \text{s} \end{aligned}$$

Therefore, the time period of the sine waveform, T is $\boxed{1000\ \text{s}}$.

Step 3 of 3

(c)

The frequency of the sine wave is, $f = 1\ \text{MHz}$.

The time period is,

$$T = \frac{1}{f}$$

Substitute $1\ \text{MHz}$ for f .

$$\begin{aligned} T &= \frac{1}{1 \times 10^{6}} \\ &= 1 \times 10^{-6}\ \text{s} \\ &= 1\ \mu\text{s} \end{aligned}$$

Therefore, the time period of the sine waveform, T is $\boxed{1\ \mu\text{s}}$.

Step 1 of 3

1.006P

Refer to Figure P1.6 in the textbook.

Draw the modified circuit of Figure P1.6.

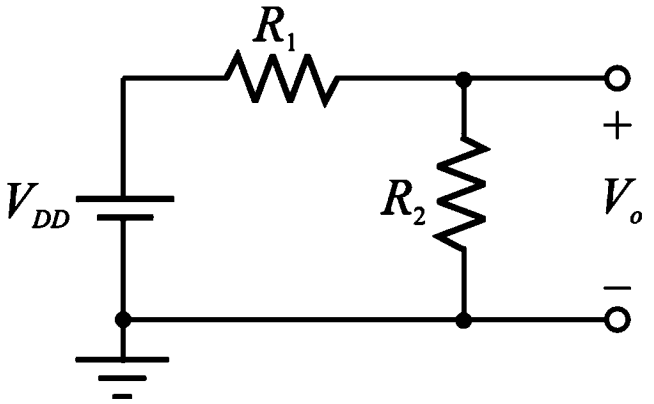


Figure 1

Step 2 of 3

Calculate the output voltage, V_o using voltage division rule.

$$V_o = \left(\frac{R_2}{R_1 + R_2}\right)(V_{DD})$$

Therefore, the output voltage, V_o is $\boxed{\left(\frac{R_2}{R_1 + R_2}\right)(V_{DD})}$.

Draw the circuit diagram, to calculate the Thevenin resistance.

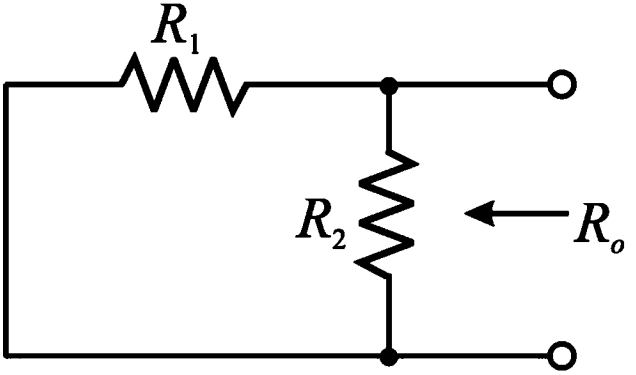


Figure 2

Step 3 of 3

From Figure 2, the equivalent output resistance is,

$$R_o = R_1 \parallel R_2$$
$$= \frac{R_1 R_2}{R_1 + R_2}$$

Therefore, the output resistance, R_o is $\boxed{\frac{R_1 R_2}{R_1 + R_2}}$.

Step 1 of 2

1.007E

Ultra High Frequency range is from $\mathbf{470\,MHz\ to\ 806\,MHz}$.

Starting channel is $\mathbf{14}$

Channel bandwidth, $\mathbf{BW = 6\,MHz}$.

Calculate the number of channels.

$$N = \frac{\Delta f}{\mathbf{BW}}$$

Substitute 6×10^6 for $\mathbf{BW}$.

$$\begin{aligned} N &= \frac{806\times10^6 - 470\times10^6}{6\times10^6} \\ &= \frac{336\times10^6}{6\times10^6} \\ &= 56 \end{aligned}$$

Therefore, the total number of channels, N is $\boxed{56}$.

Step 2 of 2

Calculate the last channel.

$$\begin{aligned} N_{\text{Last}} &= \textbf{before starting channel + total number of channels} \\ &= 13 + 56 \\ &= 69 \end{aligned}$$

Therefore, the channels that can be accommodated are $\boxed{\textbf{14 to 69}}$.

Step 1 of 5

1.007P

Voltage divider circuit shown in the figure 1

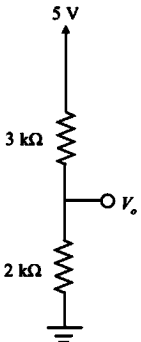


Figure 1

Step 2 of 5

Power supply voltage V_{DD} is $5\ \text{V}$

Voltage across $2\ \text{k}\Omega$ resistor is,

$$\begin{aligned} V_o &= \frac{V_{DD}}{(3+2)\ \text{k}\Omega}(2\ \text{k}\Omega) \\ &= \frac{5\ \text{V}}{(3+2)\ \text{k}\Omega}(2\ \text{k}\Omega) \\ &= 2\ \text{V} \end{aligned}$$

Output voltage V_o is $\boxed{2\ \text{V}}$

Step 3 of 5

Resistors have a manufacturing tolerance of $\pm 5\%$

Minimum value of $2\ \text{k}\Omega$ resistance is,

$$2\ \text{k}\Omega - \left(\frac{2\times 5}{100}\right)\text{k}\Omega = 1.9\ \text{k}\Omega$$

Maximum value of $2\ \text{k}\Omega$ resistance is,

$$2\ \text{k}\Omega + \left(\frac{2\times 5}{100}\right)\text{k}\Omega = 2.1\ \text{k}\Omega$$

Minimum value of $3\ \text{k}\Omega$ resistance is,

$$3\ \text{k}\Omega - \left(\frac{3\times 5}{100}\right)\text{k}\Omega = 2.85\ \text{k}\Omega$$

Maximum value of $3\ \text{k}\Omega$ resistance is,

$$2\ \text{k}\Omega + \left(\frac{2\times 5}{100}\right)\text{k}\Omega = 3.15\ \text{k}\Omega$$

Step 4 of 5

Minimum output voltages of the circuit is obtained when we use the minimum value of resistances $1.9\ \text{k}\Omega$ and maximum resistance $3.15\ \text{k}\Omega$

Voltage across $1.9\ \text{k}\Omega$ resistor is,

$$\begin{aligned} V_o &= \frac{V_{DD}}{(3.15+1.9)\ \text{k}\Omega}(1.9\ \text{k}\Omega) \\ &= \frac{5\ \text{V}}{(3.15+1.9)\ \text{k}\Omega}(1.9\ \text{k}\Omega) \\ &= 1.88\ \text{V} \end{aligned}$$

Output voltage V_o is $\boxed{1.88\ \text{V}}$

Step 5 of 5

Minimum output voltages of the circuit is obtained when we use the maximum value of resistances $2.1\ \text{k}\Omega$ and minimum value of resistance $2.85\ \text{k}\Omega$

Voltage across $2.1\ \text{k}\Omega$ resistor is,

$$\begin{aligned} V_o &= \frac{V_{DD}}{(2.85+2.1)\ \text{k}\Omega}(2.1\ \text{k}\Omega) \\ &= \frac{5\ \text{V}}{(2.85+2.1)\ \text{k}\Omega}(2.1\ \text{k}\Omega) \\ &= 2.12\ \text{V} \end{aligned}$$

Output voltage V_o is $\boxed{2.12\ \text{V}}$

Extreme output voltages of voltage divider circuit is $1.88\ \text{V}$ and $2.12\ \text{V}$

Step 1 of 6

Refer to Figure 1.5 in the textbook for square wave signal.

The Fourier series for the signal is,

$$v(t)=\frac{4V}{\pi}\left(\sin\omega_0 t+\frac{1}{3}\sin 3\omega_0 t+\frac{1}{5}\sin 5\omega_0 t+\cdots\right)$$

Calculate the total power, P.

$$\begin{aligned}P&=\frac{1}{T}\int_0^T\frac{v^2}{R}dt\\&=\frac{1}{RT}\int_0^T(V)^2\,dt\\&=\frac{V^2}{RT}\int_0^T(1)dt\\&=\frac{V^2}{RT}[t]_0^T\\&=\frac{V^2}{RT}[T-0]\\&=\frac{V^2}{RT}(T)\end{aligned}$$

$$P=\frac{V^2}{R}\quad\text{..... (1)}$$

Step 2 of 6

Calculate the total power by summing the contribution of each harmonic component.

$$\begin{aligned}P&=\frac{v^2}{R}\\&=P_1+P_3+P_5+\cdots\\&=\frac{\left(\frac{V_1}{\sqrt{2}}\right)^2}{R}+\frac{\left(\frac{V_3}{\sqrt{2}}\right)^2}{R}+\frac{\left(\frac{V_5}{\sqrt{2}}\right)^2}{R}+\cdots\\&=\frac{1}{2R}\left[\frac{4V}{\pi}\right]^2+\frac{1}{2R}\left[\frac{4V}{3\pi}\right]^2+\frac{1}{2R}\left[\frac{4V}{5\pi}\right]^2+\cdots\\&=\frac{16V^2}{2\pi^2R}\left(1+\frac{1}{9}+\frac{1}{25}+\cdots\right)\\&=\frac{V^2}{R}\left[\frac{8}{\pi^2}\left(1+\frac{1}{9}+\frac{1}{25}+\cdots\right)\right]\end{aligned}$$

Step 3 of 6

The value of $\frac{8}{\pi^2}\left(1+\frac{1}{9}+\frac{1}{25}+\cdots\right)$ is nearly equal to 1.

Consider the calculation of the infinite series in the parentheses has a sum that approaches $\frac{\pi^2}{8}$.

Therefore, the total power will be as follows:

$$P=\frac{V^2}{R}\left[\frac{8}{\pi^2}\left(\frac{\pi^2}{8}\right)\right]$$

$$P=\frac{V^2}{R}\quad\text{..... (2)}$$

From equations (1) and (2), it is clear that the two approaches are equal.

Step 4 of 6

Calculate the fraction of energy in its fundamental.

$$\begin{aligned}E_1&=\frac{8}{\pi^2}(1)\\&=0.81\end{aligned}$$

Therefore, the fraction of energy in its fundamental, E_1 is $\boxed{0.81}$.

Calculate the fraction of energy in its first five harmonics.

$$\begin{aligned}E_5&=\frac{8}{\pi^2}\left(1+\frac{1}{9}+\frac{1}{25}\right)\\&=0.93\end{aligned}$$

Therefore, the fraction of energy in its first five harmonics, E_5 is $\boxed{0.93}$.

Step 5 of 6

Calculate the fraction of energy in its first seven harmonics.

$$\begin{aligned}E_7&=\frac{8}{\pi^2}\left(1+\frac{1}{9}+\frac{1}{25}+\frac{1}{49}\right)\\&=0.95\end{aligned}$$

Therefore, the fraction of energy in its first seven harmonics, E_7 is $\boxed{0.95}$.

Calculate the fraction of energy in its first nine harmonics.

$$\begin{aligned}E_9&=\frac{8}{\pi^2}\left(1+\frac{1}{9}+\frac{1}{25}+\frac{1}{49}+\frac{1}{81}\right)\\&=0.96\end{aligned}$$

Therefore, the fraction of energy in its first nine harmonics, E_9 is $\boxed{0.96}$.

Step 6 of 6

Calculate the fraction of energy in its first three harmonics.

$$\begin{aligned}E_3&=\frac{8}{\pi^2}\left(1+\frac{1}{9}\right)\\&=0.9\\&=90\%\end{aligned}$$

Hence, the fraction of energy in its first three harmonics, E_3 is **90%**.

Therefore, **90%** of the energy of the square wave is in the first $\boxed{3}$ harmonics.

Step 1 of 8

Each resistor value is $10\ \text{k}\Omega$

Battery voltage is $9\ \text{V}$.

(i)

Connect all the resistors in series as shown in Figure 1.

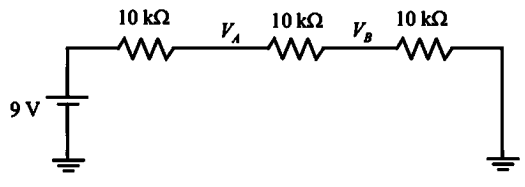


Figure 1

Step 2 of 8

Calculate the voltage, V_A using voltage divider rule.

$$V_A = \left(\frac{10\ \text{k}\Omega + 10\ \text{k}\Omega}{10\ \text{k}\Omega + 10\ \text{k}\Omega + 10\ \text{k}\Omega}\right)(9\ \text{V})$$
$$= 6\ \text{V}$$

Calculate the equivalent output resistance.

$$R_o = (10\text{k}\Omega + 10\text{k}\Omega) \| (10\text{k}\Omega)$$
$$= 20\text{k}\Omega \| 10\text{k}\Omega$$
$$= \frac{(20\text{k}\Omega)(10\text{k}\Omega)}{20\text{k}\Omega + 10\text{k}\Omega}$$
$$= 6.667\ \text{k}\Omega$$

Therefore, the output resistance, R_o is $\boxed{6.667\ \text{k}\Omega}$.

Step 3 of 8

Calculate the voltage, V_B using voltage divider rule.

$$V_B = \left(\frac{10\text{k}\Omega}{10\text{k}\Omega + 10\text{k}\Omega + 10\text{k}\Omega}\right)(9\text{V})$$
$$= \left(\frac{1}{3}\right)(9)$$
$$= 3\ \text{V}$$

Calculate the equivalent output resistance.

$$R_o = (10\text{k}\Omega) \| (10\text{k}\Omega + 10\text{k}\Omega)$$
$$= 20\text{k}\Omega \| 10\text{k}\Omega$$
$$= \frac{(20\text{k}\Omega)(10\text{k}\Omega)}{20\text{k}\Omega + 10\text{k}\Omega}$$
$$= 6.667\text{k}\Omega$$

Therefore, the output resistance, R_o is $\boxed{6.667\ \text{k}\Omega}$.

Step 4 of 8

(ii)

Draw the circuit diagram with series parallel resistors.

These are two cases,

Case (a):

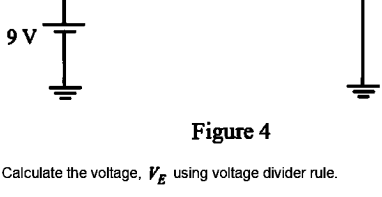


Figure 2

Step 5 of 8

Calculate the voltage, V_C using voltage divider rule.

$$V_C = \left(\frac{10\text{k}\Omega \| 10\text{k}\Omega}{10\text{k}\Omega + (10\text{k}\Omega \| 10\text{k}\Omega)}\right)(9\text{V})$$
$$= \left(\frac{5\ \text{k}\Omega}{10\ \text{k}\Omega + 5\ \text{k}\Omega}\right)(9\text{V})$$
$$= \left(\frac{5\ \text{k}\Omega}{15\ \text{k}\Omega}\right)(9)$$
$$= 3\ \text{V}$$

Calculate the equivalent output resistance.

$$R_o = (10\text{k}\Omega) \| (10\text{k}\Omega \| 10\text{k}\Omega)$$
$$= (10\text{k}\Omega) \| \left(\frac{(10\text{k}\Omega)(5\text{k}\Omega)}{10\text{k}\Omega + 5\text{k}\Omega}\right)$$
$$= (10\text{k}\Omega) \| (5\text{k}\Omega)$$
$$= \frac{(10\text{k}\Omega)(5\text{k}\Omega)}{10\text{k}\Omega + 5\text{k}\Omega}$$
$$= 3.333\ \text{k}\Omega$$

Therefore, the output resistance, R_o is $\boxed{3.333\ \text{k}\Omega}$.

Step 6 of 8

Case (b):
Draw the circuit diagram with series parallel resistors.

Figure 3

Step 7 of 8

Calculate the voltage, V_D using voltage divider rule.

$$V_D = \left(\frac{10\text{k}\Omega}{10\text{k}\Omega + (10\text{k}\Omega \| 10\text{k}\Omega)}\right)(9\text{V})$$
$$= \left(\frac{10\text{k}\Omega}{10\text{k}\Omega + \left(\frac{10\text{k}\Omega \times 10\text{k}\Omega}{10\text{k}\Omega + 10\text{k}\Omega}\right)}\right)(9\text{V})$$
$$= \left(\frac{10\text{k}\Omega}{10\text{k}\Omega + 5\ \text{k}\Omega}\right)(9\text{V})$$
$$= \left(\frac{10\text{k}\Omega}{15\text{k}\Omega}\right)(9\text{V})$$
$$= 6\ \text{V}$$

Calculate the equivalent output resistance.

$$R_o = (10\text{k}\Omega \| 10\text{k}\Omega) \| (10\text{k}\Omega)$$
$$= \left(\frac{10\text{k}\Omega \times 10\text{k}\Omega}{10\text{k}\Omega + 10\text{k}\Omega}\right) \| (10\text{k}\Omega)$$
$$= 5\text{k}\Omega \| 10\text{k}\Omega$$
$$= \frac{(10\text{k}\Omega)(5\text{k}\Omega)}{10\text{k}\Omega + 5\text{k}\Omega}$$
$$= 3.333\ \text{k}\Omega$$

Therefore, the output resistance, R_o is $\boxed{3.333\ \text{k}\Omega}$.

Step 8 of 8

(iii)

Connect two resistors in series as shown in Figure 4.

Figure 4

Calculate the voltage, V_E using voltage divider rule.

$$V_E = \left(\frac{10\text{k}\Omega}{10\text{k}\Omega + 10\text{k}\Omega}\right)(9\text{V})$$
$$= \left(\frac{10\text{k}\Omega}{20\text{k}\Omega}\right)(9\text{V})$$
$$= 4.5\ \text{V}$$

Calculate the equivalent output resistance.

$$R_o = (10\text{k}\Omega \| 10\text{k}\Omega)$$
$$= \frac{(10\text{k}\Omega)(10\text{k}\Omega)}{10\text{k}\Omega + 10\text{k}\Omega}$$
$$= 5\text{k}\Omega$$

Therefore, the output resistance, R_o is $\boxed{5\ \text{k}\Omega}$.

Therefore, the list of positive sources are $\boxed{3\text{V}, 4.5\text{V} \text{ and } 6\text{V}}$.

For generating 3V and 6V , the case (ii) is better than case (i), because it has low output resistance.

Step 1 of 5

The 4-bit digital word is, $D = b_3 b_2 b_1 b_0$

The analog signal, v_A varies from $0\,V$ to $15\,V$.

The digital word represents the different voltage levels are,

Table 1:

b_3	b_2	b_1	b_0	$D(V)$
0	0	0	0	0
0	0	0	1	1
0	0	1	0	2
0	0	1	1	3
0	1	0	0	4
0	1	0	1	5
0	1	1	0	6
0	1	1	1	7
1	0	0	0	8
1	0	0	1	9
1	0	1	0	10
1	0	1	1	11
1	1	0	0	12
1	1	0	1	13
1	1	1	0	14
1	1	1	1	15

Step 2 of 5

(a)

From table 1,

The digital word, D represent the voltage, $v_A = 0\,V$ is $\boxed{0000}$.

The digital word, D represent the voltage, $v_A = 1\,V$ is $\boxed{0001}$.

The digital word, D represent the voltage, $v_A = 2\,V$ is $\boxed{0010}$.

The digital word, D represent the voltage, $v_A = 15\,V$ is $\boxed{1111}$.

Step 3 of 5

(b)

Initially the digital word D represents, $D(b_3 b_2 b_1 b_0) = 0000$.

Change the bit b_0 from 0 to 1.

That is,

$$D(b_3 b_2 b_1 b_0) = 0001$$

The decimal equivalent of 0001 is,

$$v_A = +1\,V$$

Therefore, the change in v_A by changing the bit b_0 from 0 to 1 is $\boxed{+1\,V}$.

Change the bit, b_1 from 0 to 1.

That is,

$$D(b_3 b_2 b_1 b_0) = 0010$$

The decimal equivalent of 0010 is,

$$v_A = +2\,V$$

Therefore, the change in v_A by changing the bit b_1 from 0 to 1 is $\boxed{+2\,V}$.

Step 4 of 5

Change the bit, b_2 from 0 to 1.

That is,

$$D(b_3 b_2 b_1 b_0) = 0100$$

The decimal equivalent of 0100 is,

$$v_A = +4\,V$$

Therefore, the change in v_A by changing the bit b_2 from 0 to 1 is $\boxed{+4\,V}$.

Change the bit, b_3 from 0 to 1.

That is,

$$D(b_3 b_2 b_1 b_0) = 1000$$

The decimal equivalent of 1000 is,

$$v_A = +8\,V$$

Therefore, the change in v_A by changing the bit b_3 from 0 to 1 is $\boxed{+8\,V}$.

Step 5 of 5

(c)

The voltage is, $v_A = 5.2\,V$.

The value $v_A = 5.2\,V$ is nearer to $5\,V$.

Thus, the binary equivalent, D is $\boxed{0101}$.

Calculate the percentage error.

$$\begin{aligned}\text{error} &= \frac{5-5.2}{5}\times 100\% \\ &= -0.04\times 100\% \\ &= -4\,\%\end{aligned}$$

Therefore, the resulting error is $\boxed{-4\,\%}$.

Step 1 of 4

1.009P

Voltage divider circuit shown in the figure 1

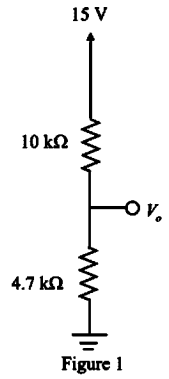


Figure 1

Step 2 of 4

Power supply voltage V_{DD} is $15\ \text{V}$

Voltage across $4.7\ \text{k}\Omega$ resistor is,

$$V_o = \frac{V_{DD}}{(10+4.7)\ \text{k}\Omega}(4.7\ \text{k}\Omega)$$
$$= \frac{15\ \text{V}}{(10+4.7)\ \text{k}\Omega}(4.7\ \text{k}\Omega)$$
$$= 4.795\ \text{V}$$

Output voltage V_o is $\boxed{4.795\ \text{V}}$

Step 3 of 4

By adding shunt resistance their equivalent resistance will decrease as the resistance is decreased voltage drop will decrease and the voltage drop in other resistor is increases.so in order to get output voltage of $5\ \text{V}$ across the $4.7\ \text{k}\Omega$ instead of $4.795\ \text{V}$ we have to shunt the $10\ \text{k}\Omega$ resistor.

Let 'x' be the shunt resistance across the $10\ \text{k}\Omega$ resistor then the equivalent resistance is,

$$\frac{10x}{10+x}$$

Voltage drop across $\frac{10x}{10+x}$ resistor is $\frac{15}{\frac{10x}{10+x}+4.7}\left(\frac{10x}{10+x}\right)$

Voltage drop across $4.7\ \text{k}\Omega$ is $5\ \text{V}$ and hence voltage drop across $\frac{10x}{10+x}$ is $10\ \text{V}$

$$10\ \text{V} = \frac{15}{\frac{10x}{10+x}+4.7}\left(\frac{10x}{10+x}\right)$$
$$10\ \text{V} = \frac{150x}{10x+47+4.7x}$$
$$(10\ \text{V})(47+14.7x) = 150x$$
$$x = \frac{470}{3}\ \text{k}\Omega$$

Shunt resistance required across $10\ \text{k}\Omega$ resistor in order to get a $5\ \text{V}$ across $4.7\ \text{k}\Omega$ resistor is $\boxed{156.7\ \text{k}\Omega}$

Step 4 of 4

In order to get output resistance of $3.33\ \text{k}\Omega$, we have to shunt the $4.7\ \text{k}\Omega$ resistor.

Let $4.7\ \text{k}\Omega$ resistor is shunted with 'y' ohm resistor in order to get output resistance of $3.33\ \text{k}\Omega$

$$\left(\frac{4.7y}{4.7+y}\right)\text{k}\Omega = 3.33\ \text{k}\Omega$$
$$4.7y = 15.651 + 3.33y$$
$$1.37y = 15.651$$
$$y = 11.42\ \text{k}\Omega$$

$11.42\ \text{k}\Omega$ resistor is shunted with $4.7\ \text{k}\Omega$ resistor in order to get output voltage of $3.33\ \text{k}\Omega$

Step 1 of 3

The voltage gain of an amplifier is, $A_v = 100\ \text{V/V}$.

The current gain of an amplifier is, $A_i = 1000\ \text{A/A}$.

Calculate the voltage gain in decibels.

$$\begin{aligned} A_V(\text{dB}) &= 20\log 100 \\ &= 20\log 10^2 && \left(\log m^n = n\log m\right) \\ &= 40\log 10 && (\log 10 = 1) \\ &= 40\,\text{dB} \end{aligned}$$

Therefore, the voltage gain in decibels, A_v is $\boxed{40\ \text{dB}}$.

Step 2 of 3

Calculate the current gain in decibels.

$$\begin{aligned} A_i(\text{dB}) &= 20\log 1000 \\ &= 20\log 10^3 && \left(\log m^n = n\log m\right) \\ &= 60\log 10 && (\log 10 = 1) \\ &= 60\,\text{dB} \end{aligned}$$

Therefore, the current gain in decibels, A_i is $\boxed{60\ \text{dB}}$.

Step 3 of 3

The power gain is,

$$\begin{aligned} A_p &= A_v A_i \\ &= (100)(1000) \\ &= 10^5\ \text{W/W} \end{aligned}$$

Calculate the power gain in decibels.

$$\begin{aligned} A_p(\text{dB}) &= 10\log(100000) \\ &= 10\log\left(10^5\right) && \left(\log m^n = n\log m\right) \\ &= 50\log(10) && (\log 10 = 1) \\ &= 50\,\text{dB} \end{aligned}$$

Therefore, the power gain in decibels, A_p is $\boxed{50\ \text{dB}}$.

Step 1 of 2

1.010P

Refer to Figure P1.10 in the textbook.

Calculate the current through resistor, R_1 using current divider rule.

$$I_1 = I\left(\frac{R_2}{R_1 + R_2}\right) \quad \text{..... (1)}$$

Therefore, the current through resistor R_1, I_1 is $\boxed{I\left(\frac{R_2}{R_1 + R_2}\right)}$.

Calculate the current through resistor, R_2 using current divider rule.

$$I_2 = I\left(\frac{R_1}{R_1 + R_2}\right) \quad \text{..... (2)}$$

Therefore, the current through resistor R_2, I_2 is $\boxed{I\left(\frac{R_1}{R_1 + R_2}\right)}$.

Step 2 of 2

Calculate the voltage across resistor, R_2.

$$V = I_2 R_2$$

Substitute $I\left(\frac{R_1}{R_1 + R_2}\right)$ for I_2.

$$V = I\left(\frac{R_1}{R_1 + R_2}\right) R_2$$

$$= I\left(\frac{R_1 R_2}{R_1 + R_2}\right)$$

Therefore, the voltage expression, V is $\boxed{I\left(\frac{R_1 R_2}{R_1 + R_2}\right)}$.

Step 1 of 2

Calculate the DC power.

$$P_{dc} = VI$$

Substitute 15 for V and 8×10^{-3} for I.

$$\begin{aligned} P_{dc} &= VI \\ &= (15)\left(8\times10^{-3}\right) \\ &= 120\ \text{mW} \end{aligned}$$

Calculate the power delivered to the load.

$$\begin{aligned} P_L &= \frac{V^2}{R} \\ &= \frac{\left(\frac{V_{max}}{\sqrt{2}}\right)^2}{R} \end{aligned}$$

Substitute $6\ \text{V}$ for V_{max} and $1\ \text{k}\Omega$ for R.

$$\begin{aligned} P_L &= \frac{\left(\frac{6}{\sqrt{2}}\right)^2}{1\ \text{k}\Omega} \\ &= 18\ \text{mW} \end{aligned}$$

Step 2 of 2

Calculate the power dissipated in the amplifier.

$$P_{dissipated} = P_{dc} - P_L$$

Substitute $18\ \text{mW}$ for P_L and $120\ \text{mW}$ for P_{dc}.

$$\begin{aligned} P_{dissipated} &= 120\ \text{m} - 18\ \text{m} \\ &= 102\ \text{mW} \end{aligned}$$

Therefore, the power dissipated in the amplifier, $P_{dissipated}$ is $\boxed{102\ \text{mW}}$.

Calculate the efficiency.

$$\eta = \frac{P_L}{P_{dc}} \times 100\%$$

Substitute $18\ \text{mW}$ for P_L and $120\ \text{mW}$ for P_{dc}.

$$\begin{aligned} \eta &= \frac{P_L}{P_{dc}} \times 100\% \\ &= \frac{18\ \text{m}}{120\ \text{m}} \times 100\% \\ &= 15\% \end{aligned}$$

Therefore, the efficiency of an amplifier, η is $\boxed{15\%}$.

Step 1 of 3

1.011P

Assume the current flowing from the source is I .

The current flowing through $10\ \text{k}\Omega$ resistor is one-third of that available from the source and hence current flowing through $10\ \text{k}\Omega$ resistor is $\frac{I}{3}$.

In order to have one-third of current flow through $10\ \text{k}\Omega$ resistor insert another resistor R in parallel to the resistor $10\ \text{k}\Omega$.

Step 2 of 3

Use current division rule to determine the current passing through the $10\ \text{k}\Omega$ resistor.

$$I\left(\frac{R}{R+10}\right)=\frac{I}{3}$$

$$3R=R+10$$

$$2R=10\ \text{k}\Omega$$

$$R=5\ \text{k}\Omega$$

Thus, the value of resistance R inserted is $\boxed{5\ \text{k}\Omega}$.

Step 3 of 3

Hence, obtain the required design as shown in Figure 1.

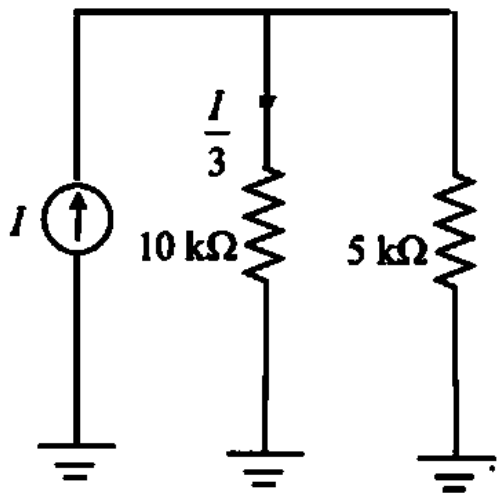


Figure 1

Step 1 of 10

Draw the circuit diagram.

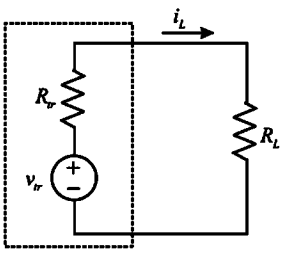


Figure 1

Step 2 of 10

Calculate the load current, i_L .

$$i_L = \frac{v_{tr}}{R_L + R_{tr}}$$

Substitute 1 for v_{tr} , $10\ \Omega$ for R_L and $1\ \text{M}\Omega$ for R_{tr} .

$$\begin{aligned} i_L &= \frac{1}{10+10^6} \\ &= 9.99\times10^{-7} \\ &= 0.99\ \mu\text{A} \end{aligned}$$

Step 3 of 10

Calculate the voltage across the load, v_L .

$$v_L = i_L R_L$$

Substitute $0.99\ \mu\text{A}$ for i_L and $10\ \Omega$ for R_L .

$$\begin{aligned} v_L &= i R_L \\ &= \left(0.99\times10^{-6}\right)(10) \\ &= 9.99\times10^{-6} \\ &= 10\ \mu\text{V} \end{aligned}$$

Therefore, the voltage across the load, v_L is $\boxed{10\ \mu\text{V}}$.

Step 4 of 10

Calculate the Power level at the load, P_L .

$$\begin{aligned} P_L &= i_L^2 R_L \\ &= \left(0.99\times10^{-6}\right)^2(10) \\ &= 9.99\times10^{-12} \\ &= 10\ \text{pW} \end{aligned}$$

Therefore, the power level at the load, P_L is $\boxed{10\ \text{pW}}$.

Step 5 of 10

Transducer connected to load via buffer amplifier is interposed between source and load as shown in Figure 2.

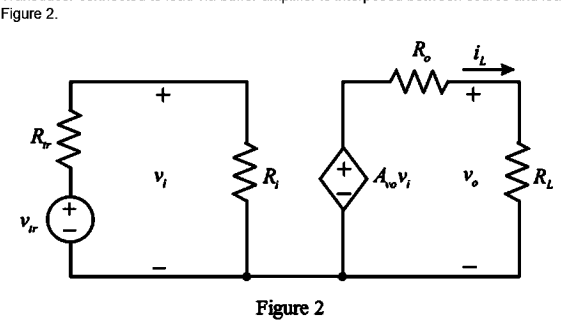


Figure 2

Step 6 of 10

Calculate the output voltage, v_o .

$$\begin{aligned} v_o &= A_{vo} v_i \left(\frac{R_L}{R_L + R_o}\right) \\ &= A_{vo}\left[v_{tr}\left(\frac{R_i}{R_i + R_{tr}}\right)\right]\left(\frac{R_L}{R_L + R_o}\right) \qquad \left(\text{Since } v_i = V_{tr}\left(\frac{R_i}{R_i + R_{tr}}\right)\right) \end{aligned}$$

Substitute 1 for v_{tr} , $1\ \text{M}\Omega$ for R_i , $10\ \Omega$ for R_L and $10\ \Omega$ for R_o .

$$\begin{aligned} v_o &= (1)\left[(1)\left(\frac{10^6}{10^6+10^6}\right)\right]\left(\frac{10}{10+10}\right) \\ &= \frac{1}{4} \\ &= 0.25\ \text{V} \end{aligned}$$

Therefore, the output voltage, v_o is $\boxed{0.25\ \text{V}}$.

Step 7 of 10

Calculate the output Power P_L' .

$$\begin{aligned} P_L' &= \frac{v_o^2}{R_L} \\ &= \frac{(0.25)^2}{10} \\ &= 0.00625 \\ &= 6.25\times10^{-3} \end{aligned}$$

Therefore, the output power, P_L' is $\boxed{6.25\ \text{mW}}$.

Step 8 of 10

Calculate the voltage gain from source to load.

$$\begin{aligned} A_v &= \frac{v_o}{v_s} \\ &= \frac{0.25}{1} \\ &= 0.25 \end{aligned}$$

Calculate the voltage gain in dB.

$$\begin{aligned} A_v\,(\text{dB}) &= 20\log(0.25) \\ &= 20(-0.602) \\ &= -12.04 \end{aligned}$$

Therefore, the voltage gain from the source to load, A_v is $\boxed{-12.04\ \text{dB}}$.

Step 9 of 10

Calculate the power supplied by the source to load.

$$\begin{aligned} P_i &= \frac{v_i^2}{R_i} \\ &= \frac{1}{R_i}\left(\frac{v_{tr} R_i}{R_i + R_{tr}}\right)^2 \\ &= \frac{v_{tr}^2 R_i}{(R_i + R_{tr})^2} \\ &= \frac{(1)^2(10^6)}{(10^6+10^6)^2} \\ &= \frac{(1)^2(10^6)}{(10^6+10^6)^2} \\ &= \frac{10^{-6}}{4} \\ &= 0.25\times10^{-6}\ \text{W} \end{aligned}$$

Step 10 of 10

Calculate the power gain in dB.

$$\begin{aligned} A_P\,(\text{dB}) &= 10\log\left(\frac{P_o}{P_i}\right) \\ &= 10\log\left(\frac{6.25\times10^{-3}}{0.25\times10^{-6}}\right) \\ &= 10\log\left(25\times10^3\right) \\ &= 10(4.398) \\ &= 43.98 \\ &\approx 44\ \text{dB} \end{aligned}$$

Therefore, the power gain from the source to load, A_P is $\boxed{44\ \text{dB}}$.

Step 1 of 5

The one third of a signal current I passes through load resistor R.

Draw the simple voltage divider circuit.

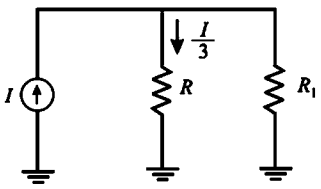


Figure 1

Step 2 of 5

Calculate the current passing through resistor R_1 using Kirchhoff's current law.

$$I_{R1} = I - \frac{I}{3}$$
$$= \frac{2I}{3}$$

In the parallel branch the voltages are equal.

Thus,

$$V_R = V_{R1}$$
$$R\left(\frac{I}{3}\right) = R_1\left(\frac{2I}{3}\right)$$
$$R_1 = \frac{R}{2}$$

Therefore, the required resistor value is $\boxed{\frac{R}{2}}$.

Calculate the input resistance, R_{in}.

$$R_{in} = R \| \frac{R}{2}$$
$$= \frac{(R)\left(\frac{R}{2}\right)}{R + \frac{R}{2}}$$
$$= \frac{R^2}{3R}$$
$$= \frac{R}{3}$$

Therefore, the input resistance, R_{in} is $\boxed{\frac{R}{3}}$.

Step 3 of 5

Connect the resistor R_2 in series with load resistor R.

Draw the simple circuit.

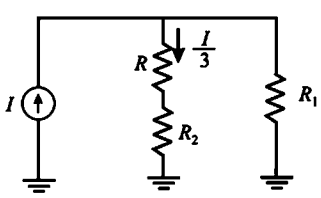


Figure 2

Step 4 of 5

Now, consider the resistor R_1 is 10% high.

$$R_1 = \frac{R}{2} + \left(\frac{R}{2}\right)\left(\frac{10}{100}\right)$$
$$R_1 = \frac{1.1R}{2}$$

In the parallel branch the voltages are equal.

Thus,

$$V_{R+R_2} = V_{R1}$$
$$\frac{I}{3}(R + R_2) = \left(1.1\frac{R}{2}\right)\frac{2I}{3}$$
$$R + R_2 = 1.1R$$
$$R_2 = 1.1R - R$$
$$R_2 = 0.1R$$

Therefore, the required resistor value is $\boxed{0.1R}$.

Step 5 of 5

Calculate the input resistance, R_{in}.

$$R_{in} = 1.1R \| \frac{1.1R}{2}$$
$$= \frac{(1.1R)\left(\frac{1.1R}{2}\right)}{1.1R + \frac{1.1R}{2}}$$
$$= \frac{\frac{1.1R}{2}}{1 + \frac{1}{2}}$$
$$= \frac{1.1R}{3}$$

Therefore, the input resistance, R_{in} is $\boxed{\frac{1.1R}{3}}$.

1.013E

The voltage gain of voltage amplifier without load is,

$$A_{vl} = A_{vo} \qquad \textbf{(Since } R_L \textbf{ is not exist)}$$

The voltage gain of voltage amplifier with load is,

$$A_v = A_{vo}\left(\frac{R_L}{R_L + R_o}\right)$$

Calculate the output resistance of the amplifier.

When a resistance $R_L = 1\ \text{k}\Omega$ is connected to a load, the output voltage is decreased by 20%.

$$A_v = \left(1 - \frac{20}{100}\right)A_{vl}$$

Substitute $A_{vo}\left(\frac{R_L}{R_L + R_o}\right)$ for A_v.

$$A_{vo}\left(\frac{R_L}{R_L + R_o}\right) = (1 - 0.2)A_{vo}$$

$$\left(\frac{10^3}{10^3 + R_o}\right) = 0.8$$

$$10^3 = \left(0.8 \times 10^3\right) + 0.8R_o$$

$$0.2 \times 10^3 = 0.8R_o$$

$$\frac{0.2 \times 10^3}{0.8} = R_o$$

$$R_o = 0.25 \times 10^3$$

$$= 250\Omega$$

Therefore, the output resistance of the amplifier, R_o is $\boxed{250\ \Omega}$.

1.013P

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1.014E

Step 1 of 4

Calculate the value of voltage gain, A_{vo} :

$$20\log(A_{vo}) = 40$$
$$\log(A_{vo}) = 2$$
$$A_{vo} = 10^2$$
$$A_{vo} = 100$$

Therefore, the value of voltage gain, A_{vo} is $\boxed{100\ \text{V/V}}$.

Step 2 of 4

Write the expression for power input of the amplifier.

$$P_i = \frac{V_i^2}{R_i}$$

Write the expression for power output of the amplifier.

$$P_o = \frac{V_o^2}{R_L}$$
$$= \frac{\left(A_{vo} V_i \frac{R_L}{R_o + R_L}\right)^2}{R_L} \qquad \left(\text{Since, } V_o = V_i\left(\frac{R_L}{R_o + R_L}\right)\right)$$
$$= \frac{A_{vo}^2 V_i^2 R_L^2}{\left(R_o + R_L\right)^2 R_L}$$
$$= \frac{A_{vo}^2 V_i^2 R_L}{\left(R_o + R_L\right)^2}$$

Step 3 of 4

Calculate the power gain, A_p .

$$A_p = \frac{P_o}{P_i}$$

Substitute $\frac{V_i^2}{R_i}$ for P_i and $\frac{A_{vo}^2 V_i^2 R_L}{\left(R_o + R_L\right)^2}$ for P_o .

$$A_p = \frac{P_o}{P_i}$$
$$= \frac{\frac{A_{vo}^2 V_i^2 R_L}{\left(R_o + R_L\right)^2}}{\left(\frac{V_i^2}{R_i}\right)}$$
$$= \frac{A_{vo}^2 R_L R_i}{\left(R_o + R_L\right)^2}$$

Substitute $10\ \text{k}\Omega$ for R_i , $1\ \text{k}\Omega$ for R_o and $1\ \text{k}\Omega$ for R_L .

$$A_p = \frac{(100)^2 \left(1\times10^3\right)\left(10\times10^3\right)}{\left(10^3 + 10^3\right)^2}$$
$$= \frac{100000\times10^6}{4\times10^6}$$
$$= 25000\ \text{W/W}$$

Step 4 of 4

Calculate the value of power gain A_p in decibels.

$$A_p\,(\text{dB}) = 10\log(25000)$$
$$= 10(4.39)$$
$$= 43.9\ \text{dB}$$

Therefore, the value of power gain, A_p is $\boxed{44\ \text{dB}}$.

Step 1 of 8

1.014P

Thevenin equivalent of a circuit is a circuit which consists of a voltage source is in series with resistance. where series resistance is the equivalent resistance of the circuit and voltage source equal to the voltage across the load terminal .

Refer to the Figure P1.14 in the textbook.

(a)

Thevenin's Voltage across the terminal 1 and 2 is calculated as follows:

Apply the voltage division rule between terminal 1 and 2.

$$V_{Th} = 1.5\left(\frac{1\ \text{k}\Omega}{1\ \text{k}\Omega + 1\ \text{k}\Omega}\right)$$
$$= 0.75\ \text{V}$$

Step 2 of 8

Calculation of Thevenin's resistance R_{Th} as follows:

Short the voltage sources and open circuit the current sources to find the equivalent resistance across the terminals 1 and 2.

$$R_{th} = 1\,\text{k}\Omega \| 1\,\text{k}\Omega$$
$$= \frac{1\,\text{k}\Omega \times 1\,\text{k}\Omega}{1\,\text{k}\Omega + 1\,\text{k}\Omega}$$
$$= 500\,\Omega$$

Therefore, the thevenin's equivalent voltage V_{Th} is $\boxed{0.75\ \text{V}}$ and thevenin resistance R_{th} is $\boxed{500\,\Omega}$.

Step 3 of 8

Thevenin's equivalent circuit between terminals 1 and 2 is shown in the figure 1.

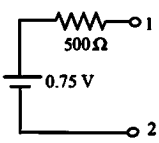


Figure 1

Step 4 of 8

(b)

Thevenin's Voltage across the terminal 2 and 3 is calculated as follows:

Apply the voltage division rule between terminal 2 and 3.

$$V_{Th} = 1.5\left(\frac{1\ \text{k}\Omega}{1\ \text{k}\Omega + 1\ \text{k}\Omega}\right)$$
$$= 0.75\ \text{V}$$

Step 5 of 8

Calculation of Thevenin's resistance R_{Th} as follows:

Short the voltage sources and open circuit the current sources then to the equivalent resistance across the terminals 2 and 3.

$$R_{th} = 1\,\text{k}\Omega \| 1\,\text{k}\Omega$$
$$= \frac{1\,\text{k}\Omega \times 1\,\text{k}\Omega}{1\,\text{k}\Omega + 1\,\text{k}\Omega}$$
$$= 500\,\Omega$$

Therefore, the thevenin's equivalent voltage V_{Th} is $\boxed{0.75\ \text{V}}$ and thevenin resistance R_{th} is $\boxed{500\,\Omega}$.

Step 6 of 8

Thevenin's equivalent circuit between terminals 2 and 3 is shown in the figure 2

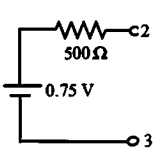


Figure 2

Step 7 of 8

(c)

Thevenin's Voltage across the terminal 1 and 3 is calculated as follows:

$$V_{Th} = 1.5\ \text{V}$$

Calculation of Thevenin's resistance R_{Th} as follows:

Short the voltage sources and open circuit the current sources then find the equivalent resistance across the terminals 1 and 3.

Since, the voltage source gets short circuited, the R_{th} is $0\,\Omega$ across 1 and 3.

Step 8 of 8

Therefore, the thevenin's equivalent voltage V_{Th} is $\boxed{1.5\ \text{V}}$ and thevenin resistance R_{th} is $\boxed{0\,\Omega}$.

Thevenin's equivalent circuit between terminals 1 and 3 is shown in the figure 3.

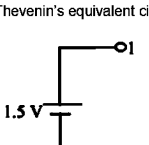


Figure 3

Step 1 of 5

1.015E

Draw the two stage amplifier circuit.

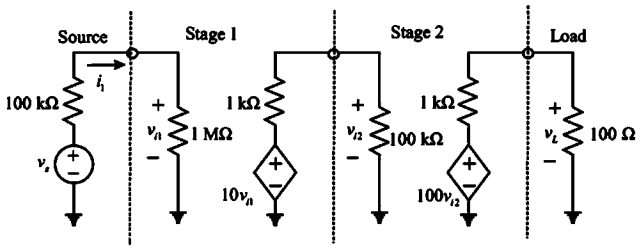


Figure 1

Step 2 of 5

The source signal appearing at the input terminal of the amplifier is,

$$v_{i1} = v_s\left(\frac{1\ \text{M}\Omega}{1\ \text{M}\Omega + 100\ \text{k}\Omega}\right)$$

$$\frac{v_{i1}}{v_s} = \left(\frac{1\ \text{M}\Omega}{1.1\ \text{M}\Omega}\right)$$

$$\frac{v_{i1}}{v_s} = 0.909\ \text{V/V}$$

Step 3 of 5

The Voltage gain of first stage is obtained by considering the input resistance of the second stage to be the load of first stage.

$$A_{v1} = \frac{v_{i2}}{v_{i1}}$$

$$= 10\left(\frac{100\ \text{k}\Omega}{100\ \text{k}\Omega + 1\ \text{k}\Omega}\right)$$

$$= 10\left(\frac{100}{101}\right)$$

$$= 9.9\ \text{V/V}$$

Step 4 of 5

Calculate the voltage gain of second stage.

$$A_{v2} = \frac{v_L}{v_{i2}}$$

$$= 100\left(\frac{100\ \Omega}{100\ \Omega + 1\ \text{k}\Omega}\right)$$

$$= 100\left(\frac{100}{1100}\right)$$

$$= 9.09\ \text{V/V}$$

Step 5 of 5

Calculate the overall voltage gain of the cascade amplifier.

$$A_v = \frac{v_L}{v_s}$$

$$= \frac{v_{i1}}{v_s} A_{v1} A_{v2}$$

$$= (0.909)(9.9)(9.09)$$

$$= 81.81\ \text{V/V}$$

Therefore, the overall voltage gain, A_v is $\boxed{81.81\ \text{V/V}}$.

Step 1 of 8

1.015P

Thevenin equivalent of a circuit is a circuit which consists of a voltage source and a series resistance . where series resistance is the equivalent resistance of the circuit .where voltage source has a voltage equal to the voltage across the load terminal.

Refer to the Figure P1.15 in the textbook.

Step 2 of 8

Consider the nodes 3 and 4 according to voltage divide rule voltage across the node 3 is twice the voltage across the node 4.

Let voltage across the node 4 is V_4 and hence voltage across the node 3 is $2V_4$. Since, two equal $20\ \text{k}\Omega$ resistors are connected to node 3(there will be equal drop of voltage at each resistor).

Similarly between the nodes 2 and 3, according to voltage division rule voltage across the node 2 is twice the voltage across the node 3. Hence, voltage across the node 2 is,

$$\begin{aligned} V_2 &= 2(V_3) \\ &= 4V_4 \end{aligned}$$

Step 3 of 8

Consider the nodes between 1 and 2, according to voltage division rule voltage across thje node 1 is twice the voltage across the node 2. Thus, voltage across the node 1 is,

$$\begin{aligned} V_1 &= 2V_2 \\ &= 2(4V_4) \\ &= 8V_4 \end{aligned}$$

Step 4 of 8

The voltage across the node 1 is $10\ \text{V}$.

$$V_1 = 10\ \text{V}$$

$$8V_4 = 10\ \text{V}$$

$$V_4 = \frac{10}{8}$$

$$V_4 = 1.25\ \text{V}$$

The voltage of node 4 is equivalent to the Thevenin's voltage. Therefore, thevenin's voltage V_{th} is $\boxed{1.25\ \text{V}}$.

Step 5 of 8

Determination of Thevenin's resistance R_{Th} as follows:

Short circuit the voltage sources and open circuit the current sources to find the equivalent resistance across the load terminal.

The resistors between nodes 1 and 2 are parallel to each other and hence equivalent resistance is,

$$\begin{aligned} \text{R}_1 &= 20\ \text{k}\Omega \| 20\ \text{k}\Omega \\ &= 10\ \text{k}\Omega \end{aligned}$$

Step 6 of 8

R_1 and $20\ \text{k}\Omega$ between nodes 2 and 3 are in series. Then equivalent resistance is,

$$\begin{aligned} \text{R}_2 &= \text{R}_1 + 20\ \text{k}\Omega \\ &= 10\ \text{k}\Omega + 20\ \text{k}\Omega \\ &= 30\ \text{k}\Omega \end{aligned}$$

R_2 $\text{k}\Omega$ and $20\ \text{k}\Omega$ between nodes 3 and ground are in parallel then equivalent resistance is ,

$$\begin{aligned} R_3 &= R_2 \| 30\ \text{k}\Omega \\ &= \frac{30\ \text{k}\Omega \times 20\ \text{k}\Omega}{30\ \text{k}\Omega + 20\ \text{k}\Omega} \\ &= 12\ \text{k}\Omega \end{aligned}$$

Step 7 of 8

R_3 and $20\ \text{k}\Omega$ are in series and their equivalent resistance is,

$$\begin{aligned} R_4 &= R_2 + 30\ \text{k}\Omega \\ &= 12\ \text{k}\Omega + 30\ \text{k}\Omega \\ &= 32\ \text{k}\Omega \end{aligned}$$

R_4 and $20\ \text{k}\Omega$ between node 4 and ground are in parallel and their equivalent resistance is simply thevenin's equivalent resistance.

$$\begin{aligned} R_{th} &= R_4 \| 20\ \text{k}\Omega \\ &= \frac{32 \times 20}{32 + 20} \\ &= 12.31\ \text{k}\Omega \end{aligned}$$

Therefore, the Thevenin's resistance R_{Th} is $\boxed{12.31\ \text{k}\Omega}$.

Consider the Load resistance of the circuit is $3\ \text{k}\Omega$.

Thevenin equivalent of a circuit is shown in Figure 1.

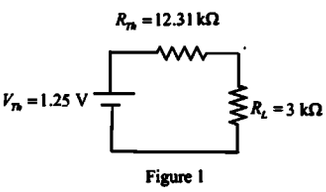


Figure 1

Step 8 of 8

Determine the current I_L is flowing through load resistance $3\ \text{k}\Omega$.

$$\begin{aligned} I_L &= \frac{V_{Th}}{(R_{Th} + R_L)} \\ &= \frac{1.25\ \text{V}}{(12.31\ \text{k}\Omega + 3\ \text{k}\Omega)} \\ &= 81.6\ \mu\text{A} \end{aligned}$$

Hence, the current I_L that flows through a load resistance of $3\ \text{k}\Omega$ is $\boxed{81.6\ \mu\text{A}}$.

Step 1 of 4

Refer to Figure 1.17 in the textbook for three stage amplifier circuit.

In Figure 1.17 in the textbook, the source signal appearing at the input terminal of the amplifier is,

$$v_{i1} = v_s\left(\frac{1\ \text{M}\Omega}{1\ \text{M}\Omega + 100\ \text{k}\Omega}\right)$$

$$\frac{v_{i1}}{v_s} = \left(\frac{1\ \text{M}\Omega}{1.1\ \text{M}\Omega}\right)$$

$$\frac{v_{i1}}{v_s} = 0.909\ \text{V/V}$$

Substitute $1\ \text{mV}$ for v_s.

$$\begin{aligned} v_{i1} &= (0.909)v_s \\ &= (0.909)(1\ \text{m}) \\ &= 0.909\ \text{mV} \end{aligned}$$

Thus, the output voltage of stage 1, v_{i1} is $\boxed{0.909\ \text{mV}}$.

Step 2 of 4

The Voltage gain of first stage is obtained by considering the input resistance of the second stage to be the load of first stage.

$$\begin{aligned} A_{v1} &= \frac{v_{i2}}{v_{i1}} \\ &= 10\left(\frac{100\ \text{k}\Omega}{100\ \text{k}\Omega + 1\ \text{k}\Omega}\right) \\ &= 10\left(\frac{100}{101}\right) \\ &= 9.9\ \text{V/V} \end{aligned}$$

Substitute $0.909\ \text{mV}$ for v_{i1}.

$$\begin{aligned} v_{i2} &= 9.9v_{i1} \\ &= (9.9)\,(0.909\ \text{mV}) \\ &= 8.999\ \text{mV} \\ &\approx 9\ \text{mV} \end{aligned}$$

Thus, the output voltage of stage 2, v_{i2} is $\boxed{9\ \text{mV}}$.

Step 3 of 4

The voltage gain of second stage is,

$$\begin{aligned} A_{v2} &= \frac{v_{i3}}{v_{i2}} \\ &= 100\left(\frac{10\ \text{k}\Omega}{10\ \text{k}\Omega + 1\ \text{k}\Omega}\right) \\ &= 100\left(\frac{10}{11}\right) \\ &= 90.9\ \text{V/V} \end{aligned}$$

Substitute $9\ \text{mV}$ for v_{i2}.

$$\begin{aligned} v_{i3} &= (90.9)(v_{i2}) \\ &= (90.9)(9\ \text{mV}) \\ &= 818.1\ \text{mV} \end{aligned}$$

Thus, the output voltage of stage 3, v_{i3} is $\boxed{818.1\ \text{mV}}$.

Step 4 of 4

The voltage gain of third stage is,

$$\begin{aligned} A_{v3} &= \frac{v_L}{v_{i3}} \\ &= 1\left(\frac{100\ \Omega}{100\ \Omega + 10\ \Omega}\right) \\ &= 1\left(\frac{100}{110}\right) \\ &= 0.909\ \text{V/V} \end{aligned}$$

Substitute $818.1\ \text{mV}$ for v_{i3}.

$$\begin{aligned} v_L &= 0.909(v_{i3}) \\ &= (0.909)(818.18\ \text{mV}) \\ &= 743.72\ \text{mV} \end{aligned}$$

Thus, the load voltage, v_L is $\boxed{743.72\ \text{mV}}$.

1.016P

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Step 1 of 6

Refer to Figure 1.17 in the textbook for three stage amplifier circuit.

(a)

The voltage amplifier model without the source and load of the three stage cascade amplifier is shown in Figure 1.

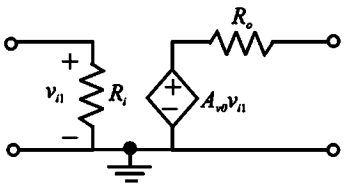


Figure 1

Step 2 of 6

From Figure 1.17 in the textbook, the input resistance of the voltage amplifier model is, $R_i = 1\text{M}\Omega$ and the output resistance is, $R_o = 10\ \Omega$.

Therefore, the input resistance, R_i of the voltage amplifier model is $\boxed{1\ \text{M}\Omega}$ and the output resistance, R_o is $\boxed{10\ \Omega}$.

The output voltage of the three stage amplifier is,

$v_0 = 1v_{i3}$ ..... (1)

From Figure 1.17 in the textbook,

$$v_{i3} = \left(\frac{100v_{i2}}{1\text{ k}+10\text{ k}}\right)(10\text{ k})$$
$$= \frac{1000v_{i2}}{11}$$

Step 3 of 6

Calculate the output of the second stage.

$$v_{i2} = \left(\frac{10(v_{i1})}{1\text{ k}+100\text{ k}}\right)(100\text{ k})$$
$$= \frac{1000v_{i1}}{101}$$

Substitute $\frac{1000v_{i2}}{11}$ for v_{i3} in equation (1).

$$v_0 = 1\left(\frac{1000v_{i2}}{11}\right)$$

Substitute $\frac{1000v_{i1}}{101}$ for v_{i2}.

$$v_0 = \frac{1000\left(\frac{1000v_{i1}}{101}\right)}{11}$$
$$= \frac{1000(9.9v_{i1})}{11}$$
$$= 1000(0.9v_{i1})$$
$$= 900v_{i1}$$

$$\frac{v_0}{v_{i1}} = 900\text{ V/V}$$

The gain of the voltage amplifier is,

$$A_{v0} = \frac{v_0}{v_{i1}}$$
$$= 900\text{ V/V}$$

Therefore, the gain of the voltage amplifier, A_{v0} is $\boxed{900\text{ V/V}}$.

Step 4 of 6

(b)

The range of the load resistance is, $R_L = 10\ \Omega$ to $1000\ \Omega$.

In Figure 1.17 in the textbook, the source signal appearing at the input terminal of the amplifier is,

$$v_{i1} = v_s\left(\frac{1\text{ M}\Omega}{1\text{ M}\Omega+100\text{ k}\Omega}\right)$$
$$\frac{v_{i1}}{v_s} = \left(\frac{1\text{ M}\Omega}{1.1\text{ M}\Omega}\right)$$
$$\frac{v_{i1}}{v_s} = 0.909\text{ V/V}$$

The gain of the first stage is,

$$A_{v1} = \frac{v_{i2}}{v_{i1}}$$
$$= 10\left(\frac{100\text{ k}}{100\text{ k}+1\text{ k}}\right)$$
$$= \frac{1000}{101}\text{ V/V}$$

Step 5 of 6

The gain of the second stage is,

$$A_{v2} = \frac{v_{i3}}{v_{i2}}$$
$$= (100)\left(\frac{10\text{ k}}{10\text{ k}+1\text{ k}}\right)$$
$$= \frac{1000}{11}\text{ V/V}$$

Calculate the gain of the third stage, when $R_L = 10\ \Omega$.

$$A_{v3} = \frac{v_L}{v_{i3}}$$
$$= 1\frac{10}{10+10}$$
$$= \frac{10}{20}\text{ V/V}$$

Calculate the total voltage gain of the three stages in cascade.

$$A_v = A_{v1}A_{v2}A_{v3}$$
$$= \left(\frac{1000}{101}\right)\left(\frac{1000}{11}\right)\left(\frac{10}{20}\right)$$
$$= 450\text{ V/V}$$

The overall voltage gain of the cascaded amplifier is,

$$\frac{v_0}{v_s} = A_v\frac{v_{i1}}{v_s}$$
$$= (450)(0.909)$$
$$= 409\text{ V/V}$$

Step 6 of 6

The gain of the third stage when $R_L = 1000\ \Omega$ is,

$$A_{v3} = \frac{v_L}{v_{i3}}$$
$$= 1\left(\frac{1000}{1000+10}\right)$$
$$= \frac{1000}{1010}\text{ V/V}$$

The total voltage gain of the three stages in cascade is

$$A_v = A_{v1}A_{v2}A_{v3}$$
$$= \left(\frac{1000}{101}\right)\left(\frac{1000}{11}\right)\left(\frac{1000}{1010}\right)$$
$$= 891.18\text{ V/V}$$

The overall voltage of the cascaded amplifier is,

$$\frac{v_0}{v_s} = A_v\left(\frac{v_{i1}}{v_s}\right)$$
$$= (891.18)(0.909)$$
$$= 810\text{ V/V}$$

Therefore, the range of the overall voltage gain is $\boxed{409\text{ V/V to }810\text{ V/V}}$.

1.017P

We don't have the solution to this problem yet.

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Step 1 of 3

Draw the current amplifier circuit.

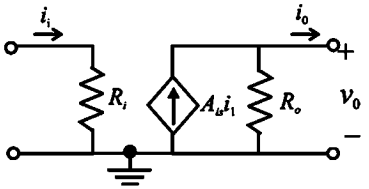


Figure 1

Step 2 of 3

The amplifier is fed from a current source i_s having a resistance R_s and the output is connected to a load resistance, R_L as shown in Figure 2.

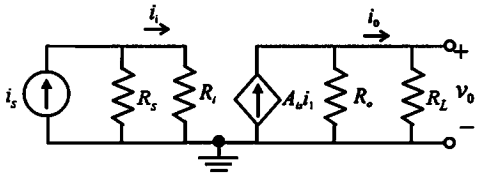


Figure 2

Step 3 of 3

Calculate the current, i_i using current division rule.

$$i_i = i_s\left(\frac{R_s}{R_s + R_i}\right) \quad \text{..... (1)}$$

Calculate the current, i_o using current division rule.

$$i_o = (A_{is}\, i_i)\left(\frac{R_o}{R_o + R_L}\right)$$

Substitute $i_s\left(\frac{R_s}{R_s + R_i}\right)$ for i_i.

$$i_o = A_{is}\left(i_s \frac{R_s}{R_s + R_i}\right)\left(\frac{R_o}{R_o + R_L}\right)$$

$$\frac{i_o}{i_s} = A_{is}\left(\frac{R_s}{R_s + R_i}\right)\left(\frac{R_o}{R_o + R_L}\right)$$

Thus, the overall current gain, $\frac{i_o}{i_s}$ is $\boxed{A_{is}\left(\frac{R_s}{R_s + R_i}\right)\left(\frac{R_o}{R_o + R_L}\right)}$.

Step 1 of 6

Refer to Figure P1.18 in the textbook.

Apply a voltage source V_x across terminals X and ground as shown in Figure 1.

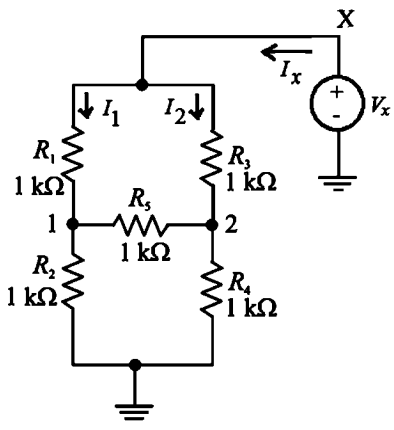


Figure 1

Step 2 of 6

Apply Kirchhoff's current law at node 1.

$$\frac{V_1 - V_x}{1\,\text{k}\Omega} + \frac{V_1 - V_2}{1\,\text{k}\Omega} + \frac{V_1}{1\,\text{k}\Omega} = 0$$

$$\frac{3V_1 - V_2 - V_x}{1\,\text{k}\Omega} = 0$$

$$3V_1 - V_2 - V_x = 0 \quad \text{..... (1)}$$

Apply Kirchhoff's current law at node 2.

$$\frac{V_2 - V_x}{1\,\text{k}\Omega} + \frac{V_2 - V_1}{1\,\text{k}\Omega} + \frac{V_2}{1\,\text{k}\Omega} = 0$$

$$\frac{3V_2 - V_1 - V_x}{1\,\text{k}\Omega} = 0$$

$$3V_2 - V_1 - V_x = 0$$

$$V_x = 3V_2 - V_1 \quad \text{..... (2)}$$

Substitute $3V_2 - V_1$ for V_x in equation (1).

$$3V_1 - V_2 - (3V_2 - V_1) = 0$$

$$4V_1 - 4V_2 = 0$$

$$V_1 = V_2 \quad \text{..... (3)}$$

Substitute V_2 for V_1 in equation (2).

$$V_x = 3V_2 - V_2$$

$$V_x = 2V_2$$

$$V_2 = \frac{V_x}{2}$$

From equation (3),

$$V_1 = \frac{V_x}{2}$$

Step 3 of 6

Apply Kirchhoff's current law at node X.

$$I_x = I_1 + I_2$$

$$= \frac{V_x - V_1}{1\,\text{k}\Omega} + \frac{V_x - V_2}{1\,\text{k}\Omega}$$

$$= \frac{V_x - V_1 + V_x - V_2}{1\,\text{k}\Omega}$$

$$I_x = \frac{2V_x - V_1 - V_2}{1\,\text{k}\Omega} \quad \text{..... (4)}$$

Substitute $\frac{V_x}{2}$ for V_1 and $\frac{V_x}{2}$ for V_2.

$$I_x = \frac{2V_x - \frac{V_x}{2} - \frac{V_x}{2}}{1\,\text{k}\Omega}$$

$$I_x = \frac{2V_x - V_x}{1\,\text{k}\Omega}$$

$$I_x = \frac{V_x}{1\,\text{k}\Omega}$$

$$\frac{V_x}{I_x} = 1\,\text{k}\Omega$$

Calculate the equivalent resistance.

$$R_{eq} = \frac{V_x}{I_x}$$

$$= 1\,\text{k}\Omega$$

Therefore, the equivalent resistance, R_{eq} is $\boxed{1\,\text{k}\Omega}$.

Step 4 of 6

Calculation of equivalent resistance, if the resistor, R_4 is raised to $1.2\,\text{k}\Omega$ as follows:

Apply a voltage source V_x across terminals X and ground as shown in Figure 2.

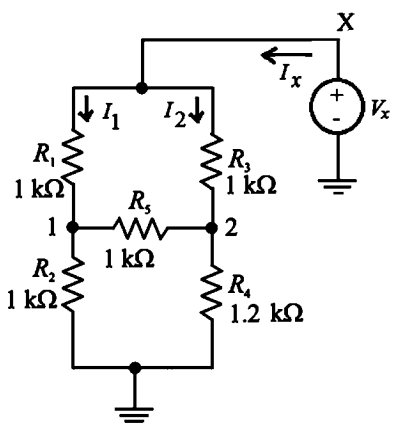


Figure 2

Step 5 of 6

Apply Kirchhoff's current law at node 1.

$$\frac{V_1 - V_x}{1\,\text{k}\Omega} + \frac{V_1 - V_2}{1\,\text{k}\Omega} + \frac{V_1}{1\,\text{k}\Omega} = 0$$

$$\frac{3V_1 - V_2 - V_x}{1\,\text{k}\Omega} = 0$$

$$3V_1 - V_2 - V_x = 0$$

$$V_x = 3V_1 - V_2 \quad \text{..... (5)}$$

Apply Kirchhoff's current law at node 2.

$$\frac{V_2 - V_x}{1\,\text{k}\Omega} + \frac{V_2 - V_1}{1\,\text{k}\Omega} + \frac{V_2}{1.2\,\text{k}\Omega} = 0$$

$$\frac{1.2V_2 - 1.2V_x + 1.2V_2 - 1.2V_1 + V_2}{1.2\,\text{k}\Omega} = 0$$

$$3.4V_2 - 1.2V_x - 1.2V_1 = 0 \quad \text{..... (6)}$$

Substitute $3V_1 - V_2$ for V_x in equation (6).

$$3.4V_2 - 1.2(3V_1 - V_2) - 1.2V_1 = 0$$

$$4.6V_2 - 4.8V_1 = 0$$

$$4.6V_2 = 4.8V_1$$

$$V_2 = 1.043V_1 \quad \text{..... (7)}$$

Substitute $1.043V_1$ for V_2 in equation (5).

$$V_x = 3V_1 - 1.043V_1$$

$$V_x = 1.957V_1$$

$$V_1 = \frac{V_x}{1.957}$$

$$V_1 = 0.51V_x$$

From equation (7),

$$V_2 = 1.043V_1$$

$$= 1.043(0.51V_x)$$

$$= 0.53V_x$$

Step 6 of 6

Apply Kirchhoff's current law at node X.

$$I_x = I_1 + I_2$$

$$= \frac{V_x - V_1}{1\,\text{k}\Omega} + \frac{V_x - V_2}{1\,\text{k}\Omega}$$

$$= \frac{V_x - V_1 + V_x - V_2}{1\,\text{k}\Omega}$$

$$I_x = \frac{2V_x - V_1 - V_2}{1\,\text{k}\Omega} \quad \text{..... (8)}$$

Substitute $0.51V_x$ for V_1 and $0.53V_x$ for V_2.

$$I_x = \frac{2V_x - 0.51V_x - 0.53V_x}{1\,\text{k}\Omega}$$

$$I_x = \frac{0.96V_x}{1\,\text{k}\Omega}$$

$$\frac{V_x}{I_x} = \frac{1\,\text{k}\Omega}{0.96}$$

$$\frac{V_x}{I_x} = 1.042\,\text{k}\Omega$$

Calculate the equivalent resistance.

$$R_{eq} = \frac{V_x}{I_x}$$

$$= 1.042\,\text{k}\Omega$$

Therefore, the equivalent resistance, R_{eq} is $\boxed{1.042\,\text{k}\Omega}$.

Step 1 of 4

Draw the transconductance amplifier circuit.

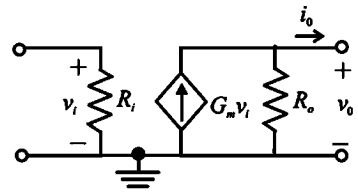


Figure 1

Step 2 of 4

The amplifier is fed from a voltage source v_s having a resistance R_s and the output is connected to a load resistance, R_L as shown in Figure 2.

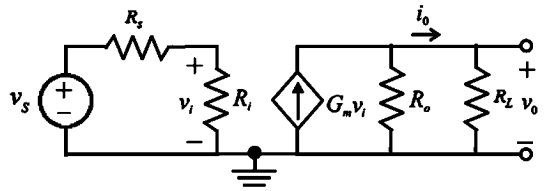


Figure 2

Step 3 of 4

Calculate the voltage across the resistance R_i using voltage division rule.

$$v_i = v_s\left(\frac{R_i}{R_i + R_s}\right) \quad \text{..... (1)}$$

Calculate the current, i_0 using current division rule.

$$i_0 = G_m v_i\left(\frac{R_0}{R_0 + R_L}\right)$$

Calculate the output voltage.

$$v_0 = i_0 R_L$$

Substitute $G_m v_i\left(\frac{R_0}{R_0 + R_L}\right)$ for i_0.

$$v_0 = i_0 R_L$$

$$v_0 = G_m v_i\left(\frac{R_0}{R_0 + R_L}\right) R_L \quad \text{..... (2)}$$

Step 4 of 4

Substitute $v_s\left(\frac{R_i}{R_i + R_s}\right)$ for v_i in equation (2).

$$v_0 = G_m v_s\left(\frac{R_i}{R_i + R_s}\right)\left(\frac{R_0}{R_0 + R_L}\right) R_L$$

$$\frac{v_0}{v_s} = G_m\left(\frac{R_i}{R_i + R_s}\right)\left(\frac{R_0 R_L}{R_0 + R_L}\right)$$

From the equation, it is known that, $R_0 \parallel R_L$.

That is,

$$R_0 \parallel R_L = \frac{R_0 R_L}{R_0 + R_L}$$

Therefore,

$$\frac{v_0}{v_s} = G_m\left(\frac{R_i}{R_i + R_s}\right)(R_0 \parallel R_L)$$

Thus, the overall voltage gain of the transconductance amplifier, $\frac{v_0}{v_s}$ is

$$\boxed{G_m\left(\frac{R_i}{R_i + R_s}\right)(R_0 \parallel R_L)}.$$

1.019P

We don't have the solution to this problem yet.

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Step 1 of 3

Draw the transresistance amplifier circuit.

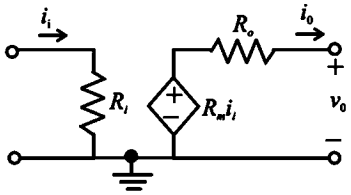


Figure 1

Step 2 of 3

The amplifier is fed from a current source i_s having a resistance R_s and the output is connected to a load resistance, R_L as shown in Figure 2.

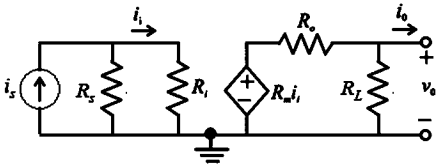


Figure 2

Step 3 of 3

Calculate the current, i_i using current division rule.

$$i_i = i_s\left(\frac{R_s}{R_s + R_i}\right) \quad \text{..... (1)}$$

Calculate the voltage, v_o using current division rule.

$$v_o = (R_m i_i)\left(\frac{R_L}{R_o + R_L}\right) \quad \text{..... (2)}$$

Substitute $i_s\left(\frac{R_s}{R_s + R_i}\right)$ for i_i equation (2).

$$v_O = R_m\left(i_s\frac{R_s}{R_s + R_i}\right)\left(\frac{R_L}{R_O + R_L}\right)$$

$$\frac{v_o}{i_s} = R_m\left(\frac{R_s}{R_s + R_i}\right)\left(\frac{R_L}{R_o + R_L}\right)$$

Thus, the overall gain, $\frac{v_o}{i_s}$ is $\boxed{R_m\left(\frac{R_s}{R_s + R_i}\right)\left(\frac{R_L}{R_o + R_L}\right)}$.

Step 1 of 7

The relation between frequency and time period is,

$$f = \frac{1}{T}$$

The angular frequency is, $\omega = 2\pi f$

(a)

The time period is,

$$T = 10^{-4}\ \text{ms}$$
$$= 10^{-7}\ \text{s}$$

The frequency is,

$$f = \frac{1}{10^{-7}}$$
$$= 10^{7}\ \text{Hz}$$

The angular frequency is,

$$\omega = 2\pi f$$
$$= 2\pi \times 10^{7}\ \text{rad/s}$$
$$= 6.283 \times 10^{7}\ \text{rad/s}$$

Thus, the time period, frequency and angular frequency in basic units are,

$$\boxed{\begin{aligned} T &= 10^{-7}\ \text{s} \\ f &= 10^{7}\ \text{Hz} \\ \omega &= 6.283 \times 10^{7}\ \text{rad/s} \end{aligned}}$$

Step 2 of 7

(b)

The frequency is,

$$f = 1\ \text{GHz}$$
$$= 10^{9}\ \text{Hz}$$

The time period is given by,

$$T = \frac{1}{f}$$
$$= 10^{-9}\ \text{s}$$

The angular frequency is,

$$\omega = 2\pi f$$
$$= 2\pi \times 10^{9}\ \text{rad/s}$$
$$= 6.283 \times 10^{9}\ \text{rad/s}$$

Thus, the frequency, time period and angular frequency in basic units are,

$$\boxed{\begin{aligned} f &= 10^{9}\ \text{Hz} \\ T &= 10^{-9}\ \text{s} \\ \omega &= 6.283 \times 10^{9}\ \text{rad/s} \end{aligned}}$$

Step 3 of 7

(c)

The angular frequency is, $\omega = 6.28 \times 10^{2}\ \text{rad/s}$.

Calculate the frequency.

$$f = \frac{\omega}{2\pi}$$
$$= \frac{6.28 \times 10^{2}}{2\pi}$$
$$= 10^{2}\ \text{Hz}$$

The time period is,

$$T = \frac{1}{f}$$
$$= 10^{-2}\ \text{s}$$

Thus, the angular frequency, frequency and time period in basic units are,

$$\boxed{\begin{aligned} \omega &= 6.28 \times 10^{2}\ \text{rad/s} \\ f &= 10^{2}\ \text{Hz} \\ T &= 10^{-2}\ \text{s} \end{aligned}}$$

Step 4 of 7

(d)

The time period is, $T = 10\ \text{s}$.

The frequency is,

$$f = \frac{1}{10}$$
$$= 0.1\ \text{Hz}$$

The angular frequency is,

$$\omega = 2\pi f$$
$$= 2\pi \times 0.1\ \text{rad/s}$$
$$= 0.6283\ \text{rad/s}$$

Thus, the time period, frequency and angular frequency in basic units are,

$$\boxed{\begin{aligned} T &= 10\ \text{s} \\ f &= 0.1\ \text{Hz} \\ \omega &= 0.6283\ \text{rad/s} \end{aligned}}$$

Step 5 of 7

(e)
Frequency, $f = 60\ \text{Hz}$

The time period is,

$$T = \frac{1}{f}$$
$$= \frac{1}{60}$$
$$= 0.0167\ \text{s}$$

The angular frequency is

$$\omega = 2\pi f$$
$$= 2\pi \times 60\ \text{rad/s}$$
$$= 377\ \text{rad/s}$$

Thus, the frequency, time period and angular frequency in basic units are,

$$\boxed{\begin{aligned} f &= 60\ \text{Hz} \\ T &= 0.0167\ \text{s} \\ \omega &= 377\ \text{rad/s} \end{aligned}}$$

Step 6 of 7

(f)

The angular frequency is,

$$\omega = 1\ \text{krad/s}$$
$$= 10^{3}\ \text{rad/s}$$

The frequency is,

$$f = \frac{\omega}{2\pi}$$
$$= \frac{10^{3}}{2\pi}$$
$$= 159.16\ \text{Hz}$$

The time period is,

$$T = \frac{1}{f}$$
$$= \frac{1}{159.16}$$
$$= 6.283 \times 10^{-3}\ \text{s}$$

Thus, the angular frequency, frequency and time period in basic units are,

$$\boxed{\begin{aligned} \omega &= 10^{3}\ \text{rad/s} \\ f &= 159.16\ \text{Hz} \\ T &= 6.283 \times 10^{-3}\ \text{s} \end{aligned}}$$

Step 7 of 7

(g)

The frequency is,

$$f = 1900\ \text{MHz}$$
$$= 1900 \times 10^{6}\ \text{Hz}$$

The time period is,

$$T = \frac{1}{f}$$
$$= \frac{1}{1900 \times 10^{6}}$$
$$= 5.26 \times 10^{-10}\ \text{s}$$

The angular frequency is,

$$\omega = 2\pi f$$
$$= 2\pi \times 1900 \times 10^{6}\ \text{rad/s}$$
$$= 11.938 \times 10^{9}\ \text{rad/s}$$

Thus, the frequency, time period and angular frequency in basic units are,

$$\boxed{\begin{aligned} f &= 1900 \times 10^{6}\ \text{Hz} \\ T &= 5.26 \times 10^{-10}\ \text{s} \\ \omega &= 11.938 \times 10^{9}\ \text{rad/s} \end{aligned}}$$

Step 1 of 2

1.021E

Refer to Figure E1.21 in the textbook.

Apply Kirchhoff's current law at node B in Figure E1.21.

$i_b = i_x$ ..... (1)

Apply Kirchhoff's current law at node E in Figure E1.21.

$i_e = i_b + \beta i_b$

$i_e = (\beta + 1) i_b$ ..... (2)

Apply Kirchhoff's voltage law to the loop BEGB.

$v_x - i_b r_\pi - i_e R_e = 0$

Substitute $(\beta + 1) i_b$ for i_e and i_x for i_b .

$v_x - i_b r_\pi + (\beta + 1) i_b R_e = 0$

$$\begin{aligned} v_x &= i_b r_\pi + (\beta + 1) i_b R_e \\ &= i_x r_\pi + (\beta + 1) i_x R_e \\ &= \left[r_\pi + (\beta + 1) R_e \right] i_x \end{aligned}$$

Therefore,

$$\frac{v_x}{i_x} = r_\pi + (\beta + 1) R_e$$

Step 2 of 2

Calculate the input resistance between the terminals B and G.

$$\begin{aligned} R_{in} &= \frac{v_x}{i_x} \\ &= r_\pi + (\beta + 1) R_e \end{aligned}$$

Thus, the input resistance, R_{in} between terminals B and G is $\boxed{r_\pi + (\beta + 1) R_e}$.

1.021P

We don't have the solution to this problem yet.

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Step 1 of 4

The dc gain of a low-pass single-time constant (STC) type voltage amplifier is,

The 3-dB frequency is, $f_0 = 1000\ \text{Hz}$.

Calculate the gain of the amplifier at $f = 10\ \text{Hz}$.

$$A_v = \frac{K}{\sqrt{1+\left(\frac{f}{f_0}\right)^2}}$$

Substitute 10^3 for K , $1000\ \text{Hz}$ for f_0 and $10\ \text{Hz}$ for f .

$$\begin{aligned} A_v &= \frac{10^3}{\sqrt{1+\left(\frac{10}{1000}\right)^2}} \\ &= \frac{10^3}{\sqrt{1.0001}} \\ &= 999.9 \end{aligned}$$

Calculate the gain of the amplifier in decibels.

$$\begin{aligned} A_{v(\text{dB})} &= 20\log(999.9) \\ &= 60\ \text{dB} \end{aligned}$$

Thus, the gain of the amplifier in dB, $A_{v(\text{dB})}$ is $\boxed{60\ \text{dB}}$.

Step 2 of 4

Calculate the gain of the amplifier at $f = 10\ \text{kHz}$.

$$A_v = \frac{K}{\sqrt{1+\left(\frac{f}{f_0}\right)^2}}$$

Substitute 10^3 for K , $1000\ \text{Hz}$ for f_0 and $10\times10^3\ \text{Hz}$ for f .

$$\begin{aligned} A_v &= \frac{10^3}{\sqrt{1+\left(\frac{10\times10^3}{1000}\right)^2}} \\ &= \frac{10^3}{\sqrt{1+100}} \\ &= \frac{10^3}{\sqrt{1+100}} \\ &= 99.5 \end{aligned}$$

Calculate the gain of the amplifier in decibels.

$$\begin{aligned} A_{v(\text{dB})} &= 20\log(99.5) \\ &= 40\ \text{dB} \end{aligned}$$

Thus, the gain of the amplifier in dB, $A_{v(\text{dB})}$ is $\boxed{40\ \text{dB}}$.

Step 3 of 4

Calculate the gain of the amplifier at $f = 100\ \text{kHz}$.

$$A_v = \frac{K}{\sqrt{1+\left(\frac{f}{f_0}\right)^2}}$$

Substitute 10^3 for K , $1000\ \text{Hz}$ for f_0 and $100\times10^3\ \text{Hz}$ for f .

$$\begin{aligned} A_v &= \frac{10^3}{\sqrt{1+\left(\frac{100\times10^3}{1000}\right)^2}} \\ &= \frac{10^3}{\sqrt{1+(100)^2}} \\ &= \frac{10^3}{\sqrt{10001}} \\ &= 10 \end{aligned}$$

Calculate the gain of the amplifier in decibels.

$$\begin{aligned} A_v &= 20\log(10) \\ &= 20\ \text{dB} \end{aligned}$$

Thus, the gain of the amplifier in dB, $A_{v(\text{dB})}$ is $\boxed{20\ \text{dB}}$.

Step 4 of 4

Calculate the gain of the amplifier at $f = 1\ \text{MHz}$.

$$A_v = \frac{K}{\sqrt{1+\left(\frac{f}{f_0}\right)^2}}$$

Substitute 10^3 for K , $1000\ \text{Hz}$ for f_0 and $1\times10^6\ \text{Hz}$ for f .

$$\begin{aligned} A_v &= \frac{10^3}{\sqrt{1+\left(\frac{10^6}{1000}\right)^2}} \\ &= \frac{10^3}{\sqrt{1+1000^2}} \\ &= \frac{10^3}{\sqrt{100001}} \\ &= 1 \end{aligned}$$

Calculate the gain of the amplifier in decibels.

$$\begin{aligned} A_v &= 20\log(1) \\ &= 0\ \text{dB} \end{aligned}$$

Thus, the gain of the amplifier in dB, $A_{v(\text{dB})}$ is $\boxed{0\ \text{dB}}$.

1.022P

We don't have the solution to this problem yet.

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Step 1 of 5

Draw the transconductance amplifier circuit.

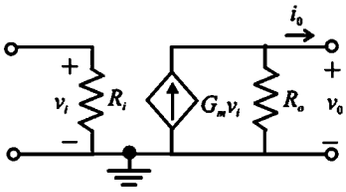


Figure 1

Step 2 of 5

The load to the amplifier consists of a resistance, R_L in parallel with a capacitance, C_L.

The output voltage of the transconductance amplifier is,

$$\begin{aligned} v_o &= G_m v_i \left[R_o \| R_L \| \frac{1}{sC_L} \right] \\ &= \frac{G_m v_i}{\frac{1}{R_o} + \frac{1}{R_L} + sC_L} \end{aligned}$$

The transfer function is,

$$\frac{v_o}{v_i} = \frac{G_m}{\frac{1}{R_o} + \frac{1}{R_L} + sC_L}$$

Take $\left(\frac{1}{R_o} + \frac{1}{R_L} \right)$ common in the denominator.

$$\frac{v_o}{v_i} = \frac{G_m}{\left(\frac{1}{R_o} + \frac{1}{R_L} \right) \left(1 + \frac{sC_L}{\frac{1}{R_o} + \frac{1}{R_L}} \right)} \quad \text{..... (1)}$$

Thus, the transfer function is of the low-pass single-time constant (STC) type voltage amplifier.

Step 3 of 5

The dc gain is,

$$20\log(K) \geq 40 \text{ dB}$$
$$\log(K) \geq 2$$
$$K \geq 10^2$$
$$K \geq 100$$

Compare with the standard STC low pass filter, the dc gain is,

$$\begin{aligned} K &= \frac{G_m}{\frac{1}{R_o} + \frac{1}{R_L}} \\ &\geq 100 \end{aligned}$$

Rearrange the terms.

$$\frac{1}{R_o} + \frac{1}{R_L} \leq \frac{G_m}{100} \quad \text{..... (2)}$$

Step 4 of 5

Substitute 50 k for R_o and 10 mA/V for G_m in equation (2).

$$\begin{aligned} \frac{1}{50 \text{ k}} + \frac{1}{R_L} &\leq \frac{G_m}{100} \\ &\leq \frac{10 \text{ mA/V}}{100} \\ &\leq 0.1 \text{ mA/V} \end{aligned}$$

Simplify further.

$$\begin{aligned} \frac{1}{R_L} &\leq 0.1 - \frac{1}{50} \\ &\leq 0.08 \text{mA/V} \end{aligned}$$

Therefore,

$$\begin{aligned} R_L &\geq \frac{1}{0.08} \text{ k}\Omega \\ &\geq 12.5 \text{k}\Omega \end{aligned}$$

Thus, the lowest value of the load resistance, R_L is $\boxed{12.5 \text{ k}\Omega}$.

Step 5 of 5

Compare the equation (1) with the standard STC low pass type.

$$\begin{aligned} \omega_o &= \frac{1}{C_L} \left(\frac{1}{R_o} + \frac{1}{R_L} \right) \\ &\geq 2\pi \times 100 \text{ kHz} \end{aligned}$$

Therefore,

$$C_L \leq \frac{1}{\omega_o} \left(\frac{1}{R_o} + \frac{1}{R_L} \right)$$

Substitute $2\pi \times 100 \text{ kHz}$ for ω_o, 50×10^3 for R_o and 12.5×10^3 for R_L.

$$\begin{aligned} C_L &\leq \left(\frac{1}{\frac{50 \times 10^3 + 12.5 \times 10^3}{2\pi \times 10^5}} \right) \\ &\leq \left(\frac{0.02 \times 10^{-3} + 0.08 \times 10^{-3}}{2\pi \times 10^5} \right) \\ &\leq \left(\frac{0.1 \times 10^{-3}}{2\pi \times 10^5} \right) \\ &\leq 159.2 \text{pF} \end{aligned}$$

Thus, the highest value of the capacitance, C_L is $\boxed{159.2 \text{ pF}}$.

1.023P

We don't have the solution to this problem yet.

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Step 1 of 2

1.024E

Refer to Figure E1.24 in the textbook.

Write the expression for voltage V_2.

$$V_2 = V_s\left(\frac{R_i}{R_s+\frac{1}{sC}+R_i}\right)$$

$$\frac{V_2}{V_s} = \frac{R_i}{R_s+\frac{1}{sC}+R_i}$$

$$= \frac{sCR_i}{sC(R_s+R_i)+1}$$

$$= \frac{s\left(\frac{R_i}{R_s+R_i}\right)}{s+\frac{1}{C(R_s+R_i)}}$$

The 3 dB frequency is,

$$\omega_0 = \frac{1}{C(R_s+R_i)}$$

$$2\pi f_0 = \frac{1}{C(R_s+R_i)}.$$

$$f_0 = \frac{1}{2\pi C(R_s+R_i)}$$

Step 2 of 2

The 3 dB frequency is not higher than 100 Hz.

Therefore,

$$\frac{1}{2\pi C(R_s+R_i)} \le 100$$

Substitute $1\ \text{k}\Omega$ for R_s and $9\ \text{k}\Omega$ for R_i.

$$\frac{1}{2\pi C\left(1\times10^3+9\times10^3\right)} \le 100$$

$$\frac{1}{2\pi C(1+9)\times10^3} \le 100$$

$$\frac{1}{2\pi\times10^4 C} \le 100$$

$$C \ge \frac{1}{2\pi\times10^4(100)}$$

Thus, the smallest value of the capacitance C is,

$$C = \frac{1}{2\pi\times10^4(100)}$$

$$= 0.16\ \mu\text{F}$$

Therefore, the smallest value of the capacitance, C is $\boxed{0.16\ \mu\text{F}}$.

1.024P

We don't have the solution to this problem yet.

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1.025P

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Step 1 of 3

1.026P

The two alternative representations of a signal source – The Thevenin form and the Norton form are shown in Figure 1.

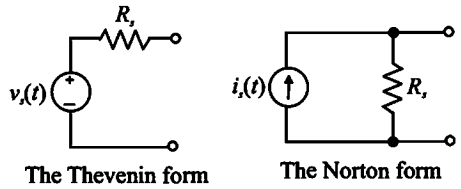


Figure 1

Step 2 of 3

The current supplied by the source is i_o.

The voltage appearing between the output terminals is v_o.

From the Thevenin form of the signal source, the voltage appearing between the output terminals is,

$$v_o = v_s - i_o R_s$$

From the Norton equivalent,

$$\begin{aligned} v_o &= R_s(i_s - i_o) \\ &= R_s i_s - R_s i_o \\ &= -R_s i_o + v_s \end{aligned}$$

This is the form of equation of the straight line which has a slope of $-R_s$ and the intercept on v_o axis is v_s and the intercept on the i_o axis is $\frac{v_s}{R_s}$.

Step 3 of 3

The sketch of output voltage, v_o versus output current i_o is shown in Figure 2.

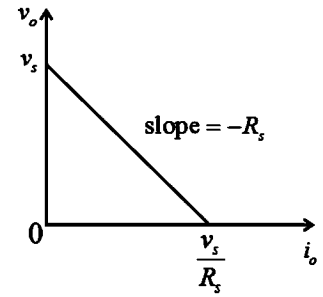


Figure 2

1.027P

We don't have the solution to this problem yet.

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1.028P

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Step 1 of 4

(a)

The rms voltage of North American household-power voltage is $V_{rms} = 117\,V_{rms}$.

Calculate the peak voltage.

$$V_{rms} = \frac{V_{peak}}{\sqrt{2}}$$
$$V_{peak} = \sqrt{2}V_{rms}$$

Substitute $117\,V_{rms}$ for V_{rms} in the equation.

$$V_{peak} = \sqrt{2}(117)$$
$$= 165.46\,V$$

Therefore, the peak voltage, V_{peak} is $\boxed{165.46\,V_{peak}}$.

Step 2 of 4

(b)

The rectifier circuit peak voltage is $V_{peak} = 33.9\,V_{peak}$.

Calculate the rms voltage.

$$V_{rms} = \frac{V_{peak}}{\sqrt{2}}$$

Substitute $33.9\,V_{peak}$ for V_{peak} in the equation.

$$V_{rms} = \frac{33.9}{\sqrt{2}}$$
$$= 23.97\,V_{rms}$$
$$\simeq 24\,V_{rms}$$

Therefore, the rms voltage, V_{rms} is $\boxed{24\,V_{rms}}$.

Step 3 of 4

(c)

The rms voltage of Europe household-power voltage is $V_{rms} = 220\,V_{rms}$.

Calculate the peak voltage.

$$V_{peak} = \sqrt{2}V_{rms}$$

Substitute $220\,V_{rms}$ for V_{rms} in the equation.

$$V_{peak} = \sqrt{2}(220)$$
$$= 311.124\,V$$

Therefore, the peak voltage, V_{peak} is $\boxed{311.124\,V_{peak}}$.

Step 4 of 4

(d)

The rms voltage of North American high-voltage transmission voltage is $V_{rms} = 220\,kV_{rms}$.

Calculate the peak voltage.

$$V_{peak} = \sqrt{2}V_{rms}$$

Substitute $220\,kV_{rms}$ for V_{rms} in the equation.

$$V_{peak} = \sqrt{2}\left(220\times10^{3}\right)$$
$$= 311.124\,kV$$

Therefore, the peak voltage, V_{peak} is $\boxed{311.124\,kV_{peak}}$.

1.030P

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Step 1 of 6

Refer to symmetrical square wave signal in Figure 1.5 in the textbook.

Write the symmetrical square-wave voltage signal expression.

$$v(t)=\frac{4V}{\pi}\left(\sin\omega_o t+\frac{1}{3}\sin 3\omega_o t+\frac{1}{5}\sin 5\omega_o t+\ldots\right) \quad \ldots\ldots (1)$$

Here,

V is the amplitude of the square wave and

$$\omega_o=\frac{2\pi}{T}$$

Here,

T is the time period of the square wave

Write the expression for the characterized signal.

$$v(t)=\frac{1}{2}+\frac{2}{\pi}\left(\sin 2000\pi t+\frac{1}{3}\sin 6000\pi t+\frac{1}{5}\sin 10000\pi t+\ldots\right) \quad \ldots\ldots (2)$$

Step 2 of 6

Compare equations (1) and (2).

Calculate the dc voltage value (or) average voltage value.

$$V_{dc}=\frac{1}{2}$$
$$=0.5$$

Therefore, the average value, V_{dc} is $\boxed{0.5\text{ V}}$.

Compare equations (1) and (2).

Calculate the voltage peak (or) amplitude value.

$$\frac{4V}{\pi}=\frac{2}{\pi}$$
$$4V=2$$
$$V=0.5\text{ V}$$

Therefore, the peak value, V is $0.5\text{ V}_{(\text{peak})}$.

Step 3 of 6

Calculate the peak-to-peak voltage.

$$V_{p\text{-}p}=2V_{(\text{peak})}$$
$$=2(0.5\text{ V}) \qquad \left(\text{Since, } V=0.5\text{ V}_{(\text{peak})}\right)$$
$$=1\text{ V}$$

Therefore, the peak-to-peak voltage, $V_{p\text{-}p}$ is $\boxed{1\text{ V}}$.

The symmetrical square-wave with its level shifted by 0.5 V.

Hence, the lowest value is $\boxed{0\text{ V}}$ and the highest value is $\boxed{1\text{ V}}$.

Step 4 of 6

Calculate the angular frequency.

$$\omega_o=2000\pi$$
$$2\pi f_o=2000\pi$$
$$f_o=\frac{2000\pi}{2\pi}$$
$$f_o=1000\text{ Hz}$$

Therefore, the frequency, f_o is $\boxed{1000\text{ Hz}}$.

Step 5 of 6

Calculate the time period.

$$T=\frac{1}{f_o}$$
$$=\frac{1}{1000\text{ Hz}} \qquad (\text{Since}, f_o=1000\text{ Hz})$$
$$=1\text{ ms}$$

Therefore, the time period, T is $\boxed{1\text{ ms}}$.

Step 6 of 6

The square-waveform is shown in Figure 1.

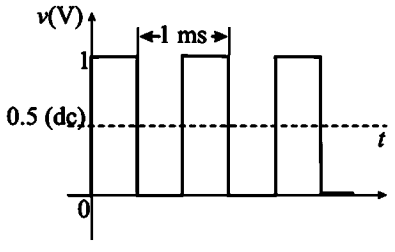


Figure 1

Step 1 of 5

Write the expression for Fourier series representation of a square wave signal.

$$v(t)=\frac{4V}{\pi}\left(\sin\omega_o t+\frac{1}{3}\sin 3\omega_o t+\frac{1}{5}\sin 5\omega_o t+...\right) \quad \text{..... (1)}$$

Here,

V is the amplitude of the square wave and

ω_o is the angular frequency $\omega_o=\frac{2\pi}{T}$

Here,

T is time period of square wave

The $i^{th}\left(i \text{ is odd}\right)$ harmonic of fundamental frequency is,

$$f_i=98\text{ kHz}$$

And $(i+2)^{th}$ harmonic frequency is,

$$f_{i+2}=126\text{ kHz}$$

Subtract i^{th} harmonic frequency from $(i+2)^{th}$ harmonic frequency.

$$\begin{aligned} f_{i+2}-f_i &= 126\text{ kHz}-98\text{ kHz} \\ &= 28\text{ kHz} \end{aligned}$$

Calculate the fundamental frequency.

$$2f_o=28\text{ kHz}$$
$$f_o=14\text{ kHz}$$

Therefore, the fundamental frequency, f_o is $\boxed{14\text{ kHz}}$.

Step 2 of 5

Write the expression for i^{th} frequency.

$$f_i=if_o$$
$$i=\frac{98\text{ kHz}}{f_o}$$

Substitute 14 kHz for f_o in the equation.

$$\begin{aligned} i &= \frac{98\text{ kHz}}{14\text{ kHz}} \\ &= 7 \end{aligned}$$

The amplitude of 7th harmonic is 63mV.

Let amplitude of fundamental be A_o.

Calculate the amplitude of 7th harmonic.

$$63\text{ mV}=\frac{A_o}{7}$$
$$A_o=(7)\left(63\times10^{-3}\right)$$
$$A_o=441\text{ mV}$$

Therefore, the amplitude of fundamental frequency component, A_o is $\boxed{441\text{ mV}}$.

Step 3 of 5

Write the expression for the rms voltage of fundamental term.

$$A_{o\text{-rms}}=\frac{A_o}{\sqrt{2}}$$

Substitute 441 mV for A_o in the equation.

$$\begin{aligned} A_{o\text{-rms}} &= \frac{A_o}{\sqrt{2}} \\ &= \frac{441\text{ mV}}{\sqrt{2}} \\ &= 311.8\text{ mV} \end{aligned}$$

Therefore, the rms voltage, $A_{o\text{-rms}}$ of fundamental term is $\boxed{311.8\text{ mV}}$.

Step 4 of 5

Calculate the peak-to-peak amplitude of square wave signal.

$$\begin{aligned} A_o &= \frac{4V}{\pi} \\ &= \frac{\frac{4V}{2}}{\pi} \qquad \left(\text{Since, } V_{p\text{-}p}=2V_{(peak)}\right) \end{aligned}$$

$$441\text{ mV}=\frac{2V}{\pi}$$

$$\begin{aligned} V &= \frac{(441\text{ mV})\pi}{2} \\ &= 692.7\text{ mV} \end{aligned}$$

Therefore, the peak-to-peak amplitude of square wave signal, V is $\boxed{692.7\text{ mV}}$.

Step 5 of 5

Write the expression for the time period of the square-wave signal.

$$T=\frac{1}{f_o}$$

Substitute 14 kHz for f_o in the equation.

$$\begin{aligned} T &= \frac{1}{14\text{ kHz}} \\ &= 71.43\ \mu\text{s} \end{aligned}$$

Therefore, the time period, T of the originating square-wave signal is $\boxed{71.43\ \mu\text{s}}$.

Step 1 of 3

1.033P

The symmetrical square-wave voltage signal is shown in Figure 1.

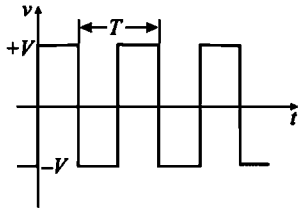


Figure 1

Step 2 of 3

The symmetrical square wave and sine wave provides same power.

$P_{square} = P_{sine}$ ..... (1)

The sine wave signal peak amplitude is $\hat{V}$

Let us consider the square wave signal peak amplitude is V_{square} .

Calculate the sine wave power delivered to *R*.

$$P_{sine} = \frac{\left(\hat{V}\right)^2}{R}$$
$$= \frac{\left(\frac{\hat{V}_{rms}}{\sqrt{2}}\right)^2}{R}$$
$$= \frac{\hat{V}_{rms}^2}{2R}$$

Step 3 of 3

Calculate the square wave power delivered to *R*.

$$P_{square} = \frac{\left(V_{square}\right)^2}{R}$$

$$P_{sine} = \frac{\left(V_{square}\right)^2}{R}$$

Substitute $\frac{\hat{V}_{rms}^2}{2R}$ for P_{sine} in equation-(1)

$$\frac{\hat{V}_{rms}^2}{2R} = \frac{\left(V_{square}\right)^2}{R}$$
$$\left(V_{square}\right)^2 = \frac{\hat{V}_{rms}^2}{2}$$
$$V_{square} = \frac{\hat{V}_{rms}}{\sqrt{2}}$$

Therefore, from the result, it is conclude that both waveforms are independent of frequency.

1.034P

We don't have the solution to this problem yet.

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Step 1 of 7

Consider a 4-bit digital word $b_3b_2b_1b_0$. The most-significant bit, b_3 is interpreted as a sign bit- 0 indicates positive and 1 for negative values. By this scheme, the values are represented as shown in table 1:

Table 1

4-bit digital word $b_3b_2b_1b_0$	$D=(1-2b_3)(b_0+2b_1+4b_2)$
0000	+0
0001	+1
0010	+2
0011	+3
0100	+4
0101	+5
0110	+6
0111	+7
1000	–0
1001	–1
1010	–2
1011	–3
1100	–4
1101	–5
1110	–6
1111	–7

As $+0$ and -0 are same (equal) hence zero is represented in this scheme in two ways 0000 and 1000.

Step 2 of 7

From the table 1,

The minimum value of D is $D_{min}=-7$ and the maximum value of D is $D_{max}=7$.

The change in voltage is,

$$\Delta V=0.5\,\text{V}$$

Write the expression for the minimum amplitude of analog signal.

$$V_{min}=D_{min}\Delta V-\frac{\Delta V}{2}$$

Substitute $0.5\ \text{V}$ for ΔV and -7 for D_{min} in the equation.

$$\begin{aligned}V_{min}&=(-7)(0.5)-\left(\frac{0.5}{2}\right)\\&=-3.5-0.25\\&=-3.75\end{aligned}$$

Step 3 of 7

Calculate the maximum amplitude of analog signal.

$$V_{max}=D_{max}\Delta V+\frac{\Delta V}{2}$$

Substitute $0.5\ \text{V}$ for ΔV and 7 for D_{max} in the equation.

$$\begin{aligned}V_{max}&=(7)(0.5)+\left(\frac{0.5}{2}\right)\\&=3.5+0.25\\&=3.75\end{aligned}$$

Therefore, the full range of the analog signal is $\boxed{-3.75\ \text{V to}\ 3.75\ \text{V}}$.

Step 4 of 7

Calculate the signed-magnitude digital code for an input of $+2.5\ \text{V}$.

$$\begin{aligned}D&=\frac{+2.5}{\Delta V}\\&=\frac{+2.5}{0.5}\qquad(\text{Since},\ \Delta V=0.5\ \text{V})\\&=+5\end{aligned}$$

Therefore, the signed-magnitude digital code, D for an input of $+2.5\,\text{V}$ is $\boxed{+5=0101}$.

Step 5 of 7

Calculate the signed-magnitude digital code for an input of $-3\ \text{V}$.

$$\begin{aligned}D&=\frac{-3}{\Delta V}\\&=\frac{-3}{0.5}\qquad(\text{Since},\ \Delta V=0.5\text{V})\\&=-6\end{aligned}$$

Therefore, the signed-magnitude digital code, D for an input of $-3\,\text{V}$ is $\boxed{-6=1110}$.

Step 6 of 7

Calculate the signed-magnitude digital code for an input of $+2.7\,\text{V}$.

$$\begin{aligned}D&=\frac{+2.7}{\Delta V}\\&=\frac{+2.7}{0.5}\qquad(\text{Since},\ \Delta V=0.5\text{V})\\&=+5.4\end{aligned}$$

Therefore, the signed-magnitude digital code, D for an input of $+2.7\,\text{V}$ is $\boxed{+5.4=0101}$.

Step 7 of 7

Calculate the signed-magnitude digital code for an input of $-2.8\,\text{V}$.

$$\begin{aligned}D&=\frac{-2.8}{\Delta V}\\&=\frac{-2.8}{0.5}\qquad(\text{Since},\ \Delta V=0.5\text{V})\\&=-5.6\end{aligned}$$

Therefore, the signed-magnitude digital code, D for an input of $-2.8\,\text{V}$ is $\boxed{-5.6=1110}$.

1.036P

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1.037P

We don't have the solution to this problem yet.

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1.038P

Calculate the speed of the system in bits/second.

$$\text{speed of the system} = \frac{N}{T}$$
$$= \frac{N}{\frac{1}{f}}$$

Substitute 16 bits for N and $44.1\ \text{kHz}$ for f in the equation.

$$\text{speed of the system} = \frac{16\ \text{bits}}{\frac{1}{44.1\ \text{kHz}}}$$
$$= (16\ \text{bits})(44.1\ \text{kHz})$$
$$= 7.056\times10^{5}\ \text{bits/second}$$

Therefore, the speed of the system in bits/second is $\boxed{7.056\times10^{5}\ \text{bits/second}}$.

1.039P

We don't have the solution to this problem yet.

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Step 1 of 6

Calculate the voltage gain.

$$A_v = \frac{v_o}{v_i}$$

Substitute $2.2\ \text{V}$ for v_o and $0.2\ \text{V}$ for v_i in the equation.

$$\begin{aligned} A_v &= \frac{2.2\ \text{V}}{0.2\ \text{V}} \\ &= 11\ \text{V/V} \end{aligned}$$

Therefore, the voltage gain, A_v is $\boxed{11\ \text{V/V}}$.

Calculate the voltage gain in decibels.

$$A_{v\text{-dB}} = 20\log|A_v|$$

Substitute $11\ \text{V/V}$ for A_v in the equation.

$$\begin{aligned} A_{v\text{-dB}} &= 20\log|11| \\ &= 20(1.0413) \\ &= 20.827\ \text{dB} \end{aligned}$$

Therefore, the voltage gain, $A_{v\text{-dB}}$ in decibels is $\boxed{20.827\ \text{dB}}$.

Step 2 of 6

Calculate the output current.

$$\begin{aligned} i_o &= \frac{v_o}{R_L} \\ &= \frac{2.2}{100} \\ &= 22\ \text{mA} \end{aligned}$$

Calculate the current gain.

$$A_i = \frac{i_o}{i_i}$$

Substitute $22\ \text{mA}$ for i_o, and $1\ \text{mA}$ for i_i in the equation.

$$\begin{aligned} A_i &= \frac{22\ \text{mA}}{1\ \text{mA}} \\ &= 22\ \text{A/A} \end{aligned}$$

Therefore, the current gain, A_i is $\boxed{22\ \text{A/A}}$.

Calculate the current gain in decibels.

$$A_{i\text{-dB}} = 20\log|A_i|$$

Substitute $22\ \text{A/A}$ for A_i in the equation.

$$\begin{aligned} A_{i\text{-dB}} &= 20\log|22| \\ &= 20(1.3424) \\ &= 26.848\ \text{dB} \end{aligned}$$

Therefore, the current gain in decibels, $A_{i\text{-dB}}$ is $\boxed{26.848\ \text{dB}}$.

Step 3 of 6

Calculate the power gain.

$$A_p = A_v A_i$$

Substitute $11\ \text{V/V}$ for A_v and $22\ \text{A/A}$ for A_i in the equation.

$$\begin{aligned} A_p &= (11)(22) \\ &= 242\ \text{W/W} \end{aligned}$$

Therefore, the power gain, A_p is $\boxed{242\ \text{W/W}}$.

Calculate the power gain in decibels.

$$A_{p\text{-dB}} = 10\log|A_p|$$

Substitute $242\ \text{W/W}$ for A_p in the equation.

$$\begin{aligned} A_{p\text{-dB}} &= 10\log|242| \\ &= 10(2.3838) \\ &= 23.838\ \text{dB} \end{aligned}$$

Therefore, the power gain, $A_{p\text{-dB}}$ in decibels is $\boxed{23.838\ \text{dB}}$.

Step 4 of 6

Calculate the power supplied by DC sources.

$$\begin{aligned} P_{dc} &= V_{CC} I_{CC} + V_{EE} I_{EE} \\ &= (3\ \text{V})(20\ \text{mA}) + (3\ \text{V})(20\ \text{mA}) \\ &= 60\ \text{mW} + 60\ \text{mW} \\ &= 0.12\ \text{W} \end{aligned}$$

Therefore, the power, P_{dc} supplied by the dc sources is $\boxed{0.12\ \text{W}}$.

Calculate the input power.

$$\begin{aligned} P_i &= V_{i_{rms}} I_{i_{rms}} \\ &= \left(\frac{0.2\ \text{V}}{\sqrt{2}}\right)\left(\frac{1\ \text{mA}}{\sqrt{2}}\right) \\ &= \frac{0.2\ \text{mW}}{2} \\ &= 0.1\ \text{mW} \end{aligned}$$

Step 5 of 6

Calculate the Power delivered to the load.

$$\begin{aligned} P_L &= V_{o_{rms}} I_{o_{rms}} \\ &= \left(\frac{2.2\ \text{V}}{\sqrt{2}}\right)\left(\frac{22\ \text{mA}}{\sqrt{2}}\right) \\ &= \frac{48.4\ \text{mW}}{2} \\ &= 24.2\ \text{mW} \end{aligned}$$

Calculate the Power dissipated in the amplifier circuit.

$$P_{dc} + P_i = P_L + P_{\text{dissipated}}$$

$$\begin{aligned} P_{\text{dissipated}} &= P_{dc} + P_i - P_L \\ &= 0.12\ \text{W} + 0.1\ \text{mW} - 24.2\ \text{mW} \\ &= 95.9\ \text{mW} \end{aligned}$$

Therefore, the power dissipated, $P_{\text{dissipated}}$ in the amplifier circuit is $\boxed{95.9\ \text{mW}}$.

Step 6 of 6

Calculate the amplifier efficiency.

$$\eta = \frac{P_L}{P_{dc}} \times 100$$

Substitute $24.2\ \text{mW}$ for P_L, and $0.12\ \text{W}$ for P_{dc} in the equation.

$$\begin{aligned} \eta &= \left(\frac{24.2\ \text{mW}}{0.12\ \text{W}}\right)(100\%) \\ &= (0.20167)(100\%) \\ &= 20.167\% \end{aligned}$$

Therefore, the amplifier efficiency, η is $\boxed{20.167\%}$.

1.041P

We don't have the solution to this problem yet.

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1.042P

We don't have the solution to this problem yet.

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Step 1 of 6

Refer to Figure 1.16(b) in the equation.

(a)

Write the expression for gain of the voltage amplifier.

$$A_v = \frac{v_o}{v_s}$$

$$A_v = \frac{v_i}{v_s} \times \frac{v_o}{v_i} \quad \text{..... (1)}$$

Apply voltage division rule and determine the expression for $\frac{v_i}{v_s}$.

$$\frac{v_i}{v_s} = \frac{R_i}{R_i + R_s}$$

Apply voltage division rule and determine the expression for $\frac{v_o}{v_i}$.

$$\frac{v_o}{v_i} = A_{vo}\left(\frac{R_L}{R_L + R_o}\right)$$

Step 2 of 6

Substitute $\frac{R_i}{R_i + R_s}$ for $\frac{v_i}{v_s}$ and $\frac{R_L}{R_L + R_o}$ for $\frac{v_o}{A_v v_i}$ in equation (1).

$$A_v = A_{vo}\left(\frac{R_i}{R_i + R_s}\right)\left(\frac{R_L}{R_L + R_o}\right) \quad \text{..... (2)}$$

Substitute 10 for A_{vo}, $10R_s$ for R_i and $10R_o$ for R_L in the equation.

$$\begin{aligned} A_v &= 10\left(\frac{10R_s}{10R_s + R_s}\right)\left(\frac{10R_o}{10R_o + R_o}\right) \\ &= 10\left(\frac{10R_s}{11R_s}\right)\left(\frac{10R_o}{11R_o}\right) \\ &= 10\left(\frac{10}{11}\right)\left(\frac{10}{11}\right) \\ &= 8.2645 \text{ V/V} \end{aligned}$$

Therefore, the overall voltage gain is $\boxed{8.2645 \text{ V/V}}$.

Step 3 of 6

Convert the overall voltage gain into decibels.

$$\begin{aligned} A_v &= 20\log(8.2645) \\ &= 20(0.91721) \\ &= 18.344 \text{ dB} \end{aligned}$$

Therefore, the overall voltage gain in decibels is $\boxed{18.344 \text{ dB}}$.

Step 4 of 6

(b)

Substitute 10 for A_{vo}, R_s for R_i and R_o for R_L in equation (2).

$$\begin{aligned} A_v &= 10\left(\frac{R_s}{R_s + R_s}\right)\left(\frac{R_o}{R_o + R_o}\right) \\ &= 10\left(\frac{R_s}{2R_s}\right)\left(\frac{R_o}{2R_o}\right) \\ &= 10\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) \\ &= 2.5 \text{ V/V} \end{aligned}$$

Therefore, the overall voltage gain is $\boxed{2.5 \text{ V/V}}$.

Convert the overall voltage gain into decibels.

$$\begin{aligned} A_v &= 20\log(2.5) \\ &= 20(0.3979) \\ &= 7.9588 \text{ dB} \end{aligned}$$

Therefore, the overall voltage gain in decibels is $\boxed{7.9588 \text{ dB}}$.

Step 5 of 6

(c)

Substitute 10 for A_{vo}, $\frac{R_s}{10}$ for R_i and $\frac{R_o}{10}$ for R_L in equation (2).

$$\begin{aligned} A_v &= 10\left(\frac{\frac{R_s}{10}}{\frac{R_s}{10} + R_s}\right)\left(\frac{\frac{R_o}{10}}{\frac{R_o}{10} + R_o}\right) \\ &= 10\left(\frac{0.1R_s}{0.1R_s + R_s}\right)\left(\frac{0.1R_o}{0.1R_o + R_o}\right) \\ &= 10\left(\frac{0.1}{1.1}\right)\left(\frac{0.1}{1.1}\right) \\ &= 0.08264 \text{ V/V} \end{aligned}$$

Therefore, the overall voltage gain is $\boxed{0.08264 \text{ V/V}}$.

Step 6 of 6

Convert the overall voltage gain into decibels.

$$\begin{aligned} A_v &= 20\log(0.08264) \\ &= 20(-1.08278) \\ &= -21.6557 \text{ dB} \end{aligned}$$

Therefore, the overall voltage gain in decibels is $\boxed{-21.6557 \text{ dB}}$.

1.044P

We don't have the solution to this problem yet.

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Step 1 of 8

Draw the circuit diagram of voltage gain amplifier.

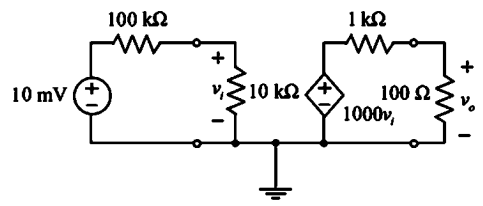


Figure 1

Step 2 of 8

Determine the overall gain for voltage amplifier.

$$A_v = \frac{v_o}{v_s}$$
$$= A_{vo}\left(\frac{R_i}{R_i + R_s}\right)\left(\frac{R_L}{R_L + R_o}\right)$$

Substitute $1000\ \text{V/V}$ for A_{vo}, $10\ \text{k}\Omega$ for R_i, $100\ \text{k}\Omega$ for R_s, $100\ \Omega$ for R_L and $1\ \text{k}\Omega$ for R_o in the equation.

$$A_v = (1000)\left(\frac{10\times10^3}{10\times10^3 + 100\times10^3}\right)\left(\frac{100}{100 + 1\times10^3}\right)$$
$$= (1000)\left(\frac{10\times10^3}{110\times10^3}\right)\left(\frac{100}{1100}\right)$$
$$= (1000)\left(\frac{1}{11}\right)\left(\frac{1}{11}\right)$$
$$= 8.2645\ \text{V/V}$$

Therefore, the overall gain for voltage amplifier is $\boxed{8.2645\ \text{V/V}}$.

Step 3 of 8

Voltage across the input resistance is calculated as

$$v_i = \left(\frac{R_i}{R_i + R_s}\right)v_s$$
$$= \left(\frac{10\times10^3}{10\times10^3 + 100\times10^3}\right)\left(10\times10^{-3}\right)$$
$$= \left(\frac{10\times10^3}{110\times10^3}\right)\left(10\times10^{-3}\right)$$
$$= \left(\frac{1}{11}\right)\left(10\times10^{-3}\right)$$

$$v_i = 0.909\ \text{mV}$$

Step 4 of 8

Calculate the percentage of voltage drop across the input resistance.

$$\%\text{Voltage drop} = \frac{v_i}{v_s}\times100$$
$$= \frac{0.909\ \text{mV}}{10\ \text{mV}}\times100$$
$$= \frac{0.909}{10}\times100$$
$$= 9.09\%$$

Determine the value of voltage drop in the internal resistance.

$$V_R = (100 - 9.09)\%$$
$$= 90.91\%$$

Therefore, remaining 90.91% of source voltage is lost in the internal resistance.

Step 5 of 8

Determine the voltage across the output resistance.

$$v_o = A_{vo}v_i\left(\frac{R_L}{R_L + R_o}\right)$$
$$= (1000)\left(0.909\times10^{-3}\right)\left(\frac{100}{100 + 1000}\right)$$
$$= (1000)\left(0.909\times10^{-3}\right)\left(\frac{100}{1100}\right)$$
$$= 82.6363\ \text{mV}$$

Step 6 of 8

Calculate the percentage of voltage drop across the output resistance.

$$\%\text{Voltage drop} = \frac{v_o}{A_{vo}v_i}\times100$$
$$= \frac{82.6363\times10^{-3}}{(1000)\left(0.909\times10^{-3}\right)}\times100$$
$$= 0.0909\times100$$
$$= 9.09\%$$

Step 7 of 8

Determine the value of voltage drop in the internal resistance.

$$V_R = (100 - 9.09)\%$$
$$= 90.91\%$$

Therefore, remaining 90.91% of source voltage is lost in the output resistance of the amplifier circuit.

Consider the source connected directly to load.

$$A_{v2} = \frac{v_o}{v_s}$$
$$= \frac{R_L}{R_L + R_s}$$
$$= \frac{100}{100 + 100\times10^3}$$
$$= \frac{100}{100.1\times10^3}$$
$$= 0.999\times10^{-3}$$

Therefore, the gain is $\boxed{0.999\ \text{mV/V}}$.

Step 8 of 8

Determine the ratio of voltage gains.

$$\frac{A_{v1}}{A_{v2}} = \frac{8.2645}{0.999\times10^{-3}}$$
$$= 8.2727\times10^3$$
$$= 8272.7$$

Therefore, the ratio of voltage gains is $\boxed{8272.7}$.

Therefore, it can be conclude that, the voltage gain due to amplifier is more beneficent than directly connected to the load.

1.046P

We don't have the solution to this problem yet.

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Step 1 of 5

Refer to Figure 1.17 in the textbook.

Apply voltage division rule and determine the value of v_{i1}.

$$v_{i1} = (v_S)\left(\frac{1\ \text{M}\Omega}{1\ \text{M}\Omega + 100\ \text{k}\Omega}\right)$$

Apply voltage division rule and determine the value of v_{i2}.

$$v_{i2} = (10v_{i1})\left(\frac{100\ \text{k}\Omega}{100\ \text{k}\Omega + 1\ \text{k}\Omega}\right)$$

Apply voltage division rule and determine the value of v_{i3}.

$$v_{i3} = (100v_{i2})\left(\frac{10\ \text{k}\Omega}{10\ \text{k}\Omega + 1\ \text{k}\Omega}\right)$$

Apply voltage division rule and determine the value of v_L.

$$v_L = v_{i3}\left(\frac{100\ \Omega}{100\ \Omega + 10\ \Omega}\right)$$

Step 2 of 5

Determine the voltage gain of the amplifier.

$$\frac{v_L}{v_s} = \frac{v_L}{v_{i3}} \times \frac{v_{i3}}{v_{i2}} \times \frac{v_{i2}}{v_{i1}} \times \frac{v_{i1}}{v_s}$$

$$= \left(\frac{100\ \Omega}{100\ \Omega + 10\ \Omega}\right) \times (100)\left(\frac{10\ \text{k}\Omega}{10\ \text{k}\Omega + 1\ \text{k}\Omega}\right) \times (10)\left(\frac{100\ \text{k}\Omega}{100\ \text{k}\Omega + 1\ \text{k}\Omega}\right) \times \left(\frac{1\ \text{M}\Omega}{1\ \text{M}\Omega + 100\ \text{k}\Omega}\right)$$

$$\frac{v_L}{v_S} = \left(\frac{100}{110}\right) \times (100)\left(\frac{10 \times 10^3}{11 \times 10^3}\right) \times (10)\left(\frac{100 \times 10^3}{101 \times 10^3}\right) \times \left(\frac{1 \times 10^6}{1.1 \times 10^6}\right)$$

$$= (10^3)\left(\frac{10}{11}\right)\left(\frac{10}{11}\right)\left(\frac{100}{101}\right)\left(\frac{1}{1.1}\right)$$

$$\frac{v_L}{v_S} = 743.8760\ \text{V/V} \quad \ldots\ldots (1)$$

Therefore, the voltage gain is $\boxed{\frac{v_o}{v_S} = 743.8760\ \text{V/V}}$.

Step 3 of 5

From figure 1.17 in the textbook, interchange first and second stage.

Draw the modified circuit diagram.

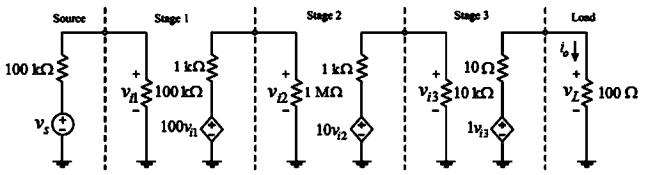


Figure 1

Step 4 of 5

Apply voltage division rule and determine the value of v_{i1}.

$$v_{i1} = (v_S)\left(\frac{100\ \text{k}\Omega}{100\ \text{k}\Omega + 100\ \text{k}\Omega}\right)$$

Apply voltage division rule and determine the value of v_{i2}.

$$v_{i2} = (100v_{i1})\left(\frac{1\ \text{M}\Omega}{1\ \text{M}\Omega + 1\ \text{k}\Omega}\right)$$

Apply voltage division rule and determine the value of v_{i3}.

$$v_{i3} = (10v_{i2})\left(\frac{10\ \text{k}\Omega}{10\ \text{k}\Omega + 1\ \text{k}\Omega}\right)$$

Apply voltage division rule and determine the value of v_L.

$$v_L = v_{i3}\left(\frac{100\ \Omega}{100\ \Omega + 10\ \Omega}\right)$$

Step 5 of 5

Determine the voltage gain of the amplifier.

$$\frac{v_L}{v_s} = \frac{v_L}{v_{i3}} \times \frac{v_{i3}}{v_{i2}} \times \frac{v_{i2}}{v_{i1}} \times \frac{v_{i1}}{v_s}$$

$$= \left(\frac{100\ \Omega}{100\ \Omega + 10\ \Omega}\right) \times (10)\left(\frac{10\ \text{k}\Omega}{10\ \text{k}\Omega + 1\ \text{k}\Omega}\right) \times (100)\left(\frac{1\ \text{M}\Omega}{1\ \text{M}\Omega + 1\ \text{k}\Omega}\right) \times \left(\frac{100\ \text{k}\Omega}{100\ \text{k}\Omega + 100\ \text{k}\Omega}\right)$$

$$= \left(\frac{10}{11}\right) \times (10)\left(\frac{10}{11}\right)(100)\left(\frac{1}{1.001}\right) \times \left(\frac{1}{2}\right)$$

$$= \frac{10^5}{242.242}$$

$$= 412.81\ \text{V/V}$$

$$\frac{v_L}{v_S} = 412.81\ \text{V/V} \quad \ldots\ldots (2)$$

Therefore, the voltage gain is $\boxed{\frac{v_o}{v_S} = 412.81\ \text{V/V}}$.

Compare equation (1) and equation (2).

The overall gain is decreased, when the first and the second stage are interchanged.

Therefore, the **gain is decreased.**

1.048P

We don't have the solution to this problem yet.

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1.049P

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Step 1 of 6

Consider the amplifier should have input resistance much higher than the source

resistance to minimize the loading on $0.5\ \text{M}\Omega$ source. Hence, amplifier stage A is to be used as first stage because $1\ \text{M}\Omega > 0.5\ \text{M}\Omega$.

Now, the middle stage should have high gain. Hence, amplifier stage B is to be used as

second stage, because $A_{vo} = 100\ \text{V/V}$.

The last stage should have low output resistance. Hence, amplifier stage C is to be used as output stage, because $20\ \Omega < 100\ \Omega$.

Draw a three stage amplifier with the three stages.

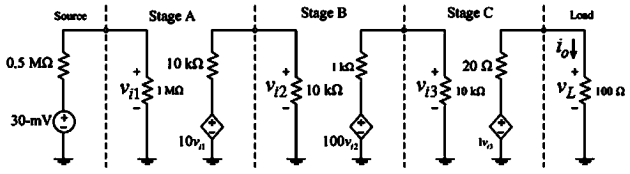


Figure 1

Step 2 of 6

Determine the input voltage of stage A.

$$\begin{aligned} v_{i1} &= \left(\frac{1\ \text{M}\Omega}{1\ \text{M}\Omega + 0.5\ \text{M}\Omega}\right)(30\ \text{mV}) \\ &= \left(\frac{1\ \text{M}\Omega}{1.5\ \text{M}\Omega}\right)(30\ \text{mV}) \\ &= 20\ \text{mV} \end{aligned}$$

Thus, the input voltage of stage A, v_{i1} is $20\ \text{mV}$.

Step 3 of 6

Determine the input of stage B or output of stage A.

$$\begin{aligned} v_{i2} &= \left(\frac{10\ \text{k}\Omega}{10\ \text{k}\Omega + 10\ \text{k}\Omega}\right)(10v_{i1}) \\ &= \left(\frac{1}{2}\right)(10 \times 20 \times 10^{-3}) \\ &= 0.1\ \text{V} \end{aligned}$$

Thus, the input voltage of stage B, v_{i2} is $0.1\ \text{V}$.

Step 4 of 6

Determine the input of stage C or output of stage B.

$$\begin{aligned} v_{i3} &= \left(\frac{10\ \text{k}\Omega}{10\ \text{k}\Omega + 1\ \text{k}\Omega}\right)(100v_{i2}) \\ &= \left(\frac{10}{11}\right)(100 \times 0.1) \\ &= 9.09\ \text{V} \end{aligned}$$

Thus, the input voltage of stage C, v_{i2} is $9.09\ \text{V}$.

Step 5 of 6

Determine the output voltage, v_L.

$$\begin{aligned} v_L &= \left(\frac{100\ \Omega}{100\ \Omega + 20\ \Omega}\right)(v_{i3}) \\ &= \left(\frac{100}{120}\right)(9.09) \\ &= 7.575\ \text{V} \end{aligned}$$

Thus, the output voltage, v_L is $\boxed{7.575\ \text{V}}$.

Step 6 of 6

Determine the output power.

$$\begin{aligned} P_L &= \frac{v_L^2}{R_L} \\ &= \frac{(7.575\ \text{V})^2}{100\ \Omega} \\ &= \frac{57.38}{100} \\ &= 0.574\ \text{W} \end{aligned}$$

Therefore, the output power, P_L is $\boxed{0.574\ \text{W}}$.

1.051P

We don't have the solution to this problem yet.

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1.052P

We don't have the solution to this problem yet.

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1.054P

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1.055P

We don't have the solution to this problem yet.

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1.056P

We don't have the solution to this problem yet.

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1.057P

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Step 1 of 13

Draw the circuit diagram of voltage amplifier.

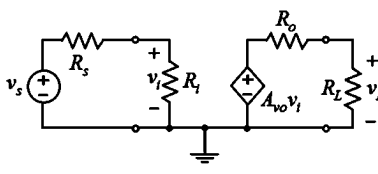


Figure 1

Step 2 of 13

Apply voltage division rule and determine the expression for v_i.

$$v_i = v_s\left[\frac{R_i}{R_i + R_s}\right]$$

Determine the expression for output voltage.

$$v_L = A_{vo}v_i\left(\frac{R_L}{R_L + R_o}\right)$$

Substitute $v_s\left[\frac{R_i}{R_i + R_s}\right]$ for v_i in the equation.

$$v_L = A_{vo}v_s\left(\frac{R_i}{R_i + R_s}\right)\left(\frac{R_L}{R_L + R_o}\right)$$

Step 3 of 13

Consider, if R_s is varied keeping R_L constant, then write the conditions for $v_{L(\max)}$ and $v_{L(\min)}$.

$$\begin{aligned} v_{L(\max)} &= A_{vo}v_s\left(\frac{R_L}{R_L + R_o}\right)\left(\frac{R_i}{R_i + R_{s(\min)}}\right) \\ &= A_{vo}v_s\left(\frac{R_L}{R_L + R_o}\right)\left(\frac{R_i}{R_i + 1\,\text{k}\Omega}\right) \end{aligned}$$

$$\begin{aligned} v_{L(\min)} &= A_{vo}v_s\left(\frac{R_L}{R_L + R_o}\right)\left(\frac{R_i}{R_i + R_{s(\max)}}\right) \\ &= A_{vo}v_s\left(\frac{R_L}{R_L + R_o}\right)\left(\frac{R_i}{R_i + 10\,\text{k}\Omega}\right) \end{aligned}$$

Step 4 of 13

Calculate the relative change in load voltage.

$$\begin{aligned} \frac{\Delta v_L}{v_L} &= \frac{v_{L(\max)} - v_{L(\min)}}{v_{L(\min)}} \\ &= \frac{v_{L(\max)}}{v_{L(\min)}} - 1 \\ &= \frac{A_{vo}v_s\left(\frac{R_L}{R_L + R_o}\right)\left(\frac{R_i}{R_i + R_{s(\min)}}\right)}{A_{vo}v_s\left(\frac{R_L}{R_L + R_o}\right)\left(\frac{R_i}{R_i + R_{s(\max)}}\right)} - 1 \\ &= \frac{R_i + R_{s(\max)}}{R_i + R_{s(\min)}} - 1 \\ &= \frac{R_i + R_{s(\max)} - R_i - R_{s(\min)}}{R_i + R_{s(\min)}} \\ &= \frac{R_{s(\max)} - R_{s(\min)}}{R_i + R_{s(\min)}} \\ &= \frac{10\,\text{k}\Omega - 1\,\text{k}\Omega}{R_i + 1\,\text{k}\Omega} \\ &= \frac{9\times10^3}{R_i + 1\times10^3} \end{aligned}$$

Step 5 of 13

Consider for maximum change in output voltage, the corresponding change in R_s is $\leq 10\%$.

That is,

$$\frac{\Delta v_L}{v_L} \leq 0.1$$

Substitute $\frac{9\times10^3}{R_i + 10^3}$ for $\frac{\Delta v_L}{v_L}$ in the equation.

$$\frac{9\times10^3}{R_i + 10^3} \leq 0.1$$

$$R_i + 10^3 \geq \frac{9\times10^3}{0.1}$$

$$R_i + 10^3 \geq 90\times10^3$$

$$R_i \geq 90\times10^3 - 10^3$$

Simplify further.

$$\begin{aligned} R_i &\geq 89\times10^3 \\ &\geq 89\,\text{k}\Omega \end{aligned}$$

Consider the value of input resistance, R_i is of the form $1\times10^m\ \Omega$.

Write the value of input resistance of the form $1\times10^m\ \Omega$.

$$\begin{aligned} R_i &\geq 89\,\text{k}\Omega \\ &\cong 1\times10^2\,\text{k}\Omega \\ &\cong 100\,\text{k}\Omega \end{aligned}$$

Therefore, the value of input resistance is $\boxed{100\,\text{k}\Omega}$.

Step 6 of 13

Consider, if R_L is varied keeping R_s constant, then write the conditions for $v_{L(\max)}$ and $v_{L(\min)}$.

$$\begin{aligned} v_{L(\max)} &= A_{vo}v_s\left(\frac{R_L}{R_L + R_o}\right)\left(\frac{R_i}{R_i + R_s}\right) \\ &= A_{vo}v_s\left(\frac{1}{1 + \frac{R_o}{R_{L(\max)}}}\right)\left(\frac{R_i}{R_i + R_s}\right) \\ &= A_{vo}v_s\left(\frac{1}{1 + \frac{R_o}{10\times10^3}}\right)\left(\frac{R_i}{R_i + R_s}\right) \end{aligned}$$

$$\begin{aligned} v_{L(\min)} &= A_{vo}v_s\left(\frac{R_L}{R_L + R_o}\right)\left(\frac{R_i}{R_i + R_s}\right) \\ &= A_{vo}v_s\left(\frac{1}{1 + \frac{R_o}{R_{L(\min)}}}\right)\left(\frac{R_i}{R_i + R_s}\right) \\ &= A_{vo}v_s\left(\frac{1}{1 + \frac{R_o}{1\times10^3}}\right)\left(\frac{R_i}{R_i + R_s}\right) \end{aligned}$$

Step 7 of 13

Step 8 of 13

Calculate the relative change in load voltage.

$$\begin{aligned} \frac{\Delta v_L}{v_L} &= \frac{v_{L(\max)} - v_{L(\min)}}{v_{L(\min)}} \\ &= \frac{v_{L(\max)}}{v_{L(\min)}} - 1 \\ &= \frac{A_{vo}v_s\left(\frac{1}{1 + \frac{R_o}{10\times10^3}}\right)\left(\frac{R_i}{R_i + R_s}\right)}{A_{vo}v_s\left(\frac{1}{1 + \frac{R_o}{1\times10^3}}\right)\left(\frac{R_i}{R_i + R_s}\right)} - 1 \\ &= \left(\frac{1 + \frac{R_o}{1\times10^3}}{1 + \frac{R_o}{10\times10^3}}\right) - 1 \end{aligned}$$

Step 9 of 13

Consider for maximum change in output voltage, the corresponding change in R_s is 10%.

$$10\% = \left(\frac{1 + \frac{R_o}{1\times10^3}}{1 + \frac{R_o}{10\times10^3}}\right) - 1$$

$$\frac{1 + \frac{R_o}{10^3}}{1 + \frac{R_o}{10^4}} = 1 + 0.1$$

$$\frac{\frac{10^3 + R_o}{10^3}}{\frac{10^4 + R_o}{10^4}} = 1.1$$

$$\frac{10^3 + R_o}{10^3} = (1.1)\left(\frac{10^4 + R_o}{10^4}\right)$$

$$10^4 + 10R_o = 1.1\times10^4 + 1.1R_o$$

$$10R_o - 1.1R_o = 1.1\times10^4 - 10^4$$

$$8.9R_o = 0.1\times10^4$$

$$\begin{aligned} R_o &= \frac{0.1\times10^4}{8.9} \\ &= 0.01123\times10^4 \\ &= 112.3\ \Omega \end{aligned}$$

Step 10 of 13

Consider the value of input resistance, R_o is of the form $1\times10^m\ \Omega$.

Write the value of input resistance of the form $1\times10^m\ \Omega$.

$$\begin{aligned} R_o &= 112\ \Omega \\ &\cong 1\times10^2\ \Omega \\ &\cong 100\ \Omega \end{aligned}$$

Therefore, the value of output resistance is $\boxed{100\ \Omega}$.

Step 11 of 13

Sketch the circuit model.

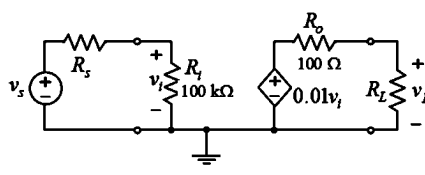


Figure 2

Step 12 of 13

Determine the intrinsic voltage gain of the circuit.

$$\begin{aligned} A_v(\min) &= \left(\frac{v_L}{v_s}\right)_{\min} \\ &= \frac{A_{vo}v_s\left(\frac{R_{L(\min)}}{R_{L(\min)} + R_o}\right)\left(\frac{R_i}{R_i + R_{s(\max)}}\right)}{v_s} \\ &= (10\times10^{-3})\left(\frac{1\times10^3}{1\times10^3 + 100}\right)\left(\frac{100\times10^3}{100\times10^3 + 10\times0^2}\right) \\ &= (10\times10^{-3})\left(\frac{1000}{1100}\right)\left(\frac{100}{110}\right) \end{aligned}$$

$$= 8.2644\times10^{-3}\ \text{V/V}$$

Therefore, the voltage gain is $\boxed{8.2644\times10^{-3}\ \text{V/V}}$.

Step 13 of 13

The input resistance is very high and output resistance is very low. The amplifier is trans-conductance amplifier.

Therefore, the amplifier is **trans-conductance amplifier**.

Step 1 of 12

Draw the circuit diagram of current amplifier.

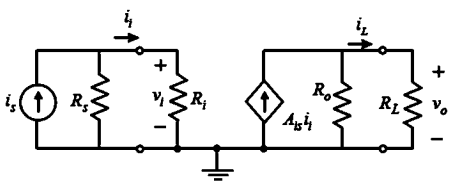


Figure 1

Step 2 of 12

Apply current division rule and determine the value of i_i.

$$i_i = i_s\left(\frac{R_s}{R_s + R_i}\right)$$

Determine the value of output current.

$$i_L = A_{is}i_i\left(\frac{R_L}{R_L + R_o}\right)$$

Substitute $i_s\left(\frac{R_s}{R_s + R_i}\right)$ for i_i in the equation.

$$i_L = A_{is}i_s\left(\frac{R_s}{R_i + R_s}\right)\left(\frac{R_L}{R_L + R_o}\right)$$

Step 3 of 12

Consider, if R_s is varied keeping R_L constant, then write the conditions for $i_{L(\max)}$ and $i_{L(\min)}$.

$$\begin{aligned} i_{L(\max)} &= A_{is}i_s\left(\frac{R_s}{R_i + R_{s(\min)}}\right)\left(\frac{R_L}{R_L + R_o}\right) \\ &= A_{is}i_s\left(\frac{R_s}{R_i + 1\,\text{k}\Omega}\right)\left(\frac{R_L}{R_L + R_o}\right) \end{aligned}$$

$$\begin{aligned} i_{L(\min)} &= A_{is}i_s\left(\frac{R_s}{R_i + R_{s(\max)}}\right)\left(\frac{R_L}{R_L + R_o}\right) \\ &= A_{is}i_s\left(\frac{R_s}{R_i + 10\,\text{k}\Omega}\right)\left(\frac{R_L}{R_L + R_o}\right) \end{aligned}$$

Step 4 of 12

Calculate the relative change in load current.

$$\begin{aligned} \frac{\Delta i_L}{i_L} &= \frac{i_{L(\max)} - i_{L(\min)}}{i_{L(\min)}} \\ &= \frac{i_{L(\max)}}{i_{L(\min)}} - 1 \\ &= \frac{A_{is}i_s\left(\frac{R_s}{R_i + 1\,\text{k}\Omega}\right)\left(\frac{R_L}{R_L + R_o}\right)}{A_{is}i_s\left(\frac{R_s}{R_i + 10\,\text{k}\Omega}\right)\left(\frac{R_L}{R_L + R_o}\right)} - 1 \\ &= \frac{R_i + 10\,\text{k}\Omega}{R_i + 1\,\text{k}\Omega} - 1 \\ &= \frac{9\,\text{k}\Omega}{R_i + 1\,\text{k}\Omega} \end{aligned}$$

Step 5 of 12

Consider for maximum change in output current, the corresponding change in R_s is $\leq 10\%$.

That is,

$$\frac{\Delta i_L}{i_L} \leq 0.1$$

Substitute $\frac{9\,\text{k}\Omega}{R_i + 1\,\text{k}\Omega}$ for $\frac{\Delta i_L}{i_L}$ in the equation.

$$\begin{aligned} &\frac{9\,\text{k}\Omega}{R_i + 1\,\text{k}\Omega} \leq 0.1 \\ &R_i + 1\,\text{k}\Omega \geq 90\,\text{k}\Omega \\ &R_i \geq 89\,\text{k}\Omega \end{aligned}$$

Step 6 of 12

Consider the value of input resistance, R_i is of the form $1\times10^m\ \Omega$.

Write the value of input resistance of the form $1\times10^m\ \Omega$.

$$\begin{aligned} R_i &\geq 89\,\text{k}\Omega \\ &\cong 1\times10^5\ \Omega \\ &\cong 100\,\text{k}\Omega \end{aligned}$$

Therefore, the value of input resistance is $\boxed{100\,\text{k}\Omega}$.

Step 7 of 12

Consider, if R_L is varied keeping R_s constant, then write the conditions for $i_{L(\max)}$ and $i_{L(\min)}$.

$$\begin{aligned} i_{L(\max)} &= A_{is}i_s\left(\frac{R_s}{R_i + R_s}\right)\left(\frac{R_L}{R_L + R_o}\right) \\ &= A_{is}i_s\left(\frac{R_s}{R_i + R_s}\right)\left(\frac{1}{1 + \frac{R_o}{R_{L(\max)}}}\right) \end{aligned}$$

$$\begin{aligned} i_{L(\min)} &= A_{is}i_s\left(\frac{R_s}{R_i + R_s}\right)\left(\frac{R_L}{R_L + R_o}\right) \\ &= A_{is}i_s\left(\frac{R_s}{R_i + R_s}\right)\left(\frac{1}{1 + \frac{R_o}{R_{L(\min)}}}\right) \end{aligned}$$

Step 8 of 12

Calculate the relative change in load voltage.

$$\begin{aligned} \frac{\Delta i_L}{i_L} &= \frac{i_{L(\max)} - i_{L(\min)}}{i_{L(\min)}} \\ &= \frac{i_{L(\max)}}{i_{L(\min)}} - 1 \\ &= \frac{A_{is}i_s\left(\frac{R_s}{R_i + R_s}\right)\left(\frac{1}{1 + \frac{R_o}{R_{L(\max)}}}\right)}{A_{is}i_s\left(\frac{R_s}{R_i + R_s}\right)\left(\frac{1}{1 + \frac{R_o}{R_{L(\min)}}}\right)} - 1 \\ &= \frac{1 + \frac{R_o}{R_{L(\min)}}}{1 + \frac{R_o}{R_{L(\max)}}} - 1 \\ &= \frac{1 + \frac{R_o}{1\times10^3}}{1 + \frac{R_o}{10\times10^3}} - 1 \end{aligned}$$

Step 9 of 12

Consider for maximum change in output current, the corresponding change in R_s is 10%.

$$\begin{aligned} 10\% &= \frac{1 + \frac{R_o}{1\times10^3}}{1 + \frac{R_o}{10\times10^3}} - 1 \\ \frac{1 + \frac{R_o}{1\times10^3}}{1 + \frac{R_o}{10\times10^3}} &= 1.1 \\ 1 + \frac{R_o}{1\times10^3} &= (1.1)\left(1 + \frac{R_o}{10\times10^3}\right) \\ 10\times10^3 + 10R_o &= (1.1)\left(10\times10^3 + R_o\right) \\ 10\times10^3 + 10R_o &= 11\times10^3 + 1.1R_o \\ 8.9R_o &= 1\times10^3 \\ R_o &= \frac{1\times10^3}{8.9} \\ &= 112.36\ \Omega \end{aligned}$$

Consider the value of input resistance, R_o is of the form $1\times10^m\ \Omega$.

Write the value of input resistance of the form $1\times10^m\ \Omega$.

$$\begin{aligned} R_o &= 112.36\ \Omega \\ &\cong 1\times10^2\ \Omega \\ &\cong 100\ \Omega \end{aligned}$$

Therefore, the value of output resistance is $\boxed{100\ \Omega}$.

Step 10 of 12

Sketch the circuit model.

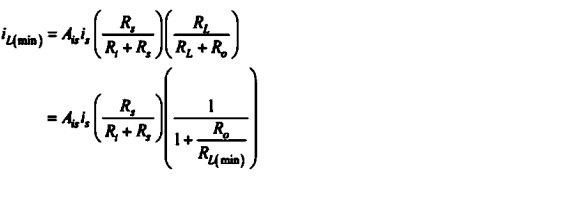


Figure 2

Step 11 of 12

Determine the intrinsic current gain of the circuit.

$$\begin{aligned} A_i(\min) &= \left(\frac{i_L}{i_s}\right)_{\min} \\ &= \frac{A_{is}i_s\left(\frac{R_{s(\min)}}{R_i + R_{s(\min)}}\right)\left(\frac{R_{L(\min)}}{R_{L(\min)} + R_o}\right)}{i_s} \\ &= \left(10\times10^{-6}\right)\left(\frac{1\times10^3}{100\times10^3 + 1\times10^3}\right)\left(\frac{1\times10^3}{1\times10^3 + 100}\right) \\ &= \left(10\times10^{-6}\right)\left(9.9\times10^{-3}\right)(0.909) \end{aligned}$$

$$\begin{aligned} &= 9\times10^{-8}\ \text{A/A} \\ &= 0.09\times10^{-6}\ \mu\text{A/A} \end{aligned}$$

Therefore, the current gain is $\boxed{0.09\ \mu\text{A/A}}$.

Step 12 of 12

The input resistance is very high and output resistance is very low. The amplifier is trans-conductance amplifier.

Therefore, the amplifier is **trans-conductance amplifier**.

Step 1 of 12

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Consider the source signal is voltage and load signal is current. Hence, a voltage to current amplifier is required.

Draw the circuit diagram.

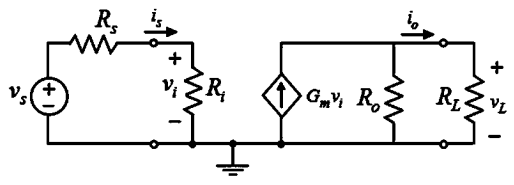


Figure 1

Step 2 of 12

Apply voltage division rule and determine the expression for v_i.

$$v_i = v_s\left[\frac{R_i}{R_i + R_s}\right]$$

Determine the output voltage.

$$\begin{aligned} v_L &= G_m v_i\left(\frac{R_o R_L}{R_L + R_o}\right) \\ &= G_m v_s\left(\frac{R_i}{R_i + R_s}\right)\left(\frac{R_o R_L}{R_L + R_o}\right) \end{aligned}$$

Step 3 of 12

Consider, if R_s is varied keeping R_L constant. Write the conditions for $i_{L(\max)}$ and $i_{L(\min)}$.

$$\begin{aligned} i_{L(\max)} &= G_m v_s\left(\frac{R_o}{R_L + R_o}\right)\left(\frac{R_i}{R_i + R_{s(\min)}}\right) \\ &= G_m v_s\left(\frac{R_o}{R_L + R_o}\right)\left(\frac{R_i}{R_i + 1\times10^3}\right) \end{aligned}$$

$$\begin{aligned} i_{L(\min)} &= G_m v_s\left(\frac{R_o}{R_L + R_o}\right)\left(\frac{R_i}{R_i + R_{s(\max)}}\right) \\ &= G_m v_s\left(\frac{R_o}{R_L + R_o}\right)\left(\frac{R_i}{R_i + 10\times10^3}\right) \end{aligned}$$

Step 4 of 12

Calculate the relative change in load current.

$$\begin{aligned} \frac{\Delta i_L}{i_L} &= \frac{i_{L(\max)} - i_{L(\min)}}{i_{L(\min)}} \\ &= \frac{i_{L(\max)}}{i_{L(\min)}} - 1 \\ &= \frac{G_m v_s\left(\frac{R_o}{R_L + R_o}\right)\left(\frac{R_i}{R_i + R_{s(\min)}}\right)}{G_m v_s\left(\frac{R_o}{R_L + R_o}\right)\left(\frac{R_i}{R_i + R_{s(\max)}}\right)} - 1 \\ &= \frac{\left(\frac{R_i}{R_i + R_{s(\min)}}\right)}{\left(\frac{R_i}{R_i + R_{s(\max)}}\right)} - 1 \\ &= \frac{R_i + R_{s(\max)}}{R_i + R_{s(\min)}} - 1 \\ &= \frac{R_i + R_{s(\max)} - R_i - R_{s(\min)}}{R_i + R_{s(\min)}} \\ &= \frac{10\times10^3 - 1\times10^3}{R_i + 1\times10^3} \\ &= \frac{10^4 - 10^3}{R_i + 10^3} \end{aligned}$$

Step 5 of 12

Consider for maximum change in output current, the corresponding change in R_s is $\leq 10\%$.

$$\begin{aligned} &\frac{10^4 - 10^3}{R_i + 10^3} \leq 0.1 \\ &R_i + 10^3 \geq \frac{9000}{0.1} \\ &R_i + 10^3 \geq 90\times10^3 \\ &R_i \geq 90\times10^3 - 10^3 \\ &R_i \geq 89\times10^3 \end{aligned}$$

Step 6 of 12

Consider the value of input resistance, R_i is of the form $1\times10^m\ \Omega$.

Write the value of input resistance of the form $1\times10^m\ \Omega$.

$$\begin{aligned} R_i &\geq 89\ \text{k}\Omega \\ &\cong 1\times10^2\ \text{k}\Omega \\ &\cong 100\ \text{k}\Omega \end{aligned}$$

Therefore, the value of input resistance is $\boxed{100\ \text{k}\Omega}$.

Step 7 of 12

Consider, if R_L is varied keeping R_s constant, then write the conditions for $i_{L(\max)}$ and $i_{L(\min)}$.

$$i_{L(\max)} = G_m v_s\left(\frac{R_o}{R_{L(\min)} + R_o}\right)\left(\frac{R_i}{R_i + R_s}\right)$$

$$i_{L(\min)} = G_m v_s\left(\frac{R_o}{R_{L(\max)} + R_o}\right)\left(\frac{R_i}{R_i + R_s}\right)$$

Calculate the relative change in load current.

$$\begin{aligned} \frac{\Delta i_L}{i_L} &= \frac{i_{L(\max)} - i_{L(\min)}}{i_{L(\min)}} \\ &= \frac{i_{L(\max)}}{i_{L(\min)}} - 1 \\ &= \frac{G_m v_s\left(\frac{R_o}{R_{L(\min)} + R_o}\right)\left(\frac{R_i}{R_i + R_s}\right)}{G_m v_s\left(\frac{R_o}{R_{L(\max)} + R_o}\right)\left(\frac{R_i}{R_i + R_s}\right)} - 1 \\ &= \frac{R_{L(\max)} + R_o}{R_{L(\min)} + R_o} - 1 \\ &= \frac{R_{L(\max)} - R_{L(\min)}}{R_{L(\min)} + R_o} \end{aligned}$$

Step 8 of 12

Consider for maximum change in output current, the corresponding change in R_s is 10%.

$$\begin{aligned} &\frac{R_{L(\max)} - R_{L(\min)}}{R_{L(\min)} + R_o} = 0.1 \\ &\frac{10\times10^3 - 1\times10^3}{1\times10^3 + R_o} = 0.1 \\ &\frac{9\times10^3}{1\times10^3 + R_o} = 0.1 \\ &1\times10^3 + R_o = 90\times10^3 \\ &R_o = 89\times10^3 \end{aligned}$$

Step 9 of 12

Consider the value of input resistance, R_o is of the form $1\times10^m\ \Omega$.

Write the value of input resistance of the form $1\times10^m\ \Omega$.

$$\begin{aligned} R_o &= 89\ \text{k}\Omega \\ &\cong 1\times10^5\ \Omega \\ &\cong 100\ \text{k}\Omega \end{aligned}$$

Therefore, the value of output resistance is $\boxed{100\ \text{k}\Omega}$.

Step 10 of 12

Sketch the circuit model.

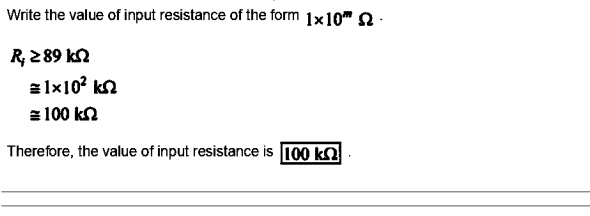


Figure 2

Step 11 of 12

The input resistance is very high and output resistance is high. The amplifier is trans-conductance amplifier.

Therefore, the amplifier is **trans-conductance amplifier.**

Step 12 of 12

Determine the trans-conductance gain of the intrinsic circuit.

$$\begin{aligned} G_m(\min) &= \left(\frac{i_L}{v_s}\right)_{\min} \\ &= \frac{G_m v_s\left(\frac{R_o}{R_{L(\min)} + R_o}\right)\left(\frac{R_i}{R_i + R_{s(\max)}}\right)}{v_s} \\ &= \left(10\times10^{-3}\right)\left(\frac{100\times10^3}{1\times10^3 + 100\times10^3}\right)\left(\frac{100\times10^3}{100\times10^3 + 10\times10^3}\right) \\ &= \left(10\times10^{-3}\right)(0.99)(0.9090) \end{aligned}$$

$$= 0.009\ \text{A/V}$$

Therefore, the trans-conductance gain of the intrinsic circuit is $\boxed{0.009\ \text{A/V}}$.

Step 1 of 12

Consider the source signal is current and load signal is voltage. Hence, a current to voltage amplifier is required.

Draw the circuit diagram.

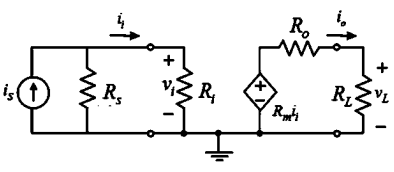


Figure 1

Step 2 of 12

Step 3 of 12

Apply current division rule and determine the value of i_i.

$$i_i = i_s\left(\frac{R_s}{R_s + R_i}\right)$$

Determine the output voltage.

$$\begin{aligned} v_L &= R_m i_i\left(\frac{R_L}{R_L + R_o}\right) \\ &= R_m i_s\left(\frac{R_s}{R_i + R_s}\right)\left(\frac{R_L}{R_L + R_o}\right) \end{aligned}$$

Step 4 of 12

Consider, if R_s is varied keeping R_L constant, then write the conditions for $v_{L(\max)}$ and $v_{L(\min)}$.

$$\begin{aligned} v_{L(\max)} &= R_m i_s\left(\frac{R_L}{R_L + R_o}\right)\left(\frac{R_s}{R_i + R_s}\right) \\ &= R_m i_s\left(\frac{R_L}{R_L + R_o}\right)\left(\frac{1}{\frac{R_i}{R_{s(\max)}} + 1}\right) \end{aligned}$$

$$\begin{aligned} v_{L(\min)} &= R_m i_s\left(\frac{R_L}{R_L + R_o}\right)\left(\frac{R_s}{R_i + R_s}\right) \\ &= R_m i_s\left(\frac{R_L}{R_L + R_o}\right)\left(\frac{1}{\frac{R_i}{R_{s(\min)}} + 1}\right) \end{aligned}$$

Step 5 of 12

Calculate the relative change in load voltage.

$$\begin{aligned} \frac{\Delta v_L}{v_L} &= \frac{v_{L(\max)} - v_{L(\min)}}{v_{L(\min)}} \\ &= \frac{v_{L(\max)}}{v_{L(\min)}} - 1 \\ &= \frac{R_m i_s\left(\frac{R_L}{R_L + R_o}\right)\left(\frac{1}{\frac{R_i}{R_{s(\max)}} + 1}\right)}{R_m i_s\left(\frac{R_L}{R_L + R_o}\right)\left(\frac{1}{\frac{R_i}{R_{s(\min)}} + 1}\right)} - 1 \\ &= \frac{\frac{R_i}{R_{s(\min)}} + 1}{\frac{R_i}{R_{s(\max)}} + 1} - 1 \\ &= \frac{\frac{R_i}{1\times10^3} + 1}{\frac{R_i}{10\times10^3} + 1} - 1 \end{aligned}$$

Step 6 of 12

Consider for maximum change in output current, the corresponding change in R_s is $\le 10\%$.

$$\begin{aligned} &\frac{\frac{R_i}{1\times10^3} + 1}{\frac{R_i}{10\times10^3} + 1} - 1 \le 0.1 \\ &\frac{\frac{R_i}{1\times10^3} + 1 - \frac{R_i}{10\times10^3} - 1}{\frac{R_i}{10\times10^3} + 1} \le 0.1 \\ &\frac{\frac{0.9R_i}{10^3}}{\frac{R_i}{10^4} + 1} \le 0.1 \\ &\frac{R_i}{10^4} + 1 \ge \frac{\frac{0.9R_i}{10^3}}{0.1} \\ &\left(0.1\times10^{-3}\right)R_i + 1 \ge \left(9\times10^{-3}\right)R_i \\ &\left(0.1\times10^{-3}\right)R_i - \left(9\times10^{-3}\right)R_i \ge -1 \\ &-\left(8.9\times10^{-3}\right)R_i \ge -1 \\ &\left(8.9\times10^{-3}\right)R_i \le 1 \\ &R_i \le \frac{1}{8.9\times10^{-3}} \\ &R_i \le 112.3595 \end{aligned}$$

Consider the value of input resistance, R_i is of the form $1\times10^m\ \Omega$.

Write the value of input resistance of the form $1\times10^m\ \Omega$.

$$\begin{aligned} R_i &\le 112.3595\ \Omega \\ &\cong 1\times10^2\ \Omega \\ &\cong 100\ \Omega \end{aligned}$$

Therefore, the value of input resistance is $\boxed{100\ \Omega}$.

Step 7 of 12

Consider, if R_L is varied keeping R_s constant, then write the conditions for $v_{L(\max)}$ and $v_{L(\min)}$.

$$\begin{aligned} v_{L(\max)} &= R_m i_s\left(\frac{R_L}{R_L + R_o}\right)\left(\frac{R_s}{R_i + R_s}\right) \\ &= R_m i_s\left(\frac{1}{1 + \frac{R_o}{R_{L(\max)}}}\right)\left(\frac{R_s}{R_i + R_s}\right) \end{aligned}$$

$$\begin{aligned} v_{L(\min)} &= R_m i_s\left(\frac{R_L}{R_L + R_o}\right)\left(\frac{R_s}{R_i + R_s}\right) \\ &= R_m i_s\left(\frac{1}{1 + \frac{R_o}{R_{L(\min)}}}\right)\left(\frac{R_s}{R_i + R_s}\right) \end{aligned}$$

Step 8 of 12

Calculate the relative change in load voltage.

$$\begin{aligned} \frac{\Delta v_L}{v_L} &= \frac{v_{L(\max)} - v_{L(\min)}}{v_{L(\min)}} \\ &= \frac{v_{L(\max)}}{v_{L(\min)}} - 1 \\ &= \frac{R_m i_s\left(\frac{R_s}{R_i + R_s}\right)\left(\frac{1}{1 + \frac{R_o}{R_{L(\max)}}}\right)}{R_m i_s\left(\frac{R_s}{R_i + R_s}\right)\left(\frac{1}{1 + \frac{R_o}{R_{L(\min)}}}\right)} - 1 \\ &= \left(\frac{1 + \frac{R_o}{R_{L(\min)}}}{1 + \frac{R_o}{R_{L(\max)}}}\right) - 1 \end{aligned}$$

Step 9 of 12

Consider for maximum change in output voltage, the corresponding change in R_s is 10%.

$$\begin{aligned} &\left(\frac{1 + \frac{R_o}{R_{L(\min)}}}{1 + \frac{R_o}{R_{L(\max)}}}\right) - 1 = 0.1 \\ &\frac{1 + \frac{R_o}{10^3}}{1 + \frac{R_o}{10^4}} = 1 + 0.1 \\ &\frac{10^3 + R_o}{\frac{10^3}{10^4 + R_o}} = 1.1 \\ &\frac{10^3 + R_o}{10^3} = 1.1\left(\frac{10^4 + R_o}{10^4}\right) \\ &10^4 + 10R_o = 1.1\times10^4 + 1.1R_o \\ &10R_o - 1.1R_o = 1.1\times10^4 - 10^4 \\ &8.9R_o = 0.1\times10^4 \\ &R_o = \frac{0.1\times10^4}{8.9} \\ &R_o = 1.123\times10^2 \end{aligned}$$

Step 10 of 12

Consider the value of input resistance, R_o is of the form $1\times10^m\ \Omega$.

Write the value of input resistance of the form $1\times10^m\ \Omega$.

$$\begin{aligned} R_o &= 112\ \Omega \\ &\cong 1\times10^2\ \Omega \\ &\cong 100\ \Omega \end{aligned}$$

Therefore, the value of output resistance is $\boxed{100\ \Omega}$.

Therefore, the amplifier is **Trans-resistance amplifier.**

Step 11 of 12

Sketch the circuit model.

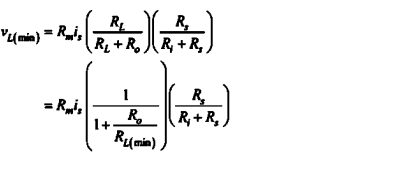


Figure 2

Step 12 of 12

Determine the intrinsic trans-resistance gain of the circuit.

$$\begin{aligned} R_{m(\min)} &= \left(\frac{v_L}{i_s}\right)_{\min} \\ &= \frac{R_m i_s\left(\frac{R_{L(\min)}}{R_{L(\min)} + R_o}\right)\left(\frac{R_{s(\min)}}{R_i + R_{s(\min)}}\right)}{i_s} \\ &= \left(10\times10^{-6}\right)\left(\frac{10^3}{10^3 + 100}\right)\left(\frac{10^3}{10^3 + 100}\right) \\ &= \left(10\times10^{-6}\right)\left(\frac{1000}{1100}\right)\left(\frac{1000}{1100}\right) \end{aligned}$$

$$= 8.2644\times10^{-6}$$

Therefore, the tans-resistance gain of the amplifier is $\boxed{8.2644\ \mu V/V}$.

Step 1 of 3

Consider the following circuit diagram:

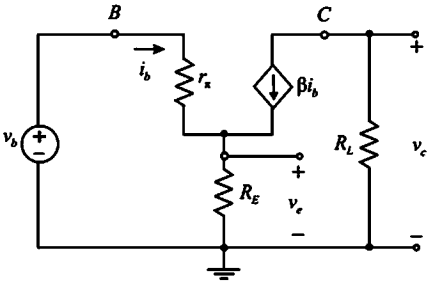


Figure 1

Step 2 of 3

From Figure 1, write the expression for the collector voltage.

$$v_c = -\beta i_b R_L$$

From Figure 1, write the expression for the emitter voltage.

$$\begin{aligned} v_e &= (i_b + i_b\beta) R_E \\ &= (\beta + 1)\, i_b R_E \end{aligned}$$

Apply Kirchhoff's voltage law at the input side of Figure 1.

$$\begin{aligned} v_b &= i_b r_\pi + v_e \\ &= i_b r_\pi + (\beta + 1)\, i_b R_E \\ &= \left[r_\pi + R_E(\beta + 1)\right] i_b \end{aligned}$$

Calculate the ratio of collector voltage to the base voltage.

$$\begin{aligned} \frac{v_c}{v_b} &= \frac{-\beta i_b R_L}{\left[r_\pi + (\beta + 1) R_E\right] i_b} \\ &= \frac{-\beta R_L}{r_\pi + (\beta + 1) R_E} \end{aligned}$$

Thus, the collector to the emitter voltage ratio is $\boxed{\frac{v_c}{v_b} = \frac{-\beta R_L}{r_\pi + (\beta + 1) R_E}}$.

Step 3 of 3

Calculate the ratio of emitter voltage to the base voltage.

$$\begin{aligned} \frac{v_e}{v_b} &= \frac{(\beta + 1) i_b R_E}{\left[r_\pi + R_E(\beta + 1)\right] i_b} \\ &= \frac{(\beta + 1) R_E}{r_\pi + R_E(\beta + 1)} \\ &= \frac{R_E}{\frac{r_\pi}{\beta + 1} + R_E} \end{aligned}$$

Thus, the emitter to base voltage ratio is $\boxed{\frac{v_e}{v_b} = \frac{R_E}{R_E + \left[\frac{r_\pi}{\beta + 1}\right]}}$.

1.064P

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Step 1 of 3

1.065P

Refer to Figure P1.63 in the textbook.

Re-draw the circuit diagram.

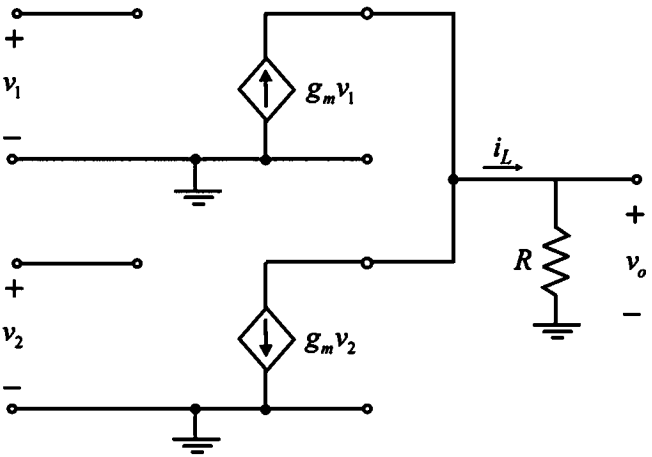


Figure 1

Step 2 of 3

Determine the current entering into the resistance.

$$i_L = g_m v_1 - g_m v_2$$
$$= g_m (v_1 - v_2)$$

Calculate the output voltage.

$$v_o = g_m (v_1 - v_2) R$$
$$= (100\times10^{-3})(5\times10^{3})(v_1 - v_2)$$
$$= 500(v_1 - v_2)$$

Therefore, the input and output voltage relationship is,

$$\boxed{v_o = 500(v_1 - v_2)}$$

Consider the value of input and output voltages,

$$v_1 = 1\ \text{V}$$
$$v_2 = 1\ \text{V}$$

Determine the output voltage.

$$v_o = 500(1-1)$$
$$= 500(0)$$
$$= 0\ \text{V}$$

Therefore, the output voltage, v_o is $\boxed{0\ \text{V}}$.

Step 3 of 3

Consider the value of input and output voltages,

$$v_1 = 1.01\ \text{V}$$
$$v_2 = 0.99\ \text{V}$$

Determine the output voltage.

$$v_o = 500(1.01-0.99)$$
$$= 500(0.02)$$
$$= 10\ \text{V}$$

Therefore, the output voltage, v_o is $\boxed{10\ \text{V}}$.

Step 1 of 3

1.066P

The most convenient representation of voltage amplifier is in terms of $g-$ parameters. The following equations describe the $g-$ parameters representation:

$I_1 = g_{11}V_1 + g_{12}I_2$ ..... (1)

$V_2 = g_{21}V_1 + g_{22}I_2$ ..... (2)

Refer to Figure P1.64 in the textbook for the equivalent circuit representation of these equations.

Refer to Figure 1.16(a) in the textbook for the voltage amplifier.

Represent the input and output voltages as V_1 and V_2. Similarly represent the input and output currents as I_1 and I_2.

Draw the equivalent circuit.

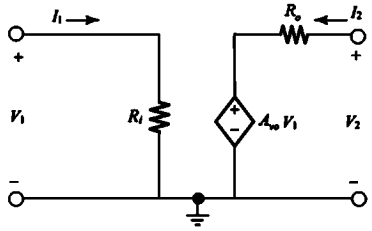


Figure 1

Step 2 of 3

From the input loop of Figure 1, write the following relation:

$V_1 = I_1 R_i$

$I_1 = \frac{1}{R_i} V_1$ ..... (3)

From the output loop of Figure 1, write the following relation:

$V_2 = A_{vo}V_1 + R_o I_2$ ..... (4)

Compare equation (1) with equation (3).

$g_{11} = \frac{1}{R_i}$

$g_{12} = 0$

Compare equation (2) with equation (4).

$g_{21} = A_{vo}$

$g_{22} = R_o$

The $g-$ parameter that correspond to R_i is,

$g_{11} = \frac{1}{R_i}$

$R_i = \frac{1}{g_{11}}$

Thus, the $g-$ parameter that corresponds to R_i is $\boxed{R_i = \frac{1}{g_{11}}}$.

Step 3 of 3

The $g-$ parameter that correspond to A_{vo} is,

$g_{21} = A_{vo}$

$A_{vo} = g_{21}$

Thus, the $g-$ parameter that correspond to A_{vo} is $\boxed{A_{vo} = g_{21}}$.

The $g-$ parameter that correspond to R_o is,

$g_{22} = R_o$

$R_o = g_{22}$

Thus, the $g-$ parameter that corresponds to R_o is $\boxed{R_o = g_{22}}$.

The g_{12} is the additional parameter with no correspondence in amplifier equivalent circuit because $g_{12} = 0$.

Hence, g_{12} signifies the reverse current gain at zero input voltage. The amplifier in the absence of $g-$ parameter is an open loop amplifier with no feedback.

Step 1 of 2

1.067P

(a)

Refer to Figure 1.22 (a) in the textbook.

Refer to Table 1.2 in the textbook.

Apply voltage division rule and determine the output voltage.

$$V_o = \left(\frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} \right) V_i$$

Substitute s for $j\omega$ in the equation.

$$V_o(s) = \left(\frac{\frac{1}{sC}}{R + \frac{1}{sC}} \right) V_i(s)$$

$$= \left(\frac{1}{sRC + 1} \right) V_i(s)$$

$$\frac{V_o(s)}{V_i(s)} = \frac{1}{1 + sRC}$$

Therefore, the transfer function is $\boxed{\frac{V_o(s)}{V_i(s)} = \frac{1}{sRC + 1}}$.

The transfer function is similar to single time constant (STC) low pass network.

The resonant frequency is, $\omega_o = \frac{1}{RC}$.

Step 2 of 2

(b)

Refer to Figure 1.22 (b) in the textbook.

Apply voltage division rule and determine the output voltage.

$$V_o = \left(\frac{R}{R + \frac{1}{j\omega C}} \right) V_i$$

Substitute s for $j\omega$ in the equation.

$$V_o(s) = \left(\frac{R}{R + \frac{1}{sC}} \right) V_i(s)$$

$$= \left(\frac{sRC}{sRC + 1} \right) V_i(s)$$

$$= \left(\frac{s}{s + \frac{1}{RC}} \right) V_i(s)$$

$$\frac{V_o(s)}{V_i(s)} = \frac{s}{s + \frac{1}{RC}}$$

Therefore, the transfer function is $\boxed{\frac{V_o(s)}{V_i(s)} = \frac{s}{s + \frac{1}{RC}}}$.

The transfer function is similar to single time constant (STC) high pass network.

The resonant frequency is, $\omega_o = \frac{1}{RC}$.

1.068P

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1.069P

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1.070P

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Step 1 of 11

Refer to Table in the textbook.

From Table in the textbook, it is a low pass filter circuit because for high frequencies, the gain becomes low.

From Table in the textbook, the maximum gain in dB is $40\ \text{dB}$.

Determine the value cut-off frequency at which the gain is $3\ \text{dB}$ less than maximum gain.

$$f_{3dB} = 10^4$$
$$= 10\ \text{kHz}$$

Step 2 of 11

Determine the low pass circuit gain.

$$\left|\frac{V_o}{V_i}\right| = \frac{A_o}{\sqrt{1+\left(\frac{f}{f_{3dB}}\right)^2}}$$

Here,

A_o is the dc gain.

Determine the value of DC gain.

$$\left|\frac{V_o}{V_i}\right| = \frac{A_o}{\sqrt{1+\left(\frac{f}{f_{3dB}}\right)^2}}$$

$$20\log\left|\frac{V_o}{V_i}\right| = 20\log\left(\frac{A_o}{\sqrt{1+\left(\frac{f}{f_{3dB}}\right)^2}}\right)$$

Substitute 0 for *f*.

From Table, the maximum value of gain in dB is $40\ \text{dB}$.

Step 3 of 11

Substitute $40\ \text{dB}$ for $20\log\left|\frac{V_o}{V_i}\right|$ in the equation.

$$40 = 20\log(A_o)$$

$$\log(A_o) = \frac{40}{20}$$

$$\log(A_o) = 2$$

$$A_o = 10^2$$
$$= 100\ \text{V/V}$$

Step 4 of 11

Therefore, the transfer function is,

$$\left|\frac{V_o}{V_i}\right| = \frac{100}{\sqrt{1+\left(\frac{f}{10^4}\right)^2}}$$

Substitute $1000\ \text{Hz}$ for f in the equation.

$$\left|\frac{V_o}{V_i}\right| = \frac{100}{\sqrt{1+\left(\frac{1000}{10^4}\right)^2}}$$
$$= \frac{100}{\sqrt{1+(0.1)^2}}$$
$$= \frac{100}{\sqrt{1+0.01}}$$
$$= \frac{100}{\sqrt{1.01}}$$
$$= \frac{100}{1.0049}$$

$$\left|\frac{V_o}{V_i}\right| = 99.5123$$

Step 5 of 11

Apply logarithm on both sides.

$$20\log\left|\frac{V_o}{V_i}\right| = 20\log(99.5123)$$
$$= 20(1.9978)$$
$$= 39.9560\ \text{dB}$$

Therefore, the magnitude at the frequency of $1000\ \text{Hz}$ is $39.9560\ \text{dB}$.

Step 6 of 11

Determine the value of phase angle.

$$\phi = -\tan^{-1}\left(\frac{f}{f_{3dB}}\right)$$
$$= -\tan^{-1}\left(\frac{1000}{10\times10^3}\right)$$
$$= -\tan^{-1}(0.1)$$
$$= -5.71059^\circ$$

Therefore, the phase angle at the frequency of $1000\ \text{Hz}$ is -5.71059° .

Step 7 of 11

Determine the phase angle at the frequency $10^5\ \text{Hz}$.

$$\phi = -\tan^{-1}\left(\frac{f}{f_{3dB}}\right)$$
$$= -\tan^{-1}\left(\frac{10^5}{10\times10^3}\right)$$
$$= -\tan^{-1}(10)$$
$$= -84.2894^\circ$$

Therefore, the phase angle at the frequency of $10^5\ \text{Hz}$ is $\boxed{-84.2894^\circ}$

Step 8 of 11

Calculate the frequency at which gain is equal to zero.

$$\left|\frac{V_o}{V_i}\right| = \frac{100}{\sqrt{1+\left(\frac{f}{10^4}\right)^2}}$$

$$20\log\left|\frac{V_o}{V_i}\right| = 20\log\left(\frac{100}{\sqrt{1+\left(\frac{f}{10^4}\right)^2}}\right)$$

$$0 = 20\log\left(\frac{100}{\sqrt{1+\left(\frac{f}{10^4}\right)^2}}\right)$$

$$\frac{100}{\sqrt{1+\left(\frac{f}{10^4}\right)^2}} = 1$$

$$\sqrt{1+\left(\frac{f}{10^4}\right)^2} = 100$$

$$1+\left(\frac{f}{10^4}\right)^2 = 10^4$$

$$\left(\frac{f}{10^4}\right)^2 = 10^4 - 1$$

$$\left(\frac{f}{10^4}\right)^2 = 9.999\times10^3$$

$$\frac{f}{10^4} = 99.994$$

$$f = 10^4(99.994)$$

$$f = 999.94\ \text{kHz}$$

Therefore, the frequency at which gain is 0 dB is $999.94\ \text{kHz}$.

Step 9 of 11

Determine the phase angle at the frequency $999.94\ \text{kHz}$.

$$\phi = -\tan^{-1}\left(\frac{f}{f_{3dB}}\right)$$
$$= -\tan^{-1}\left(\frac{999.94\times10^3}{10\times10^3}\right)$$
$$= -\tan^{-1}(99.994)$$
$$= -89.4270^\circ$$

Therefore, the phase angle at the frequency of $999.94\ \text{kHz}$ is -89.4270° .

Step 10 of 11

Therefore, fill the values in the table.

$f(\text{Hz})$	$\lvert T\rvert(\text{dB})$	$\angle T(^\circ)$
0	40	0
100	40	0
1000	39.9560 dB	-5.71059°
10^4	37	-45°
10^5	20	-84.2894°
999.94×10^3	0	-89.4270°

Step 11 of 11

Draw the magnitude frequency response plot.

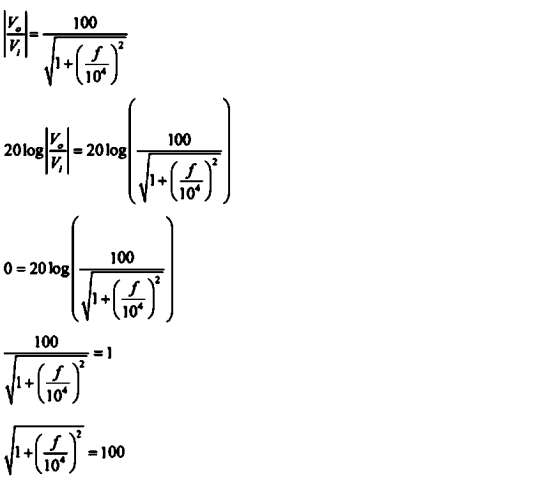


Figure 1

Thus, the Bode plot is drawn.

Step 1 of 13

Refer to Table in question 1.70 in the textbook.

From the values in the table, the circuit is low pass circuit at high frequencies and high pass circuit at low frequencies.

From the tabie, the maximum gain is $40\ \text{dB}$.

Cut-off frequency is the frequency at which gain is less than $3\ \text{dB}$ from the maximum value.

Determine the gain at cut-off frequency.

$$\begin{aligned}\textbf{Gain} &= 40-3\\ &= 37\ \text{dB}\end{aligned}$$

Thus, $3\ \text{dB}$ frequency of the high pass circuit is,

$$\begin{aligned}f_1 &= 10^2\\ &= 100\ \text{Hz}\end{aligned}$$

Step 2 of 13

Determine the expression for gain of high pass circuit.

$$\left|\frac{V_o}{V_i}\right| = \frac{2\pi f A_o}{\sqrt{1+\left(\frac{f}{f_1}\right)^2}}$$

A_o is the dc gain.

Determine the DC gain.

$$\left|\frac{V_o}{V_i}\right| = \frac{2\pi f A_o}{\sqrt{1+\left(\frac{f}{f_1}\right)^2}}$$

$$20\log\left|\frac{V_o}{V_i}\right| = 20\log\left(\frac{2\pi f A_o}{\sqrt{1+\left(\frac{f}{f_1}\right)^2}}\right)$$

Substitute $10^3\ \text{Hz}$ for f.

$$40 = 20\log\left(\frac{2000\pi A_o}{\sqrt{1+\left(\frac{1000}{100}\right)^2}}\right)$$

$$\frac{40}{20} = \log\left(\frac{2000\pi A_o}{\sqrt{1+(10)^2}}\right)$$

$$\log\left(\frac{2000\pi A_o}{\sqrt{101}}\right) = 2$$

$$\frac{2000\pi A_o}{10.0498} = 10^2$$

$$\begin{aligned}A_o &= \frac{1004.98}{2000\pi}\\ &= 0.1599\end{aligned}$$

Therefore, the dc gain, A_o is $0.1599\ \text{V/V}$.

Step 3 of 13

Determine the frequency at which gain is equal to zero.

$$\left|\frac{V_o}{V_i}\right| = \frac{2\pi f(0.1599)}{\sqrt{1+\left(\frac{f}{100}\right)^2}}$$

$$20\log\left|\frac{V_o}{V_i}\right| = 20\log\left(\frac{2\pi f(0.1599)}{\sqrt{1+\left(\frac{f}{100}\right)^2}}\right)$$

$$0 = 20\log\left(\frac{1.00468f}{\sqrt{1+\left(\frac{f}{100}\right)^2}}\right)$$

$$\frac{1.00468f}{\sqrt{1+\left(\frac{f}{100}\right)^2}} = 1$$

Step 4 of 13

Simplify the equation further.

$$1.00468f = \sqrt{1+\left(\frac{f}{100}\right)^2}$$

$$1.00938f^2 = 1+(0.01f)^2$$

$$1.00938f^2 = 1+(0.0001)f^2$$

$$1.00938f^2-(0.0001)f^2 = 1$$

Step 5 of 13

$$1.00938f^2-(0.0001)f^2 = 1$$

$$1.00928f^2 = 1$$

$$f^2 = \frac{1}{1.00928}$$

$$f^2 = 0.9908$$

$$f = \sqrt{0.9908}$$

$$f = 0.9953\ \text{Hz}$$

Therefore, frequency at which gain is $0\ \text{dB}$ is $0.9953\ \text{Hz}$.

Determine the expression for gain of low pass circuit.

$$\left|\frac{V_o}{V_i}\right| = \frac{A_o}{\sqrt{1+\left(\frac{f}{f_{MB}}\right)^2}}$$

A_o is the dc gain

Determine the gain at cut-off frequency.

$$\begin{aligned}\textbf{Gain} &= 40-3\\ &= 37\ \text{dB}\end{aligned}$$

Thus, $3\ \text{dB}$ frequency of the low pass circuit is,

$$\begin{aligned}f_2 &= 10^6\\ &= 1\ \text{MHz}\end{aligned}$$

Step 6 of 13

Determine the DC gain.

$$\left|\frac{V_o}{V_i}\right| = \frac{A_o}{\sqrt{1+\left(\frac{f}{f_2}\right)^2}}$$

$$20\log\left|\frac{V_o}{V_i}\right| = 20\log\left(\frac{A_o}{\sqrt{1+\left(\frac{f}{f_2}\right)^2}}\right)$$

Substitute $10^7\ \text{Hz}$ for f.

$$20 = 20\log\left(\frac{A_o}{\sqrt{1+\left(\frac{10^7}{10^6}\right)^2}}\right)$$

$$\frac{A_o}{\sqrt{1+(10)^2}} = 10$$

$$\frac{A_o}{\sqrt{1+100}} = 10$$

$$\frac{A_o}{\sqrt{101}} = 10$$

$$\begin{aligned}A_o &= 10(10.04998)\\ &= 100.4998\ \text{V/V}\end{aligned}$$

Step 7 of 13

Therefore, the transfer function is,

$$\left|\frac{V_o}{V_i}\right| = \frac{100.4998}{\sqrt{1+\left(\frac{f}{10^6}\right)^2}} \quad \ldots\ldots (1)$$

Substitute $10^4\ \text{Hz}$ for f in the equation (1).

$$\begin{aligned}\left|\frac{V_o}{V_i}\right| &= \frac{100.4998}{\sqrt{1+\left(\frac{10^4}{10^6}\right)^2}}\\ &= \frac{100.4998}{\sqrt{1+(0.01)^2}}\\ &= \frac{100.4998}{\sqrt{1+0.0001}}\\ &= \frac{100.4998}{\sqrt{1.0001}}\\ &= \frac{100.4998}{1}\end{aligned}$$

$$\left|\frac{V_o}{V_i}\right| = 100.4998$$

Apply logarithm on both sides.

$$\begin{aligned}20\log\left|\frac{V_o}{V_i}\right| &= 20\log(100.4998)\\ &= 20(2)\\ &= 40\ \text{dB}\end{aligned}$$

Thus, the magnitude at the frequency of $f = 10^4\ \text{Hz}$ is $40\ \text{dB}$.

Step 8 of 13

Thus, the transfer function is,

$$\left|\frac{V_o}{V_i}\right| = \frac{100.4998}{\sqrt{1+\left(\frac{f}{10^6}\right)^2}}$$

Substitute $10^5\ \text{Hz}$ for f in the equation.

$$\begin{aligned}\left|\frac{V_o}{V_i}\right| &= \frac{100.4998}{\sqrt{1+\left(\frac{10^5}{10^6}\right)^2}}\\ &= \frac{100.4998}{\sqrt{1+(0.1)^2}}\\ &= \frac{100.4998}{\sqrt{1+0.01}}\\ &= \frac{100.4998}{\sqrt{1.01}}\\ &= \frac{100.4998}{1.0049}\end{aligned}$$

$$\left|\frac{V_o}{V_i}\right| = 100.0097$$

Step 9 of 13

Apply logarithm on both sides.

$$\begin{aligned}20\log\left|\frac{V_o}{V_i}\right| &= 20\log(100.0097)\\ &= 20(2)\\ &= 40\ \text{dB}\end{aligned}$$

Therefore, the magnitude at the frequency of $f = 10^5\ \text{Hz}$ is $40\ \text{dB}$.

Step 10 of 13

Determine the frequency at which gain is equal to zero.

$$\left|\frac{V_o}{V_i}\right| = \frac{100.4998}{\sqrt{1+\left(\frac{f}{10^6}\right)^2}}$$

$$20\log\left|\frac{V_o}{V_i}\right| = 20\log\left(\frac{100.4998}{\sqrt{1+\left(\frac{f}{10^6}\right)^2}}\right)$$

$$0 = 20\log\left(\frac{100.4998}{\sqrt{1+\left(\frac{f}{10^6}\right)^2}}\right)$$

$$\frac{100.4998}{\sqrt{1+\left(\frac{f}{10^6}\right)^2}} = 1$$

Step 11 of 13

Simplify the equation further.

$$100.4998 = \sqrt{1+\left(\frac{f}{10^6}\right)^2}$$

$$10100.2098 = 1+\left(\frac{f}{10^6}\right)^2$$

$$\left(\frac{f}{10^6}\right)^2 = 10100.2098-1$$

$$\frac{f^2}{10^{12}} = 10099.2098$$

$$f^2 = (10099.2098)10^{12}$$

$$f = \sqrt{(10099.2098)10^{12}}$$

$$f = 100.4948\times10^6$$

$$f = 100.4948\ \text{MHz}$$

Thus, the frequency at which gain is $0\ \text{dB}$ is $100.4948\ \text{MHz}$.

Step 12 of 13

Write the modified tabie of the frequency response.
Table 1

$f\,(\text{Hz})$	$\lvert T\rvert\,(\text{dB})$
$0.9953\ \text{Hz}$	0
10	20
10^2	37
10^3	40
10^4	40
10^5	40
10^6	37
10^7	20
100.4948×10^6	0

Step 13 of 13

Sketch the magnitude frequency response plot.

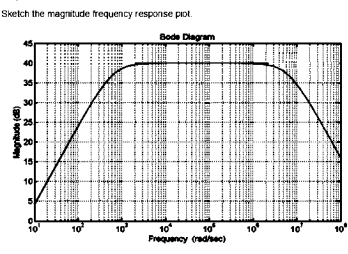


Figure 1

Thus, the magnitude frequency response is drawn.

1.073P

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1.074P

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Step 1 of 7

Consider the circuit diagram.

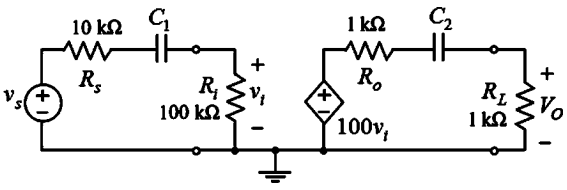


Figure 1

Step 2 of 7

Calculate the corner frequency at the input side of the circuit.

$$f_A = \frac{1}{2\pi C_1 (R_s + R_i)}$$
$$= \frac{1}{2\pi C_1 (10\times10^3 + 100\times10^3)}$$

$$\frac{1}{2\pi C_1 (110\times10^3)} \le 100 \text{(since, } f_A \le 100)$$

$$C_1 \ge \frac{1}{200\pi (110\times10^3)}$$
$$\ge \frac{1}{69.115038\times10^6}$$
$$\ge 14.4686\times10^{-9}$$
$$\ge 0.145\times10^{-7}$$

Consider the input capacitance in the form 1×10^{-n}.

$$C_1 = 1\times10^{-7}$$
$$= 100 \text{ nF}$$

Step 3 of 7

Determine the corner frequency at the input frequency.

$$f_A = \frac{1}{2\pi C_1 (R_s + R_i)}$$
$$= \frac{1}{2\pi (100\times10^{-9})(10\times10^3 + 100\times10^3)}$$
$$= \frac{1}{2\pi (100\times10^{-9})(110\times10^3)}$$
$$= \frac{1}{69.1150\times10^{-3}}$$
$$= 14.4686 \text{ Hz}$$

Step 4 of 7

Determine the frequency at the output side of the circuit.

$$f_B = \frac{1}{2\pi C_2 (R_o + R_L)}$$
$$= \frac{1}{2\pi C_2 (1\times10^3 + 1\times10^3)}$$

$$\frac{1}{2\pi C_2 (2\times10^3)} \le 100 \qquad (\text{since, } f_B \le 100)$$

$$C_2 \ge \frac{1}{200\pi (2\times10^3)}$$

$$C_2 \ge \frac{1}{1.2566\times10^6}$$
$$\ge 796\times10^{-9}$$

Therefore, the output capacitance in the form 1×10^{-n}.

$$C_2 = 1000\times10^{-9}$$
$$= 1000 \text{ nF}$$

Step 5 of 7

Determine the corner frequency at the output frequency.

$$f_B = \frac{1}{2\pi C_2 (R_o + R_L)}$$
$$= \frac{1}{2\pi (1000\times10^{-9})(1\times10^3 + 1\times10^3)}$$
$$= \frac{1}{2\pi (1000\times10^{-9})(2\times10^3)}$$
$$= \frac{1}{12.5663\times10^{-3}}$$
$$= 79.5779 \text{ Hz}$$

Step 6 of 7

Determine the transfer function.

$$T(s) = \frac{v_o}{v_s}$$
$$= \frac{1}{v_s}\left((100v_i)\left(\frac{R_L}{R_L + R_o + \frac{1}{j\omega C_2}}\right)\right)$$
$$= 100\left(\frac{R_L}{R_L + R_o + \frac{1}{j\omega C_2}}\right)\left(\frac{v_i}{v_s}\right)$$
$$= 100\left(\frac{R_L}{R_L + R_o + \frac{1}{j\omega C_2}}\right)\left(\frac{R_i}{R_s + R_s + \frac{1}{j\omega C_1}}\right)$$

Step 7 of 7

Simplify the expression further,

$$T(s) = 100\left(\frac{j\omega C_2 R_L}{j\omega C_2 (R_L + R_o) + 1}\right)\left(\frac{j\omega C_1 R_i}{j\omega C_1 (R_i + R_s) + 1}\right)$$
$$= 100\left(\frac{sC_2 R_L}{sC_2 (R_L + R_o) + 1}\right)\left(\frac{sC_1 R_i}{sC_1 (R_i + R_s) + 1}\right) \qquad (\text{Let } s = j\omega)$$
$$= 100\left(\frac{\frac{sC_2 R_L}{s}}{\frac{\left(\frac{1}{C_2 (R_L + R_o)}\right)}{s} + 1}\right)\left(\frac{\frac{sC_1 R_i}{s}}{\frac{\left(\frac{1}{C_1 (R_i + R_s)}\right)}{s} + 1}\right)$$
$$= 100\left(\frac{sC_2 R_L}{\frac{s}{2\pi f_B} + 1}\right)\left(\frac{sC_1 R_i}{\frac{s}{2\pi f_A} + 1}\right)$$
$$= 100\left(\frac{s(1000\times10^{-9})(1\times10^3)}{\frac{s}{2\pi f_B} + 1}\right)\left(\frac{s(100\times10^{-9})(100\times10^3)}{\frac{s}{2\pi f_A} + 1}\right)$$
$$= (1\times10^{-3})\left(\frac{s}{\frac{s}{2\pi f_B} + 1}\right)\left(\frac{s}{\frac{s}{2\pi f_A} + 1}\right)$$

Therefore, the transfer function is,

$$\boxed{T(s) = (1\times10^{-3})\left(\frac{s}{\frac{s}{2\pi f_B} + 1}\right)\left(\frac{s}{\frac{s}{2\pi f_A} + 1}\right)}.$$

1.077P

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1.078P

Step 1 of 8

Refer to Figure P1.76 in the textbook.

Refer to Table 1.2 in the textbook.

Determine the transfer function for the left side part of the circuit.

$$T_i(s)=\frac{V_i(s)}{V_s(s)}$$
$$=\frac{\frac{1}{sC_1}}{R_1+\frac{1}{sC_1}}$$
$$=\frac{\frac{1}{sC_1}}{\frac{sC_1R_1+1}{sC_1}}$$
$$=\frac{1}{sC_1R_1+1}$$

Therefore, the transfer function is,

$$\boxed{T_i(s)=\frac{1}{sC_1R_1+1}}.$$

Step 2 of 8

This transfer function shows, it is a low pass filter. Thus, the cut-off frequency is,

$$f_1=\frac{1}{2\pi R_1C_1}$$
$$=\frac{1}{2\pi\times1\times10^6\times10\times10^{-12}}$$
$$=0.1591\times10^5$$
$$=15.91\text{kHz}$$

Therefore, the corner frequency is $\boxed{15.91\text{kHz}}$.

Step 3 of 8

Use source transform technique to re-draw the right side circuit diagram.

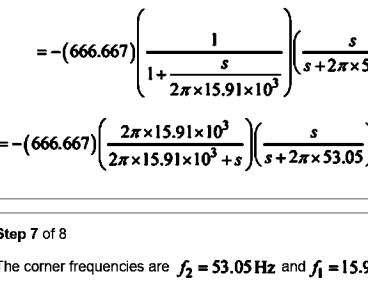


Figure 1

Step 4 of 8

Apply voltage division rule and determine the output voltage.

$$V_o(s)=-\left(\frac{G_mR_2V_i(s)}{R_2+\frac{1}{sC_2}+R_3}\right)(R_3)$$

(negative sign because, the current in opposite direction)

$$\frac{V_o(s)}{V_i(s)}=-\frac{R_2R_3}{(R_2+R_3)+\frac{1}{sC_2}}G_m$$
$$\frac{V_o(s)}{V_i(s)}=-\frac{R_2R_3}{(R_2+R_3)+\frac{1}{sC_2}}G_m$$
$$=-\left(\frac{R_2R_3}{R_2+R_3}\right)\left(\frac{1}{1+\frac{1}{sC_2(R_2+R_3)}}\right)G_m$$
$$=-\left(\frac{R_2R_3}{R_2+R_3}\right)\left(\frac{s}{s}\right)\left(\frac{1}{1+\frac{1}{sC_2(R_2+R_3)}}\right)G_m$$
$$=-\left(\frac{R_2R_3}{R_2+R_3}\right)\left(\frac{s}{s+\frac{1}{C_2(R_2+R_3)}}\right)G_m$$

Therefore, the transfer function is,

$$\boxed{T_o(s)=\frac{V_o(s)}{V_i(s)}=-\left(\frac{R_2R_3}{R_2+R_3}\right)\left(\frac{s}{s+\frac{1}{C_2(R_2+R_3)}}\right)G_m}.$$

Step 5 of 8

Determine the cut-off frequency.

$$f_2=\frac{1}{2\pi C_2(R_2+R_3)}$$
$$=\frac{1}{2\pi\times100\times10^{-9}\times\left(10\times10^3+20\times10^3\right)}$$
$$=\frac{1}{2\pi\times10^{-7}\times30\times10^3}$$
$$=\frac{1}{2\pi\times3\times10^{-3}}$$
$$=0.05305\times10^3$$
$$=53.05\text{Hz}$$

Therefore, the corner frequency is $\boxed{53.05\text{Hz}}$.

Step 6 of 8

Determine the over-all transfer function.

$$T(s)=T_o(s)T_i(s)$$
$$=\left(\frac{1}{1+\frac{s}{2\pi f_1}}\right)\left(-G_m\left(\frac{R_2R_3}{R_2+R_3}\right)\right)\left(\frac{s}{s+2\pi f_2}\right)$$
$$=\left(\frac{1}{1+\frac{s}{2\pi\times15.91\times10^3}}\right)\left(-100\times10^{-3}\left(\frac{10\times10^3\times20\times10^3}{10\times10^3+20\times10^3}\right)\right)\left(\frac{s}{s+2\pi\times53.05}\right)$$
$$=-(666.667)\left(\frac{1}{1+\frac{s}{2\pi\times15.91\times10^3}}\right)\left(\frac{s}{s+2\pi\times53.05}\right)$$
$$=-(666.667)\left(\frac{2\pi\times15.91\times10^3}{2\pi\times15.91\times10^3+s}\right)\left(\frac{s}{s+2\pi\times53.05}\right)$$

Step 7 of 8

The corner frequencies are $f_2=53.05\text{Hz}$ and $f_1=15.91\text{kHz}$.

Determine the band width of the bode plot.

$$\text{B.W}=15910-53.05$$
$$=15.857\text{kHz}$$

Therefore, the band width is $\boxed{15.857\text{kHz}}$.

Step 8 of 8

The circuit is combination of low pass and high pass filter, thus the circuit has a plot like band pass.

The constant magnitude is,

$$|T(j\omega)|=20\log(666.667)$$
$$=56.478\text{dB}$$

The bode plot increases its magnitude $+20\text{ dB}$ below the corner frequency $f_2=53.05\text{Hz}$ and it maintains constant upto the corner frequency, $f_1=15.91\text{kHz}$.

Draw the bode plot for the transfer function.

Figure 2

Thus, the bode plot is shown in Figure 2.

1.079P

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Step 1 of 3

1.080P

Refer to Figure P1.78 in the textbook.

Apply voltage division rule and determine the transfer function.

$$\frac{V_o(s)}{V_i(s)} = \frac{\left(\frac{1}{sC_2} \| R_2\right)}{\left(\frac{1}{sC_2} \| R_2\right) + \left(\frac{1}{sC_1} \| R_1\right)}$$

$$= \frac{\left(\frac{\frac{R_2}{sC_2}}{R_2 + \frac{1}{sC_2}}\right)}{\left(\frac{\frac{R_2}{sC_2}}{R_2 + \frac{1}{sC_2}}\right) + \left(\frac{\frac{R_1}{sC_1}}{R_1 + \frac{1}{sC_1}}\right)}$$

$$= \frac{\left(\frac{R_2}{1+sR_2C_2}\right)}{\left(\frac{R_2}{1+sR_2C_2}\right) + \left(\frac{R_1}{1+sR_1C_1}\right)}$$

$$= \frac{(R_2)(1+sR_1C_1)}{(R_2)(1+sR_1C_1) + (R_1)(1+sR_2C_2)}$$

$$= \frac{(R_1R_2)\left(\frac{1}{R_1} + sC_1\right)}{(R_2R_1)\left(\frac{1}{R_1} + sC_1\right) + (R_2R_1)\left(\frac{1}{R_2} + sC_2\right)}$$

$$= \frac{\left(\frac{1}{R_1} + sC_1\right)}{\left(\frac{1}{R_1} + sC_1\right) + \left(\frac{1}{R_2} + sC_2\right)}$$

$$= \frac{\frac{1}{R_1} + sC_1}{\frac{1}{R_1} + \frac{1}{R_2} + s(C_1 + C_2)}$$

$$= \frac{C_1\left(\frac{1}{R_1C_1} + s\right)}{\left[\left(\frac{1}{R_1} + \frac{1}{R_2}\right)\left(\frac{1}{C_1 + C_2}\right) + s\right](C_1 + C_2)}$$

$$\frac{V_o(s)}{V_i(s)} = \frac{C_1}{C_1 + C_2}\left(\frac{\frac{1}{R_1C_1} + s}{\left(\frac{1}{R_1} + \frac{1}{R_2}\right)\left(\frac{1}{C_1 + C_2}\right) + s}\right) \quad \text{..... (1)}$$

Step 2 of 3

From Equation (1), the transfer function can be made independent of frequency, by making the term $\frac{\frac{1}{R_1C_1} + s}{\left(\frac{1}{R_1} + \frac{1}{R_2}\right)\left(\frac{1}{C_1 + C_2}\right) + s}$ to unity.

$$\frac{\frac{1}{R_1C_1} + s}{\left(\frac{1}{R_1} + \frac{1}{R_2}\right)\left(\frac{1}{C_1 + C_2}\right) + s} = 1$$

$$\frac{1}{R_1C_1} + s = \left(\frac{1}{R_1} + \frac{1}{R_2}\right)\left(\frac{1}{C_1 + C_2}\right) + s$$

$$\frac{1}{R_1C_1} = \left(\frac{1}{R_1} + \frac{1}{R_2}\right)\left(\frac{1}{C_1 + C_2}\right)$$

$$\frac{C_1 + C_2}{C_1} = R_1\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

$$1 + \frac{C_2}{C_1} = 1 + \frac{R_1}{R_2}$$

$$\frac{C_2}{C_1} = \frac{R_1}{R_2}$$

$$C_1R_1 = C_2R_2$$

Therefore, the condition for the transfer function to be independent of frequency is,

$\boxed{C_1R_1 = C_2R_2}$.

Step 3 of 3

Consider the condition for compensated attenuator is $C_1R_1 = C_2R_2$.

Determine the expression for transfer function.

$$\frac{V_o(s)}{V_i(s)} = \frac{C_1}{C_1 + C_2}$$

$$= \frac{\frac{C_2R_2}{R_1}}{\frac{C_2R_2}{R_1} + C_2}$$

$$= \frac{\frac{R_2}{R_1}}{\frac{R_2}{R_1} + 1}$$

$$= \frac{R_2}{R_2 + R_1}$$

Therefore, the transmission of the compensated attenuator in terms of R_1 and R_2 is

$\boxed{\frac{R_2}{R_1 + R_2}}$.

1.081P

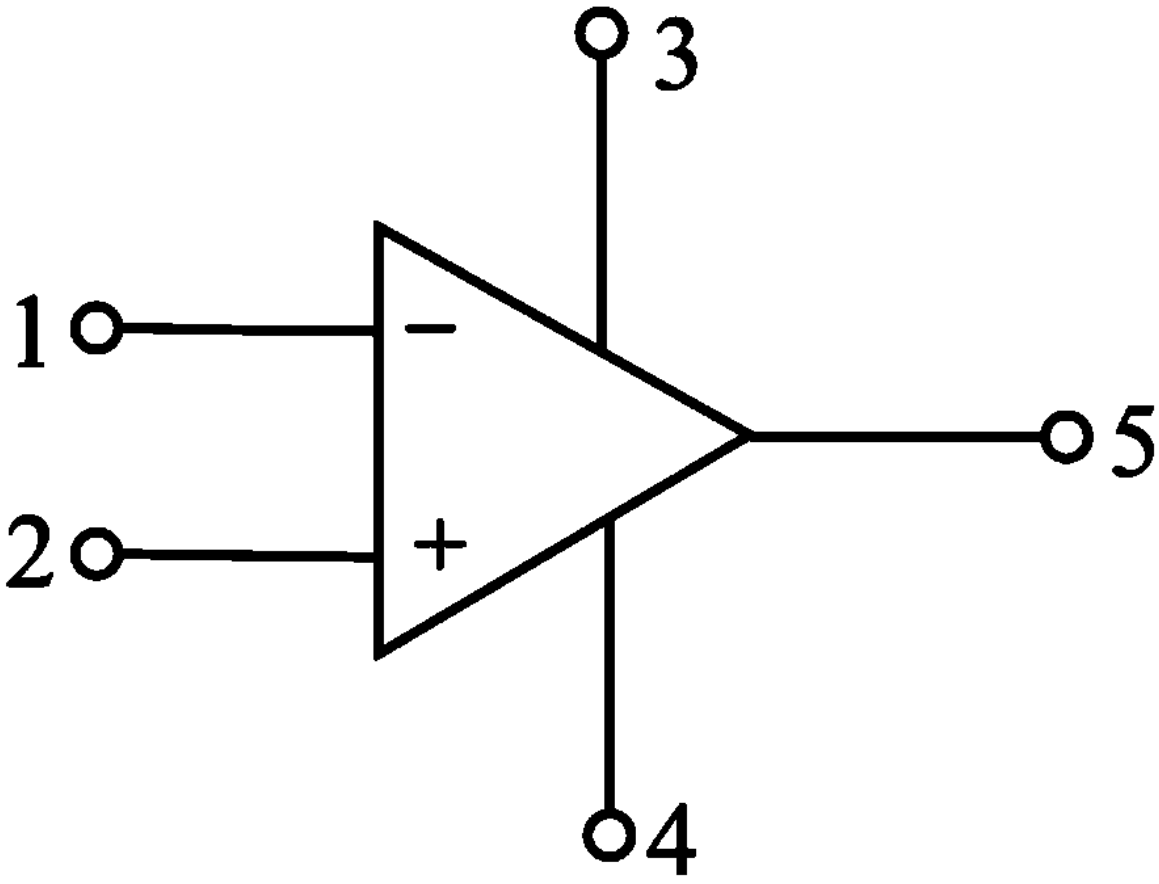
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Step 1 of 2

Draw the general representation of op-amp.



From the figure, it is observed that the number terminals in a single op-amp is 5.

Therefore, the minimum number of terminals required for single op-amp is $\boxed{5}$.

Step 2 of 2

Calculate the minimum number of terminals required on an integrated circuit package containing four op-amps (Quad op-amp).

In a quad op-amp circuit, the number of op-amps is 4. Each op-amp has two inputs and one output.

Therefore,

Number of outputs is 4.

Number of inputs is 8.

And the circuit has one V_{CC} terminal and one V_{EE} terminal.

Therefore, the total number terminals is,

$$4+8+1+1=14$$

Thus, the minimum number of terminals required for four op amps (quad op-amp) is $\boxed{14}$.

Step 1 of 2

2.001P

Determine the minimum number of pins required for dual op-amp IC package.

The number of inputs is, $2\times 2=4$

The number of output is, $2\times 1=2$

The required power pins are,

$V_{CC}=1$

$V_{EE}=1$

Find the total number of pins required.

$\Rightarrow 4+2+1+1$

$\Rightarrow 8$

Therefore, total number of pins required for dual op-amp IC package is $\boxed{8}$.

Step 2 of 2

Determine the minimum number of pins required for quad op-amp IC package:

The number of inputs is, $4\times 2=8$

The number of output is, $4\times 1=4$

The required power pins are,

$V_{CC}=1$

$V_{EE}=1$

Find the total number of pins required.

$\Rightarrow 8+4+1+1$

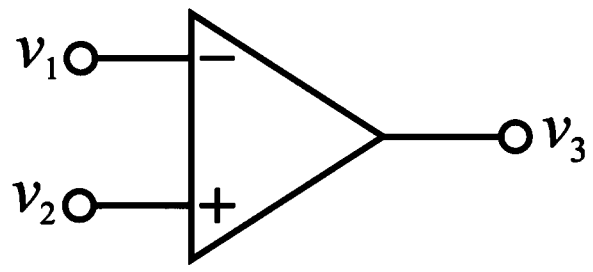
$\Rightarrow 14$

Therefore, total number of pins required for quad op-amp IC package is $\boxed{14}$.

Step 1 of 6

2.002E

Consider the following open loop op-amp circuit:



The output voltage in the open-loop is,

$v_3 = A(v_2 - v_1)$ ..... (1)

The differential-mode input is,

$v_{Id} = v_2 - v_1$ ..... (2)

The common-mode input is,

$v_{Icm} = \frac{1}{2}(v_1 + v_2)$ ..... (3)

Step 2 of 6

(a)

Calculate the voltage v_1 by substituting 10^3 for A, $0\,\text{V}$ for v_2 and $2\,\text{V}$ for v_3 in equation (1).

$$2 = 10^3(0 - v_1)$$
$$-v_1 = \frac{2}{10^3}$$
$$v_1 = -0.002\ \text{V}$$

Therefore, the voltage at the third terminal, v_1 is $\boxed{-0.002\,\text{V}}$.

Calculate the differential-mode input voltage.

$$\begin{aligned} v_{Id} &= v_2 - v_1 \\ &= 0 - (-0.002) \\ &= 0.002 \\ &= 2\ \text{mV} \end{aligned}$$

Therefore, the differential-mode input voltage, v_{Id} is $\boxed{2\,\text{mV}}$.

Step 3 of 6

Calculate the common-mode input voltage by substituting -0.002 for v_1 and 0 for v_2 in equation (3).

$$\begin{aligned} v_{Icm} &= \frac{1}{2}(v_1 + v_2) \\ &= \frac{1}{2}(-0.002 + 0) \\ &= -0.001 \\ &= -1\ \text{mV} \end{aligned}$$

Therefore, the common-mode input voltage, v_{Icm} is $\boxed{-1\,\text{mV}}$.

Step 4 of 6

(b)

Calculate the voltage v_1 by substituting 10^3 for A, $5\ \text{V}$ for v_2 and $-10\,\text{V}$ for v_3 in equation (1).

$$-10 = 10^3(5 - v_1)$$
$$5 - v_1 = \frac{-10}{10^3}$$

$$\begin{aligned} v_1 &= 5 + \frac{10}{10^3} \\ &= 5 + 0.01 \\ &= 5.01\,\text{V} \end{aligned}$$

Therefore, the voltage at the third terminal, v_1 is $\boxed{5.01\ \text{V}}$.

Calculate the differential-mode input voltage.

$$\begin{aligned} v_{Id} &= v_2 - v_1 \\ &= 5 - 5.01 \\ &= -0.01 \\ &= -10\ \text{mV} \end{aligned}$$

Therefore, the differential-mode input voltage, v_{Id} is $\boxed{-10\,\text{mV}}$.

Calculate the common-mode input voltage by substituting $5.01\,\text{V}$ for v_1 and $5\ \text{V}$ for v_2 in equation (3).

$$\begin{aligned} v_{Icm} &= \frac{1}{2}(v_1 + v_2) \\ &= \frac{1}{2}(5.01 + 5) \\ &= 5.005\ \text{V} \end{aligned}$$

Therefore, the common-mode input voltage, v_{Icm} is $\boxed{5.005\ \text{V}}$.

Step 5 of 6

(c)

Calculate the voltage v_3 by substituting 10^3 for A, $0.998\ \text{V}$ for v_2 and $1.002\ \text{V}$ for v_1 in equation (1).

$$\begin{aligned} v_3 &= 10^3(0.998 - 1.002) \\ &= 10^3(-0.004) \\ &= -4\ \text{V} \end{aligned}$$

Therefore, the voltage at the third terminal, v_3 is $\boxed{-4\ \text{V}}$.

Calculate the differential-mode input voltage.

$$\begin{aligned} v_{Id} &= v_2 - v_1 \\ &= 0.998 - 1.002 \\ &= -0.004 \\ &= -4\ \text{mV} \end{aligned}$$

Therefore, the differential-mode input voltage, v_{Id} is $\boxed{-4\,\text{mV}}$.

Calculate the common-mode input voltage by substituting $1.002\ \text{V}$ for v_1 and $0.998\ \text{V}$ for v_2 in equation (3).

$$\begin{aligned} v_{Icm} &= \frac{1}{2}(v_1 + v_2) \\ &= \frac{1}{2}(1.002 + 0.998) \\ &= 1\ \text{V} \end{aligned}$$

Therefore, the common-mode input voltage, v_{Icm} is $\boxed{1\ \text{V}}$.

Step 6 of 6

(d)

Calculate the voltage v_2 by substituting 10^3 for A, $-3.6\ \text{V}$ for v_3 and $-3.6\ \text{V}$ for v_1 in equation (1).

$$-3.6 = 10^3(v_2 - (-3.6))$$
$$v_2 + 3.6 = -0.0036$$
$$v_2 = -3.6036$$

Therefore, the voltage at the third terminal, is $\boxed{-3.6036\ \text{V}}$.

Calculate the differential-mode input voltage.

$$\begin{aligned} v_{Id} &= v_2 - v_1 \\ &= -3.6036 - (-3.6) \\ &= -0.0036 \\ &= -3.6\ \text{mV} \end{aligned}$$

Therefore, the differential-mode input voltage, v_{Id} is $\boxed{-3.6\,\text{mV}}$.

Calculate the common-mode input voltage by substituting $-3.6\ \text{V}$ for v_1 and $-3.6036\ \text{V}$ for v_2 in equation (3).

$$\begin{aligned} v_{Icm} &= \frac{1}{2}(v_1 + v_2) \\ &= \frac{1}{2}(-3.6 + (-3.6036)) \\ &= -3.6018\ \text{V} \end{aligned}$$

Therefore, the common-mode input voltage, v_{Icm} is $\boxed{-3.6\ \text{V}}$.

Step 1 of 3

2.002P

Refer to the Figure P2.2 in the textbook.

Redraw the circuit diagram by indicating with node voltages.

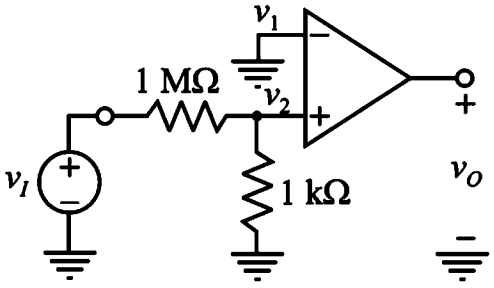


Figure 1

Step 2 of 3

Write the output voltage formula for open loop configuration.

$$v_o = A(v_2 - v_1)$$

Here,

A is gain of op-amp.

v_2 is non-inverting terminal voltage.

v_1 is inverting terminal voltage.

From the Figure 1, the inverting terminal of the op-amp is connected to the ground. Hence, $v_1 = 0\ \text{V}$.

Substitute $0\ \text{V}$ for v_1 in the output voltage equation.

$$\begin{aligned} v_o &= A(v_2 - v_1) \\ &= A(v_2 - 0) \end{aligned}$$

$$v_o = Av_2 \quad \text{..... (1)}$$

Step 3 of 3

Write the expression for voltage at the non-inverting terminal, v_2, of the op-amp by using the voltage division rule.

$$v_2 = v_I\left(\frac{1\times10^3}{(1\times10^6)+(1\times10^3)}\right)$$

Substitute v_2 value in the output voltage v_o of equation (1).

$$v_o = Av_I\left(\frac{1\times10^3}{(1\times10^6)+(1\times10^3)}\right)$$

$$\begin{aligned} A &= \left(\frac{v_o}{v_I}\right)\left(\frac{(1\times10^6)+(1\times10^3)}{1\times10^3}\right) \\ &= \left(\frac{4}{1}\right)\left(\frac{1{,}001\times10^3}{1\times10^3}\right) \\ &= 4(1{,}001) \\ &= 4004\ \text{V/V} \end{aligned}$$

Therefore, the open-loop voltage gain, A of the op-amp is **4004 V/V**.

Step 1 of 2

2.003E

Refer to Figure E2.3 in the textbook.

From the figure the output voltage is,

$v_3 = \mu v_d$ ..... (1)

Apply Kirchhoff's current law at node v_d.

$$\frac{v_d}{R} + G_m v_1 - G_m v_2 = 0$$

$$\frac{v_d}{R} + G_m (v_1 - v_2) = 0$$

$v_d = G_m R (v_2 - v_1)$ ..... (2)

Substitute $G_m R (v_2 - v_1)$ for v_d in equation (1).

$$v_3 = \mu \left[G_m R (v_2 - v_1) \right]$$

$v_3 = \mu G_m R (v_2 - v_1)$ ..... (3)

Therefore, the expression for v_3 in terms of v_1 and v_2 is $\boxed{\mu G_m R (v_2 - v_1)}$.

Step 2 of 2

From equation (3), the open loop gain is,

$$A = \frac{v_3}{v_2 - v_1}$$
$$= \mu G_m R$$

Substitute 100 for μ, 10×10^{-3} for G_m and 10×10^3 for R.

$$A = \mu G_m R$$
$$= (100)\left(10 \times 10^{-3}\right)\left(10 \times 10^3\right)$$
$$= 10{,}000 \text{ V/V}$$

Therefore, the open loop gain, A is $\boxed{10{,}000 \text{ V/V}}$.

Calculate the gain in decibels.

$$A|_{\text{dB}} = 20 \log (10{,}000)$$
$$= 20(4)$$
$$= 80 \text{ dB}$$

Therefore, the open loop gain in decibels is $\boxed{80 \text{ dB}}$.

Step 1 of 3

Consider an ideal op amp.

Output voltage $v_0 = -2.000\ \text{V}$

Input voltage on negative terminal $v_- = -1.000\ \text{V}$

Relationship between op amp-gain and output voltage is,

$$v_0 = A(v_+ - v_-)$$

Here,

v_+ is input voltage at positive terminal of op amp.

v_- is input voltage at negative terminal of op amp.

A is gain of op amp.

v_0 is output of op amp.

Step 2 of 3

Since the op-amp is ideal, the voltage at the non-inverting terminal is equal to the voltage at the inverting terminal. Therefore,

$$\begin{aligned} v_+ &= v_- \\ &= -1.000\ \text{V} \end{aligned}$$

Hence, the measured voltage value at the positive terminal, v_+ is $\boxed{-1.000\ \text{V}}$.

Step 3 of 3

Consider the new measured value at the positive terminal is, $v_+ = -1.005\ \text{V}$

The expression for voltage gain of the amplifier is

$$A_v = \frac{v_o}{v_+ - v_-}$$

Substitute $-2.000\ \text{V}$ for v_o, $-1.005\ \text{V}$ for v_+ and $-1.000\ \text{V}$ for v_-.

$$\begin{aligned} A_v &= \frac{-2.000}{-1.005-(-1.000)} \\ &= \frac{-2.000}{-0.005} \\ &= 400\ \text{V/V} \end{aligned}$$

Therefore, the actual voltage gain, A_v of the amplifier is $\boxed{400\ \text{V/V}}$.

Step 1 of 2

2.004E

Refer to Figure 2.5 in the textbook.

Assume that the op-amp is an ideal one.

According to virtual ground concept, the voltage at the inverting terminal is equal to the voltage at non-inverting terminal.

$$V_- = V_+$$
$$= 0\,\text{V}$$

Apply Kirchhoff's current law at the inverting terminal.

$$\frac{V_- - v_I}{R_1} + \frac{V_- - v_O}{R_2} = 0$$

$$\frac{0 - v_I}{R_1} + \frac{0 - v_O}{R_2} = 0 \qquad (\text{since } V_- = V_+ = 0\,\text{V})$$

$$\frac{v_O}{R_2} = -\frac{v_I}{R_1}$$

$$\frac{v_O}{v_I} = -\frac{R_2}{R_1}$$

Therefore, the closed-loop gain of the inverting amplifier is,

$$A_{CL} = -\frac{R_2}{R_1} \quad \ldots\ldots (1)$$

Step 2 of 2

For an inverting op-amp input resistance is, R_1.

Consider the value of input resistance, $R_{in} = 100\,\text{k}\Omega$.

Therefore, the value of resistor R_1 is $100\,\text{k}\Omega$.

Substitute -10 for A_{CL} and $100\,\text{k}\Omega$ for R_1 in equation (1).

$$-10 = -\frac{R_2}{100\times10^3}$$

$$R_2 = (10)\left(100\times10^3\right)$$
$$= 1\times10^6$$
$$= 1\,\text{M}\Omega$$

Therefore, input and feedback resistors of inverting op-amp are $\boxed{100\,\text{k}\Omega}$ and $\boxed{1\,\text{M}\Omega}$ respectively.

Step 1 of 7

It is given that, the op-amp is ideal except of having a finite gain A.

Write down the results obtained in various experiments.

Table 1

Experiment #	V_1	V_2	V_o
1	0.00	0.00	0.00
2	1.00	1.00	0.00
3		1.00	1.00
4	1.00	1.10	10.1
5	2.01	2.00	−0.99
6	1.99	2.00	1.00
7	5.10		−5.10

Step 2 of 7

Write the expression for output voltage of an op-amp in open-loop configuration.

$$V_o = A(V_2 - V_1)$$

Use this expression to write the equation for gain *A*.

$$A = \frac{V_o}{V_2 - V_1} \quad \text{..... (1)}$$

In experiments 1 and 2, the difference of voltages V_2 and V_1 is zero, hence the output voltage is zero.

Take the experiment 4 values and calculate the value of gain *A*.

Substitute $1.00\,\text{V}$ for V_1, $1.10\,\text{V}$ for V_2 and $10.1\,\text{V}$ for V_o in equation (1).

$$\begin{aligned} A &= \frac{10.1}{1.10 - 1.00} \\ &= \frac{10.1}{0.1} \\ &= 101\ \text{V/V} \end{aligned}$$

Step 3 of 7

Take the experiment 5 values and calculate the value of gain *A*.

Substitute $2.01\,\text{V}$ for V_1, $2.00\,\text{V}$ for V_2 and $-0.99\,\text{V}$ for V_o in equation (1).

$$\begin{aligned} A &= \frac{-0.99}{2.00 - 2.01} \\ &= \frac{-0.99}{-0.01} \\ &= 99\ \text{V/V} \end{aligned}$$

Take the experiment 6 values and calculate the value of gain *A*.

Substitute $1.99\,\text{V}$ for V_1, $2.00\,\text{V}$ for V_2 and $1\,\text{V}$ for V_o in equation (1).

$$\begin{aligned} A &= \frac{1}{2.00 - 1.99} \\ &= \frac{1}{0.01} \\ &= 100\ \text{V/V} \end{aligned}$$

Step 4 of 7

From the calculated values of open-loop gains in three different experiments, it is clear that, the results are inconsistent. Since the voltage gains obtained in three experiments are different from each other; the results are not in consistence form. Hence, it is concluded that, there is a possibility of an experiment error.

The three voltage gains are, 101, 99 and 100.

Consider that the open-loop gain of an op-amp is the average of these three voltage gains. Therefore,

$$A = 100\ \text{V/V}$$

Step 5 of 7

Determine the missing term, voltage V_1 in the experiment 3.

Re-call voltage gain, A expression from the equation (1).

$$A = \frac{V_o}{V_2 - V_1}$$

Substitute $100\ \text{V/V}$ for A, $1.00\,\text{V}$ for V_2 and $1.00\,\text{V}$ for V_o.

$$\begin{aligned} 100 &= \frac{1.00}{1.00 - V_1} \\ 100 - 100V_1 &= 1 \\ 100V_1 &= 99 \\ V_1 &= 0.99\ \text{V} \end{aligned}$$

Therefore, the voltage V_1 in experiment 3 is $\boxed{0.99\ \text{V}}$.

Step 6 of 7

Determine the missing term, voltage V_2 in the experiment 7.

Substitute $100\ \text{V/V}$ for A, $5.10\,\text{V}$ for V_1 and $-5.10\ \text{V}$ for V_o.

$$\begin{aligned} A &= \frac{V_o}{V_2 - V_1} \\ 100 &= \frac{-5.10}{V_2 - 5.10} \\ 100V_2 - 510 &= -5.10 \\ 100V_2 &= 504.9 \\ V_2 &= 5.049\ \text{V} \end{aligned}$$

Therefore, the voltage V_2 in experiment 7 is $\boxed{5.049\ \text{V}}$.

Step 7 of 7

Tabulate the results obtained in all experiments as follows:

Table 2

Experiment	V_1	V_2	V_d	V_o	$\frac{V_o}{V_d}$
1	0.00	0.00	0.00	0.00	Undefined
2	1.00	1.00	0.00	0.00	Undefined
3	0.99	1.00	0.01	1.00	100
4	1.00	1.10	0.10	10.1	101
5	2.01	2.00	−0.01	−0.99	99
6	1.99	2.00	0.01	1.00	100
7	5.10	5.049	−0.051	−5.10	100

2.005E

Step 1 of 3

Refer to Figure E2.5 (a) in the textbook.

Refer to Table 1.1 in the textbook.

Draw the circuit diagram of trans-resistance amplifier.

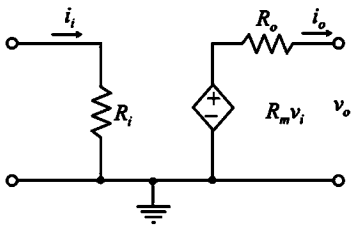


Figure 1

Step 2 of 3

Write the expression for open-loop trans-resistance.

$$R_m = \left.\frac{v_o}{i_i}\right|_{i_o=0} \text{(V/A)}$$

Ideal characteristics of trans-resistance amplifiers are,

$R_i = 0$ and $R_o = 0$

Voltage at the inverting terminal is zero.

Determine the output voltage of the op-amp.

$$v_o = -i_i\left(10\times10^3\right)$$

$$\frac{v_o}{i_i} = -10\times10^3$$

$$R_m = -10\times10^3$$

Therefore, input resistance of the circuit is $\boxed{R_i = 0\ \Omega}$.

Therefore, the output resistance of the circuit is $\boxed{R_o = 0\ \Omega}$

Trans-resistance of the op-amp is $\boxed{R_m = -10\text{k}\Omega}$.

Step 3 of 3

Refer to Figure E2.5 (b) in the textbook.

Determine trans-resistance of the amplifier.

$$\frac{v_o}{i_i} = -10\times10^3$$

$$\begin{aligned} v_o &= -\left(10\times10^3\right)(i_i) \\ &= -\left(10\times10^3\right)\left(0.5\times10^{-3}\right) \\ &= -5\text{ V} \end{aligned}$$

Therefore, output voltage, v_o of the amplifier is $\boxed{-5\text{ V}}$.

Step 1 of 3

2.005P

Consider the relation of trans-conductance and trans-resistance of an op-amp is given by, $i = G_m(v_2 - v_1)$ ..... (1)

Here,

i is the output current.

G_m is trans-conductance.

v_2 and v_1 are the input terminal voltages.

Step 2 of 3

Substitute $\frac{R_m}{v_0}$ for i in the equation (1).

$$v_0 = R_m G_m (v_2 - v_1)$$

$$A = \frac{v_0}{(v_2 - v_1)}$$

$$A = R_m G_m \quad \text{..... (2)}$$

Here,

A is open loop amplifier gain.

G_m is trans-conductance.

R_m is trans-resistance.

Step 3 of 3

Substitute $40\times10^{-3}\frac{\text{mA}}{\text{V}}$ for G_m , $1\times10^{6}\Omega$ for R_m in equation (2).

$$A = 40\times10^{-3}\frac{\text{A}}{\text{V}}\times1\times10^{6}\Omega$$
$$= 40000$$

Therefore, the open loop voltage gain, A of the amplifier is $\boxed{40000\ \text{V/V}}$.

Step 1 of 13

Refer to Figure E2.6 in the textbook.

In an ideal op-amp, the inverting and non-inverting terminal currents are zero.

Re-draw the circuit diagram.

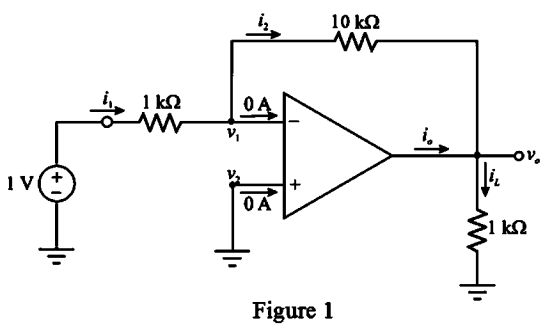


Figure 1

Step 2 of 13

Form Figure 1, the non-inverting node voltage is zero.

That is,

$v_2 = 0\,\text{V}$ (Since v_2 is connected to ground)

Using virtual ground concept, the inverting node voltage becomes,

$$v_1 = 0\,\text{V} \qquad (\text{Since}, v_1 = v_2)$$

Therefore, the inverting node voltage, v_1 is, $\boxed{0\ \text{V}}$.

Step 3 of 13

Use Ohm's law to Figure 1, the input current (i_1) becomes,

$$i_1 = \frac{1 - v_1}{1\times10^3}$$

Substitute $0\,\text{V}$ for v_1 in the equation.

$$\begin{aligned} i_1 &= \frac{1-0}{1\times10^3} \\ &= \frac{1}{1\times10^3} \\ &= 1\times10^{-3} \end{aligned}$$

Therefore, the input current (i_1) is, $\boxed{1\,\text{mA}}$.

Step 4 of 13

Use Kirchhoff's current law to at inverting node, determine the feedback current (i_2).

$$\begin{aligned} i_1 &= 0 + i_2 \\ i_2 &= i_1 \\ i_2 &= 1\times10^{-3} \qquad (\text{Since}, i_1 = 1\,\text{mA}) \end{aligned}$$

Therefore, the feedback current (i_2) is, $\boxed{1\,\text{mA}}$.

Step 5 of 13

Apply Kirchhoff's voltage law to Figure 1, determine the output voltage (v_o).

$$-v_1 + i_2(10\times10^3) + v_o = 0$$

Substitute $0\,\text{V}$ for v_1 and $1\,\text{mA}$ for i_2 in the equation.

$$\begin{aligned} -0 + (1\times10^{-3})(10\times10^3) + v_o &= 0 \\ 10 + v_o &= 0 \\ v_o &= -10 \end{aligned}$$

Therefore, the output voltage is, $\boxed{-10\,\text{V}}$.

Step 6 of 13

Use Ohm's law to Figure 1, the load current (i_L) becomes,

$$i_L = \frac{v_o}{1\times10^3}$$

Substitute $-10\,\text{V}$ for v_o in the equation.

$$\begin{aligned} i_L &= \frac{-10}{1\times10^3} \\ &= -10\times10^{-3} \end{aligned}$$

Therefore, the load current (i_L) is $\boxed{-10\,\text{mA}}$.

Step 7 of 13

Using Kirchhoff's current law to the above circuit, the output current (i_o) becomes,

$$i_o = i_L - i_2 \qquad (\text{Since}, i_o + i_2 = i_L)$$

Substitute $-10\,\text{mA}$ for i_L and $1\,\text{mA}$ for i_2 in the equation.

$$\begin{aligned} i_o &= -10\times10^{-3} - 1\times10^{-3} \\ &= -11\times10^{-3} \end{aligned}$$

Therefore, the output current (i_o) is $\boxed{-11\,\text{mA}}$.

Step 8 of 13

The voltage gain (A_v) is,

$$A_v = \frac{v_o}{v_i}$$

Substitute $-10\,\text{V}$ for v_o and $1\,\text{V}$ for v_i in the equation.

$$\begin{aligned} A_v &= \frac{-10}{1} \\ &= -10 \end{aligned}$$

Therefore, the voltage gain (A_v) is $\boxed{-10\,\text{V/V}}$.

Step 9 of 13

The voltage gain $(A_{v(\text{dB})})$ in decibels is,

$$\begin{aligned} A_{v(\text{dB})} &= 20\log|A_v| \\ &= 20\log|-10| \qquad (\text{Since}, A_v = -10\,\text{V/V}) \\ &= 20 \end{aligned}$$

Therefore, the voltage gain $(A_{v(\text{dB})})$ in decibels is $\boxed{20\,\text{dB}}$.

Step 10 of 13

The current gain (A_i) is,

$$A_i = \frac{i_L}{i_1}$$

Substitute $-10\,\text{mA}$ for i_L and $1\,\text{mA}$ for i_1 in the equation.

$$\begin{aligned} A_i &= \frac{-10\times10^{-3}}{1\times10^{-3}} \\ &= -10 \end{aligned}$$

Therefore, the current gain (A_i) is $\boxed{-10\,\text{A/A}}$.

Step 11 of 13

The current gain $(A_{i(\text{dB})})$ in decibels is,

$$\begin{aligned} A_{i(\text{dB})} &= 20\log|A_i| \\ &= 20\log|-10| \qquad (\text{Since}, A_i = -10\,\text{A/A}) \\ &= 20 \end{aligned}$$

Therefore, the current gain $(A_{i(\text{dB})})$ in decibels is $\boxed{20\,\text{dB}}$.

Step 12 of 13

The power gain $\left(\frac{P_o}{P_i}\right)$ is,

$$\frac{P_o}{P_i} = A_v A_i$$

Substitute $-10\,\text{V/V}$ for A_v and $-10\,\text{A/A}$ for A_i in the equation.

$$\begin{aligned} \frac{P_o}{P_i} &= (-10)(-10) \\ &= 100 \end{aligned}$$

Therefore, the power gain $\left(\frac{P_o}{P_i}\right)$ is $\boxed{100\,\text{W/W}}$.

Step 13 of 13

The power gain $\left(\frac{P_o}{P_i}\right)_{\text{dB}}$ in decibels is,

$$\begin{aligned} \left(\frac{P_o}{P_i}\right)_{\text{dB}} &= 10\log\left|\frac{P_o}{P_i}\right| \\ &= 10\log|100| \qquad \left(\text{Since}, \frac{P_o}{P_i} = 100\,\text{W/W}\right) \\ &= 20 \end{aligned}$$

Therefore, the power gain $\left(\frac{P_o}{P_i}\right)_{\text{dB}}$ in decibels is $\boxed{20\,\text{dB}}$.

2.006P

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Step 1 of 3

2.007E

Draw a weighted summer with two inputs.

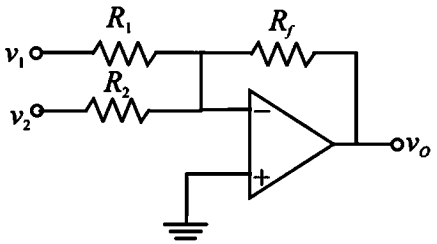


Figure 1

Step 2 of 3

The output of the inverting summing amplifier is,

$$v_o = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2\right)$$

Consider the required output is, $v_o = -(v_1 + 5v_2)$.

Thus,

$$\frac{R_f}{R_1} = 1 \quad \text{..... (1)}$$

$$\frac{R_f}{R_2} = 5 \quad \text{..... (2)}$$

Step 3 of 3

For a maximum output voltage of $10\ \text{V}$, the current in the feedback resistor will not exceed $1\ \text{mA}$.

Thus,

$$R_f = \frac{10}{1\times10^{-3}}$$
$$= 10\ \text{k}\Omega$$

From equation (1), determine the value of resistance, R_1.

$$R_1 = R_f$$
$$= 10\ \text{k}\Omega$$

From equation (2), determine the value of resistance, R_2.

$$R_2 = \frac{R_f}{5}$$
$$= \frac{10}{5}$$
$$= 2\ \text{k}\Omega$$

Thus the design parameters are:

$$\boxed{\begin{aligned} R_f &= 10\ \text{k}\Omega \\ R_1 &= 10\ \text{k}\Omega \\ R_2 &= 2\ \text{k}\Omega \end{aligned}}.$$

Step 1 of 8

Refer to Figure 2.4 and for signal representation in the text book.

The output voltage of non-ideal operational amplifiers response to both the differential and common-mode components of their input signals, which can be written as,

$v_0 = A_d v_{id} + A_{cm} v_{icm}$ ..... (1)

Here,

A_d is the differential gain and

A_{cm} is the common-mode gain.

Refer to Figure E2.3 in the text book.

The operational amplifier's effectiveness in rejecting common mode signals is measured by its CMRR. Mathematically, it can be written as follows.

$$\text{CMRR} = 20\log\left|\frac{A_d}{A_{cm}}\right|$$

Write the expressions of the transconductance of the two channels.

$$G_{m1} = G_m - \frac{1}{2}\Delta G_m$$
$$G_{m2} = G_m + \frac{1}{2}\Delta G_m$$

Step 2 of 8

Write the expression for input to the two channels.

$$v_1 = v_{icm} - \frac{1}{2}v_{id}$$
$$v_2 = v_{icm} + \frac{1}{2}v_{id}$$

Here,

v_{icm} is the common-mode input signal

v_{id} is the differential input signal.

Consider the following expressions for input currents.

$$i_1 = G_m v_1$$
$$i_2 = G_m v_2$$

Step 3 of 8

The current in the resistor R is,

$$\begin{aligned} i &= i_2 - i_1 \\ &= G_{m2}v_2 - G_{m1}v_1 \end{aligned}$$

The voltage across the resistor R is,

$$\begin{aligned} v_d &= iR \\ &= \left(G_{m2}v_2 - G_{m1}v_1\right)R \end{aligned}$$

Step 4 of 8

From the circuit, the output voltage is,

$$\begin{aligned} v_3 &= \mu v_d \\ &= \mu R\left(G_{m2}v_2 - G_{m1}v_1\right) \end{aligned}$$

Substitute the values of $G_m - \frac{1}{2}\Delta G_m$ for G_{m1}, $G_m + \frac{1}{2}\Delta G_m$ for G_{m2}, $v_{icm} - \frac{1}{2}v_{id}$ for v_1 and $v_{icm} + \frac{1}{2}v_{id}$ for v_2 in the equation.

$$\begin{aligned} v_3 &= \mu R\left\{\left(G_m + \frac{1}{2}\Delta G_m\right)\left(v_{icm} + \frac{1}{2}v_{id}\right) - \left(G_m - \frac{1}{2}\Delta G_m\right)\left(v_{icm} - \frac{1}{2}v_{id}\right)\right\} \\ &= \mu R\left\{\left(G_m + \frac{1}{2}\Delta G_m - G_m + \frac{1}{2}\Delta G_m\right)v_{icm} + \frac{1}{2}\left(G_m + \frac{1}{2}\Delta G_m + G_m - \frac{1}{2}\Delta G_m\right)v_{id}\right\} \\ &= \mu R\left(\Delta G_m v_{icm} + G_m v_{id}\right) \end{aligned}$$

From the circuit diagram, the voltage v_3 is nothing but output voltage. So, replace v_3 with v_0. Therefore,

$$v_0 = \left(\mu R G_m\right)v_{id} + \left(\mu R \Delta G_m\right)v_{icm}$$ ..... (2)

Step 5 of 8

Compare equation (2) with equation (1).

$$A_d = \mu R G_m$$
$$A_{cm} = \mu R \Delta G_m$$

Therefore, the expression for differential gain, A_d is $\boxed{\mu R G_m}$ and the expression for common mode gain, A_{cm} is $\boxed{\mu R \Delta G_m}$.

Step 6 of 8

Derive the common-mode rejection ratio (CMRR).

$$\begin{aligned} \text{CMRR} &= 20\log\left|\frac{A_d}{A_{cm}}\right| \\ &= 20\log\left|\frac{\mu R G_m}{\mu R \Delta G_m}\right| \\ &= 20\log\left|\frac{G_m}{\Delta G_m}\right| \end{aligned}$$

Therefore, the expression for CMRR of the operational amplifier is $\boxed{20\log\left|\frac{G_m}{\Delta G_m}\right|}$.

Step 7 of 8

The value of differential gain, A_d in dB is, $A_d = 60\text{dB}$

Calculate the value of A_d.

$$20\log|A_d| = 60$$
$$\log|A_d| = 3$$
$$A_d = 10^3 \text{ V/V}$$

Step 8 of 8

Determine maximum permitted percentage mismatched between the two channels of Tran's conductance.

$$\frac{A_d}{A_{cm}} = \frac{G_m}{\Delta G_m}$$

$$\begin{aligned} \Delta G_m &= \frac{A_{cm}}{A_d}G_m \\ &= \frac{100}{1000} \times 100 \\ &= 10\% \end{aligned}$$

Thus, the required mismatch permitted percentage is, $\boxed{\Delta G_m = 10\%}$.

Step 1 of 4

2.008E

Refer to Figure 2.11 in the textbook.

Draw a weighted summer capable of implementing summing coefficients of both signs.

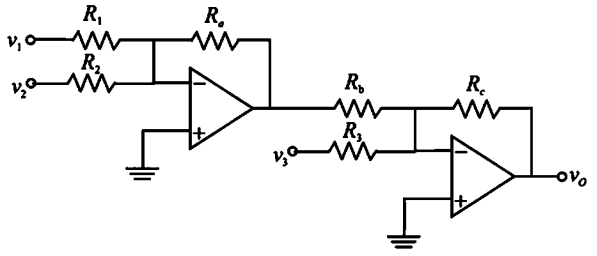


Figure 1

Step 2 of 4

Consider the required output equation is,

$$v_O = (2v_1 + v_2 - 4v_3)$$

The output of the first amplifier circuit is $-(2v_1 + v_2)$. This is inverted again at the second stage to get the desired output $(2v_1 + v_2)$.

Thus,

$$\frac{R_a}{R_1} = 2$$

$$\frac{R_a}{R_2} = 1$$

$$\frac{R_c}{R_b} = 1$$

Step 3 of 4

The non-inverting input to the second stage is $-4v_3$.

Thus,

$$\frac{R_c}{R_3} = 4$$

Consider the value of $R_c = 10\,\text{k}\Omega$.

Consider the value of $R_a = 10\,\text{k}\Omega$.

Substitute $10\,\text{k}\Omega$ for R_c in equation $\frac{R_c}{R_3} = 4$.

$$\frac{10\,\text{k}\Omega}{R_3} = 4$$

$$R_3 = \frac{R_c}{4}$$
$$= \frac{10}{4}$$
$$= 2.5\,\text{k}\Omega$$

Step 4 of 4

Determine the value of resistance, R_2.

$$R_2 = R_a$$

Substitute $10\,\text{k}\Omega$ for R_a in the equation.

$$R_2 = 10\,\text{k}\Omega$$

Determine the value of resistance, R_1.

$$R_1 = \frac{R_a}{2}$$

Substitute $10\,\text{k}\Omega$ for R_a in the equation.

$$R_1 = \frac{10\,\text{k}\Omega}{2}$$
$$= 5\,\text{k}\Omega$$

Thus, the design parameters are:

$$\boxed{\begin{aligned} R_1 &= 5\,\text{k}\Omega \\ R_2 &= 10\,\text{k}\Omega \\ R_3 &= 2.5\,\text{k}\Omega \end{aligned}}$$

$$\boxed{\begin{aligned} R_a &= 10\,\text{k}\Omega \\ R_b &= 10\text{k}\Omega \\ R_c &= 10\,\text{k}\Omega \end{aligned}}.$$

Step 1 of 15

(a)

Refer to Figure P2.8 (a) for an inverting op-amp circuit in the textbook.

Draw the circuit diagram depicting with node voltages and branch currents.

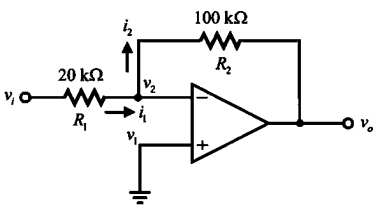


Figure 1

Step 2 of 15

From the concept of virtual ground, the potential at inverting terminal is equal to the potential at non-inverting terminal.

The voltage at non-inverting terminal is at zero potential. Therefore,

$$v_1 = v_2$$
$$= 0\ \text{V}$$

Step 3 of 15

Apply Kirchhoff's current law at node v_2.

$$i_1 = i_2 + 0$$
$$\frac{v_i - v_2}{R_1} = \frac{v_2 - v_o}{R_2}$$
$$\frac{v_i - 0}{R_1} = \frac{0 - v_o}{R_2} \qquad (\text{since } v_2 = 0)$$
$$\frac{v_i}{R_1} = \frac{-v_o}{R_2}$$
$$\frac{v_o}{v_i} = -\frac{R_2}{R_1}$$
$$G = -\frac{R_2}{R_1}$$

Substitute $20\ \text{k}\Omega$ for R_1 and $100\ \text{k}\Omega$ for R_2 in the expression of voltage gain, G.

$$G = -\frac{100\times10^3}{20\times10^3}$$
$$= -5\ \text{V/V}$$

Therefore, the voltage gain of the amplifier is $\boxed{-5\ \text{V/V}}$.

Step 4 of 15

Determine the current flowing through resistor R_1 using Ohms law.

$$i_1 = \frac{v_i - v_2}{R_1}$$
$$i_1 = \frac{v_i - 0}{R_1}$$
$$\frac{v_i}{i_1} = R_1$$
$$R_{in} = R_1$$

Substitute $20\ \text{k}\Omega$ for R_1.

$$R_{in} = R_1$$
$$= 20\ \text{k}\Omega$$

Therefore, the value of input resistance, R_{in} is $\boxed{20\ \text{k}\Omega}$.

Step 5 of 15

(b)

Refer to Figure P2.8 (b) for an inverting op-amp circuit in the textbook.

Draw the circuit diagram depicting with node voltages and branch currents.

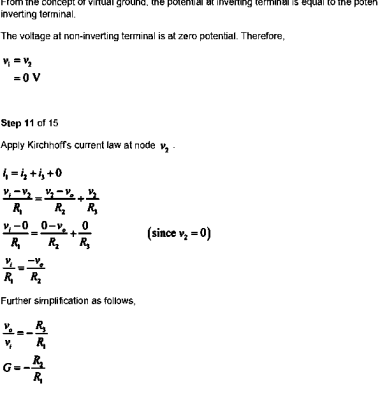


Figure 2

Step 6 of 15

From the concept of virtual ground, the potential at inverting terminal is equal to the potential at non-inverting terminal.

The voltage at non-inverting terminal is at zero potential. Therefore,

$$v_1 = v_2$$
$$= 0\ \text{V}$$

Step 7 of 15

Apply Kirchhoff's current law at node v_2.

$$i_1 = i_2 + 0$$
$$\frac{v_i - v_2}{R_1} = \frac{v_2 - v_o}{R_2}$$
$$\frac{v_i - 0}{R_1} = \frac{0 - v_o}{R_2} \qquad (\text{since } v_2 = 0)$$
$$\frac{v_i}{R_1} = \frac{-v_o}{R_2}$$
$$\frac{v_o}{v_i} = -\frac{R_2}{R_1}$$
$$G = -\frac{R_2}{R_1}$$

Substitute $20\ \text{k}\Omega$ for R_1 and $100\ \text{k}\Omega$ for R_2 in the expression of voltage gain, G.

$$G = -\frac{100\times10^3}{20\times10^3}$$
$$= -5\ \text{V/V}$$

Therefore, the voltage gain of the amplifier is $\boxed{-5\ \text{V/V}}$.

Step 8 of 15

Determine the current flowing through resistor R_1 using Ohms law.

$$i_1 = \frac{v_i - v_2}{R_1}$$
$$i_1 = \frac{v_i - 0}{R_1}$$
$$\frac{v_i}{i_1} = R_1$$
$$R_{in} = R_1$$

Substitute $20\ \text{k}\Omega$ for R_1.

$$R_{in} = R_1$$
$$= 20\ \text{k}\Omega$$

Therefore, the value of input resistance, R_{in} is $\boxed{20\ \text{k}\Omega}$.

Step 9 of 15

(c)

Refer to Figure P2.8 (c) for an inverting op-amp circuit in the textbook.

Draw the circuit diagram depicting with node voltages and branch currents.

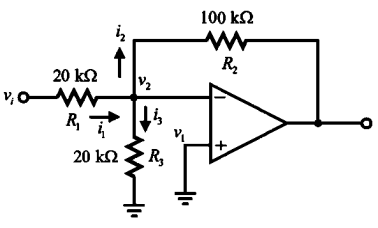


Figure 3

Step 10 of 15

From the concept of virtual ground, the potential at inverting terminal is equal to the potential at non-inverting terminal.

The voltage at non-inverting terminal is at zero potential. Therefore,

$$v_1 = v_2$$
$$= 0\ \text{V}$$

Step 11 of 15

Apply Kirchhoff's current law at node v_2.

$$i_1 = i_2 + i_3 + 0$$
$$\frac{v_i - v_2}{R_1} = \frac{v_2 - v_o}{R_2} + \frac{v_2}{R_3}$$
$$\frac{v_i - 0}{R_1} = \frac{0 - v_o}{R_2} + \frac{0}{R_3} \qquad (\text{since } v_2 = 0)$$
$$\frac{v_i}{R_1} = \frac{-v_o}{R_2}$$

Further simplification as follows,

$$\frac{v_o}{v_i} = -\frac{R_2}{R_1}$$
$$G = -\frac{R_2}{R_1}$$

Substitute $20\ \text{k}\Omega$ for R_1 and $100\ \text{k}\Omega$ for R_2 in the expression of voltage gain, G.

$$G = -\frac{100\times10^3}{20\times10^3}$$
$$= -5\ \text{V/V}$$

Therefore, the voltage gain of the amplifier is $\boxed{-5\ \text{V/V}}$.

Determine the current flowing through resistor R_1 using Ohms law.

$$i_1 = \frac{v_i - v_2}{R_1}$$
$$i_1 = \frac{v_i - 0}{R_1}$$
$$\frac{v_i}{i_1} = R_1$$
$$R_{in} = R_1$$

Substitute $20\ \text{k}\Omega$ for R_1.

$$R_{in} = R_1$$
$$= 20\ \text{k}\Omega$$

Therefore, the value of input resistance, R_{in} is $\boxed{20\ \text{k}\Omega}$.

Step 12 of 15

(d)

Refer to Figure P2.8 (d) for an inverting op-amp circuit in the textbook.

Draw the circuit diagram depicting with node voltages and branch currents.

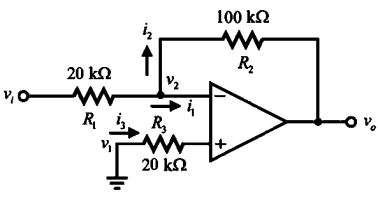


Figure 4

Step 13 of 15

From the concept of virtual ground, the potential at inverting terminal is equal to the potential at non-inverting terminal

From Figure 4, the current i_3 is,

$$i_3 = 0\ \text{A}$$

Determine the voltage et non-inverting terminal.

$$v_1 = i_3 R_3$$
$$= (0) R_3$$
$$= 0\ \text{V}$$

The voltage at non-inverting terminal is at zero potential. Therefore,

$$v_1 = v_2$$
$$= 0\ \text{V}$$

Step 14 of 15

Apply Kirchhoff's current law at node v_2.

$$i_1 = i_2 + 0$$
$$\frac{v_i - v_2}{R_1} = \frac{v_2 - v_o}{R_2}$$
$$\frac{v_i - 0}{R_1} = \frac{0 - v_o}{R_2} \qquad (\text{since } v_2 = 0)$$
$$\frac{v_i}{R_1} = \frac{-v_o}{R_2}$$
$$\frac{v_o}{v_i} = -\frac{R_2}{R_1}$$
$$G = -\frac{R_2}{R_1}$$

Substitute $20\ \text{k}\Omega$ for R_1 and $100\ \text{k}\Omega$ for R_2.

$$G = -\frac{100\times10^3}{20\times10^3}$$
$$= -5\ \text{V/V}$$

Therefore, the voltage gain of the amplifier is $\boxed{-5\ \text{V/V}}$.

Step 15 of 15

Determine the current flowing through resistor R_1 using Ohms law.

$$i_1 = \frac{v_i - v_2}{R_1}$$
$$i_1 = \frac{v_i - 0}{R_1}$$
$$\frac{v_i}{i_1} = R_1$$
$$R_{in} = R_1$$

Substitute $20\ \text{k}\Omega$ for R_1.

$$R_{in} = R_1$$
$$= 20\ \text{k}\Omega$$

Therefore, the value of input resistance, R_{in} is $\boxed{20\ \text{k}\Omega}$.

Step 1 of 6

2.009E

Refer to Figure E2.9 in the textbook.

Draw the operational amplifier circuit.

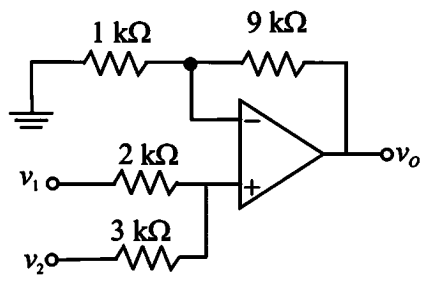


Figure 1

Step 2 of 6

By using superposition theorem, first consider the voltage source v_1 and replace the other source v_2 by $0\ \mathrm{V}$.

Re-draw the circuit diagram.

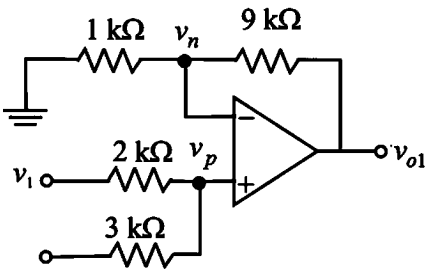


Figure 2

Step 3 of 6

Determine the voltage at the non-inverting input.

$$v_p = \frac{3\times10^3}{5\times10^3}\times v_1$$
$$= \frac{3}{5}\times v_1$$

According to virtual ground concept, the voltages at inverting and non-inverting terminals are equal.

$$v_p = v_n$$

$$v_n = \frac{3}{5}\times v_1$$

Step 4 of 6

Apply nodal analysis at the inverting input.

$$\frac{v_n}{1\,\mathrm{k}} + \frac{v_n - v_{o1}}{9\,k} = 0.$$

Substitute $\frac{3}{5}\times v_1$ for v_n in the equation.

$$\frac{\frac{3}{5}\times v_1}{1\,\mathrm{k}} + \frac{\frac{3}{5}\times v_1 - v_{o1}}{9\,\mathrm{k}} = 0$$

$$\frac{v_{o1}}{9\mathrm{k}} = \frac{3v_1}{5\,\mathrm{k}} + \frac{3v_1}{5\times9\,\mathrm{k}}$$

$$\frac{v_{o1}}{9} = \frac{3v_1}{5} + \frac{3v_1}{5\times9}$$

$$= \frac{10v_1}{15}$$

$$v_{o1} = \frac{2v_1}{3}\times9$$

$$v_{o1} = 6v_1 \quad \text{..... (1)}$$

Step 5 of 6

Now consider the voltage source v_2 and replace the other source v_1 by $0\ \mathrm{V}$.
Re-draw the circuit diagram.

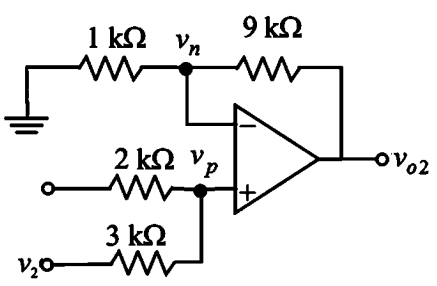


Figure 3

Step 6 of 6

Determine the voltage at the non-inverting input.

$$v_p = \frac{2\times10^3}{5\times10^3}\times v_1$$
$$= \frac{2}{5}\times v_1$$

Apply nodal analysis at the inverting input.

$$\frac{v_n}{1\,\mathrm{k}} + \frac{v_n - v_{o2}}{9\,\mathrm{k}} = 0.$$

Substitute $\frac{2}{5}\times v_1$ for v_n in the equation.

$$\frac{2v_2}{5} + \frac{\frac{2v_2}{5} - v_{o2}}{9} = 0$$

$$\frac{v_{o2}}{9} = \frac{2v_2}{45} + \frac{2v_2}{5}$$

$$\frac{v_{o2}}{9} = \frac{20v_2}{45}$$

$$v_{o2} = 4v_2$$

$$\frac{v_{o2}}{9\times10^3} = \frac{20v_2}{5\times9\times10^3}$$

$$v_{o2} = 4v_2 \quad \text{..... (2)}$$

The output voltage is obtained by adding equation (1) and equation (2).

$$v_o = v_{o1} + v_{o2}$$
$$= 6v_1 + 4v_2$$

Therefore, the output voltage of the circuit is $\boxed{v_o = 6v_1 + 4v_2}$.

Step 1 of 6

It is given that the particular inverting operational amplifier circuit is an ideal one and having two $10\ \text{k}\Omega$ resistors. Assume that one $10\ \text{k}\Omega$ resistor is in feedback path and one is at input path. Draw the topology of the expected inverting op-amp circuit diagram.

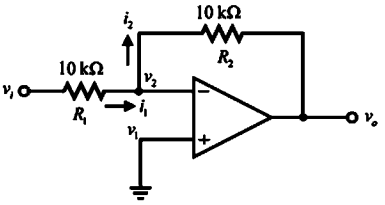


Figure 1

Step 2 of 6

From the concept of virtual ground, the potential at inverting terminal is equal to the potential at non-inverting terminal. Since, the given op-amp is an ideal; currents entering into both the terminals are zero.

The voltage at non-inverting terminal is at zero potential. Therefore,

$$v_1 = v_2$$
$$= 0\ \text{V}$$

Apply Kirchhoff's current law at node v_2.

$$i_1 = i_2 + 0$$
$$\frac{v_i - v_2}{R_1} = \frac{v_2 - v_o}{R_2}$$
$$\frac{v_i - 0}{R_1} = \frac{0 - v_o}{R_2} \qquad \text{since } v_2 = 0$$
$$\frac{v_i}{R_1} = \frac{-v_o}{R_2}$$
$$\frac{v_o}{v_i} = -\frac{R_2}{R_1}$$
$$G = -\frac{R_2}{R_1}$$

Step 3 of 6

Substitute $10\ \text{k}\Omega$ for R_1 and $10\ \text{k}\Omega$ for R_2 in the expression of closed loop voltage gain, G.

$$G = -\frac{10\times10^3}{10\times10^3}$$
$$= -1\ \text{V/V}$$

Therefore, the expected closed loop voltage gain, G of the amplifier is $\boxed{-1\ \text{V/V}}$.

Step 4 of 6

It is given that; the input dc voltage is, $v_i = 1\ \text{V}$

Write the expression for output voltage using the closed loop gain expression.

$$v_o = -\frac{R_2}{R_1} v_i$$
$$v_o = G v_i$$

Substitute $-1\ \text{V/V}$ for G and 1 V for v_i.

$$v_o = (-1\ \text{V/V})(1\ \text{V})$$
$$= -1\ \text{V}$$

Therefore, the resulted output voltage, v_o is $\boxed{-1\ \text{V}}$.

Step 5 of 6

The $10\ \text{k}\Omega$ resistors are said to be 1% resistors having values somewhere in the range (1 ± 0.01) times the nominal value.

The value of input voltage is, $v_i = 1\ \text{V}$

Write the expression for voltage gain.

$$\frac{v_o}{v_i} = \frac{-R_2}{R_1}$$

Substitute $1\ \text{V}$ for v_i.

$$\frac{v_o}{1} = \frac{-R_2}{R_1}$$
$$v_o = \frac{-R_2}{R_1}$$

Step 6 of 6

Determine the expected range of outputs as follows:

$$\frac{-(1-0.01)}{1+0.01} < v_o < \frac{-(1+0.01)}{1-0.01}$$
$$-\frac{0.99}{1.01} < v_o < \frac{-1.01}{0.99}$$
$$-0.98 < v_o < -1.020$$

Therefore, the expected range of outputs is $\boxed{-0.98 < v_o < -1.020}$.

Step 1 of 6

2.010E

Refer to Figure E2.9 in the textbook.

By using superposition theorem, the total response by the input is equal to sum individual response by each input.

Determine the response only due to v_{O1}.

Set the voltages $v_2 = 0$ and $v_3 = 0$.

Re-draw the modified circuit diagram.

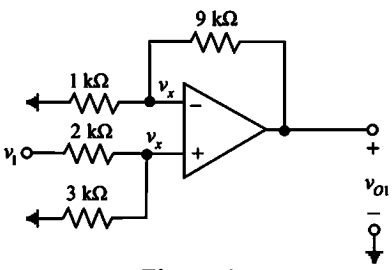


Figure 1

Step 2 of 6

Apply Kirchhoff's current law at node v_x (positive end).

$$\frac{v_x - v_1}{2\text{ k}} + \frac{v_x}{3\text{ k}} = 0$$
$$(3\text{ k})(v_x - v_1) + (2\text{ k})v_x = 0$$
$$v_x = \frac{3}{5}v_1$$

Apply Kirchhoff's current law at node v_x (negative end).

$$\frac{v_x}{1\text{ k}} + \frac{v_x - v_{O1}}{9\text{ k}} = 0$$
$$(9\text{ k})v_x + (1\text{ k})v_x - (1\text{ k})v_{O1} = 0$$
$$v_{O1} = \frac{(10\text{ k})v_x}{(1\text{ k})}$$

Substitute $\frac{3}{5}v_1$ for v_x in the equation.

$$v_{O1} = (10)\left(\frac{3}{5}v_1\right)$$

$$v_{O1} = 6v_1 \quad \text{..... (1)}$$

Step 3 of 6

Determine the response only due to v_{O2}.

Set the voltages $v_1 = 0$ and $v_3 = 0$.

Re-draw the modified circuit diagram.

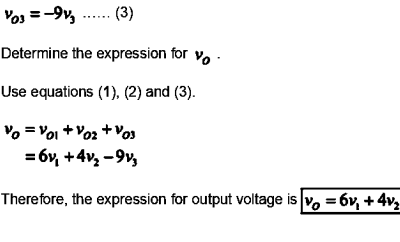


Figure 2

Step 4 of 6

Apply Kirchhoff's current law at node v_x (positive end).

$$\frac{v_x - v_2}{3\text{ k}} + \frac{v_x}{2\text{ k}} = 0$$
$$(2\text{ k})(v_x - v_2) + (3\text{ k})v_x = 0$$
$$v_x = \frac{2}{5}v_2$$

Apply Kirchhoff's current law at node v_x (negative end).

$$\frac{v_x}{1\text{ k}} + \frac{v_x - v_{O2}}{9\text{ k}} = 0$$
$$(9\text{ k})v_x + (1\text{ k})v_x - (1\text{ k})v_{O2} = 0$$
$$v_{O2} = \frac{(10\text{ k})v_x}{(1\text{ k})}$$

Substitute $\frac{2}{5}v_2$ for v_x in the equation.

$$v_{O2} = (10)\left(\frac{2}{5}v_2\right)$$

$$v_{O2} = 4v_2 \quad \text{..... (2)}$$

Step 5 of 6

Determine the response only due to v_{O3}.

Set the voltages $v_1 = 0$ and $v_2 = 0$.

Re-draw the modified circuit diagram.

Figure 3

Step 6 of 6

For the inverting amplifier, write the expression for v_{O3}.

$$v_{O3} = -\frac{R_f}{R_3}v_3$$
$$= -\frac{(9\text{ k})}{(1\text{ k})}v_3$$

$$v_{O3} = -9v_3 \quad \text{..... (3)}$$

Determine the expression for v_O.

Use equations (1), (2) and (3).

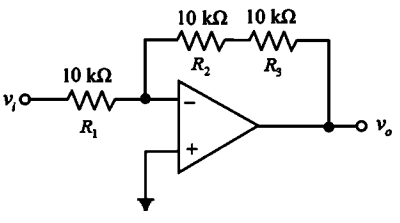
$$v_O = v_{O1} + v_{O2} + v_{O3}$$
$$= 6v_1 + 4v_2 - 9v_3$$

Therefore, the expression for output voltage is $\boxed{v_O = 6v_1 + 4v_2 - 9v_3}$.

Step 1 of 16

Using series and parallel combinations with three $10\ \text{k}\Omega$ resistors, **four types of inverting amplifier circuit topologies** are possible.

Topology 1: Two resistors are connected in series in feedback path and one resistor is at input. Draw the circuit diagram.



Step 2 of 16

Figure 1

Step 3 of 16

Calculate the equivalent resistance in the feedback path.

$$\begin{aligned} R_{eq} &= R_2 + R_3 \\ &= 10\ \text{k}\Omega + 10\ \text{k}\Omega \\ &= 20\ \text{k}\Omega \end{aligned}$$

Determine the voltage gain of the amplifier circuit.

$$\begin{aligned} G &= \frac{v_o}{v_i} \\ &= -\frac{R_{eq}}{R_1} \\ &= -\frac{20\ \text{k}\Omega}{10\ \text{k}\Omega} \\ &= -2 \end{aligned}$$

Therefore, the voltage gain of the op-amp circuit is $\boxed{-2\ \text{V/V}}$.

Step 4 of 16

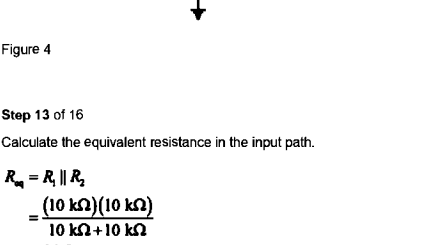
Determine the input resistance of the circuit, R_{in}.

$$\begin{aligned} R_{in} &= R_1 \\ &= 10\ \text{k}\Omega \end{aligned}$$

Therefore, the input resistance, R_{in} of the op-amp circuit is $\boxed{10\ \text{k}\Omega}$.

Step 5 of 16

Topology 2: Two resistors are connected in series at input path and one resistor is connected in feedback path. Draw the circuit diagram.



Step 6 of 16

Figure 2

Step 7 of 16

Calculate the equivalent resistance in the input path.

$$\begin{aligned} R_{eq} &= R_1 + R_2 \\ &= 10\ \text{k}\Omega + 10\ \text{k}\Omega \\ &= 20\ \text{k}\Omega \end{aligned}$$

Determine the voltage gain of the amplifier circuit.

$$\begin{aligned} G &= \frac{v_o}{v_i} \\ &= -\frac{R_3}{R_{eq}} \\ &= -\frac{10\ \text{k}\Omega}{20\ \text{k}\Omega} \\ &= -0.5\ \text{V/V} \end{aligned}$$

Therefore, the voltage gain of the op-amp circuit is $\boxed{-0.5\ \text{V/V}}$.

Step 8 of 16

Determine the input resistance of the circuit, R_{in}.

$$\begin{aligned} R_{in} &= R_{eq} \\ &= 20\ \text{k}\Omega \end{aligned}$$

Therefore, the input resistance, R_{in} of the op-amp circuit is $\boxed{20\ \text{k}\Omega}$.

Step 9 of 16

Topology 3: Two resistors are connected in parallel in feedback path and one resistor is at input. Draw the circuit diagram.

Figure 3

Step 10 of 16

Calculate the equivalent resistance in the feedback path.

$$\begin{aligned} R_{eq} &= R_2 \parallel R_3 \\ &= \frac{(10\ \text{k}\Omega)(10\ \text{k}\Omega)}{10\ \text{k}\Omega + 10\ \text{k}\Omega} \\ &= 5\ \text{k}\Omega \end{aligned}$$

Determine the voltage gain of the amplifier circuit.

$$\begin{aligned} G &= \frac{v_o}{v_i} \\ &= -\frac{R_{eq}}{R_1} \\ &= -\frac{5\ \text{k}\Omega}{10\ \text{k}\Omega} \\ &= -0.5\ \text{V/V} \end{aligned}$$

Therefore, the voltage gain of the op-amp circuit is $\boxed{-0.5\ \text{V/V}}$.

Step 11 of 16

Determine the input resistance of the circuit, R_{in}.

$$\begin{aligned} R_{in} &= R_1 \\ &= 10\ \text{k}\Omega \end{aligned}$$

Therefore, the input resistance, R_{in} of the op-amp circuit is $\boxed{10\ \text{k}\Omega}$.

Step 12 of 16

Topology 4: Two resistors are connected in parallel at input path and one resistor is connected in feedback path. Draw the circuit diagram.

Figure 4

Step 13 of 16

Calculate the equivalent resistance in the input path.

$$\begin{aligned} R_{eq} &= R_1 \parallel R_2 \\ &= \frac{(10\ \text{k}\Omega)(10\ \text{k}\Omega)}{10\ \text{k}\Omega + 10\ \text{k}\Omega} \\ &= 5\ \text{k}\Omega \end{aligned}$$

Determine the voltage gain of the amplifier circuit.

$$\begin{aligned} G &= \frac{v_o}{v_i} \\ &= -\frac{R_3}{R_{eq}} \\ &= -\frac{10\ \text{k}\Omega}{5\ \text{k}\Omega} \\ &= -2\ \text{V/V} \end{aligned}$$

Therefore, the voltage gain of the op-amp circuit is $\boxed{-2\ \text{V/V}}$.

Step 14 of 16

Determine the input resistance of the circuit, R_{in}.

$$\begin{aligned} R_{in} &= R_{eq} \\ &= 5\ \text{k}\Omega \end{aligned}$$

Therefore, the input resistance, R_{in} of the op-amp circuit is $\boxed{5\ \text{k}\Omega}$.

Step 15 of 16

From the gain calculations of all the topologies,
The largest voltage gain is possible in two topologies namely, topology 1 and topology 4.

The largest available voltage gain is $\boxed{-2\ \text{V/V}}$.

The input resistance in topology 1 is $\boxed{10\ \text{k}\Omega}$.

The input resistance in topology 4 is $\boxed{5\ \text{k}\Omega}$.

Step 16 of 16

The smallest gain is possible in two topologies namely, topology 2 and topology 3.

The largest available voltage gain is $\boxed{-0.5\ \text{V/V}}$.

The input resistance in topology 2 is $\boxed{20\ \text{k}\Omega}$.

The input resistance in topology 3 is $\boxed{10\ \text{k}\Omega}$.

Step 1 of 4

2.011E

Draw the basic model of a non-inverting op-amp.

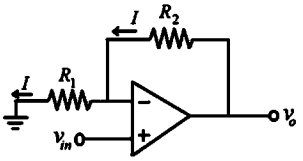


Figure 1

Step 2 of 4

For non-inverting amplifier, determine the gain.

$$A = \frac{v_o}{v_{in}}$$

$$2 = 1 + \frac{R_2}{R_1}$$

$$2R_1 = R_1 + R_2$$

$$2R_1 = 2R_1$$

$$R_1 = R_2 \quad \text{..... (1)}$$

Step 3 of 4

At maximum output voltage of $10\ \text{V}$, write the expression for current in the circuit.

$$I = \frac{v_o - 0}{R_1 + R_2}$$

$$\left(10 \times 10^{-6}\right) = \frac{10}{R_1 + R_2}$$

$$R_1 + R_2 = \frac{10}{\left(10 \times 10^{-6}\right)}$$

$$R_1 + R_2 = 1 \times 10^6$$

$$R_1 + R_2 = 1\ \text{M}\Omega \quad \text{..... (2)}$$

Step 4 of 4

Determine the values of resistors R_1 and R_2.

From equation (1) and equation (2), solve the values of R_1 and R_2.

$$R_1 + R_2 = 1\ \text{M}\Omega$$

Substitute R_1 for R_2 in the equation.

$$R_1 + R_1 = 1\ \text{M}\Omega$$

$$2R_1 = 1\ \text{M}\Omega$$

$$R_1 = 0.5\ \text{M}\Omega$$

From equation (1), $R_1 = R_2$.

$$R_2 = 0.5\ \text{M}\Omega$$

Therefore, the value of resistor R_1 is $\boxed{0.5\ \text{M}\Omega}$ and the value of resistor R_2 is $\boxed{0.5\ \text{M}\Omega}$.

Step 1 of 6

Draw the circuit diagram of inverting op-amp configuration.

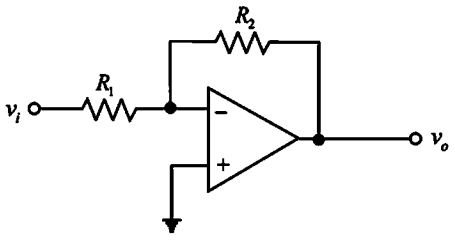


Figure 1

Step 2 of 6

(a)

Determine the closed loop gain, G.

$$G = -\frac{R_2}{R_1}$$

Substitute $10\ \text{k}\Omega$ for R_1 and $10\ \text{k}\Omega$ for R_2 in the equation.

$$G = -\frac{10\text{k}}{10\text{k}}$$
$$= -1\ \text{V/V}$$

Therefore, the value of closed loop gain, G is $\boxed{-1\ \text{V/V}}$.

Step 3 of 6

(b)

Determine the closed loop gain, G.

$$G = -\frac{R_2}{R_1}$$

Substitute $10\ \text{k}\Omega$ for R_1 and $100\ \text{k}\Omega$ for R_2 in the equation.

$$G = -\frac{100\text{k}}{10\text{k}}$$
$$= -10\ \text{V/V}$$

Therefore, the value of closed loop gain, G is $\boxed{-10\ \text{V/V}}$.

Step 4 of 6

(c)

Determine the closed loop gain, G.

$$G = -\frac{R_2}{R_1}$$

Substitute $10\ \text{k}\Omega$ for R_1 and $1\ \text{k}\Omega$ for R_2 in the equation.

$$G = -\frac{1\text{k}}{10\text{k}}$$
$$= -0.1\ \text{V/V}$$

Therefore, the value of closed loop gain, G is $\boxed{-0.1\ \text{V/V}}$.

Step 5 of 6

(d)

Determine the closed loop gain, G.

$$G = -\frac{R_2}{R_1}$$

Substitute $100\ \text{k}\Omega$ for R_1 and $10\ \text{M}\Omega$ for R_2 in the equation.

$$G = -\frac{10\text{M}}{100\text{k}}$$
$$= -100\ \text{V/V}$$

Therefore, the value of closed loop gain, G is $\boxed{-100\ \text{V/V}}$.

Step 6 of 6

(e)

Determine the closed loop gain, G.

$$G = -\frac{R_2}{R_1}$$

Substitute $100\ \text{k}\Omega$ for R_1 and $1\ \text{M}\Omega$ for R_2 in the equation.

$$G = -\frac{1\text{M}}{100\text{k}}$$
$$= -10\ \text{V/V}$$

Therefore, the value of closed loop gain, G is $\boxed{-10\ \text{V/V}}$.

Step 1 of 9

Refer to Figure 2.12 in the textbook.

Re-draw the circuit diagram.

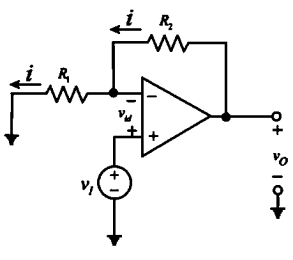


Figure 1

Step 2 of 9

(a)

If the open-loop gain of the op-amp is finite, then the input difference voltage is,

$$v_{id} = \frac{v_O}{A}$$

At the input node, determine the expression for current, i .

$$i = \frac{v_I - \frac{v_O}{A}}{R_1} \quad \text{..... (1)}$$

Determine the closed loop gain of non-inverting amplifier.

Write the loop equation from output to input.

$$v_O = \left(v_I - \frac{v_O}{A}\right) + i R_2$$

Substitute $\frac{v_I - \frac{v_O}{A}}{R_1}$ for i in the equation.

$$v_O = v_I - \frac{v_O}{A} + R_2 \frac{\left[v_I - \frac{v_O}{A}\right]}{R_1}$$

$$v_O = v_I - \frac{v_O}{A} + \frac{R_2}{R_1}\left[v_I - \frac{v_O}{A}\right]$$

$$v_O\left[1 + \frac{1}{A} + \frac{R_2}{R_1}\frac{1}{A}\right] = v_I\left[1 + \frac{R_2}{R_1}\right]$$

$$\frac{v_O}{v_I} = \frac{\left[1 + \frac{R_2}{R_1}\right]}{\left[1 + \frac{1}{A} + \frac{R_2}{R_1}\frac{1}{A}\right]}$$

$$G = \frac{1 + (R_2/R_1)}{1 + \frac{1 + (R_2/R_1)}{A}}$$

Therefore, the closed-loop gain is $\boxed{\frac{1 + (R_2/R_1)}{1 + \frac{1 + (R_2/R_1)}{A}}}$.

Step 3 of 9

(b)

Write the expression for closed loop gain for ideal op-amp.

$$\begin{aligned} G &= 1 + \frac{R_2}{R_1} \\ &= 1 + \frac{9\,\text{k}}{1\,\text{k}} \\ &= 10 \end{aligned}$$

Determine the closed loop gain for normal op-amp.

$$\begin{aligned} G &= \frac{1 + (R_2/R_1)}{1 + \frac{1 + (R_2/R_1)}{A}} \\ &= \frac{1 + (9\,\text{k}/1\,\text{k})}{1 + \frac{1 + (9\,\text{k}/1\,\text{k})}{A}} \end{aligned}$$

$$G = \frac{10}{1 + \frac{10}{A}} \quad \text{..... (2)}$$

Step 4 of 9

Consider for $A = 10^3$ and $v_I = 1\ \text{V}$.

Substitute 10^3 for A in equation (2).

$$\begin{aligned} G &= \frac{10}{1 + \frac{10}{A}} \\ &= \frac{10}{1 + \frac{10}{10^3}} \\ &= \frac{10}{1 + 0.01} \\ &= 9.9 \end{aligned}$$

Step 5 of 9

Determine the Percentage deviation of closed loop gain from ideal loop gain.

$$\begin{aligned} \varepsilon &= \frac{9.9 - 10}{10} \times 100\% \\ &= -\frac{1}{100} \times 100\% \\ &= -1\% \end{aligned}$$

Therefore, the value of percentage deviation of closed loop gain from ideal loop gain is $\boxed{-1\%}$.

Determine the output voltage, v_O .

$$\begin{aligned} v_O &= G v_I \\ &= (9.9)(1\ \text{V}) \\ &= 9.9\ \text{V} \end{aligned}$$

Therefore, the output voltage, v_O is $\boxed{9.9\ \text{V}}$.

Determine the value of difference voltage of the op amp, v_{id} .

$$\begin{aligned} v_{id} &= \frac{v_O}{A} \\ &= \frac{9.9}{10^3} \\ &= 9.9\ \text{mV} \end{aligned}$$

Therefore, the value of difference voltage of the op amp, v_{id} is $\boxed{9.9\ \text{mV}}$.

Step 6 of 9

Consider for $A = 10^4$ and $v_I = 1\ \text{V}$.

Substitute 10^3 for A in equation (2).

$$\begin{aligned} G &= \frac{10}{1 + \frac{10}{A}} \\ &= \frac{10}{1 + \frac{10}{10^4}} \\ &= 9.99 \end{aligned}$$

Determine the Percentage deviation of closed loop gain from ideal loop gain.

$$\begin{aligned} \varepsilon &= \frac{9.99 - 10}{10} \times 100\% \\ &= -\frac{1}{1000} \times 100\% \\ &= -0.1\% \end{aligned}$$

Therefore, the value of percentage deviation of closed loop gain from ideal loop gain is $\boxed{-0.1\%}$.

Step 7 of 9

Determine the output voltage, v_O .

$$\begin{aligned} v_O &= G v_I \\ &= (9.99)(1\ \text{V}) \\ &= 9.99\ \text{V} \end{aligned}$$

Therefore, the output voltage, v_O is $\boxed{9.99\ \text{V}}$.

Determine the value of difference voltage of the op amp, v_{id} .

$$\begin{aligned} v_{id} &= \frac{v_O}{A} \\ &= \frac{9.99}{10^4} \\ &= 0.99\ \text{mV} \\ &\cong 1\ \text{mV} \end{aligned}$$

Therefore, the value of difference voltage of the op amp, v_{id} is $\boxed{1\ \text{mV}}$.

Step 8 of 9

Consider for $A = 10^5$ and $v_I = 1\ \text{V}$.

Substitute 10^3 for A in equation (2).

$$\begin{aligned} G &= \frac{10}{1 + \frac{10}{A}} \\ &= \frac{10}{1 + \frac{10}{10^5}} \\ &= 9.999 \end{aligned}$$

Step 9 of 9

Determine the Percentage deviation of closed loop gain from ideal loop gain.

$$\begin{aligned} \varepsilon &= \frac{9.999 - 10}{10} \times 100\% \\ &= -\frac{1}{10000} \times 100\% \\ &= -0.01\% \end{aligned}$$

Therefore, the value of percentage deviation of closed loop gain from ideal loop gain is $\boxed{-0.01\%}$.

Determine the output voltage, v_O .

$$\begin{aligned} v_O &= G v_I \\ &= (9.999)(1\ \text{V}) \\ &= 9.999\ \text{V} \end{aligned}$$

Therefore, the output voltage, v_O is $\boxed{9.999\ \text{V}}$.

Determine the value of difference voltage of the op amp, v_{id} .

$$\begin{aligned} v_{id} &= \frac{v_O}{A} \\ &= \frac{9.999}{10^5} \\ &= 0.0999\ \text{mV} \\ &\cong 0.1\ \text{mV} \end{aligned}$$

Therefore, the value of difference voltage of the op amp, v_{id} is $\boxed{0.1\ \text{mV}}$.

Step 1 of 4

Refer to the inverting closed-loop configuration in Figure 2.5 from the textbook.

(a)

Consider the closed-loop gain of inverting amplifier, $G = -1\ \text{V/V}$ and resistor R_1, as $R_1 = 10\ \text{k}\Omega$.

Determine the unknown resistor value.

Write the expression for the closed-loop gain of an ideal inverting op-amp.

$$G = -\frac{R_2}{R_1}$$

Substitute $-1\ \text{V/V}$ for G, and $10\ \text{k}\Omega$ for resistance R_1 in the equation.

$$G = -\frac{R_2}{R_1}$$

$$-1\ \text{V/V} = -\frac{R_2}{10\ \text{k}\Omega}$$

$$\begin{aligned} R_2 &= (-1)\left(-10\times10^3\right) \\ &= 10\times10^3 \\ &= 10\ \text{k}\Omega \end{aligned}$$

Thus, the value of resistor, R_1 is $\boxed{10\ \text{k}\Omega}$ and the resistor R_2 is $\boxed{10\ \text{k}\Omega}$.

Step 2 of 4

(b)

Consider that the closed-loop gain, $G = -2\ \text{V/V}$ and resistor R_1, as $R_1 = 10\ \text{k}\Omega$.

Determine the unknown resistor value.

Write the expression for the closed loop gain of an ideal op-amp.

$$G = -\frac{R_2}{R_1}$$

Substitute $-2\ \text{V/V}$ for G, and $10\ \text{k}\Omega$ for resistance R_1 in the equation.

$$G = -\frac{R_2}{R_1}$$

$$-2\ \text{V/V} = -\frac{R_2}{10\ \text{k}\Omega}$$

$$\begin{aligned} R_2 &= (-2)\left(-10\times10^3\right) \\ &= 20\times10^3 \\ &= 20\ \text{k}\Omega \end{aligned}$$

Thus, the value of resistor, R_1 is $\boxed{10\ \text{k}\Omega}$ and the resistor, R_2 is $\boxed{20\ \text{k}\Omega}$.

Step 3 of 4

(c)

Consider that the closed-loop gain, $G = -5\ \text{V/V}$ and resistor R_1, as $R_1 = 10\ \text{k}\Omega$.

Write the expression for the closed loop gain of an ideal op-amp.

$$G = -\frac{R_2}{R_1}$$

Substitute $-5\ \text{V/V}$ for G, and $10\ \text{k}\Omega$ for resistance R_1 in the equation.

$$G = -\frac{R_2}{R_1}$$

$$-5\ \text{V/V} = -\frac{R_2}{10\ \text{k}\Omega}$$

$$\begin{aligned} R_2 &= (-5)\left(-10\times10^3\right) \\ &= 50\times10^3 \\ &= 50\ \text{k}\Omega \end{aligned}$$

Thus, the value of resistor, R_1 is $\boxed{10\ \text{k}\Omega}$ and the resistor, R_2 is $\boxed{50\ \text{k}\Omega}$.

Step 4 of 4

(d)

Consider that the closed-loop gain, $G = -100\ \text{V/V}$ and resistor R_1, as $R_1 = 10\ \text{k}\Omega$.

Determine the unknown resistor value.

Write the expression for the closed loop gain of an ideal op-amp.

$$G = -\frac{R_2}{R_1}$$

Substitute $-100\ \text{V/V}$ for G, and $10\ \text{k}\Omega$ for resistance R_1 in the equation.

$$G = -\frac{R_2}{R_1}$$

$$-100\ \text{V/V} = -\frac{R_2}{10\ \text{k}\Omega}$$

$$\begin{aligned} R_2 &= (-100)\left(-10\times10^3\right) \\ &= 1000\times10^3 \\ &= 1\ \text{M}\Omega \end{aligned}$$

Thus, the value of resistor, R_1 is $\boxed{10\ \text{k}\Omega}$ and the resistor, R_2 is $\boxed{1\ \text{M}\Omega}$.

Step 1 of 10

Refer to Figure E2.13 in the textbook.

Determine the value of current, i_I .

Since, the op- amp is an ideal, it has infinite input resistance.

Therefore, the value of input resistance, i_I is $\boxed{0\ \text{A}}$.

Step 2 of 10

Determine the value of voltage, v_1 .

Due to the virtual short circuit between two terminals, the voltage at the inverting terminal and non-inverting terminal are equal.

$$\begin{aligned} v_1 &= v_I \\ &= 1\ \text{V} \end{aligned}$$

Therefore, the value of voltage v_1 is $\boxed{1\ \text{V}}$.

Step 3 of 10

Determine the value of current, i_1 .

$$\begin{aligned} i_1 &= \frac{v_1 - 0}{1\ \text{k}} \\ &= \frac{1}{1\ \text{k}} \\ &= 1\ \text{mA} \end{aligned}$$

Therefore, the value of current, i_1 is $\boxed{1\ \text{mA}}$.

Step 4 of 10

Determine the value of current, i_2 .

Since, no current flows in and out of op-amp at input terminal.

Therefore,

$$\begin{aligned} i_2 &= i_1 \\ &= 1\ \text{mA} \end{aligned}$$

Therefore, the value of current, i_2 is $\boxed{1\ \text{mA}}$.

Step 5 of 10

Determine the value of output voltage, v_O .

$$\begin{aligned} v_O &= i_1(1\ \text{k}) + i_2(9\ \text{k}) \\ &= (1\ \text{m})(1\ \text{k}) + (1\ \text{m})(9\ \text{k}) \\ &= 1 + 9 \\ &= 10\ \text{V} \end{aligned}$$

Determine the value of output voltage, v_O is $\boxed{10\ \text{V}}$.

Step 6 of 10

Determine the value of current, i_L .

$$\begin{aligned} i_L &= \frac{v_O - 0}{1 \times 10^3} \\ &= \frac{10 - 0}{1 \times 10^3} \\ &= 10\ \text{mA} \end{aligned}$$

Therefore, the value of current, i_L is $\boxed{10\ \text{mA}}$.

Step 7 of 10

Determine the value of output current, i_O .

Apply Kirchhoff's current law at outer node.

$$\begin{aligned} i_O &= i_2 + i_L \\ &= (1 + 10)\ \text{mA} \\ &= 11\ \text{mA} \end{aligned}$$

Therefore, the value of output current, i_O is $\boxed{11\ \text{mA}}$.

Step 8 of 10

Determine the value of voltage gain v_O / v_I .

$$\begin{aligned} \frac{v_O}{v_I} &= \frac{10}{1} \\ &= 10\ \text{V/V} \end{aligned}$$

Therefore, the value of voltage gain, v_O / v_I is $\boxed{10\ \text{V/V}}$.

Step 9 of 10

Determine the value of current gain, i_L / i_I .

$$\begin{aligned} \frac{i_L}{i_I} &= \frac{10}{0}\ \text{mA} \\ &= \infty \end{aligned}$$

Therefore, the value of current gain, i_L / i_I is $\boxed{\infty}$.

Step 10 of 10

Determine the value of power gain, P_L / P_I .

$$\begin{aligned} \frac{P_L}{P_I} &= \frac{v_O i_L}{v_I i_I} \\ &= \frac{10\left(10 \times 10^{-3}\right)}{1(0)} \\ &= \infty \end{aligned}$$

Therefore, the value of power gain, $\frac{P_L}{P_I}$ is $\boxed{\infty}$.

Step 1 of 3

2.013P

Refer to Figure 2.5 from the textbook for closed-loop configuration of inverting op-amp.

Consider the gain G of the op-amp is $-10\ \text{V/V}$.

The voltage gain of the inverting op-amp is,

$$G = -\frac{R_2}{R_1}$$

$$-10 = -\frac{R_2}{R_1}$$

$$\frac{R_2}{R_1} = 10$$

$$R_2 = 10R_1 \quad \text{..... (1)}$$

Step 2 of 3

Consider the total resistance is $110\text{k}\Omega$.

$$R_1 + R_2 = 110\ \text{k}\Omega \quad \text{..... (2)}$$

Substitute equation (1) for R_2 in equation (2).

$$R_1 + 10R_1 = 110\times10^3$$

$$11R_1 = 110\times10^3$$

$$R_1 = 10\times10^3$$

$$= 10\text{k}\Omega$$

Step 3 of 3

Substitute $10\text{k}\Omega$ for R_1 in equation (1).

$$R_2 = 10\left(10\times10^3\right)$$

$$= 100\times10^3$$

$$= 100\text{k}\Omega$$

Therefore, the determined values of resistor R_1 is $\boxed{10\text{k}\Omega}$ and resistor R_1 is $\boxed{100\text{k}\Omega}$.

The designed inverting op-amp is shown in Figure 1.

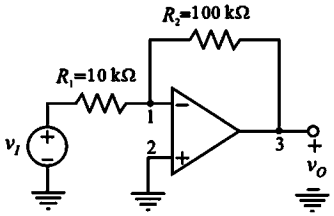


Figure 1

Step 1 of 4

(a)

Draw the transducer circuit when load is directly connected to source.

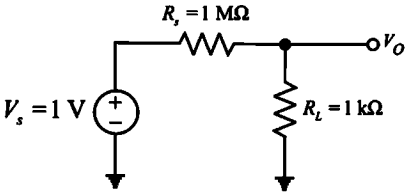


Figure 1

Step 2 of 4

Determine the load voltage, V_O.

$$V_O = \frac{V_s R_L}{R_s + R_L}$$
$$= \frac{1(10^3)}{10^6 + 10^3}$$
$$\cong 1\ \text{mV}$$

Therefore, the load voltage, V_O is $\boxed{1\ \text{mV}}$.

Step 3 of 4

(b)

Draw the transducer circuit when a voltage follower is connected between source and load.

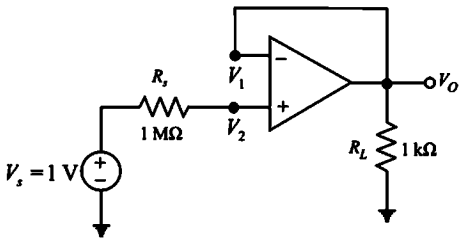


Figure 2

Step 4 of 4

Determine the load voltage, if the connection is through a unity gain voltage follower.

Since, input resistance is infinite because no current flows.

Therefore, $V_2 = V_s$

Due to the virtual short circuit, the voltage at the inverting terminal and non-inverting terminal are equal.

$V_1 = V_2$

$V_O = V_s$
$= 1\ \text{V}$

Therefore, the load voltage, V_O is $\boxed{1\ \text{V}}$.

Step 1 of 4

Refer to the Figure 2.5 in the textbook.

Gain of inverting op-amp is,

$$\frac{v_O}{v_i} = -\frac{R_2}{R_1}$$

Determine the gain of the op-amp in decibels.

$$\left.\frac{v_O}{v_i}\right|_{dB} = 20\log(|G|)$$
$$= 20\log\left(\left|-\frac{R_2}{R_1}\right|\right)$$

Step 2 of 4

Consider the voltage gain of the amplifier is $46\,\text{dB}$.

$$46 = 20\log\left(\frac{R_2}{R_1}\right)$$
$$\log\left(\frac{R_2}{R_1}\right) = 2.3$$
$$\frac{R_2}{R_1} = 199.53 \quad \text{..... (1)}$$

Step 3 of 4

Consider the largest possible resistors not greater than $1\,\text{M}\Omega$.

Consider the value of resistance, $R_2 \le 1\text{M}\Omega$.

From equation (1), $R_2 = 199.53R_1$.

Therefore,

$$R_2 \le 1\text{M}\Omega$$
$$199.53R_1 \le 1\text{M}\Omega$$

Thus, maximum possible value of R_2 resistance is $1\text{M}\Omega$.

Step 4 of 4

Determine the value of resistance, R_1.

$$R_1 = \frac{R_2}{199.53}$$

Substitute $1\,\text{M}\Omega$ for R_2 in the equation.

$$R_1 = \frac{1\times10^6}{199.53}$$
$$= 5.01\times10^3$$
$$= 5.01\ \text{k}\Omega$$

Determine the input resistance of simple inverting op-amp.

$$R_{in} = R_1$$
$$= 5.01\,\text{k}\Omega$$

Therefore, the input resistance of the inverting amplifier is $\boxed{5.01\,\text{k}\Omega}$.

Step 1 of 6

Refer to Figure 2.16 in the textbook.

(a)

Determine the value of the differential gain, A_d.

$$\begin{aligned} A_d &= \frac{R_2}{R_1} \\ &= \frac{200\times10^3}{2\times10^3} \\ &= 100 \text{ V/V} \end{aligned}$$

Therefore, the differential gain of amplifier, A_d is $\boxed{100 \text{ V/V}}$.

Step 2 of 6

Determine the value in dB.

$$\begin{aligned} A_d\big|_{dB} &= 20\log 100 \\ &= 20\log 10^2 \\ &= 20(2) \\ &= 40 \text{ dB} \end{aligned}$$

Step 3 of 6

(b)

Determine the value of the differential input resistance, R_{id}.

$$\begin{aligned} R_{id} &= 2R_1 \\ &= 2\left(2\times10^3\right) \\ &= 4\times10^3 \end{aligned}$$

Therefore, the differential input resistance, R_{id} is $\boxed{4 \text{ k}\Omega}$.

Determine the value of the differential output resistance, R_{id}.

It is known that, for an ideal op amp output resistance is zero.

Therefore, the differential output resistance, R_O is $\boxed{0 \ \Omega}$.

Step 4 of 6

(c)

Determine the value of common mode gain, A_{cm}.

Consider the resistance has 1% tolerance.

Determine the value of resistance R_1 and R_3.

$$\begin{aligned} R_1 &= 2 \text{ k}\Omega + (2 \text{ k}\Omega)(\pm1\%) \\ &= 2.02 \text{ k}\Omega \text{ or } 1.98 \text{ k}\Omega \end{aligned}$$

Determine the value of resistor R_3.

$$\begin{aligned} R_3 &= R_1 \\ &= 2.02 \text{ k}\Omega \text{ or } 1.98 \text{ k}\Omega \end{aligned}$$

Determine the value of resistance R_2 and R_4.

$$\begin{aligned} R_2 &= 200 \text{ k}\Omega + (200 \text{ k}\Omega)(\pm1\%) \\ &= 202 \text{ k}\Omega \text{ or } 198 \text{ k}\Omega \end{aligned}$$

Determine the value of resistor R_4.

$$\begin{aligned} R_4 &= R_2 \\ &= 202 \text{ k}\Omega \text{ or } 198 \text{ k}\Omega \end{aligned}$$

Step 5 of 6

Determine the common mode gain.

$$\begin{aligned} A_{cm} &\cong \frac{v_o}{v_{icm}} \\ &= \left(\frac{R_4}{R_4+R_3}\right)\left(1-\frac{R_2}{R_1}\frac{R_3}{R_4}\right) \end{aligned}$$

The worst case of common mode gain is its maximum possible value. In worst case, choose resistance value such that the equation make maximum.

Substitute $2.02 \text{ k}\Omega$ for R_1, $198 \text{ k}\Omega$ for R_2, $1.98 \text{ k}\Omega$ for R_3 and $202 \text{ k}\Omega$ for R_4 in the equation.

$$\begin{aligned} A_{cm} &= \left(\frac{202}{202+1.98}\right)\left(1-\frac{198}{2.02}\frac{1.98}{202}\right) \\ &= \left(\frac{202}{203.98}\right)(1-0.961) \\ &= 0.03 \\ &\cong 0.04 \text{ V/V} \end{aligned}$$

Therefore, the common mode gain is $\boxed{0.04 \text{ V/V}}$.

Step 6 of 6

Determine the corresponding value of CMRR.
Determine the common mode rejection ratio.

$$\begin{aligned} \text{CMRR} &= 20\log\left(\frac{A_d}{A_{cm}}\right) \\ &= 20\log\left(\frac{100}{0.04}\right) \\ &= 20(3.4) \\ &= 68 \text{ dB} \end{aligned}$$

Therefore, the common mode rejection ratio is $\boxed{68 \text{ dB}}$.

Step 1 of 4

2.015P

Refer to Figure 2.5 in the textbook.

Determine the gain of the circuit, G .

$$G = -\frac{R_2}{R_1}$$

Substitute $10\ \text{k}\Omega$ for R_1 and $100\ \text{k}\Omega$ for R_2 in the above expression.

$$G = -\frac{100\ \text{k}}{10\ \text{k}}$$
$$= -10\ \text{V/V}$$

Therefore, the gain of the op amp is $-10\ \text{V/V}$.

Step 2 of 4

The output voltage is,

$$v_O = Gv_I$$
$$= -10v_I$$

If $v_I = 0\ \text{V}$ then $v_O = 0\ \text{V}$,

$v_I = -1\ \text{V}$ then $v_O = 10\ \text{V}$.

Thus, the output square wave levels are $0\ \text{V}$ and $10\ \text{V}$.

Hence, the highest value, v_{OH} is $\boxed{10\ \text{V}}$ and the lowest value, v_{OL} is $\boxed{0\ \text{V}}$.

Step 3 of 4

Determine the average value of voltage, v_{Oavg} .

$$v_{Oavg} = \frac{0 + (10)}{2}$$
$$= 5\ \text{V}$$

Therefore, the average value of voltage, v_{Oavg} is $\boxed{5\ \text{V}}$.

Step 4 of 4

Tthe sketch of output voltage is shown in Figure 1.

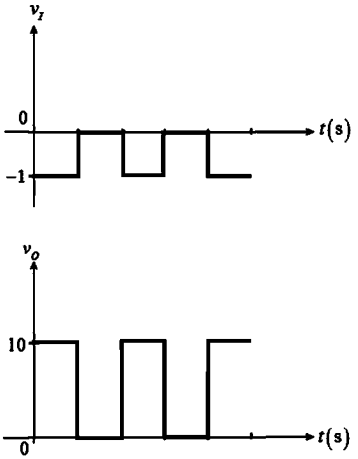


Figure 1

Step 1 of 3

2.016E

Refer to Figure 2.16 in the textbook.

Determine the input resistance of differential op-amp.

$$R_{id} = 2R_1$$
$$20\ \text{k}\Omega = 2R_1$$

$$R_1 = \frac{20\ \text{k}\Omega}{2}$$
$$= 10\ \text{k}\Omega$$

Therefore, the value of resistor R_1 is $\boxed{10\ \text{k}\Omega}$.

Step 2 of 3

For a differential amplifier, the common mode gain is zero.

Write the condition.

$$1 - \frac{R_2 R_3}{R_1 R_4} = 0$$

$$\frac{R_2}{R_1} = \frac{R_4}{R_3}$$

Consider the resistors $R_1 = R_3$ and $R_2 = R_4$.

Therefore, the value of resistor R_3 is $\boxed{10\ \text{k}\Omega}$.

Step 3 of 3

Determine the value of differential op amp gain.

$$A = \frac{R_2}{R_1}$$
$$R_2 = R_1 A$$
$$R_2 = \left(10 \times 10^3\right)(10)$$

$$R_2 = 100 \times 10^3$$
$$= 100\ \text{k}\Omega$$

Therefore, the value of resistor R_2 is $\boxed{100\ \text{k}\Omega}$.

Since, $R_2 = R_4$.

Therefore, the value of resistor R_4 is $\boxed{100\ \text{k}\Omega}$.

2.016P

We don't have the solution to this problem yet.

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Step 1 of 8

Refer to Figure 2.20 (b) in the textbook.

Determine the expression for voltage, v_{I1}.

$$\begin{aligned} v_{I1} &= v_{cm} - \frac{v_{id}}{2} \\ &= 5 - \frac{0.010\sin\omega t}{2} \\ &= 5 - 0.005\sin\omega t \end{aligned}$$

Therefore, the voltage v_{I1} is $\boxed{5 - 0.005\sin\omega t\ (\text{V})}$.

Step 2 of 8

Determine the expression for voltage, v_{I2}.

$$\begin{aligned} v_{I2} &= v_{cm} + \frac{v_{id}}{2} \\ &= 5 + \frac{0.010\sin\omega t}{2} \\ &= 5 + 0.005\sin\omega t \end{aligned}$$

Therefore, the voltage v_{I2} is $\boxed{5 + 0.005\sin\omega t\ (\text{V})}$.

Step 3 of 8

Due to virtual short circuit, the voltages at inverting terminals of both op-amps are,

$$v_-(\text{op-amp } A_1) = v_{I1}$$

Therefore, the voltage at the $v_-(\text{op-amp } A_1)$ is $\boxed{5 - 0.005\sin\omega t\ (\text{V})}$.

Therefore, the voltage at the $v_-(\text{op-amp } A_2)$ is $\boxed{5 + 0.005\sin\omega t\ (\text{V})}$.

Step 4 of 8

Determine the output voltage of A_1 operational amplifier.

$$\begin{aligned} v_{O1} &= v_{I1}\left(1 + \frac{R_2}{2R_1}\right) - v_{I2}\left(\frac{R_2}{2R_1}\right) \\ &= v_{I1} + (v_{I1} - v_{I2})\left(\frac{R_2}{2R_1}\right) \\ &= v_{I1} - v_{id}\left(\frac{R_2}{2R_1}\right) \\ &= (5 - 0.005\sin\omega t) - (0.010\sin\omega t)\left(\frac{0.5\times10^6}{1\times10^3}\right) \\ &= 5 - 0.005\sin\omega t - 5\sin\omega t \\ &= 5 - 5.005\sin\omega t \end{aligned}$$

Therefore, the output voltage of A_1 operational amplifier is $\boxed{5 - 5.005\sin\omega t\ (\text{V})}$.

Step 5 of 8

Determine the output voltage of A_2 operational amplifier:

$$\begin{aligned} v_{O2} &= v_{I2}\left(1 + \frac{R_2}{2R_1}\right) - v_{I1}\left(\frac{R_2}{2R_1}\right) \\ &= v_{I2} + (v_{I2} - v_{I1})\left(\frac{R_2}{2R_1}\right) \\ &= v_{I2} + v_{id}\left(\frac{R_2}{2R_1}\right) \end{aligned}$$

Substitute $5 + 0.005\sin\omega t$ for v_{I2}, $0.01\sin\omega t$ for v_{id}, $0.5\ \text{M}\Omega$ for R_2 and $1\ \text{k}\Omega$ for $2R_1$ in the equation.

$$\begin{aligned} v_{O2} &= v_{I2} + v_{id}\left(\frac{R_2}{2R_1}\right) \\ &= (5 + 0.005\sin\omega t) + (0.01\sin\omega t)\left(\frac{0.5\times10^6}{10^3}\right) \\ &= 5 + 0.005\sin\omega t + 5\sin\omega t \\ &= 5 + 5.005\sin\omega t \end{aligned}$$

Therefore, the output voltage of A_2 operational amplifier is $\boxed{5 + 5.005\sin\omega t\ \text{V}}$.

Step 6 of 8

Determine the input voltage at the non-inverting terminal of op-amp A_3.

$$\begin{aligned} v_+(A_3) &= v_{O2}\left(\frac{R_4}{R_3 + R_4}\right) \\ &= (5 + 5.005\sin\omega t)\left(\frac{10\text{k}}{10\text{k} + 10\text{k}}\right) \\ &= (5 + 5.005\sin\omega t)\left(\frac{1}{2}\right) \\ &= (2.5 + 2.5025\sin\omega t) \end{aligned}$$

Therefore, the input voltage at the non-inverting terminal of op-amp, $v_+(A_3)$ is

$\boxed{2.5 + 2.5025\sin\omega t\ (\text{V})}$.

Step 7 of 8

Due to virtual short voltage at the inverting terminal of op-amp A_3, the voltage at the inverting terminal and non-inverting terminal are equal.

$$v_-(A_3) = v_+(A_3)$$

Therefore, the input voltage at the non-inverting terminal of op-amp, $v_-(A_3)$ is

$\boxed{2.5 + 2.5025\sin\omega t\ (\text{V})}$.

Step 8 of 8

Determine the Output voltage, v_O.

$$\begin{aligned} v_O &= (v_{O2} - v_{O1})\left(\frac{R_4}{R_3}\right) \\ &= (10.01\sin\omega t)\left(\frac{10\text{k}}{10\text{k}}\right) \\ &= 10.01\sin\omega t \end{aligned}$$

Therefore, the Output voltage, v_O is $\boxed{10.01\sin\omega t\ (\text{V})}$.

Step 1 of 4

Consider the op-amp is ideal type.

Write the expression for closed loop gain of inverting op-amp.

$$G = \left| -\frac{R_2}{R_1} \right|$$
$$= \frac{R_2}{R_1}$$

Step 2 of 4

Consider a small tolerance is there in two resistors.

Determine the closed loop gain of the amplifier by consider $x\%$ tolerance.

$$G = \frac{R_2\left(1+\frac{x}{100}\right)}{R_1\left(1+\frac{x}{100}\right)}$$
$$= \frac{R_2}{R_1}\left(1 \pm \frac{2x}{100}\right)$$

Therefore, the tolerance on closed loop gain is $\boxed{2x}$.

Step 3 of 4

Consider the closed loop gain is $-100\,\mathrm{V/V}$.

Consider tolerance value on resistance is $x = 1$.

Here $x = 1$, so tolerance value in closed loop gain is changed by ± 2.

The range of closed loop gain is,

$$-100-2 \le G \le -100+2$$
$$-102 \le G \le -98$$

Therefore, the range of closed loop gain is $\boxed{-102 \le G \le -98}$.

Step 4 of 4

Determine the accurate range of closed loop gain.

$$-100\left(\frac{100+2x}{100-2x}\right) < G < -100\left(\frac{100-2x}{100+2x}\right)$$

Substitute 1 for x in the equation.

$$-100\left(\frac{100+2(1)}{100-2(1)}\right) < G < -100\left(\frac{100-2(1)}{100+2(1)}\right)$$
$$-100\left(\frac{102}{98}\right) < G < -100\left(\frac{98}{102}\right)$$
$$-104.08 < G < -96.07$$

Therefore, the accurate voltage gain range of the amplifier is $\boxed{-104.08 < G < -96.07}$.

Step 1 of 4

Draw the square waveform.

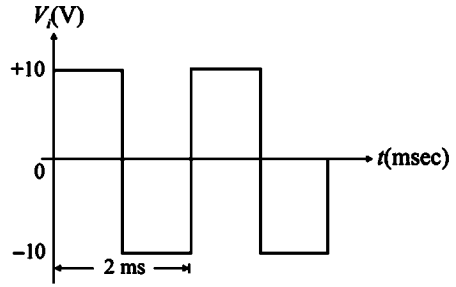


Figure 1

Step 2 of 4

The square wave is applied to the Miller integrator.

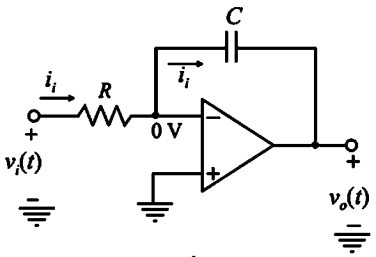


Figure 2

Step 3 of 4

Design an integrator to get a triangular waveform at the output with $20\,\mathrm{V}$ peak-to-peak voltage.

Consider in the first half-cycle, the amplitude of the input voltage is $10\,\mathrm{V}$.

Determine the current flowing through input resistor.

$$i_i = \frac{10}{R}$$

Consider the output voltage is changing from $+10\,\mathrm{V}$ to $-10\,\mathrm{V}$.

The total voltage change is $V = 20\,\mathrm{V}$.

The capacitor charges to $20\,\mathrm{V}$ in $\frac{T}{2}$.

Step 4 of 4

Write the expression for charge across the capacitor.

$$Q = CV$$

$$i_i t = CV$$

$$\left(\frac{10}{R}\right)\left(\frac{T}{2}\right) = C(20)$$

$$\left(\frac{10}{R}\right)\left(\frac{2\times10^{-3}}{2}\right) = C(20)$$

$$RC = \left(\frac{10}{20}\right)\left(\frac{2\times10^{-3}}{2}\right)$$

$$= 0.5\times10^{-3}$$

$$= 0.5\ \mathrm{ms}$$

Therefore, time constant of the Miller integrator is $\boxed{0.5\,\mathrm{ms}}$.

Step 1 of 7

Consider an ideal op amp with,

Resistors $5\ \text{k}\Omega$ and $15\ \text{k}\Omega$ is used create a $+5\ \text{V}$ supply from a $-15\ \text{V}$ reference.

Determine the gain of the inverting op-amp.

$$\text{Gain} = -\frac{5}{15}$$

$$\frac{v_o}{v_i} = -\frac{1}{3}$$

$$-\frac{R_2}{R_1} = -\frac{1}{3}$$

$$\frac{R_2}{R_1} = \frac{1}{3}$$

Therefore, $R_1 = 15\ \text{k}\Omega$ and $R_2 = 5\ \text{k}\Omega$.

Step 2 of 7

Sketch the ideal op amp.

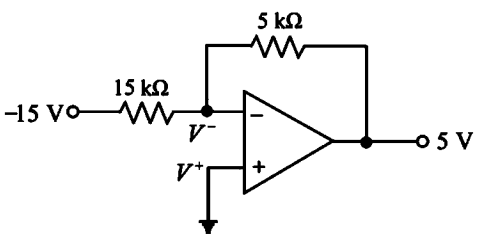


Figure 1

Step 3 of 7

From Figure 1, the voltage at the inverting terminal and non-inverting terminal are equal.

$$V^+ = V^-$$
$$= 0\ \text{V}$$

Determine the voltages at the ends of the $5\ \text{k}\Omega$ resistor.

From Figure 1, the voltage at the starting end of $5\ \text{k}\Omega$ resistor is $0\ \text{V}$.

The voltage at the ending end of $5\ \text{k}\Omega$ resistor is $5\ \text{V}$.

Therefore, the voltage at the starting end of $5\ \text{k}\Omega$ resistor is $\boxed{0\ \text{V}}$.

The voltage at the ending end of $5\ \text{k}\Omega$ resistor is $\boxed{5\ \text{V}}$.

Step 4 of 7

Consider the resistors are 1% resistors, whose actual values are bounded by nominal values by 1%.

Determine the resistance with $\pm 1\ \%$ tolerance to $15\ \text{k}\Omega$ variation.

$$\left(15 - \frac{15\times 1}{100}\right)\text{k}\Omega \text{ and } \left(15 + \frac{15\times 1}{100}\right)\text{k}\Omega$$
$$(15 - 0.15)\text{k}\Omega \text{ and } (15 + 0.15)\text{k}\Omega$$
$$14.85\ \text{k}\Omega \text{ and } 15.15\ \text{k}\Omega$$

Determine the resistance with $1\ \%$ tolerance to $5\ \text{k}\Omega$ variation.

$$\left(5 - \frac{5\times 1}{100}\right)\text{k}\Omega \text{ and } \left(5 + \frac{5\times 1}{100}\right)\text{k}\Omega$$
$$(5 - 0.05)\text{k}\Omega \text{ and } (5 + 0.05)\text{k}\Omega$$
$$4.95\ \text{k}\Omega \text{ and } 5.05\ \text{k}\Omega$$

Step 5 of 7

Determine the limits of the output voltage produced.

$$v_O = v_i\left(-\frac{R_2}{R_1}\right)$$
$$= 15\frac{R_2}{R_1}$$
$$= (15)\left(\frac{4.95}{15.15}\right) \le v_o \le (15)\left(\frac{5.05}{14.85}\right)$$
$$= 4.9\ \text{V} \le v_o \le 5.1\ \text{V}$$

Therefore, output voltage range is $\boxed{4.9\ \text{V and } 5.1\ \text{V}}$.

Step 6 of 7

Consider the tolerance to the input supply is 1%.

Determine the input voltage variation with 1% tolerance.

$$\left(15 - \frac{15\times 1}{100}\right) \text{ and } \left(15 + \frac{15\times 1}{100}\right)$$
$$(15 - 0.15) \text{ and } (15 + 0.15)$$
$$14.85\ \text{V and } 15.15\ \text{V}$$

Step 7 of 7

Determine the range of the output voltages.

$$\left(14.85\times\frac{4.95\ \text{k}\Omega}{15.15\ \text{k}\Omega}\right) \text{ to } \left(15.15\times\frac{5.05\ \text{k}\Omega}{14.85\ \text{k}\Omega}\right)$$
$$4.85\ \text{V to } 5.15\ \text{V}$$

Therefore, the range of output voltages is $\boxed{4.85\ \text{V} \le v_o \le 5.15\ \text{V}}$.

Step 1 of 7

2.019E

Draw the circuit diagram of integrator.

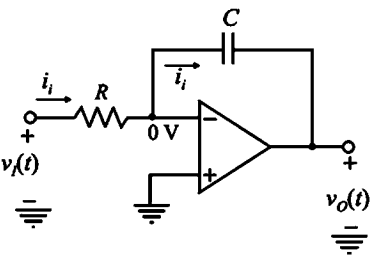


Figure 1

Step 2 of 7

Consider the op-amp is ideal one.

Consider in frequency domain, the resistor and capacitor are represented as R and $\frac{1}{sC}$.

Apply Kirchhoff's current law at the inverting terminal of the op-amp.

$$\frac{0-V_I(s)}{R}+\frac{0-V_O(s)}{\frac{1}{sC}}=0$$

$$sCV_O(s)=-\frac{V_I(s)}{R}$$

$$\frac{V_O(s)}{V_I(s)}=-\frac{1}{sCR} \quad \text{..... (1)}$$

Step 3 of 7

Write the expression or time constant of the integrator.

$$\tau = RC$$

Substitute $10\,\text{k}\Omega$ for R and $10^{-3}\,\text{s}$ for τ in the equation.

$$10^{-3}=\left(10\times10^{3}\right)C$$

$$C=\frac{10^{-3}}{10\times10^{3}}$$
$$=0.1\times10^{-6}$$

Therefore, the designed integrator values of resistor and capacitor are $\boxed{10\,\text{k}\Omega}$ and $\boxed{0.1\,\mu\text{F}}$ respectively.

Step 4 of 7

Substitute $j\omega$ for s in equation (1).

$$\frac{V_O(j\omega)}{V_I(j\omega)}=-\frac{1}{j\omega CR}$$

Magnitude of the integrator is,

$$\left|\frac{V_O(j\omega)}{V_I(j\omega)}\right|=\frac{1}{\omega CR}$$

Phase of the integrator is,

$$\phi=90^\circ \quad \text{(independent of frequency)}$$

Step 5 of 7

Consider the value of frequency is, $\omega=10\,\text{rad/s}$.

Determine the magnitude of the integrator.

$$\left|\frac{V_O(j\omega)}{V_I(j\omega)}\right|=\frac{1}{\omega CR}$$

Substitute $10\,\text{k}\Omega$ for R, $10\times10^{3}\,\text{F}$ for C and 10 for ω in the equation.

$$\left|\frac{V_O(j\omega)}{V_I(j\omega)}\right|=\frac{1}{(10)\left(0.1\times10^{-6}\right)\left(10\times10^{3}\right)}$$
$$=100\,\text{V/V}$$

Therefore, magnitude of the integrator at, $\omega=10\,\text{rad/s}$ is $\boxed{100\,\text{V/V}}$.

The phase angle of the integrator is $\boxed{\phi=90^\circ}$.

Step 6 of 7

Consider the value of frequency is, $\omega=1\,\text{rad/s}$.

Determine the magnitude of the integrator.

$$\left|\frac{V_O(j\omega)}{V_I(j\omega)}\right|=\frac{1}{\omega CR}$$

Substitute $10\,\text{k}\Omega$ for R, $10\times10^{3}\,\text{F}$ for C and 1 for ω in the equation.

$$\left|\frac{V_O(j\omega)}{V_I(j\omega)}\right|=\frac{1}{(1)\left(0.1\times10^{-6}\right)\left(10\times10^{3}\right)}$$
$$=1{,}000\,\text{V/V}$$

Therefore, magnitude of the integrator at $\omega=1\,\text{rad/s}$ is $\boxed{1{,}000\,\text{V/V}}$.

The phase angle of the integrator is $\boxed{\phi=90^\circ}$.

Step 7 of 7

Gain magnitude of the integrator is,

$$\left|\frac{V_O(j\omega)}{V_I(j\omega)}\right|=\frac{1}{\omega CR} \quad \text{..... (2)}$$

Need to determine frequency at which gain magnitude is unity.

Substitute 1 for $\left|\frac{V_O(j\omega)}{V_I(j\omega)}\right|$ in equation (2).

$$1=\frac{1}{\omega CR}$$

$$\omega=\frac{1}{CR}$$

Substitute $10\,\text{k}\Omega$ for R, $10\times10^{3}\,\text{F}$ for C and 10 for ω in the equation.

$$\omega=\frac{1}{\left(0.1\times10^{-6}\right)\left(10\times10^{3}\right)}$$
$$=\frac{1}{1\times10^{-3}}$$
$$=1{,}000\,\text{rad/s}$$

Therefore, the value of frequency at which the gain magnitude is unity is $\boxed{1{,}000\,\text{rad/s}}$.

2.019P

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Step 1 of 8

Draw the differentiator circuit.

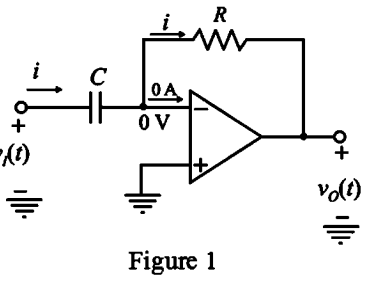


Figure 1

Step 2 of 8

Consider the op-amp is ideal one.

Consider in frequency domain, the resistor and capacitor are represented as R and $\frac{1}{sC}$.

Apply Kirchhoff's current law at the inverting terminal of the op-amp.

$$\frac{0-V_i(s)}{\frac{1}{sC}}+\frac{0-V_o(s)}{R}=0$$

$$-sCV_i(s)-\frac{V_o(s)}{R}=0$$

$$\frac{V_o(s)}{V_i(s)}=-sRC$$

Therefore, gain of the differentiator is,

$$\frac{V_o(s)}{V_i(s)}=-sRC \quad \text{..... (1)}$$

Step 3 of 8

Write the expression for time constant of the differentiator.

$$\tau = RC$$

Substitute $0.01\,\mu\text{F}$ for C and $10^{-2}\ \text{s}$ for τ in the equation.

$$10^{-2}=R\left(0.01\times10^{-6}\right)$$

$$\begin{aligned}R&=\frac{10^{-2}}{0.01\times10^{-6}}\\&=1\times10^{6}\\&=1\,\text{M}\Omega\end{aligned}$$

Therefore, designed resistor and capacitor values of the differentiator are $\boxed{1\,\text{M}\Omega}$ and $\boxed{0.01\,\mu\text{F}}$ respectively.

Step 4 of 8

Substitute $j\omega$ for s in equation (1).

$$\frac{V_o(j\omega)}{V_i(j\omega)}=-j\omega RC$$

Magnitude of the differentiator is,

$$\left|\frac{V_o(j\omega)}{V_i(j\omega)}\right|=\omega RC$$

Phase of the differentiator is,

$$\phi=-90^\circ$$

Step 5 of 8

Consider the value of frequency is, $\omega=10\,\text{rad/s}$.

Determine the magnitude of the differentiator.

$$\left|\frac{V_o(j\omega)}{V_i(j\omega)}\right|=\omega RC$$

Substitute $1\,\text{M}\Omega$ for R, $0.01\times10^{-6}\ \text{F}$ for C and 10 for ω in the equation.

$$\begin{aligned}\left|\frac{V_o(j\omega)}{V_i(j\omega)}\right|&=(10)\left(1\times10^{6}\right)\left(0.01\times10^{-6}\right)\\&=0.1\,\text{V/V}\end{aligned}$$

Therefore, magnitude at $\omega=10\,\text{rad/sec}$ is $\boxed{0.1\,\text{V/V}}$.

The phase is independent of frequency and is equal to $\boxed{\phi=-90^\circ}$.

Step 6 of 8

Consider the value of frequency is, $\omega=1{,}000\,\text{rad/s}$.

Determine the magnitude of the differentiator.

$$\left|\frac{V_o(j\omega)}{V_i(j\omega)}\right|=\omega RC$$

Step 7 of 8

Substitute $1\,\text{M}\Omega$ for R, $0.01\times10^{-6}\ \text{F}$ for C and 10 for ω in the equation.

$$\begin{aligned}\left|\frac{V_o(j\omega)}{V_i(j\omega)}\right|&=(1{,}000)\left(1\times10^{6}\right)\left(0.01\times10^{-6}\right)\\&=10\,\text{V/V}\end{aligned}$$

Therefore, magnitude at $\omega=1{,}000\,\text{rad/sec}$ is $\boxed{10\,\text{V/V}}$.

The phase is independent of frequency and is equal to $\boxed{\phi=-90^\circ}$.

Step 8 of 8

At high frequencies, input capacitor acts like short circuit and to limit the gain at high frequencies one resistor is added in series with capacitor.

At high frequencies, the gain of the amplifier is,

$$G=-\frac{R}{R_s}$$

Substitute 100 for G and $1\,\text{M}\Omega$ for R in the equation.

$$100=\frac{1\times10^{6}}{R_s}\quad\text{(neglect polarity of the gain)}$$

$$\begin{aligned}R_s&=\frac{1\times10^{6}}{100}\\&=10\times10^{3}\\&=10\,\text{k}\Omega\end{aligned}$$

Therefore, resistor connected in series with the capacitor is $\boxed{10\,\text{k}\Omega}$.

Step 1 of 3

The general expression for the closed-loop gain is,

$$G = \frac{v_0}{v_i} = -\frac{R_2}{R_1}$$

Consider that the closed-loop gain is $-200\ \text{V/V}$ and the input resistance is $1\ \text{k}\Omega$ for the amplifier.

Substitute the known values to find the value of R_2.

$$-200 = -\frac{R_2}{1000}$$
$$R_2 = 200\ \text{k}\Omega$$

The design of required inverting amplifier is,

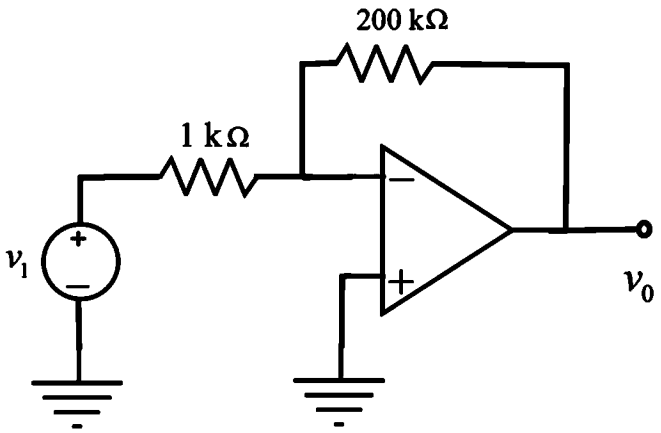


Figure 1

Step 2 of 3

(b)

Consider that the open-loop gain is $A = 5000\ \text{V/V}$.

The expression for the closed-loop gain is,

$$G = \frac{-\frac{R_2}{R_1}}{1+\frac{\left(1+\frac{R_2}{R_1}\right)}{A}}$$

Substitute the known values in the equation.

$$G = \frac{-\frac{200\times10^3}{1\times10^3}}{1+\frac{\left[1+\frac{200\times10^3}{1\times10^3}\right]}{5000}}$$
$$= \frac{-200}{1+\frac{201}{5000}}$$
$$= \frac{-200\times5000}{5000+201}$$
$$= -192.27\ \text{V/V}$$

Thus, the closed-loop gain is $\boxed{-192.27\ \text{V/V}}$.

Step 3 of 3

(c)

The value of R_e is to be selected in the following range to maintain closed-loop gain of $G = -200\ \text{V/V}$ and $R_1 = 1\ \text{k}\Omega$, $R_2 = 200\ \text{k}\Omega$.

$$\frac{R_e}{R_1} = \frac{A-G}{1+G}$$

Substitute the known values to find the value of R_e.

$$\frac{R_e}{1000} = \frac{5000-200}{1+200}$$
$$= \frac{4800}{201}$$
$$= 23.88$$
$$R_e \simeq 24\ \text{k}\Omega$$

Hence, the value of shunt resistor R_e is $\boxed{25\ \text{k}\Omega}$.

Step 1 of 2

2.021E

Refer to Figure 2.28 in the textbook.

From Figure 2.28 in the textbook, if the two input terminals of the op amp are tied together and connoted to ground, then a finite dc voltage exists at the output. If the

Op-amp has a high dc gain, and then the output will have either the positive or negative saturation level. The op amp output can be brought back to its ideal value of $\mathbf{0\ V}$ by connecting a dc voltage source of appropriate polarity and magnitude between the two input terminals of the op amp. It follows that the input offset voltage must be of equal magnitude and of opposite polarity to the voltage we applied externally.

Therefore,

$$\begin{aligned} v_{id} &= 0 - V_{os} \\ &= 0 - 5 \\ &= -5\ \text{V} \end{aligned}$$

Step 2 of 2

Draw the transfer characteristic v_o versus v_{id}.

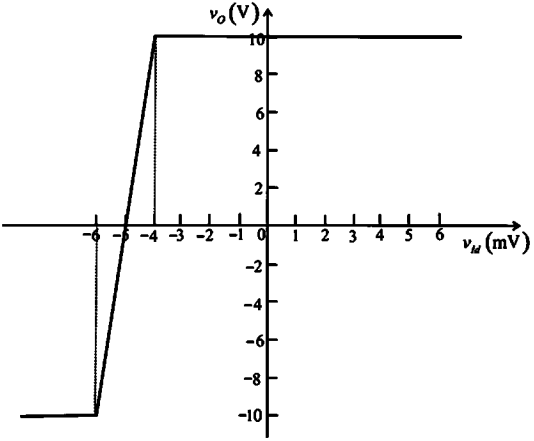


Figure 1

Therefore, the sketch of transfer characteristic v_o versus v_{id} is shown in Figure 1.

2.021P

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Step 1 of 4

(a)

Determine the peak sine wave input signal that can be applied without output clipping.

Calculate the output dc voltage due to V_{os}.

$$V_o = V_{os}\left(1+\frac{R_2}{R_1}\right)$$
$$= \left(3\times10^{-3}\right)\left(1+1000\right)$$
$$= \left(3\times10^{-3}\right)\left(1001\right)$$
$$= 3003\times10^{-3}\ \text{V}$$
$$V_o = 3\ \text{V}$$

Calculate the maximum output signal before saturation.

$$V_{max} = 10-3$$
$$= 7\ \text{V}$$

The peak sine wave input siganl is,

$$V_{inpeak} = \frac{V_{max}}{1000}$$

Substitute $7\ \text{mV}$ for V_{max} in the equation.

$$V_{inpeak} = \frac{7}{1000}$$
$$= 7\ \text{mV}$$

Therefore, the peak sine-wave input signal is $\boxed{7\ \text{mV}}$.

Step 2 of 4

(b)

(i)

Consider if $3\ \text{mV}$ input offset is nulled at room temperature.

Calculate the peak sine wave input signal at a constant temperature.

$$V_{inpeak} = \frac{V_{max}}{1000}$$

Substitute $10\ \text{mV}$ for V_{max} in the equation.

$$V_{inpeak} = \frac{10}{1000}$$
$$= 10\times10^{-3}$$
$$= 10\ \text{mV}$$

Therefore, the peak sine wave input signal is $\boxed{10\ \text{mV}}$.

Step 3 of 4

(ii)

Consider the circuit is to operate at a temperature in the range $0°\text{C to } 75°\text{C}$.

The temperature coefficient of V_{os} is $10\mu\text{V}/°\text{C}$.

Calculate the maximum change in temperature from the room temperature.

$$T_{max} = 75-25$$
$$= 50°\text{C}$$

Calculate the output offset voltage.

$$V_o = (50)\left(10\times10^{-6}\right)(1000)$$
$$= 0.5\ \text{V}$$

Thus, the maximum output voltage before saturation is,

$$V_{o(max)} = 10-0.5$$
$$= 9.5\ \text{V}$$

Step 4 of 4

Calculate the peak sine wave input signal at a range $0°\text{C to } 75°\text{C}$.

$$V_{inpeak} = \frac{V_{o(max)}}{1000}$$
$$= \frac{9.5}{1000}$$
$$= 9.5\times10^{-3}$$
$$= 9.5\ \text{mV}$$

Therefore, the peak sine wave input signal is $\boxed{9.5\ \text{mV}}$.

Step 1 of 7

Refer to Figure P2.22 in the textbook.

Re-draw the circuit diagram.

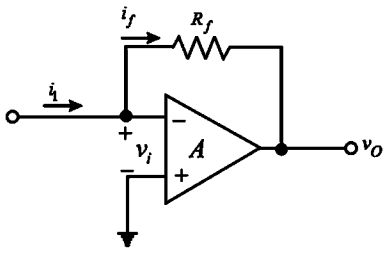


Figure 1

Step 2 of 7

(a)

Consider the op-amp gain, A is infinite.

Write the voltage between the two input terminals of the op-amp.

$$v_i = -\frac{v_O}{A}$$
$$= -\frac{v_O}{\infty}$$
$$= 0$$

Apply Kirchhoff's voltage Law in the circuit.

$$v_O = v_i - i_f R_f$$

Substitute 0 for v_i in the equation.

$$v_O = 0 - i_f R_f$$
$$= -i_f R_f$$
$$= -i_1 R_f \qquad (\text{Since } i_f = i_1)$$

Step 3 of 7

Write the expression for trans-resistance, R_m.

$$R_m = \frac{v_O}{i_1}$$

Substitute $-i_1 R_f$ for v_O in the equation.

$$R_m = \frac{-i_1 R_f}{i_1}$$
$$= -R_f$$

Therefore, the expression for the trans-resistance is $\boxed{R_m = -R_f}$.

Step 4 of 7

Determine the value of input resistance, R_i.

$$R_i = \frac{v_i}{i_i}$$

Substitute 0 for v_i in the equation.

$$R_i = \frac{0}{i_i}$$
$$= 0\ \Omega$$

Therefore, the value of input resistance is $\boxed{0\ \Omega}$.

Step 5 of 7

(b)

Consider the op-amp gain, A is finite.

Consider a certain input voltage exists at the input terminals.

Write the voltage between the two input terminals of the op-amp.

$$v_i = -\frac{v_O}{A}$$
$$v_O = -v_i A$$

Step 6 of 7

Apply Kirchhoff's voltage Law in the circuit.

$$v_O = v_i - i_f R_f$$

Substitute $-\frac{v_O}{A}$ for v_i in the equation.

$$v_O = -\frac{v_O}{A} - i_f R_f$$
$$v_O\left(1 + \frac{1}{A}\right) = -i_f R_f$$
$$v_O\left(1 + \frac{1}{A}\right) = -i_1 R_f \qquad (\text{Since } i_1 = i_f)$$
$$\frac{v_O}{i_i} = \frac{-R_f}{\left(1 + \frac{1}{A}\right)}$$

Substitute R_m for $\frac{v_O}{i_i}$ in the equation.

$$R_m = \frac{-R_f}{\left(1 + \frac{1}{A}\right)}$$

Therefore, the expression for the trans-resistance is $\boxed{R_m = \frac{-R_f}{\left(1 + \frac{1}{A}\right)}}$.

Step 7 of 7

Determine the expression of input resistance, R_i.

$$v_O = v_i - i_f R_f$$

Substitute $-v_i A$ for v_O in the equation.

$$-Av_i = v_i - i_i R_f$$
$$v_i(1 + A) = i_i R_f$$
$$\frac{v_i}{i_i} = \frac{R_f}{(1 + A)}$$

Substitute R_i for $\frac{v_i}{i_i}$ in the equation.

$$R_i = \frac{R_f}{(1 + A)}$$

Therefore, the expression of input resistance is $\boxed{R_i = \frac{R_f}{(1 + A)}}$.

Step 1 of 2

2.023E

Refer to Figure 2.31 in the textbook for a capacitive coupled inverting amplifier and the equivalent circuit for determining its dc output offset voltage.

(a)

Consider the input off set voltage, $V_{os} = 3\ \text{mV}$.

The output saturation levels of $\pm 10\text{V}$.

From the circuit, determine the dc offset voltage at the output.

$$V_o = V_{os}$$
$$= 3\ \text{mV}$$

Therefore, the dc offset voltage at the output is $\boxed{3\ \text{mV}}$.

Determine the peak sine wave signal that can be applied at the input without output clipping.

From Figure 2.31 (b) in the textbook, the circuit is a voltage follower.

So, the maximum peak sine wave signal can be applied is $10\ \text{mV}$.

Therefore, the peak sine wave signal that can be applied at the input without output clipping is $\boxed{10\ \text{mV}}$.

Therefore, there is no need for offset trimming.

Step 2 of 2

(b)

Determine the value of the coupling capacitor C_1 that will ensure that the gain will be greater than $57\ \text{dB}$ down to $100\ \text{Hz}$.

Whenever input is coupled through a capacitor it will acts as a high pass circuit.

For a high pass circuit, determine the 3dB frequency.

$$\omega_0 = \frac{1}{R_1 C_1}$$
$$2\pi f_0 = \frac{1}{R_1 C_1}$$

Substitute $100\ \text{Hz}$ for f_0 and $1\ \text{k}\Omega$ for R_1 in the equation.

$$2\pi(100) = \frac{1}{(10^3)C_1}$$

$$C_1 = \frac{1}{2\pi(100)(10^3)}$$
$$= 0.159\times 10^{-4}$$
$$= 1.59\times 10^{-6}$$
$$= 1.6\ \mu\text{F}$$

Therefore, the value of the coupling capacitor is $\boxed{1.59\ \mu\text{F}}$.

Step 1 of 6

Draw the circuit diagram of inverting amplifier.

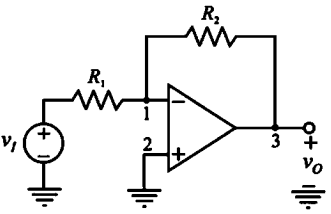


Figure 1

Step 2 of 6

Re-draw the circuit diagram with node voltages and branch currents.

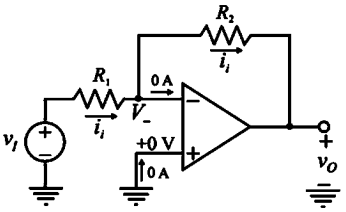


Figure 2

Step 3 of 6

From Figure 2, determine the open-loop gain of the amplifier.

$$A = \frac{v_o}{V_+ - V_-}$$

$$= \frac{v_O}{0 - V_-}$$

$$= \frac{-v_O}{V_-}$$

$$v_O = -AV_- \quad \text{..... (1)}$$

Step 4 of 6

Apply Kirchhoff's voltage law to the feedback resistor loop.

$$-V_- + i_f R_2 + v_O = 0$$

$$v_O = V_- - i_f R_2$$

Substitute $-AV_-$ for v_O in the equation.

$$-AV_- = V_- - i_f R_2$$

$$-AV_- - V_- = -i_f R_2$$

$$V_-(A+1) = i_f R_2$$

$$V_- = \frac{i_f R_2}{A+1} \quad \text{..... (2)}$$

Step 5 of 6

Apply Kirchhoff's voltage law to the input loop.

$$-v_I + i_i R_1 + V_- = 0$$

Substitute $\frac{i_f R_2}{A+1}$ for V_- in the equation.

$$-v_I + i_i R_1 + \frac{i_f R_2}{A+1} = 0$$

$$v_I = i_i R_1 + \frac{i_f R_2}{A+1}$$

$$v_I = i_i \left(R_1 + \frac{R_2}{A+1} \right)$$

$$\frac{v_I}{i_i} = R_1 + \frac{R_2}{A+1} \quad \text{..... (3)}$$

Step 6 of 6

Consider the input resistance of the op-amp amplifier is,

$$R_{in} = \frac{v_I}{i_i}$$

Substitute R_{in} for $\frac{v_I}{i_i}$ in equation (3).

$$R_{in} = R_1 + \frac{R_2}{A+1}$$

Therefore, the input resistance, R_{in} of the inverting amplifier is $\boxed{R_1 + \frac{R_2}{A+1}}$.

Step 1 of 4

2.024E

Draw the circuit diagram of inverting amplifier to analyze the effect of input bias current on output voltage.

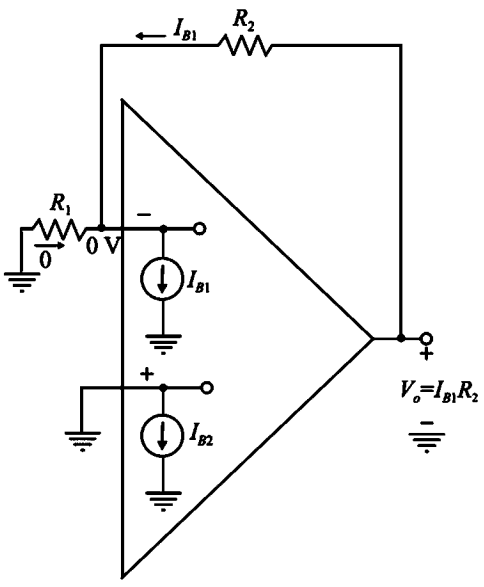


Figure 1

Step 2 of 4

From Figure 1, it is known that $I_B = I_{B1} = I_{B2}$.

Determine the output voltage.

$$V_o = I_{B1}R_2$$
$$= (100\times10^{-9})(1\times10^{6})$$
$$= 0.1\text{ V}$$

Therefore, the dc offset output voltage is, $\boxed{0.1\text{ V}}$.

This offset voltage is eliminated by introducing one resistor at non-inverting terminal as shown in Figure 2.

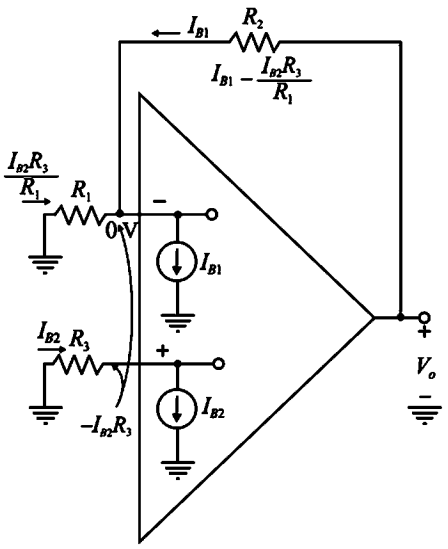


Figure 2

Step 3 of 4

Determine the value of resistor, R_3.

$$R_3 = \frac{R_2}{\left(1+\frac{R_2}{R_1}\right)}$$
$$= \frac{1\times10^{6}}{1+\frac{1\times10^{6}}{10\times10^{3}}}$$
$$= \frac{(1\times10^{6})(10\times10^{3})}{10\times10^{3}+1\times10^{6}}$$
$$= \frac{1\times10^{10}}{1.01\times10^{6}}$$
$$= 9900.9$$
$$= 9.9\text{ k}\Omega$$

Therefore, value of resistor R_3 is $\boxed{9.9\text{k}\Omega}$.

Step 4 of 4

Determine the new value of output voltage.

$$V_o = I_{OS}R_2$$
$$= (10\times10^{-9})(1\times10^{6})$$
$$= 10\times10^{-3}$$
$$= 0.01\text{ V}$$

Therefore, new output voltage is $\boxed{0.01\text{V}}$.

Step 1 of 8

Draw the diagram of inverting amplifier.

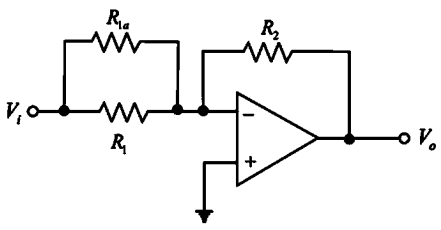


Figure 1

Step 2 of 8

Refer to equation (2.5) in the textbook.

Write the expression for closed loop gain of the inverting amplifier.

$$G=\frac{-R_2/R_1}{1+\frac{(1+R_2/R_1)}{A}}$$

From Figure 1, the resistor R_{1a} is in shunt with a resistor R_1. Determine the equivalent resistance.

$$\frac{1}{R_{eq}}=\frac{1}{R_1}+\frac{1}{R_{1a}}$$

$$R_{eq}=\frac{1}{\frac{1}{R_1}+\frac{1}{R_{1a}}}$$

Write the closed loop gain of inverting amplifier.

$$G=\frac{\left[-R_2\left(\frac{1}{R_1}+\frac{1}{R_{1a}}\right)\right]}{1+\frac{(1+R_2/R_1)}{A}}$$

$$\cong\frac{-\frac{R_2}{R_1}\left(1+\frac{R_1}{R_{1a}}\right)}{1+\frac{(1+R_2/R_1)}{A}}$$

Step 3 of 8

Consider the gain error is 0.1% .

Determine the value of open loop gain.

$$\varepsilon=\frac{(1+R_2/R_1)}{A}\times 100$$

$$A=\frac{(1+R_2/R_1)}{\varepsilon}\times 100$$

$$=\frac{(1+R_2/R_1)}{0.1}\times 100$$

$$=1000\left(1+\frac{R_2}{R_1}\right)$$

Therefore, the gain error is $\boxed{1000\left(1+\frac{R_2}{R_1}\right)}$.

Step 4 of 8

Consider the gain error is 1% .

Determine the value of open loop gain.

$$\varepsilon=\frac{(1+R_2/R_1)}{A}\times 100$$

$$A=\frac{(1+R_2/R_1)}{\varepsilon}\times 100$$

$$=\frac{(1+R_2/R_1)}{1}\times 100$$

$$=100\left(1+\frac{R_2}{R_1}\right)$$

Therefore, the gain error is $\boxed{100\left(1+\frac{R_2}{R_1}\right)}$.

Step 5 of 8

Consider the gain error is 10% .
Determine the value of open loop gain.

$$\varepsilon=\frac{(1+R_2/R_1)}{A}\times 100$$

$$A=\frac{(1+R_2/R_1)}{\varepsilon}\times 100$$

$$=\frac{(1+R_2/R_1)}{10}\times 100$$

$$=10\left(1+\frac{R_2}{R_1}\right)$$

Therefore, the gain error is $\boxed{10\left(1+\frac{R_2}{R_1}\right)}$.

Step 6 of 8

Determine the value of R_{1a} for a gain error 0.1% .

Write the formula for gain error.

$$\varepsilon=\frac{R_1}{R_{1a}}\times 100$$

$$R_{1a}=\frac{R_1}{\varepsilon}\times 100$$

$$=\frac{R_1}{0.1}\times 100$$

$$=1000R_1$$

Therefore, the value of R_{1a} is $\boxed{1000R_1}$.

Step 7 of 8

Determine the value of R_{1a} for a gain error 1% .

Write the formula for gain error.

$$\varepsilon=\frac{R_1}{R_{1a}}\times 100$$

$$R_{1a}=\frac{R_1}{\varepsilon}\times 100$$

$$=\frac{R_1}{1}\times 100$$

$$=100R_1$$

Therefore, the value of R_{1a} is $\boxed{100R_1}$.

Step 8 of 8

Determine the value of R_{1a} for a gain error 10% .

Write the formula for gain error.

$$\varepsilon=\frac{R_1}{R_{1a}}\times 100$$

$$R_{1a}=\frac{R_1}{\varepsilon}\times 100$$

$$=\frac{R_1}{10}\times 100$$

$$=10R_1$$

Therefore, the value of R_{1a} is $\boxed{10R_1}$.

Step 1 of 6

Draw the circuit diagram of Miller integrator.

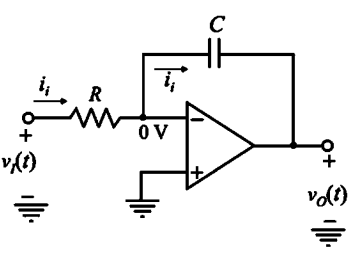


Figure 1

Step 2 of 6

Re-draw the Miller integrator circuit diagram with input offset voltage.

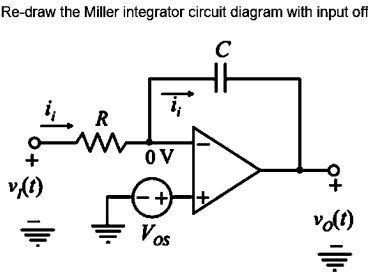


Figure 2

Step 3 of 6

(a)

From Figure 2, determine the output voltage of the integrator.

$$v_o = V_{os} + \frac{V_{os}}{CR}t$$

Substitute $12\,\text{V}$ for v_o, $2\,\text{mV}$ for V_{os}, $0.1\times10^{-6}\,\text{F}$ for C and $10\times10^{3}\,\Omega$ for R in the equation.

$$12 = 2\times10^{-3} + \frac{2\times10^{-3}}{(0.1\times10^{-6})(10\times10^{3})}t$$

$$12 = 2\times10^{-3} + 2t$$

$$2t = 12 - 0.002$$

$$2t = 11.998$$

$$t = \frac{11.998}{2}$$

$$= 5.999\ \text{s}$$

$$\cong 6\ \text{s}$$

Therefore, the output voltage reach its saturation value at $\boxed{6\text{s}}$.

Step 4 of 6

(b)

Consider a largest possible feedback resistor (R_F) is connected across capacitor.

Draw the circuit diagram.

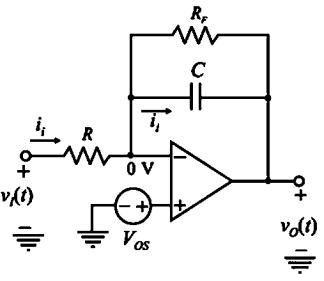


Figure 3

Step 5 of 6

Consider at least $\pm10\,\text{V}$ of output signal swing should remain.

Therefore, the output voltage must be limited to $2\,\text{V}$ $(\text{since } 12\,\text{V} - 10\,\text{V} = 2\,\text{V})$.

Write the expression for output voltage of the integrator.

$$v_o = V_{os}\left(1 + \frac{R_F}{R}\right)$$

$$2 = \left(2\times10^{-3}\right)\left(1 + \frac{R_F}{10\times10^{3}}\right)$$

$$1 + \frac{R_F}{10\times10^{3}} = \frac{2}{2\times10^{-3}}$$

$$1 + \frac{R_F}{10\times10^{3}} = 1{,}000$$

$$\frac{R_F}{10\times10^{3}} = 999$$

$$R_F = 9.99\times10^{6}$$

$$= 10\ \text{M}\Omega$$

Therefore, the feedback resistor is $\boxed{10\text{M}\Omega}$.

Step 6 of 6

Calculate the corner frequency of the integrator.

$$\omega_c = \frac{1}{CR_F}$$

$$2\pi f_c = \frac{1}{CR_F}$$

$$f_c = \frac{1}{2\pi CR_F}$$

Substitute $10\,\text{M}\Omega$ for R_F and $0.1\times10^{-6}\,\text{F}$ for C in the equation.

$$f_c = \frac{1}{2\pi(0.1\times10^{-6})(10\times10^{6})}$$

$$= \frac{1}{2\pi}$$

$$= 0.159$$

$$= 0.16\,\text{Hz}$$

Therefore, the corner frequency is $\boxed{0.16\,\text{Hz}}$.

Step 1 of 3

2.025P

Refer to Figure P2.25 in the textbook.

Re-draw the circuit diagram.

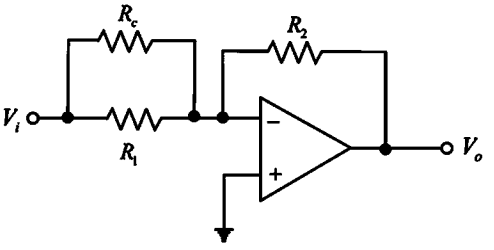


Figure 1

Step 2 of 3

Refer to equation (2.5) in the textbook.

Write the expression for closed loop gain of the inverting amplifier.

$$G = \frac{-R_2/R_1}{1+\frac{(1+R_2/R_1)}{A}}$$

Consider a resistor is added in parallel with R_1 .

Determine the equivalent resistance, R_1' .

$$R_1' = \frac{R_1 R_C}{R_1 + R_C}$$

Determine the gain due to the parallel combination.

$$G' = -\frac{\frac{R_2}{R_1'}}{1+\frac{\left(1+\frac{R_2}{R_1'}\right)}{A}}$$

$$G = \frac{A\left(\frac{R_2}{R_1'}\right)}{A+\left(1+\frac{R_2}{R_1'}\right)} \qquad \left(\text{Since } G' = G = \frac{R_2}{R_1}\right)$$

$$= \frac{AR_2}{AR_1' + R_1' + R_2}$$

$$= \frac{AR_2}{(A+1)R_1' + R_2}$$

Step 3 of 3

Simplify further,

$$GR_2 + GR_1'(1+A) = AR_2$$

$$GR_1'(1+A) = (A-G)R_2$$

$$GR_1'\frac{(1+A)}{R_2} = (A-G)$$

Substitute $\frac{R_1 R_C}{R_1 + R_C}$ for R_1' in the equation.

$$G\left(\frac{R_1 R_C}{R_1 + R_C}\right)\left(\frac{1+A}{R_2}\right) = A-G$$

$$\frac{R_2}{R_1}\left(\frac{R_1 R_C}{R_1 + R_C}\right)\left(\frac{1+A}{R_2}\right) = A-G$$

$$\left(\frac{R_C}{R_1 + R_C}\right)(1+A) = A-G$$

$$\frac{R_1 + R_C}{R_C} = \frac{1+A}{A-G}$$

$$\frac{R_1}{R_C} + 1 = \frac{1+A}{A-G}$$

$$\frac{R_1}{R_C} = \frac{1+A}{A-G} - 1$$

$$\frac{R_1}{R_C} = \frac{1+A-A+G}{A-G}$$

$$\frac{R_1}{R_C} = \frac{1+G}{A-G}$$

$$\frac{R_C}{R_1} = \frac{A-G}{1+G}$$

Therefore, the perfect compensation is achieved, when R_c is selected according to $\boxed{\frac{R_C}{R_1} = \frac{A-G}{1+G}}$.

Step 1 of 9

Convert the open loop dc gain in dB.

$$20 \log A_O = 106$$

$$\log A_O = \frac{106}{20}$$
$$= 5.3$$

$$A_O = 10^{5.3}$$
$$= 1.99 \times 10^5 \text{ V/V}$$

Step 2 of 9

Determine the break frequency, f_b.

$$f_b = \frac{f_t}{A_O}$$

Substitute 3×10^6 for f_t and 1.99×10^5 V/V for A_O in the equation.

$$f_b = \frac{3 \times 10^6}{1.99 \times 10^5}$$
$$= 15.07 \text{ Hz}$$

Therefore, the break frequency is $\boxed{15 \text{ Hz}}$.

Step 3 of 9

Write the expression for open loop gain.

$$A(j\omega) = \frac{A_O}{1 + j(\omega/\omega_b)}$$

$$A(jf) = \frac{A_O}{1 + j\left(\frac{f}{f_b}\right)} \quad \text{..... (1)}$$

Step 4 of 9

Determine the open loop gain at $f = f_b$.

Substitute f_b for f in equation (1).

$$A(jf_b) = \frac{A_O}{1 + j\left(\frac{f_b}{f_b}\right)}$$

$$A(jf_b) = \frac{A_O}{1 + j(1)}$$

$$|A(f_b)| = \frac{A_O}{\sqrt{2}}$$

Determine the open loop gain at $f = f_b$ in dB.

$$|A_{dB}(f_b)|_{dB} = 20 \log\left(\frac{A_O}{\sqrt{2}}\right)$$
$$= 20 \log A_O - 20 \log\left(\sqrt{2}\right)$$
$$= 106 - 3$$
$$= 103 \text{ dB}$$

Therefore, the open loop gain at $f = f_b$ is $\boxed{103 \text{ dB}}$.

Step 5 of 9

For $f \gg f_b$, Determine the open loop gain.

$$|A(j\omega)| = \frac{A_O \omega_b}{\omega}$$

The gain reaches to 0 dB at frequency denoted by ω_t.

$$\omega_t = A_O \omega_b$$

Therefore, $|A(f)| = \frac{f_t}{f}$

Determine the gain in dB.

$$|A(f)|_{dB} = 20 \log\left(\frac{f_t}{f}\right) \quad \text{..... (2)}$$

Step 6 of 9

Determine the open loop gain at $f = 300$ Hz.

Substitute 3 MHz for f_t and 300 Hz for f in equation (2).

$$|A(300)|_{dB} = 20 \log\left(\frac{3 \times 10^6}{300}\right)$$
$$= 20 \log\left(10^4\right)$$
$$= (20)(4)$$
$$= 80$$

Therefore, the open loop gain at $f = 300$ Hz is $\boxed{80 \text{ dB}}$.

Step 7 of 9

Determine the open loop gain at $f = 3$ kHz.

Substitute 3 MHz for f_t and 3 kHz for f in equation (2).

$$|A(300)|_{dB} = 20 \log\left(\frac{3 \times 10^6}{3 \times 10^3}\right)$$
$$= 20 \log\left(10^3\right)$$
$$= (20)(3)$$
$$= 60 \text{ dB}$$

Therefore, the open loop gain at $f = 3$ kHz is $\boxed{60 \text{ dB}}$.

Step 8 of 9

Determine the open loop gain at $f = 12$ kHz.

Substitute 3 MHz for f_t and 12 kHz for f in equation (2).

$$|A(300)|_{dB} = 20 \log\left(\frac{3 \times 10^6}{12 \times 10^3}\right)$$
$$= 20 \log(250)$$
$$= (20)(2.398)$$
$$\cong 48 \text{ dB}$$

Therefore, the open loop gain at $f = 12$ kHz is $\boxed{48 \text{ dB}}$.

Step 9 of 9

Determine the open loop gain at $f = 60$ kHz

Substitute 3 MHz for f_t and 60 kHz for f in equation (2).

$$|A(300)|_{dB} = 20 \log\left(\frac{3 \times 10^6}{60 \times 10^3}\right)$$
$$= 20 \log(50)$$
$$= 20\ (1.699)$$
$$\cong 34 \text{ dB}$$

Therefore, the open loop gain at $f = 60$ kHz is $\boxed{34 \text{ dB}}$.

Step 1 of 3

(a)

The expression for the gain error is,

$$\varepsilon \equiv \left| \frac{G - G_{nominal}}{G_{nominal}} \right|$$

Here, the expression for the nominal gain is,

$$G_{normal} = -\frac{R_2}{R_1}$$

the closed-loop gain of the system is,

$$G = \frac{G_{nominal}}{\frac{1+\frac{R_2}{R_1}}{A}}$$

$$1 + \frac{1+\frac{R_2}{R_1}}{A} = \frac{G_{nominal}}{G}$$

$$\frac{1+\frac{R_2}{R_1}}{A} = \frac{G_{nominal}}{G} - 1$$

$$A = \frac{1+\frac{R_2}{R_1}}{\frac{G_{nominal}}{G} - 1}$$

$$= \frac{\left(1+\frac{R_2}{R_1}\right)G}{G_{nominal} - G}$$

Hence, the open-loop gain is $\boxed{A = \frac{\left(1+\frac{R_2}{R_1}\right)G}{G_{nominal} - G}}$.

Step 2 of 3

(b)

Consider that the nominal gain, $G_{normal} = -100$ and $R_1 = 1\ \text{k}\Omega$.

Note that the gain error is $\varepsilon \le 10\%$. Therefore,

$$\varepsilon \equiv \left| \frac{G - G_{nominal}}{G_{nominal}} \right| \le 10\%$$

$$0.1 \ge \frac{G - G_{nominal}}{G_{nominal}}$$

$$0.1 \ge \frac{G + 100}{-100}$$

$$-10 \ge G + 100$$

$$G \le -110\ \text{V/V}$$

The closed-loop gain is $-110\ \text{V/V}$. Therefore,

$$G_{nominal} = -\frac{R_2}{R_1}$$

$$-100 = -\frac{R_2}{1000}$$

$$R_2 = 100\ \text{k}\Omega$$

Hence, the value of resistance R_2 is $\boxed{100\ \text{k}\Omega}$.

Step 3 of 3

The open-loop gain is,

$$A_{min} = \frac{\left(1+\frac{R_2}{R_1}\right)G}{G_{nominal} - G}$$

$$= \frac{\left(1+\frac{100}{1}\right)(-110)}{-100-(-110)}$$

$$= \frac{101(-110)}{-100+110}$$

$$= -1111\ \text{V/V}$$

Hence, the minimum value of open-loop gain A_{min} is $\boxed{-1111\ \text{V/V}}$.

Step 1 of 4

Consider the ac open-loop gain, $A = 40\ \text{dB}$.

Determine the value of gain, A .

$$20 \log A = 40\ \text{dB}$$
$$\log A = 2$$

$$A = 10^2$$
$$= 100$$

Determine the unity gain frequency, f_t .

$$|A| = \frac{f_t}{f}$$

$$100 = \frac{f_t}{10 \times 10^3}$$

$$f_t = 100\left(10 \times 10^3\right)$$
$$= 1 \times 10^6$$
$$= 1\ \text{MHz}$$

Therefore, the unity gain frequency is $\boxed{1\ \text{MHz}}$.

Step 2 of 4

Determine the break frequency f_b.

$$f_b = \frac{f_t}{A_o}$$
$$= \frac{10^6}{10^6}$$
$$= 1$$

Therefore, the break frequency is $\boxed{1\ \text{Hz}}$.

Step 3 of 4

Determine the gain band width product.

$$\text{Gain band width produt} = A_o f_b$$
$$= (10^6)(1)$$
$$= 1 \times 10^6$$

Therefore, the gain band width product is $\boxed{1\ \text{MHz}}$.

Step 4 of 4

Determine the open loop gain, when gain at $f = 1\ \text{kHz}$.

$$|A| = \frac{f_t}{f}$$
$$= \frac{10^6}{10^3}$$
$$= 10^3\ \text{V/V}$$

Determine the gain in dB.

$$20 \log A = 20 \log 10^3$$
$$= (20)(3)$$
$$= 60\ \text{dB}$$

Therefore, the open loop gain is $\boxed{60\ \text{dB}}$.

2.027P

We don't have the solution to this problem yet.

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2.028E

Determine the 3-dB frequency of the closed loop gain for non-inverting amplifier.

$$f_{3dB} = \frac{f_t}{1+\frac{R_2}{R_1}}$$

$$= \frac{f_t}{A} \qquad \left(\text{Since } A = 1+\frac{R_2}{R_1}\right)$$

Substitute $2\ \text{MHz}$ for f_t and 100 for A in the equation.

$$f_{3dB} = \frac{2\times10^6}{100}$$

$$= 2\times10^4$$

Therefore, 3-db frequency of closed loop gain for the non-inverting amplifier is $\boxed{20\ \text{kHz}}$.

2.028P

We don't have the solution to this problem yet.

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Step 1 of 3

2.029E

Refer to equation 2.58 in the textbook.

Refer to Figure 2.43 in the textbook.

Write the expression for voltage step response.

$$v_o(t) = V\left(1 - e^{-\omega_t t}\right)$$

The initial slope of this exponentially rising function is $(\omega_t V)$. Thus, as long as V is sufficiently small so that $\omega_t V \le SR$.

Slew rate is maximum rate of change of output voltage, it is possible when the time constant (τ) is minimum.

Therefore,

$$SR = V_{max}\omega_t$$

$$V_{max} = \frac{SR}{\omega_t}$$
$$= \frac{SR}{2\pi f_t}$$
$$= \frac{\left(\frac{1}{1\times10^{-6}}\right)}{2\pi\left(1\times10^{6}\right)}$$
$$= \frac{1}{2\pi}$$
$$= 0.159$$
$$\cong 0.16\ \text{V}$$

Therefore, the largest possible input voltage, V_{max} is $\boxed{0.16\ \text{V}}$.

Step 2 of 3

The time required to the output to reach from 10% to 90% is called rise time.

$$t_r = 2.2\tau$$
$$= \frac{2.2}{\omega_t}$$
$$= \frac{2.2}{2\pi f_t}$$
$$= \frac{2.2}{2\pi\left(1\times10^{6}\right)}$$
$$= 0.35\times10^{-6}\ \text{s}$$
$$= 0.35\ \mu\text{s}$$

Therefore, the rise time is $\boxed{0.35\ \mu\text{s}}$.

Step 3 of 3

Determine 10% to 90% rise time of the output wave form.

$$t_r = \frac{(0.9\times1.6) - (0.1\times1.6)}{\text{SR}}$$
$$= \frac{(0.9\times1.6) - (0.1\times1.6)}{10^{6}}$$
$$= \frac{1.44 - 0.16}{10^{6}}$$
$$= 1.28\times10^{-6}$$
$$= 1.28\ \mu\text{s}$$

Therefore, the rise time, t_r is $\boxed{1.28\ \mu\text{s}}$.

2.029P

We don't have the solution to this problem yet.

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Step 1 of 2

2.030E

Determine the full-power band width, f_M .

Write the relation between maximum frequency and maximum voltage.

$$\begin{aligned} f_M &= \frac{SR}{2\pi V_{O\max}} \\ &= \frac{1\times10^6}{(2\pi)(10)} \\ &= \frac{10^5}{6.283} \\ &= 15.9\times10^3 \end{aligned}$$

Therefore, the full-power band width is $\boxed{\mathbf{15.9\ kHz}}$.

Step 2 of 2

Determine the maximum possible amplitude without *SR* distortion.

$$\begin{aligned} V_{O\max} &= \frac{SR}{2\pi(5f_M)} \\ &= \frac{1\times10^6}{(2\pi)(5\times15.9\times10^3)} \\ &= \frac{10^3}{(2\pi)(5\times15.9)} \\ &= 2\ \mathrm{V_{peak}} \end{aligned}$$

Therefore, the maximum possible amplitude without *SR* distortion is $\boxed{\mathbf{2\ V}}$.

Step 1 of 4

2.030P

Refer to Figure P2.30 in the textbook.

Consider no current enters into op-amp.

Determine the voltage at the junction, v_x (which is the meeting place of the resistors, R_2, R_3 and R_4).

$v_x = 0 - i_1 R_2$ ..... (1)

Determine the current i_1 which flows through the resistor, R_1 .

$$i_1 = \frac{v_1}{R_1}$$

Substitute $\frac{v_1}{R_1}$ for i_1 in equation (1).

$$\begin{aligned} v_x &= 0 - i_1 R_2 \\ &= -\left(\frac{v_1}{R_1}\right) R_2 \\ &= -v_1\left(\frac{R_2}{R_1}\right) \end{aligned}$$

$$\frac{v_x}{v_1} = -\left(\frac{R_2}{R_1}\right)$$

Therefore, the expression for $\frac{v_x}{v_1}$ is $\boxed{\frac{v_x}{v_1} = -\left(\frac{R_2}{R_1}\right)}$.

Step 2 of 4

From Figure P2.30 in the textbook, the resistor R_2 and resistor R_3 are in parallel.

Apply voltage division rule and determine the voltage, v_x .

$$v_x = v_0 \frac{R_2 \parallel R_3}{R_2 \parallel R_3 + R_4}$$

$$\begin{aligned} \frac{v_x}{v_0} &= \frac{\frac{R_2 R_3}{R_2 + R_3}}{\frac{R_2 R_3}{R_2 + R_3} + R_4} \\ &= \frac{\frac{R_2 R_3}{R_2 + R_3}}{\frac{R_2 R_3 + R_4(R_2 + R_3)}{R_2 + R_3}} \\ &= \frac{R_2 R_3}{R_2 R_3 + R_4(R_2 + R_3)} \end{aligned}$$

Step 3 of 4

Take the reciprocal of the ratio.

$$\begin{aligned} \frac{v_0}{v_x} &= \frac{R_2 R_3 + R_4(R_2 + R_3)}{R_2 R_3} \\ &= \frac{R_2 R_3 + R_4 R_2 + R_4 R_3}{R_2 R_3} \\ &= 1 + \frac{R_4}{R_3} + \frac{R_4}{R_2} \end{aligned}$$

Therefore, the expression for $\frac{v_0}{v_x}$ is $\boxed{\frac{v_0}{v_x} = 1 + \frac{R_4}{R_3} + \frac{R_4}{R_2}}$.

Step 4 of 4

Divide the ratio $\frac{v_0}{v_x}$ by $\frac{v_1}{v_x}$ and determine the value of the ratio $\frac{v_0}{v_1}$.

$$\begin{aligned} \frac{\frac{v_0}{v_x}}{\frac{v_1}{v_x}} &= \frac{1 + \frac{R_4}{R_3} + \frac{R_4}{R_2}}{-\left(\frac{R_1}{R_2}\right)} \\ &= -\left(\frac{R_2}{R_1}\right)\left(1 + \frac{R_4}{R_3} + \frac{R_4}{R_2}\right) \end{aligned}$$

$$\frac{v_0}{v_1} = -\left(\frac{R_2}{R_1}\right)\left(1 + \frac{R_4}{R_3} + \frac{R_4}{R_2}\right)$$

Therefore, the expression for $\frac{v_0}{v_1}$ is $\boxed{\frac{v_0}{v_1} = -\left(\frac{R_2}{R_1}\right)\left(1 + \frac{R_4}{R_3} + \frac{R_4}{R_2}\right)}$.

Step 1 of 12

Refer to the circuit in Figure P2.31 in the textbook.

Consider that the op-amp is ideal. Therefore, the voltage at the inverting and non-inverting terminal is same and equal to zero volts.

The resistance network looking from the node 1 is,

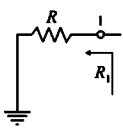


Figure 1

Calculate the resistance seen from node 1.

$$R_1 = R$$

Therefore, the resistance seen from the node 1, R_1 is, $\boxed{R}$.

Step 2 of 12

The resistance network looking from the node 2 is,

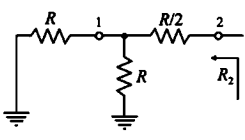


Figure 2

Calculate the resistance seen from node 2.

$$\begin{aligned} R_2 &= (R \| R) + \frac{R}{2} \\ &= \frac{(R)(R)}{R+R} + \frac{R}{2} \\ &= \frac{R}{2} + \frac{R}{2} \\ &= R \end{aligned}$$

Therefore, the resistance seen from the node 2, R_2 is, $\boxed{R}$.

Step 3 of 12

The resistance network looking from the node 3 is,

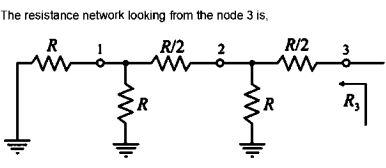


Figure 3

Step 4 of 12

Calculate the resistance seen from node 3.

$$\begin{aligned} R_3 &= \left[\left((R \| R) + \frac{R}{2}\right) \| R\right] + \frac{R}{2} \\ &= \left[\left(\frac{(R)(R)}{R+R} + \frac{R}{2}\right) \| R\right] + \frac{R}{2} \\ &= \left(\frac{R}{2} + \frac{R}{2}\right) \| R + \frac{R}{2} \\ &= (R \| R) + \frac{R}{2} \end{aligned}$$

Simplify further.

$$\begin{aligned} R_3 &= \frac{(R)(R)}{R+R} + \frac{R}{2} \\ &= \frac{R}{2} + \frac{R}{2} \\ &= R \end{aligned}$$

Therefore, the resistance seen from the node 3, R_3 is, $\boxed{R}$.

Step 5 of 12

The resistance network looking from the node 4 is,

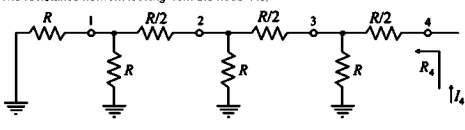


Figure 4

Step 6 of 12

Calculate the resistance seen from node 4.

$$\begin{aligned} R_4 &= \left(\left\{\left[\left((R \| R) + \frac{R}{2}\right) \| R\right] + \frac{R}{2}\right\} \| R\right) + \frac{R}{2} \\ &= \left(\left\{\left[\left(\frac{(R)(R)}{R+R} + \frac{R}{2}\right) \| R\right] + \frac{R}{2}\right\} \| R\right) + \frac{R}{2} \\ &= \left\{\left[\left(\left(\frac{R}{2} + \frac{R}{2}\right) \| R\right) + \frac{R}{2}\right] \| R\right\} + \frac{R}{2} \\ &= \left\{\left((R \| R) + \frac{R}{2}\right) \| R\right\} + \frac{R}{2} \end{aligned}$$

Simplify further.

$$\begin{aligned} R_4 &= \left\{\left(\frac{(R)(R)}{R+R} + \frac{R}{2}\right) \| R\right\} + \frac{R}{2} \\ &= \left[\left(\frac{R}{2} + \frac{R}{2}\right) \| R\right] + \frac{R}{2} \\ &= (R \| R) + \frac{R}{2} \\ &= \frac{(R)(R)}{R+R} + \frac{R}{2} \\ R_4 &= \frac{R}{2} + \frac{R}{2} \\ &= R \end{aligned}$$

Therefore, the resistance seen from the node 4, R_4 is, $\boxed{R}$.

Step 7 of 12

(b)

Apply Kirchhoff's current law at the inverting terminal of the op-amp.

$$-I + \frac{0 - V_1}{R} = 0$$

$$\frac{-V_1}{R} = I$$

$$V_1 = -IR \quad \text{..... (1)}$$

From the circuit in Figure P2.31, the voltage at node 1 is,

$$V_1 = -I_1 R \quad \text{..... (2)}$$

Equate equations (1) and (2).

$$-IR = -I_1 R$$

$$I_1 = I$$

Therefore, the current, I_1 in terms of input current is $\boxed{I}$.

Step 8 of 12

Apply Kirchhoff's current law at node 1.

$$-I - I_1 + \frac{V_1 - V_2}{\frac{R}{2}} = 0$$

$$-I - I + \frac{2(V_1 - V_2)}{R} = 0 \qquad (\text{since } I_1 = I)$$

$$\frac{-IR - V_2}{R} = I \quad (\text{since, } V_1 = -IR \text{ from equation (1)})$$

$$-I - \frac{V_2}{R} = I$$

$$V_2 = -2IR \quad \text{..... (3)}$$

From the circuit in Figure P2.31, the voltage at node 2 is,

$$V_2 = -I_2 R \quad \text{..... (4)}$$

Equate equations (3) and (4).

$$-I_2 R = -2IR$$

$$I_2 = 2I$$

Therefore, the current, I_2 in terms of input current is $\boxed{2I}$.

Step 9 of 12

Apply Kirchhoff's current law at node 2.

$$\frac{V_2 - V_1}{\frac{R}{2}} - I_2 + \frac{V_2 - V_3}{\frac{R}{2}} = 0$$

$$\frac{2(V_2 - V_1)}{R} - I_2 + \frac{2(V_2 - V_3)}{R} = 0$$

Substitute $-2IR$ for V_2, $-IR$ for V_1 and $2I$ for I_2 in the equation.

$$\frac{2(-2IR - (-IR))}{R} - 2I + \frac{2(-2IR - V_3)}{R} = 0$$

$$-2I + I - I - 2I - \frac{V_3}{R} = 0$$

$$-4I - \frac{V_3}{R} = 0$$

$$V_3 = -4IR \quad \text{..... (5)}$$

From the circuit in Figure P2.31, the voltage at node 3 is,

$$V_3 = -I_3 R \quad \text{..... (6)}$$

Equate equations (5) and (6).

$$-I_3 R = -4IR$$

$$I_3 = 4I$$

Therefore, the current, I_3 in terms of input current is $\boxed{4I}$.

Step 10 of 12

Apply Kirchhoff's current law at node 3.

$$\frac{V_3 - V_2}{\frac{R}{2}} - I_3 - I_4 = 0$$

$$\frac{2(V_3 - V_2)}{R} - I_3 - I_4 = 0$$

Substitute $-2IR$ for V_2, $-4IR$ for V_3 and $4I$ for I_3 in the equation.

$$\frac{2[-4IR - (-2IR)]}{R} - 4I - I_4 = 0$$

$$-8I + 4I - 4I - I_4 = 0$$

$$-8I - I_4 = 0$$

$$I_4 = -8I$$

Therefore, the current, I_4 in terms of input current is $\boxed{-8I}$.

Step 11 of 12

(c)

The voltage at node 1 in terms of IR is,

$$V_1 = -IR$$

Therefore, the voltage at node 1, V_1 in terms of IR is $\boxed{-IR}$.

The voltage at node 2 in terms of IR is,

$$V_2 = -2IR$$

Therefore, the voltage at node 2, V_2 in terms of IR is $\boxed{-2IR}$.

The voltage at node 3 in terms of IR is,

$$V_3 = -4IR$$

Therefore, the voltage at node 3, V_3 in terms of IR is $\boxed{-4IR}$.

Step 12 of 12

Apply Kirchhoff's current law at node 3.

$$\frac{V_3 - V_2}{\frac{R}{2}} + \frac{V_3}{R} + \frac{V_3 - V_4}{\frac{R}{2}} = 0$$

$$\frac{2(V_3 - V_2)}{R} + \frac{V_3}{R} + \frac{2(V_3 - V_4)}{R} = 0$$

$$-2V_2 + 5V_3 - 2V_4 = 0$$

Substitute $-2IR$ for V_2, and $-4IR$ for V_3 in the equation.

$$-2(-2IR) + 5(-4IR) - 2V_4 = 0$$

$$4IR - 20IR - 2V_4 = 0$$

$$2V_4 = -16IR$$

$$V_4 = -8IR$$

Therefore, the voltage at node 4, V_4 in terms of IR is $\boxed{-8IR}$.

Step 1 of 5

(a)

Refer to op-amp circuit in Figure P2.32 in the textbook.

The voltage at non-inverting terminal of the op-amp is zero. Since it is directly connected to ground.

The voltages at inverting and non-inverting terminals of the op-amp are zero for ideal op-amp.

Calculate the current through source.

$$I_1 = \frac{1\text{ V}}{10\times10^3\ \Omega}$$
$$= 0.1\text{ mA}$$

Therefore, the current, I_1 is $\boxed{0.1\text{ mA}}$.

Apply Kirchhoff current law at inverting terminal of the op-amp.

$$I_1 - I_2 = 0$$
$$I_2 = 0.1\text{ mA}$$

Therefore, the current, I_2 is $\boxed{0.1\text{ mA}}$.

Step 2 of 5

Calculate the voltage V_X using ohm's law.

$$I_2 = \frac{0 - V_X}{10\text{ k}\Omega}$$
$$-V_X = (10\text{ k}\Omega)(0.1\text{ mA})$$
$$V_X = -1\text{ V}$$

Therefore, the voltage, V_X is $\boxed{-1\text{ V}}$.

Calculate the current I_3.

$$I_3 = \frac{-V_X}{100\ \Omega}$$
$$= \frac{-(-1\text{ V})}{100\ \Omega}$$
$$= 10\text{ mA}$$

Therefore, the current, I_3 is $\boxed{10\text{ mA}}$.

Apply Kirchhoff's current law at node V_X.

$$I_L = I_2 + I_3$$
$$= 0.1\text{ mA} + 10\text{ mA}$$
$$= 10.1\text{ mA}$$

Therefore, the current, I_L is $\boxed{10.1\text{ mA}}$.

Step 3 of 5

(b)

Minimum output voltage is -13 V.

$$V_{O(\min)} = -13\text{ V}$$

Write the expression for current I_L.

$$I_L = \frac{V_X - V_O}{R_L}$$

$$R_{L(\max)} = \frac{V_X - V_{O(\min)}}{I_L}$$

Substitute -1 V for V_X, -13 V for $V_{O(\min)}$ and 10.1 mA for I_L in the equation.

$$R_{L(\max)} = \frac{-1\text{ V} - (-13\text{ V})}{10.1\text{ mA}}$$
$$= \frac{12\text{ V}}{10.1\text{ mA}}$$
$$= 1.188\text{ k}\Omega$$

Therefore, the maximum allowable load resistance, $R_{L(\max)}$ is $\boxed{1.188\text{ k}\Omega}$.

Step 4 of 5

(c)

The values of current, I_2 and current, I_3 are independent of load resistance. But current through the load resistor is sum of current, I_2 and current, I_3.

So, when the value of load resistance is varied, the current through the load resistor remains same.

Therefore, the load current, I_L is, $\boxed{10.1\text{ mA}}$.

Write the expression for current I_L.

$$I_L = \frac{V_x - V_O}{R_L}$$

$$V_O = V_X - I_L R_L \quad \text{..... (1)}$$

Substitute -1 V for V_X, 10.1 mA for I_L and $100\ \Omega$ for R_L in the equation.

$$V_O = -1\text{ V} - (10.1\text{ mA})(100\ \Omega)$$
$$= -1\text{ V} - 1.01\text{ V}$$
$$= -2.01\text{ V}$$

Step 5 of 5

Recall equation (1).

$$V_O = V_X - I_L R_L$$

Substitute -1 V for V_X, 10.1 mA for I_L and $1\text{ k}\Omega$ for R_L in the equation.

$$V_O = -1\text{ V} - (10.1\text{ mA})(1\text{ k}\Omega)$$
$$= -1\text{ V} - 10.1\text{ V}$$
$$= -11.1\text{ V}$$

Therefore, the range of output voltage is $\boxed{-11.1\text{ V}}$ to $\boxed{-2.01\text{ V}}$.

2.033P

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2.034P

We don't have the solution to this problem yet.

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2.035P

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2.036P

Refer to Figure 2.10 in the textbook.

The output of the weighted summer circuit is,

$$v_0 = -\left(\frac{R_f}{R_1}v_{01} + \frac{R_f}{R_2}v_{02} + \cdots + \frac{R_f}{R_n}v_{0n}\right)$$

The op-amp has three inputs using $10\ \text{k}\Omega$ resistors and feedback resistor of 50 k.

$$\begin{aligned} v_0 &= -\left(\frac{R_f}{R_1}v_{01} + \frac{R_f}{R_2}v_{02} + \frac{R_f}{R_3}v_{03}\right) \\ &= -\left(\frac{50}{10}v_{01} + \frac{50}{10}v_{02} + \frac{50}{10}v_{03}\right) \\ &= -5\left(v_{01} + v_{02} + v_{03}\right) \end{aligned}$$

The inputs v_{01} **and** v_{02} are tied to same input v_1 , and v_{03} is provided by v_2 .

Substitute v_1 for v_{01} **and** v_{02} and v_2 for v_{03} in output equation.

$$\begin{aligned} v_0 &= -5\left(v_1 + v_1\right) - 5v_2 \\ &= -10v_1 - 5v_2 \end{aligned}$$

Substitute 1 V for v_1 and $-1\ \text{V}$ for v_2 .

$$\begin{aligned} v_0 &= -10v_1 - 5v_2 \\ &= -10(1) - 5(-1) \\ &= -5\ \text{V} \end{aligned}$$

Hence, the output voltage v_0 is $\boxed{-5\ \text{V}}$.

Step 1 of 3

2.037P

Represent the op-amp circuit with the output voltage equation.

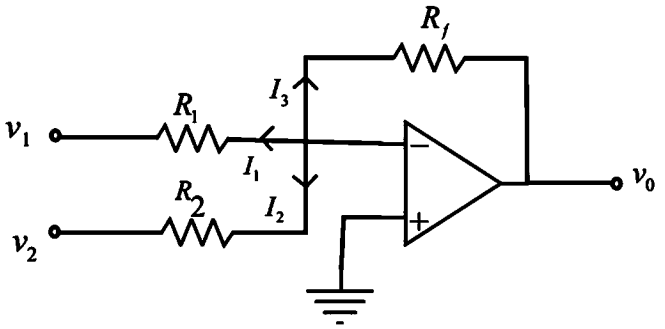


Figure 1

Step 2 of 3

The output voltage is,

$$v_0 = \frac{-R_f}{R_1} v_1 - \frac{R_f}{R_2} v_2$$

For $R_f = 2R_1$ and $R_f = \frac{R_2}{2}$, the output voltage becomes $v_0 = -\left(2v_1 + \frac{v_2}{2}\right)$.

The op-amp draws no current. Therefore,

$$I_1 + I_2 + I_3 = 0$$
$$I_3 = -(I_1 + I_2)$$
$$0 - \frac{v_0}{R_f} = -\left(\frac{0 - v_1}{R_1} + \frac{0 - v_2}{R_2}\right)$$
$$\frac{v_0}{R_f} = -\left(\frac{v_1}{R_1} + \frac{v_2}{R_2}\right)$$

Substitute 1 V for v_1 and v_2.

$$\frac{v_0}{R_f} = -\left[\frac{1}{R_1} + \frac{1}{R_2}\right]$$

Step 3 of 3

Consider that $\{I_1, I_2, I_3\} \le 50\ \mu A$.

Choose the value of resistance to satisfy these current conditions.

$$R_1 = 0.1\ M\Omega, R_2 = 0.4\ M\Omega, R_f = 0.2\ M\Omega$$

Find the values of the currents.

$$I_1 = \frac{1}{10^6}$$
$$= 10 \times 10^{-6}\ A$$
$$= 10\ \mu A$$

$$I_2 = \frac{1}{0.4 \times 10^6}$$
$$= 2.5 \times 10^{-6}\ A$$
$$= 2.5\ \mu A$$

$$I_3 = -(I_1 + I_2)$$
$$= -(10 + 2.5)\ \mu A$$
$$= 12.5\ \mu A$$

Therefore, the currents from each input signal source I_1, I_2, and I_3 values less than or equal to $50\ \mu A$ condition satisfied for possible values of resistances R_1, R_2, R_f are chosen.

Hence, $\boxed{R_1 = 0.1\ M\Omega, R_2 = 0.4\ M\Omega, \text{and } R_f = 0.2\ M\Omega}$.

2.038P

We don't have the solution to this problem yet.

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Step 1 of 10

2.039P

Use the circuit diagram of Figure 2.10 in the textbook. Note that both feedback resistor and input resistors are of same value that is 100 k.

(a)

The output is,

$$v_0 = -(v_1 + 2v_2 + 3v_3)$$

To implement the above v_0 the weighted summer circuit can be represented as shown below:

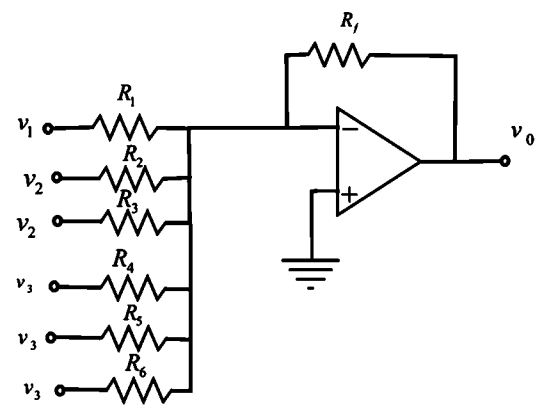


Figure 1

Step 2 of 10

Write the expression for the output voltage of the circuit.

$$v_0 = \frac{-R_f}{R_1}v_1 - \frac{R_f}{R_2}v_2 - \frac{R_f}{R_3}v_2 - \frac{R_f}{R_4}v_3 - \frac{R_f}{R_5}v_3 - \frac{R_f}{R_6}v_3$$

Substitute 100 k for all resistors.

$$v_0 = -v_1 - 2v_2 - 3v_3$$

Hence required output voltage is obtained.

The input resistances seen by the signal sources v_1, v_2, v_3 are as follows:

From source v_1 input resistance seen is 100 k.

From source v_2 input resistance seen is parallel combination of 100 k and 100 k

$$100\ \text{k}\Omega \| 100\ \text{k}\Omega = 50\ \text{k}\Omega$$

From source v_3 input resistance seen is parallel combination of 100 k, 100 k, and 100 k is,

$$100\ \text{k}\Omega \| 100\ \text{k}\Omega \| 100\ \text{k}\Omega = \frac{100}{3}\ \text{k}\Omega$$

Step 3 of 10

(b)

The output is,

$$v_0 = -(v_1 + v_2 + 2v_3 + 3v_4)$$

To implement the above v_0 the weighted summer circuit can be represented as shown below:

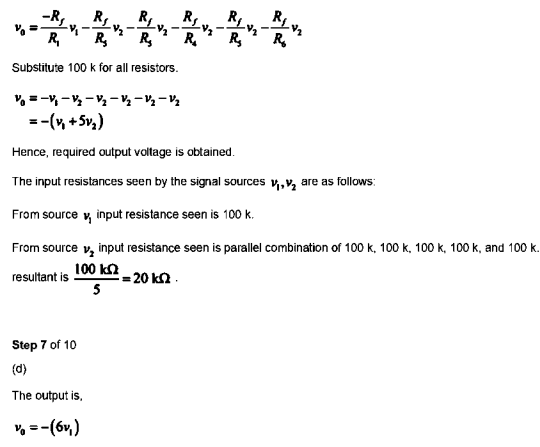


Figure 2

Step 4 of 10

Write the expression for the output voltage of the circuit.

$$v_0 = \frac{-R_f}{R_1}v_1 - \frac{R_f}{R_2}v_2 - \frac{R_f}{R_3}v_3 - \frac{R_f}{R_4}v_3 - \frac{R_f}{R_5}v_4 - \frac{R_f}{R_6}v_4$$

Substitute 100 k for all resistors.

$$\begin{aligned} v_0 &= -v_1 - v_2 - v_3 - v_3 - v_4 - v_4 \\ &= -(v_1 + v_2 + 2v_3 + 2v_4) \end{aligned}$$

Hence, required output voltage is obtained.

The input resistances seen by the signal sources v_1, v_2, v_3 are as follows:

From source v_1 input resistance seen is 100 k.

From source v_2 input resistance seen is 100 k.

From source v_3 input resistance seen is parallel combination of 100 k and 100 k is,

$$100\ \text{k}\Omega \| 100\ \text{k}\Omega = 50\ \text{k}\Omega$$

From source v_4 input resistance seen is parallel combination of 100 k and 100 k is,

$$100\ \text{k}\Omega \| 100\ \text{k}\Omega = 50\ \text{k}\Omega$$

Step 5 of 10

(c)

The output is,

$$v_0 = -(v_1 + 5v_2)$$

To implement the above v_0 the weighted summer circuit can be represented as shown below:

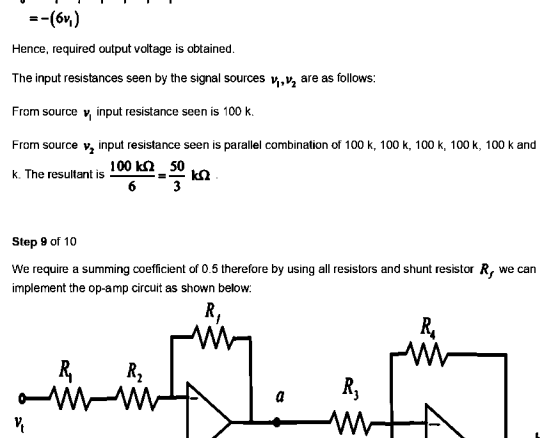


Figure 3

Step 6 of 10

Write the expression for the output voltage of the circuit.

$$v_0 = \frac{-R_f}{R_1}v_1 - \frac{R_f}{R_2}v_2 - \frac{R_f}{R_3}v_2 - \frac{R_f}{R_4}v_2 - \frac{R_f}{R_5}v_2 - \frac{R_f}{R_6}v_2$$

Substitute 100 k for all resistors.

$$\begin{aligned} v_0 &= -v_1 - v_2 - v_2 - v_2 - v_2 - v_2 \\ &= -(v_1 + 5v_2) \end{aligned}$$

Hence, required output voltage is obtained.

The input resistances seen by the signal sources v_1, v_2 are as follows:

From source v_1 input resistance seen is 100 k.

From source v_2 input resistance seen is parallel combination of 100 k, 100 k, 100 k, 100 k, and 100 k. The resultant is $\frac{100\ \text{k}\Omega}{5} = 20\ \text{k}\Omega$.

Step 7 of 10

(d)

The output is,

$$v_0 = -(6v_1)$$

To implement the above v_0 the weighted summer circuit can be represented as shown below:

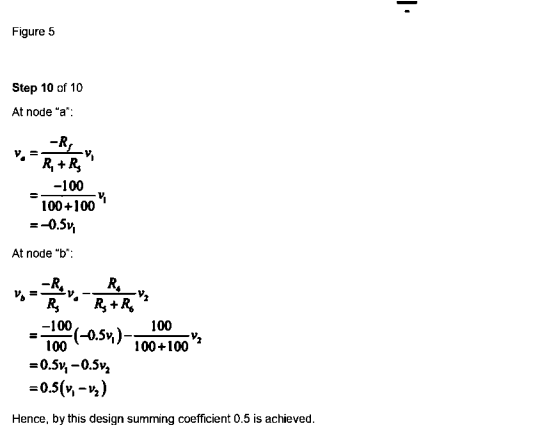


Figure 4

Step 8 of 10

Write the expression for the output voltage of the circuit.

$$v_0 = \frac{-R_f}{R_1}v_1 - \frac{R_f}{R_2}v_1 - \frac{R_f}{R_3}v_1 - \frac{R_f}{R_4}v_1 - \frac{R_f}{R_5}v_1 - \frac{R_f}{R_6}v_1$$

Substitute 100 k for all resistors.

$$\begin{aligned} v_0 &= -v_1 - v_1 - v_1 - v_1 - v_1 - v_1 \\ &= -(6v_1) \end{aligned}$$

Hence, required output voltage is obtained.

The input resistances seen by the signal sources v_1, v_2 are as follows:

From source v_1 input resistance seen is 100 k.

From source v_2 input resistance seen is parallel combination of 100 k, 100 k, 100 k, 100 k, 100 k and 100 k. The resultant is $\frac{100\ \text{k}\Omega}{6} = \frac{50}{3}\ \text{k}\Omega$.

Step 9 of 10

We require a summing coefficient of 0.5 therefore by using all resistors and shunt resistor R_f we can implement the op-amp circuit as shown below:

Figure 5

Step 10 of 10

At node "a":

$$\begin{aligned} v_a &= \frac{-R_f}{R_1 + R_2}v_1 \\ &= \frac{-100}{100+100}v_1 \\ &= -0.5v_1 \end{aligned}$$

At node "b":

$$\begin{aligned} v_b &= \frac{-R_4}{R_5}v_a - \frac{R_4}{R_5 + R_6}v_2 \\ &= \frac{-100}{100}(-0.5v_1) - \frac{100}{100+100}v_2 \\ &= 0.5v_1 - 0.5v_2 \\ &= 0.5(v_1 - v_2) \end{aligned}$$

Hence, by this design summing coefficient 0.5 is achieved.

Step 1 of 6

The output signal is,

$$v_O = 3\sin(\omega t) - 3$$

Take $v_1 = 3\sin(\omega t)$ and $v_2 = 1.5\ \text{V}$

The output voltage is,

$$v_O = v_1 - 2v_2 \quad \text{..... (1)}$$

One inverting amplifier and one two-input summing amplifier are required to meet the output voltage expression in equation (1).

The circuit is shown in Figure 1.

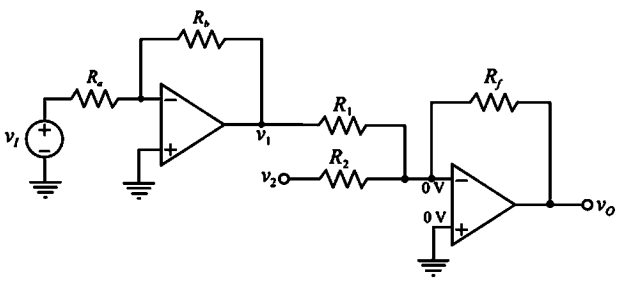


Figure 1

Step 2 of 6

Assume $R_a = 10\ \text{k}\Omega$ and $R_b = 10\ \text{k}\Omega$.

Apply Kirchhoff's current law at inverting terminal of the op-amp 1.

$$\frac{0 - v_1}{R_a} + \frac{0 - v_1}{R_b} = 0$$

$$\frac{0 - v_1}{10\ \text{k}\Omega} + \frac{0 - v_1}{10\ \text{k}\Omega} = 0$$

$$v_1 = -v_1$$
$$= -3\sin(\omega t)$$

Apply Kirchhoff's current law at inverting terminal of op-amp 2.

$$\frac{0 - v_1}{R_1} + \frac{0 - v_2}{R_2} + \frac{0 - v_O}{R_f} = 0$$

$$\frac{v_O}{R_f} = -\frac{v_1}{R_1} - \frac{v_2}{R_2}$$

$$v_O = -\left(\frac{R_f}{R_1}\right)v_1 - \left(\frac{R_f}{R_2}\right)v_2 \quad \text{..... (2)}$$

Step 3 of 6

Equate equation (1) and equation (2).

$$\frac{R_f}{R_1} = 1 \quad \text{..... (3)}$$

And,

$$\frac{R_f}{R_2} = 2 \quad \text{..... (4)}$$

Assume, $R_f = 20\ \text{k}\Omega$.

Recall equation (3).

$$\frac{R_f}{R_1} = 1$$

Substitute $20\ \text{k}\Omega$ for R_f in the equation.

$$\frac{20\ \text{k}\Omega}{R_1} = 1$$
$$R_1 = 20\ \text{k}\Omega$$

Recall equation (4).

$$\frac{R_f}{R_2} = 2$$

Substitute $20\ \text{k}\Omega$ for R_f in the equation.

$$\frac{20\ \text{k}\Omega}{R_2} = 2$$
$$R_2 = 10\ \text{k}\Omega$$

Step 4 of 6

The designed circuit with resultant resistor values is shown in Figure 2.

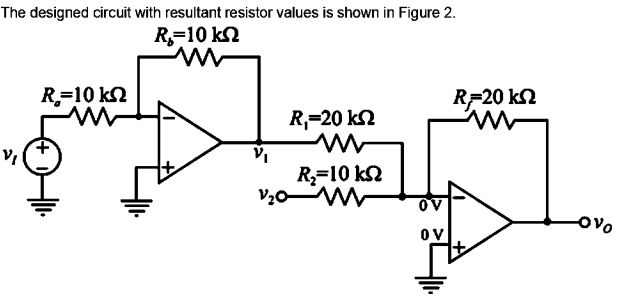


Figure 2

Step 5 of 6

Recall equation (2).

$$v_O = -\left(\frac{R_f}{R_1}\right)v_1 - \left(\frac{R_f}{R_2}\right)v_2$$

Substitute $20\ \text{k}\Omega$ for R_f, $20\ \text{k}\Omega$ for R_1, and $20\ \text{k}\Omega$ for R_1 in the equation.

$$v_O = -\left(\frac{20\ \text{k}\Omega}{20\ \text{k}\Omega}\right)v_1 - \left(\frac{20\ \text{k}\Omega}{10\ \text{k}\Omega}\right)v_2$$
$$= -v_1 - 2v_2$$

Substitute $-3\sin(\omega t)\ \text{V}$ for v_1 and $1.5\ \text{V}$ for v_2 in the equation.

$$v_O = -\left[-3\sin(\omega t)\right] - 2(1.5)$$
$$= 3\sin(\omega t) - 3$$

Step 6 of 6

The output voltage waveform is shown in Figure 3.

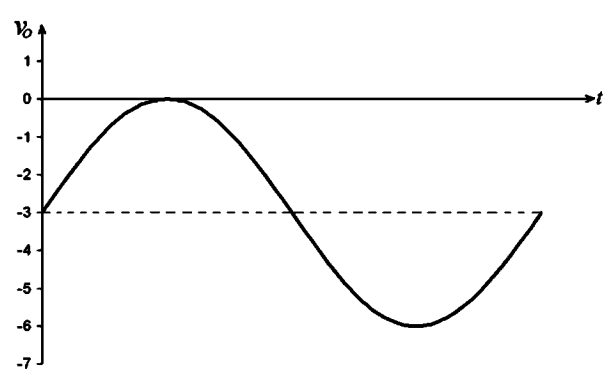


Figure 3

Step 1 of 4

The circuit to implement the summing function using two ideal op-amps and resistors is,

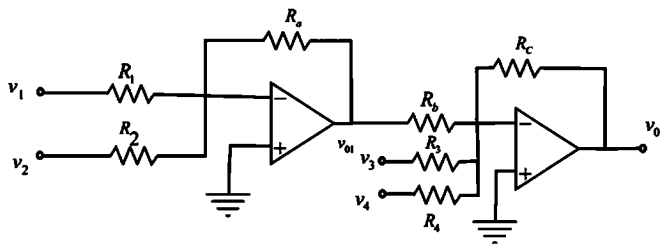


Figure 1

Step 2 of 4

The output of the weighted summer circuit is,

$$v_0 = -\left[\frac{R_C}{R_b}v_{01} + \frac{R_C}{R_3}v_3 + \frac{R_C}{R_4}v_4\right] \quad \text{..... (1)}$$

$$v_0 = -\left[\frac{R_a}{R_1}v_1 + \frac{R_a}{R_2}v_2\right] \quad \text{..... (2)}$$

The required output is,

$$v_0 = \left[v_1 + 2v_2 - 3v_3 - 5v_4\right] \quad \text{..... (3)}$$

Compare equations (1), (2) and (3).

The output, $v_{01} = -(v_1 + 2v_2)$ is possible only when $R_a = R_1$ and $R_2 = \frac{R_a}{2}$.

Step 3 of 4

The final output is,

$$v_0 = \left[v_1 + 2v_2 - 3v_3 - 5v_4\right]$$

Consider the following relations along with R_a, R_1 and R_2 equalities to produce first part of v_0 as $v_1 + 2v_2$:

$$R_C = R_b$$

$$R_a = R_1$$
$$= 2\ \text{k}\Omega$$

$$R_2 = 1\ \text{k}\Omega$$

The value of R_C also effects coefficients of v_3, v_4. So second part of v_0 is $-3v_3 - 5v_4$.

This expression is obtained under following conditions:

$$\frac{R_C}{R_3} = 3 \text{ and } R_C = 3R_3$$

$$\frac{R_C}{R_4} = 5 \text{ and } R_C = 5R_4$$

$$R_C = R_b$$
$$= 3\ \text{k}\Omega$$

$$R_3 = 1\ \text{k}\Omega$$

$$R_4 = \frac{3}{5}$$
$$= 0.6\ \text{k}\Omega$$

Step 4 of 4

Cross check these resistor values by using output voltage expression.

$$v_0 = -\left[\frac{R_C}{R_b}v_{01} + \frac{R_C}{R_3}v_3 + \frac{R_C}{R_4}v_4\right]$$
$$= -\left[(v_1 + 2v_2)\right] - \frac{3R_3}{R_3}v_3 - \frac{5R_4}{R_4}v_4$$
$$= v_1 + 2v_2 - 3v_3 - 5v_4$$

Therefore, for a required output voltage the possible choice of resistances is,

$$\boxed{\begin{array}{l} R_a = R_1 = 2\ \text{k}\Omega, \\ R_2 = 1\ \text{k}\Omega, \\ R_C = R_b = 3\ \text{k}\Omega, \\ R_3 = 1\ \text{k}\Omega, \text{ and } R_4 = 0.6\ \text{k}\Omega \end{array}}$$

Step 1 of 6

Draw the circuit of the difference amplifier using two op-amps.

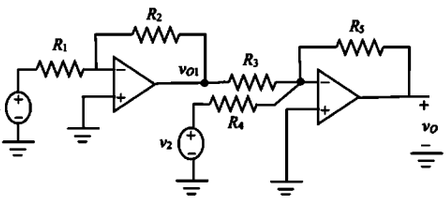


Figure 1

Step 2 of 6

Apply Kirchhoff's current law at inverting terminal of the op-amp 1.

$$\frac{0-v_1}{R_1}+\frac{0-v_{O1}}{R_2}=0$$

$$\frac{v_{O1}}{R_2}=\frac{-v_1}{R_1}$$

The output of the first op-amp circuit is,

$$v_{O1}=-\frac{R_2}{R_1}v_1$$

Step 3 of 6

The second amplifier is a summing amplifier, which provides the sum of the two voltages, the output of first op-amp circuit, v_{O1} and the non-inverting input voltage, v_2.

Apply Kirchhoff's current law at inverting terminal of the op-amp 2.

The output of the summing amplifier is,

$$\frac{0-v_{O1}}{R_3}+\frac{0-v_2}{R_4}+\frac{0-v_O}{R_5}=0$$

$$\frac{v_O}{R_5}=\frac{-v_{O1}}{R_3}+\frac{-v_2}{R_4}$$

$$v_O=-\frac{R_5}{R_3}v_{O1}-\frac{R_5}{R_4}v_2$$

Substitute $-\frac{R_2}{R_1}v_1$ for v_{O1}.

$$v_O=\left(-\frac{R_5}{R_3}\right)\left(-\frac{R_2}{R_1}v_1\right)-\frac{R_5}{R_4}v_2$$

$$=\left(\frac{R_5}{R_3}\right)\left(\frac{R_2}{R_1}\right)v_1-\left(\frac{R_5}{R_4}\right)v_2$$

The required gain of the difference between the two voltages is 100.

Thus,

$$\left(\frac{R_5}{R_3}\right)\left(\frac{R_2}{R_1}\right)=100$$

$$\frac{R_5}{R_4}=100 \quad \text{..... (1)}$$

Step 4 of 6

Select the resistance value, $R_4=100\ \text{k}\Omega$.

Calculate the value of resistance R_5.

$$R_5=(100)R_4$$
$$=(100)(100\ \text{k}\Omega)$$
$$=10^4\ \text{k}\Omega$$

Select the resistance values, $R_2=R_1=100\ \text{k}\Omega$

Thus, the ratio, $\frac{R_5}{R_3}$ from equation (1) is,

$$\frac{R_5}{R_3}=100$$

Substitute $10^4\ \text{k}\Omega$ for R_5 in the equation.

$$R_3=\left(\frac{R_5}{100}\right)$$
$$=\frac{100\ \text{k}\Omega}{100}$$
$$=1\ \text{k}\Omega$$

Step 5 of 6

The difference between the two input voltages is,

$$v_1-v_2=0.02\sin(2\pi\times1000t)\ \text{V}$$

For a gain of 100, the output voltage is,

$$v_O=100\times0.02\sin(2\pi\times1000t)$$
$$=2\sin(2\pi\times1000t)\ \text{V}$$

The magnitude of the output voltage is $\pm2\ \text{V}$.

Hence, the output voltage swing is within the limit $(\pm10\ \text{V})$.

Step 6 of 6

Draw the designed op-amp circuit that determines the difference between the two signals, v_1 and v_2.

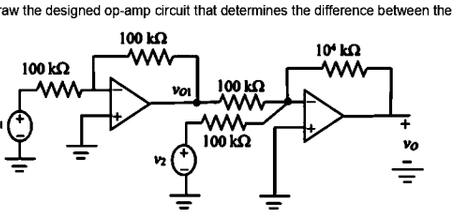


Figure 2

Thus, the required op-amp circuit is designed.

Step 1 of 2

2.043P

Refer to digital to analog converter shown in Figure P2.43 in the textbook.

The output voltage expressed for weighted summing amplifier is,

$$v_O = -\left(\frac{R_f}{R_0}v_0 + \frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \frac{R_f}{R_3}v_3\right)$$

Here,

$v_0 = 5a_0$,

$v_1 = 5a_1$,

$v_2 = 5a_2$,

$v_3 = 5a_3$

Substitute $5a_0$ for v_0, $5a_1$ for v_1, $5a_2$ for v_2, and $5a_3$ for v_3 in the eqution.

$$v_O = -\left(\frac{R_f}{R_0}(5a_0) + \frac{R_f}{R_1}(5a_1) + \frac{R_f}{R_2}(5a_2) + \frac{R_f}{R_3}(5a_3)\right)$$

Now substitute $80\ \text{k}\Omega$ for R_0, $40\ \text{k}\Omega$ for R_1, $20\ \text{k}\Omega$ for R_2, $10\ \text{k}\Omega$ for R_3 in the equation.

$$\begin{aligned} v_O &= -\left(\frac{R_f}{80\times10^3}(5a_0) + \frac{R_f}{40\times10^3}(5a_1) + \frac{R_f}{20\times10^3}(5a_2) + \frac{R_f}{10\times10^3}(5a_3)\right) \\ &= -\left(\frac{R_f}{80}(5a_0) + \frac{R_f}{40}(5a_1) + \frac{R_f}{20}(5a_2) + \frac{R_f}{10}(5a_3)\right) \quad (\text{since } R_f \text{ in k}\Omega) \\ &= -R_f\left(\frac{1}{16}a_0 + \frac{1}{8}a_1 + \frac{1}{4}a_2 + \frac{1}{2}a_3\right) \\ &= -\frac{R_f}{16}\left(a_0 + 2a_1 + 2^2a_2 + 2^3a_3\right) \end{aligned}$$

Therefore, the output voltage expression, v_O is $\boxed{-\frac{R_f}{16}\left[2^0a_0 + 2^1a_1 + 2^2a_2 + 2^3a_3\right]}$.

Step 2 of 2

The output voltage range, v_O is $0\ \text{V}$ to $-12\ \text{V}$.

The output voltage, v_O is $-12\ \text{V}$ when $a_3a_2a_1a_0 = 1111$.

Substitute 1 for a_0, a_1, a_2 and a_3 in the output voltage expression.

$$-12 = -\frac{R_f}{16}\left[2^0(1) + 2^1(1) + 2^2(1) + 2^3(1)\right]$$

$$12 = \frac{R_f}{16}\left[1 + 2 + 4 + 8\right]$$

$$\begin{aligned} R_f &= \frac{(12)(16)}{15} \\ &= 12.8\ \text{k}\Omega \quad (\text{since } R_f \text{ in k}\Omega) \end{aligned}$$

Therefore, the value of feedback resistor, R_f is $\boxed{12.8\ \text{k}\Omega}$.

2.044P

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Step 1 of 5

Refer to non-inverting amplifier in Figure 2.12 in the textbook.

The voltage gain of the non-inverting amplifier is,

$$\frac{v_o}{v_i} = 1 + \frac{R_2}{R_1}$$

Step 2 of 5

Case 1:

Take two $10\ \text{k}\Omega$ resistors in series at input and take one $10\ \text{k}\Omega$ resistor at feedback path, the modified circuit is shown in Figure 1.

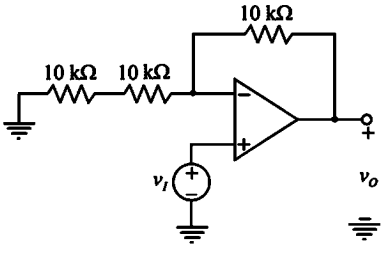


Figure 1

Step 3 of 5

Calculate the voltage gain for non-inverting amplifier shown in Figure 1.

$$\begin{aligned}\frac{v_o}{v_i} &= 1 + \frac{10\ \text{k}\Omega}{10\ \text{k}\Omega + 10\ \text{k}\Omega} \\ &= 1 + \frac{10\ \text{k}\Omega}{20\ \text{k}\Omega} \\ &= 1 + 0.5 \\ &= 1.5\ \text{V/V}\end{aligned}$$

Therefore, the voltage gain is $\boxed{1.5\ \text{V/V}}$.

Now short the one of the two series connected resistors in Figure 1.

Calculate the voltage gain.

$$\begin{aligned}\frac{v_o}{v_i} &= 1 + \frac{10\ \text{k}\Omega}{10\ \text{k}\Omega} \\ &= 1 + 1 \\ &= 2\ \text{V/V}\end{aligned}$$

By short circuiting one $10\ \text{k}\Omega$ resistor at input, the voltage gain obtained is $2\ \text{V/V}$.

Step 4 of 5

Case 2:

Take one $10\ \text{k}\Omega$ resistor at input and take two $10\ \text{k}\Omega$ resistors in parallel at feedback path, the modified circuit is shown in Figure 2.

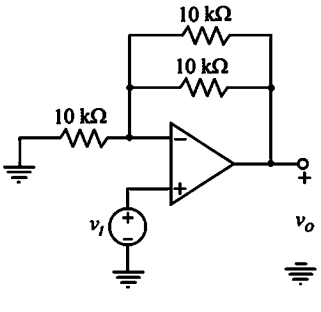


Figure 2

Step 5 of 5

Calculate the voltage gain for non-inverting amplifier shown in Figure 2.

$$\begin{aligned}\frac{v_o}{v_i} &= 1 + \frac{(10\ \text{k}\Omega) \| (10\ \text{k}\Omega)}{10\ \text{k}\Omega} \\ &= 1 + \frac{(10\ \text{k}\Omega)(10\ \text{k}\Omega)}{(10\ \text{k}\Omega + 10\ \text{k}\Omega)(10\ \text{k}\Omega)} \\ &= 1 + 0.5 \\ &= 1.5\ \text{V/V}\end{aligned}$$

Therefore, the voltage gain is $\boxed{1.5\ \text{V/V}}$.

Now short the one of the two parallel connected resistors in Figure 2.

Calculate the voltage gain.

$$\begin{aligned}\frac{v_o}{v_i} &= 1 + \frac{(10\ \text{k}\Omega) \| (0\ \Omega)}{10\ \text{k}\Omega} \\ &= 1 + \frac{0}{(10\ \text{k}\Omega + 0\ \text{k}\Omega)(10\ \text{k}\Omega)} \\ &= 1 + 0 \\ &= 1\ \text{V/V}\end{aligned}$$

By short circuiting one $10\text{k}\Omega$ resistor in feedback, we can get a gain of $1\ \text{V/V}$.

2.046P

Refer to op-am circuit in Figure P2.46 in the textbook.

Assume op-amp is ideal. For ideal op-amp, the voltages at the inverting and non-inverting terminals are same.

$$\begin{aligned} V_+ &= V_- \\ &= V \end{aligned}$$

The current flowing through resistor R is,

$$I = \frac{V}{R}$$

Current flowing through the inverting terminal is zero. Therefore, the current I is flowing through the ammeter.

Substitute $100\ \mu\text{A}$ for I and $10\ \text{V}$ for V in the equation.

$$100\ \mu\text{A} = \frac{10\ \text{V}}{R}$$

$$\begin{aligned} R &= \frac{10\ \text{V}}{100\ \mu\text{A}} \\ &= 10^5\ \Omega \\ &= 100\ \text{k}\Omega \end{aligned}$$

Therefore, input resistance, R of the amplifier is $\boxed{100\ \text{k}\Omega}$.

2.047P

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2.048P

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Step 1 of 2

2.049P

Refer to circuit diagram in Figure P2.49 in the textbook.

Assume op-amp is ideal one. Voltage at inverting and non-inverting thermals is same.

$v_+ = v_-$

Current flowing through inverting and non-inverting terminals is zero.

Apply Kirchhoff's current law at non-inverting terminal of the op-amp.

$$\frac{v_+ - v_I}{R_3} + \frac{v_+}{R_4} = 0$$

$$\left(\frac{1}{R_3} + \frac{1}{R_4}\right) v_+ = \frac{v_I}{R_3}$$

$$\left(\frac{R_3 + R_4}{R_3 R_4}\right) v_+ = \frac{v_I}{R_3}$$

$$v_+ = \frac{R_4}{R_3 + R_4} v_I$$

$$v_- = \left(\frac{R_4}{R_3 + R_4}\right) v_I \qquad (\text{since } v_- = v_+) \ldots\ldots (1)$$

Step 2 of 2

Apply Kirchhoff's current law at inverting terminal of the op-amp.

$$\frac{v_- - 0}{R_1} + \frac{v_- - v_O}{R_2} = 0$$

$$\frac{v_-}{R_1} + \frac{v_-}{R_2} - \frac{v_O}{R_2} = 0$$

$$\frac{v_O}{R_2} = v_- \left(\frac{1}{R_1} + \frac{1}{R_2}\right)$$

$$v_O = v_- \left(\frac{R_2}{R_1} + 1\right)$$

Substitute $\left(\frac{R_4}{R_3 + R_4}\right) v_I$ for v_- in the equation.

$$v_O = \left(\frac{R_4}{R_3 + R_4}\right) v_I \left(\frac{R_2}{R_1} + 1\right)$$

$$= \left(1 + \frac{R_2}{R_1}\right)\left(\frac{R_4}{R_3 + R_4}\right) v_I$$

Or,

$$v_O = \frac{1 + \frac{R_2}{R_1}}{1 + \frac{R_3}{R_4}}$$

Therefore, expression for voltage gain, $\frac{v_O}{v_I}$ of the amplifier is $\boxed{\left(1 + \frac{R_2}{R_1}\right)\left(\frac{R_4}{R_3 + R_4}\right)}$ or $\boxed{\frac{1 + \frac{R_2}{R_1}}{1 + \frac{R_3}{R_4}}}$.

2.050P

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Step 1 of 5

2.051P

Refer to Figure P2.51 in the textbook.

Write the relation of the gain of the operational amplifier, $\frac{V_0}{V_i}$ with the connected resistors.

$$\frac{V_0}{V_i} = 1 + \frac{R_2}{R_1} \quad \text{..... (1)}$$

Here,

R_2 **and** R_1 are the resistors connected to the output and input of the terminals of an operational amplifier.

From Figure P2.51 in the textbook, write the expression for output resistance.

$$R_2 = 1 - x$$

Write the input resistance.

$$R_1 = x$$

Substitute $1 - x$ for R_2 and x for R_1 in equation (1).

$$\begin{aligned}\frac{V_0}{V_i} &= 1 + \frac{R_2}{R_1} \\ &= 1 + \frac{1 - x}{x} \\ &= 1 + \frac{1}{x} - 1 \\ &= \frac{1}{x}\end{aligned}$$

Step 2 of 5

Assuming the operational amplifier to be ideal, the value of x lies between 0 and 1, that is, $0 \le x \le 1$.

The range of the gain of the operational amplifier is,

$$\frac{1}{0} \ge \frac{1}{x} \ge 1$$
$$\infty \ge A_v \ge 1$$

Thus the range of the gain of the operational amplifier is $\boxed{1 \le A_v \le \infty}$.

Step 3 of 5

Consider the circuit diagram.

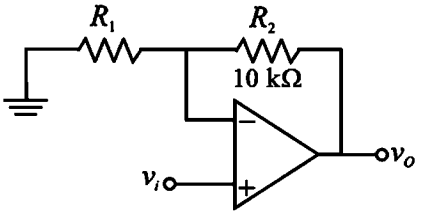


Figure 1

Step 4 of 5

Write the gain of the amplifier.

$$A_v = 1 + \frac{R_2}{R_1}$$
$$11 = 1 + \frac{10\text{ k}}{R_1}$$
$$\frac{10\text{ k}}{R_1} = 10$$
$$R_1 = 1\text{ k}\Omega$$

Therefore, the value of resistor, R_1 is $\boxed{1\text{ k}\Omega}$.

Step 5 of 5

Draw the operational amplifier circuit with fixed resistor.

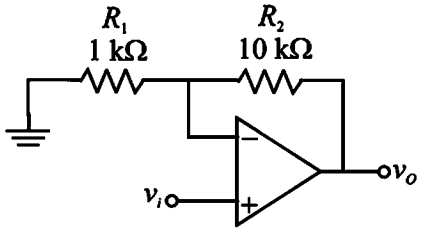


Figure 2

2.052P

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2.053P

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Step 1 of 7

2.054P

Refer to Figure 2.14 in the textbook.

Re-draw the circuit diagram.

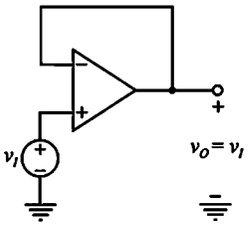


Figure 1

Step 2 of 7

Consider the op-amp is ideal except for having a finite gain A.

Determine the voltage gain of non-inverting amplifier for ideal case.

$$\frac{v_O}{v_I} = 1 + \frac{R_2}{R_1}$$

Determine the voltage gain of non-inverting amplifier, when op-amp having a finite gain A.

$$\frac{v_O}{v_I} = \frac{\left(1 + \frac{R_2}{R_1}\right)}{1 + \frac{\left(1 + \frac{R_2}{R_1}\right)}{A}} \quad \text{..... (1)}$$

Step 3 of 7

From Figure 1, the value of feedback and input resistance are zero.

$$1 + \frac{R_2}{R_1} = 1 + \frac{0}{0}$$
$$= 1$$

Substitute 1 for $1 + \frac{R_2}{R_1}$ in equation (1).

$$\frac{v_O}{v_I} = \frac{1}{1 + \frac{1}{A}} \quad \text{..... (2)}$$

Step 4 of 7

For $A = 1{,}000$, determine the voltage gain of the amplifier.

$$\frac{v_O}{v_I} = \frac{1}{1 + \frac{1}{1{,}000}}$$
$$= \frac{1}{1 + 0.001}$$
$$= 0.999\ \text{V/V}$$

Therefore, the voltage gain of the amplifier is $\boxed{0.999\ \text{V/V}}$.

Determine the percentage error in gain.

$$A_{error} = \left(\frac{0.999 - 1}{1}\right) \times 100$$
$$= -0.1\%$$

Therefore, the percentage error in voltage gain is $\boxed{-0.1\%}$.

Step 5 of 7

For $A = 100$, determine the voltage gain of the amplifier.

$$\frac{v_O}{v_I} = \frac{1}{1 + \frac{1}{100}}$$
$$= \frac{1}{1 + 0.01}$$
$$= 0.99\ \text{V/V}$$

Therefore, the voltage gain of the amplifier is $\boxed{0.99\ \text{V/V}}$.

Determine the percentage error in gain.

$$A_{error} = \left(\frac{0.99 - 1}{1}\right) \times 100$$
$$= -1\%$$

Therefore, the percentage error in voltage gain is $\boxed{-1\%}$.

Step 6 of 7

For $A = 10$, determine the voltage gain of the amplifier.

$$\frac{v_O}{v_I} = \frac{1}{1 + \frac{1}{10}}$$
$$= \frac{1}{1 + 0.1}$$
$$= 0.909\ \text{V/V}$$

Therefore, the voltage gain of the amplifier is $\boxed{0.909\ \text{V/V}}$.

Step 7 of 7

Determine the percentage error in gain.

$$A_{error} = \left(\frac{0.909 - 1}{1}\right) 100$$
$$= -9.1\%$$

Therefore, the percentage error in voltage gain is $\boxed{-9.1\%}$.

2.055P

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Step 1 of 5

Represent the nominal gain for op-amp in non-inverting configuration.

$$G_{nominal} = 1 + \frac{R_2}{R_1}$$
$$= 10 \text{ V/V}$$

The closed-loop gain is,

$$G = \frac{v_0}{v_{in}}$$
$$= \frac{1 + \frac{R_2}{R_1}}{1 + \frac{R_2}{R_1} \cdot \frac{1}{A}}$$
$$= \frac{G_{nominal}}{1 + \frac{G_{nominal}}{A}}$$

Substitute 10 V/V for $G_{nominal}$ and 100 for A.

$$G = \frac{10}{1 + \frac{10}{100}}$$
$$= \frac{10}{1 + 0.1}$$
$$= 9.09$$

Thus, the closed-loop gain G is $\boxed{9.09}$.

Step 2 of 5

Consider that the lowest value of $R_1 = 10\ \text{k}\Omega$. If R_e is shunted across R_1, the perfect compensation is achieved.

$$\frac{R_e}{R_1} = \frac{A - G}{1 + G}$$

Substitute $10\ \text{k}\Omega$ for R_1, 100 for A, and 9.09 for G.

$$R_e = \left(\frac{100 - 9.09}{1 + 9.09}\right)\left(10 \times 10^3\right)$$
$$= 9\left(10 \times 10^3\right)$$
$$= 90\ \text{k}\Omega$$

Thus, the value of shunt resistance R_e is $\boxed{90\ \text{k}\Omega}$.

Step 3 of 5

Consider the uncompensated situation.

Consider that the value of open-loop gain, $A = 200$ V/V.

Substitute 10 V/V for $G_{nominal}$ and 200 for A to find the value of G.

$$G = \frac{10}{1 + \frac{10}{200}}$$
$$= \frac{10}{1 + 0.05}$$
$$= 9.52$$

Thus, the nominal gain without compensation is $\boxed{9.52}$.

Step 4 of 5

Consider the compensated situation (with R_e).

Find the value of R_2.

$$\left(1 + \frac{R_2}{R_1}\right) = 10$$
$$\frac{R_2}{R_1} = 10 - 1$$

$$R_2 = 9R_1$$
$$= 90\ \text{k}\Omega \qquad (\text{since, } R_1 = 10\ \text{k}\Omega)$$

Find the value of nominal gain.

$$G_{nominal} = 1 + \frac{R_2}{R_1 \| R_e}$$
$$= 1 + \frac{90}{10 \| 90}$$
$$= 1 + \frac{90}{9}$$
$$= 11$$

Thus, the nominal gain with compensation is $\boxed{11}$.

Step 5 of 5

The closed-loop gain is,

$$G = \frac{G_{nominal}}{1 + \frac{G_{nominal}}{A}}$$
$$= \frac{11}{1 + \frac{11}{200}}$$
$$= 10.426$$

Hence, the closed-loop gain is $\boxed{10.426}$.

Hence, both with R_e and without R_e, the closed-loop gains are obtained.

Step 1 of 9

Write the expression for closed-loop nominal gain of non-inverting amplifier.

$$G_0 = 1 + \frac{R_2}{R_1}$$

Open-loop gain of the op-amp is A.

Refer to equation (2.11) in the textbook.

Determine the overall gain of the amplifier.

$$G = \frac{1 + \frac{R_2}{R_1}}{1 + \frac{1 + \frac{R_2}{R_1}}{A}}$$

Substitute $1 + \frac{R_2}{R_1}$ for G_0 in the equation.

$$G = \frac{G_0}{1 + \frac{G_0}{A}} \quad \text{..... (1)}$$

Step 2 of 9

Consider the reduction in the closed-loop gain G from the nominal value G_0 is to be kept less than $x\%$ of G_0.

Write the condition.

$$100\left(\frac{G_0 - G}{G_0}\right) \le x$$

$$100\left(1 - \frac{G}{G_0}\right) \le x$$

$$100\left(1 - \frac{1}{1 + \frac{G_0}{A}}\right) \le x \quad \textbf{(from equation (1))}$$

$$100\left(\frac{\frac{G_0}{A}}{1 + \frac{G_0}{A}}\right) \le x$$

Step 3 of 9

Simplify further,

$$\frac{\frac{G_0}{A}}{1 + \frac{G_0}{A}} \le \frac{x}{100}$$

$$\frac{1 + \frac{G_0}{A}}{\frac{G_0}{A}} \ge \frac{100}{x}$$

$$\frac{A}{G_0} + 1 \ge \frac{100}{x}$$

$$\frac{A}{G_0} \ge \frac{100}{x} - 1$$

$$A \ge G_0\left(\frac{100}{x} - 1\right) \quad \text{..... (2)}$$

Step 4 of 9

Consider $F = \frac{100}{x} - 1 \approx \frac{100}{x}$.

Substitute F for $\left(\frac{100}{x} - 1\right)$ in equation (2).

$$A \ge G_0 F$$

Therefore, the open-loop of the op-amp is exceed G_0 by $\approx \frac{100}{x}$.

Step 5 of 9

Consider, for $x = 0.01$, determine the value of F.

$$F = \frac{100}{0.01}$$
$$= 10^4$$

Therefore, the value of F for $x = 0.01$ is $\boxed{10^4}$.

For $x = 0.1$, determine the value of F.

$$F = \frac{100}{0.1}$$
$$= 10^3$$

Therefore, the value of F for $x = 0.1$ is $\boxed{10^3}$.

For $x = 1$, determine the value of F.

$$F = \frac{100}{1}$$
$$= 10^2$$

Therefore, the value of F for $x = 1$ is $\boxed{10^2}$.

Step 6 of 9

For $x = 10$, determine the value of F.

$$F = \frac{100}{10}$$
$$= 10$$

Therefore, the value of F for $x = 10$ is $\boxed{10}$.

Step 7 of 9

Consider the values of closed loop gains $1, 10, 10^2, 10^3$, **and** 10^4V/V.

Write the expression for open-loop gain.

$A = G_0 F$.

For $x = 0.01$, determine the value of open-loop gain.

G_0	1	10	10^2	10^3	10^4
F	10^4	10^4	10^4	10^4	10^4
$A = G_0 F$	10^4	10^5	10^6	10^7	10^8

For $x = 0.1$, determine the value of open-loop gain.

G_0	1	10	10^2	10^3	10^4
F	10^3	10^3	10^3	10^3	10^3
$A = G_0 F$	10^3	10^4	10^5	10^6	10^7

Step 8 of 9

For $x = 1$, determine the value of open-loop gain.

G_0	1	10	10^2	10^3	10^4
F	10^2	10^2	10^2	10^2	10^2
$A = G_0 F$	10^2	10^3	10^4	10^5	10^6

Step 9 of 9

For $x = 10$, determine the value of open-loop gain.

G_0	1	10	10^2	10^3	10^4
F	10	10	10	10	10
$A = G_0 F$	10	10^2	10^3	10^4	10^5

Step 1 of 16

Consider the open-loop gain of the op-amp is A.

Consider the nominal closed-loop gain of non-inverting op-amp is G_0.

Write the expression for actual closed-loop gain of the inverting op-amp.

$$G=\frac{G_0}{1+\frac{1-G_0}{A}}$$

Here,

$$G_0=-\frac{R_2}{R_1}$$

Write the expression for actual closed-loop gain of the non-inverting op-amp.

$$G=\frac{G_0}{1+\frac{G_0}{A}}$$

Here,

$$G_0=1+\frac{R_2}{R_1}$$

Write the expression for percentage of change in $|G|$ with respect to $|G_0|$. $\%e=\left(\frac{G_0-G}{G_0}\right)100$

Step 2 of 16

(a)

Consider the value of closed loop gain, $G_0=-1\,V/V$

The value of open loop gain, $A=10\,V/V$

The nominal gain is negative, so op-amp is operating in inverting configuration.

Determine the overall gain of the op-amp.

$$\begin{aligned}G&=\frac{G_0}{1+\frac{1-G_0}{A}}\\&=\frac{-1}{1+\frac{1+1}{10}}\\&=\frac{-1}{1+0.2}\\&=-0.83\ V/V\end{aligned}$$

Therefore, the overall gain of the op-amp is $\boxed{-0.83\ V/V}$.

Step 3 of 16

Determine the percentage of change in $|G|$ with respect to $|G_0|$.

$$\begin{aligned}\%e&=\left(\frac{G_0-G}{G_0}\right)100\\&=\left(\frac{-1+0.83}{-1}\right)100\\&=17\%\end{aligned}$$

Therefore, the percentage of change in $|G|$ with respect to $|G_0|$ is $\boxed{17\%}$.

Step 4 of 16

(b)

Consider the value of closed loop gain, $G_0=+1\,V/V$

The value of open loop gain, $A=10\,V/V$

The nominal gain is positive, so op-amp is operating in non-inverting configuration.

Determine the overall gain of the op-amp.

$$\begin{aligned}G&=\frac{G_0}{1+\frac{G_0}{A}}\\&=\frac{1}{1+\frac{1}{10}}\\&=\frac{1}{1+0.1}\\&=0.91\,V/V\end{aligned}$$

Therefore, the overall gain of the op-amp is $\boxed{0.91\ V/V}$.

Step 5 of 16

Determine the percentage of change in $|G|$ with respect to $|G_0|$.

$$\begin{aligned}\%e&=\left(\frac{G_0-G}{G_0}\right)100\\&=\left(\frac{1-0.91}{1}\right)100\\&=9\%\end{aligned}$$

Therefore, the percentage of change in $|G|$ with respect to $|G_0|$ is $\boxed{9\%}$.

Step 6 of 16

(c)

Consider the value of closed loop gain, $G_0=-1\,V/V$

The value of open loop gain, $A=100\,V/V$

The nominal gain is negative, so op-amp is operating in inverting configuration.

Determine the overall gain of the op-amp.

$$\begin{aligned}G&=\frac{G_0}{1+\frac{1-G_0}{A}}\\&=\frac{-1}{1+\frac{1+1}{100}}\\&=\frac{-1}{1+0.02}\\&=-0.98\,V/V\end{aligned}$$

Therefore, the overall gain of the op-amp is $\boxed{-0.98\ V/V}$.

Step 7 of 16

Determine the percentage of change in $|G|$ with respect to $|G_0|$.

$$\begin{aligned}\%e&=\left(\frac{G_0-G}{G_0}\right)100\\&=\left(\frac{-1+0.98}{-1}\right)100\\&=2\%\end{aligned}$$

Therefore, the percentage of change in $|G|$ with respect to $|G_0|$ is $\boxed{2\%}$.

Step 8 of 16

(d)

Consider the value of closed loop gain, $G_0=+10\,V/V$

The value of open loop gain, $A=10\,V/V$

The nominal gain is positive, so op-amp is operating in non-inverting configuration.

Determine the overall gain of the op-amp.

$$\begin{aligned}G&=\frac{G_0}{1+\frac{G_0}{A}}\\&=\frac{10}{1+\frac{10}{10}}\\&=\frac{10}{1+1}\\&=5\,V/V\end{aligned}$$

Therefore, the overall gain of the op-amp is $\boxed{5\ V/V}$.

Step 9 of 16

Determine the percentage of change in $|G|$ with respect to $|G_0|$.

$$\begin{aligned}\%e&=\left(\frac{G_0-G}{G_0}\right)100\\&=\left(\frac{10-5}{10}\right)100\\&=50\%\end{aligned}$$

Therefore, the percentage of change in $|G|$ with respect to $|G_0|$ is $\boxed{50\%}$.

Step 10 of 16

(e)

Consider the value of closed loop gain, $G_0=-10\,V/V$

The value of open loop gain, $A=100\,V/V$

The nominal gain is negative, so op-amp is operating in inverting configuration.

Determine the overall gain of the op-amp.

$$\begin{aligned}G&=\frac{G_0}{1+\frac{1-G_0}{A}}\\&=\frac{-10}{1+\frac{1+10}{100}}\\&=\frac{-10}{1+0.11}\\&=-9\,V/V\end{aligned}$$

Therefore, the overall gain of the op-amp is $\boxed{-9\ V/V}$.

Step 11 of 16

Determine the percentage of change in $|G|$ with respect to $|G_0|$.

$$\begin{aligned}\%e&=\left(\frac{G_0-G}{G_0}\right)100\\&=\left(\frac{-10+9}{-10}\right)100\\&=10\%\end{aligned}$$

Therefore, the percentage of change in $|G|$ with respect to $|G_0|$ is $\boxed{10\%}$.

Step 12 of 16

(f)

Consider the value of closed loop gain, $G_0=-10\,V/V$

The value of open loop gain, $A=1000\,V/V$

The nominal gain is negative, so op-amp is operating in inverting configuration.

Determine the overall gain of the op-amp.

$$\begin{aligned}G&=\frac{G_0}{1+\frac{1-G_0}{A}}\\&=\frac{-10}{1+\frac{1+10}{1000}}\\&=\frac{-10}{1+0.011}\\&=-9.89\,V/V\end{aligned}$$

Therefore, the overall gain of the op-amp is $\boxed{-9.89\ V/V}$.

Step 13 of 16

Determine the percentage of change in $|G|$ with respect to $|G_0|$.

$$\begin{aligned}\%e&=\left(\frac{G_0-G}{G_0}\right)100\\&=\left(\frac{-10+9.89}{-10}\right)100\\&=1.1\%\end{aligned}$$

Therefore, the percentage of change in $|G|$ with respect to $|G_0|$ is $\boxed{1.1\%}$.

Step 14 of 16

(g)

Consider the value of closed loop gain, $G_0=+1\,V/V$

The value of open loop gain, $A=2\,V/V$

The nominal gain is positive, so op-amp is operating in non-inverting configuration.

Determine the overall gain of the op-amp.

$$\begin{aligned}G&=\frac{G_0}{1+\frac{G_0}{A}}\\&=\frac{1}{1+\frac{1}{2}}\\&=\frac{1}{1+0.5}\\&=0.67\,V/V\end{aligned}$$

Step 15 of 16

Determine the percentage of change in $|G|$ with respect to $|G_0|$.

$$\begin{aligned}\%e&=\left(\frac{G_0-G}{G_0}\right)100\\&=\left(\frac{1-0.67}{1}\right)100\\&=33\%\end{aligned}$$

Step 16 of 16

Write the values in the table.

Case	$G_0(V/V)$	$A(V/V)$	$G(V/V)$	$\%e$
a	−1	10	−0.83	17
b	+1	10	0.91	9
c	−1	100	−0.98	2
d	+10	10	5	50
e	−10	100	−9	10
f	−10	1000	−9.89	1.1
g	+1	2	0.67	33

2.059P

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Step 1 of 3

Refer to Figure 2.16 in the textbook.

The expression for the differential voltage gain is,

$$\frac{v_0}{v_{id}} = \frac{R_2}{R_1}$$

Substitute $100\ \text{k}\Omega$ for R_2, and $5\ \text{k}\Omega$ for R_1.

$$\begin{aligned}\frac{v_0}{v_{id}} &= \frac{R_2}{R_1}\\ &= \frac{100}{5}\\ &= 20\end{aligned}$$

Find the differential input resistance.

$$\begin{aligned}R_{id} &= 2R_1\\ &= 2(5\ \text{k}\Omega)\\ &= 10\ \text{k}\Omega\end{aligned}$$

Step 2 of 3

Consider that the resistances ratios $\frac{R_2}{R_1}$ and $\frac{R_4}{R_3}$ are different from each other by 1%.

Consider that $\frac{R_4}{R_3} = 100$, therefore, $\frac{R_2}{R_1} = 99$.

Find the common-mode gain.

$$\begin{aligned}A_{cm} &= \frac{R_4}{R_3 + R_4}\left[1 - \frac{R_2 R_3}{R_1 R_4}\right]\\ &= \frac{100R_3}{100R_3 + R_3}\left[1 - \left(\frac{99R_1}{R_1}\right)\left(\frac{R_3}{100R_3}\right)\right]\\ &= \frac{100}{101}\left(\frac{100-99}{100}\right)\\ &= \frac{1}{101}\end{aligned}$$

Find the differential gain.

$$\begin{aligned}A_d &= \frac{v_0}{v_{id}}\\ &= \frac{R_2}{R_1}\\ &= \frac{99R_1}{R_1}\\ &= 99\end{aligned}$$

Step 3 of 3

Find the common-mode rejection ratio.

$$CMRR = \frac{A_d}{A_{cm}}$$

Substitute 99 for A_d, and $\frac{1}{101}$ for A_{cm}.

$$\begin{aligned}CMRR &= \frac{A_d}{A_{cm}}\\ &= \frac{99}{\frac{1}{101}}\\ &= 99(101)\\ &= 9999\end{aligned}$$

Therefore, the difference amplifier voltage gain $\frac{v_o}{v_{id}}$ is $\boxed{20}$.

The differential input resistance R_{id} is $\boxed{10\ \text{k}\Omega}$.

The common mode gain A_{cm} is $\boxed{\frac{1}{101}}$.

The CMMR value is $\boxed{9999}$.

2.061P

We don't have the solution to this problem yet.

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Step 1 of 13

Refer to Figure to Figure P2.62 in the textbook.

The currents flowing into the inverting and the non-inverting terminals of an ideal operational amplifier are zero.

Redraw the circuit with the currents into the inverting and the non-inverting terminals.

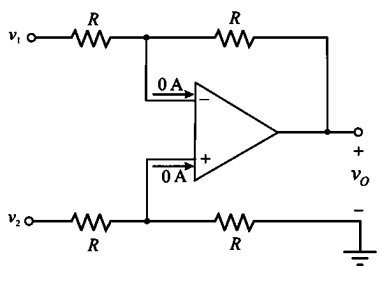


Figure 1

Step 2 of 13

Assume that v_{O1} is the output due to input, v_1 and v_{O2} is the output due to the input, v_2.

The output voltage using super position principle is, $v_O = v_{O1} + v_{O2}$.

Redraw the circuit with only one source, v_1

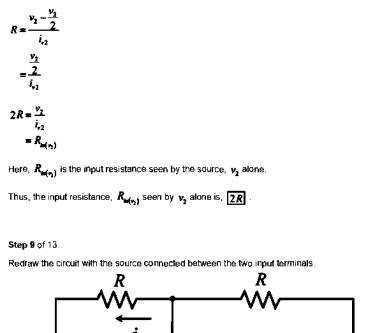


Figure 2

Here, v_n is the voltage at the inverting terminal (negative terminal), v_p is the voltage at the non-inverting terminal (positive terminal), and i_{v1} is the current due to source, v_1.

Step 3 of 13

Apply Kirchhoff's current law at non-inverting node of circuit in Figure 2.

$$\frac{v_p - v_1}{R} + \frac{v_p}{R} = 0$$
$$\frac{v_p + v_p}{R} = 0$$
$$2v_p = 0$$
$$v_p = 0$$

The voltage at the inverting terminal due to virtual ground concept is, $v_n = 0\ \text{V}$.

Apply Kirchhoff's current law at inverting node.

$$\frac{v_n - v_1}{R} + \frac{v_n - v_{O1}}{R} = 0$$
$$\frac{v_n - v_1 + v_n - v_{O1}}{R} = 0$$
$$v_n - v_1 + v_n - v_{O1} = 0$$
$$2v_n - v_1 - v_{O1} = 0$$

Substitute 0 V for v_n in the equation.

$$2(0) - v_1 - v_{O1} = 0$$
$$0 - v_1 - v_{O1} = 0$$
$$v_{O1} = -v_1$$

Step 4 of 13

Use Ohm's law to calculate the input resistance seen by source, v_1 alone.

$$i_{v1} = \frac{v_1 - v_n}{R}$$
$$R = \frac{v_1 - v_n}{i_{v1}}$$

Substitute 0 V for v_n in the equation.

$$R = \frac{v_1 - 0}{i_{v1}}$$
$$= \frac{v_1}{i_{v1}}$$
$$= R_{in(1)}$$

Here, $R_{in(1)}$ is the input resistance seen by the source, v_1 alone.

Thus, the input resistance, $R_{in(1)}$ seen by v_1 alone is, $\boxed{R}$.

Step 5 of 13

Redraw the circuit with only one source, v_2.

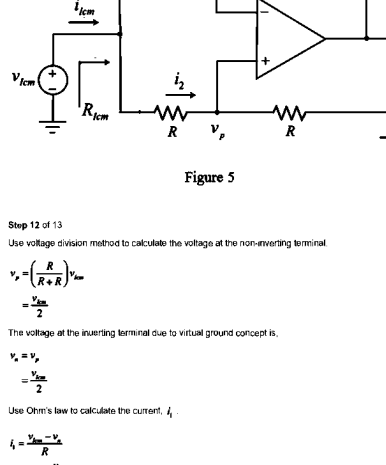


Figure 3

Here, v_{n1} is the voltage at the inverting terminal (negative terminal), v_{p1} is the voltage at the non-inverting terminal (positive terminal), and i_{v2} is the current due to source, v_2.

Step 6 of 13

Apply Kirchhoff's current law at non-inverting node of circuit in Figure 3.

$$\frac{v_{p1} - v_2}{R} + \frac{v_{p1}}{R} = 0$$
$$\frac{v_{p1} - v_2 + v_{p1}}{R} = 0$$
$$v_{p1} - v_2 + v_{p1} = 0$$
$$2v_{p1} - v_2 = 0$$
$$2v_{p1} = v_2$$
$$v_{p1} = \frac{v_2}{2}$$

The voltage at the inverting terminal due to virtual ground concept is,

$$v_{n1} = \frac{v_2}{2} \qquad (\textbf{since, } v_{p1} = v_{n1})$$

Step 7 of 13

Apply Kirchhoff's current law at inverting node.

$$\frac{v_{n1} - v_1}{R} + \frac{v_{n1} - v_{O2}}{R} = 0$$
$$\frac{v_{n1} + v_{n1} - v_{O2}}{R} = 0$$
$$v_{n1} + v_{n1} - v_{O2} = 0$$
$$v_{O2} = 2v_{n1}$$

Substitute $\frac{v_2}{2}$ for v_{n1} in the equation.

$$v_{O2} = 2\left(\frac{v_2}{2}\right)$$
$$= v_2$$

Substitute $-v_1$ for v_{O1} and v_2 for v_{O2} in equation for v_O.

$$v_O = v_{O1} + v_{O2}$$
$$= -v_1 + v_2$$
$$= v_2 - v_1$$

Thus, the total output voltage, v_O as a function of v_1 and v_2 is, $\boxed{v_2 - v_1}$.

Step 8 of 13

Use Ohm's law to calculate the input resistance seen by source, v_2 alone.

$$i_{v2} = \frac{v_2 - v_{p1}}{R}$$
$$R = \frac{v_2 - v_{p1}}{i_{v2}}$$

Substitute $\frac{v_2}{2}$ for v_{p1} in the equation.

$$R = \frac{v_2 - \frac{v_2}{2}}{i_{v2}}$$
$$= \frac{\frac{v_2}{2}}{i_{v2}}$$
$$2R = \frac{v_2}{i_{v2}}$$
$$= R_{in(2)}$$

Here, $R_{in(2)}$ is the input resistance seen by the source, v_2 alone.

Thus, the input resistance, $R_{in(2)}$ seen by v_2 alone is, $\boxed{2R}$.

Step 9 of 13

Redraw the circuit with the source connected between the two input terminals.

Figure 4

Step 18 of 13

Apply Kirchhoff's voltage law to the input loop to find the input resistance seen by source, v_M connected between the two input terminals.

$$v_M - i_1 R - i_1 R = 0$$
$$v_M - 2i_1 R = 0$$
$$v_M = 2i_1 R$$
$$2R = \frac{v_M}{i_1}$$
$$= R_M$$

Thus, the input resistance, R_M seen by source connected between the two terminals is, $\boxed{2R}$.

Step 11 of 13

Redraw the circuit with the source connected to both input terminals simultaneously.

Figure 5

Step 12 of 13

Use voltage division method to calculate the voltage at the non-inverting terminal.

$$v_p = \left(\frac{R}{R+R}\right)v_{Icm}$$
$$= \frac{v_{Icm}}{2}$$

The voltage at the inverting terminal due to virtual ground concept is,

$$v_n = v_p$$
$$= \frac{v_{Icm}}{2}$$

Use Ohm's law to calculate the current, i_1.

$$i_1 = \frac{v_{Icm} - v_n}{R}$$
$$= \frac{v_{Icm} - \frac{v_{Icm}}{2}}{R} \qquad \left(\textbf{since, } v_n = \frac{v_{Icm}}{2}\right)$$
$$= \frac{v_{Icm}}{2R}$$

Step 13 of 13

Use Ohm's law to calculate the current, i_2.

$$i_2 = \frac{v_{Icm}}{R+R}$$
$$= \frac{v_{Icm}}{2R}$$

Calculate the input resistance seen by the source, v_{Icm}.

$$R_{Icm} = \frac{v_{Icm}}{i_{Icm}}$$
$$= \frac{v_{Icm}}{i_1 + i_2}$$
$$= \frac{v_{Icm}}{\left(\frac{v_{Icm}}{2R} + \frac{v_{Icm}}{2R}\right)}$$
$$= \frac{v_{Icm}}{\left(\frac{v_{Icm}}{R}\right)}$$
$$R_{Icm} = R$$

Thus, the input resistance, R_{Icm} with the source connected to both input terminals simultaneously is, $\boxed{R}$.

Step 1 of 6

2.063P

Refer to Figure 2.16 in the textbook for a differential amplifier.

Connect the two input terminals together to a common-mode signal source.

The current through the inverting and non-inverting terminals of an ideal op-amp are zero.

Draw the modified circuit and represent the currents and voltages.

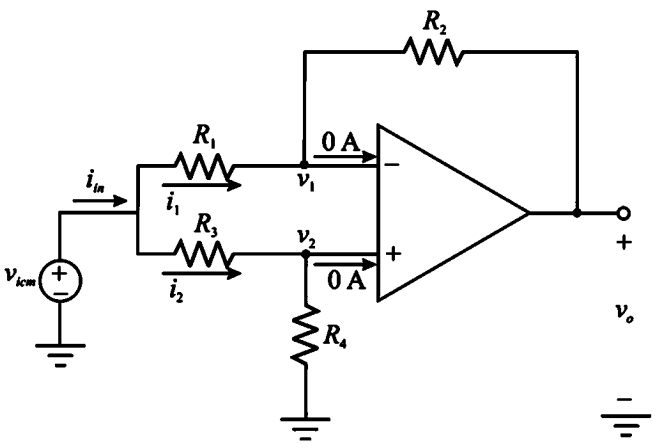


Figure 1

Step 2 of 6

The common-mode input resistance of the circuit is,

$$R_{icm} = \frac{v_{icm}}{i_{in}}$$

Apply Kirchhoff's current law at non-inverting node of the circuit.

$$\frac{v_2 - v_{icm}}{R_3} + \frac{v_2}{R_4} = 0$$

$$\frac{R_4(v_2 - v_{icm}) + R_3 v_2}{R_3 R_4} = 0$$

$$R_4(v_2 - v_{icm}) + R_3 v_2 = 0$$

$$R_4 v_2 - R_4 v_{icm} + R_3 v_2 = 0$$

$$R_4 v_{icm} = R_4 v_2 + R_3 v_2$$

$$R_4 v_{icm} = (R_4 + R_3) v_2$$

$$v_2 = \left(\frac{R_4}{R_4 + R_3}\right) v_{icm}$$

The voltage at inverting node due to virtual ground concept is,

$$v_1 = \left(\frac{R_4}{R_4 + R_3}\right) v_{icm} \qquad (\text{since, } v_1 = v_2)$$

Step 3 of 6

Use Ohm's law to find the current through the resistor, R_1 in the circuit.

$$i_1 = \frac{v_{icm} - v_1}{R_1}$$

Substitute the expression for v_1 in the equation.

$$i_1 = \frac{v_{icm} - \left(\frac{R_4}{R_4 + R_3}\right) v_{icm}}{R_1}$$

$$= v_{icm}\left[1 - \left(\frac{R_4}{R_4 + R_3}\right)\right]\frac{1}{R_1}$$

$$= v_{icm}\left(\frac{R_4 + R_3 - R_4}{R_4 + R_3}\right)\frac{1}{R_1}$$

$$= v_{icm}\left[\frac{R_3}{(R_4 + R_3)R_1}\right]$$

Step 4 of 6

Use Ohm's law to find the current through the resistor, R_3 in the circuit.

$$i_2 = \frac{v_{icm} - v_2}{R_3}$$

Substitute the expression for v_2 in the equation.

$$i_2 = \frac{v_{icm} - \left(\frac{R_4}{R_4 + R_3}\right) v_{icm}}{R_3}$$

$$= v_{icm}\left[1 - \left(\frac{R_4}{R_4 + R_3}\right)\right]$$

$$= v_{icm}\left(\frac{R_4 + R_3 - R_4}{R_4 + R_3}\right)\frac{1}{R_3}$$

$$= v_{icm}\left(\frac{1}{R_4 + R_3}\right)$$

Step 5 of 6

Apply Kirchhoff's current law at input node to find the input current.

$$i_{in} = i_1 + i_2$$

Substitute the expressions of i_1 **and** i_2 in the equation.

$$i_{in} = v_{icm}\left[\frac{R_3}{(R_4 + R_3)R_1}\right] + v_{icm}\left(\frac{1}{R_4 + R_3}\right)$$

$$= v_{icm}\left[\frac{R_3}{(R_4 + R_3)R_1} + \frac{1}{R_4 + R_3}\right]$$

$$= v_{icm}\left[\frac{1}{R_1}\left(\frac{1}{\frac{R_4}{R_3} + 1}\right) + \frac{1}{R_4 + R_3}\right]$$

Step 6 of 6

Consider that $\frac{R_2}{R_1} = \frac{R_4}{R_3}$. Therefore,

$$i_{in} = v_{icm}\left[\frac{1}{R_1}\left(\frac{1}{\frac{R_2}{R_1} + 1}\right) + \frac{1}{R_4 + R_3}\right]$$

$$= v_{icm}\left[\frac{1}{R_1}\left(\frac{1}{\frac{R_2 + R_1}{R_1}}\right) + \frac{1}{R_4 + R_3}\right]$$

$$= v_{icm}\left(\frac{1}{R_3 + R_1} + \frac{1}{R_4 + R_3}\right)$$

Simplify the function to find the input common-mode resistance, R_{icm}.

$$R_{icm} = \frac{v_{icm}}{i_{in}}$$

$$= \frac{1}{\frac{1}{R_3 + R_1} + \frac{1}{R_4 + R_3}}$$

$$= (R_3 + R_4) \| (R_1 + R_3)$$

Hence, it is shown that the input common-mode resistance is, $\boxed{R_{icm} = (R_3 + R_4) \| (R_1 + R_3)}$.

Step 1 of 5

2.064P

Refer to Figure 2.16 in the textbook for a differential amplifier.

Connect a resistance, R_s in series with each of the inputs, v_{I1} **and** v_{I2} .

Draw the modified circuit.

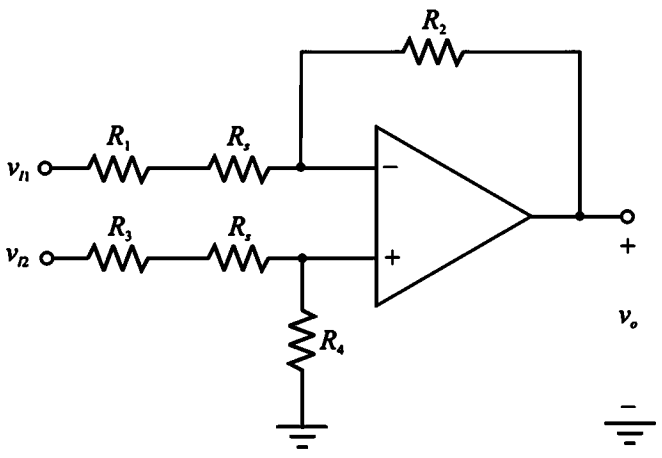


Figure 1

Step 2 of 5

Apply super position principle on the circuit.

Assume the voltage source, $v_{I2} = 0$, to find the expression for the inverting gain.

$$\frac{v_{I1} - v_-}{R_1 + R_s} = \frac{v_- - v_{o1}}{R_3}$$

Due to virtual ground concept, the voltages at inverting and non-inverting terminals are same. Hence, $v_- = 0$. Therefore,

$$\frac{-v_{I1}}{R_1 + R_s} = \frac{v_{o1}}{R_3}$$

$$\frac{v_{o1}}{v_{I1}} = -\frac{R_3}{R_1 + R_s}$$

Assume the voltage source, $v_{I1} = 0$, to find the expression for the non-inverting gain.

$$\frac{v_-}{R_1 + R_s} = \frac{v_{o2} - v_-}{R_3}$$

Due to virtual ground concept, the voltages at inverting and non-inverting terminals are same. Hence, $v_- = v_+$. Therefore,

$$\frac{v_+}{R_1 + R_s} = \frac{v_{o2} - v_+}{R_3}$$

Step 3 of 5

The voltage at non-inverting terminal due to voltage division is,

$$v_+ = \left(\frac{R_4}{R_4 + R_3 + R_s}\right) v_{I2}$$

Therefore,

$$\frac{1}{R_1 + R_s}\left(\frac{R_4}{R_4 + R_3 + R_s}\right) v_{I2} = \frac{v_{o2}}{R_3} - \frac{1}{R_3}\left(\frac{R_4}{R_4 + R_3 + R_s}\right) v_{I2}$$

$$\frac{1}{R_1 + R_s}\left(\frac{R_4}{R_4 + R_3 + R_s}\right) v_{I2} + \frac{1}{R_3}\left(\frac{R_4}{R_4 + R_3 + R_s}\right) v_{I2} = \frac{v_{o2}}{R_3}$$

$$\left(\frac{R_4}{R_4 + R_3 + R_s}\right)\left(\frac{R_3 + R_1 + R_s}{R_3 (R_1 + R_s)}\right) v_{I2} = \frac{v_{o2}}{R_3}$$

$$\left(\frac{R_4}{R_4 + R_3 + R_s}\right)\left(1 + \frac{R_3}{R_1 + R_s}\right) v_{I2} = v_{o2}$$

$$\frac{v_{o2}}{v_{I2}} = \left(\frac{R_4}{R_4 + R_3 + R_s}\right)\left(1 + \frac{R_3}{R_1 + R_s}\right)$$

Step 4 of 5

Consider that the circuit is an ideal differential amplifier. Therefore, $v_O = 0$.

$$v_O (v_{I2} + v_{I1}) = 0$$

$$v_O v_{I2} + v_O v_{I1} = 0$$

$$\frac{v_O v_{I2} + v_O v_{I1}}{v_{I2} v_{I1}} = 0$$

$$\frac{v_O}{v_{I1}} + \frac{v_O}{v_{I2}} = 0$$

Combine the two configuration gains together.

$$-\frac{R_2}{R_1 + R_s} + \left(\frac{R_4}{R_4 + R_3 + R_s}\right)\left(1 + \frac{R_2}{R_1 + R_s}\right) = 0$$

$$\left(\frac{R_4}{R_4 + R_3 + R_s}\right)\left(1 + \frac{R_2}{R_1 + R_s}\right) = \frac{R_2}{R_1 + R_s}$$

$$\left(\frac{R_4}{R_4 + R_3 + R_s}\right)\left(\frac{R_1 + R_s + R_2}{R_1 + R_s}\right) = \frac{R_2}{R_1 + R_s}$$

$$\left(\frac{R_4}{R_4 + R_3 + R_s}\right)(R_1 + R_s + R_2) = R_2$$

Step 5 of 5

Simplify the function further.

$$\frac{R_4}{R_4 + R_3 + R_s} = \frac{R_2}{R_1 + R_s + R_2}$$

$$\frac{R_4}{R_3\left(\frac{R_4}{R_3} + 1 + \frac{R_s}{R_3}\right)} = \frac{R_2}{R_1\left(1 + \frac{R_s}{R_1} + \frac{R_2}{R_1}\right)}$$

$$\frac{\frac{R_4}{R_3}}{\frac{R_4}{R_3} + 1 + \frac{R_s}{R_3}} = \frac{\frac{R_2}{R_1}}{1 + \frac{R_s}{R_1} + \frac{R_2}{R_1}}$$

Consider that $\frac{R_4}{R_3} = \frac{R_2}{R_1}$. Therefore,

$$\frac{\frac{R_2}{R_1}}{\frac{R_2}{R_1} + 1 + \frac{R_s}{R_3}} = \frac{\frac{R_2}{R_1}}{1 + \frac{R_s}{R_1} + \frac{R_2}{R_1}} \quad \left(\text{since, } \frac{R_4}{R_3} = \frac{R_2}{R_1}\right)$$

$$\frac{R_2}{R_1} + 1 + \frac{R_s}{R_3} = 1 + \frac{R_s}{R_1} + \frac{R_2}{R_1}$$

$$\frac{R_s}{R_1} = \frac{R_s}{R_3}$$

$$R_3 = R_1$$

Therefore, the condition for the ideal differential amplifier is, $\boxed{R_1 = R_3}$.

Step 1 of 6

Refer to Figure P2.62 in the textbook.

The expression for the common mode gain, A_{cm} of the differential amplifier is,

$$\begin{aligned} A_{cm} &= \frac{v_o}{v_{icm}} \\ &= \left(\frac{R_4}{R_4+R_3}\right)\left(1-\frac{R_3}{R_1}\frac{R_3}{R_4}\right) \end{aligned}$$

The worst case common-mode gain is obtained when it has maximum value.

Simplify the expression further.

$$A_{cm} = \left(\frac{1}{\frac{R_3}{R_4}+1}\right)\left(1-\frac{R_3}{R_1}\frac{R_3}{R_4}\right)$$

Consider that all the resistors are $100 \pm x\%$.

The range of $\frac{R_3}{R_4}$ is,

$$\frac{100-x}{100+x} \le \frac{R_3}{R_4} \le \frac{100+x}{100-x}$$

Similarly, the range of $\frac{R_2}{R_1}$ is,

$$\frac{100-x}{100+x} \le \frac{R_2}{R_1} \le \frac{100+x}{100-x}$$

Step 2 of 6

The maximum value of A_{cm} is obtained when the values of $\frac{R_2}{R_1}$ **and** $\frac{R_3}{R_4}$ are minimum. Select minimum values of $\frac{R_3}{R_4}$ **and** $\frac{R_2}{R_1}$, to get worst-case common-mode gain.

Substitute these values in the equation of A_{cm} to find the value of $A_{cm\text{-}max}$.

$$A_{cm\text{-}max} = \frac{1}{\left(\frac{100-x}{100+x}+1\right)}\left[1-\left(\frac{100-x}{100+x}\right)\left(\frac{100-x}{100+x}\right)\right]$$

Rearrange the equation to find the value of $A_{cm\text{-}max}$.

$$\begin{aligned} A_{cm\text{-}max} &= \frac{1}{\left(\frac{100-x+100+x}{100+x}\right)}\left[1-\frac{(100-x)^2}{(100+x)^2}\right] \\ &= \left(\frac{100+x}{200}\right)\left[\frac{(100+x)^2-(100-x)^2}{100+x}\right] \\ &= \left(\frac{1}{200}\right)\left[\frac{(100+x)^2-(100-x)^2}{(100+x)}\right] \end{aligned}$$

Simplify the expression further.

$$\begin{aligned} A_{cm\text{-}max} &= \left(\frac{1}{200}\right)\left(\frac{400x}{100+x}\right) \\ &= \frac{2x}{100+x} \\ &\simeq \frac{2x}{100} \\ &\simeq \frac{x}{50} \end{aligned}$$

Step 3 of 6

Consider that the value of $x = 0.1$.

Calculate the worst-case common-mode gain.

$$\begin{aligned} A_{cm\text{-}max} &\simeq \frac{x}{50} \\ &= \frac{0.1}{50} \\ &= 0.002 \end{aligned}$$

Consider that the value of $x = 1$.

Calculate the worst-case common-mode gain.

$$\begin{aligned} A_{cm\text{-}max} &\simeq \frac{x}{50} \\ &= \frac{1}{50} \\ &= 0.02 \end{aligned}$$

Consider that the value of $x = 5$.

Calculate the worst-case common-mode gain.

$$\begin{aligned} A_{cm\text{-}max} &\simeq \frac{x}{50} \\ &= \frac{5}{50} \\ &= 0.1 \end{aligned}$$

Thus, the values of worst-case common-mode gain, $A_{cm\text{-}max}$ for x values of 0.1, 1, and 5 are respectively, $\boxed{0.002}$, $\boxed{0.02}$, and $\boxed{0.1}$.

Step 4 of 6

The expression for the common mode rejection ratio, CMRR is,

$$\text{CMRR} = 20\log\frac{|A_d|}{|A_{cm}|}$$

Find the differential gain, A_d , assuming all the resistances are equal to R .

The output voltage of a differential amplifier using super position principle is,

$$\begin{aligned} v_o &= v_{o1} + v_{o2} \\ &= -\frac{R_2}{R_1}v_1 + v_2\left(\frac{R_4}{R_3+R_4}\right)\left(1+\frac{R_2}{R_1}\right) \end{aligned}$$

Substitute R for all resistance values.

$$\begin{aligned} v_o &= -\frac{R}{R}v_1 + v_2\left(\frac{R}{R+R}\right)\left(1+\frac{R}{R}\right) \\ &= -v_1 + v_2\left(\frac{1}{2}\right)(2) \\ &= v_2 - v_1 \end{aligned}$$

Step 5 of 6

Therefore, the differential mode gain, A_d is,

$$\begin{aligned} A_d &= \frac{v_o}{v_2 - v_1} \\ &= 1 \end{aligned}$$

Substitute 1 for A_d , and $\frac{x}{50}$ in the expression for CMRR.

$$\begin{aligned} \text{CMRR} &= 20\log\frac{1}{\frac{x}{50}} \\ &= 20\log\frac{50}{x} \end{aligned}$$

Consider that the value of $x = 0.1$.

Calculate the common-mode rejection ratio.

$$\begin{aligned} \text{CMRR} &= 20\log\frac{50}{x} \\ &= 20\log\frac{50}{0.1} \\ &= 20(2.698) \\ &\simeq 54 \text{ dB} \end{aligned}$$

Step 6 of 6

Consider that the value of $x = 1$.

Calculate the worst-case common-mode gain.

$$\begin{aligned} \text{CMRR} &= 20\log\frac{50}{x} \\ &= 20\log\frac{50}{1} \\ &= 20(1.698) \\ &\simeq 34 \text{ dB} \end{aligned}$$

Consider that the value of $x = 5$.

Calculate the worst-case common-mode gain.

$$\begin{aligned} \text{CMRR} &= 20\log\frac{50}{x} \\ &= 20\log\frac{50}{5} \\ &= 20(1) \\ &= 20 \text{ dB} \end{aligned}$$

Thus, the values of common-mode rejection ratio for x values of 0.1, 1, and 5 are respectively, $\boxed{54 \text{ dB}}$, $\boxed{34 \text{ dB}}$, and $\boxed{20 \text{ dB}}$.

2.066P

We don't have the solution to this problem yet.

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2.067P

Step 1 of 3

Refer to Figure 2.16 in the textbook for the circuit of a differential amplifier.

Consider that the differential input resistance, $R_{id} = 2\ \text{k}\Omega$.

The expression for the differential input resistance is,

$$R_{id} = 2R_1$$

Simplify the equation to find the value of R_1.

$$R_1 = \frac{R_{id}}{2}$$

Substitute $2\ \text{k}\Omega$ for R_{id}.

$$\begin{aligned} R_1 &= \frac{2}{2} \\ &= 1\ \text{k}\Omega \end{aligned}$$

Consider that the differential gain, $A_d = 1000$.

The general expression for the differential gain is,

$$A_d = \frac{R_2}{R_1}$$

Simplify the equation to find the value of R_2.

$$R_2 = A_d R_1$$

Substitute 1000 for A_d and $1\ \text{k}\Omega$ for R_1.

$$\begin{aligned} R_2 &= 1000(1\ \text{k}\Omega) \\ &= 1\ \text{M}\Omega \end{aligned}$$

Step 2 of 3

Consider that the value of common mode rejection ratio (CMRR) is 88 dB.

The general expression for the common mode rejection ratio (CMRR) is,

$$CMRR = \frac{A_d}{A_{cm}}$$

Convert the value from decibels to normal.

$$20\log\left(\frac{A_d}{A_{cm}}\right) = 88$$

$$\begin{aligned} \frac{A_d}{A_{cm}} &= 10^{\left(\frac{88}{20}\right)} \\ &= 10^{4.4} \end{aligned}$$

Substitute 1000 for A_d to find the value of common-mode gain, A_{cm}.

$$\begin{aligned} A_{cm} &= \frac{A_d}{10^{(4.4)}} \\ &= \frac{1000}{10^{4.4}} \\ &= 0.0398 \\ &\simeq 0.04 \end{aligned}$$

Step 3 of 3

The general expression for the common mode gain is,

$$A_{cm} = \frac{R_4}{R_3 + R_4}\left[1 - \frac{R_2 R_3}{R_1 R_4}\right]$$

Substitute 1000 for $\frac{R_2}{R_1}$ and 0.04 for A_{cm}.

$$0.04 = \frac{R_4}{R_3 + R_4}\left[1 - \frac{1000R_3}{R_4}\right]$$

$$0.04(R_3 + R_4) = R_4\left(\frac{R_4 - 1000R_3}{R_4}\right)$$

$$0.04R_3 + 0.04R_4 = R_4 - 1000R_3$$

$$(0.04 + 1000)R_3 = R_4(1 - 0.04)$$

$$\begin{aligned} \frac{R_3}{R_4} &= \frac{0.96}{1000.04} \\ &= 0.00096 \end{aligned}$$

$$\begin{aligned} \frac{R_4}{R_3} &= 1041.6 \\ &\simeq 1042 \\ &= (1000 + 42) \end{aligned}$$

$$R_4 = (1000 + 42)R_3$$

The resistances R_3 and R_4 have a tolerance of 42.

Hence, these resistances have tolerance of $\boxed{4.2\%}$.

Step 1 of 11

(a)

Refer to Figure P2.68 in the textbook.

The expression for the differential gain is,

$$A_d = \frac{R_2}{R_1}$$

Substitute $100\ \text{k}\Omega$ for R_1 and R_2.

$$\begin{aligned} A_d &= \frac{R_2}{R_1} \\ &= \frac{100\ \text{k}\Omega}{100\ \text{k}\Omega} \\ &= 1 \end{aligned}$$

Therefore, the differential gain, A_d is, $\boxed{1\ \text{V/V}}$.

Step 2 of 11

The expression for the common-mode gain is,

$$A_{cm} = \left(\frac{R_4}{R_4 + R_3}\right)\left(1 - \frac{R_2}{R_1}\frac{R_3}{R_4}\right)$$

Substitute $100\ \text{k}\Omega$ for R_1, R_2, R_3, and R_4.

$$\begin{aligned} A_{cm} &= \left(\frac{100\ \text{k}\Omega}{100\ \text{k}\Omega + 100\ \text{k}\Omega}\right)\left[1 - \left(\frac{100\ \text{k}\Omega}{100\ \text{k}\Omega}\right)\left(\frac{100\ \text{k}\Omega}{100\ \text{k}\Omega}\right)\right] \\ &= \left(\frac{1}{2}\right)\left[1 - (1)(1)\right] \\ &= \left(\frac{1}{2}\right)(0) \\ &= 0 \end{aligned}$$

Therefore, the common-mode gain, A_{cm} is, $\boxed{0\ \text{V/V}}$.

Step 3 of 11

(b)

Connect a common-mode voltage to the inputs of the op-amp and draw the modified circuit.

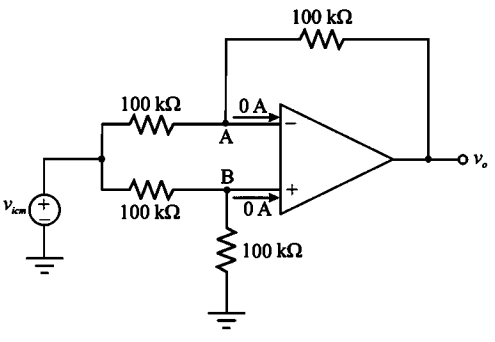


Figure 1

Step 4 of 11

Consider that the positive and negative terminals voltage is, $\pm 2.5\ \text{V}$.

Apply Kirchhoff's current law at positive terminal.

$$\begin{aligned} &\frac{V_B}{100\ \text{k}\Omega} + \frac{V_B - v_{icm}}{100\ \text{k}\Omega} + 0 = 0 \\ &\frac{V_B + V_B - v_{icm}}{100\ \text{k}\Omega} = 0 \\ &V_B + V_B - v_{icm} = 0 \\ &v_{icm} = 2V_B \end{aligned}$$

At $V_B = -2.5\ \text{V}$, the minimum common-mode input voltage is,

$$\begin{aligned} v_{icm\text{-}min} &= 2(-2.5\ \text{V}) \\ &= -5\ \text{V} \end{aligned}$$

At $V_B = 2.5\ \text{V}$, the maximum common-mode input voltage is,

$$\begin{aligned} v_{icm\text{-}max} &= 2(2.5\ \text{V}) \\ &= 5\ \text{V} \end{aligned}$$

Therefore, the range of common-mode input voltage, v_{icm} is, $\boxed{-5\ \text{V} < v_{icm} < 5\ \text{V}}$.

Step 5 of 11

(c)

Connect a $10\ \text{k}\Omega$ resistor between node A and ground, another $10\ \text{k}\Omega$ resistor between node B and ground.

Draw the modified circuit.

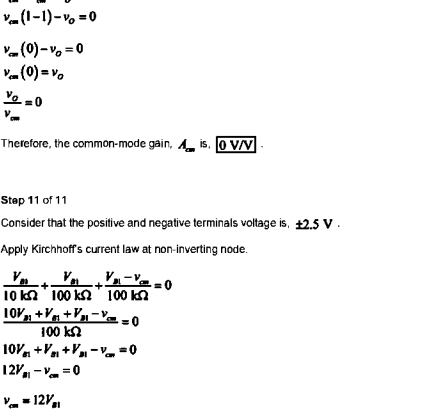


Figure 2

Step 6 of 11

Apply Kirchhoff's current law at non-inverting node.

$$\begin{aligned} &\frac{V_B}{10\ \text{k}\Omega} + \frac{V_B}{100\ \text{k}\Omega} + \frac{V_B - v_{i2}}{100\ \text{k}\Omega} = 0 \\ &\frac{10V_B + V_B + V_B - v_{i2}}{100\ \text{k}\Omega} = 0 \\ &10V_B + V_B + V_B - v_{i2} = 0 \\ &12V_B - v_{i2} = 0 \\ &12V_B = v_{i2} \\ &V_B = \frac{v_{i2}}{12} \end{aligned}$$

The inverting node voltage due to virtual ground concept is,

$$V_A = \frac{v_{i2}}{12} \qquad (\text{since, } V_A = V_B)$$

Step 7 of 11

Apply Kirchhoff's current law at inverting node.

$$\begin{aligned} &\frac{V_A}{10\ \text{k}\Omega} + \frac{V_A - v_{i1}}{100\ \text{k}\Omega} + \frac{V_B - v_O}{100\ \text{k}\Omega} = 0 \\ &\frac{10V_A + V_A - v_{i1} + V_B - v_O}{100\ \text{k}\Omega} = 0 \\ &10V_A + V_A - v_{i1} + V_B - v_O = 0 \\ &10V_A + V_A - v_{i1} + V_A - v_O = 0 \quad (\text{since, } V_A = V_B) \\ &12V_A - v_{i1} - v_O = 0 \\ &12\left(\frac{v_{i2}}{12}\right) - v_{i1} - v_O = 0 \quad \left(\text{since, } V_A = \frac{v_{i2}}{12}\right) \\ &v_{i2} - v_{i1} - v_O = 0 \\ &v_O = v_{i2} - v_{i1} \end{aligned}$$

Find the value of differential gain, A_d.

$$\begin{aligned} A_d &= \frac{v_O}{v_{i2} - v_{i1}} \\ &= 1 \end{aligned}$$

Therefore, the differential gain, A_d is, $\boxed{1\ \text{V/V}}$.

Step 8 of 11

Connect a common-mode voltage to the inputs of the op-amp and draw the modified circuit.

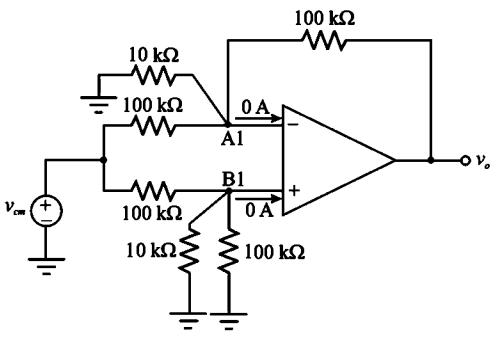


Figure 3

Step 9 of 11

Apply Kirchhoff's current law at non-inverting node.

$$\begin{aligned} &\frac{V_{B1}}{10\ \text{k}\Omega} + \frac{V_{B1}}{100\ \text{k}\Omega} + \frac{V_{B1} - v_{cm}}{100\ \text{k}\Omega} = 0 \\ &\frac{10V_{B1} + V_{B1} + V_{B1} - v_{cm}}{100\ \text{k}\Omega} = 0 \\ &10V_{B1} + V_{B1} + V_{B1} - v_{cm} = 0 \\ &12V_{B1} - v_{cm} = 0 \\ &12V_{B1} = v_{cm} \\ &V_{B1} = \frac{v_{cm}}{12} \end{aligned}$$

The inverting node voltage due to virtual ground concept is,

$$V_{A1} = \frac{v_{cm}}{12} \qquad (\text{since, } V_{A1} = V_{B1})$$

Step 10 of 11

Apply Kirchhoff's current law at inverting node.

$$\begin{aligned} &\frac{V_{A1}}{10\ \text{k}\Omega} + \frac{V_{A1} - v_{cm}}{100\ \text{k}\Omega} + \frac{V_{A1} - v_O}{100\ \text{k}\Omega} = 0 \\ &\frac{10V_{A1} + V_{A1} - v_{cm} + V_{B1} - v_O}{100\ \text{k}\Omega} = 0 \\ &10V_{A1} + V_{A1} - v_{cm} + V_{B1} - v_O = 0 \\ &10V_{A1} + V_{A1} - v_{cm} + V_{A1} - v_O = 0 \ (\text{since, } V_{A1} = V_{B1}) \\ &12V_{A1} - v_{cm} - v_O = 0 \\ &12\left(\frac{v_{cm}}{12}\right) - v_{cm} - v_O = 0 \quad \left(\text{since, } V_{A1} = \frac{v_{cm}}{12}\right) \\ &v_{cm} - v_{cm} - v_O = 0 \\ &v_{cm}(1 - 1) - v_O = 0 \\ &v_{cm}(0) - v_O = 0 \\ &v_{cm}(0) = v_O \\ &\frac{v_O}{v_{cm}} = 0 \end{aligned}$$

Therefore, the common-mode gain, A_{cm} is, $\boxed{0\ \text{V/V}}$.

Step 11 of 11

Consider that the positive and negative terminals voltage is, $\pm 2.5\ \text{V}$.

Apply Kirchhoff's current law at non-inverting node.

$$\begin{aligned} &\frac{V_{B1}}{10\ \text{k}\Omega} + \frac{V_{B1}}{100\ \text{k}\Omega} + \frac{V_{B1} - v_{cm}}{100\ \text{k}\Omega} = 0 \\ &\frac{10V_{B1} + V_{B1} + V_{B1} - v_{cm}}{100\ \text{k}\Omega} = 0 \\ &10V_{B1} + V_{B1} + V_{B1} - v_{cm} = 0 \\ &12V_{B1} - v_{cm} = 0 \\ &v_{cm} = 12V_{B1} \end{aligned}$$

At $V_{B1} = -2.5\ \text{V}$, the minimum common-mode input voltage is,

$$\begin{aligned} v_{icm\text{-}min} &= 12(-2.5) \\ &= -30\ \text{V} \end{aligned}$$

At $V_{B1} = 2.5\ \text{V}$, the maximum common-mode input voltage is,

$$\begin{aligned} v_{icm\text{-}max} &= 12(2.5) \\ &= 30\ \text{V} \end{aligned}$$

Therefore, the range of common-mode input voltage, v_{icm} is, $\boxed{-30\ \text{V} < v_{icm} < 30\ \text{V}}$.

Step 1 of 7

2.069P

Refer to Figure P2.69 in the textbook.

The inverting and non-inverting terminal currents are zero in an ideal op-amp. Draw the circuit with the node voltages and currents.

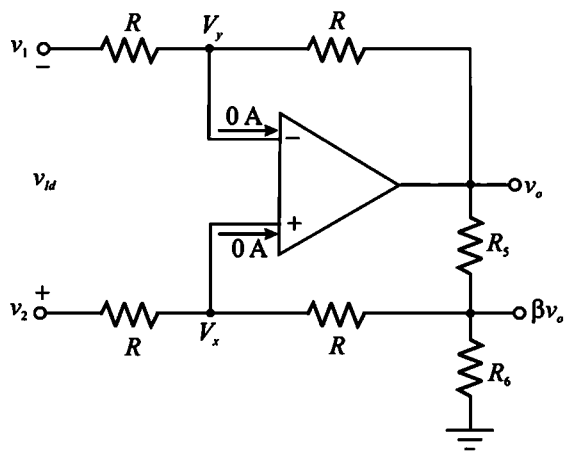


Figure 1

Step 2 of 7

Apply Kirchhoff's current law at non-inverting node, V_x.

$$\frac{V_x - v_2}{R} + \frac{V_x - \beta v_o}{R} = 0$$
$$\frac{V_x - v_2 + V_x - \beta v_o}{R} = 0$$
$$V_x - v_2 + V_x - \beta v_o = 0$$
$$2V_x - v_2 - \beta v_o = 0$$

$$2V_x = v_2 + \beta v_o$$
$$V_x = \frac{v_2 + \beta v_o}{2}$$

The inverting voltage, V_y due to virtual ground concept is,

$$V_y = \frac{v_2 + \beta v_o}{2} \quad (\text{since, } V_x = V_y)$$

Step 3 of 7

Apply Kirchhoff's current law at inverting node, V_y.

$$\frac{V_y - v_1}{R} + \frac{V_y - v_o}{R} = 0$$
$$\frac{V_y - v_1 + V_y - v_o}{R} = 0$$
$$V_y - v_1 + V_y - v_o = 0$$
$$2V_y - v_1 - v_o = 0$$

Substitute the expression for V_y in the equation.

$$2\left(\frac{v_2 + \beta v_o}{2}\right) - v_1 - v_o = 0$$
$$v_2 + \beta v_o - v_1 - v_o = 0$$
$$v_2 - v_1 - v_o(1 - \beta) = 0$$

$$v_2 - v_1 = v_o(1 - \beta)$$
$$\frac{v_o}{v_2 - v_1} = \frac{1}{1 - \beta}$$

The differential voltage gain is,

$$A_d = \frac{v_o}{v_{id}}$$
$$= \frac{v_o}{v_2 - v_1}$$
$$= \frac{1}{1 - \beta}$$

Therefore, the differential gain, A_d is, $\boxed{A_d = \frac{1}{1 - \beta}}$.

Step 4 of 7

Consider the following circuit to determine the resistor values.

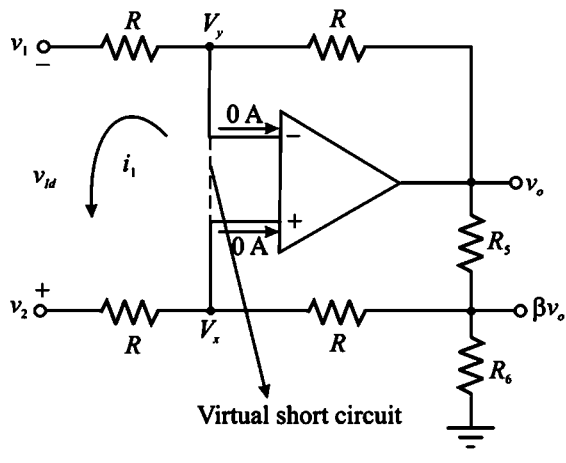


Figure 2

Step 5 of 7

Apply Kirchhoff's voltage law to the circuit to find the differential input resistance.

$$-0 + i_1 R - v_{id} + i_1 R = 0$$
$$i_1(R + R) - v_{id} = 0$$
$$v_{id} = i_1(R + R)$$
$$v_{id} = i_1 2R$$

Therefore, the differential input resistance, R_{id} is,

$$R_{id} = \frac{v_{id}}{i_1}$$
$$= 2R$$

Substitute $2\ \text{M}\Omega$ for R_{id} in the equation.

$$R = \frac{2\ \text{M}\Omega}{2}$$
$$= 1\ \text{M}\Omega$$

Therefore, the value of resistance, R is, $\boxed{1\ \text{M}\Omega}$.

Step 6 of 7

Recall the expression for the differential gain.

$$A_d = \frac{1}{1 - \beta}$$

Substitute 10 V/V for A_d in the equation.

$$10 = \frac{1}{1 - \beta}$$
$$1 - \beta = 0.1$$

$$\beta = 1 - 0.1$$
$$= 0.9$$

Consider the following due to voltage division method applied at resistance, R_6.

$$\beta = \frac{R_6}{R_5 + R_6}$$

Substitute 0.9 for β in the equation.

$$0.9 = \frac{R_6}{R_5 + R_6}$$
$$0.9(R_5 + R_6) = R_6$$
$$0.9R_5 + 0.9R_6 = R_6$$
$$0.9R_5 = R_6 - 0.9R_6$$

$$0.9R_5 = 0.1R_6$$
$$R_6 = 9R_5$$

Step 7 of 7

Consider the following condition to find the resistor values:

$$(R_5 + R_6) \le \frac{R}{100}$$

Substitute $1\ \text{M}\Omega$ for R, and the expression for R_6.

$$(R_5 + 9R_5) \le \frac{1\ \text{M}\Omega}{100}$$
$$10R_5 \le 10\ \text{k}\Omega$$
$$R_5 \le 1\ \text{k}\Omega$$

Substitute $1\ \text{k}\Omega$ for R_5 in the equation for R_6.

$$R_6 = 9R_5$$
$$= 9(1\ \text{k}\Omega)$$
$$= 9\ \text{k}\Omega$$

Therefore, the values of the resistors in the design of the circuit are,

$$\boxed{\begin{aligned} R &= 1\ \text{M}\Omega \\ R_5 &= 1\ \text{k}\Omega \\ R_6 &= 9\ \text{k}\Omega \end{aligned}}.$$

Step 1 of 6

2.070P

Refer to Figure P6.70 in the textbook for a modified version of a differential amplifier.

The inverting and non-inverting terminal currents are zero in an ideal op-amp, and due to virtual short circuit the inverting and non-inverting node voltages are equal.

Draw the circuit with the node voltages and currents.

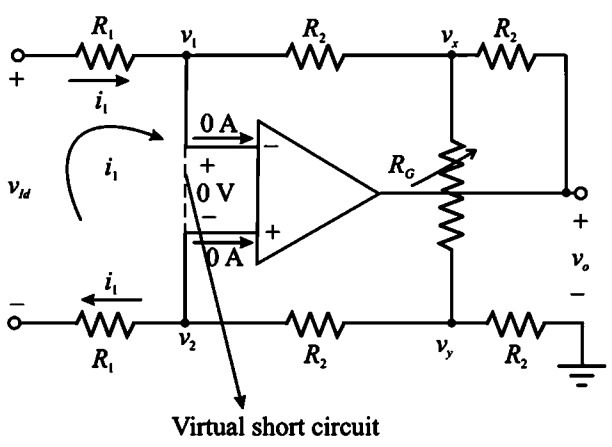


Figure 1

Step 2 of 6

Apply Kirchhoff's voltage law to left side mesh in the circuit.

$$-v_{id}+i_1R_1+i_1R_1+0=0$$
$$-v_{id}+i_1\left(R_1+R_1\right)=0$$

$$-v_{id}+i_1\left(2R_1\right)=0$$
$$i_1\left(2R_1\right)=v_{id}$$
$$i_1=\frac{v_{id}}{2R_1}$$

Therefore, the current through the resistance, R_1 is,

$$i_1=\frac{v_{id}}{2R_1}$$

Apply Kirchhoff's current law at node, v_2.

$$i_1+\frac{v_2-v_y}{R_2}+0=0$$
$$\frac{i_1R_2+v_2-v_y}{R_2}=0$$
$$i_1R_2+v_2-v_y=0$$

Substitute the expression for the current, i_1 in the equation.

$$\left(\frac{v_{id}}{2R_1}\right)R_2+v_2-v_y=0$$
$$v_2=v_y-\left(\frac{v_{id}}{2R_1}\right)R_2$$

Step 3 of 6

Apply Kirchhoff's current law at node, v_1.

$$-i_1+\frac{v_1-v_x}{R_2}+0=0$$
$$\frac{-i_1R_2+v_1-v_x}{R_2}=0$$
$$-i_1R_2+v_1-v_x=0$$

Substitute the expression for the current, i_1 in the equation.

$$-\left(\frac{v_{id}}{2R_1}\right)R_2+v_1-v_x=0$$
$$v_1=v_x+\left(\frac{v_{id}}{2R_1}\right)R_2$$

According to virtual ground concept, the inverting node voltage is equal to the non-inverting node voltage.

$$v_1=v_2$$
$$v_x+\left(\frac{v_{id}}{2R_1}\right)R_2=v_y-\left(\frac{v_{id}}{2R_1}\right)R_2$$
$$v_x-v_y=-\left(\frac{v_{id}}{2R_1}\right)R_2-\left(\frac{v_{id}}{2R_1}\right)R_2$$
$$v_x-v_y=-2\left(\frac{v_{id}}{2R_1}\right)R_2$$
$$v_x-v_y=-v_{id}\left(\frac{R_2}{R_1}\right)$$

Step 4 of 6

Apply Kirchhoff's current law at node, v_y.

$$\frac{v_y-v_x}{R_G}+\frac{v_y-v_2}{R_2}+\frac{v_y}{R_2}=0$$
$$\frac{v_{id}\left(\frac{R_2}{R_1}\right)}{R_G}+\frac{v_y}{R_2}-\frac{v_2}{R_2}+\frac{v_y}{R_2}=0 \quad \left(\text{since, } v_x-v_y=-v_{id}\left(\frac{R_2}{R_1}\right)\right)$$
$$v_{id}\left(\frac{R_2}{R_1R_G}\right)+2\frac{v_y}{R_2}-\frac{1}{R_2}\left(v_y-\left(\frac{v_{id}}{2R_1}\right)R_2\right)=0 \quad \left(\text{since, } v_2=v_y-\left(\frac{v_{id}}{2R_1}\right)R_2\right)$$
$$v_{id}\left(\frac{R_2}{R_1R_G}\right)+2\frac{v_y}{R_2}-\frac{v_y}{R_2}+\frac{v_{id}}{2R_1}=0$$

$$v_{id}\left(\frac{R_2}{R_1R_G}+\frac{1}{2R_1}\right)+\frac{v_y}{R_2}=0$$
$$\frac{v_y}{R_2}=-v_{id}\left(\frac{R_2}{R_1R_G}+\frac{1}{2R_1}\right)$$
$$v_y=-v_{id}R_2\left(\frac{R_2}{R_1R_G}+\frac{1}{2R_1}\right)$$

Step 5 of 6

Apply Kirchhoff's current law at node, v_x.

$$\frac{v_x-v_1}{R_2}+\frac{v_x-v_y}{R_G}+\frac{v_x-v_o}{R_2}=0$$
$$\frac{v_x}{R_2}-\frac{v_1}{R_2}+\frac{v_x-v_y}{R_G}+\frac{v_x}{R_2}-\frac{v_o}{R_2}=0$$
$$2\frac{v_x}{R_2}-\frac{v_1}{R_2}-\frac{v_{id}\left(\frac{R_2}{R_1}\right)}{R_G}-\frac{v_o}{R_2}=0 \qquad \left(\text{since, } v_x-v_y=-v_{id}\left(\frac{R_2}{R_1}\right)\right)$$
$$2\frac{v_x}{R_2}-\frac{1}{R_2}\left(v_x+\left(\frac{v_{id}}{2R_1}\right)R_2\right)-v_{id}\left(\frac{R_2}{R_1R_G}\right)-\frac{v_o}{R_2}=0 \qquad \left(\text{since, } v_1=v_x+\left(\frac{v_{id}}{2R_1}\right)R_2\right)$$
$$2\frac{v_x}{R_2}-\frac{v_x}{R_2}-\left(\frac{v_{id}}{2R_1}\right)-v_{id}\left(\frac{R_2}{R_1R_G}\right)-\frac{v_o}{R_2}=0$$

$$\frac{v_o}{R_2}=\frac{v_x}{R_2}-\left(\frac{v_{id}}{2R_1}\right)-v_{id}\left(\frac{R_2}{R_1R_G}\right)$$
$$=\frac{v_x}{R_2}-v_{id}\left(\frac{R_2}{R_1R_G}+\frac{1}{2R_1}\right)$$

$$\frac{v_o}{R_2}=\frac{v_y-v_{id}\left(\frac{R_2}{R_1}\right)}{R_2}-v_{id}\left(\frac{R_2}{R_1R_G}+\frac{1}{2R_1}\right) \qquad \left(\text{since, } v_x-v_y=-v_{id}\left(\frac{R_2}{R_1}\right)\right)$$
$$=\frac{v_y}{R_2}-v_{id}\left(\frac{1}{R_1}\right)-v_{id}\left(\frac{R_2}{R_1R_G}+\frac{1}{2R_1}\right)$$
$$=\frac{v_y}{R_2}-v_{id}\left(\frac{R_2}{R_1R_G}+\frac{1}{2R_1}+\frac{1}{R_1}\right)$$

Step 6 of 6

Substitute the expression for v_y in the equation.

$$\frac{v_o}{R_2}=\frac{-v_{id}R_2\left(\frac{R_2}{R_1R_G}+\frac{1}{2R_1}\right)}{R_2}-v_{id}\left(\frac{R_2}{R_1R_G}+\frac{1}{2R_1}+\frac{1}{R_1}\right)$$
$$=-v_{id}\left(\frac{R_2}{R_1R_G}+\frac{1}{2R_1}\right)-v_{id}\left(\frac{R_2}{R_1R_G}+\frac{1}{2R_1}+\frac{1}{R_1}\right)$$
$$=-v_{id}\left[\left(\frac{R_2}{R_1R_G}+\frac{1}{2R_1}\right)+\left(\frac{R_2}{R_1R_G}+\frac{1}{2R_1}+\frac{1}{R_1}\right)\right]$$
$$=-v_{id}\left[2\frac{R_2}{R_1R_G}+\frac{1}{2R_1}+\frac{1}{2R_1}+\frac{1}{R_1}\right]$$

$$\frac{v_o}{R_2}=-v_{id}\left(2\frac{R_2}{R_1R_G}+\frac{1+1+2}{2R_1}\right)$$
$$=-v_{id}\left(2\frac{R_2}{R_1R_G}+\frac{4}{2R_1}\right)$$
$$=-v_{id}\left(2\frac{R_2}{R_1R_G}+\frac{2}{R_1}\right)$$
$$=-v_{id}2\frac{1}{R_1}\left(1+\frac{R_2}{R_G}\right)$$

Simplify the expression further to get an expression for the differential gain, A_d.

$$A_d=\frac{v_o}{v_{id}}$$
$$=-2\frac{R_2}{R_1}\left(1+\frac{R_2}{R_G}\right)$$

Therefore, the differential voltage gain, A_d is, $\boxed{A_d=-2\frac{R_2}{R_1}\left(1+\frac{R_2}{R_G}\right)}$.

Step 1 of 13

(a)

Refer to circuit diagram in Figure P2.71 in the textbook.

Short circuit the terminals, C and O and connect terminal D to ground. The modified circuit is shown in Figure 1.

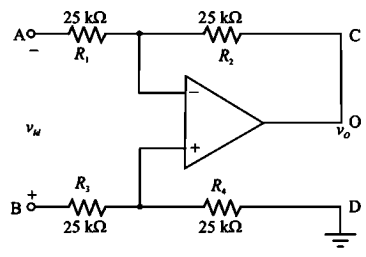


Figure 1

Step 2 of 13

Calculate the differential gain.

$$A_d = \frac{v_O}{v_M}$$
$$= \frac{R_2}{R_1}$$
$$= \frac{25\ \text{k}\Omega}{25\ \text{k}\Omega}$$
$$= 1\ \text{V/V}$$

The differential gain is unity gain.

Hence, it is proved.

Step 3 of 13

(b)

(i)

Inverting amplifier provides negative gain, so connect voltage source to terminal A, short circuit the terminals C and O and connect terminals B and D to ground, the modified circuit is shown in Figure 2.

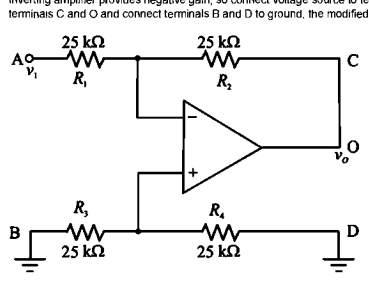


Figure 2

Step 4 of 13

Calculate the inverting amplifier gain.

$$A_d = \frac{v_o}{v_1}$$
$$= -\frac{R_2}{R_1}$$
$$= -\frac{25\ \text{k}\Omega}{25\ \text{k}\Omega}$$
$$= -1\ \text{V/V}$$

Therefore, the single-ended differential gain, A_d is $\boxed{-1\ \text{V/V}}$.

Step 5 of 13

(ii)

Non-inverting amplifier circuit provides positive gain, so connect voltage source to B terminal, short circuited the terminals C and O and connect terminals A and D to ground, the modified circuit is shown in Figure 3.

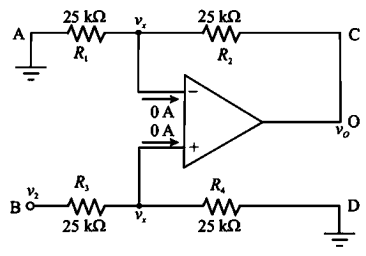


Figure 3

Step 6 of 13

In an ideal op-amp, the inverting and non-inverting terminal currents are zero, and the voltages at inverting and non-inverting node are equal.

Calculate the non-inverting node voltage, v_x using voltage division rule.

$$v_x = \left[\frac{R_4}{R_3 + R_4}\right] v_2$$
$$= \left[\frac{25\ \text{k}\Omega}{25\ \text{k}\Omega + 25\ \text{k}\Omega}\right] v_2$$
$$= \left[\frac{25\ \text{k}\Omega}{50\ \text{k}\Omega}\right] v_2$$
$$= 0.5 v_2$$

Step 7 of 13

Apply Kirchhoff's current law at inverting node.

$$\frac{v_x}{25\ \text{k}\Omega} + \frac{v_x - v_o}{25\ \text{k}\Omega} = 0$$
$$2v_x - v_o = 0$$
$$v_o = 2v_x$$

Substitute $0.5v_2$ for v_x in the equation.

$$v_o = 2(0.5v_2)$$
$$v_o = v_2$$
$$\frac{v_o}{v_2} = 1\ \text{V/V}$$
$$A_d = 1\ \text{V/V}$$

Therefore, the single-ended differential gain, A_d is $\boxed{+1\ \text{V/V}}$.

Step 8 of 13

(iii)

Non-inverting amplifier circuit provides positive gain, so connect voltage source to terminal B and terminal D, short circuited the terminals C and O and connect terminal A to ground, the modified circuit is shown in Figure 4.

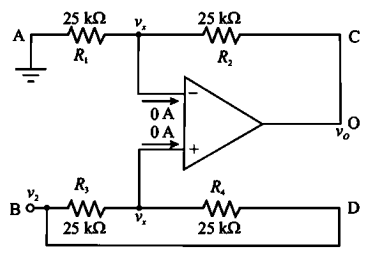


Figure 4

Step 9 of 13

In an ideal op-amp, the inverting and non-inverting terminal currents are zero, and the voltages at inverting and non-inverting node are equal.

Apply Kirchhoff's current law at non-inverting terminal.

$$\frac{v_x - v_2}{25\ \text{k}\Omega} + \frac{v_x - v_2}{25\ \text{k}\Omega} = 0$$
$$2v_x - 2v_2 = 0$$
$$2v_x = 2v_2$$
$$v_x = v_2$$

Step 10 of 13

Apply Kirchhoff's current law inverting node.

$$\frac{v_x}{25\ \text{k}\Omega} + \frac{v_x - v_o}{25\ \text{k}\Omega} = 0$$
$$v_x + v_x - v_o = 0$$
$$2v_x - v_o = 0$$
$$v_o = 2v_x$$

Substitute v_2 for v_x in the equation.

$$v_o = 2v_2$$
$$\frac{v_o}{v_2} = 2$$
$$A_d = 2\ \text{V/V}$$

Therefore, the single-ended differential gain, A_d is $\boxed{+2\ \text{V/V}}$.

Step 11 of 13

(iv)

Non-inverting amplifier circuit provides positive gain, so connect voltage source to terminal B, short circuited the terminals C and O and connect terminals A and D to ground, the modified circuit is shown in Figure 5.

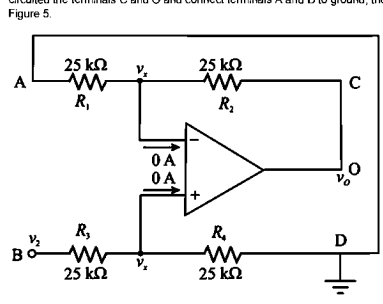


Figure 5

Step 12 of 13

In an ideal op-amp, the inverting and non-inverting terminal currents are zero, and the voltages at inverting and non-inverting node are equal.

Calculate the non-inverting node voltage, v_x using voltage division rule.

$$v_x = \left[\frac{R_4}{R_3 + R_4}\right] v_2$$
$$= \left[\frac{25\ \text{k}\Omega}{25\ \text{k}\Omega + 25\ \text{k}\Omega}\right] v_2$$
$$= \left[\frac{25\ \text{k}\Omega}{50\ \text{k}\Omega}\right] v_2$$
$$= 0.5 v_2$$

Step 13 of 13

Apply Kirchhoff's current law at inverting node.

$$\frac{v_x - v_2}{25\ \text{k}\Omega} + \frac{v_x - v_2}{25\ \text{k}\Omega} = 0$$
$$2v_x - 2v_2 = 0$$
$$2v_x = 2v_2$$
$$v_x = v_2$$

Substitute $0.5v_2$ for v_x in the equation.

$$v_o = 0.5v_2$$
$$\frac{v_o}{v_2} = 0.5$$
$$A_d = \frac{1}{2}\ \text{V/V}$$

Therefore, the single-ended differential gain, A_d is $\boxed{+\frac{1}{2}\ \text{V/V}}$.

2.072P

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Step 1 of 13

(a)

Refer to instrumentation amplifier in Figure 2.20 (a) in the textbook.

Write the expression for the input signals, v_{i1} and v_{i2} in terms of differential and common-mode components.

$$v_{i1} = v_{icm} - \frac{v_{id}}{2}$$

And,

$$v_{i2} = v_{icm} + \frac{v_{id}}{2}$$

Write the expression for the output voltage of first op-amp.

$$v_{O1} = \left(1 + \frac{R_2}{R_1}\right) v_{i1}$$

Substitute $v_{icm} - \frac{v_{id}}{2}$ for v_{i1} in the equation.

$$v_{O1} = \left(1 + \frac{R_2}{R_1}\right)\left(v_{icm} - \frac{v_{id}}{2}\right)$$

Therefore, the output voltage, v_{O1} is $\boxed{\left(1 + \frac{R_2}{R_1}\right)\left(v_{icm} - \frac{v_{id}}{2}\right)}$

Step 2 of 13

Write the expression for the output voltage of second op-amp.

$$v_{O2} = \left(1 + \frac{R_2}{R_1}\right) v_{i2}$$

Substitute $v_{icm} + \frac{v_{id}}{2}$ for v_{i2} in the equation.

$$v_{O2} = \left(1 + \frac{R_2}{R_1}\right)\left(v_{icm} + \frac{v_{id}}{2}\right)$$

Therefore, the output voltage, v_{O2} is $\boxed{\left(1 + \frac{R_2}{R_1}\right)\left(v_{icm} + \frac{v_{id}}{2}\right)}$

Step 3 of 13

Calculate the differential component $v_{O2} - v_{O1}$.

$$\begin{aligned} v_{O2} - v_{O1} &= \left[\left(1 + \frac{R_2}{R_1}\right)\left(v_{icm} + \frac{v_{id}}{2}\right)\right] - \left[\left(1 + \frac{R_2}{R_1}\right)\left(v_{icm} - \frac{v_{id}}{2}\right)\right] \\ &= \left(1 + \frac{R_2}{R_1}\right)\left[\left(v_{icm} + \frac{v_{id}}{2}\right) - \left(v_{icm} - \frac{v_{id}}{2}\right)\right] \\ &= \left(1 + \frac{R_2}{R_1}\right)\left[v_{icm} + \frac{v_{id}}{2} - v_{icm} + \frac{v_{id}}{2}\right] \\ &= \left(1 + \frac{R_2}{R_1}\right)\left[\frac{v_{id}}{2} + \frac{v_{id}}{2}\right] \\ &= \left(1 + \frac{R_2}{R_1}\right)\left[2\frac{v_{id}}{2}\right] \\ &= \left(1 + \frac{R_2}{R_1}\right) v_{id} \end{aligned}$$

Therefore, the differential component, $v_{O2} - v_{O1}$ is $\boxed{\left(1 + \frac{R_2}{R_1}\right) v_{id}}$.

Step 4 of 13

Calculate the common-mode component.

$$\begin{aligned} \frac{1}{2}(v_{O1} + v_{O2}) &= \frac{1}{2}\left(\left[\left(1 + \frac{R_2}{R_1}\right)\left(v_{icm} - \frac{v_{id}}{2}\right)\right] + \left[\left(1 + \frac{R_2}{R_1}\right)\left(v_{icm} + \frac{v_{id}}{2}\right)\right]\right) \\ &= \frac{1}{2}\left(1 + \frac{R_2}{R_1}\right)\left(\left(v_{icm} - \frac{v_{id}}{2}\right) + \left(v_{icm} + \frac{v_{id}}{2}\right)\right) \\ &= \frac{1}{2}\left(1 + \frac{R_2}{R_1}\right)(v_{icm} + v_{icm}) \\ &= \left(1 + \frac{R_2}{R_1}\right) v_{icm} \end{aligned}$$

Therefore, the common-mode component, $\frac{1}{2}(v_{O1} + v_{O2})$ is $\boxed{\left(1 + \frac{R_2}{R_1}\right) v_{icm}}$.

Step 5 of 13

Calculate the differential gain.
Recall differential component expression.

$$v_{O2} - v_{O1} = \left(1 + \frac{R_2}{R_1}\right) v_{id}$$

$$\frac{v_{O2} - v_{O1}}{v_{id}} = 1 + \frac{R_2}{R_1}$$

$$A_d = 1 + \frac{R_2}{R_1}$$

Therefore, the differential gain, A_d is $\boxed{1 + \frac{R_2}{R_1}}$.

Calculate the common mode gain.

Recall the common-mode component expression.

$$\frac{1}{2}(v_{O1} + v_{O2}) = \left(1 + \frac{R_2}{R_1}\right) v_{icm}$$

$$\frac{1}{2}\left(\frac{v_{O1} + v_{O2}}{v_{icm}}\right) = \left(1 + \frac{R_2}{R_1}\right)$$

$$A_{cm} = 1 + \frac{R_2}{R_1}$$

Therefore, the common-mode gain, A_{cm} is $\boxed{1 + \frac{R_2}{R_1}}$.

Step 6 of 13

Calculate the common-mode rejection ratio.

$$\mathbf{CMRR = 20\log\left(\frac{A_d}{A_{cm}}\right)}$$

Substitute $1 + \frac{R_2}{R_1}$ for A_d, and $1 + \frac{R_2}{R_1}$ for A_{cm} in the equation.

$$\begin{aligned} \mathbf{CMRR} &= \mathbf{20\log\left(\frac{1 + \frac{R_2}{R_1}}{1 + \frac{R_2}{R_1}}\right)} \\ &= \mathbf{20\log(1)} \\ &= \mathbf{0} \end{aligned}$$

Therefore, the common-mode rejection ration is $\boxed{\mathbf{0\ dB}}$.

Step 7 of 13

(b)

Refer to instrumentation amplifier in Figure 2.20 (b) in the textbook.

In an ideal op-amp, the inverting and non-inverting terminal currents are zero and the inverting and non-inverting node voltages are equal. The modified circuit is shown in Figure 1.

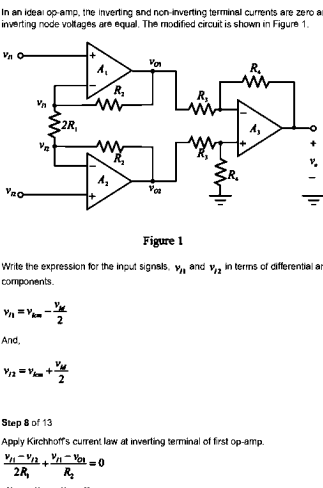


Figure 1

Write the expression for the input signals, v_{i1} and v_{i2} in terms of differential and common-mode components.

$$v_{i1} = v_{icm} - \frac{v_{id}}{2}$$

And,

$$v_{i2} = v_{icm} + \frac{v_{id}}{2}$$

Step 8 of 13

Apply Kirchhoff's current law at inverting terminal of first op-amp.

$$\begin{aligned} \frac{v_{i1} - v_{i2}}{2R_1} + \frac{v_{i1} - v_{O1}}{R_2} &= 0 \\ \frac{v_{i1}}{2R_1} - \frac{v_{i2}}{2R_1} + \frac{v_{i1}}{R_2} - \frac{v_{O1}}{R_2} &= 0 \\ \left(\frac{1}{2R_1} + \frac{1}{R_2}\right) v_{i1} - \frac{v_{i2}}{2R_1} - \frac{v_{O1}}{R_2} &= 0 \end{aligned}$$

Substitute $v_{icm} - \frac{v_{id}}{2}$ for v_{i1}, and $v_{icm} + \frac{v_{id}}{2}$ for v_{i2} in the equation.

$$\begin{aligned} \left(\frac{1}{2R_1} + \frac{1}{R_2}\right)\left(v_{icm} - \frac{v_{id}}{2}\right) - \left(\frac{1}{2R_1}\right)\left(v_{icm} + \frac{v_{id}}{2}\right) &= \frac{v_{O1}}{R_2} \\ \frac{v_{icm}}{2R_1} + \frac{v_{icm}}{R_2} - \frac{v_{id}}{4R_1} - \frac{v_{id}}{2R_2} - \frac{v_{icm}}{2R_1} - \frac{v_{id}}{4R_1} &= \frac{v_{O1}}{R_2} \\ \frac{v_{icm}}{R_2} - \left(\frac{1}{R_1} + \frac{1}{R_2}\right)\frac{v_{id}}{2} &= \frac{v_{O1}}{R_2} \\ v_{O1} &= v_{icm} - \frac{v_{id}}{2}\left(1 + \frac{R_2}{R_1}\right) \end{aligned}$$

Therefore, the output voltage of first op-amp, v_{O1} is $\boxed{v_{icm} - \frac{v_{id}}{2}\left(1 + \frac{R_2}{R_1}\right)}$.

Step 9 of 13

Apply Kirchhoff's current law at inverting terminal of first op-amp.

$$\begin{aligned} \frac{v_{i2} - v_{i1}}{2R_1} + \frac{v_{i2} - v_{O2}}{R_2} &= 0 \\ \frac{v_{i2}}{2R_1} - \frac{v_{i1}}{2R_1} + \frac{v_{i2}}{R_2} - \frac{v_{O2}}{R_2} &= 0 \\ \left(\frac{1}{2R_1} + \frac{1}{R_2}\right) v_{i2} - \frac{v_{i1}}{2R_1} - \frac{v_{O2}}{R_2} &= 0 \end{aligned}$$

Substitute $v_{icm} - \frac{v_{id}}{2}$ for v_{i1}, and $v_{icm} + \frac{v_{id}}{2}$ for v_{i2} in the equation.

$$\begin{aligned} \frac{v_{O2}}{R_2} &= \left(\frac{1}{2R_1} + \frac{1}{R_2}\right)\left(v_{icm} + \frac{v_{id}}{2}\right) - \left(\frac{1}{2R_1}\right)\left(v_{icm} - \frac{v_{id}}{2}\right) \\ \frac{v_{O2}}{R_2} &= \frac{v_{icm}}{2R_1} + \frac{v_{icm}}{R_2} + \frac{v_{id}}{4R_1} + \frac{v_{id}}{2R_2} - \frac{v_{icm}}{2R_1} + \frac{v_{id}}{4R_1} \\ \frac{v_{O2}}{R_2} &= \frac{v_{icm}}{R_2} + \frac{v_{id}}{2R_1} + \frac{v_{id}}{2R_2} \\ v_{O2} &= v_{icm} + \frac{v_{id}}{2}\left(1 + \frac{R_2}{R_1}\right) \end{aligned}$$

Therefore, the output voltage of second op-amp, v_{O2} is $\boxed{v_{icm} + \frac{v_{id}}{2}\left(1 + \frac{R_2}{R_1}\right)}$.

Step 10 of 13

Calculate the differential component output voltage $(v_{O2} - v_{O1})$.

$$\begin{aligned} v_{O2} - v_{O1} &= \left[v_{icm} + \frac{v_{id}}{2}\left(1 + \frac{R_2}{R_1}\right)\right] - \left[v_{icm} - \frac{v_{id}}{2}\left(1 + \frac{R_2}{R_1}\right)\right] \\ &= v_{icm} + \frac{v_{id}}{2}\left(1 + \frac{R_2}{R_1}\right) - v_{icm} + \frac{v_{id}}{2}\left(1 + \frac{R_2}{R_1}\right) \\ &= 2\left(\frac{v_{id}}{2}\left(1 + \frac{R_2}{R_1}\right)\right) \\ &= v_{id}\left(1 + \frac{R_2}{R_1}\right) \end{aligned}$$

Therefore, the differential component output voltage, $(v_{O2} - v_{O1})$ is $\boxed{v_{id}\left(1 + \frac{R_2}{R_1}\right)}$.

Step 11 of 13

Calculate the common-mode component $\left(\frac{1}{2}(v_{O1} + v_{O2})\right)$

$$\begin{aligned} \frac{1}{2}(v_{O1} + v_{O2}) &= \frac{1}{2}\left(\left[v_{icm} - \frac{v_{id}}{2}\left(1 + \frac{R_2}{R_1}\right)\right] + \left[v_{icm} + \frac{v_{id}}{2}\left(1 + \frac{R_2}{R_1}\right)\right]\right) \\ &= \frac{1}{2}\left(v_{icm} - \frac{v_{id}}{2}\left(1 + \frac{R_2}{R_1}\right) + v_{icm} + \frac{v_{id}}{2}\left(1 + \frac{R_2}{R_1}\right)\right) \\ &= \frac{1}{2}(2v_{icm}) \\ &= v_{icm} \end{aligned}$$

Therefore, the common-mode component, $\frac{1}{2}(v_{O1} + v_{O2})$ is $\boxed{v_{icm}}$.

Step 12 of 13

Calculate the differential gain.
Recall differential component expression.

$$v_{O2} - v_{O1} = \left(1 + \frac{R_2}{R_1}\right) v_{id}$$

$$\frac{v_{O2} - v_{O1}}{v_{id}} = 1 + \frac{R_2}{R_1}$$

$$A_d = 1 + \frac{R_2}{R_1}$$

Therefore, the differential gain, A_d is $\boxed{1 + \frac{R_2}{R_1}}$.

Calculate the common mode gain.

Recall the common-mode component expression.

$$\frac{1}{2}(v_{O1} + v_{O2}) = v_{icm}$$

$$\frac{1}{2}\left(\frac{v_{O1} + v_{O2}}{v_{icm}}\right) = 1$$

$$A_{cm} = 1$$

Therefore, the common-mode gain, A_{cm} is $\boxed{1}$.

Step 13 of 13

Calculate the common-mode rejection ratio.

$$\mathbf{CMRR = 20\log\left(\frac{A_d}{A_{cm}}\right)}$$

Substitute $1 + \frac{R_2}{R_1}$ for A_d, and 1 for A_{cm} in the equation.

$$\begin{aligned} \mathbf{CMRR} &= \mathbf{20\log\left(\frac{1 + \frac{R_2}{R_1}}{1}\right)} \\ &= \mathbf{20\log\left(1 + \frac{R_2}{R_1}\right)\ dB} \end{aligned}$$

Therefore, the common-mode rejection ration is $\boxed{\mathbf{20\log\left(1 + \frac{R_2}{R_1}\right)\ dB}}$.

Therefore, the common mode voltage is not amplified only it is propagated to the outputs of the first stage.

Step 1 of 11

Refer to instrumentation amplifier in Figure 2.20 (b) in the textbook.

Write the expression for differential gain of the instrumentation amplifier.

$$A_d = \frac{R_4}{R_3}\left(1+\frac{2R_2}{2R_1}\right)$$

Substitute $10\ \text{k}\Omega$ for $2R_1$, $100\ \text{k}\Omega$ for R_2, $100\ \text{k}\Omega$ for R_3, and $100\ \text{k}\Omega$ for R_4 in the equation.

$$\begin{aligned} A_d &= \left(\frac{100\ \text{k}\Omega}{100\ \text{k}\Omega}\right)\left(1+\frac{2(100\ \text{k}\Omega)}{10\ \text{k}\Omega}\right) \\ &= 1+2(10) \\ &= 1+20 \\ &= 21\ \text{V/V} \end{aligned}$$

Therefore, the differential gain, A_d of the instrumentation amplifier is $\boxed{21\ \text{V/V}}$.

Step 2 of 11

From the circuit shown in Figure 2.20 (b) in the textbook, equal voltage appears at the both negative input terminals of op-amp A_1 and op-amp A_2, causing common-mode output voltage equal to zero.

$$A_{cm} = 0\ \text{V/V}$$

Therefore, the common-mode gain, A_{cm} of the instrumentation amplifier is $\boxed{0\ \text{V/V}}$.

Step 3 of 11

Write the expression for the common-mode-rejection-ratio (CMRR).

$$\text{CMRR} = 20\log\left(\frac{A_d}{A_{cm}}\right)$$

Substitute $21\ \text{V/V}$ for A_d and $0\ \text{V/V}$ for A_{cm} in the equation.

$$\begin{aligned} \text{CMRR} &= 20\log\left(\frac{21}{0}\right) \\ &= \infty \end{aligned}$$

Therefore, the common-mode-rejection-ratio (CMRR) is $\boxed{\infty}$.

Step 4 of 11

Now, reduce resistor values R_3 and R_4 by 1% and increase resistor values $2R_1$ and R_2 by 1%.

The resistor values are,

$$\begin{aligned} R_2 &= 0.99R_2 \\ R_4 &= 0.99R_4 \\ R_1 &= 1.01R_1 \\ R_3 &= 1.01R_3 \end{aligned}$$

Calculate the differential gain of the instrumentation amplifier.

$$\begin{aligned} A_d &= \frac{R_4}{R_3}\left(1+\frac{2R_2}{2R_1}\right) \\ &= \frac{0.99R_4}{1.01R_3}\left(1+\frac{2(0.99R_2)}{2(1.01R_1)}\right) \\ &= \frac{(0.99)(100\ \text{k}\Omega)}{(1.01)(100\ \text{k}\Omega)}\left(1+\frac{2(0.99)(100\ \text{k}\Omega)}{(1.01)(10\ \text{k}\Omega)}\right) \\ &= (0.98)\left(1+\frac{198}{10.1}\right) \\ &= (0.98)(1+19.6) \\ &= 20.19\ \text{V/V} \end{aligned}$$

Therefore, the differential gain, A_d of the instrumentation amplifier is $\boxed{20.19\ \text{V/V}}$.

Step 5 of 11

Consider the second state amplifier, the circuit is shown in Figure 1.

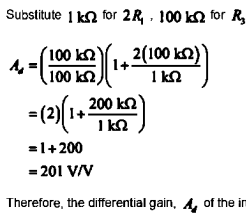


Figure 1

Step 6 of 11

Write the general common-mode gain formula.

$$A_{cm} = \frac{v_o}{v_{icm}} = \left(\frac{R_4}{R_4+R_3}\right)\left(1-\frac{R_2}{R_1}\frac{R_3}{R_4}\right)$$

$$A_{cm} = \left(\frac{1}{1+\frac{R_3}{R_4}}\right)\left(1-\frac{R_2}{R_1}\frac{R_3}{R_4}\right) \quad \ldots\ldots (1)$$

Assume all resistors be $100\ \Omega \pm x\%$.

The range of $\frac{R_2}{R_4}$ is $\frac{100\ \Omega - x\%}{100\ \Omega + x\%} < \frac{R_2}{R_4} < \frac{100\ \Omega + x\%}{100\ \Omega - x\%}$.

Similarly, the range of $\frac{R_2}{R_1}$ is $\frac{100\ \Omega - x\%}{100\ \Omega + x\%} < \frac{R_2}{R_1} < \frac{100\ \Omega + x\%}{100\ \Omega - x\%}$.

Step 7 of 11

To obtain maximum common mode gain, select the minimum values of $\frac{R_3}{R_4}$ and $\frac{R_2}{R_1}$.

Recall equation (1).

$$A_{cm-max} = \left(\frac{1}{1+\frac{R_3}{R_4}}\right)\left(1-\frac{R_2}{R_1}\frac{R_3}{R_4}\right)$$

Substitute $\frac{100\ \Omega - x\%}{100\ \Omega + x\%}$ for $\frac{R_2}{R_1}$ and $\frac{R_3}{R_4}$ in the equation.

$$\begin{aligned} A_{cm-max} &= \left(\frac{1}{1+\frac{100\ \Omega - x\%}{100\ \Omega + x\%}}\right)\left(1-\left(\frac{100\ \Omega - x\%}{100\ \Omega + x\%}\right)\left(\frac{100\ \Omega - x\%}{100\ \Omega + x\%}\right)\right) \\ &= \left(\frac{100\ \Omega + x\%}{100\ \Omega + x\% + 100\ \Omega - x\%}\right)\left(1-\left(\frac{100\ \Omega - x\%}{100\ \Omega + x\%}\right)^2\right) \\ &= \left(\frac{100\ \Omega + x\%}{200\ \Omega}\right)\left(\frac{(100\ \Omega + x\%)^2 - (100\ \Omega - x\%)^2}{(100\ \Omega + x\%)^2}\right) \\ &= \left(\frac{1}{200\ \Omega}\right)\left(\frac{200x\% + 200x\%}{100\ \Omega + x\%}\right) \\ &= \frac{2x\%}{100\ \Omega + x\%} \\ &\simeq \frac{2x\%}{100} \\ &\simeq \frac{x\%}{50} \end{aligned}$$

Therefore, the worst case common-mode gain, A_{cm-max} is $\boxed{\frac{x\%}{50}}$.

Step 8 of 11

Calculate the worst-case common-mode gain for $x = 1\%$.

$$\begin{aligned} A_{cm-max} &\simeq \frac{x\%}{50} \\ &= \frac{1}{50} \\ &= 0.02\ \text{V/V} \end{aligned}$$

Therefore, the worst-case common-mode gain is $\boxed{0.02\ \text{V/V}}$.

Step 9 of 11

Calculate the common-mode-rejection-ratio.

$$\text{CMRR} = 20\log\left(\frac{A_d}{A_{cm}}\right)$$

Substitute $20.19\ \text{V/V}$ for A_d, and $0.02\ \text{V/V}$ for A_{cm} in the equation.

$$\begin{aligned} \text{CMRR} &= 20\log\left(\frac{20.19}{0.02}\right) \\ &= 20\log(1009.5) \\ &= 20(3.004) \\ &= 60.08\ \text{dB} \end{aligned}$$

Therefore, the common-mode-rejection-ratio, CMRR is $\boxed{60.08\ \text{dB}}$.

Step 10 of 11

The value of resistor, $2R_1$ is reduced to $1\ \text{k}\Omega$.

Calculate the differential gain of the instrumentation amplifier.

$$A_d = \frac{R_4}{R_3}\left(1+\frac{2R_2}{2R_1}\right)$$

Substitute $1\ \text{k}\Omega$ for $2R_1$, $100\ \text{k}\Omega$ for R_3, $100\ \text{k}\Omega$ for R_2, and $100\ \text{k}\Omega$ for R_4 in the equation.

$$\begin{aligned} A_d &= \left(\frac{100\ \text{k}\Omega}{100\ \text{k}\Omega}\right)\left(1+\frac{2(100\ \text{k}\Omega)}{1\ \text{k}\Omega}\right) \\ &= (2)\left(1+\frac{200\ \text{k}\Omega}{1\ \text{k}\Omega}\right) \\ &= 1+200 \\ &= 201\ \text{V/V} \end{aligned}$$

Therefore, the differential gain, A_d of the instrumentation amplifier is $\boxed{201\ \text{V/V}}$.

Step 11 of 11

When, the value of resistor, $2R_1$ is reduced, the common-mode gain is same. That is,

$$A_{cm} = 0.02\ \text{V/V}$$

Therefore, the common-mode gain, A_{cm} is $\boxed{0.02\ \text{V/V}}$.

Calculate the common-mode-rejection-ratio.

$$\text{CMRR} = 20\log\left(\frac{A_d}{A_{cm}}\right)$$

Substitute $201\ \text{V/V}$ for A_d, and $0.02\ \text{V/V}$ for A_{cm} in the equation.

$$\begin{aligned} \text{CMRR} &= 20\log\left(\frac{201}{0.02}\right) \\ &= 20\log(10050) \\ &= 20(4) \\ &= 80\ \text{dB} \end{aligned}$$

Therefore, the common-mode-rejection-ratio, CMRR is $\boxed{80\ \text{dB}}$.

Therefore, for high differential gain in the stage 1, high CMRR is achieved.

2.076P

We don't have the solution to this problem yet.

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Step 1 of 6

2.077P

(a)

Refer to circuit diagram in Figure P2.77 in the textbook.

In an ideal op-amp, the inverting and non-inverting terminal currents are zero and the inverting and non-inverting node voltages are equal. The modified circuit is shown in Figure 1.

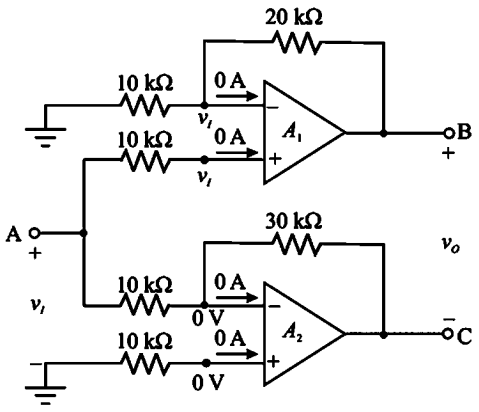


Figure 1

Step 2 of 6

The input voltage is,

$$v_I = 1\sin\omega t \text{ V}_{p\text{-}p}$$

The voltage at node A is,

$$V_A = v_I$$
$$= 1\sin\omega t \text{ V}_{p\text{-}p}$$

Apply Kirchhoff's current law at inverting terminal of the first op-amp.

$$\frac{v_I}{10\times10^3} + \frac{v_I - V_B}{20\times10^3} = 0$$
$$\frac{2v_I + v_I - V_B}{20\times10^3} = 0$$
$$3v_I - V_B = 0$$
$$V_B = 3v_I$$

Substitute $1\sin\omega t$ for v_I in the equation.

$$V_B = 3(\sin\omega t)$$
$$= 3\sin\omega t \text{ V}_{p\text{-}p}$$

Therefore, the node voltage, V_B is $\boxed{3\sin\omega t \text{ V}_{p\text{-}p}}$.

Step 3 of 6

Apply Kirchhoff's current law at inverting terminal of the second op-amp.

$$\frac{0 - v_I}{10\times10^3} + \frac{0 - V_C}{30\times10^3} = 0$$
$$\frac{-3v_I - V_C}{30\times10^3} = 0$$
$$V_C = -3v_I$$

Substitute $1\sin\omega t$ for v_I in the equation.

$$V_C = -3(\sin\omega t)$$
$$= -3\sin\omega t \text{ V}_{p\text{-}p}$$

Therefore, the node voltage, V_C is $\boxed{-3\sin\omega t \text{ V}_{p\text{-}p}}$.

Calculate the output voltage.

$$v_O = V_B - V_C$$
$$= 3\sin\omega t - (-3\sin\omega t)$$
$$= 3\sin\omega t + 3\sin\omega t$$
$$= 6\sin\omega t \text{ V}_{p\text{-}p}$$

Therefore, the output voltage, v_O is $\boxed{6\sin\omega t \text{ V}_{p\text{-}p}}$.

Step 4 of 6

The input node voltage, V_A and output node voltages V_B, V_C and v_O waveforms are shown in Figure 2.

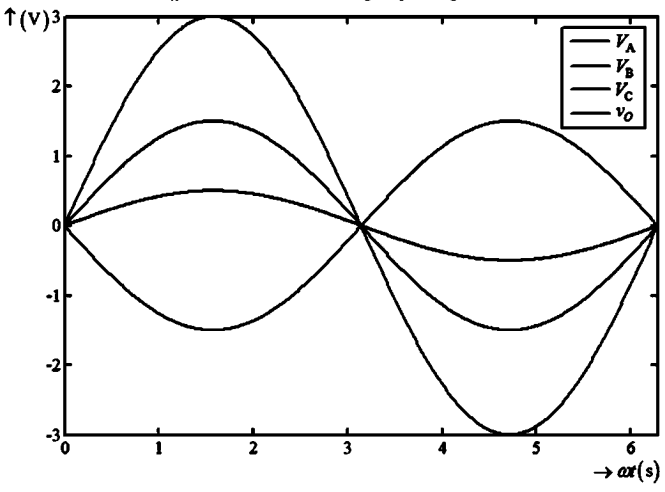


Figure 2

Step 5 of 6

(b)

Calculate the voltage gain.

$$A_v = \frac{v_O}{v_I}$$
$$= \frac{6\sin\omega t}{1\sin\omega t}$$
$$= 6 \text{ V/V}$$

Therefore, the voltage gain, A_v is $\boxed{6 \text{ V/V}}$.

Step 6 of 6

(c)

The op-amps outputs saturates at ± 14 V.

Let node voltage, V_B can be as large as 28 V peak to peak.

Similarly, node voltage, V_C is also 28 V peak to peak.

Calculate the largest sine wave output peak-to-peak voltage.

$$v_{O(p\text{-}p)} = 56 \text{ V}_{p\text{-}p}$$

Therefore, the largest sine wave output peak-to-peak voltage is $\boxed{56 \text{ V}_{p\text{-}p}}$.

Calculate the largest sine wave output rms voltage.

$$v_{O(rms)} = \frac{v_{(peak)}}{\sqrt{2}}$$
$$= \frac{28}{\sqrt{2}}$$
$$= 19.8 \text{ V}_{rms}$$

Therefore, the largest sine wave output rms voltage, $v_{O(rms)}$ is $\boxed{19.8 \text{ V}_{rms}}$.

Step 1 of 7

2.078P

(a)

Refer to circuit diagram in Figure P2.78 (a) in the textbook.

For an ideal op-map, the inverting and non-inverting terminal currents are zero and the inverting and non-inverting node voltages are equal.

Redraw the circuit in Figure P2.78 (a) to represent the currents and voltages.

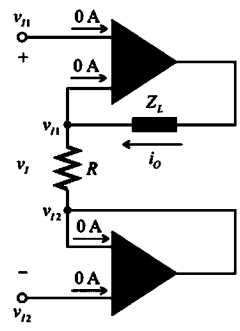


Figure 1

Step 2 of 7

Consider the following from the circuit in Figure 1.

$$v_I = v_{I1} - v_{I2}$$

Apply Kirchhoff's current law at the inverting terminal of first operational amplifier to get the output current.

$$i_O = \frac{v_{I1} - v_{I2}}{R} + 0$$

$$= \frac{v_{I1} - v_{I2}}{R}$$

Substitute $v_{I1} - v_{I2}$ for v_I.

$$i_O = \frac{v_I}{R} \quad \text{..... (1)}$$

Therefore, the output current, i_O is $\boxed{\frac{v_I}{R}}$.

Step 3 of 7

(b)

Refer to circuit diagram in Figure P2.78 (b) in the textbook.

For an ideal op-map, the inverting and non-inverting terminal currents are zero and the inverting and non-inverting node voltages are equal.

Redraw the circuit in Figure P2.78 (b) to represent the currents and voltages.

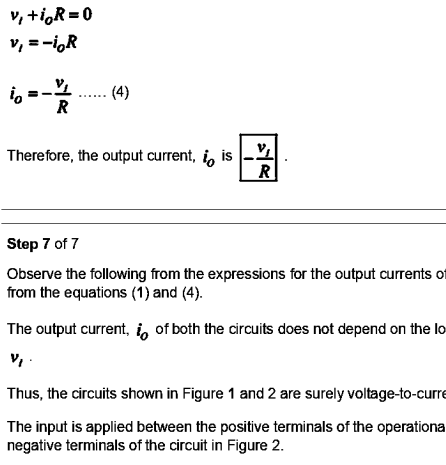


Figure 2

Step 4 of 7

Consider the following from the circuit in Figure 2.

$$v_I = v_{I1} - v_{I2}$$

Apply Kirchhoff's current law at the inverting terminal of the first operational amplifier.

$$\frac{v_A - v_{O1}}{R_1} + \frac{v_A - v_{I1}}{R_1} + 0 = 0$$

$$\frac{v_A - v_{O1} + v_A - v_{I1}}{R_1} = 0$$

$$v_A - v_{O1} + v_A - v_{I1} = 0$$

$$2v_A - v_{O1} - v_{I1} = 0$$

$$2v_A = v_{O1} + v_{I1}$$

$$v_A = \frac{v_{O1} + v_{I1}}{2} \quad \text{..... (2)}$$

Step 5 of 7

Similarly, apply Kirchhoff's current law at the non-inverting terminal of the first operational amplifier.

$$\frac{v_A - v_B}{R_1} + \frac{v_A - v_{I2}}{R_1} + 0 = 0$$

$$\frac{v_A - v_B + v_A - v_{I2}}{R_1} = 0$$

$$v_A - v_B + v_A - v_{I2} = 0$$

$$2v_A - v_B - v_{I2} = 0$$

Substitute $\frac{v_{O1} + v_{I1}}{2}$ for v_A in the equation.

$$2\left(\frac{v_{O1} + v_{I1}}{2}\right) - v_B - v_{I2} = 0$$

$$v_{O1} + v_{I1} - v_B - v_{I2} = 0$$

$$v_B = v_{O1} + v_{I1} - v_{I2}$$

Since, $v_I = v_{I1} - v_{I2}$,

$$v_B = v_{O1} + v_I \quad \text{..... (3)}$$

Step 6 of 7

Apply Kirchhoff's current law at the non-inverting terminal of the second operational amplifier:

$$\frac{v_B - v_{O1}}{R} + i_O + 0 = 0$$

$$\frac{v_B - v_{O1} + i_O R}{R} = 0$$

$$v_B - v_{O1} + i_O R = 0$$

Substitute $v_{O1} + v_I$ for v_B in the equation.

$$v_{O1} + v_I - v_{O1} + i_O R = 0$$

$$v_I + i_O R = 0$$

$$v_I = -i_O R$$

$$i_O = -\frac{v_I}{R} \quad \text{..... (4)}$$

Therefore, the output current, i_O is $\boxed{-\frac{v_I}{R}}$.

Step 7 of 7

Observe the following from the expressions for the output currents of the circuits in Figure 1 and 2 that are from the equations (1) and (4).

The output current, i_O of both the circuits does not depend on the load impedance but on the input voltage, v_I.

Thus, the circuits shown in Figure 1 and 2 are surely voltage-to-current converters.

The input is applied between the positive terminals of the operational amplifiers of the circuit in Figure 1 and negative terminals of the circuit in Figure 2.

Thus, the output current obtained for the circuit in Figure 1 is positive and for Figure 2 is negative.

Hence, the circuit in Figure 1 is a non-inverting amplifier, and the circuit in Figure 2 is an inverting amplifier.

2.079P

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2.080P

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2.081P

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Step 1 of 3

2.082P

Refer to the circuit diagram 2.24(a) in text book

Unity gain frequency =10 krad/sec

$$\left|\frac{v_0}{v_{in}}\right| = \left|-\frac{1}{j\omega RC}\right|$$

$$1 = \frac{1}{\omega RC}$$

$$C = \frac{1}{\omega R}$$

Substitute R=100 k and =10 krad/sec

$$C = \frac{1}{\omega R}$$

$$= \frac{1}{100\times10^3\times10\times10^3}$$

$$= 0.001\ \mu F$$

$$= 1\ nF$$

Input pulse of 2 V at 100 μs is applied to the Miller integrator circuit that is shown below figure 1:

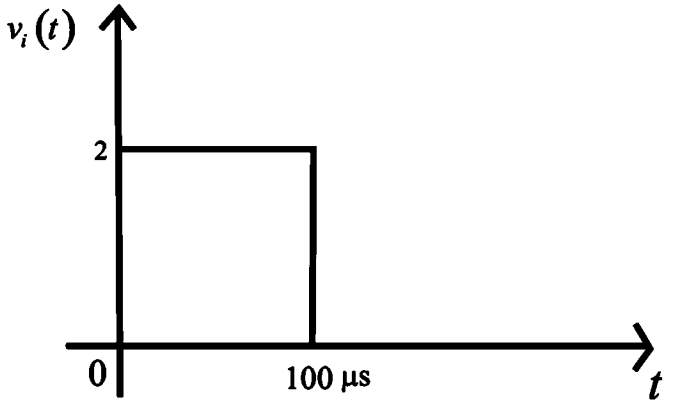


Figure: 1

Step 2 of 3

Output of the circuit is given by

$$v_0(t) = \frac{-1}{RC}\int_0^t v_i(t)\,dt$$

put $R = 100\ k\Omega, C = 1\ nF, t = 100\ \mu s$ and $v_i(t) = 2\ V$

$$v_0(t) = \frac{-1}{10^5\times10^{-9}}\int_0^{10^{-4}} 2\,dt$$

$$= \frac{-2}{10^{-4}}[t]_0^{10^{-4}}$$

$$= -2\times\frac{10^{-4}-0}{10^{-4}}$$

$$= -2\ V$$

The output waveform is as shown in below figure: 2

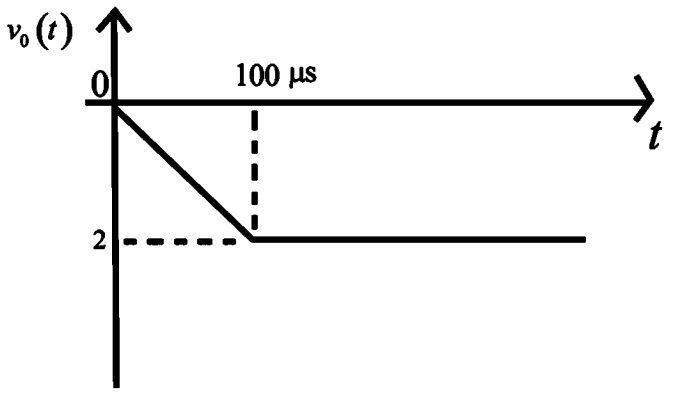


Figure: 2

Step 3 of 3

When a sine wave input is $2\sin 10^4 t$ is applied to the Miller integrator

$$v_0(t) = \frac{-1}{RC}\int_0^t v_i(t)\,dt$$

put $R = 100\ k\Omega, C = 1\ nF$, and $v_i(t) = 2\ V$

$$v_0(t) = \frac{-1}{10^5\times10^{-9}}\int_0^t 2\sin 10^4 t\,dt$$

$$= \frac{-2}{10^{-4}}\left[\frac{-\cos 10^4 t}{10^4}\right]_0^t$$

$$= 2\cos 10^4 t - 2\cos(0)$$

$$= 2\cos 10^4 t - 2$$

$$= 2\left[\cos 10^4 t - 1\right]\ V$$

Hence the output voltage is $\boxed{2\left[\cos 10^4 t - 1\right]\ V}$

Step 1 of 7

Refer to Figure 2.24(a) in textbook.

$$\left|\frac{v_0}{v_{in}}\right| = \left|\frac{1}{j\omega RC}\right|$$

$$1 = \frac{1}{\omega RC}$$

$$C = \frac{1}{\omega R}$$

$$= \frac{1}{2\pi fR}$$

Substitute R =10 k and f =100 kHz.

$$C = \frac{1}{2\pi fR}$$

$$= \frac{1}{2\pi \times 10^4 \times 10^3}$$

$$= \frac{1}{2\pi} \times 10^{-9}$$

$$= 1 \text{ nF}$$

Step 2 of 7

When a feedback resistor of R_f is introduced across the capacitor, then the circuit will changes to figure 2.25 in text book. With R_f the transfer function can be represented as shown below:

$$\frac{v_0}{v_{in}} = \frac{\left(-\frac{R_f}{R}\right)}{1 + sCR_f}$$

$$\text{Dc gain} = \frac{R_f}{R} = 40 \text{ dB}$$

$$20\log\left(\frac{R_f}{R}\right) = 40$$

$$\log\left(\frac{R_f}{R}\right) = 20$$

$$\frac{R_f}{R} = 10^2$$

$$R_f = 100R$$

Substitute $R = 10 \text{ k}\Omega$.

$$R_f = 100R$$

$$= 100(10 \text{ k}\Omega)$$

$$= 1000 \text{ k}\Omega$$

Step 3 of 7

Find the transfer function.

$$\frac{v_0(s)}{v_{in}(s)} = \frac{K}{1 + j2\pi fR_fC}$$

$$= \frac{-\frac{R_f}{R}}{1 + j\left(\frac{f}{f_{3dB}}\right)}$$

Here, $f_{3dB} = \frac{1}{2\pi RCf}$.

Substitute $C = \frac{1}{2\pi} \times 10^{-9}$ F , and $R_f = 1000 \text{ k}\Omega$.

$$f_{3dB} = \frac{1}{2\pi R_fC}$$

$$= \frac{1}{2\pi(10^6)(10^{-9})}$$

$$= 1000 \text{ Hz}$$

$$= 1 \text{ kHz}$$

Step 4 of 7

(a)

Calculate the output of a Miller integrator circuit without R_f .

$$v_0(t) = \frac{-1}{RC}\int_0^t v_i(t)\,dt$$

Substitute $R = 10 \text{ k}\Omega$, $C = \frac{1}{2\pi}$ nF, $t = 10\ \mu\text{s}$ and $v_i(t) = 1$ V .

$$v_0(t) = \frac{-1}{10^4 \times \frac{1}{2\pi} \times 10^{-9}} \int_0^{10^{-5}} 1\,dt$$

$$= \frac{-2\pi}{10^{-5}}[t]_0^{10^{-5}}$$

$$= -2\pi \times \frac{10^{-5} - 0}{10^{-5}}$$

$$= -6.28 \text{ V}$$

Step 5 of 7

Sketch the output voltage waveform.

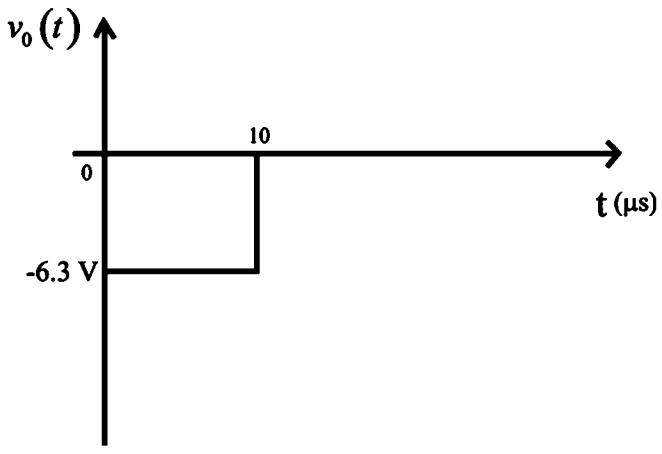


Figure 1

Step 6 of 7

(b)

Calculate the output with R_f .

$$\frac{v_0(s)}{v_{in}(s)} = \frac{-\frac{R_f}{R}}{CR_f\left[s + \frac{1}{CR_f}\right]}$$

$$= \frac{-1}{RC\left(s + \frac{1}{CR_f}\right)}$$

Apply inverse Laplace transform.

$$v_0(t) = \frac{-1}{RC} e^{-\frac{t}{CR_f}} v_i(t)$$

Substitute $R = 10 \text{ k}\Omega$, and $C = \frac{1}{2\pi}$ nF , and $t = 10\ \mu\text{s}$, $R_f = 1000 \text{ k}\Omega$, and $v_i(t) = 1$ V .

$$v_0(t) = \frac{-1}{RC} e^{-\frac{t}{CR_f}} v_i(t)$$

$$= \frac{-1}{10^{-4}\left(\frac{1}{2\pi}\right)(10^{-9})} e^{-\left(\frac{10^{-5}}{\frac{1}{2\pi} \times 10^{-9} \times 10^6}\right)}(1)$$

$$= \frac{-2\pi}{10^{-5}}\left(e^{-2\pi \times 10^{-2}}\right)$$

$$= -590054.79 \text{ V}$$

Step 7 of 7

Sketch the output voltage.

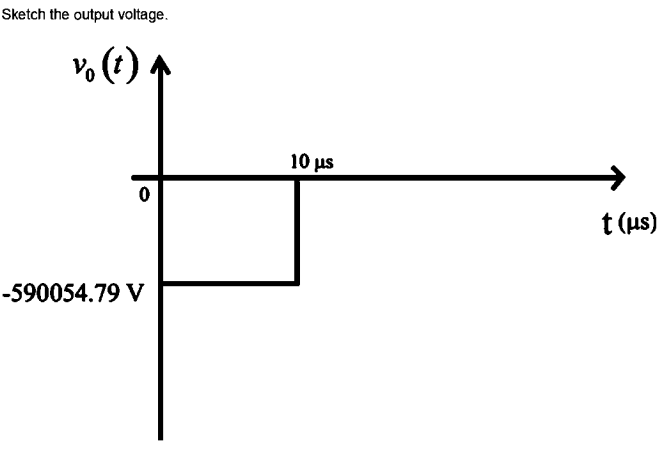


Figure 2

Step 1 of 5

2.084P

Refer to input voltage waveform in Figure P2.84 in the textbook.

Calculate the output voltage for $0 \le t \le 0.5\,\text{ms}$.

$$V_o(t) = V_o(0) - \frac{1}{CR}\int_0^t V_i\,dt$$
$$= 0 - \frac{1}{1\times10^{-3}}\int_0^{0.5\text{ ms}} 1\,dt$$
$$= -(1000)(0.5\times10^{-3} - 0)$$
$$= -0.5\text{ V}$$

Calculate the output voltage for $0.5 \le t \le 1\,\text{ms}$.

$$V_o(t) = V_o(0.5\,\text{ms}) - \frac{1}{CR}\int_{0.5}^t V_i\,dt$$
$$= -0.5 - \frac{1}{1\times10^{-3}}\int_{0.5\text{ ms}}^{1\text{ ms}} (-1)\,dt$$
$$= -0.5 + (1000)(1\text{ ms} - 0.5\text{ ms})$$
$$= -0.5 + 0.5$$
$$= 0\text{ V}$$

Step 2 of 5

The sketch of the output waveform is shown in Figure 1.

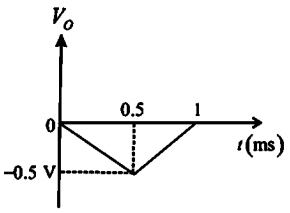


Figure 1

Step 3 of 5

For, $V_i = \pm 2$ V and $CR = 1$ ms , the miller integrator circuit is shown in Figure 2.

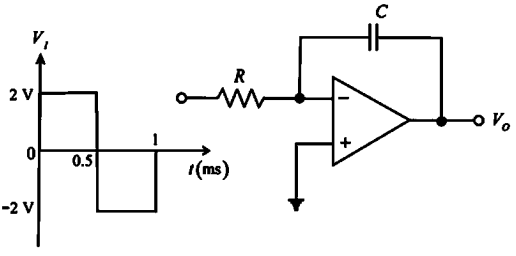


Figure 2

Step 4 of 5

Calculate the output voltage for $0 \le t \le 0.5\,\text{ms}$.

$$V_o(t) = V_o(0) - \frac{1}{CR}\int_0^t V_i\,dt$$
$$= 0 - \frac{1}{1\times10^{-3}}\int_0^{0.5\text{ ms}} 2\,dt$$
$$= 0 - (2000)(0.5\times10^{-3} - 0)$$
$$= -1\text{ V}$$

Calculate the output voltage for $0.5 \le t \le 1\,\text{ms}$.

$$V_o(t) = V_o(0.5\,\text{ms}) - \frac{1}{CR}\int_{0.5}^t V_i\,dt$$
$$= -1 - \frac{1}{1\times10^{-3}}\int_{0.5\text{ ms}}^{1\text{ ms}} (-2)\,dt$$
$$= -1 + (2000)(1\text{ ms} - 0.5\text{ ms})$$
$$= -1 + 1$$
$$= 0\text{ V}$$

Step 5 of 5

The sketch of the output waveform is shown in Figure 3.

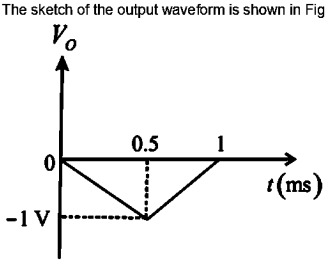


Figure 3

Step 1 of 3

2.085P

Refer to input voltage waveform in Figure P2.85 in the textbook.

Calculate the change in output voltage of miller integrator circuit.

$$\Delta V_o = -\frac{1}{CR}\int_0^{10\mu s} 1\,dt$$
$$= -\frac{t}{CR}\Big|_0^{10\mu s}$$
$$= -\frac{10\times10^{-6}-0}{1\times10^{-3}}$$
$$= -10\text{ mV}$$

Step 2 of 3

The sketch of the output voltage is shown in Figure 1.

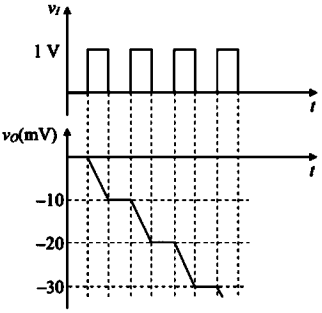


Figure 1

Step 3 of 3

From the output voltage sketch, for 10 mV change $10\ \mu s$ is required.

Calculate the required pulses for an output voltage change of 1 V.

$$\text{Pulses} = \left(\frac{1\text{ V}}{10\text{ mV}}\right)(10\ \mu s)$$
$$= (100)(10\ \mu s)$$

Therefore, total of hundred $10\ \mu s$ pulses are required for an output voltage change of 1 V .

Step 1 of 5

The voltage gain of the op-amp in inverting configuration is,

$$A_v = \frac{v_0}{v_i}$$
$$= -\frac{R_f}{R_1}$$

Here,

$$R_f = R_2 \| \frac{1}{sC_2}$$
$$= \frac{\frac{R_2}{sC}}{R_2 + \frac{1}{sC}}$$
$$= \frac{sR_2C}{sC(1+sR_2C)}$$
$$= \frac{R_2}{1+sR_2C}$$

Step 2 of 5

The transfer function is,

$$\frac{v_0}{v_i} = \frac{-\left[\frac{R_2}{1+R_2sC}\right]}{R_1}$$
$$= \frac{-\left(\frac{R_2}{R_1}\right)}{1+R_2j\omega C}$$
$$= \frac{K}{1+j2\pi fR_2C}$$
$$= \frac{K}{1+j\frac{f}{f_{3dB}}}$$

Where $K = -\left(\frac{R_2}{R_1}\right), f_{3dB} = \frac{1}{2\pi R_2C}$

In terms of ω, the equation is $\omega_0 = \frac{1}{R_2C}$

Step 3 of 5

Input resistance is $R_1 = 10\ \text{k}\Omega$

DC gain is

$$K = 40\ \text{dB}$$
$$20\log K = 40$$
$$K = 10^{\left(\frac{40}{20}\right)}$$
$$= 100$$

The general expression for K is,

$$K = -\left(\frac{R_2}{R_1}\right)$$

Here, negative sign is due to op-amp in inverting configuration.

Substitute $R_1 = 10\ \text{k}\Omega$, and $K = 100$.

$$R_2 = KR_1$$
$$= 100\times10\times10^3$$
$$= 1\ \text{M}\Omega$$

Step 4 of 5

Substitute $f_{3dB} = 1\ \text{kHz}$ and $R_2 = 1\ \text{M}\Omega$ in the equation of f_{3dB}.

$$f_{3dB} = \frac{1}{2\pi R_2C}$$
$$C = \frac{1}{2\pi R_2f_{3dB}}$$
$$= \frac{1}{2\pi\times10^6\times1\times10^3}$$
$$= 1.57\times10^{-9}$$
$$= 1.57\ \text{nF}$$

Consider that the transfer function magnitude should be equal to 1.

$$\left|\frac{v_0}{v_i}\right| = \left|\frac{K}{1+j2\pi fR_2C}\right|$$
$$1 = \frac{|K|}{\sqrt{1+(2\pi fR_2C)^2}}$$
$$K^2 = \left(\sqrt{1+(2\pi fR_2C)^2}\right)^2$$
$$f = \sqrt{\frac{K^2-1}{4\pi R_2{}^2C^2}}$$

Step 5 of 5

Substitute $K = 100$, $R_2 = 1\ \text{M}\Omega$, and $C = \frac{1}{2\pi}\ \text{nF}$.

$$f = \sqrt{\frac{10^4-1}{4\pi^2\left(10^{12}\right)\left(10^{-18}\right)\left(\frac{1}{2\pi}\right)\left(\frac{1}{2\pi}\right)}}$$
$$= \sqrt{\frac{9999}{10^{-6}}}$$
$$= \sqrt{9999\times10^6}$$
$$\approx 100\ \text{Hz}$$

Thus, the magnitude of the transfer function is unity at the frequency of **100 Hz**.

Step 1 of 7

Draw the non-ideal op-amp integrator circuit.

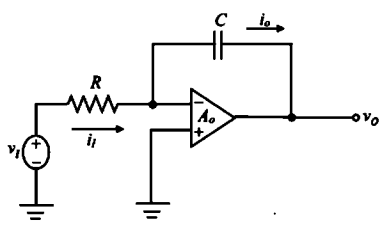


Figure 1

Step 2 of 7

The open loop gain of the op-amp is A_0 .

$$\frac{v_O}{v_-} = -A_0$$

$$v_- = -\frac{v_O}{A_0} \quad v_I$$

Calculate the current through the source.

$$i_I(s) = \frac{V_I(s) - V_-}{R}$$

Substitute $-\frac{V_O(s)}{A_0}$ for $V_-(s)$ in the equation.

$$I_I(s) = \frac{V_I(s) - \left[-\frac{V_O(s)}{A_0}\right]}{R}$$

$$= \frac{V_I(s) + \frac{V_O(s)}{A_0}}{R}$$

$$= \frac{V_I(s)}{R} + \frac{V_O(s)}{A_0 R}$$

Step 3 of 7

Apply Kirchhoff's voltage law.

$$V_-(s) = \frac{I_o(s)}{Cs} + V_O(s)$$

$$= \frac{I_I(s)}{Cs} + V_O(s) \qquad (\text{since } i_I = i_o)$$

Substitute $-\frac{V_O(s)}{A_0}$ for $V_-(s)$ and $\frac{V_I(s)}{R} + \frac{V_O(s)}{A_0 R}$ for $I_I(s)$ in the equation.

$$-\frac{V_O(s)}{A_0} = \left(\frac{V_I(s)}{R} + \frac{V_O(s)}{A_0 R}\right)\left(\frac{1}{Cs}\right) + V_O(s)$$

$$-\frac{V_O(s)}{A_0} - \frac{V_O(s)}{A_0 RCs} - V_O(s) = \frac{V_I(s)}{RCs}$$

$$\left(\frac{1}{A_0} + \frac{1}{A_0 RCs} + 1\right) V_O(s) = -\frac{V_I(s)}{RCs}$$

$$\left(\frac{RCs + 1 + A_0 RCs}{A_0 RCs}\right) V_O(s) = -\frac{V_I(s)}{RCs}$$

$$\frac{V_O(s)}{V_I(s)} = \frac{-1}{\frac{RCs + 1 + A_0 RCs}{A_0}}$$

$$= -\frac{A_0}{RCs(1 + A_0) + 1}$$

The transfer function of the integrator is $-\frac{A_0}{RCs(1 + A_0) + 1}$, which represents a low-pass STC transfer function.

Step 4 of 7

The transfer function of the non-ideal Miller integrator is,

$$\frac{V_O(s)}{V_I(s)} = -\frac{A_0}{RCs(1 + A_0) + 1}$$

Determine the pole frequency of the transfer function by substitute $j\omega$ for s in the transfer function.

$$\frac{V_O(j\omega)}{V_I(j\omega)} = -\frac{A_0}{RCj\omega(A_0 + 1) + 1}$$

$$= -\frac{A_0}{RC(A_0 + 1)\left(j\omega + \frac{1}{RC(A_0 + 1)}\right)}$$

$$= -\frac{\frac{A_0}{RC(A_0 + 1)}}{j\omega + \frac{1}{RC(A_0 + 1)}}$$

Therefore, the pole frequency of the transfer function is $\boxed{\frac{1}{RC(A_0 + 1)}}$.

Step 5 of 7

Write the transfer function of an ideal integrator.

$$\frac{V_O(j\omega)}{V_I(j\omega)} = -\frac{1}{j\omega CR}$$

Therefore, an ideal Miller integrator has a pole frequency at the origin whereas the pole frequency of a non-ideal integrator is at $\frac{1}{RC(A_0 + 1)}$, which makes the integrator more stable.

Step 6 of 7

The magnitude of the transfer function is,

$$\left|\frac{V_O}{V_I}\right| = -\frac{1}{\omega CR}$$

The angular frequency is,

$$\omega = \frac{2\pi}{T}$$

$$= \frac{2\pi}{RC}$$

Substitute $\frac{2\pi}{RC}$ for ω in the magnitude of the transfer function.

$$\left|\frac{V_O}{V_I}\right| = \frac{1}{\left(\frac{2\pi}{RC}\right)CR}$$

$$= \frac{1}{2\pi}$$

$$= 0.159$$

Substitute $-1\ \text{V}$ for V_I in the equation.

$$V_O = (0.159)(-1\ \text{V})$$

$$= -0.159\ \text{V}$$

Therefore, the output voltage of an ideal Miller integrator is $\boxed{-0.159\ \text{V}}$.

Step 7 of 7

The magnitude of the transfer function of non-ideal integrator circuit is,

$$\left|\frac{V_O}{V_I}\right| = \left|-\frac{A_0}{RCj\omega(A_0 + 1) + 1}\right|$$

$$= \frac{A_0}{\sqrt{1 + \omega^2 R^2 C^2 (A_0 + 1)^2}}$$

Substitute $\frac{2\pi}{RC}$ for ω in the magnitude of the transfer function.

$$\left|\frac{V_O}{V_I}\right| = \frac{A_0}{\sqrt{1 + \left(\frac{2\pi}{RC}\right)^2 R^2 C^2 (A_0 + 1)^2}}$$

$$= \frac{A_0}{\sqrt{1 + 4\pi^2 (A_0 + 1)^2}}$$

Substitute 1000 for A_0 in the equation.

$$\left|\frac{V_O}{V_I}\right| = \frac{1000}{\sqrt{1 + 4\pi^2 (1000 + 1)^2}}$$

$$= \frac{1000}{\sqrt{39557414.92}}$$

$$= \frac{1000}{6289.468}$$

$$= 0.159$$

Substitute $-1\ \text{V}$ for V_I in the equation.

$$V_o = (0.159)(-1\ \text{V})$$

$$= -0.159\ \text{V}$$

Thus, the output voltage of an non-ideal Miller integrator is $\boxed{-0.159\ \text{V}}$.

Step 1 of 2

Represent the transfer function of a differentiator circuit is,

$$\begin{aligned}\frac{v_0(s)}{v_i(s)} &= -j\omega RC \\ &= \left|-j\omega RC\right| \\ &= \sqrt{(-j\omega RC)^2} \\ &= \omega RC\end{aligned}$$

When the input

$$|v_i| = |v_0|$$

$$\left|\frac{v_0}{v_i}\right| = 1$$

Therefore,

$$\omega RC = 1$$

Here, $\omega_0 = 2\pi f_0$. Therefore,

$$2\pi f_0 RC = 1$$

$$f_0 = \frac{1}{2\pi RC}$$

Step 2 of 2

Substitute $R = 10\ \text{k}\Omega$ and $C = 1\ \text{nF}$.

$$\begin{aligned}f_0 &= \frac{1}{2\pi \times 10 \times 10^3 \times 10^{-9}} \\ &= 1.57 \times 10^5\ \text{Hz}\end{aligned}$$

Therefore, the output voltage is,

$$\begin{aligned}v_0 &= v_i \omega RC \\ &= v_i (2\pi f_0 RC)\end{aligned}$$

Substitute $v_{i(p-p)} = 1\ \text{V}$ and $f_0 = 1.57 \times 10^5\ \text{Hz}$.

$$\begin{aligned}v_{0(p-p)} &= 1 \times 2\pi \times 1.57 \times 10^5 \times 10 \times 10^3 \times 10^{-9} \\ &= 98.8646 \\ &\approx 98.86\ \text{V}\end{aligned}$$

Therefore, the values of f_0 and $v_{0(p-p)}$ is $\boxed{1.57 \times 10^5\ \text{Hz},\ 98.86\ \text{V}}$.

Step 1 of 3

2.089P

The time constant of an op-amp differentiator is,

$$\tau = RC$$
$$= 1 \text{ ms}$$

The input of the differentiator is the rate-controlled step shown in Figure 2.89 in the textbook.

The output of differentiator is,

$$v_0 = -RC\frac{d}{dt}v_i(t)$$

The input voltage $v_i(t)$ has slope and is calculated as,

$$v_i(t) = \frac{1-0}{0.2\times10^{-3}-0}(t-0)$$
$$= \frac{10000}{2}t$$
$$= 5000t$$

$$\frac{dv_i(t)}{dt} = 5000 \text{ up to } t = 0 \text{ to } 0.2 \text{ ms}$$
$$= 0, \quad t > 0.2$$

Step 2 of 3

The output voltage is,

$$v_0(t) = -RC\frac{d}{dt}v_i(t)$$

Substitute $RC = 10^{-3}$ s

$$\frac{d}{dt}v_i(t) = 5000 \text{ V/s}$$
$$v_0(t) = -10^{-3}\left(5\times10^{3}\right)$$
$$v_0(t) = -5 \text{ V}$$

Here, the negative sign indicates op-amp in inverting configuration.

Step 3 of 3

The representation of $v_0(t)$ with negative sign is,

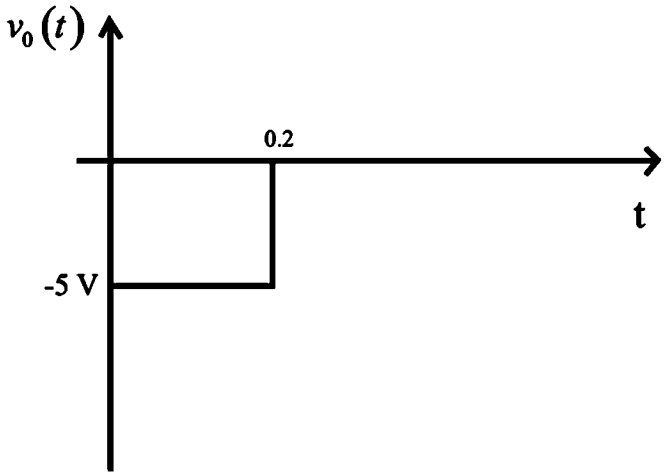


Figure 1

Hence, the output voltage $v_o(t)$ is $\boxed{-5 \text{ V}}$.

Step 1 of 5

The differentiation of triangular waveform results a square waveform. This is because differentiation of ramp signal results in a square wave.

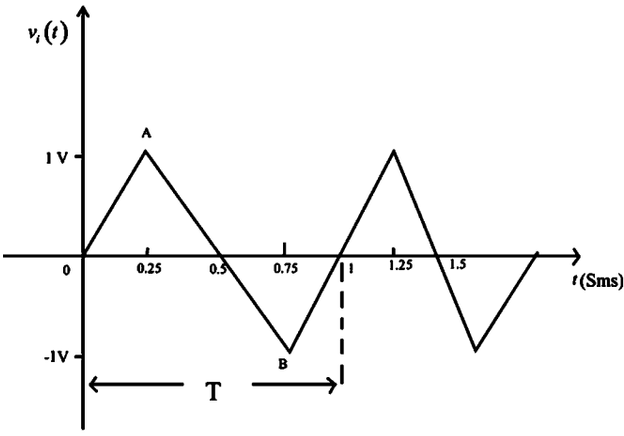


Figure 1

Step 2 of 5

The input voltage equation between 0 to A points (0, 0) and (0.00025, 1) is,

$$v_i(t)-0=\frac{1-0}{0.00025-0}(t-0)$$
$$v_i(t)=4000t$$

The input voltage equation between A and B points (0.00025, 1) and (0.00075,-1) is,

$$v_i(t)-1=\frac{-1-1}{0.00075-0.00025}\left(t-\frac{10^{-3}}{4}\right)$$
$$v_i(t)=-4000t+2$$

The input voltage equation between points B and C (0.00075,-1) and (0.001,0) is,

$$v_i(t)+1=\frac{0+1}{0.00025}\left(t-\frac{3\times10^{-3}}{4}\right)$$
$$v_i(t)+1=4000\left(t-\frac{3\times10^{-3}}{4}\right)$$
$$v_i(t)=4000t-4$$

Step 3 of 5

Calculate the total input equation for full cycle $T=1\text{ ms}$.

$$v_i(t)=4000t-4000t+2+4000t-4$$
$$=4000t-2$$

Calculate the output voltage v_0 for $t=0$ to $t=1\text{ ms}$.

$$v_0=-RC\frac{dv_i(t)}{dt}$$

Substitute $20\text{ k}\Omega$ for R and 10^{-7} F for C.

$$v_0=-\left(20\times10^3\right)\left(10^{-7}\right)\frac{d}{dt}(4000t-2)$$
$$=-\left(20\times10^{-4}\right)\left(4\times10^3\right)$$
$$=-8\text{ V}$$

Step 4 of 5

Sketch the output waveform.

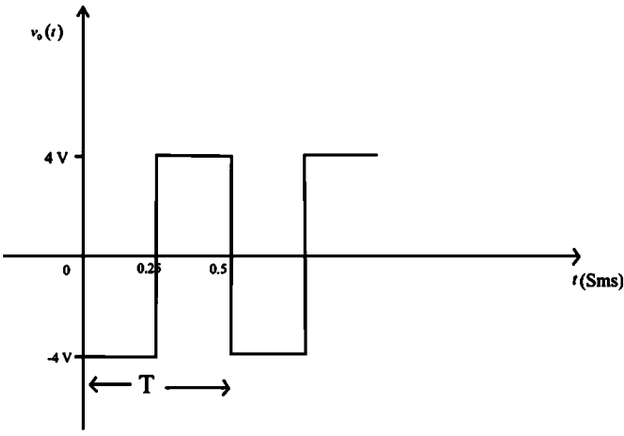


Figure 2

Step 5 of 5

Consider the following for a square wave:

$$I_{avg}=I_{rms}=I_{max}$$

$$v_{avg}=v_{rms}=v_{max}$$

Here, $v_{max}=4\text{ V}$. Hence,

$$v_{avg}=v_{rms}$$
$$=v_{max}$$
$$=4\text{ V}$$

Find the value of R for $v_0=12\text{ V}$.

$$v_0=-RC\frac{dv_i(t)}{dt}$$

$$R=\frac{-v_0}{C\dfrac{dv_i(t)}{dt}}$$
$$=\frac{-12}{\left(10^{-7}\right)(4000)}$$
$$=30\text{ k}\Omega$$

Hence, the required resistance R is $\boxed{30\text{ k}\Omega}$.

Step 1 of 9

Refer to op-amp differentiator circuit in Figure 2.27 (a) in the textbook.

Write the expression for the time constant.

$$\tau = RC$$

Substitute 10^{-3} s for τ and 10 nF for C in the equation.

$$10^{-3}\ \text{s} = R(10\ \text{nF})$$

$$\begin{aligned} R &= \frac{10^{-3}\ \text{s}}{10\ \text{nF}} \\ &= 10^{5}\ \Omega \\ &= 100\ \text{k}\Omega \end{aligned}$$

Therefore, the value of resistance, R is $\boxed{100\ \text{k}\Omega}$.

Step 2 of 9

Write the expression for gain of an active differentiator.

$$\frac{V_o(s)}{V_i(s)} = -sCR$$

Substitute $j\omega$ for s in the equation.

$$\begin{aligned} \frac{V_o(j\omega)}{V_i(j\omega)} &= -j\omega CR \\ &= -j\left(\frac{\omega}{\omega_c}\right) \quad \left(\text{since } \omega_c = \frac{1}{RC}\right) \end{aligned}$$

At any frequency, the phase shift is $-90°$.

Unity gain is obtained when, $\omega = \omega_c$.

Calculate the unity gain frequency.

$$\begin{aligned} \omega_{\text{unity-gain}} &= \frac{1}{RC} \\ &= \frac{1}{(100\ \text{k}\Omega)(10\ \text{nF})} \\ &= \frac{1}{10^{-3}} \\ &= 1\ \text{krad/sec} \end{aligned}$$

Therefore, the unity-gain frequency, $\omega_{\text{unity-gain}}$ is $\boxed{1\ \text{krad/sec}}$.

Step 3 of 9

Find the frequency for one-tenth of the unity-gain frequency.

$$\omega = \frac{\omega_c}{10}$$

Substitute $1\ \text{krad/sec}$ for ω_c in the equation.

$$\begin{aligned} \omega &= \frac{1\ \text{krad/sec}}{10} \\ &= 100\ \text{rad/sec} \end{aligned}$$

Calculate the gain of differentiator circuit at one-tenth of the unity-gain frequency.

$$\begin{aligned} \left|\frac{V_o(j\omega)}{V_i(j\omega)}\right|_{\omega=\frac{\omega_c}{10}} &= \frac{\omega}{\omega_c} \\ &= \frac{100\ \text{rad/sec}}{1\ \text{krad/sec}} \\ &= 0.1\ \text{V/V} \end{aligned}$$

Therefore, the gain of the differentiator at one-tenth of unity-gain frequency is $\boxed{0.1\ \text{V/V}}$.

Step 4 of 9

Find the frequency for ten times of the unity-gain frequency.

$$\omega = 10\omega_c$$

Substitute $1\ \text{krad/sec}$ for ω_c in the equation.

$$\begin{aligned} \omega &= 10(1\ \text{krad/sec}) \\ &= 10\ \text{krad/sec} \end{aligned}$$

Calculate the gain of differentiator circuit at ten times of the unity-gain frequency.

$$\begin{aligned} \left|\frac{V_o(j\omega)}{V_i(j\omega)}\right|_{\omega=10\omega_c} &= \frac{\omega}{\omega_c} \\ &= \frac{10\ \text{krad/sec}}{1\ \text{krad/sec}} \\ &= 10\ \text{V/V} \end{aligned}$$

Therefore, the gain of the differentiator at ten times of unity-gain frequency is $\boxed{10\ \text{V/V}}$.

Step 5 of 9

A series input resistance is added to the differentiator circuit, the modified circuit is shown in Figure 1.

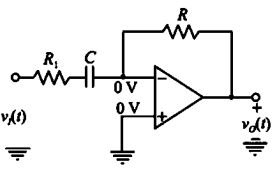


Figure 1

Step 6 of 9

Write the expression for gain of the amplifier at high frequencies.

$$\frac{V_o}{V_i} = -\frac{R}{R_1}$$

The Magnitude of this amplifier is,

$$\left|\frac{V_o}{V_i}\right| = \frac{R}{R_1}$$

Substitute $100\ \text{V/V}$ for $\left|\frac{V_o}{V_i}\right|$ and $100\ \text{k}\Omega$ for R in the equation.

$$\begin{aligned} 100\ \text{V/V} &= \frac{100\ \text{k}\Omega}{R_1} \\ R_1 &= \frac{100\ \text{k}\Omega}{100\ \text{V/V}} \\ R_1 &= 1\ \text{k}\Omega \end{aligned}$$

Therefore, the input resistance, R_1 of new differentiator is $\boxed{1\ \text{k}\Omega}$.

Step 7 of 9

Calculate the 3-dB frequency of new differentiator.

$$\begin{aligned} f_{3\text{-dB}} &= \frac{1}{2\pi CR_1} \\ &= \frac{1}{2\pi(10\ \text{nF})(1\ \text{k}\Omega)} \\ &= \frac{1}{62.83\times10^{-6}} \\ &= 15.915\ \text{kHz} \end{aligned}$$

Therefore, the 3-dB frequency, $f_{3\text{-dB}}$ of the new differentiator is $\boxed{15.915\ \text{kHz}}$.

Step 8 of 9

Calculate the transfer function of new differentiator circuit.

$$\begin{aligned} \frac{V_o(s)}{V_i(s)} &= -\frac{R}{R_1 + \frac{1}{sC}} \\ &= -\frac{sCR}{sCR_1 + 1} \end{aligned}$$

Substitute $j\omega$ for s in the equation.

$$\begin{aligned} \frac{V_o(j\omega)}{V_i(j\omega)} &= -\frac{j\omega CR}{j\omega CR_1 + 1} \\ \left|\frac{V_o(j\omega)}{V_i(j\omega)}\right| &= \frac{\omega CR}{\sqrt{(\omega CR_1)^2 + 1}} \end{aligned}$$

Substitute $100\ \text{k}\Omega$ for R, $1\ \text{k}\Omega$ for R_1, 10 nF for C, and $10\ \text{krad/sec}$ for ω in the equation.

$$\begin{aligned} \left|\frac{V_o(j\omega)}{V_i(j\omega)}\right| &= \frac{(10\ \text{krad/sec})(10\ \text{nF})(100\ \text{k}\Omega)}{\sqrt{[(10\ \text{krad/sec})(10\ \text{nF})(1\ \text{k}\Omega)]+1}} \\ &= \frac{10}{\sqrt{0.1+1}} \\ &= \frac{10}{1.0488} \\ &= 9.534\ \text{V/V} \end{aligned}$$

Therefore, the voltage gain of the new differentiator at $\omega = 10\ \text{krad/sec}$ is $\boxed{9.534\ \text{V/V}}$.

Step 9 of 9

Calculate the Phase shift of the new differentiator at $\omega = 10\ \text{krad/sec}$.

$$\phi = -90° - \tan^{-1}\left(\frac{\omega CR_1}{1}\right)$$

Substitute $1\ \text{k}\Omega$ for R_1, 10 nF for C, and $10\ \text{krad/sec}$ for ω in the equation.

$$\begin{aligned} \phi &= -90° - \tan^{-1}\left(\frac{(10\ \text{krad/sec})(10\ \text{nF})(1\ \text{k}\Omega)}{1}\right) \\ &= -90° - \tan^{-1}(0.1) \\ &= -90° - 5.71° \\ &= -95.71° \end{aligned}$$

Therefore, the phase shift, ϕ of the new differentiator at $\omega = 10\ \text{krad/sec}$ is $\boxed{-95.71°}$.

Step 1 of 4

The op-amp is in inverting configuration. Therefore, the transfer function is,

$$\frac{v_0}{v_{in}} = -\frac{R_f}{R_1}$$

Here $R_f = -R_2$ and R_1 is a series combination of R_1 and C.

$$R_1 = R_1 + \frac{1}{sC}$$
$$= \frac{1 + R_1 sC}{sC}$$

Substitute R_2 for R_f and $\frac{1 + R_1 sC}{sC}$ for R_1 in the equation.

$$\frac{v_0}{v_{in}} = -\frac{R_f}{R_1}$$
$$= \frac{-R_2}{\frac{1 + R_1 sC}{sC}}$$
$$= \frac{-\frac{R_2}{R_1}}{1 + \frac{1}{j2\pi f R_1 C}}$$
$$= \frac{K}{1 - j\left(\frac{f_{3dB}}{f}\right)}$$

Here, $K = -\frac{R_2}{R_1}$, $f_{3dB} = \frac{1}{2\pi R_1 C}$.

Step 2 of 4

The 3 dB frequency can also be represented in terms of as,

$$\omega_0 = 2\pi f_{3dB}$$
$$= \frac{1}{R_1 C}$$

High frequency input resistance is 1 k.

Here, $R_{in} = R_1 + \frac{1}{sC}$

At high frequencies $\frac{1}{sC}$ term vanishes. Therefore,

$$R_{in} = R_1$$
$$= 1\ \text{k}\Omega$$

The Dc gain is 40 dB.

$$20 \log K = 40\ \text{dB}$$

$$K = (10)^{\frac{40}{20}}$$
$$= 100$$

Step 3 of 4

Find the value of R_2 from the expression of K.

$$K = \frac{R_2}{R_1}$$
$$R_2 = KR_1$$

Substitute $K = 100$, and $R_1 = 1\ \text{k}\Omega$.

$$R_2 = KR_1$$
$$= 100(1\ \text{k}\Omega)$$
$$= 100\ \text{k}\Omega$$

Find the other component values.

$$C = \frac{1}{2\pi f_{3dB} R_1}$$
$$= \frac{1}{2\pi(2000)(1000)}$$
$$= \frac{1}{4\pi}\ \mu\text{F}$$

Step 4 of 4

The magnitude of the transfer function must be equal to 1. Therefore,

$$1 = \frac{K}{\sqrt{1 + \left(\frac{f_{3dB}}{f}\right)^2}}$$

$$K^2 = 1 + 1 + \left(\frac{f_{3dB}}{f}\right)^2$$

$$f = \frac{1}{\sqrt{\frac{K^2 - 1}{f_{3dB}^2}}}$$

Substitute $K = 100, f_{3dB} = 2\ \text{kHz}$.

$$f = \frac{1}{\sqrt{\frac{100^2 - 1}{(2000)^2}}}$$
$$= 0.02 \times 10^3\ \text{Hz}$$
$$= 20\ \text{Hz}$$

Hence magnitude of transfer function reduce to unity at frequency of **20 Hz**.

Step 1 of 12

Calculate the transfer function of the op-amp circuit.

$$\frac{v_o}{v_{in}} = -\frac{R_f}{R_1}$$

Here, the resistances are,

$$\begin{aligned} R_f &= R_2 \| \frac{1}{sC_2} \\ &= \frac{\frac{R_2}{sC_2}}{R_2 + \frac{1}{sC_2}} \\ &= \frac{sR_2C_2}{sC_2(1+sR_2C_2)} \\ &= \frac{R_2}{1+R_2sC_2} \end{aligned}$$

$$\begin{aligned} R_1 &= R_1 + \frac{1}{sC_1} \\ &= \frac{1+R_1sC_1}{sC_1} \end{aligned}$$

Step 2 of 12

Substitute the expressions of R_1 and R_f in the equation of gain.

$$\begin{aligned} \frac{v_o}{v_{in}} &= -\frac{\left[\frac{R_2}{1+R_2sC_2}\right]}{\left[\frac{1+R_1sC_1}{sC_1}\right]} \\ &= \frac{-R_2sC_1}{(1+R_1sC_1)(1+sR_2C_2)} \\ &= \frac{-R_2sC_1}{sR_1C_1\left[1+\frac{1}{sR_1C_1}\right]\left[1+sR_2C_2\right]} \\ &= \frac{K}{\left[1+\frac{\omega_1}{j\omega}\right]\left[1+\frac{j\omega}{\omega_2}\right]} \end{aligned}$$

Here, $K = \frac{-R_2}{R_1}$, $\omega_1 = \frac{1}{R_1C_1}$, and $\omega_2 = \frac{1}{R_2C_2}$.

If $\omega_2 \gg \omega_1$, $1+\frac{j\omega}{\omega_2} \approx 1$ and $\frac{v_o}{v_{in}} = \frac{\frac{-R_2}{R_1}}{\left[1+\frac{\omega_1}{j\omega}\right]}$

Step 3 of 12

(a)

Condition 1: When $\omega \ll \omega_1$, $1+\frac{\omega_1}{j\omega} \approx \frac{\omega_1}{j\omega}$:

$$\begin{aligned} \frac{v_o}{v_{in}} &= \frac{-\left(\frac{R_2}{R_1}\right)}{\left[\frac{\omega_1}{j\omega}\right]\left[1+\frac{j\omega}{\omega_2}\right]} \\ &= \frac{K}{\left[\frac{\omega_1}{j\omega}\right]\left[1+\frac{j\omega}{\omega_2}\right]} \\ &= \frac{K}{\left[\frac{\omega_1}{s}\right]\left[1+\frac{s}{\omega_2}\right]} \end{aligned}$$

Substitute $\omega_1 = 200\ \text{kHz}$, and $\omega_2 = 200\ \text{Hz}$.

$$\begin{aligned} \frac{v_o}{v_{in}} &= \frac{K}{\left[\frac{200\times10^3}{s}\right]\left[1+\frac{s}{200}\right]} \\ &= \frac{Ks}{200\times10^3\left(1+\frac{s}{200}\right)} \\ &= \frac{200s}{200\times10^3\left(1+\frac{s}{200}\right)} \\ &= \frac{0.001s}{\left(1+\frac{s}{200}\right)} \end{aligned}$$

Step 4 of 12

Sketch the Bode plot.

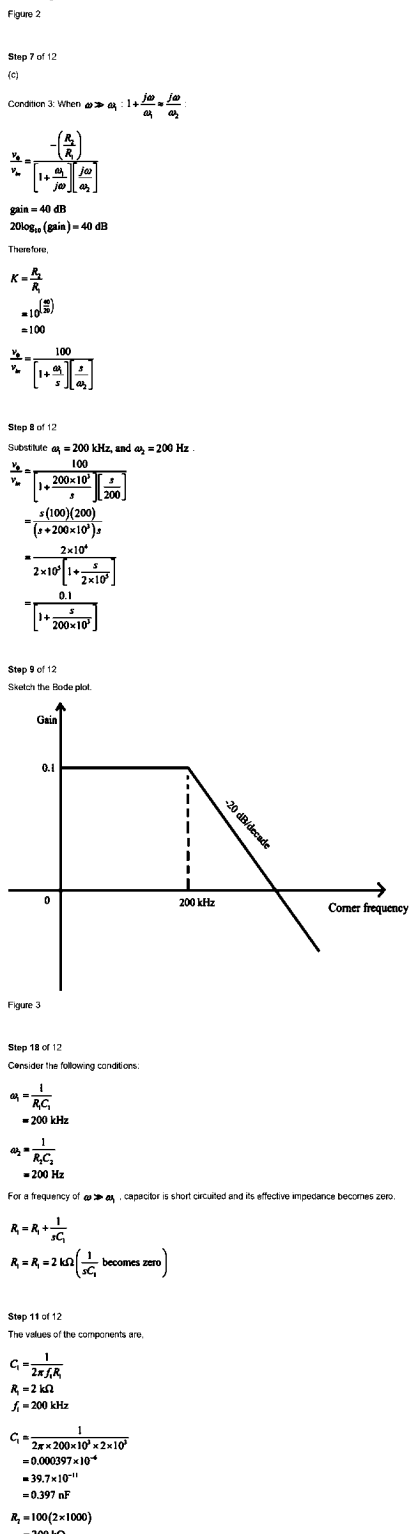


Figure 1

Step 5 of 12

(b)

Condition 2: When $\omega_1 \ll \omega \ll \omega_2$:

$$\omega \gg \omega_1$$

$$1+\frac{\omega_1}{j\omega} \approx 1$$

$$\omega \ll \omega_2$$

$$1+\frac{\omega}{j\omega_2} \approx 1$$

Therefore,

$$\frac{v_o}{v_{in}} = -\frac{R_2}{R_1}$$

Negative sign is due to inverting configuration of an op-amp.

Step 6 of 12

Sketch the Bode plot.

Figure 2

Step 7 of 12

(c)

Condition 3: When $\omega \gg \omega_1$: $1+\frac{j\omega}{\omega_1} \approx \frac{j\omega}{\omega_2}$:

$$\frac{v_o}{v_{in}} = \frac{-\left(\frac{R_2}{R_1}\right)}{\left[1+\frac{\omega_1}{j\omega}\right]\left[\frac{j\omega}{\omega_2}\right]}$$

$$\text{gain} = 40\ \text{dB}$$

$$20\log_{10}(\text{gain}) = 40\ \text{dB}$$

Therefore,

$$\begin{aligned} K &= \frac{R_2}{R_1} \\ &= 10^{\left(\frac{40}{20}\right)} \\ &= 100 \end{aligned}$$

$$\frac{v_o}{v_{in}} = \frac{100}{\left[1+\frac{\omega_1}{s}\right]\left[\frac{s}{\omega_2}\right]}$$

Step 8 of 12

Substitute $\omega_1 = 200\ \text{kHz}$, and $\omega_2 = 200\ \text{Hz}$.

$$\begin{aligned} \frac{v_o}{v_{in}} &= \frac{100}{\left[1+\frac{200\times10^3}{s}\right]\left[\frac{s}{200}\right]} \\ &= \frac{s(100)(200)}{(s+200\times10^3)s} \\ &= \frac{2\times10^4}{2\times10^5\left[1+\frac{s}{2\times10^5}\right]} \\ &= \frac{0.1}{\left[1+\frac{s}{200\times10^3}\right]} \end{aligned}$$

Step 9 of 12

Sketch the Bode plot.

Figure 3

Step 10 of 12

Consider the following conditions:

$$\begin{aligned} \omega_1 &= \frac{1}{R_1C_1} \\ &= 200\ \text{kHz} \end{aligned}$$

$$\begin{aligned} \omega_2 &= \frac{1}{R_2C_2} \\ &= 200\ \text{Hz} \end{aligned}$$

For a frequency of $\omega \gg \omega_1$, capacitor is short circuited and its effective impedance becomes zero.

$$R_1 = R_1 + \frac{1}{sC_1}$$

$$R_1 = R_1 = 2\ \text{k}\Omega\left(\frac{1}{sC_1}\ \text{becomes zero}\right)$$

Step 11 of 12

The values of the components are,

$$C_1 = \frac{1}{2\pi f_1 R_1}$$

$$R_1 = 2\ \text{k}\Omega$$

$$f_1 = 200\ \text{kHz}$$

$$\begin{aligned} C_1 &= \frac{1}{2\pi\times200\times10^3\times2\times10^3} \\ &= 0.000397\times10^{-6} \\ &= 39.7\times10^{-11} \\ &= 0.397\ \text{nF} \end{aligned}$$

$$\begin{aligned} R_2 &= 100(2\times1000) \\ &= 200\ \text{k}\Omega \end{aligned}$$

$$R_2 = 100R_1$$

$$R_1 = 2\ \text{k}\Omega$$

$$\omega_2 = \frac{1}{R_2C_2}$$

$$C_2 = \frac{1}{2\pi f_2 R_2}$$

Substitute $R_2 = 200\ \text{k}\Omega$, and $f_2 = 200\ \text{Hz}$.

$$\begin{aligned} C_2 &= \frac{1}{2\pi\times200\times200\times10^3} \\ &= 3.97\times10^{-4} \\ &= 3.97\ \text{nF} \end{aligned}$$

Step 12 of 12

Sketch the Bode plot.

Figure 4

Hence, the circuit design is obtained.

2.094P

We don't have the solution to this problem yet.

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2.095P

We don't have the solution to this problem yet.

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2.096P

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2.097P

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2.098P

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2.099P

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Step 1 of 3

2.100P

Refer to Figure 2.36 in the textbook for the op-amp amplifier circuit.

For DC analysis, the circuit acts as a non-inverting amplifier.

The non-inverting amplifier voltage gain is,

$$A = 1 + \frac{R_2}{R_1}$$

Assume $R_2 = 100\,\text{k}\Omega$

Substitute 100 V/V for *A* and $100\ \text{k}\Omega$ for R_2.

$$100 = 1 + \frac{100\,\text{k}}{R_1}$$

$$R_1 = 0.99\ \text{k}\Omega$$

The value of the resistance, R_3 is equal to the R_2.

$$R_3 = R_2 = 100\,\text{k}\Omega.$$

Therefore, the resistance values are,

R_1 is $\boxed{0.99\,\text{k}\Omega}$

R_2 is $\boxed{100\text{k}\Omega}$

R_3 is $\boxed{100\text{k}\Omega}$.

Step 2 of 3

Determine the value of C_1 at break frequency 100 Hz.

$$C_1 = \frac{1}{2\pi f_b R_1}$$

Substitute 100 Hz for f_b and $1.01\ \text{k}\Omega$ for R_1.

$$C_1 = \frac{1}{2\pi(100)(1.01\,\text{k})}$$
$$= 1.6\,\mu\text{F}$$

Therefore, the value of C_1 is $\boxed{1.6\,\mu\text{F}}$.

Step 3 of 3

Determine the value of C_2 at break frequency 10 Hz.

$$C_2 = \frac{1}{2\pi f_b R_2}$$

Substitute 10 Hz for f_b and $100\,\text{k}\Omega$ for R_2.

$$C_2 = \frac{1}{2\pi(10)(100\,\text{k})}$$
$$= 0.16\,\mu\text{F}$$

Therefore, the value of C_2 is $\boxed{0.16\,\mu\text{F}}$.

2.101P

We don't have the solution to this problem yet.

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2.102P

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2.103P

We don't have the solution to this problem yet.

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2.104P

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2.105P

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2.106P

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Step 1 of 5

The relation between the open-loop gain, band width and unity gain frequency is,

$f_t = A_o f_b$ ..... (1)

Calculate the unity gain frequency by substitute 10^5 for A_o and 10^2 for f_b in equation (1).

$$\begin{aligned} f_t &= A_o f_b \\ &= (10^5)(10^2) \\ &= 10^7 \text{ Hz} \end{aligned}$$

Step 2 of 5

Calculate the bandwidth by substitute 10^6 for A_o and 10^6 for f_t in equation (1).

$$\begin{aligned} f_b &= \frac{f_t}{A_o} \\ &= \frac{10^6}{10^6} \\ &= 1 \text{ Hz} \end{aligned}$$

Step 3 of 5

Calculate the open loop gain by substitute 10^3 for f_b and 10^8 for f_t in equation (1).

$$\begin{aligned} A_o &= \frac{f_t}{f_b} \\ &= \frac{10^8}{10^3} \\ &= 10^5 \end{aligned}$$

Step 4 of 5

Calculate the open loop gain by substitute 10^{-1} for f_b and 10^6 for f_t in equation (1).

$$\begin{aligned} A_o &= \frac{f_t}{f_b} \\ &= \frac{10^6}{10^{-1}} \\ &= 10^7 \end{aligned}$$

Step 5 of 5

Calculate the unity gain frequency by substitute 2×10^5 for A_o and 10 for f_b in equation (1).

$$\begin{aligned} f_t &= A_o f_b \\ &= (2\times10^5)(10) \\ &= 2\times10^6 \text{ Hz} \end{aligned}$$

Therefore, the complete table is

A_o	f_b (Hz)	f_t (Hz)
10^5	10^2	10^7
10^6	1	10^6
10^5	10^3	10^8
10^7	10^{-1}	10^6
2×10^5	10	2×10^6

Step 1 of 3

The gain equation for ingernally compensated op-amp is,

$$A(j\omega) = \frac{A_o}{1 + j\frac{\omega}{\omega_b}}$$

The magnitude equation is,

$$|A(\omega)| = \frac{|A_o|}{\sqrt{1 + \left(\frac{f}{f_b}\right)^2}}$$

$$\sqrt{1 + \left(\frac{f}{f_b}\right)^2} = \frac{|A_o|}{|A(\omega)|}$$

Multiply with $20\log_{10}$ to both sides

$$20\log_{10}\sqrt{1 + \left(\frac{f}{f_b}\right)^2} = 20\log_{10}\frac{|A_o|}{|A(\omega)|}$$

$$20\log_{10}\sqrt{1 + \left(\frac{f}{f_b}\right)^2} = 20\log_{10}|A_o| - 20\log_{10}|A(\omega)|$$

Substitute 98 dB for $20\log_{10}|A_o|$, 40 dB for $20\log_{10}|A(\omega)|$ and 100 kHz for *f*.

$$20\log_{10}\sqrt{1 + \left(\frac{100k}{f_b}\right)^2} = 98 - 40$$

$$20\log_{10}\sqrt{1 + \left(\frac{100k}{f_b}\right)^2} = 58$$

$$\sqrt{1 + \left(\frac{100k}{f_b}\right)^2} = 10^{2.9}$$

$$\left(\frac{100k}{f_b}\right)^2 = 630.96k$$

$$\frac{100k}{f_b} = 794.33$$

$$f_b = 125.9\,\text{Hz}$$

Therefore, the break frequency f_b, is $\boxed{125.9\,\text{Hz}}$.

Step 2 of 3

Convert DC gain from decibels to normal.

$$20\log_{10}|A_o| = 98$$

$$\log_{10}|A_o| = \frac{98}{20}$$

$$\log_{10}|A_o| = 4.9$$

$$A_o = 10^{4.9}$$

$$A_o = 79,432.82\,\text{V/V}$$

Therefore , the DC gain A_o, is $\boxed{79,432.82\,\text{V/V}}$.

Step 3 of 3

The unity gain frequency is,

$$f_t = A_o f_b$$

Substitute $79,432.82\,\text{V/V}$ for A_o and $125.9\,\text{Hz}$ for f_b.

$$f_t = (79,432.82)(125.9)$$
$$= 10\times10^6\,\text{Hz}$$

Therefore, the unity gain frequency f_t, is $\boxed{10\text{MHz}}$.

Step 1 of 2

2.109P

The dc gain is occurs at low frequencies.

So, the dc gain A_o is, $\boxed{20\times10^3\ \text{V/V}}$.

The gain equation for in genarally compensated op-amp is,

$$A(j\omega)=\frac{A_o}{1+j\dfrac{\omega}{\omega_b}}$$

The magnitude equation is,

$$|A(\omega)|=\frac{|A_o|}{\sqrt{1+\left(\dfrac{f}{f_b}\right)^2}}$$

$$\sqrt{1+\left(\frac{f}{f_b}\right)^2}=\frac{|A_o|}{|A(\omega)|}$$

Multiply with $20\log_{10}$ to both sides

$$20\log_{10}\sqrt{1+\left(\frac{f}{f_b}\right)^2}=20\log_{10}\frac{|A_o|}{|A(\omega)|}$$

$$20\log_{10}\sqrt{1+\left(\frac{f}{f_b}\right)^2}=20\log_{10}|A_o|-20\log_{10}|A(\omega)|$$

Substitute $20\times10^3\ \text{V/V}$ for A_o, $4\times10^3\ \text{V/V}$ for $A(\omega)$ and 100 kHz for *f*.

$$20\log_{10}\sqrt{1+\left(\frac{100\text{k}}{f_b}\right)^2}=20\log_{10}|20\times10^3|-20\log_{10}|4\times10^3|$$

$$20\log_{10}\sqrt{1+\left(\frac{100\text{k}}{f_b}\right)^2}=86-72$$

$$20\log_{10}\sqrt{1+\left(\frac{100\text{k}}{f_b}\right)^2}=14$$

$$\sqrt{1+\left(\frac{100\text{k}}{f_b}\right)^2}=10^{0.7}$$

$$\frac{100\text{k}}{f_b}=4.9$$

$$f_b=20.36\,\text{kHz}$$

Therefore, the break frequency f_b, is $\boxed{20.36\,\text{kHz}}$.

Step 2 of 2

The unity gain frequency is,

$$f_t=A_o f_b$$

Substitute $20\times10^3\ \text{V/V}$ for A_o and $20.36\,\text{kHz}$ for f_b.

$$f_t=(20\times10^3)(20.36)$$
$$=0.41\times10^9\,\text{Hz}$$

Therefore, the unity gain frequency f_t, is $\boxed{0.41\times10^9\,\text{Hz}}$.

2.110P

We don't have the solution to this problem yet.

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2.111P

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2.112P

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2.113P

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2.114P

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2.115P

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2.116P

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2.117P

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Step 1 of 6

The stable gain of the amplifier is, 100 V/V at 5 MHz.

If single operational amplifier is used, then the unity-gain frequency is,

$$\begin{aligned} f_t &= Af \\ &= (100)\left(5\times10^6\right) \\ &= 500\ \text{MHz} \end{aligned}$$

Thus, the unity-gain frequency of a single operational amplifier, f_t is $\boxed{500\ \text{MHz}}$.

Step 2 of 6

The best available operational amplifier has a unity gain frequency of $f_t = 40\ \text{MHz}$.

The closed loop gain of the amplifier is,

$$\begin{aligned} \left|A_{(5\ \text{MHz})}\right| &= \frac{40\times10^6}{5\times10^6} \\ &= 8 \end{aligned}$$

To obtain overall gain of 100, three amplifiers connected in cascade would be required.

Step 3 of 6

The 3-dB frequency of each stage is, $f_{3\text{-dB}} = \dfrac{40}{K}\ \text{MHz}$.

The closed loop gain of each stage is,

$$|A| = \frac{K}{\sqrt{1+\left(\dfrac{f}{\left(\dfrac{40}{K}\right)}\right)^2}}$$

Calculate the gain at 5MHz.

$$\begin{aligned} |A| &= \frac{K}{\sqrt{1+\left(\dfrac{5}{\left(\dfrac{40}{K}\right)}\right)^2}} \\ &= \frac{K}{\sqrt{1+\left(\dfrac{K}{8}\right)^2}} \end{aligned}$$

Step 4 of 6

To achieve a gain of 100 three stages must be cascaded.

Therefore,

$$100 = \left(\frac{K}{\sqrt{1+\left(\dfrac{K}{8}\right)^2}}\right)^3$$

$$4.642 = \frac{K}{\sqrt{1+\left(\dfrac{K}{8}\right)^2}}$$

$$21.5443 = \frac{K^2}{1+\dfrac{K^2}{64}}$$

$$21.5443 + 0.3366K^2 = K^2$$

$$K^2 - 0.3366K^2 = 21.5443$$

$$0.6634K^2 = 21.5443$$

Simplify further.

$$K^2 = \frac{21.5443}{0.6634}$$

$$K = \sqrt{\frac{21.5443}{0.6634}}$$

$$K = 5.7$$

Step 5 of 6

The 3 – dB frequency of each stage is,

$$\begin{aligned} f_{3\text{dB}} &= \frac{f_t}{K} \\ &= \frac{40\times10^6}{5.7} \\ &= 7.02\ \text{MHz} \end{aligned}$$

Thus, the 3 dB frequency of each stage of the operational amplifier, $f_{3\text{dB}}$ is $\boxed{7.02\ \text{MHz}}$.

Step 6 of 6

The overall 3 dB frequency of the amplifier is,

$$A = \frac{K}{\sqrt{1+\left(\dfrac{f}{f_{3\text{dB}}}\right)^2}}$$

$$\frac{5.7^3}{\sqrt{2}} = \left[\frac{5.7}{\sqrt{1+\left(\dfrac{f}{7.02\times10^6}\right)^2}}\right]^3$$

$$\frac{1}{\sqrt{1+\left(\dfrac{f}{7.02\times10^6}\right)^2}} = \left(\frac{1}{\sqrt{2}}\right)^{\frac{1}{3}}$$

$$\frac{1}{\sqrt{1+\left(\dfrac{f}{7.02\times10^6}\right)^2}} = 0.89$$

$$1+\left(\frac{f}{7.02\times10^6}\right)^2 = \frac{1}{0.89^2}$$

Simplify further.

$$1+\left(\frac{f}{7.02\times10^6}\right)^2 = \frac{1}{0.793}$$

$$\left(\frac{f}{7.02\times10^6}\right)^2 = 1.26-1$$

$$\left(\frac{f}{7.02\times10^6}\right)^2 = 0.26$$

$$\frac{f}{7.02\times10^6} = \sqrt{0.26}$$

$$\frac{f}{7.02\times10^6} = 0.509$$

$$f = 3.58\ \text{MHz}$$

Thus, the overall 3-dB frequency of the cascaded non-inverting stages, f is $\boxed{3.58\ \text{MHz}}$.

Step 1 of 2

The unity gain frequency of operational amplifier is, f_t .

(a)

The dc gain of the inverting amplifier is,

$$\frac{R_2}{R_1} = K$$

It is know that,

$$f_t = |A_o| \times f_{3dB}$$

$$f_{3dB} = \frac{f_t}{1 + \frac{R_2}{R_1}}$$

$$= \frac{f_t}{1+K}$$

The 3-dB frequency of the operational amplifier, f_{3dB} is $\boxed{\frac{f_t}{1+K}}$.

Calculate the gain-band width product.

$$\text{GBP} = |Gain|(f_{3dB})$$

$$= K\left(\frac{f_t}{1+K}\right)$$

Thus, the gain-band width product of the inverting amplifier, **GBP** is $\boxed{K\frac{f_t}{1+K}}$.

Step 2 of 2

(b)

The dc gain of the non-inverting amplifier is, $1 + \frac{R_2}{R_1} = K$

It is know that,

$$f_t = |A_o| \times f_{3dB}$$

$$f_{3dB} = \frac{f_t}{1 + \frac{R_2}{R_1}}$$

$$= \frac{f_t}{K}$$

The 3-dB frequency of the operational amplifier, f_{3dB} is, $\boxed{\frac{f_t}{K}}$.

Calculate the gain band width product.

$$\text{GBP} = |Gain| f_{3dB}$$

$$= K\left(\frac{f_t}{K}\right)$$

$$= f_t$$

Thus, the gain-band width product of the inverting amplifier, **GBP** is $\boxed{f_t}$.

Therefore, it is conclude that non-inverting configuration realizes a higher gain-bandwidth product.

2.120P

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2.121P

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2.122P

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Ask an expert

2.123P

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Step 1 of 2

2.124P

The required output is, $v_o = 10\,\text{V}$.

The width of the pulses is, $W = 2\ \mu\text{s}$

The sum of rise time and fall time is equal to 20% of pulse width (W).

That is,

$$\begin{aligned} t_r + t_f &= 0.2W \\ &= 0.2(2\ \mu\text{s}) \\ &= 0.4\ \mu\text{s} \end{aligned}$$

It is known that the rise time and fall times of pulse signal are usually measured between the 10% and 90% height points.

Step 2 of 2

Calculate the slew rate.

$$\begin{aligned} \text{SR} &= \frac{dv_o}{dt} \\ &= \frac{\Delta v}{\Delta t} \\ &= \frac{\Delta v}{t_r} \end{aligned}$$

$$\begin{aligned} &= \frac{(0.9-0.1)v_o}{0.2\times10^{-6}} \\ &= \frac{(0.9-0.1)(10)}{0.2\times10^{-6}} \\ &= \frac{40}{10^{-6}}\ \text{V/s} \\ &= 40\ \text{V/}\mu\text{s} \end{aligned}$$

Thus, the slew rate requirement for an operational amplifier, SR is $\boxed{40\ \text{V/}\mu\text{s}}$.

2.125P

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2.126P

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2.127P

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Step 1 of 2

3.001E

Calculate the intrinsic carrier density for silicon at $T = 50\,\mathrm{K}$.

$$n_i = BT^{\frac{3}{2}} e^{\frac{-E_g}{2kT}}$$

Here,

B is a material dependent parameter,

E_g is bandgap energy ,

k is Boltzmann's constant.

Substitute $7.3 \times 10^{15}\,\mathrm{cm^{-3}K^{-3/2}}$ for B , $1.12\,\mathrm{eV}$ for E_g , $50\,\mathrm{K}$ for T and $8.62 \times 10^{-5}\,\mathrm{eV/K}$ for k in the equation.

$$n_i = \left(7.3 \times 10^{15}\right)\left(50^{\frac{3}{2}}\right) e^{\left(\frac{-1.12}{2 \times 8.62 \times 10^{-5} \times 50}\right)}$$

$$= \left(7.3 \times 10^{15}\right)(353.5534)\, e^{-129.93}$$

$$= \left(2.581 \times 10^{18}\right)\left(3.733 \times 10^{-57}\right)$$

$$= 9.63 \times 10^{-39}$$

$$\simeq 9.6 \times 10^{-39}\ \mathrm{carriers/cm^3}$$

Thus, the intrinsic carrier density for silicon at $T = 50\,\mathrm{K}$, n_i is $\boxed{9.6 \times 10^{-39}\ \mathrm{carriers/cm^3}}$.

Step 2 of 2

Calculate the intrinsic carrier density for silicon at $T = 350\,\mathrm{K}$.

$$n_i = BT^{\frac{3}{2}} e^{\frac{-E_g}{2kT}}$$

Substitute $7.3 \times 10^{15}\,\mathrm{cm^{-3}K^{-3/2}}$ for B , $1.12\,\mathrm{eV}$ for E_g , $8.62 \times 10^{-5}\,\mathrm{eV/K}$ for k and $350\,\mathrm{K}$ for T in the equation.

$$n_i = \left(7.3 \times 10^{15}\right)\left(350^{\frac{3}{2}}\right) e^{\left(\frac{-1.12}{2 \times 8.62 \times 10^{-5} \times 350}\right)}$$

$$= \left(7.3 \times 10^{15}\right)(6547.9)\, e^{-18.5615}$$

$$= \left(4.7799 \times 10^{19}\right)\left(8.69 \times 10^{-9}\right)$$

$$= 4.15 \times 10^{11}\ \mathrm{carriers/cm^3}$$

Thus, the intrinsic carrier density for silicon at $T = 350\,\mathrm{K}$, n_i is $\boxed{4.15 \times 10^{11}\ \mathrm{carriers/cm^3}}$.

3.001P

We don't have the solution to this problem yet.

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Step 1 of 2

3.002E

Since the material is *n*-type silicon, the concentration of the majority electrons is,

$$n_n \approx N_D$$
$$= 10^{17} / \text{cm}^3$$

Therefore, the concentration of the majority electrons is $\boxed{10^{17} / \text{cm}^3}$.

Step 2 of 2

The intrinsic carrier density for Silicon at $T = 350\text{K}$ is,

$$n_i = 4.154 \times 10^{11} / \text{cm}^3$$

The concentration of the minority holes is,

$$p_n = \frac{n_i^2}{N_D}$$

Here, n_i is the intrinsic carrier density for silicon, that is $n_i = 4.154 \times 10^{11} / \text{cm}^3$

Substitute $4.154 \times 10^{11} / \text{cm}^3$ for n_i and $10^{17} / \text{cm}^3$ for N_D in the equation.

$$p_n = \frac{\left(4.154 \times 10^{11}\right)^2}{10^{17}}$$
$$= \frac{17.256 \times 10^{22}}{10^{17}}$$
$$= 17.256 \times 10^5 / \text{cm}^3$$

Therefore, the concentration of the minority holes is $\boxed{17.256 \times 10^5 / \text{cm}^3}$.

3.002P

Write the formula for intrinsic carrier density of gallium arsenide at the temperature, T .

$$n_i = BT^{\frac{3}{2}} e^{\frac{-E_g}{2kT}}$$

Here,

B is a material dependent parameter that is B

The bandgap energy E_g

And k is Boltzmann's constant, that is k

The temperature is $T(\text{in K})$

Substitute $3.56\times10^{14}\,\text{cm}^{-3}\text{K}^{-3/2}$ for B , $1.42\,\text{eV}$ for E_g , $8.62\times10^{-5}\,\text{eV/K}$ for k , $300\,\text{K}$ for T to calculate the carrier concentration, n_i .

$$\begin{aligned} n_i &= BT^{\frac{3}{2}} e^{\frac{-E_g}{2kT}} \\ &= \left(3.56\times10^{14}\right)\left(300^{\frac{3}{2}}\right) e^{\frac{-1.42}{2\left(8.62\times10^{-5}\right)300}} \\ &= \left(18498.30262\times10^{14}\right) e^{-27.45552978} \\ &= \left(18498.30262\times10^{14}\right)\left(1.191831668\times10^{-12}\right) \end{aligned}$$

Simplify further.

$$\begin{aligned} n_i &= 2204686.286 \\ &= 2.20469\times10^{6}\ \text{carriers/cm}^3 \end{aligned}$$

Therefore, the carrier concentration n_i at $T = 300\,\text{K}$ is, $\boxed{2.20469\times10^{6}\ \text{carriers/cm}^3}$.

Step 1 of 2

3.003E

The intrinsic carrier density for Silicon at temperature $T = 300\,\mathrm{K}$ is,

$$n_i = 1.5\times10^{10}\ \mathrm{carriers/cm^3}$$

Here, the silicon semiconductor is doped with a trivalent *p-type* impurity (boron).

The electron concentration drops below the **intrinsic level** by a factor of 10^6 . Therefore, the new electron concentration is simply ratio of the intrinsic carrier density n_i to 10^6 .

Calculate the new concentration of free electrons in the *p*-type boron.

$$\begin{aligned} n_p &= \frac{n_i}{10^6} \\ &= \frac{1.5\times10^{10}}{10^6} \\ &= 1.5\times10^4\ \mathrm{electrons/cm^3} \end{aligned}$$

Step 2 of 2

Use the below relation to find the concentration of minority electrons.

$$p_p n_p = n_i^2$$

$$p_p = \frac{n_i^2}{n_p}$$

Substitute 1.5×10^{10} for n_i and 1.5×10^4 for n_p in the equation.

$$\begin{aligned} p_p &= \frac{\left(1.5\times10^{10}\right)^2}{1.5\times10^4} \\ &= 1.5\times10^{16}\ /\mathrm{cm^3} \end{aligned}$$

Here, the silicon semiconductor is doped with a trivalent *p-type* impurity (boron).

Thus, the concentration of the ***majority*** carriers (holes) is equal to N_A (that is, if the acceptor doping concentration is N_A , where $N_A \gg n_i$, the hole concentration becomes $p_p \approx N_A$).

The acceptor concentration is,

$$\begin{aligned} N_A &\approx p_p \\ &= 1.5\times10^{16}\ /\mathrm{cm^3} \end{aligned}$$

Therefore, the acceptor doping concentration N_A is $\boxed{1.5\times10^{16}\ /\mathrm{cm^3}}$.

Step 1 of 2

3.003P

The majority carriers are holes and minority carriers are electrons in a p type semiconductor. The intrinsic carrier density for Silicon at $T = 300\ \text{K}$ is,

$$n_i = 1.5\times10^{10}\ \text{carriers/cm}^3$$

The acceptor doping concentration or doping concentration of the p side is,

$$N_A = 5\times10^{18}\ \text{carriers/cm}^3$$

If the acceptor doping concentration is N_A, where $N_A \gg n_i \left(5\times10^{18} \gg 1.5\times10^{10}\right)$, then the majority hole concentration becomes $p_p \approx N_A$.

Therefore, the hole concentration at $T = 300\ \text{K}$ is, $\boxed{5\times10^{18}\ \text{carriers/cm}^3}$.

Step 2 of 2

The concentration of minority electrons is,

$$p_p n_p = n_i^2$$

Substitute $5\times10^{18}\ \text{carriers/cm}^3$ for p_p to calculate the minority electron concentration.

$$\begin{aligned} n_p &= \frac{n_i^2}{p_p} \\ &= \frac{\left(1.5\times10^{10}\right)^2}{5\times10^{18}} \\ &= 45\ \text{carriers/cm}^3 \end{aligned}$$

Therefore, the electron concentration at $T = 300\ \text{K}$ is, $\boxed{45\ \text{carriers/cm}^3}$.

Step 1 of 4

(a)

The electron drift velocity is,

$$\begin{aligned} v_{n\text{-drift}} &= -\mu_n E \\ &= -\mu_n \left(\frac{V}{l}\right) \end{aligned}$$

The negative sign indicates the direction of electrons movement opposite to electric field.

Substitute $1350\ \text{cm}^2/\text{V.s}$ for μ_n, $1\ \text{V}$ for V, and $2\ \mu\text{m}$ for l in the equation.

$$\begin{aligned} v_{n\text{-drift}} &= 1350\left(\frac{1}{2\times 10^{-6}\ \text{m}}\right) \\ &= 1350\left(\frac{1}{2\times 10^{-4}\ \text{cm}}\right) \\ &= 6.75\times 10^6\ \text{cm/s} \end{aligned}$$

Thus, the electron drift velocity is, $\boxed{v_{n\text{-drift}} = 6.75\times 10^6\ \text{cm/s}}$.

Step 2 of 4

(b)

The time taken by the electron to cross the $2\ \mu\text{m}$ length is,

$$T = \frac{\text{length of the semiconductor}(l)}{\text{Drift velocity}(v_{n\text{-drift}})}$$

Substitute $2\ \mu\text{m}$ for l and $6.75\times 10^6\ \text{cm/s}$ for $v_{n\text{-drift}}$ in the equation.

$$\begin{aligned} T &= \frac{2\times 10^{-6}\ \text{m}}{6.75\times 10^6\ \text{cm/s}} \\ &= \frac{2\times 10^{-4}\ \text{cm}}{6.75\times 10^6\ \text{cm/s}} \\ &= 29.63\ \text{ps} \end{aligned}$$

Thus, the time taken by the electron to cross the length is, $\boxed{T = 29.63\ \text{ps}}$.

Step 3 of 4

(c)

The drift current density is,

$$\begin{aligned} J_n &= qn\mu_n E \\ &= qn\mu_n \left(\frac{V}{l}\right) \end{aligned}$$

Substitute $1.6\times 10^{-19}\ \text{C}$ for q, $10^{16}/\text{cm}^3$ for n, $1350\ \text{cm}^2/\text{V.s}$ for μ_n, $1\ \text{V}$ for V, and $2\ \mu\text{m}$ for l in the equation.

$$\begin{aligned} J_n &= \left(1.6\times 10^{-19}\right)\left(10^{16}\right)(1350)\left(\frac{1}{2\times 10^{-6}\ \text{m}}\right) \\ &= \left(1.6\times 10^{-19}\right)\left(10^{16}\right)(1350)\left(\frac{1}{2\times 10^{-4}\ \text{cm}}\right) \\ &= 1.08\times 10^4\ \text{A/cm}^2 \end{aligned}$$

Thus, the drift current density is, $\boxed{J_n = 1.08\times 10^4\ \text{A/cm}^2}$.

Step 4 of 4

(d)

The drift current is,

$$I_n = J_n A$$

Substitute $1.08\times 10^4\ \text{A/cm}^2$ for J_n and $0.25\ \mu\text{m}^2$ for A in the equation.

$$\begin{aligned} I_n &= \left(1.08\times 10^4\right)\left(0.25\times 10^{-12}\ \text{m}^2\right) \\ &= \left(1.08\times 10^4\right)\left(0.25\times 10^{-12}\times 10^4\ \text{cm}^2\right) \\ &= 27\ \mu\text{A} \end{aligned}$$

Thus, the drift current is, $\boxed{I_n = 27\ \mu\text{A}}$.

Step 1 of 4

Intrinsic concentration η_i of silicon can be calculated as follows:

$$\eta_i = BT^{\frac{3}{2}} e^{-\frac{E_g}{2kT}} \quad \text{..... (1)}$$

Here,

B is a material dependent parameter,

E_g is bandgap energy,

k is Boltzmann's constant

T is the temperature in Kelvin.

Step 2 of 4

Substitute $7.3\times10^{15}\,\text{cm}^{-3}\text{K}^{-3/2}$ for B, $1.12\,\text{eV}$ for E_g, $8.62\times10^{-5}\,\text{eV/K}$ for k, $300\,\text{K}$ for T in equation (1).

$$\begin{aligned} n_i &= BT^{\frac{3}{2}} e^{\frac{-E_g}{2kT}} \\ &= \left(7.3\times10^{15}\right)\left(300^{\frac{3}{2}}\right) e^{\frac{-1.12}{2\left(8.62\times10^{-5}\right)300}} \\ &= \left(7.3\times10^{15}\right)\left(5196.152423\right) e^{-21.65506574} \\ &= \left(37931.91269\times10^{15}\right)\left(3.938441912\times10^{-10}\right) \end{aligned}$$

$\eta_i = 1.5\times10^{10}$ /cm^3.

Therefore, the intrinsic concentration of silicon is $\boxed{1.5\times10^{10}/\text{cm}^3}$.

Step 3 of 4

Silicon crystal doped with phosphorus, Hence, n-type semiconductor is formed. For a n-type semiconductor concentration of electrons is equal to the donor concentration. $n_n = N_D$.

Form the Mass-action law,

$$np = \eta_i^2$$
$$N_D p_n = \eta_i^2$$
$$N_D = \frac{\eta_i^2}{p_n} \quad \text{..... (2)}$$

Step 4 of 4

Consider the hole concentration drops below the intrinsic level by a factor of 10^8 because of Si doped with Phosphorus.

Thus, the hole concentration is ,

$$p_n = \frac{\eta_i}{10^8}$$

Substitute $1.5\times10^{10}/\text{cm}^3$ for η_i and $\frac{\eta_i}{10^8}$ for p_n in equation (2).

$$\begin{aligned} N_D &= \frac{\eta_i^2}{\frac{\eta_i}{10^8}} \frac{\left(1.5\times10^{10}\right)}{\left(10^8\right)}\left(\frac{1}{\text{cm}^3}\right) \\ &= \eta_i \times10^8/\text{cm}^3 \\ &= 1.5\times10^{10}\times10^8/\text{cm}^3 \\ &= 1.5\times10^{18}/\text{cm}^3 \end{aligned}$$

Therefore, the concentration of N_D is must be $\boxed{1.5\times10^{18}/\text{cm}^3}$.

Step 1 of 3

3.005E

Refer to Figure E3.5 in the textbook for the linear electron-concentration profile for silicon semiconductor.

We have

$n_0 = 10^{17}\ /\text{cm}^3$

$W = 1\ \mu\text{m}$

The electron current density is,

$$J_n = qD_n \frac{dn(x)}{dx},$$

Here, q is the charge of an electron, $q = 1.6 \times 10^{-19}\text{C}$,

D_n is the diffusion constant of the holes, $D_n = 35\ \text{cm}^2/\text{s}$ for silicon and

$\frac{dn(x)}{dx}$ is the concentration gradient or the slope of the concentration profile.

From the linear electron-concentration profile for silicon semiconductor, write the following relation:

$$\frac{dn(x)}{dx} = \frac{n_0}{W}$$

Substitute $10^{17}\ /\text{cm}^3$ for n_0 and $1\ \mu\text{m}$ for W in the equation.

$$\begin{aligned}\frac{dn(x)}{dx} &= \frac{10^{17}\ /\text{cm}^3}{1\ \mu\text{m}} \\ &= \frac{10^{17} \times 10^6}{1 \times 10^{-6}} \\ &= 10^{29}\end{aligned}$$

Step 2 of 3

The current density is,

$$J_n = qD_n \frac{dn(x)}{dx}$$

Substitute $1.6\times10^{-19}\ \text{C}$ for q, $35\ \text{cm}^2/\text{s}$ for D_n, and 10^{29} for $\frac{dn(x)}{dx}$ in the equation.

$$\begin{aligned}J_n &= \left(1.6\times10^{-19}\right)\left(35\times10^{-4}\right)\left(10^{29}\right) \\ &= 56\ \mu\text{A}/\mu\text{m}^2\end{aligned}$$

Thus, the current density is, $\boxed{J_n = 56\ \mu\text{A}/\mu\text{m}^2}$.

Step 3 of 3

The diffusion current is,

$$I_n = J_n \times A$$

$$A = \frac{I_n}{J_n}$$

Substitute $1\ \text{mA}$ for I_n and $56\ \mu\text{A}/\mu\text{m}^2$ for J_n in the equation.

$$\begin{aligned}A &= \frac{1 \times 10^{-3}}{56 \times 10^6} \\ &= 17.86\ \mu\text{m}^2 \\ &\approx 18\ \mu\text{m}^2\end{aligned}$$

Thus, the cross-sectional area is $\boxed{A \approx 18\ \mu\text{m}^2}$.

Step 1 of 6

Convert the temperature T from centigrade to kelvins.

$$T_{(\text{kelvin})} = T_{(\text{centigrade})} + 273$$

For 27°C temperature,

$$\begin{aligned} T_{(\text{kelvin})} &= 27 + 273 \\ &= 300\ \text{K} \end{aligned}$$

For 125°C temperature,

$$\begin{aligned} T_{(\text{kelvin})} &= 125 + 273 \\ &= 398\ \text{K} \end{aligned}$$

Step 2 of 6

Consider a phosphorus-doped silicon layer with the impurity concentration of $10^{17}/\text{cm}^3$. Since, this donates electrons. Here, donor concentration N_D is much greater than the intrinsic concentration, n_i.

The concentration of free electrons in the *n*-type silicon is,

$$n_n \approx N_D$$

Therefore, the donor concentration is, $N_D = 10^{17}/\text{cm}^3$.

Therefore, the concentration of majority electrons n_n at $27°\text{C and } 125°\text{C}$ is $\boxed{10^{16}/\text{cm}^3}$.

Step 3 of 6

Intrinsic concentration η_i of silicon can be calculated as follows:

$$\eta_i = BT^{3/2} e^{-E_g/2kT} \quad \text{..... (1)}$$

Here,

B is material dependent constant,

T is temperature in Kelvin,

E_g is energy bandgap of Si,

k is Boltzmann's constant.

Substitute $7.3\times10^{15}\,\text{cm}^{-3}\text{K}^{-3/2}$ for B, $300\ \text{K}$ for T, $1.12\ \text{eV}$ for E_g for silicon, $8.62\times10^{-5}\ \text{eV/K}$ for k in equation (1).

$$\eta_i = \left(7.3\times10^{15}\ \text{cm}^{-3}\text{K}^{-3/2}\right)(300\ \text{K})^{\frac{3}{2}} e^{\frac{-1.12\ \text{eV}}{2(8.62\times10^{-5}\ \text{eV/K})(300\ \text{K})}}$$

$$\begin{aligned} \eta_i &= \left(7.3\times10^{15}\right)(5196.152423)e^{-21.65506574} \\ &= 1.5\times10^{10}/\text{cm}^3 \end{aligned}$$

Step 4 of 6

Under thermal equilibrium, the equation for the mass action law for the semiconductors is,

$$n_n p_n = n_i^2$$

Since $n_n \approx N_D$

$$N_D p_n = n_i^2$$

Calculate the hole concentration at $T = 27°\text{C or } 300\ \text{K}$.

$$\begin{aligned} p_n &= \frac{n_i^2}{N_D} \\ &= \frac{\left(1.5\times10^{10}\right)^2}{10^{17}} \\ &= \frac{2.25\times10^{20}}{10^{17}} \\ &= 2.25\times10^3/\text{cm}^3 \end{aligned}$$

Thus, the hole concentration p_n at $27°\text{C}$ is $\boxed{2.25\times10^3/\text{cm}^3}$.

Step 5 of 6

Calculate the intrinsic concentration of a material at $T = 125°\ \text{C or } 398\ \text{K}$.

$$\begin{aligned} n_i &= BT^{\frac{3}{2}} e^{\frac{-E_g}{2kT}} \\ &= \left(7.3\times10^{15}\right)\left(398^{\frac{3}{2}}\right)e^{\left(\frac{-1.12}{(2)(8.62\times10^{-5})(398)}\right)} \\ &= \left(57962.54796\times10^{15}\right)e^{-16.32291387} \\ &= 4.722762\times10^{12}/\text{cm}^3 \end{aligned}$$

Step 6 of 6

Calculate the hole concentration at $T = 125°\ \text{C or } 398\ \text{K}$.

$$\begin{aligned} p_n &= \frac{n_i^2}{N_D} \\ &= \frac{\left(4.722762\times10^{12}\right)^2}{10^{17}} \\ &= \frac{22.304481\times10^{24}}{10^{17}} \\ &= 22.3045\times10^7/\text{cm}^3 \end{aligned}$$

Thus, the hole concentration p_n at $125°\ \text{C}$ is $\boxed{22.3045\times10^7/\text{cm}^3}$.

Step 1 of 3

3.006E

The following is the equation that defines the relationship between diffusion constant with mobility:

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = V_T$$

Here, V_T is known as the thermal voltage, $V_T = \frac{kT}{q}$

At room temperature, $T = 300\ \text{K}$, the thermal voltage is $V_T = 25.9\ \text{mV}$

Step 2 of 3

The diffusion constant of electrons is,

$$D_n = \mu_n V_T$$

Substitute $1350\ \text{cm}^2/\text{V.s}$ for μ_n and $25.9\ \text{mV}$ for V_T in the equation.

$$D_n = 1350\left(25.9 \times 10^{-3}\right)$$
$$= 34.96\ \text{cm}^2/\text{s}$$

Therefore, the diffusion constant of electrons is $\boxed{D_n = 34.96\ \text{cm}^2/\text{s}}$.

Step 3 of 3

The diffusion constant of holes is,

$$D_p = \mu_p V_T$$

Substitute $480\ \text{cm}^2/\text{V.s}$ for μ_p and $25.9\ \text{mV}$ for V_T in the equation.

$$D_p = 480\left(25.9 \times 10^{-3}\right)$$
$$= 12.432\ \text{cm}^2/\text{s}$$

Therefore, the diffusion constant of holes is $\boxed{D_p = 12.432\ \text{cm}^2/\text{s}}$.

Step 1 of 13

(a)

Consider the material is intrinsic silicon. Under thermal equilibrium, the concentration of free electrons n is equal to the concentration of holes p. That is,

$$n = p = n_i$$

Thus, the resistivity ρ can be calculated as follows.

$$\rho = \frac{1}{\eta_i q(\mu_n + \mu_p)} \quad \text{..... (1)}$$

Here,

q is charge of an electron,

η_i is intrinsic concentraion of silicon,

μ_n is mobility of electrons,

μ_p is mobility of holes.

Substitute $1.6\times10^{-19}\text{C}$ for q, $1.5\times10^{10}/\text{cm}^3$ for η_i, $1350\ \text{cm}^2/\text{V.s}$ for μ_n, $480\ \text{cm}^2/\text{V.s}$ for μ_p in equation (1).

$$\begin{aligned}\rho &= \frac{1}{(1.6\times10^{-19}\text{C})(1.5\times10^{10}/\text{cm}^3)(1350\ \text{cm}^2/\text{V.s} + 480\ \text{cm}^2/\text{V.s})}\\ &= 227686.703\ \Omega\cdot\text{cm}\end{aligned}$$

Step 2 of 13

The resistance of intrinsic silicon can be calculated as follows:

$$R = \frac{\rho\ell}{A}$$

Here,

R is resistance,

ρ is resistivity,

ℓ length of the bar,

A cross sectional area.

Calculate the area of bar.

$$A = wt$$

Substitute 3 μm for w (width), 1 μm for t (thickness).

$$\begin{aligned}A &= (3\ \mu\text{m})(1\ \mu\text{m})\\ &= 3\ \mu\text{m}^2\end{aligned}$$

Step 3 of 13

Resistance is,

$$R = \frac{\rho l}{A}$$

Substitute $227686.703\ \Omega\cdot\text{cm}$ for ρ, $10\ \mu\text{m}$ for l, $3\ \mu\text{m}^2$ for A in the above expression.

$$\begin{aligned}R &= \frac{(227686.703\ \Omega\cdot\text{cm})(10\ \mu\text{m})}{(3\ \mu\text{m}^2)}\\ &= \frac{(227686.703\times10^4\ \Omega.\mu\text{m})(10\ \mu\text{m})\Omega}{(3\ \mu\text{m}^2)}\\ &= 7.58\times10^9\ \Omega\\ &= 7.6\ \text{G}\Omega\end{aligned}$$

Therefore, the resistance of an intrinsic silicon is $\boxed{7.6\ \text{G}\Omega}$.

Step 4 of 13

(b)

The dopant concentration level for n-doped silicon, $N_D = 5\times10^{16}/\text{cm}^3$.

The free electron concentration is approximately equal to the donors doping concentration. The electron concentration is determined by the doping concentration. Therefore, for the n-doped silicon,

$$\begin{aligned}n_n &\simeq N_D \qquad \{\text{since } N_D = 5\times10^{16}/\text{cm}^3\}\\ n_n &= 5\times10^{16}/\text{cm}^3\end{aligned}$$

Under thermal equilibrium, the equation for the mass action law for the semiconductors is,

$$n_n p_n = n_i^2$$

Since $n_n \simeq N_D$

$$N_D p_n = n_i^2$$

Step 5 of 13

The doping concentration N_A can be calculated as shown below.

$$\begin{aligned}p &= N_A\\ &= \frac{\eta_i^2}{N_D}\end{aligned}$$

Substitute $1.5\times10^{10}/\text{cm}^3$ for η_i, $5\times10^{16}/\text{cm}^3$ for N_D in the above expression.

$$\begin{aligned}N_A &= \frac{(1.5\times10^{10}/\text{cm}^3)^2}{5\times10^{16}/\text{cm}^3}\\ &= 0.45\times10^4/\text{cm}^3\end{aligned}$$

Step 6 of 13

Resistivity ρ can be calculated as follows:

$$\rho = \frac{1}{qN_D\mu_n + qN_A\mu_p}$$

Substitute $1.6\times10^{-19}\text{C}$ for q, $5\times10^{14}/\text{cm}^3$ for N_D, $1200\ \text{cm}^2/\text{V.s}$ for μ_n, $0.45\times10^4/\text{cm}^3$ for N_A, $400\ \text{cm}^2/\text{V.s}$ for μ_p

$$\begin{aligned}\rho &= \frac{1}{(1.6\times10^{-19}\text{C})(5\times10^{14}/\text{cm}^3)(1200\ \text{cm}^2/\text{V.s}) + (1.6\times10^{-19}\text{C})(0.45\times10^4/\text{cm}^3)(400\ \text{cm}^2/\text{V.s})}\\ &= 0.1042\ \Omega\cdot\text{cm}\end{aligned}$$

Step 7 of 13

Determine the resistance of an n-doped silicon.

$$R = \frac{\rho\ell}{A}$$

Substitute $0.1042\ \Omega\cdot\text{cm}$ for ρ, $10\ \mu\text{m}$ for l, $3\ \mu\text{m}^2$ for A in the above expression.

$$\begin{aligned}R &= \frac{(0.1042\ \Omega\cdot\text{cm})(10\mu\text{m})}{3\mu\text{m}^2}\\ &= \frac{(0.1042\times10^4\ \mu\text{m})(10\mu\text{m})\Omega}{3\mu\text{m}^2}\\ &= 3473\ \Omega\end{aligned}$$

Therefore, the resistance value is $\boxed{3473\ \Omega}$.

Step 8 of 13

(c)

The doping concentration N_A value can be calculated as shown below.

$$\begin{aligned}p &= N_A\\ &= \frac{\eta_i^2}{N_D}\end{aligned}$$

Substitute $1.5\times10^{10}/\text{cm}^3$ for η_i, $5\times10^{18}/\text{cm}^3$ for N_D in the above expression.

$$\begin{aligned}N_A &= \frac{(1.5\times10^{10}/\text{cm}^3)^2}{5\times10^{18}/\text{cm}^3}\\ &= 0.45\times10^2/\text{cm}^3\end{aligned}$$

Step 9 of 13

The resistivity ρ can be calculated as follows:

$$\rho = \frac{1}{qN_D\mu_n + qN_A\mu_p}$$

Substitute $1.6\times10^{-19}\text{C}$ for q, $5\times10^{18}/\text{cm}^3$ for N_D, $380\ \text{cm}^2/\text{V.s}$ for μ_n, $0.45\times10^2/\text{cm}^3$ for N_A, $160\ \text{cm}^2/\text{V.s}$ for μ_p

$$\begin{aligned}\rho &= \frac{1}{(1.6\times10^{-19}\text{C})(5\times10^{18}/\text{cm}^3)(380\ \text{cm}^2/\text{V.s}) + (1.6\times10^{-19}\text{C})(0.45\times10^2/\text{cm}^3)(160\ \text{cm}^2/\text{V.s})}\\ &= 0.003289\ \Omega\cdot\text{cm}\end{aligned}$$

Step 10 of 13

Calculate the resistance of an n-doped silicon.

$$R = \frac{\rho l}{A}$$

Substitute $0.003289\ \Omega\cdot\text{cm}$ for ρ, $10\ \mu\text{m}$ for l, $3\ \mu\text{m}^2$ for A in the above expression.

$$\begin{aligned}R &= \frac{(0.003289\ \Omega\cdot\text{cm})(10\mu\text{m})}{3\mu\text{m}^2}\\ &= \frac{(0.003289\times10^4\ \mu\text{m})(10\mu\text{m})\Omega}{3\mu\text{m}^2}\\ &= 109.63\ \Omega\end{aligned}$$

Therefore, the resistance value is $\boxed{109.63\ \Omega}$.

Step 11 of 13

(d)

The doping concentration N_D can be calculated as shown below.

$$\begin{aligned}n &= N_D\\ &= \frac{\eta_i^2}{N_A}\end{aligned}$$

Substitute $1.5\times10^{10}/\text{cm}^3$ for η_i, $5\times10^{16}/\text{cm}^3$ for N_A in the above expression.

$$\begin{aligned}N_D &= \frac{(1.5\times10^{10}/\text{cm}^3)^2}{5\times10^{16}/\text{cm}^3}\\ &= 0.45\times10^4/\text{cm}^3\end{aligned}$$

Resistivity ρ for a p-doped silicon can be calculated as shown below.

$$\rho = \frac{1}{qN_A\mu_p + qN_D\mu_n}$$

Substitute $1.6\times10^{-19}\text{C}$ for q, $5\times10^{16}/\text{cm}^3$ for N_A, $1200\ \text{cm}^2/\text{V.s}$ for μ_n, $0.45\times10^4/\text{cm}^3$ for N_D, $400\ \text{cm}^2/\text{V.s}$ for μ_p

$$\begin{aligned}\rho &= \frac{1}{(1.6\times10^{-19}\text{C})(5\times10^{16}/\text{cm}^3)(400\ \text{cm}^2/\text{V.s}) + (1.6\times10^{-19}\text{C})(0.45\times10^4/\text{cm}^3)(1200\ \text{cm}^2/\text{V.s})}\\ &= 0.3125\ \Omega\cdot\text{cm}\end{aligned}$$

Step 12 of 13

Determine the resistance of a p-doped silicon.

$$R = \frac{\rho\ell}{A}$$

Substitute $0.3125\ \Omega\cdot\text{cm}$ for ρ, $10\ \mu\text{m}$ for l, $3\ \mu\text{m}^2$ for A in the above expression.

$$\begin{aligned}R &= \frac{(0.3125\ \Omega\cdot\text{cm})(10\mu\text{m})}{3\mu\text{m}^3}\\ &= \frac{(0.3125\times10^4\ \mu\text{m})(10\mu\text{m})\Omega}{3\mu\text{m}^2}\\ &= 10{,}417\ \Omega\end{aligned}$$

Therefore, the resistance value is $\boxed{10{,}417\ \Omega}$.

Step 13 of 13

(e)

Resistance of Aluminum can be calculated as follows:

$$R = \frac{\rho\ell}{A}$$

Substitute $2.8\ \mu\Omega\cdot\text{cm}$ for ρ, $10\ \mu\text{m}$ for l, $3\ \mu\text{m}^2$ for A in the above expression.

$$\begin{aligned}R &= \frac{(2.8\ \mu\Omega\cdot\text{cm})(10\mu\text{m})}{(3\mu\text{m}^2)}\\ &= \frac{(2.8\times10^{-4}\times10^4\ \mu\text{m})(10\mu\text{m})\Omega}{(3\mu\text{m}^2)}\\ &= 0.0933\ \Omega\end{aligned}$$

Therefore, the resistance value is $\boxed{0.0933\ \Omega}$.

Step 1 of 2

3.007E

The magnitude of charge on the n side of the *pn* junction is,

$$|Q_+| = qAx_nN_D$$

The magnitude of the charge on the *p* side of the junction is,

$$|Q_-| = qAx_pN_A.$$

Here, q is the charge of the electron $q = 1.6 \times 10^{-19}\text{C}$,

A is the cross sectional area of the junction,

x_n is the built in depletion region of the n side,

x_p is the depletion region of the p side,

N_D is the donor concentration, and

N_A is the acceptor concentration.

The charge equality condition is,

$$qAx_nN_D = qAx_pN_A,$$

This can be re arranged as follows:

$$\frac{x_n}{x_p} = \frac{N_A}{N_D}.$$

Step 2 of 2

The width W of the depletion layer is,

$$W = x_n + x_p$$
$$= \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)V_0}$$

Where ε_s is the electrical permittivity of silicon

Obtain the value of the V_0 from the given relation.

$$W = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)V_0}$$
$$= \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{N_A + N_D}{N_AN_D}\right)V_0}$$

Square on both sides of the equation.

$$W^2 = \frac{2\varepsilon_s}{q}\left(\frac{N_A + N_D}{N_AN_D}\right)V_0$$

$$V_0 = \frac{W^2}{\frac{2\varepsilon_s}{q}\left(\frac{N_A + N_D}{N_AN_D}\right)}$$
$$= \frac{W^2q}{2\varepsilon_s}\left(\frac{N_AN_D}{N_A + N_D}\right)$$
$$= \frac{1}{2}\left(\frac{q}{\varepsilon_s}\right)\left(\frac{N_AN_D}{N_A + N_D}\right)W^2$$

Therefore, the barrier voltage is, $\boxed{V_0 = \frac{1}{2}\left(\frac{q}{\varepsilon_s}\right)\left(\frac{N_AN_D}{N_A + N_D}\right)W^2}$.

Step 1 of 3

3.007P

The drift velocity of electron is,

$$V_d = \mu_n E \quad \text{..... (1)}$$

The electric field E is,

$$E = \frac{V}{l}$$

Substitute 3 V for voltage V and 10 μm for length l.

$$E = \frac{3\ \text{V}}{10\ \mu\text{m}}$$
$$= 0.3\ \text{V/}\mu\text{m}$$

Step 2 of 3

The electron mobility μ_n is 1350 cm^2/V.s

Substitute 1350 cm^2/V.s for μ_n and 0.3 V/μm for E in equation (1)

$$V_d = (0.3\ \text{V/}\mu\text{m})(1350\ \text{cm}^2/\text{V.s})$$
$$= 405\left(\frac{\text{cm}^2}{\mu\text{m.s}}\right)$$

$$1\ \text{cm} = 10^{-2}\ \text{m}$$
$$1\ \mu\text{m} = 10^{-6}\ \text{m}$$

Convert all units into meters to represent velocity in meter/second.

$$V_d = 405\left(\frac{10^{-4}\text{m}^2}{10^{-6}\text{m.s}}\right)$$
$$= 405\times10^{(-4+6)}\,(\text{m/s})$$
$$= 40{,}500\ \text{m/s}$$

Therefore, the drift velocity of the electrons is **40,500 m/s**.

Step 3 of 3

The drift velocity of hole is,

$$V_d = \mu_p E \quad \text{..... (2)}$$

Substitute 480 cm^2/V.s for μ_p and 0.3 V/μm for E in equation (2)

$$V_d = (0.3\ \text{V/}\mu\text{m})(480\ \text{cm}^2/\text{V.s})$$
$$= 144\left(\frac{\text{cm}^2}{\mu\text{m.s}}\right)$$

$$1\ \text{cm} = 10^{-2}\ \text{m}$$
$$1\ \mu\text{m} = 10^{-6}\ \text{m}$$

Convert all units into meters to represent velocity in meter/second.

$$V_d = 144\left(\frac{10^{-4}\text{m}^2}{10^{-6}\text{m.s}}\right)$$
$$= 144\times10^{(-4+6)}\,(\text{m/s})$$
$$= 14{,}400\ \text{m/s}$$

Therefore, the drift velocity of hole is **14,400 m/s**.

Hence, from the values of electron and hole drift velocities it is clear that the drift velocity of electrons is more than the drift velocity of holes because the mobility of electrons is more than the mobility of holes.

Step 1 of 5

The width of the depletion layer is,

$$W = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)V_0}.$$

Since $N_A >> N_D$, $\frac{1}{N_A} << \frac{1}{N_D}$ and can be neglected.

Thus the equivalent expression is,

$$W = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{1}{N_D}\right)V_0}$$
$$= \sqrt{\frac{2\varepsilon_s}{qN_D}V_0}$$

Thus, the width of the depletion layer is, $\boxed{W = \sqrt{\frac{2\varepsilon_s}{qN_D}V_0}}$.

Step 2 of 5

The width of the depletion region in the *p* side is

$$x_n = W\frac{N_A}{N_A + N_D}.$$

Since $N_A >> N_D$, $N_A + N_D \cong N_A$.

Now, the width of the depletion region in the *p* side is,

$$x_n = W\frac{N_A}{N_A}$$
$$= W$$

Thus, the width of the depletion region in the *p* side is, $\boxed{x_n = W}$.

Step 3 of 5

The width of the depletion region in the *n* side is,

$$x_p = W\frac{N_D}{N_A + N_D}.$$

Since $N_A >> N_D$, $N_A + N_D \cong N_A$.

Now, the width of the depletion region in the *n* side is,

$$x_p = W\frac{N_D}{N_A + N_D}$$
$$= W\frac{N_D}{N_A}$$
$$= \frac{W}{\left(\frac{N_A}{N_D}\right)}$$

Thus, the width of the depletion region in the *n* side is, $\boxed{x_p = \frac{W}{\left(\frac{N_A}{N_D}\right)}}$.

Step 4 of 5

The charge stored on either side of the depletion region is,

$$Q_J = Aq\left(\frac{N_A N_D}{N_A + N_D}\right)W.$$

Since $N_A >> N_D$, $N_A + N_D \cong N_A$.

Now, the charge stored on either side of the depletion region is,

$$Q_J = Aq\left(\frac{N_A N_D}{N_A}\right)W$$
$$= AqN_D W$$

Thus, the charge stored on either side of the depletion region is, $\boxed{Q_J = AqN_D W}$.

Step 5 of 5

The charge stored on either side of the depletion region is,

$$Q_J = A\sqrt{2\varepsilon_s q\left(\frac{N_A N_D}{N_A + N_D}\right)V_0}.$$

Since $N_A >> N_D$, $N_A + N_D \cong N_A$.

Now the charge stored on either side of the depletion region is,

$$Q_J = A\sqrt{2\varepsilon_s q\left(\frac{N_A N_D}{N_A}\right)V_0}$$
$$= A\sqrt{2\varepsilon_s qN_D V_0}$$

Thus, the charge stored on either side of the depletion region is, $\boxed{Q_J = A\sqrt{2\varepsilon_s qN_D V_0}}$.

3.008P

Step 1 of 36

The current flow due to both electron and hole is,

$I = JA$ ..... (1)

The current density J is,

$$J = \sigma \mathbf{E}$$
$$\sigma = q\left(n\mu_n + p\mu_p\right)$$

Substitute 1.6×10^{-19} C for q which represents charge of an electron

1200 $cm^2/V.s$ for μ_n which represents mobility of an electron

500 $cm^2/V.s$ for μ_p which represents mobility of a hole

$10^4/cm^3$ for n which represents free electron density

$10^{16}/cm^3$ for p which represents hole density

$$\sigma = 1.6\times10^{-19} C\left(10^4/\text{cm}^3\left(1200\ \text{cm}^2/\text{V.s}\right)+10^{16}/\text{cm}^3\left(500\ \text{cm}^2/\text{V.s}\right)\right)$$
$$= 0.8\ \text{mho/cm}$$

Step 2 of 36

The electric field E is,

$$E = \frac{\mathrm{V}}{\ell}$$

Substitute 1 V for voltage V and 10 μm for length l.

$$E = \frac{1\ \mathrm{V}}{10\ \mu\text{m}}$$
$$= 0.1\ \text{V/}\mu\text{m}$$

$$J = \sigma \mathbf{E}$$

Substitute 0.1 V/μm for E and 0.8 mho/cm for σ .

$$J = 0.1\ \text{V/}\mu\text{m}\left(0.8\ \text{mho/m}\right)$$
$$= 0.08\left(\frac{\text{V.mho}}{10^{-6}\text{m}^2}\right)$$
$$= 0.08\times10^6\ \text{A/m}^2$$

Step 3 of 36

Calculate the area of cross section.

$$A = 5\mu\text{m}\times4\mu\text{m}$$
$$= 20\mu\text{m}^2$$

Determine the current I.

$$I = JA$$

Substitute 0.08×10^6 A/m^2 for J and 20 μm^2 for A

$$I = \left(0.08\times10^6\ \text{A/m}^2\right)\left(20\ \mu\text{m}^2\right)$$
$$= 1.6\times10^6\left(\frac{\text{A}\left(10^{-6}\text{m}\right)^2}{\text{m}^2}\right)$$
$$= 1.6\times10^{-6}\text{A}$$
$$= 1.6\ \mu\text{A}$$

Therefore, the current flow due to both electrons and holes is $\boxed{1.6\ \mu\text{A}}$.

Step 4 of 36

Step 5 of 36

Step 6 of 36

Step 7 of 36

Step 8 of 36

Step 9 of 36

Step 10 of 36

Step 11 of 36

Step 12 of 36

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Step 27 of 36

Step 28 of 36

Step 29 of 36

Step 30 of 36

Step 31 of 36

Step 32 of 36

Step 33 of 36

Step 34 of 36

Step 35 of 36

Step 36 of 36

Step 1 of 1

The hole or minority carrier concentration is,

$$p_{n01} = \frac{n_i^2}{N_{D1}}.$$

Since, p_{n01} is inversely proportional to N_{D1} .

$$\frac{p_{n01}}{p_{n02}} = \frac{N_{D2}}{N_{D1}}$$

It is required to increase the minority carrier concentration in the *n* region by a factor of 2. Substitute $2p_{n01}$ for p_{n02} in the equation.

$$\frac{p_{n01}}{2p_{n01}} = \frac{N_{D2}}{N_{D1}}$$

$$\frac{N_{D2}}{N_{D1}} = \frac{1}{2}$$

$$N_{D2} = \frac{N_{D1}}{2}$$

Substitute 10^{16}cm^{-3} for N_{D1} in the equation.

$$N_{D2} = \frac{10^{16}\text{cm}^{-3}}{2}$$

To increase the minority carrier concentration by 2, lower the doping concentration by a factor of 2.

Therefore, the doping concentration to increase the minority carrier concentration by a factor of 2 is,

$$\boxed{N_D = \frac{1}{2} \times 10^{16}\text{cm}^{-3}}.$$

3.009P

Step 1 of 3

The current density J for a doped silicon is,

$$J = nq\mu_n E \quad \text{..... (1)}$$

The electric field E is,

$$E = \frac{V}{l}$$

Substitute 1 V for voltage V and 10 μm for length l.

$$E = \frac{1\ \text{V}}{10\ \mu\text{m}}$$
$$= 0.1\ \text{V}/\mu\text{m}$$

Step 2 of 3

The electron mobility μ_n is 1350 cm^2/V.s

For a doped silicon the electron concentration 'n' approximately equal to donor doping concentration N_D.

$$n \approx N_D$$

The current density J value is 2 mA/μm^2.

Charge of an electron 'q' is 1.6×10^{-19} coulombs.

From equation (1) calculate the donor doping N_D.

$$N_D = n$$
$$= \frac{J}{q\mu_n E}$$

Substitute 1.6×10^{-19} C for q , 2 mA/μm^2 for J ,0.1 V/μm for E and 1350 cm^2/V.s for μ_n.

$$N_D = \frac{2\ \text{mA}/\mu\text{m}^2}{(1.6\times10^{-19}C)(1350\ \text{cm}^2/\text{V.s})(0.1\ \text{V}/\mu\text{m})}$$
$$= 9.26\times10^{16}\left(\frac{\text{mA}/\mu\text{m}}{C(\text{cm}^2/\text{s})}\right)$$
$$= 9.26\times10^{16}\left(\frac{\text{mA}/\mu\text{m}}{\text{A.s}(\text{cm}^2/\text{s})}\right)$$

Step 3 of 3

Consider the following conversions:

$$1\ \text{cm} = 10^{-2}\ \text{m}$$
$$1\ \mu\text{m} = 10^{-6}\ \text{m}$$
$$1\ \text{mA} = 10^{-3}\ \text{A}$$

Convert all units into meters to represent doping concentration in per $meter^3$.

$$N_D = 9.26\times10^{16}\left(\frac{10^{-3}\text{A}/10^{-6}\text{m}}{\text{A.s}(10^{-4}\text{m}^2/\text{s})}\right)$$
$$= 9.26\times10^{16}\left(\frac{10^{(-3+10)}}{\text{m}^3}\right)$$
$$= 9.26\times10^{(16+7)}\left(\frac{1}{\text{m}^3}\right)$$
$$= 9.26\times10^{23}/\text{m}^3$$

Therefore, the donor concentration is $\boxed{9.26\times10^{23}/\text{m}^3}$.

Step 1 of 3

3.010E

The current flowing through the junction of a semiconductor is,

$$I = Aq\left(\frac{D_p}{L_p}p_{n0} + \frac{D_n}{L_n}n_{p0}\right)\left(e^{V/V_T} - 1\right).$$

Where J_p is the current density of holes,

J_n is the current density of electrons,

A is the area of the junction,

D_p is the diffusion constant of holes,

D_n is the diffusion constant of electrons,

L_p is the diffusion length of holes,

L_n is the diffusion length of electrons,

V_T is the terminal voltage,

p_{n0} and n_{p0} are the carrier concentration levels.

Step 2 of 3

Substitute $\frac{n_i^2}{N_D}$ for p_{n0} and $\frac{n_i^2}{N_A}$ for n_{p0}, where n_i is the carrier concentration in the intrinsic silicon, N_A is the acceptor dopant concentration and N_D is the donor dopant concentration.

$$I = Aq\left(\frac{D_p}{L_p}p_{n0} + \frac{D_n}{L_n}n_{p0}\right)\left(e^{\frac{V}{V_T}} - 1\right)$$

$$= Aq\left(\frac{D_p}{L_p}\left(\frac{n_i^2}{N_D}\right) + \frac{D_n}{L_n}\left(\frac{n_i^2}{N_A}\right)\right)\left(e^{\frac{V}{V_T}} - 1\right)$$

$$= Aqn_i^2\left(\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right)\left(e^{\frac{V}{V_T}} - 1\right)$$

When a negative voltage, V (reverse biased) is applied with a magnitude of a few times $V_T\,(25.9\text{mV})$, then the exponential term would essentially be zero, and the current across the junction becomes negative and constant.

$$I = I_S\left(e^{\frac{V}{V_T}} - 1\right)$$

Step 3 of 3

Therefore, the term $Aqn_i^2\left(\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right)$ can be substituted with the term I_S. The relation for the current I_S is,

$$I_S = Aqn_i^2\left(\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right).$$

It is given that the acceptor dopant concentration is very much greater than the donor dopant concentration, $N_A \gg N_D$.

The term $\frac{D_n}{L_nN_A}$ would yield a very low value when compared with the term $\frac{D_p}{L_pN_D}$ because the as the denominator (consisting of N_A) value increases, the term $\frac{D_n}{L_nN_A}$ decreases. So neglect the term $\frac{D_n}{L_nN_A}$ then the equation is,

$$I_S \simeq Aqn_i^2\left(\frac{D_p}{L_pN_D}\right)$$

Therefore, the saturation current is,

$$\boxed{I_S = Aqn_i^2\left(\frac{D_p}{L_pN_D}\right)}.$$

3.010P

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Step 1 of 3

The relation for the current I_S for a *pn* junction semiconductor configuration is,

$$I_S = Aqn_i^2\left(\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right).$$

Substitute 10^{-4}cm^2 for A, 1.6×10^{-19} for q, $1.5\times10^{10}\ /\text{cm}^3$ for n_i, $10\text{cm}^2/\text{V.s}$ for D_p, $18\text{cm}^2/\text{V.s}$ for D_n, $5\ \mu\text{m}$ for L_p, $10\ \mu\text{m}$ for L_n, $\frac{10^{16}}{2}\ /\text{cm}^2$ for N_D, and $10^{18}\ /\text{cm}^3$ for N_A in the equation.

$$I_S = 10^{-4}\left(1.6\times10^{-19}\right)\left(1.5\times10^{10}\right)^2\left[\frac{10}{5\times10^{-4}\times\frac{10^{16}}{2}} + \frac{18}{10\times10^{-4}\times10^{18}}\right]$$

$$= 3.6\times10^{-3}\left(4\times10^{-12} + 18\times10^{-15}\right)$$

$$= 3.6\times10^{-3}\times\left(4.018\times10^{-12}\right)$$

$$= 1.4465\times10^{-14}\ \text{A}$$

Therefore, the current, I_S is $\boxed{1.4465\times10^{-14}\ \text{A}}$.

Step 2 of 3

The relation for finding the current *I* is,

$$I = I_S\left(e^{\frac{V}{V_T}} - 1\right)$$

Where V_T is the terminal voltage whose value is obtained from the relation $V_T = \frac{kT}{q}$,

k is Boltzmann's constant, $k = 8.62\times10^{-5}\ \text{eV/K}$,

Room temperature, T = 300 K and

charge of an electron, $q = 1.6\times10^{-19}\text{C}$.

Substitute corresponding values in the equation $V_T = \frac{kT}{q}$ and find the value of V_T.

$$V_T = \frac{kT}{q}$$

$$= \frac{8.62\times10^{-5}\times300\ (\text{eV})}{q\ (\text{e})}$$

$$= 25.9\ \text{mV}$$

Step 3 of 3

Find the total current *I*.

$$I = I_S\left(e^{\frac{V}{V_T}} - 1\right)$$

Substitute $25.9\ \text{mV}$ for V_T, $1.46\times10^{-14}\ \text{A}$ for I_S, and $0.605\ \text{V}$ for V in the equation.

$$I = 1.4465\times10^{-14}\left(e^{\frac{0.605}{25.9\times10^{-3}}} - 1\right)$$

$$= 1.4465\times10^{-14}\left(1.39545\times10^{10} - 1\right)$$

$$= 0.202\ \text{mA}$$

Therefore, the total current is, $\boxed{I = 0.202\ \text{mA}}$.

Step 1 of 5

The Einstein relationship to calculate diffusion constant for electrons and holes is,

$$\frac{D_p}{\mu_p} = \frac{D_n}{\mu_n} = V_T = 26\ \text{mV}$$

Here,

μ_n represents mobility of electrons

D_n represents diffusion constant for electrons

V_T represents thermal voltage at room temperature

μ_p represents mobility of holes

D_p represents diffusion constant for holes

$$D_n = \mu_n V_T$$

Case(i):

Substitute 1350 $cm^2/V.s$ for μ_n, 26 mV for V_T

$$\begin{aligned} D_n &= (1350\ \text{cm}^2/\text{V.s})(26\ \text{mV}) \\ &= (35100\ \text{cm}^2/\text{V.s})(10^{-3}\ \text{V}) \\ &= 35.1\ \text{cm}^2/\text{s} \end{aligned}$$

Step 2 of 5

Case(ii):

Substitute 1200 $cm^2/V.s$ for μ_n and 26 mV for V_T.

$$\begin{aligned} D_n &= (1200\ \text{cm}^2/\text{V.s})(26\ \text{mV}) \\ &= (31200\ \text{cm}^2/\text{V.s})(10^{-3}\ \text{V}) \\ &= 31.2\ \text{cm}^2/\text{s} \end{aligned}$$

Case(iii):

Substitute 750 $cm^2/V.s$ for μ_n and 26 mV for V_T.

$$\begin{aligned} D_n &= (750\ \text{cm}^2/\text{V.s})(26\ \text{mV}) \\ &= (19500\ \text{cm}^2/\text{V.s})(10^{-3}\ \text{V}) \\ &= 19.5\ \text{cm}^2/\text{s} \end{aligned}$$

Step 3 of 5

Case(iv):

Substitute 380 $cm^2/V.s$ for μ_n and 26 mV for V_T.

$$\begin{aligned} D_n &= (380\ \text{cm}^2/\text{V.s})(26\ \text{mV}) \\ &= (9880\ \text{cm}^2/\text{V.s})(10^{-3}\ \text{V}) \\ &= 9.88\ \text{cm}^2/\text{s} \end{aligned}$$

Step 4 of 5

The diffusion constant for holes, $D_p = \mu_p V_T$

Case(i):

Substitute 480 $cm^2/V.s$ for μ_p and 26 mV for V_T.

$$\begin{aligned} D_p &= (480\ \text{cm}^2/\text{V.s})(26\ \text{mV}) \\ &= (12480\ \text{cm}^2/\text{V.s})(10^{-3}\ \text{V}) \\ &= 12.48\ \text{cm}^2/\text{s} \end{aligned}$$

Case(ii):

Substitute 400 $cm^2/V.s$ for μ_p and 26 mV for V_T.

$$\begin{aligned} D_p &= (400\ \text{cm}^2/\text{V.s})(26\ \text{mV}) \\ &= (10400\ \text{cm}^2/\text{V.s})(10^{-3}\ \text{V}) \\ &= 10.4\ \text{cm}^2/\text{s} \end{aligned}$$

Step 5 of 5

Case(iii):

Substitute 260 $cm^2/V.s$ for μ_p and 26 mV for V_T.

$$\begin{aligned} D_p &= (260\ \text{cm}^2/\text{V.s})(26\ \text{mV}) \\ &= (6760\ \text{cm}^2/\text{V.s})(10^{-3}\ \text{V}) \\ &= 6.7\ \text{cm}^2/\text{s} \end{aligned}$$

Case(iv):

Substitute 160 $cm^2/V.s$ for μ_p and 26 mV for V_T.

$$\begin{aligned} D_p &= (160\ \text{cm}^2/\text{V.s})(26\ \text{mV}) \\ &= (4160\ \text{cm}^2/\text{V.s})(10^{-3}\ \text{V}) \\ &= 4.16\ \text{cm}^2/\text{s} \end{aligned}$$

Therefore, the diffusion constant for electrons for the four cases are,

$$\boxed{35.1\ \text{cm}^2/\text{s}, 31.2\ \text{cm}^2/\text{s}, 19.5\ \text{cm}^2/\text{s}, 9.88\ \text{cm}^2/\text{s}}.$$

The diffusion constant for holes for the four cases are,

$$\boxed{12.48\ \text{cm}^2/\text{s}, 10.4\ \text{cm}^2/\text{s}, 6.7\ \text{cm}^2/\text{s}, 4.16\ \text{cm}^2/\text{s}}.$$

3.012E

Consider the following expression for the width of the depletion region, W corresponding to the forward voltage, V_F :

$$W = \sqrt{\frac{2\varepsilon_s}{q}\left[\frac{1}{N_A}+\frac{1}{N_D}\right](V_0 - V_F)}$$

Here, the electrical permittivity of silicon, $\varepsilon_s = 1.04\times10^{-12}\ \text{F/cm}$, the charge of electron, $q = 1.6\times10^{-19}\ \text{C}$, the acceptor doping concentration or doping concentration of the p side, $N_A = 10^{18}\ \text{carriers/cm}^3$, the donor doping concentration, $N_D = 10^{16}\ \text{carriers/cm}^3$, the barrier voltage, $V_o = 0.814\ \text{V}$, and the forward bias voltage, $V_F = 0.605\ \text{V}$.

Substitute these values in the equation for W .

$$\begin{aligned} W &= \sqrt{\frac{2\left(1.04\times10^{-12}\right)}{1.6\times10^{-19}}\left[\frac{1}{10^{18}}+\frac{1}{10^{16}}\right](0.814-0.605)} \\ &= \sqrt{\left(1.3\times10^{7}\right)\left(1.01\times10^{-16}\right)(0.209)} \\ &= \sqrt{0.274\times10^{-9}} \\ &= 1.656\times10^{-5}\ \text{cm} \end{aligned}$$

Convert centimetres into meters.

$$\begin{aligned} W &= 1.656\times10^{-7}\ \text{m} \\ &= 0.1656\ \mu\text{m} \end{aligned}$$

Therefore, the width of the depletion region, W is, $\boxed{0.1656\ \mu\text{m}}$.

Step 1 of 5

The built in voltage V_o of a junction is,

$$V_o = V_T \ln\left(\frac{N_A N_D}{\eta_i^2}\right) \quad \text{..... (1)}$$

The *p* and *n* regions are equally doped.

$$N_A = N_D$$
$$= 5\times10^{16} \text{ atoms/cm}^3$$

Intrinsic concentration η_i value is $1.5\times10^{10}/\text{cm}^3$

Thermal voltage V_T value is 26 mV.

Substitute $5\times10^{16}/\text{cm}^3$ for N_A and N_D ,$1.5\times10^{10}/\text{cm}^3$ for η_i ,26 mV for V_T in equation (1).

$$V_o = 26 \text{ mV} \ln\left(\frac{\left(5\times10^{16}/\text{cm}^3\right)^2}{\left(1.5\times10^{10}/\text{cm}^3\right)^2}\right)$$
$$= 781 \text{ mV}$$
$$1 \text{ mV} = 0.001 \text{ V}$$
$$V_o = 0.781 \text{ V}$$

Step 2 of 5

The permittivity of the depletion region is,

$$\varepsilon = 11.7\varepsilon_o$$

Substitute $\varepsilon_o = 8.854\times10^{-14} \text{ F/cm}$ for the permittivity of free space.

$$\varepsilon = 11.7\left(8.854\times10^{-14} \text{ F/cm}\right)$$
$$= 1.036\times10^{-12} \text{ F/cm}$$

Determine the width of the depletion region *W*.

$$W = \sqrt{\frac{2\varepsilon V_o}{q}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)}$$

Substitute 1.6×10^{-19} C for '*q*' charge of an electron, 0.781 V for 'V_o',$5\times10^{16}/\text{cm}^3$ for N_A and N_D , 1.036×10^{-12} F/cm for ε

$$W = \sqrt{\frac{2\left(1.036\times10^{-12} \text{ F/cm}\right)\left(0.781 \text{ V}\right)}{1.6\times10^{-19} \text{ C}}\left(\frac{1}{\left(5\times10^{16}/\text{cm}^3\right)}+\frac{1}{\left(5\times10^{16}/\text{cm}^3\right)}\right)}$$
$$= 2\times10^{-5} \text{ cm}$$
$$1 \text{ cm} = 10^{-2} \text{ m}$$
$$W = 2\times10^{-5}\left(10^{-2} \text{ m}\right)$$
$$W = 2\times10^{-7} \text{ m}$$
$$= 0.2\times10^{-6} \text{ m}$$
$$= 0.2 \ \mu\text{m}$$

Step 3 of 5

The width of a depletion region is,

$$W = x_n + x_p$$
$$\frac{x_n}{x_p} = \frac{N_A}{N_D}$$
$$x_n = x_p\left(\frac{N_A}{N_D}\right)$$

x_n and x_p represent the widths of a depletion region that extends into the *p* and *n* regions.

Substitute x_n equivalent in *W* equation.

$$W = x_p\left(\frac{N_A}{N_D}\right) + x_p$$
$$= x_p\left(\frac{N_A}{N_D}+1\right)$$
$$x_p = \frac{W}{\left(\frac{N_A}{N_D}+1\right)}$$

Substitute 0.2 μm for '*W* ,5×10^{16} /cm^3 for N_A and N_D

$$x_p = \frac{0.2 \ \mu\text{m}}{\left(\frac{5\times10^{16}/\text{cm}^3}{5\times10^{16}/\text{cm}^3}+1\right)}$$
$$= \frac{0.2 \ \mu\text{m}}{2}$$
$$= 0.1 \ \mu\text{m}$$

Step 4 of 5

Determine x_n .

$$x_n = W - x_p$$

Substitute 0.1 μm for x_p and 0.2 μm for *W*

$$x_n = 0.2\mu\text{m} - 0.1\mu\text{m}$$
$$= 0.1\mu\text{m}$$

Step 5 of 5

Determine the charge stored on either side of the junction Q_j .

Cross sectional area of the junction *A* is 20 μm^2

$$Q_j = q\left(\frac{N_A N_D}{N_A + N_D}\right)AW$$

Substitute 1.6×10^{-19} C for *q* ,$5\times10^{16}/\text{cm}^3$ for N_A and N_D ,20 μm^2 for *A* ,0.2 μm for *W*

$$Q_j = 1.6\times10^{-19}\left(\frac{\left(5\times10^{16}/\text{cm}^3\right)^2}{\left(5\times10^{16}/\text{cm}^3\right)+\left(5\times10^{16}/\text{cm}^3\right)}\right)\left(20\mu\text{m}^2\right)\left(0.2\mu\text{m}\right)$$
$$= 1.6\times10^{-19}\left(2.5\times10^{16}/\text{cm}^3\right)\left(4\times\left(10^{-4}\text{m}\right)^3\right)$$
$$= 6.4\times10^{-3}\left(\frac{2.5\times10^{-18}\text{m}^3}{10^{-6}\text{m}^3}\right)$$
$$= 16\times10^{-15}\text{C}$$

Therefore, the charge stored on either side of the junction is $\boxed{16\times10^{-15}\text{C}}$.

Step 1 of 3

Consider the following expression for the width of the depletion region, W corresponding to the forward voltage, V_F :

$$W=\sqrt{\frac{2\varepsilon_s}{q}\left[\frac{1}{N_A}+\frac{1}{N_D}\right](V_0+V_R)}$$

Here, the electrical permittivity of silicon, $\varepsilon_s=1.04\times10^{-12}\ \text{F/cm}$, the charge of electron, $q=1.6\times10^{-19}\ \text{C}$, the acceptor doping concentration or doping concentration of the p side, $N_A=10^{18}\ \text{carriers/cm}^3$, the donor doping concentration, $N_D=10^{16}\ \text{carriers/cm}^3$, the barrier voltage, $V_0=0.814\ \text{V}$, and the reverse bias voltage, $V_R=2\ \text{V}$.

Substitute these values in the equation for W .

$$\begin{aligned}W&=\sqrt{\frac{2\left(1.04\times10^{-12}\right)}{1.6\times10^{-19}}\left[\frac{1}{10^{18}}+\frac{1}{10^{16}}\right](0.814+2)}\\&=\sqrt{\left(13\times10^{6}\right)\left(0.101\times10^{-15}\right)(2.814)}\\&=\sqrt{3.695\times10^{-9}}\\&=60.784\times10^{-6}\ \text{cm}\end{aligned}$$

Convert centimetres into meters.

$$\begin{aligned}W&=60.784\times10^{-8}\ \text{m}\\&=0.60785\ \mu\text{m}\end{aligned}$$

Therefore, the width of the depletion region, W is, $\boxed{0.60785\ \mu\text{m}}$.

Step 2 of 3

Consider the following expression for the charge stored in the depletion region:

$$Q_J=A\sqrt{2\varepsilon_s q\left[\frac{N_AN_D}{N_A+N_D}\right](V_o+V_R)}$$

Here, the area of the pn junction, $A=10^{-4}\text{cm}^2$.

Substitute the known values in the equation for Q_J .

$$\begin{aligned}Q_J&=\left(10^{-4}\right)\sqrt{2\left(1.04\times10^{-12}\right)\left(1.6\times10^{-19}\right)\left[\frac{10^{16}10^{18}}{10^{16}+10^{18}}\right](0.814+2)}\\&=\left(10^{-4}\right)\sqrt{\left(3.328\times10^{-31}\right)\left[\frac{10^{16}10^{18}}{10^{16}+10^{18}}\right](2.814)}\\&=\left(10^{-4}\right)\sqrt{\left(9.635\times10^{-31}\right)\left[9.901\times10^{15}\right]}\\&=\left(10^{-4}\right)\sqrt{9.2723\times10^{-15}}\end{aligned}$$

Simplify further.

$$\begin{aligned}Q_J&=9.63\times10^{-12}\\&=9.63\ \text{pC}\end{aligned}$$

Therefore, the charge stored in the depletion region, Q_J is, $\boxed{9.63\ \text{pC}}$.

Step 3 of 3

Consider the following expression for the saturation current, I_S for a pn junction semiconductor configuration:

$$I_S=Aqn_i^2\left(\frac{D_p}{L_pN_D}+\frac{D_n}{L_nN_A}\right)$$

Here, the intrinsic carrier density, $n_i=1.5\times10^{10}\ \text{carriers/cm}^3$, the diffusion length of the holes in the n region, $L_p=5\ \mu\text{m}$, the diffusion length of the electrons in the p region, $L_n=10\ \mu\text{m}$, the diffusion constant of the holes, D_p (in the n region) is $10\ \text{cm}^2/\text{V}\cdot\text{s}$, and the diffusion constant of the electron D_n (in the p region) is $18\ \text{cm}^2/\text{V}\cdot\text{s}$.

in the above equation for I_S.

Substitute the known values in the equation for I_S .

$$\begin{aligned}I_S&=\left(10^{-4}\right)\left(1.6\times10^{-19}\right)\left(1.5\times10^{10}\right)^2\left[\frac{10}{\left(5\times10^{-4}\right)\left(10^{16}\right)}+\frac{18}{\left(10\times10^{-4}\right)\left(10^{18}\right)}\right]\\&=\left(3.6\times10^{-3}\right)\left(2\times10^{-12}+18\times10^{-15}\right)\\&=\left(3.6\times10^{-3}\right)\left(2.018\times10^{-12}\right)\\&=7.265\times10^{-15}\ \text{A}\end{aligned}$$

Therefore, the reverse saturation current, I_S is, $\boxed{7.265\times10^{-15}\ \text{A}}$.

Step 1 of 5

The built in voltage V_o of a junction is,

$$V_o = V_T \ln\left(\frac{N_A N_D}{\eta_i^2}\right) \quad \text{..... (1)}$$

the *p* and *n* regions are unequally doped.

$N_A = 10^{17}/\text{cm}^3$ which represents acceptor concentration.

$N_D = 10^{16}/\text{cm}^3$ which represents donor concentration.

Intrinsic concentration η_i value is $1.5\times10^{10}/\text{cm}^3$

thermal voltage V_T value is 26 mV.

Substitute $10^{17}/\text{cm}^3$ for N_A and $10^{16}/\text{cm}^3$ for N_D ,$1.5\times10^{10}/\text{cm}^3$ for η_i ,26 mV for V_T in equation (1).

$$V_o = 26\ \text{mV}\ln\left(\frac{\left(10^{16}/\text{cm}^3\right)\left(10^{17}/\text{cm}^3\right)}{\left(1.5\times10^{10}/\text{cm}^3\right)^2}\right)$$
$$= 757\ \text{mV}$$
$$1\ \text{mV} = 0.001\ \text{V}$$
$$V_o = 0.757\ \text{V}$$

Step 2 of 5

The Width of a depletion region '*W* is,

$$W = \sqrt{\frac{2\varepsilon V_o}{q}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)}$$

for an intrinsic silicon

$$\varepsilon = 11.7\varepsilon_o$$

Substitute $\varepsilon_o = 8.854\times10^{-14}\ \text{F/cm}$ permittivity of free space

$$\varepsilon = 11.7\left(8.854\times10^{-14}\ \text{F/cm}\right)$$
$$= 1.036\times10^{-12}\ \text{F/cm}$$

Substitute 1.6×10^{-19} C for '*q*' charge of an electron, 0.757 V for 'V_o','$10^{17}/\text{cm}^3$ for N_A and $10^{16}/\text{cm}^3$ for N_D , 1.036×10^{-12} F/cm for ε .

$$W = \sqrt{\frac{2\left(1.036\times10^{-12}\ \text{F/cm}\right)\left(0.757\ \text{V}\right)}{1.6\times10^{-19}\ \text{C}}\left(\frac{1}{\left(10^{17}/\text{cm}^3\right)}+\frac{1}{\left(10^{16}/\text{cm}^3\right)}\right)}$$
$$= 3.3\times10^{-5}\ \text{cm}$$
$$1\ \text{cm} = 10^{-2}\ \text{m}$$
$$W = 3.3\times10^{-5}\left(10^{-2}\ \text{m}\right)$$
$$W = 3.3\times10^{-7}\ \text{m}$$
$$= 0.33\times10^{-6}\ \text{m}$$
$$= 0.33\ \mu\text{m}$$

Step 3 of 5

The width of a depletion region is,

$$W = x_n + x_p$$
$$\frac{x_n}{x_p} = \frac{N_A}{N_D}$$
$$x_n = x_p\left(\frac{N_A}{N_D}\right)$$

x_n and x_p represent the widths of a depletion region extends into *p* and *n* region

Substitute x_n equivalent in *W* equation.

$$W = x_p\left(\frac{N_A}{N_D}\right)+x_p$$
$$= x_p\left(\frac{N_A}{N_D}+1\right)$$
$$x_p = \frac{W}{\left(\frac{N_A}{N_D}+1\right)}$$

Substitute 0.33 μm for '*W* ,10^{16} /cm^3 for N_D and $10^{17}/\text{cm}^3$ for N_A

$$x_p = \frac{0.33\ \mu\text{m}}{\left(\frac{10^{17}/\text{cm}^3}{10^{16}/\text{cm}^3}+1\right)}$$
$$= \frac{0.33\ \mu\text{m}}{11}$$
$$= 0.03\ \mu\text{m}$$

Step 4 of 5

Determine x_n .

$$x_n = W - x_p$$

Substitute 0.03 μm for x_p and 0.33 μm for *W*

$$x_n = 0.33\mu\text{m} - 0.03\mu\text{m}$$
$$= 0.3\mu\text{m}$$

Step 5 of 5

Determine the charge stored on either side of the junction, Q_j .

$$Q_j = q\left(\frac{N_A N_D}{N_A + N_D}\right)AW$$

Substitute 1.6×10^{-19} C for *q* ,$10^{17}/\text{cm}^3$ for N_A and $10^{16}/\text{cm}^3$ for N_D ,100 μm^2 for *A* ,

0.33 μm for *W*

$$Q_j = 1.6\times10^{-19}\left(\frac{\left(10^{17}/\text{cm}^3\right)\left(10^{16}/\text{cm}^3\right)}{\left(10^{17}/\text{cm}^3\right)+\left(10^{16}/\text{cm}^3\right)}\right)\left(100\mu\text{m}^2\right)\left(0.33\mu\text{m}\right)$$
$$= 1.6\times10^{-19}\left(9.1\times10^{15}/\text{cm}^3\right)\left(33\mu\text{m}^3\right)$$
$$= 480.5\times10^{-4}\left(\frac{10^{-18}\text{m}^3}{10^{-6}\text{m}^3}\right)$$
$$= 4.805\times10^{-14}\text{C}$$

Therefore, the charge stored on either side of the junction is $\boxed{4.805\times10^{-14}\ \text{C}}$.

Step 1 of 2

3.014E

Consider the following expression for the depletion region or junction capacitance, C_j at zero reverse bias:

$$C_{j0} = A\sqrt{\left(\frac{q\varepsilon_s}{2}\right)\left[\frac{N_A N_D}{N_A + N_D}\right]\left(\frac{1}{V_o}\right)}$$

Here, the area of the *pn* junction, $A = 10^{-4}\,\text{cm}^2$, the electrical permittivity of silicon, $\varepsilon_s = 1.04\times10^{-12}\ \text{F/cm}$, the charge of electron, $q = 1.6\times10^{-19}\ \text{C}$, the acceptor doping concentration or doping concentration of the *p* side, $N_A = 10^{18}\ \text{carriers/cm}^3$, the donor doping concentration, $N_D = 10^{16}\ \text{carriers/cm}^3$, and the barrier voltage, $V_o = 0.814\ \text{V}$.

Substitute these values in the equation for C_{j0}.

$$\begin{aligned}
C_{j0} &= (10^{-4})\sqrt{\left[\frac{(1.6\times10^{-19})(1.04\times10^{-12})}{2}\right]\left[\frac{10^{18}\times10^{16}}{10^{18}+10^{16}}\right]\left(\frac{1}{0.814}\right)} \\
&= (10^{-4})\sqrt{(8.32\times10^{-32})(9.901\times10^{15})(1.2285)} \\
&= (10^{-4})\left(\sqrt{1.01\times10^{-15}}\right) \\
&= (10^{-4})(3.1812\times10^{-8})
\end{aligned}$$

Simplify further.

$$\begin{aligned}
C_{j0} &= 3.18\times10^{-12} \\
&= 3.18\ \text{pF}
\end{aligned}$$

Therefore, the junction capacitance at zero reverse bias, C_{j0} is, $\boxed{3.18\ \text{pF}}$.

Step 2 of 2

Consider the following relation between the depletion capacitance, C_j and junction capacitance at zero reverse bias, C_{j0}:

$$C_j = \frac{C_{j0}}{\left(1 + \frac{V_R}{V_0}\right)^m}$$

Here, the value of gradient coefficient, *m* is, $\frac{1}{3}$ to $\frac{1}{2}$.

Consider that the reverse bias voltage, $V_R = 2\ \text{V}$.

Substitute 3.18 pF for C_{j0}, 2 V for V_R, 0.814 V for V_0, and $\frac{1}{2}$ for m in the equation for C_j.

$$\begin{aligned}
C_j &= \frac{3.18\times10^{-12}}{\left(1 + \frac{2}{0.814}\right)^{\frac{1}{2}}} \\
&= \frac{3.18\times10^{-12}}{\sqrt{3.457}} \\
&= \frac{3.18\times10^{-12}}{1.8593} \\
&= 1.71\times10^{-12}
\end{aligned}$$

Simplify further.

$$C_j = 1.71\ \text{pF}$$

Therefore, the junction capacitance, C_j is, $\boxed{1.71\ \text{pF}}$.

The cross sectional area of the junction A is calculated as follows.

$$\begin{aligned} A &= 10\mu m \times 10\mu m \\ &= 100\mu m^2 \end{aligned}$$

The charge stored in one side of the junction Q_j is,

$$Q_j = AqN_DW$$

Substitute 1.6×10^{-19} C for q, which represents charge of an electron, $10^{18}/cm^3$ for N_D , which represents doping concentration on one side of the junction, 100 μm^2 for A , 0.1 μm for W, which represents width of the depletion region.

$$\begin{aligned} Q_j &= \left(1.6\times10^{-19}\right)\left(100\mu m^2\right)\left(0.1\mu m\right)\left(10^{18}/cm^3\right) \\ &= (1.6)\left(\frac{\left(10^{-6}m\right)^3}{10^{-6}m^3}\right) \\ &= 1.6\times10^{-12}C \end{aligned}$$

Therefore, the charge stored in one side of the junction is $\boxed{1.6\times10^{-12}\ C}$.

Consider the following expression for the incremental diffusion capacitance, C_d :

$$C_d = \frac{dQ}{dV}$$

Express the charge in terms of the diode current, I.

$$Q = \tau_T I$$

Here, τ_T is the mean transit-time of the junction.

The following is the relation between the saturation current and the diode current:

$$I = I_S\left(e^{\frac{V}{V_T}} - 1\right)$$

Here, V_T is the thermal equivalent voltage.

Substitute the expression for the diode current, I in the expression for the charge, Q .

$$\begin{aligned} Q &= \tau_T I \\ &= \tau_T I_S\left(e^{\frac{V}{V_T}} - 1\right) \end{aligned}$$

Substitute the expression for the charge, Q in the expression for the incremental diffusion capacitance, C_d .

$$\begin{aligned} C_d &= \frac{dQ}{dV} \\ &= \frac{d}{dV}\left[\tau_T I_S\left(e^{\frac{V}{V_T}} - 1\right)\right] \\ &= \tau_T I_S \frac{d}{dV}\left(e^{\frac{V}{V_T}} - 1\right) \\ &= \tau_T I_S\left(\frac{1}{V_T}\right)e^{\frac{V}{V_T}} \end{aligned}$$

Rewrite the equation.

$$\begin{aligned} C_d &= \frac{\tau_T}{V_T} I_S e^{\frac{V}{V_T}} \\ &= \left(\frac{\tau_T}{V_T}\right) I \end{aligned}$$

Here, I is the forward bias current.

Therefore, the incremental diffusion capacitance, C_d is expressed as $\boxed{C_d = \left(\frac{\tau_T}{V_T}\right) I}$.

Step 1 of 2

The charge stored in one side of the junction Q_j is,

$$Q_j \simeq AqN_DW$$

Substitute 1.6×10^{-19} C for q, which represents charge of an electron

$10^{16}/cm^3$ for N_D , which represents doping concentration on one side of the junction

10 μm^2 for A , which represents cross sectional area of the junction

0.2 μm for W, which represents width of the depletion region

$$\begin{aligned} Q_j &= \left(1.6\times10^{-19}\right)\left(10\mu m^2\right)\left(0.2\mu m\right)\left(10^{16}/cm^3\right) \\ &= \left(3.2\times10^{-3}\right)\left(\frac{\left(10^{-6}m\right)^3}{10^{-6}m^3}\right) \\ &= 3.2\times10^{-15}C \end{aligned}$$

Therefore, the charge stored in one side of the junction is $\boxed{3.2\times10^{-15}\ C}$.

Step 2 of 2

The built in voltage of the junction V_o is,

$$Q_j \simeq A\sqrt{2\varepsilon_s qN_DV_o}$$

Square the equation on both sides.

$$Q_j^2 \simeq A^2\left(\sqrt{2\varepsilon_s qN_DV_o}\right)^2$$

$$V_o = \frac{Q_j^2}{A^2\left(2\varepsilon_s qN_D\right)}$$

$$\varepsilon_s = 11.7\varepsilon_o$$

Substitute $\varepsilon_o = 8.854\times10^{-14}$ F/cm permittivity of free space

$$\begin{aligned} \varepsilon_s &= 11.7\left(8.854\times10^{-14}\ F/cm\right) \\ &= 1.036\times10^{-12}\ F/cm \end{aligned}$$

$$V_o = \frac{Q_j^2}{A^2\left(2\varepsilon_s qN_D\right)}$$

Substitute 3.2×10^{-15} C for Q_j ,10 μm^2 for A ,1.6×10^{-19} C for q ,$10^{16}/cm^3$ for N_D ,

1.036×10^{-12}F/cm for ε_s

$$\begin{aligned} V_o &= \frac{\left(3.2\times10^{-15}C\right)^2}{\left(10\mu m^2\right)^2\left(2\left(1.036\times10^{-12}\ F/cm\right)\left(1.6\times10^{-19}C\right)\left(10^{16}/cm^3\right)\right)} \\ &= 0.31\times10^{-15}\left(\frac{\left(10^{-8}m^4\right)C}{F\left(10^{-12}m^2\right)^2}\right) \\ &= 3.1\ V \end{aligned}$$

Therefore, the built in voltage value is $\boxed{3.1\ V}$

Step 1 of 3

Consider the following expression for the excess minority carrier (hole) lifetime, τ_p :

$$\tau_p = \frac{L_p^2}{D_p}$$

Here, L_p is the diffusion length of the holes and D_p is the diffusion constant of the holes. Consider that the diffusion constant of the holes, D_p in the *n* region is $10\ \text{cm}^2/\text{V}\cdot\text{s}$, and the diffusion length of the holes in the *n* region, L_p is $5\ \mu\text{m}$.

Find the value of τ_p .

$$\begin{aligned}\tau_p &= \frac{L_p^2}{D_p}\\ &= \frac{\left(5\times10^{-6}\times10^2\right)^2}{10}\\ &= 25\times10^{-9}\\ &= 25\ \text{ns}\end{aligned}$$

Therefore, the minority carrier (hole) life time, τ_p is, $\boxed{25\ \text{ns}}$.

Step 2 of 3

Consider the following expression for the incremental diffusion capacitance, C_d :

$$C_d = \left(\frac{\tau_T}{V_T}\right)I$$

Here, τ_T is the mean transit time of the junction, the forward bias current, I is 0.1 mA, and V_T is the thermal voltage whose value is obtained from the relation, $V_T = \frac{kT}{q}$.

Here, Boltzmann's constant, $k = 8.62\times10^{-5}\ \text{eV/K}$, room temperature, $T = 300\ \text{K}$, and charge of an electron, $q = 1.6\times10^{-19}\ \text{C}$.

Find the value of thermal equivalent voltage, V_T .

$$\begin{aligned}V_T &= \frac{kT}{q}\\ &= \frac{\left(8.62\times10^{-5}\right)(300)}{1.69\times10^{-19}}\\ &\approx 26\ \text{mV}\end{aligned}$$

Step 3 of 3

Consider that for this *pn* junction, $I_p \approx I$.

For instance, if $N_A \gg N_D$, one can show that, $I_p \approx I$ and $I_p \gg I_n$.

Consider that the acceptor doping concentration or doping concentration of the *p* side, $N_A = 10^{18}\ \text{carriers/cm}^3$ and the donor doping concentration, $N_D = 10^{16}\ \text{carriers/cm}^3$.

Therefore, $N_A \gg N_D\left(10^{18} \gg 10^{16}\right)$ and thus, $\tau_T \approx \tau_p$.

Substitute 25 ns for τ_T , 26 mV for V_T , 0.1 mA for I to calculate the value of incremental diffusion capacitance, C_d .

$$\begin{aligned}C_d &= \left(\frac{\tau_T}{V_T}\right)I\\ &= \left(\frac{25\times10^{-9}}{26\times10^{-3}}\right)\left(0.1\times10^{-3}\right)\\ &= \left(0.965\times10^{-6}\right)\left(0.1\times10^{-3}\right)\\ &= 96.5\times10^{-12}\end{aligned}$$

Rewrite the equation.

$$C_d = 96.5\ \text{pF}$$

Therefore, the incremental diffusion capacitance, C_d is, $\boxed{96.5\ \text{pF}}$.

Step 1 of 2

3.016P

Write the formula of voltage for pen circuit terminals of diode.

$$V_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

If N_A or N_D is increased by a factor of 10 then,

$$V_0' = V_T \ln\left(\frac{(10N_A)N_D}{n_i^2}\right)$$

$$= V_T \ln\left((10)\left(\frac{N_A N_D}{n_i^2}\right)\right)$$

$$= V_T \ln 10 + V_T \ln\left(\frac{N_A N_D}{n_i^2}\right) \qquad \left[\text{Since } \ln(xy) = \ln(x) + \ln(y)\right]$$

Substitute 25.8 mV for V_T.

$$V_0' = (25.8\,\text{mV})\ln(10) + V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

$$= (25.8\,\text{mV})(2.305) + V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

$$= 0.0594 + V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

Step 2 of 2

Determine the change in V_0.

$$V_0' - V_0 = \left(0.0594 + V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)\right) - V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

$$= 0.0594$$

$$= 59.4\,\text{mV}$$

Therefore, the output voltage increases by **59.4 mV** when N_A or N_D is increased by a factor of 10.

Step 1 of 3

The built in voltage V_0 of a junction is,

$$V_0 = V_T \ln\left(\frac{N_A N_D}{\eta_i^2}\right) \quad \text{..... (1)}$$

the *p* and *n* regions are unequally doped.

$N_A = 10^{17}/\text{cm}^3$ which represents acceptor concentration.

$N_D = 10^{16}/\text{cm}^3$ which represents donor concentration.

Intrinsic concentration η_i value is $1.5\times10^{10}/\text{cm}^3$

thermal voltage V_T value is 26 mV.

Substitute $10^{17}/\text{cm}^3$ for N_A and $10^{16}/\text{cm}^3$ for N_D ,$1.5\times10^{10}/\text{cm}^3$ for η_i ,26 mV for V_T in equation (1)

$$V_0 = 26\ \text{mV}\ln\left(\frac{(10^{16}/\text{cm}^3)(10^{17}/\text{cm}^3)}{(1.5\times10^{10}/\text{cm}^3)^2}\right)$$

$$= 757\ \text{mV}$$

$$1\ \text{mV} = 0.001\ \text{V}$$

$$V_o = 0.757\ \text{V}$$

Step 2 of 3

The width of a depletion region *W* for intrinsic silicon is,

$$W = \sqrt{\frac{2\varepsilon(V_o + V_R)}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)}$$

$$\varepsilon = 11.7\varepsilon_o$$

Substitute $\varepsilon_o = 8.854\times10^{-14}\ \text{F/cm}$ permittivity of free space

$$\varepsilon = 11.7(8.854\times10^{-14}\ \text{F/cm})$$

$$= 1.036\times10^{-12}\ \text{F/cm}$$

Substitute 1.6×10^{-19} C for '*q*' charge of an electron, 0.757 V for 'V_o',$10^{17}/\text{cm}^3$ for N_A and $10^{16}/\text{cm}^3$ for N_D , 3 V for reverse bias voltage 'V_R' ,1.036×10^{-12} F/cm for ε

$$W = \sqrt{\frac{2(1.036\times10^{-12}\ \text{F/cm})(0.757\ \text{V}+3\ \text{V})}{1.6\times10^{-19}\ \text{C}}\left(\frac{1}{(10^{17}/\text{cm}^3)} + \frac{1}{(10^{16}/\text{cm}^3)}\right)}$$

$$= 7.3\times10^{-5}\ \text{cm}$$

$$1\ \text{cm} = 10^{-2}\ \text{m}$$

$$W = 7.3\times10^{-5}(10^{-2}\ \text{m})$$

$$W = 7.3\times10^{-7}\ \text{m}$$

$$= 0.73\times10^{-6}\ \text{m}$$

$$= 0.73\ \mu\text{m}$$

Step 3 of 3

Charge stored on either side of the junction Q_j is,

$$Q_j = q\left(\frac{N_A N_D}{N_A + N_D}\right)AW$$

Substitute 1.6×10^{-19} C for *q* ,$10^{17}/\text{cm}^3$ for N_A and $10^{16}/\text{cm}^3$ for N_D ,100 μm^2 for *A* ,

0.73 μm for *W*

$$Q_j = 1.6\times10^{-19}\left(\frac{(10^{17}/\text{cm}^3)(10^{16}/\text{cm}^3)}{(10^{17}/\text{cm}^3) + (10^{16}/\text{cm}^3)}\right)(100\mu\text{m}^2)(0.73\mu\text{m})$$

$$= 1.6\times10^{-19}(9.1\times10^{15}/\text{cm}^3)(73\mu\text{m}^3)$$

$$= 1063\times10^{-4}\left(\frac{10^{-18}\text{m}^3}{10^{-6}\text{m}^3}\right)$$

$$= 10.63\times10^{-14}\text{C}$$

Therefore, the charge stored on either side of the junction is $\boxed{10.63\times10^{-14}\ \text{C}}$.

Step 1 of 4

3.018P

Write the expression for width of depletion layer when the reverse-bias voltage V_R is applied.

$$W = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)(V_0+V_R)} \quad \text{..... (1)}$$

Determine the width of depletion layer for the PN junction with open circuit terminals (at equilibrium).

Substitute $0\,\text{V}$ for V_R in equation (1).

$$W = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)(V_0+0)}$$

$$= \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)(V_0)}$$

Step 2 of 4

Simplify equation (1) is as follows:

$$W = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)(V_0+V_R)}$$

$$= \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)V_0\left(1+\frac{V_R}{V_0}\right)}$$

$$= \left(\sqrt{\frac{2\varepsilon_s}{q}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)V_0}\right)\left(\sqrt{\left(1+\frac{V_R}{V_0}\right)}\right)$$

Substitute $\sqrt{\frac{2\varepsilon_s}{q}\left(\frac{1}{N_A}+\frac{1}{N_D}\right)V_0}$ for W_0.

$$W = W_0\left(\sqrt{\left(1+\frac{V_R}{V_0}\right)}\right)$$

$$= W_0\sqrt{1+\frac{V_R}{V_0}}$$

Therefore, the width of the depletion region $\boxed{W = W_0\sqrt{1+\frac{V_R}{V_0}}}$ is proved.

Step 3 of 4

Write the expression of charge stored on either side of the depletion region.

$$Q_J = A\sqrt{2\varepsilon_s q\left(\frac{N_A N_D}{N_A+N_D}\right)(V_0+V_R)} \quad \text{..... (2)}$$

Determine the charge stored on either side of the depletion region for open circuit terminals (at equilibrium).

Substitute $0\,\text{V}$ for V_R in equation (2).

$$Q_{J0} = A\sqrt{2\varepsilon_s q\left(\frac{N_A N_D}{N_A+N_D}\right)(V_0+0\,\text{V})}$$

$$Q_{J0} = A\sqrt{2\varepsilon_s q\left(\frac{N_A N_D}{N_A+N_D}\right)V_0}$$

Step 4 of 4

Simplify equation (2) is as follows:

$$Q_J = A\sqrt{2\varepsilon_s q\left(\frac{N_A N_D}{N_A+N_D}\right)(V_0+V_R)}$$

$$Q_J = A\sqrt{2\varepsilon_s q\left(\frac{N_A N_D}{N_A+N_D}\right)V_0\left(1+\frac{V_R}{V_0}\right)}$$

$$Q_J = \left(A\sqrt{2\varepsilon_s q\left(\frac{N_A N_D}{N_A+N_D}\right)V_0}\right)\left(\sqrt{\left(1+\frac{V_R}{V_0}\right)}\right)$$

Substitute $A\sqrt{2\varepsilon_s q\left(\frac{N_A N_D}{N_A+N_D}\right)V_0}$ for Q_{J0}.

$$Q_J = Q_{J0}\left(\sqrt{\left(1+\frac{V_R}{V_0}\right)}\right)$$

$$Q_J = Q_{J0}\sqrt{1+\frac{V_R}{V_0}}$$

Hence, the charge stored on the either side of the junction $\boxed{Q_J = Q_{J0}\sqrt{1+\frac{V_R}{V_0}}}$ is proved.

Step 1 of 4

Write the expression of current component due to hole injection.

$$I_p = Aqn_i^2\left(\frac{D_p}{L_pN_D}\right)\left(e^{\frac{V}{V_T}}-1\right) \quad \text{..... (1)}$$

Write the expression of current component due to electron injection.

$$I_n = Aqn_i^2\left(\frac{D_n}{L_nN_A}\right)\left(e^{\frac{V}{V_T}}-1\right) \quad \text{..... (2)}$$

Divide equation (1) with equation (2).

$$\frac{I_p}{I_n} = \frac{Aqn_i^2\left(\frac{D_p}{L_pN_D}\right)\left(e^{\frac{V}{V_T}}-1\right)}{Aqn_i^2\left(\frac{D_n}{L_nN_A}\right)\left(e^{\frac{V}{V_T}}-1\right)}$$

$$= \frac{\left(\frac{D_p}{L_pN_D}\right)}{\left(\frac{D_n}{L_nN_A}\right)}$$

$$\frac{I_p}{I_n} = \frac{D_p}{D_n}\frac{L_n}{L_p}\frac{N_A}{N_D} \quad \text{..... (3)}$$

Hence, the ratio of current components $\boxed{\frac{I_p}{I_n} = \frac{D_p}{D_n}\frac{L_n}{L_p}\frac{N_A}{N_D}}$ is proved.

Step 2 of 4

Substitute $10\ \text{cm}^2/\text{s}$ for D_p, $20\ \text{cm}^2/\text{s}$ for D_n, $10\ \mu\text{m}$ for L_n, $5\ \mu\text{m}$ for L_p, $10^{16}\ \text{carriers}/\text{cm}^3$ for N_D and $10^{18}\ \text{carriers}/\text{cm}^3$ for N_A in the equation (3).

$$\frac{I_p}{I_n} = \frac{D_p}{D_n}\frac{L_n}{L_p}\frac{N_A}{N_D}$$

$$= \frac{(10\ \text{cm}^2/\text{s})(10\ \mu\text{m})(10^{18}\ \text{carriers}/\text{cm}^3)}{(20\ \text{cm}^2/\text{s})(5\ \mu\text{m})(10^{16}\ \text{carriers}/\text{cm}^3)}$$

$$= \frac{(10)(10)(10^{18})}{(20)(5)(10^{16})}$$

$$= 100$$

Step 3 of 4

Note that the total current flowing through the configuration is the summation of the current due to the holes and the current due to the electrons.

$$I = I_p + I_n$$

Substitute $100I_n$ for I_p.

$$I = 100I_n + I_n$$

$$= 101I_n$$

Substitute $100\ \mu\text{A}$ for I.

$$100\times10^{-6} = 101I_n$$

$$I_n = \frac{100\times10^{-6}}{101}$$

$$I_n = 0.99\ \mu\text{A}$$

Therefore, the value of electron current I_n is $\boxed{0.99\ \mu\text{A}}$.

Step 4 of 4

Determine the value of current due to holes.

$$I_p = 100I_n$$

Substitute $0.99\ \mu\text{A}$ for I_n.

$$I_p = 100(0.99\ \mu\text{A})$$

$$= 99\ \mu\text{A}$$

Therefore, the value of hole current I_p is $\boxed{99\ \mu\text{A}}$.

Step 1 of 2

The saturation current I_s is,

$$I_s = Aq\eta_i^2\left(\frac{D_p}{L_pN_D}+\frac{D_n}{L_nN_A}\right)$$

Substitute 1.6×10^{-19} C for q, which represents charge of an electron

100 μm^2 for A, which represents cross sectional area

5 μm for L_p , which represents distance travelled by holes during their life time

10 μm for L_n, which represents distance travelled by electrons during their life time

10 cm^2/s for D_P, which represents diffusion constant for holes

18 cm^2/s for D_n, which represents diffusion constant for electrons

10^{17}/cm^3 for N_A, which represents acceptor concentration

10^{16}/cm^3 for N_D, which represents donor concentration

1.5×10^{10}/cm^3 for η_i ,which represents intrinsic carrier concentration

$$I_s = \left(100\mu m^2\right)\left(1.6\times10^{-19}\ \text{C}\right)\left(1.5\times10^{10}/\text{cm}^3\right)^2\left(\frac{10\ \text{cm}^2/s}{(5\mu m)\left(10^{16}/\text{cm}^3\right)}+\frac{18\ \text{cm}^2/s}{(10\mu m)\left(10^{17}/\text{cm}^3\right)}\right)$$

$$= 3600\ \text{C}\left(\frac{\mu m^2}{\text{cm}^6}\right)\left(\frac{2.18\times10^{-16}\ \text{cm}^5}{s(\mu m)}\right)$$

$$= 7848\times10^{-16}\left(\frac{\text{C}}{\text{s}}\right)\left(\frac{\mu m}{\text{cm}}\right)$$

$$= 7848\times10^{-16}\left(\frac{\text{C}}{\text{s}}\right)\left(\frac{10^{-4}\text{cm}}{\text{cm}}\right)$$

$$I_s = 0.7848\times10^{-16}\ \text{A}$$

Therefore, the saturation current value is $\boxed{I_s = 0.7848\times10^{-16}\ \text{A}}$.

Step 2 of 2

The forward current I is,

$$I = I_s\left(e^{\left(\frac{V}{V_T}\right)}-1\right)$$

Substitute 0.7848×10^{-16} A for I_s ,which represents saturation current

750 mV for V, which represents applied voltage

26 mV for V_T , which represents thermal voltage at room temperature

$$I = \left(0.7848\times10^{-16}\ \text{A}\right)\left(e^{\left(\frac{750\ \text{mV}}{26\ \text{mV}}\right)}-1\right)$$

$$= 0.264\times10^{-3}\ \text{A}$$

Therefore, the forward current value is $\boxed{0.264\times10^{-3}\ \text{A}}$.

Step 1 of 3

Consider the current is directly proportional to the square of the carrier concentration n_i^2. $I_S \propto n_i^2$

Calculate the intrinsic carrier density at $T = 300\,\text{K}$.

$$n_i = BT^{\frac{3}{2}} e^{\frac{-E_g}{2kT}}$$

Here,

B is a material dependent parameter,

E_g is bandgap energy ,

k is Boltzmann's constant.

Substitute $7.3 \times 10^{15}\,\text{cm}^{-3}\text{K}^{-3/2}$ for B , $1.12\,\text{eV}$ for E_g , $8.62 \times 10^{-5}\,\text{eV/K}$ for k and $300\,\text{K}$ for T in the equation.

$$\begin{aligned} n_i &= \left(7.3\times10^{15}\right)\left(300^{\frac{3}{2}}\right) e^{\left(\frac{-1.12}{2\times8.62\times10^{-5}\times300}\right)} \\ &= \left(7.3\times10^{15}\right)(5196.15)e^{-21.655} \\ &= \left(3.7932\times10^{19}\right)\left(3.939\times10^{-10}\right) \\ &= 1.49\times10^{10} \\ &\approx 1.5\times10^{10}\,\text{carriers/cm}^3 \end{aligned}$$

Calculate the value of n_i^2 .

$$\begin{aligned} n_i^2 &= \left(1.5\times10^{10}\right)^2 \\ &= 2.25\times10^{20}\,\text{carriers/cm}^3 \end{aligned}$$

Step 2 of 3

Calculate the intrinsic carrier density at $T = 305\,\text{K}$.

$$n_i = BT^{\frac{3}{2}} e^{\frac{-E_g}{2kT}}$$

Substitute $7.3 \times 10^{15}\,\text{cm}^{-3}\text{K}^{-3/2}$ for B , $1.12\,\text{eV}$ for E_g , $8.62 \times 10^{-5}\,\text{eV/K}$ for k and $305\,\text{K}$ for T in the equation.

$$\begin{aligned} n_i &= \left(7.3\times10^{15}\right)\left(305^{\frac{3}{2}}\right) e^{\left(\frac{-1.12}{2\times8.62\times10^{-5}\times305}\right)} \\ &= \left(7.3\times10^{15}\right)(5326.596)e^{-21.3} \\ &= \left(3.8884\times10^{19}\right)\left(5.6173\times10^{-10}\right) \\ &= 2.18\times10^{10}\,\text{carriers/cm}^3 \end{aligned}$$

Calculate the value of n_i^2 .

$$\begin{aligned} n_i^2 &= \left(2.18\times10^{10}\right)^2 \\ &= 4.75\times10^{20}\,\text{carriers/cm}^3 \end{aligned}$$

Step 3 of 3

Determine the value of the factor by which n_i^2 changes as T changes from $300\,\text{K}$ to $305\,\text{K}$.

$$\begin{aligned} \frac{n_{i(305\,\text{k})}^2}{n_{i(300\,\text{k})}^2} &= \frac{4.75\times10^{20}}{2.25\times10^{20}} \\ &= 2.1 \end{aligned}$$

Thus, the value of factor by which intrinsic carrier density n_i^2 and the current, I_S changes as the temperature changes from $300\,\text{K}$ to $305\,\text{K}$ is $\boxed{2.1}$.

Step 1 of 4

Write the expression for current-voltage relationship of the pn-junction

$$I = I_S\left(e^{V/V_T} - 1\right) \quad \text{..... (1)}$$

Write the expression for saturation current.

$$I_S = Aqn_i^2\left(\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right)$$

In p^+n junction, the acceptor atom concentration is much greater than the donor atom concentration.

$$N_A \gg N_D$$

Step 2 of 4

Derive the expression for current-voltage relationship of the p^+n -junction.

Substitute $Aqn_i^2\left(\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right)$ for I_S in equation (1).

$$\begin{aligned} I &= Aqn_i^2\left(\frac{D_p}{L_pN_D} + \frac{D_n}{L_nN_A}\right)\left(e^{V/V_T} - 1\right) \\ &= Aqn_i^2\left(\frac{D_p}{L_pN_D} + 0\right)\left(e^{V/V_T} - 1\right) \qquad (\text{since } N_A \gg N_D) \\ &= Aqn_i^2\left(\frac{D_p}{L_pN_D}\right)\left(e^{V/V_T} - 1\right) \\ &\approx I_p \qquad \left(\text{since } I_p = Aqn_i^2\left(\frac{D_p}{L_pN_D}\right)\left(e^{V/V_T} - 1\right)\right) \end{aligned}$$

Thus, the relation for current-voltage relationship of the p^+n -junction is,

$$\boxed{I \simeq I_p = Aqn_i^2\frac{D_p}{L_pN_D}\left(e^{V/V_T} - 1\right)}.$$

Step 3 of 4

Determine the value of saturation current for p^+n -junction.

$$I_S = Aqn_i^2\left(\frac{D_p}{L_pN_D}\right)$$

Substitute $10^4\ \mu m^2$ for A, 1.6×10^{-19} C for q, $1.5\times10^{10}/cm^3$ for n_i, $10\ cm^2/s$ for D_p, $10^{17}/cm^3$ for N_D and $10\ \mu m$ for L_p in the equation.

$$\begin{aligned} I_S &= \left(10^4\times10^{-12}\ m^2\right)\left(1.6\times10^{-19}\right)\left(1.5\times10^{10}/cm^3\right)^2\left(\frac{10\ cm^2/s}{\left(10\times10^{-6}\ m\right)\left(10^{17}/cm^3\right)}\right) \\ &= \left(10^4\times10^{-12}\times10^4\ cm^2\right)\left(1.6\times10^{-19}\right)\left(1.5\times10^{10}/cm^3\right)^2\left(\frac{10\ cm^2/s}{\left(10\times10^{-6}\times10^2\ cm\right)\left(10^{17}/cm^3\right)}\right) \\ &= \left(10^{-4}\right)\left(1.6\times10^{-19}\right)\left(1.5\times10^{10}\right)^2\left(10^{-13}\right) \\ &= 3.6\times10^{-16}\ A \end{aligned}$$

Hence, the value of saturation current for p^+n junction I_S is $\boxed{3.6\times10^{-16}\ A}$.

Step 4 of 4

Determine the value of voltage, V when $I = 1\ mA$.

$$I = I_S\left(e^{V/V_T} - 1\right)$$

Substitute $1\ mA$ for I, $3.6\times10^{-16}\ A$ for I_S and $25.9\ mV$ for V_T in the equation.

$$\begin{aligned} \left(1\times10^{-3}\right) &= \left(3.6\times10^{-16}\right)\left(e^{V/25.9\times10^{-3}} - 1\right) \\ e^{V/25.9\times10^{-3}} &= \frac{1\times10^{-3}}{3.6\times10^{-16}} + 1 \\ e^{V/25.9\times10^{-3}} &= 2.777\times10^{12} \\ \frac{V}{25.9\times10^{-3}} &= \ln\left(2.777\times10^{12}\right) \\ V &= \left(25.9\times10^{-3}\right)(28.65) \\ &= 0.742\ V \end{aligned}$$

Hence, the value of voltage, V is $\boxed{0.742\ V}$.

Step 1 of 3

3.023P

Calculate the power dissipation at break down region.

In the break down region, the power dissipation is equals to the half of the rated value.

$$P_z = \frac{0.25\,\text{W}}{2}$$
$$= 0.125\,\text{W}$$

Determine the value of continuous current in the breakdown region.

$$P_z = V_z I_z$$

Substitute $0.125\,\text{W}$ for P_z and $12\,\text{V}$ for V_z in the equation.

$$0.125 = (12) I_z$$

$$I_z = \frac{0.125}{12}$$
$$= 0.01042$$
$$= 10.42\,\text{mA}$$

Thus, the value of continuous current in the breakdown region, I_z is $\boxed{10.42\,\text{mA}}$.

Step 2 of 3

Determine the value of breakdown current when the power dissipation rated at $0.25\,\text{W}$.

$$P_z = V_z I_z$$

Substitute $0.25\,\text{W}$ for P_z and $12\,\text{V}$ for V_z in the equation.

$$0.25 = (12) I_z$$

$$I_z = \frac{0.25}{12}$$
$$= 0.02083$$
$$= 20.83\,\text{mA}$$

Step 3 of 3

Determine the value of allowed average breakdown current when breakdown occurs for only $10\,\text{ms}$ in every $20\,\text{ms}$.

$$I_{z(\text{avg})} = \left(\frac{20\times10^{-3}}{10\times10^{-3}}\right)\left(20.83\times10^{-3}\right)$$
$$= (2)\left(20.83\times10^{-3}\right)$$
$$= 41.66\times10^{-3}$$
$$= 41.66\,\text{mA}$$

Thus, the allowed average breakdown current, $I_{z(\text{avg})}$ is $\boxed{41.66\,\text{mA}}$.

Step 1 of 3

The equation of the built in voltage V_0 of a junction is,

$$V_0 = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

Here, the acceptor concentration, $N_A = 10^{17}/\text{cm}^3$, donor concentration, $N_D = 10^{16}/\text{cm}^3$, and the intrinsic concentration, $n_i = 1.5\times10^{10}/\text{cm}^3$.

Note that the thermal equivalent voltage V_T is 26 mV.

Substitute $10^{17}/\text{cm}^3$ for N_A, $10^{16}/\text{cm}^3$ for N_D, $1.5\times10^{10}/\text{cm}^3$ for n_i and 26 mV for V_T.

$$\begin{aligned} V_0 &= (26\text{ mV})\ln\left[\frac{(10^{16})(10^{17})}{(1.5\times10^{10})^2}\right] \\ &= (26\text{ mV})\ln\left(\frac{10^{33}}{2.25\times10^{20}}\right) \\ &= (26\text{ mV})(29.12) \\ &= 757\text{ mV} \end{aligned}$$

Step 2 of 3

Consider that for an intrinsic silicon the permittivity, $\varepsilon_s = 11.7\varepsilon_o$.

Substitute 8.854×10^{-14} F/cm for permittivity of free space ε_o.

$$\begin{aligned} \varepsilon_s &= (11.7)(8.854\times10^{-14}\text{ F/cm}) \\ &= 1.036\times10^{-12}\text{ F/cm} \end{aligned}$$

Calculate the capacitance C_{j0}.

$$C_{j0} = A\sqrt{\frac{\varepsilon_s q}{2V_0}\left(\frac{N_A N_D}{N_A + N_D}\right)}$$

Here, q is the charge of an electron, and A is the cross sectional area.

Substitute 1.036×10^{-12} F/cm for ε_s, 1.6×10^{-19} C for q, 0.757 V for V_0, $10^{17}/\text{cm}^3$ for N_A, $10^{16}/\text{cm}^3$ for N_D, and $100\ \mu\text{m}^2$ for A.

$$\begin{aligned} C_{j0} &= (100\ \mu\text{m}^2)\sqrt{\left[\frac{(1.036\times10^{-12}\text{ F/cm})(1.6\times10^{-19}\text{ C})}{2(0.757\text{ V})}\right]\left[\frac{(10^{17})(10^{16})}{10^{17}+10^{16}}\right]} \\ &= (100\times10^{-8}\text{ cm}^2)\sqrt{\frac{(1.036)(17.6\times10^{-14})}{2(0.757)}\left(\frac{\text{F}^2}{\text{cm}^4}\right)} \\ &= 0.0315\times10^{-12}\text{ F} \end{aligned}$$

Therefore, the value of the capacitance C_{j0} is $\boxed{0.0315\times10^{-12}\text{ F}}$.

Step 3 of 3

Calculate the depletion capacitance C_j for the junction.

$$C_j = \frac{C_{j0}}{\left(1+\frac{V_R}{V_0}\right)^m}$$

Here, C_{j0} is the capacitance at zero reverse bias, V_R is the applied reverse bias voltage, V_0 is the built in voltage for the junction, and m is the grading coefficient.

Substitute 0.0315×10^{-12} F for C_{j0}, 3 V for V_R, 0.757 V for V_0, and $\frac{1}{3}$ for m.

$$\begin{aligned} C_j &= \frac{0.0315\times10^{-12}\text{ F}}{\left(1+\frac{3\text{ V}}{0.757\text{ V}}\right)^{\frac{1}{3}}} \\ &= \frac{0.0315\times10^{-12}\text{ F}}{1.7} \\ &= 0.0185\times10^{-12}\text{ F} \end{aligned}$$

Therefore, the depletion capacitance of the junction C_j is $\boxed{0.0185\times10^{-12}\text{ F}}$.

Step 1 of 2

3.025P

Write the formula for the Depletion capacitance C_j for a particular junction.

$$C_j = \frac{C_{jo}}{\left(1+\frac{V_R}{V_o}\right)^m} \quad \text{..... (1)}$$

Here, '*m*' is constant called the grading coefficient

C_{jo} represents the value of C_j at zero reverse bias

V_R represents reverse bias voltage ,

V_o represents open circuit voltage.

Substitute 0.4pF for C_{jo} , 1V for V_R , 0.75V for V_o and $\frac{1}{3}$ for *m* in equation (1).

$$\begin{aligned} C_j &= \frac{0.4\text{ pF}}{\left(1+\frac{1\text{V}}{0.75\text{V}}\right)^{\frac{1}{3}}} \\ &= 0.301578976\text{ pF} \\ &\approx 0.3\text{ pF} \end{aligned}$$

Thus, the junction capacitance at 1 V reverse bias voltage is $\boxed{0.3\text{ pF}}$.

Step 2 of 2

The reverse bias voltage V_R changed to 10 V .

Substitute 0.4 pF for C_{jo} , 10 V for V_R , 0.75 V for V_o and $\frac{1}{3}$ for *m* in equation (1).

$$\begin{aligned} C_j &= \frac{0.4\text{ pF}}{\left(1+\frac{10\text{V}}{0.75\text{V}}\right)^{\frac{1}{3}}} \\ &= 0.164668649\text{ pF} \\ &\approx 0.165\text{ pF} \end{aligned}$$

Thus, the value of capacitance C_j for V_R of 10 V is $\boxed{0.165\text{ pF}}$.

Step 1 of 5

Write the expression for junction capacitance.

$$C_j = \frac{\alpha}{2\sqrt{V_0 + V_R}}$$

Here, the value of constant α is,

$$\alpha = A\sqrt{2\varepsilon_s q \frac{N_A N_D}{N_A + N_D}}$$

Substitute $A\sqrt{2\varepsilon_s q \frac{N_A N_D}{N_A + N_D}}$ for α in the equation.

$$C_j = \frac{A\sqrt{2\varepsilon_s q \frac{N_A N_D}{N_A + N_D}}}{2\sqrt{V_0 + V_R}}$$

$$C_j^2 = \frac{A^2 2\varepsilon_s q\left(\frac{N_A N_D}{N_A + N_D}\right)}{4(V_0 + V_R)}$$

$$C_j^2 = \frac{A^2 \varepsilon_s q\left(\frac{N_A N_D}{N_A + N_D}\right)}{2(V_0 + V_R)} \quad \text{..... (1)}$$

Step 2 of 5

Write the expression for width of the depletion region.

$$W = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)(V_0 + V_R)}$$

$$W = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{N_A + N_D}{N_A N_D}\right)(V_0 + V_R)}$$

$$W^2 = \frac{2\varepsilon_s}{q}\left(\frac{N_A + N_D}{N_A N_D}\right)(V_0 + V_R)$$

$$(V_0 + V_R) = \frac{W^2}{\frac{2\varepsilon_s}{q}\left(\frac{N_A + N_D}{N_A N_D}\right)}$$

Step 3 of 5

Substitute $\frac{W^2}{\frac{2\varepsilon_s}{q}\left(\frac{N_A + N_D}{N_A N_D}\right)}$ for $(V_0 + V_R)$ in equation (1).

$$C_j^2 = \frac{A^2 \varepsilon_s q\left(\frac{N_A N_D}{N_A + N_D}\right)}{2\left(\frac{W^2}{\frac{2\varepsilon_s}{q}\left(\frac{N_A + N_D}{N_A N_D}\right)}\right)}$$

$$= \frac{A^2 \varepsilon_s q\left(\frac{N_A N_D}{N_A + N_D}\right)}{\frac{q}{\varepsilon_s} W^2\left(\frac{N_A N_D}{N_A + N_D}\right)}$$

$$C_j^2 = \frac{A^2 \varepsilon_s^2}{W^2}$$

$$C_j = \sqrt{\frac{A^2 \varepsilon_s^2}{W^2}}$$

$$= \frac{A\varepsilon_s}{W}$$

Thus, the expression for junction capacitance for a parallel-plate capacitance, C_j is $\boxed{\frac{A\varepsilon_s}{W}}$.

Step 4 of 5

Alternate method:

Write the expression for junction capacitance.

$$C_j = \frac{C_{j0}}{\sqrt{1 + \frac{V_R}{V_0}}} \quad \text{..... (2)}$$

Here,

V_R is the reverse voltage.

C_{j0} is the junction capacitance at zero reverse bias.

$$C_{j0} = A\sqrt{\left(\frac{\varepsilon_s q}{2}\right)\left(\frac{N_A N_D}{N_A + N_D}\right)\left(\frac{1}{V_0}\right)}$$

Step 5 of 5

Substitute $A\sqrt{\left(\frac{\varepsilon_s q}{2}\right)\left(\frac{N_A N_D}{N_A + N_D}\right)\left(\frac{1}{V_0}\right)}$ for C_{j0} in equation (2).

$$C_j = \frac{A\sqrt{\left(\frac{\varepsilon_s q}{2}\right)\left(\frac{N_A N_D}{N_A + N_D}\right)\left(\frac{1}{V_0}\right)}}{\sqrt{1 + \frac{V_R}{V_0}}}$$

$$C_j = \frac{A\sqrt{\left(\frac{\varepsilon_s q}{2}\right)\left(\frac{N_A N_D}{N_A + N_D}\right)\left(\frac{1}{V_0}\right)}}{\sqrt{\frac{V_0 + V_R}{V_0}}}$$

Substitute $\frac{W^2}{\frac{2\varepsilon_s}{q}\left(\frac{N_A + N_D}{N_A N_D}\right)}$ for $(V_0 + V_R)$ in the equation.

$$C_j = \frac{A\sqrt{\left(\frac{\varepsilon_s q}{2}\right)\left(\frac{N_A N_D}{N_A + N_D}\right)\left(\frac{1}{V_0}\right)}}{\sqrt{\frac{\frac{W^2}{\frac{2\varepsilon_s}{q}\left(\frac{N_A + N_D}{N_A N_D}\right)}}{V_0}}}$$

$$= \frac{A\sqrt{\left(\frac{\varepsilon_s q}{2}\right)\left(\frac{N_A N_D}{N_A + N_D}\right)\left(\frac{1}{V_0}\right)}}{\sqrt{\frac{\frac{q}{2\varepsilon_s} W^2\left(\frac{N_A N_D}{N_A + N_D}\right)}{V_0}}}$$

$$= A\sqrt{\left(\frac{\varepsilon_s^2}{W^2}\right)}$$

$$= \frac{A\varepsilon_s}{W}$$

Thus, the expression for junction capacitance for a parallel-plate capacitance, C_j is $\boxed{\frac{A\varepsilon_s}{W}}$.

Step 1 of 2

3.027P

The diffusion capacitance C_d and the forward current I is,

$$C_d = \left(\frac{\tau_T}{V_T}\right) I \quad \text{..... (1)}$$

The diffusion capacitance C_d is directly proportional to forward current I

$$\frac{C_{d1}}{C_{d2}} = \frac{I_1}{I_2}$$

$$C_{d2} = \left(\frac{I_2}{I_1}\right) C_{d1}$$

Substitute 0.1 mA for I_2 ,1 mA for I_1 and 5 pF for C_{d1}

$$C_{d2} = \left(\frac{0.1\ \text{mA}}{1\ \text{mA}}\right) 5\ \text{pF}$$
$$= 0.5\ \text{pF}$$

Step 2 of 2

From equation (1) the mean transit time of the junction is,

$$\tau_T = C_d \left(\frac{V_T}{I}\right)$$

$$C_d = C_{d2}$$
$$= 0.5\ \text{pF}$$

Substitute 0.5 pF for C_d , 0.1 mA for I, 26 mV for V_T which represents thermal voltage at room temperature in τ_T equation.

$$\tau_T = 0.5\ \text{pF}\left(\frac{26\ \text{mV}}{0.1\ \text{mA}}\right)$$
$$= 130\ \text{ps}$$

Therefore, the mean transit time of the junction is **130 ps**.

Step 1 of 4

3.028P

Refer to the problem 3.22 in the textbook.

A $p^+ n$ junction is one in which doping concentration in the p-region is much greater than that of the n-region. In such a junction the forward current is mostly due to hole injection.

Hence, $I \simeq I_p$.

Here, I is forward current and I_p is current due to holes.

Hole life time τ_p is given by,

$$\tau_p = \frac{L_p^2}{D_p} \quad \text{..... (1)}$$

Here,

L_p is diffusion length of hole,

D_p is diffusion constant.

Step 2 of 4

Substitute $10\,\mu\text{m}$ for L_p , $10\ \text{cm}^2/\text{s}$ for D_p in the equation (1).

$$\begin{aligned}\tau_p &= \frac{(10\,\mu\text{m})^2}{10\ \text{cm}^2/\text{s}} \\ &= \frac{\left(10\times10^{-4}\,\text{cm}\right)^2}{10\ \text{cm}^2/\text{s}} \\ &= 10^{-7}\,\text{s} \\ &= 0.1\,\mu\text{s}\end{aligned}$$

Therefore, the hole life time is τ_p is $\boxed{0.1\,\mu\text{s}}$.

Step 3 of 4

The excess minority carrier charge Q is,

$$\begin{aligned}Q &= I\tau_T \\ &= I_p\tau_p\end{aligned}$$

Substitute $10^{-7}\ \text{s}$ for τ_p , $0.1\ \text{mA}$ for I_p in the above equation.

$$\begin{aligned}Q &= \left(0.1\times10^{-3}\ \text{A}\right)\left(10^{-7}\ \text{s}\right) \\ &= 10^{-11}\,\text{C} \\ &= 10\times10^{-12}\ \text{C} \\ &= 10\,\text{pC}\end{aligned}$$

Therefore, the stored minority carrier charge Q is $\boxed{10\,\text{pC}}$.

Step 4 of 4

Diffusion capacitance C_d can be calculated as follows:

$$\begin{aligned}C_d &= \frac{\tau_T}{V_T}(I) \\ &= \frac{\tau_p}{V_T}(I_p)\end{aligned}$$

Here,

V_T is thermal voltage, I_p is current due to holes and τ_p is hole life time.

Substitute $10^{-7}\ \text{s}$ for τ_p , $0.1\ \text{mA}$ for I_p , $26\ \text{mV}$ for V_T in the diffusion capacitance expression.

$$\begin{aligned}C_d &= \frac{10^{-7}\ \text{s}}{26\ \text{mV}}(0.1\ \text{mA}) \\ &= 3.846\times10^{-10}\,\text{F} \\ &= 384.6\times10^{-12}\,\text{F} \\ &= 384.6\,\text{pF}\end{aligned}$$

Therefore, the values of C_d is $\boxed{384.6\,\text{pF}}$.

Step 1 of 13

(a)

Refer to Figure 3.12 in the text book.

The slope of $n_p(x)$ and $p_n(x)$ are straight lines due to the widths of the p and n regions are much smaller than L_n and L_p respectively. The doping concentration in the p region is much greater than that in the n region $(N_A \gg N_D)$.

Sketch the minority-carrier distribution in a forward biased pn junction.

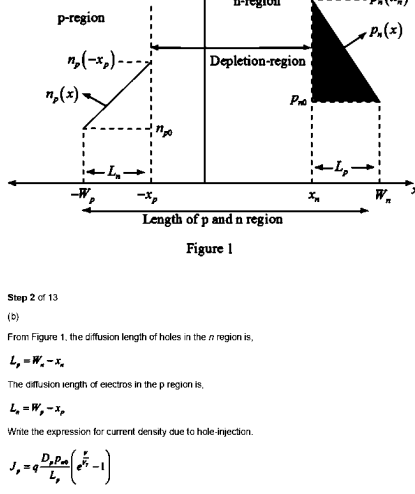


Figure 1

Step 2 of 13

(b)

From Figure 1, the diffusion length of holes in the n region is,

$$L_p = W_n - x_n$$

The diffusion length of electros in the p region is,

$$L_n = W_p - x_p$$

Write the expression for current density due to hole-injection.

$$J_p = q\frac{D_p p_{n0}}{L_p}\left(e^{\frac{v}{V_T}} - 1\right)$$

Substitute $(W_n - x_n)$ for L_p in the equation.

$$J_p = q\frac{D_p p_{n0}}{W_n - x_n}\left(e^{\frac{v}{V_T}} - 1\right)$$

Write the expression for current density due to electron-injection.

$$J_n = q\frac{D_n n_{p0}}{L_n}\left(e^{\frac{v}{V_T}} - 1\right)$$

Substitute $(W_p - x_p)$ for L_n in the equation.

$$J_n = q\frac{D_n n_{p0}}{W_p - x_p}\left(e^{\frac{v}{V_T}} - 1\right)$$

Step 3 of 13

Write the expression for total current density.

$$J = J_p + J_n$$

Substitute $q\frac{D_p p_{n0}}{W_n - x_n}\left(e^{\frac{v}{V_T}} - 1\right)$ for J_p and $q\frac{D_n n_{p0}}{W_p - x_p}\left(e^{\frac{v}{V_T}} - 1\right)$ for J_n in the equation.

$$\begin{aligned} J &= q\frac{D_p p_{n0}}{W_n - x_n}\left(e^{\frac{v}{V_T}} - 1\right) + q\frac{D_n n_{p0}}{W_p - x_p}\left(e^{\frac{v}{V_T}} - 1\right) \\ &= q\left(e^{\frac{v}{V_T}} - 1\right)\left[\frac{D_p p_{n0}}{W_n - x_n} + \frac{D_n n_{p0}}{W_p - x_p}\right] \\ &= q\left[\frac{D_p p_{n0}}{W_n - x_n} + \frac{D_n n_{p0}}{W_p - x_p}\right]\left(e^{\frac{v}{V_T}} - 1\right) \end{aligned}$$

Step 4 of 13

Write the expression for total current.

$$I = A(J_p + J_n)$$
$$I = AJ$$

Here,

J_p is the hole current density,

J_n is the electron current density, and

A = Junction area

Substitute $q\left[\frac{D_p p_{n0}}{W_n - x_n} + \frac{D_n n_{p0}}{W_p - x_p}\right]\left(e^{\frac{v}{V_T}} - 1\right)$ for J in the equation.

$$I = Aq\left[\frac{D_p p_{n0}}{W_n - x_n} + \frac{D_n n_{p0}}{W_p - x_p}\right]\left(e^{\frac{v}{V_T}} - 1\right) \quad \ldots\ldots (1)$$

Step 5 of 13

Substitute for $\frac{n_i^2}{N_D}$ for p_{n0} and $\frac{n_i^2}{N_A}$ for n_{p0} in equation (1), to derive the expression for total current.

$$\begin{aligned} I &= Aq\left[\frac{D_p\left(\frac{n_i^2}{N_D}\right)}{W_n - x_n} + \frac{D_n\left(\frac{n_i^2}{N_A}\right)}{W_p - x_p}\right]\left(e^{\frac{v}{V_T}} - 1\right) \\ &= Aqn_i^2\left[\frac{D_p}{N_D(W_n - x_n)} + \frac{D_n}{N_A(W_p - x_p)}\right]\left(e^{v/V_T} - 1\right) \end{aligned}$$

Thus, the expression for total current is,

$$\boxed{I = Aqn_i^2\left[\frac{D_p}{N_D(W_n - x_n)} + \frac{D_n}{N_A(W_p - x_p)}\right]\left(e^{v/V_T} - 1\right)}$$

Step 6 of 13

The expression for shaded area under $p_n(x)$ curve is,

$$\textbf{Shaded area under } p_n(x) = \frac{1}{2}\left[P_n x(n) - P_{n0}\right]L_p$$

Substitute $(W_n - x_n)$ for L_p in the equation.

$$\textbf{Shaded area under } p_n(x) = \frac{1}{2}\left[P_n x(n) - P_{n0}\right](W_n - x_n).$$

Derive the expression for excess minority-carrier stored charge.

$$\begin{aligned} Q_p &= Aq \times \textbf{Shaded area under } p_n(x) \\ &= Aq\left[\frac{1}{2}\left[p_n(x_n) - p_{n0}\right](W_n - x_n)\right] \end{aligned}$$

Substitute $p_{n0}e^{\frac{v}{V_T}}$ for $p_n(x_n)$ in the equation

$$\begin{aligned} Q_p &= Aq\left\{\frac{1}{2}\left[p_{n0}e^{\frac{v}{V_T}} - p_{n0}\right](W_n - x_n)\right\} \\ &= \frac{1}{2}Aq\left\{p_{n0}\left[e^{\frac{v}{V_T}} - 1\right]\right\}(W_n - x_n) \end{aligned}$$

$$Q_p = \frac{1}{2}Aqp_{n0}(W_n - x_n)\left[e^{\frac{v}{V_T}} - 1\right] \quad \ldots\ldots (2)$$

Step 7 of 13

The current density due to hole-injection is,

$$J_p = q\frac{D_p p_{n0}}{L_p}\left(e^{\frac{v}{V_T}} - 1\right)$$

Write the expression for hole-injection component of current.

$$I_p = AJ_p$$

Substitute $q\frac{D_p p_{n0}}{L_p}\left(e^{\frac{v}{V_T}} - 1\right)$ for J_p in the equation.

$$I_p = Aq\frac{D_p p_{n0}}{L_p}\left(e^{\frac{v}{V_T}} - 1\right)$$

$$e^{\frac{v}{V_T}} - 1 = \frac{I_p L_p}{AqD_p p_{n0}}$$

Step 8 of 13

Substitute $\left(\frac{I_p L_p}{AqD_p p_{n0}}\right)$ for $\left(e^{\frac{v}{V_T}} - 1\right)$ in equation (2).

$$\begin{aligned} Q_p &= \frac{1}{2}Aqp_{n0}(W_n - x_n)\left[\frac{I_p L_p}{AqD_p p_{n0}}\right] \\ &= \frac{1}{2}(W_n - x_n)\left[\frac{I_p L_p}{D_p}\right] \\ &= \frac{1}{2}(W_n - x_n)\left[\frac{I_p(W_n - x_n)}{D_p}\right] \qquad \left(\text{Since } L_p = W_n - x_n\right) \\ &= \frac{1}{2}\frac{(W_n - x_n)^2}{D_p}I_p \end{aligned}$$

Thus, the expression for excess minority-carrier stored charge is,

$$\boxed{Q_p = \frac{1}{2}\frac{(W_n - x_n)^2}{D_p}I_p}$$

Step 8 of 13

If $W_n >> x_n$, the term x_n is neglected. So, the expression for excess minority-carrier stored charge is,

$$\begin{aligned} Q_p &\approx \frac{1}{2}\frac{(W_n - 0)^2}{D_p}I_p \\ &\approx \frac{1}{2}\frac{W_n^2}{D_p}I_p \end{aligned}$$

Thus, the expression for excess minority-carrier stored charge when $W_n >> x_n$ is.

$$\boxed{Q_p \approx \frac{1}{2}\frac{W_n^2}{D_p}I_p}$$

Step 10 of 13

(c)

The expression for excess minority-carrier charge is,

$$Q_p = \frac{L_p^2}{D_p}I_p$$

$$Q_p = \tau_p I_p \qquad \left(\text{Since, } \tau_p = \frac{L_p^2}{D_p}\right)$$

$$Q \approx \tau_p I \qquad \left(\text{Since, } I \approx I_p \text{ and } Q \approx Q_p\right)$$

$$Q = \tau_T I \qquad \left(\text{Since, } \tau_T = \tau_p\right)$$

Step 11 of 13

Derive the expression for incremental diffusion capacitance.

$$C_d = \frac{dQ}{dV}$$

Substitute $\tau_T I$ for Q in the equation.

$$\begin{aligned} C_d &= \frac{d(\tau_T I)}{dV} \\ &= \tau_T\frac{dI}{dV} \\ &= \tau_T\frac{d\left(I_S\left(e^{\frac{v}{V_T}} - 1\right)\right)}{dV} \qquad \left(\text{Since } I = I_S\left(e^{\frac{v}{V_T}} - 1\right)\right) \\ &= \tau_T\frac{I_S e^{\frac{v}{V_T}}}{V_T} \end{aligned}$$

$$C_d \approx \tau_T\frac{I}{V_T} \qquad \left(\text{Since } I \approx I_S e^{\frac{v}{V_T}}\right) \quad \ldots\ldots (3)$$

Thus, the expression for incremental diffusion capacitance is,

$$\boxed{C_d = \left(\frac{\tau_T}{V_T}\right)I}$$

Step 12 of 13

The expression for excess minority-carrier stored charge when $W_n >> x_n$ is,

$$Q_p \approx \frac{1}{2}\frac{W_n^2}{D_p}I_p$$

$$Q \approx \frac{1}{2}\frac{W_n^2}{D_p}I \qquad \left(\text{Since } I \approx I_p \text{ and } Q \approx Q_p\right)$$

$$\frac{Q}{V_T} \approx \frac{\frac{1}{2}\frac{W_n^2}{D_p}I}{V_T}$$

$$C_d \approx \left(\frac{\frac{1}{2}\frac{W_n^2}{D_p}}{V_T}\right)I \qquad \left(\text{Since } C_d \approx \frac{Q}{V_T}\right) \quad \ldots\ldots (4)$$

Compare the equation (3) and equation (4).

$$\tau_T = \frac{1}{2}\frac{W_n^2}{D_p}$$

Thus, the expression for mean transit time is,

$$\boxed{\tau_T = \frac{1}{2}\frac{W_n^2}{D_p}}$$

Step 13 of 13

(d)

Determine the value of width of the n-region.

$$C_d = \frac{\left(\frac{1}{2}\frac{W_n^2}{D_p}\right)}{V_T}I$$

Substitute $8\ \text{pF}$ for C_d , $1\ \text{mA}$ for I , $10\ \text{cm}^2/\text{sec}$ for D_p and $25\ \text{mV}$ for V_T in the equation.

$$\left(8\times10^{-12}\right) = \frac{1}{2}\left(\frac{\frac{W_n^2}{10}}{25\times10^{-3}}\right)\left(1\times10^{-3}\right)$$

$$\left(8\times10^{-12}\right) = \frac{W_n^2}{(20)(25)}$$

$$W_n^2 = 4000\times10^{-12}$$

$$\begin{aligned} W_n &= \sqrt{4000\times10^{-12}} \\ &= 6.32\times10^{-5}\ \text{cm} \\ &= 6.32\times10^{-7}\ \text{m} \\ &= 0.632\ \mu\text{m} \end{aligned}$$

Thus, the value of width of the n-region, W_n is $\boxed{0.632\ \mu\text{m}}$.

Step 1 of 3

Refer to Figure 4.3 (a) for a rectifier circuit in the text book.

Refer to Figure 4.3 (b) for an input wave form in the text book.

Case (i):

During the positive half cycle $(v_I \geq 0)$, the diode is in forward bias condition. The positive v_I causes the current flows through the diode in the forward direction. In forward bias condition the ideal diode acts as a short circuit. So, the output voltage is same as the input voltage.

The equivalent circuit is shown in Figure 1.

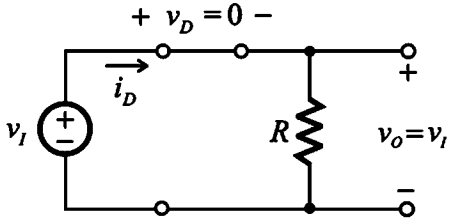


Figure 1

Step 2 of 3

Case (ii):

During the negative half cycle $(v_I \leq 0)$, the diode is in reverse bias condition. In reverse bias condition, the ideal diode acts as an open circuit and the current flows through the diode is zero. Hence, the output voltage is equals to zero.

The equivalent circuit is shown in Figure 2.

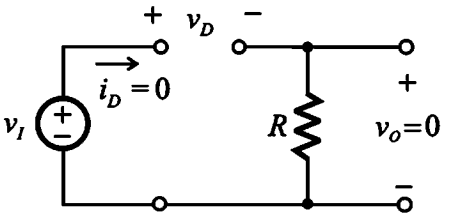


Figure 2

Step 3 of 3

Hence, the final conclusion is,

When $v_I \geq 0$, $v_o = v_I$.

That means during the positive half cycle, the diode is on and the output voltage is same as the input voltage.

When $v_I \leq 0$, $v_o = 0$ V.

That means during the negative half cycle, the diode is cut-off and the output voltage is equals to zero.

Draw the transfer characteristic curve v_o versus v_I.

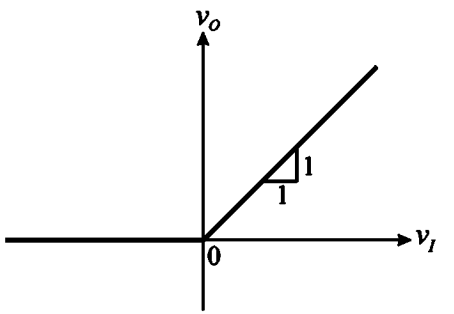


Figure 3

Thus, the transfer characteristic curve v_o versus v_I is shown in Figure 3.

Step 1 of 5

4.001P

(a)

Connect the cathode of the diode to the positive terminal of the battery and draw the equivalent circuit.

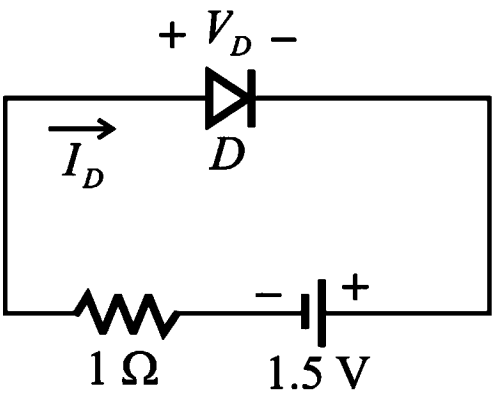


Figure 1

Since the cathode $(-ve)$ of the diode is connected to the positive of the battery, the diode will be in reverse bias.

An ideal diode in reverse bias, acts as open circuit.

Thus, the diode current I_D is $\boxed{0\,A}$.

Step 2 of 5

Replace the diode with open circuit and draw the equivalent circuit.

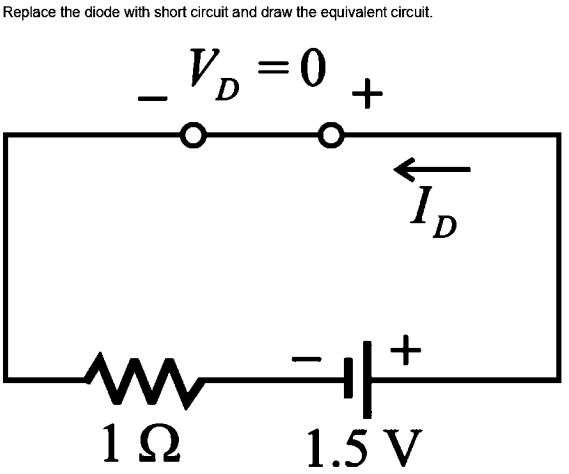


Figure 2

Apply Kirchhoff's voltage law in the circuit as shown in Figure 2.

$$V_D + 1.5 + I_D(1) = 0$$
$$V_D + 1.5 + (0)(1) = 0$$
$$V_D = -1.5\ V$$

Thus, the terminal voltage across the diode V_D is $\boxed{-1.5\,V}$.

Step 3 of 5

(b)

Connect the anode of the diode to the positive terminal of the battery and draw the equivalent circuit.

Figure 3

Since the anode $(+ve)$ of the diode is connected to the positive of the battery, the diode will be in forward bias.

An ideal diode in forward bias, acts as short circuit.

Thus, the terminal voltage across the diode V_D is $\boxed{0\,V}$.

Step 4 of 5

Replace the diode with short circuit and draw the equivalent circuit.

Figure 4

Step 5 of 5

Apply Kirchhoff's Voltage Law in the circuit as shown in Figure 4.

$$1.5 - V_D - I_D(1) = 0$$
$$1.5 - 0 - I_D = 0$$
$$1.5 - I_D = 0$$
$$I_D = 1.5\ A$$

Thus, the diode current I_D is $\boxed{1.5\,A}$.

Step 1 of 3

Refer to Figure 4.3 (a) for a rectifier circuit in the text book.

Refer to Figure 4.3 (b) for an input wave form in the text book.

Case (i):

During the positive half cycle $(v_I \geq 0)$, the diode is in forward bias condition. The positive v_I causes the current flows through the diode in the forward direction. In forward bias condition the ideal diode acts as a short circuit. Thus, the output voltage is same as the input voltage and the diode voltage is equals to zero.

The equivalent circuit is shown in Figure 1.

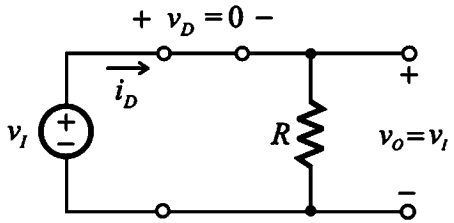


Figure 1

Step 2 of 3

Case (ii):

During the negative half cycle $(v_I \leq 0)$, the diode is in reverse bias condition. In reverse bias condition, the ideal diode acts as an open circuit and the current flows through the diode is zero. Thus, the output voltage is equals to zero and the diode voltage is same as the input voltage.

The equivalent circuit is shown in Figure 2.

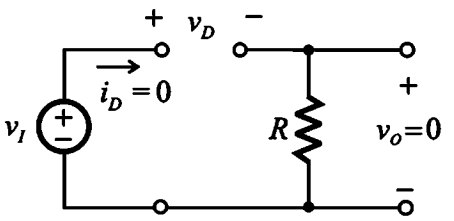


Figure 2

Step 3 of 3

Hence, the final conclusion is,

when $v_I \geq 0 \Rightarrow v_D = 0$ V.

That means during the positive half cycle, the diode is on and the diode voltage is equals to zero.

when $v_I \leq 0 \Rightarrow v_D = v_I$.

That means during the negative half cycle, the diode is cut-off and the diode voltage is same as the input voltage.

Sketch the waveform of diode voltage, v_D.

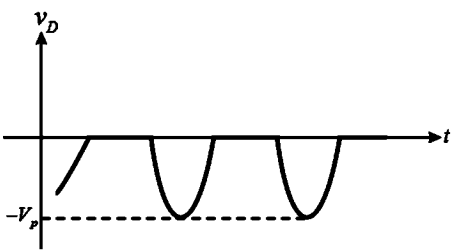


Figure 3

Thus, the waveform of diode voltage is drawn as shown in Figure 3.

Step 1 of 4

(a)

Refer to Figure P4.2 (a) in the textbook.

The cathode is connected to $-3\ \text{V}$ source and the anode is connected to $+3\ \text{V}$ source.

Therefore, the diode is in forward bias and the voltage drop across the diode is 0 V.

Find the voltage at the output node.

$$\begin{aligned} V &= 0-3 \\ &= -3\ \text{V} \end{aligned}$$

Therefore, the voltage across the diode V is $\boxed{-3\ \text{V}}$.

Find the current I flows through the diode.

The current flows through the resistor and the diode is same.

Use Ohm's law to find the current in the circuit.

$$\begin{aligned} I &= \frac{3-(-3)}{10\ \text{k}\Omega} \\ &= \frac{3+3}{10\times10^{3}} \\ &= \frac{6}{10\times10^{3}} \\ &= 0.6\ \text{mA} \end{aligned}$$

Therefore, the current flows through the diode I is $\boxed{0.6\ \text{mA}}$.

Step 2 of 4

(b)

Refer to Figure P4.2 (b) in the textbook.

The cathode is connected to $+3\ \text{V}$ source and the anode is connected to $-3\ \text{V}$ source.

Therefore, the diode is in reverse bias and the current passing through the diode is 0 A.

Therefore, the current flows through the diode I is $\boxed{0\ \text{A}}$.

Use Ohm's law to find the output voltage.

$$\begin{aligned} I &= \frac{3\ \text{V}-V}{10\ \text{k}\Omega} \\ \frac{3\ \text{V}-V}{10\times10^{3}\ \Omega} &= 0 \\ 3\ \text{V}-V &= 0 \\ V &= 3\ \text{V} \end{aligned}$$

Therefore, the voltage across the diode V is $\boxed{3\ \text{V}}$.

Step 3 of 4

(c)

Refer to Figure P4.2 (c) in the textbook.

The cathode is connected to $-3\ \text{V}$ source and the anode is connected to $+3\ \text{V}$ source.

Therefore, the diode is in forward bias and the voltage drop across the diode is 0 V.

Find the voltage at the output node.

$$V = 3\ \text{V}$$

Therefore, the voltage across the diode V is $\boxed{3\ \text{V}}$.

Find the current I flows through the diode.

The current flows through the resistor and the diode is same.

Use Ohm's law to find the current in the circuit.

$$\begin{aligned} I &= \frac{3-(-3)}{10\ \text{k}\Omega} \\ &= \frac{6}{10\times10^{3}} \\ &= 0.6\times10^{-3} \\ &= 0.6\ \text{mA} \end{aligned}$$

Therefore, the current flows through the diode I is $\boxed{0.6\ \text{mA}}$.

Step 4 of 4

(d)

Refer to Figure P4.2 (d) in the textbook.

The cathode is connected to $+3\ \text{V}$ source and the anode is connected to $-3\ \text{V}$ source.

Therefore, the diode is in reverse bias and the current passing through the diode is 0 A.

Therefore, the current flows through the diode I is $\boxed{0\ \text{A}}$.

Use Ohm's law to find the output voltage.

$$\begin{aligned} I &= \frac{V-(-3\ \text{V})}{10\ \text{k}\Omega} \\ \frac{3\ \text{V}+V}{10\times10^{3}\ \Omega} &= 0 \\ 3\ \text{V}+V &= 0 \\ V &= -3\ \text{V} \end{aligned}$$

Therefore, the voltage across the diode V is $\boxed{-3\ \text{V}}$.

Step 1 of 3

4.003E

Refer to Figure 4.3 (a) for a rectifier circuit in the text book.

Assume the diode is ideal. During the positive half cycle, the diode is in forward bias condition. The positive v_I causes the current flows through the diode in the forward direction. In forward bias condition, the ideal diode acts as a short circuit. Thus, the output voltage is same as the input voltage.

The equivalent circuit is shown in Figure 1.

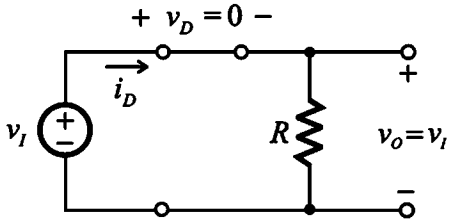


Figure 1

Step 2 of 3

Apply Ohm's law to the circuit and determine the value of peak current.

$$i_{D(peak)} = \frac{v_{I(peak)}}{R}$$

Substitute $10\ \text{V}$ for $v_{I(peak)}$ and $1\ \text{k}\Omega$ for R in the equation.

$$\begin{aligned} i_{D(peak)} &= \frac{10}{1\times10^{3}} \\ &= 10\times10^{-3} \\ &= 10\ \text{mA} \end{aligned}$$

Thus, the peak value of diode current, i_D is $\boxed{10\ \text{mA}}$.

Step 3 of 3

Determine the dc component of output voltage.

$$v_{O(dc)} = \frac{v_{I(peak)}}{\pi}$$

Substitute $10\ \text{V}$ for $v_{I(peak)}$ in the equation.

$$\begin{aligned} v_{O(dc)} &= \frac{10}{\pi} \\ &= 3.18\ \text{V} \end{aligned}$$

Thus, the dc component of output voltage, v_O is $\boxed{3.18\ \text{V}}$.

Step 1 of 4

(a)

Refer to Figure P4.3 (a) in the textbook.

The diode D_1 is cut-off, and the diode D_2 is in the conducting mode in the circuit.

The modified circuit diagram is,

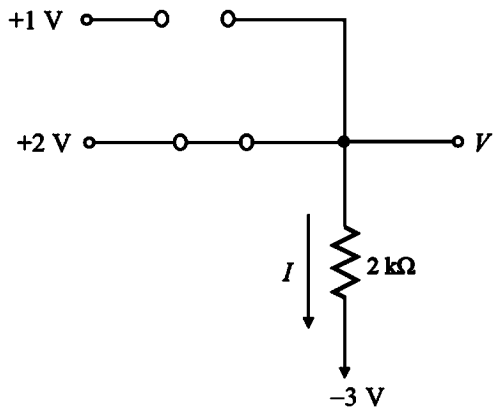


Figure 1

Step 2 of 4

The voltage, V at the output terminal of Figure 1 is obtained as the voltage at the receiving end of the diode D_2.

$$V = 2\text{ V}$$

Therefore, the value of voltage V is $\boxed{2\text{ V}}$.

Calculate the value of the current that flows through the resistance.

$$I = \frac{V-(-3)}{2\text{ k}\Omega}$$
$$= \frac{V+5}{2\times10^3}$$

Substitute 2 V for V in the expression.

$$I = \frac{2+3}{2\times10^3}$$
$$= \frac{5}{2\times10^3}$$
$$= 2.5\text{ mA}$$

Therefore, the value of current I is $\boxed{2.5\text{ mA}}$.

Step 3 of 4

(b)

Refer to Figure P4.3 (b) in the textbook.

The diode D_1 is conducting, and the diode D_2 is in the cut-off mode in the circuit.

The modified circuit diagram is,

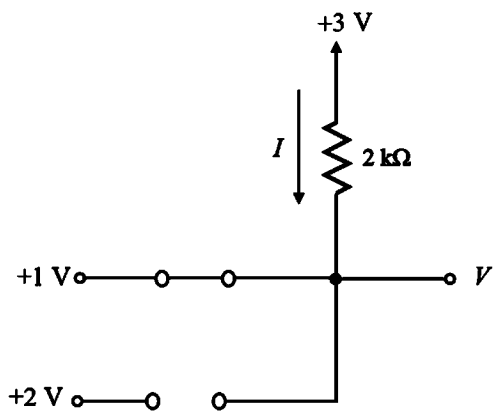


Figure 2

Step 4 of 4

The output voltage, V is obtained as the voltage at the receiving end at the diode, D_1.

The voltage at the output terminal, $V = 1\text{ V}$.

Therefore, the value of voltage V is $\boxed{1\text{ V}}$.

Calculate the value of the current that flows through the resistance.

$$I = \frac{3-V}{2\text{ k}\Omega}$$

Substitute 1 V for V in the expression.

$$I = \frac{3-1}{2\times10^3}$$
$$= \frac{2}{2\times10^3}$$
$$= 1\text{ mA}$$

Therefore, the value of current I is $\boxed{1\text{ mA}}$.

Step 1 of 13

The ideal diode circuit is shown in Figure 1.

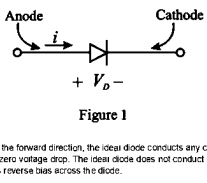


Figure 1

In the forward direction, the ideal diode conducts any current forced by the external circuit while displaying a zero voltage drop. The ideal diode does not conduct in the reverse direction any applied voltage appears as reverse bias across the diode.

Step 2 of 13

(a)

Refer to Figure E4.4 (a) in the textbook for the circuit.

Assume diode is conducting, and then the equivalent circuit is shown in Figure 2.

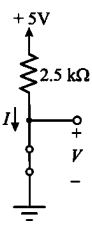


Figure 2

Step 3 of 13

Determine the value of output voltage.

The voltage across short circuit is zero.

$V = 0\ \text{V}$

Thus, the value of output voltage, V is $\boxed{0\ \text{V}}$.

Apply ohm's law to find the current in the circuit.

$$I = \frac{5 - V}{2.5 \times 10^3}$$

Substitute $0\ \text{V}$ for V in the equation.

$$\begin{aligned} I &= \frac{5 - 0}{2.5 \times 10^3} \\ &= \frac{5}{2.5 \times 10^3} \\ &= 2 \times 10^{-3} \\ &= 2\ \text{mA} \end{aligned}$$

Since, the current $I > 0$, the diode is conducting. So, the assumption is correct.

Thus, the value of current in the circuit, I is $\boxed{2\ \text{mA}}$.

Step 4 of 13

(b)

Refer to Figure E4.4 (b) in the textbook for the circuit.

Assume the diode is operated in reverse biased, then the diode is open (cut off).

The equivalent circuit is shown in Figure 3.

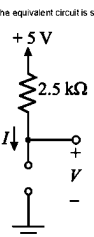


Figure 3

Step 5 of 13

Determine the value of current in the circuit.

In reverse bias condition, the ideal diode acts as an open circuit and the current flows through the diode is zero.

$I = 0\ \text{mA}$

Thus, the value of current in the circuit, I is $\boxed{0\ \text{mA}}$.

Apply ohm's law to determine the value of output voltage.

$$I = \frac{5 - V}{2.5 \times 10^3}$$

Substitute $0\ \text{mA}$ for I in the equation.

$$\begin{aligned} 0 &= \frac{5 - V}{2.5 \times 10^3} \\ 0 &= 5 - V \\ V &= 5\ \text{V} \end{aligned}$$

Since, $V = 5\ \text{V}$, diode is open. So, the assumption is correct.

Thus, the value of output voltage in the circuit, V is $\boxed{5\ \text{V}}$.

Step 6 of 13

(c)

Refer to Figure E4.4 (c) in the textbook for the circuit.

Assume the diode is operated in reverse biased, then the diode is open (cut off).

The equivalent circuit is shown in Figure 4.

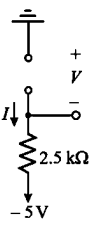


Figure 4

Step 7 of 13

Determine the value of current in the circuit.

In reverse bias condition, the ideal diode acts as an open circuit and the current flows through the diode is zero.

$I = 0\ \text{mA}$

Thus, the value of current in the circuit, I is $\boxed{0\ \text{mA}}$.

Apply ohm's law to determine the value of output voltage.

$$I = \frac{-V - (-5)}{2.5 \times 10^3}$$

Substitute $0\ \text{mA}$ for I in the equation.

$$\begin{aligned} 0 &= \frac{-V + 5}{2.5 \times 10^3} \\ 0 &= -V + 5 \\ V &= 5\ \text{V} \end{aligned}$$

Since, $V = 5\ \text{V}$, diode is open. So, the assumption is correct.

Thus, the value of output voltage, V is $\boxed{5\ \text{V}}$.

Step 8 of 13

(d)

Refer to Figure E4.4 (d) in the textbook for the circuit.

Assume the diode is conducting. The equivalent circuit is shown in Figure 5.

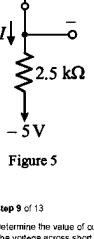


Figure 5

Step 9 of 13

Determine the value of output voltage.

The voltage across short circuit is zero.

$V = 0\ \text{V}$

Thus, the value of output voltage, V is $\boxed{0\ \text{V}}$.

Apply ohm's law to find the current in the circuit.

$$I = \frac{-V - (-5)}{2.5 \times 10^3}$$

Substitute $0\ \text{V}$ for V in the equation.

$$\begin{aligned} I &= \frac{0 + 5}{2.5 \times 10^3} \\ &= \frac{5}{2.5 \times 10^3} \\ &= 2 \times 10^{-3} \\ &= 2\ \text{mA} \end{aligned}$$

Since, the current $I > 0$, the diode is conducting. So, the assumption is correct.

Thus, the value of current in the circuit, I is $\boxed{2\ \text{mA}}$.

Step 10 of 13

(e)

Refer to Figure E4.4 (e) in the textbook for the circuit.

From the circuit, the lower two diodes $(+1\ \text{V and } +2\ \text{V})$ are cut off, whereas the upper $(+3\ \text{V})$ diode is in conducting mode. The equivalent circuit is shown in Figure 6.

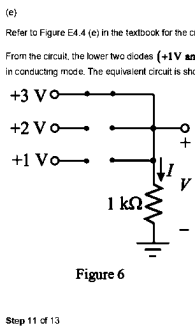


Figure 6

Step 11 of 13

Determine the value of output voltage.

The voltage at the output terminal is same as the voltage at the receiving end of the conducting diode.

$V = +3\ \text{V}$

Thus, the value of output voltage, V is $\boxed{3\ \text{V}}$.

Calculate the value of the current flows through the $1\ \text{k}\Omega$ resistor.

$$I = \frac{V}{1 \times 10^3}$$

Substitute $3\ \text{V}$ for V in the equation.

$$\begin{aligned} I &= \frac{3}{1 \times 10^3} \\ &= 3 \times 10^{-3} \\ &= 3\ \text{mA} \end{aligned}$$

Thus, the value of the current flows through the $1\ \text{k}\Omega$ resistor, I is $\boxed{3\ \text{mA}}$.

Step 12 of 13

(f)

Refer to Figure E4.4 (f) in the textbook for the circuit.

The maximum input current occurs only when one input is low and the other two are high. From the circuit, the upper two diodes $(+2\ \text{V and } +3\ \text{V})$ are cut off, whereas the lower $(+1\ \text{V})$ diode is in the conducting mode.

The equivalent circuit is shown in Figure 7.

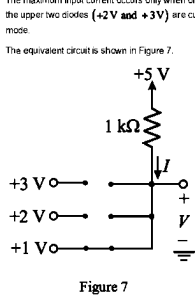


Figure 7

Step 13 of 13

Determine the value of output voltage.

The voltage at the output terminal is same as the voltage at the receiving end of the conducting diode.

$V = +1\ \text{V}$

Thus, the value of output voltage, V is $\boxed{1\ \text{V}}$.

Calculate the value of the current flows through the $1\ \text{k}\Omega$ resistor.

$$I = \frac{5 - V}{1 \times 10^3}$$

Substitute $1\ \text{V}$ for V in the equation.

$$\begin{aligned} I &= \frac{5 - 1}{1 \times 10^3} \\ &= \frac{4}{1 \times 10^3} \\ &= 4 \times 10^{-3} \\ &= 4\ \text{mA} \end{aligned}$$

Thus, the value of the current flows through the $1\ \text{k}\Omega$ resistor, I is $\boxed{4\ \text{mA}}$.

4.004P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

Step 1 of 3

4.005E

Refer to Figure E4.5 for an ac voltmeter in the text book.

Assume the diode is ideal. During the positive half cycle of the input, the diode is in forward bias condition. The positive v_I causes the current flows through the diode in the forward direction. In forward bias condition, the ideal diode acts as a short circuit. Thus, the output voltage is same as the input voltage.

The equivalent circuit is shown in Figure 1.

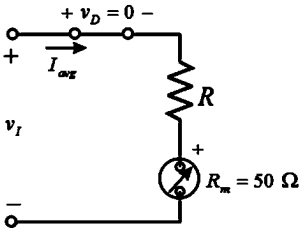


Figure 1

Step 2 of 3

Determine the average value of the input voltage. $v_{I(avg)} = \frac{v_{I(peak)}}{\pi}$

Substitute $10\ \text{V}$ for $v_{I(peak)}$ in the equation.

$$v_{I(avg)} = \frac{10}{\pi}$$
$$v_{I(avg)} = 3.183\ \text{V}$$

Step 3 of 3

Apply Ohm's law to the circuit and determine the value of resistor.

$$I_{avg} = \frac{v_{I(avg)}}{R + R_m}$$

Substitute $1\ \text{mA}$ for I_{avg}, $50\ \Omega$ for R_m and $3.183\ \text{V}$ for $v_{I(avg)}$ in the equation.

$$1\times10^{-3} = \frac{3.183}{R+50}$$
$$R+50 = \frac{3.183}{1\times10^{-3}}$$
$$R = 3183 - 50$$
$$= 3133$$
$$= 3.133\times10^{3}$$
$$= 3.133\ \text{k}\Omega$$

Thus, the value of resistor in the circuit, R is $\boxed{3.133\ \text{k}\Omega}$.

4.005P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

The Difference voltage is,

$$V_2 - V_1 = V_T \ln\left(\frac{I_2}{I_1}\right)$$

Here, $V_T = 25.8\,\text{mV}$

$I_1 = 0.1\,\text{mA}$ is the current flowing at voltage V_1 .

$I_2 = 10\,\text{mA}$ is current flowing at voltage V_2 .

Substitute $25\,\text{mV}$ for V_T , $0.1\,\text{mA}$ for I_1 , and $10\,\text{mA}$ for I_2 in the equation.

$$\begin{aligned} V_2 - V_1 &= \left(25.8\times10^{-3}\right)\ln\left(\frac{10\times10^{-3}}{0.1\times10^{-3}}\right) \\ &= \left(25.8\times10^{-3}\right)(4.605) \\ &= 118.8\times10^{-3} \\ &\approx 119\ \text{mV} \end{aligned}$$

Therefore, the change in diode voltage, when the change in current is from $0.1\,\text{mA}$ to $10\,\text{mA}$ is,

$\boxed{V_2 - V_1 = 119\ \text{mV}}$.

Step 1 of 3

4.006P

Refer to Figure P4.6 (a) in the text book.

The connection is such that if any one of the input $(A \text{ or } B)$ is low then the output (X) is low.

The output (X) is high only if both the inputs A and B are high.

Refer to Figure P4.6 (b) in the text book.

The connection is such that if any one of the input $(A \text{ or } B)$ is high, then the output (Y) is high.

The output (Y) is low only if both the inputs A and B are low.

Step 2 of 3

Write the table for possible input and output combinations.

A	B	X	Y
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

The logic function of X is same as that of an **AND** gate.

The logic function of Y is same as that of an **OR** gate.

Step 3 of 3

For $A = 0$ and $B = 0$ (or) $A = 1$ and $B = 1$, the X and Y have the same values.

For $A = 0, B = 1$ and $A = 1, B = 0$, the X and Y have the opposite values.

Step 1 of 3

4.007E

The $i-v$ relation of the diode is,

$$i = I_s e^{\frac{v}{V_T}}$$

Where, V_T is thermal voltage, $V_T = 25\,\text{mV}$

And I_s is reverse saturation current

Substitute $0.7\ \text{V}$ for v, $1\,\text{mA}$ for i, and $25\,\text{mV}$ for V_T in the equation.

$$i = I_s e^{\frac{v}{V_T}}$$

$$1\times10^{-3} = I_s e^{\frac{0.7}{25\times10^{-3}}}$$

$$I_S = \frac{10^{-3}}{e^{28}}$$

$$= \frac{10^{-3}}{1.446\times10^{12}}$$

$$= 6.91\times10^{-16}\,\text{A}$$

Step 2 of 3

Find the diode voltage drop at $i = 0.1\,\text{mA}$.

$$v = V_T \ln\left(\frac{i}{I_s}\right)$$

$$= \left(25\times10^{-3}\right)\ln\left(\frac{0.1\times10^{-3}}{6.91\times10^{-16}}\right)$$

$$= \left(25\times10^{-3}\right)\ln\left(14.47\times10^{10}\right)$$

$$= 642.45\times10^{-3}$$

$$= 0.6425\,\text{V}$$

Therefore, the diode voltage drop at $i = 0.1\,\text{mA}$ is, $\boxed{v = 0.6425\,\text{V}}$

Step 3 of 3

Find the diode voltage drop at $i = 10\,\text{mA}$.

$$v = V_T \ln\left(\frac{i}{I_s}\right)$$

$$= \left(25\times10^{-3}\right)\ln\left(\frac{10\times10^{-3}}{6.91\times10^{-16}}\right)$$

$$= \left(25\times10^{-3}\right)\ln\left(1.447\times10^{13}\right)$$

$$= 757.58\times10^{-3}$$

$$= 0.7576\,\text{V}$$

Therefore, the diode voltage drop at $i = 10\,\text{mA}$ is, $\boxed{v = 0.7576\,\text{V}}$

Write the expression of the current passing through the resistor R .

$$I = \frac{V_Y - 0}{R}$$

Substitute $5\ \text{V}$ for V_Y .

$$I = \frac{5-0}{R}$$
$$= \frac{5}{R}$$

The current should not exceed the value 0.2 mA.

$$I \le 0.2\ \text{mA}$$
$$\frac{5}{R} \le 0.2\ \text{mA}$$
$$\frac{5}{0.2\times10^{-3}} \le R$$
$$25\ \text{k}\Omega \le R$$

Thus, the value of R is $\boxed{25\ \text{k}\Omega}$.

4.008E

The silicon diode has saturation current of $I_s = 10^{-14}\ \text{A}$ at the temperature $25°\text{C}$.

And this current increased by 15% per $°\text{C}$ rise in temperature.

The increase in temperature is,

$$\begin{aligned} T &= 125°\text{C} - 25°\text{C} \\ &= 100°\text{C} \end{aligned}$$

The saturation current at $125°\text{C}$ is,

$$\begin{aligned} I_{S(125°\text{C})} &= I_{S(25°\text{C})} \times \left(1 + \frac{15}{100}\right)^T \\ &= \left(10^{-14}\ \text{A}\right)(1.15)^{100} \\ &= 1174313.451 \times 10^{-14} \\ &= 1.174 \times 10^{-8} \end{aligned}$$

Therefore, the saturation current at $125°\text{C}$ is,

$$\boxed{I_s = 1.174 \times 10^{-8}\ \text{A}}.$$

4.008P

Refer to Figure 4.5 (b) in the text book.

The current from the input signal is,

$$I = \frac{V_Y - V_A}{R}$$

The current does not exceed 0.2 mA.

$$\frac{V_Y - V_A}{R} < 0.2 \text{ mA}$$

Substitute 5 V for V_Y and 0 V for V_A.

$$\frac{5 - 0}{R} \leq 0.2 \text{ mA}$$

$$\frac{5}{R} \leq 0.2 \text{ mA}$$

$$R \geq \frac{5}{0.2}$$

$$R \geq 25 \text{ k}\Omega$$

Thus, the suitable value for the resistance R so that the current required from each of the input signal does not exceed 0.2 mA is $\boxed{25 \text{ k}\Omega}$.

Step 1 of 3

4.009E

Refer to Figure E4.9 in the textbook for the circuit.

The diode in the circuit is a large high current device whose reverse leakage is reasonably independent of voltage.

At $T = 20°C$, the voltage $V = 1$.

At $V = 1$, the diode is reverse biased. So the current flowing in the circuit is reverse saturation current I_s.

$$i = I_s$$

Calculate the current in the circuit.

$$i = \frac{V}{R}$$

Substitute I_s for i , $1\ \text{V}$ for V , and $1\ \text{M}\Omega$ for R in the equation.

$$\begin{aligned} I_{s(20°C)} &= \frac{1}{1\times10^{6}} \\ &= 1\ \mu\text{A} \end{aligned}$$

Thus, the reverse current at $20°C$ is $1\ \mu\text{A}$.

Step 2 of 3

Since the reverse current doubles for every $10°C$ rise in temperature, the reverse current at $30°C$ is,

$$\begin{aligned} I_{s(30°C)} &= 2I_{s(20°C)} \\ &= 2\left(1\times10^{-6}\right) \\ &= 2\ \mu\text{A} \end{aligned}$$

Similarly, the reverse current at $40°C$ is,

$$\begin{aligned} I_{s(40°C)} &= 2I_{s(30°C)} \\ &= 2\left(2\times10^{-6}\right) \\ &= 4\ \mu\text{A} \end{aligned}$$

The voltage V at $40°C$ is,

$$\begin{aligned} V &= iR \\ &= I_{s(20°C)}R \\ &= \left(4\times10^{-6}\right)\left(1\times10^{6}\right) \\ &= 4\ \text{V} \end{aligned}$$

Therefore, the voltage V at $40°C$ is $\boxed{4\ \text{V}}$.

Step 3 of 3

Since the reverse current doubles for every $10°C$ rise in temperature, the reverse current at $10°C$ is,

$$I_{s(20°C)} = 2I_{s(10°C)}$$

$$\begin{aligned} I_{s(10°C)} &= \frac{I_{s(20°C)}}{2} \\ &= \frac{1\times10^{-6}}{2} \\ &= 0.5\ \mu\text{A} \end{aligned}$$

Similarly, the reverse current at $0°C$ is,

$$I_{s(10°C)} = 2I_{s(0°C)}$$

$$\begin{aligned} I_{s(0°C)} &= \frac{I_{s(10°C)}}{2} \\ &= \frac{0.5\times10^{-6}}{2} \\ &= 0.25\ \mu\text{A} \end{aligned}$$

The voltage V at $0°C$ is,

$$\begin{aligned} V &= iR \\ &= I_{s(0°C)}R \\ &= \left(0.25\times10^{-6}\right)\left(1\times10^{6}\right) \\ &= 0.25\ \text{V} \end{aligned}$$

Therefore, the voltage V at $0°C$ is $\boxed{0.25\ \text{V}}$.

4.009P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

Step 1 of 4

4.010E

Refer to Figure 4.10 in the textbook.

Redraw the circuit by representing the value of resistor and voltage source as shown in Figure 1.

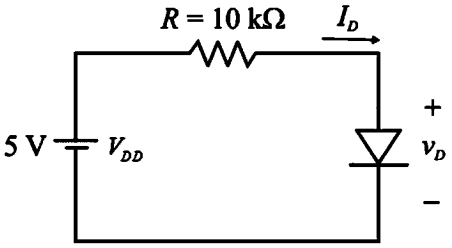


Figure 1

Step 2 of 4

The diode has a voltage of $0.7\,\text{V}$ at $I_D = 1\,\text{mA}$.

Apply Kirchhoff's voltage law to the circuit.

$$-5 + I_D\left(10\times10^3\right) + v_D = 0$$

$$\begin{aligned} I_D &= \frac{5 - v_D}{10\times10^3} \\ &= \frac{5 - 0.7}{10\times10^3} \\ &= 0.43\times10^{-3} \\ &= 0.43\,\text{mA} \end{aligned}$$

Therefore, the diode current is, $\boxed{I = 0.43\,\text{mA}}$.

Step 3 of 4

The thermal voltage equivalent is, $V_T = 25.9\,\text{mA}$.

Consider V_2 is the voltage at $I_2 = 0.43\,\text{mA}$ and $V_1 = 0.7\,\text{V}$ at $I_1 = 1\,\text{mA}$.

Equation for difference voltage is,

$$V_2 - V_1 = 2.3V_T \log\left(\frac{I_2}{I_1}\right)$$

$$V_2 - 0.7 = 2.3\left(25\times10^{-3}\right)\log\left(\frac{0.43\times10^{-3}}{1\times10^{-3}}\right)$$

$$V_2 - 0.7 = 2.3\left(25\times10^{-3}\right)\left(-0.3665\right)$$

$$V_2 - 0.7 = -0.021$$

$$\begin{aligned} V_2 &= 0.679 \\ &\approx 0.68\ \text{V} \end{aligned}$$

Therefore, the voltage across the diode is, $\boxed{V_2 = 0.68\,\text{V}}$.

Step 4 of 4

(b)

The relation for the current I_D is,

$$I_D = \frac{V_{DD} - V_D}{R}$$

Where, V_{DD} is the dc source voltage,

V_D is the diode voltage, and

R is the resistance connected across the diode, $R = 10\,\text{k}\Omega$.

Substitute $5\,\text{V}$ for V_{DD}, $0.7\,\text{V}$ for V_D, and $10\,\text{k}\Omega$ for R in the equation.

$$\begin{aligned} I_D &= \frac{V_{DD} - V_D}{R} \\ &= \frac{5 - 0.7}{10\times10^3} \\ &= 0.43\ \text{mA} \end{aligned}$$

Therefore, the diode current and voltage are, $\boxed{I_D = 0.43\,\text{mA}}$ and $\boxed{V_D = 0.7\text{V}}$.

4.010P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

Step 1 of 3

4.011E

Refer to Figure E4.11 in the textbook.

Calculate the corresponding voltage drop across each of the diodes.

$$V_{D2} = \frac{V_O}{3}$$

Substitute $2.4\ \text{V}$ for V_O in the equation.

$$V_{D2} = \frac{2.4\ \text{V}}{3}$$
$$= 0.8\ \text{V}$$

Step 2 of 3

Consider the exponential diode current and voltage relation.

$$V_2 - V_1 = 2.3V_T \log\left(\frac{I_2}{I_1}\right)$$

here,

V_2, the actual voltage drop across the diode

V_1, the assumed or initial voltage across the diode

I_2, the resulting current in the diode

I_1, the assumed current in the diode

Substitute $57.5\ \text{mV}$ for $2.3V_T$, $1\ \text{mA}$ for I_1, $0.7\ \text{V}$ for V_1, $0.8\ \text{V}$ for V_2 in the equation.

$$0.8 - 0.7 = (57.5\ \text{mV})\left(\log\left(\frac{I_2}{1\ \text{mA}}\right)\right)$$
$$0.1 = \left(57.5\times10^{-3}\right)\log\left(\frac{I_2}{1\times10^{-3}}\right)$$
$$I_2 = \left(10^{-3}\right)\left(10^{\frac{100}{57.5}}\right)\text{A}$$
$$I_2 = 54.84\ \text{mA}$$

Step 3 of 3

Apply Ohm's law to calculate the value of resistance to obtain the current I_2 at an output voltage of $2.4\ \text{V}$.

$$R = \frac{10 - V_O}{I_2}$$
$$= \frac{10 - 2.4}{54.84\ \text{mA}}$$
$$= \frac{7.6}{54.84\ \text{mA}}$$
$$= 139\ \Omega$$

Thus, the value of resistance to obtain the output voltage of $2.4\ \text{V}$ is $\boxed{139\ \Omega}$.

4.011P

We don't have the solution to this problem yet.

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Step 1 of 21

The ideal diode circuit is shown in Figure 1.

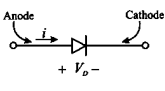


Figure 1

Step 2 of 21

In the forward direction, the ideal diode conducts any current forced by the external circuit while displaying a zero voltage drop. The ideal diode does not conduct in the reverse direction; any applied voltage appears as reverse bias across the diode.

Step 3 of 21

(a)

Refer to Figure E4.4 (a) in the textbook for the circuit.

Redraw the circuit by representing diode voltage in the circuit.

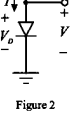


Figure 2

Step 4 of 21

Assume diode is conducting, and then the equivalent circuit is shown in Figure 3.

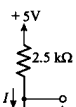


Figure 3

Step 5 of 21

From the circuit, the output voltage is,

$V = V_D$

$= 0.7$

Therefore, the voltage V is $\boxed{0.7\text{ V}}$

Step 6 of 21

Use ohm's law to find the current in the circuit.

$$I = \frac{5 - V}{2.5\times10^3}$$

$$= \frac{5 - 0.7}{2.5\times10^3} \qquad (\text{Since, } V = 0.7\text{ V})$$

$$= \frac{4.3}{2.5\times10^3}$$

$$= 1.72\times10^{-3}$$

Since, here, $I > 0$, so diode is conducting. So, the assumption is correct.

Therefore, the current, I is, $\boxed{1.72\text{ mA}}$

Step 7 of 21

(b)

Refer to Figure E4.4 (b) in the textbook for the circuit.

Redraw the circuit by representing diode voltage in the circuit.

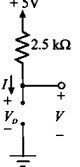


Figure 4

Assume diode is operated in reverse biased then the diode is open (cut off), the equivalent circuit is shown in Figure 5.

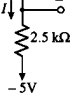


Figure 5

Step 8 of 21

From the circuit, the current is,

$I = 0\text{ A}$

Therefore, the current I is $\boxed{0\text{ A}}$

Apply ohm's law to the circuit

$$I = \frac{5 - V}{2.5\times10^3}$$

$$0 = \frac{5 - V}{2.5\times10^3} \qquad (\text{Since, } I = 0\text{ A})$$

$$0 = \frac{5 - V}{2.5\times10^3}$$

$$V = 5$$

Since, $V = 5\text{ V}$, diode is open. So, the assumption is correct.

Therefore, the voltage V is $\boxed{5\text{ V}}$.

Step 9 of 21

(c)

Refer to Figure E4.4 (c) in the textbook for the circuit.

Redraw the circuit by representing diode voltage in the circuit.

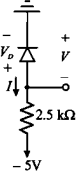


Figure 6

Step 10 of 21

Assume diode is operated in reverse biased then the diode is open (cut off). The equivalent circuit is shown in Figure 7.

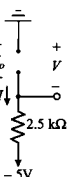


Figure 7

Step 11 of 21

From the circuit, the current is,

$I = 0\text{ A}$

Therefore, the current I is $\boxed{0\text{ A}}$.

Apply ohm's law to the circuit.

$$I = \frac{-V - (-5)}{2.5\times10^3}$$

$$0 = \frac{-V + 5}{2.5\times10^3} \qquad (\text{Since, } I = 0\text{ A})$$

$$0 = -V + 5$$

$$V = 5$$

Since, $V = 5\text{ V}$, diode is open. So, the assumption is correct.

Therefore, the voltage V is $\boxed{5\text{ V}}$

Step 12 of 21

(d)

Refer to Figure E4.4 (d) in the textbook for the circuit.

Redraw the circuit by representing diode voltage in the circuit.

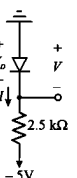


Figure 8

Step 13 of 21

Assume diode is conducting. The equivalent circuit is shown in Figure 9.

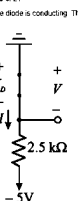


Figure 9

Step 14 of 21

From the circuit, the output voltage is,

$V = V_D$

$= 0.7$

Therefore, the voltage V is $\boxed{0.7\text{ V}}$.

Step 15 of 21

Use ohm's law to find the current in the circuit.

$$I = \frac{-V - (-5)}{2.5\times10^3}$$

$$= \frac{-0.7 + 5}{2.5\times10^3} \qquad (\text{Since, } V = 0.7\text{ V})$$

$$= \frac{4.3}{2.5\times10^3}$$

$$= 1.72\times10^{-3}$$

Therefore, the current I is $\boxed{1.72\text{ mA}}$

Step 16 of 21

(e)

Refer to Figure E4.4 (e) in the textbook for the circuit.

Redraw the circuit by representing diode voltage in the circuit.

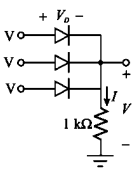


Figure 10

From the circuit, the lower two diodes $(+1\text{V and } +2\text{V})$ are cut off, whereas the upper $(+3\text{V})$ diode is in conducting mode. The equivalent circuit is shown in Figure 11

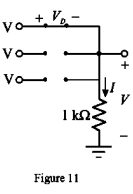


Figure 11

Step 17 of 21

Apply Kirchhoff's voltage law to the above circuit.

$$V = 3 - V_D$$

$$= 3 - 0.7 \qquad (\text{Since, } V_D = 0.7\text{ V})$$

$$= 2.3$$

Therefore, the output voltage V is $\boxed{2.3\text{ V}}$.

The value of the current that flows through the $1\text{ k}\Omega$ resistance is,

$$I = \frac{V}{1\times10^3}$$

$$= \frac{2.3}{1\times10^3} \qquad (\text{Since, } V = +2.3\text{ V})$$

$$= 2.3\times10^{-3}$$

Therefore, the value of the current that flows through the resistance is, $\boxed{I = 2.3\text{ mA}}$

Step 18 of 21

(f)

Refer to Figure E4.4 (f) in the textbook for the circuit.

Redraw the circuit by representing diode voltage in the circuit.

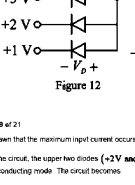


Figure 12

Step 19 of 21

It is known that the maximum input current occurs when one input is low and the other two are high.

From the circuit, the upper two diodes $(+2\text{V and } +3\text{V})$ are cut off, whereas the lower $(+1\text{V})$ diode is in the conducting mode. The circuit becomes

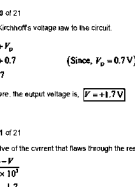


Figure 13

Step 20 of 21

Apply Kirchhoff's voltage law to the circuit.

$$V = 1 + V_D$$

$$= 1 + 0.7 \qquad (\text{Since, } V_D = 0.7\text{ V})$$

$$= 1.7$$

Therefore, the output voltage is, $\boxed{V = +1.7\text{ V}}$

Step 21 of 21

The value of the current that flows through the resistance $1\text{ k}\Omega$ is,

$$I = \frac{5 - V}{1\times10^3}$$

$$= \frac{5 - 1.7}{1\times10^3} \qquad (\text{Since, } V = +1.7\text{ V})$$

$$= \frac{3.3}{1\times10^3}$$

$$= 3.3\times10^{-3}$$

Therefore, the value of the current that flows through the resistance is, $\boxed{I = 3.3\text{ mA}}$

Step 1 of 4

Draw the rectifier circuit with a source resistance, R_s .

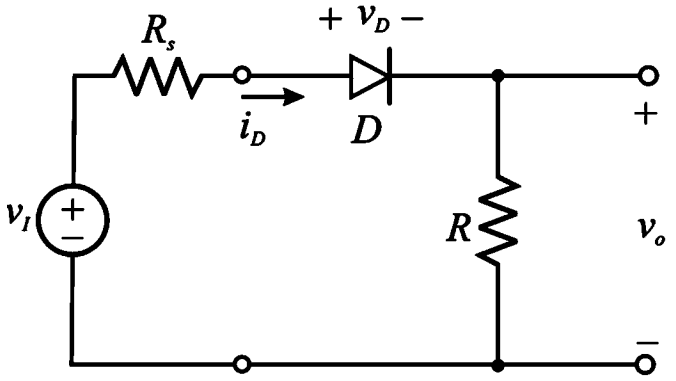


Figure 1

Step 2 of 4

During positive half cycle of the input, the diode is forward biased. Replace the diode with a short circuit, assuming it to be ideal. Draw the diode circuit.

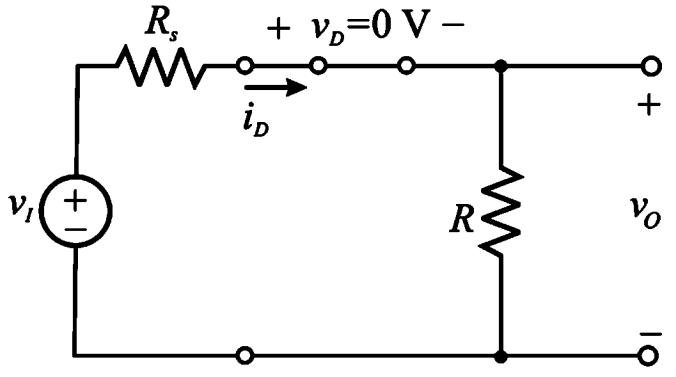


Figure 2

Step 3 of 4

Apply voltage division rule to find the output voltage, v_o .

$$v_O = \left(\frac{R}{R+R_s}\right)v_I$$

For $R_s = R$ the output voltage is,

$$v_O = \left(\frac{R}{R+R}\right)v_I$$
$$= \frac{v_I}{2}$$
$$= 0.5v_I$$

Thus, during positive half cycle of the input voltage, the output voltage is $0.5v_I$. It is represented by a straight line with a slope of 0.5 in the transfer characteristics.

During negative half cycle the diode is reverse biased and can be replaced with an open circuit, assuming it to be ideal. As the diode is open, no current flows through the output resistance. Hence, the output voltage is zero.

Step 4 of 4

Draw the transfer characteristic, v_o versus v_I .

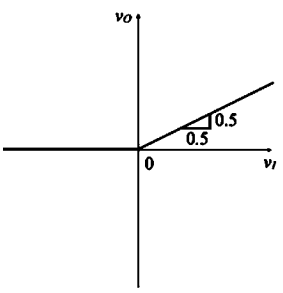


Figure 3

Thus, the transfer characteristic of the rectifier circuit is drawn.

Step 1 of 4

The relation of the diode small signal resistance, r_d with the bias current, I is,

$$r_d = \frac{V_T}{I}$$

Here, V_T is the thermal voltage.

$$V_T = \frac{kT}{q}$$

Here, Boltzmann's constant, $k = 8.62 \times 10^{-5}\ \text{eV/K}$,

Room temperature, $T = 300\ \text{K}$ and

Charge of an electron, $q = 1.6 \times 10^{-19}\text{C}$.

Substitute $8.62 \times 10^{-5}\ \text{eV/K}$ for k, $300\ \text{K}$ for T, and $1.6 \times 10^{-19}\text{C}$ for q in the equation and find the value of V_T.

$$\begin{aligned} V_T &= \frac{kT}{q} \\ &= \frac{8.62 \times 10^{-5} \times 300\ (\text{eV})}{q\ (\text{e})} \\ &= 25.9\ \text{mV} \end{aligned}$$

For convenience, the approximate value of V_T is taken as 25 mV.

Step 2 of 4

The small-signal resistance r_d at bias current of $0.1\ \text{mA}$ is,

$$r_d = \frac{V_T}{I}$$

Substitute $25\ \text{mV}$ for V_T and $0.1\ \text{mA}$ for I_D in the equation.

$$\begin{aligned} r_d &= \frac{25 \times 10^{-3}}{0.1 \times 10^{-3}} \\ &= 250\ \Omega \end{aligned}$$

Therefore, the value of diode small signal resistance r_d at bias current of $0.1\ \text{mA}$ is $\boxed{250\ \Omega}$.

Step 3 of 4

The small-signal resistance r_d at bias current of $1\ \text{mA}$ is,

$$r_d = \frac{V_T}{I}$$

Substitute $25\ \text{mV}$ for V_T and $1\ \text{mA}$ for I_D in the equation.

$$\begin{aligned} r_d &= \frac{25 \times 10^{-3}}{1 \times 10^{-3}} \\ &= 25\ \Omega \end{aligned}$$

Therefore, the value of diode small signal resistance r_d at bias current of $1\ \text{mA}$ is $\boxed{25\ \Omega}$

Step 4 of 4

The small-signal resistance r_d at bias current of $10\ \text{mA}$ is,

$$r_d = \frac{V_T}{I}$$

Substitute $25\ \text{mV}$ for V_T and $10\ \text{mA}$ for I_D in the equation.

$$\begin{aligned} r_d &= \frac{25 \times 10^{-3}}{10 \times 10^{-3}} \\ &= 2.5\ \Omega \end{aligned}$$

Therefore, the value of diode small signal resistance r_d at bias current of $10\ \text{mA}$ is $\boxed{2.5\ \Omega}$.

4.013P

We don't have the solution to this problem yet.

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Step 1 of 13

(a) (i) We know that

$$r_d = \frac{\Delta v_d}{\Delta i_d}$$

$$= \frac{V_T}{I_D}$$

The value of r_d is

$$r_d = \frac{V_T}{I_D}$$

$$= \frac{0.025}{1 \times 10^{-3}}$$

$$= 25\ \Omega$$

Step 2 of 13

Find Δi_d as

$$\Delta i_d = \frac{\Delta v_d}{r_d}$$

$$= \frac{-10 \times 10^{-3}}{25}$$

$$= -0.4 \times 10^{-3}$$

Thus, using the small signal model, we get

$$\boxed{\Delta i_d = -0.4\ \text{mA}}$$

Step 3 of 13

(ii) We know that

$$\Delta v_d = 2.3V_T \log\left(\frac{I_2}{I_1}\right)$$

$$\frac{\Delta v_d}{2.3V_T} = \log\left(\frac{I_2}{I_1}\right)$$

$$\frac{I_2}{I_1} = 10^{\left(\frac{\Delta v_d}{2.3V_T}\right)}$$

$$\frac{I_1}{I_2} = -\frac{1}{2}10^{\left(\frac{-10 \times 10^{-3}}{2.3 \times 25 \times 10^{-3}}\right)}$$

$$\Delta I = -\frac{1}{2}10^{(-0.174)}$$

$$= -\frac{1}{2}(0.67)$$

$$= -0.335$$

Thus, using the exponential method, we get

$$\boxed{\Delta I = -0.335\ \text{A}}$$

Step 4 of 13

(b) (i) We know that

$$r_d = \frac{\Delta v_d}{\Delta i_d}$$

$$= \frac{V_T}{I_D}$$

The value of r_d is

$$r_d = \frac{V_T}{I_D}$$

$$= \frac{0.025}{1 \times 10^{-3}}$$

$$= 25\ \Omega$$

Step 5 of 13

Find Δi_d as

$$\Delta i_d = \frac{\Delta v_d}{r_d}$$

$$= \frac{-5 \times 10^{-3}}{25}$$

$$= -0.2\ \text{mA}$$

Thus, using the small signal model, we get

$$\boxed{\Delta i_d = -0.2\ \text{mA}}$$

Step 6 of 13

(ii) We know that

$$\Delta v_d = 2.3V_T \log\left(\frac{I_2}{I_1}\right)$$

$$\frac{\Delta v_d}{2.3V_T} = \log\left(\frac{I_2}{I_1}\right)$$

$$\frac{I_2}{I_1} = 10^{\left(\frac{\Delta v_d}{2.3V_T}\right)}$$

$$= -\frac{1}{2}10^{\left(\frac{-5 \times 10^{-3}}{2.3 \times 25 \times 10^{-3}}\right)}$$

$$\Delta I = -\frac{1}{2}10^{(-0.0869)}$$

$$= -\frac{1}{2}(0.8185)$$

$$= -0.41$$

Thus, using the exponential method, we get

$$\boxed{\Delta I = -0.41\ \text{A}}$$

Step 7 of 13

(c) (i) We know that

$$r_d = \frac{\Delta v_d}{\Delta i_d}$$

$$= \frac{V_T}{I_D}$$

The value of r_d is

$$r_d = \frac{V_T}{I_D}$$

$$= \frac{0.025}{1 \times 10^{-3}}$$

$$= 25\ \Omega$$

Step 8 of 13

Find Δi_d as

$$\Delta i_d = \frac{\Delta v_d}{r_d}$$

$$= \frac{10 \times 10^{-3}}{25}$$

$$= 0.4 \times 10^{-3}$$

Thus, using the small signal model, we get

$$\boxed{\Delta i_d = 0.4\ \text{mA}}$$

Step 9 of 13

(ii) We know that

$$\Delta v_d = 2.3V_T \log\left(\frac{I_2}{I_1}\right)$$

$$\frac{\Delta v_d}{2.3V_T} = \log\left(\frac{I_2}{I_1}\right)$$

$$\frac{I_2}{I_1} = 10^{\left(\frac{\Delta v_d}{2.3V_T}\right)}$$

$$\frac{I_1}{I_2} = -\frac{1}{2}10^{\left(\frac{10 \times 10^{-3}}{2.3 \times 25 \times 10^{-3}}\right)}$$

$$\Delta I = -\frac{1}{2}10^{(0.174)}$$

$$= -\frac{1}{2}(1.4925)$$

$$= -0.746$$

Thus, using the exponential method, we get

$$\boxed{\Delta I = -0.746\ \text{A}}$$

Step 10 of 13

(ii) We know that

$$\Delta v_d = 2.3V_T \log\left(\frac{I_2}{I_1}\right)$$

$$\frac{\Delta v_d}{2.3V_T} = \log\left(\frac{I_2}{I_1}\right)$$

$$\frac{I_2}{I_1} = 10^{\left(\frac{\Delta v_d}{2.3V_T}\right)}$$

$$= -\frac{1}{2}10^{\left(\frac{5 \times 10^{-3}}{2.3 \times 25 \times 10^{-3}}\right)}$$

$$\Delta I = -\frac{1}{2}10^{(0.0869)}$$

$$= -\frac{1}{2}(1.22)$$

$$= -0.611$$

Thus, using the exponential method, we get

$$\boxed{\Delta I = -0.611\ \text{A}}$$

Step 11 of 13

(d) (i) We know that

$$r_d = \frac{\Delta v_d}{\Delta i_d}$$

$$= \frac{V_T}{I_D}$$

The value of r_d is

$$r_d = \frac{V_T}{I_D}$$

$$= \frac{0.025}{1 \times 10^{-3}}$$

$$= 25\ \Omega$$

Step 12 of 13

Find Δi_d as

$$\Delta i_d = \frac{\Delta v_d}{r_d}$$

$$= \frac{10 \times 10^{-3}}{25}$$

$$= 0.4 \times 10^{-3}$$

Thus, using the small signal model, we get

$$\boxed{\Delta i_d = 0.4\ \text{mA}}$$

Step 13 of 13

(ii) We know that

$$\Delta v_d = 2.3V_T \log\left(\frac{I_2}{I_1}\right)$$

$$\frac{\Delta v_d}{2.3V_T} = \log\left(\frac{I_2}{I_1}\right)$$

$$\frac{I_2}{I_1} = 10^{\left(\frac{\Delta v_d}{2.3V_T}\right)}$$

$$\frac{I_1}{I_2} = -\frac{1}{2}10^{\left(\frac{10 \times 10^{-3}}{2.3 \times 25 \times 10^{-3}}\right)}$$

$$\Delta I = -\frac{1}{2}10^{(0.174)}$$

$$= -\frac{1}{2}(1.4925)$$

$$= -0.746$$

Thus, using the exponential method, we get

$$\boxed{\Delta I = -0.746\ \text{A}}$$

4.014P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

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Step 1 of 7

4.015E

(a)

Refer to Figure E4.15 in the text book for the diode circuit to be designed. It consists of a string of four diodes.

Draw the small signal equivalent circuit.

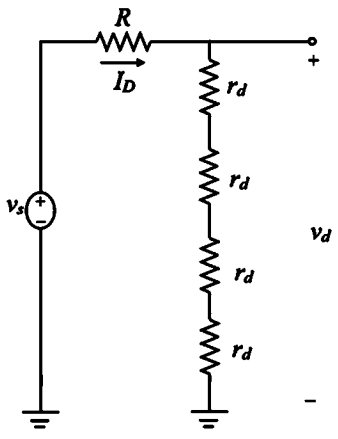


Figure 1

Step 2 of 7

The diode is replaced by a resistance, r_d .

As the load current is zero, the change in output voltage is equal to the change in diode voltage, ∂v_D and the change in load current is equal to change in diode current, ∂i_D .

Determine the total diode resistance (resistance of the combination of four diodes in series), $4r_d$, which is equal to the change in output voltage to the change in current.

$$4r_d = \frac{\partial v_D}{\partial i_D}$$

Substitute 20 mV for ∂v_D and 1 mA for ∂i_D .

$$4r_d = \frac{\partial v_D}{\partial i_D}$$
$$4r_d = \frac{20}{1}$$
$$r_d = \frac{20}{4}$$
$$= 5\ \Omega$$

Step 3 of 7

Determine the diode current, I_D .

$$I_D = \frac{V_T}{r_d}$$

Here,

Thermal voltage V_T is 25 mV.

Substitute 25 mV for V_T and $5\ \Omega$ for r_d .

$$I_D = \frac{25\times10^{-3}}{5}$$
$$= 5\ \text{mA}$$

The string of four diodes should provide an output voltage of $V_O = 3\ \text{V}$ at no load.

Determine the nominal value of the current in the diode string, I_D .

$$I_D = \frac{15-3}{R}$$
$$= \frac{12}{R}$$

Substitute $5\ \text{mA}$ for I_D in the equation.

$$5\times10^{-3} = \frac{12}{R}$$
$$R = \frac{12}{5\times10^{-3}}$$
$$= 2.4\ \text{k}\Omega$$

Thus, the value of resistance, *R* is $\boxed{2.4\ \text{k}\Omega}$.

Step 4 of 7

(b)

Write the formula for the saturation current, I_S .

$$I_S = i_D e^{-\frac{v}{V_T}}$$

For output voltage, $V_O = 3\ \text{V}$ voltage drop across each diode is,

$$\frac{3}{4} = 0.75\ \text{V}$$

Step 5 of 7

Substitute 5 mA for i_D , 0.75 V for v , and 25 mV for V_T in the formula of saturation current.

$$I_S = 5\times10^{-3} e^{-\frac{0.75}{25\times10^{-3}}}$$
$$= 4.7\times10^{-16}\ \text{A}$$

Thus, the saturation current of the diodes is $\boxed{4.7\times10^{-16}\ \text{A}}$.

Step 6 of 7

(c)

Determine the diode current with load.

$$I_D = I_{D(\text{no load})} - I_L$$

Substitute $5\ \text{mA}$ for $I_{D(\text{no load})}$ and $1\ \text{mA}$ for I_L in the equation.

$$I_D = 5\times10^{-3} - 1\times10^{-3}$$
$$= 4\ \text{mA}$$

Write the formula for voltage across each diode.

$$V_D = V_T \ln\left(\frac{I_D}{I_S}\right)$$

Substitute $25\times10^{-3}\ \text{V}$ for V_T , $4\times10^{-3}\ \text{A}$ for I_D and $4.7\times10^{-16}\ \text{A}$ for I_S in the equation.

$$V_D = 25\times10^{-3} \ln\left(\frac{4\times10^{-3}}{4.7\times10^{-16}}\right)$$
$$= 0.74431\ \text{V}$$

Determine the voltage drop across four diodes which is equal to 4 times each diode voltage as the diodes are connected in series.

$$4V_D = 4\times0.74431$$
$$= 2.97723\ \text{V}$$

Thus, the output voltage with a load current of 1 mA is 2.99723 V.

Step 7 of 7

The output voltage at no-load is 3 V

Determine the change in output voltage from no-load to load.

$$\nabla V_O = 2.97723 - 3$$
$$= -0.02277\ \text{V}$$
$$= -22.77\ \text{mV}$$

Thus, the change in output voltage at a load current of 1 mA to no-load is $\boxed{-22.77\ \text{mV}}$.

4.015P

We don't have the solution to this problem yet.

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Step 1 of 3

4.016E

Consider a zener diode with nominal voltage, 10 V at zener current, (I_z) of 10 mA.

The diode has an incremental resistance of $r_z = 50\ \Omega$.

Consider the following expression for the change in zener voltage for the corresponding change in current:

$$\Delta V = r_z \Delta I$$

Here, r_z is the incremental resistance.

Consider that the diode current is halved (That is, decreased by 5 mA).

Substitute 5 mA for change in zener current, ΔI , $50\ \Omega$ for r_z .

$$\begin{aligned}\Delta V &= (50\ \Omega)(5\ \text{mA}) \\ &= 0.25\ \text{V}\end{aligned}$$

Refer to Figure 4.17 in the textbook for the diode $i-v$ characteristics. Observe that the relation between diode current and voltage is linear. If the diode current decreases, the diode voltage also decreases.

Substitute 10 V for nominal voltage, V_z , and 0.25 V for change in voltage, ΔV to find the expected voltage.

$$\begin{aligned}V &= V_z - \Delta V \\ &= 10 - 0.25 \\ &= 9.75\ \text{V}\end{aligned}$$

Therefore, the expected voltage, V is, $\boxed{9.75\ \text{V}}$.

Step 2 of 3

Consider that the diode current is doubled (That is, increased by 10 mA).

Substitute 10 mA for change in zener current, ΔI , $50\ \Omega$ for r_z .

$$\begin{aligned}\Delta V &= (50\ \Omega)(10\ \text{mA}) \\ &= 0.5\ \text{V}\end{aligned}$$

The diode voltage increases with increase in diode current due to linear behavior of diode.

Substitute 10 V for nominal voltage, V_z , and 0.5 V for change in voltage, ΔV to find the expected voltage.

$$\begin{aligned}V &= V_z + \Delta V \\ &= 10 + 0.5 \\ &= 10.5\ \text{V}\end{aligned}$$

Therefore, the expected voltage, V is, $\boxed{10.5\ \text{V}}$.

Step 3 of 3

Refer to Figure 4.18 in the textbook for the model of zener diode.

Consider the following expression for the zener diode:

$$V_z = V_{z0} + r_z I_z$$

Modify the expression for the knee voltage, V_{z0} .

$$V_{z0} = V_z - r_z I_z$$

Substitute $50\ \Omega$ for r_z , 10 mA for I_z , and 10 V for V_z .

$$\begin{aligned}V_{z0} &= 10 - (50\ \Omega)(10\ \text{mA}) \\ &= 10 - 0.5 \\ &= 9.5\ \text{V}\end{aligned}$$

Therefore, the value of knee voltage, V_{z0} is $\boxed{9.5\ \text{V}}$.

4.016P

Refer to Figure P4.16 in the text book for the circuit used in signaling system.

Assume ideal diodes.

For the input voltage equal to +3 V, the diode D_1 is forward biased and the diode D_2 is reverse biased. Replace D_1 with a short circuit and D_2 with an open circuit.

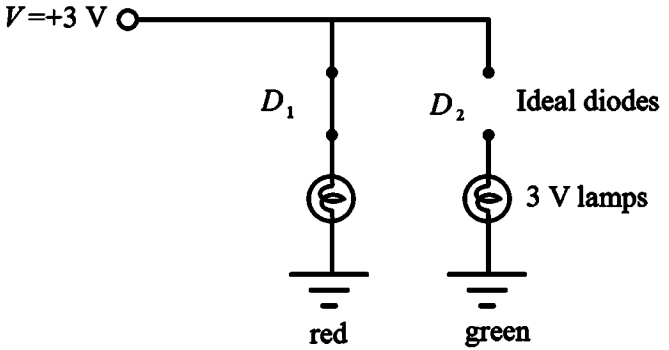


Figure 1

Thus, when $V = +3$ V the red lamp is ON and the green lamp is OFF.

For the input voltage equal to 0 V, both the diodes are reverse biased. Replace both the diodes, D_1 and D_2 with an open circuit.

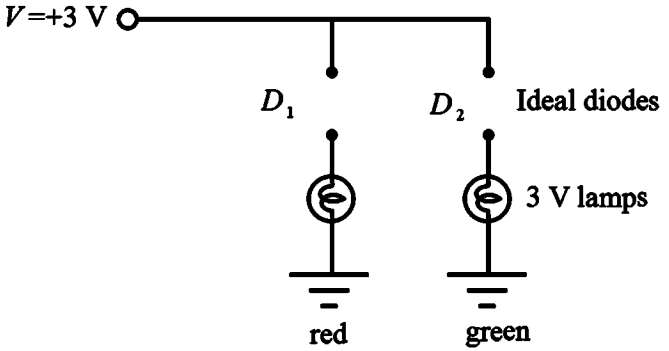


Figure 2

Thus, when $V = 0$ V the red lamp is OFF and the green lamp is OFF.

For the input voltage equal to ◊3 V, the diode D_2 is forward biased and the diode D_1 is reverse biased. Replace D_2 with a short circuit and D_1 with an open circuit.

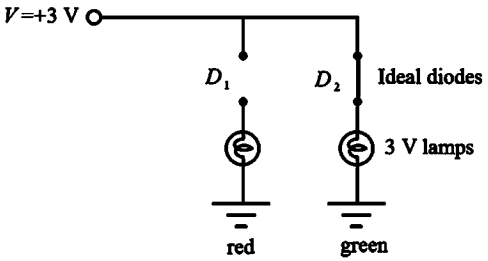


Figure 3

Thus, when $V = -3$ V the green lamp is ON and the red lamp is OFF.

Step 1 of 3

4.017E

Consider that a zener diode is used in the design of a shunt regulator.

Refer to Figure 4.19 in the textbook for the arrangement of zener diode in the design of shunt regulator.

The zener knee current, I_{ZK} is 1 mA.

Consider that the zener diode current is greater than five times the knee current.

$$\begin{aligned} I_Z &= 5I_{ZK} \\ &= 5(1\text{ mA}) \\ &= 5\text{ mA} \end{aligned}$$

Draw the zener diode circuit with the known values.

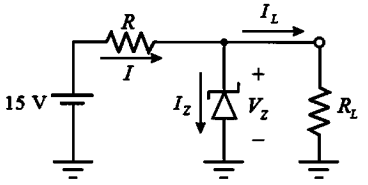


Figure 1

Step 2 of 3

Use Ohm's law to find the unknown resistor value.

$$\begin{aligned} R &= \frac{15 - V_Z}{I} \\ &= \frac{15 - V_Z}{I_{L\max} + I_Z} \end{aligned}$$

Consider that the zener voltage, V_Z , is 5.6 V, and the maximum value of load current, $I_{L\max}$ is 15 mA. Therefore,

$$\begin{aligned} R &= \frac{15 - 5.6}{15\text{ mA} + 5\text{ mA}} \\ &= \frac{9.4}{20 \times 10^{-3}} \\ &= 470\ \Omega \end{aligned}$$

Therefore, the unknown resistor value, R is, $\boxed{470\ \Omega}$.

Step 3 of 3

The maximum power dissipation of the zener diode is,

$$\begin{aligned} P_Z &= (V_Z)(I_{\max}) \\ &= (5.6\text{ V})(15\text{ mA} + 5\text{ mA}) \\ &= (5.6)(20 \times 10^{-3}) \\ &= 112\text{ mW} \end{aligned}$$

Thus, the maximum power dissipation of the zener diode, P_Z is, $\boxed{112\text{ mW}}$.

4.017P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

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Step 1 of 5

4.018E

Consider that a zener diode is used in the design of a shunt regulator.

Refer to Figure 4.19 in the textbook for the arrangement of zener diode in the design of shunt regulator.

Draw the zener diode circuit with the known values.

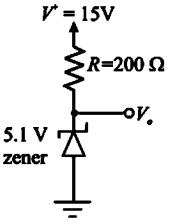


Figure 1

Step 2 of 5

Replace the zener diode with its equivalent circuit model.

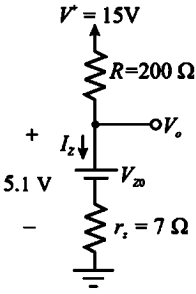


Figure 2

Step 3 of 5

Apply Kirchhoff's voltage law to the circuit in Figure 2.

$$-V_Z + V_{Z0} + I_Z r_Z = 0$$

Substitute 5.1 V for V_Z, 50 mA for I_Z, and $7\ \Omega$ for r_z to find the value of V_{Z0}.

$$\begin{aligned} -5.1\text{ V} + V_{Z0} + (50\text{ mA})(7\ \Omega) &= 0 \\ -5.1\text{ V} + V_{Z0} + 0.35\text{ V} &= 0 \\ -4.75\text{ V} + V_{Z0} &= 0 \\ V_{Z0} &= 4.75\text{ V} \end{aligned}$$

The output voltage at no load is,

$$V_{O\text{-NL}} = I_Z r_z + V_{Z0}$$

Substitute 4.75 V for V_{Z0}, 50 mA for I_Z, and $7\ \Omega$ for r_z.

$$\begin{aligned} V_{O\text{-NL}} &= (50\text{ mA})(7\ \Omega) + 4.75\text{ V} \\ &= 0.35\text{ V} + 4.75\text{ V} \\ &= 5.1\text{ V} \end{aligned}$$

Thus, the output voltage at no load, $V_{O\text{-NL}}$ is, $\boxed{5.1\text{ V}}$.

Step 4 of 5

The line regulation is,

$$\text{Line regulation} = \frac{r_z}{R + r_z}$$

Substitute $200\ \Omega$ for R, and $7\ \Omega$ for r_z.

$$\begin{aligned} \text{Line regulation} &= \frac{7}{200 + 7} \\ &= 0.0338 \\ &= 33.8\text{ mV/V} \end{aligned}$$

Therefore, the line regulation is, $\boxed{33.8\text{ mV/V}}$.

Step 5 of 5

The load regulation is,

$$\begin{aligned} \text{Load regulation} &= -(r_z \parallel R) \\ &= -\frac{r_z R}{r_z + R} \end{aligned}$$

Substitute $200\ \Omega$ for R, and $7\ \Omega$ for r_z.

$$\begin{aligned} \text{Load regulation} &= -\frac{(7)(200)}{7 + 200} \\ &= -\frac{1400}{207} \\ &\approx -7 \end{aligned}$$

Therefore, the load regulation is, $\boxed{-7\text{ mV/mA}}$.

4.018P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

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Step 1 of 7

(a)

Refer to Figure 4.21 (a) in the textbook for a half-wave rectifier circuit.

Replace the diode, D in the circuit with its piece-wise linear model. That is, replace the diode, D with V_{D0} and r_D in series.

Neglect the effect of diode resistance, r_D, and draw the equivalent circuit.

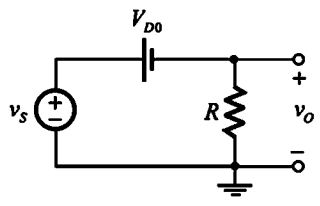


Figure 1

Step 2 of 7

Apply Kirchhoff's voltage law to the circuit in Figure 1.

$$v_S - V_{D0} - v_O = 0$$

Simplify the expression for the output voltage, v_O.

$$v_O = v_S - V_{D0}$$

Consider that the input to the half-wave rectifier is, $v_S = V_S \sin \omega t$. The diode does not conduct during the negative half cycle.

Draw the input/output waveforms together.

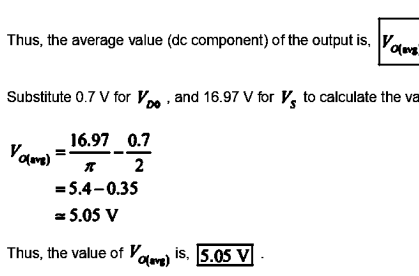


Figure 2

Step 3 of 7

In the waveform in Figure 2, θ is the angle at which the input signal reaches V_{D0} and the diode starts conducting. The diode conduction terminates at $(\pi - \theta)$ and hence the total conduction angle is, $(\pi - \theta - \theta) = (\pi - 2\theta)$. Therefore,

$$V_{D0} = V_S \sin\theta$$
$$\sin\theta = \frac{V_{D0}}{V_S}$$
$$\theta = \sin^{-1}\left(\frac{V_{D0}}{V_S}\right)$$

Therefore, for the half-cycles during which the diode conducts, conduction begins at an angle, $\boxed{\theta = \sin^{-1}\left(\frac{V_{D0}}{V_S}\right)}$ and terminates at $\boxed{(\pi - \theta)}$ for a total conduction angle of $\boxed{(\pi - 2\theta)}$.

Step 4 of 7

Consider that the sinusoidal input is 12 V (rms).

Find the peak value.

$$\begin{aligned} V_{peak} &= \sqrt{2} V_{rms} \\ &= \sqrt{2}(12\text{ V}) \\ &= 16.97\text{ V} \end{aligned}$$

From the waveform in Figure 2, $V_{peak} = V_S$. Therefore, $V_S = 16.97$ V.

Recall the expression for the angle, θ.

$$\theta = \sin^{-1}\left(\frac{V_{D0}}{V_S}\right)$$

Substitute 0.7 V for V_{D0}, and 16.97 V for V_S.

$$\begin{aligned} \theta &= \sin^{-1}\left(\frac{0.7}{16.97}\right) \\ &= \sin^{-1}(0.04) \\ &\simeq 2.4° \end{aligned}$$

Therefore, the value of angle, θ is, $\boxed{2.4°}$.

The total conduction angle is,

$$\begin{aligned} (\pi - 2\theta) &= 180° - 2(2.4) \\ &= 175° \end{aligned}$$

Therefore, the total conduction angle, $(\pi - \theta)$ is, $\boxed{175°}$.

Step 5 of 7

(b)

The average value (dc component) of the output is,

$$V_{O(avg)} = \frac{1}{T}\int_0^T v_O(t)\,dt$$

Observe from the waveform that the time period, $T = 2\pi$.

The limits of the integration are from $\phi = \theta$ to $\phi = \pi - \theta$, since the output exists from θ to $(\pi - \theta)$.

Calculate the output.

$$v_O = v_S - V_{D0}$$

Here, $v_S = V_S \sin\phi$ in the interval, $\phi = \theta$ to $\phi = \pi - \theta$. Therefore,

$$\begin{aligned} V_{O(avg)} &= \frac{1}{2\pi}\int_{\phi=\theta}^{\pi-\theta}(V_S \sin\phi - V_{D0})\,d\phi \\ &= \frac{1}{2\pi}\left[V_S\left(-\cos\phi\Big|_\theta^{\pi-\theta} - V_{D0}\left(\phi\Big|_\theta^{\pi-\theta}\right)\right)\right] \\ &= \frac{1}{2\pi}\left[V_S\left(\cos(\theta) - \cos(\pi-\theta) - V_{D0}(\pi - \theta - \theta)\right)\right] \end{aligned}$$

Assume that, $\cos\theta \simeq 1$, then $\cos(\pi - \theta) \simeq -1$ and $\pi - 2\theta \simeq \pi$. Therefore,

$$\begin{aligned} V_{O(avg)} &\simeq \frac{1}{2\pi}\left[V_S(2 - V_{D0}\pi)\right] \\ &\simeq \frac{2V_S}{2\pi} - \frac{V_{D0}\pi}{2\pi} \\ &\simeq \frac{V_S}{\pi} - \frac{V_{D0}}{2} \end{aligned}$$

Thus, the average value (dc component) of the output is, $\boxed{V_{O(avg)} \simeq \frac{1}{\pi}V_S - \frac{1}{2}V_{D0}}$.

Substitute 0.7 V for V_{D0}, and 16.97 V for V_S to calculate the value of $V_{O(avg)}$.

$$\begin{aligned} V_{O(avg)} &= \frac{16.97}{\pi} - \frac{0.7}{2} \\ &= 5.4 - 0.35 \\ &\simeq 5.05\text{ V} \end{aligned}$$

Thus, the value of $V_{O(avg)}$ is, $\boxed{5.05\text{ V}}$.

Step 6 of 7

(c)

The peak diode current occurs at the peak diode voltage, V_D.

Draw the diode circuit to find the peak diode current.

Figure 3

Step 7 of 7

Apply Kirchhoff's voltage law to the circuit in Figure 3.

$$V_S - V_{D0} - \hat{i}_D R = 0$$
$$V_S - V_{D0} = \hat{i}_D R$$
$$\hat{i}_D = \frac{V_S - V_{D0}}{R}$$

Therefore, the peak diode current is, $\boxed{\hat{i}_D = \frac{V_S - V_{D0}}{R}}$.

Substitute 0.7 V for V_{D0}, 16.97 V for V_S, and $100\ \Omega$ for R to calculate the value of $\hat{i}_D$.

$$\begin{aligned} \hat{i}_D &= \frac{16.97 - 0.7}{100} \\ &= 0.1627\text{ A} \end{aligned}$$

Thus, the value of current, $\hat{i}_D$ is, $\boxed{163\text{ mA}}$.

Calculate the peak inverse voltage (PIV).

$$\text{PIV} = V_S$$

Substitute 16.97 V for V_S.

$$\text{PIV} = 16.97\text{ V}$$

Therefore, the peak inverse voltage, PIV is, $\boxed{17\text{ V}}$.

Step 1 of 2

The *i–v* relationship of a diode for the constant $n = 1$ is,

$$i = I_S\left(e^{\frac{v}{V_T}} - 1\right)$$

Here,

I_S is the saturation current of the diode

V_T is the thermal temperature

The diode current at 0.7 V forward voltage is,

$$i_1 = I_S\left(e^{\frac{0.7}{V_T}} - 1\right)$$

The diode current at 0.5 V forward voltage is,

$$i_2 = I_S\left(e^{\frac{0.5}{V_T}} - 1\right)$$

Step 2 of 2

Determine the ratio of currents, $\frac{i_2}{i_1}$.

$$\frac{i_2}{i_1} = \frac{e^{\frac{0.5}{V_T}} - 1}{e^{\frac{0.7}{V_T}} - 1}$$

The diode current i_2 is,

$$i_2 = \left(\frac{e^{\frac{0.5}{V_T}} - 1}{e^{\frac{0.7}{V_T}} - 1}\right) i_1$$

Substitute $25\times10^{-3}\ \text{V}$ for V_T and $10^{-3}\ \text{A}$ for i_1.

$$\begin{aligned} i_2 &= \left(\frac{e^{\frac{0.5}{25\times10^{-3}}} - 1}{e^{\frac{0.7}{25\times10^{-3}}} - 1}\right)\left(10^{-3}\right) \\ &= \left(\frac{4.8516\times10^{8}}{1.44626\times10^{12}}\right)\left(10^{-3}\right) \\ &= 0.335\times10^{-6}\ \text{A} \\ &= 0.335\ \mu\text{A} \end{aligned}$$

Thus, the diode current when the diode is operated at 0.5 V is $\boxed{0.335\ \mu\text{A}}$.

(a)

Refer to Figure 4.22 (a) in the textbook for a full-wave rectifier circuit.

Replace the diode, D_1 in the circuit with its piece-wise linear model. That is, replace the diode, D_1 with V_D and r_D in series.

Neglect the effect of r_D and draw the equivalent circuit for the right side upper portion of the center tap.

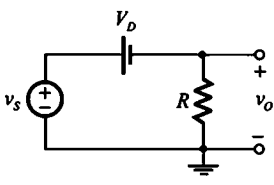


Figure 1

Step 2 of 10

Apply Kirchhoff's voltage law to the circuit.

$$v_S - V_D - v_D = 0$$

Simplify the expression for the output voltage, v_D.

$$v_D = v_S - V_D$$

Consider that the input to the full-wave rectifier is, $v_S = V_S \sin \omega t$. The diode conducts during both the half cycles.

Draw the input/output waveforms together.

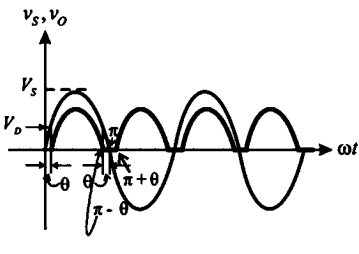


Figure 2

Step 3 of 10

In the waveform in Figure 2, θ is the angle at which the input signal reaches V_D and the diode starts conducting. The diode conduction terminates at $(\pi - \theta)$ and hence the total conduction angle is, $(\pi - \theta - \theta) = (\pi - 2\theta)$. Therefore,

$$V_D = V_S \sin\theta$$
$$\sin\theta = \frac{V_D}{V_S}$$
$$\theta = \sin^{-1}\left(\frac{V_D}{V_S}\right)$$

From the waveform in Figure 2, it is clear that the output is zero from $(\pi - \theta)$ to $(\pi + \theta)$.

Hence, the output is zero for an angle of $\boxed{2\sin^{-1}\left(\frac{V_D}{V_S}\right)}$, centered around the zero-crossing points of the sine-wave input.

Step 4 of 10

(b)
The average value (dc component) of the output is,

$$V_{O(avg)} = \frac{1}{T}\int_0^T v_D(t)\,dt$$

Observe from the waveform that the time period, $T = 2\pi$.

The limits of the integration are from $\phi = \theta$ to $\phi = \pi - \theta$ and $\phi = \pi + \theta$ to $\phi = 2\pi - \theta$.

Calculate the output.

$$v_D = v_S - V_D$$

Here, $v_S = V_S \sin\phi$ in the interval, $\phi = \theta$ to $\phi = \pi - \theta$ and $\phi = \pi + \theta$ to $\phi = 2\pi - \theta$. Therefore, evaluate the integral over the first interval and make it twice to get the required value.

$$V_{O(avg)} = \frac{1}{2\pi}\int_{\phi=\theta}^{\pi-\theta} 2\left(V_S \sin\phi - V_D\right)d\phi$$
$$= \frac{2}{2\pi}\left[V_S\left(-\cos\phi\Big|_\theta^{\pi-\theta}\right) - V_D\left(\phi\Big|_\theta^{\pi-\theta}\right)\right]$$
$$= \frac{1}{\pi}\left[V_S\left(\cos(\theta) - \cos(\pi-\theta)\right) - V_D(\pi - \theta - \theta)\right]$$

Assume that, $\cos\theta = 1$, then $\cos(\pi - \theta) = -1$ and $\pi - 2\theta = \pi$. Therefore,

$$V_{O(avg)} = \frac{1}{\pi}\left[V_S(2) - V_D\pi\right]$$
$$= \frac{2V_S}{\pi} - \frac{V_D\pi}{\pi}$$
$$= \frac{2V_S}{\pi} - V_D$$

Thus, the average value (dc component) of the output is, $\boxed{V_{O(avg)} \simeq \frac{2}{\pi}V_S - V_D}$.

Step 5 of 10

(c)

The peak diode current occurs at the peak diode voltage, V_D.

Draw the diode circuit to find the peak diode current.

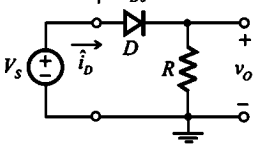


Figure 3

Apply Kirchhoff's voltage law to the circuit in Figure 3.

$$V_S - V_D - \hat{i}_D R = 0$$
$$V_S - V_D = \hat{i}_D R$$
$$\hat{i}_D = \frac{V_S - V_D}{R}$$

Therefore, the peak diode current is, $\boxed{\hat{i}_D = \frac{V_S - V_D}{R}}$.

Step 6 of 10

Determine the percentage of each cycle during which $v_D > 0$.

The total duration of the input for one complete cycle is 2π.

The duration for which the diode conducts is,

$$T_C = \left[(\pi - \theta) - \theta\right] + \left[(2\pi - \theta) - (\pi + \theta)\right]$$
$$= \left[\pi - \theta - \theta\right] + \left[2\pi - \theta - \pi - \theta\right]$$
$$= \left[\pi - 2\theta\right] + \left[\pi - 2\theta\right]$$
$$= 2\left[\pi - 2\theta\right]$$

The fraction of each cycle during which $v_D > 0$ is,

$$\%T_C = \frac{2\left[\pi - 2\theta\right]}{2\pi} \times 100$$

The expression for θ is, $\theta = \sin^{-1}\left(\frac{V_D}{V_S}\right)$. Therefore,

$$\%T_C = \frac{\pi - 2\sin^{-1}\left(\frac{V_D}{V_S}\right)}{\pi} \times 100$$

Step 7 of 10

Consider that the sinusoidal input is 12 V (rms).
Find the peak value.

$$V_{peak} = \sqrt{2}V_{rms}$$
$$= \sqrt{2}(12\ \text{V})$$
$$= 16.97\ \text{V}$$

From the waveform in Figure 2, $V_{peak} = V_S$. Therefore, $V_S = 16.97\ \text{V}$.

Substitute 0.7 V for V_D, and 16.97 V for V_S in the expression for $\%T_C$.

$$\%T_C = \frac{\pi - 2\sin^{-1}\left(\frac{0.7}{16.97}\right)}{\pi} \times 100$$
$$= \frac{180^\circ - 2(2.36^\circ)}{180^\circ} \times 100 \qquad \left(\text{since, } \pi^c = 180^\circ\right)$$
$$\simeq 97.4\%$$

Thus, the fraction (percentage) of each cycle during which $v_D > 0$ is, $\boxed{97.4\%}$.

Step 8 of 10

Determine the average value of V_D.

Recall the expression for the average value of output voltage, $V_{O(avg)}$.

$$V_{O(avg)} = \frac{2}{\pi}V_S - V_D$$

Substitute 0.7 V for V_D, and 16.97 V for V_S in the expression for $V_{O(avg)}$.

$$V_{O(avg)} = \frac{2}{\pi}(16.97) - (0.7)$$
$$= (0.637)(16.97) - (0.7)$$
$$= 10.8 - 0.7$$
$$= 10.1$$

Therefore, the average value of output voltage, V_D is, $\boxed{10.1\ \text{V}}$.

Step 9 of 10

Calculate the peak diode current.

Recall the expression for the diode current, $\hat{i}_D$.

$$\hat{i}_D = \frac{V_S - V_D}{R}$$

Substitute 0.7 V for V_D, and 16.97 V for V_S in the expression for $\hat{i}_D$.

$$\hat{i}_D = \frac{16.97 - 0.7}{100}$$
$$= 0.1627$$

Thus, the value of current, $\hat{i}_D$ is, $\boxed{163\ \text{mA}}$.

Step 10 of 10

Determine the peak inverse voltage, PIV.

The peak inverse voltage of a full-wave rectifier is,

$$\text{PIV} = 2V_S - V_D$$

Substitute 0.7 V for V_D, and 16.97 V for V_S in the expression for PIV.

$$\text{PIV} = 2(16.97) - (0.7)$$
$$= 33.94 - 0.7$$
$$= 33.24$$

Therefore, the peak inverse voltage, PIV is, $\boxed{33.24\ \text{V}}$.

4.020P

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Step 1 of 7

(a)

Refer to Figure 4.23 (a) in the textbook for a bridge rectifier circuit.

During the positive half-cycle of input, the diodes, D_1 and D_2 are ON and the diodes, D_3 and D_4 are OFF. During the negative half-cycle of input, the diodes, D_3 and D_4 are ON and the diodes, D_1 and D_2 are OFF.

Replace the diodes which are conducting with their constant-voltage-drop diode model.

Consider the following circuit for positive half-cycle of the input signal:

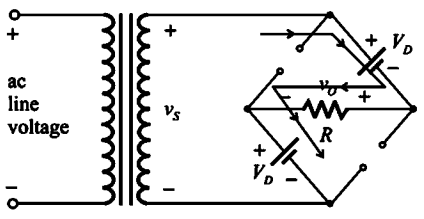


Figure 1

Step 2 of 7

Apply Kirchhoff's voltage law to the circuit in Figure 1.

$$v_S - V_D - v_O - V_D = 0$$
$$v_S - 2V_D - v_O = 0$$

Simplify the expression for the output voltage, v_O.

$$v_O = v_S - 2V_D$$

Consider that the input to the bridge rectifier is, $v_S = V_S \sin \omega t$. The diodes conduct during both the half-cycles.

Draw the input/output waveforms together.

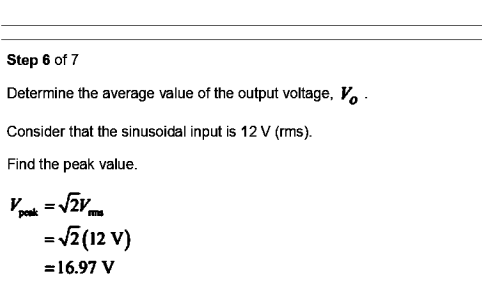


Figure 2

Step 3 of 7

In the waveform in Figure 2, θ is the angle at which the input signal reaches V_{D0} and the diode starts conducting. The diode conduction terminates at $(\pi - \theta)$ and hence the total conduction angle is, $(\pi - \theta - \theta) = (\pi - 2\theta)$. Therefore,

$$2V_D = V_S \sin \theta$$
$$\sin \theta = \frac{2V_D}{V_S}$$
$$\theta = \sin^{-1}\left(\frac{2V_D}{V_S}\right)$$

The average value (dc component) of the output is,

$$V_{O(avg)} = \frac{1}{T}\int_0^T v_O(t)\,dt$$

Observe from the waveform that the time period, $T = 2\pi$.

The limits of the integration are from $\phi = \theta$ to $\phi = \pi - \theta$, since the output exists from θ to $(\pi - \theta)$.

Step 4 of 7

Calculate the output.

$$v_O = v_S - 2V_D$$

Here, $v_S = V_S \sin \phi$ in the interval, $\phi = \theta$ to $\phi = \pi - \theta$ and $\phi = \pi + \theta$ to $\phi = 2\pi - \theta$. Therefore, evaluate the integral over the first interval and make it twice to get the required value.

$$V_{O(avg)} = \frac{1}{2\pi}\int_{\phi=\theta}^{\pi-\theta} 2(V_S \sin\phi - 2V_D)\,d\phi$$
$$= \frac{2}{2\pi}\left[V_S\left(-\cos\phi\big|_\theta^{\pi-\theta}\right) - 2V_D\left(\phi\big|_\theta^{\pi-\theta}\right)\right]$$
$$= \frac{1}{\pi}\left[V_S\left(\cos(\theta) - \cos(\pi-\theta)\right) - 2V_D(\pi - \theta - \theta)\right]$$

Assume that, $\cos\theta \simeq 1$, then $\cos(\pi - \theta) \simeq -1$ and $\pi - 2\theta \simeq \pi$. Therefore,

$$V_{O(avg)} = \frac{1}{\pi}\left[V_S(2) - 2V_D\pi\right]$$
$$= \frac{2V_S}{\pi} - \frac{2V_D\pi}{\pi}$$
$$\simeq \frac{2V_S}{\pi} - 2V_D$$

Thus, the average value (dc component) of the output is, $\boxed{V_{O(avg)} \simeq \frac{2}{\pi}V_S - 2V_D}$.

Step 5 of 7

(b)

The peak diode current occurs at the peak diode voltage, V_D.

Draw the diode circuit to find the peak diode current.

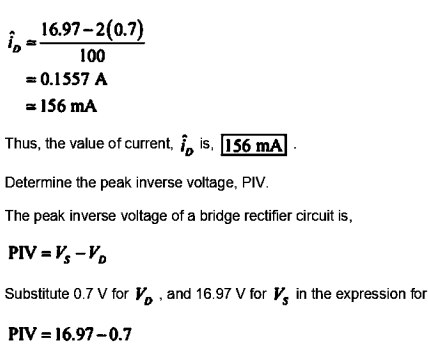


Figure 3

Apply Kirchhoff's voltage law to the circuit in Figure 3.

$$V_S - V_D - \hat{i}_D R - V_D = 0$$
$$V_S - 2V_D = \hat{i}_D R$$
$$\hat{i}_D = \frac{V_S - 2V_D}{R}$$

Therefore, the peak diode current is, $\boxed{\hat{i}_D = \frac{V_S - 2V_D}{R}}$.

Step 6 of 7

Determine the average value of the output voltage, V_O.

Consider that the sinusoidal input is 12 V (rms).

Find the peak value.

$$V_{peak} = \sqrt{2}V_{rms}$$
$$= \sqrt{2}(12\text{ V})$$
$$= 16.97\text{ V}$$

From the waveform in Figure 2, $V_{peak} = V_S$. Therefore, $V_S = 16.97$ V.

Recall the expression for the out[put voltage, V_O.

$$V_{O(avg)} = \frac{2V_S}{\pi} - 2V_D$$

Substitute 0.7 V for V_D, and 16.97 V for V_S in the expression for $V_{O(avg)}$.

$$V_{O(avg)} = \frac{2}{\pi}(16.97) - 2(0.7)$$
$$= (0.637)(16.97) - 1.4$$
$$= 10.8 - 1.4$$
$$= 9.4\text{ V}$$

Therefore, the average value of output voltage, V_O is, $\boxed{9.4\text{ V}}$.

Step 7 of 7

Calculate the value of peak diode current, $\hat{i}_D$.

Recall the expression for the peak diode current, $\hat{i}_D$.

$$\hat{i}_D = \frac{V_S - 2V_D}{R}$$

Substitute 0.7 V for V_D, and 16.97 V for V_S in the expression for $\hat{i}_D$.

$$\hat{i}_D = \frac{16.97 - 2(0.7)}{100}$$
$$= 0.1557\text{ A}$$
$$= 156\text{ mA}$$

Thus, the value of current, $\hat{i}_D$ is, $\boxed{156\text{ mA}}$.

Determine the peak inverse voltage, PIV.

The peak inverse voltage of a bridge rectifier circuit is,

$$\text{PIV} = V_S - V_D$$

Substitute 0.7 V for V_D, and 16.97 V for V_S in the expression for PIV.

$$\text{PIV} = 16.97 - 0.7$$
$$= 16.27$$

Therefore, the peak inverse voltage, PIV is, $\boxed{16.3\text{ V}}$.

Step 1 of 8

(a)

The *i–v* relationship of a diode is,

$$i = I_s\left(e^{\frac{v}{V_T}} - 1\right)$$

Here,

I_s is the saturation current of the diode

V_T is the thermal temperature

Substitute 1 A for *i*, 25×10^{-3} V for V_T and 0.7 V for *v*.

$$1 = I_s\left(e^{\frac{0.7}{25\times10^{-3}}} - 1\right)$$

$$I_s = \frac{1}{e^{\frac{0.7}{25\times10^{-3}}} - 1}$$

$$= 0.691\times10^{-12}\ \text{A}$$

Thus, the saturation current I_s of the diode is $\boxed{0.691\times10^{-12}\ \text{A}}$.

Step 2 of 8

Determine 10% of the diode current.

$$i = (0.1)(1)$$

$$= 0.1\ \text{A}$$

Determine the terminal voltage at 10% of the diode current.

$$i = I_s\left(e^{\frac{v}{V_T}} - 1\right)$$

Substitute 0.1 A for *i*, 25×10^{-3} V for V_T and 0.691×10^{-12} A for I_s.

$$0.1 = \left(0.691\times10^{-12}\right)\left(e^{\frac{v}{25\times10^{-3}}} - 1\right)$$

$$e^{\frac{v}{25\times10^{-3}}} - 1 = 1.447\times10^{11}$$

$$e^{\frac{v}{25\times10^{-3}}} = 1.447\times10^{11}$$

$$\frac{v}{25\times10^{-3}} = \ln\left(1.447\times10^{11}\right)$$

$$v = 0.642\ \text{V}$$

Thus, the junction voltage at 10% of the measured diode current is $\boxed{0.642\ \text{V}}$.

Step 3 of 8

(b)

Determine the saturation current at 0.65 V of diode voltage and 1 A of diode current.

$$i = I_s\left(e^{\frac{v}{V_T}} - 1\right)$$

$$10^{-3} = I_s\left(e^{\frac{0.65}{25\times10^{-3}}} - 1\right)$$

$$I_s = \frac{10^{-3}}{e^{\frac{0.65}{25\times10^{-3}}} - 1}$$

$$= 5.11\times10^{-15}\ \text{A}$$

Thus, the saturation current I_s of the diode is $\boxed{5.11\times10^{-15}\ \text{A}}$.

Step 4 of 8

Determine 10% of the diode current.

$$i = (0.1)(1\ \text{mA})$$

$$= 0.1\ \text{mA}$$

Determine the terminal voltage at 10% of the diode current.

$$i = I_s\left(e^{\frac{v}{V_T}} - 1\right)$$

Substitute 0.1×10^{-3} A for *i*, 25×10^{-3} V for V_T and 5.11×10^{-15} A for I_s.

$$0.1\times10^{-3} = \left(5.11\times10^{-15}\right)\left(e^{\frac{v}{25\times10^{-3}}} - 1\right)$$

$$e^{\frac{v}{25\times10^{-3}}} - 1 = 1.957\times10^{10}$$

$$e^{\frac{v}{25\times10^{-3}}} = 1.957\times10^{10}$$

$$\frac{v}{25\times10^{-3}} = \ln\left(1.957\times10^{10}\right)$$

$$v = 0.592\ \text{V}$$

Thus, the junction voltage at 10% of the measured diode current is $\boxed{0.592\ \text{V}}$.

Step 5 of 8

(c)

Determine the saturation current at 0.65 V of diode voltage and $10\ \mu\text{A}$ of diode current.

$$i = I_s\left(e^{\frac{v}{V_T}} - 1\right)$$

$$10\times10^{-6} = I_s\left(e^{\frac{0.65}{25\times10^{-3}}} - 1\right)$$

$$I_s = \frac{10\times10^{-6}}{e^{\frac{0.65}{25\times10^{-3}}} - 1}$$

$$= 5.11\times10^{-17}\ \text{A}$$

Thus, the saturation current I_s of the diode is $\boxed{5.11\times10^{-17}\ \text{A}}$.

Step 6 of 8

Determine 10% of the diode current.

$$i = (0.1)(1\ \mu\text{A})$$

$$= 0.1\ \mu\text{A}$$

Determine the terminal voltage at 10% of the diode current.

$$i = I_s\left(e^{\frac{v}{V_T}} - 1\right)$$

Substitute 0.1×10^{-6} A for *i*, 25×10^{-3} V for V_T and 5.11×10^{-17} A for I_s.

$$0.1\times10^{-3} = \left(5.11\times10^{-17}\right)\left(e^{\frac{v}{25\times10^{-3}}} - 1\right)$$

$$e^{\frac{v}{25\times10^{-3}}} - 1 = 1.957\times10^{12}$$

$$e^{\frac{v}{25\times10^{-3}}} = 1.957\times10^{12}$$

$$\frac{v}{25\times10^{-3}} = \ln\left(1.957\times10^{12}\right)$$

$$v = 0.707\ \text{V}$$

Thus, the junction voltage at 10% of the measured diode current is $\boxed{0.707\ \text{V}}$.

Step 7 of 8

(d)

Determine the saturation current at 0.7 V of diode voltage and $10\ \text{mA}$ of diode current.

$$i = I_s\left(e^{\frac{v}{V_T}} - 1\right)$$

$$10\times10^{-3} = I_s\left(e^{\frac{0.7}{25\times10^{-3}}} - 1\right)$$

$$I_s = \frac{10\times10^{-3}}{e^{\frac{0.7}{25\times10^{-3}}} - 1}$$

$$= 6.91\times10^{-15}\ \text{A}$$

Thus, the saturation current I_s of the diode is $\boxed{6.91\times10^{-15}\ \text{A}}$.

Step 8 of 8

Determine 10% of the diode current.

$$i = (0.1)(10\ \text{mA})$$

$$= 1\ \text{mA}$$

Determine the terminal voltage at 10% of the diode current.

$$i = I_s\left(e^{\frac{v}{V_T}} - 1\right)$$

Substitute 10^{-3} A for *i*, 25×10^{-3} V for V_T and 6.91×10^{-15} A for I_s.

$$10^{-3} = \left(6.91\times10^{-15}\right)\left(e^{\frac{v}{25\times10^{-3}}} - 1\right)$$

$$e^{\frac{v}{25\times10^{-3}}} - 1 = 1.447\times10^{11}$$

$$e^{\frac{v}{25\times10^{-3}}} = 1.447\times10^{11}$$

$$\frac{v}{25\times10^{-3}} = \ln\left(1.447\times10^{11}\right)$$

$$v = 0.642\ \text{V}$$

Thus, the junction voltage at 10% of the measured diode current is $\boxed{0.642\ \text{V}}$.

Step 1 of 8

Consider the following circuit for a full-wave peak rectifier circuit:

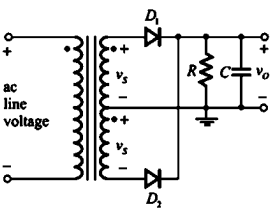


Figure 1

Step 2 of 8

Draw the input waveform and the full-wave rectifier output waveform.

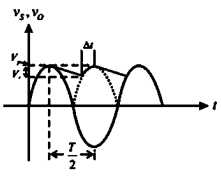


Figure 2

Step 3 of 8

The diodes, D_1 and D_2 conduct for a brief interval, Δt near the peak of the input sinusoid and supplies the capacitor with the charge equal to that lost during the much longer discharge interval.

The discharge time is approximately equal to the period, $\frac{T}{2}$.

The ripple voltage, (V_r) is the amount of discharge that occurs when the diodes are not conducting.

Express the output, v_O during the diode OFF interval.

$$v_O = V_p e^{-\frac{t}{CR}}$$

Since, the discharge time is equal to only half of the period, T, consider the following at the end of the discharge interval:

$$V_p - V_r = V_p e^{-\frac{\left(\frac{T}{2}\right)}{CR}}$$

$$V_r = V_p - V_p e^{-\frac{\left(\frac{T}{2}\right)}{CR}}$$

$$V_r = V_p\left(1 - e^{\frac{-T}{2CR}}\right)$$

Step 4 of 8

Consider the following approximation, since $CR \gg \frac{T}{2}$:

$$e^{\frac{-T}{2CR}} \simeq 1 - \frac{T}{2CR}$$

Use this approximation to simplify the expression for V_r.

$$V_r = V_p\left[1 - \left(1 - \frac{T}{2CR}\right)\right]$$

$$V_r = V_p\left(\frac{T}{2CR}\right)$$

The expression for the time period is, $T = \frac{1}{f}$. Therefore,

$$V_r = \frac{V_p}{2fCR}$$

Thus, the ripple voltage, V_r is, $\boxed{V_r = \frac{V_p}{2fCR}}$.

Step 5 of 8

Determine the average current, i_{Dav}.

Note that the charge supplied during the conduction is same as the charge lost during the discharge. That is, $Q_{Supplied} = Q_{Lost}$. Therefore,

$$i_{Cav}\Delta t = CV_r$$

Consider the following from the circuit:

$$i_{Cav} = i_{Dav} - I_L$$

Hence,

$$(i_{Dav} - I_L)\Delta t = CV_r$$

Substitute the expression for V_r.

$$(i_{Dav} - I_L)\Delta t = C\left(\frac{V_p}{2fCR}\right)$$

$$(i_{Dav} - I_L)\Delta t = \frac{V_p}{2fR}$$

The angular frequency, ω is, $2\pi f$. Therefore,

$$2f = \frac{\omega}{\pi}$$

Substitute this in the equation.

$$(i_{Dav} - I_L)\Delta t = \frac{V_p}{\left(\frac{\omega}{\pi}\right)R}$$

$$= \frac{V_p \pi}{\Delta t \omega R}$$

$$i_{Dav} = I_L + \frac{V_p \pi}{\omega \Delta t R}$$

Here, $\omega\Delta t$ is the conduction angle.

Step 6 of 8

The expression for the conduction angle is,

$$\omega\Delta t = \sqrt{\frac{2V_r}{V_p}}$$

Substitute this in equation.

$$i_{Dav} = I_L + \frac{V_p \pi}{\left(\sqrt{\frac{2V_r}{V_p}}\right)R}$$

Approximate the load current, since the output is approximately held at V_p.

$$I_L = \frac{V_p}{R}$$

Hence, modify the equation for i_{Dav}.

$$i_{Dav} = I_L + \frac{\pi I_L}{\left(\sqrt{\frac{2V_r}{V_p}}\right)}$$

$$= I_L\left[1 + \pi\left(\sqrt{\frac{V_p}{2V_r}}\right)\right]$$

Thus, the average diode current is, $\boxed{i_{Dav} = I_L\left(1 + \pi\sqrt{\frac{V_p}{2V_r}}\right)}$.

Step 7 of 8

Determine the maximum diode current.

If $t = 0$ at the peak, the maximum diode current occurs at the onset of conduction or at $t = \omega\Delta t$.

The diode current during the conduction is,

$$i_D = i_C + i_L$$

The current, i_D is maximum when the capacitor charges to v_S.

$$i_{Dmax} = C\frac{dv_S}{dt} + i_L$$

Assume that i_L is constant.

$$i_L \approx \frac{V_p}{R}$$

$$i_L \approx I_L$$

Therefore,

$$i_{Dmax} \approx C\frac{dv_S}{dt} + I_L$$

Here, $v_S = V_p \cos\omega t$. Hence,

$$i_{Dmax} = C\frac{d}{dt}\left(V_p \cos\omega t\right) + I_L$$

$$= -C\omega V_p\left(\sin\omega t\right) + I_L$$

At $t = \Delta t$, $i_{Dmax} = -C\omega V_p \sin\left(-\omega\Delta t\right) + I_L$

Consider that for small conduction angle, $\sin\left(-\omega\Delta t\right) \approx -\omega\Delta t$. Therefore,

$$i_{Dmax} = C\omega V_p\left(\omega\Delta t\right) + I_L$$

Step 8 of 8

Consider that, $\omega\Delta t \simeq \sqrt{\frac{2V_r}{V_p}}$.

Substitute this in the equation.

$$i_{Dmax} = C\omega V_p\left(\sqrt{\frac{2V_r}{V_p}}\right) + I_L$$

Substitute $\omega = 2\pi f$ in the equation.

$$i_{Dmax} = C(2\pi f)V_p\left(\sqrt{\frac{2V_r}{V_p}}\right) + I_L$$

Note that, $V_r = \frac{V_p}{2fCR}$. Therefore,

$$f = \frac{V_p}{2V_r CR}$$

Substitute this in equation.

$$i_{Dmax} = C\left(2\pi\left(\frac{V_p}{2V_r CR}\right)\right)V_p\left(\sqrt{\frac{2V_r}{V_p}}\right) + I_L$$

$$i_{Dmax} = \left(\frac{\pi V_p}{V_r R}\right)V_p\left(\sqrt{\frac{2V_r}{V_p}}\right) + I_L$$

Recall that, $\frac{V_p}{R} = I_L$.

Substitute this in the equation.

$$i_{Dmax} = \left(\frac{\pi}{V_r}I_L\right)V_p\left(\sqrt{\frac{2V_r}{V_p}}\right) + I_L$$

$$= I_L\left(1 + \pi\sqrt{\frac{2V_p}{V_r}}\right)$$

$$= I_L\left(1 + 2\pi\sqrt{\frac{V_p}{2V_r}}\right)$$

Therefore, the maximum diode current or peak current is, $\boxed{i_{Dmax} = I_L\left(1 + 2\pi\sqrt{\frac{V_p}{2V_r}}\right)}$.

Step 1 of 15

(a)

The *i–v* relationship of a diode is,

$$i = I_S\left(e^{\frac{v}{V_T}} - 1\right)$$

Here,

I_S is the saturation current of the diode

V_T is the thermal temperature

Substitute $10\times10^{-3}\ \text{A}$ for i, $25\times10^{-3}\ \text{V}$ for V_T and $700\times10^{-3}\ \text{V}$ for v.

$$10\times10^{-3} = I_S\left(e^{\frac{700\times10^{-3}}{25\times10^{-3}}} - 1\right)$$

$$I_S = \frac{10\times10^{-3}}{e^{\frac{700\times10^{-3}}{25\times10^{-3}}} - 1}$$
$$= 6.91\times10^{-15}\ \text{A}$$

Thus, the saturation current I_S of the diode is $\boxed{6.91\times10^{-15}\ \text{A}}$

Step 2 of 15

Determine the terminal voltage at 10 times the diode current.

$$i = I_S\left(e^{\frac{v}{V_T}} - 1\right)$$

Substitute $100\times10^{-3}\ \text{A}$ for i, $25\times10^{-3}\ \text{V}$ for V_T and $6.91\times10^{-15}\ \text{A}$ for I_S.

$$100\times10^{-3} = \left(6.91\times10^{-15}\right)\left(e^{\frac{v}{25\times10^{-3}}} - 1\right)$$

$$e^{\frac{v}{25\times10^{-3}}} - 1 = 1.447\times10^{13}$$

$$e^{\frac{v}{25\times10^{-3}}} = 1.447\times10^{13}$$

$$\frac{v}{25\times10^{-3}} = \ln\left(1.447\times10^{13}\right)$$

$$v = 0.76\ \text{V}$$

Thus, the junction voltage at 10 times the measured diode current is $\boxed{0.76\ \text{V}}$.

Step 3 of 15

Determine the terminal voltage at $\frac{1}{10}$ times the diode current.

$$i = I_S\left(e^{\frac{v}{V_T}} - 1\right)$$

Substitute $10^{-3}\ \text{A}$ for i, $25\times10^{-3}\ \text{V}$ for V_T and $6.91\times10^{-15}\ \text{A}$ for I_S.

$$10^{-3} = \left(6.91\times10^{-15}\right)\left(e^{\frac{v}{25\times10^{-3}}} - 1\right)$$

$$e^{\frac{v}{25\times10^{-3}}} - 1 = 1.447\times10^{11}$$

$$e^{\frac{v}{25\times10^{-3}}} = 1.447\times10^{11}$$

$$\frac{v}{25\times10^{-3}} = \ln\left(1.447\times10^{11}\right)$$

$$v = 0.642\ \text{V}$$

Thus, the junction voltage at $\frac{1}{10}$ times the diode current is $\boxed{0.642\ \text{V}}$.

Step 4 of 15

(b)

Determine the saturation current by substituting $10^{-3}\ \text{A}$ for i, $25\times10^{-3}\ \text{V}$ for V_T and $700\times10^{-3}\ \text{V}$ for v.

$$10^{-3} = I_S\left(e^{\frac{700\times10^{-3}}{25\times10^{-3}}} - 1\right)$$

$$I_S = \frac{10^{-3}}{e^{\frac{700\times10^{-3}}{25\times10^{-3}}} - 1}$$
$$= 6.91\times10^{-16}\ \text{A}$$

Thus, the saturation current I_S of the diode is $\boxed{6.91\times10^{-16}\ \text{A}}$.

Step 5 of 15

Determine the terminal voltage at 10 times the diode current.

$$i = I_S\left(e^{\frac{v}{V_T}} - 1\right)$$

Substitute $100\times10^{-3}\ \text{A}$ for i, $25\times10^{-3}\ \text{V}$ for V_T and $6.91\times10^{-16}\ \text{A}$ for I_S.

$$10\times10^{-3} = \left(6.91\times10^{-16}\right)\left(e^{\frac{v}{25\times10^{-3}}} - 1\right)$$

$$e^{\frac{v}{25\times10^{-3}}} - 1 = 1.447\times10^{13}$$

$$e^{\frac{v}{25\times10^{-3}}} = 1.447\times10^{13}$$

$$\frac{v}{25\times10^{-3}} = \ln\left(1.447\times10^{13}\right)$$

$$v = 0.76\ \text{V}$$

Thus, the junction voltage at 10 times the measured diode current is $\boxed{0.76\ \text{V}}$.

Step 6 of 15

Determine the terminal voltage at $\frac{1}{10}$ times the diode current.

$$i = I_S\left(e^{\frac{v}{V_T}} - 1\right)$$

Substitute $0.1\times10^{-3}\ \text{A}$ for i, $25\times10^{-3}\ \text{V}$ for V_T and $6.91\times10^{-15}\ \text{A}$ for I_S.

$$0.1\times10^{-3} = \left(6.91\times10^{-16}\right)\left(e^{\frac{v}{25\times10^{-3}}} - 1\right)$$

$$e^{\frac{v}{25\times10^{-3}}} - 1 = 1.447\times10^{11}$$

$$e^{\frac{v}{25\times10^{-3}}} = 1.447\times10^{11}$$

$$\frac{v}{25\times10^{-3}} = \ln\left(1.447\times10^{11}\right)$$

$$v = 0.642\ \text{V}$$

Thus, the junction voltage at $\frac{1}{10}$ times the diode current is $\boxed{0.642\ \text{V}}$.

Step 7 of 15

(c)

Determine the saturation current by substituting $10\ \text{A}$ for i, $25\times10^{-3}\ \text{V}$ for V_T and $800\times10^{-5}\ \text{V}$ for v.

$$10 = I_S\left(e^{\frac{800\times10^{-5}}{25\times10^{-3}}} - 1\right)$$

$$I_S = \frac{10}{e^{\frac{800\times10^{-5}}{25\times10^{-3}}} - 1}$$
$$= 1.266\times10^{-13}\ \text{A}$$

Thus, the saturation current I_S of the diode is $\boxed{1.266\times10^{-13}\ \text{A}}$.

Step 8 of 15

Determine the terminal voltage at 10 times the diode current.

$$i = I_S\left(e^{\frac{v}{V_T}} - 1\right)$$

Substitute $100\ \text{A}$ for i, $25\times10^{-3}\ \text{V}$ for V_T and $1.266\times10^{-13}\ \text{A}$ for I_S.

$$100 = \left(1.266\times10^{-13}\right)\left(e^{\frac{v}{25\times10^{-3}}} - 1\right)$$

$$e^{\frac{v}{25\times10^{-3}}} - 1 = 7.898\times10^{14}$$

$$e^{\frac{v}{25\times10^{-3}}} = 7.898\times10^{14}$$

$$\frac{v}{25\times10^{-3}} = \ln\left(7.898\times10^{14}\right)$$

$$v = 0.86\ \text{V}$$

Thus, the junction voltage at 10 times the measured diode current is $\boxed{0.86\ \text{V}}$.

Step 9 of 15

Substitute $1\ \text{A}$ for i, $25\times10^{-3}\ \text{V}$ for V_T and $1.266\times10^{-13}\ \text{A}$ for I_S.

$$1 = \left(1.266\times10^{-13}\right)\left(e^{\frac{v}{25\times10^{-3}}} - 1\right)$$

$$e^{\frac{v}{25\times10^{-3}}} - 1 = 7.898\times10^{12}$$

$$e^{\frac{v}{25\times10^{-3}}} = 7.898\times10^{14}$$

$$\frac{v}{25\times10^{-3}} = \ln\left(7.898\times10^{14}\right)$$

$$v = 0.74\ \text{V}$$

Thus, the junction voltage at $\frac{1}{10}$ times the diode current is $\boxed{0.74\ \text{V}}$.

Step 10 of 15

(d)

Determine the saturation current by substituting $1\ \text{mA}$ for i, $25\times10^{-3}\ \text{V}$ for V_T and $700\times10^{-3}\ \text{V}$ for v.

$$10^{-3} = I_S\left(e^{\frac{700\times10^{-3}}{25\times10^{-3}}} - 1\right)$$

$$I_S = \frac{10^{-3}}{e^{\frac{700\times10^{-3}}{25\times10^{-3}}} - 1}$$
$$= 6.91\times10^{-16}\ \text{A}$$

Thus, the saturation current I_S of the diode is $\boxed{6.91\times10^{-16}\ \text{A}}$.

Step 11 of 15

Determine the terminal voltage at 10 times the diode current.

$$i = I_S\left(e^{\frac{v}{V_T}} - 1\right)$$

Substitute $10\times10^{-3}\ \text{A}$ for i, $25\times10^{-3}\ \text{V}$ for V_T and $6.91\times10^{-16}\ \text{A}$ for I_S.

$$10\times10^{-3} = \left(6.91\times10^{-16}\right)\left(e^{\frac{v}{25\times10^{-3}}} - 1\right)$$

$$e^{\frac{v}{25\times10^{-3}}} - 1 = 1.447\times10^{13}$$

$$e^{\frac{v}{25\times10^{-3}}} = 1.447\times10^{13}$$

$$\frac{v}{25\times10^{-3}} = \ln\left(1.447\times10^{13}\right)$$

$$v = 0.76\ \text{V}$$

Thus, the junction voltage at 10 times the measured diode current is $\boxed{0.76\ \text{V}}$.

Step 12 of 15

Determine the terminal voltage at $\frac{1}{10}$ times the diode current.

$$i = I_S\left(e^{\frac{v}{V_T}} - 1\right)$$

Substitute $0.1\times10^{-3}\ \text{A}$ for i, $25\times10^{-3}\ \text{V}$ for V_T and $6.91\times10^{-16}\ \text{A}$ for I_S.

$$0.1\times10^{-3} = \left(6.91\times10^{-16}\right)\left(e^{\frac{v}{25\times10^{-3}}} - 1\right)$$

$$e^{\frac{v}{25\times10^{-3}}} - 1 = 1.447\times10^{11}$$

$$e^{\frac{v}{25\times10^{-3}}} = 1.447\times10^{11}$$

$$\frac{v}{25\times10^{-3}} = \ln\left(1.447\times10^{11}\right)$$

$$v = 0.64\ \text{V}$$

Thus, the junction voltage at $\frac{1}{10}$ times the measured diode current is $\boxed{0.64\ \text{V}}$.

Step 13 of 15

(e)

Determine the saturation current by substituting $10\times10^{-6}\ \text{A}$ for i, $25\times10^{-3}\ \text{V}$ for V_T and $700\times10^{-3}\ \text{V}$ for v.

$$10\times10^{-6} = I_S\left(e^{\frac{700\times10^{-3}}{25\times10^{-3}}} - 1\right)$$

$$I_S = \frac{10\times10^{-6}}{e^{\frac{700\times10^{-3}}{25\times10^{-3}}} - 1}$$
$$= 6.91\times10^{-18}\ \text{A}$$

Thus, the saturation current I_S of the diode is $\boxed{6.91\times10^{-18}\ \text{A}}$

Step 14 of 15

Determine the terminal voltage at 10 times the diode current.

$$i = I_S\left(e^{\frac{v}{V_T}} - 1\right)$$

Substitute $100\times10^{-6}\ \text{A}$ for i, $25\times10^{-3}\ \text{V}$ for V_T and $6.91\times10^{-18}\ \text{A}$ for I_S.

$$100\times10^{-6} = \left(6.91\times10^{-18}\right)\left(e^{\frac{v}{25\times10^{-3}}} - 1\right)$$

$$e^{\frac{v}{25\times10^{-3}}} - 1 = 1.447\times10^{13}$$

$$e^{\frac{v}{25\times10^{-3}}} = 1.447\times10^{13}$$

$$\frac{v}{25\times10^{-3}} = \ln\left(1.447\times10^{13}\right)$$

$$v = 0.76\ \text{V}$$

Thus, the junction voltage at 10 times the measured diode current is $\boxed{0.76\ \text{V}}$.

Step 15 of 15

Determine the terminal voltage at $\frac{1}{10}$ times the diode current.

$$i = I_S\left(e^{\frac{v}{V_T}} - 1\right)$$

Substitute $10^{-6}\ \text{A}$ for i, $25\times10^{-3}\ \text{V}$ for V_T and $6.91\times10^{-18}\ \text{A}$ for I_S.

$$10^{-6} = \left(6.91\times10^{-18}\right)\left(e^{\frac{v}{25\times10^{-3}}} - 1\right)$$

$$e^{\frac{v}{25\times10^{-3}}} - 1 = 1.447\times10^{11}$$

$$e^{\frac{v}{25\times10^{-3}}} = 1.447\times10^{11}$$

$$\frac{v}{25\times10^{-3}} = \ln\left(1.447\times10^{11}\right)$$

$$v = 0.64\ \text{V}$$

Thus, the junction voltage at $\frac{1}{10}$ times the measured diode current is $\boxed{0.64\ \text{V}}$.

Step 1 of 10

4.023E

Consider the following circuit for a bridge rectifier circuit with a filter capacitor, C placed across the load resistor, R :

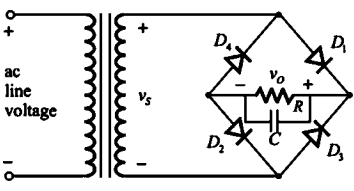


Figure 1

Step 2 of 10

Consider that the peak-to-peak ripple voltage, $V_r = 1\ \text{V}$.

Determine the value of C using the following expression for the ripple voltage:

$$V_r = \frac{V_p}{2fCR}$$

Consider that the sinusoidal input is 12 V (rms). Find the peak value.

$$\begin{aligned} V_{peak} &= \sqrt{2}V_{rms} \\ &= \sqrt{2}(12\ \text{V}) \\ &= 16.97\ \text{V} \end{aligned}$$

The expression for the output voltage, v_O is,

$$v_O = (V_p - 2V_D)e^{-\frac{t}{RC}}$$

The output voltage at the end of the discharge interval is,

$$v_O = (V_p - 2V_D - V_r)$$

Compare the expressions for v_O .

$$(V_p - 2V_D)e^{-\left(\frac{t}{RC}\right)} = (V_p - 2V_D - V_r)$$

Step 3 of 10

The discharge occurs almost over half of the time period, $\frac{T}{2}$.

Use the following approximation, since the time constant is very large compared to the discharge time:

$$e^{-\left(\frac{t}{RC}\right)} \simeq 1 - \frac{t}{RC}$$

For $RC \gg \frac{T}{2}$,

$$e^{-\left(\frac{t}{RC}\right)} \simeq 1 - \left(\frac{T}{2}\right)\left(\frac{1}{RC}\right)$$

Use this approximation in the equation.

$$(V_p - 2V_D)\left[1 - \left(\frac{T}{2}\right)\left(\frac{1}{RC}\right)\right] = (V_p - 2V_D - V_r)$$

$$(V_p - 2V_D) - (V_p - 2V_D)\left(\frac{T}{2}\right)\left(\frac{1}{RC}\right) = V_p - 2V_D - V_r$$

$$V_r = (V_p - 2V_D)\left(\frac{T}{2RC}\right)$$

The expression for the time period is, $T = \frac{1}{f}$. Therefore,

$$V_r = (V_p - 2V_D)\left(\frac{1}{2fRC}\right)$$

Step 4 of 10

Simplify the expression to get an expression for the capacitor, C .

$$C = \frac{V_p - 2V_D}{2V_r fR}$$

Substitute 16.97 V for V_p , 0.8 V for V_D , 1 V for V_r , 60 Hz for f , and $100\ \Omega$ for R .

$$\begin{aligned} C &= \frac{16.97 - 2(0.8)}{2(1)(60)(100)} \\ &= \frac{15.37}{12 \times 10^3} \\ &\simeq 1281 \times 10^{-6} \\ &= 1281\ \mu\text{F} \end{aligned}$$

Thus, the value of the capacitor required is, $\boxed{1281\ \mu\text{F}}$.

Step 5 of 10

The dc voltage at the output is,

$$\begin{aligned} v_{O(dc)} &= V_p - 2V_D \\ &= 16.97 - 2(0.8) \\ &= 15.37\ \text{V} \end{aligned}$$

Therefore, the dc voltage at the output is, $\boxed{15.37\ \text{V}}$.

Consider the ripple voltage to calculate the dc voltage at the output.

$$\begin{aligned} v_{O(dc)} &= V_p - 2V_D - \frac{V_r}{2} \\ &= 16.97 - 2(0.8) - \frac{1}{2} \\ &= 15.37 - 0.5 \\ &= 14.87\ \text{V} \end{aligned}$$

Therefore, the dc voltage at the output for better estimate is, $\boxed{14.87\ \text{V}}$.

Step 6 of 10

Calculate the load current without considering the ripple voltage.

$$I_L = \frac{v_{O(dc)}}{R}$$

Substitute 15.37 V for $v_{O(dc)}$, and $100\ \Omega$ for R .

$$\begin{aligned} I_L &= \frac{15.37\ \text{V}}{100\ \Omega} \\ &= 0.1537\ \text{A} \end{aligned}$$

Therefore, the load current, I_L is, $\boxed{0.15\ \text{A}}$.

Step 7 of 10

Use the following equation to find the diode's conduction angle:

$$\omega\Delta t \simeq \sqrt{\frac{2V_r}{V_p}}$$

The effect of diode voltage is introduced in the peak voltage, V_p . Hence,

$$\omega\Delta t \simeq \sqrt{\frac{2V_r}{V_p - 2V_D}}$$

Substitute 16.97 V for V_p , 0.8 V for V_D , and 1 V for V_r .

$$\begin{aligned} \omega\Delta t &= \sqrt{\frac{2(1)}{16.97 - 2(0.8)}} \\ &= \sqrt{\frac{2}{15.37}} \\ &\simeq 0.36\ \text{rad} \\ &= 20.7^\circ \end{aligned}$$

Thus, the diode's conduction angle is, $\boxed{0.36\ \text{rad or } 20.7^\circ}$.

Step 8 of 10

The average diode current is,

$$i_{Dav} = I_L\left(1 + \pi\sqrt{\frac{V_p}{2V_r}}\right)$$

Here, the load current and the peak voltage are affected by the ripple voltage and hence,

$$I_L = \frac{v_{O(dc)}}{R}$$

But here, $v_{O(dc)} = 14.87\ \text{V}$. That implies,

$$\begin{aligned} I_L &= \frac{14.87}{100} \\ &\simeq 0.149 \end{aligned}$$

Therefore,

$$\begin{aligned} i_{Dav} &= (0.149)\left(1 + (3.14)\sqrt{\frac{16.97 - 2(0.8)}{2(1)}}\right) \\ &= (0.149)(9.7) \\ &= 1.44\ \text{A} \end{aligned}$$

Therefore, the average diode current, i_{Dav} is, $\boxed{1.44\ \text{A}}$.

Step 9 of 10

The peak diode current is,

$$i_{Dmax} = I_L\left(1 + 2\pi\sqrt{\frac{V_p}{2V_r}}\right)$$

Here, the load current and the peak voltage are affected by the ripple voltage and hence,

$$I_L = \frac{v_{O(dc)}}{R}$$

But here, $v_{O(dc)} = 14.87\ \text{V}$. That implies,

$$\begin{aligned} I_L &= \frac{14.87}{100} \\ &\simeq 0.149 \end{aligned}$$

Therefore,

$$\begin{aligned} i_{Dmax} &= (0.149)\left(1 + 2(3.14)\sqrt{\frac{16.97 - 2(0.8)}{2(1)}}\right) \\ &= (0.149)(18.4) \\ &= 2.74\ \text{A} \end{aligned}$$

Therefore, the peak diode current, i_{Dmax} is, $\boxed{2.74\ \text{A}}$.

Step 10 of 10

The peak inverse voltage of the diodes is,

$$\text{PIV} = V_S - V_D$$

Note that, $V_S = V_p$.

Substitute 16.97 V for V_p , and 0.8 V for V_D .

$$\begin{aligned} \text{PIV} &= V_p - V_D \\ &= 16.97 - 0.8 \\ &= 16.17 \end{aligned}$$

Therefore, the peak inverse voltage is, $\boxed{16.17\ \text{V}}$.

Thus select the diode with 3.5 A to 4 A peak current and a 20 V PIV rating.

4.023P

We don't have the solution to this problem yet.

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Step 1 of 3

4.024E

Refer to Figure 4.27 (a) in the textbook for a "super node" precision half-wave rectifier.

Case 1: For $v_I = 10\ \text{mV}$.

The voltage at positive terminal of op-amp is greater than the voltage at the negative terminal, that is, $V^+ > V^-$. Therefore, output of the op-amp is, $+V_{sat}$. Then the diode is in conduction state. Due to diode conduction, there is a possibility of closed loop path. At this condition, due to virtual ground concept, the output voltage is equal to input voltage.

Therefore, the rectifier output, v_O is, $\boxed{10\ \text{mV}}$.

The output of the op-amp with respect to ground is,

$$\begin{aligned} v_A &= V_D + v_O \\ &= 0.7 + 10\ \text{mV} \\ &= 0.71\ \text{V} \end{aligned}$$

Therefore, the output of the op-amp, v_A is, $\boxed{0.71\ \text{V}}$.

Step 2 of 3

Case 2: For $v_I = 1\ \text{V}$.

The voltage at positive terminal of op-amp is greater than the voltage at the negative terminal, that is $V^+ > V^-$. Therefore, output of the op-amp is, $+V_{sat}$. Then the diode is in conduction state. Due to diode conduction, there is a possibility of closed loop path. At this condition, due to virtual ground concept, the output voltage is equal to input voltage.

Therefore, the rectifier output, v_O is, $\boxed{1\ \text{V}}$.

The output of the op-amp with respect to ground is,

$$\begin{aligned} v_A &= V_D + v_O \\ &= 0.7 + 1 \\ &= 1.7\ \text{V} \end{aligned}$$

Therefore, the output of the op-amp, v_A is, $\boxed{1.7\ \text{V}}$.

Step 3 of 3

Case 3: For $v_I = -1\ \text{V}$.

The voltage at negative terminal of op-amp is greater than the voltage at the positive terminal, that is, $V^+ < V^-$. Therefore, the output of the op-amp is, $-V_{sat}$. Then the diode is in cutoff condition and hence there is an open connection between the output of the op-amp and the output of the rectifier.

Therefore, the rectifier output, v_O is, $\boxed{0\ \text{V}}$.

Then, the voltage, $-V_{sat}$ directly appears at the output of the opamp. Therefore, the output of the op-amp, v_A is $\boxed{-12\ \text{V}}$.

Step 1 of 3

4.024P

Draw the circuit of a junction diode supplied with a constant current I and an identical diode connected in parallel with it.

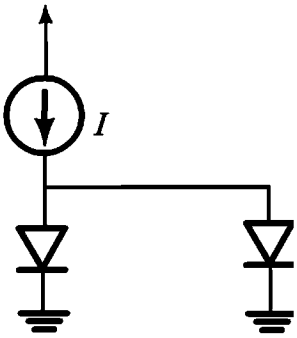


Figure 1

Step 2 of 3

The current through the diode without parallel diode is equal to I.

Write the formula for exponential relationship between the current and the voltage of the diode.

$$i = I_s e^{\frac{v}{V_T}}$$

Here,

i is the diode current

V_T is the thermal voltage

Equate the current to I.

$$I = I_s e^{\frac{v}{V_T}} \quad \text{..... (1)}$$

As the two diodes are identical, the current through each diode of parallel connection in Figure 1 is $\frac{I}{2}$.

The current through the diode is,

$$\frac{I}{2} = I_s e^{\frac{v_1}{V_T}} \quad \text{..... (2)}$$

Step 3 of 3

Divide equation (2) by equation (1).

$$\frac{\frac{I}{2}}{I} = \frac{I_s e^{\frac{v_1}{V_T}}}{I_s e^{\frac{v}{V_T}}}$$

$$e^{\frac{v_1 - v}{V_T}} = \frac{1}{2}$$

$$\frac{v_1 - v}{V_T} = \ln\left(\frac{1}{2}\right)$$

Substitute 25×10^{-3} V for V_T.

$$\frac{v_1 - v}{25 \times 10^{-3}} = \ln\left(\frac{1}{2}\right)$$

$$v_1 - v = -0.0173$$

$$v_1 = v - 0.0173$$

Thus, the forward voltage of the diode is decreased by **0.0173 V**.

Step 1 of 4

4.025E

Refer to Figure 4.27 (a) in the textbook for a "super node" precision half-wave rectifier.

Consider that the diode in the circuit is reversed. Draw the modified circuit.

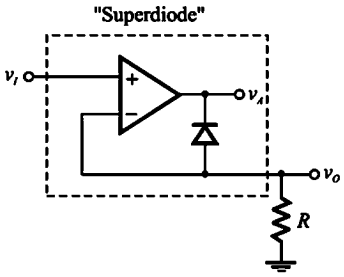


Figure 1

Step 2 of 4

The operation of the super diode in the circuit in Figure 1:

If $v_I \geq 0$, since the output of the op-amp is connected to the negative terminal of the op-amp, the diode works in the reverse bias.

Note that the diode in reverse bias is cutoff.

Since the op-amp is ideal, there is no current flow through the op-amp terminals.

Hence, in the circuit, the current through the resistor is nothing but the current through the diode.

But since, the diode is cutoff (open circuit), the current through the diode is zero and hence, the current through the resistor is zero.

Since the current through the resistor is zero, there is no voltage drop across the resistor.

Therefore, $v_O = 0$.

Step 3 of 4

If $v_I \leq 0$, since the output of the op-amp is connected to the negative terminal of the op-amp, the diode works in the forward bias.

Note that the diode in forward bias is ON (conducts).

Since the op-amp is ideal, there is no current flow through the op-amp terminals.

Hence, in the circuit, the current through the resistor is nothing but the current through the diode.

Since, the diode is conducting, there is some current flow through the diode and hence there is a current flow through the resistor.

From the circuit, it is clear that, v_O connected to the negative terminal of the op-amp. Since the op-amp is ideal, the voltage at the inverting terminal of the op-amp is same as the voltage at the non-inverting terminal of the op-amp.

Hence, $v_O = v_I$.

Thus, the output voltage in the circuit is,

$$v_O = \begin{cases} 0, & \text{for } v_I \geq 0, \\ v_I, & \text{for } v_I \leq 0. \end{cases}$$

Step 4 of 4

Draw the transfer characteristics of the circuit.

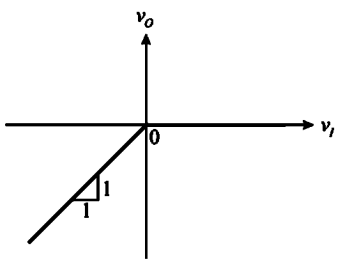


Figure 2

Hence, the transfer characteristic for v_O as function of v_I is shown in Figure 2.

4.025P

We don't have the solution to this problem yet.

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Step 1 of 5

4.026E

Refer to Figure E4.26 in the textbook.

Case (1): for $v_I \le -5\,\text{V}$

The diode D_1 is in forward bias and the diode D_2 is in reverse bias.

An ideal diode acts as short circuit if forward biased and acts as an open circuit if reverse biased.

Draw the equivalent circuit.

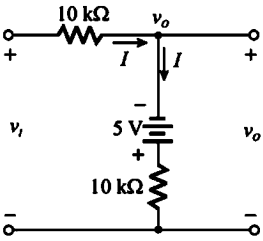


Figure 1

Step 2 of 5

Apply Kirchhoff's Voltage Law at the node voltage v_o in the circuit as shown in Figure 1.

$$\frac{v_I - v_o}{10\times10^3} = \frac{v_o - (-5)}{10\times10^3}$$

$$v_I - v_o = v_o + 5$$

$$v_I = 2v_o + 5$$

Simplify further.

$$v_o = \frac{v_I - 5}{2}$$

$$v_o = \frac{1}{2}v_I - 2.5 \quad \text{..... (1)}$$

Step 3 of 5

Case (2): for $-5\,\text{V} \le v_I \le +5\,\text{V}$

Both the diodes D_1 and D_2 are in reverse bias.

Open the diodes D_1 and D_2 as shown in Figure 1 and draw the equivalent circuit.

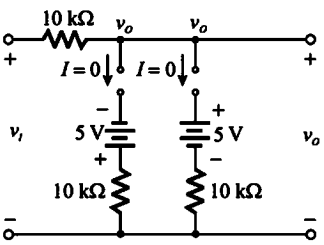


Figure 2

There is no current flow in the circuit and the input will appear across the output.

$$v_o = v_I \quad \text{..... (2)}$$

Step 4 of 5

Case (3): for $v_I \ge +5\,\text{V}$

The diode D_1 is in reverse bias and the diode D_2 is in forward bias.

The diode D_1 acts as open and the diode D_2 acts as short.

Draw the equivalent circuit.

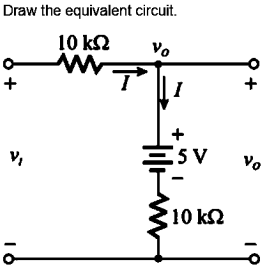


Figure 3

Step 5 of 5

Apply Kirchhoff's Voltage Law at the node voltage v_o as shown in Figure 4.

$$\frac{v_I - v_o}{10\times10^3} = \frac{v_o - (5)}{10\times10^3}$$

$$v_I - v_o = v_o - 5$$

$$v_I = 2v_o - 5$$

Simplify further.

$$v_o = \frac{v_I + 5}{2}$$

$$v_o = \frac{1}{2}v_I + 2.5 \quad \text{..... (3)}$$

Use equations (1), (2) and (3) to obtain the transfer characteristics.

Therefore, the transfer characteristic of the circuit is,

$$\begin{cases} v_o = \frac{1}{2}v_I - 2.5 & \text{for } v_I \le -5\,\text{V} \\ v_o = v_I & \text{for } -5\,\text{V} \le v_I \le +5\,\text{V} \\ v_o = \frac{1}{2}v_I + 2.5 & \text{for } v_I \ge +5\,\text{V} \end{cases}.$$

4.026P

We don't have the solution to this problem yet.

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Step 1 of 3

4.027E

Refer to the diode circuit in Figure 4.32 in the text book.

Reverse the diode and draw the circuit.

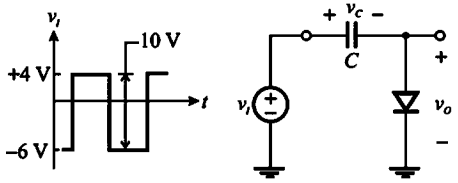


Figure 1

Step 2 of 3

In the circuit shown in Figure 1, because of the polarity in which the diode is connected, the capacitor charges to a voltage, v_C with the polarity indicated in the Figure 1 and equal to the magnitude of the most positive peak of the input signal. Subsequently, the diode turns off and the capacitor retains its voltage indefinitely.

The voltage across the capacitor is,

$$v_C = +4 \text{ V}$$

Apply Kirchhoff's voltage law to the circuit in Figure 1.

$$v_i - v_C - v_o = 0$$
$$v_o = v_i - v_C$$
$$v_o = v_i - 4$$

Step 3 of 3

Draw the output waveform.

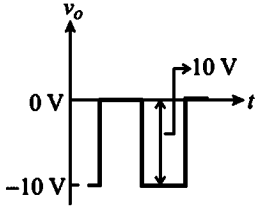


Figure 2

Determine the dc component of v_o.

$$v_{o(dc)} = \frac{v_{o(min)} + v_{o(max)}}{2}$$
$$= \frac{-10 + 0}{2}$$
$$= -5 \text{ V}$$

Therefore, the dc component of v_o is, $\boxed{-5 \text{ V}}$.

Step 1 of 2

4.027P

Refer to Figure P4.27 in the textbook.

The saturation current I_S of a diode is directly proportional to the junction area.

Hence, the saturation current of diode D_1 is 10 times that of D_2.

Consider that the current across diode D_2 be I_1 and I_2 be the current across diode D_1.

The diode current I_2 is,

$$I_2 = I_S e^{\frac{V_{D1}}{V_T}} \quad \text{..... (1)}$$

The current through the diode D_2 is,

$$I_1 - I_2 = \frac{I_S}{10} e^{\frac{V_{D2}}{V_T}} \quad \text{..... (2)}$$

Divide equation (2) by equation (1).

$$\frac{I_1 - I_2}{I_2} = \frac{\frac{I_S}{10} e^{\frac{V_{D2}}{V_T}}}{I_S e^{\frac{V_{D1}}{V_T}}}$$

$$\frac{I_1 - I_2}{I_2} = \frac{1}{10} e^{\frac{V_{D2} - V_{D1}}{V_T}}$$

Substitute 10 mA for I_1, 3 mA for I_2 and 0.025 V for V_T.

$$\frac{10-3}{3} = \frac{1}{10} e^{\frac{V_{D2} - V_{D1}}{0.025}}$$

$$e^{\frac{V_{D2} - V_{D1}}{0.025}} = 23.34$$

$$\frac{V_{D2} - V_{D1}}{0.025} = \ln(23.34)$$

$$V_{D2} - V_{D1} = 0.0787 \text{ V}$$

The output voltage is,

$$V = V_{D2} - V_{D1}$$
$$= 78.7 \text{ mV}$$

Thus, the value of V is **78.7 mV**.

Step 2 of 2

Consider the following relation:

$$\frac{I_1 - I_2}{I_2} = \frac{1}{10} e^{\frac{V_{D2} - V_{D1}}{V_T}}$$

$$\frac{I_1 - I_2}{I_2} = \frac{1}{10} e^{\frac{V}{V_T}}$$

Substitute 10 mA for I_1, 60 mV for V and 0.025 V for V_T.

$$\frac{10 - I_2}{I_2} = \frac{1}{10} e^{\frac{60 \text{ mV}}{0.025}}$$

$$\frac{10 - I_2}{I_2} = 1.1$$

$$10 - I_2 = 1.1 I_2$$

$$I_2 = 4.75 \text{ mA}$$

Thus, the current I_2 to get a voltage of 60 mV is **4.75 mA**.

Step 1 of 5

4.028P

Refer to Figure P4.28 in the textbook.

Represent the current directions and draw the circuit.

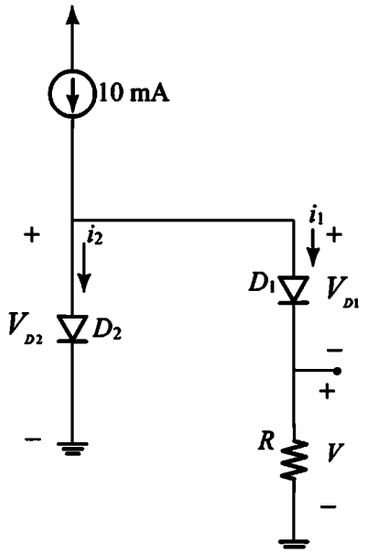


Figure 1

Step 2 of 5

Consider that the voltage across resistor R is, $V = 50\ \text{mV}$.

Apply Kirchhoff's voltage law to the circuit.

$$-V_{D2} + V_{D1} + V = 0$$
$$V = V_{D2} - V_{D1}$$

Apply Kirchhoff's current law to the circuit.

$$i_1 + i_2 = 10\ \text{mA}$$

$$i_2 = 10\ \text{mA} - i_1 \quad \text{..... (1)}$$

Use the following formula to calculate the value of current i_1 .

$$i_1 = I_s e^{\left(\frac{V_{D1}}{V_T}\right)} \quad \text{..... (2)}$$

Use the following formula to calculate the value of current i_2 .

$$i_2 = I_s e^{\left(\frac{V_{D2}}{V_T}\right)} \quad \text{..... (3)}$$

Step 3 of 5

Divide equation (2) by equation (1).

$$\frac{i_2}{i_1} = \frac{I_s e^{\left(\frac{V_{D2}}{V_T}\right)}}{I_s e^{\left(\frac{V_{D1}}{V_T}\right)}}$$
$$= \frac{e^{\left(\frac{V_{D2}}{V_T}\right)}}{e^{\left(\frac{V_{D1}}{V_T}\right)}}$$
$$= e^{\left(\frac{V_{D2}}{V_T}\right)\left(\frac{V_{D1}}{V_T}\right)}$$
$$= e^{\left(\frac{V_{D2} - V_{D1}}{V_T}\right)}$$

Apply logarithm on both sides.

$$\ln\left(\frac{i_2}{i_1}\right) = \left(\frac{V_{D2} - V_{D1}}{V_T}\right)$$
$$\ln\left(\frac{i_2}{i_1}\right) = \left(\frac{V}{V_T}\right) \qquad (\text{since}, V = V_{D2} - V_{D1})$$

$$V = V_T \ln\left(\frac{i_2}{i_1}\right) \quad \text{..... (4)}$$

Step 4 of 5

Substitute 50 mV for V and 0.025 mV for V_T .

$$\left(50 \times 10^{-3}\right) = \left(25 \times 10^{-3}\right) \ln\left(\frac{i_2}{i_1}\right)$$

$$\ln\left(\frac{i_2}{i_1}\right) = 2$$

$$\frac{i_2}{i_1} = e^2$$

$$\frac{i_2}{i_1} = 7.389$$

Substitute $i_2 = 10\ \text{mA} - i_1$ from equation (1).

$$\frac{10\ \text{mA} - i_1}{i_1} = 7.389$$
$$0.01 - i_1 = 7.389 i_1$$
$$0.01 = 8.389 i_1$$

$$i_1 = \frac{0.01}{8.839}$$
$$= 1.192\ \text{mA}$$

Step 5 of 5

Use Ohm's law to calculate the value of R .

$$R = \frac{V}{i_1}$$

Substitute 50 mV for V and 1.192 mA for i_1 .

$$R = \frac{50\ \text{mV}}{1.1932\ \text{mA}}$$
$$= 42\ \Omega$$

Therefore, the value of resistance R is $\boxed{42\ \Omega}$.

Step 1 of 3

Write the expression for the current through the diode.

$$I = I_S e^{\frac{V}{V_T}}$$

Here,

I_S is the saturation current of the diode

V_T is the thermal voltage

I is the current through the diode

V is the diode voltage

At $20°\ \mathrm{C}$, the thermal voltage, $V_T = 25\ \mathrm{mV}$

Substitute 1 mA for I, 690 mV for V and 25 mV for V_T.

$$1 = I_S e^{\frac{690}{25}}$$

$$I_S = \frac{1}{e^{\frac{690}{25}}}$$

$$= 1.03\times10^{-12}\ \mathrm{mA}$$

$$= 1.03\times10^{-15}\ \mathrm{A}$$

Step 2 of 3

Write the formula for diode voltage v.

$$v = V_T \ln\left(\frac{i}{I_S}\right)$$

$$= \frac{kT}{q}\ln\left(\frac{i}{I_S}\right)$$

Substitute $1.38\times10^{-23}\ \mathrm{joules/kelvin}$ for k, $1.6\times10^{-19}\ \mathrm{coulomb}$ for q, $10^{-3}\ \mathrm{A}$ for i, $1.03\times10^{-15}\ \mathrm{A}$ for I_S and $(273-20)\ \mathrm{K}$ for T.

$$v = \frac{(1.38\times10^{-23})(273-20)}{1.6\times10^{-19}}\ln\left(\frac{10^{-3}}{1.03\times10^{-15}}\right)$$

$$= 0.0218\ln\left(0.971\times10^{12}\right)$$

$$= 0.6017\ \mathrm{V}$$

$$= 601.7\ \mathrm{mV}$$

Thus, the diode voltage at $-20°\ \mathrm{C}$ is $\boxed{601.7\ \mathrm{mV}}$.

Step 3 of 3

Determine the diode voltage v at $85°\ \mathrm{C}$.

$$v = V_T \ln\left(\frac{i}{I_S}\right)$$

$$= \frac{kT}{q}\ln\left(\frac{i}{I_S}\right)$$

Substitute $1.38\times10^{-23}\ \mathrm{joules/kelvin}$ for k, $1.6\times10^{-19}\ \mathrm{coulomb}$ for q, $10^{-3}\ \mathrm{A}$ for i, $1.03\times10^{-15}\ \mathrm{A}$ for I_S and $(273+85)\ \mathrm{K}$ for T.

$$v = \frac{(1.38\times10^{-23})(273+85)}{1.6\times10^{-19}}\ln\left(\frac{10^{-3}}{1.03\times10^{-15}}\right)$$

$$= 0.0309\ln\left(0.971\times10^{12}\right)$$

$$= 0.8529\ \mathrm{V}$$

$$= 852.9\ \mathrm{mV}$$

Thus, the diode voltage at $85°\ \mathrm{C}$ is $\boxed{852.9\ \mathrm{mV}}$.

Step 1 of 3

Refer to Figure P4.28 for the circuit.

The voltage across the resistance is,

$$V_{R1} = 520 \text{ mV}$$

$$IR_1 = 520 \times 10^{-3}$$

Substitute $520 \times 10^3\ \Omega$ for R_1.

$$I\left(520 \times 10^3\right) = 520 \times 10^{-3}$$

$$I = 10^{-6} \text{ A}$$

$$= 1\ \mu\text{A}$$

For reverse-biased diode, the diode current is equal to the saturation current.

$$I = I_S = 1\ \mu\text{A}$$

The reverse voltage across the diode D_1 is,

$$V_1 = 10 - \left(V_{R1} + V_2\right)$$

$$= 10 - 0.52 - 0.52$$

$$= 8.96 \text{ V}$$

Step 2 of 3

As the reverse current doubles for every $10°\text{C}$ rise in temperature, the reverse current at $0°\text{C}$ is,

$$I_{S(0°\text{C})} = \frac{I_S}{4}$$

$$= \frac{1\ \mu\text{A}}{4}$$

$$= 0.25\ \mu\text{A}$$

Determine the voltage across the resistance, V_{R1}.

$$V_{R1} = R_1 I_{S(0°\text{C})}$$

$$= \left(520 \times 10^3\right)\left(0.25 \times 10^{-6}\right)$$

$$= 0.13 \text{ V}$$

$$= 130 \text{ mV}$$

Thus, the voltage V_{R1} at $0°\text{C}$ is $\boxed{130 \text{ mV}}$.

At a constant diode current, the voltage drop across the diode decreases by approximately 2 mV for every $1°\text{C}$ increase in the temperature. Hence, at a temperature of $0°\text{C}$ the voltage drop is across D_2 is,

$$V_2 = 520 + 40$$

$$= 560 \text{ mV}$$

Thus, the voltage V_2 at $0°\text{C}$ is $\boxed{560 \text{ mV}}$.

Step 3 of 3

As the reverse current doubles for every $10°\text{C}$ rise in temperature, the reverse current at $40°\text{C}$ is,

$$I_{S(40°\text{C})} = 4I_S$$

$$= (4)(1\ \mu\text{A})$$

$$= 4\ \mu\text{A}$$

Determine the voltage across the resistance, V_{R1}.

$$V_{R1} = R_1 I_{S(40°\text{C})}$$

$$= \left(520 \times 10^3\right)\left(4 \times 10^{-6}\right)$$

$$= 2.08 \text{ V}$$

Thus, the voltage V_{R1} at $40°\text{C}$ is $\boxed{2.08 \text{ V}}$.

At a constant diode current, the voltage drop across the diode decreases by approximately 2 mV for every $1°\text{C}$ increase in the temperature. Hence, at a temperature of $0°\text{C}$ the voltage drop is across D_2 is,

$$V_2 = 520 - (20)(2)$$

$$= 520 - 40$$

$$= 480 \text{ mV}$$

Thus, the voltage V_2 at $40°\text{C}$ is $\boxed{480 \text{ mV}}$.

4.031P

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Step 1 of 3

Determine the thermal voltage at $20\ ^\circ\text{C}$.

$$\begin{aligned} V_T &= \frac{kT}{q} \\ &= \frac{(1.38\times10^{-23})(273+20)}{1.60\times10^{-19}} \\ &= 0.02527\ \text{V} \\ &= 25.27\ \text{mV} \end{aligned}$$

The *i*–*v* relationship of a diode is,

$$i = I_S\left(e^{\frac{v}{V_T}} - 1\right)$$

Here,

I_S is the saturation current of the diode

V_T is the thermal temperature

Substitute 0.8 V for *v*, 10 A for *i*, $25.27\times10^{-3}\ \text{V}$ for V_T.

$$\begin{aligned} 10 &= I_S\left(e^{\frac{0.8}{25.27\times10^{-3}}} - 1\right) \\ I_S &= \frac{10}{e^{\frac{0.8}{25.27\times10^{-3}}} - 1} \\ &= 1.783\times10^{-13}\ \text{A} \end{aligned}$$

Step 2 of 3

Write the formula for diode voltage, *V*.

$$v = V_T \ln\left(\frac{i}{I_S}\right)$$

Substitute $25.27\times10^{-3}\ \text{V}$ for V_T, $1.783\times10^{-13}\ \text{A}$ for I_S and $1\times10^{-3}\ \text{A}$ for *i*.

$$\begin{aligned} v &= (25.27\times10^{-3})\ln\left(\frac{1\times10^{-3}}{1.783\times10^{-13}}\right) \\ &= 0.5673\ \text{V} \end{aligned}$$

The diode voltage at $20\ ^\circ\text{C}$ and at diode current equal to 1 mA is 0.5673 V.

Determine the diode voltage at diode current equal to 3 mA and at $25\ ^\circ\text{C}$.

$$\begin{aligned} v &= (25.27\times10^{-3})\ln\left(\frac{3\times10^{-3}}{1.783\times10^{-13}}\right) \\ &= 0.5950\ \text{V} \end{aligned}$$

Step 3 of 3

The temperature changes to $-20\ ^\circ\text{C}$.

Determine the thermal voltage at $-20\ ^\circ\text{C}$.

$$\begin{aligned} V_T &= \frac{kT}{q} \\ &= \frac{(1.38\times10^{-23})(273-20)}{1.60\times10^{-19}} \\ &= 0.0218\ \text{V} \\ &= 21.8\ \text{mV} \end{aligned}$$

Determine the diode voltage at diode current equal to 3 mA and at $-20\ ^\circ\text{C}$.

Substitute $21.8\times10^{-3}\ \text{V}$ for V_T, $1.783\times10^{-13}\ \text{A}$ for I_S and $3\times10^{-3}\ \text{A}$ for *I*.

$$\begin{aligned} v &= (21.8\times10^{-3})\ln\left(\frac{3\times10^{-3}}{1.783\times10^{-13}}\right) \\ &= 0.5133\ \text{V} \end{aligned}$$

Thus, the expected range of the diode voltage is $\boxed{0.5133\ \text{V} \le v \le 0.5950\ \text{V}}$.

Step 1 of 3

The two diodes operate for a forward voltage of 700 mV.

One diode D_1 conducts with a saturation current $i_{s_1} = 0.1\ \text{mA}$.

Other diode D_2 with a saturation current $i_{s_2} = 1\ \text{A}$.

Consider the expression for diode current in forward region.

$$i = I_S e^{\frac{v}{nV_T}}$$

Let us assume the voltage across the diode D_1 is v_1.

$$i_1 = i_{s_1} e^{\frac{v_1}{nV_T}}$$

Let us assume the voltage across the diode D_2 is v_2.

$$i_2 = i_{s_2} e^{\frac{v_2}{nV_T}}$$

Designer proposed to use the idea to use supply two diodes of different junction areas with equal currents.

$$i_1 = i_2$$

$$i_{S_1} e^{\frac{v_1}{nV_T}} = i_{S_2} e^{\frac{v_2}{nV_T}}$$

$$\frac{e^{\frac{v_1}{nV_T}}}{e^{\frac{v_2}{nV_T}}} = \frac{i_{S_2}}{i_{S_1}}$$

$$e^{\frac{v_1}{nV_T} - \frac{v_2}{nV_T}} = \frac{i_{S_2}}{i_{S_1}}$$

$$e^{\frac{v_1 - v_2}{nV_T}} = \frac{i_{S_2}}{i_{S_1}}$$

Assume $n = 1$ and $V_T = 25\ \text{mV}$.

$$e^{\frac{v_1 - v_2}{(0.025)}} = \frac{1}{0.1 \times 10^{-3}}$$

$$e^{\frac{v_1 - v_2}{(0.025)}} = 10^4$$

$$\frac{v_1 - v_2}{0.025} = \ln 10^4$$

$$v_1 - v_2 = 0.025 \times \ln 10^4$$

$$v_1 - v_2 = 0.23\ \text{V}$$

Therefore, the voltage difference between two diodes is $\boxed{0.23\ \text{V}}$ which is independent of current.

Step 2 of 3

Now the current diodes D_1 and D_2 with saturation currents of $i_{s_1} = 1\ \text{mA}$ to $i_{s_2} = 3\ \text{mA}$.

Find the difference in the voltage range.

$$e^{\frac{v_1 - v_2}{nV_T}} = \frac{I_{S_2}}{I_{S_1}}$$

$$\frac{v_1 - v_2}{(1)0.025} = \ln \frac{I_{S_2}}{I_{S_1}}$$

$$v_1 - v_2 = 0.025 \ln \frac{3}{1}$$

$$v_1 - v_2 = 0.02746$$

$$v_1 - v_2 = 27.46\ \text{mV}.$$

Difference voltage is $0.23 \pm 2.75\ \text{mV}$.

Therefore, the voltage difference between the diodes is 0.230 V independent of the current. However the two currents can vary by a factor of 3 (for the identical currents in the range of 0.5 mA to 1.5 mA supplied to each).

Therefore, the difference in the voltage will be within the limit $0.23\ \text{V} \pm nV_T \ln(3)$.

Step 3 of 3

The relation for finding the range of the difference in the voltages is,

$$\begin{aligned}\textbf{Range of the voltage} &= 0.23\ \text{V} \pm nV_T \ln(3) \\ &= 0.23 \pm 1 \times 25 \times 10^{-3} \times 1.0986 \\ &= (0.23 \pm 0.027)\ \text{V}\end{aligned}$$

Therefore, the difference in the voltage lies in the range of 0.257 and 0.203. It provides the range of the voltage in between $\boxed{0.257\ \text{V and } 0.203\ \text{V}}$.

Since, the temperature changes affect both the diodes similarly the difference voltage remains constant even the temperature changes by $\pm 25°\text{C}$.

Step 1 of 18

Refer to Figure 4.10 in the text book.

Consider the value of n is,

$n = 1$

Consider the diode current is much greater than I_s , and the diode $i-v$ characteristic by the exponential relationship is,

$$I_D = I_s e^{\frac{V_D}{nV_T}}$$

Substitute $10^{-15}\,\text{A}$ for I_s , 1 for n and $25\,\text{mV}$ for V_T in equation.

$$I_D = (10^{-15}) e^{\frac{V_D}{(1)(25\times10^{-3})}}$$

$$I_D = (10^{-15}) e^{\frac{V_D}{25\times10^{-3}}}\ \text{A} \quad \ldots\ldots (1)$$

Step 2 of 18

Apply Kirchhoff's voltage law.

$$-V_{DD} + I_D R + V_D = 0$$
$$I_D R = V_{DD} - V_D$$
$$I_D = \frac{V_{DD} - V_D}{R}$$

Substitute $1\,\text{V}$ for V_{DD} , and $1\,\text{k}\Omega$ for R in equation.

$$I_D = \frac{1 - V_D}{1\times10^3} \quad \ldots\ldots (2)$$

Step 3 of 18

Calculate small number of points on the diode characteristics in the vicinity.

Varying diode voltage (V_D) from 0 to $0.8\,\text{V}$, the corresponding diode current (I_D) values are:

Substitute $0\,\text{V}$ for V_D in equation (1).

$$I_D = (10^{-15}) e^{\frac{V_D}{25\times10^{-3}}}$$
$$= (10^{-15}) e^{\frac{0}{25\times10^{-3}}}$$
$$= (10^{-15}) e^{0}$$
$$= (10^{-15})(1)$$
$$I_D = 10^{-15}$$
$$\approx 0\,\text{A}$$

So the diode current is approximately zero.

Step 4 of 18

Substitute $0.2\,\text{V}$ for V_D in equation (1).

$$I_D = (10^{-15}) e^{\frac{V_D}{25\times10^{-3}}}$$
$$= (10^{-15}) e^{\frac{0.2}{25\times10^{-3}}}$$
$$= (10^{-15}) e^{8}$$
$$= (10^{-15})(2980.96)$$
$$I_D = 2.98\times10^{-12}$$
$$\approx 0\,\text{A}$$

So the diode current is approximately zero.

Step 5 of 18

Substitute $0.4\,\text{V}$ for V_D in equation (1).

$$I_D = (10^{-15}) e^{\frac{V_D}{25\times10^{-3}}}$$
$$= (10^{-15}) e^{\frac{0.4}{25\times10^{-3}}}$$
$$= (10^{-15}) e^{16}$$
$$= (10^{-15})(8.886\times10^{6})$$
$$I_D = 8.89\times10^{-9}$$
$$\approx 0\,\text{A}$$

So the diode current is approximately zero.

Step 6 of 18

Substitute $0.6\,\text{V}$ for V_D in equation (1).

$$I_D = (10^{-15}) e^{\frac{V_D}{25\times10^{-3}}}$$
$$= (10^{-15}) e^{\frac{0.6}{25\times10^{-3}}}$$
$$= (10^{-15}) e^{24}$$
$$= (10^{-15})(2.65\times10^{10})$$
$$I_D = 0.026\times10^{-3}$$
$$= 0.026\,\text{mA}$$

Step 7 of 18

Substitute $0.8\,\text{V}$ for V_D in equation (1).

$$I_D = (10^{-15}) e^{\frac{V_D}{25\times10^{-3}}}$$
$$= (10^{-15}) e^{\frac{0.8}{25\times10^{-3}}}$$
$$= (10^{-15}) e^{32}$$
$$= (10^{-15})(7.896\times10^{13})$$
$$I_D = 0.0789$$

Step 8 of 18

Substitute $V_D = 0\,\text{V}$ in equation-(2).

$$I_D = \frac{1 - V_D}{1\times10^3}$$
$$= \frac{1 - 0}{1\times10^3}$$
$$= 1\times10^{-3}$$
$$I_D = 1\,\text{mA}.$$

Step 9 of 18

Substitute $I_D = 0\ \text{A}$ in equation-(2)

$$I_D = \frac{1 - V_D}{1\times10^3}$$
$$0 = \frac{1 - V_D}{1\times10^3}$$
$$0 = 1 - V_D$$
$$V_D = 1$$

Step 10 of 18

The graphical representation of diode characteristics and load line is shown in Figure 1.

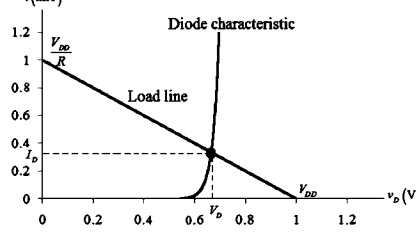


Figure 1

Step 11 of 18

From Figure 1, the operating point must lie between $V_D = 0.65$ to $0.7\,\text{V}$ and corresponding diode currents are $I_D \approx 0.3\,\text{mA}$ and $I_D \approx 0.4\,\text{mA}$.

Consider the diode voltage equation.

$$V_D = nV_T \ln\frac{I_D}{I_s} \quad \ldots\ldots (3)$$

Substitute $10^{-15}\,\text{A}$ for I_s , 1 for n and $25\,\text{mV}$ for V_T in equation.

$$V_D = (1)(25\times10^{-3}) \ln\frac{I_D}{10^{-15}}$$

$$V_D = (25\times10^{-3}) \ln\frac{I_D}{10^{-15}}$$

Step 12 of 18

Substitute $0.3\,\text{mA}$ for I_D in equation-(3).

$$V_D = (25\times10^{-3}) \ln\frac{0.3\times10^{-3}}{10^{-15}}$$
$$= (25\times10^{-3}) \ln(3\times10^{11})$$
$$= (25\times10^{-3})(26.427)$$
$$= 0.6606\ \text{V}$$

Step 13 of 18

Substitute $0.4\,\text{mA}$ for I_D in equation-(3).

$$V_D = (25\times10^{-3}) \ln\frac{0.4\times10^{-3}}{10^{-15}}$$
$$= (25\times10^{-3}) \ln(4\times10^{11})$$
$$= (25\times10^{-3})(26.715)$$
$$= 0.6678\ \text{V}$$

Step 14 of 18

Using equations-(1) and (2), varying drain current (I_D) values from $0.3\,\text{mA}$ to $0.4\,\text{mA}$, the diode voltage (V_D) and load line values are

I_D	V_D	Load line
0.3	0.660676	0.339324
0.31	0.661496	0.338504
0.32	0.66229	0.33771
0.33	0.663059	0.336941
0.34	0.663805	0.336195
0.35	0.66453	0.33547
0.36	0.665234	0.334766
0.37	0.665919	0.334081
0.38	0.666586	0.333414
0.39	0.667235	0.332765
0.4	0.667868	0.332132

Step 15 of 18

Table 1

Step 16 of 18

The graphical representation of diode characteristics and load line plot is shown in Figure 2

Figure 2

Step 17 of 18

From Figure 2, the operating voltage V_D is $0.6634\,\text{V}$ and corresponding diode current I_D is $0.337\,\text{mA}$.

Consider the diode voltage equation.

$$V_D = nV_T \ln\frac{I_D}{I_s}$$

Substitute $10^{-15}\,\text{A}$ for I_s , 1 for n , $0.34\,\text{mA}$ for I_D and $25\,\text{mV}$ for V_T in equation (3).

$$V_D = (1)(25\times10^{-3}) \ln\frac{0.337\times10^{-3}}{10^{-15}}$$
$$= (25\times10^{-3}) \ln(3.37\times10^{11})$$
$$= (25\times10^{-3})(26.543)$$
$$V_D = 0.6636$$

Therefore, the operating voltage V_D is $0.6636\,\text{V}$ and corresponding diode current I_D is $0.337\,\text{mA}$.

Step 18 of 18

The difference between obtained V_D value and graphical representation V_D value is

$$V_{D\text{-diff}} = V_{D\text{-obtained}} - V_{D\text{-graphical}}$$
$$= 0.6636 - 0.6634$$
$$V_{D\text{-diff}} = 0.2\times10^{-3}$$

Therefore, difference voltage value is $\boxed{V_{D\text{-diff}} = 0.2\,\text{mV}}$.

Step 1 of 6

Consider the circuit diagram is shown in Figure 1.

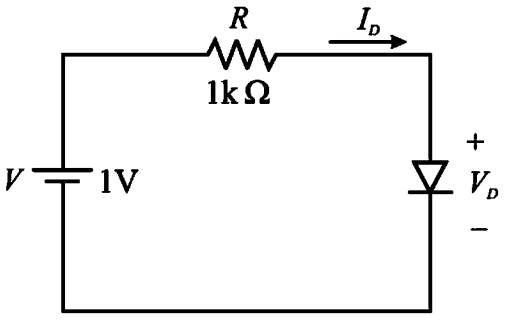


Figure 1

Step 2 of 6

Consider the voltage drop across diode is,

$V_D = 0.7\ \text{V}$

From Figure 1, the current through the diode is,

$$I_D = \frac{V_{DD} - V_D}{R} \quad \text{..... (1)}$$

Substitute the values $1\ \text{V}$ for V_{DD}, $1\ \text{k}\Omega$ for R and $0.7\ \text{V}$ for V_D.

$$I_D = \frac{1 - 0.7}{1 \times 10^3}$$
$$= 0.3\ \text{mA}$$

Step 3 of 6

the relation for the diode current is,

$$I_D = I_S e^{\frac{V_D}{V_T}}$$

Rearrange the equation.

$$V = 0.025 \ln\left(\frac{I_D}{I_S}\right) \quad \text{..... (2)}$$

Substituting the values $0.3\ \text{mA}$ for I_D, $10^{-12}\ \text{mA}$ for I_S in equation.

$$V_D = 0.025 \ln\left(\frac{I_D}{I_S}\right)$$
$$= 0.025 \times \ln\left(\frac{0.3}{10^{-12}}\right)$$
$$= 0.66\ \text{V}$$

Step 4 of 6

Substitute $0.66\ \text{V}$ for V_D, 1V for V_{DD} and $1\ \text{k}\Omega$ for R in equation (1).

$$I_D = \frac{V_{DD} - V_D}{R}$$
$$= \frac{1 - 0.66}{1 \times 10^3}$$
$$I_D = 0.3392\ \text{mA}$$

Step 5 of 6

Substitute $0.34\ \text{mA}$ for I_D and $I_S 10^{-12}\ \text{mA}$ for I_S in equation (2).

$$V_D = 0.025 \ln\left(\frac{I_D}{I_S}\right)$$
$$= 0.025 \times \ln\left(\frac{0.34}{10^{-12}}\right)$$
$$V_D = 0.6638\ \text{V}$$

Step 6 of 6

Substitute $0.6638\ \text{V}$ for V_D, 1V for V_{DD} and $1\ \text{k}\Omega$ for R in equation (1).

$$I_D = \frac{V_{DD} - V_D}{R}$$
$$= \frac{1 - 0.6638}{1 \times 10^3}$$
$$I_D = 0.3362\ \text{mA}$$

The same values of currents and voltages are repeating.

Therefore the voltage across the diode is $\boxed{0.6638\ \text{V}}$ and current through the diode is $\boxed{0.3362\ \text{mA}}$.

4.036P

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4.037P

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Ask an expert

4.038P

Step 1 of 4

The diode operates in series circuit with resistance R and dc voltage V.

Constant-Voltage model:

The diode current is,

$$I_D = \frac{V_{DD} - v_D}{R}$$

Here,

V_{DD} is dc source voltage.

R is the resistor and

v_D is the diode voltage.

For $v_D = 0.7$:

The diode current is,

$$I_{D1} = \frac{V_{DD} - 0.7}{R}$$

For $v_D = 0.6$:

The diode current is,

$$I_{D2} = \frac{V_{DD} - 0.6}{R}$$

Step 2 of 4

Determine the supply voltage for 1% difference of obtained current.

$$I_{D2} = \left(1 + \frac{1}{100}\right) I_{D1}$$

$$\frac{V_{DD} - 0.6}{R} = 1.01\left[\frac{V_{DD} - 0.7}{R}\right]$$

$$V_{DD} - 0.6 = 1.01V_{DD} - 0.707$$

$$0.01V_{DD} = 0.707 - 0.6$$

$$0.01V_{DD} = 0.107$$

$$V_{DD} = 10.7$$

Therefore, the dc source voltage is $\boxed{10.7\ \text{V}}$.

Step 3 of 4

Determine the value of I_{D1} when $v_D = 0.7\ \text{V}$:

$$I_{D1} = \frac{V_{DD} - v_D}{R}$$

$$I_{D1} = \frac{3 - 0.7}{1}$$

$$I_{D1} = 2.3\ \text{mA}.$$

Determine the value of I_{D2} when $v_D = 0.6\ \text{V}$:

$$I_{D2} = \frac{V_{DD} - v_D}{R}$$

$$I_{D2} = \frac{3 - 0.6}{1}$$

$$I_{D2} = 2.4\ \text{mA}$$

Step 4 of 4

The percentage difference of the current is,

$$\%I_D = \left(\frac{I_{D2} - I_{D1}}{I_{D1}}\right) \times 100$$

$$= \left(\frac{2.4 \times 10^{-3} - 2.3 \times 10^{-3}}{2.3 \times 10^{-3}}\right) \times 100$$

$$= \left(\frac{0.1 \times 10^{-3}}{2.3 \times 10^{-3}}\right) \times 100$$

$$= 4.35\%$$

Therefore, the percentage difference in diode currents is $\boxed{4.35\%}$.

4.039P

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Step 1 of 10

(a)

Refer to Figure 4.6(a) in the text book.

Redraw the circuit diagram as shown in Figure 1.

Consider the voltage at node B is V_B .

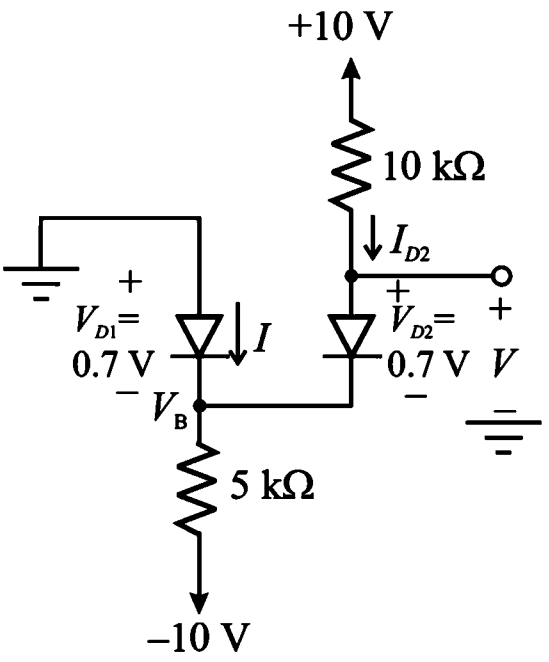


Figure 1

Step 2 of 10

Apply Kirchhoff's voltage law .

The voltage across diode D_1 is,

$$V_{D1} = 0 - V_B$$
$$0.7 = 0 - V_B$$
$$V_B = -0.7$$

Step 3 of 10

Applying Kirchhoff's current law at node V_B :

$$I = \frac{V_B - (-10)}{5\times10^3} + \frac{V_B - (-V_{D2}) - 10}{10\times10^3}$$
$$= \frac{V_B + 10}{5\times10^3} + \frac{V_B + V_{D2} - 10}{10\times10^3}$$
$$= \frac{2(V_B + 10) + (V_B + V_{D2} - 10)}{10\times10^3}$$
$$= \frac{2V_B + 20 + V_B + V_{D2} - 10}{10\times10^3}$$
$$I = \frac{3V_B + V_{D2} + 10}{10\times10^3}$$

Substitute $-0.7\,\text{V}$ for V_B and $V_{D2} = 0.7\,\text{V}$ in equation.

$$I = \frac{3(-0.7) + 0.7 + 10}{10\times10^3}$$
$$= \frac{-2.1 + 0.7 + 10}{10\times10^3}$$
$$= \frac{8.6}{10\times10^3}$$
$$I = 0.86\times10^{-3}$$

Therefore, the value of current *I* is $\boxed{0.86\,\text{mA}}$.

Step 4 of 10

Apply Kirchhoff's voltage law.

The voltage across diode D_2 is

$$V_{D2} = V - V_B$$
$$V = V_{D2} + V_B$$

Substitute $-0.7\,\text{V}$ for V_B and $0.7\,\text{V}$ for V_{D2} in equation.

$$V = 0.7 - 0.7$$
$$V = 0$$

Therefore, the value of *V* is $\boxed{0\,\text{V}}$.

Step 5 of 10

(b)
Refer to Figure 4.6(b) in the text book.

Redraw the circuit diagram as shown in Figure 2.

Consider the voltage at node B is V_B .

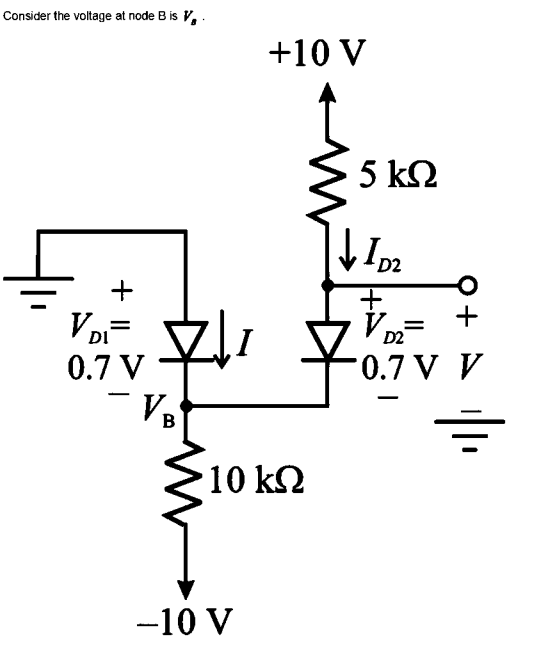


Figure 2

Step 6 of 10

Assuming both diodes are conducting.

Apply Kirchhoff's voltage law.

The voltage across diode D_1 is

$$V_{D1} = 0 - V_B$$
$$0.7 = 0 - V_B$$
$$V_B = -0.7$$

Applying Kirchhoff's current law at node V_B :

$$I = \frac{V_B - (-10)}{10\times10^3} + \frac{V_B + V_{D2} - 10}{5\times10^3}$$
$$= \frac{V_B + 10}{10\times10^3} + \frac{V_B + V_{D2} - 10}{5\times10^3}$$
$$= \frac{(V_B + 10) + 2(V_B + V_{D2} - 10)}{10\times10^3}$$
$$= \frac{V_B + 10 + 2V_B + 2V_{D2} - 20}{10\times10^3}$$
$$I = \frac{3V_B + 2V_{D2} - 10}{10\times10^3}$$

Substitute $-0.7\,\text{V}$ for V_B and $0.7\,\text{V}$ for V_{D2} in equation.

$$I = \frac{3(-0.7) + 2(0.7) - 10}{10\times10^3}$$
$$= \frac{-2.1 + 1.4 - 10}{10\times10^3}$$
$$= \frac{-10.7}{10\times10^3}$$
$$I = -1.07\times10^{-3}$$

Therefore, the value of current *I* is $-1.07\,\text{mA}$.

Since, this is not possible, original assumption is not correct.

Step 7 of 10

Consider that D_1 is off and D_2 is on. Then draw the modified circuit in Figure 2 as shown in Figure 3.

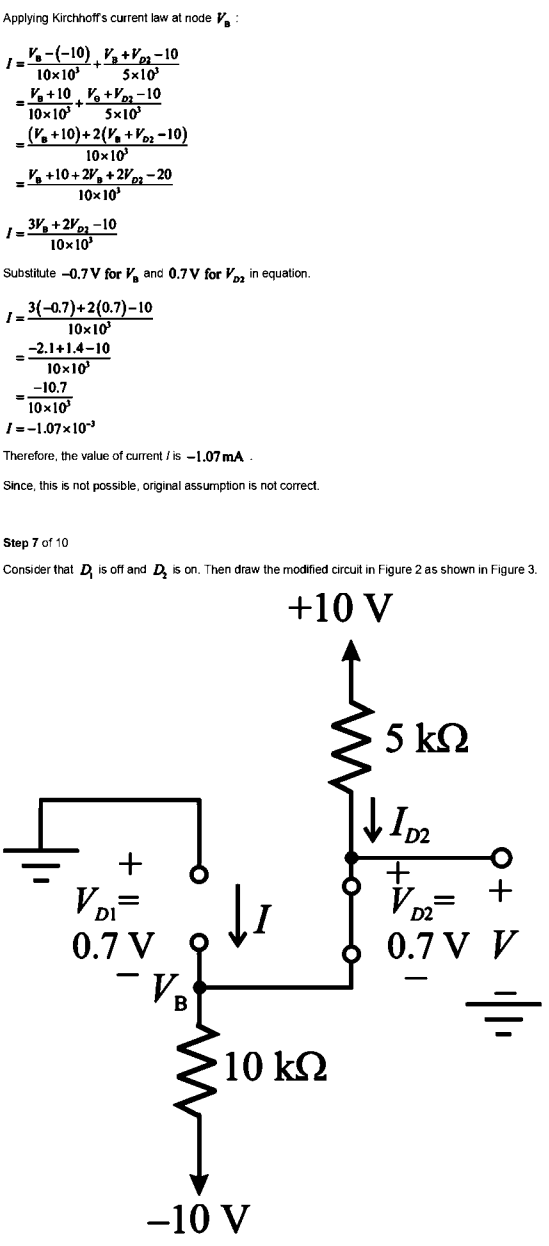


Figure 3

Step 8 of 10

From the above circuit,

The value of current *I*, $I = 0\,\text{A}$.

Applying Kirchhoff's current law at node V_B :

$$I = \frac{V_B - (-10)}{10\times10^3} + \frac{V_B + V_{D2} - 10}{5\times10^3}$$
$$0 = \frac{V_B + 10}{10\times10^3} + \frac{V_B + V_{D2} - 10}{5\times10^3}$$
$$0 = \frac{(V_B + 10) + 2(V_B + V_{D2} - 10)}{10\times10^3}$$
$$0 = \frac{V_B + 10 + 2V_B + 2V_{D2} - 20}{10\times10^3}$$
$$0 = 3V_B + 2V_{D2} - 10$$

Step 9 of 10

Substitute $0.7\,\text{V}$ for V_{D2} in the equation.

$$0 = 3V_B + 2V_{D2} - 10$$
$$0 = 3V_B + 2(0.7) - 10$$
$$0 = 3V_B - 8.6$$
$$3V_B = 8.6$$
$$V_B = \frac{8.6}{3}$$
$$V_B = 2.8667$$

Step 10 of 10

Apply Kirchhoff's voltage law ,the voltage across diode D_2 is

$$V_{D2} = V - V_B$$
$$V = V_{D2} + V_B$$

Substitute $V_B = 2.8667\,\text{V}$ and $V_{D2} = 0.7\,\text{V}$ in the above equation,

$$V = 0.7 + 2.8667$$
$$V = 3.5667$$

Therefore, the current through the D1 is $\boxed{0\,\text{A}}$. and voltage across output is $\boxed{3.6\,\text{V}}$.

Step 1 of 7

(a)

The constant-voltage drop diode model:

The diode current is,

$$I_D = \frac{V_{DD} - V_D}{R}.$$

Here,

V_{DD} is the supply voltage,

V_D is the voltage across the diode.

R is the resistor.

Refer to Figure P4.2(a) in the text book.

Find the voltage across the diode.

$$V_D = -3 + 0.7$$
$$V_D = -2.3\ \text{V}.$$

Hence, the voltage across the diode $\boxed{V = -2.3\ \text{V}}$.

Step 2 of 7

Determine the value of diode current:

$$I_D = \frac{V_{DD} - V_D}{R}$$
$$= \frac{3 - (-2.3)}{10 \times 10^3}$$
$$= \frac{5.3}{10 \times 10^3}$$
$$= 0.53 \times 10^{-3}$$
$$I_D = 0.53\ \text{mA}$$

Thus, the value of diode current is, $\boxed{I = 0.53\ \text{mA}}$.

Step 3 of 7

(b)

Refer to Figure P4.2(b) in the text book.

Find the voltage across the diode.

The diode is in cut-off region, hence $V_D = 3\ \text{V}.$

Voltage across the diode $\boxed{V = 3\ \text{V}}$.

Determine the value of diode current:

$$I_D = \frac{V_{DD} - V_D}{R}$$
$$= \frac{3 - 3}{10 \times 10^3}$$
$$= 0\ \text{A}$$

Thus, the value of diode current is, $\boxed{I = 0\ \text{A}}$.

Step 4 of 7

(c)

Refer to Figure P4.2(c) in the text book.

Find the voltage across the diode.

$$V_D = 3 - 0.7$$
$$V_D = 2.3\ \text{V}.$$

Thus, the voltage across the diode $\boxed{V = 2.3\ \text{V}}$.

Step 5 of 7

Determine the value of diode current:

$$I_D = \frac{V_{DD} - V_D}{R}$$
$$= \frac{3 + 2.3}{10 \times 10^3}$$
$$= 0.53 \times 10^{-3}$$
$$= 0.53\ \text{mA}$$

Thus, the value of diode current is, $\boxed{I = 0.53\ \text{mA}}$.

Step 6 of 7

(d)

Refer to Figure P4.2(d) in the text book.

Find the voltage across the diode.

The diode is in cut-off region, hence $V_D = -3\ \text{V}.$

Thus, the voltage across the diode $\boxed{V = -3\ \text{V}}$.

Step 7 of 7

Determine the value of diode current:

$$I_D = \frac{V_{DD} - V_D}{R}$$
$$= \frac{3 + (-3)}{10 \times 10^3}$$
$$= 0\ \text{A}$$

Thus, the value of diode current is, $\boxed{I = 0\ \text{A}}$.

4.042P

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4.043P

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Step 1 of 14

Consider the voltage drop across the diode is, $V_D = 0.7\,\text{V}$.

Consider the following circuit diagram:

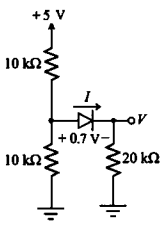


Figure 1

Step 2 of 14

Consider the diode and $20\,\text{k}\Omega$ resistor is the load and determines the Thevenin's voltage and the Thevenin's resistor.

Replace the load with an open circuit as shown in Figure 2.

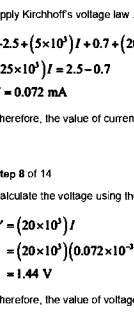


Figure 2

Step 3 of 14

Calculate the Thevenin's voltage by using the voltage division formula.

$$\begin{aligned} V_{Th} &= \left(\frac{10\times10^3}{10\times10^3+10\times10^3}\right)(5) \\ &= \left(\frac{10\times10^3}{20\times10^3}\right)(5) \\ &= \left(\frac{1}{2}\right)(5) \\ &= 2.5\,\text{V} \end{aligned}$$

Therefore, the Thevenin's voltage, V_{Th} is $2.5\,\text{V}$.

Step 4 of 14

Redraw the circuit in Figure 1 to find the Thevenin's resistance is shown in Figure 3

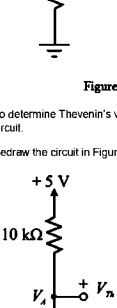


Figure 3

Step 5 of 14

Calculate the Thevenin's resistance.

$$\begin{aligned} R_{Th} &= 10\times10^3 \,\|\, 10\times10^3 \\ &= \frac{\left(10\times10^3\right)\left(10\times10^3\right)}{10\times10^3+10\times10^3} \\ &= \frac{100\times10^6}{20\times10^3} \\ &= 5\,\text{k}\Omega \end{aligned}$$

Therefore, the Thevenin's resistance R_{Th} is $5\,\text{k}\Omega$.

Step 6 of 14

The Thevenin's equivalent circuit is shown in Figure 4.

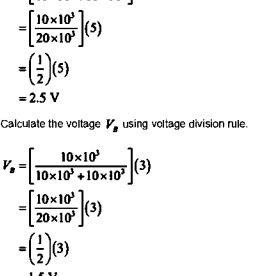


Figure 4

Redraw the circuit in Figure 4 by replacing the diode with the a dc voltage source of magnitude 0.7 V.

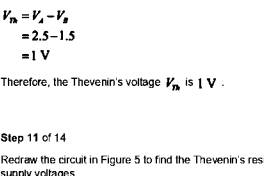


Figure 5

Step 7 of 14

Apply Kirchhoff's voltage law .

$$\begin{aligned} &-2.5+\left(5\times10^3\right)I+0.7+\left(20\times10^3\right)I=0 \\ &\left(25\times10^3\right)I=2.5-0.7 \\ &I=0.072\,\text{mA} \end{aligned}$$

Therefore, the value of current I is $\boxed{0.072\,\text{mA}}$.

Step 8 of 14

Calculate the voltage using the Ohm's law.

$$\begin{aligned} V &= \left(20\times10^3\right)I \\ &= \left(20\times10^3\right)\left(0.072\times10^{-3}\right) \\ &= 1.44\,\text{V} \end{aligned}$$

Therefore, the value of voltage V, is $\boxed{1.44\,\text{V}}$.

Step 9 of 14

(b)

Consider the circuit diagram shown in Figure 6.

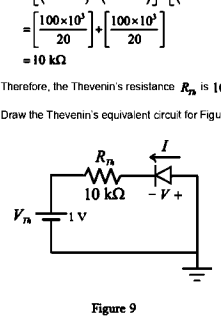


Figure 6

To determine Thevenin's voltage and Thevenin's resistor across the diode, replace the diode with the open circuit.

Redraw the circuit in Figure 6 to find the Thevenin's voltage as shown in Figure 7.

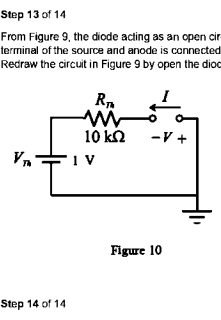


Figure 7

Step 18 of 14

Calculate the voltage V_A using voltage division rule.

$$\begin{aligned} V_A &= \left[\frac{10\times10^3}{10\times10^3+10\times10^3}\right](5) \\ &= \left[\frac{10\times10^3}{20\times10^3}\right](5) \\ &= \left(\frac{1}{2}\right)(5) \\ &= 2.5\,\text{V} \end{aligned}$$

Calculate the voltage V_B using voltage division rule.

$$\begin{aligned} V_B &= \left[\frac{10\times10^3}{10\times10^3+10\times10^3}\right](3) \\ &= \left[\frac{10\times10^3}{20\times10^3}\right](3) \\ &= \left(\frac{1}{2}\right)(3) \\ &= 1.5\,\text{V} \end{aligned}$$

Calculate the Thevenin's voltage.

$$\begin{aligned} V_{Th} &= V_A - V_B \\ &= 2.5-1.5 \\ &= 1\,\text{V} \end{aligned}$$

Therefore, the Thevenin's voltage V_{Th} is $1\,\text{V}$.

Step 11 of 14

Redraw the circuit in Figure 5 to find the Thevenin's resistance by opening the diode and shorting the supply voltages.

Figure 8

Step 12 of 14

Calculate the Thevenin's resistance.

$$\begin{aligned} R_{Th} &= \left[\left(10\times10^3\right)\|\left(10\times10^3\right)\right]+\left[\left(10\times10^3\right)\|\left(10\times10^3\right)\right] \\ &= \left[\frac{\left(10\times10^3\right)\left(10\times10^3\right)}{\left(10\times10^3\right)+\left(10\times10^3\right)}\right]+\left[\frac{\left(10\times10^3\right)\left(10\times10^3\right)}{\left(10\times10^3\right)+\left(10\times10^3\right)}\right] \\ &= \left[\frac{100\times10^6}{20}\right]+\left[\frac{100\times10^6}{20}\right] \\ &= 10\,\text{k}\Omega \end{aligned}$$

Therefore, the Thevenin's resistance R_{Th} is $10\,\text{k}\Omega$.

Draw the Thevenin's equivalent circuit for Figure 6.

Figure 9

Step 13 of 14

From Figure 9, the diode acting as an open circuit, as cathode of the diode is connected to the positive terminal of the source and anode is connected to the negative terminal of the source.
Redraw the circuit in Figure 9 by open the diode.

Figure 10

Step 14 of 14

As the diode is acting as an open circuit, the current I , flowing through the diode is $0\,\text{A}$. Hence, the current I, labeled in the Figure 6 is $\boxed{0\,\text{A}}$.

Apply the Kirchhoff's voltage law to the Figure 10.

$$\begin{aligned} &V+\left(10\times10^3\right)I+1=0 \\ &V+\left(11.667\times10^3\right)(0)+1=0 \qquad (\text{Since, } I=0\,\text{A}) \\ &V+0+1=0 \\ &V=-1\,\text{V} \end{aligned}$$

Therefore, the value of voltage V is $\boxed{-1\,\text{V}}$.

Step 1 of 3

The diode RMS voltage is 120 V.

Calculate peak voltage.

$$V_{rms} = \frac{V_p}{\sqrt{2}}$$

$$V_p = \sqrt{2} V_{rms}$$

$$V_p = 120\sqrt{2} \text{ V}.$$

Diode current should not exceed 40 mA.

Step 2 of 3

By constant-voltage drop model:

$$I_D = \frac{120\sqrt{2} - 0.7}{R} \le 40 \text{ mA}$$

$$\frac{R}{120\sqrt{2} - 0.7} \ge \frac{1}{40 \text{ mA}}$$

$$R \ge \frac{120\sqrt{2} - 0.7}{40 \text{ mA}}$$

$$R \ge \frac{120\sqrt{2} - 0.7}{40 \times 10^{-3}}$$

$$\ge \frac{169.70 - 0.7}{40 \times 10^{-3}}$$

$$\ge \frac{169}{40 \times 10^{-3}}$$

$$\ge 4.22 \text{ k}\Omega$$

Therefore, to maintain the diode current below 40 mA, the resistance $\boxed{R \ge 4.22 \text{ k}\Omega}$.

Step 3 of 3

The largest reverse voltage appearing across the diode is the peak voltage.

$$V_p = 120\sqrt{2}$$

$$V_p \approx 169.70 \text{ V}$$

The largest reverse voltage is $\boxed{V_p \approx 169.70 \text{ V}}$.

Since the voltage $169.70 > 0.7$, the diode operates in forward bias condition.

Hence, the circuit is same.

Step 1 of 6

The variations in the voltage ΔV is $5\ \text{mV}$.

The diode current is,

$$i_D = I_S e^{\frac{\Delta V}{nV_T}} \quad \text{..... (1)}$$

Consider the diode current is i_{D2} for the voltage variation of 5 mV.

Find the value of the ratio $\frac{i_{D2}}{i_{D1}}$.

Substitute $5\ \text{mV}$ for ΔV, 1 for n and $0.025\ \text{V}$ for V_T in equation (1).

$$\frac{i_{D2}}{i_{D1}} = e^{\frac{\Delta V}{nV_T}}$$
$$= e^{\frac{0.005}{1\times 0.025}}$$
$$\frac{i_{D2}}{i_{D1}} = 1.22$$

Step 2 of 6

The percentage change of the current is ,

$$\%\ \text{change} = \frac{i_{D2} - i_{D1}}{i_{D1}} \times 100.$$

Substitute 1.22 for $\frac{i_{D2}}{i_{D1}}$ in equation.

$$\%\ \text{change} = \frac{1.22\, i_{D1} - i_{D1}}{i_{D1}} \times 100$$
$$= 0.22 \times 100$$
$$= 22\%$$

Step 3 of 6

Substitute $-5\ \text{mV}$ for ΔV, 1 for n and $0.025\ \text{V}$ for V_T in equation (1).

$$\frac{i_{D2}}{i_{D1}} = e^{\frac{\Delta V}{nV_T}}$$
$$= e^{\frac{-0.005}{1\times 0.025}}$$
$$\frac{i_{D2}}{i_{D1}} = 0.82$$

Step 4 of 6

The percentage change of the current is ,

$$\%\ \text{change} = \frac{i_{D2} - i_{D1}}{i_{D1}} \times 100.$$

Substitute 0.82 for $\frac{i_{D2}}{i_{D1}}$ in equation.

$$\%\ \text{change} = \frac{0.82 i_{D1} - i_{D1}}{i_{D1}} \times 100$$
$$= -0.18 \times 100$$
$$= 18\%$$

Hence, the percentage change in the current for both the positive and negative signal is $\boxed{22\%\ \text{and}\ -18\%}$.

Step 5 of 6

Given that the current change is limited to 10%. Then the change in the current would be 90% to 110%. Therefore, the current variation would be within 0.9 to 1.1.

Substitute the values 0.9 for $\frac{i_{D2}}{i_{D1}}$, 1 for n and $0.025\ \text{V}$ for V_T in equation.

$$0.9 = e^{\frac{\Delta V}{nV_T}}$$
$$\frac{\Delta V}{1\times 0.025} = \ln(0.9)$$
$$\Delta V = -2.6\ \text{mV}$$

Step 6 of 6

Substitute the values 1.1 for $\frac{i_{D2}}{i_{D1}}$, 1 for n and $0.025\ \text{V}$ for V_T in equation.

$$1.1 = e^{\frac{\Delta V}{nV_T}}$$
$$\frac{\Delta V}{1\times 0.025} = \ln(1.1)$$
$$\Delta V = 2.4\ \text{mV}$$

Therefore, the maximum allowable voltage signal if the current change is to be limited to 10% is $\boxed{-2.6\ \text{mV}}$ to $\boxed{2.4\ \text{mV}}$.

Step 1 of 8

All the diodes are in parallel combination and characteristics are also same.

Calculate the current through each diode.

$$I_D = \frac{0.1}{10}$$
$$= 0.01\ \text{A}$$

Therefore, the current through the each diode is $\boxed{0.01\ \text{A}}$.

Step 2 of 8

Calculate the small signal resistance , r_d .

$$r_d = \frac{nV_T}{I_D}$$

Substitute $0.01\ \text{A}$ for I_D, 1 for n and $0.025\ \text{V}$ for V_T in equation.

$$r_d = \frac{nV_T}{I_D}$$
$$= \frac{1 \times 0.025}{0.01}$$

Therefore, the small signal for single diode is $\boxed{2.5\ \Omega}$.

Step 3 of 8

Calculate the equivalent resistance, R_{equiv} .

$$R_{equiv} = \frac{r_d}{N}$$

Substitute $2.5\ \Omega$ for r_d and 10 for N in equation.

$$R_{equiv} = \frac{2.5}{10}$$
$$= 0.25\ \Omega$$

Therefore, the equivalent resistance is $\boxed{0.25\ \Omega}$.

Step 4 of 8

When one diode is conducting the current through the diode is $0.1\ \text{A}$.

Calculate the small signal resistance.

$$r_d = \frac{nV_T}{I_D}$$
$$= \frac{1 \times 0.025}{0.1}$$

Therefore, the small signal resistance is $\boxed{0.25\ \Omega}$.

Step 5 of 8

The parallel combination of the diodes can be considered as equivalent to a single diode having the junction area ten times as that of a single diode. So in both the cases the small signal resistances are same.

Step 6 of 8

Consider the resistance $0.2\ \Omega$ is connected in series with the diode.

Since the small signal resistance, r_d is obtained as $2.5\ \Omega$ which is to be connected in series with the resistance $0.2\ \Omega$.

The equivalent resistance of the 10 parallel-connected diodes is,

$$R_{equiv} = \frac{(R + r_d)}{N}$$

Substitute the values $2.5\ \Omega$ for r_d , 10 for N and $0.2\ \Omega$ for R in equation. $R_{equiv} = \frac{0.2 + 2.5}{10}$
$= 0.27\ \Omega$

Therefore the equivalent resistance is $\boxed{0.27\ \Omega}$.

Step 7 of 8

The small signal resistance connected, r_d in series with the resistance, r_S is shown in Figure 1.

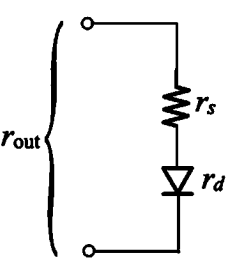


Figure 1

Step 8 of 8

From Figure 1, the expression for series resistance is,

$$r_{out} = r_d + r_S$$

Substitute $0.27\ \Omega$ for r_{out} and $0.25\ \Omega$ for r_d in equation.

$$0.27 = 0.25 + r_S$$
$$r_S = 0.27 - 0.25.$$
$$r_S = 0.02\ \Omega$$

Therefore, the equivalent resistance of the 10 parallel connected diodes is $0.27\ \Omega$ and the series resistance is $\boxed{0.02\ \Omega}$.

Step 1 of 5

For small signal model C_1 and C_2 are short circuited, dc current source is replaced by open circuit and diode is replaced with r_d.

The modified circuit is shown in Figure 1.

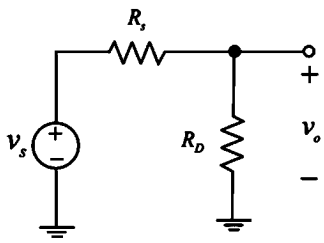


Figure 1

Step 2 of 5

The dynamic resistance of the diode is,

$$r_d = \frac{V_T}{I}$$

By using the voltage division rule,

$$v_o = v_s\left(\frac{r_d}{r_d + R_s}\right) \quad \text{..... (1)}$$

Substitute $\frac{V_T}{I}$ for r_d in equation.

$$v_o = v_s\left(\frac{(1)V_T}{(1)V_T + IR_s}\right)$$
$$= v_s\left(\frac{V_T}{V_T + IR_s}\right)$$

Therefore the output voltage is $\boxed{v_s\left(\frac{V_T}{V_T + IR_s}\right)}$.

Step 3 of 5

Find the value of v_o for $I = 1\ \text{mA}$:

Substitute 1 mA for I, 25 mV for V_T and $1\ \text{k}\Omega$ for R_s in v_o.

$$v_o = (10\times10^{-3})\left(\frac{(0.025)}{(0.025)+(10^{-3}\times10^{3})}\right)$$
$$= (10\times10^{-3})\left(\frac{0.025}{1.025}\right)$$
$$= (10)(0.02439)\times10^{-3}$$
$$= 0.2439\ \text{mV}$$

Thus, the output voltage is $\boxed{0.2439\ \text{mV}}$.

Step 4 of 5

Find the value of v_o for $I = 1\ \text{mA}$:

Substitute 0.1 mA for I, 25 mV for V_T and $1\ \text{k}\Omega$ for R_s in v_o.

$$v_o = (10\times10^{-3})\left(\frac{(0.025)}{(0.025)+(0.1\times10^{-3}\times10^{3})}\right)$$
$$= (10\times10^{-3})\left(\frac{0.025}{0.125}\right)$$
$$= (10)(0.2)\times10^{-3}$$
$$= 2\ \text{mV}$$

Therefore, the output voltage is $\boxed{2\ \text{mV}}$.

Step 5 of 5

Find the value of v_o for $I = 1\ \text{mA}$:

Substitute 1 μ A for I, 25 mV for V_T and $1\ \text{k}\Omega$ for R_s in v_o.

$$v_o = v_s\left(\frac{V_T}{V_T + IR_s}\right)$$
$$= (10\times10^{-3})\left(\frac{(0.025)}{(0.025)+(1\times10^{-6}\times10^{3})}\right)$$
$$= (10\times10^{-3})\left(\frac{0.025}{0.026}\right)$$
$$= (10)(0.9615)\times10^{-3}$$

Therefore, the output voltage is $\boxed{9.615\ \text{mV}}$

Find the dc current I:

Substitute 25 mV for V_T and $1\ \text{k}\Omega$ for R_s and $\frac{1}{2}v_s$ for v_o in expression of v_o.

$$\frac{1}{2}v_s = v_s\left(\frac{(0.025)}{(0.025)+(I\times10^{3})}\right)$$
$$\frac{1}{2} = \frac{(0.025)}{0.025+(I\times10^{3})}$$
$$0.025+(I\times10^{3}) = (0.025)2$$
$$(I\times10^{3}) = 0.05-0.025$$
$$I = 0.025\times10^{-3}$$

Therefore the resultant current is $\boxed{25\ \mu\text{A}}$.

For small signal model C_1 and C_2 are short circuited, dc current source is replaced by open circuit and diode is replaced with r_d.

The modified circuit is shown in Figure 1.

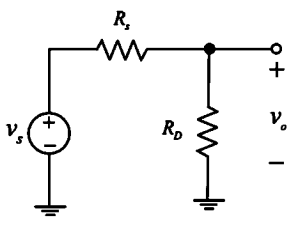


Figure 1

Step 2 of 11

The dynamic resistance of the diode is,

$$r_d = \frac{V_T}{I}$$

By using the voltage division rule,

$$v_o = v_s\left(\frac{r_d}{r_d + R_s}\right)$$

Substitute $\frac{V_T}{I}$ for r_d in equation.

$$v_o = v_s\left(\frac{(1)V_T}{(1)V_T + IR_s}\right)$$

$$= v_s\left(\frac{V_T}{V_T + IR_s}\right)$$

$$\frac{v_o}{v_s} = \left(\frac{V_T}{V_T + IR_s}\right)$$

The small-signal voltage gain of the amplifier is,

$$\frac{v_o}{v_s} = \frac{V_T}{V_T + IR_s} \quad \text{..... (1)}$$

Substitute $25\,\text{mV}$ for V_T and $10\,\text{k}\Omega$ for R_s in equation (1).

$$\frac{v_o}{v_s} = \frac{25\times10^{-3}}{25\times10^{-3} + (10\times10^{3})I} \quad \text{..... (2)}$$

Step 3 of 11

It is given that dc diode current is limited to $\pm 10\%$ variations.

Thus, the range of diode current is from $0.9\,\text{mA}$ to $1.1\,\text{mA}$.

Equation for diode current is,

$$I_D = I_S e^{\frac{v_D}{V_T}}$$

From diode current equation the change in diode voltage is,

$$\Delta V = V_T \ln\left(\frac{I_{D2}}{I_{D1}}\right)$$

Here $I_{D1} = 1\,\text{mA}$ and I_{D2} is varying from $0.9\,\text{mA}$ to $1.1\,\text{mA}$.

Step 4 of 11

The voltage V for I_{D2} is $0.9\,\text{mA}$.

$$V_{\text{at }0.9\,\text{mA}} = 25\times10^{-3}\ln\left(\frac{0.9\times10^{-3}}{1\times10^{-3}}\right)$$

$$= 25\times10^{-3}\ln(0.9)$$

$$= 25\times10^{-3}(-0.105360512)$$

$$= -2.634\times10^{-3}\ \text{V}$$

$$= -2.634\,\text{mV}$$

Step 5 of 11

The voltage V for I_{D2} is $1.1\,\text{mA}$.

$$V_{\text{at }1.1\,\text{mV}} = 25\times10^{-3}\ln\left(\frac{1.1\times10^{-3}}{1\times10^{-3}}\right)$$

$$= 25\times10^{-3}\ln(1.1)$$

$$= 25\times10^{-3}(0.0953101798)$$

$$= 2.383\times10^{-3}$$

$$V_{\text{at }1.1\,\text{mV}} = 2.383\,\text{mV}$$

The diode voltage varies from $-2.634\,\text{mV}$ to $2.383\,\text{mV}$.

Diode is across the output terminals , hence the range of output voltage change is,

$$\Delta v_{o(\text{range})} : -2.634\,\text{mV to } 2.383\,\text{mV}$$

Step 6 of 11

By using gain formula the range of input signal is ,

$$\Delta v_s = \frac{\Delta v_o}{\left(\frac{v_o}{v_s}\right)}$$

Thus, the range of input signal is,

$$\Delta v_{s(\text{range})} : -\frac{2.634\,\text{mV}}{\left(\frac{v_o}{v_s}\right)} \quad \text{to} \quad \frac{2.383\,\text{mV}}{\left(\frac{v_o}{v_s}\right)} \quad \text{..... (3)}$$

Step 7 of 11

Consider the voltage gain $\frac{v_o}{v_s}$ is 0.5.

Substitute 0.5 for $\frac{v_o}{v_s}$ in equation (1).

$$0.5 = \frac{25\times10^{-3}}{25\times10^{-3} + (10\times10^{3})I}$$

$$25\times10^{-3} + (10\times10^{3})I = 50\times10^{-3}$$

$$I = \frac{25\times10^{-3}}{10\times10^{3}}$$

$$= 0.0025\times10^{-3}$$

$$= 0.0025\,\text{mA}$$

Substitute 0.5 for $\frac{v_o}{v_s}$ in equation (3).

$$\Rightarrow \Delta v_{s(\text{range})} : -\frac{2.634\,\text{mV}}{0.5} \quad \text{to} \quad \frac{2.383\,\text{mV}}{0.5}$$

$$\Rightarrow \Delta v_{s(\text{range})} : -5.268\,\text{mV} \quad \text{to} \quad 4.766\,\text{mV}$$

Step 8 of 11

Consider the voltage gain $\frac{v_o}{v_s}$ is 0.10.

Substitute 0.10 for $\frac{v_o}{v_s}$ in equation (1).

$$0.1 = \frac{25\times10^{-3}}{25\times10^{-3} + (10\times10^{3})I}$$

$$25\times10^{-3} + (10\times10^{3})I = 250\times10^{-3}$$

$$I = \frac{225\times10^{-3}}{10\times10^{3}}$$

$$= 0.0225\times10^{-3}$$

$$= 0.0225\,\text{mA}$$

Substitute 0.1 for $\frac{v_o}{v_s}$ in equation (3).

$$\Rightarrow \Delta v_{s(\text{range})} : -\frac{2.634\,\text{mV}}{0.1} \quad \text{to} \quad \frac{2.383\,\text{mV}}{0.1}$$

$$\Rightarrow \Delta v_{s(\text{range})} : -26.34\,\text{mV} \quad \text{to} \quad 23.83\,\text{mV}$$

Step 9 of 11

Consider the voltage gain $\frac{v_o}{v_s}$ is 0.01.

Substitute 0.01 for $\frac{v_o}{v_s}$ in equation (1).

$$0.01 = \frac{25\times10^{-3}}{25\times10^{-3} + (10\times10^{3})I}$$

$$25\times10^{-3} + (10\times10^{3})I = 2{,}500\times10^{-3}$$

$$I = \frac{2{,}475\times10^{-3}}{10\times10^{3}}$$

$$= 0.2475\times10^{-3}$$

$$= 0.2475\,\text{mA}$$

Substitute 0.01 for $\frac{v_o}{v_s}$ in equation (3).

$$\Rightarrow \Delta v_{s(\text{range})} : -\frac{2.634\,\text{mV}}{0.01} \quad \text{to} \quad \frac{2.383\,\text{mV}}{0.01}$$

$$\Rightarrow \Delta v_{s(\text{range})} : -263.4\,\text{mV} \quad \text{to} \quad 238.3\,\text{mV}$$

Step 10 of 11

Consider the voltage gain $\frac{v_o}{v_s}$ is 0.001.

Substitute 0.001 for $\frac{v_o}{v_s}$ in equation (1).

$$0.001 = \frac{25\times10^{-3}}{25\times10^{-3} + (10\times10^{3})I}$$

$$25\times10^{-3} + (10\times10^{3})I = 25{,}000\times10^{-3}$$

$$I = \frac{24{,}975\times10^{-3}}{10\times10^{3}}$$

$$= 2.4975\times10^{-3}$$

$$= 2.4975\,\text{mA}$$

Substitute 0.01 for $\frac{v_o}{v_s}$ in equation (3).

$$\Rightarrow \Delta v_{s(\text{range})} : -\frac{2.634\,\text{mV}}{0.001} \quad \text{to} \quad \frac{2.383\,\text{mV}}{0.001}$$

$$\Rightarrow \Delta v_{s(\text{range})} : -2634\,\text{mV} \quad \text{to} \quad 2383\,\text{mV}$$

Step 11 of 11

Therefore, current flowing through the current source and input voltage range at different gains are shown in Table 1.

Table 1

$\frac{v_o}{v_s}$	I (mA)	v_s (mV)
0.5	0.0025	−5.268 to 4.766
0.1	0.0225	−26.34 to 23.83
0.01	0.2475	−263.4 to 238.3
0.001	2.4975	−2634 to 2383

Therefore, current flowing through the current source and input voltage range at different gains are obtained.

Step 1 of 13

Refer to Figure P4.48 in the text book.

For small signal analysis current source acts as an open circuit and capacitors acts an short circuit.

The Small-signal equivalent circuit is shown in Figure 1.

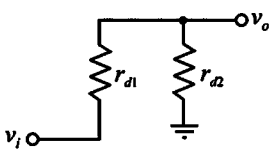


Figure 1

Step 2 of 13

Apply potential divider rule.

$$v_o = \left(\frac{r_{d2}}{r_{d1}+r_{d2}}\right) v_i \quad \frac{v_o}{v_i} = \frac{r_{d2}}{r_{d1}+r_{d2}} \quad \text{..... (1)}$$

Thus, the small-signal voltage gain of the given circuit is: $\boxed{\frac{r_{d2}}{r_{d1}+r_{d2}}}$.

Step 3 of 13

From the circuit shown in Figure 1, the current flowing through diode D_1 is I.

The current flowing through diode D_2 is $1\,\text{mA} - I$.

The Internal resistance of Diode1 is,

$$r_{d1} = \frac{V_T}{I}$$

The internal resistance of Diode 2 is,

$$r_{d2} = \frac{V_T}{1\,\text{mA} - I}$$

Step 4 of 13

Calculate the voltage gain.

Substitute $\frac{V_T}{1\,\text{mA}-I}$ for r_{d2} and $\frac{V_T}{I}$ for r_{d1} in equation (1)

$$\frac{v_o}{v_i} = \frac{\frac{V_T}{1\text{m}-I}}{\frac{V_T}{I}+\frac{V_T}{1\text{m}-I}}$$

$$= \frac{\frac{V_T}{1\text{m}-I}}{\frac{V_T(1\text{m})-V_T I+V_T I}{I(1\text{m}-I)}} \qquad \frac{v_o}{v_i} = 10^3 I$$

$$= \frac{V_T I}{V_T(1\text{mA})}$$

Thus, the voltage gain of the circuit is $\boxed{10^3 I}$.

Step 5 of 13

For $I = 0\,\text{mA}$, calculate the voltage gain.

Substitute 0 mA for I in equation.

$$\frac{v_o}{v_i} = 10^3(0)$$
$$= 0$$

Step 6 of 13

For $I = 1\,\mu\text{A}$, calculate the voltage gain.

Substitute $1\,\mu\text{A}$ for I in equation.

$$\frac{v_o}{v_i} = 10^3(10^{-6})$$
$$= 10^{-3}$$

Step 7 of 13

For $I = 10\,\mu\text{A}$, calculate the voltage gain.

$$\frac{v_o}{v_i} = 10^3 \times 10 \times 10^{-6}$$
$$= 10^{-2}$$
$$= 0.01$$

Step 8 of 13

For $I = 100\,\mu\text{A}$, calculate the voltage gain.

$$\frac{v_o}{v_i} = 10^3 \times 100 \times 10^{-6}$$
$$= 10^{-1}$$
$$= 0.1$$

For $I = 500\,\mu\text{A}$, calculate the voltage gain.

$$\frac{v_o}{v_i} = 10^3 \times 500 \times 10^{-6}$$
$$= 5 \times 10^{-1}$$
$$= 0.5$$

Step 9 of 13

For $I = 600\,\mu\text{A}$, calculate the voltage gain.

$$\frac{v_o}{v_i} = 10^3 \times 600 \times 10^{-6}$$
$$= 6 \times 10^{-1}$$
$$= 0.6$$

Step 10 of 13

For $I = 900\,\mu\text{A}$, calculate the voltage gain.

$$\frac{v_o}{v_i} = 10^3 \times 900 \times 10^{-6}$$
$$= 9 \times 10^{-1}$$
$$= 0.9$$

Step 11 of 13

For $I = 990\,\mu\text{A}$, calculate the voltage gain.

$$\frac{v_o}{v_i} = 10^3 \times 990 \times 10^{-6}$$
$$= 9.9 \times 10^{-1}$$
$$= 0.99$$

Step 12 of 13

For $I = 1\,\text{mA}$, calculate the voltage gain.

$$\frac{v_o}{v_i} = 10^3 \times 1\,\text{m}$$
$$= 1$$

Step 13 of 13

Tabulate the gain at different currents as shown in Table 1.

I	$\frac{v_o}{v_i}$
0mA	0
1μA	1m
10μA	0.01
100μA	0.1
500μA	0.5
600μA	0.6
900μA	0.9
990μA	0.99
1mA	1

Table 1

Therefore , for different values of current gain is tabulated.

Step 1 of 9

(a)

For small signal analysis replace the DC current source by open circuit and diode with dynamic resistance.

Assume the diode is conducting.

(i) Open circuit the load resistor.

(ii) Replace the current source as open circuit.

(iii) Replace the voltage source and diode with equivalent resistance r_d.

Draw the small signal equivalent circuit.

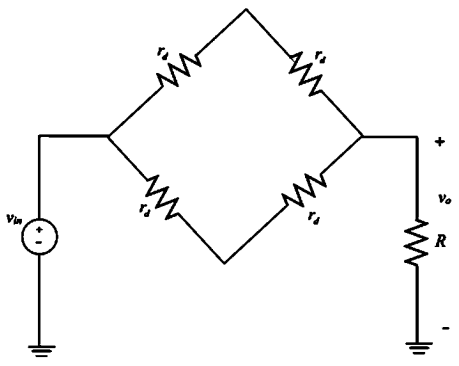


Figure 1

Step 2 of 9

Calculate the expression for the equivalent resistance.

$$\begin{aligned} R_{eq} &= R + (2r_d \| 2r_d) \\ &= R + \frac{(2r_d)(2r_d)}{2r_d + 2r_d} \\ &= R + \frac{4r_d^2}{4r_d} \\ &= R + r_d \end{aligned}$$

Step 3 of 9

Apply voltage division rule to calculate the gain of the circuit.

$$v_o = \frac{R}{R_{eq}} v_i$$

$$v_o = \frac{R}{R + r_d} v_i$$

$$\frac{v_o}{v_i} = \frac{R}{R + r_d}$$

Calculate the value of diode resistance.

$$\begin{aligned} r_d &= \frac{v_T}{\frac{I}{2}} \\ &= \frac{2(0.025)}{I} \\ &= \frac{0.05}{I} \end{aligned}$$

Step 4 of 9

Write the table to find $\frac{v_o}{v_i}$ for different values of I.

Table 1

$I\,(\text{mA})$	$r_d = \frac{0.05}{I}$	$\frac{v_o}{v_i} = \frac{R}{R + r_d}$
0	∞	0
10^{-3}	50 kΩ	0.167 V
10^{-2}	5 kΩ	0.667 V
10^{-1}	500 Ω	0.952 V
1	50 Ω	0.995 V
10	5 Ω	0.9995 V

Step 5 of 9

(b)

If the signal current is to be limited to $\pm 10\ \%$, change in diode voltage is given by Δv_d.

$\frac{i_D}{I} = e^{\frac{\Delta v_D}{nV_T}}$ whose value is between 0.9 and 1.1.

Calculate the change in diode voltage for $\frac{i_D}{I}$ is 0.9.

$$0.9 = e^{\frac{\Delta v_D}{(25)}}$$

$$\ln\left(\frac{i_D}{I}\right) = \frac{\Delta v_D}{(25)}$$

$$\begin{aligned} \Delta v_D &= 25 \ln(0.9) \\ &= -2.63\ \text{mV} \end{aligned}$$

Step 6 of 9

Calculate the change in diode voltage for $\frac{i_D}{I}$ is 1.1.

$$1.1 = e^{\frac{\Delta v_D}{(25)}}$$

$$\ln(1.1) = \frac{\Delta v_D}{(25)}$$

$$\begin{aligned} \Delta v_D &= 25 \ln(1.1) \\ &= 2.38\ \text{mV} \end{aligned}$$

Where, the input voltage varies in between $-2.63\ \text{mV}$ to $+2.38\ \text{mV}$ for signal current is limited to 10% of dc bias current.

Thus, the largest signal voltage is $\boxed{\pm 2.5\ \text{mV}}$.

Step 7 of 9

For input $v_{i\,peak} = 10\ \text{mV}$, the output current is,

$$\begin{aligned} i_o &= \frac{5\ \text{mV}}{10\ \text{k}} \\ &= 0.5\ \mu\text{A} \end{aligned}$$

Each diode is carrying a current of $0.25\ \mu\text{A}$, for this to be at most 10% of dc current, dc current in each diode must be at least $2.5\ \mu\text{A}$.

Thus, the minimum value of I is $\boxed{5\ \mu\text{A}}$.

Step 8 of 9

(c)

For $I = 1\ \text{mA}$, $I_D = 0.5\ \text{mA}$ and for maximum signal of 10% $I_D = 0.05\ \text{mA}$.

$$\begin{aligned} i_D &= 2I_D \\ &= 0.1\ \text{mA} \end{aligned}$$

The total current in each diode is $\boxed{0.1\ \text{mA}}$.

The corresponding maximum v_o is,

$$\begin{aligned} v_o &= 0.1\ \text{mA} \times 10\ \text{k} \\ &= 1\ \text{V} \end{aligned}$$

Thus, the largest possible output signal is $\boxed{1\ \text{V}}$.

Step 9 of 9

The corresponding peak input can be found by dividing v_o by transmission factor of 0.995.

Calculate the peak input voltage.

$$\begin{aligned} V_{i\,max} &= \frac{1\ \text{V}}{0.995\ \text{V}} \\ &= 1.005\ \text{V} \end{aligned}$$

Thus, the peak in put voltage is $\boxed{1.005\ \text{V}}$.

Step 1 of 22

Refer to Figure P4.49 in the text book.

Redraw the circuit is shown in Figure 1.

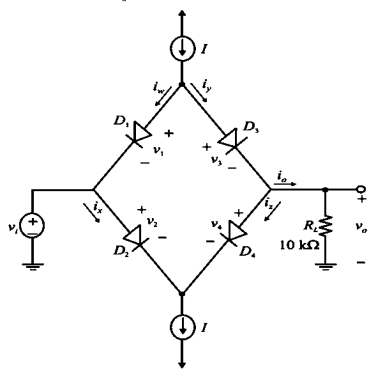


Figure 1

Step 2 of 22

The current flowing through the constant-current source is , $I = 1\ \text{mA}$.

Each of the diode has a voltage drop of $0.7\ \text{V}$, at a diode current of $1\ \text{mA}$.

Equation for the diode current is,

$$I_D = I_S e^{\frac{v_D}{V_T}} \quad \ldots\ldots (1)$$

Because of circuit symmetry same results will occur in positive and negative portion of the input.

Consider that I_{D1} and I_{D2} are currents in a diode and the corresponding voltages across the diode are V_1 and V_2

Now by using diode current equation we obtain

$$\frac{I_{D2}}{I_{D1}} = \frac{I_S e^{\frac{v_2}{V_T}}}{I_S e^{\frac{v_1}{V_T}}}$$

$$\frac{I_{D2}}{I_{D1}} = e^{\frac{v_2 - v_1}{V_T}}$$

$$V_2 = V_1 + V_T \ln\left(\frac{I_{D2}}{I_{D1}}\right) \quad \ldots\ldots (2)$$

Step 3 of 22

Substitute $0.7\ \text{V}$ for V_1 , $25\ \text{mA}$ for V_T , and $1\ \text{mA}$ for I_{D1} in equation (2).

$$V_2 = 0.7 + 25\times10^{-3} \ln\left(\frac{I_{D2}}{1\times10^{-3}}\right)$$

$$V_2 = 0.7 + 0.025\ln\left(\frac{I_{D2}}{1}\right) \quad \ldots\ldots (3)$$

From the circuit shown in Figure 2 , the relation between currents are,

$$I = i_w + i_y = i_x + i_z$$

$$i_y = i_z + i_x$$

Step 4 of 22

For $v_o = 0\ \text{V}$:

Current flowing through the load is,

$$\begin{aligned} i_o &= \frac{v_o}{10\times10^3} \\ &= \frac{0}{10\times10^3} \\ &= 0\ \text{A} \end{aligned}$$

Step 5 of 22

As the diodes are identical. So Current through the each diode is same.

Current passing through the diode D1 is,

$$\begin{aligned} i_w &= \frac{1\times10^{-3}}{2} \\ &= 0.5\ \text{mA} \end{aligned}$$

Current passing through the diode D2 is,

$$\begin{aligned} i_x &= \frac{1\times10^{-3}}{2} \\ &= 0.5\ \text{mA} \end{aligned}$$

Current passing through the diode D3 is,

$$\begin{aligned} i_y &= \frac{1\times10^{-3}}{2} \\ &= 0.5\ \text{mA} \end{aligned}$$

Current passing through the diode D4 is.

$$\begin{aligned} i_z &= \frac{1\times10^{-3}}{2} \\ &= 0.5\ \text{mA} \end{aligned}$$

Step 6 of 22

Substitute $0.5\ \text{mA}$ in equation (3)

$$\begin{aligned} V_2 &= 0.7 + 0.025\ln\left(\frac{0.5}{1}\right) \\ &= 0.7 + 0.025(-0.69315) \\ &= 0.7 - 0.01732875 \\ &= 0.68267125 \\ &= 683\ \text{mV} \end{aligned}$$

Here $683\ \text{mV}$ is each diode drop when the output voltage is zero.

$$v_1 = 0\ \text{V}$$
$$v_2 = 0\ \text{V}$$
$$v_3 = 0\ \text{V}$$
$$v_4 = 0\ \text{V}$$

Step 7 of 22

By applying Kirchhoff's voltage law to the circuit shown in Figure 2 .

$$-v_1 - v_4 + v_3 = 0$$
$$v_3 = -v_4 + v_1$$
$$v_3 = -683\ \text{mV} + 683\ \text{mV}$$
$$v_3 = 0\ \text{V}$$

Thus, the input voltage is $\boxed{0\ \text{V}}$

Step 8 of 22

For $v_o = 1\ \text{V}$:

Current flowing through the load is,

$$\begin{aligned} i_o &= \frac{v_o}{10\times10^3} \\ &= \frac{1}{10\times10^3} \\ &= 0.1\ \text{mA} \end{aligned}$$

Circuit is symmetry, so the currents are

$$\begin{aligned} i_x &= \frac{1\ \text{mA}}{2} + \frac{0.1\ \text{mA}}{2} \\ &= 0.5\ \text{mA} + 0.05\ \text{mA} \\ &= 0.55\ \text{mA} \end{aligned}$$

So the current through the D3 is same as D2.

$$i_y = 0.55\ \text{mA}$$

Step 9 of 22

Similarly the currents through the diode D1 is,

$$\begin{aligned} i_w &= 1 - 0.55 \\ &= 0.45\ \text{mA} \end{aligned}$$

The current through the diode D3 is,

$$i_z = 1 - 0.55$$
$$= 0.45\ \text{mA}$$

Step 10 of 22

Substitute $0.55\ \text{mA}$ for I_{D2} in equation (3)

$$\begin{aligned} V_2 &= 0.7 + 0.025\ln\left(\frac{0.55}{1}\right) \\ &= 0.7 + 0.025(-0.59784) \\ &= 0.7 - 0.014946 \\ &= 0.68267125 \\ &= 685\ \text{mV} \end{aligned}$$

Therefore the diode drops $v_2 = 685\ \text{mV}$ and $v_3 = 685\ \text{mV}$.

Step 11 of 22

Substitute $0.45\ \text{mA}$ for I_{D2} in equation (3)

$$\begin{aligned} V_2 &= 0.7 + 0.025\ln\left(\frac{0.45}{1}\right) \\ &= 0.7 + 0.025(-0.7985) \\ &= 0.7 - 0.0199625 \\ &= 0.6800375 \\ &= 680\ \text{mV} \end{aligned}$$

Therefore the diode drops $v_1 = 680\ \text{mV}$ and $v_4 = 680\ \text{mV}$.

Step 12 of 22

Apply Kirchhoff's voltage law to the circuit shown in Figure 2 .

$$-v_i - v_1 + v_3 + v_o = 0$$

$$\begin{aligned} v_i &= -v_1 + v_3 + v_o \\ &= -680\ \text{mV} + 685\ \text{mV} + 1 \\ &= 1.005\ \text{V} \end{aligned}$$

Thus, for $v_o = 1\ \text{V}$, the input voltage is, $1.005\ \text{V}$.

Step 13 of 22

For $v_o = 2\ \text{V}$:

Current flowing through the load is,

$$\begin{aligned} i_o &= \frac{v_o}{10\times10^3} \\ &= \frac{2}{10\times10^3} \\ &= 0.2\ \text{mA} \end{aligned}$$

Step 14 of 22

Circuit is symmetry, so the currents are

$$\begin{aligned} i_x = i_y &= \frac{1\ \text{mA}}{2} + \frac{0.2\ \text{mA}}{2} \\ &= 0.5\ \text{mA} + 0.1\ \text{mA} \\ &= 0.6\ \text{mA} \end{aligned}$$

$$i_w = i_z = 0.4\ \text{mA}$$

For $v_o = 1\ \text{V}$:

Current flowing through the load is,

$$\begin{aligned} i_o &= \frac{v_o}{10\times10^3} \\ &= \frac{1}{10\times10^3} \\ &= 0.1\ \text{mA} \end{aligned}$$

Circuit is symmetry, so the currents are

$$\begin{aligned} i_x &= \frac{1\ \text{mA}}{2} + \frac{0.2\ \text{mA}}{2} \\ &= 0.5\ \text{mA} + 0.1\ \text{mA} \\ &= 0.6\ \text{mA} \end{aligned}$$

So the current through the D3 is same as D2.

$$i_y = 0.6\ \text{mA}$$

Step 15 of 22

Similarly the currents through the diode D1 is,

$$\begin{aligned} i_w &= 1 - 0.6 \\ &= 0.4\ \text{mA} \end{aligned}$$

The current through the diode D3 is,

$$i_z = 1 - 0.6$$
$$= 0.4\ \text{mA}$$

Step 16 of 22

Substitute $0.6\ \text{mA}$ in equation (3).

$$\begin{aligned} V_2 &= 0.7 + 0.025\ln\left(\frac{0.6}{1}\right) \\ &= 0.7 + 0.025(-0.51083) \\ &= 0.7 - 0.01277075 \\ &= 0.68722925 \\ &= 687\ \text{mV} \end{aligned}$$

Therefore the diode drops $v_2 = 687\ \text{mV}$ and $v_3 = 687\ \text{mV}$.

Step 17 of 22

Substitute $0.4\ \text{mA}$ in equation (3).

$$\begin{aligned} V_2 &= 0.7 + 0.025\ln\left(\frac{0.4}{1}\right) \\ &= 0.7 + 0.025(-0.91629) \\ &= 0.7 - 0.02290725 \\ &= 0.67709275 \\ &= 677\ \text{mV} \end{aligned}$$

Therefore the diode drops $v_1 = 677\ \text{mV}$ and $v_2 = 677\ \text{mV}$.

Step 18 of 22

By applying Kirchhoff's voltage law to the circuit shown in Figure 2 .

$$-v_i - v_1 + v_3 + v_o = 0$$

$$\begin{aligned} v_i &= -v_1 + v_3 + v_o \\ &= -677\ \text{mV} + 687\ \text{mV} + 2 \\ &= 2.01\ \text{V} \end{aligned}$$

Thus, the input voltage is, $2.01\ \text{V}$.

Step 19 of 22

Similarly we can find the other values in positive side.

All values in positive side shown in Table 1

Table 1

v_o(V)	i_o(mA)	$i_x = i_y$ (mA)	$i_w = i_z$ (mA)	$v_2 = v_3$ (mV)	$v_1 = v_4$ (mV)	v_i(V)
0	0	0.5	0.5	683	683	0
1	0.1	0.55	0.45	685	680	1.005
2	0.2	0.6	0.4	687	677	2.010
5	0.5	0.75	0.25	693	665	5.028
9	0.9	0.95	0.05	699	625	9.074
9.9	0.99	0.995	0.005	700	568	10.09
9.99	0.999	0.9995	0.0005	700	510	10.18
10	1	1	0	700	0	10.7

Step 20 of 22

For output voltage greater than $10\ \text{V}$ given circuit is not operate properly. This is limited by $1\ \text{mA}$ current source.

Thus, the possible maximum positive and negative output voltage amplitudes are $+10\ \text{V}$ and $-10\ \text{V}$.

Step 21 of 22

In another words , for $v_i > +10\ \text{V}$ and $v_i < -10\ \text{V}$, output voltage amplitude is constant at $+10\ \text{V}$ and $-10\ \text{V}$ respectively.

Voltage transfer characteristics plot for $1\ \text{mA}$ current source is shown in Figure 2.

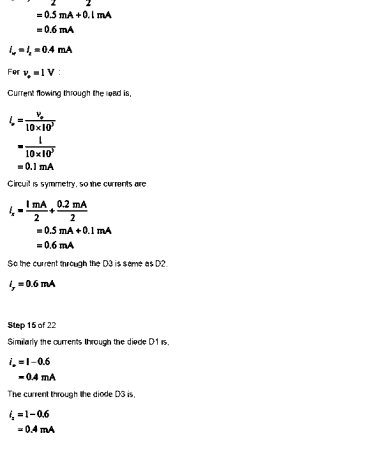


Figure 2

Step 22 of 22

Voltage transfer characteristics plot for $0.5\ \text{mA}$ source current is shown in Figure 3

Figure 3

Step 1 of 4

Refer to Figure P4.51 in the textbook.

Calculate the value of diode small signal resistance, r_d .

$$r_d = \frac{V_T}{i_D}$$

Substitute $26\ \text{mV}$ for V_T

$$r_d = \frac{0.026}{i_D}$$

Draw the small signal circuit with opened current source.

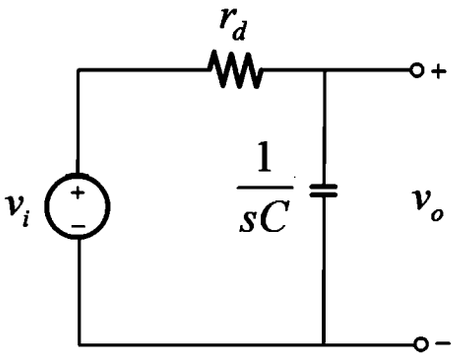


Figure 1

Step 2 of 4

Apply voltage division rule to calculate the input output relation.

$$\frac{v_o}{v_i} = \frac{\frac{1}{sC}}{\frac{1}{sC} + r_d}$$
$$= \frac{1}{1 + sCr_d}$$

Substitute $j\omega$ for s .

$$\frac{v_o}{v_i} = \frac{1}{(1 + j\omega Cr_d)}$$

Calculate the phase between input and output voltages.

$$\theta = 0 - \tan^{-1}\left(\frac{\omega Cr_d}{1}\right)$$
$$= -\tan^{-1}(2\pi fCr_d)$$

Substitute $100\ \text{kHz}$ for f , $10\ \text{nF}$ for C and $\frac{0.026}{i_D}$ for r_d .

$$\theta = -\tan^{-1}\left(2\pi \times 10^5 \times 10 \times 10^{-9} \times \frac{0.025}{i_D}\right)$$
$$= -\tan^{-1}\left(\frac{0.157 \times 10^{-3}}{i_D}\right)$$
$$= -\tan^{-1}\left(\frac{0.157 \times 10^{-3}}{I}\right) \qquad (I = i_D)$$

Thus, the phase between the input and output voltages is given by the relation,

$$\boxed{-\tan^{-1}\left(\frac{0.157 \times 10^{-3}}{I}\right)}.$$

Step 3 of 4

Calculate the value of the current, I .

$$\theta = -\tan^{-1}\left(\frac{0.157 \times 10^{-3}}{I}\right)$$
$$\frac{0.157 \times 10^{-3}}{I} = -\tan\theta$$
$$I = -\frac{0.157 \times 10^{-3}}{\tan\theta}$$

Substitute $-45°$ for θ .

$$I = -\frac{0.157 \times 10^{-3}}{\tan(-45°)}$$
$$= 0.157 \times 10^{-3}$$
$$= 157\ \mu\text{A}$$

Thus, the value of the current is $\boxed{157\ \mu\text{A}}$.

Step 4 of 4

The value of the current is varied over the range of 0.1 times to 10 times, then the range of the current becomes $15.7\ \mu\text{A}$ to $1570\ \mu\text{A}$.
Calculate the phase angle for $I = 15.7\ \mu\text{A}$.

$$\theta = -\tan^{-1}\left(\frac{0.157 \times 10^{-3}}{15.7 \times 10^{-6}}\right)$$
$$= -\tan^{-}(0.1)$$
$$= -84.3°$$

Calculate the phase angle for $I = 1570\ \mu\text{A}$.

$$\theta = -\tan^{-1}\left(\frac{0.157 \times 10^{-3}}{1570 \times 10^{-6}}\right)$$
$$= -\tan^{-}(0.1)$$
$$= -5.71°$$

Thus, the range of the phase shift achieved as I varies over the range of 0.1 times to 10 times the value of the current, $i_D = 157\ \mu\text{A}$ is $\boxed{-84.3° \text{ to } -5.71°}$.

4.054P

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Step 1 of 9

Draw the circuit for two-diode voltage regulator.

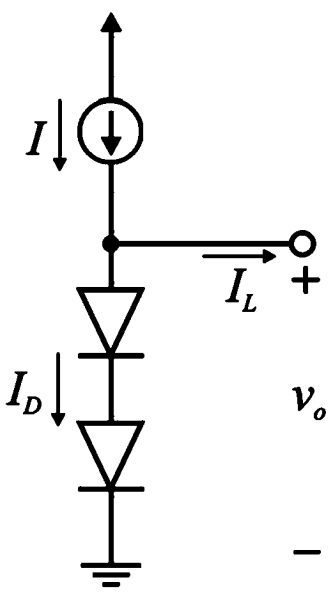


Figure 1

Step 2 of 9

Apply Kirchhoff's current law to the circuit.

$I = I_D + I_L$

The range of load current is $2\ \text{mA}$ to $7\ \text{mA}$.

Therefore, to meet the maximum load current condition available $5\ \text{mA}$ current case not fits to this circuit.

Thus, $10\ \text{mA}$ and $15\ \text{mA}$ cases can fit to this circuit, but further analyze these two cases to get the best results.

The expression for the diode current is,

$I_D = I_S e^{\frac{v_D}{V_T}}$ ..... (1)

Here,

I_D - the current flowing through the diode

I_S - the diode reverse bias current

v_D - the voltage across the diode

V_T - the thermal equivalent of voltage

Step 3 of 9

Let us take the current supplied by the constant-current source as $10\ \text{mA}$.

Now, the range of current flowing through the diodes branch is $(10-2)\ \text{mA}$ to $(10-7)\ \text{mA}$ that is $8\ \text{mA}$ to $3\ \text{mA}$, since the load current range $2\ \text{mA}$ to $7\ \text{mA}$.

Let us take diode current as I_1 when diode drop as V_1 and I_2 when diode drop as V_2.

From the diode current equation,

$$\begin{aligned}\frac{I_2}{I_1} &= \frac{I_S e^{\frac{V_1}{V_T}}}{I_S e^{\frac{V_2}{V_T}}} \\ &= e^{\frac{V_1}{V_T}} e^{-\frac{V_2}{V_T}} \\ &= e^{\frac{V_1 - V_2}{V_T}} \\ &= e^{\frac{\Delta V}{V_T}}\end{aligned}$$

In Figure 1, two diodes are across the output voltage, current ration equation becomes

$\frac{I_2}{I_1} = e^{\frac{\Delta V}{2V_T}}$ ..... (2)

Step 4 of 9

Without load the current in the diodes is $10\ \text{mA}$ and with load the range of diode current is $8\ \text{mA}$ to $3\ \text{mA}$.

Substitute $8\ \text{mA}$ for I_2, $10\ \text{mA}$ for I_1 and $25\ \text{mV}$ for V_T in equation (2).

$$\frac{8\times10^{-3}}{10\times10^{-3}} = e^{\frac{\Delta V_1}{2V_T}}$$

$$0.8 = e^{\frac{\Delta V_1}{2(25\times10^{-3})}}$$

$$-0.223 = \frac{\Delta V_1}{2(25\times10^{-3})}$$

$$\begin{aligned}\Delta V_1 &= -11.15\times10^{-3} \\ &= -11.15\ \text{mV}\end{aligned}$$

Step 5 of 9

Now substitute $3\ \text{mA}$ for I_2, $10\ \text{mA}$ for I_1 and $25\ \text{mV}$ for V_T in equation (2).

$$\frac{3\times10^{-3}}{10\times10^{-3}} = e^{\frac{\Delta V_2}{2V_T}}$$

$$0.3 = e^{\frac{\Delta V_2}{2(25\times10^{-3})}}$$

$$-1.204 = \frac{\Delta V_2}{2(25\times10^{-3})}$$

$$\begin{aligned}\Delta V_2 &= -60.2\times10^{-3} \\ &= -60.2\ \text{mV}\end{aligned}$$

Step 6 of 9

Calculate the change in output voltage.

$$\begin{aligned}\Delta V_o &= \Delta V_2 - \Delta V_1 \\ &= -60.2\,\text{m} - (-11.15\,\text{m}) \\ &= -40.09\,\text{m}\end{aligned}$$

Calculate the change in load current.

$$\begin{aligned}\Delta I_L &= 7\,\text{m} - 2\,\text{m} \\ &= 5\,\text{m}\end{aligned}$$

Calculate the value of load regulation of the circuit.

$$\begin{aligned}\frac{\Delta V_o}{\Delta I_L} &= \frac{-40.09\times10^{-3}}{5\times10^{-3}} \\ &= -9.81\,\text{mV/mA}\end{aligned}$$

Therefore, load regulation of the circuit when $10\ \text{mA}$ current supplied by the constant-current source is, $\frac{\Delta V_o}{\Delta I_L} = -9.81\,\text{mV/mA}$.

Step 7 of 9

Let us take the current supplied by the constant-current source as $15\ \text{mA}$.

Now, the range of current flowing through the diodes branch is $(15-2)\ \text{mA}$ to $(15-7)\ \text{mA}$ that is $13\ \text{mA}$ to $8\ \text{mA}$, since the load current range $2\ \text{mA}$ to $7\ \text{mA}$.

Without load the current in the diodes is $10\ \text{mA}$ and with load the range of diode current is $13\ \text{mA}$ to $8\ \text{mA}$.

Substitute $13\ \text{mA}$ for I_2, $10\ \text{mA}$ for I_1 and $25\ \text{mV}$ for V_T in equation (2).

$$\frac{13\times10^{-3}}{10\times10^{-3}} = e^{\frac{\Delta V_1}{2V_T}}$$

$$1.3 = e^{\frac{\Delta V_1}{2(25\times10^{-3})}}$$

$$-0.2624 = \frac{\Delta V_1}{2(25\times10^{-3})}$$

$$\begin{aligned}\Delta V_1 &= 13.12\times10^{-3} \\ &= 13.12\ \text{mV}\end{aligned}$$

Step 8 of 9

Substitute $8\ \text{mA}$ for I_2, $10\ \text{mA}$ for I_1 and $25\ \text{mV}$ for V_T in equation (2).

$$\frac{8\times10^{-3}}{10\times10^{-3}} = e^{\frac{\Delta V_2}{2V_T}}$$

$$0.8 = e^{\frac{\Delta V_1}{2(25\times10^{-3})}}$$

$$-0.223 = \frac{\Delta V_2}{2(25\times10^{-3})}$$

$$\begin{aligned}\Delta V_2 &= -11.15\times10^{-3} \\ &= -11.15\ \text{mV}\end{aligned}$$

Calculate the change in output voltage.

$$\begin{aligned}\Delta V_o &= \Delta V_2 - \Delta V_1 \\ &= -11.15\,\text{m} - 13.12\,\text{m} \\ &= -24.27\,\text{m}\end{aligned}$$

Calculate the change in load current.

$$\begin{aligned}\Delta I_L &= 7\,\text{m} - 2\,\text{m} \\ &= 5\,\text{m}\end{aligned}$$

Step 9 of 9

Calculate the value of load regulation of the circuit.

$$\begin{aligned}\frac{\Delta V_o}{\Delta I_L} &= \frac{-24.27\times10^{-3}}{5\times10^{-3}} \\ &= -4.854\,\text{mV/mA}\end{aligned}$$

Therefore, load regulation of the circuit when $15\ \text{mA}$ current supplied by the constant-current source is, $\frac{\Delta V_o}{\Delta I_L} = -4.854\,\text{mV/mA}$.

In $15\ \text{mA}$ current case, load regulation is less compared to $10\ \text{mA}$ and power dissipation is more because of higher current rate .

In $10\ \text{mA}$ current case, power dissipation is less and output voltage vary by $-40.09\,\text{mV}$ over a load current variation of $2\ \text{mA}$ to $7\ \text{mA}$.

Thus, choose constant current supplies of $10\ \text{mA}$ for better performance.

4.058P

We don't have the solution to this problem yet.

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Step 1 of 10

Refer to Figure 4.17 in the textbook.

(a)

Consider the expression for diode voltage.

$$V_z = V_{z0} + r_z I_{ZT}$$
$$r_z = \frac{V_z - V_{z0}}{I_{ZT}}$$

Substitute $10\ \text{V}$ for V_z, $9.6\ \text{V}$ for V_{zk} and $50\ \text{mA}$ for I_{ZT}.

$$r_z = \frac{10 - 9.6}{50 \times 10^{-3}}$$
$$= \frac{400}{50}$$
$$= 8\ \Omega$$

Calculate the value of new diode current.

$$I'_{ZT} = 2 \times I_{ZT}$$

Substitute $50\ \text{mA}$ for I_{ZT}.

$$I'_{ZT} = 2(50\ \text{mA})$$
$$= 100\ \text{mA}$$

Step 2 of 10

Calculate the value of new diode voltage, V'_z.

$$V'_z = V_{z0} + r_z I'_{ZT}$$

Substitute $9.6\ \text{V}$ for V_{zk}, $100\ \text{mA}$ for I'_{ZT} and $8\ \Omega$ for r_z.

$$V'_z = 9.6 + (8)(100 \times 10^{-3})$$
$$= 10.4\ \text{V}$$

Calculate the power rating of the diode.

$$P = V'_z I'_{ZT}$$

Substitute $10.4\ \text{V}$ for V'_z and $100\ \text{mA}$ for I'_{ZT}.

$$P = 10.4 \times 100 \times 10^{-3}$$
$$= 1.04\ \text{W}$$

Thus, the power rating of the diode is $\boxed{1.04\ \text{W}}$ and the missing resistance is $\boxed{8\ \Omega}$.

Step 3 of 10

(b)

Consider the expression for diode voltage.

$$V_z = V_{z0} + r_z I_{ZT}$$
$$V_{z0} = V_z - r_z I_{ZT}$$

Substitute $9.1\ \text{V}$ for V_z, $30\ \Omega$ for r_z and $10\ \text{mA}$ for I_{ZT}.

$$V_{z0} = 9.1 - (30)(10 \times 10^{-3})$$
$$= 9.1 - 0.3$$
$$= 8.8\ \text{V}$$

Calculate the value of new diode current.

$$I'_{ZT} = 2 \times I_{ZT}$$

Substitute $10\ \text{mA}$ for I_{ZT}.

$$I'_{ZT} = 2(10\ \text{mA})$$
$$= 20\ \text{mA}$$

Step 4 of 10

Calculate the value of new diode voltage, V'_z.

$$V'_z = V_{z0} + r_z I'_{ZT}$$

Substitute $8.8\ \text{V}$ for V_{zk}, $20\ \text{mA}$ for I'_{ZT} and $30\ \Omega$ for r_z.

$$V'_z = 8.8 + (30)(20 \times 10^{-3})$$
$$= 8.8 + 0.6$$
$$= 9.4\ \text{V}$$

Calculate the power rating of the diode.

$$P = V'_z I'_{ZT}$$

Substitute $9.4\ \text{V}$ for V'_z and $20\ \text{mA}$ for I'_{ZT}.

$$P = 9.4 \times 20 \times 10^{-3}$$
$$= 188\ \text{mW}$$

Thus, the power rating of the diode is $\boxed{188\ \text{mW}}$ and the missing voltage is $\boxed{8.8\ \text{V}}$.

Step 5 of 10

(c)
Consider the expression for diode voltage.

$$V_z = V_{z0} + r_z I_{ZT}$$
$$I_{ZT} = \frac{V_z - V_{z0}}{r_z}$$

Substitute $6.8\ \text{V}$ for V_z, $6.6\ \text{V}$ for V_{zk} and $2\ \Omega$ for r_z.

$$I_z = \frac{6.8 - 6.6}{2}$$
$$= 100\ \text{mA}$$

Calculate the value of new diode current.

$$I'_{ZT} = 2 \times I_{ZT}$$

Substitute $100\ \text{mA}$ for I_{ZT}.

$$I'_{ZT} = 2(100\ \text{mA})$$
$$= 200\ \text{mA}$$

Step 6 of 10

Calculate the value of new diode voltage, V'_z.

$$V'_z = V_{z0} + r_z I'_{ZT}$$

Substitute $6.6\ \text{V}$ for V_{zk}, $200\ \text{mA}$ for I'_{ZT} and $2\ \Omega$ for r_z.

$$V'_z = 6.6 + (2)(200 \times 10^{-3})$$
$$= 6.6 + 0.4$$
$$= 7\ \text{V}$$

Calculate the power rating of the diode.

$$P = V'_z I'_{ZT}$$

Substitute $7\ \text{V}$ for V'_z and $200\ \text{mA}$ for I'_{ZT}.

$$P = 7 \times 200 \times 10^{-3}$$
$$= 1.4\ \text{W}$$

Thus, the power rating of the diode is $\boxed{1.4\ \text{W}}$ missing current is $\boxed{200\ \text{mA}}$.

Step 7 of 10

(d)

Consider the expression for diode voltage.

$$V_z = V_{z0} + r_z I_{ZT}$$
$$r_z = \frac{V_z - V_{z0}}{I_{ZT}}$$

Substitute $18\ \text{V}$ for V_z, $17.6\ \text{V}$ for V_{zk} and $5\ \text{mA}$ for I_{ZT}.

$$r_z = \frac{18 - 17.6}{5 \times 10^{-3}}$$
$$= \frac{400}{5}$$
$$= 80\ \Omega$$

Calculate the value of new diode current.

$$I'_{ZT} = 2 \times I_{ZT}$$

Substitute $5\ \text{mA}$ for I_{ZT}.

$$I'_{ZT} = 2(5\ \text{mA})$$
$$= 10\ \text{mA}$$

Step 8 of 10

Calculate the value of new diode voltage, V'_z.

$$V'_z = V_{z0} + r_z I'_{ZT}$$

Substitute $17.6\ \text{V}$ for V_{zk}, $10\ \text{mA}$ for I'_{ZT} and $80\ \Omega$ for r_z.

$$V'_z = 17.6 + (80)(10 \times 10^{-3})$$
$$= 17.6 + 0.8$$
$$= 18.4\ \text{V}$$

Calculate the power rating of the diode.

$$P = V'_z I'_{ZT}$$

Substitute $18.4\ \text{V}$ for V'_z and $10\ \text{mA}$ for I'_{ZT}.

$$P = 18.4 \times 10 \times 10^{-3}$$
$$= 184\ \text{mW}$$

Thus, the power rating of the diode is $\boxed{184\ \text{mW}}$ missing resistance is $\boxed{80\ \Omega}$.

Step 9 of 10

(e)

Consider the expression for diode voltage.

$$V_z = V_{z0} + r_z I_{ZT}$$
$$V_{z0} = V_z - r_z I_{ZT}$$

Substitute $7.5\ \text{V}$ for V_z, $1.5\ \Omega$ for r_z and $200\ \text{mA}$ for I_{ZT}.

$$V_{z0} = 7.5 - (1.5)(200 \times 10^{-3})$$
$$= 7.5 - 0.3$$
$$= 7.2\ \text{V}$$

Calculate the value of new diode current.

$$I'_{ZT} = 2 \times I_{ZT}$$

Substitute $200\ \text{mA}$ for I_{ZT}.

$$I'_{ZT} = 2(200\ \text{mA})$$
$$= 400\ \text{mA}$$

Step 10 of 10

Calculate the value of new diode voltage, V'_z.

$$V'_z = V_{z0} + r_z I'_{ZT}$$

Substitute $7.2\ \text{V}$ for V_{zk}, $400\ \text{mA}$ for I'_{ZT} and $1.5\ \Omega$ for r_z.

$$V'_z = 7.2 + (1.5)(400 \times 10^{-3})$$
$$= 7.2 + 0.6$$
$$= 7.8\ \text{V}$$

Calculate the power rating of the diode.

$$P = V'_z I'_{ZT}$$

Substitute $7.8\ \text{V}$ for V'_z and $400\ \text{mA}$ for I'_{ZT}.

$$P = 7.8 \times 400 \times 10^{-3}$$
$$= 3.12\ \text{W}$$

Thus, the power rating of the diode is $\boxed{3.12\ \text{W}}$ missing voltage is $\boxed{7.2\ \text{V}}$.

4.060P

We don't have the solution to this problem yet.

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4.061P

We don't have the solution to this problem yet.

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Ask an expert

4.062P

We don't have the solution to this problem yet.

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Ask an expert

4.063P

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Step 1 of 7

Draw the circuit diagram for the shunt regulator with **1N5235** Zener diode.

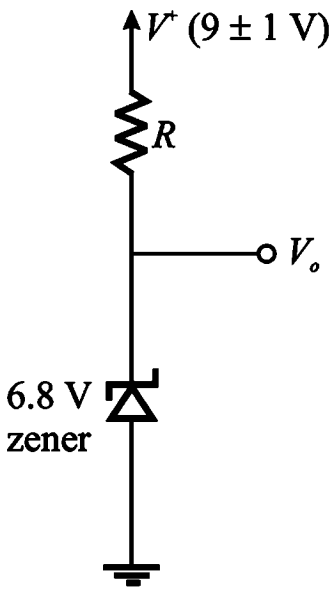


Figure 1

Step 2 of 7

Design 1:

Assume that the availability of supply current is not a problem, thus the diode operate at **20mA**

Redraw the circuit with notations.

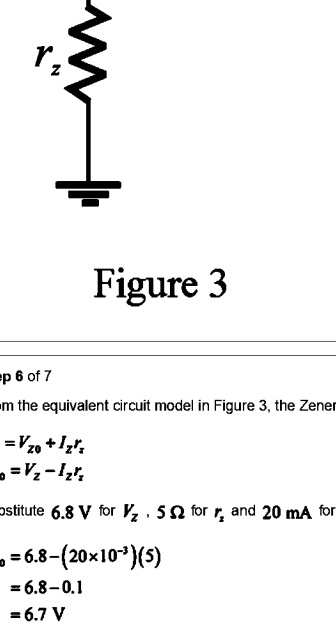


Figure 2

Step 3 of 7

Apply Kirchhoff's Current Law at node V_o .

$$I = I_z$$

$$\frac{V^+ - V_o}{R} = I_z$$

$$R = \frac{V^+ - V_o}{I_z}$$

Substitute **20mA** for I_z , **9 V** for V^+ and **6.8 V** for V_o .

$$R = \frac{9 - 6.8}{20 \times 10^{-3}}$$
$$= 110\ \Omega$$

Thus, the value of the resistor, R is $\boxed{110\ \Omega}$.

Step 4 of 7

Calculate the value of line regulation.

$$\frac{\Delta V_o}{\Delta V^+} = \frac{r_z}{R + r_z}$$

Substitute **110 Ω** for R and **5 Ω** for r_z .

$$\frac{\Delta V_o}{\Delta V^+} = \frac{5}{110 + 5}$$
$$= 43.478 \times 10^{-3}$$
$$= 43.478\ \text{mV/V}$$

Thus, the value of line regulation is $\boxed{43.478\ \text{mV/V}}$.

Step 5 of 7

Design 2:
Draw the circuit with its equivalent circuit model.

Figure 3

Step 6 of 7

From the equivalent circuit model in Figure 3, the Zener voltage is,

$$V_z = V_{z0} + I_z r_z$$
$$V_{z0} = V_z - I_z r_z$$

Substitute **6.8 V** for V_z , **5 Ω** for r_z and **20 mA** for I_z .

$$V_{z0} = 6.8 - (20 \times 10^{-3})(5)$$
$$= 6.8 - 0.1$$
$$= 6.7\ \text{V}$$

Now, the diode is forced to operate at **0.25mA** .

Calculate the value of the resistance, R .

$$R = \frac{V^+ - V_{z0}}{I_z}$$

Substitute **0.25 mA** for I_z , **9 V** for V^+ and **6.7 V** for V_o .

$$R = \frac{9 - 6.7}{0.25 \times 10^{-3}}$$
$$= 9200$$
$$= 9.2\ \text{k}\Omega$$

Thus, the value of the resistor, R is $\boxed{9.2\ \text{k}\Omega}$.

Step 7 of 7

Calculate the value of line regulation.

$$\frac{\Delta V_o}{\Delta V^+} = \frac{r_z}{R + r_z}$$

Substitute **9.2 kΩ** for R and **750 Ω** for r_z .

$$\frac{\Delta V_o}{\Delta V^+} = \left(\frac{750}{9200 + 750}\right)$$
$$= 75.376 \times 10^{-3}$$
$$= 75.376\ \text{mV/V}$$

Thus, the value of line regulation is $\boxed{75.376\ \text{mV/V}}$.

4.065P

We don't have the solution to this problem yet.

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Step 1 of 15

Draw the circuit for Zener shunt regulator.

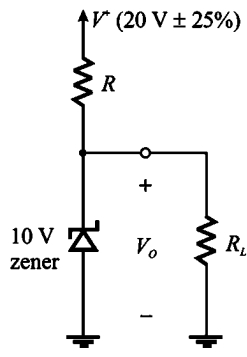


Figure 1

Step 2 of 15

(a)

Redraw the circuit with its equivalent circuit model.

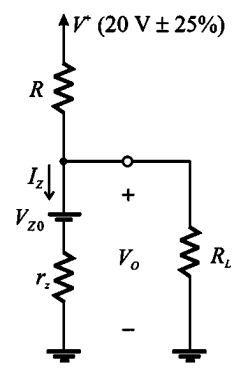


Figure 2

Step 3 of 15

From the equivalent circuit model, the Zener voltage is,

$$V_{ZT} = V_{Z0} + r_z I_{ZT}$$
$$V_{Z0} = V_{ZT} - r_z I_{ZT}$$

Substitute $10\ \text{V}$ for V_{ZT}, $25\ \text{mA}$ for I_{ZT} and $7\ \Omega$ for r_z.

$$V_{Z0} = 10 - (7)(0.025)$$
$$= 9.825\ \text{V}$$

Therefore, the Zener voltage, V_{Z0} is $\boxed{9.825\ \text{V}}$

Step 4 of 15

(b)

The regulator is required to supply a load current of $0\ \text{mA}$ to $20\ \text{mA}$, which means the maximum load current is $20\ \text{mA}$.

The minimum Zener current is possible if the current through the load is maximum.

$$I_{L(\max)} = 20\ \text{mA}$$
$$I_{Z(\min)} = 5\ \text{mA}$$

Also, the supply voltage is at its minimum.

The maximum supply voltage is,

$$V_{S(\max)} = V_S + (25\%)(V_S)$$
$$= 20 + (25\%)(20)$$
$$= 25\ \text{V}$$

Therefore, the supply voltage is,

$$V_{S(\max)} = 25\ \text{V}$$

Draw the circuit to illustrate the process.

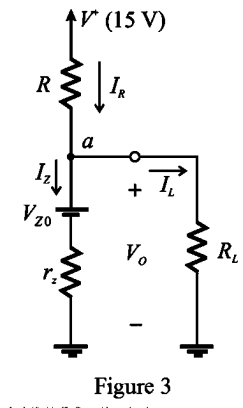


Figure 3

Apply Kirchhoff's Current Law at node a.

$$I_R = I_Z + I_L$$
$$= 5\ \text{mA} + 20\ \text{mA}$$
$$= 25\ \text{mA}$$

From the circuit, the current through the supply resistor is,

$$I_R = \frac{15 - V_Z}{R}$$
$$R = \frac{15 - (V_{Z0} + r_z I_z)}{I_R}$$

Substitute $25\ \text{mA}$ for I_R, $5\ \text{mA}$ for I_Z, $9.825\ \text{V}$ for V_{Z0} and $7\ \Omega$ for r_z.

$$R = \frac{15 - 9.825 - (7)(0.005)}{0.025}$$
$$= 205.6\ \Omega$$

Therefore, the required value of the supply resistor is $\boxed{205\ \Omega}$.

Step 7 of 15

(c)

Calculate the value of line regulation.

$$\frac{\Delta V_o}{\Delta V^+} = \frac{r_z}{R + r_z}$$

Substitute $205\ \Omega$ for R and $7\ \Omega$ for r_z.

$$\frac{\Delta V_o}{\Delta V^+} = \frac{7}{205 + 7}$$
$$= 33.02 \times 10^{-3}$$
$$= 33.02\ \text{mV/V}$$

Therefore, the value of line regulation is $\boxed{33.02\ \text{mV/V}}$

Step 8 of 15

The maximum variation in the supply voltage is,

$$\Delta V_S = \pm 25\%(V_S)$$
$$= \pm 0.25 \times 20\ \text{V}$$
$$= \pm 5\ \text{V}$$

The change in V_o expressed as a percentage, corresponding to $\pm 25\%$ change in V_S is,

$$\frac{\Delta V_o}{\Delta V^+} \times \frac{\Delta V_S}{V_S} \times 100\% = (33 \times 10^{-3})\left(\frac{\pm 5}{10}\right)(100\%)$$
$$= 1.65\%$$

Therefore, the required percentage change in the output is $\boxed{1.65\%}$

Step 9 of 15

(d)

The load regulation is,

$$\frac{\Delta V_o}{\Delta I_L} = -(r_z \parallel R)$$
$$= -\frac{r_z R}{r_z + R}$$

Substitute $205\ \Omega$ for R and $7\ \Omega$ for r_z.

$$\frac{\Delta V_o}{\Delta I_L} = -\left(\frac{7 \times 205}{7 + 205}\right)$$
$$= -6.768\ \Omega$$
$$= -6.768\ \text{mV/mA}$$

Therefore, the value of the load regulation is $\boxed{-6.768\ \text{mV/mA}}$

Step 10 of 15

The change in the output voltage is,

$$\Delta V_o = (-6.768)\Delta I_L$$
$$= (-6.768\ \text{V/A})(20\ \text{mA} - 0\ \text{mA})$$
$$= (-6.768\ \text{V/A})(20\ \text{mA})$$
$$= -135.36\ \text{mV}$$

The design output of shunt regulator should be $V_o = 10\ \text{V}$.

Calculate the percentage change in the output voltage.

$$\frac{\Delta V_o}{V_o} \times 100\% = \frac{-135.36\ \text{mV}}{10\ \text{V}} \times 100\%$$
$$= \frac{-135.36\ \text{mV}}{10\ \text{V}} \times 100\%$$
$$= 1.3536\%$$

Therefore, the percentage change in the output voltage from no-load to the full-load condition is $\boxed{1.3536\%}$.

Step 11 of 15

(e)

The maximum Zener current in the design is possible, when the load current is minimum.

The range of the load current is $0 - 20\ \text{mA}$

So, the minimum load current is $0\ \text{mA}$.

Also, the supply voltage is at its maximum.

The maximum supply voltage is,

$$V_{S(\max)} = V_S + (25\%)(V_S)$$
$$= 20 + (25\%)(20)$$
$$= 25\ \text{V}$$

Therefore, the supply voltage is,

$$V_{S(\max)} = 25\ \text{V}$$

Draw the circuit to illustrate the process.

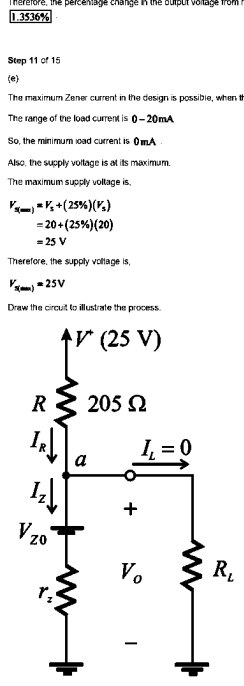


Figure 4

Step 12 of 15

Apply Kirchhoff's Current Law at node a.

$$I_R = I_Z + I_L$$
$$I_R = I_Z + 0$$
$$I_R = I_Z$$

Calculate the value of Zener current, I_Z

$$I_Z = \frac{V^+ - V_Z}{R}$$
$$= \frac{25 - V_Z}{205}$$

Step 13 of 15

Calculate the value of Zener voltage

$$V_Z = V_{Z0} + r_z I_z$$

Substitute $9.825\ \text{V}$ for V_{Z0}, $\frac{25 - V_Z}{205}$ for I_Z and $7\ \Omega$ for r_z.

$$V_Z = 9.825 + 7\left(\frac{25 - V_Z}{205}\right)$$
$$V_Z = 9.825 + 0.853 - 0.034V_Z$$
$$1.034V_Z = 10.678$$
$$V_Z = 10.326\ \text{V}$$

Step 14 of 15

Calculate the current through the supply resistor

$$I_R = \frac{V^+ - V_Z}{R}$$

Substitute $25\ \text{V}$ for V^+, $10.326\ \text{V}$ for V_Z and $205\ \Omega$ for R.

$$I_R = \frac{25\ \text{V} - 10.326\ \text{V}}{205\ \Omega}$$
$$= 0.07158\ \text{A}$$
$$= 71.58\ \text{mA}$$

The Zener current is,

$$I_Z = I_R$$
$$= 71.58\ \text{mA}$$

Therefore, the maximum current that the Zener is required to conduct is $\boxed{71.58\ \text{mA}}$.

Step 15 of 15

Calculate the power dissipated by the Zener diode.

$$P_Z = V_Z I_Z$$

Substitute $71.58\ \text{mA}$ for I_Z and $10.326\ \text{V}$ for V_Z.

$$P_Z = (10.326)(71.58 \times 10^{-3})$$
$$= 0.739\ \text{W}$$

Therefore, the Zener power dissipation is $\boxed{0.739\ \text{W}}$.

4.067P

We don't have the solution to this problem yet.

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Step 1 of 3

4.068P

Refer to Figure 4.21 (a) in the textbook.

The expression for exponential diode current equation is,

$$I_D = I_S e^{\frac{v_D}{V_T}} \quad \text{..... (1)}$$

Here,

I_D - the current flowing through the diode

I_S - the diode reverse bias current

v_D - the voltage across the diode

V_T - the thermal equivalent of voltage

Let us take the current flowing through the diode is $i_D(\text{at } 1 \text{ mA})$ when voltage across the diode is $v_D(\text{at } 1 \text{ mA})$ and also assume that the current flowing through the diode is i_D when voltage across the diode is v_D .

Step 2 of 3

Consider the relation between different diode currents from equation (1).

$$\frac{i_D}{i_D(1 \text{ mA})} = \frac{I_S e^{\frac{v_D}{V_T}}}{I_S e^{\frac{v_D(\text{at } 1 \text{ mA})}{V_T}}}$$

$$\frac{i_D}{i_D(1 \text{ mA})} = e^{\frac{v_D - v_D(\text{at } 1 \text{ mA})}{V_T}}$$

$$\frac{v_D - v_D(\text{at } 1 \text{ mA})}{V_T} = \ln\left(\frac{i_D}{i_D(1 \text{ mA})}\right)$$

$$v_D = v_D(\text{at } 1 \text{ mA}) + V_T \ln\left(\frac{i_D}{i_D(1 \text{ mA})}\right)$$

Step 3 of 3

Apply Kirchhoff's voltage law in the circuit

$$-v_S + v_D + v_O = 0$$

$$v_O = v_S - v_D$$

Substitute $v_D(\text{at } 1 \text{ mA}) + V_T \ln\left(\frac{i_D}{i_D(1 \text{ mA})}\right)$ for v_D .

$$v_O = v_S - v_D(\text{at } 1 \text{ mA}) - V_T \ln\left(\frac{i_D}{1 \times 10^{-3}}\right)$$

$$= v_S - v_D(\text{at } 1 \text{ mA}) - V_T \ln\left(\frac{\frac{v_O}{R}}{1 \times 10^{-3}}\right) \quad \left(\text{since } i_D = \frac{v_o}{R}\right)$$

$$= v_S - v_D(\text{at } 1 \text{ mA}) - V_T \ln\left(\frac{v_O}{R}\right) \quad (\text{since } R \text{ is in kilo Ohms})$$

Thus, the output voltage is verified and same.

Step 1 of 4

Refer to Figure 4.21 (a) in the textbook.

Draw the modified circuit with notations.

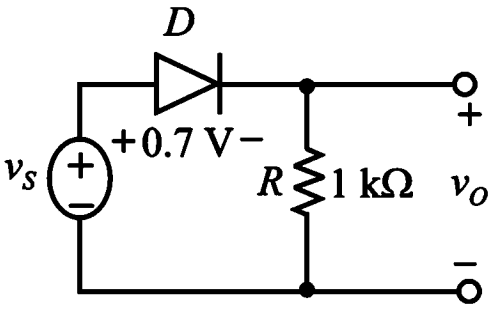


Figure 1

Step 2 of 4

In diode ON condition voltage drop across the diode is $+0.7\ \text{V}$.

Thus, the diode is ON mode when the input voltage is $v_s \geq +0.7\ \text{V}$.

Output voltage is available when $v_s \geq +0.7\ \text{V}$ and for $v_s < +0.7\ \text{V}$ output voltage is zero.

Draw the input-output triangular waveform.

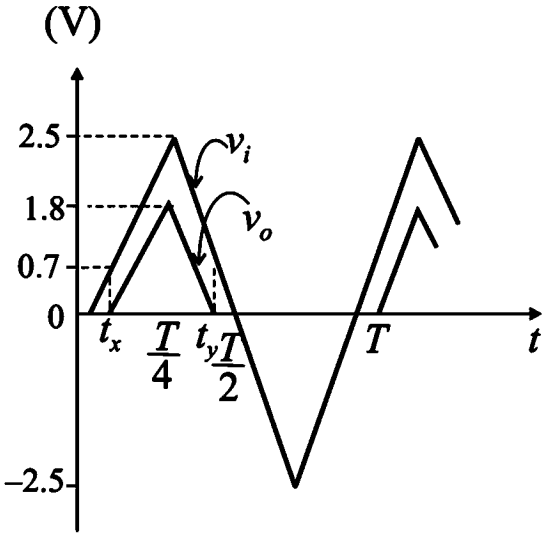


Figure 2

Step 3 of 4

Apply Kirchhoff's voltage law in the circuit.

$$v_S = 0.7 + v_O$$
$$v_O = v_S - 0.7$$

Thus, the output voltage is equal to the 0.7 V less than to the input voltage.

From the input-output waveform in Figure 2,

$$\frac{2.5}{\frac{T}{4}} = \frac{0.7}{t_x}$$

$$t_x = \frac{0.7T}{(2.5)4}$$
$$= 0.07T$$

From the input-output waveform in Figure 2,

$$t_y = \frac{T}{2} - t_x$$
$$= \frac{T}{2} - 0.07T$$
$$= 0.43T$$

Step 4 of 4

Calculate the output average voltage, $v_{O(\text{avg})}$.

$$v_{O(\text{avg})} = \frac{1}{T} \times (\text{area of the small triangle})$$
$$= \frac{1}{T} \times (2.5 - 0.7)\left(\frac{T}{4} - t_x\right)$$
$$= \frac{1}{T}(1.8)\left(\frac{1}{4} - 0.07\right)T$$
$$= (1.8)(0.25 - 0.07)$$
$$= 0.324\ \text{V}$$

Thus, the average output voltage of the circuit, $v_{O(\text{avg})}$ is $\boxed{0.324\ \text{V}}$.

4.070P

We don't have the solution to this problem yet.

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4.071P

We don't have the solution to this problem yet.

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4.072P

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Step 1 of 7

4.073P

Refer to Figure 4.22 in the textbook for the full wave rectifier circuit.

In this full wave rectifier, one diode, either D_1 or D_2 conducts at any time.

During positive half cycle, the diode D_1 is forward biased or conducts and the diode D_2 is reverse biased or do not conduct. During negative half cycle, the diode D_2 is forward biased or conducts and the diode D_1 is reverse biased or do not conduct.

Here, each diode has a voltage drop of 0.7 V and in either cycle, one diode conducts.

The diode D_1 in the circuit can be replaced with its piece-wise linear model, that is, the diode D_1 is replaced by V_D **and** r_D in series.

Neglecting the effect of r_D , the equivalent circuit for the right side upper portion of the Center tap is shown in Figure 1.

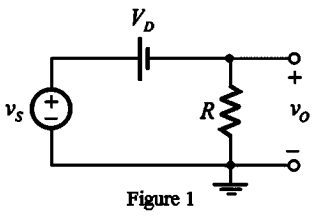


Figure 1

Step 2 of 7

Apply Kirchhoff's Voltage Law to the circuit.

$$v_S - V_D - v_O = 0$$

$$v_O = v_S - V_D \quad \text{..... (1)}$$

The input to the full-wave rectifier is, $v_S = V_S \sin \omega t$.

The output is obtained with the help of equation (1). The diode conducts during both the half cycles.

The input/output waveforms together are shown in Figure 2.

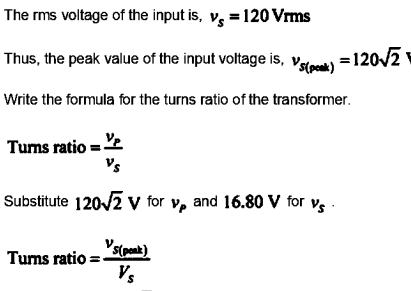


Figure 2

Step 3 of 7

In the waveform in Figure 2, θ is the angle at which the input signal reaches V_D and the diode starts conducting. The diode conduction terminates at $(\pi - \theta)$ and the hence the total conduction angle is $(\pi - \theta - \theta)$ or $(\pi - 2\theta)$.

Consider the following by observing the waveform:

$$V_D = V_S \sin \theta$$

$$\sin \theta = \frac{V_D}{V_S}$$

$$\theta = \sin^{-1}\left(\frac{V_D}{V_S}\right)$$

The average value (dc component) of the output is,

$$v_{O(avg)} = \frac{1}{T}\int_0^T v_O(t)\,dt$$

From the waveform, $T = 2\pi$

The limits of the integration are from $\phi = \theta$ **to** $\phi = \pi - \theta$ **and** $\phi = \pi + \theta$ **to** $\phi = 2\pi - \theta$

Recall equation (1).

$$v_O = v_S - V_D$$

Here, $v_S = V_S \sin \phi$ in the interval $\phi = \theta$ **to** $\phi = \pi - \theta$ and also the same in the interval $\phi = \pi + \theta$ **to** $\phi = 2\pi - \theta$

Step 4 of 7

Evaluate the integral over the first interval and make it twice.

$$v_{O(avg)} = \frac{1}{2\pi}\int_{\phi=\theta}^{\pi-\theta} 2(V_S \sin\phi - V_D)\,d\phi$$

$$= \frac{2}{2\pi}\left[V_S\left(-\cos\phi\Big|_\theta^{\pi-\theta} - V_D\left(\phi\Big|_\theta^{\pi-\theta}\right)\right)\right]$$

$$v_{O(avg)} = \frac{1}{\pi}\left[V_S\left(\cos(\theta) - \cos(\pi-\theta) - V_D(\pi-\theta-\theta)\right)\right]$$

Now let $\cos\theta \approx 1$, then $\cos(\pi - \theta) \approx -1$ and $\pi - 2\theta \approx \pi$ as θ is very small.

Therefore,

$$v_{O(avg)} \simeq \frac{1}{\pi}\left[V_S(2 - V_D\pi)\right]$$

$$\simeq \frac{2V_S}{\pi} - \frac{V_D\pi}{\pi}$$

$$v_{O(avg)} \simeq \frac{2V_S}{\pi} - V_D \quad \text{..... (2)}$$

Step 5 of 7

(a)

The average output voltage is $v_{o(avg)} = 10\ \text{V}$

Substitute 10 V for $v_{o(avg)}$ in equation (2) to calculate the value of v_S .

$$10 = \frac{2V_S}{\pi} - 0.7$$

$$\frac{2V_S}{\pi} = 10 + 0.7$$

$$V_S = 16.80\ \text{V}$$

Step 6 of 7

The rms voltage of the input is, $v_S = 120\ \text{Vrms}$

Thus, the peak value of the input voltage is, $v_{S(peak)} = 120\sqrt{2}\ \text{V}$

Write the formula for the turns ratio of the transformer.

$$\text{Turns ratio} = \frac{v_P}{v_S}$$

Substitute $120\sqrt{2}\ \text{V}$ for v_P and $16.80\ \text{V}$ for v_S .

$$\text{Turns ratio} = \frac{v_{S(peak)}}{V_S}$$

$$= \frac{120\sqrt{2}}{16.80}$$

$$= \frac{10.095}{1}$$

Therefore, the transformer turns ratio for each half of the secondary is $\boxed{10.095:1}$.

Step 7 of 7

(b)

The average output voltage is, $v_{o(avg)} = 100\ \text{V}$

Substitute 100 V for $v_{o(avg)}$ in equation (2) to calculate the value of v_S .

$$100 = \frac{2V_S}{\pi} - 0.7$$

$$\frac{2V_S}{\pi} = 100 + 0.7$$

$$V_S = 158.179\ \text{V}$$

Write the formula for the turns ratio of the transformer.

$$\text{Turns ratio} = \frac{v_P}{v_S}$$

Substitute $120\sqrt{2}\ \text{V}$ for v_P and $158.179\ \text{V}$ for v_S .

$$\text{Turns ratio} = \frac{v_{S(peak)}}{V_S}$$

$$= \frac{120\sqrt{2}}{158.179}$$

$$= \frac{1.0728}{1}$$

Therefore, the transformer turns ratio for each half of the secondary is $\boxed{1.0728:1}$.

Step 1 of 6

Refer to Figure 4.23 in the textbook for the bridge rectifier circuit.

In this bridge rectifier, two diodes either D_1 and D_2 or D_3 and D_4 conduct at any time.

During the positive half cycle of the input, the diodes D_1 and D_2 are on and the diodes D_3 and D_4 are off. During the negative half cycle of the input, the diodes D_3 and D_4 are on and the diodes D_1 and D_2 are off. The diodes which are conducting are replaced with their constant-voltage-drop diode model.

Consider the circuit for positive half cycle.

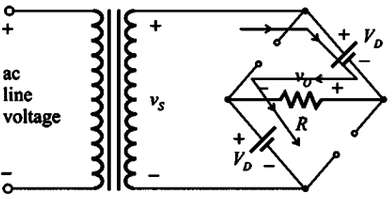


Figure 1

Step 2 of 6

Apply Kirchhoff's Voltage Law to the circuit.

$$v_S - V_D - v_O - V_D = 0$$
$$v_S - 2V_D - v_O = 0$$

$$v_O = v_S - 2V_D$$

The ac line voltage is $120\ \text{Vrms}$

Thus, the primary voltage is,

$$v_P = 120\sqrt{2}\ \text{V}$$

The average output voltage when $2V_D \ll v_S$ is,

$$\begin{aligned} v_{o(avg)} &= \frac{1}{2\pi}\int_{\phi=0}^{\pi-0} 2(V_S \sin\phi - 2V_D)\,d\phi \\ &= \frac{2V_m}{\pi} - 2V_D \\ &= \frac{2v_S}{\pi} - 2(0.7) \\ &= \frac{2v_S}{\pi} - 1.4 \end{aligned}$$

Step 3 of 6

(a)

The average output voltage is $v_{o(avg)} = 10\ \text{V}$

The average output voltage is,

$$v_{o(avg)} = \frac{2v_S}{\pi} - 1.4$$

Substitute 10 V for $v_{o(avg)}$ to calculate the value of v_S.

$$10 = \frac{2v_S}{\pi} - 1.4$$
$$\frac{2v_S}{\pi} = 11.4$$

$$\begin{aligned} v_S &= \frac{11.4\pi}{2} \\ &= 17.9071\ \text{V} \end{aligned}$$

Step 4 of 6

Write the formula for the turns ratio of the transformer.

$$\frac{N_P}{N_S} = \frac{v_P}{v_S}$$

Substitute $120\sqrt{2}\ \text{V}$ for v_P and $17.9071\ \text{V}$ for v_S.

$$\begin{aligned} \frac{N_P}{N_S} &= \frac{120\sqrt{2}}{17.9071} \\ &= 9.477 \end{aligned}$$

Therefore, the required turns ratio for the transformer is $\boxed{9.477:1}$.

Step 5 of 6

(b)
The average output voltage is,

$$v_{o(avg)} = 100\ \text{V}$$

The average output voltage is,

$$v_{o(avg)} = \frac{2v_S}{\pi} - 1.4$$

Substitute 100 V for $v_{o(avg)}$ to calculate the value of v_S.

$$100 = \frac{2v_S}{\pi} - 1.4$$
$$\frac{2v_S}{\pi} = 101.4$$

$$\begin{aligned} v_S &= \frac{101.4\pi}{2} \\ &= 159.279\ \text{V} \end{aligned}$$

Step 6 of 6

Write the formula for the turns ratio of the transformer.

$$\frac{N_P}{N_S} = \frac{v_P}{v_S}$$

Substitute $120\sqrt{2}\ \text{V}$ for v_P and $159.279\ \text{V}$ for v_S.

$$\begin{aligned} \frac{N_P}{N_S} &= \frac{120\sqrt{2}}{159.279} \\ &= 1.0655 \end{aligned}$$

Therefore, the required turns ratio of the transformer is $\boxed{1.0655:1}$.

4.075P

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4.076P

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4.077P

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4.078P

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4.079P

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4.080P

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4.081P

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4.082P

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Ask an expert

4.088P

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Ask an expert

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Ask an expert

4.090P

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4.091P

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Step 1 of 6

(a)

Consider the diode voltage drop in conduction as, $V_D = 0.7\,\text{V}$

Draw the following limiter circuit with the input signal in above $-0.7\ \text{V}$ range.

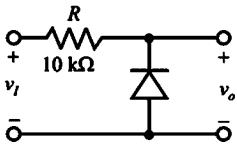


Figure 1

Step 2 of 6

From Figure 1 observe that the input voltage is below $-0.7\,\text{V}$ range, the diode in cut-off state. thus the output follows the input. When the input crosses $-0.7\,\text{V}$ then the diode is in conduction mode. Thus, the output is $-0.7\,\text{V}$.

Write the expression of output voltage.

$$v_o = \begin{cases} v_I & v_I > -0.7\,\text{V} \\ -0.7\,\text{V} & v_I < -0.7\,\text{V} \end{cases}$$

Therefore, the required limiter circuit is shown in Figure 1.

Step 3 of 6

(b)

Consider the diode voltage drop in conduction as $V_D = 0.7\,\text{V}$

Draw the following limiter circuit with the input signal limited to above $-2.1\,\text{V}$ range. Connect the three diodes in series.

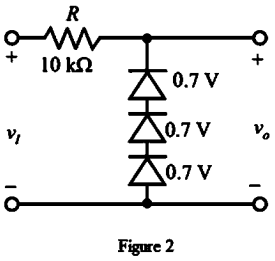


Figure 2

Step 4 of 6

For the input voltage below $-2.1\,\text{V}$ range, the diodes are in cut-off state. Thus, the output follows the input. When the input crosses $-0.7\,\text{V}$, the diodes are in conduction mode. Thus, the output is $-2.1\,\text{V}$.

Write the expression for the output voltage.

$$v_o = \begin{cases} v_I & v_I > -2.1\,\text{V} \\ -2.1\,\text{V} & v_I < -2.1\,\text{V} \end{cases}$$

Therefore, the required limiter circuit is shown in Figure 2.

Step 5 of 6

(c)

Consider the diode voltage drop in conduction as $V_D = 0.7\,\text{V}$

For the required limiter circuit the signal should be limited to above $\pm 1.4\,\text{V}$ range.

For limiting to $+1.4\,\text{V}$, connect two diodes in series in forward direction to the input.

For limiting to $-1.4\,\text{V}$, connect two diodes in series in reverse direction to the input.

Draw the following circuit diagram:

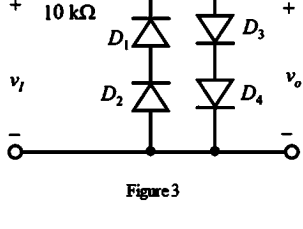


Figure 3

Step 6 of 6

If the input is greater than $+1.4\,\text{V}$, the diodes D_3 and D_4 are in conduction. Thus, the output is $+1.4\,\text{V}$.

If the input is less than $-1.4\,\text{V}$, the diodes D_1 and D_2 are in conduction. Thus, the output is $-1.4\,\text{V}$.

If the input is in between $+1.4\,\text{V}$ and $-1.4\,\text{V}$ then all the diodes are in cutoff region. Thus, the output follows the input.

Write the expression for the output voltage.

$$v_o = \begin{cases} +1.4\,\text{V} & v_I > 1.4\,\text{V} \\ v_I & -1.4\,\text{V} < v_I < 1.4\,\text{V} \\ -1.4\,\text{V} & v_I < -1.4\,\text{V} \end{cases}$$

Therefore, the required limiter circuit is drawn.

4.093P

We don't have the solution to this problem yet.

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4.094P

We don't have the solution to this problem yet.

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4.095P

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4.096P

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Step 1 of 23

Refer to Figure P4.95 for the input square wave in the text book.

(a)

Refer to Figure P4.95 (a) in the text book.

For an AC signal, the capacitor acts as short circuit. So, the total input voltage is dropped across the resistor. The input is simply passes to the output.

$v_o = v_i$

Draw the output waveform.

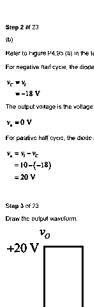


Figure 1

Step 2 of 23

(b)

Refer to Figure P4.95 (b) in the text book.

For negative half cycle, the diode acts as forward bias (short) and the capacitor starts charging.

$v_C = v_i$

$= -18$ V

The output voltage is the voltage across short circuited diode.

$v_o = 0$ V

For positive half cycle, the diode acts as reverse bias (open) and the capacitor starts discharging.

$v_o = v_i - v_C$

$= 10 - (-18)$

$= 20$ V

Step 3 of 23

Draw the output waveform.

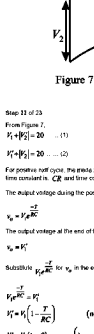


Figure 2

Step 4 of 23

(c)

Refer to Figure P4.95 (c) in the text book.

For positive half cycle, the diode acts as forward bias (short) and the capacitor starts charging.

$v_C = v_i$

$= 10$ V

The output voltage is the voltage across short circuited diode.

$v_o = 0$ V

For negative half cycle, the diode acts as reverse bias (open) and the capacitor starts discharging.

$v_o = v_i - v_C$

$= -10 - (18)$

$= -20$ V

Step 5 of 23

Draw the output waveform.

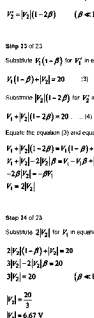


Figure 3

Step 6 of 23

(d)

Refer to Figure P4.95 (d) in the text book.

The time constant $CR \gg T$. Thus, it rectifies as a part (c).

For positive half cycle, the diode acts as forward bias (short) and the capacitor starts charging.

$v_C = v_i$

$= 10$ V

The output voltage is the voltage across short circuited diode.

$v_o = 0$ V

For negative half cycle, the diode acts as reverse bias (open) and the capacitor starts discharging.

$v_o = v_i - v_C$

$= -10 - (18)$

$= -20$ V

Step 7 of 23

Draw the output waveform.

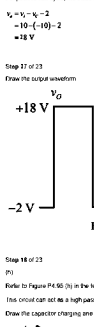


Figure 4

Step 8 of 23

(e)

Refer to Figure P4.95 (e) in the text book.

The time constant $CR \gg T$. So, the $2CR \gg T$.

For positive half cycle, left side diode is on and for negative half cycle, the right side diode is on.

Draw the modified circuit for either positive or negative half cycle input square wave.

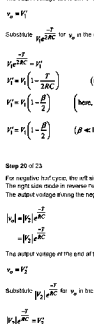


Figure 5

Step 9 of 23

(f)

Refer to Figure P4.95 (f) in the text book.

This circuit can act as a high pass filter and it has two different time constants.

Draw the capacitor charging and discharging voltage waveforms.

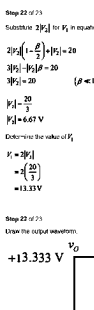


Figure 7

Step 11 of 23

From Figure 7,

$V_1 + |V_2| = 20$ ... (1)

$V_1' + |V_2| = 20$ ... (2)

For positive half cycle, the diode acts as reverse bias (open) and the capacitor starts discharging. Thus, the time constant is CR and time constant $CR \gg T$.

The output voltage during the positive half cycle is,

$v_o = V_1 e^{\frac{-T}{RC}}$

The output voltage at the end of the positive half cycle is

$v_o = V_1'$

Substitute $V_1 e^{\frac{-T}{RC}}$ for v_o in the equation.

$V_1 e^{\frac{-T}{RC}} = V_1'$

$V_1' = V_1\left(1 - \frac{T}{RC}\right)$ **(neglect higher order terms)**

$V_1' = V_1(1-\beta)$ $\left(\text{here, } \beta = \frac{T}{RC}\right)$

$V_1' = V_1(1-\beta)$ $(\beta \ll 1)$

Step 12 of 23

For negative half cycle, the diode acts as forward bias (short) and the capacitor starts charging. Thus, the time constant is,

$\tau = C(R \| R)$

$= \frac{RC}{2}$

The output voltage during the negative half cycle is,

$|v_o| = |V_2| e^{\frac{-T}{\frac{RC}{2}}}$

$= |V_2| e^{\frac{-2T}{RC}}$

The output voltage at the end of the negative half cycle is,

$v_o = V_2'$

Substitute $|V_2| e^{\frac{-2T}{RC}}$ for v_o in the equation

$|V_2| e^{\frac{-2T}{RC}} = V_2'$

$V_2' = |V_2| e^{\frac{-2T}{RC}}$

$= |V_2|\left(1 - \frac{2T}{RC}\right)$ **(neglect higher order terms)**

$= |V_2|(1-2\beta)$ $\left(\text{here, } \beta = \frac{T}{RC}\right)$

$V_2' = |V_2|(1-2\beta)$ $(\beta \ll 1)$

Step 13 of 23

Substitute $V_1(1-\beta)$ for V_1' in equation (2)

$V_1(1-\beta) + |V_2| = 20$ ... (3)

Substitute $|V_2|(1-2\beta)$ for V_2' in equation (1)

$V_1 + |V_2|(1-2\beta) = 20$ ... (4)

Equate the equation (3) and equation (4).

$V_1 + |V_2|(1-2\beta) = V_1(1-\beta) + |V_2|$

$V_1 + |V_2| - 2|V_2|\beta = V_1 - V_1\beta + |V_2|$

$-2\beta|V_2| = -\beta V_1$

$V_1 = 2|V_2|$

Step 14 of 23

Substitute $2|V_2|$ for V_1 in equation (3), to determine the value of $|V_2|$

$2|V_2|(1-\beta) + |V_2| = 20$

$3|V_2| - 2|V_2|\beta = 20$

$3|V_2| = 20$ $(\beta \ll 1, \text{so neglect } \beta|V_2|)$

$|V_2| = \frac{20}{3}$

$|V_2| = 6.67$ V

Determine the value of V_1

$V_1 = 2|V_2|$

$= 2\left(\frac{20}{3}\right)$

$= 13.33$ V

Step 15 of 23

Draw the output waveform.

Figure 8

Step 16 of 23

(g)

Refer to Figure P4.95 (g) in the text book.

For the negative input greater than -2 V, the diode acts as forward bias (short) and the capacitor starts charging.

$v_C = v_i$

$= -18$ V

The output voltage is the equals to the DC supply.

$v_o = -2$ V

For positive half cycle, the diode acts as reverse bias (open) and the capacitor starts discharging.

$v_o = v_i - v_C - 2$

$= 10 - (-10) - 2$

$= 18$ V

Step 17 of 23

Draw the output waveform

Figure 9

Step 18 of 23

(h)

Refer to Figure P4.95 (h) in the text book.

This circuit can act as a high pass filter and it has two different time constants.

Draw the capacitor charging and discharging voltage waveforms

Figure 10

Step 19 of 23

From Figure 10,

$V_1 + |V_2| = 20$ ... (5)

$V_1' + |V_2| = 20$ ... (6)

For positive half cycle, the left side diode acts as reverse bias (open) and the capacitor starts discharging. The right side diode in forward bias (short). Thus, the time constant is, $2CR$ and time constant $2CR \gg T$.

The output voltage during the positive half cycle is,

$v_o = V_1 e^{\frac{-T}{2RC}}$

The output voltage at the end of the positive half cycle is

$v_o = V_1'$

Substitute $V_1 e^{\frac{-T}{2RC}}$ for v_o in the equation

$V_1 e^{\frac{-T}{2RC}} = V_1'$

$V_1' = V_1\left(1 - \frac{T}{2RC}\right)$ **(neglect higher order terms)**

$V_1' = V_1\left(1 - \frac{\beta}{2}\right)$ $\left(\text{here, } \beta = \frac{T}{RC}\right)$

$V_1' = V_1\left(1 - \frac{\beta}{2}\right)$ $(\beta \ll 1)$

Step 20 of 23

For negative half cycle, the left side diode acts as forward bias (short) and the capacitor starts charging. The right side diode in reverse bias (open). Thus, the time constant is, RC

The output voltage during the negative half cycle is,

$|v_o| = |V_2| e^{\frac{-T}{RC}}$

$= |V_2| e^{\frac{-T}{RC}}$

The output voltage at the end of the negative half cycle is,

$v_o = V_2'$

Substitute $|V_2| e^{\frac{-T}{RC}}$ for v_o in the equation

$|V_2| e^{\frac{-T}{RC}} = V_2'$

$V_2' = |V_2| e^{\frac{-T}{RC}}$

$= |V_2|\left(1 - \frac{T}{RC}\right)$ **(neglect higher order terms)**

$= |V_2|(1-\beta)$ $\left(\text{here, } \beta = \frac{T}{RC}\right)$

$V_2' = |V_2|(1-\beta)$ $(\beta \ll 1)$

Step 21 of 23

Substitute $V_1\left(1 - \frac{\beta}{2}\right)$ for V_1' in equation (6)

$V_1\left(1 - \frac{\beta}{2}\right) + |V_2| = 20$ ... (7)

Substitute $|V_2|(1-\beta)$ for V_2' in equation (5)

$V_1 + |V_2|(1-\beta) = 20$ ..... (8)

Equate the equation (7) and equation (8).

$V_1 + |V_2|(1-\beta) = V_1\left(1 - \frac{\beta}{2}\right) + |V_2|$

$V_1 + |V_2| - |V_2|\beta = V_1 - V_1\frac{\beta}{2} + |V_2|$

$-\beta|V_2| = -\frac{\beta}{2}V_1$

$V_1 = 2|V_2|$

Step 22 of 23

Substitute $2|V_2|$ for V_1 in equation (7), to determine the value of $|V_2|$

$2|V_2|\left(1 - \frac{\beta}{2}\right) + |V_2| = 20$

$3|V_2| - |V_2|\beta = 20$

$3|V_2| = 20$ $(\beta \ll 1, \text{so neglect } \beta|V_2|)$

$|V_2| = \frac{20}{3}$

$|V_2| = 6.67$ V

Determine the value of V_1

$V_1 = 2|V_2|$

$= 2\left(\frac{20}{3}\right)$

$= 13.33$ V

Step 23 of 23

Draw the output waveform.

Figure 11

Step 1 of 2

5.001E

The permittivity of silicon dioxide is,

$$\varepsilon_{ox} = 3.45 \times 10^{-11}\ \text{F/m}$$

The oxide capacitance is,

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$$

Substitute $3.45 \times 10^{-11}\ \text{F/m}$ for ε_{ox} and $4\ \text{nm}$ for t_{ox} in the equation.

$$C_{ox} = \frac{3.45 \times 10^{-11}}{4 \times 10^{-9}}$$

$$= 0.8625 \times 10^{-2}\ \text{F/m}^2$$

The Trans conductance parameter is,

$$k_n' = \mu_n C_{ox}$$

Substitute $450\ \text{cm}^2/\text{Vs}$ for μ_n and $0.8625 \times 10^{-2}\ \text{F/m}^2$ for C_{ox} in the equation.

$$k_n' = \left(450 \times 10^{-4}\right)\left(0.8625 \times 10^{-2}\right)$$

$$= 388.125\ \mu\text{A/V}^2$$

Hence, the value of process transconductance parameter is, $\boxed{k_n' = 388.125\ \mu\text{A/V}^2}$.

Step 2 of 2

The relation for the linear resistance is,

$$r_{DS} = \frac{1}{\left(\mu_n C_{ox}\right)\left(\frac{W}{L}\right)\left(V_{GS} - V_t\right)}$$

Substitute $1\ \text{k}\Omega$ for r_{DS}, $450\ \text{cm}^2/\text{Vs}$ for μ_n, $0.8625 \times 10^{-2}\ \text{F/m}^2$ for C_{ox}, $0.18\ \mu\text{m}$ for L, 1V for V_{GS}, and $0.5\ \text{V}$ for V_t in the equation.

$$\left(1 \times 10^3\right) = \frac{1}{\left(450 \times 10^{-4}\right)\left(0.8625 \times 10^{-2}\right)\left(\frac{W}{0.18 \times 10^{-6}}\right)(1 - 0.5)}$$

$$\frac{W}{0.18 \times 10^{-6}} = \frac{1}{\left(388.125 \times 10^{-6}\right)(0.5)\left(1 \times 10^3\right)}$$

$$W = \frac{0.18 \times 10^{-6}}{194.0625 \times 10^{-3}}$$

$$W = 0.93 \times 10^{-6}$$

Hence, value of W is, $\boxed{W = 0.93\,\mu\text{m}}$.

Step 1 of 3

Write the formula for oxide capacitance, C_{ox}.

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$$

Here,

ε_{ox} is the permittivity of silicon dioxide

t_{ox} is the oxide thickness

Substitute $3.45\times10^{-11}\ \text{F/m}$ for ε_{ox} and $2\times10^{-9}\ \text{m}$ for t_{ox}.

$$C_{ox} = \frac{3.45\times10^{-11}}{2\times10^{-9}}$$
$$= 0.01725\ \text{F/m}^2$$

The oxide capacitance is defined as the capacitance of the parallel-plate capacitor per unit gate area.

$$C_{ox} = \frac{C}{WL}$$

Substitute $10^{-12}\ \text{F}$ for C and $0.01725\ \text{F/m}^2$ for C_{ox}.

$$0.01725 = \frac{10^{-12}}{WL}$$
$$WL = 57.97\times10^{-12}\ \text{m}^2$$
$$= 57.97\ \mu\text{m}^2$$

Step 2 of 3

Determine the oxide capacitance for an oxide thickness of 10 nm.

$$C_{ox} = \frac{3.45\times10^{-11}}{10\times10^{-9}}$$
$$= 3.45\times10^{-3}\ \text{F/m}^2$$

Determine the area required.

$$C_{ox} = \frac{C}{WL}$$
$$3.45\times10^{-3} = \frac{10^{-12}}{WL}$$
$$WL = 289.86\times10^{-12}\ \text{m}^2$$
$$= 289.86\ \mu\text{m}^2$$

Thus, the range of area required for 1 pF capacitance is $\boxed{57.97\ \mu\text{m}^2}$ to $\boxed{289.86\ \mu\text{m}^2}$.

Step 3 of 3

Determine the area required for 10 pF capacitance and at a thickness of $2\ \text{nm}$.

$$C_{ox} = \frac{C}{WL}$$
$$0.01725 = \frac{10\times10^{-12}}{WL}$$
$$WL = 579.71\times10^{-12}\ \text{m}^2$$
$$= 579.71\ \mu\text{m}^2$$

Determine the area required for 10 pF capacitance and at a thickness of $10\ \text{nm}$.

$$C_{ox} = \frac{C}{WL}$$
$$3.45\times10^{-3} = \frac{10\times10^{-12}}{WL}$$
$$WL = 2898.6\times10^{-12}\ \text{m}^2$$
$$= 2898.6\ \mu\text{m}^2$$

Thus, the range of area required for 1 pF capacitance is $\boxed{579.71\ \mu\text{m}^2}$ to $\boxed{2898.6\ \mu\text{m}^2}$.

5.002E

Step 1 of 3

Permittivity of the silicon dioxide ε_{ox} is $3.45\times10^{-11}\ \frac{F}{m}$

Oxide thickness t_{ox} is $4\ nm$

Oxide capacitance C_{ox} is $\frac{\varepsilon_{ox}}{t_{ox}}$

$$C_{ox}=\frac{\varepsilon_{ox}}{t_{ox}}$$
$$=\frac{3.45\times10^{-11}}{4\times10^{-9}}$$
$$=\left(\frac{8.6\times10^{-15}}{10^{-6}}\right)\frac{F}{m^2}$$
$$=8.6\ \frac{fF}{m^2}$$

Oxide capacitance C_{ox} is $\boxed{8.6\ \frac{fF}{m^2}}$.

Step 2 of 3

Mobility of electron μ_n is $450\ \frac{cm^2}{V\cdot s}$

k_n' is $\mu_n C_{ox}$

Substitute μ_n and C_{ox} in $k_n'=\mu_n C_{ox}$

$$k_n'=\left(450\ \frac{cm^2}{V\cdot s}\right)\times\left(8.6\ \frac{fF}{\mu m^2}\right)$$
$$=\left(450\times10^8\ \frac{\mu m^2}{V\cdot s}\right)\left(8.6\times10^{-15}\ \frac{F}{\mu m^2}\right)$$
$$=387\times10^{-6}\ \frac{F}{V\cdot s}$$
$$=387\ \frac{\mu A}{V^2}$$

Process trans-conductance k_n' is $\boxed{387\ \frac{\mu A}{V^2}}$

Step 3 of 3

Aspect ratio $\frac{W}{L}$ is 20

Drain current i_D flowing through transistor when the transistor is in saturation is $0.3\ mA$

For operation of transistor in saturation region $i_D=\frac{1}{2}k_n'\frac{W}{L}V_{ov}^2$

$$i_D=\frac{1}{2}k_n'\frac{W}{L}V_{ov}^2$$
$$0.3\times10^{-3}=\frac{1}{2}\left(387\times10^{-6}\right)(20)V_{ov}^2$$
$$0.0775=V_{ov}^2$$
$$0.28=V_{ov}$$

Over drive voltage V_{ov} is $\boxed{0.28\ V}$

For transistor in saturation minimum value of V_{DS} is equal to over drive voltage V_{ov} and hence minimum value of V_{DS} is equal to $\boxed{0.28\ V}$.

Step 1 of 2

5.002P

Write the formula for charge stored in the channel of an NMOS.

$$|Q| = C_{ox}(WL)v_{ov}$$

Here,

C_{ox} is the oxide capacitance

W is the width of the channel

L is the length of the channel

v_{ov} is the overdrive voltage

Step 2 of 2

Substitute $9\ \text{fF}/\mu\text{m}^2$ for C_{ox}, $0.36\ \mu\text{m}$ for L, $3.6\ \mu\text{m}$ for W and $0.2\ \text{V}$ for V_{ov} in the expression for charge stored in the channel of an NMOS.

$$|Q| = \left(9\ \text{fF}/\mu\text{m}^2\right)(0.36\ \mu\text{m})(3.6\ \mu\text{m})(0.2\ \text{V})$$
$$= 2.3328\ \text{fF}$$

Thus, the total charge stored in the channel of an NMOS transistor is **2.3328 fF** .

Step 1 of 7

5.003E

The relation for the drain current for a MOSFET operating in the saturation region is, $i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)V_{OV}^2$ ..... (1)

Here, k_n' is the process transconductance parameter,

V_{OV} is the overdrive voltage,

W is the channel width, and

L is the channel length.

Step 2 of 7

(a)

When the channel length is doubled, the corresponding drain current is,

$$i_{D1} = \frac{1}{2}k_n'\left(\frac{W}{2L}\right)V_{OV}^2 \quad \text{..... (2)}$$

Divide equation (1) by equation (2).

$$\frac{i_D}{i_{D1}} = \frac{\frac{1}{2}k_n'\left(\frac{W}{L}\right)V_{OV}^2}{\frac{1}{2}k_n'\left(\frac{W}{2L}\right)V_{OV}^2}$$

$$\frac{i_D}{i_{D1}} = \frac{\frac{1}{L}}{\frac{1}{2L}}$$

$$\frac{i_D}{i_{D1}} = 2$$

$$i_{D1} = \frac{1}{2}i_D$$

Therefore, the value of drain current becomes **halved** when the length is doubled.

Step 3 of 7

(b)

When the channel width is doubled, the relation for the drain current i_D is, $i_{D1} = \frac{1}{2}k_n'\left(\frac{2W}{L}\right)V_{OV}^2$ ..... (3)

Divide equation (1) by equation (3).

$$\frac{i_D}{i_{D1}} = \frac{\frac{1}{2}k_n'\left(\frac{W}{L}\right)V_{OV}^2}{\frac{1}{2}k_n'\left(\frac{2W}{L}\right)V_{OV}^2}$$

$$\frac{i_D}{i_{D1}} = \frac{W}{2W}$$

$$\frac{i_D}{i_{D1}} = \frac{1}{2}$$

$$i_{D1} = 2i_D$$

Hence, the value of drain current becomes **doubled** when the width is doubled.

Step 4 of 7

(c)

When the overdrive voltage is doubled, the relation for the drain current i_D is,

$$i_{D1} = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(2V_{OV})^2 \quad \text{..... (4)}$$

Divide equation (1) by equation (4).

$$\frac{i_D}{i_{D1}} = \frac{\frac{1}{2}k_n'\left(\frac{W}{L}\right)V_{OV}^2}{\frac{1}{2}k_n'\left(\frac{W}{L}\right)(2V_{OV})^2}$$

$$\frac{i_D}{i_{D1}} = \frac{V_{OV}^2}{4V_{OV}^2}$$

$$\frac{i_D}{i_{D1}} = \frac{1}{4}$$

$$i_{D1} = 4i_D$$

Hence, the value of drain current becomes **four times** when the width is doubled.

Step 5 of 7

(d)

In the saturation region, the drain source voltage, V_{DSsat} is equal to the overdrive voltage, V_{OV} of the MOSFET.

The drain-to- source voltage is doubled. The relation of the current in saturation region is,

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)V_{OV}^2$$

Since the drain current is independent of drain-to- source voltage v_{DS} , there is no change in the drain current i_D even when the drain-to-source voltage is doubled.

Therefore, the drain current i_D **remains same**, when the drain-to-source voltage is doubled.

Step 6 of 7

(e)
When the channel length, channel width, overdrives voltage, drain-to-source voltage is doubled simultaneously; the relation for the drain current is,

$$i_{D1} = \frac{1}{2}k_n'\left(\frac{2W}{2L}\right)(2V_{OV})^2 \quad \text{..... (5)}$$

Divide equation (1) by equation (5).

$$\frac{i_D}{i_{D1}} = \frac{\frac{1}{2}k_n'\left(\frac{W}{L}\right)V_{OV}^2}{\frac{1}{2}k_n'\left(\frac{2W}{2L}\right)(2V_{OV})^2}$$

$$\frac{i_D}{i_{D1}} = \frac{1}{4}$$

$$i_{D1} = 4i_D$$

Hence, the value of drain current becomes **four times** when the width is doubled.

Step 7 of 7

The MOSFET is said to be in the edge of saturation region, when the overdrive voltage, V_{OV} is equal to the drain to source voltage $V_{OV} = V_{DS}$.

The MOSFET leave the saturation region, when the overdrive voltage is doubled, and the drain to source voltage, V_{DS} is smaller than twice the value of the overdrive voltage.

Hence, the case (c), when the overdrive voltage is doubled and the voltage, V_{DS} being less than twice the overdrive voltage is the case that the MOSFET leave the saturation region.

Step 1 of 3

5.003P

Write the formula for the transconductance parameter, k_n' .

$$k_n' = \mu_n C_{ox}$$

Here,

μ_n is the mobility

C_{ox} is the oxide capacitance

μ_n has the dimensions of $\mathbf{m^2/V \cdot s}$

C_{ox} has the dimensions of $\mathbf{F/m^2}$

Write the formula for capacitance.

$$C = \frac{q}{V}$$
$$= \frac{it}{V}$$

Here,

q is the charge stored in the capacitor

V is the applied voltage

i is the current flowing through the capacitor

t is the time of flow of current

The unit of capacitance, F has the dimensions of $\mathbf{A \cdot s/V}$.

Step 2 of 3

Determine the dimensions of k_n' .

$$k_n' = \left(\mathrm{m^2/V \cdot s}\right)\left(\frac{\mathrm{A \cdot s/V}}{\mathrm{m^2}}\right)$$
$$= \left(\frac{\mathrm{m^2}}{\mathrm{V \cdot s}}\right)\left(\frac{\mathrm{A \cdot s}}{\mathrm{V \cdot m^2}}\right)$$
$$= \left(\frac{1}{\mathrm{V}}\right)\left(\frac{\mathrm{A}}{\mathrm{V}}\right)$$
$$= \mathrm{A/V^2}$$

Thus, the units of transconductance parameter are $\boxed{\mathrm{A/V^2}}$.

Step 3 of 3

Write the formula for MOSFET transconductance parameter, k_n .

$$k_n = k_n'\left(\frac{W}{L}\right)$$

As the width W is expressed in meters and the length L is also expressed in meters, the ratio, $\frac{W}{L}$ is dimensionless.

Hence, k_n and k_n' have the same units.

Thus, the units of MOSFET transconductance parameter are $\boxed{\mathrm{A/V^2}}$.

Step 1 of 2

5.004E

NMOS transistor is operating at the edge of saturation, with an overdrive voltage V_{OV} and drain current I_D.

Since the NMOS transistor works in the edge of saturation, the drain source voltage is equal to the overdrive voltage, $V_{DS} = V_{OV}$. If the overdrive voltage, V_{OV} is doubled, the drain source voltage, V_{DS} should become twice the value of former overdrive voltage to maintain operation at the edge of saturation.

Therefore, the voltage V_{DS} changed to $\boxed{2\,V_{OV}}$.

Step 2 of 2

The relation of the drain current I_D with the overdrive voltage V_{OV} is,

$$I_D = \frac{1}{2} k_n V_{OV}^2 \quad \text{..... (1)}$$

The new drain current I_D' , when V_{OV} is doubled is,

$$I_D' = \frac{1}{2} k_n (2V_{OV})^2 \quad \text{..... (2)}$$

Divide equation (1) by equation (2).

$$\frac{I_D}{I_D'} = \frac{\frac{1}{2} k_n V^2{}_{OV}}{\frac{1}{2} k_n (2V_{OV})^2}$$

$$\frac{I_D}{I_D'} = \frac{V^2{}_{OV}}{(2V_{OV})^2}$$

$$\frac{I_D}{I_D'} = \frac{1}{4}$$

$$I_D' = 4I_D$$

Therefore, the new value of drain current is $\boxed{4I_D}$.

Step 1 of 6

Write the formula for current flowing through the induced *n* channel.

$$i_D = \left[(\mu_n C_{OX})\left(\frac{W}{L}\right)(v_{GS} - V_t)\right] v_{DS}$$

Here,

μ_n is the mobility of the electrons at the surface of the channel

C_{OX} is the oxide capacitance

$\frac{W}{L}$ is the width to length ratio of the channel

v_{GS} is the gate-to-source voltage

v_{DS} is the drain-to-source voltage

V_t is the threshold voltage

The oxide thickness indicates the electrical insulating capacity between the source and drain regions.

For small v_{DS} , the channel behaves as a linear resistance.

$$\frac{v_{DS}}{i_D} = \frac{1}{(\mu_n C_{OX})\left(\frac{W}{L}\right)(v_{GS} - V_t)}$$

The linear resistance is,

$$r_{DS} = \frac{1}{(\mu_n C_{OX})\left(\frac{W}{L}\right)(v_{GS} - V_t)}$$

Substitute v_{OV} for $v_{GS} - V_t$.

$$r_{DS} = \frac{1}{(\mu_n C_{OX})\left(\frac{W}{L}\right)v_{OV}}$$

Here,

v_{OV} is the overdrive voltage

Step 2 of 6

(a)

When v_{OV} is doubled, say $2v_{DV}$, the new linear resistance is,

$$r'_{DS} = \frac{1}{(\mu_n C_{OX})\left(\frac{W}{L}\right)2v_{OV}}$$

$$= \frac{r_{DS}}{2}$$

Thus, when v_{OV} is doubled, the linear resistance r_{DS} is **halved.**

Hence, the resistance r_{DS} is reduced by a factor of $\boxed{2}$.

Step 3 of 6

(b)

When the width *W* is doubled, the new linear resistance is,

$$r'_{DS} = \frac{1}{(\mu_n C_{OX})\left(\frac{2W}{L}\right)v_{OV}}$$

$$= \frac{r_{DS}}{2}$$

Thus, when *W* is doubled, the linear resistance r_{DS} is **halved.**

Hence, the resistance r_{DS} is reduced by a factor of $\boxed{2}$.

Step 4 of 6

(c)

When both the width *W* and the Length *L* are doubled, the new linear resistance is,

$$r'_{DS} = \frac{1}{(\mu_n C_{OX})\left(\frac{2W}{2L}\right)v_{OV}}$$

$$= r_{DS}$$

Thus, there is no change in the resistance r_{DS} when both *L* and *W* are doubled.

Hence, the resistance r_{DS} remains $\boxed{\textbf{unchanged}}$.

Step 5 of 6

(d)

The oxide capacitance in terms of oxide thickness is,

$$C_{OX} = \frac{\varepsilon_{OX}}{t_{OX}}$$

Here,

ε_{OX} is the permittivity of silicon dioxide

t_{OX} is the oxide thickness

When oxide thickness is halved, the oxide capacitance becomes doubled.

Step 6 of 6

When both the width *W* and the Length *L* are halved, and the oxide thickness is halved, the new linear resistance is,

$$r'_{DS} = \frac{1}{(\mu_n 2C_{OX})\left(\frac{\frac{W}{2}}{\frac{L}{2}}\right)v_{OV}}$$

$$= \frac{r_{DS}}{2}$$

Thus, when the oxide thickness is halved, the linear resistance r_{DS} is **halved.**

Hence, the resistance r_{DS} is reduced by a factor of $\boxed{2}$.

Step 1 of 3

The device transconductance parameter k_n is,

$$k_n = \mu_n C_{ox}\left(\frac{W}{L}\right)$$

Here, μ_n is the mobility of electrons and C_{ox} is the oxide capacitance

The linear resistance r_{DS} is,

$$r_{DS} = \frac{1}{\mu_n C_{ox}\left(\frac{W}{L}\right)V_{OV}}$$

$$r_{DS} = \frac{1}{k_n V_{OV}}$$

Substitute $1\ \text{k}\Omega$ for r_{DS} and $0.5\ \text{V}$ for V_{OV} in the equation.

$$\left(1\times 10^3\right) = \frac{1}{k_n(0.5)}$$

$$k_n = \frac{1}{0.5\times 10^3}$$

$$k_n = 2\ \text{mA/V}^2$$

Hence, the value of transconductance parameter is $\boxed{k_n = 2\ \text{mA/V}^2}$.

Step 2 of 3

When the voltage v_{DS} is increased to $0.5\ \text{V}$, then $v_{DS} = V_{OV}$. So, the transistor operates at the edge of saturation.

The relation for the drain current is,

$$I_D = \frac{1}{2}k_n V_{OV}^2$$

Substitute $2\ \text{mA/V}^2$ for k_n, $0.5\ \text{V}$ for V_{OV}, and $0.5\ \text{V}$ for v_{DS} in the equation.

$$\begin{aligned} I_D &= \frac{1}{2}\left(2\times 10^{-3}\right)(0.5)^2 \\ &= \left(2\times 10^{-3}\right)(0.125) \\ &= 0.25\ \text{mA} \end{aligned}$$

Hence, the value of drain current, when the voltage V_{DS} is increased to $0.5\ \text{V}$ is, $\boxed{I_D = 0.25\ \text{mA}}$.

Step 3 of 3

When the voltage v_{DS} is increased to $1\ \text{V}$, then $v_{DS} > V_{OV}$. So, the transistor operates in saturation.

The relation for the drain current is,

$$I_D = \frac{1}{2}k_n V_{OV}^2$$

Substitute $2\ \text{mA/V}^2$ for k_n, $0.5\ \text{V}$ for V_{OV}, and $0.5\ \text{V}$ for v_{DS} in the equation.

$$\begin{aligned} I_D &= \frac{1}{2}\left(2\times 10^{-3}\right)(0.5)^2 \\ &= \left(2\times 10^{-3}\right)(0.125) \\ &= 0.25\ \text{mA} \end{aligned}$$

Hence, the value of drain current, when the voltage V_{DS} is increased to $1\ \text{V}$ is, $\boxed{I_D = 0.25\ \text{mA}}$.

When transistor is operating in saturation, the drain current is $I_D = \frac{1}{2}k_n V_{OV}^2$, which is independent of v_{DS}. So, the drain current remains same even when the v_{DS} increases.

Step 1 of 3

Linear resistance r_{DS} range is $250\ \Omega$ to $1000\ \Omega$.

Control voltage V_{GS} is inversely proportional to linear resistance r_{DS} . Therefore for maximum allowable voltage of $1.8\ \text{V}$ linear resistance r_{DS} is minimum value and hence resistance r_{DS} is $250\ \Omega$.

Therefore MOSFET has a linear resistance of $250\ \Omega$ when voltage V_{GS} is equal to $1.8\ \text{V}$

Step 2 of 3

Linear resistance r_{DS} is $\dfrac{1}{k_n'(W/L)(v_{GS}-V_t)}$

Let MOSFET has a linear resistance of $1000\ \Omega$ when voltage v_{GS} is equal to V_{GS}

MOSFET has a linear resistance of $250\ \Omega$ when voltage V_{GS} is equal to $1.8\ \text{V}$

k_n' Value is $400\ \mu\text{A/V}^2$

Threshold voltage V_t is $0.5\ \text{V}$

Therefore,

$$250=\frac{1}{k_n'(W/L)(1.8-0.5)} \quad \text{..... (1)}$$

$$1000=\frac{1}{k_n'(W/L)(V_{GS}-0.5)} \quad \text{..... (2)}$$

Divide equation (1) with (2) as follows:

$$\frac{250}{1000}=\frac{k_n'(W/L)(V_{GS}-0.5)}{k_n'(W/L)(1.8-0.5)}$$

$$\frac{1}{4}=\frac{(V_{GS}-0.5)}{1.3}$$

$$0.325=(V_{GS}-0.5)$$

$$0.825=V_{GS}$$

Hence, the Minimum allowable voltage is $0.825\ \text{V}$

The range of control voltage V_{GS} required for r_{DS} range is between $250\ \Omega$ to $1000\ \Omega$ is

$\boxed{0.825\ \text{V to}\ 1.8\ \text{V}}$.

Step 3 of 3

It is required to use the smallest possible device and hence channel length L of the MOSFET is L_{min} .

Channel length L_{min} is $0.18\ \mu\text{m}$

Substitute L_{min} in place of L in equation (1).

$$\frac{1}{k_n'(W/L)(1.8-0.5)}=250$$

$$\frac{1}{\left(400\ \frac{\mu\text{A}}{\text{V}^2}\right)\left(\frac{W}{0.18\ \mu\text{m}}\right)(1.3)}=250$$

$$W=\frac{0.18\times10^{-6}}{(400\times10^{-6})(1.3)(250)}$$

$$W=1.38\ \mu\text{m}$$

Thus, the transistor width W is $\boxed{1.38\ \mu\text{m}}$.

Step 1 of 5

The Device parameter V_A is,

$$\begin{aligned} V_A &= V_A' L \\ &= \left(\frac{50\,\text{V}}{1\times10^{-6}\,\text{m}}\right)\left(0.8\times10^{-6}\,\text{m}\right) \\ &= 40 \end{aligned}$$

Therefore, the value of V_A is, $\boxed{V_A = 40\,\text{V}}$.

Step 2 of 5

The Device parameter λ is,

$$\begin{aligned} \lambda &= \frac{1}{V_A} \\ &= \frac{1}{40} \\ &= 0.025 \end{aligned}$$

Therefore, the value of λ is, $\boxed{\lambda = 0.025\,\text{V}^{-1}}$.

Step 3 of 5

The overdrive voltage of the device is, $V_{OV} = 0.5\,\text{V}$.

The Drain to source voltage is, $v_{DS} = 1\,\text{V}$.

Hence, $v_{DS} > V_{OV}$. So, the device is operating in saturation region.

In saturation region, the drain current is,

$$i_D = \frac{1}{2}\left(\mu_n C_{ox}\right)\left(\frac{W}{L}\right)\left(V_{OV}\right)^2\left(1+\lambda v_{DS}\right)$$

Substitute $200\,\mu\text{A}/\text{V}^2$ for $\mu_n C_{ox}$, $16\,\mu\text{m}$ for W , $0.8\,\mu\text{m}$ for L , $0.5\,\text{V}$ for V_{OV} , $1\,\text{V}$ for v_{DS} and $0.025\,\text{V}^{-1}$ for λ .

$$\begin{aligned} i_D &= \frac{1}{2}\left(200\times10^{-6}\right)\left(\frac{16\times10^{-6}}{0.8\times10^{-6}}\right)\left(0.5\right)^2\left[1+0.025(1)\right] \\ &= \left(100\times10^{-6}\right)(20)(0.25)(1.025) \\ &= 0.5125\times10^{-3}\ \text{A} \end{aligned}$$

Therefore, the drain current is, $\boxed{i_D = 0.51\,\text{mA}}$.

Step 4 of 5

The Output resistance is,

$$\begin{aligned} r_o &= \frac{V_A}{i_D} \\ &= \frac{40}{0.5125\times10^{-3}} \\ &= 78.43\times10^3\ \Omega \\ &= 80\,\text{k}\Omega \end{aligned}$$

Therefore, the output resistance is, $\boxed{r_o = 80\,\text{k}\Omega}$.

Step 5 of 5

The voltage v_{DS} is increased of by $2\,\text{V}$. So, the new drain to source voltage is,

$$v_{DS} = 3\,\text{V}$$

Since, $v_{DS} > V_{OV}$, the device is operating in saturation region.

In saturation region drain current is,

$$i_D = \frac{1}{2}\left(\mu_n C_{ox}\right)\left(\frac{W}{L}\right)\left(V_{OV}\right)^2\left(1+\lambda v_{DS}\right)$$

Substitute $200\,\mu\text{A}/\text{V}^2$ for $\mu_n C_{ox}$, $16\,\mu\text{m}$ for W , $0.8\,\mu\text{m}$ for L , $0.5\,\text{V}$ for V_{OV} , $3\,\text{V}$ for v_{DS} and $0.025\,\text{V}^{-1}$ for λ .

$$\begin{aligned} i_D &= \frac{1}{2}\left(200\times10^{-6}\right)\left(\frac{16\times10^{-6}}{0.8\times10^{-6}}\right)\left(0.5\right)^2\left[1+0.025(3)\right] \\ &= \left(100\times10^{-6}\right)(20)(0.25)(1.075) \\ &= 537.5\times10^{-6} \\ &= 0.5375\times10^{-3} \end{aligned}$$

The increase in drain current is,

$$\begin{aligned} i_{D(\text{increment})} &= 0.5375\times10^{-3} - 0.5125\times10^{-3} \\ &= 0.025\times10^{-3} \end{aligned}$$

Therefore, the drain current increment is, $\boxed{i_{D(\text{increment})} = 0.025\,\text{mA}}$.

Step 1 of 5

The drain current is,

$$i_D = \left[k_n'\left(\frac{W}{L}\right)\left(v_{GS} - V_t\right)\right]v_{DS}$$

Here,

k_n' is transconductance parameter

$\frac{W}{L}$ is the width to length ratio of the channel

v_{GS} is the gate-to-source voltage

V_t is the threshold voltage

The MOSFET transconductance parameter is,

$$k_n = k_n'\left(\frac{W}{L}\right)$$

$$i_D = \left[k_n\left(v_{GS} - V_t\right)\right]v_{DS}$$

Substitute 5×10^{-3} A/V² for k_n , 0.5 V for V_t .

$$i_D = \left[5\times10^{-3}\left(v_{GS} - 0.5\right)\right]v_{DS}$$

Step 2 of 5

Enter the following code in MATLAB to plot i_D versus v_{DS} for different values of v_{GS} .

VGS=0.5;vDS=0:0.001:50E-3;iD=(5E-3).*(VGS-0.5).*vDS;plot(vDS,iD)>> hold on

VGS=1;vDS=0:0.001:50E-3;iD=(5E-3).*(VGS-0.5).*vDS;plot(vDS,iD,'r')

VGS=1.5;vDS=0:0.001:50E-3;iD=(5E-3).*(VGS-0.5).*vDS;plot(vDS,iD,'g')

VGS=2;vDS=0:0.001:50E-3;iD=(5E-3).*(VGS-0.5).*vDS;plot(vDS,iD,'y')

VGS=2.5;vDS=0:0.001:50E-3;iD=(5E-3).*(VGS-0.5).*vDS;plot(vDS,iD,'b')

xlabel vDsylabel iD

Step 3 of 5

The set of i_D versus v_{DS} for different values of v_{GS} is shown in Figure 1.

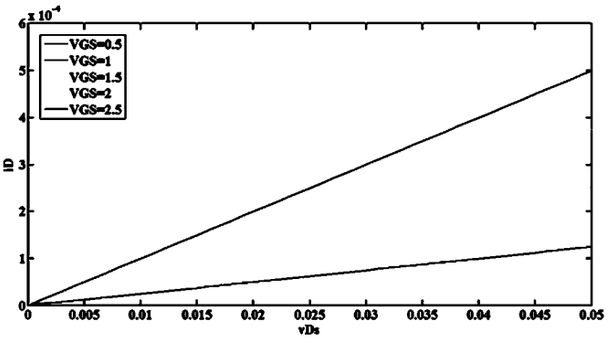


Figure 1

Write the formula for drain-to-source resistance, r_{DS} for small v_{DS} .

$$r_{DS} = \frac{1}{k_n'\left(\frac{W}{L}\right)\left(v_{GS} - V_t\right)}$$

The MOSFET transconductance parameter is,

$$k_n = k_n'\left(\frac{W}{L}\right)$$

The linear resistance is,

$$r_{DS} = \frac{1}{k_n\left(v_{GS} - V_t\right)}$$

Substitute 5×10^{-3} A/V² for k_n , 0.5 V for V_t .

$$r_{DS} = \frac{1}{5\times10^{-3}\left(v_{GS} - 0.5\right)}$$

Step 4 of 5

Determine r_{DS} for $v_{GS} = 0.5$ V .

$$r_{DS} = \frac{1}{5\times10^{-3}\left(0.5 - 0.5\right)}$$
$$= \infty$$

Thus, the value of r_{DS} for $v_{GS} = 0.5$ V is $\boxed{\infty}$.

Determine r_{DS} for $v_{GS} = 1$ V .

$$r_{DS} = \frac{1}{5\times10^{-3}\left(1 - 0.5\right)}$$
$$= 2.5\ \text{k}\Omega$$

Thus, the value of r_{DS} for $v_{GS} = 1$ V is $\boxed{2.5\ \text{k}\Omega}$.

Determine r_{DS} for $v_{GS} = 1.5$ V .

$$r_{DS} = \frac{1}{5\times10^{-3}\left(1.5 - 0.5\right)}$$
$$= 200\ \Omega$$

Thus, the value of r_{DS} for $v_{GS} = 1.5$ V is $\boxed{200\ \Omega}$.

Step 5 of 5

Determine r_{DS} for $v_{GS} = 2$ V .

$$r_{DS} = \frac{1}{5\times10^{-3}\left(2 - 0.5\right)}$$
$$= 133.33\ \Omega$$

Thus, the value of r_{DS} for $v_{GS} = 2$ V is $\boxed{133.33\ \Omega}$.

Determine r_{DS} for $v_{GS} = 2.5$ V .

$$r_{DS} = \frac{1}{5\times10^{-3}\left(2.5 - 0.5\right)}$$
$$= 100\ \Omega$$

Thus, the value of r_{DS} for $v_{GS} = 2.5$ V is $\boxed{100\ \Omega}$.

Step 1 of 11

5.007E

Refer to Figure E5.7 in the textbook for the circuit.

(a)

From the circuit, the source voltage is, $V_S = 5\text{V}$.

The condition for transistor conduction is,

$$V_{GS} \le V_{tp}$$

$$V_G - V_S \le V_{tp}$$

$$\begin{aligned} V_G &\le V_{tp} + V_S \\ &\le -1 + 5 \\ &\le +4\text{V} \end{aligned}$$

Therefore, range of V_G for which transistor conducts is, $\boxed{V_G \le +4\text{V}}$.

Step 2 of 11

(b)

Condition for triode region is $V_{DS} \ge V_{GS} - V_t$.

$$V_{DS} \ge V_{GS} - V_t$$
$$V_D - V_S \ge V_G - V_S - V_t$$

$$\begin{aligned} V_D &\ge V_G - V_t \\ &\ge V_G - (-1) \\ &\ge V_G + 1 \end{aligned}$$

Therefore, the range of V_D in terms of V_G is, $\boxed{V_D \ge V_G + 1}$.

Step 3 of 11

(c)

Condition for saturation is $V_{DS} \le V_{GS} - V_t$.

$$V_{DS} \le V_{GS} - V_t$$
$$V_D - V_S \le V_G - V_S - V_t$$

$$\begin{aligned} V_D &\le V_G - V_t \\ &\le V_G - (-1) \\ &\le V_G + 1 \end{aligned}$$

Therefore, the range of V_D in terms of V_G is, $\boxed{V_D \le V_G + 1}$.

Step 4 of 11

(d)

The equation for drain current is,

$$I_D = \frac{1}{2} k_p' \frac{W}{L} V_{OV}^2$$

Substitute $60\,\mu\text{A/V}^2$ for k_p' , 10 for $\frac{W}{L}$, and $75\,\mu\text{A}$ for I_D in the equation.

$$75 \times 10^{-6} = \frac{1}{2}(60 \times 10^{-6})(10)(|V_{OV}|)^2$$

$$75 = (30)(10)(|V_{OV}|)^2$$

$$(|V_{OV}|)^2 = \frac{75}{(30)(10)}$$

$$(|V_{OV}|)^2 = 0.25$$

$$|V_{OV}| = 0.5\text{V}$$

Therefore, the overdrive voltage is, $\boxed{|V_{OV}| = 0.5\text{V}}$.

Step 5 of 11

Given MOSFET is p- channel type, and we have

$$|V_{OV}| = -V_{GS} + V_{tp}$$

$$V_{GS} = -|V_{OV}| + V_{tp}$$
$$V_{GS} = -0.5 + (-1)$$
$$V_{GS} = -1.5$$
$$V_G - V_S = -1.5 \quad (\text{Since, } V_{GS} = V_G - V_S)$$

Step 6 of 11

Simplify further.

$$\begin{aligned} V_G &= -1.5 + V_S \\ &= -1.5 + 5 \\ &= 3.5\text{V} \end{aligned}$$

Therefore, the gate voltage is, $\boxed{V_G = 3.5\text{V}}$.

Step 7 of 11

The condition for saturation is,

$$V_{DS} \le V_{GS} - V_t$$

$$\begin{aligned} V_{DS} &\le -1.5 - (-1) \\ &\le -1.5 + 1 \\ &\le -0.5 \end{aligned}$$

Substitute $V_D - V_S$ for V_{DS} in the equation.

$$V_D - V_S \le -0.5$$

$$\begin{aligned} V_D &\le V_S - 0.5 \\ &\le 5 - 0.5 \\ &\le 4.5\text{V} \end{aligned}$$

Therefore, the range of drain voltage is, $\boxed{V_D \le 4.5\text{V}}$.

Step 8 of 11

(e)

The output resistance of the device in saturation region is,

$$\begin{aligned} r_o &= \frac{1}{\lambda I_D} \\ &= \frac{1}{(0.02)(75 \times 10^{-6})} \\ &= 0.67 \times 10^6\ \Omega \\ &= 0.67\ \text{M}\Omega \end{aligned}$$

Therefore, the output resistance of the device in saturation region is, $\boxed{r_o = 0.67\,\text{M}\Omega}$

Step 9 of 11

(f)

The Drain to source voltage is,

$$\begin{aligned} V_{DS} &= V_D - V_S \\ &= 3 - 5 \\ &= 2\text{V} \end{aligned}$$

The Overdrive voltage is, $|V_{OV}| = 0.5\text{V}$.

Since $V_{DS} > |V_{OV}|$, the transistor is operating in triode region and equation for drain current is,

$$\begin{aligned} I_D &= \frac{1}{2} k_p' \frac{W}{L} |V_{OV}|^2 (1 + |\lambda||V_{DS}|) \\ &= \frac{1}{2}(60 \times 10^{-6})(10)(0.5)^2 [1 + (0.02)(2)] \\ I_D &= 78 \times 10^{-6} \end{aligned}$$

Therefore, the drain current at $V_D = +3\text{V}$ is, $\boxed{I_D = 78\,\mu\text{A}}$.

Step 10 of 11

The Drain to source voltage is,

$$\begin{aligned} V_{DS} &= V_D - V_S \\ &= 0 - 5 \\ &= -5\text{V} \end{aligned}$$

Since $V_{DS} > |V_{OV}|$, the transistor is operating in triode region and equation for drain current is,

$$\begin{aligned} I_D &= \frac{1}{2} k_p' \frac{W}{L} |V_{OV}|^2 (1 + |\lambda||V_{DS}|) \\ &= \frac{1}{2}(60 \times 10^{-6})(10)(0.5)^2 [1 + (0.02)(5)] \\ &= (30 \times 10^{-6})(10)(0.25)(1.1) \\ &= 82.5 \times 10^{-6} \end{aligned}$$

Therefore, the drain current at $V_D = 0\text{V}$ is, $\boxed{I_D = 82.5\,\mu\text{A}}$.

Step 11 of 11

Apparent output resistance of the device is

$$\begin{aligned} r_o &= \frac{\partial V_{DS}}{\partial I_D} \\ &= \frac{5 - 2}{(82.5 - 78) \times 10^{-6}} \\ &= \frac{3}{4.5 \times 10^{-6}} \\ &= 0.67 \times 10^6 \Omega \end{aligned}$$

Therefore, the apparent output resistance in saturation is, $\boxed{r_o = 0.67\,\text{M}\Omega}$.

Thus, the output resistance obtained in part (e) is same as the output resistance obtained in this sub-part.

Step 1 of 2

5.007P

N- Channel MOS device operates in triode region.

Control voltage V_{GS} is inversely proportional to linear resistance r_{DS}. Therefore for maximum allowable voltage of $1.8\ \text{V}$ linear resistance r_{DS} is Minimum.

Minimum available resistance is $1\ \text{k}\Omega$

Linear resistance r_{DS} is $\dfrac{1}{k_n'(W/L)(v_{GS}-V_t)}$

k_n' Value is $400\left(\mu\text{A}/\text{V}^2\right)$

Threshold voltage V_t is $0.5\ \text{V}$

Step 2 of 2

Channel length L_{min} is $0.18\ \mu\text{m}$

Substitute value of L_{min} in L in the expression of linear resistance.

$$1000=\frac{1}{k_n'(W/L_{min})(1.8-0.5)}$$

$$W=\frac{0.18\ \mu\text{m}}{\left(400\ \mu\text{A}/\text{V}^2\right)(1.8-0.5)(1000\ \Omega)}$$

$$=\frac{0.18\times10^{-6}\ \text{m}}{\left(400\times10^{-6}\ \frac{\text{A}}{\text{V}^2}\right)(1.3\ \text{V})(1000\ \Omega)}$$

$$=3.46\times10^{-7}\ \text{m}$$

$$=0.346\ \mu\text{m}$$

Thus, the width of channel needed to ensure that the minimum available resistance of $1\ \text{k}\Omega$ is

$\boxed{0.346\ \mu\text{m}}$

Step 1 of 6

5.008E

Refer to Figure 5.21 in the textbook for the circuit

Redraw the circuit representing voltages and currents as shown in Figure 1.

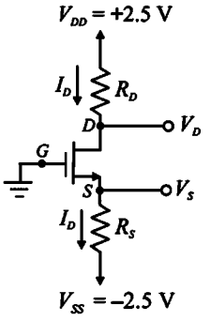


Figure 1

Step 2 of 6

From the circuit, the gate voltage is,

$$V_G = 0\,\text{V}$$

The drain voltage is, $V_D = +0.4\,\text{V}$

Here, V_D is greater than V_G . Therefore, MOSFET is operating in saturation region.

Under saturation condition, the drain current is,

$$I_D = \frac{1}{2}\left(\mu_n C_{ox}\right)\left(\frac{W}{L}\right)V_{OV}^2.$$

Substitute $0.3\,\text{mA}$ for I_D , $60\,\mu\text{A}/\text{V}^2$ for $\mu_n C_{ox}$, and $\frac{120\,\mu\text{m}}{3\,\mu\text{m}}$ for $\frac{W}{L}$ in the equation

$$0.3\times10^{-3} = \frac{1}{2}\left(60\times10^{-6}\right)\left(\frac{120\times10^{-6}}{3\times10^{-6}}\right)\left(V_{OV}\right)^2$$

$$0.3\times10^{-3} = \left(30\times10^{-6}\right)\left(40\right)\left(V_{OV}\right)^2$$

$$\left(V_{OV}\right)^2 = \frac{0.3\times10^{-3}}{\left(30\times10^{-6}\right)\left(40\right)}$$

$$\left(V_{OV}\right)^2 = \frac{10}{40}$$

$$\left(V_{OV}\right)^2 = 0.25$$

$$V_{OV} = 0.5\,\text{V}$$

Step 3 of 6

The Overdrive voltage is,

$$V_{OV} = V_{GS} - V_t$$

$$0.5 = V_{GS} - 1$$

$$V_{GS} = 1.5\,\text{V}$$

From gate to source voltage, determine voltage at source terminal.

$$V_{GS} = 1.5$$

$$V_G - V_S = 1.5$$

$$0 - V_S = 1.5 \quad \text{(Since, gate voltage is 0 V)}$$

$$V_S = -1.5\,\text{V}$$

Step 4 of 6

Apply Kirchhoff's voltage law to the circuit.

$$-V_S + I_D R_S + V_{SS} = 0$$

$$-(-1.5) + \left(0.3\times10^{-3}\right)R_S + (-2.5) = 0$$

$$1.5 + \left(0.3\times10^{-3}\right)R_S - 2.5 = 0$$

$$\left(0.3\times10^{-3}\right)R_S = 1$$

$$R_S = \frac{1}{0.3\times10^{-3}}$$

$$= 3.3\,\text{k}\Omega$$

Therefore, the source resistance is, $\boxed{R_S = 3.3\,\text{k}\Omega}$.

Step 5 of 6

Apply Kirchhoff's voltage law to the circuit.

$$-V_{DD} + I_D R_D + V_D = 0$$

$$-2.5 + \left(0.3\times10^{-3}\right)R_D + 0.4 = 0$$

$$\left(0.3\times10^{-3}\right)R_D = 2.1$$

$$R_D = \frac{2.1}{0.3\times10^{-3}}$$

$$R_D = 7\,\text{k}\Omega$$

Therefore, the drain resistance is, $\boxed{R_D = 7\,\text{k}\Omega}$.

Step 6 of 6

The designed circuit is shown in Figure 2.

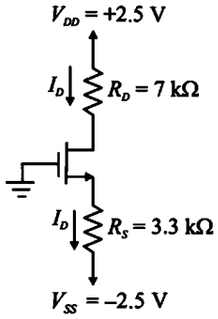


Figure 2

Step 1 of 7

Write the expression for drain current i_D for NMOS transistor operating in the triode region as v_{DS} is increased.

$$i_D = \left[k_n'\left(\frac{W}{L}\right)\left(V_{OV}v_{DS} - \frac{1}{2}v_{DS}^2\right)\right]$$

Here,

Transconductance parameter k_n is $k_n'\frac{W}{L}$

Therefore,

$$i_D = k_n\left[V_{OV}v_{DS} - \frac{1}{2}v_{DS}^2\right]$$

Step 2 of 7

Incremental resistance r_{ds} is defined as follows:

$$\begin{aligned} r_{ds} &\equiv 1\Big/\left.\frac{\partial i_D}{\partial v_{DS}}\right|_{v_{DS}=V_{DS}} \\ &= \frac{1}{\left.\dfrac{\partial\left(\left[k_n\left(V_{OV}v_{DS} - \frac{1}{2}v_{DS}^2\right)\right]\right)}{\partial v_{DS}}\right|_{v_{DS}=V_{DS}}} \\ &= \frac{1}{k_n\left(V_{OV} - v_{DS}\right)_{v_{DS}=V_{DS}}} \\ &= \frac{1}{\left[k_n\left(V_{OV} - V_{DS}\right)\right]} \end{aligned}$$

Thus, the expression for Incremental resistance r_{ds} is $\boxed{\frac{1}{\left[k_n\left(V_{OV} - V_{DS}\right)\right]}}$

Step 3 of 7

Determine the internal resistance when $V_{DS} = 0\ \text{V}$:

$$\begin{aligned} r_{ds} &= \frac{1}{\left[k_n\left(V_{OV} - V_{DS}\right)\right]} \\ &= \frac{1}{\left[k_n\left(V_{OV} - 0\right)\right]} \\ &= \frac{1}{k_nV_{OV}} \end{aligned}$$

Thus, the value of r_{ds} is $\boxed{\frac{1}{k_nV_{OV}}}$

Step 4 of 7

Determine the internal resistance when $V_{DS} = 0.2V_{OV}$:

$$\begin{aligned} r_{ds} &= \frac{1}{\left[k_n\left(V_{OV} - V_{DS}\right)\right]} \\ &= \frac{1}{\left[k_n\left(V_{OV} - 0.2V_{OV}\right)\right]} \\ &= \frac{1}{0.8k_nV_{OV}} \end{aligned}$$

Thus, the value of r_{ds} is $\boxed{\frac{1}{0.8k_nV_{OV}}\ \Omega}$

Step 5 of 7

Determine the internal resistance when $V_{DS} = 0.5V_{OV}$:

$$\begin{aligned} r_{ds} &= \frac{1}{\left[k_n\left(V_{OV} - V_{DS}\right)\right]} \\ &= \frac{1}{\left[k_n\left(V_{OV} - 0.5V_{OV}\right)\right]} \\ &= \frac{1}{0.5k_nV_{OV}} \end{aligned}$$

Thus, the value of r_{ds} is $\boxed{\frac{1}{0.5k_nV_{OV}}\ \Omega}$

Step 6 of 7

Determine the internal resistance when $V_{DS} = 0.8V_{OV}$:

$$\begin{aligned} r_{ds} &= \frac{1}{\left[k_n\left(V_{OV} - V_{DS}\right)\right]} \\ &= \frac{1}{\left[k_n\left(V_{OV} - 0.8V_{OV}\right)\right]} \\ &= \frac{1}{0.2k_nV_{OV}} \end{aligned}$$

Thus, the value of r_{ds} is $\boxed{\frac{1}{0.2k_nV_{OV}}\ \Omega}$

Step 7 of 7

Determine the internal resistance when $V_{DS} = V_{OV}$:

$$\begin{aligned} r_{ds} &= 1\Big/\left[k_nV_{OV} - V_{DS}\right] \\ &= 1\Big/\left[k_nV_{OV} - V_{OV}\right] \\ &= \infty \end{aligned}$$

Thus, the value of r_{ds} is $\boxed{\infty\ \Omega}$

Step 1 of 2

5.009E

Refer to Figure E5.9 in the text book.

Drain and gate terminals are short circuited and hence MOSFET in the figure **E5.9** will be operate in saturation region.

The drain current I_D for transistor in saturation region is,

$$\frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)(V_G - V_{tn})^2$$

Gate voltage V_G is $0.7\ \text{V}$

Threshold voltage V_{tn} is $0.5\ \text{V}$

Value of $\mu_n C_{ox}$ is $0.4\ \text{mA/V}^2$

$$I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)(V_G - V_{tn})^2$$

$$= \frac{1}{2}\left(0.4\times10^{-3}\ \frac{\text{A}}{\text{V}^2}\right)\left(\frac{0.72\ \mu\text{m}}{0.18\ \mu\text{m}}\right)(0.7\ \text{V} - 0.5\ \text{V})^2$$

$$= 3.2\times10^{-5}\ \text{A}$$

Step 2 of 2

The drain voltage V_D is $0.7\ \text{V}$.

Apply Kirchhoff's voltage law to the drain loop.

$$-1.8 + I_D R + V_D = 0$$

$$I_D = \frac{1.8 - 0.7}{R}$$

$$32\times10^{-6} = \frac{1.1}{R}$$

$$R = 34.4\ \text{k}\Omega$$

Thus, the value of resistance R is $\boxed{34.4\ \text{k}\Omega}$

Step 1 of 2

5.009P

NMOS transistor operated with gate source voltage V_{GS} is $1\ \text{V}$

NMOS transistor enters into saturation region when $V_{DS} \geq V_{OV}$

V_{OV} is the over drive voltage and is equal to $V_{GS} - V_t$.

$$\begin{aligned} V_{OV} &= V_{GS} - V_t \\ &= 1 - 0.5 \\ &= 0.5\ \text{V} \end{aligned}$$

$V_{DS} \geq V_{OV}$ NMOS transistor operated in saturation region. Hence for $V_{DS} = 0.5\ \text{V}$ transistor enters into saturation region.

Step 2 of 2

The drain current in saturation mode I_D is, $\frac{1}{2} k_n (V_{GS} - V_t)^2$

Transconductance for NMOS transistor k_n is $4\ \left(\text{mA}/\text{V}^2\right)$

Threshold voltage V_t is $0.5\ \text{V}$

$$\begin{aligned} I_D &= \frac{1}{2} k_n (V_{GS} - V_t)^2 \\ &= \frac{1}{2}\left(4 \times 10^{-3}\ \text{A}/\text{V}^2\right)(1 - 0.5)^2 \\ &= 0.5\ \text{mA} \end{aligned}$$

Thus, the drain current in saturation mode I_D is $\boxed{0.5\ \text{mA}}$

Step 1 of 2

5.010E

Refer to Figure E5.10 in the textbook.

In the circuit, the gate and drain of transistor Q_1 are short. So, the gate to source voltage of Q_1 is,

$$\begin{aligned} V_{GS1} &= V_{D1} \\ &= 0.8\ \text{V} \end{aligned}$$

From the circuit, the gate source voltage of transistor Q_2 is,

$$\begin{aligned} V_{GS2} &= V_{GS1} \\ &= 0.8\ \text{V} \end{aligned}$$

The transistor Q_2 is operating at the edge of saturation. Hence, the drain current is,

$$i_{D2} = \frac{1}{2}\mu_n C_{ox}\frac{W_2}{L_2}\left(V_{GS2} - V_{tn}\right)^2$$

Since transistor Q_2 is identical to Q_1, Substitute $0.4\ \text{mA/V}^2$ for $\mu_n C_{ox}$, $\frac{0.72\ \mu\text{m}}{0.18\ \mu\text{m}}$ for $\frac{W_2}{L_2}$, $0.8\ \text{V}$ for V_{GS2}, and $0.5\ \text{V}$ for V_{tn} in the equation.

$$\begin{aligned} i_{D2} &= \frac{1}{2}\left(0.4\times10^{-3}\right)\left(\frac{0.72\times10^{-6}}{0.18\times10^{-6}}\right)\left(0.8-0.5\right)^2 \\ &= \left(0.2\times10^{-3}\right)(4)(0.3)^2 \\ &= \left(0.8\times10^{-3}\right)(0.09) \\ &= 72\ \mu\text{A} \end{aligned}$$

The transistor Q_2 is operating at the edge of saturation. Hence, the condition for drain to source voltage of transistor Q_2 is,

$$V_{DS2} = V_{GS2} - V_{tn}$$

Substitute $0.8\ \text{V}$ for V_{GS2} and $0.5\ \text{V}$ for V_{tn} in the equation.

$$\begin{aligned} V_{DS2} &= 0.8 - 0.5 \\ &= 0.3\ \text{V} \end{aligned}$$

Step 2 of 2

From the circuit, the drain voltage of transistor Q_2 is, $V_{D2} = V_{DS2} = 0.3\ \text{V}$

From the circuit, the resistance R_2 is,

$$\begin{aligned} R_2 &= \frac{V_{DD} - V_{D2}}{i_{D2}} \\ &= \frac{1.8 - 0.3}{72\times10^{-6}} \\ &= 20.8\ \text{k}\Omega \end{aligned}$$

Therefore, the resistance R_2 is, $\boxed{R_2 = 20.8\ \text{k}\Omega}$.

Step 1 of 6

(a)

Oxide capacitance C_{ox} is $\frac{\varepsilon_{ox}}{t_{ox}}$

Permittivity of the silicon dioxide ε_{ox} is $3.45\times10^{-11}\ \mathrm{F/m}$

Oxide thickness t_{ox} is $6\ \mathrm{nm}$

$$\begin{aligned}C_{ox}&=\frac{\varepsilon_{ox}}{t_{ox}}\\&=\frac{3.45\times10^{-11}\ \mathrm{F/m}}{6\times10^{-9}\ \mathrm{m}}\\&=5.75\times10^{-3}\ \mathrm{F/m^2}\\&=5.75\ \mathrm{mF/m^2}\end{aligned}$$

Thus, the Oxide capacitance C_{ox} is $\boxed{5.75\ \mathrm{mF/m^2}}$.

Step 2 of 6

Mobility of electron μ_n is $460\ \mathrm{cm^2/V\cdot s}$

Value of k_n' is $\mu_n C_{ox}$

Substitute μ_n and C_{ox} in $k_n'=\mu_n C_{ox}$

$$\begin{aligned}k_n'&=\left(460\ \mathrm{cm^2/V\cdot s}\right)\times\left(5.75\ \mathrm{mF/m^2}\right)\\&=460\times10^{-4}\left(\mathrm{m^2/V\cdot s}\right)\times5.75\times10^{-3}\left(\mathrm{F/m^2}\right)\\&=264.5\times10^{-6}\ \left(\mathrm{F/V\cdot s}\right)\\&=264.5\ \left(\mu\mathrm{A/V^2}\right)\end{aligned}$$

Thus, the value of k_n' is $\boxed{264.5\ \left(\mu\mathrm{A/V^2}\right)}$.

Step 3 of 6

(b)

Aspect ratio $\frac{W}{L}$ is $\frac{20\ \mu\mathrm{m}}{0.25\ \mu\mathrm{m}}$.

DC Drain current I_D flowing through transistor when the transistor is in saturation region is $0.5\ \mathrm{mA}$

DC Drain current I_D for transistor operated in saturation region $I_D=\frac{1}{2}k_n'\frac{W}{L}V_{ov}^2$

$$\begin{aligned}I_D&=\frac{1}{2}k_n'\frac{W}{L}V_{ov}^2\\0.5\times10^{-3}&=\frac{1}{2}\left(264.5\times10^{-6}\right)\left(\frac{20}{0.25}\right)V_{ov}^2\\0.04726&=V_{ov}^2\\0.22&=V_{ov}\end{aligned}$$

Thus, the over drive voltage V_{ov} is $\boxed{0.22\ \mathrm{V}}$.

Step 4 of 6

The relation between gate source voltage V_{GS}, over drive voltage V_{DV} and threshold voltage V_t is $V_{GS}=V_{OV}+V_t$

$$\begin{aligned}V_{GS}&=V_{OV}+V_t\\&=0.22+0.5\\&=0.72\ \mathrm{V}\end{aligned}$$

Gate source voltage V_{GS} is $\boxed{0.72\ \mathrm{V}}$

NMOS transistor operated in the saturation region, condition for minimum drain sourse voltage is $V_{DS\min}=V_{OV}$

Thus, the minimum drain source voltage $V_{DS\min}$ is $\boxed{0.22\ \mathrm{V}}$

Step 5 of 6

(c)

Linear resistance r_{DS} for over drive voltage V_{OV} is $\frac{1}{k_n'(W/L)V_{OV}}$

NMOS acts as a resistor having linear resistance value equal to $100\ \Omega$

$$100=\frac{1}{k_n'(W/L)V_{OV}}$$

$$\begin{aligned}V_{OV}&=\frac{0.25\ \mu\mathrm{m}}{\left(264.5\ \mu\mathrm{A/V^2}\right)(20\ \mu\mathrm{m})(100)}\\&=\frac{0.25\times10^{-6}}{5.29\times10^{-7}}\ \mathrm{V}\\&=0.472\ \mathrm{V}\end{aligned}$$

Thus, the over drive voltage V_{ov} when the device operates as a resistor is $\boxed{0.472\ \mathrm{V}}$

Step 6 of 6

The relation between gate source voltage V_{GS}, over drive voltage V_{OV} and threshold voltage V_t is $V_{GS}=V_{OV}+V_t$

$$\begin{aligned}V_{GS}&=V_{OV}+V_t\\&=0.472+0.5\\&=0.972\ \mathrm{V}\end{aligned}$$

Thus, the gate source voltage V_{GS} when the device operates as a resistor is $\boxed{0.972\ \mathrm{V}}$.

Step 1 of 6

Consider the following circuit diagram:

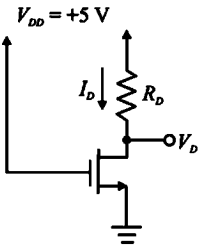


Figure 1

Step 2 of 6

The equation for drain current in triode region is,

$$I_D = k_n'\left(\frac{W}{L}\right)\left[(V_{GS} - V_{tn})V_{DS} - \frac{1}{2}V_{DS}^2\right]$$

Substitute $1\,\text{mA/V}^2$ for $k_n'\left(\frac{W}{L}\right)$ and $1\,\text{V}$ for V_{tn} in the equation.

$$I_D = \left(1\times10^{-3}\right)\left[(V_{GS} - 1)V_{DS} - \frac{1}{2}V_{DS}^2\right]$$

Consider drain current is in mA, now the equation is as follows:

$$I_D = (V_{GS} - 1)V_{DS} - \frac{1}{2}V_{DS}^2$$

$$I_D = (V_G - V_S - 1)(V_D - V_S) - \frac{1}{2}(V_D - V_S)^2 \quad \text{..... (1)}$$

Step 3 of 6

In the circuit, source is connected to ground. Hence, the source voltage is,

$V_S = 0\,\text{V}$.

Substitute $0\,\text{V}$ for V_S in the equation.

$$I_D = (V_G - 0 - 1)(V_D - 0) - \frac{1}{2}(V_D - 0)^2$$
$$= (V_G - 1)V_D - \frac{1}{2}V_D^2$$

From the circuit, the gate voltage is, $V_G = 5\,\text{V}$. Substitute $5\,\text{V}$ for V_G in the equation.

$$I_D = (5 - 1)V_D - \frac{1}{2}V$$

$$I_D = 4V_D - \frac{1}{2}V_D^2 \quad \text{..... (2)}$$

Step 4 of 6

Apply Kirchhoff's voltage law to the circuit.

$$-V_{DD} + I_D R_D + V_D = 0$$

$$-5 + I_D(24.8) + V_D = 0 \qquad \begin{pmatrix}\text{Since } R_D = 24.8\,\text{k}\Omega\text{, and consider}\\ I_D \text{ in mA}\end{pmatrix}$$

$$I_D(24.8) + V_D = 5$$

$$V_D = 5 - I_D(24.8)$$

Step 5 of 6

Substitute $4V_D - \frac{1}{2}V_D^2$ for I_D in the equation.

$$V_D = 5 - \left(4V_D - \frac{1}{2}V_D^2\right)(24.8)$$

$$V_D = 5 - 99.2V_D + 12.4V_D^2$$

$$12.4V_D^2 - 100.2V_D + 5 = 0$$

$$124V_D^2 - 1002V_D + 50 = 0$$

Solutions for the equation are,

$$V_D = \frac{1002 \pm \sqrt{1002^2 - 4\times124\times50}}{2(124)}$$
$$= \frac{1002 \pm 989.547}{248}$$
$$= \frac{1002 + 989.547}{248} \text{ or } V_D = \frac{1002 - 989.547}{248}$$

$$V_D = 8.03\,\text{V} \text{ or } V_D = 0.05\,\text{V}$$

The accepted value is $0.05\,\text{V}$ because the drain voltage cannot be more than supply voltage.

Therefore, the drain voltage is, $\boxed{V_D = 0.05\,\text{V}}$.

Step 6 of 6

Substitute $0.05\,\text{V}$ for V_D in equation (2).

$$I_D = 4(0.05) - \frac{1}{2}(0.05)^2$$
$$= 0.2 - 0.00125$$
$$= 0.19875$$
$$\approx 0.2\ \text{mA}$$

Therefore, the drain current is, $\boxed{I_D = 0.2\,\text{mA}}$.

Step 1 of 5

(a)

Thresh hold voltage of P-channel MOSFET V_{tp} is $-0.7\ \text{V}$

Magnitude of overdrive voltage $|V_{OV}|$ is $0.4\ \text{V}$

Relation between gate source voltage thresh hold voltage V_{tp} and overdrive voltage for P-channel MOSFET is $V_{GS} - V_{tp} = V_{OV}$.

For P-channel MOSFET applied gate source voltage is negative and satisfies the condition $v_{GS} \le V_{tp}$ or $|v_{GS}| \ge |V_{tp}|$.

$v_{GS} \le V_{tp}$ or $|v_{GS}| \ge |V_{tp}|$ condition is satisfied when

$$V_{GS} - V_{tp} = -0.4$$
$$V_{GS} - (-0.7) = -0.4$$
$$V_{GS} = -1.1\ \text{V}$$

Thus, the gate voltage required for the device to operate with an overdrive voltage of $|V_{OV}| = 0.4\ \text{V}$ is $\boxed{-1.1\ \text{V}}$.

Step 2 of 5

(b)

The voltage at which saturation occurs is denoted by V_{Dsat} .

$$\begin{aligned} V_{Dsat} &= V_{GS} - V_t \\ &= -1.1 - (-0.7) \\ &= -0.4\ \text{V} \end{aligned}$$

Hence, the highest voltage allowed at the drain while the device operates in saturation region is $\boxed{-0.4\ \text{V}}$.

Step 3 of 5

(c)

Applied drain voltage greater than $-0.4\ \text{V}$ then P-channel MOSFET will operate in triode region if drain voltage less than $-0.4\ \text{V}$ then P-channel MOSFET will operate in saturation region.

Drain current when P-channel MOSFET operates in saturation mode is $0.5\ \text{mA}$

For drain voltage $V_D = -2\ \text{V}$, P-channel MOSFET operates in saturation mode and hence drain current is $\boxed{0.5\ \text{mA}}$

Step 4 of 5

The drain current i_D when P-channel MOSFET operates in saturation mode is,

$$\frac{1}{2} k_p' \left(\frac{W}{L}\right) v_{OV}^2$$

The drain current i_D when P-channel MOSFET operates in triode region is,

$$k_p' \left(\frac{W}{L}\right)\left(v_{OV} - \frac{1}{2} v_{DS}\right) v_{DS}$$

For drain voltage $V_D = -20\ \text{mV}$ P-channel MOSFET operates in triode region.

For $V_D = -2\ \text{V}$

$$0.5 \times 10^{-3} = \frac{1}{2} k_p' \left(\frac{W}{L}\right) V_{OV}^2 \quad \text{..... (1)}$$

For $V_D = -20\ \text{mV}$

$$i_D = k_p' \left(\frac{W}{L}\right)\left(v_{OV} - \frac{1}{2} v_{DS}\right) v_{DS} \quad \text{..... (2)}$$

Step 5 of 5

Divide equation (2) and (1) as follows:

$$\frac{i_D}{0.5 \times 10^{-3}} = \frac{k_p' \left(\frac{W}{L}\right)\left(v_{OV} - \frac{1}{2} v_{DS}\right) v_{DS}}{\frac{1}{2} k_p' \left(\frac{W}{L}\right) V_{OV}^2}$$

$$\begin{aligned} i_D &= \frac{\left(v_{OV} - \frac{1}{2} v_{DS}\right)\left(0.5 \times 10^{-3}\right) v_{DS}}{V_{OV}^2} \\ &= \frac{\left(-0.4 - 0.5\left(-20 \times 10^{-3}\right)\right)\left(0.5 \times 10^{-3}\right)\left(-20 \times 10^{-3}\right)}{(-0.4)^2} \\ &= \frac{(-0.39)\left(-1 \times 10^{-5}\right)}{0.16} \\ &= 24.4\ \mu\text{A} \end{aligned}$$

Thus, the drain current for P-channel MOSFET when $V_D = -20\ \text{mV}$ is $\boxed{24.4\ \mu\text{A}}$.

Step 1 of 4

Refer to Figure 5.24 in the textbook for the circuit.

Redraw the circuit by representing node voltages in the circuit.

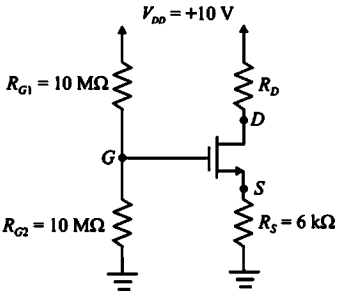


Figure 1

Step 2 of 4

Apply voltage division rule to the circuit.

$$V_G = V_{DD}\left(\frac{R_{G2}}{R_{G1}+R_{G2}}\right)$$
$$= 10\left(\frac{10\times10^6}{10\times10^6+10\times10^6}\right)$$
$$= 10\left(\frac{1}{2}\right)$$
$$= 5\,\text{V}$$

Step 3 of 4

The condition for saturation is,

$$V_{DS} \geq V_{GS} - V_t$$
$$V_D - V_S \geq V_G - V_S - V_t$$
$$V_D \geq V_G - V_t$$

For minimum value of V_D and for transistor to be in saturation the condition is,

$$V_{D(\min)} = V_G - V_t$$
$$= 5 - 1$$
$$= 4\,\text{V}$$

Step 4 of 4

Apply Kirchhoff's voltage law to the drain branch.

$$-V_{DD} + I_D R_D + V_D = 0$$
$$R_D = \frac{V_{DD} - V_D}{I_D}$$

Maximum possible value of drain is,

$$R_{D(\max)} = \frac{V_{DD} - V_{D(\min)}}{I_D}$$

Substitute $10\,\text{V}$ for V_{DD}, $4\,\text{V}$ for $V_{D(\min)}$, and $0.5\,\text{mA}$ for I_D in the equation.

$$R_{D(\max)} = \frac{10-4}{0.5\times10^{-3}}$$
$$= \frac{10-4}{0.5\times10^{-3}}$$
$$= 12\,\text{k}\Omega$$

Thus, the maximum possible value of drain resistance is, $\boxed{R_{D(\max)} = 12\,\text{k}\Omega}$.

Step 1 of 2

5.012P

N-channel and P-channel devices have same width, equal channel length, and equal drain current and equal over drive voltages.

Write the expression for drain current for P-channel device operates in saturation mode.

$$i_D = \frac{1}{2}\mu_p C_{ox}\left(\frac{W_p}{L}\right)v_{OV}^2$$

Here, mobility of hole is $0.4\mu_n$

Substitute $0.4\mu_n$ for μ_p in the equation.

$$i_D = \frac{1}{2}(0.4\mu_n)C_{ox}\left(\frac{W_p}{L}\right)v_{OV}^2 \quad \text{..... (1)}$$

Write the expression for drain current for N-channel device operates in saturation mode.

$$i_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W_n}{L}\right)v_{OV}^2 \quad \text{..... (2)}$$

Step 2 of 2

Divide equation (1) with (2) as follows:

$$\frac{i_D}{i_D} = \frac{\frac{1}{2}(0.4\mu_n)C_{ox}\left(\frac{W_p}{L}\right)v_{OV}^2}{\frac{1}{2}(\mu_n)C_{ox}\left(\frac{W_n}{L}\right)v_{OV}^2}$$

$$1 = \frac{0.4W_p}{W_n}$$

$$W_n = 0.4W_p$$

$$\frac{W_n}{W_p} = 0.4$$

Here, the width of N-channel is 0.4 times the width of P-channel.

Thus, the relative width of the *n*-channel and *p*-channel devices is $\boxed{\frac{W_n}{W_p} = 0.4}$.

Step 1 of 6

Refer to Figure 5.24 in the textbook for the circuit.

Redraw the circuit by representing node voltages in the circuit.

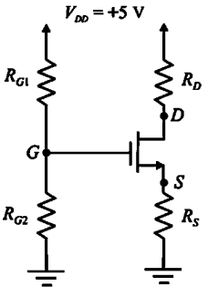


Figure 1

Step 2 of 6

The transistor is operating in saturation. Hence, the drain current is,

$$I_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)V_{OV}^2$$

Substitute $0.32\,\text{mA}$ for I_D and $1\,\text{mA/V}^2$ for $k_n'\left(\frac{W}{L}\right)$ in the equation.

$$0.32\times10^{-3} = \frac{1}{2}\left(1\times10^{-3}\right)V_{OV}^2$$

$$V_{OV}^2 = 0.64$$

$$V_{OV} = 0.8$$

The overdrive voltage is,

$$V_{OV} = V_{GS} - V_t$$

$$0.8 = V_{GS} - 1$$

$$V_{GS} = 1.8$$

$$V_G - V_S = 1.8$$

$$V_G = 1.8 + V_S$$

Step 3 of 6

Substitute $1.6\,\text{V}$ for V_S in the equation.

$$V_G = 1.8 + 1.6$$

$$= 3.4\,\text{V}$$

From the circuit, the gate voltage is,

$$V_G = (1\mu)(R_{G2})$$

$$3.4 = \left(1\times10^{-6}\right)(R_{G2})$$

$$R_{G2} = 3.4\times10^{6}\,\Omega$$

$$= 3.4\,\text{M}\Omega$$

Thus, the value of R_{G2} is $\boxed{3.4\,\text{M}\Omega}$

Step 4 of 6

Apply Kirchhoff's voltage law to the e circuit .

$$-V_{DD} + (1\mu)R_{G1} + V_G = 0$$

$$-5 + \left(1\times10^{-6}\right)R_{G1} + 3.4 = 0$$

$$\left(1\times10^{-6}\right)R_{G1} = 1.6$$

$$R_{G1} = 1.6\times10^{6}\,\Omega$$

Thus, the value of R_{G1} is $\boxed{1.6\,\text{M}\Omega}$

Apply Kirchhoff's voltage law to gate to source loop

$$-(1\mu)(R_{G2}) + V_{GS} + (I_D)(R_S) = 0$$

$$-\left(1\times10^{-6}\right)\left(3.4\times10^{6}\right) + 1.8 + \left(0.32\times10^{-3}\right)(R_S) = 0$$

$$-3.4 + 1.8 + \left(0.32\times10^{-3}\right)(R_S) = 0$$

$$\left(0.32\times10^{-3}\right)(R_S) = 1.6$$

$$R_S = \frac{1.6}{0.32\times10^{-3}}$$

$$= 5\times10^{3}\,\Omega$$

Thus, the value of R_S is $\boxed{5\,\text{k}\Omega}$.

Step 5 of 6

Apply Kirchhoff's voltage law to drain loop.

$$-V_{DD} + I_D R_D + V_D = 0$$

$$-5 + \left(0.32\times10^{-3}\right)R_D + 3.4 = 0$$

$$\left(0.32\times10^{-3}\right)R_D = 1.6$$

$$R_D = \frac{1.6}{0.32\times10^{-3}}$$

$$= 5\,\text{k}\Omega$$

Thus, the value of R_D is $\boxed{5\,\text{k}\Omega}$.

Step 6 of 6

The designed circuit is shown in Figure 2.

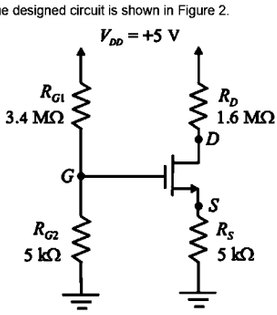


Figure 2

5.013P

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Step 1 of 4

5.014E

Refer to Figure E5.14 in the text book.

Consider PMOSFET is operating in saturation.

Condition for saturation is $V_{SD} \geq V_{SG} - |V_{tp}|$ ..... (1)

Where gate and drain are connected to ground.

Rearrange the equation (1).

$$V_S - V_D \geq V_S - V_G - |V_{tp}|$$

Substitute $\mathbf{0\ V}$ for V_G and $\mathbf{0\ V}$ for V_D in the equation.

$V_S \geq V_S - |V_{tp}|$ ..... (2)

equation (2) is satisfied, therefore transistor is operating in saturation region.

Step 2 of 4

The Overdrive voltage is,

$$|V_{ov}| = 0.6$$
$$V_{SG} - |V_{tp}| = 0.6$$
$$V_{SG} - 0.4 = 0.6$$

$$V_{SG} = 1\ \text{V}$$
$$V_S - V_G = 1$$
$$V_S - 0 = 1 \quad (\text{since } V_G = 0)$$
$$V_S = 1\ \text{V}$$

Step 3 of 4

The MOSFET is operating in saturation and equation for drain current is,

$$I_D = \frac{1}{2} k_p' \frac{W}{L} \left(V_{SG} - |V_{tp}|\right)^2$$
$$= \frac{1}{2}\left(0.1\times10^{-3}\right)\left(\frac{10\times10^{-6}}{0.18\times10^{-6}}\right)(1-0.4)^2$$
$$= \left(0.05\times10^{-3}\right)\left(\frac{10}{0.18}\right)(0.6)^2$$

Simplify further.

$$I_D = \left(0.05\times10^{-3}\right)\left(\frac{10}{0.18}\right)(0.36)$$
$$= 1\times10^{-3}$$
$$= 1\ \text{mA}$$

Therefore, the drain current is $\mathbf{1\ mA}$.

Step 4 of 4

Apply Kirchhoff's voltage law to the source to gate loop

$$-1.8 + I_D R + V_{SG} = 0$$

$$R = \frac{1.8 - V_{SG}}{I_D}$$
$$= \frac{1.8-1}{1\times10^{-3}}$$
$$= 0.8\times10^{3}$$
$$= 800\Omega$$

Therefore, the resistance in source branch is $\boxed{800\Omega}$.

5.014P

We don't have the solution to this problem yet.

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Step 1 of 11

Consider the input voltage, $v_I = 0\,\mathrm{V}$.

Therefore gate voltage for both the MOSFETs is zero.

Under this condition output voltage in the above circuit is zero.

$$v_O = 0\,\mathrm{V}$$

Therefore gate to source voltage V_{GS} for both the transistor is zero.

Both the MOSFETs are at OFF state and no drain current is flowing through the circuit.

$$i_{DN} = i_{DP}$$
$$= 0\,\mathrm{mA}$$

Therefore for $v_I = 0\,\mathrm{V}$, drain current of two MOSFETs and output voltage are

$\boxed{0\,\mathrm{mA}}$, $\boxed{0\,\mathrm{mA}}$ and $\boxed{0\,\mathrm{V}}$ respectively.

Step 2 of 11

Consider the input voltage $v_I = +2.5\,\mathrm{V}$.

For NMOS transistor gate and drain are at the same potential.

$$V_{DG} = 0\,\mathrm{V}$$

For NMOS transistor take saturation condition

$$V_{DS} \geq V_{GS} - V_{tn}$$
$$V_{DG} + V_{GS} \geq V_{GS} - V_{tn}$$
$$V_{DG} \geq -V_{tn}$$
$$V_{GD} \leq V_{tn}$$

Substitute $0\,\mathrm{V}$ for V_{GD} and $1\,\mathrm{V}$ for V_{tn} in equation.

$$0 \leq 1$$

Condition satisfied, hence NMOS transistor is in saturation condition.

Step 3 of 11

Positive gate voltage is applied for PMOS transistor , hence it is in OFF state

$i_{DP} = 0\,\mathrm{mA}$.

At this condition, modified circuit diagram is

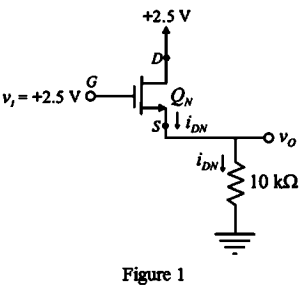


Figure 1

Step 4 of 11

Equation for drain current of NMOS transistor is

$$I_{DN} = \frac{1}{2}k_n'\left(\frac{W_n}{L_n}\right)\left(V_{GS} - V_t\right)^2$$
$$= \frac{1}{2}k_n'\left(\frac{W_n}{L_n}\right)\left(V_G - V_S - V_t\right)^2$$
$$I_{DN} = \frac{1}{2}\left(1\times10^{-3}\right)\left(2.5 - V_S - 1\right)^2$$
$$I_{DN} = \left(0.5\times10^{-3}\right)\left(1.5 - V_S\right)^2 \quad \ldots\ldots (1)$$

Step 5 of 11

Apply Kirchhoff's voltage law to the output circuit

$$v_O = I_{DN}R_L$$
$$v_O = I_{DN}\left(10\times10^3\right)$$

From Figure 1, it is clear that is $v_O = V_S$.

Now above equation becomes

$$V_S = I_{DN}\left(10\times10^3\right)$$

Substitute V_S value in equation (1)

$$I_{DN} = \left(0.5\times10^{-3}\right)\left[1.5 - I_{DN}\left(10\times10^3\right)\right]^2$$
$$I_{DN} = \left(0.5\times10^{-3}\right)\left[2.25 - 30I_{DN}\times10^3 + 10^8 I_{DN}^2\right]$$
$$I_{DN} = 1.125\times10^{-3} - 15I_{DN} + 50\times10^3 I_{DN}^2$$
$$0 = 1.125\times10^{-3} - 16I_{DN} + 50\times10^3 I_{DN}^2$$

Solutions for above equation are

$$I_{DN} = 0.000104322\,\mathrm{A} \quad \text{or} \quad 0.000215678\,\mathrm{A}$$

If $I_{DN} = 0.215678\,\mathrm{A}$, $V_{GS} < V_t$.

Therefore drain current of NMOS transistor is $\boxed{0.10432\,\mathrm{mA}}$.

Step 6 of 11

The Output voltage is,

$$v_O = \left(0.10432\times10^{-3}\right)\left(10\times10^3\right)$$
$$= 1.0432\,\mathrm{V}$$

Therefore for $v_I = +2.5\,\mathrm{V}$, drain current of two MOSFETs and output voltage are

$\boxed{0.104\,\mathrm{mA}}$, $\boxed{0\,\mathrm{mA}}$ and $\boxed{1.04\,\mathrm{V}}$.

Step 7 of 11

Consider the input voltage, $v_I = -2.5\,\mathrm{V}$.

For PMOS transistor gate and drain are at the same potential

$$\Rightarrow V_{DG} = 0\,\mathrm{V}$$

For PMOS transistor take saturation condition

$$V_{SD} \geq V_{SG} - |V_t|$$
$$V_{SG} + V_{GD} \geq V_{SG} - |V_t|$$
$$V_{GD} \geq -|V_t|$$

Substitute $V_{DG} = V_{GD} = 0$ and $|V_t| = 1\,\mathrm{V}$ values in the above equation.

$$0 \geq -1$$

Condition satisfied, hence PMOS transistor is in saturation condition.

Step 8 of 11

Negative gate voltage is applied for NMOS transistor, hence it is in OFF state

$i_{DN} = 0\,\mathrm{mA}$.

At this condition, modified circuit diagram is

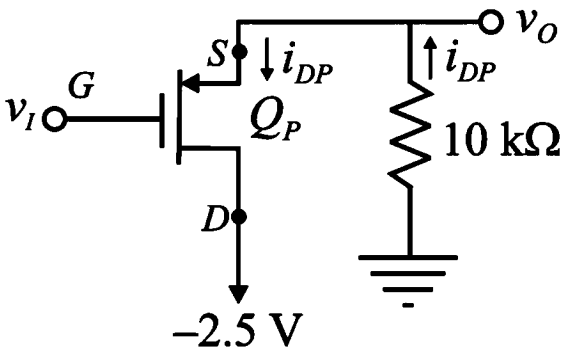


Figure 2

Step 9 of 11

Equation for drain current of NMOS transistor is

$$I_{DP} = \frac{1}{2}k_p'\left(\frac{W_p}{L_p}\right)\left(V_{GS} - |V_t|\right)^2$$
$$= \frac{1}{2}k_p'\left(\frac{W_p}{L_p}\right)\left(V_G - V_S - V_t\right)^2$$
$$= \frac{1}{2}\left(1\times10^{-3}\right)\left(-2.5 - V_S - 1\right)^2$$
$$I_{DP} = \frac{1}{2}\left(1\times10^{-3}\right)\left(3.5 + V_S\right)^2$$
$$I_{DP} = \left(0.5\times10^{-3}\right)\left(3.5 + V_S\right)^2 \quad \ldots\ldots (2)$$

Step 10 of 11

Apply Kirchhoff's voltage law to the output circuit

$$v_O = -I_{DP}R_L$$
$$v_O = -I_{DP}\left(10\times10^3\right)$$

In the above circuit $v_O = V_S$.

Now above equation becomes

$$V_S = -I_{DP}\left(10\times10^3\right)$$

Substitute V_S value in equation (1)

$$I_{DP} = \left(0.5\times10^{-3}\right)\left[3.5 - I_{DP}\left(10\times10^3\right)V_S\right]^2$$
$$= \left(0.5\times10^{-3}\right)\left(12.25 - 70\times10^3 I_{DP} + 10^8 I_{DP}^2\right)$$
$$= 6.125\times10^{-3} - 35I_{DP} + 50\times10^3 I_{DP}^2$$
$$0 = 6.125\times10^{-3} - 36I_{DP} + 50\times10^3 I_{DP}^2$$

Solutions for above equation are

$$I_{DP} = -0.000104322\,\mathrm{A} \quad \text{or} \quad -0.000215678\,\mathrm{A}$$

Step 11 of 11

If $I_{DP} = -0.215678\,\mathrm{A}$, $V_{SG} < |V_t|$.

Therefore drain current of PMOS transistor is $I_{DP} = -0.10432\,\mathrm{mA}$.

Output voltage is

$$v_O = \left(-0.10432\times10^{-3}\right)\left(10\times10^3\right)$$
$$v_O = -1.0432\,\mathrm{V}$$

Therefore for $v_I = +2.5\,\mathrm{V}$, drain current of two MOSFETs and output voltage are

$\boxed{0\,\mathrm{mA}}$ and $\boxed{-0.104\,\mathrm{mA}}$, $\boxed{-1.04\,\mathrm{V}}$.

Step 1 of 20

Relax to the fields of practice 5.1 to let that book

Let the parameters L and $\mathbf{W}$ are stored by the same factor to keep the magnitudes of the electrical fields unchanged.

This has ASCENT-less constant feed coding.

Table 1

Parameters	Before scaling	After scaling
Channel width	W	$\frac{W}{s}$
Channel length	L	$\frac{L}{s}$
Area	WL	$\frac{WL}{s^2}$
Oxide thickness	t_{ox}	$\frac{t_{ox}}{s}$
Oxide capacitance	C_{ox}	sC_{ox}
Interdevice voltage	V_t	V_t
Supply voltage	V_{DD}	$\frac{V_{DD}}{s}$
Gate voltage	V_{GS}	$\frac{V_{GS}}{s}$
Drain voltage	V_{DS}	$\frac{V_{DS}}{s}$
Carrying density	N_A	sN_A

Step 2 of 20

Choose 8 for the index to the text book that L were cranks from 0.5 μm to 0.25 μm

Obtain the scaling factor, s_1

$$\frac{L}{s} = 0.25\ \mu\text{m}$$

Substitute $0.5\ \mu\text{m}$ for L.

$$\frac{0.5\ \mu\text{m}}{s_1} = 0.25\ \mu\text{m}$$

$$s_1 = \frac{0.5\ \mu\text{m}}{0.25\ \mu\text{m}}$$

$$s_1 = 2$$

Thus, the scaling factor, s_1 is 2

Update the W value cranks from 0.25 μm to 0.18 μm

Obtain the scaling factor, s_2

$$\frac{L}{s_2} = 0.18\ \mu\text{m}$$

Substitute $0.25\ \mu\text{m}$ for L.

$$\frac{0.25\ \mu\text{m}}{s_2} = 0.18\ \mu\text{m}$$

$$s_2 = \frac{0.25\ \mu\text{m}}{0.18\ \mu\text{m}}$$

$$s_2 = 1.388$$

Thus, the scaling factor, s_2 is 1.388

Step 3 of 20

Observe that W value changes from $0.18\ \mu\text{m}$ to $0.13\ \mu\text{m}$

Obtain the scaling factor, s_3

$$\frac{W}{s_3} = 0.13\ \mu\text{m}$$

Substitute $0.18\ \mu\text{m}$ for L.

$$\frac{0.18\ \mu\text{m}}{s_3} = 1.3\ \mu\text{m}$$

$$s_3 = \frac{0.18\ \mu\text{m}}{0.13\ \mu\text{m}}$$

$$s_3 = 1.385$$

Thus, the scaling factor, s_3 is 1.385

Step 4 of 20

Choose the parameters L and $\mathbf{x_d}$ are stored by same factor; assess the value of $\mathbf{x_d}$ after scaling.

Value of $\mathbf{x_d}$ before scaling is $5\ \mu\text{m}$

$$\mathbf{x_d} = 5\ \mu\text{m}$$

Value of $\mathbf{x_d}$ after scaling is $3.6\ \mu\text{m}$ is 1.388.

$$\frac{\mathbf{x_d}}{s_4} = \frac{5\ \mu\text{m}}{1.388}$$

$$\frac{\mathbf{x_d}}{s_4} = 3.6\ \mu\text{m}$$

Step 5 of 20

Value of s_4 after scaling is factor 1.385.

Since the $\frac{W}{L}$ ratio is 10, obtain the value of $\frac{W}{L}$ for t'm value of $\mathbf{t = 0.5\ \mu m}$. After scaling

$$\frac{\mathbf{t_m}}{s_4} = \frac{0.5\ \mu\text{m}}{1.385}$$

$$\frac{\mathbf{t_m}}{s_4} = 2.6\ \mu\text{m}$$

$$\frac{W}{L} = 10$$

$$W = 5\ \mu\text{m}$$

Thus, device area was is

$$A = WL$$

$$= (5\ \mu\text{m})(0.5\ \mu\text{m})$$

$$= 2.5\ \mu\text{m}^2$$

Step 6 of 20

The device area after scaling is factor 2 is.

$$\frac{WL}{s_4^2} = \frac{2.5\ \mu\text{m}^2}{1.388^2}$$

$$= 0.823\ \mu\text{m}^2$$

The device area after scaling is factor 1.385 is.

$$\frac{WL}{s_4^2} = \frac{0.823\ \mu\text{m}^2}{1.388^2}$$

$$= 0.324\ \mu\text{m}^2$$

Step 7 of 20

Observe the threshold voltage V_{DS} are stored by same factor

Value of V_{DS} after scaling is factor 2 is.

$$\frac{V_{DS}}{s} = \frac{5}{2}$$

$$\frac{V_{DS}}{s} = 2.5\ \text{V}$$

Step 8 of 20

Choose the number of devices per chip as 81.3 bits ; constructing Gate the false area is $2.5\mu\ \text{m}^2$.

Area per device is $0.823\ \mu\text{m}$ is $0.8167\ \mu\text{m}^2$

Calculate the number of devices per chip as 81.3 bits ; constructing Gate the false area is $2.5\mu\ \text{m}^2$.

Area per device is $0.324\ \mu\text{m}$ is $0.8167\ \mu\text{m}^2$

Step 9 of 20

Choose the number of devices per chip must be integer multiple of $\mathbf{n}$. Thus, the possible number of devices per chip is $\mathbf{81.3}$ bits technology is $\mathbf{1\mu}$

Step 10 of 20

Calculate the number of devices per chip must be integer multiple of $\mathbf{n}$. Thus, the possible number of devices per chip is $\mathbf{81.3}$ bits technology is $\mathbf{1\mu}$

Step 11 of 20

Choose the false number of devices per chip must be integer multiple of $\mathbf{n}$. Thus, the possible number of devices per chip is $\mathbf{81.3}$ bits technology is $\mathbf{1\mu}$

Step 12 of 20

Choose the false number of devices per chip must be integer multiple of $\mathbf{n}$. Thus, the possible number of devices per chip is $\mathbf{81.3}$ bits technology is $\mathbf{1\mu}$

Step 13 of 20

Calculate the power dissipation at $\mathbf{V_{DS} = 1.8\ V}$

$$P = V_{DS} I_D$$

$$= (1.8)(4.69\ \text{mA})$$

$$= 8.44\ \text{mW}$$

Calculate the power density at $\mathbf{V_{DS} = 1.8\ V}$

$$\text{Power density} = \frac{P}{A}$$

Substitute $\mathbf{8.44\ mW}$ for P and $\mathbf{0.324\mu m^2}$ for A

$$\text{Power density} = \frac{8.44\ \text{mW}}{0.324\ \mu\text{m}^2}$$

$$= 26.04\ \text{mW}/\mu\text{m}^2$$

Step 14 of 20

Calculate I_D at $\mathbf{V_{DS} = 1.3\ V}$

Substitute $\mathbf{65.3\ \mu A/V^2}$ for K_n ' , 10 for $\frac{W}{L}$ and $\mathbf{0.4\ V}$ for V_t.

$$I_D = \frac{1}{2}(65.3)(10)(1.3-0.4)^2$$

$$= 265.6\ \text{mA}$$

Calculate the power dissipation at $\mathbf{V_{DS} = 1.3\ V}$

$$P = V_{DS} I_D$$

$$= 1.3(265.6\ \text{mA})$$

$$= 3.5\ \text{mW}$$

Calculate the power density at $\mathbf{V_{DS} = 1.3\ V}$

$$\text{Power density} = \frac{P}{A}$$

Substitute $\mathbf{3.5\ mW}$ for P and $\mathbf{0.169\ \mu m^2}$ for A

$$\text{Power density} = \frac{3.5\ \text{mW}}{0.169\ \mu\text{m}^2}$$

$$= 20.7\ \text{mW}/\mu\text{m}^2$$

Step 15 of 20

Choose $\mathbf{8.3}$ bits Intensity under these $\mathbf{n}$. $\mathbf{A = 2.5\ \mu m^2}$. Bit in the inverter chip area, it covers capacitive area $\mathbf{A\ mV}$ threshold.

Thus, the bias Intensity is $\mathbf{2.5\ m}$ technology.

Since, the final number of devices is to be calculated considering $\mathbf{0.3\ \mu m}$ technology consider the false area $\mathbf{2.5\mu m^2}$.

Calculate the number of devices per chip as $\mathbf{81.3}$ bits ; constructing Gate the false area is $\mathbf{2.5\mu\ \text{m}^2}$.

Area per device is $\mathbf{0.823\ \mu m}$ or $\mathbf{0.8167\ \mu m^2}$

Calculate the number of devices per chip as $\mathbf{81.3}$ bits ; constructing Gate the false area is $\mathbf{2.5\mu\ \text{m}^2}$.

Area per device is $\mathbf{0.324\ \mu m}$.

$$\begin{bmatrix}\text{number of}\\ \text{devices per chip}\\ \text{in } \mathbf{0.25\mu m\ technology}\end{bmatrix} = \frac{\mathbf{2.5\ m^2}}{\mathbf{0.625}}$$

$$\begin{bmatrix}\text{number of}\\ \text{devices per chip}\\ \text{in } \mathbf{0.25\mu m\ technology}\end{bmatrix} = \mathbf{4n}$$

Step 16 of 20

Find value of C_n after scaling by a factor 1.385 is.

$$s_1 C_n = 1.388(9.58)\ \text{fF}/\mu\text{m}^2$$

$$= 9.58\ \text{fF}/\mu\text{m}^2$$

Calculate the transconductance parameter, K_n as $\mathbf{C_n = 9.58\ fF/\mu m^2}$.

$$K_n' = 500 \times 10^{-4} \times 9.58 \times 10^{-15} \times 10^{12}$$

$$= \mathbf{479\ \mu A/V^2}$$

Calculate the transconductance parameter, K_n as $\mathbf{K_n' = 479\ \mu A/V^2}$

$$K_n = \mathbf{479(10)}$$

$$= \mathbf{-4.79\ mA/V^2}$$

Step 17 of 20

Find value of C_n after scaling by a factor 1.385 is.

$$s_1 C_n = 1.388(9.58)\ \text{fF}/\mu\text{m}^2$$

$$= \mathbf{13.27\ fF}/\mu\text{m}^2$$

Calculate the transconductance parameter, K_n as $\mathbf{C_n = 13.27\ fF/\mu m^2}$.

$$K_n' = 500 \times 10^{-4} \times 13.27 \times 10^{-15} \times 10^{12}$$

$$= \mathbf{663\ \mu A/V^2}$$

Calculate the transconductance parameter, K_n as $\mathbf{K_n' = 663\ \mu A/V^2}$

$$K_n = \mathbf{663(10)}$$

$$= \mathbf{-6.63\ mA/V^2}$$

Step 18 of 20

Choose the number of devices per chip as $\mathbf{0.1\ m}$. Constructing Bit the false area is $\mathbf{2.5\mu\ \text{m}^2}$.

Area per device is $\mathbf{0.25\ \mu m}$ or $\mathbf{0.0625\ \mu m^2}$

Calculate the number of devices per chip as $\mathbf{0.1\ m}$. Constructing Bit the false area is $\mathbf{2.5\mu\ \text{m}^2}$.

Area per device is $\mathbf{0.324\ \mu m}$.

$$\begin{bmatrix}\text{number of}\\ \text{devices per chip}\\ \text{in } \mathbf{0.25\mu m\ technology}\end{bmatrix} = \frac{\mathbf{1.5n}}{\mathbf{0.324}}$$

$$\begin{bmatrix}\text{number of}\\ \text{devices per chip}\\ \text{in } \mathbf{0.25\mu m\ technology}\end{bmatrix} = \mathbf{4.7n}$$

Step 19 of 20

Choose the number of devices per chip must be integer multiple of $\mathbf{n}$. Thus, the possible number of devices per chip is $\mathbf{81.3}$ bits technology is $\mathbf{1\mu}$

Step 20 of 20

Table 2 Adjustments for values of each bit of parameters.

Table 2

1	I_n (mm)	0.25	0.18	0.13	
2	r_n (mm)	10	6	3.6	2.6
3	C_n $(fF/\mu m^2)$	3.45	9.58	13.26	
4	K_n $(\mu A/V^2)$ $(\sigma_n = 500\ \text{cm}^2/V.s)$	172.5	345	479	663.5
5	$\frac{W}{L} = 10$	1.725	1.45	4.79	6.635
6	K_p $(\mu A/V^2)$	2.85	8.324	8.169	
7	V_{tn} (V)	0.5	2.25	1.8	1.3
8	V_t (V)	0.7	0.5	0.4	0.4
9	t_{ox} (nm)	15.5	8.9	4.55	2.58
10	r_t (mV)	7.5	0.2	8.44	3.5
11	P_d $(mW/\mu m^2)$	31.8	27.6	26.04	20.7
12	Device per chip	n	$4n$	$7n$	$14n$

5.016E

Find the value of threshold voltage, V_t of an NMOS transistor.

$$V_t = V_{to} + \gamma\left[\sqrt{2\phi_f + V_{SB}} - \sqrt{2\phi_f}\right]$$

Here, the threshold voltage, V_{to} when the reverse bias voltage between source and body, $V_{SB} = 0$ is, 0.8 V, the fabrication process parameter, γ is, $0.4\ \text{V}^{\frac{1}{2}}$, the physical parameter, $2\phi_f$ is, 0.7 V, and the reverse substrate bias voltage, V_{SB} is, 3 V.

Substitute the values in the expression of V_t.

$$\begin{aligned} V_t &= 0.8\ \text{V} + \left(0.4\sqrt{\text{V}}\right)\left(\sqrt{0.7\ \text{V} + 3\ \text{V}} - \sqrt{0.7\ \text{V}}\right) \\ &= 0.8\ \text{V} + \left(0.4\sqrt{\text{V}}\right)\left(1.924\sqrt{\text{V}} - 0.837\sqrt{\text{V}}\right) \\ &= 0.8\ \text{V} + \left(0.4\sqrt{\text{V}}\right)\left(1.086\sqrt{\text{V}}\right) \\ &= 0.8\ \text{V} + 0.435\ \text{V} \end{aligned}$$

Therefore,

$$V_t = 1.234\ \text{V}$$

Thus, the threshold voltage, V_t is, $\boxed{1.234\ \text{V}}$.

Step 1 of 9

Consider the expression of drain current of MOSFET, i_D, including the channel length modulation.

$$i_D = \frac{1}{2}k_n'\frac{W}{L}(V_{GS}-V_m)^2(1+\lambda V_{DS})$$

Neglect channel length modulation, that is, substitute 0 for λ.

$$i_D = \frac{1}{2}k_n'\frac{W}{L}(V_{GS}-V_m)^2$$

The expression for the drain current shows that the NMOS transistor is operating in the saturation region.

The expression for k_n is,

$$k_n = k_n'\frac{W}{L}$$

Substitute v_{OV} for $V_{GS}-V_m$ and $k_n'\frac{W}{L}$ for k_n in the expression for drain current.

$$i_D = \frac{1}{2}k_n v_{OV}^2$$

Thus, the relation between $\frac{i_D}{k_n}$ and v_{OV} is,

$$\frac{i_D}{k_n} = \frac{1}{2}v_{OV}^2$$

Step 2 of 9

To plot $\frac{i_D}{k_n}$ versus v_{DS} for different values of v_{OV}, the relation is,

$$\frac{i_D}{k_n} = \frac{1}{2}(v_{OV})^2 v_{DS}$$

Substitute 0.5 V for v_{OV}.

$$\frac{i_D}{k_n} = \frac{1}{2}(0.5)^2 v_{DS}$$
$$= 0.125 v_{DS}$$

Table 1 shows the value of the parameter $\frac{i_D}{k_n}$ for different values of v_{DS}, at v_{OV} equal to 0.5 V.

Table 1

v_{DS}	0	0.1	0.2	0.3	0.4	0.5
$\frac{i_D}{k_n}$	0	0.0125	0.025	0.0375	0.05	0.0625

The equation of $\frac{i_D}{k_n}$ for $v_{OV} = 0.3$ V is,

$$\frac{i_D}{k_n} = \frac{1}{2}(0.3)^2 v_{DS}$$
$$= 0.045 v_{DS}$$

Table 2 shows the value of the parameter $\frac{i_D}{k_n}$ for different values of v_{DS}, at v_{OV} equal to 0.3 V.

Table 2

v_{DS}	0	0.1	0.2	0.3
$\frac{i_D}{k_n}$	0	0.0045	0.009	0.0135

Step 3 of 9

For $v_{DS} \geq v_{OV}$, the NMOS transistor enters into saturation. Thus, the value of $\frac{i_D}{k_n}$ becomes independent of v_{DS}.

$$\frac{i_D}{k_n} = \frac{1}{2}v_{OV}^2$$

Determine the value of $\frac{i_D}{k_n}$ at $v_{OV} = 0.5$ V.

$$\frac{i_D}{k_n} = \frac{1}{2}(0.5)^2$$
$$= 0.125$$

Thus, the value of $\frac{i_D}{k_n}$ at $v_{OV} = 0.3$ V is,

$$\frac{i_D}{k_n} = \frac{1}{2}(0.3)^2$$
$$= 0.045$$

Step 4 of 9

Thus, use all the obtained values to plot $\frac{i_D}{k_n}$ versus v_{DS} for different values of v_{OV}.

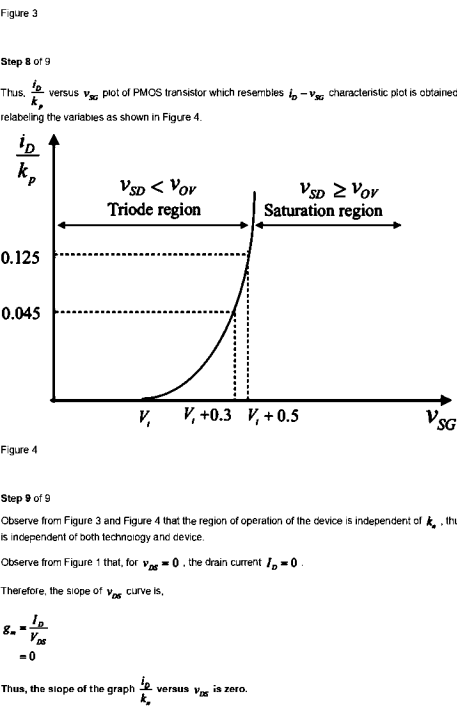


Figure 2

Observe from the plot that $\frac{i_D}{k_n}$ versus v_{DS} resembles the plot of $i_D - v_{DS}$ characteristics.

Since, the region of operation of a device is independent of k_n, thus, it is independent of both technology and device.

Thus, plot of $\frac{i_D}{k_n}$ versus v_{DS} results in unique representation of $i_D - v_{DS}$ characteristics of NMOS transistor which is independent of both technology and device.

Step 5 of 9

Consider the expression for $\frac{i_D}{k_n}$.

$$\frac{i_D}{k_n} = \frac{1}{2}(V_{GS}-V_t)^2$$

For $V_{GS} = V_t$, the ratio is,

$$\frac{i_D}{k_n} = 0$$

For $V_{GS} = V_t + 0.3$, the ratio is,

$$\frac{i_D}{k_n} = \frac{1}{2}(V_t + 0.3 - V_t)^2$$
$$= 0.045$$

For $V_{GS} = V_t + 0.5$, the ratio is,

$$\frac{i_D}{k_n} = \frac{1}{2}(V_t + 0.5 - V_t)^2$$
$$= 0.0125$$

Step 6 of 9

Observe from the obtained values that, $i_D - v_{GS}$ characteristics plot can be obtained from the $i_D - V_{GS}$ by plotting the saturated values at different values of v_{OV} and shifting the origin of the plot to $v_{GS} = V_t$.

Sketch $\frac{i_D}{k_n}$ versus v_{GS} characteristics.

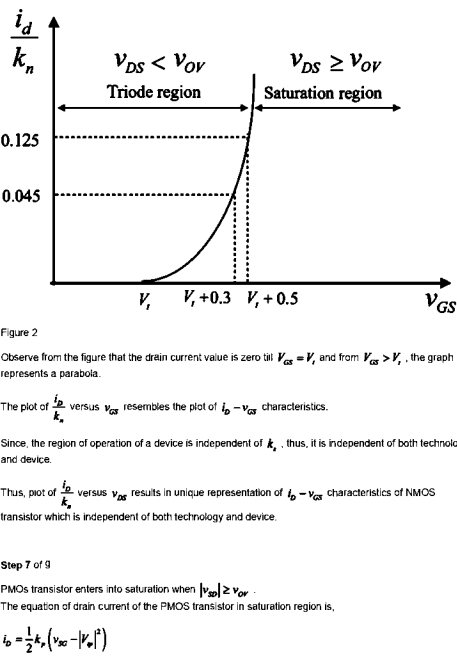


Figure 2

Observe from the figure that the drain current value is zero till $V_{GS} = V_t$ and from $V_{GS} > V_t$, the graph represents a parabola.

The plot of $\frac{i_D}{k_n}$ versus v_{GS} resembles the plot of $i_D - v_{GS}$ characteristics.

Since, the region of operation of a device is independent of k_n, thus, it is independent of both technology and device.

Thus, plot of $\frac{i_D}{k_n}$ versus v_{DS} results in unique representation of $i_D - v_{GS}$ characteristics of NMOS transistor which is independent of both technology and device.

Step 7 of 9

PMOs transistor enters into saturation when $|v_{SD}| \geq v_{OV}$.

The equation of drain current of the PMOS transistor in saturation region is,

$$i_D = \frac{1}{2}k_p\left(v_{SG}-|V_e|^2\right)$$

Thus, the expression for $\frac{i_D}{k_p}$ is,

$$\frac{i_D}{k_p} = \frac{1}{2}\left(v_{SG}-|V_e|^2\right)$$
$$= \frac{1}{2}(v_{OV})^2$$

Compare the expression of drain current of PMOS transistor with equation (2).

Thus, $\frac{i_D}{k_p}$ versus v_{SD} plot of PMOS transistor which resembles $i_D - v_{SD}$ characteristic plot is obtained by relabeling the variables.

Figure 3

Step 8 of 9

Thus, $\frac{i_D}{k_p}$ versus v_{SG} plot of PMOS transistor which resembles $i_D - v_{SG}$ characteristic plot is obtained by relabeling the variables as shown in Figure 4.

Figure 4

Step 9 of 9

Observe from Figure 3 and Figure 4 that the region of operation of the device is independent of k_n, thus, it is independent of both technology and device.

Observe from Figure 1 that, for $v_{DS} = 0$, the drain current $I_D = 0$.

Therefore, the slope of v_{DS} curve is,

$$g_m = \frac{I_D}{V_{DS}}$$
$$= 0$$

Thus, the slope of the graph $\frac{i_D}{k_n}$ versus v_{DS} is zero.

V_{OV} represents the overdrive voltage and at this value of $v_{OV} = V_{OV}$, the device saturates. So, the ratio, $\frac{I_D}{k_n}$ remains constant and for a constant value, the slope is zero.

Thus, the slope at $v_{OV} = V_{OV}$ is zero.

5.017E

The condition for a depletion-type NMOS transistor to operate in saturation region is,

$$v_{DS} \geq v_{GS} - V_t$$

Therefore, the minimum value of drain-source voltage, v_{DS} is,

$$v_{DS(\min)} = v_{GS} - V_t$$

Substitute 1 V for the gate-source voltage, v_{GS} , and $-2\ \text{V}$ for the threshold voltage, V_t .

$$\begin{aligned} v_{DS(\min)} &= 1\ \text{V} - (-2\ \text{V}) \\ &= 3\ \text{V} \end{aligned}$$

Therefore, the minimum value of drain to source voltage, $v_{DS(\min)}$ is, $\boxed{3\ \text{V}}$.

The equation for drain current at this drain to source voltage is,

$$i_D = \frac{1}{2} k_n' \frac{W}{L} (v_{GS} - V_t)^2$$

Substitute $2\ \text{mA/V}^2$ for the process trans-conductance parameter, $k_n' \frac{W}{L}$, and 3 V for $v_{GS} - V_t$.

$$\begin{aligned} i_D &= \frac{1}{2}(2\ \text{mA/V}^2)(3\ \text{V})^2 \\ &= (1\ \text{mA/V}^2)(9\ \text{V}^2) \\ &= 9\ \text{mA} \end{aligned}$$

Therefore, the drain current, i_D is $\boxed{9\ \text{mA}}$.

5.017P

Step 1 of 3

NMOS transistor is operated in Triode Region.

Threshold voltage V_t is $0.8\ \text{V}$

Linear resistance r_{DS} is $\dfrac{1}{k_n'(W/L)(V_{GS}-V_t)}$

For V_{GS} is equal to $1.2\ \text{V}$ it is found to have a resistance r_{DC} is $1000\ \Omega$

$$1000=\frac{1}{k_n'(W/L)(1.2-0.8)} \quad \text{..... (1)}$$

For r_{DC} equal to $200\ \Omega$ gate source voltage applied is V_{GS}

$$200=\frac{1}{k_n'(W/L)(V_{GS}-0.8)} \quad \text{..... (2)}$$

Step 2 of 3

Divide equation (1) with (2) as follows:

$$\frac{1000}{200}=\frac{\dfrac{1}{k_n'(W/L)(1.2-0.8)}}{\dfrac{1}{k_n'(W/L)(V_{GS}-0.8)}}$$

$$5=\frac{V_{GS}-0.8}{0.4}$$

$$2=V_{GS}-0.8$$

$$V_{GS}=2.8\ \text{V}$$

Thus, the gate source voltage V_{GS} is $\boxed{2.8\ \text{V}}$

Step 3 of 3

Let R is the resistance of the device when width of the channel is doubled and V_{GS} is equal to $2.8\ \text{V}$

$$R=\frac{1}{k_n'(2W/L)(V_{GS}-0.8)} \quad \text{..... (3)}$$

Divide equation (3) with (2) as follows:

$$\frac{R}{200}=\frac{\dfrac{1}{k_n'(2W/L)(V_{GS}-0.8)}}{\dfrac{1}{k_n'(W/L)(V_{GS}-0.8)}}$$

$$\frac{R}{200}=\frac{1}{2}$$

$$R=100\ \Omega$$

Thus, the value of resistance R is $\boxed{100\ \Omega}$.

Step 1 of 4

MOSFET operated in saturation region and Drain current i_D is $50\ \mu A$

Drain current i_D for transistor in saturation region is $\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_{tn})^2$

Threshold voltage V_{tn} is $0.5\ V$

The value of $k_n'\left(\frac{W}{L}\right)$ is $1.6\ \frac{mA}{V^2}$

$$i_D=\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_{tn})^2$$

$$50\times10^{-6}=\frac{1}{2}\times1.6\times10^{-3}\times(v_{GS}-0.5)^2$$

$$0.0625=(v_{GS}-0.5)^2$$

$$0.25=(v_{GS}-0.5)$$

Thus, the value of gate source voltage v_{GS} is $\boxed{0.75\ V}$.

Step 2 of 4

If $v_{DS}\geq v_{GS}-V_{tn}$ then the N-channel transistor will be in saturation region.

$v_{DS\,min}$ Required for N-channel transistor operate in saturation region is $v_{GS}-V_{tn}$

$$\begin{aligned}v_{DS\,min}&=v_{GS}-V_{tn}\\&=0.75-0.5\\&=0.25\end{aligned}$$

Thus, the minimum Drain source voltage $v_{DS(min)}$ is $\boxed{0.25\ V}$

Step 3 of 4

MOSFET operated in saturation region and Drain current i_D is $200\ \mu A$

The drain current i_D for transistor in saturation region is $\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_{tn})^2$

$$i_D=\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_{tn})^2$$

$$200\times10^{-6}=\frac{1}{2}\times1.6\times10^{-3}\times(v_{GS}-0.5)^2$$

$$0.25=(v_{GS}-0.5)^2$$

$$0.5=(v_{GS}-0.5)$$

Thus, the value of gate source voltage v_{GS} is $\boxed{1\ V}$

Step 4 of 4

$v_{DS}\geq v_{GS}-V_{tn}$ Then N-channel transistor will be in saturation region.

Minimum v_{DS} required for N-channel transistor will be in saturation region is $v_{GS}-V_{tn}$

$$\begin{aligned}v_{DS\,min}&=v_{GS}-V_{tn}\\&=1-0.5\\&=0.5\end{aligned}$$

Thus, the drain source voltage $v_{DS\,min}$ is $\boxed{0.5\ V}$.

5.019P

Step 1 of 4

For $v_{GS} = v_{DS}$ MOSFET will act in saturation mode.

Drain current i_D for transistor in saturation region is $\frac{1}{2}k_n(v_{GS} - V_t)^2$

Drain current i_D is $0.4\ \text{mA}$ for $v_{GS} = v_{DS} = 1\ \text{V}$

$$0.4\times10^{-3} = \frac{1}{2}k_n(1 - V_t)^2 \quad \text{..... (1)}$$

Drain current i_D is $0.1\ \text{mA}$ for $v_{GS} = v_{DS} = 0.8\ \text{V}$

$$0.1\times10^{-3} = \frac{1}{2}k_n(0.8 - V_t)^2 \quad \text{..... (2)}$$

Step 2 of 4

Divide equation (1) with (2)

$$\frac{0.4\times10^{-3}}{0.1\times10^{-3}} = \frac{\frac{1}{2}k_n(1 - V_t)^2}{\frac{1}{2}k_n(0.8 - V_t)^2}$$

$$4 = \left(\frac{1 - V_t}{0.8 - V_t}\right)^2$$

$$2 = \frac{1 - V_t}{0.8 - V_t}$$

Step 3 of 4

$$\frac{1 - V_t}{0.8 - V_t} = 2$$

$$1 - V_t = 1.6 - 2V_t$$

$$V_t = 0.6$$

Threshold voltage V_t is $\boxed{0.6\ \text{V}}$

Step 4 of 4

Substitute V_t in equation (1)

$$0.4\times10^{-3} = \frac{1}{2}k_n(1 - V_t)^2$$

$$0.8\times10^{-3} = k_n(1 - 0.6)^2$$

$$5\times10^{-3} = k_n$$

Value of k_n is $\boxed{5\ \frac{\text{mA}}{\text{V}^2}}$

Step 1 of 2

5.020P

Trans-conductance parameter k_n' is $400\ \frac{\mu A}{V^2}$

Threshold voltage V_t is $0.5\ V$

For $v_{GS} = v_{DS}$ MOSFET will act in saturation mode.

Drain current i_D is $0.4\ mA$ for $v_{GS} = v_{DS} = V_{supply} = 1.8\ V$

Minimum length of the channel L_{min} is $0.18\ \mu m$

Drain current i_D for transistor in saturation region is $\frac{1}{2}k_n'\left(\frac{W}{L_{min}}\right)(v_{GS} - V_t)^2$

Step 2 of 2

Substitute corresponding values in the equation.

$$0.4\ (mA) = \frac{1}{2}\left(400\ \frac{\mu A}{V^2}\right)\left(\frac{W}{(0.18)\ \mu m}\right)(1.8-0.5)^2$$

$$0.4\times10^{-3}\ (A) = 1877.8W\ (Am)$$

$$2.1\times10^{-7} = W$$

$$W = 0.21\ \mu m$$

Channel width of NMOS transistor for IC fabrication process is $\boxed{0.21\ \mu m}$.

5.021P

Step 1 of 4

NMOS transistor operating in linear region with v_{DS} equal to $50\ \text{mV}$

Drain current i_D for $v_{GS} = 1\ \text{V}$ is $25\ \mu\text{A}$

Drain current i_D for $v_{GS} = 1.5\text{V}$ is $50\ \mu\text{A}$

Drain current i_D for transistor in triode region for small v_{DS} is $k_n'\left(\frac{W}{L}\right)\left((v_{GS} - V_t)\right)v_{DS}$

For $v_{GS} = 1\ \text{V}$ and $v_{DS} = 50\ \text{mV}$ is,

$$\left(25\times10^{-6}\ \text{A}\right) = k_n'\left(\frac{W}{L}\right)\left((1 - V_t)\right)\left(50\times10^{-3}\ \text{V}\right) \quad \text{..... (1)}$$

For $v_{GS} = 1.5\ \text{V}$ and $v_{DS} = 50\ \text{mV}$ is,

$$\left(50\times10^{-6}\ \text{A}\right) = k_n'\left(\frac{W}{L}\right)\left((1 - V_t)\right)\left(50\times10^{-3}\ \text{V}\right) \quad \text{..... (2)}$$

Step 2 of 4

Divide equation (1) with (2)

$$\frac{\left(25\times10^{-6}\ \text{A}\right)}{\left(50\times10^{-6}\ \text{A}\right)} = \frac{k_n'\left(\frac{W}{L}\right)\left((1 - V_t)\right)\left(50\times10^{-3}\ \text{V}\right)}{k_n'\left(\frac{W}{L}\right)\left((1 - V_t)\right)\left(50\times10^{-3}\ \text{V}\right)}$$

$$\frac{1}{2} = \frac{(1 - V_t)}{(1 - V_t)}$$

$$(1 - V_t) = 2 - 2V_t$$

$$V_t = 1\text{V}$$

Threshold voltage V_t is $\boxed{1\ \text{V}}$

Step 3 of 4

Value of k_n' is $50\ \mu\text{A}/\text{V}^2$

Substitute k_n' in equation (2)

$$(50\ \mu\text{A}) = \left(50\ \mu\text{A}/\text{V}^2\right)\left(\frac{W}{L}\right)\left((1.5 - 1)\right)\left(50\times10^{-3}\ \text{V}\right)$$

$$\left(25\times10^{-6}\ \text{A}\right) = \left(50\times10^{-6}\ \text{A}/\text{V}^2\right)\left(\frac{W}{L}\right)(0.5)\left(50\times10^{-3}\ \text{V}\right)$$

$$\left(\frac{W}{L}\right) = 0.05$$

Aspect ratio $\frac{W}{L}$ is 0.05

Step 4 of 4

For $v_{GS} = 2\ \text{V}$ and $v_{DS} = 0.1\ \text{V}$, $v_{GS} - V_t = 1\ \text{V}$ which is greater than v_{DS} and hence for $v_{GS} = 2\ \text{V}$ and $v_{DS} = 0.1\ \text{V}$ NMOS transistor operated in triode region.

Drain current i_D for transistor in triode region for large v_{DS} is,

$$k_n'\left(\frac{W}{L}\right)\left((v_{GS} - V_t) - \frac{1}{2}v_{DS}\right)v_{DS}$$

$$i_D = \left(50\ \mu\text{A}/\text{V}^2\right)(0.05)\left((2 - 1) - \frac{1}{2}(0.1)\right)(0.1)$$

$$= 0.245\ \mu\text{A}$$

Drain current i_D for transistor For $v_{GS} = 2\ \text{V}$ and $v_{DS} = 0.1\ \text{V}$ is $\boxed{0.245\ \mu\text{A}}$.

Step 1 of 2

5.022P

Threshold voltage V_t is $0.4\ \text{V}$

Range of v_{GS} is $1\ \text{V}$ to $1.8\ \text{V}$

$v_{DS} \geq v_{GS} - V_t$ Then N-channel transistor will be in saturation mode otherwise in triode region.

Step 2 of 2

Therefore maximum value of v_{DS} required for NMOS transistor channel remains constant is

$$v_{DS} = v_{GS} - V_t$$

$$\begin{aligned} v_{DS} &= v_{GS} - V_t \\ &= 1.8 - 0.4 \\ &= 1.4\ \text{V} \end{aligned}$$

Maximum value of v_{DS} required for NMOS transistor channel remains constant is $\boxed{1.4\ \text{V}}$

Step 1 of 9

Threshold voltage V_t is $0.8\ \text{V}$

Range of v_{GS} is $1\ \text{V}$ to $4.8\ \text{V}$

Width of the (W) channel is $20\ \mu\text{m}$

Length of the (L) channel is $1\ \mu\text{m}$

Trans-conductance parameter k_n' is $50\ \mu\text{A}/\text{V}^2$

Linear resistance r_{DS} is $\dfrac{1}{k_n'(W/L)(V_{GS}-V_t)}$

Linear resistance r_{DS} for v_{GS} equal to $1\ \text{V}$ is,

$$r_{DS} = \frac{1}{50\left(\mu\text{A}/\text{V}^2\right)\left(\frac{20\ \mu\text{m}}{1\ \mu\text{m}}\right)(1-0.8)}$$
$$= \frac{1}{\left(50\times10^{-6}\ \text{A}/\text{V}^2\right)(20)(0.2)}$$
$$= 5000\ \Omega$$

Linear resistance r_{DS} for v_{GS} equal to $1\ \text{V}$ is $\boxed{5000\ \Omega}$

Step 2 of 9

Linear resistance r_{DS} for v_{GS} equal to $4.8\ \text{V}$ is,

$$r_{DS} = \frac{1}{50\left(\mu\text{A}/\text{V}^2\right)\left(\frac{20\ \mu\text{m}}{1\ \mu\text{m}}\right)(4.8-0.8)}$$
$$= \frac{1}{\left(50\times10^{-6}\ \text{A}/\text{V}^2\right)(20)(4)}$$
$$= 250\ \Omega$$

Linear resistance r_{DS} for v_{GS} equal to 4.8V is $\boxed{250\ \Omega}$

Range of resistor values is $\boxed{250\Omega \text{ to } 5000\ \Omega}$

Step 3 of 9

(a)

Device width of the channel is halved and hence width of the channel is $10\ \mu\text{m}$

Linear resistance r_{DS} for v_{GS} equal to $1\ \text{V}$ is,

$$r_{DS} = \frac{1}{50\left(\mu\text{A}/\text{V}^2\right)\left(\frac{10\ \mu\text{m}}{1\ \mu\text{m}}\right)(1-0.8)}$$
$$= \frac{1}{\left(50\times10^{-6}\ \text{A}/\text{V}^2\right)(10)(0.2)}$$
$$= 10000\ \Omega$$

Linear resistance r_{DS} for v_{GS} equal to $1\ \text{V}$ is $\boxed{10000\ \Omega}$

Step 4 of 9

Linear resistance r_{DS} for v_{GS} equal to $4.8\ \text{V}$ is,

$$r_{DS} = \frac{1}{50\left(\mu\text{A}/\text{V}^2\right)\left(\frac{10\ \mu\text{m}}{1\ \mu\text{m}}\right)(4.8-0.8)}$$
$$= \frac{1}{\left(50\times10^{-6}\ \text{A}/\text{V}^2\right)(10)(4)}$$
$$= 500\ \Omega$$

Linear resistance r_{DS} for v_{GS} equal to 4.8V is $\boxed{500\ \Omega}$

Range of resistor values after width of the channel is halved is $\boxed{500\Omega \text{ to } 10000\ \Omega}$

Step 5 of 9

(b)

Device length of the channel is halved and hence length of the channel is $0.5\ \mu\text{m}$

Linear resistance r_{DS} for v_{GS} equal to $1\ \text{V}$ is,

$$r_{DS} = \frac{1}{50\left(\mu\text{A}/\text{V}^2\right)\left(\frac{20\ \mu\text{m}}{0.5\ \mu\text{m}}\right)(1-0.8)}$$
$$= \frac{1}{\left(50\times10^{-6}\ \text{A}/\text{V}^2\right)(40)(0.2)}$$
$$= 2500\ \Omega$$

Linear resistance r_{DS} for v_{GS} equal to $1\ \text{V}$ is $\boxed{2500\ \Omega}$

Step 6 of 9

Linear resistance r_{DS} for v_{GS} equal to $4.8\ \text{V}$ is,

$$r_{DS} = \frac{1}{50\left(\mu\text{A}/\text{V}^2\right)\left(\frac{20\ \mu\text{m}}{0.5\ \mu\text{m}}\right)(4.8-0.8)}$$
$$= \frac{1}{\left(50\times10^{-6}\ \text{A}/\text{V}^2\right)(40)(4)}$$
$$= 125\ \Omega$$

Step 7 of 9

Linear resistance r_{DS} for v_{GS} equal to 4.8V is $\boxed{125\ \Omega}$

Range of resistor values after length of the channel is halved is $\boxed{125\Omega \text{ to } 2500\ \Omega}$

Step 8 of 9

(c)

Device length and width of the channel is halved and hence length of the channel is $0.5\ \mu\text{m}$, width of the channel is $10\ \mu\text{m}$.

Linear resistance r_{DS} for v_{GS} equal to $1\ \text{V}$ is,

$$r_{DS} = \frac{1}{50\left(\mu\text{A}/\text{V}^2\right)\left(\frac{10\ \mu\text{m}}{0.5\ \mu\text{m}}\right)(1-0.8)}$$
$$= \frac{1}{\left(50\times10^{-6}\ \text{A}/\text{V}^2\right)(20)(0.2)}$$
$$= 5000\ \Omega$$

Linear resistance r_{DS} for v_{GS} equal to $1\ \text{V}$ is $\boxed{5000\ \Omega}$

Step 9 of 9

Linear resistance r_{DS} for v_{GS} equal to $4.8\ \text{V}$ is,

$$r_{DS} = \frac{1}{50\left(\mu\text{A}/\text{V}^2\right)\left(\frac{10\ \mu\text{m}}{0.5\ \mu\text{m}}\right)(4.8-0.8)}$$
$$= \frac{1}{\left(50\times10^{-6}\ \text{A}/\text{V}^2\right)(20)(4)}$$
$$= 250\ \Omega$$

Range of resistor values after width and length of the channel are halved is $\boxed{250\Omega \text{ to } 5000\ \Omega}$

Step 1 of 2

5.024P

Refer to Figure P5.24 in the text book for the drain and gate connections of MOSFET.

(a)

The drain and source of the two transistors in the figures are short-circuited. This configuration results in diode-connected transistor model.

The voltage *v* in the figure is,

$$\begin{aligned} v &= v_{DS} \\ &= v_{DG} + v_{GS} \\ &= 0 + v_{GS} \\ &= v_{GS} \end{aligned}$$

As v_{DG} is 0 V, $v_{DG} \le V_t$. Hence, the transistor is operating in the saturation region.

Write the formula for the drain current, i of a MOSFET in the saturation mode.

$$i = \frac{1}{2}k'\frac{W}{L}\left(v_{GS} - |V_t|\right)^2$$

Here,

k' is the process transconductance parameter

$\frac{W}{L}$ is the width to length ratio of the channel

v_{GS} is the gate-to-source voltage

V_t is the threshold voltage

Substitute *v* for v_{GS} .

$$i = \frac{1}{2}k'\frac{W}{L}\left(v - |V_t|\right)^2$$

Thus, the *i-v* relationship of the MOSFET is,

$$\boxed{i = \frac{1}{2}k'\frac{W}{L}\left(v - |V_t|\right)^2}.$$

Step 2 of 2

(b)

Write the formula for incremental resistance of a MOSFET.

$$r = \frac{1}{\left(\frac{\partial i}{\partial v}\right)}$$

Determine $\frac{\partial i}{\partial v}$ using the *i-v* relationship of the MOSFET.

$$\begin{aligned} i &= \frac{1}{2}k'\frac{W}{L}\left(v - |V_t|\right)^2 \\ \frac{\partial i}{\partial v} &= \frac{1}{2}k'\frac{W}{L}2\left(v - |V_t|\right) \\ &= k'\frac{W}{L}\left(v - |V_t|\right) \end{aligned}$$

Thus, the incremental resistance is,

$$\boxed{r = \frac{1}{\left(k'\frac{W}{L}\left(v - |V_t|\right)\right)}}.$$

Step 1 of 3

5.025P

Refer to Figure P5.25 in the text book for the transistor circuit.

Observe from the figure that $v_{GD} = 0\ \text{V}$.

And the gate-to-source voltage is,

$$v_{GS} = V_{DD} - v_S$$

For the transistor to operate in saturation region, v_{GD} should be less than or equal to the threshold voltage, V_{tn} .

As $v_{GD} < V_{tn}$, the transistor is in saturation mode.

Write the formula for the drain current, i_D in the saturation mode of operation. $i_D = \frac{1}{2}k_n(v_{GS} - V_t)^2$

Here,

k_n is the transistor transconductance parameter

v_{GS} is the gate-to-source voltage

V_t is the threshold voltage

Substitute $V_{DD} - v_S$ for v_{GS} .

$$\begin{aligned} i_D &= \frac{1}{2}k_n(V_{DD} - v_S - V_t)^2 \\ &= \frac{1}{2}k_n\left[(V_{DD} - V_t)^2 - 2v_S(V_{DD} - V_t) + v_S^2\right] \end{aligned}$$

Step 2 of 3

For $v_{GS} < V_t$ the transistor is in cut-off region.

$$\begin{aligned} &v_{GS} < V_t \\ &\Rightarrow V_{DD} - v_S < V_t \\ &v_S > V_{DD} - V_t \end{aligned}$$

For $v_S > V_{DD} - V_t$ the drain current is zero.

Thus, the relation, $i_D = \frac{1}{2}k_n\left[(V_{DD} - V_t)^2 - 2v_S(V_{DD} - V_t) + v_S^2\right]$ holds in the range,

$0 \le v_S \le V_{DD} - V_t$.

Determine i_D when $v_S = 0$.

$$\begin{aligned} i_D &= \frac{1}{2}k_n\left[(V_{DD} - V_t)^2 - 2(0)(V_{DD} - V_t) + 0\right] \\ &= \frac{1}{2}k_n(V_{DD} - V_t)^2 \end{aligned}$$

Step 3 of 3

Plot i_D versus v_S characteristics for v_S varying from 0 to V_{DD} .

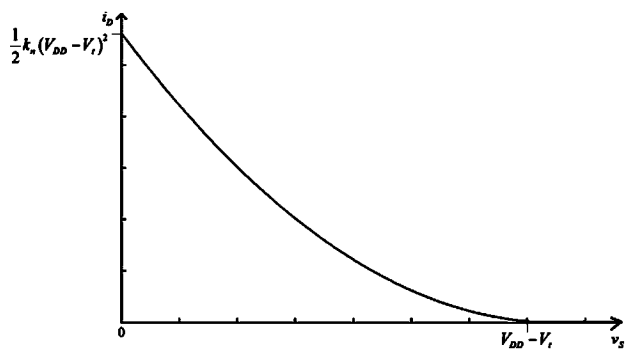


Figure 1

Thus, the plot of i_D versus v_S is sketched.

Step 1 of 2

5.026P

Refer to Figure P5.26 in the text book for the circuit.

The gate and drain in the circuit are short circuited.

The drain-to-source voltage is,

$$\begin{aligned} v_{DS} &= v_{DG} + v_{GS} \\ &= 0 + v_{GS} \\ &= v_{GS} \end{aligned}$$

Thus, $v_{DS} = v_{GS}$

As v_{DG} is 0 V, $v_{DG} \le V_t$. Hence, the transistor is operating in the saturation region.

Write the formula for the drain current, i_D of a MOSFET in the saturation mode.

$$i_D = \frac{1}{2} k_n (v_{GS} - V_t)^2$$

Here,

k_n is the transistor transconductance parameter

v_{GS} is the gate-to-source voltage

V_t is the threshold voltage

$$(v_{GS} - V_t)^2 = \frac{2i_D}{k_n}$$

$$v_{GS} - V_t = \sqrt{\frac{2i_D}{k_n}}$$

$$v_{GS} = \sqrt{\frac{2i_D}{k_n}} + V_t$$

As $v_{DS} = v_{GS}$,

$$v_{DS} = \sqrt{\frac{2i_D}{k_n}} + V_t$$

When $i_D = 0$, $v_{DS} = V_t$.

Step 2 of 2

Sketch v_{DS} versus i_D plot.

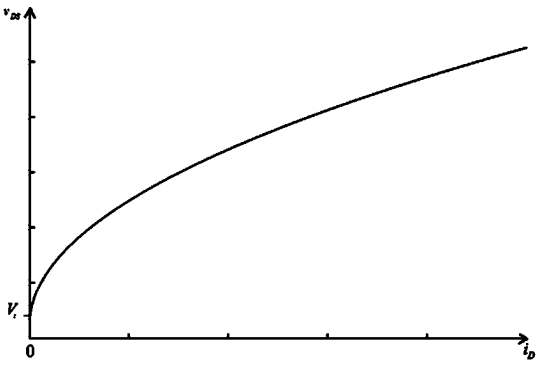


Figure 1

Thus, the graph for v_{DS} versus i_D is sketched.

Step 1 of 12

The threshold voltage of an NMOS transistor, $V_t = 1\ \text{V}$.

(a)

The gate-to-source voltage V_{GS} is,

$$\begin{aligned} V_{GS} &= V_G - V_S \\ &= 1 - 1 \\ &= 0\ \text{V} \end{aligned}$$

The over-drive voltage V_{OV} is,

$$\begin{aligned} V_{OV} &= V_{GS} - V_t \\ &= 0 - 1 \\ &= -1\ \text{V} \end{aligned}$$

The drain-to-source voltage V_{DS} is,

$$\begin{aligned} V_{DS} &= V_D - V_S \\ &= 2 - 1 \\ &= 1\ \text{V} \end{aligned}$$

As V_{DS} is equal to 0 V, the transistor is operating in the cut-off region.

Step 2 of 12

(b)

The gate-to-source voltage V_{GS} is,

$$\begin{aligned} V_{GS} &= V_G - V_S \\ &= 2.5 - 1 \\ &= 1.5\ \text{V} \end{aligned}$$

The over-drive voltage V_{OV} is,

$$\begin{aligned} V_{OV} &= V_{GS} - V_t \\ &= 1.5 - 1 \\ &= 0.5\ \text{V} \end{aligned}$$

The drain-to-source voltage V_{DS} is,

$$\begin{aligned} V_{DS} &= V_D - V_S \\ &= 2 - 1 \\ &= 1\ \text{V} \end{aligned}$$

As $V_{GS} \geq V_{GS} - V_t$, the transistor is operating in the saturation region.

Step 3 of 12

(c)

The gate-to-source voltage V_{GS} is,

$$\begin{aligned} V_{GS} &= V_G - V_S \\ &= 2.5 - 1 \\ &= 1.5\ \text{V} \end{aligned}$$

The over-drive voltage V_{OV} is,

$$\begin{aligned} V_{OV} &= V_{GS} - V_t \\ &= 1.5 - 1 \\ &= 0.5\ \text{V} \end{aligned}$$

The drain-to-source voltage V_{GS} is,

$$\begin{aligned} V_{DS} &= V_D - V_S \\ &= 1.5 - 1 \\ &= 0.5\ \text{V} \end{aligned}$$

As $V_{GS} \geq V_{GS} - V_t$, the transistor is operating in the edge of saturation region.

Step 4 of 12

(d)

The gate-to-source voltage V_{GS} is,

$$\begin{aligned} V_{GS} &= V_G - V_S \\ &= 1.5 - 1 \\ &= 0.5\ \text{V} \end{aligned}$$

The over-drive voltage V_{OV} is,

$$\begin{aligned} V_{OV} &= V_{GS} - V_t \\ &= 0.5 - 1 \\ &= -0.5\ \text{V} \end{aligned}$$

The drain-to-source voltage V_{DS} is,

$$\begin{aligned} V_{DS} &= V_D - V_S \\ &= 0 - 1 \\ &= -1\ \text{V} \end{aligned}$$

As V_{DS} is negative, drain and source can be exchanged.

As $V_{DS} \geq V_{GS} - V_t$, the transistor is operating in the saturation region.

Step 5 of 12

(e)

The gate-to-source voltage V_{GS} is,

$$\begin{aligned} V_{GS} &= V_G - V_S \\ &= 2.5 - 0 \\ &= 2.5\ \text{V} \end{aligned}$$

The over-drive voltage V_{OV} is,

$$\begin{aligned} V_{OV} &= V_{GS} - V_t \\ &= 2.5 - 1 \\ &= 1.5\ \text{V} \end{aligned}$$

The drain-to-source voltage V_{DS} is,

$$\begin{aligned} V_{DS} &= V_D - V_S \\ &= 1 - 0 \\ &= 1\ \text{V} \end{aligned}$$

As $V_{DS} < V_{GS} - V_t$, the transistor is operating in the triode region.

Step 6 of 12

(f)

The gate-to-source voltage V_{GS} is,

$$\begin{aligned} V_{GS} &= V_G - V_S \\ &= 1 - 1 \\ &= 0\ \text{V} \end{aligned}$$

The over-drive voltage V_{OV} is,

$$\begin{aligned} V_{OV} &= V_{GS} - V_t \\ &= 0 - 1 \\ &= -1\ \text{V} \end{aligned}$$

The drain-to-source voltage V_{GS} is,

$$\begin{aligned} V_{DS} &= V_D - V_S \\ &= 1 - 1 \\ &= 0\ \text{V} \end{aligned}$$

As V_{GS} is equal to 0 V, the transistor is operating in the cut-off region.

Step 7 of 12

(g)

The gate-to-source voltage V_{GS} is,

$$\begin{aligned} V_{GS} &= V_G - V_S \\ &= 0 + 1 \\ &= 1\ \text{V} \end{aligned}$$

The over-drive voltage V_{OV} is,

$$\begin{aligned} V_{OV} &= V_{GS} - V_t \\ &= 1 - 1 \\ &= 0\ \text{V} \end{aligned}$$

The drain-to-source voltage V_{DS} is,

$$\begin{aligned} V_{DS} &= V_D - V_S \\ &= 0 - (-1) \\ &= 1\ \text{V} \end{aligned}$$

As $V_{DS} \geq V_{GS} - V_t$, the transistor is operating in the saturation region.

Step 8 of 12

(h)

The gate-to-source voltage V_{GS} is,

$$\begin{aligned} V_{GS} &= V_G - V_S \\ &= 0 - (-1.5) \\ &= 1.5\ \text{V} \end{aligned}$$

The over-drive voltage V_{OV} is,

$$\begin{aligned} V_{OV} &= V_{GS} - V_t \\ &= 1.5 - 1 \\ &= 0.5\ \text{V} \end{aligned}$$

The drain-to-source voltage V_{GS} is,

$$\begin{aligned} V_{DS} &= V_D - V_S \\ &= 0 - (-1.5) \\ &= 1.5\ \text{V} \end{aligned}$$

As $V_{DS} \geq V_{GS} - V_t$, the transistor is operating in the saturation region.

Step 9 of 12

(i)

The gate-to-source voltage V_{GS} is,

$$\begin{aligned} V_{GS} &= V_G - V_S \\ &= 0 - (-1) \\ &= 1\ \text{V} \end{aligned}$$

Step 10 of 12

The over-drive voltage V_{OV} is,

$$\begin{aligned} V_{OV} &= V_{GS} - V_t \\ &= 1 - 1 \\ &= 0\ \text{V} \end{aligned}$$

The drain-to-source voltage V_{DS} is,

$$\begin{aligned} V_{DS} &= V_D - V_S \\ &= 1 - (-1) \\ &= 2\ \text{V} \end{aligned}$$

As $V_{DS} \geq V_{GS} - V_t$, the transistor is operating in the saturation region.

Step 11 of 12

(j)

The gate-to-source voltage V_{GS} is,

$$\begin{aligned} V_{GS} &= V_G - V_S \\ &= 2 - 0.5 \\ &= 1.5\ \text{V} \end{aligned}$$

The over-drive voltage V_{OV} is,

$$\begin{aligned} V_{OV} &= V_{GS} - V_t \\ &= 1.5 - 1 \\ &= 0.5\ \text{V} \end{aligned}$$

The drain-to-source voltage V_{GS} is,

$$\begin{aligned} V_{DS} &= V_D - V_S \\ &= 0.5 - 0.5 \\ &= 0\ \text{V} \end{aligned}$$

As $V_{GS} < V_{GS} - V_t$, the transistor is operating in the triode region.

Step 12 of 12

Fill the entries in the table.

Case	V_S	V_G	V_D	V_{GS}	V_{OV}	V_{DS}	Region of operation
a	+1.0	+1.0	+2.0	0	−1	+1	Cut-off
b	+1.0	+2.5	+2.0	+1.5	+0.5	+1	Saturation
c	+1.0	+2.5	+1.5	+1.5	+0.5	+0.5	Edge of saturation
d	+1.0	+1.5	0	+0.5	−0.5	−1	Saturation
e	0	+2.5	+1.0	+2.5	+1.5	+1	Triode
f	+1.0	+1.0	+1.0	0	−1	0	Cut-off
g	−1.0	0	0	+1	0	+1	Saturation
h	−1.5	0	0	+1.5	+0.5	+1.5	Saturation
i	−1.0	0	+1.0	+1	0	+2	Saturation
j	+0.5	+2.0	+0.5	+1.5	+0.5	0	Triode

Thus, the entries in the table are calculated and filled.

5.028P

Refer to Figure P5.28 in the text book for the NMOS transistor circuit.

For $v_{GS} < V_t$ the transistor is in cut-off region.

For $v_{GS} < 0.4\ \text{V}$, $i_D = 0$

For $v_{GS} \geq 0.4\ \text{V}$, the transistor is in saturation region.

Write the formula for the drain current i_D.

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - V_t)^2$$

Here,

k_n' is the process transconductance parameter

$\frac{W}{L}$ is the width to length ratio of the channel

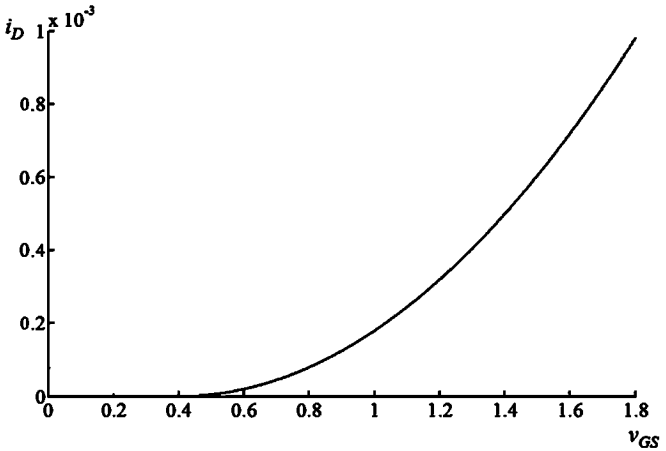
v_{GS} is the gate-to-source voltage

V_t is the threshold voltage

Substitute $1\times10^{-3}\ \text{A/V}^2$ for $k_n'\left(\frac{W}{L}\right)$ and $0.4\ \text{V}$ for V_t.

$$\begin{aligned} i_D &= \frac{1}{2}\left(10^{-3}\right)(v_{GS} - 0.4)^2 \\ &= \left(0.5\times10^{-3}\right)(v_{GS} - 0.4)^2 \end{aligned}$$

Plot i_D versus v_{GS}.



The equations of the two regions of the plot are,

$$\begin{array}{|ll|} \hline i_D = 0 & \text{for } v_{GS} < 0.4\ \text{V} \\ i_D = \left(0.5\times10^{-3}\right)(v_{GS} - 0.4)^2 & \text{for } v_{GS} \geq 0.4\ \text{V} \\ \hline \end{array}$$

Step 1 of 5

(a)

For $v_{GS} = v_{DS}$ MOSFET will act in saturation mode.

Drain current i_D for transistor in saturation region is $\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - V_t)^2$

Let $\frac{W}{L} = x$ and the drain current flowing through transistor Q_1 for $\frac{W}{L} = x$ is i_{D1}

$$i_{D1} = \frac{1}{2}k_n'(x)(v_{GS} - V_t)^2 \quad \text{..... (1)}$$

Step 2 of 5

$\frac{W}{L}$ ratio is 3% more for transistor Q_2, then the new $\frac{W}{L}$ ratio is,

$$\frac{W}{L}\left(\frac{103}{100}\right) = 1.03\frac{W}{L}$$
$$= 1.03x$$

New $\frac{W}{L}$ ratio is $1.03x$

Drain current flowing through transistor Q_2 for $\frac{W}{L} = 1.03x$ is i_{D2}

$$i_{D2} = \frac{1}{2}k_n'(1.03x)(v_{GS} - V_t)^2 \quad \text{..... (2)}$$

Step 3 of 5

Divide equation (2) with (1)

$$\frac{i_{D2}}{i_{D1}} = \frac{\frac{1}{2}k_n'(1.03x)(v_{GS} - V_t)^2}{\frac{1}{2}k_n'(x)(v_{GS} - V_t)^2}$$
$$= \frac{1.03}{1}$$

i_{D2} in terms of i_{D1} is $1.03i_{D1}$

Mismatch percent in drain current is $\frac{(i_{D2} - i_{D1})}{i_{D1}} \times 100$

$$\frac{(i_{D2} - i_{D1})}{i_{D1}} \times 100 = \frac{1.03i_{D1} - i_{D1}}{i_{D1}} \times 100$$
$$= \frac{i_{D1}(1.03 - 1)}{i_{D1}} \times 100$$
$$= 3\%$$

Maximum mismatch in drain current for maximum mismatch $\frac{W}{L}$ ratio of 3% is $\boxed{3\%}$

Step 4 of 5

(b)

Nominal value of V_t is $0.6\ \text{V}$

Mismatch in V_t value is $10\ \text{mV}$

Let V_t value is increased by $10\ \text{mV}$ then new V_t value is $0.61\ \text{mV}$

For V_t is $0.6\ \text{V}$, drain current flowing through transistor Q_1 is i_{D1}

$$i_{D1} = \frac{1}{2}k_n'(x)(v_{GS} - 0.6)^2 \quad \text{..... (3)}$$

For V_t is $0.61\ \text{V}$, drain current flowing through transistor Q_2 is i_{D2}

$$i_{D2} = \frac{1}{2}k_n'(x)(v_{GS} - 0.61)^2 \quad \text{..... (4)}$$

Step 5 of 5

Divide equation (4) with (3)

$$\frac{i_{D2}}{i_{D1}} = \frac{\frac{1}{2}k_n'(x)(v_{GS} - 0.61)^2}{\frac{1}{2}k_n'(x)(v_{GS} - 0.6)^2}$$
$$= \left[\frac{(v_{GS} - 0.61)}{(v_{GS} - 0.6)}\right]^2$$

From the figure P5.29 v_{GS} is $1\ \text{V}$

$$\frac{i_{D2}}{i_{D1}} = \left[\frac{(1 - 0.61)}{(1 - 0.6)}\right]^2$$
$$= 0.951$$

i_{D2} in terms of i_{D1} is $0.951i_{D1}$

Mismatch percent in drain current is $\frac{(i_{D2} - i_{D1})}{i_{D1}} \times 100$

$$\frac{(i_{D2} - i_{D1})}{i_{D1}} \times 100 = \frac{0.951i_{D1} - i_{D1}}{i_{D1}} \times 100$$
$$= \frac{i_{D1}(0.951 - 1)}{i_{D1}} \times 100$$
$$= -4.9\%$$

Maximum mismatch in drain current is $\boxed{-4.9\%}$

5.030P

Step 1 of 4

Drain current for $v_{DS} = 1\ \text{V}$ and for a constant v_{GS} is $0.5\ \text{mA}$

Drain current for $v_{DS} = 2\ \text{V}$ and for a constant v_{GS} is $0.52\ \text{mA}$

Drain current with channel-length modulation effect is considered is ,

$$\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-v_{tn})^2(1+\lambda v_{DS})$$

$$0.5\ \text{mA} = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-v_{tn})^2(1+\lambda 1) \quad \text{..... (1)}$$

$$0.52\ \text{mA} = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-v_{tn})^2(1+\lambda 2) \quad \text{..... (2)}$$

Step 2 of 4

Divide equation (2) with equation (1)

$$\frac{0.52\ \text{mA}}{0.50\ \text{mA}} = \frac{\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-v_{tn})^2(1+\lambda 2)}{\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-v_{tn})^2(1+\lambda)}$$

$$\frac{52}{50} = \frac{1+\lambda 2}{1+\lambda 1}$$

$$52+52\lambda = 50+100\lambda$$

$$\lambda = \frac{1}{24}$$

Value of λ is $\boxed{\frac{1}{24}}$.

Step 3 of 4

Output resistance r_0 is $\frac{1}{\lambda i_D}$

Output resistance for i_D equal to $0.5\ \text{mA}$

$$r_0 = \frac{1}{\lambda i_D}$$

$$= \frac{1}{\left(\frac{1}{24}\right)\left(0.5\times 10^{-3}\right)}$$

$$= \frac{24\times 10^3}{0.5}$$

$$= 48\ \text{k}\Omega$$

Output resistance for i_D equal to $0.5\ \text{mA}$ is $\boxed{48\ \text{k}\Omega}$.

Step 4 of 4

Output resistance for i_D equal to $0.52\ \text{mA}$

$$r_0 = \frac{1}{\lambda i_D}$$

$$= \frac{1}{\left(\frac{1}{24}\right)\left(0.52\times 10^{-3}\right)}$$

$$= \frac{24\times 10^3}{0.52}$$

$$= 46.15\ \text{k}\Omega$$

Output resistance for i_D equal to $0.5\ \text{mA}$ is $\boxed{46.15\ \text{k}\Omega}$

Value of Early voltage V_A is $\frac{1}{\lambda}$

Early voltage V_A is $\boxed{24\ \text{V}}$.

Step 1 of 8

Value of V_A is $20\ \text{V}$

Value of λ is $\dfrac{1}{V_A}$

Therefore value of λ is $\dfrac{1}{20}$

Output resistance r_0 is $\dfrac{1}{\lambda I_D}$

Output resistance for I_D equal to $0.1\ \text{mA}$

$$r_0 = \frac{1}{\lambda I_D}$$
$$= \frac{1}{\left(\frac{1}{20}\right)\left(0.1\times10^{-3}\right)}$$
$$= \frac{20\times10^{3}}{0.1}$$
$$= 200\ \text{k}\Omega$$

Output resistance for I_D equal to $0.1\ \text{mA}$ is $\boxed{200\ \text{k}\Omega}$

Step 2 of 8

Output resistance for I_D equal to $1\ \text{mA}$

$$r_0 = \frac{1}{\lambda I_D}$$
$$= \frac{1}{\left(\frac{1}{20}\right)\left(1\times10^{-3}\right)}$$
$$= \frac{20\times10^{3}}{1}$$
$$= 200\ \text{k}\Omega$$

Output resistance for I_D equal to $1\ \text{mA}$ is $\boxed{20\ \text{k}\Omega}$

Step 3 of 8

Drain current flowing through transistor when $(v_{DS})\ \text{V}$ applied across drain and source is $0.1\ \text{mA}$

Therefore $\frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\left(1+\lambda\left(v_{DS}\right)\right)=0.1\ \text{mA}$

Change in v_{DS} is $1\ \text{V}$ and hence new drain source voltage applied to MOSFET is $(v_{DS}+1)\ \text{V}$ and drain current is i_D

Drain current flowing through transistor when $(v_{DS}+1)\ \text{V}$ applied across drain and source is $\frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\left(1+\lambda\left(v_{DS}+1\right)\right)$

Therefore $\frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\left(1+\lambda\left(v_{DS}+1\right)\right)=i_D$

Step 4 of 8

Output resistance r_0 is $\left[\lambda\frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\right]^{-1}$

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\left(1+\lambda\left(v_{DS}+1\right)\right)$$
$$= \frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\left(1+\lambda\left(v_{DS}\right)\right)+\frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\left(\lambda\right)$$
$$= 0.1\times10^{-3}+\frac{1}{r_0}$$
$$= 0.1\times10^{-3}+\frac{1}{200\times10^{3}}$$

Current i_D is $0.105\ \text{mA}$

Step 5 of 8

Percentage change in drain current is $\left(\dfrac{i_D-0.1}{0.1}\times100\right)$

$$\left(\frac{i_D-0.1}{0.1}\times100\right)=\frac{0.105-0.1}{0.1}\times100$$
$$=5\%$$

Percentage change in drain current for a change in v_{DS} of $1\ \text{V}$ is $\boxed{5\%}$

Step 6 of 8

Drain current flowing through transistor when $(v_{DS})\ \text{V}$ applied across drain and source is $1\ \text{mA}$

Therefore $\frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\left(1+\lambda\left(v_{DS}\right)\right)=1\ \text{mA}$

Change in v_{DS} is $1\ \text{V}$ and hence new drain source voltage applied to MOSFET is $(v_{DS}+1)\ \text{V}$ and drain current is i_D

Drain current flowing through transistor when $(v_{DS}+1)\ \text{V}$ applied across drain and source is $\frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\left(1+\lambda\left(v_{DS}+1\right)\right)$

Therefore $\frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\left(1+\lambda\left(v_{DS}+1\right)\right)=i_D$

Step 7 of 8

Output resistance r_0 is $\left[\lambda\frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\right]^{-1}$

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\left(1+\lambda\left(v_{DS}+1\right)\right)$$
$$= \frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\left(1+\lambda\left(v_{DS}\right)\right)+\frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\left(\lambda\right)$$
$$= 0.1\times10^{-3}+\frac{1}{r_0}$$
$$= 1\times10^{-3}+\frac{1}{20\times10^{3}}$$

Current i_D is $1.05\ \text{mA}$

Step 8 of 8

Percentage change in drain current is $\left(\dfrac{i_D-0.1}{0.1}\times100\right)$

$$\left(\frac{i_D-0.1}{0.1}\times100\right)=\frac{1.05-1}{1}\times100$$
$$=5\%$$

Percentage change in drain current for a change in v_{DS} of $1\ \text{V}$ is $\boxed{5\%}$.

5.032P

We don't have the solution to this problem yet.

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Step 1 of 7

Minimum length of the MOS technology n- channel device is $0.5\ \mu m$

Value of λ is $0.03\ V^{-1}$

Channel length and the value of λ are inversely related to each other,

$\frac{1}{\lambda} = V_A' L$ Where V_A' is constant which depends on type of technology used.

$$\frac{1}{\lambda} = V_A' L$$
$$\frac{1}{0.03}\ V = V_A' (0.5\ \mu m)$$
$$V_A' = 66.7 \times 10^6\ \frac{V}{m}$$

Step 2 of 7

For a device length of the channel is $1.5\ \mu m$ and the corresponding value of λ is λ_1

$$\frac{1}{\lambda_1} = V_A' (1.5 \times 10^{-6})$$

Substitute value of V_A' in the above equation

$$\frac{1}{\lambda_1} = (66.7 \times 10^6)(1.5 \times 10^{-6})$$
$$= 100$$

Value of is λ_1 is 0.01

Step 3 of 7

Drain current with channel-length modulation effect is considered is,

$$\frac{1}{2} k_n' \left(\frac{W}{L}\right)(v_{GS} - v_m)^2 (1 + \lambda v_{DS})$$

For v_{DS} equal to $1\ V$ drain current is $100\ \mu A$

$$100 \times 10^{-6} = \frac{1}{2} k_n' \left(\frac{W}{L}\right)(v_{GS} - v_m)^2 (1 + \lambda 1) \quad \text{..... (1)}$$

v_{DS} is raised to $5\ V$ new drain current is i_D

$$i_D = \frac{1}{2} k_n' \left(\frac{W}{L}\right)(v_{GS} - v_m)^2 (1 + \lambda 5) \quad \text{..... (2)}$$

Step 4 of 7

Divide equation (2) with (1) and substitute value of λ_1 in λ

$$\frac{i_D}{100 \times 10^{-6}} = \frac{\frac{1}{2} k_n' \left(\frac{W}{L}\right)(v_{GS} - v_m)^2 (1 + \lambda 5)}{\frac{1}{2} k_n' \left(\frac{W}{L}\right)(v_{GS} - v_m)^2 (1 + \lambda)}$$
$$\frac{i_D}{100 \times 10^{-6}} = \frac{(1 + \lambda 5)}{(1 + \lambda 1)}$$
$$\frac{i_D}{100 \times 10^{-6}} = \frac{(1 + (0.01)5)}{(1 + (0.01)1)}$$
$$i_D = 104\ \mu A$$

Drain current for v_{DS} is raised to $5\ V$ is $\boxed{104\ \mu A}$

Step 5 of 7

Percentage change in drain current is,

$$\left(\frac{104 - 100}{100}\right) \times 100 = 4\%$$

Percentage change in drain current for v_{DS} is raised to $5\ V$ is 4%

Step 6 of 7

Drain current percentage is reduced by a factor of 2, therefore new drain current percentage is 2%

Let new drain current is i_D

$$\left(\frac{i_D - 100}{100}\right) \times 100 = 2$$
$$i_D - 100 = 2$$
$$i_D = 102\ mA$$

New drain current i_D is $\boxed{102\ mA}$

Step 7 of 7

In order to reduce percentage of drain current by a factor of 2, we have to reduce v_{DS}

$$102 \times 10^{-6} = \frac{1}{2} k_n' \left(\frac{W}{L}\right)(v_{GS} - v_m)^2 (1 + \lambda v_{DS}) \quad \text{..... (3)}$$

Divide equation (3) with (1) and substitute value of λ_1 in λ

$$\frac{102 \times 10^{-6}}{100 \times 10^{-6}} = \frac{\frac{1}{2} k_n' \left(\frac{W}{L}\right)(v_{GS} - v_m)^2 (1 + \lambda v_{DS})}{\frac{1}{2} k_n' \left(\frac{W}{L}\right)(v_{GS} - v_m)^2 (1 + \lambda)}$$
$$\frac{102}{100} = \frac{(1 + \lambda v_{DS})}{(1 + \lambda)}$$
$$\frac{102}{100} = \frac{(1 + 0.01 v_{DS})}{(1 + 0.01)}$$
$$v_{DS} = 3.02\ V$$

In order to reduce drain current by a factor of 2, we have to reduce v_{DS} to $\boxed{3.02\ V}$

Step 1 of 5

Value of k_n' is $200\ \mu A/V^2$

Value of V_A' is $20\ V/\mu m$

Channel length L is $1.5\ \mu m$

Early voltage V_A is $V_A'L$

$$V_A = V_A'L$$
$$= (20\ V/\mu m)(1.5\ \mu m)$$
$$= 30\ V$$

Early voltage V_A is $\boxed{30\ V}$

Step 2 of 5

Relation between V_A and λ is,

$$\lambda = \frac{1}{V_A}$$
$$= \frac{1}{30}$$
$$= 0.033\ V^{-1}$$

Value of λ is $\boxed{0.033\ V^{-1}}$

Step 3 of 5

Over drive voltage $(v_{GS} - V_m)$ is $0.5\ V$

Drain source voltage v_{DS} is $2\ V$

Drain current with channel-length modulation effect is considered is ,

$$\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - v_m)^2(1 + \lambda v_{DS})$$

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - v_m)^2(1 + \lambda v_{DS})$$
$$= \frac{1}{2}\left(200\ \mu A/V^2\right)\left(\frac{15\ \mu m}{1.5\ \mu m}\right)(0.5)^2(1 + 0.033(2))$$
$$= 266.5\ \mu A$$

Drain current i_D is $\boxed{266.5\ \mu A}$

Step 4 of 5

Output resistance r_0 is $\frac{1}{\lambda i_D}$

Output resistance for i_D equal to $266.6\ \mu A$

$$r_0 = \frac{1}{\lambda i_D}$$
$$= \frac{1}{0.033\left(266.6 \times 10^{-6}\right)}$$
$$= \frac{24 \times 10^3}{0.5}$$
$$= 113.7\ k\Omega$$

Output resistance for i_D equal to $266.6\ \mu A$ is $\boxed{113.7\ k\Omega}$

Step 5 of 5

Drain source voltage v_{DS} increased by $1\ V$, therefore new drain source voltage v_{DS} is $3\ V$

Drain current for v_{DS} equal to $3\ V$

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - v_m)^2(1 + \lambda v_{DS})$$
$$= \frac{1}{2}\left(200\ \mu A/V^2\right)\left(\frac{15\ \mu m}{1.5\ \mu m}\right)(0.5)^2(1 + 0.033(3))$$
$$= 274.75\ \mu A$$

Drain current i_D for $v_{DS} = 3V$ is $\boxed{274.75\ \mu A}$

Change in i_D is ,

$$274.75 - 266.6 = 8.15\ \mu A$$

Change in i_D is $\boxed{8.15\ \mu A}$.

5.035P

The dc bias current is,

$$I_D = \frac{1}{2}k_n' \frac{W}{L}(V_{OV})^2$$

Here,

k_n' is the transconductance parameter

W is the width of the channel

L is the channel length

V_{OV} is the overdrive voltage

The output resistance, r_o is,

$$r_o = \frac{1}{\lambda I_D}$$

Substitute $\frac{1}{2}k_n' \frac{W}{L}(V_{OV})^2$ for I_D .

$$r_o = \frac{1}{\lambda\left[\frac{1}{2}k_n' \frac{W}{L}(V_{OV})^2\right]}$$

As both W and L are quadrupled , there is no change in the ratio, $\frac{W}{L}$.

The overdrive voltage V_{OV} is halved.

Determine the new output resistance.

$$\begin{aligned} r_{o(new)} &= \frac{1}{\lambda\left[\frac{1}{2}k_n' \frac{W}{L}\left(\frac{V_{OV}}{2}\right)^2\right]} \\ &= \frac{4}{\lambda\left[\frac{1}{2}k_n' \frac{W}{L}(V_{OV})^2\right]} \\ &= 4r_o \end{aligned}$$

Thus, when both W and L are quadrupled and overdrive voltage is halved, the output resistance increases by a factor of $\boxed{4}$.

Step 1 of 4

Drain voltage for transistor Q_1 is $2\ \text{V}$

Drain voltage for transistor Q_2 is $2.5\ \text{V}$

Drain current with channel-length modulation effect is considered is ,

$$\frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\left(1+\lambda v_{DS}\right)$$

Drain current for v_{DS} equal to $2\ \text{V}$ is i_{D1}

$$i_{D1}=\frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\left(1+\lambda 2\right) \quad \text{..... (1)}$$

Drain current for v_{DS} equal to $2.5\ \text{V}$ is i_{D2}

$$i_{D1}=\frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\left(1+\lambda 2.5\right) \quad \text{..... (2)}$$

Step 2 of 4

Maximum difference allowed between i_{D1} and i_{D2} is 1%

$$\left(\frac{i_{D2}-i_{D1}}{i_{D1}}\right)\times 100=1 \quad \text{..... (3)}$$

Substitute equation (1) and (2) in (3)

$$\left(\frac{\frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\left(1+\lambda 2.5\right)-\frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\left(1+\lambda 2\right)}{\frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS}-v_{tn}\right)^2\left(1+\lambda 2\right)}\right)\times 100=1$$

$$\left(\frac{\left(1+\lambda 2.5\right)-\left(1+\lambda 2\right)}{\left(1+\lambda 2\right)}\right)\times 100=1$$

$$50\lambda=1+2\lambda$$

$$\lambda=\frac{1}{48}$$

Value of λ is $\frac{1}{48}\ \text{V}^{-1}$

Step 3 of 4

Relation between V_A and λ is,

$$V_A=\frac{1}{\lambda}$$

$$=\frac{1}{\frac{1}{48}}$$

$$=48\ \text{V}$$

Early voltage V_A is $\boxed{48\ \text{V}}$

Step 4 of 4

Value of V_A' is $100\ \text{V}/\mu\text{m}$

Early voltage V_A is $V_A'L$

$$V_A=V_A'L$$

$$48\ \text{V}=\left(20\ \text{V}/\mu\text{m}\right)(L)$$

$$L=2.4\ \mu\text{m}$$

Minimum channel length L is $\boxed{2.5\ \mu\text{m}}$

Step 1 of 9

From column 1, V_A is $20\ \text{V}$, I_D is $0.5\ \text{mA}$

Relation between V_A and λ is,

$$\lambda = \frac{1}{V_A}$$
$$= \frac{1}{20}$$
$$= 0.05\ \text{V}^{-1}$$

Value of λ is $\boxed{0.05\ \text{V}^{-1}}$

Step 2 of 9

Output resistance r_0 is $\frac{1}{\lambda I_D}$

Output resistance for i_D equal to $0.5\ \text{mA}$

$$r_0 = \frac{1}{\lambda I_D}$$
$$= \frac{1}{0.05\left(0.5\times10^{-3}\right)}$$
$$= 40\ \text{k}\Omega$$

Output resistance is $\boxed{40\ \text{k}\Omega}$

Step 3 of 9

From column 2, λ is $0.02\ \text{V}^{-1}$, r_0 is $25\ \text{k}\Omega$

Relation between V_A and λ is,

$$V_A = \frac{1}{\lambda}$$
$$= \frac{1}{0.02}$$
$$= 50\ \text{V}$$

Value of V_A is $\boxed{50\ \text{V}}$

Step 4 of 9

Output resistance r_0 is $\frac{1}{\lambda I_D}$

$$r_0 = \frac{1}{\lambda I_D}$$
$$25\times10^3 = \frac{1}{0.02 I_D}$$
$$I_D = 2\ \text{mA}$$

Drain current I_D is $2\ \text{mA}$

Step 5 of 9

From column 3, I_D is $0.1\ \text{mA}$, r_0 is $100\ \text{k}\Omega$

Output resistance r_0 is $\frac{1}{\lambda I_D}$

$$r_0 = \frac{1}{\lambda I_D}$$
$$100\times10^3 = \frac{1}{\lambda\left(0.1\times10^{-3}\right)}$$
$$\lambda = \frac{1}{10}$$

Value of λ is $\frac{1}{10}$

Step 6 of 9

Relation between V_A and λ is,

$$V_A = \frac{1}{\lambda}$$
$$= \frac{1}{0.1}$$
$$= 10\ \text{V}$$

Value of V_A is $\boxed{10\ \text{V}}$

Step 7 of 9

From column 4, V_A is $100\ \text{V}$, r_0 is $500\ \text{k}\Omega$

Relation between V_A and λ is,

$$\lambda = \frac{1}{V_A}$$
$$= \frac{1}{100}$$
$$= 0.01$$

Value of λ is $\boxed{0.01\ \text{V}^{-1}}$

Step 8 of 9

Output resistance r_0 is $\frac{1}{\lambda I_D}$

$$r_0 = \frac{1}{\lambda I_D}$$
$$100\times10^3 = \frac{1}{(0.01) I_D}$$
$$I_D = 1\ \text{mA}$$

Value of I_D is $1\ \text{mA}$

Step 9 of 9

MOS	1	2	3	4
$\lambda\left(\text{V}^{-1}\right)$	0.05	0.02	0.1	0.01
$V_A(\text{V})$	20	50	10	100
$I_D(\text{mA})$	0.5	2	0.1	1
$r_0(\text{k}\Omega)$	40	25	100	500

Step 1 of 5

Aspect ratio of PMOS transistor $k_p'\left(W/L\right)$ is $100\ \mu A/V^2$

Threshold voltage V_t is $-1\ V$ and hence $|V_t|$ is $1\ V$

Value of λ is $-0.02\ V^{-1}$

Gate connected to ground and the source to $+5\ V$ and hence V_{SG} is $5\ V$

Applied V_D is $4\ V$ and hence V_{DG} is $4\ V$

For $V_{DG} > |V_t|$ transistor operated in triode region

$4 > |-1|$ and hence PMOS transistor operate in triode region.

$$\begin{aligned} V_{SD} &= V_S - V_D \\ &= (5-4) \\ &= 1 \end{aligned}$$

Value of V_{SD} is $1\ V$

Drain current for PMOS transistor operated in triode region is,

$$i_D = k_p'\left(\frac{W}{L}\right)\left(v_{SG} - |v_t|\right)v_{SD} - \frac{1}{2}v_{SD}^2$$

$$\begin{aligned} i_D &= k_p'\left(\frac{W}{L}\right)\left[\left(v_{SG} - |v_t|\right)v_{SD} - \frac{1}{2}v_{SD}^2\right] \\ &= \left(100\ \mu A/V^2\right)\left[(5-1)(1) - \frac{1}{2}(1)^2\right] \\ &= 350\ \mu A \end{aligned}$$

Drain current for V_D equal to $+4\ V$ is $\boxed{350\ \mu A}$

Step 2 of 5

Applied V_D is $2\ V$ and hence V_{DG} is $2\ V$

For $V_{DG} > |V_t|$ transistor operated in triode region

$2 > |-1|$ and hence PMOS transistor operate in triode region.

$$\begin{aligned} V_{SD} &= V_S - V_D \\ &= (5-2) \\ &= 3 \end{aligned}$$

Value of V_{SD} is $3\ V$

Drain current for PMOS transistor operated in triode region is,

$$i_D = k_p'\left(\frac{W}{L}\right)\left(v_{SG} - |v_t|\right)v_{SD} - \frac{1}{2}v_{SD}^2$$

$$\begin{aligned} i_D &= k_p'\left(\frac{W}{L}\right)\left[\left(v_{SG} - |v_t|\right)v_{SD} - \frac{1}{2}v_{SD}^2\right] \\ &= \left(100\ \mu A/V^2\right)\left[(5-1)(3) - \frac{1}{2}(3)^2\right] \\ &= 750\ \mu A \end{aligned}$$

Drain current for V_D equal to $+2\ V$ is $\boxed{750\ \mu A}$

Step 3 of 5

Applied V_D is $1\ V$ and hence V_{DG} is $1\ V$

For $V_{DG} \le |V_t|$ transistor operated in saturation region

$1 \le |-1|$ and hence PMOS transistor operate in saturation region.

$$\begin{aligned} V_{SD} &= V_S - V_D \\ &= (5-1) \\ &= 4 \end{aligned}$$

Value of V_{SD} is $4\ V$

Drain current for PMOS transistor operated in saturation region is,

$$\frac{1}{2}k_p'\left(\frac{W}{L}\right)\left(v_{SG} - v_{tn}\right)^2\left(1 + \lambda v_{SD}\right)$$

$$\begin{aligned} i_D &= \frac{1}{2}k_p'\left(\frac{W}{L}\right)\left(v_{SG} - |v_t|\right)^2\left(1 + |\lambda| v_{SD}\right) \\ &= \frac{1}{2}\left(100\ \mu A/V^2\right)(5-1)^2\left(1 + (0.02)(4)\right) \\ &= 864\ \mu A \end{aligned}$$

Drain current for V_D equal to $1\ V$ is $\boxed{864\ \mu A}$

Step 4 of 5

Applied V_D is $0\ V$ and hence V_{DG} is $0\ V$

For $V_{DG} \le |V_t|$ transistor operated in saturation region

$0 \le |-1|$ and hence PMOS transistor operate in saturation region.

$$\begin{aligned} V_{SD} &= V_S - V_D \\ &= (5-0) \\ &= 5 \end{aligned}$$

Value of V_{SD} is $5\ V$

Drain current for PMOS transistor operated in saturation region is,

$$\frac{1}{2}k_p'\left(\frac{W}{L}\right)\left(v_{SG} - v_{tn}\right)^2\left(1 + \lambda v_{SD}\right)$$

$$\begin{aligned} i_D &= \frac{1}{2}k_p'\left(\frac{W}{L}\right)\left(v_{SG} - |v_t|\right)^2\left(1 + |\lambda| v_{SD}\right) \\ &= \frac{1}{2}\left(100\ \mu A/V^2\right)(5-1)^2\left(1 + (0.02)(5)\right) \\ &= 880\ \mu A \end{aligned}$$

Drain current for V_D equal to $0\ V$ is $\boxed{880\ \mu A}$

Step 5 of 5

Applied V_D is $-5\ V$ and hence V_{DG} is $-5\ V$

For $V_{DG} \le |V_t|$ transistor operated in saturation region

$-5 \le |-1|$ and hence PMOS transistor operate in saturation region.

$$\begin{aligned} V_{SD} &= V_S - V_D \\ &= (5-(-5)) \\ &= 10 \end{aligned}$$

Value of V_{SD} is $10\ V$

Drain current for PMOS transistor operated in saturation region is,

$$\frac{1}{2}k_p'\left(\frac{W}{L}\right)\left(v_{SG} - v_{tn}\right)^2\left(1 + \lambda v_{SD}\right)$$

$$\begin{aligned} i_D &= \frac{1}{2}k_p'\left(\frac{W}{L}\right)\left(v_{SG} - |v_t|\right)^2\left(1 + |\lambda| v_{SD}\right) \\ &= \frac{1}{2}\left(100\ \mu A/V^2\right)(5-1)^2\left(1 + (0.02)(10)\right) \\ &= 960\ \mu A \end{aligned}$$

Drain current for V_D equal to $-5\ V$ is $\boxed{960\ \mu A}$

Step 1 of 3

5.039P

For p- channel transistor threshold voltage V_t is negative by convention.

Voltages v_{SG} and v_{SD} are positive in order to operate P-channel transistor operated in saturation mode, which means voltage applied across the source is greater than the voltage applied across the gate and drain terminals.

Given $|v_{GS}| = 3\ \text{V}$

Gate voltage is less than source voltage and hence v_{GS} is negative and hence v_{GS} is $\boxed{-3\ \text{V}}$

Gate voltage is less than source voltage and hence v_{SG} is positive and hence v_{SG} is $\boxed{3\ \text{V}}$

Step 2 of 3

Given $|v_{DS}| = 4\ \text{V}$

v_{SD} are positive in order to operate P-channel transistor operated in saturation mode and hence value of v_{SD} is $4\ \text{V}$

v_{SD} is positive and hence v_{DS} is negative therefore value of v_{DS} is $-4\ \text{V}$

For p- channel transistor threshold voltage V_t is negative by convention.

Given $|v_t|$ is $0.8\ \text{V}$, and hence value of V_t is $\boxed{-0.8\ \text{V}}$

PMOS transistor by convention λ and V_A are negative.

Given $|v_A|$ is $40\ \text{V}$ and hence value of V_A is $\boxed{-40\ \text{V}}$

Value of λ is $\boxed{-\frac{1}{40}\ \text{V}^{-1}}$

Step 3 of 3

Drain current for PMOS transistor is $3\ \text{mA}$

$$3\times10^{-3} = \frac{1}{2}k_p'\left(\frac{W}{L}\right)\left(v_{SG} - |v_t|\right)^2\left(1+|\lambda| v_{SD}\right)$$

$$3\times10^{-3} = \frac{1}{2}k_p'\left(\frac{W}{L}\right)(3-0.8)^2\left(1+\left(\frac{1}{40}\right)4\right)$$

$$3\times10^{-3} = 2.662k_p'\left(\frac{W}{L}\right)$$

$$k_p'\left(\frac{W}{L}\right) = 1.13\times10^{-3}$$

Value of $k_p'\left(\frac{W}{L}\right)$ is $\boxed{1.13\ \mu\text{A}/\text{V}^2}$

Step 1 of 8

(a)

Determine the voltage, V_{SG} .

$$\begin{aligned} V_{SG} &= V_S - V_G \\ &= +2 - 2 \\ &= 0\ \text{V} \end{aligned}$$

Thus, the voltage V_{SG} is $\boxed{0\ \text{V}}$.

Determine the overdrive voltage, $|V_{OV}|$.

$$\begin{aligned} |V_{OV}| &= V_{SG} - |V_\phi| \\ &= 0 - 1 \\ &= -1\ \text{V} \end{aligned}$$

Thus, the overdrive voltage $|V_{OV}|$ is $\boxed{-1\ \text{V}}$.

Determine the source-to-drain voltage, V_{SD} .

$$\begin{aligned} V_{SD} &= V_S - V_D \\ &= 2 - 0 \\ &= 2\ \text{V} \end{aligned}$$

Thus, the source-to-drain voltage, V_{SD} is $\boxed{2\ \text{V}}$.

As V_{SG} is 0 V, the transistor is operating in the **cutoff region**.

Step 2 of 8

(b)

Determine the voltage, V_{SG} .

$$\begin{aligned} V_{SG} &= V_S - V_G \\ &= +2 - 1 \\ &= 1\ \text{V} \end{aligned}$$

Thus, the voltage V_{SG} is $\boxed{1\ \text{V}}$.

Determine the overdrive voltage, $|V_{OV}|$.

$$\begin{aligned} |V_{OV}| &= V_{SG} - |V_\phi| \\ &= 1 - 1 \\ &= 0\ \text{V} \end{aligned}$$

Thus, the overdrive voltage $|V_{OV}|$ is $\boxed{0\ \text{V}}$.

Determine the source-to-drain voltage, V_{SD} .

$$\begin{aligned} V_{SD} &= V_S - V_D \\ &= 2 - 0 \\ &= 2\ \text{V} \end{aligned}$$

Thus, the source-to-drain voltage, V_{SD} is $\boxed{2\ \text{V}}$.

As $V_{SD} = V_{SG} - V_\phi$, the transistor is operating at the **edge of saturation region.**

Step 3 of 8

(c)

Determine the voltage, V_{SG} .

$$\begin{aligned} V_{SG} &= V_S - V_G \\ &= +2 - 0 \\ &= 2\ \text{V} \end{aligned}$$

Thus, the voltage V_{SG} is $\boxed{2\ \text{V}}$.

Determine the overdrive voltage, $|V_{OV}|$.

$$\begin{aligned} |V_{OV}| &= V_{SG} - |V_\phi| \\ &= 2 - 1 \\ &= 1\ \text{V} \end{aligned}$$

Thus, the overdrive voltage $|V_{OV}|$ is $\boxed{1\ \text{V}}$.

Determine the source-to-drain voltage, V_{SD} .

$$\begin{aligned} V_{SD} &= V_S - V_D \\ &= 2 - 0 \\ &= 2\ \text{V} \end{aligned}$$

Thus, the source-to-drain voltage, V_{SD} is $\boxed{2\ \text{V}}$.

As $V_{SD} \geq V_{SG} - V_\phi$, the transistor is operating in the **saturation region.**

Step 4 of 8

(d)

Determine the voltage, V_{SG} .

$$\begin{aligned} V_{SG} &= V_S - V_G \\ &= +2 - 0 \\ &= 2\ \text{V} \end{aligned}$$

Thus, the voltage V_{SG} is $\boxed{2\ \text{V}}$.

Step 5 of 8

Determine the overdrive voltage, $|V_{OV}|$.

$$\begin{aligned} |V_{OV}| &= V_{SG} - |V_\phi| \\ &= 2 - 1 \\ &= 1\ \text{V} \end{aligned}$$

Thus, the overdrive voltage $|V_{OV}|$ is $\boxed{1\ \text{V}}$.

Determine the source-to-drain voltage, V_{SD} .

$$\begin{aligned} V_{SD} &= V_S - V_D \\ &= 2 - 1 \\ &= 1\ \text{V} \end{aligned}$$

Thus, the source-to-drain voltage, V_{SD} is $\boxed{1\ \text{V}}$.

As $V_{SD} = V_{SG} - V_\phi$, the transistor is operating in the **edge of saturation region.**

Step 6 of 8

(e)

Determine the voltage, V_{SG} .

$$\begin{aligned} V_{SG} &= V_S - V_G \\ &= +2 - 0 \\ &= 2\ \text{V} \end{aligned}$$

Thus, the voltage V_{SG} is $\boxed{2\ \text{V}}$.

Determine the overdrive voltage, $|V_{OV}|$.

$$\begin{aligned} |V_{OV}| &= V_{SG} - |V_\phi| \\ &= 2 - 1 \\ &= 1\ \text{V} \end{aligned}$$

Thus, the overdrive voltage $|V_{OV}|$ is $\boxed{1\ \text{V}}$.

Determine the source-to-drain voltage, V_{SD} .

$$\begin{aligned} V_{SD} &= V_S - V_D \\ &= 2 - 1.5 \\ &= 0.5\ \text{V} \end{aligned}$$

Thus, the source-to-drain voltage, V_{SD} is $\boxed{0.5\ \text{V}}$.

As $V_{SD} < V_{SG} - V_\phi$, the transistor is operating in the **triode region.**

Step 7 of 8

(f)

Determine the voltage, V_{SG} .

$$\begin{aligned} V_{SG} &= V_S - V_G \\ &= +2 - 0 \\ &= 2\ \text{V} \end{aligned}$$

Thus, the voltage V_{SG} is $\boxed{2\ \text{V}}$.

Determine the overdrive voltage, $|V_{OV}|$.

$$\begin{aligned} |V_{OV}| &= V_{SG} - |V_\phi| \\ &= 2 - 1 \\ &= 1\ \text{V} \end{aligned}$$

Thus, the overdrive voltage $|V_{OV}|$ is $\boxed{1\ \text{V}}$.

Determine the source-to-drain voltage, V_{SD} .

$$\begin{aligned} V_{SD} &= V_S - V_D \\ &= 2 - 2 \\ &= 0\ \text{V} \end{aligned}$$

Thus, the source-to-drain voltage, V_{SD} is $\boxed{0\ \text{V}}$.

As $V_{SD} < V_{SG} - V_\phi$, the transistor is operating in the **triode region.**

Step 8 of 8

Fill the entries in the table with the calculated values.

Table 1

	V_S	V_G	V_D	V_{SG}	$\lvert V_{OV}\rvert$	V_{SD}	**Region of operation**
a	+2	+2	0	0	−1	+2	Cutoff
b	+2	+1	0	+1	0	+2	Edge of saturation
c	+2	0	0	+2	+1	+2	Saturation
d	+2	0	+1	2	1	1	Edge of saturation
e	+2	0	+1.5	2	1	0.5	Triode
f	+2	0	+2	2	1	0	Triode

Thus, the table entries are determined.

5.041P

Step 1 of 4

Threshold voltage of PMOS transistor V_{tp} is $-0.5\ \text{V}$

From the figure P5.41 drain voltage v_D is $1\ \text{V}$, source voltage v_S is $3\ \text{V}$.

PMOS transistor operated in three modes of operation

$v_{SG} < |V_{tp}|$ Then PMOS transistor operated in Cutoff region.

$v_{SD} < (v_{SG} - |V_{tp}|)$ Then PMOS transistor operated in Triode region.

$v_{SD} \geq (v_{SG} - |V_{tp}|)$ Then PMOS transistor operated in Saturation region.

Step 2 of 4

For cutoff region

$$v_{SG} < |V_{tp}|$$
$$v_S - v_G < 0.5$$
$$-v_G < 0.5 - 3$$
$$v_G > 2.5\ \text{V}$$

Value of v_G at which PMOS transistor operated in cutoff region is $\boxed{v_G > 2.5\ \text{V}}$

Step 3 of 4

For triode region,

$$v_{SD} < (v_{SG} - |V_{tp}|)$$
$$v_S - v_D < (v_S - v_G - 0.5)$$
$$3 - 1 < 3 - v_G - 0.5$$
$$v_G < 0.5\ \text{V}$$

Value of v_G at which PMOS transistor operated in triode region is $v_G < 0.5\ \text{V}$

Step 4 of 4

For saturation region,

$$v_{SD} \geq (v_{SG} - |V_{tp}|)$$
$$v_S - v_D \geq (v_S - v_G - 0.5)$$
$$3 - 1 \geq 3 - v_G - 0.5$$
$$v_G \geq 0.5\ \text{V}$$

Value of v_G at which PMOS transistor operated in saturation region is $v_G \geq 0.5\ \text{V}$

At $v_G = 0.5\ \text{V}$ PMOS transistor is changed from triode region to saturation region.

At $v_G = 2.5\ \text{V}$ PMOS transistor is changed from saturation region to cutoff region.

Step 1 of 18

The value of λ is 0

Case a:

case	transistor	V_s	V_G	V_D	I_D	$V_{SG} = V_S - V_G$	$V_{SD} = V_S - V_D$
a	1	0	1	2.5	100	-1	-2.5
	1	0	1.5	2.5	400	-1.5	-2.5

The V_{SG} and V_{SD} are negative then the type of transistor is NMOS and hence for case (a) transistor type is NMOS.

Step 2 of 18

Drain current I_D for NMOS when $\lambda = 0$ is,

$$\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - V_t)^2$$

$$100 = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(1 - V_t)^2 \quad \text{..... (1)}$$

$$400 = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(1.5 - V_t)^2 \quad \text{..... (2)}$$

Divide equation (1) with (2)

$$\frac{100}{400} = \frac{\frac{1}{2}k_n'\left(\frac{W}{L}\right)(1 - V_t)^2}{\frac{1}{2}k_n'\left(\frac{W}{L}\right)(1.5 - V_t)^2}$$

$$\frac{1}{4} = \left[\frac{1 - V_t}{1.5 - V_t}\right]^2$$

$$\frac{1}{2} = \frac{1 - V_t}{1.5 - V_t}$$

$$V_t = 0.5$$

Thus, the threshold voltage V_t is $\boxed{0.5\ \text{V}}$

Step 3 of 18

$v_{DS} < (v_{GS} - V_t)$ Then NMOS transistor operated in triode mode otherwise in saturation mode.

For $v_{DS} = 2.5$, $v_{GS} = 1$ and $V_t = 0.5$

$v_{GS} - V_t = 0.5$. $v_{DS} > (v_{GS} - V_t)$ and hence for $v_{GS} = 1$. $v_{DS} = 2.5$ NMDS transistor operated in saturation region.

For $v_{DS} = 2.5$, $v_{GS} = 1.5$ and $V_t = 0.5$

$v_{GS} - V_t = 1$. $v_{DS} > (v_{GS} - V_t)$ and hence for $v_{GS} = 1.5$, $v_{DS} = 2.5$ NMOS transistor operated in saturation region.

Step 4 of 18

For drain current $100\ \mu\text{A}$:

$$100 \times 10^{-6} = \frac{1}{2}\mu C_{ox}\left(\frac{W}{L}\right)(V_{GS} - V_t)^2$$

$$100 \times 10^{-6} = \frac{1}{2}\mu C_{ox}\left(\frac{W}{L}\right)(1 - 0.5)^2$$

$$\frac{2 \times 100 \times 10^{-6}}{(1 - 0.5)^2} = \mu C_{ox}\left(\frac{W}{L}\right)$$

$$\mu C_{ox}\left(\frac{W}{L}\right) = 800\ \frac{\mu\text{A}}{\text{V}^2}$$

Thus, the value of $\mu C_{ox}\left(\frac{W}{L}\right)$ is $\boxed{800\ \frac{\mu\text{A}}{\text{V}^2}}$

Step 5 of 18

Case b:

case	transistor	V_s	V_G	V_D	I_D	$V_{SG} = V_S - V_G$	$V_{SD} = V_S - V_D$
b	2	5	3	-4.5	50	2	9.5
	2	5	2	-0.5	450	3	5.5

The V_{SG} and V_{SD} are positive then the type of transistor is PMOS and hence for case (b) transistor type is PMDS.

Step 6 of 18

The drain current I_D for PMOS when $\lambda = 0$ is,

$$\frac{1}{2}k_p'\left(\frac{W}{L}\right)(v_{SG} - |V_t|)^2$$

$$50 = \frac{1}{2}k_p'\left(\frac{W}{L}\right)(2 - |V_t|)^2 \quad \text{..... (3)}$$

$$450 = \frac{1}{2}k_p'\left(\frac{W}{L}\right)(3 - |V_t|)^2 \quad \text{..... (4)}$$

Step 7 of 18

Divide equation (3) with (4) as follows:

$$\frac{50}{450} = \frac{\frac{1}{2}k_p'\left(\frac{W}{L}\right)(2 - |V_t|)^2}{\frac{1}{2}k_p'\left(\frac{W}{L}\right)(3 - |V_t|)^2}$$

$$\frac{1}{9} = \left[\frac{2 - |V_t|}{3 - |V_t|}\right]^2$$

$$\frac{1}{3} = \frac{2 - |V_t|}{3 - |V_t|}$$

$$|V_t| = 1.5$$

Threshold voltage V_t is negative for PMOS transistor

Thus, the threshold voltage V_t is $\boxed{-1.5\ \text{V}}$

Step 8 of 18

If $v_{SD} < \left|(v_{SG} - |V_t|)\right|$ then the PMOS transistor operated in triode mode otherwise in saturation mode.

For $v_{SD} = 9.5$, $v_{SG} = 2$ and $|V_t| = 1.5$

$v_{SG} - |V_t| = 0.5$. $v_{SD} > \left|(v_{SG} - |V_t|)\right|$ and hence for $v_{SG} = 2$. $v_{SD} = 2.5$ PMOS transistor operated in saturation region.

For $v_{SD} = 5.5$, $v_{SG} = 3$ and $V_t = 0.5$

$v_{SG} - |V_t| = 2.5$. $v_{SD} > \left|(v_{SG} - |V_t|)\right|$ and hence for $v_{SG} = 3$. $v_{SD} = 5.5$ PMOS transistor operated in saturation region

Step 9 of 18

For drain current $50\ \mu\text{A}$:

$$50 \times 10^{-6} = \frac{1}{2}\mu C_{ox}\left(\frac{W}{L}\right)(V_{SG} - |V_t|)^2$$

$$50 \times 10^{-6} = \frac{1}{2}\mu C_{ox}\left(\frac{W}{L}\right)(2 - |1.5|)^2$$

$$\frac{2 \times 50 \times 10^{-6}}{(2 - 1.5)^2} = \mu C_{ox}\left(\frac{W}{L}\right)$$

$$\mu C_{ox}\left(\frac{W}{L}\right) = 400\ \frac{\mu\text{A}}{\text{V}^2}$$

Thus, the value of $\mu C_{ox}\left(\frac{W}{L}\right)$ is $\boxed{400\ \frac{\mu\text{A}}{\text{V}^2}}$

Step 10 of 18

Case c.

case	transistor	V_s	V_G	V_D	I_D	$V_{SG} = V_S - V_G$	$V_{SD} = V_S - V_D$
c	3	5	3	4	200	2	1
	3	5	2	0	800	3	5

The V_{SG} and V_{SD} are positive then the type of transistor is PMOS and hence for case (c) transistor type is PMOS.

Step 11 of 18

Drain current I_D for PMOS when $\lambda = 0$ is,

$$\frac{1}{2}k_p'\left(\frac{W}{L}\right)(v_{SG} - |V_t|)^2$$

$$200 = \frac{1}{2}k_p'\left(\frac{W}{L}\right)(2 - |V_t|)^2 \quad \text{..... (5)}$$

$$800 = \frac{1}{2}k_p'\left(\frac{W}{L}\right)(3 - |V_t|)^2 \quad \text{..... (6)}$$

Divide equation (3) with (4) as follows:

$$\frac{200}{800} = \frac{\frac{1}{2}k_p'\left(\frac{W}{L}\right)(2 - |V_t|)^2}{\frac{1}{2}k_p'\left(\frac{W}{L}\right)(3 - |V_t|)^2}$$

$$\frac{1}{4} = \left[\frac{2 - |V_t|}{3 - |V_t|}\right]^2$$

$$\frac{1}{2} = \frac{2 - |V_t|}{3 - |V_t|}$$

$$|V_t| = 1$$

Threshold voltage V_t is negative for PMOS transistor

Thus, the threshold voltage V_t is $\boxed{-1\ \text{V}}$

Step 12 of 18

$v_{SD} < \left|(v_{SG} - |V_t|)\right|$ Then PMOS transistor operated in triode mode otherwise in saturation mode.

For $v_{SD} = 1$, $v_{SG} = 2$ and $|V_t| = 1$

$v_{SG} - |V_t| = 1$. $v_{SD} = \left|(v_{SG} - |V_t|)\right|$ and hence for $v_{SG} = 2$. $v_{SD} = 1$ PMOS transistor operated in saturation region.

For $v_{SD} = 5$, $v_{SG} = 3$ and $V_t = 1$

$v_{SG} - |V_t| = 2$. $v_{SD} > \left|(v_{SG} - |V_t|)\right|$ and hence for $v_{SG} = 3$. $v_{SD} = 5$ PMOS transistor operated in saturation region.

Step 13 of 18

For drain current $200\ \mu\text{A}$:

$$200 \times 10^{-6} = \frac{1}{2}\mu C_{ox}\left(\frac{W}{L}\right)(V_{SG} - |V_t|)^2$$

$$200 \times 10^{-6} = \frac{1}{2}\mu C_{ox}\left(\frac{W}{L}\right)(2 - |1|)^2$$

$$\frac{2 \times 200 \times 10^{-6}}{(2 - 1)^2} = \mu C_{ox}\left(\frac{W}{L}\right)$$

$$\mu C_{ox}\left(\frac{W}{L}\right) = 400\ \frac{\mu\text{A}}{\text{V}^2}$$

Thus, the value of $\mu C_{ox}\left(\frac{W}{L}\right)$ is $\boxed{400\ \frac{\mu\text{A}}{\text{V}^2}}$

Step 14 of 18

Case d:

case	transistor	V_s	V_G	V_D	I_D	$V_{SG} = V_S - V_G$	$V_{SD} = V_S - V_D$
d	1	-2	0	0	72	-2	-2
	1	-4	0	-3	270	-4	-1

Thus, V_{SG} and V_{SD} are negative then the type of transistor is NMOS and hence for case (d) transistor type is NMOS.

Step 15 of 18

Drain current I_D for NMOS when $\lambda = 0$ is,

$$\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{DS} - V_t)^2$$

$$72 = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(2 - V_t)^2 \quad \text{..... (7)}$$

$$270 = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(4 - V_t)^2 \quad \text{..... (8)}$$

Divide equation (7) with (8) as follows:

$$\frac{72}{270} = \frac{\frac{1}{2}k_n'\left(\frac{W}{L}\right)(2 - V_t)^2}{\frac{1}{2}k_n'\left(\frac{W}{L}\right)(4 - V_t)^2}$$

$$\frac{4}{15} = \left[\frac{2 - V_t}{4 - V_t}\right]^2$$

$$0.52 = \frac{2 - V_t}{4 - V_t}$$

$$V_t = -0.17$$

Thus, the threshold voltage V_t is $-0.17\ \text{V}$

Step 16 of 18

$v_{DS} < (v_{GS} - V_t)$ Then NMOS transistor operated in triode mode otherwise in saturation mode.

For $v_{DS} = 2$. $v_{GS} = 2$ and $V_t = -0.17$

$v_{GS} - V_t = 2.17$. $v_{DS} < (v_{DS} - V_t)$ and hence for $v_{GS} = 2$. $v_{DS} = 2$ NMOS transistor operated in triode region.

For $v_{DS} = 1$. $v_{GS} = 4$ and $V_t = -0.17$

$v_{GS} - V_t = 4.17$. $v_{DS} < (v_{GS} - V_t)$ and hence for $v_{GS} = 4$. $v_{DS} = 1$ NMOS transistor operated in triode region

Step 17 of 18

For drain current $72\ \mu\text{A}$:

Drain current i_D for transistor in triode region is $\mu C_{ox}\left(\frac{W}{L}\right)\left(v_{GS} - V_t - \frac{1}{2}v_{DS}\right)v_{DS}$

$$72 \times 10^{-6} = \mu C_{ox}\left(\frac{W}{L}\right)\left(V_{GS} - V_t - \frac{1}{2}v_{DS}\right)v_{DS}$$

$$72 \times 10^{-6} = \mu C_{ox}\left(\frac{W}{L}\right)\left(4 + 0.17 - \frac{1}{2}(2)\right)2$$

$$\frac{72 \times 10^{-6}}{6.34} = \mu C_{ox}\left(\frac{W}{L}\right)$$

$$\mu C_{ox}\left(\frac{W}{L}\right) = 11.36\ \frac{\mu\text{A}}{\text{V}^2}$$

Thus, the value of $\mu C_{ox}\left(\frac{W}{L}\right)$ is $\boxed{11.36\ \frac{\mu\text{A}}{\text{V}^2}}$

Step 18 of 18

The required complete table is shown in Table 1:

Table 1

case	transistor	V_s	V_G	V_D	I_D	type	mode	$\mu C_{ox}\frac{W}{L}$	V_t
a	1	0	1	2.5	100	NMOS	saturation	800	0.5
	1	0	1.5	2.5	400	NMOS	saturation	800	0.5
b	2	5	3	-4.5	50	PMOS	saturation	400	1.5
	2	5	2	-0.5	450	PMOS	saturation	400	1.5
c	3	5	3	4	200	PMOS	Saturation	400	1
	3	5	2	0	800	PMOS	saturation	400	1
d	4	-2	0	0	72	NMOS	triode	11.36	-0.17
	4	-4	3	-3	270	NMOS	triode	19.62	-0.17

5.043P

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Step 1 of 3

5.044P

Refer to Figure P5.44 in the text book.

The drain current for N-channel MOSFET transistor is $0.1\ \text{mA}$.

The drain voltage is $0.3\ \text{V}$

The gate voltage V_G is $0\ \text{V}$

$$R_D = \frac{V_{DD} - V_D}{I_D}$$
$$= \frac{1 - 0.3}{0.1 \times 10^{-3}}$$
$$= \frac{0.7 \times 10^3}{0.1}$$
$$= 7\ \text{k}\Omega$$

Thus, the drain resistance R_D is $\boxed{7\ \text{k}\Omega}$

Step 2 of 3

The threshold voltage V_t is $0.5\ \text{V}$

The Length of the channel L is $0.4\ \mu\text{m}$

The Width of the channel W is $5\ \mu\text{m}$

To determine the value required for R_S, we need to know the voltage at the source, which can be easily found if we know V_{GS}. This in turn can be determined from V_{OV}. Toward that end, we note that since $V_D = 0.3\ \text{V}$ is greater than V_G, the N-channel MOSFET transistor is operating in the saturation region, and we can use the saturation- region expression of i_D to determine the required value of V_{OV} .

$$I_D = \frac{1}{2}\mu_n C_{ox} \frac{W}{L} V_{OV}^2$$
$$0.1 \times 10^{-3} = \frac{1}{2}\left(400\ \mu\text{A}/\text{V}^2\right)\left(\frac{5\ \mu\text{m}}{0.4\ \mu\text{m}}\right)\left(V_{OV}^2\right)$$
$$0.04 = \left(V_{OV}^2\right)$$
$$V_{OV} = 0.2\ \text{V}$$

The over drive voltage V_{OV} is $0.2\ \text{V}$

Step 3 of 3

The relation between v_{GS}, V_t and V_{OV} is,

$$v_{GS} = V_t + V_{OV}$$
$$= 0.5 + 0.2$$
$$= 0.7\ \text{V}$$

The gate source voltage v_{GS} is $0.7\ \text{V}$ and hence source voltage is $-0.7\ \text{V}$

$$R_S = \frac{V_S - V_{SS}}{I_D}$$
$$= \frac{-0.7 - (-1)}{0.1 \times 10^{-3}}$$
$$= 3\ \text{k}\Omega$$

Thus, the value of R_S is $\boxed{3\ \text{k}\Omega}$.

Step 1 of 4

5.045P

Refer to Figure P5.44 in the text book.

The drain voltage is $0.2\ \text{V}$

The gate voltage V_G is $0\ \text{V}$

The threshold voltage V_t is $0.4\ \text{V}$

The source voltage is $-0.6\ \text{V}$

$v_{DS} < (v_{GS} - V_t)$ Then NMOS transistor operated in triode mode otherwise in saturation mode.

$v_{DS} = 0.8\ \text{V}$, $v_{GS} = 0.6\ \text{V}$ and $V_t = 0.4$

$v_{GS} - V_t = 0.2$, $v_{DS} > (v_{GS} - V_t)$ and hence for $v_{GS} = 0.6$, $v_{DS} = 0.8$ NMOS transistor operated in saturation region.

Step 2 of 4

The drain current i_D for NMOS in saturation is,

$$\begin{aligned} i_D &= \frac{1}{2} k_n (v_{GS} - V_t)^2 \\ &= \frac{1}{2}\left(4\ \text{mA/V}^2\right)(0.6 - 0.4)^2 \\ &= \frac{1}{2}(4)(0.04) \\ &= 0.08\ \text{mA} \end{aligned}$$

Thus, the drain current i_D is $\boxed{0.08\ \text{mA}}$

Step 3 of 4

From the figure P5.44 V_{SS} is $1\ \text{V}$, V_{DD} is $-1\ \text{V}$.

$$\begin{aligned} R_S &= \frac{V_S - V_{SS}}{I_D} \\ &= \frac{-0.6 - (-1)}{0.08 \times 10^{-3}} \\ &= 5\ \text{k}\Omega \end{aligned}$$

Thus, the value of R_S is $\boxed{5\ \text{k}\Omega}$

Determine the value of R_D :

$$\begin{aligned} R_D &= \frac{V_{DD} - V_D}{I_D} \\ &= \frac{1 - 0.2}{0.08 \times 10^{-3}} \\ &= \frac{0.8 \times 10^3}{0.08} \\ &= 10\ \text{k}\Omega \end{aligned}$$

Thus, the value of R_D is $\boxed{10\ \text{k}\Omega}$

Step 4 of 4

The drain current i_D remains unchanged in saturation mode.

The largest value of R_D is possible for minimum drain voltage.

In saturation mode, minimum drain voltage is possible when $v_{DS} = (v_{GS} - V_t)$

$v_{DS} = (v_{GS} - V_t)$ is possible when $v_D = 0.2\ \text{V}$

$$\begin{aligned} R_D &= \frac{V_{DD} - V_D}{I_D} \\ &= \frac{1 - 0.2}{0.08 \times 10^{-3}} \\ &= \frac{0.8 \times 10^3}{0.08} \\ &= 10\ \text{k}\Omega \end{aligned}$$

Thus, the largest value of R_D for which I_D remains unchanged is $\boxed{10\ \text{k}\Omega}$.

Step 1 of 4

5.046P

Refer to Figure E5.10 in the text book.

From the figure E5.10 for Q_1 transistor drain voltage v_D is equal to the gate voltage of Q_1 transistor. Hence transistor Q_1 will operate in saturation mode.

Drain current i_D for transistor Q_1 is $50\ \mu A$

Source is grounded.

Drain current i_D for NMOS in saturation is,

$$\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_t)^2$$

$$50\times10^{-6}=\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_t)^2$$

$$50\times10^{-6}=\frac{1}{2}\left(0.4\times10^{-3}\ \text{A/V}^2\right)\left(\frac{1.44\ \mu\text{m}}{0.36\ \mu\text{m}}\right)(v_G-0-0.5)^2$$

$$62.5\times10^{-3}=(v_G-0.5)^2$$

$$0.75=v_G$$

Thus, the gate voltage of Q_1 transistor is $0.75\ \text{V}$

Step 2 of 4

The v_D is equal to the gate voltage of Q_1 transistor and hence v_D is $0.75\ \text{V}$

From the figure E5.10 V_{DD} is $1.8\ \text{V}$

$$R=\frac{V_{DD}-V_D}{I_D}$$
$$=\frac{1.8-0.75}{50\times10^{-6}}$$
$$=21\ \text{k}\Omega$$

Thus, the value of resistance R is $\boxed{21\ \text{k}\Omega}$.

Step 3 of 4

(b)

The gate voltage of Q_1 transistor is equal to Gate voltage of Q_2 transistor and hence gate voltage of Q_2 transistor is $0.75\ \text{V}$.

Q_2 Transistor is at the edge of saturation and hence $v_{DS}=(v_{GS}-V_t)$

$$v_{DS}=(v_{GS}-V_t)$$
$$v_D-0=(0.75-0-0.5)$$
$$v_D=0.25\ \text{V}$$

The drain current of Q_2 Transistor is $0.5\ \text{mA}$

The drain current i_D for NMOS in saturation is,

$$\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_t)^2$$

$$0.5\times10^{-3}=\frac{1}{2}k_n'\left(\frac{W_2}{L}\right)(v_{GS}-V_t)^2$$

$$0.5\times10^{-3}=\frac{1}{2}\left(0.4\times10^{-3}\ \text{A/V}^2\right)\left(\frac{W_2}{0.36\ \mu\text{m}}\right)(0.75-0.5)^2$$

$$14.4\ \mu\text{m}=W_2$$

Thus, the width of the channel of the transistor W_2 is $\boxed{14.4\ \mu\text{m}}$

Step 4 of 4

From the figure E5.10 V_{DD} is $1.8\ \text{V}$

$$R_2=\frac{V_{DD}-V_D}{I_D}$$
$$=\frac{1.8-0.25}{0.5\times10^{-3}}$$
$$=3.1\ \text{k}\Omega$$

Thus, the value of resistance R_2 is $\boxed{3.1\ \text{k}\Omega}$.

Step 1 of 2

5.047P

Refer to Figure P5.47 in the text book.

The threshold voltage V_t is $0.4\ \text{V}$

The value of k_n' is $0.4\ \text{mA/V}^2$

The source voltage is zero.

The power voltage V_{DD} is $1.3\ \text{V}$

The gate voltage is $1.3\ \text{V}$

The drain current I_D for NMOS when $\lambda = 0$ is,

$$\begin{aligned} i_D &= \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_t)^2 \\ &= \frac{1}{2}\left(0.4\ \text{mA/V}^2\right)\left(\frac{W}{L}\right)(1.3-0-0.4)^2 \\ &= \frac{1}{2}\left(0.4\times10^{-3}\right)\left(\frac{W}{L}\right)(0.81) \\ &= 0.162\times10^{-3}\left(\frac{W}{L}\right)\text{A} \end{aligned}$$

Thus, the value of drain current is $0.162\times10^{-3}\left(\frac{W}{L}\right)\text{A}$.

Step 2 of 2

At the edge of saturation $v_{DS} = (v_{GS} - V_t)$

$$\begin{aligned} v_{DS} &= (v_{GS}-V_t) \\ v_D - 0 &= 1.3-0-0.4 \\ v_D &= 0.9\ \text{V} \end{aligned}$$

The drain voltage V_D is $0.9\ \text{V}$

$$R_D = \frac{V_{DD}-V_D}{I_D}$$

$$R_D = \frac{1.3-0.9}{0.162\times10^{-3}\times\left(\frac{W}{L}\right)}$$

$$R_D\left(\frac{W}{L}\right) = 2469.1\ \Omega$$

$$R_D\left(\frac{W}{L}\right) \simeq 2.5\ \text{k}\Omega$$

Thus, the transistor operated at the edge of saturation is obtained when $\boxed{R_D\left(\frac{W}{L}\right) \simeq 2.5\ \text{k}\Omega}$.

5.048P

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5.049P

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Step 1 of 3

Refer to Figure P5.50 in the text book.

The threshold voltage V_t is $0.5\ \text{V}$

The value of $\mu_n C_{ox}$ is $250\ \mu\text{A}/\text{V}^2$

The current flowing through the resistor R is $0.5\ \text{mA}$

From the Figure $\mathbf{P5.50}$,

$$R = \frac{2.5-1.8}{0.5\times10^{-3}}$$
$$= 1.4\ \text{k}\Omega$$

Thus, the value of resistance R is $\boxed{1.4\ \text{k}\Omega}$

Step 2 of 3

The drain voltage of NMOS transistor Q_2 is $1.8\ \text{V}$

The drain and gate voltage of transistor Q_2 are equal and hence transistor Q_2 will operate in saturation region.

The source voltage of Q_2 is $1\ \text{V}$

The drain current i_D for NMOS transistor Q_2 in saturation is,

$$\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_t)^2$$

i_D for NMOS transistor Q_2 is $0.5\ \text{mA}$

$$i_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W_2}{L_2}\right)(v_{GS}-V_t)^2$$

$$0.5\times10^{-3}\ \text{A} = \frac{1}{2}\left(250\times10^{-6}\ \text{A}/\text{V}^2\right)\left(\frac{W_2}{0.25\ \mu\text{m}}\right)\left((1.8-1)-0.5\right)^2$$

$$W_2 = 11.1\ \mu\text{m}$$

Thus, the width of the channel NMOS transistor Q_2 is $\boxed{11.1\ \mu\text{m}}$

Step 3 of 3

The drain voltage of NMOS transistor Q_1 is $1\ \text{V}$

The drain and gate voltage of transistor Q_1 are equal and hence transistor Q_1 will operate in saturation region.

The source voltage of Q_1 is $0\ \text{V}$

The drain current i_D for NMOS transistor Q_1 in saturation is,

$$\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_t)^2$$

i_D For NMOS transistor Q_1 is $0.5\ \text{mA}$, since Q_2 and Q_1 are in series and are operated in saturation mode.

$$i_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W_1}{L_1}\right)(v_{GS}-V_t)^2$$

$$0.5\times10^{-3}\ \text{A} = \frac{1}{2}\left(250\times10^{-6}\ \text{A}/\text{V}^2\right)\left(\frac{W_1}{0.25\ \mu\text{m}}\right)\left((1-0)-0.5\right)^2$$

$$W_1 = 4\ \mu\text{m}$$

Thus, the width of the channel NMOS transistor Q_2 is $\boxed{4\ \mu\text{m}}$.

Step 1 of 3

5.051P

Refer to Figure P5.51 in the textbook.

The drain voltage of NMOS transistor Q_3 is $2.5\ \text{V}$.

The drain and gate voltage of transistor Q_3 are equal and hence transistor Q_3 will operate in saturation region.

The source voltage of Q_3 is $1.5\ \text{V}$

Write the expression for drain current i_D for NMOS transistor Q_3 in saturation.

$$i_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W_3}{L_3}\right)(v_{GS} - V_t)^2$$

Substitute $0.5\ \text{V}$ for V_t, $90\ \mu\text{A}/\text{V}^2$ for $\mu_n C_{ox}$, $0.5\ \mu\text{m}$ for L_3 and $90\ \mu\text{A}$ for i_D in the equation.

$$90\times10^{-6}\ \text{A} = \frac{1}{2}\left(90\times10^{-6}\ \text{A}/\text{V}^2\right)\left(\frac{W_3}{0.5\ \mu\text{m}}\right)\left((2.5-1.5)-0.5\right)^2$$

$$W_3 = 4\ \mu\text{m}$$

Thus, the width of the channel of NMOS transistor Q_3 is $\boxed{4\ \mu\text{m}}$.

Step 2 of 3

The drain voltage of NMOS transistor Q_2 is $1.5\ \text{V}$

Drain and gate voltage of transistor Q_2 are equal and hence transistor Q_2 will operate in saturation region.

The source voltage of Q_2 is $0.8\ \text{V}$.

Write the expression for drain current i_D for NMOS transistor Q_2 in saturation.

$$i_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W_2}{L_2}\right)(v_{GS} - V_t)^2$$

Substitute $0.5\ \text{V}$ for V_t, $90\ \mu\text{A}/\text{V}^2$ for $\mu_n C_{ox}$, $0.5\ \mu\text{m}$ for L_2 and $90\ \mu\text{A}$ for i_D in the equation.

$$90\times10^{-3}\ \text{A} = \frac{1}{2}\left(90\times10^{-6}\ \text{A}/\text{V}^2\right)\left(\frac{W_1}{0.5\ \mu\text{m}}\right)\left((1.5-0.8)-0.5\right)^2$$

$$W_2 = 25\ \mu\text{m}$$

Thus, the width of the channel NMOS transistor Q_2 is $\boxed{25\ \mu\text{m}}$

Step 3 of 3

The drain voltage of NMOS transistor Q_1 is $0.8\ \text{V}$.

The drain and gate voltage of transistor Q_1 are equal and hence transistor Q_1 will operate in saturation region.

The source voltage of Q_1 is $0\ \text{V}$.

Write the expression for drain current i_D for NMOS transistor Q_1 in saturation.

$$i_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W_1}{L_1}\right)(v_{GS} - V_t)^2$$

Substitute $0.5\ \text{V}$ for V_t, $90\ \mu\text{A}/\text{V}^2$ for $\mu_n C_{ox}$, $0.5\ \mu\text{m}$ for L_1 and $90\ \mu\text{A}$ for i_D in the equation.

$$90\times10^{-6}\ \text{A} = \frac{1}{2}\left(90\times10^{-6}\ \text{A}/\text{V}^2\right)\left(\frac{W_1}{0.5\ \mu\text{m}}\right)\left((0.8-0)-0.5\right)^2$$

$$W_1 = 11.1\ \mu\text{m}$$

Thus, the width of the channel W_1 for NMOS transistor Q_1 is $\boxed{11.1\ \mu\text{m}}$.

5.052P

We don't have the solution to this problem yet.

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Step 1 of 5

Refer to Figure 5.24 (a) in the text book.

The threshold voltage of PMOS transistor V_t is $-1.5\ \text{V}$

The value of V_{DD} is $10\ \text{V}$

The drain current for PMOS transistor for $\lambda = 0$ is,

$$\frac{1}{2}k_p'\left(\frac{W}{L}\right)\left(v_{SG}-|v_t|\right)^2$$

The drain current I_D for PMOS transistor is $0.5\ \text{mA}$

$$0.5\times10^{-3} = \frac{1}{2}k_p'\left(\frac{W}{L}\right)\left(v_{SG}-|v_t|\right)^2$$

$$0.5\times10^{-3} = \frac{1}{2}\left(1.5\times10^{-3}\ \frac{\text{A}}{\text{V}^2}\right)\left(v_{SG}-|-1.5|\right)^2$$

$$0.82 = \left(v_{SG}-|-1.5|\right)$$

$$v_{SG} = 2.32\ \text{V}$$

Thus, the source and gate voltage v_{SG} is $2.32\ \text{V}$

Step 2 of 5

The gate voltage v_G is $6\ \text{V}$

$$v_{SG} = v_S - v_G$$

$$2.32 = v_S - 6$$

$$8.32 = v_S$$

The source voltage v_S is $8.32\ \text{V}$

The drain voltage is $5\ \text{V}$

$$R_D = \frac{V_D}{I_D}$$

$$= \frac{5}{0.5\times10^{-3}}$$

$$= 10\ \text{k}\Omega$$

Thus, the value of resistance R_D is $\boxed{10\ \text{k}\Omega}$

Step 3 of 5

The value of resistance R_S is $\frac{v_{DD}-V_S}{I_D}$

$$R_S = \frac{v_{DD}-V_S}{I_D}$$

$$= \frac{10-5}{0.5\times10^{-3}}$$

$$= 10\ \text{k}\Omega$$

Thus, the value of source resistance R_s is $\boxed{10\ \text{k}\Omega}$

Step 4 of 5

The current flowing through divider circuit I_{div} is $1\ \mu\text{A}$

Since the gate current is zero,

$$R_{G1} = \frac{V_{DD}-V_G}{I_{div}}$$

$$R_{G1} = \frac{V_{DD}-V_G}{I_{div}}$$

$$= \frac{10-6}{1\times10^{-6}}$$

$$= 4\ \text{M}\Omega$$

Thus, the value of resistance R_{G1} is $\boxed{4\ \text{M}\Omega}$

Step 5 of 5

The required circuit diagram is shown in Figure 1.

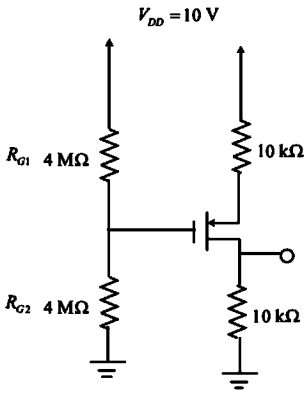


Figure 1

Step 1 of 3

Refer to Figure P5.54 in the text book.

The drain source resistance r_{DS} is $50\ \Omega$

The threshold voltage V_t is $0.4\ \text{V}$

The gate voltage v_I is $1.3\ \text{V}$

The source is grounded and hence v_{GS} is $1.3\ \text{V}$

$$r_{DS} = \frac{1}{k_n'\left(\frac{W}{L}\right)(v_{GS} - V_t)}$$

$$50\ \Omega = \frac{1}{\left(500\times10^{-6}\ \mu\text{A}/\text{V}^2\right)\left(\frac{W}{L}\right)(1.3-0.4)}$$

$$\left(\frac{W}{L}\right) = \frac{1}{\left(500\times10^{-6}\ \mu\text{A}/\text{V}^2\right)50(1.3-0.4)}$$

$$\left(\frac{W}{L}\right) = 44.44$$

Thus, the value of $\left(\frac{W}{L}\right)$ is $\boxed{44.44}$

Step 2 of 3

The drain voltage v_0 is $50\ \text{mV}$

$v_{DS} < (v_{GS} - V_t)$ And hence NMOS transistor operated in triode region.

The drain current in triode region,

$$i_D = k_n'\left(\frac{W}{L}\right)\left(v_{GS} - V_t - \frac{1}{2}v_{DS}\right)v_{DS}$$

$$= \left(500\times10^{-6}\ \frac{\text{A}}{\text{V}^2}\right)(44.44)\left(1.3-0.4-\frac{1}{2}\left(50\times10^{-3}\right)\right)\left(50\times10^{-3}\right)$$

$$= 0.97\ \text{mA}$$

Thus, the drain current I_D is $0.97\ \text{mA}$

Step 3 of 3

From the Figure **P5.54** Value of resistance R is,

$$R = \frac{(V_{DD} - v_0)}{i_D}$$

$$= \frac{1.3-\left(50\times10^{-3}\right)}{0.97\times10^{-3}}$$

$$= 1288.6\ \text{k}\Omega$$

Thus, the value of resistance R is $\boxed{1288.6\ \text{k}\Omega}$.

5.055P

We don't have the solution to this problem yet.

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Step 1 of 6

5.056P

Refer to Figure P5.56 (a) in the text book.

The threshold voltage V_t is $0.8\ \text{V}$

The value of $k_n'\left(W/L\right)$ is $0.5\ \text{mA/V}^2$

The gate voltage is zero.

Drain voltage is $5\ \text{V}$

The source voltage is V_1

The current flowing through gate is zero and hence drain current is $10\ \mu\text{A}$

The drain current I_D for NMOS when $\lambda = 0$ is,

$$\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_t)^2$$

$$10\times10^{-6} = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_t)^2$$

$$10\times10^{-6} = \frac{1}{2}\left(0.5\ \text{mA/V}^2\right)(0-V_1-0.8)^2$$

$$0.2 = (V_1+0.8)$$

$$V_1 = -0.6\ \text{V}$$

Thus, the voltage V_1 for the circuit shown in the figure P5.56 (a) is $\boxed{-0.6\ \text{V}}$

Step 2 of 6

Refer to Figure P5.56 (b) in the text book.

The drain voltage is $5\ \text{V}$

The source voltage is V_2

The gate voltage is zero.

The current flowing through gate is zero and hence drain current is $100\ \mu\text{A}$

The drain current I_D for NMOS when $\lambda = 0$ is,

$$\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_t)^2$$

$$100\times10^{-6} = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_t)^2$$

$$100\times10^{-6} = \frac{1}{2}\left(0.5\ \text{mA/V}^2\right)(0-V_2-0.8)^2$$

$$0.632 = (V_2+0.8)$$

$$V_2 = -0.17\ \text{V}$$

Thus, the voltage V_2 for the circuit shown in the figure P5.56 (b) is $\boxed{-0.17\ \text{V}}$

Step 3 of 6

Refer to Figure P5.56 (c) in the text book.

The drain voltage is $5\ \text{V}$

The source voltage is V_3

The gate voltage is zero.

The drain current is $1\ \text{mA}$

The drain current I_D for NMOS when $\lambda = 0$ is,

$$\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_t)^2$$

$$1\times10^{-3} = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_t)^2$$

$$1\times10^{-3} = \frac{1}{2}\left(0.5\ \text{mA/V}^2\right)(0-V_3-0.8)^2$$

$$2 = (V_3+0.8)$$

$$V_3 = 1.4\ \text{V}$$

Thus, the voltage V_3 for the circuit shown in the figure P5.56 (c) is $\boxed{1.4\ \text{V}}$

Step 4 of 6

Refer to Figure P5.56 (d) in the text book.

The drain voltage is V_4 V

The source voltage is zero

The gate voltage is V_4 V

The drain current is $10\ \mu\text{A}$

The drain current I_D for NMOS when $\lambda = 0$ is,

$$\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_t)^2$$

$$10\times10^{-6} = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_t)^2$$

$$10\times10^{-6} = \frac{1}{2}\left(0.5\ \text{mA/V}^2\right)(V_4-0.8)^2$$

$$0.2 = (V_4-0.8)$$

$$V_4 = 1\ \text{V}$$

Thus, the voltage V_4 for the circuit shown in the figure P5.56 (d) is $\boxed{1\ \text{V}}$

Step 5 of 6

Refer to Figure P5.56 (e) in the text book.

The drain voltage is V_5 V

The source voltage is zero

The gate voltage is V_5 V

The drain current is $1\ \text{mA}$

The drain current I_D for NMOS when $\lambda = 0$ is,

$$\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_t)^2$$

$$1\times10^{-3} = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_t)^2$$

$$1\times10^{-3} = \frac{1}{2}\left(0.5\ \text{mA/V}^2\right)(V_5-0.8)^2$$

$$2 = (V_5-0.8)$$

$$V_5 = 2.8\ \text{V}$$

Thus, the voltage V_5 for the circuit shown in the figure P5.56 (e) is $\boxed{2.8\ \text{V}}$.

Step 6 of 6

Refer to Figure P5.56 (f) in the text book.

$$I_D = \frac{5-V_6}{400\times10^3}$$

The drain voltage is V_6 V

The source voltage is zero

The gate voltage is V_6 V

The drain current I_D for NMOS when $\lambda = 0$ is,

$$\frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_t)^2$$

$$\frac{5-V_6}{400\times10^3} = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS}-V_t)^2$$

$$0.01(5-V_6) = (V_6-0.8)^2$$

$$V_6^2 - 1.59V_6 + 0.59 = 0$$

$$V_6 = 0.59\ \text{V}$$

Thus, the voltage V_6 for the circuit shown in the figure P5.56 (f) is $\boxed{0.59\ \text{V}}$.

Step 1 of 10

Refer to Figure P5.57 (a) in the text book.

Observed that transistors Q_1 and Q_2 are in series and hence drain current of Q_1 is equal to the drain current of Q_2

The threshold voltages of two NMOS transistors is $0.9\ \text{V}$

The value of $k_n'\frac{W}{L}$ of two transistors is $1.5\ \text{mA/V}^2$

The source voltage of Q_1 is $V_1\ \text{V}$

The gate voltage of Q_1 is $2.5\ \text{V}$

The gate source voltage V_{GS} is $V_G - V_S$ and hence V_{GS} is equal to $(2.5 - V_1)$

The drain current for NMOS Q_1 transistor is,

$$\begin{aligned} i_{D1} &= \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - V_t)^2 \\ &= \frac{1}{2}\left(1.5\times10^{-3}\ \frac{\text{A}}{\text{V}^2}\right)\left[(2.5 - V_1) - 0.9\right]^2 \\ &= 0.75(1.6 - V_1)^2 \end{aligned}$$

Thus, the drain current for NMOS Q_1 transistor is,

$$i_{D1} = 0.75(1.6 - V_1)^2\ \text{mA} \quad \text{..... (1)}$$

Step 2 of 10

The Source voltage of Q_2 is $V_2\ \text{V}$

The gate voltage of Q_2 is $0\ \text{V}$

The drain current for NMOS Q_2 transistor is,

$$\begin{aligned} i_{D2} &= \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - V_t)^2 \\ &= \frac{1}{2}\left(1.5\times10^{-3}\ \frac{\text{A}}{\text{V}^2}\right)\left[(0 - V_2) - 0.9\right]^2 \\ &= 0.75(-0.9 - V_2)^2 \end{aligned}$$

Thus, the drain current for NMOS Q_2 transistor is,

$$i_{D2} = 0.75(-0.9 - V_2)^2\ \text{mA} \quad \text{..... (2)}$$

Step 3 of 10

Q_1 and Q_2 are in series current flowing through gate terminals is zero and hence drain current of Q_1 is equal to the drain current of Q_2 .

Equate equation (1) and (2) as follows:

$$\begin{aligned} 0.75(-0.9 - V_2)^2 &= 0.75(1.6 - V_1)^2 \\ -0.9 - V_2 &= 1.6 - V_1 \\ V_1 - V_2 &= 2.5\ \text{V} \end{aligned}$$

The relation between V_1 and V_2 is $V_1 - V_2 = 2.5\ \text{V}$

Step 4 of 10

From the Figure P5.57 (a) for Q_2 transistor,

$$\begin{aligned} i_{D2} &= \frac{V_2 - (-2.5)}{1000} \\ &= V_2 + 2.5\ \text{mA} \end{aligned}$$

Drain current i_{D2} is $V_2 + 2.5\ \text{mA}$ ..... (3)

Equate equation (3) with equation (2) as follows:

$$\begin{aligned} 0.75(-0.9 - V_2)^2 &= V_2 + 2.5 \\ 0.75(0.81 + V_2^2 + 1.8V_2) &= V_2 + 2.5 \\ 0.6075 + 0.75V_2^2 + 1.35V_2 - V_2 - 2.5 &= 0 \\ 0.75V_2^2 + 0.35V_2 - 1.8925 &= 0 \end{aligned}$$

Step 5 of 10

Solve for the equation $0.75V_2^2 + 0.35V_2 - 1.8925 = 0$

After solving the equation value of V_2 is $\boxed{1.37\ \text{V}}$

From the equation $V_1 - V_2 = 2.5\ \text{V}$

$$\begin{aligned} V_1 - V_2 &= 2.5\ \text{V} \\ V_1 &= 2.5 + 1.37 \\ V_1 &= 3.87\ \text{V} \end{aligned}$$

Thus, the drain voltage V_1 of Q_2 NMOS transistor is $\boxed{3.87\ \text{V}}$

Step 6 of 10

Refer to Figure P5.57 (b) in the text book.

The source voltage of Q_1 is $V_4\ \text{V}$

The gate voltage of Q_1 is $V_3\ \text{V}$

The drain current for NMOS Q_1 transistor is,

$$\begin{aligned} i_{D1} &= \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - V_t)^2 \\ &= \frac{1}{2}\left(1.5\times10^{-3}\ \frac{\text{A}}{\text{V}^2}\right)\left[(V_3 - V_4) - 0.9\right]^2 \\ &= 0.75(V_3 - V_4 - 0.9)^2 \end{aligned}$$

Thus, the drain current for NMOS Q_1 transistor is,

$$I_{D1} = 0.75(V_3 - V_4 - 0.9)^2\ \text{mA} \quad \text{..... (4)}$$

Step 7 of 10

The source voltage of Q_2 is $V_5\ \text{V}$

The gate voltage of Q_2 is $V_4\ \text{V}$

The drain current for NMOS Q_2 transistor is,

$$\begin{aligned} i_{D2} &= \frac{1}{2}k_n'\left(\frac{W}{L}\right)(v_{GS} - V_t)^2 \\ &= \frac{1}{2}\left(1.5\times10^{-3}\ \frac{\text{A}}{\text{V}^2}\right)\left[(V_4 - V_5) - 0.9\right]^2 \\ &= 0.75(V_4 - V_5 - 0.9)^2 \end{aligned}$$

Thus, the drain current for NMOS Q_2 transistor is,

$$i_{D2} = 0.75(V_4 - V_5 - 0.9)^2\ \text{mA} \quad \text{..... (5)}$$

Step 8 of 10

Q_1 and Q_2 are operated in saturation region since gate and drain terminals are short circuited.

Q_1 and Q_2 are in series hence drain current of Q_1 is equal to the drain current of Q_2 .

Equate equation (4) and (5)

$$\begin{aligned} 0.75(V_3 - V_4 - 0.9)^2 &= 0.75(V_4 - V_5 - 0.9)^2 \\ \frac{V_3 + V_5}{2} &= V_4 \end{aligned}$$

Thus, the relation between V_4 , V_3 and V_5 is,

$$\frac{V_3 + V_5}{2} = V_4 \quad \text{..... (6)}$$

Step 9 of 10

From the Figure P5.57 (b) for Q_2 transistor,

$$\begin{aligned} i_{D2} &= \frac{V_5 - (0)}{1000} \\ &= V_5\ \text{mA} \end{aligned}$$

The drain current i_{D2} is $V_5\ \text{mA}$ ..... (7)

From the figure P5.57 (b) for Q_1 transistor,

$$\begin{aligned} i_{D1} &= \frac{5 - (V_3)}{1000} \\ &= 5 - V_3\ \text{mA} \end{aligned}$$

The drain current i_{D1} is $5 - V_3\ \text{mA}$ ..... (8)

From the equation (7) and (8)

$$\begin{aligned} 5 - V_3 &= V_5 \\ V_3 + V_5 &= 5 \end{aligned}$$

Substitute $V_3 + V_5 = 5$ in equation (6)

$$\begin{aligned} V_4 &= \frac{V_3 + V_5}{2} \\ &= \frac{5}{2} \\ &= 2.5\ \text{V} \end{aligned}$$

Thus, the drain voltage V_4 of Q_2 NMOS transistor is $\boxed{2.5\ \text{V}}$

Step 10 of 10

From equation (4) and (7)

$$\begin{aligned} 0.75(V_3 - V_4 - 0.9)^2\ \text{mA} &= V_5\ \text{mA} \\ 0.75(V_3 - 2.5 - 0.9)^2 &= V_5 \end{aligned}$$

Substitute $V_5 = 0.75(V_3 - 3.4)^2$ in $V_3 + V_5 = 5\ \text{V}$

$$\begin{aligned} V_3 + V_5 &= 5\ \text{V} \\ V_3 + 0.75(V_3 - 3.4)^2 &= 5 \\ V_3 + 0.75(V_3^2 + 11.56 - 6.8V_3) &= 5 \\ 0.75V_3^2 - 5.1V_3 + V_3 + 8.67 - 5 &= 0 \\ 0.75V_3^2 - 4.1V_3 + 3.67 &= 0 \end{aligned}$$

Solve the equation $0.75V_3^2 - 4.1V_3 + 3.67 = 0$

The drain voltage V_3 of Q_1 NMOS transistor is $\boxed{1.13\ \text{V}}$

$$\begin{aligned} V_3 + V_5 &= 5 \\ V_5 &= 5 - V_3 \\ V_5 &= 3.87\ \text{V} \end{aligned}$$

Thus, the voltage V_5 is $\boxed{3.87\ \text{V}}$.

Step 1 of 7

5.058P

(a)

From the Fig.P5.58, it is clear that voltage across R is V_{DG}.

Current flowing through gate is zero and hence current flowing through R is I and hence $IR = V_{DG}$ ..... (1)

Condition for PMOS transistor operate in saturation is,

$$V_{SD} \geq V_{SG} - |V_{tp}|$$
$$V_S - V_D \geq V_S - V_G - |V_{tp}|$$
$$-V_D \geq -V_G - |V_{tp}|$$
$$V_D - V_G \leq |V_{tp}|$$

Condition for PMOS transistor Operate in saturation is $V_{DG} \leq |V_{tp}|$ ..... (2)

From equation (1) and (2)

$$V_{DG} \leq |V_{tp}|$$
$$IR \leq |V_{tp}|$$

Condition for PMOS transistor Operate in saturation is $\boxed{IR \leq |V_{tp}|}$

Step 2 of 7

Value of I is $0.1\ \text{mA}$ and hence drain current is $0.1\ \text{mA}$.

Drain current i_D for PMOS in saturation is,

$$\frac{1}{2}k_p\left(V_{SG} - |V_t|\right)^2$$

Value of k_p is $0.2\ \text{mA/V}^2$

$$i_D = \frac{1}{2}k_p\left(V_{SG} - |V_t|\right)^2$$
$$0.1\times10^{-3}\ \text{A} = \frac{1}{2}\left(0.2\times10^{-3}\ \frac{\text{A}}{\text{V}^2}\right)\left(V_{SG} - 1\right)^2$$
$$V_{SG} - V_t = 1$$
$$V_{SG} = 2\ \text{V}$$

Source gate V_{SG} voltage is $\boxed{2\ \text{V}}$

Step 3 of 7

From the Fig.P5.58 source voltage V_S is $10\ \text{V}$

$$V_{SG} = 2\ \text{V}$$
$$V_S - V_G = 2$$
$$10 - 2 = V_G$$
$$8 = V_G$$

Gate voltage V_G is $8\ \text{V}$

From equation (1) $IR = V_{DG}$

$$IR = V_{DG}$$
$$\left(0.1\times10^{-3}\ \text{A}\right)(R) = V_D - V_G$$
$$\left(0.1\times10^{-3}\ \text{A}\right)(R) = V_D - 8$$

Step 4 of 7

For $R = 0\ \Omega$,

$$\left(0.1\times10^{-3}\ \text{A}\right)(R) = V_D - 8$$
$$\left(0.1\times10^{-3}\ \text{A}\right)(0) = V_D - 8$$
$$V_D = 8\ \text{V}$$

Therefore value of V_{SD} is,

$$V_{SD} = V_S - V_D$$
$$= 10 - 8$$
$$= 2\ \text{V}$$

Source drain voltage for $R = 0\ \Omega$ is $\boxed{2\ \text{V}}$

Step 5 of 7

For $R = 10\ \text{k}\Omega$,

$$\left(0.1\times10^{-3}\ \text{A}\right)(R) = V_D - 8$$
$$\left(0.1\times10^{-3}\ \text{A}\right)\left(10\times10^{3}\right) = V_D - 8$$
$$V_D = 9\ \text{V}$$

Therefore value of V_{SD} is,

$$V_{SD} = V_S - V_D$$
$$= 10 - 9$$
$$= 1\ \text{V}$$

Source drain voltage for $R = 10\ \text{k}\Omega$ is $\boxed{1\ \text{V}}$

Step 6 of 7

For $R = 30\ \text{k}\Omega$,

$$\left(0.1\times10^{-3}\ \text{A}\right)\left(30\times10^{3}\right) = V_D - 8$$
$$V_D = 11\ \text{V}$$

$$V_{SD} = V_S - V_D$$
$$= 10 - 11$$
$$= -1\ \text{V}$$

Source drain voltage for $R = 30\ \text{k}\Omega$ is $\boxed{-1\ \text{V}}$.

Step 7 of 7

For $R = 100\ \text{k}\Omega$,

$$\left(0.1\times10^{-3}\ \text{A}\right)(R) = V_D - 8$$
$$\left(0.1\times10^{-3}\ \text{A}\right)\left(100\times10^{3}\right) = V_D - 8$$
$$V_D = 18\ \text{V}$$

Therefore value of V_{SD} is,

$$V_{SD} = V_S - V_D$$
$$= 10 - 18$$
$$= -8\ \text{V}$$

Source drain voltage for $R = 100\ \text{k}\Omega$ is $\boxed{-8\ \text{V}}$.

5.059P

We don't have the solution to this problem yet.

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Step 1 of 6

5.060P

Refer to Figure P5.60 in the textbook for the circuit.

Redraw the circuit by representing currents and voltages as shown in Figure 1.

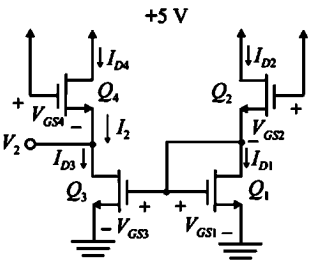


Figure 1

Step 2 of 6

Drain and gate terminals of transistor Q_1 are short circuited. Hence, the transistor Q_1 is in saturation region. Assume transistor Q_2 is also in saturation region. Drain currents of both the transistors are same. So, the gate to source voltages of both the transistors is same.

Apply Kirchhoff's voltage law from supply to ground for Q_2 and Q_1 loop.

$$-5+V_{GS2}+V_{GS1}=0$$
$$-5+2V_{GS1}=0 \qquad (\text{since } V_{GS1}=V_{GS2})$$
$$V_{GS1}=\frac{5}{2}$$
$$V_{GS1}=2.5\,\text{V}$$

Therefore, $V_{GS1}=V_{GS2}=2.5\,\text{V}$

Step 3 of 6

The equation for drain current of transistor Q_1 is,

$$I_{D1}=\frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS1}-V_t)^2$$

Substitute $50\,\mu\text{A/V}^2$ for $\mu_n C_{ox}$, $10\,\mu\text{m}$ for W, $1\,\mu\text{m}$ for L, $1\,\text{V}$ for $|V_t|$, and $2.5\,\text{V}$ for V_{GS1} in the equation.

$$I_{D1}=\frac{1}{2}\left(50\times10^{-6}\right)\left(\frac{10\times10^{-6}}{1\times10^{-6}}\right)(2.5-1)^2$$
$$=\left(25\times10^{-6}\right)(10)(1.5)^2$$
$$=\left(25\times10^{-6}\right)(10)(2.25)$$
$$=562.5\times10^{-6}\,\text{A}$$

The current I_{D2} is,

$$I_{D2}=I_{D1}$$
$$=562.5\,\mu\text{A}$$

Step 4 of 6

From the circuit, the voltage V_{GS3} is,

$$V_{GS3}=V_{GS1}$$
$$=2.5\,\text{V}$$

In the circuit arrangement, Q_1 and Q_3 combination acts like current mirror circuit.

Hence, the current I_{D3} is,

$$I_{D3}=I_{D1}$$
$$=562.5\,\mu\text{A}$$

The current I_2 is,

$$I_2=I_{D3}$$
$$=562.5\,\mu\text{A}$$

Therefore, the current I_2 is, $\boxed{I_2=562.5\,\mu\text{A}}$.

Transistor Q_4 is also in saturation. The current I_{D4} is,

$$I_{D4}=I_{D3}$$
$$=562.5\,\mu\text{A}$$

Gate to source voltage of Q_1 and Q_4 transistors are same.

$$V_{GS4}=V_{GS1}$$
$$=2.5\,\text{V}$$

From the circuit, write the following relation:

$$-5+V_{GS4}+V_2=0$$
$$-5+2.5+V_2=0$$
$$V_2=2.5\,\text{V}$$

Therefore, the voltage V_2 is, $\boxed{V_2=2.5\,\text{V}}$.

Step 5 of 6

The new width of the channel is, $W=100\,\mu\text{A}$.

The new drain current of transistor Q_1 is,

$$I_{D1}=\frac{1}{2}\left(50\times10^{-6}\right)\left(\frac{100\times10^{-6}}{1\times10^{-6}}\right)(2.5-1)^2$$
$$=\left(25\times10^{-6}\right)(100)(1.5)^2$$
$$=\left(25\times10^{-6}\right)(100)(2.25)$$
$$=5.625\,\text{mA}$$

The current I_{D2} is,

$$I_{D2}=I_{D1}$$
$$=5.625\,\text{mA}$$

Step 6 of 6

From the circuit, the voltage V_{GS3} is,

$$V_{GS3}=V_{GS1}$$
$$=2.5\,\text{V}$$

In the circuit arrangement, Q_1 and Q_3 combination acts like current mirror circuit.

Hence, the current I_{D3} is,

$$I_{D3}=I_{D1}$$
$$=5.625\,\text{mA}$$

The current I_2 is,

$$I_2=I_{D3}$$
$$=5.625\,\text{mA}$$

Therefore, the current I_2 is, $\boxed{I_2=5.625\,\text{mA}}$.

Transistor Q_4 is also in saturation. The current I_{D4} is,

$$I_{D4}=I_{D3}$$
$$=5.625\,\text{mA}$$

Gate to source voltage of Q_1 and Q_4 transistors are same.

$$V_{GS4}=V_{GS1}$$
$$=2.5\,\text{V}$$

From the circuit, write the following relation:

$$-5+V_{GS4}+V_2=0$$
$$-5+2.5+V_2=0$$
$$V_2=2.5\,\text{V}$$

Therefore, the voltage V_2 is, $\boxed{V_2=2.5\,\text{V}}$.

Hence, there is no change in voltage V_2.

Step 1 of 13

(a)

From the Fig.P5.61,

Drain current flowing through transistor Q_1 is,

$$i_{D1} = \frac{2.5 - V_1}{20 \times 10^3} \text{ A}$$

Drain current flowing through transistor Q_2 is,

$$i_{D2} = \frac{2.5 - V_2}{20 \times 10^3} \text{ A}$$

Step 2 of 13

Drain current flowing through NMOS transistor is,

$$i_D = \frac{1}{2} k_n' \left(W/L\right)(v_{GS} - V_t)^2$$

Trans-conductance parameter k_n' is $125 \ \mu\text{A}/\text{V}^2$

Threshold voltage V_t is 0.7 V

$$\left(W/L\right)_1 = \left(W/L\right)_2 = 20$$

Drain current flowing through Q_1 NMOS transistor is,

$$\begin{aligned} i_{D1} &= \frac{1}{2} k_n' \left(W/L\right)_1 (v_{GS} - V_t)^2 \\ &= \frac{1}{2}\left(125 \times 10^{-6} \ \text{A}/\text{V}^2\right)(20)(0 - V_3 - 0.7)^2 \\ &= 1.25(V_3 + 0.7)^2 \text{ mA} \end{aligned}$$

Drain current i_{D1} is $1.25(V_3 + 0.7)^2$ mA ..... (1)

Step 3 of 13

Drain current flowing through Q_2 NMOS transistor is,

$$\begin{aligned} i_{D2} &= \frac{1}{2} k_n' \left(W/L\right)_2 (v_{GS} - V_t)^2 \\ &= \frac{1}{2}\left(125 \times 10^{-6} \ \text{A}/\text{V}^2\right)(20)(0 - V_3 - 0.7)^2 \\ &= 1.25(V_3 + 0.7)^2 \text{ mA} \end{aligned}$$

Drain current i_{D2} is $1.25(V_3 + 0.7)^2$ mA ..... (2)

Step 4 of 13

From equation (1) and (2) we observed that $i_{D1} = i_{D2}$

$$i_{D1} + i_{D2} = 200 \ \mu\text{A}$$

Therefore Drain current i_{D1} is $100 \ \mu$A

Drain current i_{D2} is $100 \ \mu$A

We observed that $i_{D1} = \frac{2.5 - V_1}{20 \times 10^3}$ A

$$i_{D1} = \frac{2.5 - V_1}{20 \times 10^3} \text{ A}$$

$$100 \times 10^{-6} \text{ A} = \frac{2.5 - V_1}{20 \times 10^3} \text{ A}$$

$$2 = 2.5 - V_1$$

$$V_1 = 0.5 \text{ V}$$

Drain voltage of Q_1 NMOS transistor V_1 is $\boxed{0.5 \text{ V}}$

Step 5 of 13

We observed that $i_{D2} = \frac{2.5 - V_2}{20 \times 10^3}$ A

$$i_{D2} = \frac{2.5 - V_2}{20 \times 10^3} \text{ A}$$

$$100 \times 10^{-6} \text{ A} = \frac{2.5 - V_2}{20 \times 10^3} \text{ A}$$

$$2 = 2.5 - V_2$$

$$V_2 = 0.5 \text{ V}$$

Drain voltage of Q_2 NMOS transistor V_2 is $\boxed{0.5 \text{ V}}$

Step 6 of 13

From equation (1) we observed that,

$$i_{D1} = 1.25(V_3 + 0.7)^2 \text{ mA}$$

$$100 \times 10^{-6} \text{ A} = 1.25(V_3 + 0.7)^2 \left(10^{-3}\right) \text{ A}$$

$$(V_3 + 0.7)^2 = 0.08$$

$$V_3 = -0.42 \text{ V}$$

Source voltage V_3 is $\boxed{-0.42 \text{ V}}$

Step 7 of 13

(b)
From the Fig.P5.61,

Drain current flowing through transistor Q_1 is,

$$i_{D1} = \frac{2.5 - V_1}{20 \times 10^3} \text{ A}$$

Drain current flowing through transistor Q_2 is,

$$i_{D2} = \frac{2.5 - V_2}{20 \times 10^3} \text{ A}$$

Step 8 of 13

Drain current flowing through NMOS transistor is,

$$i_D = \frac{1}{2} k_n' \left(W/L\right)(v_{GS} - V_t)^2$$

Trans-conductance parameter k_n' is $125 \ \mu\text{A}/\text{V}^2$

Threshold voltage V_t is 0.7 V

$$\left(W/L\right)_1 = 1.5\left(W/L\right)_2 = 20$$

Drain current flowing through Q_1 NMOS transistor is,

$$\begin{aligned} i_{D1} &= \frac{1}{2} k_n' \left(W/L\right)_1 (v_{GS} - V_t)^2 \\ &= \frac{1}{2}\left(125 \times 10^{-6} \ \text{A}/\text{V}^2\right)(20)(0 - V_3 - 0.7)^2 \\ &= 1.25(V_3 + 0.7)^2 \text{ mA} \end{aligned}$$

Drain current i_{D1} is $1.25(V_3 + 0.7)^2$ mA ..... (3)

Step 9 of 13

Drain current flowing through Q_2 NMOS transistor is,

$$\begin{aligned} i_{D2} &= \frac{1}{2} k_n' \left(W/L\right)_2 (v_{GS} - V_t)^2 \\ &= \frac{1}{2}\left(125 \times 10^{-6} \ \text{A}/\text{V}^2\right)(13.33)(0 - V_3 - 0.7)^2 \\ &= 0.833(V_3 + 0.7)^2 \text{ mA} \end{aligned}$$

Drain current i_{D2} is $0.833(V_3 + 0.7)^2$ mA ..... (4)

Step 10 of 13

Divide equation (3) with (4)

$$\frac{i_{D1}}{i_{D2}} = \frac{1.25(V_3 + 0.7)^2 \text{ mA}}{0.833(V_3 + 0.7)^2 \text{ mA}}$$

$$\frac{i_{D1}}{i_{D2}} = 1.5$$

Relation between i_{D1} and i_{D2} is $i_{D1} = 1.5 i_{D2}$

$$i_{D1} + i_{D2} = 200 \ \mu\text{A}$$

$$1.5 i_{D2} + i_{D2} = 200 \times 10^{-6} \text{ A}$$

$$2.5 i_{D2} = 200 \times 10^{-6} \text{ A}$$

$$i_{D2} = 80 \ \mu\text{A}$$

Drain current i_{D2} is $80 \ \mu$A

Therefore Drain current i_{D1} is $120 \ \mu$A

Step 11 of 13

We observed that $i_{D1} = \frac{2.5 - V_1}{20 \times 10^3}$ A

$$i_{D1} = \frac{2.5 - V_1}{20 \times 10^3} \text{ A}$$

$$120 \times 10^{-6} \text{ A} = \frac{2.5 - V_1}{20 \times 10^3} \text{ A}$$

$$2.4 = 2.5 - V_1$$

$$V_1 = 0.1 \text{ V}$$

Drain voltage of Q_1 NMOS transistor V_1 is $\boxed{0.1 \text{ V}}$

Step 12 of 13

We observed that $i_{D2} = \frac{2.5 - V_2}{20 \times 10^3}$ A

$$i_{D2} = \frac{2.5 - V_2}{20 \times 10^3} \text{ A}$$

$$80 \times 10^{-6} \text{ A} = \frac{2.5 - V_2}{20 \times 10^3} \text{ A}$$

$$1.6 = 2.5 - V_2$$

$$V_2 = 0.9 \text{ V}$$

Drain voltage of Q_2 NMOS transistor V_2 is $\boxed{0.9 \text{ V}}$

Step 13 of 13

From equation (3) we observed that,

$$i_{D1} = 1.25(V_3 + 0.7)^2 \text{ mA}$$

$$120 \times 10^{-6} \text{ A} = 1.25(V_3 + 0.7)^2 \left(10^{-3}\right) \text{ A}$$

$$(V_3 + 0.7)^2 = 0.096$$

$$V_3 = -0.39 \text{ V}$$

Source voltage V_3 is $\boxed{-0.39 \text{ V}}$

Step 1 of 2

5.062P

Consider the following expression for the threshold voltage, V_t in an n -channel MOSFET:

$$V_t = V_{t0} + \gamma\left[\sqrt{2\phi_f + V_{SB}} - \sqrt{2\phi_f}\right]$$

Here, V_{t0} is the threshold voltage at $V_{SB} = 0$, ϕ_f is a physical parameter, γ is the body-effect parameter, and V_{SB} is the reverse substrate bias voltage.

Substitute $0.5\sqrt{\text{V}}$ for γ , 0.6 V for $2\phi_f$, and 1 V for V_{t0} .

Take minimum value of V_{SB} , that is 0 to calculate the minimum value of V_t .

$$\begin{aligned} V_t &= 1\text{ V} + \left(0.5\sqrt{\text{V}}\right)\left[\sqrt{0.6\text{ V} + 0} - \sqrt{0.6\text{ V}}\right] \\ &= 1\text{ V} + \left(0.5\sqrt{\text{V}}\right)(0) \\ &= 1\text{ V} \end{aligned}$$

Take maximum value of V_{SB} , that is 4 V to calculate the maximum value of V_t .

$$\begin{aligned} V_t &= 1\text{ V} + \left(0.5\sqrt{\text{V}}\right)\left[\sqrt{0.6\text{ V} + 4\text{ V}} - \sqrt{0.6\text{ V}}\right] \\ &= 1\text{ V} + \left(0.5\sqrt{\text{V}}\right)\left(1.37\sqrt{\text{V}}\right) \\ &= 1\text{ V} + 0.685\text{ V} \\ &= 1.685\text{ V} \end{aligned}$$

Therefore, the range of threshold voltage, V_t is, $\boxed{1\text{ V to }1.685\text{ V}}$.

Step 2 of 2

Consider that the gate oxide thickness is increased by 4 times.

$$t_{ox(new)} = 4t_{ox}$$

The expression for the fabrication-process parameter is,

$$\begin{aligned} \gamma &= \frac{\sqrt{2qN_A\varepsilon_s}}{C_{ox}} \\ &= \frac{\sqrt{2qN_A\varepsilon_s}}{\left(\frac{\varepsilon_{ox}}{t_{ox}}\right)} \\ &= t_{ox}\left(\frac{\sqrt{2qN_A\varepsilon_s}}{\varepsilon_{ox}}\right) \end{aligned}$$

The new value of threshold voltage is,

$$\begin{aligned} \frac{V_{t(new)}}{V_t} &= \frac{V_{t0} + t_{ox(new)}\left(\frac{\sqrt{2qN_A\varepsilon_s}}{\varepsilon_{ox}}\right)\left[\sqrt{2\phi_f + V_{SB}} - \sqrt{2\phi_f}\right]}{V_{t0} + t_{ox}\left(\frac{\sqrt{2qN_A\varepsilon_s}}{\varepsilon_{ox}}\right)\left[\sqrt{2\phi_f + V_{SB}} - \sqrt{2\phi_f}\right]} \\ &= \frac{t_{ox(new)}}{t_{ox}} \\ &= \frac{4t_{ox}}{t_{ox}} \\ &= 4 \end{aligned}$$

Therefore,

$$V_{t(new)} = 4V_t$$

Thus, the threshold voltage, V_t increases by 4 times, if the gate oxide thickness increases by 4 times.

5.063P

Consider the following expression for threshold voltage, V_t in a p -channel transistor:

$$V_t = V_{t0} - \gamma\left[\sqrt{2\phi_f + V_{SB}} - \sqrt{2\phi_f}\right]$$

Here,

V_{t0} is the threshold voltage at $V_{SB} = 0$,

ϕ_f is a physical parameter,

γ is the body-effect parameter, and

V_{SB} is the reverse substrate bias voltage.

Substitute $0.5\sqrt{\text{V}}$ for γ , 0.75 V for $2\phi_f$, 3 V for V_{SB} , and $-0.7\ \text{V}$ for V_{t0} .

$$\begin{aligned} V_t &= -0.7\ \text{V} - \left(0.5\sqrt{\text{V}}\right)\left[\sqrt{0.75\ \text{V} + 3\ \text{V}} - \sqrt{0.75\ \text{V}}\right] \\ &= -0.7\ \text{V} - \left(0.5\sqrt{\text{V}}\right)\left(1.07\sqrt{\text{V}}\right) \\ &= -0.7\ \text{V} - 0.535\ \text{V} \\ &= -1.235\ \text{V} \end{aligned}$$

Therefore, the threshold voltage, V_t is, $\boxed{-1.235\ \text{V}}$.

5.064P

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Step 1 of 11

5.065P

Consider the following circuit diagram with source and gate of the transistor are grounded:

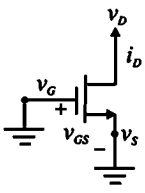


Figure 1

Step 2 of 11

In the circuit gate and source are connected to ground $v_G = 0\,\mathrm{V}$ and $v_S = 0\,\mathrm{V}$. So, the gate to source voltage is,

$$v_{GS} = 0\,\mathrm{V}$$

For $v_D = 0.1\,\mathrm{V}$:

The Drain to source voltage is,

$$\begin{aligned} v_{DS} &= v_D - v_S \\ &= 0.1 - 0 \\ &= 0.1\,\mathrm{V} \end{aligned}$$

Step 3 of 11

The condition for triode region is,

$$v_{DS} \le v_{GS} - V_t$$
$$0.1 \le 0 - (-3)$$
$$0.1 \le 3$$

Since the condition is satisfied, the transistor is operating in **triode region** .

Step 4 of 11

The equation for drain current is,

$$\begin{aligned} i_D &= k_n' \frac{W}{L}\left[(v_{GS} - V_t)v_{DS} - \frac{1}{2}v_{DS}^2\right] \\ &= \left(2\times10^{-3}\right)\left[(0-(-3))(0.1) - \frac{1}{2}(0.1)^2\right] \\ &= \left(2\times10^{-3}\right)(0.3 - 0.005) \\ &= \left(2\times10^{-3}\right)(0.295) \\ &= 0.59\times10^{-3} \\ &= 0.59\,\mathrm{mA} \end{aligned}$$

Therefore, the drain current is, $\boxed{i_D = 0.59\,\mathrm{mA}}$.

Step 5 of 11

For $v_D = 1\,\mathrm{V}$:

The Drain to source voltage is,

$$\begin{aligned} v_{DS} &= v_D - v_S \\ &= 1 - 0 \\ &= 1\,\mathrm{V} \end{aligned}$$

Step 6 of 11

The condition for triode region is,

$$v_{DS} \le v_{GS} - V_t$$
$$1 \le 0 - (-3)$$
$$1 \le 3$$

Since the condition is satisfied, the transistor is operating in **triode region** .

Step 7 of 11

The equation for drain current is,

$$\begin{aligned} i_D &= k_n' \frac{W}{L}\left[(v_{GS} - V_t)v_{DS} - \frac{1}{2}v_{DS}^2\right] \\ &= \left(2\times10^{-3}\right)\left[(0-(-3))(1) - \frac{1}{2}(1)^2\right] \\ &= \left(2\times10^{-3}\right)(3 - 0.5) \\ &= \left(2\times10^{-3}\right)(2.5) \\ &= 5\times10^{-3} \\ &= 5\,\mathrm{mA} \end{aligned}$$

Therefore, the drain current is, $\boxed{i_D = 5\,\mathrm{mA}}$.

Step 8 of 11

For $v_D = 3\,\mathrm{V}$:

The Drain to source voltage is,

$$\begin{aligned} v_{DS} &= v_D - v_S \\ &= 3 - 0 \\ &= 3\,\mathrm{V} \end{aligned}$$

Step 9 of 11

The condition for saturation is,

$$v_{DS} \ge v_{GS} - V_t$$
$$3 \ge 0 - (-3)$$
$$3 = 3$$

Since $v_{DS} = v_{GS} - V_t$, the transistor is operating at the **edge of saturation** .

Step 10 of 11

The equation for drain current is,

$$\begin{aligned} i_D &= \frac{1}{2}k_n' \frac{W}{L}(v_{GS} - V_t)^2 \\ &= \frac{1}{2}\left(2\times10^{-3}\right)(0-(-3))^2 \\ &= \left(1\times10^{-3}\right)(3)^2 \\ &= \left(1\times10^{-3}\right)(9) \\ &= 9\times10^{-3} \\ &= 9\,\mathrm{mA} \end{aligned}$$

Therefore, the drain current is, $\boxed{i_D = 9\,\mathrm{mA}}$.

Step 11 of 11

For $v_D = 5\,\mathrm{V}$:

The Drain to source voltage is,

$$\begin{aligned} v_{DS} &= v_D - v_S \\ &= 5 - 0 \\ &= 5\,\mathrm{V} \end{aligned}$$

The condition for saturation is,

$$v_{DS} \ge v_{GS} - V_t$$
$$5 \ge 0 - (-3)$$
$$5 \ge 3$$

Since the condition is satisfied, the transistor is operating in **saturation region** .

The equation for drain current is,

$$\begin{aligned} i_D &= \frac{1}{2}k_n' \frac{W}{L}(v_{GS} - V_t)^2 \\ &= \frac{1}{2}\left(2\times10^{-3}\right)(0-(-3))^2 \\ &= \left(1\times10^{-3}\right)(3)^2 \\ &= \left(1\times10^{-3}\right)(9) \\ &= 9\times10^{-3} \\ &= 9\,\mathrm{mA} \end{aligned}$$

Therefore, the drain current is, $\boxed{i_D = 9\,\mathrm{mA}}$.

Step 1 of 6

Calculate the overdrive voltage.

$$\begin{aligned} V_{OV} &= v_{GS} - V_t \\ &= 0 - (-2) \\ &= 2\ \text{V} \end{aligned}$$

Step 2 of 6

Case (i):

For $v_{DS} = 1\text{V}$, the voltage $v_{DS} < V_{OV}\left(v_{GS} - V_t\right)$. So, the transistor is operating in triode region. The equation for drain current is,

$$i_D = k_n'\left(\frac{W}{L}\right)\left[\left(v_{GS} - V_t\right)v_{DS} - \frac{1}{2}v_{DS}^2\right]$$

Substitute $200\,\mu\text{A/V}^2$ for $k_n'\left(\frac{W}{L}\right)$, 0V for v_{GS}, -2V for V_t, and 1V for v_{DS} in the equation.

$$\begin{aligned} i_D &= \left(200\times10^{-6}\right)\left[\left(0-(-2)\right)(1) - \frac{1}{2}(1)^2\right] \\ &= \left(200\times10^{-6}\right)[2-0.5] \\ &= 300\times10^{-6} \end{aligned}$$

Therefore, for $v_{DS} = 1\text{V}$, value of drain current is, $\boxed{i_D = 300\,\mu\text{A}}$.

Step 3 of 6

Case (ii):

For $v_{DS} = 2\text{V}$, the voltage $v_{DS} = V_{OV}\left(v_{GS} - V_t\right)$. So, the transistor is operating at edge of saturation region. The equation for drain current is,

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS} - V_t\right)^2\left[1 + \lambda v_{DS}\right]$$

Substitute $200\,\mu\text{A/V}^2$ for $k_n'\left(\frac{W}{L}\right)$, 0V for v_{GS}, -2V for V_t, $0.02\ \text{V}^{-1}$ for λ, and 2V for v_{DS} in the equation.

$$\begin{aligned} i_D &= \frac{1}{2}\left(200\times10^{-6}\right)\left(0-(-2)\right)^2\left[1+0.02(2)\right] \\ &= \left(100\times10^{-6}\right)(4)(1.04) \\ &= 416\times10^{-6} \end{aligned}$$

Therefore, for $v_{DS} = 2\text{V}$, value of drain current is $\boxed{i_D = 416\,\mu\text{A}}$.

Step 4 of 6

Case (iii):

For $v_{DS} = 3\text{V}$, the voltage $v_{DS} > V_{OV}\left(v_{GS} - V_t\right)$. So, the transistor is operating in saturation region. The equation for drain current is,

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS} - V_t\right)^2\left[1 + \lambda v_{DS}\right]$$

Substitute $200\,\mu\text{A/V}^2$ for $k_n'\left(\frac{W}{L}\right)$, 0V for v_{GS}, -2V for V_t, $0.02\ \text{V}^{-1}$ for λ, and 3V for v_{DS} in the equation.

$$\begin{aligned} i_D &= \frac{1}{2}\left(200\times10^{-6}\right)\left(0-(-2)\right)^2\left[1+0.02(3)\right] \\ &= \left(100\times10^{-6}\right)(4)(1.06) \\ &= 424\times10^{-6} \end{aligned}$$

Therefore, for $v_{DS} = 3\text{V}$, value of drain current is $\boxed{i_D = 424\,\mu\text{A}}$.

Step 5 of 6

Case (iv):

For $v_{DS} = 10\text{V}$, the voltage $v_{DS} > V_{OV}\left(v_{GS} - V_t\right)$. So, the transistor is operating in saturation region. The equation for drain current is,

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS} - V_t\right)^2\left[1 + \lambda v_{DS}\right]$$

Substitute $200\,\mu\text{A/V}^2$ for $k_n'\left(\frac{W}{L}\right)$, 0V for v_{GS}, -2V for V_t, $0.02\ \text{V}^{-1}$ for λ, and 10V for v_{DS} in the equation.

$$\begin{aligned} i_D &= \frac{1}{2}\left(200\times10^{-6}\right)\left(0-(-2)\right)^2\left[1+0.02(10)\right] \\ &= \left(100\times10^{-6}\right)(4)(1.2) \\ &= 480\times10^{-6} \end{aligned}$$

Therefore, for $v_{DS} = 10\text{V}$, value of drain current is $\boxed{i_D = 480\,\mu\text{A}}$.

Step 6 of 6

In all the four cases, the relation between drain current and width-to-length ratio is

$$i_D \propto \frac{W}{L}.$$

If width is doubled with L the same, then drain current also doubled in all the four cases.

If width and length are doubled, then there is no change in drain current in all the four cases.

Step 1 of 6

Refer to Figure P5.123 in the textbook for the depletion-type NMOS transistor.

From the circuit, the gate to source voltage is,

$$v_{GS} = v_{DS} = v$$

The threshold voltage V_t is negative. Hence, $v_{DS} < v_{GS} - V_t$.

Step 2 of 6

For $v > V_t$, the condition is,

$$v < v - V_t$$
$$v_{DS} < v_{GS} - V_t$$

Since $v_{DS} < v_{GS} - V_t$, transistor is operating in the triode region and the equation for drain current is,

$$i_D = k_n' \frac{W}{L}\left[(v_{GS} - V_t)v_{DS} - \frac{1}{2}v_{DS}^2\right]$$

Replace v for v_{GS} **and** v_{DS}, i for i_D in the equation.

$$\begin{aligned} i_D &= k_n' \frac{W}{L}\left[(v - V_t)v - \frac{1}{2}v^2\right] \\ &= k_n' \frac{W}{L}\left[v^2 - vV_t - \frac{1}{2}v^2\right] \\ &= k_n' \frac{W}{L}\left[\frac{1}{2}v^2 - vV_t\right] \\ &= \frac{1}{2}k_n' \frac{W}{L}\left[v^2 - 2vV_t\right] \end{aligned}$$

Step 3 of 6

For $v < V_t$, the condition is,

$$v > v - V_t$$
$$v_{DS} > v_{GS} - V_t$$

Hence, transistor is operating in saturation region and current flows in reverse direction.

Source acts like drain and drain acts like source. Therefore, source and gate are short circuited and $v_{GS} = 0$.

The resultant circuit configuration is shown in Figure 1.

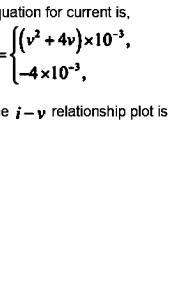


Figure 1

Step 4 of 6

The equation for drain current is,

$$i_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS} - V_t^2\right)$$

Replace i for i_D in the equation.

$$i = \frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS} - V_t^2\right)$$
$$i = \frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(0 - V_t^2\right) \qquad (\text{since } v_{GS} = 0)$$
$$i = -\frac{1}{2}k_n'\left(\frac{W}{L}\right)V_t^2$$

Therefore, $i - v$ relationship is,

$$\boxed{\begin{aligned} i &= \frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v^2 - 2V_t v\right), && \text{for } v \geq V_t \\ i &= -\frac{1}{2}k_n'\left(\frac{W}{L}\right)V_t^2, && \text{for } v \leq V_t \end{aligned}}$$

Step 5 of 6

The parameters are,

$$V_t = -2\,\text{V} \text{ and } k_n'\left(\frac{W}{L}\right) = 2\,\text{mA/V}^2$$

Substitute $-2\,\text{V}$ for V_t and $2\,\text{mA/V}^2$ for $k_n'\left(\frac{W}{L}\right)$ in drain current equations for $v \geq V_t$.

$$\begin{aligned} i &= \frac{1}{2}\left(2\times10^{-3}\right)\left(v^2 - 2(-2)v\right), && \text{for } v \geq -2\,\text{V} \\ &= \left(v^2 + 4v\right)\times10^{-3}, && \text{for } v \geq -2\,\text{V} \end{aligned}$$

Substitute $-2\,\text{V}$ for V_t and $2\,\text{mA/V}^2$ for $k_n'\left(\frac{W}{L}\right)$ in drain current equations for $v \leq V_t$.

$$\begin{aligned} i &= -\frac{1}{2}\left(2\times10^{-3}\right)(-2)^2, && \text{for } v \leq -2\,\text{V} \\ &= -4\times10^{-3}, && \text{for } v \leq -2\,\text{V} \end{aligned}$$

Step 6 of 6

Equation for current is,

$$i = \begin{cases} \left(v^2 + 4v\right)\times10^{-3}, & \text{for } v \geq -2\,\text{V} \\ -4\times10^{-3}, & \text{for } v \leq -2\,\text{V} \end{cases}$$

The $i - v$ relationship plot is shown in Figure 2.

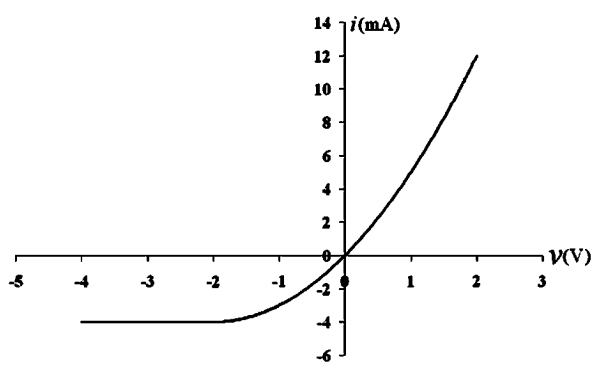


Figure 2

Step 1 of 3

Write the expression of the collector current, i_C .

$$i_C = I_S e^{\frac{v_{BE}}{V_T}}$$

Simplify further.

$$v_{BE} = V_T \ln \frac{i_C}{I_S}$$

Substitute 0.7 V for v_{BE} , 25 mV for V_T and 1 mA for i_C .

$$0.7 = (25\times10^{-3}) \ln\left(\frac{1\times10^{-3}}{I_S}\right) \quad \text{..... (1)}$$

Step 2 of 3

The voltage v_{BE} at $i_C = 0.1\ \text{mA}$ is,

$$v_{BE} = (25\times10^{-3}) \ln\left(\frac{0.1\times10^{-3}}{I_S}\right) \quad \text{..... (2)}$$

Subtract equation (1) from equation (2).

$$v_{BE} - 0.7 = (25\times10^{-3}) \ln\left(\frac{0.1\times10^{-3}}{I_S}\right) - (25\times10^{-3}) \ln\left(\frac{1\times10^{-3}}{I_S}\right)$$

$$v_{BE} - 0.7 = (25\times10^{-3}) \left[\ln\left(\frac{0.1\times10^{-3}}{I_S}\right) - \ln\left(\frac{1\times10^{-3}}{I_S}\right)\right]$$

$$v_{BE} - 0.7 = (25\times10^{-3}) \ln\left(\frac{\frac{0.1\times10^{-3}}{I_S}}{\frac{1\times10^{-3}}{I_S}}\right)$$

Simplify further.

$$v_{BE} - 0.7 = (25\times10^{-3}) \ln(0.1)$$

$$v_{BE} = 0.7 - 0.06$$

$$v_{BE} = 0.64\ \text{V}$$

Thus, the voltage v_{BE} at $i_C = 0.1\ \text{mA}$ is $\boxed{0.64\ \text{V}}$.

Step 3 of 3

The voltage v_{BE} at $i_C = 10\ \text{mA}$ is,

$$v_{BE} = (25\times10^{-3}) \ln\left(\frac{10\times10^{-3}}{I_S}\right) \quad \text{..... (3)}$$

Subtract equation (1) from equation (3).

$$v_{BE} - 0.7 = (25\times10^{-3}) \ln\left(\frac{10\times10^{-3}}{I_S}\right) - (25\times10^{-3}) \ln\left(\frac{1\times10^{-3}}{I_S}\right)$$

$$v_{BE} - 0.7 = (25\times10^{-3}) \left[\ln\left(\frac{10\times10^{-3}}{I_S}\right) - \ln\left(\frac{1\times10^{-3}}{I_S}\right)\right]$$

$$v_{BE} - 0.7 = (25\times10^{-3}) \ln\left(\frac{\frac{10\times10^{-3}}{I_S}}{\frac{1\times10^{-3}}{I_S}}\right)$$

Simplify further.

$$v_{BE} - 0.7 = (25\times10^{-3}) \ln(10)$$

$$v_{BE} = 0.7 + 0.06$$

$$v_{BE} = 0.76\ \text{V}$$

Thus, the voltage v_{BE} at $i_C = 10\ \text{mA}$ is $\boxed{0.76\ \text{V}}$.

Step 1 of 4

6.001P

In case of *npn* transistors, the emitter-base junction is forward biased when the voltage across the Emitter base junction (EBJ) is negative, that is, $V_{EB} < 0$

The emitter-base junction to be reverse biased when the voltage across the emitter-base junction (EBJ) is positive, that is, $V_{EB} > 0$

In case of *npn* transistors collector base junction is forward biased when the voltage across the collector base junction (CBJ) is negative, that is, $V_{CB} < 0$

The collector base junction is reverse biased when the voltage across the collector base junction (CBJ) is positive, that is, $V_{CB} > 0$

Step 2 of 4

If the emitter-base junction (EBJ) and collector-base junction (CBJ) are reverse biased then the BJT operates in Cutoff mode.

If the emitter-base junction (EBJ) is forward biased collector-base junctions (CBJ) are reverse biased then the BJT operates in Active mode.

If the emitter-base junction (EBJ) is forward biased and the collector base junction (CBJ) is forward biased then the BJT operates in saturation mode.

Step 3 of 4

Fill the table with the condition and the mode of operation.

case	E	B	C	V_{EB}	V_{CB}	Condition for NPN transistor	Mode
1	0	0.7	0.7	**–0.7**	0	$V_{EB} < 0$ EBJ is forward biased $V_{CB} \geq 0$ CBJ is reverse biased	Active mode
2	0	0.8	0.1	**–0.8**	**–0.7**	$V_{EB} < 0$ EBJ is forward biased $V_{CB} < 0$ CBJ is forward biased	Saturation mode
3	**–0.7**	0	1.0	**–0.7**	1.0	$V_{EB} < 0$ EBJ is forward biased $V_{CB} > 0$ CBJ is reverse biased	Active mode

Step 4 of 4

case	E	B	C	V_{EB}	V_{CB}	Condition for NPN transistor	Mode
4	**–0.7**	0	**–0.6**	**–0.7**	**–0.6**	$V_{EB} < 0$ EBJ is forward biased $V_{CB} < 0$ CBJ is forward biased	Saturation mode
5	1.3	2.0	5.0	**–0.7**	3	$V_{EB} < 0$ EBJ is forward biased $V_{CB} > 0$ CBJ is reverse biased	Active mode
6	0	0	5.0	0	5	$V_{EB} \geq 0$ EBJ is reverse biased $V_{CB} > 0$ CBJ is reverse biased	Cut off mode

Thus, the mode of operation for each case is identified.

Step 1 of 2

6.002E

Write the expression of α .

$$\alpha = \frac{\beta}{\beta + 1}$$

The value of α when $\beta = 50$ is,

$$\begin{aligned}\alpha &= \frac{\beta}{\beta + 1} \\ &= \frac{50}{50 + 1} \\ &= 0.980\end{aligned}$$

Step 2 of 2

The value of α when $\beta = 150$ is,

$$\begin{aligned}\alpha &= \frac{\beta}{\beta + 1} \\ &= \frac{150}{150 + 1} \\ &= 0.993\end{aligned}$$

Thus, the range of α values is $\boxed{0.980 \text{ to } 0.993}$.

Step 1 of 3

The relation between collector current i_C and saturation current I_S is,

$$i_C = I_S e^{v_{BE}/V_T}$$

Base-Emitter voltages v_{BE} of two transistors is $0.75\ \text{V}$

Thermal voltage V_T is $25\ \text{mV}$ approximately at room temperature.

Determine the saturation current of the first transistor by substituting 0.5 mA for the collector current.

$$i_{C1} = I_{S1} e^{v_{BE}/V_T}$$
$$0.5\times10^{-3} = I_{S1} e^{0.75/(25\times10^{-3})}$$
$$0.5\times10^{-3} = I_{S1} e^{30}$$
$$I_{S1} = 4.68\times10^{-17}\ \text{A}$$

Thus, the saturation current I_{S1} for the first transistor is $\boxed{4.68\times10^{-17}\ \text{A}}$.

Step 2 of 3

Determine the saturation current of the first transistor by substituting 2 mA for the collector current.

$$i_{C2} = I_{S2} e^{v_{BE}/V_T}$$
$$2\times10^{-3} = I_{S2} e^{0.75/(25\times10^{-3})}$$
$$2\times10^{-3} = I_{S2} e^{30}$$
$$I_{S2} = 1.87\times10^{-16}\ \text{A}$$

Thus, the saturation current I_{S2} for the second transistor is $\boxed{1.87\times10^{-16}\ \text{A}}$.

Step 3 of 3

The relation between saturation current and junction area is,

$$I_S = A_E q D_n n_{P0}/W$$

Two transistors are fabricated with the same technology, but with different junction areas.

The saturation current for the first transistor is,

$$I_{S1} = A_{E1} q D_n n_{P0}/W \quad \text{..... (1)}$$

The saturation current for the second transistor is,

$$I_{S2} = A_{E2} q D_n n_{P0}/W \quad \text{..... (2)}$$

Divide equation (1) by equation (2).

$$\frac{I_{S1}}{I_{S2}} = \frac{A_{E1} q D_n n_{P0}/W}{A_{E2} q D_n n_{P0}/W}$$
$$\frac{4.68\times10^{-17}\ \text{A}}{1.87\times10^{-16}\ \text{A}} = \frac{A_{E1}}{A_{E2}}$$
$$A_{E2} = 4A_{E1}$$

Thus, the junction area of second transistor is **four times** the junction area of the first transistor.

Step 1 of 3

The emitter current is,

$$i_E = i_C + i_B$$
$$i_E = \beta i_B + i_B \qquad (\text{Since } i_C = \beta i_B)$$
$$i_E = (\beta + 1)\, i_B$$
$$\beta + 1 = \frac{i_E}{i_B}$$

Simplify further.

$$\beta = \frac{i_E}{i_B} - 1$$
$$\beta = \frac{1.460 \times 10^{-3}}{14.46 \times 10^{-6}} - 1$$
$$\beta = 101 - 1$$
$$\beta = 100$$

Hence the value of β is $\boxed{100}$.

Step 2 of 3

The value of α is

$$\alpha = \frac{\beta}{\beta + 1}$$
$$= \frac{100}{100 + 1}$$
$$\alpha = 0.99$$

Hence the value of α is $\boxed{0.99}$

Step 3 of 3

Write the expression of the emitter current i_E.

$$i_E = \frac{I_S}{\alpha} e^{\frac{V_{BE}}{V_T}}$$
$$I_S = \frac{\alpha i_E}{e^{\frac{V_{BE}}{V_T}}}$$

Substitute 0.99 for α , $1.46\ \text{mA}$ for I_E , 0.7 V for V_{BE} and $25\ \text{mV}$ for V_T .

$$i_s = \frac{0.99\left(1.460 \times 10^{-3}\right)}{e^{\frac{0.7}{25 \times 10^{-3}}}}$$
$$i_s = \frac{0.99\left(1.460 \times 10^{-3}\right)}{1.446 \times 10^{12}}$$
$$i_s = 10^{-15}\ \text{A}$$

Hence the value of i_S is $\boxed{10^{-15}\ \text{A}}$.

Step 1 of 4

6.003P

The relation between collector current i_C and base-emitter voltage v_{BE} for active mode operated transistor is,

$$i_C = I_S e^{v_{BE}/V_T}$$

The collector current of smaller BJT operating at v_{BE} equal to $30V_T$ is $200\ \mu A$

$$i_{C1} = I_{S1} e^{v_{BE}/V_T}$$
$$200\times10^{-6} = I_{S1} e^{30V_T/V_T}$$
$$I_{S1} e^{30} = 200\times10^{-6}$$
$$I_{S1} = 1.87\times10^{-17}\ \text{A}$$

Thus, the saturation current for smaller BJT is $\boxed{1.87\times10^{-17}\ \text{A}}$

Step 2 of 4

The relation between saturation current and junction area is,

$$I_S = A_E q D_n n_{P0}/W$$

Another transistor manufactured with the same technology but has a emitter junction cross section equal to 32 times the emitter junction area of smaller BJT.

Foe smaller transistor saturation current is,

$$I_{S1} = A_E q D_n n_{P0}/W \quad \text{..... (1)}$$

The saturation current for later transistor having a emitter junction cross section 32 times the emitter junction area of smaller BJT is,

$$I_{S2} = 32 A_E q D_n n_{P0}/W \quad \text{..... (2)}$$

Divide equation (2) by (1).

$$\frac{I_{S2}}{I_{S1}} = \frac{32 A_E q D_n n_{P0}/W}{A_E q D_n n_{P0}/W}$$
$$I_{S2} = 32 I_{S1}$$
$$I_{S2} = 32\times\left(1.87\times10^{-17}\ \text{A}\right)$$
$$I_{S2} = 59.84\times10^{-17}\ \text{A}$$

Thus, the saturation current for the transistor (I_{S2}) having a emitter junction cross section 32 times the emitter junction area of smaller BJT is $\boxed{59.84\times10^{-17}\ \text{A}}$

Step 3 of 4

The collector current of latter BJT operating at v_{BE} equal to $30V_T$ is,

$$\begin{aligned} i_{C2} &= I_{S2} e^{v_{BE}/V_T} \\ &= 59.84\times10^{-17} e^{30V_T/V_T} \\ &= 6.4\times10^{-3}\ \text{A} \\ &= 6.4\ \text{mA} \end{aligned}$$

Thus, the collector current of the BJT operating at v_{BE} equal to $30V_T$ is $\boxed{6.4\ \text{mA}}$

Step 4 of 4

The relation between collector current i_C and base-emitter voltage v_{BE} for active mode operated latter transistor is,

$$i_{C2} = I_{S2} e^{v_{BE}/V_T}$$

The transistor has a collector current of $1\ \text{mA}$ for base- emitter voltage of v_{BE}.

$$i_{C2} = I_{S2} e^{v_{BE}/V_T}$$
$$1\times10^{-3} = \left(59.84\times10^{-17}\ \text{A}\right) e^{v_{BE}/V_T}$$
$$1.67\times10^{12} = e^{v_{BE}/V_T}$$

Apply natural log on both sides

$$\ln\left(1.67\times10^{12}\right) = \ln\left(e^{v_{BE}/V_T}\right)$$
$$28.14 = \frac{v_{BE}}{V_T}$$
$$v_{BE} = 28.14 V_T$$

Thus, the transistor has a collector current of $1\ \text{mA}$ for base- emitter voltage v_{BE} equal to $\boxed{28.14V_T}$.

6.004E

Step 1 of 4

Write the expression of β.

$$\beta = \frac{\alpha}{1-\alpha}$$

Substitute 0.99 for α.

$$\begin{aligned}\beta &= \frac{0.99}{1-0.99}\\ &= 99\end{aligned}$$

Thus, the value of β for which $\alpha = 0.99$ is $\boxed{99}$.

Step 2 of 4

The base current of the transistor for $\alpha = 0.99$ is,

$$\begin{aligned}i_B &= \frac{i_C}{\beta}\\ &= \frac{10\times10^{-3}}{99}\\ &= 0.1\times10^{-3}\\ &= 0.1\ \text{mA}\end{aligned}$$

Thus, the base current for $\alpha = 0.99$ is $\boxed{0.1\ \text{mA}}$.

Step 3 of 4

Write the expression of β.

$$\beta = \frac{\alpha}{1-\alpha}$$

Substitute 0.98 for α.

$$\begin{aligned}\beta &= \frac{0.98}{1-0.98}\\ &= 49\end{aligned}$$

Thus, the value of β for which $\alpha = 0.98$ is $\boxed{49}$.

Step 4 of 4

The base current of the transistor for $\alpha = 0.98$ is,

$$\begin{aligned}i_B &= \frac{i_C}{\beta}\\ &= \frac{10\times10^{-3}}{49}\\ &= 0.2\times10^{-3}\\ &= 0.2\ \text{mA}\end{aligned}$$

Thus, the base current for $\alpha = 0.98$ is $\boxed{0.2\ \text{mA}}$.

Step 1 of 2

6.004P

The relation between saturation current and junction area is,

$$I_S = A_E q D_n n_{P0}/W$$

Two transistors are fabricated with the same technology, but having different junction areas.

The saturation current for the first transistor is,

$$I_{S1} = A_{E1} q D_n n_{P0}/W \quad \text{..... (1)}$$

The saturation current for the second transistor is,

$$I_{S2} = A_{E2} q D_n n_{P0}/W \quad \text{..... (2)}$$

Active mode operated first transistor has an Emitter-Base junction (**EBJ**) area, A_{E1} equal to $200\ \mu m \times 200\ \mu m$.

Active mode operated second transistor has Emitter-Base junction (**EBJ**) area, A_{E2} equal to $0.4\ \mu m \times 0.4\ \mu m$.

Divide equation (1) by (2).

$$\frac{I_{S1}}{I_{S2}} = \frac{A_{E1} q D_n n_{P0}/W}{A_{E2} q D_n n_{P0}/W}$$

$$\frac{I_{S1}}{I_{S2}} = \frac{A_{E1}}{A_{E2}}$$

$$\frac{I_{S1}}{I_{S2}} = \frac{200\ \mu m \times 200\ \mu m}{0.4\ \mu m \times 0.4\ \mu m}$$

$$\frac{I_{S1}}{I_{S2}} = 25 \times 10^4$$

Hence, the saturation current, I_{S1} in terms of I_{S2} is $25 \times 10^4 I_{S2}$

Step 2 of 2

The relation between collector current i_C and base-emitter voltage v_{BE} for active mode operated transistor is,

$$i_C = I_S e^{v_{BE}/V_T}$$

Collector current for first transistor is $i_{C1} = I_{S1} e^{v_{BE1}/V_T}$ ..... (3)

Collector current for second transistor is $i_{C2} = I_{S2} e^{v_{BE2}/V_T}$ ..... (4)

Two transistors operate in active mode and conduct equal collector currents and hence $i_{C1} = i_{C2}$

Divide equation (4) by equation (3).

$$\frac{i_{C2}}{i_{C1}} = \frac{I_{S2} e^{v_{BE2}/V_T}}{I_{S1} e^{v_{BE1}/V_T}}$$

$$1 = \frac{I_{S2} e^{v_{BE2}/V_T}}{25 \times 10^4 I_{S2} e^{v_{BE1}/V_T}}$$

$$25 \times 10^4 = e^{(v_{BE2} - v_{BE1})}$$

Apply natural log to the equation $25 \times 10^4 = e^{(v_{BE2} - v_{BE1})}$

$$\ln\left(25 \times 10^4\right) = \ln e^{(v_{BE2} - v_{BE1})}$$

$$12.43 = \left(V_{BE2} - V_{BE1}\right)$$

Thus, the difference in the V_{BE} values of two transistors is $\boxed{12.43\ V}$.

Step 1 of 3

Write the expression of the base-emitter voltage of the transistor.

$$V_{BE} = V_T \ln\left(\frac{\frac{I_B}{I_S}}{\beta}\right)$$

Substitute $I_B = \frac{I_C}{\beta}$.

$$V_{BE} = V_T \ln\left(\frac{\frac{\frac{I_C}{\beta}}{I_S}}{\beta}\right)$$
$$= V_T \ln\left(\frac{I_C}{I_S}\right)$$

Substitute $1\ \text{mA}$ for I_C, $10^{-16}\ \text{A}$ for I_S and $25\ \text{mV}$ for V_T.

$$V_{BE} = (25\times10^{-3})\ln\left(\frac{1\times10^{-3}}{10^{-16}}\right)$$
$$= (25\times10^{-3})\ln(10^{13})$$
$$= (25\times10^{-3})(29.934)$$
$$= 748.3\ \text{mV}$$

Thus, the value of V_{BE} is $\boxed{748.3\ \text{mV}}$.

Step 2 of 3

Write the expression of the current I_{SE}.

$$I_{SE} = \frac{I_S}{\alpha}$$
$$= \frac{I_S}{\frac{\beta}{\beta+1}}$$

Substitute $10^{-16}\ \text{A}$ for I_S and 100 for β.

$$I_{SE} = \frac{10^{-16}}{\frac{100}{101}}$$
$$= \frac{101\times10^{-16}}{100}$$
$$= 1.01\times10^{-16}\ \text{A}$$

Thus, the value of the current I_{SE} is $\boxed{1.01\times10^{-16}\ \text{A}}$.

Step 3 of 3

Write the expression of the current I_{SB}.

$$I_{SB} = \frac{I_S}{\beta}$$

Substitute $10^{-16}\ \text{A}$ for I_S and 100 for β.

$$I_{SB} = \frac{10^{-16}}{100}$$
$$= 10^{-18}\text{A}$$

Thus, the value of the current I_{SB} is $\boxed{10^{-18}\ \text{A}}$.

Step 1 of 4

The relation between collector current (i_C), saturation current (I_S) and base – emitter voltage v_{BE} for active mode operated transistor is,

$$i_C = I_S e^{v_{BE}/V_T}$$

Thermal voltage V_T is $25\ \text{mV}$ approximately at room temperature

Determine the collector current for Base – emitter voltage v_{BE} equal to $700\ \text{mV}$ and saturation current I_S equal to $10^{-13}\ \text{A}$.

$$\begin{aligned} i_C &= I_S e^{v_{BE}/V_T} \\ &= 10^{-13}\left(e^{\frac{700\times10^{-3}}{25\times10^{-3}}}\right) \\ &= 10^{-12}e^{28} \\ &= 1.45\ \text{A} \end{aligned}$$

Thus, the collector current for Base – emitter voltage v_{BE} equal to $700\ \text{mV}$ and saturation current I_S equal to $10^{-13}\ \text{A}$ is $\boxed{1.45\ \text{A}}$.

Step 2 of 4

Determine the collector current for Base – emitter voltage v_{BE} equal to $700\ \text{mV}$ and saturation current I_S equal to $10^{-18}\ \text{A}$.

$$\begin{aligned} i_C &= I_S e^{v_{BE}/V_T} \\ &= 10^{-18}\left(e^{\frac{700\times10^{-3}}{25\times10^{-3}}}\right) \\ &= 10^{-18}e^{28} \\ &= 1.45\ \mu\text{A} \end{aligned}$$

Thus, the collector current for Base – emitter voltage v_{BE} equal to $700\ \text{mV}$ and saturation current I_S equal to $10^{-18}\ \text{A}$ is $\boxed{1.45\ \mu\text{A}}$

Step 3 of 4

The relation between saturation current and junction area is $I_S = A_E q D_n n_{P0}/W$. Therefore, saturation current is directly proportional to junction area of transistor.

Higher is the saturation current, larger is the junction area and hence the transistor having a saturation current of $10^{-13}\ \text{A}$ has larger junction area.

Transistor having saturation current of $10^{-13}\ \text{A}$ is the larger transistor and transistor having saturation current of $10^{-18}\ \text{A}$ is the smaller transistor.

Step 4 of 4

Observe that the Collector current for smaller transistor having v_{BE} equal to $700\ \text{mV}$ is $1.45\ \mu\text{A}$.

Larger transistor collector current can be made equal to smaller transistor collector current by adjusting v_{BE} value of the larger transistor.

Collector current of smaller transistor is equal to larger transistor and hence collector current for larger transistor is $1.45\ \mu\text{A}$

For larger transistor,

$$i_C = I_S e^{v_{BE}/V_T}$$

$$1.45\times10^{-6} = 10^{-13}\left(e^{\frac{v_{BE}}{25\times10^{-3}}}\right)$$

$$145\times10^{5} = e^{\frac{v_{BE}}{25\times10^{-3}}}$$

Apply natural log to the above equation

$$\ln\left(145\times10^{5}\right) = \ln\left(e^{\frac{v_{BE}}{25\times10^{-3}}}\right)$$

$$16.49 = \frac{v_{BE}}{25\times10^{-3}}$$

$$0.412 = v_{BE}$$

Thus, the voltage v_{BE} of the larger transistor in order that the collector current of larger transistor is equal to that of the smaller transistor is $\boxed{0.412\ \text{V}}$

Refer to Figure 6.6(a) in the text book.

Calculate the value of resistor, R_C .

$$R_C = \frac{V_{CC} - V_{CE}}{I_C}$$

Substitute $5\ \text{V}$ for V_{CC} and $0.69\ \text{V}$ for V_{CE} and $1\ \text{mA}$ for I_C .

$$\begin{aligned} R_C &= \frac{5 - 0.69}{1 \times 10^{-3}} \\ &= 4.31 \times 10^{3} \\ &= 4.31\ \text{k}\Omega \end{aligned}$$

Thus, the value of the resistor, R_C is $\boxed{4.31\ \text{k}\Omega}$.

Step 1 of 2

The relation between collector current i_C and base-emitter voltage v_{BE} for active mode operated transistor is $i_C = I_S e^{v_{BE}/V_T}$

The saturation current of old technology fabricated transistor I_S is $2\times10^{-15}\ \text{A}$

The collector current of old technology fabricated transistor is $1\ \text{mA}$

The thermal voltage V_T is $25\ \text{mV}$ approximately at room temperature

$$i_C = I_S e^{v_{BE}/V_T}$$
$$1\times10^{-3}\ \text{A} = \left(2\times10^{-15}\ \text{A}\right)e^{v_{BE}/V_T}$$
$$5\times10^{11} = e^{v_{BE}/V_T}$$

Apply natural log on both sides

$$\ln\left(5\times10^{11}\right) = \ln e^{\frac{v_{BE}}{V_T}}$$
$$26.94 = \frac{v_{BE}}{V_T}$$
$$v_{BE} = 26.94\left(25\times10^{-3}\right)$$
$$v_{BE} = 0.67\ \text{V}$$

Thus, the base-emitter voltage required for old technology fabricated transistor is $\boxed{0.67\ \text{V}}$.

Step 2 of 2

The saturation current of new technology fabricated transistor I_S is $2\times10^{-18}\ \text{A}$

The collector current of new technology fabricated transistor is $1\ \text{mA}$

Thermal voltage V_T is $25\ \text{mV}$ approximately at room temperature

$$i_C = I_S e^{v_{BE}/V_T}$$
$$1\times10^{-3}\ \text{A} = \left(2\times10^{-18}\ \text{A}\right)e^{v_{BE}/V_T}$$
$$5\times10^{14} = e^{v_{BE}/V_T}$$

Apply natural log on both sides

$$\ln\left(5\times10^{14}\right) = \ln e^{v_{BE}/V_T}$$
$$33.845 = v_{BE}/V_T$$
$$v_{BE} = 33.845\times\left(25\times10^{-3}\right)$$
$$v_{BE} = 0.85\ \text{V}$$

Thus, the base-emitter voltage required for new technology fabricated transistor is $\boxed{0.85\ \text{V}}$.

Determine the current, I_{SE} .

$$\begin{aligned} I_{SE} &= \frac{I_S}{\alpha} \\ &= \frac{10^{-15}}{1} \\ &= 10^{-15}\,\text{A} \end{aligned}$$

The current I_{SC} is 10 to 100 times larger than I_{SE} .

Determine the current I_{SC}.

$$\begin{aligned} I_{SC} &= 100 I_{SE} \\ &= 100 \times 10^{-15} \\ &= 10^{-13}\,\text{A} \end{aligned}$$

Thus, the current, I_{SC} is $\boxed{10^{-13}\,\text{A}}$.

Step 1 of 3

6.007P

The relation between collector current i_C and base-emitter voltage v_{BE} is,

$$i_C = I_S e^{v_{BE}/V_T}$$

The base-emitter (v_{BE}) drop of *npn* transistor is $0.76\ \text{V}$

Thermal voltage V_T is $25\ \text{mV}$ approximately at room temperature

The collector current i_C for v_{BE} equal to $0.76\ \text{V}$ is $5\ \text{mA}$

$$i_C = I_S e^{v_{BE}/V_T}$$

$$5\times10^{-3} = I_S e^{\frac{0.76}{25\times10^{-3}}}$$

$$I_S = \frac{5\times10^{-3}}{e^{\frac{0.76}{25\times10^{-3}}}}\ \text{A}$$

$$I_S = 3.14\times10^{-16}\ \text{A}$$

Thus, the saturation current I_S for *npn* transistor is $3.14\times10^{-16}\ \text{A}$

Step 2 of 3

The collector current for v_{BE} equal to $0.70\ \text{V}$ is,

$$i_C = I_S e^{v_{BE}/V_T}$$

$$= 3.14\times10^{-16} e^{\frac{0.70}{25\times10^{-3}}}\ \text{A}$$

$$= 3.14\times10^{-16} e^{28}\ \text{A}$$

$$= 0.45\times10^{-3}\ \text{A}$$

$$= 0.45\ \text{mA}$$

Thus, the collector current for v_{BE} equal to $0.70\ \text{V}$ is $\boxed{0.45\ \text{mA}}$.

Step 3 of 3

Determine the base-emitter voltage v_{BE} for collector current i_C equal to $5\ \mu\text{A}$.

$$i_C = I_S e^{v_{BE}/V_T}$$

$$5\times10^{-6} = I_S e^{\frac{V_{BE}}{25\times10^{-3}}}$$

$$5\times10^{-6} = 3.14\times10^{-16} e^{\frac{V_{BE}}{25\times10^{-3}}}$$

$$1.59\times10^{10} = e^{\frac{V_{BE}}{25\times10^{-3}}}$$

$$\ln\left(1.59\times10^{10}\right) = \ln\left(e^{\frac{V_{BE}}{25\times10^{-3}}}\right)$$

$$23.49 = \frac{V_{BE}}{25\times10^{-3}}$$

$$0.59 = V_{BE}$$

Thus, the base-emitter voltage v_{BE} for collector current i_C equal to $5\ \mu\text{A}$ is $\boxed{0.59\ \text{V}}$.

Step 1 of 2

6.008E

Write the expression for the collector current i_C in the saturation mode.

$$i_C = I_S e^{\frac{v_{BE}}{V_T}} - I_{SC} e^{\frac{v_{BC}}{V_T}}$$

Here,

I_S is the saturation current

V_T is the thermal voltage

v_{BE} is the base-to-emitter voltage

v_{BC} is the base-to-collector voltage

I_{SC} is the collector scale current

Equate i_C to 0.

$$I_S e^{\frac{v_{BE}}{V_T}} - I_{SC} e^{\frac{v_{BC}}{V_T}} = 0$$

$$I_S e^{\frac{v_{BE}}{V_T}} = I_{SC} e^{\frac{v_{BC}}{V_T}}$$

$$\frac{e^{\frac{v_{BE}}{V_T}}}{e^{\frac{v_{BC}}{V_T}}} = \frac{I_{SC}}{I_S}$$

$$e^{\frac{v_{BE}-v_{BC}}{V_T}} = \frac{I_{SC}}{I_S}$$

Apply natural logarithm on both sides.

$$\frac{v_{BE}-v_{BC}}{V_T} = \ln\left(\frac{I_{SC}}{I_S}\right)$$

$$v_{BE}-v_{BC} = V_T \ln\left(\frac{I_{SC}}{I_S}\right)$$

$$v_{CE} = V_T \ln\left(\frac{I_{SC}}{I_S}\right)$$

(As $v_{BE}-v_{BC} = v_{CE}$ in saturation mode)

Thus, the collector current reaches zero at $v_{CE} = V_T \ln\left(\frac{I_{SC}}{I_S}\right)$. Hence proved.

Step 2 of 2

The saturation current of a transistor is proportional to the area.

So, $I_{SC} = 100 I_S$

Calculate V_{CE} for $I_{SC} = 100 I_S$.

$$v_{CE} = V_T \ln\left(\frac{100 I_S}{I_S}\right)$$

Substitute 25×10^{-3} V for V_T.

$$v_{CE} = \left(25\times10^{-3}\right)\ln\left(\frac{100 I_S}{I_S}\right)$$

$$= 115 \text{ mV}$$

Thus, the voltage V_{CE} is $\boxed{115 \text{ mV}}$.

Step 1 of 2

6.008P

The common emitter current gain β is $\frac{i_C}{i_B}$

The collector current i_C is $800\ \mu A$

The base current i_B is $10\ \mu A$

Determine the gain β .

$$\begin{aligned}\beta &= \frac{i_C}{i_B} \\ &= \frac{800\ \mu A}{10\ \mu A} \\ &= 80\end{aligned}$$

Thus, the common emitter current gain β is $\boxed{80}$.

Step 2 of 2

Write the formula for α in terms of β .

$$\alpha = \frac{\beta}{1+\beta}$$

Substitute 80 for β .

$$\begin{aligned}\alpha &= \frac{\beta}{1+\beta} \\ &= \frac{80}{1+80} \\ &= 0.9876\end{aligned}$$

Thus, the value of α for $\beta = 80$ is $\boxed{0.9876}$.

Step 1 of 3

The collector current of the transistor in saturation mode is,

$$i_C = I_S e^{\frac{V_{BE}}{V_T}} - I_{SC} e^{\frac{V_{BC}}{V_T}} \quad \text{..... (1)}$$

The base current of the transistor in saturation mode is,

$$i_B = \frac{I_S}{\beta} e^{\frac{V_{BE}}{V_T}} + I_{SC} e^{\frac{V_{BC}}{V_T}} \quad \text{..... (2)}$$

The forced common emitter current gain is,

$$\beta_{forced} = \left.\frac{i_C}{i_B}\right|_{saturation} \le \beta \quad \text{..... (3)}$$

Substitute the expressions for i_C and i_B from equations (1) and (2) in equation (3).

$$\begin{aligned}\beta_{forced} &= \left.\frac{i_C}{i_B}\right|_{saturation} \\ &= \frac{I_S e^{\frac{V_{BE}}{V_T}} - I_{SC} e^{\frac{V_{BC}}{V_T}}}{\frac{I_S}{\beta} e^{\frac{V_{BE}}{V_T}} + I_{SC} e^{\frac{V_{BC}}{V_T}}} \\ &= \beta \frac{I_S e^{\frac{V_{BE}}{V_T}} - I_{SC} e^{\frac{V_{BC}}{V_T}}}{I_S e^{\frac{V_{BE}}{V_T}} + \beta I_{SC} e^{\frac{V_{BC}}{V_T}}}\end{aligned}$$

Divide both numerator and denominator by $e^{\frac{V_{BC}}{V_T}}$.

$$\begin{aligned}\beta_{forced} &= \beta \frac{e^{\frac{V_{BC}}{V_T}} \left(I_S e^{\frac{V_{BE}}{V_T}} - I_{SC} e^{\frac{V_{BC}}{V_T}} \right)}{e^{\frac{V_{BC}}{V_T}} \left(I_S e^{\frac{V_{BE}}{V_T}} + \beta I_{SC} e^{\frac{V_{BC}}{V_T}} \right)} \\ &= \beta \frac{\left(I_S e^{\frac{V_{BE}-V_{BC}}{V_T}} - I_{SC} \right)}{\left(I_S e^{\frac{V_{BE}-V_{BC}}{V_T}} + \beta I_{SC} \right)}\end{aligned}$$

Step 2 of 3

The collector-to-emitter voltage of a saturated transistor is,

$$V_{CEsat} = V_{BE} - V_{BC}$$

Substitute V_{CEsat} for $V_{BE} - V_{BC}$.

$$\begin{aligned}\beta_{forced} &= \beta \frac{\left(I_S e^{\frac{V_{CEsat}}{V_T}} - I_{SC} \right)}{\left(I_S e^{\frac{V_{CEsat}}{V_T}} + \beta I_{SC} \right)} \\ &= \beta \frac{I_S \left(e^{\frac{V_{CEsat}}{V_T}} - \frac{I_{SC}}{I_S} \right)}{I_S \left(e^{\frac{V_{CEsat}}{V_T}} + \beta \frac{I_{SC}}{I_S} \right)} \\ &= \beta \frac{e^{\frac{V_{CEsat}}{V_T}} - \frac{I_{SC}}{I_S}}{e^{\frac{V_{CEsat}}{V_T}} + \beta \frac{I_{SC}}{I_S}}\end{aligned}$$

Thus the forced β is, $\boxed{\beta_{forced} = \beta \dfrac{e^{\frac{V_{CEsat}}{V_T}} - \frac{I_{SC}}{I_S}}{e^{\frac{V_{CEsat}}{V_T}} + \beta \frac{I_{SC}}{I_S}}}$.

Step 3 of 3

Determine the value of β_{forced} by substituting 100 for β, 100 for $\frac{I_{SC}}{I_S}$, 25×10^{-3} V for V_T and 0.2 V for V_{CEsat}.

$$\begin{aligned}\beta_{forced} &= 100 \frac{\left(e^{\frac{0.2}{25\times10^{-3}}} - 100 \right)}{\left(e^{\frac{0.2}{25\times10^{-3}}} + 100 \times 100 \right)} \\ &= 100 \frac{\left(e^{8} - 100 \right)}{\left(e^{8} + 100 \times 100 \right)} \\ &= 100 \frac{(2980.96 - 100)}{(2980.96 + 100 \times 100)} \\ &= 22.2\end{aligned}$$

Thus, the value of forced β is $\boxed{22.2}$.

6.009P

Step 1 of 8

Write the formula for β in terms of α.

$$\beta = \frac{\alpha}{1-\alpha}$$

Substitute 0.5 for α.

$$\begin{aligned}\beta &= \frac{\alpha}{1-\alpha}\\ &= \frac{0.5}{1-0.5}\\ &= 1\end{aligned}$$

Thus, the value of β for $\alpha = 0.5$ is $\boxed{1}$.

Step 2 of 8

Determine the value of β for $\alpha = 0.8$.

$$\begin{aligned}\beta &= \frac{\alpha}{1-\alpha}\\ &= \frac{0.8}{1-0.8}\\ &= 4\end{aligned}$$

Thus, the value of β for $\alpha = 0.8$ is $\boxed{4}$.

Step 3 of 8

Determine the value of β for $\alpha = 0.9$.

$$\begin{aligned}\beta &= \frac{\alpha}{1-\alpha}\\ &= \frac{0.9}{1-0.9}\\ &= 9\end{aligned}$$

Thus, the value of β for $\alpha = 0.9$ is $\boxed{9}$.

Step 4 of 8

Determine the value of β for $\alpha = 0.95$.

$$\begin{aligned}\beta &= \frac{\alpha}{1-\alpha}\\ &= \frac{0.95}{1-0.95}\\ &= 19\end{aligned}$$

Thus, the value of β for $\alpha = 0.95$ is $\boxed{19}$.

Step 5 of 8

Determine the value of β for $\alpha = 0.98$.

$$\begin{aligned}\beta &= \frac{\alpha}{1-\alpha}\\ &= \frac{0.98}{1-0.98}\\ &= 49\end{aligned}$$

Thus, the value of β for $\alpha = 0.98$ is $\boxed{49}$.

Step 6 of 8

Determine the value of β for $\alpha = 0.99$.

$$\begin{aligned}\beta &= \frac{\alpha}{1-\alpha}\\ &= \frac{0.99}{1-0.99}\\ &= 99\end{aligned}$$

Thus, the value of β for $\alpha = 0.99$ is $\boxed{99}$.

Step 7 of 8

Determine the value of β for $\alpha = 0.995$.

$$\begin{aligned}\beta &= \frac{\alpha}{1-\alpha}\\ &= \frac{0.995}{1-0.995}\\ &= 199\end{aligned}$$

Thus, the value of β for $\alpha = 0.995$ is $\boxed{199}$.

Step 8 of 8

Determine the value of β for $\alpha = 0.999$.

$$\begin{aligned}\beta &= \frac{\alpha}{1-\alpha}\\ &= \frac{0.999}{1-0.999}\\ &= 999\end{aligned}$$

Thus, the value of β for $\alpha = 0.999$ is $\boxed{999}$.

Step 1 of 3

6.010E

Refer to Figure 6.11 in the text book for the large-signal model of the pnp transistor.

Determine the transistor parameter α using the following relation:

$$\begin{aligned}\alpha &= \frac{\beta}{\beta+1}\\ &= \frac{50}{50+1}\\ &= 0.98\end{aligned}$$

Write the expression for the emitter current, i_E.

$$i_E = \left(\frac{I_S}{\alpha}\right)e^{\frac{v_{EB}}{V_T}}$$

$$2\times10^{-3} = \left(\frac{10^{-14}}{0.98}\right)e^{\frac{v_{EB}}{25\times10^{-3}}}$$

$$e^{\frac{v_{EB}}{25\times10^{-3}}} = 1.96\times10^{11}$$

$$\frac{v_{EB}}{25\times10^{-3}} = \ln\left(1.96\times10^{11}\right)$$

$$\frac{v_{EB}}{25\times10^{-3}} = 26.0014$$

$$v_{EB} = 26.0014\left(25\times10^{-3}\right)$$

$$v_{EB} = 0.65\ \text{V}$$

Hence, the emitter voltage is $\boxed{0.65\ \text{V}}$.

Step 2 of 3

The base current is,

$$\begin{aligned}i_B &= \frac{i_E}{1+\beta}\\ &= \frac{2\times10^{-3}}{1+50}\\ &= 39.2\times10^{-6}\\ i_B &= 39.2\ \mu\text{A}\end{aligned}$$

Hence, the base current i_B is $\boxed{39.2\ \mu\text{A}}$.

Step 3 of 3

The collector current is,

$$\begin{aligned}i_C &= \beta i_B\\ &= (50)\left(39.2\times10^{-6}\right)\\ &= 1.96\times10^{-3}\\ i_C &= 1.96\ \text{mA}\end{aligned}$$

Hence, the collector current i_C is $\boxed{1.96\ \text{mA}}$.

Step 1 of 6

Write the formula for α in terms of β.

$$\alpha = \frac{\beta}{1+\beta}$$

Substitute 1 for β.

$$\begin{aligned}\alpha &= \frac{\beta}{1+\beta}\\ &= \frac{1}{1+1}\\ &= 0.5\end{aligned}$$

Thus, the value of α for $\beta = 1$ is $\boxed{0.5}$.

Step 2 of 6

Determine the value of α for $\beta = 2$.

$$\begin{aligned}\alpha &= \frac{\beta}{1+\beta}\\ &= \frac{2}{1+2}\\ &= 0.67\end{aligned}$$

Thus, the value of α for $\beta = 2$ is $\boxed{0.67}$.

Determine the value of α for $\beta = 10$.

$$\begin{aligned}\alpha &= \frac{\beta}{1+\beta}\\ &= \frac{10}{1+10}\\ &= 0.91\end{aligned}$$

Thus, the value of α for $\beta = 10$ is $\boxed{0.91}$.

Step 3 of 6

Determine the value of α for $\beta = 20$.

$$\begin{aligned}\alpha &= \frac{\beta}{1+\beta}\\ &= \frac{20}{1+20}\\ &= 0.95\end{aligned}$$

Thus, the value of α for $\beta = 20$ is $\boxed{0.95}$.

Determine the value of α for $\beta = 50$.

$$\begin{aligned}\alpha &= \frac{\beta}{1+\beta}\\ &= \frac{50}{1+50}\\ &= 0.98\end{aligned}$$

Thus, the value of α for $\beta = 50$ is $\boxed{0.98}$.

Step 4 of 6

Determine the value of α for $\beta = 100$.

$$\begin{aligned}\alpha &= \frac{\beta}{1+\beta}\\ &= \frac{100}{1+100}\\ &= 0.99\end{aligned}$$

Thus, the value of α for $\beta = 100$ is $\boxed{0.99}$.

Step 5 of 6

Determine the value of α for $\beta = 200$.

$$\begin{aligned}\alpha &= \frac{\beta}{1+\beta}\\ &= \frac{200}{1+200}\\ &= 0.995\end{aligned}$$

Thus, the value of α for $\beta = 200$ is $\boxed{0.995}$.

Step 6 of 6

Determine the value of α for $\beta = 500$.

$$\begin{aligned}\alpha &= \frac{\beta}{1+\beta}\\ &= \frac{500}{1+500}\\ &= 0.998\end{aligned}$$

Thus, the value of α for $\beta = 500$ is $\boxed{0.998}$.

Determine the value of α for $\beta = 1000$.

$$\begin{aligned}\alpha &= \frac{\beta}{1+\beta}\\ &= \frac{1000}{1+1000}\\ &= 0.999\end{aligned}$$

Thus, the value of α for $\beta = 1000$ is $\boxed{0.999}$.

Write the formula for the collector current, i_C .

$$i_C = I_S e^{\frac{v_{EB}}{V_T}}$$

$$v_{EB} = V_T \ln\left(\frac{i_C}{I_S}\right)$$

Here,

v_{EB} is the base-emitter voltage

I_S is the saturation current of the transistor

V_T is the thermal voltage

Substitute 1.5 A for i_C , 25×10^{-3} V for V_T and 10^{-11} A for I_S .

$$v_{EB} = \left(25\times10^{-3}\right)\ln\left(\frac{1.5}{10^{-11}}\right)$$

$$= 0.643 \text{ V}$$

Thus, the base-emitter voltage v_{EB} is $\boxed{0.643 \text{ V}}$.

Step 1 of 3

The relation between α and β for a transistor is,

$$\alpha = \frac{\beta}{1+\beta}$$

Generally most of the transistors have β value is greater than 80

Consider β value to be 100 .

$$\begin{aligned}\alpha &= \frac{\beta}{1+\beta}\\ &= \frac{100}{1+100}\\ &= 0.99\\ &\simeq 1\end{aligned}$$

Hence the transistor has α close to unity.

Step 2 of 3

If α changes by a small amount $\Delta\alpha$, the new value of α is

$$\begin{aligned}\beta_1 &= \frac{\alpha}{1-\alpha}\\ &= \frac{\alpha+\Delta\alpha}{1-\alpha-\Delta\alpha}\end{aligned}$$

Determine the change in β .

$$\begin{aligned}\frac{\Delta\beta}{\beta} &= \frac{\beta_1-\beta}{\beta}\\ &= \frac{\dfrac{\alpha+\Delta\alpha}{1-\alpha-\Delta\alpha}-\dfrac{\alpha}{1-\alpha}}{\dfrac{\alpha}{1-\alpha}}\\ &\approx \beta\frac{\Delta\alpha}{\alpha}\end{aligned}$$

Thus, per unit change in β is $\boxed{\frac{\Delta\beta}{\beta} = \beta\frac{\Delta\alpha}{\alpha}}$.

Step 3 of 3

The relation between small change in α and small change in β is,

$$\frac{\Delta\beta}{\beta} \simeq \beta\left(\frac{\Delta\alpha}{\alpha}\right)$$

Drop in β value is 10% hence new value of β is $90, \Delta\beta = 10$

Percent change in α is $\left(\frac{\Delta\alpha}{\alpha}\times 100\right)$

Nominal value of β is 100

$$\begin{aligned}\frac{\Delta\beta}{\beta} &\simeq \beta\left(\frac{\Delta\alpha}{\alpha}\right)\\ \frac{\Delta\beta}{\beta}\times 100 &= \beta\left(\frac{\Delta\alpha}{\alpha}\times 100\right)\\ \frac{10}{100}\times 100 &= 100\left(\frac{\Delta\alpha}{\alpha}\times 100\right)\\ \left(\frac{\Delta\alpha}{\alpha}\times 100\right) &= 0.1\%\end{aligned}$$

Percent change in α value corresponds to 10% drop in β is $\boxed{0.1\%}$.

Step 1 of 4

Refer to Figure 6.14 in the text book for the transistor circuit.

The dc power supply utilized is $\pm 1.5\ \text{V}$.

The common-emitter current gain of the transistors is,

$\beta = 100$.

The base to emitter voltage of the transistors is,

$V_{BE} = 0.8\ \text{V}$.

This V_{BE} exists at the collector current of $i_C = 1\ \text{mA}$.

To obtain a collector voltage V_C of $+0.5\ \text{V}$, the voltage drop across the resistor R_C must be equal to $1.5 - 0.5 = 1\ \text{V}$.

Determine the resistance R_C.

$$\begin{aligned} R_C &= \frac{1}{I_C} \\ &= \frac{1}{2\times 10^{-3}} \\ &= 500\ \Omega \end{aligned}$$

Thus the value of resistance R_C is $\boxed{500\ \Omega}$.

Step 2 of 4

Determine the base to emitter voltage of the transistor.

$$\begin{aligned} V_{BE} &= v_{BE} + V_T \ln\left(\frac{I_C}{i_C}\right) \\ &= 0.8 + 25\times 10^{-3} \ln\left(\frac{2}{1}\right) \\ &= 0.817\ \text{V} \end{aligned}$$

Since the base is at $0\ \text{V}$, the emitter voltage is,

$$\begin{aligned} V_E &= -V_{BE} \\ &= -0.817\ \text{V} \end{aligned}$$

Step 3 of 4

The common-base current gain for $\beta = 100$ is,

$$\begin{aligned} \alpha &= \frac{\beta}{1+\beta} \\ &= \frac{100}{101} \\ &= 0.99 \end{aligned}$$

The emitter current is,

$$\begin{aligned} I_E &= \frac{I_C}{\alpha} \\ &= \frac{2}{0.99} \\ &= 2.02\ \text{mA} \end{aligned}$$

Step 4 of 4

Determine the required value of the emitter resistance.

$$R_E = \frac{V_E - (-1.5)}{I_E}.$$

Substitute $-0.817\ \text{V}$ for V_E and $2.02\times 10^{-3}\ \text{A}$ for I_E.

$$\begin{aligned} R_E &= \frac{-0.817 + 1.5}{2.02\times 10^{-3}} \\ &= \frac{0.683}{2.02\times 10^{-3}} \\ &= 338\ \Omega \end{aligned}$$

Thus, the emitter resistance R_E is $\boxed{338\ \Omega}$.

Step 1 of 4

The current injected into the base I_B is $10\ \mu A$

The common emitter current gain β is $\dfrac{\text{collector current } (I_C)}{\text{base current } (I_B)}$

The range of β is 50 to 300.

Consider value of β as 50, then collector current I_C is,

$$\beta = \frac{\text{collector current } (I_C)}{\text{base current } (I_B)}$$

$$50 = \frac{I_C}{(10 \times 10^{-6})}$$

$$I_C = 500\ \mu A$$

Thus, the collector current for β equal to 50 is $\boxed{500\ \mu A}$.

Step 2 of 4

Consider the value of β as 300, then collector current I_C is,

$$\beta = \frac{\text{collector current } (I_C)}{\text{base current } (I_B)}$$

$$300 = \frac{I_C}{(10 \times 10^{-6})}$$

$$I_C = 3\ mA$$

Thus, the collector current for β equal to 300 is $\boxed{3\ mA}$

Thus, the range of collector current for β between (50,300) is $\boxed{(500\ \mu A, 3\ mA)}$.

Step 3 of 4

The relation between collector current, base current and emitter current is,

$$I_E = I_B + I_C$$

Determine the emitter current for β equal to 50.

$$I_E = I_B + I_C$$
$$= 10\ \mu A + 500\ \mu A$$
$$= 510\ \mu A$$

Thus, the emitter current for β equal to 50 is $\boxed{510\ \mu A}$.

Step 4 of 4

Determine the emitter current for β equal to 300.

$$I_E = I_B + I_C$$
$$= 10\ \mu A + 3\ mA$$
$$= 3.01\ mA$$

Thus, the emitter current for β equal to 300 is $\boxed{3.01\ mA}$.

Thus, the range of the emitter current for β between (50,300) is $\boxed{(510\ \mu A, 3.01\ mA)}$.

Step 1 of 3

Refer to Figure E6.13 in the text book for the transistor circuit.

From the circuit, the emitter voltage V_E is,

$$V_E = I_E R_E - 10$$

Substitute $-0.7\ \text{V}$ for V_E and $10\times10^3\ \Omega$ for R_E.

$$-0.7 = I_E\left(10\times10^3\right) - 10$$

$$I_E = \frac{10-0.7}{10\times10^3}$$

$$= 0.93\ \text{mA}$$

Thus, the emitter current I_E is $\boxed{0.93\ \text{mA}}$.

Step 2 of 3

Determine the base current I_B using the following relation:

$$I_B = \frac{I_E}{\beta+1}$$

$$= \frac{0.93\times10^{-3}}{50+1}$$

$$= 18.2\times10^{-6}\ \text{A}$$

$$= 18.2\ \mu\text{A}$$

Thus, the base current I_B is $\boxed{18.2\ \mu\text{A}}$.

Determine the collector current I_C using the following relation:

$$I_C = \beta I_B$$

$$= (50)\left(18.2\times10^{-6}\right)$$

$$= 0.91\ \text{mA}$$

Thus, the collector current I_C is $\boxed{0.91\ \text{mA}}$.

Step 3 of 3

Determine the collector voltage, V_C.

$$V_C = V_{CC} - I_C R_C$$

Substitute 10 V for V_{CC}, $0.91\ \text{mA}$ for I_C and $5\ \text{k}\Omega$ for R_C.

$$V_C = 10 - (0.91)(5)$$

$$= 5.45\ \text{V}$$

Thus, the collector voltage V_C is $\boxed{5.45\ \text{V}}$.

Step 1 of 5

The relation between base current (i_B), saturation current (I_S) and base – emitter voltage v_{BE} for active mode operated transistor is,

$$i_B = \left(\frac{I_S}{\beta}\right) e^{v_{BE}/V_T}$$

Thermal voltage V_T is $25\ \text{mV}$ approximately at room temperature

Determine the base current for base-emitter voltage v_{BE} equal to $0.700\ \text{V}$ and saturation current I_S equal to $5\times10^{-15}\ \text{A}$ and for β equal to 50.

$$\begin{aligned} i_B &= \left(\frac{I_S}{\beta}\right) e^{v_{BE}/V_T} \\ &= \left(\frac{5\times10^{-15}}{50}\right)\left(e^{\frac{0.700}{25\times10^{-3}}}\right) \\ &= \left(\frac{5\times10^{-15}}{50}\right) e^{28} \\ &= 0.145\ \text{mA} \end{aligned}$$

The base current for base-emitter voltage v_{BE} equal to $0.700\ \text{V}$ and saturation current I_S equal to $5\times10^{-15}\ \text{A}$ and for β equal to 50 is $\boxed{0.145\ \text{mA}}$

Step 2 of 5

Determine the base current for base-emitter voltage v_{BE} equal to $0.700\ \text{V}$ and saturation current I_S equal to $5\times10^{-15}\ \text{A}$ and for β equal to 200.

$$\begin{aligned} i_B &= \left(\frac{I_S}{\beta}\right) e^{v_{BE}/V_T} \\ &= \left(\frac{5\times10^{-15}}{200}\right)\left(e^{\frac{0.700}{25\times10^{-3}}}\right) \\ &= \left(\frac{5\times10^{-15}}{200}\right) e^{28} \\ &= 36\ \mu\text{A} \end{aligned}$$

The base current for Base – emitter voltage v_{BE} equal to $0.700\ \text{V}$ and saturation current I_S equal to $5\times10^{-15}\ \text{A}$ and for β equal to 200 is $\boxed{36\ \mu\text{A}}$

Thus, the range of base current for β between (50,200) is $\boxed{(0.145\ \text{mA}, 36\ \mu\text{A})}$

Step 3 of 5

The common emitter current gain β is $\dfrac{\text{collector current } (i_C)}{\text{base current } (i_B)}$

For β equal to 50, base current I_B is $0.145\ \text{mA}$ the collector current is,

$$\begin{aligned} \beta &= \frac{\text{collector current } (i_C)}{\text{base current } (i_B)} \\ 50 &= \frac{i_C}{0.145\times10^{-3}\ \text{A}} \\ I_C &= 7.2\ \text{mA} \end{aligned}$$

Thus, the collector current for β equal to 50 is $7.2\ \text{mA}$

Step 4 of 5

For β equal to 200 base current I_B is $36\ \mu\text{A}$ then collector current is,

$$\begin{aligned} \beta &= \frac{\text{collector current } (i_C)}{\text{base current } (i_B)} \\ 200 &= \frac{i_C}{36\times10^{-6}\ \text{A}} \\ I_C &= 7.2\ \text{mA} \end{aligned}$$

Thus, the collector current for β equal to 200 is $7.2\ \text{mA}$

Hence, the collector current is a fixed value for different values of β .

Step 5 of 5

The relation between collector current, base current and emitter current is,.

$$I_E = I_B + I_C$$

The emitter current for β equal to 50 is,

$$\begin{aligned} I_E &= I_B + I_C \\ &= 0.145\ \text{mA} + 7.2\ \text{mA} \\ &= 7.345\ \text{mA} \end{aligned}$$

The emitter current for β equal to 50 is $7.345\ \text{mA}$

The emitter current for β equal to 200 is,

$$\begin{aligned} I_E &= I_B + I_C \\ &= 36\ \mu\text{A} + 7.2\ \text{mA} \\ &= 7.236\ \text{mA} \end{aligned}$$

Thus, the emitter current for β equal to 200 is $7.236\ \text{mA}$

Hence, the range of the emitter current for β between (50,200) is $\boxed{(7.345\ \text{mA}, 7.236\ \text{mA})}$.

Step 1 of 4

Refer to Figure E6.14 in the text book for the PNP transistor circuit.

Apply Kirchhoff's voltage law to the output loop.

$$V_{cc} = I_E R_E + V_E$$

Substitute 10 V for V_{cc}, $5\times10^3\ \Omega$ for R_E and 1.7 V for V_E.

$$10 = I_E(5\times10^3) + 1.7$$
$$I_E = \frac{10-1.7}{5\times10^3}$$
$$= 1.66\times10^{-3}\ \text{A}$$
$$= 1.66\ \text{mA}$$

Determine the base current, I_B.

$$I_B = \frac{V_B}{100\ \text{k}}$$
$$= \frac{1}{100\times10^3}$$
$$= 0.01\ \text{mA}$$

Step 2 of 4

Determine the transistor parameter, α.

$$\alpha = \frac{I_C}{I_E}$$
$$= \frac{I_E - I_B}{I_E}$$
$$= 1 - \frac{I_B}{I_E}$$

Substitute 1.66 mA for I_E and 0.01 mA for I_B.

$$\alpha = 1 - \frac{0.01}{1.66}$$
$$= 0.994$$

Thus, the value of α is $\boxed{0.994}$.

Step 3 of 4

Determine the transistor parameter, β.

$$\beta = \frac{\alpha}{1-\alpha}$$

Substitute 0.994 for α.

$$\beta = \frac{0.994}{1-0.994}$$
$$= 165.7$$
$$\approx 165$$

Thus, the value of β is $\boxed{165}$.

Step 4 of 4

Determine the collector voltage V_C.

$$V_C = I_C(5\ \text{k}) - 10$$
$$= (I_E - I_B)(5\ \text{k}) - 10$$

Substitute 1.66 mA for I_E and 0.01 mA for I_B.

$$V_C = (1.66\ \text{m} - 0.01\ \text{m})(5\ \text{k}) - 10$$
$$= -1.75\ \text{V}$$

Thus, the voltage V_C is $\boxed{-1.75\ \text{V}}$.

Step 1 of 9

Different transistors operating in active mode with emitter current i_E equal to $1\ \text{mA}$

First transistor has base current equal to $10\ \mu\text{A}$.

Collector current in terms of base current and emitter current is $i_C = i_E - i_B$

$$\begin{aligned} i_C &= i_E - i_B \\ &= 1\times10^{-3}\ \text{A} - 10\times10^{-6}\ \text{A} \\ &= 0.99\ \text{mA} \end{aligned}$$

Thus, the collector current (i_C) for first transistor is $\boxed{0.99\ \text{mA}}$.

Step 2 of 9

The common emitter current gain β is $\dfrac{\text{collector current } (i_C)}{\text{base current } (i_B)}$

$$\begin{aligned} \beta &= \frac{i_C}{i_B} \\ &= \frac{0.99\times10^{-3}\ \text{A}}{10\times10^{-6}\ \text{A}} \\ &= 99 \end{aligned}$$

Thus, the common emitter current gain β for the first transistor is $\boxed{99}$.

Step 3 of 9

The common base current gain α is $\dfrac{\text{collector current } (i_C)}{\text{emitter current } (i_E)}$

$$\begin{aligned} \alpha &= \frac{i_C}{i_E} \\ &= \frac{0.99\times10^{-3}\ \text{A}}{1\times10^{-3}\ \text{A}} \\ &= 0.99 \end{aligned}$$

Thus, the common base current gain α for the first transistor is $\boxed{0.99}$.

Step 4 of 9

The second transistor has base current equal to $20\ \mu\text{A}$.

The collector current in terms of base current and emitter current is,

$$i_C = i_E - i_B$$

$$\begin{aligned} i_C &= i_E - i_B \\ &= 1\times10^{-3}\ \text{A} - 20\times10^{-6}\ \text{A} \\ &= 0.98\ \text{mA} \end{aligned}$$

Thus, the collector current (i_C) for the second transistor is $\boxed{0.98\ \text{mA}}$.

Step 5 of 9

The common emitter current gain β is $\dfrac{\text{collector current } (i_C)}{\text{base current } (i_B)}$

$$\begin{aligned} \beta &= \frac{i_C}{i_B} \\ &= \frac{0.98\times10^{-3}\ \text{A}}{10\times10^{-6}\ \text{A}} \\ &= 98 \end{aligned}$$

Thus, the common emitter current gain β for the second transistor is $\boxed{98}$.

Step 6 of 9

The common base current gain α is $\dfrac{\text{collector current } (i_C)}{\text{emitter current } (i_E)}$

$$\begin{aligned} \alpha &= \frac{i_C}{i_E} \\ &= \frac{0.98\times10^{-3}\ \text{A}}{1\times10^{-3}\ \text{A}} \\ &= 0.98 \end{aligned}$$

Thus, the common base current gain α for the second transistor is $\boxed{0.98}$.

Step 7 of 9

Third transistor has base current equal to $50\ \mu\text{A}$

The collector current in terms of base current and emitter current is,

$$i_C = i_E - i_B$$

$$\begin{aligned} i_C &= i_E - i_B \\ &= 1\times10^{-3}\ \text{A} - 50\times10^{-6}\ \text{A} \\ &= 0.95\ \text{mA} \end{aligned}$$

Thus, the collector current (i_C) for the third transistor is $\boxed{0.95\ \text{mA}}$.

Step 8 of 9

The common emitter current gain β is $\dfrac{\text{collector current } (i_C)}{\text{base current } (i_B)}$

$$\begin{aligned} \beta &= \frac{i_C}{i_B} \\ &= \frac{0.95\times10^{-3}\ \text{A}}{10\times10^{-6}\ \text{A}} \\ &= 95 \end{aligned}$$

Thus, the common emitter current gain β for the third transistor is $\boxed{95}$.

Step 9 of 9

Common base current gain α is $\dfrac{\text{collector current } (i_C)}{\text{emitter current } (i_E)}$

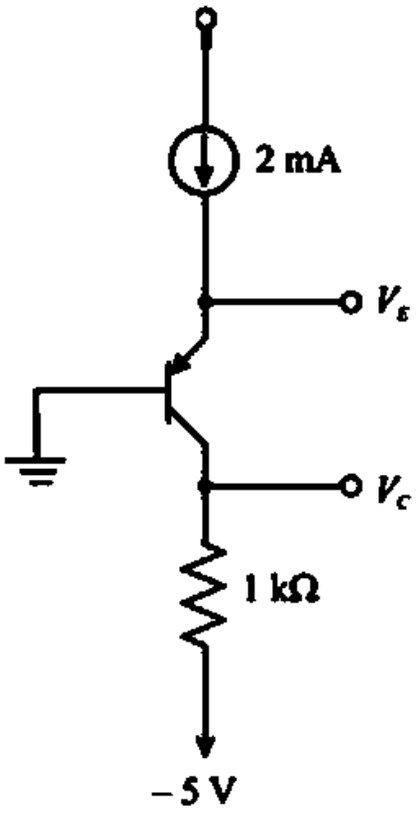
$$\begin{aligned} \alpha &= \frac{i_C}{i_E} \\ &= \frac{0.95\times10^{-3}\ \text{A}}{1\times10^{-3}\ \text{A}} \\ &= 0.95 \end{aligned}$$

Common base current gain α for third transistor is $\boxed{0.95}$

Step 1 of 2

6.015E

Draw the transistor circuit for the data.



Step 2 of 2

The voltage across the emitter-base junction decreases by about 2 mV for every $1°C$ rise of temperature provided the junction is operating at a constant current.

Hence, the emitter voltage decreases by $2\ mV/°C$ rise of temperature.

Determine the change in emitter voltage if the temperature increases by $30\ °C$.

$$\nabla v_{BE} = (-2\ mV/°C)(30°C)$$
$$= -60\ mV$$

Thus, the change in emitter voltage is $\boxed{-60\ mV}$.

As the emitter is fed by a constant current source, the emitter current I_E is constant. Hence, the collector current is constant. As the collector current is constant, there is no change in the collector voltage. Therefore, the change in collector voltage is $\boxed{0\ V}$.

Step 1 of 19

The common emitter current gain β is $\dfrac{\text{collector current } (i_C)}{\text{base current } (i_B)}$

The collector current for transistor (a) is $1.000\ \text{mA}$

The base current for transistor (a) is $10\ \mu\text{A}$

$$\begin{aligned}\beta &= \frac{i_C}{i_B}\\ &= \frac{1\times10^{-3}\ \text{A}}{10\times10^{-6}\ \text{A}}\\ &= 100\end{aligned}$$

Thus, the common emitter current gain for transistor (a) is $\boxed{100}$.

Step 2 of 19

The relation between collector current, base current and emitter current is,

$$I_E = I_B + I_C$$

$$\begin{aligned}I_E &= I_B + I_C\\ &= 10\times10^{-6}\ \text{A} + 1\times10^{-3}\ \text{A}\\ &= 1.01\ \text{mA}\end{aligned}$$

Thus, the emitter current for transistor (a) is $\boxed{1.01\ \text{mA}}$.

Step 3 of 19

The common base current gain α is $\dfrac{\text{collector current } (i_C)}{\text{Emitter current } (i_E)}$

$$\begin{aligned}\alpha &= \frac{i_C}{i_E}\\ &= \frac{1\times10^{-3}\ \text{A}}{1.01\times10^{-3}\ \text{A}}\\ &= 0.99\end{aligned}$$

Thus, the common base current gain for transistor (a) is $\boxed{0.99}$.

Step 4 of 19

The relation between collector current i_C and saturation current I_S is,

$$i_C = I_S e^{v_{BE}/V_T}$$

The base-emitter voltages v_{BE} of transistor (a) is $700\ \text{mV}$

Thermal voltage V_T is $25\ \text{mV}$ approximately at room temperature.

$$\begin{aligned}i_C &= I_S e^{v_{BE}/V_T}\\ 1\times10^{-3}\ \text{A} &= I_S e^{700\times10^{-3}/(25\times10^{-3})}\\ 1\times10^{-3}\ \text{A} &= I_S e^{28}\\ I_S &= 6.9\times10^{-16}\ \text{A}\end{aligned}$$

Thus, the saturation current I_S for transistor (a) is $\boxed{4.68\times10^{-17}\ \text{A}}$.

Step 5 of 19

The relation between collector current, base current and emitter current is,

$$I_E - I_C = I_B$$

The emitter current for transistor (b) is $1.020\ \text{mA}$

The collector current for transistor (b) is $1.000\ \text{mA}$

$$\begin{aligned}I_B &= I_E - I_C\\ &= 1.020\times10^{-3}\ \text{A} - 1.000\times10^{-3}\ \text{A}\\ &= 20\ \mu\text{A}\end{aligned}$$

Thus, the base current for transistor (b) is $\boxed{20\ \mu\text{A}}$

Step 6 of 19

The common base current gain α is $\dfrac{\text{collector current } (i_C)}{\text{Emitter current } (i_E)}$

$$\begin{aligned}\alpha &= \frac{i_C}{i_E}\\ &= \frac{1.000\times10^{-3}\ \text{A}}{1.020\times10^{-3}\ \text{A}}\\ &= 0.98\end{aligned}$$

Thus, the common base current gain for transistor (b) is $\boxed{0.98}$.

Step 7 of 19

The common emitter current gain β is $\dfrac{\text{collector current } (i_C)}{\text{base current } (i_B)}$

$$\begin{aligned}\beta &= \frac{i_C}{i_B}\\ &= \frac{1.000\times10^{-3}\ \text{A}}{20\times10^{-6}\ \text{A}}\\ &= 50\end{aligned}$$

Thus, the common emitter current gain for transistor (b) is $\boxed{50}$.

Step 8 of 19

Relation between collector current i_C and saturation current I_S is $i_C = I_S e^{v_{BE}/V_T}$

The base-Emitter voltages v_{BE} of transistor (b) is $690\ \text{mV}$.

$$\begin{aligned}i_C &= I_S e^{v_{BE}/V_T}\\ 1\times10^{-3}\ \text{A} &= I_S e^{690\times10^{-3}/(25\times10^{-3})}\\ 1\times10^{-3}\ \text{A} &= I_S e^{27.6}\\ I_S &= 1.03\times10^{-15}\ \text{A}\end{aligned}$$

The saturation current I_S for transistor (b) is $\boxed{1.03\times10^{-15}\ \text{A}}$.

The relation between collector current, base current and emitter current is,

$$I_B - I_B = I_C$$

The emitter current for transistor (c) is $0.235\ \text{mA}$

The base current for transistor (c) is $5\ \mu\text{A}$

$$\begin{aligned}I_C &= I_E - I_B\\ &= 0.235\times10^{-3}\ \text{A} - 5\times10^{-4}\ \text{A}\\ &= 0.23\ \text{mA}\end{aligned}$$

Thus, the collector current for transistor (c) is $\boxed{0.23\ \text{mA}}$

Step 9 of 19

The common base current gain α is $\dfrac{\text{collector current } (i_C)}{\text{Emitter current } (i_E)}$

$$\begin{aligned}\alpha &= \frac{i_C}{i_E}\\ &= \frac{0.23\times10^{-3}\ \text{A}}{0.235\times10^{-3}\ \text{A}}\\ &= 0.979\end{aligned}$$

Thus, the common base current gain for transistor (c) is $\boxed{0.979}$.

Step 10 of 19

The common emitter current gain β is $\dfrac{\text{collector current } (i_C)}{\text{base current } (i_B)}$

$$\begin{aligned}\beta &= \frac{i_C}{i_B}\\ &= \frac{0.23\times10^{-3}\ \text{A}}{5\times10^{-6}\ \text{A}}\\ &= 46\end{aligned}$$

Thus, the common emitter current gain for transistor (c) is $\boxed{46}$.

Step 11 of 19

The relation between collector current i_C and saturation current I_S is,

$$i_C = I_S e^{v_{BE}/V_T}$$

The base-emitter voltages v_{BE} of transistor (c) is $580\ \text{mV}$.

$$\begin{aligned}i_C &= I_S e^{v_{BE}/V_T}\\ 0.23\times10^{-3}\ \text{A} &= I_S e^{580\times10^{-3}/(25\times10^{-3})}\\ 0.23\times10^{-3}\ \text{A} &= I_S e^{23.2}\\ I_S &= 1.93\times10^{-14}\ \text{A}\end{aligned}$$

Thus, the saturation current I_S for transistor (c) is $\boxed{1.93\times10^{-14}\ \text{A}}$.

Step 12 of 19

The common emitter current gain β is $\dfrac{\text{collector current } (i_C)}{\text{base current } (i_B)}$

The collector current for transistor (d) is $10.10\ \text{mA}$

The base current for transistor (d) is $120\ \mu\text{A}$

$$\begin{aligned}\beta &= \frac{i_C}{i_B}\\ &= \frac{10.10\times10^{-3}\ \text{A}}{120\times10^{-6}\ \text{A}}\\ &= 84.17\end{aligned}$$

Thus, the common emitter current gain for transistor (d) is $\boxed{84.17}$.

Step 13 of 19

The relation between collector current, base current and emitter current is,

$$I_B = I_B + I_C$$

$$\begin{aligned}I_E &= I_B + I_C\\ &= 120\times10^{-6}\ \text{A} + 10.10\times10^{-3}\ \text{A}\\ &= 10.22\ \text{mA}\end{aligned}$$

Thus, the emitter current for transistor (d) is $\boxed{10.22\ \text{mA}}$

Step 14 of 19

The common base current gain α is $\dfrac{\text{collector current } (i_C)}{\text{Emitter current } (i_E)}$

$$\begin{aligned}\alpha &= \frac{i_C}{i_E}\\ &= \frac{10.10\times10^{-3}\ \text{A}}{10.22\times10^{-3}\ \text{A}}\\ &= 0.988\end{aligned}$$

Thus, the common base current gain for transistor (d) is $\boxed{0.998}$

Step 15 of 19

The relation between collector current i_C and saturation current I_S is,

$$i_C = I_S e^{v_{BE}/V_T}$$

The base-emitter voltages v_{BE} of transistor (d) is $780\ \text{mV}$

Thermal voltage V_T is $25\ \text{mV}$ approximately at room temperature.

$$\begin{aligned}i_C &= I_S e^{v_{BE}/V_T}\\ 10.10\times10^{-3}\ \text{A} &= I_S e^{780\times10^{-3}/(25\times10^{-3})}\\ 10.10\times10^{-3}\ \text{A} &= I_S e^{31.2}\\ I_S &= 2.85\times10^{-16}\ \text{A}\end{aligned}$$

Thus, the saturation current I_S for transistor (d) is $\boxed{2.85\times10^{-16}\ \text{A}}$.

Step 16 of 19

The relation between collector current, base current and emitter current is,

$$I_E - I_B = I_C$$

The emitter current for transistor (e) is $75\ \text{mA}$

The base current for transistor (e) is $1050\ \mu\text{A}$

$$\begin{aligned}I_E &= I_B - I_B\\ &= 75\times10^{-3}\ \text{A} - 1050\times10^{-6}\ \text{A}\\ &= 73.95\ \text{mA}\end{aligned}$$

Thus, the collector current for transistor (e) is $\boxed{73.95\ \text{mA}}$.

Step 17 of 19

The common base current gain α is $\dfrac{\text{collector current } (i_C)}{\text{Emitter current } (i_E)}$

$$\begin{aligned}\alpha &= \frac{i_C}{i_E}\\ &= \frac{73.95\times10^{-3}\ \text{A}}{75\times10^{-3}\ \text{A}}\\ &= 0.986\end{aligned}$$

Thus, the common base current gain for transistor (e) is $\boxed{0.979}$.

Step 18 of 19

The common emitter current gain β is $\dfrac{\text{collector current } (i_C)}{\text{base current } (i_B)}$

$$\begin{aligned}\beta &= \frac{i_C}{i_B}\\ &= \frac{73.95\times10^{-3}\ \text{A}}{1050\times10^{-6}\ \text{A}}\\ &= 70.43\end{aligned}$$

Thus, the common emitter current gain for transistor (d) is $\boxed{70.43}$.

Step 19 of 19

The relation between collector current i_C and saturation current I_S is,

$$i_C = I_S e^{v_{BE}/V_T}$$

The base-emitter voltages v_{BE} of transistor (d) is $820\ \text{mV}$.

$$\begin{aligned}i_C &= I_S e^{v_{BE}/V_T}\\ 73.95\times10^{-3}\ \text{A} &= I_S e^{820\times10^{-3}/(25\times10^{-3})}\\ 73.95\times10^{-3}\ \text{A} &= I_S e^{32.8}\\ I_S &= 4.21\times10^{-16}\ \text{A}\end{aligned}$$

Thus, the saturation current I_S for transistor (e) is $\boxed{4.21\times10^{-16}\ \text{A}}$.

6.016E

Refer to Figure 6.18 in the text book for the large signal equivalent circuits of an *npn* transistor operating in the active mode.

The output resistance r_o is,

$$r_o = \frac{V_A}{I_S e^{\frac{v_{BE}}{V_T}}}$$

Here,

V_A is the early voltage of the transistor

V_T is the thermal voltage

I_S is the saturation current of the transistor

V_{BE} is the base-emitter voltage

Apply Kirchhoff's current law to the node C.

$$i_C = \frac{v_{CE}}{r_o} + I_S e^{\frac{v_{BE}}{V_T}}$$

Here,

V_{CE} is the collector-to-emitter voltage

Substitute $\frac{V_A}{I_S e^{\frac{v_{BE}}{V_T}}}$ for r_o .

$$\begin{aligned} i_C &= \frac{v_{CE}}{\left(\frac{V_A}{I_S e^{\frac{v_{BE}}{V_T}}}\right)} + I_S e^{\frac{v_{BE}}{V_T}} \\ &= I_S e^{\frac{v_{BE}}{V_T}} \frac{v_{CE}}{V_A} + I_S e^{\frac{v_{BE}}{V_T}} \\ &= I_S e^{\frac{v_{BE}}{V_T}} \left(1 + \frac{v_{CE}}{V_A}\right) \end{aligned}$$

Hence, the circuit model is a direct representation of the equation, $\boxed{i_C = I_S e^{\frac{v_{BE}}{V_T}} \left(1 + \frac{v_{CE}}{V_A}\right)}$.

6.016P

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Step 1 of 2

6.017E

Determine the output resistance r_o of the BJT using the formula,

$$r_o = \frac{V_A}{I_C}$$

Here,

V_A is the early voltage of the transistor

I_C is the collector current.

Substitute 100 v for V_A and $0.1\times10^{-3}\ \text{A}$ for I_C .

$$\begin{aligned} r_o &= \frac{100}{0.1\times10^{-3}} \\ &= 10^6\ \Omega \\ &= 1\ \text{M}\Omega \end{aligned}$$

Thus, the output resistance of the BJT at 0.1 mA of collector current is $\boxed{1\ \text{M}\Omega}$.

Step 2 of 2

Determine the output resistance of the BJT at 1 mA of collector current.

Substitute 100 V for V_A and $10^{-3}\ \text{A}$ for I_C .

$$\begin{aligned} r_o &= \frac{100}{10^{-3}} \\ &= 10^5\ \Omega \\ &= 100\ \text{k}\Omega \end{aligned}$$

Thus, the output resistance of the BJT at 1 mA of collector current is $\boxed{100\ \text{k}\Omega}$.

Determine the output resistance of the BJT at 10 mA of collector current.

Substitute 100 V for V_A and $10\times10^{-3}\ \text{A}$ for I_C .

$$\begin{aligned} r_o &= \frac{100}{10\times10^{-3}} \\ &= 10^4\ \Omega \\ &= 10\ \text{k}\Omega \end{aligned}$$

Thus, the output resistance of the BJT at 10 mA of collector current is $\boxed{10\ \text{k}\Omega}$.

Step 1 of 5

Refer to the Figure 6.5 (b) in the test book.

Draw the modified circuit diagram as shown in Figure 1.

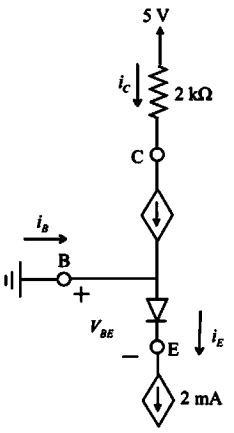


Figure 1

Step 2 of 5

Write the emitter current using Figure 1.

$$i_E = 2 \text{ mA}$$

Write the expression for common emitter current gain.

$$\beta = \frac{\text{collector current } (i_C)}{\text{base current } (i_B)}$$

Base current in terms of collector current is, $i_B = \frac{i_C}{\beta}$

Write the relation between collector current, base current and emitter current.

$$i_E = i_B + i_C$$
$$2 \text{ mA} = \frac{i_C}{\beta} + i_C$$
$$2 \text{ mA} = i_C\left(\frac{1}{100} + 1\right)$$
$$i_C = 1.98 \text{ mA}$$

Determine the value of base current.

$$i_B = \frac{i_C}{\beta}$$
$$= \frac{1.98 \text{ mA}}{100}$$
$$= 0.0198 \text{ mA}$$
$$\approx 0.02 \text{ mA}$$

Therefore, the value of base current is, $\boxed{0.02 \text{ mA}}$.

Step 3 of 5

Determine the value of collector voltage.

$$i_C \text{ A} = \frac{5 - V_C}{2 \text{ k}\Omega}$$
$$1.98 \times 10^{-3} \text{ A} = \frac{5 - V_C}{2 \text{ k}\Omega}$$
$$5 - V_C = 3.96$$
$$V_C = 1.04 \text{ V}$$

Therefore, the value of collector voltage is, $\boxed{1.04 \text{ V}}$.

Step 4 of 5

The Saturation current for npn transistor is, $I_S = 5 \times 10^{-15}$ A

Determine the value of base emitter voltage.

$$i_C = I_S e^{V_{BE}/V_T}$$
$$1.98 \times 10^{-3} = I_S e^{\frac{V_{BE}}{25 \times 10^{-3}}}$$
$$1.98 \times 10^{-3} = 5 \times 10^{-15} e^{\frac{V_{BE}}{25 \times 10^{-3}}}$$
$$3.96 \times 10^{11} = e^{\frac{V_{BE}}{25 \times 10^{-3}}}$$

Apply natural log on both sides.

$$\ln\left(3.96 \times 10^{11}\right) = \ln\left(e^{\frac{V_{BE}}{25 \times 10^{-3}}}\right)$$
$$26.70 = \frac{V_{BE}}{25 \times 10^{-3}}$$
$$0.67 = V_{BE}$$

Therefore, the value of base emitter voltage is $\boxed{0.67 \text{ V}}$.

Step 5 of 5

Base is connected to the ground and hence V_B is zero volts.
Determine the value of emitter voltage.

$$V_{BE} = V_B - V_E$$
$$0.67 \text{ V} = 0 - V_E$$
$$V_E = -0.67 \text{ V}$$

Therefore, the value of emitter voltage is $\boxed{-0.67 \text{ V}}$.

Step 1 of 2

6.018E

Refer to Figure 6.17 in the text book for the bipolar transistor circuit.

Write the formula for collector current, I_C .

$$I_C = I_S e^{\frac{V_{BE}}{V_T}}\left(1+\frac{V_{CE}}{V_A}\right)$$

Here,

I_S is the saturation current of the transistor

V_{BE} is the base-emitter voltage

V_T is the thermal voltage

V_{CE} is the collector-to-emitter voltage

V_A is the early voltage

Substitute 1 V for V_{CE} , 10^{-3} A for I_C and 100 V for V_A .

$$10^{-3} = I_S e^{\frac{V_{BE}}{V_T}}\left(1+\frac{1}{100}\right)$$

$$10^{-3} = 1.01 I_S e^{\frac{V_{BE}}{V_T}}$$

$$I_S e^{\frac{V_{BE}}{V_T}} = 0.99\times 10^{-3}$$

Step 2 of 2

V_{BE} is kept constant and the voltage V_{CE} is increased to 11 V.

The new collector current is,

$$I_{C(new)} = I_S e^{\frac{V_{BE}}{V_T}}\left(1+\frac{11}{100}\right)$$

Substitute 0.99×10^{-3} for $I_S e^{\frac{V_{BE}}{V_T}}$.

$$I_{C(new)} = \left(0.99\times 10^{-3}\right)\left(1+\frac{11}{100}\right)$$

$$= 1.1\times 10^{-3}\ \text{A}$$

$$= 1.1\ \text{mA}$$

Thus, the collector current for V_{CE} raised to 11 v is $\boxed{1.1\ \text{mA}}$.

Step 1 of 5

Refer to the Figures 6.5 (d) and 6.6 (b) in the test book.

Draw the modified circuit diagram as shown in Figure 1.

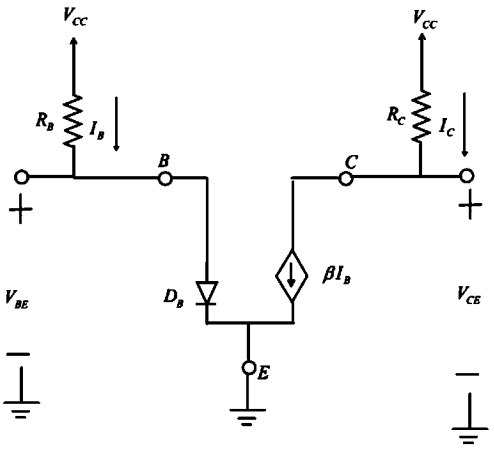


Figure 1

Step 2 of 5

Apply Kirchhoff's voltage law at output loop of Figure 1.

$$V_{CC} - I_C R_C = V_{CE}$$
$$5 - \left(0.5\times10^{-3}\right)R_C = 1$$
$$-\left(0.5\times10^{-3}\right)R_C = -4$$
$$R_C = 8\ \text{k}\Omega$$

Therefore, the value of resistor R_C is $\boxed{8\ \text{k}\Omega}$.

Step 3 of 5

Write the expression of collector current.

$$I_C = I_S e^{V_{BE}/V_T}$$
$$0.5\times10^{-3} = \left(5\times10^{-15}\ \text{A}\right)e^{V_{BE}/\left(25\times10^{-3}\right)}$$
$$1\times10^{11} = e^{V_{BE}/\left(25\times10^{-3}\right)}$$

Apply natural log on both sides

$$\ln\left(1\times10^{11}\right) = \ln e^{V_{BE}/\left(25\times10^{-3}\right)}$$
$$25.33 = \frac{V_{BE}}{25\times10^{-3}}$$
$$V_{BE} = 0.633\ \text{V}$$

Step 4 of 5

Write the expression for Common emitter current gain.

$$\beta = \frac{\text{collector current } (i_C)}{\text{base current } (i_B)}$$
$$\beta = \frac{0.5\times10^{-3}}{i_B}$$
$$i_B = \frac{0.5\times10^{-3}}{50}$$
$$i_B = 10\ \mu\text{A}$$

Step 5 of 5

Apply Kirchhoff's voltage law at input loop of Figure 1.

$$V_{CC} - I_B R_B = V_{BE}$$
$$5 - \left(10\times10^{-6}\right)R_B = 0.633\ \text{V}$$
$$-\left(10\times10^{-6}\right)R_B = -4.367$$
$$R_B = 43.67\ \text{k}\Omega$$

Therefore, the value of resistor R_B is $\boxed{43.67\ \text{k}\Omega}$.

Step 1 of 5

6.019E

Redraw the circuit in Figure 6.21 in the textbook by replacing R_C with $10\ \text{k}\Omega$ resistor.

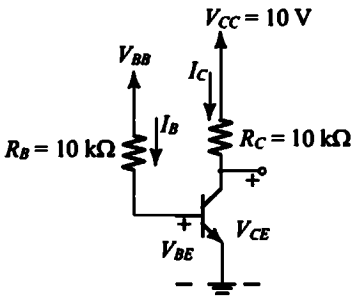


Figure 1

Step 2 of 5

(a)

The transistor is operating in the active mode with $V_{CE} = 5$ V.

Determine the collector current.

$$I_C = \frac{V_{CC} - V_{CE}}{R_C}$$
$$= \frac{10 - 5}{10\ \text{k}\Omega}$$
$$= 0.5\ \text{mA}$$

The base current is,

$$I_B = \frac{I_C}{\beta}$$
$$= \frac{0.5}{50}$$
$$= 0.01\ \text{mA}$$

Determine the supply voltage, V_{BB}.

$$V_{BB} = I_B R_B + V_{BE}$$
$$= 0.01 \times 10 + 0.7$$
$$= 0.8\ \text{V}$$

Thus, the voltage V_{BB} is $\boxed{0.8\ \text{V}}$.

Step 3 of 5

(b)

The operation at the end of saturation is obtained with $V_{CE} = 0.3$ V.

Determine the collector current.

$$I_C = \frac{V_{CC} - V_{CE}}{R_C}$$
$$= \frac{10 - 0.3}{10\ \text{k}\Omega}$$
$$= 0.97\ \text{mA}$$

Since, at the edge of saturation, I_C and I_B are still related by β, determine the base current.

$$I_B = \frac{I_C}{\beta}$$
$$= \frac{0.97}{50}$$
$$= 0.0194\ \text{mA}$$

Determine the supply voltage, V_{BB}.

$$V_{BB} = I_B R_B + V_{BE}$$
$$= 0.0194 \times 10 + 0.7$$
$$= 0.894\ \text{V}$$

Thus, the voltage V_{BB} is $\boxed{0.894\ \text{V}}$.

Step 4 of 5

(c)

The operation at deep in saturation with $\beta_{forced} = 10$ is obtained with $V_{CE} = V_{CEsat} = 0.2$ V.

Determine the collector current.

$$I_C = \frac{V_{CC} - V_{CE}}{R_C}$$
$$= \frac{10 - 0.2}{10\ \text{k}\Omega}$$
$$= 0.98\ \text{mA}$$

The base current is,

$$I_B = \frac{I_C}{\beta_{forced}}$$
$$= \frac{0.98}{10}$$
$$= 0.098\ \text{mA}$$

Step 5 of 5

Determine the supply voltage, V_{BB}.

$$V_{BB} = I_B R_B + V_{BE}$$
$$= 0.098 \times 10 + 0.7$$
$$= 1.68\ \text{V}$$

Thus, the voltage V_{BB} is $\boxed{1.68\ \text{V}}$.

Step 1 of 7

Write the relation between saturation currents I_S and I_{SC} .

$$\frac{I_S}{I_{SC}} = \frac{\text{area of EBJ junction}}{\text{area of CBJ junction}}$$

npn transistor has a CBJ with an area 100 times that of the EBJ.

$$\frac{I_S}{I_{SC}} = \frac{\text{area of EBJ junction}}{\text{area of CBJ junction}}$$

$$\frac{I_S}{I_{SC}} = \frac{\text{area of EBJ junction}}{100(\text{area of EBJ junction})}$$

$$I_{SC} = 100 I_S$$

Step 2 of 7

Saturation current emitter base junction is, $I_S = 10^{-15}\ \text{A}$

Determine the saturation current for collector base junction.

$$\begin{aligned} I_{SC} &= 100 I_S \\ &= 100\left(10^{-15}\ \text{A}\right) \\ &= 10^{-13}\ \text{A} \end{aligned}$$

Step 3 of 7

Both EBJ and CBJ junction are forward biased and hence npn transistor is operated in saturation mode.

Current flowing between collector and emitter junction is, $I_S e^{v_{BE}/V_T}$

Current flowing between base and collector junction is, $I_{SC} e^{v_{BC}/V_T}$

Given that Conducting current for each junction is $1\ \text{mA}$ for npn transistor hence

$I_S e^{v_{BE}/V_T} = 1\times10^{-3}\ \text{A}$ ..... (1)

$I_{SC} e^{v_{BC}/V_T} = 1\times10^{-3}\ \text{A}$ ..... (2)

Step 4 of 7

Thermal voltage V_T is $25\ \text{mV}$ approximately at room temperature.

From equation (1),

$$I_S e^{v_{BE}/V_T} = 1\times10^{-3}\ \text{A}$$

$$10^{-15} e^{v_{BE}/V_T} = 1\times10^{-3}$$

$$e^{v_{BE}/V_T} = 10^{12}$$

Apply natural log on both sides,

$$\ln\left(e^{v_{BE}/V_T}\right) = \ln 10^{12}$$

$$V_{BE}/V_T = 27.63$$

$$V_{BE} = 27.63\times25\times10^{-3}$$

$$V_{BE} = 0.69\ \text{V}$$

Therefore, the voltage drop across EBJ V_{BE} for npn transistor is $\boxed{0.69\ \text{V}}$.

Step 5 of 7

From equation (2),

$$I_{SC} e^{v_{BC}/V_T} = 1\times10^{-3}\ \text{A}$$

$$10^{-13} e^{v_{BC}/V_T} = 1\times10^{-3}$$

$$e^{v_{BC}/V_T} = 10^{10}$$

Apply natural log on both sides,

$$\ln\left(e^{v_{BC}/V_T}\right) = \ln 10^{10}$$

$$V_{BE}/V_T = 23.02$$

$$V_{BC} = 23.02\times25\times10^{-3}$$

$$V_{BC} = 0.575\ \text{V}$$

Therefore, the voltage drop across CBJ V_{BC} for npn transistor is $\boxed{0.575\ \text{V}}$.

Step 6 of 7

Forward bias voltage applied across EBJ V_{BE} is $0.5\ \text{V}$

Current flowing between emitter and base junction i_{EB} is $I_S e^{v_{BE}/V_T}$

$$\begin{aligned} i_{EB} &= I_S e^{v_{BE}/V_T} \\ &= 10^{-15}\left(e^{0.5/25\times10^{-3}}\right) \\ &= 10^{-15}\left(e^{20}\right) \\ &= 0.48\ \mu\text{A} \end{aligned}$$

Therefore, the current flowing between emitter and base junction i_{EB} is $\boxed{0.48\ \mu\text{A}}$.

Step 7 of 7

Forward bias voltage applied across CBJ V_{BE} is $0.5\ \text{V}$

Current flowing between base and collector junction i_{BC} is $I_{SC} e^{v_{BC}/V_T}$

$$\begin{aligned} i_{BC} &= I_{SC} e^{v_{BC}/V_T} \\ &= 10^{-13}\left(e^{0.5/25\times10^{-3}}\right) \\ &= 10^{-13}\left(e^{20}\right) \\ &= 48.5\ \mu\text{A} \end{aligned}$$

Therefore, the current flowing between base and collector junction i_{BC} is $\boxed{48.5\ \mu\text{A}}$.

6.020E

Refer to Figure 6.21 in the text book for the transistor circuit.

The base voltage, $V_{BB} = 0\ \mathbf{V}$.

From the figure, the supply voltage, $V_{CC} = 10\ \mathbf{V}$.

As $V_{BB} = 0\ \mathbf{V}$, the base current is,

$I_B = 0\ \mathbf{A}$.

The collector current is,

$$\begin{aligned} I_C &= \beta I_B \\ &= 0\ \mathbf{A} \end{aligned}$$

Determine the collector to emitter voltage.

$$\begin{aligned} V_{CE} &= V_{CC} - R_C I_C \\ &= V_{CC} \\ &= +10\ \mathbf{V} \end{aligned}$$

Thus, collector to emitter voltage V_{CE} is $\boxed{+10\ \mathbf{V}}$.

6.020P

We don't have the solution to this problem yet.

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Step 1 of 3

Refer to Figure 6.21 in the text book for the circuit.

Determine the base current of the transistor. $I_B = \frac{V_{BB} - V_{BE}}{R_B}$

Substitute 1.7 V for V_{BB} , 0.7 V for V_{BE} and $10\ \text{k}\Omega$ for R_B .

$$I_B = \frac{1.7 - 0.7}{10 \times 10^3}$$
$$= 0.1\ \text{mA}$$

Determine the collector current.

$$I_C = \beta I_B$$
$$= 50 \times 0.1\ \text{m}$$
$$= 5\ \text{mA}$$

The collector to emitter voltage is,

$$V_{CE} = V_{CC} - R_C I_C$$

Substitute 10 V for V_{CC} , $1\ \text{k}\Omega$ for R_C and 5 mA for I_C .

$$V_{CE} = 10 - 5 \times 1$$
$$= 5\ \text{V}$$

For the transistor to operate in active region, the collector to emitter voltage should be greater than the base to emitter voltage.

As $V_{CE} > V_{BE}$, the transistor is operating in the active mode.

Step 2 of 3

(a)

For operation at the edge of saturation, the collector to emitter voltage must be $V_{CE} = 0.3\ \text{V}$.

Determine the required collector resistance at the edge of saturation.

$$R_C = \frac{V_{CC} - V_{CE}}{I_C}$$
$$= \frac{10 - 0.3}{5 \times 10^{-3}}$$
$$= \frac{9.7}{5 \times 10^{-3}}$$
$$= 1.94\ \text{k}\Omega$$

Thus the value to which collector resistance should be increased is $\boxed{1.94\ \text{k}\Omega}$.

Step 3 of 3

(b)

The common-emitter current gain is,

$\beta_{forced} = 10$.

For operation deep in saturation, the collector to emitter voltage is $V_{CE} = 0.2\ \text{V}$.

The collector current is,

$$I_C = \beta I_B$$
$$= 10 \times 0.1\ \text{m}$$
$$= 1\ \text{mA}$$

Determine the collector resistance.

$$R_C = \frac{V_{CC} - V_{CE}}{I_C}$$
$$= \frac{10 - 0.2}{1 \times 10^{-3}}$$
$$= \frac{9.8}{10^{-3}}$$
$$= 9.8\ \text{k}\Omega$$

Thus the value to which the collector resistance R_C should be increased for operation deep in saturation is $\boxed{9.8\ \text{k}\Omega}$.

Step 1 of 10

Write the expression for collector current for the BJT operated in saturation mode.

$$i_C = I_S e^{v_{BE}/V_T} - I_{SC} e^{v_{BC}/V_T}$$

Write the expression for base current for the BJT operated in saturation mode.

$$i_B = \frac{I_S}{\beta} e^{v_{BE}/V_T} + I_{SC} e^{v_{BC}/V_T}$$

The value of the ratio i_C/i_B for a saturated transistor is less than or equal to the value of β.

$$\left.\frac{i_C}{i_B}\right|_{\text{saturation}} \le \beta$$

Step 2 of 10

Write the expression for forced β.

$$\beta_{\text{forced}} = \left.\frac{i_C}{i_B}\right|_{\text{saturation}} \le \beta$$

Substitute $\left(I_S e^{v_{BE}/V_T} - I_{SC} e^{v_{BC}/V_T}\right)$ for i_C and $\left(\frac{I_S}{\beta} e^{v_{BE}/V_T} + I_{SC} e^{v_{BC}/V_T}\right)$ for i_B in the equation.

$$\beta_{\text{forced}} = \frac{I_S e^{\frac{v_{BE}}{V_T}} - I_{SC} e^{\frac{v_{BC}}{V_T}}}{\frac{I_S}{\beta} e^{\frac{v_{BE}}{V_T}} + I_{SC} e^{\frac{v_{BC}}{V_T}}}$$

$$= \frac{I_{SC} e^{\frac{v_{BC}}{V_T}} \left(\frac{I_S e^{\frac{v_{BE}}{V_T}}}{I_{SC} e^{\frac{v_{BC}}{V_T}}} - 1 \right)}{I_{SC} e^{\frac{v_{BC}}{V_T}} \left(\frac{\frac{I_S}{\beta} e^{\frac{v_{BE}}{V_T}}}{I_{SC} e^{\frac{v_{BC}}{V_T}}} + 1 \right)}$$

$$\beta_{\text{forced}} = \frac{\frac{I_S}{I_{SC}} \left(e^{\frac{v_{BE}-v_{BC}}{V_T}} \right) - 1}{\frac{I_S}{\beta I_{SC}} \left(e^{\frac{v_{BE}-v_{BC}}{V_T}} \right) + 1} \quad \text{..... (1)}$$

Step 3 of 10

The expression for collector-to-emitter voltage of saturated transistor is,

$$v_{CE(\text{sat})} = v_{BE} - v_{BC}$$

Substitute $\left(v_{BE} - v_{BC}\right)$ for $v_{CE(\text{sat})}$ in equation (1).

$$\beta_{\text{forced}} = \frac{\frac{I_S}{I_{SC}} \left(e^{\frac{v_{CE(\text{sat})}}{V_T}} \right) - 1}{\frac{I_S}{\beta I_{SC}} \left(e^{\frac{v_{CE(\text{sat})}}{V_T}} \right) + 1}$$

$$\beta_{\text{forced}} \left(\frac{I_S}{\beta I_{SC}} e^{\frac{v_{CE(\text{sat})}}{V_T}} + 1 \right) = \frac{I_S}{I_{SC}} e^{\frac{v_{CE(\text{sat})}}{V_T}} - 1$$

$$\beta_{\text{forced}} \frac{I_S}{\beta I_{SC}} e^{\frac{v_{CE(\text{sat})}}{V_T}} + \beta_{\text{forced}} = \frac{I_S}{I_{SC}} e^{\frac{v_{CE(\text{sat})}}{V_T}} - 1$$

$$1 + \beta_{\text{forced}} = \frac{I_S}{I_{SC}} e^{\frac{v_{CE(\text{sat})}}{V_T}} - \beta_{\text{forced}} \frac{I_S}{\beta I_{SC}} e^{\frac{v_{CE(\text{sat})}}{V_T}}$$

Step 4 of 10

Further simplification as follows,

$$1 + \beta_{\text{forced}} = \frac{I_S}{I_{SC}} \left(1 - \frac{\beta_{\text{forced}}}{\beta} \right) e^{\frac{v_{CE(\text{sat})}}{V_T}}$$

$$\frac{1 + \beta_{\text{forced}}}{1 - \frac{\beta_{\text{forced}}}{\beta}} = \frac{I_S}{I_{SC}} e^{\frac{v_{CE(\text{sat})}}{V_T}}$$

$$e^{\frac{v_{CE(\text{sat})}}{V_T}} = \frac{I_{SC}}{I_S} \left(\frac{1 + \beta_{\text{forced}}}{1 - \frac{\beta_{\text{forced}}}{\beta}} \right)$$

$$\frac{v_{CE(\text{sat})}}{V_T} = \ln \left[\frac{I_{SC}}{I_S} \left(\frac{1 + \beta_{\text{forced}}}{1 - \frac{\beta_{\text{forced}}}{\beta}} \right) \right]$$

$$v_{CE(\text{sat})} = V_T \ln \left[\frac{I_{SC}}{I_S} \left(\frac{1 + \beta_{\text{forced}}}{1 - \frac{\beta_{\text{forced}}}{\beta}} \right) \right]$$

Thus, the expression for collector-to-emitter voltage of saturated transistor is,

$$\boxed{v_{CE(\text{sat})} = V_T \ln \left[\frac{I_{SC}}{I_S} \left(\frac{1 + \beta_{\text{forced}}}{1 - \frac{\beta_{\text{forced}}}{\beta}} \right) \right]}.$$

Step 5 of 10

Write the expression for emitter scale current.

$$I_{SE} = \frac{I_S}{\alpha}$$

$$= \frac{I_S}{\frac{\beta}{\beta + 1}}$$

Substitute 100 for β in the equation.

$$I_{SE} = \frac{\frac{I_S}{100}}{\frac{101}{100}}$$

$$= \left(\frac{101}{100} \right) I_S$$

Step 6 of 10

CBJ area is 100 times of the EBJ area.

$$I_{SC} = 100 I_{SE}$$

Substitute $\left(\frac{101}{100}\right) I_S$ for I_{SE} in the equation.

$$I_{SC} = (100) \left(\frac{101}{100} \right) I_S$$

$$I_{SC} = (101) I_S$$

Step 7 of 10

Determine the value of saturated collector-to-emitter voltage for $\beta_{\text{forced}} = 50$.

$$v_{CE(\text{sat})} = V_T \ln \left[\frac{I_{SC}}{I_S} \left(\frac{1 + \beta_{\text{forced}}}{1 - \frac{\beta_{\text{forced}}}{\beta}} \right) \right]$$

Substitute $25\ \text{mV}$ for V_T, $(101) I_S$ for I_{SC}, 50 for β_{forced} and 100 for β in the equation.

$$v_{CE(\text{sat})} = \left(25 \times 10^{-3}\right) \ln \left[\frac{101 I_S}{I_S} \left(\frac{1 + 50}{1 - \frac{50}{100}} \right) \right]$$

$$= \left(25 \times 10^{-3}\right) \ln \left[(101)(102) \right]$$

$$= 0.231\ \text{V}$$

Thus, the value of saturated collector-to-emitter voltage for $\beta_{\text{forced}} = 50$ is,

$$\boxed{v_{CE(\text{sat})} = 0.231\ \text{V}}.$$

Step 8 of 10

Determine the value of saturated collector-to-emitter voltage for $\beta_{\text{forced}} = 10$.

$$v_{CE(\text{sat})} = V_T \ln \left[\frac{I_{SC}}{I_S} \left(\frac{1 + \beta_{\text{forced}}}{1 - \frac{\beta_{\text{forced}}}{\beta}} \right) \right]$$

Substitute $25\ \text{mV}$ for V_T, $(101) I_S$ for I_{SC}, 10 for β_{forced} and 100 for β in the equation.

$$v_{CE(\text{sat})} = \left(25 \times 10^{-3}\right) \ln \left[\frac{101 I_S}{I_S} \left(\frac{1 + 10}{1 - \frac{10}{100}} \right) \right]$$

$$= \left(25 \times 10^{-3}\right) \ln \left[(101) \left(\frac{110}{9} \right) \right]$$

$$= 0.178\ \text{V}$$

Thus, the value of saturated collector-to-emitter voltage for $\beta_{\text{forced}} = 10$ is,

$$\boxed{v_{CE(\text{sat})} = 0.178\ \text{V}}.$$

Step 9 of 10

Determine the value of saturated collector-to-emitter voltage for $\beta_{\text{forced}} = 5$.

$$v_{CE(\text{sat})} = V_T \ln \left[\frac{I_{SC}}{I_S} \left(\frac{1 + \beta_{\text{forced}}}{1 - \frac{\beta_{\text{forced}}}{\beta}} \right) \right]$$

Substitute $25\ \text{mV}$ for V_T, $(101) I_S$ for I_{SC}, 5 for β_{forced} and 100 for β in the equation.

$$v_{CE(\text{sat})} = \left(25 \times 10^{-3}\right) \ln \left[\frac{101 I_S}{I_S} \left(\frac{1 + 5}{1 - \frac{5}{100}} \right) \right]$$

$$= \left(25 \times 10^{-3}\right) \ln \left[(101) \left(\frac{120}{19} \right) \right]$$

$$= 0.161\ \text{V}$$

Thus, the value of saturated collector-to-emitter voltage for $\beta_{\text{forced}} = 5$ is,

$$\boxed{v_{CE(\text{sat})} = 0.161\ \text{V}}.$$

Step 10 of 10

Determine the value of saturated collector-to-emitter voltage for $\beta_{\text{forced}} = 1$.

$$v_{CE(\text{sat})} = V_T \ln \left[\frac{I_{SC}}{I_S} \left(\frac{1 + \beta_{\text{forced}}}{1 - \frac{\beta_{\text{forced}}}{\beta}} \right) \right]$$

Substitute $25\ \text{mV}$ for V_T, $(101) I_S$ for I_{SC}, 1 for β_{forced} and 100 for β in the equation.

$$v_{CE(\text{sat})} = \left(25 \times 10^{-3}\right) \ln \left[\frac{101 I_S}{I_S} \left(\frac{1 + 1}{1 - \frac{1}{100}} \right) \right]$$

$$= \left(25 \times 10^{-3}\right) \ln \left[(101) \left(\frac{200}{99} \right) \right]$$

$$= 0.133\ \text{V}$$

Thus, the value of saturated collector-to-emitter voltage for $\beta_{\text{forced}} = 1$ is,

$$\boxed{v_{CE(\text{sat})} = 0.133\ \text{V}}$$

Step 1 of 2

6.022E

Refer to Figure 6.22 (a) in the text book for the transistor circuit.

The base of the transistor can be raised to the highest voltage with transistor remaining in the active mode at the edge of saturation, that is, at $V_{CE} = 0.3\ \text{V}$.

Apply Kirchhoff's voltage law to the collector-emitter loop.

$$10 = I_C R_C + V_{CE} + I_E R_E$$

Substitute αI_E for I_C , 1 for α , $4.7\times10^3\ \Omega$ for R_C , $3.3\times10^3\ \Omega$ for R_E and 0.3 V for V_{CE} .

$$10 = \alpha I_E R_C + V_{CE} + I_E R_E$$
$$10 = I_E(\alpha R_C + R_E) + V_{CE}$$
$$10 = I_E\left[1\left(4.7\times10^3\right) + 3.3\times10^3\right] + 0.3$$
$$10 = I_E\left(8\times10^3\right) + 0.3$$

$$I_E = \frac{10 - 0.3}{8\times10^3}$$
$$= 1.2125\times10^{-3}\ \text{A}$$
$$= 1.2125\ \text{mA}$$

Step 2 of 2

Determine the highest base voltage.

$$V_B = 0.7 + I_E R_E$$

Substitute $1.2125\times10^{-3}\ \text{A}$ for I_E and $3.3\times10^3\ \Omega$ for R_E .

$$V_B = 0.7 + \left(1.2125\times10^{-3}\right)\left(3.3\times10^3\right)$$
$$= +4.7\ \text{V}$$

Thus, the highest base voltage for the transistor to remain in the active mode is $\boxed{+4.7\ \text{V}}$.

Step 1 of 5

6.022P

Emitter of the pnp transistor is connected to the ground, collector is connected to $-5\ \text{V}$ DC source through a $8.2\ \text{k}\Omega$ resistors and a current source of $2\ \text{mA}$ is connected to the base with the polarity so that current is drawn out of the base terminal is shown in the below figure 1.

Step 2 of 5

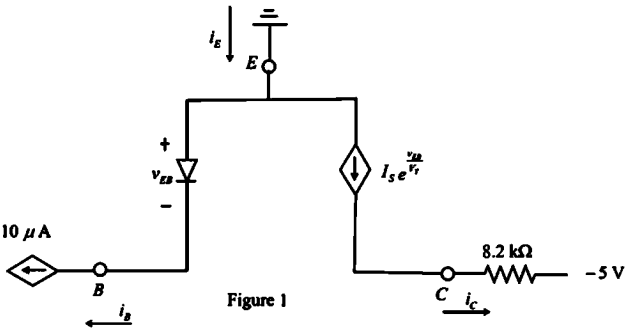


Figure 1

Step 3 of 5

From the figure 1 we observed that base current i_B is equal to $10\ \text{mA}$

Common emitter current gain β is $\dfrac{\text{collector current } (i_C)}{\text{base current } (i_B)}$

β for pnp transistor is 50

Collector current in terms of base current is βi_B

Relation between collector current, base current and emitter current is $i_E = i_B + i_C$

$$\begin{aligned} i_E &= i_B + i_C \\ &= i_B + \beta i_B \\ &= (10\ \mu\text{A}) + (50 \times 10\ \mu\text{A}) \\ &= 60\ \mu\text{A} \end{aligned}$$

Thus,Emitter current for pnp transistor is $\boxed{60\ \mu\text{A}}$

Step 4 of 5

Voltage across the collector terminal is V_C

Refer figure 1,we observed that $V_C - (-5) = i_C (8.2\ \text{k}\Omega)$

Collector current i_C is $50\ \mu\text{A}$

$$\begin{aligned} V_C - (-5) &= i_C (8.2\ \text{k}\Omega) \\ V_C + 5 &= (50\ \mu\text{A})(8.2 \times 10^3\ \Omega) \\ V_C + 5 &= 0.41 \\ V_C &= -4.59\ \text{V} \end{aligned}$$

Thus, Voltage across the collector terminal V_C is $\boxed{-4.59\ \text{V}}$

Step 5 of 5

Saturation current for pnp transistor I_S is $10^{-14}\ \text{A}$

Emitter-base voltage v_{EB} for collector current i_C equal to $50\ \mu\text{A}$ is,

$$\begin{aligned} i_C &= I_S e^{v_{EB}/V_T} \\ 50 \times 10^{-6} &= (10^{-14}) e^{\frac{V_{EB}}{25 \times 10^{-3}}} \\ 50 \times 10^{8} &= e^{\frac{V_{EB}}{25 \times 10^{-3}}} \end{aligned}$$

Apply natural log on both sides

$$\begin{aligned} \ln(50 \times 10^8) &= \ln\left(e^{\frac{V_{EB}}{25 \times 10^{-3}}}\right) \\ 22.33 &= \frac{V_{EB}}{25 \times 10^{-3}} \\ 0.56 &= V_{EB} \end{aligned}$$

Emitter – base voltage V_{EB} is $0.56\ \text{V}$

Emitter terminal is connected to ground hence V_E is $0\ \text{V}$

$$\begin{aligned} V_E - V_B &= 0.56 \\ 0 - V_B &= 0.56 \\ V_B &= -0.56\ \text{V} \end{aligned}$$

Thus , Base voltage V_B is $\boxed{-0.56\ \text{V}}$

Step 1 of 2

6.023E

Refer to Figure 6.22 (a) in the text book for the transistor circuit.

Write the formula for the base voltage, V_B .

$$V_B = 0.7 + I_E R_E$$

Substitute $\frac{I_C}{\alpha}$ for I_E .

$$V_B = 0.7 + \frac{I_C}{\alpha} R_E$$

Substitute $0.5\times10^{-3}\ \text{A}$ for I_C , 1 for α and $4\ \text{V}$ for V_B .

$$4 = 0.7 + \frac{0.5\times10^{-3}}{1} R_E$$

$$R_E = \frac{4-0.7}{0.5\times10^{-3}}$$

$$= 6.6\ \text{k}\Omega$$

Thus, the new value of emitter resistance, R_E is $\boxed{6.6\ \text{k}\Omega}$.

Step 2 of 2

Apply Kirchhoff's voltage law to the collector-emitter loop.

$$10 = I_C R_C + V_{CB} + 4$$

Substitute $0.5\times10^{-3}\ \text{A}$ for I_C and 2 V for V_{CB} .

$$10 = \left(0.5\times10^{-3}\right) R_C + 2 + 4$$

$$R_C = \frac{10-2-4}{0.5\times10^{-3}}$$

$$= 8\ \text{k}\Omega$$

Thus, the new value of collector resistance, R_C is $\boxed{8\ \text{k}\Omega}$.

Step 1 of 3

6.023P

The relation between collector current (i_C), saturation current (I_S) and emitter- base voltage v_{EB} for active PNP transistor is,

$$i_C = I_S e^{v_{EB}/V_T}$$

Thermal voltage V_T is $25\ \text{mV}$ approximately at room temperature

The collector current for Emitter – base voltage v_{EB} equal to $0.7\ \text{V}$ is $1\ \text{mA}$

$$i_C = I_S e^{v_{EB}/V_T}$$
$$1\times10^{-3} = I_S e^{0.7/25\times10^{-3}}$$
$$1\times10^{-3} = I_S e^{28}$$
$$I_S = 6.91\times10^{-17}\ \text{A}$$

Thus, the saturation current for PNP transistor is $\boxed{6.91\times10^{-17}\ \text{A}}$.

Step 2 of 3

For different values of v_{EB} the PNP transistor has different collector currents.

The collector current of the transistor is $10\ \text{mA}$

$$i_C = I_S e^{v_{EB}/V_T}$$
$$10\times10^{-3} = \left(6.91\times10^{-17}\ \text{A}\right)e^{v_{EB}/25\times10^{-3}}$$
$$1.45\times10^{14} = e^{v_{EB}/25\times10^{-3}}$$

Apply natural log on both sides.

$$\ln\left(1.45\times10^{14}\right) = \ln\left(e^{v_{EB}/25\times10^{-3}}\right)$$
$$32.61 = \frac{v_{BE}}{25\times10^{-3}}$$
$$v_{BE} = 0.815\ \text{V}$$

Thus, the emitter base voltage for transistor having collector current of $10\ \text{mA}$ is $\boxed{0.815\ \text{V}}$.

Step 3 of 3

The collector current of the transistor is $100\ \text{mA}$

$$i_C = I_S e^{v_{EB}/V_T}$$
$$100\times10^{-3} = \left(6.91\times10^{-17}\ \text{A}\right)e^{v_{EB}/25\times10^{-3}}$$
$$1.45\times10^{15} = e^{v_{EB}/25\times10^{-3}}$$

Apply natural log on both sides.

$$\ln\left(1.45\times10^{15}\right) = \ln\left(e^{v_{EB}/25\times10^{-3}}\right)$$
$$34.91 = \frac{v_{BE}}{25\times10^{-3}}$$
$$v_{BE} = 0.873\ \text{V}$$

Thus, the emitter base voltage for transistor having collector current of $100\ \text{mA}$ is $\boxed{0.873\ \text{V}}$.

Step 1 of 2

6.024E

Refer to Figure 6.23(a) in the text book for the transistor circuit.

The transistor parameter β_{forced} is,

$$\beta_{\text{forced}} = 5$$
$$\frac{I_C}{I_B} = 5$$
$$I_B = \frac{I_C}{5}$$

The emitter current is,

$I_E = I_B + I_C$ Substitute $\frac{I_C}{5}$ for I_B.

$$I_E = \frac{I_C}{5} + I_C$$
$$I_E = \frac{6}{5} I_C$$
$$I_C = \frac{5I_E}{6}$$

Step 2 of 2

Apply Kirchhoff's voltage law to the circuit.

$$10 = I_C\left(4.7\times10^3\right) + V_{CE} + I_E\left(3.3\times10^3\right)$$

Substitute $\frac{5I_E}{6}$ for I_C.

$$10 = \frac{5I_E}{6}\left(4.7\times10^3\right) + V_{CE} + I_E\left(3.3\times10^3\right)$$
$$10 = I_E\left[\frac{5}{6}\left(4.7\times10^3\right) + 3.3\times10^3\right] + 0.2$$
$$I_E\left(7.2167\times10^3\right) + 0.2 = 10$$
$$I_E = \frac{10 - 0.2}{7.2167\times10^3}$$
$$= 1.358\ \text{mA}$$

Determine the base voltage.

$$V_B = 0.7 + I_E R_E$$
$$= 0.7 + \left(1.358\times10^{-3}\right)\left(3.3\times10^3\right)$$
$$= 0.7 + 4.48$$
$$= +5.18\ \text{V}$$

Thus, the base voltage V_B for the transistor to operate in saturation region is **+5.18 V**.

Step 1 of 7

Base of the pnp transistor is connected to the ground, collector is connected to $-2\ \text{V}$ DC source and a current of $1\ \text{mA}$ is injected into its emitter is shown in the below figure 1.

Step 2 of 7

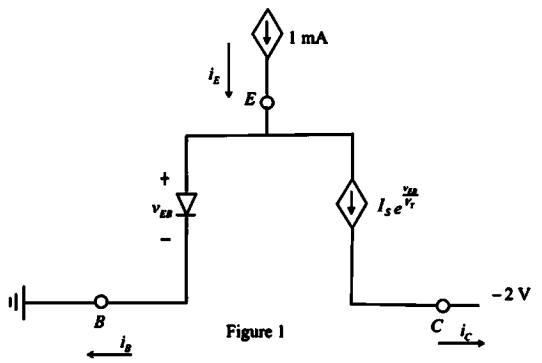


Figure 1

Step 3 of 7

Refer the figure 1 we observed that emitter current i_E is equal to $1\ \text{mA}$

Common emitter current gain β is 10

Collector current in terms of base current is βi_B

Relation between collector current, base current and emitter current is $i_E = i_B + i_C$

$$i_E = i_B + i_C$$
$$1\ \text{mA} = i_B + \beta i_B$$
$$1\ \text{mA} = i_B(1+10)$$
$$i_B = 91\ \mu\text{A}$$

Thus, Base current for pnp transistor is $\boxed{91\ \mu\text{A}}$

Base current flow from out of the base terminal.

Step 4 of 7

Collector current in terms of base current is βi_B

$$i_C = \beta i_B$$
$$= 10(91\ \mu\text{A})$$
$$= 910\ \mu\text{A}$$

Thus, Collector current for pnp transistor is $\boxed{910\ \mu\text{A}}$

collector current flow from out of the collector terminal

Step 5 of 7

Saturation current for pnp transistor I_S is $10^{-15}\ \text{A}$

Emitter-base voltage v_{EB} for collector current i_C equal to $50\ \mu\text{A}$ is,

$$i_C = I_S e^{v_{EB}/V_T}$$
$$910\times10^{-6} = \left(10^{-15}\right)e^{\frac{V_{EB}}{25\times10^{-3}}}$$
$$9.1\times10^{11} = e^{\frac{V_{EB}}{25\times10^{-3}}}$$

Apply natural log on both sides

$$\ln\left(9.1\times10^{11}\right) = \ln\left(e^{\frac{V_{EB}}{25\times10^{-3}}}\right)$$
$$27.54 = \frac{V_{EB}}{25\times10^{-3}}$$
$$0.69 = V_{EB}$$

Thus, Emitter – base voltage V_{EB} is $0.69\ \text{V}$

Base terminal is connected to ground hence V_B is $0\ \text{V}$

$$V_E - V_B = 0.69$$
$$V_E - 0 = 0.69$$
$$V_E = 0.69\ \text{V}$$

Thus Emitter voltage V_E for pnp transistor is $\boxed{0.69\ \text{V}}$

Step 6 of 7

Common emitter current gain β is 1000

Collector current in terms of base current is βi_B

Relation between collector current, base current and emitter current is $i_E = i_B + i_C$

$$i_E = i_B + i_C$$
$$1\ \text{mA} = i_B + \beta i_B$$
$$1\ \text{mA} = i_B(1+1000)$$
$$i_B = 0.999\ \mu\text{A}$$

Thus, Base current for pnp transistor is $\boxed{0.999\ \mu\text{A}}$

Step 7 of 7

Collector current in terms of base current is βi_B

$$i_C = \beta i_B$$
$$= 1000(0.999\ \mu\text{A})$$
$$= 999\ \mu\text{A}$$

Collector current for pnp transistor is $\boxed{999\ \mu\text{A}}$

Step 1 of 2

6.025E

Refer to Figure 6.25(a) in the text book for the transistor circuit.

Apply Kirchhoff's voltage law to the emitter-base loop.

$$10 = I_E R_E + 0.7 + 0$$
$$10 = I_E\left(2\times10^3\right) + 0.7 + 0$$

$$I_E = \frac{10-0.7}{2\times10^3}$$
$$= 4.65\times10^{-3}\ \text{A}$$
$$= 4.65\ \text{mA}$$

Assume $\beta = 100$.

Determine α .

$$\alpha = \frac{\beta}{\beta+1}$$
$$= \frac{100}{101}$$
$$= 0.99$$

Determine the collector current, I_C .

$$I_C = \alpha I_E$$
$$= (0.99)\left(4.65\times10^{-3}\right)$$
$$= 4.6\times10^{-3}\ \text{A}$$
$$= 4.6\ \text{mA}$$

Step 2 of 2

Apply Kirchhoff's voltage law to the collector-base loop.

$$0 + V_{BC} + I_C R_C - 10 = 0$$
$$R_C = \frac{10 - V_{BC}}{I_C}$$

Substitute $V_{EC} - V_{EB}$ for V_{BC} .

$$R_C = \frac{10 - \left(V_{EC} - V_{EB}\right)}{I_C}$$

Substitute 0.3 V for V_{EC} as the transistor is at the edge of saturation and 0.7 V for V_{EB} .

$$R_C = \frac{10 - \left(V_{EC} - V_{EB}\right)}{I_C}$$

$$R_C = \frac{10 - (0.3 - 0.7)}{4.6\times10^{-3}}$$
$$= \frac{10 + 0.4}{4.6\times10^{-3}}$$
$$= 2.26\times10^3\ \Omega$$
$$= 2.26\ \text{k}\Omega$$

Thus, the largest value of R_C for the transistor to remain in the active mode is $\boxed{2.26\ \text{k}\Omega}$.

Step 1 of 6

The relation between collector current (i_C), saturation current (I_S) and emitter- base voltage v_{EB} for active PNP transistor is,

$$i_C = I_S e^{v_{EB}/V_T}$$

Thermal voltage V_T is $25\ \text{mV}$ approximately at room temperature

The common emitter current gain β is $\dfrac{\text{collector current } (i_C)}{\text{base current } (i_B)}$

Hence, the base current in terms of β and collector current (i_c) is,

$$i_B = \frac{i_C}{\beta} \quad \text{..... (1)}$$

Step 2 of 6

The relation between collector current, base current and emitter current is,

$$i_E = i_B + i_C$$

Emitter current for *pnp* power transistor is $5\ \text{A}$

Substitute emitter current and base current in terms of collector current in the equation $i_E = i_B + i_C$

$$5\ \text{A} = \frac{i_C}{\beta} + i_C$$

$$5 = i_C\left(\frac{1+\beta}{\beta}\right)$$

Substitute 20 for β .

$$i_C\left(\frac{1+\beta}{\beta}\right) = 5\ \text{A}$$

$$i_C\left(\frac{1+20}{20}\right) = 5\ \text{A}$$

$$i_C = 5\left(\frac{20}{21}\right)\ \text{A}$$

$$i_C = 4.762\ \text{A}$$

Thus, the collector current for the *pnp* power transistor is $4.762\ \text{A}$

Step 3 of 6

The base current i_B is $\dfrac{i_C}{\beta}$

$$i_\beta = \frac{i_C}{\beta}$$
$$= \frac{4.762}{20}$$
$$= 0.24\ \text{A}$$

Thus, the base current i_B for pnp power transistor is $\boxed{0.24\ \text{A}}$.

Step 4 of 6

The relation between collector current i_C and saturation current I_S is,

$$i_C = I_S e^{v_{EB}/V_T}$$

The collector current for *pnp* power transistor is $4.762\ \text{A}$

The emitter-base voltage for *pnp* power transistor is $0.8\ \text{V}$

$$i_C = I_S e^{v_{EB}/V_T}$$

$$4.762\ \text{A} = I_S e^{0.8/(25\times10^{-3})}$$

$$4.762 = I_S e^{32}$$

$$I_S = 6.03\times10^{-14}\ \text{A}$$

Thus, the saturation current I_S for power transistor is $\boxed{6.03\times10^{-32}\ \text{A}}$.

Step 5 of 6

The collector current for small signal transistor is $1\ \text{mA}$

The emitter – base voltage for pnp small signal transistor is $0.70\ \text{V}$

$$i_C = I_S e^{v_{EB}/V_T}$$

$$1\times10^{-3}\ \text{A} = I_S e^{0.70/(25\times10^{-3})}$$

$$1\times10^{-3} = I_S e^{28}$$

$$I_S = 6.91\times10^{-16}\ \text{A}$$

The saturation current I_S for small signal transistor is $\boxed{6.91\times10^{-16}\ \text{A}}$.

Step 6 of 6

The relation between saturation current and junction area is,

$$I_S = A_E q D_n n_{P0}/W$$

For power transistor

$$6.03\times10^{-14} = A_{Ep} q D_n n_{P0}/W \quad \text{..... (2)}$$

For small signal transistor

$$6.91\times10^{-16} = A_{Es} q D_n n_{P0}/W \quad \text{..... (3)}$$

Divide equation (2) by (3).

$$\frac{6.03\times10^{-14}}{6.91\times10^{-16}} = \frac{A_{Ep} q D_n n_{P0}/W}{A_{Es} q D_n n_{P0}/W}$$

$$87.26 = \frac{A_{Ep}}{A_{Es}}$$

$$A_{Ep} = 87.26 A_{Es}$$

Thus, the junction area of power transistor is **87.26 times** the junction area of small signal transistor

Step 1 of 2

6.026E

Refer to Figure 6.25(a) in the text book for the transistor circuit.

Apply Kirchhoff's voltage law to the emitter-base loop.

$$10 = I_E R_E + 0.7 + 0$$

$$10 = \frac{I_C}{\alpha} R_E + 0.7$$

Substitute 1 for α , 10^{-3} A for I_C .

$$10 = \frac{10^{-3}}{1} R_E + 0.7$$

$$R_E = \frac{10 - 0.7}{10^{-3}}$$

$$= 9.3\ \text{k}\Omega$$

Thus, the emitter resistance R_E to establish a collector current of 1 mA is $\boxed{9.3\ \text{k}\Omega}$.

Step 2 of 2

Apply Kirchhoff's voltage law to the collector-base loop.

$$0 + V_{BC} + I_C R_C - 10 = 0$$

$$R_C = \frac{10 - V_{BC}}{I_C}$$

Substitute 4 V for V_{BC} and 10^{-3} A for I_C

$$R_C = \frac{10 - (V_{EC} - V_{EB})}{I_C}$$

Substitute 0.3 V for V_{EC} as the transistor is at the edge of saturation and 0.7 V for V_{EB} .

$$R_C = \frac{10 - 4}{I_C}$$

$$= \frac{10 - 4}{10^{-3}}$$

$$= 6 \times 10^3\ \Omega$$

$$= 6\ \text{k}\Omega$$

Thus, the collector resistance R_C is $\boxed{6\ \text{k}\Omega}$.

6.026P

Refer to Figure 6.5 for four possible large-signal equivalent circuit models of the *npn* BJT operating in the forward active mode in the text book.

Refer to Figure 6.11 for two possible large-signal equivalent circuit models of the *pnp* BJT operating in the active mode in the text book.

The current-voltage relationship of the *pnp* transistor is identical to the *npn* transistor except that v_{BE} has to be replaced by v_{EB}.

Draw the remaining two large-signal models of a ***pnp*** transistor in the active mode.

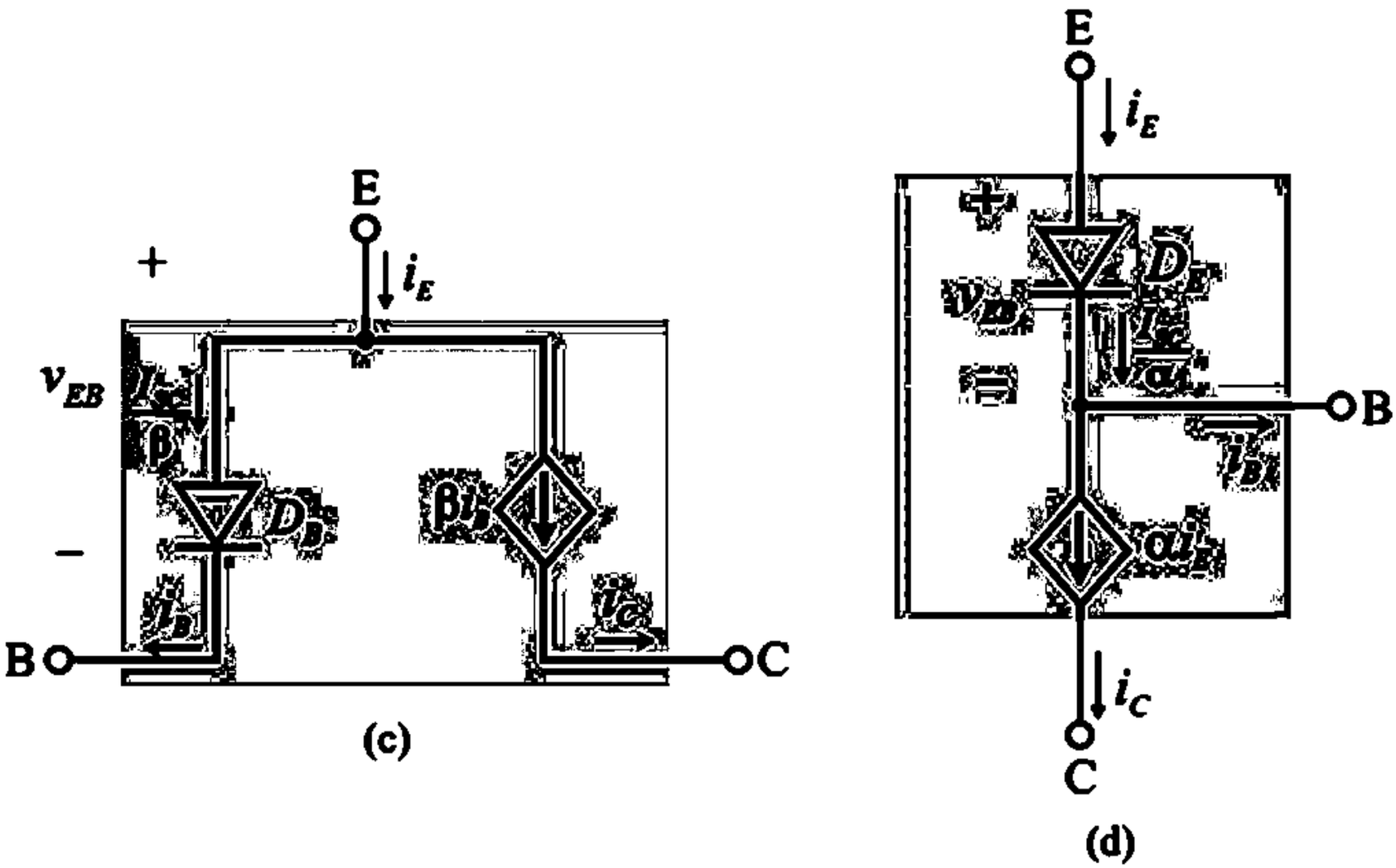


Figure 1

Thus, the remaining two large-signal models of a ***pnp*** transistor in the active mode are drawn as shown in Figure 1.

Step 1 of 3

Refer to Figure 6.26 in the text book for the transistor circuit.

Apply Kirchhoff's voltage law to the emitter-base circuit.

$$I_B R_B + V_{BE} = V_{BB}$$

$$I_B = \frac{V_{BB} - V_{BE}}{R_B}$$
$$= \frac{5 - 0.7}{100 \times 10^3}$$
$$= 43 \times 10^{-6} \text{ A}$$
$$= 43 \ \mu\text{A}$$

Determine the collector current for $\beta = 150$.

$$I_C = \beta I_B$$
$$= 150\left(43 \times 10^{-6}\right)$$
$$= 6.45 \text{ mA}$$

Step 2 of 3

Transistor remains in the active region up to $V_{CE} = 0.3 \text{ V}$.

Apply Kirchhoff's voltage law to the collector-emitter circuit.

$$10 = I_C R_C + V_{CE}$$
$$R_C = \frac{10 - V_{CE}}{I_C}$$
$$R_C = \frac{10 - 0.3}{6.45 \times 10^{-3}}$$
$$R_C = 1.5 \text{ k}\Omega$$

Therefore the value of R_C for which all fabricated circuits are guaranteed to be in the active mode is $\boxed{1.5 \text{ k}\Omega}$.

Step 3 of 3

Determine the collector current for $\beta = 50$.

$$I_C = \beta I_B$$
$$= 50\left(43 \times 10^{-6}\right)$$
$$= 2.15 \text{ mA}$$

Determine the collector voltage.

$$V_C = V_{CC} - I_C R_C$$
$$= 10 - \left(2.15 \times 10^{-3}\right)\left(1.5 \times 10^3\right)$$
$$= 6.8 \text{ V}$$

Determine the collector voltage for $\beta = 150$.

$$V_C = V_{CC} - I_C R_C$$
$$= 10 - \left(6.45 \times 10^{-3}\right)\left(1.5 \times 10^3\right)$$
$$= 0.3 \text{ V}$$

Thus, the range of collector voltage V_C is $\boxed{0.3 \text{ V to } 6.8 \text{ V}}$.

6.027P

Refer to Figure 6.9 for modeling the operation of an *npn* transistor in saturation in the text book.

The current-voltage relationship of the *pnp* transistor will be identical to that of the *npn* transistor except that v_{BE} has to be replaced by v_{EB}.

Draw the equivalent circuit of a *pnp* transistor in saturation is shown in Figure 1.

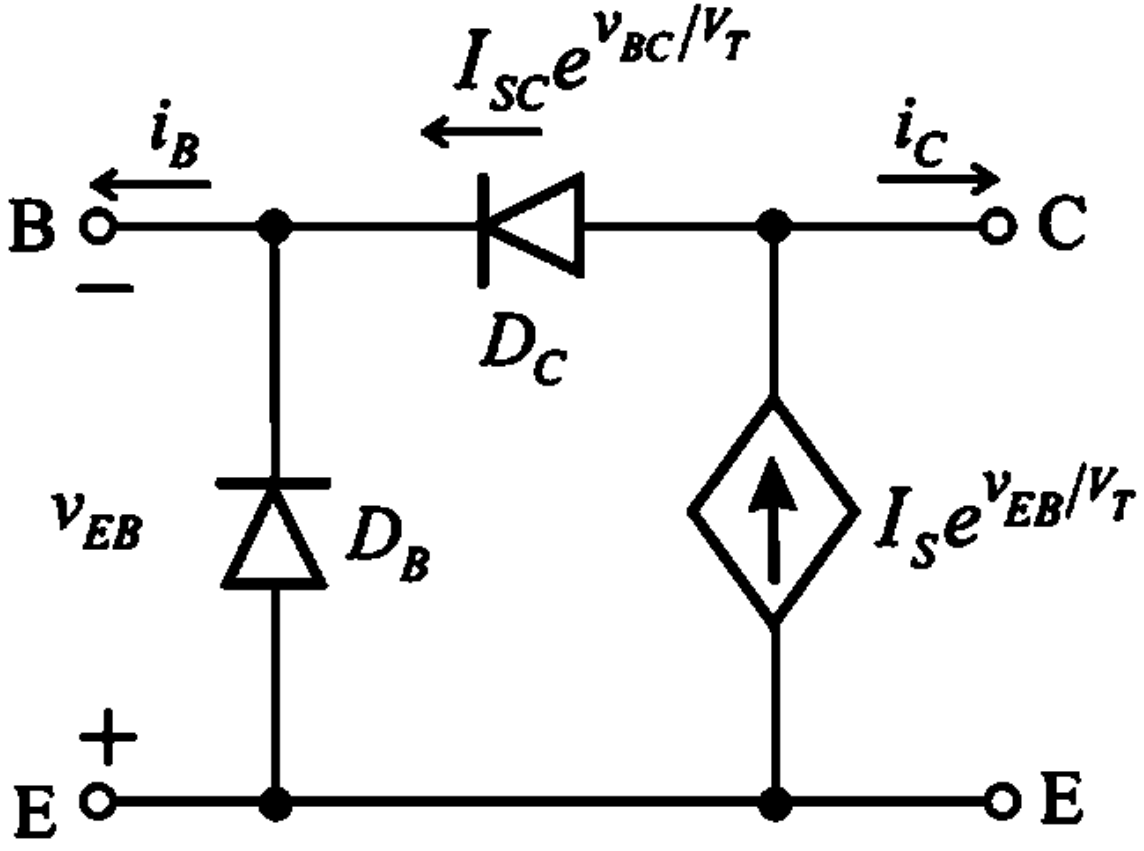


Figure 1

Thus, the equivalent circuit of modeling the operation of an *pnp* transistor in saturation is drawn as shown in Figure 1.

Step 1 of 8

Refer to Figure 6.28 (a) in the text book.

Simplify the base circuit using Thevenin's theorem.

Determine the Thevenin voltage.

$$V_{BB} = (+15)\left(\frac{R_{B2}}{R_{B1} + R_{B2}}\right)$$

Substitute $100\,\text{k}\Omega$ for R_{B1} and $50\,\text{k}\Omega$ for R_{B2}.

$$\begin{aligned} V_{BB} &= 15\left(\frac{50\times10^3}{100\times10^3 + 50\times10^3}\right) \\ &= 15\left(\frac{50\times10^3}{150\times10^3}\right) \\ &= +5\text{ V} \end{aligned}$$

Step 2 of 8

Determine the base resistance, R_{BB}.

$$R_{BB} = R_{B1} \parallel R_{B2}$$

Substitute $100\,\text{k}\Omega$ for R_{B1} and $50\,\text{k}\Omega$ for R_{B2}.

$$\begin{aligned} R_{BB} &= \left(100\times10^3\right) \parallel \left(50\times10^3\right) \\ &= \frac{\left(100\times10^3\right)\left(50\times10^3\right)}{\left(100\times10^3\right) + \left(50\times10^3\right)} \\ &= 33.3\text{ k}\Omega \end{aligned}$$

Step 3 of 8

Draw the resultant circuit.

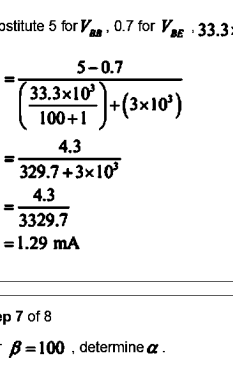


Figure 1

Step 4 of 8

From the circuit in Figure 1, apply Kirchhoff's voltage law to the input loop.

$$V_{BB} = I_B R_{BB} + V_{BE} + I_E R_E$$

Assume active-mode operation, and replace I_B with $\frac{I_E}{\beta+1}$.

$$V_{BB} = \left(\frac{I_E}{\beta+1}\right) R_{BB} + V_{BE} + I_E R_E$$

$$V_{BB} - V_{BE} = \left(\frac{R_{BB}}{\beta+1} + R_E\right) I_E$$

$$I_E = \frac{V_{BB} - V_{BE}}{\left(\frac{R_{BB}}{\beta+1} + R_E\right)}$$

Substitute 5 for V_{BB}, 0.7 for V_{BE}, 33.3×10^3 for R_{BB} and 3×10^3 for R_E.

$$\begin{aligned} I_E &= \frac{5 - 0.7}{\left(\frac{33.3\times10^3}{50+1}\right) + \left(3\times10^3\right)} \\ &= \frac{4.3}{652.9 + 3\times10^3} \\ &= \frac{4.3}{3652.94} \\ &= 1.177\text{ mA} \end{aligned}$$

Step 5 of 8

Determine collector current from the following formula:

$$I_C = \alpha I_E$$

Here,

$$\begin{aligned} \alpha &= \frac{\beta}{\beta+1} \\ &= \frac{50}{50+1} \\ &= 0.98 \end{aligned}$$

Thus,

$$\begin{aligned} I_C &= 0.98\left(1.177\times10^{-3}\right) \\ &= 1.15\times10^{-3}\text{ A} \\ &= 1.15\text{ mA} \end{aligned}$$

Therefore, the new value of collector current I_C is $\boxed{1.15\text{ mA}}$.

Step 6 of 8

Determine I_E for $\beta = 100$

$$I_E = \frac{V_{BB} - V_{BE}}{\left(\frac{R_{BB}}{\beta+1} + R_E\right)}$$

Substitute 5 for V_{BB}, 0.7 for V_{BE}, 33.3×10^3 for R_{BB}, 3×10^3 for R_E and 100 for β.

$$\begin{aligned} I_E &= \frac{5 - 0.7}{\left(\frac{33.3\times10^3}{100+1}\right) + \left(3\times10^3\right)} \\ &= \frac{4.3}{329.7 + 3\times10^3} \\ &= \frac{4.3}{3329.7} \\ &= 1.29\text{ mA} \end{aligned}$$

Step 7 of 8

For $\beta = 100$, determine α.

$$\begin{aligned} \alpha &= \frac{\beta}{\beta+1} \\ &= \frac{100}{100+1} \\ &= 0.99 \end{aligned}$$

Determine collector current.

$$\begin{aligned} I_C &= 0.99\left(1.29\times10^{-3}\right) \\ &= 1.28\times10^{-3} \\ &= 1.28\text{ mA} \end{aligned}$$

Step 8 of 8

Determine the percentage change in I_C.

$$\begin{aligned} \%\,I_C &= \left(\frac{1.15\times10^{-3} - 1.28\times10^{-3}}{1.28\times10^{-3}}\right)\times100 \\ &= -10\% \end{aligned}$$

Therefore, the percentage change in I_C is $\boxed{-10\%}$.

Step 1 of 9

(a)

Refer to the Figure P6.28 (a) in the text book,

The common emitter current gain β is,

$$\beta = \frac{\text{collector current } (i_C)}{\text{base current } (i_B)}$$

Transistors have very large β hence base current is negligible when compared with the collector current therefore $i_B + i_C \approx i_C$.

Relation between collector current, base current and emitter current is,

$$i_E = i_B + i_C$$
$$i_E \simeq i_C$$

For large value of β collector current is equal to the emitter current.

From Figure P6.28 (a), the emitter current, I_1 is,

$$I_1 = \frac{(10.7\text{ V}) - (0.7\text{ V})}{(5\text{ k}\Omega)}$$
$$= \frac{10\text{ V}}{5\times10^3\ \Omega}$$
$$= 2\text{ mA}$$

Thus, value of Emitter current I_1 is $\boxed{2\text{ mA}}$.

Step 2 of 9

Determine the value of collector voltage:

The collector current is equal to emitter current for very large value of β and hence I_C is equal to 2 mA.

$$I_C = \frac{(V_2) - (-10.7)}{5\text{ k}\Omega}$$
$$2\times10^{-3} = \frac{(V_2 + 10.7)\text{ V}}{5000\ \Omega}$$
$$10 = (V_2 + 10.7)\text{ V}$$
$$V_2 = -0.7\text{ V}$$

Thus, value of Collector voltage V_2 is $\boxed{-0.7\text{ V}}$

Step 3 of 9

(b)

Refer to the figure P6.28 (b) in the text book,

The collector current I_C is,

$$I_C = \frac{(-4) - (-10)}{2.4\text{ k}\Omega}$$
$$= \frac{(-4 + 10)\text{ V}}{2.4\times10^3\ \Omega}$$
$$= 2.5\text{ mA}$$

Thus, value of Collector current I_C is $\boxed{2.5\text{ mA}}$.

Step 4 of 9

The collector current is equal to emitter current for very large value of β and hence I_E is equal to 2.5 mA

Refer to the figure P6.28 (b) in the text book,

$$I_E = \frac{12 - (V_3)}{5.6\text{ k}\Omega}$$
$$2.5\times10^{-3} = \frac{(12 - V_3)\text{ V}}{5.6\times10^3\ \Omega}$$
$$12 - V_3 = 14$$
$$V_3 = -2\text{ V}$$

Thus, value of Emitter voltage V_3 is $\boxed{-2\text{ V}}$

Step 5 of 9

(c)

Refer to the figure P6.28 (c) in the textbook

The collector current I_C is,

$$I_C = \frac{(0) - (-10)}{20\text{ k}\Omega}$$
$$= \frac{(10)\text{ V}}{20\times10^3\ \Omega}$$
$$= 0.5\text{ mA}$$

Collector current I_C is 0.5 mA

Collector current is equal to emitter current for very large value of β.

Hence the current I_5 is equal to $\boxed{0.5\text{ mA}}$,

Step 6 of 9

For very large value of β base current I_B is zero.

$$I_B = \frac{V_4 - 2}{15\text{ k}\Omega}$$
$$0 = \frac{V_4 - 2}{15\text{ k}\Omega}$$
$$V_4 = 2\text{ V}$$

Thus, value of Base voltage V_4 is $\boxed{2\text{ V}}$

Step 7 of 9

(d)
Refer to the figure P6.28 (d) in the textbook.

In Figure P6.28 (d), the transistor has very large value of β.

$$I_C \simeq I_E$$
$$= I_6 \quad (\text{since } I_E = I_6)$$

and

$$I_B \simeq \frac{I_C}{\beta}$$
$$\simeq 0\text{ A} \quad (\text{since } \beta >> 0)$$

Step 8 of 9

Determine the value of current, I_6.

Apply Kirchhoff's voltage law in Figure P6.28 (d).

$$-10 + (9.1\times10^3)I_C + (10\times10^3)I_B + 0.7 + (3\times10^3)I_6 - 10 = 0$$
$$(9.1\times10^3)I_C + (10\times10^3)I_B + (3\times10^3)I_6 = 19.3$$

Substitute 0 A for I_B and I_6 for I_C in above expression.

$$(9.1\times10^3)I_6 + (10\times10^3)(0) + (3\times10^3)I_6 = 19.3$$
$$(12.1\times10^3)I_6 = 19.3$$
$$I_6 = \frac{19.3}{12.1\times10^3}$$
$$I_6 = 1.595\text{ mA}$$

Thus, the value of current, I_6 is $\boxed{1.595\text{ mA}}$.

Step 9 of 9

Determine the value of voltage, V_7.

$$I_C = \frac{10 - V_7}{9.1\times10^3}$$
$$V_7 = 10 - (9.1\times10^3)I_C$$
$$V_7 = 10 - (9.1\times10^3)I_6 \quad (\text{since } I_C = I_6)$$

Substitute 1.595 mA for I_6 in the equation.

$$V_7 = 10 - (9.1\times10^3)(1.595\times10^{-3})$$
$$= 10 - 14.5145$$
$$= -4.5145\text{ V}$$

Thus, the value of voltage, V_7 is $\boxed{-4.5145\text{ V}}$.

Step 1 of 2

6.029E

Refer to Figure 6.29 (b) in the text book.

The currents flowing from the power supply are 0.103 mA, 1.252 mA and 2.78 mA.

Determine the total current drawn from the power supply using Kirchhoff's current law.

$$\begin{aligned} I_s &= 0.103\,\text{mA} + 1.252\,\text{mA} + 2.78\,\text{mA} \\ &= \left(0.103\times10^{-3}\right) + \left(1.252\times10^{-3}\right) + \left(2.78\times10^{-3}\right) \\ &= 4.135\times10^{-3}\ \text{A} \\ &= 4.135\ \text{mA} \end{aligned}$$

Therefore, the current drawn from the power supply is $\boxed{4.135\ \text{mA}}$.

Step 2 of 2

Determine the power dissipated in the circuit.

$$P = V_{CC} I_s$$

Substitute 15 V for V_{CC} and 4.135 mA for I_s .

$$\begin{aligned} P &= (15)\left(4.135\times10^{-3}\right) \\ &= 62\ \text{mW} \end{aligned}$$

Therefore, the power dissipated in the circuit is $\boxed{62\ \text{mW}}$.

Step 1 of 12

(a)

Refer to Figure P6.29 (a) in the text book.

The modified circuit diagram is shown in Figure 1.

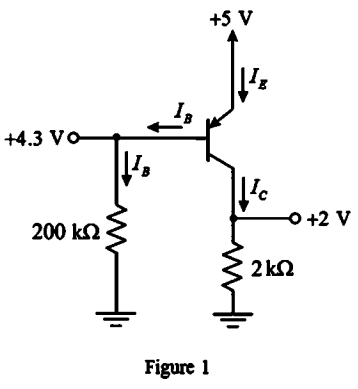


Figure 1

Step 2 of 12

Determine the value of base current.

$$I_B = \frac{4.3}{200\times10^3}$$
$$= 2.15\times10^{-5}$$
$$= 21.5\ \mu A$$

Determine the value of collector current.

$$I_C = \frac{2}{2\times10^3}$$
$$= 1\times10^{-3}$$
$$= 1\ mA$$

Step 3 of 12

Determine the value of common emitter current gain.

$$\beta = \frac{I_C}{I_B}$$

Substitute $21.5\ \mu A$ for I_B and $1\ mA$ for I_C in the equation.

$$\beta = \frac{1\times10^{-3}}{21.5\times10^{-6}}$$
$$= 46.51$$
$$\approx 47$$

Thus, the value of common emitter current gain, β is $\boxed{47}$.

Step 4 of 12

(b)

Refer to Figure P6.29 (b) in the text book.

The modified circuit diagram is shown in Figure 2.

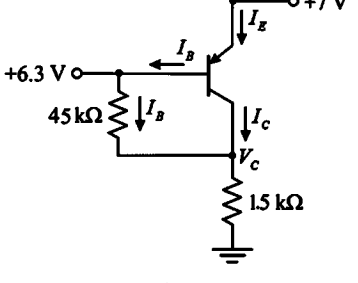


Figure 2

Step 5 of 12

Determine the value of base current.

$$I_B = \frac{4.3-3}{27\times10^3}$$
$$= 0.048\times10^{-3}$$
$$= 0.048\ mA$$

Determine the value of current through the $230\ \Omega$ resistor.

$$I_{750\,\Omega} = \frac{3-0}{750}$$
$$= 4\ mA$$

Step 6 of 12

Determine the value of collector current.

$$I_B + I_C = I_{750\,\Omega}$$
$$I_C = I_{750\,\Omega} - I_B$$

Substitute $0.048\ mA$ for I_B and $4\ mA$ for $I_{750\,\Omega}$ in the equation.

$$I_C = \left(4\times10^{-3}\right) - \left(0.048\times10^{-3}\right)$$
$$= 3.95\times10^{-3}$$
$$= 3.95\ mA$$

Step 7 of 12

Determine the value of common emitter current gain.

$$\beta = \frac{I_C}{I_B}$$

Substitute $0.1\ mA$ for I_B and $9.9\ mA$ for I_C in the equation.

$$\beta = \frac{3.95\times10^{-3}}{0.048\times10^{-3}}$$
$$= 82.3$$
$$\approx 82$$

Thus, the value of common emitter current gain, β is $\boxed{82}$.

Step 8 of 12

(c)

Refer to Figure P6.29 (c) in the text book.

The modified circuit diagram is shown in Figure 3.

Figure 3

Step 9 of 12

Determine the value of emitter current.

$$I_E = \frac{10-7}{1\times10^3}$$
$$= 3\times10^{-3}$$
$$= 3\ mA$$

Determine the value of voltage, V_C.

$$V_C = I_{1\,k\Omega}\left(1.5\times10^3\right)$$
$$= \left(I_C + I_B\right)\left(1.5\times10^3\right) \quad \left(\text{since } I_{1\,k\Omega} = I_C + I_B\right)$$
$$V_C = \left(I_E\right)\left(1.5\times10^3\right) \quad \left(\text{since } I_E = I_C + I_B\right)$$

Substitute $3\ mA$ for I_E in the equation.

$$V_C = \left(3\times10^{-3}\right)\left(1.5\times10^3\right)$$
$$= 4.5\ V$$

Step 10 of 12

Determine the value of base current.

$$I_B = \frac{6.3-V_C}{45\times10^3}$$

Substitute $4.5\ V$ for V_C in the equation.

$$I_B = \frac{6.3-4.5}{45\times10^3}$$
$$= 0.04\times10^{-3}$$
$$= 0.04\ mA$$

Step 11 of 12

Determine the value of collector current.

$$I_B + I_C = I_{1\,k\Omega}$$
$$I_C = I_E - I_B \quad \left(\text{since } I_{1\,k\Omega} = I_E\right)$$

Substitute $0.04\ mA$ for I_B and $3\ mA$ for I_E in the equation.

$$I_C = \left(3\times10^{-3}\right) - \left(0.04\times10^{-3}\right)$$
$$= 2.96\times10^{-3}$$
$$= 2.96\ mA$$

Step 12 of 12

Determine the value of common emitter current gain.

$$\beta = \frac{I_C}{I_B}$$

Substitute $0.04\ mA$ for I_B and $2.96\ mA$ for I_C in the equation.

$$\beta = \frac{2.967\times10^{-3}}{0.04\times10^{-3}}$$
$$= 74$$

Thus, the value of common emitter current gain, β is $\boxed{74}$.

Step 1 of 7

6.030E

Refer to Figure 6.29 (a) and Figure E6.30 in the text book. The collector of transistor Q_2 is connected to the base of transistor Q_1.

Draw the following diagram:

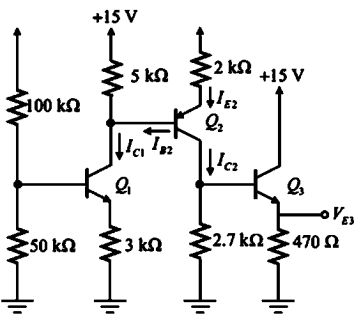


Figure 1

Step 2 of 7

Refer to Figure 6.29 (b) in the text book.

Draw the following diagram with current directions.

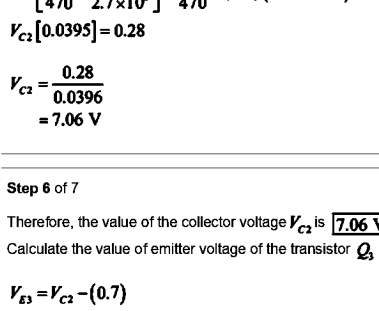


Figure 2

Step 3 of 7

Calculate the value of I_{B3}.

Apply Kirchhoff's current law at node V_{C2}.

$$I_{B3} = \left(2.75\times10^{-3}\right) - \frac{V_{C2}}{2.7\times10^{3}}$$

Apply Kirchhoff's voltage law for the circuit in Figure 2.

$$V_{C2} - V_{BE3} - V_{E3} = 0$$
$$V_{E3} = V_{C2} - (0.7) \qquad (\text{since} \quad V_{BE3} = 0.7\ \text{V})$$

Calculate the emitter current of the transistor Q_3.

$$I_{E3} = \frac{V_{E3}}{470}$$

Substitute $V_{C2} - (0.7)$ for V_{E3} in I_{E3}.

$$I_{E3} = \frac{V_{C2} - (0.7)}{470} \quad \text{..... (1)}$$

Step 4 of 7

Apply Kirchhoff's current law at the node V_{C2} for the circuit in Figure 2.

$$\left(2.75\times10^{-3}\right) = I_{B3} + \frac{V_{C2}}{2.7\times10^{3}}$$
$$I_{B3} = \left(2.75\times10^{-3}\right) - \frac{V_{C2}}{2.7\times10^{3}}$$

The expression of emitter current of transistor Q_3 is,

$$I_{E3} = (\beta+1) I_{B3}$$

Substitute 100 for β and $\left(2.75\times10^{-3}\right) - \frac{V_{C2}}{2.7\times10^{3}}$ for I_{B3} in I_{E3}.

$$I_{E3} = (101)\left[\left(2.75\times10^{-3}\right) - \frac{V_{C2}}{2.7\times10^{3}}\right] \quad \text{..... (2)}$$

Step 5 of 7

Equate equations (1) and (2).

$$\frac{V_{C2} - (0.7)}{470} = (101)\left[\left(2.75\times10^{-3}\right) - \frac{V_{C2}}{2.7\times10^{3}}\right]$$
$$V_{C2}\left[\frac{1}{470} + \frac{101}{2.7\times10^{3}}\right] = \frac{0.7}{470} + (101)\left(2.75\times10^{-3}\right)$$
$$V_{C2}\left[0.0395\right] = 0.28$$

$$V_{C2} = \frac{0.28}{0.0396}$$
$$= 7.06\ \text{V}$$

Step 6 of 7

Therefore, the value of the collector voltage V_{C2} is $\boxed{7.06\ \text{V}}$.

Calculate the value of emitter voltage of the transistor Q_3.

$$V_{E3} = V_{C2} - (0.7)$$

Substitute $7.06\ \text{V}$ for V_{C2}.

$$V_{E3} = 7.06 - (0.7)$$
$$= 6.36\ \text{V}$$

Therefore, the emitter voltage V_{E3} is $\boxed{6.36\ \text{V}}$.

Step 7 of 7

Calculate the collector current, I_{C3}.

$$I_{C3} = \alpha I_{E3}$$
$$= \left(\frac{\beta}{\beta+1}\right)\left(\frac{V_{E3}}{470}\right)$$

Substitute 100 for β and $6.36\ \text{V}$ for V_{E3}.

$$I_{C3} = \left(\frac{100}{100+1}\right)\left(\frac{6.36}{470}\right)$$
$$= 0.0134$$
$$= 13.4\ \text{mA}$$

Therefore, the collector current, I_{C3} is $\boxed{13.4\ \text{mA}}$.

Step 1 of 7

6.030P

Refer to Figure P6.30 in the text book.

In Figure P6.30, the base terminal and the collector terminal are connected by a short circuit. Thus, the base-emitter voltage is equals to the collector-emitter voltage.

$$\begin{aligned} V_{CE} &= V_{BE} \\ &= 0.7\ \text{V} \end{aligned}$$

Apply Kirchhoff's voltage law straight down the line of the resistance and the transistor in Figure P6.30.

$$V_{CC} - I_{R_C} R_C - V_{CE} = 0$$
$$V_{CE} = V_{CC} - I_{R_C} R_C$$

Substitute $0.7\ \text{V}$ for V_{CE}, $1\ \text{mA}$ for I_{R_C} and $9\ \text{V}$ for V_{CC} in the equation.

$$0.7 = 9 - \left(1\times10^{-3}\right)R_C$$

$$\begin{aligned} R_C &= \frac{9-0.7}{1\times10^{-3}} \\ &= \frac{8.3\ \text{V}}{\left(1\times10^{-3}\right)} \\ &= 8.3\times10^{3} \\ &= 8.3\ \text{k}\Omega \end{aligned}$$

Thus, the value of resistor, R_C is $\boxed{8.3\ \text{k}\Omega}$.

Step 2 of 7

Determine the value of base current with full scale reading of the meter.

$$\begin{aligned} I_B &= I_{FSD} \\ &= 50\ \mu\text{A} \end{aligned}$$

Determine the value of collector current with full scale reading of the meter.

The emitter current is the summation of the base current and the collector current.

$$I_E = I_C + I_B$$
$$I_C = I_E - I_B$$

Substitute $50\ \mu\text{A}$ for I_B and $1\ \text{mA}$ for I_E in the equation.

$$\begin{aligned} I_C &= \left(1\times10^{-3}\right) - \left(50\times10^{-6}\right) \\ &= 9.5\times10^{-4} \\ &= 950\ \mu\text{A} \end{aligned}$$

Step 3 of 7

Determine the value of common-emitter current gain with full scale reading of the meter.

$$\beta = \frac{I_C}{I_B}$$

Substitute $50\ \mu\text{A}$ for I_B and $950\ \mu\text{A}$ for I_C in the equation.

$$\begin{aligned} \beta &= \frac{950\times10^{-6}}{50\times10^{-6}} \\ &= 19 \end{aligned}$$

Thus, the value of common-emitter current gain for a full scale meter reading, β is $\boxed{19}$.

Step 4 of 7

Determine the value of base current with $1/5\text{th}$ of full scale meter reading.

$$I_B = \frac{1}{5} I_{FSD}$$

Substitute $50\ \mu\text{A}$ for I_{FSD} in the equation.

$$\begin{aligned} I_B &= \left(\frac{1}{5}\right)\left(50\times10^{-6}\right) \\ &= 10\ \mu\text{A} \end{aligned}$$

Determine the value of collector current with $1/5\text{th}$ of full scale meter reading.

$$I_E = I_C + I_B$$
$$I_C = I_E - I_B$$

Substitute $10\ \mu\text{A}$ for I_B and $1\ \text{mA}$ for I_E in the equation.

$$\begin{aligned} I_C &= \left(1\times10^{-3}\right) - \left(10\times10^{-6}\right) \\ &= 9.9\times10^{-4} \\ &= 990\ \mu\text{A} \end{aligned}$$

Step 5 of 7

Determine the value of common-emitter current gain with $1/5\text{th}$ of full scale meter reading.

$$\beta = \frac{I_C}{I_B}$$

Substitute $10\ \mu\text{A}$ for I_B and $990\ \mu\text{A}$ for I_C in the equation.

$$\begin{aligned} \beta &= \frac{990\times10^{-6}}{10\times10^{-6}} \\ &= 99 \end{aligned}$$

Thus, the value of common-emitter current gain for $1/5\text{th}$ of the full scale meter reading, β is $\boxed{99}$.

Step 6 of 7

Determine the value of base current with $1/10\text{th}$ of full scale meter reading.

$$I_B = \frac{1}{5} I_{FSD}$$

Substitute $50\ \mu\text{A}$ for I_{FSD} in the equation.

$$\begin{aligned} I_B &= \left(\frac{1}{10}\right)\left(50\times10^{-6}\right) \\ &= 5\ \mu\text{A} \end{aligned}$$

Determine the value of collector current with $1/10\text{th}$ of full scale meter reading.

$$I_E = I_C + I_B$$
$$I_C = I_E - I_B$$

Substitute $5\ \mu\text{A}$ for I_B and $1\ \text{mA}$ for I_E in the equation.

$$\begin{aligned} I_C &= \left(1\times10^{-3}\right) - \left(5\times10^{-6}\right) \\ &= 9.95\times10^{-4} \\ &= 995\ \mu\text{A} \end{aligned}$$

Step 7 of 7

Determine the value of common-emitter current gain with $1/10\text{th}$ of full scale meter reading.

$$\beta = \frac{I_C}{I_B}$$

Substitute $5\ \mu\text{A}$ for I_B and $995\ \mu\text{A}$ for I_C in the equation.

$$\begin{aligned} \beta &= \frac{995\times10^{-6}}{5\times10^{-6}} \\ &= 199 \end{aligned}$$

Thus, the value of common-emitter current gain for $1/10\text{th}$ of the full scale meter reading, β is $\boxed{199}$.

Step 1 of 4

6.031E

Refer to Figure 6.30 in the text book.

Draw the circuit diagram as shown in Figure 1.

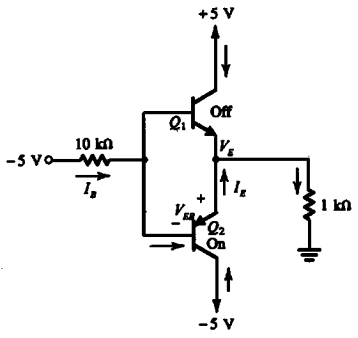


Figure 1

Step 2 of 4

Apply Kirchhoff's voltage law to the base-emitter loop of the circuit in Figure 1.

$$-(-5)+I_B(10\times10^3)-V_{EB}+I_E(1\times10^3)=0$$
$$5+I_B(10\times10^3)-V_{EB}+I_E(1\times10^3)=0$$

Substitute $(1+\beta)I_B$ for I_E.

$$5+I_B(10\times10^3)-0.7+(1+\beta)I_B(1\times10^3)=0$$

$$5+I_B(10\times10^3)-0.7+(1+100)I_B(1\times10^3)=0$$
$$4.3+I_B(10\times10^3)+(101)I_B(1\times10^3)=0$$
$$I_B(111\times10^3)=-4.3$$

$$I_B=-0.039\,\text{mA}$$

Step 3 of 4

Determine the value of emitter current.

$$I_E=(1+\beta)I_B$$

Substitute 100 for β and $-0.039\,\text{mA}$ for I_B.

$$I_E=(1+100)(-0.039\times10^{-3})$$
$$=-3.9\,\text{mA}$$

Step 4 of 4

Use Ohm's law to determine the value of emitter voltage.

$$V_E=I_E(1\text{k}\Omega)$$
$$=I_E(1\times10^3)$$

Substitute $-3.9\,\text{mA}$ for I_E.

$$V_E=-(3.9\times10^{-3})(1\times10^3)$$
$$=-3.9\,\text{V}$$

Therefore, the value of emitter voltage is $\boxed{-3.9\,\text{V}}$.

Step 1 of 5

Refer to Figure E6.13 in the textbook.

The modified circuit diagram is shown in Figure 1.

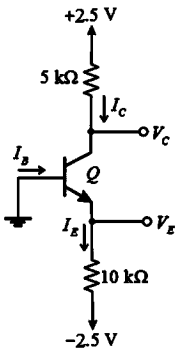


Figure 1

Step 2 of 5

Determine the value of emitter current.

$$I_E = \frac{V_E - (-2.5)}{10 \times 10^3}$$

Substitute $-0.7\ \text{V}$ for V_E in the equation.

$$\begin{aligned} I_E &= \frac{-0.7 + 2.5}{10 \times 10^3} \\ &= 0.18 \times 10^{-3} \\ &= 180\ \mu\text{A} \end{aligned}$$

Thus, the value of emitter current, I_E is $\boxed{180\ \mu\text{A}}$.

Step 3 of 5

Determine the value of base current.

$$I_B = \frac{I_E}{\beta + 1}$$

Substitute 50 for β and $180\ \mu\text{A}$ for I_E in the equation.

$$\begin{aligned} I_B &= \frac{180 \times 10^{-6}}{50 + 1} \\ &= 3.52 \times 10^{-6} \\ &= 3.52\ \mu\text{A} \end{aligned}$$

Thus, the value of base current, I_B is $\boxed{3.52\ \mu\text{A}}$.

Step 4 of 5

Determine the value of collector current.

$$I_C = \beta I_B$$

Substitute 50 for β and $3.52\ \mu\text{A}$ for I_B in the equation.

$$\begin{aligned} I_C &= (50)\left(3.52 \times 10^{-6}\right) \\ &= 176.47 \times 10^{-6} \\ &= 176.47\ \mu\text{A} \end{aligned}$$

Thus, the value of collector current, I_C is $\boxed{176.47\ \mu\text{A}}$.

Step 5 of 5

Determine the value of collector voltage.

$$I_C = \frac{2.5 - V_C}{5 \times 10^3}$$

$$V_C = 2.5 - I_C\left(5 \times 10^3\right)$$

Substitute $176.47\ \mu\text{A}$ for I_C in the equation.

$$\begin{aligned} V_C &= 2.5 - \left(176.47 \times 10^{-6}\right)\left(5 \times 10^3\right) \\ &= 2.5 - 0.882 \\ &= 1.617 \\ &\approx 1.62\ \text{V} \end{aligned}$$

Thus, the value of collector voltage, V_C is $\boxed{1.62\ \text{V}}$.

Step 1 of 3

6.032E

Refer to the Figure 6.30 in the text book.

For the input voltage +10 V, the transistor Q_1 is in **saturation** and the transistor Q_2 in **cutoff region**.

In saturation, the collector to emitter voltage of transistor Q_1 is 0.2 V.

$$V_{CE(sat)} = 0.2\,\text{V}$$
$$V_C - V_E = 0.2\,\text{V}$$

Substitute 5 V for V_C.

$$5\,\text{V} - V_E = 0.2\,\text{V}$$
$$V_E = 5\,\text{V} - 0.2\,\text{V}$$
$$V_E = +4.8\,\text{V}$$

Therefore, the value of emitter voltage is $\boxed{+4.8\,\text{V}}$.

Step 2 of 3

Determine the value of base voltage.

$$V_{BE} = 0.7\,\text{V}$$
$$V_B - V_E = 0.7\,\text{V}$$

Substitute 4.8 V for V_E.

$$V_B - 4.8\,\text{V} = 0.7\,\text{V}$$
$$V_B = 4.8\,\text{V} + 0.7\,\text{V}$$
$$V_B = 5.5\,\text{V}$$

Therefore, the value of base voltage is $\boxed{5.5\,\text{V}}$.

Step 3 of 3

Use Ohm's law to determine the value of emitter current.

$$V_E = I_{E1}\left(1\,\text{k}\Omega\right)$$
$$4.8\,\text{V} = I_{E1}\left(1\times10^3\right)$$
$$I_{E1} = 4.8\,\text{mA}$$

Determine the value of collector current.

$$I_{C1} = \frac{I_{E1}}{\left(1+\frac{1}{\beta}\right)}$$

Substitute 4.8 mA for I_{E1} and 30 for β.

$$I_{C1} = \frac{4.8\,\text{mA}}{\left(1+\frac{1}{30}\right)}$$
$$= \frac{4.8\,\text{mA}}{1.033}$$
$$= 4.6\,\text{mA}$$

Therefore, the value of collector current of transistor Q_1 is $\boxed{4.6\,\text{mA}}$.

The transistor Q_2 is in **cutoff region** and its collector current is equal to zero.

Therefore, the value of collector current of transistor Q_2 is $\boxed{0\,\text{A}}$.

Step 1 of 5

6.032P

Refer to Figure P6.32 in the text book.

The modified circuit diagram is shown in Figure 1.

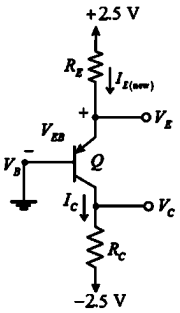


Figure 1

Step 2 of 5

Determine the value of emitter-base voltage.

$$V_{EB} = v_{EB} + V_T \ln\left(\frac{I_{E(new)}}{I_E}\right)$$

Substitute $0.64\ \text{V}$ for v_{EB}, $25\ \text{mV}$ for V_T, $0.5\ \text{mA}$ for $I_{E(new)}$ and $0.1\ \text{mA}$ for I_E in the equation.

$$\begin{aligned} V_{EB} &= 0.64 + \left(25\times10^{-3}\right)\ln\left(\frac{0.5\times10^{-3}}{0.1\times10^{-3}}\right) \\ &= 0.64 + 0.04 \\ &= 0.68\ \text{V} \end{aligned}$$

Thus, the value of emitter-base voltage, V_{EB} is $\boxed{0.68\ \text{V}}$.

Step 3 of 5

Determine the value of emitter resistance.

$$-2.5 + I_{E(new)}R_E + V_{EB} = 0$$

Substitute $0.68\ \text{V}$ for V_{EB} and $0.5\ \text{mA}$ for $I_{E(new)}$ in the equation.

$$-2.5 + \left(0.5\times10^{-3}\right)R_E + 0.68 = 0$$

$$\begin{aligned} R_E &= \frac{2.5 - 0.68}{0.5\times10^{-3}} \\ &= 3.64\times10^{3} \\ &\approx 3.6\ \text{k}\Omega \end{aligned}$$

Thus, the value of emitter resistance, R_E is $\boxed{3.6\ \text{k}\Omega}$.

Step 4 of 5

Determine the value of collector current.

$$I_C = \left(\frac{\beta}{\beta+1}\right)I_{E(new)}$$

Substitute 100 for β and $0.5\ \text{mA}$ for $I_{E(new)}$ in the equation.

$$\begin{aligned} I_C &= \left(\frac{100}{100+1}\right)\left(0.5\times10^{-3}\right) \\ &= 0.49\times10^{-3} \\ &= 0.49\ \text{mA} \end{aligned}$$

Thus, the value of collector current, I_C is $\boxed{0.49\ \text{mA}}$.

Step 5 of 5

Determine the value of collector resistance.

$$R_C = \frac{V_C - (-2.5)}{I_C}$$

Substitute $0.49\ \text{mA}$ for I_C and $-0.5\ \text{V}$ for V_C in the equation.

$$\begin{aligned} R_C &= \frac{-0.5 + 2.5}{0.49\times10^{-3}} \\ &= \frac{2}{0.49\times10^{-3}} \\ &= 4.08\times10^{3} \\ &= 4.08\ \text{k}\Omega \end{aligned}$$

Thus, the value of collector resistance can be increased while the collector current remains unchanged, R_C is $\boxed{4.08\ \text{k}\Omega}$.

Step 1 of 2

6.033E

Refer to Figure E6.56 in the textbook for the circuit diagram.

The breakdown voltage (base to collector) of the transistor is, $BV_{BCO} = 70\ \text{V}$

The modified circuit is shown in Figure 1.

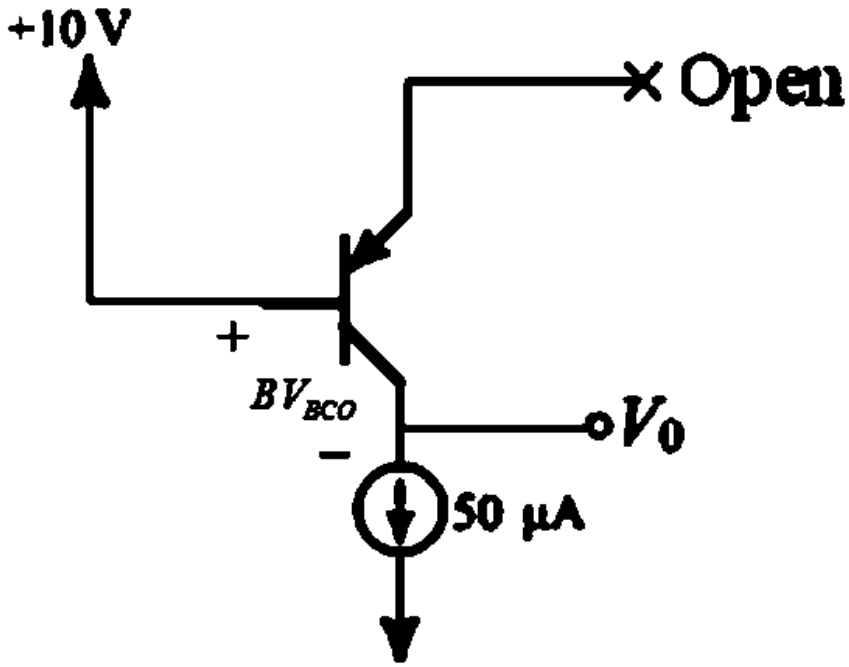


Figure 1

Step 2 of 2

From the circuit, the base voltage is, $V_B = 10\ \text{V}$

Apply Kirchhoff's voltage law left bottom loop to calculate the output voltage of the transistor.

$$-V_B + BV_{BCO} + V_o = 0$$
$$V_o = V_B - BV_{BCO}$$

Substitute 70 V for BV_{BCO} and 10 V for V_B .

$$V_o = 10 - 70$$
$$= -60\ \text{V}$$

Thus, the output voltage of the transistor circuit is, $\boxed{-60\ \text{V}}$.

6.033P

We don't have the solution to this problem yet.

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Step 1 of 6

Refer to Figure P6.34 in the textbook.

The modified circuit diagram is shown in Figure 1.

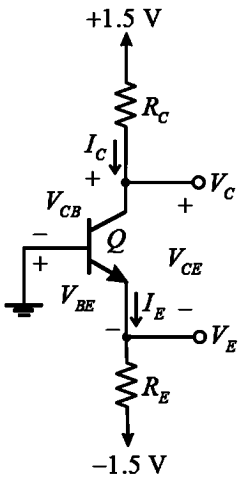


Figure 1

Step 2 of 6

Determine the value of base-emitter voltage.

$$V_{BE} = v_{BE} + V_T \ln\left(\frac{I_C}{i_C}\right)$$

Substitute $0.8\ \text{V}$ for v_{BE}, $25\ \text{mV}$ for V_T, $0.2\ \text{mA}$ for I_C and $1\ \text{mA}$ for i_C in the equation.

$$\begin{aligned} V_{BE} &= 0.8 + \left(25\times10^{-3}\right)\ln\left(\frac{0.2\times10^{-3}}{1\times10^{-3}}\right) \\ &= 0.8 - 0.040 \\ &= 0.76\ \text{V} \end{aligned}$$

Thus, the value of base-emitter voltage, V_{BE} is $\boxed{0.76}$.

Step 3 of 6

Determine the value of collector resistance.

$$R_C = \frac{1.5 - V_C}{I_C}$$

Substitute $0.5\ \text{V}$ for V_C and $0.2\ \text{mA}$ for I_C in the equation.

$$\begin{aligned} R_C &= \frac{1.5 - 0.5}{0.2\times10^{-3}} \\ &= 5\times10^{3} \\ &= 5\ \text{k}\Omega \end{aligned}$$

Thus, the value of collector resistance, R_C is $\boxed{5\ \text{k}\Omega}$.

Step 4 of 6

Determine the value of emitter current.

$$I_E = \left(\frac{\beta+1}{\beta}\right)I_C$$

Substitute 100 for β and $0.2\ \text{mA}$ for I_C in the equation.

$$\begin{aligned} I_E &= \left(\frac{100+1}{100}\right)\left(0.2\times10^{-3}\right) \\ &= 0.202\times10^{-3} \\ &= 0.202\ \text{mA} \end{aligned}$$

Thus, the value of emitter current, I_E is $\boxed{0.202\ \text{mA}}$.

Step 5 of 6

Determine the value of collector-emitter voltage.

$$\begin{aligned} V_{CE} &= V_{BE} + V_{CB} \\ V_{CE} &= V_{BE} + V_C - V_B \end{aligned}$$

Substitute $0.76\ \text{V}$ for V_{BE}, $0.5\ \text{V}$ for V_C and $0\ \text{V}$ for V_B in the equation.

$$\begin{aligned} V_{CE} &= 0.76 + 0.5 - 0 \\ &= 1.26\ \text{V} \end{aligned}$$

Thus, the value of collector-emitter voltage, V_{CE} is $\boxed{1.26\ \text{V}}$.

Step 6 of 6

Determine the value of emitter resistance.

Apply the Kirchhoff's voltage law for the collector to emitter in the transistor configuration.

$$-1.5 + I_C R_C + V_{CE} + I_E R_E - 1.5 = 0$$

Substitute $0.2\ \text{mA}$ for I_C, $5\ \text{k}\Omega$ for R_C, $1.26\ \text{V}$ for V_{CE} and $0.202\ \text{mA}$ for I_E in the equation.

$$\begin{aligned} &-1.5 + \left(0.2\times10^{-3}\right)\left(5\times10^{3}\right) + 1.26 - \left(0.202\times10^{-3}\right)R_E - 1.5 = 0 \\ &1 + \left(0.101\times10^{-3}\right)R_E = 3 - 1.26 \end{aligned}$$

$$\begin{aligned} R_E &= \frac{1.74 - 1}{0.202\times10^{-3}} \\ &\approx 3.66\times10^{3} \\ &\approx 3.7\ \text{k}\Omega \end{aligned}$$

Thus, the value of emitter resistance, R_E is $\boxed{3.7\ \text{k}\Omega}$.

Step 1 of 24

(a)

Refer to Figure P6.35 (a) in the text book.

The modified circuit diagram is shown in Figure 1.

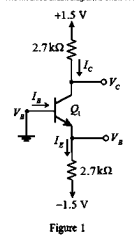


Figure 1

Step 2 of 24

Determine the value of base voltage.

From Figure 1, the base voltage is,

$$V_B = 0\ \text{V}$$

Thus, the value of base voltage, V_B is $\boxed{0\ \text{V}}$.

Determine the value of emitter voltage.

$$|V_{BE}| = 0.8$$
$$V_B - V_E = 0.8$$

Substitute $0\ \text{V}$ for V_B in the equation.

$$0 - V_E = 0.8$$
$$V_E = -0.8\ \text{V}$$

Thus, the value of emitter voltage, V_E is $\boxed{-0.8\ \text{V}}$.

Step 3 of 24

Determine the value of emitter current.

$$I_E = \frac{V_E - (-1.5)}{2.7\times10^3}$$

Substitute $-0.8\ \text{V}$ for V_E in the equation.

$$I_E = \frac{-0.8+1.5}{2.7\times10^3}$$
$$= \frac{0.7}{2.7\times10^3}$$
$$= 2.592\times10^{-4}$$
$$= 259.2\ \mu\text{A}$$

Thus, the value of emitter current, I_E is $\boxed{259.2\ \mu\text{A}}$.

Step 4 of 24

Determine the value of collector current.

$$I_C = \left(\frac{\beta}{\beta+1}\right)I_E$$

$$I_C = \left(\frac{50}{50+1}\right)\left(259.2\times10^{-6}\right)$$
$$= \left(\frac{50}{51}\right)\left(259.2\times10^{-6}\right)$$
$$= 254.11\times10^{-6}$$
$$= 254.11\ \mu\text{A}$$

Thus, the value of collector current, I_C is $\boxed{254.11\ \mu\text{A}}$.

Step 5 of 24

Determine the value of collector voltage.

$$I_C = \frac{1.5 - V_C}{2.7\times10^3}$$
$$V_C = 1.5 - I_C\left(2.7\times10^3\right)$$

Substitute $311.96\ \mu\text{A}$ for I_C in the equation.

$$V_C = 1.5 - \left(254.11\times10^{-6}\right)\left(2.7\times10^3\right)$$
$$= 0.813\ \text{V}$$

Thus, the value of collector voltage, V_C is $\boxed{0.813\ \text{V}}$.

Step 6 of 24

Determine the value of base current.

$$I_B = \frac{I_C}{\beta}$$

$$I_B = \frac{254.11\times18^{-6}}{50}$$
$$= 5.08\times10^{-6}$$
$$= 5.88\ \mu\text{A}$$

Thus, the value of base current, I_B is $\boxed{5.08\ \mu\text{A}}$.

Step 7 of 24

(b)

Refer to Figure P6.35 (b) in the text book.

The modified circuit diagram is shown in Figure 2.

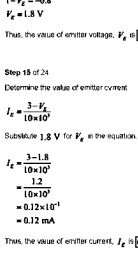


Figure 2

Step 8 of 24

Determine the value of base voltage.

From Figure 2, the base voltage is,

$$V_B = 0\ \text{V}$$

Thus, the value of base voltage, V_B is $\boxed{0\ \text{V}}$.

Determine the value of emitter voltage.

$$|V_{BE}| = 0.8$$
$$V_B - V_E = -0.8 \quad \textbf{(Since pnp transistor)}$$

Substitute $0\ \text{V}$ for V_B in the equation.

$$0 - V_E = -0.8$$
$$V_E = 0.8\ \text{V}$$

Thus, the value of emitter voltage, V_E is $\boxed{0.8\ \text{V}}$.

Step 9 of 24

Determine the value of emitter current.

$$I_E = \frac{1.5 - V_E}{2\times10^3}$$

Substitute $0.8\ \text{V}$ for V_E in the equation.

$$I_E = \frac{1.5-0.8}{2\times10^3}$$
$$= 0.35\times10^{-3}$$
$$= 0.35\ \text{mA}$$

Thus, the value of emitter current, I_E is $\boxed{0.35\ \text{mA}}$.

Step 10 of 24

Determine the value of collector current.

$$I_C = \left(\frac{\beta}{\beta+1}\right)I_E$$

Substitute 50 for β and $8.7\ \text{mA}$ for I_E in the equation.

$$I_C = \left(\frac{50}{50+1}\right)\left(0.35\times10^{-3}\right)$$
$$= 343.13\times18^{-6}$$
$$= 343.13\ \mu\text{A}$$

Thus, the value of collector current, I_C is $\boxed{343.13\ \mu\text{A}}$.

Step 11 of 24

Determine the value of collector voltage.

$$I_C = \frac{V_C - (-1.5)}{2\times10^3}$$
$$V_C = -1.5 - I_C\left(2\times10^3\right)$$
$$V_C = -1.5 + \left(343.13\times10^{-6}\right)\left(1\times10^3\right)$$
$$= -1.15\ \text{V}$$

Thus, the value of collector voltage, V_C is $\boxed{-1.15\ \text{V}}$.

Step 12 of 24

Determine the value of base current.

$$I_B = \frac{I_C}{\beta}$$

Substitute 50 for β and $686.3\ \mu\text{A}$ for I_C in the equation.

$$I_B = \frac{343.13\times10^{-6}}{50}$$
$$= 6.86\times10^{-6}$$
$$= 6.86\ \mu\text{A}$$

Thus, the value of base current, I_B is $\boxed{6.86\ \mu\text{A}}$.

Step 13 of 24

(c)

Refer to Figure P6.35 (c) in the text book.

The modified circuit diagram is shown in Figure 3.

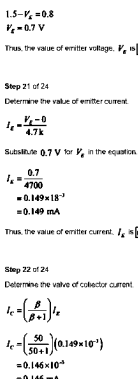


Figure 3

Step 14 of 24

Determine the value of base voltage.

From Figure 3, the base voltage is,

$$V_B = 1\ \text{V}$$

Thus, the value of base voltage, V_B is $\boxed{1\ \text{V}}$.

Determine the value of emitter voltage.

$$|V_{BE}| = 0.8$$
$$V_B - V_E = -0.8 \quad \textbf{(Since pnp transistor)}$$

Substitute $1\ \text{V}$ for V_B in the equation.

$$1 - V_E = -0.8$$
$$V_E = 1.8\ \text{V}$$

Thus, the value of emitter voltage, V_E is $\boxed{1.8\ \text{V}}$.

Step 15 of 24

Determine the value of emitter current.

$$I_E = \frac{3 - V_E}{10\times10^3}$$

Substitute $1.8\ \text{V}$ for V_E in the equation.

$$I_E = \frac{3-1.8}{10\times10^3}$$
$$= \frac{1.2}{10\times10^3}$$
$$= 0.12\times10^{-1}$$
$$= 0.12\ \text{mA}$$

Thus, the value of emitter current, I_E is $\boxed{0.12\ \text{mA}}$.

Step 16 of 24

Determine the value of collector current.

$$I_C = \left(\frac{\beta}{\beta+1}\right)I_E$$

$$I_C = \left(\frac{50}{50+1}\right)\left(0.12\times10^{-3}\right)$$
$$= 0.117\times10^{-3}$$
$$= 0.117\ \text{mA}$$

Thus, the value of collector current, I_C is $\boxed{0.117\ \text{mA}}$.

Step 17 of 24

Determine the value of collector voltage.

$$I_C = \frac{V_C - 0}{5.6\ \text{k}\Omega}$$
$$V_C = I_C(5600)$$

$$V_C = \left(0.117\times10^{-3}\right)(5600)$$
$$= 0.655\ \text{V}$$

Thus, the value of collector voltage, V_C is $\boxed{0.655\ \text{V}}$.

Step 18 of 24

Determine the value of base current.

$$I_B = \frac{I_C}{\beta}$$

$$I_B = \frac{0.117\times18^{-3}}{50}$$
$$= 2.34\times18^{-6}$$
$$= 2.34\ \mu\text{A}$$

Thus, the value of base current, I_B is $\boxed{2.34\ \mu\text{A}}$.

Step 19 of 24

(d)

Refer to Figure P6.35 (d) in the text book.

The modified circuit diagram is shown in Figure 4.

Figure 4

Step 20 of 24

Determine the value of base voltage.

From Figure 4, the base voltage is,

$$V_B = 1.5\ \text{V}$$

Thus, the value of base voltage, V_B is $\boxed{1.5\ \text{V}}$.

Determine the value of emitter voltage.

$$|V_{BE}| = 0.8$$
$$V_B - V_E = 0.8$$

Substitute $1.5\ \text{V}$ for V_B in the equation.

$$1.5 - V_E = 0.8$$
$$V_E = 0.7\ \text{V}$$

Thus, the value of emitter voltage, V_E is $\boxed{0.7\ \text{V}}$.

Step 21 of 24

Determine the value of emitter current.

$$I_E = \frac{V_E - 0}{4.7\ \text{k}}$$

Substitute $0.7\ \text{V}$ for V_E in the equation.

$$I_E = \frac{0.7}{4700}$$
$$= 0.149\times18^{-3}$$
$$= 0.149\ \text{mA}$$

Thus, the value of emitter current, I_E is $\boxed{0.149\ \text{mA}}$.

Step 22 of 24

Determine the value of collector current.

$$I_C = \left(\frac{\beta}{\beta+1}\right)I_E$$

$$I_C = \left(\frac{50}{50+1}\right)\left(0.149\times10^{-3}\right)$$
$$= 0.146\times10^{-6}$$
$$= 0.146\ \text{mA}$$

Thus, the value of collector current, I_C is $\boxed{0.146\ \text{mA}}$.

Step 23 of 24

Determine the value of collector voltage.

$$I_C = \frac{3 - V_C}{8.2\times10^3}$$
$$V_C = 3 - I_C\left(8.2\times10^3\right)$$

$$V_C = 3 - \left(0.146\times10^{-3}\right)\left(8.2\times10^3\right)$$
$$= 3 - 1.46$$
$$= 1.8\ \text{V}$$

Thus, the value of collector voltage, V_C is $\boxed{1.8\ \text{V}}$.

Step 24 of 24

Determine the value of base current.

$$I_B = \frac{I_C}{\beta}$$

Substitute 50 for β and $1.46\ \text{mA}$ for I_C in the equation.

$$I_B = \frac{0.146\times10^{-3}}{50}$$
$$= 2.92\times10^{-6}$$
$$= 2.92\ \mu\text{A}$$

Thus, the value of base current, I_B is $\boxed{2.92\ \mu\text{A}}$.

6.036P

The collector-base reverse current of a small transistor at $25^\circ\,C$ is, $I_{CBO} = 10\ nA$.

The collector-base reverse current strongly depends on temperature and approximately double for every $10^\circ\,C$ rise.

Determine the expected value of collector-base reverse current of the transistor at $125^\circ\,C$.

$$I_{CBO} = 2^{\left(\frac{\Delta T}{10}\right)}\left(10\times10^{-9}\right)$$

$$= 2^{\left(\frac{125^\circ - 25^\circ}{10^\circ}\right)}\left(10\times10^{-9}\right)$$

$$= \left(2^{10}\right)\left(10\times10^{-9}\right)$$

$$= 10.24\times10^{-6}$$

$$= 10.24\ \mu A$$

Thus, the expected value of collector-base reverse current of the transistor at $125^\circ\,C$, I_{CBO} is $\boxed{10.24\ \mu A}$.

Step 1 of 5

6.037P

Refer to Figure 6.18 (a) for a large-signal equivalent circuit model of *npn* BJT in the text book.

Augmenting the model of the ***npn*** BJT shown in Figure 6.18 (a), by a current source representing I_{CBO} and neglecting the output resistance r_o is shown in Figure 1.

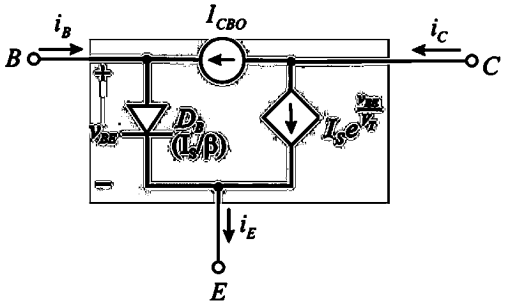


Figure 1

Step 2 of 5

From Figure 1, to write the expressions for terminal current, i_B .

$$\boxed{i_B = \frac{I_S}{\beta} e^{v_{BE}/V_T} - I_{CBO}} \quad \text{..... (1)}$$

Write the expressions for terminal current, i_C .

$$\boxed{i_C = I_S e^{v_{BE}/V_T} + I_{CBO}} \quad \text{..... (2)}$$

Write the expressions for terminal current, i_E .

$$i_E = \frac{I_S}{\beta} e^{v_{BE}/V_T} + I_S e^{v_{BE}/V_T}$$

$$\boxed{i_E = I_S \left(1 + \frac{1}{\beta}\right) e^{v_{BE}/V_T}} \quad \text{..... (3)}$$

Step 3 of 5

If the base lead is open-circuited, while the emitter is connected to ground and the collector is connected to a positive power supply. Then, the base current is

$$i_B = 0 \text{ A}$$

Substitute 0 A for i_B in equation (1).

$$0 = \frac{I_S}{\beta} e^{v_{BE}/V_T} - I_{CBO}$$

$$I_{CBO} = \frac{I_S}{\beta} e^{v_{BE}/V_T}$$

$$\beta I_{CBO} = I_S e^{v_{BE}/V_T}$$

$$e^{v_{BE}/V_T} = \frac{\beta I_{CBO}}{I_S}$$

Step 4 of 5

Determine the expression for collector current.

Substitute $\frac{\beta I_{CBO}}{I_S}$ for e^{v_{BE}/V_T} in equation (2).

$$i_C = I_S \left(\frac{\beta I_{CBO}}{I_S}\right) + I_{CBO}$$
$$= \beta I_{CBO} + I_{CBO}$$
$$= (\beta + 1) I_{CBO}$$

Thus, the expression for collector current, i_C is $\boxed{(\beta + 1) I_{CBO}}$.

Step 5 of 5

Determine the expression for emitter current.

Substitute $\frac{\beta I_{CBO}}{I_S}$ for e^{v_{BE}/V_T} in equation (3).

$$i_E = I_S \left(1 + \frac{1}{\beta}\right)\left(\frac{\beta I_{CBO}}{I_S}\right)$$
$$= \left(1 + \frac{1}{\beta}\right)(\beta I_{CBO})$$
$$= \beta I_{CBO} + I_{CBO}$$
$$= (\beta + 1) I_{CBO}$$

Thus, the expression for collector current, i_E is $\boxed{(\beta + 1) I_{CBO}}$.

Step 1 of 2

6.038P

Determine the value of base-emitter voltage at $0\ ^\circ\text{C}$.

The voltage across the base-emitter junction, V_{BE} decreases by about $2\ \text{mV}/^\circ\text{C}$ for each rise of 1°C in temperature.

$$V_{BE(0^\circ\text{C})} = V_{BE(25^\circ\text{C})} - \left(2\times10^{-3}\right)(0^\circ - 25^\circ)$$

Substitute $0.70\ \text{V}$ for $V_{BE(25^\circ\text{C})}$ in the equation.

$$\begin{aligned} V_{BE(0^\circ\text{C})} &= 0.70 + \left(2\times10^{-3}\right)(25) \\ &= 0.70 + 0.05 \\ &= 0.75\ \text{V} \end{aligned}$$

Thus, the value of base-emitter voltage, V_{BE} at $0\ ^\circ\text{C}$ is $\boxed{0.75\ \text{V}}$.

Step 2 of 2

Determine the value of base-emitter voltage at $100\ ^\circ\text{C}$.

The voltage across the base-emitter junction, V_{BE} decreases by about $2\ \text{mV}/^\circ\text{C}$ for each rise of 1°C in temperature.

$$V_{BE(100^\circ\text{C})} = V_{BE(25^\circ\text{C})} - \left(2\times10^{-3}\right)(100^\circ - 25^\circ)$$

Substitute $0.70\ \text{V}$ for $V_{BE(25^\circ\text{C})}$ in the equation.

$$\begin{aligned} V_{BE(100^\circ\text{C})} &= 0.70 - \left(2\times10^{-3}\right)(75) \\ &= 0.70 - 0.15 \\ &= 0.55\ \text{V} \end{aligned}$$

Thus, the value of base-emitter voltage, V_{BE} at $100\ ^\circ\text{C}$ is $\boxed{0.55\ \text{V}}$.

Step 1 of 5

(a)

Determine the value of emitter-base voltage at $50\ °C$.

The voltage across the emitter-base junction, v_{EB} decreases by about $2\ mV/°C$ for each rise of $1°C$ in temperature.

$$v_{EB(50°C)} = v_{EB(20°C)} - (2\times10^{-3})(50° - 20°)$$

Substitute $692\ mV$ for $v_{EB(20°C)}$ in the equation.

$$\begin{aligned} v_{EB(50°C)} &= (692\times10^{-3}) - (2\times10^{-3})(30) \\ &= (692\times10^{-3}) - (60\times10^{-3}) \\ &= 632\times10^{-3} \\ &= 0.632\ V \end{aligned}$$

Thus, the value of emitter-base voltage, v_{EB} at $50\ °C$ is $\boxed{0.632\ V}$.

Step 2 of 5

(b)

Write the expression emitter current for a particular *pnp* transistor operated at fixed emitter-bias voltage of $700\ mV$.

$$i_E = \frac{I_S}{\alpha} e^{\frac{v_{EB}}{V_T}}$$

Substitute $25\ mA$ for V_T and $700\ mV$ for v_{EB} in the equation.

$$i_E = \frac{I_S}{\alpha} e^{\left(\frac{700\times10^{-3}}{25\times10^{-3}}\right)} \quad \text{..... (1)}$$

Write the expression emitter current for a particular *pnp* transistor at $20°C$.

$$i_{E(20°C)} = \frac{I_S}{\alpha} e^{\frac{v_{EB}}{V_T}}$$

Substitute $0.5\ mA$ for $i_{E(20°C)}$, $25\ mA$ for V_T and $692\ mV$ for v_{EB} in the equation.

$$(0.5\times10^{-3}) = \frac{I_S}{\alpha} e^{\left(\frac{692\times10^{-3}}{25\times10^{-3}}\right)} \quad \text{..... (2)}$$

Step 3 of 5

Determine the new value of emitter current flows at $20°C$.

Divide equation (1) by equation (2).

$$\frac{i_E}{0.5\times10^{-3}} = \frac{\frac{I_S}{\alpha} e^{\left(\frac{700\times10^{-3}}{25\times10^{-3}}\right)}}{\frac{I_S}{\alpha} e^{\left(\frac{692\times10^{-3}}{25\times10^{-3}}\right)}}$$

$$\begin{aligned} i_E &= (0.5\times10^{-3}) \frac{e^{\left(\frac{700\times10^{-3}}{25\times10^{-3}}\right)}}{e^{\left(\frac{692\times10^{-3}}{25\times10^{-3}}\right)}} \\ &= (0.5\times10^{-3}) e^{\left(\frac{700}{25} - \frac{692}{25}\right)} \\ &= 6.89\times10^{-4} \\ &= 0.689\ mA \end{aligned}$$

Thus, the new value of emitter current flows at $20°C$, i_E is $\boxed{0.689\ mA}$.

Step 4 of 5

Write the expression emitter current for a particular *pnp* transistor at $50°C$.

$$i_{E(50°C)} = \frac{I_S}{\alpha} e^{\frac{v_{EB}}{V_T}}$$

Substitute $0.5\ mA$ for $i_{E(50°C)}$, $25\ mA$ for V_T and $632\ mV$ for v_{EB} in the equation.

$$(0.5\times10^{-3}) = \frac{I_S}{\alpha} e^{\left(\frac{632\times10^{-3}}{25\times10^{-3}}\right)} \quad \text{..... (3)}$$

Step 5 of 5

Determine the new value of emitter current flows at $50°C$.

Divide equation (1) by equation (3).

$$\frac{i_E}{0.5\times10^{-3}} = \frac{\frac{I_S}{\alpha} e^{\left(\frac{700\times10^{-3}}{25\times10^{-3}}\right)}}{\frac{I_S}{\alpha} e^{\left(\frac{632\times10^{-3}}{25\times10^{-3}}\right)}}$$

$$\begin{aligned} i_E &= (0.5\times10^{-3}) \frac{e^{\left(\frac{700\times10^{-3}}{25\times10^{-3}}\right)}}{e^{\left(\frac{632\times10^{-3}}{25\times10^{-3}}\right)}} \\ &= (0.5\times10^{-3}) e^{\left(\frac{700}{25} - \frac{632}{25}\right)} \\ &= 7.59\times10^{-3} \\ &= 7.59\ mA \end{aligned}$$

Thus, the new value of emitter current flows at $50°C$, i_E is $\boxed{7.59\ mA}$.

Step 1 of 5

Write the expression collector current for a particular *npn* transistor operated at fixed base-emitter voltage of $0.7\ \text{V}$.

$$i_{C1} = I_S e^{\left(\frac{v_{BE1}}{V_T}\right)}$$

Substitute $10\ \text{mA}$ for i_{C1}, $25\ \text{mV}$ for V_T and $0.7\ \text{V}$ for v_{BE1} in the equation.

$$\left(10\times10^{-3}\right) = I_S e^{\left(\frac{0.7}{25\times10^{-3}}\right)} \quad \text{..... (1)}$$

Write the expression collector current for a particular *npn* transistor operated at fixed base-emitter voltage of $0.5\ \text{V}$.

$$i_{C2} = I_S e^{\frac{v_{BE2}}{V_T}}$$

Substitute $25\ \text{mV}$ for V_T and $0.5\ \text{V}$ for v_{BE2} in the equation.

$$i_{C2} = I_S e^{\left(\frac{0.5}{25\times10^{-3}}\right)} \quad \text{..... (2)}$$

Step 2 of 5

Determine the value of collector current for a base-emitter voltage of $0.5\ \text{V}$.

Divide equation (1) by equation (2).

$$\frac{10\times10^{-3}}{i_{C2}} = \frac{I_S e^{\left(\frac{0.7}{25\times10^{-3}}\right)}}{I_S e^{\left(\frac{0.5}{25\times10^{-3}}\right)}}$$

$$\begin{aligned} i_{C2} &= \left(10\times10^{-3}\right) e^{\left(\frac{0.5-0.7}{25\times10^{-3}}\right)} \\ &= \left(10\times10^{-3}\right) e^{\left(\frac{-0.2}{25\times10^{-3}}\right)} \\ &= 3.355\times10^{-6} \\ &= 3.355\ \mu\text{A} \end{aligned}$$

Thus, the new value of collector current, i_{C2} is $\boxed{3.355\ \mu\text{A}}$.

Step 3 of 5

Determine the slope of the $i_C - v_{BE}$ curve at $v_{BE1} = 700\ \text{mV}$.

$$\begin{aligned} A_{v1} &= -\left(\frac{i_{C1}}{V_T}\right) R_C \\ &= -\left(\frac{I_S e^{\frac{v_{BE1}}{V_T}}}{V_T}\right) R_C \qquad \left(\text{since } i_{C1} = I_S e^{\frac{v_{BE1}}{V_T}}\right) \end{aligned}$$

Substitute $25\ \text{mV}$ for V_T and $700\ \text{mV}$ for v_{BE1} in the equation.

$$A_{v1} = -\left(\frac{I_S e^{\left(\frac{700\times10^{-3}}{25\times10^{-3}}\right)}}{25\times10^{-3}}\right) R_C \quad \text{..... (3)}$$

Step 4 of 5

Determine the slope of the $i_C - v_{BE}$ curve at $v_{BE2} = 500\ \text{mV}$.

$$\begin{aligned} A_{v2} &= -\left(\frac{i_{C2}}{V_T}\right) R_C \\ &= -\left(\frac{I_S e^{\frac{v_{BE2}}{V_T}}}{V_T}\right) R_C \qquad \left(\text{since } i_{C2} = I_S e^{\frac{v_{BE2}}{V_T}}\right) \end{aligned}$$

Substitute $25\ \text{mV}$ for V_T and $500\ \text{mV}$ for v_{BE2} in the equation.

$$A_{v2} = -\left(\frac{I_S e^{\left(\frac{500\times10^{-3}}{25\times10^{-3}}\right)}}{25\times10^{-3}}\right) R_C \quad \text{..... (4)}$$

Step 5 of 5

Evaluate the ratio of slopes of the $i_C - v_{BE}$ curves at $700\ \text{mV}$ and $500\ \text{mV}$.

Divide equation (3) by equation (4).

$$\begin{aligned} \frac{A_{v1}}{A_{v2}} &= \frac{-\left(\frac{I_S e^{\left(\frac{700\times10^{-3}}{25\times10^{-3}}\right)}}{25\times10^{-3}}\right) R_C}{-\left(\frac{I_S e^{\left(\frac{500\times10^{-3}}{25\times10^{-3}}\right)}}{25\times10^{-3}}\right) R_C} \\ &= e^{\left(\frac{700\times10^{-3}}{25\times10^{-3}} - \frac{500\times10^{-3}}{25\times10^{-3}}\right)} \\ &= e^{\left(\frac{700-500}{25}\right)} \\ &= 2980.9 \end{aligned}$$

Thus, the ratio of the slopes of the curves at $700\ \text{mV}$ and $500\ \text{mV}$ is, $\boxed{\frac{A_{v1}}{A_{v2}} = 2980.9}$.

Step 1 of 7

Consider the equation.

$$i_C = I_S e^{\frac{V_{BE}}{V_T}}\left(1+\frac{V_{CE}}{V_A}\right)$$
$$= 10^{-15}\, e^{\frac{V_{BE}}{0.025}}\left(1+\frac{V_{CE}}{100}\right)$$

Substitute $V_{CE} = 0$ in the above equation to find the intercept of the straight line characteristics on the i_C axis.

$$i_C = 10^{-15}\, e^{\frac{V_{BE}}{0.025}}\left(1+\frac{0}{100}\right)$$
$$= 10^{-15}\, e^{\frac{V_{BE}}{0.025}}$$

for $V_{BE} = 0.65\ \text{V}$, the intercept is

$$i_C = 10^{-15}\, e^{\frac{0.65}{0.025}}$$
$$= 0.2\ \text{mA}$$

The slope of the curve $V_{BE} = 0.65\ \text{V}$ is

$$\text{Slope} = \frac{i_C}{V_A}$$
$$= \frac{0.2\ \text{mA}}{100}$$
$$\text{Slope} = 0.002\ \text{mA/V}$$

Step 2 of 7

For $V_{BE} = 0.70\ \text{V}$, the intercept is

$$i_C = 10^{-15}\, e^{\frac{0.70}{0.025}}$$
$$= 1.45\ \text{mA}$$

The slope of the curve $V_{BE} = 0.70\ \text{V}$ is

$$\text{Slope} = \frac{i_C}{V_A}$$
$$= \frac{1.45\ \text{mA}}{100}$$
$$\text{Slope} = 0.0145\ \text{mA/V}$$

Step 3 of 7

For $V_{BE} = 0.72\ \text{V}$, the intercept is

$$i_C = 10^{-15}\, e^{\frac{0.72}{0.025}}$$
$$= 3.22\ \text{mA}$$

The slope of the curve $V_{BE} = 0.72\ \text{V}$ is

$$\text{Slope} = \frac{i_C}{V_A}$$
$$= \frac{3.22\ \text{mA}}{100}$$
$$\text{Slope} = 0.0322\ \text{mA/V}$$

Step 4 of 7

For $V_{BE} = 0.73\ \text{V}$, the intercept is

$$i_C = 10^{-15}\, e^{\frac{0.73}{0.025}}$$
$$= 4.80\ \text{mA}$$

The slope of the curve $V_{BE} = 0.73\ \text{V}$ is

$$\text{Slope} = \frac{i_C}{V_A}$$
$$= \frac{4.8\ \text{mA}}{100}$$
$$\text{Slope} = 0.048\ \text{mA/V}$$

Step 5 of 7

For $V_{BE} = 0.74\ \text{V}$, the intercept is

$$i_C = 10^{-15}\, e^{\frac{0.74}{0.025}}$$
$$= 7.16\ \text{mA}$$

The slope of the curve $V_{BE} = 0.74\ \text{V}$ is

$$\text{Slope} = \frac{i_C}{V_A}$$
$$= \frac{7.16\ \text{mA}}{100}$$
$$\text{Slope} = 0.0716\ \text{mA/V}$$

Step 6 of 7

Tabulate the values as shown in Table 1.

V_{BE} (V)	0.65	0.70	0.72	0.73	0.74
i_C intercept (mA)	0.20	1.46	3.21	4.80	7.16
Slope (mA/V)	0.002	0.0146	0.0321	0.048	0.0716

Table 1

Step 7 of 7

The plot of i_C versus v_{CE} is shown in Figure 1.

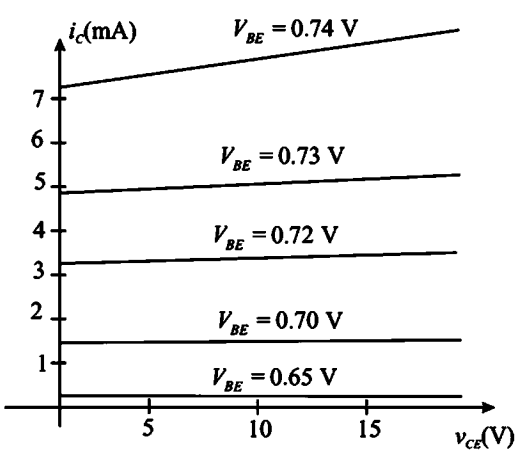


Figure 1

We don't have the solution to this problem yet.

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Step 1 of 2

6.043P

The slope of $i_C - v_{CE}$ characteristics is,

$$\frac{\partial i_C}{\partial v_{CE}} = 0.8\times10^{-5}\text{℧}$$

Determine the output resistance.

$$\begin{aligned} r_0 &= \left[\frac{\partial i_C}{\partial v_{CE}}\right]^{-1} \\ &= \frac{1}{0.8\times10^{-5}} \\ &= 125\,\text{k}\Omega \end{aligned}$$

Determine the value of early voltage.

$$\begin{aligned} V_A &= r_0 I_C \\ &= (125\,\text{k}\Omega)(1\,\text{mA}) \\ &= 125\,\text{V} \end{aligned}$$

Step 2 of 2

When operating at current $I_C = 10\,mA$

Determine the value of output resistance.

$$\begin{aligned} r_0 &= \frac{V_A}{I_C} \\ &= \frac{125}{10\text{ mA}} \\ &= 12.5\,\text{k}\Omega \end{aligned}$$

Therefore, the output resistance when current is 10 mA is $\boxed{12.5\text{ k}\Omega}$.

Step 1 of 2

6.044P

The output resistance at $I_C' = 1\ \text{mA}$ is given by

$$r_0 = \frac{V_A}{I_C'}$$

Substitute $150\ \text{V}$ for V_A and $1\ \text{mA}$ for I_C' in equation.

$$r_0 = \frac{150}{10^{-3}}$$
$$= 150\ \text{k}\Omega$$

Therefore, the output resistance at $I_C' = 1\ \text{mA}$ is $\boxed{150\ \text{k}\Omega}$.

Step 2 of 2

The output resistance at $I_C' = 100\ \mu\text{A}$ is given by

$$r_0 = \frac{V_A}{I_C'}$$

Substitute $150\ \text{V}$ for V_A and $100\ \mu\text{A}$ for I_C' in equation.

$$r_0 = \frac{150}{100 \times 10^{-6}}$$
$$= 1500\ \text{k}\Omega$$

Therefore, the output resistance at $I_C' = 100\ \mu\text{A}$ is $\boxed{1500\ \text{k}\Omega}$.

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The *pnp* transistor equivalent circuit models are shown in Figure 1 and Figure 2.

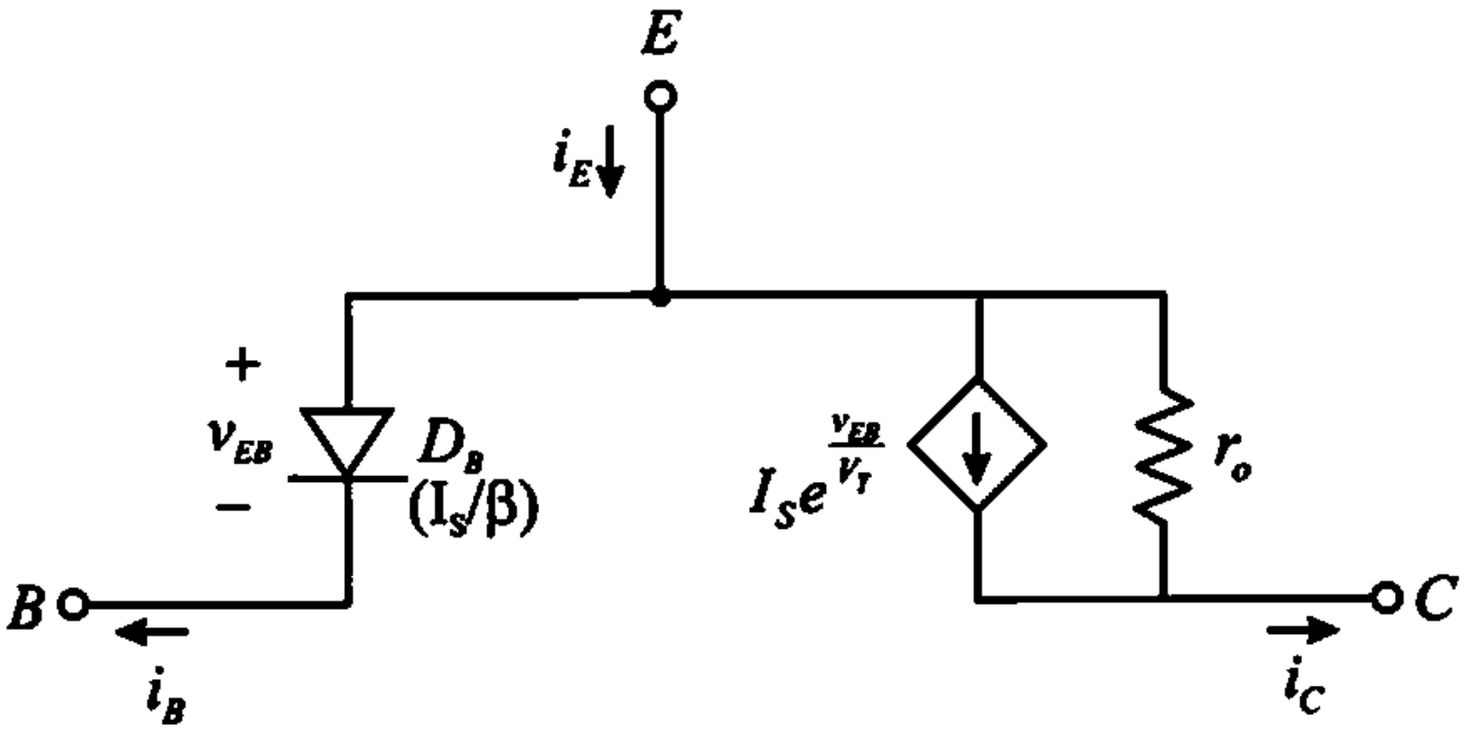


Figure 1

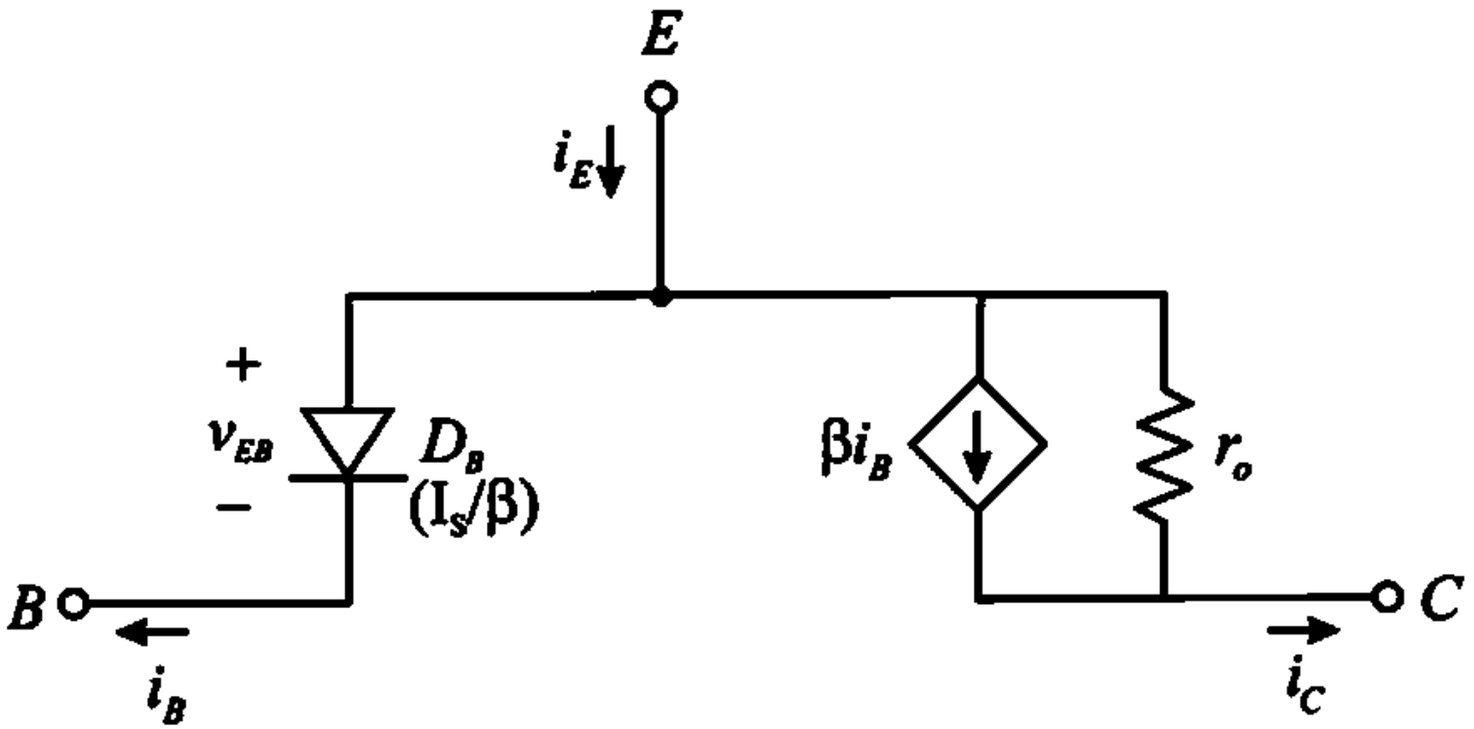


Figure 2

Therefore , the small signal equivalents of *pnp* BJT are drawn.

6.047P

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Step 1 of 6

(a)

The transistor is in active mode $V_{CE} = 2\ \text{V}$.

The base emitter voltage is $V_{BE} = 0.7\ \text{V}$.

Calculate the collector current I_C.

$$\begin{aligned} I_C &= \frac{V_{CC} - V_{CE}}{R_C} \\ &= \frac{10-2}{1\ \text{k}\Omega} \\ &= 8\ \text{mA} \end{aligned}$$

Calculate the base current.

$$\begin{aligned} I_B &= \frac{I_C}{\beta} \\ &= \frac{8\ \text{mA}}{50} \\ &= 0.16\ \text{mA} \end{aligned}$$

Step 2 of 6

Calculate the supply voltage V_{BB}.

$$V_{BB} - I_B R_B - V_{BE} = 0$$

$$\begin{aligned} V_{BB} &= I_B R_B + V_{BE} \\ &= \left(0.16\times10^{-3}\right)\left(10\times10^{3}\right) + 0.7 \\ &= 2.3\ \text{V} \end{aligned}$$

Therefore, the supply voltage V_{BB} is $\boxed{2.3\ \text{V}}$.

Step 3 of 6

(b)

The transistor is at edge saturation $\left(V_{CE} = 0.3\ \text{V}\right)$.

Calculate the collector current:

$$\begin{aligned} I_C &= \frac{V_{CC} - V_{CE}}{R_C} \\ &= \frac{10-0.3}{1\ \text{k}\Omega} \\ &= 9.7\ \text{mA} \end{aligned}$$

Calculate the base current.

$$\begin{aligned} I_B &= \frac{I_C}{\beta} \\ &= \frac{9.7\ \text{mA}}{50} \\ &= 0.194\ \text{mA} \end{aligned}$$

Step 4 of 6

Calculate the supply voltage V_{BB}.

$$V_{BB} - I_B R_B - V_{BE} = 0$$

$$\begin{aligned} V_{BB} &= I_B R_B + V_{BE} \\ &= \left(0.194\times10^{-3}\right)\left(10\times10^{3}\right) + 0.7 \\ &= 2.64\ \text{V} \end{aligned}$$

Therefore, the supply voltage V_{BB} is $\boxed{2.64\ \text{V}}$.

Step 5 of 6

(c)

At deep saturation $V_{CE} = 0.2\ \text{V}$, and $\beta_{\text{forced}} = 10$

Calculate the collector current:

$$\begin{aligned} I_C &= \frac{V_{CC} - V_{CE}}{R_C} \\ &= \frac{10-0.2}{1\ \text{k}\Omega} \\ &= 9.8\ \text{mA} \end{aligned}$$

Calculate the base current.

$$\begin{aligned} I_B &= \frac{I_C}{\beta} \\ &= \frac{9.8\ \text{mA}}{10} \\ &= 0.98\ \text{mA} \end{aligned}$$

Step 6 of 6

Calculate the supply voltage V_{BB}.

$$V_{BB} - I_B R_B - V_{BE} = 0$$

$$\begin{aligned} V_{BB} &= I_B R_B + V_{BE} \\ &= \left(0.98\times10^{-3}\right)\left(10\times10^{3}\right) + 0.7 \\ &= 10.5\ \text{V} \end{aligned}$$

Therefore, the supply voltage V_{BB} is $\boxed{10.5\ \text{V}}$.

Step 1 of 9

Refer to Figure P6.49 in the text book.

Redraw the circuit is shown in Figure 1.

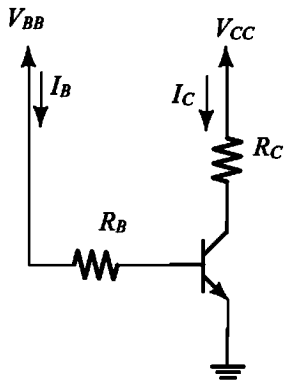


Figure 1

Step 2 of 9

The equivalent circuit is shown in Figure 2.

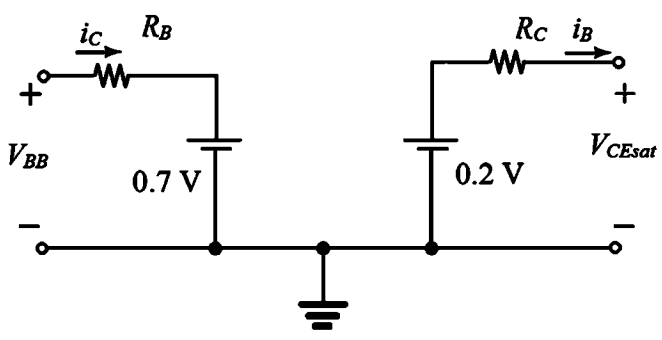


Figure 2

Step 3 of 9

Given that $V_{BB} = V_{CC}$.

Find i_C .

$$V_{CC} - i_C R_C - 0.2 = 0$$
$$i_C R_C = V_{CC} - 0.2$$
$$i_C = \frac{V_{CC} - 0.2}{R_C}$$

Step 4 of 9

Obtain the expression for i_B .

$$V_{CC} - i_B R_B - 0.7 = 0$$
$$i_B R_B = V_{CC} - 0.7$$
$$i_B = \frac{V_{CC} - 0.7}{R_B}$$

Step 5 of 9

Now find β .

$$\beta = \frac{i_C}{i_B}$$
$$= \frac{\frac{V_{CC} - 0.2}{R_C}}{\frac{V_{CC} - 0.7}{R_B}}$$
$$= \frac{R_C}{R_B}\left[\frac{V_{CC} - 0.7}{V_{CC} - 0.2}\right]$$

Thus, the forced β is $\boxed{\frac{R_C}{R_B}\left[\frac{V_{CC} - 0.7}{V_{CC} - 0.2}\right]}$.

Step 6 of 9

Obtain the power dissipation.

$$P = i_B^2 R_B + i_C^2 R_C$$
$$= i_B^2 R_B + \beta^2 i_B^2 R_C$$
$$= i_B^2 \left(R_B + \beta^2 R_C\right)$$

Therefore the expression of power dissipation is $\boxed{i_B^2 \left(R_B + \beta^2 R_C\right)}$.

Step 7 of 9

Obtain the value of $\frac{R_C}{R_B}$, if $V_{CC} = 5\ \text{V}$, $\beta_{\text{forced}} \le 10$ and $P \le 20\ \text{mW}$.

$$\beta \ge \frac{R_C}{R_B}\left[\frac{V_{CC} - 0.7}{V_{CC} - 0.2}\right]$$
$$10 \ge \frac{R_C}{R_B}\left[\frac{5 - 0.7}{5 - 0.2}\right]$$
$$\frac{R_C}{R_B} \le 11.2$$

Step 8 of 9

Find i_B .

$$P \ge i_B^2 \left(R_B + \beta^2 R_C\right)$$
$$0.02 \ge i_B^2 \left(R_B + 1120 R_B\right)$$
$$i_B^2 \le \frac{0.02}{1121 R_B}$$

$$i_B^2 \le \frac{0.02}{1121 R_B}$$

Now using KVL on Base-Emitter side, and find R_B .

$$5 - i_B R_B - 0.7 = 0$$
$$i_B R_B = 4.3$$
$$\left(i_B R_B\right)^2 = 18.49$$

Substitute $i_B^2 = \frac{0.02}{1121 R_B}$ in equation.

$$\left(i_B R_B\right)^2 = 18.49$$
$$\frac{0.02}{1121 R_B} R_B^2 = 18.49$$
$$R_B = 1.03 \times 10^6$$

Therefore , the base resistance is $\boxed{1.03\ \text{M}\Omega}$.

Step 9 of 9

Calculate R_C .

$$\frac{R_C}{R_B} = 11.2$$
$$R_C = 11.2\left(1.03 \times 10^6\right)$$
$$R_C = 11.53 \times 10^6$$

Therefore the collector resistance is $\boxed{11.53\ \text{M}\Omega}$.

6.050P

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Step 1 of 6

(a)

Since the base is connected to +2 V and emitter is connected to ground through a resistance R_E=1 K?, hence the base-emitter junction is forward biased.

Let us assume that the voltage V_{BE} is approximately equal to 0.7 V

The emitter voltage is,

$$\begin{aligned} V_E &= V_B - V_{BE} \\ &= 2 - 0.7 \\ &= 1.3\,\text{V} \end{aligned}$$

The emitter current is,

$$\begin{aligned} I_E &= \frac{V_E - 0}{R_E} \\ &= \frac{1.3\,\text{V} - 0\,\text{V}}{1\,\text{k}\Omega} \\ &= 1.3\,\text{mA} \end{aligned}$$

Step 2 of 6

The collector is connected to +3 V through a resistance $R_C = 1\,\text{k}\Omega$

Let us assume the active mode of operation

Collector current is

$$I_C = \alpha I_E$$

Since the value for β is not given assume $\beta = 100$

$$\begin{aligned} \alpha &= \frac{\beta}{\beta + 1} \\ &= \frac{100}{100 + 1} \\ &= 0.99 \end{aligned}$$

Determine the value of emitter current.

$$\begin{aligned} I_C &= \alpha I_E \\ &= 0.99(1.3\,\text{mA}) \\ &= 1.28\,\text{mA} \end{aligned}$$

Collector voltage is,

$$\begin{aligned} V_C &= 3 - I_C R_C \\ &= 3 - (1.28 \times 1) \\ V_C &= 1.72\ \text{V} \end{aligned}$$

Since the collector voltage is less than the base voltage, the collector base junction is forward biased .hence the assumption of active mode is incorrect.

Assuming saturation mode

$$\begin{aligned} V_C &= V_E + V_{CEsat} \\ &= 1.3 + 0.2 \\ V_C &= 1.5\ \text{V} \end{aligned}$$

Therefore, the emitter and collector voltages are $\boxed{V_E = 1.3V}$ $\boxed{V_C = 1.5V}$.

Step 3 of 6

(b)

Since the base is connected to +1.7 V and emitter is connected to ground through a resistance $R_E = 1\,\text{k}\Omega$, hence the base-emitter junction is forward biased.

Let us assume that the voltage V_{BE} is approximately equal to 0.7 V

The emitter voltage is given as

$$\begin{aligned} V_E &= V_B - V_{BE} \\ &= 1.7 - 0.7 \\ V_E &= 1V \end{aligned}$$

Emitter current is,

$$\begin{aligned} I_E &= \frac{V_E - 0}{R_E} \\ &= \frac{1 - 0}{1\,\text{k}\Omega} \\ I_E &= 1\,\text{mA} \end{aligned}$$

Step 4 of 6

Let us assume the active mode of operation

Collector current is

$$I_C = \alpha I_E$$

Since the value for β is not given assume $\beta = 100$

$$\begin{aligned} \alpha &= \frac{\beta}{\beta + 1} \\ &= \frac{100}{100 + 1} \\ &= 0.99 \end{aligned}$$

The collector current is,

$$\begin{aligned} I_C &= \alpha I_E \\ &= (0.99)(1\,\text{mA}) \\ &= 0.99\,\text{mA} \end{aligned}$$

Step 5 of 6

The collector voltage is,

$$\begin{aligned} V_C &= 3 - I_C R_C \\ &= 3 - (0.99 \times 1) \\ &= 2.01\,\text{V} \end{aligned}$$

Since the collector voltage is greater than the base voltage, the collector base junction is reverse biased .hence the assumption of active mode is correct.

Therefore, the emitter and collector voltages are $\boxed{V_E = 1\,\text{V}}$ $\boxed{V_C = 2.01\,\text{V}}$.

Step 6 of 6

(c)

Finding the voltages V_E and V_C when $V_B = 0\,\text{V}$

Since the base is connected to 0 V and emitter is connected to ground through a resistance $R_E = 1\,\text{k}\Omega$, the base-emitter junction cannot conduct and emitter current is zero.

Emitter voltage is zero.

The collector current is zero. The collector is connected to +3 V. The collector voltage is +3 V.

Therefore, the emitter and collector voltages are $\boxed{V_E = 0V}$ $\boxed{V_C = 3V}$.

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Step 1 of 18

Refer to Figure P6.61 in the text book.

Draw the modified circuit.

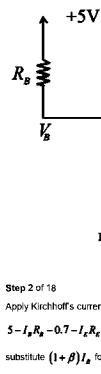


Figure 1

Step 2 of 18

Apply Kirchhoff's current law across the base emitter circuit.

$$5 - I_B R_B - 0.7 - I_E R_E = 0$$

substitute $(1+\beta) I_B$ for I_E.

$$5 - I_B R_B - 0.7 - I_B(\beta+1) R_E = 0$$
$$4.3 - I_B R_B - I_B(\beta+1) R_E = 0$$
$$4.3 - I_B\left[R_B + \beta R_E + R_E\right] = 0$$
$$I_B = \frac{4.3}{R_B + \beta R_E + R_E}$$

Step 3 of 18

The voltage V_B is

$$V_B = 5 - I_B R_B$$
$$= 5 - \left(\frac{4.3}{R_B + \beta R_E + R_E}\right) R_B$$

$$V_B = 5 - \frac{4.3 R_B}{R_B + \beta R_E + R_E} \quad \text{..... (1)}$$

The voltage V_E is

$$V_E = I_E R_E$$
$$= (1+\beta) I_B R_E$$
$$= (1+\beta)\left(\frac{4.3}{R_B + \beta R_E + R_E}\right) R_E$$

$$V_E = \frac{4.3(1+\beta) R_E}{R_B + \beta R_E + R_E} \quad \text{..... (2)}$$

Step 4 of 18

The voltage V_C is

$$V_C = 5 - I_C R_C$$
$$V_C = 5 - \beta I_B R_C$$
$$V_C = 5 - \beta\left(\frac{4.3}{R_B + \beta R_E + R_E}\right) R_C$$

$$V_C = 5 - \frac{4.3 \beta R_C}{R_B + \beta R_E + R_E} \quad \text{..... (3)}$$

Step 5 of 18

Substitute $\beta = 100$, $R_E = 1\ \text{k}\Omega$ and $R_C = 1\ \text{k}\Omega$ in equation (1).

$$V_B = 5 - \frac{4.3 R_B}{R_B + (1000)(100) + 1000}$$

$$V_B = 5 - \frac{4.3 R_B}{R_B + 10^3 + 10^2} \quad \text{..... (4)}$$

Substitute $\beta = 100$, $R_E = 1\ \text{k}\Omega$ and $R_C = 1\ \text{k}\Omega$ in equation (2).

$$V_E = \frac{4.3(101)(1000)}{R_B + (100)(1000) + 1000}$$

$$V_E = \frac{434300}{R_B + 10^5 + 10^3} \quad \text{..... (5)}$$

Substitute $\beta = 100$, $R_E = 1\ \text{k}\Omega$ and $R_C = 1\ \text{k}\Omega$ in equation (3).

$$V_C = 5 - \frac{4.3(100)(1000)}{R_B + (100)(1000) + 1000}$$

$$V_C = 5 - \frac{430000}{R_B + 10^5 + 10^2} \quad \text{..... (6)}$$

Step 6 of 18

Determine the voltages V_B, V_C and V_E for $R_B = 100\ \text{k}\Omega$.

From equation (4);

$$V_B = 5 - \frac{4.3\left(10^5\right)}{10^5 + 10^3 + 10^2}$$
$$= 5 - 2.139$$

Thus, the voltage V_B for $100\ \text{k}\Omega$ is $\boxed{2.861\text{V}}$.

From equation (5);

$$V_E = \frac{434300}{10^5 + 10^3 + 10^2}$$

Thus, the emitter voltage for $100\ \text{k}\Omega$ is $\boxed{2.161\text{V}}$

From equation (6);

$$V_C = 5 - \frac{430000}{10^5 + 10^3 + 10^2}$$
$$= 5 - 2.139$$

Thus, the voltage V_C $\boxed{2.861\text{V}}$.

Step 7 of 18

The voltage V_{CE} is,

$$V_{CE} = V_C - V_E$$
$$= 2.861 - 2.161$$
$$V_{CE} = 0.7\ \text{V}$$

Since V_{CE} is positive, the transistor is in active mode and the obtained voltages are correct.

Step 8 of 18

Determine the voltages V_B, V_C and V_E for $R_B = 10\ \text{k}\Omega$.

$$V_B = 5 - \frac{4.3 R_B}{R_B + 10^3 + 10^2}$$
$$= 5 - \frac{4.3\left(10^4\right)}{10^4 + 10^3 + 10^2}$$
$$= 5 - 0.38738$$
$$V_B = 4.612\text{V}$$

Step 9 of 18

From equation (5);

$$V_E = \frac{434300}{10^4 + 10^3 + 10^2}$$
$$= \frac{434300}{111000}$$
$$V_E = 3.91\text{V}$$

From equation (6);

$$V_C = 5 - \frac{430000}{10^4 + 10^3 + 10^2}$$
$$= 5 - 3.8738$$
$$V_C = 1.126\text{V}$$

The voltage V_{CE} is,

$$V_{CE} = V_C - V_E$$
$$= 1.126 - 3.91$$
$$V_{CE} = -2.784\ \text{V}$$

Since V_{CE} is negative, the transistor is in saturation mode and hence the obtained voltages are not correct.

Step 10 of 18

In saturation, the voltage $V_{CE} = 0.2\ \text{V}$ and $V_{BE} = 0.7\ \text{V}$

From the given circuit,

$$I_B = \frac{5 - (V_E + V_{BE})}{R_B}$$
$$= \frac{5 - (V_E + 0.7)}{10 \times 10^3}$$

and

$$I_C = \frac{5 - (V_E + V_{CE})}{R_C}$$
$$= \frac{5 - (V_E + 0.2)}{1 \times 10^3}$$

The expression for the emitter current is,

$$I_E = I_B + I_C$$
$$\frac{V_E}{R_E} = \frac{5 - (V_E + 0.7)}{10 \times 10^3} + \frac{5 - (V_E + 0.2)}{1 \times 10^3}$$
$$\frac{V_E}{1 \times 10^3} = \frac{5 - (V_E + 0.7)}{10 \times 10^3} + \frac{5 - (V_E + 0.2)}{1 \times 10^3}$$
$$10 V_E = 5 - (V_E + 0.7) + 10\left(5 - (V_E + 0.2)\right)$$

Simplify further.

$$10 V_E = 5 - V_E - 0.7 + 50 - 10 V_E - 2$$
$$21 V_E = 52.3$$
$$V_E = 2.49\ \text{V}$$

Hence the emitter voltage when $R_B = 10\text{k}\Omega$ is $\boxed{2.49\ \text{V}}$.

Step 11 of 18

The voltage V_C is,

$$V_C = V_E + V_{CE}$$
$$= 2.49 + 0.2$$
$$V_C = 2.69\ \text{V}$$

Hence the voltage V_C when $R_B = 10\text{k}\Omega$ is $\boxed{2.69\ \text{V}}$.

Step 12 of 18

The voltage V_B is,

$$V_B = V_E + V_{CE}$$
$$= 2.49 + 0.7$$
$$V_B = 3.19\ \text{V}$$

Hence the voltage V_B when $R_B = 10\text{k}\Omega$ is $\boxed{3.19\ \text{V}}$.

Step 13 of 18

Determine the voltages V_B, V_C and V_E for $R_B = 1\ \text{k}\Omega$.

$$V_B = 5 - \frac{4.3 R_B}{R_B + 10^3 + 10^2}$$
$$= 5 - \frac{4.3\left(10^3\right)}{10^3 + 10^3 + 10^2}$$
$$= 5 - 0.04215$$
$$V_B = 4.957\text{V}$$

Step 14 of 18

From equation (5);

$$V_E = \frac{434300}{10^3 + 10^3 + 10^1}$$
$$= \frac{434300}{102000}$$
$$V_E = 4.257\text{V}$$

From equation (6);

$$V_C = 5 - \frac{430000}{10^3 + 10^3 + 10^3}$$
$$= 5 - 4.2156$$
$$V_C = 0.7843\text{V}$$

The voltage V_{CE} is,

$$V_{CE} = V_C - V_E$$
$$= 0.7843 - 4.257$$
$$V_{CE} = -3.4727\ \text{V}$$

Since V_{CE} is negative, the transistor is in saturation mode and hence the obtained voltages are not correct.

Step 15 of 18

In saturation, the voltage $V_{CE} = 0.2\ \text{V}$ and $V_{BE} = 0.7\ \text{V}$

From the given circuit,

$$I_B = \frac{5 - (V_E + V_{BE})}{R_B}$$
$$= \frac{5 - (V_E + 0.7)}{1 \times 10^3}$$

and

$$I_C = \frac{5 - (V_E + V_{CE})}{R_C}$$
$$= \frac{5 - (V_E + 0.2)}{1 \times 10^3}$$

Step 16 of 18

The expression for the emitter current is,

$$I_E = I_B + I_C$$
$$\frac{V_E}{R_E} = \frac{5 - (V_E + 0.7)}{1 \times 10^3} + \frac{5 - (V_E + 0.2)}{1 \times 10^3}$$
$$\frac{V_E}{1 \times 10^3} = \frac{5 - (V_E + 0.7)}{1 \times 10^3} + \frac{5 - (V_E + 0.2)}{1 \times 10^3}$$
$$V_E = 5 - (V_E + 0.7) + \left(5 - (V_E + 0.2)\right)$$

Simplify further.

$$V_E = 5 - V_E - 0.7 + 5 - V_E - 0.2$$
$$3 V_E = 9.1$$
$$V_E = 3.03\ \text{V}$$

Hence the emitter voltage when $R_B = 1\ \text{k}\Omega$ is $\boxed{3.03\ \text{V}}$.

Step 17 of 18

The voltage V_C is,

$$V_C = V_E + V_{CE}$$
$$= 3.03 + 0.2$$
$$V_C = 3.23\ \text{V}$$

Hence the voltage V_C when $R_B = 1\ \text{k}\Omega$ is $\boxed{3.23\ \text{V}}$.

Step 18 of 18

The voltage V_B is,

$$V_B = V_E + V_{CE}$$
$$= 3.03 + 0.7$$
$$V_B = 3.73\ \text{V}$$

Hence the voltage V_B when $R_B = 10\text{k}\Omega$ is $\boxed{3.73\ \text{V}}$.

6.061P

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e resistor R and draw the matched circuit.

32



Figure 2

32

r I_{C_1} is very small compared to I_A.

$$\frac{0}{2.18}$$

$$0^{\circ}$$

mA

the current I.

37

$$\frac{I_{C_1}}{2\times10^{-2}} = 50\,I_{C_1}$$

$$I_{C_1} = \frac{9-V_{C_1}}{2\times10^{3}}$$

$$V_{C_1} = 9 - (2\times10^{3})I_{C_1} \quad \text{..... (1)}$$

$$V_{C_1} = 9 - 0.7$$

$$= 8.3\ \text{V}$$

$$I_{C_2} = \frac{9-0.7}{10^{3}+40\times10^{3}}$$

$$= 198\ \mu\text{A}$$

$$V_{B_2} = (40\times10^{3})(198\times10^{-6})$$

$$= 7.92\ \text{V}$$

$$V_{C_2} = 9 - 0.7$$

$$= 8.3\ \text{V}$$

$$I_{C_2} = \frac{9-8.3}{10^{3}+40\times10^{3}}$$

$$= 17.5\ \mu\text{A}$$

$$V_{B_2} = (40\times10^{3})(17.5\times10^{-6})$$

$$= 0.7\ \text{V}$$

$$V_{C_1} = 9 - (2\times10^{3})(2\times10^{-3})$$

$$= 5\ \text{V}$$

$$I_{C_1} = \frac{9-5}{2\times10^{3}}$$

$$= 2\ \text{mA}$$

$$I_{B_1} = \frac{2\times10^{-3}}{100}$$

$$= 20\ \mu\text{A}$$

$$V_{B_1} = (40\times10^{3})(20\times10^{-6})$$

$$= 0.8\ \text{V}$$

$$V_{C_1} = 9 - 0.8$$

$$= 8.2\ \text{V}$$

$$I_{C_2} = \frac{9-8.2}{10^{3}+40\times10^{3}}$$

$$= 19.5\ \mu\text{A}$$

$$V_{B_2} = (40\times10^{3})(19.5\times10^{-6})$$

$$= 0.78\ \text{V}$$

$$V_{C_2} = 9 - 0.78$$

$$= 8.22\ \text{V}$$

$$I_{C_1} = \frac{9-8.22}{2\times10^{3}}$$

$$= 390\ \mu\text{A}$$

$$I_{B_1} = \frac{390\times10^{-6}}{100}$$

$$= 3.9\ \mu\text{A}$$

$$V_{B_1} = (40\times10^{3})(3.9\times10^{-6})$$

$$= 0.156\ \text{V}$$

$$V_{C_1} = 9 - 0.156$$

$$= 8.844\ \text{V}$$

$$I_{C_2} = \frac{9-8.844}{10^{3}+40\times10^{3}}$$

$$= 3.9\ \mu\text{A}$$

$$V_{B_2} = (40\times10^{3})(3.9\times10^{-6})$$

$$= 0.156\ \text{V}$$

$$V_{C_2} = 9 - 0.156$$

$$= 8.844\ \text{V}$$

$$I_{C_1} = \frac{9-8.844}{2\times10^{3}}$$

$$= 78\ \mu\text{A}$$

$$I_{B_1} = \frac{78\times10^{-6}}{100}$$

$$= 0.78\ \mu\text{A}$$

$$V_{B_1} = (40\times10^{3})(0.78\times10^{-6})$$

$$= 0.031\ \text{V}$$

$$V_{C_1} = 9 - 0.031$$

$$= 8.969\ \text{V}$$

$$I_{C_2} = \frac{9-8.969}{10^{3}+40\times10^{3}}$$

$$= 0.78\ \mu\text{A}$$

$$V_{B_2} = (40\times10^{3})(0.78\times10^{-6})$$

$$= 0.031\ \text{V}$$

$$V_{C_2} = 9 - 0.031$$

$$= 8.969\ \text{V}$$

$$I_{C_1} = \frac{9-8.969}{2\times10^{3}}$$

$$= 15.5\ \mu\text{A}$$

$$I_{B_1} = \frac{15.5\times10^{-6}}{100}$$

$$= 0.155\ \mu\text{A}$$

$$V_{B_1} = (40\times10^{3})(0.155\times10^{-6})$$

$$= 0.006\ \text{V}$$

$$V_{C_1} = 9 - 0.006$$

$$= 8.994\ \text{V}$$

$$I_{C_2} = \frac{9-8.994}{10^{3}+40\times10^{3}}$$

$$= 0.155\ \mu\text{A}$$

$$V_{B_2} = (40\times10^{3})(0.155\times10^{-6})$$

$$= 0.006\ \text{V}$$

$$V_{C_2} = 9 - 0.006$$

$$= 8.994\ \text{V}$$

$$I_{C_1} = \frac{9-8.994}{2\times10^{3}}$$

$$= 3\ \mu\text{A}$$

$$I_{B_1} = \frac{3\times10^{-6}}{100}$$

$$= 0.03\ \mu\text{A}$$

$$V_{B_1} = (40\times10^{3})(0.03\times10^{-6})$$

$$= 0.0012\ \text{V}$$

$$V_{C_1} = 9 - 0.0012$$

$$= 8.9988\ \text{V}$$

$$I_{C_2} = \frac{9-8.9988}{10^{3}+40\times10^{3}}$$

$$= 0.03\ \mu\text{A}$$

$$V_{B_2} = (40\times10^{3})(0.03\times10^{-6})$$

$$= 0.0012\ \text{V}$$

$$V_{C_2} = 9 - 0.0012$$

$$= 8.9988\ \text{V}$$

$$I_{C_1} = \frac{9-8.9988}{2\times10^{3}}$$

$$= 0.6\ \mu\text{A}$$

$$I_{B_1} = \frac{0.6\times10^{-6}}{100}$$

$$= 0.006\ \mu\text{A}$$

$$V_{B_1} = (40\times10^{3})(0.006\times10^{-6})$$

$$= 0.00024\ \text{V}$$

$$V_{C_1} = 9 - 0.00024$$

$$= 8.99976\ \text{V}$$

$$I_{C_2} = \frac{9-8.99976}{10^{3}+40\times10^{3}}$$

$$= 0.006\ \mu\text{A}$$

$$V_{B_2} = (40\times10^{3})(0.006\times10^{-6})$$

$$= 0.00024\ \text{V}$$

$$V_{C_2} = 9 - 0.00024$$

$$= 8.99976\ \text{V}$$

$$I_{C_1} = \frac{9-8.99976}{2\times10^{3}}$$

$$= 0.12\ \mu\text{A}$$

$$I_{B_1} = \frac{0.12\times10^{-6}}{100}$$

$$= 0.0012\ \mu\text{A}$$

$$V_{B_1} = (40\times10^{3})(0.0012\times10^{-6})$$

$$= 0.000048\ \text{V}$$

$$V_{C_1} = 9 - 0.000048$$

$$= 8.999952\ \text{V}$$

$$I_{C_2} = \frac{9-8.999952}{10^{3}+40\times10^{3}}$$

$$= 0.0012\ \mu\text{A}$$

$$V_{B_2} = (40\times10^{3})(0.0012\times10^{-6})$$

$$= 0.000048\ \text{V}$$

$$V_{C_2} = 9 - 0.000048$$

$$= 8.999952\ \text{V}$$

$$I_{C_1} = \frac{9-8.999952}{2\times10^{3}}$$

$$= 0.024\ \mu\text{A}$$

$$I_{B_1} = \frac{0.024\times10^{-6}}{100}$$

$$= 0.00024\ \mu\text{A}$$

$$V_{B_1} = (40\times10^{3})(0.00024\times10^{-6})$$

$$= 0.0000096\ \text{V}$$

$$V_{C_1} = 9 - 0.0000096$$

$$= 8.9999904\ \text{V}$$

$$I_{C_2} = \frac{9-8.9999904}{10^{3}+40\times10^{3}}$$

$$= 0.00024\ \mu\text{A}$$

$$V_{B_2} = (40\times10^{3})(0.00024\times10^{-6})$$

$$= 0.0000096\ \text{V}$$

$$V_{C_2} = 9 - 0.0000096$$

$$= 8.9999904\ \text{V}$$

$$I_{C_1} = \frac{9-8.9999904}{2\times10^{3}}$$

$$= 0.0048\ \mu\text{A}$$

$$I_{B_1} = \frac{0.0048\times10^{-6}}{100}$$

$$= 0.000048\ \mu\text{A}$$

$$V_{B_1} = (40\times10^{3})(0.000048\times10^{-6})$$

$$= 0.0000019\ \text{V}$$

$$V_{C_1} = 9 - 0.0000019$$

$$= 8.9999981\ \text{V}$$

$$I_{C_2} = \frac{9-8.9999981}{10^{3}+40\times10^{3}}$$

$$= 0.000048\ \mu\text{A}$$

$$V_{B_2} = (40\times10^{3})(0.000048\times10^{-6})$$

$$= 0.0000019\ \text{V}$$

$$V_{C_2} = 9 - 0.0000019$$

$$= 8.9999981\ \text{V}$$

$$I_{C_1} = \frac{9-8.9999981}{2\times10^{3}}$$

$$= 0.00096\ \mu\text{A}$$

$$I_{B_1} = \frac{0.00096\times10^{-6}}{100}$$

$$= 0.0000096\ \mu\text{A}$$

$$V_{B_1} = (40\times10^{3})(0.0000096\times10^{-6})$$

$$= 0.00000038\ \text{V}$$

$$V_{C_1} = 9 - 0.00000038$$

$$= 8.99999962\ \text{V}$$

$$I_{C_2} = \frac{9-8.99999962}{10^{3}+40\times10^{3}}$$

$$= 0.0000096\ \mu\text{A}$$

$$V_{B_2} = (40\times10^{3})(0.0000096\times10^{-6})$$

$$= 0.00000038\ \text{V}$$

$$V_{C_2} = 9 - 0.00000038$$

$$= 8.99999962\ \text{V}$$

$$I_{C_1} = \frac{9-8.99999962}{2\times10^{3}}$$

$$= 0.00019\ \mu\text{A}$$

$$I_{B_1} = \frac{0.00019\times10^{-6}}{100}$$

$$= 0.0000019\ \mu\text{A}$$

$$V_{B_1} = (40\times10^{3})(0.0000019\times10^{-6})$$

$$= 0.000000076\ \text{V}$$

$$V_{C_1} = 9 - 0.000000076$$

$$= 8.999999924\ \text{V}$$

$$I_{C_2} = \frac{9-8.999999924}{10^{3}+40\times10^{3}}$$

$$= 0.0000019\ \mu\text{A}$$

$$V_{B_2} = (40\times10^{3})(0.0000019\times10^{-6})$$

$$= 0.000000076\ \text{V}$$

$$V_{C_2} = 9 - 0.000000076$$

$$= 8.999999924\ \text{V}$$

$$I_{C_1} = \frac{9-8.999999924}{2\times10^{3}}$$

$$= 0.000038\ \mu\text{A}$$

$$I_{B_1} = \frac{0.000038\times10^{-6}}{100}$$

$$= 0.00000038\ \mu\text{A}$$

$$V_{B_1} = (40\times10^{3})(0.00000038\times10^{-6})$$

$$= 0.000000015\ \text{V}$$

$$V_{C_1} = 9 - 0.000000015$$

$$= 8.999999985\ \text{V}$$

$$I_{C_2} = \frac{9-8.999999985}{10^{3}+40\times10^{3}}$$

$$= 0.00000038\ \mu\text{A}$$

$$V_{B_2} = (40\times10^{3})(0.00000038\times10^{-6})$$

$$= 0.000000015\ \text{V}$$

$$V_{C_2} = 9 - 0.000000015$$

$$= 8.999999985\ \text{V}$$

$$I_{C_1} = \frac{9-8.999999985}{2\times10^{3}}$$

$$= 0.0000076\ \mu\text{A}$$

$$I_{B_1} = \frac{0.0000076\times10^{-6}}{100}$$

$$= 0.000000076\ \mu\text{A}$$

$$V_{B_1} = (40\times10^{3})(0.000000076\times10^{-6})$$

$$= 0.000000003\ \text{V}$$

$$V_{C_1} = 9 - 0.000000003$$

$$= 8.999999997\ \text{V}$$

$$I_{C_2} = \frac{9-8.999999997}{10^{3}+40\times10^{3}}$$

$$= 0.000000076\ \mu\text{A}$$

$$V_{B_2} = (40\times10^{3})(0.000000076\times10^{-6})$$

$$= 0.000000003\ \text{V}$$

$$V_{C_2} = 9 - 0.000000003$$

$$= 8.999999997\ \text{V}$$

$$I_{C_1} = \frac{9-8.999999997}{2\times10^{3}}$$

$$= 0.0000015\ \mu\text{A}$$

$$I_{B_1} = \frac{0.0000015\times10^{-6}}{100}$$

$$= 0.000000015\ \mu\text{A}$$

$$V_{B_1} = (40\times10^{3})(0.000000015\times10^{-6})$$

$$= 0.000000001\ \text{V}$$

$$V_{C_1} = 9 - 0.000000001$$

$$= 8.999999999\ \text{V}$$

$$I_{C_2} = \frac{9-8.999999999}{10^{3}+40\times10^{3}}$$

$$= 0.000000015\ \mu\text{A}$$

$$V_{B_2} = (40\times10^{3})(0.000000015\times10^{-6})$$

$$= 0.000000001\ \text{V}$$

$$V_{C_2} = 9 - 0.000000001$$

$$= 8.999999999\ \text{V}$$

$$I_{C_1} = \frac{9-8.999999999}{2\times10^{3}}$$

$$= 0.0000003\ \mu\text{A}$$

$$I_{B_1} = \frac{0.0000003\times10^{-6}}{100}$$

$$= 0.000000003\ \mu\text{A}$$

$$V_{B_1} = (40\times10^{3})(0.000000003\times10^{-6})$$

$$= 0.000000000\ \text{V}$$

$$V_{C_1} = 9 - 0.000000000$$

$$= 9.000000000\ \text{V}$$

$$I_{C_2} = \frac{9-9.000000000}{10^{3}+40\times10^{3}}$$

$$= 0.000000003\ \mu\text{A}$$

$$V_{B_2} = (40\times10^{3})(0.000000003\times10^{-6})$$

$$= 0.000000000\ \text{V}$$

$$V_{C_2} = 9 - 0.000000000$$

$$= 9.000000000\ \text{V}$$

$$I_{C_1} = \frac{9-9.000000000}{2\times10^{3}}$$

$$= 0\ \mu\text{A}$$

$$I_{B_1} = \frac{0\times10^{-6}}{100}$$

$$= 0\ \mu\text{A}$$

$$V_{B_1} = (40\times10^{3})(0\times10^{-6})$$

$$= 0\ \text{V}$$

$$V_{C_1} = 9 - 0$$

$$= 9\ \text{V}$$

$$I_{C_2} = \frac{9-9}{10^{3}+40\times10^{3}}$$

$$= 0\ \mu\text{A}$$

$$V_{B_2} = (40\times10^{3})(0\times10^{-6})$$

$$= 0\ \text{V}$$

$$V_{C_2} = 9 - 0$$

$$= 9\ \text{V}$$

$$I_{C_1} = \frac{9-9

6.066P

We don't have the solution to this problem yet.

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6.067P

We don't have the solution to this problem yet.

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6.068P

We don't have the solution to this problem yet.

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Ask an expert

6.069P

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Step 1 of 5

7.001E

Refer to the Figure 7.2 in the textbook.

End points of the active-region segment of the voltage –transfer characteristics are point "A" and point "B".

Point "A" is the point at which transistor is changed from cutoff region to saturation region.

$v_{DS} < V_t$ MOSFET will be in cutoff region. For $v_{DS} \geq V_t$ transistor enter into saturation region and hence $v_{GS} = V_t$ is the x-co-ordinate of the point "A".

Gate-source voltage of the point "A" is V_T

Threshold voltage of MOSFET is $0.4\ \text{V}$

Gate-source voltage of the point "A" is $\boxed{0.4\ \text{V}}$

Step 2 of 5

Output voltage is given by,

$$v_{DS} = V_{DD} - i_D R_D$$

$i_D = 0$ in cutoff. Thus, $v_{DS} = V_{DD}$.

Hence, y-coordinate of the point "A" is $\boxed{1.8\ \text{V}}$.

Therefore, the co-ordinates of A is,

$$\begin{aligned} A &= (V_t, V_{DD}) \\ &= (0.4\ \text{V}, 1.8\ \text{V}) \end{aligned}$$

Step 3 of 5

Point "B" is the point at which transistor is changed from saturation region to triode region.

The co-ordinates of B is $\left(V_{GS}\big|_B, V_{DS}\big|_B\right)$.

Determine the value of $V_{GS}\big|_B$.

$$V_{GS}\big|_B = V_t + \frac{\sqrt{2k_n R_D V_{DD} + 1}}{k_n R_D} - 1$$

Substitute $4\ \text{mA/V}^2$ for k_n , $17.5\ \text{k}\Omega$ for R_D , $1.8\ \text{V}$ for V_{DD} and $0.4\ \text{V}$ for V_t in equation.

$$\begin{aligned} V_{GS}\big|_B &= 0.4 + \frac{\sqrt{2 \times 4 \times 10^{-3} \times 17.5 \times 10^3 \times 1.8 + 1} - 1}{4 \times 10^{-3} \times 17.5 \times 10^3} \\ &= 0.4 + \frac{\sqrt{252 + 1} - 1}{70} \\ &= 0.4 + \frac{15.906 - 1}{70} \\ &= 0.4 + 0.213 \\ &= 0.613\ \text{V} \end{aligned}$$

Step 4 of 5

Calculate the value of $V_{DS}\big|_B$.

$$\begin{aligned} V_{DS}\big|_B &= V_{DD} - \frac{1}{2} k_n R_D (V_{GS} - V_t)^2 \\ &= 1.8 - \frac{1}{2} \times 4 \times 10^{-3} \times 17.5 \times 10^3 (0.613 - 0.4)^2 \\ &= 1.8 - 35(0.213)^2 \\ &= 1.8 - 35(0.0454) \\ &= 1.8 - 1.588 \\ &= 0.213\ \text{V} \end{aligned}$$

Therefore, the co-ordinates of B is $(0.613\ \text{V}, 0.213\ \text{V})$.

Step 5 of 5

Calculate the value of $V_{DS}\big|_C$.

$$\begin{aligned} V_{DS}\big|_C &= \frac{V_{DD}}{100} \\ &= \frac{1.8}{100} \\ &= 18\ \text{mV} \end{aligned}$$

Thus, the co-ordinates of A is $\boxed{(0.4\ \text{V}, 1.8\ \text{V})}$, B is $\boxed{(0.613\ \text{V}, 0.213\ \text{V})}$ and $V_{DS}\big|_C$ is $\boxed{18\ \text{mV}}$.

Step 1 of 5

7.001P

Consider the Figure 7.2(a) and Figure 7.2(b) in the textbook.

Segment AB is indicating the active-region segment of the voltage–transfer characteristics of MOS amplifier.

If $v_{DS} < V_T$ MOSFET will be in cutoff region. That is,

$$i_D = 0$$

Hence, $v_{DS} = V_{DD}$.

Hence y-coordinate of the point "A" is V_{DD}

Step 2 of 5

For $v_{DS} \geq V_T$ transistor enter into saturation region and hence $v_{GS} = V_T$ is the x-co-ordinate of the point "A".

$$\begin{aligned} v_{GS} &= V_T \\ &= 0.5\ \text{V} \end{aligned}$$

Therefore, the X and Y co-ordinates of the point "A" is $0.5\ \text{V}$ and $5\ \text{V}$ respectively.

Step 3 of 5

Point "B" is the point at which transistor is changed from saturation region to triode region.

Relation between v_{DS} and v_{GS} for MOSFET in active region is,

$$v_{DS} = V_{DD} - i_D R_D$$

$$v_{DS} = V_{DD} - \frac{1}{2} k_n R_D \left(v_{GS} - V_T\right)^2 \quad \text{..... (1)}$$

Substitute $20\ \text{k}\Omega$ for R_D , $10\ \text{mA/V}^2$ for k_n and $v_{GS}|_B - V_T$ for $v_{DS}|_B$ in the equation (1).

$$\left(v_{GS}|_B - V_T\right) = V_{DD} - \frac{1}{2} k_n R_D \left(v_{GS}|_B - V_T\right)^2$$

Determine the $v_{GS}|_B - V_T$.

$$\left(v_{GS}|_B - V_T\right) = \frac{\sqrt{2 k_n R_D V_{DD} + 1} - 1}{k_n R_D} \quad \text{..... (2)}$$

Step 4 of 5

Substitute k_n , R_D and V_{DD} in equation (2)

$$\begin{aligned} \left(v_{GS}|_B - V_T\right) &= \frac{\sqrt{2 k_n R_D V_{DD} + 1} - 1}{k_n R_D} \\ &= \frac{\sqrt{2\left(10 \times 10^{-3}\ \text{A/V}^2\right)\left(20 \times 10^3\ \Omega\right)\left(5\ \text{V}\right) + 1} - 1}{\left(10 \times 10^{-3}\ \text{A/V}^2\right)\left(20 \times 10^3\ \Omega\right)} \\ &= \frac{43.73}{200} \\ &= 0.22\ \text{V} \end{aligned}$$

Value of $\left(v_{GS}|_B - V_T\right)$ is $0.22\ \text{V}$. Hence, $v_{DS}|_B$ is equal to $\boxed{0.22\ \text{V}}$

Thus, the y-coordinate of point B is, $\boxed{0.22\ \text{V}}$.

Step 5 of 5

Relation between $v_{DS}|_B$ and $v_{GS}|_B$ is,

$$\begin{aligned} v_{DS}|_B &= v_{GS}|_B - V_T \\ 0.22\ \text{V} &= v_{GS}|_B - 0.5 \\ v_{GS}|_B &= 0.72\ \text{V} \end{aligned}$$

Thus, the x-coordinate of point B is, $\boxed{0.72\ \text{V}}$.

Step 1 of 9

(a)

The resistor R_D is changing and V_{OV} is constant and voltage gain (A_v) is -10.

The Over drive voltage V_{OV} of the amplifier studied in Example 7.1 is $0.2\ \text{V}$

Calculate the value of V_{GS} :

$$V_{OV} = V_{GS} - V_t$$

Substitute $0.2\ \text{V}$ for V_{OV} in equation.

$$\begin{aligned} V_{GS} &= V_{OV} - V_t \\ &= 0.2 + 0.4 \end{aligned}$$

Therefore, the value of V_{GS} is $\boxed{0.6\ \text{V}}$.

Step 2 of 9

Calculate the value of I_D :

$$I_D = \frac{1}{2} k_n' \frac{W}{L} (V_{OV})^2$$

Substitute 10 for $\frac{W}{L}$, $0.4\ \text{mA/V}$ for k_n', $0.2\ \text{V}$ for V_{OV} in equation.

$$I_D = \frac{1}{2}\left(0.4 \times 10^{-3}\right)(10)(0.2)^2$$

Therefore, the drain current is $\boxed{0.08\ \text{mA}}$.

Step 3 of 9

Determine the value of drain resistance:

$$A_v = -k_n V_{OV} R_D$$

$$A_v = -k_n' \frac{W}{L} V_{OV} R_D$$

$$R_D = -\frac{A_v}{k_n' \frac{W}{L} V_{OV}}$$

Substitute 10 for $\frac{W}{L}$, $0.4\ \text{mA/V}$ for k_n', $0.2\ \text{V}$ for V_{OV} and -10 for A_v in equation.

$$\begin{aligned} R_D &= -\left(\frac{-10}{\left(0.4 \times 10^{-3}\right)(10)(0.2)}\right) \\ &= 12.5\ \text{k}\Omega \end{aligned}$$

Therefore, the drain resistance is $\boxed{12.5\ \text{k}\Omega}$.

Step 4 of 9

Calculate the value of V_{DS}.

$$V_{DS} = V_{DD} - I_D R_D$$

Substitute $0.08\ \text{mA}$ for I_D, $1.8\ \text{V}$ for V_{DD} and $12.5\ \text{k}\Omega$ for R_D in equation.

$$\begin{aligned} V_{DS} &= (1.8\ \text{V}) - \left((0.08\ \text{mA})(12.5\ \text{k}\Omega)\right) \\ &= 1.8\ \text{V} - 1\ \text{V} \\ &= 0.8\ \text{V} \end{aligned}$$

Therefore, the drain to source voltage is $\boxed{0.8\ \text{V}}$.

Step 5 of 9

(b)

The resistor, R_D is constant and V_{OV} is changing and voltage gain (A_v) is -10

Drain resistance R_D of the amplifier studied in Example 7.1 is $17.5\ \text{k}\Omega$

Drain resistance of the circuit for case (b) is $\boxed{17.5\ \text{k}\Omega}$

Step 6 of 9

Determine the value of Over drive voltage:

$$A_v = -k_n V_{OV} R_D$$

$$A_v = -k_n' \frac{W}{L} V_{OV} R_D$$

$$-10 = -\left(0.4 \times 10^{-3}\right)(10)(V_{OV})(17.5\ \text{k}\Omega)$$

$$V_{OV} = \frac{10}{\left(4 \times 10^{-3}\right)\left(17.5 \times 10^{3}\right)}$$

$$V_{OV} = 0.143\ \text{V}$$

Step 7 of 9

The threshold voltage V_t is $0.4\ \text{V}$.

Calculate the value of V_{GS}.

$$V_{OV} = V_{GS} - V_t$$

$$0.143\ \text{V} = V_{GS} - 0.4\ \text{V}$$

$$V_{GS} = 0.543\ \text{V}$$

$$V_{GS} \approx 0.54\ \text{V}$$

Therefore, the gate to source voltage is $\boxed{0.54\ \text{V}}$.

Step 8 of 9

Calculate the drain current, I_D.

$$\begin{aligned} I_D &= \frac{1}{2} k_n' \frac{W}{L} (V_{OV})^2 \\ &= \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2 \end{aligned}$$

Substitute 10 for $\frac{W}{L}$, $0.4\ \text{mA/V}$ for k_n', $0.4\ \text{V}$ for V_t and $0.54\ \text{V}$ for V_{GS} in equation.

$$\begin{aligned} I_D &= \frac{1}{2}\left(0.4 \times 10^{-3}\right)(10)(0.54 - 0.4)^2 \\ &= 0.04\ \text{mA} \end{aligned}$$

Therefore, the drain current is $\boxed{0.04\text{mA}}$.

Step 9 of 9

Calculate the value of V_{DS}.

$$V_{DS} = V_{DD} - I_D R_D$$

Substitute $0.04\ \text{mA}$ for I_D, $1.8\ \text{V}$ for V_{DD} and $17.5\ \text{k}\Omega$ for R_D in equation.

$$\begin{aligned} V_{DS} &= (1.8\ \text{V}) - \left((0.04\ \text{mA})(17.5\ \text{k}\Omega)\right) \\ &= 1.8\ \text{V} - 0.7\ \text{V} \\ &= 1.1\ \text{V} \end{aligned}$$

Therefore, the value of V_{DS} is $\boxed{1.1\ \text{V}}$.

Step 1 of 2

7.002P

Point "B" is the point at which transistor is changed from saturation region to triode region.

Relation between v_{DS} and v_{GS} for MOSFET in active region is,

$$v_{DS} = V_{DD} - \frac{1}{2}k_n R_D (v_{GS} - V_T)^2 \quad \text{..... (1)}$$

Voltage at point "B" $v_{DS}|_B$ is $0.5\ \text{V}$

Value of k_n is $5\ \text{mA/V}^2$

Relation between $v_{DS}|_B$ and $v_{GS}|_B$ is,

$$v_{DS}|_B = v_{GS}|_B - V_T \quad \text{..... (2)}$$

Substitute equation (2) in equation (1)

$$v_{DS} = V_{DD} - \frac{1}{2}k_n R_D (v_{GS} - V_T)^2$$

$$(v_{DS}|_B) = V_{DD} - \frac{1}{2}k_n R_D (v_{GS}|_B - V_T)^2$$

$$(v_{DS}|_B) = V_{DD} - \frac{1}{2}k_n R_D (v_{DS}|_B)^2$$

$$0.5\ \text{V} = 5 - \frac{1}{2}\left(5\ \text{mA/V}^2\right)(R_D)(0.5\ \text{V})^2$$

$$0.5\ \text{V} = 5 - \left(0.625\times10^{-3}\ \text{V}\right)R_D$$

$$R_D = \frac{4.5}{0.625\times10^{-3}}\ \Omega$$

$$R_D = 7.2\ \text{k}\Omega$$

Thus, Drain resistance R_D is $\boxed{7.2\ \text{k}\Omega}$

Step 2 of 2

Value of k_n is doubled and hence value of k_n is $10\ \text{mA/V}^2$

$$(v_{DS}|_B) = V_{DD} - \frac{1}{2}k_n R_D (v_{GS}|_B - V_T)^2$$

$$(v_{DS}|_B) = V_{DD} - \frac{1}{2}k_n R_D (v_{DS}|_B)^2$$

$$0.5\ \text{V} = 5 - \frac{1}{2}\left(10\ \text{mA/V}^2\right)(R_D)(0.5\ \text{V})^2$$

$$0.5\ \text{V} = 5 - \left(1.25\times10^{-3}\ \text{V}\right)R_D$$

$$R_D = \frac{4.5}{1.25\times10^{-3}}\ \Omega$$

$$R_D = 3.6\ \text{k}\Omega$$

Thus, Drain resistance for doubling the k_n is $\boxed{3.6\ \text{k}\Omega}$

Step 1 of 3

7.003E

Collector current I_C is $1\ \text{mA}$

For BJT relation between voltage gain A_v, I_C and R_C is,

$$A_v = \frac{-I_C R_C}{V_T}$$

Value of V_T for BJT is $0.025\ \text{V}$

Voltage gain A_v is $-320\ \text{V/V}$

$$A_v = \frac{-I_C R_C}{V_T}$$

$$-320\ \text{V/V} = \frac{-(1\ \text{mA})R_C}{0.025}$$

$$R_C = \frac{320 \times 0.025}{1 \times 10^{-3}}\ \Omega$$

$$R_C = 8\ \text{k}\Omega$$

Thus, Collector resistance R_C for voltage gain of $-320\ \text{V/V}$ is $\boxed{8\ \text{k}\Omega}$

Step 2 of 3

Minimum value of v_{CE} is $0.3\ \text{V}$ below this v_{CE} should not decrease.

Voltage gain A_v is $-\left(\frac{V_{CC} - v_{CE}}{V_T}\right)$

Power supply V_{CC} is $10\ \text{V}$

$$A_v = -\frac{V_{CC} - V_{CE}}{V_T}$$

$$-320\ \text{V/V} = -\frac{10 - V_{CE}}{0.025}$$

$$8 = 10 - V_{CE}$$

$$V_{CE} = 2\ \text{V}$$

Collector emitter voltage at the bias point is $2\ \text{V}$

Transistor is operated in linear region

In order to reduce V_{CE} from $2\ \text{V}$ to $0.3\ \text{V}$ largest negative signal swing allowed at the output is $(2 - 0.3)\ \text{V}$

Thus, largest negative signal swing allowed at the output is $\boxed{1.7\ \text{V}}$

Step 3 of 3

Voltage gain A_v is $\dfrac{\text{output voltage}}{\text{input voltage}}$

Output voltage is $1.7\ \text{V}$

$$A_v = \frac{\text{output voltage}}{\text{input voltage}}$$

$$320 = \frac{1.7}{\text{input voltage}}$$

$$\text{input voltage} = 5.3\ \text{mV}$$

Thus, Input signal amplitude for largest negative signal swing of $1.7\ \text{V}$ is $\boxed{5.3\ \text{mV}}$

Step 1 of 8

Over drive voltage at point "Q" $V_{OV}|_Q$ is $0.2\ \text{V}$

Value of V_t is $0.5\ \text{V}$

Relation between over drive voltage and gate-source voltage at "Q" point is,

$$V_{OV}|_Q = V_{GS}|_Q - V_t$$
$$0.2\ \text{V} = V_{GS}|_Q - 0.5$$
$$V_{GS}|_Q = 0.7\ \text{V}$$

Thus, Gate source voltage at "Q" point is $0.7\ \text{V}$

Step 2 of 8

Drain current i_D for MOSFET is $\frac{1}{2}k_n(V_{OV})^2$

Value of k_n is $10\ \text{mA/V}^2$

$$i_D = \frac{1}{2}k_n(V_{OV})^2$$
$$= \frac{1}{2}\left(10\ \text{mA/V}^2\right)(0.2\ \text{V})^2$$
$$= 0.2\ \text{mA}$$

Thus, Drain current i_D at "Q" point is $0.2\ \text{mA}$

Step 3 of 8

Refer to the figure 7.4(a) in the text book,

Relation between Drain-Source voltage, Drain current (i_D) and drain resistance R_D is,

$$V_{DS} = V_{DD} - I_D R_D$$

Drain –Source voltage V_{DS} values at "Q" point is $1\ \text{V}$

Substitute drain current and V_{DS} values at "Q" point in the above equation

$$V_{DS} = V_{DD} - I_D R_D$$
$$1\ \text{V} = 5 - (0.2\ \text{mA})(R_D)$$
$$R_D = \frac{4}{0.2\ \text{mA}}$$
$$R_D = 20\ \text{k}\Omega$$

Thus, Drain resistance R_D is $\boxed{20\ \text{k}\Omega}$

Step 4 of 8

Point "B" is the point at which transistor is changed from saturation region to triode region.

Relation between v_{DS} and v_{GS} for MOSFET in active region is,

$$v_{DS} = V_{DD} - \frac{1}{2}k_n R_D (v_{GS} - V_T)^2 \quad \text{..... (1)}$$

In order to find boundary point "B" substitute v_{GS} equal to $v_{GS}|_B$ and $v_{DS}|_B = v_{GS}|_B - V_T$ in the equation (1)

$$v_{DS} = V_{DD} - \frac{1}{2}k_n R_D (v_{GS} - V_T)^2$$
$$(v_{GS}|_B - V_T) = V_{DD} - \frac{1}{2}k_n R_D (v_{GS}|_B - V_T)^2$$

Solve for $v_{GS}|_B - V_T$

$$(v_{GS}|_B - V_T) = \frac{\sqrt{2k_n R_D V_{DD} + 1} - 1}{k_n R_D} \quad \text{..... (2)}$$

Step 5 of 8

Substitute k_n, R_D and V_{DD} in equation (2)

$$(v_{GS}|_B - V_T) = \frac{\sqrt{2k_n R_D V_{DD} + 1} - 1}{k_n R_D}$$
$$= \frac{\sqrt{2(10\times10^{-3}\ \text{A/V}^2)(20\times10^3\ \Omega)(5\ \text{V}) + 1} - 1}{(10\times10^{-3}\ \text{A/V}^2)(20\times10^3\ \Omega)}$$
$$= \frac{43.73}{200}$$
$$= 0.22\ \text{V}$$

Value of $(v_{GS}|_B - V_T)$ is $0.22\ \text{V}$ and hence $v_{DS}|_B$ is equal to $\boxed{0.22\ \text{V}}$

Step 6 of 8

Relation between $v_{DS}|_B$ and $v_{GS}|_B$ is,

$$v_{DS}|_B = v_{GS}|_B - V_T$$
$$0.22\ \text{V} = v_{GS}|_B - 0.5$$
$$v_{GS}|_B = 0.72\ \text{V}$$

Gate source voltage at point "B" $v_{GS}|_B$ is $\boxed{0.72\ \text{V}}$

X-coordinate of the point "B" $v_{GS}|_B$ is $\boxed{0.72\ \text{V}}$

Y-coordinate of the point "B" $v_{DS}|_B$ is $\boxed{0.22\ \text{V}}$

Step 7 of 8

Voltage gain (A_v) of the MOS amplifier is $-k_n V_{OV} R_D$

$$A_v = -k_n V_{OV} R_D$$
$$= -\left(10\ \text{mA/V}^2\right)(0.2\ \text{V})(20\ \text{k}\Omega)$$
$$= -40$$

Small signal voltage gain of the amplifier (A_v) is $\boxed{-40}$

Maximum allowable negative signal swing at the output in order to keep the transistor in active region is difference between the drain source voltage at "Q" and "B" points.

$v_{DS}|_B$ is $0.22\ \text{V}$

$v_{DS}|_Q$ is $1\ \text{V}$

Maximum allowable negative signal swing at the output is $\boxed{0.78\ \text{V}}$

Step 8 of 8

Voltage gain A_v is $\frac{\text{output voltage}}{\text{input voltage}}$

Output swing voltage is $0.78\ \text{V}$

$$A_v = \frac{\text{output voltage}}{\text{input voltage}}$$
$$40 = \frac{0.78}{\text{input voltage}}$$
$$\text{input voltage} = 19.5\ \text{mV}$$

Peak input signal is $\boxed{19.5\ \text{mV}}$

Step 1 of 8

7.004E

Refer to the circuit diagram shown in figure 7.15(a) in the textbook.

Consider the circuit without load resistor R_L and with neglected channel length modulation. Hence, R_L and r_0 are negligible.

Step 2 of 8

The small signal equivalent circuit for the given circuit is shown in Figure 1.

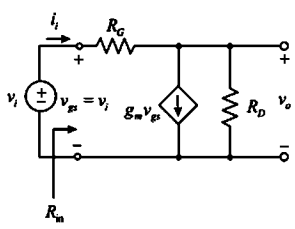


Figure 1

Step 3 of 8

Write the input current expression from figure 1.

$$i_i = \frac{v_i - v_o}{R_G} \quad \text{..... (1)}$$

Write the formula for input resistance,

$$R_{in} = \frac{v_i}{i_i} \quad \text{..... (2)}$$

Substitute equation (1) in equation (2).

$$R_{in} = \frac{v_i}{\left(\frac{v_i - v_o}{R_G}\right)}$$

$$R_{in} = \frac{v_i}{(v_i - v_o)} R_G$$

$$R_{in} = \frac{1}{\left(1 - \frac{v_o}{v_i}\right)} R_G$$

$$R_{in} = \frac{1}{(1 - A_v)} R_G$$

$$R_G = (1 - A_v) R_{in}$$

Substitute -25 V/V for A_v and 0.5 MΩ for R_{in} in equation.

$$R_G = (1 - (-25))(0.5 \times 10^6)$$

$$R_G = 13 \text{ MΩ}$$

Hence, the value of gate resistance is $\boxed{13 \text{ MΩ}}$.

Step 4 of 8

Overdrive voltage is defined as,

$$V_{OV} = V_{GS} - V_t \quad \text{..... (3)}$$

From the circuit, V_{GS} is,

$$V_{GS} = V_{DD} - I_D R_D \quad \text{..... (4)}$$

Substitute equation (4) in equation (3).

$$V_{OV} = V_{DD} - I_D R_D - V_t$$

Substitute 0.7 V for V_t . 5 V for V_{DD} in equation.

$$V_{OV} = 5 - 12.5 V_{OV} - 0.7$$

$$13.5 V_{OV} = 4.3$$

$$V_{OV} = \frac{4.3}{13.5}$$

$$V_{OV} = 0.3185 \text{ V}$$

Therefore, the value of V_{OV} is $\boxed{0.319 \text{ V}}$.

Step 5 of 8

The trans-conductance is,

$$g_m = k_n V_{OV}$$
$$= (1 \times 10^{-3})(0.3185)$$
$$= 318.5 \text{ μA/V}$$

Step 6 of 8

Drain resistance can be calculated as follows:

$$A_v = -g_m R_D$$

$$-25 = -(318.5 \times 10^{-6}) R_D$$

$$R_D = \frac{25}{318.5 \times 10^{-6}}$$

$$R_D = 78.5 \text{ kΩ}$$

Thus, the value of R_D is $\boxed{78.5 \text{ kΩ}}$.

Step 7 of 8

Drain current is,

$$I_D = \frac{1}{2} k_n V_{OV}^2$$

Substitute 1 mA/V^2 for k_n . 0.319 V for V_{OV} in the drain current expression.

$$I_D = \frac{1}{2} k_n V_{OV}^2$$
$$= \frac{1}{2}(1 \times 10^{-3} \text{ A/V}^2)(0.319 \text{ V})^2$$
$$= 50.9 \text{ μA}$$

Thus, Drain current I_D is $\boxed{50.9 \text{ μA}}$

Step 8 of 8

The maximum allowable input signal is

$$\hat{v}_i = \frac{V_t}{|A_v| + 1}$$
$$= \frac{0.7}{|-25| + 1}$$
$$= \frac{0.7}{26}$$
$$= 27 \text{ mV}$$

Hence the maximum allowable input signal is $\boxed{\hat{v}_i = 27 \text{ mV}}$.

7.004P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

7.005E

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

Step 1 of 13

(a)

End points of the active-region segment of the voltage –transfer characteristics are point "A" and point "B".

Point "A" is the point at which transistor is changed from cutoff region to saturation region.

$v_{GS} < V_t$ MOSFET will be in cutoff region. For $v_{GS} \geq V_t$ transistor enter into saturation region and hence $v_{GS} = V_t$ is the x-co-ordinate of the point "A".

Gate-source voltage of the point "A" is V_t

Threshold voltage of MOSFET is $1\ \text{V}$

Gate-source voltage of the point "A" is $\boxed{1\ \text{V}}$

Step 2 of 13

Refer to the figure 7.4 (a) in the text book,

$$v_{DS} = V_{DD} - i_D R_D$$

For MOSFET in cutoff $i_D = 0$ and hence $v_{DS} = V_{DD}$

Hence y-coordinate of the point "A" is V_{DD}

Value of V_{DD} is $5\ \text{V}$

Hence y-coordinate of the point "A" is $\boxed{5\ \text{V}}$

X and Y co-ordinates of the point "A" is $1\ \text{V}$ and $5\ \text{V}$ respectively.

Step 3 of 13

Point "B" is the point at which transistor is changed from saturation region to triode region.

Relation between v_{DS} and v_{GS} for MOSFET in active region is,

$$v_{DS} = V_{DD} - \frac{1}{2} k_n R_D (v_{GS} - V_t)^2 \quad \text{..... (1)}$$

Drain resistance R_D is $24\ \text{k}\Omega$

Value of $k_n' \left(W/L\right)$ is $1\ \text{mA/V}^2$

$k_n' \left(W/L\right) = k_n$ and hence k_n is equal to $1\ \text{mA/V}^2$

In order to find boundary point "B" substitute v_{GS} equal to $v_{GS}|_B$ and $v_{DS}|_B = v_{GS}|_B - V_t$ in the equation (1)

$$v_{DS} = V_{DD} - \frac{1}{2} k_n R_D (v_{GS} - V_t)^2$$

$$\left(v_{GS}|_B - V_t\right) = V_{DD} - \frac{1}{2} k_n R_D \left(v_{GS}|_B - V_t\right)^2$$

Solve for $v_{GS}|_B - V_t$

$$\left(v_{GS}|_B - V_t\right) = \frac{\sqrt{2k_n R_D V_{DD} + 1} - 1}{k_n R_D} \quad \text{..... (2)}$$

Step 4 of 13

Substitute k_n, R_D and V_{DD} in equation (2)

$$\begin{aligned}\left(v_{GS}|_B - V_t\right) &= \frac{\sqrt{2k_n R_D V_{DD} + 1} - 1}{k_n R_D} \\ &= \frac{\sqrt{2\left(1\times10^{-3}\ \text{A/V}^2\right)\left(24\times10^3\ \Omega\right)(5\ \text{V}) + 1} - 1}{\left(1\times10^{-3}\ \text{A/V}^2\right)\left(24\times10^3\ \Omega\right)} \\ &= \frac{14.52}{24} \\ &= 0.605\ \text{V}\end{aligned}$$

Thus, Value of $\left(v_{GS}|_B - V_t\right)$ is $0.605\ \text{V}$ and hence $v_{DS}|_B$ is equal to $\boxed{0.605\ \text{V}}$

Step 5 of 13

Relation between $v_{DS}|_B$ and $v_{GS}|_B$ is,

$$v_{DS}|_B = v_{GS}|_B - V_t$$
$$0.605\ \text{V} = v_{GS}|_B - 1$$
$$v_{GS}|_B = 1.605\ \text{V}$$

Thus, Gate source voltage at point "B" $v_{GS}|_B$ is $\boxed{1.605\ \text{V}}$

Step 6 of 13

(b)

Amplifier is biased with an Over drive voltage V_{OV} $0.5\ \text{V}$

Relation between over drive voltage and gate-source voltage is $V_{OV} = V_{GS} - V_t$

$$V_{OV} = V_{GS} - V_t$$
$$0.5\ \text{V} = V_{GS} - (1\text{V})$$
$$V_{GS} = 1.5\ \text{V}$$

Gate –Source voltage at point "Q" is $V_{GS}|_Q$ is $1.5\ \text{V}$

Step 7 of 13

For $V_{GS}|_Q$ equal to $1.5\ \text{V}$ MOSFET will operate in saturation mode.

Relation between v_{DS} and v_{GS} for MOSFET in saturation region is,

$$v_{DS} = V_{DD} - \frac{1}{2} k_n R_D (v_{GS} - V_t)^2$$

Relation between v_{DS} and v_{GS} for MOSFET at "Q" point is,

$$v_{DS}|_Q = V_{DD} - \frac{1}{2} k_n R_D \left(v_{GS}|_Q - V_t\right)^2$$

Step 8 of 13

Substitute $V_{GS}|_Q$, V_t in the above equation

$$\begin{aligned}v_{DS}|_Q &= V_{DD} - \frac{1}{2} k_n R_D \left(v_{GS}|_Q - V_t\right)^2 \\ &= (5\ \text{V}) - \frac{1}{2}\left(1\times10^{-3}\ \frac{\text{A}}{\text{V}^2}\right)\left(24\times10^3\ \Omega\right)(1.5-1)^2 \\ &= (5-3)\ \text{V} \\ &= 2\ \text{V}\end{aligned}$$

Drain-Source voltage at point "Q" $v_{DS}|_Q$ is $2\ \text{V}$

X-coordinate of point "Q" $v_{GS}|_Q$ is $\boxed{1.5\ \text{V}}$

Y-coordinate of point "Q" $v_{DS}|_Q$ is $\boxed{2\ \text{V}}$

Step 9 of 13

Drain current I_D is $\frac{1}{2} k_n V_{OV}^2$

Drain current I_D at "Q" point is,

$$\begin{aligned}I_D &= \frac{1}{2} k_n V_{OV}^2 \\ &= \frac{1}{2}\left(1\times10^{-3}\ \frac{\text{A}}{\text{V}^2}\right)(0.5)^2 \\ &= 0.125\ \text{mA}\end{aligned}$$

Thus, Drain current I_D at "Q" point is $\boxed{0.125\ \text{mA}}$

Step 10 of 13

Voltage gain A_v is $-\frac{I_D R_D}{V_{OV}/2}$

Voltage gain at "Q" point is,

$$\begin{aligned}A_v &= -\frac{I_D R_D}{V_{OV}/2} \\ &= -\frac{(0.125\ \text{mA})\left(24\times10^3\ \Omega\right)}{0.5/2} \\ &= -12\end{aligned}$$

Voltage gain at "Q" point is -12

Step 11 of 13

(c)

Gate source voltage at point "Q" is $1.5\ \text{V}$

Gate source voltage at point "B" is $1.605\ \text{V}$

Difference between gate-source voltages of "Q" and "B" will give the largest amplitude of sine wave voltage applied to the input in order to keep the transistor in saturation.

Largest amplitude of sine wave voltage applied to the input $\hat{v}_{gs}$ is $\boxed{0.105\ \text{V}}$

Amplitude of output is obtained by multiplying the largest amplitude of input signal with voltage gain at point "Q"

$$\begin{aligned}\hat{v}_{ds} &= |A_v| \hat{v}_{gs} \\ &= 12(0.105) \\ &= 1.26\ \text{V}\end{aligned}$$

Largest amplitude of output $\hat{v}_{ds}$ is $\boxed{1.26\ \text{V}}$

$$\begin{aligned}v_{DS\max} &= V_{DS} + \hat{v}_{gs} \\ &= 2\ \text{V} + 1.26\ \text{V} \\ &= 3.26\ \text{V}\end{aligned}$$

Maximum drain source voltage $v_{DS\max}$ is $\boxed{3.26\ \text{V}}$

Step 12 of 13

Over drive voltage V_{OV} at point "B" is $0.605\ \text{V}$

Drain current at point "B" is,

$$\begin{aligned}I_D &= \frac{1}{2} k_n V_{OV}^2 \\ &= \frac{1}{2}\left(1\times10^{-3}\ \frac{\text{A}}{\text{V}^2}\right)(0.605)^2 \\ &= 0.183\ \text{mA}\end{aligned}$$

Drain current I_D at "B" point is $\boxed{0.183\ \text{mA}}$

Step 13 of 13

Voltage gain at "B" point is,

$$\begin{aligned}A_v &= -\frac{I_D R_D}{V_{OV}/2} \\ &= \frac{(0.183\ \text{mA})\left(24\times10^3\ \Omega\right)}{0.605/2} \\ &= 14.52\end{aligned}$$

Voltage gain at "B" point is 14.52

Percentage voltage gain difference between "Q" and "B" point is $\left(\frac{A_v|_Q - A_v|_B}{A_v|_B}\right)\times100$

$$\left(\frac{A_v|_Q - A_v|_B}{A_v|_B}\right)\times100 = \left(\frac{14.52-12}{12}\right)\times100$$
$$= \frac{2.52}{12}\times100$$
$$= 21\%$$

Percentage voltage gain difference between "Q" and "B" point is $\boxed{21\%}$

Step 1 of 9

7.006E

Refer to Figure 7.4 in the textbook

(a)

Determine the dc current I_D and dc voltage V_{DS}.

To find the dc parameters put $v_{gs} = 0$.

Dc drain current I_D is,

$$I_D = \frac{1}{2}k_n'(W/L)(V_{GS} - V_t)^2 \quad \text{..... (1)}$$

Here,

V_t is threshold voltage,

V_{GS} is gate source voltage,

k_n' is transconductance ,

(W/L) is width to length ratio.

Substitute $1\ \text{V}$ for threshold voltage , $2\ \text{V}$ for V_{GS} , $20\ \mu\text{A/V}^2$ for k_n' and 20 for (W/L) in the equation (1)

$$I_D = \frac{1}{2}\left(20\times10^{-6}\ \text{A/V}^2\right)(20)(2\ \text{V} - 1\ \text{V})^2$$
$$= 200\ \mu\text{A}$$

Thus, Drain current I_D is $\boxed{200\ \mu\text{A}}$

Step 2 of 9

Apply Kirchhoff's voltage law to the drain loop.

$$-V_{DD} + I_D R_D + V_{DS} = 0$$

$$V_{DS} = V_{DD} - I_D R_D$$

Substitute $10\ \text{k}\Omega$ for drain resistance R_D , $5\ \text{V}$ for power voltage V_{DD} and $200\ \mu\text{A}$ for drain current I_D in the above equation.

$$V_{DS} = V_{DD} - I_D R_D$$
$$= 5\ \text{V} - (200\ \mu\text{A})(10\ \text{k}\Omega)$$
$$= 5 - 2$$
$$= 3\ \text{V}$$

Thus, Drain source voltage V_{DS} is $\boxed{3\ \text{V}}$

Step 3 of 9

(b)

The trans-conductance g_m is,

$$g_m = \frac{2I_D}{(V_{GS} - V_t)}$$

Substitute $1\ \text{V}$ for V_t , $2\ \text{V}$ for V_{GS} and $200\ \mu\text{A}$ for I_D in the above equation.

$$g_m = \frac{2(200\ \mu\text{A})}{(2\ \text{V} - 1\ \text{V})}$$
$$= \frac{2\left(200\times10^{-6}\right)}{1}$$
$$= 0.4\ \text{mA/V}$$

Thus, trans-conductance g_m at the bias point is $\boxed{0.4\ \text{mA/V}}$.

Step 4 of 9

(c)

The voltage gain A_v is,

$$A_v = -g_m R_D$$

Substitute $10\ \text{k}\Omega$ for R_D , $0.4\ \text{mA/V}$ for g_m in the above equation

$$A_v = -\left(0.4\ \text{mA/V}^2\right)(10\ \text{k}\Omega)$$
$$= -4\ \text{V/V}$$

Therefore, the voltage gain A_v is $\boxed{-4\ \text{V/V}}$.

Step 5 of 9

(d)

Small signal approximation holds good. Hence, relation between A_v , v_{ds} and v_{gs} is, $A_v = \frac{v_{ds}}{v_{gs}}$

Consider gate-Source signal v_{gs} is $0.2\sin\omega t\ \text{V}$.

Substitute $-4\ \text{V}$ for A_v and $0.2\sin\omega t\ \text{V}$ for v_{gs} in the equation.

$$-4\ \text{V/V} = \frac{v_{ds}}{0.2\sin\omega t}$$
$$v_{ds} = -4(0.2\sin\omega t)$$
$$v_{ds} = -0.8\sin\omega t$$

Hence, small signal voltage v_{ds} is $\boxed{-0.8\sin\omega t\ \text{V}}$.

Step 6 of 9

Minimum and maximum values obtain by combining ac and dc values at the same time.
Minimum value of the v_{DS} is,

$$v_{DS(\min)} = 3 - 0.8\sin\omega t$$
$$= 3 - 0.8(1)$$
$$= 3 - 0.8$$
$$= 2.2\ \text{V}$$

Step 7 of 9

Maximum value of the v_{DS} is,

$$v_{DS(\max)} = 3 + 0.8\sin\omega t$$
$$= 3 + 0.8(1)$$
$$= 3 + 0.8$$
$$= 3.8\ \text{V}$$

Therefore, the minimum and maximum values of drain to source voltage are $\boxed{2.2\ \text{V}}$ and $\boxed{3.8\ \text{V}}$.

Step 8 of 9

(e)

Consider the situation with the input signal v_{gs} applied. The total instantaneous gate-to-source voltage will be:

$$v_{GS} = V_{GS} + v_{gs}$$

Instantaneous drain current i_D from equation (7.28) in the textbook is,

$$\frac{1}{2}k_n(V_{GS} - V_t)^2 + k_n(V_{GS} - V_t)v_{gs} + \frac{1}{2}k_n v_{gs}^2 \quad \text{..... (2)}$$

Substitute $\left(20\times10^{-6}\right)(20)\ \mu\text{A/V}^2$ for k_n , $2\ \text{V}$ for V_{GS} , $1\ \text{V}$ for V_t and $0.2\sin\omega t\ \text{V}$ for v_{gs} in equation (2).

$$i_D = \frac{1}{2}\left(400\ \mu\text{A/V}^2\right)\left[(2-1)^2 + 2(2-1)0.2\sin\omega t + (0.2)^2\sin^2(\omega t)\right]$$
$$= \frac{1}{2}\left(400\ \mu\text{A/V}^2\right)\left[(2-1)^2 + 2(2-1)0.2\sin\omega t + (0.2)^2\left[\frac{1}{2} - \frac{1}{2}\cos 2\omega t\right]\right]$$
$$= 200\left[1 + 0.4\sin\omega t + 0.04\left[\frac{1}{2} - \frac{1}{2}\cos 2\omega t\right]\right]$$
$$= 200 + 80\sin\omega t + 8\left[\frac{1}{2} - \frac{1}{2}\cos 2\omega t\right]$$
$$= (204 + 80\sin\omega t - 4\cos 2\omega t)\ \mu\text{A}$$

Therefore, various component drain current is $\boxed{(204 + 80\sin\omega t - 4\cos 2\omega t)\ \mu\text{A}}$.

Step 9 of 9

Consider the drain current equation is,

$$i_D = (204 + 80\sin\omega t - 4\cos 2\omega t)\,\mu\text{A}$$

In the above equation amplitude of second harmonic is $4\ \mu\text{A}$.

Amplitude of fundament component is $80\ \mu\text{A}$.

Second harmonics amplitude percentage with respect to the fundamental component is, $= \frac{4\times10^{-6}}{80\times10^{-6}}\times100$

$$= \frac{4\times10^{-6}}{80\times10^{-6}}\times100$$
$$= 5\%$$

Therefore second harmonics amplitude percentage with respect to the fundamental component is $\boxed{5\%}$.

Step 1 of 3

Voltage across the drain resistance (V_{RD}) is $1.5\ \text{V}$

Voltage gain A_v of NMOS amplifier is $-\dfrac{2(V_{DD}-V_{DS})}{V_{OV}}$

$(V_{DD}-V_{DS})$ is the voltage across the drain resistor R_D and hence A_v is $-\dfrac{2V_{RD}}{V_{OV}}$

Over drive voltage V_{OV} is $(V_{GS}-V_t)$

Gate-Source voltage V_{GS} is $0.7\ \text{V}$

$$A_v=-\frac{2V_{RD}}{V_{GS}-V_t}$$
$$-10=-\frac{2(1.5\ \text{V})}{0.7-V_t}$$
$$7-10V_t=3$$
$$V_t=0.4\ \text{V}$$

Thus, Value of V_t for the NMOS transistor is $\boxed{0.4\ \text{V}}$

Step 2 of 3

Drain resistance R_D is $20\ \text{k}\Omega$

Voltage gain for NMOS in another form is $A_v=-k_nV_{OV}R_D$

$$A_v=-k_nV_{OV}R_D$$
$$-10=-k_n(0.3)(20\ \text{k}\Omega)$$
$$k_n=\frac{10}{0.3(20\times10^3)}$$
$$k_n=1.67\ \frac{\text{mA}}{\text{V}^2}$$

Thus value of k_n is $1.67\ \dfrac{\text{mA}}{\text{V}^2}$

Step 3 of 3

Relation between k_n and transconductance parameter k_n' is $k_n=k_n'\left(W/L\right)$

Transconductance parameter k_n' is $200\ \dfrac{\mu\text{A}}{\text{V}^2}$

$$k_n=k_n'\left(W/L\right)$$
$$1.67\ \frac{\text{mA}}{\text{V}^2}=\left(200\ \frac{\mu\text{A}}{\text{V}^2}\right)\left(W/L\right)$$
$$\left(W/L\right)=\frac{1.67\times10^{-3}\ \frac{\text{A}}{\text{V}^2}}{200\times10^{-6}\ \frac{\text{A}}{\text{V}^2}}$$
$$\left(W/L\right)=8.35$$

MOSFET $\left(W/L\right)$ value is $\boxed{8.35}$

Step 1 of 6

(a)

The drain current, i_D for NMOS is,

$$i_D = \frac{1}{2}k_n' \frac{W}{L}(V_{GS} - V_t)^2$$

Here,

k_n' is process trans-conductance parameter,

$\frac{W}{L}$ is width to length ratio of the channel,

V_{GS} is gate-to-source voltage,

V_t is threshold voltage.

Substitute $1\ \text{V}$ for V_t, $1.5\ \text{V}$ for V_{GS}, $60\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$, 40 for W/L in equation.

$$\begin{aligned} I_D &= \frac{1}{2}(\mu_n C_{ox})(W/L)(V_{GS} - V_t)^2 \\ &= \frac{1}{2}\left(60\times10^{-6}\ \text{A/V}^2\right)(40)(1.5\ \text{V} - 1\ \text{V})^2 \\ &= \frac{1}{2}\left(60\times10^{-6}\right)(40)(0.5)^2 \\ &= 300\ \mu\text{A} \end{aligned}$$

Thus, Drain current I_D is $\boxed{300\ \mu\text{A}}$.

Step 2 of 6

The trans-conductance g_m is,

$$g_m = \frac{2I_D}{(V_{GS} - V_t)}$$

Substitute $1\ \text{V}$ for V_t, $1.5\ \text{V}$ for V_{GS} and $300\ \mu\text{A}$ for I_D in the above equation.

$$\begin{aligned} g_m &= \frac{2(300\ \mu\text{A})}{(1.5\ \text{V} - 1\ \text{V})} \\ &= \frac{2\left(300\times10^{-6}\right)}{0.5} \\ &= 1.2\ \text{mA/V} \end{aligned}$$

Hence, trans-conductance g_m at the bias point is $\boxed{1.2\ \text{mA/V}}$.

Step 3 of 6

Determine the finite output resistance r_o.

The output resistance is,

$$r_0 = \frac{|V_A|}{I_D}$$

Substitute $15\ \text{V}$ for early voltage V_A, $300\ \mu\text{A}$ for I_D in the above expression.

$$\begin{aligned} r_0 &= \frac{15\ \text{V}}{300\ \mu\text{A}} \\ &= 50\ \text{k}\Omega \end{aligned}$$

Hence, Finite output resistance r_o between drain and source is $\boxed{50\ \text{k}\Omega}$.

Step 4 of 6

(b)

Determine the gate to source voltage V_{GS}.

The drain current, i_D is,

$$i_D = \frac{1}{2}k_n' \frac{W}{L}(V_{GS} - V_t)^2$$

Substitute $1\ \text{V}$ for V_t, $0.5\ \text{mA}$ for I_D, $60\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$, 40 for W/L in equation.

$$\begin{aligned} I_D &= \frac{1}{2}(\mu_n C_{ox})(W/L)(V_{GS} - V_t)^2 \\ \left(0.5\times10^{-3}\right)\text{A} &= \frac{1}{2}\left(60\times10^{-6}\ \text{A/V}^2\right)(40)(V_{GS} - 1)^2 \\ (V_{GS} - 1)^2 &= 0.416 \\ V_{GS} &= 1.645\ \text{V} \end{aligned}$$

Step 5 of 6

The trans-conductance g_m is,

$$g_m = \frac{2I_D}{(V_{GS} - V_t)}$$

Substitute $1\ \text{V}$ for V_t, $1.645\ \text{V}$ for V_{GS} and $0.5\ \text{mA}$ for I_D in the above equation.

$$\begin{aligned} g_m &= \frac{2(0.5\ \text{mA})}{(1.645\ \text{V} - 1\ \text{V})} \\ &= \frac{2\left(0.5\times10^{-3}\right)}{0.645} \\ &= 1.55\ \text{mA/V} \end{aligned}$$

Hence, trans-conductance g_m at the bias point is $\boxed{1.55\ \text{mA/V}}$.

Step 6 of 6

Determine the finite output resistance r_o.

The output resistance is,

$$r_0 = \frac{|V_A|}{I_D}$$

Substitute $15\ \text{V}$ for early voltage V_A, $0.5\ \text{mA}$ for I_D in the above expression

$$\begin{aligned} r_0 &= \frac{|V_A|}{I_D} \\ &= \frac{15\ \text{V}}{0.5\ \text{mA}} \\ &= 30\ \text{k}\Omega \end{aligned}$$

Hence, finite output resistance r_o between drain and source is $\boxed{30\ \text{k}\Omega}$.

Step 1 of 17

V_{DS} is greater than V_{OV} ,the transistor indeed operate in saturation.

Gain is negative for MOS amplifier and hence there is a 180° phase shift between the input and output signal and hence for positive peak of input signal we get negative peak output signal.

MOSFET to remain in saturation at the negative peak drain source voltage, we must ensure that

$$v_{DS\min} \geq v_{GS\min} - V_t$$

Largest allowable output signal swing in the negative direction is $\hat{v}_o$ and hence $v_{DS\min}$ is equal to $V_{DS} - \hat{v}_o$

Step 2 of 17

MOSFET circuit is operated in linear operation around the bias point.

Gate-Source signal applied to the circuit in terms of output signal $\hat{v}_o$ is $\frac{\hat{v}_o}{|A_v|}$ and hence $v_{GS\min}$ value is

$V_{GS} + \frac{\hat{v}_o}{|A_v|}$.

Substitute $v_{GS\min}$ and $v_{DS\min}$ in the equation $v_{DS\min} \geq v_{GS\min} - V_t$

$$V_{DS} - \hat{v}_o \geq \left(V_{GS} + \frac{\hat{v}_o}{|A_v|}\right) - V_t$$

$$V_{DS} - \hat{v}_o \geq \left(V_{GS} - V_t + \frac{\hat{v}_o}{|A_v|}\right)$$

$$V_{DS} - \hat{v}_o \geq \left(V_{OV} + \frac{\hat{v}_o}{|A_v|}\right)$$

$$-\frac{\hat{v}_o}{|A_v|} - \hat{v}_o \geq V_{OV} - V_{DS}$$

Multiply above equation with negative sign.

$$\frac{\hat{v}_o}{|A_v|} + \hat{v}_o \leq -V_{OV} + V_{DS}$$

$$\left(1 + \frac{1}{|A_v|}\right)\hat{v}_o \leq V_{DS} - V_{OV}$$

$$\hat{v}_o = (V_{DS} - V_{OV}) \Big/ \left(1 + \frac{1}{|A_v|}\right)$$

Thus, the largest possible negative output signal peak $\hat{v}_O = \frac{V_{DS} - V_{OV}}{1 + \frac{1}{|A_v|}}$ is proved.

Step 3 of 17

For the bias voltage at the drain terminal, $V_{DS} = 1\ \text{V}$:

Write the formula for the voltage gain of the amplifier.

$$A_v = -\frac{2(V_{DD} - V_{DS})}{V_{OV}}$$

Substitute 5 V for V_{DD} , 1 V for V_{DS} , and 0.5 V for V_{OV} .

$$A_v = -2\frac{(5-1)\ \text{V}}{0.5\ \text{V}}$$
$$= -16$$

Thus, the voltage gain A_v for V_{DS} equal to $1\ \text{V}$ is $\boxed{-16}$

Step 4 of 17

The largest possible negative output signal peak is,

$$\hat{v}_o = (V_{DS} - V_{OV}) \Big/ \left(1 + \frac{1}{|A_v|}\right)$$

$$\hat{v}_o = (V_{DS} - V_{OV}) \Big/ \left(1 + \frac{1}{|A_v|}\right)$$
$$= (1\ \text{V} - 0.5\ \text{V}) \Big/ 1 + \frac{1}{|16|}$$
$$= 0.5/1.0625$$
$$= 0.47\ \text{V}$$

Thus, Negative output signal peak $\hat{v}_o$ that is achievable while the transistor remains saturated is $\boxed{0.47\ \text{V}}$

Step 5 of 17

The relation between input signal and output signal is,

$$\hat{v}_i = \frac{\hat{v}_o}{A_v}$$
$$= \frac{0.47\ \text{V}}{16}$$
$$= 29.375\ \text{mV}$$

Thus, the input signal peak, $\hat{v}_i$ is $29.375\ \text{mV}$

Step 6 of 17

For the bias voltage at the drain terminal, $V_{DS} = 1.5\ \text{V}$:
Calculate the voltage gain of the amplifier.

$$A_v = -\frac{2(V_{DD} - V_{DS})}{V_{OV}}$$
$$= -2\frac{(5-1.5)\ \text{V}}{0.5\ \text{V}}$$
$$= -14$$

Thus, the voltage gain A_v for V_{DS} equal to $1.5\ \text{V}$ is $\boxed{-14}$

Step 7 of 17

Calculate the output signal peak.

$$\hat{v}_o = (V_{DS} - V_{OV}) \Big/ \left(1 + \frac{1}{|A_v|}\right)$$

$$\hat{v}_o = (V_{DS} - V_{OV}) \Big/ \left(1 + \frac{1}{|A_v|}\right)$$
$$= (1.5\ \text{V} - 0.5\ \text{V}) \Big/ 1 + \frac{1}{|14|}$$
$$= 1/1.071$$
$$= 0.93\ \text{V}$$

Thus, Negative output signal peak $\hat{v}_o$ that is achievable while the transistor remains saturated is $\boxed{0.93\ \text{V}}$

Step 8 of 17

The relation between input signal and output signal is,

$$\hat{v}_i = \frac{\hat{v}_o}{A_v}$$
$$= \frac{0.93\ \text{V}}{14}$$
$$= 66.44\ \text{mV}$$

Thus, the input signal peak, $\hat{v}_i$ is $66.44\ \text{mV}$

Step 9 of 17

For the bias voltage at the drain terminal, $V_{DS} = 2\ \text{V}$:

Calculate the voltage gain of the amplifier.

$$A_v = -\frac{2(V_{DD} - V_{DS})}{V_{OV}}$$
$$= -2\frac{(5-2)\ \text{V}}{0.5\ \text{V}}$$
$$= -12$$

Thus, the voltage gain A_v for V_{DS} equal to $2\ \text{V}$ is $\boxed{-12}$.

Step 10 of 17

Calculate the output signal peak.

$$\hat{v}_o = (V_{DS} - V_{OV}) \Big/ \left(1 + \frac{1}{|A_v|}\right)$$

$$\hat{v}_o = (V_{DS} - V_{OV}) \Big/ \left(1 + \frac{1}{|A_v|}\right)$$
$$= (2\ \text{V} - 0.5\ \text{V}) \Big/ 1 + \frac{1}{|12|}$$
$$= 1.5/1.083$$
$$= 1.385\ \text{V}$$

Thus, the negative output signal peak $\hat{v}_o$ that is achievable while the transistor remains saturated is $\boxed{1.385\ \text{V}}$.

Step 11 of 17

The relation between input signal and output signal is,

$$\hat{v}_i = \frac{\hat{v}_o}{A_v}$$
$$= \frac{1.385\ \text{V}}{12}$$
$$= 115.42\ \text{mV}$$

Thus, the input signal $\hat{v}_i$ is $115.42\ \text{mV}$.

Step 12 of 17

For the bias voltage at the drain terminal, $V_{DS} = 2.5\ \text{V}$:

Calculate the voltage gain of the amplifier.

$$A_v = -\frac{2(V_{DD} - V_{DS})}{V_{OV}}$$
$$= -2\frac{(5-2.5)\ \text{V}}{0.5\ \text{V}}$$
$$= -10$$

Thus, the voltage gain A_v for V_{DS} equal to $2.5\ \text{V}$ is $\boxed{-10}$.

Step 13 of 17

Calculate the output signal peak.

$$\hat{v}_o = (V_{DS} - V_{OV}) \Big/ \left(1 + \frac{1}{|A_v|}\right)$$

$$\hat{v}_o = (V_{DS} - V_{OV}) \Big/ \left(1 + \frac{1}{|A_v|}\right)$$
$$= (2.5\ \text{V} - 0.5\ \text{V}) \Big/ 1 + \frac{1}{|10|}$$
$$= 2/1.1$$
$$= 1.818\ \text{V}$$

Thus, the negative output signal peak $\hat{v}_o$ that is achievable while the transistor remains saturated is $\boxed{1.818\ \text{V}}$

Step 14 of 17

The relation between input signal and output signal is,

$$\hat{v}_i = \frac{\hat{v}_o}{A_v}$$
$$= \frac{1.818\ \text{V}}{10}$$
$$= 181.8\ \text{mV}$$

Thus, the Input signal $\hat{v}_i$ is $181.8\ \text{mV}$

Step 15 of 17

The tabulation of A_v, $\hat{v}_o$ and the corresponding v_i values for $V_{DS} = 1, 1.5, 2, \text{and } 2.5\ \text{V}$ is shown in Table 1.

Table 1

V_{DS}	A_v	v_o	v_i
1.0 V	−16	0.47	29.375mV
1.5 V	−14	0.93 V	66.44mV
2.0 V	−12	1.385 V	115.42mV
2.5 V	−10	1.818 V	181.8mV

Tabular form for different values of V_{DS} is constructed.

Step 16 of 17

The value of $k_n'(W/L)$ is $1\ \text{mA/V}^2$.

The voltage gain A_v for V_{DS} equal to $1\ \text{V}$ is -16

Calculate the drain resistance.

$$A_v = -k_n'(W/L)V_{OV}R_D$$
$$-16 = -\left(1\ \text{mA/V}^2\right)(0.5\ \text{V})R_D$$
$$R_D = \frac{16}{\left(1\ \text{mA/V}^2\right)(0.5\ \text{V})}$$
$$R_D = 32\ \text{k}\Omega$$

Thus, the drain resistance for V_{DS} equal to $1\ \text{V}$ is $\boxed{32\ \text{k}\Omega}$.

Step 17 of 17

Write the formula for the drain current.

$$I_D = \frac{1}{2}k_n'\frac{W}{L}V_{OV}^2$$
$$= \frac{1}{2}\left(k_n'(W/L)\right)(V_{OV})^2$$
$$= \frac{1}{2}\left(1\ \text{mA/V}^2\right)(0.5)^2$$
$$= 0.125\ \text{mA}$$

Thus, the drain current I_D for V_{DS} equal to $1\ \text{V}$ is $\boxed{0.125\ \text{mA}}$.

Step 1 of 2

7.008E

The trans-conductance g_m is,

$$g_m = \frac{2I_D}{(V_{GS} - V_t)}$$
$$= \frac{2I_D}{V_{OV}}$$

Here,

I_D is drain current I_D ,

V_{OV} is overdrive voltage.

Substitute $1\,\text{mA/V}$ for g_m , $0.1\,\text{mA}$ for I_D in equation.

$$g_m = \frac{2I_D}{V_{OV}}$$
$$1\ \text{mA/V} = \frac{2(0.1\ \text{mA})}{V_{OV}}$$
$$V_{OV} = \frac{2(0.1\times10^{-3})}{1\times10^{-3}}$$
$$V_{OV} = 0.2\ \text{V}$$

Hence, the overdrive voltage V_{OV} is $\boxed{0.2\ \text{V}}$.

Step 2 of 2

Determine the (W/L) ratio from drain current expression.

$$I_D = \frac{1}{2}k_n'(W/L)(V_{OV})^2$$

Substitute $1\,\text{mA/V}$ for g_m , $0.1\,\text{mA}$ for I_D, $50\,\mu\text{A/V}^2$ for k_n' in the above equation.

$$I_D = \frac{1}{2}k_n'(W/L)(V_{OV})^2$$
$$(0.1\times10^{-3})\ \text{A} = \frac{1}{2}(50\times10^{-6})(W/L)(0.2)^2$$
$$(W/L) = \frac{(0.1\times10^{-3})2}{(0.2)^2(50\times10^{-6})}$$
$$(W/L) = 100$$

Hence, the obtained (W/L) ratio is $\boxed{100}$.

Step 1 of 6

(a)

Maximum symmetrical signal swing allowable at drain end is $\pm 0.5\ \text{V}$

Over drive voltage V_{OV} is $0.2\ \text{V}$

In order to keep transistor in saturation region $V_{DS} - v_o \geq V_{OV}$ where v_o is maximum negative signal swing

Negative signal swing at the drain is $-0.5\ \text{V}$ and hence $V_{DS} - 0.5 \geq V_{OV}$

$$V_{DS} - 0.5 \geq V_{OV}$$
$$V_{DS} \geq 0.2\ \text{V} + 0.5\ \text{V}$$
$$V_{DS} \geq 0.7\ \text{V}$$

Condition for transistor in saturation region for negative signal swing is,

$V_{DS} \geq 0.7\ \text{V}$ ..... (1)

Step 2 of 6

Positive signal swing at the drain is greater than V_{DD} then transistor enters into cutoff region in order to keep transistor in saturation region $V_{DS} + v_o \leq V_{DD}$ where v_o is maximum positive signal swing

Positive signal swing at the drain is $0.5\ \text{V}$ and hence $V_{DS} + 0.5 \leq V_{DD}$

Power Voltage V_{DD} is $5\ \text{V}$

$$V_{DS} + 0.5 \leq V_{DD}$$
$$V_{DS} \leq 5\ \text{V} - 0.5\ \text{V}$$
$$V_{DS} \leq 4.5\ \text{V}$$

Condition for transistor in saturation region for positive signal swing is,

$V_{DS} \leq 4.5\ \text{V}$ ..... (2)

Equation (1) and (2) satisfied simultaneously when $V_{DS} = 4.5\ \text{V}$

Hence V_{DS} at the bias point is $\boxed{4.5\ \text{V}}$

Step 3 of 6

(b)

Voltage gain A_v of the MOS amplifier is $-\dfrac{2(V_{DD} - V_{DS})}{V_{OV}}$

Substitute values of V_{DD}, V_{DS} and V_{OV} in the equation $-\dfrac{2(V_{DD} - V_{DS})}{V_{OV}}$

$$A_v = -\frac{2(V_{DD} - V_{DS})}{V_{OV}}$$
$$= -\frac{2(5 - 4.5)}{0.2}$$
$$= -5$$

Thus, Voltage gain A_v is $\boxed{-5}$

Step 4 of 6

Amplitude of output signal $(\hat{v}_{ds})$ is $0.5\ \text{V}$

$$|A_v| = \frac{\text{ampliude of output signal}(\hat{v}_{ds})}{\text{ampliude of input signal}(\hat{v}_{gs})}$$

$$|A_v| = \frac{\hat{v}_{ds}}{\hat{v}_{gs}}$$
$$5 = \frac{0.5\ \text{V}}{\hat{v}_{gs}}$$
$$\hat{v}_{gs} = 0.1\ \text{V}$$

Amplitude of input signal $(\hat{v}_{gs})$ is $\boxed{0.1\ \text{V}}$

Step 5 of 6

(c)

DC bias current in the drain is $100\ \mu\text{A}$

Voltage drop across drain resistance is $V_{DD} - V_{DS}$ and hence voltage drop across drain resistance is $0.5\ \text{V}$

Voltage drop across the drain resistance in terms of drain current is $I_D R_D$

$$I_D R_D = 0.5\ \text{V}$$
$$R_D = \frac{0.5\ \text{V}}{100\ \mu\text{A}}$$
$$R_D = 5\ \text{k}\Omega$$

Thus, Drain resistance R_D of MOSFET is $\boxed{5\ \text{k}\Omega}$

Step 6 of 6

(d)

Value of k_n' is $200\ \text{mA/V}^2$

Voltage gain A_v is $-k_n'(W/L)V_{OV}R_D$

Voltage gain A_v for V_{DS} equal to $4.5\ \text{V}$ is -5

$$A_v = -k_n'(W/L)V_{OV}R_D$$
$$-5 = -\left(200\ \mu\text{A}/\text{V}^2\right)(W/L)(0.2\ \text{V})(5\ \text{k}\Omega)$$
$$(W/L) = \frac{5}{\left(200\ \mu\text{A}/\text{V}^2\right)(0.2\ \text{V})(5000)}$$
$$(W/L) = 25$$

W/L Ratio required for MOSFET is $\boxed{25}$

Step 1 of 2

7.009E

Consider PMOS and NMOS transistors have equal g_m for the same bias condition.

As the bias conditions are same, then trans-conductance for PMOS and NMOS are equal, since, drain currents are equal.

Drain current for P-channel device operates in saturation mode is,

$$i_D = \frac{1}{2}\mu_p C_{ox}\left(\frac{W_p}{L}\right)v_{OV}^2$$

Substitute $0.4\mu_n$ for μ_p in the above expression.

$$i_D = \frac{1}{2}(0.4\mu_n)C_{ox}\left(\frac{W_p}{L}\right)v_{OV}^2 \quad \ldots\ldots (1)$$

Drain current for n-channel device operates in saturation mode is,

$$i_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W_n}{L}\right)v_{OV}^2 \quad \ldots\ldots (2)$$

Step 2 of 2

Divide equation (1) with (2)

$$\frac{i_D}{i_D} = \frac{\frac{1}{2}(0.4\mu_n)C_{ox}\left(\frac{W_p}{L}\right)v_{OV}^2}{\frac{1}{2}(\mu_n)C_{ox}\left(\frac{W_n}{L}\right)v_{OV}^2}$$

$$1 = \frac{0.4W_p}{W_n}$$

$$\frac{W_p}{W_n} = 2.5$$

Therefore, the ratio of the width of a PMOS transistor to an NMOS transistor is $\boxed{2.5}$.

Step 1 of 6

7.009P

From the figure P7.9 in the textbook Q_1 NMOs transistor is operated in saturation region if gate-source voltage of Q_1 is greater than threshold voltage V_{t1} and drain source voltage of Q_1 is greater than the over drive voltage.

Gate-source voltage of transistor Q_1 is v_I.

Gate-source voltage of Q_1 is greater than threshold voltage V_{t1} then,

$v_I \geq V_{t1}$ ..... (1)

Over drive voltage V_{OV} is the difference between gate source voltage and threshold voltage V_{t1} and hence V_{OV} is equal to $v_I - V_{t1}$

Drain source voltage of Q_1 is shown in the figure P7.9

Drain source voltage of Q_1 is greater than the over drive voltage. Then,

$v_o \geq V_{OV}$

Substitute V_{OV} in terms of gate-source voltage in the above equation

$v_o \geq v_I - V_{t1}$ ..... (2)

Step 2 of 6

For NMOS transistor operated in saturation region both equation (1) and (2) should satisfy.

From equation (1) $v_I \geq V_{t1}$ and hence $V_{t1} \leq v_I$

$V_{t1} \leq v_I$ ..... (3)

From equation (2) $v_o \geq v_I - V_{t1}$ and hence $v_I \leq v_o + V_{t1}$

$v_I \leq v_o + V_{t1}$ ..... (4)

Combine equations (3) and (4)

$V_{t1} \leq v_I \leq v_o + V_{t1}$

NMOS Q_1 transistor operated in saturation then $V_{t1} \leq v_I \leq v_o + V_{t1}$ condition should be satisfied.

Step 3 of 6

Refer to the figure P7.9 in the textbook gate source voltage of NMOS transistor Q_2 is $(V_{DD} - v_0)$

We assumed that $V_{t1} = V_{t2} = V_t$

Drain current for transistor Q_2 is i_{D2}

$$i_{D2} = \frac{1}{2} k_n' \left(W/L\right)_2 (V_{DD} - v_0 - V_t)^2 \text{ ..... (5)}$$

Drain current for transistor Q_1 is i_{D1}

$$i_{D1} = \frac{1}{2} k_n' \left(W/L\right)_1 (v_I - V_t)^2 \text{ ..... (6)}$$

Step 4 of 6

NMOS transistors both have same drain currents and hence equate equations (5) and (6)

$$i_{D2} = i_{D1}$$

$$\frac{1}{2} k_n' \left(W/L\right)_2 (V_{DD} - v_0 - V_t)^2 = \frac{1}{2} k_n' \left(W/L\right)_1 (v_I - V_t)^2$$

$$\frac{(V_{DD} - v_0 - V_t)}{(v_I - V_t)} = \sqrt{\frac{\left(W/L\right)_1}{\left(W/L\right)_2}}$$

$$(V_{DD} - v_0 - V_t) = (v_I - V_t)\sqrt{\frac{\left(W/L\right)_1}{\left(W/L\right)_2}}$$

$$v_0 = V_{DD} - V_t + V_t\sqrt{\frac{\left(W/L\right)_1}{\left(W/L\right)_2}} - v_I\sqrt{\frac{\left(W/L\right)_1}{\left(W/L\right)_2}}$$

Output voltage v_0 is $\boxed{V_{DD} - V_t + V_t\sqrt{\frac{\left(W/L\right)_1}{\left(W/L\right)_2}} - v_I\sqrt{\frac{\left(W/L\right)_1}{\left(W/L\right)_2}}}$

Step 5 of 6

Value of $\left(W/L\right)_2$ is $50\ \mu m / 0.5\ \mu m$

Value of $\left(W/L\right)_1$ is $5\ \mu m / 0.5\ \mu m$

Substitute $\left(W/L\right)_1, \left(W/L\right)_2$ in the equation $V_{DD} - V_t + V_t\sqrt{\frac{\left(W/L\right)_1}{\left(W/L\right)_2}} - v_I\sqrt{\frac{\left(W/L\right)_1}{\left(W/L\right)_2}}$

Circuit functions as a linear amplifier, even for large input signals and hence from the equation

$V_{DD} - V_t + V_t\sqrt{\frac{\left(W/L\right)_1}{\left(W/L\right)_2}} - v_I\sqrt{\frac{\left(W/L\right)_1}{\left(W/L\right)_2}}$ neglect $V_{DD} - V_t + V_t\sqrt{\frac{\left(W/L\right)_1}{\left(W/L\right)_2}}$ and hence $v_o = -v_I\sqrt{\frac{\left(W/L\right)_1}{\left(W/L\right)_2}}$

Step 6 of 6

Substitute $\left(W/L\right)_1$ and $\left(W/L\right)_2$ value in $v_o = -v_I\sqrt{\frac{\left(W/L\right)_1}{\left(W/L\right)_2}}$

$$v_o = -v_I\sqrt{\frac{\left(W/L\right)_1}{\left(W/L\right)_2}}$$

$$\frac{v_o}{v_I} = -\sqrt{\frac{\frac{50\ \mu m}{0.5\ \mu m}}{\frac{5\ \mu m}{0.5\ \mu m}}}$$

$$\frac{v_o}{v_I} = -\sqrt{10}$$

$$A_v = -3.16$$

Thus, voltage gain A_v is $\boxed{-3.16}$

Step 1 of 4

The Drain current in saturation region is ,

$$I_D = \frac{1}{2}k_p'\left(\frac{W}{L}\right)\left(V_{SG} - |V_t|\right)^2$$

Substitute $\frac{16\,\mu m}{0.8\,\mu m}$ for $\frac{W}{L}$, $-1.6\,V$ for V_{GS} , $60\,\mu A/V^2$ for k_p' and $-1V$ for V_t in the above equation.

$$\begin{aligned} I_D &= \frac{1}{2}\left(60\times10^{-6}\right)\left(\frac{16\times10^{-6}}{0.8\times10^{-6}}\right)(1.6-1)^2 \\ &= \left(30\times10^{-6}\right)(20)(0.36) \\ &= 216\times10^{-6}\ A \\ &= 216\ \mu A \end{aligned}$$

Therefore, the drain current is $\boxed{216\ \mu A}$.

Step 2 of 4

Trans-conductance g_m for PMOS transistor is,

$$g_m = \frac{2I_D}{\left(V_{SG} - |V_t|\right)}$$

Substitute $216\,\mu A$ for I_D , $-1.6\,V$ for V_{GS} and $-1V$ for V_t in the above equation.

$$\begin{aligned} g_m &= \frac{2I_D}{\left(V_{SG} - |V_t|\right)} \\ &= \frac{2(216\ \mu A)}{(1.6\ -\ |-1|)\,V} \\ &= 0.72\ mA/V \end{aligned}$$

Hence, trans-conductance g_m at the bias point is $\boxed{0.72\ mA/V}$.

Step 3 of 4

The value of λ is inversely proportional to the length of the channel.

Consider λ for PMOS at $L = 1\ \mu m$ is $-0.04\ V^{-1}$.

Determine λ for PMOS at $L = 0.8\ \mu m$ is as follows:

$$\frac{-0.04}{0.8}(1) = -0.05$$

Hence, the λ for PMOS at $L = 0.8\ \mu m$ is $-0.05\ V^{-1}$

Early voltage V_A and λ is given by,

$$V_A = \frac{1}{\lambda}$$

Substitute $-0.05\ V^{-1}$ for λ in equation.

$$\begin{aligned} V_A &= \frac{1}{-0.05\ V^{-1}} \\ &= 20\ V \end{aligned}$$

Therefore, early voltage V_A is $-20\ V$

Step 4 of 4

Determine the finite output resistance r_o .

$$r_0 = \frac{|V_A|}{I_D}$$

Substitute $216\,\mu A$ for I_D , $-20\,V$ for V_A in equation.

$$\begin{aligned} r_0 &= \frac{|-20|\ V}{216\ \mu A} \\ &= 92.6\ k\Omega \end{aligned}$$

Therefore, finite output resistance r_o between drain and source is $\boxed{92.6\ k\Omega}$.

Step 1 of 3

7.010P

Voltage gain A_v of the circuit show in the figure 7.6 in the textbook is $-\frac{(V_{CC}-V_{CE})}{V_T}$

Thermal voltage of BJT at room temperature is $25\ \text{mV}$

Power Voltage V_{CC} is $5\ \text{V}$

Collector-Emitter Voltage V_{CE} is $1\ \text{V}$

Substitute V_{CC} and V_{CE} in the equation $-\frac{(V_{CC}-V_{CE})}{V_T}$

$$\begin{aligned} A_v &= -\frac{(V_{CC}-V_{CE})}{V_T} \\ &= -\frac{(5-1)\ \text{V}}{25\ \text{mV}} \\ &= \frac{4}{25\times 10^{-3}} \\ &= 160 \end{aligned}$$

Thus, Voltage gain A_v of the circuit show in the figure 7.6 in the textbook is $\boxed{160}$

Step 2 of 3

BJT is biased at V_{CE} equal to $+1\ \text{V}$

Minimum Value of V_{CE} for which BJT will enter into saturation region from linear region is $0.3\ \text{V}$

Hence maximum allowed output negative swing without the transistor entering into

Saturation is difference between biased voltage V_{CE} and minimum value of V_{CE} for which BJT will enter into saturation region

Maximum allowed output negative swing without the transistor entering into

Saturation is $(1-0.3)\ \text{V}$ that is $0.7\ \text{V}$

Thus, Maximum allowed output negative swing without the transistor entering into

Saturation is $\boxed{0.7\ \text{V}}$

Step 3 of 3

Maximum input signal permitted is ratio of maximum allowed output negative swing and voltage gain

$$\begin{aligned} v_{in} &= \frac{(0.7)}{A_v} \\ &= \frac{0.7}{160} \\ &= 4.375\ \text{mV} \end{aligned}$$

Thus, Maximum input signal permitted is $\boxed{4.375\ \text{mV}}$

7.011E

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Step 1 of 15

V_{cc} of the amplifier circuit shown in the figure. 7.6 in the textbook is $5\ V$.

Collector resistance R_c is $1\ k\Omega$.

Refer to the Figure. 7.6 in the textbook, we observed that collector-Emitter voltage (V_{ce}) is $(V_{cc} - I_c R_c)$

Collector –Emitter voltage V_{ce} for I_c equal to $0.5\ mA$ is $V_{ce} = (V_{cc} - I_c R_c)$

$$\begin{aligned} V_{ce} &= (V_{cc} - I_c R_c) \\ &= \left(5 - \left(0.5 \times 10^{-3}\ A\right)(1000\ \Omega)\right) \\ &= (5 - 0.5)\ v \\ &= 4.5\ V \end{aligned}$$

Thus, Bias voltage V_{ce} for I_c equal to $0.5\ mA$ is $\boxed{4.5\ V}$

Step 2 of 15

Voltage gain for BJT amplifier is $-\left(\frac{V_{cc} - V_{ce}}{V_T}\right)$

Thermal voltage V_T at room temperature is $25\ mV$

Voltage gain A_v for V_{ce} equal to $4.5\ V$ is,

$$\begin{aligned} A_v &= -\left(\frac{V_{cc} - V_{ce}}{V_T}\right) \\ &= -\left(\frac{(5 - 4.5)\ V}{25 \times 10^{-3}\ V}\right) \\ &= -20 \end{aligned}$$

Thus, Voltage gain A_v for V_{ce} equal to $4.5\ V$ is $\boxed{-20}$

Step 3 of 15

Amplifier circuit shown in the figure 7.6 in the textbook will operate in the active region then collector-Emitter voltage v_{ce} should lies between V_{cc} and $V_{ce_{sat}}$

$V_{ce_{sat}}$ is $0.3\ V$

Maximum possible positive output signal swing in order to keep the transistor in active region is $V_{cc} - V_{ce}$

$V_{cc} - V_{ce}$ is $(5 - 4.5)\ V$ that is $\boxed{0.5\ V}$

Maximum possible negative output signal swing in order to keep the transistor in active region is $V_{ce} - V_{ce_{sat}}$

$V_{ce} - V_{ce_{sat}}$ is $(4.5 - 0.3)\ V$ that is $\boxed{4.2\ V}$

Step 4 of 15

V_{ce} for I_c equal to $1\ mA$ is.

$$\begin{aligned} V_{ce} &= (V_{cc} - I_c R_c) \\ &= \left(5 - \left(1 \times 10^{-3}\ A\right)(1000\ \Omega)\right) \\ &= (5 - 1)\ v \\ &= 4V \end{aligned}$$

Thus, Bias voltage V_{ce} for I_c equal to $1\ mA$ is $\boxed{4\ V}$

Step 5 of 15

Voltage gain for BJT amplifier is $-\left(\frac{V_{cc} - V_{ce}}{V_T}\right)$

Thermal voltage V_T at room temperature is $25\ mV$

Voltage gain A_v for V_{ce} equal to $4\ V$ is,

$$\begin{aligned} A_v &= -\left(\frac{V_{cc} - V_{ce}}{V_T}\right) \\ &= -\left(\frac{(5 - 4)\ V}{25 \times 10^{-3}\ V}\right) \\ &= -40 \end{aligned}$$

Thus, Voltage gain A_v for V_{ce} equal to $4\ V$ is $\boxed{-40}$

Step 6 of 15

Amplifier circuit shown in the figure 7.6 will operate in the active region then collector-Emitter voltage V_{ce} should lies between V_{cc} and $V_{ce_{sat}}$

$V_{ce_{sat}}$ is $0.3\ V$

Maximum possible positive output signal swing in order to keep the transistor in active region is $V_{cc} - V_{ce}$

$V_{cc} - V_{ce}$ is $(5 - 4)\ V$ that is $\boxed{1\ V}$

Maximum possible negative output signal swing in order to keep the transistor in active region is $V_{ce} - V_{ce_{sat}}$

$V_{ce} - V_{ce_{sat}}$ is $(4 - 0.3)\ V$ that is $\boxed{3.7\ V}$

Step 7 of 15

V_{ce} for I_c equal to $2.5\ mA$ is.

$$\begin{aligned} V_{ce} &= (V_{cc} - I_c R_c) \\ &= \left(5 - \left(2.5 \times 10^{-3}\ A\right)(1000\ \Omega)\right) \\ &= (5 - 2.5)\ v \\ &= 2.5\ V \end{aligned}$$

Thus, Bias voltage V_{ce} for I_c equal to $2.5\ mA$ is $\boxed{2.5\ V}$

Step 8 of 15

Voltage gain for BJT amplifier is $-\left(\frac{V_{cc} - V_{ce}}{V_T}\right)$

Thermal voltage V_T at room temperature is $25\ mV$

Voltage gain A_v for V_{ce} equal to $2.5\ V$ is,

$$\begin{aligned} A_v &= -\left(\frac{V_{cc} - V_{ce}}{V_T}\right) \\ &= -\left(\frac{(5 - 2.5)\ V}{25 \times 10^{-3}\ V}\right) \\ &= -100 \end{aligned}$$

Thus, Voltage gain A_v for V_{ce} equal to $2.5\ V$ is $\boxed{-100}$

Step 9 of 15

Amplifier circuit shown in the figure 7.6 will operate in the active region then collector-Emitter voltage V_{ce} should lies between V_{cc} and $V_{ce_{sat}}$

$V_{ce_{sat}}$ is $0.3\ V$

Maximum possible positive output signal swing in order to keep the transistor in active region is $V_{cc} - V_{ce}$

$V_{cc} - V_{ce}$ is $(5 - 2.5)\ V$ that is $\boxed{2.5\ V}$

Maximum possible negative output signal swing in order to keep the transistor in active region is $V_{ce} - V_{ce_{sat}}$

$V_{ce} - V_{ce_{sat}}$ is $(2.5 - 0.3)\ V$ that is $\boxed{2.2\ V}$

Step 10 of 15

V_{ce} for I_c equal to $4\ mA$ is.

$$\begin{aligned} V_{ce} &= (V_{cc} - I_c R_c) \\ &= \left(5 - \left(4 \times 10^{-3}\ A\right)(1000\ \Omega)\right) \\ &= (5 - 4)\ v \\ &= 1\ V \end{aligned}$$

Thus, Bias voltage V_{ce} for I_c equal to $4\ mA$ is $\boxed{1\ V}$

Step 11 of 15

Voltage gain for BJT amplifier is $-\left(\frac{V_{cc} - V_{ce}}{V_T}\right)$

Thermal voltage V_T at room temperature is $25\ mV$

Voltage gain A_v for V_{ce} equal to $1\ V$ is,

$$\begin{aligned} A_v &= -\left(\frac{V_{cc} - V_{ce}}{V_T}\right) \\ &= -\left(\frac{(5 - 1)\ V}{25 \times 10^{-3}\ V}\right) \\ &= -160 \end{aligned}$$

Thus, Voltage gain A_v for V_{ce} equal to $1\ V$ is $\boxed{-160}$

Step 12 of 15

Amplifier circuit shown in the figure 7.6 will operate in the active region then collector-Emitter voltage V_{ce} should lies between V_{cc} and $V_{ce_{sat}}$

$V_{ce_{sat}}$ is $0.3\ V$

Maximum possible positive output signal swing in order to keep the transistor in active region is $V_{cc} - V_{ce}$

$V_{cc} - V_{ce}$ is $(5 - 1)\ V$ that is $\boxed{4\ V}$

Maximum possible negative output signal swing in order to keep the transistor in active region is $V_{ce} - V_{ce_{sat}}$

$V_{ce} - V_{ce_{sat}}$ is $(1 - 0.3)\ V$ that is $\boxed{0.7\ V}$

Step 13 of 15

V_{ce} for I_c equal to $4.5\ mA$ is.

$$\begin{aligned} V_{ce} &= (V_{cc} - I_c R_c) \\ &= \left(5 - \left(4.5 \times 10^{-3}\ A\right)(1000\ \Omega)\right) \\ &= (5 - 4.5)\ v \\ &= 0.5\ V \end{aligned}$$

Thus, Bias voltage V_{ce} for I_c equal to $4.5\ mA$ is $\boxed{0.5\ V}$

Step 14 of 15

Voltage gain for BJT amplifier is $-\left(\frac{V_{cc} - V_{ce}}{V_T}\right)$

Thermal voltage V_T at room temperature is $25\ mV$

Voltage gain A_v for V_{ce} equal to $0.5\ V$ is,

$$\begin{aligned} A_v &= -\left(\frac{V_{cc} - V_{ce}}{V_T}\right) \\ &= -\left(\frac{(5 - 0.5)\ V}{25 \times 10^{-3}\ V}\right) \\ &= -180 \end{aligned}$$

Thus, Voltage gain A_v for V_{ce} equal to $0.5\ V$ is -180

Amplifier circuit shown in the figure 7.6 will operate in the active region then collector-Emitter voltage V_{ce} should lies between V_{cc} and $V_{ce_{sat}}$

$V_{ce_{sat}}$ is $0.3\ V$

Maximum possible positive output signal swing in order to keep the transistor in active region is $V_{cc} - V_{ce}$

$V_{cc} - V_{ce}$ is $(5 - 0.5)\ V$ that is $\boxed{4.5\ V}$

Maximum possible negative output signal swing in order to keep the transistor in active region is $V_{ce} - V_{ce_{sat}}$

$V_{ce} - V_{ce_{sat}}$ is $(0.5 - 0.3)\ V$ that is $\boxed{0.2\ V}$

Step 15 of 15

I_c mA	V_{cc} V	(V_{ce}) V	A_v	Positive output swing	Negative output swing
0.5	5	4.5	20	0.5	4.2
1	5	4	40	1	3.7
2.5	5	2.5	100	2.5	2.2
4	5	1	160	4	0.7
4.5	5	0.5	180	4.5	0.2

Table 1

7.012E

Refer to the equation (7.63) in the textbook.

The trans-conductance g_m is defined as $g_m = \left.\frac{\partial i_c}{\partial v_{BE}}\right|_{i_c = I_C}$.

Substitute $I_s e^{v_{BE}/V_T}$ for i_C in the above expression.

$$g_m = \left.\frac{\partial\left(I_S e^{v_{BE}/V_T}\right)}{\partial v_{BE}}\right|_{i_c = I_C}$$

$$= I_S e^{v_{BE}/V_T}\left(\frac{1}{V_T}\right)\Bigg|_{i_c = I_C}$$

$$= i_C\left(\frac{1}{V_T}\right)\Bigg|_{i_c = I_C}$$

$$= \frac{I_C}{V_T}$$

Therefore, the trans-conductance g_m is $\boxed{\frac{I_C}{V_T}}$.

Step 1 of 3

7.012P

V_{CC} of the CE amplifier is $+5\ \mathbf{V}$

In order to obtain the maximum possible output amplitude for a given input transistor should bias at the edge of saturation as possible without entering saturation at any time.

Saturated collector emitter voltage V_{CEsat} is $0.3\ \mathbf{V}$ below this voltage transistor enters into saturation region from active region.

Maximum Voltage gain for V_{CEsat} equal to $0.3\ \mathbf{V}$ is,

$$\begin{aligned} A_{v\max} &= -\left(\frac{V_{CC}-V_{CEsat}}{V_T}\right) \\ &= -\left(\frac{(5-0.3)\ \mathbf{V}}{25\times10^{-3}\ \mathbf{V}}\right) \\ &= -188 \end{aligned}$$

Maximum Voltage gain for V_{CEsat} equal to $0.3\ \mathbf{V}$ is $\boxed{-188}$

Step 2 of 3

Input sine wave signal v_{be} is $5\ \mathbf{mv}$

Output sine wave signal v_{ce} is $A_v v_{be}$

$$\begin{aligned} v_{ce} &= A_v v_{be} \\ &= -188(5\ \mathrm{mV}) \\ &= 940\ \mathrm{mV} \end{aligned}$$

Thus, Output sine wave signal v_{ce} is $\boxed{940\ \mathrm{mV}}$

Step 3 of 3

Output sine wave signal v_{ce} is $940\ \mathbf{mv}$ and hence maximum allowed output negative swing without the transistor enters into saturation is $940\ \mathbf{mv}$

Maximum allowed output negative swing is added with V_{CEsat} gives the biased collector voltage of transistor without entering into saturation region from active region.

$$\begin{aligned} V_{CE} &= V_{CEsat} + v_{ce} \\ &= 0.3\ \mathrm{V} + 0.940\ \mathrm{V} \\ &= 1.24\ \mathrm{V} \end{aligned}$$

Thus, Point at which transistor should be biased is $\boxed{1.24\ \mathrm{v}}$

Step 1 of 2

7.013E

Determine the trans-conductance g_m.

$$g_m = \frac{I_C}{V_T}$$

Here,

I_C is collector current.

V_T is thermal voltage.

Step 2 of 2

Substitute $0.5\ \text{mA}$ for I_C , $25\ \text{mV}$ for V_T in equation.

$$\begin{aligned} g_m &= \frac{I_C}{V_T} \\ &= \frac{0.5\ \text{mA}}{25\ \text{mV}} \\ &= 20\ \text{mA/V} \end{aligned}$$

Therefore, the trans-conductance, g_m is $\boxed{20\ \text{mA/V}}$.

Step 1 of 12

Saturated collector emitter voltage of BJT is $0.3\ \text{V}$

BJT will have a maximum voltage gain when transistor operated near the saturation region.

Power voltage V_{CC} is $1\ \text{V}$

Thermal voltage V_T at room temperature is $25\ \text{mV}$

Voltage gain A_v is $-\left(\frac{V_{CC}-V_{CE}}{V_T}\right)$

$$A_v = -\left(\frac{V_{CC}-V_{CE}}{V_T}\right)$$
$$= -\left(\frac{(1-0.3)\ \text{V}}{25\times10^{-3}\ \text{V}}\right)$$
$$= -28$$

Thus, Voltage gain A_v for $V_{CC}=1\ \text{V}$ is $\boxed{-28}$

Step 2 of 12

Power voltage V_{CC} is $1.5\ \text{V}$

Thermal voltage V_T at room temperature is $25\ \text{mV}$

Voltage gain A_v is $-\left(\frac{V_{CC}-V_{CE}}{V_T}\right)$

$$A_v = -\left(\frac{V_{CC}-V_{CE}}{V_T}\right)$$
$$= -\left(\frac{(1.5-0.3)\ \text{V}}{25\times10^{-3}\ \text{V}}\right)$$
$$= -48$$

Thus, Voltage gain A_v for $V_{CC}=1\ \text{V}$ is $\boxed{-48}$

Step 3 of 12

Power voltage V_{CC} is $2\ \text{V}$

Thermal voltage V_T at room temperature is $25\ \text{mV}$

Voltage gain A_v is $-\left(\frac{V_{CC}-V_{CE}}{V_T}\right)$

$$A_v = -\left(\frac{V_{CC}-V_{CE}}{V_T}\right)$$
$$= -\left(\frac{(2-0.3)\ \text{V}}{25\times10^{-3}\ \text{V}}\right)$$
$$= -68$$

Thus, Voltage gain A_v for $V_{CC}=2\ \text{V}$ is $\boxed{-68}$

Step 4 of 12

Power voltage V_{CC} is $3\ \text{V}$

Thermal voltage V_T at room temperature is $25\ \text{mV}$

Voltage gain A_v is $-\left(\frac{V_{CC}-V_{CE}}{V_T}\right)$

$$A_v = -\left(\frac{V_{CC}-V_{CE}}{V_T}\right)$$
$$= -\left(\frac{(3-0.3)\ \text{V}}{25\times10^{-3}\ \text{V}}\right)$$
$$= -108$$

Thus, Voltage gain A_v for $V_{CC}=3\ \text{V}$ is $\boxed{-108}$

Step 5 of 12

Collector emitter voltage of BJT is $\frac{V_{CC}}{2}\ \text{V}$

Power voltage V_{CC} is $1\ \text{V}$

Thermal voltage V_T at room temperature is $25\ \text{mV}$

Voltage gain A_v is $-\left(\frac{V_{CC}-V_{CE}}{V_T}\right)$

$$A_v = -\left(\frac{V_{CC}-V_{CE}}{V_T}\right)$$
$$= -\left(\frac{\left(1-\frac{1}{2}\right)\ \text{V}}{25\times10^{-3}\ \text{V}}\right)$$
$$= -20$$

Thus, Voltage gain A_v for $V_{CC}=1\ \text{V}$ is $\boxed{-20}$

Step 6 of 12

Maximum allowed output negative swing without the transistor enters into saturation is $0.4\ \text{v}$

Maximum allowed output negative swing is added with V_{CEsat} gives the biased collector voltage of transistor without entering into saturation region from active region.

$$V_{CE} = V_{CEsat} + v_{ce}$$
$$= 0.3\ \text{V} + 0.4\ \text{V}$$
$$= 0.7\ \text{V}$$

Thus, Point at which transistor should be biased is $\boxed{0.7\ \text{v}}$

Step 7 of 12

Collector emitter voltage of BJT is $\frac{V_{CC}}{2}\ \text{V}$

Power voltage V_{CC} is $1.5\ \text{V}$

Thermal voltage V_T at room temperature is $25\ \text{mV}$

Voltage gain A_v is $-\left(\frac{V_{CC}-V_{CE}}{V_T}\right)$

$$A_v = -\left(\frac{V_{CC}-V_{CE}}{V_T}\right)$$
$$= -\left(\frac{\left(1.5-\frac{1.5}{2}\right)\ \text{V}}{25\times10^{-3}\ \text{V}}\right)$$
$$= -30$$

Thus, Voltage gain A_v for $V_{CC}=1.5\ \text{V}$ is $\boxed{-30}$

Step 8 of 12

Maximum allowed output negative swing without the transistor enters into saturation is $0.4\ \text{v}$

Maximum allowed output negative swing is added with V_{CEsat} gives the biased collector voltage of transistor without entering into saturation region from active region.

$$V_{CE} = V_{CEsat} + v_{ce}$$
$$= 0.3\ \text{V} + 0.4\ \text{V}$$
$$= 0.7\ \text{V}$$

Thus, Point at which transistor should be biased is $\boxed{0.7\ \text{v}}$

Step 9 of 12

Collector emitter voltage of BJT is $\frac{V_{CC}}{2}\ \text{V}$

power voltage V_{CC} is $2\ \text{V}$

Thermal voltage V_T at room temperature is $25\ \text{mV}$

Voltage gain A_v is $-\left(\frac{V_{CC}-V_{CE}}{V_T}\right)$

$$A_v = -\left(\frac{V_{CC}-V_{CE}}{V_T}\right)$$
$$= -\left(\frac{\left(2-\frac{2}{2}\right)\ \text{V}}{25\times10^{-3}\ \text{V}}\right)$$
$$= -40$$

Thus, Voltage gain A_v for $V_{CC}=2\ \text{V}$ is $\boxed{-40}$

Step 10 of 12

Maximum allowed output negative swing without the transistor enters into saturation is $0.4\ \text{v}$

Maximum allowed output negative swing is added with V_{CEsat} gives the biased collector voltage of transistor without entering into saturation region from active region.

$$V_{CE} = V_{CEsat} + v_{ce}$$
$$= 0.3\ \text{V} + 0.4\ \text{V}$$
$$= 0.7\ \text{V}$$

Thus, Point at which transistor should be biased is $\boxed{0.7\ \text{v}}$

Step 11 of 12

Collector emitter voltage of BJT is $\frac{V_{CC}}{2}\ \text{V}$

Power voltage V_{CC} is $3\ \text{V}$

Thermal voltage V_T at room temperature is $25\ \text{mV}$

Voltage gain A_v is $-\left(\frac{V_{CC}-V_{CE}}{V_T}\right)$

$$A_v = -\left(\frac{V_{CC}-V_{CE}}{V_T}\right)$$
$$= -\left(\frac{\left(3-\frac{3}{2}\right)\ \text{V}}{25\times10^{-3}\ \text{V}}\right)$$
$$= -60$$

Thus, Voltage gain A_v for $V_{CC}=3\ \text{V}$ is $\boxed{-60}$

Step 12 of 12

Maximum allowed output negative swing without the transistor enters into saturation is $0.4\ \text{v}$

Maximum allowed output negative swing is added with V_{CEsat} gives the biased collector voltage of transistor without entering into saturation region from active region.

$$V_{CE} = V_{CEsat} + v_{ce}$$
$$= 0.3\ \text{V} + 0.4\ \text{V}$$
$$= 0.7\ \text{V}$$

Point at which transistor should be biased is $\boxed{0.7\ \text{v}}$

Step 1 of 5

7.014E

Determine the transconductance g_m of BJT.

$$g_m = \frac{I_C}{V_T}$$

Substitute 0.5 mA for I_C and 25 mV for V_T in equation.

$$\begin{aligned} g_m &= \frac{I_C}{V_T} \\ &= \frac{0.5\ \text{mA}}{25\ \text{mV}} \\ &= 20\ \text{mA/V} \end{aligned}$$

Hence, transconductance g_m for BJT amplifier is $\boxed{20\ \text{mA/V}}$.

Step 2 of 5

Determine the value of base current I_B for $\beta = 50$.

$$I_B = \frac{I_C}{\beta}$$

Substitute 50 for β and 0.5 mA for I_C in equation.

$$\begin{aligned} I_B &= \frac{I_C}{\beta} \\ &= \frac{0.5\ \text{mA}}{50} \\ &= 10\ \mu\text{A} \end{aligned}$$

Hence, the base current I_B for $\beta = 50$, is $\boxed{10\ \mu\text{A}}$.

Step 3 of 5

Determine the value of base current for $\beta = 200$.

$$I_B = \frac{I_C}{\beta}$$

Substitute 200 for β and 0.5 mA for I_C in equation.

$$\begin{aligned} I_B &= \frac{I_C}{\beta} \\ &= \frac{0.5\ \text{mA}}{200} \\ &= 2.5\ \mu\text{A} \end{aligned}$$

Thus, the base current I_B for $\beta = 200$, is $\boxed{2.5\ \mu\text{A}}$.

Therefore, the range of base current I_B is $\boxed{10\ \mu\text{A to } 2.5\ \mu\text{A}}$.

Step 4 of 5

Determine the small signal input resistance between the base and emitter for $\beta = 50$.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute 50 for β and $20\ \text{mA/V}$ for g_m in equation.

$$\begin{aligned} r_\pi &= \frac{50}{20\ \text{mA/V}} \\ &= 2.5\ \text{k}\Omega \end{aligned}$$

Hence, the small signal input resistance r_π for $\beta = 50$, is $\boxed{2.5\ \text{k}\Omega}$.

Step 5 of 5

Determine the small signal input resistance between the base and emitter for $\beta = 200$.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute 200 for β and $20\ \text{mA/V}$ for g_m in equation.

$$\begin{aligned} r_\pi &= \frac{200}{20 \times 10^{-3}} \\ &= 10\ \text{k}\Omega \end{aligned}$$

Hence, the small signal input resistance r_π for $\beta = 200$, is $10\ \text{k}\Omega$.

Therefore, the range of small signal input resistance r_π is $\boxed{2.5\ \text{k}\Omega\text{ to }10\ \text{k}\Omega}$.

Step 1 of 9

7.014P

Refer to Figure 7.6 in the text book.

In order to get undistorted sine wave output from a BJT amplifier, BJT amplifier should operate in active region it will not enter into either cutoff or saturation region.

For BJT amplifier circuit largest voltage gain is occurred when BJT operated near to saturation region.

Output sine wave peak amplitude v_{ce} is "P" and hence maximum allowed output negative swing without the transistor enters into saturation is "P"

Maximum allowed output negative swing is added with V_{CEsat} gives the biased collector voltage of transistor without entering into saturation region from active region.

Point at which transistor should be biased is $V_{CEsat} + P$

Step 2 of 9

A_v is the largest possible voltage gain

$$|A_v| = \frac{V_{CC} - (V_{CEsat} + P)}{V_T}$$

$$|A_v| V_T = V_{CC} - (V_{CEsat} + P)$$

$$V_{CC} = |A_v| V_T + V_{CEsat} + P$$

Thus, Minimum supply voltage V_{CC} needed is $\boxed{V_{CC} = |A_v| V_T + V_{CEsat} + P}$

Step 3 of 9

(a)

Determine the Minimum supply voltage when $A_v = -20\ \text{V/V}$ and $P = 0.2\ \text{V}$:

$$V_{CC} = |A_v| V_T + V_{CEsat} + P$$

Substitute $-20\ \text{V/V}$ for A_v, $0.5\ \text{V}$ for V_{CEsat}, for V_T and $0.2\ \text{V}$ for P in the equation.

$$\begin{aligned} V_{CC} &= |-20|(25\times10^{-3}\ \text{V}) + 0.5\ \text{V} + 0.2\ \text{V} \\ &= 0.5\ \text{V} + 0.5\ \text{V} + 0.2\ \text{V} \\ &= 1.2\ \text{V} \end{aligned}$$

Thus, the minimum supply voltage V_{CC} is $\boxed{1.2\ \text{V}}$.

Step 4 of 9

(b)

Determine the Minimum supply voltage when $A_v = -50\ \text{V/V}$ and $P = 0.5\ \text{V}$:

$$V_{CC} = |A_v| V_T + V_{CEsat} + P$$

Substitute $-50\ \text{V/V}$ for A_v, $0.5\ \text{V}$ for V_{CEsat}, for V_T and $0.2\ \text{V}$ for P in the equation.

$$\begin{aligned} V_{CC} &= |-50|(25\times10^{-3}\ \text{V}) + 0.5\ \text{V} + 0.5\ \text{V} \\ &= 1.25\ \text{V} + 0.5\ \text{V} + 0.5\ \text{V} \\ &= 2.25\ \text{V} \end{aligned}$$

Thus, the minimum supply voltage V_{CC} is $\boxed{2.25\ \text{V}}$.

Step 5 of 9

(c)

Determine the Minimum supply voltage when $A_v = -100\ \text{V/V}$ and $P = 0.5\ \text{V}$:

$$V_{CC} = |A_v| V_T + V_{CEsat} + P$$

Substitute $-100\ \text{V/V}$ for A_v, $0.5\ \text{V}$ for V_{CEsat}, for V_T and $0.2\ \text{V}$ for P in the equation.

$$\begin{aligned} V_{CC} &= |-100|(25\times10^{-3}\ \text{V}) + 0.5\ \text{V} + 0.5\ \text{V} \\ &= 2.5\ \text{V} + 0.5\ \text{V} + 0.5\ \text{V} \\ &= 3.5\ \text{V} \end{aligned}$$

Thus, the minimum supply voltage V_{CC} is $\boxed{3.5\ \text{V}}$.

Step 6 of 9

(d)

Determine the Minimum supply voltage when $A_v = -100\ \text{V/V}$ and $P = 1.0\ \text{V}$:

$$V_{CC} = |A_v| V_T + V_{CEsat} + P$$

Substitute $-100\ \text{V/V}$ for A_v, $0.5\ \text{V}$ for V_{CEsat}, for V_T and $1.0\ \text{V}$ for P in the equation.

$$\begin{aligned} V_{CC} &= |-100|(25\times10^{-3}\ \text{V}) + 0.5\ \text{V} + 1\ \text{V} \\ &= 2.5\ \text{V} + 0.5\ \text{V} + 1\ \text{V} \\ &= 4\ \text{V} \end{aligned}$$

Thus, the minimum supply voltage V_{CC} is $\boxed{4\ \text{V}}$

Step 7 of 9

(e)

Determine the Minimum supply voltage when $A_v = -200\ \text{V/V}$ and $P = 1.0\ \text{V}$:

$$V_{CC} = |A_v| V_T + V_{CEsat} + P$$

Substitute $-200\ \text{V/V}$ for A_v, $0.5\ \text{V}$ for V_{CEsat}, for V_T and $1.0\ \text{V}$ for P in the equation.

$$\begin{aligned} V_{CC} &= |-200|(25\times10^{-3}\ \text{V}) + 0.5\ \text{V} + 1\ \text{V} \\ &= 5\ \text{V} + 0.5\ \text{V} + 1\ \text{V} \\ &= 5.6\ \text{V} \end{aligned}$$

Thus, the minimum supply voltage V_{CC} is $\boxed{5.6\ \text{V}}$.

Step 8 of 9

(f)

Determine the Minimum supply voltage when $A_v = -500\ \text{V/V}$ and $P = 1.0\ \text{V}$:

$$V_{CC} = |A_v| V_T + V_{CEsat} + P$$

Substitute $-500\ \text{V/V}$ for A_v, $0.5\ \text{V}$ for V_{CEsat}, for V_T and $1.0\ \text{V}$ for P in the equation.

$$\begin{aligned} V_{CC} &= |-500|(25\times10^{-3}\ \text{V}) + 0.5\ \text{V} + 1\ \text{V} \\ &= 12.5\ \text{V} + 0.5\ \text{V} + 1\ \text{V} \\ &= 14\ \text{V} \end{aligned}$$

Thus, the minimum supply voltage V_{CC} is $\boxed{14\ \text{V}}$.

Step 9 of 9

(g)

Determine the Minimum supply voltage when $A_v = -500\ \text{V/V}$ and $P = 2.0\ \text{V}$:

$$V_{CC} = |A_v| V_T + V_{CEsat} + P$$

Substitute $-500\ \text{V/V}$ for A_v, $0.5\ \text{V}$ for V_{CEsat}, for V_T and $2.0\ \text{V}$ for P in the equation.

$$\begin{aligned} V_{CC} &= |-500|(25\times10^{-3}\ \text{V}) + 0.5\ \text{V} + 2\ \text{V} \\ &= 12.5\ \text{V} + 0.5\ \text{V} + 2\ \text{V} \\ &= 15\ \text{V} \end{aligned}$$

Thus, the minimum supply voltage V_{CC} is $\boxed{15\ \text{V}}$.

Step 1 of 3

7.015E

Determine the value of trans-conductance g_m.

$$g_m = \frac{I_C}{V_T}$$

Substitute $1\ \text{mA}$ for I_C and 25 mV for V_T in equation.

$$\begin{aligned} g_m &= \frac{I_C}{V_T} \\ &= \frac{1\ \text{mA}}{25\ \text{mV}} \\ &= 40\ \text{mA/V} \end{aligned}$$

Thus, trans-conductance g_m for BJT amplifier is $\boxed{40\ \text{mA/V}}$.

Step 2 of 3

Determine the emitter resistance r_e.

$$r_e = \frac{1}{g_m}$$

Substitute $40\times10^{-3}\ \text{A/V}$ for g_m in equation.

$$\begin{aligned} r_e &= \frac{1}{g_m} \\ &= \frac{1}{40\ \text{mA/V}} \\ &= 25\ \Omega \end{aligned}$$

Therefore, the emitter resistance r_e is $\boxed{25\ \Omega}$.

Step 3 of 3

Determine the input resistance r_π.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute 100 for β and $40\times10^{-3}\ \text{A/V}$ for g_m in equation.

$$\begin{aligned} r_\pi &= \frac{\beta}{g_m} \\ &= \frac{100}{40\ \text{mA/V}} \\ &= 2.5\ \text{k}\Omega \end{aligned}$$

Therefore, the input resistance r_π is $\boxed{2.5\ \text{k}\Omega}$.

7.015P

We don't have the solution to this problem yet.

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Step 1 of 3

Refer to Figure 7.20(a) in the textbook.

Determine the voltage gain A_v.

$$\begin{aligned} A_v &= \frac{v_{ce}}{v_{be}} \\ &= -g_m R_C \\ &= -\frac{I_C R_C}{V_T} \qquad \left(\text{since } g_m = \frac{I_C}{V_T}\right) \end{aligned}$$

Substitute $1\ \text{mA}$ for I_C, $10\ \text{k}\Omega$ for R_C and $25\ \text{mV}$ for V_T in equation.

$$\begin{aligned} A_v &= -\frac{I_C R_C}{V_T} \\ &= -\frac{(1\ \text{mA})(10\ \text{k}\Omega)}{25\ \text{mV}} \\ &= -400\ \text{V/V} \end{aligned}$$

Thus, the voltage gain A_v, is $\boxed{-400\ \text{V/V}}$.

Step 2 of 3

Consider the base emitter voltage v_{be} is $0.005\sin\omega t\ \text{V}$.

From the circuit, the voltage $v_C(t)$ is,

$$\begin{aligned} v_C(t) &= V_{CC} - i_C R_C \\ &= V_{CC} - (I_C + i_c) R_C \qquad (\text{Since } i_C = I_C + i_c) \\ &= V_{CC} - I_C R_C - i_c R_C \\ &= V_{CC} - I_C R_C + v_{ce} \qquad (\text{Since } v_c = -i_c R_C) \end{aligned}$$

Therefore,

$$v_C(t) = V_{CC} - I_C R_C - 400 v_{be} \qquad \left(\text{since } \frac{v_{ce}}{v_{be}} = -400\ \text{V/V}\right)$$

Substitute the corresponding values.

$$\begin{aligned} v_C(t) &= 15 - \left(1\times10^{-3}\right)\left(10\times10^{3}\right) - 400\left(0.005\sin\omega t\right) \\ &= 5 - 2\sin\omega t\ \text{V} \end{aligned}$$

Hence, the voltage $v_C(t)$ is $\boxed{5 - 2\sin\omega t\ \text{V}}$.

Step 3 of 3

Write the expression for the current $i_B(t)$

$$\begin{aligned} i_B(t) &= I_B + i_b(t) \\ &= \frac{I_C}{\beta} + \frac{i_c(t)}{\beta} \\ &= \frac{I_C}{\beta} + \frac{g_m v_{be}(t)}{\beta} \\ &= \frac{I_C}{\beta} + \frac{\left(\frac{I_C}{V_T}\right) v_{be}(t)}{\beta} \end{aligned}$$

Substitute the corresponding values.

$$\begin{aligned} i_B(t) &= \frac{1\times10^{-3}}{100} + \frac{\left(\frac{1\times10^{-3}}{25\times10^{-3}}\right)(0.005\sin\omega t)}{100} \\ &= 10\times10^{-6} + \frac{(0.005\sin\omega t)}{2500} \\ &= 10\times10^{-6} + 2\times10^{-6}\sin\omega t \\ &= (10 + 2\sin\omega t)\ \mu\text{A} \end{aligned}$$

Hence, the current $i_B(t)$ is $\boxed{(10 + 2\sin\omega t)\ \mu\text{A}}$.

7.016P

We don't have the solution to this problem yet.

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Step 1 of 2

7.017E

Refer to the figure 7.24(b) in the textbook.

The collector current of the transistor is,

$i_c = \beta i_b$ ..... (1)

The small signal input resistance between the base and emitter, looking into the base is,

$$r_\pi = \frac{\beta}{g_m}$$

$\beta = r_\pi g_m$ ..... (2)

Substitute equation (2) in equation (1).

$i_c = r_\pi g_m i_b$ ..... (3)

The voltage across the resistance r_π is v_{be} and current through the resistance r_π is i_b. Therefore,

$v_{be} = r_\pi i_b$ ..... (4)

Substitute equation (4) in equation (3).

$i_c = g_m v_{be}$

Hence, the collector current $\boxed{i_c = g_m v_{be}}$ is proved.

Step 2 of 2

Relation between i_e and i_c is,

$$i_e = \frac{i_c}{\alpha}$$ ..... (5)

Substitute $i_c = i_e \alpha$ in the equation (5).

$i_c = g_m v_{be}$

$i_e \alpha = g_m v_{be}$

$$i_e = \frac{g_m v_{be}}{\alpha}$$ ..... (6)

Refer to the equation (7.74) in the textbook.

The emitter resistance is $r_e = \frac{\alpha}{g_m}$.

Substitute $g_m = \frac{\alpha}{r_e}$ in the equation (6).

$$i_e = \frac{g_m v_{be}}{\alpha}$$

$$= \frac{\alpha v_{be}}{\alpha r_e}$$

$$= \frac{v_{be}}{r_e}$$

Hence, the emitter current $\boxed{i_e = v_{be}/r_e}$ is proved.

Step 1 of 6

Collector current i_C including early effect is $I_s e^{v_{BE}/V_T}\left(1+\frac{v_{CE}}{V_A}\right)$

Relation between output voltage v_{CE} and collector current i_C is,

$$v_{CE} = V_{CC} - R_C i_C$$

Substitute i_C in the above equation

$$v_{CE} = V_{CC} - R_C I_s e^{v_{BE}/V_T}\left(1+\frac{v_{CE}}{V_A}\right)$$

Relation between output voltage v_{CE} and input voltage v_{BE} is ,

$$v_{CE} = V_{CC} - R_C I_s e^{v_{BE}/V_T}\left(1+\frac{v_{CE}}{V_A}\right)$$

Step 2 of 6

Small signal voltage gain of BJT amplifier is,

$$A_v = \left.\frac{dv_{CE}}{dv_{BE}}\right|_{v_{BE}=V_{BE}}$$

$$A_v = \left.\frac{d\left(V_{CC} - R_C I_s e^{v_{BE}/V_T}\left(1+\frac{v_{CE}}{V_A}\right)\right)}{dv_{BE}}\right|_{v_{BE}=V_{BE}}$$

$$= 0 - R_C I_s e^{V_{BE}/V_T}\left(0+\frac{1}{V_A}\left(\frac{dv_{CE}}{dv_{BE}}\right)_{v_{BE}=V_{BE}}\right) - R_C\left(1+\frac{v_{CE}}{V_A}\right) I_s e^{V_{BE}/V_T}\left(\frac{1}{V_T}\right)$$

Step 3 of 6

DC Collector current I_C is $I_s e^{V_{BE}/V_T}\left(1+\frac{V_{CE}}{V_A}\right)$

Substitute $\frac{I_C}{\left(1+\frac{V_{CE}}{V_A}\right)}$ in the place of $I_s e^{V_{BE}/V_T}$ for above equation

$$A_v = 0 - R_C \frac{I_C}{\left(1+\frac{V_{CE}}{V_A}\right)}\frac{1}{V_A}(A_v) - R_C I \frac{I_C}{\left(1+\frac{V_{CE}}{V_A}\right)}\left(\frac{V_A+v_{CE}}{V_A}\right)\left(\frac{1}{V_T}\right)$$

$$A_v\left[1+\frac{R_C I_C}{V_A\left(1+\frac{V_{CE}}{V_A}\right)}\right] = -R_C I_C\left(\frac{1}{V_T}\right)$$

$$A_v = -\frac{\frac{R_C I_C}{V_T}}{1+\frac{R_C I_C}{V_A\left(1+\frac{V_{CE}}{V_A}\right)}}$$

$$A_v = -\frac{\frac{R_C I_C}{V_T}}{\left[1+\frac{R_C I_C}{V_A+V_{CE}}\right]}$$

Thus, Voltage gain expression for BJT including early effect is $\boxed{A_v = -\frac{\frac{R_C I_C}{V_T}}{\left[1+\frac{R_C I_C}{V_A+V_{CE}}\right]}}$

Step 4 of 6

Voltage drop across the collector resistance $I_C R_C$ is $(V_{CC} - V_{CE})$

Instead of $I_C R_C$ substitute $(V_{CC} - V_{CE})$ in the equation $A_v = -\frac{\frac{R_C I_C}{V_T}}{\left[1+\frac{R_C I_C}{V_A+V_{CE}}\right]}$

$$A_v = -\frac{\frac{R_C I_C}{V_T}}{\left[1+\frac{R_C I_C}{V_A+V_{CE}}\right]}$$

$$= -\frac{\frac{(V_{CC}-V_{CE})}{V_T}}{\left[1+\frac{V_{CC}-V_{CE}}{V_A+V_{CE}}\right]}$$

$$= \frac{(V_{CC}-V_{CE})/V_T}{\left[1+\frac{V_{CC}-V_{CE}}{V_A+V_{CE}}\right]}$$

Thus, Voltage gain $\boxed{A_v = -\frac{I_C R_C/V_T}{\left[1+\frac{R_C I_C}{V_A+V_{CE}}\right]} = \frac{(V_{CC}-V_{CE})/V_T}{\left[1+\frac{(V_{CC}-V_{CE})}{V_A+V_{CE}}\right]}}$

Step 5 of 6

Voltage gain A_v without early effect is $-\frac{(V_{CC}-V_{CE})}{V_T}$

Supply voltage V_{CC} is $+5\ \text{V}$

Collector emitter voltage V_{CE} is $+3\ \text{V}$

$$A_v = -\frac{(V_{CC}-V_{CE})}{V_T}$$

$$= -\frac{(5-3)\ \text{V}}{\left(25\times10^{-3}\right)\ \text{V}}$$

$$= -80\ \text{V/V}$$

Thus, Voltage gain A_v without early effect is $\boxed{-80\ \text{V/V}}$

Step 6 of 6

Early voltage V_A is $100\ \text{V}$

Voltage gain A_v with early effect is $-\frac{(V_{CC}-V_{CE})/V_T}{\left[1+\frac{(V_{CC}-V_{CE})}{V_A+V_{CE}}\right]}$

$$A_v = -\frac{(V_{CC}-V_{CE})/V_T}{\left[1+\frac{(V_{CC}-V_{CE})}{V_A+V_{CE}}\right]}$$

$$= -\frac{(5-3)\ \text{V}/\left(25\times10^{-3}\ \text{V}\right)}{\left[1+\frac{(5-3)\ \text{V}}{100\ \text{V}+3\ \text{V}}\right]}$$

$$= \frac{80}{1.019}$$

$$= 78.48\ \text{V/V}$$

Thus, Voltage gain A_v with early effect is $\boxed{-78.48\ \text{V/V}}$

Step 1 of 2

7.018E

Refer to the figure 7.24(a) in the textbook.

Collector current i_c and emitter current i_e are defined as follows:

$$i_c = g_m v_{be},$$

$$i_e = \frac{v_{be}}{r_e}.$$

Apply Kirchhoff's current law at the middle node.

$$i_e = i_b + i_c$$

$$i_b = i_e - i_c$$

Substitute $g_m v_{be}$ for i_c and $\frac{v_{be}}{r_e}$ for i_e in above equation.

$$i_b = \frac{v_{be}}{r_e} - g_m v_{be}$$

Refer to the equation (7.66) in the textbook $g_m v_{be} = i_b \beta$

$$i_b = \frac{v_{be}}{r_e} - g_m v_{be}$$

$$i_b = \frac{v_{be}}{r_e} - i_b \beta$$

$$i_b (1+\beta) = \frac{v_{be}}{r_e}$$

Step 2 of 2

Refer to the equation (7.75) in the textbook.

$$r_\pi = (1+\beta) r_e$$

Substitute $r_\pi = (1+\beta) r_e$ in the equation $i_b (1+\beta) = \frac{v_{be}}{r_e}$

$$i_b (1+\beta) = \frac{v_{be}}{r_e}$$

$$i_b = \frac{v_{be}}{r_e (1+\beta)}$$

$$i_b = \frac{v_{be}}{r_\pi}$$

Hence, the based current expression $\boxed{i_b = \frac{v_{be}}{r_\pi}}$ is proved.

Step 1 of 6

Refer to Figure 7.6 in the textbook.

The collector current I_C is,

$$\begin{aligned} I_C &= \frac{V_{CC} - V_C}{R_C} \\ &= \frac{5-2}{1\times10^3} \\ &= 3\ \text{mA} \end{aligned}$$

Thus, the collector current I_C is $\boxed{3\ \text{mA}}$.

Step 2 of 6

The small signal voltage gain is,

$$\begin{aligned} A_v &= -\frac{I_C R_C}{V_T} \\ &= \frac{-\left(3\times10^{-3}\right)\left(1\times10^3\right)}{25\times10^{-3}} \\ A_v &= -120 \end{aligned}$$

Thus, the small signal voltage gain A_v is $\boxed{-120}$.

First method: Using transistor exponential characteristics:

Determine the collector current i_C for $\Delta v_{BE} = 5\ \text{mV}$.

$$\begin{aligned} i_C &= \left(3\times10^{-3}\right)e^{\frac{\Delta v_{BE}}{V_T}} \\ &= \left(3\times10^{-3}\right)e^{\frac{5\times10^{-3}}{25\times10^{-3}}} \\ &= 3.66\ \text{mA} \end{aligned}$$

Determine the output voltage v_O.

$$\begin{aligned} v_O &= V_{CC} - i_C R_C \\ &= 5 - \left(3.66\times10^{-3}\right)\left(1\times10^3\right) \\ &= 1.34\ \text{V} \end{aligned}$$

The value of Δv_O for a change of $\Delta v_{BE} = +5\ \text{mV}$ is,

$$\begin{aligned} \Delta v_O &= v_O - V_C \\ &= 1.34 - 2 \\ &= -0.66\ \text{V} \end{aligned}$$

Thus, the value of Δv_O for a change of $\Delta v_{BE} = +5\ \text{mV}$ is $\boxed{-0.66\ \text{V}}$.

Step 3 of 6

Determine the collector current i_C for $\Delta v_{BE} = -5\ \text{mV}$.

$$\begin{aligned} i_C &= \left(3\times10^{-3}\right)e^{\frac{\Delta v_{BE}}{V_T}} \\ &= \left(3\times10^{-3}\right)e^{\frac{-5\times10^{-3}}{25\times10^{-3}}} \\ &= 2.46\ \text{mA} \end{aligned}$$

Determine the output voltage.

$$\begin{aligned} v_O &= V_{CC} - i_C R_C \\ &= 5 - \left(2.46\times10^{-3}\right)\left(1\times10^3\right) \\ &= 2.54\ \text{V} \end{aligned}$$

The value of Δv_O for a change of $\Delta v_{BE} = -5\ \text{mV}$ is,

$$\begin{aligned} \Delta v_O &= v_O - V_C \\ &= 2.54 - 2 \\ &= 0.54\ \text{V} \end{aligned}$$

Thus, the value of Δv_O for a change of $\Delta v_{BE} = -5\ \text{mV}$ is $\boxed{0.54\ \text{V}}$.

Step 4 of 6

Second method: Using small-signal voltage gain:

The value of Δv_O for a change of $\Delta v_{BE} = +5\ \text{mV}$ is,

$$\begin{aligned} \Delta v_O &= A_v\left(\Delta v_i\right) \\ &= A_v\left(\Delta v_{BE}\right) \\ &= -120\left(5\times10^{-3}\right) \\ &= -0.6\ \text{V} \end{aligned}$$

Thus, the value of Δv_O for a change of $\Delta v_{BE} = +5\ \text{mV}$ is $\boxed{-0.6\ \text{V}}$.

Step 5 of 6

The value of Δv_O for a change of $\Delta v_{BE} = -5\ \text{mV}$ is,

$$\begin{aligned} \Delta v_O &= A_v\left(\Delta v_i\right) \\ &= A_v\left(\Delta v_{BE}\right) \\ &= -120\left(-5\times10^{-3}\right) \\ &= 0.6\ \text{V} \end{aligned}$$

Thus, the value of Δv_O for a change of $\Delta v_{BE} = -5\ \text{mV}$ is $\boxed{0.6\ \text{V}}$.

Step 6 of 6

Summarize the results obtained in a table.
Table 1

Δv_{BE}	Δv_O using transistor exponential characteristic	Δv_O using small-signal voltage gain
−5 mV	−0.66 V	0.6 V
5 mV	0.54 V	−0.6 V

Thus, the change in the output voltage is determined using the two methods.

Step 1 of 4

Voltage across the collector V_C is,

$$V_C = V^- + I_C R_C$$

Power voltage V^- is $-10\ \text{V}$

Find the emitter current using ohm's law.

$$\begin{aligned} I_E &= \frac{10 - V_E}{R_E} \\ &= \frac{10 - 0.7}{10 \times 10^3} \\ &= 0.93 \times 10^{-3} \\ &= 0.93\ \text{mA} \end{aligned}$$

Step 2 of 4

For $\beta = 100$, the current gain α is,

$$\begin{aligned} \alpha &= \frac{\beta}{\beta + 1} \\ &= \frac{100}{100 + 1} \\ &= \frac{100}{101} \\ &= 0.99 \end{aligned}$$

Collector current is,

$$\begin{aligned} I_C &= \alpha I_E \\ &= (0.99)\left(0.93 \times 10^{-3}\right) \\ &= 0.92\ \text{mA} \end{aligned}$$

Collector resistance R_C is $7.5\ \text{k}\Omega$

$$\begin{aligned} V_C &= V^- + I_C R_C \\ &= -10 + \left(0.92 \times 10^{-3}\ \text{A}\right)\left(7.5 \times 10^3\ \Omega\right) \\ &= -3.1\ \text{V} \end{aligned}$$

Thus, collector voltage V_C is $\boxed{-3.1\ \text{V}}$.

Step 3 of 4

Emitter resistance r_e is,

$$\begin{aligned} r_e &= \frac{V_T}{I_E} \\ &= \frac{0.025}{0.92 \times 10^3} \\ &= 27.2\ \Omega \end{aligned}$$

The voltage gain A_v is,

$$A_v = \frac{\alpha R_C}{r_e}$$

Substitute corresponding values in the equation.

$$\begin{aligned} A_v &= \frac{0.99(7.5\ \text{k}\Omega)}{27.2\ \Omega} \\ &= 273\ \text{V/V} \end{aligned}$$

Thus, voltage gain A_v is $\boxed{273\ \text{V/V}}$.

Step 4 of 4

Peak input sine wave v_i is $10\ \text{mV}$

Peak amplitude of the output sine wave is,

$$v_o = A_v v_i$$

Substitute 273 for v_i and $10\ \text{mV}$ for v_i

$$\begin{aligned} v_o &= (273\ \text{V/V})(10\ \text{mV}) \\ &= 2.73\ \text{V} \end{aligned}$$

Thus, peak amplitude of the output sine wave corresponding to an input sine wave v_i of $10\ \text{mV}$ peak is

Step 1 of 11

(a)

Refer to the Figure 7.6 in the textbook

Power supply voltage V_{CC} is $+3\ \text{V}$

Write the expression for the theoretical maximum voltage gain.

$$A_v = -\frac{I_C R_C}{v_T}$$

Substitute $V_{CC} - V_{CE(\text{sat})}$ for $I_C R_C$.

$$\begin{aligned} A_v &= -\frac{V_{CC} - V_{CE\,SAT}}{V_T} \\ &= -\frac{3 - 0.3}{0.025} \\ &= -108 \end{aligned}$$

Thus, Theoretical maximum voltage gain is $\boxed{-108\ \text{V/V}}$

Step 2 of 11

(b)

Write the expression for the voltage gain.

$$A_v = \frac{V_{CC} - V_{CE}}{V_T}$$

Substitute $-60\ \text{V/V}$ for A_v .

$$\begin{aligned} A_v &= -\frac{V_{CC} - V_{CE}}{V_T} \\ -60\ \text{V/V} &= -\frac{(3 - V_{CE})\ \text{V}}{(25\times10^{-3})\ \text{V}} \\ 1.5\ \text{V} &= 3 - V_{CE} \\ V_{CE} &= 1.5\ \text{V} \end{aligned}$$

Thus, In order to get voltage gain of $-60\ \text{V/V}$, V_{CE} of amplifier should be biased to a voltage at $\boxed{1.5\ \text{V}}$.

Step 3 of 11

(c)

The DC collector current I_C is $0.5\ \text{mA}$

Collector emitter voltage V_{CE} is $1.5\ \text{V}$

From the Fig. 7.6, $V_{CC} - V_{CE} = I_C R_C$. Therefore,

$$\begin{aligned} V_{CC} - V_{CE} &= I_C R_C \\ (3 - 1.5)\ \text{V} &= (0.5\ \text{mA}) R_C \\ R_C &= \frac{1.5\ \text{V}}{0.5\ \text{mA}} \\ R_C &= 3\ \text{k}\Omega \end{aligned}$$

Thus Collector resistance R_C is $\boxed{3\ \text{k}\Omega}$.

Step 4 of 11

(d)

Relation between saturation current I_S and DC collector current I_C is $I_C = I_S e^{V_{BE}/V_T}$

Saturation current I_S is $10^{-15}\ \text{A}$

Collector current I_C is $0.5\ \text{mA}$

$$\begin{aligned} I_C &= I_S e^{V_{BE}/V_T} \\ 0.5\ \text{mA} &= (10^{-15}\ \text{A}) e^{V_{BE}/(25\times10^{-3}\ \text{V})} \\ (5\times10^{11}) &= e^{V_{BE}/(25\times10^{-3}\ \text{V})} \end{aligned}$$

Apply natural log on both sides to the above equation

$$\begin{aligned} \ln(5\times10^{11}) &= \ln e^{\frac{V_{BE}}{(25\times10^{-3})}} \\ 26.94 &= \frac{V_{BE}}{(25\times10^{-3})} \\ V_{BE} &= 0.67\ \text{V} \end{aligned}$$

Thus, Base-Emitter voltage of BJT is $\boxed{0.67\ \text{V}}$.

Step 5 of 11

(e)

Peak amplitude of sine wave signal v_{be} is $5\ \text{mV}$

Voltage gain A_v is $-\frac{V_{CC} - V_{CE}}{V_T}$

$$\begin{aligned} A_v &= -\frac{V_{CC} - V_{CE}}{V_T} \\ &= -\frac{(3 - 1.5)\ \text{V}}{(25\times10^{-3})\ \text{V}} \\ &= -60\ \frac{\text{V}}{\text{V}} \end{aligned}$$

Step 6 of 11

The relation between output voltage signal v_{ce} , A_v and input sine wave signal v_{be} is,

$$v_{ce} = A_v v_{be}$$

Substitute the values in the equation.

$$\begin{aligned} v_{ce} &= \left(-60\ \frac{\text{V}}{\text{V}}\right)(5\ \text{mV}) \\ &= 0.3\ \text{V} \end{aligned}$$

Thus, Output voltage signal v_{ce} superimposed on V_{CE} around the bias point is $\boxed{0.3\ \text{V}}$.

Step 7 of 11

(f)

Output voltage signal v_{ce} superimposed on V_{CE} around the bias point is the voltage drop across the collector resistance R_C due to signal current i_c which is superimposed on the dc bias current I_C

Hence relation between V_{CE} , i_c and R_C is $v_{ce} = i_c R_C$

Output voltage signal v_{ce} superimposed on V_{CE} around the bias point is $0.3\ \text{V}$

$$\begin{aligned} v_{ce} &= i_c R_C \\ 0.3\ \text{V} &= i_c\,(1000\ \Omega) \\ i_c &= \frac{0.3\ \text{V}}{1000\ \Omega} \\ i_c &= 0.3\ \text{mA} \end{aligned}$$

Thus, Signal current i_c that will be superimposed on the dc bias current I_C is $\boxed{0.3\ \text{mA}}$

Step 8 of 11

(g)

Common emitter current gain β is 100

DC collector current I_C is $0.5\ \text{mA}$

Relation between I_C , I_B and β is $\beta = \frac{I_C}{I_B}$

$$\begin{aligned} \beta &= \frac{I_C}{I_B} \\ 100 &= \frac{0.5\ \text{mA}}{I_B} \\ I_B &= \frac{0.5\ \text{mA}}{100} \\ I_B &= 50\ \mu\text{A} \end{aligned}$$

Thus, DC bias current I_B is $\boxed{50\ \mu\text{A}}$

Step 9 of 11

Relation between signal current i_b , signal current i_c and β is $\beta = \frac{i_c}{i_b}$

$$\begin{aligned} \beta &= \frac{i_c}{i_b} \\ 100 &= \frac{0.3\ \text{mA}}{i_b} \\ i_b &= \frac{0.3\ \text{mA}}{100} \\ i_b &= 3\ \mu\text{A} \end{aligned}$$

Thus, Signal current i_b that will be superimposed on the dc base current i_c is $\boxed{3\ \mu\text{A}}$

Step 10 of 11

(h)

Small signal input resistance r_i of the amplifier is $\frac{v_{be}}{i_b}$

Amplitude of i_b is $3\ \mu\text{A}$

Amplitude of v_{be} is $5\ \text{mV}$

$$\begin{aligned} r_i &= \frac{v_{be}}{i_b} \\ &= \frac{5\ \text{mV}}{3\ \mu\text{A}} \\ &= 1.67\ \text{k}\Omega \end{aligned}$$

Thus, Input resistance r_i of the amplifier is $\boxed{1.67\ \text{k}\Omega}$

Step 11 of 11

(i)

Sketch the waveforms for v_{BE}, v_{CE}, i_C, and i_B .

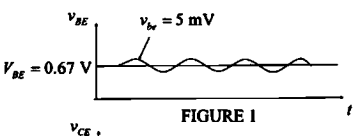


FIGURE 1

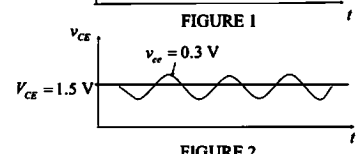


FIGURE 2

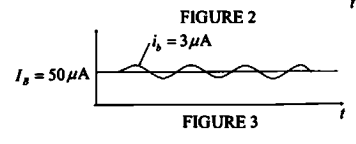


FIGURE 3

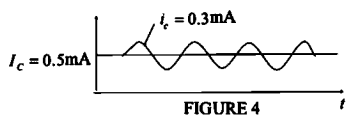


FIGURE 4

Thus, the graphs are sketched.

7.020E

We don't have the solution to this problem yet.

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Step 1 of 3

Write the expression of voltage gain.

$$A_v = \frac{-I_C R_C}{V_T} \quad \text{..... (1)}$$

It is known that, the voltage gain A_v is ratio of output voltage to input voltage.

$$A_v = \frac{\Delta v_O}{\Delta v_{BE}} \quad \text{..... (2)}$$

Equate equations (1) and (2).

$$\frac{\Delta v_O}{\Delta v_{BE}} = -\frac{I_C R_C}{V_T} \quad \text{..... (3)}$$

Step 2 of 3

Write the expression of output voltage.

$$\begin{aligned} \Delta v_O &= -\Delta i_C R_C \\ &= -\left(g_m \Delta v_{BE}\right) R_C \qquad \left(\text{Since } \Delta i_C = g_m \Delta V_{BE}\right) \end{aligned}$$

$$\Delta v_O = -\left(g_m \Delta v_{BE}\right) R_C \quad \text{..... (4)}$$

Substitute equation (4) in equation (3).

$$\frac{-\left(g_m \Delta v_{BE}\right) R_C}{\Delta v_{BE}} = -\frac{I_C R_C}{V_T}$$

$$g_m R_C = \frac{I_C R_C}{V_T}$$

$$g_m = \frac{I_C R_C}{V_T R_C}$$

$$g_m = \frac{I_C}{V_T}$$

Therefore, the expression for transconductance $\left(g_m\right)$ is $\boxed{\frac{I_C}{V_T}}$.

Step 3 of 3

Determine the value of transconductance at $I_C = 0.5\ \text{mA}$.

$$g_m = \frac{I_C}{V_T}$$

Substitute 0.5 mA for I_C and 25 mV for V_T.

$$\begin{aligned} g_m &= -\frac{I_C}{V_T} \\ &= -\frac{0.5\ \text{mA}}{25\ \text{mV}} \\ &= 0.02\ \text{A/V} \end{aligned}$$

Thus, Transistor transconductance g_m for $I_C = 0.5\ \text{mA}$ is $\boxed{0.02\ \text{A/V}}$.

Step 1 of 6

7.021E

Refer to the Equivalent circuit of common source amplifier of Figure 7.35 is shown in the textbook.

The input resistance R_{in} is $\boxed{\infty}$.

The gain A_{vo} is,

$$A_{vo} = -g_m R_D \quad \text{..... (1)}$$

Here,

g_m is transconductance,

R_D is drain resistance.

Substitute $\frac{2I_D}{V_{OV}}$ for g_m in the equation (1).

$$A_{vo} = -\frac{2I_D}{V_{OV}} R_D \quad \text{..... (2)}$$

Step 2 of 6

Substitute $0.25\ \text{mA}$ for I_D , $0.25\ \text{V}$ for V_{OV} , $20\ \text{k}\Omega$ for R_D in the equation (2).

$$\begin{aligned} A_{vo} &= -\frac{2I_D}{V_{OV}} R_D \\ &= -\left[\frac{2(0.25\ \text{mA})}{0.25\ \text{V}}\right][20\ \text{k}\Omega] \\ &= -40\ \text{V/V} \end{aligned}$$

Thus, voltage gain A_{vo} is $\boxed{-40\ \text{V/V}}$.

Step 3 of 6

Drain resistance is equal to the output resistance.

$$R_o = R_D$$

Substitute $20\ \text{k}\Omega$ for R_D in equation.

Therefore, the output resistance R_o of the CS amplifier is $\boxed{20\ \text{k}\Omega}$.

Step 4 of 6

The voltage gain A_v is,

$$A_v = A_{vo}\frac{R_L}{R_L + R_o} \quad R_L \text{ is } 20\ \text{k}\Omega$$

Substitute $20\ \text{k}\Omega$ for R_L , $20\ \text{k}\Omega$ for R_o and $-40\ \text{V/V}$ for A_{vo} in the equation.

$$\begin{aligned} A_v &= [-40\ \text{V/V}]\left[\frac{20\ \text{k}\Omega}{20\ \text{k}\Omega + 20\ \text{k}\Omega}\right] \\ &= -20\ \text{V/V} \end{aligned}$$

Therefore, voltage gain A_v is $\boxed{-20\ \text{V/V}}$.

Step 5 of 6

The overall voltage gain G_v is,

$$\begin{aligned} G_v &= -g_m(R_D \parallel R_L) \\ &= -\frac{2I_D}{V_{OV}}(R_D \parallel R_L) \end{aligned}$$

Substitute $0.25\ \text{mA}$ for I_D , $0.25\ \text{V}$ for V_{OV} , $20\ \text{k}\Omega$ for R_D , $20\ \text{k}\Omega$ for R_L in the equation.

$$\begin{aligned} G_v &= -\left(\frac{2(0.25\ \text{mA})}{0.25\ \text{V}}\right)\left(\frac{20\ \text{k}\Omega \times 20\ \text{k}\Omega}{20\ \text{k}\Omega + 20\ \text{k}\Omega}\right) \\ &= -(2\ \text{mA/V})(10\ \text{k}\Omega) \\ &= -20\ \text{V/V} \end{aligned}$$

Therefore, the overall voltage gain G_v is $\boxed{-20\ \text{V/V}}$.

Step 6 of 6

Peak of the input sine wave signal is 10% of $2V_{OV}$ and hence peak of the input signal is,

$$\begin{aligned} v_{sig} &= \left(\frac{10}{100}\right)(2(0.25\ \text{V})) \\ &= 50\ \text{mV} \end{aligned}$$

Thus, maximum amplitude of input sine wave signal is $50\ \text{mV}$.

Overall voltage gain G_v is,

$$\begin{aligned} G_v &= \frac{v_o}{v_{sig}} \\ (20\ \text{V/V}) &= \frac{v_o}{50\ \text{mV}} \\ v_o &= 1\ \text{V} \end{aligned}$$

Therefore, output voltage v_o is $\boxed{1\ \text{V}}$.

7.021P

We don't have the solution to this problem yet.

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Step 1 of 9

7.022E

Refer to the Example 7.8 in the textbook for data.

Base current is reduced to half of the value of base current of the amplifier in Example 7.8. Hence, collector current is halved for new amplifier circuit since common emitter current gain β is constant.

Collector current of the amplifier circuit in example 7.8 is $1\ \text{mA}$. Hence, the collector current of new designed amplifier circuit is $0.5\ \text{mA}$.

Step 2 of 9

Determine transconductance g_m .

The transconductance g_m is,

$$g_m = \frac{I_C}{V_T}$$

Substitute $0.5\ \text{mA}$ for I_C , $25\ \text{mV}$ for V_T in equation.

$$\begin{aligned} g_m &= \frac{0.5\ \text{mA}}{25\ \text{mV}} \\ &= 20\ \text{mA/V} \end{aligned}$$

Thus, transconductance g_m is $20\ \text{mA/V}$

Step 3 of 9

Determine resistance r_π .

The resistance r_π is,

$$r_\pi = \frac{\beta}{g_m}$$

Substitute 100 for β , $20\ \text{mA/V}$ for g_m in equation.

$$\begin{aligned} r_\pi &= \frac{100}{20\ \text{mA/V}} \\ &= 5\ \text{k}\Omega \end{aligned}$$

Determine input resistance R_{in} .

$$R_{in} = r_\pi$$

Substitute $5\ \text{k}\Omega$ for r_π in equation.

$$R_{in} = 5\ \text{k}\Omega$$

Therefore, the input resistance R_{in} is $\boxed{5\ \text{k}\Omega}$.

Step 4 of 9

Collector resistance is doubled. New collector resistance is $10\ \text{k}\Omega$.

Determine the voltage gain A_{vo} .

The voltage gain A_{vo} is,

Substitute $20\ \text{mA/V}$ for g_m and $10\ \text{k}\Omega$ for R_C in equation.

$$\begin{aligned} A_{vo} &= -(20\ \text{mA/V})(10\ \text{k}\Omega) \\ &= -200\ \text{V/V} \end{aligned}$$

Therefore, the voltage gain A_{vo} is $\boxed{-200\ \text{V/V}}$.

Step 5 of 9

Determine the output resistance R_o .

Output resistance R_o is equal to collector resistance R_C .

$$R_o = R_C$$

Substitute $10\ \text{k}\Omega$ for R_C in equation.

$$R_o = 10\ \text{k}\Omega$$

Therefore, the output resistance R_o is $\boxed{10\ \text{k}\Omega}$.

Step 6 of 9

Determine the voltage gain A_v .

Voltage gain A_v is given by,

Substitute $20\ \text{mA/V}$ for g_m , $10\ \text{k}\Omega$ for R_C , $5\ \text{k}\Omega$ for R_L in equation.

$$\begin{aligned} A_v &= -[20\ \text{mA/V}][(10\ \text{k}\Omega) \| (5\ \text{k}\Omega)] \\ &= -[20\ \text{mA/V}][3.33\ \text{k}\Omega] \\ &= -66.7\ \text{V/V} \end{aligned}$$

Therefore, the voltage gain A_v is $\boxed{-66.7\ \text{V/V}}$

Step 7 of 9

Determine the overall voltage gain G_v .

$$G_v = \left[\frac{R_{in}}{R_{in} + R_{stg}}\right][A_v]$$

Substitute $5\ \text{k}\Omega$ for R_{in} , $5\ \text{k}\Omega$ for R_{stg} , $-66.7\ \text{V/V}$ for A_v in equation.

$$\begin{aligned} G_v &= \left[\frac{5\ \text{k}\Omega}{5\ \text{k}\Omega + 5\ \text{k}\Omega}\right][-66.7\ \text{V/V}] \\ &= -33.3\ \text{V/V} \end{aligned}$$

Therefore, the overall voltage gain G_v is $\boxed{-33.3\ \text{V/V}}$.

Step 8 of 9

Determine signal voltage $\hat{v}_{stg}$.

$$\hat{v}_{stg} = \left(\frac{R_{in} + R_{stg}}{R_{in}}\right)\hat{v}_\pi$$

Substitute $5\ \text{mV}$ for v_π , $5\ \text{k}\Omega$ for R_{in} , $5\ \text{k}\Omega$ for R_{stg} in equation.

$$\begin{aligned} \hat{v}_{stg} &= \left(\frac{5\ \text{k}\Omega + 5\ \text{k}\Omega}{5\ \text{k}\Omega}\right)(5\ \text{mV}) \\ &= 10\ \text{mV} \end{aligned}$$

Therefore, the signal voltage $\hat{v}_{stg}$ is $\boxed{10\ \text{mV}}$.

Step 9 of 9

Determine the output $\hat{v}_o$.

$$\hat{v}_o = G_v \hat{v}_{stg}$$

Substitute $33.3\ \text{V/V}$ for G_v and $10\ \text{mV}$ for $\hat{v}_{stg}$ in equation.

$$\begin{aligned} \hat{v}_o &= (33.3\ \text{V/V})(10\ \text{mV}) \\ &= 0.33\ \text{V} \end{aligned}$$

Therefore, the output voltage $\hat{v}_o$ is $\boxed{0.33\ \text{V}}$.

7.022P

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Step 1 of 5

Refer to the exercise 7.21 in the textbook for common source MOSFET amplifier.

The input signal v_{sig} is equal to gate source voltage v_{gs} .

Thus, $v_{gs} = v_{sig}$.

Substitute $50\ \text{mV}$ for v_{sig} in the above equation.

$$v_{gs} = 50\ \text{V}$$

Hence, the gate source voltage v_{gs} of modified circuit is $50\ \text{mV}$.

Step 2 of 5

Determine source resistance R_s .

Common source amplifier with a source resistance has input voltage v_i equal to $0.2\ \text{V}$

Gate source voltage for common source amplifier with a source resistance is,

$$v_{gs} = \frac{v_i}{1 + g_m R_s} \quad \text{..... (1)}$$

The Transconductance g_m obtained from Exercise 7.21 is $2\ \text{mA/V}$.

Substitute $2\ \text{mA/V}$ for g_m , $50\ \text{mV}$ for v_{gs} , and $0.2\ \text{V}$ for v_i in equation (1).

$$v_{gs} = \frac{v_i}{1 + g_m R_s}$$

$$50\ \text{mV} = \frac{0.2\ \text{V}}{1 + (2\ \text{mA/V})(R_s)}$$

$$1 + (2\ \text{mA/V})(R_s) = 4$$

$$R_s = 1.5\ \text{k}\Omega$$

Therefore, the source resistance R_s is $\boxed{1.5\ \text{k}\Omega}$.

Step 3 of 5

Determine the voltage gain $\frac{v_o}{v_i}$.

$$\frac{v_o}{v_i} = -\frac{g_m (R_D \| R_L)}{1 + g_m R_s}$$

Substitute $2\ \text{mA/V}$ for g_m , $20\ \text{k}\Omega$ for R_D , $20\ \text{k}\Omega$ for R_L , $1.5\ \text{k}\Omega$ for R_s in equation.

$$\frac{v_o}{v_i} = -\frac{(2\ \text{mA/V})(20\ \text{k}\Omega \| 20\ \text{k}\Omega)}{1 + (2\ \text{mA/V})(1.5\ \text{k}\Omega)}$$

$$= -\frac{(2\ \text{mA/V})}{1 + (2\ \text{mA/V})(1.5\ \text{k}\Omega)} \left(\frac{20\ \text{k}\Omega \times 20\ \text{k}\Omega}{20\ \text{k}\Omega + 20\ \text{k}\Omega} \right)$$

$$= -\left(\frac{2 \times 10^{-3}}{1 + 3} \right) (10 \times 10^{3})$$

$$= -5$$

Therefore, the voltage $\frac{v_o}{v_i}$ is $-5\ \text{V/V}$.

Step 4 of 5

Determine the overall voltage gain G_v .

$$G_v = \frac{v_o}{v_{sig}}$$

$$= \left(\frac{v_i}{v_{sig}} \right) \left(\frac{v_o}{v_i} \right)$$

For Common source amplifier with a source resistance has voltage gain A_v is equal to total gain G_v since $v_i = v_{sig}$.

$$G_v = \frac{v_i}{v_{sig}} \frac{v_o}{v_i}$$

$$= \frac{v_o}{v_i}$$

Thus, overall voltage gain G_v is $\boxed{-5\ \text{V/V}}$.

Step 5 of 5

Calculate peak signal output.

$$G_v = \frac{v_o}{v_i}$$

Substitute $0.2\ \text{V}$ for v_i , $-5\ \text{V/V}$ for G_v in equation.

$$-5\ \text{V/V} = \frac{v_o}{0.2\ \text{V}}$$

$$v_o = 1\ \text{V/V}$$

Therefore, the peak signal at the output is $\boxed{1\ \text{V/V}}$.

Step 1 of 3

7.023P

Consider the instantaneous drain current i_D from equation (7.28) in the textbook is,

$$i_D = \frac{1}{2}k_n(V_{GS}-V_t)^2 + k_n(V_{GS}-V_t)v_{gs} + \frac{1}{2}k_n v_{gs}^2 \quad \text{..... (1)}$$

Substitute $V_{gs}\sin\omega t$ in place of v_{gs} in the equation (1)

$$\begin{aligned} i_D &= \frac{1}{2}k_n(V_{GS}-V_t)^2 + k_n(V_{GS}-V_t)V_{gs}\sin\omega t + \frac{1}{2}k_n V_{gs}^2\sin^2(\omega t) \\ &= \frac{1}{2}k_n(V_{GS}-V_t)^2 + k_n(V_{GS}-V_t)V_{gs}\sin\omega t + \frac{1}{2}k_n V_{gs}^2\left[\frac{1}{2}-\frac{1}{2}\cos 2\omega t\right] \\ &= \frac{1}{2}k_n(V_{GS}-V_t)^2 + \frac{1}{4}k_n V_{gs}^2 + k_n(V_{GS}-V_t)V_{gs}\sin\omega t + \frac{1}{2}k_n V_{gs}^2\left[-\frac{1}{2}\cos 2\omega t\right] \end{aligned}$$

Step 2 of 3

Amplitude of signal having frequency 2ω is $\frac{1}{4}k_n V_{gs}^2$

Amplitude of signal having frequency ω is $k_n(V_{GS}-V_t)V_{gs}$

Second harmonic distortion is defined as ratio of signal at frequency 2ω to that at frequency ω

$$\begin{aligned} \text{second-harmonic distortion} &= \frac{\text{Amplitude of signal having frequency } 2\omega}{\text{Amplitude of signal having frequency } \omega}\times 100 \\ &= \frac{\frac{1}{4}k_n V_{gs}^2}{k_n(V_{GS}-V_t)V_{gs}}\times 100 \\ &= \frac{\frac{1}{4}V_{gs}}{(V_{GS}-V_t)}\times 100 \\ &= \frac{1}{4}\frac{V_{gs}}{V_{OV}}\times 100 \end{aligned}$$

Thus, percent second harmonic distortion is $\frac{1}{4}\frac{V_{gs}}{V_{OV}}\times 100$

Step 3 of 3

In a particular operation Gate source voltage V_{gs} is 10 mV

Second harmonic distortion is 1%

$$\text{second-harmonic distortion} = \frac{1}{4}\frac{V_{gs}}{V_{OV}}\times 100$$

$$1\% = \frac{1}{4}\frac{10\text{ mV}}{V_{OV}}\times 100$$

$$V_{OV} = \frac{10\times 10^{-3}\times 100}{4\times 1}$$

$$V_{OV} = 0.25\text{ V}$$

Thus, minimum over drive voltage at which the transistor should be operated so that the second harmonic distortion is kept so less than 1% is $\boxed{0.25\text{ V}}$

Step 1 of 7

7.024E

Refer to Figure 7.38 (a) and Figure 7.38 (b) the textbook.

Consider inclusion of the resistance R_e in the emitter for CE amplifier. Then, the voltage v_π is,

$$v_\pi = v_i\left(\frac{r_e}{r_e + R_e}\right) \quad \text{..... (1)}$$

The voltage v_i is,

$$v_i = v_{sig}\left(\frac{R_{in}}{R_{in} + R_{sig}}\right) \quad \text{..... (2)}$$

Substitute equation (2) in equation (1).

$$v_\pi = v_{sig}\left(\frac{R_{in}}{R_{in} + R_{sig}}\right)\left(\frac{r_e}{r_e + R_e}\right)$$

$$\frac{v_\pi\left(R_{in} + R_{sig}\right)\left(r_e + R_e\right)}{R_{in} r_e} = v_{sig}$$

$$v_{sig} = \frac{v_\pi\left(R_{in} + R_{sig}\right)\left(r_e + R_e\right)}{R_{in} r_e}$$

Step 2 of 7

The maximum input signal $\hat{v}_{sig}$ is,

Substitute $\hat{v}_\pi$ for v_π, $\hat{v}_{sig}$ for v_{sig} in the equation

$$\hat{v}_{sig} = \frac{\hat{v}_\pi\left(R_{in} + R_{sig}\right)\left(r_e + R_e\right)}{R_{in} r_e}$$

$$= \hat{v}_\pi\left[\frac{R_{in} r_e + R_{in} R_e + R_{sig} r_e + R_{sig} R_e}{R_{in} r_e}\right]$$

$$= \hat{v}_\pi\left[1 + \frac{R_e}{r_e} + \frac{R_{sig}}{R_{in}} + \frac{R_{sig} R_e}{R_{in} r_e}\right]$$

$$\hat{v}_{sig} = \hat{v}_\pi\left[1 + \frac{R_e}{r_e} + \frac{R_{sig}}{R_{in}}\left(1 + \frac{R_e}{r_e}\right)\right] \quad \text{..... (3)}$$

Step 3 of 7

Refer to the equation 7.107 in the textbook .

$$R_{in} = (1+\beta)\left(r_e + R_e\right) \quad \text{..... (4)}$$

Relation between r_e and r_π is $r_\pi = (1+\beta) r_e$.

Substitute equation (4) in the equation (3).

$$\hat{v}_{sig} = \hat{v}_\pi\left[1 + \frac{R_e}{r_e} + \frac{R_{sig}}{(1+\beta)\left(r_e + R_e\right)}\left(1 + \frac{R_e}{r_e}\right)\right]$$

$$= \hat{v}_\pi\left[1 + \frac{R_e}{r_e} + \frac{R_{sig}}{(1+\beta) r_e}\right] \qquad \left(\text{since } r_\pi = (1+\beta) r_e\right)$$

$$\hat{v}_{sig} = \hat{v}_\pi\left[1 + \frac{R_e}{r_e} + \frac{R_{sig}}{r_\pi}\right] \quad \text{..... (5)}$$

Therefore, the maximum input signal $\hat{v}_{sig}$ is $\boxed{\hat{v}_\pi\left[1 + \frac{R_e}{r_e} + \frac{R_{sig}}{r_\pi}\right]}$.

Step 4 of 7

Determine transconductance g_m .

$$g_m = \frac{I_C}{V_T}$$

Substitute $0.5\ \text{mA}$ for I_C , $25\ \text{mV}$ for V_T in equation.

$$g_m = \frac{0.5\ \text{mA}}{25\ \text{mV}}$$

$$= \frac{1}{50}$$

Resistance r_e is,

$$r_e = \frac{1}{g_m}$$

$$= \frac{1}{\left(\frac{1}{50}\right)}$$

$$= 50$$

Hence, the resistance r_e is $50\ \Omega$.

Step 5 of 7

Relation between r_e and r_π is,

$$r_\pi = (1+\beta) r_e$$

Substitute $50\ \Omega$ for r_e , 100 for β in equation.

$$r_\pi = (1+\beta) r_e$$

$$= (1+100)50$$

$$= 5050\ \Omega$$

Therefore, the resistance r_π is $5050\ \Omega$.

Substitute $100\ \text{mV}$ for $\hat{v}_{sig}$, $10\ \text{k}\Omega$ for R_{sig} , $10\ \text{mV}$ for $\hat{v}_\pi$, $50\ \Omega$ for r_e , $5050\ \Omega$ for r_π in the equation (5).

$$\hat{v}_{sig} = \hat{v}_\pi\left[1 + \frac{R_e}{r_e} + \frac{R_{sig}}{r_\pi}\right]$$

$$100\ \text{mV} = 10\ \text{mV}\left[1 + \frac{R_e}{50\ \Omega} + \frac{10\ \text{k}\Omega}{5050\ \Omega}\right]$$

$$\frac{R_e}{50\ \Omega} + \frac{10\ \text{k}\Omega}{5050\ \Omega} = 9$$

$$R_e = 350.9\ \Omega$$

Therefore, the resistance R_e is $\boxed{350\ \Omega}$.

Step 6 of 7

Determine input resistance R_{in} .

$$R_{in} = (1+\beta)\left(r_e + R_e\right)$$

Substitute 100 for β , $50\ \Omega$ for r_e , $350\ \Omega$ for R_e in equation.

$$R_{in} = (1+100)(50+350)$$

$$= 40.4\ \text{k}\Omega$$

Therefore, the input resistance R_{in} is $\boxed{40.4\ \text{k}\Omega}$.

Step 7 of 7

Determine the overall voltage gain G_v .

$$G_v = -\beta\frac{R_C}{R_{sig} + (1+\beta)\left(r_e + R_e\right)}$$

Substitute $10\ \text{k}\Omega$ for R_C , $10\ \text{k}\Omega$ for R_{sig} , 100 for β , $50\ \Omega$ for r_e , $350\ \Omega$ for R_e in equation.

$$G_v = (-100)\left(\frac{10\ \text{k}\Omega}{10\ \text{k}\Omega + (1+100)(50+350)}\right)$$

$$= -100\left(\frac{10\ \text{k}\Omega}{40.4\ \text{k}\Omega}\right)$$

$$= -19.8\ \text{V/V}$$

Therefore, the overall voltage gain G_v is $\boxed{-19.8\ \text{V/V}}$.

Step 1 of 8

To ensure saturation region operation, we must have $V_{DS} > V_{OV}$.

DC bias current I_D for NMOS transistor operated in saturation region is,

$$i_D = \frac{1}{2}k_n V_{OV}^2$$

Substitute $10\ \frac{\text{mA}}{\text{V}^2}$ for k_n , $0.2\ \text{V}$ for V_{OV} in the equation.

$$\begin{aligned} I_D &= \frac{1}{2}\left(10\ \text{mA/V}^2\right)(0.2)^2 \\ &= \frac{1}{2}\left(10\times10^{-3}\right)(0.04) \\ &= 0.2\ \text{mA} \end{aligned}$$

DC bias current I_D for NMOS transistor is $\boxed{0.2\ \text{mA}}$.

Step 2 of 8

Input signal v_{gs} is superimposed on V_{GS} , then the total instantaneous gate-to-source voltage is $v_{GS} = V_{GS} + v_{gs}$

Total instantaneous drain current i_D is,

$$\begin{aligned} i_D &= \frac{1}{2}k_n\left(V_{GS} + v_{gs} - V_t\right)^2 \\ &= \frac{1}{2}k_n\left[\left(V_{GS} - V_t\right)^2 + 2\left(V_{GS} - V_t\right)v_{gs} + v_{gs}^2\right] \\ &= \frac{1}{2}k_n\left(V_{GS} - V_t\right)^2 + k_n\left(V_{GS} - V_t\right)v_{gs} + \frac{1}{2}k_n v_{gs}^2 \end{aligned}$$

$$i_D = \frac{1}{2}k_n\left(V_{OV}\right)^2 + k_n\left(V_{OV}\right)v_{gs} + \frac{1}{2}k_n v_{gs}^2 \quad \ldots\ldots (1)$$

Step 3 of 8

Substitute $0.02\ \text{V}$ for v_{gs} in the equation (1).

Instantaneous drain current i_D is,

$$\begin{aligned} i_D &= \frac{1}{2}k_n\left(V_{OV}\right)^2 + k_n\left(V_{OV}\right)v_{gs} + \frac{1}{2}k_n v_{gs}^2 \\ &= \frac{1}{2}\left(10\ \frac{\text{mA}}{\text{V}^2}\right)(0.2)^2 + \left(10\ \frac{\text{mA}}{\text{V}^2}\right)(0.2)(0.02) + \frac{1}{2}\left(10\ \frac{\text{mA}}{\text{V}^2}\right)(0.02)^2 \\ &= 0.2\ \text{mA} + 0.04\ \text{mA} + 2\ \mu\text{A} \\ &\simeq 0.242\ \text{mA} \end{aligned}$$

Instantaneous total drain current i_D for v_{gs} equal to $0.02\ \text{V}$ is $\boxed{0.242\ \text{mA}}$.

Step 4 of 8

Incremental in collector current is evaluated by subtracting the dc bias current I_D from total collector current i_D

$$\begin{aligned} \text{incremental current}\left(i_d\right) &= \left(i_D - I_D\right) \\ &= 0.242\ \text{mA} - 0.2\ \text{mA} \\ &= 0.042\ \text{mA} \end{aligned}$$

Thus, incremental current for v_{gs} equal to $0.02\ \text{V}$ is $\boxed{0.042\ \text{mA}}$.

Step 5 of 8

v_{gs} Superimposed on V_{GS} is $-0.02\ \text{V}$

Instantaneous drain current i_D is,

$$\begin{aligned} i_D &= \frac{1}{2}k_n\left(V_{OV}\right)^2 + k_n\left(V_{DV}\right)v_{gs} + \frac{1}{2}k_n v_{gs}^2 \\ &= \frac{1}{2}\left(10\ \frac{\text{mA}}{\text{V}^2}\right)(0.2)^2 + \left(10\ \frac{\text{mA}}{\text{V}^2}\right)(0.2)(-0.02) + \frac{1}{2}\left(10\ \frac{\text{mA}}{\text{V}^2}\right)(-0.02)^2 \\ &= 0.2\ \text{mA} - 0.04\ \text{mA} + 2\ \mu\text{A} \\ &\simeq 0.162\ \text{mA} \end{aligned}$$

Instantaneous total drain current i_D for v_{gs} equal to $-0.02\ \text{V}$ is $\boxed{0.162\ \text{mA}}$.

Step 6 of 8

Incremental in collector current is evaluated by subtracting the dc bias current I_D from total collector current i_D

$$\begin{aligned} \text{incremental current}\left(i_d\right) &= \left(i_D - I_D\right) \\ &= 0.162\ \text{mA} - 0.2\ \text{mA} \\ &= -0.038\ \text{mA} \end{aligned}$$

Thus, incremental current for v_{gs} equal to $-0.02\ \text{V}$ is $\boxed{-0.04\ \text{mA}}$.

Step 7 of 8

Determine the Trans-conductance as follows:

$$\begin{aligned} g_m &= \frac{i_d}{v_{gs}} \\ &= \frac{0.042\ \text{mA}}{0.02\ \text{V}} \\ &= 2.1\ \frac{\text{mA}}{\text{V}} \end{aligned}$$

Thus, g_m of FET at the bias point is $\boxed{2.1\ \frac{\text{mA}}{\text{V}}}$.

Step 8 of 8

Consider the equation (7.33) in the textbook is given by,

$$g_m = k_n V_{OV}$$

Substitute $10\ \frac{\text{mA}}{\text{V}^2}$ for k_n , $0.2\ \text{V}$ for V_{OV} in the equation.

$$\begin{aligned} g_m &= \left(10\times10^{-3}\right)(0.2) \\ &= 2\ \frac{\text{mA}}{\text{V}} \end{aligned}$$

Thus, g_m of FET at the bias point is $\boxed{2\ \frac{\text{mA}}{\text{V}}}$

Step 1 of 3

Refer to Figure 7.39 (a) in the textbook.

Consider input resistance of common gate amplifier is equal to the signal source resistance.

Hence,

$$R_{in} = R_{sig}$$
$$= 100\,\Omega$$

Input resistance R_{in} in terms of transconducantce g_m is,

$$R_{in} = \frac{1}{g_m}$$

Step 2 of 3

Calculation of drain current I_D is as follows:

Over drive voltage V_{OV} is $0.20\ \text{V}$

$$R_{in} = \frac{1}{g_m}$$
$$= \frac{1}{\left(\frac{2I_D}{V_{OV}}\right)} \qquad \left(\text{since } g_m = \frac{2I_D}{V_{OV}}\right)$$
$$= \frac{V_{OV}}{2I_D}$$
$$I_D = \frac{V_{OV}}{2R_{in}}$$

Substitute $0.20\ \text{V}$ for V_{OV} , $100\,\Omega$ for R_{in} in equation.

$$I_D = \frac{0.20}{2(100)}$$
$$= 1\ \text{mA}$$

Therefore, the drain current I_D is $\boxed{1\ \text{mA}}$.

Step 3 of 3

Determine the voltage gain G_v .

$$G_v = \frac{R_D}{R_{sig} + \frac{1}{g_m}}$$
$$= \frac{R_D}{R_{sig} + R_{in}} \qquad \left(\text{since } R_{in} = \frac{1}{g_m}\right)$$

Substitute $2\ \text{k}\Omega$ for R_D , $100\,\Omega$ for R_{in} in equation.

$$G_v = \frac{2\ \text{k}\Omega}{100\,\Omega + 100\,\Omega}$$
$$= \frac{2\ \text{k}\Omega}{200\,\Omega}$$
$$= 10\ \text{V}$$

Therefore, the overall voltage gain G_v is $\boxed{10\ \text{V}}$.

Step 1 of 8

7.025P

Refer to the Fig. 7.10 in the textbook

(a)

The dc drain current I_D is given by,

$$I_D = \frac{1}{2}k_n(V_{GS} - V_t)^2$$

Here,

V_t is Threshold voltage,

V_{GS} is the gate to source volage,

k_n is process transconductance parameter.

Step 2 of 8

Substitute $0.4\ \text{V}$ for V_t, $0.6\ \text{V}$ for V_{GS} and $5\ \frac{\text{mA}}{\text{V}^2}$ for k_n.

$$\begin{aligned} I_D &= \frac{1}{2}\left(5\times10^{-3}\right)(0.6-0.4)^2 \\ &= \frac{1}{2}\left(5\times10^{-3}\right)(0.2)^2 \\ &= 0.1\ \text{mA} \end{aligned}$$

Thus, the drain current I_D is $\boxed{0.1\ \text{mA}}$.

Step 3 of 8

Determine the drain to source voltage V_{DS}.

$$V_{DS} = V_{DD} - I_D R_D$$

Substitute $10\ \text{k}\Omega$ for R_D, $1.8\ \text{V}$ for V_{DD} in the above equation.

$$\begin{aligned} V_{DS} &= 1.8\ \text{V} - \left(0.1\times10^{-3}\ \text{A}\right)\left(10\times10^{3}\ \Omega\right) \\ &= 1.8 - 1 \\ &= 0.8\ \text{V} \end{aligned}$$

Thus, Gate-Source voltage V_{DS} is $\boxed{0.8\ \text{V}}$.

Step 4 of 8

(b)

Determine the transconductance g_m.

$$g_m = k_n(V_{GS} - V_t)$$

Substitute $5\ \text{mA/V}^2$ for k_n, $0.6\ \text{V}$ for V_{GS}, and $0.4\ \text{V}$ for V_t.

$$\begin{aligned} g_m &= \left(5\times10^{-3}\right)(0.6-0.4) \\ &= 1\times10^{-3}\ \text{S} \\ &= 1\ \text{mS} \end{aligned}$$

Thus, transconductance g_m at the bias point is $\boxed{1\ \text{mS}}$.

Step 5 of 8

(c)

Determine voltage gain A_v.

$$A_v = -g_m R_D$$

Substitute $1\ \text{mS}$ for g_m, $10\ \Omega$ for R_D in the above equation.

$$\begin{aligned} A_v &= -g_m R_D \\ &= -(1\ \text{mS})(10\ \text{k}\Omega) \\ &= -10\ \text{V/V} \end{aligned}$$

Thus, voltage gain A_v is $\boxed{-10}$.

Step 6 of 8

(d)

Relation between early voltage V_A and λ is,

$$V_A = \frac{1}{\lambda}$$

Substitute $0.1\ \text{V}^{-1}$ for λ in the above expression.

$$\begin{aligned} V_A &= \frac{1}{0.1\ \text{V}^{-1}} \\ &= 10\ \text{V} \end{aligned}$$

Thus, early voltage V_A is $10\ \text{V}$.

Step 7 of 8

Output resistance r_o is given by,

$$r_0 = \frac{|V_A|}{I_D}$$

Substitute $10\ \text{V}$ for V_A and $0.1\ \text{mA}$ for I_D in the above expression.

$$\begin{aligned} r_0 &= \frac{|V_A|}{I_D} \\ &= \frac{10\ \text{V}}{0.1\ \text{mA}} \\ &= 100\ \text{k}\Omega \end{aligned}$$

Thus, Finite output resistance r_o between drain and source is $100\ \text{k}\Omega$

Step 8 of 8

Determine the gain A_v.

$$A_v = -g_m(R_D \| r_o)$$

Substitute $1\ \text{mS}$ for g_m, $10\ \Omega$ for R_D and $100\ \text{k}\Omega$ for R_D in the above equation.

$$\begin{aligned} A_v &= -(1\ \text{mS})(10\ \text{k}\Omega \| 0.1\ \text{M}\Omega) \\ &= -\left(1\times10^{-3}\right)\left[\frac{(10\times10^{3})(100\times10^{3})}{(10\times10^{3}+100\times10^{3})}\right] \\ &= -9.09\ \frac{\text{V}}{\text{V}} \end{aligned}$$

Thus, the voltage gain A_v is $\boxed{-9.09}$.

Step 1 of 4

7.026E

Find transconductance g_m.

$$g_m = \frac{I_C}{V_T}$$

Substitute $1\ \text{mA}$ for I_C, $25\ \text{mV}$ for V_T in equation.

$$\begin{aligned} g_m &= \frac{1\ \text{mA}}{25\ \text{mV}} \\ &= 40\ \text{mA/V} \end{aligned}$$

Therefore, the transconductance g_m is $40\ \text{mA/V}$.

Calculate the input resistance R_{in}.

$$R_{in} = \frac{1}{g_m}$$

Substitute $40\ \text{mA/V}$ for g_m in equation.

$$\begin{aligned} R_{in} &= \frac{1}{40\ \text{mA/V}} \\ &= 25\ \Omega \end{aligned}$$

Hence, input resistance R_{in} is $\boxed{25\ \Omega}$.

Step 2 of 4

Determine the open circuit voltage gain A_{vo}.

$$A_{vo} = g_m R_C$$

Substitute $5\ \text{k}\Omega$ for R_C, $40\ \text{mA/V}$ for g_m in equation.

$$\begin{aligned} A_{vo} &= (40\ \text{mA/V})(5\ \text{k}\Omega) \\ &= 200\ \text{V/V} \end{aligned}$$

Therefore, the open circuit voltage gain A_{vo} is $\boxed{200\ \text{V/V}}$.

Step 3 of 4

Calculate the voltage gain A_v.

$$A_v = g_m (R_C \| R_L)$$

Substitute $5\ \text{k}\Omega$ for R_L, $5\ \text{k}\Omega$ for R_C, $40\ \text{mA/V}$ for g_m in equation.

$$\begin{aligned} A_v &= (40\ \text{mA/V})(5\ \text{k}\Omega \| 5\ \text{k}\Omega) \\ &= (40\ \text{mA/V})(2.5\ \text{k}\Omega) \\ &= 100\ \text{V/V} \end{aligned}$$

Thus, the voltage gain A_v is $\boxed{100\ \text{V/V}}$.

Step 4 of 4

Determine the overall voltage gain G_v.

$$G_v = \alpha \frac{R_C \| R_L}{R_{sig} + r_e}$$

Substitute $5\ \text{k}\Omega$ for R_{sig}, $5\ \text{k}\Omega$ for R_L, $5\ \text{k}\Omega$ for R_C, $25\ \Omega$ for r_e in equation.

$$\begin{aligned} G_v &= 1\left(\frac{5\ \text{k}\Omega \| 5\ \text{k}\Omega}{5\ \text{k}\Omega + 25\ \Omega}\right) \qquad (\text{since } \alpha = 1) \\ &= 1\left(\frac{2.5\ \text{k}\Omega}{5025\ \Omega}\right) \\ &= 0.5\ \text{V} \end{aligned}$$

Therefore, the overall voltage gain G_v for common base amplifier is $\boxed{0.5\ \text{V}}$.

7.026P

We don't have the solution to this problem yet.

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Step 1 of 2

7.027E

Input resistance of CB amplifier is equal to the characteristic resistance of co-axial cable and hence $R_{in} = 50\ \Omega$.

Calculation of collector current I_C is as follows:

input resistance is,

$$R_{in} = r_e$$
$$= \frac{1}{g_m} \quad R_{in} = \frac{V_T}{I_C}$$

Substitute $25\ \text{mV}$ for V_T, $50\ \Omega$ for R_{in} in equation.

$$50\ \Omega = \frac{25\ \text{mV}}{I_C}$$
$$I_C = 0.5\ \text{mA}$$

Thus, the collector current I_C is $\boxed{0.5\ \text{mA}}$.

Step 2 of 2

Co-axial cable supply signal to the CB amplifier and hence signal resistance R_{sig} for CB amplifier is equal to Co-axial cable resistance and hence $R_{sig} = 50\ \Omega$.

Determine the overall gain.

$$G_v = \alpha \frac{R_C \parallel R_L}{R_{sig} + r_e}$$

Substitute $50\ \Omega$ for R_{sig}, $50\ \Omega$ for r_e, $40\ \text{V/V}$ for G_v, 1 for α in equation.

$$40\ \text{V/V} = 1\left(\frac{R_C \parallel R_L}{50\ \Omega + 50\ \Omega}\right)$$
$$R_C \parallel R_L = (40\ \text{V/V})(100)$$
$$R_C \parallel R_L = 4\ \text{k}\Omega$$

Therefore, the total resistance in the collector $(R_C \parallel R_L)$ is equal to $\boxed{4\ \text{k}\Omega}$.

7.027P

We don't have the solution to this problem yet.

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Step 1 of 3

Refer to the Figure 7.41(c) in the textbook.

output resistance R_o is $100\ \Omega$, load resistance R_L is $1\ \text{k}\Omega$

The output resistance R_o is,

$$R_o = \frac{1}{g_m}$$
$$= \frac{1}{\left(\frac{2I_D}{V_{ov}}\right)} \qquad \left(\text{since } g_m = \frac{2I_D}{V_{ov}}\right)$$
$$R_o = \frac{V_{ov}}{2I_D} \quad \text{..... (1)}$$

Calculate the drain current I_D.

Substitute $0.25\ \text{V}$ for V_{ov}, $100\ \Omega$ for R_o in equation (1).

$$100\ \Omega = \frac{0.25\ \text{V}}{2(I_D)}$$
$$I_D = 1.25\ \text{mA}$$

Therefore, the drain current I_D is $\boxed{1.25\ \text{mA}}$.

Step 2 of 3

Determine the output voltage v_o.

$$v_o = v_i\left(\frac{R_L}{R_L + \frac{1}{g_m}}\right)$$
$$= v_i\left(\frac{R_L}{R_L + R_o}\right) \qquad \left(\text{since } \frac{1}{g_m} = R_o\right)$$

Substitute $1\ \text{V}$ for v_i, $1\ \text{k}\Omega$ for R_L, $100\ \Omega$ for R_o in equation.

$$v_o = (1\ \text{V})\left[\frac{1\ \text{k}\Omega}{1\ \text{k}\Omega + 100\ \Omega}\right]$$
$$= (1\ \text{V})\left[\frac{1000\ \Omega}{1100\ \Omega}\right]$$
$$= 0.91\ \text{V}$$

Therefore, the output signal amplitude v_o is $\boxed{0.91\ \text{V}}$.

Step 3 of 3

Refer to the figure 7.42(b) in the textbook.

Signal amplitude between gate and source is ,

$$v_{gs} = v_i\left(\frac{\frac{1}{g_m}}{R_L + \frac{1}{g_m}}\right)$$
$$= v_i\left(\frac{R_o}{R_L + R_o}\right) \qquad \left(\text{since } \frac{1}{g_m} = R_o\right)$$

Substitute $1\ \text{V}$ for v_i, $1\ \text{k}\Omega$ for R_L, $100\ \Omega$ for R_o in equation.

$$v_{gs} = (1\ \text{V})\left[\frac{100\ \Omega}{1000\ \Omega + 100\ \Omega}\right]$$
$$= (1\ \text{V})\left[\frac{100\ \Omega}{1100\ \Omega}\right]$$
$$= 91\ \text{mV}$$

Therefore, signal amplitude between gate and source is $\boxed{91\ \text{mV}}$.

7.028P

We don't have the solution to this problem yet.

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Step 1 of 4

7.029E

Determine the drain current I_D.

The output resistance R_o is,

$$R_o = \frac{1}{g_m}$$

$$= \frac{1}{\left(\frac{2I_D}{V_{ov}}\right)} \qquad \left(\text{since } g_m = \frac{2I_D}{V_{ov}}\right)$$

$$R_o = \frac{V_{ov}}{2I_D}$$

$$I_D = \frac{V_{ov}}{2R_o}$$

Substitute $200\ \Omega$ for R_o, $0.25\ \text{V}$ for V_{ov} in equation.

$$I_D = \frac{0.25\ \text{V}}{2(200\ \Omega)}$$

$$= 0.625\ \text{mA}$$

Therefore, the drain current is $\boxed{0.625\ \text{mA}}$.

Step 2 of 4

Calculation of the $\left(\frac{W}{L}\right)$ ratio is as follows:

Drain current I_D is,

$$I_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)(V_{ov})^2$$

Substitute $0.4\ \text{mA/V}^2$ for k_n', $0.625\ \text{mA}$ for I_D, $0.25\ \text{V}$ for V_{ov} in equation.

$$0.625\ \text{mA} = \frac{1}{2}(0.4\ \text{mA/V}^2)\left(\frac{W}{L}\right)(0.25)^2$$

$$\left(\frac{W}{L}\right) = \frac{\frac{1}{2}(0.4\ \text{mA/V}^2)(0.25)^2}{0.625\ \text{mA}}$$

$$= 50$$

Therefore, the $\left(\frac{W}{L}\right)$ ratio of MOSFET is $\boxed{50}$.

Step 3 of 4

Determine the overall voltage gain G_v.

$$G_v = \frac{R_L}{R_L + \frac{1}{g_m}}$$

$$= \frac{R_L}{R_L + R_o} \qquad \left(\text{since } R_o = \frac{1}{g_m}\right)$$

Substitute for $1\ \text{k}\Omega$ for R_L, $200\ \Omega$ for R_o in equation.

$$G_v = \frac{1\ \text{k}\Omega}{1\ \text{k}\Omega + 200\ \Omega}$$

$$= \frac{1000\ \Omega}{1200\ \Omega}$$

$$= 0.83\ \text{V/V}$$

Thus, overall voltage gain G_v for R_L equal to $1\ \text{k}\Omega$ is $0.83\ \text{V/V}$.

Step 4 of 4

Determine the overall voltage gain G_v.

$$G_v = \frac{R_L}{R_L + \frac{1}{g_m}}$$

$$= \frac{R_L}{R_L + R_o}$$

Substitute for $1\ \text{k}\Omega$ for R_L, $200\ \Omega$ for R_o in equation.

$$G_v = \frac{10\ \text{k}\Omega}{10\ \text{k}\Omega + 200\ \Omega}$$

$$= \frac{10000\ \Omega}{10200\ \Omega}$$

$$= 0.98\ \text{V/V}$$

Thus, overall voltage gain G_v for R_L equal to $10\ \text{k}\Omega$ is $0.98\ \text{V/V}$.

Thus, range of G_v of the source follower is $\boxed{0.83\ \text{V/V to } 0.98\ \text{V/V}}$.

Step 1 of 3

Consider the expression for the Trans-conductance, g_m.

$$g_m = \sqrt{2k_n' \frac{W}{L} I_D}$$

Rearrange the expression for $\frac{W}{L}$.

$$\frac{W}{L} = \frac{g_m^2}{2k_n' I_D} \quad \{\text{since } \mu_n C_{ox} = k_n'\}$$

$$\frac{W}{L} = \frac{g_m^2}{2\mu_n C_{ox} I_D}$$

Step 2 of 3

Substitute $0.5\,\mu\text{m}$ for L, $250\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$, 0.25 mA for I_D, and $2\,\frac{\text{mA}}{\text{V}}$ for g_m.

$$\frac{W}{0.5\times10^{-6}} = \frac{\left(2\times10^{-3}\right)^2}{2\left(250\times10^{-6}\right)\left(0.25\times10^{-3}\right)}$$

$$\frac{W}{0.5} = \frac{4\times10^{-3}}{(125)}$$

$$W = \frac{2\times10^{-3}}{(125)}$$

$$W = 16\times10^{-6}\ \text{m}$$

$$= 16\ \mu\text{m}$$

Therefore, the value of width of transistor is $\boxed{16\,\mu\text{m}}$.

Step 3 of 3

Write the expression for gate to source voltage, V_{GS}.

$$V_{GS} = V_{OV} + V_t \qquad \left\{\text{since } V_{OV} = \frac{2I_D}{g_m}\right\}$$

$$= \frac{2I_D}{g_m} + V_t$$

Substitute 0.25 mA for I_D, and $2\ \text{mA/V}$ for g_m, and 0.5 V for V_t.

$$V_{GS} = \left(\frac{2\left(0.25\times10^{-3}\right)}{2\times10^{-3}}\right) + 0.5$$

$$= 0.25 + 0.5$$

$$V_{GS} = 0.75\ \text{V}$$

Therefore, the value of gate to source voltage V_{GS} is $\boxed{0.75\ \text{V}}$.

7.030E

We don't have the solution to this problem yet.

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Step 1 of 5

Refer to Figure P7.30 in the textbook for the NMOS amplifier.

It is observed that v_d is the voltage at the drain terminal and v_s is the voltage at the source terminal of the NMOS. Refer to Figure 5.40 (d) and 5.41 (a) for the T equivalent circuit model with output resistance, r_o.

Consider the expression for the output resistance.

$$r_o = \frac{1}{\lambda I_D} \quad \{\text{since } \lambda = 0\}$$
$$= \frac{1}{(0) I_D}$$
$$= \infty \left(\text{the resistance } r_o \text{ between drain and source is open}\right)$$

Step 2 of 5

Draw the NMOS equivalent circuit by replacing the transistor by its T-equivalent model.

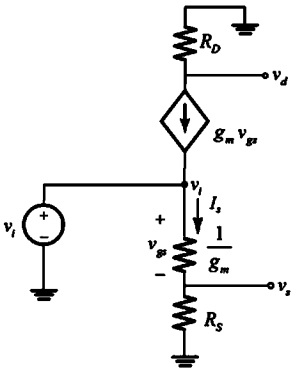


Figure 1

Step 3 of 5

Apply voltage divider rule (bottom).

$$v_s = \left(\frac{R_S}{R_S + \frac{1}{g_m}}\right) v_i$$

$$\frac{v_s}{v_i} = \frac{R_S}{\frac{R_S g_m + 1}{g_m}}$$
$$= \frac{R_S g_m}{R_S g_m + 1}$$

Therefore, the expression for voltage gain, $\frac{v_s}{v_i}$ is $\boxed{\frac{R_S g_m}{1 + R_S g_m}}$.

Step 4 of 5

From Figure 1, the input voltage (by Ohm's law) is,

$$v_i = I_s \left(\frac{1}{g_m} + R_S\right)$$

Apply Kirchhoff's current law at node, v_d

$$\frac{v_d}{R_D} + g_m v_{gs} = 0$$
$$\frac{v_d}{R_D} = -g_m v_{gs}$$
$$v_d = -g_m R_D v_{gs}$$

Divide the expression with v_i

$$\frac{v_d}{v_i} = -\frac{g_m R_D v_{gs}}{v_i}$$

Step 5 of 5

Substitute $I_s \left(\frac{1}{g_m} + R_S\right)$ for v_i and $I_s \left(\frac{1}{g_m}\right)$ for v_{gs}.

$$\frac{v_d}{v_i} = -g_m R_D \left[\frac{I_s \left(\frac{1}{g_m}\right)}{I_s \left(\frac{1}{g_m} + R_S\right)}\right]$$
$$= -g_m R_D \left[\frac{\left(\frac{1}{g_m}\right)}{\left(\frac{1}{g_m} + R_S\right)}\right]$$
$$= -\frac{g_m R_D}{1 + g_m R_S}$$

Therefore, the expression for the voltage gain, $\frac{v_d}{v_i}$ is $\boxed{-\frac{g_m R_D}{1 + g_m R_S}}$.

7.031E

We don't have the solution to this problem yet.

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7.031P

We don't have the solution to this problem yet.

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7.032E

We don't have the solution to this problem yet.

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7.032P

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7.033E

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7.033P

We don't have the solution to this problem yet.

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Ask an expert

7.034E

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Step 1 of 26

(a)

While the relation between collector current i_C and dc collector current I_C is $i_C = I_C e^{v_{be}/V_T}$.

Rewrite the equation.

$$\frac{i_C}{I_C} = e^{v_{be}/V_T}$$

Thus, exponential relation between fraction of I_C and input signal v_{be} is $\frac{i_C}{I_C} = e^{v_{be}/V_T}$.

Step 2 of 26

Input signal v_{be} is $+1$ mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = e^{v_{be}/V_T}$$

Substitute corresponding values.

$$\frac{i_C}{I_C} = e^{+1 \text{ mV}/25 \text{ mV}}$$

$$= e^{+1/25}$$

$$= 0.961$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of $+1$ mV is $\boxed{0.961}$

Step 3 of 26

Input signal v_{be} is -1 mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = e^{v_{be}/V_T}$$

$$= e^{-1 \text{ mV}/25 \text{ mV}}$$

$$= e^{-1/25}$$

$$= 0.961$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of -1 mV is $\boxed{0.961}$

Step 4 of 26

Input signal v_{be} is $+2$ mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = e^{v_{be}/V_T}$$

$$= e^{+2 \text{ mV}/25 \text{ mV}}$$

$$= e^{+2/25}$$

$$= 1.083$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of $+2$ mV is $\boxed{1.083}$

Step 5 of 26

Input signal v_{be} is -2 mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = e^{v_{be}/V_T}$$

$$= e^{-2 \text{ mV}/25 \text{ mV}}$$

$$= 0.9231$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of -2 mV is $\boxed{0.9231}$

Step 6 of 26

Input signal v_{be} is $+5$ mV .

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = e^{v_{be}/V_T}$$

$$= e^{+5 \text{ mV}/25 \text{ mV}}$$

$$= 1.221$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of $+5$ mV is $\boxed{1.221}$

Step 7 of 26

Input signal v_{be} is -5 mV .

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = e^{v_{be}/V_T}$$

$$= e^{-5 \text{ mV}/25 \text{ mV}}$$

$$= e^{-5/25}$$

$$= 0.82$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of -5 mV is $\boxed{0.82}$

Step 8 of 26

Input signal v_{be} is $+8$ mV .

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = e^{v_{be}/V_T}$$

$$= e^{+8 \text{ mV}/25 \text{ mV}}$$

$$= e^{+8/25}$$

$$= 1.377$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of $+8$ mV is $\boxed{1.377}$

Step 9 of 26

Input signal v_{be} is -8 mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = e^{v_{be}/V_T}$$

$$= e^{-8 \text{ mV}/25 \text{ mV}}$$

$$= e^{-8/25}$$

$$= 0.726$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of -8 mV is $\boxed{0.726}$.

Step 10 of 26

Input signal v_{be} is $+10$ mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = e^{v_{be}/V_T}$$

$$= e^{+10 \text{ mV}/25 \text{ mV}}$$

$$= e^{+10/25}$$

$$= 1.492$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of $+10$ mV is $\boxed{1.492}$.

Step 11 of 26

Input signal v_{be} is -10 mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = e^{v_{be}/V_T}$$

$$= e^{-10 \text{ mV}/25 \text{ mV}}$$

$$= e^{-10/25}$$

$$= 0.67$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of -10 mV is $\boxed{0.67}$

Step 12 of 26

Input signal v_{be} is $+12$ mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = e^{v_{be}/V_T}$$

$$= e^{+12 \text{ mV}/25 \text{ mV}}$$

$$= e^{+12/25}$$

$$= 1.616$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of $+12$ mV is $\boxed{1.616}$.

Step 13 of 26

Input signal v_{be} is -12 mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = e^{v_{be}/V_T}$$

$$= e^{-12 \text{ mV}/25 \text{ mV}}$$

$$= 0.618$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of -12 mV is $\boxed{0.618}$.

Step 14 of 26

(b)

Refer the equation (7.57) in the textbook small signal approximation relation between collector current i_c and dc collector current I_C is $i_c = I_C\left[\frac{v_{be}}{V_T}\right]$

$$i_c = I_C\left[1 + \frac{v_{be}}{V_T}\right]$$

Input signal v_{be} is $+1$ mV

$$\frac{i_C}{I_C} = \left[1 + \frac{v_{be}}{V_T}\right]$$

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = \left[1 + \frac{+1 \text{ mV}}{25 \text{ mV}}\right]$$

$$= \left[1 + \frac{+1}{25}\right]$$

$$= 1.04$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of $+1$ mV is $\boxed{1.04}$

Step 15 of 26

Input signal v_{be} is -1 mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = \left[1 + \frac{v_{be}}{V_T}\right]$$

$$= \left[1 + \frac{-1 \text{ mV}}{25 \text{ mV}}\right]$$

$$= 0.96$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of -1 mV is $\boxed{0.96}$

Step 16 of 26

Input signal v_{be} is $+2$ mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = \left[1 + \frac{v_{be}}{V_T}\right]$$

$$= \left[1 + \frac{+2 \text{ mV}}{25 \text{ mV}}\right]$$

$$= 1.08$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of $+2$ mV is $\boxed{1.08}$

Step 17 of 26

Input signal v_{be} is -2 mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = \left[1 + \frac{v_{be}}{V_T}\right]$$

$$= \left[1 + \frac{-2 \text{ mV}}{25 \text{ mV}}\right]$$

$$= 0.92$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of -2 mV is $\boxed{0.92}$

Step 18 of 26

Input signal v_{be} is $+5$ mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = \left[1 + \frac{v_{be}}{V_T}\right]$$

$$= \left[1 + \frac{+5 \text{ mV}}{25 \text{ mV}}\right]$$

$$= 1.2$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of $+5$ mV is $\boxed{1.2}$

Step 19 of 26

Input signal v_{be} is -5 mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = \left[1 + \frac{v_{be}}{V_T}\right]$$

$$= \left[1 + \frac{-5 \text{ mV}}{25 \text{ mV}}\right]$$

$$= 0.8$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of -5 mV is $\boxed{0.8}$

Step 20 of 26

Input signal v_{be} is $+8$ mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = \left[1 + \frac{v_{be}}{V_T}\right]$$

$$= \left[1 + \frac{8 \text{ mV}}{25 \text{ mV}}\right]$$

$$= 1.32$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of $+8$ mV is $\boxed{1.32}$

Step 21 of 26

Input signal v_{be} is -8 mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = \left[1 + \frac{v_{be}}{V_T}\right]$$

$$= \left[1 + \frac{-8 \text{ mV}}{25 \text{ mV}}\right]$$

$$= 0.68$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of -8 mV is $\boxed{0.68}$

Step 22 of 26

Input signal v_{be} is $+10$ mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = \left[1 + \frac{v_{be}}{V_T}\right]$$

$$= \left[1 + \frac{10 \text{ mV}}{25 \text{ mV}}\right]$$

$$= 1.4$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of $+10$ mV is $\boxed{1.4}$

Step 23 of 26

Input signal v_{be} is -10 mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = \left[1 + \frac{v_{be}}{V_T}\right]$$

$$= \left[1 + \frac{-10 \text{ mV}}{25 \text{ mV}}\right]$$

$$= 0.6$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of -10 mV is $\boxed{0.6}$

Step 24 of 26

Input signal v_{be} is $+12$ mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = \left[1 + \frac{v_{be}}{V_T}\right]$$

$$= \left[1 + \frac{12 \text{ mV}}{25 \text{ mV}}\right]$$

$$= 1.48$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of $+12$ mV is $\boxed{1.48}$

Step 25 of 26

Input signal v_{be} is -12 mV

Thermal voltage V_T is 25 mV approximately at room temperature

$$\frac{i_C}{I_C} = \left[1 + \frac{v_{be}}{V_T}\right]$$

$$= \left[1 + \frac{-12 \text{ mV}}{25 \text{ mV}}\right]$$

$$= 0.52$$

Thus, collector signal current as a fraction of I_C for input signal v_{be} of -12 mV is $\boxed{0.52}$

Step 26 of 26

List the collector current in table

Table 1

v_{be}	$\frac{i_C}{I_C}$ using Exponential characteristic	$\frac{i_C}{I_C}$ using Small signal approximation	error
1	**1.041**	**1.04**	**0.001**
−1	**0.961**	**0.96**	**0.001**
2	**1.083**	**1.08**	**0.003**
−2	**0.923**	**0.92**	**0.003**
5	**1.221**	**1.2**	**0.021**
−5	**0.82**	**0.8**	**0.02**
8	**1.377**	**1.32**	**0.057**
−8	**0.726**	**0.68**	**0.046**
10	**1.492**	**1.4**	**0.092**
−10	**0.67**	**0.6**	**0.07**
12	**1.616**	**1.48**	**0.136**
−12	**0.618**	**0.52**	**0.098**

7.035E

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Step 1 of 7

Base emitter voltage V_{BE} is $0.700\ \text{V}$

Collector current I_C is $0.5\ \text{mA}$

Collector resistance R_C is $5\ \text{k}\Omega$

Power supply voltage V_{CC} is $+5\ \text{V}$

Determine the collector voltage V_C.

$$\begin{aligned} V_C &= V_{CC} - I_C R_C \\ &= 5 - (0.5\ \text{mA})(5\ \text{k}\Omega) \\ &= 5 - 2.5 \\ &= 2.5\ \text{V} \end{aligned}$$

Thus, the collector voltage V_C is $\boxed{2.5\ \text{V}}$.

Step 2 of 7

The signal current applied to base v_{BE} is raised to $0.705\ \text{V}$

The total instantaneous base emitter voltage v_{BE} is $(V_{BE} + v_{be})$.

$$\begin{aligned} v_{BE} &= V_{BE} + v_{be} \\ 0.705\ \text{V} &= 0.700\ \text{V} + v_{be} \\ v_{be} &= 5\ \text{mV} \end{aligned}$$

Thus, the value of the signal v_{be} is $\boxed{5\ \text{mV}}$.

Step 3 of 7

Refer to the equation (7.56) in the textbook.

Write the expression for the total instantaneous collector current i_C.

$$i_C = I_C e^{v_{be}/V_T}$$

Substitute $0.5\ \text{mA}$ for I_C, $25\ \text{mV}$ for V_T and $5\ \text{mV}$ for v_{be}.

$$\begin{aligned} i_C &= (0.5\ \text{mA}) e^{\frac{5\ \text{mV}}{25\ \text{mV}}} \\ &= 0.61\ \text{mA} \end{aligned}$$

Thus, the total instantaneous collector current i_C is $\boxed{0.61\ \text{mA}}$.

Step 4 of 7

Determine the total Collector voltage after base emitter voltage increased.

$$\begin{aligned} v_C &= V_{CC} - i_C R_C \\ &= 5\ \text{V} - (0.61\ \text{mA})(5\ \text{k}\Omega) \\ &= 5\ \text{V} - 3.05\ \text{V} \\ &= 1.95\ \text{V} \end{aligned}$$

Thus, the collector voltage v_C after base emitter voltage increased to $0.705\ \text{V}$ is $\boxed{1.95\ \text{V}}$.

Step 5 of 7

The increased voltage v_c across the collector resistance is $v_C - V_C$

$$\begin{aligned} v_c &= v_C - V_C \\ &= 1.95\ \text{V} - 2.5\ \text{V} \\ &= -0.55\ \text{V} \end{aligned}$$

Thus, the voltage v_c is $\boxed{-0.55\ \text{V}}$.

Determine the voltage gain A_v.

$$\begin{aligned} A_v &= v_c / v_{be} \\ &= \frac{-0.55\ \text{V}}{5\ \text{mV}} \\ &= -110 \end{aligned}$$

Thus, the voltage gain A_v using exponential characteristic is $\boxed{-110\ \frac{\text{V}}{\text{V}}}$.

Step 6 of 7

The dc collector current I_C is $0.5\ \text{mA}$.

Refer to the equation (7.62) in the textbook.

Write the expression for the transconductance g_m for BJT.

$$g_m = \frac{I_C}{V_T}$$

Substitute $25\ \text{mV}$ for V_T and $0.5\ \text{mA}$ for I_C.

$$\begin{aligned} g_m &= \frac{0.5\ \text{mA}}{25\ \text{mV}} \\ &= 20\ \text{mA/V} \end{aligned}$$

Thus, the transconductance g_m for BJT amplifier is $\boxed{20\ \text{mA/V}}$.

Step 7 of 7

Write the expression for the voltage gain A_v using small signal approximation.

$$A_v = -g_m R_C$$

Substitute $-20\ \text{mA/V}$ for g_m and $5\ \text{k}\Omega$ for R_C.

$$\begin{aligned} A_v &= (-20\ \text{mA/V})(5\ \text{k}\Omega) \\ &= 100\ \frac{\text{V}}{\text{V}} \end{aligned}$$

Thus, the voltage gain A_v is $\boxed{100\ \text{V/V}}$.

Hence, the voltage gain A_v obtain is equal to the voltage gain in small signal approximation.

7.036E

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7.036P

Step 1 of 6

Refer to the equation (7.62) in the textbook.

Write the expression of the transconductance g_m for BJT.

$$g_m = \frac{I_C}{V_T}$$

Substitute $25\ \text{mV}$ for V_T and $0.5\ \text{mA}$ for I_C.

$$\begin{aligned} g_m &= \frac{0.5\ \text{mA}}{25\ \text{mV}} \\ &= 20\ \text{mA/V} \end{aligned}$$

Thus, the transconductance g_m for BJT amplifier is $\boxed{20\ \text{mA/V}}$.

Step 2 of 6

Refer to the equation (7.74) in the textbook.

Determine the emitter resistance r_e.

$$\begin{aligned} r_e &= \frac{1}{g_m} \\ &= \frac{1}{20\ \text{mA/V}} \\ &= 50\ \Omega \end{aligned}$$

Thus, the Emitter resistance r_e is $\boxed{50\ \Omega}$.

Step 3 of 6

Refer to the equation (7.68) in the textbook.

Determine the input resistance r_π.

$$\begin{aligned} r_\pi &= \frac{\beta}{g_m} \\ &= \frac{100}{20\ \text{mA/V}} \\ &= 5\ \text{k}\Omega \end{aligned}$$

Thus, the input resistance r_π is $\boxed{5\ \text{k}\Omega}$.

Step 4 of 6

Consider the collector current I_C is $50\ \mu\text{A}$.

Refer to the equation (7.62) in the textbook.

Determine the transconductance g_m for BJT.

$$\begin{aligned} g_m &= \frac{I_C}{V_T} \\ &= \frac{50\ \mu\text{A}}{25\ \text{mV}} \\ &= 2\ \text{mA/V} \end{aligned}$$

Thus, the transconductance g_m for BJT amplifier is $\boxed{2\ \text{mA/V}}$.

Step 5 of 6

Refer to the equation (7.74) in the textbook.

Determine the emitter resistance r_e.

$$\begin{aligned} r_e &= \frac{1}{g_m} \\ &= \frac{1}{2\ \text{mA/V}} \\ &= 500\ \Omega \end{aligned}$$

Thus, the Emitter resistance r_e is $\boxed{500\ \Omega}$.

Step 6 of 6

Refer to the equation (7.68) in the textbook.

Determine the input resistance r_π.

$$\begin{aligned} r_\pi &= \frac{\beta}{g_m} \\ &= \frac{100}{2\ \text{mA/V}} \\ &= 50\ \text{k}\Omega \end{aligned}$$

Thus, the Input resistance r_π is $\boxed{50\ \text{k}\Omega}$.

7.037E

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7.037P

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7.038E

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7.039E

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7.040E

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7.040P

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7.041E

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7.041P

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7.042E

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7.042P

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7.043E

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Step 1 of 15

Refer to the figure 7.25(a) and 7.25(b) in the textbook.

The transconductance g_m is,

$$g_m = I_C / V_T$$

Substitute $1\ \text{mA}$ for I_C, 100 for β in the equation.

$$\begin{aligned} g_m &= \frac{1\ \text{mA}}{25\ \text{mV}} \\ &= 40\ \text{mA/V} \end{aligned}$$

Thus, value of g_m is $\boxed{40\ \text{mA/V}}$

Step 2 of 15

The input resistance r_π is,

$$r_\pi = \frac{\beta}{g_m}$$

Substitute $40\ \text{mA/V}$ for g_m, 100 for β in the equation.

$$\begin{aligned} r_\pi &= \frac{100}{40\ \text{mA/V}} \\ &= 2500\ \Omega \\ &= 2.5\ \text{k}\Omega \end{aligned}$$

Thus, value of resistance r_π is $\boxed{2.5\ \text{k}\Omega}$.

Step 3 of 15

Output resistance r_o is,

$$r_o = \frac{V_A}{I_C}$$

Substitute $100\ \text{V}$ for V_A, $1\ \text{mA}$ for I_C in the equation.

$$\begin{aligned} r_o &= \frac{100\ \text{V}}{1\ \text{mA}} \\ &= 0.1\ \text{M}\Omega \end{aligned}$$

Thus, output resistance r_o is $\boxed{0.1\ \text{M}\Omega}$.

Step 4 of 15

Determine the based current I_B.

Common emitter current gain β is,

$$\beta = \frac{I_C}{I_B}$$

Substitute 100 for β, $1\ \text{mA}$ for I_C in the equation.

$$\begin{aligned} 100 &= \frac{1\ \text{mA}}{I_B} \\ I_B &= 10\ \mu\text{A} \end{aligned}$$

Thus, the base current I_B is $\boxed{10\ \mu\text{A}}$

Step 5 of 15

Determine the voltage v_π.

$$v_\pi = I_B r_\pi$$

Substitute $10\ \mu\text{A}$ for I_B, $2.5\ \text{k}\Omega$ for r_π in the equation.

$$\begin{aligned} v_\pi &= (10\ \mu\text{A})(2.5\ \text{k}\Omega) \\ &= (10\times10^{-6})(2.5\times10^{3}) \\ &= 25\ \text{mV} \end{aligned}$$

Thus, the voltage v_π is $\boxed{25\ \text{mV}}$

Step 6 of 15

Determine the product of $g_m v_\pi$.

Substitute $40\ \text{mA/V}$ for g_m, $25\ \text{mV}$ for v_π in expression.

$$\begin{aligned} g_m v_\pi &= \left(40\ \frac{\text{mA}}{\text{V}}\right)(25\ \text{mV}) \\ &= 1\ \text{mA} \end{aligned}$$

Thus, value of $g_m v_\pi$ is $\boxed{1\ \text{mA}}$

Step 7 of 15

Hybrid - π model of the BJT is shown in the Figure1 and Figure 2.

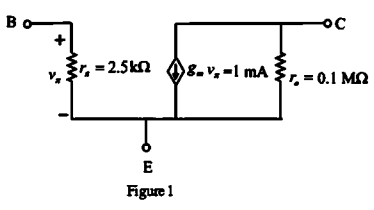


Figure 1

Step 8 of 15

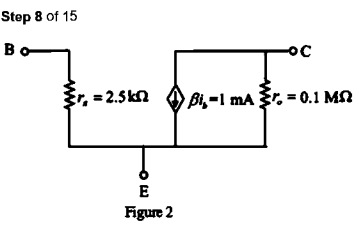


Figure 2

Step 9 of 15

Determination the T-Model equivalent circuit is as follows:

Determine the emitter current I_E.

$$I_E = I_B + I_C$$

Substitute $10\ \mu\text{A}$ for I_B, $1\ \text{mA}$ for I_C in equation.

$$\begin{aligned} I_E &= 10\ \mu\text{A} + 1\ \text{mA} \\ &= 10\times10^{-6} + 1\times10^{-3} \\ &= 0.01\times10^{-3} + 1\times10^{-3} \\ &= 1.01\ \text{mA} \end{aligned}$$

Thus, the emitter current I_E is $\boxed{1.01\ \text{mA}}$.

Step 10 of 15

The emitter resistance r_e is,

$$r_e = \frac{V_T}{I_E}$$

Substitute $25\ \text{mV}$ for V_T, $1.01\ \text{mA}$ for I_E in the equation.

$$\begin{aligned} r_e &= \frac{25\ \text{mV}}{1.01\ \text{mA}} \\ &= 24.75\ \Omega \end{aligned}$$

Thus, the value of emitter resistance r_e is $\boxed{24.75\ \Omega}$.

Step 11 of 15

Determine the value of α.

$$\alpha = \frac{\beta}{\beta + 1}$$

Substitute 100 for β in equation.

$$\begin{aligned} \alpha &= \frac{100}{100+1} \\ &= 0.99 \end{aligned}$$

Thus, the value of α is $\boxed{0.99}$.

Step 12 of 15

Determine the current i_e.

$$i_e = \frac{v_\pi}{r_e}$$

Substitute $25\ \text{mV}$ for v_π and $24.75\ \Omega$ for r_e in equation.

$$\begin{aligned} i_e &= \frac{25\ \text{mV}}{24.75\ \Omega} \\ &= 1.01\ \text{mA} \end{aligned}$$

Therefore, the value of i_e is $\boxed{1.01\ \text{mA}}$.

Step 13 of 15

Determine the product of αi_e.

Substitute 0.99 for α and $1.01\ \text{mA}$ for i_e in expression.

$$\begin{aligned} \alpha i_e &= (0.99)(1.01\ \text{mA}) \\ &= 1\ \text{mA} \end{aligned}$$

Therefore, the value of αi_e is $\boxed{1\ \text{mA}}$.

Step 14 of 15

T-modal of BJT is shown in the Figure 3 and Figure 4.

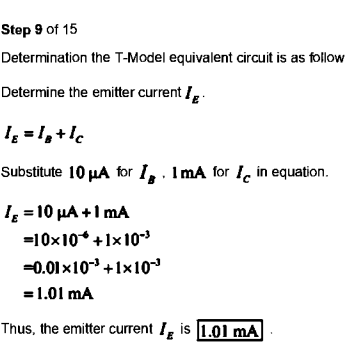


Figure 3

Step 15 of 15

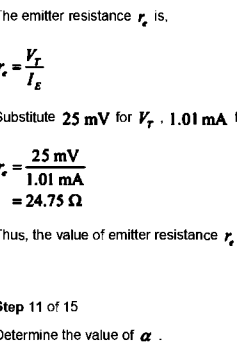


Figure 4

Therefore, the designs of four small-signal models of BJT with their parameters are done.

7.044E

We don't have the solution to this problem yet.

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Ask an expert

Step 1 of 3

7.044P

Refer to the figure 7.26(a) in the textbook.

Apply Kirchhoff's current law at node B.

$i_b = -g_m v_{be} + i_e$ ..... (1)

Relation between i_e, i_b and i_c is,

$i_e = i_b + i_c$

Substitute $i_b + i_c$ for i_e in the equation (1).

$i_b = -g_m v_{be} + i_b + i_c$

$i_c = g_m v_{be}$ ..... (2)

Step 2 of 3

The collector current i_c is,

$i_c = \beta i_b$

Substitute $i_c = \beta i_b$ in the equation (2).

$\beta i_b = g_m v_{be}$

$\frac{\beta}{g_m} = \frac{v_{be}}{i_b}$ ..... (3)

Step 3 of 3

Determine the input resistance between base and emitter.

Refer to the equation 7.67 in the textbook.

$r_\pi = \frac{\beta}{g_m}$

Substitute r_π for $\frac{\beta}{g_m}$ in the equation (3).

$r_\pi = \frac{v_{be}}{i_b}$

Thus, input resistance between the base and emitter, looking into the base is equal to $\boxed{r_\pi}$.

Step 1 of 3

7.045P

Refer to the figure 7.26(b) in the textbook.

The collector current i_c is,

$i_c = \alpha i_e$ ..... (1)

Voltage drop across the resistance r_e is v_{be}.

$v_{be} = i_e r_e$

$i_e = \frac{v_{be}}{r_e}$

Substitute $\frac{v_{be}}{r_e}$ for i_e in the equation (1).

$i_c = \alpha \frac{v_{be}}{r_e}$

$i_c = \frac{\alpha}{r_e} v_{be}$ ..... (2)

Step 2 of 3

Recall equation (7.74) in the textbook for transconductance g_m.

$g_m = \frac{\alpha}{r_e}$

Substitute g_m for $\frac{\alpha}{r_e}$ in the equation (2).

Thus, the collector current i_c is,

$i_c = g_m v_{be}$ ..... (3)

Step 3 of 3

Refer to the figure 7.26(a) in the textbook for collector current i_c.

$i_c = g_m v_{be}$ ..... (4)

Therefore, from equation (3) and (4), the collector current provided by the model in Figure 7.26(b) is equal to that of the modal in the Figure 7.26(a).

7.046P

We don't have the solution to this problem yet.

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7.047P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

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Step 1 of 5

7.048P

Refer to the figure P7.48 in the textbook.

Consider value of α is 1. Since, β is very large.

Determine collector current I_C.

$$I_C = \alpha I_E$$

Substitute 1 for α and $0.2\ \text{mA}$ for I_E.

$$\begin{aligned} I_C &= (1)\left(0.2\times10^{-3}\right) \\ &= 0.2\times10^{-3} \\ &= 0.2\,\text{mA} \end{aligned}$$

Determine collector voltage V_C.

$$V_C = V_{CC} - I_C R_C$$

Substitute 3 V for V_{CC}, $0.2\ \text{mA}$ for I_C and $10\,\text{k}\Omega$ for R_C.

$$\begin{aligned} V_C &= \left(3-(0.2\ \text{mA})(10\ \text{k}\Omega)\right) \\ &= 3-2 \\ &= 1\ \text{V} \end{aligned}$$

Thus, the value of collector voltage V_C is $\boxed{1\ \text{V}}$.

Step 2 of 5

T- Equivalent of the amplifier is shown in the Figure 1

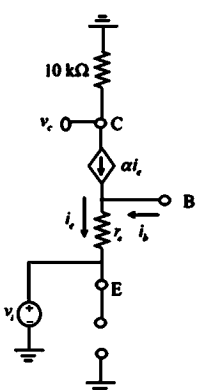


Figure1

Step 3 of 5

From Figure1, the value of v_c is,

$$v_c = -\alpha i_e (10\ \text{k}\Omega) \quad \text{.....(1)}$$

Potential drop across base and emitter is,

$$v_{be} = i_e r_e$$

Base is grounded and hence v_b value is zero. Emitter voltage is v_i.

$$\begin{aligned} v_{be} &= i_e r_e \\ v_b - v_e &= i_e r_e \\ 0 - v_i &= i_e r_e \end{aligned}$$

$$v_i = -i_e r_e \quad \text{.....(2)}$$

Step 4 of 5

The emitter resistance r_e is,

$$r_e = \frac{V_T}{i_e}$$

Substitute $25\ \text{mV}$ for V_T, $0.2\ \text{mA}$ for i_e in equation.

$$\begin{aligned} r_e &= \frac{25\ \text{mV}}{0.2\ \text{mA}} \\ &= 125\ \Omega \end{aligned}$$

Emitter resistance r_e is $\boxed{125\ \Omega}$.

Step 5 of 5

Substitute equation (2) in equation (1)

$$v_c = -\alpha i_e (10\ \text{k}\Omega)$$

$$v_c = -\alpha\left(-\frac{v_i}{r_e}\right)(10\ \text{k}\Omega)$$

$$\frac{v_c}{v_i} = \frac{\alpha}{r_e}(10\ \text{k}\Omega)$$

$$\frac{v_c}{v_i} = \frac{1}{r_e}(10\ \text{k}\Omega) \qquad (\text{since } \beta \text{ is very large})$$

Substitute $125\ \Omega$ for r_e in the equation.

$$\begin{aligned} \frac{v_c}{v_i} &= \frac{1}{125\ \Omega}(10\ \text{k}\Omega) \\ &= 80 \end{aligned}$$

Therefore, the voltage gain $\frac{v_c}{v_i}$ is $\boxed{80\ \text{V/V}}$.

Step 1 of 2

7.049P

Determine the base to emitter voltage.

$$\frac{v_{ce}}{v_{be}} = -g_m R_C$$

Substitute $50\ \text{mA/V}$ for g_m, $2\ \text{k}\Omega$ for R_C and $1\ \text{V}$ for v_{be}.

$$\frac{1\ \text{V}}{v_{be}} = -(50\ \text{mA/V})(2\ \text{k}\Omega)$$

$$v_{be} = -10\ \text{mV}$$

Thus, peak-to-peak value of v_{be} is $\boxed{10\ \text{mV}}$.

Step 2 of 2

Determine the peak-to-peak value of the base current.

$$i_b = \frac{g_m v_{be}}{\beta}$$

Substitute $50\ \text{mA/V}$ for g_m, 100 for β and $-10\ \text{V}$ for v_{be}.

$$i_b = \frac{(50\ \text{mA/V})(-10\ \text{mV})}{100}$$

$$= -5\ \mu\text{A}$$

Therefore, the peak-to peak value of i_b is $\boxed{5\ \mu\text{A}}$.

7.050P

Step 1 of 5

Refer to the Figure P7.50 in the textbook.

Hybrid-π equivalent of the circuit is shown in the figure 1

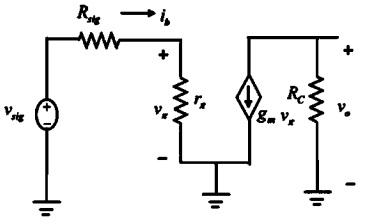


Figure 1

Step 2 of 5

Determine the input resistance R_{in}.

The voltage v_π is,

$$v_\pi = i_b r_\pi$$

$$\frac{v_\pi}{i_b} = r_\pi$$

$$R_{in} = r_\pi$$

Therefore, the input resistance R_{in} is $\boxed{r_\pi}$.

Step 3 of 5

Apply Kirchhoff's voltage law to the left side loop in Figure 1.

$$-v_{sig} + i_b R_{sig} + i_b r_\pi = 0$$

$$-v_{sig} + i_b (R_{sig} + r_\pi) = 0$$

$$-v_{sig} + \left(\frac{v_\pi}{r_\pi}\right)(R_{sig} + r_\pi) = 0 \qquad \left(\text{since } i_b = \frac{v_\pi}{r_\pi}\right)$$

$$v_{sig} = v_\pi \left(\frac{R_{sig} + r_\pi}{r_\pi}\right)$$

$$\frac{v_\pi}{v_{sig}} = \frac{r_\pi}{R_{sig} + r_\pi} \quad \text{..... (1)}$$

Therefore, the voltage transmission from source to the amplifier input is $\boxed{\frac{r_\pi}{R_{sig} + r_\pi}}$.

Step 4 of 5

The voltage v_o is,

$$v_o = -i_c R_C$$

$$= -g_m v_\pi R_C \qquad (\text{since } i_c = g_m v_\pi)$$

$$\frac{v_o}{v_\pi} = -g_m R_C$$

Therefore, the value of $\frac{v_o}{v_\pi}$ is $\boxed{-g_m R_C}$ ..... (2)

Step 5 of 5

Determine the overall gain $\frac{v_o}{v_{sig}}$.

Multiply equation (1) with equation (2).

$$\left(\frac{v_\pi}{v_{sig}}\right)\left(\frac{v_o}{v_\pi}\right) = \left(\frac{1}{1 + \frac{R_{sig}}{r_\pi}}\right)(-g_m R_C)$$

$$\frac{v_o}{v_{sig}} = \frac{r_\pi (-g_m R_C)}{(r_\pi + R_{sig})}$$

Substitute $g_m R_\pi$ for β in the equation.

$$\frac{v_o}{v_{sig}} = \frac{-\beta R_C}{(r_\pi + R_{sig})}$$

$$= -\frac{\beta R_C}{(r_\pi + R_{sig})}$$

Thus, voltage gain from base to collector $\frac{v_o}{v_\pi}$ is $\boxed{-\frac{\beta R_C}{(r_\pi + R_{sig})}}$.

7.051P

Refer to the Figure P7.51 in the textbook.

The base and collector short circuited, voltage $v_{BC} = 0\,\text{V}$ implies the transistor is operating in the active mode.

BJT in the figure P7.51 is represented with current controlled current source representation and the resultant equivalent circuit is represented in the Figure 1.

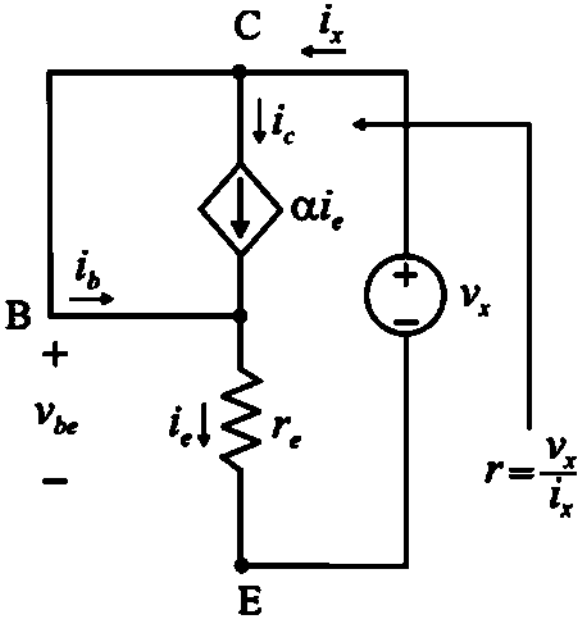


Figure 1

Apply Kirchhoff's current law at node C in Figure 1.

$$i_x = i_b + i_c \quad \text{..... (1)}$$

Emitter current i_e is,

$$i_e = i_b + i_c \quad \text{..... (2)}$$

Since collector and base are short circuited,

$$\begin{aligned} v_x &= i_e r_e \\ &= (i_b + i_c) r_e \qquad (\text{since } i_e = i_b + i_c) \\ &= i_x r_e \qquad (\text{since } i_x = i_b + i_c) \end{aligned}$$

$$\frac{v_x}{i_x} = r_e$$

Therefore, the small-signal resistance r is $\boxed{r_e}$.

Step 1 of 4

7.052P

Refer to the figure P7.52 in the text book.

The T equivalent-circuit is shown in Figure 1.

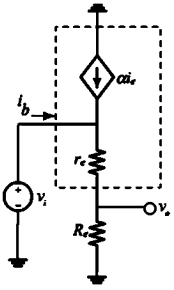


Figure 1

Step 2 of 4

Determine the voltage v_i.

Apply Kirchhoff's voltage law in at input loop in Figure 1.

$$-V_i + i_e(r_e + R_e) = 0$$
$$V_i = i_e(r_e + R_e)$$

Substitute $(\beta + 1)i_b$ for i_e.

$$V_i = (\beta + 1)i_b(r_e + R_e) \quad \text{..... (1)}$$

Therefore, the value of v_i is $i_b(1+\beta)(r_e + R_e)$.

Step 3 of 4

Calculate input resistance R_{in}.

$$R_{in} = \frac{v_i}{i_b}$$

Substitute $i_b(1+\beta)(r_e + R_e)$ for v_i in equation.

$$R_{in} = \frac{(1+\beta)(r_e + R_e)i_b}{i_b}$$
$$= (1+\beta)(r_e + R_e)$$

Therefore, the input resistance R_{in} is $\boxed{(1+\beta)(r_e + R_e)}$.

Step 4 of 4

The output voltage v_o is,

$$v_o = (i_b + \alpha i_e)R_e \quad \text{..... (2)}$$

Manipulate the equation (1) as follows:

$$\frac{v_i}{(r_e + R_e)} = (i_b + \alpha i_e)$$

Substitute $\frac{v_i}{(r_e + R_e)}$ for $(i_b + \alpha i_e)$ in equation (2).

$$v_o = (i_b + \alpha i_e)R_e$$
$$v_o = \frac{v_i R_e}{(r_e + R_e)}$$
$$\frac{v_o}{v_i} = \frac{R_e}{(r_e + R_e)}$$

Therefore, the voltage gain $\frac{v_o}{v_i}$ is $\boxed{\frac{R_e}{(r_e + R_e)}}$.

Step 1 of 7

Refer to Figure P7.53 in the textbook.

Draw the small signal equivalent circuit as shown in Figure 1:

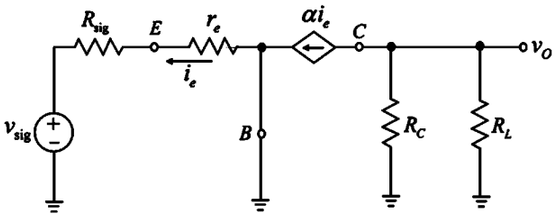


Figure 1

Step 2 of 7

The relation between emitter resistance r_e and emitter current is,

$$r_e = \frac{V_T}{I_E}$$

Substitute 0.33 mA for I_E and 25 mV for V_T.

$$r_e = \frac{25\text{ mV}}{0.33\text{ mA}}$$
$$= 75.76\ \Omega$$

Thus, emitter resistance r_e is $\boxed{75.76\ \Omega}$.

Step 3 of 7

The relation between g_m and r_e is,

$$g_m = \frac{\alpha}{r_e}$$

Substitute 0.99 for α and $75.76\ \Omega$ for r_e.

$$g_m = \frac{0.99}{75.76\ \Omega}$$
$$= 13\ \frac{\text{mA}}{\text{V}}$$

Thus, transconductance g_m is $\boxed{13\ \frac{\text{mA}}{\text{V}}}$

Step 4 of 7

Refer to the figure 1 emitter resistance r_e is equal to internal resistance R_{in} thus, internal resistance R_{in} is $\boxed{75.76\ \Omega}$

The small-signal equivalent of the circuit with T modal for the BJT is,

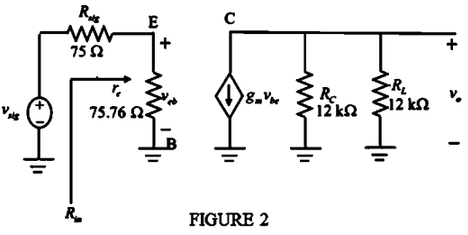


FIGURE 2

Step 5 of 7

Refer to the figure 1 voltage v_{eb} is $\frac{v_{sig} r_e}{r_e + R_{sig}}$

$$v_{eb} = \frac{v_{sig} r_e}{r_e + R_{sig}} \quad \text{..... (1)}$$

Refer to the figure 1 resistances R_C and R_L are parallel to each other and hence equivalent resistance is $\frac{R_L R_C}{R_C + R_L}$ and current flowing through $\frac{R_L R_C}{R_C + R_L}$ is $g_m v_{be}$

Hence, Output voltage v_o is $-\frac{R_L R_C}{R_C + R_L}(g_m v_{be})$

$$v_o = -\left(\frac{R_L R_C}{R_C + R_L}\right)(g_m v_{be}) \quad \text{..... (2)}$$

Step 6 of 7

Substitute v_{eb} from equation (1) into the equation (2)

$$v_o = -\left(\frac{R_L R_C}{R_C + R_L}\right)(g_m v_{be})$$
$$= -\left(\frac{R_L R_C}{R_C + R_L}\right)\left(-g_m \frac{v_{sig} r_e}{r_e + R_{sig}}\right)$$
$$= \left(\frac{R_L R_C}{R_C + R_L}\right)\left(\frac{g_m v_{sig} r_e}{r_e + R_{sig}}\right)$$

Step 7 of 7

Substitute $R_C = 12\ \text{k}\Omega, R_L = 12\ \text{k}\Omega, r_e = 75.76\ \Omega$ in the equation

$$\frac{v_o}{v_{sig}} = \left(\frac{R_L R_C}{R_C + R_L}\right)\left(\frac{g_m r_e}{r_e + R_{sig}}\right)$$

$$\frac{v_o}{v_{sig}} = \left(\frac{R_L R_C}{R_C + R_L}\right)\left(\frac{g_m r_e}{r_e + R_{sig}}\right)$$
$$= \left(\frac{(12\ \text{k}\Omega)(12\ \text{k}\Omega)}{(12\ \text{k}\Omega) + (12\ \text{k}\Omega)}\right)\left(\frac{\left(13\ \frac{\text{mA}}{\text{V}}\right)(75.76\ \Omega)}{(75.76\ \Omega) + (75\ \Omega)}\right)$$
$$= 39.20\ \frac{\text{V}}{\text{V}}$$

Overall voltage gain $\frac{v_o}{v_{sig}}$ is $\boxed{39.20\ \frac{\text{V}}{\text{V}}}$.

7.054P

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Step 1 of 5

7.055P

Refer to Figure 7.25 (a) in the text book.

The modified circuit diagram is shown in Figure 1.

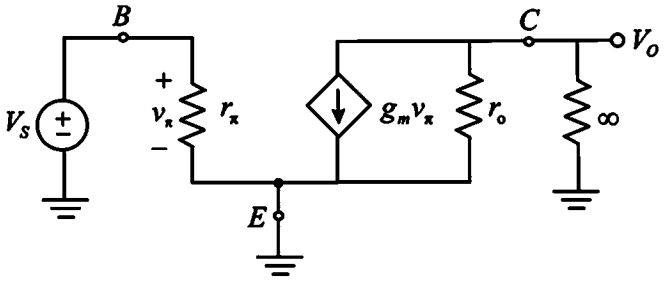


Figure 1

Step 2 of 5

The signal source (v_{sig}) is directly connected to base of the circuit, hence voltage v_π is equal to signal voltage v_{sig}.

Resistance load r_o is $\frac{V_A}{I_C}$

Refer to the figure 7.25(a) in the textbook output voltage v_o is $-g_m v_\pi r_o$

Substitute v_{sig} in the place of v_π in the equation $v_o = -g_m v_\pi r$ since $v_{sig} = v_\pi$

$$\begin{aligned} v_o &= -g_m v_\pi r_o \\ &= -g_m v_{sig} r_o \end{aligned}$$

Thus, voltage gain $\frac{v_o}{v_{sig}}$ is $\boxed{-g_m r_o}$

Step 3 of 5

Relation between g_m and I_C is $g_m = \frac{I_C}{V_T}$

Resistance load r_o is $\frac{V_A}{I_C}$

Substitute $r_o = \frac{V_A}{I_C}$ and $g_m = \frac{I_C}{V_T}$ in the equation $\frac{v_o}{v_{sig}} = -g_m r_o$

$$\begin{aligned} \frac{v_o}{v_{sig}} &= -g_m r_o \\ &= -\left(\frac{I_C}{V_T}\right)\left(\frac{V_A}{I_C}\right) \\ &= -\left(\frac{V_A}{V_T}\right) \end{aligned}$$

Thus, voltage gain $\frac{v_o}{v_{sig}}$ is $\boxed{-\left(\frac{V_A}{V_T}\right)}$.

Step 4 of 5

Voltage gain for early voltage V_A equal to $25\ \text{V}$ is,

$$\begin{aligned} \frac{v_o}{v_{sig}} &= -\left(\frac{V_A}{V_T}\right) \\ &= -\frac{25\ \text{V}}{25\ \text{mV}} \\ &= 1000\ \frac{\text{V}}{\text{V}} \end{aligned}$$

Thus, maximum possible voltage gain for $V_A = 25\ \text{V}$ is $\boxed{1000\ \frac{\text{V}}{\text{V}}}$.

Step 5 of 5

Voltage gain for early voltage V_A equal to $125\ \text{V}$ is,

$$\begin{aligned} \frac{v_o}{v_{sig}} &= -\left(\frac{V_A}{V_T}\right) \\ &= -\frac{125\ \text{V}}{25\ \text{mV}} \\ &= 5000\ \frac{\text{V}}{\text{V}} \end{aligned}$$

Thus, maximum possible voltage gain for $V_A = 125\ \text{V}$ is $\boxed{5000\ \frac{\text{V}}{\text{V}}}$.

7.056P

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7.057P

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7.083P

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7.084P

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Get help from a Chegg subject expert.

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Step 1 of 2

9.001E

Refer to Figure 8.2 in the textbook for a metal oxide semiconductor (MOS) differential pair with a common-mode input voltage, V_{CM}.

Consider that the drain resistances are increases by a factor of 2. Therefore,

$$\begin{aligned} R_D &= 2(2.5\ \text{k}\Omega) \\ &= 5\ \text{k}\Omega \end{aligned}$$

Find the values of v_{D1}, and v_{D2} in the circuit.

$$\begin{aligned} v_{D1} &= v_{D2} \\ &= V_{DD} - \frac{I}{2}R_D \end{aligned}$$

Here, V_{DD} is the supply voltage, I is the bias current, and R_D is the drain resistance.

Substitute 1.5 V for V_{DD}, 0.4 mA for I, and $5\ \text{k}\Omega$ for R_D.

$$\begin{aligned} v_{D1} &= v_{D2} \\ &= 1.5\ \text{V} - \left(\frac{0.4\ \text{mA}}{2}\right)(5\ \text{k}\Omega) \\ &= 1.5\ \text{V} - 1\ \text{V} \\ &= 0.5\ \text{V} \end{aligned}$$

Step 2 of 2

The maximum value of common-mode input, $V_{CM\,max}$ is,

$$V_{CM\,max} = V_t + V_D$$

Here, V_t is the threshold voltage, and V_D is the drain voltage.

Substitute 0.5 V for V_t, and 0.5 V for V_D.

$$\begin{aligned} V_{CM\,max} &= 0.5\ \text{V} + 0.5\ \text{V} \\ &= 1\ \text{V} \end{aligned}$$

The minimum value of common-mode input, $V_{CM\,min}$ is,

$$V_{CM\,min} = -V_{DD} + V_{CS} + V_{GS}$$

Here, V_{CS} is the minimum voltage required by the current source for proper operation, and V_{GS} is the gate-source voltage.

Substitute 1.5 V for V_{DD}, 0.4 V for V_{CS}, and 0.82 V for V_{GS}.

$$\begin{aligned} V_{CM\,min} &= -1.5\ \text{V} + 0.4\ \text{V} + 0.82\ \text{V} \\ &= -1.5\ \text{V} + 1.22\ \text{V} \\ &= -0.28\ \text{V} \end{aligned}$$

Thus, the range of the common-mode input, V_{CM} is, $\boxed{-0.28\ \text{V} \le V_{CM} \le 1\ \text{V}}$.

9.001P

We don't have the solution to this problem yet.

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Step 1 of 8

(a)

Refer to Figure 8.2 in the textbook for a MOS differential pair.

Draw the circuit if Q_1 conducts and Q_2 turn off.

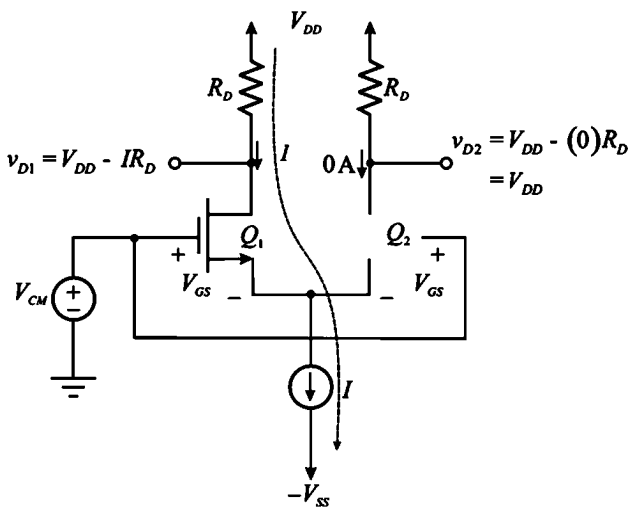


Figure 1

Step 2 of 8

The value of differential input, v_{id} at which the entire bias current, I is steered into Q_1 is,

$$v_{id\,max} = v_{GS1} + v_S$$

Find the value of gate-source voltage, v_{GS1}.

$$I = \frac{1}{2}\left(k_n' \frac{W}{L}\right)\left(v_{GS1} - V_t\right)^2$$

$$\frac{I}{\frac{1}{2}\left(k_n' \frac{W}{L}\right)} = \left(v_{GS1} - V_t\right)^2$$

$$v_{GS1} - V_t = \sqrt{\frac{I}{\frac{1}{2}\left(k_n' \frac{W}{L}\right)}}$$

$$v_{GS1} = V_t + \sqrt{\frac{I}{\frac{1}{2}\left(k_n' \frac{W}{L}\right)}}$$

$$v_{GS1} = V_t + \sqrt{\frac{2I}{k_n' \frac{W}{L}}}$$

$$v_{GS1} = V_t + \sqrt{2}\sqrt{\frac{I}{k_n' \frac{W}{L}}}$$

$$v_{GS1} = V_t + \sqrt{2}V_{OV} \qquad \left(\text{Since, } V_{OV} = \sqrt{\frac{I}{k_n' \frac{W}{L}}}\right)$$

Hence, the gate-source voltage is, $v_{GS1} = V_t + \sqrt{2}V_{OV}$.

Step 3 of 8

Substitute the expression of v_{GS1} in the equation of $v_{id\,max}$.

$$v_{id\,max} = v_{GS1} + v_S$$
$$= V_t + \sqrt{2}V_{OV} + v_S$$
$$= V_t + \sqrt{2}V_{OV} - V_t \qquad (\text{since, } v_S = -V_t)$$
$$= \sqrt{2}V_{OV}$$

Simplify the function further.

$$v_{id\,max} = \sqrt{2}\sqrt{\frac{I}{k_n' \frac{W}{L}}} \qquad \left(\text{since, } V_{OV} = \sqrt{\frac{I}{k_n' \frac{W}{L}}}\right)$$
$$= \sqrt{\frac{2I}{k_n' \frac{W}{L}}}$$

Here, the transistor parameter, $k_n'\left(\frac{W}{L}\right) = 4\,\text{mA/V}^2$, the bias current, $I = 0.4\,\text{mA}$.

$$v_{id\,max} = \sqrt{\frac{2(0.4\text{ mA})}{(4\text{ mA/V}^2)}}$$
$$= \sqrt{0.2\text{ V}^2}$$
$$= 0.45\text{ V}$$

Therefore, the value of v_{id} is, $\boxed{0.45\text{ V}}$.

Step 4 of 8

The drain voltage of transistor, Q_1 in the circuit is,

$$v_{D1} = V_{DD} - IR_D$$

Here, the supply voltage, $V_{DD} = 1.5\text{ V}$, the bias current, $I = 0.4\text{ mA}$, and the drain resistance, $R_D = 2.5\,\text{k}\Omega$.

$$v_{D1} = 1.5\text{ V} - (0.4\text{ mA})(2.5\,\text{k}\Omega)$$
$$= 1.5 - 1$$
$$= 0.5\text{ V}$$

Therefore, the drain voltage of transistor, Q_1 is, $\boxed{0.5\text{ V}}$.

The drain voltage of transistor, Q_2 is,

$$v_{D2} = V_{DD}$$

Substitute 1.5 V for V_{DD}.

$$v_{D2} = 1.5\text{ V}$$

Therefore, the drain voltage of transistor, Q_2 is, $\boxed{1.5\text{ V}}$.

Step 5 of 8

(b)

Redraw the circuit if Q_2 conducts and Q_1 turn off.

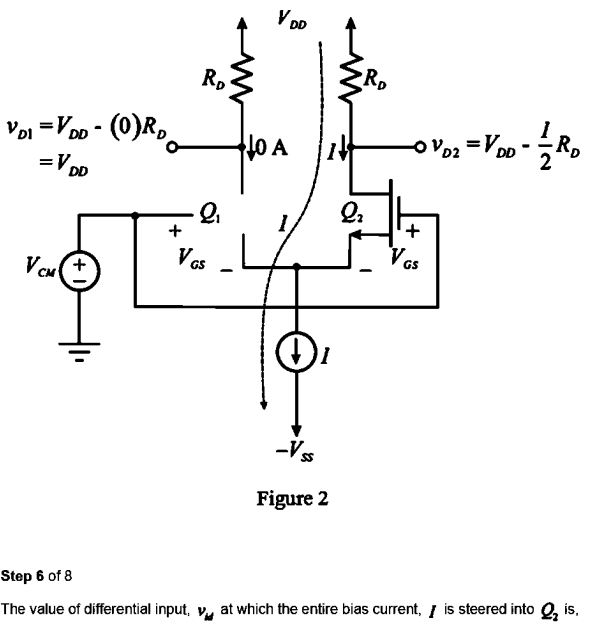


Figure 2

Step 6 of 8

The value of differential input, v_{id} at which the entire bias current, I is steered into Q_2 is,

$$v_{id\,max} = -\sqrt{2}V_{OV}$$
$$= -\sqrt{2}\left(\sqrt{\frac{I}{k_n' \frac{W}{L}}}\right) \qquad \left(\text{since, } V_{OV} = \sqrt{\frac{I}{k_n' \frac{W}{L}}}\right)$$
$$= -\sqrt{\frac{2I}{k_n' \frac{W}{L}}}$$

Here, the transistor parameter, $k_n'\left(\frac{W}{L}\right) = 4\,\text{mA/V}^2$, the bias current, $I = 0.4\,\text{mA}$.

$$v_{id\,max} = -\sqrt{\frac{2(0.4\text{ mA})}{(4\text{ mA/V}^2)}}$$
$$= -\sqrt{0.2\text{ V}^2}$$
$$= -0.45\text{ V}$$

Therefore, the value of v_{id} is, $\boxed{-0.45\text{ V}}$.

Step 7 of 8

The drain voltage of transistor, Q_1 is,

$$v_{D1} = V_{DD}$$

Substitute 1.5 V for V_{DD}.

$$v_{D1} = 1.5\text{ V}$$

Therefore, the drain voltage of transistor, Q_1 is, $\boxed{1.5\text{ V}}$.

The drain voltage of transistor, Q_2 is,

$$v_{D2} = V_{DD} - IR_D$$

Substitute 1.5 V for V_{DD}, 0.4 mA for I, and $2.5\,\text{k}\Omega$ for R_D.

$$v_{D2} = 1.5 - (0.4\text{ mA})(2.5\,\text{k}\Omega)$$
$$= 1.5 - 1$$
$$= 0.5\text{ V}$$

Therefore, the drain voltage of transistor, Q_2 is, $\boxed{0.5\text{ V}}$.

Step 8 of 8

(c)
The differential output voltage is,

Case-(1):

$$v_D = v_{D2} - v_{D1}$$
$$= 1.5 - 0.5$$
$$= 1\text{ V}$$

Hence, the differential output, $v_D = 1\text{ V}$.

Case-(2):

$$v_D = v_{D2} - v_{D1}$$
$$= 0.5 - 1.5$$
$$= -1\text{ V}$$

Hence, the differential output, $v_D = -1\text{ V}$.

Therefore, the range of differential output voltage is, $\boxed{-1\text{ V to }1\text{ V}}$.

9.002P

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Step 1 of 4

Refer to Figure 8.5 in the textbook for a MOS differential pair.

Consider the following expressions for the drain currents of transistors:

$$i_{D1} = \frac{I}{2} + \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)\sqrt{1 - \left(\frac{\frac{v_{id}}{2}}{V_{OV}}\right)^2}$$

$$i_{D2} = \frac{I}{2} - \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)\sqrt{1 - \left(\frac{\frac{v_{id}}{2}}{V_{OV}}\right)^2}$$

Here, I is the bias current, V_{OV} is the overdrive voltage, and v_{id} is the differential input.

Consider that the term involving v_{id}^2 in the equations is limited to 0.1.

$$\left(\frac{\frac{v_{id}}{2}}{V_{OV}}\right)^2 = 0.1$$

$$\frac{\frac{v_{id}}{2}}{V_{OV}} = \sqrt{0.1}$$

$$\frac{v_{id}}{2} = 0.3162 V_{OV}$$

$$\begin{aligned} v_{id} &= V_{OV}(2)(0.3162) \\ &= 0.6324 V_{OV} \end{aligned}$$

Therefore, the maximum value of differential input is, $|v_{id}|_{max} = 0.632 V_{OV}$.

The transconductance is,

$$g_m = \frac{I}{V_{OV}}$$

Consider the following expression for the drain current.

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}V_{OV}^2$$

Simplify the expression for the width to length ratio, $\frac{W}{L}$.

$$\begin{aligned} \frac{W}{L} &= \frac{2I_D}{\mu_n C_{ox} V_{OV}^2} \\ &= \frac{2\left(\frac{I}{2}\right)}{\mu_n C_{ox} V_{OV}^2} \quad \left(\text{since, } I_D = \frac{I}{2}\right) \\ &= \frac{I}{\mu_n C_{ox} V_{OV}^2} \end{aligned}$$

Here, the process transconductance parameter is $\mu_n C_{ox}$.

Step 2 of 4

Consider that the overdrive voltage, V_{OV} is, 0.2 V.

Substitute 0.2 V for V_{OV} in the equation of $|v_{id}|_{max}$.

$$\begin{aligned} |v_{id}|_{max} &= 0.632 V_{OV} \\ &= 0.632(0.2\text{ V}) \\ &= 0.1264\text{ V} \\ &\approx 126\text{ mV} \end{aligned}$$

Substitute 0.4 mA for I_D , and 0.2 V for V_{OV} in the equation of g_m .

$$\begin{aligned} g_m &= \frac{I}{V_{OV}} \\ &= \frac{0.4\text{ mA}}{0.2\text{ V}} \\ &= 2\text{ mA/V} \end{aligned}$$

Substitute 0.4 mA for I_D , 0.2 mA/V^2 for $\mu_n C_{ox}$, and 0.2 V for V_{OV} in the equation of $\frac{W}{L}$.

$$\begin{aligned} \frac{W}{L} &= \frac{I}{\mu_n C_{ox} V_{OV}^2} \\ &= \frac{0.4\text{ mA}}{(0.2\text{ mA/V}^2)(0.2\text{ V})^2} \\ &= \frac{0.4\text{ mA}}{0.008\text{ mA}} \\ &= 50 \end{aligned}$$

Thus, the width to length ratio, transconductance, and the maximum value of differential input respectively for the overdrive voltage, $V_{OV} = 0.2$ V are, $\boxed{50}$, $\boxed{2\text{ mA/V}}$, and $\boxed{126\text{ mV}}$.

Step 3 of 4

Consider that the overdrive voltage, V_{OV} is, 0.3 V.

Substitute 0.3 V for V_{OV} in the equation of $|v_{id}|_{max}$.

$$\begin{aligned} |v_{id}|_{max} &= 0.632 V_{OV} \\ &= 0.632(0.3\text{ V}) \\ &= 0.1897\text{ V} \\ &\approx 190\text{ mV} \end{aligned}$$

Substitute 0.4 mA for I_D , and 0.3 V for V_{OV} in the equation of g_m .

$$\begin{aligned} g_m &= \frac{I}{V_{OV}} \\ &= \frac{0.4\text{ mA}}{0.3\text{ V}} \\ &= 1.34\text{ mA/V} \end{aligned}$$

Substitute 0.4 mA for I_D , 0.2 mA/V^2 for $\mu_n C_{ox}$, and 0.3 V for V_{OV} in the equation of $\frac{W}{L}$.

$$\begin{aligned} \frac{W}{L} &= \frac{I}{\mu_n C_{ox} V_{OV}^2} \\ &= \frac{0.4\text{ mA}}{(0.2\text{ mA/V}^2)(0.3\text{ V})^2} \\ &= \frac{0.4\text{ mA}}{0.018\text{ mA}} \\ &= 22.23 \end{aligned}$$

Thus, the width to length ratio, transconductance, and the maximum value of differential input respectively for the overdrive voltage, $V_{OV} = 0.3$ V are, $\boxed{22.23}$, $\boxed{1.34\text{ mA/V}}$, and $\boxed{190\text{ mV}}$.

Step 4 of 4

Consider that the overdrive voltage, V_{OV} is, 0.4 V.

Substitute 0.4 V for V_{OV} in the equation of $|v_{id}|_{max}$.

$$\begin{aligned} |v_{id}|_{max} &= 0.632 V_{OV} \\ &= 0.632(0.4\text{ V}) \\ &= 0.2529\text{ V} \\ &\approx 253\text{ mV} \end{aligned}$$

Substitute 0.4 mA for I_D , and 0.4 V for V_{OV} in the equation of g_m .

$$\begin{aligned} g_m &= \frac{I}{V_{OV}} \\ &= \frac{0.4\text{ mA}}{0.4\text{ V}} \\ &= 1\text{ mA/V} \end{aligned}$$

Substitute 0.4 mA for I_D , 0.2 mA/V^2 for $\mu_n C_{ox}$, and 0.4 V for V_{OV} in the equation of $\frac{W}{L}$.

$$\begin{aligned} \frac{W}{L} &= \frac{I}{\mu_n C_{ox} V_{OV}^2} \\ &= \frac{0.4\text{ mA}}{(0.2\text{ mA/V}^2)(0.4\text{ V})^2} \\ &= \frac{0.4\text{ mA}}{0.032\text{ mA}} \\ &= 12.5 \end{aligned}$$

Thus, the width to length ratio, transconductance, and the maximum value of differential input respectively for the overdrive voltage, $V_{OV} = 0.2$ V are, $\boxed{12.5}$, $\boxed{1\text{ mA/V}}$, and $\boxed{253\text{ mV}}$.

9.003P

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Step 1 of 4

9.004E

Refer to Figure 8.4 in the textbook for a MOS differential amplifier.

The overdrive voltage is,

$$V_{OV} = \sqrt{\frac{I}{\mu_n C_{ox}\left(\frac{W}{L}\right)}}$$

Here, the bias current, $I = 0.8\ \text{mA}$, the process transconductance parameter, $\mu_n C_{ox} = 0.2\ \text{mA/V}^2$, and the width to length ratio, $\frac{W}{L} = 100$.

Substitute these values in the equation of V_{OV}.

$$\begin{aligned} V_{OV} &= \sqrt{\frac{0.8\ \text{mA}}{(0.2\ \text{mA/V}^2)(100)}} \\ &= \sqrt{\frac{0.8\ \text{mA}}{20\ \text{mA/V}^2}} \\ &= \sqrt{0.04\ \text{V}^2} \\ &= 0.2\ \text{V} \end{aligned}$$

Therefore, the overdrive voltage, V_{OV} is, $\boxed{0.2\ \text{V}}$.

Step 2 of 4

The transconductance is,

$$g_m = \frac{I}{V_{OV}}$$

Substitute 0.8 mA for I, and 0.2 V for V_{OV} in the equation.

$$\begin{aligned} g_m &= \frac{0.8\ \text{mA}}{0.2\ \text{V}} \\ &= 4\ \text{mA/V} \end{aligned}$$

Therefore, the transconductance, g_m is, $\boxed{4\ \text{mA/V}}$.

Step 3 of 4

The small-signal output resistance is,

$$\begin{aligned} r_o &= \frac{V_A}{I_D} \\ &= \frac{V_A}{\frac{I}{2}} \quad \left(\text{since, } I_D = \frac{I}{2}\right) \end{aligned}$$

Here, V_A is the early voltage.

Substitute 0.8 mA for I, and 20 V for V_A in the equation.

$$\begin{aligned} r_o &= \frac{20\ \text{V}}{\frac{0.8\ \text{mA}}{2}} \\ &= \frac{20\ \text{V}}{0.4\ \text{mA}} \\ &= 50\ \text{k}\Omega \end{aligned}$$

Therefore, the small-signal output resistance, r_o is, $\boxed{50\ \text{k}\Omega}$.

Step 4 of 4

The differential gain is,

$$\begin{aligned} A_D &= g_m(r_o \parallel R_D) \\ &= g_m\left(\frac{r_o R_D}{r_o + R_D}\right) \end{aligned}$$

Here, R_D is the drain resistance.

Substitute 4 mA/V for g_m, $50\ \text{k}\Omega$ for r_o, and $5\ \text{k}\Omega$ for R_D in the equation.

$$\begin{aligned} A_D &= (4\ \text{mA/V})\left[\frac{(50\ \text{k}\Omega)(5\ \text{k}\Omega)}{50\ \text{k}\Omega + 5\ \text{k}\Omega}\right] \\ &= (4\ \text{mA/V})\left(\frac{250\ \text{M}\Omega}{55\ \text{k}\Omega}\right) \\ &= (4\ \text{mA/V})(4.55\ \text{k}\Omega) \\ &= 18.18\ \text{V/V} \end{aligned}$$

Therefore, the differential gain, A_D is, $\boxed{18.18\ \text{V/V}}$.

Step 1 of 8

Refer to the circuit diagram in Figure P8.2 in the textbook.

Threshold voltage for PMOS is, $V_{tp} = -0.8\,\text{V}$

Trans-conductance parameter with width-to-length ratio is, $k_p'\left(\frac{W}{L}\right) = 4\,\text{mA/V}^2$.

Step 2 of 8

If transistor Q_1 is conducts and transistor Q_2 is turn off, the value of v_{id} at which the entire bias current I is steered into Q_1 is,

$$v_{id} = v_{GS1} + v_S \quad \text{..... (1)}$$

Write the expression for current, I.

$$I = \frac{1}{2}\left(k_p'\frac{W}{L}\right)\left(v_{GS1} - V_{tp}\right)^2$$

$$\frac{I}{\frac{1}{2}\left(k_p'\frac{W}{L}\right)} = \left(v_{GS1} - V_{tp}\right)^2$$

Rewrite the equation.

$$v_{GS1} - V_{tp} = \sqrt{\frac{I}{\frac{1}{2}\left(k_p'\frac{W}{L}\right)}}$$

$$= V_{tp} + \sqrt{\frac{I}{\frac{1}{2}\left(k_p'\frac{W}{L}\right)}}$$

Simplify further.

$$v_{GS1} = V_{tp} + \sqrt{\frac{2I}{k_p'\frac{W}{L}}}$$

$$= V_{tp} + \sqrt{2}\sqrt{\frac{I}{k_p'\frac{W}{L}}}$$

$$= V_{tp} - \sqrt{2}V_{OV} \qquad \left(\text{Since}, V_{OV} = -\sqrt{\frac{I}{k_n'\frac{W}{L}}}\right)$$

Hence, gate to source voltage in terms of threshold voltage and over driven voltage $v_{GS1} = V_{tp} - \sqrt{2}V_{OV}$

Step 3 of 8

Substitute v_{GS1} in equation (1)

$$v_{id} = v_{GS1} + v_S$$

$$= V_{tp} - \sqrt{2}V_{OV} + v_S$$

$$= V_{tp} - \sqrt{2}V_{OV} - V_{tp} \qquad \left(\text{Since}, v_S = -V_{tp}\right)$$

$$= -\sqrt{2}V_{OV}$$

$$= -\sqrt{2}\left(\sqrt{\frac{I}{k_p'\frac{W}{L}}}\right) \qquad \left(\text{Since}, V_{OV} = \sqrt{\frac{I}{k_n'\frac{W}{L}}}\right)$$

Rewrite the equation.

$$v_{id} = -\sqrt{\frac{2I}{k_p'\frac{W}{L}}} \quad \text{..... (2)}$$

Substitute I and $k_p'\frac{W}{L}$ in equation (2).

$$v_{id} = -\sqrt{\frac{2\left(0.5\times10^{-3}\right)}{\left(4\times10^{-3}\right)}}$$

$$= -\sqrt{\frac{1}{4}}$$

$$= -\sqrt{0.25}$$

$$= -0.5$$

Hence, we get $v_{id} = -0.5\,\text{V}$

Step 4 of 8

If Q_2 conducts and Q_1 turn off, In the similar manner we can show that in the negative direction, as v_{id} reaches $-\sqrt{2}V_{OV}$.

That is,

$$v_{id} = -\sqrt{2}V_{OV}$$

$$= \sqrt{2}\left(\sqrt{\frac{I}{k_p'\frac{W}{L}}}\right) \qquad \left(\text{Since}, V_{OV} = -\sqrt{\frac{I}{k_p'\frac{W}{L}}}\right)$$

$$= \sqrt{\frac{2I}{k_n'\frac{W}{L}}}$$

Step 5 of 8

Substitute I and $k_p'\frac{W}{L}$ in the equation.

$$v_{id} = \sqrt{\frac{2\left(0.5\times10^{-3}\right)}{\left(4\times10^{-3}\right)}}$$

$$= \sqrt{\frac{1}{4}}$$

$$= \sqrt{0.25}$$

$$= 0.5$$

Hence, we get $v_{id} = 0.5\,\text{V}$

Therefore, the range of v_{id} is $\boxed{-0.5 \le v_{id} \le 0.5}$

Step 6 of 8

When Q_1 conducts and Q_2 turn off, the entire bias current I is steered into Q_1:

The common-source voltage is,

$$v_S = -V_{tp}$$

$$= -(-0.8) \qquad \left(\text{Since}, V_{tp} = -0.8\,\text{V}\right)$$

$$= 0.8$$

Therefore, the common-source voltage is $\boxed{v_S = 0.8\,\text{V}}$.

Step 7 of 8

The drain voltage for Q_1 is,

$$v_{D1} = IR_D - V_{DD}$$

Substitute $V_{DD} = 2.5\,\text{V}, I = 0.5\,\text{mA}$ and $R_D = 4\,\text{k}\Omega$ in the equation.

$$v_{D1} = \left(0.5\times10^{-3}\right)\left(4\times10^{3}\right) - 2.5$$

$$= 2 - 2.5$$

$$= -0.5$$

Therefore, the drain voltage for Q_1 is $\boxed{v_{D1} = -0.5\,\text{V}}$.

Step 8 of 8

The drain voltage for Q_2 is,

$$v_{D1} = -V_{DD}$$

Substitute $V_{DD} = 2.5\,\text{V}$ in the equation.

$$v_{D2} = -2.5$$

$$= -2.5$$

$$= -2.5$$

Therefore, the drain voltage for Q_2 is $\boxed{v_{D2} = -2.5\,\text{V}}$.

Step 1 of 4

Refer to Figure 8.11 (a) in the textbook for a differential amplifier with current-source loads.

Find the width to length ratios for the transistors, Q_1 **and** Q_2 .

Write the expression for the drain current in these transistors.

$$I_D = \frac{1}{2}\left(\mu_n C_{ox}\right)\left(\frac{W}{L}\right)V_{OV}^2$$

Simplify the expression for the width to length ratio.

$$\begin{aligned}\left(\frac{W}{L}\right)_{1,2} &= \frac{2I_D}{\left(\mu_n C_{ox}\right)V_{OV}^2} \\ &= \frac{2\left(\frac{I}{2}\right)}{\left(\mu_n C_{ox}\right)V_{OV}^2} \quad \left(\text{since, } I_D = \frac{I}{2}\right) \\ &= \frac{I}{\left(\mu_n C_{ox}\right)V_{OV}^2}\end{aligned}$$

Here, I is the bias current, $\mu_n C_{ox}$ is the process transconductance parameter of an NMOS device, V_{OV} is the overdrive voltage.

Substitute $200\ \mu\text{A}$ for I , $400\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$, and 0.2 V for V_{OV} .

$$\begin{aligned}\left(\frac{W}{L}\right)_{1,2} &= \frac{200\ \mu\text{A}}{\left(400\ \mu\text{A/V}^2\right)\left(0.2\ \text{V}\right)^2} \\ &= \frac{200\ \mu\text{A}}{\left(400\ \mu\text{A/V}^2\right)\left(0.04\ \text{V}^2\right)} \\ &= \frac{200\ \mu\text{A}}{16\ \mu\text{A}} \\ &= 12.5\end{aligned}$$

Thus, the width to length ratios, $\left(\frac{W}{L}\right)_{1,2}$ for the transistors, Q_1, **and** Q_2 are, $\boxed{12.5}$.

Step 2 of 4

Find the width to length ratios for the transistors, Q_3, **and** Q_4 .

Write the expression for the drain current in these transistors.

$$I_D = \frac{1}{2}\left(\mu_p C_{ox}\right)\left(\frac{W}{L}\right)V_{OV}^2$$

Substitute $\frac{I}{2}$ for I_D , and simplify the expression for the width to length ratio.

$$\left(\frac{W}{L}\right)_{3,4} = \frac{I}{\left(\mu_p C_{ox}\right)V_{OV}^2}$$

The process transconductance parameter of a PMOS device is,

$$\begin{aligned}\mu_p C_{ox} &= \frac{\mu_n C_{ox}}{4} \\ &= \frac{400\ \mu\text{A/V}^2}{4} \\ &= 100\ \mu\text{A/V}^2\end{aligned}$$

Substitute $200\ \mu\text{A}$ for I , $100\ \mu\text{A/V}^2$ for $\mu_p C_{ox}$, and 0.2 V for V_{OV} .

$$\begin{aligned}\left(\frac{W}{L}\right)_{3,4} &= \frac{200\ \mu\text{A}}{\left(100\ \mu\text{A/V}^2\right)\left(0.2\ \text{V}\right)^2} \\ &= \frac{200\ \mu\text{A}}{\left(100\ \mu\text{A/V}^2\right)\left(0.04\ \text{V}^2\right)} \\ &= \frac{200\ \mu\text{A}}{4\ \mu\text{A}} \\ &= 50\end{aligned}$$

Thus, the width to length ratios, $\left(\frac{W}{L}\right)_{3,4}$ for the transistors, Q_3, **and** Q_4 are, $\boxed{50}$.

Step 3 of 4

The output resistance of transistor, Q_1 is,

$$\begin{aligned}r_{o1} &= \frac{V_A' L}{I_D} \\ &= \frac{2V_A' L}{I}\left(\text{since, } I_D = \frac{I}{2}\right)\end{aligned}$$

Here, V_A' is the early voltage, and L is the length of the technology.

Substitute $10\ \text{V/}\mu\text{m}$ for V_A' , $0.18\ \mu\text{m}$ for L , and $200\ \mu\text{A}$ for I .

$$\begin{aligned}r_{o1} &= \frac{2\left(10\ \text{V/}\mu\text{m}\right)\left(0.18\ \mu\text{m}\right)}{200\ \mu\text{A}} \\ &= \frac{3.6\ \text{V}}{200\ \mu\text{A}} \\ &= 18\ \text{k}\Omega\end{aligned}$$

Here, the output resistance, r_o for all the transistors is equal, since the output resistance does not depend on the process transconductance parameter.

The transconductance of transistor, Q_1 is,

$$g_m = \frac{I}{V_{OV}}$$

Substitute $200\ \mu\text{A}$ for I , and 0.2 V for V_{OV} .

$$\begin{aligned}g_m &= \frac{200\ \mu\text{A}}{0.2\ \text{V}} \\ &= 1\ \text{mA/V}\end{aligned}$$

Here, the transconductance, g_m for all the transistors is equal, since the transconductance does not depend on the process transconductance parameter.

Step 4 of 4

The differential gain of the amplifier is,

$$\begin{aligned}A_d &= 2g_{m1}\left(r_{o1} \parallel r_{o3}\right) \\ &= 2g_{m1}\left(\frac{r_{o1} r_{o3}}{r_{o1} + r_{o3}}\right)\end{aligned}$$

Substitute 1 mA/V for g_{m1} , $18\ \text{k}\Omega$ for r_{o1} **and** r_{o3} to find the differential gain, A_d .

$$\begin{aligned}A_d &= 2\left(1\ \text{mA/V}\right)\left[\frac{\left(18\ \text{k}\Omega\right)\left(18\ \text{k}\Omega\right)}{18\ \text{k}\Omega + 18\ \text{k}\Omega}\right] \\ &= 2\left(1\ \text{mA/V}\right)\left(9\ \text{k}\Omega\right) \\ &= 18\ \text{V/V}\end{aligned}$$

Thus, the differential gain, A_d for the cascode differential amplifier is, $\boxed{18\ \text{V/V}}$.

9.005P

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Step 1 of 4

Refer to Figure 8.12 (a) in the textbook for a CMOS (complementary metal oxide semiconductor) cascode differential amplifier.

Find the width to length ratios for the transistors, Q_1, Q_2, Q_3, and Q_4.

Write the expression for the drain current in these transistors.

$$I_D = \frac{1}{2}\left(\mu_n C_{ox}\right)\left(\frac{W}{L}\right)V_{OV}^2$$

Simplify the expression for the width to length ratio.

$$\begin{aligned}\left(\frac{W}{L}\right)_{1,2,3,4} &= \frac{2I_D}{\left(\mu_n C_{ox}\right)V_{OV}^2} \\ &= \frac{2\left(\frac{I}{2}\right)}{\left(\mu_n C_{ox}\right)V_{OV}^2} \quad \left(\text{since, } I_D = \frac{I}{2}\right) \\ &= \frac{I}{\left(\mu_n C_{ox}\right)V_{OV}^2}\end{aligned}$$

Here, I is the bias current, $\mu_n C_{ox}$ is the process transconductance parameter of an NMOS device, V_{OV} is the overdrive voltage.

Substitute $200\ \mu\text{A}$ for I, $400\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$, and 0.2 V for V_{OV}.

$$\begin{aligned}\left(\frac{W}{L}\right)_{1,2,3,4} &= \frac{200\ \mu\text{A}}{\left(400\ \mu\text{A/V}^2\right)\left(0.2\ \text{V}\right)^2} \\ &= \frac{200\ \mu\text{A}}{\left(400\ \mu\text{A/V}^2\right)\left(0.04\ \text{V}^2\right)} \\ &= \frac{200\ \mu\text{A}}{16\ \mu\text{A}} \\ &= 12.5\end{aligned}$$

Thus, the width to length ratios, $\left(\frac{W}{L}\right)_{1,2,3,4}$ for the transistors, Q_1, Q_2, Q_3, and Q_4 are, $\boxed{12.5}$.

Step 2 of 4

Find the width to length ratios for the transistors, Q_5, Q_6, Q_7, and Q_8.

Write the expression for the drain current in these transistors.

$$I_D = \frac{1}{2}\left(\mu_p C_{ox}\right)\left(\frac{W}{L}\right)V_{OV}^2$$

Substitute $\frac{I}{2}$ for I_D, and simplify the expression for the width to length ratio.

$$\left(\frac{W}{L}\right)_{5,6,7,8} = \frac{I}{\left(\mu_p C_{ox}\right)V_{OV}^2}$$

Here, I is the bias current, and V_{OV} is the overdrive voltage.

The process transconductance parameter of a PMOS device is,

$$\begin{aligned}\mu_p C_{ox} &= \frac{\mu_n C_{ox}}{4} \\ &= \frac{400\ \mu\text{A/V}^2}{4} \\ &= 100\ \mu\text{A/V}^2\end{aligned}$$

Substitute $200\ \mu\text{A}$ for I, $100\ \mu\text{A/V}^2$ for $\mu_p C_{ox}$, and 0.2 V for V_{OV}.

$$\begin{aligned}\left(\frac{W}{L}\right)_{5,6,7,8} &= \frac{200\ \mu\text{A}}{\left(100\ \mu\text{A/V}^2\right)\left(0.2\ \text{V}\right)^2} \\ &= \frac{200\ \mu\text{A}}{\left(100\ \mu\text{A/V}^2\right)\left(0.04\ \text{V}^2\right)} \\ &= \frac{200\ \mu\text{A}}{4\ \mu\text{A}} \\ &= 50\end{aligned}$$

Thus, the width to length ratios, $\left(\frac{W}{L}\right)_{5,6,7,8}$ for the transistors, Q_5, Q_6, Q_7, and Q_8 are, $\boxed{50}$.

Step 3 of 4

The output resistance of transistor, Q_1 is,

$$\begin{aligned}r_{o1} &= \frac{V_A' L}{I_D} \\ &= \frac{2V_A' L}{I} \left(\text{since, } I_D = \frac{I}{2}\right)\end{aligned}$$

Here, V_A' is the early voltage, and L is the length of the technology.

Substitute $10\ \text{V/}\mu\text{m}$ for V_A', $0.18\ \mu\text{m}$ for L, and $200\ \mu\text{A}$ for I.

$$\begin{aligned}r_{o1} &= \frac{2\left(10\ \text{V/}\mu\text{m}\right)\left(0.18\ \mu\text{m}\right)}{200\ \mu\text{A}} \\ &= \frac{3.6\ \text{V}}{200\ \mu\text{A}} \\ &= 18\ \text{k}\Omega\end{aligned}$$

Here, the output resistance, r_o for all the transistors is equal, since the output resistance does not depend on the process transconductance parameter.

The transconductance of transistor, Q_1 is,

$$g_m = \frac{I}{V_{OV}}$$

Substitute $200\ \mu\text{A}$ for I, and 0.2 V for V_{OV}.

$$\begin{aligned}g_m &= \frac{200\ \mu\text{A}}{0.2\ \text{V}} \\ &= 1\ \text{mA/V}\end{aligned}$$

Here, the transconductance, g_m for all the transistors is equal, since the transconductance does not depend on the process transconductance parameter.

Step 4 of 4

The differential gain of the amplifier is,

$$\begin{aligned}A_d &= 2g_{m1}\left(R_{on} \parallel R_{op}\right) \\ &= 2g_{m1}\left(\frac{R_{on}R_{op}}{R_{on} + R_{op}}\right)\end{aligned}$$

Here,

$$R_{on} = 2\left(g_{m3}r_{o3}\right)r_{o1}$$

Substitute 1 mA/V for g_{m3}, $18\ \text{k}\Omega$ for r_{o3} and r_{o1}.

$$\begin{aligned}R_{on} &= 2\left(1\ \text{mA/V}\right)\left(18\ \text{k}\Omega\right)\left(18\ \text{k}\Omega\right) \\ &= 648\ \text{k}\Omega\end{aligned}$$

Similarly,

$$R_{op} = 2\left(g_{m5}r_{o5}\right)r_{o7}$$

Substitute 1 mA/V for g_{m5}, $18\ \text{k}\Omega$ for r_{o5} and r_{o7}.

$$\begin{aligned}R_{op} &= 2\left(1\ \text{mA/V}\right)\left(18\ \text{k}\Omega\right)\left(18\ \text{k}\Omega\right) \\ &= 648\ \text{k}\Omega\end{aligned}$$

Substitute 1 mA/V for g_{m1}, $648\ \text{k}\Omega$ for r_{on} and r_{op} to find the differential gain, A_d.

$$\begin{aligned}A_d &= 2\left(1\ \text{mA/V}\right)\left[\frac{\left(648\ \text{k}\Omega\right)\left(648\ \text{k}\Omega\right)}{648\ \text{k}\Omega + 648\ \text{k}\Omega}\right] \\ &= 2\left(1\ \text{mA/V}\right)\left(324\ \text{k}\Omega\right) \\ &= 648\ \text{V/V}\end{aligned}$$

Thus, the differential gain, A_d for the cascode differential amplifier is, $\boxed{648\ \text{V/V}}$.

9.006P

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9.007E

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9.007P

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9.008E

Consider a bipolar differential amplifier.

Consider that it is to provide a certain minimum voltage, V_{CS} across the current source, I to ensure its proper operation.

Determine the lower end of common-mode voltage, V_{CM} .

$$V_{CM\,min} = -V_{EE} + V_{CS} + V_{BE}$$

Here, the operating voltages of differential amplifier are, $\pm 2.5\ \text{V}$, the required voltage for proper operation is 0.3 V, and the base-emitter voltage is 0.7 V.

Substitute 2.5 V for V_{EE} , 0.3 V for V_{CS} , and 0.7 V for V_{BE} .

$$\begin{aligned} V_{CM\,min} &= -2.5\ \text{V} + 0.3\ \text{V} + 0.7\ \text{V} \\ &= -1.5\ \text{V} \end{aligned}$$

Note that the transistors, Q_1 **and** Q_2 leave the active mode and enters into saturation.

Assume that $|v_{BE}|$ of conducting transistor is approximately 0.7 V and that current gain, $\alpha \simeq 1$.

Determine the upper end of common-mode voltage, V_{CM} .

$$V_{CM\,max} = V_{CC} - \alpha \frac{I}{2} R_C + 0.4\ \text{V}$$

Substitute 2.5 V for V_{CC} , 1 for α , 0.4 mA for bias current, I , $5\ \text{k}\Omega$ for the collector resistance, R_C .

$$\begin{aligned} V_{CM\,max} &= 2.5\ \text{V} - (1)\left(\frac{0.4\ \text{mA}}{2}\right)(5\ \text{k}\Omega) + 0.4\ \text{V} \\ &= 2.9\ \text{V} - (0.2\ \text{mA})(5\ \text{k}\Omega) \\ &= 2.9\ \text{V} - 1\ \text{V} \\ &= 1.9\ \text{V} \end{aligned}$$

Thus, the range of input common mode input, V_{CM} is, $\boxed{-1.5\ \text{V to } +1.9\ \text{V}}$.

Step 1 of 5

Write the drain current, i_{D1} in terms of the input differential voltage v_{id}.

$$i_{D1} = \frac{I}{2} + \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)\sqrt{1 - \left(\frac{\frac{v_{id}}{2}}{V_{OV}}\right)^2}$$

Assume term $\left(\frac{\frac{v_{id}}{2}}{V_{OV}}\right)^2$ as k.

$$\left(\frac{\frac{v_{id}}{2}}{V_{OV}}\right)^2 = k$$

$$\frac{\frac{v_{id}}{2}}{V_{OV}} = \pm\sqrt{k}$$

$$v_{id} = 2\left(\pm\sqrt{k}\right)V_{OV}$$

Therefore, the maximum value of v_{id} is,

$$v_{id\,max} = 2\sqrt{k}V_{OV} \quad \text{..... (1)}$$

Step 2 of 5

Rewrite the drain current equation.

$$i_{D1} = \frac{I}{2} + \left(\frac{I}{V_{OV}}\right)\left(\frac{v_{id}}{2}\right)\sqrt{1 - \left(\frac{\frac{v_{id}}{2}}{V_{OV}}\right)^2}$$

Substitute $\left(\frac{\frac{v_{id}}{2}}{V_{OV}}\right)^2$ for k and $2\left(\pm\sqrt{k}\right)V_{OV}$ for v_{id}.

$$i_{D1} = \frac{I}{2} + \left(\frac{I}{V_{OV}}\right)\left(\frac{2\left(\pm\sqrt{k}\right)V_{OV}}{2}\right)\sqrt{1-k}$$

$$= \frac{I}{2} + \left(\frac{I}{V_{OV}}\right)\left(\frac{2\sqrt{k}V_{OV}}{2}\right)\sqrt{1-k}$$

$$= \frac{I}{2} + I\left(\sqrt{k}\right)\sqrt{1-k}$$

$$= \frac{I}{2} + I\sqrt{k(1-k)}$$

The maximum possible fractional change in the transistor current is,

$$\Delta I_{max} = i_{D1} - i_{D1\text{-original}}$$

$$= \left(\frac{I}{2} + I\sqrt{k(1-k)}\right) - \left(\frac{I}{2}\right) \qquad \left\{\begin{array}{c}\textbf{For matched transistors}\\ \textbf{since } i_{D1} = i_{D2} = \frac{I}{2}\end{array}\right\}$$

$$= I\sqrt{k(1-k)}$$

$$= I\left(\frac{2}{2}\right)\sqrt{k(1-k)}$$

Rewrite the equation.

$$\frac{\Delta I_{max}}{\frac{I}{2}} = 2\sqrt{k(1-k)} \quad \text{..... (2)}$$

Step 3 of 5

Recall equation (1).

$$v_{id\,max} = 2\sqrt{k}V_{OV}$$

Find maximum differential voltage for $k = 0.01$.

$$v_{id\,max} = 2\sqrt{0.01}V_{OV}$$

$$= 2(0.1)V_{OV}$$

$$= 0.2V_{OV}$$

Therefore, maximum differential voltage for $k = 0.01$ is $\boxed{v_{id\,max} = 0.2V_{OV}}$.

Recall equation (2).

$$\frac{\Delta I_{max}}{\frac{I}{2}} = 2\sqrt{k(1-k)}$$

Find maximum possible fractional change in the transistor current for $k = 0.01$.

$$\frac{\Delta I_{max}}{\frac{I}{2}} = 2\sqrt{0.01(1-0.01)}$$

$$= 2\sqrt{0.01(0.99)}$$

$$= 2\sqrt{0.0099}$$

$$= 0.199$$

Thus, maximum possible fractional change in the transistor for $k = 0.01$ is $\boxed{\frac{\Delta I_{max}}{\frac{I}{2}} = 0.199}$.

Step 4 of 5

Recall equation (1).

$$v_{id\,max} = 2\sqrt{k}V_{OV}$$

Find maximum differential voltage for $k = 0.1$.

$$v_{id\,max} = 2\sqrt{0.1}V_{OV}$$

$$= 2(0.316)V_{OV}$$

$$= 0.632V_{OV}$$

Therefore, maximum differential voltage for $k = 0.01$ is $\boxed{v_{id\,max} = 0.632V_{OV}}$.

Recall equation (2).

$$\frac{\Delta I_{max}}{\frac{I}{2}} = 2\sqrt{k(1-k)}$$

Find maximum possible fractional change in the transistor current for $k = 0.1$.

$$\frac{\Delta I_{max}}{\frac{I}{2}} = 2\sqrt{0.1(1-0.1)}$$

$$= 2\sqrt{0.1(0.9)}$$

$$= 2\sqrt{0.09}$$

$$= 0.6$$

Thus, maximum possible fractional change in the transistor for $k = 0.1$ is $\boxed{\frac{\Delta I_{max}}{\frac{I}{2}} = 0.6}$.

Step 5 of 5

Recall equation (1).

$$v_{id\,max} = 2\sqrt{k}V_{OV}$$

Find maximum differential voltage for $k = 0.2$.

$$v_{id\,max} = 2\sqrt{0.2}V_{OV}$$

$$= 2(0.447)V_{OV}$$

$$= 0.894V_{OV}$$

Therefore, maximum differential voltage for $k = 0.2$ is $\boxed{v_{id\,max} = 0.894V_{OV}}$.

Recall equation (2).

$$\frac{\Delta I_{max}}{\frac{I}{2}} = 2\sqrt{k(1-k)}$$

Find maximum possible fractional change in the transistor current for $k = 0.2$.

$$\frac{\Delta I_{max}}{\frac{I}{2}} = 2\sqrt{0.2(1-0.2)}$$

$$= 2\sqrt{0.2(0.8)}$$

$$= 0.8$$

Thus, maximum possible fractional change in the transistor for $k = 0.2$ is $\boxed{\frac{\Delta I_{max}}{\frac{I}{2}} = 0.8}$.

9.009E

Refer to Figure 5.18 in the textbook for basic bipolar junction transistor (BJT) differential-pair configuration.

Consider that the emitter current, i_{E1} for transistor, Q_1 is $0.99I$.

Here, I is the bias current.

The exponential relation between the emitter current and input differential voltage is,

$$i_{E1} = \frac{I}{1+e^{\frac{-v_{id}}{V_T}}}$$

$$0.99I = \frac{I}{1+e^{\frac{-v_{id}}{V_T}}} \qquad (\text{since, } i_{E1} = 0.99I)$$

$$0.99 = \frac{1}{1+e^{\frac{-v_{id}}{V_T}}}$$

$$0.99\left(1+e^{\frac{-v_{id}}{V_T}}\right) = 1$$

$$1+e^{\frac{-v_{id}}{V_T}} = \frac{1}{0.99}$$

$$1+e^{\frac{-v_{id}}{V_T}} = 1.0101$$

$$e^{\frac{-v_{id}}{V_T}} = 1.0101-1$$

$$e^{\frac{-v_{id}}{V_T}} = 0.0101$$

$$\frac{-v_{id}}{V_T} = \ln(0.0101)$$

$$\frac{-v_{id}}{V_T} = -4.595$$

$$\frac{v_{id}}{V_T} = 4.595$$

$$v_{id} = 4.595V_T$$

Substitute 25 mV for the thermal equivalent voltage, V_T .

$$v_{id} = 4.595(25\text{ mV})$$
$$= 115\text{ mV}$$

Therefore, the input differential signal voltage, v_{id} is, $\boxed{115\text{ mV}}$.

9.009P

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Step 1 of 8

(a)

Refer to Figure 8.19 in the textbook for the differential amplifier.

Consider that the base-to-emitter voltage, v_{BE} is 0.7 V at the collector current, i_C of 1 mA.

Find the dc value of base-emitter voltage, V_{BE}.

$$\begin{aligned} V_{BE} &= v_{be} + V_T \ln\left(\frac{I_C}{i_C}\right) \\ &= v_{be} + V_T \ln\left(\frac{\alpha I_E}{i_C}\right) \quad (\text{since, } I_C = \alpha I_E) \\ &= v_{be} + V_T \ln\left(\frac{\alpha \frac{I}{2}}{i_C}\right) \quad \left(\text{since, for matched transistors, } I_E = \frac{I}{2}\right) \end{aligned}$$

Here, V_T is the thermal equivalent voltage, α is the current gain, and I is the bias current.

Substitute 0.7 V for v_{BE}, 25 mV for V_T, 1 mA for I and i_C, and 1 for α.

$$\begin{aligned} V_{BE} &= 0.7\text{ V} + (25\text{ mV}) \ln\left[\frac{(1)\left(\frac{1\text{ mA}}{2}\right)}{1\text{ mA}}\right] \\ &= 0.7\text{ V} + (25\text{ mV})(-0.69) \\ &= 0.7\text{ V} - 0.0173\text{ V} \\ &= 0.683\text{ V} \end{aligned}$$

Find the emitter voltage.

$$V_E = V_B - V_{BE} \quad (\text{since, } V_{BE} = V_B - V_E)$$

Substitute 5 V for V_B, and 0.683 V for V_{BE}.

$$\begin{aligned} V_E &= 5\text{ V} - 0.683\text{ V} \quad (\text{since, for DC, } V_B = 5\text{ V}) \\ &= 4.317\text{ V} \end{aligned}$$

Therefore, the emitter voltage, V_E is, $\boxed{4.317\text{ V}}$.

Step 2 of 8

(b)

The transconductance is,

$$\begin{aligned} g_m &= \frac{I_C}{V_T} \\ &= \frac{\alpha I_E}{V_T} \ (\text{since, } I_C = \alpha I_E) \\ &= \frac{\alpha \frac{I}{2}}{V_T} \\ &= \frac{I}{2V_T} \ (\text{since, } \alpha = 1) \end{aligned}$$

Substitute 1 mA for I, and 25 mV for V_T.

$$\begin{aligned} g_m &= \frac{1\text{ mA}}{2(25\text{ mV})} \\ &= \frac{1\text{ mA}}{50\text{ mV}} \\ &= 20\text{ mA/V} \end{aligned}$$

Therefore, the transconductance, g_m for each transistor is, $\boxed{20\text{ mA/V}}$.

Step 3 of 8

(c)

The collector current for transistor, Q_1 is,

$$\begin{aligned} i_{C1} &= \frac{\alpha I}{2} + g_m \frac{v_{id}}{2} \\ &= \frac{\alpha I}{2} + g_m \left(\frac{v_{B1} - v_{B2}}{2}\right) \ (\text{since, } v_{id} = v_{B1} - v_{B2}) \end{aligned}$$

Here, the input voltages, v_{B1} and v_{B2} are,

$$\begin{aligned} v_{B1} &= 5 + 0.005\sin(2\pi 1000t)\text{ V} \\ v_{B2} &= 5 - 0.005\sin(2\pi 1000t)\text{ V} \end{aligned}$$

Substitute all the values in the equation of i_{C1}.

$$\begin{aligned} i_{C1} &= \frac{(1)(1\text{ mA})}{2} + (20\text{ mA/V})\left[\frac{(5 + 0.005\sin(2\pi 1000t)) - (5 - 0.005\sin(2\pi 1000t))}{2}\right] \\ &= \frac{1\text{ mA}}{2} + (20\text{ mA/V})\left[\frac{5 + 0.005\sin(2\pi 1000t) - 5 + 0.005\sin(2\pi 1000t)}{2}\right] \\ &= 0.5\text{ mA} + (20\text{ mA/V})\left[0.005\sin(2\pi 1000t)\right] \\ &= 0.5 + 0.1\sin(2\pi 1000t)\text{ mA} \end{aligned}$$

Therefore, the collector current for transistor, Q_1 is, $\boxed{i_{C1} = 0.5 + 0.1\sin(2\pi 1000t)\text{ mA}}$.

Step 4 of 8

The collector current for transistor, Q_2 is,

$$\begin{aligned} i_{C2} &= \frac{\alpha I}{2} - g_m \frac{v_{id}}{2} \\ &= \frac{\alpha I}{2} - g_m \left(\frac{v_{B1} - v_{B2}}{2}\right) \end{aligned}$$

Substitute all the values in the equation of i_{C2}.

$$\begin{aligned} i_{C2} &= \frac{(1)(1\text{ mA})}{2} - (20\text{ mA/V})\left[\frac{(5 + 0.005\sin(2\pi 1000t)) - (5 - 0.005\sin(2\pi 1000t))}{2}\right] \\ &= \frac{1\text{ mA}}{2} - (20\text{ mA/V})\left[\frac{5 + 0.005\sin(2\pi 1000t) - 5 + 0.005\sin(2\pi 1000t)}{2}\right] \\ &= 0.5\text{ mA} - (20\text{ mA/V})\left[0.005\sin(2\pi 1000t)\right] \\ &= 0.5 - 0.1\sin(2\pi 1000t)\text{ mA} \end{aligned}$$

Therefore, the collector current for transistor, Q_2 is, $\boxed{i_{C2} = 0.5 - 0.1\sin(2\pi 1000t)\text{ mA}}$.

Step 5 of 8

(d)

The collector voltage for transistor, Q_1 is,

$$\begin{aligned} v_{C1} &= \left(V_{CC} - \frac{\alpha I}{2} R_C\right) - g_m R_C \frac{v_{id}}{2} \\ &= \left(V_{CC} - \frac{\alpha I}{2} R_C\right) - g_m R_C \left(\frac{v_{B1} - v_{B2}}{2}\right) \ (\text{since, } v_{id} = v_{B1} - v_{B2}) \end{aligned}$$

Here, the supply voltage, V_{CC} is 15 V, and the collector resistance, R_C is $10\text{ k}\Omega$.

Substitute all the values in the equation of v_{C1}.

$$\begin{aligned} v_{C1} &= \left[15\text{ V} - \frac{(1)(1\text{ mA})(10\text{ k}\Omega)}{2}\right] - (20\text{ mA/V})(10\text{ k}\Omega)\left\{\frac{\left[5 + 0.005\sin(2\pi 1000t)\right] - \left[5 - 0.005\sin(2\pi 1000t)\right]}{2}\right\} \\ &= \left[15\text{ V} - \frac{10\text{ V}}{2}\right] - (200)\left[\frac{5 + 0.005\sin(2\pi 1000t) - 5 + 0.005\sin(2\pi 1000t)}{2}\right] \\ &= (15\text{ V} - 5\text{ V}) - (200)\left[0.005\sin(2\pi 1000t)\right] \\ &= 10 - \sin(2\pi 1000t)\text{ V} \end{aligned}$$

Therefore, the collector voltage for transistor, Q_1 is, $\boxed{v_{C1} = 10 - \sin(2\pi 1000t)\text{ V}}$.

Step 6 of 8

The collector voltage for transistor, Q_2 is,

$$\begin{aligned} v_{C2} &= \left(V_{CC} - \frac{\alpha I}{2} R_C\right) + g_m R_C \frac{v_{id}}{2} \\ &= \left(V_{CC} - \frac{\alpha I}{2} R_C\right) + g_m R_C \left(\frac{v_{B1} - v_{B2}}{2}\right) \end{aligned}$$

Substitute all the values in the equation of v_{C2}.

$$\begin{aligned} v_{C2} &= \left[15\text{ V} - \frac{(1)(1\text{ mA})(10\text{ k}\Omega)}{2}\right] + (20\text{ mA/V})(10\text{ k}\Omega)\left\{\frac{\left[5 + 0.005\sin(2\pi 1000t)\right] - \left[5 - 0.005\sin(2\pi 1000t)\right]}{2}\right\} \\ &= \left[15\text{ V} - \frac{10\text{ V}}{2}\right] + (200)\left[\frac{5 + 0.005\sin(2\pi 1000t) - 5 + 0.005\sin(2\pi 1000t)}{2}\right] \\ &= (15\text{ V} - 5\text{ V}) + (200)\left[0.005\sin(2\pi 1000t)\right] \\ &= 10 + \sin(2\pi 1000t)\text{ V} \end{aligned}$$

Therefore, the collector voltage for transistor, Q_2 is, $\boxed{v_{C2} = 10 + \sin(2\pi 1000t)\text{ V}}$.

Step 7 of 8

(e)

The differential voltage between the two collector voltages is,

$$v_{Cdiff} = v_{C2} - v_{C1}$$

Substitute $10 + \sin(2\pi 1000t)\text{ V}$ for v_{C2}, and $10 - \sin(2\pi 1000t)\text{ V}$ for v_{C1}.

$$\begin{aligned} v_{Cdiff} &= \left[10 + \sin(2\pi 1000t)\right] - \left[10 - \sin(2\pi 1000t)\right] \\ &= 10 + \sin(2\pi 1000t) - 10 + \sin(2\pi 1000t) \\ &= 2\sin(2\pi 1000t)\text{ V} \end{aligned}$$

Therefore, the differential voltage between the two collector voltages is, $\boxed{v_{Cdiff} = 2\sin(2\pi 1000t)\text{ V}}$.

Step 8 of 8

(f)
The voltage gain is,

$$A_v = \frac{v_{C2} - v_{C1}}{v_{B1} - v_{B2}}$$

Substitute all the values in the equation of A_v.

$$\begin{aligned} A_v &= \frac{\left[10 + \sin(2\pi 1000t)\right] - \left[10 - \sin(2\pi 1000t)\right]}{\left[5 + 0.005\sin(2\pi 1000t)\right] - \left[5 - 0.005\sin(2\pi 1000t)\right]} \\ &= \frac{10 + \sin(2\pi 1000t) - 10 + \sin(2\pi 1000t)}{5 + 0.005\sin(2\pi 1000t) - 5 + 0.005\sin(2\pi 1000t)} \\ &= \frac{2\sin(2\pi 1000t)}{0.01\sin(2\pi 1000t)} \\ &= 200 \end{aligned}$$

Therefore, the voltage gain, A_v is, $\boxed{200\text{ V/V}}$.

9.010P

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9.011E

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9.011P

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9.012E

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9.012P

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9.013E

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9.013P

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9.014E

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9.014P

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9.015E

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9.015P

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9.016E

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9.016P

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9.017E

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Step 1 of 3

9.017P

Draw the common-source (CS) amplifier is shown in Figure 1:

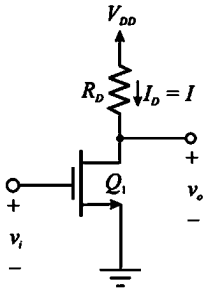


Figure 1

From Figure 1, the drain current for CS amplifier is,

$$I_D = I$$
$$I = I_D$$

The differential voltage gain for CS amplifier is,

$$A_{vo} = -g_m R_D$$

Step 2 of 3

Draw The MOS differential amplifier is shown in Figure 2:

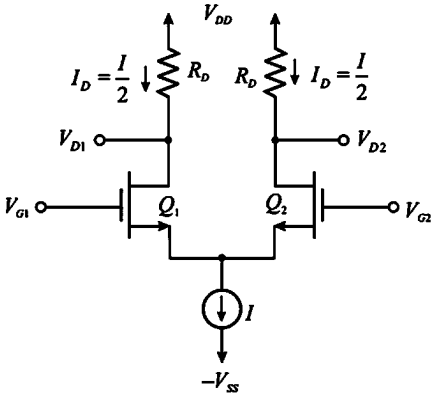


Figure 2

Step 3 of 3

From Figure 2, the drain current for differential amplifier is,

$$I_D = \frac{I}{2}$$
$$I = 2I_D$$

The differential voltage gain for differential amplifier is,

$$A_d = g_m R_D$$

Therefore, the differential amplifier requires twice bias current (I) compared to CS amplifier.

The power dissipation of the differential amplifier is also high, because the bias current is twice the drain current.

9.018E

We don't have the solution to this problem yet.

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Step 1 of 6

The common-source (CS) amplifier is shown in Figure 1:

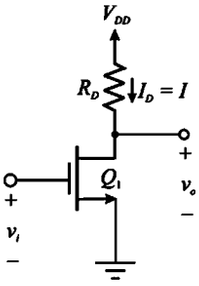


Figure 1

Step 2 of 6

From Figure 1, the drain current for CS amplifier is,

$$I = I_{CS}$$

$$I_D = I_{CS}$$
$$I_{CS} = I_D$$

The differential voltage gain for CS amplifier is,

$A_{vo} = -g_{m\text{-}CS}R_D$ or $|A_{vo}| = g_{m\text{-}CS}R_D$.

Step 3 of 6

The MOS differential amplifier is shown in Figure 2:

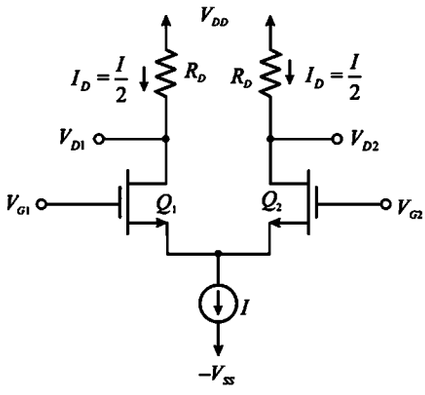


Figure 2

Step 4 of 6

From Figure 2, the drain current for differential amplifier is,

Assume bias current I is equal to I_d .

$$I = I_d$$

So,

$$I_D = \frac{I_d}{2}$$
$$I_d = 2I_D$$

The differential voltage gain for differential amplifier is,

$$|A_d| = g_{m\text{-}d}R_D$$

Step 5 of 6

Equating differential amplifier voltage gain and common-source amplifier voltage gain,

$$|A_d| = |A_{vo}|$$
$$g_{m\text{-}d}R_D = g_{m\text{-}CS}R_D$$
$$\frac{I_d}{V_{OV\text{-}d}}R_D = \frac{I_{CS}}{V_{OV\text{-}CS}}R_D$$
$$\frac{2I_D}{V_{OV\text{-}d}} = \frac{I_D}{V_{OV\text{-}CS}} \qquad (\text{Since, } I_d = 2I_D \text{ and } I_{CS} = I_D)$$
$$\frac{2}{V_{OV\text{-}d}} = \frac{1}{V_{OV\text{-}CS}}$$
$$V_{OV\text{-}d} = 2V_{OV\text{-}CS}$$

Substitute $\sqrt{\frac{I}{k_n'\left(\frac{W}{L}\right)}}$ for V_{OV} .

$$\sqrt{\frac{I_d}{k_n'\left(\frac{W_d}{L_d}\right)}} = 2\sqrt{\frac{I_{CS}}{k_n'\left(\frac{W_{CS}}{L_{CS}}\right)}}$$
$$\sqrt{\frac{2I_D}{k_n'\left(\frac{W_d}{L_d}\right)}} = 2\sqrt{\frac{I_D}{k_n'\left(\frac{W_{CS}}{L_{CS}}\right)}} \qquad (\text{Since, } I_d = 2I_D \text{ and } I_{CS} = I_D)$$

Step 6 of 6

Since all the transistors have same channel length. Thus

$$\sqrt{\frac{2I_D}{W_d}} = 2\sqrt{\frac{I_D}{W_{CS}}}$$

Squaring on both sides

$$\frac{2I_D}{W_d} = 4\frac{I_D}{W_{CS}}$$
$$\frac{1}{W_d} = \frac{2}{W_{CS}}$$
$$W_{CS} = 2W_d$$

Thus, the common-source amplifier width is high compared to differential amplifier width.

9.019E

We don't have the solution to this problem yet.

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Step 1 of 7

9.019P

Refer to the circuit diagram of MOS differential amplifier in Figure P8.20 in the textbook.

Replace the PMOS transistors Q_3 and Q_4 with drain resistance R_D'

Draw the differential half circuit of MOS differential amplifier.

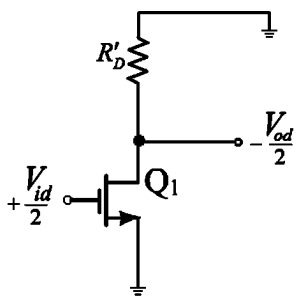


Figure 1

Step 2 of 7

(a)

Write the general expression for differential gain of a differential amplifier.

$$A_d = g_{m1,2}\left(r_{o1} \| R_o\right)$$

Find the output resistance R_o .

Apply test source to the half circuit to find the output resistance.

Draw the equivalent circuit for the half circuit of a MOS differential amplifier.

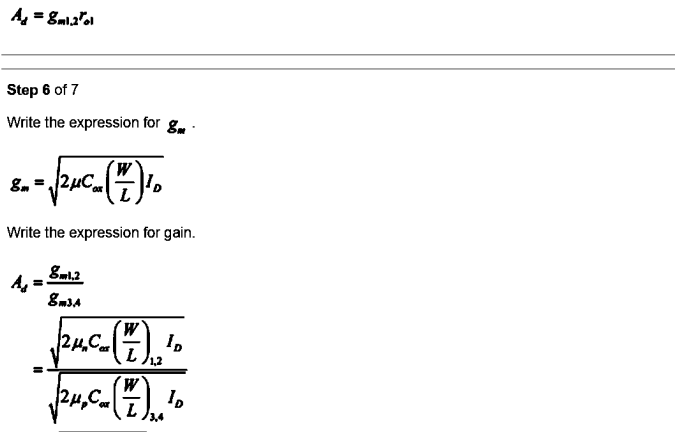


Figure 2

Step 3 of 7

From Figure 2, it is clear that applied test source is equal to gate-to-source voltage of the transistor.

$$V_x = V_{gs}$$

Define output resistance.

$R_o = \frac{v_x}{i_x}$ Apply Kirchhoff's current law at node '*D*'.

$$i_x = g_{m_3}V_{gs3} + \frac{V_x}{r_{o3}}$$

Substitute V_x for V_{gs3} in the equation.

$$i_x = g_{m_3}v_x + \frac{v_x}{r_{o3}}$$

$$i_x = \left(g_{m_3} + \frac{1}{r_{o3}}\right)v_x$$

$$\frac{v_x}{i_x} = \frac{1}{g_{m_3} + \frac{1}{r_{o3}}}$$

Step 4 of 7

Substitute $\frac{v_x}{i_x}$ for R_o in the equation.

$$R_o = \frac{1}{g_{m_3} + \frac{1}{r_{o3}}}$$

$$= \frac{r_{o3}}{1 + g_{m_3}r_{o3}}$$

Substitute $\frac{r_{o3}}{1 + g_{m_3}r_{o3}}$ for R_o in the equation.

$$A_d = g_{m1,2}\left(r_{o1} \| \left(\frac{r_{o3}}{1 + g_{m_3}r_{o3}}\right)\right)$$

Thus, differential gain of the half circuit A_d is $\boxed{g_{m1,2}\left(r_{o1} \| \left(\frac{r_{o3}}{1 + g_{m_3}r_{o3}}\right)\right)}$.

Step 5 of 7

(b)
Recall the output resistance expression.

$$R_o = \frac{1}{g_{m_3} + \frac{1}{r_{o3}}}$$

Ignore r_{o3} term in the expression.

$$R_o = \frac{1}{g_{m_3}}$$

Differential gain becomes:

$$A_d = g_{m1,2}\left(r_{o1} \| \frac{1}{g_{m_3}}\right)$$

Term $\frac{1}{g_{m_3}}$ is very small.

$$A_d = g_{m1,2}r_{o1}$$

Step 6 of 7

Write the expression for g_m .

$$g_m = \sqrt{2\mu C_{ox}\left(\frac{W}{L}\right)I_D}$$

Write the expression for gain.

$$A_d = \frac{g_{m1,2}}{g_{m3,4}}$$

$$= \frac{\sqrt{2\mu_n C_{ox}\left(\frac{W}{L}\right)_{1,2} I_D}}{\sqrt{2\mu_p C_{ox}\left(\frac{W}{L}\right)_{3,4} I_D}}$$

$$= \sqrt{\frac{\mu_n}{\mu_p} \cdot \frac{\left(\frac{W}{L}\right)_{1,2}}{\left(\frac{W}{L}\right)_{3,4}}}$$

Step 7 of 7

(c)

Recall the gain expression.

$$A_d = \sqrt{\frac{\mu_n}{\mu_p} \cdot \frac{\left(\frac{W}{L}\right)_{1,2}}{\left(\frac{W}{L}\right)_{3,4}}}$$

Substitute μ_n for $4\mu_p$, L for L_1, L_2, L_3 and 10 for A_d

$$10 = \sqrt{\frac{4\mu_p}{\mu_p} \cdot \frac{W_{1,2}}{W_{3,4}}}$$

$$10 = \sqrt{4 \cdot \frac{W_{1,2}}{W_{3,4}}}$$

$$\frac{10}{2} = \sqrt{\frac{W_{1,2}}{W_{3,4}}}$$

$$5 = \sqrt{\frac{W_{1,2}}{W_{3,4}}}$$

Solve for $\frac{W_{1,2}}{W_{3,4}}$.

$$\frac{W_{1,2}}{W_{3,4}} = 25$$

Thus, channel width ratio of transistor $Q_{1,2}$ to $Q_{3,4}$ is $\boxed{25}$.

9.020E

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Step 1 of 5

9.020P

Refer to the circuit diagram of differential amplifier in Figure P8.21 in the textbook.

Draw the differential half circuit for the differential amplifier.

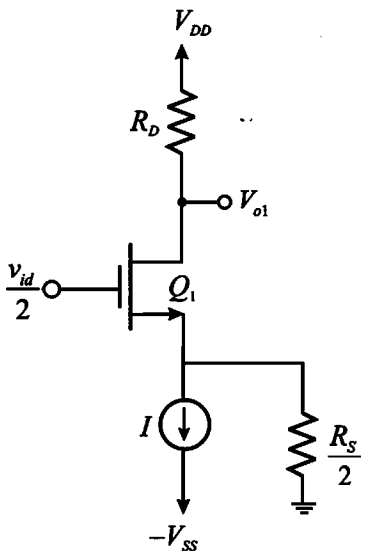


Figure 1

Step 2 of 5

Draw the small signal equivalent circuit.

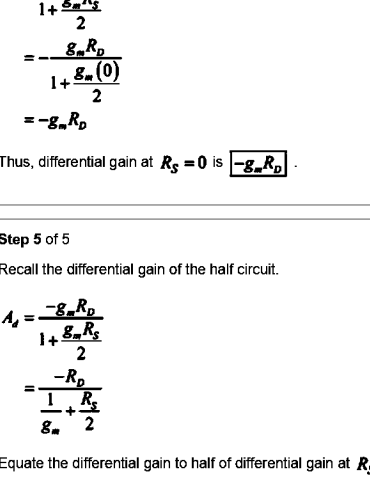


Figure 2

Step 3 of 5

Define differential gain.

$$A_d = \frac{V_{o1}}{\frac{v_{id}}{2}}$$

Apply Kirchhoff's voltage law around input loop.

$$\frac{v_{id}}{2} = V_{gs} + g_m V_{gs} \frac{R_S}{2}$$

$$V_{gs}\left(1 + g_m \frac{R_S}{2}\right) = \frac{v_{id}}{2}$$

Solve for V_{gs}.

$$V_{gs} = \frac{\frac{v_{id}}{2}}{1 + g_m \frac{R_S}{2}}$$

Output voltage of the half circuit is,

$$V_{o1} = -g_m V_{gs} R_D$$

Substitute $\frac{\frac{v_{id}}{2}}{1 + g_m \frac{R_S}{2}}$ for V_{gs} in the equation.

$$V_{o1} = -g_m \left(\frac{\frac{v_{id}}{2}}{1 + g_m \frac{R_S}{2}}\right) R_D$$

$$\frac{V_{o1}}{\frac{v_{id}}{2}} = -\left(\frac{g_m R_D}{1 + \frac{g_m R_S}{2}}\right)$$

$$A_d = \frac{-g_m R_D}{1 + \frac{g_m R_S}{2}} \qquad \text{since } A_d = \frac{V_{o1}}{\frac{v_{id}}{2}}$$

Thus, differential gain of the half circuit for the differential amplifier A_d is $\boxed{\frac{-g_m R_D}{1 + \frac{g_m R_S}{2}}}$.

Step 4 of 5

Gain at $R_S = 0$ is,

$$A_d = \frac{-g_m R_D}{1 + \frac{g_m R_S}{2}}$$

$$= -\frac{g_m R_D}{1 + \frac{g_m (0)}{2}}$$

$$= -g_m R_D$$

Thus, differential gain at $R_S = 0$ is $\boxed{-g_m R_D}$.

Step 5 of 5

Recall the differential gain of the half circuit.

$$A_d = \frac{-g_m R_D}{1 + \frac{g_m R_S}{2}}$$

$$= \frac{-R_D}{\frac{1}{g_m} + \frac{R_S}{2}}$$

Equate the differential gain to half of differential gain at $R_S = 0$.

$$\frac{-g_m R_D}{2} = \frac{-R_D}{\frac{1}{g_m} + \frac{R_S}{2}}$$

$$\frac{g_m}{2} = \frac{1}{\frac{1}{g_m} + \frac{R_S}{2}}$$

$$\frac{1}{g_m} + \frac{R_S}{2} = \frac{2}{g_m}$$

Solve for R_S.

$$R_S = \frac{2}{g_m}(2-1)$$

$$R_S = \frac{2}{g_m}$$

Thus, source resistance of the differential half circuit to reduce the differential gain at $R_S = 0$ is

$\boxed{R_S = \frac{2}{g_m}}$.

9.021E

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9.021P

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Step 1 of 16

Refer to Figure 8.41 in the textbook.

(a)

The value of the drain current of the transistor Q_6 is,

$$\boldsymbol{I_6 = 100\ \mu A}$$

Calculate the value of $\left(\frac{W}{L}\right)_6$.

$$\begin{aligned}\frac{W}{L}\Big)_6 &= \frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_4}\left(\frac{I}{2}\right)\\ I_6 &= \frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_4}\left(\frac{2}{I}\right)(I_6)\end{aligned}$$

Substitute **100** for $\left(\frac{W}{L}\right)_5$, **100 µA** for I and **100 µA** for I_6 in $\left(\frac{W}{L}\right)_6$.

$$\begin{aligned}\left(\frac{W}{L}\right)_6 &= (100)\left(\frac{2}{100\ \mu A}\right)(100\ \mu A)\\ &= 200\end{aligned}$$

Therefore, the value of $\left(\frac{W}{L}\right)_6$ is $\boxed{200}$.

Step 2 of 16

Calculate the value of $\left(\frac{W}{L}\right)_7$.

$$\begin{aligned}I_7 &= \frac{\left(\frac{W}{L}\right)_3}{\left(\frac{W}{L}\right)_2}(I)\\ \left(\frac{W}{L}\right)_7 &= \left(\frac{W}{L}\right)_3\left(\frac{1}{I}\right)(I_6) \qquad (\text{since}\quad I_7 = I_6)\end{aligned}$$

Substitute **200** for $\left(\frac{W}{L}\right)_7$ and **100 µA** for I_6 in the expression of $\left(\frac{W}{L}\right)_7$.

$$\begin{aligned}\left(\frac{W}{L}\right)_7 &= (200)\left(\frac{1}{100\ \mu A}\right)(100\ \mu A)\\ &= 200\end{aligned}$$

Therefore, the value of $\left(\frac{W}{L}\right)_7$ is $\boxed{200}$.

Step 3 of 16

(b)

Calculate the overdrive voltage $|V_{ov}|$ when Q_5 operates.

$$I_D = \frac{1}{2}(\mu_n C_{ox})\left(\frac{W}{L}\right)_5 (V_{ov1})^2$$

$$\begin{aligned}|V_{ov1}| &= \sqrt{\frac{2\left(\frac{I}{2}\right)}{(\mu_n C_{ox})\left(\frac{W}{L}\right)_5}} \qquad \left(\text{since}\quad I_D = \frac{I}{2}\ \text{for}\ Q_1, Q_2, Q_3\ \text{and}\ Q_5\right)\\ &= \sqrt{\frac{I}{(\mu_n C_{ox})\left(\frac{W}{L}\right)_5}}\end{aligned}$$

Substitute **100 µA** for I and **30 µA/V²** for $\mu_n C_{ox}$ in the expression of V_{ov1}.

$$\begin{aligned}V_{ov1} &= \sqrt{\frac{100\times10^{-6}}{(30\times10^{-6})(200)}}\\ &= \sqrt{\frac{1}{60}}\\ &= 0.129\ \text{V}\end{aligned}$$

Therefore, the overdrive voltage $|V_{ov}|$ when Q_5 operates is $\boxed{0.129\ \text{V}}$.

Step 4 of 16

Calculate the overdrive voltage $|V_{ov}|$ when Q_2 operates.

$$I_D = \frac{1}{2}(\mu_n C_{ox})\left(\frac{W}{L}\right)_2 (V_{ov2})^2$$

$$\begin{aligned}|V_{ov2}| &= \sqrt{\frac{2\left(\frac{I}{2}\right)}{(\mu_n C_{ox})\left(\frac{W}{L}\right)_2}} \qquad \left(\text{since}\quad I_D = \frac{I}{2}\ \text{for}\ Q_1, Q_2, Q_3\ \text{and}\ Q_5\right)\\ &= \sqrt{\frac{I}{(\mu_n C_{ox})\left(\frac{W}{L}\right)_2}}\end{aligned}$$

Step 5 of 16

Substitute **200** for $\left(\frac{W}{L}\right)_2$, **100 µA** for I and **30 µA/V²** for $\mu_n C_{ox}$ in the expression of V_{ov2}.

$$\begin{aligned}V_{ov2} &= \sqrt{\frac{100\times10^{-6}}{(30\times10^{-6})(200)}}\\ &= 0.129\ \text{V}\end{aligned}$$

Therefore, the overdrive voltage $|V_{ov}|$ when Q_2 operates is $\boxed{0.129\ \text{V}}$.

Step 6 of 16

Calculate the overdrive voltage $|V_{ov}|$ when Q_6 operates.

$$I_D = \frac{1}{2}(\mu_n C_{ox})\left(\frac{W}{L}\right)_6 (V_{ov6})^2$$

$$\begin{aligned}V_{ov6} &= \sqrt{\frac{2(I)}{(\mu_n C_{ox})\left(\frac{W}{L}\right)_6}} \qquad (\text{since}\quad I_D = I\ \text{for}\ Q_3, Q_4\ \text{and}\ Q_6)\\ &= \sqrt{\frac{2I}{(\mu_n C_{ox})\left(\frac{W}{L}\right)_6}}\end{aligned}$$

Substitute **200** for $\left(\frac{W}{L}\right)_6$, **100 µA** for I and **90 µA/V²** for $\mu_n C_{ox}$ in the expression of V_{ov6}.

$$\begin{aligned}V_{ov6} &= \sqrt{\frac{2(100\times10^{-6})}{(90\times10^{-6})(200)}}\\ &= \sqrt{\frac{1}{90}}\\ &= 0.105\ \text{V}\end{aligned}$$

Therefore, the overdrive voltage $|V_{ov}|$ when Q_6 operates is $\boxed{0.105\ \text{V}}$.

Step 7 of 16

(c)

Calculate the transconductance for Q_5.

$$\begin{aligned}g_{m5} &= \frac{2I_D}{V_{ov5}}\\ &= \frac{2\left(\frac{I}{2}\right)}{V_{ov5}} \qquad \left(\text{since}\quad I_D = \frac{I}{2}\ \text{for}\ Q_1, Q_2, Q_3\ \text{and}\ Q_5\right)\\ &= \frac{I}{V_{ov5}}\end{aligned}$$

Substitute **100 µA** for I and **0.129 V** for V_{ov1} in the expression of g_{m5}.

$$\begin{aligned}g_{m5} &= \frac{100\times10^{-6}}{0.129}\\ &= 0.775\times10^{-3}\\ &= 0.775\ \text{mA/V}\end{aligned}$$

Therefore, the transconductance for Q_5 is $\boxed{0.775\ \text{mA/V}}$.

Step 8 of 16

Calculate the transconductance for Q_2.

$$\begin{aligned}g_{m2} &= \frac{2I_D}{V_{ov2}}\\ &= \frac{2\left(\frac{I}{2}\right)}{V_{ov2}} \qquad \left(\text{since}\quad I_D = \frac{I}{2}\ \text{for}\ Q_1, Q_2, Q_3\ \text{and}\ Q_5\right)\\ &= \frac{I}{V_{ov2}}\end{aligned}$$

Substitute **100 µA** for I and **0.129 V** for V_{ov2} in the expression of g_{m2}.

$$\begin{aligned}g_{m2} &= \frac{100\times10^{-6}}{0.129}\\ &= 0.775\times10^{-3}\\ &= 0.775\ \text{mA/V}\end{aligned}$$

Therefore, the transconductance for Q_2 is $\boxed{0.775\ \text{mA/V}}$.

Step 9 of 16

Calculate the transconductance for Q_6.

$$\begin{aligned}g_{m6} &= \frac{2I_D}{V_{ov6}}\\ &= \frac{2(I)}{V_{ov6}} \qquad (\text{since}\quad I_D = I\ \text{for}\ Q_3, Q_4\ \text{and}\ Q_6)\\ &= \frac{2I}{V_{ov6}}\end{aligned}$$

Substitute **100 µA** for I and **0.105 V** for V_{ov6} in the expression of g_{m6}.

$$\begin{aligned}g_{m6} &= \frac{100\times10^{-6}}{0.105}\\ &= 1.9\times10^{-3}\\ &= 1.9\ \text{mA/V}\end{aligned}$$

Therefore, the transconductance for Q_6 is $\boxed{1.9\ \text{mA/V}}$.

Step 10 of 16

(d)

Calculate the value of r_2 for the transistor Q_2.

$$\begin{aligned}r_{o2} &= \frac{|V_A|}{I_D}\\ &= \frac{|V_A|}{\left(\frac{I}{2}\right)} \qquad \left(\text{since}\quad I_D = \frac{I}{2}\ \text{for}\ Q_1, Q_2, Q_3\ \text{and}\ Q_5\right)\\ &= \frac{2|V_A|}{I}\end{aligned}$$

Substitute **100 µA** for I and **10 V** for $|V_A|$ in the expression of r_{o2}.

$$\begin{aligned}r_{o2} &= \frac{2(10)}{100\times10^{-6}}\\ &= 0.2\times10^{6}\\ &= 200\ \text{k}\Omega\end{aligned}$$

Therefore, the value of r_{o2} is $\boxed{200\ \text{k}\Omega}$.

Step 11 of 16

Calculate the value of r_5 for the transistor Q_5.

$$\begin{aligned}r_{o5} &= \frac{|V_A|}{I_D}\\ &= \frac{|V_A|}{\left(\frac{I}{2}\right)} \qquad \left(\text{since}\quad I_D = \frac{I}{2}\ \text{for}\ Q_1, Q_2, Q_3\ \text{and}\ Q_5\right)\\ &= \frac{2|V_A|}{I}\end{aligned}$$

Substitute **100 µA** for I and **10 V** for $|V_A|$ in the expression of r_{o5}.

$$\begin{aligned}r_{o5} &= \frac{2(10)}{100\times10^{-6}}\\ &= 0.2\times10^{6}\\ &= 200\ \text{k}\Omega\end{aligned}$$

Therefore, the value of r_{o5} is $\boxed{200\ \text{k}\Omega}$.

Step 12 of 16

Calculate the value of r_7 for the transistor Q_6.

$$\begin{aligned}r_{o6} &= \frac{|V_A|}{I_D}\\ &= \frac{|V_A|}{\left(\frac{I}{2}\right)} \qquad \left(\text{since}\quad I_D = \frac{I}{2}\ \text{for}\ Q_1, Q_2, Q_3\ \text{and}\ Q_5\right)\\ &= \frac{2|V_A|}{I}\end{aligned}$$

Substitute **100 µA** for I and **10 V** for $|V_A|$ in the expression of r_{o6}.

$$\begin{aligned}r_{o6} &= \frac{2(10)}{100\times10^{-6}}\\ &= 0.2\times10^{6}\\ &= 200\ \text{k}\Omega\end{aligned}$$

Therefore, the value of r_{o6} is $\boxed{200\ \text{k}\Omega}$.

Step 13 of 16

Calculate the value of r_7 for the transistor Q_7.

$$\begin{aligned}r_{o7} &= \frac{|V_A|}{I_D}\\ &= \frac{|V_A|}{\left(\frac{I}{2}\right)} \qquad \left(\text{since}\quad I_D = \frac{I}{2}\ \text{for}\ Q_1, Q_2, Q_3\ \text{and}\ Q_5\right)\\ &= \frac{2|V_A|}{I}\end{aligned}$$

Substitute **100 µA** for I and **10 V** for $|V_A|$ in the expression of r_{o7}.

$$\begin{aligned}r_{o7} &= \frac{2(10)}{100\times10^{-6}}\\ &= 0.2\times10^{6}\\ &= 200\ \text{k}\Omega\end{aligned}$$

Therefore, the value of r_{o7} is $\boxed{200\ \text{k}\Omega}$.

Step 1 of 6

(a)

Refer to the circuit diagram in Figure P8.23 in the textbook.

Apply 0 V to gate terminals of all the transistors.

$$V_{GS} = V_t + V_{OV}$$
$$V_G - V_S = V_t + V_{OV}$$

Substitute 0 for V_G.

$$0 - V_S = V_t + V_{OV}$$
$$V_S = -(V_t + V_{OV})$$

Source voltage, V_{S1} of transistor Q_1 is $\boxed{-(V_t + V_{OV})}$.

Similarly, source voltage V_{S2} of transistor Q_2 is $\boxed{-(V_t + V_{OV})}$.

Step 2 of 6

The source voltages are same for the transistor Q_1 and transistor Q_2.

There is no potential difference exist at V_{S1} and V_{S2}. Thus, current flow the transistors Q_3 and Q_4 is $\boxed{0\ A}$.

Step 3 of 6

Gate to source voltage of the transistors Q_3 and Q_4 are,

$$V_{GS} = V_t + V_{OV}$$

Here V_{GS} is same for all the transistors.

V_t is same for all the transistors.

From the equation it is clear that At $v_{G1} = v_{G2} = 0\ V$ the overdrive voltage of transistors Q_3 and Q_4 are same as overdrive voltage of the transistor of Q_1 and Q_2.

Drain current I is,

$$I = \left[k_n' \left(\frac{W}{L} \right)_3 V_{OV3}^2 \right]$$

Drain to source resistance of transistor Q_3 is,

$$r_{DS3} = \frac{V_{OV}}{I}$$

Substitute $k_n' \left(\frac{W}{L} \right)_3 V_{OV3}^2$ for I in the equation.

$$r_{DS3} = \frac{V_{OV}}{k_n' \left(\frac{W}{L} \right)_3 V_{OV3}^2}$$

$$= \left[k_n' \left(\frac{W}{L} \right)_3 V_{OV3} \right]^{-1} \quad \text{..... (1)}$$

Step 4 of 6

Need find the expression for V_{OV3}.

The overdrive voltage of transistor Q_3 and Q_4 are same as overdrive voltage of transistor Q_1 and Q_2.

Define trans-conductance of transistor Q_1.

$$g_{m1} = \frac{1}{2} k_n' \left(\frac{W}{L} \right)_1 V_{OV1}$$

Rewrite the equation.

$$V_{OV1} = \frac{g_{m1}}{\frac{1}{2} k_n' \left(\frac{W}{L} \right)_1}$$

All transistor have the same overdrive voltage.

$$V_{OV3} = V_{OV1}$$
$$= \frac{g_{m1}}{\frac{1}{2} k_n' \left(\frac{W}{L} \right)_1}.$$

Recall equation (1).

$$r_{DS3} = \left[k_n' \left(\frac{W}{L} \right)_3 V_{OV3} \right]^{-1}$$

Substitute $\frac{g_{m1}}{\frac{1}{2} k_n' \left(\frac{W}{L} \right)_1}$ for V_{OV3} in equation (1).

$$r_{DS3} = \left[k_n' \left(\frac{W}{L} \right)_3 \left(\frac{g_{m1}}{\frac{1}{2} k_n' \left(\frac{W}{L} \right)_1} \right) \right]^{-1}$$
$$= \left[\frac{2 k_n' g_{m1} \left(\frac{W}{L} \right)_3}{\left(\frac{W}{L} \right)_1 k_n'} \right]^{-1}$$
$$= \frac{\left(\frac{W}{L} \right)_1}{2 g_{m1} \left(\frac{W}{L} \right)_3}$$

Step 5 of 6

Source resistance R_S is sum of drain to source resistances.

$$R_S = r_{DS3} + r_{DS4}$$
$$= 2 r_{DS3,4} \qquad \text{since } r_{DS4} = r_{DS3}$$

Substitute $\frac{\left(\frac{W}{L} \right)_1}{2 g_{m1} \left(\frac{W}{L} \right)_3}$ for $r_{DS3,4}$ in the equation.

$$R_S = 2 \left(\frac{\left(\frac{W}{L} \right)_1}{2 g_{m1} \left(\frac{W}{L} \right)_3} \right)$$
$$= \frac{\left(\frac{W}{L} \right)_1}{g_{m1} \left(\frac{W}{L} \right)_3}$$

Thus, source resistance of the amplifier R_S is, $\boxed{\frac{\left(\frac{W}{L} \right)_{1,2}}{g_{m1,2} \left(\frac{W}{L} \right)_{3,4}}}$.

Step 6 of 6

(b)

Write the general expression for differential voltage gain A_d of the differential amplifier is,

$$A_d = \frac{V_{OV}}{V_{id}}$$
$$= \frac{g_{m_{1,2}} R_D}{1 + \frac{g_{m1,2} R_S}{2}}$$

Substitute $\frac{\left(\frac{W}{L} \right)_{1,2}}{g_{m1,2} \left(\frac{W}{L} \right)_{3,4}}$ for R_S.

$$A_d = \frac{g_{m_{1,2}} R_D}{1 + \frac{g_{m_{1,2}}}{2} \left(\frac{\left(\frac{W}{L} \right)_{1,2}}{g_{m1,2} \left(\frac{W}{L} \right)_{3,4}} \right)}$$
$$= \frac{g_{m_{1,2}} R_D}{1 + \frac{\left(\frac{W}{L} \right)_{1,2}}{2 \left(\frac{W}{L} \right)_{3,4}}}$$

Thus, differential voltage of the amplifier is $\boxed{\frac{g_{m_{1,2}} R_D}{1 + \frac{\left(\frac{W}{L} \right)_{1,2}}{2 \left(\frac{W}{L} \right)_{3,4}}}}$.

Step 1 of 3

Refer to Figure 8.43 in the textbook.

Refer to Example 8.7 in the textbook.

The differential input resistance is,

$R_{id} = 20.2\,\text{k}\Omega$

The overall voltage gain of the circuit in Figure 8.43 in the textbook is,

$A_{vo} = 8513\,\text{V/V}$

The output resistance of the amplifier circuit is,

$R_o = 152\,\Omega$

Consider the source resistance as $10\,\text{k}\Omega$ and the load resistance as $1\,\text{k}\Omega$.

The amplifier model of the circuit in Figure 8.43 is as follows:

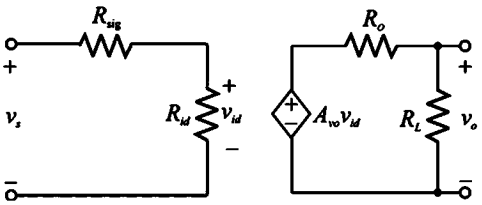


Figure 1

Here, v_s is the source voltage

v_{id} is the input differential voltage

v_o is the output voltage

R_{sig} is the source resistance

R_L is the load resistance

Step 2 of 3

Apply voltage division rule to input loop of Figure 1.

$$v_{id} = \left[\frac{R_{id}}{R_{id} + R_{sig}}\right] v_s \quad \text{..... (1)}$$

Similarly, apply voltage division rule to output loop.

$$v_o = \left[\frac{R_L}{R_L + R_o}\right] (A_{vo} v_{id}) \quad \text{..... (2)}$$

Substitute the value of v_{id} in equation (2) from equation (1).

$$v_o = \left[\frac{R_L}{R_L + R_o}\right] \left(A_{vo} \left(\left[\frac{R_{id}}{R_{id} + R_{sig}}\right] v_s\right)\right)$$

$$\frac{v_o}{v_s} = \left[\frac{R_L}{R_L + R_o}\right] \left[\frac{R_{id}}{R_{id} + R_{sig}}\right] (A_{vo})$$

Step 3 of 3

The overall voltage gain of the amplifier is,

$$A_v = \frac{v_o}{v_s}$$

$$= \left[\frac{R_L}{R_L + R_o}\right] \left[\frac{R_{id}}{R_{id} + R_{sig}}\right] (A_{vo})$$

$$= \left[\frac{1\times10^3}{1\times10^3 + 152}\right] \left[\frac{20.2\times10^3}{20.2\times10^3 + 10\times10^3}\right] (8513)$$

$$= \left[\frac{1\times10^3}{1.152\times10^3}\right] \left[\frac{20.2\times10^3}{30.2\times10^3}\right] (8513)$$

$$= [0.8681][0.6689](8513)$$

$$= 4943.3$$

Therefore, the overall voltage gain, A_v, is $\boxed{4943.3\ \text{V/V}}$.

9.023P

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Step 1 of 10

Refer to Figure 8.49 in the textbook.

Refer to Example 8.7 in the textbook.

The input resistance is, $R_{i1} = 20.2\,\text{k}\Omega$

The input resistance of second stage is, $R_{i2} = 5.05\,\text{k}\Omega$

The input resistance of the third stage is, $R_{i3} = 234.8\,\text{k}\Omega$

The input resistance of the fourth stage is, $R_{i4} = 303.5\,\text{k}\Omega$

The gain of every transistor is, $\beta = 100$

Step 2 of 10

The current-transmission factors are calculated as follows:

Calculate the value of $\frac{i_{e8}}{i_{b8}}$.

$$\begin{aligned}\frac{i_{e8}}{i_{b8}} &= \beta_8 + 1\\ &= 100 + 1\\ &= 101\end{aligned}$$

Therefore, the value of $\frac{i_{e8}}{i_{b8}}$ is $\boxed{101\ \text{A/A}}$.

Step 3 of 10

Calculate the value of $\frac{i_{b8}}{i_{c7}}$.

$$\begin{aligned}\frac{i_{b8}}{i_{c7}} &= \frac{R_5}{R_5 + R_{i4}}\\ &= \frac{15.7\times10^3}{15.7\times10^3 + 303.5\times10^3}\\ &= \frac{15.7\times10^3}{319.2\times10^3}\\ &= 0.0492\end{aligned}$$

Therefore, the value of $\frac{i_{b8}}{i_{c7}}$ is $\boxed{0.0492\ \text{A/A}}$.

Step 4 of 10

Calculate the value of $\frac{i_{c7}}{i_{b7}}$.

$$\begin{aligned}\frac{i_{c7}}{i_{b7}} &= \beta_7\\ &= 100\end{aligned}$$

Therefore, the value of $\frac{i_{c7}}{i_{b7}}$ is $\boxed{100\ \text{A/A}}$.

Step 5 of 10

Calculate the value of $\frac{i_{b7}}{i_{c5}}$.

$$\begin{aligned}\frac{i_{b7}}{i_{c5}} &= \frac{R_3}{R_3 + R_{i3}}\\ &= \frac{3\times10^3}{3\times10^3 + 234.8\times10^3}\\ &= \frac{3\times10^3}{237.8\times10^3}\\ &= 0.0126\end{aligned}$$

Therefore, the value of $\frac{i_{b7}}{i_{c5}}$ is $\boxed{0.0126\ \text{A/A}}$.

Step 6 of 10

Calculate the value of $\frac{i_{c5}}{i_{b5}}$.

$$\begin{aligned}\frac{i_{c5}}{i_{b5}} &= \beta_5\\ &= 100\end{aligned}$$

Therefore, the value of $\frac{i_{c5}}{i_{b5}}$ is $\boxed{100\ \text{A/A}}$.

Step 7 of 10

Calculate the value of $\frac{i_{b5}}{i_{c2}}$.

$$\begin{aligned}\frac{i_{b5}}{i_{c2}} &= \frac{(R_1 + R_2)}{(R_1 + R_2) + R_{i2}}\\ &= \frac{\left(20\times10^3 + 20\times10^3\right)}{\left(20\times10^3 + 20\times10^3\right) + 5.05\times10^3}\\ &= \frac{40\times10^3}{40\times10^3 + 5.05\times10^3}\\ &= 0.8879\end{aligned}$$

Therefore, the value of $\frac{i_{b5}}{i_{c2}}$ is $\boxed{0.8879\ \text{A/A}}$.

Step 8 of 10

Calculate the value of $\frac{i_{c2}}{i_i}$.

$$\begin{aligned}\frac{i_{c2}}{i_i} &= \beta_2\\ &= 100\end{aligned}$$

Thus, the value of $\frac{i_{c2}}{i_i}$ is $\boxed{100\ \text{A/A}}$.

Step 9 of 10

The overall current gain of the circuit is,

$$\begin{aligned}\frac{i_{e8}}{i_i} &= \left(\frac{i_{e8}}{i_{b8}}\right)\left(\frac{i_{b8}}{i_{c7}}\right)\left(\frac{i_{c7}}{i_{b7}}\right)\left(\frac{i_{b7}}{i_{c5}}\right)\left(\frac{i_{c5}}{i_{b5}}\right)\left(\frac{i_{b5}}{i_{c2}}\right)\left(\frac{i_{c2}}{i_i}\right)\\ &= (101)(0.0492)(100)(0.0126)(100)(0.8879)(100)\\ &= 55593.124\end{aligned}$$

Therefore, the overall current gain, $\frac{i_{e8}}{i_i}$, is $\boxed{55593.124\ \text{A/A}}$.

Step 10 of 10

Calculate the overall voltage gain, $\frac{v_o}{v_{id}}$ by substituting the known values.

$$\begin{aligned}\frac{v_o}{v_{id}} &= \frac{R_6}{R_{i1}}\frac{i_{e8}}{i_i}\\ &= \frac{3\times10^3}{20.2\times10^3}(55593.124)\\ &= 8256.4\end{aligned}$$

Therefore, the overall voltage gain, $\frac{v_o}{v_{id}}$, is $\boxed{8256.4\ \text{V/V}}$.

Step 1 of 6

Refer to the circuit diagram in Figure P8.25 in the textbook.

(a)

The drain current for Q_1 is,

$$i_{D1}=\frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS1}-V_t\right)^2 \quad \text{..... (1)}$$

The drain current for Q_2 is,

$$i_{D2}=\frac{1}{2}k_n'\left(2\frac{W}{L}\right)\left(v_{GS2}-V_t\right)^2 \quad \text{..... (2)}$$

Step 2 of 6

From the circuit, the gate voltage, source voltage and threshold voltages are same for both Q_1 and Q_2 transistors.

So,

$$v_{GS1}-V_t=v_{GS2}-V_t.$$

Recall equation (2).

$$i_{D2}=\frac{1}{2}k_n'\left(2\frac{W}{L}\right)\left(v_{GS2}-V_t\right)^2$$

Rewrite the equation.

$$\left(v_{GS2}-V_t\right)^2=\frac{i_{D2}}{\frac{1}{2}k_n'\left(2\frac{W}{L}\right)}$$

Substitute $\frac{i_{D2}}{\frac{1}{2}k_n'\left(2\frac{W}{L}\right)}$ for $\left(v_{GS1}-V_t\right)^2$ in equation (1)

$$i_{D1}=\frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(\frac{i_{D2}}{\frac{1}{2}k_n'\left(2\frac{W}{L}\right)}\right)$$

$$i_{D1}=\frac{i_{D2}}{2}$$

$$\frac{i_{D1}}{i_{D2}}=\frac{1}{2} \quad \text{..... (3)}$$

Step 3 of 6

From the above circuit, the constant bias current imposes the constraint

$$i_{D1}+i_{D2}=I$$

$$i_{D2}=I-i_{D1}$$

Substitute $I-i_{D1}$ for i_{D2} in equation (3).

$$\frac{i_{D1}}{I-i_{D1}}=\frac{1}{2}$$

$$i_{D1}=\frac{I-i_{D1}}{2}$$

$$i_{D1}=\frac{I}{2}-\frac{i_{D1}}{2}$$

Simplify further.

$$i_{D1}+\frac{i_{D1}}{2}=\frac{I}{2}$$

$$\frac{3i_{D1}}{2}=\frac{I}{2}$$

$$i_{D1}=\frac{I}{3}$$

Therefore, the drain current for Q_1 is $\boxed{i_{D1}=\frac{I}{3}}$.

Therefore, the drain current for Q_2 is $\boxed{i_{D2}=\frac{2I}{3}}$ $\left(\text{Since, } \frac{i_{D1}}{i_{D2}}=\frac{1}{2}\right)$.

Step 4 of 6

(b)

The overdrive voltage for Q_1 can be determined form equation (1).

$$i_{D1}=\frac{1}{2}k_n'\left(\frac{W}{L}\right)\left(v_{GS1}-V_t\right)^2$$

$$i_{D1}=\frac{1}{2}k_n'\left(\frac{W}{L}\right)V_{OV1}^2 \qquad \left(\text{Since, } V_{OV}=v_{GS}-V_t\right)$$

$$\frac{I}{3}=\frac{1}{2}k_n'\left(\frac{W}{L}\right)V_{OV1}^2 \qquad \left(\text{Since, } i_{D1}=\frac{I}{3}\right)$$

$$\frac{2}{3}\frac{I}{k_n'\left(\frac{W}{L}\right)}=V_{OV1}^2$$

$$V_{OV1}=\sqrt{\frac{2}{3}\frac{I}{k_n'\left(\frac{W}{L}\right)}}$$

Therefore, the overdrive voltage for Q_1 is $\boxed{V_{OV1}=\sqrt{\frac{2}{3}\frac{I}{k_n'\left(\frac{W}{L}\right)}}}$.

Step 5 of 6

Recall equation (2).

$$i_{D2}=\frac{1}{2}k_n'\left(2\frac{W}{L}\right)\left(v_{GS2}-V_t\right)^2$$

The overdrive voltage for Q_2 can be determined form equation-

$$i_{D2}=\frac{1}{2}k_n'\left(2\frac{W}{L}\right)\left(v_{GS2}-V_t\right)^2$$

$$i_{D2}=\frac{1}{2}k_n'\left(2\frac{W}{L}\right)V_{OV2}^2 \qquad \left(\text{Since, } V_{OV}=v_{GS}-V_t\right)$$

$$\frac{2I}{3}=\frac{1}{2}k_n'\left(2\frac{W}{L}\right)V_{OV2}^2 \qquad \left(\text{Since, } i_{D2}=\frac{2I}{3}\right)$$

$$\frac{2}{3}\frac{I}{k_n'\left(\frac{W}{L}\right)}=V_{OV2}^2$$

Solve for V_{DV2}.

$$V_{OV2}=\sqrt{\frac{2}{3}\frac{I}{k_n'\left(\frac{W}{L}\right)}}$$

Therefore, the overdrive voltage for Q_2 is $\boxed{V_{OV2}=\sqrt{\frac{2}{3}\frac{I}{k_n'\left(\frac{W}{L}\right)}}}$.

Step 6 of 6

(c)
The differential gain can be determined from differential output.

$$v_{o2}-v_{o1}=\left(g_{m2}\frac{v_{id}}{2}R_D\right)-\left(-g_{m1}\frac{v_{id}}{2}R_D\right)$$

$$v_{o2}-v_{o1}=g_{m2}\frac{v_{id}}{2}R_D+g_{m1}\frac{v_{id}}{2}R_D$$

$$v_{o2}-v_{o1}=\frac{v_{id}R_D}{2}\left(g_{m1}+g_{m2}\right)$$

$$\frac{v_{o2}-v_{o1}}{v_{id}}=\frac{R_D}{2}\left(g_{m1}+g_{m2}\right)$$

simplify further.

$$\frac{v_{o2}-v_{o1}}{v_{id}}=\frac{R_D}{2}\left(\frac{2i_{D1}}{V_{OV}}+\frac{2i_{D2}}{V_{OV}}\right) \qquad \left(\text{Since, } g_m=\frac{2I_D}{V_{OV}}\right)$$

$$=\frac{R_D}{V_{OV}}\left(i_{D1}+i_{D2}\right)$$

$$=\frac{R_D}{V_{OV}}\left(\frac{I}{3}+\frac{2I}{3}\right)$$

$$=\frac{R_DI}{V_{OV}}$$

Therefore, the differential gain A_d is $\boxed{\frac{IR_D}{V_{OV}}}$.

9.025P

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9.026P

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Step 1 of 5

(a)

Refer to Figure 8.16 (a) in the text book for the differential pair with a common-mode input voltage, V_{CM} .

When $v_{B1} = v_{B2} = V_{CM} = 0\ \text{V}$, the emitter voltage is,

$$\begin{aligned} V_E &= V_{CM} - v_{BE} \\ &= 0 - 0.69 \\ &= -0.69\ \text{V} \end{aligned}$$

Thus, the common emitter voltage V_E is $\boxed{-0.69\ \text{V}}$.

The collector voltage is,

$$\begin{aligned} V_{C1} = V_{C2} &= V_{CC} - \frac{\alpha I}{2} R_C \\ &= V_{CC} - \left(\frac{\beta}{\beta+1}\right)\left(\frac{I}{2} R_C\right) \end{aligned}$$

Substitute 1.2 V for V_{CC} , 50 for β , $20\times10^{-6}\ \text{A}$ for I and $82\times10^{3}\ \Omega$ for R_C .

$$\begin{aligned} V_{C1} = V_{C2} &= 1.2 - \left(\frac{50}{51}\right)\left(\frac{20\times10^{-6}}{2}\right)\left(82\times10^{3}\right) \\ &= 1.2 - 0.804 \\ &= 0.396\ \text{V} \end{aligned}$$

Thus, the collector voltages V_{C1} and V_{C2} are $\boxed{0.396\ \text{V}}$.

Step 2 of 5

(b)

Write the formula for maximum common-mode voltage, $V_{CM\,\max}$.

$$\begin{aligned} V_{CM\,\max} &= V_{CC} - \frac{I}{2}\alpha R_C + 0.3 \\ &= V_{CC} - \frac{I}{2}\left(\frac{\beta}{\beta+1}\right) R_C + 0.3 \end{aligned}$$

Substitute 1.2 V for V_{CC} , $20\times10^{-6}\ \text{A}$ for I ,50 for β and $82\times10^{3}\ \Omega$ for R_C .

$$\begin{aligned} V_{CM\,\max} &= 1.2 - \left(\frac{20\times10^{-6}}{2}\right)\left(\frac{50}{50+1}\right)\left(82\times10^{3}\right) + 0.3 \\ &= 1.2 - 0.804 + 0.3 \\ &= 0.696\ \text{V} \end{aligned}$$

Step 3 of 5

Write the formula for minimum common-mode voltage, $V_{CM\,\min}$.

$$V_{CM\,\min} = -V_{EE} + V_{CS} + V_{BE}$$

The voltage across the current source by is,

$$V_{CS} = V_{CC} + V_{EE} - i_C R_C - V_{CE}$$

$$V_{CM\,\min} = -V_{EE} + \left(V_{CC} + V_{EE} - i_C R_C - V_{CE}\right) + V_{BE}$$

Substitute the values $1.2\ \text{V}$ for V_{CC} , $20\times10^{-6}\ \text{A}$ for i_C , $1.2\ \text{V}$ for V_{EE} , $0.69\ \text{V}$ for v_{BE} and $82\times10^{3}\ \Omega$ for R_C .

$$\begin{aligned} V_{CM\,\min} &= -1.2 + \left[1.2 + 1.2 - \left(20\times10^{-6}\right)\left(82\times10^{3}\right) - 0.3\right] + 0.69 \\ &= -1.2 + 0.46 + 0.69 \\ &= -0.05\ \text{V} \end{aligned}$$

Thus, the input common mode range is $\boxed{-0.05\ \text{V} \le V_{CM} \le 0.696\text{V}}$.

Step 4 of 5

(c)

Refer to Figure 8.16 (b) in the text book for the BJT differential pair configuration with

$v_{B2} = 0$.

Transistor Q_2 is OFF and Q_1 is ON. So, the transistor Q_1 conducts all of the current *I*.

When $v_{B1} = 0$, the current in Q_1 is,

$$\begin{aligned} I_C &= \alpha I \\ &= \left(\frac{\beta}{\beta+1}\right) I \\ &= \left(\frac{50}{50+1}\right)(20) \\ &= 19.6\ \mu\text{A} \end{aligned}$$

The current in Q_1 is increased by 10% .

$$\begin{aligned} I_C &= 19.6 + (0.1)(19.6) \\ &= 21.56\ \mu\text{A} \end{aligned}$$

The common emitter voltage is,

$$\begin{aligned} V_E &= V_{CC} - I_C R_C + 0.3 \\ &= 1.2 - \left(21.56\times10^{-6}\right)\left(82\times10^{3}\right) + 0.3 \\ &= -0.27\ \text{V} \end{aligned}$$

Step 5 of 5

The emitter voltage to increase the collector current by 10% is,

$V_E = v_{B1} - 0.69$

Substitute $-0.27\ \text{V}$ for V_E .

$$\begin{aligned} -0.27 &= v_{B1} - 0.69 \\ v_{B1} &= 0.42\ \text{V} \end{aligned}$$

Thus, the base voltage v_{B1} to increase the current by 10% is $\boxed{0.42\ \text{V}}$.

Step 1 of 2

Refer to the circuit in Figure 8.16(a) in the textbook for the BJT differential pair configuration.

The common mode input voltage provided to the base terminal of the transistors Q_1 and Q_2 is V_{CM}. Thus, the voltage at the base of the transistor Q_1 is v_{B1} and Q_2 is v_{B2}.

The voltages, v_{B1} and v_{B2} are,

$$v_{B1} = v_{B2} = V_{CM}$$

The supply voltage V_{CC} is not pure. It consists of ripple component v_r of small amplitude and a frequency of 120 Hz.

Thus the supply voltage is $V_{CC} + v_r$.

The voltage at the collector of the transistor of Q_1 is v_{C1} and Q_2 is v_{C2}. The transistors Q_1 and Q_2 are identical transistors. Hence, the current divides equally between both of them. The current flowing through the collector is,

$$I_C = \frac{\alpha I}{2}$$

Apply Kirchhoff's voltage law across the collector of the transistor, Q_1.

$$-(V_{CC} + v_r) + \alpha\frac{I}{2}R_C - v_{C1} = 0$$

$$v_{C1} = (V_{CC} + v_r) - \alpha\frac{I}{2}R_C$$

As the transistors are identical, the voltages developed across the collector terminal of both the transistors are the same.

Hence, the voltage at the collector terminal of the transistors is,

$$\boxed{v_{C2} = v_{C1} = (V_{CC} + v_r) - \alpha\frac{I}{2}R_C}.$$

Step 2 of 2

Determine the differential voltage, v_{od}.

The differential voltage as the name suggests is the difference between the collector voltages of the two transistors, Q_1 and Q_2.

$$\begin{aligned} v_{od} &= v_{C1} - v_{C2} \\ &= \left[(V_{CC} + v_r) - \alpha\frac{I}{2}R_C\right] - \left[(V_{CC} + v_r) - \alpha\frac{I}{2}R_C\right] \\ &= 0 \end{aligned}$$

Thus, the differential output voltage v_{od} is $\boxed{0}$.

Observe from the calculation of differential voltage v_{od} that the terms in the expression cancel each other. This occurs due to the symmetry and the choice of identical transistors.

Thus, even though there is ripple present in the supply voltage, it does not affect the differential voltage, but it reflects in the collector voltage of the individual transistors Q_1 and Q_2. The differential voltage is not affected because it is computed by taking the difference between the collector voltages of the two transistors. Since the collector voltages of the two transistors are equal due to the identical nature of the transistors, the voltages cancel each other and the differential output is zero.

Therefore, the response of the differential amplifier to the ripple present inherently in the supply voltage is determined.

9.032P

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Step 1 of 4

Refer to Figure 8.15 in the text book for the basic BJT differential pair configuration.

The differential input signal is,

$$v_{id} = V_{B1} - V_{B2}$$
$$v_{id} = 10\ \text{mV}$$

Determine the emitter current i_{E1} of transistor, Q_1.

$$i_{E1} = \frac{I}{1 + e^{-\left(\frac{v_{id}}{V_T}\right)}}$$

Substitute $10\ \text{mV}$ for v_{id} and 25 mV for V_T.

$$i_{E1} = \frac{I}{1 + e^{-\left(\frac{10}{25}\right)}}$$
$$= \frac{I}{1 + e^{-0.4}}$$
$$= \frac{I}{1.67}$$
$$= 0.5987I$$

Thus, the emitter current i_{E1} of transistor, Q_1 is $\boxed{0.5987I}$.

Step 2 of 4

Determine the emitter current i_{E2} of transistor, Q_2.

$$I = i_{E1} + i_{E2}$$

Substitute $0.5987I$ for i_{E1}.

$$I = 0.5987I + i_{E2}$$
$$i_{E2} = 0.4013I$$

Thus, the emitter current i_{E2} of transistor, Q_2 is $\boxed{0.4013I}$.

Step 3 of 4

Determine the output voltage V_o.

$$V_o = V_{C1} - V_{C2}$$
$$= (V_{CC} - i_{C1}R_C) - (V_{CC} - i_{C2}R_C)$$
$$= (i_{C1} - i_{C2})R_C$$

Substitute αi_{E1} for i_{C1} and αi_{E2} for i_{C2}.

$$V_o = \alpha(i_{E1} - i_{E2})R_C$$

Substitute $0.5987I$ for i_{E1} and $0.4013I$ for i_{E2}.

$$V_o = (0.5987 - 0.4013)IR_C$$
$$= 0.1974IR_C$$

Thus, the output voltage V_o is $\boxed{0.1974IR_C\ \text{V}}$.

Step 4 of 4

Determine the value of R_C from the equation,

$$V_o = 0.1974IR_C$$

Substitute $10^{-3}\ \text{A}$ for I and 1 V for V_o

$$1 = (0.1974)\left(10^{-3}\right)R_C$$

$$R_C = \frac{V_o}{0.1974I}$$
$$= \frac{1}{0.1974 \times 1 \times 10^{-3}}$$
$$= 5.1 \times 10^3\ \Omega$$
$$= 5.1\ \text{k}\Omega$$

Thus, the collector resistance R_C is $\boxed{5.1\ \text{k}\Omega}$.

Hence, all the design parameters are determined.

9.035P

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Step 1 of 13

Refer to Figure P9.54 in the textbook for emitter-degenerated differential amplifier

Ground the dc sources V_{CC} and $-V_{EE}$ to draw the differential half-circuit.

Draw the differential half-circuit of the emitter-degenerated differential amplifier.

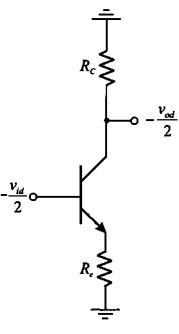


Figure 1

Step 2 of 13

Draw the ac equivalent circuit of emitter-degenerated differential amplifier.

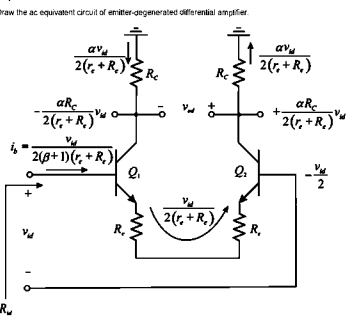


Figure 2

Step 3 of 13

The differential output voltage is,

$$v_{od}=v_{C2}-v_{C1}$$
$$=\frac{\alpha R_C}{2(r_e+R_e)}v_{id}-\left(-\frac{\alpha R_C}{2(r_e+R_e)}v_{id}\right)$$
$$=\left(\frac{\alpha R_C}{r_e+R_e}\right)v_{id}$$
$$\frac{v_{od}}{v_{id}}=\frac{\alpha R_C}{r_e+R_e}$$

Therefore, the differential gain is $\boxed{\frac{\alpha R_C}{r_e+R_e}}$.

Step 4 of 13

Write the expression for base current, i_b.

$$i_b=\frac{v_{id}}{2(\beta+1)(r_e+R_e)}$$

Write the expression for input resistance, R_{id}.

$$R_{id}=\frac{v_{id}}{i_b}$$
$$=\frac{v_{id}}{\frac{v_{id}}{2(\beta+1)(r_e+R_e)}}$$
$$=2(\beta+1)(r_e+R_e)$$

Therefore, the input resistance is $\boxed{2(\beta+1)(r_e+R_e)}$.

The resistance seen between the two bases of the transistors is the product of the total resistance in the circuit and the term $(\beta+1)$.

Step 5 of 13

Draw the modified circuit of emitter-degenerated differential amplifier with notations.

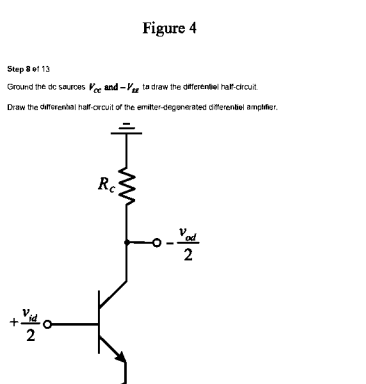


Figure 3

Step 6 of 13

Apply Kirchhoff's voltage law to left side loop.

$$-\left(V_{CM}+\frac{v_{id}}{2}\right)+V_{BE}+\frac{I}{2}R_e+V_s=0$$

At quiescent state, the differential input voltage is zero (That is, $v_{id}=0$).

$$-\left(V_{CM}+\frac{0}{2}\right)+V_{BE}+\frac{I}{2}R_e+V_s=0$$
$$-V_{CM}+V_{BE}+\frac{I}{2}R_e+V_s=0$$
$$V_s=V_{CM}-V_{BE}-\frac{I}{2}R_e$$

Therefore, the dc voltage across the current source is $\boxed{V_{CM}-V_{BE}-\frac{I}{2}R_e}$.

Step 7 of 13

Draw the modified circuit of emitter-degenerated differential amplifier

Figure 4

Step 8 of 13

Ground the dc sources V_{CC} and $-V_{EE}$ to draw the differential half-circuit.

Draw the differential half-circuit of the emitter-degenerated differential amplifier.

Figure 5

Step 9 of 13

Draw the ac equivalent circuit of in Figure 4.

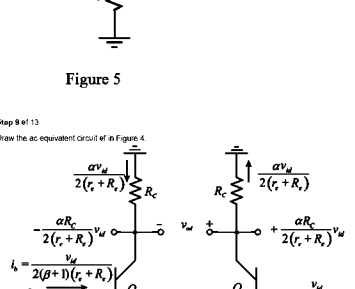


Figure 6

Step 10 of 13

The differential output voltage is,

$$v_{od}=v_{C2}-v_{C1}$$
$$=\frac{\alpha R_C}{2(r_e+R_e)}v_{id}-\left(-\frac{\alpha R_C}{2(r_e+R_e)}v_{id}\right)$$
$$=\left(\frac{\alpha R_C}{r_e+R_e}\right)v_{id}$$
$$\frac{v_{od}}{v_{id}}=\frac{\alpha R_C}{r_e+R_e}$$

Therefore, the differential gain is $\boxed{\frac{\alpha R_C}{r_e+R_e}}$.

Step 11 of 13

Write the expression for base current, i_b.

$$i_b=\frac{v_{id}}{2(\beta+1)(r_e+R_e)}$$

Write the expression for input resistance, R_{id}.

$$R_{id}=\frac{v_{id}}{i_b}$$
$$=\frac{v_{id}}{\frac{v_{id}}{2(\beta+1)(r_e+R_e)}}$$
$$=2(\beta+1)(r_e+R_e)$$

Therefore, the input resistance is $\boxed{2(\beta+1)(r_e+R_e)}$.

The resistance seen between the two bases of the transistors is the product of the total resistance in the circuit and the term $(\beta+1)$.

Step 12 of 13

Draw the modified circuit of emitter-degenerated differential amplifier.

Figure 7

Step 13 of 13

Apply Kirchhoff's voltage law to left side loop.

$$-\left(V_{CM}+\frac{v_{id}}{2}\right)+V_{BE}+V_s=0$$

At quiescent state, the differential input voltage is zero (That is, $v_{id}=0$).

$$-\left(V_{CM}+\frac{0}{2}\right)+V_{BE}+V_s=0$$
$$-V_{CM}+V_{BE}+V_s=0$$
$$V_s=V_{CM}-V_{BE}$$

Therefore, the dc voltage across the current source is $\boxed{V_s=V_{CM}-V_{BE}}$

Thus, in second part, there is no voltage drop in $2R_e$ resistor in the circuit. So, this circuit has lower negative V_{CM} value compared to first part circuit.

Therefore, the $\boxed{\text{Circuit in first part}}$ allows a larger negative value V_{CM}

Step 1 of 4

Draw the circuit for differential amplifier.

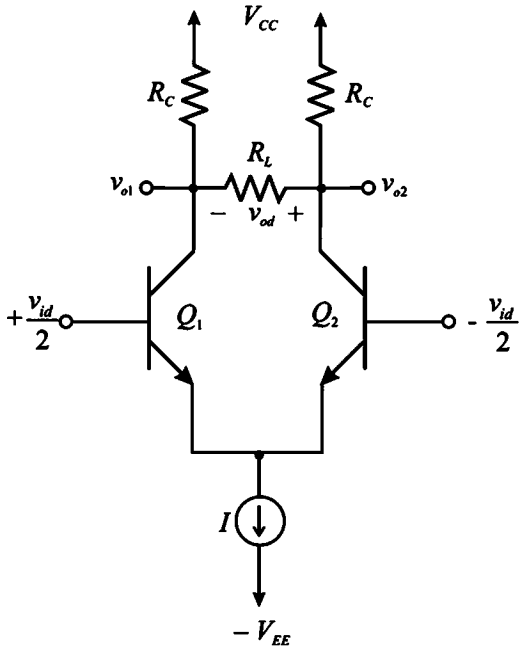


Figure 1

Step 2 of 4

Draw the differential half circuit of Figure 1.

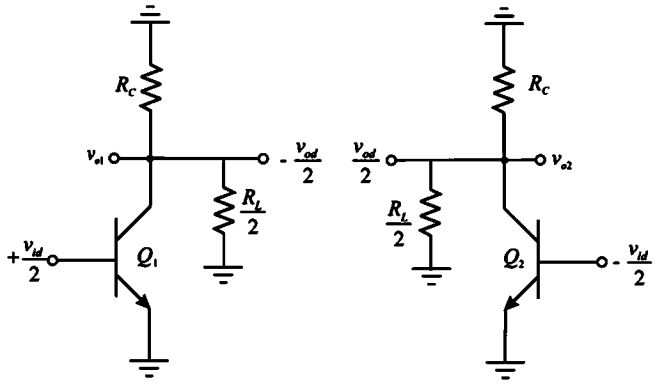


Figure 2

Step 3 of 4

Draw the equivalent circuit model of the differential half-circuit formed by Q_1.

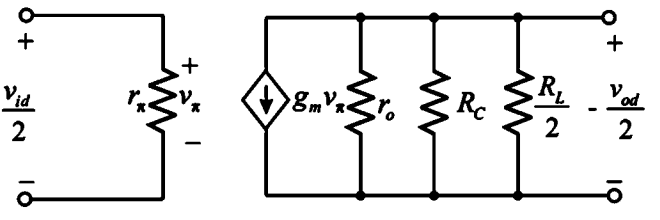


Figure 3

Step 4 of 4

The voltage across the resistor, r_π is,

$$v_\pi = \frac{v_{id}}{2}$$

Apply nodal analysis at the output.

$$\frac{-\frac{v_{od}}{2}}{\frac{R_L}{2}} + \frac{-\frac{v_{od}}{2}}{R_C} + \frac{-\frac{v_{od}}{2}}{r_o} + g_m v_\pi = 0$$

$$\frac{v_{od}}{2}\left(\frac{1}{\frac{R_L}{2}} + \frac{1}{R_C} + \frac{1}{r_o}\right) = g_m v_\pi$$

$$\frac{v_{od}}{2}\left(\frac{1}{\frac{R_L}{2} \| R_C \| r_o}\right) = g_m\left(\frac{v_{id}}{2}\right)$$

$$\frac{v_{od}}{v_{id}} = g_m\left(\frac{R_L}{2} \| R_C \| r_o\right)$$

Therefore, the differential voltage gain is $\boxed{g_m\left(\frac{R_L}{2} \| R_C \| r_o\right)}$.

9.049P

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Step 1 of 7

Refer to Figure P8.57 in the textbook.

Draw the modified differential amplifier circuit.

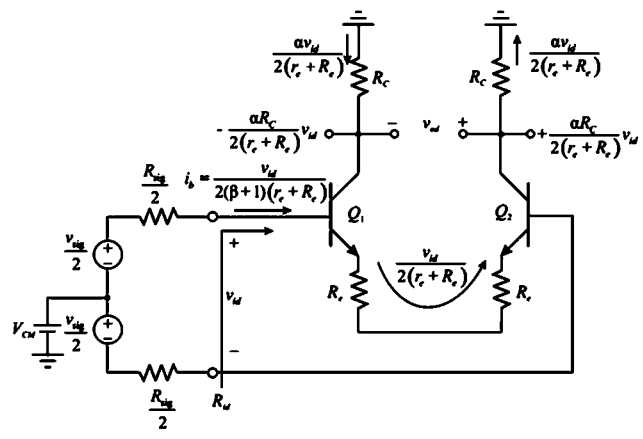


Figure 1

Step 2 of 7

Write the expression for base current, i_b.

$$i_b = \frac{v_{id}}{2(\beta+1)(r_e+R_e)}$$

Write the expression for input resistance, R_{id}.

$$\begin{aligned} R_{id} &= \frac{v_{id}}{i_b} \\ &= \frac{v_{id}}{\left[\frac{v_{id}}{2(\beta+1)(r_e+R_e)}\right]} \\ &= 2(\beta+1)(r_e+R_e) \end{aligned}$$

Step 3 of 7

Apply Kirchhoff's voltage law to left side loop in Figure 1.

$$\begin{aligned} -\frac{v_{sig}}{2} - \frac{v_{sig}}{2} + i_b\frac{R_{sig}}{2} + v_{id} + i_b\frac{R_{sig}}{2} &= 0 \\ -v_{sig} - v_{sig} + i_bR_{sig} + 2v_{id} + i_bR_{sig} &= 0 \\ -2v_{sig} + 2i_bR_{sig} + 2v_{id} &= 0 \\ -v_{sig} + i_bR_{sig} + v_{id} &= 0 \end{aligned}$$

$$-v_{sig} + \left(\frac{v_{id}}{R_{id}}\right)R_{sig} + v_{id} = 0$$

$$v_{sig} = \left(\frac{R_{sig}}{R_{id}} + 1\right)v_{id}$$

$$\frac{v_{id}}{v_{sig}} = \frac{R_{id}}{R_{sig}+R_{id}}$$

Thus, the gain for left side circuit is, $\frac{v_{id}}{v_{sig}} = \frac{R_{id}}{R_{sig}+R_{id}}$.

Step 4 of 7

The differential output voltage is,

$$\begin{aligned} v_{od} &= v_{C2} - v_{C1} \\ &= \frac{\alpha R_C}{2(r_e+R_e)}v_{id} - \left(-\frac{\alpha R_C}{2(r_e+R_e)}v_{id}\right) \\ &= \left(\frac{\alpha R_C}{r_e+R_e}\right)v_{id} \end{aligned}$$

$$\frac{v_{od}}{v_{id}} = \frac{\alpha R_C}{r_e+R_e}$$

Thus, the gain for right side circuit is, $\frac{v_{od}}{v_{id}} = \frac{\alpha(2R_C)}{2(r_e+R_e)}$.

Calculate the expression for overall differential voltage gain.

$$\begin{aligned} G_v &= \frac{v_{od}}{v_{sig}} \\ &= \left(\frac{v_{od}}{v_{id}}\right)\left(\frac{v_{id}}{v_{sig}}\right) \\ &= \left(\frac{\alpha R_C}{r_e+R_e}\right)\left(\frac{R_{id}}{R_{sig}+R_{id}}\right) \end{aligned}$$

Thus, the overall differential voltage gain, G_v is $\boxed{\left(\frac{\alpha R_C}{r_e+R_e}\right)\left(\frac{R_{id}}{R_{sig}+R_{id}}\right)}$.

Step 5 of 7

Calculate the expression for overall differential voltage gain when $v_{id} = 0.5v_{sig}$.

$$\begin{aligned} G_v &= \frac{v_{od}}{v_{sig}} \\ &= \left(\frac{v_{od}}{v_{id}}\right)\left(\frac{v_{id}}{v_{sig}}\right) \\ &= \left(\frac{v_{od}}{v_{id}}\right)\left(\frac{0.5v_{sig}}{v_{sig}}\right) \\ &= \frac{1}{2}\frac{v_{od}}{v_{id}} \end{aligned}$$

Substitute $\frac{\alpha R_C}{r_e+R_e}$ for $\frac{v_{od}}{v_{id}}$.

$$G_v = \frac{\alpha R_C}{2(r_e+R_e)}$$

Thus, the expression for overall differential voltage gain when $v_{id} = 0.5v_{sig}$ is $\boxed{G_v = \frac{\alpha R_C}{2(r_e+R_e)}}$.

Step 6 of 7

Calculate the expression for overall differential voltage gain in terms of β.

$$G_v = \frac{\alpha R_C}{2(r_e+R_e)}$$

Substitute $\frac{\beta}{\beta+1}$ for α.

$$\begin{aligned} G_{v1} &= \frac{\left(\frac{\beta}{\beta+1}\right)R_C}{2(r_e+R_e)} \\ &= \frac{\beta R_C}{2(\beta+1)(r_e+R_e)} \end{aligned}$$

Consider the expression for overall differential voltage gain when β is doubled.

$$\begin{aligned} G_{v2} &= \frac{\left(\frac{2\beta}{2\beta+1}\right)R_C}{2(r_e+R_e)} \\ &= \frac{\beta R_C}{(2\beta+1)(r_e+R_e)} \end{aligned}$$

Step 7 of 7

Calculate the ratio of two gain expressions.

$$\begin{aligned} \frac{G_{v1}}{G_{v2}} &= \frac{\frac{\beta R_C}{2(\beta+1)(r_e+R_e)}}{\frac{\beta R_C}{(2\beta+1)(r_e+R_e)}} \\ &= \frac{(2\beta+1)}{2(\beta+1)} \qquad (\beta = 100) \\ &= \frac{201}{202} \end{aligned}$$

$$G_{v2} = 1.005G_{v1}$$

Calculate the percentage increase in G_v when β is doubled.

$$\begin{aligned} \%G_v &= \frac{G_{v2} - G_{v1}}{G_{v1}} \times 100 \\ &= \frac{1.005G_{v1} - G_{v1}}{G_{v1}} \times 100 \\ &= 0.5\% \end{aligned}$$

Thus, the factor of G_v is increased by $\boxed{0.5\%}$.

9.051P

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9.052P

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Step 1 of 7

Refer to Figure P8.61 in the textbook.

(a)

Draw the differential amplifier circuit with notations.

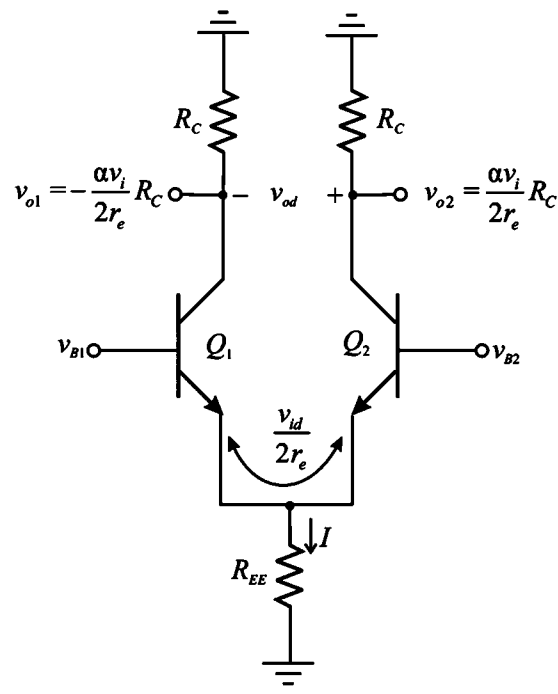


Figure 1

Step 2 of 7

Calculate the emitter current of half circuit of differential amplifier.

$$I_E = \frac{I}{2}$$

The expression for small signal emitter resistance is,

$$\begin{aligned} r_e &= \frac{V_T}{I_E} \\ &= \frac{V_T}{\frac{I}{2}} \\ &= \frac{2V_T}{I} \end{aligned}$$

The expression for the small signal emitter current is,

$$i_e = \frac{v_i}{2r_e}$$

Calculate the expression for collector current.

$$\begin{aligned} i_c &= \alpha i_e \\ &= \alpha\left(\frac{v_i}{2r_e}\right) \end{aligned}$$

Step 3 of 7

Calculate the expression for output voltage.

$$\begin{aligned} v_{od} &= v_{o2} - v_{o1} \\ &= v_{o2} - 0 \\ &= i_c R_C \\ &= \left(\frac{\alpha v_i}{2r_e}\right) R_C \end{aligned}$$

Calculate the small signal voltage gain.

$$\begin{aligned} A_v &= \frac{v_{od}}{v_i} \\ &= \frac{\frac{\alpha v_i}{2r_e} R_C}{v_i} \\ &= \frac{\alpha}{2r_e} R_C \end{aligned}$$

Therefore, the expression for small signal voltage gain of differential amplifier is $\boxed{\frac{\alpha}{2r_e} R_C}$.

Step 4 of 7

(b)

As a cascade of a common-collector stage Q_1 and common-base stage Q_2

Draw the cascaded circuit diagram.

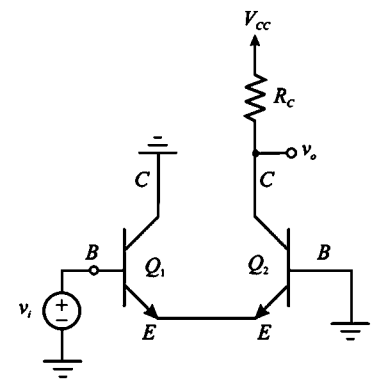


Figure 2

Step 5 of 7

Draw the small signal equivalent circuit of Figure 2.

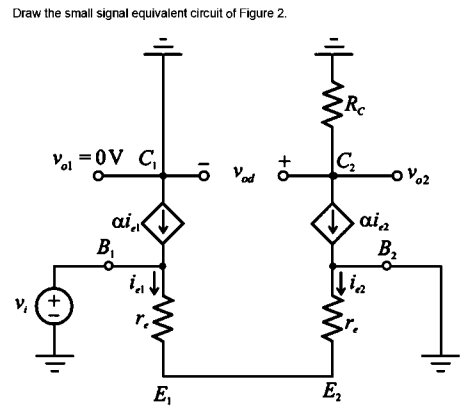


Figure 3

Step 6 of 7

Apply Kirchhoff's voltage law to bottom loop.

$$\begin{aligned} &-v_i + i_{e1} r_e - i_{e2} r_e = 0 \qquad (i_{e2} = -i_{e1}) \\ &-v_i + 2i_{e1} r_e = 0 \\ &2i_{e1} r_e = v_i \\ &i_{e1} = \frac{v_i}{2r_e} \end{aligned}$$

Apply Kirchhoff's current law at node v_{o2}.

$$\begin{aligned} &\alpha i_{e2} + \frac{v_{o2}}{R_C} = 0 \\ &\frac{v_{o2}}{R_C} = -\alpha i_{e2} \\ &v_{o2} = -\alpha i_{e2} R_C \\ &v_{o2} = -\alpha(-i_{e1}) R_C \qquad (i_{e2} = -i_{e1}) \\ &\quad = \alpha i_{e1} R_C \\ &\quad = \alpha\left(\frac{v_i}{2r_e}\right) R_C \end{aligned}$$

Step 7 of 7

Calculate the expression for output voltage.

$$\begin{aligned} v_{od} &= v_{o2} - v_{o1} \\ &= \alpha\left(\frac{v_i}{2r_e}\right) R_C - 0 \\ &= \alpha\left(\frac{v_i}{2r_e}\right) R_C \end{aligned}$$

Calculate the small signal voltage gain.

$$\begin{aligned} A_v &= \frac{v_{od}}{v_i} \\ &= \frac{\alpha\left(\frac{v_i}{2r_e}\right) R_C}{v_i} \\ &= \frac{\alpha}{2r_e} R_C \end{aligned}$$

Therefore, the expression for small signal voltage gain of cascaded amplifier is $\boxed{\frac{\alpha}{2r_e} R_C}$.

9.055P

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9.056P

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Step 1 of 7

Refer to Figure P8.62 in the textbook.

(a)

Draw the circuit diagram with notations.

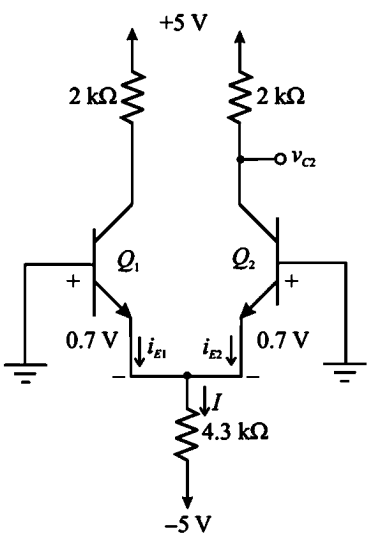


Figure 1

Step 2 of 7

Apply Kirchhoff's voltage law to left side loop.

$$0.7 + I\left(4.3\times10^{3}\right) + \left(-5\right) = 0$$

$$I\left(4.3\times10^{3}\right) = 5 - 0.7$$

$$I = \frac{4.3}{4.3\times10^{3}}$$
$$= 1\times10^{-3}$$
$$= 1\ \text{mA}$$

Calculate the value of emitter current.

$$I_E = \frac{I}{2}$$
$$= \frac{1\times10^{-3}}{2}$$
$$= 0.5\times10^{-3}$$
$$= 0.5\ \text{mA}$$

Step 3 of 7

Calculate the value of small signal emitter resistance.

$$r_e = \frac{V_T}{I_E}$$

Substitute $25\ \text{mV}$ for V_T and $0.5\ \text{mA}$ for I_E.

$$r_e = \frac{25\times10^{-3}}{0.5\times10^{-3}}$$
$$= 50\ \Omega$$

Calculate the value of small signal differential voltage gain, A_d.

$$A_d = \frac{\alpha}{2r_e}R_C$$

Substitute 1 for α, $50\ \Omega$ for r_e and $2\ \text{k}\Omega$ for R_C.

$$A_d = \frac{1}{2(50)}\left(2\times10^{3}\right)$$
$$= 20\ \text{V/V}$$

Therefore, the differential gain is $\boxed{20\ \text{V/V}}$.

Step 4 of 7

(b)

Draw the circuit diagram with common mode input.

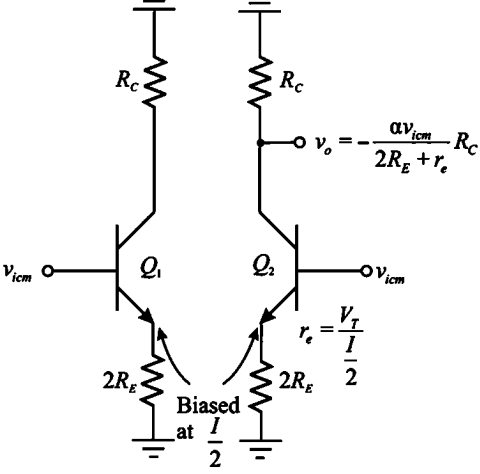


Figure 2

Step 5 of 7

The expression for the common mode voltage gain is,

$$v_{ocl} = -\frac{\alpha v_{icm}}{2R_E + r_e}R_C$$

$$\frac{v_{ocl}}{v_{icm}} = -\frac{\alpha R_C}{2R_E + r_e}$$

$$A_{cm} = -\frac{\alpha R_C}{2R_E + r_e}$$

Substitute 1 for α, $50\ \Omega$ for r_e and $2\ \text{k}\Omega$ for R_C.

$$A_{cm} = \frac{1\left(2\times10^{3}\right)}{2\left(4.3\times10^{3}\right) + 50}$$
$$= \frac{2\times10^{3}}{8.65\times10^{3}}$$
$$= 0.23\ \text{V/V}$$

Therefore, the common-mode gain is $\boxed{0.23\ \text{V/V}}$.

Step 6 of 7

(c)

Calculate the value of common-mode rejection ratio.

$$\text{CMRR} = 20\log_{10}\left(\frac{A_d}{A_{cm}}\right)$$

Substitute $0.23\ \text{V/V}$ for A_{cm} and $20\ \text{V/V}$ for A_d.

$$\text{CMRR} = 20\log_{10}\left(\frac{20}{0.23}\right)$$
$$= 20\log_{10}(86.96)$$
$$= 20(1.94)$$
$$= 38.8\ \text{dB}$$

Therefore, the common-mode rejection ration is $\boxed{38.8\ \text{dB}}$.

Step 7 of 7

(d)

Calculate the output voltage.

$$v_O = A_d\, v_{id} + A_{cm} v_{icm}$$
$$= A_d\left(v_{B1} - v_{B2}\right) + A_{cm}\left(\frac{v_{B1} + v_{B2}}{2}\right)$$

Substitute $0.1\sin120\pi t + 0.005\sin2000\pi t$ for v_{B1} and $0.1\sin120\pi t - 0.005\sin2000\pi t$ for v_{B2}.

$$v_O = \left[\begin{array}{l}(20)\left(\begin{array}{l}(0.1\sin120\pi t + 0.005\sin2000\pi t) - \\ (0.1\sin120\pi t - 0.005\sin2000\pi t)\end{array}\right) + \\ \left(\frac{0.23}{2}\right)\left\{\begin{array}{l}(0.1\sin120\pi t + 0.005\sin2000\pi t) + \\ (0.1\sin120\pi t - 0.005\sin2000\pi t)\end{array}\right\}\end{array}\right]$$
$$= (20)(0.01\sin2\pi\times1000t) + (0.23)(0.1\sin2\pi\times60t)$$
$$= 0.2\sin2000\pi t + 0.023\sin120\pi t\ \text{V}$$

Therefore, the output voltage is $\boxed{0.2\sin2000\pi t + 0.023\sin120\pi t\ \text{V}}$.

Step 1 of 7

Refer to Figure P8.63 in the textbook.

Draw the differential half circuit for the differential amplifier.

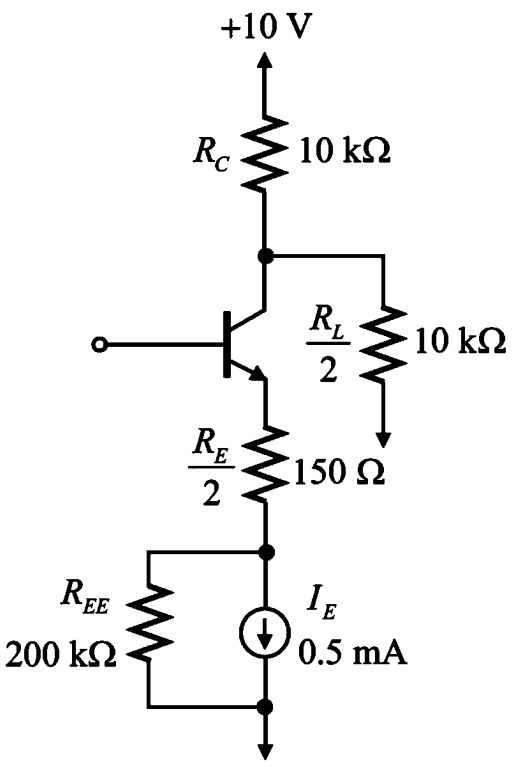


Figure 1

Step 2 of 7

Calculate the emitter current of half circuit of differential amplifier.

$I_E = I$

Substitute $0.5\ \text{mA}$ for I. $I_E = 0.5\ \text{mA}$

Calculate the value of resistor r_e.

$$r_e = \frac{V_T}{I_E}$$

Substitute $25\ \text{mV}$ for V_T and $0.5\ \text{mA}$ for I_E.

$$r_e = \frac{25\times10^{-3}}{0.5\times10^{-3}}$$
$$= 50\ \Omega$$

Calculate the value of α.

$$\alpha = \frac{\beta}{\beta+1}$$

Substitute 100 for β.

$$\alpha = \frac{100}{100+1}$$
$$= 0.99$$
$$\approx 1$$

Step 3 of 7

Calculate the value of the differential gain of the circuit.

$$A_d = \frac{\alpha\left(R_C \parallel \frac{R_L}{2}\right)}{r_e + \frac{R_E}{2}}$$

Substitute $50\ \Omega$ for r_e, $300\ \Omega$ for R_E, 1 for α, $10\ \text{k}\Omega$ for R_C and $20\ \text{k}\Omega$ for R_L.

$$A_d = \frac{(1)\left[\left(10\times10^3\right) \parallel \frac{\left(20\times10^3\right)}{2}\right]}{50+\frac{300}{2}}$$
$$= \frac{\left[\frac{\left(10\times10^3\right)\left(10\times10^3\right)}{\left(10\times10^3\right)+\left(10\times10^3\right)}\right]}{50+150}$$
$$= \frac{5\times10^3}{200}$$
$$= 25\ \text{V/V}$$

Thus, the value of the differential gain of the circuit is $\boxed{25\ \text{V/V}}$.

Step 4 of 7

Calculate the value of differential input resistance of the amplifier, R_{id}.

$$R_{id} = (\beta+1)(2r_e + R_e)$$

Substitute $50\ \Omega$ for r_e, $300\ \Omega$ for R_e and 100 for β.

$$R_{id} = (100+1)\left[2(50)+300\right]$$
$$= (101)(400)$$
$$= 40400$$
$$= 40.4\ \text{k}\Omega$$

Thus, the value of differential input resistance of the amplifier, R_{id} is $\boxed{40.4\ \text{k}\Omega}$.

Step 5 of 7

Calculate the common-mode gain, A_{icm}.

$$A_{icm} = \frac{R_C}{R_{EE}}\frac{\Delta R_C}{R_C}$$

Where, $\Delta R_C = 0.02R_C$ in the worst case

Substitute $0.02R_C$ for ΔR_C, $10\ \text{k}\Omega$ for R_C and $200\ \text{k}\Omega$ for R_{EE}.

$$A_{icm} = \frac{10\ \text{k}\Omega}{200\ \text{k}\Omega}\left(\frac{0.02R_C}{R_C}\right)$$
$$= \frac{0.2}{200}$$
$$= 0.001\ \text{V/V}$$

Thus, the common-mode gain, A_{icm} is $\boxed{0.001\ \text{V/V}}$.

Step 6 of 7

Calculate the value of r_o.

$$r_o = \frac{V_A}{I_C}$$
$$= \frac{V_A}{\alpha I_E}$$

Substitute 1 for α, $100\ \text{V}$ for V_A and $0.5\ \text{mA}$ for I_E.

$$r_o = \frac{100}{(1)\left(0.5\times10^{-3}\right)}$$
$$= 200\times10^3$$
$$= 200\ \text{k}\Omega$$

Step 7 of 7

Calculate the value of common-mode input resistance of the amplifier, R_{icm}.

$$R_{icm} = (\beta R_{EE})\left(\frac{1+\frac{R_C}{\beta r_o}}{1+\frac{R_C+2R_{EE}}{r_o}}\right)$$

Substitute $200\ \text{k}\Omega$ for r_o, $200\ \text{k}\Omega$ for R_{EE}, $10\ \text{k}\Omega$ for R_C and 100 for β.

$$R_{icm} = (100)\left(200\times10^3\right)\left[\frac{1+\frac{10\times10^3}{(100)\left(200\times10^3\right)}}{1+\frac{\left(10\times10^3\right)+2\left(200\times10^3\right)}{\left(200\times10^3\right)}}\right]$$
$$= \left(2\times10^7\right)\left(\frac{1}{1+\frac{410\times10^3}{200\times10^3}}\right)$$
$$= 6.56\times10^6$$
$$= 6.56\ \text{M}\Omega$$

Thus, the value of common-mode input resistance of the amplifier, R_{icm} is $\boxed{6.56\ \text{M}\Omega}$.

9.063P

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Step 1 of 4

9.069P

Draw the following circuit diagram:

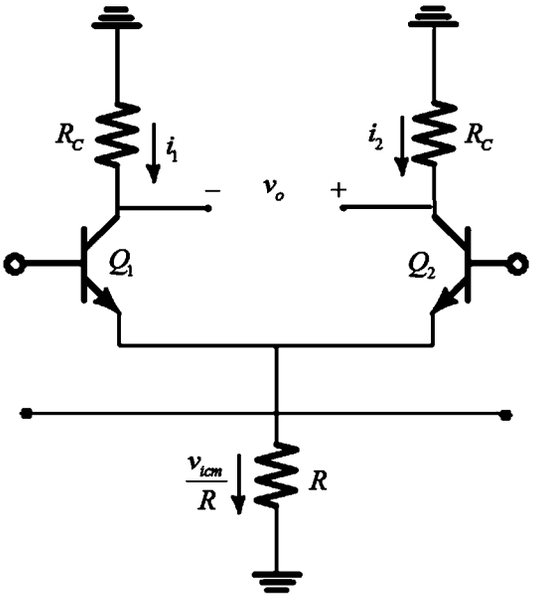


Figure 1

Step 2 of 4

Calculate the expression for total emitter current, I .

$$I = \frac{v_{icm}}{R}$$

The base emitter junction area of one transistor is twice the other. So, the emitter bias current will split between the two transistors according to their area ratio.

$$i_{E2} = 2i_{E1}$$
$$i_2 = 2i_1$$

Calculate the expression for i_1 .

$$i_1 + i_2 = I$$

Substitute $2i_1$ for i_2 and $\frac{v_{icm}}{R}$ for I .

$$i_1 + 2i_1 = \frac{v_{icm}}{R}$$
$$3i_1 = \frac{v_{icm}}{R}$$
$$i_1 = \frac{v_{icm}}{3R}$$

Step 3 of 4

Calculate the expression for i_2 .

$$i_2 = 2i_1$$

Substitute $\frac{v_{icm}}{3R}$ for i_1 .

$$i_2 = 2\left(\frac{v_{icm}}{3R}\right)$$
$$= \frac{2v_{icm}}{3R}$$

Calculate the expression for differential output voltage, v_{od} .

$$v_{od} = -i_2R_C - \left(-i_1R_C\right)$$
$$= R_C\left(i_1 - i_2\right)$$

Substitute $\frac{v_{icm}}{3R}$ for i_1 and $\frac{2v_{icm}}{3R}$ for i_2 .

$$v_{od} = -i_2R_C - \left(-i_1R_C\right)$$
$$= R_C\left(\frac{v_{icm}}{3R} - \frac{2v_{icm}}{3R}\right)$$
$$= -\frac{R_Cv_{icm}}{3R}$$

Step 4 of 4

Calculate the value of common-mode gain of the circuit, A_{cm} .

$$A_{cm} = \left|\frac{v_{od}}{v_{icm}}\right|$$

Substitute $-\frac{R_Cv_{icm}}{3R}$ for v_{od} .

$$A_{cm} = \left|\frac{-\frac{R_Cv_{icm}}{3R}}{v_{icm}}\right|$$
$$= \frac{R_C}{3R}$$

Substitute $12\ \text{k}\Omega$ for R_C and $500\ \text{k}\Omega$ for R .

$$A_{cm} = \frac{12\ \text{k}\Omega}{3(500\ \text{k}\Omega)}$$
$$= \frac{4}{500}$$
$$= 0.008\ \text{V/V}$$

Thus, the value of common-mode gain of the circuit, A_{cm} is $\boxed{0.008\ \text{V/V}}$.

Step 1 of 7

Case (i): The bias current at the emitter junction, I is $160\ \mu A$

Calculate the drain current of half circuit of differential amplifier.

$$I_D = \frac{I}{2}$$

Substitute $160\ \mu A$ for I. $I_D = \frac{160\ \mu A}{2}$

$$= 80\ \mu A$$

Calculate the value of trans-conductance, g_m.

$$g_m = \sqrt{2k_n' \frac{W}{L} I_D}$$

Substitute $4\ mA/V^2$ for $k_n' \frac{W}{L}$ and $80\ \mu A$ for I_D.

$$g_m = \sqrt{2\left(4 \times 10^{-3}\right)\left(80 \times 10^{-6}\right)}$$
$$= \sqrt{64 \times 10^{-8}}$$
$$= 8 \times 10^{-4}$$
$$= 0.8\ mA/V$$

Step 2 of 7

Calculate the value of differential gain, A_d.

$$A_d = g_m R_D$$

Substitute $0.8\ mA/V$ for g_m and $10\ k\Omega$ for R_D.

$$A_d = \left(10 \times 10^3\right)\left(0.8 \times 10^{-3}\right)$$
$$= 8\ V/V$$

Thus, the value of differential gain, A_v for $I = 160\ \mu A$ is $\boxed{8\ V/V}$.

Consider the expression for trans-conductance to calculate overdrive voltage, V_{ov}.

$$g_m = \frac{2I_D}{V_{ov}}$$
$$V_{ov} = \frac{2I_D}{g_m}$$

Substitute $0.8\ mA/V$ for g_m and $80\ \mu A$ for I_D.

$$V_{ov} = \frac{2\left(80 \times 10^{-6}\right)}{0.8 \times 10^{-3}}$$
$$= 0.2\ V$$

Step 3 of 7

Calculate the value of input offset voltage, V_{os}.

$$V_{os} = \left(\frac{V_{ov}}{2}\right)\left(\frac{\Delta R_D}{R_D}\right)$$

Where $\Delta R_D = 0.02 R_D$ is the worst case

Substitute $0.02R_D$ for ΔR_D and $0.2\ V$ for V_{ov}.

$$V_{os} = \left(\frac{0.2}{2}\right)\left(\frac{0.02R_D}{R_D}\right)$$
$$= (0.1)(0.02)$$
$$= 2 \times 10^{-3}$$
$$= 2\ mV$$

Thus, the value of input offset voltage, V_{os} for $I = 160\ \mu A$ is $\boxed{2\ mV}$.

Step 4 of 7

Case (ii): The bias current at the emitter junction, I is $360\ \mu A$

Calculate the drain current of half circuit of differential amplifier.

$$I_D = \frac{I}{2}$$

Substitute $360\ \mu A$ for I. $I_D = \frac{360\ \mu A}{2}$

$$= 180\ \mu A$$

Calculate the value of trans-conductance, g_m.

$$g_m = \sqrt{2k_n' \frac{W}{L} I_D}$$

Substitute $4\ mA/V^2$ for $k_n' \frac{W}{L}$ and $180\ \mu A$ for I_D.

$$g_m = \sqrt{2\left(4 \times 10^{-3}\right)\left(180 \times 10^{-6}\right)}$$
$$= \sqrt{144 \times 10^{-8}}$$
$$= 12 \times 10^{-4}$$
$$= 1.2\ mA/V$$

Step 5 of 7

Calculate the value of differential gain, A_d.

$$A_d = g_m R_D$$

Substitute $1.2\ mA/V$ for g_m and $10\ k\Omega$ for R_D.

$$A_d = \left(10 \times 10^3\right)\left(1.2 \times 10^{-3}\right)$$
$$= 12\ V/V$$

Thus, the value of differential gain, A_v for $I = 360\ \mu A$ is $\boxed{12\ V/V}$.

Consider the expression for trans-conductance to calculate overdrive voltage, V_{ov}.

$$g_m = \frac{2I_D}{V_{ov}}$$
$$V_{ov} = \frac{2I_D}{g_m}$$

Substitute $1.2\ mA/V$ for g_m and $180\ \mu A$ for I_D.

$$V_{ov} = \frac{2\left(180 \times 10^{-6}\right)}{1.2 \times 10^{-3}}$$
$$= 0.3\ V$$

Step 6 of 7

Calculate the value of input offset voltage, V_{os}.

$$V_{os} = \left(\frac{V_{ov}}{2}\right)\left(\frac{\Delta R_D}{R_D}\right)$$

Where $\Delta R_D = 0.02 R_D$ is the worst case

Substitute $0.02R_D$ for ΔR_D and $0.3\ V$ for V_{ov}.

$$V_{os} = \left(\frac{0.3}{2}\right)\left(\frac{0.02R_D}{R_D}\right)$$
$$= (0.15)(0.02)$$
$$= 3 \times 10^{-3}$$
$$= 3\ mV$$

Thus, the value of input offset voltage, V_{os} for $I = 360\ \mu A$ is $\boxed{3\ mV}$.

Step 7 of 7

Calculate the ratio of two gains.

$$\frac{A_{d2}}{A_{d1}} = \frac{12}{8}$$
$$= 1.5$$

Calculate the ratio of two input offset voltages.

$$\frac{V_{os2}}{V_{os1}} = \frac{3\ mV}{2\ mV}$$
$$= 1.5$$

Thus, both the ratios of A_d and V_{os} increase by the same factor, since both are proportional to $\sqrt{I}$. So, the any bias currents can choose depending on the requirement.

9.071P

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9.072P

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9.075P

The scale currents are differ by 10% .

$$\begin{aligned}\Delta I_S &= 10\% I_S \\ &= \left(\frac{10}{100}\right) \times I_S \\ &= (0.1) I_S\end{aligned}$$

Calculate the value of resulting input offset voltage, V_{OS} .

$$V_{OS} = \frac{\Delta I_S}{I_S} \times V_T$$

Substitute $(0.1) I_S$ for ΔI_S and $25\ \text{mV}$ for V_T .

$$\begin{aligned}V_{OS} &= \frac{(0.1) I_S}{I_S} \times \left(25 \times 10^{-3}\right) \\ &= 2.5 \times 10^{-3} \\ &= 2.5\ \text{mV}\end{aligned}$$

Thus, the value of resulting input offset voltage, V_{OS} is $\boxed{2.5\ \text{mV}}$.

Step 1 of 4

Consider the effect of the load mismatch.

$$R_{C1} = R_C + \frac{\Delta R_C}{2}$$

$$R_{C2} = R_C - \frac{\Delta R_C}{2}$$

The transistors Q_1 and Q_2 are perfectly matched. It follows that current, I will divide equally between Q_1 and Q_2, and thus collector voltages are,

$$V_{C1} = V_{CC} - \left(\frac{\alpha I}{2}\right)\left(R_C + \frac{\Delta R_C}{2}\right)$$

$$V_{C2} = V_{CC} - \left(\frac{\alpha I}{2}\right)\left(R_C - \frac{\Delta R_C}{2}\right)$$

Step 2 of 4

Calculate the differential output voltage, V_{od}.

$$\begin{aligned} V_{od} &= V_{C2} - V_{C1} \\ &= \left[V_{CC} - \left(\frac{\alpha I}{2}\right)\left(R_C - \frac{\Delta R_C}{2}\right)\right] - \left[V_{CC} - \left(\frac{\alpha I}{2}\right)\left(R_C + \frac{\Delta R_C}{2}\right)\right] \\ &= -\left(\frac{\alpha I}{2}\right)\left(R_C - \frac{\Delta R_C}{2}\right) + \left(\frac{\alpha I}{2}\right)\left(R_C + \frac{\Delta R_C}{2}\right) \\ &= \left(\frac{\alpha I}{2}\right)\left[-R_C + \frac{\Delta R_C}{2} + R_C + \frac{\Delta R_C}{2}\right] \end{aligned}$$

$$V_{od} = \left(\frac{\alpha I}{2}\right)\Delta R_C$$

Step 3 of 4

The expression for differential gain including the emitter resistance is,

$$A_d = \frac{R_C}{r_e + R_e} \qquad \left(r_e = \frac{V_T}{I_C}\right)$$

$$= \frac{R_C}{\frac{V_T}{I_C} + R_e} \qquad \left(I_C = \frac{I}{2}\right)$$

$$= \frac{R_C}{\frac{2V_T}{I} + R_e}$$

$$= \frac{IR_C}{2V_T + IR_e}$$

Step 4 of 4

Calculate the expression for the input offset voltage.

$$|V_{OS}| = \frac{V_{od}}{A_d}$$

Substitute $\frac{IR_C}{2V_T + IR_e}$ for A_d and $\left(\frac{\alpha I}{2}\right)\Delta R_C$ for V_{od}.

$$\begin{aligned} |V_{OS}| &= \frac{\left(\frac{\alpha I}{2}\right)\Delta R_C}{\left[\frac{IR_C}{2V_T + IR_e}\right]} \\ &= \left(\frac{\alpha I}{2}\right)\left(\frac{2V_T + IR_e}{IR_C}\right)\Delta R_C \\ &= \frac{\alpha\left(2V_T + IR_e\right)}{2}\frac{\Delta R_C}{R_C} \\ &= \alpha\left(V_T + \frac{IR_e}{2}\right)\frac{\Delta R_C}{R_C} \end{aligned}$$

Therefore, the input offset voltage is $\boxed{\alpha\left(V_T + \frac{IR_e}{2}\right)\frac{\Delta R_C}{R_C}}$.

9.077P

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Step 1 of 5

Draw the following circuit diagram;

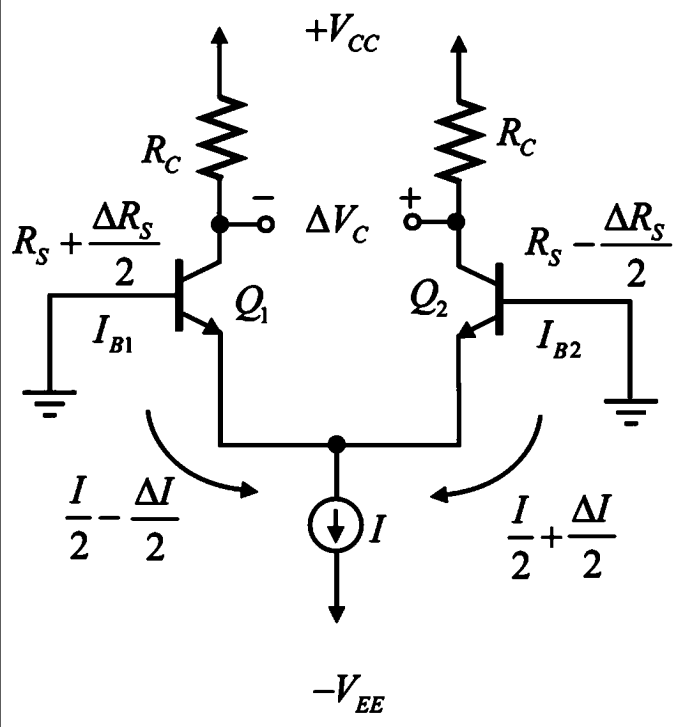


Figure 1

Step 2 of 5

The emitter currents due to mismatch are,

$$I_{E1} = \frac{I}{2} - \frac{\Delta I}{2}$$
$$I_{E2} = \frac{I}{2} + \frac{\Delta I}{2}$$

The base currents are,

$$I_{B1} = \frac{I_{E1}}{\beta + 1}$$
$$= \frac{1}{\beta + 1}\left(\frac{I}{2} - \frac{\Delta I}{2}\right)$$

$$I_{B2} = \frac{I_{E2}}{\beta + 1}$$
$$= \frac{1}{\beta + 1}\left(\frac{I}{2} + \frac{\Delta I}{2}\right)$$

Step 3 of 5

Apply Kirchhoff's voltage law at the input side.

$$I_{B1}\left(R_S + \frac{\Delta R_S}{2}\right) + I_{E1}r_e = I_{B2}\left(R_S - \frac{\Delta R_S}{2}\right) + I_{E2}r_e$$

Substitute $\frac{1}{\beta+1}\left(\frac{I}{2} + \frac{\Delta I}{2}\right)$ for I_{B2}, $\frac{1}{\beta+1}\left(\frac{I}{2} - \frac{\Delta I}{2}\right)$ for I_{B1}, $\frac{I}{2} + \frac{\Delta I}{2}$ for I_{E2} and $\frac{I}{2} - \frac{\Delta I}{2}$ for I_{E1}.

$$\frac{1}{\beta+1}\left(\frac{I}{2} - \frac{\Delta I}{2}\right)\left(R_S + \frac{\Delta R_S}{2}\right) + \left(\frac{I}{2} - \frac{\Delta I}{2}\right)r_e = \frac{1}{\beta+1}\left(\frac{I}{2} + \frac{\Delta I}{2}\right)\left(R_S - \frac{\Delta R_S}{2}\right) + \left(\frac{I}{2} + \frac{\Delta I}{2}\right)r_e$$

$$\frac{1}{\beta+1}\left(\frac{IR_S}{2} - \frac{\Delta IR_S}{2} + \frac{I\Delta R_S}{4} - \frac{\Delta I\Delta R_S}{4}\right) - \frac{\Delta I}{2}r_e = \frac{1}{\beta+1}\left(\frac{IR_S}{2} + \frac{\Delta IR_S}{2} - \frac{I\Delta R_S}{4} - \frac{\Delta I\Delta R_S}{4}\right) + \frac{\Delta I}{2}r_e$$

$$\frac{\Delta I}{2}r_e + \frac{\Delta I}{2}r_e = \frac{1}{\beta+1}\left(\frac{I\Delta R_S}{2} - \Delta IR_S\right)$$

$$\Delta I\left(r_e + \frac{R_S}{\beta+1}\right) = \frac{I\Delta R_S}{2(\beta+1)}$$

$$\Delta I = \frac{I\Delta R_S}{2(\beta+1)} \times \frac{(\beta+1)}{r_e(\beta+1) + R_S}$$
$$= \frac{I\Delta R_S}{2\left[r_e(\beta+1) + R_S\right]}$$

Step 4 of 5

Calculate the expression for ΔV_C.

$$\Delta V_C = -\Delta IR_C$$

Substitute $\frac{I\Delta R_S}{2\left[r_e(\beta+1) + R_S\right]}$ for ΔI.

$$\Delta V_C = \frac{I\Delta R_S R_C}{2\left[r_e(\beta+1) + R_S\right]}$$

Calculate the expression for differential voltage gain, A_d.

$$A_d = g_m R_C$$

Calculate the expression for the input offset voltage, V_{OS}.

$$V_{OS} = \frac{\Delta V_C}{A_d}$$

Step 5 of 5

Substitute $g_m R_C$ for A_d and $\frac{I\Delta R_S R_C}{2\left[r_e(\beta+1) + R_S\right]}$ for ΔV_C.

$$V_{OS} = \frac{I\Delta R_S R_C}{2\left[r_e(\beta+1) + R_S\right]} \times \frac{1}{g_m R_C} \qquad \left[r_e(\beta+1) = r_\pi\right]$$
$$= \frac{I\Delta R_S}{2g_m(r_\pi + R_S)}$$
$$= \frac{I\Delta R_S}{2(g_m r_\pi + g_m R_S)} \qquad \left[g_m r_\pi = \beta\right]$$
$$= \frac{I\Delta R_S}{2(\beta + g_m R_S)}$$

$$V_{OS} = \frac{I\Delta R_S}{2\beta\left(1 + \frac{g_m R_S}{\beta}\right)}$$
$$= \frac{\frac{I\Delta}{2\beta}R_S}{1 + \frac{g_m R_S}{\beta}}$$

Thus, the expression for overdrive voltage, V_{OS} is proved.

9.081P

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Step 1 of 5

The base emitter junction area of one transistor is twice the other.

The emitter bias current splits between the two transistors according to their area ratio.

Consider Q_2 as the larger device.

$$I_{E2} = 2I_{E1}$$

Find I_{E1} in terms of I

$$I_{E1} + I_{E2} = I$$
$$I_{E1} + 2I_{E1} = I$$
$$I_{E1} = \frac{I}{3}$$

Obtain I_{E2} in terms of I.

$$I_{E2} = I - I_{E1}$$
$$= I - \frac{I}{3}$$
$$= \frac{2}{3}I$$

Step 2 of 5

Find the expression for ΔV_C .

$$\Delta V_C = V_{C2} - V_{C1}$$
$$= I_{C2}R_C - I_{C1}R_C$$

Since, $\beta >> 1$,

$$I_{C2} = I_{E2}$$
$$I_{C1} = I_{E1}$$

The modified expression for ΔV_C is,

$$\Delta V_C = I_{E2}R_C - I_{E1}R_C$$
$$= \left(\frac{2I}{3} - \frac{I}{3}\right)R_C$$
$$= \frac{1}{3}IR_C$$

Step 3 of 5

The expression for the differential gain is,

$$A_d = \frac{R_C}{r_e}$$
$$= \frac{IR_C}{2V_T}$$

Find the dc offset voltage, V_{OS} .

$$V_{OS} = \frac{\Delta V_C}{A_d}$$
$$= \frac{\frac{1}{3}IR_C}{\frac{IR_C}{2V_T}}$$
$$= \frac{2}{3}V_T$$
$$= \frac{2}{3} \times 25 \times 10^{-3}$$

Simplify further.

$$V_{OS} = 16.7 \times 10^{-3}$$
$$= 16.7 \text{ mV}$$

Thus, the input voltage that would restore current balance to the differential pair in small-signal is $\boxed{16.7 \text{ mV}}$.

Hence, in small signal analysis, if a dc signal of 16.7 mV is applied as $V_{B2} - V_{B1}$, then it would restore the current balance in the transistors and reduce ΔV_C to zero.

Step 4 of 5

Consider the large-signal analysis.

The expression for the current i_{E1} is,

$$i_{E1} = I_{S1}e^{\frac{V_{B1} - V_E}{V_T}}$$

The expression for the current i_{E2} is,

$$i_{E2} = I_{S2}e^{\frac{V_{B2} - V_E}{V_T}}$$

Find the ratio $\frac{i_{E2}}{i_{E1}}$.

$$\frac{i_{E2}}{i_{E1}} = \frac{I_{S2}e^{\frac{(V_{B2} - V_E)}{V_T}}}{I_{S1}e^{\frac{(V_{B1} - V_E)}{V_T}}}$$

Step 5 of 5

Obtain the value of $V_{B2} - V_{B1}$.

$$\frac{I_{e2}}{I_{e1}} = \frac{I_{S2}e^{\frac{V_{B2} - V_E}{V_T}}}{I_{S1}e^{\frac{V_{B1} - V_E}{V_T}}}$$
$$1 = \frac{I_{S2}}{I_{S1}}e^{\frac{V_{B2} - V_E - V_{B1} + V_E}{V_T}}$$
$$1 = \frac{1}{2}e^{\frac{V_{B2} - V_{B1}}{V_T}}$$

Simplify further.

$$V_{B2} - V_{B1} = V_T \ln(2)$$
$$= 25 \times 10^{-3} \times 0.693$$
$$= 17.3 \times 10^{-3}$$

Thus, the input voltage that would restore current balance to the differential pair in small-signal is $\boxed{17.3 \text{ mV}}$.

Hence, similar results are obtained in both large-signal analysis and small-signal analysis.

9.084P

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Step 1 of 6

9.099P

Refer to Figure P8.102 in the text book.

(a)

The largest possible output voltage signal is,

$v_{o\text{-}max} = V_{CB7} + V_{B7}$ $\quad (\text{Since, } V_{CB7} = v_{o\text{-}max} - V_{B7})$

Assume that the collector-to-base junction voltage $V_{CB} = 0.4\text{V}$.

$v_{o\text{-}max} = 0.4 + V_{B7}$ ..... (1)

Step 2 of 6

The base voltage for Q_7 is,

$$\begin{aligned} V_{B7} &= V_{E7} - V_{EB7} && (\text{Since, } V_{EB7} = V_{E7} - V_{B7}) \\ &= V_{B6} - V_{EB7} && (\text{Since, } V_{E7} = V_{B6}) \\ &= (5 - V_{EB6}) - V_{EB7} && (\text{Since, } V_{EB6} = 5 - V_{B6}) \\ &= (5 - 0.7) - 0.7 && (\text{Since, } v_{EB} = 0.7\,\text{V for all } npn \text{ transistors}) \end{aligned}$$

Simplify further.

$$\begin{aligned} V_{B7} &= 5 - 1.4 \\ &= 3.6 \end{aligned}$$

Substitute $V_{B7} = 3.6\text{V}$ in equation (1).

$$\begin{aligned} v_{o\text{-}max} &= 0.4 + 3.6 \\ v_{o\text{-}max} &= 4 \end{aligned}$$

Therefore, the largest possible output voltage signal $v_{o\text{-}max}$ is $\boxed{4\text{V}}$.

Step 3 of 6

(b)

The dc bias voltage at output voltage is,

$v_{odc} = v_{o\text{-}max} - V_{+swing}$

Substitute $V_{+swing} = 1.5\text{V}$ and $v_{o\text{-}max} = 4\text{V}$.

$$\begin{aligned} v_{odc} &= 4 - 1.5 \\ &= 2.5\text{ V} \end{aligned}$$

Therefore, the dc bias voltage at output voltage is $\boxed{2.5\text{V}}$.

Step 4 of 6

(c)

Use Kirchhoff's voltage law to the circuit shown in Figure P8.102 in the text book.

$V_{BIAS} = v_{o\text{-}min} - V_{CB4}$ $\quad (\text{Since, } V_{CB4} = v_{o\text{-}min} - V_{BIAS})$

Obtain the minimum output voltage at negative output voltage swing.

The expression for the minimum output voltage is,

$v_{o\text{-}min} = v_{odc} - V_{swing}$

Substitute $v_{odc} = 2.5\text{V}$ and $V_{-swing} = 1.5\text{V}$ in the above equation

$$\begin{aligned} v_{o\text{-}min} &= 2.5 - 1.5 \\ v_{o\text{-}min} &= 1 \end{aligned}$$

Step 5 of 6

Determine the bias voltage.

$$\begin{aligned} V_{BIAS} &= 1 - 0.4 \\ V_{BIAS} &= 0.6\text{ V} \end{aligned}$$

Therefore, the bias voltage V_{BIAS} is $\boxed{0.6\text{V}}$.

Step 6 of 6

(d)

Apply Kirchhoff's voltage law to the circuit shown in Figure P8.102 in the text book.

The input common-mode voltage is,

$$\begin{aligned} -V_{BIAS} + V_{BE4} + V_{CB2} + v_{icm} &= 0 \\ v_{icm} &= V_{BIAS} - V_{BE4} - V_{CB2} \\ v_{icm} &= 0.6 - 0.7 - 0.4 \\ v_{icm} &= -0.5 \end{aligned}$$

Therefore, the common-mode voltage v_{icm} is $\boxed{-0.5\text{V}}$.

Step 1 of 13

Refer to Figure P8.104 in the text book.

(a)

Draw the equivalent circuit for $v_i = 0\,\mathrm{V}\,\mathrm{dc}$.

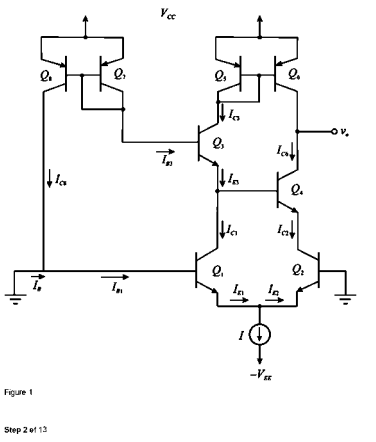


Figure 1

Step 2 of 13

From the above circuit, the base voltages of Q_1 and Q_2 are same (That is, 0 V). Since Q_1 and Q_2 are matched transistors, and assuming an ideal bias current source / with infinite output resistance, it follows that the current / was remain constant and from symmetry that / wa divide equaay between the two devices.

$$I_{B1} = I_{E1} = \frac{I}{2}$$

The collector currens for Q_1 and Q_2 are,

$$I_{C1} = I_{C2} = \alpha\frac{I}{2} \qquad (\text{Since}, I_C = \alpha I_E)$$

Step 3 of 13

From the above circuit, the emitter current of Q_4 is equal to the collector current of Q_2. The collector current of Q_4 is,

$$\begin{aligned} I_{C4} &= \alpha I_{E4} \\ &= \alpha I_{C2} \\ &= \alpha\left(\alpha\frac{I}{2}\right) \\ &= \alpha^2\frac{I}{2} \end{aligned}$$

The transistors Q_3 and Q_4 are act as a simple current mirror circuit, the collector current of Q_3 is,

$$\begin{aligned} I_{C3=4} &= I_{C4} \\ &= \alpha^2\frac{I}{2} \end{aligned}$$

Step 4 of 13

The base current of Q_3 is,

$$\begin{aligned} I_{B3} &= \frac{I_{E3}}{\beta+1} \\ &= \frac{I_{C3}}{\frac{\alpha}{\beta+1}} \qquad (\text{Since}, I_C = \alpha I_E) \\ &= \frac{\alpha^2\frac{I}{2}}{\frac{\alpha}{\beta+1}} \qquad \left(\text{Since}, I_{C3} = \alpha^2\frac{I}{2}\right) \\ &= \alpha\frac{I}{2}\frac{1}{\beta+1} \end{aligned}$$

The transistors Q_3 and Q_4 are act as a simple current mirror circuit, the collector current of Q_5 is,

$$\begin{aligned} I_{C5} &= I_{B3} \\ &= \alpha\frac{I}{2}\frac{1}{\beta+1} \end{aligned}$$

Step 5 of 13

The base current of Q_5 is,

$$\begin{aligned} I_{B5} &= \frac{I_{E5}}{\beta+1} \\ &= \frac{\frac{I}{2}}{\beta+1} \\ &= \frac{I}{2}\frac{1}{\beta+1} \end{aligned}$$

Step 6 of 13

Using Kirchhoff's current law to the above circuit, the current I_B becomes

$$\begin{aligned} I_B &= I_{B1} - I_{C5} \qquad (\text{Since}, I_{C5} + I_B = I_{B1}) \\ &= \frac{I}{2}\frac{1}{\beta+1} - \alpha\frac{I}{2}\frac{1}{\beta+1} \\ &= \frac{I}{2}\frac{1}{\beta+1}(1-\alpha) \\ &= \frac{I}{2}\frac{1}{\beta+1}\left(1-\frac{\beta}{\beta+1}\right) \qquad \left(\text{Since}, \alpha = \frac{\beta}{\beta+1}\right) \end{aligned}$$

Simpaify further.

$$\begin{aligned} I_B &= \frac{I}{2}\frac{1}{\beta+1}\left(\frac{\beta+1-\beta}{\beta+1}\right) \\ &= \frac{I}{2}\frac{1}{(\beta+1)^2} \end{aligned}$$

Therefore, the current I_B is $\boxed{I_B = \frac{I}{2}\frac{1}{(\beta+1)^2}}$.

Step 7 of 13

Without connections Q_3 and Q_4 the current I_B is,

$$\begin{aligned} I_B &= I_{B1} \\ &= \frac{I}{2}\frac{1}{\beta+1} \end{aligned}$$

Therefore, without connections Q_3 and Q_4 the current I_B is $\boxed{I_B = \frac{I}{2}\frac{1}{(\beta+1)}}$.

Hence, the current I_B is reduced by factor $\frac{1}{\beta+1}$.

Step 8 of 13

(b)

Draw the equivalent circuit for $v_i = 0\,\mathrm{V}\,(\mathrm{dc}) + v_{id}$.

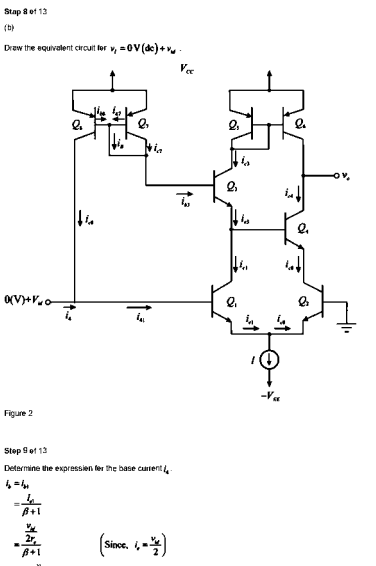


Figure 2

Step 9 of 13

Determine the expression for the base current i_b.

$$\begin{aligned} i_b &= i_{b1} \\ &= \frac{i_{e1}}{\beta+1} \\ &= \frac{\frac{v_{id}}{2r_e}}{\beta+1} \qquad \left(\text{Since}, \ i_e = \frac{v_{id}}{2}\right) \\ &= \frac{v_{id}}{2r_e(\beta+1)} \end{aligned}$$

Simpaify further.

$$\begin{aligned} i_b &= \frac{v_{id}}{2\left(\frac{V_T}{\frac{I}{2}}\right)(\beta+1)} \qquad \left(\text{Since}, r_e = \frac{V_T}{\frac{I}{2}}\right) \\ &= \frac{v_{id}I}{4V_T(\beta+1)} \end{aligned}$$

Therefore, the input signal current i_b is $\boxed{\frac{v_{id}I}{4V_T(\beta+1)}}$.

Step 10 of 13

The input differential resistance is,

$$R_{id} = \frac{v_{id}}{i_b}$$

Substitute $\frac{v_{id}I}{4V_T(\beta+1)}$ for i_b.

$$\begin{aligned} &\frac{v_{id}}{i_b} = \frac{4V_T(\beta+1)}{I} \\ &\Rightarrow R_{id} = \frac{v_{id}}{i_b} = \frac{4V_T(\beta+1)}{I} \end{aligned}$$

Therefore, the input differential resistance R_{id} is $\boxed{\frac{4V_T(\beta+1)}{I}}$.

Step 11 of 13

Determine the signal current i_b with connections Q_3 and Q_4.

Apply Kirchhoff's current law at node v_{id}.

$$\begin{aligned} &i_b + i_{c4} = i_{b1} \\ &i_b = i_{b1} - i_{c4} \end{aligned}$$

The transistors Q_3 and Q_4 are act as a simple current mirror circuit, then the collector current of Q_1 becomes

$$i_{c4} = i_{b3}$$

Hence, the signal current i_b becomes

$$\begin{aligned} i_b &= i_{b1} - i_{b3} \\ &= i_{b1} - (i_{c3} + i_B) \qquad (\text{Since}, i_{c3} + i_B = i_{b3}) \\ &= i_{b1} - i_{c3} - i_B \\ &= i_{b1} - i_{c3} - (i_{b3} + i_{b8}) \qquad (\text{Since}, i_{b7} + i_{b8} = i_B) \end{aligned}$$

Simpaify further.

$$\begin{aligned} i_b &= i_{b1} - i_{c3} - (2i_{b8}) \qquad (\text{Since}, i_{c7} = i_{b8} \text{ in simple current mirror circuit}) \\ &= i_{b1} - i_{c3} - 2i_{b8} \\ &= i_{b1} - \beta i_{b7} - 2i_{b8} \qquad (\text{Since}, i_c = \beta i_b) \\ &= i_{b1} - \beta i_{b8} - 2i_{b8} \end{aligned}$$

Simpaify further.

$$\begin{aligned} i_b &= i_{b1} - i_{b8}(\beta+2) \\ &= i_{b1} - \frac{i_{E8}}{\beta}(\beta+2) \\ &= i_{b1} - \frac{i_{b3}}{\beta}(\beta+2) \qquad (\text{Since}, i_{c8} = i_{b3}) \\ &= i_{b1} - \frac{i_{b1}}{\beta}(\beta+2) \qquad (\text{Since, ignoring the effect of } r_o) \end{aligned}$$

Simpaify further.

$$\begin{aligned} i_b &= i_{b1}\left(\frac{\beta-\beta+2}{\beta}\right) \\ &= i_{b1}\left(\frac{2}{\beta}\right) \\ &= \frac{i_{e1}}{\beta+1}\left(\frac{2}{\beta}\right) \\ &= \frac{v_{id}}{2r_e(\beta+1)}\left(\frac{2}{\beta}\right) \end{aligned}$$

Step 12 of 13

Simpaify further.

$$\begin{aligned} i_b &= \frac{v_{id}}{2r_e(\beta+1)}\left(\frac{2}{\beta}\right) \\ &= \frac{v_{id}}{2\left(\frac{V_T}{\frac{I}{2}}\right)(\beta+1)}\left(\frac{2}{\beta}\right) \\ &= \frac{v_{id}I}{4V_T(\beta+1)}\left(\frac{2}{\beta}\right) \end{aligned}$$

Therefore, the input signal current i_b is $\boxed{\frac{v_{id}I}{4V_T(\beta+1)}\left(\frac{2}{\beta}\right)}$.

Step 13 of 13

Determine the input differential resistance R_{id}.

$$R_{id} = \frac{v_{id}}{i_b}$$

Substitute $\frac{v_{id}I}{4V_T(\beta+1)}\left(\frac{2}{\beta}\right)$ for i_b.

$$\begin{aligned} &\frac{v_{id}}{i_b} = \frac{4V_T(\beta+1)}{I}\left(\frac{\beta}{2}\right) \\ &R_{id} = \frac{4V_T(\beta+1)}{I}\left(\frac{\beta}{2}\right) \end{aligned}$$

Therefore, the input differential resistance R_{id} with connections Q_3 and Q_4 is

$$\boxed{\frac{4V_T(\beta+1)}{I}\left(\frac{\beta}{2}\right)}$$

9.101P

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9.108P

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9.109P

We don't have the solution to this problem yet.

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9.110P

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9.111P

We don't have the solution to this problem yet.

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9.112P

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Step 1 of 12

9.113P

Refer to Figure 8.41 in the text book.

From the circuit shown in Figure 8.41. it is clear that the reference current is,

$$\begin{aligned} I &= I_{REF} \\ &= 225\ \mu A \end{aligned}$$

The reference current flowing through Q_8 is mirrored in Q_5 and Q_7.

The bias current flowing in Q_6 is equal to the current flowing in Q_7 that is,

$$I_{REF} = 225\ \mu A$$

Hence the bias currents flowing in Q_5, Q_6, Q_7 and Q_8 are equal that is

I_5, I_6, I_7 **and** I_8 **are** $\boxed{225\ \mu A}$.

Step 2 of 12

The bias current flowing through Q_5 is then divided equally in Q_1, Q_2, Q_3 and Q_4, that is

$$\frac{I}{2} = \frac{I_{REF}}{2}$$

Substitute $225\ \mu A$ **for** I_{REF} in equation.

$$\begin{aligned} \frac{I}{2} &= \frac{225\ \mu A}{2} \\ &= 112.5\ \mu A \end{aligned}$$

Hence the bias currents flowing in Q_1, Q_2, Q_3 and Q_4 is

I_5, I_6, I_7 **and** I_8 **are** $\boxed{112.5\ \mu A}$.

Step 3 of 12

Now for the Operational amplifier to not have a systemic offset voltage, we know that the ratio of W/L of the transistors must be related by the expression

$$\frac{\left(\frac{W}{L}\right)_6}{\left(\frac{W}{L}\right)_4} = \frac{2\left(\frac{W}{L}\right)_7}{\left(\frac{W}{L}\right)_5}$$

Substitute $10/0.5$ for $\left(\frac{W}{L}\right)_4$, $60/0.5$ for $\left(\frac{W}{L}\right)_7$, $60/0.5$ for $\left(\frac{W}{L}\right)_5$ and $W/0.5$ for $\left(\frac{W}{L}\right)_6$ in equation.

Substituting the values of the W/L of various devices from the table above, we get

$$\frac{\left(\frac{W}{0.5}\right)}{\left(\frac{10}{0.5}\right)} = \frac{2\left(\frac{60}{0.5}\right)}{\left(\frac{60}{0.5}\right)}$$

$$\frac{W}{0.5} = 2\left(\frac{10}{0.5}\right)$$

Hence the W/L of the transistor Q_6 is $\boxed{\frac{20}{0.5}}$.

The bias current is related to the over drive voltage by the expression

$$I_D = \frac{1}{2}\mu C_{ox}\left(\frac{W}{L}\right)V_{OV}^2$$

Where, V_{OV} is the over drive voltage of a transistor, which has a certain μC_{ox} according to the material used for preparing the device.

Substitute $\left(\frac{30}{0.5}\right)$ of W/L, $112.5\ \mu A$ for I_D **and** 60 **for** μC_{ox} in equation.

$$\begin{aligned} 112.5 &= \frac{1}{2}(60)\left(\frac{30}{0.5}\right)V_{OV_1}^2 \\ V_{OV_1}^2 &= 0.0625\ V \\ V_{OV_1} &= 0.25\ V \end{aligned}$$

Hence, the over drive voltage for the device Q_1 is $\boxed{0.25\ V}$.

Similarly, the values of V_{OV_2}, V_{OV_3}, V_{OV_4}, V_{OV_5}, V_{OV_6}, V_{OV_7} **and** V_{OV_8} **are** $\boxed{0.25\ V}$.

The over drive voltage V_{OV} is related to the gate source voltage and the thermal voltage by the expression

$$|V_{GS}| = |V_{OV}| + |V_t|,$$

where, V_{GS} is the gate source voltage, and V_t is the thermal voltage of the device.

For the Q_1, the gate source voltage is,

$$\begin{aligned} |V_{GS_1}| &= |V_{OV_1}| + |V_t| \\ &= 0.25 + 0.75 \\ &= 1\ V \end{aligned}$$

The gate source $|V_{GS_1}|$ **is** $\boxed{1\ V}$.

Step 4 of 12

As the overdrive voltages of all the devices are equal the gate source voltages will also be equal, that is,

V_{GS_2}, V_{GS_3}, V_{GS_4}, V_{GS_5}, V_{GS_6}, V_{GS_7} **and** V_{GS_8} **is** $\boxed{1\ V}$.

The transconductance g_m of a device is given by the expression,

$$g_m = \frac{2I_D}{|V_{OV}|}$$

Step 5 of 12

The transconductance of Q_1 is,

$$\begin{aligned} g_{m_1} &= \frac{2I_1}{|V_{OV_1}|} \\ &= \frac{2(112.5\ \mu)}{0.25} \\ &= 4(225\ \mu) \\ &= 0.9\ mA/V \end{aligned}$$

Hence, the transconductance of the device Q_1 is $\boxed{0.9\ mA/V}$.

As the currents of Q_1, Q_2, Q_3 and Q_4 are equal, their transconductance will be the same,

g_{m_2}, g_{m_3} **and** g_{m_4} **are** $\boxed{0.9\ mA/V}$.

Step 6 of 12

The transconductance of Q_5 is,

$$\begin{aligned} g_{m_5} &= \frac{2I_5}{|V_{OV_5}|} \\ &= \frac{2(225\ \mu)}{0.25} \\ &= 8(225\ \mu) \\ &= 1.8\ mA/V \end{aligned}$$

Hence, the transconductance of the device Q_5 is $\boxed{1.8\ mA/V}$.

As the currents of Q_5, Q_6, Q_7 and Q_8 are equal, their transconductance will be the same,

g_{m_6}, g_{m_7} **and** g_{m_8} **are** $\boxed{1.8\ mA/V}$.

Step 7 of 12

The output resistance (r_O) of a device is given by the expression,

$$r_O = \frac{|V_A|}{I_D}$$

For the transistor Q_1 the output resistance is,

$$\begin{aligned} r_{O_1} &= \frac{|V_A|}{I_1} \\ &= \frac{9}{112.5\ \mu} \\ &= \frac{9000,000}{112.5} \\ &= 80\ k\Omega \end{aligned}$$

Hence the output resistance is $\boxed{80\ k\Omega}$.

As the currents of Q_1, Q_2, Q_3 and Q_4 are equal, their output resistances will be equal, that is

r_{O_2}, r_{O_3} **and** r_{O_4} **are** $\boxed{80\ k\Omega}$.

Step 8 of 12

For the transistor Q_5 the output resistance is,

$$\begin{aligned} r_{O_5} &= \frac{|V_A|}{I_5} \\ &= \frac{9}{225\ \mu} \\ &= \frac{9000,000}{225} \\ &= 40\ k\Omega \end{aligned}$$

Hence the output resistance is $\boxed{40\ k\Omega}$.

As the currents of Q_5, Q_6, Q_7 and Q_8 are equal, their output resistances will be equal, that is

r_{O_6}, r_{O_7} **and** r_{O_8} **are** $\boxed{40\ k\Omega}$.

Step 9 of 12

The gain of the first stage of the two stage operational amplifier is,

$$\begin{aligned} A_1 &= -g_{m_1}(r_{O_1} \| r_{O_2}) \\ &= -\left(0.9\ \frac{mA}{V}\right)\left(\left(\frac{(80)(80)}{80+80}\right)k\Omega\right) \\ &= (0.9)(40) \\ &= -36\ V/V \end{aligned}$$

Similarly, the gain of the second stage of the two stage operational amplifier is,

$$\begin{aligned} A_2 &= -g_{m_5}(r_{O_5} \| r_{O_7}) \\ &= -\left(1.8\ \frac{mA}{V}\right)\left(\left(\frac{(40)(40)}{40+40}\right)k\Omega\right) \\ &= (1.8)(20) \\ &= -36\ V/V \end{aligned}$$

The overall open loop DC gain is,

$$\begin{aligned} A_0 &= A_1 A_2 \\ &= (-36)(-36) \\ &= 1296\ V/V \end{aligned}$$

Hence the open loop DC gain of the device is $\boxed{1296\ V/V}$.

Step 10 of 12

The lower limit of the input common mode is the value of the input voltage at which the transistors Q_1 and Q_2 leave the saturation region, this happens when the output at the drain falls by one volt below the $|V_{tp}|$ of the device.
The drain voltage of the Q_1 is,

$$\begin{aligned} V_{D_1} &= -V_{DD} + V_{GS_1} \\ &= -1.5 + 1 \\ &= -0.5\ V \end{aligned}$$

Hence the lower limit of the input common mode range is,

$$\begin{aligned} V_{in_{CM_{MIN}}} &= -0.5 - |V_{tp}| \\ &= -0.5 - 0.75\ ^S \\ &= -1.25\ V \end{aligned}$$

The lower limit of the input common mode range is $\boxed{-1.25\ V}$.

Step 11 of 12

The upper limit of the input common mode is the value of the input voltage at which the transistor Q_3 leaves the saturation region. Since for Q_3 to operate in saturation, the voltage across it must at least be equal to the over drive voltage at which it is operating, that is,

$$\begin{aligned} V_{in_{CM_{MAX}}} &= 1.25 - 1 \\ &= +0.25\ V \end{aligned}$$

Hence the input range of the operational amplifier is $\boxed{-1.25\ V \text{ to } 0.25\ V}$.

Step 12 of 12

The highest allowable output voltage is the value at which Q_7 leaves saturation, that is

$$\begin{aligned} V_{O_{CM_{MAX}}} &= V_{DD} - V_{OV_7} \\ &= 1.5 - 0.25 \\ &= 1.25\ V \end{aligned}$$

The lowest value of the output voltage is the value at which Q_6 leaves saturation, that is

$$\begin{aligned} V_{O_{CM_{MIN}}} &= V_{SS} + V_{OV_6} \\ &= -1.5 + 0.25 \\ &= -1.25\ V \end{aligned}$$

Hence, the output voltage range is,

Hence the output range of the operational amplifier is $\boxed{-1.25\ V \text{ to } 1.25\ V}$.

9.114P

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Step 1 of 5

9.115P

Refer to Figure 8.41 in the text book.

(a)

The change in overdrive voltage can be obtained by using drain current

$$I_D = \frac{1}{2}k_n'\left(\frac{W}{L}\right)V_{DV}^2$$

$$2I_D = k_n'\left(\frac{W}{L}\right)V_{OV}^2$$

$$\frac{2I_D}{k_n'\left(\frac{W}{L}\right)} = V_{OV}^2$$

$$\frac{2I_D}{K} = V_{OV}^2 \qquad \left(\text{since } K = k_n'\left(\frac{W}{L}\right)\right)$$

$$\Rightarrow V_{OV} = \sqrt{\frac{2I_D}{K}}$$

If the factor $K = 4$, then the overdrive voltage becomes

$$V_{OV} = \sqrt{\frac{2I_D}{4}}$$
$$= \sqrt{\frac{I_D}{2}}$$

Therefore, if *K* is increases by a factor of 4, then the overdrive voltage is decreases by a factor of $\frac{1}{2}$.

Step 2 of 5

The trans conductance is,

$$g_m = \frac{2I_D}{V_{DV}}$$

Substitute $V_{OV} = \sqrt{\frac{2I_D}{K}}$ in the above equation

$$g_m = \frac{2I_D}{\sqrt{\frac{2I_D}{K}}}$$
$$= \frac{\sqrt{2I_D}\sqrt{2I_D}}{\frac{\sqrt{2I_D}}{\sqrt{K}}}$$
$$= \frac{\sqrt{2I_D}}{\frac{1}{\sqrt{K}}}$$
$$= \sqrt{2I_D}\sqrt{K}$$

Therefore, the value of g_m is $\boxed{\sqrt{2I_D}\sqrt{K}}$.

Step 3 of 5

If the factor $K = 4$, then the trans conductance becomes

$$g_m = \sqrt{2I_D}\sqrt{4}$$
$$= 2\sqrt{2I_D}$$

Therefore, if *K* is increases by a factor of 4, then the trans conductance is increases by a multiplier of 2.

Step 4 of 5

(b)

The voltage gain of the first stage is ,

$$A_1 = -g_{m1}\left(r_{o2} \| r_{o4}\right)$$
$$= -g_{m1}\left(\frac{V_A}{I_D} \| \frac{V_A}{I_D}\right)$$
$$= -g_{m1}\left(\frac{V_A}{2I_D}\right)$$

Substitute $\sqrt{2I_D}\sqrt{K}$ for g_m in the equation.

$$A_1 = -\left(\sqrt{2I_D}\sqrt{K}\right)\left(\frac{V_A}{2I_D}\right)$$
$$= -\left(\sqrt{K}\right)\left(\frac{V_A}{\sqrt{2I_D}}\right)$$

Substitute **4 for** K in equation.

$$A_1 = -\left(\sqrt{4}\right)\left(\frac{V_A}{\sqrt{2I_D}}\right)$$
$$= -2\left(\frac{V_A}{\sqrt{2I_D}}\right)$$

Therefore, if *K* is increases by a factor of 4, then the voltage gain of the first stage is increases by a multiplier of 2.

Similarly the overall voltage is also increases by a multiplier of 2.

Step 5 of 5

(c)

The offset voltage for mismatch threshold voltage is, $V_{os} = \Delta V_t$

The offset voltage for mismatch drain resistance is, $V_{os} = \left(\frac{V_{OV}}{2}\right)\left(\frac{\Delta R_D}{R_D}\right)$

The offset voltage for mismatch width-to-length ratio is , $V_{os} = \left(\frac{V_{DV}}{2}\right)\left(\frac{\Delta\left(\frac{W}{L}\right)}{\left(\frac{W}{L}\right)}\right)$

The offset voltage for mismatch threshold voltage is unchanged remaining all are changed (reduced) by a factor of $\frac{1}{2}$.

9.116P

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Step 1 of 3

9.117P

Refer to Figure 8.41 in the text book.

Redraw the circuit diagram as shown in Figure 1.

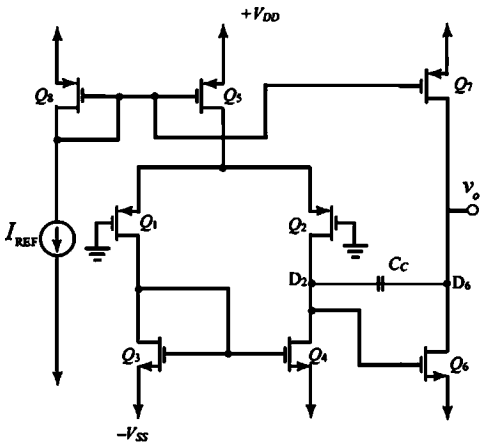


Figure 1

Step 2 of 3

Form the circuit shown in Figure1, the offset current is,

$$I_O = I_{D3} - I_{D4}$$

$$= \frac{1}{2}k_n'\left(\frac{W}{L}\right)(V_{GS} - V_t)^2 - \frac{1}{2}k_n'\left(\frac{W}{L}\right)(V_{GS} - (V_t + \Delta V_t))^2$$

$$= \frac{1}{2}k_n'\left(\frac{W}{L}\right)\left[(V_{GS} - V_t)^2 - (V_{GS} - (V_t + \Delta V_t))^2\right]$$

$$= \frac{1}{2}k_n'\left(\frac{W}{L}\right)\left[V_{GS}^2 + V_t^2 - 2V_{GS}V_t - V_{GS}^2 - (V_t + \Delta V_t)^2 + 2V_{GS}(V_t + \Delta V_t)\right]$$

$$= \frac{1}{2}k_n'\left(\frac{W}{L}\right)\left[V_t^2 - (V_t + \Delta V_t)^2 + 2V_{GS}\Delta V_t\right]$$

$$= \frac{1}{2}k_n'\left(\frac{W}{L}\right)\left[V_t^2 - (V_t^2 + \Delta V_t^2 + 2V_t\Delta V_t) + 2V_{GS}\Delta V_t\right]$$

$$= \frac{1}{2}k_n'\left(\frac{W}{L}\right)\left[-\Delta V_t^2 - 2V_t\Delta V_t + 2V_{GS}\Delta V_t\right]$$

$$= \frac{1}{2}k_n'\left(\frac{W}{L}\right)\left[-\Delta V_t(\Delta V_t + 2V_t - 2V_{GS})\right]$$

$$= \frac{1}{2}k_n'\left(\frac{W}{L}\right)\Delta V_t\left[(2V_{GS} - 2V_t - \Delta V_t)\right]$$

$$\simeq \frac{1}{2}k_n'\left(\frac{W}{L}\right)\Delta V_t\left[(2V_{GS} - 2V_t)\right]$$

$$\simeq k_n'\left(\frac{W}{L}\right)[V_{GS} - V_t]\Delta V_t$$

$$\simeq k_n'\left(\frac{W}{L}\right)V_{OV}\,\Delta V_t$$

$$= g_m \Delta V_t \qquad \left(\text{Since, } g_m = \frac{2I_D}{V_{OV}} = \frac{2\left(\frac{1}{2}\right)k_n'\left(\frac{W}{L}\right)V_{OV}^2}{V_{OV}} = k_n'\left(\frac{W}{L}\right)V_{OV}\right)$$

$$I_O = g_m \Delta V_t$$

Therefore, the output current at $Q_3 - Q_4$ is $\boxed{g_m \Delta V_t}$.

Step 3 of 3

The offset voltage is,

$$I_O = g_{m1}V_{OS}$$

$$V_{OS} = \frac{I_O}{g_{m1}}$$

Substitute $I_O = g_m \Delta V_t$ in equation.

$$V_{OS} = \frac{g_m \Delta V_t}{g_{m1}}$$

Therefore, the offset voltage is $\boxed{\frac{g_m}{g_{m1}}\Delta V_t}$.

9.118P

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9.119P

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9.120P

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Step 1 of 4

The Input resistance of the first stage is,

$$R_{id_1} = (\beta+1)(R_{E_1} + r_{e_1})2$$

Substitute 100 for β , $100\ \Omega$ for r_{e1} in equation.

$$R_{id_1} = (101)(100+100)2$$
$$= 40.4\ \text{k}\Omega$$

The input resistance of the second stage R_{id_1} is $\boxed{40.4\ \text{k}\Omega}$.

Hence the internal resistance of the first stage has increased.

Step 2 of 4

Now in the second stage the input resistance is,

$$R_{id_2} = (\beta+1)(2)(R_{E_2} + r_{e_2})$$

Substitute 100 for β , $25\ \Omega$ for r_{e2} in equation.

$$R_{id_2} = (101)(25+25)2$$
$$= 10.0\ \text{k}\Omega$$

Hence the voltage gain in the first stage is,

$$A_1 = \frac{\text{Total resistance in the collector circuit}}{\text{Total resistance in the emitter circuit}}$$
$$= \frac{2(R_{C_1}) \| R_{id_2}}{2(R_{E_1} + r_{e_1})}$$
$$= 2\frac{(20\ \text{k}) \| (10\ \text{k})}{2(100+100)}$$
$$= 20.2\ \text{V}$$

Hence the voltage gain in the first stage is $\boxed{20.2\ \text{V}}$.

Therefore, the voltage gain in the first stage has decreased.

Step 3 of 4

The voltage gain of the second stage of the circuit is,

$$A_2 = \frac{-R_3 \| R_{i3}}{2(R_{E_2} + r_{e_2})}$$
$$= -\frac{(3\ \text{k}) \| (234.8\ \text{k})}{2(25+25)}$$
$$= -29.6\ \text{V/V}$$

Step 4 of 4

Now, the circuit in the third stage and fourth stage are not affected by the changes made in the first stage, hence we notice that A_3 and A_4 , have not changed.

Hence the over all gain of the circuit is

$$A = A_1 A_2 A_3 A_4$$
$$= (20.2)(-29.2)(-6.42)(1)$$
$$= 3823\ \text{V/V}$$

Hence the over all gain is $\boxed{A = 3843\ \text{V/V}}$.

Therefore, the overall gain, has decreased.

Step 1 of 4

Refer to Figure 8.43 in the textbook for a four-stage bipolar op-amp.

Consider the following expression for the output resistance:

$$R_o = R_6 \| \left[r_{e8} + \frac{R_5}{\beta + 1} \right]$$

The term, $\frac{R_5}{\beta+1}$ value depends on the value of current gain, β. If the value of β changes the value of this term, the value of the term with series combination of r_{e8} changes and hence the entire value of output resistance changes.

Therefore, the value of resistor, $\boxed{R_5}$ effects the output resistance.

Step 2 of 4

Consider that the output resistance reduced by a factor 2.

Consider that the output resistance, $R_o = 76\ \Omega$. Therefore,

$$R_6 \| R_o' = 76\ \Omega$$
$$\frac{R_6 R_o'}{R_6 + R_o'} = 76\ \Omega$$

Here,

$$R_o' = r_{e8} + \frac{R_5}{\beta + 1}$$

Substitute $3\ \text{k}\Omega$ for R_6.

$$\frac{(R_o')(3000)}{R_o' + 3000} = 76$$
$$3000R_o' = 76R_o' + 228000$$
$$2924R_o' = 228000$$

$$R_o' = \frac{228000}{2924}$$
$$= 77.975\ \Omega$$

The resistance, R_5 is,

$$R_5 = (\beta + 1)(R_o' - r_{e8})$$

Substitute 100 for the current gain, β, $77.97\ \Omega$ for R_o', and $5\ \Omega$ for r_{e8}.

$$R_5 = (77.97\ \Omega - 5\ \Omega)(100 + 1)$$
$$= (72.975\ \Omega)(101)$$
$$= 7.37\ \text{k}\Omega$$

Therefore, the new value of resistor, R_5 is, $\boxed{7.37\ \text{k}\Omega}$.

Step 3 of 4

Calculate the change in gain at third stage of the amplifier due to the change in resistance.

$$A_3 = -\frac{(R_5 \| R_{14})}{r_{e7} + R_4}$$
$$= -\frac{R_5 R_{14}}{(R_5 + R_{14})(r_{e7} + R_4)}$$

Substitute $7.37\ \text{k}\Omega$ for R_5, $303.5\ \text{k}\Omega$ for R_{14}, $25\ \Omega$ for r_{e7}, and $2300\ \Omega$ for R_4.

$$A_{3a} = -\frac{(7.37\ \text{k}\Omega)(303.5\ \text{k}\Omega)}{(7.37\ \text{k}\Omega + 303.5\ \text{k}\Omega)(25\ \Omega + 2300\ \Omega)}$$
$$= -\frac{2236.795\ \text{M}\Omega}{(310.87\ \text{k}\Omega)(2.325\ \text{k}\Omega)}$$
$$= -\frac{2236.795}{722.77}$$
$$= -3.094\ \text{V/V}$$

Calculate the third stage gain with $R_5 = 15.7\ \text{k}\Omega$.

Substitute $15.7\ \text{k}\Omega$ for R_5, $303.5\ \text{k}\Omega$ for R_{14}, $25\ \Omega$ for r_{e7}, and $2300\ \Omega$ for R_4.

$$A_{3o} = -\frac{(15.7\ \text{k}\Omega)(303.5\ \text{k}\Omega)}{(15.7\ \text{k}\Omega + 303.5\ \text{k}\Omega)(25\ \Omega + 2300\ \Omega)}$$
$$= -\frac{4764.95\ \text{M}\Omega}{(319.2\ \text{k}\Omega)(2.325\ \text{k}\Omega)}$$
$$= -\frac{4764.95}{742.14}$$
$$= -6.42\ \text{V/V}$$

Step 4 of 4

Calculate the value of overall gain.

$$A = A_{old}\left(\frac{A_{3a}}{A_{3o}}\right)$$

Substitute 8513 V/V for A_{old}, $-3.094\ \text{V/V}$ for A_{3a}, and $-6.42\ \text{V/V}$ for A_{3o}.

$$A = (8513\ \text{V/V})\left(\frac{-3.094\ \text{V/V}}{-6.42\ \text{V/V}}\right)$$
$$= (8513\ \text{V/V})(0.482)$$
$$= 4103.3\ \text{V/V}$$

Thus, the value of overall gain is $\boxed{4103.3\ \text{V/V}}$.

The factor of gain change is,

$$\Delta A = \frac{8513}{4103.3}$$
$$= 2.075$$

Therefore, the gain reduced by factor of $\boxed{2.075}$ for the change in output resistance by a factor of 2.

To maintain the amplifier as same value, reduce the resistor, R_4 by a factor 2.075.

Thus, the resistor R_4 value is,

$$R_4 = \frac{2.3}{2.07}$$
$$= 1.11\ \text{k}\Omega$$

Therefore, by reducing R_4 value to $\boxed{1.11\ \text{k}\Omega}$, it is possible to maintain the gain at 8513.

9.123P

Refer to Figure 8.43 in the textbook for a four-stage bipolar op-amp.

Replace the resistor, R_5 with a constant current source of 1 mA.

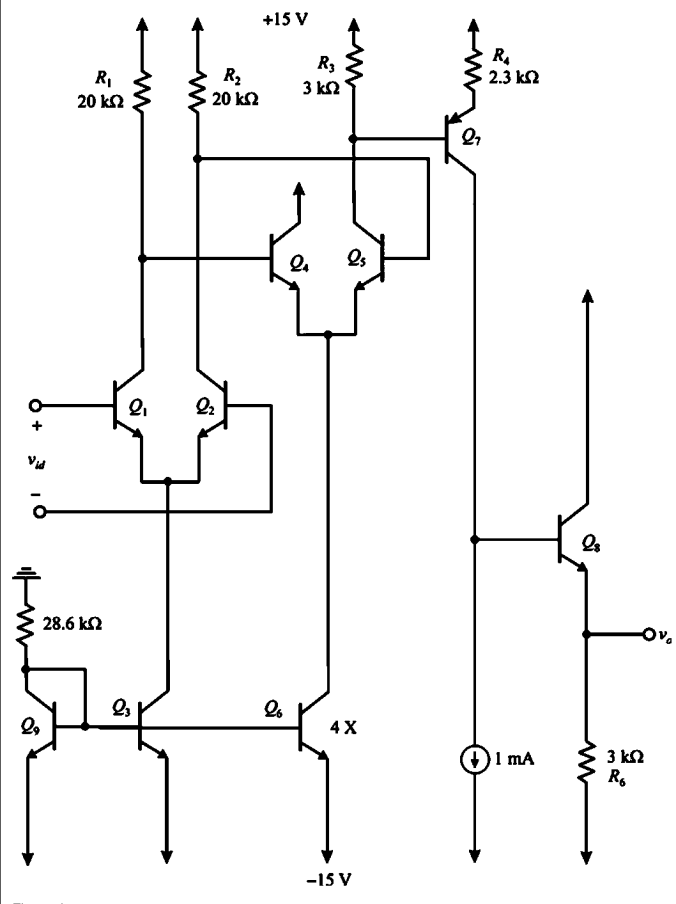


Figure 1

(a)

Calculate the third stage gain with $R_5 = 15.7\ \text{k}\Omega$.

$$A_3 = -\frac{(R_5 \parallel R_{i4})}{r_{e7} + R_4}$$
$$= -\frac{R_5 R_{i4}}{(R_5 + R_{i4})(r_{e7} + R_4)}$$

Substitute $15.7\ \text{k}\Omega$ for R_5, $303.5\ \text{k}\Omega$ for R_{i4}, $25\ \Omega$ for r_{e7}, and $2300\ \Omega$ for R_4.

$$A_{3o} = -\frac{(15.7\ \text{k}\Omega)(303.5\ \text{k}\Omega)}{(15.7\ \text{k}\Omega + 303.5\ \text{k}\Omega)(25\ \Omega + 2300\ \Omega)}$$
$$= -\frac{4764.95\ \text{M}\Omega}{(319.2\ \text{k}\Omega)(2.325\ \text{k}\Omega)}$$
$$= -\frac{4764.95}{742.14}$$
$$= -6.42\ \text{V/V}$$

Consider that the resistance of the current source is very high.

Calculate the new value of the gain at third stage of the amplifier.

$$A_{3o} = -\frac{(R_5 \parallel R_{i4})}{r_{e7} + R_4}$$
$$= -\frac{R_{i4}}{r_{e7} + R_4} \quad (\text{since, } R_5 \parallel R_{i4} \simeq R_{i4})$$

Substitute $303.5\ \text{k}\Omega$ for R_{i4}, $25\ \Omega$ for r_{e7}, and $2300\ \Omega$ for R_4.

$$A_{3o} = -\frac{303.5\ \text{k}\Omega}{25\ \Omega + 2300\ \Omega}$$
$$= -\frac{303.5}{2.325}$$
$$= -130.5\ \text{V/V}$$

Calculate the change in the overall gain.

$$A = A_{old}\left(\frac{A_{3o}}{A_{3o}}\right)$$

Substitute 8513 V/V for A_{old}, $-130.5\ \text{V/V}$ for A_{3o}, and $-6.42\ \text{V/V}$ for A_{3o}.

$$A = (8513\ \text{V/V})\left(\frac{-130.5\ \text{V/V}}{-6.42\ \text{V/V}}\right)$$
$$= (8513\ \text{V/V})(20.33)$$
$$= 173079.5\ \text{V/V}$$

Therefore, the overall gain, A is, $\boxed{173079.5\ \text{V/V}}$.

(b)

Consider that the load resistance, $R_L = 100\ \Omega$.

Calculate the amplifier the gain.

$$A_L = A\frac{R_L}{R_o + R_L}$$

Here, the output resistance, $R_o = 152\ \Omega$.

Substitute $152\ \Omega$ for R_o, $100\ \Omega$ for R_L, and 8513 V/V for A.

$$A_L = (8513\ \text{V/V})\left(\frac{100\ \Omega}{152\ \Omega + 100\ \Omega}\right)$$
$$= (8513\ \text{V/V})(0.396)$$
$$= 3378.17\ \text{V/V}$$

Therefore, the original amplifier gain when loaded with $100\ \Omega$ is, $\boxed{3378.17\ \text{V/V}}$.

The output resistance is,

$$R_o = R_6 \parallel \left[r_{e8} + \frac{R_5}{\beta + 1}\right]$$
$$= R_6 \quad (\text{since, the resistance, } R_5 \text{ of current source is very high})$$
$$= 3\ \text{k}\Omega$$

Therefore, calculate the new value of the gain.

Substitute $3000\ \Omega$ for R_o, $100\ \Omega$ for R_L, and 173079.5 V/V for A.

$$A_L = (173079.5\ \text{V/V})\left(\frac{100\ \Omega}{3000\ \Omega + 100\ \Omega}\right)$$
$$= (173079.5\ \text{V/V})(0.0322)$$
$$= 5583.2\ \text{V/V}$$

Therefore, the gain when the output resistance modified with $100\ \Omega$ load is, $\boxed{5583.2\ \text{V/V}}$.

9.124P

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9.125P

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Step 1 of 11

(a)

Refer to Figure P8.119 in the textbook for a metal oxide semi-conductor (MOS) differential amplifier.

The bias current, I flows through two equal branches, one of which consists transistor, Q_1 and a current mirror, $Q_3 - Q_5$, and another one consists transistor, Q_2 and a current mirror, $Q_4 - Q_6$. The current equally divides into the two branches.

Consider that the current mirror, $Q_3 - Q_5$ has a transmission factor of 1, therefore, the same one-half of bias current passes through the current mirror and transistor, Q_7.

Consider that the current mirror, $Q_4 - Q_6$ has a transmission factor of 2, therefore, the other half of bias current does not pass through the current mirror. The current doubles and flows through transistors, Q_8 **and** Q_6.

Consider that all the transistors have same overdrive voltage, V_{OV}. Use this value to find the transconductance of each transistor.

$$g_m = \frac{2I_D}{|V_{OV}|}$$

Consider that the early voltage is V_A. Use this value to find the small-signal output resistance, r_o of each transistor.

$$r_o = \frac{|V_A|}{I_D}$$

Step 2 of 11

Tabulate the results of drain current, transconductance, and small signal output resistance for each of the transistor.

Table 1

Transistor	Drain current, I_D	Trans-conductance, g_m	Small-signal output resistance, r_o
Q_1	$\frac{I}{2}$	$\frac{I}{\lvert V_{OV}\rvert}$	$\frac{2\lvert V_A\rvert}{I}$
Q_2	$\frac{I}{2}$	$\frac{I}{\lvert V_{OV}\rvert}$	$\frac{2\lvert V_A\rvert}{I}$
Q_3	$\frac{I}{2}$	$\frac{I}{\lvert V_{OV}\rvert}$	$\frac{2\lvert V_A\rvert}{I}$
Q_4	$\frac{I}{2}$	$\frac{I}{\lvert V_{OV}\rvert}$	$\frac{2\lvert V_A\rvert}{I}$
Q_5	$\frac{I}{2}$	$\frac{I}{\lvert V_{OV}\rvert}$	$\frac{2\lvert V_A\rvert}{I}$
Q_6	I	$\frac{2I}{\lvert V_{OV}\rvert}$	$\frac{\lvert V_A\rvert}{I}$
Q_7	$\frac{I}{2}$	$\frac{I}{\lvert V_{OV}\rvert}$	$\frac{2\lvert V_A\rvert}{I}$
Q_8	I	$\frac{2I}{\lvert V_{OV}\rvert}$	$\frac{\lvert V_A\rvert}{I}$

Step 3 of 11

(b)

Find the differential gain of the amplifier.

Apply two voltages to the two input transistors, Q_1 **and** Q_2. The input differential voltage is v_{id}. Consider that the two input voltages are, $-\frac{v_{id}}{2}$ **and** $\frac{v_{id}}{2}$.

Find the output voltage at transistor, Q_1 in one branch.

$$v_{o1} = g_{m1}\left(r_{o1} \| r_{o3} \| \frac{1}{g_{m3}}\right)\left(\frac{v_{id}}{2}\right)$$

Here, $\frac{1}{g_{m3}} \ll (r_{o1} \| r_{o7})$, therefore, simplify the function.

$$v_{o1} = \left(\frac{g_{m1}}{g_{m3}}\right)\left(\frac{v_{id}}{2}\right)$$

Find the current through the transistor, Q_3.

$$\begin{aligned} i_{o3} &= -v_{o3}g_{m3} \\ &= -\left(\frac{g_{m1}}{g_{m3}}\right)\left(\frac{v_{id}}{2}\right)g_{m3} \\ &= \left(\frac{-g_{m1}g_{m3}}{g_{m3}}\right)\left(\frac{v_{id}}{2}\right) \end{aligned}$$

Step 4 of 11

The current, i_{o3} passes through the gates of transistors, Q_3 **and** Q_4. Find the gate voltages.

$$\begin{aligned} v_{g8} &= v_{g7} \\ &= i_{o3}\left(r_{o5} \| r_{o7} \| \frac{1}{g_{m7}}\right) \\ &= \left(\frac{-g_{m1}g_{m3}}{g_{m3}g_{m7}}\right)\left(\frac{v_{id}}{2}\right) \quad \left[\text{since, } \frac{1}{g_{m3}} \ll (r_{o3} \| r_{o7})\right] \\ &= \left(\frac{-g_{m1}}{g_{m7}}\right)\left(\frac{v_{id}}{2}\right) \quad [\text{since, } g_{m3} = g_{m5}] \end{aligned}$$

Find the current through the transistor, Q_8.

$$\begin{aligned} i_{o8} &= -v_{g8}g_{m8} \\ &= -\left(\frac{-g_{m1}}{g_{m7}}\right)\left(\frac{v_{id}}{2}\right)g_{m8} \\ &= \left(\frac{g_{m1}}{g_{m7}}\right)\left(\frac{v_{id}}{2}\right)2g_{m7} \quad (\text{since, } g_{m8} = 2g_{m7}) \\ &= g_{m1}v_{id} \end{aligned}$$

Step 5 of 11

Find the output voltage at transistor, Q_4 in other branch.

$$v_{o4} = -g_{m2}\left(r_{o3} \| r_{o4} \| \frac{1}{g_{m4}}\right)\left(\frac{v_{id}}{2}\right)$$

Here, $\frac{1}{g_{m4}} \ll (r_{o2} \| r_{o4})$, therefore, simplify the function.

$$v_{o4} = -\left(\frac{g_{m1}}{g_{m4}}\right)\left(\frac{v_{id}}{2}\right)$$

Find the current through the transistor, Q_4.

$$\begin{aligned} i_{o4} &= -v_{o6}g_{m6} \\ &= -v_{o4}(2g_{m4}) \quad (\text{since, } v_{o5} = v_{o4} \text{ and } g_{m6} = 2g_{m4}) \\ &= -\left(\frac{-g_{m1}}{g_{m4}}\right)\left(\frac{v_{id}}{2}\right)2g_{m4} \\ &= g_{m1}v_{id} \end{aligned}$$

Step 6 of 11

Find the total current, i_o.

$$\begin{aligned} i_o &= i_{o4} + i_{o8} \\ &= g_{m1}v_{id} + g_{m1}v_{id} \\ &= 2g_{m1}v_{id} \end{aligned}$$

Find the differential output voltage, v_o.

$$\begin{aligned} v_o &= i_o R_o \\ &= (2g_{m1}v_{id})(r_{o4} \| r_{o8}) \\ &= 2g_{m1}(r_{o4} \| r_{o8})v_{id} \end{aligned}$$

Find the differential gain, A_d.

$$\begin{aligned} A_d &= \frac{v_o}{v_{id}} \\ &= \frac{2g_{m1}(r_{o4} \| r_{o8})v_{id}}{v_{id}} \\ &= 2g_{m1}(r_{o4} \| r_{o8}) \end{aligned}$$

Substitute the values of g_{m1}, r_{o4}, **and** r_{o8} from Table 1.

$$\begin{aligned} A_d &= 2\left(\frac{I}{|V_{OV}|}\right)\left(\frac{|V_A|}{I} \| \frac{|V_A|}{I}\right) \\ &= 2\left(\frac{I}{|V_{OV}|}\right)\frac{|V_A|}{2I} \\ &= \frac{|V_A|}{|V_{OV}|} \end{aligned}$$

Thus, the differential voltage gain, A_d is, $\boxed{g_{m1}(r_{o4} \| r_{o8}) = \frac{V_A}{V_{OV}}}$.

Step 7 of 11

(c)

Replace each of the input transistors, Q_1 **and** Q_2 together with their source resistance, $2R_{SS}$ with a controlled current source, $\frac{v_{icm}}{2R_{SS}}$ and an output resistance.

Here, the resistance, R_{SS} is the output resistance of the bias current source, I.

Consider that the following is the current transfer ratio for each current mirror.

$$A_i = A_i(ideal)\left(1 - \frac{1}{g_m r_o}\right)$$

Find the current through the transistor, Q_6 through the current mirror, $Q_6 - Q_8$ with the specified current transfer ratio.

$$\begin{aligned} i_{o6} &= -\left(\frac{v_{icm}}{2R_{SS}}\right)(2)\left(1 - \frac{1}{g_{m4}r_{o4}}\right) \\ &= -\left(\frac{v_{icm}}{R_{SS}}\right)\left(1 - \frac{1}{g_{m4}r_{o4}}\right) \end{aligned}$$

Find the current through the transistor, Q_4 through the current mirrors, $Q_3 - Q_5$, and $Q_7 - Q_8$ with the specified current transfer ratios.

$$\begin{aligned} i_{o6} &= \left(\frac{v_{icm}}{2R_{SS}}\right)(1)\left(1 - \frac{1}{g_{m3}r_{o3}}\right)(2)\left(1 - \frac{1}{g_{m7}r_{o7}}\right) \\ &= \left(\frac{v_{icm}}{R_{SS}}\right)\left(1 - \frac{1}{g_{m3}r_{o3}}\right)\left(1 - \frac{1}{g_{m7}r_{o7}}\right) \end{aligned}$$

Step 8 of 11

Find the total current, i_o.

$$\begin{aligned} i_o &= i_{o6} + i_{o8} \\ &= \left(\frac{v_{icm}}{R_{SS}}\right)\left(1 - \frac{1}{g_{m4}r_{o4}}\right) + \left(\frac{v_{icm}}{R_{SS}}\right)\left(1 - \frac{1}{g_{m3}r_{o3}}\right)\left(1 - \frac{1}{g_{m7}r_{o7}}\right) \\ &= -\left(\frac{v_{icm}}{R_{SS}}\right)\left[\left(1 - \frac{1}{g_{m4}r_{o4}}\right) - \left(1 - \frac{1}{g_{m7}r_{o3}}\right)\left(1 - \frac{1}{g_{m7}r_{o7}}\right)\right] \\ &= -\left(\frac{v_{icm}}{R_{SS}}\right)\left[\left(1 - \frac{1}{g_{m3}r_{o3}}\right) - \left(1 - \frac{1}{g_{m3}r_{o3}}\right)\left(1 - \frac{1}{g_{m7}r_{o7}}\right)\right] \quad \begin{pmatrix}\text{since, } g_{m3} = g_{m4}, \\ \text{and } r_{o3} = r_{o4}\end{pmatrix} \end{aligned}$$

Simplify the expression further.

$$\begin{aligned} i_o &= -\left(\frac{v_{icm}}{R_{SS}}\right)\left(1 - \frac{1}{g_{m7}r_{o3}}\right)\left[1 - \left(1 - \frac{1}{g_{m7}r_{o7}}\right)\right] \\ &= -\left(\frac{v_{icm}}{R_{SS}}\right)\left(1 - \frac{1}{g_{m3}r_{o3}}\right)\left(\frac{1}{g_{m7}r_{o7}}\right) \\ &= -\left(\frac{v_{icm}}{R_{SS}}\right)\left(1 - \frac{1}{g_{m3}r_{o3}}\right)\left(\frac{1}{g_{m3}r_{o7}}\right) \\ &\approx -\left(\frac{v_{icm}}{R_{SS}}\right)\left(\frac{1}{g_{m7}r_{o7}}\right) \quad \left(\text{since, } \frac{1}{g_{m3}r_{o3}} \ll 1\right) \end{aligned}$$

Step 9 of 11

Find the differential output voltage, v_o.

$$\begin{aligned} v_o &= i_o R_o \\ &= \left[-\left(\frac{v_{icm}}{R_{SS}}\right)\left(\frac{1}{g_{m7}r_{o7}}\right)\right](r_{o4} \| r_{o8}) \\ &= v_{icm}\left(\frac{r_{o4} \| r_{o8}}{R_{SS}}\right)\left(\frac{1}{g_{m7}r_{o7}}\right) \end{aligned}$$

Find the common mode gain, A_{cm}.

$$\begin{aligned} |A_{cm}| &= \frac{v_o}{v_{icm}} \\ &= \frac{v_{icm}\left(\frac{r_{o4} \| r_{o8}}{R_{SS}}\right)\left(\frac{1}{g_{m7}r_{o3}}\right)}{v_{icm}} \\ &\approx \left(\frac{r_{o4} \| r_{o8}}{R_{SS}}\right)\left(\frac{1}{g_{m7}r_{o3}}\right) \end{aligned}$$

Thus, the common mode gain, $|A_{cm}|$ is $\boxed{\left(\frac{r_{o4} \| r_{o8}}{R_{SS}}\right)\left(\frac{1}{g_{m7}r_{o7}}\right)}$.

Step 10 of 11

(d)

Find the common mode rejection ratio (CMRR).

$$\begin{aligned} \text{CMRR} &= \frac{|A_d|}{|A_{cm}|} \\ &= \frac{\frac{V_A}{V_{OV}}}{\left(\frac{r_{o4} \| r_{o8}}{R_{SS}}\right)\left(\frac{1}{g_{m7}r_{o7}}\right)} \end{aligned}$$

Here, the resistance, R_{SS} is the output resistance of the bias current source, I. Consider that the bias current source is implemented as a simple mirror. Therefore,

$$R_{SS} = \frac{V_A}{I}$$

Substitute the values of g_{m1}, g_{m3}, r_{o4}, r_{o7}, **and** r_{o8} from Table 1.

$$\begin{aligned} \text{CMRR} &= \frac{\frac{V_A}{V_{OV}}}{\left(\frac{\frac{V_A}{I} \| \frac{V_A}{I}}{\frac{V_A}{I}}\right)\frac{1}{\left(\frac{2I}{V_{OV}}\right)\left(\frac{V_A}{I}\right)}} \\ &= \frac{\frac{V_A}{V_{OV}}}{\left(\frac{1}{2}\right)\frac{V_{OV}I}{2IV_A}} \\ &= \frac{4V_A^2}{V_{OV}^3} \\ &= 4\left(\frac{V_A}{V_{OV}}\right)^3 \end{aligned}$$

Thus, the common mode rejection ratio (CMRR) is, $\boxed{4\left(\frac{V_A}{V_{OV}}\right)^3}$.

Step 11 of 11

(e)

Find the input common mode range of the amplifier.

Consider the upper and lower limits of the input voltage.

The lower limit of the common mode voltage reduces the voltage across the bias current source, I to reach the minimum required value.

$$\begin{aligned} V_{CM\,min} &= V_{GS1} + V_{OV} + V_{SS} \\ &= V_t + 2V_{OV} - V_{DD} \end{aligned}$$

The upper limit of the common mode voltage causes the transistors, Q_1 **and** Q_2 to leave the saturation region and enter the triode region.

$$\begin{aligned} V_{CM\,max} &= V_t + V_{OV} + V_{t1} \quad (V_{t1} = V_{DD} - V_{SG3} - V_{OV1}) \\ &= V_t + V_{OV} + (V_{DD} - V_t - 2V_{OV}) \\ &= V_{DD} - V_{OV} \end{aligned}$$

Thus, the common mode range is, $\boxed{(V_t + 2V_{OV} - V_{DD}) \le v_{icm} \le (V_{DD} - V_{OV})}$.

Step 1 of 10

(a)

Refer to Figure P8.120 in the textbook.

Consider that the base-emitter voltage of all the transistors is, $V_{BE} = 0.7\ \text{V}$, the current gain, β is very high, and the ground voltage is, $-5\ \text{V}$.

Write the nodal equation at node, A .

$$\begin{aligned} V_A &= -5\ \text{V} + 0.7\ \text{V} \\ &= -4.3\ \text{V} \end{aligned}$$

Write the nodal equation at node, B .

$$\begin{aligned} V_B &= 0 - 0.7\ \text{V} \\ &= -0.7\ \text{V} \end{aligned}$$

Write the nodal equation at node, C .

$$\begin{aligned} V_C &= 0 + 0.7\ \text{V} \\ &= 0.7\ \text{V} \end{aligned}$$

The node, D is grounded. Therefore,

$$V_D = 0$$

Write the nodal equation at node, E .

The ground voltage is $+5\ \text{V}$ for both transistors. So, the node equation is,

$$\begin{aligned} V_E &= 5\ \text{V} - 2(0.7\ \text{V}) \\ &= 5\ \text{V} - 1.4\ \text{V} \\ &= 3.6\ \text{V} \end{aligned}$$

Step 2 of 10

Write the nodal equation at node, F .

$$\begin{aligned} V_F &= 5\ \text{V} - 0.7\ \text{V} \\ &= 4.3\ \text{V} \end{aligned}$$

Write the nodal equation at node, G .

$$\begin{aligned} V_G &= 5\ \text{V} - 0.7\ \text{V} \\ &= 4.3\ \text{V} \end{aligned}$$

Apply Ohm's law to find the resistance, R so that the current, $I_{REF} = 100\ \mu\text{A}$.

$$\begin{aligned} R &= \frac{V_E - V_A}{I_{REF}} \\ &= \frac{3.6\ \text{V} - (-4.3\ \text{V})}{100\ \mu\text{A}} \\ &= \frac{7.9\ \text{V}}{100\ \mu\text{A}} \\ &= 7.9\ \text{k}\Omega \end{aligned}$$

The value of resistance, R is $\boxed{7.9\ \text{k}\Omega}$, the node voltages at node A to G are respectively, $\boxed{-4.3\ \text{V}}$, $\boxed{-0.7\ \text{V}}$, $\boxed{0.7\ \text{V}}$, $\boxed{0}$, $\boxed{3.6\ \text{V}}$, $\boxed{4.3\ \text{V}}$, and $\boxed{4.3\ \text{V}}$.

Step 3 of 10

(b)

The reference current, I_{REF} flows through transistor, Q_A matches with the currents in Q_1 , Q_2 , Q_3 , Q_C , Q_F and Q_E . Hence, the current flow in these transistors is $100\ \mu\text{A}$.

The current in transistor, Q_B and Q_G are current, I_{REF} multiplied by a factor of 2. Hence, the current flow through these transistors is $200\ \mu\text{A}$.

Similarly, the current in transistor, Q_D is 10 times the current, I_{REF} . Hence, the current flow in this transistor is $1000\ \mu\text{A}$.

The thermal equivalent voltage is, $V_T = 25\ \text{mV}$.

The transconductance of the high gain transistor Q_1 is,

$$\begin{aligned} g_{m1} &= \frac{I_C}{V_T} \\ &= \frac{0.1\ \text{mA}}{25\ \text{mV}} \\ &= 4\ \text{mA/V} \end{aligned}$$

Similarly, for the other transistors the transconductance are,

$$\begin{aligned} g_{m2} &= 4\ \text{mA/V} \\ g_{m3} &= 4\ \text{mA/V} \\ g_{m4} &= 40\ \text{mA/V} \\ g_{m5} &= 0\ \text{mA/V} \end{aligned}$$

Step 4 of 10

The intrinsic output resistance is,

$$\begin{aligned} r_{o1} &= \frac{V_A}{I_C} \\ &= \frac{200\ \text{V}}{100\ \mu\text{A}} \\ &= 2\ \text{M}\Omega \end{aligned}$$

Similarly, for the other transistor the intrinsic output resistance are,

$$\begin{aligned} r_o &= 2\ \text{M}\Omega \\ r_{o3} &= 2\ \text{M}\Omega \\ r_{o4} &= 0.2\ \text{M}\Omega \\ r_{o5} &= \infty\ \Omega \end{aligned}$$

The output resistance for the remaining transistors are,

$$\begin{aligned} r_{oC} &= 2\ \text{M}\Omega \\ r_{oD} &= 0.2\ \text{M}\Omega \\ r_{oG} &= 1\ \text{M}\Omega \end{aligned}$$

Step 5 of 10

(c)

The total resistance of the collector of transistor, Q_3 is,

$$\begin{aligned} R_{Q3} &= \beta r_{o3} \| r_{oC} \| (\beta + 1)(r_{o4} + r_{oD}) \\ &= (100)(2\ \text{M}\Omega) \| (2\ \text{M}\Omega) \| (100 + 1)(0.2\ \text{M}\Omega + 0.2\ \text{M}\Omega) \\ &= (200 \| 2 \| 40.4)\ \text{M}\Omega \\ &= 1.648\ \text{M}\Omega \end{aligned}$$

The voltage gain ratio at the transistor, Q_3 is,

$$\begin{aligned} \frac{v_{C3}}{v_i} &= \left(\frac{1}{2}\right)(g_{m1})(R_{Q3}) \\ &= (4\ \text{mV/A})\left(\frac{1}{2}\right)(1.65\ \text{M}\Omega) \\ &= 3300\ \text{V/V} \end{aligned}$$

And

$$\frac{v_o}{v_{C3}} \approx 1\ \text{V/V}$$

Hence, the gain is, $\boxed{\frac{v_0}{v_+ - v_-} = 3300\ \text{V/V}}$.

Step 6 of 10

(d)

The input resistance is,

$$\begin{aligned} R_{in} &= 2r_\pi \\ &= 2\frac{\beta}{g_{m1}} \\ &= 2\left(\frac{100\ \text{V/V}}{4\ \text{mA/V}}\right) \\ &= 50\ \text{k}\Omega \end{aligned}$$

The input resistance of the resistor, R_{in} is, $\boxed{50\ \text{k}\Omega}$.

The output resistance is,

$$\begin{aligned} R_{out} &= r_{oD} \| r_{o4} \| \left[r_o + \frac{r_{o2} \| \beta r_{o2}}{\beta + 1} \right] \\ &= (0.2\ \text{M}\Omega) \| (0.2\ \text{M}\Omega) \| \left[25\left(10^{-6}\right)\ \text{M}\Omega + \frac{2 \| 100(2\ \text{M}\Omega)}{101} \right] \\ &= 16.39\ \text{k}\Omega \end{aligned}$$

The output resistance of the resistor, R_{out} is, $\boxed{16.39\ \text{k}\Omega}$.

Step 7 of 10

(e)

The input common mode range is,

$$\begin{aligned} |v_{ICM}|_{min} &= (-4.3 - 0.4 + 0.7)\ \text{V} \\ &= -4\ \text{V} \end{aligned}$$

$$\begin{aligned} |v_{ICM}|_{max} &= V_G + 0.4\ \text{V} \\ &= 4.7\ \text{V} \end{aligned}$$

Thus, the input common mode range is, $\boxed{-4\ \text{V} < v_{ICM} < 4.7\ \text{V}}$.

Step 8 of 10

(f)

The maximum voltage at the base of transistor, Q_4 can go to V_{B3} before Q_3 goes into saturation, that is,

$$\begin{aligned} V_{max} &= V_E + 0.4 - 0.7 \\ &= 2.2 + 0.4 - 0.7 \\ &= 3.3\ \text{V} \end{aligned}$$

The output of the amplifier can go down to a voltage of V_{knee} of transistor, Q_D , that is,

$$\begin{aligned} V_{min} &= -4.3\ \text{V} - 0.4\ \text{V} \\ &= -4.7\ \text{V} \end{aligned}$$

Hence the linear output range is, $\boxed{-4.7\ \text{V to } 3.3\ \text{V}}$.

Step 9 of 10

(g)

The following is the circuit diagram of the amplifier with Q_2 just cut-off.

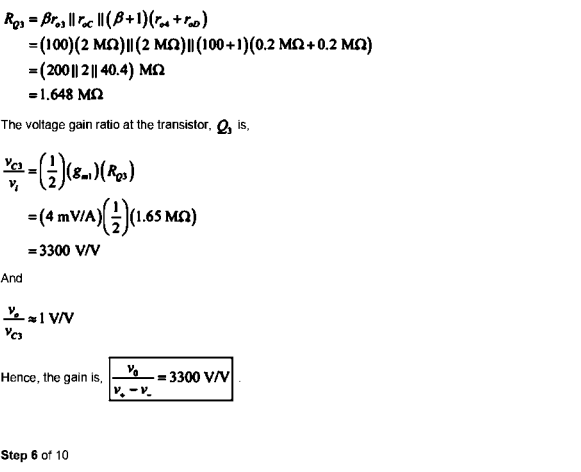


Figure 1

Step 10 of 10

Consider that the positive maximum of the output is $V_L = 4.3\ \text{V}$.

The minimum resistance, R_L in the circuit such that the transistor, Q_2 is cut-off is,

$$\begin{aligned} R_L &= \frac{V_L}{I_L} \\ &= \frac{4.7\ \text{V}}{9.1\ \text{mA}} \\ &= 363\ \Omega \end{aligned}$$

Thus, the minimum resistance, R_L is, $\boxed{363\ \Omega}$.

At the negative maximum of v_o , the transistor, Q_1 , Q_3 and Q_4 are cut-off.

Similarly, the minimum resistance, R_L , such that the transistor, Q_3 is cut-off is,

$$\begin{aligned} R_L &= \frac{V_L}{I_L} \\ &= \frac{3.3\ \text{V}}{11.1\ \text{mA}} \\ &= 423\ \Omega \end{aligned}$$

Thus, the minimum resistance, R_L is, $\boxed{423\ \Omega}$.

Step 1 of 15

9.128P

(a)

Refer to Figure P8.121 in the textbook.

The expression for the drain current in transistors, Q_5 and Q_7 is,

$$I_D = I_{REF} = \frac{1}{2}\mu_p C_{ox}\left(\frac{W}{L}\right)\left(V_{GS_{5,7}} - V_t\right)^2$$

Here, the intrinsic transconductance for the PMOS is, $\mu_p C_{ox} = 20\ \mu A/V^2$, the length of each device is, $L = 5\ \mu m$, the width of the devices as multiples of W is, $W = 5\ \mu m$, the drain current is $10\ \mu A$, the threshold voltage, $V_t = 1\ V$.

Substitute all the values in the expression of I_D to find the value of gate-source voltage, $V_{GS_{5,7}}$.

$$10\ \mu A = \frac{1}{2}(20\ \mu A/V^2)\frac{5\ \mu m}{5\ \mu m}\left(V_{GS_{5,7}} - 1\ V\right)^2$$

Simplify the expression further.

$$\left(V_{GS_{5,7}} - 1\ V\right)^2 = \frac{20}{20}$$

$$V_{GS_{5,7}} = 1\ V + 1\ V$$
$$= 2\ V$$

Therefore, the voltages across the drain and source terminals (gate and drain are shorted) of transistors, Q_5 and Q_7 are same and equal to 2 V.

Step 2 of 15

The expression for the drain current in transistor, Q_A is,

$$I_D = \frac{1}{2}(\mu_n C_{ox})\left(\frac{W}{L}\right)\left(V_{GS_A} - V_t\right)^2$$

Here, the intrinsic transconductance for the NMOS is, $\mu_n C_{ox} = 40\ \mu A/V^2$.

Substitute all the values in the expression of I_D to find the value of V_{GS}.

$$10\ \mu A = \frac{1}{2}(40\ \mu A)\left(\frac{5\ \mu m}{5\ \mu m}\right)\left(V_{GS_A} - 1\right)^2$$

$$\left(V_{GS_A} - 1\right)^2 = \frac{20}{40}$$

$$V_{GS_A} = 0.707 + 1$$
$$= 1.707\ V$$

Therefore, the voltages across the drain and source terminals (gate and drain are shorted) of transistor, Q_A is 1.707 V.

Step 3 of 15

Write the nodal equation at node, A.

$$V_A = -5 + V_{GS_A}$$
$$= -5 + 1.7$$
$$= -3.3\ V$$

Write the nodal equation at node, E.

$$V_E = 5 - V_{GS_5} - V_{GS_7}$$
$$5\ V - 2\ V - 2\ V$$
$$= 1\ V$$

The resistance R required to get a reference current of $10\ \mu A$, is,

$$R = \frac{V_A - V_E}{I_{REF}}$$
$$= \frac{1\ V - (-3.3)\ V}{10\ \mu A}$$
$$= \frac{4.3\ V}{10\ \mu A}$$
$$= 430\ k\Omega$$

Thus, the value of resistance, R is $\boxed{430\ k\Omega}$.

Step 4 of 15

(b)

The gate source voltages of the transistors Q_1 and Q_2 equal in the circuit.

$$V_{GS_1} = V_{GS_2}$$
$$= V_{GS_3}$$
$$= 1.7\ V$$

The gate source voltage at Q_3 is,

$$V_{GS_3} = V_t + \sqrt{\frac{2I_3}{\mu_p C_{ox}\left(\frac{W}{L}\right)}}$$
$$= 1 + \sqrt{\frac{2(10\ \mu A)}{(20\ \mu A/V^2)\left(\frac{10}{5}\right)}}$$
$$= 1.7\ V$$

Step 5 of 15

The gate source voltage for the transistor Q_4 is,

$$V_{GS_4} = V_t + \sqrt{\frac{2I_4}{\mu_n C_{ox}\left(\frac{W}{L}\right)}}$$
$$= 1 + \sqrt{\frac{2(50\ \mu A)}{(40\ \mu A/V^2)\left(\frac{50}{5}\right)}}$$
$$= 1.5\ V$$

Write the nodal equation at node, B.

Consider that v_+ and v_- are equal to 0 V.

$$V_B = 0 - 1.7$$
$$= -1.7\ V$$

Write the nodal equation at node, C.

$$V_C = 5 + 3.5\ V$$
$$= 1.5\ V$$

Write the nodal equation at node, D.

$$V_D = v_O$$
$$= 0\ V$$

Step 6 of 15

Write the nodal equation at node, F.

$$V_F = 5\ V - 2\ V$$
$$= 3\ V$$

Write the nodal equation at node, G.

$$V_G = 5\ V - 1.7\ V$$
$$= 3.3\ V$$

Write the nodal equation at node, H.

$$V_H = 1\ V + 1.7\ V$$
$$= 2.7\ V$$

Therefore, the voltages at various nodes are,

$$V_A = -3.3\ V$$
$$V_B = -1.7\ V$$
$$V_C = 1.5\ V$$
$$V_D = 0\ V$$
$$V_E = 1\ V$$
$$V_F = 3\ V$$
$$V_G = 3.3\ V$$
$$V_H = 2.7\ V$$

Step 7 of 15

Draw the circuit with bias point voltages at all the nodes in the circuit.

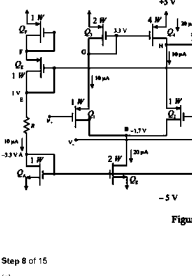


Figure 1

Step 8 of 15

(c)

The following table represent all the transistor parameters for all the transistors:

Table 1

Transistor	V_{GS}	$(I_D)\ \mu A$	(g_m) mA/V	r_o MΩ
Q_1	1.7 V	10	28.3	5
Q_2	1.7 V	10	28.3	5
Q_3	1.7 V	10	28.3	5
Q_4	1.7 V	20	56.6	2.5
Q_5	1.7 V	10	28.3	5
Q_6	1.5 V	50	200	1
Q_7	−1.5 V	0	0	∞
Q_8	1.7 V	10	28.3	5
Q_9	1.7 V	20	56.6	2.5
Q_C	1.7 V	10	28.3	5
Q_D	1.7 V	50	141.4	1
Q_E	2 V	10	20	5
Q_F	2 V	10	20	5

Step 8 of 15

(d)

The total resistance at the drain of transistor, Q_5 is,

$$R_{Q_5} = g_{m_5} r_{o5}\left(r_{o4} \| r_{o2}\right) \| r_{o6}$$
$$= \left[(28.3\ mA/V)(5\ M\Omega)(2.5\ M\Omega \| 2\ M\Omega)\right] \| (5\ M\Omega)$$
$$= \left[(141.5\ k\Omega)(1.12\ M\Omega)\right] \| (5\ k\Omega)$$
$$= 4.9\ M\Omega$$

The voltage gain at the transistor, Q_5 is,

$$\frac{v_{o_1}}{v_i} = g_{m_5} R_{Q_5}$$
$$= (28.3\ mV/A)(4.9\ M\Omega)$$
$$= 138.7\ kV/V$$

Step 10 of 15

The ratio between the output and v_{o2} is,

$$\frac{v_o}{v_{o_1}} = \frac{\left(r_{o8} \| r_{o9}\right)}{\left(r_{o8} \| r_{o9}\right) + \frac{1}{g_{m8}}}$$
$$= \frac{(1\ M\Omega \| 1\ M\Omega)}{(1\ M\Omega \| 1\ M\Omega) + \frac{1}{200\ mA/V}}$$
$$= \frac{500\ k\Omega}{500\ k\Omega + 5\ \Omega}$$
$$= 1\ V/V$$

The gain of the amplifier is,

$$A_v = \frac{v_o}{v_i}$$
$$= \frac{v_o}{v_{o_1}} \frac{v_{o_1}}{v_i}$$
$$= (138.7)(1)$$
$$= 138.7\ kV/V$$

Thus, the gain of the amplifier, A_v is, $\boxed{138.7\ V/V}$.

Step 11 of 15

The input resistance of the amplifier, R_{in} is, $\boxed{\infty}$.

The output resistance is,

$$R_{out} = r_{o9} \| r_{o8} \| \frac{1}{g_{m8}}$$
$$= (1\ M\Omega \| 1\ M\Omega) + \frac{1}{200\ mA/V}$$
$$= 500\ k\Omega + 5\ \Omega$$
$$= 500\ k\Omega$$

Thus, the output resistance of the amplifier, R_{out} is, $\boxed{500\ k\Omega}$.

Step 12 of 15

(e)

The input common mode range is,

$$v_{CM\,max} = -V_t + V_A + V_{GS_1}$$
$$= -1\ V - 3.3\ V + 1.7\ V$$
$$= -2.6\ V$$

$$v_{CM\,max} = V_G + V_t$$
$$= 3.3 + 1$$
$$= 4.3\ V$$

Thus, the range of input common-mode, v_{CM} is, $\boxed{-2.6\ V \le v_{CM} \le 4.3\ V}$.

(f)

The maximum voltage at the base of the transistor, Q_4 can go to V_{B_1} before transistor, Q_3 goes into saturation.

$$v_{o\,max} = V_{C\,max} - V_{GS_5}$$
$$= V_B + V_t - V_{GS_5}$$
$$= 1 + 1 - 1.5$$
$$= 0.5\ V$$

$$v_{o\,min} = V_A - V_t$$
$$= -3.3 - 1$$
$$= -4.3\ V$$

Hence, the range of linear output, v_o is, $\boxed{-4.3\ V \le v_o \le 0.5\ V}$.

Step 13 of 15

(g)

Consider that a load resistance, R_L is connected

Consider also that the output is limited to $-1\ V$.

Observe the circuit that the current through the transistor, Q_D is $50\ \mu A$. Therefore,

$$R_L = \frac{1\ V}{50\ \mu A}$$
$$= 20\ k\Omega$$

Thus, the load resistance, R_L is, $\boxed{20\ k\Omega}$.

Step 14 of 15

(h)

Consider that the new value of load resistance is one-tenth of $20\ k\Omega$.

$$R_L = \frac{20\ k\Omega}{10}$$
$$= 2\ k\Omega$$

The maximum possible value of drain voltage at transistor, Q_5 is 2 V.

Consider that the transistor, Q_7 is in cutoff, and transistor, Q_4 conducts. Therefore,

$$I_D = 50\ \mu A + \frac{v_o}{2\ k\Omega}$$

The general expression for the drain current of transistor, Q_4 is,

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}\left(2 - v_o - V_t\right)^2$$

Equate the expression to the value of I_D, and substitute all the values.

$$50\ \mu A + \frac{v_o}{2\ k\Omega} = \frac{1}{2}(40\ \mu A/V^2)(10)(2 - v_o - 1)^2$$

$$(1 - v_o)^2 = \frac{0.00005 + 0.0005 v_o}{0.0002}$$

$$(1 - v_o)^2 - \frac{v_o}{2} = 0.25$$

$$1 + v_o^2 - 2v_o - 2.5v_o = 0.25$$

$$v_o^2 - 4.5v_o + 0.75 = 0$$

Solve the equation to get the value of output voltage.

$$v_o = 0.17\ V$$

Step 15 of 15

The minimum possible value of drain voltage at transistor, Q_5 is $-4.3\ V$.

Consider that the transistor, Q_4 is in cutoff, and transistor, Q_7 conducts. Therefore,

$$I_D = 50\ \mu A + \frac{v_o}{2\ k\Omega}$$

The general expression for the drain current of transistor, Q_7 is,

$$I_D = \frac{1}{2}\mu_p C_{ox}\frac{W}{L}\left(-v_o + 4.3 - V_t\right)^2$$

Equate the expression to the value of I_D, and substitute all the values.

$$50\ \mu A + \frac{v_o}{2\ k\Omega} = \frac{1}{2}(20\ \mu A/V^2)(20)(-v_o + 4.3 - 1)^2$$

$$(3.3 - v_o)^2 = \frac{0.00005 + 0.0005 v_o}{0.0002}$$

$$(3.3 - v_o)^2 - \frac{v_o}{2} = 0.25$$

$$10.89 + v_o^2 - 6.6v_o - 2.5v_o = 0.25$$

$$v_o^2 - 9.1v_o + 10.64 = 0$$

Solve the equation to get the value of output voltage.

$$v_o = 1.37\ V$$

Thus, the range of output signal swing is, $\boxed{0.17\ V \text{ to } 1.45\ V}$.

Step 1 of 4

Refer to Figure 9.2 in the textbook for the common source (CS) amplifier circuit.

Write the expression for the mid-band gain.

$$A_M = -\frac{R_G}{R_G + R_{sig}}\left[g_m\left(R_D \| R_L\right)\right]$$

Substitute $10\ \text{M}\Omega$ for R_G, $100\ \text{k}\Omega$ for R_{sig}, 2 mA/V for g_m, $10\ \text{k}\Omega$ for R_D and R_L.

$$\begin{aligned}
A_M &= -\frac{10\times10^6}{10\times10^6+100\times10^3}\left[2\times10^{-3}\left(10\times10^3 \| 10\times10^3\right)\right] \\
&= -\frac{10\times10^6}{10\times10^6+100\times10^3}\left[2\times10^{-3}\left(\frac{\left(10\times10^3\right)\left(10\times10^3\right)}{10\times10^3+10\times10^3}\right)\right] \\
&= -\frac{10\times10^6}{10^5(100+1)}\left[2\times10^{-3}\left(\frac{100\times10^6}{20\times10^3}\right)\right] \\
&= -\frac{10\times10^6}{10^5(100+1)}\left[\left(2\times10^{-3}\right)\left(5\times10^3\right)\right]
\end{aligned}$$

Simplify further.

$$\begin{aligned}
A_M &= -\frac{100}{101}[10] \\
&= -9.9\ \text{V/V}
\end{aligned}$$

Thus, the mid-band gain of the CS amplifier is, $\boxed{-9.9\ \text{V/V}}$.

Step 2 of 4

Write the expression for the bypass capacitor.

$$C_S = \frac{g_m}{2\pi f_L}$$

Rearrange the expression for the lower 3-dB frequency.

$$f_L = \frac{g_m}{2\pi C_S}$$

Substitute 2 mA/V for g_m and $1\ \mu\text{F}$ for C_S.

$$\begin{aligned}
f_L &= \frac{2\times10^{-3}}{2\pi\left(1\times10^{-6}\right)} \\
&= 318.3\ \text{Hz}
\end{aligned}$$

Thus, the lower 3-dB frequency is, $\boxed{318.3\ \text{Hz}}$.

Determine the value of dominant pole frequency.

It is known that, the dominant pole frequency is equal to the lower 3-dB frequency.

Thus, the dominant pole frequency, f_{P2} is, $\boxed{318.3\ \text{Hz}}$.

Step 3 of 4

Write the expression for the break frequency.

$$f_{P1} = \frac{1}{2\pi C_{C1}\left(R_G + R_{sig}\right)}$$

Substitute $1\ \mu\text{F}$ for C_{C1}, $10\ \text{M}\Omega$ for R_G, and $100\ \text{k}\Omega$ for R_{sig}.

$$\begin{aligned}
f_{P1} &= \frac{1}{2\pi\left(1\times10^{-6}\right)\left(10\times10^6+100\times10^3\right)} \\
&= \frac{1}{2\pi\left(1\times10^{-6}\right)\left(101\times10^5\right)} \\
&= \frac{1}{2\pi(0.1)(101)} \\
&= 0.016\ \text{Hz}
\end{aligned}$$

Thus, the break frequency, f_{P1} is $\boxed{0.016\ \text{Hz}}$.

Step 4 of 4

Write the expression for the pole frequency, f_{P3}.

$$f_{P3} = \frac{1}{2\pi C_{C2}\left(R_D + R_L\right)}$$

Substitute $1\ \mu\text{F}$ for C_{C2}, $10\ \text{k}\Omega$ for R_D, and $10\ \text{k}\Omega$ for R_L.

$$\begin{aligned}
f_{P3} &= \frac{1}{2\pi\left(1\times10^{-6}\right)\left(10\times10^3+10\times10^3\right)} \\
&= \frac{1}{2\pi\left(1\times10^{-6}\right)\left(20\times10^3\right)} \\
&\approx 8\ \text{Hz}
\end{aligned}$$

Thus, the pole frequency, f_{P3} is $\boxed{8\ \text{Hz}}$.

Step 1 of 3

10.001P

Refer to Figure 10.3 (a) from the text book.

Calculate the value of the resistance R_G.

$$R_G = R_{G1} \| R_{G2}$$

Substitute $2\ \text{M}\Omega$ for R_{G1} and $1\ \text{M}\Omega$ for R_{G2}.

$$\begin{aligned} R_G &= 2\ \text{M} \| 1\ \text{M} \\ &= \frac{(2\ \text{M})(1\ \text{M})}{2\ \text{M} + 1\ \text{M}} \\ &= \frac{(2\ \text{M})(1\ \text{M})}{3\ \text{M}} \\ &= \frac{2}{3}\ \text{M} \end{aligned}$$

$$R_G = 0.67\ \text{M}\Omega$$

Therefore, the value of the resistance R_G is $0.67\ \text{M}\Omega$.

Step 2 of 3

Consider the expression for the pole frequency associated with the capacitor C_{C1}.

$$\omega_{P1} = \frac{1}{C_{C1}\left(R_{sig} + R_G\right)}$$

Substitute $2\pi f_{P1}$ for ω_{P1}.

$$2\pi f_{P1} = \frac{1}{C_{C1}\left(R_{sig} + R_G\right)}$$

Solve for C_{C1}.

$$C_{C1} = \frac{1}{2\pi f_{P1}\left(R_{sig} + R_G\right)} \quad \text{..... (1)}$$

Therefore, the expression for capacitance C_{C1} is $\dfrac{1}{2\pi f_{P1}\left(R_{sig} + R_G\right)}$.

Step 3 of 3

Recall equation (1).

$$C_{C1} = \frac{1}{2\pi f_{P1}\left(R_{sig} + R_G\right)}$$

Substitute $10\ \text{Hz}$ for f_{P1}, $0.67\ \text{M}\Omega$ for R_G and $200\ \text{k}\Omega$ for R_{sig}.

$$\begin{aligned} C_{C1} &= \frac{1}{2\pi(10)(200\ \text{k} + 0.67\ \text{M})} \\ &= \frac{1}{(62.86)(870\ \text{k})} \\ &= \frac{1}{54.69\ \text{M}} \\ &= 18.3\ \mu\text{F} \end{aligned}$$

Therefore, the value of the capacitance C_{C1} is $\boxed{18.3\ \mu\text{F}}$.

Step 1 of 8

Calculate the value of the resistance R_{C1}.

$$R_{C1} = (R_B \| r_\pi) + R_{sig}$$

Substitute $100\,\text{k}\Omega$ for R_B, $2.5\,\text{k}\Omega$ for r_π and $5\,\text{k}\Omega$ for R_{sig}.

$$\begin{aligned} R_{C1} &= (100\,\text{k} \| 2.5\,\text{k}) + 5\,\text{k} \\ &= \frac{(100\,\text{k})(2.5\,\text{k})}{100\,\text{k} + 2.5\,\text{k}} + 5\,\text{k} \\ &= \frac{250\,\text{M}}{102.5\,\text{k}} + 5\,\text{k} \\ &= 2.44\,\text{k} + 5\,\text{k} \end{aligned}$$

$$R_{C1} = 7.44\,\text{k}\Omega$$

Therefore, the value of the resistance R_{C1} is $7.44\,\text{k}\Omega$.

Step 2 of 8

Calculate the value of the resistance R_{CE}.

$$R_{CE} = R_E \| \left[r_e + \frac{R_B \| R_{sig}}{\beta + 1} \right]$$

Substitute 100 for β, $100\,\text{k}\Omega$ for R_B, $5\,\text{k}\Omega$ for R_E, $25\,\Omega$ for r_e and $5\,\text{k}\Omega$ for R_{sig}.

$$\begin{aligned} R_{CE} &= 5\,\text{k}\Omega \| \left[25 + \frac{100\,\text{k} \| 5\,\text{k}}{100 + 1} \right] \\ &= 5\,\text{k}\Omega \| \left[25 + \frac{1}{101} \left(\frac{(100\,\text{k})(5\,\text{k})}{100\,\text{k} + 5\,\text{k}} \right) \right] \\ &= 5\,\text{k}\Omega \| \left[25 + \frac{1}{101} \left(\frac{500\,\text{M}}{105\,\text{k}} \right) \right] \\ &= 5\,\text{k}\Omega \| [25 + 47.15] \end{aligned}$$

$$\begin{aligned} R_{CE} &= 5\,\text{k} \| 72.15 \\ &= \frac{(5\,\text{k})(72.15)}{5\,\text{k} + 72.15} \\ &= \frac{(5\,\text{k})(72.15)}{5.72\,\text{k}} \\ &= 71.13\,\Omega \end{aligned}$$

Therefore, the value of the resistance R_{CE} is $71.13\,\Omega$.

Step 3 of 8

Calculate the value of the resistance R_{C2}.

$$R_{C2} = R_C + R_L$$

Substitute $8\,\text{k}\Omega$ for R_C and $5\,\text{k}\Omega$ for R_L.

$$\begin{aligned} R_{C2} &= 8\,\text{k} + 5\,\text{k} \\ &= 13\,\text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance R_{C2} is $13\,\text{k}\Omega$.

Step 4 of 8

Calculate the value of the time constant τ_{C1}.

$$\tau_{C1} = C_{C1} R_{C1}$$

Substitute $7.44\,\text{k}\Omega$ for R_{C1} and $1\,\mu\text{F}$ for C_{C1}.

$$\begin{aligned} \tau_{C1} &= (1\,\mu)(7.44\,\text{k}) \\ &= 7.44\,\text{ms} \end{aligned}$$

Therefore, the value of the time constant τ_{C1} is $\boxed{7.44\,\text{ms}}$.

Step 5 of 8

Calculate the value of the time constant τ_{CE}.

$$\tau_{CE} = C_E R_{CE}$$

Substitute $71.13\,\Omega$ for R_E and $1\,\mu\text{F}$ for C_E.

$$\begin{aligned} \tau_{CE} &= (1\,\mu)(71.13) \\ &= 0.071\,\text{ms} \end{aligned}$$

Therefore, the value of the time constant τ_{C1} is $\boxed{0.071\,\text{ms}}$.

Step 6 of 8

Calculate the value of the time constant τ_{C1}.

$$\tau_{C2} = C_{C2} R_{C2}$$

Substitute $13\,\text{k}\Omega$ for R_{C2} and $1\,\mu\text{F}$ for C_{C2}.

$$\begin{aligned} \tau_{C2} &= (1\,\mu)(13\,\text{k}) \\ &= 13\,\text{ms} \end{aligned}$$

Therefore, the value of the time constant τ_{C2} is $\boxed{13\,\text{ms}}$.

Step 7 of 8

Calculate the value of the frequency f_L.

$$f_L = \frac{1}{2\pi} \left(\frac{1}{\tau_{C1}} + \frac{1}{\tau_{CE}} + \frac{1}{\tau_{C2}} \right)$$

Substitute $13\,\text{ms}$ for τ_{C2}, $0.071\,\text{ms}$ for τ_{CE} and $7.44\,\text{ms}$ for τ_{C1}.

$$\begin{aligned} f_L &= \frac{1}{2\pi} \left(\frac{1}{7.44\,\text{m}} + \frac{1}{0.071\,\text{m}} + \frac{1}{13\,\text{m}} \right) \\ &= \frac{1}{2\pi} (134.41 + 14084.51 + 76.92) \\ &= \frac{1}{2\pi} (14295.84) \\ &= 2.28\,\text{kHz} \end{aligned}$$

Therefore, the value of the frequency f_L is $\boxed{2.28\,\text{kHz}}$.

Step 8 of 8

Calculate the value of the frequency f_Z.

$$f_Z = \frac{1}{2\pi C_E R_E}$$

Substitute $5\,\text{k}\Omega$ for R_E and $1\,\mu\text{F}$ for C_E.

$$\begin{aligned} f_Z &= \frac{1}{2\pi (1\,\mu)(5\,\text{k})} \\ &= \frac{1}{31.42\,\text{m}} \\ &= 31.8\,\text{Hz} \end{aligned}$$

Therefore, the value of the frequency f_Z is $\boxed{31.8\,\text{Hz}}$.

The value of the frequency f_Z is much smaller compared to the value of the frequency f_L. Hence, the value of the frequency f_Z does not effect the value of f_L.

Step 1 of 2

10.002P

Refer to Figure 10.3 (a) from the text book.

Consider the expression for the pole frequency associated with the capacitor C_{C2}.

$$\omega_{P3} = \frac{1}{C_{C2}(R_D + R_L)}$$

Substitute $2\pi f_{P3}$ for ω_{P3}.

$$2\pi f_{P3} = \frac{1}{C_{C2}(R_{sig} + R_G)}$$

Solve for C_{C2}.

$$C_{C2} = \frac{1}{2\pi f_{P3}(R_D + R_L)} \quad \text{..... (1)}$$

Therefore, the expression for capacitance C_{C2} is $\frac{1}{2\pi f_{P3}(R_D + R_L)}$.

Step 2 of 2

Recall equation (1).

$$C_{C2} = \frac{1}{2\pi f_{P3}(R_D + R_L)}$$

Substitute $10\ \text{Hz}$ for f_{P1}, $10\ \text{k}\Omega$ for R_D and R_L.

$$\begin{aligned} C_{C2} &= \frac{1}{2\pi(10)(10\ \text{k} + 10\ \text{M})} \\ &= \frac{1}{(62.86)(20\ \text{k})} \\ &= \frac{1}{1.26\ \text{M}} \\ &= 0.8\ \mu\text{F} \end{aligned}$$

Therefore, the value of the capacitance C_{C2} is $\boxed{0.8\ \mu\text{F}}$.

Step 1 of 6

Consider the *n*-channel MOSFET (transistor is in saturation region).

Write the expression for the gate body model capacitance.

$$C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$$

Here,

$$\varepsilon_{ox} = 3.9\varepsilon_0 \quad \{\text{For silicon } K_{ox} = 3.9\}$$
$$\varepsilon_0 = 8.854\times10^{-12}\ \text{F/m}$$

Calculate the gate body model capacitance.

$$C_{ox} = \frac{3.9\left(8.854\times10^{-12}\right)}{10\times10^{-9}}$$
$$= 3.45\times10^{-3}\ \text{F/m}^2$$

Adjust the units to get the gate body model capacitance in $\text{fF}/\mu\text{m}^2$

$$C_{ox} = \frac{34.5\times10^{-4}\times10^{-12}}{10^{-12}}$$
$$= \frac{3.45\times10^{-15}}{10^{-12}}$$
$$= 3.45\ \text{fF}/\mu\text{m}^2$$

Thus, the gate body model capacitance is, $\boxed{3.45\ \text{fF}/\mu\text{m}^2}$.

Step 2 of 6

Write the expression for the overlap capacitance.

$$C_{ov} = WL_{ov}C_{ox}$$

Substitute $10\ \mu\text{m}$ for W, $0.05\ \mu\text{m}$ for L_{ov}, and $3.45\ \text{fF}/\mu\text{m}^2$ for C_{ox}.

$$C_{ov} = \left(10\times10^{-6}\right)\left(0.05\times10^{-6}\right)\frac{\left(3.45\times10^{-15}\right)}{10^{-12}}$$
$$= 1.72\ \text{fF}$$

Thus, the overlap capacitance is, $\boxed{1.72\ \text{fF}}$.

Step 3 of 6

Consider the following gate capacitances.

$$C_{gs} = \frac{2}{3}WLC_{ox} + WL_{ov}C_{ox}$$
$$C_{gd} = WL_{ov}C_{ox}$$

Substitute $10\ \mu\text{m}$ for W, $1\ \mu\text{m}$ for L, and $3.45\ \text{fF}/\mu\text{m}^2$ for C_{ox} to find the value of C_{gs}.

$$C_{gs} = \frac{2}{3}\left(10\times10^{-6}\right)\left(1\times10^{-6}\right)\left(\frac{3.45\times10^{-15}}{10^{-12}}\right) + \left(10\times10^{-6}\right)\left(0.05\times10^{-6}\right)\left(\frac{3.45\times10^{-15}}{10^{-12}}\right)$$
$$= 24.7\ \text{fF}$$

Thus, the gate to source capacitance is, $\boxed{24.7\ \text{fF}}$.

Step 4 of 6

Substitute $10\ \mu\text{m}$ for W, $1\ \mu\text{m}$ for L, and $3.45\ \text{fF}/\mu\text{m}^2$ for C_{ox} to find the value of C_{gd}.

$$C_{gd} = \left(10\times10^{-6}\right)\left(0.05\times10^{-6}\right)\left(\frac{3.45\times10^{-15}}{10^{-12}}\right)$$
$$= 1.72\ \text{fF}$$

Thus, the gate to drain capacitance is, $\boxed{1.72\ \text{fF}}$.

Step 5 of 6

Write the expression for the source body capacitance.

$$C_{sb} = \frac{C_{sb0}}{\sqrt{1+\frac{V_{SB}}{V_0}}}$$

Substitute 10 fF for C_{sb0}, 1 V for V_{SB}, 0.6 V for V_0.

$$C_{sb} = \frac{10\times10^{-15}}{\sqrt{1+\frac{1}{0.6}}}$$
$$= 6.12\ \text{fF}$$

Thus, the source body capacitance is, $\boxed{6.12\ \text{fF}}$.

Step 6 of 6

Write the expression for the drain-body capacitance.

$$C_{db} = \frac{C_{db0}}{\sqrt{1+\frac{V_{DB}}{V_0}}}$$
$$= \frac{C_{db0}}{\sqrt{1+\frac{V_{SB}+V_{DS}}{V_0}}}$$

Substitute 10 fF for C_{db0}, 2 V for V_{DS}, 1 V for V_{SB}, and 0.6 V for V_0.

$$C_{db} = \frac{10\times10^{-15}}{\sqrt{1+\frac{1+2}{0.6}}}$$
$$= 4.1\ \text{fF}$$

Thus, the drain-body capacitance is, $\boxed{4.1\ \text{fF}}$.

Step 1 of 4

Refer to Figure 10.3 (a) from the text book.

Consider the expression for the pole frequency associated with the capacitor C_{C2}.

$$\omega_{P2} = \frac{g_m + \frac{1}{R_S}}{C_S}$$

Substitute $2\pi f_{P2}$ for ω_{P2}.

$$2\pi f_{P2} = \frac{g_m + \frac{1}{R_S}}{C_S}$$

Solve for C_S.

$$C_S = \frac{g_m + \frac{1}{R_S}}{2\pi f_{P2}} \quad \text{..... (1)}$$

Therefore, the expression for capacitance C_S is $\frac{g_m + \frac{1}{R_S}}{2\pi f_{P2}}$.

Step 2 of 4

Recall equation (1).

$$C_S = \frac{g_m + \frac{1}{R_S}}{2\pi f_{P2}}$$

Substitute $100\ \text{Hz}$ for f_{P2}, $1.8\ \text{k}\Omega$ for R_S and $5\ \text{mA/V}$ for g_m.

$$\begin{aligned} C_S &= \frac{5\ \text{m} + \frac{1}{1.8\ \text{k}}}{2\pi(100)} \\ &= \frac{5\ \text{m} + 0.56\ \text{m}}{628.32} \\ &= \frac{5.56\ \text{m}}{628.32} \\ &= 8.85\ \mu\text{F} \end{aligned}$$

Therefore, the value of the capacitance C_S is $\boxed{8.85\ \mu\text{F}}$.

The frequency of the pole f_{P2} is $\boxed{100\ \text{Hz}}$.

Step 3 of 4

Calculate the frequency of the zero f_Z.

$$\omega_Z = \frac{1}{C_S R_S}$$

Substitute $2\pi f_Z$ for ω_Z.

$$2\pi f_Z = \frac{1}{C_S R_S}$$

$$f_Z = \frac{1}{2\pi C_S R_S} \quad \text{..... (2)}$$

Therefore, the expression for the frequency f_Z is $\frac{1}{2\pi C_S R_S}$.

Step 4 of 4

Recall equation (2).

$$f_Z = \frac{1}{2\pi C_S R_S}$$

Substitute C_S for $8.85\ \mu\text{F}$ and $1.8\ \text{k}\Omega$ for R_S.

$$\begin{aligned} f_Z &= \frac{1}{2\pi(8.85\ \mu)(1.8\ \text{k})} \\ &= \frac{1}{0.1} \\ &= 10\ \text{Hz} \end{aligned}$$

Therefore, the value of the frequency f_Z is $\boxed{10\ \text{Hz}}$.

Step 1 of 3

Consider the *n*-channel MOSFET.

Write the expression for the gate body model capacitance.

$$C_{ox}=\frac{\varepsilon_{ox}}{t_{ox}}$$

Here,

$$\varepsilon_{ox}=3.9\varepsilon_0 \quad \{\text{For silicon } K_{ox}=3.9\}$$
$$\varepsilon_0=8.854\times10^{-12}\ \text{F/m}$$

Calculate the gate body model capacitance.

$$C_{ox}=\frac{3.9\left(8.854\times10^{-12}\right)}{10\times10^{-9}}$$
$$=3.45\times10^{-3}\ \text{F/m}^2$$

Adjust the units to get the gate body model capacitance in $\text{fF}/\mu\text{m}^2$

$$C_{ox}=\frac{34.5\times10^{-4}\times10^{-12}}{10^{-12}}$$
$$=\frac{3.45\times10^{-15}}{10^{-12}}$$
$$=3.45\ \text{fF}/\mu\text{m}^2$$

Thus, the gate body model capacitance is, $3.45\ \text{fF}/\mu\text{m}^2$

Step 2 of 3

Consider the following gate capacitances.

$$C_{gs}=\frac{2}{3}WLC_{ox}+WL_{ov}C_{ox}$$
$$C_{gd}=WL_{ov}C_{ox}$$

Substitute $10\ \mu\text{m}$ for W, $1\ \mu\text{m}$ for L, and $3.45\ \text{fF}/\mu\text{m}^2$ for C_{ox} to find the value of C_{gs}.

$$C_{gs}=\frac{2}{3}\left(10\times10^{-6}\right)\left(1\times10^{-6}\right)\left(\frac{3.45\times10^{-15}}{10^{-12}}\right)+\left(10\times10^{-6}\right)\left(0.05\times10^{-6}\right)\left(\frac{3.45\times10^{-15}}{10^{-12}}\right)$$
$$=24.7\ \text{fF}$$

Substitute $10\ \mu\text{m}$ for W, $1\ \mu\text{m}$ for L, and $3.45\ \text{fF}/\mu\text{m}^2$ for C_{ox} to find the value of C_{gd}.

$$C_{gd}=\left(10\times10^{-6}\right)\left(0.05\times10^{-6}\right)\left(\frac{3.45\times10^{-15}}{10^{-12}}\right)$$
$$=1.72\ \text{fF}$$

Step 3 of 3

The transistor is in the saturation region and the drain current is, $I_D=100\ \mu\text{A}$ and $k_n'=160\ \mu\text{A/V}$.

Write the expression for the trans-conductance, g_m.

$$g_m=\sqrt{2k_n'\frac{W}{L}I_D}$$

Substitute $160\ \mu\text{A/V}$ for k_n', $10\ \mu\text{m}$ for W, $1\ \mu\text{m}$ for L, and $100\ \mu\text{A}$ for I_D.

$$g_m=\sqrt{2\left(160\times10^{-6}\right)\left(\frac{10\times10^{-6}}{1\times10^{-6}}\right)\left(100\times10^{-6}\right)}$$
$$=5.66\times10^{-4}\ \text{A/V}$$

Write the expression for the unity-gain frequency.

$$f_T=\frac{g_m}{2\pi\left(C_{gs}+C_{gd}\right)}$$

Substitute $23\ \text{fF}$ for C_{gs}, $5.66\times10^{-4}\ \text{A/V}$ for g_m, and $17.25\ \text{fF}$ for C_{gd}

$$f_T=\frac{5.66\times10^{-4}}{2\pi\left(24.7\times10^{-15}+1.72\times10^{-15}\right)}$$
$$=3.41\ \text{GHz}$$

Thus, the unity-gain frequency is, $\boxed{3.41\ \text{GHz}}$.

Step 1 of 7

Refer to Figure 10.3 (a) from the text book.

Calculate the value of the resistance R_G.

$$R_G = R_{G1} \| R_{G2}$$

Substitute $47\ \text{M}\Omega$ for R_{G1} and $10\ \text{M}\Omega$ for R_{G2}.

$$\begin{aligned} R_G &= 47\ \text{M} \| 10\ \text{M} \\ &= \frac{(47\ \text{M})(10\ \text{M})}{47\ \text{M} + 10\ \text{M}} \\ &= \frac{(47\ \text{M})(10\ \text{M})}{57\ \text{M}} \\ &= 8.25\ \text{M}\Omega \end{aligned}$$

Therefore, the value of the resistance R_G is $8.25\ \text{M}\Omega$.

Step 2 of 7

Calculate the value of the gain A_M.

$$A_M = -\frac{R_G}{R_G + R_{sig}} g_m (R_D \| R_L)$$

Substitute $8.25\ \text{M}\Omega$ for R_G, $4.7\ \text{k}\Omega$ for R_D, $10\ \text{k}\Omega$ for R_L, $100\ \text{k}\Omega$ for R_{sig} and $5\ \text{mA/V}$ for g_m.

$$\begin{aligned} A_M &= -\left(\frac{8.25\ \text{M}}{8.25\ \text{M} + 100\ \text{k}}\right)(5\ \text{m})(4.7\ \text{k} \| 10\ \text{k}) \\ &= -\left(\frac{8.25\ \text{M}}{8.35\ \text{M}}\right)(5\ \text{m})\left(\frac{(4.7\ \text{k})(10\ \text{k})}{4.7\ \text{k} + 10\ \text{k}}\right) \\ &= -(0.99)(5\ \text{m})\left(\frac{(4.7\ \text{k})(10\ \text{k})}{14.7\ \text{k}}\right) \\ &= -(4.95\ \text{m})(3.2\ \text{k}) \end{aligned}$$

$$A_M = -15.8\ \text{V/V}$$

Therefore, the value of the gain A_M is $\boxed{-15.8\ \text{V/V}}$.

Step 3 of 7

Calculate the value of the frequency f_{P1}.

$$f_{P1} = \frac{1}{2\pi C_{C1}(R_{sig} + R_G)}$$

Substitute $8.25\ \text{M}\Omega$ for R_G, $100\ \text{k}\Omega$ for R_{sig} and $0.01\ \mu\text{F}$ for C_{C1}.

$$\begin{aligned} f_{P1} &= \frac{1}{2\pi(0.01\ \mu)(100\ \text{k} + 8.25\ \text{M})} \\ &= \frac{1}{(0.063\ \mu)(8.35\ \text{M})} \\ &= \frac{1}{0.526} \\ &= 1.9\ \text{Hz} \end{aligned}$$

Therefore, the value of the frequency is f_{P1} is $\boxed{1.9\ \text{Hz}}$.

Step 4 of 7

Calculate the value of the frequency f_{P2}.

$$f_{P2} = \frac{g_m + \frac{1}{R_S}}{2\pi C_S}$$

Substitute $2\ \text{k}\Omega$ for R_S, $5\ \text{mA/V}$ for g_m and $10\ \mu\text{F}$ for C_S.

$$\begin{aligned} f_{P2} &= \frac{5\ \text{m} + \frac{1}{2\ \text{k}}}{2\pi(10\ \mu)} \\ &= \frac{5.5\ \text{m}}{62.83\ \mu} \\ &= 87.54\ \text{Hz} \end{aligned}$$

Therefore, the value of the frequency is f_{P2} is $\boxed{87.54\ \text{Hz}}$.

Step 5 of 7

Calculate the value of the frequency f_{P3}.

$$f_{P3} = \frac{1}{2\pi C_S(R_D + R_L)}$$

Substitute $4.7\ \text{k}\Omega$ for R_D, $10\ \text{k}\Omega$ for R_L and $1\ \mu\text{F}$ for C_{C2}.

$$\begin{aligned} f_{P3} &= \frac{1}{2\pi(1\ \mu)(4.7\ \text{k} + 10\ \text{k})} \\ &= \frac{1}{(6.3\ \mu)(5.7\ \text{M})} \\ &= \frac{1}{35.91} \\ &= 0.03\ \text{Hz} \end{aligned}$$

Therefore, the value of the frequency is f_{P3} is $\boxed{0.03\ \text{Hz}}$.

Step 6 of 7

Calculate the value of the frequency f_Z.

$$f_Z = \frac{1}{2\pi C_S R_S}$$

Substitute C_S for $10\ \mu\text{F}$ and $2\ \text{k}\Omega$ for R_S.

$$\begin{aligned} f_Z &= \frac{1}{2\pi(10\ \mu)(2\ \text{k})} \\ &= \frac{1}{0.126} \\ &= 7.94\ \text{Hz} \end{aligned}$$

Therefore, the value of the frequency f_Z is $\boxed{7.94\ \text{Hz}}$.

Step 7 of 7

The value of the frequency f_L.

$$f_L = \sqrt{f_{P1}^2 + f_{P2}^2 + f_{P3}^2 - 2f_Z^2}$$

Substitute $7.94\ \text{Hz}$ for f_Z, $0.03\ \text{Hz}$ for f_{P3}, $87.54\ \text{Hz}$ for f_{P2} and $1.9\ \text{Hz}$ for f_{P1}.

$$\begin{aligned} f_L &= \sqrt{(1.9)^2 + (87.54)^2 + (0.03)^2 - 2(7.94)^2} \\ &= \sqrt{3.61 + 7663.25 + 0.0009 - 2(63.04)} \\ &= \sqrt{3.61 + 7663.25 + 0.0009 - 126.08} \\ &= \sqrt{7666.86 - 126.08} \end{aligned}$$

$$\begin{aligned} f_L &= \sqrt{7540.78} \\ &= 86.63\ \text{Hz} \end{aligned}$$

Therefore, he value of the frequency f_L is $\boxed{86.63\ \text{Hz}}$.

Step 1 of 6

Refer to Table 9.2 in the textbook for the BJT-high frequency model.

Write the expression for the small signal diffusion capacitance.

$$C_{de} = \tau_F g_m$$
$$= \tau_F \frac{I_C}{V_T}$$

Here,

The bias or collector current, I_C

The thermal voltage, V_T

The device constant with the dimension of time, τ_F .

Substitute $20\ \text{ps}$ for τ_F , 1 mA for I_C , and 25 mV for V_T .

$$C_{de} = 20\times10^{-12}\left(\frac{1\times10^{-3}}{25\times10^{-3}}\right)$$
$$= 0.8\ \text{pF}$$

Thus, the small signal diffusion capacitance is, $\boxed{0.8\ \text{pF}}$.

Step 2 of 6

Write the expression for the EBJ (emitter base junction) depletion layer capacitance.

$$C_{je} = 2C_{je0}$$

Here,

The common-emitter Junction (EBJ) depletion layer capacitance at zero EBJ voltage, C_{je0}

Substitute $20\ \text{fF}$ for C_{je0} .

$$C_{je} = 2(20\ \text{fF})$$
$$= 40\ \text{fF}$$

Thus, the EBJ depletion layer capacitance is, $\boxed{40\ \text{fF}}$.

Step 3 of 6

Write the expression for the emitter-base capacitance.

$$C_\pi = C_{de} + C_{je}$$

Here,

The small signal diffusion capacitance, C_{de}

The EBJ (emitter base junction) depletion layer capacitance, C_{je}

Substitute 0.8 pF for C_{de} and 40 fF for C_{je} .

$$C_\pi = \left(0.8\times10^{-12}\right)+\left(40\times10^{-15}\right)$$
$$= \left(0.8\times10^{-12}\right)+\left(0.04\times10^{-12}\right)$$
$$= 0.84\ \text{pF}$$

Thus, the emitter-base capacitance is, $\boxed{0.84\ \text{pF}}$.

Step 4 of 6

Write the expression for the depletion capacitance of the CBJ.

$$C_\mu = \frac{C_{\mu0}}{\left(1+\frac{|V_{CB}|}{V_{0c}}\right)^m}$$

Here,

The common-base Junction (CBJ) depletion capacitance at zero voltage, $C_{\mu0}$

The common-base Junction (CBJ) reverse bias voltage, V_{CB}

The CBJ built in voltage, V_{0c}

The grading coefficient of the CBJ, m

Substitute 20 fF for $C_{\mu0}$, 2 V for V_{CB} , 0.5 V for V_{0c} , and 0.33 for m .

$$C_\mu = \frac{20\times10^{-15}}{\left(1+\frac{2}{0.5}\right)^{0.33}}$$
$$= \frac{20\times10^{-15}}{(5)^{0.33}}$$
$$= 11.76\ \text{fF}$$

Thus, the depletion capacitance of the CBJ is, $\boxed{11.76\ \text{fF}}$.

Step 5 of 6

Write the expression for the transition frequency.

$$f_T = \frac{g_m}{2\pi\left(C_\pi + C_\mu\right)} \quad \left\{\text{here}, g_m = \frac{I_C}{V_T}\right\}$$

Here,

The trans-conductance, g_m

The depletion capacitance of the CBJ, C_μ

The emitter-base capacitance, C_π

Step 6 of 6

Substitute 0.04 A/V for g_m , 11.76 fF for C_μ , and 0.84 pF for C_π .

$$f_T = \frac{0.04}{2\pi\left(0.84\times10^{-12}+11.76\times10^{-15}\right)}$$
$$= \frac{0.04}{2\pi\left(0.84\times10^{-12}+0.01176\times10^{-12}\right)}$$
$$= \frac{0.04}{2\pi\left(0.85176\times10^{-12}\right)}$$
$$= 7.47\ \text{GHz}$$

Thus, the transition frequency is, $\boxed{7.47\ \text{GHz}}$.

Step 1 of 25

Refer to Figure P18.5 from the text book.

(a)

Draw the small signal equivalent circuit diagram to calculate the value of mid band gain. Capacitor acts as short circuit.

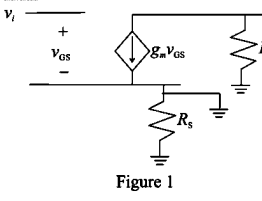


Figure 1

Step 2 of 25

Consider the expression for the output voltage v_o.

$$v_o = -g_m v_{GS} R_0$$

$$v_{GS} = \frac{v_o}{-g_m R_0}$$

Therefore, the expression for the voltage v_{GS} is $\frac{v_o}{-g_m R_0}$.

Step 3 of 25

Consider the expression for the input voltage v_{in}.

$$v_{in} = v_{GS}$$

Substitute $\frac{v_o}{-g_m R_0}$ for v_{GS}.

$$v_{in} = \frac{v_o}{-g_m R_0}$$

$$\frac{v_o}{v_{in}} = -g_m R_0 \quad \ldots (1)$$

Therefore, the expression for the voltage gain $\frac{v_o}{v_{in}}$ is $-g_m R_0$.

Step 4 of 25

Recall equation (1).

$$\frac{v_o}{v_{in}} = -g_m R_0$$

Substitute -20 V/V for $\frac{v_o}{v_{in}}$ and 2 mA/V for g_m.

$$-20 = -(2\text{ m})R_0$$

$$R_0 = \frac{20}{(2\text{ m})}$$

$$R_0 = 10\text{ k}\Omega$$

Therefore, the value of the resistance R_0 is $\boxed{10\text{ k}\Omega}$.

Step 5 of 25

(b)

Draw the small signal equivalent circuit diagram including the capacitance C_S.

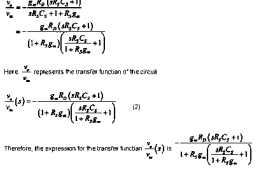


Figure 2

Step 6 of 25

Consider the expression for the output voltage v_o.

$$v_o = -g_m v_{GS} R_0$$

$$v_{GS} = \frac{v_o}{-g_m R_0}$$

Therefore, the expression for the voltage v_{GS} is $\frac{v_o}{-g_m R_0}$.

Step 7 of 25

Consider the expression for the input voltage v_{in}.

$$v_{in} = v_{GS} + g_m v_{GS}(R_S \| sC_S)$$

$$= v_{GS}(1 + g_m(R_S \| sC_S))$$

Substitute $\frac{v_o}{-g_m R_0}$ for v_{GS}.

$$v_{in} = \frac{v_o}{-g_m R_0}(1 + g_m(R_S \| sC_S))$$

$$\frac{v_o}{v_{in}} = -\frac{g_m R_0}{1 + g_m\left(R_S \| \frac{1}{sC_S}\right)}$$

$$\frac{v_o}{v_{in}} = -\frac{g_m R_0}{1 + g_m\left(\frac{R_S \cdot \frac{1}{sC_S}}{R_S + \frac{1}{sC_S}}\right)}$$

$$\frac{v_o}{v_{in}} = -\frac{g_m R_0}{1 + g_m\left(\frac{R_S}{sR_S C_S + 1}\right)}$$

$$\frac{v_o}{v_{in}} = -\frac{g_m R_0(sR_S C_S + 1)}{sR_S C_S + 1 + R_S g_m}$$

$$= -\frac{g_m R_0(sR_S C_S + 1)}{(1 + R_S g_m)\left(\frac{sR_S C_S}{1 + R_S g_m} + 1\right)}$$

Here $\frac{v_o}{v_{in}}$ represents the transfer function of the circuit

$$\frac{v_o}{v_{in}}(s) = -\frac{g_m R_0(sR_S C_S + 1)}{(1 + R_S g_m)\left(\frac{sR_S C_S}{1 + R_S g_m} + 1\right)} \quad (2)$$

Therefore, the expression for the transfer function $\frac{v_o}{v_{in}}(s)$ is $-\frac{g_m R_0(sR_S C_S + 1)}{(1 + R_S g_m)\left(\frac{sR_S C_S}{1 + R_S g_m} + 1\right)}$.

Step 8 of 25

Recall equation (2).

$$\frac{v_o}{v_{in}}(s) = -\frac{g_m R_0(sR_S C_S + 1)}{(1 + R_S g_m)\left(\frac{sR_S C_S}{1 + R_S g_m} + 1\right)}$$

Equate denominator to zero.

$$g_m R_0(sR_S C_S + 1) = 0$$

$$sR_S C_S + 1 = 0$$

$$s + \frac{1}{R_S C_S} = 0$$

Here, the term $\frac{1}{R_S C_S}$ represents the pole frequency ω_p due to the components R_S and C_S.

$$\omega_p = \frac{1}{R_S C_S}$$

Substitute $2\pi f_p$ for ω_p.

$$2\pi f_p = \frac{1}{R_S C_S} \quad \ldots (3)$$

$$f_p = \frac{1}{2\pi R_S C_S}$$

Therefore, the expression for the pole frequency f_p is $\frac{1}{2\pi R_S C_S}$.

Step 9 of 25

Recall equation (3).

$$f_p = \frac{1}{2\pi R_S C_S}$$

Substitute 100 Hz for f_p, 2 mA/V for g_m and 4.5 kΩ for R_S.

$$100 = \frac{1}{2\pi(4.5\text{ k})C_S}$$

$$C_S = \frac{1}{2\pi(4.5\text{ k})(100)}$$

$$C_S = \frac{10}{2.83\text{ M}}$$

$$C_S = 3.53\ \mu\text{F}$$

Therefore, the required value of the capacitance C_S is $\boxed{3.53\ \mu\text{F}}$.

Step 10 of 25

(c)

Recall equation (2)

$$\frac{v_o}{v_{in}}(s) = -\frac{g_m R_0(sR_S C_S + 1)}{(1 + R_S g_m)\left(\frac{sR_S C_S}{1 + R_S g_m} + 1\right)}$$

Equate numerator to zero.

$$g_m R_0(sR_S C_S + 1) = 0$$

$$sR_S C_S + 1 = 0$$

$$s + \frac{1}{R_S C_S} = 0$$

Here, the term $\frac{1}{R_S C_S}$ represents the zero frequency ω_z due to the components R_S and C_S.

$$\omega_z = \frac{1}{R_S C_S}$$

Substitute $2\pi f_z$ for ω_z.

$$2\pi f_z = \frac{1}{R_S C_S} \quad (4)$$

$$f_z = \frac{1}{2\pi R_S C_S}$$

Therefore, the expression for the zero frequency f_z is $\frac{1}{2\pi R_S C_S}$.

Step 11 of 25

Recall equation (4).

$$f_z = \frac{1}{2\pi R_S C_S}$$

Substitute 3.53 μF for C_S and 4.5 kΩ for R_S.

$$f_z = \frac{1}{2\pi(4.5\text{ k})(3.53\ \mu)}$$

$$= \frac{1}{99.81\text{ m}}$$

$$= 10\text{ Hz}$$

Therefore, the value of the zero frequency f_z is $\boxed{10\text{ Hz}}$.

Step 12 of 25

(d)

The value of the 3-dB frequency f_L is approximately equal to the pole frequency f_p of the circuit.

Therefore, the value of the frequency is $\boxed{100\text{ Hz}}$.

Step 13 of 25

(e)

Recall equation (2)

$$\frac{v_o}{v_{in}}(s) = -\frac{g_m R_0(sR_S C_S + 1)}{(1 + R_S g_m)\left(\frac{sR_S C_S}{1 + R_S g_m} + 1\right)}$$

Substitute $j\omega$ for s.

$$\frac{v_o}{v_{in}}(j\omega) = -\frac{g_m R_0(j\omega R_S C_S + 1)}{(1 + R_S g_m)\left(\frac{j\omega R_S C_S}{1 + R_S g_m} + 1\right)}$$

Substitute 2 mA/V for g_m, 4.5 kΩ for R_S, 3.53 μF for C_S and 10 kΩ for R_0.

$$\frac{v_o}{v_{in}}(j\omega) = -\frac{(2\text{ m})(10\text{ k})(j\omega(4.5\text{ k})(3.53\ \mu)+1)}{(1+(2\text{ m})(4.5\text{ k}))\left(\frac{j\omega(4.5\text{ k})(3.53\ \mu)}{1+(2\text{ m})(4.5\text{ k})}+1\right)}$$

$$= \frac{(20)\left(j\omega(15.885\text{ m})+1\right)}{(1+9)\left(j\omega\left(\frac{15.885\text{ m}}{1+9}\right)+1\right)}$$

$$= \frac{(20)(j\omega(15.885\text{ m})+1)}{(10)(j\omega(1.5885\text{ m})+1)}$$

$$= \frac{(2)(j\omega(15.885\text{ m})+1)}{(j\omega(1.5885\text{ m})+1)}$$

Therefore, the expression for the transfer function $\frac{v_o}{v_{in}}(j\omega)$ is $\frac{(2)(j\omega(15.885\text{ m})+1)}{(j\omega(1.5885\text{ m})+1)}$.

Step 14 of 25

Consider the absolute value of the function

$$\left|\frac{v_o}{v_{in}}(j\omega)\right| = \frac{(2)(j\omega(15.885\text{ m})+1)}{(j\omega(1.5885\text{ m})+1)}$$

Here, 2 represents the dc gain of the amplifier.

Step 15 of 25

Consider Table 1 representing the slopes and change in slopes for the frequencies.

Table 1

function	Corner Frequency (rad/sec)	Slope (dB/dec)	Change in slope (dB/dec)
2	.	0	.
$j\omega(15.885\text{ m})+1$	$\omega_{c1} = \frac{1}{15.885\text{ m}} = 62.95$	20	$\Delta\omega_1 = 0+20 = 20$
$\frac{1}{j\omega(1.5885\text{ m})+1}$	$\omega_{c2} = \frac{1}{1.585\text{ m}} = 629.55$	-20	$\Delta\omega_2 = 20-20 = 0$

Step 16 of 25

Assume the lower and the higher corner frequencies to be $\omega_1 = 10$ rad/sec and $\omega_2 = 1000$ rad/sec respectively.

Step 17 of 25

Calculate the gain at ω_1.

$$A_1 = 20\log(2)$$

$$= 6.02\text{ dB}$$

Therefore, the value of the gain at ω_1 is 6.02 dB.

Step 18 of 25

Calculate the gain at ω_{c1}.

$$A_2 = 20\log(40)$$

$$= 6.02\text{ dB}$$

Therefore, the value of the gain at ω_{c1} is 6.02 dB.

Step 19 of 25

Calculate the gain at ω_{c2}.

$$A_3 = \Delta\omega_1 \times \log\left(\frac{\omega_{c2}}{\omega_{c1}}\right) + A_2$$

Substitute 6.02 dB for A_2, -20 dB/dec for $\Delta\omega_1$, 62.95 Hz for ω_{c1} and 629.55 Hz for ω_{c2}.

$$A_3 = 20 \times \log\left(\frac{629.55}{62.95}\right) + 6.82$$

$$= 20 \times \log(10) + 6.02$$

$$= 20 + 6.02$$

$$= 26.02\text{ dB}$$

Therefore, the value of the gain at ω_{c2} is 26.02 dB.

Step 20 of 25

Calculate the gain at ω_2.

$$A_4 = \Delta\omega_2 \times \log\left(\frac{\omega_2}{\omega_{c2}}\right) + A_3$$

Substitute 26.02 dB for A_3, 0 dB/dec for $\Delta\omega_2$, 629.55 Hz for ω_{c2} and 1000 Hz for ω_2.

$$A_4 = 0 \times \log\left(\frac{1000}{629.55}\right) + 26.82$$

$$= 0 \times \log(10) + 26.02$$

$$= 0 + 26.02$$

$$= 26.02\text{ dB}$$

Therefore, the value of the gain at ω_2 is 26.02 dB.

Step 21 of 25

Plot the bode plot using the values of the gains against their corner frequencies

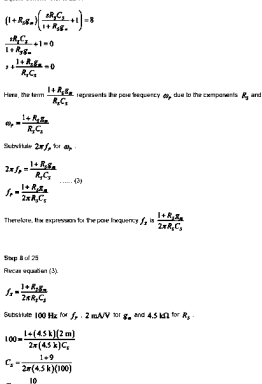


Figure 3

Step 22 of 25

Draw the small signal equivalent circuit diagram to calculate the value of dc gain

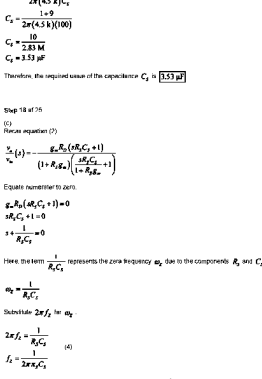


Figure 4

Step 23 of 25

Consider the expression for the output voltage v_o.

$$v_o = -g_m v_{GS} R_0$$

$$v_{GS} = \frac{v_o}{-g_m R_0}$$

Therefore, the expression for the voltage v_{GS} is $\frac{v_o}{-g_m R_0}$.

Step 24 of 25

Consider the expression for the input voltage v_{in}.

$$v_{in} = v_{GS}$$

Substitute $\frac{v_o}{-g_m R_0}$ for v_{GS}.

$$v_{in} = \frac{-v_o(1 + g_m R_S)}{g_m R_0}$$

$$\frac{v_o}{v_{in}} = -\frac{g_m R_0}{1 + g_m R_S} \quad \ldots (5)$$

Therefore, the expression for the voltage gain $\frac{v_o}{v_{in}}$ is $-\frac{g_m R_0}{1 + g_m R_S}$.

Step 25 of 25

Recall equation (1).

$$\frac{v_o}{v_{in}} = -\frac{g_m R_0}{1 + g_m R_S}$$

Consider the absolute value.

$$\left|\frac{v_o}{v_{in}}\right| = \frac{g_m R_0}{1 + g_m R_S}$$

Substitute 10 kΩ for R_0, 4.5 kΩ for R_S and 2 mA/V for g_m.

$$\left|\frac{v_o}{v_{in}}\right| = \frac{(2\text{ m})(10\text{ k})}{1 + (2\text{ m})(4.5\text{ k})}$$

$$= \frac{20}{1+9}$$

$$= \frac{20}{10}$$

$$= 2\text{ V/V}$$

Therefore, the value of the dc gain $\left|\frac{v_o}{v_{in}}\right|$ is 2 V/V.

The value at the dc gain is 2 V/V equivalent to 6.02 dB in the plot and occurs as the capacitor acts as open circuit at dc powers.

Step 1 of 4

Write the expression for the transition frequency.

$$f_T = \frac{g_m}{2\pi\left(C_\pi + C_\mu\right)}$$

Calculate the trans-conductance.

$$\begin{aligned} g_m &= \frac{I_C}{V_T} \\ &= \frac{1\times10^{-3}}{25\times10^{-3}} \\ &= 0.04\ \text{A/V} \end{aligned}$$

Write the expression for the common-emitter short circuit gain.

$$h_{fe} = \frac{g_m r_\pi}{\sqrt{1+\left(C_\pi + C_\mu\right)^2 r_\pi^2}}$$

$$10 = \frac{g_m r_\pi}{\sqrt{1+\left(C_\pi + C_\mu\right)^2 r_\pi^2}} \quad \text{..... (1)}$$

Write the expression for the 3 dB frequency.

$$\omega_\beta = \frac{1}{\left(C_\pi + C_\mu\right) r_\pi}$$

Substitute 50 MHz for f_β

$$\frac{1}{\left(C_\pi + C_\mu\right) r_\pi} = 2\pi\left(50\times10^6\right)$$

$$\left(C_\pi + C_\mu\right) r_\pi = \frac{1}{100\pi\times10^6}$$

$$\left(C_\pi + C_\mu\right) r_\pi = 3.183\times10^{-9} \quad \text{..... (2)}$$

Step 2 of 4

Substitute equation (2) in equation (1).

$$10 = \frac{0.04 r_\pi}{\sqrt{1+\left(3.183\times10^{-9}\right)^2}}$$

$$\begin{aligned} r_\pi &= \frac{10\sqrt{1+\left(3.183\times10^{-9}\right)^2}}{0.04} \\ &= 250\ \Omega \end{aligned}$$

Substitute $250\ \Omega$ for r_π in equation (2).

$$\left(C_\pi + C_\mu\right)250 = 3.183\times10^{-9}$$

$$C_\pi + C_\mu = 12.73\times10^{-12}$$

Step 3 of 4

Substitute $12.73\times10^{-12}\text{F}$ for $C_\pi + C_\mu$ to calculate the transition frequency.

$$\begin{aligned} f_T &= \frac{0.04}{2\pi\left(12.73\times10^{-12}\right)} \\ &= 500\ \text{MHz} \end{aligned}$$

Thus, the transition frequency is, $\boxed{500\ \text{MHz}}$.

Step 4 of 4

Consider the following expression.

$$C_\pi + C_\mu = 12.74\times10^{-12}$$

Substitute 2 pF for C_μ.

$$\begin{aligned} C_\pi + 2\times10^{-12} &= 12.73\times10^{-12} \\ C_\pi &= 12.73\times10^{-12} - 2\times10^{-12} \\ C_\pi &= 10.73\times10^{-12} \\ C_\pi &= 10.73\ \text{pF} \end{aligned}$$

Thus, the emitter-base capacitance is, $\boxed{10.73\ \text{pF}}$.

Step 1 of 14

Refer to Figure P10.6 from the text book.

Draw the small signal equivalent circuit diagram to calculate the value of the mid band gain.

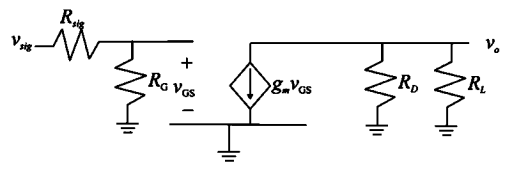


Figure 1

Step 2 of 14

Consider the expression for the voltage v_{GS} in terms of the voltage v_{sig}.

$$v_{GS} = v_{sig}\left(\frac{R_G}{R_G + R_{sig}}\right)$$

Therefore, the expression fro the voltage v_{GS} is $v_{sig}\left(\frac{R_G}{R_G + R_{sig}}\right)$.

Step 3 of 14

Consider the expression for the output voltage v_o.

$$v_o = -g_m v_{GS}\left(R_D \parallel R_L\right)$$

Substitute $v_{sig}\left(\frac{R_G}{R_G + R_{sig}}\right)$ for v_{GS}.

$$v_o = -g_m v_{sig}\left(\frac{R_G}{R_G + R_{sig}}\right)\left(R_D \parallel R_L\right)$$

$$\frac{v_o}{v_{sig}} = -g_m\left(\frac{R_G}{R_G + R_{sig}}\right)\left(\frac{R_D R_L}{R_D + R_L}\right)$$

Here, the term $\frac{v_o}{v_{sig}}$ represents the mid band gain A_M of the amplifier.

$$A_M = -\frac{g_m R_G R_D R_L}{\left(R_G + R_{sig}\right)\left(R_D + R_L\right)} \quad \text{..... (1)}$$

Therefore, the expression for the mid-band gain is $-\frac{g_m R_G R_D R_L}{\left(R_G + R_{sig}\right)\left(R_D + R_L\right)}$.

Step 4 of 14

Recall equation (1).

$$A_M = -\frac{g_m R_G R_D R_L}{\left(R_G + R_{sig}\right)\left(R_D + R_L\right)}$$

Substitute $3\ \text{mA/V}$ for g_m, $2\ \text{M}\Omega$ for R_G, $20\ \text{k}\Omega$ for R_D, $0.5\ \text{M}\Omega$ for R_{sig} and $10\ \text{k}\Omega$ for R_L.

$$\begin{aligned} A_M &= -\frac{(3\ \text{m})(2\ \text{M})(20\ \text{k})(10\ \text{k})}{(2\ \text{M} + 0.5\ \text{M})(20\ \text{k} + 10\ \text{k})} \\ &= -\frac{1.2\times10^{12}}{(2.5\ \text{M})(30\ \text{k})} \\ &= -\frac{1.2\times10^{12}}{7.5\times10^{10}} \\ &= -16\ \text{V/V} \end{aligned}$$

Therefore, the value of the voltage gain A_M is $\boxed{-16\ \text{V/V}}$.

Step 5 of 14

Consider the small signal equivalent circuit to calculate the low frequency poles.

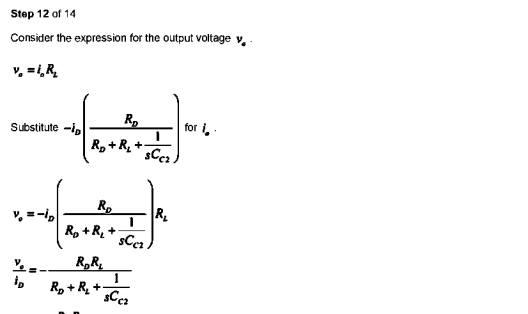


Figure 2

Step 6 of 14

The highest frequency in the low frequency amplifier is the frequency due to the capacitor C_{C2} followed by the capacitors C_{C3} and C_{C1}. Hence, the values of the frequencies are $f_1 = 1\ \text{Hz}$, $f_2 = 100\ \text{Hz}$ and $f_3 = 10\ \text{Hz}$.

Step 7 of 14

Consider the expression for the voltage v_G.

$$v_G = v_{sig}\left(\frac{R_G}{R_G + R_{sig} + \frac{1}{sC_{C1}}}\right)$$

$$\frac{v_G}{v_{sig}} = \frac{R_G}{R_G + R_{sig} + \frac{1}{sC_{C1}}}$$

$$\frac{v_G}{v_{sig}} = \frac{R_G}{R_G + R_{sig}} \cdot \frac{s}{s + \frac{1}{C_{C1}\left(R_G + R_{sig}\right)}}$$

Consider the expression for the pole ω_{P1} due to the capacitor C_{C1}.

$$\omega_{P1} = \frac{1}{C_{C1}\left(R_G + R_{sig}\right)} \quad \text{..... (2)}$$

Therefore, the expression for the pole ω_{P1} is $\frac{1}{C_{C1}\left(R_G + R_{sig}\right)}$.

Step 8 of 14

Recall equation (2).

$$\omega_{P1} = \frac{1}{C_{C1}\left(R_G + R_{sig}\right)}$$

Substitute $2\pi f_1$ for ω_{P1}.

$$2\pi f_1 = \frac{1}{C_{C1}\left(R_G + R_{sig}\right)}$$

$$C_{C1} = \frac{1}{2\pi f_1\left(R_G + R_{sig}\right)}$$

Substitute $1\ \text{Hz}$ for f_1, $2\ \text{M}\Omega$ for R_G and $0.5\ \text{M}\Omega$ for R_{sig}.

$$\begin{aligned} C_{C1} &= \frac{1}{2\pi(1)(2\ \text{M} + 0.5\ \text{M})} \\ &= \frac{1}{(6.28)(2.5\ \text{M})} \\ &= \frac{1}{15.7\ \text{M}} \\ &= 63.69\ \text{nF} \end{aligned}$$

Therefore, the value of the capacitor C_{C1} is $\boxed{63.69\ \text{nF}}$.

Step 9 of 14

Consider the expression for the voltage v_s.

$$v_s = i_D\left(\frac{1}{g_m} + \frac{1}{sC_s}\right)$$

$$v_s = i_D\frac{1}{g_m}\left(s + \frac{g_m}{C_s}\right)$$

$$\frac{i_D}{v_s} = \frac{g_m}{s + \frac{g_m}{C_s}}$$

Consider the pole due to the capacitor C_s.

$$\omega_{P2} = \frac{g_m}{C_s} \quad \text{..... (3)}$$

Therefore, the expression for the pole ω_{P2} is $\frac{g_m}{C_s}$.

Step 10 of 14

Recall equation (3).

$$\omega_{P2} = \frac{g_m}{C_s}$$

Substitute $2\pi f_2$ for ω_{P2}.

$$2\pi f_2 = \frac{g_m}{C_s}$$

$$C_s = \frac{g_m}{2\pi f_2}$$

Substitute $3\ \text{mA/V}$ for g_m and $100\ \text{Hz}$ for f_{P2}.

$$\begin{aligned} C_s &= \frac{3\ \text{m}}{2\pi(100)} \\ &= \frac{3\ \text{m}}{628.32} \\ &= 4.78\ \mu\text{F} \end{aligned}$$

Therefore, the value of the capacitor C_S is $\boxed{4.78\ \mu\text{F}}$.

Step 11 of 14

Consider the expression for the output current i_o.

$$i_o = -i_D\left(\frac{R_D}{R_D + R_L + \frac{1}{sC_{C2}}}\right)$$

Therefore, the expression for the output current i_o is $-i_D\left(\frac{R_D}{R_D + R_L + \frac{1}{sC_{C2}}}\right)$.

Step 12 of 14

Consider the expression for the output voltage v_o.

$$v_o = i_o R_L$$

Substitute $-i_D\left(\frac{R_D}{R_D + R_L + \frac{1}{sC_{C2}}}\right)$ for i_o.

$$v_o = -i_D\left(\frac{R_D}{R_D + R_L + \frac{1}{sC_{C2}}}\right)R_L$$

$$\frac{v_o}{i_D} = -\frac{R_D R_L}{R_D + R_L + \frac{1}{sC_{C2}}}$$

$$\frac{v_o}{i_D} = -\frac{R_D R_L}{R_D + R_L} \cdot \frac{s}{s + \frac{1}{C_{C2}\left(R_D + R_L\right)}}$$

Consider the pole ω_{P3} due to the capacitor C_{C2}.

$$\omega_{P3} = \frac{1}{C_{C2}\left(R_D + R_L\right)} \quad \text{..... (4)}$$

Therefore, the expression for the pole ω_{P3} is $\frac{1}{C_{C2}\left(R_D + R_L\right)}$.

Step 13 of 14

Recall equation (4).

$$\omega_{P3} = \frac{1}{C_{C2}\left(R_D + R_L\right)}$$

Substitute $2\pi f_3$ for ω_{P2}.

$$2\pi f_3 = \frac{1}{C_{C2}\left(R_D + R_L\right)}$$

$$C_{C2} = \frac{1}{2\pi f_3\left(R_D + R_L\right)}$$

Substitute $10\ \text{Hz}$ for f_3, $20\ \text{k}\Omega$ for R_D and $10\ \text{k}\Omega$ for R_L.

$$\begin{aligned} C_{C2} &= \frac{1}{2\pi(10)(20\ \text{k} + 10\ \text{k})} \\ &= \frac{1}{(62.83)(30\ \text{k})} \\ &= \frac{1}{1.89\ \text{M}} \\ &= 0.53\ \mu\text{F} \end{aligned}$$

Therefore, the value of the capacitor C_{C2} is $\boxed{0.53\ \mu\text{F}}$.

Step 14 of 14

The value of the dominant pole frequency is approximately equal to the pole frequency due to the capacitor C_s. That is the frequency f_2. Therefore, the value of the dominant pole frequency is $\boxed{100\ \text{Hz}}$.

10.007E

Step 1 of 3

Write the expression for the trans-conductance.

$$g_m = \frac{I_C}{V_T}$$

Substitute 0.1 mA for I_C and 25 mV for V_T .

$$g_m = \frac{0.1\times10^{-3}}{25\times10^{-3}}$$
$$= 0.004 \text{ A/V}$$

The common-emitter short circuit gain is,

$$h_{fe} = \frac{g_m r_\pi}{\sqrt{1+\left(C_\pi + C_\mu\right)^2 r_\pi^2}}$$

$$10 = \frac{g_m r_\pi}{\sqrt{1+\left(C_\pi + C_\mu\right)^2 r_\pi^2}} \quad \text{..... (1)}$$

The 3 dB frequency is,

$$\omega_\beta = \frac{1}{\left(C_\pi + C_\mu\right) r_\pi}$$

$$\frac{1}{\left(C_\pi + C_\mu\right) r_\pi} = 100\pi\times10^6$$

$$\left(C_\pi + C_\mu\right) r_\pi = \frac{1}{100\pi\times10^6}$$

$$\left(C_\pi + C_\mu\right) r_\pi = 3.185\times10^{-9} \quad \text{..... (2)}$$

Step 2 of 3

Substitute equation (2) in equation (1).

$$10 = \frac{0.004 r_\pi}{\sqrt{1+\left(3.185\times10^{-9}\right)^2}}$$

$$r_\pi = \frac{10\sqrt{1+\left(3.185\times10^{-9}\right)^2}}{0.004}$$
$$= 2.5 \text{ k}\Omega$$

Substitute $2.5 \text{ k}\Omega$ for r_π in equation (2).

$$\left(C_\pi + C_\mu\right)2.5\times10^3 = 3.185\times10^{-9}$$
$$C_\pi + C_\mu = 1.274\times10^{-12}$$

Step 3 of 3

Write the expression for the transition frequency.

$$f_T = \frac{g_m}{2\pi\left(C_\pi + C_\mu + 2\text{pF}\right)}$$

Here,

The trans-conductance is, g_m

Calculate the unity band width frequency.

$$f_T = \frac{0.004}{2\pi\left(3.274\times10^{-12}\right)}$$
$$= 194 \text{ MHz}$$

Thus, the transition frequency f_T is, **194 MHz** .

Step 1 of 10

10.007P

Refer to Figure P10.7 from the text book.

The circuit is similar to the Figure 10.9 from the text book with the resistor $R_E = \infty$ and $R_{B1} \| R_{B2}$ as the resistor R_B.

Consider the expression for the time constant associated with capacitor C_E.

$$\tau_{CE} = C_E R_{CE}$$

Substitute $R_E \| \left[r_e + \frac{R_B \| R_{sig}}{\beta + 1} \right]$ for R_{CE} and $\frac{1}{2\pi f_2}$ for τ_{CE}.

$$\begin{aligned}
\frac{1}{2\pi f_2} &= C_E \left(R_E \| \left[r_e + \frac{R_B \| R_{sig}}{\beta + 1} \right] \right) \\
&= \frac{C_E \cdot \left(R_E \left[r_e + \frac{R_B \| R_{sig}}{\beta + 1} \right] \right)}{\left(R_E + \left[r_e + \frac{R_B \| R_{sig}}{\beta + 1} \right] \right)} \\
&= \frac{C_E \cdot \left(R_E \left[r_e + \frac{R_B \| R_{sig}}{\beta + 1} \right] \right)}{R_E \left(1 + \frac{1}{R_E} \left[r_e + \frac{R_B \| R_{sig}}{\beta + 1} \right] \right)} \\
&= \frac{C_E \cdot \left(\left[r_e + \frac{R_B \| R_{sig}}{\beta + 1} \right] \right)}{\left(1 + \frac{1}{R_E} \left[r_e + \frac{R_B \| R_{sig}}{\beta + 1} \right] \right)}
\end{aligned}$$

Substitute ∞ for R_E.

$$\begin{aligned}
\frac{1}{2\pi f_2} &= \frac{C_E \cdot \left(\left[r_e + \frac{R_B \| R_{sig}}{\beta + 1} \right] \right)}{\left(1 + \frac{1}{\infty} \left[r_e + \frac{R_B \| R_{sig}}{\beta + 1} \right] \right)} \\
&= \frac{C_E \cdot \left(\left[r_e + \frac{R_B \| R_{sig}}{\beta + 1} \right] \right)}{\left(1 + (0) \left[r_e + \frac{R_B \| R_{sig}}{\beta + 1} \right] \right)} \\
&= \frac{C_E \cdot \left(\left[r_e + \frac{R_B \| R_{sig}}{\beta + 1} \right] \right)}{(1 + 0)} \\
&= C_E \left[r_e + \frac{R_B \| R_{sig}}{\beta + 1} \right]
\end{aligned}$$

$$\frac{1}{2\pi f_2} = C_E \left[r_e + \frac{R_B R_{sig}}{(\beta + 1)(R_B + R_{sig})} \right]$$

$$C_E = \frac{1}{2\pi f_2 \left[r_e + \frac{R_B R_{sig}}{(\beta + 1)(R_B + R_{sig})} \right]} \quad \text{..... (1)}$$

Therefore, the expression for the capacitor C_E is $\frac{1}{2\pi f_2 \left[r_e + \frac{R_B R_{sig}}{(\beta + 1)(R_B + R_{sig})} \right]}$.

Step 2 of 10

Calculate the value of the transconductance g_m.

$$g_m = \frac{I_C}{V_T}$$

Substitute $25\ \text{mV}$ for V_T and $100\ \mu\text{A}$ for I_C.

$$\begin{aligned}
g_m &= \frac{100\ \mu}{25\ \text{m}} \\
&= 4\ \text{mA/V}
\end{aligned}$$

Therefore, the value of the transconductance g_m is $4\ \text{mA/V}$.

Step 3 of 10

Calculate the value of the resistance r_π.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute $4\ \text{mA/V}$ for g_m and 100 for β.

$$\begin{aligned}
r_\pi &= \frac{100}{4\ \text{m}} \\
&= 25\ \text{k}\Omega
\end{aligned}$$

Therefore, the value of the resistance r_π is $25\ \text{k}\Omega$.

Step 4 of 10

Calculate the value of the resistance r_e.

$$r_e = \frac{\beta}{(\beta + 1) g_m}$$

Substitute 100 for β and $4\ \text{mA/V}$ for g_m.

$$\begin{aligned}
r_e &= \frac{100}{(100 + 1)(4\text{m})} \\
&= \frac{100}{404\ \text{m}} \\
&= 247.53\ \Omega
\end{aligned}$$

Therefore, the value of the resistance r_e is $247.53\ \Omega$.

Step 5 of 10

Recall equation (1).

$$C_E = \frac{1}{2\pi f_2 \left[r_e + \frac{R_B R_{sig}}{(\beta + 1)(R_B + R_{sig})} \right]}$$

Substitute $80\% f_L$ for f_2, $200\ \text{k}\Omega$ for R_B, $20\ \text{k}\Omega$ for R_{sig}, 100 for β and $247.53\ \Omega$ for r_e.

$$C_E = \frac{1}{2\pi (80\%) f_L \left[247.53 + \frac{(200\ \text{k})(20\ \text{k})}{(100 + 1)(200\ \text{k} + 20\ \text{k})} \right]}$$

Substitute $100\ \text{Hz}$ for f_L.

$$\begin{aligned}
C_E &= \frac{1}{2\pi (80\%)(100) \left[247.53 + \frac{4000\ \text{M}}{(101)(220\ \text{k})} \right]} \\
&= \frac{1}{2\pi (0.8)(100) \left[247.53 + \frac{4000\ \text{M}}{22.22\ \text{M}} \right]} \\
&= \frac{1}{(502.66)[247.53 + 180.02]} \\
&= \frac{1}{(502.66)(427.55)}
\end{aligned}$$

$$\begin{aligned}
C_E &= \frac{1}{214.91\ \text{k}} \\
&= 4.65\ \mu\text{F} \\
&= 5\ \mu\text{F}
\end{aligned}$$

Therefore, the value of the capacitor C_E is $\boxed{5\ \mu\text{F}}$.

Step 6 of 10

Consider the expression for the time constant associated with the capacitor C_{C1}.

$$\tau_{C1} = C_{C1} R_{C1}$$

Substitute $(R_B \| r_\pi) + R_{sig}$ for R_{C1} and $\frac{1}{2\pi f_1}$ for τ_{C1}.

$$\frac{1}{2\pi f_1} = C_{C1} \left((R_B \| r_\pi) + R_{sig} \right)$$

$$C_{C1} = \frac{1}{2\pi f_1 \left((R_B \| r_\pi) + R_{sig} \right)} \quad \text{..... (2)}$$

Therefore, the expression for the capacitor C_{C1} is $\frac{1}{2\pi f_1 \left((R_B \| r_\pi) + R_{sig} \right)}$.

Step 7 of 10

Recall equation (2).

$$C_{C1} = \frac{1}{2\pi f_1 \left((R_B \| r_\pi) + R_{sig} \right)}$$

Substitute $10\% f_L$ for f_2, $200\ \text{k}\Omega$ for R_B, $20\ \text{k}\Omega$ for R_{sig}, and $25\ \text{k}\Omega$ for r_π.

$$C_{C1} = \frac{1}{2\pi (10\%) f_L \left((200\ \text{k} \| 25\ \text{k}) + 20\ \text{k} \right)}$$

Substitute $100\ \text{Hz}$ for f_L.

$$\begin{aligned}
C_{C1} &= \frac{1}{2\pi (10\%)(100) \left(\frac{(200\ \text{k})(25\ \text{k})}{200\ \text{k} + 25\text{k}} + 20\ \text{k} \right)} \\
&= \frac{1}{2\pi (10) \left(\frac{(200\ \text{k})(25\ \text{k})}{225\text{k}} + 20\ \text{k} \right)} \\
&= \frac{1}{(62.83)(22.22\ \text{k} + 20\ \text{k})} \\
&= \frac{1}{(62.83)(42.22\ \text{k})}
\end{aligned}$$

$$\begin{aligned}
C_{C1} &= \frac{1}{2.65\ \text{M}} \\
&= 0.38\ \mu\text{F} \\
&= 0.4\ \mu\text{F}
\end{aligned}$$

Therefore, the value of the capacitor C_{C1} is $\boxed{0.4\ \mu\text{F}}$.

Consider the expression for the time constant associated with the capacitor C_{C2}.

$$\tau_{C2} = C_{C2} R_{C2}$$

Substitute $R_C + R_L$ for R_{C2} and $\frac{1}{2\pi f_3}$ for τ_{C2}.

$$\frac{1}{2\pi f_3} = C_{C2} (R_C + R_L)$$

$$C_{C2} = \frac{1}{2\pi f_3 (R_C + R_L)} \quad \text{..... (3)}$$

Therefore, the expression fro the capacitor C_{C2} is $\frac{1}{2\pi f_3 (R_C + R_L)}$.

Step 8 of 10

Recall equation (3).

$$C_{C2} = \frac{1}{2\pi f_3 (R_C + R_L)}$$

Substitute $10\% f_L$ for f_3, $20\ \text{k}\Omega$ for R_C and $10\ \text{k}\Omega$ for R_L.

$$C_{C2} = \frac{1}{2\pi (10\%) f_L (10\ \text{k} + 20\ \text{k})}$$

Substitute $100\ \text{Hz}$ for f_L.

$$\begin{aligned}
C_{C2} &= \frac{1}{2\pi (10\%)(100)(30\ \text{k})} \\
&= \frac{1}{1.89\ \text{M}} \\
&= 0.53\ \mu\text{F} \\
&= 0.5\ \mu\text{F}
\end{aligned}$$

Therefore, the value of the capacitor C_{C2} is $\boxed{0.5\ \mu\text{F}}$.

Step 9 of 10

Consider the expression for the dominant pole frequency f_L.

$$f_L = \frac{1}{2\pi} \left[\frac{1}{C_{C1} R_{C1}} + \frac{1}{C_E R_{CE}} + \frac{1}{C_{C2} R_{C2}} \right]$$

Substitute $r_e + \frac{R_B \| R_{sig}}{\beta + 1}$ for R_{CE}, $(R_B \| r_\pi) + R_{sig}$ for R_{C1} and $R_C + R_L$ for R_{C2}.

$$f_L = \frac{1}{2\pi} \left[\frac{1}{C_{C1} \left((R_B \| r_\pi) + R_{sig} \right)} + \frac{1}{C_E \left(r_e + \frac{R_B \| R_{sig}}{\beta + 1} \right)} + \frac{1}{C_{C2} (R_C + R_L)} \right]$$

Substitute $200\ \text{k}\Omega$ for R_B, $20\ \text{k}\Omega$ for R_{sig}, $20\ \text{k}\Omega$ for R_C and $10\ \text{k}\Omega$ for R_L, 100 for β and $247.53\ \Omega$ for r_e.

$$f_L = \frac{1}{2\pi} \left[\frac{1}{(0.4\ \mu) \left((200\ \text{k} \| 25\ \text{k}) + 20\ \text{k} \right)} + \frac{1}{(5\ \mu) \left(247.5 + \frac{200\ \text{k} \| 20\ \text{k}}{100 + 1} \right)} + \frac{1}{(0.5\ \mu)(30\ \text{k} + 10\ \text{k})} \right]$$

Step 1 of 4

10.008E

The actual internal resistance of the signal source fed to a common source amplifier is $100\ \text{k}\Omega$. But the signal source resistance is reduced to $10\ \text{k}\Omega$. That is, $R_{sig} = 10\ \text{k}\Omega$

Write the expression for the mid-band gain.

$$A_M = -\left(\frac{R_G}{R_G + R_{sig}}\right) g_m R_L'$$

Here,

$$R_L' = r_o \| R_D \| R_L$$

Substitute $150\ \text{k}\Omega$ for r_o, $15\ \text{k}\Omega$ for R_D, and $15\ \text{k}\Omega$ for R_L to calculate R_L'.

$$\begin{aligned} R_L' &= 150\times10^3 \| 15\times10^3 \| 15\times10^3 \\ &= \left(\frac{(150\times10^3)(15\times10^3)}{150\times10^3 + 15\times10^3}\right) \| 15\times10^3 \\ &= 13.64\times10^3 \| 15\times10^3 \end{aligned}$$

Further simplification yields,

$$\begin{aligned} R_L' &= \frac{(13.64\times10^3)(15\times10^3)}{13.64\times10^3 + 15\times10^3} \\ &= 7.144\ \text{k}\Omega \end{aligned}$$

Step 2 of 4

Substitute $4.7\ \text{M}\Omega$ for R_G, $10\ \text{k}\Omega$ for R_{sig}, $7.144\ \text{k}\Omega$ for R_L', and 1 mA/V for g_m to calculate R_L'.

$$\begin{aligned} A_M &= -\left(\frac{4.7\times10^6}{4.7\times10^6 + 10\times10^3}\right)(1\times10^{-3})(7.143\times10^3) \\ &= -\left(\frac{4.7\times10^6}{4.71\times10^6}\right)(1\times10^{-3})(7.143\times10^3) \\ &= -7.128\ \text{V/V} \end{aligned}$$

Thus, the modified gain of the CS amplifier is, $\boxed{-7.128\ \text{V/V}}$.

Step 3 of 4

The upper 3-dB frequency is,

$$f_H = \frac{1}{2\pi C_{in}\left(R_{sig} \| R_G\right)}$$

Here, the total input capacitance is, $C_{in} = C_{gs} + C_{eq}$.

The equivalent capacitance is,

$$C_{eq} = \left(1 + g_m R_L'\right) C_{gd}$$

Substitute 1 mA/V for g_m, $7.144\ \text{k}\Omega$ for R_L', and 0.4 pF for C_{gd}.

$$\begin{aligned} C_{eq} &= \left[1 + (10^{-3})(7.143\times10^3)\right](0.4\times10^{-12}) \\ &= (8.143)(0.4\times10^{-12}) \\ &= 3.2572\ \text{pF} \end{aligned}$$

Now calculate the total input capacitance.

$$\begin{aligned} C_{in} &= 1\ \text{pF} + 3.2572\ \text{pF} \\ &= 4.2572\ \text{pF} \end{aligned}$$

Step 4 of 4

Calculate the expression, $R_{sig} \| R_G$

$$\begin{aligned} R_{sig} \| R_G &= \frac{R_{sig} R_G}{R_{sig} + R_G} \\ &= \frac{(10\times10^3)(4.7\times10^6)}{10\times10^3 + 4.7\times10^6} \\ &= 9.98\ \text{k}\Omega \end{aligned}$$

Calculate the upper 3-dB frequency.

$$\begin{aligned} f_H &= \frac{1}{2\pi C_{in}\left(R_{sig} \| R_G\right)} \\ &= \frac{1}{2\pi\left(4.2572\times10^{-12}\right)\left(9.98\times10^3\right)} \\ &= 3.74\ \text{MHz} \end{aligned}$$

Thus, the upper 3-dB frequency of the CS amplifier is, $\boxed{3.74\ \text{MHz}}$.

10.008P

Step 1 of 19

Refer to Figure 10.9 (a) from the text book.

Draw the small signal equivalent circuit diagram.

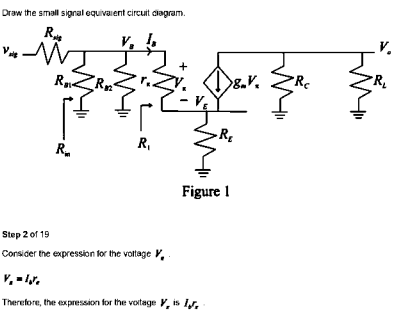


Figure 1

Step 2 of 19

Consider the expression for the voltage V_π.

$$V_\pi = I_b r_\pi$$

Therefore, the expression for the voltage V_π is $I_b r_\pi$.

Step 3 of 19

Consider the expression for the voltage V_E.

$$V_E = (I_b + g_m V_\pi) R_E$$

Substitute $I_b r_\pi$ for V_π.

$$\begin{aligned} V_E &= (I_b + g_m I_b r_\pi) R_E \\ &= I_b (1 + g_m r_\pi) R_E \end{aligned}$$

Substitute β for $g_m r_\pi$.

$$V_E = I_b (1+\beta) R_E$$

Therefore, the expression for the voltage V_E is $I_b(1+\beta)R_E$.

Step 4 of 19

Consider the expression fro the voltage V_B.

$$V_B = V_\pi + V_E$$

Substitute $I_b(1+\beta)R_E$ for V_E and $I_b r_\pi$ for V_π.

$$V_B = I_b r_\pi + I_b (1+\beta) R_E$$

$$V_B = I_b \left[r_\pi + (1+\beta) R_E \right] \quad \text{..... (1)}$$

$$\frac{V_B}{I_b} = r_\pi + (1+\beta) R_E$$

Here, the term $\frac{V_B}{I_b}$ represents the resistance R_1.

$$R_1 = r_\pi + (1+\beta) R_E$$

Therefore, the expression for the resistance R_1 is $r_\pi + (1+\beta)R_E$.

Step 5 of 19

Consider the expression for the input resistance R_{in}.

$$R_{in} = R_{B1} \parallel R_{B2} \parallel R_1$$

Substitute $r_\pi + (1+\beta)R_E$ for R_1.

$$R_{in} = R_{B1} \parallel R_{B2} \parallel \left[r_\pi + (1+\beta) R_E \right] \quad \text{..... (2)}$$

Therefore, the expression for the input resistance R_{in} is $R_{B1} \parallel R_{B2} \parallel \left[r_\pi + (1+\beta) R_E \right]$

Step 6 of 19

Calculate the value of the transconductance g_m.

$$g_m = \frac{I_C}{V_T}$$

Substitute $\frac{I_E \beta}{1+\beta}$ for I_C.

$$g_m = \frac{I_E \beta}{(1+\beta) V_T}$$

Substitute $0.3\ \text{mA}$ for I_E, $25\ \text{mV}$ for V_T and 120 for β.

$$\begin{aligned} g_m &= \frac{(0.3\ \text{m})(120)}{(120+1)(25\ \text{m})} \\ &= \frac{36\ \text{m}}{(121)(25\ \text{m})} \\ &= \frac{36\ \text{m}}{3.025} \\ &= 11.9\ \text{mA/V} \end{aligned}$$

Therefore, the value of the transconductance g_m is $11.9\ \text{mA/V}$.

Step 7 of 19

Calculate the value of the resistance r_π.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute $11.9\ \text{mA/V}$ for g_m and 120 for β.

$$\begin{aligned} r_\pi &= \frac{120}{11.9\ \text{m}} \\ &= 10.08\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance r_π is $10.08\ \text{k}\Omega$.

Step 8 of 19

Recall equation (1).

$$R_{in} = R_{B1} \parallel R_{B2} \parallel \left[r_\pi + (1+\beta) R_E \right]$$

Substitute $22\ \text{k}\Omega$ for R_{B2}, $33\ \text{k}\Omega$ for R_{B1}, 120 for β, $3.9\ \text{k}\Omega$ for R_E and $10.08\ \text{k}\Omega$ for r_π.

$$\begin{aligned} R_{in} &= 33\ \text{k} \parallel 22\ \text{k} \parallel \left[10.08\ \text{k} + (1+120)(3.9\ \text{k}) \right] \\ &= 33\ \text{k} \parallel 22\ \text{k} \parallel \left[10.08\ \text{k} + (121)(3.9\ \text{k}) \right] \\ &= 33\ \text{k} \parallel 22\ \text{k} \parallel \left[10.08\ \text{k} + 471.9\ \text{k} \right] \\ &= 33\ \text{k} \parallel 22\ \text{k} \parallel 481.98\ \text{k} \end{aligned}$$

$$\begin{aligned} R_{in} &= \frac{(33\ \text{k})(22\ \text{k})(481.98\ \text{k})}{(33\ \text{k})(22\ \text{k}) + (481.98\ \text{k})(33\ \text{k}) + (22\ \text{k})(481.98\ \text{k})} \\ &= \frac{349917.8\times10^9}{726\ \text{M} + 15905.34\ \text{M} + 10603.56\ \text{M}} \\ &= \frac{349917.8\times10^9}{2.72349\times10^{19}} \\ &= 12.85\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the input resistance R_{in} is $\boxed{12.85\ \text{k}\Omega}$.

Step 8 of 19

Consider the expression for the base voltage V_B in terms of the voltage V_{sig}.

$$V_B = V_{sig} \left(\frac{R_{in}}{R_{in} + R_{sig}} \right)$$

Therefore, the expression for the base voltage V_B is $V_{sig}\left(\frac{R_{in}}{R_{in}+R_{sig}}\right)$

Step 10 of 19

Recall equation (1).

$$V_B = I_b \left[r_\pi + (1+\beta) R_E \right]$$

Substitute $\frac{V_\pi}{r_\pi}$ for I_b and $V_{sig}\left(\frac{R_{in}}{R_{in}+R_{sig}}\right)$ for V_B.

$$V_{sig} \left(\frac{R_{in}}{R_{in} + R_{sig}} \right) = \frac{V_\pi}{r_\pi} \left[r_\pi + (1+\beta) R_E \right]$$

$$V_\pi = V_{sig} \left(\frac{R_{in}}{R_{in} + R_{sig}} \right) \left(\frac{r_\pi}{r_\pi + (1+\beta) R_E} \right)$$

Therefore, the expression for the voltage V_π is $V_{sig}\left(\frac{R_{in}}{R_{in}+R_{sig}}\right)\left(\frac{r_\pi}{r_\pi+(1+\beta)R_E}\right)$

Step 11 of 19

Consider the expression for the output voltage V_o.

$$V_o = -g_m V_\pi (R_C \parallel R_L)$$

Substitute $V_{sig}\left(\frac{R_{in}}{R_{in}+R_{sig}}\right)\left(\frac{r_\pi}{r_\pi+(1+\beta)R_E}\right)$ for V_π.

$$V_o = -g_m V_{sig} \left(\frac{R_{in}}{R_{in} + R_{sig}} \right) \left(\frac{r_\pi}{r_\pi + (1+\beta) R_E} \right) (R_C \parallel R_L)$$

$$\frac{V_o}{V_{sig}} = -g_m \left(\frac{R_{in}}{R_{in} + R_{sig}} \right) \left(\frac{r_\pi}{r_\pi + (1+\beta) R_E} \right) (R_C \parallel R_L)$$

Here, the term $\frac{V_o}{V_{sig}}$ represents the voltage gain A_M.

$$A_M = -g_m \left(\frac{R_{in}}{R_{in} + R_{sig}} \right) \left(\frac{r_\pi}{r_\pi + (1+\beta) R_E} \right) \left(\frac{R_C R_L}{R_C + R_L} \right) \quad \text{..... (3)}$$

Therefore, the expression for the voltage gain A_M is

$$-g_m \left(\frac{R_{in}}{R_{in} + R_{sig}} \right) \left(\frac{r_\pi}{r_\pi + (1+\beta) R_E} \right) \left(\frac{R_C R_L}{R_C + R_L} \right).$$

Step 12 of 19

Recall equation (3).

$$A_M = -g_m \left(\frac{R_{in}}{R_{in} + R_{sig}} \right) \left(\frac{r_\pi}{r_\pi + (1+\beta) R_E} \right) \left(\frac{R_C R_L}{R_C + R_L} \right)$$

Substitute $12.1\ \text{mA/V}$ for g_m, $12.85\ \text{k}\Omega$ for R_{in}, $5\ \text{k}\Omega$ for R_{sig}, 120 for β, $3.9\ \text{k}\Omega$ for R_E, $4.7\ \text{k}\Omega$ for R_C, $5.6\ \text{k}\Omega$ for R_L, $0.99\ \text{k}\Omega$ for r_π.

$$\begin{aligned} A_M &= -(12.1\ \text{m}) \left(\frac{12.85\ \text{k}}{12.85\ \text{k} + 5\ \text{k}} \right) \left(\frac{10.08\ \text{k}}{10.08\ \text{k} + (1+120)(3.9\ \text{k})} \right) \left(\frac{(4.7\ \text{k})(5.6\ \text{k})}{4.7\ \text{k} + 5.6\ \text{k}} \right) \\ &= -(12.1\ \text{m}) \left(\frac{12.85\ \text{k}}{17.85\ \text{k}} \right) \left(\frac{10.08\ \text{k}}{10.08\ \text{k} + (121)(3.9\ \text{k})} \right) \left(\frac{(4.7\ \text{k})(5.6\ \text{k})}{10.3\ \text{k}} \right) \\ &= -(12.1\ \text{m})(0.72) \left(\frac{10.08\ \text{k}}{10.08\ \text{k} + 471.9\ \text{k}} \right) (2.56\ \text{k}) \\ &= -(22.3) \left(\frac{10.08\ \text{k}}{481.98\ \text{k}} \right) \\ A_M &= -(22.3)(20.91\ \text{m}) \\ &= -0.47\ \text{V/V} \end{aligned}$$

Therefore, the value of the voltage gain A_M is $\boxed{-0.47\ \text{V/V}}$.

Step 13 of 19

Calculate the value of the resistance R_{C1}.

$$R_{C1} = (R_{B1} \parallel R_{B2} \parallel r_\pi) + R_{sig}$$

Substitute $22\ \text{k}\Omega$ for R_{B2}, $33\ \text{k}\Omega$ for R_{B1}, $5\ \text{k}\Omega$ for R_{sig} and $10.08\ \text{k}\Omega$ for r_π.

$$\begin{aligned} R_{C1} &= \frac{(22\ \text{k})(33\ \text{k})(10.08\ \text{k})}{(22\ \text{k})(33\ \text{k}) + (10.08\ \text{k})(22\ \text{k}) + (10.08\ \text{k})(33\ \text{k})} + 5\ \text{k} \\ &= \frac{7.31808\times10^{12}}{726\ \text{M} + 221.76\ \text{M} + 332.64\ \text{M}} + 5\ \text{k} \\ &= \frac{7.31808\times10^{12}}{1.2804\times10^9} + 5\ \text{k} \\ &= 5.72\ \text{k} + 5\ \text{k} \\ R_{C1} &= 10.72\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance R_{C1} is $10.72\ \text{k}\Omega$.

Step 14 of 19

Calculate the value of the time constant associated with the capacitance C_{C1}.

$$\tau_{C1} = C_{C1} R_{C1}$$

Substitute $10.72\ \text{k}\Omega$ for R_{C1} and $1\ \mu\text{F}$ for C_{C1}.

$$\begin{aligned} \tau_{C1} &= (1\ \mu)(10.72\ \text{k}) \\ &= 10.72\ \text{ms} \end{aligned}$$

Therefore, the value of the time constant τ_{C1} is $\boxed{10.72\ \text{ms}}$.

Step 15 of 19

Calculate the value of the resistance R_{CE}.

$$R_{CE} = R_E \parallel \left[r_\pi + \frac{R_{B1} \parallel R_{B2} \parallel R_{sig}}{\beta + 1} \right]$$

Substitute $22\ \text{k}\Omega$ for R_{B2}, $33\ \text{k}\Omega$ for R_{B1}, $5\ \text{k}\Omega$ for R_{sig}, $25\ \Omega$ for r_π, $3.9\ \text{k}\Omega$ for R_E and 120 for β.

$$\begin{aligned} R_{CE} &= 3.9\ \text{k} \parallel \left[25 + \frac{22\ \text{k} \parallel 33\ \text{k} \parallel 5\ \text{k}}{120+1} \right] \\ &= 3.9\ \text{k} \parallel \left[25 + \frac{1}{121} \left(\frac{(22\ \text{k})(33\ \text{k})(5\ \text{k})}{(22\ \text{k})(33\ \text{k}) + (5\ \text{k})(22\ \text{k}) + (33\ \text{k})(5\ \text{k})} \right) \right] \\ &= 3.9\ \text{k} \parallel \left[25 + \frac{1}{121} \left(\frac{3.63\times10^{12}}{726\ \text{M} + 110\ \text{M} + 165\ \text{M}} \right) \right] \\ &= 3.9\ \text{k} \parallel \left[25 + \frac{1}{121} \left(\frac{3.63\times10^{12}}{1001\ \text{M}} \right) \right] \\ R_{CE} &= 3.9\ \text{k} \parallel [25 + 29.97] \\ &= 3.9\ \text{k} \parallel 54.97 \\ &= \frac{(3.9\ \text{k})(54.97)}{3.9\ \text{k} + 54.97} \\ &= \frac{(3.9\ \text{k})(54.97)}{3.96\ \text{k}} \\ R_{CE} &= 54.14\ \Omega \end{aligned}$$

Therefore, the value of the resistance R_{CE} is $54.14\ \Omega$.

Step 16 of 19

Calculate the value of the time constant associated with the capacitance C_{CE}.

$$\tau_{CE} = C_{CE} R_{CE}$$

Substitute $54.14\ \Omega$ for R_{CE} and $20\ \mu\text{F}$ for C_{CE}.

$$\begin{aligned} \tau_{CE} &= (20\ \mu)(54.14) \\ &= 1.08\ \text{ms} \end{aligned}$$

Therefore, the value of the time constant τ_{CE} is $\boxed{1.08\ \text{ms}}$.

Step 17 of 17

Calculate the value of the resistance R_{C2}.

$$R_{C2} = R_C + R_L$$

Substitute $4.7\ \text{k}\Omega$ for R_C and $5.6\ \text{k}\Omega$ for R_L.

$$\begin{aligned} R_{C2} &= 4.7\ \text{k} + 5.6\ \text{k} \\ &= 10.3\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance R_{C2} is $10.3\ \text{k}\Omega$.

Step 18 of 19

Calculate the value of the time constant associated with the capacitance C_{C2}.

$$\tau_{C2} = C_{C2} R_{C2}$$

Substitute $10.3\ \text{k}\Omega$ for R_{C2} and $1\ \mu\text{F}$ for C_{C2}.

$$\begin{aligned} \tau_{C2} &= (1\ \mu)(10.3\ \text{k}) \\ &= 10.3\ \text{ms} \end{aligned}$$

Therefore, the value of the time constant τ_{C2} is $\boxed{10.3\ \text{ms}}$.

Step 19 of 19

Calculate the value of the dominant frequency f_L.

$$f_L = \frac{1}{2\pi} \left[\frac{1}{\tau_{E1}} + \frac{1}{\tau_{CE}} + \frac{1}{\tau_{C2}} \right]$$

Substitute $10.3\ \text{ms}$ for τ_{C2}, $1.08\ \text{ms}$ for τ_{CE} and $10.72\ \text{ms}$ for τ_{C1}.

$$\begin{aligned} f_L &= \frac{1}{2\pi} \left[\frac{1}{10.72\ \text{m}} + \frac{1}{1.08\ \text{m}} + \frac{1}{10.3\ \text{m}} \right] \\ &= \frac{1}{2\pi} [93.28 + 925.93 + 97.09] \\ &= \frac{1116.3}{6.28} \\ &= 177.76\ \text{Hz} \end{aligned}$$

Therefore, the value of the frequency f_L is $\boxed{177.76\ \text{Hz}}$.

Step 1 of 2

10.009E

Consider the common source amplifier circuit.

Write the expression for the 3-dB frequency of a common source amplifier.

$$f_H = \frac{1}{2\pi\left[C_{gs} + (1 + g_m R_L')C_{gd}\right](R_{sig} \| R_G)}$$

Here,

$$R_L' = r_o \| R_D \| R_L$$

Substitute $150\ \text{k}\Omega$ for r_o, $15\ \text{k}\Omega$ for R_D, and $15\ \text{k}\Omega$ for R_L to calculate R_L'.

$$\begin{aligned} R_L' &= 150\times10^3 \| 15\times10^3 \| 15\times10^3 \\ &= \left(\frac{(150\times10^3)(15\times10^3)}{150\times10^3 + 15\times10^3}\right) \| 15\times10^3 \\ &= 13.64\times10^3 \| 15\times10^3 \end{aligned}$$

Further simplification yields,

$$\begin{aligned} R_L' &= \frac{(13.64\times10^3)(15\times10^3)}{13.64\times10^3 + 15\times10^3} \\ &= 7.144\ \text{k}\Omega \end{aligned}$$

Step 2 of 2

Recall the expression for 3-dB frequency.

$$f_H = \frac{1}{2\pi\left[C_{gs} + (1 + g_m R_L')C_{gd}\right](R_{sig} \| R_G)}$$

Rearrange the expression for the bridging capacitance.

$$C_{gd} = \frac{\dfrac{1}{2\pi f_H (R_{sig} \| R_G)} - C_{gs}}{1 + g_m R_L'}$$

Substitute 1 pF for C_{gs}, 1 mA/V for g_m, $7.144\ \text{k}\Omega$ for R_L', $100\ \text{k}\Omega$ for R_{sig}, and $4.7\ \text{M}\Omega$ for R_G.

$$\begin{aligned} C_{gd} &= \frac{\dfrac{1}{2\pi(1\times10^6)(100\times10^3 \| 4.7\times10^6)} - 10^{-12}}{1 + (1\times10^{-3})(7.144\times10^3)} \\ &= \frac{\dfrac{1}{2\pi(1\times10^6)(100\times10^3 \| 4.7\times10^6)} - 10^{-12}}{1 + 7.144} \\ &= \frac{\dfrac{1}{2\pi(1\times10^6)\left(\dfrac{(100\times10^3)(4.7\times10^6)}{100\times10^3 + 4.7\times10^6}\right)} - 10^{-12}}{8.144} \end{aligned}$$

Further simplification yields,

$$\begin{aligned} C_{gd} &= \frac{1.625\times10^{-12} - 10^{-12}}{8.14} \\ &= \frac{0.625\times10^{-12}}{8.14} \\ &= 0.077\ \text{pF} \\ &\approx 0.08\ \text{pF} \end{aligned}$$

Thus, the maximum value of bridging capacitance required is, $\boxed{\approx 0.08\ \text{pF}}$.

Step 1 of 8

Refer to Figure 10.9 (a) from the text book.

Calculate the value of the transconductance g_m.

$$g_m = \frac{I_C}{V_T}$$

Substitute $\frac{I_E\beta}{1+\beta}$ for I_C.

$$g_m = \frac{I_E\beta}{(1+\beta)V_T}$$

Substitute $0.3\ \text{mA}$ for I_E, $25\ \text{mV}$ for V_T and 120 for β.

$$\begin{aligned} g_m &= \frac{(0.3\ \text{m})(120)}{(120+1)(25\ \text{m})} \\ &= \frac{36\ \text{m}}{(121)(25\ \text{m})} \\ &= \frac{36\ \text{m}}{3.025} \\ &= 11.9\ \text{mA/V} \end{aligned}$$

Therefore, the value of the transconductance g_m is $11.9\ \text{mA/V}$.

Step 2 of 8

Calculate the value of the resistance r_π.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute $11.9\ \text{mA/V}$ for g_m and 120 for β..

$$\begin{aligned} r_\pi &= \frac{120}{11.9\ \text{m}} \\ &= 10.08\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance r_π is $10.08\ \text{k}\Omega$.

Step 3 of 8

Calculate the value of the resistance R_{CE}.

$$R_{CE} = R_E \| \left[r_e + \frac{R_{B1} \| R_{B2} \| R_{sig}}{\beta+1} \right]$$

Substitute $22\ \text{k}\Omega$ for R_{B2}, $33\ \text{k}\Omega$ for R_{B1}, $5\ \text{k}\Omega$ for R_{sig}, $25\ \Omega$ for r_e, $3.9\ \text{k}\Omega$ for R_E and 120 for β.

$$\begin{aligned} R_{CE} &= 3.9\ \text{k} \| \left[25 + \frac{22\ \text{k} \| 33\ \text{k} \| 5\ \text{k}}{120+1} \right] \\ &= 3.9\ \text{k} \| \left[25 + \frac{1}{121} \left(\frac{(22\ \text{k})(33\ \text{k})(5\ \text{k})}{(22\ \text{k})(33\ \text{k})+(5\ \text{k})(22\ \text{k})+(33\ \text{k})(5\ \text{k})} \right) \right] \\ &= 3.9\ \text{k} \| \left[25 + \frac{1}{121} \left(\frac{3.63\times10^{12}}{726\ \text{M}+110\ \text{M}+165\ \text{M}} \right) \right] \\ &= 3.9\ \text{k} \| \left[25 + \frac{1}{121} \left(\frac{3.63\times10^{12}}{1001\ \text{M}} \right) \right] \end{aligned}$$

$$\begin{aligned} R_{CE} &= 3.9\ \text{k} \| [25+29.97] \\ &= 3.9\ \text{k} \| 54.97 \\ &= \frac{(3.9\ \text{k})(54.97)}{3.9\ \text{k}+54.97} \\ &= \frac{(3.9\ \text{k})(54.97)}{3.96\ \text{k}} \end{aligned}$$

$$R_{CE} = 54.14\ \Omega$$

Therefore, the value of the resistance R_{CE} is $54.14\ \Omega$.

Step 4 of 8

Assume capacitor C_E contributes only 80% of the dominant frequency f_L.

Calculate the value of the time constant associated with the capacitance C_E.

$$\tau_{CE} = C_E R_{CE}$$

Substitute $\frac{1}{2\pi f_2}$ for τ_{CE}.

$$\frac{1}{2\pi f_2} = C_E R_{CE}$$

$$C_E = \frac{1}{2\pi f_2 R_{CE}}$$

Substitute $80\% f_L$ for f_2.

$$C_E = \frac{1}{2\pi(80\%) f_L R_{CE}}$$

Substitute $54.14\ \Omega$ for R_{CE} and $50\ \text{Hz}$ for f_L.

$$\begin{aligned} C_E &= \frac{1}{2\pi(80\%)(50)(54.14)} \\ &= \frac{1}{2\pi(40)(54.14)} \\ &= \frac{1}{13.61\ \text{k}} \\ &= 73.48\ \mu\text{F} \end{aligned}$$

Therefore, the value of the capacitor C_g is $\boxed{73.48\ \mu\text{F}}$.

Step 5 of 8

Calculate the value of the resistance R_{C1}.

$$R_{C1} = (R_{B1} \| R_{B2} \| r_\pi) + R_{sig}$$

Substitute $22\ \text{k}\Omega$ for R_{B2}, $33\ \text{k}\Omega$ for R_{B1}, $5\ \text{k}\Omega$ for R_{sig} and $10.08\ \text{k}\Omega$ for r_π.

$$\begin{aligned} R_{C1} &= \frac{(22\ \text{k})(33\ \text{k})(10.08\ \text{k})}{(22\ \text{k})(33\ \text{k})+(10.08\ \text{k})(22\ \text{k})+(10.08\ \text{k})(33\ \text{k})} + 5\ \text{k} \\ &= \frac{7.31808\times10^{12}}{726\ \text{M}+221.76\ \text{M}+332.64\ \text{M}} + 5\ \text{k} \\ &= \frac{7.31808\times10^{12}}{1.2804\times10^{9}} + 5\ \text{k} \\ &= 5.72\ \text{k}+5\ \text{k} \end{aligned}$$

$$R_{C1} = 10.72\ \text{k}\Omega$$

Therefore, the value of the resistance R_{C1} is $10.72\ \text{k}\Omega$.

Step 6 of 8

Calculate the value of the time constant associated with the capacitance C_{C1}.

$$\tau_{C1} = C_{C1} R_{C1}$$

Substitute $\frac{1}{2\pi f_1}$ for τ_{C1}.

$$\frac{1}{2\pi f_1} = C_{C1} R_{C1}$$

$$C_{C1} = \frac{1}{2\pi f_1 R_{C1}}$$

Substitute $10\% f_L$ for f_1.

$$C_{C1} = \frac{1}{2\pi(10\%) f_L R_{C1}}$$

Substitute $10.72\ \text{k}\Omega$ for R_{C1} and $50\ \text{Hz}$ for f_L.

$$\begin{aligned} C_{C1} &= \frac{1}{2\pi(10\%)(50)(10.72\ \text{k})} \\ &= \frac{1}{2\pi(5)(10.72\ \text{k})} \\ &= \frac{1}{0.336\ \text{M}} \\ &= 2.98\ \mu\text{F} \end{aligned}$$

Therefore, the value of the capacitor C_{C1} is $\boxed{2.98\ \mu\text{F}}$.

Step 7 of 8

Calculate the value of the resistance R_{C2}.

$$R_{C2} = R_C + R_L$$

Substitute $4.7\ \text{k}\Omega$ for R_C and $5.6\ \text{k}\Omega$ for R_L.

$$\begin{aligned} R_{C2} &= 4.7\ \text{k}+5.6\ \text{k} \\ &= 10.3\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance R_{C2} is $10.3\ \text{k}\Omega$.

Step 8 of 8

Calculate the value of the time constant associated with the capacitance C_{C2}.

$$\tau_{C2} = C_{C2} R_{C2}$$

Substitute $\frac{1}{2\pi f_2}$ for τ_{C2}.

$$\frac{1}{2\pi f_2} = C_{C2} R_{C2}$$

$$C_{C2} = \frac{1}{2\pi f_2 R_{C2}}$$

Substitute $10\% f_L$ for f_3.

$$C_{C2} = \frac{1}{2\pi(10\%) f_L R_{C2}}$$

Substitute $10.3\ \text{k}\Omega$ for R_{C2} and $50\ \text{Hz}$ for f_L.

$$\begin{aligned} C_{C2} &= \frac{1}{2\pi(10\%)(50)(10.3\ \text{k})} \\ &= \frac{1}{2\pi(5)(10.3\ \text{k})} \\ &= \frac{1}{0.324\ \text{M}} \\ &= 3.09\ \mu\text{F} \end{aligned}$$

Therefore, the value of the capacitor C_{C2} is $\boxed{3.09\ \mu\text{F}}$.

Step 1 of 3

Consider the expression for the mid band gain A_M.

$$A_M = -\frac{R_B}{R_B + R_{sig}} \frac{r_\pi}{r_\pi + r_x + \left(R_B \parallel R_{sig}\right)} g_m R_L'$$

The value of the mid band gain is directly proportional to the value of the resistance R_L', keeping the other components constant. That is,

$$A_M \alpha R_L'$$

As the value of the mid band decreases to half, the value of the resistance R_L' also decreases to half. Hence, calculate the new value of the resistance R_L'.

$$\begin{aligned} R_L' &= \frac{3\text{ k}}{2} \\ &= 1.5\text{ k}\Omega \end{aligned}$$

Therefore, the new value of the resistance R_L' is $1.5\text{ k}\Omega$.

Step 2 of 3

Consider the expression for the resistance R_L'.

$$R_L' = r_o \parallel R_C \parallel R_L$$

Substitute $1.5\text{ k}\Omega$ for R_L', $8\text{ k}\Omega$ for R_C and $100\text{ k}\Omega$ for r_o.

$$1.5\text{ k} = 100\text{ k} \parallel 8\text{ k} \parallel R_L$$

$$1.5\text{ k} = \frac{(100\text{ k})(8\text{ k})R_L}{(100\text{ k})(8\text{ k}) + R_L(100\text{ k}) + (8\text{ k})R_L}$$

$$1.5 = \frac{(800\text{ k})R_L}{800\text{ M} + R_L(108\text{ k})}$$

$$(1.5)(800\text{ M} + R_L(108\text{ k})) = (800\text{ k})R_L$$

$$(1.5)(800\text{ M} + R_L(108\text{ k})) = (800\text{ k})R_L$$

$$1.2\text{ G} + R_L(162\text{ k}) = (800\text{ k})R_L$$

$$(800\text{ k} - 162\text{ k})R_L = 1.2\text{ G}$$

$$(638\text{ k})R_L = 1.2\text{ G}$$

$$\begin{aligned} R_L &= \frac{1.2\text{ G}}{638\text{ k}} \\ &= 1.9\text{ k}\Omega \end{aligned}$$

Therefore, the value of the resistance R_L is $\boxed{1.9\text{ k}\Omega}$.

Step 3 of 3

Calculate the value of the frequency f_H.

$$f_H = \frac{1}{2\pi C_{in} R_{sig}'}$$

Substitute 68 pF for C_{in} and $1.65\text{ k}\Omega$ for R_{sig}'.

$$\begin{aligned} f_H &= \frac{1}{2\pi(68\text{ p})(1.65\text{ k})} \\ &= \frac{1}{704.97\text{ n}} \\ &= 1.42\text{ MHz} \end{aligned}$$

Therefore, the value of the frequency f_H is $\boxed{1.42\text{ MHz}}$.

Step 1 of 3

Refer to Figure 10.9 (a) from the text book.

Calculate the value of the resistance R_{CE}.

$$R_{CE} = R_E \| \left[r_e + \frac{R_B \| R_{sig}}{\beta + 1} \right]$$

Substitute $10\ \text{k}\Omega$ for R_B, $5\ \text{k}\Omega$ for R_{sig}, $25\ \Omega$ for r_e, $1.5\ \text{k}\Omega$ for R_E and 100 for β.

$$\begin{aligned} R_{CE} &= 1.5\ \text{k} \| \left[25 + \frac{10\ \text{k} \| 5\ \text{k}}{100+1} \right] \\ &= 1.5\ \text{k} \| \left[25 + \frac{1}{101} \left(\frac{(10\ \text{k})(5\ \text{k})}{10\ \text{k} + 5\ \text{k}} \right) \right] \\ &= 1.5\ \text{k} \| \left[25 + \frac{1}{101} \left(\frac{50\ \text{M}}{15\ \text{k}} \right) \right] \\ &= 1.5\ \text{k} \| \left[25 + \frac{1}{101} (3.33\ \text{k}) \right] \end{aligned}$$

$$\begin{aligned} R_{CE} &= 1.5\ \text{k} \| [25 + 32.97] \\ &= 1.5\ \text{k} \| 57.97 \\ &= \frac{(1.5\ \text{k})(57.97)}{1.5\ \text{k} + 57.97} \\ &= 55.81\ \Omega \end{aligned}$$

Therefore, the value of the resistance R_{CE} is $55.81\ \Omega$.

Step 2 of 3

Calculate the value of the resistance R_{C1}.

$$R_{C1} = (R_B \| r_\pi) + R_{sig}$$

Substitute $10\ \text{k}\Omega$ for R_B, $5\ \text{k}\Omega$ for R_{sig} and $1\ \text{k}\Omega$ for r_π.

$$\begin{aligned} R_{C1} &= \frac{(10\ \text{k})(1\ \text{k})}{10\ \text{k} + 1\ \text{k}} + 5\ \text{k} \\ &= \frac{10\ \text{M}}{10.1\ \text{k}} + 5\ \text{k} \\ &= 990.1 + 5\ \text{k} \\ &= 5.99\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance R_{C1} is $5.99\ \text{k}\Omega$.

Step 3 of 3

For the capacitors C_E and C_{C1} to contribute equal values of the frequencies, their respective time constants should be equal. That is,

$$\tau_{CE} = \tau_{C1}$$

Substitute $C_E R_{CE}$ for τ_{CE} and $C_{C1} R_{C1}$ for τ_{C1}.

$$\begin{aligned} C_E R_{CE} &= C_{C1} R_{C1} \\ \frac{C_E}{C_{C1}} &= \frac{R_{C1}}{R_{CE}} \end{aligned}$$

Substitute $5.99\ \text{k}\Omega$ for R_{C1} and $55.81\ \Omega$ for R_{CE}.

$$\begin{aligned} \frac{C_E}{C_{C1}} &= \frac{5.99\ \text{k}}{55.81} \\ &= 107.33 \end{aligned}$$

Therefore, the value of the ratio $\frac{C_E}{C_{C1}}$ is $\boxed{107.33}$.

Step 1 of 2

10.011E

Calculate the value of the frequency f_H .

$$f_H = \frac{f_L}{|A_M|}$$

Substitute $2\ \text{GHz}$ for f_H and 12.5 for $|A_M|$.

$$f_H = \frac{2\ \text{G}}{12.5}$$
$$= 160\ \text{MHz}$$

Therefore, the value of the frequency f_H is $160\ \text{MHz}$.

Step 2 of 2

Consider the expression for the frequency f_H .

$$f_H = \frac{1}{2\pi\left(C_L + C_{wi}\right)R_L'}$$

Substitute $160\ \text{MHz}$ for f_H , $10\ \text{k}\Omega$ for R_L' and $5\ \text{fF}$ for C_{wi} .

$$160\ \text{M} = \frac{1}{2\pi(C_L + 5\ \text{f})(10\ \text{k})}$$

$$C_L + 5\ \text{f} = \frac{1}{2\pi(160\ \text{M})(10\ \text{k})}$$

$$C_L + 5\ \text{f} = \frac{1}{10.05\ \text{T}}$$

$$C_L + 5\ \text{f} = 99.5\ \text{f}$$

$$C_L = 99.5\ \text{f} - 5\ \text{f}$$
$$= 94.5\ \text{fF}$$

Therefore, the value of the capacitance C_L is $\boxed{94.5\ \text{fF}}$.

10.011P

We don't have the solution to this problem yet.

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Step 1 of 2

10.012E

The direct-coupled amplifier has a dc gain, $\mathbf{1000\ V/V}$

Write the expression for the transfer function of the direct-coupled amplifier.

$$\mathbf{TF} = \frac{A_M}{1 + \frac{s}{2\pi f_H}}$$

Here,

The dc voltage gain is, A_M

The 3-dB frequency is, f_H

Substitute $\mathbf{1000\ V/V}$ for A_M and $\mathbf{100\ kHz}$ for f_H .

$$\mathbf{TF} = \frac{1000}{1 + \frac{s}{2\pi\left(100\times10^3\right)}}$$

$$= \frac{1000}{1 + \frac{s}{2\pi\times10^5}}$$

Thus, the transfer function of the direct-coupled amplifier is, $\boxed{\frac{1000}{1 + \frac{s}{2\pi\times10^5}}}$.

Step 2 of 2

Write the expression for the gain bandwidth product of the direct-coupled amplifier.

$$f_t = \left|A_M\right| f_H$$

Substitute $\mathbf{1000\ V/V}$ for A_M and $\mathbf{100\ kHz}$ for f_H

$$f_t = (1000)\left(100\times10^3\right)$$

$$= 10^8\ \mathbf{Hz}$$

Thus, the gain bandwidth product of the direct-coupled amplifier is, $\boxed{10^8\ \mathbf{Hz}}$.

10.012P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

Step 1 of 3

10.013E

Consider the following 3-dB frequency expression that can be extended to any number of poles and zero.

$$\omega_H \approx \frac{1}{\sqrt{\left(\frac{1}{\omega_{P1}^2}+\frac{1}{\omega_{P2}^2}+....\right)-\left(\frac{1}{\omega_{Z1}^2}+\frac{1}{\omega_{Z2}^2}+...\right)}}$$

The amplifier is characterized by two zeros at $s=\infty$ and two poles ω_{P1} and ω_{P2}, so,

$$\omega_H \approx \frac{1}{\sqrt{\frac{1}{\omega_{p1}^2}+\frac{1}{\omega_{p2}^2}}}$$

The transfer function for this amplifier is,

$$H(s)=\frac{A_M}{\left(1+\frac{s}{\omega_{P1}}\right)\left(1+\frac{s}{\omega_{P2}}\right)}$$

Substitute $j\omega$ for s.

$$H(j\omega)=\frac{A_M}{\left(1+\frac{j\omega}{\omega_{P1}}\right)\left(1+\frac{j\omega}{\omega_{P2}}\right)}$$

Take the magnitude expression.

$$|H(j\omega)|=\frac{A_M}{\sqrt{\left(1+\left(\frac{\omega}{\omega_{P1}}\right)^2\right)\left(1+\left(\frac{\omega}{\omega_{P2}}\right)^2\right)}}$$

Step 2 of 3

The 3-dB frequency ω_H occurs at $|H(j\omega_H)|=\frac{1}{\sqrt{2}}$. Therefore,

$$\frac{A_M^2}{\left(1+\left(\frac{\omega_H}{\omega_{P1}}\right)^2\right)\left(1+\left(\frac{\omega_H}{\omega_{P2}}\right)^2\right)}=\frac{A_M^2}{2}$$

$$\left[1+\left(\frac{\omega_H}{\omega_{P1}}\right)^2\right]\left[1+\left(\frac{\omega_H}{\omega_{P2}}\right)^2\right]=2$$

Substitute $k\omega_{P1}$ for ω_{P2} and $0.9\omega_{P1}$ for ω_H

$$\left[1+\left(\frac{0.9\omega_{P1}}{\omega_{P1}}\right)^2\right]\left[1+\left(\frac{0.9\omega_{P1}}{k\omega_{P1}}\right)^2\right]=2$$

$$\left[1+(0.9)^2\right]\left[1+\left(\frac{0.9}{k}\right)^2\right]=2$$

$$[1.81]\left[1+\frac{0.81}{k^2}\right]=2$$

$$1+\frac{0.81}{k^2}=\frac{2}{1.81}$$

$$k=\sqrt{\frac{0.81}{\frac{2}{1.81}-1}}$$

$$=2.78$$

Therefore, the value of k for $\omega_H=0.9\omega_{P1}$ is, $\boxed{2.78}$.

Step 3 of 3

Substitute $k\omega_{P1}$ for ω_{P2} and $0.99\omega_{P1}$ for ω_H

$$\left[1+\left(\frac{0.99\omega_{P1}}{\omega_{P1}}\right)^2\right]\left[1+\left(\frac{0.99\omega_{P1}}{k\omega_{P1}}\right)^2\right]=2$$

$$\left[1+(0.99)^2\right]\left[1+\left(\frac{0.99}{k}\right)^2\right]=2$$

$$[1.9801]\left[1+\frac{0.9801}{k^2}\right]=2$$

$$1+\frac{0.9801}{k^2}=\frac{2}{1.9801}$$

$$k=\sqrt{\frac{0.9801}{\frac{2}{1.9801}-1}}$$

$$=9.88$$

Therefore, the value of k for $\omega_H=0.99\omega_{P1}$ is, $\boxed{9.88}$.

Step 1 of 11

Refer to Figure 10.12 (a) from the text book.

Calculate the value of C_{ox}.

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

Substitute $3.45\times10^{-11}\,\text{F/m}$ for ϵ_{ox} and $8\,\text{nm}$ for t_{ox}.

$$C_{ox} = \frac{3.45\times10^{-11}}{8\,\text{n}}$$
$$= 4.31\,\text{mF/m}^2$$

Therefore, the value of C_{ox} is $\boxed{4.31\,\text{mF/m}^2}$.

Step 2 of 11

Calculate the value of the over drive voltage V_{OV}.

$$V_{DV} = \sqrt{\frac{2I_D}{\mu_n C_{ox}\left(\frac{W}{L}\right)}}$$

Substitute $200\,\mu\text{A}$ for I_D, $450\,\text{cm}^2/\text{V}\cdot\text{s}$ for μ_n, $4.31\,\text{mF/m}^2$ for C_{ox}, $20\,\mu\text{m}$ for W and $1\,\mu\text{m}$ for L.

$$V_{OV} = \sqrt{\frac{2(200\,\mu)}{\left(450\times10^{-4}\right)(4.31\,\text{m})\left(\frac{20\,\mu}{1\,\mu}\right)}}$$
$$= \sqrt{\frac{0.4\,\text{m}}{3.88\,\text{m}}}$$
$$= \sqrt{103.09\,\text{m}}$$
$$= 0.32\,\text{V}$$

Therefore, the value of the overdrive voltage V_{DV} is $0.32\,\text{V}$.

Step 3 of 11

The device is operating in saturation region since the value of the drain to source voltage V_{DS} is greater than the overdrive voltage V_{DV}.

Calculate the value of the transconductance g_m.

$$g_m = \frac{2I_D}{V_{OV}}$$

Substitute $200\,\mu\text{A}$ for I_D and $0.32\,\text{V}$ for V_T.

$$g_m = \frac{2(200\,\mu)}{0.32}$$
$$= \frac{0.4\,\text{m}}{0.32}$$
$$= 1.25\,\text{mA/V}$$

Therefore, the value of the transconductance g_m is $\boxed{1.25\,\text{mA/V}}$.

Step 4 of 11

Calculate the value of χ.

$$\chi = \frac{\gamma}{2\sqrt{2\phi_f + V_{SB}}}$$

Substitute $0.5\,\text{V}^{1/2}$ for γ, $0.65\,\text{V}$ for $2\phi_f$ and $1\,\text{V}$ for V_{SB}.

$$\chi = \frac{0.5}{2\sqrt{0.65+1}}$$
$$= \frac{0.5}{2\sqrt{1.65}}$$
$$= \frac{0.5}{2(1.29)}$$
$$= \frac{0.5}{2.58}$$
$$\chi = 0.19$$

Therefore, the value of the χ is $\boxed{0.19}$.

Step 5 of 11

Calculate the value of g_{mb}.

$$g_{mb} = \chi g_m$$

Substitute 0.19 for χ and $1.25\,\text{mA/V}$ for g_m.

$$g_{mb} = (0.19)(1.25\,\text{m})$$
$$= 0.24\,\text{mA/V}$$

Therefore, the value of g_{mb} is $\boxed{0.24\,\text{mA/V}}$.

Step 6 of 11

Calculate the value of the resistance r_o.

$$r_o = \frac{1}{\lambda I_D}$$

Substitute $200\,\mu\text{A}$ for I_D and $0.05\,\text{V}^{-1}$ for λ.

$$r_o = \frac{1}{(0.05)(200\,\mu)}$$
$$= \frac{1}{10\,\mu}$$
$$= 100\,\text{k}\Omega$$

Therefore, the value of the resistance r_o is $\boxed{100\,\text{k}\Omega}$.

Step 7 of 11

Calculate the value of the capacitance C_{ov}.

$$C_{ov} = WL_{ov}C_{ox}$$

Substitute $4.31\,\text{F/m}^2$ for C_{ox}, $20\,\mu\text{m}$ for W and $0.05\,\mu\text{m}$ for L_{ov}.

$$C_{ov} = (20\,\mu)(0.05\,\mu)(4.31\,\text{m})$$
$$= 4.31\,\text{fF}$$

Therefore, the value of C_{ov} is $4.31\,\text{fF}$.

The value of the capacitance C_{gd} is equal to the value of the capacitance C_{ov}, that is $\boxed{4.31\,\text{fF}}$.

Step 8 of 11

Calculate the value of the capacitance C_{gs}.

$$C_{gs} = \frac{2}{3}WLC_{ox} + C_{ov}$$

Substitute $4.31\,\text{fF}$ for C_{ov}, $4.31\,\text{F/m}^2$ for C_{ox}, $20\,\mu\text{m}$ for W and $1\,\mu\text{m}$ for L.

$$C_{gs} = \frac{2}{3}(20\,\mu)(1\,\mu)(4.31\,\text{m}) + 4.31\,\text{f}$$
$$= 57.47\,\text{f} + 4.31\,\text{f}$$
$$= 61.78\,\text{fF}$$

Therefore, the value of the capacitance C_{gs} is $\boxed{61.78\,\text{fF}}$.

Step 9 of 11

Calculate the value of the capacitance C_{sb}.

$$C_{sb} = \frac{C_{sb0}}{\sqrt{1+\frac{V_{SB}}{V_0}}}$$

Substitute $20\,\text{fF}$ for C_{sb0}, $1\,\text{V}$ for V_{SB} and $0.7\,\text{V}$ for V_0.

$$C_{sb} = \frac{20\,\text{f}}{\sqrt{1+\frac{1}{0.7}}}$$
$$= \frac{20\,\text{f}}{\sqrt{1+1.43}}$$
$$= \frac{20\,\text{f}}{\sqrt{2.43}}$$
$$= \frac{20\,\text{f}}{1.56}$$
$$C_{sb} = 12.82\,\text{fF}$$

Therefore, the value of the capacitance C_{sb} is $\boxed{12.82\,\text{fF}}$.

Step 10 of 11

Calculate the value of the capacitance C_{db}.

$$C_{db} = \frac{C_{db0}}{\sqrt{1+\frac{V_{DS}+V_{SB}}{V_0}}}$$

Substitute $20\,\text{fF}$ for C_{db0}, $1\,\text{V}$ for V_{SB}, $1.5\,\text{V}$ for V_{DS} and $0.7\,\text{V}$ for V_0.

$$C_{db} = \frac{20\,\text{f}}{\sqrt{1+\frac{1+1.5}{0.7}}}$$
$$= \frac{20\,\text{f}}{\sqrt{1+\frac{2.5}{0.7}}}$$
$$= \frac{20\,\text{f}}{\sqrt{1+3.57}}$$
$$= \frac{20\,\text{f}}{\sqrt{4.57}}$$
$$C_{db} = \frac{20\,\text{f}}{2.14}$$
$$= 9.34\,\text{fF}$$

Therefore, the value of the capacitance C_{db} is $\boxed{9.34\,\text{fF}}$.

Step 11 of 11

Calculate the value of the unity gain frequency f_T.

$$f_T = \frac{g_m}{2\pi(C_{gs}+C_{gd})}$$

Substitute $4.31\,\text{fF}$ for C_{gd}, $61.78\,\text{fF}$ for C_{gs} and $1.25\,\text{mA/V}$ for g_m.

$$f_T = \frac{1.25\,\text{m}}{2\pi(61.78\,\text{f}+4.31\,\text{f})}$$
$$= \frac{1.25\,\text{m}}{2\pi(66.09\,\text{f})}$$
$$= \frac{1.25\,\text{m}}{415.26\,\text{f}}$$
$$= 3.01\,\text{GHz}$$

Therefore, the value of the unity gain frequency f_T is $\boxed{3.01\,\text{GHz}}$.

Step 1 of 7

Consider the following 3-dB frequency expression that can be extended to any number of poles and zero.

$$\omega_H \approx \frac{1}{\sqrt{\left(\frac{1}{\omega_{P1}^2}+\frac{1}{\omega_{P2}^2}+....\right)-\left(\frac{1}{\omega_{Z1}^2}+\frac{1}{\omega_{Z2}^2}+...\right)}}$$

For the exact value:

The transfer function for this amplifier is,

$$H(s)=\frac{A_M}{\left(1+\frac{s}{\omega_{P1}}\right)\left(1+\frac{s}{\omega_{P2}}\right)}$$

Substitute $j\omega$ for s.

$$H(j\omega)=\frac{A_M}{\left(1+\frac{j\omega}{\omega_{P1}}\right)\left(1+\frac{j\omega}{\omega_{P2}}\right)}$$

Take the magnitude expression.

$$|H(j\omega)|=\frac{A_M}{\sqrt{\left(1+\left(\frac{\omega}{\omega_{P1}}\right)^2\right)\left(1+\left(\frac{\omega}{\omega_{P2}}\right)^2\right)}}$$

The 3-dB frequency ω_H occurs at $|H(j\omega_H)|=\frac{1}{\sqrt{2}}$. Therefore,

$$\frac{A_M^2}{\left(1+\left(\frac{\omega_H}{\omega_{P1}}\right)^2\right)\left(1+\left(\frac{\omega_H}{\omega_{P2}}\right)^2\right)}=\frac{A_M^2}{2}$$

$$\left[1+\left(\frac{\omega_H}{\omega_{P1}}\right)^2\right]\left[1+\left(\frac{\omega_H}{\omega_{P2}}\right)^2\right]=2$$

Step 2 of 7

Substitute $k\omega_{P1}$ for ω_{P2}

$$\left[1+\left(\frac{\omega_H}{\omega_{P1}}\right)^2\right]\left[1+\left(\frac{\omega_H}{k\omega_{P1}}\right)^2\right]=2 \qquad \{\text{since } k=1\}$$

$$\left[1+\left(\frac{\omega_H}{\omega_{P1}}\right)^2\right]^2=2$$

$$1+\left(\frac{\omega_H}{\omega_{P1}}\right)^2=\sqrt{2}$$

$$\omega_H=\left(\sqrt{\sqrt{2}-1}\right)\omega_{P1}$$
$$=0.64\omega_{P1}$$

Therefore, the exact value of ω_H (function of ω_{P1}) is, $\boxed{0.64}$.

Step 3 of 7

For approximate value:

The amplifier is characterized by two zeros at $s=\infty$ and two poles ω_{P1} and ω_{P2}, so,

$$\omega_H \approx \frac{1}{\sqrt{\frac{1}{\omega_{P1}^2}+\frac{1}{\omega_{P2}^2}}}$$

Substitute $k\omega_{P1}$ for ω_{P2}

$$\omega_H=\frac{1}{\sqrt{\frac{1}{\omega_{P1}^2}+\frac{1}{k^2\omega_{P1}^2}}} \qquad \{\text{since } k=1\}$$
$$=\frac{1}{\sqrt{\frac{1}{\omega_{P1}^2}+\frac{1}{(1)^2\omega_{P1}^2}}}$$
$$=\sqrt{\frac{\omega_{P1}^2}{2}}$$
$$=0.71\omega_{P1}$$

Therefore, the approximate value of ω_H (function of ω_{P1}) is, $\boxed{0.71}$.

Step 4 of 7

For the exact value:
Consider the expression.

$$\left[1+\left(\frac{\omega_H}{\omega_{P1}}\right)^2\right]\left[1+\left(\frac{\omega_H}{\omega_{P2}}\right)^2\right]=2$$

Substitute $k\omega_{P1}$ for ω_{P2}

$$\left[1+\left(\frac{\omega_H}{\omega_{P1}}\right)^2\right]\left[1+\left(\frac{\omega_H}{k\omega_{P1}}\right)^2\right]=2 \qquad \{\text{since } k=2\}$$

$$\left[1+\left(\frac{\omega_H}{\omega_{P1}}\right)^2\right]\left[1+\left(\frac{\omega_H}{2\omega_{P1}}\right)^2\right]=2 \qquad \left\{\text{let } m=\frac{\omega_H}{\omega_{P1}}\right\}$$

$$\left[1+m^2\right]\left[1+\left(\frac{m}{2}\right)^2\right]=2$$

$$m^4+5m^2+4=8$$

$$\frac{\omega_H}{\omega_{P1}}=-0.84, 0.84$$

$$\omega_H=0.84\omega_{P1}$$

Therefore, the exact value of ω_H (function of ω_{P1}) is, $\boxed{0.84}$.

Step 5 of 7

For approximate value:

The amplifier is characterized by two zeros at $s=\infty$ and two poles ω_{P1} and ω_{P2}, so,

$$\omega_H \approx \frac{1}{\sqrt{\frac{1}{\omega_{P1}^2}+\frac{1}{\omega_{P2}^2}}}$$

Substitute $k\omega_{P1}$ for ω_{P2}

$$\omega_H=\frac{1}{\sqrt{\frac{1}{\omega_{P1}^2}+\frac{1}{k^2\omega_{P1}^2}}} \qquad \{\text{since } k=2\}$$
$$=\frac{1}{\sqrt{\frac{1}{\omega_{P1}^2}+\frac{1}{(2)^2\omega_{P1}^2}}}$$
$$=\frac{1}{\sqrt{\frac{1}{\omega_{P1}^2}+\frac{1}{4\omega_{P1}^2}}}$$

Simplify further.

$$\omega_H=\frac{1}{\sqrt{\frac{1}{\omega_{P1}^2}(1+0.25)}}$$
$$=\frac{1}{\sqrt{1.25}}\omega_{P1}$$
$$=0.89\omega_{P1}$$

Therefore, the approximate value of ω_H (function of ω_{P1}) is, $\boxed{0.89}$.

Step 6 of 7

For the exact value:

Consider the expression.

$$\left[1+\left(\frac{\omega_H}{\omega_{P1}}\right)^2\right]\left[1+\left(\frac{\omega_H}{\omega_{P2}}\right)^2\right]=2$$

Substitute $k\omega_{P1}$ for ω_{P2}

$$\left[1+\left(\frac{\omega_H}{\omega_{P1}}\right)^2\right]\left[1+\left(\frac{\omega_H}{k\omega_{P1}}\right)^2\right]=2 \qquad \{\text{since } k=4\}$$

$$\left[1+\left(\frac{\omega_H}{\omega_{P1}}\right)^2\right]\left[1+\left(\frac{\omega_H}{4\omega_{P1}}\right)^2\right]=2 \qquad \left\{\text{let } m=\frac{\omega_H}{\omega_{P1}}\right\}$$

$$\left[1+m^2\right]\left[1+\left(\frac{m}{4}\right)^2\right]=2$$

$$m^4+17m^2+16=32$$

$$\frac{\omega_H}{\omega_{P1}}=-0.95, 0.95$$

$$\omega_H=0.95\omega_{P1}$$

Therefore, the exact value of ω_H (function of ω_{P1}) is, $\boxed{0.95}$.

Step 7 of 7

For approximate value:
The amplifier is characterized by two zeros at $s=\infty$ and two poles ω_{P1} and ω_{P2}, so,

$$\omega_H \approx \frac{1}{\sqrt{\frac{1}{\omega_{P1}^2}+\frac{1}{\omega_{P2}^2}}}$$

Substitute $k\omega_{P1}$ for ω_{P2}

$$\omega_H=\frac{1}{\sqrt{\frac{1}{\omega_{P1}^2}+\frac{1}{k^2\omega_{P1}^2}}} \qquad \{\text{since } k=4\}$$
$$=\frac{1}{\sqrt{\frac{1}{\omega_{P1}^2}+\frac{1}{(4)^2\omega_{P1}^2}}}$$
$$=\frac{1}{\sqrt{\frac{1}{\omega_{P1}^2}+\frac{1}{16\omega_{P1}^2}}}$$

Simplify further.

$$\omega_H=\frac{1}{\sqrt{\frac{1}{\omega_{P1}^2}(1+0.0625)}}$$
$$=\frac{1}{\sqrt{1.0625}}\omega_{P1}$$
$$=0.97\omega_{P1}$$

Therefore, the approximate value of ω_H (function of ω_{P1}) is, $\boxed{0.97}$.

Step 1 of 2

10.014P

Calculate the value of the transconductance g_m.

$$g_m = \frac{2I_D}{V_{OV}}$$

Substitute $200\ \mu A$ for I_D and $0.32\ V$ for V_T.

$$\begin{aligned} g_m &= \frac{2(200\ \mu)}{0.3} \\ &= \frac{0.4\ m}{0.3} \\ &= 1.33\ mA/V \end{aligned}$$

Therefore, the value of the transconductance g_m is $\boxed{1.33\ mA/V}$.

Step 2 of 2

Calculate the value of the unity gain frequency f_T.

$$f_T = \frac{g_m}{2\pi(C_{gs} + C_{gd})}$$

Substitute $5\ fF$ for C_{gd}, $25\ fF$ for C_{gs} and $1.33\ mA/V$ for g_m.

$$\begin{aligned} f_T &= \frac{1.33\ m}{2\pi(25\ f + 5\ f)} \\ &= \frac{1.33\ m}{2\pi(30\ f)} \\ &= \frac{1.33\ m}{188.5\ f} \\ &= 7.06\ GHz \end{aligned}$$

Therefore, the value of the unity gain frequency f_T is $\boxed{7.06\ GHz}$.

Step 1 of 3

Calculate the value of the input capacitance C_{in}.

$$C_{in} = C_{gs} + C_{gd}\left(1 + g_m R_L'\right)$$

Substitute $20\ \text{fF}$ for C_{gs}, $5\ \text{fF}$ for C_{gd}, $1.25\ \text{mA/V}$ for g_m and $10\ \text{k}\Omega$ for R_L'.

$$\begin{aligned} C_{in} &= 20\ \text{f} + (5\ \text{f})(1 + (1.25\ \text{m})(10\ \text{k})) \\ &= 20\ \text{f} + (5\ \text{f})(1 + 12.5) \\ &= 20\ \text{f} + (5\ \text{f})(13.5) \\ &= 20\ \text{f} + 67.5\ \text{f} \end{aligned}$$

$$C_{in} = 87.5\ \text{fF}$$

Therefore, the value of the input capacitance C_{in} is $87.5\ \text{fF}$.

Step 2 of 3

Calculate the value of the frequency f_H.

$$f_H = \frac{1}{2\pi C_{in} R_{sig}'}$$

Substitute $87.5\ \text{fF}$ for C_{in} and $10\ \text{k}\Omega$ for R_{sig}'.

$$\begin{aligned} f_H &= \frac{1}{2\pi(87.5\ \text{f})(10\ \text{k})} \\ &= \frac{1}{5.498\ \text{n}} \\ &= 181.9\ \text{MHz} \end{aligned}$$

Therefore, the value of the frequency f_H is $\boxed{181.9\ \text{MHz}}$.

Step 3 of 3

Calculate the value of the percentage change in the value of f_H in Miller method compared to open circuit time constants method.

$$f_H\% = \frac{f_{H,OC} - f_{H,MILLER}}{f_{H,OC}} \times 100$$

Substitute $181.9\ \text{MHz}$ for $f_{H,MILLER}$ and $135.5\ \text{MHz}$ for $f_{H,OC}$.

$$\begin{aligned} f_H\% &= \frac{181.9\ \text{M} - 135.5\ \text{M}}{135.5\ \text{M}} \times 100 \\ &= \frac{46.4\ \text{M}}{135.5\ \text{M}} \times 100 \\ &= 0.34 \times 100 \\ &= 34\% \end{aligned}$$

Therefore, the value of the percentage change in the value of the frequency $f_H\%$ is $\boxed{34\%}$.

Consider the equation.

$$f_T = \frac{g_m}{2\pi\left(C_{gs} + C_{gd}\right)}$$

Where, $C_{gs} \gg C_{gd}$.

Consider the equations.

$$g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}$$

$$C_{gs} = \frac{2}{3} WLC_{ox}$$

From the given data C_{gd} is negligible in the expression of f_T. Now the f_T becomes,

$$f_T = \frac{g_m}{2\pi C_{gs}}$$

Substitute $\sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}$ for g_m and $\frac{2}{3} WLC_{ox}$ for C_{gs} in equation.

$$f_T = \frac{\sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}}{2\pi \times \frac{2}{3} WLC_{ox}}$$

$$= \frac{1}{\pi} \sqrt{\frac{2\mu_n C_{ox} \frac{W}{L} I_D}{\left(\frac{4}{3} WLC_{ox}\right)^2}}$$

$$= \frac{1}{\pi} \sqrt{\frac{2\mu_n C_{ox} \frac{W}{L} I_D}{\frac{4}{9} W^2 L^2 C_{ox}^2}}$$

$$= \frac{1}{\pi} \sqrt{\frac{9\mu_n C_{ox} WI_D}{2W^2 L^3 C_{ox}^2}}$$

$$= \frac{1}{\pi} \sqrt{\frac{9\mu_n I_D}{2WL^3 C_{ox}}}$$

Therefore, the expression for f_T is $\boxed{\frac{1.5}{\pi L} \sqrt{\frac{\mu_n I_D}{2WLC_{ox}}}}$.

Calculate the value of the Gain Band Width product $\mathbf{GBW}$.

$$\mathbf{GBW} = |A_M| f_H$$

Substitute $\mathbf{12.5}$ for $|A_M|$ and $\mathbf{135.5\ MHz}$ for f_H .

$$\begin{aligned}\mathbf{GBW} &= (12.5)(135.5\ \mathrm{M}) \\ &= 1.69\ \mathrm{GHz}\end{aligned}$$

Therefore, the value of the Gain Band Width product $\mathbf{GBW}$ is $\boxed{1.69\ \mathrm{GHz}}$.

Step 1 of 4

Consider the expression for the unity gain frequency f_T.

$$f_T = \frac{g_m}{2\pi\left(C_{gs} + C_{gd}\right)}$$

Here, C_{gd} can be neglected since $C_{gs} >> C_{gd}$.

$$f_T = \frac{g_m}{2\pi C_{gs}}$$

Substitute $\mu_n C_{ox}\left(\frac{W}{L}\right)V_{OV}$ for g_m and $\frac{2}{3}WLC_{ox}$ for C_{gs}.

$$f_T = \frac{\mu_n C_{ox}\left(\frac{W}{L}\right)V_{OV}}{2\pi\left(\frac{2}{3}WLC_{ox}\right)}$$

$$f_T = \frac{3\mu_n V_{OV}}{4\pi L^2} \quad \text{..... (1)}$$

Therefore, the expression for the unity gain frequency f_T is $\boxed{\frac{3\mu_n V_{OV}}{4\pi L^2}}$.

Step 2 of 4

Recall equation (1).

$$f_T = \frac{3\mu_n V_{OV}}{4\pi L^2}$$

Here, unity gain frequency f_T is directly proportional to the over drive voltage V_{OV} where

$\frac{3\mu_n}{4\pi L^2}$ is a constant.

Step 3 of 4

Recall equation (1).

$$f_T = \frac{3\mu_n V_{OV}}{4\pi L^2}$$

Substitute $450\ \text{cm}^2/\text{V}\times\text{s}$ for μ_n, $0.2\ \text{V}$ for V_{OV} and $0.5\ \mu\text{m}$ for L.

$$f_T = \frac{3\left(450\times10^{-4}\right)(0.2)}{4\pi\left(0.5\,\mu\right)^2}$$

$$= \frac{27\ \text{m}}{3.14\ \text{p}}$$

$$= 8.6\ \text{GHz}$$

Therefore, the value of the unity gain frequency f_T for $V_{OV} = 0.2\ \text{V}$ is $\boxed{8.6\ \text{GHz}}$.

Step 4 of 4

Recall equation (1).

$$f_T = \frac{3\mu_n V_{OV}}{4\pi L^2}$$

Substitute $450\ \text{cm}^2/\text{V}\times\text{s}$ for μ_n, $0.4\ \text{V}$ for V_{OV} and $0.5\ \mu\text{m}$ for L.

$$f_T = \frac{3\left(450\times10^{-4}\right)(0.4)}{4\pi\left(0.5\,\mu\right)^2}$$

$$= \frac{54\ \text{m}}{3.14\ \text{p}}$$

$$= 17.2\ \text{GHz}$$

Therefore, the value of the unity gain frequency f_T for $V_{OV} = 0.2\ \text{V}$ is $\boxed{17.2\ \text{GHz}}$.

Step 1 of 8

Consider the expression for the mid band gain A_M .

$$A_M = -\frac{R_B}{R_B + R_{sig}} \frac{r_o}{r_o + r_z + \left(R_B \parallel R_{sig}\right)} g_m R_L'$$

The value of the mid band gain is directly proportional to the value of the resistance R_L' , keeping the other components constant. That is,

$$A_M \alpha R_L'$$

As the value of the resistance R_L' decreases to half, the value of the mid band gain A_M also decreases to half. Hence, calculate the new value of the mid band gain A_M .

$$\begin{aligned} |A_M| &= \frac{12.5}{2} \\ &= 6.25 \text{ V/V} \end{aligned}$$

Therefore, the new value of the mid band gain A_M is $\boxed{6.25 \text{ V/V}}$.

Step 2 of 8

Calculate the value of the resistance R_{gd} .

$$R_{gd} = R_{sig}' \left(1 + g_m R_L'\right) + R_L'$$

Substitute $5\ \text{k}\Omega$ for R_L' , $10\ \text{k}\Omega$ for R_L' and $1.25\ \text{mA/V}$ for g_m .

$$\begin{aligned} R_{gd} &= (10\text{ k})\left(1 + (1.25\text{ m})(5\text{ k})\right) + 5\text{ k} \\ &= (10\text{ k})(1 + 6.25) + 5\text{ k} \\ &= (10\text{ k})(7.25) + 5\text{ k} \\ &= 72.5\text{ k} + 5\text{ k} \\ R_{gd} &= 77.5\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance R_{gd} is $77.5\ \text{k}\Omega$.

Step 3 of 8

Calculate the value of the time constant τ_{gs} .

$$\tau_{gs} = C_{gs} R_{gs}$$

Substitute $77.5\ \text{k}\Omega$ for R_{gs} and $20\ \text{fF}$ for C_{gs} .

$$\begin{aligned} \tau_{gs} &= (20\text{ f})(10\text{ k}) \\ &= 200\text{ ps} \end{aligned}$$

Therefore, the value of the time constant τ_{gs} is $200\ \text{ps}$.

Step 4 of 8

Calculate the value of the time constant τ_{gd} .

$$\tau_{gd} = C_{gd} R_{gd}$$

Substitute $77.5\ \text{k}\Omega$ for R_{gd} and $5\ \text{fF}$ for C_{gd} .

$$\begin{aligned} \tau_{gd} &= (5\text{ f})(77.5\text{ k}) \\ &= 387.5\text{ ps} \end{aligned}$$

Therefore, the value of the time constant τ_{gd} is $387.5\ \text{ps}$.

Step 5 of 8

Calculate the value of the time constant τ_{C_L} .

$$\tau_{C_L} = C_L R_{C_L}$$

Substitute $5\ \text{k}\Omega$ for R_{C2} and $25\ \text{fF}$ for C_{C2} .

$$\begin{aligned} \tau_{C_L} &= (25\text{ f})(5\text{ k}) \\ &= 125\text{ ps} \end{aligned}$$

Therefore, the value of the time constant τ_{C_L} is $125\ \text{ps}$.

Step 6 of 8

Calculate the value of the total time constant τ_H .

$$\tau_H = \tau_{gs} + \tau_{gd} + \tau_{C_L}$$

Substitute $125\ \text{ps}$ for τ_{C_L} , $387.5\ \text{ps}$ for τ_{gd} and $200\ \text{ps}$ for τ_{gs} .

$$\begin{aligned} \tau_H &= 200\text{ p} + 387.5\text{ p} + 125\text{ p} \\ &= 712.5\text{ ps} \end{aligned}$$

Therefore, the value of the total time constant τ_H is $712.5\ \text{ps}$.

Step 7 of 8

Calculate the value of the frequency is f_H .

$$f_H = \frac{1}{2\pi\tau_H}$$

Substitute $712.5\ \text{ps}$ for τ_H .

$$\begin{aligned} f_H &= \frac{1}{2\pi(712.5\text{ p})} \\ &= \frac{1}{4.476\text{ n}} \\ &= 223.4\ \text{MHz} \end{aligned}$$

Therefore, the value of the frequency f_H is $\boxed{223.4\ \text{MHz}}$.

Step 8 of 8

Calculate the value of the Gain Band Width product GBW .

$$\text{GBW} = |A_M| f_H$$

Substitute $6.25\ \text{V/V}$ for $|A_M|$ and $223.4\ \text{MHz}$ for f_H .

$$\begin{aligned} \text{GBW} &= (6.25)(223.4\text{ M}) \\ &= 1.4\ \text{GHz} \end{aligned}$$

Therefore, the value of the Gain Band Width product GBW is $\boxed{1.4\ \text{GHz}}$.

10.017P

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10.018E

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Step 1 of 3

10.018P

Calculate the value of the transconductance g_m.

$$g_m = \frac{I_C}{V_T}$$

Substitute $0.5\ \text{mA}$ for I_C and $25\ \text{mV}$ for V_T.

$$g_m = \frac{0.5\ \text{m}}{25\ \text{m}}$$
$$= 20\ \text{mA/V}$$

Therefore, the value of the transconductance g_m is $20\ \text{mA/V}$.

Step 2 of 3

Calculate the value of unity gain frequency f_T.

$$f_T = \frac{g_m}{2\pi(C_\pi + C_\mu)}$$

Substitute $20\ \text{mA/V}$ for g_m, $8\ \text{pF}$ for C_π and $1\ \text{pF}$ for C_μ.

$$f_T = \frac{20\ \text{m}}{2\pi(8\ \text{p} + 1\text{p})}$$
$$= \frac{20\ \text{m}}{2\pi(9\ \text{p})}$$
$$= \frac{20\ \text{m}}{56.55\text{p}}$$
$$= 353.67\ \text{MHz}$$

Therefore, the value of the unity gain frequency f_T is $\boxed{353.67\ \text{MHz}}$.

Step 3 of 3

Calculate the value of the frequency f_β.

$$f_\beta = \frac{f_T}{\beta}$$

Substitute $353.67\ \text{MHz}$ for f_T and 100 for β.

$$f_\beta = \frac{353.67\ \text{M}}{100}$$
$$= 3.54\ \text{MHz}$$

Therefore, the value of f_β is $\boxed{3.54\ \text{MHz}}$.

10.019E

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10.019P

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Step 1 of 5

Refer to Figure 9.28 in the textbook for the common gate amplifier.

Write the expression for the input resistance.

$$R_{in} = \frac{r_o + R_L}{1 + g_m r_o}$$

Substitute $20\ \text{k}\Omega$ for r_o, $500\ \text{k}\Omega$ for R_L and 25 for $g_m r_o$.

$$\begin{aligned} R_{in} &= \frac{20\times10^3 + 500\times10^3}{1+25} \\ &= \frac{520\times10^3}{26} \\ &= 20\times10^3 \end{aligned}$$

Therefore, the input resistance is, $\boxed{20\ \text{k}\Omega}$.

Compare to old input resistance, $R_{\text{in-old}} = 1.54\ \text{k}\Omega$, the new input resistance is increased by factor of 12.9

Step 2 of 5

Write the expression for the mid-band gain.

$$G_v = \frac{R_L}{R_{sig} + R_{in}}$$

Substitute $500\ \text{k}\Omega$ for R_L, $10\ \text{k}\Omega$ for R_{sig}, and $20\ \text{k}\Omega$ for R_{in}.

$$\begin{aligned} G_v &= \frac{500\times10^3}{10\times10^3 + 20\times10^3} \\ &= \frac{500\times10^3}{30\times10^3} \\ &= 16.7\ \text{V/V} \end{aligned}$$

Therefore, the mid-band gain is, $\boxed{16.7\ \text{V/V}}$.

Compare to old mid-band gain, $G_{v\text{-old}} = 1.73\ \text{V/V}$, the new mid-band gain is increased by factor of 9.7

Step 3 of 5

Write the expression for the upper 3-dB frequency.

$$\begin{aligned} f_H &= \frac{1}{2\pi\tau_H} \\ &= \frac{1}{2\pi\left[C_{gs}R_{gs} + \left(C_{gd} + C_L\right)R_{gd}\right]} \end{aligned}$$

Here,

The gate-to-source resistance is, $R_{gs} = R_{sig} \parallel R_{in}$

Substitute $10\ \text{k}\Omega$ for R_{sig} and $20\ \text{k}\Omega$ for R_{in} to calculate the gate to source resistance

$$\begin{aligned} R_{gs} &= 10\times10^3 \parallel 20\times10^3 \\ &= \frac{\left(10\times10^3\right)\left(20\times10^3\right)}{10\times10^3 + 20\times10^3} \\ &= \frac{200\times10^6}{30\times10^3} \\ &= 6.67\times10^3 \end{aligned}$$

Thus, the gate-to-source resistance is, $R_{gs} = 6.67\ \text{k}\Omega$.

Step 4 of 5

The gate-to-drain resistance is,

$$\begin{aligned} R_{gd} &= R_L \parallel R_o \\ &= R_L \parallel \left(r_o + R_{sig} + \left(g_m r_o\right)R_{sig}\right) \end{aligned}$$

Substitute $500\ \text{k}\Omega$ for R_L, $20\ \text{k}\Omega$ for r_o, $10\ \text{k}\Omega$ for R_{sig}, and 25 for $g_m r_o$.

$$\begin{aligned} R_{gd} &= \left(500\times10^3\right) \parallel \left(20\times10^3 + 10\times10^3 + (25)\left(10\times10^3\right)\right) \\ &= \left(500\times10^3\right) \parallel \left(280\times10^3\right) \\ &= \frac{\left(500\times10^3\right)\left(280\times10^3\right)}{\left(500\times10^3\right) + \left(280\times10^3\right)} \\ &= 179.487\times10^3 \end{aligned}$$

Thus, the gate-to-drain resistance is $R_{gd} = 179.487\ \text{k}\Omega$

Step 5 of 5

Recall the expression for the upper 3-dB frequency.

$$f_H = \frac{1}{2\pi\left[C_{gs}R_{gs} + \left(C_{gd} + C_L\right)R_{gd}\right]}$$

Substitute $20\ \text{fF}$ for C_{gs}, $6.67\ \text{k}\Omega$ for R_{gs}, $5\ \text{fF}$ for C_{gd}, $15\ \text{fF}$ for C_L, and $179.487\ \text{k}\Omega$ for R_{gd} in the upper 3-dB frequency equation.

$$\begin{aligned} f_H &= \frac{1}{2\pi\left[\left(20\times10^{-15}\right)\left(6.67\times10^3\right) + \left(5\times10^{-15} + 15\times10^{-15}\right)\left(179.487\times10^3\right)\right]} \\ &= \frac{1}{2\pi\left[\left(20\times10^{-15}\right)\left(6.67\times10^3\right) + \left(20\times10^{-15}\right)\left(179.487\times10^3\right)\right]} \\ &= \frac{1}{2\pi\left[\left(133.4\times10^{-12}\right) + \left(3589.74\times10^{-12}\right)\right]} \\ &= 42.73\times10^6 \end{aligned}$$

Therefore, the upper 3-dB frequency is, $\boxed{42.73\ \text{MHz}}$.

Compare to old upper 3-dB frequency, $f_H = 397.3\ \text{MHz}$, the new upper 3-dB frequency is decreased by factor of 9.3

Step 1 of 10

10.020P

Consider Figure 1.

Step 2 of 10

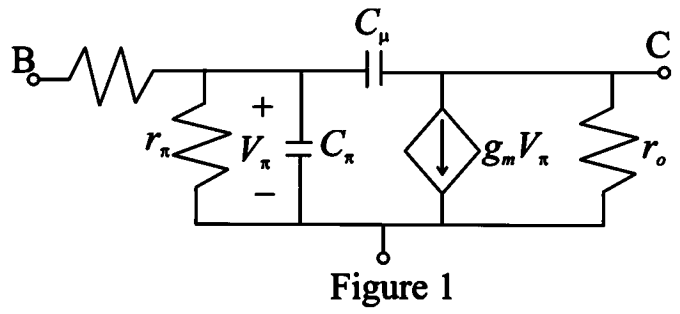


Figure 1

Step 3 of 10

Calculate the value of the transconductance g_m.

$$g_m = \frac{I_C}{V_T}$$

Substitute $1\ \text{mA}$ for I_C and $25\ \text{mV}$ for V_T.

$$g_m = \frac{1\ \text{m}}{25\ \text{m}}$$
$$= 40\ \text{mA/V}$$

Therefore, the value of the transconductance g_m is $\boxed{40\ \text{mA/V}}$.

Step 4 of 10

Calculate the value of the resistance r_π.

$$r_\pi = \frac{\beta_o}{g_m}$$

Substitute $40\ \text{mA/V}$ for g_m and 100 for β.

$$r_\pi = \frac{100}{40\ \text{m}}$$
$$= 2.5\ \text{k}\Omega$$

Therefore, the value of the resistance r_π is $\boxed{2.5\ \text{k}\Omega}$.

Step 5 of 10

Calculate the value of the resistance r_o.

$$r_o = \frac{V_A}{I_C}$$

Substitute $50\ \text{V}$ for V_A and $1\ \text{mA}$ for I_C.

$$r_o = \frac{50}{1\ \text{m}}$$
$$= 50\ \text{k}\Omega$$

Therefore, the value of the resistance r_o is $\boxed{50\ \text{k}\Omega}$.

Step 6 of 10

Calculate the value of the capacitance C_μ.

$$C_\mu = \frac{C_{\mu o}}{\left(1 + \frac{V_{CB}}{V_{0c}}\right)^m}$$

Substitute $2\ \text{V}$ for V_{CB}, $0.75\ \text{V}$ for V_{0c}, 0.5 FOR m and $30\ \text{fF}$ for $C_{\mu o}$.

$$C_\mu = \frac{30\text{f}}{\left(1 + \frac{2}{0.75}\right)^{0.5}}$$
$$= \frac{30\ \text{f}}{(1+2.67)^{0.5}}$$
$$= \frac{30\ \text{f}}{(3.67)^{0.5}}$$
$$= \frac{30\ \text{f}}{1.92}$$
$$C_\mu = 15.63\ \text{fF}$$

Therefore, the value of the capacitance C_μ is $\boxed{15.63\ \text{fF}}$.

Step 7 of 10

Calculate the value of the capacitance C_{je}.

$$C_{je} = 2C_{je0}$$

Substitute $20\ \text{fF}$ for C_{je0}.

$$C_{je} = 2(20\ \text{f})$$
$$= 40\ \text{fF}$$

Therefore, the value of the capacitance C_{je} is $40\ \text{fF}$.

Step 8 of 10

Calculate the value of the capacitance C_{de}.

$$C_{de} = \tau_F g_m$$

Substitute $30\ \text{pF}$ for τ_F and $40\ \text{mA/V}$ for g_m.

$$C_{de} = (30\ \text{p})(40\ \text{m})$$
$$= 1.2\ \text{pF}$$

Therefore, the value of the capacitance C_{de} is $1.2\ \text{pF}$.

Step 9 of 10

Calculate the value of the capacitance C_π.

$$C_\pi = C_{de} + C_{je}$$

Substitute $1.2\ \text{pF}$ for C_{de} and $40\ \text{fF}$ for C_{je}.

$$C_\pi = 1.2\ \text{p} + 40\ \text{f}$$
$$= 1.24\ \text{pF}$$

Therefore, the value of the capacitance C_π is $\boxed{1.24\ \text{pF}}$.

Step 10 of 10

Calculate the value of unity gain frequency f_T.

$$f_T = \frac{g_m}{2\pi(C_\pi + C_\mu)}$$

Substitute $40\ \text{mA/V}$ for g_m, $1.24\ \text{pF}$ for C_π and $15.63\ \text{fF}$ for C_μ.

$$f_T = \frac{40\ \text{m}}{2\pi(15.63\ \text{f} + 1.24\text{p})}$$
$$= \frac{40\ \text{m}}{2\pi(1.26\ \text{p})}$$
$$= \frac{40\ \text{m}}{7.92\text{p}}$$
$$= 5.05\ \text{GHz}$$

Therefore, the value of the unity gain frequency f_T is $\boxed{5.05\ \text{GHz}}$.

Step 1 of 6

(a)

Determine the low frequency gain of the common-source (CS) amplifier.

$$\begin{aligned} A_{M(\text{CS})} &= -g_m\left(r_o \parallel R_L\right) \\ &= -g_m\left(r_o \parallel r_o\right) \ \ (\text{since, } R_L = r_o) \\ &= -\frac{g_m r_o}{2} \end{aligned}$$

Here, g_m is the transconductance, and r_o is the small-signal output resistance.

Substitute 40 for $g_m r_o$.

$$\begin{aligned} A_{M(\text{CS})} &= -\frac{40}{2} \\ &= -20 \end{aligned}$$

Determine the low frequency gain of the CS amplifier.

$$\begin{aligned} A_{M(\text{Cascode})} &= -g_m\left(R_o \parallel R_L\right) \\ &= -g_m\left(\left[2r_o + (g_m r_o) r_o\right] \parallel r_o\right) \\ &= -g_m\left(\left[2r_o + 40r_o\right] \parallel r_o\right) \\ &= -g_m\left(42r_o \parallel r_o\right) \end{aligned}$$

$$\begin{aligned} A_{M(\text{Cascode})} &= -g_m\left(\frac{42r_o^2}{42r_o + r_o}\right) \\ &= -g_m\left(\frac{42r_o^2}{43r_o}\right) \\ &= -0.977 g_m r_o \\ &= -0.977(40) \end{aligned}$$

$$A_{M(\text{Cascode})} = -39.1 \text{ V/V}$$

The ratio of the low frequency gain of the cascade amplifier to that of CS amplifier is,

$$\frac{A_{M(\text{Cascode})}}{A_{M(\text{CS})}} = \frac{-39.1}{-20}$$

$$\frac{A_{M(\text{Cascode})}}{A_{M(\text{CS})}} \simeq 2$$

Hence, the ratio of the low frequency gain of the cascade amplifier to that of CS amplifier is $\boxed{2}$.

Step 2 of 6

(b)

Determine the time constant, τ_H.

$$\begin{aligned} \tau_H = R_{sig}\left[C_{gs1} + C_{gd1}\left(1 + g_{m1}R_{d1}\right)\right] \\ + R_{d1}\left(C_{gd1} + C_{db1} + C_{gs2}\right) + \left(R_L \parallel R_o\right)\left(C_L + C_{db2} + C_{gd2}\right) \end{aligned}$$

Neglect components without R_{sig}.

$$\begin{aligned} \tau_H &= R_{sig}\left[C_{gs1} + C_{gd1}\left(1 + g_{m1}R_{d1}\right)\right] \\ &= R_{sig}\left[C_{gs1} + 0.25C_{gs1}\left(1 + g_{m1}\frac{r_o + R_L}{g_m r_o}\right)\right] \quad \left(\text{Since } R_{d1} = \frac{r_o + R_L}{g_m r_o}\right) \\ &= R_{sig}C_{gs1}\left[1 + 0.25\left(1 + g_{m1}\frac{2r_o}{g_{m1}r_o}\right)\right] \quad (\text{Since } R_L = r_o) \\ &= R_{sig}C_{gs1}\left[1 + 0.25\left(1 + g_{m1}\frac{2r_o}{g_{m1}r_o}\right)\right] \end{aligned}$$

$$\begin{aligned} \tau_H &= R_{sig}C_{gs1}\left[1 + 0.25(1 + 2)\right] \\ &= 1.75R_{sig}C_{gs1} \\ &= 1.75R_{sig}C_{gs} \end{aligned}$$

Step 3 of 6

Determine the upper 3 dB frequency of cascade amplifier, $f_{H(\text{Cascode})}$.

$$\begin{aligned} f_{H(\text{Cascode})} &= \frac{1}{2\pi\tau_H} \\ &= \frac{1}{2\pi\left(1.75R_{sig}C_{gs}\right)} \end{aligned}$$

For CS amplifier R_L' is

$$\begin{aligned} R_L' &= r_o \parallel R_L \\ &= r_o \parallel r_o \\ &= \frac{r_o}{2} \end{aligned}$$

Determine the time constant, τ_H.

$$\begin{aligned} \tau_H &= C_{gs}R_{sig} + C_{gd}\left[\left(1 + g_m R_L'\right)R_{sig}\right] \\ &= C_{gs}R_{sig} + 0.25C_{gs}\left[\left(1 + g_m\frac{r_o}{2}\right)R_{sig}\right] \\ &= R_{sig}C_{gs}\left[1 + 0.25\left(1 + \frac{40}{2}\right)\right] \\ &= 6.25R_{sig}C_{gs} \end{aligned}$$

Step 4 of 6

The upper 3 dB frequency of cascade amplifier, $f_{H(\text{Cascode})}$ is,

$$\begin{aligned} f_{H(\text{CS})} &= \frac{1}{2\pi\tau_H} \\ &= \frac{1}{2\pi\left(6.25R_{sig}C_{gs}\right)} \end{aligned}$$

Determine the ratio f_H of the cascade amplifier to that of CS amplifier.

$$\begin{aligned} \frac{f_{H(\text{Cascode})}}{f_{H(\text{CS})}} &= \frac{\frac{1}{2\pi\left(1.75R_{sig}C_{gs}\right)}}{\frac{1}{2\pi\left(6.25R_{sig}C_{gs}\right)}} \\ &= \frac{6.25}{1.75} \\ &= 3.57 \\ &\approx 3.6 \end{aligned}$$

Hence, the ratio of f_H for the cascade amplifier to that of CS amplifier is $\boxed{3.6}$.

Step 5 of 6

(c)

The unity gain frequency of cascade amplifier is,

$$\begin{aligned} f_{t(\text{Cascode})} &= \left|A_M\right| f_H \\ &= 40\left(\frac{1}{2\pi\left(1.75R_{sig}C_{gs}\right)}\right) \\ &= \frac{40}{2\pi\left(1.75R_{sig}C_{gs}\right)} \end{aligned}$$

The unity gain frequency of CS amplifier is

$$\begin{aligned} f_{t(\text{CS})} &= \left|A_M\right| f_H \\ &= 20\left(\frac{1}{2\pi\left(6.25R_{sig}C_{gs}\right)}\right) \\ &= \frac{20}{2\pi\left(6.25R_{sig}C_{gs}\right)} \end{aligned}$$

Step 6 of 6

Determine the ratio of f_t for the cascade to that of CS amplifier.

$$\begin{aligned} \frac{f_{t(\text{Cascode})}}{f_{t(\text{CS})}} &= \frac{\frac{40}{2\pi\left(1.75R_{sig}C_{gs}\right)}}{\frac{20}{2\pi\left(6.25R_{sig}C_{gs}\right)}} \\ &= \frac{40(6.25)}{20(1.75)} \\ &= 7.2 \end{aligned}$$

Hence, the ratio of f_t for the cascade to that of CS amplifier is $\boxed{7.2}$.

Step 1 of 4

Write the formula for Common-Emitter short-circuit current gain, $|h_{fe}|$.

$$|h_{fe}| = \frac{f_T}{f} \quad \text{..... (1)}$$

Here,

f_T is the transition frequency

f is the frequency of operation of npn transistor

Determine the transition frequency at collector current of 0.2 mA.

Substitute $50\times10^6\ \text{Hz}$ for f and 10 for $|h_{fe}|$ in equation (1).

$$10 = \frac{f_T}{50\times10^6\ \text{Hz}}$$
$$f_T = 500\times10^6\ \text{Hz}$$
$$= 500\ \text{MHz}$$

Therefore, the transition frequency at collector current of 0.2 mA is **500 MHz**.

Step 2 of 4

Determine the transition frequency at collector current of 1 mA.

Substitute $50\times10^6\ \text{Hz}$ for f and 12 for $|h_{fe}|$ in equation (1).

$$12 = \frac{f_T}{50\times10^6\ \text{Hz}}$$
$$f_T = 600\times10^6\ \text{Hz}$$
$$= 600\ \text{MHz}$$

Thus, the transition frequency at collector current of 1 mA is **600 MHz**.

Step 3 of 4

Write the expression for transition frequency, f_T

$$f_T = \frac{g_m}{2\pi(C_\pi + C_\mu)}$$

Here,

C_μ is the collector-base capacitance

C_π is the emitter-base capacitance

g_m is the trans-conductance

Re-arrange the terms to get the expression for C_π.

$$2\pi(C_\pi + C_\mu) = \frac{g_m}{f_T}$$
$$C_\pi + C_\mu = \frac{g_m}{2\pi f_T}$$
$$C_\pi = \frac{g_m}{2\pi f_T} - C_\mu$$

Determine C_π at a collector current of 0.2 mA.

Substitute 8×10^{-3} for g_m, $500\times10^6\ \text{Hz}$ for f_T and $0.1\times10^{-12}\ \text{F}$ for C_μ.

$$C_\pi = \frac{8\times10^{-3}}{(2\pi)(500\times10^6\ \text{Hz})} - (0.1\times10^{-12}\ \text{F})$$
$$= 2.446\times10^{-12}\ \text{F}$$

Determine C_π at a collector current of 1 mA.

Substitute 40×10^{-3} for g_m, $600\times10^6\ \text{Hz}$ for f_T and $0.1\times10^{-12}\ \text{F}$ for C_μ.

$$C_\pi = \frac{40\times10^{-3}}{(2\pi)(600\times10^6\ \text{Hz})} - 0.1\times10^{-12}\ \text{F}$$
$$= 10.51\times10^{-12}\ \text{F}$$

Step 4 of 4

Consider the following formula for the emitter-base capacitance, C_π in terms of base-emitter junction capacitance, C_{je} and forward base-transit time, τ_F :

$$C_\pi = C_{je} + \tau_F g_m$$

Substitute 8×10^{-3} for g_m and $2.446\times10^{-12}\ \text{F}$ for C_π.

$$C_{je} + \tau_F(8\times10^{-3}) = 2.446\times10^{-12} \quad \text{..... (2)}$$

Substitute 40×10^{-3} for g_m and $10.51\times10^{-12}\ \text{F}$ for C_π.

$$C_{je} + \tau_F(40\times10^{-3}) = 10.51\times10^{-12} \quad \text{..... (3)}$$

Solve equations (1) and (2) to get the values of C_{je} and τ_F.

$$C_{je} = 0.43\times10^{-12}\ \text{F}$$
$$= 0.43\ \text{pF}$$

$$\tau_F = 2.52\times10^{-10}\ \text{F}$$
$$= 252\ \text{ps}$$

Therefore, the forward base-transit time, τ_F is **252 ps**.

Therefore, the base-emitter junction capacitance C_{je} is **0.43 pF**.

Step 1 of 12

Refer to Figure 9.31 in the text book for the BJT cascade amplifier.

The transconductance is,

$$\begin{aligned} g_{m1} &= g_{m2} \\ &= g_m \\ &= \frac{I}{V_T} \end{aligned}$$

Here, I is the collector current and V_T thermal equivalent voltage.

Substitute $1\,\text{mA}$ for I and $25\,\text{mV}$ for V_T.

$$\begin{aligned} g_m &= \frac{1}{25} \\ &= 40\ \text{mA/V} \\ &= 40\times10^{-3}\ \text{A/V} \end{aligned}$$

Determine the resistance r_π.

$$\begin{aligned} r_{\pi1} &= r_{\pi2} \\ &= r_\pi \\ &= \frac{\beta}{g_m} \end{aligned}$$

Here, β is the current gain.

Substitute 200 for β and $40\times10^{-3}\ \text{A/V}$ for g_m.

$$\begin{aligned} r_\pi &= \frac{200}{40\times10^{-3}} \\ &= 5\times10^3\ \Omega \end{aligned}$$

Step 2 of 12

Determine the input resistance, R_{in}.

$$R_{in} = r_{\pi1} + r_\pi$$

Substitute $5\times10^3\ \Omega$ for $r_{\pi1}$ and $200\ \Omega$ for r_π.

$$\begin{aligned} R_{in} &= 5\times10^3 + 200 \\ &= 5.2\times10^3\ \Omega \\ &= 5.2\ \text{k}\Omega \end{aligned}$$

Thus, the input resistance is $\boxed{5.2\ \text{k}\Omega}$.

Step 3 of 12

Determine the output gain, A_0.

$$A_0 = g_m r_o$$

Substitute $40\times10^{-3}\ \text{A/V}$ for g_m and $130\times10^3\ \Omega$ for r_o.

$$\begin{aligned} A_0 &= \left(40\times10^{-3}\right)\left(130\times10^3\right) \\ &= 5200\ \text{V/V} \end{aligned}$$

Thus, the output gain A_0 is $\boxed{5200\ \text{V/V}}$.

The output resistance of the transistor Q_1 is,

$$\begin{aligned} R_{o1} &= r_o \\ &= 130\ \text{k}\Omega \end{aligned}$$

Thus, the output resistance R_{o1} is $\boxed{130\ \text{k}\Omega}$.

Step 4 of 12

Calculate the output resistance of the transistor, Q_2.

$$R_{in2} = \frac{r_o + R_L}{g_m r_o}$$

Substitute $130\times10^3\ \Omega$ for r_o, $40\times10^{-3}\ \text{A/V}$ for g_m and $50\times10^3\ \Omega$ for R_L.

$$\begin{aligned} R_{in2} &= \frac{130\times10^3 + 50\times10^3}{\left(40\times10^{-3}\right)\left(130\times10^3\right)} \\ &= \frac{180\times10^3}{5200} \\ &= 34.62\ \Omega \\ &\approx 35\ \Omega \end{aligned}$$

Therefore, the input resistance R_{in2} is $\boxed{35\Omega}$.

Step 5 of 12

Calculate the output resistance.

$$\begin{aligned} R_o &= \beta_2 r_{o2} \\ &= \beta r_o \end{aligned}$$

Substitute 200 for β and $130\times10^3\ \Omega$ for r_o.

$$\begin{aligned} R_o &= (200)\left(130\times10^3\right) \\ &= 26\times10^6\ \Omega \\ &= 26\ \text{M}\Omega \end{aligned}$$

Therefore, the total output resistance is $\boxed{26\text{M}\Omega}$.

Step 6 of 12

The midband gain is,

$$A_M = -\frac{r_\pi}{r_\pi + r_x + R_{sig}}\, g_m \left(\beta r_o \,\|\, R_L\right)$$

Substitute $5\times10^3\ \Omega$ for $r_{\pi1}$, $200\ \Omega$ for r_x, $36\times10^3\Omega$ for R_{sig}, 200 for β, $40\times10^{-3}\ \text{A/V}$ for g_m, $130\times10^3\ \Omega$ for r_o, and $50\times10^3\ \Omega$ for R_L.

$$\begin{aligned} A_M &= -\frac{5\times10^3}{5\times10^3 + 200 + 36\times10^3}\left(40\times10^{-3}\right)\left[(200)\left(130\times10^3\right)\|\left(50\times10^3\right)\right] \\ &= -\frac{5\times10^3}{41.2\times10^3}\left(40\times10^{-3}\right)\left[\left(26\times10^6\right)\|\left(50\times10^3\right)\right] \\ &= -\left(\frac{25}{206}\right)\left(40\times10^{-3}\right)\left(\frac{\left(26\times10^6\right)\left(50\times10^3\right)}{\left(26\times10^6\right)+\left(50\times10^3\right)}\right) \\ &\approx -241\ \text{V/V} \end{aligned}$$

Thus, the midband gain A_M is $\boxed{-241\ \text{V/V}}$.

Step 7 of 12

Determine the value of the resistance, $R_{\pi1}$.

$$\begin{aligned} R_{\pi1} &= R'_{sig} \\ &= r_{\pi1}\,\|\left(r_{x1} + R_{sig}\right) \\ &= \left(5\times10^3\right)\|\left(200 + 36\times10^3\right) \\ &= \left(5\times10^3\right)\|\left(36.2\times10^3\right) \end{aligned}$$

$$\begin{aligned} R_{\pi1} &= \frac{\left(5\times10^3\right)\left(36.2\times10^3\right)}{\left(5\times10^3\right)+\left(36.2\times10^3\right)} \\ &= 4.39\times10^3\ \Omega \end{aligned}$$

Determine value of the resistance, R_{c1}.

$$\begin{aligned} R_{c1} &= r_{o1}\,\|\,R_{in2} \\ &= \left(130\times10^3\right)\|(35) \\ &= \frac{\left(130\times10^3\right)(35)}{\left(130\times10^3\right)+(35)} \\ &\approx 35\ \Omega \end{aligned}$$

Step 8 of 12

Determine the value of the resistance, $R_{\mu1}$.

$$\begin{aligned} R_{\mu1} &= R'_{sig}\left(1 + g_{m1}R_{c1}\right) + R_{c1} \\ &= R_{\pi1}\left(1 + g_{m1}R_{c1}\right) + R_{c1} \\ &= \left(4.39\times10^3\right)\left[1 + \left(40\times10^{-3}\right)(34.99)\right] + 35 \\ &= \left(4.39\times10^3\right)\left(1 + 1.3996\right) + 35 \end{aligned}$$

$$\begin{aligned} &= \left(4.39\times10^3\right)(2.3996) + 35 \\ &= \left(10.534\times10^3\right) + 35 \\ &= 10.57\times10^3\ \Omega \end{aligned}$$

Step 9 of 12

Calculate the upper 3-dB frequency, f_H.

$$\begin{aligned} f_H &= \frac{1}{2\pi\tau_H} \\ &= \frac{1}{2\pi\left[C_{\pi1}R_{\pi1} + C_{\mu1}R_{\mu1} + \left(C_{\pi1} + C_{\pi2}\right)R_{c1} + \left(C_L + C_{\pi2} + C_{\mu2}\right)\left(R_L \,\|\, R_o\right)\right]} \end{aligned}$$

Substitute the known values of capacitances and resistances.

$$\begin{aligned} f_H &= \frac{1}{2\pi\left[\begin{array}{l}\left(16\times10^{-12}\right)\left(4.39\times10^3\right) + \left[\left(0.3\times10^{-12}\right)\left(10.57\times10^3\right)\right] + \\ \left[\left(0 + 16\times10^{-12}\right)(35)\right] + \\ \left[\left(5\times10^{-12} + 0 + 0.3\times10^{-12}\right)\left(50\times10^3 \,\|\, 26\times10^6\right)\right]\end{array}\right]} \\ &= \frac{1}{2\pi\left[\begin{array}{l}70.24\times10^{-9} + 3.171\times10^{-9} + 0.56\times10^{-9} + \\ \left\{\left(5.3\times10^{-9}\right)\left[\frac{\left(50\times10^3\right)\left(26\times10^6\right)}{50\times10^3 + 26\times10^6}\right]\right\}\end{array}\right]} \\ &= \frac{1}{2\pi\left[70.24\times10^{-9} + 3.171\times10^{-9} + 0.56\times10^{-9} + 264.49\times10^{-9}\right]} \end{aligned}$$

$$\begin{aligned} &= 470\times10^3\ \text{Hz} \\ &= 470\ \text{kHz} \end{aligned}$$

Thus, the upper 3-dB frequency, f_H is $\boxed{470\ \text{kHz}}$.

Step 10 of 12

Calculate the gain bandwidth product.

$$f_t = |A_M|\, f_N$$

Substitute $-241\ \text{V/V}$ for A_M and $470\ \text{kHz}$ for f_H.

$$\begin{aligned} f_t &= |-241|\left(470\times10^3\right) \\ &= 113.3\times10^6\ \text{Hz} \\ &= 113.3\ \text{MHz} \end{aligned}$$

Therefore, the gain bandwidth product f_t is $\boxed{113.3\ \text{MHz}}$.

Step 11 of 12

The original value of midband gain, $A_{M\text{-original}} = -175\ \text{V/V}$.

The original value of upper3-dB frequency, $f_{H\text{-original}} = 75.1\ \text{kHz}$.

The original value of gain bandwidth product, $f_{t\text{-original}} = 13.1\ \text{MHz}$.

The obtained midband gain is increased from $-175\ \text{V/V}$ to $-241.52\ \text{V/V}$ when compared to the original midband gain.

The obtained upper3-dB frequency is increased from $75.1\ \text{kHz}$ to $470\ \text{kHz}$ when compared to original upper3-dB frequency.

The obtained gain bandwidth product is increased from $13.1\ \text{MHz}$ to $113.5\ \text{MHz}$ when compared to the original gain bandwidth product.

If the 3-dB frequency is $f_N = 1\ \text{MHz}$, then the time constant is,

$$\begin{aligned} \tau_H &= \frac{1}{2\pi f_H} \\ &= \frac{1}{2\pi\left(1\times10^6\right)} \\ &= 159\times10^{-9} \end{aligned}$$

Thus, the time constant τ_H is $\boxed{159\ \text{ns}}$.

Step 12 of 12

Calculate the value of load capacitance by subsuming the known values.

$$\tau_H = C_{\pi1}R_{\pi1} + C_{\mu1}R_{\mu1} + \left(C_{\pi1} + C_{\pi2}\right)R_{c1} + \left(C_L + C_{\pi2} + C_{\mu2}\right)\left(R_L \,\|\, R_o\right)$$

$$159\times10^{-9} = \left[\begin{array}{l}\left\{\left(16\times10^{-12}\right)\left(4.39\times10^3\right)\right\} + \left\{\left(0.3\times10^{-12}\right)\left(10.57\times10^3\right)\right\} + \\ \left\{\left(0 + 16\times10^{-12}\right)(35)\right\} + \\ \left\{\left(C_L + 0 + 0.3\times10^{-12}\right)\left(50\times10^3 \,\|\, 26\times10^6\right)\right\}\end{array}\right]$$

$$159\times10^{-9} = \left[\begin{array}{l}70.24\times10^{-9} + 3.171\times10^{-9} + 0.56\times10^{-9} + \\ \left\{\left(C_L + 0.3\times10^{-12}\right)\left(\frac{\left(50\times10^3\right)\left(26\times10^6\right)}{50\times10^3 + 26\times10^6}\right)\right\}\end{array}\right]$$

$$159\times10^{-9} = \left[\begin{array}{l}70.24\times10^{-9} + 3.171\times10^{-9} + 0.56\times10^{-9} + \\ \left(C_L + 0.3\times10^{-12}\right)\left(50\times10^3\right)\end{array}\right]$$

$$159\times10^{-9} = \left(73.971\times10^{-9}\right) + \left\{\left(C_L + 0.3\times10^{-12}\right)\left(50\times10^3\right)\right\}$$

$$159\times10^{-9} - \left(73.971\times10^{-9}\right) = \left\{\left(C_L + 0.3\times10^{-12}\right)\left(50\times10^3\right)\right\}$$

$$85.029\times10^{-9} = \left(C_L + 0.3\times10^{-12}\right)\left(50\times10^3\right)$$

$$\begin{aligned} C_L + 0.3\times10^{-12} &= \frac{85.029\times10^{-9}}{50\times10^3} \\ C_L + 0.3\times10^{-12} &= 1.7\times10^{-12} \\ C_L &= 1.4\times10^{-12}\ \text{F} \\ &= 1.4\ \text{pF} \end{aligned}$$

Therefore, the load capacitance is reduced to $\boxed{1.4\ \text{pF}}$.

Step 1 of 4

Calculate the value of the transconductance g_m.

$$g_m = \frac{I_C}{V_T}$$

Substitute $1\ \text{mA}$ for I_C and $25\ \text{mV}$ for V_T.

$$g_m = \frac{1\ \text{m}}{25\ \text{m}}$$
$$= 40\ \text{mA/V}$$

Therefore, the value of the transconductance g_m is $40\ \text{mA/V}$.

Step 2 of 4

Calculate the value of unity gain frequency f_T.

$$f_T = \frac{g_m}{2\pi(C_\pi + C_\mu)}$$

Substitute $40\ \text{mA/V}$ for g_m, $0.1\ \text{pF}$ for C_μ and $10\ \text{GHz}$ for f_T.

$$10\ \text{G} = \frac{40\ \text{m}}{2\pi(C_\pi + 0.1\text{p})}$$
$$C_\pi + 0.1\text{p} = \frac{40\ \text{m}}{2\pi(10\ \text{G})}$$
$$C_\pi = \frac{40\ \text{m}}{62.83\text{G}} - 0.1\ \text{p}$$
$$C_\pi = 636.64\ \text{f} - 0.1\ \text{p}$$
$$C_\pi = 536.64\ \text{fF}$$

Therefore, the value of the capacitance C_π is $\boxed{536.64\ \text{fF}}$.

Step 3 of 4

Calculate the value of the resistance r_π.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute $40\ \text{mA/V}$ for g_m and 120 for β.

$$r_\pi = \frac{120}{40\ \text{m}}$$
$$= 3\ \text{k}\Omega$$

Therefore, the value of the resistance r_π is $\boxed{3\ \text{k}\Omega}$.

Step 4 of 4

Calculate the value of the frequency f_β.

$$f_\beta = \frac{f_T}{\beta}$$

Substitute $10\ \text{GHz}$ for f_T and 120 for β.

$$f_\beta = \frac{10\ \text{G}}{120}$$
$$= 83.33\ \text{MHz}$$

Therefore, the value of f_β is $\boxed{83.33\ \text{MHz}}$.

Step 1 of 3

Consider the expression for the time constant τ_H .

$$\tau_H = \left(C_{gd} + \frac{C_{gs}}{g_m R'_L + 1} \right) R_{sig} + \left(\frac{C_{gs} + C_L}{g_m R'_L + 1} \right) R'_L \quad \text{..... (1)}$$

Substitute 0 for C_{gd} and C_L to calculate the value of the resistance R_{gs} .

$$\begin{aligned}
\tau_H &= \left(0 + \frac{C_{gs}}{g_m R'_L + 1} \right) R_{sig} + \left(\frac{C_{gs} + 0}{g_m R'_L + 1} \right) R'_L \\
&= \left(\frac{C_{gs}}{g_m R'_L + 1} \right) R_{sig} + \left(\frac{C_{gs}}{g_m R'_L + 1} \right) R'_L \\
&= \frac{C_{gs} R_{sig}}{g_m R'_L + 1} + \frac{C_{gs} R'_L}{g_m R'_L + 1} \\
&= C_{gs} \left(\frac{R_{sig} + R'_L}{g_m R'_L + 1} \right)
\end{aligned}$$

$$\frac{\tau_H}{C_{gs}} = \frac{R_{sig} + R'_L}{g_m R'_L + 1}$$

Here, $\frac{\tau_H}{C_{gs}}$ represents the resistance R_{gs} .

$$R_{gs} = \frac{R_{sig} + R'_L}{g_m R'_L + 1}$$

Therefore, the value of the resistance R_{gs} is $\boxed{\frac{R_{sig} + R'_L}{g_m R'_L + 1}}$.

Step 2 of 3

Recall equation (1).

$$\tau_H = \left(C_{gd} + \frac{C_{gs}}{g_m R'_L + 1} \right) R_{sig} + \left(\frac{C_{gs} + C_L}{g_m R'_L + 1} \right) R'_L$$

Substitute 0 for C_{gs} and C_L to calculate the value of the resistance R_{gd} .

$$\begin{aligned}
\tau_H &= \left(C_{gd} + \frac{0}{g_m R'_L + 1} \right) R_{sig} + \left(\frac{0 + 0}{g_m R'_L + 1} \right) R'_L \\
&= C_{gd} R_{sig} + 0 \\
&= C_{gd} R_{sig}
\end{aligned}$$

$$\frac{\tau_H}{C_{gd}} = R_{sig}$$

Here, $\frac{\tau_H}{C_{gs}}$ represents the resistance R_{gd} .

$$R_{gd} = R_{sig}$$

Therefore, the value of the resistance R_{gd} is $\boxed{R_{sig}}$.

Step 3 of 3

Recall equation (1).

$$\tau_H = \left(C_{gd} + \frac{C_{gs}}{g_m R'_L + 1} \right) R_{sig} + \left(\frac{C_{gs} + C_L}{g_m R'_L + 1} \right) R'_L$$

Substitute 0 for C_{gd} and C_{gs} to calculate the value of the resistance R_{C_L} .

$$\begin{aligned}
\tau_H &= \left(0 + \frac{0}{g_m R'_L + 1} \right) R_{sig} + \left(\frac{0 + C_L}{g_m R'_L + 1} \right) R'_L \\
&= 0 + \left(\frac{0 + C_L}{g_m R'_L + 1} \right) R'_L \\
&= \frac{C_L R'_L}{g_m R'_L + 1}
\end{aligned}$$

$$\frac{\tau_H}{C_L} = \frac{R'_L}{g_m R'_L + 1}$$

Here, $\frac{\tau_H}{C_L}$ represents the resistance R_{C_L} .

$$R_{C_L} = \frac{R'_L}{g_m R'_L + 1}$$

Therefore, the value of the resistance R_{C_L} is $\boxed{\frac{R'_L}{g_m R'_L + 1}}$.

10.023P

Step 1 of 3

Write the expression of unity gain frequency.

$$\omega_T = \beta_o \omega_\beta$$

$$\omega_\beta = \frac{\omega_T}{\beta_o}$$

Substitute 2 GHz for ω_T and 200 for β_o.

$$\omega_\beta = \frac{2 \times 10^9}{200}$$
$$= 0.01 \text{ GHz}$$

Write the general expression of ω_β.

$$\omega_\beta = \frac{1}{(C_\pi + C_\mu) r_\pi}$$

$$\omega = \frac{1}{(C_\pi + C_\mu) r_\pi}$$

$$(C_\pi + C_\mu) r_\pi = \frac{1}{0.01 \times 10^9}$$
$$= 100 \times 10^{-9}$$

Step 2 of 3

Refer to the equation 9.39 in the text book.

Write the expression of h_{fe}.

$$h_{fe} = \frac{\beta_o}{1 + 2\pi f (C_\pi + C_\mu) r_\pi}$$

Substitute 100×10^{-9} for $(C_\pi + C_\mu) r_\pi$ and 40 for h_{fe}.

$$40 = \frac{200}{1 + 2\pi f (100 \times 10^{-9})}$$

$$1 + 2\pi f (100 \times 10^{-9}) = 5$$

$$f = \frac{4}{2\pi (100 \times 10^{-9})}$$

$$f = 6.36 \text{ MHz}$$

Therefore, the value of frequency f is $\boxed{6.36 \text{ MHz}}$.

Step 3 of 3

Determine the value of frequency f_β.

$$2\pi f_\beta = \frac{1}{\omega_\beta}$$

$$f_\beta = \frac{1}{2\pi \omega_\beta}$$

Substitute 0.01 GHz for ω_β.

$$f_\beta = \frac{1}{2\pi (0.01 \text{ GHz})}$$
$$= \frac{1}{0.01 \times 10^9 \times 2\pi}$$
$$= 1.59 \text{ GHz}$$

Therefore, the value of f_β is $\boxed{1.59 \text{ GHz}}$.

Step 1 of 2

Consider the expression for the time constant τ_H .

$$\tau_H = \left(C_{gd} + \frac{C_{gs}}{g_m R_L' + 1} \right) R_{sig} + \left(\frac{C_{gs} + C_L}{g_m R_L' + 1} \right) R_L'$$

Substitute $5\ \text{fF}$ for C_{gd} , $20\ \text{fF}$ for C_{gs} , $15\ \text{fF}$ for C_L , $2\ \text{mA/V}$ for g_m , $2\ \text{k}\Omega$ for R_L'

and $10\ \text{k}\Omega$ for R_{sig} .

$$\begin{aligned}
\tau_H &= \left(5\ \text{f} + \frac{20\ \text{f}}{(2\ \text{m})(2\ \text{k}) + 1} \right)(10\ \text{k}) + \left(\frac{20\ \text{f} + 15\ \text{f}}{(2\ \text{m})(2\ \text{k}) + 1} \right)(2\ \text{k}) \\
&= \left(5\ \text{f} + \frac{20\ \text{f}}{4 + 1} \right)(10\ \text{k}) + \left(\frac{35\ \text{f}}{4 + 1} \right)(2\ \text{k}) \\
&= \left(5\ \text{f} + \frac{20\ \text{f}}{5} \right)(10\ \text{k}) + \left(\frac{35\ \text{f}}{5} \right)(2\ \text{k}) \\
&= (5\ \text{f} + 4\ \text{f})(10\ \text{k}) + (7\ \text{f})(2\ \text{k})
\end{aligned}$$

$$\begin{aligned}
\tau_H &= (9\ \text{f})(10\ \text{k}) + 14\ \text{p} \\
&= 90\ \text{p} + 14\ \text{p} \\
&= 104\ \text{ps}
\end{aligned}$$

Therefore, the value of the time constant τ_H is $104\ \text{ps}$.

Step 2 of 2

Calculate the value of the frequency is f_H .

$$f_H = \frac{1}{2\pi\tau_H}$$

Substitute $104\ \text{pF}$ for τ_H .

$$\begin{aligned}
f_H &= \frac{1}{2\pi(104\ \text{p})} \\
&= \frac{1}{653.45\ \text{p}} \\
&= 1.53\ \text{GHz}
\end{aligned}$$

Therefore, the value of the frequency f_H is $\boxed{1.53\ \text{GHz}}$.

Step 1 of 5

Draw the following high frequency hybrid-π model.

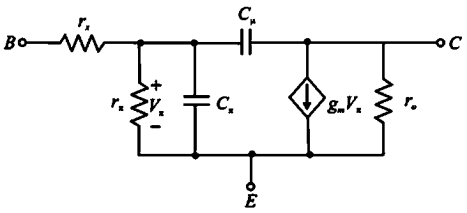


Figure 1

Step 2 of 5

Draw the following modified circuit diagram with emitter and collector terminals are grounded.

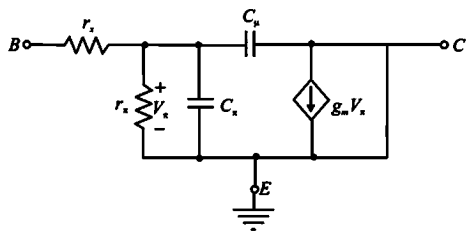


Figure 2

Step 3 of 5

From Figure 2, it is clear that two capacitors C_μ **and** C_π are in parallel.

$$C = C_\mu + C_\pi$$

Write the expression of input impedance.

$$r_x + \left(r_\pi \| \frac{1}{sC}\right)$$

Simplify the expression of $\left(r_\pi \| \frac{1}{sC}\right)$.

$$\left(r_\pi \| \frac{1}{sC}\right) = \left(r_\pi \| \frac{1}{sC}\right)$$
$$= \frac{r_\pi\left(\frac{1}{sC}\right)}{r_\pi + \frac{1}{sC}}$$
$$= \frac{r_\pi}{1 + r_\pi sC}$$

The magnitude is,

$$\left|r_\pi \| \frac{1}{sC}\right| = \frac{r_\pi}{\sqrt{1 + (r_\pi sC)^2}}$$

Step 4 of 5

The frequency is defined in terms of ω_β, is such an estimate of r_x good to within 10% under the condition that $r_x \le r_\pi / 10$.

For satisfying the above condition $\sqrt{1 + (r_\pi sC)^2}$ should be greater than or equal to $0.1\, r_\pi$.

$$\left[1 + \left(r_\pi^2 s^2 C^2\right)\right]^{\frac{1}{2}} \ge 0.1 r_\pi$$
$$\left[1 + \left(r_\pi^2 s^2 C^2\right)\right]^{\frac{1}{2}} = 0.1 r_\pi$$
$$\left[1 + \left(r_\pi^2 s^2 C^2\right)\right]^{\frac{1}{2}} = 0.1 r_\pi$$
$$1 + \left(r_\pi^2 s^2 C^2\right) = 0.01 (r_\pi)^2$$
$$r_\pi^2 s^2 C^2 = 0.01 (r_\pi)^2 - 1 \quad \text{..... (1)}$$

Step 5 of 5

Consider the resistance r_π is,

$$r_\pi = 0.1 r_x$$

Simplify the equation (1) as follows:

$$r_\pi^2 s^2 C^2 = 0.01 (r_\pi)^2 - 1$$
$$0.01 (r_\pi)^2 \left(1 - s^2 C^2\right) = 1$$
$$1 - s^2 C^2 = \frac{100}{(r_\pi)^2}$$

Substitute $2\pi f$ for s.

$$s^2 C^2 = 1 - \frac{100}{(r_\pi)^2}$$
$$(2\pi f)^2 C^2 = 1 - \frac{100}{(r_\pi)^2}$$
$$f = \left[\frac{1}{4\pi^2 C^2} - \frac{100}{4\pi^2 C^2 r_\pi^2}\right]^{\frac{1}{2}}$$

Therefore, the expression of frequency is $\boxed{f = \left[\frac{1}{4\pi^2 C^2} - \frac{100}{4\pi^2 C^2 r_\pi^2}\right]^{\frac{1}{2}}}$.

10.025E

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10.025P

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Step 1 of 7

Refer to Figure 9.34 (a) for a resistively loaded MOS differential amplifier.

(a)

Write the formula for over-drive voltage of the transistor.

$$V_{OV} = \sqrt{\frac{I}{k_n'\left(\frac{W}{L}\right)}}$$

Here,

I is the bias current

k_n' is the constant

W is the width of the channel

L is the length of the channel

Substitute 0.8×10^{-3} A for I, 0.2×10^{-3} A/V^2 for k_n' and 100 for $\frac{W}{L}$.

$$\begin{aligned} V_{OV} &= \sqrt{\frac{0.8\times10^{-3}}{\left(0.2\times10^{-3}\right)(100)}} \\ &= \sqrt{\frac{1}{25}} \\ &= 0.2 \end{aligned}$$

Thus, the over-drive voltage V_{OV} for each transistor is $\boxed{0.2\text{ V}}$.

Step 2 of 7

Write the formula for transconductance, g_m.

$$g_m = \frac{I}{V_{DV}}$$

Substitute 0.8×10^{-3} A for I and 0.2 V for V_{OV}.

$$\begin{aligned} g_m &= \frac{0.8\times10^{-3}}{0.2} \\ &= 4\times10^{-3}\text{ A/V} \\ &= 4\text{ mA/V} \end{aligned}$$

Thus, the transconductance g_m of each transistor is $\boxed{4\text{ mA/V}}$.

Step 3 of 7

(b)

Write the formula for differential gain, A_d.

$$A_d = g_m\left(R_D \| r_o\right)$$

Substitute $\frac{V_A}{I_D}$ for r_o.

$$A_d = g_m\left(R_D \| \frac{V_A}{I_D}\right)$$

Substitute $\frac{I}{2}$ for I_D.

$$A_d = g_m\left(R_D \| \frac{V_A}{\frac{I}{2}}\right)$$

Substitute 4×10^{-3} A/V for g_m, $5\times10^3\ \Omega$ for R_D, 20 V for V_A and 0.8×10^{-3} A for I.

$$\begin{aligned} A_d &= \left(4\times10^{-3}\right)\left[\left(5\times10^3\right) \| \left(\frac{20}{\frac{0.8\times10^{-3}}{2}}\right)\right] \\ &= \left(4\times10^{-3}\right)\left[\left(5\times10^3\right) \| \left(50\times10^3\right)\right] \\ &= \left(4\times10^{-3}\right)\left[\frac{\left(5\times10^3\right)\left(50\times10^3\right)}{\left(5\times10^3\right)+\left(50\times10^3\right)}\right] \\ &= \left(4\times10^{-3}\right)\left(4.55\times10^3\right) \\ A_d &= 18.2\text{ V/V} \end{aligned}$$

Thus, the differential gain, A_d is $\boxed{18.2\text{ V/V}}$.

Step 4 of 7

(c)

Determine the resistance, R_L'.

$$R_L' = R_D \| r_o$$

Substitute $\frac{V_A}{\left(\frac{I}{2}\right)}$ for r_o.

$$\begin{aligned} R_L' &= R_D \| \frac{V_A}{\left(\frac{I}{2}\right)} \\ &= R_D \| \frac{2V_A}{I} \end{aligned}$$

Step 5 of 7

Substitute $5\times10^3\ \Omega$ for R_D, 20 V for V_A and 0.8×10^{-3} A for I.

$$\begin{aligned} R_L' &= R_D \| \frac{2V_A}{I} \\ &= \left(5\times10^3\right) \| \left(\frac{2\times20}{0.8\times10^{-3}}\right) \\ &= \left(5\times10^3\right) \| \left(50\times10^3\right) \\ &= \frac{\left(5\times10^3\right)\left(50\times10^3\right)}{\left(5\times10^3\right)+\left(50\times10^3\right)} \\ R_L' &= 4.55\times10^3\ \Omega \\ &= 4.55\text{ k}\Omega \end{aligned}$$

Determine the capacitance, C_L.

$$C_L = C_{db} + C_{\text{drain-ground}}$$

Substitute 10×10^{-15} F for C_{db} and 100×10^{-15} F for $C_{\text{drain-ground}}$.

$$\begin{aligned} C_L &= (10+100)\times10^{-15} \\ &= 110\times10^{-15}\text{ F} \end{aligned}$$

Step 6 of 7

Calculate the upper 3-dB frequency, f_H.

$$\begin{aligned} f_H &= \frac{1}{2\pi\left(C_L + C_{gd}\right)R_L'} \\ &= \frac{1}{2\pi(110+10)\times10^{-15}\left(4.55\times10^3\right)} \\ &\approx 291\text{ MHz} \end{aligned}$$

Thus, the upper 3-dB frequency f_H is $\boxed{291\text{ MHz}}$.

Step 7 of 7

(d)

The signal source resistance, $R_{sig} = 20\text{k}\Omega$ or $R_{sig}' = 10\text{k}\Omega$ (connected to the gate terminal)

The open-circuit time constant equation is,

$$\begin{aligned} \tau_H &= C_{gs}R_{sig}' + C_{gd}\left[R_{sig}'\left(1+g_mR_L'\right)+R_L'\right]+C_LR_L' \\ &= \left\{\left(50\times10^{-15}\right)\left(10\times10^3\right)\right\}+\left(10\times10^{-15}\right)\left[\left(10\times10^3\right)\left(1+\left[\frac{\left(4\times10^{-3}\right)}{\left(4.55\times10^3\right)}\right]\right)+\left(4.55\times10^3\right)\right]+ \\ &\qquad \left(110\times10^{-15}\right)\left(4.55\times10^3\right) \\ &= 500\times10^{-12}+1965.5\times10^{-12}+500.5\times10^{-12} \\ &= 2.966\times10^{-9}\text{ s} \end{aligned}$$

Calculate the upper 3-dB frequency.

$$\begin{aligned} f_H &= \frac{1}{2\pi\tau_H} \\ &= \frac{1}{2\pi\left(2.966\times10^{-9}\right)} \\ &= 53.7\times10^6\text{ Hz} \\ &= 53.7\text{ MHz} \end{aligned}$$

Thus, the upper 3-dB frequency f_H is $\boxed{53.7\text{ MHz}}$.

Step 1 of 3

Calculate the value of the capacitance C_{eq}.

$$C_{eq} = C_{gd}\left(1 + g_m R_L'\right)$$

Substitute $0.1\ \text{pF}$ for C_{gd} and $39\ \text{V/V}$ for $g_m R_L'$.

$$\begin{aligned} C_{eq} &= (0.1\ \text{p})(1+39) \\ &= (0.1\ \text{p})(40) \\ &= 4\ \text{pF} \end{aligned}$$

Therefore, the value of the capacitance C_{eq} is $4\ \text{pF}$.

Step 2 of 3

Calculate the value of the input capacitance C_{in}.

$$C_{in} = C_{gs} + C_{eq}$$

Substitute $4\ \text{pF}$ for C_{eq} and $1\ \text{pF}$ for C_{gs}.

$$\begin{aligned} C_{in} &= 1\ \text{p} + 4\ \text{p} \\ &= 5\ \text{pF} \end{aligned}$$

Therefore, the value of the input capacitance C_{in} is $\boxed{5\ \text{pF}}$.

Step 3 of 3

Consider the expression for the 3dB frequency f_H.

$$f_H = \frac{1}{2\pi C_{in} R_{sig}'}$$

Substitute $1\ \text{MHz}$ for f_H and $5\ \text{pF}$ for C_{in}.

$$\begin{aligned} 1\ \text{M} &= \frac{1}{2\pi(5\ \text{p})R_{sig}'} \\ R_{sig}' &= \frac{1}{2\pi(5\ \text{p})(1\ \text{M})} \\ R_{sig}' &= \frac{1}{31.42\ \mu} \\ R_{sig}' &= 31.83\ \text{k}\Omega \end{aligned}$$

Neglect the value of resistance R_G, the value of the R_{sig} is equal to the value R_{sig}'.

Therefore, for the value of the 3dB frequency to be greater than $1\ \text{MHz}$, the value of R_{sig} should be lower than the value of $\boxed{31.83\ \text{k}\Omega}$ since f_H is inversely proportional to R_{sig}.

10.027E

Calculate the value of the frequency is f_z .

$$f_z = \frac{1}{2\pi C_{ss} R_{ss}}$$

Substitute $0.4\ \text{pF}$ for C_{ss} and $75\ \text{k}\Omega$ for R_{ss} .

$$\begin{aligned} f_z &= \frac{1}{2\pi(0.4\ \text{p})(75\ \text{k})} \\ &= \frac{1}{188.5\ \text{n}} \\ &= 5.3\ \text{MHz} \end{aligned}$$

Therefore, the value of the frequency f_z is $\boxed{5.3\ \text{MHz}}$.

10.027P

We don't have the solution to this problem yet.

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Step 1 of 3

Determine the transconductance, g_m.

$$g_m = \frac{\left(\frac{I}{2}\right)}{V_T}$$

Substitute 10^{-3} A for I and 25×10^{-3} V for V_T.

$$g_m = \frac{\frac{1\times10^{-3}}{2}}{25\times10^{-3}}$$
$$= 0.02 \text{ A/V}$$
$$= 20 \text{ mA/V}$$

Determine the output resistance r_o.

$$r_o = \frac{V_A}{\left(\frac{I}{2}\right)}$$

Substitute 100 V for V_A and 10^{-3} A for I.

$$r_o = \frac{100}{\left(\frac{10^{-3}}{2}\right)}$$
$$= 200\times10^3 \ \Omega$$
$$= 200 \text{ k}\Omega$$

Step 2 of 3

Determine the differential voltage gain, A_d.

$$A_d = \frac{1}{2}g_m r_o$$

Substitute 0.02 A/V for g_m and $200\times10^3\ \Omega$ for r_o.

$$A_d = \frac{1}{2}(0.02)\left(200\times10^3\right)$$
$$= 2000 \text{ V/V}$$

Thus, the differential gain A_d is $\boxed{2000 \text{ V/V}}$.

Step 3 of 3

The domination pole is at the location of the common-mode gain zero introduced by C_{SS} and R_{SS}.

Determine the dominant pole frequency, f_Z.

$$f_Z = \frac{1}{2\pi C_L\left(r_{o2} \| r_{o4}\right)}$$
$$= \frac{1}{2\pi C_L\left(\frac{r_{o2}}{2}\right)}$$
$$= \frac{1}{\pi C_L r_{o2}}$$

Substitute 2×10^{-12} F for C_L and $200\times10^3\ \Omega$ for r_{o2}.

$$f_Z = \frac{1}{\pi\left(2\times10^{-12}\right)\left(200\times10^3\right)}$$
$$\approx 0.8 \text{ MHz}$$

Thus, the dominant pole frequency is $\boxed{0.8 \text{ MHz}}$.

10.028P

We don't have the solution to this problem yet.

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Step 1 of 7

(a)

Determine the voltage gain, A_M .

$$A_M = -g_m\left(R_L \| r_o\right)$$

Substitute 2×10^{-3} mA/V for g_m , $20\times10^3\ \Omega$ for R_L and $20\times10^3\ \Omega$ for r_o .

$$\begin{aligned} A_M &= -\left(2\times10^{-3}\right)\left(20\times10^3 \| 20\times10^3\right) \\ &= -\left(2\times10^{-3}\right)\left(10\times10^3\right) \\ &= -20\ \text{V/V} \end{aligned}$$

Thus, the voltage gain A_M is $\boxed{-20\ \text{V/V}}$.

Determine the resistance, R_L' .

$$\begin{aligned} R_L' &= r_o \| R_L \\ &= \left(20\times10^3\right) \| \left(20\times10^3\right) \\ &= 10\times10^3\ \Omega \\ &= 10\ \text{k}\Omega \end{aligned}$$

Step 2 of 7

Determine the time constant, τ_H .

$$\begin{aligned} \tau_H &= C_{gs}R_{sig} + C_{gd}\left[R_{sig}\left(1+g_mR_L'\right)+R_L'\right] + C_LR_L' \\ &= \left\{\begin{array}{l} \left(20\times10^{-15}\right)\left(20\times10^3\right) + \\ 5\times10^{-15}\left[20\times10^3\left(1+\left(2\times10^{-3}\right)\left(10\times10^3\right)\right)+10\times10^3\right]+ \\ \left(5\times10^{-15}\right)\left(10\times10^3\right) \end{array}\right\} \\ &= 400\times10^{-12} + \left(5\times10^{-15}\right)\left(430\times10^3\right) + \left(5\times10^{-15}\right)\left(10\times10^3\right) \\ &= 2.6\times10^{-9}\ \text{s} \end{aligned}$$

Determine the upper 3-dB frequency, f_H .

$$\begin{aligned} f_H &= \frac{1}{2\pi\tau_H} \\ &= \frac{1}{2\pi\times2.6\times10^{-9}} \\ &= 61.2\ \text{MHz} \end{aligned}$$

Thus, the upper 3-dB frequency, f_H is $\boxed{61.2\ \text{MHz}}$.

Step 3 of 7

Determine the gain bandwidth product.

$$\begin{aligned} \text{Gain}\times\text{BW} &= \left|A_M\right| f_H \\ &= (20)\left(61.2\times10^6\right) \\ &= 1224\times10^6\ \text{Hz} \\ &= 1.22\ \text{GHz} \end{aligned}$$

Thus, the gain bandwidth product is $\boxed{1.22\ \text{GHz}}$.

Step 4 of 7

(b)

Determine the value of G_M .

$$\begin{aligned} G_M &= \frac{g_m}{1+g_mR_s} \\ &= \frac{2\times10^{-3}}{1+2} \\ &= 0.667\ \text{mA/V} \end{aligned}$$

Determine the value of the output resistance, R_o .

$$\begin{aligned} R_o &= r_o\left(1+g_mR_s\right) \\ &= \left(20\times10^3\right)(1+2) \\ &= 60\ \text{k}\Omega \end{aligned}$$

Determine the voltage gain.

$$\begin{aligned} A_M &= -G_M\left(R_o \| R_L\right) \\ &= -\left(0.667\times10^{-3}\right)\left(60\times10^3 \| 20\times10^3\right) \\ &= -\left(0.667\times10^{-3}\right)\left(15\times10^3\right) \\ &= -10.005\ \text{V/V} \\ &\approx -10\ \text{V/V} \end{aligned}$$

Thus, the voltage gain A_M is $\boxed{-10\ \text{V/V}}$.

Step 5 of 7

Determine the resistance R_s

$$\begin{aligned} R_s &= \frac{2}{g_m} \\ &= \frac{2}{2\times10^{-3}} \\ &= 1\ \text{k}\Omega \end{aligned}$$

Determine the value of R_{gd} .

$$\begin{aligned} R_{gd} &= R_{sig}\left[1+G_M\left(R_L \| R_o\right)\right] + \left(R_L \| R_o\right) \\ &= \left(20\times10^3\right)\left[1+\left(0.667\times10^{-3}\right)\left(20\times10^3 \| 60\times10^3\right)\right] + \left(20\times10^3 \| 60\times10^3\right) \\ &= \left(20\times10^3\right)(1+10)+15\times10^3 \\ &= 235\ \text{k}\Omega \end{aligned}$$

Step 6 of 7

Determine the value of the resistance, R_{gs} .

$$\begin{aligned} R_{gs} &= \frac{R_{sig}+R_s}{1+g_mR_s\left(\dfrac{r_o}{r_o+R_L}\right)} \\ &= \frac{20\times10^3+1\times10^3}{1+(2)\left(\dfrac{20\times10^3}{20\times10^3+20\times10^3}\right)} \\ &= 10.5\ \text{k}\Omega \end{aligned}$$

Determine the time constant, τ_H .

$$\begin{aligned} \tau_H &= C_{gs}R_{gs} + C_{gd}R_{gd} + C_LR_{C_L} \\ &= C_{gs}R_{gs} + C_{gd}R_{gd} + C_L\left(R_L \| R_o\right) \\ &= \left[\begin{array}{l} \left(20\times10^{-15}\times10.5\times10^3\right) + \left(5\times10^{-15}\times233\times10^3\right) + \\ 5\times10^{-15}\left(60\times10^3 \| 20\times10^3\right) \end{array}\right] \\ &= 1.45\times10^{-9}\ \text{s} \end{aligned}$$

Step 7 of 7

Determine the upper 3-dB frequency, f_H .

$$\begin{aligned} f_H &= \frac{1}{2\pi\tau_H} \\ &= \frac{1}{2\pi\times1.45\times10^{-9}} \\ &= 109.8\ \text{MHz} \end{aligned}$$

Thus, the upper 3-dB frequency, f_H is $\boxed{109.8\ \text{MHz}}$.

Determine the gain bandwidth product.

$$\begin{aligned} \text{Gain}\times\text{BW} &= \left|A_M\right| f_H \\ &= (10)\left(109.8\times10^6\right) \\ &= 1098\times10^6\ \text{Hz} \\ &= 1.1\ \text{GHz} \end{aligned}$$

Thus, the gain bandwidth product is $\boxed{1.1\ \text{GHz}}$.

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Step 1 of 8

Refer to Figure 9.41 (a) in the text book for the CC-CB amplifier.

Write the formula for transconductance, g_m.

$$g_m = \frac{I_C}{V_T}$$
$$= \frac{I}{V_T}$$

Substitute $0.5\times10^{-3}\ \text{A}$ for I and $25\times10^{-3}\ \text{V}$ for V_T.

$$g_m = \frac{0.5\times10^{-3}}{25\times10^{-3}}$$
$$= 20\times10^{-3}\ \text{A/V}$$

Write the formula for the low-frequency input resistance, R_{in}.

$$R_{in} = (\beta_1 + 1)(r_{e1} + r_{e2})$$

Here,

$$r_{e1} = r_{e2}$$
$$= r_\pi$$

$$\beta_1 = \beta_2$$
$$= \beta$$

$$r_\pi = \frac{\beta}{g_m}$$

$$R_{in} = 2r_\pi$$
$$= 2\left(\frac{\beta}{g_m}\right)$$

Substitute 100 for β and $20\times10^{-3}\ \text{A/V}$ for g_m.

$$R_{in} = 2\left(\frac{100}{20\times10^{-3}}\right)$$
$$= 10\times10^{3}\ \Omega$$

Step 2 of 8

Calculate the low-frequency overall voltage gain, A_M.

$$A_M = \frac{V_o}{V_{sig}} = \frac{1}{2}\left(\frac{R_{in}}{R_{in}+R_{sig}}\right)(g_m R_L)$$

Substitute $10\times10^{3}\ \Omega$ for R_{sig}, $10\times10^{3}\ \Omega$ for R_L, $20\times10^{-3}\ \text{A/V}$ for g_m and $10\times10^{3}\ \Omega$ for R_{in}.

$$A_M = \frac{1}{2}\left(\frac{R_{in}}{R_{in}+R_{sig}}\right)(g_m R_L)$$
$$= \frac{1}{2}\left(\frac{10\times10^{3}}{10\times10^{3}+10\times10^{3}}\right)\left[\left(20\times10^{-3}\right)\left(10\times10^{3}\right)\right]$$
$$= \frac{1}{2}\left(\frac{10}{20}\right)(200)$$
$$= 50$$

Therefore, the low- frequency overall voltage gain A_M is $\boxed{50\ \text{V/V}}$.

Step 3 of 8

Calculate the first pole frequency.

$$f_{p1} = \frac{1}{2\pi\left(\frac{C_\pi}{2}+C_\mu\right)\left(R_{sig}\parallel 2r_\pi\right)}$$
$$= \frac{1}{2\pi\left(\frac{C_\pi}{2}+C_\mu\right)\left(R_{sig}\parallel 2\left(\frac{\beta}{g_m}\right)\right)} \qquad \left(\text{Since, } r_\pi = \frac{\beta}{g_m}\right)$$

Substitute $6\times10^{-12}\ \text{F}$ for C_π, $2\times10^{-12}\ \text{F}$ for C_μ, $20\times10^{-3}\ \text{A/V}$ for g_m, $10\times10^{3}\ \Omega$ for R_{sig} and 100 for β

$$f_{p1} = \frac{1}{2\pi\left(\frac{6\times10^{-12}}{2}+2\times10^{-12}\right)\left(10\times10^{3}\parallel 2\left(\frac{100}{20\times10^{-3}}\right)\right)}$$
$$= \frac{1}{2\pi\left(3\times10^{-12}+2\times10^{-12}\right)\left(10\times10^{3}\parallel 10\times10^{3}\right)}$$
$$= \frac{1}{2\pi\left(5\times10^{-12}\right)\left(\frac{\left(10\times10^{3}\right)\left(10\times10^{3}\right)}{\left(10\times10^{3}\right)+\left(10\times10^{3}\right)}\right)}$$
$$= \frac{1}{2\pi\left(5\times10^{-12}\right)\left(5\times10^{3}\right)}$$

$$f_{p1} = 6.366\times10^{6}\ \text{Hz}$$
$$\approx 6.4\ \text{MHz}$$

Therefore, the first pole frequency f_{p1} is $\boxed{6.4\ \text{MHz}}$.

Step 4 of 8

Calculate the second pole frequency.

$$f_{p2} = \frac{1}{2\pi C_\mu R_L}$$

Substitute $6\times10^{-12}\ \text{F}$ for C_π and $10\times10^{3}\ \Omega$ for R_L.

$$f_{p2} = \frac{1}{2\pi\left(2\times10^{-12}\right)\left(10\times10^{3}\right)}$$
$$= \frac{1}{2\pi\left(20\times10^{-9}\right)}$$
$$= 7.9577\times10^{6}\ \text{Hz}$$
$$\approx 8\ \text{MHz}$$

Therefore, the second pole frequency f_{p2} is $\boxed{8\ \text{MHz}}$.

Step 5 of 8

Use exact analysis to calculate the upper 3 dB frequency.

The case of a circuit having two poles in the high-frequency band is,

$$F_H(s) = \frac{1}{\left(1+\frac{s}{\omega_{p1}}\right)\left(1+\frac{s}{\omega_{p2}}\right)}$$

Substitute $j\omega$ for s.

$$F_H(j\omega) = \frac{1}{\left(1+\frac{j\omega}{\omega_{p1}}\right)\left(1+\frac{j\omega}{\omega_{p2}}\right)}$$

Determine the magnitude of $F_H(j\omega)$.

$$\left|F_H(j\omega)\right| = \frac{1}{\sqrt{\left(1+\frac{\omega^2}{\omega_{p1}^2}\right)\left(1+\frac{\omega^2}{\omega_{p2}^2}\right)}}$$
$$\left|F_H(j\omega)\right|^2 = \frac{1}{\left(1+\frac{\omega^2}{\omega_{p1}^2}\right)\left(1+\frac{\omega^2}{\omega_{p2}^2}\right)}$$

By definition, the magnitude of $\left|F_H\right|^2$ at $\omega = \omega_H$ is $\frac{1}{2}$.

$$\frac{1}{2} = \frac{1}{\left(1+\frac{\omega_H^2}{\omega_{p1}^2}\right)\left(1+\frac{\omega_H^2}{\omega_{p2}^2}\right)}$$
$$\frac{1}{2} = \frac{1}{\left(1+\frac{f_H^2}{f_{p1}^2}\right)\left(1+\frac{f_H^2}{f_{p2}^2}\right)} \qquad (\text{Since, } \omega = 2\pi f)$$
$$\left(1+\frac{f_H^2}{f_{p1}^2}\right)\left(1+\frac{f_H^2}{f_{p2}^2}\right) = 2$$

Step 6 of 8

Substitute $6.4\times10^{6}\ \text{Hz}$ for f_{p1} and $8\ \text{MHz}$ for f_{p2}.

$$\left(1+\frac{f_H^2}{\left(6.4\times10^{6}\right)^2}\right)\left(1+\frac{f_H^2}{\left(8\times10^{6}\right)^2}\right) = 2$$
$$\left(1+\frac{f_H^2}{40.96\times10^{12}}\right)\left(1+\frac{f_H^2}{64\times10^{12}}\right) = 2$$
$$\left(\frac{40.96\times10^{12}+f_H^2}{\left(40.96\times10^{12}\right)}\right)\left(\frac{64\times10^{12}+f_H^2}{\left(64\times10^{12}\right)}\right) = 2$$
$$\left(40.96\times10^{12}+f_H^2\right)\left(64\times10^{12}+f_H^2\right) = 5242.88\times10^{24}$$

$$2621.44\times10^{24} + 40.96\times10^{12} f_H^2 + 64\times10^{12} f_H^2 + f_H^4 = 5242.88\times10^{24}$$
$$104.96\times10^{12} f_H^2 + f_H^4 = 2621.44\times10^{24}$$

Let us assume that $f = f_H^2$.

$$f^2 + 104.96\times10^{12} f - 2621.44\times10^{24} = 0$$

Use formula for the roots of the quadratic equation to determine the high-frequency.

$$f = \frac{-\left(104.96\times10^{12}\right)\pm\sqrt{\left(104.96\times10^{12}\right)^2 - 4(1)\left(-2621.44\times10^{24}\right)}}{2(1)}$$
$$= \frac{-\left(104.96\times10^{12}\right)\pm\sqrt{11016.6016\times10^{24}+10485.76\times10^{24}}}{2}$$
$$= \frac{-\left(104.96\times10^{12}\right)\pm\sqrt{21502.3616\times10^{24}}}{2}$$
$$= \frac{-\left(104.96\times10^{12}\right)\pm\left(146.64\times10^{12}\right)}{2}$$

Step 7 of 8

The high-frequency by considering the positive term is,

$$f = \frac{-\left(104.96\times10^{12}\right)+\left(146.64\times10^{12}\right)}{2}$$
$$= \frac{41.68\times10^{12}}{2}$$
$$= 20.86\times10^{12}$$

Since, $f = f_H^2$

$$f_H^2 = 20.86\times10^{12}$$
$$f_H = \sqrt{20.86\times10^{12}}$$
$$= 4.567\times10^{6}\ \text{Hz}$$
$$= 4.6\ \text{MHz}$$

Therefore, the upper 3 dB frequency f_H by exact analysis is $\boxed{4.6\ \text{MHz}}$.

Step 8 of 8

Calculate the upper 3 dB frequency using the approximate formula.

$$f_H = \frac{1}{\sqrt{\frac{1}{f_{p1}^2}+\frac{1}{f_{p2}^2}}}$$

Substitute $6.4\times10^{6}\ \text{Hz}$ for f_{p1} and $8\ \text{MHz}$ for f_{p2}.

$$f_H = \frac{1}{\sqrt{\frac{1}{\left(6.4\times10^{6}\right)^2}+\frac{1}{\left(8\times10^{6}\right)^2}}}$$
$$= \frac{1}{\sqrt{\frac{1}{40.96\times10^{12}}+\frac{1}{64\times10^{12}}}}$$
$$= \frac{1}{\sqrt{24.4\times10^{-15}+15.63\times10^{-15}}}$$
$$= \frac{1}{\sqrt{40.03\times10^{-15}}}$$
$$= 4.997\times10^{6}\ \text{Hz}$$
$$\approx 5\ \text{Hz}$$

Therefore, the upper 3 dB frequency f_H by approximation analysis is $\boxed{5\ \text{MHz}}$.

Step 1 of 18

Determine the value of load resistor.

$R_L' = r_o \| R_D \| R_L$

Substitute $150\,\text{k}\Omega$ for r_o, $15\,\text{k}\Omega$ for R_D, and $15\,\text{k}\Omega$ for R_L.

$$\begin{aligned}R_L' &= 150\times10^3 \| 15\times10^3 \| 15\times10^3 \\ &= 150\times10^3 \| \left(\frac{(15\times10^3)(15\times10^3)}{15\times10^3+15\times10^3}\right) \\ &= 150\times10^3 \| (7.5\times10^3) \\ &= \frac{(150\times10^3)(7.5\times10^3)}{150\times10^3+7.5\times10^3} \\ &= 7.143\times10^3\end{aligned}$$

Step 2 of 18

Determine the value of midband gain (A_M).

$$A_M = -\left(\frac{R_G}{R_G+R_{sig}}\right) g_m R_L'$$

Substitute $4.7\,\text{M}\Omega$ for R_G, $100\,\text{k}\Omega$ for R_{sig}, $1\,\text{mA/V}$ for g_m and $7.143\,\text{k}\Omega$ for R_L'.

$$\begin{aligned}A_M &= -\left(\frac{4.7\times10^6}{4.7\times10^6+100\times10^3}\right)(1\times10^{-3})(7.143\times10^3) \\ &= -\left(\frac{4.7\times10^6}{4.8\times10^6}\right)(7.143) \\ &= -(0.9792)(7.143) \\ &= -6.99\end{aligned}$$

Therefore, the value of midband gain is $\boxed{-6.99\,\text{V/V}}$.

Step 3 of 18

Determine the value of equivalent capacitance.

$C_{eq} = (1+g_m R_L')C_{gd}$

Substitute $0.4\,\text{pF}$ for C_{gd}, $1\,\text{mA/V}$ for g_m and $7.143\,\text{k}\Omega$ for R_L'.

$$\begin{aligned}C_{eq} &= \left(1+(1\times10^{-3})(7.143\times10^3)\right)(0.4\times10^{-12}) \\ &= (1+7.143)(0.4\times10^{-12}) \\ &= (8.143)(0.4\times10^{-12}) \\ &= 3.26\times10^{-12}\end{aligned}$$

Step 4 of 18

Determine the value of input capacitance.

$C_{in} = C_{gs} + C_{eq}$

Substitute $1\,\text{pF}$ for C_{gs} and $3.26\,\text{pF}$ for C_{eq}.

$$\begin{aligned}C_{in} &= (1\times10^{-12}) + (3.26\times10^{-12}) \\ &= 4.26\times10^{-12} \\ &= 4.26\,\text{pF}\end{aligned}$$

Determine the value of upper 3-dB frequency (f_H).

$$f_H = \frac{1}{2\pi C_{in}(R_{sig} \| R_G)}$$

Substitute $4.26\,\text{pF}$ for C_{in}, $100\,\text{k}\Omega$ for R_{sig} and $4.7\,\text{M}\Omega$ for R_G.

$$\begin{aligned}f_H &= \frac{1}{2\pi(4.26\times10^{-12})(100\times10^3 \| 4.7\times10^6)} \\ &= \frac{1}{2\pi(4.26\times10^{-12})\left(\frac{(100\times10^3)(4.7\times10^6)}{100\times10^3+4.7\times10^6}\right)} \\ &= \frac{1}{2\pi(4.26\times10^{-12})\left(\frac{4.7\times10^{11}}{4.8\times10^6}\right)} \\ &= \frac{1}{2\pi(4.26\times10^{-12})(97.92\times10^3)} \\ &= 381.54\times10^3\end{aligned}$$

Therefore, the upper 3-dB frequency (f_H) is $\boxed{381.54\,\text{kHz}}$.

Step 5 of 18

Determine the value of gain-bandwidth product.

$\text{GBW} = |A_M| f_H$

Substitute $-7\,\text{V/V}$ for A_M and $382\,\text{kHz}$ for f_H.

$$\begin{aligned}\text{GBW} &= (7)(382\times10^3) \\ &= 2.674\times10^6\end{aligned}$$

Therefore, the gain-bandwidth product is $\boxed{\text{GBW} = 2.674\,\text{MHz}}$.

Step 6 of 18

The new width is half of the original transistor width.

$W_{new} = \frac{1}{2}W$

$\frac{W_{new}}{W} = \frac{1}{2}$

Therefore, the ratio of new width to old width is $\boxed{\frac{W_{new}}{W} = \frac{1}{2}}$.

Step 7 of 18

The transconductance is,

$$\begin{aligned}g_m &= \frac{I}{V_{ov}} \\ &= \frac{I}{\sqrt{\frac{I}{k_n'\left(\frac{W}{L}\right)}}} \qquad \left(\text{Since, } V_{ov} = \sqrt{\frac{I}{k_n'\left(\frac{W}{L}\right)}}\right) \\ &= \frac{I}{\sqrt{\frac{I}{k_n'\left(\frac{W_{new}}{L}\right)}}} \qquad \left(\text{Since, } W_{new} = \frac{1}{2}W\right) \\ &= \frac{I}{\sqrt{\frac{I}{k_n'\left(\frac{\frac{W}{2}}{L}\right)}}} \\ &= \frac{I\sqrt{k_n'\left(\frac{\frac{W}{2}}{L}\right)}}{\sqrt{I}} \\ &= \sqrt{I}\sqrt{k_n'\left(\frac{W}{2L}\right)} \\ &= \frac{1}{\sqrt{2}}\sqrt{Ik_n'\left(\frac{W}{L}\right)}\end{aligned}$$

$g_{m\,new} = \frac{1}{\sqrt{2}} g_m \qquad \left(\text{Since, } g_m = \sqrt{Ik_n'\left(\frac{W}{L}\right)}\right)$

Therefore, the ratio of new transconductance to old transconductance is $\boxed{\frac{g_{m\,new}}{g_m} = \frac{1}{\sqrt{2}}}$.

Step 8 of 18

The new transconductance is,

$$\begin{aligned}g_{m\,new} &= \frac{I}{V_{ov}} \\ &= \frac{I}{\sqrt{\frac{I}{k_n'\left(\frac{W_{new}}{L}\right)}}} \qquad \left(\text{Since, } V_{ov} = \sqrt{\frac{I}{k_n'\left(\frac{W}{L}\right)}}\right) \\ &= \frac{I}{\sqrt{\frac{I}{k_n'\left(\frac{\frac{W}{2}}{L}\right)}}} \qquad \left(\text{Since, } W_{new} = \frac{1}{2}W\right) \\ &= \frac{I\sqrt{k_n'\left(\frac{\frac{W}{2}}{L}\right)}}{\sqrt{I}} \\ &= \sqrt{I}\sqrt{k_n'\left(\frac{W}{2L}\right)} \\ &= \frac{1}{\sqrt{2}}\sqrt{Ik_n'\left(\frac{W}{L}\right)}\end{aligned}$$

$g_{m\,new} = \frac{1}{\sqrt{2}} g_m \qquad \left(\text{Since, } g_m = \sqrt{Ik_n'\left(\frac{W}{L}\right)}\right)$

Therefore, the ratio of new transconductance to old transconductance is $\boxed{\frac{g_{m\,new}}{g_m} = \frac{1}{\sqrt{2}}}$.

Step 9 of 18

The new overdrive voltage is,

$$\begin{aligned}V_{OV\,new} &= \sqrt{\frac{I}{k_n'\left(\frac{W_{new}}{L}\right)}} \\ &= \sqrt{\frac{I}{k_n'\left(\frac{\frac{W}{2}}{L}\right)}} \qquad \left(\text{Since, } W_{new} = \frac{1}{2}W\right) \\ &= \sqrt{\frac{I}{k_n'\left(\frac{W}{2L}\right)}} \\ &= \sqrt{2}\sqrt{\frac{I}{k_n'\left(\frac{W}{L}\right)}}\end{aligned}$$

$V_{OV\,new} = \sqrt{2}V_{OV} \qquad \left(\text{Since, } V_{OV} = \sqrt{\frac{I}{k_n'\left(\frac{W}{L}\right)}}\right)$

Therefore, the ratio of new overdrive voltage to old overdrive voltage is $\boxed{\frac{V_{OV\,new}}{V_{OV}} = \sqrt{2}}$.

Step 10 of 18

The new gate-to-source capacitance is half of the original gate-to-source capacitance
That is

$C_{gs\,new} = \frac{1}{2}C_{gs}$

Substitute $1\,\text{pF}$ for C_{gs}.

$$\begin{aligned}C_{gs\,new} &= \frac{1}{2}(1\times10^{-12}) \\ &= 0.5\times10^{-12} \\ &= 0.5\,\text{pF}\end{aligned}$$

Therefore, the ratio of the new gate-to-source capacitance to gate-to-source capacitance is

$\boxed{\frac{C_{gs\,new}}{C_{gs}} = \frac{1}{2}}$.

Step 11 of 18

The new gate-to-drain capacitance is half of the original gate-to-drain capacitance

$C_{gd\,new} = \frac{1}{2}C_{gd}$

Substitute $0.4\,\text{pF}$ for C_{gd}.

$$\begin{aligned}C_{gd\,new} &= \frac{1}{2}(0.4\times10^{-12}) \\ &= 0.2\times10^{-12} \\ &= 0.2\,\text{pF}\end{aligned}$$

Therefore, the ratio of the new gate-to-drain capacitance to gate-to- drain capacitance is

$\boxed{\frac{C_{gd\,new}}{C_{gd}} = \frac{1}{2}}$

Step 12 of 18

Determine the equivalent capacitance.

$$\begin{aligned}C_{eq\,new} &= (1+g_{m\,new}R_L')C_{gd\,new} \\ &= \left(1+\frac{1}{\sqrt{2}}g_m R_L'\right)C_{gd\,new}\end{aligned}$$

Substitute $0.2\,\text{pF}$ for $C_{gd\,new}$, $1\,\text{mA/V}$ for g_m and $7.143\,\text{k}\Omega$ for R_L'.

$$\begin{aligned}C_{eq\,new} &= \left(1+\frac{1}{\sqrt{2}}(1\times10^{-3})(7.143\times10^3)\right)(0.2\times10^{-12}) \\ &= \left(1+\frac{7.143}{\sqrt{2}}\right)(0.2\times10^{-12}) \\ &= (1+5.051)(0.2\times10^{-12}) \\ &= (6.051)(0.2\times10^{-12}) \\ &= 1.2102\times10^{-12} \\ &= 1.2102\,\text{pF}\end{aligned}$$

Step 13 of 18

Determine the value of new input capacitance.

$C_{in\,new} = C_{gs\,new} + C_{eq\,new}$

Substitute $1.2102\,\text{pF}$ for $C_{eq\,new}$ and $0.5\,\text{pF}$ for $C_{gs\,new}$.

$$\begin{aligned}C_{in\,new} &= 0.5\times10^{-12} + 1.2102\times10^{-12} \\ &= 1.7102\times10^{-12}\end{aligned}$$

Therefore, the new input capacitance is $C_{in\,new} = 1.7\,\text{pF}$

Determine the ratio of new input capacitance to old input capacitance.

$$\frac{C_{in\,new}}{C_{in}} = \frac{1.7\times10^{-12}}{4.26\times10^{-12}}$$

$$\frac{C_{in\,new}}{C_{in}} = 0.399$$

Therefore, the ratio of new input capacitance to old input capacitance is $\boxed{\frac{C_{in\,new}}{C_{in}} = 0.4}$.

Step 14 of 18

The new midband gain $(A_{M\,new})$ is,

$$\begin{aligned}A_{M\,new} &= -\frac{R_G}{R_G+R_{sig}} g_{m\,new} R_L' \\ &= -\left(\frac{R_G}{R_G+R_{sig}}\right)\left(\frac{1}{\sqrt{2}}g_m\right)R_L'\end{aligned}$$

Substitute $4.7\,\text{M}\Omega$ for R_G, $100\,\text{k}\Omega$ for R_{sig}, $1\,\text{mA/V}$ for g_m and $7.143\,\text{k}\Omega$ for R_L'.

$$\begin{aligned}A_{M\,new} &= -\left(\frac{4.7\times10^6}{4.7\times10^6+100\times10^3}\right)\left(\frac{1}{\sqrt{2}}(1\times10^{-3})\right)(7.143\times10^3) \\ &= -\left(\frac{4.7\times10^6}{4.8\times10^6}\right)\left(\frac{7.143}{\sqrt{2}}\right) \\ &= -(0.9792)(5.051) \\ &= -4.95\end{aligned}$$

Step 15 of 18

Determine the ratio of new midband gain to old midband gain.

$$\frac{A_{M\,new}}{A_M} = \frac{-5}{-7}$$

$$= 0.7143$$

Therefore, the ratio of new midband gain to old midband gain is $\boxed{0.7143}$.

Step 16 of 18

Determine the value of new upper 3-dB frequency.

$$f_{H\,new} = \frac{1}{2\pi C_{in\,new}(R_{sig} \| R_G)}$$

Substitute $1.7\,\text{pF}$ for $C_{in\,new}$, $100\,\text{k}\Omega$ for R_{sig} and $4.7\,\text{M}\Omega$ for R_G.

$$\begin{aligned}f_{H\,new} &= \frac{1}{2\pi(1.7\times10^{-12})(100\times10^3 \| 4.7\times10^6)} \\ &= \frac{1}{2\pi(1.7\times10^{-12})\left(\frac{(100\times10^3)(4.7\times10^6)}{100\times10^3+4.7\times10^6}\right)} \\ &= \frac{1}{2\pi(1.7\times10^{-12})\left(\frac{4.7\times10^{11}}{4.8\times10^6}\right)} \\ &= \frac{1}{2\pi(1.7\times10^{-12})(97.92\times10^3)} \\ &= 956.1\times10^3\end{aligned}$$

Step 17 of 18

Determine the ratio of new upper 3-dB frequency to old upper 3-dB frequency.

$$\frac{f_{H\,new}}{f_H} = \frac{956\times10^3}{382\times10^3}$$

$$\frac{f_{H\,new}}{f_H} = 2.5$$

Therefore, the ratio of new upper 3-dB frequency to old upper 3-dB frequency is

$\boxed{\frac{f_{H\,new}}{f_H} = 2.5}$

Step 18 of 18

Determine the new gain-bandwidth product.

$\text{GBW}_{new} = |A_{M\,new}| f_{H\,new}$

Substitute $956\,\text{kHz}$ for $f_{H\,new}$ and $-5\,\text{V/V}$ for $A_{M\,new}$.

$$\begin{aligned}\text{GBW}_{new} &= (5)(956\times10^3) \\ &= 4.78\times10^6\end{aligned}$$

Determine the ratio of new gain-bandwidth product to old gain-bandwidth product.

$$\frac{\text{GBW}_{new}}{\text{GBW}} = \frac{4.78\times10^6}{2.674\times10^6}$$

$$= 1.788$$

Therefore, the ratio of new gain-bandwidth product to old gain-bandwidth product is

$\boxed{\frac{\text{GBW}_{new}}{\text{GBW}} = 1.788}$

10.031P

We don't have the solution to this problem yet.

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Step 1 of 6

Calculate the value of the resistance R'_{sig}.

$$R'_{sig} = R_{sig} \parallel R_G$$

Substitute $500\ \text{k}\Omega$ for R_{sig} and $2\ \text{M}\Omega$ for R_G.

$$\begin{aligned} R'_{sig} &= 500\ \text{k} \parallel 2\ \text{M} \\ &= \frac{(500\ \text{k})(2\text{M})}{500\ \text{k} + 2\ \text{M}} \\ &= \frac{1\ \text{T}}{2.5\ \text{M}} \\ &= 400\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance R'_{sig} is $400\ \text{k}\Omega$.

Step 2 of 6

Calculate the value of the resistance R'_L.

$$\begin{aligned} R'_L &= r_o \parallel R_D \parallel R_L \\ &= \frac{r_o R_D R_L}{r_o R_D + R_L r_o + R_D R_L} \end{aligned}$$

Substitute $100\ \text{k}\Omega$ for r_o, $20\ \text{k}\Omega$ for R_D and $20\ \text{k}\Omega$ for R_L.

$$\begin{aligned} R'_L &= \frac{(100\ \text{k})(20\ \text{k})(20\ \text{k})}{(100\ \text{k})(20\ \text{k}) + (20\ \text{k})(100\ \text{k}) + (20\ \text{k})(20\ \text{k})} \\ &= \frac{40\ \text{T}}{2\ \text{G} + 2\ \text{G} + 0.4\ \text{G}} \\ &= \frac{40\ \text{T}}{4.4\ \text{G}} \\ &= 9.09\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance R'_L is $9.09\ \text{k}\Omega$.

Step 3 of 6

(a)

Calculate he value of the mid band gain A_M.

$$A_M = -\frac{R_G}{R_G + R_{sig}}(g_m R'_L)$$

Substitute $5\ \text{mA/V}$ for g_m, $500\ \text{k}\Omega$ for R_{sig}, $9.09\ \text{k}\Omega$ for R'_L and $2\ \text{M}\Omega$ for R_G.

$$\begin{aligned} A_M &= -\left(\frac{2\ \text{M}}{2\ \text{M} + 500\ \text{k}}\right)(5\ \text{m})(9.09\ \text{k}) \\ &= -\left(\frac{2\ \text{M}}{2.5\ \text{M}}\right)(45.45) \\ &= -(0.8)(45.45) \\ &= -36.36\ \text{V/V} \end{aligned}$$

Therefore, the value of the mid band gain A_M is $\boxed{-36.36\ \text{V/V}}$.

Step 4 of 6

(b)

Calculate the value of the input capacitance C_{in}.

$$C_{in} = C_{gs} + C_{gd}(1 + g_m R'_L)$$

Substitute $3\ \text{pF}$ for C_{gs} and $0.5\ \text{pF}$ for C_{gd}, $5\ \text{mA/V}$ for g_m and $9.09\ \text{k}\Omega$ for R'_L.

$$\begin{aligned} C_{in} &= 3\ \text{p} + (0.5\ \text{p})(1 + (5\ \text{m})(9.09\ \text{k})) \\ &= 3\ \text{p} + (0.5\ \text{p})(1 + 45.45) \\ &= 3\ \text{p} + (0.5\ \text{p})(46.45) \\ &= 3\ \text{p} + 23.23\ \text{p} \end{aligned}$$

$$C_{in} = 26.23\ \text{pF}$$

Therefore, the value of the input capacitance C_{in} is $26.23\ \text{pF}$.

Step 5 of 6

Calculate the value of the frequency f_H.

$$f_H = \frac{1}{2\pi C_{in} R'_{sig}}$$

Substitute $26.23\ \text{pF}$ for C_{in} and $400\ \text{k}\Omega$ for R'_{sig}.

$$\begin{aligned} f_H &= \frac{1}{2\pi(26.23\ \text{p})(400\ \text{k})} \\ &= \frac{1}{65.92\ \mu} \\ &= 15.17\ \text{kHz} \end{aligned}$$

Therefore, the value o the frequency f_H is $\boxed{15.17\ \text{kHz}}$.

Step 6 of 6

(c)

Calculate the value of the frequency f_Z.

$$f_Z = \frac{g_m}{2\pi C_{gd}}$$

Substitute $0.5\ \text{pF}$ for C_{gd} and $5\ \text{mA/V}$ for g_m.

$$\begin{aligned} f_Z &= \frac{5\ \text{m}}{2\pi(0.5\ \text{p})} \\ &= \frac{5\ \text{m}}{3.14\ \text{p}} \\ &= 1.59\ \text{GHz} \end{aligned}$$

Therefore, the value of the frequency f_Z is $\boxed{1.59\ \text{GHz}}$.

Step 1 of 6

10.033P

Refer to Figure 10.3 (a) from the text book.

Calculate the value of the resistance R_G.

$$R_G = R_{G1} \| R_{G2}$$

Substitute $47\ \text{M}\Omega$ for R_{G1} and $10\ \text{M}\Omega$ for R_G.

$$\begin{aligned} R_G &= 47\ \text{M} \| 10\ \text{M} \\ &= \frac{(47\ \text{M})(10\ \text{M})}{47\ \text{M} + 10\ \text{M}} \\ &= \frac{470\ \text{T}}{57\ \text{M}} \\ &= 8.25\ \text{M}\Omega \end{aligned}$$

Therefore, the value of the resistance R_G is $8.25\ \text{M}\Omega$.

Step 2 of 6

Calculate the value of the resistance R'_{sig}.

$$R'_{sig} = R_{sig} \| R_G$$

Substitute $100\ \text{k}\Omega$ for R_{sig} and $8.25\ \text{M}\Omega$ for R_G.

$$\begin{aligned} R'_{sig} &= 100\ \text{k} \| 8.25\ \text{M} \\ &= \frac{(100\ \text{k})(8.25\ \text{M})}{100\ \text{k} + 8.25\ \text{M}} \\ &= \frac{825\ \text{G}}{8.35\ \text{M}} \\ &= 98.8\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance R'_{sig} is $98.8\ \text{k}\Omega$.

Step 3 of 6

Calculate the value of the resistance R'_L.

$$\begin{aligned} R'_L &= r_o \| R_D \| R_L \\ &= \frac{r_o R_D R_L}{r_o R_D + R_L r_o + R_D R_L} \end{aligned}$$

Substitute $100\ \text{k}\Omega$ for r_o, $4.7\ \text{k}\Omega$ for R_D and $10\ \text{k}\Omega$ for R_L.

$$\begin{aligned} R'_L &= \frac{(100\ \text{k})(4.7\ \text{k})(10\ \text{k})}{(100\ \text{k})(4.7\ \text{k}) + (10\ \text{k})(100\ \text{k}) + (4.7\ \text{k})(10\ \text{k})} \\ &= \frac{4.7\ \text{T}}{0.47\ \text{G} + 1\ \text{G} + 0.047\ \text{G}} \\ &= \frac{4.7\ \text{T}}{1.517\ \text{G}} \\ &= 3.1\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistance R'_L is $3.1\ \text{k}\Omega$.

Step 4 of 6

Calculate he value of the mid band gain A_M.

$$A_M = -\frac{R_G}{R_G + R_{sig}}(g_m R'_L)$$

Substitute $3\ \text{mA/V}$ for g_m, $100\ \text{k}\Omega$ for R_{sig}, $3.1\ \text{k}\Omega$ for R'_L and $8.25\ \text{M}\Omega$ for R_G.

$$\begin{aligned} A_M &= -\left(\frac{8.25\ \text{M}}{8.25\ \text{M} + 100\ \text{k}}\right)(3\ \text{m})(3.1\ \text{k}) \\ &= -\left(\frac{8.25\ \text{M}}{8.35\ \text{M}}\right)(9.3) \\ &= -(0.99)(9.3) \\ &= -9.21\ \text{V/V} \end{aligned}$$

Therefore, the value of the mid band gain A_M is $\boxed{-9.21\ \text{V/V}}$.

Step 5 of 6

Calculate the value of the input capacitance C_{in}.

$$C_{in} = C_{gs} + C_{gd}(1 + g_m R'_L)$$

Substitute $1\ \text{pF}$ for C_{gs} and $0.2\ \text{pF}$ for C_{gd}, $3\ \text{mA/V}$ for g_m and $3.1\ \text{k}\Omega$ for R'_L.

$$\begin{aligned} C_{in} &= 1\ \text{p} + (0.2\ \text{p})(1 + (3\ \text{m})(3.1\ \text{k})) \\ &= 1\ \text{p} + (0.2\ \text{p})(1 + 9.3) \\ &= 1\ \text{p} + (0.2\ \text{p})(10.3) \\ &= 1\ \text{p} + 2.06\ \text{p} \end{aligned}$$

$$C_{in} = 3.06\ \text{pF}$$

Therefore, the value of the input capacitance C_{in} is $3.06\ \text{pF}$.

Step 6 of 6

Calculate the value of the frequency f_H.

$$f_H = \frac{1}{2\pi C_{in} R'_{sig}}$$

Substitute $3.06\ \text{pF}$ for C_{in} and $98.8\ \text{k}\Omega$ for R'_{sig}.

$$\begin{aligned} f_H &= \frac{1}{2\pi(3.06\ \text{p})(98.8\ \text{k})} \\ &= \frac{1}{1.9\ \mu} \\ &= 526.32\ \text{kHz} \end{aligned}$$

Therefore, the value o the frequency f_H is $\boxed{526.32\ \text{kHz}}$.

10.034P

We don't have the solution to this problem yet.

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Step 1 of 2

10.035P

Refer to Figure 10.19 (b) from the text book.

Calculate the value of the input capacitance C_{in} .

$$C_{in} = C_\pi + C_\mu (1 + g_m R_L')$$

Substitute $10\ \text{pF}$ for C_π and $1\ \text{pF}$ for C_μ A and $50\ \text{V/V}$ for $g_m R_L'$.

$$\begin{aligned} C_{in} &= 10\ \text{p} + (1\ \text{p})(1+50) \\ &= 10\ \text{p} + (1\ \text{p})(51) \\ &= 10\ \text{p} + 51\ \text{p} \\ &= 61\ \text{pF} \end{aligned}$$

Therefore, the value of the input capacitance C_{in} is $\boxed{61\ \text{pF}}$.

Step 2 of 2

Calculate the value of the frequency f_H .

$$f_H = \frac{1}{2\pi C_{in} R_{sig}'}$$

Substitute $61\ \text{pF}$ for C_{in} and $5\ \text{k}\Omega$ for R_{sig}' .

$$\begin{aligned} f_H &= \frac{1}{2\pi (61\ \text{p})(5\ \text{k}\Omega)} \\ &= \frac{1}{1.92\ \mu} \\ &= 520.83\ \text{kHz} \end{aligned}$$

Therefore, the value o the frequency f_H is $\boxed{520.83\ \text{kHz}}$.

Step 1 of 6

Refer to Figure 10.19 (a) from the text book.

Calculate the value of the resistance r_π .

$$r_\pi = \frac{\beta}{g_m}$$

Substitute $40\ \text{mA/V}$ for g_m and 100 for β .

$$r_\pi = \frac{100}{40\ \text{m}}$$
$$= 2.5\ \text{k}\Omega$$

Therefore, the value of the resistance r_π is $2.5\ \text{k}\Omega$.

Step 2 of 6

Calculate the value of the resistance R_L' .

$$R_L' = r_o \parallel R_D \parallel R_L$$
$$= \frac{r_o R_D R_L}{r_o R_D + R_L r_o + R_D R_L}$$

Substitute $100\ \text{k}\Omega$ for r_o , $10\ \text{k}\Omega$ for R_C and $10\ \text{k}\Omega$ for R_L .

$$R_L' = \frac{(100\ \text{k})(10\ \text{k})(10\ \text{k})}{(100\ \text{k})(10\ \text{k}) + (10\ \text{k})(100\ \text{k}) + (10\ \text{k})(10\ \text{k})}$$
$$= \frac{10\ \text{T}}{1\ \text{G} + 1\ \text{G} + 0.1\ \text{G}}$$
$$= \frac{10\ \text{T}}{2.1\ \text{G}}$$
$$= 4.76\ \text{k}\Omega$$

Therefore, the value of the resistance R_L' is $4.76\ \text{k}\Omega$.

Step 3 of 6

Calculate the value of the mid band gain A_M .

$$A_M = -\frac{R_B}{R_B + R_{sig}} \frac{r_\pi}{r_\pi + r_x + \left(R_B \parallel R_{sig}\right)} g_m R_L'$$

Substitute $4.76\ \text{k}\Omega$ for R_L' , $40\ \text{mA/V}$ for g_m , $2.5\ \text{k}\Omega$ for r_π , $100\ \Omega$ for r_x , $100\ \text{k}\Omega$ for R_B and $10\ \text{k}\Omega$ for R_{sig} .

$$A_M = -\left(\frac{100\ \text{k}}{100\ \text{k} + 10\ \text{k}}\right)\left(\frac{2.5\ \text{k}}{2.5\ \text{k} + 100 + (100\ \text{k} \parallel 10\ \text{k})}\right)(40\ \text{m})(4.76\ \text{k})$$
$$= -\left(\frac{100\ \text{k}}{110\ \text{k}}\right)\left(\frac{2.5\ \text{k}}{2.6\ \text{k} + \frac{(100\ \text{k})(10\ \text{k})}{100\ \text{k} + 10\ \text{k}}}\right)(190.4)$$
$$= -(0.91)\left(\frac{2.5\ \text{k}}{2.6\ \text{k} + \frac{(100\ \text{k})(10\ \text{k})}{110\ \text{k}}}\right)(190.4)$$
$$= -\left(\frac{2.5\ \text{k}}{2.6\ \text{k} + 9.09\ \text{k}}\right)(173.26)$$

$$A_M = -\left(\frac{2.5\ \text{k}}{11.69\ \text{k}}\right)(173.26)$$
$$= -(0.21)(173.26)$$
$$= -36.39\ \text{V/V}$$

Therefore, the value of the mid band gain A_M is $\boxed{-36.39\ \text{V/V}}$.

Step 4 of 6

Calculate the value of the input capacitance C_{in} .

$$C_{in} = C_\pi + C_\mu\left(1 + g_m R_L'\right)$$

Substitute $10\ \text{pF}$ for C_π and $1\ \text{pF}$ for C_μ , $40\ \text{mA/V}$ for g_m and $4.76\ \text{k}\Omega$ for R_L' .

$$C_{in} = 10\ \text{p} + (1\ \text{p})\left(1 + (40\ \text{m})(4.76\ \text{k})\right)$$
$$= 10\ \text{p} + (1\ \text{p})(1 + 190.4)$$
$$= 10\ \text{p} + (1\ \text{p})(191.4)$$
$$= 10\ \text{p} + 191.4\ \text{p}$$

$$C_{in} = 201.4\ \text{pF}$$

Therefore, the value of the input capacitance C_{in} is $201.4\ \text{pF}$.

Step 5 of 6

Calculate the value of the resistance R_{sig}' .

$$R_{sig}' = r_\pi \parallel \left[r_x + R_B \parallel R_{sig}\right]$$

Substitute $100\ \Omega$ for r_x , $2.5\ \text{k}\Omega$ for r_π , $100\ \text{k}\Omega$ for R_B and $10\ \text{k}\Omega$ for R_{sig} .

$$R_{sig}' = 2.5\ \text{k} \parallel \left[100 + (100\ \text{k} \parallel 10\ \text{k})\right]$$
$$= 2.5\ \text{k} \parallel \left[100 + \frac{(100\ \text{k})(10\ \text{k})}{100\ \text{k} + 10\ \text{k}}\right]$$
$$= 2.5\ \text{k} \parallel \left[100 + \frac{(100\ \text{k})(10\ \text{k})}{110\ \text{k}}\right]$$
$$= 2.5\ \text{k} \parallel \left[100 + 9.09\ \text{k}\right]$$

$$R_{sig}' = 2.5\ \text{k} \parallel 9.19\ \text{k}$$
$$= \frac{(2.5\ \text{k})(9.19\ \text{k})}{2.5\ \text{k} + 9.19\ \text{k}}$$
$$= \frac{22.98\ \text{M}}{11.69\ \text{k}}$$
$$= 1.97\ \text{k}\Omega$$

Therefore, the value of the resistance R_{sig}' is $1.97\ \text{k}\Omega$.

Step 6 of 6

Calculate the value of the frequency f_H .

$$f_H = \frac{1}{2\pi C_{in} R_{sig}'}$$

Substitute $201.4\ \text{pF}$ for C_{in} and $1.97\ \text{k}\Omega$ for R_{sig}' .

$$f_H = \frac{1}{2\pi(201.4\ \text{p})(1.97\ \text{k}\Omega)}$$
$$= \frac{1}{2.49\ \mu}$$
$$= 401.61\ \text{kHz}$$

Therefore, the value o the frequency f_H is $\boxed{401.61\ \text{kHz}}$.

10.037P

We don't have the solution to this problem yet.

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10.038P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

10.039P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

10.040P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

10.041P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

10.042P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

10.043P

We don't have the solution to this problem yet.

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Step 1 of 12

(a)

Refer to the Figure P9.41 in the text book.

Draw the following circuit diagram:

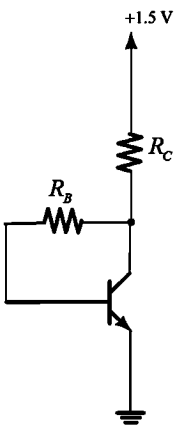


Figure 1

Step 2 of 12

Apply Kirchhoff's voltage law in Figure 1.

$$-1.5\,\text{V}+(I_C+I_B)R_C+I_BR_B+V_{BE}=0$$
$$-1.5\,\text{V}+(\beta I_B+I_B)R_C+I_BR_B+V_{BE}=0$$
$$I_B\left[(\beta+1)R_C+R_B\right]=1.5-V_{BE}$$
$$I_B=\frac{1.5-V_{BE}}{\left[(\beta+1)R_C+R_B\right]}$$

Substitute 0.7 V for V_{BE}, $1\,\text{k}\Omega$ for R_C, $47\,\text{k}\Omega$ for R_B and 100 for β.

$$\begin{aligned}I_B&=\frac{1.5-0.7}{\left[(100+1)(1\,\text{k}\Omega)+(47\,\text{k}\Omega)\right]}\\&=\frac{0.8}{148\,\text{k}\Omega}\\&=0.0054\,\text{mA}\end{aligned}$$

Step 3 of 12

Determine the value of collector current (I_C).

$$I_C=\beta I_B$$

Substitute 100 for β and $0.0054\,\text{mA}$ for I_B.

$$\begin{aligned}I_C&=100I_B\\&=(100)\left(0.0054\times10^{-3}\right)\\&=0.54\,\text{mA}\end{aligned}$$

Therefore, the value of collector current is $\boxed{0.54\,\text{mA}}$.

Step 4 of 12

(b)

Determine the value of transconductance.

$$g_m=\frac{I_C}{V_T}$$

Substitute 0.54 mA for I_C and 25 mV for V_T.

$$\begin{aligned}g_m&=\frac{0.54\times0^{-3}}{0.025}\\&=21.6\,\text{mA/V}\end{aligned}$$

Therefore, the value of transconductance is $\boxed{21.6\ \text{mA/V}}$.

Step 5 of 12

Determine the value of resistance r_π.

$$r_\pi=\frac{\beta}{g_m}$$

Substitute 100 for β and 21.6 mA/V for g_m.

$$\begin{aligned}r_\pi&=\frac{100}{21.6\times10^{-3}}\\&=4.62\,\text{k}\Omega\end{aligned}$$

Therefore, the value of resistance r_π is $\boxed{4.62\,\text{k}\Omega}$.

Step 6 of 12

(c)

Determine the value of voltage gain.

$$\begin{aligned}v_o&=-I_C\left(R_L\parallel R_C\right)\\&=-\left(\beta I_B\right)\left(R_L\parallel R_C\right)\end{aligned}$$
$$v_o=-\left(\beta\left(\frac{v_{in}}{r_\pi}\right)\right)\left(R_L\parallel R_C\right)$$
$$\frac{v_o}{v_{in}}=-\left(\frac{\beta}{r_\pi}\right)\left(R_L\parallel R_C\right)$$

Substitute 100 for β, $4.62\,\text{k}\Omega$ for r_π, $1\,\text{k}\Omega$ for R_C and $1\,\text{k}\Omega$ for R_L.

$$\begin{aligned}\frac{v_o}{v_{in}}&=-\left(\frac{100}{4.62\times10^3}\right)\left(\frac{10^3\times10^3}{10^3+10^3}\right)\\&=-\left(\frac{100}{4.62\times10^3}\right)(500)\\&=-10.82\,\text{V/V}\end{aligned}$$

Therefore, the value of voltage gain is $\boxed{-10.82\ \text{V/V}}$.

Step 7 of 12

(d)

Determine the value of input resistance (R_{in}).

$$R_{in}=r_\pi\parallel\left[R_B+\left(R_L\parallel R_C\right)\right]$$

Substitute $47\,\text{k}\Omega$ for R_B, $4.62\,\text{k}\Omega$ for r_π, $1\,\text{k}\Omega$ for R_C and $1\,\text{k}\Omega$ for R_L.

$$\begin{aligned}R_{in}&=(4.62\,\text{k}\Omega)\parallel\left[(47\,\text{k}\Omega)+\left((1\,\text{k}\Omega)\parallel(1\,\text{k}\Omega)\right)\right]\\&=(4.62\,\text{k}\Omega)\parallel\left[47\,\text{k}\Omega+0.5\,\text{k}\Omega\right]\\&=(4.62\,\text{k}\Omega)\parallel(47.5\,\text{k}\Omega)\\&=\frac{(4.62\,\text{k}\Omega)(47.5\,\text{k}\Omega)}{(4.62\,\text{k}\Omega)+(47.5\,\text{k}\Omega)}\\&=4.21\,\text{k}\Omega\end{aligned}$$

Therefore, the value of input resistance, R_{in} is $\boxed{4.21\,\text{k}\Omega}$.

Step 8 of 12

(e)

Determine the value of midband voltage gain.

$$\begin{aligned}A_v&=\left(\frac{r_\pi}{r_\pi+R_{sig}}\right)g_mR_L'\\&=\left(\frac{r_\pi}{r_\pi+R_{sig}}\right)g_m\left(R_L\parallel R_C\right)\end{aligned}$$

Substitute $1\,\text{k}\Omega$ for R_{sig}, 21.6 mA/V for g_m, $4.62\,\text{k}\Omega$ for r_π, $1\,\text{k}\Omega$ for R_C and $1\,\text{k}\Omega$ for R_L.

$$\begin{aligned}A_v&=\left(\frac{4.62\,\text{k}\Omega}{4.62\,\text{k}\Omega+1\,\text{k}\Omega}\right)\left(21.6\times10^{-3}\right)\left(\frac{10^3\times10^3}{10^3+10^3}\right)\\&=\left(\frac{4.62\,\text{k}\Omega}{5.62\,\text{k}\Omega}\right)\left(21.6\times10^{-3}\right)(500)\\&=8.87\,\text{V/V}\end{aligned}$$

Therefore, the value of voltage gain is $\boxed{8.87\ \text{V/V}}$.

Step 9 of 12

Determine the voltage gain with negative feedback.

$$A_v\,(\text{with negative feedback})=\frac{A_v}{1+A_v\left(\frac{\left(r_\pi\parallel R_{sig}\right)}{\left(r_\pi\parallel R_{sig}\right)+R_B}\right)}$$

Substitute 8.87 V/V for A_v, $1\,\text{k}\Omega$ for R_{sig}, $4.62\,\text{k}\Omega$ for r_π, $47\,\text{k}\Omega$ for R_B.

$$\begin{aligned}A_v\,(\text{with negative feedback})&=\frac{8.87}{1+8.87\left(\frac{(4.62\,\text{k}\Omega\parallel1\,\text{k}\Omega)}{(4.62\,\text{k}\Omega\parallel1\,\text{k}\Omega)+47\,\text{k}\Omega}\right)}\\&=\frac{8.87}{1+8.87\left(\frac{0.822\,\text{k}\Omega}{0.822\,\text{k}\Omega+47\,\text{k}\Omega}\right)}\\&=\frac{8.87}{1.152}\\&=7.69\,\text{V/V}\end{aligned}$$

Therefore, the overall gain is $\boxed{7.69\ \text{V/V}}$.

Step 10 of 12

(f)

Determine the value of capacitance C_π.

$$C_\pi+C_\mu=\frac{g_m}{\omega_T}$$
$$C_\pi+C_\mu=\frac{g_m}{2\pi f_T}$$

Substitute 0.8 pF for C_μ, 21.6 mA/V for g_m and 600 MHz for f_T.

$$\begin{aligned}C_\pi+0.8\,\text{pF}&=\frac{21.6\times10^{-3}}{2\pi\left(600\times10^6\right)}\\C_\pi+0.8\,\text{pF}&=5.72\,\text{pF}\\C_\pi&=4.92\,\text{pF}\end{aligned}$$

Step 11 of 12

Determine the value of input capacitance.

$$C_{in}=C_\pi+C_\mu\left(1+g_mR_L'\right)$$

Substitute 0.8 pF for C_μ, 21.6 mA/V for g_m, 4.92 pF for C_π and $0.5\,\text{k}\Omega$ for R_L'.

$$\begin{aligned}C_{in}&=0.8\,\text{pF}+(4.92\,\text{pF})\left(1+\left(21.6\times10^{-3}\right)\left(0.5\times10^3\right)\right)\\&=0.8\,\text{pF}+58.056\,\text{pF}\\&=58.856\,\text{pF}\end{aligned}$$

Therefore, the value of input capacitance is $\boxed{58.856\,\text{pF}}$.

Step 12 of 12

(g)

Determine the value of resistance (R_{sig}').

$$R_{sig}'=r_\pi\parallel R_{sig}$$

Substitute $1\,\text{k}\Omega$ for R_{sig} and $4.62\,\text{k}\Omega$ for r_π.

$$\begin{aligned}R_{sig}'&=(1\,\text{k}\Omega)\parallel(4.62\,\text{k}\Omega)\\&=\frac{(1\,\text{k}\Omega)(4.62\,\text{k}\Omega)}{(1\,\text{k}\Omega)+(4.62\,\text{k}\Omega)}\\&=0.822\,\text{k}\Omega\end{aligned}$$

Determine the value of 3 dB frequency.

$$f_H=\frac{1}{2\pi C_{in}R_{sig}'}$$

Substitute $0.822\,\text{k}\Omega$ for R_{sig}' and 58.856 pf for C_{in}.

$$\begin{aligned}f_H&=\frac{1}{2\pi\left(0.822\times10^3\right)\left(58.856\times10^{-12}\right)}\\&=\frac{1}{3.03\times10^{-7}}\\&=3.3\,\text{MHz}\end{aligned}$$

Therefore, the value of 3dB frequency is $\boxed{3.3\text{MHz}}$.

Step 1 of 8

Refer to the high-frequency hybrid-π model of the BJT circuit in Figure 9.8 in the text book.

The resistance, r_x that models the resistance of the silicon material of the base region between the base terminal B and a fictitious internal base terminal, B' is zero.

The output resistance, $r_o = \infty$

Redraw the high-frequency hybrid-π model of the transistor of the BJT circuit by replacing resistance, r_x with a short-circuit and the resistance, r_o with an open-circuit.

Consider the circuit in Figure P9.42 in the text book which has a diode connected to the BJT.

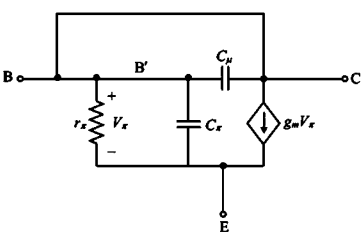


Figure 1

Step 2 of 8

Write the formula for input admittance.

$$Y_i(s) = \left(g_m + \frac{1}{r_\pi}\right) + sC_\pi$$

Here,

g_m is the transconductance of the BJT

r_π is the emitter-base resistance

C_π is the emitter-base capacitance

The input impedance is the reciprocal of the input admittance.

$$Z_i(s) = \frac{1}{Y_i(s)}$$
$$= \frac{1}{\left(g_m + \frac{1}{r_\pi}\right) + sC_\pi}$$

The effect of r_π is negligible at high frequencies.

Thus,

$$Z_i(s) = \frac{1}{g_m + sC_\pi}$$

The emitter resistance, r_e is,

$$r_e = \frac{1}{g_m}$$

Substitute $\frac{1}{r_e}$ for g_m in the expression for input impedance.

$$Z_i(s) = \frac{1}{\frac{1}{r_e} + sC_\pi}$$

Thus, the expression for $Z_i(s)$ is $\boxed{\frac{1}{\frac{1}{r_e} + sC_\pi}}$.

Step 3 of 8

Consider the expression for transition frequency, f_T.

$$f_T = \frac{g_m}{2\pi\left(C_\pi + C_\mu\right)}$$

At high currents, C_π is very much greater than C_μ.

So, $C_\pi + C_\mu = C_\pi$

Replace $C_\pi + C_\mu$ with C_π in the expression for f_T.

$$f_T = \frac{g_m}{2\pi C_\pi}$$

Substitute $\frac{1}{r_e}$ for g_m.

$$f_T = \frac{1}{2\pi C_\pi r_e}$$
$$2\pi f_T = \frac{1}{C_\pi r_e}$$

Step 4 of 8

Consider the expression for input impedance,

$$Z_i(s) = \frac{1}{\frac{1}{r_e} + sC_\pi}$$
$$= \frac{r_e}{1 + sC_\pi r_e}$$

Substitute $\frac{1}{2\pi f_T}$ for $C_\pi r_e$.

$$Z_i(s) = \frac{r_e}{1 + s\left(\frac{1}{2\pi f_T}\right)}$$

Substitute $j\omega$ for s.

$$Z_i(s) = \frac{r_e}{1 + j\omega\left(\frac{1}{2\pi f_T}\right)}$$

Step 5 of 8

The phase angle of the input impedance is,

$$\angle Z_i(s) = -\tan^{-1}\left(\frac{\omega}{2\pi f_T}\right)$$

Equate the phase angle to $-45°$.

$$-\tan^{-1}\left(\frac{\omega}{2\pi f_T}\right) = -45°$$
$$\frac{\omega}{2\pi f_T} = \tan 45°$$
$$\frac{\omega}{2\pi f_T} = 1$$
$$\omega = 2\pi f_T$$

Substitute 400×10^6 for f_T.

$$\omega = 2\pi\left(400 \times 10^6\right)$$

$$f = \frac{\omega}{2\pi}$$
$$= \frac{2\pi\left(400 \times 10^6\right)}{2\pi}$$
$$= 400 \times 10^6 \text{ Hz}$$
$$= 400 \text{ MHz}$$

Therefore, the frequency at which the phase angle of the input impedance $-45°$ is $\boxed{400 \text{ MHz}}$.

Step 6 of 8

Consider the expression for transition frequency, f_T.

$$f_T = \frac{g_m}{2\pi\left(C_\pi + C_\mu\right)}$$

Where,

Transition frequency $C_\pi = C_\mu$

$$f_T = \frac{g_m}{4\pi C_\pi}$$

Substitute $\frac{1}{r_e}$ for g_m.

$$f_T = \frac{1}{4\pi C_\pi r_e}$$
$$4\pi f_T = \frac{1}{C_\pi r_e}$$

Step 7 of 8

Consider the expression for input impedance,

$$Z_i(s) = \frac{1}{\frac{1}{r_e} + sC_\pi}$$
$$= \frac{r_e}{1 + sC_\pi r_e}$$

Substitute $\frac{1}{4\pi f_T}$ for $C_\pi r_e$.

$$Z_i(s) = \frac{r_e}{1 + s\left(\frac{1}{4\pi f_T}\right)}$$

Substitute $j\omega$ for s.

$$Z_i(s) = \frac{r_e}{1 + j\omega\left(\frac{1}{4\pi f_T}\right)}$$

The phase angle of the input impedance is,

$$\angle Z_i(s) = -\tan^{-1}\left(\frac{\omega}{4\pi f_T}\right)$$

Step 8 of 8

Equate the phase angle to $-45°$.

$$-\tan^{-1}\left(\frac{\omega}{4\pi f_T}\right) = -45°$$
$$\frac{\omega}{4\pi f_T} = \tan 45°$$
$$\frac{\omega}{4\pi f_T} = 1$$
$$\omega = 4\pi f_T$$

Substitute 400×10^6 for f_T.

$$\omega = 4\pi\left(400 \times 10^6\right)$$

$$f = \frac{\omega}{2\pi}$$
$$= \frac{4\pi\left(400 \times 10^6\right)}{2\pi}$$
$$= 800 \times 10^6 \text{ Hz}$$
$$= 800 \text{ MHz}$$

Therefore, the frequency at which the phase angle of the input impedance is $-45°$ for $C_\pi = C_\mu$ is $\boxed{800 \text{ MHz}}$.

10.046P

We don't have the solution to this problem yet.

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10.047P

We don't have the solution to this problem yet.

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10.048P

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10.049P

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Step 1 of 9

Determine the value of transcondutance (g_m).

$$g_m = \frac{I_C}{V_T}$$

Substitute $2\,\text{mA}$ for I_C and $25\,\text{mV}$ for V_T.

$$\begin{aligned} g_m &= \frac{2\,\text{mA}}{25\,\text{mV}} \\ &= \frac{2\,\text{mA}}{25\times10^{-3}\,\text{V}} \\ &= 80\,\text{mA/V} \end{aligned}$$

Step 2 of 9

Determine the value of resistance (r_π).

$$r_\pi = \frac{\beta}{g_m}$$

Substitute 120 for β and $80\,\text{mA/V}$ for g_m.

$$\begin{aligned} r_\pi &= \frac{120}{80\,\text{mA}} \\ &= 1.5\times10^3\,\Omega \\ &= 1.5\,\text{k}\Omega \end{aligned}$$

Step 3 of 9

Determine the value of capacitance (C_π).

$$C_\mu + C_\pi = \frac{g_m}{\omega_T}$$

$$C_\pi = \frac{g_m}{\omega_T} - C_\mu$$

$$C_\pi = \frac{g_m}{2\pi f_T} - C_\mu$$

Substitute the values $80\,\text{mA/V}$ for g_m, $2\,\text{GHz}$ for f_T and $1\,\text{pF}$ for C_μ

$$\begin{aligned} C_\pi &= \frac{80\,\text{mA/V}}{2\pi\times2\,\text{GHz}} - (1\,\text{pF}) \\ &= \frac{80\times10^{-3}}{2\pi\times2\times10^9} - \left(10^{-12}\,\text{F}\right) \\ &= \left(6.37\times10^{-12}\right) - \left(10^{-12}\,\text{F}\right) \\ &= 5.37\times10^{-12} \end{aligned}$$

$$= 5.37\,\text{pF}$$

Step 4 of 9

(a)

Write expression of midband voltage gain.

$$A_M = -g_m R_L' \left(\frac{r_\pi}{r_\pi + r_s}\right)$$

Substitute $100\,\Omega$ for r_s, $1.5\,\text{k}\Omega$ for r_π and $-10\,\text{V/V}$ for A_M.

$$-10 = -g_m R_L' \left(\frac{1.5\,\text{k}\Omega}{1.5\,\text{k}\Omega + 100}\right)$$

$$-10 = -g_m R_L' \left(\frac{1.5\,\text{k}\Omega}{1.5\,\text{k}\Omega + 0.1\,\text{k}\Omega}\right)$$

$$-10 = -g_m R_L' (0.9375)$$

$$g_m R_L' = \frac{-10}{-0.9375}$$

$$g_m R_L' = 10.66$$

Step 5 of 9

Determine the value of input capacitance.

$$C_{in} = C_\pi + C_\mu (1 + g_m R_L')$$

Substitute $1\,\text{pF}$ for C_μ, $5.37\,\text{pF}$ for C_π and 10.66 for $g_m R_L'$.

$$\begin{aligned} C_{in} &= 5.37\,\text{pF} + (1\,\text{pF})(1 + 10.66) \\ &= 5.37\,\text{pF} + (1\,\text{pF})(11.66) \\ &= 5.37\,\text{pF} + 11.66\,\text{pF} \\ &= 17.03\,\text{pF} \end{aligned}$$

Determine the value of resistance R_{sig}'.

$$R_{sig}' = r_\pi \parallel r_s$$

Substitute $100\,\Omega$ for r_s and $1.5\,\text{k}\Omega$ for r_π.

$$\begin{aligned} R_{sig}' &= 1.5\,\text{k}\Omega \parallel 100\,\Omega \\ &= \frac{(1.5\,\text{k}\Omega)(100\,\Omega)}{1.5\,\text{k}\Omega + 100\,\Omega} \\ &= \frac{(1.5\,\text{k}\Omega)(0.1\,\text{k}\Omega)}{1.5\,\text{k}\Omega + 0.1\,\text{k}\Omega} \\ &= 93.75\,\Omega \end{aligned}$$

Step 6 of 9

Determine the value of frequency (f_H).

$$f_H = \frac{1}{2\pi C_{in} R_{sig}'}$$

Substitute $17.03\,\text{pF}$ for C_{in} and $93.75\,\Omega$ for R_{sig}'.

$$\begin{aligned} f_H &= \frac{1}{2\pi(17.03\,\text{pF})(93.75\,\Omega)} \\ &= \frac{1}{2\pi\times\left(17.03\times10^{-12}\right)(93.75)} \\ &= 9.973\times10^{-5}\times10^{12} \\ &= 99.73\times10^6 \end{aligned}$$

$$= 99.73\,\text{MHz}$$

Therefore, the value of f_H is $\boxed{99.73\,\text{MHz}}$.

Step 7 of 9

(b)

Write the expression of midband gain.

$$A_M = -(g_m R_L')\left(\frac{r_\pi}{r_\pi + r_s}\right)$$

Substitute $100\,\Omega$ for r_s, $1.5\,\text{k}\Omega$ for r_π and $-1\,\text{V/V}$ for A_M.

$$-1 = -g_m R_L' \left(\frac{1.5\,\text{k}\Omega}{1.5\,\text{k}\Omega + 100}\right)$$

$$-1 = -g_m R_L' \left(\frac{1.5\,\text{k}\Omega}{1.5\,\text{k}\Omega + 0.1\,\text{k}\Omega}\right)$$

$$-1 = -g_m R_L' (0.9375)$$

$$g_m R_L' = \frac{-1}{-0.9375}$$

$$g_m R_L' = 1.066$$

Step 8 of 9

Determine the value of input capacitance C_{in}

$$C_{in} = C_\pi + C_\mu (1 + g_m R_L')$$

Substitute $1\,\text{pF}$ for C_μ, $5.37\,\text{pF}$ for C_π and 1.066 for $g_m R_L'$.

$$\begin{aligned} C_{in} &= 5.37\,\text{pF} + (1\,\text{pF})(1 + 1.066) \\ &= 5.37\,\text{pF} + (1\,\text{pF})(2.066) \\ &= 5.37\,\text{pF} + 2.066\,\text{pF} \\ &= 7.436\,\text{pF} \end{aligned}$$

Step 9 of 9

Determine the value of resistance R_{sig}'.

$$R_{sig}' = r_\pi \parallel r_s$$

Substitute $100\,\Omega$ for r_s and $1.5\,\text{k}\Omega$ for r_π.

$$\begin{aligned} R_{sig}' &= 1.5\,\text{k}\Omega \parallel 100\,\Omega \\ &= \frac{(1.5\,\text{k}\Omega)(100\,\Omega)}{1.5\,\text{k}\Omega + 100\,\Omega} \\ &= \frac{(1.5\,\text{k}\Omega)(0.1\,\text{k}\Omega)}{1.5\,\text{k}\Omega + 0.1\,\text{k}\Omega} \\ &= 93.75\,\Omega \end{aligned}$$

Determine the value of frequency (f_H).

$$f_H = \frac{1}{2\pi C_{in} R_{sig}'}$$

Substitute $7.436\,\text{pF}$ for C_{in} and $93.75\,\Omega$ for R_{sig}'.

$$\begin{aligned} f_H &= \frac{1}{2\pi(7.436\,\text{pF})(93.75\,\Omega)} \\ &= \frac{1}{2\pi\left(7.436\times10^{-12}\right)(93.75)} \\ &= 2.28\times10^{-4}\times10^{12} \\ &= 2.284\times10^8 \end{aligned}$$

$$= 228.4\times10^6$$

$$= 228.4\,\text{MHz}$$

Therefore, the value of f_H is $\boxed{228.4\,\text{MHz}}$.

10.051P

We don't have the solution to this problem yet.

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10.052P

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Step 1 of 2

10.053P

(a)

Refer to the equation 9.68 in the text book.

Write the expression of ω_H.

$$\omega_H = \frac{1}{\sqrt{\frac{1}{\omega_{P1}^2} + \frac{1}{\omega_{P2}^2}}}$$

Substitute $0.9\omega_{P1}$ for ω_H.

$$0.9\omega_{P1} = \frac{1}{\sqrt{\frac{1}{\omega_{P1}^2} + \frac{1}{\omega_{P2}^2}}}$$

$$\omega_{p1} = 1.1\left(\frac{1}{\sqrt{\frac{1}{\omega_{P1}^2} + \frac{1}{\omega_{P2}^2}}}\right)$$

$$\omega_{p1}^2 = 1.21\frac{\omega_{P1}^2\omega_{P2}^2}{\omega_{P1}^2 + \omega_{P2}^2}$$

$$\omega_{P1}^2 + \omega_{P2}^2 = 1.21\omega_{P2}^2$$

$$\omega_{P1}^2 = 0.21\omega_{P2}^2$$

$$\frac{\omega_{P1}^2}{\omega_{P2}^2} = 0.21$$

$$\frac{\omega_{P2}}{\omega_{P1}} = 2.18$$

Therefore, the ratio of $\frac{\omega_{P2}}{\omega_{P1}}$ is $\boxed{2.18}$.

Step 2 of 2

(b)

Write the expression of ω_H.

$$\omega_H = \frac{1}{\sqrt{\frac{1}{\omega_{P1}^2} + \frac{1}{\omega_{P2}^2}}}$$

Substitute $0.99\omega_{P1}$ for ω_H.

$$0.99\omega_{P1} = \frac{1}{\sqrt{\frac{1}{\omega_{P1}^2} + \frac{1}{\omega_{P2}^2}}}$$

$$\omega_{p1} = 1.01\left(\frac{1}{\sqrt{\frac{1}{\omega_{P1}^2} + \frac{1}{\omega_{P2}^2}}}\right)$$

$$\omega_{p1}^2 = 1.02\frac{\omega_{P1}^2\omega_{P2}^2}{\omega_{P1}^2 + \omega_{P2}^2}$$

$$\omega_{P1}^2 + \omega_{P2}^2 = 1.02\omega_{P2}^2$$

$$\omega_{P1}^2 = 0.02\omega_{P2}^2$$

$$\frac{\omega_{P1}^2}{\omega_{P2}^2} = 0.02$$

$$\frac{\omega_{P2}}{\omega_{P1}} = 7.07$$

Therefore, the ratio of $\frac{\omega_{P2}}{\omega_{P1}}$ is $\boxed{7.07}$.

10.054P

We don't have the solution to this problem yet.

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Step 1 of 2

10.055P

A direct coupled amplifier has a dominant pole at $1000\ \text{rad/sec}$ and three coincident poles at much higher frequency. These nondominent poles cause the phase lag of the amplifier at high frequencies to exceed the $90°$ angle due to the dominant pole.

Determine the value of pole.

$$\tan^{-1}(\omega T) = 30°$$

Substitute $10^7\ \text{rad/s}$ for ω.

$$\tan^{-1}\left((10^7)T\right) = 30°$$

$$(10^7)T = \tan(30°)$$

$$(10^7)\left(\frac{1}{\omega_p}\right) = 0.577$$

$$\omega_p = 1.73 \times 10^7\ \text{rad/s}$$

Therefore, the value of nondominant pole is $\boxed{1.73 \times 10^7\ \text{rad/s}}$.

Step 2 of 2

Determine the value of frequency (f_p).

$$\omega_p = 2\pi f_p$$

$$f_p = \frac{\omega_p}{2\pi}$$

Substitute $1.73 \times 10^7\ \text{rad/s}$ for ω_p.

$$f_p = \frac{1.73 \times 10^7\ \text{rad/s}}{2\pi}$$
$$= 2.75\ \text{MHz}$$

Therefore, the value of frequency is $\boxed{2.75\ \text{MHz}}$.

10.056P

We don't have the solution to this problem yet.

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10.057P

We don't have the solution to this problem yet.

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10.058P

We don't have the solution to this problem yet.

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10.059P

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10.060P

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10.061P

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10.062P

We don't have the solution to this problem yet.

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10.063P

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Step 1 of 3

Consider a common-source amplifier.

The 3-dB frequency, f_H is,

$$f_H = \frac{1}{2\pi C_{in} R'_{sig}}$$

Here, the source signal input resistance, R'_{sig} is,

$$R'_{sig} = \frac{r_o}{2}$$

Here, the small-signal output resistance, $r_o = 20\ \text{k}\Omega$. Therefore,

$$\begin{aligned} R'_{sig} &= \frac{r_o}{2} \\ &= \frac{20\ \text{k}\Omega}{2} \\ &= 10\ \text{k}\Omega \end{aligned}$$

Find the value of C_{in}.

$$C_{in} = C_{gs} + C_{gd}\left(1 + g_m R'_L\right)$$

Here, g_m is transconductance, and capacitances are, C_{gs} and C_{gd}.

Here, find the equivalent load resistance.

$$\begin{aligned} R'_L &= r_o \parallel r_o \\ &= 20\ \text{k}\Omega \parallel 20\ \text{k}\Omega \\ &= 10\ \text{k}\Omega \end{aligned}$$

Substitute 0.1 pF for C_{gs} and C_{gd}, 2 mA/V for g_m, and $10\ \text{k}\Omega$ for R'_{sig}.

$$\begin{aligned} C_{in} &= 0.1\ \text{pF} + 0.1\ \text{pF}\left[1 + (2\ \text{mA/V})(10\ \text{k}\Omega)\right] \\ &= 0.1\ \text{pF} + (0.1\ \text{pF})(21) \\ &= 2.2\ \text{pF} \end{aligned}$$

Find 3-dB frequency, f_H using Miller approximation.

$$f_H = \frac{1}{2\pi C_{in} R'_{sig}}$$

Substitute 2.2 pF for C_{in}, $10\ \text{k}\Omega$ for R'_{sig}.

$$\begin{aligned} f_H &= \frac{1}{2\pi (2.2\ \text{pF})(10\ \text{k}\Omega)} \\ &= \frac{1}{138.23\ \text{ns}} \\ &= 7.23\ \text{MHz} \end{aligned}$$

Hence, the 3-dB frequency, f_H is, $\boxed{7.23\ \text{MHz}}$.

Step 2 of 3

(i)

Consider that the bias current is reduced by a factor of 4.

Calculate the small-signal output resistance.

$$\begin{aligned} r_o &= \frac{V_A}{\left(\frac{I_D}{4}\right)} \\ &= 4\left(\frac{V_A}{I_D}\right) \\ &= 4(20\ \text{k}\Omega) \quad \left(\text{since, } r_o = \frac{V_A}{I_D}\right) \\ &= 80\ \text{k}\Omega \end{aligned}$$

Find the transconductance.

$$\begin{aligned} g_m &= \frac{2\left(\frac{I_D}{4}\right)}{\left(\frac{V_{OV}}{2}\right)} \quad \left(\text{since, } I_D \ \alpha\ V_{OV}^2\right) \\ &= \frac{1}{2}(2\ \text{mA/V}) \left(\text{since, } g_m = \frac{2I_D}{V_{OV}}\right) \\ &= 1\ \text{mA/V} \end{aligned}$$

Find the source signal input resistance, R'_{sig}.

$$\begin{aligned} R'_L &= R'_{sig} \\ &= \frac{r_o}{2} \\ &= \frac{80\ \text{k}\Omega}{2} \\ &= 40\ \text{k}\Omega \end{aligned}$$

The input capacitance is,

$$\begin{aligned} C_{in} &= C_{gs} + C_{gd}\left(1 + g_m R'_L\right) \\ &= 0.1\ \text{pF} + 0.1\ \text{pF}\left[1 + (1\ \text{mA/V})(40\ \text{k}\Omega)\right] \\ &= 0.1\ \text{pF} + (0.1\ \text{pF})(11) \\ &= 4.2\ \text{pF} \end{aligned}$$

The 3-dB frequency is,

$$\begin{aligned} f_H &= \frac{1}{2\pi C_{in} R'_{sig}} \\ &= \frac{1}{2\pi (4.2\ \text{pF})(40\ \text{k}\Omega)} \\ &= 947\ \text{kHz} \end{aligned}$$

Thus, the 3-dB frequency, f_H is, $\boxed{947\ \text{kHz}}$.

Step 3 of 3

(ii)

Consider that the bias current is increased by a factor of 4.

Calculate the small-signal output resistance.

$$\begin{aligned} r_o &= \frac{V_A}{4I_D} \\ &= \frac{1}{4}(20\ \text{k}\Omega) \quad \left(\text{since, } r_o = \frac{V_A}{I_D}\right) \\ &= 5\ \text{k}\Omega \end{aligned}$$

Find the transconductance.

$$\begin{aligned} g_m &= \frac{2(4I_D)}{16V_{OV}} \quad \left(\text{since, } I_D \alpha V_{OV}^2\right) \\ &= 2(2\ \text{mA/V}) \left(\text{since, } g_m = \frac{2I_D}{V_{OV}}\right) \\ &= 4\ \text{mA/V} \end{aligned}$$

Find the source signal input resistance, R'_{sig}.

$$\begin{aligned} R'_L &= R'_{sig} \\ &= \frac{r_o}{2} \\ &= \frac{5\ \text{k}\Omega}{2} \\ &= 2.5\ \text{k}\Omega \end{aligned}$$

The input capacitance is,

$$\begin{aligned} C_{in} &= C_{gs} + C_{gd}\left(1 + g_m R'_L\right) \\ &= 0.1\ \text{pF} + 0.1\ \text{pF}\left[1 + (4\ \text{mA/V})(2.5\ \text{k}\Omega)\right] \\ &= 0.1\ \text{pF} + (0.1\ \text{pF})(11) \\ &= 1.2\ \text{pF} \end{aligned}$$

The 3-dB frequency is,

$$\begin{aligned} f_H &= \frac{1}{2\pi C_{in} R'_{sig}} \\ &= \frac{1}{2\pi (1.2\ \text{pF})(2.5\ \text{k}\Omega)} \\ &= 53\ \text{MHz} \end{aligned}$$

Thus, the 3-dB frequency, f_H is, $\boxed{53\ \text{MHz}}$.

Step 1 of 4

(a)

The 3-dB frequency, f_H is,

$$f_H = \frac{1}{2\pi\tau_H}$$

Here, $\tau_H = C_{gs}R_{sig} + C_{gd}\left[R_{sig}\left(1+g_m R_L'\right)+R_L'\right]+C_L R_L'$.

Find the equivalent load resistance.

$$\begin{aligned} R_L' &= R_L \| r_o \\ &= \frac{R_L r_o}{R_L + r_o} \\ &= \frac{(12\ \text{k}\Omega)(20\ \text{k}\Omega)}{12\ \text{k}\Omega + 20\ \text{k}\Omega} \\ &= 7.5\ \text{k}\Omega \end{aligned}$$

Consider that the load capacitance, $C_L = 0$.

Substitute 0.2 pF for C_{gs} and C_{gd}, 100 kΩ for R_{sig}, 1.5 mA/V for g_m, 7.5 kΩ for R_L', and 0 for C_L in the expression of τ_H.

$$\begin{aligned} \tau_H &= (0.2\ \text{pF})(100\ \text{k}\Omega) + (0.2\ \text{pF})\left\{(100\ \text{k}\Omega)\left[1+(1.5\ \text{mA/V})(7.5\ \text{k}\Omega)\right]+7.5\ \text{k}\Omega\right\}+0 \\ &= 20\ \text{ns} + 246.5\ \text{ns} \\ &= 266.5\ \text{ns} \end{aligned}$$

Substitute 402.4 ns for τ_H to find the value of f_H.

$$\begin{aligned} f_H &= \frac{1}{2\pi(266.5\ \text{ns})} \\ &= \frac{1}{1.67\ \mu\text{s}} \\ &= 597\ \text{kHz} \end{aligned}$$

Thus, the 3-dB frequency, f_H is, **597 kHz**.

Step 2 of 4

(b)

Consider that the load capacitance, $C_L = 10\ \text{pF}$.

Substitute 0.2 pF for C_{gs} and C_{gd}, 100 kΩ for R_{sig}, 1.5 mA/V for g_m, 7.5 kΩ for R_L', and 10 pF for C_L in the expression of τ_H.

$$\begin{aligned} \tau_H &= \left[\begin{array}{l}(0.2\ \text{pF})(100\ \text{k}\Omega)+(0.2\ \text{pF})\left\{(100\ \text{k}\Omega)\left[1+(1.5\ \text{mA/V})(7.5\ \text{k}\Omega)\right]+7.5\ \text{k}\Omega\right\} \\ +(10\ \text{pF})(7.5\ \text{k}\Omega)\end{array}\right] \\ &= 20\ \text{ns} + 246.5\ \text{ns} + 75\ \text{ns} \\ &= 341.5\ \text{ns} \end{aligned}$$

Substitute 341.5 ns for τ_H to find the value of f_H.

$$\begin{aligned} f_H &= \frac{1}{2\pi(341.5\ \text{ns})} \\ &= \frac{1}{2.145\ \mu\text{s}} \\ &= 466\ \text{kHz} \end{aligned}$$

Thus, the 3-dB frequency, f_H is, **466 kHz**.

Step 3 of 4

(c)

Consider that the load capacitance, $C_L = 50\ \text{pF}$.

Substitute 0.2 pF for C_{gs} and C_{gd}, 100 kΩ for R_{sig}, 1.5 mA/V for g_m, 7.5 kΩ for R_L', and 50 pF for C_L in the expression of τ_H.

$$\begin{aligned} \tau_H &= \left[\begin{array}{l}(0.2\ \text{pF})(100\ \text{k}\Omega)+(0.2\ \text{pF})\left\{(100\ \text{k}\Omega)\left[1+(1.5\ \text{mA/V})(7.5\ \text{k}\Omega)\right]+7.5\ \text{k}\Omega\right\} \\ +(50\ \text{pF})(7.5\ \text{k}\Omega)\end{array}\right] \\ &= 20\ \text{ns} + 246.5\ \text{ns} + 375\ \text{ns} \\ &= 641.5\ \text{ns} \end{aligned}$$

Substitute 641.5 ns for τ_H to find the value of f_H.

$$\begin{aligned} f_H &= \frac{1}{2\pi(641.5\ \text{ns})} \\ &= \frac{1}{4.03\ \mu\text{s}} \\ &= 248\ \text{kHz} \end{aligned}$$

Thus, the 3-dB frequency, f_H is, **248 kHz**.

Step 4 of 4

Use Miller approximation to find the 3-dB frequency, f_H.

$$f_H = \frac{1}{2\pi\left[C_{gs} + C_{gd}\left(1+g_m R_L'\right)\right]R_{sig}'}$$

Substitute 0.2 pF for C_{gs} and C_{gd}, 100 kΩ for R_{sig}, 1.5 mA/V for g_m, 7.5 kΩ for R_L'.

$$\begin{aligned} f_H &= \frac{1}{2\pi\left\{0.2\ \text{pF}+(0.2\ \text{pF})\left[1+(1.5\ \text{mA/V})(7.5\ \text{k}\Omega)\right]\right\}(100\ \text{k}\Omega)} \\ &= \frac{1}{2\pi(2.65\ \text{pF})(100\ \text{k}\Omega)} \\ &= \frac{1}{1.665\ \mu\text{s}} \\ &= 600\ \text{kHz} \end{aligned}$$

Thus, the 3-dB frequency, f_H using Miller approximation is, **600 kHz**.

Neglecting the effect of C_L, the 3-dB frequency, f_H is approximately same when $C_L = 0$. In other two cases C_L affects the f_H value.

The value obtained for Miller approximation is larger than all cases and almost equal to the case where $C_L = 0$.

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Step 1 of 10

Consider the following common-gate amplifier circuit:

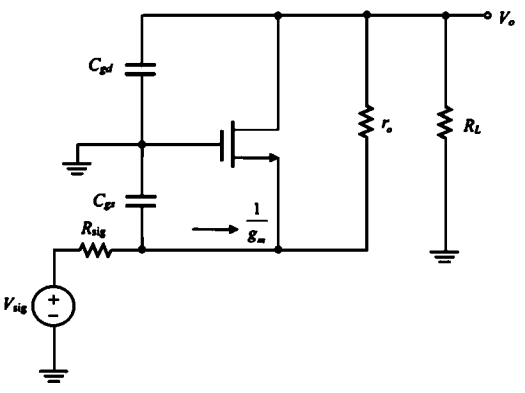


Figure 1

Step 2 of 10

Consider that the common-gate amplifier have load resistance, $R_L = r_o$ and signal source have the resistance, $R_{sig} = \frac{r_o}{2}$. The mid band equivalent circuit is shown in Figure 2.

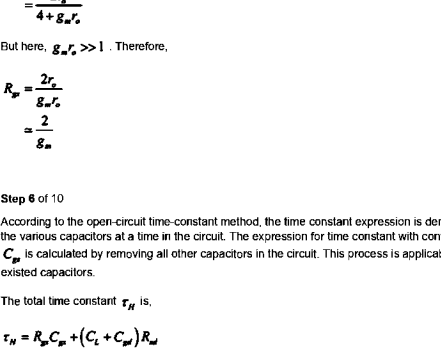


Figure 2

Step 3 of 10

Open circuit time-constant method:

The open-circuit time constant method is widely used for determining the bandwidth of the analog circuit. It is not simple to determine the poles and zeros of the transfer function by quick hand analysis. Instead of this, the method of the open-circuit time constant is used to avoid the complexity in deriving the transfer function of the circuit.

To calculate the gate to source impedance and input impedance, set the input signal source to zero and open all the capacitors. Find the gate to source impedance seen by the open capacitor, C_L or C_{gs}.

The modified circuit for finding the impedances is shown in Figure 3.

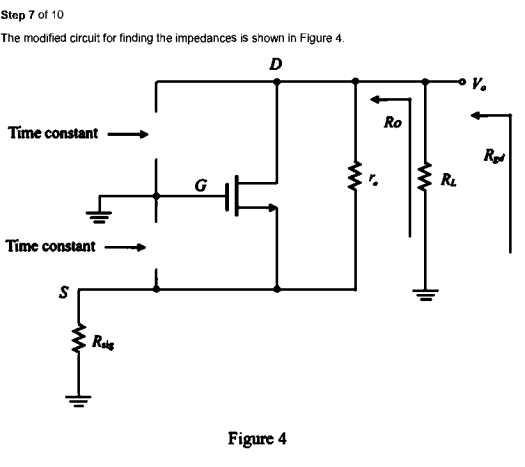


Figure 3

Step 4 of 10

By inspection, the gate to source impedance seen by the open capacitor, C_L or C_{gs} is,

$$R_{gs} = R_i \| R_{sig}$$

From Figure 1 and 3, the input resistance is,

$$\begin{aligned} R_i &= \frac{1}{g_m} + \frac{R_L}{1+g_m r_o} \\ &= \frac{R_L + r_o}{1+g_m r_o} \qquad \text{since } R_L = r_o \\ &= \frac{r_o + r_o}{1+g_m r_o} \\ &= \frac{2r_o}{1+g_m r_o} \end{aligned}$$

Step 5 of 10

But here, $g_m r_o >> 1$. Therefore,

$$\begin{aligned} R_i &= \frac{2r_o}{g_m r_o} \\ &= \frac{2}{g_m} \end{aligned}$$

Therefore, the gate to source impedance is,

$$\begin{aligned} R_{gs} &= \frac{2}{g_m} \| \frac{r_o}{2} \\ &= \frac{\frac{2}{g_m}\frac{r_o}{2}}{\frac{2}{g_m}+\frac{r_o}{2}} \\ &= \frac{2r_o}{4+g_m r_o} \end{aligned}$$

But here, $g_m r_o >> 1$. Therefore,

$$\begin{aligned} R_{gs} &= \frac{2r_o}{g_m r_o} \\ &\simeq \frac{2}{g_m} \end{aligned}$$

Step 6 of 10

According to the open-circuit time-constant method, the time constant expression is derived by considering the various capacitors at a time in the circuit. The expression for time constant with contribution of capacitor C_{gs} is calculated by removing all other capacitors in the circuit. This process is applicable to all other existed capacitors.

The total time constant τ_H is,

$$\tau_H = R_{gs}C_{gs} + (C_L + C_{gd})R_{gd}$$

The pole at the input side is,

$$\omega_H = \frac{1}{\tau_H}$$

$$\begin{aligned} f_H &= \frac{1}{2\pi\tau_H} \\ &= \frac{1}{2\pi\left[R_{gs}C_{gs} + (C_L + C_{gd})R_{gd}\right]} \end{aligned}$$

Here,Upper 3dB frequency, f_H.

Gate to source resistance, R_{gs}.

Gate to source capacitance, C_{gs}.

Gate to drain capacitance, C_{gd}.

Gate to drain resistance, R_{gd}.

Step 7 of 10

The modified circuit for finding the impedances is shown in Figure 4.

Figure 4

Step 8 of 10

By inspection, the gate to source impedance seen by the open capacitor, C_{gd} is,

$$R_{gd} = R_o \| R_L$$

From Figure 1 and 3, the input resistance is,

$$\begin{aligned} R_o &= r_o + R_{sig} + g_m r_o R_{sig} \qquad \left(\text{Since } R_{sig} = \frac{r_o}{2}\right) \\ &= r_o + \frac{r_o}{2} + g_m r_o \frac{r_o}{2} \\ &= \frac{3r_o}{2} + g_m r_o \frac{r_o}{2} \qquad (\text{Since } g_m R_o >> 1) \\ &\approx g_m r_o \frac{r_o}{2} \end{aligned}$$

Step 9 of 10

Therefore, the gate to drain impedance is,

$$\begin{aligned} R_{gd} &= g_m r_o \frac{r_o}{2} \| r_o \\ &= \frac{g_m r_o \frac{r_o}{2} r_o}{g_m r_o \frac{r_o}{2} + r_o} \quad (\text{Since } g_m r_o >> 1) \\ &= \frac{g_m r_o \frac{r_o}{2} r_o}{g_m r_o \frac{r_o}{2}} \\ &= r_o \end{aligned}$$

Substitute r_o for R_{gd}, C_{gs} for C_L, and $\frac{2}{g_m}$ for R_{gs} in the expression of f_H.

$$\begin{aligned} \omega_H &= \frac{1}{\frac{2}{g_m}C_{gs} + (C_{gs} + C_{gd})r_o} \\ &= \frac{1}{\frac{2}{g_m}C_{gs} + C_{gs}r_o + r_o C_{gd}} \\ &= \frac{1}{\left[\frac{2}{g_m} + r_o\right]C_{gs} + r_o C_{gd}} \\ &= \frac{1}{\left[\frac{2+g_m r_o}{g_m}\right]C_{gs} + r_o C_{gd}} \end{aligned}$$

Since, $g_m r_o >> 1$, therefore,

$$\begin{aligned} \omega_H &= \frac{1}{\left[\frac{g_m r_o}{g_m}\right]C_{gs} + r_o C_{gd}} \\ &= \frac{1}{C_{gs}r_o + r_o C_{gd}} \\ f_H &= \frac{1}{2\pi r_o (C_{gs} + C_{gd})} \end{aligned}$$

Step 10 of 10

The unity current gain frequency of MOSFET is,

$$f_T = \frac{g_m}{2\pi(C_{gs} + C_{gd})}$$

$$2\pi(C_{gs} + C_{gd}) = \frac{g_m}{f_T}$$

The upper 3dB frequency related with unity current gain frequency of MOSFET is,

$$\begin{aligned} f_H &= \frac{1}{r_o\left(\frac{g_m}{f_T}\right)} \\ &= \frac{f_T}{(g_m r_o)} \end{aligned}$$

Hence, the expression $\boxed{f_H = \frac{f_T}{(g_m r_o)}}$ is proved.

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Step 1 of 7

(a)

Consider a common-source amplifier.

The low frequency gain is,

$$\begin{aligned} A_M &= -g_m\left(R_L \| r_o\right) \\ &= -g_m\left(\frac{R_L r_o}{R_L + r_o}\right) \end{aligned}$$

Here, g_m is the transconductance, R_L is the load resistance, r_o is the small-signal output resistance.

Substitute 4 mA/V for g_m, $20\ \text{k}\Omega$ for R_L and r_o.

$$\begin{aligned} A_M &= -(4\ \text{mA/V})\left[\frac{(20\ \text{k}\Omega)(20\ \text{k}\Omega)}{20\ \text{k}\Omega + 20\ \text{k}\Omega}\right] \\ &= -(4\ \text{mA/V})(10\ \text{k}\Omega) \\ &= -40\ \text{V/V} \end{aligned}$$

Therefore, the low frequency gain, A_M is, $\boxed{-40\ \text{V/V}}$.

Step 2 of 7

Calculate the value of high frequency.

$$f_H = \frac{1}{2\pi\tau_H}$$

Here,

$$\tau_H = C_{gs}R'_{sig} + C_{gd}\left[R'_{sig}\left(1 + g_m R'_L\right) + R'_L\right] + C_L R'_L$$

Calculate the resistances in the expression.

$$\begin{aligned} R_{sig} &= R_{sig} \\ &= 20\ \text{k}\Omega \end{aligned}$$

$$\begin{aligned} R'_L &= R_L \| r_o \\ &= 20\ \text{k}\Omega \| 20\ \text{k}\Omega \\ &= 10\ \text{k}\Omega \end{aligned}$$

The values of capacitances in the expression are,

$$\begin{aligned} C_{gd} &= 0.2\ \text{pF}, \\ C_{gs} &= 2\ \text{pF}, \\ C_L &= 1\ \text{pF}, \\ C_{db} &= 0.2\ \text{pF} \end{aligned}$$

Substitute all the resistances and capacitances values in the expression of τ_H.

$$\begin{aligned} \tau_H &= \left[\begin{array}{l}(2\ \text{pF})(20\ \text{k}\Omega) + (0.2\ \text{pF})\{(20\ \text{k}\Omega)[1 + (4\ \text{mA/V})(10\ \text{k}\Omega)] + 10\ \text{k}\Omega\} \\ + (1\ \text{pF})(10\ \text{k}\Omega)\end{array}\right] \\ &= 40\ \text{ns} + 166\ \text{ns} + 10\ \text{ns} \\ &= 216\ \text{ns} \end{aligned}$$

The 3-dB frequency, f_H is,

$$\begin{aligned} f_H &= \frac{1}{2\pi\tau_H} \\ &= \frac{1}{2\pi(216\ \text{ns})} \\ &= 0.736\ \text{MHz} \end{aligned}$$

Thus, the value of high frequency, f_H is, $\boxed{0.786\ \text{MHz}}$.

Step 3 of 7

The gain-bandwidth product is,

$$f_t = A_M f_H$$

Substitute $-40\ \text{V/V}$ for A_M, and 0.786 MHz for f_H.

$$\begin{aligned} f_t &= (-40\ \text{V/V})(0.786\ \text{MHz}) \\ &= -29.47\ \text{MHz} \end{aligned}$$

Therefore, the gain-bandwidth product is, $\boxed{-29.47\ \text{MHz}}$.

Step 4 of 7

(b)

Consider that common-gate stage is cascaded to the common source stage to form a cascode amplifier.

The overall output resistance is,

$$R_o = r_o\left(2 + g_m r_o\right)$$

Substitute $20\ \text{k}\Omega$ for r_o, 4 mA/V for g_m.

$$\begin{aligned} R_o &= (20\ \text{k}\Omega)\left[2 + (4\ \text{mA/V})(20\ \text{k}\Omega)\right] \\ &= (20\ \text{k}\Omega)(82) \\ &= 1640\ \text{k}\Omega \end{aligned}$$

The gain of the cascaded stage is,

$$\begin{aligned} A_{vo} &= -g_m R_o \\ &= -(4\ \text{mA/V})(1640\ \text{k}\Omega) \\ &= 6560\ \text{V/V} \end{aligned}$$

The low frequency gain of the overall cascode amplifier is,

$$\begin{aligned} A_M &= A_{vo}\left(\frac{R_L}{R_L + R_o}\right) \\ &= (-5650\ \text{V/V})\left(\frac{20\ \text{k}\Omega}{20\ \text{k}\Omega + 1640\ \text{k}\Omega}\right) \\ &= 79\ \text{V/V} \end{aligned}$$

Thus, the low frequency gain, A_M is, $\boxed{-79\ \text{V/V}}$.

Step 5 of 7

Calculate the value of high frequency of the cascode amplifier.

$$f_H = \frac{1}{2\pi\tau_H}$$

Here,

$$\tau_H = R_{d1}\left(C_{gd1} + C_{db1} + C_{gs2}\right) + R'_L\left(C_L + C_{gd2} + C_{db2}\right)$$

Calculate the resistances in the expression.

$$\begin{aligned} R'_L &= R_L \| R_o \\ &= \frac{R_L R_o}{R_L + R_o} \\ &= \frac{(1640\ \text{k}\Omega)(20\ \text{k}\Omega)}{1640\ \text{k}\Omega + 20\ \text{k}\Omega} \\ &= 19.76\ \text{k}\Omega \end{aligned}$$

$$\begin{aligned} R_{d1} &= r_{o1} \| \left(\frac{1}{g_{m2}} + \frac{R_L}{1 + g_{m2}r_{o2}}\right) \\ &= (20\ \text{k}\Omega) \| \left(\frac{1}{4\ \text{mA/V}} + \frac{20\ \text{k}\Omega}{1 + (4\ \text{mA/V})(20\ \text{k}\Omega)}\right) \\ &= (20\ \text{k}\Omega) \| (250\ \Omega + 247\ \Omega) \\ &= (20\ \text{k}\Omega) \| (500\ \Omega) \end{aligned}$$

Simplify further.

$$\begin{aligned} R_{d1} &= \frac{(20\ \text{k}\Omega)(500\ \Omega)}{20\ \text{k}\Omega + 500\ \Omega} \\ &= \frac{10 \times 10^6}{20.5 \times 10^3} \\ &\simeq 485\ \Omega \end{aligned}$$

Step 6 of 7

Substitute all the resistances and capacitances values in the following expression of τ_H of a cascode amplifier:

$$\begin{aligned} \tau_H &= R'_{sig}\left[C_{gs1} + C_{gd1}\left(1 + g_{m1}R_{d1}\right)\right] + R_{d1}\left(C_{gd1} + C_{db1} + C_{gs2}\right) + R'_L\left(C_L + C_{gd2}\right) \\ &= \left[\begin{array}{l}(20\ \text{k}\Omega)\{2\ \text{pF} + \{(0.2\ \text{pF})[1 + (4\ \text{mA/V})(485\ \Omega)]\}\} \\ + (485\ \Omega)(0.2\ \text{pF} + 0.2\ \text{pF} + 2\ \text{pF}) + (19.76\ \text{k}\Omega)(1 + 0.2\ \text{pF})\end{array}\right] \\ &= 51.76\ \text{ns} + 1.164\ \text{ns} + 23.712\ \text{ns} \\ &= 76.636\ \text{ns} \end{aligned}$$

The 3-dB frequency, f_H is,

$$\begin{aligned} f_H &= \frac{1}{2\pi\tau_H} \\ &= \frac{1}{2\pi(76.636\ \text{ns})} \\ &= 2.07\ \text{MHz} \end{aligned}$$

Thus, the value of high frequency, f_H is, $\boxed{2.07\ \text{MHz}}$.

Step 7 of 7

The gain-bandwidth product is,

$$f_t = A_M f_H$$

Substitute $-79\ \text{V/V}$ for A_M, and 2.07 MHz for f_H.

$$\begin{aligned} f_t &= (-79\ \text{V/V})(2.07\ \text{MHz}) \\ &= -164\ \text{MHz} \end{aligned}$$

Therefore, the gain-bandwidth product is, $\boxed{-164\ \text{MHz}}$.

Step 1 of 5

Write the expression for transconductance of the MOSFET high frequency model.

$$g_m = \mu_n C_{ox} \frac{W}{L} |V_{OV}|$$

Modify the expression for the overdrive voltage.

$$|V_{OV}| = \frac{g_m}{\mu_n C_{ox} \frac{W}{L}}$$

Here, $\mu_n C_{ox}$ is the process transconductance parameter, and $\frac{W}{L}$ is the width to length ratio.

Substitute 1 mA/V for g_m, $200\ \mu A/V^2$ for $\mu_n C_{ox}$, and 50 for $\frac{W}{L}$ in the equation.

$$\begin{aligned} |V_{OV}| &= \frac{1\ \text{mA/V}}{(200\ \mu\text{A/V}^2)(50)} \\ &= \frac{1\ \text{mA/V}}{10\ \text{mA/V}^2} \\ &= 0.1\ \text{V} \end{aligned}$$

Thus, the overdrive voltage, V_{OV} is, $\boxed{0.1\ \text{V}}$.

Step 2 of 5

Write the expression for drain current.

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} V_{OV}^2$$

Substitute 0.1 V for V_{OV}, $200\ \mu A/V^2$ for $\mu_n C_{ox}$, and 50 for $\frac{W}{L}$ in the equation.

$$\begin{aligned} I_D &= \frac{1}{2}(200\ \mu\text{A/V}^2)(50)(0.1\ \text{V})^2 \\ &= \frac{1}{2}(200\ \mu\text{A/V}^2)(50)(0.01\ \text{V}^2) \\ &= 50\ \mu\text{A} \end{aligned}$$

Thus, the drain current, I_D is, $\boxed{50\ \mu\text{A}}$.

Step 3 of 5

Write the expression for 3-dB frequency.

$$f_H = \frac{1}{2\pi(C_L + C_{gd}) A_o R_L'}$$

Here, $R_L' = R_L \| R_o$.

The load resistance, $R_L = R_o$.

Write the expression for output resistance, R_o.

$$\begin{aligned} R_o &= g_m r_o \\ &= g_m \left(\frac{V_A}{I_D} \right) \end{aligned}$$

V_A is the early voltage.

Substitute $1\ \text{mA/V}$ for g_m, $50\ \mu\text{A}$ for I_D and $10\ \Omega$ for V_A in the equation.

$$\begin{aligned} R_o &= (1\ \text{mA/V})(200\ \text{k}\Omega) \\ &= 200\ \Omega \end{aligned}$$

Equivalent load resistance, R_L' is,

$$\begin{aligned} R_L' &= R_L \| R_o \\ &= 200\ \Omega \| 200\ \Omega \\ &= 100\ \Omega \end{aligned}$$

The dc gain of the cascade amplifier is $74\ \text{dB}$ or 5011.87 V/V.

Substitute $1\ \text{pF}$ for C_L, $0.1\ \text{pF}$ for C_{gd} and $100\ \Omega$ for R_L' in the equation of f_H.

$$\begin{aligned} f_H &= \frac{1}{2\pi(1\ \text{pF} + 0.1\ \text{pF})(100\ \Omega)(5011.87)} \\ &= 0.28\ \text{MHz} \end{aligned}$$

Thus, 3-dB frequency of the cascade amplifier, f_H is, $\boxed{0.28\ \text{MHz}}$.

Step 4 of 5

Write the expression for unity gain frequency, f_t.

$$f_t = |A_M| f_H$$

Write the expression for dc gain.

$$|A_M| = A_o g_m R_L'$$

Unity gain frequency f_t is,

$$f_t = A_o g_m R_L' f_H$$

The dc gain of the cascade amplifier is $74\ \text{dB}$ or 5011.87 V/V

Find the unity gain frequency by substituting corresponding values.

$$\begin{aligned} f_t &= (5011.87)(1 \times 10^{-3})(100\ \Omega)(0.28\ \text{MHz}) \\ &= (5011.87)(1 \times 10^{-3})(100)(0.28 \times 10^6) \\ &= 0.14\ \text{GHz} \end{aligned}$$

Thus, the unity gain frequency f_t is $\boxed{0.14\ \text{GHz}}$.

Step 5 of 5

Cascade transistor is removed and load resistance R_L remains unchanged.

Usually, Cascading increases the dc gain by factor A_0 while keeping the unity gain frequency constant.

$$A_o' = A_o g_m R_L'$$

Substitute $g_m r_o$ for A_o in the equation.

$$\begin{aligned} A_o' &= A_o g_m R_L' \\ &= (g_m r_o) g_m (R_o \| R_L) \quad (\text{since, } A_o = g_m r_o, \text{ and } R_L' = R_o \| R_L) \\ &= g_m^2 r_o (R_o \| R_L) \\ &= \frac{g_m^2 r_o R_o}{2} \qquad (\text{since, } R_o = R_L) \end{aligned}$$

Therefore, the dc voltage gain at desired configuration is, $\boxed{\frac{g_m^2 r_o R_o}{2}}$.

10.074P

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10.091P

Consider the differential amplifier biased with a current source.

The output resistance of a differential amplifier is $\mathbf{1\ M\Omega}$

The output capacitance of a differential amplifier is $\mathbf{1\ pF}$

The differential gain exhibits a dominant pole at $\mathbf{2\ MHz}$

The CMRR have poles at $f_z = \mathbf{2\ MHz}$ and at $f_{P1} = \dfrac{1}{2\pi r_o C_o}$

Calculate the value of the pole, f_{P1} .

$$f_{P1} = \frac{1}{2\pi r_o C_o}$$

$$= \frac{1}{2\pi\left(1\times10^{6}\right)\left(1\times10^{-12}\right)}$$

$$= 159.15\ \mathbf{kHz}$$

Therefore, the poles of the CMRR (common mode rejection ratio) are, $\boxed{\mathbf{2\ MHz\ and\ 159.15\ kHz}}$ respectively.

Step 1 of 4

Consider a current mirror loaded MOS differential amplifier.

The differentia amplifier is biased with $I = 0.2\ \text{mA}$ and all resistors are identical. The early voltage, $V_{An} = |V_{Ap}| = 10\ \text{V}$.

Calculate the value of r_O.

$$\begin{aligned} r_O &= \frac{V_A}{I_D} \\ &= \frac{10\ \text{V}}{\left(\frac{0.2\ \text{mA}}{2}\right)} \\ &= 100\ \text{k}\Omega \end{aligned}$$

The two NMOS transistors of the differential pair operating at the over drive voltage $V_{OV} = 0.2\ \text{V}$. The two PMOS transistors of the differential pair operating at the over drive voltage $V_{OV} = 0.2\ \text{V}$.

All transconductances are identical. Calculate the value of g_m.

$$\begin{aligned} g_m &= \frac{2I_D}{V_{OVN}} \\ &= \frac{2\left(\frac{0.2\ \text{mA}}{2}\right)}{0.2\ \text{V}} \\ &= 1\ \text{mA/V} \end{aligned}$$

Step 2 of 4

Write the formula for the low frequency differential gain.

$$A_d = g_{m12}\left(r_{O2} \| r_{O4}\right)$$

Substitute the corresponding value.

$$\begin{aligned} A_d &= (1\ \text{mA/V})(100\ \text{k}\Omega \| 100\ \text{k}\Omega) \\ &= (1\ \text{mA/V})\left(\frac{(100\ \text{k}\Omega)(100\ \text{k}\Omega)}{100\ \text{k}\Omega + 100\ \text{k}\Omega}\right) \\ &= (1\ \text{mA/V})(50\ \text{k}\Omega) \\ &= 50\ \text{V/V} \end{aligned}$$

Therefore, the DC value of the voltage gain is, $\boxed{50\ \text{V/V}}$.

Step 3 of 4

Write the expression for the pole frequency.

$$f_{p1} = \frac{1}{2\pi C_L R_O}$$

Here,

The output resistance is, $R_O = r_{D2} \| r_{O4}$.

Calculate the value of R_O.

$$\begin{aligned} R_O &= (100\ \text{k}\Omega \| 100\ \text{k}\Omega) \\ &= \left(\frac{(100\ \text{k}\Omega)(100\ \text{k}\Omega)}{100\ \text{k}\Omega + 100\ \text{k}\Omega}\right) \\ &= 50\ \text{k}\Omega \end{aligned}$$

The total capacitance at the output node of the mirror is $0.2\ \text{pF}$.

Substitute the corresponding values.

$$\begin{aligned} f_{p1} &= \frac{1}{2\pi\left(0.2\times10^{-12}\right)\left(50\times10^{3}\right)} \\ &= 15.9\ \text{MHz} \end{aligned}$$

Therefore, the frequency of the first pole is, $\boxed{15.9\ \text{MHz}}$.

Step 4 of 4

The total capacitance at the input node of the mirror is $0.1\ \text{pF}$.

Write the expression for the pole frequency.

$$f_{p2} = \frac{g_{m3}}{2\pi C_m}$$

Substitute the corresponding values.

$$\begin{aligned} f_{p2} &= \frac{1\times10^{-3}}{2\pi\left(0.1\times10^{-12}\right)} \\ &= 1.6\ \text{GHz} \end{aligned}$$

Therefore, the frequency of the second pole is, $\boxed{1.6\ \text{GHz}}$.

Write the expression for zero frequency.

$$f_L = 2f_{p2}$$

Substitute the corresponding values.

$$\begin{aligned} f_L &= 2\left(1.6\times10^{9}\right) \\ &= 3.2\ \text{GHz} \end{aligned}$$

Therefore, the frequency of the pole is, $\boxed{3.2\ \text{GHz}}$.

10.093P

We don't have the solution to this problem yet.

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Step 1 of 20

10.100

(a)

Refer to the common source amplifier circuit in Figure P9.102 (a) in the textbook.

Find dc small signal equivalent circuit of the common source amplifier. Refer to Figure 9.13 in the textbook for the equivalent circuit of high-frequency response of the common source amplifier

Neglect all the capacitances and resistances. R_G, R_S, and R_L to obtain the small-signal equivalent circuit of the common source amplifier.

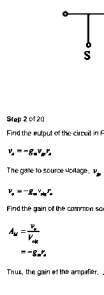


Figure 1

Step 2 of 20

Find the output of the circuit in Figure 1.

$$v_o = -g_m v_{gs} r_o$$

The gate to source voltage, v_{gs} is nothing but the input voltage, V_{sig}. Therefore,

$$v_o = -g_m v_{sig} r_o$$

Find the gain of the common source amplifier.

$$\begin{aligned} A_{sg} &= \frac{v_o}{V_{sig}} \\ &= -g_m r_o \end{aligned}$$

Thus, the gain of the amplifier, A_{sg} is, $\boxed{-g_m r_o}$

Step 3 of 20

Refer to Figure 9.19 in the textbook for the generalized high-frequency equivalent circuit of the common source amplifier.

The voltage source in the circuit is V_{sig} and the input resistance is R_{sig}. Also the load resistance in the circuit is r_o only.

Apply the method of open circuit time constants on the high-frequency equivalent circuit of common source amplifier.

$$\begin{aligned} \tau_H &= \sum_i C_i R_i \\ &= C_{gs} R_{sig} + C_{gd} R_{gd} + C_L R_{C_L} \end{aligned}$$

Neglect the capacitances, C_{gd} and C_L to get the resistance, R_{gs} equal to R_{sig}.

Neglect the capacitances, C_{gs} and C_L to get the resistance, R_{gd} equal to r_o.

Neglect the capacitances, C_{gs} and C_L to get the resistance, R_{C_L} equal to r_o.

Step 4 of 20

Refer to Figure 9.21 in the textbook for the application of the open-circuit time constants method to the common source amplifier

Find the resistance, r_{gs}.

$$\begin{aligned} r_{gs} &= R_{sig} \\ &= \frac{V_{sig}}{I_x} \end{aligned}$$

Find the voltage at node G.

$$V_{sig} = -I_x R_{sig}$$

Find the current at node D.

$$I_x = g_m V_{gs} + \frac{V_x + V_{sig}}{r_o}$$

Substitute the expression for V_{gs}.

$$\begin{aligned} I_x &= g_m\left(-I_x R_{sig}\right) + \frac{V_x - I_x R_{sig}}{r_o} \\ I_x + I_x R_{sig} g_m &= \frac{V_x - I_x R_{sig}}{r_o} \\ r_o I_x \left(1 + g_m R_{sig}\right) &= V_x - I_x R_{sig} \\ r_o I_x \left(1 + g_m R_{sig}\right) + I_x R_{sig} &= V_x \end{aligned}$$

Step 5 of 20

Simplify the expression further.

$$\begin{aligned} I_x\left[r_o\left(1 + g_m R_{sig}\right) + R_{sig}\right] &= V_x \\ I_x\left[r_o + r_o g_m R_{sig} + R_{sig}\right] &= V_x \\ I_x\left[r_o + R_{sig}\left(1 + g_m r_o\right)\right] &= V_x \end{aligned}$$

Find the resistance, r_{gs}.

$$\frac{V_x}{I_x} = r_o + R_{sig}\left(1 + g_m r_o\right)$$

Step 6 of 20

Substitute the resistances values in the expression for τ_H.

$$\begin{aligned} \tau_H &= C_{gs} R_{sig} + C_{gd} R_{gd} + C_L R_{C_L} \\ &= C_{gs} R_{sig} + C_{gd}\left[r_o + R_{sig}\left(1 + g_m r_o\right)\right] + C_L r_o \end{aligned}$$

Thus, the expression for the effective time constant, τ_H is,

$$\boxed{C_{gs} R_{sig} + C_{gd}\left[r_o + R_{sig}\left(1 + g_m r_o\right)\right] + C_L r_o}$$

Step 7 of 20

Substitute 1 mA/V for g_m, and $20\ \text{k}\Omega$ for r_o in the expression for A_M.

$$\begin{aligned} A_M &= -\left(1\ \text{mA/V}\right)\left(20\ \text{k}\Omega\right) \\ &= -\left(1\times10^{-3}\right)\left(20\times10^{3}\right) \\ &= -20 \end{aligned}$$

Thus, the value of gain, A_M is, $\boxed{-20}$

Substitute 1 mA/V for g_m, $20\ \text{k}\Omega$ for r_o, $28\ \text{k}\Omega$ for R_{sig}, 20 fF for C_{gs}, 5 fF for C_{gd}, and 10 fF for C_L in the expression for τ_H.

$$\begin{aligned} \tau_H &= \left\{\begin{array}{l} \left(28\ (\text{F})\right)\left(20\ \text{k}\Omega\right) + \left(5\ (\text{F})\right)\left[28\ \text{k}\Omega + \left(20\ \text{k}\Omega\right)\left(1 + \left(1\ \text{mA/V}\right)\left(20\ \text{k}\Omega\right)\right)\right] \\ + \left(18\ (\text{F})\right)\left(20\ \text{k}\Omega\right) \end{array}\right\} \\ &= \left\{\begin{array}{l} \left(28\times10^{-15}\right)\left(20\times10^{3}\right) + \left(5\times10^{-15}\right)\left[\left(28\times10^{3}\right) + \left(20\times10^{3}\right)\left(1+20\right)\right] \\ + \left(18\times10^{-15}\right)\left(20\times10^{3}\right) \end{array}\right\} \\ &= \left(400\times10^{-12}\right) + \left(5\times10^{-15}\right)\left[\left(28\times10^{3}\right) + \left(428\times10^{3}\right)\right] + \left(200\times10^{-12}\right) \\ &= \left(400\times10^{-12}\right) + \left(2.2\times10^{-9}\right) + \left(200\times10^{-12}\right) \end{aligned}$$

$$\begin{aligned} \tau_H &= 2.8\times10^{-9} \\ &= 2.8\ \text{ns} \end{aligned}$$

Step 8 of 20

Find the upper 3-dB frequency.

$$f_H = \frac{1}{2\pi\tau_H}$$

Substitute 2.8 ns for τ_H.

$$\begin{aligned} f_H &= \frac{1}{2\pi\left(2.8\ \text{ns}\right)} \\ &= \frac{1}{17.6\times10^{-9}} \\ &= 56.84\times10^{6} \end{aligned}$$

$$f_H = 56.84\ \text{MHz}$$

Thus, the upper 3-dB frequency, f_H is, $\boxed{56.84\ \text{MHz}}$

Step 9 of 20

Find the gain-bandwidth product.

$$\text{GBP} = \left|A_M\right| f_H$$

Substitute -20 for A_M and 56.84 MHz for f_H.

$$\begin{aligned} \text{GBP} &= \left|A_M\right| f_H \\ &= \left|-20\right|\left(56.84\ \text{MHz}\right) \\ &= \left(20\right)\left(56.84\times10^{6}\right) \\ &= 1.136\times10^{9} \end{aligned}$$

$$\text{GBP} = 1.136\ \text{GHz}$$

Thus, the gain-bandwidth product is, $\boxed{1.136\ \text{GHz}}$

Step 10 of 20

(b)

Refer to the common drain-common source amplifier circuit in Figure P9.102 (b) in the textbook

It is a source follower circuit placed between the signal source and the input of the common source amplifier.

Consider that the fraction is the expression for the gain of the common-drain amplifier.

$$A_{df} = \frac{R_L'}{R_L' + \frac{1}{g_m}}$$

Here, the load resistance is,

$$R_L' = R_1 \| r_{o1}$$

There is no load resistance in the circuit. Neglect the capacitances, C_{gs1} and C_{L1} to get the resistance, R_L' equal to r_{o1}. Therefore,

$$A_{df1} = \frac{r_{o1}}{r_{o1} + \frac{1}{g_{m1}}}$$

Find the output of the common-drain amplifier.

$$\begin{aligned} A_{df1} &= \frac{v_{o1}}{v_{sig1}} \\ &= \frac{r_{o1}}{r_{o1} + \frac{1}{g_{m1}}} \end{aligned}$$

Step 11 of 20

The gate to source voltage of transistor, Q_1, v_{gs1} is nothing but the input voltage, V_{sig}. Therefore,

$$v_{o1} = \frac{r_{o1}}{r_{o1} + \frac{1}{g_{m1}}} v_{sig}$$

Consider the small signal equivalent of the common source amplifier from Figure 1.

Find the output of the circuit in Figure 1.

$$v_{o2} = -g_{m2} v_{gs2} r_{o2}$$

The gate to source voltage of transistor Q_2, v_{gs2} is nothing but the output voltage of transistor, Q_1. Therefore,

$$v_{o1} = -g_{m2} v_{o1} r_{o2}$$

Substitute the expression for the output voltage of transistor, Q_1

$$\begin{aligned} v_{o2} &= -g_{m2} r_{o2}\left(\frac{r_{o1}}{r_{o1} + \frac{1}{g_{m1}}} v_{sig}\right) \\ &= -\left(\frac{r_{o1}}{r_{o1} + \frac{1}{g_{m1}}}\right)\left(g_{m2} r_{o2}\right) v_{sig} \end{aligned}$$

Find the gain of the common drain-common source amplifier.

$$\begin{aligned} A_{df} &= \frac{v_{o2}}{V_{sig}} \\ &= -\left(\frac{r_{o1}}{r_{o1} + \frac{1}{g_{m1}}}\right)\left(g_{m2} r_{o2}\right) \end{aligned}$$

Thus, the gain of the amplifier, A_{df} is, $\boxed{-\left(\frac{r_{o1}}{r_{o1} + \frac{1}{g_{m1}}}\right)\left(g_{m2} r_{o2}\right)}$

Step 12 of 20

Combine the both circuits to obtain the overall high-frequency equivalent circuit of the common drain-common source amplifier. Refer to Figure 9.32 in the textbook for the generalized high-frequency equivalent circuit of the common drain amplifier. Refer to Figure 9.19 in the textbook for the generalized high-frequency equivalent circuit of the common source amplifier.

Here, the load resistance, R_L' is r_o and neglect all capacitances.

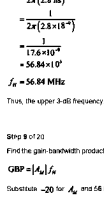


Figure 2

Step 13 of 20

Apply the method of open circuit time constants on the high-frequency equivalent circuit of common drain-common source amplifier

$$\begin{aligned} \tau_H &= \sum_i C_i R_i \\ &= C_{gs1} R_{gs1} + C_{gs2} R_{gs2} + C_{gd1} R_{gd1} + C_{gd2} R_{gd2} + C_{L1} R_{C_{L1}} + C_{L2} R_{C_{L2}} \end{aligned}$$

Neglect the capacitances, C_{gs1} and C_{L2} to get the resistance, R_{gs1} equal to R_{sig}.

Neglect the capacitances, C_{gs2} and C_{L2} to get the resistance, R_{gd2} equal to r_o.

Neglect the capacitances, C_{gs2} and C_{L2} to get the resistance, $R_{C_{L2}}$ equal to r_o.

Find the resistance, r_{gs1} seen by C_{gs1} by straightforward analysis

$$r_{gs1} = \frac{R_{sig} + r_{o1}}{1 + g_{m1} r_{o1}} \left(\text{neglecting load resistance}\right)$$

Find the resistance, r_{gs2}, it is nothing but the output resistance of the common-drain amplifier.

$$r_{gs2} = \frac{1}{g_{m2}} \| r_{o1}$$

Step 14 of 20

Find the resistance, r_{gd1} seen by C_{gd1}.

$$r_{gd1} = \frac{V_x}{I_x}$$

Find the voltage at node G

$$V_x = -I_x R_{sig}$$

Substitute r_{gs1} for R_{sig}

$$V_x = -I_x \left(\frac{1}{g_{m1}} \| r_{o1}\right)$$

Find the current at node D

$$I_x = g_{m1} V_{gs1} + \frac{V_x + V_{gs1}}{r_{o1}}$$

Step 15 of 20

Substitute the expression for V_x

$$\begin{aligned} I_x &= g_{m1}\left[-I_x\left(\frac{1}{g_{m1}} \| r_{o1}\right)\right] + \frac{V_x - I_x R_{sig}}{r_{o1}} \\ I_x + I_x\left(\frac{1}{g_{m1}} \| r_{o1}\right) g_{m1} &= \frac{V_x - I_x\left(\frac{1}{g_{m1}} \| r_{o1}\right)}{r_{o1}} \\ r_{o1} I_x\left[1 + g_{m1}\left(\frac{1}{g_{m1}} \| r_{o1}\right)\right] &= V_x - I_x\left(\frac{1}{g_{m1}} \| r_{o1}\right) \\ r_{o1} I_x\left[1 + g_{m1}\left(\frac{1}{g_{m1}} \| r_{o1}\right)\right] + I_x\left(\frac{1}{g_{m1}} \| r_{o1}\right) &= V_x \end{aligned}$$

Simplify the expression further

$$\begin{aligned} I_x\left[r_{o1}\left(1 + g_{m1}\left(\frac{1}{g_{m1}} \| r_{o1}\right)\right) + \left(\frac{1}{g_{m1}} \| r_{o1}\right)\right] &= V_x \\ I_x\left[r_{o1} + r_{o1} g_{m1}\left(\frac{1}{g_{m1}} \| r_{o1}\right) + \left(\frac{1}{g_{m1}} \| r_{o1}\right)\right] &= V_x \\ I_x\left[r_{o1} + \left(\frac{1}{g_{m1}} \| r_{o1}\right)\left(1 + g_{m1} r_{o1}\right)\right] &= V_x \end{aligned}$$

Find the resistance, r_{gd1}.

$$\frac{V_x}{I_x} = r_{o1} + \left(\frac{1}{g_{m1}} \| r_{o1}\right)\left(1 + g_{m1} r_{o1}\right)$$

Step 15 of 20

Substitute the resistances values in the expression for τ_H.

$$\tau_H = \left\{\begin{array}{l} C_{gs1} R_{sig} + C_{gs1}\left[\frac{R_{sig} + r_{o1}}{1 + g_{m1} r_{o1}} + C_{gd1}\left[\left(\frac{1}{g_{m1}} \| r_{o1}\right)\left(1 + g_{m1} r_{o2}\right) + r_{o2}\right]\right] \\ + C_{gd2}\left(\frac{1}{g_{m1}} \| r_{o1}\right) + C_L r_{o2} \end{array}\right\}$$

Thus, the expression for the effective time constant, τ_H is,

$$\boxed{\left\{\begin{array}{l} C_{gs1} R_{sig} + C_{gs1}\left[\frac{R_{sig} + r_{o1}}{1 + g_{m1} r_{o1}} + C_{gd1}\left[\left(\frac{1}{g_{m1}} \| r_{o1}\right)\left(1 + g_{m1} r_{o2}\right) + r_{o2}\right]\right] \\ + C_{gd2}\left(\frac{1}{g_{m1}} \| r_{o1}\right) + C_L r_{o2} \end{array}\right\}}$$

Step 17 of 20

Substitute 1 mA/V for g_{m1}, $20\ \text{k}\Omega$ for r_{o1}, 1 mA/V for g_{m2}, and $20\ \text{k}\Omega$ for r_{o2} in the expression for A_M

$$\begin{aligned} A_M &= -\left(\frac{r_{o1}}{r_{o1} + \frac{1}{g_{m1}}}\right)\left(g_{m2} r_{o2}\right) \\ &= -\left(\frac{20\ \text{k}\Omega}{28\ \text{k}\Omega + \frac{1}{1\ \text{mA/V}}}\right)\left(1\ \text{mA/V}\right)\left(20\ \text{k}\Omega\right) \\ &= -\left(\frac{28\times10^{3}}{28\times10^{3} + \frac{1}{1\times10^{-3}}}\right)\left(1\times10^{-3}\right)\left(20\times10^{3}\right) \\ &= -\left(8.952\right)\left(1\times18^{-3}\right)\left(20\times10^{3}\right) \end{aligned}$$

$$A_M \approx -19$$

Thus, the value of gain A_M is, $\boxed{-19}$

Step 18 of 20

Substitute 1 mA/V for g_{m1}, $20\ \text{k}\Omega$ for r_{o1}, 1 mA/V for g_{m2}, and $28\ \text{k}\Omega$ for r_{o2}, $20\ \text{k}\Omega$ for R_{sig}, 20 fF for C_{gs1}, 5 fF for C_{gd1}, 20 fF for C_{gd2}, 5 fF for C_{gd2}, and 10 fF for C_L in the expression for τ_H.

$$\begin{aligned} \tau_H &= \left\{\begin{array}{l} C_{gs1} R_{sig} + C_{gs1}\left[\frac{R_{sig} + r_{o1}}{1 + g_{m1} r_{o1}} + C_{gd1}\left[\left(\frac{1}{g_{m1}} \| r_{o1}\right)\left(1 + g_{m1} r_{o2}\right) + r_{o2}\right]\right] \\ + C_{gd2}\left(\frac{1}{g_{m1}} \| r_{o1}\right) + C_L r_{o2} \end{array}\right\} \\ &= \left\{\begin{array}{l} \left(5\times18^{-15}\right)\left(20\times10^{3}\right) + \left(20\times18^{-15}\right)\left[\frac{\left(28\times10^{3}\right) + \left(20\times10^{3}\right)}{1+28}\right] \\ + \left(5\times10^{-15}\right)\left[\left(10^{3}\ 28\times10^{3}\right)\left(1+28\right) + \left(28\times18^{3}\right)\right] \\ + \left(20\times10^{-15}\right)\left(10^{3}\ 28\times10^{3}\right) + \left(18\times10^{-15}\right)\left(20\times10^{3}\right) \end{array}\right\} \\ &= \left\{\begin{array}{l} \left(100\times10^{-12}\right) + \left(28\times10^{-15}\right)\left(1.9\times10^{3}\right) + \left(5\times10^{-15}\right)\left(40\times10^{3}\right) \\ + \left(20\times10^{-15}\right)\left(952.38\right) + \left(200\times10^{-12}\right) \end{array}\right\} \\ &= \left(100\times10^{-12}\right) + \left(38\times10^{-12}\right) + \left(200\times18^{-12}\right) + \left(19\times10^{-12}\right) + \left(200\times10^{-12}\right) \end{aligned}$$

$$\begin{aligned} \tau_H &= 557\times10^{-12} \\ &= 557\ \text{ps} \end{aligned}$$

Step 19 of 20

Find the upper 3-dB frequency.

$$f_H = \frac{1}{2\pi\tau_H}$$

Substitute 557 ps for τ_H

$$\begin{aligned} f_H &= \frac{1}{2\pi\left(557\ \text{ps}\right)} \\ &= \frac{1}{2\pi\left(557\times10^{-12}\right)} \\ &= \frac{1}{3.49\times10^{-9}} \\ &= 285.7\times10^{6} \end{aligned}$$

$$f_H = 285.7\ \text{MHz}$$

Thus, the upper 3-dB frequency, f_H is, $\boxed{285.7\ \text{MHz}}$

Step 20 of 20

Find the gain-bandwidth product.

$$\text{GBP} = \left|A_M\right| f_H$$

Substitute -19 for A_M and 285.7 MHz for f_H.

$$\begin{aligned} \text{GBP} &= \left|A_M\right| f_H \\ &= \left|-19\right|\left(285.7\ \text{MHz}\right) \\ &= \left(19\right)\left(285.7\times10^{6}\right) \\ &= 5.43\times10^{9} \end{aligned}$$

$$\text{GBP} = 5.43\ \text{GHz}$$

Thus, the gain-bandwidth product is, $\boxed{5.43\ \text{GHz}}$

Hence, it is observe that the band extension is obtained for the circuit in Figure P9.102 (b) than the circuit in Figure P9.102 (a).

10.101P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

10.102P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

10.103P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

Step 1 of 6

10.104P

Refer to Figure 9.41 (c) in the textbook for the common drain-common gate amplifier circuit diagram.

Draw the small signal equivalent circuit.

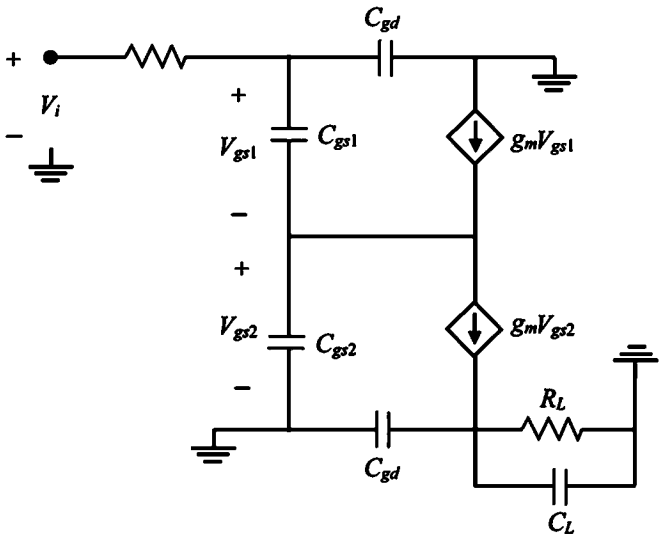


Figure 1

Step 2 of 6

From the circuit in Figure 1, the input voltage is the sum of the gate to source voltage of the MOSFET 1 and gate to source voltage of the MOSFET 2. That is,

$$\begin{aligned} V_i &= V_{gs1} + V_{gs2} \quad \{\text{since } V_{gs1} = V_{gs2} = V_{gs}\} \\ &= 2V_{gs} \end{aligned}$$

The current flowing through the load is, g_mV_{gs2} . Therefore, the output voltage is,

$$\begin{aligned} V_o &= g_mV_{gs2}R_L \\ &= g_mV_{gs}R_L \end{aligned}$$

Determine the ratio of output voltage to input voltage to get the voltage gain.

$$\begin{aligned} \frac{V_o}{V_i} &= \frac{g_mV_{gs}R_L}{2V_{gs}} \\ &= \frac{g_mR_L}{2} \end{aligned}$$

Thus, the mid-band voltage gain is, $A_M = \frac{g_mR_L}{2}$

Step 3 of 6

Substitute $5\ \text{mA/V}$ for g_m and $20\ \text{k}\Omega$ for R_L to calculate the voltage gain

$$\begin{aligned} A_M &= \frac{\left(5\times10^{-3}\right)\left(20\times10^{3}\right)}{2} \\ &= \frac{100}{2} \\ &= 50\ \text{V/V} \end{aligned}$$

Thus, the mid-band gain is, $\boxed{50\ \text{V/V}}$.

Step 4 of 6

The pole at the input, with a frequency f_{P1} by neglecting r_o is,

$$f_{P1} = \frac{1}{2\pi\left(\frac{C_{gs}}{2} + C_{gd}\right)R_{sig}}$$

Here,

The gate to source capacitance, C_{gs}

The gate to drain capacitance, C_{gd}

Substitute 2 pF for C_{gs} and 0.1 pF for C_{gd} .

$$\begin{aligned} f_{P1} &= \frac{1}{2\pi\left(\frac{2\times10^{-12}}{2} + 0.1\times10^{-12}\right)\left(20\times10^{3}\right)} \\ &= \frac{1}{2\pi\left(1.1\times10^{-12}\right)\left(20\times10^{3}\right)} \\ &= 7.23\ \text{MHz} \end{aligned}$$

Step 5 of 6

The pole at the input, with a frequency f_{P2} is,

$$f_{P2} = \frac{1}{2\pi\left(C_{gd} + C_L\right)R_L}$$

Here,

The gate to drain capacitance, C_{gd}

The load capacitance, C_L

The load resistance, R_L

Step 6 of 6

Substitute 0.1 pF for C_{gd} , 1 pF for C_L , and $20\ \text{k}\Omega$ for R_L .

$$\begin{aligned} f_{P2} &= \frac{1}{2\pi\left(0.1\times10^{-12} + 1\times10^{-12}\right)\left(20\times10^{3}\right)} \\ &= \frac{1}{2\pi\left(1.1\times10^{-12}\right)\left(20\times10^{3}\right)} \\ &= 7.23\ \text{MHz} \end{aligned}$$

Write the expression for the frequency at which the gain downs by 3 dB.

$$f_H = \frac{1}{\sqrt{\left(\frac{1}{f_{P1}}\right)^2 + \left(\frac{1}{f_{P2}}\right)^2}}$$

Substitute 7.23 MHz for f_{P1} and 7.23 MHz for f_{P2} .

$$\begin{aligned} f_H &= \frac{1}{\sqrt{\left(\frac{1}{7.23\times10^{6}}\right)^2 + \left(\frac{1}{7.23\times10^{6}}\right)^2}} \\ &= \frac{1}{\sqrt{2\left(\frac{1}{7.23\times10^{6}}\right)^2}} \\ &= \frac{1}{\sqrt{2\left(1.913\times10^{-14}\right)}} \\ &= 5.11\ \text{MHz} \end{aligned}$$

Thus, the 3 dB frequency is, $\boxed{5.11\ \text{MHz}}$.

10.105P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

10.106P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

10.107P

We don't have the solution to this problem yet.

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Step 1 of 1.23

Refer to Figure P3.108 (a) in the textbook to the circuit.

The transistors $\mathbf{T_1}$ **and** $\mathbf{T_2}$ are identical complementary MOSFET devices. $\mathbf{r_o}$ **= ∞ and** $\mathbf{r_\pi}$ **=** ∞ . From the circuit in Figure P3.108 (a) it is clear that the transistor acts as common emitter amplifier. The quiescent collector current of the amplifier is $\mathbf{0.5\ mA}$.

Calculate the value of trans-conductance $\mathbf{g_m}$.

$$\begin{aligned} g_m &= \frac{I_C}{V_T} \\ &= \frac{0.5\times10^{-3}}{25\times10^{-3}} \\ &= \mathbf{20\ mA/V} \end{aligned}$$

Calculate the value of r_π.

$$\begin{aligned} r_\pi &= \frac{\beta}{g_m} \\ &= \frac{100}{20\times10^{-3}} \\ &= \mathbf{5k\Omega} \end{aligned}$$

Calculate the value of r_o.

$$\begin{aligned} r_o &= \frac{V_A}{I_C} \\ &= \frac{5000}{101} \\ &= \mathbf{49.5\ \Omega} \end{aligned}$$

Consider the following expression to determine the value of C_π.

$$\begin{aligned} C_\pi &= C_T - \frac{g_m}{\omega_T} \\ &= \frac{g_m}{2\pi f_T} \end{aligned}$$

Substitute 2 pF for C_T, 20 mA/V for g_m and 400 MHz for f_T.

$$\begin{aligned} C_\pi &= 2\times10^{-12} - \frac{20\times10^{-3}}{2\pi(400\times10^6)} \\ C_\pi &= 2\times10^{-12} - 7.9577\times10^{-12} \\ C_\pi &= 2\times10^{-12} - 8\times10^{-12} \\ C_\pi &= \mathbf{-6\ pF} \end{aligned}$$

Write the expression for the mid band gain.

$$A_M = \frac{v_o}{v_s} = \frac{r_\pi}{r_\pi + R_s}\left(-g_m R_C\right)$$

Substitute $\mathbf{10\ k\Omega}$ for $\mathbf{R_s}$, $\mathbf{10\ k\Omega}$ for $\mathbf{R_C}$, 20 mA/V for $\mathbf{g_m}$, and $\mathbf{5\ k\Omega}$ for $\mathbf{r_\pi}$.

$$\begin{aligned} A_M &= \frac{5\times10^3}{(5\times10^3)+(10\times10^3)}\left[-(20\times10^{-3})(10\times10^3)\right] \\ &= \frac{(10\times10^3)(5\times10^3)}{(10\times10^3)+(5\times10^3)}\left[-(20\times10^{-3})(10\times10^3)\right] \\ &= \frac{-200}{3} \\ &= \mathbf{-66.67\ V/V} \end{aligned}$$

Therefore, the mid band gain is, $\boxed{\mathbf{-66.67\ V/V}}$.

Step 4 of 23

Write the expression for the 3-dB frequency.

$$f_L = \frac{1}{2\pi\left[R_{eq}\|r_\pi\right]\left(C_\pi + \left[1+g_m R_C\right]C_\mu\right)}$$

Substitute $\mathbf{10\ k\Omega}$ for $\mathbf{R_{eq}}$, $\mathbf{5\ k\Omega}$ for $\mathbf{r_\pi}$, 6 pF for C_π, and 2 fF for C_μ.

$$\begin{aligned} f_L &= \frac{1}{2\pi\left[(10\times10^3)\|(5\times10^3)\right]\left(6\times10^{-12}+\left[1+(20\times10^{-3})(10\times10^3)\right](2\times10^{-15})\right]} \\ &= \frac{1}{2\pi\left[\left(\frac{(10\times10^3)(5\times10^3)}{(10\times10^3)+(5\times10^3)}\right)\left(6\times10^{-12}+404\times10^{-15}\right)\right]} \\ &= \mathbf{117\ kHz} \end{aligned}$$

Therefore, the 3 dB frequency is, $\boxed{\mathbf{117\ kHz}}$.

Step 5 of 23

(b)

Refer to Figure P3.108 (b) in the textbook to the circuit in the circuit. It is clear that the transistor acts as common emitter amplifier. The quiescent collector current of the amplifier is $\mathbf{0.5\ mA}$.

Substitute the corresponding values.

$$\begin{aligned} A_M &= \frac{-g_m R_C}{R_{eq}+r_\pi} \\ &= \frac{-(100)(0.99)(10\times10^3)}{(10\times10^3)+(5\times10^3)} \\ &= \mathbf{-66\ V/V} \end{aligned}$$

Therefore, the mid-band gain is, $\boxed{\mathbf{-66\ V/V}}$.

Step 6 of 23

The 3-dB frequency f_L is,

$$f_L = \frac{1}{2\pi\tau}$$

Here,

The time constant, $\tau = C_\pi R_{in} + C_\mu R_1 + C_\mu R_2 + C_\mu R_3$

Calculate the value of R_{in}.

$$\begin{aligned} R_{in} &= \left(R_{eq}\|r_\pi\right) \\ &= \frac{(10\times10^3)(5\times10^3)}{(10\times10^3)+(5\times10^3)} \\ &= \mathbf{3.33\ k\Omega} \end{aligned}$$

Calculate the value of R_1.

$$\begin{aligned} R_1 &= r_o\|\left(\frac{R_{eq}+r_\pi}{1+\beta}\right) \\ &= \left\{(5\times10^3)\|\frac{(10\times10^3)+(5\times10^3)}{1+(20\times10^{-3})(5\times10^3)}\right\} \\ &= \mathbf{144\ \Omega} \end{aligned}$$

Calculate the value of R_2.

$$\begin{aligned} R_2 &= r_o\|\left(\frac{R_{eq}+r_\pi}{1+\beta}\right) \\ &= \mathbf{144\ \Omega} \end{aligned}$$

Calculate the value of R_3.

$$\begin{aligned} R_3 &= r_o\|\left(\frac{R_{eq}+r_\pi}{1+\beta}\right) \\ &= \mathbf{144\ \Omega} \end{aligned}$$

Calculate the value of C_π.

$$\begin{aligned} C_\pi &= C_T + C_T\left(1+g_m R_C R_1^{-1}\right) \\ &= 6\times10^{-12} + (2\times10^{-15})(1+20\times10^{-3}\times10^3) \\ &= 6\times10^{-12} + 402\times10^{-12} \\ &= \mathbf{488\ pF} \end{aligned}$$

And, $R_{eq} = R_s \cdot R_B = \mathbf{10\ k\Omega}$.

Step 18 of 23

Transcribe the Bode line constant

$$\begin{aligned} \tau_o &= \left[C_\mu R_C + C_\mu R_1 + C_T R_1 + C_\mu R_2\right] \\ &= \left[(2\times10^{-15}\times9.81\times10^3)+(6\times10^{-12}\times144)+(402\times18\times144)\right] \end{aligned}$$

Step 19 of 23

Calculate the Bode corner frequency.

$$\begin{aligned} f_{o1} &= \frac{1}{2\pi C_{o1} R_o} \\ &= \frac{1}{2\pi(2\times10^{-12})(10\times10^3)} \\ &= \mathbf{7.96\ MHz} \end{aligned}$$

Since, f_{o1} is very high. The term $\left(\frac{1}{f_{o1}}\right)^2$ can be neglected.

Substitute $\mathbf{10\ k\Omega}$ for $\mathbf{R_{eq}}$, $\mathbf{10\ k\Omega}$ for $\mathbf{R_C}$, 20 mA/V for $\mathbf{g_m}$, and $\mathbf{5\ k\Omega}$ for $\mathbf{r_\pi}$.

$$\begin{aligned} A_M &= \frac{5\times10^3}{(10\times10^3+5\times10^3)}\left[-(20\times10^{-3})(10\times10^3)\right] \\ &= \frac{-200}{3} \\ &= \mathbf{-66.67\ V/V} \end{aligned}$$

Therefore, the mid band gain is, $\boxed{\mathbf{-66.67\ V/V}}$.

Step 20 of 23

Refer to Figure P3.108 (b) in the textbook to CC-CE Cascode amplifier. From the circuit, it is clear that the transistor acts as CC-CE cascode amplifier. For a CC-CE Cascode amplifier, the mid-band gain, $\mathbf{A_M}$ is,

$$\begin{aligned} A_M &= \frac{-\beta R_C}{R_{eq}+r_\pi\left(\beta+1\right)} \\ &= \frac{-(101)(100\times10^3)}{10\times10^3+5\times10^3+(101)(5\times10^3)} \\ &= \mathbf{-194\ V/V} \end{aligned}$$

Therefore, the mid-band gain is, $\boxed{\mathbf{-194\ V/V}}$.

Step 21 of 23

The 3-dB frequency f_L is,

$$f_L = \frac{1}{2\pi\tau}$$

Here, f_{o1}, f_{o2} **and** f_{o3} are frequency poles.

Calculate the value of R_{in}.

$$\begin{aligned} R_{in} &= \left(R_{eq}\|r_\pi\right) \\ &= \frac{(10\times10^3)(5\times10^3)}{(10\times10^3)+(5\times10^3)} \\ &= \mathbf{3.33\ k\Omega} \end{aligned}$$

Calculate the value of R_1.

$$\begin{aligned} R_1 &= r_o\|\left(\frac{R_{eq}+r_\pi}{1+\beta}\right) \\ &= \left\{(5\times10^3)\|\frac{(10\times10^3)+(5\times10^3)}{1+(20\times10^{-3})(5\times10^3)}\right\} \\ &= \mathbf{144\ \Omega} \end{aligned}$$

Calculate the value of R_2.

$$\begin{aligned} R_2 &= r_o\|\left(\frac{R_{eq}+r_\pi}{1+\beta}\right) \\ &= \mathbf{144\ \Omega} \end{aligned}$$

Calculate the value of R_3.

$$\begin{aligned} R_3 &= r_o\|\left(\frac{R_{eq}+r_\pi}{1+\beta}\right) \\ &= \mathbf{144\ \Omega} \end{aligned}$$

Calculate the value of C_π.

$$\begin{aligned} C_\pi &= C_T + C_T\left(1+g_m R_C R_1^{-1}\right) \\ &= 6\times10^{-12} + (2\times10^{-15})(1+20\times10^{-3}\times10^3) \\ &= 6\times10^{-12} + 402\times10^{-12} \\ &= \mathbf{488\ pF} \end{aligned}$$

And, $R_{eq} = R_s \cdot R_B = \mathbf{10\ k\Omega}$.

Step 22 of 23

Transcribe the Bode line constant

$$\begin{aligned} \tau_o &= \left[C_\mu R_C + C_\mu R_1 + C_T R_1 + C_\mu R_2\right] \\ &= \left[(2\times10^{-15}\times9.81\times10^3)+(6\times10^{-12}\times144)+(402\times18\times144)\right] \end{aligned}$$

Step 23 of 23

Write the expression for the mid band gain.

$$A_M = \frac{v_o}{v_s} = \frac{r_\pi}{r_\pi + R_s}\left(-g_m R_C\right)$$

Substitute $\mathbf{10\ k\Omega}$ for $\mathbf{R_s}$, $\mathbf{10\ k\Omega}$ for $\mathbf{R_C}$, 20 mA/V for $\mathbf{g_m}$, and $\mathbf{5\ k\Omega}$ for $\mathbf{r_\pi}$.

$$\begin{aligned} A_M &= \frac{5\times10^3}{(10\times10^3+5\times10^3)}\left[-(20\times10^{-3})(10\times10^3)\right] \\ &= \frac{-200}{3} \\ &= \mathbf{-66.67\ V/V} \end{aligned}$$

Therefore, the mid-band gain is, $\boxed{\mathbf{-66.67\ V/V}}$.

Step 7 of 23

Calculate the output pole, f_{o1}.

$$\begin{aligned} f_{o1} &= \frac{1}{2\pi C_{o1} R_o} \\ &= \frac{1}{2\pi(2\times10^{-12})(10\times10^3)} \\ &= \mathbf{7.96\ MHz} \end{aligned}$$

Step 8 of 23

Consider the mid band gain

$$A_M = \frac{v_o}{v_s} = \frac{r_\pi}{r_\pi + R_s}\left(-g_m R_C\right)$$

Substitute $\mathbf{10\ k\Omega}$ for $\mathbf{R_s}$, $\mathbf{10\ k\Omega}$ for $\mathbf{R_C}$, 20 mA/V for $\mathbf{g_m}$, and $\mathbf{5\ k\Omega}$ for $\mathbf{r_\pi}$.

$$\begin{aligned} A_M &= \frac{5\times10^3}{(10\times10^3+5\times10^3)}\left[-(20\times10^{-3})(10\times10^3)\right] \\ &= \frac{-200}{3} \\ &= \mathbf{-66.67\ V/V} \end{aligned}$$

Therefore, the mid-band gain is, $\boxed{\mathbf{-66.67\ V/V}}$.

Step 9 of 23

Consider the 3-dB frequency.

$$f_L = \frac{1}{2\pi\left[\sqrt{\left(\frac{1}{f_{o1}}\right)^2+\left(\frac{1}{f_{o2}}\right)^2+\left(\frac{1}{f_{o3}}\right)^2}\right]}$$

Here, f_{o1}, f_{o2} **and** f_{o3} are frequency poles.

Calculate the input pole, f_{o1}.

$$\begin{aligned} f_{o1} &= \frac{1}{2\pi\left[R_{eq}\|r_\pi\right]\left(C_\pi + 2C_\mu\right)} \\ &= \frac{1}{2\pi\left[(10\times10^3)\|(5\times10^3)\right]\left(6\times10^{-12}+2\times10^{-15}\right)} \\ &= \mathbf{117\ kHz} \end{aligned}$$

Therefore, the 3 dB frequency is, $\boxed{\mathbf{117\ kHz}}$.

Step 10 of 23

(c)

Refer to Figure P3.108 (c) in the textbook to CC-CE Cascode amplifier. It is clear that the transistor acts as CC-CE Cascode compound-Darlington amplifier. For a paired Cascode amplifier, the mid-band gain, $\mathbf{A_M}$ is,

$$\begin{aligned} A_M &= \frac{\beta R_C}{R_{eq}+r_\pi} \\ &= \frac{-(100)(99)(10\times10^3)}{10\times10^3+5\times10^3} \\ &= \mathbf{-66\ V/V} \end{aligned}$$

Therefore, the mid-band gain is, $\boxed{\mathbf{-66\ V/V}}$.

Step 11 of 23

Consider the 3-dB frequency.

$$f_L = \frac{1}{2\pi\left[\sqrt{\left(\frac{1}{f_{o1}}\right)^2+\left(\frac{1}{f_{o2}}\right)^2+\left(\frac{1}{f_{o3}}\right)^2}\right]}$$

Here, f_{o1}, f_{o2} **and** f_{o3} are frequency poles.

Calculate the input pole, f_{o1}.

$$\begin{aligned} f_{o1} &= \frac{1}{2\pi\left[R_{eq}\|r_\pi\right]\left(C_\pi + 2C_\mu\right)} \\ &= \frac{1}{2\pi\left[(10\times10^3)\|(5\times10^3)\right]\left(6\times10^{-12}+2\times10^{-15}\right)} \\ &= \mathbf{117\ kHz} \end{aligned}$$

Therefore, the 3 dB frequency is, $\boxed{\mathbf{117\ kHz}}$.

Step 12 of 23

(d)

Refer to Figure P3.108 (d) in the textbook to the circuit. It is clear that the transistor acts as common emitter amplifier. For a paired Darlington amplifier, the mid-band gain, $\mathbf{A_M}$ is,

$$\begin{aligned} A_M &= \frac{\beta R_C}{R_{eq}+r_\pi\left(\beta+1\right)} \\ &= \frac{-(100)(100)(10\times10^3)}{10\times10^3+5\times10^3+(101)(5\times10^3)} \\ &= \mathbf{-194\ V/V} \end{aligned}$$

Therefore, the mid-band gain is, $\boxed{\mathbf{-194\ V/V}}$.

Step 13 of 23

The 3-dB frequency f_L is,

$$f_L = \frac{1}{2\pi\tau}$$

Here, f_{o1}, f_{o2} **and** f_{o3} are frequency poles.

Calculate the input pole, f_{o1}.

$$\begin{aligned} f_{o1} &= \frac{1}{2\pi\left[R_{eq}\|r_\pi\right]\left(C_\pi + 2C_\mu\right)} \\ &= \frac{1}{2\pi\left[(10\times10^3)\|(5\times10^3)\right]\left(6\times10^{-12}+2\times10^{-15}\right)} \\ &= \mathbf{117\ kHz} \end{aligned}$$

Therefore, the 3 dB frequency is, $\boxed{\mathbf{117\ kHz}}$.

Step 14 of 23

(e)

Refer to Figure P3.108 (e) in the textbook to CC-CE Cascode amplifier. From the circuit, it is clear that the transistor acts as CC-Darlington Cascode amplifier. For a paired Cascode amplifier, the mid-band gain, $\mathbf{A_M}$ is,

$$\begin{aligned} A_M &= \frac{\beta R_C}{R_{eq}+r_\pi} \\ &= \frac{-(100)(99)(10\times10^3)}{10\times10^3+5\times10^3} \\ &= \mathbf{-66\ V/V} \end{aligned}$$

Therefore, the mid-band gain is, $\boxed{\mathbf{-66\ V/V}}$.

Step 15 of 23

Consider the 3-dB frequency.

$$f_L = \frac{1}{2\pi\left[\sqrt{\left(\frac{1}{f_{o1}}\right)^2+\left(\frac{1}{f_{o2}}\right)^2+\left(\frac{1}{f_{o3}}\right)^2}\right]}$$

Here, f_{o1}, f_{o2} **and** f_{o3} are frequency poles.

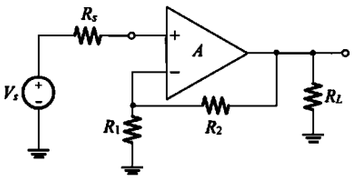
Calculate the input pole, f_{o1}.

$$\begin{aligned} f_{o1} &= \frac{1}{2\pi\left[R_{eq}\|r_\pi\right]\left(C_\pi + 2C_\mu\right)} \\ &= \frac{1}{2\pi\left[(10\times10^3)\|(5\times10^3)\right]\left(6\times10^{-12}+2\times10^{-15}\right)} \\ &= \mathbf{117\ kHz} \end{aligned}$$

Therefore, the 3 dB frequency is, $\boxed{\mathbf{117\ kHz}}$.

Step 1 of 8

The non-inverting op-amp circuit is



Step 2 of 8

c)

The closed loop gain is: $A_f = 10\ \text{V/V}$.

The open loop gain of the amplifier is: $A = 100\ \text{V/V}$.

The closed loop gain is given by,

$$A_f = \frac{A}{1 + A\beta}$$
$$10 = \frac{100}{1 + 100\beta}$$
$$10 + 1000\beta = 100$$
$$1000\beta = 90$$

The feedback factor is,

$$\beta = \frac{90}{1000}$$
$$= 0.09$$

Step 3 of 8

Hence, we have

$$\frac{1}{\beta} = 1 + \frac{R_2}{R_1}$$
$$\frac{R_2}{R_1} = \frac{1}{0.09} - 1$$
$$= 10.1$$

Therefore, the ratio R_2 / R_1 of the non-inverting op-amp is $\boxed{R_2 / R_1 = 10.1}$.

Step 4 of 8

d)

The amount of feedback is,

$$1 + A\beta = \frac{A}{A_f}$$
$$= \frac{100}{10}$$
$$= 10\ \text{s}$$

The amount of feedback in decibels is,

$$20\log(1 + A\beta) = 20\log(10)$$
$$= 20\ \text{dB}$$

Therefore, the amount of feedback in decibels is $\boxed{20\ \text{dB}}$.

Step 5 of 8

e)

The relation of the output voltage with the input voltage is given by

$$V_0 = A_f V_s$$
$$= 10 \times 1$$
$$= 10\ \text{V}$$

Thus the output voltage of the amplifier is $\boxed{V_0 = 10\ \text{V}}$.

Step 6 of 8

The feed forward voltage is given by

$$V_f = \beta V_0$$
$$= 0.09 \times 10$$
$$= 0.9\ \text{V}$$

Thus, the feed forward voltage of the amplifier is $\boxed{V_f = 0.9\ \text{V}}$.

Step 7 of 8

The input voltage is given by

$$V_i = \frac{V_0}{A}$$
$$= \frac{10}{100}$$
$$= 0.1\ \text{V}$$

Thus the input voltage of the amplifier is $\boxed{V_i = 0.1\ \text{V}}$.

Step 8 of 8

(f)
If A decreases by 20%, thus we have, $A = 80\ \text{V/V}$,

The value of A_f becomes

$$A_f = \frac{A}{1 + A\beta}$$
$$= \frac{80}{1 + 80 \times 0.09}$$
$$= 9.756\ \text{V/V}$$

The percentage decrease in A_f is,

$$A_f = \left(\frac{10 - 9.756}{10}\right) \times 100$$
$$= \frac{0.244}{10} \times 100$$
$$= 2.44\%$$

Therefore, the percentage decrease in A_f is: $\boxed{2.44\%}$.

Step 1 of 2

The closed loop gain of negative feedback amplifier is: $A_f = 200$

The open loop gain of negative feedback amplifier is: $A = 10^4$

The closed loop gain is: $A_f = \dfrac{A}{1 + A\beta}$

Substitute, A_f **and** A values in closed loop gain expression.

$$200 = \frac{10^4}{1 + 10^4 \beta}$$
$$200 + (2000000)\beta = 10000$$
$$(2000000)\beta = 10000 - 200$$
$$(2000000)\beta = 9800$$

The feedback factor is,

$$\beta = \frac{9800}{2000000}$$
$$= 0.0049$$
$$= 4.9 \times 10^{-3}$$

Thus the feedback factor of a negative-feedback amplifier is $\boxed{4.9 \times 10^{-3}}$.

Step 2 of 2

Now the open loop gain of negative feedback amplifier is: $A = 10^3$

The closed loop gain is,

$$A_f = \frac{A}{1 + A\beta}$$
$$= \frac{10^3}{1 + 10^3 \times 4.9 \times 10^{-3}}$$
$$= \frac{10^3}{1 + 4.9}$$
$$= 169.49$$

The percentage reduction in the closed loop gain is,

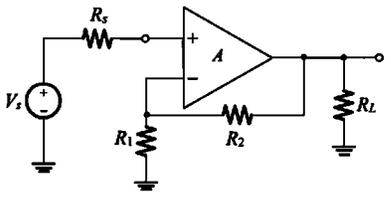
$$\frac{\Delta A_f}{A_f} = \left(\frac{169.49 - 200}{200}\right) \times 100$$
$$= \left(\frac{-30.51}{200}\right) \times 100$$
$$= -15.255\%$$

Thus the percentage change in the closed loop gain of a negative-feedback amplifier is

$\boxed{-15.255\%}$.

Step 1 of 5

The non-inverting op-amp circuit is,



The closed loop gain is: $A_f = 10^3\ V/V$.

The gain of the amplifier is: $A = 10^4\ V/V$.

The source voltage is: $V_s = 0.01\ V$.

Step 2 of 5

a)

The closed loop gain of non-inverting operational amplifier is,

$$A_f = \frac{A}{1 + A\beta}$$

$$10^3 = \frac{10^4}{1 + 10^4 \beta}$$

$$\beta = 9 \times 10^{-4}$$

From the given information,

$$\frac{1}{\beta} = 1 + \frac{R_2}{R_1}$$

$$\frac{R_2}{R_1} = \frac{1}{9 \times 10^{-4}} - 1$$

$$= 1110.11$$

Hence, the ratio R_2 / R_1 of the non-inverting operational amplifier is $\boxed{1110.11}$.

Step 3 of 5

b)

From the given information: $1 + A\beta = \frac{A}{A_f}$.

The amount of feedback is,

$$1 + A\beta = \frac{A}{A_f}$$

$$= \frac{10^4}{10^3}$$

$$= 10\ s$$

The amount of feedback in decibel is,

$$20\log(1 + A\beta) = 20\log(10)$$

$$= 20\ dB$$

Hence, the amount of feedback in decibel is $\boxed{20\ dB}$.

Step 4 of 5

c)

The relation of the output voltage with the input voltage is,

$$V_0 = A_f V_s$$

$$= 10^3 \times 0.01$$

$$= 10\ V$$

Hence, the output voltage of the amplifier is $\boxed{10\ V}$.

The feedforward voltage is,

$$V_f = \beta V_0$$

$$= 9 \times 10^{-4} \times 10$$

$$= 0.009\ V$$

Hence, the feedforward voltage of the amplifier is $\boxed{0.009\ V}$.

The input voltage is,

$$V_i = \frac{V_0}{A}$$

$$= \frac{10}{10^4}$$

$$= 0.001\ V$$

Hence, the input voltage of the amplifier is $\boxed{0.001\ V}$.

d)

If A decreases by 20%, thus we have, $A = 8000\ V/V$,

Step 5 of 5

The value of A_f becomes

$$A_f = \frac{A}{1 + A\beta}$$

$$= \frac{8000}{1 + 8000 \times 9 \times 10^{-4}}$$

$$= 975.6\ V/V$$

The percentage decrease in A_f is,

$$= \left(\frac{1000 - 975.6}{1000}\right) \times 100$$

$$= \frac{24.4}{1000} \times 100$$

$$= 2.44\%$$

Hence, the percentage decrease in A_f is $\boxed{2.44\%}$.

Step 1 of 9

Refer to the circuit diagram in Figure P11.2 in the textbook.

Reconstructing given circuit by replacing op-amp with its equivalent circuit.

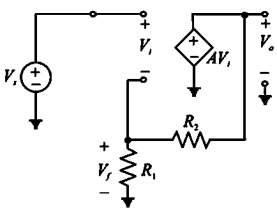


Figure 1

Step 2 of 9

a)

The feedback factor is: $\beta = \frac{V_f}{V_o}$

By Applying voltage division rule to find the required voltage is,

$$V_f = \frac{R_1}{R_1 + R_2} V_o$$

$$\frac{V_f}{V_o} = \frac{R_1}{R_1 + R_2}$$

$$\beta = \frac{R_1}{R_1 + R_2}$$

Therefore, the required feedback factor is $\boxed{\frac{R_1}{R_1 + R_2}}$

Step 3 of 9

b)

i)

Calculate R_2 at $A = 1000\ \text{V/V}$.

The closed loop gain A_f is,

$$A_f = \frac{A}{1 + A\beta}$$

Substitute $A_f = 10\ \text{V/V}$ and $A = 1000\ \text{V/V}$ in the closed loop gain equation

$$10 = \frac{10^3}{1 + 10^3 \beta}$$

$$10 + 10^4 \beta = 10^3$$

$$10^4 \beta = 10^3 - 10$$

$$\beta = \frac{990}{10000}$$
$$= 0.099$$

Evaluating R_2

$$\beta = \frac{R_1}{R_1 + R_2}$$

Substitute $R_1 = 10\ \text{k}\Omega$

$$0.099 = \frac{10 \times 10^3}{(10 \times 10^3) + R_2}$$

$$R_2 = 91.01\ \text{k}\Omega$$

Therefore, the required value of R_2 is $\boxed{91.01\ \text{k}\Omega}$

Step 4 of 9

(ii)

Calculate R_2 at $A = 200\ \text{V/V}$.

The closed loop gain is $A_f = \frac{A}{1 + A\beta}$

Substituting A_f and A values in the closed loop gain expression.

$$A_f = \frac{A}{1 + A\beta}$$

$$10 = \frac{200}{1 + 200\beta}$$

$$10(1 + 200\beta) = 200$$

$$10 + 2000\beta = 200$$

$$2000\beta = 200 - 10$$

$$2000\beta = 190$$

$$\beta = \frac{190}{2000}$$

$$\beta = 0.095$$

Step 5 of 9

Evaluating R_2

$$\beta = \frac{R_1}{R_1 + R_2}$$

$$0.095 = \frac{10 \times 10^3}{(10 \times 10^3) + R_2}$$

$$950 + 0.095 R_2 = 10000$$

$$0.095 R_2 = 10000 - 950$$

$$R_2 = \frac{9050}{0.095}$$

$$R_2 = 95.26\ \text{k}\Omega$$

Therefore, the required value of R_2 is $\boxed{95.26\ \text{k}\Omega}$

Step 6 of 9

(iii)

Calculate the R_2 at $A = 15\ \text{V/V}$

Substituting A_f and A values in equation $A_f = \frac{A}{1 + A\beta}$

$$10 = \frac{15}{1 + 15\beta}$$

$$10(1 + 15\beta) = 15$$

$$10 + 150\beta = 15$$

$$150\beta = 15 - 10$$

$$150\beta = 15 - 10$$

$$\beta = \frac{5}{150}$$

$$\beta = 0.0333$$

Evaluating R_2

$$\beta = \frac{R_1}{R_1 + R_2}$$

$$0.0333 = \frac{10 \times 10^3}{(10 \times 10^3) + R_2}$$

$$0.0333(10000 + R_2) = 10000$$

$$333 + 0.0333 R_2 = 10000$$

$$0.0333 R_2 = 10000 - 333$$

$$R_2 = \frac{9667}{0.0333}$$

$$R_2 = 290.3\ \text{k}\Omega$$

Therefore, the required value of R_2 is $\boxed{290.3\ \text{k}\Omega}$

Step 7 of 9

(c)

(i)

Find the percentage change in A_f, if A is reduced by 20 %.

From the given information, $A = 1000\ \text{V/V}$

Reduced to **20%,** new gain will be

$$A_{(\text{new})} = A - A \times \frac{20}{100}$$
$$= 1000 - 1000 \times \frac{20}{100}$$
$$= 1000 - 200$$
$$= 800$$

Calculating A_f

$$A_f = \frac{A}{1 + A\beta}$$
$$= \frac{800}{1 + 0.095 \times 800}$$
$$= 9.975\ \text{V/V}$$

Now find the percentage change in A_f

$$\text{Percentage change in } A_f = \frac{\Delta A_f}{A_f} \times 100$$
$$= \frac{9.975 - 10}{10} \times 100$$
$$= -0.25\ \%$$

Therefore, the required percentage change in A_f is $\boxed{-0.25\ \%}$

Step 8 of 9

(ii)

For $A = 200\ \text{V/V}$

Reduced to **20%,** new gain will be

$$A_{(\text{new})} = A - A \times \frac{20}{100}$$
$$= 200 - 200 \times \frac{20}{100}$$
$$= 200 - 40$$
$$= 160$$

Calculate A_f.

$$A_f = \frac{A}{1 + A\beta}$$
$$= \frac{160}{1 + 0.095 \times 160}$$
$$= 9.876\ \text{V/V}$$

Now find the percentage change in A_f.

$$\text{Percentage change in } A_f = \frac{\Delta A_f}{A_f} \times 100$$
$$= \frac{9.876 - 10}{10} \times 100$$
$$= -1.24\ \%$$

Thus, percentage change in A_f is $\boxed{-1.24\%}$.

Step 9 of 9

(iii)

For $A = 15\ \text{V/V}$

Reduced to **20%,** new gain will be

$$A_{(\text{new})} = A - A \times \frac{20}{100}$$
$$= 15 - 15 \times \frac{20}{100}$$
$$= 15 - 3$$
$$= 12$$

Calculate A_f

$$A_f = \frac{A}{1 + A\beta}$$
$$= \frac{12}{1 + 0.095 \times 12}$$
$$= 5.60747\ \text{V/V}$$

Now find the percentage change in A_f.

$$\text{Percentage change in } A_f = \frac{\Delta A_f}{A_f} \times 100$$
$$= \frac{5.60747 - 10}{10} \times 100$$
$$= -0.4392\ \%$$
$$= -43.92\%$$

Therefore, the percentage change in A_f is $\boxed{-43.92\%}$.

Step 1 of 4

11.003E

The gain change is 10% as the operating temperature changes from $25\ ^\circ C$ to $75\ ^\circ C$.

$$\frac{dA}{A} = 10\%$$

It is required to constrain the change to 0.1% by applying negative feedback.

$$\frac{dA_f}{A_f} = 0.1\%$$

Refer to equation 10.9 in the text book.

$$\frac{dA_f}{A_f} = \left(\frac{1}{1+A\beta}\right)\frac{dA}{A}$$

Substitute 0.1% for $\frac{dA_f}{A_f}$ and 10% for $\frac{dA}{A}$.

$$0.1 = \left(\frac{1}{1+A\beta}\right)10$$
$$1+A\beta = 100$$

Step 2 of 4

Determine the value of closed loop gain of the amplifier.

$$A_f = \frac{A}{1+A\beta}$$

Substitute 1000 for A and 100 for $1+A\beta$.

$$A_f = \frac{1000}{100}$$
$$= 10\ \text{V/V}$$

Therefore, the value of largest possible closed loop gain is $\boxed{10\ \text{V/V}}$.

Step 3 of 4

Determine the overall gain of three feedback amplifiers in cascade.

$$A = (A_f)(A_f)(A_f)$$

Substitute 10 for A_f.

$$A = (10)(10)(10)$$
$$= 1000\ \text{V/V}$$

Therefore, the overall gain of three feedback amplifiers in cascade is $\boxed{1000\ \text{V/V}}$.

Step 4 of 4

The maximum variability for a single stage amplifier is, 0.1%

Determine the maximum variability for three feedback amplifiers in cascade.

$$\textbf{Maximum variability} = 3(0.1\%)$$
$$= 0.3\%$$

Therefore, the maximum variability is $\boxed{0.3\%}$.

Step 1 of 3

The amount of feedback is: $FB = 20\log(1 + A\beta)$

Substitute, A **and** β values in amount of feedback equation.

$$\begin{aligned} FB &= 20\log(1 + 1000 \times 1) \\ &= 20\log(1001) \\ &= 60 \text{ dB} \end{aligned}$$

Therefore, the amount of feedback of non-inverting buffer op-amp FB is $\boxed{60 \text{ dB}}$.

Source voltage (V_s) is: 1 V

Calculate the output voltage.

$$\begin{aligned} V_0 &= V_s \times A_f \\ &= 1 \times 0.999 \\ &= 0.999 \text{ V} \end{aligned}$$

Therefore, the output voltage of feedback of non-inverting buffer op-amp V_0 is $\boxed{0.999 \text{ V}}$.

Step 2 of 3

The input voltage is,

$$\begin{aligned} V_i &= V_s - V_0\beta \\ &= 1 - 0.999 \times 1 \\ &= 10^{-3} \text{ V} \\ &= 1 \text{ mV} \end{aligned}$$

Hence, the required input voltage is $\boxed{1 \text{ mV}}$.

The open loop gain is decreased by 10%, So open loop gain becomes:

$$\begin{aligned} A &= 1000 - 100 \\ &= 900 \end{aligned}$$

Step 3 of 3

The closed loop gain is,

$$\begin{aligned} A_f &= \frac{A}{1 + A\beta} \\ &= \frac{900}{1 + (900 \times 1)} \\ &= 0.9989 \end{aligned}$$

The percentage decrease in the closed loop gain is,

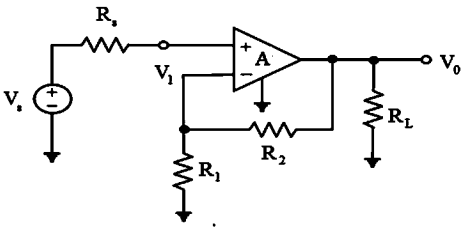
$$\begin{aligned} \frac{\Delta A_f}{A_f} &= \left(\frac{0.999 - 0.9989}{0.999}\right) \times 100 \\ &= \frac{10^{-4}}{0.999} \times 100 \\ &= 0.01\% \end{aligned}$$

Hence, the required gain percentage of closed loop is $\boxed{0.01\%}$.

Step 1 of 3

11.004E

Refer to the Example 11.1in the text book.



The open-loop gain A_M is: 10^4

Upper **3-dB** frequency of before negative feedback is: $\omega_H = 100$ **Hz**

$R_1 = 1\ \text{k}\Omega$ **and** $R_2 = 9\ \text{k}\Omega$

Step 2 of 3

Calculate the value of feedback factor β

$$\beta = \frac{R_1}{R_1 + R_2}$$
$$= \frac{1}{1+9}$$
$$= 0.1$$

Now, calculate the Low-frequency gain $\mathbf{A_f}$ of a closed loop amplifier

$$A_f = \frac{A_M}{1 + A_M\beta}$$
$$= \frac{10^4}{1+(10^4)(0.1)}$$
$$= \frac{10000}{1001}$$
$$= 9.99\ \text{V/V}$$

Hence, the required answer is: **9.99 V/V**.

Step 3 of 3

Calculate the Upper 3-dB frequency of a closed-loop amplifier (ω_{Hf})

$$\omega_{Hf} = \omega_H(1 + A_M\beta)$$
$$= (100)\left[1+(10^4)(0.1)\right]$$
$$= 100.1\ \text{kHz}$$

Therefore, the required answer is **100.1 kHz**.

Step 1 of 2

11.004P

Refer to the circuit diagram in Figure 10.1 in the textbook.

Find open-loop voltage gain A.

$$x_0 = Ax_i$$

Rewrite the equation.

$$A = \frac{x_0}{x_i}$$

Substitute 10 for x_0 and 10×10^{-3} for x_i in the equation.

$$A = \frac{10}{\left(10\times10^{-3}\right)}$$
$$= 1000$$

Therefore, the open loop voltage gain A is $\boxed{1000}$.

Step 2 of 2

Find feedback factor β.

$$x_f = \beta x_o$$

Rewrite the equation.

$$\beta = \frac{x_f}{x_o}$$
$$= \frac{x_s - x_i}{x_o}$$

Substitute 1 for x_s, 0.01 for x_i and 10 for x_o in the equation.

$$\beta = \frac{1-0.01}{10}$$
$$= 0.099$$

Therefore, the feedback factor β is $\boxed{0.099}$.

Step 1 of 3

Determine the value of signal voltage V_{sf}.

$$V_{sf} = \frac{V_s A_1 A_2}{1 + A_1 A_2 \beta}$$

Substitute 1 V for V_s, 1 for A_1 and 1 for A_2.

$$\begin{aligned} V_{sf} &= \frac{(1)(1)(100)}{1+(1)(100)(1)} \\ &= \frac{100}{101} \\ &= 0.99 \text{ V} \end{aligned}$$

Therefore, the value of signal voltage is $\boxed{1 \text{ V}}$.

Step 2 of 3

Determine the value of noise voltage V_{nf}.

$$V_{nf} = \frac{V_n A_1}{1 + A_1 A_2 \beta}$$

Substitute 1 V for V_n, 1 for A_1 and 1 for A_2.

$$\begin{aligned} V_{nf} &= \frac{(1)(1)}{1+(1)(100)(1)} \\ &= \frac{1}{101} \\ &= 0.0099 \text{ V} \\ &= 0.01 \text{ V} \end{aligned}$$

Therefore, the value of noise voltage is $\boxed{0.01 \text{ V}}$.

Step 3 of 3

Determine the value of signal to interference ratio at the output.

$$\frac{S}{I} = \left(\frac{V_s}{V_n}\right) A_2$$

Substitute 1 V for V_n, 1 V for V_s and 1 for A_2.

$$\begin{aligned} \frac{S}{I} &= \left(\frac{1}{1}\right)(100) \\ &= 100 \end{aligned}$$

Determine the value of signal to interference ratio in dB.

$$\begin{aligned} \left.\frac{S}{I}\right|_{\text{dB}} &= 20\log(100) \\ &= 20\log(10^2) \\ &= (20)(2)\log(10) \\ &= 40 \text{ dB} \end{aligned}$$

Therefore, the value of signal to interference ratio in dB is $\boxed{40 \text{ dB}}$.

Step 1 of 5

(a)

Consider the closed loop amplifier gain.

$$A_f = \frac{A}{1+A\beta}$$

Rewrite the closed loop gain A_f further.

$$A_f = \frac{A}{A\beta\left(\frac{1}{A\beta}+1\right)}$$
$$= \frac{1}{\beta\left(\frac{1}{A\beta}+1\right)}$$

$$\frac{A_f}{\beta} = \frac{1}{\frac{1}{A\beta}+1}$$

Step 2 of 5

Consider that the closed-loop gain A_f and feedback factor $\frac{1}{\beta}$ differ by $y\%$. Therefore,

$$\frac{1}{A\beta} = \frac{y}{100}$$
$$\frac{1}{A\beta} = 0.01y$$
$$y = \frac{100}{A\beta}$$

Hence, the closed loop gain A_f is lower than its ideal value of $\frac{1}{\beta}$ is by $\boxed{\left(\frac{100}{A\beta}\right)\%}$.

Step 3 of 5

(b)

Consider the closed-loop gain A_f and β differ by 0.1%. Therefore,

$$y = \left(\frac{100}{A\beta}\right)\%$$

Substitute 0.1% for y in the equation.

$$0.1\% = \frac{100}{A\beta}\%$$
$$A\beta = \frac{100}{0.1}$$
$$= 1000$$

Thus, the loop gain $A\beta$ at 0.1% difference is $\boxed{1000}$.

Step 4 of 5

Consider the closed-loop gain A_f and β differ by 1%. Therefore,

$$y = \left(\frac{100}{A\beta}\right)\%$$

Substitute 1% for y in the equation.

$$1\% = \frac{100}{A\beta}\%$$
$$A\beta = \frac{100}{1}$$
$$= 100$$

Hence, the loop gain $A\beta$ at 1% difference is $\boxed{100}$.

Step 5 of 5

Consider the closed-loop gain A_f and β differ by 5%. Therefore,

$$y = \left(\frac{100}{A\beta}\right)\%$$

Substitute 0.5% for y in the equation.

$$5\% = \frac{100}{A\beta}\%$$
$$A\beta = \frac{100}{5}$$
$$= 20$$

Hence, the loop gain $A\beta$ at 5% difference is $\boxed{20}$.

Step 1 of 9

Refer to the Figure 11.8 (c) in the text book.

From the circuit, $R_1 + R_2 \gg R_D$

The output voltage of the amplifier is,

$$V_0 = g_m V_i R_D$$
$$= g_m V_i (R_1 + R_2)$$

$$V_0 = g_m V_i (R_1 + R_2) \quad \text{..... (1)}$$

The feedback voltage is: $V_f = g_m V_i R_1$ .....(2)

Step 2 of 9

(a)

Find the expression for β.

The feedback factor is,

$$\beta = \frac{V_f}{V_0}$$
$$= \frac{g_m V_i R_1}{g_m V_i (R_1 + R_2)} \qquad \text{(From equation(1) and (2))}$$
$$= \frac{R_1}{(R_1 + R_2)}$$

Hence, the required expression is $\boxed{\beta = \frac{R_1}{R_1 + R_2}}$.

Step 3 of 9

(b)

Find the expression for the loop gain $A\beta$.

The loop gain of the amplifier is,

$$A\beta = \frac{V_0}{V_i}$$
$$= \frac{g_m V_i [R_D \| R_1]}{V_i}$$
$$= \frac{g_m [R_D \| R_1]}{V_i}$$

$$= g_m \left[\frac{R_D \times R_1}{R_D + (R_1 + R_2)} \right]$$
$$= \frac{g_m R_D R_1}{R_D + R_1 + R_2}$$

Hence, the required expression is, $\boxed{A\beta = g_m \left[\frac{R_D R_1}{R_D + R_1 + R_2} \right]}$.

Step 4 of 9

(c)

Find an expression for the open-loop gain A.

The open loop gain A is,

$$A = \frac{V_0}{V_i}$$
$$= \frac{g_m V_i [R_D \| (R_1 + R_2)]}{V_i}$$
$$= g_m [R_D \| (R_1 + R_2)]$$

$$= g_m \left[\frac{R_D \times (R_1 + R_2)}{R_D + (R_1 + R_2)} \right]$$
$$= g_m \left[\frac{R_D (R_1 + R_2)}{R_D + R_1 + R_2} \right]$$

Hence, the required expression for A is $\boxed{g_m \left[\frac{R_D (R_1 + R_2)}{R_D + R_1 + R_2} \right]}$.

Step 5 of 9

(d)

(i)

Calculate the value of β.

$$\beta = \frac{R_1}{R_1 + R_2}$$
$$= \frac{20}{20 + 80}$$
$$= 0.2$$

Hence, the required value of β is $\boxed{0.2}$.

Step 6 of 9

(ii)
Calculate the value of $A\beta$.

$$A\beta = g_m \left[\frac{R_D R_1}{R_D + R_1 + R_2} \right]$$
$$= 4 \left[\frac{10 \times 20}{10 + 20 + 80} \right]$$
$$= \frac{800}{110}$$
$$= 7.27$$

Hence, the required value of $A\beta$ is $\boxed{7.27}$.

Step 7 of 9

(iii)

Calculate the value of A.

$$A = g_m \left[\frac{R_D (R_1 + R_2)}{R_D + R_1 + R_2} \right]$$
$$= 4 \left[\frac{10(20 + 80)}{10 + 20 + 80} \right]$$
$$= \frac{4000}{110}$$
$$= 36.36$$

Hence, the required value of A is $\boxed{36.36}$.

Step 8 of 9

(iv)

Calculate the value of A_f.

$$A_f = \frac{g_m R_D}{1 + g_m R_D \frac{R_1}{(R_1 + R_2)}}$$
$$= \frac{4 \times 10}{1 + 4 \times 10 \left(\frac{20}{20 + 80} \right)}$$

$$= \frac{40}{(100 + 800)/100}$$
$$= \frac{4000}{900}$$
$$= 4.4 \text{ V/V}$$

Hence, the required value of A is $\boxed{4.4 \text{ V/V}}$.

Step 9 of 9

(v)

Consider the following from the part (ii) and part (iii) calculations:

$$\frac{A}{A\beta} = \frac{36.36}{7.27}$$
$$= 5 \text{ V/V}$$

Hence, the 5 V/V much greater than the unity.

Step 1 of 16

(a)

The gain of closed loop amplifier is: $A_f = \frac{A}{1+A\beta}$

Substitute, $\beta = 0$ and $A = 1$ in A_f equation.

$$A_f = \frac{1}{1+(1)(0)}$$
$$= \frac{1}{1}$$
$$= 1 \text{ V/V}$$

Hence, the amplifier open-loop gain is $\boxed{1 \text{ V/V}}$.

Step 2 of 16

Find the closed loop gain at $\beta = 0.5$ and $A = 1$ V/V.

Substitute, $\beta = 0.5$ and $A = 1$ in A_f equation.

$$A_f = \frac{1}{1+(1)(0.5)}$$
$$= \frac{1}{1.5}$$

Hence, the amplifier open-loop gain is $\boxed{0.667 \text{ V/V}}$.

Step 3 of 16

Find the closed loop gain at $\beta = 1$ and $A = 1$ V/V.

Substitute, $\beta = 1$ and $A = 1$ in A_f equation.

$$A_f = \frac{1}{1+(1)(1.0)}$$
$$= \frac{1}{2}$$

Hence, the amplifier open-loop gain is $\boxed{0.5 \text{ V/V}}$.

Step 4 of 16

(b)

Find the closed loop gain at $\beta = 0$ and $A = 10$ V/V

Substitute, $\beta = 0$ and $A = 10$ in A_f equation.

$$A_f = \frac{10}{1+(10)(0)}$$
$$= \frac{10}{1}$$

Hence, the amplifier open-loop gain is $\boxed{10 \text{ V/V}}$.

Step 5 of 16

Find the closed loop gain at $\beta = 0.5$ and $A = 10$ V/V

Substitute, $\beta = 0.5$ and $A = 10$ V/V in A_f equation.

$$A_f = \frac{10}{1+(10)(0.5)}$$
$$= \frac{10}{6}$$
$$= 1.667 \text{ V/V}$$

Hence, the amplifier open-loop gain is $\boxed{1.667 \text{ V/V}}$.

Step 6 of 16

Find the closed loop gain at $\beta = 1$ and $A = 10$ V/V

Substitute, $\beta = 1$ and $A = 10$ V/V in A_f equation.

$$A_f = \frac{10}{1+(10)(1)}$$
$$= \frac{10}{11}$$
$$\approx 0.91 \text{ V/V}$$

Hence, the amplifier open-loop gain is $\boxed{0.91 \text{ V/V}}$.

Step 7 of 16

(c)

Find the closed loop gain at $\beta = 0$ and $A = 100$ V/V

Substitute, $\beta = 0$ and $A = 100$ V/V in A_f equation.

$$A_f = \frac{100}{1+(100)(0)}$$
$$= 100 \text{ V/V}$$

Hence, the amplifier open-loop gain is $\boxed{100 \text{ V/V}}$.

Step 8 of 16

Find the closed loop gain at $\beta = 0.5$ and $A = 100$ V/V

Substitute, $\beta = 0.5$ and $A = 100$ V/V in A_f equation.

$$A_f = \frac{100}{1+(100)(0.5)}$$
$$= \frac{100}{51}$$
$$= 1.96 \text{ V/V}$$

Hence, the amplifier open-loop gain is $\boxed{1.96 \text{ V/V}}$.

Step 9 of 16

Find the closed loop gain at $\beta = 1$ and $A = 100$ V/V.

Substitute, $\beta = 0.5$ and $A = 100$ V/V in A_f equation.

$$A_f = \frac{100}{1+(100)(1)}$$
$$= \frac{100}{101}$$
$$= 0.99 \text{ V/V}$$

Hence, the amplifier open-loop gain is $\boxed{0.99 \text{ V/V}}$.

Step 10 of 16

(d)

Find the closed loop gain at $\beta = 0$ and $A = 1000$ V/V

Substitute, $\beta = 0$ and $A = 1000$ V/V in A_f equation.

$$A_f = \frac{1000}{1+(1000)(0)}$$
$$= 1000 \text{ V/V}$$

Hence, the amplifier open-loop gain is $\boxed{1000 \text{ V/V}}$.

Step 11 of 16

Find the closed loop gain at $\beta = 0.5$ and $A = 1000$ V/V

Substitute, $\beta = 0.5$ and $A = 1000$ V/V in A_f equation.

$$A_f = \frac{1000}{1+(1000)(0.5)}$$
$$= \frac{1000}{501}$$
$$= 1.996 \text{ V/V}$$

Hence, the amplifier open-loop gain is $\boxed{1.996 \text{ V/V}}$.

Step 12 of 16

Find the closed loop gain at $\beta = 1$ and $A = 1000$ V/V.

Substitute, $\beta = 0.5$ and $A = 1000$ V/V in A_f equation.

$$A_f = \frac{1000}{1+(1000)(1)}$$
$$= \frac{1000}{1001}$$
$$= 0.999 \text{ V/V}$$

Hence, the amplifier open-loop gain is $\boxed{0.999 \text{ V/V}}$.

Step 13 of 16

(e)

Find the closed loop gain at $\beta = 0$ and $A = 10{,}000$ V/V

Substitute, $\beta = 0$ and $A = 10{,}000$ V/V in A_f equation.

$$A_f = \frac{10{,}000}{1+(10{,}000)(0)}$$
$$= 10{,}000$$

Hence, the amplifier open-loop gain is $\boxed{10{,}000 \text{ V/V}}$.

Step 14 of 16

Find the closed loop gain at $\beta = 0.5$ and $A = 10{,}000$ V/V.

Substitute, $\beta = 0$ and $A = 10{,}000$ V/V in A_f equation.

$$A_f = \frac{10{,}000}{1+(10{,}000)(0.5)}$$
$$= \frac{10{,}000}{5001}$$
$$= 1.99 \text{ V/V}$$

Hence, the amplifier open-loop gain is $\boxed{1.99 \text{ V/V}}$.

Step 15 of 16

Find the closed loop gain at $\beta = 1$ and $A = 10{,}000$ V/V

Substitute, $\beta = 1$ and $A = 10{,}000$ V/V in A_f equation.

$$A_f = \frac{10{,}000}{1+(10{,}000)(1)}$$
$$= \frac{10{,}000}{10{,}001}$$
$$= 0.99 \text{ V/V}$$

Hence, the amplifier open-loop gain is $\boxed{0.99 \text{ V/V}}$.

Step 16 of 16

The table representation as shown below,

A \ β	0	0.5	1
1	1	0.667	0.5
10	10	1.667	0.91
100	100	1.96	0.99
1000	1000	1.996	0.999
10000	10000	1.99	0.99

Step 1 of 3

11.007E

Refer to the Figure 10.16 (a) in the text book.

Determine the value of voltage gain.

$$A = \frac{V_o}{V_s}$$

$$= \left(\mu \frac{R_L \| (R_1 + R_2)}{\left[R_L \| (R_1 + R_2)\right] + r_0}\right)\left(\frac{R_{id}}{R_{id} + R_s + (R_1 \| R_2)}\right)$$

Substitute $2\ \text{k}\Omega$ for R_L, $1\ \text{k}\Omega$ for R_1, $1\ \text{M}\Omega$ for R_2, $10\ \text{k}\Omega$ for R_s, $1\ \text{k}\Omega$ for r_0, 10^4 for μ and $100\ \text{k}\Omega$ for R_{id}.

$$A = (10^4)\left(\frac{2000 \| (10^3 + 10^6)}{\left[2000 \| (10^5 + 10^6)\right] + 10^3}\right)\left(\frac{10^5}{10^5 + 10^4 + \left(\frac{10^3 \times 10^6}{10^3 + 10^6}\right)}\right)$$

$$= (10^4)\left(\frac{2000 \| (1001000)}{\left[2000 \| (1001000)\right] + 10^3}\right)\left(\frac{10^5}{10^5 + 10^4 + \left(\frac{10^9}{1001000}\right)}\right)$$

$$= (10^4)\left(\frac{\frac{(2000)(1001000)}{(2000) + (1001000)}}{\left[\frac{(2000)(1001000)}{(2000) + (1001000)}\right] + 10^3}\right)\left(\frac{10^5}{10^5 + 10^4 + \left(\frac{10^9}{1001000}\right)}\right)$$

$$\simeq 6000\ \text{V/V}$$

Step 2 of 3

Determine the value of feedback factor β.

$$\beta = \frac{V_f}{V_o}$$

$$= \frac{R_1}{R_1 + R_2}$$

Substitute $1\ \text{k}\Omega$ for R_1 and $1\ \text{M}\Omega$ for R_2.

$$\beta = \frac{10^3}{10^3 + 10^6}$$

$$= \frac{1000}{1001000}$$

$$\simeq 10^{-3}\ \text{V/V}$$

Step 3 of 3

Determine the value of upper **3-dB** frequency of the closed loop gain.

$$f_{Hf} = f_H (1 + A\beta)$$

Substitute 6000 for A and 10^{-3} for β.

$$f_{Hf} = 10^3 \left(1 + (6000)(10^{-3})\right) \qquad (\text{Since } f_H = 1\ \text{kHz})$$

$$= 10^3 (1 + 6)$$

$$= 7\ \text{kHz}$$

Therefore, the value of 3-dB frequency of the closed loop gain is $\boxed{7\ \text{kHz}}$.

Step 1 of 5

Without feedback:

The source signal of a feedback amplifier is $V_s = 2\ \text{mV}$.

The output voltage is $V_0 = 5\ \text{V}$

With feedback:

The source signal of a feedback amplifier is $V_s = 100\ \text{mA}$.

The output voltage is $V_0 = 5\ \text{V}$.

The gain of open loop amplifier is,

$$A = \frac{5\ \text{V}}{2 \times 10^{-3}\ \text{V}}$$
$$= 2500$$

Hence, the required gain is $\boxed{2500}$.

Step 2 of 5

The gain of open loop in dB is,

$$A = 20 \log 2500$$
$$= 20 \times 3.398$$
$$= 67.96\ \text{dB}$$

Hence, the required gain in dB is $\boxed{67.96\ \text{dB}}$.

Step 3 of 5

The gain of closed loop is,

$$A_f = \frac{V_0}{V_s}$$
$$= \frac{5}{100 \times 10^{-3}}$$
$$= 50$$

Hence, the required gain is $\boxed{50}$.

The gain of closed loop in dB is,

$$A_f = 20 \log A_f$$
$$= 20 \log 50$$
$$= 33.98\ \text{dB}$$

Hence, the required gain in dB is $\boxed{33.98\ \text{dB}}$.

Step 4 of 5

The gain of closed loop is,

$$A_f = \frac{A}{1 + A\beta}$$
$$50 = \frac{2500}{1 + (2500\beta)}$$
$$50 + 125000\beta = 2500$$
$$125000\beta = 2500 - 50$$
$$\beta = \frac{2450}{125000}$$
$$= 0.0196$$

Hence, the required feedback factor is $\boxed{0.0196}$.

Step 5 of 5

The feedback factor in dB is,

$$\beta = 20 \log 0.0196$$
$$= 20 \times (-1.7077)$$
$$= -34.15\ \text{dB}$$

Hence, the required feedback factor β in dB is $\boxed{-34.15\ \text{dB}}$.

The value of $A\beta$ is,

$$A\beta = 2500 \times 0.0196$$
$$= 49$$

Therefore, the value of $(A\beta)$ is $\boxed{49}$.

Step 1 of 8

Refer to the Figure E10.11 in the text book.

The emitter current of transistors Q_1 and Q_2 is,

$$I_{E1} = \frac{1}{2}\text{mA}$$

$$I_{E2} = \frac{1}{2}\text{mA}$$

Determine the value of emitter resistors of transistors Q_1 and Q_2.

$$\begin{aligned} r_{e1} &= r_{e2} \\ &= \frac{V_T}{I_E} \\ &= \frac{25\,\text{mV}}{0.5\,\text{mA}} \\ &= 50\,\Omega \end{aligned}$$

Determine the value of emitter resistors of transistor Q_3.

$$\begin{aligned} r_{e3} &= \frac{V_T}{I_E} \\ &= \frac{25\,\text{mV}}{5\,\text{mA}} \\ &= 5\,\Omega \end{aligned}$$

Step 2 of 8

Determine the value of gain of the feedback amplifier.

$$A = \left[\frac{R_{D1} \| (\beta+1)\left[r_{e2} + R_L \| (R_1 + R_2)\right]}{r_{e1} + r_{e2} + \frac{R_x}{\beta+1} + \frac{(R_1 \| R_2)}{\beta+1}}\right]\left[\frac{R_L \| (R_1+R_2)}{r_{e3} + (R_L \| (R_1+R_2))}\right]$$

Substitute $20\,\text{k}\Omega$ for R_{D1}, 100 for β, $2\,\text{k}\Omega$ for R_L, $1\,\text{k}\Omega$ for R_1, $9\,\text{k}\Omega$ for R_2, $50\,\Omega$ for r_{e1} and $50\,\Omega$ for r_{e2}, $5\,\Omega$ for r_{e3} and $10\,\text{k}\Omega$ for R_x.

$$\begin{aligned} A &= \left[\frac{(20\times10^3) \| \left\{(100+1)\left[50 + \{2 \| (1+9)\} \times 10^3\right]\right\}}{50 + 50 + \frac{10\times10^3}{100+1} + \frac{(1 \| 9)}{100+1}\times10^3}\right]\left[\frac{\{2 \| (1+9)\}\times10^3}{5 + \{2 \| (1+9)\}\times10^3}\right] \\ &= \left[\frac{(20\times10^3) \| (173.4\times10^3)}{50+50+99+8.9}\right]\left[\frac{1.67\times10^3}{5+1.67\times10^3}\right] \\ &= \left[\frac{17.93\times10^3}{207.9}\right](0.997) \\ &= 85.9\ \text{V/V} \end{aligned}$$

Therefore, the gain of the amplifier is $\boxed{85.9\ \text{V/V}}$.

Step 3 of 8

Determine the value of feedback factor.

$$\beta = \frac{R_1}{R_1 + R_2}$$

Substitute $1\,\text{k}\Omega$ for R_1 and $9\,\text{k}\Omega$ for R_2.

$$\begin{aligned} \beta &= \frac{1\,\text{k}\Omega}{1\,\text{k}\Omega + 9\,\text{k}\Omega} \\ &= \frac{1}{10} \\ &= 0.1\ \text{V/V} \end{aligned}$$

Therefore, the value of feedback factor of the amplifier is $\boxed{0.1\ \text{V/V}}$.

Step 4 of 8

Determine the voltage gain feedback amplifier.

$$A_f = \frac{A}{1 + A\beta}$$

Substitute $85.9\ \text{V/V}$ for A and 0.1 for β.

$$\begin{aligned} A_f &= \frac{85.9}{1 + (85.9)(0.1)} \\ &= \frac{85.9}{1+8.59} \\ &= 8.96\ \text{V/V} \end{aligned}$$

Thus, the voltage gain of the amplifier with feedback is $\boxed{8.96\ \text{V/V}}$.

Step 5 of 8

Determine the value of input impedance without feedback.

$$R_i = R_x + (\beta+1)(r_{e1} + r_{e2}) + (R_1 \| R_2)$$

Substitute $10\,\text{k}\Omega$ for R_x, 100 for β, $50\,\Omega$ for r_{e1} and $50\,\Omega$ for r_{e2}, $1\,\text{k}\Omega$ for R_1 and $9\,\text{k}\Omega$ for R_2.

$$\begin{aligned} R_i &= (10\times10^3) + (100+1)(50+50) + (1\,\text{k}\Omega \| 9\,\text{k}\Omega) \\ &= (10\times10^3) + (100+1)(100) + (0.9\,\text{k}\Omega) \\ &= 10\times10^3 + 10100 + 0.9\times10^3 \\ &= 21\,\text{k}\Omega \end{aligned}$$

Step 6 of 8

Determine the input resistance of the feedback amplifier.

$$R_{in} = R_i(1 + A\beta) - R_x$$

Substitute $21\,\text{k}\Omega$ for R_i, 85.9 for A, 0.1for β and $10\,\text{k}\Omega$ for R_x.

$$\begin{aligned} R_{in} &= (21\times10^3)\left[1 + (85.9)(0.1)\right] - (10\times10^3) \\ &= (201-10)\times10^3 \\ &= 191\times10^3\ \Omega \\ &= 191\ \text{k}\Omega \end{aligned}$$

Therefore, the value of input resistance of the feedback amplifier is $\boxed{191\ \text{k}\Omega}$.

Step 7 of 8

Determine the output resistance amplifier without feedback.

$$R_o = R_L \| (R_1 + R_2) \| \left[r_{e3} + \frac{R_{D1}}{\beta+1}\right]$$

Substitute $20\,\text{k}\Omega$ for R_{D1}, 100 for β, 5 for r_{e3}, $2\,\text{k}\Omega$ for R_L, $1\,\text{k}\Omega$ for R_1 and $9\,\text{k}\Omega$ for R_2.

$$\begin{aligned} R_o &= \left[2\,\text{k}\Omega \| (1\,\text{k}\Omega + 9\,\text{k}\Omega)\right] \| \left[5 + \frac{2\,\text{k}\Omega}{100+1}\right] \\ &= \left[2\,\text{k}\Omega \| (10\,\text{k}\Omega)\right] \| [203.03] \\ &= 181\,\Omega \end{aligned}$$

Step 8 of 8

Determine the output resistance of the feedback amplifier.

$$R_{out} = \left(\frac{R_o}{1 + A\beta}\right) \| R_L$$

Substitute $24.43\,\Omega$ for R_o, 85.9 for A, 0.1for β and $2\,\text{k}\Omega$ for R_L.

$$\begin{aligned} R_{out} &= \left(\frac{181}{1 + (85.9)(0.1)}\right) \| (2\,\text{k}\Omega) \\ &= (18.87\,\Omega) \| (2\,\text{k}\Omega) \\ &= 18.87\,\Omega \end{aligned}$$

Therefore, the value of output resistance of the feedback amplifier is $\boxed{18.87\,\Omega}$.

11.008P

Step 1 of 3

Consider the open loop gain A is,

$$A = 1000$$

Change in open loop gain is,

$$\frac{\Delta A}{A} = \pm 50\%$$

Consider the change in closed loop gain is,

$$\frac{\Delta A_f}{A_f} = \frac{\left(\frac{\Delta A}{A}\right)}{1 + A\beta}$$

The closed loop gain is,

$$\frac{\Delta A_f}{A_f} = \left(\frac{\left(\frac{\Delta A}{A}\right)}{1 + A\beta}\right)$$

Substitute 1% for $\frac{\Delta A_f}{A_f}$, 50% for $\frac{\Delta A}{A}$ in equation.

$$\frac{1}{100} = \frac{\left(\frac{50}{100}\right)}{1 + 1000\beta}$$
$$1 + 1000\beta = 50$$
$$\beta = 0.049$$

Step 2 of 3

Consider the 50% of A as 500.

Therefore, the nominal value of open loop gain varies 1500 to 500.

Determine the closed loop gain.

$$A_f = \frac{A}{1 + A\beta}$$

Substitute 1500 for A, 0.049 for β in equation.

$$A_f = \frac{1500}{1 + 1500(0.049)}$$
$$= \frac{1500}{74.5}$$
$$= 20.13$$

Determine the closed loop gain.

$$A_f = \frac{A}{1 + A\beta}$$

Substitute 500 for A, 0.049 for β in equation.

$$A_f = \frac{500}{1 + 500(0.049)}$$
$$= \frac{500}{25.5}$$
$$= 19.6$$

Therefore, the maximum nominal value of closed loop gain A_f is $\boxed{20.13}$.

Step 3 of 3

Consider the three such amplifiers are placed in cascaded. Then the total gain G is,

$$G = (A_f)^3$$
$$= (20.13)^3$$
$$= 8157.01$$

Therefore, the nominal value of gain G is $\boxed{8157.01}$.

Step 1 of 5

11.009E

Refer to the Figure 11.8 (c) in the text book, the expression for the output voltage can be written as:

$$V_0 = g_m V_i \left[R_D \| (R_1 + R_2) \right]$$

$V_0 \rightarrow$ Output voltage

$V_i \rightarrow$ Input voltage

$g_m \rightarrow$ Trans-conductance

$R_D \rightarrow$ Drain Resistance

The feedback signal expression can write as: $V_f = g_m V_i R_1$

Find the value of open-loop gain (A).

$$\begin{aligned} A &= \frac{V_0}{V_i} \\ &= \frac{g_m V_i \left[R_D \| (R_1 + R_2) \right]}{V_i} \\ &= g_m \left[R_D \| (R_1 + R_2) \right] \end{aligned}$$

Therefore, the required expression for open loop gain is $\boxed{g_m \left[R_D \| (R_1 + R_2) \right]}$.

Step 2 of 5

Find the expression for β.

The series-shunt feedback voltage amplifier for the common-emitter current gain is,

$$\begin{aligned} \beta &= \frac{V_f}{V_0} \\ &= \frac{g_m V_i R_1}{g_m V_i \left[R_D \| (R_1 + R_2) \right]} \end{aligned}$$

If R_D is consider the resultant of the feedback is negative, so we neglect R_D.

$$\begin{aligned} \beta &= \frac{g_m V_i}{g_m V_i} \frac{R_1}{(R_1 + R_2)} \\ &= \frac{R_1}{R_1 + R_2} \end{aligned}$$

Therefore, the required expression is $\boxed{\frac{R_1}{(R_1 + R_2)}}$.

Step 3 of 5

Find the expression for A_f.

$$\begin{aligned} A_f &= \frac{A}{1 + A\beta} \\ &= \frac{g_m R_D}{1 + g_m R_D \left(\frac{R_1}{R_1 + R_2} \right)} \end{aligned} \qquad \left(\begin{array}{l} A = g_m R_D \text{ Since } R_1 + R_2 \gg R_D \\ \text{and } A\beta = g_m R_D \left(\frac{R_1}{R_1 + R_2} \right) \end{array} \right)$$

Therefore, the required expression for the A_f is $\boxed{(g_m R_D) / \left(1 + g_m R_D \left(\frac{R_1}{R_1 + R_2} \right) \right)}$.

Step 4 of 5

Find the expression for the input resistance (R_{in}).

$$\begin{aligned} R_{in} &= \frac{1}{g_m} \left[1 + (g_m R_D) \frac{R_1}{(R_1 + R_2)} \right] \\ &= \frac{1}{g_m} [1 + A\beta] \qquad \left(\text{Since, } A\beta = (g_m R_D) \frac{R_1}{(R_1 + R_2)} \right) \\ &= \frac{1 + A\beta}{g_m} \end{aligned}$$

Therefore, the required expression is input resistance is $\boxed{\frac{1 + A\beta}{g_m}}$.

Step 5 of 5

Find the expression for the output resistance (R_{out}).

$$\begin{aligned} R_{out} &= R_{of} \\ &= \frac{R_0}{1 + A\beta} \\ &= \frac{R_D \| (R_1 + R_2)}{(1 + A\beta)} \end{aligned} \qquad \left(\text{From the referred figure } R_0 = R_D \| (R_1 + R_2) \right)$$

Therefore, the required expression for R_{out} is $\boxed{\frac{R_D \| (R_1 + R_2)}{(1 + A\beta)}}$.

Step 1 of 6

Consider the Figure P11.9 in the textbook.

The equivalent circuit diagram is shown in Figure 1.

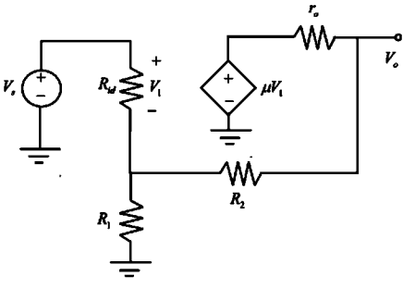


Figure 1

Step 2 of 6

The loading effect of the feedback network at the input side is can be obtained by short-circuiting at output side and looking into input side.

Then, the resultant resistance is $R_1 \| R_2$.

The loading effect of the feedback network at the output side is can be obtained by open-circuiting at input side and looking into output side.

Then, the resultant resistance is $R_1 + R_2$.

The equivalent circuit diagram is shown in Figure 2.

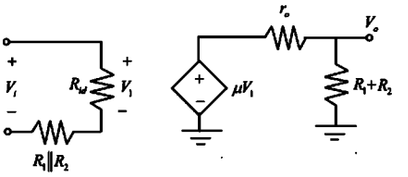


Figure 2

Step 3 of 6

Determine the value of open loop gain A .

Open loop gain A is defined as,

$$A = \frac{V_o}{V_i}$$

It can be also written as follows:

$$A = \left(\frac{V_o}{V_1}\right)\left(\frac{V_1}{V_i}\right) \quad \text{..... (1)}$$

Calculate the ratio of $\frac{V_o}{V_1}$ from the Figure 2.

Apply Voltage division rule at output.

$$V_o = V_1\left(\frac{R_1 + R_2}{R_1 + R_2 + r_o}\right)$$

$$\frac{V_o}{V_1} = \mu\frac{(R_1 + R_2)}{(R_1 + R_2) + r_o} \quad \text{..... (2)}$$

Step 4 of 6

Calculate the ratio of $\frac{V_1}{V_i}$ from the Figure 2.

Apply Voltage division rule at output.

$$V_1 = V_i\left(\frac{R_{id}}{R_{id} + (R_1 \| R_2)}\right) R \quad \text{..... (3)}$$

Step 5 of 6

Substitute the equatin 2 and 3 in equation 1.

$$A = \left(\frac{V_o}{V_1}\right)\left(\frac{V_1}{V_i}\right)$$

$$= \left(\mu\frac{(R_1 + R_2)}{(R_1 + R_2) + r_o}\right)\left(\frac{R_{id}}{R_{id} + (R_1 \| R_2)}\right)$$

Thus, the open loop gain A is obtained as, $\left(\mu\frac{(R_1 + R_2)}{(R_1 + R_2) + r_o}\right)\left(\frac{R_{id}}{R_{id} + (R_1 \| R_2)}\right)$.

The feedback factor β is obtained from the feedback network.

$$\beta = \frac{R_1}{R_1 + R_2}$$

Step 6 of 6

Determine the loop gain $A\beta$.

Substitute $\left(\mu\frac{(R_1 + R_2)}{(R_1 + R_2) + r_o}\right)\left(\frac{R_{id}}{R_{id} + (R_1 \| R_2)}\right)$ for A and $\frac{R_1}{R_1 + R_2}$ for β in equation.

$$A\beta = \left[\left(\mu\frac{(R_1 + R_2)}{(R_1 + R_2) + r_o}\right)\left(\frac{R_{id}}{R_{id} + (R_1 \| R_2)}\right)\right]\left(\frac{R_1}{R_1 + R_2}\right)$$

$$= \left(\mu\frac{R_1}{R_1 + R_2 + r_o}\right)\left(\frac{R_{id}}{R_{id} + (R_1 \| R_2)}\right)$$

Consider the resistance R_1 is greater than $R_2 + r_o$. Thus, $R_1 >> R_2 + r_o$.

$$A\beta = \left(\mu\frac{R_1}{R_1}\right)\left(\frac{R_{id}}{R_{id} + (R_1 \| R_2)}\right)$$

$$= (\mu)\left(\frac{R_{id}}{R_{id} + (R_1 \| R_2)}\right)$$

Therefore, the loop gain is $\boxed{\mu\left(\frac{R_{id}}{R_{id} + (R_1 \| R_2)}\right)}$.

Step 1 of 4

Refer to the Example 11.6 in the textbook.

Gain of op-amp μ is decrease by 10%.

Thus, gain is

$$\begin{aligned}\mu_{new} &= 1000-\left(1000\times\frac{10}{100}\right)\\ &= 1000-100\\ &= 900\end{aligned}$$

Step 2 of 4

The loop gain $A\beta$ is given by,

$$\begin{aligned}A\beta &= -\frac{V_r}{V_t}\\ &= \mu\frac{g_m\left(R_F\|R_{id}\|r_{o2}\right)}{1+g_m\left(R_F\|R_{id}\|r_{o2}\right)}\end{aligned}$$

Substitute 900 for μ, $2\ \text{mA/V}$ for g_m, $20\ \text{k}\Omega$ for r_{o2}, $1\ \text{k}\Omega$ for R_F, $10\ \text{k}\Omega$ for R_{id} in equation.

$$\begin{aligned}A\beta &= 900\left[\frac{2\left(1\|100\|20\right)}{1+2\left(1\|100\|20\right)}\right]\\ &= 900\left[\frac{1.886}{1+1.886}\right]\\ &= 900\left[\frac{1.886}{2.886}\right]\end{aligned}$$

$$A\beta = 588.14$$

Step 3 of 4

Open loop gain A is,

$$\begin{aligned}A &= \frac{A\beta}{\beta}\\ &= \frac{588.14}{1\times10^3}\\ &= 588.14\times10^{-3}\end{aligned}$$

The closed loop gain A_f is,

$$A_f = \frac{A}{1+A\beta}$$

Substitute 0.986 for A, 492.82 for $A\beta$ in equation.

$$\begin{aligned}A_{f(new)} &= \frac{588.14\ \text{m}}{1+588.14}\\ &= 0.9983\times10^{-3}\ \text{A/V}\\ &= 0.9983\ \text{mA/V}\end{aligned}$$

Step 4 of 4

The percentage change in A_f is,

$$\%\text{change in } A_f = \frac{A_{f(new)}-A_f}{A_f}\times100$$

Substitute $0.9983\ \text{mA/V}$ for $A_{f(new)}$, $0.9985\ \text{mA/V}$ for A_f in equation.

$$\begin{aligned}\%\text{changein } A_f &= \frac{0.9983-0.9985}{0.9985}\times100\\ &= \frac{-0.0002}{0.9985}\times100\\ &= -0.02\end{aligned}$$

Therefore, the percentage change in A_f is $\boxed{-0.02\%}$.

11.010P

Step 1 of 3

Refer to the Figure 11.1 in the text book.

The sensitivity of closed-loop gain to open loop gain is,

$$\frac{\left(\frac{dA_f}{A_f}\right)}{\left(\frac{dA}{A}\right)} = -40 \text{ dB} \quad \text{..... (1)}$$

We know that, the sensitivity of closed loop gain is,

$$\frac{dA_f}{A_f} = \left(\frac{1}{1 + A\beta}\right) \times \left(\frac{dA}{A}\right) \quad \text{..... (2)}$$

Step 2 of 3

By equating equation 1 and 2, the obtained expression is,

$$\frac{1}{1 + A\beta} = -40 \text{ dB}$$

$$20 \log\left(\frac{1}{1 + A\beta}\right) = -40$$

$$\log\left(\frac{1}{1 + A\beta}\right) = \frac{-40}{20}$$

Simplify further.

$$\log\left(\frac{1}{1 + A\beta}\right) = -2$$

$$\frac{1}{1 + A\beta} = 10^{-2}$$

$$1 + A\beta = 100$$

$$A\beta = 100 - 1$$

$$A\beta = 99$$

Hence, the loop gain of the feedback amplifier $(A\beta)$ is $\boxed{99}$.

Step 3 of 3

The sensitivity of closed-loop gain to open loop gain is,

$$\frac{\left(\frac{dA_f}{A_f}\right)}{\left(\frac{dA}{A}\right)} = \frac{1}{5}$$

$$\frac{1}{1 + A\beta} = \frac{1}{5}$$

$$1 + A\beta = 5$$

$$A\beta = 4$$

Hence, the loop gain of the feedback amplifier $(A\beta)$ is $\boxed{4}$.

Step 1 of 3

11.011E

Refer to the Figure 11.17(a) and Figure 11.17(b) in the textbook.

The feedback factor β of trans-conductance amplifier is,

$$\begin{aligned}\beta &= \frac{V_f}{I_o}\\ &= R_F\end{aligned}$$

The closed-loop gain A_f is,

$$A_f = \frac{A}{1+A\beta}$$

Here, A is the open-loop gain.

The expression of A_f for $A\beta >> 1$ is,

$$\begin{aligned}A_f &= \frac{1}{\beta}\\ &= \frac{1}{R_F}\end{aligned}$$

Substitute $2\ \text{mA/V}$ for A_f to determine resistor R_F.

$$2\times10^{-3} = \frac{1}{R_F}$$

$$\begin{aligned}R_F &= \frac{1}{2\times10^{-3}}\\ &= 0.5\times10^{3}\\ &= 500\,\Omega\end{aligned}$$

Therefore, the resistance R_F is $\boxed{500\,\Omega}$.

Step 2 of 3

The expression for the loop gain $A\beta$ is,

$$\begin{aligned}A\beta &= -\frac{V_r}{V_t}\\ &= \mu\frac{g_m\left(R_F \parallel R_{id} \parallel r_{o2}\right)}{1+g_m\left(R_F \parallel R_{id} \parallel r_{o2}\right)}\end{aligned}$$

Here, μ is the open-circuit voltage gain, g_m is the transconductance, R_{id} is the differential input resistance, and r_{o2} is the small-signal output resistance.

Substitute 1000 for μ, $2\ \text{mA/V}$ for g_m, $20\ \text{k}\Omega$ for r_{o2}, $500\,\Omega$ for R_F, $100\ \text{k}\Omega$ for R_{id} in equation.

$$\begin{aligned}A\beta &= 1000\left[\frac{2\times10^{-3}\left(500 \parallel 100\times10^{3} \parallel 20\times10^{3}\right)}{1+2\times10^{-3}\left(500 \parallel 100\times10^{3} \parallel 20\times10^{3}\right)}\right]\\ &= 1000\left[\frac{2\times10^{-3}(487.43)}{1+2\times10^{-3}(487.43)}\right]\\ &= \left[\frac{970.87}{1+970.87\times10^{-3}}\right]\\ &= \left[\frac{970.87}{1.970}\right]\end{aligned}$$

$$A\beta = 492.82$$

Step 3 of 3

Open loop gain A is,

$$\begin{aligned}A &= \frac{A\beta}{\beta}\\ &= \frac{492.82}{500}\\ &= 0.986\end{aligned}$$

The closed loop gain A_f is,

$$A_f = \frac{A}{1+A\beta}$$

Substitute 0.986 for A, 492.82 for $A\beta$ in equation.

$$\begin{aligned}A_f &= \frac{0.986}{1+492.82}\\ &= 1.9959\times10^{-3}\ \text{A/V}\\ &= 1.996\ \text{mA/V}\end{aligned}$$

Therefore, the closed loop gain A_f is $\boxed{1.996\ \text{mA/V}}$.

Step 1 of 8

The closed loop gain of the amplifier expression is,

$$A_f = \frac{A}{1 + A\beta}$$

Substitute A and A_f values in the expression of A_f to calculate Feedback factor (β).

$$10 = \frac{1000}{1 + 1000\beta}$$
$$10(1 + 1000\beta) = 1000$$
$$10 + 10000\beta = 1000$$
$$10000\beta = 1000 - 10$$
$$\beta = \frac{990}{10000}$$
$$\beta = 0.099$$

Thus, the required feedback factor β is $\boxed{0.099}$.

Step 2 of 8

Calculate the de-sensitivity factor.

$$D = 1 + A\beta$$

Substitute $1000\ \text{V/V}$ for A and 0.099 for β.

$$D = 1 + 1000 \times 0.099$$
$$= 1 + 99$$
$$= 100$$

Therefore, the required de-sensitivity factor D is $\boxed{100}$.

Step 3 of 8

From the given information: $A = 500\ \text{V/V}$ and $A_f = 10\ \text{V/V}$

Substitute A and A_f values in the expression of A_f to calculate Feedback factor (β).

$$10 = \frac{500}{1 + 500\beta}$$
$$10(1 + 500\beta) = 500$$
$$10 + 500\beta = 500$$
$$500\beta = 500 - 10$$
$$\beta = \frac{490}{500}$$
$$\beta = 0.98$$

Thus, the required feedback factor β is $\boxed{0.098}$.

Step 4 of 8

Calculate the de-sensitivity factor.

$$D = 1 + A\beta$$

Substitute $500\ \text{V/V}$ for A and 0.098 for β.

$$D = 1 + 500 \times 0.098$$
$$= 1 + 49$$
$$= 50$$

Therefore, the required de-sensitivity factor D is $\boxed{50}$.

Step 5 of 8

Calculate the open loop gain uncertainty for the gain uncertainty of $\pm 10\ \%$.

$$A = 1000\ \text{V/V}$$
$$dA = (1000)\left(\frac{10}{100}\right)$$
$$= 100\ \text{V/V}$$

The closed loop gain uncertainty for the gain uncertainty of $\pm 10\ \%$ is,

$$dA_f = \frac{dA}{(1 + A\beta)^2}$$

Substitute corresponding values.

$$dA_f = \frac{100}{(1 + 1000 \times 0.099)^2}$$
$$= \frac{100}{10000}$$
$$= 0.01\ \text{V/V}$$

Step 6 of 8

Calculate the maximum allowable uncertainty for closed loop gain $A_f = 10\ \text{V/V}$.

$$dA_f = 10\left(\frac{x}{100}\right)$$
$$0.01 = 10\left(\frac{x}{100}\right) \qquad (\text{since, } dA_f = 0.01)$$
$$1 = 10x$$
$$x = \frac{1}{10}$$
$$= 0.1$$

Thus, the required closed loop gain uncertainty x is $\boxed{0.1\ \%}$.

Step 7 of 8

Calculate the open loop gain uncertainty for the gain uncertainty of $\pm 10\ \%$.

$$A = 500\ \text{V/V}$$
$$dA = 500\left(\frac{10}{100}\right)$$
$$= 50\ \text{V/V}$$

Calculate the closed loop gain uncertainty for the gain uncertainty of $\pm 10\ \%$.

$$dA_f = \frac{dA}{(1 + A\beta)^2}$$
$$= \frac{50}{(1 + 500 \times 0.098)^2}$$
$$= \frac{50}{2500}$$
$$= 0.02\ \text{V/V}$$

Step 8 of 8

Calculate the maximum allowable uncertainty for closed loop gain $A_f = 10\ \text{V/V}$.

$$dA_f = 10\left(\frac{x}{100}\right)$$
$$0.02 = \frac{10x}{100}$$
$$2 = 10x$$
$$x = \frac{2}{10}$$
$$= 0.2$$

Therefore, the required closed loop gain uncertainty x is $\boxed{0.2\%}$.

Step 1 of 5

Refer to the example 10.5 in the text book.

Determine the value of feedback resistance.

$$R_F = \frac{1}{A_f}$$

Substitute 5 mA/V for A_f.

$$R_F = \frac{1}{5\times10^{-3}}$$
$$= 200\ \Omega$$

Therefore, the feedback resistance of the feedback transcondutance amplifier is

$\boxed{200\ \Omega}$.

Step 2 of 5

Determine the value of closed loop gain of the feedback transcondutance amplifier . $A_f = \frac{A_i g_{m2}}{1+A_i g_{m2} R_F}$

Substitute 200 V/V for A_i, 2 mA/V for g_{m2} and $200\ \Omega$ for R_F.

$$A_f = \frac{(200)(2\times10^{-3})}{1+(200)(2\times10^{-3})(200)}$$
$$= \frac{0.4}{1+80}$$
$$= 4.938\ \text{mA/V}$$

Therefore, the closed loop gain of the feedback transcondutance amplifier is $\boxed{4.938\ \text{mA/V}}$.

Step 3 of 5

Determine the input resistance of the feedback transcondutance amplifier.

$$R_{in} = R_{id} + R_F + A_i g_{m2} R_F R_{id}$$

Substitute $100\,\text{k}\Omega$ for R_{id}, 200 V/V for A_i, 2 mA/V for g_{m2} and $200\ \Omega$ for R_F.

$$R_{in} = (100\times10^3) + 200 + (200(2\times10^{-3})(200)(100\times10^3))$$
$$= (100\times10^3) + 200 + (8\times10^6)$$
$$= (100\times10^3) + 200 + (8\times10^6)$$
$$\approx 8.1\ \text{M}\Omega$$

Therefore, the input resistance of the feedback transcondutance amplifier is $\boxed{8.1\ \text{M}\Omega}$.

Step 4 of 5

Determine the output resistance of the feedback transcondutance amplifier.

$$R_{out} = r_{o2} + R_F + A_i g_{m2} R_F r_{o2}$$

Substitute $20\,\text{k}\Omega$ for r_{02}, 200 V/V for A_i, 2 mA/V for g_{m2} and $200\ \Omega$ for R_F.

$$R_{out} = (20\times10^3) + 200 + ((200)(2\times10^{-3})(200)(20\times10^3))$$
$$= (20\times10^3) + 200 + (16\times10^5)$$
$$= 1.62\ \text{M}\Omega$$

Therefore, the output resistance of the feedback transcondutance amplifier is $\boxed{1.62\ \text{M}\Omega}$.

Step 5 of 5

The transconductance of the feedback transconductance amplifier is given by

If the value of g_{m2} drops by 50% then,

$$g_{m2} = 1\ \text{mA/V}$$

Determine the closed loop gain of the feedback transconductance amplifier.

$$A_f = \frac{A_i g_{m2}}{1+A_i g_{m2} R_f}$$
$$= \frac{200(1\times10^{-3})}{1+(200)(1\times10^{-3})(200)}$$
$$= \frac{0.2}{1+40}$$
$$= 4.878\ \text{mA/V}$$

Determine the percentage change in the closed loop gain.

$$\%A_f = \left(\frac{4.878-4.938}{4.878}\right)\times100$$
$$= -\left(\frac{0.06}{4.878}\right)\times100$$
$$= -1.23\%$$

Therefore, the percentage change in the closed loop gain is $\boxed{-1.23\%}$.

Step 1 of 3

Consider the following data:

$$\frac{dA_f}{A_f} = 0.1\% \text{ and } \frac{dA}{A} = \pm 10\%$$

The closed loop system de-sensitivity expression is,

$$\frac{dA_f}{A_f} = \left(\frac{1}{1 + A\beta}\right)\left(\frac{dA}{A}\right)$$

$$0.1 = \left(\frac{1}{1 + A\beta}\right)(10)$$

$$0.1(1 + A\beta) = 10$$

$$0.1 + 0.1A\beta = 10$$

$$0.1A\beta = 10 - 0.1$$

$$A\beta = \frac{9.9}{0.1}$$

$$A\beta = 99$$

Hence, the required value of $A\beta$ is $\boxed{99}$.

Step 2 of 3

For $\pm 0.1\%$ change in closed loop gain A_f leads to $\pm 10\%$ change in open loop gain A.

When closed loop A_f is 10 leads to open loop gain A is 100.

Calculate the lowest gain.

$$\begin{aligned}\text{Lowest gain} &= A - 10\%A \\ &= 100 - 0.1(100) \\ &= 100 - 10 \\ &= 90\end{aligned}$$

Therefore, the required lowest gain is $\boxed{90\text{ V}}$.

Step 3 of 3

Calculate the feedback gain β.

$$\begin{aligned}\beta &= \frac{A\beta}{A} \\ &= \frac{99}{100} \\ &= 0.99\end{aligned}$$

Therefore, the required feedback gain β is $\boxed{0.99}$.

Step 1 of 3

11.013E

Break the feedback loop at G_2 in Figure 11.20(a) in the text book and apply voltage V_{g2}.

$$V_{id} = -V_i \frac{R_{id}}{R_{id} + R_s + R_F}$$

$$V_{g2} = A_1 V_{id}$$

$$I_0 = -g_{m2} V_{g2} \frac{r_{o2}}{r_{o2} + R_L + R_F}$$

Step 2 of 3

Determine the gain *A*.

$$A = \frac{I_0}{V_i}$$

$$= \frac{-g_{m2} V_{g2} \dfrac{r_{o2}}{r_{o2} + R_L + R_F}}{\dfrac{-V_{g2}}{A_1 R_{id}} \left(R_{id} + R_s + R_F \right)}$$

$$= \frac{g_{m2} \left(A_1 R_{id} \right) \dfrac{r_{o2}}{r_{o2} + R_L + R_F}}{\left(R_{id} + R_s + R_F \right)}$$

$$= \left(A_1 g m_2 \right) \left(\frac{R_{id}}{R_{id} + R_s + R_F} \right) \left(\frac{r_{o2}}{r_{o2} + R_{L+} R_F} \right)$$

Step 3 of 3

Apply the conditions, $R_F \ll \left(R_{id} + R_s \right)$ and $r_{o2} \gg R_L + R_F$ to get the gain.

$$A \simeq A_1 g_{m2}$$

The output resistance R_0 by setting $V_s = 0$.

$$R_0 = r_{02} + R_L + R_F$$

The input resistance R_i is,

$$R_i = R_s + R_{id} + R_F$$

Thus, the loop gain $(A\beta)$ is,

$$A\beta = \left(A_1 g m_2 \right) \left(\frac{R_{id}}{R_{id} + R_s + R_F} \right) \left(\frac{r_{o2}}{r_{o2} + R_{L+} R_F} \right)$$

$$\simeq A_1 g_{m2} R_F$$

Therefore, the expression is, $\boxed{A\beta \simeq A_1 g_{m2} R_F}$.

Step 1 of 3

11.013P

Consider the following data:

$$\frac{dA_f}{A_f} = 1\%$$

$$\frac{dA}{A} = 10$$

The formula for the change in closed loop gain is,

$$\frac{dA_f}{A_f} = \left(\frac{1}{1+A\beta}\right)\left(\frac{dA}{A}\right)$$

$$1 = \left(\frac{1}{1+A\beta}\right)10$$

$$1 + A\beta = 10$$

$$A\beta = 9$$

Step 2 of 3

Calculate the closed loop gain.

Consider that $A_f = 25$. Therefore,

$$A_f = \frac{A}{1+A\beta}$$

$$25 = \frac{A}{1+9}$$

$$A = 25(10)$$

$$= 250 \text{ V/V}$$

Now, calculate the lowest gain.

$$A - 10\% = 250 - \left(\frac{10}{100}\right)250$$

$$= 250 - 25$$

$$= 225 \text{ V/V}$$

Hence, the lowest gain of the amplifier is $\boxed{225 \text{ V/V}}$.

The nominal gain of the amplifier is $\boxed{250 \text{ V/V}}$.

Step 3 of 3

Calculate the feedback factor (β).

$$\beta = \frac{A\beta}{A}$$

$$= \frac{9}{250}$$

$$= 0.036$$

Therefore, the required feedback factor β is $\boxed{0.036}$.

Step 1 of 3

Write the equation for the loop gain, $A\beta$.

$$A\beta = (A_1 gm_2)\left(\frac{R_{id}}{R_{id} + R_s + R_F}\right)\left(\frac{r_{o2}}{r_{o2} + R_{L+} R_F}\right)$$

From the conditions, $r_{o2} \gg R_L + R_F$ and $RF \ll R_{id}$

The input resistance is,

$$R_i = R_s + R_{id} + R_F$$

Step 2 of 3

The input resistance with feedback is,

$$R_{if} = R_i(1 + A\beta)$$

Substitute, R_i result in R_{if} expression.

$$R_{if} = R_s + R_{id} + R_F(1 + A\beta)$$

$$= R_s + R_{id} + R_F + A_1 g_{m2} R_F R_{id} \frac{r_{o2}}{r_{o2} + R_L + R_F}$$

Consider the condition, $r_{o2} \gg R_L + R_F$.

$$R_{if} = R_s + R_{id} + R_F + A_1 g_{m2} R_F R_{id}$$

Step 3 of 3

Subtract R_s from R_{if} to obtain R_{in}.

$$R_{in} = R_{id} + R_F + A_1 g_{m2} R_F R_{id}$$

By considering $RF \ll R_{id}$ condition that results,

$$R_{in} = R_{id} + A_1 g_{m2} R_F R_{id}$$

$$= R_{id}(1 + A_1 g_{m2} R_F)$$

Therefore, the required expression is, $\boxed{R_{in} = R_{id}(1 + A_1 g_{m2} R_F)}$.

Step 1 of 14

The gain of a power amplifier employing an output stage varies from 2 to 12.

The overall gain with feedback, $A_f = 100$ V/V

The change in overall gain, $\frac{dA_f}{A_f} = 5\%$

An ideal amplifier is connected to drive the power amplifier.

Step 2 of 14

Calculate the average value of the gain of the power amplifier.

$$A = \frac{2+12}{2}$$
$$= \frac{14}{2}$$
$$= 7$$

Calculate the average change in gain.

$$dA = \frac{12-2}{2}$$
$$= 5$$

Calculate the percentage change in gain.

$$\frac{dA}{A} = \left(\frac{5}{7}\right)(100)$$
$$= 71.43\%$$

Step 3 of 14

Write the formula for the change in closed loop gain.

$$\frac{dA_f}{A_f} = \left(\frac{1}{1+A\beta}\right)\left(\frac{dA}{A}\right)$$

Substitute the known values in the formula.

$$5 = \left(\frac{1}{1+A\beta}\right)(71.43)$$
$$1 + A\beta = 14.286$$
$$A\beta = 13.286$$

Step 4 of 14

Write the formula for closed loop gain.

$$A_f = \frac{A}{1+A\beta}$$

Substitute the known values in the formula.

$$100 = \frac{A}{1+13.286} \qquad A = (14.286)(100)$$
$$= 1428.6 \text{ V/V}$$

Thus, the nominal gain of the ideal amplifier A is $\boxed{1428.6 \text{ V/V}}$.

Step 5 of 14

Calculate the feedback factor from the loop gain value, $A\beta = 13.286$

$$\beta = \frac{13.286}{A}$$
$$= \frac{13.286}{1428.6}$$
$$= 0.0093$$

Thus, the value of the feedback factor β is $\boxed{0.0093}$.

Step 6 of 14

The change in overall gain, $\frac{dA_f}{A_f} = 0.5\%$

Write the formula for the change in closed loop gain.

$$\frac{dA_f}{A_f} = \left(\frac{1}{1+A\beta}\right)\left(\frac{dA}{A}\right)$$

Substitute the known values in this formula.

$$0.5 = \left(\frac{1}{1+A\beta}\right)(71.43)$$
$$1 + A\beta = 142.86$$
$$A\beta = 141.86$$

Step 7 of 14

Write the formula for closed loop gain.

$$A_f = \frac{A}{1+A\beta}$$

Substitute the known values in this formula.

$$100 = \frac{A}{1+141.86}$$
$$A = (142.86)(100)$$
$$A = 14286 \text{ V/V}$$

Thus, the nominal gain of the ideal amplifier A is $\boxed{14286 \text{ V/V}}$.

Step 8 of 14

Calculate the feedback factor from the loop gain value, $A\beta = 141.86$

$$\beta = \frac{141.86}{A}$$
$$= \frac{141.86}{14286}$$
$$= 0.00993$$

Thus, the value of the feedback factor β is $\boxed{0.00993}$.

Step 9 of 14

The overall gain with feedback, $A_f = 10$ V/V

Write the formula for the change in the closed loop gain.

$$\frac{dA_f}{A_f} = \left(\frac{1}{1+A\beta}\right)\left(\frac{dA}{A}\right)$$

Substitute the known values in this formula.

$$5 = \left(\frac{1}{1+A\beta}\right)(71.43)$$
$$1 + A\beta = 14.286$$
$$A\beta = 13.286$$

Step 10 of 14

Write the formula for the closed loop gain.

$$A_f = \frac{A}{1+A\beta}$$

$$10 = \frac{A}{1+13.286}$$
$$A = (14.286)(10)$$
$$A = 142.86 \text{ V/V}$$

Calculate the pre-amplifier gain by dividing the gain by the average value of gain.

$$A_1 = \frac{142.86}{7}$$
$$= 20.4 \text{ V/V}$$

Thus, the required pre-amplifier gain v is $\boxed{20.4 \text{ V/V}}$.

Step 11 of 14

Calculate the feedback factor from the loop gain value, $A\beta = 13.286$

$$\beta = \frac{13.286}{A}$$
$$= \frac{13.286}{142.86}$$
$$= 0.093$$

Thus, the value of the feedback factor β is $\boxed{0.093}$.

Step 12 of 14

For the change in overall gain, $\frac{dA_f}{A_f} = 0.5\%$:

Write the formula for the change in closed loop gain.

$$\frac{dA_f}{A_f} = \left(\frac{1}{1+A\beta}\right)\left(\frac{dA}{A}\right)$$

Substitute the known values in this formula.

$$0.5 = \left(\frac{1}{1+A\beta}\right)(71.43)$$
$$1 + A\beta = 142.86$$
$$A\beta = 141.86$$

Step 13 of 14

Write the formula for closed loop gain.

$$A_f = \frac{A}{1+A\beta}$$

$$10 = \frac{A}{1+141.86}$$
$$A = (142.86)(10)$$
$$A = 1428.6 \text{ V/V}$$

Calculate the pre-amplifier gain by dividing the gain, by the average value of gain.

$$A_1 = \frac{1428.6}{7}$$
$$= 204.086 \text{ V/V}$$

Thus, the required pre-amplifier gain A_1 is $\boxed{204.086 \text{ V/V}}$.

Step 14 of 14

Calculate the feedback factor from the loop gain value, $A\beta = 141.86$

$$\beta = \frac{141.86}{A}$$
$$= \frac{141.86}{1428.6}$$
$$= 0.0993$$

Thus, the value of the feedback factor β is $\boxed{0.0993}$.

Step 1 of 4

The amplifier gain is,

$$A_o = 1000 \pm 30\%$$

The required amplifier gain is,

$$A_f = 100 \pm 1\%$$

To reduce the percentage change in the gain A_o, equate the term $\frac{1}{1+A\beta}$ to 30.

$$\frac{1}{1+A\beta} = \frac{1}{30}$$

The gain of the feedback amplifier is,

$$A_f = \frac{A_o}{1+A\beta}$$

Substitute 1000 for A_o and $1+A\beta$ for 30.

$$\begin{aligned} A_f &= \frac{1000}{30} \\ &= \frac{100}{3} < 100 \end{aligned}$$

Step 2 of 4

The closed loop gain for the single stage is,

$$A_f = \frac{A_o}{1+A\beta_1}$$

$$\begin{aligned} \frac{1}{1+A\beta_1} &= \frac{A_f}{A_0} \\ &= \frac{100}{1000} \\ &= \frac{1}{10} \end{aligned}$$

If the two stages (identical) are connected in cascade form, the gain is,

$$\begin{aligned} \frac{1}{1+A\beta_2} &= \left(\frac{1}{10}\right)\left(\frac{1}{10}\right) \\ &= \frac{1}{100} \end{aligned}$$

Step 3 of 4

Determine the percentage accuracy of the gain variation.

$$\begin{aligned} \textbf{Gain variation} &= \pm\frac{30}{100} \\ &= \pm 0.3\% \end{aligned}$$

The two identical cascade stages have a gain variation of

$$\textbf{Gain variation} = 2 \times (\pm 0.3\%)$$

Thus, the gain variation is $\boxed{\pm 0.6\%}$.

Step 4 of 4

If three stages (identical) are connected in cascade form,

$$\begin{aligned} A_3 &= A_f^{1/3} \\ &= 100^{1/3} \\ &= \sqrt[3]{100} \end{aligned}$$

Thus,

$$1 + A\beta_3 = \frac{1000}{\sqrt[3]{100}}$$

Therefore, each state is accurate with in,

$$\begin{aligned} \textbf{Gain variation} &= \pm\frac{30}{\left(\frac{1000}{\sqrt[3]{100}}\right)} \\ &= \pm 0.14\% \end{aligned}$$

Thus, the gain variation is $\boxed{\pm 0.14\%}$.

Step 1 of 2

11.016E

Refer to the Figure 11.21 (a) in the text book.

The A_f can be expressed as: $A_f = \frac{1}{R_{E2}}\left(1+\frac{R_{E2}+R_F}{R_{E1}}\right)$

Substitute, $R_{E1} = 100\ \Omega, R_{E2} = 100\ \Omega,$ and $A_f = 100\ \text{mA/V}$ in above A_f equation.

$$100\times10^{-3} = \frac{1}{100}\left(1+\frac{100+R_F}{100}\right)$$

$$0.1 = 0.01\left(\frac{100+100+R_F}{100}\right)$$

$$10 = 0.01\left(200+R_F\right)$$

Solve for R_F .

$$10 = 2+0.01R_F$$

$$\Rightarrow 0.01R_F = 10-2$$

$$\Rightarrow R_F = \frac{10-2}{0.01}$$

$$= 800\ \Omega$$

Therefore, the value of resistor R_F is $\boxed{800\Omega}$.

Step 2 of 2

The voltage gain at the output is: $\frac{V_0}{V_S} = -A_f R_{C3}$

Substitute, $A_f = 100\ \text{mA/V}$ and $R_{C3} = 600\ \Omega$.

$$\frac{V_0}{V_S} = -\left(100\times10^{-3}\right)(600)$$

$$= -0.1\times600$$

$$= -60\ \text{V/V}$$

Therefore, the required voltage gain is $\boxed{-60\ \text{V/V}}$.

Step 1 of 5

With the desired drop in closed-loop gain, calculate the closed loop gain of the amplifier.

$$A_{f1} = \frac{A_o}{1 + A_o\beta}$$
$$= 10 - (2\% \text{ of } 10)$$
$$= 10 - 0.2$$
$$= 9.8 \text{ V/V}$$

Thus, the nominal open-loop amplifier gain in the design is $\boxed{9.8 \text{ V/V}}$.

Step 2 of 5

The closed loop gain of the amplifier is,

$$A_{f1} = \frac{A_o}{1 + A_o\beta}$$
$$A_o = A_{f1}(1 + A_o\beta)$$

Substitute 9.8 for A_f in the equation.

$$A_o = 9.8(1 + A_o\beta)$$
$$1 + A_o\beta = \frac{A_o}{9.8}$$
$$A_o\beta = \frac{A_o}{9.8} - 1$$
$$\beta = \frac{A_o}{9.8A_o} - \frac{1}{A_o}$$
$$\beta = \frac{1}{9.8} - \frac{1}{A_o} \quad \text{..... (1)}$$

Step 3 of 5

Calculate the gain at normal full-battery value over the life of the battery.

$$A_{f2} = \frac{2A}{1 + 2A\beta} = 10$$

Substitute β from equation (1) in this equation.

$$A_{f2} = \frac{2A_o}{1 + 2A_o\left(\frac{1}{9.8} - \frac{1}{A_o}\right)} = 10$$
$$10 + 20A_o\left(\frac{1}{9.8} - \frac{1}{A_o}\right) = 2A_o$$
$$1 + 2A_o\left(\frac{1}{9.8} - \frac{1}{A_o}\right) = \frac{2A_o}{10}$$

Rewrite the equation.

$$2A_o\left(\frac{1}{9.8} - \frac{1}{10}\right) = 2 - 1$$

Simplify further to get the new gain.

$$A_o = \frac{10}{0.04082}$$
$$= 245$$

Thus, nominal open loop amplifier gain A_o at 2% drop in closed loop gain is $\boxed{245 \text{ V/V}}$.

Step 4 of 5

Closed loop gain of an amplifier.

$$A_{f1} = \frac{A_o}{1 + A_o\beta}$$

Substitute 9.8 for A_{f1} in the equation.

$$9.8 = \frac{A_o}{1 + A_o\beta_{new}}$$

If β varies by $\pm 1\%$, the worst case for A_f becomes:

$$9.8 = \frac{A_o}{1 + A_o\beta 1.01}$$

So, $\beta_{new} = 1.01\beta$

Step 5 of 5

Recall equation (1).

$$\beta = \frac{1}{9.8} - \frac{1}{A_o}$$

Substitute 245 for A_o in the equation.

$$\beta = \frac{1}{9.8} - \frac{1}{245}$$
$$= \frac{48}{490}$$

Calculate the nominal open loop gain at $\beta = 0.0979$.

$$A_o = 9.8(1 + A_o\beta 1.01)$$

Substitute $\frac{48}{490}$ for β in the equation.

$$A_o = 9.8\left(1 + A_o\left(\frac{48}{490}\right)1.01\right)$$

Solve for A_o .

$$A_o = 322 \text{ V/V}$$

Thus, nominal open loop amplifier gain A_o at $+1\%$ variation in feedback factor β is $\boxed{322 \text{ V/V}}$.

11.017E

We don't have the solution to this problem yet.

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Step 1 of 4

The closed loop gain is,

$$A_f = \frac{A}{1+A\beta}$$

Substitute 100 for A_f .

$$\frac{A}{1+A\beta} = 100$$

$$A = 100(1+A\beta) \quad \text{..... (1)}$$

The feedback amplifier should provide a reduction in A_f to $99\,V/V$ for a reduction in A to one-tenth its nominal value.

That is,

$$\frac{\frac{A}{10}}{1+\frac{A}{10}\beta} = 99$$

$$\frac{A}{10+A\beta} = 99$$

$$A = 99(10+A\beta) \quad \text{..... (2)}$$

Step 2 of 4

Compare equations (1) and (2).

$$100(1+A\beta) = 99(10+A\beta)$$
$$100+100A\beta = 990+99A\beta$$
$$A\beta = 890$$

Hence, the required loop gain $A\beta$ is $\boxed{890}$.

Recall equation (1).

$$A = 100(1+A\beta)$$

Substitute 890 for $A\beta$.

$$A = 100(1+890)$$
$$= 89,100$$

Therefore, the required nominal A is $\boxed{89,100}$.

Step 3 of 4

We need to determine the value of β that should be used.

$$\beta = \frac{\text{loop gain } A\beta}{\text{nominal value of } A}$$
$$= \frac{890}{89,100}$$
$$= 0.01$$

Therefore, the value of β is $\boxed{0.01}$.

Step 4 of 4

The closed loop gain is,

$$A_f = \frac{A}{1+A\beta}$$

Now if A were increased ten times, determine the closed loop gain.

$$A_f = \frac{10A}{1+10A\beta}$$

Substitute $89,100$ for A and 890 for $A\beta$.

$$A_f = \frac{10\times 89,100}{1+10\times 890}$$
$$= \frac{891,000}{8901}$$
$$= 100.10$$

Therefore, the closed loop gain A_f is $\boxed{100.10}$.

The closed loop gain is,

$$A_f = \frac{A}{1+A\beta}$$

This can be re-written as,

$$A_f = \frac{A}{A\left(\frac{1}{A}+\beta\right)}$$
$$= \frac{1}{\left(\frac{1}{A}+\beta\right)}$$

Now if A were made infinity, need to determine the closed loop gain.

Hence,

$$A_f = \frac{1}{\left(\frac{1}{\infty}+\beta\right)}$$
$$= \frac{1}{\beta}$$

Substitute 0.01 for β in the equation.

That implies,

$$A_f = \frac{1}{0.01}$$
$$= 100$$

Hence, the closed loop gain A_f is $\boxed{100}$.

11.018E

We don't have the solution to this problem yet.

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Step 1 of 3

11.018P

Amplifier is having a mid-band gain A_M and a low-frequency response characterized by a pole at $s=-\omega_L$ and a zero at $s=0$. The amplifier is connected in a negative feedback loop with a feedback factor β.

The open loop gain of given amplifier is,

$$A(s)=A_M\frac{(s-0)}{(s+\omega_L)}$$
$$=A_M\frac{s}{s+\omega_L}$$

Step 2 of 3

The closed-loop gain of negative feedback amplifier is,

$$A_f(s)=\frac{A(s)}{1+A(s)\beta}$$
$$=\frac{\frac{A_M s}{s+\omega_L}}{1+\frac{A_M s\beta}{s+\omega_L}}$$
$$=\frac{A_M s}{(s+\omega_L)+A_M s\beta}$$
$$=\frac{A_M s}{s(1+A_M\beta)+\omega_L}$$

Simplify further.

$$A_f(s)=\frac{1}{(1+A_M\beta)}\frac{A_M s}{s+\left(\frac{\omega_L}{1+A_M\beta}\right)}$$
$$=\left(\frac{A_M}{1+A_M\beta}\right)\frac{s}{s+\left(\frac{\omega_L}{1+A_M\beta}\right)}$$

Step 3 of 3

Thus, the feedback amplifier has a mid-band gain of $A_M/(1+A_M\beta)$ and if the open loop gain is characterized by a dominant low frequency pole giving rise to a lower **3-dB** frequency ω_L then the feedback amplifier has a lower **3-dB** frequency ω_{Lf} defined as,

$$\omega_{Lf}=\frac{\omega_L}{1+A_M\beta}$$

Therefore, the amplifier bandwidth is increased by the same factor by which its mid-band gain is decreased to maintain the gain band width product at a constant value.

Step 1 of 15

(a)

Refer to the Figure E10.15 in the text book.

Write the expression of output voltage for large open loop gain.

$$V_o = I_s R_f$$

$$V_o = -I_f R_f \qquad (\text{Since } I_f = -I_s)$$

$$\frac{V_o}{I_f} = -R_f$$

$$A_f = -R_f$$

The feedback gain is,

$$\beta = \frac{1}{A_f}$$

$$= -\frac{1}{R_f}$$

Therefore, the open loop gain is $\boxed{-R_f}$

Step 2 of 15

(b)

Draw the equivalent circuit diagram.

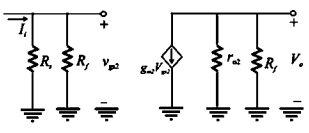


Figure 1

Step 3 of 15

Let us first find the open-loop gain

$$A = \frac{V_o}{I_s}$$

Write the expression of gate source voltage.

$$V_{gs2} = I_s\left(R_f \parallel R_s\right) \quad \ldots\ldots (1)$$

Write the expression of output voltage.

$$V_o = -g_{m2}V_{gs2}\left(R_f \parallel r_o\right) \quad \ldots\ldots (2)$$

Substitute equation (1) in equation (2).

$$V_o = -g_{m2}I_s\left(R_f \parallel R_s\right)\left(R_f \parallel r_o\right)$$

$$V_o = -\left(R_s \parallel R_f\right)g_{m2}I_s\left(R_f \parallel r_o\right)$$

$$\frac{V_o}{I_s} = -\left(R_s \parallel R_f\right)g_{m2}\left(R_f \parallel r_o\right)$$

$$A = -\left(R_s \parallel R_f\right)g_{m2}\left(R_f \parallel r_o\right)$$

Step 4 of 15

Determine the closed loop gain.

$$A_f = \frac{A}{1 + A\beta}$$

Substitute $-\left(R_s \parallel R_f\right)g_{m2}\left(R_f \parallel r_o\right)$ for A and $-\frac{1}{R_f}$ for β.

$$A_f = \frac{-\left(R_s \parallel R_f\right)g_{m2}\left(R_f \parallel r_o\right)}{1+\left(-\left(R_s \parallel R_f\right)g_{m2}\left(R_f \parallel r_o\right)\right)\left(-\frac{1}{R_f}\right)}$$

Therefore, the expression of closed loop gain is $\boxed{\frac{-\left(R_s \parallel R_f\right)g_{m2}\left(R_f \parallel r_o\right)}{1+\left(\left(R_s \parallel R_f\right)g_{m2}\left(R_f \parallel r_o\right)\right)\left(\frac{1}{R_f}\right)}}$.

Step 5 of 15

(c)

Write the expression of input resistance.

The input resistance with feedback R_{if}, is obtained by dividing R_i with $(1 + A\beta)$.

$$R_{if} = \frac{R_i}{1 + A\beta}$$

$$\frac{1}{R_{if}} = \frac{1}{R_i} + \frac{A\beta}{R_i}$$

Substitute $R_s \parallel R_f$ for R_i, $-\left(R_s \parallel R_f\right)g_{m2}\left(R_f \parallel r_o\right)$ for A and $-\frac{1}{R_f}$ for β.

$$\frac{1}{R_{if}} = \frac{1}{R_s \parallel R_f} + \frac{-\left(R_s \parallel R_f\right)g_{m2}\left(R_f \parallel r_o\right)}{\left(R_s \parallel R_f\right)\left(-R_f\right)}$$

Step 6 of 15

Assume the following expressions.

$$g_{m2}\left(R_f \parallel r_o\right) = g'_{m2}$$

Where,

g'_{m2} is lower than the value of g_{m2}, **results in**

$$R_{if} = R_{sf} \parallel R_f \parallel R_s \parallel \left(R_f / g'_{m2}\right)$$

The two terms containing R_f can be combined and substitute g'_{m2} value

$$R_{if} = R_s \parallel \left(\frac{R_f}{\left(1 + g_{m2}\left(R_f \parallel r_o\right)\right)}\right)$$

Since $R_{if} = R_s \parallel R_{in}$, we see that

$$\frac{R_f}{\left(1 + g_{m2}\left(R_f \parallel r_o\right)\right)}$$

Therefore, the expression of input resistance is $\boxed{\frac{R_f}{\left(1 + g_{m2}\left(R_f \parallel r_o\right)\right)}}$.

Step 7 of 15

(d)

Determine the expression of output resistance with feedback.

$$R_{of} = \frac{R_o}{1 + A\beta}$$

$$\frac{1}{R_{of}} = \frac{1}{R_o} + \frac{A\beta}{R_o}$$

Substitute $r_o \parallel R_f$ for R_o.

$$\frac{1}{R_{of}} = \frac{1}{r_o \parallel R_f} + \frac{\left(\left(R_s \parallel R_f\right)g_{m2}\left(R_f \parallel r_o\right)\right)\left(\frac{1}{R_f}\right)}{r_o \parallel R_f}$$

$$R_{of} = R_L \parallel r_0 \parallel \frac{R_f}{1 + g_m\left(R_s + R_f\right)}$$

$$R_{of} = R_L \parallel R_{out}$$

Here,

$$R_{out} = r_0 \parallel \frac{R_f}{1 + g_m\left(R_s + R_f\right)}$$

Therefore, the expression of output resistance is $\boxed{r_0 \parallel \frac{R_f}{1 + g_m\left(R_s + R_f\right)}}$.

Step 8 of 15

(e)

Determine the value of open loop gain.

$$A = -\left(R_s \parallel R_f\right)g_{m2}\left(R_f \parallel r_o\right)$$

Substitute $1\,\text{k}\Omega$ for R_s, $10\,\text{k}\Omega$ for R_f, 5 mA/V for g_{m2} and $20\,\text{k}\Omega$ for r_0.

$$A = -(1000 \parallel 10000)\left(5\times10^{-3}\right)(10000 \parallel 20000)$$

$$= -\left(\frac{10^7}{10^4+10^4}\right)\left(5\times10^{-3}\right)\left(\frac{2\times10^8}{10^4+\left(2\times10^4\right)}\right)$$

$$= -\left(\frac{10^7}{10^3+10^4}\right)\left(5\times10^{-3}\right)\left(\frac{2\times10^8}{10^4+\left(2\times10^4\right)}\right)$$

$$= -30.3\times10^3$$

Therefore, the value of open loop voltage gain is $\boxed{-30.3\times10^3}$.

Determine the value of feedback gain.

$$\beta = -\frac{1}{R_f}$$

Substitute $10\,\text{k}\Omega$ for R_f.

$$\beta = -\frac{1}{10\ \text{k}\Omega}$$

$$= -0.1\ \text{mA/V}$$

Therefore, the value of feedback gain is $\boxed{-0.1\ \text{mA/V}}$.

Step 9 of 15

Determine the value of $A\beta$.

$$A\beta = \left(-30.3\times10^3\right)\left(-0.1\times10^{-3}\right)$$

$$= 3.03$$

Therefore, the value of $A\beta$ is $\boxed{3.03}$.

Step 10 of 15

Determine the value of gain with feedback.

$$A_f = \frac{A}{1 + A\beta}$$

Substitute 30.3×10^3 for A and 3.03 for $A\beta$.

$$A_f = \frac{\left(30.3\times10^3\right)}{1 + 3.03}$$

$$= 7.52\ \text{k}\Omega$$

Therefore, the value of gain with feedback is $\boxed{7.52\ \text{k}\Omega}$.

Step 11 of 15

Determine the value of input impedance.

$$R_i = R_s \parallel R_f$$

Substitute $1\,\text{k}\Omega$ for R_s and $10\,\text{k}\Omega$ for R_f.

$$R_i = 1\,\text{k}\Omega \parallel 10\,\text{k}\Omega$$

$$= \frac{10^7}{10^3+10^7}$$

$$= 909\,\Omega$$

Therefore, the value of input impedance is $\boxed{909\,\Omega}$.

Step 12 of 15

Determine the value of output resistance.

$$R_o = r_o \parallel R_f$$

Substitute $10\,\text{k}\Omega$ for R_f and $20\,\text{k}\Omega$ for r_0.

$$R_o = 20\ \text{k}\Omega \parallel 10\ \text{k}$$

$$= \frac{\left(20\times10^3\right)\left(10\times10^3\right)}{\left(20\times10^3\right)+\left(10\times10^3\right)}$$

$$= 6.67\,\text{k}\Omega$$

Therefore, the value of output resistance is $\boxed{6.67\ \text{k}\Omega}$.

Step 13 of 15

Determine the value of input resistance with feedback amplifier.

$$R_{if} = \frac{R_i}{1 + A\beta}$$

Substitute $909\,\Omega$ for R_i and 3.03 for $A\beta$.

$$R_{if} = \frac{909}{1 + 3.03}$$

$$= 226\ \Omega$$

Therefore, the value of input resistance with feedback amplifier is $\boxed{226\ \Omega}$.

Determine the value of resistance R_{in}.

$$R_{in} = \frac{R_f}{\left[1 + g_m\left(r_o \parallel R_f\right)\right]}$$

Substitute $10\,\text{k}\Omega$ for R_f, 5 mA/V for g_{m2} and $20\,\text{k}\Omega$ for r_0.

$$R_{in} = \frac{10\ \text{k}}{\left[1 + \left(5\times10^{-3}\right)\left(6.67\ \text{k}\right)\right]}$$

$$= \frac{10^4}{1 + 33.35}$$

$$= 291\,\Omega$$

Therefore, the value of resistance R_{in} is $\boxed{291\Omega}$.

Step 14 of 15

Determine the value of resistance R_{of}.

$$R_{of} = \frac{R_o}{1 + A\beta}$$

Substitute $6.67\,\text{k}\Omega$ for R_o, and 3.03 for $A\beta$.

$$R_{of} = \frac{6.67\ \text{k}}{1 + 3.03}$$

$$= \frac{6.67\ \text{k}}{4.03}$$

$$= 1.66\,\text{k}\Omega$$

Therefore, the value of resistance R_{of} is $\boxed{1.66\ \text{k}\Omega}$.

Step 15 of 15

Determine the value of output resistance R_{out}.

$$R_{out} = r_o \parallel \frac{R_f}{1 + g_m\left(R_s \parallel R_f\right)}$$

Substitute $1\,\text{k}\Omega$ for R_f, $10\,\text{k}\Omega$ for R_f, 5 mA/V for g_{m2} and $20\,\text{k}\Omega$ for r_0.

$$R_{out} = 20\ \text{k}\Omega \parallel \frac{10\ \text{k}}{1 + 5\times10^{-3}\left(1\ \text{k} \parallel 10\ \text{k}\right)}$$

$$= \left(20\times10^3\right) \parallel \frac{10^4}{1 + \left(5\times10^{-3}\right)\left(\frac{10}{11}\times10^4\right)}$$

$$= \left(20\times10^3\right) \parallel \left(1.8\times10^3\right)$$

$$= 1.66\,\text{k}\Omega$$

Therefore, the value of output resistance is $\boxed{1.66\ \text{k}\Omega}$.

Step 1 of 3

The closed loop gain of capacitive coupled amplifier is,

$$A_f = \frac{A_0}{1 + A_0\beta}.$$

The band gain of a capacitive coupled amplifier is,

$$A_0 = 1000 \text{ V/V}.$$

Substitute 1000 for A_0 and 10 for A_f.

$$10 = \frac{1000}{1 + A_0\beta}$$
$$1 + A_0\beta = 100$$

Step 2 of 3

The lower 3 dB frequency is,

$$f_L' = \frac{f_L}{1 + A_0\beta}.$$

It has a single band high-frequency pole at 10 kHz and a single low-frequency pole at

100 Hz.

So,

$$f_H = 10 \text{ kHz}$$

$$f_L = 100 \text{ Hz}$$

Substitute 100 for $1 + A_0\beta$ and 100 for f_L.

$$f_L' = \frac{f_L}{1 + A_0\beta}$$
$$= \frac{100}{100} \text{ Hz}$$
$$= 1 \text{ Hz}$$

Thus the lower 3 dB frequency of capacitive coupled amplifier f_L' is $\boxed{1 \text{ Hz}}$.

Step 3 of 3

The upper dB frequency is,

$$f_H' = f_H(1 + A_0\beta)$$
$$= 10 \times 10^3 \times 100$$
$$= 1 \text{ MHz}$$

Thus, the upper 3 dB frequency of capacitive coupled amplifier f_H' is $\boxed{1 \text{ MHz}}$.

11.020E

Step 1 of 5

Refer to the Example 10.8 in the text book.

Determine the value of resistance R_i.

$$R_i = R_s \| R_{id} \| (R_1 + R_2)$$

Substitute ∞ for R_s, ∞ for R_{id}, $10\,\text{k}\Omega$ for R_1 and $90\,\text{k}\Omega$ for R_2.

$$\begin{aligned} R_i &= \infty \| \infty \| (10\,\text{k}\Omega + 90\,\text{k}\Omega) \\ &= 100\,\text{k}\Omega \end{aligned}$$

Determine the value of voltage gain.

$$A = -\mu \left(\frac{R_i}{R_1 \| R_2} \right)$$

Substitute -100 for μ, $100\,\text{k}\Omega$ for R_i, $10\,\text{k}\Omega$ for R_1 and $90\,\text{k}\Omega$ for R_2.

$$\begin{aligned} A &= -100 \left(\frac{100\,\text{k}\Omega}{10\,\text{k}\Omega \| 90\,\text{k}\Omega} \right) \\ &= -100 \left(\frac{100\,\text{k}\Omega}{\left(\frac{900\,\text{k}\Omega}{100} \right)} \right) \\ &= -100 \left(\frac{100\,\text{k}\Omega}{9\,\text{k}\Omega} \right) \\ &= -1111.11\ \text{A/A} \end{aligned}$$

Step 2 of 5

Determine the value of feedback factor.

$$\beta = -\frac{R_1}{R_1 + R_2}$$

Substitute $10\,\text{k}\Omega$ for R_1 and $90\,\text{k}\Omega$ for R_2.

$$\begin{aligned} \beta &= -\frac{10\,\text{k}\Omega}{10\,\text{k}\Omega + 90\,\text{k}\Omega} \\ &= -\frac{10\,\text{k}\Omega}{100\,\text{k}\Omega} \\ &= -0.1\ \text{A/A} \end{aligned}$$

Step 3 of 5

Determine the value of closed loop gain.

$$A_f = \frac{A}{1 + A\beta}$$

Substitute $-1111.11\ \text{A/A}$ for A and -0.1 A/A for β.

$$\begin{aligned} A_f &= -\frac{1111.11}{1 + (1111.11)(0.1)} \\ &= -\frac{1111.11}{1 + (111.111)} \\ &= -9.91\ \text{A/A} \end{aligned}$$

Therefore, the closed loop gain of the feedback current amplifier is $\boxed{-9.91\ \text{A/A}}$.

Step 4 of 5

Determine the value of input resistance.

$$R_{in} = \frac{R_2}{\mu}$$

Substitute -100 for μ and $90\,\text{k}\Omega$ for R_2.

$$\begin{aligned} R_{in} &= \frac{90\,\text{k}\Omega}{100} \\ &= 900\ \Omega \end{aligned}$$

Therefore, the value of input resistance of the feedback current amplifier is $\boxed{900\ \Omega}$.

Step 5 of 5

Determine the value of output resistance.

$$R_{out} = (1 + A\beta)(g_m r_{02})(R_1 \| R_2)$$

Substitute $-1111.11\ \text{A/A}$ for A, -0.1 A/A for β, 5 mA/V for g_m, $20\,\text{k}\Omega$ for r_{02}, $10\,\text{k}\Omega$ for R_1 and $90\,\text{k}\Omega$ for R_2.

$$\begin{aligned} R_{out} &= (111.11 + 1)\left(5 \times 10^{-3}\right)\left(20 \times 10^{3}\right)\left(\frac{10 \times 90}{10 + 90} \right)\left(10^{3}\right) \\ &= (112.11)(100)(9)\left(10^{3}\right) \\ &= 100.899\,\text{M}\Omega \end{aligned}$$

Therefore, the output resistance of the feedback current amplifier is $\boxed{100.899\ \text{M}\Omega}$.

Step 1 of 5

11.020P

Consider the following circuit diagram:

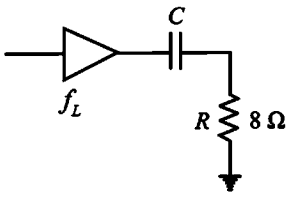


Figure 1

Step 2 of 5

Find the value of capacitor C .

The lower cutoff frequency is,

$$f_L = \frac{1}{2\pi RC}$$

Substitute 8 for R and 100 for f_L .

$$100 = \frac{1}{2\pi(8)(C)}$$

Solve for C .

$$C = \frac{1}{2\pi(8)(100)}$$
$$= 199\times10^{-6}\ \text{F}$$
$$= 199\ \mu\text{F}$$

Therefore, the value of capacitor size C is $\boxed{199\ \mu\text{F}}$

Step 3 of 5

Find the value of f_{Lf} .

$$f_{Lf} = \frac{f_L}{1+A\beta}$$
$$= \frac{f_L}{\left(\frac{A}{A_f}\right)}$$
$$= \frac{100}{\left(\frac{1000}{10}\right)}$$
$$= 1\ \text{Hz}$$

Therefore, the value of f_{Lf} is $\boxed{1\ \text{Hz}}$.

Step 4 of 5

Find the frequency f_L for cut-off frequency, $f_{Lf} = 50\ \text{Hz}$:

$$f_{Lf} = \frac{f_L}{1+A\beta}$$
$$f_{Lf} = \frac{f_L}{\left(\frac{A}{A_f}\right)}$$

Rewrite the equation.

$$f_L = f_{Lf}\left(\frac{A}{A_f}\right)$$
$$= (50)\left(\frac{1000}{10}\right)$$
$$= 5000$$

Therefore, the value of frequency f_L is $\boxed{5\ \text{kHz}}$.

Step 5 of 5

Find the value of the capacitance C .

$$f_L = \frac{1}{2\pi RC}$$

Rewrite the equation.

$C = \frac{1}{2\pi Rf_L}$ Substitute corresponding values.

$$C = \frac{1}{2\pi(8)\left(5\times10^3\right)}$$
$$= \frac{1}{2\pi(8)\left(5\times10^3\right)}$$
$$= 3.98\times10^{-6}\ \text{F}$$
$$= 3.98\ \mu\text{F}$$

Therefore, the final value of capacitor size C is $\boxed{3.98\ \mu\text{F}}$.

Step 1 of 8

Refer to Figure 10.31 (a) in the text book.

Draw the following modified circuit diagram.

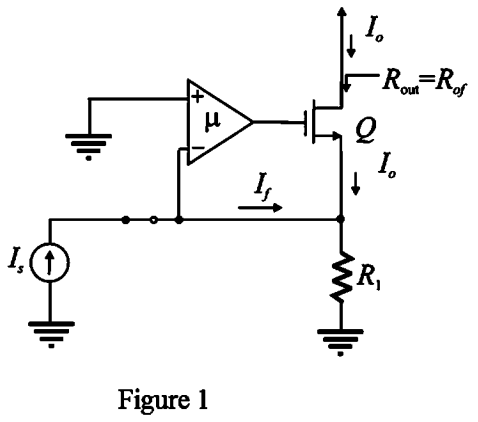


Figure 1

Step 2 of 8

Write the following expressions using Figure 1.

$I_s = I_f$ ..... (1)

$I_f = -I_o$ $\quad$ (Since $V_- = 0\,\text{V}$) ..... (2)

Determine the value of feedback gain.

$$A_f = \frac{I_o}{I_s}$$
$$= -1\,\text{A/A}$$

Therefore, the value of A_f is $\boxed{-1\,\text{A/A}}$.

Step 3 of 8

Draw the following equivalent circuit for $R_s = R_{id} = \infty$.

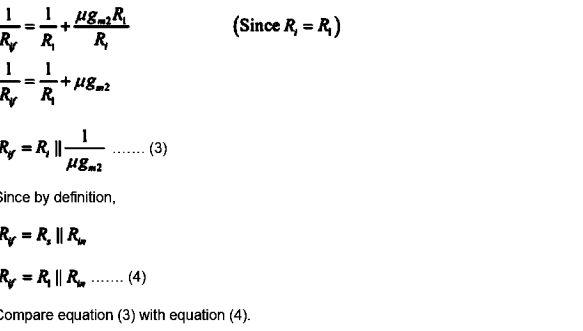


Figure 2

Step 4 of 8

Write the expression of input resistance using Figure 2.

$R_i = R_1$

Write the expression of output resistance using Figure 2.

$R_o = r_{o2}$

Write the expression of A.

$$A = \frac{I_o}{I_i}$$

The expression of output current is,

$I_o = -g_{m2}V_{gs2}$

Substitute μV_{id} for v_{gs2}.

$$I_o = -\mu g_{m2} V_{id}$$
$$= -\mu g_{m2}(I_i R_1) \quad \text{(Since } V_{id} = I_i R_1\text{)}$$
$$\frac{I_o}{I_i} = -\mu g_{m2} R_1$$

Therefore, the expression of current gain is $\boxed{-\mu g_{m2} R_1}$.

Step 5 of 8

Determine the expression of feedback gain.

$I_f = -I_o$

$$\beta = \frac{I_f}{I_o}$$
$$= -1$$

Therefore, the expression of feedback gain is $\boxed{-1}$.

Step 6 of 8

Determine the expression of closed loop gain.

$$A_f = \frac{A}{1+\beta A}$$

Substitute $-\mu g_{m2} R_1$ for A and -1 for β.

$$A_f = \frac{(-\mu g_{m2} R_1)}{1+(-1)(-\mu g_{m2} R_1)}$$
$$= \frac{-\mu g_{m2} R_1}{1+\mu g_{m2} R_1}$$

Therefore, the expression of A_f is $\boxed{-\dfrac{\mu g_{m2} R_1}{1+\mu g_{m2} R_1}}$.

Step 7 of 8

Determine the value of feedback input resistance.

$$R_{if} = \frac{R_i}{1+A\beta}$$
$$\frac{1}{R_{if}} = \frac{1}{R_i} + \frac{A\beta}{R_i}$$

Substitute $-\mu g_{m2} R_1$ for A.

$$\frac{1}{R_{if}} = \frac{1}{R_i} + \frac{\mu g_{m2} R_1}{R_i}$$
$$\frac{1}{R_{if}} = \frac{1}{R_i} + \frac{\mu g_{m2} R_1}{R_i} \quad (\text{Since } R_i = R_1)$$
$$\frac{1}{R_{if}} = \frac{1}{R_1} + \mu g_{m2}$$

$R_{if} = R_1 \| \dfrac{1}{\mu g_{m2}}$ ..... (3)

Since by definition,

$R_{if} = R_s \| R_{in}$

$R_{if} = R_1 \| R_{in}$ ..... (4)

Compare equation (3) with equation (4).

$$R_{in} = \frac{1}{\mu g_{m2}}$$

Therefore, the expression of input resistance is $\boxed{\dfrac{1}{\mu g_{m2}}}$.

Step 8 of 8

Determine the expression of output resistance.

$R_{out} = (1+A\beta)R_o$

Substitute $-\mu g_{m2} R_1$ for A and r_{o2} for R_o.

$$R_{out} = (1+\mu g_{m2} R_1) r_{o2}$$
$$\approx \mu (g_{m2} r_{o2}) R_1$$

Therefore, the expression of output resistance is $\boxed{\mu (g_{m2} r_{o2}) R_1}$.

Step 1 of 8

Consider the negative feedback amount as,

$$1+A\beta=\frac{f_{Hf}}{f_H} \quad \text{..... (1)}$$

Here,

A is the low frequency gain of the dc amplifier.

f_{Hf} is the 3-dB frequency,

f_H is the high frequency pole frequency.

Substitute $1\,\text{MHz}$ for f_{Hf} , $20\,\text{kHz}$ for f_H in equation (1).

$$\begin{aligned}1+A\beta&=\frac{1\times10^6}{20\times10^3}\\&=50\end{aligned}$$

Step 2 of 8

The gain of the feedback amplifier at the first stage is,

$$A_{f_1}=\frac{A}{1+A\beta}$$

Substitute 1000 for A , 50 for $1+A\beta$ in equation.

$$\begin{aligned}A_{f_1}&=\frac{1000}{50}\\&=20\end{aligned}$$

Hence, the feedback gain A_{f_1} , is $\boxed{20}$.

Step 3 of 8

The feedback factor of the first stage is,

$$1+(1000)\beta_1=50$$

$$\beta_1=\frac{(50-1)}{1000}$$

$$\beta_1=49\times10^{-3}$$

The gain of the subsequent stage is,

$$\frac{A_{\text{Total}}}{A_{f_1}}=\frac{1000}{20} \qquad (\text{Since } A_{\text{Total}}=A)$$

$$\frac{A_{\text{Total}}}{A_{f_1}}=50$$

Consider frequency f_{H_2} is as,

$$\begin{aligned}f_{H_2}&=3\times f_{Hf}\\&=3\times1\,\text{MHz}\\&=3\,\text{MHz}\end{aligned}$$

Therefore, the frequency f_{H_2} is $\boxed{3\,\text{MHz}}$.

Step 4 of 8

Therefore, the negative feedback amount is,

$$1+A\beta_2=\frac{f_{H_2}}{f_H}$$

Substitute $3\,\text{MGz}$ for f_{H_2} and $20\,\text{kHz}$ for f_H in equation.

$$\begin{aligned}1+A\beta_2&=\frac{3\,\text{MHz}}{20\,\text{kHz}}\\&=150\end{aligned}$$

Thus the gain of the feedback amplifier at the second stage is,

$$A_{f_2}=\frac{A}{1+A\beta_2}$$

Substitute 1000 for A , 150 for $1+A\beta_2$ in equation.

$$\begin{aligned}A_{f_2}&=\frac{1000}{150}\\&=6.67\end{aligned}$$

Therefore, gain of the feedback amplifier A_{f_2} at second state is, $\boxed{6.67}$.

Step 5 of 8

Now the feedback factor of the second stage is,

$$1+(1000)\beta_2=150$$

$$\beta_2=\frac{150-1}{1000}$$

$$\beta_2=0.149$$

The gain of stage 3 is,

$$A_{f_3}=\frac{\left(\frac{A_{\text{Total}}}{A_{f_1}}\right)}{A_{f_2}}$$

Substitute 50 for $\frac{A_{\text{Total}}}{A_{f_1}}$, 6.67 for A_{f_2} in equation.

$$\begin{aligned}A_{f_3}&=\frac{50}{6.67}\\&=7.5\end{aligned}$$

Therefore, the gain of the third stage is,

$$\boxed{A_{f_3}=7.5}$$

Step 6 of 8

The negative feedback gain is,

$$\begin{aligned}1+A\beta&=\frac{A_{\text{Total}}}{A_{f_3}}\\&=\frac{1000}{7.5}\\1+A\beta&=133.33\end{aligned}$$

The frequency f_{H_3} is,

$$\begin{aligned}f_{H_3}&=f_H(1+A\beta)\\&=(20\times10^3)(133.33)\end{aligned}$$

Therefore, the frequency f_{H_3} is $\boxed{2.66\,\text{MHz}}$.

This frequency is safe and well above $1\,\text{MHz}$.

Step 7 of 8

The feedback factor of the third stage is,

$$1+(1000)\beta_3=133.33$$

That implies,

$$\begin{aligned}\beta_3&=\frac{(133.33-1)}{(1000)}\\&=0.13233\end{aligned}$$

Step 8 of 8

The actual 3-dB frequency, f_{Hf} is,

$$f_{Hf}=\frac{1}{\left[\frac{1}{f_{H_1}^2}+\frac{1}{f_{H_2}^2}+\frac{1}{f_{H_3}^2}\right]}$$

Substitute $1\,\text{MGz}$ for f_{H_1} , $3\,\text{MGz}$ for f_{H_2} and $2.66\,\text{MHz}$ for f_{H_3} in equation

$$\begin{aligned}f_{Hf}&=\frac{1}{\left[\left(\frac{1}{1\times10^6}\right)^2+\left(\frac{1}{3\times10^6}\right)^2+\left(\frac{1}{2.66\times10^6}\right)^2\right]}\\&=\frac{1}{10^{-12}+0.11\times10^{-12}+0.141\times10^{-12}}\\&=\frac{1}{1.251\times10^{-12}}\\&=799.36\,\text{GHz}\end{aligned}$$

Therefore, the frequency f_{Hf} is $\boxed{799.36\,\text{GHz}}$.

Thus, using a four stage amplifier would allow lower stage gains and higher pole frequencies.

Step 1 of 4

Consider the open loop transfer function of a feedback amplifier as,

$$A(s)=\left(\frac{10}{1+\frac{s}{10^4}}\right)^3$$

Substitute $j\omega$ for s in the transfer function.

$$A(j\omega)=\left(\frac{10}{1+\frac{j\omega}{10^4}}\right)^3$$

The phase angle is given by,

$$\begin{aligned}\phi &= -\tan^{-1}\left(\frac{\omega}{10^4}\right)^3 \\ &= -3\tan^{-1}\left(\frac{\omega}{10^4}\right)\end{aligned}$$

Step 2 of 4

Consider the phase shift, $\phi = 180°$

Equate the phase angle to $180°$.

$$180° = -3\tan^{-1}\left(\frac{\omega_{180°}}{10^4}\right)$$

$$\tan^{-1}\left(\frac{\omega_{180°}}{10^4}\right) = -60°$$

$$\frac{\omega_{180°}}{10^4} = \tan(-60°)$$

$$\frac{\omega_{180°}}{10^4} = -\sqrt{3}$$

Consider positive value of frequency. Hence, the frequency is,

$$\omega_{180°} = \sqrt{3}\times 10^4 \ \frac{\text{rad}}{\text{s}}$$

Therefore, the frequency at which the phase shift is $180°$ is $\boxed{\omega_{180°} = \sqrt{3}\times 10^4 \ \frac{\text{rad}}{\text{s}}}$.

Step 3 of 4

Determine the feedback factor.

The feedback amplifier is stable if the loop gain, $A\beta < 1$ and is unstable if $A\beta \geq 1$.

Equate $A(j\omega_{180°})\beta$ to 1 to get the critical value of the feedback factor.

The critical value of feedback factor is given by,

$$\begin{aligned}\beta_{cr} &= \frac{1}{\left|A(j\omega_{180°})\right|} \\ &= \frac{1}{\left|\left(\frac{10}{1+\frac{j\omega_{180°}}{10^4}}\right)^3\right|}\end{aligned}$$

Substitute $\sqrt{3}\times 10^4 \ \frac{\text{rad}}{\text{s}}$ for $\omega_{180°}$ in equation.

$$= \frac{1}{\left|\left(\frac{10}{1+\frac{j\left(\sqrt{3}\times 10^4\right)}{10^4}}\right)^3\right|}$$

$$\begin{aligned}\beta_{cr} &= \frac{1}{\left|\left(\frac{10}{1+j\sqrt{3}}\right)^3\right|} \\ &= \frac{1}{\left\{\frac{10^3}{\left|\left[1+\left(\sqrt{3}\right)^2\right]^{\frac{3}{2}}\right|}\right\}} \\ &= \frac{4^{\frac{3}{2}}}{10^3} \\ &= 0.008\end{aligned}$$

Thus, the critical value of the feedback factor of feedback amplifier is $\boxed{\beta_{cr} = 0.008}$.

Step 4 of 4

The closed loop gain $A_{f_{\text{clos}}}$ is given by,

$$A_{f_{\text{clos}}} = \frac{1}{\beta}$$

Substitute 0.009 for β in equation.

$$\begin{aligned}A_{f_{\text{clos}}} &= \frac{1}{0.009} \\ &= 111.1\end{aligned}$$

Therefore, the closed loop gain $A_{f_{\text{clos}}}$ is $\boxed{111.1}$.

Step 1 of 7

Refer to the block diagram of negative feedback amplifier in Figure 10.4 in the textbook.

The overall output voltage of the amplifier is,

$$V_0 = V_S \frac{A_1 A_2}{1 + A_1 A_2 \beta} + V_N \frac{A_1}{1 + A_1 A_2 \beta}$$ Here,

$$= V_{Sf} + V_{Nf}$$

The voltage V_{Sf} is the closed loop voltage due to the source voltage V_S.

The voltage V_{Nf} is the closed loop voltage due to the interference voltage V_N.

The gain, $A_1 = 0.9\ \text{V/V}$.

The power supply ripple, $V_N = +1\ \text{V}$.

The closed loop gain, $A_f = 10\ \text{V/V}$.

Step 2 of 7

Write the expression for closed loop gain due to the interference voltage.

$$V_0 = V_N \frac{A_1}{1 + A_1 A_2 \beta}$$

Substitute 100×10^{-3} for V_0, 1 for V_N, 1000 for A_1 and 0.9 for A_2.

$$100 \times 10^{-3} = 1 \times \frac{0.9}{1 + (0.9 A_2 \beta)}$$

$$100 \times 10^{-3} (1 + 0.9 A_2 \beta) = 0.9$$

$$0.1 + (0.09) A_2 \beta = 0.9$$

Solve for $A_2\beta$.

$$A_2 \beta = \frac{0.8}{0.09}$$

$$= 8.88$$

Step 3 of 7

Write the expression for closed loop gain due to source voltage.

$$V_0 = V_S \frac{A_1 A_2}{1 + A_1 A_2 \beta}$$

$$\frac{V_0}{V_S} = \frac{A_1 A_2}{1 + A_1 A_2 \beta}$$

$$A_f = \frac{A_1 A_2}{1 + A_1 A_2 \beta}$$

Substitute 10 for A_f, 8.88 for $A_2\beta$ and 0.9 for A_1.

$$10 = \frac{(0.9) A_2}{1 + (0.9)(8.88)}$$

$$10 + 79.92 = 0.9 A_2$$

Solve for A_2.

$$A_2 \approx 100$$

Thus, the gain of the low-ripple preamplifier A_2 at $100\ \text{mV}$ is $\boxed{100}$.

The loop gain is,

$$A_2 \beta = 8.88.$$

Substitute 100 for A_2.

$$\beta = \frac{8.88}{100}$$

$$= 0.0888$$

Thus, feedback factor β at $100\ \text{mV}$ is $\boxed{0.0888}$.

Step 4 of 7

Write the expression for the closed loop gain due to the interference voltage.

$$V_0 = V_N \frac{A_1}{1 + A_1 A_2 \beta}$$

Substitute 10×10^{-3} for V_0, 1 for V_N, 1000 for A_1 and 0.9 for A_2 in the equation.

$$10 \times 10^{-3} = 1 \times \frac{0.9}{1 + (0.9 A_2 \beta)}$$

$$10 \times 10^{-3} (1 + 0.9 A_2 \beta) = 0.9$$

$$0.01 + (9 \times 10^{-3}) A_2 \beta = 0.9$$

Solve for $A_2\beta$.

$$A_2 \beta = \frac{0.89}{9 \times 10^{-3}}$$

$$= 98.88$$

Step 5 of 7

Write the expression for closed loop gain due to source voltage.

$$V_0 = V_S \frac{A_1 A_2}{1 + A_1 A_2 \beta}$$

$$\frac{V_0}{V_S} = \frac{A_1 A_2}{1 + A_1 A_2 \beta}$$

$$A_f = \frac{A_1 A_2}{1 + A_1 A_2 \beta}$$

Substitute 10 for A_f, 98.88 for $A_2\beta$ and 0.9 for A_1 in the equation.

$$10 = \frac{(0.9) A_2}{1 + (0.9)(98.88)}$$

Solve for A_2.

$$A_2 \approx 1000$$

Thus, the gain of the low-ripple preamplifier A_2 at $10\ \text{mV}$ is $\boxed{1000}$.

The loop gain is,

$$A_2 \beta = 98.88.$$

Substitute 1000 for A_2.

$$\beta = \frac{98.88}{1000}$$

$$= 0.09888$$

Thus, feedback factor β at $10\ \text{mV}$ is $\boxed{0.09888}$.

Step 6 of 7

Write the expression for closed loop gain due to the interference voltage.

$$V_0 = V_N \frac{A_1}{1 + A_1 A_2 \beta}$$

Substitute 1×10^{-3} for V_0, 1 for V_N, 1000 for A_1 and 0.9 for A_2.

$$1 \times 10^{-3} = 1 \times \frac{0.9}{1 + (0.9 A_2 \beta)}$$

$$1 \times 10^{-3} (1 + 0.9 A_2 \beta) = 0.9$$

$$1 \times 10^{-3} + (9 \times 10^{-4}) A_2 \beta = 0.9$$

Solve for $A_2\beta$.

$$A_2 \beta = \frac{0.899}{9 \times 10^{-4}}$$

$$= 998.88$$

Step 7 of 7

Write the expression for closed loop gain due to source voltage.

$$V_0 = V_S \frac{A_1 A_2}{1 + A_1 A_2 \beta}$$

$$\frac{V_0}{V_S} = \frac{A_1 A_2}{1 + A_1 A_2 \beta}$$

$$A_f = \frac{A_1 A_2}{1 + A_1 A_2 \beta}$$

Substitute 10 for A_f, 998.88 for $A_2\beta$ and 0.9 for A_1.

$$10 = \frac{(0.9) A_2}{1 + (0.9)(988.88)}$$

Solve for A_2.

$$A_2 \approx 10000$$

Thus, the gain of the low-ripple preamplifier A_2 at $1\ \text{mV}$ is $\boxed{10000}$.

The loop gain is,

$$A_2 \beta = 998.88.$$

Substitute 10000 for A_2.

$$\beta = \frac{998.88}{10000}$$

$$= 0.099888$$

Thus, feedback factor β at $1\ \text{mV}$ is $\boxed{0.099888}$.

Step 1 of 3

11.023E

Calculate the factor by which the pole is shifted by the feedback.

$$\begin{aligned} 1+A_0\beta &= 1+10^5(0.01) \\ &= 1+10^3 \\ &= 1001 \end{aligned}$$

Therefore, the factor by which the pole is shifted by the feedback is $\boxed{1001}$.

Step 2 of 3

Calculate the frequency to which the feedback is shifted.

$$f_{Pf} = f_P(1+A_0\beta)$$

Substitute 100 Hz for f_P and 1001 for $1+A_0\beta$.

$$\begin{aligned} f_{Pf} &= (100)(1001) \\ &= 100{,}100 \text{ Hz} \\ &= 100.1 \text{ kHz} \end{aligned}$$

Therefore, the frequency to which the feedback is shifted is $\boxed{100.1 \text{ kHz}}$.

Step 3 of 3

Consider the feedback factor is changed to a value such that the closed-loop gain is +1.

Equate closed-loop gain to 1. Thus the feedback factor, $\beta = 1$.

Determine the frequency to which the feedback is shifted.

$$f'_{Pf} = f_P(1+A_0\beta)$$

Substitute 100 Hz for f_P and 10^5 for A_0 and 1 for β.

$$\begin{aligned} f'_{Pf} &= (100)\left[1+10^5(1)\right] \\ &= 10{,}000{,}100 \text{ Hz} \\ &= 10 \text{ MHz} \end{aligned}$$

Therefore, the frequency to which the feedback is shifted is $\boxed{10 \text{ MHz}}$.

Step 1 of 5

11.023P

Design a feedback amplifier using a feedback loop connected around a two-stage amplifier. First stage is a direct-coupled small-signal amplifier with a high upper 3-dB frequency. The second stage is a power output stage.

The upper 3 dB frequency is,

$$\omega_{Hf} = \omega_H (1 + A_M \beta)$$

The upper 3 dB frequency is $\omega_H = 8\,\text{kHz}$

The upper 3-dB frequency of the feedback amplifier is $\omega_{Hf} = 40\,\text{kHz}$

Substitute corresponding values.

$$40 \times 10^3 = (8 \times 10^3)(1 + A_M \beta)$$

Rewrite the equation.

$$(1 + A_M \beta) = \frac{40 \times 10^3}{8 \times 10^3}$$
$$(1 + A_M \beta) = 5$$

Step 2 of 5

The mid-band gain of the feedback amplifier is,

$$A_f(s) = \frac{A_M}{1 + A_M \beta}$$

The mid-band gain of the feedback amplifier should be $100\,\text{V/V}$

$$100 = \frac{A_M}{1 + A_M \beta}$$

Find mid-band gain.

$$A_M = 100(1 + A_M \beta)$$

Substitute 5 for $1 + A_M \beta$ in the equation.

$$A_M = (100)(5)$$
$$= 500$$

Step 3 of 5

Overall voltage gain of two-stage amplifier is,

$$A_M = A_1 \cdot A_2$$

The mid-band gain is $A_2 = 10\,\text{V/V}$

Substitute 500 for A_M and 10 for A_2 in the equation.

$$500 = A_1 \cdot (10)$$

Solve for A_1.

$$A_1 = \frac{500}{10}$$
$$= 50$$

Hence, the required gain A_1 of small signal amplifier is $\boxed{50\,\text{V/V}}$.

Step 4 of 5

Determine the value of β.

$$(1 + A_M \beta) = 5$$

Substitute $A_1 \cdot A_2$ for A_M.

$$(1 + A_1 A_2 \beta) = 5$$
$$A_1 A_2 \beta = 4$$

Solve for feedback factor β

$$\beta = \frac{4}{A_1 A_2}$$

Substitute 50 for A_1 and 10 for A_2 in the equation.

$$\beta = \frac{4}{(50)(10)}$$
$$= 0.008$$

Hence, the value of β is $\boxed{0.008}$.

Step 5 of 5

Determine the lower 3-dB frequency of overall amplifier.

The lower 3-dB frequency of the feedback amplifier is,

$$\omega_{Lf} = \frac{\omega_L}{(1 + A_M \beta)}$$

The lower 3 dB frequency is $\omega_L = 80\,\text{Hz}$

Substitute 5 for $(1 + A_M \beta)$ and 80 for ω_L in the equation.

$$\omega_{Lf} = \frac{80}{5}$$
$$= 16$$

Hence, the lower 3-dB frequency of overall amplifier ω_{Lf} is $\boxed{16\,\text{Hz}}$.

Step 1 of 4

Refer to equation 10.92 in the text book.

Write the condition for the poles of the closed-loop amplifier coincide.

$$(\omega_{P1}+\omega_{P2})^2-4(1+A_0\beta)\omega_{P1}\omega_{P2}=0$$

Substitute 10^4 rad/s for ω_{P1} and 10^6 rad/s for ω_{P2} and 100 for A_0.

$$(10^4+10^6)^2-4(1+100\beta)(10^4)(10^6)=0$$
$$4(1+100\beta)(10^4)(10^6)=(10^4+10^6)^2$$
$$1+100\beta=\frac{(10^4+10^6)^2}{4(10^4)(10^6)}$$
$$1+100\beta=25.5025$$
$$100\beta=24.5025$$
$$\beta=\frac{24.5025}{100}$$
$$\beta=0.245$$

Therefore, the feedback factor at which the poles of the closed-loop amplifier coincide is $\boxed{0.245}$.

Step 2 of 4

Refer to equation 10.94 in the text book.

Write the expression of quality factor.

$$Q=\frac{(1+A_0\beta)\omega_{P1}\omega_{P2}}{\omega_{P1}+\omega_{P2}}$$

Substitute 10^4 rad/s for ω_{P1} and 10^6 rad/s for ω_{P2} and 100 for A_0.

$$Q=\frac{\sqrt{[1+100(0.245)](10^4)(10^6)}}{10^4+10^6}$$
$$=\frac{\sqrt{(25.5025)(10^4)(10^6)}}{10^4+10^6}$$
$$=0.5$$

Therefore, the quality factor at which the poles of the closed-loop amplifier coincide is $\boxed{0.5}$.

Step 3 of 4

For a maximally flat response the quality factor, $Q=0.707$.

$$Q=\frac{(1+A_0\beta)\omega_{P1}\omega_{P2}}{\omega_{P1}+\omega_{P2}}$$

Substitute 0.707 for Q, 10^4 rad/s for ω_{P1}, 10^6 rad/s for ω_{P2} and 100 for A_0.

$$0.707=\frac{\sqrt{[1+100\beta](10^4)(10^6)}}{10^4+10^6}$$
$$0.707=\frac{\left(\sqrt{1+100\beta}\right)10^5}{10^4+10^6}$$
$$\sqrt{1+100\beta}=\frac{0.707(10^4+10^6)}{10^5}$$
$$\sqrt{1+100\beta}=7.1407$$
$$1+100\beta=50.9895$$
$$100\beta=49.9895$$
$$\beta=0.499895$$
$$\beta\approx 0.5$$

Therefore, the feedback factor for a maximally flat response is $\boxed{0.5}$.

Step 4 of 4

Calculate the low-frequency closed-loop gain for maximally flat response.

$$A=\frac{A_0}{1+A_0\beta}$$

Substitute 100 for A_0, 0.5 for β.

$$A=\frac{100}{1+(100)(0.5)}$$
$$=\frac{100}{1+50}$$
$$=1.96\ \text{V/V}$$

Therefore, the low-frequency closed-loop gain for maximally flat response is $\boxed{1.96\ \text{V/V}}$.

Step 1 of 5

11.024P

Refer to the circuit diagram of complementary BJT follower in Figure P10.22 (a) in the textbook.

Refer to the transfer characteristics of complementary BJT follower in Figure P10.22 (b) in the textbook.

The BJT follower is driven by the output of a differential amplifier of gain 100. Differential amplifier positive input terminal is connected to the input signal source v_s. Differential amplifier negative-input terminal is connected to the emitters of the follower.

Draw the desired circuit diagram.

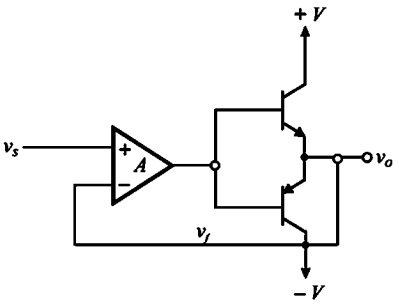


Figure 1

Step 2 of 5

The transfer characteristics, the output is zero during the input range $-0.7\,\text{V} \le v_i \le 0.7\,\text{V}$ and hence is called the "dead band".

This dead band leads to the crossover distortion and due to the feedback; the dead band will be narrowed by a factor equal to the amount of feedback.

Since, the amount of feedback is $1+A\beta$ and hence the dead band will be narrowed by a factor equal to $1+A\beta$.

The feedback factor β is,

$$\beta = \frac{v_f}{v_o}$$

But, from the circuit, it is clear that $v_f = v_o$.

That implies,

$$\beta = \frac{v_o}{v_o}$$
$$= 1$$

Step 3 of 5

Therefore,

$$1+A\beta = 1+A$$

Given, $A(=100) \gg 1$

Hence,

$$1+A \approx A$$

Hence the dead band will be narrowed by a factor equal to A.

We have the input range as $-0.7\,\text{V} \le v_i \le 0.7\,\text{V}$

Now due to the feedback, the input is v_s and the range of v_s is:

$$\frac{-0.7\,\text{V}}{A} \le v_s \le \frac{0.7\,\text{V}}{A}$$
$$\frac{-0.7\,\text{V}}{100} \le v_s \le \frac{0.7\,\text{V}}{100}$$
$$-0.7\times10^{-2}\,\text{V} \le v_s \le 0.7\times10^{-2}\,\text{V}$$
$$-7\,\text{mV} \le v_s \le 7\,\text{mV}$$

Hence the limits of the dead band are $\boxed{\pm 7\,\text{mV}}$.

Step 4 of 5

The slope of the steepest segment is,

$$A_f = \frac{A}{1+A\beta}$$
$$= \frac{100}{1+(100)(1)}$$
$$= 0.99$$

Since the slope of the steepest segment is equal to the closed loop gain or gain of the feedback amplifier, hence

$$\text{gain} = \text{slope}$$
$$= 0.99$$

Therefore, the gains outside the dead band are: $\boxed{0.99}$

Step 5 of 5

Draw the transfer function characteristics, v_o **versus** v_s is shown in Figure 2.

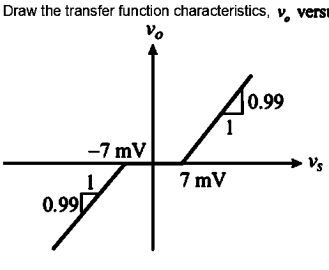


Figure 2

Thus, transfercharacteristics of resulting feedback amplifier is drawn.

Step 1 of 3

11.025E

Write the expression of open loop transfer function of a feedback amplifier.

$$A(s)=\left(\frac{10}{1+\frac{s}{10^4}}\right)^3$$

Determine the closed loop poles.

$$1+A(s)\beta=0$$

$$1+\left[\frac{10^3}{\left(1+\frac{s}{10^4}\right)^3}\right]\beta=0$$

$$\frac{\left(1+\frac{s}{10^4}\right)^3+10^3\beta}{\left(1+\frac{s}{10^4}\right)^3}=0$$

$$\left(1+\frac{s}{10^4}\right)^3+10^3\beta=0$$

Expand the term with cubic power.

$$1+\frac{s^3}{10^{12}}+\frac{3s^2}{10^8}+\frac{3s}{10^4}+10^3\beta=0$$

Take $s_1=\frac{s}{10^4}$ then,

Now write the equation with this assumption.

$$s_1^3+3s_1+3s_1+(1+100\beta)=0$$

$$s_1=-1-10\beta^{\frac{1}{3}}$$

$$s_{2,3}=-1+5\beta^{\frac{1}{3}}\pm j5\sqrt{3}\beta^{\frac{1}{3}}$$

Therefore, the roots of the cubic equation are $\boxed{\left(-1-10\beta^{\frac{1}{3}}\right)\text{ and }\left(-1+5\beta^{\frac{1}{3}}\pm j5\sqrt{3}\beta^{\frac{1}{3}}\right)}$.

Step 2 of 3

The root locus of the feedback amplifier is shown in Figure 1.

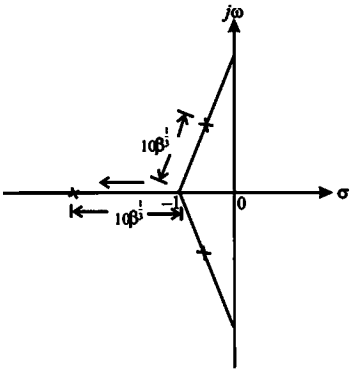


Figure 1

Step 3 of 3

The amplifier becomes unstable when the complex poles lie on the imaginary axis, that is, when $\beta=\beta_{critical}$ .The real part of the complex roots should be zero.

Equate the real part to zero.

$$-1+5\beta_{critical}^{\frac{1}{3}}=0$$

$$5\beta_{critical}^{\frac{1}{3}}=1$$

$$\beta_{critical}^{\frac{1}{3}}=0.2$$

$$\beta_{critical}=(0.2)^3$$

$$\beta_{critical}=0.008$$

Therefore, the feedback factor at which the amplifier becomes unstable is $\boxed{0.008}$.

Step 1 of 5

A certain amplifier has a nonlinear transfer characteristic with the following approximations:

The input voltage for small inputs signals, $|v_i| \le 10\ \text{mV}$

The gain of the open loop amplifier for small inputs,

$$A_1 = \frac{v_o}{v_i}$$
$$= 10^3$$

For intermediate input signals, $10\ \text{mV} \le |v_i| \le 60\ \text{mV}$, the open loop gain,

$$A_2 = \frac{\Delta v_o}{\Delta v_i}$$
$$= 10^2$$

For large signals, $|v_i| \ge 60\ \text{mV}$, the output saturates.

Step 2 of 5

From the data,

$$A_2 = \frac{A_1}{10}$$

With negative feedback loop, there should be a change in gain of 10%

The requirement of gains with feedback is:

$$1.1A_{f2} = A_{f1}$$
$$\frac{A_{f1}}{A_{f2}} = 1.1$$

Write the formulae for feedback gains.

$$A_{f1} = \frac{A_1}{1+\beta A_1}$$
$$A_{f2} = \frac{A_2}{1+\beta A_2}$$

Now calculate their ratio.

$$\frac{A_{f1}}{A_{f2}} = \frac{A_1(1+\beta A_2)}{A_2(1+\beta A_1)}$$

Substitute the known values in the ratio.

$$1.1 = \frac{10^3\left(1+10^2\beta\right)}{10^2\left(1+10^3\beta\right)}$$
$$1.1 = \frac{10\left(1+10^2\beta\right)}{\left(1+10^3\beta\right)}$$
$$1.1+1100\beta = 10+1000\beta$$
$$100\beta = 8.9$$
$$\beta = 0.089$$

Thus, the feedback factor β that reduces the factor-of-10 change in gain to only a 10% change is: $\boxed{0.089}$

.

Step 3 of 5

Calculate the feedback gains of the amplifier.

$$A_{f1} = \frac{A_1}{1+\beta A_1}$$
$$= \frac{10^3}{1+(0.089)(10^3)}$$
$$= 11.11$$

Similarly,

$$A_{f2} = \frac{A_2}{1+\beta A_2}$$
$$= \frac{10^2}{1+(0.089)(10^2)}$$
$$= 10.1$$

Step 4 of 5

Calculate the intersection points (P_1, Q_1) and (P_2, Q_2) on the transfer characteristic.

We have, $Q_1 = 10\ \text{mV}$

$$P_1 = Q_1(1+A_1\beta)$$
$$= 10\times10^{-3}\left[1+10^3(0.089)\right]$$
$$= 0.9\ \text{V}$$
$$= 900\ \text{mV}$$

Calculate the other intersection point.

$$Q_2 = 60\ \text{mV}$$

$$P_2 = P_1+(Q_2-Q_1)(1+\beta A_2)$$
$$= 900\ \text{mV}+(60-10)\times10^{-3}\left[1+10^2(0.089)\right]$$
$$= 900\ \text{mV}+50\times10^{-3}(9.9)$$
$$= 1395\ \text{mV}$$

Step 5 of 5

Draw the transfer characteristic of the amplifier with feedback.

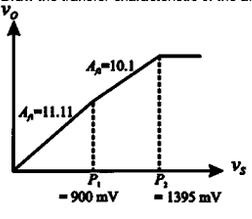


Figure 1

Thus, transfer characteristic of the feedback amplifier is plotted.

Step 1 of 3

Determine the closed loop gain.

$$A = \frac{A_0}{1 + j\left(\frac{\omega}{\omega_P}\right)}$$

$$= \frac{A_0}{1 + j\left(\frac{f}{f_P}\right)}$$

Substitute 10^5 for A_0 and 10 Hz for f_P.

$$A = \frac{10^5}{1 + j\left(\frac{f}{10}\right)}$$

Determine the value of feedback factor β .

$$A\beta = 1$$

Substitute 100 for *A*.

$$(100)\beta = 1$$

$$\beta = \frac{1}{100}$$

$$\beta = 0.01$$

Step 2 of 3

Equate the magnitude of $|A\beta| = 1$.

$$\frac{10^5(0.01)}{\sqrt{1 + \left(\frac{f}{10}\right)^2}} = 1$$

$$\left[10^5(0.01)\right]^2 = 1 + \left(\frac{f}{10}\right)^2$$

$$10^6 = 1 + \left(\frac{f}{10}\right)^2$$

$$\left(\frac{f}{10}\right)^2 = 10^6 - 1$$

$$\frac{f}{10} = \sqrt{10^6 - 1}$$

$$\frac{f}{10} = 999.99$$

$$f = 9999.999$$

$$f \approx 10^4 \text{ Hz}$$

Therefore, the frequency at which $|A\beta| = 1$ is $\boxed{10^4 \text{ Hz}}$.

Step 3 of 3

Determine the phase angle.

$$\phi = -\tan^{-1}\left(\frac{10^4}{10}\right)$$

$$\approx -90°$$

Calculate the phase margin.

$$\begin{aligned}\textbf{Phase margin} &= 180° + \phi \\ &= 180° - 90° \\ &= 90°\end{aligned}$$

Therefore, the phase margin at which $|A\beta| = 1$ is $\boxed{90°}$.

Step 1 of 5

Consider the circuit diagram which is shown in Figure 11.8(a) in the textbook.

The feedback factor β is defined as,

$$\beta = \frac{V_f}{V_o}$$

Apply voltage division rule to find voltage V_f .

$$V_f = V_o\left(\frac{R_1}{R_1 + R_2}\right)$$

$$\frac{V_f}{V_o} = \frac{R_1}{R_1 + R_2}$$

Therefore, feedback factor definition,

$$\beta = \frac{R_1}{R_1 + R_2} \quad \text{..... (1)}$$

Step 2 of 5

The closed loop gain A_f is,

$$A_f = \frac{A}{1 + \beta A}$$

Substitute 10 for A_f , $1000\ \text{V/V}$ for A in the equation.

$$10 = \frac{1000}{1 + \beta(1000)}$$

$$10 + \beta(10000) = 1000$$

$$\beta = \frac{1000 - 10}{10000}$$

$$= 0.099$$

Step 3 of 5

Calculation of resistance R_2 is as follows:

Substitute 0.099 for β , $10\ \text{k}\Omega$ for R_1 in equation (1).

$$0.099 = \frac{10\ \text{k}\Omega}{10\ \text{k}\Omega + R_2}$$

$$990 + 0.099R_2 = 10000$$

$$R_2 = \frac{10000 - 990}{0.099}$$

$$= 91010.10\ \Omega$$

$$= 91.01\ \text{k}\Omega$$

Therefore, the resistance R_2 , is $\boxed{91.01\ \text{k}\Omega}$.

Step 4 of 5

The loop gain $A\beta$ is defined as,

$$A\beta = 1000(0.099)$$

$$= 99$$

Determine the closed loop gain.

The closed loop gain A_f is given by,

$$A_f = 1 + \frac{R_2}{R_1}$$

Substitute $10\ \text{k}\Omega$ for R_1 , $90.01\ \text{k}\Omega$ for R_2 in equation.

$$A_f = 1 + \frac{91.01\ \text{k}\Omega}{10\ \text{k}\Omega}$$

$$= 1 + 9.101$$

$$= 10.101$$

Step 5 of 5

If the voltage gain is to be exactly 10, then the resistance R_2 is selected as follows:

$$A_f = 1 + \frac{R_2}{R_1}$$

Substitute $10\ \text{k}\Omega$ for R_1 , 10 for A_f in equation.

$$10 = 1 + \frac{R_2}{10\ \text{k}}$$

$$R_2 = 90\ \text{k}\Omega$$

Therefore, the resistance R_2 , is $\boxed{90\ \text{k}\Omega}$.

11.027E

We don't have the solution to this problem yet.

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Step 1 of 4

Refer to the Figure 11.11(a) and Example 11.3 in the textbook.

(a)

The ideal closed loop gain A_f is given by,

$$A_f = 1 + \frac{R_2}{R_1}$$

Substitute $10\,\text{k}\Omega$ for R_1, 10 for A_f, in equation

$$10 = 1 + \frac{R_2}{R_1}$$
$$= 1 + \frac{R_2}{10\,\text{k}\Omega}$$

$$\frac{R_2}{10\,\text{k}\Omega} = 10 - 1$$

$$R_2 = 9(10\,\text{k}\Omega)$$
$$= 90\,\text{k}\Omega$$

Therefore, the resistor R_2 is $\boxed{90\,\text{k}\Omega}$.

Step 2 of 4

(b)

Refer to the figure 11.11(b) in the textbook.

Determine the loop gain $A\beta$.

Consider the loop gain $A\beta$ derived in Example 11.3 from the textbook.

$$A\beta = -\frac{V_r}{V_t}$$
$$= \mu\left(\frac{\left\{R_L \left\| \left[R_1 + R_2 \left\| \left(R_{id} + R_s\right)\right.\right]\right\}}{R_L \left\| \left[R_1 + R_2 \left\| \left(R_{id} + R_s\right)\right.\right] + r_o}\right)\left(\frac{R_1 \left\| \left(R_{id} + R_s\right)\right.}{\left(R_1 \left\| R_{id} + R_s\right.\right) + R_2}\right)\left(\frac{R_{id}}{R_{id} + R_1}\right)$$

Substitute 1000 for μ, $100\,\text{k}\Omega$ for R_{id}, $100\,\text{k}\Omega$ for R_s, $1\,\text{k}\Omega$ for r_o, $10\,\text{k}\Omega$ for R_L, $10\,\text{k}\Omega$ for R_1, $1.11\,\text{k}\Omega$ for R_2 in equation.

$$A\beta = \left((1000)\times\left(\frac{\left\{10\,\text{k}\left\|\left[10\,\text{k}+90\,\text{k}\right\|(100\,\text{k}+100\,\text{k})\right]\right\}}{10\,\text{k}\left\|\left[10\,\text{k}+90\,\text{k}\right\|(100\,\text{k}+100\,\text{k})\right]+1\,\text{k}}\right)\right.$$
$$\left.\times\left(\frac{10\,\text{k}\left\|(100\,\text{k}+100\,\text{k})\right.\right.}{\left(10\,\text{k}\left\|(100\,\text{k}+100\,\text{k})\right.\right)+90\,\text{k}}\right)\times\left(\frac{100\,\text{k}}{100\,\text{k}+10\,\text{k}}\right)\Bigg)$$
$$= \left((1000)\times\left(\frac{\left\{10\,\text{k}\left\|\left[90\,\text{k}\right\|(200\,\text{k})\right]\right\}}{10\,\text{k}\left\|\left[90\,\text{k}\right\|(200\,\text{k})\right]+1\,\text{k}}\right)\right.$$
$$\left.\times\left(\frac{10\,\text{k}\left\|(200\,\text{k})\right.\right.}{\left(10\,\text{k}\left\|(200\,\text{k})\right.\right)+90\,\text{k}}\right)\times\left(\frac{100\,\text{k}}{110\,\text{k}}\right)\Bigg)$$

$$A\beta = \left((1000)\times\left(\frac{10\,\text{k}\left\|[62\,\text{k}]\right.}{10\,\text{k}\left\|[62\,\text{k}]\right.+1\,\text{k}}\right)\times\left(\frac{9.52\,\text{k}}{(9.52\,\text{k})+90\,\text{k}}\right)\times(0.909)\right)$$
$$= (1000)\times\left(\frac{8.6\,\text{k}}{8.6\,\text{k}+1\,\text{k}}\right)\times\left(\frac{9.52\,\text{k}}{99.52\,\text{k}}\right)\times(0.909)$$
$$= (1000)(0.07789)$$
$$= 77.89$$

Therefore, the loop gain $A\beta$ is $\boxed{77.89}$.

Step 3 of 4

Feedback factor β is,

$$\beta = \frac{R_1}{R_1 + R_2}$$

Substitute $10\,\text{k}\Omega$ for R_1, $90\,\text{k}\Omega$ for R_2 in equation.

$$\beta = \frac{10}{10+90}$$
$$= 0.1$$

The value of A can be calculated as follows:

$$A\beta = 77.89$$

$$A = \frac{77.89}{\beta}$$
$$= \frac{77.89}{0.1}$$
$$= 778.9$$

Determine the closed loop gain A_f.

$$A_f = \frac{A}{1 + A\beta}$$

Substitute 778.9 for A and 77.89 for $A\beta$ in equation.

$$A_f = \frac{778.9}{1+77.89}$$
$$= 9.87$$

Therefore, the closed loop gain A_f is $\boxed{9.87}$.

Step 4 of 4

(c)

If A_f is of 1% of 10, then closed loop gain is $A_f = 0.1$.

The loop gain can written as follows:

$$A\beta = 77.89$$
$$= 0.07789\mu$$

Thus, the closed loop gain is,

$$0.1 = \frac{A}{1 + A\beta}$$

Substitute 778.9 for A and 0.07789μ in equation.

$$0.1 = \frac{778.9}{1+0.07789\mu}$$
$$0.1 + 0.07789\mu = 77.89$$
$$0.07789\mu = 77.79$$

$$\mu = \frac{77.79}{0.07789}$$
$$= 998.71$$

Thus, In order to have A_f to 1% of 10, the μ should increase to,

$$\mu = \frac{998.71}{1000}$$
$$= 0.998$$

Therefore, the value of μ should be increased to factor of $\boxed{0.998}$.

Step 1 of 2

11.028E

The feedback factor of an ideal operational amplifier connected as a differentiator is,

$$\beta = \frac{\left(\frac{1}{sC}\right)}{R+\left(\frac{1}{sC}\right)}$$
$$= \frac{1}{1+sCR}$$

Substitute $j\omega$ for s.

$$\beta = \frac{1}{1+j\omega CR}$$
$$\frac{1}{\beta} = 1+j\omega CR$$

Calculate the magnitude of $\frac{1}{\beta}$.

$$\left|\frac{1}{\beta}\right| = \sqrt{1+(\omega CR)^2}$$

Step 2 of 2

For stable performance of the differentiator, the frequency should be less than 1 Hz.

$$\omega CR = 1$$
$$2\pi fCR = 1$$
$$f = \frac{1}{2\pi RC} \le 1$$

Rearrange the terms.

$$\frac{1}{RC} \le 2\pi$$
$$RC \ge \frac{1}{2\pi}$$
$$RC \ge 0.159$$
$$\tau \ge 0.159 \text{ s}$$
$$\tau \ge 159 \text{ ms}$$

Therefore, for stable performance, the differentiator time constant should be greater than **159 ms**.

Step 1 of 4

Refer to Figure 11.8(b) in the textbook.

(a)

The feedback factor β is,

$$\beta = \frac{V_f}{V_o}$$
$$= \frac{R_1}{R_1 + R_2}$$

The closed-loop gain A_f is,

$$A_f = \frac{A}{1 + A\beta}$$

The closed-loop gain for $A\beta >> 1$ is,

$$A_f = \frac{1}{\beta}$$
$$= \frac{1}{\left(\frac{R_1}{R_1 + R_2}\right)}$$

$$A_f = 1 + \frac{R_2}{R_1}$$

Substitute $5\ \text{V/V}$ for A_f, $1\ \text{k}\Omega$ for R_1 to determine the resistance R_2.

$$5 = 1 + \frac{R_2}{1\ \text{k}\Omega}$$
$$R_2 = 4\ \text{k}\Omega$$

Therefore, the resistance value of R_2 is $\boxed{4\ \text{k}\Omega}$.

Step 2 of 4

(b)

Determine the value of $A\beta$.

Consider the expression of $A\beta$ in the Example 11.2 from the textbook.

$$A\beta = -\frac{V_r}{V_t}$$
$$= (g_{m1}R_{D1})(g_{m2}R_{D2})\left(\frac{1}{1 + g_{m1}R_1}\right)\left(\frac{R_1}{R_{D2} + R_2 + \left(R_1 \| \frac{1}{g_{m1}}\right)}\right)$$

Substitute $4\ \text{mA/V}$ for g_{m1} and g_{m2}, $10\ \text{k}\Omega$ for R_{D1} and R_{D2}, $1\ \text{k}\Omega$ for R_1, $4\ \text{k}\Omega$ for R_1 in equation.

$$A\beta = (4 \times 10)(4 \times 10)\left(\frac{1}{1 + 4(1)}\right)\left(\frac{1}{10 + 4 + \left(1 \| \frac{1}{4}\right)}\right)$$
$$= 1600\left(\frac{1}{5}\right)\left(\frac{1}{14 + 0.2}\right)$$
$$= 22.53$$

Step 3 of 4

The feedback factor is,

$$\beta = \frac{R_1}{R_1 + R_2}$$

Substitute $1\ \text{k}\Omega$ for R_1, $4\ \text{k}\Omega$ for R_1 in equation.

$$\beta = \frac{1}{1 + 4}$$
$$= 0.2$$

Open loop gain A is,

$$A = \frac{A\beta}{\beta}$$
$$= \frac{22.53}{0.2}$$
$$= 112.65\ \text{V/V}$$

Step 4 of 4

Determine the closed loop gain A_f.

$$A_f = \frac{A}{1 + A\beta}$$

Substitute 112.65 for A is, 22.53 for $A\beta$ in equation.

$$A_f = \frac{112.65}{1 + 22.53}$$
$$= 4.78\ \text{V/ V}$$

Therefore, the closed loop gain A_f is $\boxed{4.78\ \text{V/ V}}$.

Step 1 of 2

11.029E

Calculate the closed-loop gain by reducing the open-loop gain to 20 dB.

$$A = 100\,\text{dB} - 20\,\text{dB}$$
$$= 80\,\text{dB}$$

Convert the open loop gain in dB to a constant.

$$20 \log A = 80$$
$$\log A = 4$$
$$A = 10^4$$

Step 2 of 2

Calculate the frequency at which the new pole must be placed.

$$f_D = \frac{f_{P1}}{A}$$

Substitute $10^6\,\text{Hz}$ for f_{P1} and 10^4 for A.

$$f_D = \frac{10^6}{10^4}$$
$$= 100\,\text{Hz}$$

Therefore, the frequency at which the new pole must be placed is $\boxed{100\,\text{Hz}}$.

11.029P

We don't have the solution to this problem yet.

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Step 1 of 3

11.030E

Calculate the closed-loop gain by reducing the open-loop gain to 20 dB.

$$A = 100\,\text{dB} - 20\,\text{dB}$$
$$= 80\,\text{dB}$$

Convert the open loop gain in dB to a constant.

$$20 \log A = 80$$
$$\log A = 4$$
$$A = 10^4$$

Step 2 of 3

Calculate the frequency to which the first pole must be lowered.

$$f_D = \frac{f_{P2}}{A}$$

Substitute 10 MHz for f_{P2} and 10^4 for *A*.

$$f_D = \frac{10 \times 10^6}{10^4}$$
$$= 1000\ \text{Hz}$$

Therefore, the frequency to which the first pole must be lowered is $\boxed{1000\ \text{Hz}}$.

Step 3 of 3

Calculate the factor by which capacitance at the controlling node be increased.

$$C_{new} = C_{old}\left(f_D\right)$$
$$\frac{C_{new}}{C_{old}} = f_D$$

Substitute 1000 Hz for f_D .

$$\frac{C_{new}}{C_{old}} = 1000$$

Therefore, the factor by which capacitance at the controlling node be increased is $\boxed{1000}$.

Step 1 of 4

Refer to the Figure 11.8(c) in the textbook.

The feedback factor β is,

$$\begin{aligned}\beta &= \frac{V_f}{V_o}\\ &= \frac{R_1}{R_1+R_2}\end{aligned}$$

Step 2 of 4

Expression of loop gain $A\beta$ is considered from the Exercise 11.6 in the textbook.

$$A\beta = g_m\left(\frac{R_D R_1}{R_D+R_1+R_2}\right)$$

Open loop gain A is,

$$A = \frac{A\beta}{\beta}$$

Substitute $g_m\left(\frac{R_D R_1}{R_D+R_1+R_2}\right)$ for $A\beta$, $\frac{R_1}{R_1+R_2}$ for β in equation.

$$\begin{aligned}A &= \frac{\left(g_m\left(\frac{R_D R_1}{R_D+R_1+R_2}\right)\right)}{\left(\frac{R_1}{R_1+R_2}\right)}\\ &= g_m\left(\frac{R_D}{R_D+R_1+R_2}\right)(R_1+R_2)\\ &= g_m\left(R_D \| (R_1+R_2)\right)\end{aligned}$$

Step 3 of 4

The closed loop gain A_f is,

$$\begin{aligned}A_f &= \frac{A}{1+A\beta}\\ &= \frac{g_m\left(R_D \| (R_1+R_2)\right)}{1+g_m\left(\frac{R_D R_1}{R_D+R_1+R_2}\right)}\\ &= \frac{g_m\left(\frac{R_D}{R_D+R_1+R_2}\right)(R_1+R_2)}{1+g_m\left(\frac{R_D R_1}{R_D+R_1+R_2}\right)}\end{aligned}$$

$$A_f = \frac{g_m R_D (R_1+R_2)}{(R_D+R_1+R_2)+g_m(R_D R_1)} \quad \text{..... (1)}$$

Step 4 of 4

Determine the resistor R_1.

Substitute $5\ \text{V/V}$ for A_f, $4\ \text{mA/V}$ for g_m, $10\ \text{k}\Omega$ for R_D, $1\ \text{M}\Omega$ for R_1+R_2 in equation 1.

$$5 = \frac{4\,\text{m}\times 10\,\text{k}\times(1\,\text{M})}{(10\,\text{k}+1\,\text{M})+4\,\text{m}(10\,\text{k}\times R_1)}$$

$$5 = \frac{4\times10^{-3}\times10\times10^3\times(1\times10^6)}{(10\times10^3+1000\times10^3)+4\times10^{-3}((10\times10^3)R_1)}$$

$$5 = \frac{40\times10^6}{(1010\times10^3)+R_1(40)}$$

$$5050\times10^3 + R_1(200) = 40\times10^6$$

Further simplification as follows,

$$R_1(200) = 40\times10^6 - 5.050\times10^6$$

$$\begin{aligned}R_1 &= 34.950\times10^6\\ &= \frac{34.950\times10^6}{200}\\ &= 174.75\times10^3\\ &= 174.75\ \text{k}\Omega\end{aligned}$$

Therefore, the resistance of resistor R_1 is $\boxed{174.75\ \text{k}\Omega}$.

11.031P

We don't have the solution to this problem yet.

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Step 1 of 14

(a)

Refer to the circuit diagram of P11.32 in the textbook.

Draw the A circuit for the Figure.

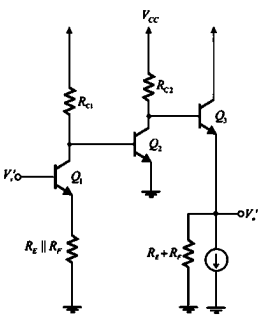


Figure 1

Step 2 of 14

Draw the circuit for determine β.

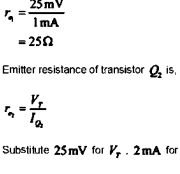


Figure 2

Step 3 of 14

The closed loop voltage gain is given by

$$A_f = \frac{V_o}{V_i} = \frac{A}{1 + A\beta}$$

Now if $A\beta$ is large, that is $A\beta \gg 1$, then

$$1 + A\beta \approx A\beta$$

Hence,

$$A_f = \frac{A}{A\beta}$$
$$= \frac{1}{\beta}$$

Step 4 of 14

From the Figure 2, feedback gain β defined as,

$$\beta = \frac{V_f}{V_o'}$$

Apply voltage division rule,

$$V_f = \left(\frac{R_E}{R_E + R_F}\right)\left(V_o'\right)$$
$$\frac{V_f}{V_o'} = \left(\frac{R_E}{R_E + R_F}\right)$$

Hence,

$$\beta = \left(\frac{R_E}{R_E + R_F}\right)$$

Step 5 of 14

Therefore,

$$A_f = \frac{1}{\beta}$$
$$= \frac{1}{\frac{R_E}{R_E + R_F}}$$
$$A_f = \frac{R_F + R_E}{R_E}$$

Hence, the closed loop voltage gain A_f is $\boxed{\frac{R_F + R_E}{R_E}}$

Step 6 of 14

Calculate the value of resistance as follows:
The closed loop voltage gain A_f is,

$$A_f = \frac{R_F + R_E}{R_E}.$$

Substitute $50\,\Omega$ for R_E, $25\,\text{V/V}$ for A_f in equation.

$$A_f = 1 + \frac{R_F}{R_E}$$
$$25 = 1 + \frac{R_F}{50}$$
$$R_F = (25 - 1)50$$
$$= 1200\,\Omega$$

Therefore, the resistance R_F is $\boxed{1.2\,\text{k}\Omega}$.

Step 7 of 14

(b)

Emitter resistance of transistor Q_1 is,

$$r_{e_1} = \frac{V_T}{I_{Q_1}}$$

Substitute $25\,\text{mV}$ for V_T, $1\,\text{mA}$ for I_{Q_1}.

$$r_{e_1} = \frac{25\,\text{mV}}{1\,\text{mA}}$$
$$= 25\,\Omega$$

Emitter resistance of transistor Q_2 is,

$$r_{e_2} = \frac{V_T}{I_{Q_2}}$$

Substitute $25\,\text{mV}$ for V_T, $2\,\text{mA}$ for I_{Q_2}.

$$r_{e_2} = \frac{25\,\text{mV}}{2\,\text{mA}}$$
$$= 12.5\,\Omega$$

Emitter resistance of transistor Q_3 is,

$$r_{e_3} = \frac{V_T}{I_{Q_3}}$$

Substitute $25\,\text{mV}$ for V_T, $5\,\text{mA}$ for I_{Q_3}.

$$r_{e_3} = \frac{25\,\text{mV}}{5\,\text{mA}}$$
$$= 5\,\Omega$$

Step 8 of 14

The resistance r_{π_1} of transistor Q_1 is,

$$r_{\pi_1} = \frac{\beta V_T}{I_{C_1}}$$
$$= \frac{100(25\,\text{mV})}{1\,\text{mA}}$$
$$= 2500$$
$$= 2.5\,\text{k}\Omega$$

The resistance r_{π_2} of transistor Q_2 is,

$$r_{\pi_2} = \frac{\beta V_T}{I_{C_2}}$$
$$= \frac{100(25\,\text{mV})}{2\,\text{mA}}$$
$$= 1250$$
$$= 1.25\,\text{k}\Omega$$

The resistance r_{π_3} of transistor Q_3 is,

$$r_{\pi_3} = \frac{\beta V_T}{I_{C_1}}$$
$$= \frac{100(25\,\text{mV})}{5\,\text{mA}}$$
$$= 500$$
$$= 0.5\,\text{k}\Omega$$

Step 9 of 14

Consider the closed loop gain of transistor Q_1 is,

$$A_{Q_1} = \frac{-R_{C1} \parallel r_{\pi_2}}{r_{e_1} + (R_E \parallel R_F)}$$

Substitute $2\,\text{k}\Omega$ for R_{C1}, $1.25\,\text{k}\Omega$ for r_{π_2}, $25\,\Omega$ for r_{e_1}, $50\,\Omega$ for R_E, $1.2\,\text{k}\Omega$ for R_F and 100 for β in equation.

$$A_{Q_1} = \frac{-(2\,\text{k} \parallel 1.25\,\text{k})}{25 + (50 \parallel 1.2\,\text{k})}$$
$$= -\frac{0.769\,\text{k}}{25 + 48}$$
$$= -\frac{0.769\,\text{k}}{73}$$
$$= -10.53$$

Therefore, the gain of transistor Q_1 is $-10.53\,\text{V/V}$.

Step 10 of 14

The closed loop gain of transistor Q_2 is,

$$A_{Q_2} = \frac{-R_{C2} \parallel \left[\beta\left(r_{e_1} + R_E + R_F\right)\right]}{r_{e_2}}$$

Substitute $1\,\text{k}\Omega$ for R_{C2}, $1.25\,\text{k}\Omega$ for r_{π_2}, $25\,\Omega$ for r_{e_1}, $12.5\,\Omega$ for r_{e_2}, $50\,\Omega$ for R_E, $1.2\,\text{k}\Omega$ for R_F and 100 for β in equation.

$$A_{Q_2} = -\left(\frac{1\,\text{k} \parallel \left[100(25 + 50 + 1200)\right]}{12.5}\right)$$
$$= -\left(\frac{1\,\text{k} \parallel \left[100(1275)\right]}{12.5}\right)$$
$$= -\left(\frac{1\,\text{k} \parallel 127.5\,\text{k}}{12.5}\right)$$
$$= -\frac{0.99\,\text{k}}{12.5}$$
$$A_{Q_2} = -79.3$$

Therefore, the gain of transistor Q_2 is $-79.3\,\text{V/V}$.

Step 11 of 14

The closed loop gain of transistor Q_3 is,

$$A_{Q_3} = \frac{R_E + R_F}{r_{e_3} + R_E + R_F}$$

Substitute $5\,\Omega$ for r_{e_3}, $50\,\Omega$ for R_E, $1.2\,\text{k}\Omega$ for R_F in equation.

$$A_{Q_3} = \frac{50 + 1.2 \times 10^3}{5 + 50 + 1.2 \times 10^3}$$
$$= \frac{1250}{1255}$$
$$= 0.996$$

Hence, the gain of transistor Q_3 is $0.996\,\text{V/V}$.

Step 12 of 14

The overall voltage gain is given by

$$A = A_{Q_1} A_{Q_2} A_{Q_3}$$
$$= (-10.53)(-79.3)(0.996)$$
$$= 831.7\,\text{V/V}$$

The feedback factor is given by

$$\beta_{FB} = \frac{R_E}{R_E + R_F}$$

Substitute $50\,\Omega$ for R_E, $1.2\,\text{k}\Omega$ for R_F in equation.

$$\beta_{FB} = \frac{50}{50 + 1200}$$
$$= 0.04$$

Step 13 of 14

Determine the loop gain $A\beta$.

$$A\beta = (831.7\,\text{V/V})0.04$$
$$= 33.26\,\text{V/V}$$

Therefore, the loop gain $A\beta$ is $\boxed{33.26\,\text{V/V}}$.

Step 14 of 14

Determine the closed loop gain.

$$A_f = \frac{A}{1 + A\beta_{FB}}$$

Substitute $831.7\,\text{V/V}$ for A, 0.04 for β in equation.

$$A_f = \frac{831.7}{1 + (831.7)(0.04)}$$
$$= \frac{831.7}{1 + 33.26}$$
$$= 24.27$$

Therefore, the closed loop gain A_f is $\boxed{24.27}$.

Step 1 of 4

Refer to the circuit diagram in Figure P11.33 in textbook.

(a)

Consider the expression for gain A.

$$A = g_m\left[r_{o2} \| r_{o4} \| (R_1 + R_2)\right] \quad \text{..... (1)}$$

Here,

g_m is trans-conductance,

r_{o2} and r_{o4} are output resistance of Q_2 and Q_4 respectively.

Substitute $\frac{2I_D}{V_{OV}}$ for g_m, $\frac{V_A}{I_D}$ for r_{o2} in the equation 1.

$$A = \frac{2I_D}{V_{OV}}\left(\frac{V_A}{I_D} \| \frac{V_A}{I_D} \| (R_1 + R_2)\right)$$

$$A = \frac{2I_D}{V_{OV}}\left(\frac{V_A}{2I_D} \| (R_1 + R_2)\right) \quad \text{..... (2)}$$

Therefore, the open loop gain A is $\boxed{\frac{2I_D}{V_{OV}}\left(\frac{V_A}{2I_D} \| (R_1 + R_2)\right)}$.

Step 2 of 4

From circuit diagram, the drain current I_D is,

$$\begin{aligned} I_D &= \frac{I_S}{2} \\ &= \frac{200\,\mu\text{A}}{2} \\ &= 100\,\mu\text{A} \end{aligned}$$

Substitute $100\ \mu\text{A}$ for I_D, $1\ \text{M}\Omega$ for $R_1 + R_2$, $0.2\ \text{V}$ for $|V_{OV}|$, $10\ \text{V}$ for $|V_A|$ in the equation 2.

$$\begin{aligned} A &= \left[\frac{(2)(100\,\mu)}{0.2}\right]\left\{\left[\frac{10}{(2)(100\,\mu)}\right] \| (1\,\text{M})\right\} \\ &= \left(10^{-3}\right)\left[50000 \| 10^6\right] \\ &= 47.61\ \text{V/V} \end{aligned}$$

Thus, the gain of the amplifier A is $\boxed{47.61\ \text{V/V}}$.

Step 3 of 4

The feedback factor β is,

$$\beta = \frac{R_2}{R_1 + R_2}$$

Substitute $1\ \text{M}\Omega$ for $R_1 + R_2$ in equation.

$$\beta = \frac{R_2}{1\,\text{M}}$$

Substitute $179\ \text{k}\Omega$ for R_2 .(Refer part (b) for resistance value R_2)

$$\begin{aligned} \beta &= \frac{R_2}{1\,\text{M}} \\ &= \frac{179\,\text{k}}{1000\,\text{k}} \\ &= 0.179 \end{aligned}$$

The loop gain $A\beta$ is,

$$\begin{aligned} A\beta &= (47.6)(0.179) \\ &= 8.52 \end{aligned}$$

Therefore, the loop gain $A\beta$ is $\boxed{8.52}$.

Step 4 of 4

(b)

Determine the resistance of resistors R_1 and R_2.

Consider the closed loop gain A_f.

$$A_f = \frac{A}{1 + A\beta}$$

Substitute $5\ \text{V/V}$ for A_f, $\frac{R_2}{1\,\text{M}}$ for β in the equation.

$$5 = \frac{47.61}{1 + (47.61)\left(\frac{R_2}{1\,\text{M}}\right)}$$

$$5 + 5\left(47.61 \times 10^{-6}\right)R_2 = 47.61$$

$$5\left(47.61 \times 10^{-6}\right)R_2 = 42.61$$

$$\begin{aligned} R_2 &= \frac{42.61}{5\left(47.61 \times 10^{-6}\right)} \\ &= 179\ \text{K}\Omega \end{aligned}$$

Determine the resistance R_1.

$$\begin{aligned} R_1 &= 1\,\text{M} - 179\,\text{k} \\ &= 1000\,\text{k} - 179\,\text{k} \\ &= 821\ \text{k}\Omega \end{aligned}$$

Therefore, the resistances of R_1 is $\boxed{821\ \text{k}\Omega}$ and R_2 is $\boxed{179\ \text{K}\Omega}$.

Step 1 of 3

The closed loop gain of the amplifier is,

$$A_f = \frac{A}{1 + A\beta}$$

The gain of the amplifier is, A = 1000 V/V.

The feedback factor is, $\beta = 0.1\ \text{V/V}$.

Substitute 0.1 for β and 1000 for A in the equation.

$$A_f = \frac{10^3}{1 + (10^3 \times 0.1)}$$
$$= 9.9$$

Thus, the closed loop gain of series shunt feedback amplifier A_f is $\boxed{9.9}$.

Step 2 of 3

The input resistance of the closed loop amplifier is,

$$R_{if} = R_i[1 + A\beta].$$

The input resistance of a series shunt feedback amplifier is $R_i = 2\ \text{k}\Omega$.

Substitute $2000\ \Omega$ for R_i , $0.1\ \text{V/V}$ for β and 1000 for A in the equation.

$$R_{if} = 2000\left[1 + (1000 \times 0.1)\right]\ \Omega$$
$$= 2000[1 + 100]\ \Omega$$
$$= 202\ \text{k}\Omega$$

Thus the input resistance of the closed loop series shunt feedback amplifier R_{if} is $\boxed{202\ \text{k}\Omega}$.

Step 3 of 3

The input resistance of the closed loop amplifier is,

$R_{of} = \frac{R_o}{(1 + A\beta)}$. The output resistance is $R_o = 2\ \text{k}\Omega$.

Substitute $2000\ \Omega$ for R_o , 0.1 for β and 1000 for A in the equation.

$$R_{of} = \frac{2000}{1 + (0.1 \times 10^3)}\ \Omega$$
$$= \frac{2000}{1 + 100}\ \Omega$$
$$= 19.8\ \Omega$$

Thus the output resistance of the closed loop series shunt feedback amplifier R_{of} is

Step 1 of 2

11.035P

Compute the loop gain $A\beta$ for a change of sampled output voltage by a factor of 200.

$$(1 + A\beta) = 200$$

$$A\beta = 199$$

Hence, the loop gain of the feedback amplifier $A\beta$ is $\boxed{199}$.

Step 2 of 2

The output feedback resistance R_{of} is $100\ \Omega$

The output resistance R_o without feedback is,

$$\begin{aligned} R_o &= R_{of}(1 + A\beta) \\ &= 100(1+199)\ \Omega \\ &= 20000\ \Omega \\ &= 20\ \text{k}\Omega \end{aligned}$$

Hence, the output resistance without feedback R_o is $\boxed{20\ \text{k}\Omega}$.

Therefore, the resistance with feedback is lower.

Step 1 of 7

Open loop gain A is,

$$A=\frac{A_0}{\left(1+\left(\frac{s}{\omega_H}\right)\right)} \quad \text{..... (1)}$$

Series-shunt amplifier has an input resistance R_i, an output resistance R_o.

Thus, input resistance is also valid for the case where open loop gain A is the function of frequency. So,

$$R_{if}=(1+A\beta)R_i$$

Output resistance is also valid for the case where open loop gain A is the function of frequency. So,

$$R_{of}=\frac{R_o}{(1+A\beta)}$$

Here,

Feedback factor β is the independent of frequency.

Step 2 of 7

Find input resistance (function of frequency).

$$Z_{if}=R_i(1+A\beta)$$

Substitute $\frac{A_0}{\left(1+\left(\frac{s}{\omega_H}\right)\right)}$ for A in the equation.

$$Z_{if}=R_i\left(1+\frac{A_0\beta}{\left(1+\left(\frac{s}{\omega_H}\right)\right)}\right)$$

$$Z_{if}=R_i+\frac{A_0\beta R_i}{\left(1+\left(\frac{s}{\omega_H}\right)\right)} \quad \text{..... (2)}$$

Step 3 of 7

Draw equivalent circuit for the input impedance Z_{if} in s -domain.

Figure 1

Step 4 of 7

Equivalent input impedance for Figure 1 is,

$$\begin{aligned}Z_{if}&=R_i+\left(R\parallel\frac{1}{sC}\right)\\&=R_i+\frac{(R)\left(\frac{1}{sC}\right)}{R+\frac{1}{sC}}\\&=R_i+\frac{\frac{R}{sC}}{\frac{sRC+1}{sC}}\end{aligned}$$

Rewrite the equation.

$$Z_{if}=R_i+\frac{R}{1+RCs} \quad \text{..... (3)}$$

Compare equation (2) and (3).

$$R=A_0\beta R_i$$

$$RC=\frac{1}{\omega_H}$$

$$C=\frac{1}{A_0\beta R_i\omega_H}$$

Replace R and C in the equivalent circuit.

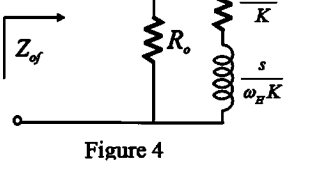


Figure 2

Step 5 of 7

Find output resistance Z_{of}

$$Z_{of}=\frac{R_o}{(1+A\beta)}$$

Substitute $\frac{A_0}{\left(1+\left(\frac{s}{\omega_H}\right)\right)}$ for A in the equation.

$$\begin{aligned}Z_{of}&=\frac{R_0}{1+\left(\frac{A_0}{1+\frac{s}{\omega_H}}\right)\beta}\\&=\frac{1}{\frac{A_0\beta}{R_0}+\frac{1}{1+\frac{s}{\omega_H}}}\end{aligned}$$

Rewrite the equation.

$$Z_{of}=\frac{1}{\frac{1}{R_0}+\frac{1}{\frac{1}{K}+\frac{S}{\omega_H K}}} \quad \text{..... (4)}$$

Here

$$K=\frac{A_0\beta}{R_0}$$

Step 6 of 7

Draw the equivalent circuit z_{of} in s domain.

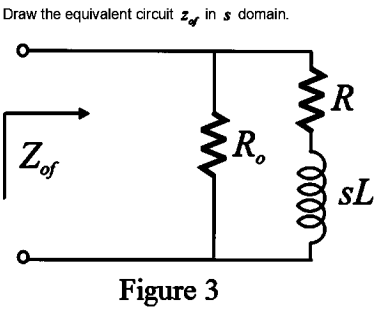


Figure 3

Step 7 of 7

Find the equivalent impedance for the circuit.

$$Z_{of}=\frac{R_o(R+sL)}{R_o+R+sL}$$

Rewrite the equation.

$$Z_{of}=\frac{1}{\frac{1}{R+sL}+\frac{1}{R_o}} \quad \text{..... (5)}$$

Equate (4) and (5) equations.

$$R=\frac{1}{K}$$

And $L=\frac{1}{\omega_H K}$

Thus the equivalent circuit for Z_{of} is,

Figure 4

Thus, equivalent circuit for the input impedance and output impedance for the feedback amplifier has been designed.

Step 1 of 3

11.037P

The closed loop input resistance of the amplifier is,

$$R_{if} = (1 + A\beta) R_i$$

The voltage gain of the basic voltage amplifier, $1000\ \text{V/V}$

The input resistance of the amplifier, $1000\ \Omega$

The closed-loop resistance of the amplifier, $10\ \text{k}\Omega$

Substitute the value of closed loop input resistance, R_{if} and the input resistance R_i.

$$10 \times 10^3 = (1 + A\beta)(1000)$$

$$1 + A\beta = \frac{10 \times 10^3}{1000}$$

$$1 + A\beta = 10$$

Substitute 1000 for A.

$$(1000)\beta = 9$$

$$\beta = \frac{9}{1000}$$

Step 2 of 3

The forward gain of the amplifier is,

$$A_F = \frac{A}{1 + A\beta}$$

Substitute 1000 for A and $\frac{9}{1000}$ for β.

$$A_F = \frac{A}{1 + A\beta}$$

$$= \frac{1000}{1+9}$$

$$= 100$$

Thus, the closed loop gain, A_F of the amplifier is, $\boxed{100}$.

Step 3 of 3

As the basic amplifier is used to implement the unity gain voltage buffer, therefore $A = 1$

The input resistance in this case is,

$$R_i = R_{if}(1 + A\beta)$$

The closed-loop resistance of the amplifier, $10\ \text{k}\Omega$

$$R_i = (10\ \text{k}\Omega)\left(1 + (1)\left(\frac{9}{1000}\right)\right)$$

$$= (10\ \text{k}\Omega)\left(\frac{1009}{1000}\right)$$

$$= 10.09\ \text{k}\Omega$$

Thus, if the amplifier is used to implement a voltage gain buffer, the input resistance R_i of the device is $\boxed{10.09\ \text{k}\Omega}$

Step 1 of 11

Refer to the Figure 11.15(a) and Example 11.4 in the textbook.

Consider the resistors $R_1 = \infty$ and $R_2 = 0$ in the feedback network.

(a)

The feedback network is drawn in Figure 1.

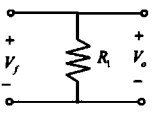


Figure 1

Determine the feedback factor β.

$$\beta = \frac{V_f}{V_o}$$
$$= 1 \qquad (\text{since } V_f = V_o)$$

Therefore, the value of β is $\boxed{1}$.

Step 2 of 11

Determine the ideal closed loop gain.

The closed loop gain A_f is given by,

$$A_f = \frac{A}{1 + A\beta}$$

Since, $A\beta >> 1$. The ideal value of closed loop gain is,

$$A_f = \frac{1}{\beta}$$

Substitute 1 for β in equation.

$$A_f = \frac{1}{1}$$
$$= 1$$

Therefore, the value of A_f is $\boxed{1}$.

Step 3 of 11

(b)

Determine the open loop gain A.

Consider the loading effect of the feedback network at input side is obtained by short-circuiting port 2 of the feedback network and looking into port 1.

Then, the resistor gets short-circuited due to short at output side.

$$R_{11} = R_1 \| 0$$
$$= 0$$

Consider the loading effect of the feedback network at the output side is found by open-circuiting port 1 and looking into the port 2.

$$R_{22} = R_1 + 0$$
$$= R_1$$

Step 4 of 11

The open loop gain A is,

$$A = \frac{V_o}{V_i}$$
$$= \frac{V_o}{V_1}\frac{V_1}{V_i}$$
$$A = \left(\frac{R_{id}}{R_{id} + R_s + 0}\right)\left(\frac{\mu(R_L \| R_1)}{(R_L \| R_1) + r_o}\right)$$

Therefore, the open loop gain A is $\boxed{\left(\frac{R_{id}}{R_{id} + R_s + 0}\right)\left(\frac{\mu(R_L \| R_1)}{(R_L \| R_1) + r_o}\right)}$.

Step 5 of 11

Determine the value of $A\beta$.

$$A\beta = \left[\left(\frac{R_{id}}{R_{id} + R_s + 0}\right)\left(\frac{\mu(R_L \| R_1)}{(R_L \| R_1) + r_o}\right)\right] \times 1$$
$$= \left(\frac{R_{id}}{R_{id} + R_s}\right)\left(\frac{\mu(R_L \| R_1)}{(R_L \| R_1) + r_o}\right) \quad (\text{since } \beta = 1)$$

Therefore, the open loop gain $A\beta$ is $\boxed{\left(\frac{R_{id}}{R_{id} + R_s}\right)\left(\frac{\mu(R_L \| R_1)}{(R_L \| R_1) + r_o}\right)}$.

Step 6 of 11

(c)
Calculate the open loop gain A.

The open loop gain A is given by,

$$A = \left(\frac{R_{id}}{R_{id} + R_s}\right)\left(\frac{\mu(R_L \| R_1)}{(R_L \| R_1) + r_o}\right)$$

Substitute 10^4 for μ, $100\ \text{k}\Omega$ for R_{id}, $10\ \text{k}\Omega$ for R_s, $1\ \text{k}\Omega$ for r_o, $2\ \text{k}\Omega$ for R_L in equation.

$$A = \left(\frac{R_{id}}{R_{id} + R_s}\right)\left(\frac{\mu(R_L)}{(R_L) + r_o}\right)$$
$$= \left(\frac{100\ \text{k}\Omega}{100\ \text{k}\Omega + 10\ \text{k}\Omega}\right)\left(\frac{10^4(2\ \text{k}\Omega)}{(2\ \text{k}\Omega) + 1\ \text{k}\Omega}\right)$$
$$= 0.909\left(0.67 \times 10^4\right)$$
$$= 6060.60\ \text{V/V}$$

Therefore, the open loop gain A is $\boxed{6060.60\ \text{V/V}}$.

Step 7 of 11

Determine the value of $A\beta$.

Substitute $6060.60\ \text{V/V}$ for A.

$$A\beta = (6060.60) \times (1)$$
$$= 6060.60$$

Step 8 of 11

Determine the value of A_f.

$$A_f = \frac{A}{1 + A\beta}$$
$$\approx \frac{6060.60}{6061.60}$$
$$= 0.999$$
$$\approx 1$$

Therefore, the closed loop gain A_f is $\boxed{1}$.

Step 9 of 11

Determine the resistance R_{in}.

The input resistance R_{if} due to feedback is given by,

$$R_{if} = R_i(1 + A\beta)$$
$$= (R_s + R_{id} + 0)(1 + A\beta)$$

Here, R_i is input resistance.

Resistance R_{in} is can be calculated as follows:

$$R_{in} = R_{if} - R_s$$
$$= \left[(R_s + R_{id})(1 + A\beta)\right] - R_s$$

Substitute $100\ \text{k}\Omega$ for R_{id}, $10\ \text{k}\Omega$ for R_s, 6061.6 for $1 + A\beta$ in equation.

$$R_{in} = \left[(10\ \text{k}\Omega + 100\ \text{k}\Omega)(6061.60)\right] - 10\ \text{k}\Omega$$
$$= \left[(110\ \text{k}\Omega)(6061.60)\right] - 10\ \text{k}\Omega$$
$$= \left[(110\ \text{k}\Omega)(6061.60)\right] - 10\ \text{k}\Omega$$

$$R_{in} \approx 666776\ \text{k}\Omega$$
$$\approx 666.77\ \text{M}\Omega$$

Therefore, the input resistance R_{in} is $\boxed{666.77\ \text{M}\Omega}$.

Step 10 of 11

Determine the resistance R_{out}.

Resistance R_o is output resistance of the A circuit. R_o can be obtained as,

$$R_o = r_o \| R_L \| R_1$$
$$= r_o \| R_L \qquad (\text{sicnce } R_1 = \infty)$$

Substitute $1\ \text{k}\Omega$ for r_o, $2\ \text{k}\Omega$ for R_L in equation.

$$R_o = 1\ \text{k} \| 2\ \text{k}$$
$$= 666.7\ \text{k}\Omega$$

The reistance R_{of} is given by the feedback equations. That is,

$$R_{of} = \frac{R_o}{1 + A\beta}$$

Substitute $666.7\ \text{k}\Omega$ for R_o, 6061.60 for $1 + A\beta$ in equation.

$$R_{of} = \frac{666.7}{(6061.60)}$$
$$= 0.1099\ \Omega$$
$$= 109.9\ \text{m}\Omega$$

Step 11 of 11

Output resistance R_{out} obtained excluding R_L as follows:

$$R_{of} = R_L \| R_{out}$$
$$= \frac{R_L R_{out}}{R_L + R_{out}}$$

$$R_{of} R_L + R_{of} R_{out} = R_L R_{out}$$

$$R_{of} R_L = (R_L - R_{of}) R_{out}$$

$$R_{out} = \frac{(R_L - R_{of})}{R_{of} R_L}$$

Substitute $109.9\ \text{m}\Omega$ for R_{of}, $2\ \text{k}\Omega$ for R_L in equation.

$$R_{out} = \frac{(2000 - 0.1099)}{2\text{k}(109.9\ \text{m})}$$
$$= \frac{1999.89}{219.8}$$
$$= 9\ \Omega$$

Therefore, the input resistance R_{out} is $\boxed{9\ \Omega}$.

11.039P

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11.040P

We don't have the solution to this problem yet.

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Ask an expert

Step 1 of 19

Refer to the circuit diagram of series-shunt amplifier in Figure P10.38 in the text book.

From the figure it is observed that:

1. MOS transistor Q_1 is connected in common source amplifier configuration.

2. MOS transistor Q_2 is also connected in common source amplifier configuration.

3. MOS transistor Q_3 is connected in common drain amplifier configuration.

Basically, the common source amplifier has reactively high small signal voltage gain. That is,

$$\frac{V_o}{V_s} = \text{high}\left(\cong g_m r_o\right)$$

It is clear that when the source voltage at the MOS transistor Q_1 increases, then the output voltage at drain terminal voltage of the transistor Q_1 increases. Drain terminal voltage of the transistor Q_1 is connected to the gate of the MOS transistor Q_2. When increasing the gate voltage of the transistor Q_2 will cause the increase in the drain terminal voltage at the transistor Q_2.

Common drain or source follower circuit has low voltage gain at its output. An increase in the gate voltage of the MOS transistor Q_3 results an increase at the output voltage V_o through the voltage dividing feedback circuit.

It is concluded that when input is high, then the output is also high. The gain is very high for negative feedback of the series shunt amplifier. Therefore, the gain of the series shunt amplifier is high.

Therefore, the feedback used in the circuit is negative.

Step 2 of 19

(b)

The feedback amplifier gain is,

$$A_f = \frac{A}{1+A\beta}$$

Here, the closed loop gain is very high, $A\beta >> 1$. Therefore,

$$\begin{aligned} A_f &= \frac{A}{A\beta} \\ &= \frac{1}{\beta} \end{aligned}$$

Here, β is called the feedback factor.

Draw the beta circuit.

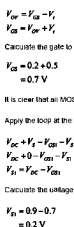


Figure 1

Step 3 of 19

Calculate the feedback factor, β.

$$\begin{aligned} \beta &= \frac{V_f}{V_o} \\ &= \frac{R_1}{R_1+R_2} \end{aligned}$$

Therefore, the closed loop voltage gain is,

$$\begin{aligned} A_f &= \frac{1}{\beta} \\ &= \frac{R_1+R_2}{R_1} \end{aligned}$$

Substitute $2\,\text{k}\Omega$ for R_1 and $18\,\text{k}\Omega$ for R_2.

$$\begin{aligned} A_f &= \frac{2\times10^3+18\times10^3}{2\times10^3} \\ &= 10 \end{aligned}$$

Therefore, closed loop voltage gain is $\boxed{10}$.

Step 4 of 19

(c)

Suppose the source voltage V_s has zero DC voltage. The general expression for overdrive voltage or effective voltage is,

$$\begin{aligned} V_{ov} &= V_{GS} - V_t \\ V_{GS} &= V_{ov} + V_t \end{aligned}$$

Calculate the gate to source voltage.

$$\begin{aligned} V_{GS} &= 0.2 + 0.5 \\ &= 0.7\ \text{V} \end{aligned}$$

It is clear that all MOS transistors are operated in enhancement mode since it obeys the condition $V_{GS} > V_t$.

Apply the loop at the left side of transistor, Q_1.

$$\begin{aligned} V_{DC} + V_s - V_{GS1} - V_{S1} &= 0 \\ V_{DC} + 0 - V_{GS1} - V_{S1} &= 0 \\ V_{S1} &= V_{DC} - V_{GS1} \end{aligned}$$

Calculate the voltage at the source terminal at the transistor Q_1.

$$\begin{aligned} V_{S1} &= 0.9 - 0.7 \\ &= 0.2\ \text{V} \end{aligned}$$

Therefore, the DC voltage at the node S_1 is $\boxed{0.2\ \text{V}}$.

Step 5 of 19

Apply the loop at the top of transistors to calculate the drain voltage of Q_2.

$$\begin{aligned} V_{G2} - V_{SD} + V_{SG2} &= 0 \\ V_{G2} &= 1.8 - 0.7 \\ V_{G2} &= 1.1\ \text{V} \end{aligned}$$

Therefore, the DC voltage at the node G_2 is $\boxed{1.1\ \text{V}}$.

Each current source has minimum DC voltage across it for proper operation. The following condition is applicable to all current sources to operate in the saturation region.

$$V_{OV} \le V_{GS}$$

Here,

$$V_{SG} = V_{SD} - V_{G2}$$

Step 6 of 19

Calculate the drain to source voltage

$$\begin{aligned} V_{DS} &= 1.8 - 1.1 \\ &= 0.7\ \text{V} \end{aligned}$$

Apply the loop to the transistor Q_3.

$$\begin{aligned} V_{G3} &= V_{GS3} + V_{S3} \end{aligned}$$

Calculate the DC voltage at the gate terminal of the transistor Q_3.

$$\begin{aligned} V_{G3} &= 0.7 + 0.2 \\ &= 0.9\ \text{V} \end{aligned}$$

Therefore, the DC voltage at the node G_3 is $\boxed{0.9\ \text{V}}$.

Step 7 of 19

Apply the loop to the transistor Q_3.

$$\begin{aligned} V_{G3} - V_{GS3} - V_{S3} &= 0 \\ V_{S3} &= V_{G3} - V_{GS3} \\ V_{S3} &= 0.9 - 0.7 \\ &= 0.2\ \text{V} \end{aligned}$$

Therefore, the DC voltage at the node S_3 is $\boxed{0.2\ \text{V}}$.

Step 8 of 19

(d)

The open loop amplifier is also known as A circuit. Ground the V_{SG2} and each current source has the output resistance equal to $r_o = r_{oc}$. So, replace the current sources with its output resistances. Split the feedback resistor R_2.

Draw the A circuit for the feedback amplifier.

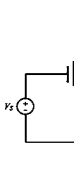


Figure 2

Step 9 of 19

Calculation of gain of MOS transistor Q_1 **in the first stage:**

In Figure 2, the output is taken from the drain terminal of the transistor Q_1 and the input of the transistor is V_s. Therefore, the voltage gain of the transistor Q_1 is,

$$\begin{aligned} A_{V1} &= \frac{V_{G2}}{V_s} \qquad (\text{since } V_{G2} = V_{o1}) \\ &= g_{m,eff}\left(r_{oc} \| r_{oc}\right) \end{aligned}$$

But the common source amplifier with source resistor R_s has effective transconductance and output resistance.

$$\begin{aligned} g_{m,eff} &= \frac{g_m}{1+g_m R_s} \\ r_{o,eff} &= r_o\left(1+g_m R_s\right) \end{aligned}$$

Calculate the value of trans-conductance.

$$\begin{aligned} g_m &= \frac{2I_D}{V_{ov}} \\ &= \frac{2\left(0.1\times10^{-3}\right)}{0.2} \\ &= 1\ \text{mA/V} \end{aligned}$$

Calculate the value of r_o.

$$\begin{aligned} r_o &= \frac{V_A}{I_D} \\ &= \frac{10}{0.1\times10^{-3}} \\ &= 100\ \text{k}\Omega \end{aligned}$$

Calculate the resistance R_s.

$$\begin{aligned} R_s &= R_1 \| R_2 \\ &= \frac{\left(2\times10^3\right)\left(18\times10^3\right)}{2\times10^3+18\times10^3} \\ &= 1.8\ \text{k}\Omega \end{aligned}$$

Step 10 of 19

Calculate the voltage gain of the transistor Q_1.

$$A_{V1} = \frac{g_m}{1+g_m R_s}\left[\left(r_o\left(1+g_m R_s\right)\right) \| r_{oc}\right]$$

Substitute $1.8\,\text{k}\Omega$ for R_s, $1\ \text{mA/V}$ for g_m, and $100\,\text{k}\Omega$ for r_o.

$$\begin{aligned} A_{V1} &= \frac{1\times10^{-3}}{1+\left(1\times10^{-3}\right)\left(1.8\times10^3\right)}\left[\left\{100\times10^3\left[1+\left(1\times10^{-3}\right)\left(1.8\times10^3\right)\right]\right\} \| 100\times10^3\right] \\ &= \frac{1\times10^{-3}}{2.8}\left\{10^3\left[2.8\right] \| 10^3\right\} \\ &= \frac{1\times10^{-3}}{2.8}\left\{\frac{\left(2.8\times10^3\right)\left(10^3\right)}{2.8\times10^3+10^3}\right\} \\ &= 26.31\ \text{V/V} \end{aligned}$$

Therefore, the gain of first stage is $\boxed{26.31\ \text{V/V}}$.

Step 11 of 19

Calculation of gain of MOS transistor Q_2 **in the second stage:**

In Figure 2, the output is taken from the drain terminal of the transistor Q_2 and the input of the transistor Q_2 is V_{G1}. Therefore, the voltage gain of the transistor Q_2 is,

$$\begin{aligned} A_{V2} &= \frac{V_{o2}}{V_{G2}} \\ &= g_{m2}\left(r_{oc} \| r_{oc}\right) \qquad (\text{since } r_{oc} = r_{o2} \text{ and } g_{m2} = g_m) \\ &= g_m\left(\frac{r_{oc}}{2}\right) \end{aligned}$$

Calculate the voltage gain of the transistor Q_2.

$$\begin{aligned} A_{V2} &= 1\times10^{-3}\left(\frac{10^3}{2}\right) \\ &= \frac{10^1}{2} \\ &= 50\ \text{V/V} \end{aligned}$$

Therefore, the gain of second stage is $\boxed{50\ \text{V/V}}$.

Step 12 of 19

Calculation of gain of MOS transistor Q_3 **in the third stage:**

In Figure 2, the output is taken from the source terminal of the transistor Q_3 and the input of the transistor Q_3 is V_{G3}. Therefore, the voltage gain of the transistor Q_3 is,

$$\begin{aligned} A_{V3} &= \frac{V_o}{V_{G3}} \\ &= \frac{r_o \| R_L}{r_o \| R_L + \frac{1}{g_m}} \end{aligned}$$

Here,

$$R_L = \left(R_1+R_2\right) \| r_{oc}$$

Step 13 of 19

Calculate the load resistor.

$$\begin{aligned} R_L &= \left(2\times10^3+18\times10^3\right) \| \left(100\times10^3\right) \\ &= \left(20\times10^3\right) \| \left(100\times10^3\right) \\ &= \frac{\left(20\times10^3\right)\left(100\times10^3\right)}{\left(20\times10^3\right)+\left(100\times10^3\right)} \\ &= 16.67\ \text{k}\Omega \end{aligned}$$

Calculate the voltage gain of the transistor Q_3.

$$\begin{aligned} A_{V3} &= \frac{100\times10^3 \| 16.67\times10^3}{100\times10^3 \| 16.67\times10^3 + \frac{1}{1\times10^{-3}}} \\ &= \frac{\frac{\left(100\times10^3\right)\left(16.67\times10^3\right)}{100\times10^3+16.67\times10^3}}{\frac{\left(100\times10^3\right)\left(16.67\times10^3\right)}{100\times10^3+16.67\times10^3} + \frac{1}{1\times10^{-3}}} \\ &= 0.99\ \text{V/V} \end{aligned}$$

Therefore, the gain of third stage is $\boxed{0.99\ \text{V/V}}$.

Step 14 of 19

Calculate the overall voltage gain.

$$\begin{aligned} A_v &= A_{V1} A_{V2} A_{V3} \\ &= (26.31)(50)(0.99) \\ &= 1302.345\ \text{V/V} \end{aligned}$$

Therefore, the overall voltage gain is $\boxed{1302.345\ \text{V/V}}$.

Step 15 of 19

(e)

The Figure 1 in part (a) is called the feedback network (voltage dividing feedback network).

Recall the expression for feedback factor, β from part (b).

$$\beta = \frac{R_1}{R_1+R_2}$$

Substitute $2\,\text{k}\Omega$ for R_1 and $18\,\text{k}\Omega$ for R_2.

$$\begin{aligned} \beta &= \frac{2\times10^3}{2\times10^3+18\times10^3} \\ &= \frac{2}{20} \\ &= 0.1 \end{aligned}$$

Therefore, the value of β is $\boxed{0.1}$.

Step 16 of 19

(f)

The expression for voltage gain is,

$$\frac{V_o}{V_s} = \frac{A}{1+A\beta}$$

Substitute $1223\ \text{V/V}$ for A and 0.1 for β.

$$\begin{aligned} \frac{V_o}{V_s} &= \frac{1302.3}{1+(1302.3)(0.1)} \\ &= \frac{1302.3}{131.23} \\ &= 9.923\ \text{V/V} \end{aligned}$$

This voltage gain $\frac{V_o}{V_s}$ is approximately equal to $\frac{1}{\beta}$ value that is calculated in part (b).

Therefore, the value of voltage gain is $\boxed{9.923\ \text{V/V}}$.

Step 17 of 19

(g)

The circuit for finding output impedance is shown in Figure 3

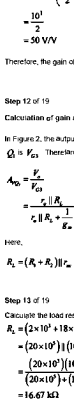


Figure 3

Clearly the output resistance R_o is,

$$R_o = \frac{1}{g_m} \| \left[\left(R_1+R_2\right) \| r_{oc}\right]$$

Step 18 of 19

Calculate the value of trans-conductance.

$$\begin{aligned} g_m &= \frac{2I_D}{V_{ov}} \\ &= \frac{2\left(0.1\times10^{-3}\right)}{0.2} \\ &= 1\ \text{mA/V} \end{aligned}$$

Calculate the value of r_o.

$$\begin{aligned} r_o &= \frac{V_A}{I_D} \\ &= \frac{10}{0.1\times10^{-3}} \\ &= 100\ \text{k}\Omega \end{aligned}$$

Step 19 of 19

Calculate the output resistance.

$$\begin{aligned} R_o &= \frac{1}{1\times10^{-3}} \| \left[\left(2\times10^3+18\times10^3\right) \| 100\times10^3\right] \\ &= 1000 \| \left[20\times10^3 \| 100\times10^3\right] \\ &= 1000 \| \left[\frac{\left(20\times10^3\right)\left(100\times10^3\right)}{20\times10^3+100\times10^3}\right] \\ &= 1\times10^3 \| \left[16.67\times10^3\right] \end{aligned}$$

Simplify further.

$$\begin{aligned} R_o &= 1\times10^3 \| 16.67\times10^3 \\ &= \frac{\left(1\times10^3\right)\left(16.67\times10^3\right)}{1\times10^3+16.67\times10^3} \\ &= \frac{16.67\times10^3}{17.67} \\ &= 943.4\ \Omega \end{aligned}$$

Therefore, the output resistance is $\boxed{943.4\ \Omega}$.

Step 1 of 13

Refer to the Figure P11.33 in the textbook.

(a)

As input voltage (V_s) increases, the output voltage also increases with a small amount of voltage. Hence, the feedback is negative.

Step 2 of 13

(b)

By neglecting the early effect, the dc voltages at gate Q_2 **and** Q_1 are equal.

Hence, the voltage at Q_2 is $\boxed{V_{G2} = V_{G1}}$.

Therefore, the output dc voltage (V_{odc}) is $\boxed{V_{G1}\left(1+\frac{R_1}{R_2}\right)}$.

Step 3 of 13

(c)

Draw the A-circuit.

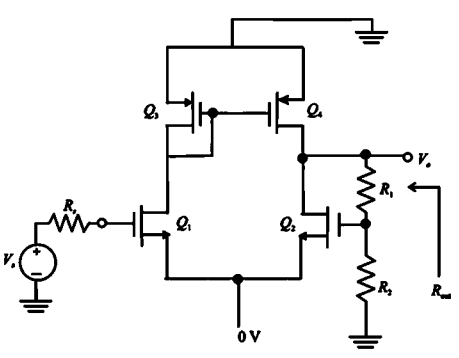


Figure-1

Step 4 of 13

The expression for the gain, A is,

$$A = g_m\left[r_{o2} \| r_{o4} \| (R_1 + R_2)\right] \quad \ldots\ldots (1)$$

Here,

$A \rightarrow$ Gain

$g_m \rightarrow$ Trans-conductance

r_{02} **and** $r_{04} \rightarrow$ Output resistance r_o

The transconductance, $g_m = \frac{2I_D}{V_{OV}}$

$$r_{o2} = r_{o4} = \frac{V_A}{I_D}$$

Substitute g_m **and** r_{02} results in equation (1).

$$A = \frac{2I_D}{V_{OV}}\left[\frac{V_A}{I_D} \| \frac{V_A}{I_D} \| (R_1 + R_2)\right]$$
$$= \frac{2I_D}{V_{OV}}\left[\frac{V_A}{2I_D} \| (R_1 + R_2)\right]$$

Hence, the gain A is $\boxed{\frac{2I_D}{V_{OV}}\left[\frac{V_A}{2I_D} \| (R_1 + R_2)\right]}$.

Step 5 of 13

The sum of resistances is,

$$R_1 + R_2 = 1\ \text{M}\Omega$$
$$|V_0| = 0.2\ \text{V}$$
$$|V_A| = 10\ \text{V}$$

Substitute the values in the equation for A.

$$A = \left[\frac{(2)(100\ \mu)}{0.2}\right]\left\{\left[\frac{10}{(2)(100\ \mu)}\right] \| (1\ \text{M})\right\}$$
$$= \left(10^{-3}\right)\left[50000 \| 10^6\right]$$
$$= 47.61\ \text{V/V}$$

Hence, the value of gain A is $\boxed{47.61\ \text{V/V}}$.

Step 6 of 13

d)

The feedback factor β is,

$$\beta = \frac{R_2}{R_1 + R_2}$$
$$= \frac{R_2}{1\ \text{M}}$$

Step 7 of 13

The gain with feedback is,

$$A_f = \frac{A}{1 + A\beta} \quad \ldots\ldots (2)$$

The closed-loop gain is,

$$\frac{V_o}{V_s} = A_f = 5\ \text{V/V}$$

Substitute the values in equation (2).

$$5 = \frac{47.61}{1 + (47.61)\left(\frac{R_2}{1\ \text{M}}\right)}$$
$$5\left(1 + (47.61 \times 10^{-6})R_2\right) = 47.61$$
$$5 + 5(47.61 \times 10^{-6})R_2 = 47.61$$
$$5(47.61 \times 10^{-6})R_2 = 42.61$$
$$R_2 = \frac{42.61}{5(47.61 \times 10^{-6})}$$
$$= 179\ \text{K}\Omega$$

Step 8 of 13

Calculate the value of the resistance, R_1.

$$R_1 = 1\ \text{M} - 179\ \text{k}$$
$$= 1000\ \text{k} - 179\ \text{k}$$
$$= 821\ \text{k}\Omega$$

Hence, the resistances R_1 **and** R_2 are $\boxed{179\ \text{k}\Omega}$ and $\boxed{821\ \text{k}\Omega}$.

Step 9 of 13

(e)

Compute, the value of the resistance, R_o.

$$R_o = r_{o2} \| r_{o4} \| (R_1 + R_2)$$
$$= \frac{V_A}{I_D} \| \frac{V_A}{I_D} \| (R_1 + R_2)$$
$$= \frac{V_A}{2I_D} \| (R_1 + R_2)$$
$$= \left[\frac{10}{(2)(100\ \mu)}\right] \| (1\ \text{M})$$
$$= \left[\frac{10}{200\ \mu}\right] \| (1\ \text{M})$$
$$= 50000 \| 10^6$$
$$= \frac{(50000)(10^6)}{50000 + 10^6}$$
$$= 47.62\ \text{k}\Omega$$

Hence, the value of the resistance R_o is $\boxed{47.62\ \text{k}\Omega}$.

Step 10 of 13

Calculate the output resistance with feedback.

$$R_{of} = \frac{R_o}{1 + A\beta}$$
$$= \frac{47.62\ \text{K}\Omega}{1 + (47.61)\left(\frac{R_2}{1\ \text{M}}\right)}$$
$$= \frac{47.62\ \text{K}\Omega}{1 + (47.61)\left(\frac{179\ \text{K}}{1\ \text{M}}\right)}$$
$$= \frac{47.62 \times 10^3}{1 + (47.61)(179 \times 10^{-3})}$$
$$= \frac{47.62 \times 10^3}{1 + 8.52219}$$
$$= \frac{47.62 \times 10^3}{9.52219}$$
$$= 5.0009 \times 10^3$$
$$= 5\ \text{K}\Omega$$

The output resistance, $R_{out} = R_{of} = 5\ \text{k}\Omega$

Hence, the output resistance of the amplifier is $\boxed{R_{out} = 5\ \text{k}\Omega}$.

Step 11 of 13

(f)

The load resistance, $R_L = 10\ \text{k}\Omega$

The expression for the gain of the amplifier is,

$$A_{TOT} = \frac{V_o}{V_i}$$
$$= \frac{A_f R_L}{R_{of} + R_L}$$
$$= \frac{(5)(10\ \text{k})}{5\ \text{k} + 10\ \text{k}}$$
$$= \frac{50\ \text{k}}{15\ \text{k}}$$
$$= 3.33\ \text{V/V}$$

Thus, the total gain of the amplifier A_{TOT} is $\boxed{3.33\ \text{V/V}}$.

Step 12 of 13

(g)
The expression for the gain of the amplifier with load resistance included in the circuit is,

$$A = g_m\left[\frac{r_o}{2} \| (R_1 + R_2) \| R_L\right]$$
$$= \frac{2I_D}{V_{OV}}\left[\frac{V_A}{2I_O} \| (R_1 + R_2) \| R_L\right]$$
$$= \left[\frac{(2)(100\ \mu)}{0.2}\right]\left\{\left[\frac{10}{(2)(100\ \mu)}\right] \| (1\ \text{M}) \| 10\ \text{k}\right\}$$
$$= \left(10^{-3}\right)\left[50000 \| 10^6 \| 10\ \text{k}\right]$$
$$= \left(10^{-3}\right)\left[50000 \| 10^6 \| 10\ \text{k}\right]$$
$$= \left(10^{-3}\right)\left[47.62\ \text{k} \| 10\ \text{k}\right]$$
$$= \left(10^{-3}\right)\left[\frac{(47.62\ \text{k})(10\ \text{k})}{(47.62\ \text{k}) + (10\ \text{k})}\right]$$
$$= 8.264$$

Hence, the value of the gain, A is $\boxed{8.624}$.

Step 13 of 13

The feedback factor (β) is,

$$\beta = \frac{R_2}{1\ \text{M}}$$
$$= \frac{179\ \text{K}}{1000}$$
$$= 0.179$$

The gain with feedback A_f is,

$$A_f = \frac{A}{1 + A\beta}$$
$$= \frac{8.264}{1 + (8.264)(0.179)}$$
$$= \frac{8.264}{2.479}$$
$$= 3.33\ \text{V/V}$$

Hence, the gain with feedback A_f is $\boxed{3.33\ \text{V/V}}$.

Therefore, the gain with feedback (A_f) is same as the part (f) total gain of the amplifier A_{TOT}.

11.043P

We don't have the solution to this problem yet.

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Step 1 of 13

(a)

Refer to the circuit diagram of P10.41 in the textbook.

Draw the A circuit for the Figure.

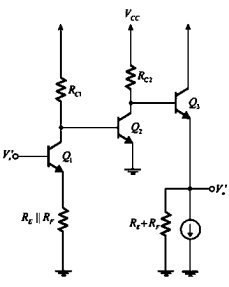


Figure 1

Step 2 of 13

Draw the circuit for determine β.

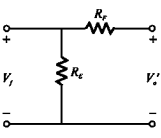


Figure 2

Step 3 of 13

(b)

The closed loop voltage gain is given by

$$A_f = \frac{V_o}{V_o'} = \frac{A}{1 + A\beta}$$

Now if $A\beta$ is large, that is $A\beta \gg 1$, then

$$1 + A\beta \simeq A\beta$$

Hence,

$$A_f = \frac{A}{A\beta} = \frac{1}{\beta}$$

From the β circuit,

$$\beta = \frac{V_f}{V_o'}$$

Apply voltage division rule,

$$V_f = \left(\frac{R_E}{R_E + R_F}\right)\left(V_o'\right)$$

$$\frac{V_f}{V_o'} = \left(\frac{R_E}{R_E + R_F}\right)$$

Hence,

$$\beta = \left(\frac{R_E}{R_E + R_F}\right)$$

Step 4 of 13

Therefore,

$$A_f = \frac{1}{\beta} = \frac{1}{\frac{R_E}{R_E + R_F}} = \frac{R_F + R_E}{R_E}$$

Hence, the closed loop voltage gain A_f is $\boxed{\frac{R_F + R_E}{R_E}}$.

Step 5 of 13

(c)

Consider the following data:

$$R_E = 50\Omega$$

$$A_f = 25\,\text{V/V}$$

Determine the value of R_F

Feedback gain is,

$$A_f = \frac{R_F + R_E}{R_E}$$

That implies,

$$A_f = 1 + \frac{R_F}{R_E}$$

$$1 + \frac{R_F}{50} = 25$$

Solve for R_F.

$$R_F = 24 \times 50 = 1200$$

Therefore,

$$\boxed{R_F = 1.2\,\text{k}\Omega}$$

Step 6 of 13

(d)

Consider the following data:

$$I_{Q_1} = 1\,\text{mA}$$

$$I_{Q_2} = 2\,\text{mA}$$

$$I_{Q_3} = 5\,\text{mA}$$

$$h_{fe} = 100$$

$$A_{Q_2} = -10$$

$$A_{Q_3} = -50$$

We need to determine the approximate values of R_{C1} and R_{C2} .

$$\beta = h_{fe} = 100$$

Step 7 of 13

Emitter resistance of transistor Q_1 is,

$$r_{e_1} = \frac{V_T}{I_{Q_1}} = \frac{25\,\text{mV}}{1\,\text{mA}} = 25\Omega$$

Emitter resistance of transistor Q_2 is,

$$r_{e_2} = \frac{V_T}{I_{Q_2}} = \frac{25\,\text{mV}}{2\,\text{mA}} = 12.5\Omega$$

Emitter resistance of transistor Q_3 is,

$$r_{e_3} = \frac{V_T}{I_{Q_3}} = \frac{25\,\text{mV}}{5\,\text{mA}} = 5\Omega$$

Step 8 of 13

The closed loop gain of transistor Q_1 is,

$$A_{Q_1} = \frac{-R_{C1} \parallel r_{e_2}}{r_{e_1} + (R_E \parallel R_F)}$$

That implies,

$$-R_{C1} \parallel r_{e_2} = A_{Q_1}\left(r_{e_1} + (R_E \parallel R_F)\right)$$

$$-\left(\frac{R_{C1} \times r_{e_2}}{R_{C1} + r_{e_2}}\right) = A_{Q_1}\left(r_{e_1} + \left(\frac{R_E \times R_F}{R_E + R_F}\right)\right)$$

We have, $r_{e_2} = 1.25\,\text{k}\Omega$

Substitute the known values in the equation.

$$-\left(\frac{R_{C1} \times 1.25\,\text{k}}{R_{C1} + 1.25\,\text{k}}\right) = -10\left(25 + \left(\frac{1.2\,\text{k} \times 50}{1.2\,\text{k} + 50}\right)\right)$$

$$\left(\frac{R_{C1} \times 1.25\,\text{k}}{R_{C1} + 1.25\,\text{k}}\right) = 730$$

$$R_{C1} = (0.584)(R_{C1} + 1.25\,\text{k})$$

$$R_{C1} = 0.584R_{C1} + 730$$

$$0.416R_{C1} = 730$$

$$R_{C1} = 1754.80$$

Therefore, collector resistance R_{C1} is $\boxed{1.75\,\text{k}\Omega}$

Step 8 of 13

The closed loop gain of transistor Q_2 is,

$$A_{Q_2} = \frac{-R_{C2} \parallel \left[\beta\left(r_{e_3} + R_E + R_F\right)\right]}{r_{e_2}}$$

That implies,

$$-R_{C2} \parallel \left[\beta\left(r_{e_3} + R_E + R_F\right)\right] = A_{Q_2} r_{e_2}$$

Substitute corresponding value.

$$-R_{C2} \parallel \left[100\left(25 + 50 + 1.2 \times 10^3\right)\right] = -50(12.5)$$

$$R_{C2} \parallel [127500] = 50(12.5)$$

$$\frac{R_{C2} \times [127500]}{R_{C2} + 127500} = 50(12.5)$$

$$\frac{R_{C2}}{R_{C2} + 127500} = 4.9 \times 10^{-3}$$

$$R_{C2} = \left(R_{C2} + 127500\right)\left(4.9 \times 10^{-3}\right)$$

$$R_{C2} = R_{C2}\left(4.9 \times 10^{-3}\right) + 625$$

$$0.9951R_{C2} = 625$$

$$R_{C2} \cong 628$$

Therefore, collector resistance R_{C2} is $\boxed{628\Omega}$.

Step 10 of 13

The closed loop gain of transistor Q_3 is,

$$A_{Q_3} = \frac{R_E + R_F}{r_{e_3} + R_E + R_F} = \frac{50 + 1.2 \times 10^3}{5 + 50 + 1.2 \times 10^3} = \frac{1250}{1255} = 0.996$$

Hence,

$$A_{Q_3} = 0.996\,\text{V/V}$$

Step 11 of 13

(e)

The overall open loop voltage gain is given by

$$A = A_{Q_1} A_{Q_2} A_{Q_3} = (-10)(-50)(0.996) = 498$$

The feedback factor is given by

$$\beta_{FB} = \frac{R_E}{R_E + R_F} = \frac{50}{50 + 1200} = 0.04$$

Hence, the overall closed loop gain is given by

$$A_f = \frac{A}{1 + A\beta_{FB}} = \frac{498}{1 + (498)(0.04)} = \frac{498}{20.92} = 23.80$$

Therefore,

$$\boxed{A_f = 23.80\,\text{V/V}}$$

Step 12 of 13

(f)

From the circuit,

The input resistance without feedback is

$$R_i = (\beta + 1)\left(r_{e_1} + R_E \parallel R_F\right) = (100 + 1)(25 + 50 \parallel 1200) = (101)(25 + 48) = 7373$$

The input resistance of the closed loop amplifier is,

$$R_{if} = R_i\left(1 + A\beta_{FB}\right) = (7373)\left(1 + (498)(0.04)\right) = (7373)(20.92) = 154.18 \times 10^3$$

Therefore, the required input resistance is,

$$\boxed{R_{if} = 154.18\,\text{k}\Omega}$$

Step 13 of 13

The output resistance without feedback is,

$$R_o = (R_E + R_F) \parallel \left(r_{e_3} + R_{C1} \parallel (\beta + 1)\right) = \left(50 + 1.2 \times 10^3\right) \parallel \left(5 + 628.1 \parallel (100 + 1)\right) = (1250) \parallel (92) = 85.70$$

The output resistance of the closed loop amplifier is,

$$R_{of} = \frac{R_o}{\left(1 + A\beta_{FB}\right)} = \frac{85.70}{(1 + 498 \times 0.04)} = \frac{85.70}{20.92} = 4.096$$

Therefore, the required output resistance R_{of} is $\boxed{4.096\Omega}$

Step 1 of 13

(b)

Calculate the value of the resistance R_2.

The resistance, $R_1 = 20\ \text{k}\Omega$.

The voltage gain is,

$$\frac{V_0}{V_s} = A_f = 5\ \text{V/V}$$

$$A_f = \frac{R_1 + R_2}{R_1}$$

$$5 = \frac{20 + R_2}{20}$$

$$100 = 20 + R_2$$

$$\Rightarrow R_2 = 100 - 20$$

$$= 80\ \text{k}\Omega$$

Hence, the value of the resistance R_2 is $\boxed{80\ \text{k}\Omega}$.

Step 2 of 13

(c)

The differential input resistance of amplifier A_1, $R_{id1} = 82\ \text{k}\Omega$

The open-circuit differential voltage gain, $A_1 = 20\ \text{V/V}$

The output resistance, $R_{o1} = 3.2\ \text{k}\Omega$

The input resistance of the amplifier A_2, $R_{i2} = 5\ \text{k}\Omega$

The short-circuit trans-conductance, $g_{m2} = 20\ \text{mA/V}$

The output resistance, $R_{o2} = 20\ \text{k}\Omega$

The input resistance of amplifier A_2, $R_{i3} = 20\ \text{k}\Omega$

The open-circuit voltage gain is unity.

The output resistance, $R_{o3} = 1\ \text{k}\Omega$

The load resistance of the amplifier, $R_L = 1\ \text{k}\Omega$

The source resistance, $R_s = 9\ \text{k}\Omega$

The feedback resistances are,

$$R_1 = 20\ \text{K}\Omega$$
$$R_2 = 80\ \text{K}\Omega$$

Step 3 of 13

Draw the small signal equivalent circuit $(A - \text{Circuit})$.

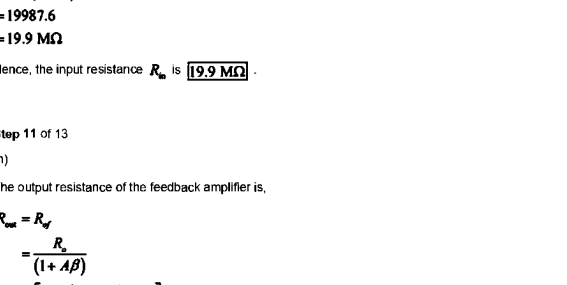


Figure 1

Step 4 of 13

d)

The *A*-circuit is shown in Figure 1.

The open loop gain of the amplifier is,

$$A = A_1 A_2 A_2 \frac{(R_1 + R_2) \| R_L}{(R_1 + R_2) \| R_L + R_L}$$

Determine the gain of the first stage of the amplifier.

$$A_1 = \left(\frac{R_{id1}}{R_{id1} + R_s + R_1 \| R_2} \right)$$

$$= \frac{82\ \text{k}}{82\ \text{k} + 9\ \text{k} + 20\ \text{k} \| 80\ \text{k}}$$

$$= \frac{82\ \text{k}}{107\ \text{k}}$$

$$= 0.766$$

Step 5 of 13

Determine the gain of the second stage of the amplifier.

$$A_2 = \frac{A_1 R_{i2}}{R_{o1} + R_{i2}}$$

$$= \frac{(20)(5\ \text{k})}{3.2\ \text{k} + 5\ \text{k}}$$

$$= 12.19$$

Step 6 of 13

Determine the gain of the third stage of the amplifier.

$$A_2 = g_{m2} (R_{o2} \| R_{i3})$$

$$= (20 \times 10^{-3})(20\ \text{k} \| 20\ \text{k})$$

$$= (20 \times 10^{-3})(10\ \text{k})$$

$$= 200$$

Step 7 of 13

Substitute the calculated values in the gain formula.

$$A = (0.766)(12.19)(200)\left[\frac{(20\ \text{k} + 80\ \text{k}) \| 1\ \text{k}}{(20\ \text{k} + 80\ \text{k}) \| 1\ \text{k} + 1\ \text{k}} \right]$$

$$= (0.766)(12.19)(200)\left[\frac{100\ \text{k} \| 1\ \text{k}}{100\ \text{k} \| 1\ \text{k} + 1\ \text{k}} \right]$$

$$= (0.766)(12.19)(200)\left[\frac{990.1}{990.1 + 1\ \text{k}} \right]$$

$$\approx 929\ \text{V/V}$$

Hence, the gain A of the three-stage amplifier is $\boxed{929\ \text{V/V}}$.

Step 8 of 13

(e)

The feedback factor is,

$$\beta = \frac{R_1}{R_1 + R_2}$$

$$= \frac{20\ \text{k}}{20\ \text{k} + 80\ \text{k}}$$

$$= \frac{20\ \text{k}}{100\ \text{k}}$$

$$= 0.2$$

Hence, the feedback factor is $\boxed{\beta = 0.2}$.

Calculate the amount of feedback.

$$1 + A\beta = 1 + (929)(0.2)$$

$$= 186.8$$

Hence, the amount of feedback, $1 + A\beta$ is $\boxed{186.8}$.

Step 9 of 13

(f)

The closed loop gain of the amplifier is,

$$A_f = \frac{A}{1 + A\beta}$$

$$= \frac{929}{186.8}$$

$$= 4.9732\ \text{V/V}$$

Hence, the closed loop gain A_f is $\boxed{4.9732\ \text{V/V}}$.

Step 10 of 13

(g)

The input resistance of the feedback amplifier is,

$$R_{in} = R_{if}$$

$$= R_i (1 + A\beta)$$

$$= \left[R_{id1} + R_s + R_1 \| R_2 \right](1 + A\beta)$$

$$= (82\ \text{k} + 9\ \text{k} + 20\ \text{k} \| 80\ \text{k})(186.8)$$

$$= 107(186.8)$$

$$= 19987.6$$

$$= 19.9\ \text{M}\Omega$$

Hence, the input resistance R_{in} is $\boxed{19.9\ \text{M}\Omega}$.

Step 11 of 13

(h)

The output resistance of the feedback amplifier is,

$$R_{out} = R_{of}$$

$$= \frac{R_o}{(1 + A\beta)}$$

$$= \frac{\left[R_L \| (R_1 + R_2) \| R_{o3} \right]}{(1 + A\beta)}$$

$$= \frac{(1\ \text{k}) \| (100\ \text{k}) \| (1\ \text{k})}{186.8}$$

$$\approx 2.66\ \Omega$$

Hence, the required input resistance R_{out} is $\boxed{2.66\ \Omega}$.

Step 12 of 13

(i)

The high frequency, $f_H = 100\ \text{Hz}$

The upper 3 dB frequency of the closed-loop gain is,

$$f_{Hf} = f_H (1 + A\beta)$$

$$= (100)(186.8)$$

$$= 18.68\ \text{kHz}$$

Hence, the required upper 3 dB frequency f_{Hf} is $\boxed{18.68\ \text{kHz}}$.

Step 13 of 13

(j)

The gain A_1 drops to half of its nominal value.

The new closed-loop gain is,

$$A_f = \frac{A}{1 + A\beta}$$

$$= \frac{\frac{929}{2}}{1 + \left(\frac{929}{2} \right)(0.2)}$$

$$= \frac{464.5}{93.3}$$

$$= 4.978\ \text{V/V}$$

Calculate the percentage change in the closed-loop gain.

$$\frac{\Delta A_f}{A_f} = \left(\frac{4.978 - 4.9732}{4.9732} \right) \times 100$$

$$= (9.651 \times 10^{-4}) \times 100$$

$$= 0.096\%$$

Therefore, the percentage change in gain A_f is $\boxed{0.096\%}$.

Step 1 of 4

11.046P

Refer to the Figure 11.17(a) and Figure 11.17(b) in the textbook.

The feedback network is fed with a current I_o at port 2 and the voltage V_f appearing across port 1. The feedback parameter is,

$$\begin{aligned}\beta &= \frac{V_f}{I_o} \\ &= R_F\end{aligned}$$

If $A\beta >> 1$,then the closed loop gain A_f is,

$$\begin{aligned}A_f &= \frac{I_o}{V_s} \\ &= \frac{1}{\beta}\end{aligned}$$

$$A_f = \frac{1}{R_F} \quad \text{..... (1)}$$

Step 2 of 4

Determine the resistance R_F.

Substitute $10\,\text{mA/V}$ for A_f in equation 1.

$$10\,\text{m} = \frac{1}{R_F}$$

$$\begin{aligned}R_F &= \frac{1}{10\times10^{-3}} \\ &= 100\,\Omega \\ &= 0.1\,\text{k}\Omega\end{aligned}$$

Therefore, the resistance R_F is $\boxed{0.1\,\text{k}\Omega}$.

Step 3 of 4

Consider the loop gain $A\beta$.

$$\begin{aligned}A\beta &= -\frac{V_r}{V_t} \\ &= \mu\frac{g_m\left(R_F \| R_{id} \| r_{o2}\right)}{1+g_m\left(R_F \| R_{id} \| r_{o2}\right)}\end{aligned}$$

Substitute 1000 for μ , $2\,\text{mA/V}$ for g_m , $100\,\text{k}\Omega$ for R_{id} , $20\,\text{k}\Omega$ for r_{o2} in equation.

$$\begin{aligned}A\beta &= (1000)\left(\frac{2\,\text{m}\left(0.1\,\text{k}\|100\,\text{k}\|20\,\text{k}\right)}{1+2\,\text{m}\left(0.1\,\text{k}\|100\,\text{k}\|20\,\text{k}\right)}\right) \\ &= 1000\left(\frac{2\,\text{m}(0.099\,\text{k})}{1+2\,\text{m}(0.099\,\text{k})}\right) \\ &= 1000\left(\frac{0.19}{1.19}\right) \\ &= 167.06\end{aligned}$$

Step 4 of 4

Consider feedback factor as $\beta = 0.1\,\text{k}\Omega$.

Open loop gain A is,

$$\begin{aligned}A &= \frac{A\beta}{\beta} \\ &= \frac{167.06}{0.1\,\text{k}} \\ &= 1.67\end{aligned}$$

The closed loop gain A_f is,

$$\begin{aligned}A_f &= \frac{A}{1+A\beta} \\ &= \frac{1.67}{1+167.06} \\ &= 9.94\,\text{mA/V}\end{aligned}$$

Therefore, the closed loop gain A_f is $\boxed{9.94\,\text{mA/V}}$.

11.047P

We don't have the solution to this problem yet.

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Step 1 of 7

(a)

Refer to the Figure P11.48 (a) and Figure P11.48 (b) in the textbook.

The feedback factor β is,

$$\beta = \frac{I_f}{V_o}$$

Substitute $-R_F I_f$ for V_o in equation.

$$\beta = \frac{I_f}{-R_F I_f}$$
$$= -\frac{1}{R_F}$$

Therefore, the feedback factor β is $\boxed{-\frac{1}{R_F}}$.

Step 2 of 7

The closed loop gain A_f is,

$$A_f = \frac{A}{1 + A\beta}$$

The expression of A_f for $A\beta >> 1$ is,

$$A_f = \frac{1}{\beta}$$
$$= \frac{1}{\left(-\frac{1}{R_F}\right)}$$
$$= -R_F$$

Therefore, the closed loop gain A_f is, $\boxed{-R_F}$.

Step 3 of 7

Determine the resistance of R_F.

Consider the closed loop gain A_f.

$$-R_F = A_f$$

Consider magnitude of the resistance R_F, since, it can't be negative.

$$|R_F| = A_f$$

Substitute $1\ \text{k}\Omega$ for A_f in equation.

$$|R_F| = 1\ \text{k}\Omega$$

Therefore, the resistance of R_F is $\boxed{1\ \text{k}\Omega}$.

Step 4 of 7

Determine the loop gain $A\beta$.

Set $I_s = 0$ and break the loop at input, such that feedback network and loop gain doesn't change. The equivalent circuit diagram is shown in Figure 1.

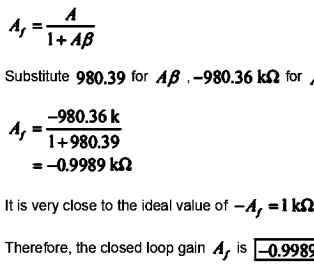


Figure 1

Step 5 of 7

The loop gain $A\beta$ is,

$$A\beta = -\frac{V_r}{V_i}$$

Apply voltage division rule across resistor R_{id}.

$$V_r = -\mu V_i \left(\frac{R_{id}}{R_{id} + R_F + r_o}\right)$$
$$-\frac{V_r}{V_i} = \mu\left(\frac{R_{id}}{R_{id} + R_F + r_o}\right)$$
$$A\beta = \mu\left(\frac{R_{id}}{R_{id} + R_F + r_o}\right)$$

Therefore, the loop gain $A\beta$ is $\boxed{\mu\left(\frac{R_{id}}{R_{id} + R_F + r_o}\right)}$.

Step 6 of 7

Determine the closed loop gain A_f.

Calculation of the loop gain $A\beta$ is as follows:

$$A\beta = \mu\left(\frac{R_{id}}{R_{id} + R_F + r_o}\right)$$

Substitute 1000 for μ, $100\ \text{k}\Omega$ for R_{id}, $1\ \text{k}\Omega$ for r_o, $1\ \text{k}\Omega$ for R_F in equation.

$$A\beta = 1000\left(\frac{100\ \text{k}}{100\ \text{k} + 1\ \text{k} + 1\ \text{k}}\right)$$
$$= 1000\left(\frac{100}{102}\right)$$
$$= 980.39$$

The feedback factor β is,

$$\beta = -\frac{1}{R_F}$$

Substitute $1\ \text{k}\Omega$ for R_F in equation.

$$\beta = -\frac{1}{1\ \text{k}}$$
$$= -0.001$$

Step 7 of 7

The open loop gain A is,

$$A = \frac{A\beta}{\beta}$$

Substitute 980.39 for $A\beta$, -0.001 for β in equation.

$$A = \frac{980.39}{-0.001}$$
$$= -980.36\ \text{k}\Omega$$

Closed loop gain A_f is,

$$A_f = \frac{A}{1 + A\beta}$$

Substitute 980.39 for $A\beta$, $-980.36\ \text{k}\Omega$ for A in equation.

$$A_f = \frac{-980.36\ \text{k}}{1 + 980.39}$$
$$= -0.9989\ \text{k}\Omega$$

It is very close to the ideal value of $-A_f = 1\ \text{k}\Omega$.

Therefore, the closed loop gain A_f is $\boxed{-0.9989\ \text{k}\Omega}$.

Step 1 of 6

Draw the series-series feedback amplifier equivalent *A* circuit.

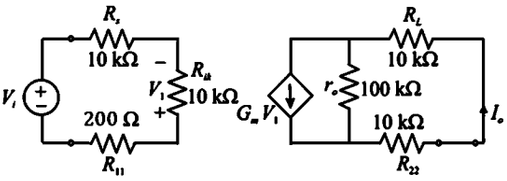


Figure 1

Step 2 of 6

From the series-series feedback amplifier of equivalent circuit in Figure-1, the voltage V_1 is,

$$V_1 = -V_i \frac{R_{it}}{R_{it} + R_s + R_{11}}$$

Substitute $10\ \text{k}\Omega$ for R_{it}, $200\ \Omega$ for R_{11} and $10\ \text{k}\Omega$ for R_s.

$$V_1 = -V_i \frac{10\ \text{k}}{200 + 10\ \text{k} + 10\ \text{k}}$$
$$= -V_i \left(\frac{10\ \text{k}}{20.2\ \text{k}}\right)$$

Step 3 of 6

From Figure 1, the output current I_o is,

$$I_o = -\frac{G_m V_1 (r_o)}{r_o + R_L + R_{22}}$$

Substitute $0.6\ \text{A/V}$ for G_m, $100\ \text{k}\Omega$ for r_o, $10\ \text{k}\Omega$ for R_{22} and $10\ \text{k}\Omega$ for R_L.

$$I_o = -\frac{0.6 V_1 (100\ \text{k})}{100\ \text{k} + 10\ \text{k} + 10\ \text{k}}$$

Substitute $-V_i \left(\frac{10\ \text{k}}{20.2\ \text{k}}\right)$ for V_1.

$$I_o = V_i \left(\frac{0.6(100\ \text{k})}{100\ \text{k} + 10\ \text{k} + 10\ \text{k}}\right)\left(\frac{10\ \text{k}}{20.2\ \text{k}}\right)$$
$$= V_i (0.2475)$$

The output equation is,

$$\frac{I_o}{V_i} = 0.2475$$
$$A = 0.2475\ \text{A/V}$$

Hence, the required open loop gain A is $\boxed{0.2475\ \text{A/V}}$.

Step 4 of 6

The closed loop gain of series-series feedback amplifier that employs a trans-conductance amplifier is,

$$A_f = \frac{A}{1 + A\beta}$$
$$= \frac{0.2475}{1 + 0.2475(200)}$$
$$= \frac{0.2475}{50.5}$$
$$= 4.9 \times 10^{-3}\ \text{A/V}$$
$$= 4.9\ \text{mA/V}$$

Hence, the required loop gain is $\boxed{A_f = 4.9\ \text{mA/V}}$

Step 5 of 6

The input resistance of the open loop amplifier is,

$$R_i = R_s + R_{11} + R_{it}$$
$$= 10\ \text{k}\Omega + 200\ \Omega + 10\ \text{k}\Omega$$
$$= 20.2\ \text{k}\Omega$$

The, input resistance of the closed loop amplifier is,

$$R_{if} = R_i (1 + A\beta)$$
$$= 20.2(1 + 0.2475 \times 200)$$
$$= 1020.1\ \text{k}\Omega$$

Obtain the input resistance, R_{in} which is the resistance seen by R_s from the circuit,

$$R_{in} = R_{if} - R_s$$
$$= 1020.1\ \text{k}\Omega - 10\ \text{k}\Omega$$
$$= 1010.1\ \text{k}\Omega$$

Hence, the required input resistance of the circuit R_{in} is $\boxed{1010.1\ \text{k}\Omega}$.

Step 6 of 6

The open loop output resistance R_o is,

$$R_o = R_L + r_o + R_{22}$$
$$= 10\ \text{k}\Omega + 100\ \text{k}\Omega + 10\ \text{k}\Omega$$
$$= 120\ \text{k}\Omega$$

The output resistance with feedback is,

$$R_{of} = R_o (1 + A\beta)$$
$$= (120\ \text{k}\Omega)(1 + 0.2475 \times 200)$$
$$= 120\ \text{k}\Omega (1 + 49.5)$$
$$= 120\ \text{k}\Omega (50.5)$$
$$= 6060\ \text{k}\Omega$$
$$= 6.06\ \text{M}\Omega$$

Determine the output resistance, R_{out}.

$$R_{out} = R_{of} - R_L$$
$$= 6.06\ \text{M}\Omega - 10\ \text{k}\Omega$$
$$= 6.050\ \text{M}\Omega$$

Hence, the required value of R_{out} is $\boxed{6.050\ \text{M}\Omega}$.

11.050P

We don't have the solution to this problem yet.

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Step 1 of 11

(a)

Refer to Figure 11.31P in the text book.

Determine the value of the feedback resistance, R_F using the formula for the feedback factor, β.

$$\frac{1}{\beta}=\frac{R_{S2}\text{ or }R_{S1}}{R_{S2}+R_{S1}+R_F}$$

Assume $\beta=10$

$$\frac{1}{10}=\frac{100}{100+100+R_F}$$

$$0.1=\frac{100}{200+R_F}$$

$$100=200\left(\frac{1}{10}\right)+(0.1)R_F$$

$$100=20+0.1R_F$$

$$100-20=0.1R_F$$

$$80=0.1R_F$$

$$R_F=\frac{80}{0.1}$$

$$=800\ \Omega$$

Hence, the resistance R_F is $\boxed{800\ \Omega}$.

Step 2 of 11

(b)

Draw the A circuit.

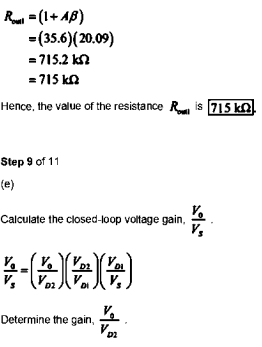


Figure 1

Step 3 of 11

Determine the value of A.

$$A=\frac{I_0}{V_i}$$

It can also be expressed as,

$$A=\left(\frac{I_0}{V_{D2}}\right)\left(\frac{V_{D2}}{V_{D1}}\right)\left(\frac{V_{D1}}{V_{Di}}\right)$$

$$\frac{V_{D1}}{V_i}=-\frac{gmR_{D1}}{1+gm\left[R_{S1}\|\left(R_F+R_{S2}\right)\right]}$$

$$=-\frac{\left(4\times10^{-2}\right)\left(10\times10^{3}\right)}{1+\left(4\times10^{-3}\right)\left[100\|\left(800+100\right)\right]}$$

$$=\frac{\left(4\times10^{-3}\right)\left(10\times10^{3}\right)}{1+\left(4\times10^{-3}\right)\left[\frac{100\times900}{100+900}\right]}$$

$$=\frac{\left(4\times10^{-3}\right)\left(10\times10^{3}\right)}{1+\left(4\times10^{-3}\right)(90)}$$

$$=-29.41\ \text{V/V}$$

Step 4 of 11

Determine the value of $\frac{V_{D2}}{V_{D1}}$.

$$\frac{V_{D2}}{V_{D1}}=-gmR_{D2}$$

$$=-\left(4\times10^{-3}\right)\left(10\times10^{3}\right)$$

$$=-40\ \text{V/V}$$

$$\frac{I_0}{V_{D2}}=\frac{1}{\frac{1}{gm}+R_{S2}\|\left[R_{S1}+R_F\right]}$$

$$=\frac{1}{\frac{1}{4\times10^{-3}}+100\|\left(100+800\right)}$$

Step 5 of 11

$$=\frac{1}{\frac{1}{4\times10^{-3}}+\frac{100\times900}{100+900}}$$

$$=\frac{1}{\frac{1}{4\times10^{-3}}+90}$$

$$=2.94\times10^{-2}\ \text{A/V}$$

Substitute the values in the gain, A.

$$A=\left(\frac{I_0}{V_{D2}}\right)\left(\frac{V_{D2}}{V_{D1}}\right)\left(\frac{V_{D1}}{V_{Di}}\right)$$

$$=\left(2.94\times10^{-3}\right)(-40)(-29.41)$$

$$=3.46\ \text{A/V}$$

Hence, the value of A is $\boxed{3.46\ \text{A/V}}$.

Step 6 of 11

(c)

Calculate the values of $\left(1+A\beta\right)$ and A_f.

$$1+A\beta=1+(3.46)(10)$$

$$=1+34.6$$

$$=35.6$$

Hence, the value of $\left(1+A\beta\right)$ is $\boxed{35.6}$.

$$A_f=\frac{A}{1+A\beta}$$

$$=\frac{3.46}{35.6}$$

$$=0.0972\ \text{A/V}$$

Hence, the feedback gain A_f is $\boxed{0.0972\ \text{A/V}}$.

Step 7 of 11

Designed A_f value is 0.1 A/V $\left(100\ \text{mA/V}\right)$

Determine the percentage difference of A_f.

$$\%A_f=\frac{0.1-0.0972}{0.1}\times100$$

$$=2.8\%$$

Hence, the percentage of A_f is $\boxed{2.8\%}$.

The resistance must be decreased to make A_f exactly equal to 100mA/V.

Step 8 of 11

(d)

The resistance, $r_{o3}=20\ \text{k}\Omega$

Calculate the values of R_{o3} and R_{out1}.

$$R_{o3}=r_{o3}+\left[R_{S2}\|\left(R_F+R_{S1}\right)\right]$$

$$=20\ \text{k}+\left[100\|\left(800+100\right)\right]$$

$$=20\ \text{k}+\left(100\|900\right)$$

$$=20\ \text{k}+\frac{(100)(900)}{1000}$$

$$=20\ \text{k}+90$$

$$=20.09\ \text{k}\Omega$$

$$R_{out1}=\left(1+A\beta\right)$$

$$=(35.6)(20.09)$$

$$=715.2\ \text{k}\Omega$$

$$=715\ \text{k}\Omega$$

Hence, the value of the resistance R_{out1} is $\boxed{715\ \text{k}\Omega}$.

Step 9 of 11

(e)

Calculate the closed-loop voltage gain, $\frac{V_0}{V_s}$.

$$\frac{V_0}{V_s}=\left(\frac{V_0}{V_{D2}}\right)\left(\frac{V_{D2}}{V_{D1}}\right)\left(\frac{V_{D1}}{V_s}\right)$$

Determine the gain, $\frac{V_0}{V_{D2}}$.

$$\frac{V_0}{V_{D2}}=\frac{R_{S2}\|\left(R_{S1}+R_F\right)}{\frac{1}{gm}+R_{S2}\|\left(R_{S1}+R_F\right)}$$

$$=\frac{100\|(100+800)}{\frac{1}{4\times10^{-3}}+100\|(100+800)}$$

$$=\frac{100\times900}{\frac{1}{4\times10^{-3}}+\left(\frac{100\times900}{100+900}\right)}$$

$$=\frac{90}{\frac{1}{4\times10^{-3}}+90}$$

$$=0.2646\ \text{V/V}$$

$$\frac{V_{D2}}{V_{D1}}=-40\ \text{V/V}$$

Step 10 of 11

Determine the value of $\frac{V_{D1}}{V_s}$.

$$\frac{V_{D1}}{V_S}=-\frac{gmR_{D1}}{1+gm\left[R_{S1}\|R_F\right]}$$

$$=-\frac{\left(4\times10^{-3}\right)\left(10\times10^{3}\right)}{1+\left(4\times10^{-3}\right)\left[100\|800\right]}$$

$$=\frac{\left(4\times10^{-3}\right)\left(10\times10^{3}\right)}{1+\left(4\times10^{-3}\right)\left(\frac{100\times800}{100+800}\right)}$$

$$=\frac{\left(4\times10^{-3}\right)\left(10\times10^{3}\right)}{1+\left(4\times10^{-2}\right)(88.89)}$$

$$=-29.51\ \text{V/V}$$

Determine the gain, $\frac{V_0}{V_s}$.

$$\frac{V_0}{V_s}=\left(\frac{V_0}{V_{D2}}\right)\left(\frac{V_{D2}}{V_{D1}}\right)\left(\frac{V_{D1}}{V_s}\right)$$

$$=(0.2646\ \text{V/V})(-40\ \text{V/V})(-29.51\ \text{V/V})$$

$$=312.3\ \text{V/V}$$

Hence, the gain $\frac{V_0}{V_s}$ is $\boxed{312.3\ \text{V/V}}$.

Step 11 of 11

(f)

The feedback factor β is,

$$\beta=\frac{100}{100+800}$$

$$=0.1111$$

The resistance $R_0=\left(1+A\beta\right)\left[R_{S2}\|\left(R_{S1}+R_F\right)\right]\|r_0\|\frac{1}{g_m}$

$$R_0=(35.7)\left[100\|900\right]\|20\ \text{k}\|\frac{1}{4\times10^{-3}}$$

$$=(35.7)\left[\frac{100\times90}{100+900}\|20\ \text{k}\|\frac{1}{4\times10^{-3}}\right]$$

$$=35.7\left[90\|20\ \text{k}\|250\right]$$

$$=35.7\left[\frac{90\times20\times10^{2}}{90+20\times10^{2}}\|250\right]$$

$$=35.7(65.958)$$

$$=2354.70$$

$$=2.36\ \text{k}\Omega$$

The value of R_{out2} is,

$$R_{out2}=35.7\times R_0$$

$$=35.7\times2.36$$

$$=84.1\ \text{k}\Omega$$

Therefore, the output resistance R_{out2} is $\boxed{84.1\ \text{k}\Omega}$.

Step 1 of 6

The trans-conductance amplifier circuit with input is V_s

The trans-conductance amplifier circuit with output is I_o.

Here, as the input voltage V_s increases, the output current I_o also increases and the feedback circuit is composed of resistor R_F.

Step 2 of 6

(b)

The *A* circuit is shown in Figure-1.

The β circuit is shown in Figure 2.

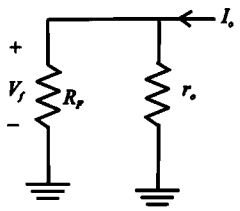


Figure 2

Step 3 of 6

(c)

The expression for the A is,

$$A = \frac{I_o}{V_i}$$
$$= g_m$$

Hence, the amplifier circuit gain is $\boxed{A = g_m}$.

Step 4 of 6

The expression for the feedback factor β is,

$$\beta = r_o \| R_F$$
$$= \frac{r_o R_F}{r_o + R_F}$$

Hence, the required expression for feedback factor is $\boxed{\beta = \frac{r_o R_F}{r_o + R_F}}$.

Step 5 of 6

The expression for the $1 + A\beta$ is,

$$1 + A\beta = 1 + g_m \left(\frac{r_o R_F}{r_o + R_F} \right)$$
$$= 1 + \frac{g_m r_o R_F}{r_o + R_F}$$

Hence, the required expression is, $\boxed{1 + A\beta = 1 + \frac{g_m r_o R_F}{r_o + R_F}}$.

Step 6 of 6

Find the feedback factor A_f.

$$A_f = \frac{A}{1 + A\beta}$$
$$= \frac{g_m}{1 + \frac{g_m r_o R_F}{r_o + R_F}}$$

Hence, the required feedback factor is $\boxed{A_f = \frac{g_m}{1 + \frac{g_m r_o R_F}{r_o + R_F}}}$.

The amplifier output resistance is $\boxed{R_o = r_o}$.

The amplifier feedback resistance is $\boxed{R_{of} = r_o (1 + g_m R_F)}$.

Step 1 of 8

The expression for feedback factor β is,

$$\beta = \frac{V_f}{I_o}$$
$$= R_F$$

The closed loop gain A_f is,

$$A_f = \frac{A}{1 + A\beta}$$

For $A\beta \gg 1$, closed loop gain A_f is,

$$A_f \approx \frac{1}{\beta} = \frac{1}{R_F}$$

Thus, the approximate expression for closed-loop trans-conductance A_f is $\boxed{\frac{1}{R_F}}$.

Step 2 of 8

The desired loop gain is,

$A_f = 5\ \text{mA/V}$.

$$A_f = \frac{1}{R_F}$$
$$R_F = \frac{1}{A_f}$$
$$= \frac{1}{5\text{mA/V}}$$

The feedback resistor, R_F is $200\ \Omega$.

Therefore, to get the desired A_f, R_F should be $\boxed{200\ \Omega}$.

Step 3 of 8

(b)

Find the voltage V_{id} by applying voltage division rule.

$$V_{id} = V_i \frac{R_{id}}{R_{id} + R_F}$$

The gate to source voltage V_{gs2} is,

$$V_{gs2} = \mu V_{id}$$

Find the output current by applying current division rule.

$$I_o = -g_{m2} V_{gs2} \frac{r_{o2}}{r_{o2} + R_F}$$

Substitute $\mu V_{id} = V_{gs2}$

$$I_o = -g_{m2} \mu V_{id} \frac{r_{o2}}{r_{o2} + R_F}$$

Substitute $V_i \frac{R_{id}}{R_{id} + R_F} = V_{id}$

$$A = \frac{I_o}{V_i} = I_o = -(g_{m2}\mu)\left(\frac{R_{id}}{R_{id} + R_F}\right)\left(\frac{r_{o2}}{r_{o2} + R_F}\right) \qquad (\text{Since}, R_{id} \gg (R_F))$$

Hence, the resulting expression for A is $\boxed{-(g_{m2}\mu)\left(\frac{r_{o2}}{r_{o2} + R_F}\right)}$.

Step 4 of 8

Substitute corresponding values in the expression for A.

$$A = -(g_{m2}\mu)\left(\frac{r_{o2}}{r_{o2} + R_F}\right)$$
$$= -\left(2 \times 10^{-3} \times 1000\right)\left(\frac{20000}{20000 + 200}\right)$$
$$= -1.98$$
$$\cong -2\ \text{A/A}$$

Thus, the open loop gain of the trans-conductance amplifier A is $\boxed{-2\ \text{A/A}}$.

Step 5 of 8

(c)
Multiply A and β terms.

$$A\beta = (g_{m2}\mu)\left(\frac{r_{o2}}{r_{o2} + R_F}\right)(R_F)$$
$$= (g_{m2}\mu)\left(\frac{r_{o2}R_F}{r_{o2} + R_F}\right)$$

The expression for feedback amount is,

$$1 + A\beta = 1 + (g_{m2}\mu)\left(\frac{r_{o2}R_F}{r_{o2} + R_F}\right)$$

Substitute corresponding values.

$$1 + A\beta = 1 + \left(2 \times 10^{-3} \times 1000\right)\left(\frac{(20000)(200)}{20000 + 200}\right)$$
$$= 1 + 396.02$$
$$\cong 397$$

Hence, the required answer for $1 + A\beta$ is $\boxed{397}$.

Step 6 of 8

(d)

The closed loop gain A_f is,

$$A_f = \frac{A}{1 + A\beta}$$
$$= \frac{2}{397}$$
$$= 0.005\ \text{A/V}$$

Thus, the closed loop gain is,

$\boxed{A_f = 5\ \text{mA/V}}$

Hence, the closed loop gain A_f is equal to the result in part (a).

Step 7 of 8

(e)

From Figure 1, the output resistance is, $R_o = r_o$

The output resistance can be increased by connecting the series connection at the output.

Hence, the output resistance is,

$$R_{of} = R_o(1 + A\beta)$$

Substitute, $g_{m2}\mu\left(\frac{r_{o2}R_F}{r_{o2} + R_F}\right)$ for $A\beta$ and r_o for R_o.

$$R_{of} \approx r_o\left(1 + g_{m2}\mu\left(\frac{r_{o2}R_F}{r_{o2} + R_F}\right)\right)$$

Thus, output resistance of the amplifier R_{of} is $\boxed{r_o\left(1 + g_{m2}\mu\left(\frac{r_{o2}R_F}{r_{o2} + R_F}\right)\right)}$

Step 8 of 8

Substitute, $R_0 = 20000$ and $1 + A\beta = 397$ in the equation for R_{of}.

$$R_{of} = R_o(1 + A\beta)$$
$$= (20000)(397)$$
$$= 7940000\ \Omega$$
$$= 7.94\ \text{M}\Omega$$

Therefore, the output resistance of the amplifier R_{of} is $\boxed{7.94\ \text{M}\Omega}$.

Step 1 of 6

11.054P

Refer to the circuit in Figure P10.48 in the textbook. Redraw the circuit by replacing the BJT with its small-signal, hybrid-π model as shown in Figure 1. Short the source V_s and apply a test voltage V_x and assume the current flowing through the circuit due to the voltage source is I_x.

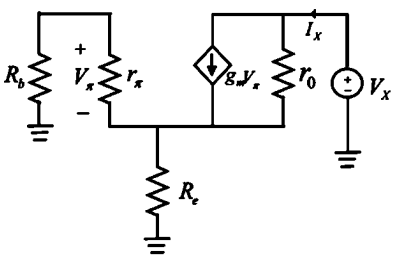


Figure 1

Step 2 of 6

Now observe from the circuit in Figure 1 that the resistors R_b and r_π are in series and they are connected in parallel to the resistor R_e.

Consider the dependent current source connected in parallel to the resistance r_o. Convert it into a voltage source $g_m V_\pi r_o$ in series with the resistance r_o. Draw the simplified circuit as shown in the following Figure 2.

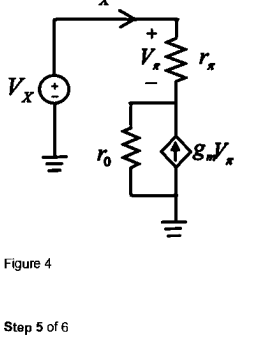


Figure 2

Step 3 of 6

Observe from Figure 2 that the current leaving the resistor r_o and entering the node following it is I_x. Apply current division principle to the resistor r_π to determine the current flowing through it. Denote it as I_1.

$$I_1 = \frac{R_e I_x}{R_e + r_\pi + R_b}$$

Determine the voltage V_π across the resistor r_π.

$$\begin{aligned} V_\pi &= -I_1 r_\pi \\ &= -\left(\frac{R_e I_x}{R_e + r_\pi + R_b}\right) r_\pi \end{aligned}$$

The resistors R_b and r_π are in series. Hence the series equivalent of the two resistors is,

$$R_1 = R_b + r_\pi$$

The series equivalent of the resistors is parallel to the resistor R_e. Hence the parallel equivalent of the two resistors is,

$$R_2 = \frac{(R_b + r_\pi)(R_e)}{(R_b + r_\pi) + R_e}$$

Since the current flowing through the elements is same the same current I_x flows throughout the circuit. Apply the Kirchhoff's voltage law around the loop of the circuit in Figure 3.

$$-V_x - g_m V_\pi r_o + I_x r_o + \left((R_b + r_\pi) \| (R_e)\right) I_x = 0$$

Substitute the expression for V_π in the Kirchhoff's voltage law equation.

$$-V_x - g_m V_\pi r_o + I_x r_o + \left((R_b + r_\pi) \| (R_e)\right) I_x = 0$$

$$-V_x - g_m \left(-\frac{r_\pi I_x R_e}{R_e + R_b + r_\pi}\right) r_o + I_x r_o + \left((R_b + r_\pi) \| (R_e)\right) I_x = 0$$

$$V_x = -g_m \left(-\frac{r_\pi I_x R_e}{R_e + R_b + r_\pi}\right) r_o + I_x r_o + \left((R_b + r_\pi) \| (R_e)\right) I_x$$

$$V_x = I_x \left\{ g_m r_o \left(\frac{r_\pi R_e}{R_e + R_b + r_\pi}\right) + r_o + \left[(R_b + r_\pi) \| (R_e)\right] \right\}$$

Simplify further.

$$\frac{V_x}{I_x} = g_m r_o \left(\frac{r_\pi R_e}{R_e + R_b + r_\pi}\right) + r_o + \left[(R_b + r_\pi) \| (R_e)\right]$$

$$\begin{aligned} R_o &= r_o + g_m r_o \left(\frac{r_\pi R_e}{R_e + R_b + r_\pi}\right) + \left[(R_b + r_\pi) \| (R_e)\right] \\ &= r_o + \left[(R_e) \| (R_b + r_\pi)\right] \left[g_m r_o \left(\frac{r_\pi R_e}{R_e + R_b + r_\pi}\right) \frac{1}{(R_e) \| (R_b + r_\pi)} + 1 \right] \\ &= r_o + \left[(R_e) \| (R_b + r_\pi)\right] \left[g_m r_o \left(\frac{r_\pi R_e}{R_e + R_b + r_\pi}\right) \frac{R_e + R_b + r_\pi}{(R_e)(R_b + r_\pi)} + 1 \right] \end{aligned}$$

Simplify further.

$$\begin{aligned} R_o &= r_o + \left[(R_e) \| (R_b + r_\pi)\right] \left[g_m r_o \left(\frac{r_\pi R_e}{(R_e)(R_b + r_\pi)}\right) + 1 \right] \\ &= r_o + \left[(R_e) \| (R_b + r_\pi)\right] \left[g_m r_o \left(\frac{r_\pi}{(R_b + r_\pi)}\right) + 1 \right] \\ &= r_o + \left[(R_e) \| (R_b + r_\pi)\right] \left[1 + g_m r_o \frac{r_\pi}{r_\pi + R_b} \right) \end{aligned}$$

Hence, the output resistance of the BJT circuit in Figure P10.48 is determined and its value R_o is

$$\boxed{r_o + \left[(R_e) \| (R_b + r_\pi)\right] \left(1 + g_m r_o \frac{r_\pi}{r_\pi + R_b} \right)}.$$

To calculate the maximum value of R_o, substitute $R_b = 0$ in the expression of R_o.

$$\begin{aligned} R_o &= r_o + \left[(R_e) \| (0 + r_\pi)\right] \left(1 + g_m r_o \frac{r_\pi}{r_\pi + 0} \right) \\ &= r_o + \left[(R_e) \| (r_\pi)\right] (1 + g_m r_o) \end{aligned}$$

The output resistance R_o is maximum when $R_e >> r_\pi$. So, approximate $(R_e) \| (r_\pi) \cong r_\pi$. Now, the output resistance is,

$$R_o = r_o + r_\pi (1 + g_m r_o)$$

Thus, the maximum value of the resistance R_o is $\boxed{r_o + r_\pi (1 + g_m r_o)}$.

Step 4 of 6

Now, set R_b to zero and choose the value of the emitter resistance so large that the current flowing through it becomes nearly zero. Hence, the resistance can be thought as open-circuited. Draw the simplified circuit as shown in the following Figure 4.

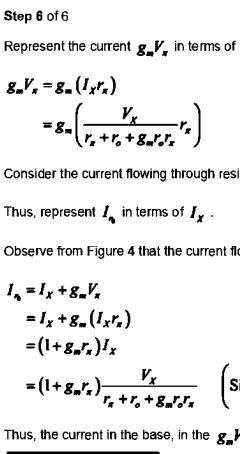


Figure 4

Step 5 of 6

Now represent the dependent current source connected in parallel to the resistance r_o as a voltage source $g_m V_\pi r_o$ in series with the resistance r_o. Draw the simplified circuit as shown in the following Figure 5.

Figure 5

Determine the value of the voltage V_π.

$$V_\pi = I_X r_\pi$$

Now, apply Kirchhoff's voltage law around the loop of the circuit in Figure 5.

$$-V_X + I_X r_\pi + g_m r_o V_\pi + I_X r_o = 0$$

$$V_X - g_m r_o V_\pi = I_X (r_\pi + r_o)$$

$$I_X = \frac{V_X}{r_\pi + r_o + g_m r_o r_\pi}$$

Thus, the current in the base in the $g_m V_\pi$ generator is I_X.

Thus, expression for I_X is $\boxed{\frac{V_X}{r_\pi + r_o + g_m r_o r_\pi}}$.

Step 6 of 6

Represent the current $g_m V_\pi$ in terms of I_X.

$$\begin{aligned} g_m V_\pi &= g_m (I_X r_\pi) \\ &= g_m \left(\frac{V_X}{r_\pi + r_o + g_m r_o r_\pi} r_\pi\right) \end{aligned}$$

Consider the current flowing through resistance r_0 as *I*.

Thus, represent I_{r_0} in terms of I_X.

Observe from Figure 4 that the current flowing through resistance, r_o is $(I_X + g_m V_\pi)$.

$$\begin{aligned} I_{r_0} &= I_X + g_m V_\pi \\ &= I_X + g_m (I_X r_\pi) \\ &= (1 + g_m r_\pi) I_X \\ &= (1 + g_m r_\pi) \frac{V_X}{r_\pi + r_o + g_m r_o r_\pi} \quad \left(\text{Since } I_X = \frac{V_X}{r_\pi + r_o + g_m r_o r_\pi}\right) \end{aligned}$$

Thus, the current in the base, in the $g_m V_\pi$ generator and in r_o in terms of I_X are

$\boxed{g_m \left(\frac{V_X}{r_\pi + r_o + g_m r_o r_\pi} r_\pi\right)}$ and $\boxed{(1 + g_m r_\pi) \frac{V_X}{r_\pi + r_o + g_m r_o r_\pi}}$ respectively.

The currents on the equivalent circuit with resistance R_E set to infinity are shown in Figure 4.

11.055P

From Figure 1 the voltage V_π is,

$$V_\pi = -I_x r_\pi$$

Apply Kirchhoff's voltage law to the loop.

$$V_x - g_m r_o r_\pi I_x = I_x (r_o + r_\pi)$$
$$V_x = I_x (r_o + r_\pi) + g_m r_o r_\pi I_x$$
$$V_x = I_x (r_o + r_\pi + g_m r_o r_\pi)$$
$$\frac{V_x}{I_x} = r_o + r_\pi + g_m r_o r_\pi$$

The output resistance R_{out} is,

$$R_{out} = \frac{V_x}{I_x}$$
$$= r_o + r_\pi + g_m r_o r_\pi$$

$$= r_\pi + (g_m r_\pi + 1) r_o \quad \text{..... (1)}$$

The feedback factor for the hybrid model,

$$h_{fe} = g_m r_\pi \quad \text{..... (2)}$$

Substitute, equation (2) in equation (1).

$$R_{out} = r_x + (h_{fe} + 1) r_o$$

The output resistance, R_{out} is limited to $R_{out} \simeq h_{fe} r_o$ for large amounts of feedback.

Therefore, the required feedback expression is $\boxed{R_{out} = h_{fe} r_o}$.

11.056P

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Step 1 of 9

11.057P

Refer to the circuit diagram of feedback trans-conductance amplifier in Figure P10.51 in the textbook.

Draw the A_- circuit.

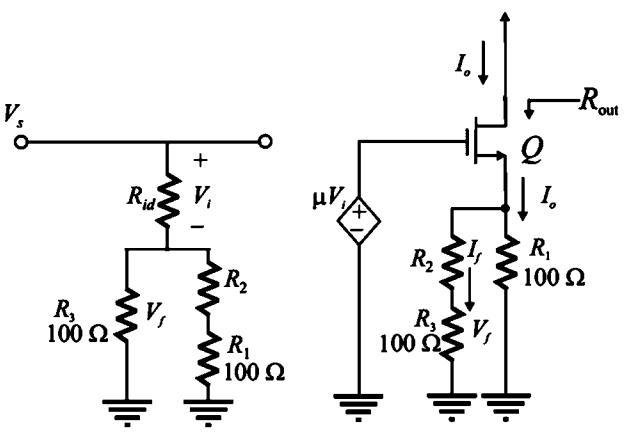


Figure 1

Step 2 of 9

Define the feedback factor β.

$$\beta = \frac{V_f}{I_o}$$

Use current division rule to write the expression for current I_f.

$$I_f = \frac{I_o R_1}{R_1 + R_2 + R_3}$$

From figure 1, feedback voltage V_f is,

$$V_f = I_f R_3$$

Substitute $\frac{I_o R_1}{R_1 + R_2 + R_3}$ for I_f.

$$V_f = \frac{I_o R_1 R_3}{R_1 + R_2 + R_3}$$

Rewrite the equation.

$$\frac{V_f}{I_o} = \frac{R_1 R_3}{R_1 + R_3 + R_3}$$

Thus feedback factor β is $\boxed{\frac{R_1 R_3}{R_1 + R_2 + R_3}}$.

Step 3 of 9

From Figure1, it is clear that gate to source V_{gs2} is,

$$V_{gs2} = \mu V_i$$

Find open-loop gain A.

$$I_o = -g_{m2} V_{gs2} \frac{r_{o2}}{r_{o2} + \left(R_1 \| \left(R_3 + R_3\right)\right)}$$

$$I_o = -g_{m2} \mu V_i \frac{r_{o2}}{r_{o2} + \left(R_1 \| \left(R_2 + R_3\right)\right)}$$

Substitute μV_i for V_{gs2} in equation. $\frac{I_o}{V_i} = -g_{m2} \mu \frac{r_{o2}}{r_{o2} + \left(R_1 \| \left(R_3 + R_3\right)\right)}$

$$A = -g_{m2} \mu \frac{r_{o2}}{r_{o2} + \left(R_1 \| \left(R_2 + R_3\right)\right)}$$

Thus, open loop gain A is $\boxed{g_{m2} \mu \frac{r_{o2}}{r_{o2} + \left(R_1 \| \left(R_3 + R_3\right)\right)}}$.

Step 4 of 9

The closed loop gain A_f is,

$$A_f = \frac{A}{1 + A\beta}$$

Substitute $g_{m2} \mu \frac{r_{o2}}{r_{o2} + \left(R_1 \| \left(R_3 + R_3\right)\right)}$ for A and $\frac{R_1 R_3}{R_1 + R_3 + R_3}$ for β.

$$A_f = \frac{g_{m2} \mu \frac{r_{o2}}{r_{o2} + \left(R_1 \| \left(R_2 + R_3\right)\right)}}{1 + \left(g_{m2} \mu \frac{r_{o2}}{r_{o2} + \left(R_1 \| \left(R_2 + R_3\right)\right)}\right)\left(\frac{R_1 R_3}{R_1 + R_2 + R_3}\right)}$$

For case $A\beta \gg 1$

$$A_f = \frac{1}{\left(\frac{R_1 R_3}{R_1 + R_2 + R_3}\right)}$$

Thus, closed loop gain A_f is $\boxed{\frac{R_1 + R_2 + R_3}{R_1 R_3}}$.

Step 5 of 9

Write the expression for closed loop gain.

$A_f = \frac{R_1 + R_2 + R_2}{R_1 R_3}$ Substitute $100\,\text{mA/V}$ for A_f, $100\ \Omega$ for R_1, $100\ \Omega$ for R_3.

$$100\ \text{mA/V} = \frac{100 + R_2 + 100}{(100)(100)}$$

$$R_2 + 200 = 1000$$

$$R_2 = 1000 - 200$$

Output resistance R_2 is $\boxed{800\ \Omega}$.

Similarly, feedback factor β is,

$$\beta = \frac{R_1 R_3}{R_1 + R_2 + R_3}$$

Substitute 100 for R_1 and 100 for R_3.

$$\beta = \frac{(100)(100)}{100 + 800 + 100}$$

$$= \frac{10000}{1000}$$

$$= 10$$

Thus, feedback factor β is $\boxed{10}$.

Step 6 of 9

Equate the feedback amount to 60 dB.

$$20 \log A\beta = 60\ \text{dB}$$

$$A\beta = 10^{\left(\frac{60}{20}\right)}$$

$$= 10^3$$

$$= 1000$$

Recall the expression for open loop gain A.

$$A = g_{m2} \mu \frac{r_{o2}}{r_{o2} + \left(R_1 \| \left(R_2 + R_3\right)\right)}$$

Trans-conductance of the amplifier g_m is $1\ \text{mA/V}$.

Substitute corresponding values.

$$A = \left(1 \times 10^{-3}\right) \mu \frac{50000}{50000 + \left(100 \| \left(800 + 100\right)\right)}$$

$$= \left(1 \times 10^{-3}\right)(\mu) \frac{50000}{50000 + 90}$$

$$= \frac{\mu}{1001.8}$$

Step 7 of 9

Equate the feedback amount to 1000.

$$A\beta = 1000$$

Substitute $\frac{\mu}{1001.8}$ for A and 10 for β.

$$\left(\frac{\mu}{1001.8}\right)(10) = 1000$$

Thus, μ is $\boxed{1001800}$.

Step 8 of 9

The output resistance R_o is,

$$R_o = R_1 \| R_3 + R_3$$

Substitute the corresponding values.

$R_o = r_o + g_m r_o \left[R_1 \| \left(R_2 + R_3\right)\right] + \left[R_1 \| \left(R_2 + R_3\right)\right]$ Substitute the corresponding values.

$$R_o = 50\ \text{k}\Omega + \left(1 \times 10^{-3}\right)(50\ \text{k}\Omega)\left[100 \| (800 + 100)\right] + \left[100 \| (800 + 100)\right]$$

$$= 50\ \text{k}\Omega + 90\ \Omega + 4.5\ \text{k}\Omega$$

$$= 54.59\ \text{k}\Omega$$

Hence, the output resistance R_o is $\boxed{54.59\ \text{k}\Omega}$.

Step 9 of 9

Obtain output resistance of an amplifier R_{out}, which is the resistance seen by R_L in the circuit implies subtract R_L from R_{of}. Thus

$$R_{out} = R_{of}$$

$$= R_o (1 + A\beta)$$

$$\approx R_o (A\beta) \qquad \text{since } 1 \gg A\beta$$

Substitute corresponding values.

$$R_{out} = 54.59\ \text{k}\Omega\ (1 + 10000)$$

$$= 54.59\ \text{M}\Omega$$

Thus, the output resistance R_{out} is $\boxed{54.59\ \text{M}\Omega}$.

Step 1 of 17

(a)

Refer to Figure P11.58 in the textbook.

Determine the dc voltage at the transistor Q_1 .

Consider the equation for the gate to source voltage V_{GS} .

$$|V_{GS}| = |V_{GP}| + |V_t|$$

Consider $|V_{GP}| = 8.2$ V and $|V_t| = 8.4$ V .

$$\begin{aligned}|V_{GS}| &= |V_{GP}| + |V_t| \\ &= 0.2 + 0.4 \\ &= 0.6 \text{ V}\end{aligned}$$

Apply Kirchhoff voltage law at V_G of transistor Q_3

$$V_{G1} = V_{DD} - V_{GS} \quad (1)$$

Replace 1.2 for V_{DD} , 0.6 for V_{GS} in equation 1.

$$\begin{aligned}V_{G1} &= V_{DD} - V_{GS} \\ &= 1.2 - 0.6 \\ &= 0.6 \text{ V}\end{aligned}$$

The voltage at drain of Q_1 is equal to the voltage at gate of Q_3

$$\begin{aligned}V_{D1} &= V_{G1} \\ &= 0.6 \text{ V}\end{aligned}$$

Therefore, the voltage at the transistor Q_1 is $\boxed{0.6 \text{ V}}$.

Step 2 of 17

Determine the voltage at the drain of transistor Q_1 .

The drain voltage at Q_2 is equal to drain voltage at Q_4 and the voltage at drain of Q_4 is equal to voltage at gate of transistor Q_3 ,so $V_{D4} = V_{G3}$.

The voltage at gate of Q_3 is equal to voltage at drain of Q_4 , so $V_{G3} = V_{D1}$

The voltage at drain of transistor Q_1 is 0.6 V .

$$V_{D1} = 0.6 \text{ V}$$

$$\begin{aligned}V_{G3} &= V_{D1} \\ &= 0.6 \text{ V}\end{aligned}$$

$$\begin{aligned}V_{D4} &= V_{G3} \\ &= 0.6 \text{ V}\end{aligned}$$

$$\begin{aligned}V_{D2} &= V_{G3} \\ &= 0.6 \text{ V}\end{aligned}$$

Therefore, the voltage at the drain of transistor Q_2 is $\boxed{0.6 \text{ V}}$.

Step 3 of 17

Determine the voltage at drain of the transistor Q_3 .

The voltage at gate of transistor Q_4 is equal to the voltage at drain of transistor Q_2 , so $V_{G3} = V_{D4}$.

Consider the voltage at drain of transistor Q_4 is $V_{D4} = 0.6$ V .

$$\begin{aligned}V_{G3} &= V_{D4} \\ &= 0.6 \text{ V}\end{aligned}$$

Apply Kirchhoff voltage law at Q_3

$$\begin{aligned}V_G &= V_{GS} + V_{D3} \\ 0.6 \text{ V} &= 0.6 \text{ V} + V_{D4} \\ V_{D4} &= 8 \text{ V}\end{aligned}$$

Therefore, the dc voltage at the output is $\boxed{V_{D4} = 0 \text{ V}}$.

Step 4 of 17

(b)

Determine the expression for close loop gain A_f

$$A_f = \frac{A}{1 + A\beta}$$

Consider $A\beta >> 1$.

$$\begin{aligned}A_f &= \frac{A}{1 + A\beta} \\ &= \frac{A}{A\beta} \\ &= \frac{1}{\beta}\end{aligned}$$

Consider $\beta = R_F$.

$$\begin{aligned}A_f &= \frac{1}{\beta} \\ &= \frac{1}{R_F}\end{aligned}$$

Consider $R_F = 10 \text{ k}\Omega$.

$$\begin{aligned}A_f &= \frac{1}{R_F} \\ &= \frac{1}{10 \text{ k}\Omega} \\ &= 10^{-4} \text{ A/V}\end{aligned}$$

Therefore, the closed loop gain A_f is $\boxed{10^{-4} \text{ A/V}}$

Step 5 of 17

(c)

Consider the following small-signal equivalent circuit of transistor Q_1 :

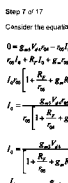


Figure 1

Consider the equation for output voltage V_{o1}

$$V_{o1} = g_{m1}(r_{o1} \| r_{o5}) V_i$$

$$\frac{V_{o1}}{V_i} = g_{m1}(r_{o1} \| r_{o5})$$

Step 6 of 17

Consider the following small-signal equivalent circuit at transistor Q_2 :

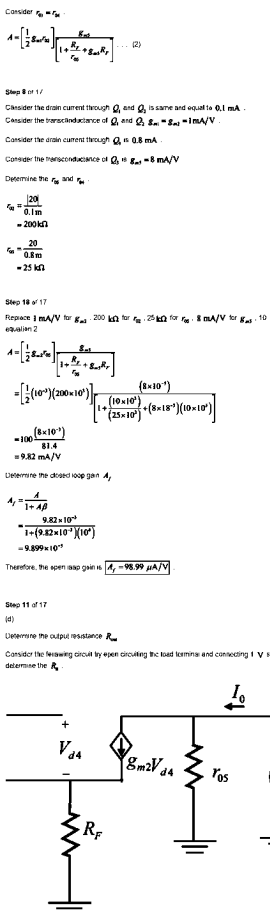


Figure 2

Consider V_{o1} is the voltage at the gate terminal at transistor Q_4 , which is equal to the voltage at drain of transistor Q_1

Step 7 of 17

Consider the equation for output current, I_o .

$$0 = g_{m1}V_{o1}r_{o1} - I_0 I_4 - R_F I_4 - g_{m1}r_{o1}R_F I_0$$

$$r_{o1}I_4 + R_F I_4 + g_{m1}r_{o1}R_F I_0 = g_{m1}V_{o1}r_{o1}$$

$$I_4(r_{o1})\left[1 + \frac{R_F}{r_{o1}} + g_{m1}R_F\right] = g_{m1}V_{o1}r_{o1}$$

$$I_4 = \frac{g_{m1}V_{o1}r_{o1}}{r_{o1}\left[1 + \frac{R_F}{r_{o1}} + g_{m1}R_F\right]}$$

$$I_0 = \frac{g_{m1}V_{o1}}{\left[1 + \frac{R_F}{r_{o1}} + g_{m1}R_F\right]}$$

$$\frac{I_0}{V_{o1}} = \frac{g_{m1}}{\left[1 + \frac{R_F}{r_{o1}} + g_{m1}R_F\right]}$$

Step 8 of 17

Determine the open loop gain

$$A_f = \frac{I_0}{V_i}$$

$$A = \left(\frac{V_{o1}}{V_i}\right)\left(\frac{I_0}{V_{o1}}\right)$$

Consider $V_{o1} = V_{o4}$.

$$A = \left(\frac{V_{o1}}{V_i}\right)\left(\frac{I_0}{V_{o1}}\right)$$

$$\left(\frac{I_0}{V_i}\right) = \left[g_{m1}(r_{o1} \| r_{o5})\right]\left[\frac{g_{m1}}{1 + \frac{R_F}{r_{o1}} + g_{m1}R_F}\right]$$

Consider $r_{o1} = r_{o5}$

$$A = \left[\frac{1}{2}g_{m1}r_{o1}\right]\left[\frac{g_{m1}}{1 + \frac{R_F}{r_{o1}} + g_{m1}R_F}\right] \quad \ldots (2)$$

Step 9 of 17

Consider the transconductance of Q_1 and Q_2 $g_{m1} = g_{m2} = 1$ mA/V .

Consider the drain current through Q_1 is 0.8 mA .

Consider the transconductance of Q_1 is $g_{m1} = 8$ mA/V

Determine the r_{o1} and r_{o4} .

$$\begin{aligned}r_{o1} &= \frac{|20|}{0.1 \text{ m}} \\ &= 200 \text{ k}\Omega\end{aligned}$$

$$\begin{aligned}r_{o4} &= \frac{20}{0.8 \text{ m}} \\ &= 25 \text{ k}\Omega\end{aligned}$$

Step 10 of 17

Replace 1 mA/V for g_{m1} , 200 kΩ for r_{o1} , 25 kΩ for r_{o4} , 8 mA/V for g_{m1} , 10 kΩ for R_F in equation 2

$$\begin{aligned}A &= \left[\frac{1}{2}g_{m1}r_{o1}\right]\left[\frac{g_{m1}}{1 + \frac{R_F}{r_{o1}} + g_{m1}R_F}\right] \\ &= \left[\frac{1}{2}(10^{-3})(200 \times 10^3)\right]\left[\frac{(8 \times 10^{-3})}{1 + \left(\frac{10 \times 10^3}{25 \times 10^3}\right) + (8 \times 10^{-3})(10 \times 10^3)}\right] \\ &= 100\left(\frac{8 \times 10^{-3}}{81.4}\right) \\ &= 9.82 \text{ mA/V}\end{aligned}$$

Determine the closed loop gain A_f

$$\begin{aligned}A_f &= \frac{A}{1 + A\beta} \\ &= \frac{9.82 \times 10^{-3}}{1 + (9.82 \times 10^{-3})(10^4)} \\ &= 9.899 \times 10^{-5}\end{aligned}$$

Therefore, the open loop gain is $\boxed{A_f = 98.99 \ \mu\text{A/V}}$

Step 11 of 17

(d)

Determine the output resistance R_{out}

Consider the following circuit by open circuiting the load terminal and connecting 1 V source to determine the R_o

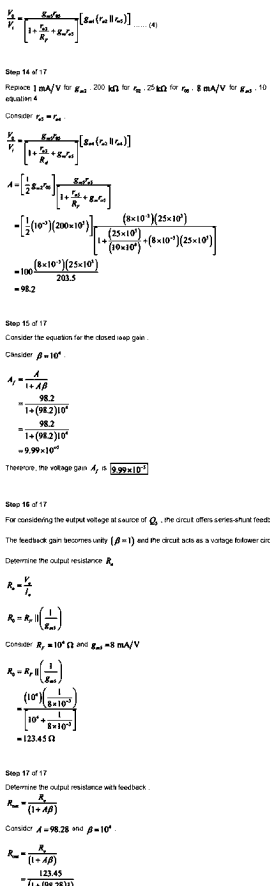


Figure 3

Apply Kirchhoff's voltage law at the output node.

$$1 \text{ V} = I_0\left[r_{o5} + R_F + g_{m1}r_{o5}R_F\right]$$

$$\frac{1 \text{ V}}{I_0} = \left[r_{o5} + R_F + g_{m1}r_{o5}R_F\right]$$

$$R_o = \frac{1 \text{ V}}{I_0}$$

$$R_o = \left[r_{o5} + R_F + g_{m1}r_{o5}R_F\right] \quad \ldots (3)$$

Replace 25 kΩ for r_{o5} , 8 mA/V for g_{m1} , 10 kΩ for R_F in equation 3.

$$\begin{aligned}R_o &= r_{o5} + R_F + g_{m1}r_{o5}R_F \\ &= (25 \times 10^3) + (10 \times 10^3) + (8 \times 10^{-3})(25 \times 10^3)(10 \times 10^3) \\ &= 2035 \text{ k}\Omega\end{aligned}$$

Step 12 of 17

Consider the equation for output resistance with feedback

$$R_{out} = R_o(1 + A\beta)$$

Consider 9.82 mA/V for A , 2035 kΩ for R_o and $\beta = 10^4$.

$$\begin{aligned}R_{out} &= R_o(1 + A\beta) \\ &= (2035 \times 10^3)\left(1 + (9.82 \times 10^{-3})(10^4)\right) \\ &= (2035 \times 10^3)(99.2) \\ &= 20.5 \text{ k}\Omega\end{aligned}$$

Therefore, the output resistance R_{out} is $\boxed{20.5 \text{ k}\Omega}$.

Step 13 of 17

(e)

Consider the output is taken as voltage across source of Q_2

From Figure 3 consider the equation for output voltage V_o .

Apply Kirchhoff's current law at the output terminal.

$$g_{m1}V_{o1}r_{o1} - r_{o1}I_0 - R_F I_4 - g_{m1}r_{o1}R_F I_4 = 0$$

$$r_{o1}I_4 + R_F I_4 + g_{m1}r_{o1}R_F I_4 = g_{m1}V_{o1}r_{o1}$$

$$I_4(r_{o1})\left[1 + \frac{r_{o1}}{R_F} + g_{m1}r_{o1}\right] = g_{m1}V_{o1}r_{o1}$$

$$V_4 = \frac{g_{m1}V_{o1}r_{o1}}{\left[1 + \frac{r_{o1}}{R_F} + g_{m1}r_{o1}\right]}$$

Consider $V_{o4} = V_4\left[g_{m1}(r_{o1} \| r_{o5})\right]$

$$V_4 = \frac{g_{m1}V_{o1}r_{o1}}{\left[1 + \frac{r_{o1}}{R_F} + g_{m1}r_{o1}\right]}\left[g_{m1}(r_{o1} \| r_{o5})\right]$$

$$\frac{V_4}{V_{o1}} = \frac{g_{m1}r_{o1}}{\left[1 + \frac{r_{o1}}{R_F} + g_{m1}r_{o1}\right]}\left[g_{m1}(r_{o1} \| r_{o5})\right] \quad \ldots (4)$$

Step 14 of 17

Replace 1 mA/V for g_{m1} , 200 kΩ for r_{o1} , 25 kΩ for r_{o5} , 8 mA/V for g_{m1} , 10 kΩ for R_F in equation 4

Consider $r_{o1} = r_{o4}$.

$$\begin{aligned}\frac{V_4}{V_{o1}} &= \frac{g_{m1}r_{o1}}{\left[1 + \frac{r_{o1}}{R_F} + g_{m1}r_{o1}\right]}\left[g_{m1}(r_{o1} \| r_{o4})\right] \\ A &= \left[\frac{1}{2}g_{m1}r_{o1}\right]\left[\frac{g_{m1}r_{o1}}{1 + \frac{r_{o1}}{R_F} + g_{m1}r_{o1}}\right] \\ &= \left[\frac{1}{2}(10^{-3})(200 \times 10^3)\right]\left[\frac{(8 \times 10^{-3})(25 \times 10^3)}{1 + \frac{(25 \times 10^3)}{(10 \times 10^3)} + (8 \times 10^{-3})(25 \times 10^3)}\right] \\ &= 100\left(\frac{(8 \times 10^{-3})(25 \times 10^3)}{203.5}\right) \\ &= 98.2\end{aligned}$$

Step 15 of 17

Consider the equation for the closed loop gain .

Consider $\beta = 10^4$.

$$\begin{aligned}A_f &= \frac{A}{1 + A\beta} \\ &= \frac{98.2}{1 + (98.2)10^4} \\ &= \frac{98.2}{1 + (98.2)10^4} \\ &= 9.99 \times 10^{-3}\end{aligned}$$

Therefore, the voltage gain A_f is $\boxed{9.99 \times 10^{-3}}$

Step 16 of 17

For considering the output voltage at source of Q_2 , the circuit offers series-shunt feedback.

The feedback gain becomes unity ($\beta = 1$) and the circuit acts as a voltage follower circuit.

Determine the output resistance R_o

$$R_o = \frac{V_x}{I_x}$$

$$R_o = R_F \| \left(\frac{1}{g_{m1}}\right)$$

Consider $R_F = 10^4 \ \Omega$ and $g_{m1} = 8$ mA/V

$$\begin{aligned}R_o &= R_F \| \left(\frac{1}{g_{m1}}\right) \\ &= \frac{(10^4)\left(\frac{1}{8 \times 10^{-3}}\right)}{10^4 + \frac{1}{8 \times 10^{-3}}} \\ &= 123.45 \ \Omega\end{aligned}$$

Step 17 of 17

Determine the output resistance with feedback .

$$R_{out} = \frac{R_o}{(1 + A\beta)}$$

Consider $A = 98.28$ and $\beta = 10^4$.

$$\begin{aligned}R_{out} &= \frac{R_o}{(1 + A\beta)} \\ &= \frac{123.45}{(1 + (98.28)1)} \\ &= \frac{123.45}{99.28} \\ &= 1.24 \ \Omega\end{aligned}$$

Step 1 of 6

11.059P

Refer to Figure P11.58 in the textbook.

Consider the source voltage $V_s = 0$.

Draw the small signal equivalent circuit for differential amplifier as shown in Figure 1.

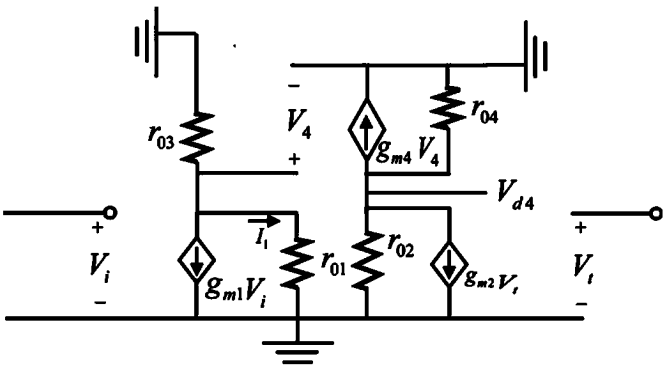


Figure 1

Step 2 of 6

Determine the open-loop gain A .

Consider the equation for output voltage V_{d4} .

Consider V_{d4} is the voltage across the darin terminals of the transistor Q_4 and Q_5 .

$$V_{d4} = g_{m2}(r_{02} \| r_{04})V_i$$

$$\frac{V_{d4}}{V_i} = g_{m2}(r_{02} \| r_{04})$$

Step 3 of 6

Consider the following diagram for determining the gain at transistor Q_5 .

Figure 2

Step 4 of 6

Consider V_0 is the voltage across R_F .

Determine the voltage across R_F .

$$V_0 = \frac{V_{d4}(R_F \| r_{05})}{1/g_{m5} + (R_F \| r_{05})}$$

$$\frac{V_0}{V_{d4}} = \frac{(R_F \| r_{05})}{1/g_{m5} + (R_F \| r_{05})}$$

Consider $A = \frac{I_0}{V_i}$.

$$A = \left(\frac{V_{d4}}{V_i}\right)\left(\frac{V_0}{V_{d4}}\right)$$

$$= \left(\frac{V_{d4}}{V_i}\right)\left(\frac{I_0(R_F \| r_{05})}{V_{d4}}\right)$$

$$\frac{I_0(R_F \| r_{05})}{V_i} = g_{m2}(r_{02} \| r_{04})\left[\frac{(R_F \| r_{05})}{1/g_{m5} + (R_F \| r_{05})}\right]$$

$$\frac{I_0}{V_i} = g_{m2}(r_{02} \| r_{04})\left[\frac{1}{1/g_{m5} + (R_F \| r_{05})}\right]$$

$$A = g_{m2}(r_{02} \| r_{04})\left[\frac{1}{1/g_{m5} + (R_F \| r_{05})}\right]$$

Step 5 of 6

Determine the feedback gain β .

$$\beta = \frac{V_t}{I_0}$$

$$V_t = I_0(R_F \| r_{05})$$

$$\frac{V_t}{I_0} = (R_F \| r_{05})$$

$$\beta = (R_F \| r_{05})$$

Step 6 of 6

Determine the loop gain $A\beta$.

Consider $A = g_{m2}(r_{02} \| r_{04})\left[\frac{1}{1/g_{m5} + (R_F \| r_{05})}\right]$ and $\beta = (R_F \| r_{05})$.

$$A\beta = g_{m2}(r_{02} \| r_{04})\left[\frac{1}{1/g_{m5} + (R_F \| r_{05})}\right](R_F \| r_{05})$$

$$= g_{m2}(r_{02} \| r_{04})\left[\frac{(R_F \| r_{05})}{1/g_{m5} + (R_F \| r_{05})}\right]$$

Therefore, the loop gain $A\beta$ is $\boxed{A\beta = g_{m2}(r_{02} \| r_{04})\left[\frac{(R_F \| r_{05})}{1/g_{m5} + (R_F \| r_{05})}\right]}$.

Step 1 of 5

The open loop gain A is,

$$A = -\mu R_i \frac{(R_F \| R_L)}{r_o + (R_F \| R_L)}$$

The amplifier gain, $\mu = 10^3 \text{ V/V}$

The input resistance, $R_{id} = \infty$

The output resistance, $r_o = 100\ \Omega$

The feedback resistance, $R_F = 10\ \text{k}\Omega$

The source resistance, $R_s = 1\ \text{k}\Omega$

The load resistance, $R_L = 1\ \text{k}\Omega$

The input resistance R_i is,

$$R_i = R_{id} \| R_F \| R_s$$

Substitute the corresponding values.

$$\begin{aligned} R_i &= \infty \| 10\ \text{k} \| 1\ \text{k} \\ &\approx 10\ \text{k} \| 1\ \text{k} \\ &= \frac{(10\ \text{k})(1\ \text{k})}{(10\ \text{k}) + (1\ \text{k})} \\ &= 909.09\ \Omega \end{aligned}$$

Recall the formula for open loop gain A.

$$A = -\mu R_i \frac{(R_F \| R_L)}{r_o + (R_F \| R_L)}$$

Substitute the corresponding values.

$$\begin{aligned} A &= -(10^3)(909.09) \frac{(10\ \text{k} \| 1\ \text{k})}{100 + (10\ \text{k} \| 1\ \text{k})} \\ &= -(10^3)(909.09) \frac{909.09}{100 + (909.09)} \\ &\approx -819 \times 10^3 \end{aligned}$$

Thus, open loop gain of the amplifier A is $\boxed{-819 \times 10^3}$.

Step 2 of 5

The loop gain $A\beta$ is,

$$\begin{aligned} A\beta &= \mu \left(\frac{R_i}{R_F} \right) \frac{(R_F \| R_L)}{r_o + (R_F \| R_L)} \\ &= \frac{\mu R_i \left(\frac{(R_F \| R_L)}{r_o + (R_F \| R_L)} \right)}{R_F} \end{aligned}$$

Substitute the corresponding values.

$$\begin{aligned} A\beta &= -\frac{(10^3)(909.09)}{10\ \text{k}} \frac{(10\ \text{k} \| 1\ \text{k})}{100 + (10\ \text{k} \| 1\ \text{k})} \\ &= -\frac{(10^3)(909.09)}{10\ \text{k}} \frac{909.09}{100 + (909.09)} \\ &= -\frac{818.9 \times 10^3}{10 \times 10^3} \\ &= 81.9 \end{aligned}$$

Thus, loop gain $A\beta$ is $\boxed{81.9}$.

Step 3 of 5

The closed loop gain A_f is,

$$A_f = \frac{A}{1 + A\beta}$$

Substitute the corresponding values.

$$\begin{aligned} A_f &= -\frac{819 \times 10^3}{1 + 81.9} \\ &= -\frac{819 \times 10^3}{82.9} \\ &= -9.88 \times 10^3 \text{ V/V} \end{aligned}$$

Therefore, the closed loop gain A_f is $\boxed{-9.88\ \text{kV/V}}$.

Step 4 of 5

Write the expression for input resistance, R_{in}.

$$R_{in} = \frac{\left[(R_F \| R_L) + r_o \right] R_F}{\mu (R_F \| R_L)}$$

Substitute corresponding values.

$$\begin{aligned} R_{in} &= \frac{\left[(10\ \text{k}\Omega \| 1\ \text{k}\Omega) + 100 \right] (10 \times 10^3)}{10^3 (10\ \text{k}\Omega \| 1\ \text{k}\Omega)} \\ &= \frac{\left[(909.09) + 100 \right] (10 \times 10^3)}{10^3 (909.09)} \\ &= \frac{[1009.09](10)}{(909.09)} \\ &= 11.1\ \Omega \end{aligned}$$

Therefore, the input resistance R_{in} is $\boxed{11.1\ \Omega}$.

Step 5 of 5

Write the expression for output resistance R_{out}.

$$R_{out} \approx \left(\frac{R_F}{R_i} \right) \left(\frac{r_o}{\mu} \right)$$

Substitute the known values in the equation.

$$\begin{aligned} R_{out} &= \left(\frac{10 \times 10^3}{909.09} \right) \left(\frac{100}{10^3} \right) \\ &= (11)(0.1) \\ &= 1.1\ \Omega \end{aligned}$$

Therefore, output resistance R_{out} is $\boxed{1.1\ \Omega}$.

Step 1 of 10

Refer to the Figure E11.19 in the textbook.

Consider the Example 11.9 from the textbook.

The circuits in Example 11.9 and in Exercise 11.19 are equivalent with following considerations.

Since, there is the load resistor, $R_L = \infty$,

Input impedance of MOSFET is, $R_{id} = \infty$,

Gain of op-amp is, $\mu = g_m r_o$.

Calculate the input resistance of the amplifier.

$$R_i = R_F \parallel R_s$$

Substitute $10\ \text{k}$ for R_F and $1\ \text{k}$ for R_s .

$$\begin{aligned} R_i &= 10\ \text{k} \parallel 1\ \text{k} \\ &= \frac{(10\ \text{k})(1\ \text{k})}{10\ \text{k}+1\ \text{k}} \\ &= 909.1\ \Omega \end{aligned}$$

Thus, the input resistance of the amplifier R_i is $\boxed{909.1\ \Omega}$.

Step 2 of 10

Determine the open loop gain of the amplifier.

$$A = -g_m r_o R_i \frac{R_F \parallel R_L}{r_o + (R_F \parallel R_L)}$$

Substitute ∞ for R_L , $5\ \text{mA/V}$ for g_m , $20\ \text{k}\Omega$ for r_o , $10\ \text{k}\Omega$ for R_F , $909.1\ \Omega$ for R_i .

$$\begin{aligned} A &= -(5\times10^{-3})(20\times10^{3})(909.1)\left[\frac{10\ \text{k}}{20\ \text{k}+10\ \text{k}}\right] \\ &= -(5\times10^{-3})(20\times10^{3})(909.1)\left[\frac{10\ \text{k}}{30\ \text{k}}\right] \\ &= -30.3\ \text{k}\Omega \end{aligned}$$

Thus, the open loop gain of the amplifier A is $\boxed{-30.3\ \text{k}\Omega}$.

Step 3 of 10

Calculate the feedback factor.

$$\beta = -\frac{1}{R_f}$$

Substitute $10\ \text{k}$ for R_f in the equation.

$$\begin{aligned} \beta &= -\frac{1}{10\times10^{3}} \\ &= -0.1\ \text{mA/V} \end{aligned}$$

Thus, the feedback factor of the amplifier β is $\boxed{-0.1\ \text{mA/V}}$.

Step 4 of 10

Calculate the loop gain of the amplifier.

$$\begin{aligned} A\beta &= (-30.3\times10^{3})(-0.1\times10^{-3}) \\ &= 3.03 \end{aligned}$$

Thus, the loop gain of the amplifier $A\beta$ is $\boxed{3.03}$.

Step 5 of 10

Calculate the gain with feedback of the amplifier.

$$A_f = \frac{A}{1+A\beta}$$

Substitute 3.03 for $A\beta$ and -30.3×10^{3} for A in equation.

$$\begin{aligned} A_f &= \frac{-30.3\times10^{3}}{1+3.03} \\ &= -7.52\ \text{k}\Omega \end{aligned}$$

Thus, the gain with feedback of the amplifier A_f is $\boxed{-7.52\ \text{k}\Omega}$.

Step 6 of 10

Calculate the output resistance of the amplifier.

$$\begin{aligned} R_o &= r_o \parallel R_F \\ &= 20\ \text{k} \parallel 10\ \text{k} \\ &= 6.67\ \text{k}\Omega \end{aligned}$$

Thus, the output resistance of the amplifier R_o is $\boxed{6.67\ \text{k}\Omega}$.

Step 7 of 10

Calculate the input resistance with feedback of the amplifier.

$$R_{if} = \frac{R_i}{1+A\beta}$$

Substitute 3.03 for $A\beta$ and $909.1\ \Omega$ for R_i in equation.

$$\begin{aligned} R_{if} &= \frac{909.1}{1+3.03} \\ &= 226\ \Omega \end{aligned}$$

Thus, the input resistance with feedback of the amplifier R_{if} is $\boxed{226\ \Omega}$.

Step 8 of 10

Calculate the output resistance with feedback of the amplifier.

$$R_{of} = \frac{R_o}{1+A\beta}$$

Substitute 3.03 for $A\beta$ and $6.67\ \text{k}\Omega$ for R_o in equation.

$$\begin{aligned} R_{of} &= \frac{6.67\ \text{k}}{1+3.03} \\ &= 1.66\ \text{k}\Omega \end{aligned}$$

Thus, the output resistance with feedback of the amplifier R_{of} is $\boxed{1.66\ \text{k}\Omega}$.

Step 9 of 10

Calculate the input resistance of the feedback amplifier.

$$R_{in} = \frac{R_f}{\mu' + 1}$$

Here,

$$\begin{aligned} \mu' &= \frac{g_m r_o (R_F \parallel R_L)}{r_o + (R_F \parallel R_L)} \\ &= \frac{g_m r_o R_F}{r_o + R_F} \qquad (\text{since } R_L = \infty) \end{aligned}$$

Substitute $5\ \text{mA/V}$ for g_m , $20\ \text{k}\Omega$ for r_o , $10\ \text{k}\Omega$ for R_F in equation.

$$\begin{aligned} \mu' &= \frac{(5\times10^{-3})(20\times10^{3})(10\ \text{k})}{20\ \text{k}+10\ \text{k}} \\ &= \frac{(5\times10^{-3})(20\times10^{3})(10\ \text{k})}{30\ \text{k}} \\ &= 33.33 \end{aligned}$$

Substitute $10\ \text{k}\Omega$ for R_F , 33.33 for μ in R_{in} .

$$\begin{aligned} R_{in} &= \frac{R_F}{\mu' + 1} \\ &= \frac{10\ \text{k}}{33.33+1} \\ &= 291\ \Omega \end{aligned}$$

Thus, the input resistance of the feedback amplifier R_{in} is $\boxed{291\ \Omega}$.

Step 10 of 10

Calculation of the output resistance as follows:

$$\begin{aligned} R_{out} &= R_f \parallel \frac{r_o}{1+g_m r_o \frac{R_i}{R_f}} \\ &= 10\ \text{k} \parallel \frac{20\ \text{k}}{1+(5\times10^{-3})(20\times10^{3})\left(\frac{909.1}{10\ \text{k}}\right)} \\ &= 10\ \text{k} \parallel \left(\frac{20\ \text{k}}{1+9.091}\right) \\ &= 10\ \text{k} \parallel 1.982\ \text{k} \\ &= \frac{(10\ \text{k})(1.982\ \text{k})}{10\ \text{k}+1.982\ \text{k}} \\ &= 1.654\ \text{k}\Omega \end{aligned}$$

Thus, the output resistance as seen by the load, of the amplifier R_{out} is $\boxed{1.654\ \text{k}\Omega}$.

Step 1 of 9

Refer to Figure E11.19 in the textbook.

Consider $I_s = 0$ and break the feedback loop.

Draw the small-signal equivalent circuit as shown in Figure 1.

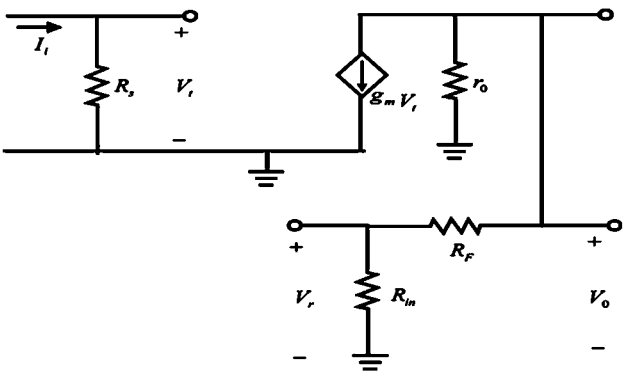


Figure 1

Step 2 of 9

Determine the value of R_{in} .

Consider R_{in} is the input resistance of the amplifier before breaking the feedback loop.

$$R_{in} = R_s \parallel R_F$$

Write the expression for V_t .

$$V_t = I_t R_s$$

Write the expression for V_0 .

$$V_0 = -g_m V_t \left(r_0 \parallel \left(R_F \parallel R_{in}\right)\right) \quad \text{..... (1)}$$

Determine the voltage across R_{in} .

$$V_r = \frac{V_0 R_{in}}{R_{in} + R_F} \quad \text{..... (2)}$$

Step 3 of 9

Substitute equation 1 in equation 2.

$$V_r = \frac{R_{in}}{R_{in} + R_F}\left[-g_m V_t \left(r_0 \parallel \left(R_F \parallel R_{in}\right)\right)\right]$$

$$\frac{V_r}{V_t} = -\frac{R_{in}}{R_{in} + R_F}\left[g_m \left(r_0 \parallel \left(R_F \parallel R_{in}\right)\right)\right]$$

Determine the loop gain $A\beta$.

$$A\beta = \frac{-V_r}{V_t}$$

$$A\beta = \frac{R_{in}}{R_{in} + R_F}\left[g_m \left(r_0 \parallel \left(R_F \parallel R_{in}\right)\right)\right] \quad \text{..... (3)}$$

Therefore, the loop-gain $A\beta$ is $\boxed{\frac{R_{in}}{R_{in} + R_F}\left[g_m \left(r_0 \parallel \left(R_F \parallel R_{in}\right)\right)\right]}$.

Step 4 of 9

Determine the open-loop gain A .

$$A = \frac{V_0}{I_t}$$

Consider the equation 1.

$$V_0 = -g_m V_t \left(r_0 \parallel \left(R_F \parallel R_{in}\right)\right)$$

Consider $V_t = I_t R_s$.

$$V_0 = -g_m \left(I_t R_s\right)\left(r_0 \parallel \left(R_F \parallel R_{in}\right)\right)$$

$$\frac{V_0}{I_t} = -g_m R_s \left(r_0 \parallel \left(R_F \parallel R_{in}\right)\right)$$

$$A = -g_m R_s \left(r_0 \parallel \left(R_F \parallel R_{in}\right)\right) \quad \text{..... (4)}$$

Therefore, the open loop gain A is $\boxed{-g_m R_s \left(r_0 \parallel \left(R_F \parallel R_{in}\right)\right)}$.

Step 5 of 9

Determine the value of loop gain $A\beta$.

Consider the equation for the loop gain $A\beta$.

Consider $R_{in} = \left(R_F \parallel R_s\right)$.

$$A\beta = \frac{R_{in}}{R_{in} + R_F}\left[g_m \left(r_0 \parallel \left(R_F \parallel R_{in}\right)\right)\right]$$

Step 6 of 9

Consider simplification of $A\beta$.

$$A\beta = \frac{\left(10^4 \parallel 10^3\right)}{\left(10^4 \parallel 10^3\right) + 10^4}\left[\left(5\times10^{-3}\right)\left(\left(20\times10^3\right) \parallel 833.33\right)\right]$$

$$= \frac{\left(10^4 \parallel 10^3\right)}{\left(10^4 \parallel 10^3\right) + 10^4}\left[\left(5\times10^{-3}\right)799.99\right]$$

$$= \frac{909.09}{909.09 + 10^4}\left[3.99\right]$$

$$= 0.333$$

Therefore, the loop gain is $\boxed{0.333}$.

Step 7 of 9

Determine the feedback gain β .

Divide equation 3 with equation 4.

$$\frac{A\beta}{A} = -\frac{\frac{R_{in}}{R_{in} + R_F}\left[g_m \left(r_0 \parallel \left(R_F \parallel R_{in}\right)\right)\right]}{g_m R_s \left(r_0 \parallel \left(R_F \parallel R_{in}\right)\right)}$$

$$\beta = -\frac{R_{in}}{R_{in} + R_F} \quad \text{..... (6)}$$

Replace 10 $\text{k}\Omega$ for R_F , and $1\ \text{k}\Omega$ for R_s in following equation.

Consider $R_{in} = R_s \parallel R_F$.

$$R_{in} = R_s \parallel R_F$$

$$= 10^3 \parallel 10^4$$

$$= 909.09\ \Omega$$

Replace 10 $\text{k}\Omega$ for R_F , and $909.09\ \Omega$ for R_{in} in equation 6.

$$\beta = -\frac{R_{in}}{R_{in} + R_F}$$

$$= -\frac{909.09}{909.09 + 10^4}$$

$$= -0.0833$$

Therefore, the feedback gain β is $\boxed{-0.0833}$.

Step 8 of 9

Determine the open-loop gain A .

Consider the equation for open-loop gain.

$$A = -g_m R_s \left(r_0 \parallel \left(R_F \parallel R_{in}\right)\right)$$

Replace $20\ \text{k}\Omega$ for r_0 , $5\ \text{mA/V}$ for g_m , 10 $\text{k}\Omega$ for R_F , $909.09\ \Omega$ for R_{in} and $1\ \text{k}\Omega$ for R_s in equation 4.

$$A = -\left(5\times10^{-3}\right)\left(10^3\right)\left(\left(20\times10^3\right) \parallel \left(10^4 \parallel 909.9\right)\right)$$

$$= -\left(5\times10^{-3}\right)\left(10^3\right)\left(\left(20\times10^3\right) \parallel 833.33\right)$$

$$= -\left[\left(5\times10^{-3}\right)799.99\right]$$

$$= -3.99$$

Therefore, the open-loop gain A is $\boxed{-3.99}$.

Step 9 of 9

Determine the closed loo gain A_f .

Consider the equation for close loop gain.

$$A_f = \frac{A}{1 + A\beta} \quad \text{..... (7)}$$

Replace -3.99 for A , 0.333 for $A\beta$ in equation 7.

$$A_f = \frac{A}{1 + A\beta}$$

$$= \frac{-3.99}{1 + (.333)}$$

$$= -2.99$$

Therefore, the closed loop gain A_f is $\boxed{-2.99}$.

The results are different from the values of exercise 11.19, because of the approximations inherent in the systematic feedback-analysis method.

Step 1 of 21

Refer to the Figure P11.63 in the textbook.

The CE BJT amplifier has shunt-shunt feedback. Since, the feedback resistor R_f senses the output voltage V_o and provides a feedback current to the base node.

Consider the modified circuit diagram which is shown in Figure 1.

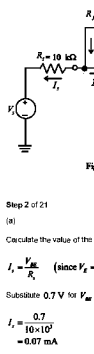


Figure 1

Step 2 of 21

(a)

Calculate the value of the current I_s.

$$I_s = \frac{V_{BE}}{R_s} \quad (\text{since } V_E = 0, V_B = V_{BE})$$

Substitute 0.7 V for V_{BE} and $10\times10^3\ \Omega$ for R_s.

$$I_s = \frac{0.7}{10\times10^3}$$
$$= 0.07 \text{ mA}$$

Step 3 of 21

Determine the dc collector current I_C.

Apply Kirchhoff's voltage law to loop which consist of feedback resistor ,output and base from Figure 2.

$$-V_o + V_{BE} + I_f R_f = 0$$
$$V_o = V_{BE} + I_f R_f$$
$$= V_{BE} + (I_B + I_s)R_f \qquad (\text{since } I_f = I_B + I_s)$$

Substitute 0.07 mA for I_s, 0.7 V for V_{BE} and 56 kΩ for R_f in V_o

$$V_o = 0.7 + (I_B + 0.07\times10^{-3})(56\times10^3)$$
$$= 0.7 + (56\times10^3)I_B + 3.92$$

$$V_o = (56\times10^3)I_B + 4.62 \quad \ldots\ldots (1)$$

Step 4 of 21

Apply Kirchhoff's current law at the collector node in Figure 2.

$$\frac{V_o - 15}{5.6\times10^3} + I_C + I_f = 0$$
$$\frac{V_o - 15}{5.6\times10^3} + \beta I_B + (I_B + I_s) = 0 \qquad \left(\text{since } \begin{matrix} I_C = \beta I_B \\ I_f = I_B + I_s \end{matrix}\right)$$

Substitute 0.07 mA for I_s and 100 for β.

$$\frac{V_o - 15}{5.6\times10^3} + (100+1)I_B + (0.07\times10^{-3}) = 0$$
$$15 - V_o = (101)(5.6\times10^3)I_B + 0.392$$
$$15 - V_o = (565.6\times10^3)I_B + 0.392$$

$$V_o = 14.608 - (565.5\times10^3)I_B \quad \ldots\ldots (2)$$

Step 5 of 21

Subtract equation (2) from equation (1).

$$(56\times10^3)I_B + 4.62 - 14.608 + (565.6\times10^3)I_B = 0$$
$$(621.6\times10^3)I_B = 9.988$$
$$I_B = \frac{9.988}{(621.6\times10^3)}$$
$$= 16.06\ \mu\text{A}$$

Calculate the dc collector current $I_C(\text{dc})$

$$I_C(\text{dc}) = \beta I_B$$

Substitute $16.06\ \mu$A for I_B and 100 for β in $I_C(\text{dc})$.

$$I_C(\text{dc}) = (100)(16.06\times10^{-6})$$
$$= 1.606 \text{ mA}$$

Therefore, the dc collector current $I_C(\text{dc})$ of the BJT is $\boxed{1.606 \text{ mA}}$.

Step 6 of 21

(b)

Draw the small signal equivalent circuit

The small- signal equivalent circuit diagram is shown in Figure 2.

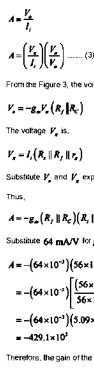


Figure 2

Step 7 of 21

(c)

Draw the A circuit for the amplifier.

Figure 3

Step 8 of 21

Calculate the trans-conductance, g_m.

$$g_m = \frac{I_C}{V_T}$$

Substitute 1.606 mA for I_C and 25 mV for V_T.

$$g_m = \frac{1.606\times10^{-3}}{25\times10^{-3}}$$
$$= 0.064 \text{ A/V}$$
$$= 64 \text{ mA/V}$$

Step 9 of 21

Calculate the value of the resistance, r_π.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute 100 for β and 64 mA/V for g_m

$$r_\pi = \frac{100}{64\times10^{-3}}$$
$$= 1.56 \text{ kΩ}$$

Step 10 of 21

Calculate the gain of the circuit A.

$$A = \frac{V_o}{I_s}$$

$$A = \left(\frac{V_\pi}{I_s}\right)\left(\frac{V_o}{V_\pi}\right) \quad \ldots\ldots (3)$$

From the Figure 3, the voltage V_π is,

$$V_\pi = -g_m V_\pi (R_f \| R_C)$$

The voltage V_π is,

$$V_\pi = I_s (R_s \| R_f \| r_\pi)$$

Substitute V_π and V_π expressions in equation 3.

Thus,

$$A = -g_m (R_f \| R_C)(R_s \| R_f \| r_\pi)$$

Substitute 64 mA/V for g_m, 56 kΩ for R_f, 5.6 kΩ for R_C, 10 kΩ for R_s and 1.56 kΩ for r_π in A.

$$A = -(64\times10^{-3})(56\times10^3 \| 5.6\times10^3)(10\times10^3 \| 56\times10^3 \| 1.56\times10^3)$$
$$= -(64\times10^{-3})\left(\frac{(56\times10^3)(5.6\times10^3)}{56\times10^3 + 5.6\times10^3}\right)\left(10\times10^3 \| \left[\frac{(56\times10^3)(1.56\times10^3)}{56\times10^3 + 1.56\times10^3}\right]\right)$$
$$= -(64\times10^{-3})(5.09\times10^3)(1.317\times10^3)$$
$$= -429.1\times10^3$$

Therefore, the gain of the amplifier is $\boxed{-429.1 \text{ kΩ}}$.

Step 11 of 21

Calculate the input resistance R_i.

$$R_i = R_s \| R_f \| r_\pi$$

Substitute 56 kΩ for R_f, 10 kΩ for R_s and 1.56 kΩ for r_π in R_i

$$R_i = \left(10\times10^3 \| 56\times10^3 \| 1.56\times10^3\right)$$
$$= 1.31 \text{ kΩ}$$

Therefore, the input resistance R_i of the amplifier is $\boxed{1.31 \text{ kΩ}}$.

Step 12 of 21

Calculate the output resistance.

$$R_o = R_f \| R_C$$

Substitute 56 kΩ for R_f, 5.6 kΩ for R_C in equation.

$$R_o = \left(56\times10^3 \| 5.6\times10^3\right)$$
$$= 5.09 \text{ kΩ}$$

Therefore, the output resistance R_o of the amplifier is $\boxed{5.09 \text{ kΩ}}$.

Step 13 of 21

(d)

Calculate the feedback factor β.

$$\beta = \frac{I_f}{V_o}$$
$$= -\frac{1}{R_f}$$

Substitute 56 kΩ for R_f in β.

$$\beta = -\frac{1}{56\times10^3}$$
$$= -17.85\times10^{-6}$$

Therefore, the feedback factor β is $\boxed{-17.85\times10^{-6}}$.

Step 14 of 21

Calculate the loop gain $A\beta$.

$$A\beta = (-429.1\times10^3)(-17.85\times10^{-6})$$
$$= 7.662$$

Therefore, the loop gain $A\beta$ of the amplifier is $\boxed{7.66}$.

Calculate the amount of feedback $1 + A\beta$.

$$1 + A\beta = 1 + 7.66$$
$$= 8.66$$

Therefore, the amount of feedback $1 + A\beta$ is $\boxed{8.66}$.

Step 15 of 21

(e)

Calculate the gain with feedback, A_f.

$$A_f = \frac{A}{1 + A\beta}$$

Substitute 8.66 for $1 + A\beta$ and -429.1×10^3 for A in A_f.

$$A_f = \frac{-429.1\times10^3}{8.66}$$
$$= -49.54\times10^3$$

Therefore, the gain with feedback A_f of the amplifier is $\boxed{-49.54 \text{ kΩ}}$.

Step 16 of 21

Calculate the input resistance with feedback R_{if}.

$$R_{if} = \frac{R_i}{1 + A\beta}$$

Substitute 8.66 for $1 + A\beta$ and 1.31 kΩ for R_i in R_{if}

$$R_{if} = \frac{1.31\times10^3}{8.66}$$
$$= 151.27\ \Omega$$

Therefore, the input resistance with feedback R_{if} of the amplifier is $\boxed{151.27\ \Omega}$.

Step 17 of 21

Calculate the output resistance with feedback R_{of}.

$$R_{of} = \frac{R_o}{1 + A\beta}$$

Substitute 8.66 for $1 + A\beta$ and 5.09 kΩ for R_o in R_{of}

$$R_{of} = \frac{5.09\times10^3}{8.66}$$
$$= 587.7\ \Omega$$

Therefore, the output resistance with feedback R_{of} of the amplifier is $\boxed{587.7\ \Omega}$.

Step 18 of 21

Calculate the input resistance, R_{in}.

$$R_{in} = \frac{1}{\frac{1}{R_f} - \frac{1}{R_s}}$$

Substitute 151.2 kΩ for R_f, 10 kΩ for R_s in equation.

$$R_{in} = \frac{1}{\frac{1}{151.2} - \frac{1}{10\times10^3}}$$
$$= \frac{1}{6.073\times10^{-3}}$$
$$= 153.52\ \Omega$$

Therefore, the input resistance R_{in} of the amplifier is $\boxed{153.52\ \Omega}$.

Step 19 of 21

(f)

Calculate the voltage gain, $\frac{V_o}{V_s}$.

$$\frac{V_o}{V_s} = \frac{V_o}{I_s R_s}$$
$$= \frac{A_f}{R_s} \qquad \left(\text{since } A_f = \frac{V_o}{I_s}\right)$$

Substitute $-49.54\times10^3\ \Omega$ for A_f and 10 kΩ for R_s in equation

$$\frac{V_o}{V_s} = \frac{-49.54\times10^3}{10\times10^3}$$
$$= -4.95 \text{ V/V}$$

Therefore, the voltage gain $\frac{V_o}{V_s}$ of the amplifier is $\boxed{-4.95 \text{ V/V}}$.

Step 20 of 21

If the loop gain is very large, then the gain with feedback A_f is as follows:

$$A_f = \frac{1}{\beta}$$

Substitute -17.85×10^{-6} for β in A_f.

$$A_f = \frac{1}{-17.85\times10^{-6}}$$
$$= -56.0\times10^3$$
$$= -56 \text{ kΩ}$$

Therefore, the closed loop gain $A_f = -R_f$.

Step 21 of 21

Calculate the voltage gain $\frac{V_o}{V_s}$.

$$\frac{V_o}{V_s} = \frac{A_f}{R_s}$$

Substitute -56×10^3 for A_f and 10 kΩ for R_s in equation.

$$\frac{V_o}{V_s} = \frac{-56\times10^3}{10\times10^3}$$
$$= -5.6 \text{ V/V}$$

Therefore, the voltage gain $\frac{V_o}{V_s}$ of the amplifier for very large loop gain is $\boxed{-5.6 \text{ V/V}}$.

Step 1 of 7

Refer to the Figure P11.64 in the textbook.

The modified circuit diagram is shown in Figure 1.

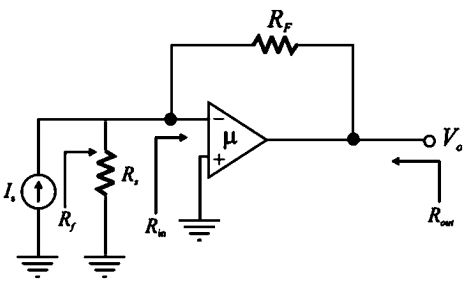


Figure 1

Step 2 of 7

(a)

Define closed loop gain of the amplifier.

$$A_f = \frac{V_o}{I_s}$$

The closed loop gain of the amplifier is,

$$A_f = \frac{A}{1 + A\beta}$$

The loop gain $A\beta$ is large.

$$A_f = \frac{A}{A\beta} \quad \text{since } 1 << A\beta$$
$$= \frac{1}{\beta}$$
$$= \frac{1}{-\frac{1}{R_F}}$$
$$= -R_F$$

Substitute $\frac{V_o}{I_s}$ for A_f in the equation.

$$\frac{V_o}{I_s} = \frac{1}{\beta} \quad \text{..... (1)}$$
$$\frac{V_o}{I_s} = -R_F$$

Step 3 of 7

From Figure 1, the voltage V_s is,

$$V_s = I_s R_s$$

$$I_s = \frac{V_s}{R_s}$$

Substitute $\frac{V_o}{R_s}$ for I_s and $-\frac{1}{R_F}$ for β in the equation 1.

$$\frac{V_o}{\left(\frac{V_s}{R_s}\right)} = \frac{1}{-\frac{1}{R_F}}$$

$$\frac{V_o}{V_s} = -\frac{R_F}{R_s}$$
$$= \frac{A_f}{R_s}$$

Substitute $-10\,\text{V/V}$ for $\frac{V_o}{V_s}$ and $1\,\text{k}\Omega$ for R_s in equation.

$$-10 = -\frac{R_F}{1000}$$

$$R_F = (1000)(10)$$
$$= 10000\,\Omega$$
$$= 10\,\text{k}\Omega$$

Hence, the value of feedback resistor R_F is $\boxed{10\,\text{k}\Omega}$.

Step 4 of 7

(b)
A circuit for the shunt-shunt feedback topology as,

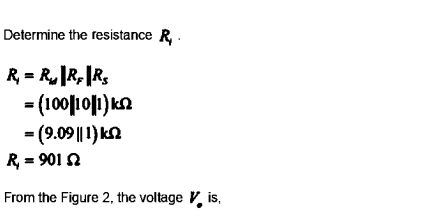


Figure 2

Determine the resistance R_i.

$$R_i = R_{id} \| R_F \| R_S$$
$$= (100 \| 10 \| 1)\,\text{k}\Omega$$
$$= (9.09 \| 1)\,\text{k}\Omega$$
$$R_i = 901\,\Omega$$

From the Figure 2, the voltage V_o is,

$$V_o = -\mu V_{id} \frac{(R_F)}{r_o + (R_F)}$$

$$V_o = -\mu I_i R_i \frac{(R_F)}{r_o + (R_F)} \qquad (\text{Since } V_{id} = I_i R_i)$$

$$\frac{V_o}{I_i} = -\mu R_i \frac{R_F}{r_o + (R_F)}$$

$$A = -\mu R_i \frac{R_F}{r_o + (R_F)}$$

Calculation of the closed loop gain A_f as follows:

$$A_f = \frac{A}{1 + A\beta}$$

Substitute $-\mu R_i \frac{(R_F)}{r_o + R_F}$ for A, $-\frac{1}{R_F}$ for β in equation.

$$A_f = \frac{-\mu R_i \frac{(R_F)}{r_o + R_F}}{1 + \mu R_i \left(\frac{(R_F)}{r_o + R_F}\right)\left(\frac{1}{R_F}\right)}$$
$$= \frac{-\mu R_i R_F}{r_o + R_F + \mu R_i}$$

Substitute $10^3\,\text{V/V}$ for μ, $100\,\text{k}\Omega$ for R_{id}, $1\,\text{k}\Omega$ for r_o, $10\,\text{k}\Omega$ for R_F, $1\,\text{k}\Omega$ for R_s, $901\,\text{k}\Omega$ for R_i in equation.

$$A_f = -\frac{10^3(901)(10\,\text{k})}{1\,\text{k} + 10\,\text{k} + 10^3(901)}$$
$$= -\frac{901 \times 10^4}{912}$$
$$= -9.8 \times 10^3$$
$$= -9.8\,\text{k}\Omega$$

Thus, the closed loop gain A_f is $-9.8\,\text{k}\Omega$.

Step 5 of 7

Determine the V_o/V_s.

$$\frac{V_o}{V_s} = \frac{A_f}{R_s}$$

Substitute $-9.8\,\text{k}\Omega$ for A_f, $1\,\text{k}\Omega$ for R_s in equation.

$$\frac{V_o}{V_s} = -\frac{9.8\,\text{k}\Omega}{1\,\text{k}\Omega}$$
$$= -9.8$$

Therefore, the actual value of V_o/V_s is $\boxed{\frac{V_o}{V_s} = -9.8\,\text{V/V}}$.

Step 6 of 7

Determine the value of input resistance R_{in}.

$$R_{in} = \frac{R_F(r_o + R_F)}{\mu(R_F)}$$

Substitute $10^3\,\text{V/V}$ for μ, $1\,\text{k}\Omega$ for r_o, $10\,\text{k}\Omega$ for R_F in equation.

$$R_{in} = \frac{10\,\text{k}(1\,\text{k} + 10\,\text{k})}{10^3(10\,\text{k})}$$
$$= \frac{11 \times 10^3}{10^3}$$

Therefore, the value of input resistance is $\boxed{R_{in} = 11\,\Omega}$.

Step 7 of 7

Determine the value of output resistance R_{out}.

$$R_{out} \simeq \frac{R_F}{R_i}\left(\frac{r_o}{\mu}\right)$$

Substitute $10^3\,\text{V/V}$ for μ, $1\,\text{k}\Omega$ for r_o, $10\,\text{k}\Omega$ for R_F, $901\,\text{k}\Omega$ for R_i in equation.

$$R_{out} = \frac{10\,\text{k}}{901}\left(\frac{1\,\text{k}}{10^3}\right)$$
$$= \frac{10\,\text{k}}{901}$$
$$= 11.09$$

Therefore, the value of output resistance is $\boxed{R_{out} = 11.09\,\Omega}$.

Step 1 of 3) Assume R_F and R_L are known. Express I_L in terms of V_o, R_F, and R_L.

$$I_L = \frac{V_o}{R_F + R_L}$$

Step 2 of 3) Express V_o in terms of V_{in}, R_F, and R_L.

$$V_o = V_{in} \cdot \frac{R_L}{R_F + R_L}$$

Step 3 of 3) Express R_L in terms of V_{in}, V_o, and R_F.

$$R_L = \frac{V_o \cdot R_F}{V_{in} - V_o}$$

$R_L = \frac{V_o \cdot R_F}{V_{in} - V_o}$

Step 1 of 2) Express I_L in terms of V_{in} and R_L.

$$I_L = \frac{V_{in}}{R_L}$$

Step 2 of 2) Express V_o in terms of V_{in}, R_F, and R_L.

$$V_o = V_{in} \cdot \frac{R_L}{R_F + R_L}$$

$V_o = V_{in} \cdot \frac{R_L}{R_F + R_L

Step 1 of 4

Refer to Figure P11.65 for the feedback transresistance amplifier in the textbook.

Redraw the circuit by setting $I_s = 0$ and breaking the feedback loop as shown in figure 1.

Connect a voltage source of voltage V_T as shown in figure 1.

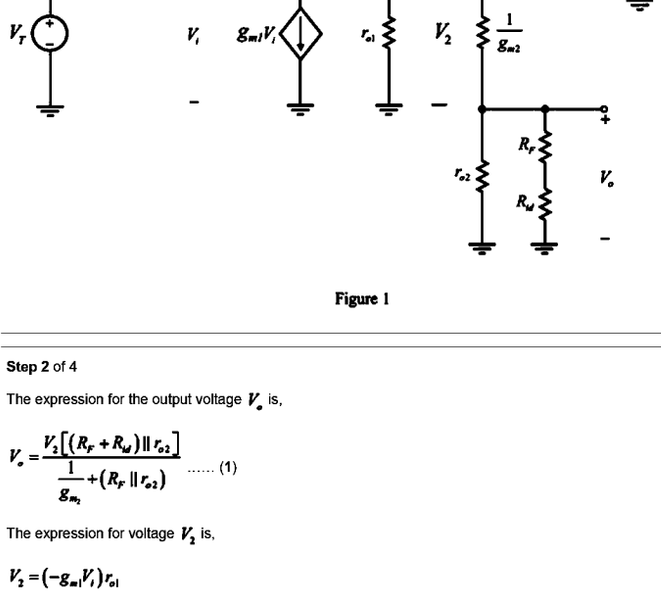


Figure 1

Step 2 of 4

The expression for the output voltage V_o is,

$$V_o = \frac{V_2\left[(R_F + R_{id}) \| r_{o2}\right]}{\frac{1}{g_{m_2}} + (R_F \| r_{o2})} \quad \text{..... (1)}$$

The expression for voltage V_2 is,

$$V_2 = (-g_{m1}V_i)r_{o1}$$

Substitute the above expression in equation (1).

$$V_o = \frac{(-g_{m1}V_i)r_{o1}\left[(R_F + R_{id}) \| r_{o2}\right]}{\frac{1}{g_{m_2}} + (R_F \| r_{o2})}$$

Substitute V_T for V_i .

$$V_o = \frac{(-g_{m1}V_T)r_{o1}\left[(R_F + R_{id}) \| r_{o2}\right]}{\frac{1}{g_{m_2}} + (R_F \| r_{o2})}$$

$$\frac{-V_o}{V_T} = \frac{g_{m1}r_{o1}\left[(R_F + R_{id}) \| r_{o2}\right]}{\frac{1}{g_{m_2}} + (R_F \| r_{o2})}$$

$$A\beta = \frac{g_{m1}r_{o1}\left[(R_F + R_{id}) \| r_{o2}\right]}{\frac{1}{g_{m_2}} + (R_F \| r_{o2})} \quad \text{..... (2)}$$

Therefore, the expression for the loop-gain of the amplifier is,

$$\boxed{A\beta = \frac{g_{m1}r_{o1}\left[(R_F + R_{id}) \| r_{o2}\right]}{\frac{1}{g_{m_2}} + (R_F \| r_{o2})}}.$$

Step 3 of 4

The expression for the loop-gain of the amplifier in problem P11.65 is,

$$A\beta = \frac{g_{m1}r_{o1}(R_F \| r_{o2})}{\frac{1}{g_{m_2}} + (R_F \| r_{o2})} \quad \text{..... (3)}$$

Recall equation (2).

$$A\beta = \frac{g_{m1}r_{o1}(R_F \| r_{o2} \| R_{in})}{\frac{1}{g_{m_2}} + (R_F \| r_{o2})}$$

Compare the equations (2) and (3).

The loop gain of the amplifier obtained by breaking the feedback loop and setting $I_s = 0$ is more accurate when compared to the loop-gain of the amplifier in problem P11.65. This is because the value of R_{id} is involved.

Step 4 of 4

Recall equation (2).

$$A\beta = \frac{g_{m1}r_{o1}\left[(R_F + R_{id}) \| r_{o2}\right]}{\frac{1}{g_{m_2}} + (R_F \| r_{o2})}$$

Substitute 4×10^{-3} for g_{m1} and g_{m2} ; and 10×10^3 for R_F , and 40×10^3 for r_{o1} and r_{o2} ; 64.04 for R_{id} .

$$\begin{aligned} A\beta &= \frac{(4\times10^{-3})(40\times10^3)\left[(10\times10^3 + 64.04) \| 40\times10^3\right]}{\frac{1}{4\times10^{-3}} + (10\times10^3 \| 40\times10^3)} \\ &= \frac{(4)(40)(8.0409\times10^3)}{\frac{10^3}{4} + (8\times10^3)} \\ &= \frac{1286.5493\times10^3}{10^3(0.25+8)} \\ &= 155.945 \end{aligned}$$

The percentage difference between the two values is,

$$\%\text{change} = \frac{(A\beta)_2 - (A\beta)_1}{(A\beta)_1} \times 100$$

Substitute 155.1515 for $(A\beta)_1$ and 155.945 for $(A\beta)_2$.

$$\begin{aligned} \%\text{change} &= \frac{155.945 - 155.1515}{155.1515} \times 100 \\ &= 0.51\% \end{aligned}$$

Therefore, the two values of loop-gain differ by $\boxed{0.51\%}$.

Step 1 of 6

11.067P

Refer to the circuit diagram of Figure E11.19 in the textbook.

The modified equivalent circuit diagram is shown in Figure 1.

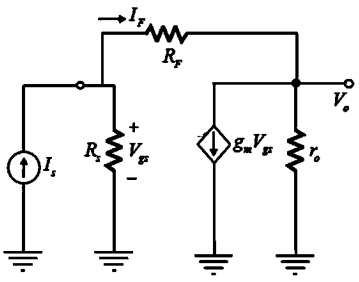


Figure 1

Step 2 of 6

Apply Kirchhoff's current law at node V_o.

$$I_F = g_m V_{gs} + \frac{V_o}{r_o}$$

$$\frac{V_{gs} - V_o}{R_F} = g_m V_{gs} + \frac{V_o}{r_o}$$

$$\frac{V_{gs}}{R_F} - g_m V_{gs} = \frac{V_o}{R_F} + \frac{V_o}{r_o}$$

$$V_{gs}\left(\frac{1}{R_F} - g_m\right) = V_o\left(\frac{1}{R_F} + \frac{1}{r_o}\right)$$

$$V_{gs}\left(\frac{1}{R_F} - g_m\right) = \frac{V_o}{r_o \parallel R_F} \quad \text{..... (1)}$$

Step 3 of 6

Applying Kirchhoff's current law at node V_{gs}

$$I_s = \frac{V_{gs}}{R_s} + I_F$$

$$I_F = I_s - \frac{V_{gs}}{R_s}$$

Substitute $\frac{V_{gs} - V_o}{R_F}$ for I_F in the equation.

$$\frac{V_{gs} - V_o}{R_F} = I_s - \frac{V_{gs}}{R_s}$$

$$V_{gs}\left(\frac{1}{R_F} + \frac{1}{R_s}\right) = I_s + \frac{V_o}{R_F}$$

$$\frac{V_{gs}}{(R_F \parallel R_s)} = I_s + \frac{V_o}{R_F}$$

$$V_{gs} = I_s(R_F \parallel R_s) + \frac{V_o(R_F \parallel R_s)}{R_F} \quad \text{..... (2)}$$

Step 4 of 6

Multiply on both sides with term $\left(g_m - \frac{1}{R_F}\right)$ to equation (2).

$$V_{gs}\left(g_m - \frac{1}{R_F}\right) = I_s(R_F \parallel R_s)\left(g_m - \frac{1}{R_F}\right) + \frac{V_o(R_F \parallel R_s)}{R_F}\left(g_m - \frac{1}{R_F}\right) \quad \text{..... (3)}$$

From equation (1), $V_{gs}\left(g_m - \frac{1}{R_F}\right) = -\frac{V_o}{r_o \parallel R_F}$.

Substitute $V_{gs}\left(g_m - \frac{1}{R_F}\right)$ for $-\frac{V_o}{r_o \parallel R_F}$ in equation 3.

$$-\frac{V_o}{r_o \parallel R_F} = I_s(R_F \parallel R_s)\left(g_m - \frac{1}{R_F}\right) + \frac{V_o(R_F \parallel R_s)}{R_F}\left(g_m - \frac{1}{R_F}\right)$$

$$-V_o\left(\frac{1}{r_o \parallel R_F} + \frac{(R_F \parallel R_s)}{R_F}\left(g_m - \frac{1}{R_F}\right)\right) = I_s(R_F \parallel R_s)\left(g_m - \frac{1}{R_F}\right)$$

$$\frac{V_o}{I_s} = -\frac{(R_F \parallel R_s)\left(g_m - \frac{1}{R_F}\right)}{\frac{1}{r_o \parallel R_F} + \frac{(R_F \parallel R_s)}{R_F}\left(g_m - \frac{1}{R_F}\right)}$$

Step 5 of 6

The closed loop gain of trans-resistance amplifier A_f is,

$$A_f = \frac{V_o}{I_s}$$

$$A_f = -\frac{(R_F \parallel R_s)\left(g_m - \frac{1}{R_F}\right)(r_o \parallel R_F)}{1 + \frac{(R_F \parallel R_s)}{R_F}\left(g_m - \frac{1}{R_F}\right)(r_o \parallel R_F)} \quad \text{..... (3)}$$

Therefore, the closed loop gain of the amplifier A_f is,

$$\boxed{\frac{V_o}{I_s} = -\frac{(R_F \parallel R_s)\left(g_m - \frac{1}{R_F}\right)(r_o \parallel R_F)}{1 + \frac{(R_F \parallel R_s)}{R_F}\left(g_m - \frac{1}{R_F}\right)(r_o \parallel R_F)}}.$$

Step 6 of 6

Comparing the expression to the given in Exercise 11.19 part (b) is as follows:

With feedback approach calculated closed loop gain for trans-resistance amplifier is,

$$A_f = \frac{V_o}{I_s} = -\frac{(R_F \parallel R_s)(g_m)(r_o \parallel R_F)}{1 + \frac{(R_F \parallel R_s)}{R_F}(g_m)(r_o \parallel R_F)}$$

Notice that the only difference is that g_m is replaced by $\left(g_m - \frac{1}{R_F}\right)$.

Hence, the condition for feedback-analysis method to be reasonably accurate for the above circuit is

$$\boxed{g_m \gg \frac{1}{R_F}}.$$

Step 1 of 12

Refer to Figure P11.68 in the textbook.

Consider the following expression:

$$A_f = \frac{A}{1+A\beta}$$

Consider the following inequality:

$$A\beta >> 1$$

Rewrite the expression as,

$$A_f = \frac{A}{A\beta}$$

$$A_f = \frac{1}{\beta} \quad \text{..... (1)}$$

Step 2 of 12

Draw the circuit for β as shown in figure 1.

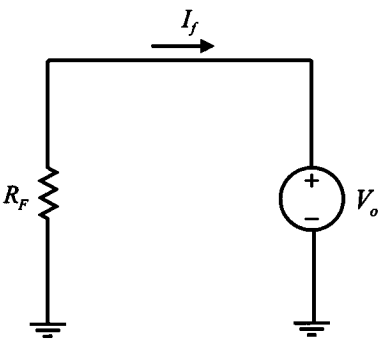


Figure 1

Step 3 of 12

Apply Kirchhoff's voltage law to the circuit shown in figure 1.

$$V_o = -I_f R_F$$

$$\frac{I_f}{V_o} = \frac{-1}{R_F}$$

Consider the expression for β.

$$\beta = \frac{I_f}{V_o}$$

Substitute $\frac{-1}{R_F}$ for $\frac{I_f}{V_o}$.

$$\beta = \frac{-1}{R_F}$$

Recall equation (1).

$$A_f = \frac{1}{\beta}$$

Substitute $\frac{-1}{R_F}$ for β.

$$A_f = \frac{1}{\left(\frac{-1}{R_F}\right)}$$

$$A_f = -R_F$$

Substitute $-10\ \text{k}\Omega$ for A_f.

$$-10\ \text{k}\Omega = -R_F$$

$$R_F = 10\ \text{k}\Omega$$

Therefore, the value of R_F is $\boxed{10\ \text{k}\Omega}$.

Step 4 of 12

Draw the circuit as shown in Figure 2.

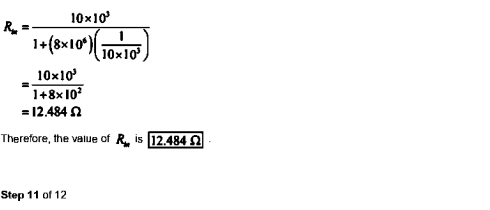


Figure 2

Step 5 of 12

Assume that the voltage across the drain terminal D_1 is V_{D_1}.

$$V_{D_1} = -g_{m1}V_{GS_1}R_{D_1} \quad \text{..... (1)}$$

As the gate terminal G_1 is grounded, consider the following equation:

$$V_{GS_1} = -V_i$$

Substitute the above expression for V_{GS_1} in equation (1).

$$V_{D_1} = -g_{m1}(-V_i)R_{D_1}$$

$$V_{D_1} = g_{m1}V_iR_{D_1} \quad \text{..... (2)}$$

Apply Source transformation technique to the current source I_s and resistance R_F.

$$V_i = I_sR_F$$

Substitute the above expression for V_i in equation (2).

$$V_{D_1} = g_{m1}I_sR_FR_{D_1} \quad \text{..... (3)}$$

Step 6 of 12

Consider the following equation:

$$V_2 = V_{D_1} \quad \text{..... (4)}$$

Recall equation (3).

$$V_{D_1} = g_{m1}I_sR_FR_{D_1}$$

Substitute the above expression for V_{D_1} in equation (4).

$$V_2 = g_{m1}I_sR_FR_{D_1} \quad \text{..... (5)}$$

Step 7 of 12

Determine the expression for the DC output voltage V_o.

$$V_o = \left(R_{D_2} \parallel R_F\right)\left(-g_{m2}V_2\right) \quad \text{..... (6)}$$

Recall equation (5).

$$V_2 = g_{m1}I_sR_FR_{D_1}$$

Substitute the above expression for V_2 in equation (6).

$$V_o = \left(R_{D_2} \parallel R_F\right)\left(-g_{m2}g_{m1}I_sR_FR_{D_1}\right)$$

$$V_o = -g_{m1}g_{m2}I_sR_FR_{D_1}\left(R_{D_2} \parallel R_F\right) \quad \text{..... (7)}$$

Determine the expression for open-loop transresistance A of the amplifier.

$$A = \frac{V_o}{I_s}$$

Consider equation (7).

Substitute $\left[-g_{m1}g_{m2}I_sR_FR_{D_1}\left(R_{D_2} \parallel R_F\right)\right]$ for V_o.

$$A = \frac{\left[-g_{m1}g_{m2}I_sR_FR_{D_1}\left(R_{D_2} \parallel R_F\right)\right]}{I_s}$$

$$= -g_{m1}g_{m2}R_FR_{D_1}\left(R_{D_2} \parallel R_F\right)$$

Substitute 4×10^{-3} for g_{m1} and g_{m2}, 10×10^3 for R_F, R_{D_1}, and R_{D_2}.

$$A = -\left(4\times10^{-3}\right)\left(4\times10^{-3}\right)\left(10\times10^3\right)\left(10\times10^3\right)\left(10\times10^3 \parallel 10\times10^3\right)$$

$$= -\left(4\times10^{-3}\right)\left(4\times10^{-3}\right)\left(10\times10^3\right)\left(10\times10^3\right)\left(5\times10^3\right)$$

$$= -(4)(4)(10)\left(5\times10^3\right)$$

$$= -8\times10^3\ \text{k}\Omega$$

Step 8 of 12

Consider the following expression:

$$A_f = \frac{A}{1+A\beta}$$

Substitute -8×10^6 for A, and $\frac{1}{10\times10^3}$ for β.

$$A_f = \frac{-8\times10^6}{1+\left(8\times10^6\right)\left(\frac{1}{10\times10^3}\right)}$$

$$= \frac{-8\times10^6}{1+\left(8\times10^3\right)}$$

$$= -10.012\ \text{k}\Omega$$

Therefore, the actual value of A_f is $\boxed{-10.012\ \text{k}\Omega}$.

Step 9 of 12

Determine the input resistance R_i.

$$R_i = R_F$$

Substitute 10×10^3 for R_F.

$$R_i = 10\times10^3$$

$$= 10\ \text{k}\Omega$$

Step 10 of 12

Determine the resistance R_{in}.

$$R_{in} = \frac{R_i}{1+A\beta}$$

Substitute 10×10^3 for R_i, 8×10^6 for A, and $\frac{-1}{10\times10^3}$ for β.

$$R_{in} = \frac{10\times10^3}{1+\left(8\times10^6\right)\left(\frac{1}{10\times10^3}\right)}$$

$$= \frac{10\times10^3}{1+8\times10^2}$$

$$= 12.484\ \Omega$$

Therefore, the value of R_{in} is $\boxed{12.484\ \Omega}$.

Step 11 of 12

Determine the resistance R_o.

$$R_o = \left(R_F \parallel R_{D_2}\right)$$

Substitute 10×10^3 for R_F, and R_{D_2}.

$$R_o = \left(10\times10^3 \parallel 10\times10^3\right)$$

$$= 5\ \text{k}\Omega$$

Step 12 of 12

Determine the resistance R_{out}.

$$R_{out} = \frac{R_o}{1+A\beta}$$

Substitute 5×10^3 for R_i, 8×10^6 for A, and $\frac{-1}{10\times10^3}$ for β.

$$R_{in} = \frac{5\times10^3}{1+\left(8\times10^6\times\frac{1}{10\times10^3}\right)}$$

$$= \frac{5\times10^3}{1+8\times10^2}$$

$$= 6.242\ \Omega$$

Therefore, the value of R_{in} is $\boxed{6.242\ \Omega}$.

Step 1 of 16

Refer to Figure P11.69 in the textbook

(a)

The source current I_1 has zero DC component.

Assume that the emitter current of transistor Q_1 is I_{E1}.

Determine the emitter current I_{E1}.

$$\begin{aligned} I_{E1} &= \frac{I_{Z1}}{\beta_{dc1}+1} \\ &= \frac{0.7\text{ mA}}{100+1} \\ &= 6.931\text{ }\mu\text{A} \end{aligned}$$

Substitute 0.7 mA for I_{Z1} and 100 for β_{dc1}.

Determine the emitter current I_{E2}.

$$\begin{aligned} I_{E2} &= \frac{I_{Z2}}{\beta_{dc2}+1} \\ &= \frac{0.7\text{ mA}}{100+1} \\ &= 6.931\text{ }\mu\text{A} \end{aligned}$$

Substitute 0.7 mA for I_{Z2} and 100 for β_{dc2}.

Determine the collector current of transistor Q_1.

$$\begin{aligned} I_{C1} &= \frac{\beta_{dc1}}{\beta_{dc1}+1} I_{E1} \\ &= \frac{100}{100+1}(6.931\text{ }\mu\text{A}) \\ &= 6.862\text{ }\mu\text{A} \end{aligned}$$

Substitute 100 for β_{dc1} and $6.931\text{ }\mu\text{A}$ for I_{E1}.

Determine the collector current of transistor Q_2.

$$\begin{aligned} I_{C2} &= \frac{\beta_{dc2}}{\beta_{dc2}+1} I_{E2} \\ &= \frac{100}{100+1}(6.931\text{ }\mu\text{A}) \\ &= 6.862\text{ }\mu\text{A} \end{aligned}$$

Substitute 100 for β_{dc2} and $6.931\text{ }\mu\text{A}$ for I_{E2}.

Determine the base current of transistor Q_1.

$$\begin{aligned} I_{B1} &= \frac{I_{C1}}{\beta_{dc1}} \\ &= \frac{6.862\text{ }\mu\text{A}}{100} \\ &= 0.069\text{ }\mu\text{A} \end{aligned}$$

Substitute $6.862\text{ }\mu\text{A}$ for I_{C1} and 100 for β_{dc1}.

Determine the base current of transistor Q_2.

$$\begin{aligned} I_{B2} &= \frac{I_{C2}}{\beta_{dc2}} \\ &= \frac{6.862\text{ }\mu\text{A}}{100} \\ &= 0.069\text{ }\mu\text{A} \end{aligned}$$

Substitute $6.862\text{ }\mu\text{A}$ for I_{C2} and 100 for β_{dc2}.

Determine the load current I_L.

$$\begin{aligned} I_L &= I_{Z1} - I_{B1} - I_{B2} \\ &= 0.7\text{ mA} - 0.069\text{ }\mu\text{A} - 0.069\text{ }\mu\text{A} \\ &= 0.7\text{ mA} \end{aligned}$$

Substitute 0.7 mA for I_{Z1}, $0.069\text{ }\mu\text{A}$ for I_{B1}, and $0.069\text{ }\mu\text{A}$ for I_{B2}.

Therefore, the load current I_L is $\boxed{0.7\text{ mA}}$.

(b)

Determine the load regulation.

$$\begin{aligned} \text{Load regulation} &= \frac{V_{NL}-V_{FL}}{V_{FL}}\times 100\% \\ &= \frac{0.7\text{ V}-0.7\text{ V}}{0.7\text{ V}}\times 100\% \\ &= 0\% \end{aligned}$$

Substitute 0.7 V for V_{NL} and 0.7 V for V_{FL}.

Therefore, the load regulation is $\boxed{0\%}$.

(c)

Determine the line regulation.

$$\begin{aligned} \text{Line regulation} &= \frac{V_{NL}-V_{FL}}{V_{FL}}\times 100\% \\ &= \frac{0.7\text{ V}-0.7\text{ V}}{0.7\text{ V}}\times 100\% \\ &= 0\% \end{aligned}$$

Substitute 0.7 V for V_{NL} and 0.7 V for V_{FL}.

Therefore, the line regulation is $\boxed{0\%}$.

(d)

Determine the output resistance.

$$\begin{aligned} R_{out} &= \frac{V_{NL}-V_{FL}}{I_{FL}} \\ &= \frac{0.7\text{ V}-0.7\text{ V}}{0.7\text{ mA}} \\ &= 0\text{ }\Omega \end{aligned}$$

Substitute 0.7 V for V_{NL}, 0.7 V for V_{FL}, and 0.7 mA for I_{FL}.

Therefore, the output resistance is $\boxed{0\text{ }\Omega}$.

Step 2 of 16

Calculate the output voltage.

$$V_o = V_A$$

Here, V_A is the voltage at the base of transistor Q_1.

Calculate the voltage at the base of transistor Q_2.

$$V_B = V_A + 0.7$$

Here, V_B is the voltage at the base of transistor Q_2.

Substitute 0.7 for V_A.

$$V_B = 0.7 + 0.7$$

$$= 1.4\text{ V}$$

Determine the current through the resistor R_C.

$$I_R = \frac{V_{CC} - V_B}{R_C}$$

Substitute 5 for V_{CC}, 1.4 for V_B, and 10×10^3 for R_C.

$$\begin{aligned} I_R &= \frac{5-1.4}{10\times10^3} \\ &= 0.36\text{ mA} \end{aligned}$$

Step 3 of 16

Determine the collector current of transistor Q_2.

$$I_C = \alpha_{dc} I_R$$

Substitute 100 for β, and 0.57×10^{-3} for I_R.

$$\begin{aligned} I_C &= \left(\frac{100}{1+100}\right)(0.57\times10^{-3}) \\ &= 0.564\text{ mA} \end{aligned}$$

Determine the collector current of transistor Q_1.

$$I_C = I_R - I_B$$

Here, I_B is the base current of transistor Q_2.

$$\begin{aligned} I_C &= I_R - \frac{I_C}{\beta} \\ &= I_R - \frac{\alpha I_R}{\beta} \\ &= I_R - \frac{\left(\frac{\beta}{1+\beta}\right)I_R}{\beta} \\ &= I_R - \frac{I_R}{1+\beta} \end{aligned}$$

Substitute 0.36×10^{-3} for I_R, 100 for β, and 0.57×10^{-3} for I_R.

$$\begin{aligned} I_C &= 0.36\times10^{-3} - \left(\frac{0.57\times10^{-3}}{1+100}\right) \\ &= (0.36 - 0.00564)\times10^{-3} \\ &= 0.3543\text{ mA} \end{aligned}$$

Hence, the DC collector currents of transistors Q_1 and Q_2 are $\boxed{0.3543\text{ mA}}$ and $\boxed{0.564\text{ mA}}$.

The DC voltage at the output is $\boxed{0.7\text{ V}}$.

Step 4 of 16

Draw the A circuit as shown in figure 1.

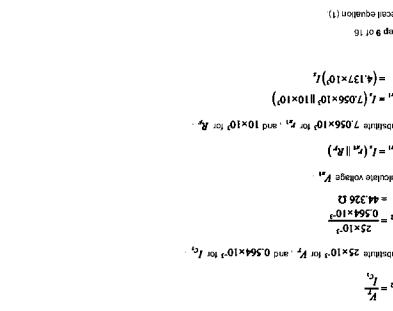


Figure 1

Step 5 of 16

Consider the following expression for the open-loop transresistance of shunt-shunt feedback amplifier.

$$A = \frac{V_o}{I_i}$$

Determine the expression for voltage V_o.

Apply Superposition theorem and obtain V_o.

$$V_o = V_{o1} + V_{o2}$$

Consider the first dependent source as shown in figure 2. Determine V_{o1}.



Figure 2

Step 6 of 16

Apply voltage division principle and obtain V_{o1}.

$$V_{o1} = \frac{-(R_E \| R_F)\beta g_{m1} V_{\pi 1} R_C}{R_C + r_{\pi 2} + (R_E \| R_F)}$$

Consider another source as shown in figure 3. Determine V_{o2}.

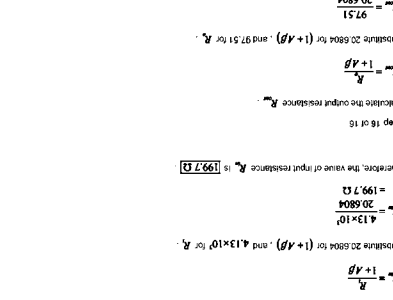


Figure 3

Step 7 of 16

Apply voltage division principle and obtain V_{o2}.

$$V_{o1} = \frac{-(R_E \| R_F) g_{m1} V_{\pi 1} R_C}{R_C + r_{\pi 2} + (R_E \| R_F)}$$

Therefore, the expression for the output voltage is.

$$V_o = V_{o1} + V_{o2}$$

Determine the transconductance g_{m1}.

$$g_{m1} = \frac{I_{C1}}{V_T}$$

Substitute 0.3543×10^{-3} for I_C and 25×10^{-3} for V_T.

$$\begin{aligned} g_{m1} &= \frac{0.3543\times10^{-3}}{25\times10^{-3}} \\ &= 14.172\text{ mA/V} \end{aligned}$$

Step 8 of 16

Calculate the resistance $r_{\pi 1}$.

$$r_{\pi 1} = \frac{\beta}{g_{m1}}$$

Substitute 100 for β and 14.172×10^{-3} for g_{m1}.

$$\begin{aligned} r_{\pi 1} &= \frac{100}{14.172\times10^{-3}} \\ &= 7.056\text{ k}\Omega \end{aligned}$$

Calculate the emitter resistance $r_{\pi 2}$.

$$r_{\pi 2} = \frac{V_T}{I_{C1}}$$

Substitute 25×10^{-3} for V_T, and 0.564×10^{-3} for I_{C1}.

$$\begin{aligned} r_{\pi 2} &= \frac{25\times10^{-3}}{0.564\times10^{-3}} \\ &= 44.326\text{ }\Omega \end{aligned}$$

Calculate voltage $V_{\pi 1}$.

$$V_{\pi 1} = I_1(r_{\pi 1} \| R_F)$$

Substitute 7.056×10^3 for $r_{\pi 1}$, and 10×10^3 for R_F.

$$V_{\pi 1} = I_1\left(7.056\times10^3 \| 10\times10^3\right)$$

$$= (4.137\times10^3)I_1$$

Step 9 of 16

Recall equation (1)

$$V_o = \frac{-(\beta)(R_E \| R_F)\beta g_{m1} V_{\pi 1} R_C}{R_C + r_{\pi 2} + (R_E \| R_F)}$$

Substitute 100 for β, 14.172×10^{-3} for g_{m1}, 10×10^3 for R_F, R_C, and R_E, and $(4.137\times10^3)I_1$ for $V_{\pi 1}$, 44.326 for $r_{\pi 2}$.

Step 1 of 13

(a)

Refer to the circuit diagram of voltage amplifier in Figure P10.61 (a) in the textbook.

Draw the equivalent circuit of voltage amplifier.

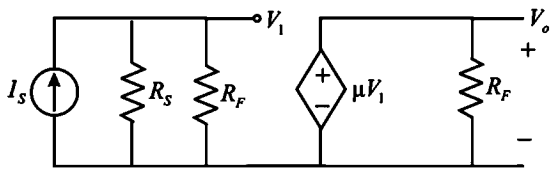


Figure 1

Step 2 of 13

From Figure 1, write the expression for output voltage V_O.

$$V_O = -\mu V_1$$

Define open loop gain of the amplifier A is,

$$A = \frac{V_O}{I_S}$$

Substitute μV_1 for V_O in the equation.

$$\begin{aligned} A &= \frac{-\mu V_1}{I_S} \\ &= \frac{-\mu\left(I_S\left(R_S \| R_F\right)\right)}{I_S} \qquad \text{since } V_1 = I_S\left(R_S \| R_F\right) \\ &= -\mu\left(R_S \| R_F\right) \end{aligned}$$

Expression for open loop gain A is $\boxed{-\mu\left(R_S \| R_F\right)}$.

Step 3 of 13

From Figure 1, the feedback factor β is,

$$\beta = -\frac{1}{R_F}$$

Define closed loop gain.

$$A_F = \frac{V_o}{I_S}$$

Closed loop gain of the amplifier is,

$$A_F = \frac{A}{1 + A\beta}$$

Substitute $-\mu\left(R_S \| R_F\right)$ for A and $-\frac{1}{R_F}$ for β in the equation.

$$\begin{aligned} A_F &= \frac{-\mu\left(R_S \| R_F\right)}{1 + \mu\left(\frac{R_S \| R_F}{R_F}\right)} \\ &= \frac{-\mu\left(\frac{R_S R_F}{R_S + R_F}\right)}{1 + \mu\left(\frac{1}{R_F}\left(\frac{R_S R_F}{R_S + R_F}\right)\right)} \\ &= \frac{-\mu R_S R_F}{R_S + R_F + \mu R_S} \\ &= \frac{-R_F}{1 + \frac{R_S + R_F}{\mu R_S}} \\ &\approx -R_F \qquad \text{since } \mu \text{ is very large.} \end{aligned}$$

Step 4 of 13

Determine the expression for the voltage gain.

$$\frac{V_o}{V_s} = \frac{V_o}{I_S R_S}$$

Substitute $-R_F$ for $\frac{V_o}{I_S}$ in the equation.

$$\begin{aligned} \frac{V_o}{V_s} &= \left(-R_F\right)\left(\frac{1}{R_S}\right) \\ &= -\frac{R_F}{R_S} \qquad (\text{For } \mu \gg 1) \end{aligned}$$

Therefore, the expression for the voltage gain $\frac{V_O}{V_S}$ is $\boxed{-\frac{R_F}{R_S}}$.

Step 5 of 13

(b)
Refer to circuit diagram of BJT amplifier in Figure P10.61 (b) in the textbook.

Find the Thevenin voltage.

$$\begin{aligned} V_{TH} &= V_{CC}\left(\frac{10\ \text{k}\Omega}{10\ \text{k}\Omega + 15\ \text{k}\Omega}\right) \\ &= 15\left(\frac{10}{25}\right) \end{aligned}$$

Find the base current I_B.

$$I_B = \left(\frac{V_{TH} - V_{BE}}{\left(15\ \text{k}\Omega \| 10\ \text{k}\Omega\right) + 4.7\ \text{k}\Omega(1+\beta)}\right)$$

Substitute $15\left(\frac{10}{25}\right)$ for V_{TH} and 100 for β in the equation.

$$\begin{aligned} I_B &= \left(\frac{15\left(\frac{10}{25}\right) - 0.7}{\left(15\ \text{k}\Omega \| 10\ \text{k}\Omega\right) + 4.7\ \text{k}\Omega(1+100)}\right) \\ &= \frac{15\left(\frac{10}{25}\right) - 0.7}{6\ \text{k}\Omega + 474.7\ \text{k}\Omega} \\ &= 11.02\ \mu\text{A} \end{aligned}$$

Step 6 of 13

Find the Trans conductance g_m.

$$\begin{aligned} g_m &= \frac{\beta}{r_\pi} \\ &= \frac{\beta}{(\beta+1)r_\pi} \\ &= \frac{100}{(100+1)(22.6)} \\ &= \frac{100}{2.286\ \text{k}\Omega} \\ &= 43.7\ \text{mA/V} \end{aligned}$$

Trans-conductance g_m of the amplifier is $\boxed{43.7\ \text{mA/V}}$

Step 7 of 13

Determine the input resistance.

$$\begin{aligned} R_{in} &= \left(R_{B1} \| R_{B2} \| r_\pi\right) \\ &= (15 \| 10 \| 2.286)\,\text{k}\Omega \\ &= \left(\left(\frac{15 \times 10}{15+10}\right) \| 2.286\right)\text{k}\Omega \\ &= 1.65\ \text{k}\Omega \end{aligned}$$

Therefore, the input resistance R_{in} is $\boxed{1.65\ \text{k}\Omega}$.

Step 8 of 13

Find the output resistance.

$$\begin{aligned} R_{out} &= R_C \| R_B \| r_\pi \\ &= R_C \| \left(R_{B1} \| R_{B2}\right) \| r_\pi \\ &= \left(7.5 \| \frac{(10)(15)}{10+15} \| 2.286\right)\text{k}\Omega \\ &= 1.35\,\text{k}\Omega \end{aligned}$$

Therefore, the output resistance R_{out} is $\boxed{1.35\,\text{k}\Omega}$.

Step 9 of 13

Calculate the voltage gain for stage-1.

$$\begin{aligned} \frac{V_1}{V_s} &= \frac{R_{in}}{R_{B1} + R_{in}} \\ &= \left(\frac{1.65\ \text{k}\Omega}{(10+1.65)\text{k}\Omega}\right) \\ &= \frac{1.65}{11.65} \\ &= 0.14 \end{aligned}$$

Step 10 of 13

Calculating the voltage gain for stage-2:

$$\begin{aligned} \frac{V_2}{V_1} &= -g_m R_{out} \\ &= -(43.7\ \text{mA})(1.35\ \text{k}\Omega) \\ &= -58.99 \end{aligned}$$

Step 11 of 13

Calculate the voltage gain for stage-3:

$$\begin{aligned} \frac{V_3}{V_2} &= -g_m R_{out} \\ &= -(43.7\ \text{mA})(1.35\ \text{k}\Omega) \\ &= -58.99 \end{aligned}$$

Step 12 of 13

Calculate the voltage gain for stage-3:

$$\begin{aligned} \frac{V_o}{V_3} &= -g_m R_C \\ &= -(43.7\ \text{mA})(7.5\ \text{k}\Omega) \\ &= -327.75 \end{aligned}$$

$$\frac{V_o}{V_3} = -327.75$$

Step 13 of 13

Determine the actual value of voltage gain:

$$\begin{aligned} \frac{V_O}{V_S} &= \left(\frac{V_1}{V_s}\right)\left(\frac{V_2}{V_1}\right)\left(\frac{V_3}{V_2}\right)\left(\frac{V_O}{V_3}\right) \\ &= (0.14)(-58.99)(-58.99)(-327.75) \\ &= -1.5 \times 10^5\ \text{V/V} \end{aligned}$$

Therefore, the voltage gain $\frac{V_O}{V_S}$ is $\boxed{-1.5 \times 10^5\ \text{V/V}}$.

11.071P

(a)

Identifying the feedback topology to be used Input resistance is to be lowered and output resistance is raised:

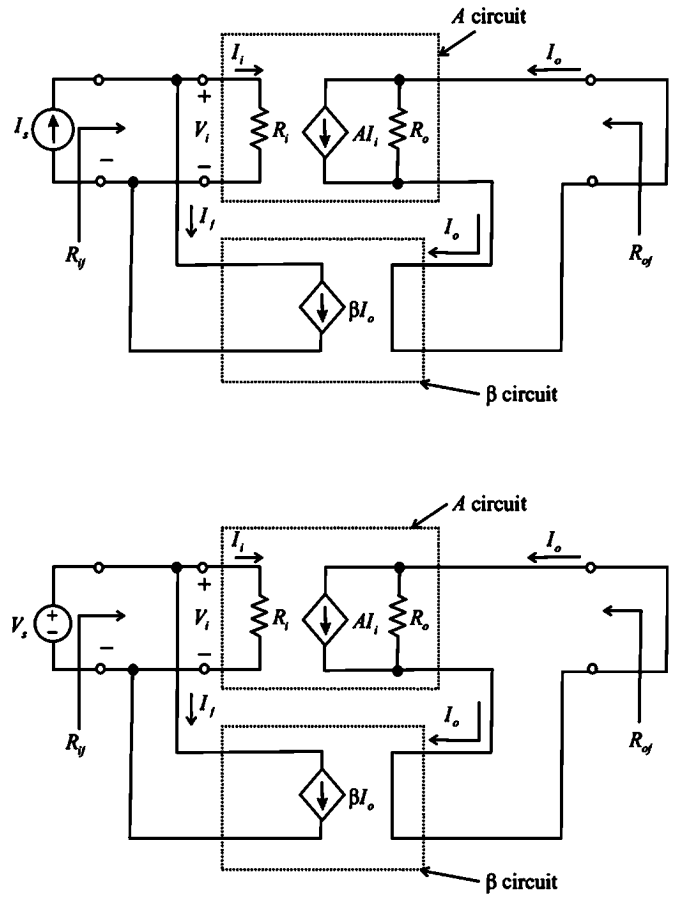


Figure 1

The input resistance with feedback is the resistance seen by the current source I_s and given by,

$$R_{if} = \frac{R_i}{1 + A\beta}$$

Where, we note that the shunt connection at the input reduces the input resistance.

The output resistance with feedback is the resistance seen by breaking the output circuit, this resistance is given by

$$R_{of} = R_o(1 + A\beta)$$

Where, we note that the increase in output resistance is due to the current (series) sampling.

Therefore, **Shunt - Series** feedback topology is to be used Input resistance is to be lowered and output resistance is raised.

(b)

Identifying the feedback topology to be used both input and output resistances are to be raised:

Figure 2

This Trans conductance with feedback is included in the equivalent circuit model of the feedback amplifier. In this model R_{if} is the input resistance with feedback.

$$R_{if} = R_i(1 + A\beta)$$

This relationship is identical to that obtained in the case of series shunt feedback. This confirms our earlier observation that the relationship between R_{if} and R_i is a function only of the method mixing. Voltage (or series) mixing therefore always increases the input resistance.

For the output resistance R_{of} of series-series feedback amplifier, we reduce

Both input and output resistances are to be raised V_s to zero and break the output circuit to apply attest current.

$$R_{of} = R_o(1 + A\beta)$$

That is, in this case the negative feedback increases the output resistance. This should have been expected, since the negative feedback tries to make I_o constant in spite of changes in the output voltage, which means increased output resistance.

Therefore, **Series - Series** feedback topology is to be used both input and output resistances are to be raised.

(c)
Identifying the feedback topology to be used both input and output resistances are to be lowered:

Figure 3

The input resistance with feedback is given by

$$R_{if} = \frac{R_i}{1 + A\beta}$$

Where we note that, the shunt connection is at the input results in a reduced input resistance. Also note that the resistance R_{if} is the resistance seen by the source I_s and it includes any source resistance.

The output resistance with feedback is given by,

$$R_{of} = \frac{R_o}{1 + A\beta}$$

Where we note that, the shunt connection is at the input results in a reduced output resistance. This resistance includes any load resistance.

Therefore, **Shunt – Shunt** feedback topology is to be used both input and output resistances are to be lowered.

Step 1 of 8

Refer to Figure P11.72 in the textbook.

Draw the equivalent circuit model as shown in Figure 1.

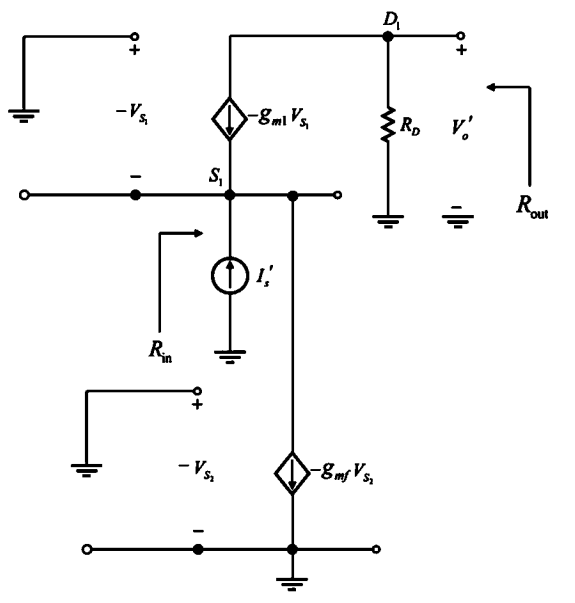


Figure 1

Step 2 of 8

Apply Nodal analysis at node-S_1.

$$g_{m1}V_{S_1} - I_s' + g_{mf}V_{S_2} = 0$$
$$I_s' = g_{m1}V_{S_1} + g_{mf}V_{S_2}$$

Refer to figure 1. Substitute 0 for V_{S_2}.

$$I_s' = g_{m1}V_{S_1} + g_{mf}(0)$$

$$I_s' = g_{m1}V_{S_1} \quad \text{..... (1)}$$

Apply Nodal analysis at node-S_2.

$$\frac{V_o'}{R_D} - g_{m1}V_{S_1} = 0$$

$$V_o' = R_D g_{m1} V_{S_1} \quad \text{..... (2)}$$

The expression for the overall trans-resistance is,

$$A_f = \frac{V_o'}{I_s'}$$

Consider equations (1) and (2). Substitute the expressions for V_o' and I_s'.

$$A_f = \frac{R_D g_{m1} V_{S_1}}{g_{m1} V_{S_1}}$$

$$A_f = R_D \quad \text{..... (3)}$$

Therefore, the expression for $A_f = \frac{V_o}{I_s}$ is, $\boxed{A_f = R_D}$.

Step 3 of 8

The expression for the input resistance R_{in} is,

$\boxed{R_{in} = R_D}$.

The expression for the output resistance R_{out} is,

$$R_{out} = \frac{R_o}{1 + A\beta}$$

Determine the resistance R_o.

$$R_o = R_D$$

Therefore, the expression for the output resistance R_{out} is,

$$\boxed{R_{out} = \frac{R_D}{1 + A\beta}}.$$

Consider the expression for the feedback factor β.

$$\beta = \frac{I_f}{V_o} \quad \text{..... (4)}$$

Step 4 of 8

Determine the value of β.
Draw the feedback β circuit as shown in Figure 2.

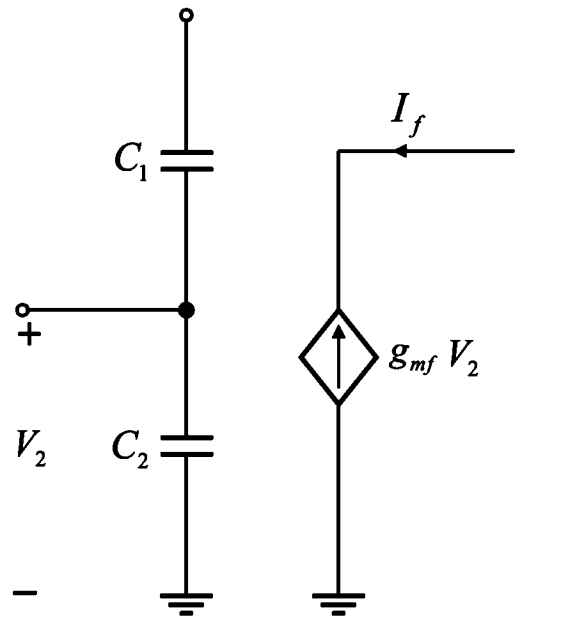


Figure 2

Step 5 of 8

Determine voltage V_2.

Apply voltage division principle.

$$V_2 = \frac{C_1}{C_1 + C_2} V_o \quad \text{..... (4)}$$

Consider figure 2.

$$I_f = -g_{mf} V_2$$

Consider equation (4). Substitute the expression for V_2 in the equation.

$$I_f = -g_{mf} \frac{C_1}{C_1 + C_2} V_o$$
$$\frac{I_f}{V_o} = -g_{mf} \frac{C_1}{C_1 + C_2}$$

Substitute 0.9×10^{-12} for C_1, 0.1×10^{-12} for C_2, and 2×10^{-3} for g_{mf}.

$$\frac{I_f}{V_o} = -(2\times10^{-3})\left(\frac{0.9\times10^{-12}}{0.9\times10^{-12} + 0.1\times10^{-12}}\right)$$
$$= -(2\times10^{-3})\left[\frac{0.9\times10^{-12}}{(0.9+0.1)\times10^{-12}}\right]$$
$$= -(2\times10^{-3})(0.9)$$
$$= -1.8 \text{ mA/V}$$

Step 6 of 8

Consider equation (4).

$$\beta = \frac{I_f}{V_o}$$

Therefore, the value of β is -1.8 mA/V.

Recall equation (3).

$$A_f = R_D$$

Substitute $10\ \text{k}\Omega$ for R_D.

$$A_f = 10\ \text{k}\Omega$$

Therefore, the value of A_f is $\boxed{10\ \text{k}\Omega}$.

Step 7 of 8

Consider the expression for the input resistance R_{in}.

$$R_{in} = R_D$$

Substitute $10\ \text{k}\Omega$ for R_D.

$$R_{in} = 10\ \text{k}\Omega$$

Therefore, the value of R_{in} is $\boxed{10\ \text{k}\Omega}$.

Step 8 of 8

Consider the expression for the output resistance R_{out}.

$$R_{out} = \frac{R_D}{1 + A\beta}$$
$$= \frac{R_D}{1 + R_D\beta}$$

Substitute 10×10^3 for R_D, and -1.8×10^{-3} for β.

$$R_{out} = \left|\frac{10\times10^3}{1 + (10\times10^3)(-1.8\times10^{-3})}\right|$$
$$= \left|\frac{10\times10^3}{1 - 18}\right|$$
$$= 588.23\ \Omega$$

Therefore, the value of R_{out} is $\boxed{588.23\ \Omega}$.

Task 1 of 22

Refer to Figure 1.

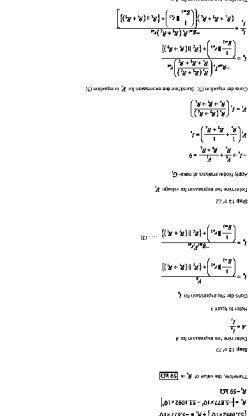
(a)

Trace the conducted circuit in an isolated figure.

(b)

Label the nodes.

Figure 1



Task 2 of 22

Clockwise the order of the error between R_{eq} and source terminals R_{eq}.

Loops the following therefore

$$R_{eq} = 6\,\Omega + 4\,\Omega$$

$$R_{eq} = 10\,\Omega$$

dissipates 0.2 W for R_{eq}. $0.2\,\text{W}$ 0.6 W P_{eq}.

$$P_{eq} = 0.8\,\text{V} \cdot 0.8\,\text{A}$$

$$P_{eq} = 0.64\,\text{W}$$

Transpose.

Task 3 of 22

Clockwise the current I_1 through R_1.

$$I_1 = \frac{\mathcal{E}}{R_1 + R_{eq}}$$

Transform. The components for R in.

$$R = \frac{1}{R_1 + R_{eq}}$$

Task 4 of 22

Apply Ohm's law to the circuit shown in figure 2.

Task 5 of 22

Apply symmetry when voltage one is not small known a figure 3.

$$I_T - I_1 - 2I_2 - 2I_3 = 0$$

$$2I_2 R_2 - I_3 R_3 = 0$$

$$I_T + I_1 + 2I_2 + 2I_3 = 0$$

Transform. Set expression into the resistance R_{eq}.

$$R_{eq} = \frac{\mathcal{E}_1}{I_T}$$

$$R_{eq} = \frac{\mathcal{E}_1 (R_1 + R_2) + \mathcal{E}_1 (R_1 + R_2)}{(R_1 + R_2)^2 + (R_1 + R_2)^2}$$

$$R_{eq} = \frac{2\mathcal{E}_1 (R_1 + R_2)}{2(R_1 + R_2)^2}$$

$$R_{eq} = \frac{\mathcal{E}_1}{R_1 + R_2}

Step 1 of 13

(a)

Refer to the circuit diagram of feedback current amplifier in Figure 10.8(b) in the textbook.

Draw the *A* –circuit for the feedback current amplifier.

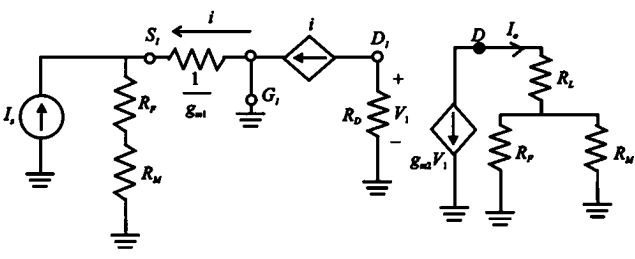


Figure 1

Step 2 of 13

Derive the expression for the input resistance.

$$R_i = \left[(R_F + R_M) \| \frac{1}{g_{m1}} \right]$$
$$= \frac{(R_F + R_M)\left(\frac{1}{g_{m1}}\right)}{R_F + R_M + \frac{1}{g_{m1}}}$$
$$= \frac{(R_F + R_M)}{1 + g_{m1}(R_F + R_M)}$$

Thus, the input resistance of the feedback current amplifier R_i is $\boxed{\frac{(R_F + R_M)}{1 + g_{m1}(R_F + R_M)}}$.

Step 3 of 13

Derive the expression for the gain of the amplifier.

$$\frac{i}{I_s} = -\frac{R_F + R_M}{R_F + R_M + \frac{1}{g_{m1}}}$$

From Figure 1, write the expression for voltage V_1 using ohm's law.

$$V_1 = -iR_D$$

From Figure 1, the current I_o using ohm's law.

$$I_o = -g_{m2}V_1$$

Calculate the expression for the gain of the amplifier.

$$A = \left(\frac{i}{I_s}\right)\left(\frac{V_1}{i}\right)\left(\frac{I_o}{V_1}\right)$$
$$= -\left(\frac{R_F + R_M}{R_F + R_M + \frac{1}{g_{m1}}}\right)(-R_D)(-g_{m2})$$
$$= -\frac{(R_F + R_M)R_D g_{m2}}{R_F + R_M + \frac{1}{g_{m1}}}$$

Since $R_F + R_M >> \frac{1}{g_{m1}}$

Write the reduced expression for the gain of the amplifier.

$$A = -\frac{(R_F + R_M)R_D g_{m2}}{R_F + R_M}$$
$$= -R_D g_{m2}$$

Thus, the gain of the feedback current amplifier A is $\boxed{-R_D g_{m2}}$.

Step 4 of 13

(b)

Draw β circuit for the current amplifier circuit.

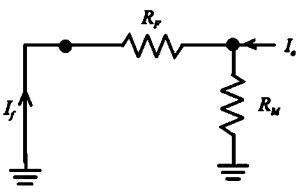


Figure 2

Step 5 of 13

Derive the expression for the feedback factor.

$$\beta = \frac{I_f}{I_o}$$
$$= -\frac{R_M}{R_F + R_M}$$

Thus, the feedback factor β of the current amplifier is $\boxed{-\frac{R_M}{R_F + R_M}}$.

Step 6 of 13

(c)

Derive the expression for the loop gain of the amplifier.

$$A\beta = (-R_D g_{m2})\left(-\frac{R_M}{R_F + R_M}\right)$$
$$= \frac{R_M R_D g_{m2}}{R_F + R_M}$$

Thus, the loop gain $A\beta$ of the amplifier is $\boxed{\frac{R_M R_D g_{m2}}{R_F + R_M}}$.

Step 7 of 13

(d)

The trans-conductance, $g_{m1} = g_{m2} = 5\ \text{mA/V}$

The resistances,

$R_D = 20\ \text{k}\Omega$
$R_M = 10\ \text{k}\Omega$
$R_F = 90\ \text{k}\Omega$

Calculate the gain of the amplifier.

$$A = -R_D g_{m2}$$
$$= -(20\ \text{k})(5\ \text{m})$$
$$= -100\ \text{A/A}$$

Thus, the gain A of the amplifier is $\boxed{-100\ \text{A/A}}$.

Step 8 of 13

Calculate the feedback factor of the amplifier.

$$\beta = -\frac{R_M}{R_F + R_M}$$
$$= -\frac{10\ \text{k}}{90\ \text{k} + 10\ \text{k}}$$
$$= -\frac{10\ \text{k}}{100\ \text{k}}$$
$$= -0.1$$

Thus, the feedback factor β of the amplifier is $\boxed{-0.1}$.

Step 9 of 13

Calculate the loop gain of the amplifier.

$$A\beta = (-100)(-0.1)$$
$$= 10$$

Thus, the loop gain $A\beta$ of the amplifier is $\boxed{10}$.

Step 10 of 13

Calculate the gain with feedback.

$$A_f = \frac{A}{1 + A\beta}$$
$$= \frac{-100}{1 + 10}$$
$$= -9.09\ \text{A/A}$$

Thus, the gain with feedback of the amplifier A_f is $\boxed{-9.09\ \text{A/A}}$.

Step 11 of 13

Calculate the input resistance of the amplifier.

$$R_i = \frac{(R_F + R_M)}{1 + g_{m1}(R_F + R_M)}$$
$$= \frac{90\ \text{k} + 10\ \text{k}}{1 + 5 \times 10^{-3}(90\ \text{k} + 10\ \text{k})}$$
$$= \frac{100\ \text{k}}{1 + 5 \times 10^{-3}(100\ \text{k})}$$
$$= 200\ \Omega$$

Thus, the input resistance R_i of the amplifier is $\boxed{200\ \Omega}$.

Calculate the input resistance with feedback of the amplifier.

$$R_{if} = \frac{R_i}{1 + A\beta}$$
$$= \frac{200}{1 + 10}$$
$$= 18.18\ \Omega$$

Thus, the input resistance with feedback of the amplifier R_{if} is $\boxed{18.18\ \Omega}$.

Step 12 of 13

(e)

The output resistance, $r_{o2} = 20\ \text{k}\Omega$

The load resistance, $R_L = 1\ \text{k}\Omega$

Determine the output resistance of the amplifier.

$$R_o = r_{o2} + R_L + (R_M \| R_F)$$

Substitute corresponding values.

$$R_o = 20\ \text{k} + 1\ \text{k} + (10\ \text{k} \| 90\ \text{k})$$
$$= 20\ \text{k} + 1\ \text{k} + 9\ \text{k}$$
$$= 30\ \text{k}\Omega$$

Step 13 of 13

Determine the output resistance with feedback.

$$R_{of} = R_o(1 + A\beta)$$
$$= (30\ \text{k})(1 + 10)$$
$$= 330\ \text{k}\Omega$$

Determine the output resistance as seen by the load resistance.

$$R_{out} = R_{of} - R_L$$
$$= 330\ \text{k} - 1\ \text{k}$$
$$= 329\ \text{k}\Omega$$

Thus, the output resistance R_{out} as seen by the load resistance is $\boxed{329\ \text{k}\Omega}$.

Step 1 of 12

Refer to the circuit diagram in Figure 10.27(a) in the textbook.

Consider the feedback amount of the amplifier is $40\ \text{dB}$.

Thus, the amount of feedback in dB is,

$$20\log(1+A\beta)=40\ \text{dB}$$

$$(1+A\beta)=10^{\frac{40}{20}}$$

$$\begin{aligned}A\beta&=10^2-1\\&=100-1\\&=99\end{aligned}$$

Thus, the loop gain $A\beta$ is 99 .

Step 2 of 12

Closed loop gain A_f of the amplifier is,

$$A_f=\frac{A}{1+A\beta}$$

Substitute 100 for A_f and 99 for $A\beta$ in the equation

$$100=\frac{A}{1+99}$$

$$\begin{aligned}A&=-100\times 100\\&=-10000\ \text{A/A}\end{aligned}$$

Therefore, the Open loop gain A of the current amplifier is $\boxed{-10000\ \text{A/A}}$.

Step 3 of 12

Determine the feedback factor β.

Loop gain of the amplifier is,

$$A\beta=99$$

Substitute -10000 for A in the equation.

$$(-10000)\beta=99$$

$$\begin{aligned}\beta&=-\frac{99}{10000}\\&=-0.0099\end{aligned}$$

Step 4 of 12

The resistance R_i is,

$$R_i=R_s\|R_{id}\|(R_1+R_2)$$

At $R_{id}\to\infty, R_s\to\infty$ the equation becomes as follows:

$$R_i=(R_1+R_2)$$

Input resistance R_{in} of the closed loop amplifier is,

$$R_{in}=R_{id}\|(R_1+R_2)\|\frac{R_2}{\mu}$$

Since, the μ is very large. Thus, the input resistance R_{in} is,

$$R_{in}\approx\frac{R_2}{\mu}$$

$$\frac{R_2}{\mu}=1\ \text{k}\Omega\ \ldots\ldots\ (1)$$

Step 5 of 12

Assume that $\frac{1}{g_m}<<(R_1\|R_2\|r_{02})$, then

$$\begin{aligned}A\beta&=\frac{\mu R_i}{R_2}\\&=\frac{R_i}{\left(\frac{R_2}{\mu}\right)}\end{aligned}$$

Substitute R_1+R_2 for R_i in the equation.

$$A\beta=\left(\frac{R_1+R_2}{\left(\frac{R_2}{\mu}\right)}\right)$$

Substitute 99 for $A\beta$ and 1×10^3 for $\frac{R_2}{\mu}$ in the equation.

$$\begin{aligned}99&=\frac{R_1+R_2}{1\times10^3}\\R_1+R_2&=(99)(1\times10^3)\end{aligned}$$

$$R_1+R_2=99000\ \ldots\ldots\ (2)$$

Step 6 of 12

Consider the feedback factor.

$$\beta=-\frac{R_1}{R_1+R_2}$$

Substitute -0.0099 for β and 99000 for R_1+R_2 in the equation.

$$-0.0099=-\frac{R_1}{99000}$$

$$R_1=980.1\ \Omega$$

Thus, the resistance of resistor R_1 is $\boxed{980.1\ \Omega}$.

Step 7 of 12

Substitute 980.1 for R_1 in the equation 2.

$$R_2+980.1=99000$$

$$\begin{aligned}R_2&=99000-980.1\\&=98019.9\ \Omega\\&\approx98.02\ \text{k}\Omega\end{aligned}$$

Thus, the resistance of resistor R_2 is $\boxed{98.02\ \text{k}\Omega}$.

Step 8 of 12

Determine the value of gain μ .

Substitute $98.02\ \text{k}\Omega$ for R_2 in the equation 1.

$$\frac{98.02\ \text{k}\Omega}{\mu}=1\ \text{k}\Omega$$

$$\begin{aligned}\mu&=\frac{98.02\ \text{k}}{1\ \text{k}}\\&=98.02\ \text{V/V}\end{aligned}$$

Thus, the value of gain μ is $\boxed{98.02\ \text{V/V}}$.

Step 9 of 12

Input resistance of open loop amplifier is,

$$R_i=R_s\|R_{id}\|(R_1+R_2)$$

For $R_S=\infty$ and $R_{id}=\infty$, the equation becomes.

$$R_i=(R_1+R_2).$$

Substitute $980.1\ \Omega$ for R_1 and $98.02\ \text{k}\Omega$ for R_2 in the equation.

$$\begin{aligned}R_i&=(980.1\ \Omega+98.02\ \text{k}\Omega)\\&\approx99\ \text{k}\Omega\end{aligned}$$

Step 10 of 12

Consider trans-conductance g_m of the transistor is $5\ \text{mA/V}$, output resistance r_o is $20\ \text{k}\Omega$.

Since $\frac{1}{g_m}<<(R_1\|R_2\|r_{02})$,

$$A\approx-\mu\frac{R_i}{\frac{1}{g_m}+(R_1\|R_2\|r_{o2})}\left(\frac{r_{o2}}{r_{o2}+(R_1\|R_2)}\right)$$

Substitute 10000 for A , 99000 for R_i , $980.1\ \Omega$ for R_1 , $98.02\ \text{k}\Omega$ for R_2 , $20\ \text{k}\Omega$ for r_{o2} and $5\times10^{-3}\ \text{A/V}$ for g_m in the equation.

$$-10000=-\mu\left(\frac{99000}{\frac{1}{5\times10^{-3}}+(980.1\|98.02\ \text{k}\|20\ \text{k})}\right)\left(\frac{20\ \text{k}}{20\ \text{k}+(980.1\|98.02\ \text{k})}\right)$$

$$\begin{aligned}10000&=\mu\left(\frac{99000}{200+925.49}\right)\left(\frac{20\ \text{k}}{20\ \text{k}+970.39}\right)\\&=\mu(87.96)(0.95)\end{aligned}$$

$$\mu=119.2$$

Step 11 of 12

The output resistance of the open loop amplifier is,

$$R_o=r_{o2}+g_mr_{o2}(R_1\|R_2)+(R_1\|R_2)$$

Substitute $5\times10^{-3}\ \text{A/V}$ for g_m , $20000\ \Omega$ for r_o and $970.29\ \Omega$ for $R_1\|R_2$ in equation.

$$\begin{aligned}R_o&=20000+5\times10^{-3}(20000)(970.29)+970.29\\&\approx119\ \text{k}\Omega\end{aligned}$$

The output resistance of the amplifier R_o is $\boxed{119\ \text{k}\Omega}$.

Step 12 of 12

The output resistance seen through the load R_L is,

$$R_{out}=R_{of}-R_L$$

Substitute $1000\ \Omega$ for R_L and $11.9\times10^6\ \Omega$ for R_{of} in equation.

$$\begin{aligned}R_{out}&=11.9\times10^6-1000\\&=11.89\ \text{M}\Omega\end{aligned}$$

Thus, output resistance seen through the load R_L is $\boxed{11.89\ \text{M}\Omega}$.

Step 1 of 10

11.076P

Refer to the circuit shown in Figure 11.27 (a) in the textbook.

Draw the circuit as shown in Figure 1.

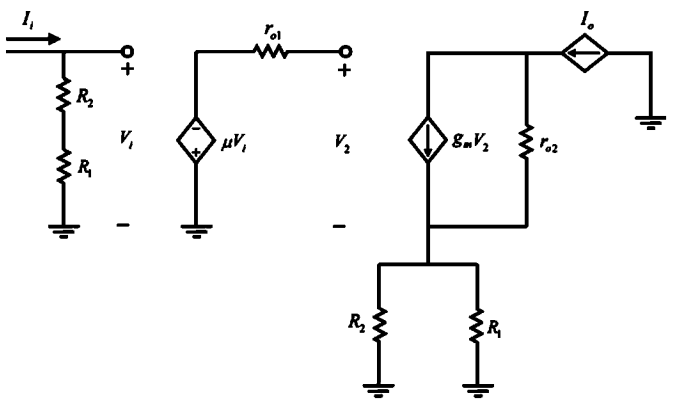


Figure 1

Step 2 of 10

Determine the expression for current I_o.

Apply Kirchhoff's voltage law at the output end of the amplifier.

$$g_m V_2 r_{o2} - (R_2 \| R_1) I_o - I_o r_{o2} = 0$$
$$I_o\left[(R_2 \| R_1) + r_{o2}\right] = g_m V_2 r_{o2}$$

$$I_o = \frac{g_m V_2 r_{o2}}{(R_2 \| R_1) + r_{o2}} \quad \text{..... (1)}$$

Step 3 of 10

The expression for the voltage V_2 is,

$$V_2 = -\mu V_i$$

Recall equation (1).

$$I_o = \frac{g_m V_2 r_{o2}}{(R_2 \| R_1) + r_{o2}}$$

Substitute $(-\mu V_i)$ for V_2.

$$I_o = \frac{g_m(-\mu V_i) r_{o2}}{(R_2 \| R_1) + r_{o2}}$$

$$I_o = \frac{-\mu g_m r_{o2} V_i}{(R_2 \| R_1) + r_{o2}} \quad \text{..... (2)}$$

Step 4 of 10

Apply Kirchhoff's voltage law at the input end of the amplifier.

The expression for the voltage V_i is,

$$V_i = (R_2 + R_1) I_i$$

Recall equation (2).

$$I_o = \frac{-\mu g_m r_{o2} V_i}{(R_2 \| R_1) + r_{o2}}$$

Substitute $(R_2 + R_1) I_i$ for V_i.

$$I_o = \frac{-\mu g_m r_{o2} (R_2 + R_1) I_i}{(R_2 \| R_1) + r_{o2}}$$
$$\frac{I_o}{I_i} = \frac{-\mu g_m r_{o2} (R_2 + R_1)}{(R_2 \| R_1) + r_{o2}}$$

$$A = \frac{-\mu g_m r_{o2} (R_2 + R_1)}{(R_2 \| R_1) + r_{o2}} \quad \text{..... (3)}$$

Step 5 of 10

Draw the feedback circuit for β as shown in figure 2.

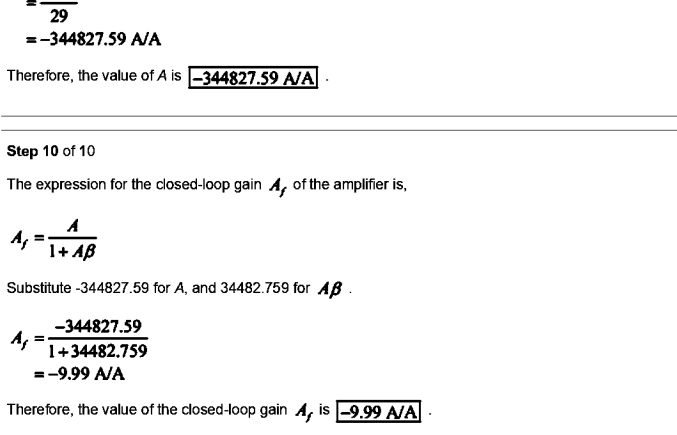


Figure 2

Step 6 of 10

Apply Kirchhoff's voltage law to the circuit shown in figure 2.

$$I_f R_1 + I_f R_2 + I_o R_1 = 0$$
$$I_f (R_1 + R_2) = -I_o R_1$$
$$\frac{I_f}{I_o} = \frac{-R_1}{R_1 + R_2}$$

$$\beta = \frac{-R_1}{R_1 + R_2} \quad \text{..... (4)}$$

Step 7 of 10

Consider equations (3) and (4).

$$A = \frac{-\mu g_m r_{o2} (R_2 + R_1)}{(R_2 \| R_1) + r_{o2}}$$

$$\beta = \frac{-R_1}{R_1 + R_2}$$

The expression for the loop gain of the amplifier is,

$$A\beta = \left[\frac{-\mu g_m r_{o2} (R_2 + R_1)}{(R_2 \| R_1) + r_{o2}}\right]\left(\frac{-R_1}{R_1 + R_2}\right)$$
$$= \frac{\mu g_m r_{o2} R_1}{\left[(R_2 \| R_1) + r_{o2}\right]}$$

Therefore, the expression for the loop gain of the amplifier $A\beta$ is,

$$\boxed{A\beta = \frac{\mu g_m r_{o2} R_1}{\left[(R_2 \| R_1) + r_{o2}\right]}}.$$

Step 8 of 10

Consider the following expression of $A\beta$:

$$A\beta = \frac{\mu g_m r_{o2} R_1}{\left[(R_2 \| R_1) + r_{o2}\right]}$$

Substitute 1000 for μ, 5×10^{-3} for g_m, 20×10^3 for r_{o2}, 10×10^3 for R_1, and 90×10^3 for R_2.

$$A\beta = \frac{1000(5\times10^{-3})(20\times10^3)(10\times10^3)}{\left[(90\times10^3 \| 10\times10^3) + (20\times10^3)\right]}$$
$$= \frac{5(20\times10^3)(10\times10^3)}{\left[(9\times10^3) + (20\times10^3)\right]}$$
$$= \frac{10^6\times10^3}{29\times10^3}$$
$$= 34482.76$$

Therefore, the value of $A\beta$ is $\boxed{34482.759}$.

Step 9 of 10

Recall equation (3).

$$A = \frac{-\mu g_m r_{o2} (R_2 + R_1)}{(R_2 \| R_1) + r_{o2}}$$

Substitute 1000 for μ, 5×10^{-3} for g_m, 20×10^3 for r_{o2}, 10×10^3 for R_1, and 90×10^3 for R_2.

$$A = \frac{-1000(5\times10^{-3})(20\times10^3)(90\times10^3 + 10\times10^3)}{(90\times10^3 \| 10\times10^3) + 20\times10^3}$$
$$= \frac{-5(20\times10^3)(100\times10^3)}{(9\times10^3) + (20\times10^3)}$$
$$= \frac{-10^7}{29}$$
$$= -344827.59 \text{ A/A}$$

Therefore, the value of *A* is $\boxed{-344827.59 \text{ A/A}}$.

Step 10 of 10

The expression for the closed-loop gain A_f of the amplifier is,

$$A_f = \frac{A}{1 + A\beta}$$

Substitute -344827.59 for *A*, and 34482.759 for $A\beta$.

$$A_f = \frac{-344827.59}{1 + 34482.759}$$
$$= -9.99 \text{ A/A}$$

Therefore, the value of the closed-loop gain A_f is $\boxed{-9.99 \text{ A/A}}$.

The results differ somewhat from those found in Example 11.10 because the resistances R_s, R_{id} are kept open and the expressions of *A* and A_f differ.

Step 1 of 17

(a)

Refer to the circuit diagram of feedback current amplifier in Figure P10.65 in the textbook.

Write the formula for V_{GS} .

$$V_{GS} = V_{OV} + V_t$$

Substitute 0.2 for V_{OV} and 0.5 for V_t in the equation.

$$V_{GS} = 0.2 + 0.5$$
$$= 0.7 \text{ V}$$

Step 2 of 17

From the Figure, Find the voltage V_{S2} using ohm's law.

$$I_{D2} = \frac{V_{S2}}{R_1}$$

Substitute 0.7 for V_{S2} and $3.5\ \text{k}\Omega$ for R_1 in the equation.

$$I_{D2} = \frac{0.7 \text{ V}}{3.5\ \text{k}\Omega}$$
$$= 0.2 \text{ mA}$$

The drain current I_{D2} is $\boxed{0.2 \text{ mA}}$.

Step 3 of 17

(b)

We know that , $V_{GS1} = V_{G1} = 0.7 \text{ V}$.

Also, $I_{GS} = 0$, so $V_{S2} = 0.7 \text{ V}$

Write the formula for trans-conductance g_m.

$$g_m = \frac{2I_D}{V_{OV}}$$

Substitute 0.4×10^{-3} for I_D and 0.2 for V_{OV} in the equation.

$$g_m = \frac{0.4\times10^{-3}}{0.2}$$
$$= 2 \text{ mA/V}$$

Step 4 of 17

Output resistance r_O is,

$$r_O = \frac{V_A}{I_D}$$

Substitute 10 for V_A and 0.2×10^{-3} for I_D in the equation

$$r_O = \frac{10}{0.2\times10^{-3}}$$
$$= 50\ \text{k}\Omega$$

Thus output resistance r_O is $\boxed{50\ \text{k}\Omega}$.

Step 5 of 17

(c)

Draw the A circuit for Figure P10.65.

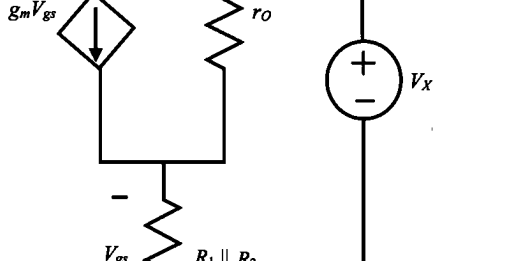


Figure 1

Use ohm's law to write the expression for voltage V_1 .

$$V_1 = I_S(R_1 + R_2)$$

Use ohm's law to write the expression for voltage V_2

$$V_2 = -g_{m1}r_{o1}V_1$$
$$= -g_{m1}r_{o1}(R_1 + R_2)I_S$$

Write the expression for current i_O flows through the resistor $\frac{1}{g_{m2}}$.

$$i_o = \frac{V_2}{1/g_m + R_2 \| R_1 \| r_{o2}}$$
$$= \frac{-g_{m1}\cdot r_{o1}(R_1 + R_2)I_S}{1/g_m + R_2 \| R_1 \| r_{o2}}$$

Step 6 of 17

Step 7 of 17

Open loop gain of the amplifier is,

$$A = \frac{i_O}{i_S}$$

Substitute $-g_{m1}\cdot r_{o1}(R_1 + R_2)$ for i_O and $1/g_m + (R_2 \| R_1 \| r_{o2})$ for I_S in the equation.

$$A = \frac{-g_{m1}\cdot r_{o1}(R_1 + R_2)}{1/g_m + (R_2 \| R_1 \| r_{o2})}$$

Substitute corresponding values.

$$A = \frac{-2 \times 50(3.5 + 14)\ \text{k}}{500 + (3.5 \| 14 \| 50)\ \text{k}}$$
$$= -555.3 \text{ A/A}$$

Input resistance of the circuit is,

$$R_i = R_2 + R_1$$
$$= 3.5 + 14$$
$$= 17.5\ \text{k}\Omega$$

Step 8 of 17

Now, find R_O.

Derive the expression for output resistance R_O.

$$V_{gs} = -I_x(R_1 \| R_2) \quad \ldots\ldots (1)$$

$$V_x + g_m r_o V_{gs} = I_x(r_o + R_1 \| R_2) \quad \ldots\ldots (2)$$

Substitute equation (1) in (2).

$$V_x + g_m r_o(-I_x(R_1 \| R_2)) = I_x(r_o + R_1 \| R_2)$$
$$\frac{V_x}{I_x} = g_m r_o(R_1 \| R_2) + (r_o + R_1 \| R_2)$$

Define output resistance R_O

$$R_O = \frac{V_x}{I_x}$$
$$= r_o + R_1 \| R_2 + gm\ r_o(R_1 \| R_2)$$

Substitute corresponding values.

$$R_O = 50\ \text{k} + (3.5 \| 14)\ \text{k} + 2 \times 50(3.5 \| 14)\ \text{k}$$
$$= 332.8\ \text{k}\Omega$$

Step 9 of 17

(d)

Draw the equivalent circuit diagram of the output side.

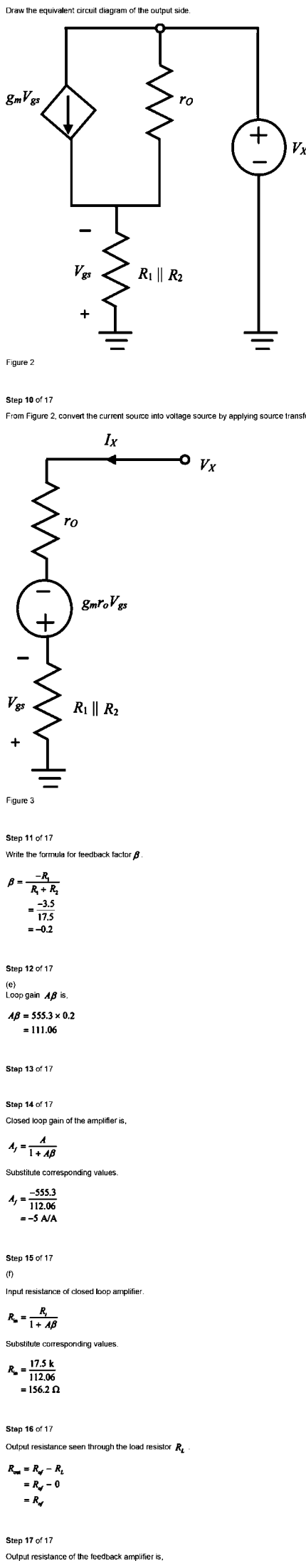


Figure 2

Step 10 of 17

From Figure 2, convert the current source into voltage source by applying source transformation technique.

Figure 3

Step 11 of 17

Write the formula for feedback factor β .

$$\beta = \frac{-R_1}{R_1 + R_2}$$
$$= \frac{-3.5}{17.5}$$
$$= -0.2$$

Step 12 of 17

(e)

Loop gain $A\beta$ is,

$$A\beta = 555.3 \times 0.2$$
$$= 111.06$$

Step 13 of 17

Step 14 of 17

Closed loop gain of the amplifier is,

$$A_f = \frac{A}{1 + A\beta}$$

Substitute corresponding values.

$$A_f = \frac{-555.3}{112.06}$$
$$= -5 \text{ A/A}$$

Step 15 of 17

(f)

Input resistance of closed loop amplifier.

$$R_{in} = \frac{R_i}{1 + A\beta}$$

Substitute corresponding values.

$$R_{in} = \frac{17.5\ \text{k}}{112.06}$$
$$= 156.2\ \Omega$$

Step 16 of 17

Output resistance seen through the load resistor R_L .

$$R_{out} = R_{of} - R_L$$
$$= R_{of} - 0$$
$$= R_{of}$$

Step 17 of 17

Output resistance of the feedback amplifier is,

$$R_{out} = R_o(1 + A\beta)$$

Substitute corresponding values.

$$R_{out} = 332.8\ \text{k} \times 112.06$$
$$= 37.3\ \text{M}\Omega$$

Thus, output resistance of the feedback amplifier is $\boxed{37.3\ \text{M}\Omega}$.

Step 1 of 12

(a)

Refer to Figure P10.66 (a) for the feedback current amplifier circuit. In this, the super common gate transistor is connected between the gate and source terminals of the MOSFET.

Replace the op-amp with its equivalent model and replace the MOSFET with its small signal equivalent model.

For ideal op-amp, the input resistance is very high (infinite) which is in the order of mega ohms (the resistor R_{id} is open). And the drain current i is approximately equal to the source current. That is

$$i_D \cong i_S$$

The closed loop gain the op-amp is,

$$A_f = \frac{\mu}{1+\mu\beta}$$

The closed loop gain is increase when the open loop gain μ is large. Therefore, from Figure 1, clearly observe that the input voltage is,

$$V_{in} = -I_S R_S$$

Therefore, the signal voltage at the input terminal V_{in} is $\boxed{-I_S R_S}$.

Step 2 of 12

Since the drain current i is approximately equal to the source current. And the current flowing through the resistor R_S is I_S since there is no current flowing through the negative terminal of the op-amp. Therefore, it is concluded that the output current I_D is flowing through the source resistance, R_S. That is,

$$I_O = I_S$$

$$\frac{I_O}{I_S} \cong 1 \text{ A/A}$$

Therefore, the current gain $\frac{I_O}{I_S}$ is $\boxed{1 \text{ A/A}}$.

Clearly observe that the input resistance is R_{id}.

$$R_i = R_{id}$$

Therefore, the input resistance R_i is $\boxed{R_{id}}$.

Step 3 of 12

(b)

For finite value of μ and the input resistance of the ideal op-amp is very high (infinite).

The open loop amplifier circuit (*A* circuit) is shown in Figure 1.

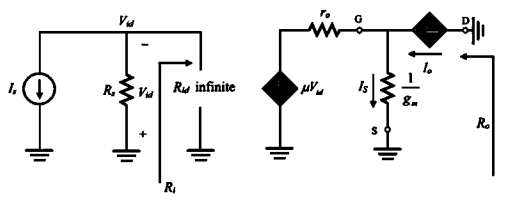


Figure 1

Step 4 of 12

For ideal op-amp the output resistance is very small. So, neglect the resistance, r_o.

From Figure 1, the voltage is,

$$V_{id} = I_S R_S$$

The source current I_S through the resistor is,

$$I_o = \frac{\mu V_{id}}{\frac{1}{g_m}} \qquad \text{since } I_o \cong I_S$$

$$= \mu g_m V_{id}$$

$$= \mu g_m I_S R_S$$

The expression for current gain A is,

$$I_o = \mu g_m I_S R_S$$

$$\frac{I_o}{I_S} = \mu g_m R_S$$

$$A = \mu g_m R_S$$

Here,

Source resistance is R_S

Trans-conductance g_m

Therefore, the gain A is $\boxed{\mu g_m R_S}$.

Step 5 of 12

From Figure 1, clearly the total input resistance is,

$$R_i = R_S \qquad \text{since } R_{id} = \infty$$

Therefore, the input resistance R_i is $\boxed{R_S}$

The output resistance is,

$$R_o = r_o$$

Therefore, the out resistance R_o is $\boxed{r_o}$.

Step 6 of 12

(c)

The closed loop gain of the op-amp is,

$$A_i = \frac{A}{1+A\beta}$$

$$A_i = \frac{A}{A\beta} \qquad \text{since } A\beta >> 1$$

Substitute 1 for A_i.

$$1 = \frac{1}{\beta}$$

$$\beta = 1$$

Therefore, the value of β is $\boxed{1}$.

Step 7 of 12

(d)

Substitute $\mu g_m R_S$ for A and 1 for β to derive the expression for $A\beta$.

$$A\beta = (\mu g_m R_S)(1)$$

$$= \mu g_m R_S$$

Therefore, the expression for $A\beta$ is $\boxed{A\beta = \mu g_m R_S}$.

The closed loop gain the op-amp is,

$$A_f = \frac{A}{1+A\beta}$$

Substitute $\mu g_m R_S$ for A and 1 for β to obtain A_f.

$$A_f = \frac{\mu g_m R_S}{1+\mu g_m R_S (1)}$$

$$= \frac{\mu g_m R_S}{1+\mu g_m R_S}$$

If μ is large, then $\mu g_m R_S >> 1$. That is,

$$A_f = \frac{\mu g_m R_S}{\mu g_m R_S}$$

$$\cong 1$$

Therefore, the value of the feedback gain A_f is $\boxed{1}$.

Step 8 of 12

(e)

Refer to Figure 10.28 in the text book for the shunt series feedback amplifier and its equivalent circuit.

The input resistance R_{if} found from the feedback amplifier is,

$$R_{if} = \frac{R_i}{1+A\beta}$$

Here,

The input resistance, R_i

A and β, which are current gains, are current gains.

Substitute R_S for R_i and $\mu g_m R_S$ for $A\beta$.

$$R_{if} = \frac{R_S}{1+\mu g_m R_S}$$

Divide right side expression with R_S.

$$R_{if} = \frac{1}{\frac{1}{R_S} + \frac{\mu g_m R_S}{R_S}}$$

$$= \frac{1}{\frac{1}{R_S} + \mu g_m}$$

Step 9 of 12

This expression can also be written as,

$$R_{if} = R_S \| \frac{1}{\mu g_m}$$

$$R_{if} = R_S \| R_m$$

Here,

$$R_m = \frac{1}{\mu g_m}$$

Since the loop gain μ is large, then inverse of μ is very less (approximately zero). Therefore,

$$R_m = 0$$

Therefore, the input resistance R_m is **very less**.

The output resistance R_{of} found from the feedback amplifier is,

$$R_{of} = (1+A\beta)R_o$$

Substitute $\frac{1}{g_m}$ for R_o and $\mu g_m R_S$ for $A\beta$.

$$R_{of} = (1+\mu g_m R_S)\frac{1}{g_m}$$

$$= \frac{1}{g_m} + \frac{\mu g_m R_S}{g_m}$$

$$= \frac{1}{g_m} + \mu R_S$$

Here, there is no load resistance. Therefore,

$$R_{of} = R_{out}$$

$$R_{out} = \frac{1}{g_m} + \mu R_S$$

Since the loop gain μ is large, and then the output resistance is,

$$R_{out} = \infty \text{ (large)}$$

Therefore, the output resistance R_{out} is **very large**.

Step 10 of 12

(f)

Refer to Figure P10.66 (b) for the cascode configuration in which the super CG transistor is used.

The small signal equivalent circuit is shown in Figure 2.

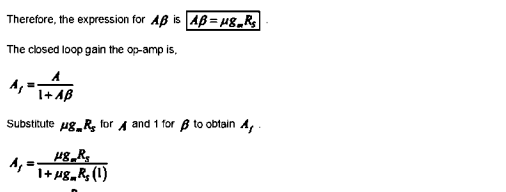


Figure 2

Step 11 of 12

From Figure 2, clearly observe that the current I_S is,

$$I_S = g_{m1} V_{gs}$$

The voltage V_i is applied to the gate terminal of the transistor Q_1. Therefore, the gate source voltage becomes V_i. That is,

$$V_i = V_{gs}$$

Therefore,

$$I_S = g_{m1} V_i$$

The closed loop gain of the op-amp is,

$$A_f = \frac{A}{1+A\beta}$$

$$\frac{I_o}{I_S} = \frac{A}{1+A\beta}$$

Substitute $\mu g_{m2} R_S$ for A and 1 for β.

$$\frac{I_o}{I_S} = \frac{\mu g_{m2} R_S}{1+\mu g_{m2} R_S}$$

$$I_o = I_S \frac{\mu g_{m2} R_S}{1+\mu g_{m2} R_S}$$

$$= \left(\frac{\mu g_{m2} R_S}{1+\mu g_{m2} R_S}\right) g_{m1} V_i$$

Since $\mu g_m R_S >> 1$, then,

$$I_o = \left(\frac{\mu g_{m2} R_S}{\mu g_{m2} R_S}\right) g_{m1} V_i$$

$$= g_{m1} V_i$$

Therefore, the expression for output current I_o is $\boxed{g_{m1} V_i}$.

Step 12 of 12

The output amplifier is,

$$R_{out} = (1+A\beta)R_o$$

Substitute $\mu g_{m2} R_S$ for $A\beta$.

$$R_{out} = (1+\mu g_{m2} R_S) R_o$$

$$\cong \mu g_{m2} R_S R_o$$

Therefore, the expression for output resistance R_{out} is $\boxed{\mu g_{m2} R_S R_o}$.

Step 1 of 5

11.079P

Refer to the circuit diagram of Figure P11.79 in the textbook.

(a)

The following is the small signal equivalent circuit of the circuit shown in Figure 1.

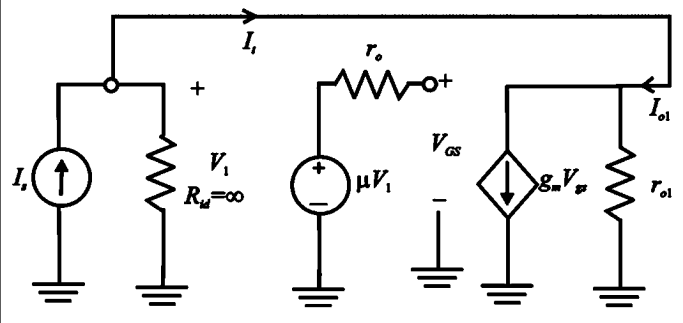


Figure 1

Step 2 of 5

From the small signal equivalent circuit in Figure 1,

$\beta = 1$ **and**

$I_s = I_f = I_o$

Thus, the feedback is negative.

Step 3 of 5

(b)

It is given that the gain of the amplifier, μ is infinite.

The dc voltage that appears at the drain of transistor, Q_1 is: $\boxed{V_{DS} \geq V_{OV}}$.

It is given that the transistor, Q_1 is operating at an overdrive voltage of 0.2 V.

Thus, the minimum value of bias voltage is: $\boxed{V_{BIAS(min)} = 0.2\ \text{V}}$.

Step 4 of 5

(c)

Redraw Figure 1 for the differential resistance $R_{id} = \infty$.

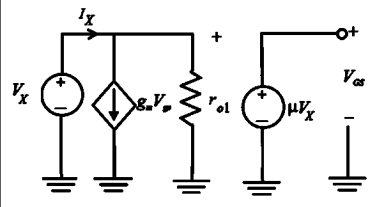


Figure 2

From Figure 2,

$V_{GS} = V_X$

Apply Kirchhoff's current law at input node.

$$I_X = g_m V_{GS} + \frac{V_X}{r_{o1}}$$

$$= g_m \mu V_X + \frac{V_X}{r_{o1}}$$

$$I_X = \left(\frac{g_m \mu r_{o1} + 1}{r_{o1}} \right) V_X$$

Calculate the input resistance.

$$R_{in} = \frac{V_X}{I_X}$$

$$= \frac{r_{o1}}{1 + g_m \mu r_{o1}}$$

Thus, the small-signal input resistance R_{in} for the amplifier is $\boxed{\frac{r_{o1}}{1 + g_m \mu r_{o1}}}$.

Step 5 of 5

(d)

From small signal equivalent circuit, the output resistance is,

$\boxed{R_{out} = r_{o2}}$.

11.080P

We don't have the solution to this problem yet.

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11.081P

Step 1 of 16

Refer to the circuit diagram of amplifier circuit in Figure P10.69 in the textbook.

Assume that V_S has zero dc components and the circuit can be represented as shown in Figure 1.

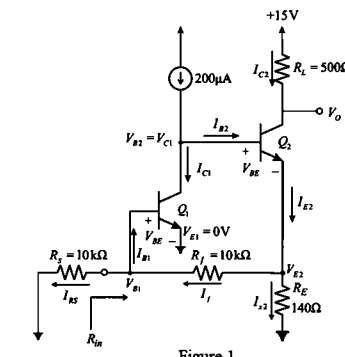


Figure 1

Step 2 of 16

From Figure1, the voltage at emitter V_{E1} of transistor Q_1 is $\boxed{0\text{ V}}$.

The current flow through the emitter terminal I_{E1} of transistor Q_1 is $\boxed{0\text{ A}}$.

Collector current of the transistor Q_1 is,

$$I_{C1} \cong 200\,\mu\text{A}$$

The voltage at base to emitter V_{BE} is,

$$V_{BE} = 0.7\text{ V and}$$
$$\beta = 100$$

The voltage at base to emitter of transistor Q_1 is,

$$V_{BE1} = V_{B1} - V_{S1}$$
$$0.7 = V_{B1} - (0)$$
$$0.7 = V_{B1}$$

Thus, the voltage at base V_{B1} of transistor Q_1 is $\boxed{0.7\text{ V}}$.

Step 3 of 16

Find current flows through resistor R_S is,

$$I_{R2} = \frac{V_{B1}}{R_2}$$
$$= \frac{0.7}{10\times10^3}$$
$$= 0.07\times10^{-3}$$

Therefore, the current I_{R2} flow through resistor R_2 is $\boxed{0.07\text{ mA}}$.

Step 4 of 16

Base current of the transistor can be expressed as,

$$I_{B1} = \frac{I_{C1}}{\beta}$$

The base current is,

$$I_{B1} = \frac{I_{C1}}{\beta}$$
$$= \frac{200\times10^{-6}}{100}$$
$$= 2\times10^{-6}$$

Therefore base current I_{B1} of transistor Q_1 is $2\,\mu\text{A}$.

Step 5 of 16

Apply Kirchhoff's current low at node V_{B1}.

$$I_F = I_{R2} + I_{B1}$$
$$= 0.07\times10^{-3} + 2\times10^{-6}$$
$$= 0.07\times10^{-3} + 0.002\times10^{-3}$$
$$= 0.072\times10^{-3}$$

Therefore, current flow through the resistor $I_F = 0.072\text{ mA}$

Find emitter voltage of Q_2.

$$V_{E2} = V_{B1} + I_F R_F$$
$$= 0.7 + \left(0.072\times10^{-3}\times10\times10^3\right)$$
$$= 0.7 + 0.72$$
$$= 1.42\text{ V}$$

Thus, emitter voltage V_{E2} of transistor Q_2 is $\boxed{1.42\text{ V}}$.

Find base voltage of Q_2.

$$V_{B2} = V_{E2} + V_{BE}$$
$$= 1.42 + 0.7$$
$$= 2.12\text{ V}$$

Thus, base voltage V_{B2} of transistor Q_2 is $\boxed{V_{C1} = V_{B2} = 2.12\text{ V}}$.

Step 6 of 16

The current I_{E2} is,

$$I_{E2} = \frac{V_{E2}}{140}$$
$$= \frac{1.42}{140}$$
$$= 10.143\times10^{-3}$$

Thus, the current I_{E2} is $\boxed{10.143\text{ mA}}$.

Step 7 of 16

Apply Kirchhoff's current law at node V_{E2}.

$$I_{E2} = I_F + I_{C2}$$
$$= 0.072\times10^{-3} + 10.143\times10^{-3}$$
$$= 10.215\times10^{-3}$$

Thus, current I_{E2} flow through emitter of transistor Q_2 is $\boxed{10.22\text{ mA}}$.

Step 8 of 16

Now obtain the collector current I_{C2}.

$$I_{C2} = I_{E2} + I_F$$
$$= 0.072\times10^{-3} + 10.143\times10^{-3}$$
$$= 10.215\times10^{-3}$$

Thus, collector current I_{C2} is $\boxed{10.22\text{ mA}}$.

Step 9 of 16

Output voltage V_0 is,

$$V_0 = 15 - \left(I_{C2}\times R_L\right)$$
$$= 15 - \left(10.22\times10^{-3}\times500\right)$$
$$= 15 - 5.11$$
$$= 9.89\text{ V}$$

Thus, output voltage V_0 of the circuit is $\boxed{10\text{ V}}$.

Step 10 of 16

Draw A-circuit for the amplifier circuit.

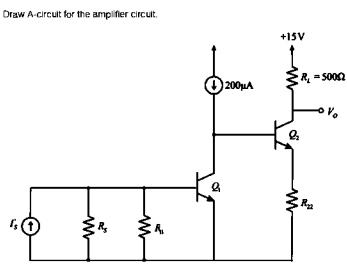


Figure 2

Step 11 of 16

The output voltage is expressed as,

$$r_0 = \frac{V_T}{I_S}$$

Substitute known values in the equation.

$$r_0 = \frac{V_T}{I_S}$$
$$= \frac{25\times10^{-3}}{0.1\times10^{-3}}$$
$$= 250\ \Omega$$

Find the feedback factor β.

$$\beta = \frac{R_E}{R_E + R_F}$$

Here,

$$R_2 = 10\text{k}\Omega,$$
$$R_{11} = 10\times10^3 + 140$$
$$R_{11} = 10.14\text{k}\Omega$$

$$R_{22} = 10\times10^2 \parallel 140$$

$$\beta = \frac{0.14}{0.14 + 10}$$
$$= 0.0138$$

Thus, feedback factor β of the amplifier is $\boxed{0.0138}$.

Step 12 of 16

Use the following equation to find the voltage V_{B1}.

$$V_{B1} = \left[10\times10^2 \parallel 10.14\times10^3 \parallel (100)(250)\right] I'_S$$

Simplify further.

$$V_{B1} \approx 4.2\times10^2 I_S$$

Thus voltage V_{B1} at base is $\boxed{4.2\times10^2 I_S}$.

Step 13 of 16

Calculate gain of the amplifier

$$A = \frac{I'_0}{I'_S}.$$

From the circuit, gain can be expressed as,

$$\frac{I'_0}{I'_S} = V_{B1}\times\frac{(\beta+1)\left(r_{e2} + 10\times10^2 \parallel 140\right)}{250}\times\frac{1}{\left(r_{e2} + 10\times10^3 \parallel 140\right)}$$
$$= 4.2\times10^3\times\frac{(\beta+1)\left(r_{e2} + 10\times10^2 \parallel 140\right)}{250}\times\frac{1}{\left(r_{e2} + 10\times10^3 \parallel 140\right)}$$
$$= 4.2\times10^3\times\frac{(\beta+1)}{250}$$

Substitute 100 for β in the equation.

$$\frac{I'_0}{I'_S} = \left(4.2\times10^3\right)\left(\frac{100+1}{250}\right)$$
$$= 1.69\times10^3\text{ A/A}$$

Thus, gain of the amplifier A is $\boxed{1.69\text{ kA}}$.

Step 14 of 16

Define closed loop gain.

$$A_F = \frac{I_0}{I_S}$$
$$= \frac{A}{1 + A\beta}$$

Substitute corresponding values.

$$A_F = \frac{1.69\times10^3}{1 + \left(1.69\times10^3\times0.0138\right)}$$
$$= 69.6\text{ A/A}$$
$$\approx 70\text{ A/A}$$

Thus, closed loop gain A_F is $\boxed{70\text{ A/A}}$.

Step 15 of 16

Voltage gain of the amplifier is,

$$\frac{V_0}{V_S} = \frac{I_0 R_L}{I_S R_S}$$

Substitute corresponding values.

$$\frac{V_0}{V_S} = \frac{500}{10^2}\times69.6$$
$$= 3.5\text{ V/V}$$

Thus, voltage gain of the amplifier $\frac{V_0}{V_S}$ is $\boxed{3.5\text{ V/V}}$.

Define closed loop input resistance of the feedback amplifier R_f.

$$R_f = \frac{R_i}{(1 + A\beta)}$$

We know that R_i is the parallel combination of 10 k, 10.14 k and 25 k.

$$R_i = 10\text{ k} \parallel 10.14\text{ k} \parallel 25\text{ k}$$
$$= 4.2\text{ k}\Omega$$

Substitute 4.2×10^3 for R_i, 1.69×10^3 for A and 0.0138 for β in the equation.

$$R_f = \frac{R_i}{(1 + A\beta)}$$
$$= \frac{4.2\times10^3}{1 + \left(1.69\times10^3\times0.0138\right)}$$
$$= \frac{4.2\times10^3}{(1 + 23.3)}$$
$$= 172.8\ \Omega$$

Thus, input resistance R_F of the feedback amplifier is $\boxed{172.8\ \Omega}$.

Step 16 of 16

Input resistance seen through the source resistor of the amplifier is,

$$R_F = \frac{1}{\left(\frac{1}{R_f} - \frac{1}{R_S}\right)}$$

Simplify further.

$$R_F = \frac{1}{\left(\frac{1}{R_f} - \frac{1}{R_S}\right)}$$
$$= \frac{R_f R_S}{R_S - R_f}$$

Substitute 172.8 for R_f and 10×10^2 for R_S.

$$R_F = \frac{(172.8)(10\times10^2)}{(10\times10^2) - (172.8)}$$
$$= 175.83\ \Omega$$

Thus, input resistance seen through the source R_F is $\boxed{175.8\ \Omega}$.

11.082P

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Step 1 of 4

The open loop transfer function characterized by a single pole is,

$$A(s)=\frac{A_0}{1+\frac{s}{\omega_{P1}}}$$

Substitute 10^5 for A_0 and 100 rad/s for ω_{P1}.

$$A(s)=\frac{10^5}{1+\frac{s}{100}}$$

Step 2 of 4

Calculate the phase angle of open loop transfer function for an op-amp having an additional pair of pole at 20000 rad/s.

$$\phi=-\tan^{-1}\left(\frac{\omega}{\omega_{P1}}\right)-2\tan^{-1}\left(\frac{\omega}{\omega_{P2}}\right)$$

Substitute $180°$ for ϕ, 2×10^4 rad/s for ω_{P2} in equation.

$$180^\circ=-\tan^{-1}\left(\frac{\omega}{100}\right)-2\tan^{-1}\left(\frac{\omega}{2\times10^4}\right)$$

Determine the frequency for the total phase shift to reach $180°$.

$$-180^\circ=-\tan^{-1}(\text{large value})-2\tan^{-1}\left(\frac{\omega_{180^\circ}}{2\times10^4}\right)\quad(\text{since }\omega_{180^\circ}>>100)$$

$$-180^\circ=-\tan^{-1}(\infty)-2\tan^{-1}\left(\frac{\omega_{180^\circ}}{2\times10^4}\right)$$

$$180^\circ=90^\circ+2\tan^{-1}\left(\frac{\omega_{180^\circ}}{2\times10^4}\right)$$

$$\tan^{-1}\frac{\omega_{180^\circ}}{2\times10^4}=\frac{90^\circ}{2}$$

$$\frac{\omega_{180^\circ}}{2\times10^4}=\tan 45^\circ$$

$$\omega_{180^\circ}=2\times10^4\text{ rad/s}$$

Therefore, the value of ω_{180° is $\boxed{2\times10^4}$.

Step 3 of 4

Determine the value of β by equating the loop gain to unity.

$$|A\beta|=\left(\frac{10^5}{\left(\sqrt{1+\left(\frac{\omega}{100}\right)^2}\right)\left(\sqrt{1+\left(\frac{\omega}{2\times10^4}\right)}\right)^2}\right)\beta$$

$$1=\left(\frac{10^5}{\left(\sqrt{1+\left(\frac{\omega}{100}\right)^2}\right)\left(\sqrt{1+\left(\frac{\omega}{2\times10^4}\right)}\right)^2}\right)\beta$$

Substitute 10^4 rad/s for ω in the equation.

$$1=\frac{10^5\beta}{\sqrt{1+\left(\frac{10^4}{100}\right)^2}}\times\frac{1}{\left(\sqrt{1+\left(\frac{10^4}{2\times10^4}\right)}\right)^2}$$

$$1=\frac{10^5\beta}{\sqrt{1+(100)^2}}\times\frac{1}{\left(\sqrt{\frac{3}{2}}\right)^2}$$

$$1=\frac{10^5\beta}{\sqrt{1+(100)^2}}\times\frac{1}{(1.5)}$$

$$\frac{150}{10^5}=\beta$$

$$\beta=0.0015$$

Therefore, the value of β is $\boxed{0.0015}$.

Step 4 of 4

Determine the closed loop gain at low frequencies.

$$A_f(0)=\frac{A}{1+A\beta}$$

Substitute 10^5 for A, 0.0015 for β in equation.

$$A_f(0)=\frac{10^5}{1+10^5(0.0015)}$$

$$=\frac{10^5}{1+150}$$

$$=\frac{10^5}{151}$$

$$=662.25\text{ V/V}$$

Therefore, the closed loop gain at low frequencies is $\boxed{662.25\text{ V/V}}$.

Step 1 of 17

Consider the Problem 11.83 in the textbook.

Calculate the phase angle for $\omega = 0$ rad/s, $\beta_1 = 1.0$ and $\beta_2 = 10^{-3}$.

$$\angle A = -\tan^{-1}\left(\frac{\omega}{100}\right) - 2\tan^{-1}\left(\frac{\omega}{2\times10^4}\right)$$

Substitute $\omega = 0$ rad/s in equation.

$$\angle A = -\tan^{-1}\left(\frac{0}{100}\right) - 2\tan^{-1}\left(\frac{0}{2\times10^4}\right)$$
$$= 0°$$

Therefore, the phase angle is $\boxed{0°}$.

Step 2 of 17

Calculate the loop gain, $|A\beta_1|$.

$$|A\beta_1| = \frac{10^2\beta_1}{\sqrt{1+\left(\frac{\omega}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{\omega}{2\times10^4}\right)^2}\right)^2}$$

Substitute 0 rad/s for ω, 1.0 for β_1 in equation.

$$|A\beta_1| = \frac{10^2(1)}{\sqrt{1+\left(\frac{0}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{0}{2\times10^4}\right)^2}\right)^2}$$
$$= 10^2$$

Therefore, the value of $|A\beta_1|$ is $\boxed{10^2}$.

Step 3 of 17

Calculate the loop gain, $|A\beta_2|$.

$$|A\beta_2| = \frac{10^2\beta_2}{\sqrt{1+\left(\frac{\omega}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{\omega}{20\times10^3}\right)^2}\right)^2}$$

Substitute 0 rad/s for ω, 10^{-2} for β_1 in equation.

$$|A\beta_2| = \frac{10^2(10^{-3})}{\sqrt{1+\left(\frac{0}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{0}{20\times10^3}\right)^2}\right)^2}$$
$$= 10^2$$

Therefore, the value of $|A\beta_2|$ is $\boxed{10^2}$

Step 4 of 17

Calculate the phase angle for $\omega = 100$ rad/s, $\beta_1 = 1.0$ and $\beta_2 = 10^{-3}$

$$\angle A = -\tan^{-1}\left(\frac{\omega}{100}\right) - 2\tan^{-1}\left(\frac{\omega}{2\times10^4}\right)$$

Substitute $\omega = 100$ rad/s in equation.

$$\angle A = -\tan^{-1}\left(\frac{100}{100}\right) - 2\tan^{-1}\left(\frac{100}{2\times10^4}\right)$$
$$= -\tan^{-1}(1) - 2\tan^{-1}(0.5\times10^{-2})$$
$$= -45° - 0.57°$$
$$= -45.57°$$

Therefore, the phase angle is $\boxed{-45.57°}$.

Step 5 of 17

The loop gain $|A\beta_1|$ is,

$$|A\beta_1| = \frac{10^2\beta_1}{\sqrt{1+\left(\frac{\omega}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{\omega}{2\times10^4}\right)^2}\right)^2}$$

Substitute 100 rad/s for ω, 1.0 for β_1 in equation.

$$|A\beta_1| = \frac{10^2(1)}{\sqrt{1+\left(\frac{100}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{100}{2\times10^4}\right)^2}\right)^2}$$
$$= \frac{10^2(1)}{\sqrt{1+1}} \times \frac{1}{\left(1+(0.5\times10^{-2})^2\right)}$$
$$= \frac{10^2(1)}{\sqrt{1+1}} \times \frac{1}{\left(1+(0.25\times10^{-4})\right)}$$
$$= \frac{10^2(1)}{\sqrt{2}} \times \frac{1}{1}$$

Therefore, the value of $|A\beta_1|$ is $\boxed{7.07\times10^1}$

Step 6 of 17

The loop gain $|A\beta_2|$ is,

$$|A\beta_2| = \frac{10^2\beta_2}{\sqrt{1+\left(\frac{\omega}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{\omega}{2\times10^4}\right)^2}\right)^2}$$

Substitute 100 rad/s for ω, 10^{-3} for β_2 in equation.

$$|A\beta_2| = \frac{10^2(10^{-3})}{\sqrt{1+\left(\frac{100}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{100}{2\times10^4}\right)^2}\right)^2}$$
$$= \frac{10^2(10^{-3})}{\sqrt{2}} \times \frac{1}{(1.000025)}$$
$$\approx 70.71$$

Therefore, the value of $|A\beta_2|$ is $\boxed{70.71}$.

Step 7 of 17

Calculate the phase angle for $\omega = 10^4$ rad/s, $\beta_1 = 1.0$ and $\beta_2 = 10^{-3}$

$$\angle A = -\tan^{-1}\left(\frac{\omega}{100}\right) - 2\tan^{-1}\left(\frac{\omega}{2\times10^4}\right)$$

Substitute $\omega = 10^4$ rad/s in equation.

$$\angle A = -\tan^{-1}\left(\frac{10^4}{100}\right) - 2\tan^{-1}\left(\frac{10^4}{2\times10^4}\right)$$
$$= -\tan^{-1}(10^2) - 2\tan^{-1}(0.5)$$
$$= -89.42° - 53.13°$$
$$= -142.55°$$

Therefore, the phase angle is $\boxed{-142.55°}$.

Step 8 of 17

The loop gain $|A\beta_1|$ is,

$$|A\beta_1| = \frac{10^2\beta_1}{\sqrt{1+\left(\frac{\omega}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{\omega}{2\times10^4}\right)^2}\right)^2}$$

Substitute 10^4 rad/s for ω, 1.0 for β_1 in equation.

$$|A\beta_1| = \frac{10^2(1)}{\sqrt{1+\left(\frac{10^4}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{10^4}{2\times10^4}\right)^2}\right)^2}$$
$$= \frac{10^2(1)}{\sqrt{1+10^4}} \times \frac{1}{\left(1+(0.5)^2\right)}$$
$$= 10^2 \times \frac{1}{1.25}$$
$$= 800$$

Therefore, the value of $|A\beta_1|$ is $\boxed{800}$

Step 9 of 17

The loop gain $|A\beta_2|$ is,

$$|A\beta_2| = \frac{10^2\beta_2}{\sqrt{1+\left(\frac{\omega}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{\omega}{2\times10^4}\right)^2}\right)^2}$$

Substitute 10^4 rad/s for ω, 10^{-3} for β_2 in equation.

$$|A\beta_2| = \frac{10^2(10^{-3})}{\sqrt{1+\left(\frac{10^4}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{10^4}{2\times10^4}\right)^2}\right)^2}$$
$$= \frac{10^2(10^{-3})}{\sqrt{1+10^4}} \times \frac{1}{\left(1+(0.5)^2\right)}$$
$$= \frac{10^2(10^{-3})}{10^2} \times \frac{1}{\left(1+(0.5)^2\right)}$$
$$= \frac{1}{1.25}$$
$$= 0.8$$

Therefore, the value of $|A\beta_2|$ is $\boxed{0.8}$

Step 10 of 17

Calculate the phase angle for $\omega = 2\times10^4$ rad/s, $\beta_1 = 1.0$ and $\beta_2 = 10^{-3}$

$$\angle A = -\tan^{-1}\left(\frac{\omega}{100}\right) - 2\tan^{-1}\left(\frac{\omega}{2\times10^4}\right)$$

Substitute $\omega = 2\times10^4$ rad/s in equation.

$$\angle A = -\tan^{-1}\left(\frac{2\times10^4}{100}\right) - 2\tan^{-1}\left(\frac{2\times10^4}{2\times10^4}\right)$$
$$= -\tan^{-1}(2\times10^2) - 2\tan^{-1}(1)$$
$$= -89.71° - 90°$$
$$= -179.71°$$

Therefore, the phase angle is $\boxed{-179.71°}$.

Step 11 of 17

The loop gain $|A\beta_1|$ is,

$$|A\beta_1| = \frac{10^2\beta_1}{\sqrt{1+\left(\frac{\omega}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{\omega}{2\times10^4}\right)^2}\right)^2}$$

Substitute 2×10^4 rad/s for ω, 1.0 for β_1 in equation.

$$|A\beta_1| = \frac{10^2(1)}{\sqrt{1+\left(\frac{2\times10^4}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{2\times10^4}{2\times10^4}\right)^2}\right)^2}$$
$$= \frac{10^2(1)}{\sqrt{1+4\times10^4}} \times \frac{1}{\left(1+(1)^2\right)}$$
$$= \frac{10^2(1)}{200} \times \frac{1}{(2)}$$
$$\approx 025\times10^2$$
$$= 250$$

Therefore, the value of $|A\beta_1|$ is $\boxed{250}$

Step 12 of 17

The loop gain $|A\beta_2|$ is,

$$|A\beta_2| = \frac{10^2\beta_2}{\sqrt{1+\left(\frac{\omega}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{\omega}{2\times10^4}\right)^2}\right)^2}$$

Substitute 2×10^4 rad/s for ω, 10^{-3} for β_2 in equation.

$$|A\beta_2| = \frac{10^2(10^{-3})}{\sqrt{1+\left(\frac{2\times10^4}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{2\times10^4}{2\times10^4}\right)^2}\right)^2}$$
$$= \frac{10^2(10^{-2})}{200} \times \frac{1}{\left(1+(1)^2\right)}$$
$$= \frac{10^2(10^{-3})}{200} \times \frac{1}{2}$$
$$= \frac{1}{4}$$
$$= 0.25$$

Therefore, the value of $|A\beta_2|$ is $\boxed{0.25}$.

Step 13 of 17

Calculate the phase angle for $\omega = \infty$ rad/s, $\beta_1 = 1.0$ and $\beta_2 = 10^{-3}$

$$\angle A = -\tan^{-1}\left(\frac{\omega}{100}\right) - 2\tan^{-1}\left(\frac{\omega}{2\times10^4}\right)$$

Substitute $\omega = \infty$ rad/s in equation.

$$\angle A = -\tan^{-1}\left(\frac{\infty}{100}\right) - 2\tan^{-1}\left(\frac{\infty}{2\times10^4}\right)$$
$$= -270°$$

Therefore, the phase angle is $\boxed{-270°}$

Step 14 of 17

The loop gain $|A\beta_1|$ is,

$$|A\beta_1| = \frac{10^2\beta_1}{\sqrt{1+\left(\frac{\omega}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{\omega}{2\times10^4}\right)^2}\right)^2}$$

Substitute ∞ rad/s for ω, 1.0 for β_1 in equation.

$$|A\beta_1| = \frac{10^2(1)}{\sqrt{1+\left(\frac{\infty}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{\infty}{2\times10^4}\right)^2}\right)^2}$$
$$= 0$$

Therefore, the value of $|A\beta_1|$ is $\boxed{0}$

Step 15 of 17

The loop gain $|A\beta_2|$ is,

$$|A\beta_2| = \frac{10^2\beta_2}{\sqrt{1+\left(\frac{\omega}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{\omega}{2\times10^4}\right)^2}\right)^2}$$

Substitute ∞ rad/s for ω, 10^{-3} for β_2 in equation.

$$|A\beta_2| = \frac{10^2(10^{-3})}{\sqrt{1+\left(\frac{\infty}{100}\right)^2}} \times \frac{1}{\left(\sqrt{1+\left(\frac{\infty}{2\times10^4}\right)^2}\right)^2}$$
$$= 0$$

Therefore, the value of $|A\beta_2|$ is $\boxed{0}$.

Step 16 of 17

The obtained values of are shown in Table 1.

ω	$\angle A$	$\lvert A\beta_1\rvert$	$\lvert A\beta_2\rvert$
0	0°	10^2	10^2
100	−45.57°	7.07×10^1	70.71
10^4	−142.55°	800	0.8
2×10^4	−179.43°	250	0.25
∞	−270°	0	0

Table 1

Step 17 of 17

Draw the Nyquist plot corresponding values of Table 1.

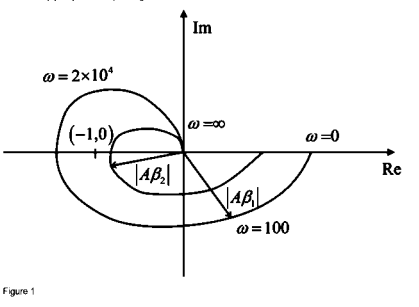


Figure 1

Hence, the Nyquist plot is drawn.

Step 1 of 3

11.085P

An amplifier whose open-loop transfer function is characterized with single pole is,

$$A(s)=\frac{A_0}{\left(1+\frac{s}{\omega_P}\right)} \quad \text{..... (1)}$$

Here,

A_0 is low frequency gain,

ω_P is pole.

Substitute 10^4 for A_0, $10^3\,\text{rad/s}$ for ω_P in equation 1.

$$A(s)=\frac{10^4}{\left(1+\frac{s}{10^3}\right)}$$

Since, the amplifier is connected in a negative feedback loop having a transmission k and a two-poll roll off at $10^3\,\text{rad/s}$, the gain is,

$$\beta(s)=\frac{k}{\left(1+\frac{s}{10^3}\right)^2}$$

Step 2 of 3

The open loop gain of feedback amplifier is,

$$A(s)\beta(s)=\left(\frac{10^4}{\left(1+\frac{s}{10^3}\right)}\right)\left(\frac{k}{\left(1+\frac{s}{10^3}\right)^2}\right)$$

$$A(j\omega)\beta(j\omega)=\left(\frac{10^4}{1+j\frac{\omega}{10^3}}\right)\left(\frac{k}{\left(1+j\frac{\omega}{10^3}\right)^2}\right)$$

$$\angle A(j\omega)\beta(j\omega)=-\tan^{-1}\left(\frac{\omega}{10^3}\right)-2\tan^{-1}\left(\frac{\omega}{10^3}\right)$$

$$\angle A(j\omega)\beta(j\omega)=-3\tan^{-1}\left(\frac{\omega}{10^3}\right)$$

$$180^\circ=-3\tan^{-1}\left(\frac{\omega_{180^\circ}}{10^3}\right)$$

$$\tan(-60^\circ)=\left(\frac{\omega_{180^\circ}}{10^3}\right)\theta$$

$$\omega_{180^\circ}=-\sqrt{3}\times10^3\ \text{rad/s}$$

Step 3 of 3

Find the value of k for which the closed loop amplifier becomes unstable.

$$|A\beta|<1$$

$$\left|\frac{10^4}{1-j\frac{\sqrt{3}\times10^3}{10^3}}\right|\times\left|\frac{k}{\left(1-j\frac{\sqrt{3}\times10^3}{10^3}\right)^2}\right|<1$$

$$\left(\frac{10^4}{\sqrt{1+\left(\sqrt{3}\right)^2}}\right)\left(\frac{k}{\left(\sqrt{1+\left(\sqrt{3}\right)^2}\right)^2}\right)<1$$

$$\left(\frac{10^4}{2}\right)\left(\frac{k}{4}\right)<1$$

$$k<\frac{8}{10^4}$$

$$k<0.0008$$

Therefore, the value of k should be $\boxed{k<0.0008}$ to become closed loop system unstable.

Step 1 of 5

The open loop gain $A(s)$ for a feedback amplifier is,

$$A(s)=\frac{1000}{\left(1+\frac{s}{10^4}\right)\left(1+\frac{s}{10^5}\right)^2}$$

Feedback factor β is independent of frequency.

Step 2 of 5

Find the frequency at which the phase shift is $180°$.

$$\angle A(s)=-\tan^{-1}\left(\frac{\omega}{10^4}\right)-2\tan^{-1}\left(\frac{\omega}{10^5}\right)$$

For $\omega=10^4$:

$$\begin{aligned}\left|\angle A(s)\right| &= -\tan^{-1}\left(\frac{10^4}{10^4}\right)-2\tan^{-1}\left(\frac{10^4}{10^5}\right)\\ &= 45°+2(5.71°)\\ \left|\angle A(s)\right| &= 56.42°\end{aligned}$$

Step 3 of 5

Determine the phase for $\omega=10^5$.

$$\begin{aligned}\left|\angle A(s)\right| &= -\tan^{-1}\left(\frac{10^5}{10^4}\right)-2\tan^{-1}\left(\frac{10^5}{10^5}\right)\\ &= 84.29°+2(45°)\\ \left|\angle A(s)\right| &= 174.29°\end{aligned}$$

Determine the phase for $\omega=10^6$.

$$\begin{aligned}\left|\angle A(s)\right| &= -\tan^{-1}\left(\frac{10^6}{10^4}\right)-2\tan^{-1}\left(\frac{10^6}{10^5}\right)\\ &= 89.43°+2(84.29°)\\ \left|\angle A(s)\right| &= 258°\end{aligned}$$

Step 4 of 5

Determine the phase for $\omega=1.5\times10^5$.

$$\begin{aligned}\left|\angle A(s)\right| &= -\tan^{-1}\left(\frac{1.5\times10^5}{10^4}\right)-2\tan^{-1}\left(\frac{1.5\times10^5}{10^5}\right)\\ &= 86.18°+2(56.3°)\\ \left|\angle A(s)\right| &= 198.8°\end{aligned}$$

Determine the phase for $\omega=1.1\times10^5$.

$$\begin{aligned}\left|\angle A(s)\right| &= -\tan^{-1}\left(\frac{1.1\times10^5}{10^4}\right)-2\tan^{-1}\left(\frac{1.1\times10^5}{10^5}\right)\\ &= 84.8°+2(47.73°)\\ \left|\angle A(s)\right| &= 180.2°\end{aligned}$$

Therefore, the frequency at which the phase shift is $180°$ is $\boxed{\omega\approx1.1\times10^5 \text{ rad/s}}$.

Step 5 of 5

Find the critical value of β at which oscillation commence.

$$\left|A\beta(\omega_{180})\right|\geq 1$$

$$\frac{1000}{\left(\sqrt{1+\left(\frac{1.1\times10^5}{10^4}\right)^2}\right)}\frac{1}{\left(\sqrt{1+\left(\frac{1.1\times10^5}{10^5}\right)^2}\right)^2}(\beta)\geq 1$$

$$\frac{1000}{(11.0454)}\frac{1}{(2.21)}(\beta)\geq 1$$

$$\beta\geq\frac{(2.21)(11.0454)}{1000}$$

$$\beta\geq 0.0244 \text{ V/V}$$

Therefore, the critical value of β is $\boxed{\beta\geq 0.0244 \text{ V/V}}$.

Step 1 of 5

The open loop transfer function of a single pole is,

$$A(s)=\frac{A_0}{1+\frac{s}{\omega_P}}$$

$$A(s)=\frac{A_0}{1+j\left(\frac{f}{f_P}\right)}$$

$$|A(s)|=\frac{A_0}{\sqrt{1+\left(\frac{f}{f_P}\right)^2}}$$

Substitute 1 for $|A(s)|$, 10 Hz for f_P and 1 MHz for f.

$$1=\frac{A_0}{\sqrt{1+\left(\frac{10^6}{10}\right)^2}}$$

$$A_0^2=1+\left(\frac{10^6}{10}\right)^2$$

$$A_0^2=1+10^{10}$$

$$A_0=10^5$$

Step 2 of 5

Calculate the low frequency gain of feedback.

$$A_f(0)=\frac{A_0}{(1+A_0\beta)}$$

Substitute 10^5 for A_0, 0.1 for β in equation.

$$A_f(0)=\frac{10^5}{\left(1+10^5(0.1)\right)}$$

$$=\frac{10^5}{1+10^4}$$

$$=9.99\ \text{V/V}$$

Therefore, the low frequency gain is $A_f(0)$ is $\boxed{9.99\ \text{V/V}}$.

Step 3 of 5

The closed loop transfer function is,

$$A_f(s)=\left(\frac{\left(\frac{A_0}{1+A_0\beta}\right)}{1+\frac{s}{\omega_P(1+A_0\beta)}}\right)$$

The 3-dB frequency is,

$$\omega_{Hf}=\omega_P(1+A_0\beta)$$

Substitute $10\ \text{Hz}$ for ω_p, 10^5 for A_0, 0.1 for β in equation.

$$\omega_{Hf}=10\left(1+(10^5)(0.1)\right)$$

$$=10\left(1+(10^4)\right)$$

$$=10(10001)$$

$$\simeq 10^5\ \text{Hz}$$

Therefore, the 3-dB frequency is ω_{Hf} is $\boxed{10^5\ \text{Hz}}$.

Step 4 of 5

Determine the unity gain frequency.

$$|A_f(s)|=\left|\frac{\frac{A_0}{(1+A_0\beta)}}{1+j\frac{f}{f_P(1+A_0\beta)}}\right|$$

$$|A_f(s)|=\frac{\frac{A_0}{(1+A_0\beta)}}{\left|\sqrt{1+\left(\frac{f}{f_P(1+A_0\beta)}\right)^2}\right|}$$

Substitute $9.99\ \text{V/V}$ for $\frac{A_0}{(1+A_0\beta)}$, 1 for $|A_f(s)|$, 10^5 for $f_P(1+A_0\beta)$ in equation.

$$1=\frac{9.99}{\sqrt{1+\left(\frac{f}{10^5}\right)^2}}$$

$$1+\left(\frac{f}{10^5}\right)^2=(9.99)^2$$

$$\left(\frac{f}{10^5}\right)^2=(9.99)^2-1$$

$$\frac{f}{10^5}=\sqrt{98.8001}$$

$$f=9.9398\times10^5$$

$$=0.99398\times10^6$$

$$=0.993\ \text{MHz}$$

Thus, the unity gain frequency f is $\boxed{0.993\ \text{MHz}}$.

Step 5 of 5

Calculate the factor of shift of a pole.

$$(1+A_0\beta)=\left(1+10^5(0.1)\right)$$

$$=(1+10000)$$

$$=10001$$

Therefore, the factor of shift of a pole is $\boxed{10001}$.

Step 1 of 5

Consider that the open-loop dc gain $A_0(0)_{dB}$ of an amplifier is 80 dB.

Convert the value to normal gain in watts.

$$A_0(0)_{dB} = 80\,dB$$
$$20\log(A_0(0)) = 80$$

$$A_0(0) = 10^{\frac{80}{20}}$$
$$= 10^4$$

Thus, the open loop dc gain $A_0(0)$ is 10^4.

Step 2 of 5

Consider that the amplifier open-loop transfer function is characterized by a single pole.

$$A(s) = \frac{A_0}{\left(1+\frac{s}{f_{p1}}\right)}$$

Here, f_{p1} is the single-pole frequency.

Substitute 10^4 for A_0, 100 Hz for f_{p1} in the equation.

$$A(s) = \frac{10^4}{\left(1+\frac{s}{100}\right)}$$

Step 3 of 5

Consider the 3-dB frequency of 10 kHz.

The 3-dB frequency of frequency f_0 is,

$$f_0 = f_{p1}(1+A_0\beta)$$

Here, β is the feedback factor.

Substitute 10^4 for A_0, 100 Hz for f_{p1} and 10 kHz for f_0 in equation.

$$10\,kHz = 100\,Hz(1+10^4\beta)$$
$$10\times10^3 = 100(1+10^4\beta)$$
$$(100-1) = 10^4\beta$$
$$\beta = 99\times10^{-4}$$

Therefore, the feedback factor β is $\boxed{99\times10^{-4}}$.

Step 4 of 5

The dc closed-loop gain $A_f(0)$ is,

$$A_f(0) = \frac{A_0}{1+A_0\beta}$$

Substitute 10^4 for A_0, 99×10^{-4} for β in equation.

$$A_f(0) = \frac{10^4}{1+10^4(99\times10^{-4})}$$
$$= \frac{10^4}{100}$$
$$= 100$$

Thus, the dc closed loop gain $A_f(0)$ is $\boxed{100}$.

Step 5 of 5

The expression of closed loop transfer function is,

$$A_f(s) = \frac{\frac{A_0}{\left(1+\frac{s}{f_{p1}}\right)}}{\left(1+\frac{A_0}{\left(1+\frac{s}{f_{p1}}\right)}\beta\right)}$$
$$= \frac{A_0}{\left(\left(1+\frac{s}{f_{p1}}\right)+A_0\beta\right)}$$

Substitute 10^4 for A_0, $100\,Hz$ for f_{p1}, 99×10^{-4} for β in equation.

$$A_f(s) = \frac{10^4}{\left(\left(1+\frac{s}{100}\right)+10^4(99\times10^{-4})\right)}$$
$$= \frac{10^4}{\left(\left(1+\frac{s}{100}\right)+99\right)}$$
$$= \frac{10^4}{\left(100+\frac{s}{100}\right)}$$

Thus, the closed loop gain expression $A_f(s)$ is, $\boxed{\frac{10^4}{\left(100+\frac{s}{100}\right)}}$.

Step 1 of 9

Consider the open loop gain $A(s)$ of the amplifier with two poles.

$$A(s)=\frac{A_0}{\left(1+\frac{s}{\omega_{p1}}\right)\left(1+\frac{s}{\omega_{p2}}\right)}$$

Substitute 10^4 for A_0, 10^4 Hz for ω_{p1} and 10^5 Hz for ω_{p2} in equation.

$$A(s)=\frac{10^4}{\left(1+\frac{s}{10^4}\right)\left(1+\frac{s}{10^5}\right)}$$

Step 2 of 9

(a)

Determine the closed loop poles of feedback-amplifier.

$$1+A(s)\beta=0$$

$$1+\frac{10^4}{\left(1+\frac{s}{10^4}\right)\left(1+\frac{s}{10^5}\right)}\beta=0$$

$$\left(1+\frac{s}{10^4}\right)\left(1+\frac{s}{10^5}\right)+10^4\beta=0$$

$$\left(10^4+s\right)\left(10^5+s\right)+\left(10^4\right)\left(10^4\right)\left(10^5\right)\beta=0$$

$$\left(s^2+\left(11\times10^4\right)s+10^9\right)+\left(10^{13}\right)\beta=0$$

$$s^2+\left(11\times10^4\right)s+10^9\left(1+10^4\beta\right)=0 \quad \text{..... (1)}$$

Compare with standard generalized second order equation.

$$s^2+2\zeta\omega_0 s+\omega_0^2=0$$

To ensure the poles coincidence, damping factor should be unity.

$$\zeta=1$$

Then, the modified standard second order equation is,

$$s^2+2\omega_0 s+\omega_0^2=0 \quad \text{..... (2)}$$

Step 3 of 9

Compare the equation (1) with equation (2).

$$2\omega_0=11\times10^4$$

$$\omega_0=\frac{11\times10^4}{2}$$

$$=5.5\times10^4 \text{ Hz}$$

Therefore, the frequency ω_0 is $\boxed{5.5\times10^4 \text{ Hz}}$.

Step 4 of 9

Determine the feedback factor β.

Compare the equation (1) with equation (2).

$$\omega_0^2=10^9\left(1+10^4\beta\right)$$

Substitute 5.5×10^4 Hz for ω_0 in equation.

$$\left(55\times10^3\right)^2=10^9\left(1+10^4\beta\right)$$

$$1+10^4\beta=\frac{3025\times10^6}{10^9}$$

$$1+10^4\beta=3.025$$

$$10^4\beta=2.025$$

$$\beta=2.025\times10^{-4}$$

Therefore, the feedback factor β of the amplifier is $\boxed{2.025\times10^{-4}}$.

Step 5 of 9

(b)

Calculate the low-frequency gain A_f.

$$A_f(0)=\frac{A_0}{1+A\beta}$$

Substitute 2.025×10^{-4} for β and 10^4 for A_0.

$$A_f(0)=\frac{10^4}{1+10^4\left(2.025\times10^{-4}\right)}$$

$$=\frac{10^4}{1+2.025}$$

$$=\frac{10^4}{3.025}$$

$$=3305.78 \text{ V/V}$$

Therefore, the low-frequency gain A_f is $\boxed{3305.78 \text{ V/V}}$.

Step 6 of 9

Calculate the closed loop gain A_f.

$$A_f=\frac{A_f(0)}{1+|A\beta|}$$

Substitute 3305.78 V/V for $A_f(0)$ and 1 for $|A\beta|$ in A_f.

$$A_f=\frac{3305.78}{1+1}$$

$$=\frac{3305.78}{2}$$

$$=1652.89 \text{ V/V}$$

Therefore, the value of closed loop gain A_f is $\boxed{1652.89 \text{ V/V}}$.

Step 7 of 9

(c)

Determine the value of quality factor Q.

The generalized form of the characteristic equation is as follows:

$$s^2+\frac{\omega_0}{Q}s+\omega_0^2=0 \quad \text{..... (3)}$$

Compare the equation (1) with equation (3).

$$\frac{\omega_0}{Q}=11\times10^4$$

$$Q=\frac{\omega_0}{11\times10^4}$$

Substitute 5.5×10^4 Hz for ω_0 in equation.

$$Q=\frac{5.5\times10^4}{11\times10^4}$$

$$=\frac{1}{2}$$

$$=0.5$$

Therefore, the quality factor Q of the amplifier is $\boxed{0.5}$.

Step 8 of 9

(d)

Calculate the new pole locations if β is increased by a factor of 10.

Thus, the new value of β is,

$$\beta=\left(2.025\times10^{-4}\right)10$$

$$=20.25\times10^{-4}$$

From equation (1), the characteristic equation is,

$$s^2+\left(11\times10^4\right)s+10^9\left(1+10^4\beta\right)=0$$

Substitute 20.25×10^{-4} for β in equation.

$$s^2+\left(11\times10^4\right)s+10^9\left(1+10^4\left(20.25\times10^{-4}\right)\right)=0$$

$$s^2+\left(11\times10^4\right)s+10^9\left(1+20.25\right)=0$$

$$s^2+\left(11\times10^4\right)s+\left(21.25\times10^9\right)=0 \quad \text{..... (4)}$$

Step 9 of 9

Compare equations (3) and (4).

$$\omega_0^2=21.25\times10^9$$

$$\omega_0=\sqrt{21.25\times10^9}$$

$$=1.46\times10^5$$

$$\frac{\omega_0}{Q}=11\times10^4$$

$$Q=\frac{\omega_0}{11\times10^4}$$

Substitute 1.46×10^5 for ω_0.

$$Q=\frac{1.46\times10^5}{11\times10^4}$$

$$=1.33$$

Therefore, the quality factor Q of the amplifier is $\boxed{1.33}$.

Step 1 of 5

Consider the closed loop gain A_f .

$$A_f = \frac{A_0}{1 + A_0\beta}$$

Obtain the value of β from the equation.

$$A_f + A_0A_f\beta = A_0$$
$$A_0A_f\beta = A_0 - A_f$$
$$\beta = \frac{A_0 - A_f}{A_fA_0}$$

Substitute the values of 1000 for A_0 , 10 for A_f in equation.

$$\beta = \frac{1000-10}{10\times1000}$$
$$= 99\times10^{-3}$$

Therefore, the feedback factor β is $\boxed{99\times10^{-3}}$.

Step 2 of 5

Consider the required expression for open loop gain A .

$$A = \frac{A_0}{\left(1+\frac{s}{\omega_{p1}}\right)\left(1+\frac{s}{\omega_{p2}}\right)}$$

The pole Q factor for the standard second order network expression is,

$$Q = \frac{\sqrt{(1+A_0\beta)f_{p1}f_{p2}}}{f_{p1}+f_{p2}}.$$

Consider $Q = \frac{1}{\sqrt{2}}$ for maximally flat response.

Step 3 of 5

Determine the pole frequency f_{p2} as follows:

$$\frac{1}{\sqrt{2}} = \frac{\sqrt{(1+A\beta)f_{p1}f_{p2}}}{f_{p1}+f_{p2}}$$

$$\frac{1}{2} = \frac{(1+A\beta)f_{p1}f_{p2}}{(f_{p1}+f_{p2})^2}$$

$$(f_{p1}+f_{p2})^2 = 2(1+A\beta)f_{p1}f_{p2}$$

Substitute the values of 1000 for A_0 , $1000\ \text{Hz}$ for f_{p1} in equation.

$$(1000+f_{p2})^2 = 2\left[1+(1000\times99\times10^{-3})\right](1000)f_{p2}$$
$$(1000+f_{p2})^2 = 2\left[1+(99)\right](1000)f_{p2}$$
$$(1000+f_{p2})^2 = 200\times10^3 f_{p2}$$
$$f_{p2}^2 + 2\times10^3 f_{p2} + 10^6 = 200\times10^3 f_{p2}$$

Further simplification is as follows:

$$f_{p2}^2 - 198\times10^3 f_{p2} + 10^6 = 0$$
$$f_{p2}^2 - 198\times10^3 f_{p2} + 10^6 = 0$$
$$f_{p2} = \frac{-(-198\times10^3)\pm\sqrt{(-198\times10^3)^2 - 4\times1\times10^6}}{2\times1}$$
$$= \frac{198\times10^3 \pm 197.99\times10^3}{2}$$
$$= 197.9\ \text{kHz (or) } 5.05\ \text{Hz}$$

Step 4 of 5

Since, the $f_{p1} = 1\ \text{kHz}$ is dominant pole, the second pole f_{p2} is must be greater than f_{p2} .

Thus,

$$f_{p2} > f_{p1}$$
$$> 1\ \text{kHz}$$

Hence, the choose the second pole frequency f_{p2} as $197.9\ \text{kHz}$.

Therefore, the second pole frequency f_{p2} is placed at $\boxed{197.9\ \text{kHz}}$.

Step 5 of 5

The 3-dB frequency of the closed loop system is,

$$\omega_0 = \omega_{p1}(1+A_0\beta)$$
$$f_0 = f_{p1}(1+A_0\beta)$$

Substitute 1000 for A_0 , 99×10^{-3} for β and $1\ \text{kHz}$ for f_{p1} in equation.

$$f_0 = 1\times10^3\left(1+1000\left(99\times10^{-3}\right)\right)$$
$$= 1\times10^3(1+99)$$
$$= 100\ \text{kHz}$$

Therefore, the 3-dB frequency f_0 is $\boxed{100\ \text{kHz}}$.

Step 1 of 8

11.091P

Refer to Example 10.9 in the text book.

Taped network of Figure 10.40 is shown in Figure 1 in which the components immediately adjacent to the amplifier inputs are raised in impedance to $C/10$ and $10R$.

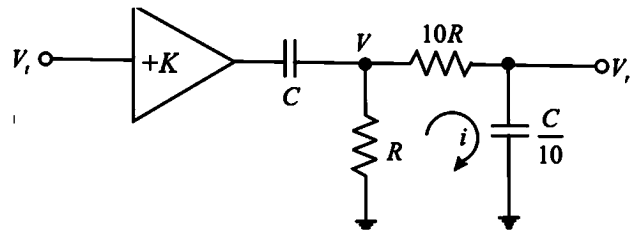


Figure 1

Step 2 of 8

Find the current from the circuit shown in Figure 1.

$$i = \frac{V_r}{X_C}$$
$$= \frac{V_r}{\frac{10}{sC}}$$
$$i = \frac{sCV_r}{10}$$

The voltage V is,

$$V_r = V - sCRV_r$$
$$V = V_r + sCRV_r$$

$$V = V_r(1 + SCR) \quad \text{..... (1)}$$

Step 3 of 8

Apply Kirchhoff's current law at node 'V'.

$$\frac{KV_r - V}{\frac{1}{sC}} - \frac{V}{R} - i = 0$$
$$sC(KV_r - V) - \frac{V}{R} - \frac{sCV_r}{10} = 0$$
$$sC(KV_r - V_r(1 + sCR)) - \frac{V_r(1 + sCR)}{R} - \frac{sCV_r}{10} = 0$$
$$KsCV_r - sCV_r(1 + sCR) - V_r\left(\frac{1 + sCR}{R} + \frac{sC}{10}\right) = 0$$

Further simplification as,

$$KsRCV_r - sRCV_r(1 + sCR) - V_r(1 + sCR + 0.1sRC) = 0$$
$$KsRCV_r = V_r\left[(sCR)^2 + 2.1sCR + 1\right]$$
$$\frac{V_r}{V_i} = \frac{KsRC}{\left[(sCR)^2 + 2.1sCR + 1\right]}$$
$$\frac{V_r}{V_i} = \frac{1}{(RC)^2}\frac{KsRC}{\left[s^2 + 2.1\frac{s}{CR} + \frac{1}{(CR)^2}\right]}$$
$$\frac{V_r}{V_i} = \frac{k\left(\frac{s}{RC}\right)}{\left[s^2 + 2.1\frac{s}{CR} + \frac{1}{(CR)^2}\right]}$$

Step 4 of 8

Find the expression for the resulting frequency ω_0 and Q factor.

The loop transmission is,

$$L(s) \triangleq -\frac{V_r}{V_i}$$
$$= \frac{-K\left(\frac{s}{RC}\right)}{\left[s^2 + 2.1\frac{s}{CR} + \frac{1}{(CR)^2}\right]}$$
$$L(s) = \frac{-\left(\frac{K}{CR}\right)s}{s^2 + \left(\frac{\omega_0}{Q}\right)s + \omega_0^2}$$

Step 5 of 8

The characteristic equation is,

$$1 + L(s) = 0$$
$$1 + \frac{-K\left(\frac{s}{RC}\right)}{\left[s^2 + 2.1\frac{s}{CR} + \frac{1}{(CR)^2}\right]} = 0$$
$$s^2 + 2.1\frac{s}{CR} + \frac{1}{(CR)^2} - K\left(\frac{s}{RC}\right) = 0$$
$$s^2 + (2.1 - K)\frac{s}{CR} + \frac{1}{(CR)^2} = 0$$
$$s^2 + \left(\frac{\omega_0}{Q}\right)s + \omega_0^2 = 0$$

Compare the two equations.

$$\omega_0 = \frac{1}{RC} \text{ and } Q = \frac{1}{2.1 - K}$$

Step 6 of 8

Find the value of K by considering the poles.

Poles coincide, when $Q = \frac{1}{2}$.

$$\frac{1}{2.1 - K} = \frac{1}{2}$$
$$2.1 - K = 2$$
$$K = 2.1 - 2$$
$$K = 0.1$$

Therefore, the value of K is $\boxed{0.1}$.

Step 7 of 8

Find the value of K for the response to become maximally flat, that is, when $Q = \frac{1}{\sqrt{2}}$

$$\frac{1}{2.1 - K} = \frac{1}{\sqrt{2}}$$
$$2.1 - K = 1.414$$
$$K = 2.1 - 1.414$$
$$K = 0.686$$

Therefore, the value of K is $\boxed{0.686}$.

Step 8 of 8

Find the value of K at which the circuit oscillates, that is, when Q tends to infinity

$$Q = \infty$$
$$\frac{1}{2.1 - K} = \infty$$
$$2.1 - K = 0$$
$$K = 2.1$$

Therefore, the value of K is $\boxed{2.1}$.

Step 1 of 8

The expression for the open loop transfer function of a single amplifier is,

$$A_1(s)=\frac{A_0}{1+\frac{s}{\omega_p}} \quad \text{..... (1)}$$

Consider the block diagram of the feedback amplifier as shown in figure 1.

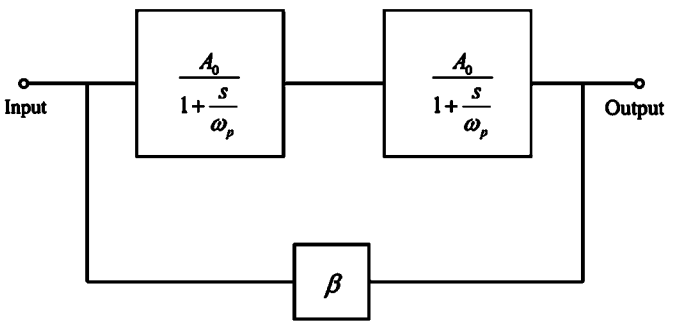


Figure 1

Step 2 of 8

The expression for the open-loop transfer function of the cascaded feedback amplifier is,

$$A(s)=\frac{A_0^2}{\left(1+\frac{s}{\omega_p}\right)^2}$$

The expression for the closed-loop transfer function of the cascaded feedback amplifier is,

$$A_f(s)=\frac{A_0^2}{\left(1+\frac{s}{\omega_p}\right)^2+A_0^2\beta}$$

$$A_f(s)=\frac{A_0^2\omega_p^2}{s^2+2\omega_p s+\omega_p^2+A_0^2\beta\omega_p^2}$$

$$A_f(s)=\frac{A_0^2\omega_p^2}{s^2+2\omega_p s+\omega_p^2\left(1+A_0^2\beta\right)} \quad \text{..... (2)}$$

Determine the expression for DC closed-loop gain.

Substitute 0 for *s* in equation (2).

$$\text{DC gain}=\frac{A_0^2\omega_p^2}{\left(1+A_0^2\beta\right)\omega_p^2}$$

$$\text{DC gain}=\frac{A_0^2}{1+A_0^2\beta}$$

Substitute 10 for DC gain.

$$\frac{A_0^2}{1+A_0^2\beta}=10 \quad \text{..... (3)}$$

Step 3 of 8

Calculate the radial frequency ω_p.

$$\omega_p=2\pi f_p$$

Substitute 10×10^3 for f_p.

$$\omega_p=2\pi\times10\times10^3$$
$$=2\times10^4\pi \text{ rad/s}$$

Step 4 of 8

Consider the following condition for maximally flat response:

$$Q=\frac{1}{\sqrt{2}}$$

Refer to equation 11.70 for the expression of quality factor Q given in the textbook.

Substitute ω_p for both ω_{p1} and ω_{p2}.

The expression for the quality factor Q is,

$$Q=\frac{\sqrt{\left(1+A_0\beta\right)\omega_p^2}}{2\omega_p}$$

Substitute $\frac{1}{\sqrt{2}}$ for Q.

$$\frac{1}{\sqrt{2}}=\frac{\sqrt{\left(1+A_0\beta\right)\omega_p^2}}{2\omega_p}$$
$$\frac{1}{2}=\frac{\left(1+A_0\beta\right)\omega_p^2}{4\omega_p^2}$$
$$2=1+A_0\beta$$
$$A_0\beta=1 \quad \text{..... (4)}$$

Step 5 of 8

Recall equation (3).

$$\frac{A_0^2}{1+A_0^2\beta}=10$$
$$\frac{A_0^2}{1+\left(A_0\beta\right)A_0}=10$$

Substitute 1 for $A_0\beta$.

$$\frac{A_0^2}{1+(1)A_0}=10$$
$$\frac{A_0^2}{1+A_0}=10$$
$$A_0^2-10A_0-10=0$$

Solve for A_0.

$$A_0=\frac{10\pm\sqrt{100+40}}{2}$$
$$=5\pm\frac{\sqrt{140}}{2}$$
$$=5\pm5.916$$

Therefore, the values for A_0 are 10.196 and ◊◊◊◊◊◊

Consider 10.196 for A_0 because A_0 is always greater than β.

Step 6 of 8

Recall equation (4).

$$A_0\beta=1$$

Substitute 10.196 for A_0.

$$10.196\beta=1$$
$$\beta=\frac{1}{10.196}$$
$$\beta=0.098$$

Therefore, the value of β is $\boxed{0.098}$.

Step 7 of 8

Recall equation (1).

$$A_1(s)=\frac{A_0}{1+\frac{s}{\omega_p}}$$

Substitute $j\omega_0$ for *s*.

$$A_1(j\omega_0)=\frac{A_0}{1+\frac{j\omega_0}{\omega_p}}$$

Determine the magnitude of the transfer function $A_1(j\omega)$.

$$\left|A_1(j\omega)\right|=\frac{A_0}{\sqrt{1+\left(\frac{\omega_0}{\omega_p}\right)^2}}$$

Substitute $\frac{1}{\sqrt{2}}$ for $\left|A_1(j\omega)\right|$.

$$\frac{1}{\sqrt{2}}=\frac{A_0}{\sqrt{1+\left(\frac{\omega_0}{\omega_p}\right)^2}}$$
$$1+\left(\frac{\omega_0}{\omega_p}\right)^2=2A_0^2$$

Step 8 of 8

Substitute $2\times10^4\pi$ for ω_p, and 10.196 for A_0.

$$1+\left(\frac{\omega_0}{2\times10^4\pi}\right)^2=2\times(10.196)^2$$
$$\left(2\times10^4\pi\right)^2+\omega_0^2=\left(2\times10^4\pi\right)^2(2)(10.196)^2$$
$$\omega_0^2=\left(2\times10^4\pi\right)^2\left[2(10.196)^2-1\right]$$
$$(2\pi f_0)^2=\left(2\times10^4\pi\right)^2(206.917)$$

Simplify further.

$$(f_0)^2=\left(10^4\right)^2(206.917)$$
$$f_0=10^4\sqrt{206.917}$$
$$f_0=143.846 \text{ kHz}$$

Therefore, the 3-dB frequency of each of the two amplifier stages is $\boxed{143.846 \text{ kHz}}$.

Recall equation (2).

$$A_f(s)=\frac{A_0^2\omega_p^2}{s^2+2\omega_p s+\omega_p^2\left(1+A_0^2\beta\right)}$$

Therefore, the expression for $A_f(s)$ is,

$$\boxed{A_f(s)=\frac{A_0^2\omega_p^2}{s^2+2\omega_p s+\omega_p^2\left(1+A_0^2\beta\right)}}$$

Step 1 of 5

Consider the circuit shown in Figure 1.

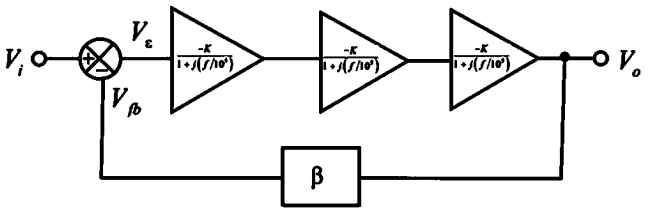


Figure 1

Step 2 of 5

For a feedback amplifier for which the open-loop transfer function of single stage $A_1(f)$ is,

$$A_1(f)=\frac{-K}{1+j(f/10^5)}$$

Feedback amplifier for which the open-loop transfer function $A(f)$ is,

$$\begin{aligned}A(f)&=A_1(f)A_2(f)A_3(f)\\&=\left(\frac{-K}{1+j(f/10^5)}\right)\left(\frac{-K}{1+j(f/10^5)}\right)\left(\frac{-K}{1+j(f/10^5)}\right)\\&=\frac{-K^3}{\left(1+j(f/10^5)\right)^3}\end{aligned}$$

Step 3 of 5

Consider the feedback factor, β as 1

The loop gain is.

$$A\beta=\frac{-K^3}{\left(1+j(f/10^5)\right)^3}(1)$$

$$A\beta=\frac{-K^3}{\left(1+j(f/10^5)\right)^3}$$

Step 4 of 5

The phase of the loop gain is,

$$\phi=3\tan^{-1}\left(1+\left(\frac{f_f}{10^5}\right)\right)$$

Find the frequency at which the phase margin is equal to $180°$.

$$180°=3\tan^{-1}\left(\left(\frac{f_f}{10^5}\right)\right)$$

$$\tan^{-1}\left(\left(\frac{f_f}{10^5}\right)\right)=\frac{180°}{3}$$

$$\tan^{-1}\left(\left(\frac{f_f}{10^5}\right)\right)=60°$$

$$\left(\frac{f_f}{10^5}\right)=\tan 60°$$

$$\left(\frac{f_f}{10^5}\right)=1.73$$

$$f_f=1.73\times10^5$$

Thus, the frequency of oscillation f_f is $\boxed{1.73\times10^5\ \text{Hz}}$.

Step 5 of 5

At this frequency (f_f), the loop gain magnitude is unity.

$$\left|\frac{K^3}{\left(1+j(f/10^5)\right)^3}\right|=|A\beta|$$

$$\frac{K^3}{\left(\sqrt{1+(f/10^5)^2}\right)^3}=1$$

$$K=\sqrt{1+(f/10^5)^2}$$

Substitute $1.73\times10^5\ \text{Hz}$ for f_f .

$$K=\sqrt{1+(1.73\times10^5/10^5)}$$

$$K=\sqrt{2.73}$$

$$K=1.65$$

Thus, the minimum value of K at which the circuit oscillates is $\boxed{1.65}$.

Step 1 of 1

11.094P

Compute, the phase margin, ϕ .

$$\phi = 180° - \tan^{-1}\left(\frac{f}{f_p}\right)$$

$$= 180° - \tan^{-1}\left(\frac{10^6}{10}\right)$$

$$= 180° - \tan^{-1}\left(10^5\right)$$

$$= 180° - 89.99^0$$

$$= 90.01°$$

Therefore, the required phase margin is $\boxed{90°}$.

Step 1 of 4

The transfer function can be characterized by two real axis poles.

$$A=\frac{10^5}{\left(1+\frac{jf}{f_{P1}}\right)\left(1+\frac{jf}{f_{P2}}\right)}$$

Substitute $f_{P2}=10^3$ and $f_{P1}=10\ \text{Hz}$.

$$A=\frac{10^5}{\left(1+\frac{jf}{10}\right)\left(1+\frac{jf}{10^3}\right)}$$

Step 2 of 4

Find the frequency at which $|A\beta|=1$.

$$|A\beta|=\left|\frac{10^5}{\left(1+\frac{jf}{10}\right)\left(1+\frac{jf}{10^3}\right)}\right|\beta$$

$$1=\left|\frac{\beta 10^5}{\left(1+\frac{jf}{10}\right)\left(1+\frac{jf}{10^3}\right)}\right|$$

$$\left|\left(1+\frac{jf}{10}\right)\left(1+\frac{jf}{10^3}\right)\right|=\beta 10^5$$

$$\sqrt{\left(1+\frac{f^2}{10^2}\right)\left(1+\frac{f^2}{10^6}\right)}=(0.01)10^5$$

$$\left(1+\frac{f^2}{10^2}\right)\left(1+\frac{f^2}{10^6}\right)=10^6$$

$$1+\frac{f^2}{10^6}+\frac{f^2}{10^2}+\frac{f^4}{10^6}=10^6$$

$$f^4+f^2\left(10^6+10^2\right)-\left(10^6\right)\left(10^2\right)\left(10^6\right)=0$$

$$f^2=\frac{-\left(10^6+10^2\right)+\sqrt{\left(10^6+10^2\right)^2+4\times10^6\times10^6\times10^2}}{2}$$

$$f^2=9.512\times10^6$$

$$f=3.084\ \text{kHz}$$

Therefore, the required frequency f is $\boxed{3.084\ \text{kHz}}$.

Step 3 of 4

Calculate the phase margin.

$$\phi=180°-\tan^{-1}\left(\frac{f}{f_{P1}}\right)-\tan^{-1}\left(\frac{f}{f_{P2}}\right)$$

$$=180°-\tan^{-1}\left(\frac{3.084\times10^3}{10}\right)-\tan^{-1}\left(\frac{3.084\times10^3}{10^3}\right)$$

$$=180°-\tan^{-1}(89.8)-\tan^{-1}(3.084)$$

$$=180°-90°-72°$$

$$=18°$$

Therefore, the phase margin is $\boxed{18°}$.

Step 4 of 4

Calculate the value of β for phase margin greater than or equal $45°$.

$$\tan^{-1}\left(\frac{f_1}{10^3}\right)\le 45°$$

$$\frac{f_1}{10^3}\le\tan(45°)\qquad(\text{Since},\tan(45°)=1)$$

$$f_1\le10^3$$

$$|A\beta|=1$$

$$1=\frac{10^5\beta}{\sqrt{\left(1+\frac{\left(10^4\right)^2}{(10)^2}\right)\left(1+\frac{\left(10^4\right)^2}{\left(10^3\right)^2}\right)}}$$

$$1=\frac{10^5\beta}{\sqrt{\left(1+\left(10^3\right)^2\right)\left(1+10^2\right)}}$$

$$\sqrt{\left(1+10^6\right)\left(1+10^2\right)}=10^5\beta$$

$$10049.88=10^5\beta$$

$$\beta=\frac{10049.88}{10^5}$$

$$=0.10049$$

Therefore, the required value of β is $\boxed{0.10049}$.

Step 1 of 6

The magnitude of the gain is

$$A_f(j\omega)=\frac{\frac{1}{\beta}}{\left|1-e^{-j\theta}\right|}$$

Where,

$$\begin{aligned}\left|1+e^{-j\theta}\right| &= \left|1+\cos\theta-j\sin\theta\right| \\ &= \left[(1+\cos\theta)^2+(\sin\theta)^2\right] \\ &= \left[1+2\cos\theta+\cos^2\theta+\sin^2\theta\right]^{1/2} \\ &= \left[1+2\cos\theta+1\right]^{1/2} \\ &= \sqrt{2}(1+\cos\theta)^{1/2}\end{aligned}$$

Step 2 of 6

Determining the phase margin does the gain peaking has a 5%:

$$\begin{aligned}1+\cos\theta &= \frac{1}{\left(1+\frac{5}{100}\right)^2(2)} \\ &= \frac{1}{(1+0.05)^2(2)} \\ &= \frac{1}{(1.05)^2(2)}\end{aligned}$$

$$\begin{aligned}1+\cos\theta &= \frac{1}{(1.05)^2(2)} \\ \cos\theta &= 0.4535 \\ \theta &= \cos^{-1}(-0.5465) \\ &= 123.13^\circ\end{aligned}$$

Step 3 of 6

The phase margin is,

$$\begin{aligned}\phi &= 180^\circ-123.13^\circ \\ &= 56.87^\circ\end{aligned}$$

Therefore, the phase margin ϕ is $\boxed{56.87^\circ}$.

Step 4 of 6

Determining the phase margin does the gain peaking has a 10%:

$$\begin{aligned}1+\cos\theta &= \frac{1}{\left(1+\frac{10}{100}\right)^2(2)} \\ &= \frac{1}{(1+0.1)^2(2)} \\ &= \frac{1}{(1.1)^2(2)} \\ &= \frac{1}{(1.1)^2(2)} \\ 1+\cos\theta &= 0.4132 \\ \cos\theta &= -0.5868 \\ \theta &= \cos^{-1}(-0.5868) \\ &= 125.93^\circ\end{aligned}$$

Therefore, the phase margin is

$$\begin{aligned}\phi &= 180^\circ-125.93^\circ \\ &= 54.07^\circ\end{aligned}$$

Therefore, the phase margin ϕ is $\boxed{54.07^\circ}$.

Step 5 of 6

Determining the phase margin does the gain peaking has a 0.1dB:

$$\begin{aligned}20\log x &= 0.1 \\ \log x &= \frac{0.1}{20} \\ x &= 10^{\left(\frac{0.1}{20}\right)} \\ &= 1.0116\end{aligned}$$

$$\begin{aligned}1+\cos\theta &= \frac{1}{(1.0116)^2 2} \\ &= 0.4886 \\ &= -1+0.4886 \\ \cos\theta &= -0.5114 \\ \theta &= \cos^{-1}(-0.5114) \\ &= 120.76^\circ\end{aligned}$$

Therefore, the phase margin is,

$$\begin{aligned}\phi &= 180^\circ-120.76^\circ \\ &= 59.24^\circ\end{aligned}$$

Therefore, the phase margin is $\boxed{59.24^\circ}$.

Step 6 of 6

Determining the phase margin does the gain peaking has a 1 dB:

$$\begin{aligned}20\log x &= 1 \\ \log x &= \frac{1}{20} \\ x &= 10^{\left(\frac{1}{20}\right)} \\ &= 1.122\end{aligned}$$

$$\begin{aligned}1+\cos\theta &= \frac{1}{(1.122)^2 2} \\ 1+\cos\theta &= 0.3972 \\ \cos\theta &= -1+0.3972 \\ &= -0.6028 \\ \theta &= \cos^{-1}(-0.6028) \\ &= 127.07^\circ\end{aligned}$$

Therefore, the phase margin is

$$\begin{aligned}\phi &= 180^\circ-127.07^\circ \\ &= 52.93^\circ\end{aligned}$$

Therefore, the phase margin ϕ is $\boxed{52.93^\circ}$.

Step 1 of 5

Consider the open loop gain the amplifier with three poles.

$$A(jf)=\frac{A_0}{\left(1+\frac{jf}{f_{P1}}\right)\left(1+\frac{jf}{f_{P2}}\right)\left(1+\frac{jf}{f_{P3}}\right)} \quad \text{..... (1)}$$

Here,

A_0 is low-frequency gain,

f_{P1}, f_{P2} and f_{P3} are poles.

Step 2 of 5

Substitute 10^5 Hz for f_{P1}, 3.16×10^5 Hz for f_{P2}, 10^6 Hz for f_{P3} and 10^5 for A_0 in equation (1).

$$A(jf)=\frac{10^5}{\left(1+\frac{jf}{10^5}\right)\left(1+\frac{jf}{3.16\times10^5}\right)\left(1+\frac{jf}{10^6}\right)}$$

Consider the phase margin of $45°$.

Thus, the phase margin is,

$$P.M=180°-\theta$$
$$\theta=180°-P.M$$

Substitute $45°$ for $P.M$ in equation.

$$\theta=180°-45°$$
$$=135°$$

Step 3 of 5

Determine the phase angle θ.

$$\theta=-\left(\tan^{-1}\frac{f_1}{10^5}+\tan^{-1}\frac{f_1}{3.16\times10^5}+\tan^{-1}\frac{f_1}{10^6}\right)$$
$$135°=\left(\tan^{-1}\frac{f_1}{10^5}+\tan^{-1}\frac{f_1}{3.16\times10^5}+\tan^{-1}\frac{f_1}{10^6}\right)$$

Apply tan on both sides of equation.

$$\tan(135°)=\tan\left(\tan^{-1}\frac{f}{10^5}+\tan^{-1}\frac{f}{3.16\times10^5}+\tan^{-1}\frac{f}{10^6}\right)$$

Solve the equation as follows:

$$\tan(a+b+c)=\frac{\tan a+\tan b+\tan c-\tan a\tan b\tan c}{1-\tan a\tan b-\tan b\tan c-\tan c\tan a}$$

$$-1=\frac{\left(\frac{f}{10^5}\right)+\left(\frac{f}{3.16\times10^5}\right)+\left(\frac{f}{10^6}\right)-\left(\frac{f}{10^5}\times\frac{f}{3.16\times10^5}\times\frac{f}{10^6}\right)}{1-\left(\frac{f}{10^5}\right)\left(\frac{f}{3.16\times10^5}\right)-\left(\frac{f}{3.16\times10^5}\right)\left(\frac{f}{10^6}\right)-\left(\frac{f}{10^5}\right)\left(\frac{f}{10^6}\right)}$$

$$\frac{f}{10^5}+\frac{f}{3.16\times10^5}+\frac{f}{10^6}-\frac{f^3}{3.16\times10^{16}}=-1+\frac{f^2}{3.16\times10^{10}}+\frac{f^2}{3.16\times10^{11}}+\frac{f^2}{10^{11}}$$

$$\left\{\begin{aligned}&f^3\left(\frac{1}{3.16\times10^{16}}\right)+f^2\left(\frac{1}{3.16\times10^{10}}+\frac{1}{3.16\times10^{11}}+\frac{1}{10^{11}}\right)\\&-f\left(\frac{1}{10^5}+\frac{1}{3.16\times10^5}+\frac{1}{10^6}\right)-1\end{aligned}\right\}=0$$

$$f^3\left(3.16\times10^{-17}\right)+f^2\left(4.48\times10^{-11}\right)-f\left(1.41\times10^{-5}\right)-1=0$$

By solving the third order equation, the obtained frequencies are,

$$f=3.16\times10^5 \text{ Hz}$$

$$f=-0.59\times10^5 \text{ Hz}$$

$$f=-1.67\times10^6 \text{ Hz}$$

Step 4 of 5

Consider the frequency $f=3.19\times10^5$ Hz. Since, for this frequency only the phase margin obtains about $45°$.

Determination of β as follows.

$$|A(jf)\beta|=1$$

$$\left|\frac{10^5\beta}{\left(1+j\frac{3.16\times10^5}{10^5}\right)\left(1+j\frac{3.16\times10^5}{3.16\times10^5}\right)\left(1+j\frac{3.16\times10^5}{10^6}\right)}\right|=1$$

$$\left|\frac{10^5\beta}{(1+j3.16)(1+j1)(1+j0.316)}\right|=1$$

$$\frac{10^5\beta}{\left(\sqrt{1+(3.16)^2}\right)\left(\sqrt{2}\right)\left(\sqrt{1+(0.316)^2}\right)}=1$$

$$10^5\beta=\left(\sqrt{1+(3.16)^2}\right)\left(\sqrt{2}\right)\left(\sqrt{1+(0.316)^2}\right)$$

$$\beta=\frac{(3.314)(1.41)(1.048)}{10^5}$$
$$\simeq4.9\times10^{-5}$$

Therefore, the value of β is $\boxed{4.9\times10^{-5}}$.

Step 5 of 5

Determining the Corresponding closed loop gain of op-amp.

$$A_f(jf)=\frac{A(jf)}{1+A(jf)\beta}$$

Substitute 10^5 for $A(jf)$, 4.9×10^{-5} for β in equation.

$$A_f(j\omega)=\frac{10^5}{1+\left(10^5\right)\left(4.9\times10^{-5}\right)}$$
$$=\frac{10^5}{5.9}$$
$$=0.169\times10^5$$

Therefore, the closed loop gain of op amp $A_f(jf)$ is $\boxed{16.9\times10^3}$.

Step 1 of 6

The Circuit of a differentiator is shown in Figure 1.

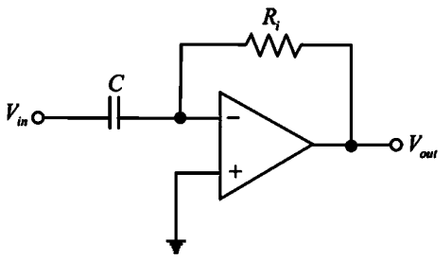


Figure 1

Step 2 of 6

The loop gain of the amplifier is,

$$A(jf) = \frac{A_0}{\left(1+\frac{jf}{f_{p1}}\right)\left(1+\frac{jf}{f_{p2}}\right)}$$

Substitute $1\ \text{MHz}$ for f_{p1}, $10\ \text{MHz}$ for f_{p2} and 10^3 for A_0 in equation.

$$A(jf) = \frac{10^3}{\left(1+\frac{jf}{10^6}\right)\left(1+\frac{jf}{10^7}\right)}$$

Step 3 of 6

Finding the smallest differentiator time constant:

$$A\beta = A_0 \frac{1}{2\pi f_{p_1} CR}$$

$$1 = A_0 \frac{1}{2\pi f_{p_1} CR}$$

$$RC = A_0\left(\frac{1}{2\pi f_{p_1}}\right)$$

Substitute 10^3 for A_0, $1\ \text{MHz}$ for f_{p1} in equation.

$$RC = (10^3)\left(\frac{1}{2\pi(10^6)}\right)$$

$$RC = 0.16\times10^{-3}$$

Therefore, the time constant of differentiator RC is $\boxed{0.16\ \text{ms}}$.

Step 4 of 6

Finding the phase margin of the differentiator:

At $f = 1\ \text{MHz}$,

$$\text{Ang}(\beta) = -90°$$

$$\begin{aligned}\text{Ang}(A) &= -\tan^{-1}\left(\frac{f}{f_{p1}}\right) - \tan^{-1}\left(\frac{f}{f_{p2}}\right)\\ &= -\tan^{-1}\left(\frac{10^6}{10^6}\right) - \tan^{-1}\left(\frac{10^6}{10^7}\right)\\ &= -\tan^{-1}(1) - \tan^{-1}(0.1)\\ &= -45° - 5.7°\end{aligned}$$

$$= -50.7°$$

$$\begin{aligned}\text{PM} &= 180° - \theta\\ &= 180° - (90° + 50.7°)\\ &= 180° - (140.7°)\\ &= 39.3°\end{aligned}$$

Therefore, the phase margin of the differentiator is $\boxed{\text{PM} = 39.3°}$.

Step 5 of 6

Finding the gain margin of the differentiator:

$$\tan^{-1}\frac{f}{10^6} + \tan^{-1}\frac{f}{10^7} = 90°$$

$$\tan\left(\tan^{-1}\frac{f}{10^6} + \tan^{-1}\frac{f}{10^7}\right) = \tan(90°)$$

$$\frac{\frac{f}{10^6} + \frac{f}{10^7}}{1-\left(\frac{f}{10^6}\right)\left(\frac{f}{10^7}\right)} = \infty$$

Further simplification as,

$$1-\left(\frac{f}{10^6}\right)\left(\frac{f}{10^7}\right) = 0$$

$$f^2\left(\frac{1}{10^6}\right)\left(\frac{1}{10^7}\right) = 1$$

$$f^2 = (10^6)(10^7)$$

$$f = \sqrt{(10^6)(10^7)}$$

$$= 3.16\ \text{MHz}$$

Step 6 of 6

Therefore, the gain margin is

$$20\log|A\beta(f_{180})| = 20\log|A| + 20\log|\beta|$$

$$\text{GM} = 20\log|A| - 20\log\left|\frac{1}{\beta}\right|$$

At a phase cross over frequency $|A|$ has falls down by $10\ \text{dB}$ and $|\beta|$ has rises by $10\ \text{dB}$.

$$\begin{aligned}\text{GM} &= 10\ \text{dB} - (-10\ \text{dB})\\ &= 20\ \text{dB}\end{aligned}$$

Therefore, the gain margin of the differentiator is $\boxed{20\ \text{dB}}$.

Step 1 of 12

Refer to Figure 10.43 in the text book.

Find the minimum closed loop voltage gain for phase margin $90°$.

Phase can be obtained as,

$$\tan^{-1}\frac{f_1}{10^5}+\tan^{-1}\frac{f_1}{10^6}+\tan^{-1}\frac{f_1}{10^7}=90°$$

For $f_1=2\times10^5\ \text{Hz}$:

$$\tan^{-1}\frac{f_1}{10^5}+\tan^{-1}\frac{f_1}{10^6}+\tan^{-1}\frac{f_1}{10^7}=90°$$
$$\tan^{-1}\frac{2\times10^5}{10^5}+\tan^{-1}\frac{2\times10^5}{10^6}+\tan^{-1}\frac{2\times10^5}{10^7}=90°$$
$$\tan^{-1}(2)+\tan^{-1}(0.2)+\tan^{-1}(0.02)=90°$$
$$63.43°+11.31°+1.15°=90°$$
$$75.89°\neq90°$$

Step 2 of 12

For $f_1=3\times10^5\ \text{Hz}$:

$$\tan^{-1}\frac{f_1}{10^5}+\tan^{-1}\frac{f_1}{10^6}+\tan^{-1}\frac{f_1}{10^7}=90°$$
$$\tan^{-1}\frac{3\times10^5}{10^5}+\tan^{-1}\frac{3\times10^5}{10^6}+\tan^{-1}\frac{3\times10^5}{10^7}=90°$$
$$\tan^{-1}(3)+\tan^{-1}(0.3)+\tan^{-1}(0.03)=90°$$
$$71.6°+16.7°+1.72°=90°$$
$$89.72°=90°$$

Step 3 of 12

For $f_1=4\times10^5\ \text{Hz}$:

$$\tan^{-1}\frac{f_1}{10^5}+\tan^{-1}\frac{f_1}{10^6}+\tan^{-1}\frac{f_1}{10^7}=90°$$
$$\tan^{-1}\frac{4\times10^5}{10^5}+\tan^{-1}\frac{4\times10^5}{10^6}+\tan^{-1}\frac{4\times10^5}{10^7}=90°$$
$$\tan^{-1}(4)+\tan^{-1}(0.4)+\tan^{-1}(0.04)=90°$$
$$75.96°+21.8°+2.29°=90°$$
$$100.05°\neq90°$$

Therefore, from the iterations frequency is $f_1=3\times10^5\ \text{Hz}$.

Step 4 of 12

Open loop gain of this amplifier is,

$$\begin{aligned}|A_f(f_1)|&=\frac{10^5}{\sqrt{1+\left(\frac{3\times10^5}{10^5}\right)^2}\sqrt{1+\left(\frac{3\times10^5}{10^6}\right)^2}\sqrt{1+\left(\frac{3\times10^5}{10^7}\right)^2}}\\&=\frac{10^5}{\sqrt{1+(3)^2}\sqrt{1+(0.3)^2}\sqrt{1+(0.03)^2}}\\&=30.28\times10^3\end{aligned}$$

Step 5 of 12

Finding the value of β:

$$|A\beta|=1$$
$$\beta=\frac{1}{|A|}$$
$$\beta=\frac{1}{30.28\times10^3}$$
$$\beta=33.0\times10^{-6}$$

Step 6 of 12

The closed loop gain of this amplifier is,

$$\begin{aligned}A_f(0)&=\frac{A}{1+A\beta}\\&=\frac{10^5}{1+10^5\left(33\times10^{-6}\right)}\\&=2.32\times10^4\end{aligned}$$

Therefore, the closed loop gain of this amplifier is $\boxed{2.32\times10^4}$.

Step 7 of 12

Finding the minimum closed loop voltage gain for phase margin $45°$:

Phase can be obtained as,

$$\tan^{-1}\frac{f_2}{10^5}+\tan^{-1}\frac{f_2}{10^6}+\tan^{-1}\frac{f_2}{10^7}=180-45°$$

For $f_2=0.5\times10^6\ \text{Hz}$

$$\tan^{-1}\frac{f_2}{10^5}+\tan^{-1}\frac{f_2}{10^6}+\tan^{-1}\frac{f_2}{10^7}=(180-45)°$$
$$\tan^{-1}\frac{0.5\times10^6}{10^5}+\tan^{-1}\frac{0.5\times10^6}{10^6}+\tan^{-1}\frac{0.5\times10^6}{10^7}=135°$$
$$\tan^{-1}(5)+\tan^{-1}(0.5)+\tan^{-1}(0.05)=135°$$
$$78.69°+26.56°+2.86°=135°$$
$$108.11°\neq135°$$

Step 8 of 12

For $f_2=1\times10^6\ \text{Hz}$:

$$\tan^{-1}\frac{f_2}{10^5}+\tan^{-1}\frac{f_2}{10^6}+\tan^{-1}\frac{f_2}{10^7}=(180-45)°$$
$$\tan^{-1}\frac{1\times10^6}{10^5}+\tan^{-1}\frac{1\times10^6}{10^6}+\tan^{-1}\frac{1\times10^6}{10^7}=135°$$
$$\tan^{-1}(10)+\tan^{-1}(1)+\tan^{-1}(0.1)=135°$$
$$84.28°+45°+5.71°=135°$$
$$134.99°=135°$$

Step 9 of 12

For $f_2=1.5\times10^6\ \text{Hz}$:

$$\tan^{-1}\frac{f_2}{10^5}+\tan^{-1}\frac{f_2}{10^6}+\tan^{-1}\frac{f_2}{10^7}=(180-45)°$$
$$\tan^{-1}\frac{1.5\times10^6}{10^5}+\tan^{-1}\frac{1.5\times10^6}{10^6}+\tan^{-1}\frac{1.5\times10^6}{10^7}=135°$$
$$\tan^{-1}(15)+\tan^{-1}(1.5)+\tan^{-1}(0.15)=135°$$
$$86.18°+56.31°+8.53°=135°$$
$$151.02°\neq135°$$

Therefore, from the iterations frequency f_2 is $\boxed{1\times10^6\ \text{Hz}}$.

Step 10 of 12

Open loop gain of this amplifier is,

$$\begin{aligned}|A_f(f_2)|&=\frac{10^5}{\sqrt{1+\left(\frac{1\times10^6}{10^5}\right)^2}\sqrt{1+\left(\frac{1\times10^6}{10^6}\right)^2}\sqrt{1+\left(\frac{1\times10^6}{10^7}\right)^2}}\\&=\frac{10^5}{\sqrt{1+(10)^2}\sqrt{1+(1)^2}\sqrt{1+(0.1)^2}}\\&=7\times10^3\end{aligned}$$

Step 11 of 12

Find the value of β.

$$|A\beta|=1$$
$$\begin{aligned}\beta&=\frac{1}{|A|}\\&=\frac{1}{7\times10^3}\\&=1.43\times10^{-4}\end{aligned}$$

Step 12 of 12

The closed loop gain of this amplifier is,

$$\begin{aligned}A_f(0)&=\frac{A}{1+A\beta}\\&=\frac{10^5}{1+10^5\left(1.43\times10^{-4}\right)}\\&=\frac{10^5}{1+(14.33)}\\&=6523.15\end{aligned}$$

Therefore, the closed loop gain of this amplifier $A_f(0)$ is $\boxed{6523.15}$.

11.100P

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The new pole frequency is,

$$f_p = \frac{f_1}{A_0}$$

Convert the open loop gain dB into normal value.

$$80 = 20 \log A_0$$

$$4 = \log A_0$$

$$A_0 = 10^4$$

Substitute the A_0 value and $f_1 = 1\ \text{MHz}$.

$$f_p = \frac{1 \times 10^6}{10^4}$$

$$= 1 \times 10^2\ \text{Hz}$$

Therefore, the value of f_p is $\boxed{100\ \text{Hz}}$.

Step 1 of 2

11.101P

Calculate the required new pole is placed at a frequency f_D'.

$$f_D' = \frac{f_{P2}}{A_0} \quad \text{..... (1)}$$

Substitute $A_0 = 10^4$ and $f_{P2} = 20\ \text{MHz}$.

$$\begin{aligned} f_D' &= \frac{20\times10^6}{10^4} \\ &= 20\times10^2\ \text{Hz} \\ &= 2\ \text{kHz} \end{aligned}$$

Step 2 of 2

The capacitance at the controlling node increased factor is,

$$\frac{f_{P1}}{f_D'} = \frac{\left(\frac{1}{2\pi RC}\right)}{\left(\frac{1}{2\pi RC_C}\right)}$$

$$\frac{f_{P1}}{f_D'} = \frac{C_C}{C} \quad \text{..... (2)}$$

Substitute $f_D' = 2\ \text{kHz}$ and $f_{P1} = 1\ \text{MHz}$.

$$\begin{aligned} \frac{C_C}{C} &= \frac{1\times10^6}{2\times10^3} \\ &= 0.5\times10^3 \\ &= C\left(0.5\times10^3\right) \\ C_C &= 500C \end{aligned}$$

Hence, the required factor is $\boxed{500C}$

Therefore, the capacitance at controlling node must be increased.

Step 1 of 5

11.102P

Refer to the Figure 11.38 in the textbook.

In the method of frequency compensation, introduction of new pole in the function of $A(s)$ at frequency, f_D is sufficiently low.

Consider the modified open loop gain $A'(s)$.

For stable operation, $A'(s)$ intersects the $20\log\left(\frac{1}{\beta}\right)$ curve with the slope difference of $20\ \text{dB/decade}$.

Step 2 of 5

(a)

For $\beta = 0.001$:

Substitute 0.001 for β in the curve $20\log\left(\frac{1}{\beta}\right)$.

$$20\log\left(\frac{1}{\beta}\right) = 20\log\left(\frac{1}{10^{-3}}\right)$$
$$= 60\ \text{dB}$$

Step 3 of 5

Determination of lower pole is as follows:

Draw the horizontal line at the $60\ \text{dB}$ level to represent $20\log\left(\frac{1}{\beta}\right)$, as shown in Figure 1.

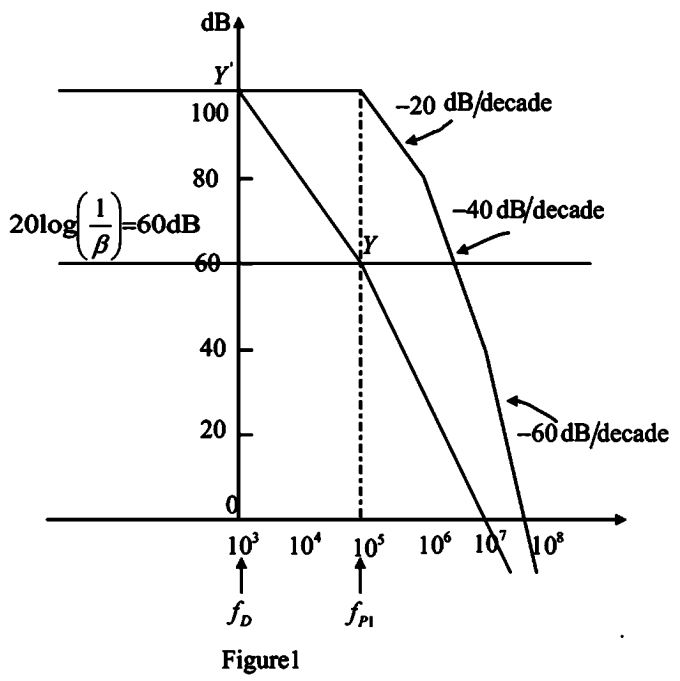


Figure1

1) Then consider the point Y on this line at the frequency of the first pole, f_{P1}.

2) Draw a line with $-20\ \text{dB/decade}$ slope and determine the point at which it intersects the dc gain line, point Y'.

3) The point Y', gives the frequency f_D of the new pole lower than frequency first pole f_{P1}.

Therefore, the lowered frequency for stable performance is $\boxed{10^3\ \text{Hz}}$.

Step 4 of 5

(b)

For $\beta = 0.1$:

Substitute 0.1 for β in the curve $20\log\left(\frac{1}{\beta}\right)$.

$$20\log\left(\frac{1}{\beta}\right) = 20\log\left(\frac{1}{10^{-1}}\right)$$
$$= 20\ \text{dB}$$

Step 5 of 5

Determination of lower pole is as follows:

Draw the horizontal line at the $20\ \text{dB}$ level to represent $20\log\left(\frac{1}{\beta}\right)$, as shown in Figure 2.

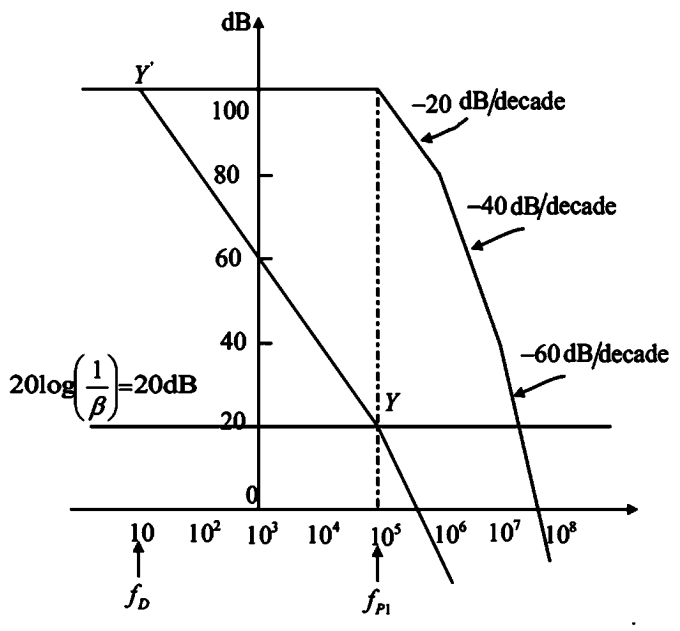


Figure2

1) Then consider the point Y on this line at the frequency of the first pole, f_{P1}.

2) Draw a line with $-20\ \text{dB/decade}$ slope and determine the point at which it intersects the dc gain line, point Y'.

3) The point Y', gives the frequency f_D of the new pole lower than frequency first pole f_{P1}.

Therefore, the lowered frequency for stable performance is $\boxed{10\ \text{Hz}}$.

Step 1 of 3

Write the equation for the second pole.

$$\omega_{p2} = \frac{1}{C_2 R_2}$$

Substitute $R_2 = R$ and $C_2 = C$.

$$\omega_{p2} = \frac{1}{RC}$$

Therefore, the required value of ω_{p2} is $\boxed{\frac{1}{RC}}$.

Step 2 of 3

Write the equation for the first pole, ω'_{p1}.

$$\omega'_{p1} = \frac{1}{g_m R_2 C_f R_1}$$

Substitute $R_2 = R_1 = R$ and $g_m = \frac{100}{R}$.

$$\omega'_{p1} = \frac{1}{\frac{100}{R} RC_f R}$$

Therefore, the required value of ω'_{p1} is $\boxed{\frac{1}{100C_f R}}$.

Step 3 of 3

Consider the ω'_{p2} expression.

$$\begin{aligned}\omega'_{p2} &\approx \frac{g_m C_f}{C_1 C_2 + C_f (C_1 + C_2)} \\ &= \frac{g_m C_f}{C_1 (C_2 + C_f) + C_2 C_f}\end{aligned}$$

From the given condition, $C_f \gg C_2$, the value of ω_{p2} is,

$$\begin{aligned}\omega'_{p2} &= \frac{g_m C_f}{(C_1 + C_2) C_f} \\ &= \frac{g_m}{(C_1 + C_2)}\end{aligned}$$

Substitute $g_m = \frac{100}{R}; C_2 = C$ and $C_1 = 10C$ in ω'_{p2} equation.

$$\begin{aligned}\omega'_{p2} &= \frac{100}{R(10C + C)} \\ &= \frac{100}{11RC} \\ &= \frac{9.09}{RC} \\ &\cong \frac{9.1}{RC}\end{aligned}$$

Therefore, the required value of ω'_{p2} is $\boxed{\frac{9.1}{RC}}$.

Observe that the poles are decreased by a factor by which the respective element value is increased.

Step 1 of 2

11.104P

The fourth pole f_{p4} is must be kept at lower than the lowest amplifier pole that is $f_{p1} = 10^6\ \text{Hz}$. Since, pole f_{p4} will become the dominant pole which is near to imaginary axis.

Determine the value of f_p .

The new pole frequency is,

$$f_{p4} = \frac{f_{p1}}{A_0}$$

Substitute 10^5 for A_0 , $10^6\ \text{Hz}$ for f_{p1} in equation.

$$f_{p4} = \frac{10^6}{10^5}$$
$$= 10\ \text{Hz}$$

Therefore, the frequency of dominant pole f_{p4} is $\boxed{10\ \text{Hz}}$.

Step 2 of 2

Consider the resistance of $1\ \text{M}\Omega$ in the RC low-pass network.

Determine the value of capacitance.

$$f_{p4} = \frac{1}{2\pi RC}$$
$$C = \frac{1}{2\pi R f_{p4}}$$

Substitute $1\ \text{M}\Omega$ for R , $100\ \text{Hz}$ for f_{p4} in equation.

$$C = \frac{1}{2\pi(10^6)(10)}$$
$$= 15.9\ \text{nF}$$

Therefore, the required capacitance of capacitor C is $\boxed{15.9\ \text{nF}}$.

Step 1 of 5

Finding the value of resistor R_1:

$$f_{P1} = \frac{1}{2\pi C_1 R_1}$$

$$R_1 = \frac{1}{2\pi f_{P1} C_1}$$

Substitute 10^5 Hz for f_{P1}, 150 pF for C_1 and in equation.

$$R_1 = \frac{1}{2\pi\left(10^5\right)\left(150\times10^{-12}\right)}$$
$$= 10.62\times10^3$$
$$= 10.62\ \text{k}\Omega$$

Step 2 of 5

Finding the value of resistor R_2:

$$f_{P2} = \frac{1}{2\pi C_2 R_2}$$

$$R_2 = \frac{1}{2\pi f_{P2} C_2}$$

Substitute 10^6 Hz for f_{P2}, 8 pF for C_2 in equation.

$$R_2 = \frac{1}{2\pi\left(10^6\right)\left(5\times10^{-12}\right)}$$
$$= 31.85\times10^3$$
$$= 31.85\ \text{k}\Omega$$

Step 3 of 5

First pole is to be located at

$$f'_{P1} = \frac{f_{P3}}{A_0}$$

Substitute 2×10^6 Hz for f_{P3}, 10^4 for A_0 in equation.

$$f'_{P1} = \frac{2\times10^6}{10^4}$$
$$= 200\ \text{Hz}$$

Step 4 of 5

Finding the required value of the compensating miller capacitance C_f:

Consider the expression.

$$f'_{P1} = \frac{1}{2\pi g_m R_1 R_2 C_f}$$

$$C_f = \frac{1}{2\pi g_m R_1 R_2 f'_{P1}}$$
$$= \frac{1}{2\pi\left(40\times10^{-3}\right)\left(10.62\times10^3\right)\left(31.85\times10^3\right)(200)}$$
$$= 5.88\times10^{-11}$$
$$= 58.8\times10^{-12}$$

Therefore, the miller capacitance C_f is $\boxed{58.8\ \text{pF}}$.

Step 5 of 5

Determining the new frequency of the output pole f'_{P2}:

$$f'_{P2} = \frac{g_m C_f}{2\pi\left[C_1C_2 + C_f\left(C_1 + C_2\right)\right]}$$
$$= \frac{1}{2\pi}\left(\frac{\left(40\times10^{-3}\right)\left(58.8\times10^{-12}\right)}{\left(150\times10^{-12}\right)\left(5\times10^{-12}\right) + \left(58.8\times10^{-12}\right)\left(\left(150\times10^{-12}\right) + \left(5\times10^{-12}\right)\right)}\right)$$
$$= \frac{1}{2\pi}\left(\frac{\left(2352\times10^{-15}\right)}{\left(150\times10^{-12}\right)\left(5\times10^{-12}\right) + \left(58.8\times10^{-12}\right)\left(155\times10^{-12}\right)}\right)$$
$$= \frac{1}{2\pi}\left(\frac{\left(2352\times10^{-15}\right)}{\left(750\times10^{-24}\right) + \left(9114\times10^{-24}\right)}\right)$$
$$= \frac{1}{2\pi}\left(\frac{\left(2352\times10^{-15}\right)}{\left(9864\times10^{-24}\right)}\right)$$
$$= 0.0379\times10^9$$

Therefore, the new frequency of the output pole f'_{P2} is $\boxed{37.9\ \text{MHz}}$.

Step 1 of 8

(a)

Refer to the Figure P10.100 in the text book.

Determine the value of feedback factor.

$$\beta(s)=\frac{\left(\frac{1}{sC}\right)}{R+\left(\frac{1}{sC}\right)}$$
$$=\frac{1}{1+sCR}$$

Substitute $0.01\,\mu\text{F}$ for C and $100\,\text{k}\Omega$ for R.

$$\beta(s)=\frac{1}{1+s\left(0.01\times10^{-6}\right)\left(100\times10^{3}\right)}$$
$$=\frac{1}{1+10^{-3}s}$$
$$=\frac{1}{1+\left(\frac{s}{10^{3}}\right)}$$

Step 2 of 8

Since the open-loop gain, $A=10^{5}$ and a single-pole roll off with $\omega_{3\text{dB}}=10\,\text{rad/s}$, the loop gain is,

$$A\beta(s)=\left(\frac{10^{5}}{1+\frac{s}{10}}\right)\left(\frac{1}{1+\frac{s}{10^{3}}}\right)$$

Step 3 of 8

Draw the bode plot as shown in Figure 1.

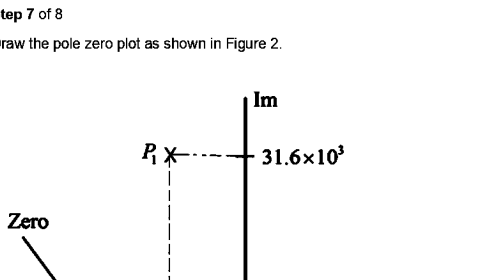


Figure 1

Step 4 of 8

(b)

From Figure 1, it is clear that,

$$20\log|A\beta|=20\text{dB at }\omega=10^{4}$$

Hence,

$|A\beta|=1$ at half the decade from $\omega=10^{4}$. That is, $|A\beta|=1$ at $\omega=3.6\ \text{krad/s}$.

Therefore, the phase margin is $\boxed{3.6\ \text{krad/s}}$.

Step 5 of 8

(c)

The closed loop transfer function is,

$$A_f(s)=\frac{A}{1+A\beta}$$
$$=\frac{\left(\frac{10^{5}}{1+\frac{s}{10}}\right)}{1+\left(\frac{10^{5}}{1+\frac{s}{10}}\right)\left(\frac{1}{1+\frac{s}{10^{3}}}\right)}$$
$$=\frac{10^{5}\left(1+\frac{s}{10^{3}}\right)}{\left(1+\frac{s}{10}\right)\left(1+\frac{s}{10^{3}}\right)+10^{5}}$$

Further simplification yields,

$$A_f(s)=\frac{10^{5}\left(1+\frac{s}{10^{3}}\right)}{10^{5}\left(1+\frac{s}{10^{6}}+\frac{s^{2}}{10^{9}}\right)}$$
$$=\frac{10^{6}s+10^{9}}{s^{2}+10^{3}s+10^{9}}$$

Therefore, the closed loop gain is $\boxed{\frac{10^{6}s+10^{9}}{s^{2}+10^{3}s+10^{9}}}$.

Step 6 of 8

From the above transfer function, the zero is at

$$10^{6}s+10^{9}=0$$
$$10^{6}s=-10^{9}$$
$$s=\frac{-10^{9}}{10^{6}}$$
$$s=-10^{3}$$

From the above transfer function, the poles are at

$$s=\frac{-10^{3}\pm\sqrt{\left(10^{3}\right)^{2}-4(1)\left(10^{9}\right)}}{2(1)}$$
$$=\frac{-10^{3}\pm j63.2\times10^{3}}{2}$$
$$=-500\pm j31.6\times10^{3}\,\text{rad/s}$$

Here,

$$\omega_0=31.6\ \text{krad/s}$$

Step 7 of 8

Draw the pole zero plot as shown in Figure 2.

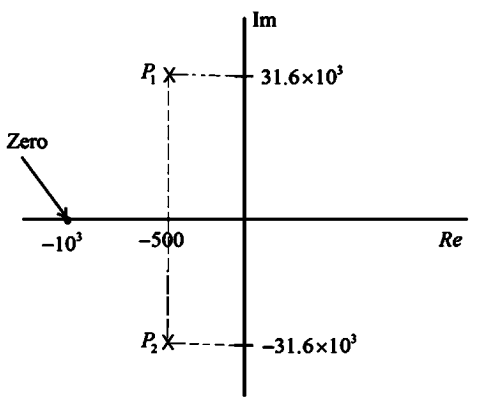


Figure 2

Step 8 of 8

The sketch of magnitude of transfer function versus frequency is shown in Figure 3.

Figure 3

Step 1 of 5

Refer to Figure11.2 in the textbook.

The two transistors have same characteristic, the collector-emitter saturation voltages are,

$$V_{CEsat} = V_{CE1sat} = V_{CE2sat} = 0.2\,\text{V}$$

The positive limit of the linear region is determined by the saturation of Q_1 ; thus the maximum output is,

$$v_{o\max} = V_{CC} - V_{CE1sat}$$

Substitute $15\ \text{V}$ for V_{CC} and $0.2\,\text{V}$ for V_{CE1sat} .

$$\begin{aligned} v_{o\max} &= 15 - 0.2 \\ &= 14.8\,\text{V} \end{aligned}$$

In the negative direction, depending on the values of I and R_L , the limit of the linear region is determined by Q_1 turning off. The minimum output is,

$$v_{O\min} = -V_{CC} + V_{CE2sat}$$

Substitute $15\ \text{V}$ for V_{CC} and $0.2\,\text{V}$ for V_{CE1sat} .

$$\begin{aligned} v_{O\min} &= -15 + 0.2 \\ &= -14.8\,\text{V} \end{aligned}$$

Thus, the output voltage swing is from $\boxed{-14.8\ \text{V to } 14.8\ \text{V}}$.

Step 2 of 5

The absolutely lowest (most negative) output voltage is that given by minimum output equation and is achieved provided the bias current I is greater than the magnitude of the corresponding load current, I .

$$I = \frac{\left|-V_{CC} + V_{CE2sat}\right|}{R_L}$$

Substitute $15\ \text{V}$ for V_{CC} , $1\ \text{k}\Omega$ for R_L and $0.2\,\text{V}$ for V_{CE1sat} .

$$\begin{aligned} I &= \frac{\left|-15 + 0.7\right|}{1000} \\ &= \frac{14.3}{1000} \\ &= 14.3\ \text{mA} \end{aligned}$$

Step 3 of 5

Calculate the value of base voltage of the transistor, Q_2 .

$$V_{B2} = -V_{CC} + V_{BE2sat}$$

Substitute $15\ \text{V}$ for V_{CC} and $0.7\,\text{V}$ for V_{BE2sat} .

$$\begin{aligned} V_{B2} &= -15 + 0.7 \\ &= -14.3\,\text{V} \end{aligned}$$

Calculate the value of the resistor, R .

$$R = \frac{|V_{B2}|}{I}$$

Substitute $-14.3\,\text{V}$ for V_{B2} and $14.3\ \text{mA}$ for I .

$$\begin{aligned} R &= \frac{|-14.3|}{14.8 \times 10^{-3}} \\ &= 0.97 \times 10^{3} \\ &= 0.97\ \text{k}\Omega \end{aligned}$$

Thus, the value of the resistance, R is $\boxed{0.97\ \text{k}\Omega}$.

Step 4 of 5

Calculate the value of maximum load current, $i_{L\max}$.

$$i_{L\max} = \frac{v_{o\max}}{R_L}$$

Substitute $14.8\,\text{V}$ for $v_{o\max}$ and $1\ \text{k}\Omega$ for R_L .

$$\begin{aligned} i_{L\max} &= \frac{14.8}{1000} \\ &= 14.8\,\text{mA} \end{aligned}$$

Calculate the value of minimum load current, $i_{L\min}$.

$$i_{L\min} = \frac{v_{o\min}}{R_L}$$

Substitute $-14.8\,\text{V}$ for $v_{o\min}$ and $1\ \text{k}\Omega$ for R_L .

$$\begin{aligned} i_{L\min} &= \frac{-14.8}{1000} \\ &= -14.8\,\text{mA} \end{aligned}$$

Step 5 of 5

Calculate the value of minimum emitter current of the transistor, Q_1 .

$$i_{E1\min} = I + i_{L\min}$$

Substitute $-14.8\,\text{mA}$ for $i_{L\min}$ and $14.3\ \text{mA}$ for I .

$$\begin{aligned} i_{E1\min} &= 14.8\,\text{mA} - 14.8\,\text{mA} \\ &= 0\ \text{A} \end{aligned}$$

Thus, the value of minimum emitter current of transistor, Q_1 is $\boxed{0\ \text{A}}$.

Calculate the value of maximum emitter current of transistor, Q_1 .

$$i_{E1\max} = I + i_{L\max}$$

Substitute $14.8\,\text{mA}$ for $i_{L\max}$ and $14.3\ \text{mA}$ for I .

$$\begin{aligned} i_{E1\max} &= 14.8\,\text{mA} + 14.8\,\text{mA} \\ &= 29.6\,\text{mA} \end{aligned}$$

Thus, the value of maximum emitter current of transistor, Q_1 is $\boxed{29.6\,\text{mA}}$.

12.001P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

Step 1 of 11

Refer to Figure11.2 in the textbook.

Calculate the load current at the output voltage of $v_O = -10\ \text{V}$.

$$i_L = \frac{v_O}{R_L}$$

Substitute $-10\ \text{V}$ for v_O and $1\ \text{k}\Omega$ for R_L.

$$\begin{aligned} i_L &= \frac{-10}{1\times10^3} \\ &= -10\times10^{-3} \\ &= -10\ \text{mA} \end{aligned}$$

Calculate the emitter current of the transistor, Q_1.

$$i_{E1} = I + i_L$$

Substitute $14.8\ \text{mA}$ for I and $-10\ \text{mA}$ for i_L.

$$\begin{aligned} i_{E1} &= 14.8\ \text{mA} - 10\ \text{mA} \\ &= 4.8\ \text{mA} \end{aligned}$$

Step 2 of 11

Calculate the base-to-emitter voltage of the transistor, Q_1.

$$v_{BE1} = v_{BE} + V_T \ln\left(\frac{i_{E1}}{i_C}\right)$$

Substitute $0.6\ \text{V}$ for v_{BE}, $25\ \text{mV}$ for V_T, $4.8\ \text{mA}$ for i_{E1} and $1\ \text{mA}$ for i_C.

$$\begin{aligned} v_{BE1} &= 0.6 + 0.025\ln\left(\frac{4.8}{1}\right) \\ &= 0.6 + (0.025)(1.569) \\ &= 0.64\ \text{V} \end{aligned}$$

Calculate the input voltage, v_I.

$$v_I = v_O + v_{BE1}$$

Substitute $-10\ \text{V}$ for v_O and $0.64\ \text{V}$ for v_{BE1}.

$$\begin{aligned} v_I &= -10 + 0.64 \\ &= -9.36\ \text{V} \end{aligned}$$

Thus, the input voltage corresponding to $v_O = -10\ \text{V}$ is $\boxed{v_I = -9.36\ \text{V}}$.

Step 3 of 11

Calculate the load current at output voltage of $v_O = 0\ \text{V}$.

$$i_L = \frac{v_O}{R_L}$$

Substitute $0\ \text{V}$ for v_O and $1\ \text{k}\Omega$ for R_L.

$$\begin{aligned} i_L &= \frac{0}{1\times10^3} \\ &= 0\ \text{A} \end{aligned}$$

Calculate the emitter current of transistor Q_1.

$$i_{E1} = I + i_L$$

Substitute $14.8\ \text{mA}$ for I and $0\ \text{A}$ for i_L.

$$\begin{aligned} i_{E1} &= 14.8\ \text{mA} - 0\ \text{A} \\ &= 14.8\ \text{mA} \end{aligned}$$

Step 4 of 11

Calculate the base-to-emitter voltage of transistor Q_1.

$$v_{BE1} = v_{BE} + V_T \ln\left(\frac{i_{E1}}{i_C}\right)$$

Substitute $0.6\ \text{V}$ for v_{BE}, $25\ \text{mV}$ for V_T, $14.8\ \text{mA}$ for i_{E1} and $1\ \text{mA}$ for i_C.

$$\begin{aligned} v_{BE1} &= 0.6 + 0.025\ln\left(\frac{14.8}{1}\right) \\ &= 0.6 + (0.025)(2.695) \\ &= 0.67\ \text{V} \end{aligned}$$

Calculate the input voltage, v_I.

$$v_I = v_O + v_{BE1}$$

Substitute $0\ \text{V}$ for v_O and $0.67\ \text{V}$ for v_{BE1}.

$$\begin{aligned} v_I &= 0 + 0.67 \\ &= 0.67\ \text{V} \end{aligned}$$

Thus, the input voltage corresponding to $v_O = 0\ \text{V}$ is $\boxed{v_I = 0.67\ \text{V}}$.

Step 5 of 11

Calculate the load current at output voltage of $v_O = 10\ \text{V}$.

$$i_L = \frac{v_O}{R_L}$$

Substitute $10\ \text{V}$ for v_O and $1\ \text{k}\Omega$ for R_L.

$$\begin{aligned} i_L &= \frac{10}{1\times10^3} \\ &= 10\times10^{-3} \\ &= 10\ \text{mA} \end{aligned}$$

Calculate the emitter current of transistor Q_1.

$$i_{E1} = I + i_L$$

Substitute $14.8\ \text{mA}$ for I and $10\ \text{mA}$ for i_L.

$$\begin{aligned} i_{E1} &= 14.8\ \text{mA} + 10\ \text{mA} \\ &= 24.8\ \text{mA} \end{aligned}$$

Calculate the base-to-emitter voltage of transistor Q_1.

$$v_{BE1} = v_{BE} + V_T \ln\left(\frac{i_{E1}}{i_C}\right)$$

Substitute $0.6\ \text{V}$ for v_{BE}, $25\ \text{mV}$ for V_T, $24.8\ \text{mA}$ for i_{E1} and $1\ \text{mA}$ for i_C.

$$\begin{aligned} v_{BE1} &= 0.6 + 0.025\ln\left(\frac{24.8}{1}\right) \\ &= 0.6 + (0.025)(3.211) \\ &= 0.68\ \text{V} \end{aligned}$$

Calculate the input voltage, v_I.

$$v_I = v_O + v_{BE1}$$

Substitute $10\ \text{V}$ for v_O and $0.68\ \text{V}$ for v_{BE1}.

$$\begin{aligned} v_I &= 10 + 0.68 \\ &= 10.68\ \text{V} \end{aligned}$$

Thus, the input voltage corresponding to $v_O = 10\ \text{V}$ is $\boxed{v_I = 10.68\ \text{V}}$.

Step 6 of 11

At $v_O = -10\ \text{V}$, the emitter current is $i_{E1} = 4.8\ \text{mA}$.

Calculate the small signal emitter resistance, r_{e1}.

$$r_{e1} = \frac{V_T}{i_{E1}}$$

Substitute $25\ \text{mV}$ for V_T and $4.8\ \text{mA}$ for i_{E1}.

$$\begin{aligned} r_{e1} &= \frac{25\ \text{mV}}{4.8\ \text{mA}} \\ &= 5.21\ \Omega \end{aligned}$$

Step 7 of 11

Calculate the voltage gain at $v_O = -10\ \text{V}$.

$$\frac{v_o}{v_i} = \frac{R_L}{R_L + r_{e1}}$$

Substitute $5.21\ \Omega$ for r_{e1} and $1\ \text{k}\Omega$ for R_L.

$$\begin{aligned} \frac{v_o}{v_i} &= \frac{1000}{1000 + 5.21} \\ &= 0.995\ \text{V/V} \end{aligned}$$

Thus, the voltage gain of the emitter follower at $v_O = -10\ \text{V}$ is $\boxed{0.995\ \text{V/V}}$.

Step 8 of 11

At $v_O = 0\ \text{V}$, the emitter current is $i_{E1} = 14.8\ \text{mA}$.

Calculate the small signal emitter resistance, r_{e1}.

$$r_{e1} = \frac{V_T}{i_{E1}}$$

Substitute $25\ \text{mV}$ for V_T and $14.8\ \text{mA}$ for i_{E1}.

$$\begin{aligned} r_{e1} &= \frac{25\ \text{mV}}{14.8\ \text{mA}} \\ &= 1.689\ \Omega \end{aligned}$$

Step 9 of 11

Calculate the voltage gain at $v_O = 0\ \text{V}$.

$$\frac{v_o}{v_i} = \frac{R_L}{R_L + r_{e1}}$$

Substitute $1.689\ \Omega$ for r_{e1} and $1\ \text{k}\Omega$ for R_L.

$$\begin{aligned} \frac{v_o}{v_i} &= \frac{1000}{1000 + 1.689} \\ &= 0.998\ \text{V/V} \end{aligned}$$

Thus, the voltage gain of the emitter follower at $v_O = 0\ \text{V}$ is $\boxed{0.998\ \text{V/V}}$.

Step 10 of 11

At $v_O = 10\ \text{V}$, the emitter current is $i_{E1} = 24.8\ \text{mA}$.

Calculate the small signal emitter resistance, r_{e1}.

$$r_{e1} = \frac{V_T}{i_{E1}}$$

Substitute $25\ \text{mV}$ for V_T and $24.8\ \text{mA}$ for i_{E1}.

$$\begin{aligned} r_{e1} &= \frac{25\ \text{mV}}{24.8\ \text{mA}} \\ &= 1.008\ \Omega \end{aligned}$$

Step 11 of 11

Calculate the voltage gain at $v_O = 10\ \text{V}$.

$$\frac{v_o}{v_i} = \frac{R_L}{R_L + r_{e1}}$$

Substitute $1.008\ \Omega$ for r_{e1} and $1\ \text{k}\Omega$ for R_L.

$$\begin{aligned} \frac{v_o}{v_i} &= \frac{1000}{1000 + 1.008} \\ &= 0.999\ \text{V/V} \end{aligned}$$

Thus, the voltage gain of the emitter follower at $v_O = 10\ \text{V}$ is $\boxed{0.999\ \text{V/V}}$.

12.002P

We don't have the solution to this problem yet.

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Step 1 of 3

Refer to Figure11.2 in the textbook.

(a)

Calculate the average power delivered to the load, P_L.

$$P_L = \frac{\left(\hat{V}_o/\sqrt{2}\right)^2}{R_L}$$
$$= \frac{1}{2} \times \frac{\hat{V}_o^2}{R_L}$$

Substitute $8\ \mathrm{V}$ for $\hat{V}_o$ and $100\ \Omega$ for R_L.

$$P_L = \frac{1}{2} \times \frac{8^2}{100}$$
$$= \frac{64}{200}$$
$$= 0.32\ \mathrm{W}$$

Therefore, the power delivered to the load, P_L is $\boxed{0.32\ \mathrm{W}}$.

Step 2 of 3

(b)

Calculate the average power drawn from the supplies, P_S

$$P_S = 2V_{CC}I$$

Substitute $10\ \mathrm{V}$ for V_{CC} and $100 \times 10^{-3}\ \mathrm{mA}$ for I.

$$P_S = 2(10)\left(100 \times 10^{-3}\right)$$
$$= (20)(0.1)$$
$$= 2\ \mathrm{W}$$

Therefore, power drawn from the supplies, P_S is $\boxed{2\ \mathrm{W}}$.

Step 3 of 3

(c)

Calculate the power conversion efficiency, η.

$$\eta = \frac{P_L}{P_S} \times 100\ \%$$

Substitute $0.32\ \mathrm{W}$ for P_L and $2\ \mathrm{W}$ for P_S.

$$\eta = \frac{0.32}{2} \times 100\%$$
$$= 16\%$$

Therefore, the power conversion efficiency, η is $\boxed{16\%}$.

12.003P

We don't have the solution to this problem yet.

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Step 1 of 5

Refer to Figure 11.5 in the text book.

(a)

Find the output power.

Write the expression for the average output power.

$$P_L = \frac{1}{2}\frac{\hat{V}_o^2}{R_L}$$

Substitute 4.5 V for $\hat{V}$ and $4\ \Omega$ for R_L.

$$\begin{aligned} P_L &= \frac{1}{2}\frac{(4.5)^2}{4} \\ &= \frac{20.25}{8} \\ &= 2.531\ \text{W} \end{aligned}$$

Therefore, the output power is $\boxed{2.531\ \text{W}}$.

Step 2 of 5

(b)

Find the power drawn from each supply.

Each supply provides a current waveform of half-sinusoid, thus average power drawn from each supply is,

$$P_{S+} = P_{S-} = \frac{\hat{I}_o}{\pi} \times V_{CC}$$

Where, the peak current drawn from each supply is

$$\hat{I}_o = \frac{\hat{V}_o}{R_L}$$

Write the expression for the power drawn from supply.

$$P_{S+} = P_{S-} = \frac{\hat{V}_o}{\pi R_L} \times V_{CC}$$

Substitute 4.5 V for $\hat{V}$ and $4\ \Omega$ for R_L.

$$P_{S+} = P_{S-} = \frac{4.5}{4\pi} \times 6$$

Therefore, the power drawn from each supply is $\boxed{2.148\ \text{W}}$.

Step 3 of 5

(c)

Find the power efficiency.

Determine the total power drawn from supply.

$$\begin{aligned} P_S &= 2P_{S+} \\ &= 2(2.148) \\ &= 4.296\ \text{W} \end{aligned}$$

Write the expression for the power conversion efficiency.

$$\eta = \frac{P_L}{P_S}$$

Substitute 2.531 W for P_L and 4.296 W for P_S.

$$\begin{aligned} \eta &= \frac{2.531}{4.296} \times 100 \\ &= 58.91\% \end{aligned}$$

Therefore, the power efficiency η is $\boxed{58.91\%}$.

Step 4 of 5

(d)

Find peak current supplied by v_I.

Given

$$\beta_N = \beta_P = 50$$

The input peak current is

$$\hat{I}_i = \frac{\hat{I}_o}{1+\beta}$$

Substitute $\frac{\hat{V}_o}{R_L}$ for $\hat{I}_o$.

$$\hat{I}_i = \frac{\hat{V}_o}{R_L(1+\beta)}$$

Substitute 4.5 V for $\hat{V}_o$ and 50 for β.

$$\begin{aligned} \hat{I}_i &= \frac{4.5}{4(1+50)} \\ &= \frac{4.5}{204} \\ &= 22.05\ \text{mA} \end{aligned}$$

Therefore, the input peak current $\hat{I}_i$ is $\boxed{22.05\ \text{mA}}$.

Step 5 of 5

(e)

Find the maximum power dissipation in each transistor.

The maximum power dissipation in each transistor is,

$$P_{DN\,\max} = P_{DP\,\max} = \frac{V_{CC}^2}{\pi^2 R_L}$$

Substitute 6 V for V_{CC} and $4\ \Omega$ for R_L.

$$P_{DN\,\max} = P_{DP\,\max} = \frac{6^2}{4\pi^2}$$

Therefore, the maximum power dissipated in each transistor is $\boxed{0.911\ \text{W}}$.

12.004P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

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Step 1 of 9

(a)

For class AB circuit, to increase the linearity of the class AB output stage, the quiescent current is increased.

The quiescent power dissipation is $I_Q V_{CC}$.

The total power dissipation in two transistors is,

$$\begin{aligned} P_D &= 2I_Q V_{CC} \\ &= (2)(2\times10^{-3})(15) \\ &= 60\ \text{mW} \end{aligned}$$

Thus, the quiescent power dissipation is $\boxed{60\ \text{mW}}$.

Step 2 of 9

(b)

The quiescent current is increased to 10 mA.

The output voltage, $v_O = 0$

At $v_O = 0$, the currents conducted by transistors Q_P and Q_N is,

$$\begin{aligned} i_N &= i_P \\ &= I_Q \\ &= 10\ \text{mA} \end{aligned}$$

The output resistance is,

$$\begin{aligned} R_{out} &= \frac{V_T}{i_P + i_N} \\ &= \frac{25}{10+10} \\ &= \frac{25}{20} \\ &= 1.25\ \Omega \end{aligned}$$

The voltage gain is,

$$\begin{aligned} \frac{v_o}{v_i} &= \frac{R_L}{R_L + R_{out}} \\ &= \frac{100}{100+1.25} \\ &= 0.988 \end{aligned}$$

Thus, the voltage gain at $v_O = 0$ is $\boxed{0.988}$.

Step 3 of 9

Determine the load current at $v_O = 10\ \text{V}$.

$$\begin{aligned} i_L &= \frac{v_O}{R_L} \\ &= \frac{10}{100} \\ &= 0.1\ \text{A} \end{aligned}$$

Use the quadratic equation,

$$\begin{aligned} i_N^2 - i_L i_N - I_Q^2 &= 0 \\ i_N^2 - (0.1) i_N - \left(10\times10^{-3}\right)^2 &= 0 \\ i_N^2 - 0.1 i_N - 10^{-4} &= 0 \end{aligned}$$

Solve this quadratic equation to obtain the current i_N.

$$i_N = 0.101\ \text{A}$$

Determine the current i_P.

$$\begin{aligned} i_P &= \frac{I_Q^2}{i_N} \\ &= \frac{\left(10\times10^{-3}\right)^2}{0.101} \\ &= 9.9\times10^{-4}\ \text{A} \end{aligned}$$

Step 4 of 9

The output resistance is,

$$\begin{aligned} R_{out} &= \frac{V_T}{i_N + i_P} \\ &= \frac{0.025}{0.101 + 9.9\times10^{-4}} \\ &= 0.245\ \Omega \end{aligned}$$

The voltage gain is,

$$\begin{aligned} \frac{v_o}{v_i} &= \frac{R_L}{R_L + R_{out}} \\ &= \frac{100}{100+0.245} \\ &= 0.998 \\ &\approx 1 \end{aligned}$$

Thus, the voltage gain at $v_O = 10\ \text{V}$ is $\boxed{1}$.

Step 5 of 9

The percentage change in the voltage gain is,

$$\begin{aligned} \%\text{Change} &= \left(\frac{1-0.988}{1}\right)100 \\ &= 1.2\% \end{aligned}$$

Thus, the percentage change in voltage gain for output voltage change from 0 to 10 V is $\boxed{1.2\%}$.

Step 6 of 9

In the example 11.3, the quiescent current is 2 mA.

The output voltage, $v_O = 0$

At $v_O = 0$, the currents conducted by transistors Q_P and Q_N is,

$$\begin{aligned} i_N &= i_P \\ &= I_Q \\ &= 2\ \text{mA} \end{aligned}$$

The output resistance is,

$$\begin{aligned} R_{out} &= \frac{V_T}{i_P + i_N} \\ &= \frac{25}{2+2} \\ &= \frac{25}{4} \\ &= 6.25\ \Omega \end{aligned}$$

The voltage gain is,

$$\begin{aligned} \frac{v_o}{v_i} &= \frac{R_L}{R_L + R_{out}} \\ &= \frac{100}{100+6.25} \\ &= 0.941 \end{aligned}$$

Step 7 of 9

At $v_O = 10\ \text{V}$, the load current is,

$$\begin{aligned} i_L &= \frac{v_O}{R_L} \\ &= \frac{10}{100} \\ &= 0.1\ \text{A} \end{aligned}$$

Use the quadratic equation,

$$\begin{aligned} i_N^2 - i_L i_N - I_Q^2 &= 0 \\ i_N^2 - (0.1) i_N - \left(2\times10^{-3}\right)^2 &= 0 \\ i_N^2 - 0.1 i_N - 4\times10^{-6} &= 0 \end{aligned}$$

Solve this quadratic equation to obtain the current i_N.

$$i_N = 0.1\ \text{A}$$

The current,

$$\begin{aligned} i_P &= \frac{I_Q^2}{i_N} \\ &= \frac{\left(2\times10^{-3}\right)^2}{0.1} \\ &= 4\times10^{-5}\ \text{A} \end{aligned}$$

Step 8 of 9

The output voltage is,

$$\begin{aligned} R_{out} &= \frac{V_T}{i_N + i_P} \\ &= \frac{0.025}{0.1 + 4\times10^{-5}} \\ &= 0.25\ \Omega \end{aligned}$$

The voltage gain is,

$$\begin{aligned} \frac{v_o}{v_i} &= \frac{R_L}{R_L + R_{out}} \\ &= \frac{100}{100+0.25} \\ &= 0.998 \\ &\approx 1 \end{aligned}$$

The percentage change in the voltage gain is,

$$\begin{aligned} \%A_v &= \left(\frac{1-0.941}{1}\right)100 \\ &= 0.06 \\ &= 6\% \end{aligned}$$

Thus, the percentage change in voltage gain for output voltage change from 0 to 10 V is $\boxed{6\%}$.

Step 9 of 9

(c)

The quiescent power dissipation in two transistors is,

$$\begin{aligned} P_D &= 2I_Q V_{CC} \\ &= (2)(10\times10^{-3})(15) \\ &= 300\ \text{mW} \end{aligned}$$

Thus, the quiescent power dissipation in the two transistors is $\boxed{300\ \text{mW}}$.

Step 1 of 14

Refer to Figure 11.2 in the text book.

Neglecting V_{CEsat}, the output range is $+V_{CC}$ to $-V_{CC}$.

The bias current is, $I = \frac{V_{CC}}{R_L}$.

Consider the sine wave form is shown in Figure 1.

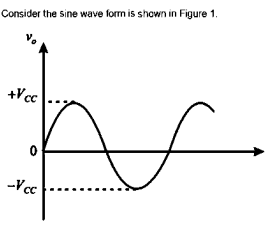


Figure 1

Step 2 of 14

The corresponding square wave with swing from $+V_{CC}$ to $-V_{CC}$ is shown in Figure 2.

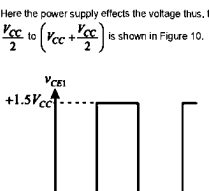


Figure 2

Step 3 of 14

Consider the waveform shown in Figure 3.

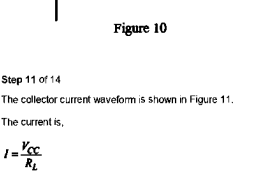


Figure 3

Step 4 of 14

The corresponding square wave with swing from $+2V_{CC}$ to 0 is shown in Figure 4.

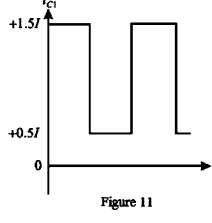


Figure 4

Step 5 of 14

The collector current waveform of transistor Q_1 is shown in Figure 5.

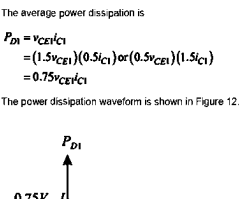


Figure 5

Step 6 of 14

The corresponding square wave form with swing 0 to $2I$ is shown in Figure 6.

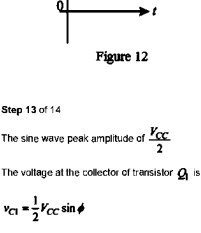


Figure 6

Step 7 of 14

The power dissipation sine waveform is shown in Figure 7.

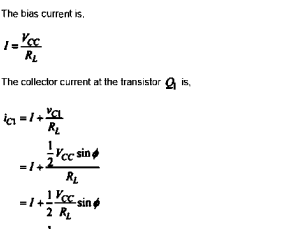


Figure 7

Step 8 of 14

The power dissipation is,

$P_{D1} = v_{CE1}i_{C1}$

Multiply these square wave forms.

The corresponding power dissipation square wave is shown in Figure 8.

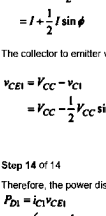


Figure 8

Step 9 of 14

If the square wave of peak amplitude is $\pm\frac{V_{CC}}{2}$ levels then the resulting wave form is shown in Figure 9.

Figure 9

Step 10 of 14

Here the power supply effects the voltage thus, the square wave swings from $\frac{V_{CC}}{2}$ to $\left(V_{CC}+\frac{V_{CC}}{2}\right)$ is shown in Figure 10.

Figure 10

Step 11 of 14

The collector current waveform is shown in Figure 11.

The current is,

$I = \frac{V_{CC}}{R_L}$

Figure 11

Step 12 of 14

The average power dissipation is

$$P_{D1} = v_{CE1}i_{C1}$$
$$= (1.5v_{CE1})(0.5i_{C1}) \text{ or } (0.5v_{CE1})(1.5i_{C1})$$
$$= 0.75v_{CE1}i_{C1}$$

The power dissipation waveform is shown in Figure 12.

Figure 12

Step 13 of 14

The sine wave peak amplitude of $\frac{V_{CC}}{2}$

The voltage at the collector of transistor Q_1 is,

$$v_{C1} = \frac{1}{2}V_{CC}\sin\phi$$

The bias current is,

$$I = \frac{V_{CC}}{R_L}$$

The collector current at the transistor Q_1 is,

$$i_{C1} = I + \frac{v_{C1}}{R_L}$$
$$= I + \frac{\frac{1}{2}V_{CC}\sin\phi}{R_L}$$
$$= I + \frac{1}{2}\frac{V_{CC}}{R_L}\sin\phi$$
$$= I + \frac{1}{2}I\sin\phi$$

The collector to emitter voltage of the transistor Q_1 is

$$v_{CE1} = V_{CC} - v_{C1}$$
$$= V_{CC} - \frac{1}{2}V_{CC}\sin\phi$$

Step 14 of 14

Therefore, the power dissipation is,

$$P_{D1} = i_{C1}v_{CE1}$$
$$= \left(V_{CC} - \frac{1}{2}V_{CC}\sin\phi\right)\left(I + \frac{1}{2}I\sin\phi\right)$$
$$= IV_{CC} - \frac{1}{2}IV_{CC}\sin\phi + \frac{1}{2}V_{CC}I\sin\phi - \frac{1}{4}V_{CC}I\sin^2\phi$$
$$= IV_{CC}\left(1 - \frac{1}{4}\sin^2\phi\right)$$
$$= IV_{CC}\left(1 - \frac{1}{4}\left(\frac{1-\cos 2\phi}{2}\right)\right)$$
$$= IV_{CC}\left(\frac{7}{8} + \frac{\cos 2\phi}{8}\right)$$

Therefore, the average power dissipation (average of the cos function is zero) is

$$\boxed{\frac{7}{8}IV_{CC}}$$

12.006E

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12.006P

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12.007E

Calculate the transconductance g_m.

$$g_m = \frac{i_C}{V_T}$$
$$= \frac{10\ \text{mA}}{25\ \text{mV}}$$
$$= 0.4\ \text{mA/mV}$$

Calculate the change in collector current.

$$\Delta I_C = (g_m \times 2\ \text{mV/°C})(5\ \text{°C})\ \text{mA}$$
$$= (0.4\ \text{mA/mV})(2\ \text{mV/°C})(5\ \text{°C})\ \text{mA}$$
$$= 4\ \text{mA}$$

Thus, the change in collector current resulting from an increase in temperature of $5°\ \text{C}$ is $\boxed{4\ \text{mA}}$.

12.007P

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12.0CBE

Refer to Figure 11.15 in the text book

The transistor is to be operated in **12 V** the

$$V_{BB} = 12 \text{ V}$$

Step 1 of 25

Calculate the base current that should be supplied to the multiplier to obtain the required minimum voltage.

$$I_{min} = I_1 + I_2$$

Substitute **8.48 mA** for I_1 and **8.37mA** for I_2.

$$I_{min} = 0.37 \text{ mA} + 0.48 \text{ mA}$$

$$= 0.85 \text{ mA}$$

Therefore, the base current that should be supplied to the multiplier to obtain the change in terminal voltage of **−50 mV** is **0.85 mA**

Step 2 of 25

Calculate the base current drawn the change in terminal voltage of **+200 mV** .

The change in the terminal is to obtain is **+200 mV** .

Step 3 of 25

Calculate the required voltage across the bias network. V_{aa} .

$$\Delta V_{aa} = +200 \text{ mV}$$

Calculate the required voltage across the bias network. V_{aa} .

$$V_{aa} = (V_{aa})_{ss} + \Delta V_{aa}$$

Substitute **1.2 V** for $(V_{aa})_{ss}$ and **+200 mV** for ΔV_{aa}.

$$V_{aa} = 1.2 \text{ V} + 200 \text{ mV}$$

$$= 1.4 \text{ V}$$

Step 11 of 25

Consider the expression for voltage across the bias network to calculate the current. I_2.

$$V_{aa} = I_2 (R_1 + R_2)$$

$$I_2 = \frac{V_{aa}}{R_1 + R_2}$$

Substitute **1.2 kΩ** for R_1, **1.2 kΩ** for R_2 and **1.4 V** for V_{aa}.

$$I_2 = \frac{1.4}{1.2 \text{ k}\Omega + 1.2 \text{ k}\Omega}$$

$$= \frac{1.4}{2.4 \times 10^3}$$

$$= 0.58 \times 10^{-3}$$

$$= 0.58 \text{ mA}$$

Step 12 of 25

Calculate the value of base-emitter voltage. V_{BE}

$$V_{BE} = \frac{V_{BB}}{2}$$

Substitute **1.4 V** for V_{BB}.

$$V_{BE} = \frac{1.4 \text{ V}}{2}$$

$$= 0.7 \text{ V}$$

Calculate the value of the emitter current. I_{E1}

$$I_{E1} = I_E e^{\frac{qV_{BE}}{\eta_2 kT}} \left[\Delta V_{BE} = V_{BE} - (V_{BE})_{ss} \right]$$

Substitute **0.55 V** for V_{BE}, **1 mA** for I_E, **25 mV** for V_T

$$I_{E1} = (1 \text{ mA}) e^{\frac{0.55 - 0.7}{25 \times 10^{-3}}}$$

$$= e^{\frac{-150}{25}} \text{ mA}$$

$$= e^{-6} \text{ mA} \quad \left(e^{-6} = 0.013 \right)$$

$$= 0.013 \text{ mA}$$

Calculate the bias current that should be supplied to the multiplier to obtain the required minimum voltage.

$$I_{min}$$

Substitute **0.046 mA** for I_B and **0.13 mA** for I_{E1}

$$I_{min} = 0.046 \text{ mA} + 0.13 \text{ mA}$$

$$= 0.176 \text{ mA}$$

Thus, the bias current that should be supplied to the multiplier to attain the required minimum voltage of **−100 mV** is **0.59 mA**

Step 13 of 25

Calculate the bias current drawn the change in terminal voltage of **−200 mV** .

The change in the terminal is to obtain is **−200 mV** .

Step 14 of 25

Calculate the required voltage across the bias network. V_{aa} .

$$\Delta V_{aa} = -200 \text{ mV}$$

Calculate the required voltage across the bias network. V_{aa} .

$$V_{aa} = (V_{aa})_{ss} + \Delta V_{aa}$$

Substitute **1.2 V** for $(V_{aa})_{ss}$ and **−200 mV** for ΔV_{aa}.

$$V_{aa} = 1.2 \text{ V} - 200 \text{ mV}$$

$$= 1.2 - 0.2$$

$$= 1 \text{ V}$$

Step 15 of 25

Consider the expression for voltage across the bias network to calculate the current. I_2.

$$V_{aa} = I_2 (R_1 + R_2)$$

$$I_2 = \frac{V_{aa}}{R_1 + R_2}$$

Substitute **1.2 kΩ** for R_1, **1.2 kΩ** for R_2 and **1 V** for V_{aa}.

$$I_2 = \frac{1}{1.2 \text{ k}\Omega + 1.2 \text{ k}\Omega}$$

$$= \frac{1}{2.4 \times 10^3}$$

$$= 0.42 \times 10^{-3}$$

$$= 0.42 \text{ mA}$$

Step 16 of 25

Calculate the value of base-emitter voltage. V_{BE}

$$V_{BE} = \frac{V_{BB}}{2}$$

Substitute **1 V** for V_{BB}

$$V_{BE} = \frac{1 \text{ V}}{2}$$

$$= 0.5 \text{ V}$$

Calculate the value of the emitter current. I_{E1}

$$I_{E1} = I_E e^{\frac{qV_{BE}}{\eta_2 kT}} \left[\Delta V_{BE} = V_{BE} - (V_{BE})_{ss} \right]$$

Substitute **0.5 V** for V_{BE}, **1 mA** for I_E, **25 mV** for V_T

$$I_{E1} = (1 \text{ mA}) e^{\frac{0.5 - 0.7}{25 \times 10^{-3}}}$$

$$= e^{\frac{-200}{25}} \text{ mA}$$

$$= e^{-8} \text{ mA} \quad \left(e^{-8} = 0.0003 \right)$$

$$= 0.0003 \text{ mA}$$

Calculate the bias current that should be supplied to the multiplier to obtain the required minimum voltage.

$$I_{min}$$

Substitute **0.046 mA** for I_B and **0.0003 mA** for I_{E1}.

$$I_{min} = 0.046 \text{ mA} + 0.0003 \text{ mA}$$

$$= 0.0463 \text{ mA}$$

Thus, the bias current that should be supplied to the multiplier to attain the change in terminal voltage of **−200 mV** is **0.43 mA**

Step 17 of 25

Calculate the bias current drawn the change in terminal voltage of **+100 mV** .

The change in the terminal is to obtain is **+100 mV** .

Step 18 of 25

Calculate the required voltage across the bias network. V_{aa} .

$$\Delta V_{aa} = +100 \text{ mV}$$

Calculate the required voltage across the bias network. V_{aa} .

$$V_{aa} = (V_{aa})_{ss} + \Delta V_{aa}$$

Substitute **1.2 V** for $(V_{aa})_{ss}$ and **+100 mV** for ΔV_{aa}.

$$V_{aa} = 1.2 \text{ V} + 100 \text{ mV}$$

$$= 1.2 + 0.1$$

$$= 1.3 \text{ V}$$

Step 19 of 25

Consider the expression for voltage across the bias network to calculate the current. I_2.

$$V_{aa} = I_2 (R_1 + R_2)$$

$$I_2 = \frac{V_{aa}}{R_1 + R_2}$$

Substitute **1.2 kΩ** for R_1, **1.2 kΩ** for R_2 and **1.15 V** for V_{aa}.

$$I_2 = \frac{1.15}{1.2 \text{ k}\Omega + 1.2 \text{ k}\Omega}$$

$$= \frac{1.15}{2.4 \times 10^3}$$

$$= 0.48 \times 10^{-3}$$

$$= 0.48 \text{ mA}$$

Step 20 of 25

Calculate the value of base-emitter voltage. V_{BE}

$$V_{BE} = \frac{V_{BB}}{2}$$

Substitute **1.15 V** for V_{BB}

$$V_{BE} = \frac{1.15}{2}$$

$$= 0.575 \text{ V}$$

Calculate the value of the emitter current. I_{E1}

$$I_{E1} = I_E e^{\frac{qV_{BE}}{\eta_2 kT}} \left[\Delta V_{BE} = V_{BE} - (V_{BE})_{ss} \right]$$

Substitute **0.575 V** for V_{BE}, **1 mA** for I_E, and **25 mV** for V_T.

$$I_{E1} = (1 \text{ mA}) e^{\frac{0.575 - 0.7}{25 \times 10^{-3}}}$$

$$= e^{\frac{-125}{25}} \text{ mA}$$

$$= e^{-5} \text{ mA} \quad \left(e^{-5} = 0.37 \right)$$

Calculate the value of the emitter current. I_{E1}

$$I_{E1} = I_E e^{\frac{qV_{BE}}{\eta_2 kT}} \left[\Delta V_{BE} = V_{BE} - (V_{BE})_{ss} \right]$$

Substitute **0.5 V** for V_{BE}, **1 mA** for I_E, and **25 mV** for V_T.

$$I_{E1} = (1 \text{ mA}) e^{\frac{0.5 - 0.7}{25 \times 10^{-3}}}$$

$$= e^{\frac{-200}{25}} \text{ mA}$$

$$= e^{-8} \text{ mA} \quad \left(e^{-8} = 0.0003 \right)$$

Calculate the bias current that should be supplied to the multiplier to obtain the required minimum voltage.

$$I_{min} = I_1 + I_2$$

Substitute **8.37 mA** for I_1 and **0.37 mA** for I_2.

$$I_{min} = 8.37 \text{ mA} + 0.37 \text{ mA}$$

$$= 8.74 \text{ mA}$$

Therefore, the base current that should be supplied to the multiplier to attain the change in terminal voltage of **+100 mV** is **8.74 mA**

Step 4 of 25

Calculate the bias current drawn the change in terminal voltage of **−100 mV** .

The change in the terminal is to obtain is **−100 mV** .

Step 5 of 25

Calculate the required voltage across the bias network. V_{aa} .

$$\Delta V_{aa} = -100 \text{ mV}$$

Calculate the required voltage across the bias network. V_{aa} .

$$V_{aa} = (V_{aa})_{ss} + \Delta V_{aa}$$

Substitute **1.2 V** for $(V_{aa})_{ss}$ and **−100 mV** for ΔV_{aa}.

$$V_{aa} = 1.2 \text{ V} - 100 \text{ mV}$$

$$= 1.2 - 0.1$$

$$= 1.1 \text{ V}$$

Step 6 of 25

Consider the expression for voltage across the bias network to calculate the current. I_2.

$$V_{aa} = I_2 (R_1 + R_2)$$

$$I_2 = \frac{V_{aa}}{R_1 + R_2}$$

Substitute **1.2 kΩ** for R_1, **1.2 kΩ** for R_2, and **1.1 V** for V_{aa}.

$$I_2 = \frac{1.1}{1.2 \text{ k}\Omega + 1.2 \text{ k}\Omega}$$

$$= \frac{1.1}{2.4 \times 10^3}$$

$$= 0.46 \times 10^{-3}$$

$$= 0.46 \text{ mA}$$

Step 7 of 25

Calculate the value of base-emitter voltage. V_{BE}

$$V_{BE} = \frac{V_{BB}}{2}$$

Substitute **1 V** for V_{BB}

$$V_{BE} = \frac{1 \text{ V}}{2}$$

$$= 0.5 \text{ V}$$

Calculate the value of the emitter current. I_{E1}

$$I_{E1} = I_E e^{\frac{qV_{BE}}{\eta_2 kT}} \left[\Delta V_{BE} = V_{BE} - (V_{BE})_{ss} \right]$$

Substitute **0.5 V** for V_{BE}, **1 mA** for I_E, and **25 mV** for V_T.

$$I_{E1} = (1 \text{ mA}) e^{\frac{0.5 - 0.7}{25 \times 10^{-3}}}$$

$$= e^{\frac{-200}{25}} \text{ mA}$$

$$= e^{-8} \text{ mA} \quad \left(e^{-8} = 0.0003 \right)$$

$$= 0.0003 \text{ mA}$$

Calculate the bias current that should be supplied to the multiplier to obtain the required minimum voltage.

$$I_{min} = I_1 + I_2$$

Substitute **8.48 mA** for I_1 and **0.37 mA** for I_2.

$$I_{min} = 8.48 \text{ mA} + 0.37 \text{ mA}$$

$$= 8.85 \text{ mA}$$

Therefore, the base current that should be supplied to the multiplier to obtain the change in terminal voltage of **−50 mV** is **8.85 mA**

Step 8 of 25

Calculate the bias current drawn the change in terminal voltage of **+200 mV** .

The change in the terminal is to obtain is **+200 mV** .

Step 9 of 25

Calculate the required voltage across the bias network. V_{aa} .

$$\Delta V_{aa} = +200 \text{ mV}$$

Calculate the required voltage across the bias network. V_{aa} .

$$V_{aa} = (V_{aa})_{ss} + \Delta V_{aa}$$

Substitute **1.2 V** for $(V_{aa})_{ss}$ and **+200 mV** for ΔV_{aa}.

$$V_{aa} = 1.2 \text{ V} + 200 \text{ mV}$$

$$= 1.4 \text{ V}$$

Step 10 of 25

Consider the expression for voltage across the bias network to calculate the current. I_2.

$$V_{aa} = I_2 (R_1 + R_2)$$

$$I_2 = \frac{V_{aa}}{R_1 + R_2}$$

Substitute **1.2 kΩ** for R_1, **1.2 kΩ** for R_2, and **1.4 V** for V_{aa}.

$$I_2 = \frac{1.4}{1.2 \text{ k}\Omega + 1.2 \text{ k}\Omega}$$

$$= \frac{1.4}{2.4 \times 10^3}$$

$$= 0.58 \times 10^{-3}$$

$$= 0.58 \text{ mA}$$

Step 21 of 25

Substitute **0.55 V** for V_{BE}, **1 mA** for I_E, **25 mV** for V_T

$$I_{E1} = (1 \text{ mA}) e^{\frac{0.55 - 0.7}{25 \times 10^{-3}}}$$

$$= e^{\frac{-150}{25}} \text{ mA}$$

$$= e^{-6} \text{ mA} \quad \left(e^{-6} = 0.013 \right)$$

$$= 0.013 \text{ mA}$$

Calculate the bias current that should be supplied to the multiplier to obtain the required minimum voltage.

$$I_{min} = I_1 + I_2$$

Substitute **0.046 mA** for I_B and **0.013 mA** for I_{E1}

$$I_{min} = 0.046 \text{ mA} + 0.013 \text{ mA}$$

$$= 0.059 \text{ mA}$$

Thus, the bias current that should be supplied to the multiplier to attain the required minimum voltage of **−100 mV** is **0.059 mA**

Step 22 of 25

Calculate the bias current drawn the change in terminal voltage of **−200 mV** .

The change in the terminal is to obtain is **−200 mV** .

Step 23 of 25

Consider the expression for voltage across the bias network to calculate the current. I_2.

$$V_{aa} = I_2 (R_1 + R_2)$$

$$I_2 = \frac{V_{aa}}{R_1 + R_2}$$

Substitute **1.2 kΩ** for R_1, **1.2 kΩ** for R_2, and **1 V** for V_{aa}.

$$I_2 = \frac{1}{1.2 \text{ k}\Omega + 1.2 \text{ k}\Omega}$$

$$= \frac{1}{2.4 \times 10^3}$$

$$= 0.41 \times 10^{-3}$$

$$= 0.41 \text{ mA}$$

Calculate the base-emitter voltage. V_{BE}

$$V_{BE} = \frac{V_{BB}}{2}$$

Substitute **1 V** for V_{BB}

$$V_{BE} = \frac{1 \text{ V}}{2}$$

$$= 0.5 \text{ V}$$

Calculate the value of the emitter current. I_{E1}

$$I_{E1} = I_E e^{\frac{qV_{BE}}{\eta_2 kT}} \left[\Delta V_{BE} = V_{BE} - (V_{BE})_{ss} \right]$$

Substitute **0.5 V** for V_{BE}, **1 mA** for I_E, and **25 mV** for V_T.

$$I_{E1} = (1 \text{ mA}) e^{\frac{0.5 - 0.7}{25 \times 10^{-3}}}$$

$$= e^{\frac{-200}{25}} \text{ mA}$$

$$= e^{-8} \text{ mA} \quad \left(e^{-8} = 0.0003 \right)$$

$$= 0.0003 \text{ mA}$$

Calculate the bias current that should be supplied to the multiplier to obtain the required minimum voltage.

$$I_{min} = I_1 + I_2$$

Substitute **8.48 mA** for I_1 and **0.02 mA** for I_{E1}.

$$I_{min} = 8.48 \text{ mA} + 8.41 \text{ mA}$$

$$= 16.89 \text{ mA}$$

Therefore, the base current that should be supplied to the multiplier to obtain the change in terminal voltage of **−50 mV** is **16.89 mA**

Step 24 of 25

Consider the expression for voltage across the bias network to calculate the current. I_2.

$$V_{aa} = I_2 (R_1 + R_2)$$

$$I_2 = \frac{V_{aa}}{R_1 + R_2}$$

Substitute **1.2 kΩ** for R_1, **1.2 kΩ** for R_2, and **1 V** for V_{aa}.

$$I_2 = \frac{1}{1.2 \text{ k}\Omega + 1.2 \text{ k}\Omega}$$

$$= \frac{1}{2.4 \times 10^3}$$

$$= 0.41 \times 10^{-3}$$

$$= 0.41 \text{ mA}$$

Calculate the base-emitter voltage. V_{BE}

$$V_{BE} = \frac{V_{BB}}{2}$$

Substitute **1 V** for V_{BB}

$$V_{BE} = \frac{1 \text{ V}}{2}$$

$$= 0.5 \text{ V}$$

Calculate the value of the emitter current. I_{E1}

$$I_{E1} = I_E e^{\frac{qV_{BE}}{\eta_2 kT}} \left[\Delta V_{BE} = V_{BE} - (V_{BE})_{ss} \right]$$

Substitute **0.5 V** for V_{BE}, **1 mA** for I_E, and **25 mV** for V_T.

$$I_{E1} = (1 \text{ mA}) e^{\frac{0.5 - 0.7}{25 \times 10^{-3}}}$$

$$= e^{\frac{-200}{25}} \text{ mA}$$

$$= e^{-8} \text{ mA} \quad \left(e^{-8} = 0.0003 \right)$$

$$= 0.0003 \text{ mA}$$

Calculate the bias current that should be supplied to the multiplier to obtain the required minimum voltage.

$$I_{min} = I_1 + I_2$$

Substitute **7.99 mA** for I_1 and **0.54 mA** for I_{E1}.

$$I_{min} = 7.99 \text{ mA} + 0.54 \text{ mA}$$

$$= 8.53 \text{ mA}$$

Therefore, the base current that should be supplied to the multiplier to obtain the change in terminal voltage of **+100 mV** is **8.53 mA**

12.008P

We don't have the solution to this problem yet.

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Step 1 of 3

12.009P

Consider the class B output stage circuit is shown in Figure 1.

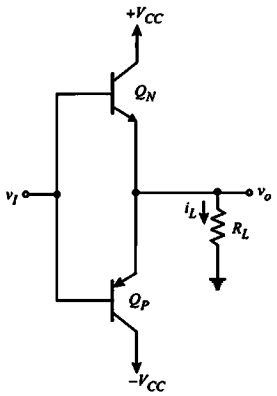


Figure 1

Step 2 of 3

Consider the peak amplitude as $\hat{V}_o$.

When the input voltage v_I is zero, both the transistors are cut off and the output voltage, v_o is zero.

As v_I goes positive and exceeds $0.5\,\mathrm{V}$, the transistor Q_N conducts as an emitter follower.

In this case, v_o follows v_I $\left(v_o = v_I - v_{BEN}\right)$.

If the input goes negative by more than about $0.5\,\mathrm{V}$, Q_P turns on and acts as an emitter follower. Again, v_o follows v_I $\left(v_o = v_I + v_{BEP}\right)$.

Sketch the transfer characteristic of the class B stage as shown in Figure 2.

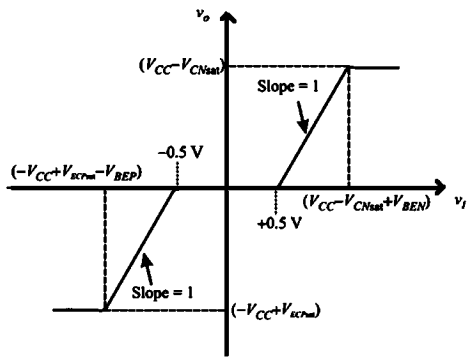


Figure 2

Step 3 of 3

The effect of crossover distortion is the most pronounced when the amplitude of the input signal is small. Crossover distortion in audio amplifier gives rise to unpleasant sounds.

From Figure 2, the loss in the peak amplitude due to cross over distortion is $0.5\,\mathrm{V}$

The percentage of loss is,

$$\text{percentage loss} = \frac{0.5}{\hat{V}_o} \times 100 \ \%$$

The given percentage of loss is 10%

Therefore,

$$\frac{0.5}{\hat{V}_o} \times 100\ \% = 10\%$$

$$50 = 10\hat{V}_o$$

$$\hat{V}_o = \frac{50}{10}$$

$$= 5\,\mathrm{V}$$

Therefore, for $5\,\mathrm{V}$ amplitude of input signal the crossover distortion represents 10% loss in peak amplitude.

Step 1 of 4

Refer to Figure 11.32 in the text book.

(a)

Determine the emitter current of transistor, Q_1 .

$$I_E = \left(\frac{\beta_P + 1}{\beta_P}\right)\left(\frac{i_C}{\beta_N + 1}\right)$$
$$\simeq \frac{i_C}{\beta_N}$$

Determine the base current of the composite transistor.

$$i_B = \frac{I_E}{\beta_P}$$
$$= \frac{i_C}{\beta_N \beta_P}$$

Thus the base current of the composite *pnp* transistor is $\boxed{i_B = \frac{i_C}{\beta_N \beta_P}}$.

Step 2 of 4

The emitter current and the collector current are nearly equal, thus, $\boxed{i_E \simeq i_C}$.

The transistor Q_1 has saturation current, I_{SP} . Determine the collector current of the composite transistor.

$$i_{C1} = I_{SP} e^{\frac{v_{EB}}{V_T}}$$
$$\frac{i_C}{\beta_N + 1} = I_{SP} e^{\frac{v_{EB}}{V_T}}$$
$$i_C = (\beta_N + 1) I_{SP} e^{\frac{v_{EB}}{V_T}}$$
$$i_C \simeq \beta_N I_{SP} e^{\frac{v_{EB}}{V_T}}$$

Thus, the collector current i_C for the composite *pnp* transistor is $\boxed{\beta_N I_{SP} e^{\frac{v_{EB}}{V_T}}}$.

Thus, the effective scale value I_S is $\boxed{\beta_N I_{SP}}$.

Step 3 of 4

(b)

Calculate the effective current gain of the compound device.

$$\frac{i_C}{i_B} = \beta_P \beta_N$$
$$= (20)(50)$$
$$= 1000$$

Thus, the effective current gain of the compound device is $\boxed{1000}$.

Step 4 of 4

Calculate the emitter to base voltage.

$$i_C \simeq \beta_N I_{SP} e^{\frac{v_{EB}}{V_T}}$$
$$100 \times 10^{-3} = (50)(10^{-14}) e^{\frac{v_{EB}}{0.025}}$$
$$e^{\frac{v_{EB}}{0.025}} = \frac{100 \times 10^{-3}}{(50)(10^{-14})}$$
$$e^{\frac{v_{EB}}{0.025}} = 2 \times 10^{11}$$

Take natural logarithm on both sides.

$$\frac{v_{EB}}{0.025} = \ln(2 \times 10^{11})$$
$$\frac{v_{EB}}{0.025} = 26.02$$
$$v_{EB} = 0.651 \text{ V}$$

Thus, the emitter-to-base voltage v_{EB} of the compound device is $\boxed{0.651 \text{ V}}$.

Step 1 of 4

Refer to Figure 11.9 in the text book.

The NPN transistor (Q_N) is conducting when positive voltage is applied, which is greater than cutoff voltage and the PNP transistor (Q_P) is conducting when negative voltage is applied, which is greater than cutoff voltage.

In the circuit shown in Figure 11.9, the transistor Q_N conducts when input voltage v_I is positive.The gain is,

$$A_o = \frac{v_o + V_{BE}}{v_I - v_o}$$

Substitute $100\,\text{V/V}$ for A_o and $0.7\,\text{V}$ for $|V_{BE}|$.

$$100(v_I - v_o) = v_o + 0.7$$
$$100v_I - 0.7 = v_o + 100v_o$$
$$101v_o = 100v_I - 0.7$$

$$v_o = \frac{100}{101}v_I - \frac{0.7}{101}$$
$$= \frac{100}{101}v_I - \frac{100}{101}(0.007)$$
$$= \frac{100}{101}(v_I - 0.007)$$

Therefore, the output is exist for $v_I \geq 0.007\,\text{V}$.

Step 2 of 4

In the circuit the transistor Q_P conducts when input voltage v_I is negative, then the gain is,

$$A_o = \frac{v_o + V_{BE}}{v_I - v_o}$$
$$100(v_I - v_o) = v_o - 0.7$$
$$100v_I + 0.7 = v_o + 100v_o$$
$$101v_o = 100v_I + 0.7$$

$$v_o = \frac{100}{101}v_I + \frac{0.7}{101}$$
$$= \frac{100}{101}v_I + \frac{100}{101}(0.007)$$
$$= \frac{100}{101}(v_I + 0.007)$$
$$= 0.99(v_I + 0.007)$$

Therefore, the output exists for $v_I \geq -0.007\,\text{V}$

Step 3 of 4

Draw the transfer characteristic v_o versus v_I with feedback as shown in Figure 1.

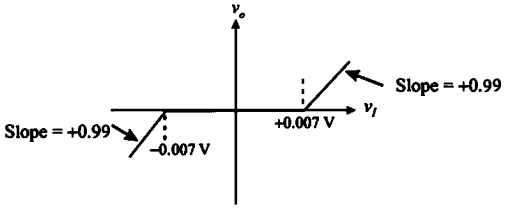


Figure 1

If there is no feedback then the dead band is,

$$DB = \pm 0.007(A_o)$$
$$= \pm 0.007(100)$$
$$= \pm 0.7\,\text{V}$$

Step 4 of 4

The transfer characteristic v_o versus v_I is shown in Figure 2.

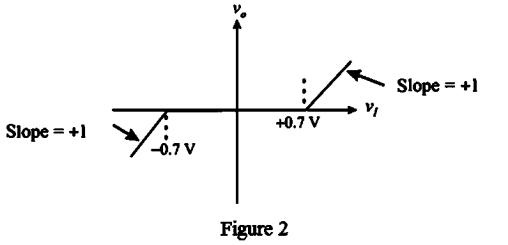


Figure 2

Therefore, the transfer characteristic without feedback is drawn.

Step 1 of 4

Refer to Figure 11.34 in the text book.

Calculate the base-emitter voltage of transistor, Q_5.

$$\begin{aligned} V_{BE5} &= V_T \ln\left(\frac{I_{C5}}{I_S}\right) \\ &= V_T \ln\left(\frac{I_{BIAS}}{I_S}\right) \\ &= 25\times10^{-3} \ln\left(\frac{2\times10^{-3}}{10^{-14}}\right) \\ &= 0.651 \text{ V} \end{aligned}$$

Step 2 of 4

Determine the value of R_{E1}.

$$\begin{aligned} V_{BE5} &= i_L R_{E1} \\ 0.651 &= \left(150\times10^{-3}\right) R_{E1} \\ R_{E1} &= \frac{0.651}{150\times10^{-3}} \\ R_{E1} &= 4.3\ \Omega \end{aligned}$$

Thus, the value of R_{E1} is $\boxed{4.3\ \Omega}$.

Step 3 of 4

Calculate the voltage drop across R_{E1}.

$$\begin{aligned} V_{BE1} &= i_{LP} R_{E1} \\ &= \left(100\times10^{-3}\right)(4.3) \\ &= 430\times10^{-3} \text{ V} \\ &= 430 \text{ mV} \end{aligned}$$

Thus, the voltage drop V_{BE1} across R_{E1} is $\boxed{430 \text{ mV}}$.

Step 4 of 4

Calculate the collector current of Q_5.

$$\begin{aligned} i_{C5} &= I_S e^{\frac{V_{BE5}}{V_T}} \\ &= \left(10^{-14}\right) e^{\frac{430\times10^{-3}}{25\times10^{-3}}} \\ &= 0.3\times10^{-6} \text{ A} \\ &= 0.3\ \mu\text{A} \end{aligned}$$

Thus, the collector current i_{C5} of transistor Q_5 is $\boxed{0.3\ \mu\text{A}}$.

Step 1 of 6

Refer to Figure P11.10 in the text book.

The peak output voltage is,

$$v_o = v_i - V_{GS1}$$

Substitute 5 V for v_i and 0.5 V for V_{GS1}.

$$\begin{aligned} v_o &= 5 - 0.5 \\ &= 4.5\text{ V} \end{aligned}$$

Therefore, the peak output voltage at high value of load resistance v_o is $\boxed{4.5\text{ V}}$.

Step 2 of 6

The sine of the interval is,

$$\sin\theta = \frac{V_{GS1}}{V_i}$$

Substitute 5 V for v_i and 0.5 V for V_{GS1}.

$$\begin{aligned} \sin\theta &= \frac{0.5}{5} \\ \sin\theta &= 0.1 \\ \theta &= 5.74^\circ \end{aligned}$$

Therefore, the value of θ is 5.74°.

Step 3 of 6

The cross over interval is,

$$\text{cross interval} = 4\theta$$

Substitute 5.74° for θ.

$$\begin{aligned} \text{cross interval} &= 4\times 5.74^\circ \\ &= 22.96^\circ \end{aligned}$$

The fraction of the cross over interval with the sine wave period is,

$$\begin{aligned} \text{Fraction of cycle} &= \frac{22.96^\circ}{360^\circ} \\ &= 0.064 \end{aligned}$$

The percentage of the cross over interval with the sine wave period is,

$$\begin{aligned} \%\text{ of fraction of cycle} &= 0.064\times 100 \\ &= 6.4\% \end{aligned}$$

Therefore, the % of fraction of cycle is $\boxed{6.4\%}$.

Step 4 of 6

The peak output voltage is,

$$\begin{aligned} v_o &= \frac{4.5}{2}\text{ V} \\ &= 2.25\text{ V} \end{aligned}$$

The gate to source voltage is,

$$V_{GS} = v_i - v_0$$

Substitute 5 V for v_i and 2.25 V for v_o.

$$\begin{aligned} V_{GS} &= 5 - 2.5 \\ &= 2.5\text{ V} \end{aligned}$$

Step 5 of 6

The drain current is,

$$i_D = \frac{1}{2}\mu C_{ox}\frac{W}{L}(V_{GS} - V_t)^2$$

Substitute 2 mA/V^2 for $\mu C_{ox}\frac{W}{L}$ and 0.5 V for $|V_t|$.

$$\begin{aligned} i_D &= \frac{1}{2}\times 2\times 10^{-3}(2.5 - 0.5)^2 \\ &= 4\text{ mA} \end{aligned}$$

Step 6 of 6

The load resistance is,

$$R_L = \frac{V_{GS}}{I_D}$$

Substitute 2.5 V for V_{GS} and 4 mA for i_D.

$$\begin{aligned} R_L &= \frac{2.5}{4\times 10^{-3}} \\ &= 625\ \Omega \end{aligned}$$

Thus, the load resistance when the peak output voltage is reduced to half the input is $\boxed{625\ \Omega}$.

Step 1 of 6

Refer to Figure 11.17 in the text book.

Write the expression for the quiescent current of the transistor.

$$I_Q = I_{BIAS}\frac{\left(\frac{W}{L}\right)_n}{\left(\frac{W}{L}\right)_p}$$

Substitute 1 mA for I_Q and 0.2 mA for I_{BIAS}.

$$1 = 0.2\frac{\left(\frac{W}{L}\right)_n}{\left(\frac{W}{L}\right)_p}$$

$$\frac{\left(\frac{W}{L}\right)_n}{\left(\frac{W}{L}\right)_p} = 5$$

Step 2 of 6

Write the expression for the bais current of transistor Q_1.

$$I_{BIAS} = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_1(V_{GS1} - V_{tn})^2$$

Substitute 0.2 mA for I_{BIAS}, $250\ \mu A/V^2$ for k_n', 0.7 V for V_{GS1} and 0.5 V for V_{tn}.

$$0.2\times10^{-3} = \frac{1}{2}\times250\times10^{-6}\left(\frac{W}{L}\right)_1(0.7-0.5)^2$$

$$\left(\frac{W}{L}\right)_1 = \frac{0.2\times10^{-3}\times2}{250\times10^{-6}\times(0.2)^2}$$

$$= 40$$

Thus, the ratio $\left(\frac{W}{L}\right)_1$ of transistor Q_1 is $\boxed{40}$.

Step 3 of 6

Write the expression for the bias current for transistor Q_2.

$$I_{BIAS} = \frac{1}{2}k_p'\left(\frac{W}{L}\right)_2(V_{GS} - |V_t|)^2$$

Substitute 0.2 mA for I_{BIAS}, $100\ \mu A/V^2$ for k_n', 0.7 V for V_{GS1} and 0.5 V for V_t.

$$0.2\times10^{-3} = \frac{1}{2}\times100\times10^{-6}\left(\frac{W}{L}\right)_2(0.7-0.5)^2$$

$$\left(\frac{W}{L}\right)_2 = \frac{0.2\times10^{-3}\times2}{100\times10^{-6}\times(0.2)^2}$$

$$\left(\frac{W}{L}\right)_2 = 100$$

Thus the ratio $\left(\frac{W}{L}\right)_2$ of transistor Q_2 is $\boxed{100}$.

Step 4 of 6

Write the expression of the quiescent current for transistor Q_N.

$$I_Q = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_N(V_{GS} - V_t)^2$$

Substitute 1 mA for I_{BIAS}, $250\ \mu A/V^2$ for k_n', 0.7 V for V_{GS1} and 0.5 V for V_t.

$$1\times10^{-3} = \frac{1}{2}\times250\times10^{-6}\left(\frac{W}{L}\right)_N(0.7-0.5)^2$$

$$\left(\frac{W}{L}\right)_N = \frac{1\times10^{-3}\times2}{250\times10^{-6}\times(0.2)^2}$$

$$\left(\frac{W}{L}\right)_N = 200$$

Thus, the ratio $\left(\frac{W}{L}\right)_N$ of transistor Q_N is $\boxed{200}$.

Step 5 of 6

Write the expression of the quiescent current for transistor Q_P.

$$I_Q = \frac{1}{2}k_p'\left(\frac{W}{L}\right)_P(V_{GS} - |V_t|)^2$$

Substitute 1 mA for I_{BIAS}, $100\ \mu A/V^2$ for k_n', 0.7 V for V_{GS1} and 0.5 V for V_t.

$$1\times10^{-3} = \frac{1}{2}\times100\times10^{-6}\left(\frac{W}{L}\right)_P(0.7-0.5)^2$$

$$\left(\frac{W}{L}\right)_P = \frac{1\times10^{-3}\times2}{100\times10^{-6}\times(0.2)^2}$$

$$\left(\frac{W}{L}\right)_P = 500$$

Thus, the ratio $\left(\frac{W}{L}\right)_P$ of transistor Q_P is $\boxed{500}$.

Step 6 of 6

Determine the gate to gate voltage of the transistors.

$$V_{GG} = V_{GS1} + V_{GS2}$$

$$= (V_{OV1} + V_t) + (V_{OV2} + |V_t|)$$

$$= (0.2 + 0.5) + (0.2 + 0.5)$$

$$= 1.4\ V$$

Thus, the gate to gate voltage V_{GG} of the transistors is $\boxed{1.4\ V}$.

12.012P

We don't have the solution to this problem yet.

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12.013E

Refer to Figure 11.17 in the text book.

The maximum load current is

Write the expression of the current passing through the transistor Q_N .

$$I_n = i_{L\max}$$
$$= \frac{1}{2}k_n'\left(\frac{W}{L}\right)_n V_{OV}^2$$

Substitute $10\ \text{mA}$ for $I_{L\max}$, $250\ \mu\text{A/V}^2$ for k_n' and 200 for $\left(\frac{W}{L}\right)_n$.

$$10\times10^{-3} = \frac{1}{2}\left(250\times10^{-6}\right)(200)\left(V_{OV}^2\right)$$
$$V_{OV}^2 = 0.4$$
$$V_{OV} = 0.632\ \text{V}$$

Determine the maximum output voltage.

$$v_{0\max} = V_{DD} - V_{OV}\big|_{Bias} - V_{tn} - V_{OVN}$$
$$= 2.5 - 0.2 - 0.5 - 0.632$$
$$= 1.168\ \text{V}$$

Thus, the maximum output voltage $v_{0\max}$ is $\boxed{1.168\ \text{V}}$.

12.013P

We don't have the solution to this problem yet.

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Step 1 of 7

Draw the circuit in quiescent state.

Step 2 of 7

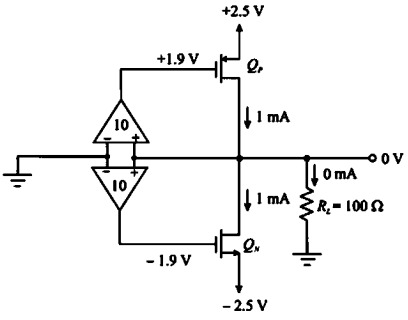


Figure 1

Step 3 of 7

Determine the width to length ratio of transistor Q_p.

$$\left(\frac{W}{L}\right)_P = \frac{2000}{2}$$
$$= 1000$$

Determine the width to length ratio of transistor Q_N.

$$\left(\frac{W}{L}\right)_N = \frac{800}{2}$$
$$= 400$$

Step 4 of 7

Write the expression for the quiescent current.

$$I_Q = \frac{1}{2}k'_p\left(\frac{W}{L}\right)_p V_{OV}^2$$

Substitute 1 mA for I_Q $0.1\ \text{mA/V}^2$ for k'_p and 1000 for $\left(\frac{W}{L}\right)_p$.

$$1\times10^{-3} = \frac{1}{2}\times0.1\times10^{-3}\times1000\times V_{OV}^2$$
$$V_{OV}^2 = \frac{2}{100}$$
$$V_{OV} = 0.14\ \text{V}$$

Thus, the value of V_{OV} is $\boxed{0.14\ \text{V}}$.

Step 5 of 7

Determine the new value of gain error.

$$\%\text{Gain error} = -\frac{V_{OV}}{4\mu I_Q R_L}\times100$$
$$= -\frac{0.14}{4(10)\left(1\times10^{-3}\right)100}\times100$$
$$= -\frac{0.14}{4}\times100$$
$$= -3.5\%$$

Therefore, the gain error is $\boxed{-3.5\ \%}$.

Step 6 of 7

The transconductance at quiescent point is,

$$g_m = g_{mp} = g_{mn}$$

Determine the transconductance at quiescent point.

$$g_m = \frac{2I_Q}{V_{OV}}$$
$$= \frac{2\times10^{-3}}{0.14}$$
$$= 14.286\times10^{-3}$$
$$= 14.286\,\text{mA/V}$$

Step 7 of 7

Write the expression of the output resistance R_{out}.

$$R_{out} = \frac{1}{\mu\left(g_{mp} + g_{mn}\right)}$$

Substitute 10 for μ, $14.286\,\text{mA/V}$ for g_{mp} and $14.286\,\text{mA/V}$ for g_{mn}.

$$R_{out} = \frac{1}{10\left(14.286\times10^{-3} + 14.286\times10^{-3}\right)}$$
$$= \frac{1}{10(2)\left(14.286\times10^{-3}\right)}$$
$$= 0.003499\times10^{3}$$
$$= 3.499\,\Omega$$

Therefore, the output resistance R_{out} is $\boxed{3.499\ \Omega}$.

12.014P

We don't have the solution to this problem yet.

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Step 1 of 3

12.015E

Refer to Figure 11.37 in the textbook for the small signal analysis of the internal circuit of the LM380 IC power amplifier.

The current directions of each transistor have indicated with the arrow marks as shown in Figure 11.37.

Consider the node 14 at the collector terminal of the transistor, Q_6. The current through the collector of the transistor, Q_6 is, $\frac{v_i}{R_2}$. The current through the collector of the transistor, Q_4 towards the node 14 is, $\frac{v_i}{R_2}+\frac{v_o}{R_2}$.

The total resistance between the collector of Q_6 and ground is R.

Apply Kirchhoff's current law at node 14 (collector terminal of Q_6).

$$\left(\frac{v_i}{R_2}+\frac{v_o}{R_2}+\frac{v_i}{R_2}\right)R=-\frac{v_o}{A}$$

$$\left(\frac{2v_i}{R_2}+\frac{v_o}{R_2}\right)R=-\frac{v_o}{A}$$

$$\frac{2Rv_i}{R_2}+\frac{Rv_o}{R_2}=-\frac{v_o}{A}$$

$$\frac{2Rv_i}{R_2}=-\frac{v_o}{A}-\frac{Rv_o}{R_2}$$

$$\frac{2Rv_i}{R_2}=-\left(\frac{1}{A}+\frac{R}{R_2}\right)v_o$$

Step 2 of 3

Rearrange the terms to determine the voltage gain.

$$\frac{v_o}{v_i}=\frac{\frac{2R}{R_2}}{-\left(\frac{1}{A}+\frac{R}{R_2}\right)}$$

$$=\frac{-\frac{2R}{R_2}}{\frac{R_2+AR}{AR_2}}$$

$$=\frac{-\left(\frac{2R}{R_2}\right)AR_2}{R_2+AR}$$

Simplify further.

$$\frac{v_o}{v_i}=\frac{-\left(\frac{2R}{R_2}\right)AR_2}{AR\left(1+\frac{R_2}{AR}\right)}$$

$$=\frac{-\left(\frac{2R_2}{R_2}\right)}{\left(1+\frac{R_2}{AR}\right)}$$

Thus, the gain of the amplifier is, $\boxed{\frac{-\left(\frac{2R_2}{R_2}\right)}{\left(1+\frac{R_2}{AR}\right)}}$.

Step 3 of 3

The condition is, $AR>>R_2$

Use this condition to reduce the gain expression.

$$\frac{R_2}{AR}<<1$$

Therefore, the gain expression is,

$$\frac{v_o}{v_i}=\frac{-\left(\frac{2R_2}{R_2}\right)}{1} \qquad \left\{\text{since } \frac{R_2}{AR}<<1\right\}$$

$$\frac{v_o}{v_i}=-\frac{2R_2}{R_2}$$

Thus, the reduced the gain expression is, $\boxed{-\frac{2R_2}{R_2}}$.

Step 1 of 8

12.015P

Consider the circuit diagram of the class B output stage is shown in Figure 1.

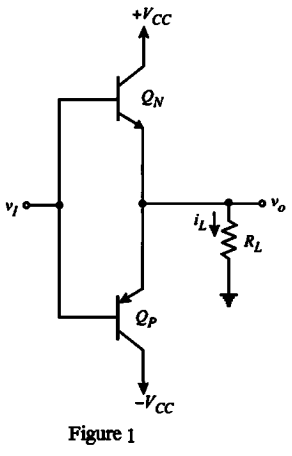


Figure 1

Step 2 of 8

The Output voltage amplitude is $\hat{V}_o$

And output is a square wave, i.e.

$$V_o = \begin{cases} \hat{V}_o & \text{For first half cycle} \\ -\hat{V}_o & \text{For second half cycle} \end{cases}$$

The power of square wave is,

$$P_{sq} = \hat{V}_o^2$$

The average load power is,

$$P_L = \frac{\hat{V}_o^2}{R_L}$$

Therefore, the average load power is $\boxed{\frac{\hat{V}_o^2}{R_L}}$.

Step 3 of 8

Each supply provides a current waveform of half-square wave, thus average power drawn from each supply is,

$$P_{S+} = P_{S-} = \frac{\hat{I}_o}{2} \times V_{CC}$$

Where, the peak current drawn from each supply is

$$\hat{I}_o = \frac{\hat{V}_o}{R_L}$$

The supply power is

$$P_{S+} = P_{S-} = \frac{\hat{V}_o}{2R_L} \times V_{CC}$$

The supply power is $\boxed{\frac{\hat{V}_o}{2R_L} \times V_{CC}}$.

Step 4 of 8

Total power supply is,

$$P_S = 2P_{S+}$$

$$P_S = 2\frac{\hat{V}_o}{2R_L} \times V_{CC}$$

The total power supply is $\boxed{\frac{\hat{V}_o}{R_L} \times V_{CC}}$.

Step 5 of 8

The power conversion efficiency is,

$$\eta = \frac{P_L}{P_S} = \frac{\frac{\hat{V}_o^2}{R_L}}{\frac{\hat{V}_o}{R_L} \times V_{CC}} = \frac{\hat{V}_o}{V_{CC}}$$

The power conversion efficiency is $\boxed{\frac{\hat{V}_o}{V_{CC}}}$.

Step 6 of 8

To attain the maximum power conversion, the efficiency is 100%

$$100\% = \frac{\hat{V}_o}{V_{CC}}$$

$$\frac{100}{100} = \frac{\hat{V}_o}{V_{CC}}$$

To attain the maximum power conversion, the corresponding value of output is

$$\boxed{\hat{V}_o = V_{CC}}$$

The maximum available load power is $\boxed{P_L = \frac{V_{CC}^2}{R_L}}$

Step 7 of 8

The power dissipation is difference between source and load power

$$P_{dis} = P_S - P_L = \frac{\hat{V}_o}{R_L} V_{CC} - \frac{\hat{V}_o^2}{R_L}$$

To get the maximum value, derivative it with respect to output voltage and then equate to zero

$$\frac{\partial P_{dis}}{\partial \hat{V}_o} = \frac{1}{R_L} V_{CC} - \frac{2\hat{V}_o}{R_L}$$

$$0 = \frac{1}{R_L} V_{CC} - \frac{2\hat{V}_o}{R_L}$$

$$V_{CC} - 2\hat{V}_o = 0$$

$$\hat{V}_o = \frac{V_{CC}}{2}$$

The maximum output voltage is $\boxed{\frac{V_{CC}}{2}}$.

Step 8 of 8

The power conversion efficiency is,

$$\eta = \frac{P_L}{P_S} = \frac{\frac{\hat{V}_o^2}{R_L}}{\frac{\hat{V}_o}{R_L} \times V_{CC}} = \frac{\hat{V}_o}{V_{CC}} = \frac{\frac{V_{CC}}{2}}{V_{CC}}$$

Therefore, the power conversion efficiency is $\boxed{50\%}$.

Step 1 of 2

12.016E

Refer to Figure 11.38 in the textbook for the curves of output power versus device dissipation.

From the curves, observe that the maximum supply voltage 20 V is possible for the maximum power dissipation limited to 2.9 W.

Therefore, the maximum supply voltage for 3% distortion is, **20 V** .

If the distortion allowed is, **THD = 3%**, then the output power at which the 20 V supply curve intersects 3% distortion line is, 4.2 W

Therefore, the output power for 3% distortion line is, **4.2 W** .

Step 2 of 2

Calculate the peak output voltage.

$$P_L = \frac{\left(\frac{\hat{V}_o}{\sqrt{2}}\right)^2}{R_L}$$

Arrange the terms for $\hat{V}_o$

$$\hat{V}_o = \sqrt{2P_L R_L}$$

Substitute 4.2 W for P_L and $8\ \Omega$ for R_L .

$$\hat{V}_o = \sqrt{(2)(4.2)(8)}$$
$$= 8.2\ \text{V}$$

The peak-to-peak output voltage is,

$$V_{pp} = 2\left(\hat{V}_o\right)$$
$$= 2(8.2)$$
$$= 16.4\ \text{V}$$

Thus, the peak-to-peak output voltage is, **16.4 V** .

12.016P

We don't have the solution to this problem yet.

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Step 1 of 5

Refer to Figure 11.40 in the textbook for the bridge amplifier circuit.

It is observed that, the output voltage of the upper op-amp (inverting amplifier) is,

$$v_{O1} = \left(1 + \frac{R_2}{R_1}\right) v_I$$

$$\text{here, } K = 1 + \frac{R_2}{R_1}$$

It is observed that, the output voltage of the lower op-amp (non-inverting amplifier) is,

$$v_{O2} = -\frac{R_4}{R_3} v_I$$

$$\text{here, } K = -\frac{R_4}{R_3}$$

Step 2 of 5

Calculate the gain of the amplifier.

$$\begin{aligned} K &= 1 + \frac{R_2}{R_1} \\ &= 1 + \frac{5\,\text{k}}{10\,\text{k}} \\ &= 1.5 \end{aligned}$$

From the circuit, the output voltage is,

$$v_O = 2K v_I$$

$$\frac{v_O}{v_I} = 2K$$

$$A_v = 2K$$

Substitute 1.5 for K.

$$\begin{aligned} A_v &= (2)(1.5) \\ &= 3\ \text{V/V} \end{aligned}$$

Thus, the voltage gain of the bridge amplifier is, $\boxed{3\ \text{V/V}}$.

Step 3 of 5

From the circuit, it it observed that the input resistance is equal to the resistance, R_3.

Thus, the input resistance of the bridge amplifier is, $\boxed{10\ \text{k}\Omega}$.

The voltage gain is,

$$A_v = \frac{v_O}{v_I}$$

Write the expression for the peak-to-peak output voltage.

$$v_O = A_v v_I$$

Substitute 20 V for v_I and $3\ \text{V/V}$ for A_v.

$$\begin{aligned} v_O &= (3)(20) \\ &= 60\ \text{V} \end{aligned}$$

Thus, the peak-to-peak output voltage is, $\boxed{60\ \text{V}}$.

Step 4 of 5

Calculate the peak load current.

$$i_L = \frac{\frac{v_O}{2}}{R_L}$$

Substitute $60\ \text{V}$ for v_O and $8\ \Omega$ for R_L.

$$\begin{aligned} i_L &= \frac{\frac{60}{2}}{8} \\ &= 3.75\ \text{A} \end{aligned}$$

Thus, the peak load current of the bridge amplifier is, $\boxed{3.75\ \text{A}}$.

Step 5 of 5

Calculate the load power.

$$P_L = \frac{\left(\frac{v_O}{2\sqrt{2}}\right)^2}{R_L}$$

Substitute $60\ \text{V}$ for v_O and $8\ \Omega$ for R_L.

$$\begin{aligned} P_L &= \frac{\left(\frac{60}{2\sqrt{2}}\right)^2}{8} \\ &= 56.25\ \text{W} \end{aligned}$$

Thus, the load power is, $\boxed{56.25\ \text{W}}$.

12.017P

We don't have the solution to this problem yet.

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12.018E

We don't have the solution to this problem yet.

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12.018P

We don't have the solution to this problem yet.

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12.019E

We don't have the solution to this problem yet.

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12.019P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

12.020E

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Step 1 of 4

Calculate the load current.

$$i_L = \frac{v_o}{R_L}$$

Calculate the input current to the amplifier.

$$i_I = \frac{i_N}{\beta_N + 1} - \frac{i_P}{\beta_P + 1}$$
$$= \frac{i_L}{\beta + 1}$$

Step 2 of 4

The input resistance, $v_I = v_o + v_{BEN} - \frac{V_{BB}}{2}$

The voltage,

$$V_{BB} = 2V_T \ln\left(\frac{I_Q}{I_S}\right)$$
$$= 2(0.025)\ln\left(\frac{2\times10^{-3}}{10^{-13}}\right)$$
$$= 2(0.025)(23.72)$$
$$= 1.186 \text{ V}$$

Step 3 of 4

Calculate the large signal input resistance.

$$R_{in} = \frac{v_I}{i_I}$$

Calculate the approximate value of the large signal input resistance.

$$R_{in} \simeq \beta R_L$$

Step 4 of 4

The table 1 shows the values calculated for different output voltages in the range,

-10 V to $+10$ V

Table 1

v_o (V)	$i_L = \frac{v_o}{R_L}$ (mA)	$i_I = \frac{i_L}{\beta+1}$ (mA)	v_I (V)	$R_{in} = \frac{v_I}{i_I}$ (kΩ)	$R_{in} \simeq \beta R_L$ (approximate kΩ)
+10	100	2	10.1	5.05	4.9
5	50	1	5.08	5.08	4.9
1	10	0.2	1.041	5.205	4.9
0.5	5	0.1	0.526	5.26	4.9
0.2	2	0.04	0.212	5.3	4.9
0.1	1	0.02	0.106	5.3	4.9
0	0	0	0	0	0
−0.1	−1	−0.02	−0.106	5.3	4.9
−0.2	−2	−0.04	−0.212	5.3	4.9
−0.5	−5	−0.1	−0.526	5.26	4.9
−1	−10	−0.2	−1.041	5.205	4.9
−5	−50	−1	−5.08	5.08	4.9
−10	−100	−2	−10.1	5.05	4.9

Thus, the input resistance is calculated, and compared to the approximated value.

12.021E

We don't have the solution to this problem yet.

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12.021P

We don't have the solution to this problem yet.

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Step 1 of 6

Refer to Figure 11.11 in the text book.

The positive V_o causes a current i_L to follow through R_L, and thus i_N must increases; that is

$$i_N = i_P + i_L \quad \text{..... (1)}$$

$$i_L = \frac{V_o}{R_L}$$

Substitute -1V for V_o and 100Ω for R_L in equation.

$$i_L = \frac{-1\text{V}}{100\Omega}$$
$$= -10\text{mA}$$

Step 2 of 6

The increase in i_N will be accompanied by a corresponding increase in v_{BEN}.

However, since the voltage between the two bases remain constant at v_{BB}, the increase in v_{BEN} will result in equal decrease in v_{BEP} and hence in i_P. The relationship between i_N and i_P can be derived as follows:

$$v_{BEN} + v_{EBP} = V_{BB}$$
$$V_T \ln\frac{i_N}{I_S} + V_T \ln\frac{i_P}{I_S} = 2V_T \ln\frac{I_Q}{I_S}$$

$$i_N i_P = I_Q^2 \quad \text{..... (2)}$$

Thus as i_N increases, i_P decreases by the same ratio where the product remain constant the equations 1 and 2 can be combined to yield i_N for a given i_L as the solution to the quadratic equation is

$$i_N^2 - i_L i_N - I_Q^2 = 0$$

Substitute the values

$$i_N^2 - (-10)i_N - (1)^2 = 0$$
$$i_N^2 + 10i_N - 1 = 0$$

Therefore, the value of i_N is

$$i_N = \frac{-10 \pm \sqrt{100 - 4(1)(-1)}}{2}$$
$$= \frac{-10 + 10.198}{2}$$
$$= 0.099\text{mA}$$

Substitute 0.099mA for i_N in equation 2

$$i_N i_P = I_Q^2$$
$$0.099\text{mA} \times i_P = 1\text{mA}$$
$$i_P = 10.099\text{A}$$

Step 3 of 6

The decrease in i_N will be accomplished by a corresponding increases in v_{BEP}.

$$v_{BEP} = V_T \ln\frac{i_P}{I_S}$$
$$v_{BEP} = \left(2.5\times10^{-3}\right)\ln\frac{10.099\text{mA}}{1\text{mA}}$$
$$v_{BEP} = \left(2.5\times10^{-3}\right)(2.3124)$$
$$v_{BEP} = 0.057811\text{V}$$

Therefore, the input step must be $\boxed{-1.057816}$.

Step 4 of 6

Calculate the value of v_{BEN}.

$$v_{BEN} = V_T \ln\frac{i_N}{I_S}$$
$$= \left(2.5\times10^{-3}\right)\ln\frac{0.099\text{mA}}{1\text{mA}}$$
$$= \left(2.5\times10^{-3}\right)(-2.3126)$$
$$= -0.057816\text{V}$$

Step 5 of 6

The single supply utilizing is 10V

Consider the transistor saturation voltage is zero

The given biased current is $V_I = 6\text{V}$

Therefore, the possible positive output is 6V to 10V

The largest possible positive-going step at the output is

$$= 10 - 6$$
$$= 4\text{V}$$

Therefore, the largest possible positive-going step at the output is $\boxed{4\text{V}}$.

Step 6 of 6

Consider the transistor saturation voltage is zero.

The given biased current is $V_I = 6\text{V}$

Therefore, the possible negative output is 6V to 0V

The largest possible negative-going step at the output is

$$= 6 - 0$$
$$= 6\text{V}$$

Therefore, the largest possible negative-going step at the output is $\boxed{6\text{V}}$.

12.023P

We don't have the solution to this problem yet.

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12.024P

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12.025P

We don't have the solution to this problem yet.

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12.026P

We don't have the solution to this problem yet.

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12.027P

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Step 1 of 6

12.028P

Refer to Figure 11.14 in the text book.

The saturation current at $20^\circ C$ is,

$$I_S = I_Q e^{-\left(\frac{V_{BE}}{V_T}\right)}$$

Substitute $0.6\ V$ for $|V_{BE}|$ and $25\ mV$ for V_T in equation.

$$I_S = 10^{-3} e^{-\left(\frac{0.6}{25\times10^{-3}}\right)}$$
$$= 3.775\times10^{-11}\ mA$$

The saturation current at $70^\circ C$ increases by $14\%/^\circ C$.

$$I_S = I_{S(20^\circ C)}(1.14)^{70^\circ - 20^\circ}$$

Substitute $3.775\times10^{-11}\ mA$ for I_S in equation.

$$I_S = 3.775\times10^{-11}\times10^{-3}(1.14)^{50}$$
$$= 2.64\times10^{-8}\ mA$$

Therefore, the saturation current at $70^\circ C$ is $\boxed{2.64\times10^{-8}\ mA}$.

Step 2 of 6

The thermal voltage at $70^\circ C$ is given by

$$V_T = \frac{25\times10^{-3}}{(273+20)}(273+70)$$
$$= \frac{25\times10^{-3}}{293}\times343$$
$$= 29.26\ mV$$

Thus the quiescent current at $70^\circ C$ is

$$I_Q = I_S e^{\left(\frac{V_{BE}}{V_T}\right)}$$
$$= 2.64\times10^{-8}\times10^{-3} e^{\left(\frac{0.6}{29.26\times10^{-3}}\right)}$$
$$= 21.15\ mA$$

Step 3 of 6

The additional current is

$$= 21.15 - 1$$
$$= 20.15\ mA$$

The additional power dissipated is,

$$P_D = 2\times20\times20.15\times10^{-3}$$
$$= 806\ mW$$

Therefore, the additional power dissipated is $\boxed{806\ mW}$.

Step 4 of 6

The additional temperature rise is

$$= 10\times806\times10^{-3}$$
$$= 8.06^\circ C$$

The additional temperature rise is $\boxed{8.06^\circ C}$.

Step 5 of 6

At $8.06^\circ C$, the thermal voltage is,

$$V_T = \frac{25\times10^{-3}}{(273+20)}(273+8.06)$$
$$= \frac{25\times10^{-3}}{293}\times281.06$$
$$= 24\ mV$$

Step 6 of 6

The quiescent current is given by

$$I_Q = I_S e^{\left(\frac{V_{BE}}{V_T}\right)}$$
$$= 3.775\times10^{-11}\times10^{-3} e^{\left(\frac{0.6}{29.26\times10^{-3}}\right)}$$
$$= 30.37\ mA$$

Therefore, the current increase is $\boxed{30.37\ mA}$.

12.029P

We don't have the solution to this problem yet.

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12.030P

We don't have the solution to this problem yet.

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12.031P

We don't have the solution to this problem yet.

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Step 1 of 17

Refer to Figure 11.30 in the text book.

Consider the results of Exercise 11.14 in the text book.

The feedback factor is

The quiescent current in each of the four transistors is for $v_I = 0$ and $R_L = \infty$.

The collector current of transistor Q_1 is $I_{C1} = 0.88\,\text{mA}$ for $v_I = +10\,\text{V}$ and $R_L = \infty$.

The collector current of transistor Q_1 is $I_{C1} = 4.87\,\text{mA}$ for $v_I = -10\,\text{V}$ and $R_L = \infty$.

The collector current of transistor Q_2 is $I_{C2} = 4.87\,\text{mA}$ for $v_I = +10\,\text{V}$ and $R_L = \infty$.

The collector current of transistor Q_2 is $I_{C2} = 0.88\,\text{mA}$ for $v_I = -10\,\text{V}$ and $R_L = \infty$.

The collector current of transistor Q_1 is $I_{C1} = 0.38\,\text{mA}$ for $v_I = +10\,\text{V}$ and $R_L = 100\Omega$.

The collector current of transistor Q_1 is $I_{C1} = 4.87\,\text{mA}$ for $v_I = -10\,\text{V}$ and $R_L = 100\Omega$.

The collector current of transistor Q_2 is $I_{C2} = 4.87\,\text{mA}$ for $v_I = +10\,\text{V}$ and $R_L = 100\Omega$.

The collector current of transistor Q_2 is $I_{C2} = 0.38\,\text{mA}$ for $v_I = -10\,\text{V}$ and $R_L = 100\Omega$.

Consider the circuit diagram for load resistance $R_L = \infty$ is shown in Figure 1.

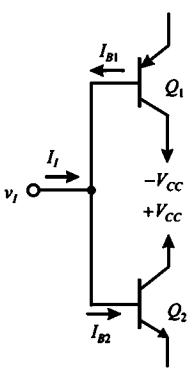


Figure 1

Step 2 of 17

Consider the input voltage, $v_I = 0\,\text{V}$.

The base current at Q_1 is,

$$I_{B1} = \frac{I_Q}{\beta}$$

Substitute $2.87\,\text{mA}$ for I_Q and 200 for β in equation.

$$I_{B1} = \frac{2.87}{200}$$
$$= 0.01435\,\text{mA}$$

Step 3 of 17

The base current at Q_2 is,

$$I_{B2} = \frac{I_Q}{\beta}$$

Substitute $2.87\,\text{mA}$ for I_Q and 200 for β in equation.

$$I_{B2} = \frac{2.87}{200}$$
$$= 0.01435\,\text{mA}$$

Step 4 of 17

The input current is,

$$I_I = I_{B2} - I_{B1}$$
$$= 0.01435 - 0.01435$$
$$= 0\,\text{A}$$

Therefore, the input current I_I is for $v_I = 0$ and $R_L = \infty$ is $\boxed{0\,\text{A}}$.

Step 5 of 17

Consider the input voltage, $v_I = +10\,\text{V}$.

The base current at Q_1 is,

$$I_{B1} = \frac{I_{C1}}{\beta}$$

Substitute $0.88\,\text{mA}$ for I_{C1} and 200 for β in equation.

$$I_{B1} = \frac{0.88}{200}$$
$$= 0.0044\,\text{mA}$$

Step 6 of 17

The base current at Q_2 is,

$$I_{B2} = \frac{I_{C2}}{\beta}$$

Substitute $4.87\,\text{mA}$ for I_{C1} and 200 for β in equation.

$$I_{B2} = \frac{4.87}{200}$$
$$= 0.02435\,\text{mA}$$

Step 7 of 17

The input current is,

$$I_I = I_{B2} - I_{B1}$$

Substitute $0.02435\,\text{mA}$ for I_{B2} and $0.0044\,\text{mA}$ for I_{B1} in equation.

$$I_I = 0.02435 - 0.0044$$
$$= 0.01995\,\text{mA}$$

Therefore, the input current for $v_I = +10\,\text{V}$ and $R_L = \infty$ is $\boxed{0.01995\,\text{mA}}$.

Step 8 of 17

Consider the input voltage, $v_I = -10\,\text{V}$.

The base current at Q_1 is,

$$I_{B1} = \frac{I_{C1}}{\beta}$$
$$= \frac{4.87}{200}$$
$$= 0.02435\,\text{mA}$$

Step 9 of 17

The base current at Q_2 is,

$$I_{B2} = \frac{I_{C2}}{\beta}$$
$$= \frac{0.88}{200}$$
$$= 0.0044\,\text{mA}$$

Step 10 of 17

The input current is,

$$I_I = I_{B2} - I_{B1}$$
$$= 0.0044 - 0.02435$$
$$= -0.01995\,\text{mA}$$

Therefore, the input current for $v_I = -10\,\text{V}$ and $R_L = \infty$ is $\boxed{-0.01995\,\text{mA}}$.

Step 11 of 17

For load resistance $R_L = 100\Omega$

For $v_I = 0\,\text{V}$

Thus, the input current also zero

Therefore, the input current for $v_I = 0$ and $R_L = 100\Omega$ is $\boxed{0\,\text{A}}$

Step 12 of 17

Consider the input voltage , $v_I = +10\,\text{V}$.

The base current at Q_1 is,

$$I_{B1} = \frac{I_{C1}}{\beta}$$
$$= \frac{0.38}{200}$$
$$= 0.0019\,\text{mA}$$

Step 13 of 17

The base current at Q_2 is,

$$I_{B2} = \frac{I_{C2}}{\beta}$$
$$= \frac{4.87}{200}$$
$$= 0.02435\,\text{mA}$$

Step 14 of 17

The input current is,

$$I_I = I_{B2} - I_{B1}$$
$$= 0.02435 - 0.0019$$
$$= 0.02245\,\text{mA}$$

Therefore, the input current for $v_I = +10\,\text{V}$ and $R_L = 100\Omega$ is $\boxed{0.02245\,\text{mA}}$.

Step 15 of 17

Consider the input voltage , $v_I = -10\,\text{V}$.

The base current at Q_1 is,

$$I_{B1} = \frac{I_{C1}}{\beta}$$
$$= \frac{4.87}{200}$$
$$= 0.02435\,\text{mA}$$

Step 16 of 17

The base current at Q_2 is,

$$I_{B2} = \frac{I_{C2}}{\beta}$$
$$= \frac{0.38}{200}$$
$$= 0.0019\,\text{mA}$$

Step 17 of 17

The input current is,

$$I_I = I_{B2} - I_{B1}$$
$$= 0.0019 - 0.02435$$
$$= -0.02245\,\text{mA}$$

Therefore, the input current for $v_I = -10\,\text{V}$ and $R_L = 100\Omega$ is $\boxed{-0.02245\,\text{mA}}$.

Step 1 of 3

12.033P

Refer to Figure 11.30 in the textbook for output stage of a class AB amplifier with an input buffer.

Draw the circuit diagram with notations.

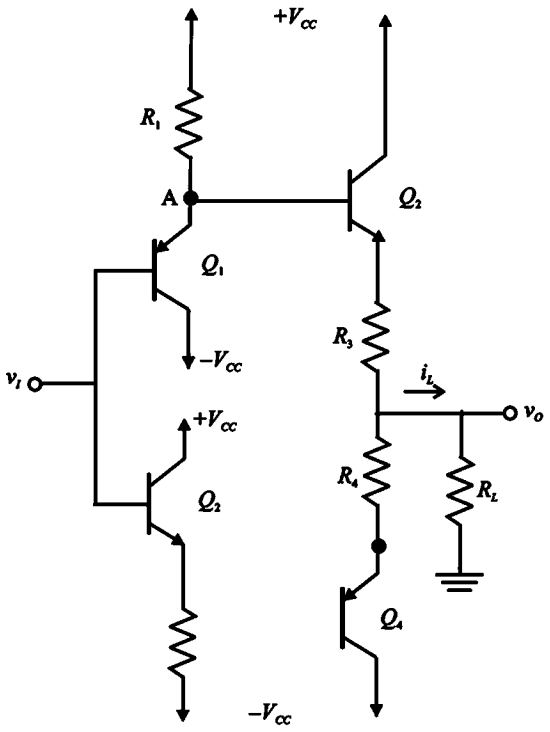


Figure 1

Step 2 of 3

The circuit in Figure 1 is operated near $v_I = 0$ and fed with a signal source having zero resistance. The top and bottom halves of the circuit are perfectly matched.

The transistor, Q_1 can be replaced with a equivalent resistance of r_{e1}.

Calculate the equivalent resistance looking the circuit from point A.

$$R_{e1} = R_1 \| r_{e1}$$

Calculate the reflected resistance from base to the emitter of the transistor, Q_3.

$$R_{e3} = \frac{R_{e1}}{\beta_3 + 1}$$
$$= \frac{R_1 \| r_{e1}}{\beta_3 + 1}$$

Calculate the resistance looking into the circuit at the output, from the upper half of the circuit.

$$R_{out1} = R_{e3} + r_{e3} + \frac{R_1 \| r_{e1}}{\beta_3 + 1}$$

Step 3 of 3

Calculate the resistance looking into the circuit at the output, from the lower half of the circuit.

$$R_{out2} = R_{out1}$$
$$= R_{e3} + r_{e3} + \frac{R_1 \| r_{e1}}{\beta_3 + 1}$$

Calculate the output resistance.

$$R_{out} = R_{out1} \| R_{out2} \qquad (R_{out2} = R_{out1})$$
$$= \frac{R_{out1} \times R_{out1}}{R_{out1} + R_{out1}}$$
$$= \frac{R_{out1}}{2}$$
$$= \frac{1}{2}\left[R_{e3} + r_{e3} + \frac{R_1 \| r_{e1}}{\beta_3 + 1}\right]$$

Thus, the output resistance of a class AB amplifier with an input buffer is,

$$\boxed{R_{out} = \frac{1}{2}\left[R_{e3} + r_{e3} + \frac{R_1 \| r_{e1}}{\beta_3 + 1}\right]}.$$

Step 1 of 13

Refer to Figure 11.30 in the textbook.

Calculate the voltage across the resistor, R_1.

$$\begin{aligned} V_{R1} &= V_{CC} - V_{BE} - v_I \\ &= 10 - 0.7 - 5 \\ &= 4.3\,\text{V} \end{aligned}$$

For $V_{CC} = 10\,\text{V}$, the values for resistors R_1 through R_4 allows a base current of at least $10\,\text{mA}$ in Q_3 and Q_4 at $v_I = 5\,\text{V}$ with at most a 2-to-1 variation current Q_1 and Q_2.

The current passing through resistor R_1 is,

$$\begin{aligned} i_{R_1} &= 2 \times 10\,\text{mA} \\ &= 20\,\text{mA} \end{aligned}$$

Step 2 of 13

Calculate the value of the resistor, R_1.

$$\begin{aligned} R_1 &= \frac{V_{R1}}{I_{R1}} \\ &= \frac{4.3}{20} \\ &= 215\,\Omega \end{aligned}$$

The emitter resisters of transistors Q_1 and Q_2 are equal, so the value of the resistor, R_2 is same as R_1.

$$\begin{aligned} R_2 &= R_1 \\ &= 215\,\Omega \end{aligned}$$

Therefore, the value of the resistors R_1 and R_2 are $\boxed{215\,\Omega}$.

Step 3 of 13

At the input voltage of $v_I = 5\,\text{V}$

The base emitter voltage of transistor Q_1 and Q_2 is $|V_{BE}| = 0.7\,\text{V}$ at $10\,\text{mA}$.

Calculate the voltage across the resistor, R_1.

$$\begin{aligned} V_{R1} &= V_{CC} - V_{BE} - v_I \\ &= 10 - 0.7 - 0 \\ &= 9.3\,\text{V} \end{aligned}$$

The current passing through resistor R_1 is,

$$\begin{aligned} i_{R1} &= \frac{V_{R1}}{R_1} \\ &= \frac{9.3\,\text{V}}{215\,\Omega} \\ &= 0.0432558 \\ &= 43.256\,\text{mA} \end{aligned}$$

Step 4 of 13

Calculate the emitter base voltage of transistor, Q_1.

$$\begin{aligned} V_{EB1} &= 0.7 + V_T \ln \frac{43.256}{10} \\ &= 0.7 + (0.025) \times \ln(4.3256) \\ &= 0.7 + 0.03661 \\ &= 0.73661\,\text{V} \end{aligned}$$

The no load quiescent current in transistor Q_3 and Q_4 is $I_Q = 40\,\text{mA}$

The feedback factor of Q_1 and Q_2 three times of Q_3 and Q_4, thus

$$\begin{aligned} I_{S3} &= 3I_{S1} \\ &= 3(10\,\text{mA}) \\ &= 30\,\text{mA} \end{aligned}$$

Step 5 of 13

Calculate the emitter base voltage of transistor, Q_3.

$$\begin{aligned} V_{EB3} &= 0.7 + V_T \ln \frac{40}{30} \\ &= 0.7 + (0.025) \times \ln(1.3333) \\ &= 0.7 + 0.00719 \\ &= 0.70719\,\text{V} \end{aligned}$$

Calculate the value of the resistor, R_3.

$$\begin{aligned} R_3 &= \frac{V_{EB1} - V_{EB3}}{I_{Q3}} \\ &= \frac{0.73661 - 0.70719}{40 \times 10^{-3}} \\ &= 0.7355\,\Omega \end{aligned}$$

The emitter resisters of transistors Q_3 and Q_4 are equal so the value of the resistor, R_3 is same as R_4.

$$\begin{aligned} R_4 &= R_3 \\ &= 0.7355\,\Omega \end{aligned}$$

Therefore, the value of the resistors R_3 and R_4 are $\boxed{0.7355\,\Omega}$.

Step 6 of 13

At the input voltage $v_I = 0\,\text{V}$, the output resistance is

$$R_{out} = \frac{1}{2}\left(R_3 + r_{e3} + \frac{r_{e1}}{\beta_3 + 1}\right)$$

Calculate the value of r_{e1}.

$$\begin{aligned} r_{e1} &\cong r_{e3} \\ &= \frac{25\,\text{mV}}{40\,\text{mA}} \\ &= 0.625\,\Omega \end{aligned}$$

Calculate the value of output resistance, R_{out}.

$$\begin{aligned} R_{out} &= \frac{1}{2}\left(0.7355 + 0.625 + \frac{0.625}{50+1}\right) \\ &= \frac{1}{2}(1.3605 + 0.01225) \\ &= 0.6864\,\Omega \end{aligned}$$

Therefore, the output resistance, R_{out} is $\boxed{0.6864\,\Omega}$.

Step 7 of 13

Calculate the value of load current for $V_O \cong 1\,\text{V}$.

$$\begin{aligned} i_L &= \frac{V_o}{R_L} \\ &= \frac{1}{2} \\ &= 0.5 \\ &= 500\,\text{mA} \end{aligned}$$

Calculate the base current of the transistor, Q_3.

$$\begin{aligned} i_{B3} &= \frac{i_L}{\beta} \\ &= \frac{500}{50} \\ &= 10\,\text{mA} \end{aligned}$$

Step 8 of 13

Calculate the emitter current of the transistor, Q_1.

$$\begin{aligned} i_{E1} &= \frac{V_{CC} - V_I - V_{BE}}{R_1} - i_{B3} \\ &= \frac{10 - 1 - 0.7}{215} - 10 \times 10^{-3} \\ &= 38.61 - 10 \\ &= 28.61\,\text{mA} \end{aligned}$$

Calculate the emitter base voltage of transistor, Q_1.

$$\begin{aligned} V_{EB1} &= 0.7 + V_T \ln \frac{28.61}{10} \\ &= 0.7 + (0.025) \times \ln(2.861) \\ &= 0.7 + 0.0263 \\ &= 0.7263\,\text{V} \end{aligned}$$

Step 9 of 13

Calculate the base voltage of transistor, Q_3.

$$\begin{aligned} V_{B3} &= V_I + V_{EB1} \\ &= 1 + 0.7263 \\ &= 1.7263\,\text{V} \end{aligned}$$

Calculate the emitter base voltage of transistor, Q_3.

$$\begin{aligned} V_{BE3} &= 0.7 + V_T \ln\left(\frac{i_L}{I_{S3}}\right) \\ &= 0.7 + 0.025 \ln \frac{500}{30} \\ &= 0.7 + 0.07033 \\ &= 0.77033\,\text{V} \end{aligned}$$

Step 10 of 13

Calculate the current passing through the load resistor.

$$\begin{aligned} I_L &= \frac{V_{B3} - V_{BE3}}{R_4 + R_L} \\ &= \frac{1.7263 - 0.77033}{0.7355 + 2} \\ &= 0.3495\,\text{A} \end{aligned}$$

Calculate the output voltage, V_O.

$$\begin{aligned} V_O &= R_L i_L \\ &= 2 \times 0.3495 \\ &= 0.699\,\text{V} \end{aligned}$$

Thus, the voltage drop across the series combination of R_4 and the EB junction of Q_4 is,

$$\begin{aligned} V_{B4} &= V_{E2} \\ &= V_I - V_{BE1} \\ &= 1 - 0.7366 \\ &= 0.2634\,\text{V} \end{aligned}$$

Step 11 of 13

After leaving a drop across the series combination of R_4 and EBJA of $0.698 - 0.26 = 0.433$. It follows that $i_{E4} \cong 0$, as assumed.

Iterate again:

$$i_L \cong 0.35\,\text{A}$$

The emitter current of Q_3 is,

$$\begin{aligned} i_{B3} &= \frac{0.35}{51} \\ &= 7\,\text{mA} \end{aligned}$$

The emitter current of Q_1 is,

$$\begin{aligned} i_{E1} &= \frac{V_{CC} - V_I - V_{EB1}}{R_1} - i_{B3} \\ &= \frac{10 - 1 - 0.73}{215} - 7 \\ &= 31.465\,\text{mA} \end{aligned}$$

Step 12 of 13

Calculate the emitter base voltage of transistor, Q_3.

$$\begin{aligned} V_{EB3} &= 0.7 + V_T \ln \frac{31.465}{10} \\ &= 0.7 + (0.025) \times \ln(3.1465) \\ &= 0.7 + 0.02866 \\ &= 0.72866\,\text{V} \end{aligned}$$

Calculate the base voltage of transistor, Q_3.

$$\begin{aligned} V_{B3} &= V_I + V_{EB1} \\ &= 1 + 0.72866 \\ &= 1.72866\,\text{V} \end{aligned}$$

Calculate the base emitter voltage of transistor, Q_3.

$$\begin{aligned} V_{BE3} &= 0.7 + V_T \ln\left(\frac{i_L}{I_{S3}}\right) \\ &= 0.7 + 0.025 \ln \frac{350}{30} \\ &= 0.7 + 0.0614 \\ &= 0.7614\,\text{V} \end{aligned}$$

Step 13 of 13

Calculate the current passing through the load resistor.

$$\begin{aligned} I_L &= \frac{V_{B3} - V_{BE3}}{R_4 + R_L} \\ &= \frac{1.72866 - 0.7614}{0.7355 + 2} \\ &= 0.3536\,\text{A} \end{aligned}$$

Calculate the value of output voltage is

$$\begin{aligned} V_O &= R_L i_L \\ &= 2 \times 0.3536 \\ &= 0.7072\,\text{V} \end{aligned}$$

Therefore, the output voltage is $\boxed{0.7072\,\text{V}}$.

12.035P

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Step 1 of 9

Refer to Figure 11.31 in the textbook.

(a)

The base current of transistor, Q_2 is,

$$I_{B2} = \frac{I_{E2}}{\beta_2 + 1}$$

The base current of transistor, Q_2 is same emitter current of transistor, Q_1.

$$\begin{aligned} I_{E1} &= I_{B2} \\ &= \frac{I_{E2}}{\beta_2 + 1} \end{aligned}$$

The base current of transistor, Q_1 is,

$$\begin{aligned} I_{B1} &= \frac{I_{E1}}{\beta_1 + 1} \\ &= \frac{1}{\beta_1 + 1}\left(\frac{I_{E2}}{\beta_2 + 1}\right) \\ &= \frac{I_{E2}}{(\beta_1 + 1)(\beta_2 + 1)} \end{aligned}$$

Step 2 of 9

The collector current of transistor, Q_2 is,

$$\begin{aligned} I_{C2} &= \alpha_2 I_{E2} \\ &= \left(\frac{\beta_2}{\beta_2 + 1}\right) I_{E2} \end{aligned}$$

Calculate the expression for β.

$$\begin{aligned} \beta &= \frac{I_{C2}}{I_{B1}} \\ &= \frac{\left(\frac{\beta_2}{\beta_2 + 1}\right) I_{E2}}{\frac{I_{E2}}{(\beta_1 + 1)(\beta_2 + 1)}} \\ &= \beta_2(\beta_1 + 1) \qquad (\beta_1, \beta_2 \gg 1) \\ &= \beta_1 \beta_2 \end{aligned}$$

Thus, the expression $\beta = \beta_1 \beta_2$ is proved.

Step 3 of 9

(b)

Calculate the operating current of transistor, Q_2.

$$\begin{aligned} I_{Q_2} &= I_{E2} \\ &= \frac{I_{C2}}{\alpha_2} \\ &= \left(\frac{\beta_2 + 1}{\beta_2}\right) I_C \qquad (\beta_2 \gg 1) \\ &\approx I_C \end{aligned}$$

Thus, the transistor, Q_2 operates at a collector current approximately.

Calculate the operating current of transistor, Q_1.

$$\begin{aligned} I_{Q_1} &= I_{E1} \\ &= \frac{I_{E2}}{\beta_2 + 1} \\ &= \frac{1}{\beta_2 + 1}\left(\frac{I_{C2}}{\alpha_2}\right) \\ &= \left(\frac{1}{\beta_2 + 1}\right)\left(\frac{\beta_2 + 1}{\beta_2}\right) I_C \end{aligned}$$

$$I_{Q_1} = \frac{I_C}{\beta_2}$$

Thus, the transistor, Q_2 operates at a collector current equal to $\frac{I_C}{\beta_2}$.

Step 4 of 9

(c)

The transistor, Q_2 operates at a collector current.

$$I_{C2} = I_C$$

The transistor, Q_2 operates at a collector current equal to I_C / β_2.

$$I_{C1} = \frac{I_C}{\beta_2}$$

Consider the expression for the collector current to calculate V_{BE}.

$$\begin{aligned} I_C &= I_S e^{\frac{V_{BE}}{V_T}} \\ e^{\frac{V_{BE}}{V_T}} &= \frac{I_C}{I_S} \\ V_{BE} &= V_T \ln\left(\frac{I_C}{I_S}\right) \end{aligned}$$

Step 5 of 9

The expression for the base-emitter voltage of transistor, Q_1 is,

$$\begin{aligned} V_{BE1} &= V_T \ln\left(\frac{I_{C1}}{I_S}\right) \qquad \left(I_{C1} = \frac{I_C}{\beta_2}\right) \\ &= V_T \ln\left(\frac{I_C}{\beta_2 I_S}\right) \\ &= V_T \ln\left(\frac{I_C}{I_S}\right) + V_T \ln\left(\frac{1}{\beta_2}\right) \\ &= V_T \ln\left(\frac{I_C}{I_S}\right) - V_T \ln(\beta_2) \end{aligned}$$

The expression for the base-emitter voltage of transistor, Q_2 is,

$$\begin{aligned} V_{BE2} &= V_T \ln\left(\frac{I_{C2}}{I_S}\right) \qquad (I_{C2} = I_C) \\ &= V_T \ln\left(\frac{I_C}{I_S}\right) \end{aligned}$$

Calculate the expression for total base-emitter voltage, V_{BE}.

$$\begin{aligned} V_{BE} &= V_{BE1} + V_{BE2} \\ &= V_T \ln\left(\frac{I_C}{I_S}\right) - V_T \ln(\beta_2) + V_T \ln\left(\frac{I_C}{I_S}\right) \\ &= 2V_T \ln\left(\frac{I_C}{I_S}\right) - V_T \ln(\beta_2) \end{aligned}$$

Thus, the expression, $V_{BE} = 2V_T \ln\left(\frac{I_C}{I_S}\right) - V_T \ln(\beta_2)$ is proved.

Step 6 of 9

(d)

Calculate the expression for small signal emitter resistance of Q_1.

$$\begin{aligned} r_{e1} &= \frac{V_T}{I_{E1}} \\ &= \frac{V_T}{I_{B2}} \\ &= \frac{V_T}{\left(\frac{I_{E2}}{\beta_2 + 1}\right)} \qquad \left(r_{e2} = \frac{V_T}{I_{E2}}\right) \\ &= (\beta_2 + 1) r_{e2} \end{aligned}$$

Calculate the equivalent small signal emitter resistance.

$$\begin{aligned} r_\pi &= (\beta_1 + 1)\left[r_{e1} + (\beta_2 + 1) r_{e2}\right] \\ &= (\beta_1 + 1)\left[(\beta_2 + 1) r_{e2} + (\beta_2 + 1) r_{e2}\right] \\ &= 2(\beta_1 + 1)(\beta_2 + 1) r_{e2} \qquad (\beta_2, \beta_2 \gg 1) \\ &= 2\beta_1 \beta_2 \frac{V_T}{I_C} \end{aligned}$$

Thus, the expression for $r_\pi = 2\beta_1 \beta_2 \frac{V_T}{I_C}$ is proved.

Step 7 of 9

(e)

Calculate the expression for small signal emitter resistance of Q_1.

$$\begin{aligned} r_{e1} &= \frac{V_T}{I_{E1}} \\ &= \frac{V_T}{I_{B2}} \\ &= \frac{V_T}{\left(\frac{I_{E2}}{\beta_2 + 1}\right)} \qquad \left(r_{e2} = \frac{V_T}{I_{E2}}\right) \\ &= (\beta_2 + 1) r_{e2} \end{aligned}$$

Consider the expression for trans-conductance of Q_2.

$$\begin{aligned} g_{m2} &= \frac{I_{C2}}{V_T} \\ &\approx \frac{I_{E2}}{V_T} \qquad \left(r_{e2} = \frac{V_T}{I_{E2}}\right) \\ &\approx \frac{1}{r_{e2}} \end{aligned}$$

Step 8 of 9

Consider the expression for small signal collector current, i_C.

$$\begin{aligned} i_C &= g_{m2} v_{be2} \\ &= g_{m2}\left[\frac{(\beta_2 + 1) r_{e2}}{r_{e1} + (\beta_2 + 1) r_{e2}}\right] v_i \end{aligned}$$

Substitute $\frac{1}{r_{e2}}$ for g_{m2} and $(\beta_2 + 1) r_{e2}$ for r_{e1}.

$$\begin{aligned} i_C &= \left(\frac{1}{r_{e2}}\right)\left[\frac{(\beta_2 + 1) r_{e2}}{(\beta_2 + 1) r_{e2} + (\beta_2 + 1) r_{e2}}\right] v_i \\ &= \left(\frac{1}{r_{e2}}\right)\left[\frac{(\beta_2 + 1) r_{e2}}{2(\beta_2 + 1) r_{e2}}\right] v_i \\ &= \frac{v_i}{2 r_{e2}} \end{aligned}$$

Step 9 of 9

Calculate the expression for overall trans-conductance.

$$\begin{aligned} g_m &= \frac{i_C}{v_i} \\ &= \frac{\frac{v_i}{2 r_{e2}}}{v_i} \\ &= \frac{1}{2 r_{e2}} \\ &= \frac{I_C}{2 V_T} \end{aligned}$$

Thus, the expression for $g_m = \frac{I_C}{2 V_T}$ is proved.

12.037P

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12.038P

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Step 1 of 19

Refer to Figure 11.33 in the textbook

The quiescent current through Q_1 **and** Q_2 is:

$$I_{C1} = I_{C4}$$
$$= 2\text{ mA}$$

Calculate the base-emitter voltage of Q_1 **and** Q_2.

$$V_{BE1} = V_{BE4}$$
$$= V_T \ln\left(\frac{I_{C1}}{I_{ES}}\right)$$

Substitute 2 mA for I_{C1} and 100 for β_F.

$$I_{C1} = \frac{I_{C1}}{\beta_F}$$
$$= \frac{2\times10^{-3}}{100}$$
$$= 0.02\times10^{-3}$$
$$= 0.02\text{ mA}$$

Step 2 of 19

The collector current of transistor Q_1 is almost same as its emitter current, which is same as base current of transistor Q_1.

Step 3 of 19

Calculate base-emitter voltage of Q_1 **and** Q_2.

$$V_{BE1} = V_{BE2}$$
$$= V_T \ln\left(\frac{I_{C1}}{I_{S2}}\right)$$

Substitute 0.7 V for V_T, 25 mV for I_{C1} and 1 mA for I_{S2}.

$$V_{BE1} = 0.7 + 0.025\ln\left(\frac{0.02}{1}\right)$$
$$= 0.7 - 0.0978$$
$$= 0.602\text{ V}$$

Calculate base current of transistor Q_1.

$$I_{B1} = \frac{I_{C1}}{\beta_F}$$

Substitute 0.02 mA for I_{C1} and 100 for β_F.

$$I_{B1} = \frac{0.02\times10^{-3}}{100}$$
$$= 0.2\times10^{-4}$$
$$= 0.2\text{ µA}$$

Step 4 of 19

The base current is 100 time the base current in transistor. Q_1

Calculate value of base current, I_{bias}.

$$I_{bias} = 10I_{B1}$$

Substitute 0.2 µA for I_{B1}.

$$I_{bias} = 100(0.2\times10^{-4})$$
$$= 20\times10^{-4}$$
$$= 20\text{ µA}$$

Thus, the value of bias current, I_{bias} is $\boxed{20\text{ µA}}$.

Step 5 of 19

The bias current is nine times the current associated with the resistors.

Calculate current through the resistor, I_R

$$9I_R = I_{bias}$$
$$I_R = \frac{I_{bias}}{9}$$

Substitute 20 µA for I_{bias}.

$$I_R = \frac{20\times10^{-6}}{9}$$
$$= 2\times10^{-6}$$
$$= 2\text{ µA}$$

Step 6 of 19

Apply nodal analysis at the base current node to calculate the collector current through the transistor, Q_1.

$$I_{C1(act)} + I_B = I_{R2}$$
$$I_{C1} + I_B = I_{bias}$$

Substitute 2 µA for I_B and 20 µA for I_{bias}.

$$I_{C1} = I_{bias} - I_B$$
$$= 20\text{ µA} - 2\text{ µA}$$
$$= 18\text{ µA}$$

Calculate the base-emitter voltage of Q_1.

$$V_{BE1} = 0.7 + 0.025\ln\left(\frac{I_{C1}}{I_{S1}}\right)$$

Substitute 0.7 V for V_{AE}, 25 mV for V_T, 18 µA for $I_{C1(act)}$ and 1 mA for I_{S1}.

$$V_{BE1} = 0.7 + 0.025\ln\left(\frac{18\times10^{-6}}{1\times10^{-3}}\right)$$
$$= 0.7 - 0.102$$
$$= 0.598\text{ V}$$

Apply voltage division rule to calculate the voltage, $V_{BE(act)}$.

$$V_A = \left(\frac{R}{R + R_E}\right)V_{BE(act)}$$

The voltage across the resistor, R_E is same as base-emitter voltage of transistor, Q_1.

Substitute $300\text{ k}\Omega$ for R_E, $632\text{ k}\Omega$ for R, 8.598 V for $V_{BE(act)}$ and 0.598 V for V_{BE1}.

$$V_{BE(act)} = \left(\frac{R_E + R_E}{R_E}\right)V_{BE1}$$
$$= \left(\frac{632 + 300}{300}\right)(0.598)$$
$$= 1.838\text{ V}$$

Step 7 of 19

Apply nodal analysis at the multiplier circuit to calculate the new collector current through the transistor, Q_2.

Refer to Figure 11.33 in the textbook

The quiescent current through Q_1 **and** Q_2 is:

$$I_{C1} = I_{C4}$$
$$= 2\text{ mA}$$

Calculate base-emitter voltage of Q_1 **and** Q_2.

$$V_{BE1} = V_{BE4}$$
$$= V_T \ln\left(\frac{I_{C1}}{I_{ES}}\right)$$

Substitute 0.7 V for V_T, 25 mV for I_{C1} and 1 mA for I_{ES}.

$$V_{BE1} = 0.7 + 0.025\ln\left(\frac{2}{10}\right)$$
$$= 0.7 - 0.04$$
$$= 0.660\text{ V}$$

Calculate the base current of transistor Q_1.

$$I_{B1} = \frac{I_{C1}}{\beta_F}$$

Substitute 2 mA for I_{C1} and 100 for β_F.

$$I_{B1} = \frac{2\times10^{-3}}{100}$$
$$= 0.02\times10^{-3}$$
$$= 0.02\text{ mA}$$

The load current is almost similar to the collector current of transistor, Q_1.

$$I_{C2} = I_L$$
$$= 10\text{ mA}$$

The currents through the transistors Q_1 and Q_2 is increased by $\frac{10}{2} = 5$ **times**.

Calculate the base-emitter voltage of Q_1.

$$V_{BE1(new)} = V_{BE1} + V_T \ln(5)$$

Substitute 0.66 V for V_{BE1} and 25 mV for V_T.

$$V_{BE(new)} = 0.66 + 0.025\ln(5)$$
$$= 0.66 + 0.04$$
$$= 0.7\text{ V}$$

Calculate the base-emitter voltage of Q_1.

$$V_{BE1(new)} = V_{BE1} + V_T \ln(5)$$

Substitute 0.602 V for V_{BE1} and 25 mV for V_T.

$$V_{BE(new)} = 0.602 + 0.025\ln(5)$$
$$= 0.642\text{ V}$$

Calculate the new base current of Q_1.

$$I_{B1(new)} = \frac{I_{C1(new)}}{\beta_F}$$

Substitute 0.7 V for V_{AE}, 25 mV for V_T, 18.2 µA for $I_{C1(new)}$ and 1 mA for I_{ES}.

$$I_{B1(new)} = \frac{0.7 + 0.025\ln\left(\frac{18.2\times10^{-6}}{1\times10^{-3}}\right)}{100}$$

Thus, the base current flows through the multiplier so that the current through the transistor, Q_1 is:

$$I_{C1} = 18\text{ µA} + 0.2\text{ µA}$$
$$= 18.2\text{ µA}$$

Thus, the voltage across the R_{E2} -multiplier remains approximately constant and the voltage $\left(V_{BE2}\right)$ decrease by the same value that V_{BE1} increases by.

$$V_{BE1} + V_{BE2} = 0.660 + 0.602 - 0.04$$

Since the current through each of Q_1 **and** Q_2 decreases by the same factor, consider as m

$$m = \frac{0.025\ln(m) + 0.025\ln(m) = -0.04\text{ V}}{0.45\times2)\text{mA}}$$
$$I_{C1} = 0.9\text{mA}$$

The current through the transistor Q_1 is:

$$I_{C1} = 0.9\text{ mA} + 0.9\text{ mA}$$
$$= 1.09\text{ mA}$$

It is increased by a factor of 5.5

Step 8 of 19

Calculate the base-emitter voltage of Q_1.

Substitute 0.602 V for V_{BE1} and 25 mV for V_T.

$$V_{BE(new)} = V_{BE} + V_T \ln(3.5)$$
$$= 0.602 + 0.025\ln(3.5)$$
$$= 0.644\text{ V}$$

Calculate the input voltage required to produce -10 V output.

$$V_I = -(V_{BE(new)} - V_{BE(new)})$$

Substitute 0.644 V for $V_{BE(new)}$

$$V_I = -10 - 0.644$$
$$= -10.644\text{ V}$$

Thus, the input voltage required to produce -10 V output is $\boxed{-10.644\text{ V}}$.

Step 9 of 19

Apply Kirchhoffs voltage law to the input loop around the resistors, R_1 **and** R_2 and the transistors Q_1 **and** Q_2.

$$V_{BE1} = V_{BE1} + V_{BE2}$$
$$= 2V_{BE1}$$
$$= 2(0.602)$$
$$= 1.844\text{ V}$$

Substitute 0.602 V for V_{BE1} and 1.844 V for V_{BE2}.

Thus, the value of the resistor, R_1 is $\boxed{632\text{ k}\Omega}$.

Step 10 of 19

Calculate the value of load current, I_L for $V_L = 10\text{ V}$.

$$I_L = \frac{V_L}{R_L}$$

Substitute 10 V for V_L and $1\text{ k}\Omega$ for R_L.

$$I_L = \frac{10}{1\times10^3}$$
$$= 10\times10^{-3}$$
$$= 10\text{ mA}$$

The load current is almost similar to the collector current of transistor, Q_2.

$$I_{C2} = I_L$$
$$= 10\text{ mA}$$

The currents through the transistors Q_1 and Q_2 is increased by $\frac{10}{2} = 5$ **times**.

Calculate the base-emitter voltage of Q_2.

$$V_{BE2(new)} = V_{BE2} + V_T \ln(5)$$

Substitute 0.66 V for V_{BE2} and 25 mV for V_T.

$$V_{BE(new)} = 0.66 + 0.025\ln(5)$$
$$= 0.66 + 0.04$$
$$= 0.7\text{ V}$$

Calculate the base-emitter voltage of Q_2.

$$V_{BE1(new)} = V_{BE1} + V_T \ln(5)$$

Substitute 0.602 V for V_{BE1} and 25 mV for V_T.

$$V_{BE(new)} = 0.602 + 0.025\ln(5)$$
$$= 0.642\text{ V}$$

Calculate the new base current of Q_2.

$$I_{B2(new)} = \frac{I_{C2(new)}}{\beta_F}$$

Substitute 0.7 V for V_{AE}, 25 mV for V_T, 18.2 µA for $I_{C2(new)}$ and 1 mA for I_{ES}.

$$I_{B2(new)} = \frac{0.7 + 0.025\ln\left(\frac{18.2\times10^{-6}}{1\times10^{-3}}\right)}{100}$$

Substitute $300\text{ k}\Omega$ for R_E, $632\text{ k}\Omega$ for R, 8.598 V for $V_{BE(new)}$ and 0.598 V for V_{BE1}.

$$V_{BE(new)} = \left(\frac{R_E + R_E}{R_E}\right)V_{BE1}$$
$$= \left(\frac{632 + 300}{300}\right)(0.598)$$
$$= 1.838\text{ V}$$

Step 11 of 19

Calculate the new base current of transistor Q_2.

$$I_{B(new)} = \frac{I_{C2}}{\beta_F}$$

Substitute 0.2 µA for I_{B1}.

$$I_{B(new)} = \frac{5(8.2\text{ µA})}{1\text{ µA}}$$

Calculate the current through the multiplier.

$$I_M = I_{BIAS} - I_{B(new)}$$

Substitute 1 µA for $I_{B(new)}$ and 20 µA for I_{BIAS}.

$$I_M = 20\text{ µA} - 1\text{ µA}$$
$$= 19\text{ µA}$$

Step 12 of 19

Apply nodal analysis at the multiplier circuit to calculate the new collector current through the transistor, Q_2.

$$I_{C1(act)} + I_B = I_M$$
$$I_{C1} + I_B = I_M$$

Substitute 2 µA for I_B and 19 µA for I_M.

$$I_{C1} = 19\text{ µA} - 2\text{ µA}$$
$$= 17\text{ µA}$$

Calculate the new base-emitter voltage of Q_2.

$$V_{BE2(new)} = V_{AE} + V_T \ln\left(\frac{I_{C1(act)}}{I_{S1}}\right)$$

Substitute 0.7 V for V_{AE}, 25 mV for V_T, 17 µA for $I_{C1(act)}$ and 1 mA for I_{S1}.

$$V_{BE2(new)} = 0.7 + 0.025\ln\left(\frac{17\times10^{-6}}{1\times10^{-3}}\right)$$
$$= 0.7 - 0.102$$
$$= 0.598\text{ V}$$

Apply voltage division rule to calculate the voltage, $V_{BE(new)}$.

$$V_A = \left(\frac{R}{R + R_E}\right)V_{BE(new)}$$

The voltage across the resistor, R_E is same as base-emitter voltage of transistor, Q_2.

Substitute $300\text{ k}\Omega$ for R_E, $632\text{ k}\Omega$ for R, 8.598 V for $V_{BE(new)}$ and 0.598 V for $V_{BE2(new)}$.

$$V_{BE(new)} = \left(\frac{R_E + R_E}{R_E}\right)V_{BE2(new)}$$
$$= \left(\frac{632 + 300}{300}\right)(0.598)$$
$$= 1.838\text{ V}$$

Thus, the value of the resistor, R_2 is $\boxed{300\text{ k}\Omega}$.

Step 13 of 19

Calculate the voltage drop, V_{R1} for I_{R1} and 2 µA for I_B

$$V_{R1} = I_{R1}(R_1 + R_2)$$
$$R_1 + R_2 = \frac{V_{R1}}{I_B}$$
$$R_1 = \frac{V_{R1}}{I_B} - R_2$$

Substitute 2 µA for I_B, $300\text{ k}\Omega$ for R_2 and 1.864 V for V_{BE}

$$R_1 = \frac{1.864}{2\times10^{-6}} - 300\times10^3$$
$$= 932\times10^3 - 300\times10^3$$
$$= 632\times10^3$$
$$= 632\text{ k}\Omega$$

Thus, the value of the resistor, R_1 is $\boxed{632\text{ k}\Omega}$.

Step 14 of 19

Calculate the input voltage required to produce $+10\text{ V}$ output

$$V_I = 10 + V_{BE(act)} + V_{BE(act)} - V_{BE(act)}$$

Substitute 1.838 V for $V_{BE(act)}$ and 0.7 V for $V_{BE(act)}$.

$$V_I = 10 + 0.642 + 0.7 - 1.838$$
$$= 9.484\text{ V}$$

Thus, the input voltage required to produce $+10\text{ V}$ output is $\boxed{9.484\text{ V}}$.

Step 15 of 19

Calculate the corresponding change in the collector current of transistor, Q_1.

$$\textbf{change in } I_{C1} = e^{\frac{-1.09}{0.025}}$$
$$= 0.035\times10^{-3}$$
$$= 0.035\text{ mA}$$

Calculate the input voltage required to produce -10 V output

$$V_I = -10 - V_{BE(new)}$$

Substitute 0.602 V for V_{BE1} and 25 mV for V_T.

$$V_{BE(new)} = 0.602 + 0.025\ln(3)$$
$$= 0.602\text{ V}$$

Calculate the input voltage required to produce -10 V output.

$$V_I = -10 - V_{BE(new)}$$

Substitute 0.644 V for $V_{BE(new)}$

$$V_I = -10 - 0.644$$
$$= -10.644\text{ V}$$

Thus, the input voltage required to produce -10 V output is $\boxed{-10.644\text{ V}}$.

12.040P

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Step 1 of 7

Refer to Figure 11.35 in the textbook.

Calculate the value of Zener voltage at $125°\ C$.

$$V_Z = V_{Z_1} + (T_2 - T_1)TC$$

Substitute $6.8\ V$ for V_{Z_1} , $125°\ C$ for T_2 , $25°\ C$ for T_1 , and $2\ mV/°C$ for TC .

$$\begin{aligned} V_Z &= 6.8 + (125-25)\left(2\times10^{-3}\right) \\ &= 6.8 + (100)\left(2\times10^{-3}\right) \\ &= 6.8 + 0.2 \\ &= 7.0\ V \end{aligned}$$

Step 2 of 7

The combination of the positive temperature coefficient of zener diode and the negative temperature coefficient of V_{BE1} causes the voltage at the emitter of Q_1 to rise.

Calculate the emitter voltage of the transistor Q_1 .

$$V_{E1} = V_Z - \left[V_{BE} + (T_2 - T_1)TC\right]$$

Substitute $7\ V$ for V_Z , $0.7\ V$ for V_{BE} , $125°\ C$ for T_2 , $25°\ C$ for T_1 , and $-2\ mV/°C$ for TC .

$$\begin{aligned} V_{E1} &= 7.0 - \left[0.7 + (125-25)\times\left(-2\times10^{-3}\right)\right] \\ &= 7.0 - (0.7 - 0.2) \\ &= 7.0 - 0.5 \\ &= 6.5\ V \end{aligned}$$

Step 3 of 7

Calculate the base emitter voltage of the transistor Q_2 .

$$V_{BE2} = V_{E1} - V_Z$$

Substitute $7\ V$ for V_Z and $6.5\ V$ for V_{E1} .

$$\begin{aligned} V_{BE2} &= 7.0 - 6.5 \\ &= 0.5\ V \end{aligned}$$

Calculate the value of the resistance, R_2 .

$$R_2 = \frac{V_{BE2}}{I_S}$$

Substitute $0.5\ V$ for V_{BE2} and $100\ \mu A$ for I_S .

$$\begin{aligned} R_2 &= \frac{0.5}{100\times10^{-6}} \\ &= 5\times10^3 \\ &= 5\ k\Omega \end{aligned}$$

Thus, the value of the resistance, R_2 is $\boxed{5\ k\Omega}$.

Step 4 of 7

Calculate the value of the resistance, R_1 .

$$R_1 = \frac{V_{E1} - V_{BE2}}{I_S}$$

Substitute $0.5\ V$ for V_{BE2} , $6.5\ V$ for V_{E1} and $100\ \mu A$ for I_S .

$$\begin{aligned} R_1 &= \frac{6.5 - 0.5}{100\times10^{-6}} \\ &= 6\times10^4 \\ &= 60\ k\Omega \end{aligned}$$

Thus, the value of the resistance, R_1 is $\boxed{60\ k\Omega}$.

Step 5 of 7

Calculate the value of Zener voltage at $25°\ C$.

$$V_Z = V_{Z_1} + (T_2 - T_1)TC$$

Substitute $6.8\ V$ for V_{Z_1} , $25°\ C$ for T_2 , $25°\ C$ for T_1 , and $2\ mV/°C$ for TC .

$$\begin{aligned} V_Z &= 6.8 + (25-25)\left(2\times10^{-3}\right) \\ &= 6.8 + (0)\left(2\times10^{-3}\right) \\ &= 6.8\ V \end{aligned}$$

Calculate the emitter voltage of the transistor Q_1 at $25°\ C$.

$$V_{E1} = V_Z - \left[V_{BE} + (T_2 - T_1)TC\right]$$

Substitute $6.8\ V$ for V_Z , $0.7\ V$ for V_{BE} , $25°\ C$ for T_2 , $25°\ C$ for T_1 , and $-2\ mV/°C$ for TC .

$$\begin{aligned} V_{E1} &= 6.8 - \left[0.7 + (25-25)\times\left(-2\times10^{-3}\right)\right] \\ &= 6.8 - 0.7 \\ &= 6.1\ V \end{aligned}$$

Step 6 of 7

Apply voltage division rule to calculate the base voltage of the transistor Q_2 .

$$V_{B2} = \left(\frac{R_2}{R_1 + R_2}\right)V_{E1}$$

Substitute $60\ k\Omega$ for R_1 , $5\ k\Omega$ for R_2 and $6.1\ V$ for V_{E1} .

$$\begin{aligned} V_{B2} &= \left(\frac{5}{60+5}\right)(6.1) \\ &= 6.1\times0.0769 \\ &= 0.4692\ V \end{aligned}$$

Step 7 of 7

Calculate the value of collector current of transistor, Q_2 .

$$I_{C2} = I_S e^{\frac{V_{B2} - V_{B1}}{V_T}}$$

Substitute $0.4692\ V$ for V_{B2} , $0.7\ V$ for V_{B1} , $25\ mV$ for V_T and $100\ \mu A$ for I_S .

$$\begin{aligned} I_{C2} &= \left(100\times10^{-6}\right)e^{\frac{0.469-0.7}{0.025}} \\ &= 100\times10^{-6}e^{-9.23} \\ &= 0.01\times10^{-6} \\ &= 0.01\ \mu A \end{aligned}$$

Therefore, the collector current of transistor, Q_2 at $25°C$ is $\boxed{0.01\ \mu A}$.

12.045P

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12.046P

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12.047P

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12.048P

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12.052P

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Step 1 of 9

12.053P

Refer to Figure 11.19 in the text book.

(a)

Write the formula for quiescent collector current.

$$I_Q = \frac{1}{2}k_n'\frac{W}{L}V_{OV}^2$$

Substitute $250\ \mu A/V^2$ for k_n', $1.5\ mA$ for I_Q, $0.15\ V$ for $|V_{OV}|$ in equation.

$$1.5\times10^{-3} = \frac{1}{2}\left(250\times10^{-6}\right)\left(\frac{W}{L}\right)_n(0.15)^2$$

$$\left(\frac{W}{L}\right)_n = \frac{2\left(1.5\times10^{-3}\right)}{\left(250\times10^{-6}\right)(0.15)^2}$$

$$\left(\frac{W}{L}\right)_n = 533.33$$

Thus, the width-to-length ratios of the NPN transistor is $\boxed{533.33}$.

Step 2 of 9

Substitute the known values for PNP transistor in this equation.

$$1.5\times10^{-3} = \frac{1}{2}\left(100\times10^{-6}\right)\left(\frac{W}{L}\right)_p(0.15)^2$$

$$\left(\frac{W}{L}\right)_p = \frac{2\left(1.5\times10^{-3}\right)}{\left(100\times10^{-6}\right)(0.15)^2}$$

$$\left(\frac{W}{L}\right)_p = 1333.33$$

Thus, the width-to-length ratios of the PNP transistor is $\boxed{1333.33}$.

Step 3 of 9

(b)

Write the formula for transconductance of the amplifier.

$$g_m = \frac{2I_Q}{V_{OV}}$$

Substitute $0.15\ V$ for $|V_{OV}|$, $1.5\ mA$ for I_Q in equation.

$$g_m = \frac{(2)\left(1.5\times10^{-3}\right)}{0.15}$$
$$= 20\times10^{-3}\ A/V$$
$$= 20\ mA/V$$

Step 4 of 9

Write the formula for the gain of the amplifier.

$$\mu = \frac{1}{2R_{out}g_m}$$

Substitute $2.5\ \Omega$ for R_{out}, $20\ mA/V$ for g_m in equation.

$$\mu = \frac{1}{(2)(2.5)\left(20\times10^{-3}\right)}$$
$$= 10$$

Thus, the gain of the amplifiers is $\boxed{10}$.

Step 5 of 9

(c)

Write the formula for expected error in gain.

$$\text{Gain error} = -\frac{V_{OV}}{4\mu I_Q R_L}$$

Substitute 10 for μ, $1.5\ mA$ for I_Q, $50\ \Omega$ for R_L and $0.15\ V$ for $|V_{OV}|$ in equation.

$$\text{Gain error} = -\frac{0.15}{(4)(10)\left(1.5\times10^{-3}\right)(50)}$$
$$= -0.05$$

Step 6 of 9

Calculate the percentage gain error.

$$\text{Gain error} = -0.05\times100$$
$$= -5\%$$

Thus, the expected gain error in the stage gain is: $\boxed{-5\%}$.

(d)

In the quiescent stage, the output voltage, $v_o = 0$

Calculate the dc voltage that must appear at the output of each of the error amplifier,

$$V_{DC} = \mu(v_O - v_I)$$
$$= -\mu v_I$$

Thus, the dc voltage that must appear at the output of each of the error amplifiers is: $\boxed{-\mu v_I}$.

Step 7 of 9

(e)

Write the formula for the current through the PNP transistor.

$$i_P = \frac{1}{2}k_p'\left(\frac{W}{L}\right)_p\left(2|V_{OV}|\right)^2$$

Substitute 1333.33 for $\left(\frac{W}{L}\right)_p$, $0.15\ V$ for $|V_{OV}|$, $100\ \mu A/V^2$ for k_p' in equation.

$$i_P = \frac{1}{2}\left(100\times10^{-6}\right)(1333.33)(4)(0.15)^2$$
$$= 6\ mA$$

Step 8 of 9

Calculate the output voltage at which the transistor Q_p will be supplying all the load current.

$$v_O = i_P R_L$$
$$= \left(6\times10^{-3}\right)(50)$$
$$= 0.3\ V$$

Thus, the output voltage at which the transistor Q_p will be supplying all the load current is $\boxed{0.3\ V}$.

For negative output voltage and the transistor Q_N supplying all the load current,

the output voltage is $\boxed{v_O > 0.3\ V}$.

Step 9 of 9

(f)

The maximum output voltage of the amplifier circuit is $2.05\ V$.

The linear range of the output voltage is $\boxed{2.05 \text{ to } -2.05\ V}$.

12.054P

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12.056P

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12.057P

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Step 1 of 3

12.058P

Refer to Figure P11.52 in the textbook.

The current through the inverting terminal of op-amp is i_{E1}.

$$i_{E1} = \frac{v_I}{R}$$

Calculate the collector current of the transistor, Q_1.

$$\begin{aligned} i_{C1} &= \alpha i_{E1} \\ &= \left(\frac{\beta}{1+\beta}\right)\left(\frac{v_I}{R}\right) \end{aligned}$$

As per current mirror concept, the current through the transistor, Q_1 is taken as reference current. So, the collector current of the transistor Q_4 is,

$$\begin{aligned} i_{C4} &= \frac{nI_{ref}}{1+\frac{1}{\beta}(n+1)} \qquad \left(\begin{matrix} n=1 \\ I_{ref}=i_{C1} \end{matrix}\right) \\ &= \frac{i_{C1}}{1+\frac{1}{\beta}(1+1)} \\ &= \frac{i_{C1}}{1+\frac{2}{\beta}} \qquad (\beta \text{ is large}) \\ &= i_{C1} \end{aligned}$$

Calculate the output current, i_o.

$$\begin{aligned} i_o &= i_{C4} \\ &= i_{C1} \\ &= \left(\frac{\beta}{1+\beta}\right)\left(\frac{v_I}{R}\right) \qquad (\beta \text{ is large} \Rightarrow \beta \approx \beta+1) \\ &= \frac{v_I}{R} \end{aligned}$$

Thus, the expression for the output current, $i_o = \frac{v_I}{R}$ is proved.

Step 2 of 3

Calculate the actual value of output current, i_o.

$$\begin{aligned} i_o &= \frac{i_{C1}}{1+\frac{2}{\beta}} \\ &= \left(\frac{1}{1+\frac{2}{\beta}}\right)\left(\frac{\beta}{1+\beta}\right)\left(\frac{v_I}{R}\right) \\ &= \left(\frac{\beta}{\beta+2}\right)\left(\frac{\beta}{1+\beta}\right)\left(\frac{v_I}{R}\right) \\ &= \frac{\beta^2}{(\beta+2)(1+\beta)}\left(\frac{v_I}{R}\right) \end{aligned}$$

Substitute 100 for β.

$$\begin{aligned} i_o &= \frac{100^2}{(100+2)(1+100)}\left(\frac{v_I}{R}\right) \\ &= 0.97\frac{v_I}{R} \end{aligned}$$

Step 3 of 3

Calculate the percentage change in output current.

$$\begin{aligned} \%\text{change}\ (i_o) &= \frac{\frac{v_I}{R} - \frac{0.97v_I}{R}}{\frac{v_I}{R}} \times 100 \\ &= (1-0.97)\times 100 \\ &= 3\% \end{aligned}$$

Thus, the percentage change in output current is $\boxed{3\%}$ that is actual output current is 3% lower than the current, i_o.

12.059P

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12.060P

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12.067P

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Step 1 of 3

(a)

The thermal resistance is,

$$\theta_{JA} = \frac{T_{J\max} - T_{A0}}{P_{D0}}$$

Substitute 100^0C for $T_{J\max}$, 25^0C for T_{A0} and 2 W for P_{D0} in equation.

$$\begin{aligned}\theta_{JA} &= \frac{100°C - 25°C}{2\text{ W}} \\ &= \frac{75°C}{2\text{ W}} \\ &= 37.5°C/\text{W}\end{aligned}$$

Therefore, the thermal resistance is $\boxed{37.5°C/\text{W}}$.

Step 2 of 3

(b)

In a steady state in which the transistor is dissipating P_D watts, the temperature rise of the junction relative to the surrounding ambience can be expressed as

$$T_J - T_A = \theta_{JA} P_D$$

Which is the inverse of the slope of the power-derating straight line. At an ambient temperature T_A, higher than T_{A0}, the maximum allowable power dissipation $P_{D\max}$ can be obtained from above equation by substituting $T_J = T_{J\max}$; thus

$$P_{D\max} = \frac{T_{J\max} - T_A}{\theta_{JA}}$$

The given ambient temperature is $T_A = 50°\text{C}$.

$$\begin{aligned}P_{D\max} &= \frac{100°\text{C} - 50°\text{C}}{37.5°\text{C/W}} \\ &= \frac{50°\text{C}}{37.5°\text{C/W}} \\ &= 1.333\text{ W}\end{aligned}$$

Therefore, the maximum power dissipation is $\boxed{1.333\text{ W}}$.

Step 3 of 3

(c)

In a steady state in which the transistor is dissipating P_D watts, the temperature rise of the junction relative to the surrounding ambience can be expressed as

$$\begin{aligned}T_J - T_A &= \theta_{JA} P_D \\ T_J &= T_A + \theta_{JA} P_D\end{aligned}$$

Finding the junction temperature of the device is operating at $T_A = 25°\text{C}$ and dissipating power is $P_D = 1\text{ W}$

Substituting the values in $T_J = T_A + \theta_{JA} P_D$,

$$\begin{aligned}T_J &= 25°\text{C} + (37.5°\text{C/W})(1\text{ W}) \\ &= 25°\text{C} + 37.5°\text{C} \\ &= 62.5°\text{ C}\end{aligned}$$

Therefore, the junction temperature of the device is $\boxed{62.5°\text{C}}$.

12.069P

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12.070P

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Step 1 of 4

13.001E

Refer to the circuit diagram of the basic two-stage CMOS op-amp configuration in FIGURE 9.1 in the textbook.

Write the formula to find the maximum input common mode voltage.

$$V_{ICM(\max)} \le V_{DD} - |V_{OV5}| - |V_{tp}| - |V_{OV1}|$$

Substitute corresponding values.

$$\begin{aligned} V_{ICM(\max)} &\le 1.65 - 0.3 - 0.5 - 0.3 \\ &\le 1.65 - 1.1 \\ &\le 0.55 \end{aligned}$$

Step 2 of 4

Write the formula to find the minimum input common mode voltage.

$$\begin{aligned} V_{ICM(\min)} &\le -V_{SS} + V_{OV3} + V_{tn} - |V_{tp}| \\ &\le -1.65 + 0.3 + 0.5 - 0.5 \\ &\le -1.35 \text{ V} \end{aligned}$$

So, the common mode input range is $\boxed{-1.35 \le V_{ICM} \le 0.55}$

Step 3 of 4

Write the formula to find the maximum output common mode voltage.

$$\begin{aligned} V_{o(\max)} &\le V_{DD} - |V_{OV7}| \\ &\le 1.65 - 0.5 \\ &\le 1.15 \text{ V} \end{aligned}$$

Step 4 of 4

Write the formula to find the minimum output common mode voltage.

$$\begin{aligned} V_{o(\min)} &\le -V_{SS} + |V_{OV6}| \\ &\le -1.65 + 0.5 \\ &\le -1.15 \text{ V} \end{aligned}$$

So, the common mode output range is $\boxed{-1.15 \le V_o \le 1.15}$.

13.001P

Refer to Figure 13.1 in the textbook for two-stage CMOS op-amp configuration.

The equation for the input common mode range is,

$$-V_{SS}+V_{OV3}+V_{tn}-\left|V_{tp}\right| \le V_{ICM} \le V_{DD}-\left|V_{tp}\right|-\left|V_{OV1}\right|-\left|V_{OV5}\right|$$

Substitute 1 V for supplies, 0.2 V for over-drive voltages, and 0.4 V for threshold voltages.

$$-1+0.2+0.4-0.4 \le V_{ICM} \le 1-0.4-0.2-0.2$$
$$-0.8 \le V_{ICM} \le 0.2$$

Therefore, the range of input common mode gain is $\boxed{-0.8 \le V_{ICM} \le 0.2}$.

The equation for the range of output voltage is,

$$-V_{SS}+V_{OV6} \le V_O \le V_{DD}-\left|V_{OV7}\right|$$

Substitute 1 V for supplies, 0.2 V for over-drive voltages, and 0.4 V for threshold voltages.

$$-1+0.2 \le V_O \le 1-0.2$$
$$-0.8 \le V_O \le 0.8$$

Therefore, the range of output voltage is $\boxed{-0.8 \le V_O \le 0.8}$.

Step 1 of 6

Determine the value of gain A_1.

$$A_1 = \frac{\left(\frac{-2}{V_{OV1}}\right)}{\left[\frac{1}{V_{A2}} + \frac{1}{V_{A4}}\right]}$$

Substitute 0.2 V for V_{OV1}, $20\,\text{V}/\mu\text{m}$ for V_{A2} and $20\,\text{V}/\mu\text{m}$ for V_{A4}.

$$A_1 = \frac{\left(\frac{-2}{0.2}\right)}{\left[\frac{1}{20\,\text{V}/\mu\text{m}} + \frac{1}{20\,\text{V}/\mu\text{m}}\right]}$$

Since the device is $1\,\mu\text{m}$ length,

$$\begin{aligned} A_1 &= \frac{\left(\frac{-2}{0.2}\right)}{\left[\frac{1}{20\,\text{V}} + \frac{1}{20\,\text{V}}\right]} \\ &= \frac{\left(\frac{-2}{0.2}\right)}{\left[\frac{1}{10\,\text{V}}\right]} \\ &= \left(\frac{-2}{0.2}\right)(10) \\ &= -100\,\text{V/V} \end{aligned}$$

Therefore, the value of gain A_1 is $\boxed{-100\,\text{V/V}}$.

Step 2 of 6

Determine the value of gain A_2.

$$A_2 = \frac{\left(\frac{-2}{V_{OV6}}\right)}{\left[\frac{1}{V_{A6}} + \frac{1}{|V_{A7}|}\right]}$$

Substitute 0.5 V for V_{OV6}, $20\,\text{V}/\mu\text{m}$ for V_{A6} and $20\,\text{V}/\mu\text{m}$ for V_{A7}.

$$A_2 = \frac{\left(\frac{-2}{0.5}\right)}{\left[\frac{1}{20\,\text{V}/\mu\text{m}} + \frac{1}{20\,\text{V}/\mu\text{m}}\right]}$$

Since the device is $1\,\mu\text{m}$ length,

$$\begin{aligned} A_2 &= \frac{\left(\frac{-2}{0.5}\right)}{\left[\frac{1}{20\,\text{V}} + \frac{1}{20\,\text{V}}\right]} \\ &= \frac{\left(\frac{-2}{0.5}\right)}{\left[\frac{1}{10\,\text{V}}\right]} \\ &= \left(\frac{-2}{0.5}\right)(10) \\ &= -40\,\text{V/V} \end{aligned}$$

Therefore, the value of gain A_2 is $\boxed{-40\,\text{V/V}}$.

Step 3 of 6

Determine the value of overall voltage gain.

$$A_v = A_1 A_2$$

Substitute $-100\,\text{V/V}$ for A_1 and $-40\,\text{V/V}$ for A_2.

$$\begin{aligned} A_v &= (-100\,\text{V/V})(-40\,\text{V/V}) \\ &= 4000\,\text{V/V} \end{aligned}$$

Therefore, the value of overall voltage gain is $\boxed{4000\,\text{V/V}}$.

Step 4 of 6

Determine the value of resistance r_{o6}.

$$r_{o6} = \frac{V_{A6}}{I_{D6}}$$

Substitute $20\,\text{V}/\mu\text{m}$ for V_{A6} and 0.5 mA for I_{D6}.

$$r_{o6} = \frac{20\,\text{V}/\mu\text{m}}{0.5\,\text{mA}}$$

Since the device is $1\,\mu\text{m}$ length,

$$\begin{aligned} r_{o6} &= \frac{20\,\text{V}}{0.5\times10^{-3}} \\ &= 40\times10^{3}\,\Omega \\ &= 40\,\text{k}\Omega \end{aligned}$$

Step 5 of 6

Determine the value of resistance r_{o7}.

$$r_{o7} = \frac{V_{A7}}{I_{D7}}$$

Substitute $20\,\text{V}/\mu\text{m}$ for V_{A7} and 0.5 mA for I_{D7}.

$$r_{o7} = \frac{20\,\text{V}/\mu\text{m}}{0.5\,\text{mA}}$$

Since the device is $1\,\mu\text{m}$ length,

$$\begin{aligned} r_{o7} &= \frac{20\,\text{V}}{0.5\times10^{-3}} \\ &= 40\times10^{3}\,\Omega \\ &= 40\,\text{k}\Omega \end{aligned}$$

Step 6 of 6

Determine the value of output resistance R_o.

$$R_o = r_{o6} \| r_{o7}$$

Substitute $40\,\text{k}\Omega$ for r_{o6} and $40\,\text{k}\Omega$ for r_{o7}.

$$\begin{aligned} R_o &= (40\,\text{k}\Omega)\|(40\,\text{k}\Omega) \\ &= \frac{(40\,\text{k}\Omega)\|(40\,\text{k}\Omega)}{(40\,\text{k}\Omega)+(40\,\text{k}\Omega)} \\ &= \frac{(40\,\text{k}\Omega)(40\,\text{k}\Omega)}{80\,\text{k}\Omega} \\ &= 20\,\text{k}\Omega \end{aligned}$$

Therefore, the value of output resistance R_o is $\boxed{20\,\text{k}\Omega}$.

Step 1 of 5

Refer to Figure 13.1 in the textbook for two-stage CMOS op-amp configuration.

The over-drive voltages are,

$$|V_{OV1}| = |V_{OV2}|$$
$$= 0.15 \text{ V}$$

$$|V_{OV6}| = 0.2 \text{ V}$$

The equation of the dc gain of first stage is,

$$A_1 = -\frac{\frac{2}{|V_{OV1}|}}{\left[\frac{1}{|V_{A2}|} + \frac{1}{V_{A4}}\right]}$$

Here,

$$|V_{A2}| = |V'_{Ap}|L$$
$$= (20)(0.3)$$
$$= 6 \text{ V}$$

$$V_{A4} = V'_{An}L$$
$$= (25)(0.3)$$
$$= 7.5 \text{ V}$$

Step 2 of 5

Substitute this values in the equation of A_1.

$$A_1 = -\frac{\frac{2}{0.15}}{\left[\frac{1}{6} + \frac{1}{7.5}\right]}$$
$$= -\frac{13.33}{0.3}$$
$$= -44.44$$

Hence, the dc gain of first stage A_1 is $\boxed{-44.44 \text{ V/V}}$.

Step 3 of 5

Similarly, find the dc gain of second stage.

$$A_2 = -\frac{\frac{2}{|V_{OV6}|}}{\left[\frac{1}{|V_{A7}|} + \frac{1}{V_{A6}}\right]}$$

Here,

$$|V_{A7}| = |V'_{Ap}|L$$
$$= (20)(0.3)$$
$$= 6 \text{ V}$$

$$V_{A6} = V'_{An}L$$
$$= (25)(0.3)$$
$$= 7.5 \text{ V}$$

Substitute this values in the equation of A_2.

$$A_2 = -\frac{\frac{2}{0.2}}{\left[\frac{1}{6} + \frac{1}{7.5}\right]}$$
$$= -\frac{100}{3}$$
$$= -33.33$$

Hence, the dc gain of second stage A_2 is $\boxed{-33.33 \text{ V/V}}$.

Step 4 of 5

Find the open loop DC gain.

$$A_v = A_1A_2$$
$$= (44.44)(33.33)$$
$$= 1481.5 \text{ V/V}$$

The equation for the op-amp output resistance is,

$$R_0 = R_2$$
$$= r_{o6} \| r_{o7}$$

Here,

$$r_{o6} = \frac{V_{A6}}{I_{D6}}$$

Therefore,

$$r_{o6} = \frac{7.5 \text{ V}}{0.3 \text{ mA}}$$
$$= 25 \text{ k}\Omega$$

$$r_{o7} = \frac{|V_{A7}|}{I_{D7}}$$
$$= \frac{6 \text{ V}}{0.3 \text{ mA}}$$
$$= 20 \text{ k}\Omega$$

Therefore, the op-amp output resistance is,

$$R_0 = \frac{r_{o6}r_{o7}}{r_{o6} + r_{o7}}$$
$$= \frac{(25)(20)}{25+20}$$
$$= 11.11 \text{ k}\Omega$$

Hence, the output resistance, R_0 is $\boxed{11.11 \text{ k}\Omega}$.

Step 5 of 5

The output resistance if op-amp is operated as unity gain buffer is,

$$A_v = 1$$
$$A_1A_2 = 1$$
$$A_2 = \frac{1}{A_1}$$

Therefore,

$$A_2 = -G_{m2}R_2$$
$$= -g_{m6}R_0$$

Here,

$$g_{m6} = \frac{2I_{D6}}{V_{ov6}}$$
$$= \frac{2(0.3)}{0.2}$$
$$= 3$$

Therefore,

$$R_0 = \frac{1}{3(44.44)}$$
$$= \frac{1}{133.33}$$
$$= 7.5 \text{ m}\Omega$$

Therefore, the output resistance R_0 is $\boxed{7.5 \text{ m}\Omega}$.

Step 1 of 2

13.003E

Refer to the circuit diagram of two stage CMOS operational-amplifier configuration in Figure 12.1 in the textbook.

The input resistance of the two stage CMOS operational-amplifier is $R_{in} = \infty$.

The two stage CMOS operational-amplifier is connected as a unity-gain buffer.

The dc gain of the first stage of the amplifier is,

$$A_1 = -g_{m1}\left(r_{o2} \parallel r_{o4}\right).$$

The dc gain of the second stage of the amplifier is,

$$\begin{aligned} A_2 &= -g_{m6}\left(r_{o6} \parallel r_{o7}\right) \\ &= -G_{m2}R_2 \end{aligned}$$

Step 2 of 2

Output resistance of the closed-loop CMOS operational amplifier is,

$$\begin{aligned} R_{out} &= \frac{R_2}{A_1 A_2} \\ &= \frac{r_{06} \parallel r_{07}}{-g_{m1}\left(r_{o2} \parallel r_{o4}\right) \times -g_{m6}\left(r_{o6} \parallel r_{o7}\right)} \\ &= \frac{r_{06} \parallel r_{07}}{g_{m1}g_{m6}\left(r_{o2} \parallel r_{o4}\right)\left(r_{o6} \parallel r_{o7}\right)} \\ &= \frac{1}{g_{m6}g_{m1}\left(r_{o2} \parallel r_{o4}\right)} \end{aligned}$$

Thus output resistance R_{out} of closed-loop CMOS operational amplifier is $\boxed{\frac{1}{g_{m6}\left[g_{m1}\left(r_{o2} \parallel r_{o4}\right)\right]}}$.

Step 1 of 3

13.003P

Refer to Figure 13.1 in the textbook for two-stage CMOS op-amp configuration.

Consider that all the over drive voltages are equal.

$$|V_A| = |V_A'|L$$
$$= (20)(0.3)$$
$$= 6\ \text{V}$$

The dc gain of first stage is,

$$A_1 = -\frac{\frac{2}{|V_{OV1}|}}{\left[\frac{1}{|V_{A2}|} + \frac{1}{V_{A4}}\right]}$$
$$= -\frac{\frac{2}{|V_{OV}|}}{\left[\frac{1}{6} + \frac{1}{6}\right]}$$
$$= -\frac{6}{V_{OV}}$$

Step 2 of 3

The dc gain of second stage is,

$$A_2 = -\frac{\frac{2}{|V_{OV6}|}}{\left[\frac{1}{|V_{A6}|} + \frac{1}{V_{A7}}\right]}$$
$$= -\frac{\frac{2}{|V_{OV}|}}{\left[\frac{1}{6} + \frac{1}{6}\right]}$$
$$= -\frac{6}{V_{OV}}$$

Step 3 of 3

Consider that the open loop DC gain, $A_V = 1600\ \text{V/V}$.

$$A_1 A_2 = 1600$$
$$\left(-\frac{6}{V_{OV}}\right)\left(-\frac{6}{V_{OV}}\right) = 1600$$
$$V_{OV}^2 = \frac{36}{1600}$$
$$|V_{OV}| = \frac{3}{200}$$
$$= 15\ \text{mV}$$

Therefore, the magnitude of over drive voltage $|V_{OV}|$ is $\boxed{15\ \text{mV}}$.

13.004E

We don't have the solution to this problem yet.

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13.004P

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Step 1 of 5

(a)

Write the expression of unity gain frequency.

$$f_t = \frac{G_{m1}}{2\pi C_C}$$

$$C_C = \frac{G_{m1}}{2\pi f_t}$$

Substitute 1 mA/V for G_{m1} and 100 MHz for f_t.

$$\begin{aligned} C_C &= \frac{1\text{ mA/V}}{2\pi(100\text{ MHz})} \\ &= \frac{10^{-3}}{2\pi\left(100\times10^{6}\right)} \\ &= 1.6\times10^{-12} \\ &= 1.6\text{ pF} \end{aligned}$$

Therefore, the value of capacitor C_C is $\boxed{1.6\text{pF}}$.

Step 2 of 5

Determine the 3 –dB frequency of the open loop gain.

$$\begin{aligned} f_{p2} &= \frac{1}{2\pi(r_{o2}\parallel r_{o4})(r_{o6}\parallel r_{o7})G_{m2}C_C} \\ &= \frac{1}{2\pi(100\text{ k}\Omega\parallel100\text{ k}\Omega)(40\text{ k}\Omega\parallel40\text{ k}\Omega)(2\text{ mA/V})\left(1.6\times10^{-12}\right)} \\ &= \frac{1}{2\pi(50\text{ k}\Omega)(20\text{ k}\Omega)(2\text{ mA/V})\left(1.6\times10^{-12}\right)} \\ &= \frac{1}{2\pi\left(50\times10^{3}\right)\left(20\times10^{3}\right)\left(2\times10^{-3}\right)\left(1.6\times10^{-12}\right)} \\ &= 50\times10^{3} \\ &= 50\text{ kHz} \end{aligned}$$

Therefore, the value of 3 –dB frequency is $\boxed{50\text{kHz}}$.

(b)

Write the expression of zero.

$$s = \frac{1}{C_C\left(\frac{1}{G_{m2}} - R\right)}$$

If the zero is located at infinity,

$$\begin{aligned} \infty &= \frac{1}{C_C\left(\frac{1}{G_{m2}} - R\right)} \\ C_C\left(\frac{1}{G_{m2}} - R\right) &= 0 \\ R &= \frac{1}{G_{m2}} \\ R &= \frac{1}{2\text{ mA/V}} \\ R &= \frac{1}{2\times10^{-3}} \\ R &= 500\,\Omega \end{aligned}$$

Therefore, the value of resistance *R* is $\boxed{500\,\Omega}$.

Step 3 of 5

(c)

Write the expression for frequency of second pole.

$$f_{p2} = \frac{G_{m2}}{2\pi C_2}$$

Substitute 2 mA/V for G_{m2} and 1 pF for C_2.

$$\begin{aligned} f_{p2} &= \frac{2\times10^{-3}}{2\pi\left(10^{-12}\right)} \\ &= 318.3\times10^{6} \\ &= 318.3\text{ MHz} \end{aligned}$$

Therefore, the frequency of second pole is $\boxed{318.3\text{ MHz}}$.

Step 4 of 5

Determine the excess phase lag at $f = f_t$.

$$\begin{aligned} \phi_{p2} &= \tan^{-1}\left(\frac{f_t}{f_{p2}}\right) \\ &= \tan^{-1}\left(\frac{100\times10^{6}}{318\times10^{6}}\right) \\ &= \tan^{-1}\left(\frac{100}{318}\right) \\ &= 17.4^{\circ} \end{aligned}$$

Therefore, the excess phase angle introduced by the second pole is $\boxed{17.4^{\circ}}$.

Step 5 of 5

Determine the value of phase margin.

$$\begin{aligned} \text{Phase margin} &= 90^{\circ} - \tan^{-1}\left(\frac{f_t}{f_{p2}}\right) - \tan^{-1}\left(\frac{f_t}{f_z}\right) \\ &= 90^{\circ} - \tan^{-1}\left(\frac{100\times10^{6}}{318\times10^{6}}\right) - \tan^{-1}\left(\frac{f_t}{\infty}\right) \\ &= 90^{\circ} - \tan^{-1}\left(\frac{100}{318}\right) - \tan^{-1}(0) \\ &= 90^{\circ} - 17.4^{\circ} - 0^{\circ} \\ &= 72.6^{\circ} \end{aligned}$$

Therefore, the value of phase margin is $\boxed{72.6^{\circ}}$.

13.005P

We don't have the solution to this problem yet.

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Step 1 of 2

13.006E

Determine the value of slew rate.

$$SR = 2\pi f_t V_{ov}$$

Substitute 100 MHz for f_t and 0.2 V for V_{ov} .

$$\begin{aligned} SR &= 2\pi\left(100\times10^{6}\right)(0.2) \\ &= 4\pi\left(10^{7}\right) \\ &= 126\left(10^{6}\right) \\ &= \frac{126}{10^{-6}} \\ &= 126\ \text{V}/\mu\text{s} \end{aligned}$$

Therefore, the value of slew rate is $\boxed{126\ \text{V}/\mu\text{s}}$.

Step 2 of 2

Determine the value of Bias current.

$$SR = \frac{I}{C_c}$$

$$I = (SR)(C_c)$$

Substitute $126\ \text{V}/\mu\text{s}$ for SR and 1.6 pF for C_c .

$$\begin{aligned} I &= (SR)(C_c) \\ &= (126\ \text{V}/\mu\text{s})(1.6\text{pF}) \\ &= \left(126\times10^{6}\right)\left(1.6\times10^{-12}\right) \\ &= 201\times10^{-6} \\ &\approx 200\,\mu\text{A} \end{aligned}$$

Therefore, the value of Bias current is $\boxed{200\,\mu\text{A}}$.

Step 1 of 3

13.006P

Determine the value of capacitance C_c using unity-gain frequency expression.

$$\omega_t = \frac{G_{m1}}{C_c}$$

$$C_c = \frac{G_{m1}}{2\pi f_t}$$

Substitute 0.8 mA/V for G_{m1} and 120 MHz for f_t.

$$C_c = \frac{0.8\times10^{-3}}{2\pi\left(120\times10^{6}\right)}$$
$$= 1.06\times10^{-12}\,\text{F}$$
$$= 1.06\,\text{pF}$$

Therefore, the value of C_c that provides unity-gain frequency of 120 MHz is $\boxed{1.06\,\text{pF}}$.

Step 2 of 3

Write the expression for frequency ω_{p2}.

$$\omega_{p2} = \frac{G_{m2}}{C_2}$$

$$2\pi f_{p2} = \frac{G_{m2}}{C_2}$$

$$f_{p2} = \frac{G_{m2}}{2\pi C_2}$$

Substitute 2.4 mA/V for G_{m2} and 1.2 pF for C_2.

$$f_{p2} = \frac{2.4\times10^{-3}}{2\pi\times1.2\times10^{-12}}$$
$$= 318.3\times10^{6}\,\text{Hz}$$
$$= 318.3\,\text{MHz}$$

Therefore, the value of f_{p2} is, $\boxed{318.3\text{MHz}}$.

Step 3 of 3

Write the expression for frequency ω_z.

$$\omega_z = \frac{G_{m2}}{C_c}$$

$$2\pi f_z = \frac{G_{m2}}{C_c}$$

$$f_z = \frac{G_{m2}}{2\pi C_c}$$

Substitute 2.4 mA/V for G_{m2} and 1.06 pF for C_c.

$$f_z = \frac{2.4\times10^{-3}}{2\pi\times1.06\times10^{-12}}$$
$$= 360.35\times10^{6}\,\text{Hz}$$
$$= 360.35\,\text{MHz}$$

Therefore, the value of f_z is $\boxed{360.35\text{ MHz}}$.

13.007E

We don't have the solution to this problem yet.

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13.007P

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13.008E

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Step 1 of 4

Determine the value of capacitance C_c using unity-gain frequency expression.

$$\omega_t = \frac{G_{m1}}{C_c}$$

$$C_c = \frac{G_{m1}}{2\pi f_t}$$

Substitute 1 mA/V for G_{m1} and 50 MHz for f_t.

$$\begin{aligned} C_c &= \frac{1\times10^{-3}}{2\pi\left(50\times10^{6}\right)} \\ &= 3.18\times10^{-12}\,\text{F} \\ &= 3.18\,\text{pF} \end{aligned}$$

Therefore, the value of capacitance C_c is $\boxed{3.18\text{ pF}}$.

Step 2 of 4

Write the expression for frequency ω_{p2}.

$$\omega_{p2} = \frac{G_{m2}}{C_2}$$

$$2\pi f_{p2} = \frac{G_{m2}}{C_2}$$

$$f_{p2} = \frac{G_{m2}}{2\pi C_2}$$

Substitute 3 mA/V for G_{m2} and 3 pF for C_2.

$$\begin{aligned} f_{p2} &= \frac{3\times10^{-3}}{2\pi\times3\times10^{-12}} \\ &= 159.15\times10^{6}\,\text{Hz} \\ &= 159.15\,\text{MHz} \end{aligned}$$

Therefore, the value of f_{p2} is, $\boxed{159.15\,\text{MHz}}$.

Step 3 of 4

Write the expression for frequency ω_z.

$$\omega_z = \frac{G_{m2}}{C_c}$$

$$2\pi f_z = \frac{G_{m2}}{C_c}$$

$$f_z = \frac{G_{m2}}{2\pi C_c}$$

Substitute 3 mA/V for G_{m2} and 3.8 pF for C_c.

$$\begin{aligned} f_z &= \frac{3\times10^{-3}}{2\pi\times3.8\times10^{-12}} \\ &= 150\times10^{6}\,\text{Hz} \\ &= 150\,\text{MHz} \end{aligned}$$

Therefore, the value of f_z is $\boxed{150\text{ MHz}}$.

Step 4 of 4

Therefore, the values of frequencies are,

$$f_t = 50\,\text{MHz}$$
$$f_{p2} = 159.15\,\text{MHz}$$
$$f_z = 150\,\text{MHz}$$

Hence, it is verified that f_t is lower than f_{p2} and f_z.

13.009E

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13.009P

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13.010E

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13.010P

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13.011E

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13.011P

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13.012E

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13.012P

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13.013P

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13.014E

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13.014P

13.015E

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13.015P

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13.016E

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13.016P

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13.017E

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13.017P

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13.018E

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13.019E

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13.019P

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13.020E

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13.020P

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13.021E

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13.021P

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13.022E

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13.022P

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13.023E

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13.023P

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13.024P

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13.025E

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13.025P

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13.026E

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Step 1 of 4

Draw the imbalance current mirror circuit diagram.

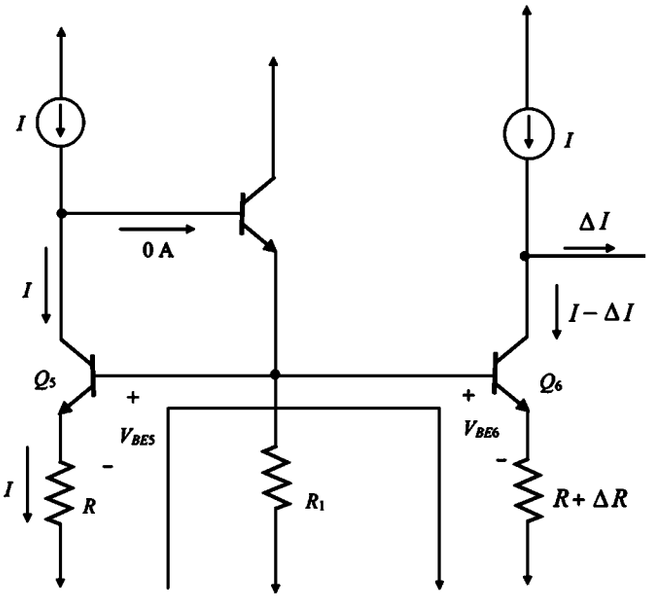


Figure 1

Apply Kirchhoff's voltage law at input loop of Figure 1.

$$-IR - V_{BE5} + V_{BE6} + (I - \Delta I)(R + \Delta R) = 0$$

$$(I - \Delta I)(R + \Delta R) = IR + (V_{BE5} - V_{BE6}) \quad \text{..... (1)}$$

Step 2 of 4

Draw the equivalent circuit diagram with emitter resistors.

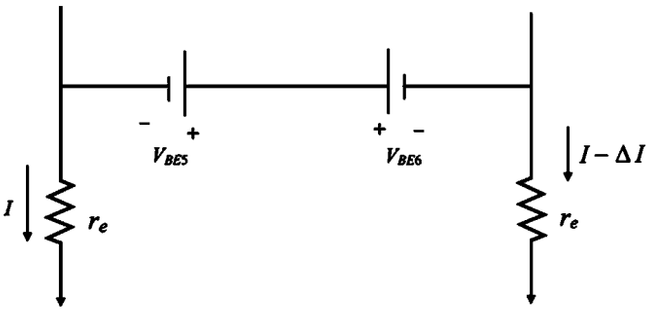


Figure 2

Step 3 of 4

Apply Kirchhoff's voltage law at input loop of Figure 2.

$$-Ir_e - V_{BE5} + V_{BE6} + (I - \Delta I)r_e = 0 \qquad (\text{Since } r_e = r_{e5} = r_{e6})$$
$$-Ir_e - V_{BE5} + V_{BE6} + Ir_e - \Delta Ir_e = 0$$
$$V_{BE6} - V_{BE5} - \Delta Ir_e = 0$$
$$V_{BE6} - V_{BE5} = \Delta Ir_e$$

The difference in emitter base voltage causes change in ΔI .

$$|V_{BE5} - V_{BE6}| = (\Delta I)r_{e5} \quad \text{..... (2)}$$

Substitute equation (2) in equation (1).

$$(I - \Delta I)(R + \Delta R) = IR + (\Delta Ir_{e5})$$
$$I(R + \Delta R) - \Delta I(R + \Delta R) = IR + (\Delta Ir_{e5})$$
$$I(R + \Delta R - R) = \Delta I(R + \Delta R + r_{e5})$$
$$I(\Delta R) = \Delta I(R + \Delta R + r_{e5})$$

$$\frac{\Delta I}{I} = \frac{\Delta R}{R + \Delta R + r_{e5}}$$

The error ε_m is, $\varepsilon_m = \frac{\Delta I}{I}$

$$\varepsilon_m = \frac{\Delta R}{R + \Delta R + r_{e5}}$$

Hence, the error $\boxed{\varepsilon_m = \frac{\Delta R}{R + \Delta R + r_{e5}}}$ is proved.

Step 4 of 4

Consider the value of resistor R is, $R = 1\,\text{k}\Omega$

The value of emitter resistor $r_{e5} = 2.63\,\text{k}\Omega$

Determine the value of ΔR .

$$\Delta R = 0.02R$$
$$= 0.02(1\,\text{k}\Omega)$$
$$= 0.02\,\text{k}\Omega$$

Determine the value of error ε_m .

$$\varepsilon_m = \frac{\Delta R}{R + r_{e5} + \Delta R}$$

Substitute $2.63\,\text{k}\Omega$ for r_{e5}, $0.02\,\text{k}\Omega$ for ΔR and $1\,\text{k}\Omega$ for R .

$$\varepsilon_m = \frac{0.02\,\text{k}\Omega}{1\,\text{k}\Omega + 2.63\,\text{k}\Omega + 0.02\,\text{k}\Omega}$$
$$= \frac{0.02\,\text{k}\Omega}{3.65\,\text{k}\Omega}$$
$$= 0.00547$$
$$= 5.5 \times 10^{-3}$$

Therefore, the value of error ε_m is $\boxed{5.5 \times 10^{-3}}$.

Step 1 of 5

13.028E

Refer to the circuit diagram of 741 operational amplifier input stage in figure 12.23 in the textbook.

Output resistance of transistor Q_9 is,

$$R_{o9} = r_{o9} = \frac{V_A}{I_9}$$

The value of early voltage V_A for *npn* transistor is, $V_A = 125\ \text{V}$

The value of early voltage V_A for *pnp* transistor is, $V_A = 50\ \text{V}$

The value of dc collector currents of the 741 circuit is $I_9 = I_{10} = 19\ \mu\text{A}$

Substitute 50 for V_A and 19×10^{-6} for I_9 in the equation.

$$R_{o9} = \frac{50}{19\times10^{-6}} = 2.63\times10^{6} = 2.63\ \text{M}\Omega$$

Thus, output resistance R_{o9} of transistor Q_9 is $\boxed{2.63\ \text{M}\Omega}$.

Step 2 of 5

Calculate the value of:

$$r_{o10} = \frac{V_A}{I_9}$$

Substitute 125 for V_A and 19×10^{-6} for I_9 in the equation.

$$r_{o10} = \frac{125}{19\times10^{-6}} = 6.57\ \text{M}\Omega$$

Step 3 of 5

Write the formula for trans-conductance.

$$g_{m10} = \frac{I_{10}}{V_T}$$

Substitute 19×10^{-6} for I_{10} and 0.025 for V_T in the equation.

$$g_{m10} = \frac{19\times10^{-6}}{0.025} = 0.76\ \text{mA/V}$$

Step 4 of 5

Calculate the value of output resistance of transistor Q_{10}.

$$R_{o10} = r_{o10}\left(1 + g_{m10}R_E\right) = \left(6.57\times10^{6}\right)\left(1 + \left(0.76\times10^{-3}\right)\left(5\times10^{3}\right)\right) = \left(6.57\times10^{6}\right)\left(1+3.8\right) = \left(6.57\times10^{6}\right)\left(4.8\right)$$

Simplify further.

$$R_{o10} = 31.53\times10^{6} = 31.53\ \text{M}\Omega$$

Thus, the value of output resistance R_{o10} of transistor Q_{10} is $\boxed{31.53\ \text{M}\Omega}$.

Step 5 of 5

Calculate the output resistance of circuit.

$$R_o = R_{o9} \parallel R_{o10} = \frac{R_{o9}\times R_{o10}}{R_{o9} + R_{o10}}$$

Substitute 2.63×10^{6} for and 31.53×10^{6} for R_{o10} in the equation.

$$R_o = \left(\frac{2.63\times31.53}{2.63+31.53}\right)\times10^{6} = 2.43\times10^{6} = 2.43\ \text{M}\Omega$$

Thus, the value of output resistance R_o is $\boxed{2.43\ \text{M}\Omega}$.

13.028P

We don't have the solution to this problem yet.

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Step 1 of 9

13.029E

Refer to the circuit diagram of the 741input stage in Figure 12.23 in the textbook.

The value of V_A for *npn* transistor is, $V_A = 125\ \text{V}$

The value offor *pnp* transistor is, $V_A = 50\ \text{V}$

The value of dc collector currents of the 741 circuit is,

$I_9 = I_{10} = 19\ \mu\text{A}$
$I_1 = 9.5\ \mu\text{A}$

The value of error is $\varepsilon_m = 5.5 \times 10^{-3}$

The value ofis, $\beta_P = 50$.

Step 2 of 9

Calculate the value of output resistance for Q_9:

Step 1 of 9

Refer to the circuit diagram of the 741input stage in Figure 12.23 in the textbook.

The value of V_A for *npn* transistor is, $V_A = 125\ \text{V}$

The value of V_A for *pnp* transistor is, $V_A = 50\ \text{V}$

The value of dc collector currents of the 741 circuit is,

$I_9 = I_{10} = 19\ \mu\text{A}$
$I_1 = 9.5\ \mu\text{A}$

The value of error is $\varepsilon_m = 5.5 \times 10^{-3}$

The value of β_P is, $\beta_P = 50$.

Step 2 of 9

Calculate the value of output resistance for Q_9:

Step 1 of 9

Refer to the circuit diagram of the 741input stage in Figure 12.23 in the textbook.

The value of V_A for *npn* transistor is, $V_A = 125\ \text{V}$

The value of V_A for *pnp* transistor is, $V_A = 50\ \text{V}$

The value of dc collector currents of the 741 circuit is,

$I_9 = I_{10} = 19\ \mu\text{A}$
$I_1 = 9.5\ \mu\text{A}$

The value of error is $\varepsilon_m = 5.5 \times 10^{-3}$

The value of β_P is, $\beta_P = 50$.

Step 2 of 9

Calculate the value of output resistance for Q_9:

Step 1 of 9

Refer to the circuit diagram of the 741input stage in Figure 12.23 in the textbook.

The value of V_A for *npn* transistor is, $V_A = 125\ \text{V}$

The value of V_A for *pnp* transistor is, $V_A = 50\ \text{V}$

The value of dc collector currents of the 741 circuit is,

$I_9 = I_{10} = 19\ \mu\text{A}$

13.030E

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13.031E

We don't have the solution to this problem yet.

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13.032E

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13.032P

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Step 1 of 13

Refer to Figure 12.13 in the textbook.

The total emitter current passing through transistor, Q_{13} is,

$$I_{EA} + I_{EB} = 0.73 \text{ mA}$$
$$I_{EA} = 0.73 \text{ mA} - I_{EB}$$

The expression for the base to emitter voltage of transistor, Q_{13A} is,

$$V_{EB13A} = V_T \ln\left(\frac{I_{C13A}}{I_{SA}}\right) \quad \text{..... (1)}$$

The expression for the base to emitter voltage of transistor, Q_{13B} is,

$$V_{EB13B} = V_T \ln\left(\frac{I_{C13B}}{I_{SB}}\right) \quad \text{..... (2)}$$

Step 2 of 13

Divide equation (1) with equation (2).

$$\frac{V_{EB13A}}{V_{EB13B}} = \frac{V_T \ln\left(\frac{I_{C13A}}{I_{SA}}\right)}{V_T \ln\left(\frac{I_{C13B}}{I_{SB}}\right)}$$

The base emitter voltage of transistor, Q_{13A} is equal to transistor, Q_{13B} since the base-emitter junctions of the transistors are parallel.

$$1 = \frac{V_T \ln\left(\frac{I_{C13A}}{I_{SA}}\right)}{V_T \ln\left(\frac{I_{C13B}}{I_{SB}}\right)}$$
$$\ln\left(\frac{I_{C13A}}{I_{SA}}\right) = \ln\left(\frac{I_{C13B}}{I_{SB}}\right)$$
$$\frac{I_{C13A}}{I_{SA}} = \frac{I_{C13B}}{I_{SB}}$$
$$\frac{I_{EA}}{I_{SA}} = \frac{I_{EB}}{I_{SB}}$$

Step 3 of 13

Substitute $0.73 \text{ mA} - I_{EB}$ for I_{EA}.

$$\frac{0.73 \text{ mA} - I_{EB}}{I_{SA}} = \frac{I_{EB}}{I_{SB}}$$

Substitute 0.25×10^{-14} A for I_{SA} and 0.75×10^{-14} A for I_{SB}.

$$\frac{0.73 \text{ mA} - I_{EB}}{0.25\times10^{-14}} = \frac{I_{EB}}{0.75\times10^{-14}}$$
$$0.73 \text{ mA} - I_{EB} = \frac{I_{EB}}{3}$$
$$2.19\times10^{-3} = 4I_{EB}$$
$$I_{EB} = 0.5475\times10^{-3} \text{ A}$$

Determine the current I_{EA}.

$$I_{EA} = 0.73 \text{ mA} - 0.5475 \text{ mA}$$
$$= 0.1825 \text{ mA}$$

Step 4 of 13

Write the expression for the voltage V_{EB}.

$$V_{EB} = V_T \ln\left(\frac{I_{C13B}}{I_{SB}}\right)$$
$$= V_T \ln\left(\frac{I_{EB}\left(\frac{\beta}{\beta+1}\right)}{I_{SB}}\right)$$

Substitute 25 mV for V_T, 0.5475×10^{-3} A for I_{EB}, 0.75×10^{-14} A for I_{SB} and 50 for β.

$$V_{EB} = \left(25\times10^{-3}\right)\ln\left(\frac{\left(0.5475\times10^{-3}\right)\left(\frac{50}{50+1}\right)}{0.75\times10^{-14}}\right)$$
$$= \left(25\times10^{-3}\right)\ln\left(\frac{(0.5475)(0.98)\left(10^{11}\right)}{(0.75)}\right)$$
$$= \left(25\times10^{-3}\right)(-24.99)$$
$$= -0.625 \text{ V}$$

Thus, the value V_{BE} for transistors, Q_{13A} and Q_{13B} is, $\boxed{0.625 \text{ V}}$.

Step 5 of 13

Determine the trans-conductance of transistor A.

$$g_{mA} = \frac{I_C}{V_T}$$
$$= \frac{I_{EA}\left(\frac{\beta}{\beta+1}\right)}{V_T}$$
$$= \frac{\left(0.1825\times10^{-3}\right)\left(\frac{50}{50+1}\right)}{25\times10^{-3}}$$
$$= 7.157 \text{ mA/V}$$

Thus, the value of g_m for transistor A is $\boxed{7.157 \text{ mA/V}}$.

Step 6 of 13

Determine the trans-conductance of transistor Q_{13B}.

$$g_{mB} = \frac{I_{EB}\left(\frac{\beta}{\beta+1}\right)}{V_T}$$
$$= \frac{\left(0.5475\times10^{-3}\right)\left(\frac{50}{50+1}\right)}{25\times10^{-3}}$$
$$= 21.47 \text{ mA/V}$$

Thus, the value of g_m for transistor B is $\boxed{21.47 \text{ mA/V}}$.

Step 7 of 13

Determine the value of α.

$$\alpha = \frac{\beta}{1+\beta}$$
$$= \frac{50}{50+1}$$
$$= \frac{50}{51}$$
$$= 0.98$$

The small signal resistance between base and emitter is,

$$r_e = \frac{V_T}{I_E}$$
$$= \frac{V_T}{(I_C/\alpha)}$$
$$= \frac{\alpha}{(I_C/V_T)}$$
$$= \frac{\alpha}{g_m}$$

Step 8 of 13

Calculate the value of r_e for transistor A.

$$r_e = \frac{\alpha}{g_{mA}}$$
$$= \frac{0.98}{7.157\times10^{-3}}$$
$$= 136.9 \ \Omega$$

Thus, the value of r_e for transistor A is $\boxed{136.9 \ \Omega}$.

Step 9 of 13

Calculate the value of r_e for transistor B.

$$r_e = \frac{\alpha}{g_{mA}}$$
$$= \frac{0.98}{21.47\times10^{-3}}$$
$$= 45.645 \ \Omega$$

Thus, the value of r_e for transistor B is $\boxed{45.645 \ \Omega}$.

Step 10 of 13

The small-signal input resistance between base and emitter of standard *npn* transistor is, $r_\pi = \frac{\beta}{g_m}$

Determine r_π for transistor A.

$$r_\pi = \frac{50}{7.157\times10^{-3}}$$
$$= 6.986 \text{ k}\Omega$$

Thus, the value of r_π for transistor A is $\boxed{6.986 \text{ k}\Omega}$.

Step 11 of 13

Determine r_π for transistor B.

$$r_\pi = \frac{50}{21.47\times10^{-3}}$$
$$= 2.329 \text{ k}\Omega$$

Thus, the value of r_π for transistor B is $\boxed{2.329 \text{ k}\Omega}$.

Step 12 of 13

The expression for the output resistance of transistor is,

$$r_0 = \frac{V_A}{I_C}$$

Determine r_o for transistor A.

$$r_0 = \frac{V_A}{I_{CA}}$$
$$= \frac{V_A}{I_{EA}\left(\frac{\beta}{\beta+1}\right)}$$
$$= \frac{50}{0.1825\times10^{-3}\left(\frac{50}{50+1}\right)}$$
$$= 279.45 \text{ k}\Omega$$

Thus, the output resistance of transistor A is $\boxed{279.45 \text{ k}\Omega}$.

Step 13 of 13

Determine r_o for transistor B.

$$r_0 = \frac{V_A}{I_{CB}}$$
$$= \frac{V_A}{I_{EB}\left(\frac{\beta}{\beta+1}\right)}$$
$$= \frac{50}{\left(0.5475\times10^{-3}\right)\left(\frac{50}{50+1}\right)}$$
$$= 93.15 \text{ k}\Omega$$

Thus, the output resistance of transistor B is $\boxed{93.15 \text{ k}\Omega}$.

13.034E

We don't have the solution to this problem yet.

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13.034P

Refer to Figure 12.13 in the text book.

The emitter-base breakdown voltage of transistors, Q_1 and Q_2 is,

$$BV_{EBO(1,2)} = 7 \text{ V}$$

The break down voltage of transistors, Q_3 and Q_4 is,

$$BV_{EBO(3,4)} = 50 \text{ V}$$

Apply Kirchhoff's voltage law to the loop from the base of transistor, Q_3 to the base of transistor, Q_2 .

The differential input voltage that would result in the breakdown of the input-stage transistors is given by

$$\begin{aligned} V_{ID} &= V_{BE1} + V_{BE2} + BV_{EBO(1,2)} + BV_{EBO(3,4)} \\ &= 0.7 + 0.7 + 7 + 50 \\ &= 58.4 \text{ V} \end{aligned}$$

Thus the differential input voltage that would result in the breakdown of the input-stage transistors is,

$\boxed{V_{ID} \geq 58.4 \text{ V}}$.

13.035E

We don't have the solution to this problem yet.

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Step 1 of 3

13.037E

Refer to the Figure 12.38 in the text book.

The area of transistors is directly proportional to the reverse saturation current.

Refer to the equation 12.129 in the text book.

Write the expression for current.

$$I = \frac{V_T}{R_2}\ln\left(\frac{I_{S2}}{I_{S1}}\right)$$

Rearrange the equation to determine the resistance R_2.

$$R_2 = \frac{V_T}{I}\ln\left(\frac{I_{S2}}{I_{S1}}\right)$$

Substitute 25 mV for V_T, $10\,\mu\text{A}$ for I and 2 for $\frac{I_{S2}}{I_{S1}}$.

$$\begin{aligned} R_2 &= \left(\frac{25\times10^{-3}}{10\times10^{-6}}\right)\ln(2) \\ &= (2500)\ln(2) \\ &= 1732.87\ \Omega \\ &= 1.732\ \text{k}\Omega \end{aligned}$$

Therefore, the value of resistance R_2 is $\boxed{1.732\ \text{k}\Omega}$.

Step 2 of 3

Write the expression of voltage drop across the resistance R_3.

$$V = IR_3$$

$$R_3 = \frac{V}{I}$$

Substitute 0.2 V for V and $10\,\mu\text{A}$ for I.

$$\begin{aligned} R_3 &= \frac{0.2}{10\times10^{-6}} \\ &= 20\ \text{k}\Omega \end{aligned}$$

Therefore, the value of resistance R_3 is $\boxed{20\ \text{k}\Omega}$.

Step 3 of 3

Write the expression of voltage drop across the resistance R_4.

$$V = IR_4$$

$$R_4 = \frac{V}{I}$$

Substitute 0.2 V for V and $10\,\mu\text{A}$ for I.

$$\begin{aligned} R_4 &= \frac{0.2}{10\times10^{-6}} \\ &= 20\ \text{k}\Omega \end{aligned}$$

Therefore, the value of resistance R_4 is $\boxed{20\ \text{k}\Omega}$.

13.037P

13.038P

We don't have the solution to this problem yet.

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Step 1 of 4

13.039E

(a)

Refer to the Figure 12.47 in the text book.

The current gain due to transistor is β_N, transistor Q_2 is β_P and transistor Q_P is β_P.

Determine the current gain from v_{IP} terminal to the output terminal.

The overall current gain is combination of three transistors Q_1, Q_2 and current gains.

$$\frac{i_P}{i_{B1}} = \beta_N \beta_P \beta_P$$
$$= \beta_N \beta_P^2$$

Therefore, the overall current gain from v_{IP} is $\boxed{\beta_N \beta_P^2}$.

Step 2 of 4

Step 1 of 4

(a)

Refer to the Figure 12.47 in the text book.

The current gain due to transistor is β_N, transistor Q_2 is β_P and transistor Q_P is β_P.

Determine the current gain from v_{IP} terminal to the output terminal.

The overall current gain is combination of three transistors Q_1, Q_2 and current gains.

$$\frac{i_P}{i_{B1}} = \beta_N \beta_P \beta_P$$
$$= \beta_N \beta_P^2$$

Therefore, the overall current gain from v_{IP} is $\boxed{\beta_N \beta_P^2}$.

Step 2 of 4

Step 1 of 4

Step 1 of 4

(a)

Step 1 of 4

Refer to Figure 12.15 in the text book.

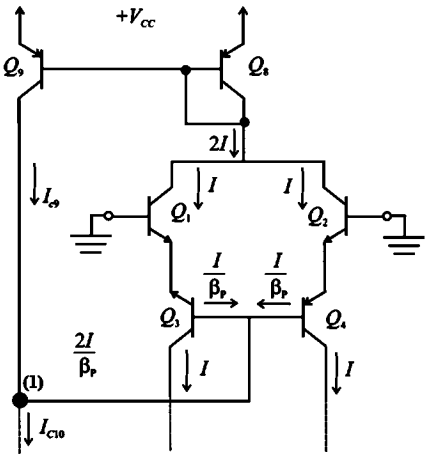


Figure 1

Step 2 of 4

Refer to equation 7.68 in the text book.

Write the relationship between I_{C9} and.

$$\frac{2I}{I_{C9}} = \frac{I_{s8}}{I_{s9}}$$

Substitute $I_{S9} = 2I_{S8}$.

$$\frac{2I}{I_{C9}} = \frac{I_{s8}}{2I_{s8}}$$
$$I_{C9} = 4I$$

Apply Kirchhoff's current law at node (1) as shown in Figure 1.

$$I_{C10} = I_{C9} + \frac{2I}{\beta_p}$$
$$= 4I + \frac{2I}{\beta_p}$$

Since, $\beta_p >> 1$,

$$I_{C10} = 4I$$

Determine the current I.

$$I = \frac{I_{C10}}{4}$$
$$= \frac{19\times10^{-6}}{4}$$
$$= 4.75\ \mu A$$

Thus, the value of I is $\boxed{4.75\ \mu A}$.

Step 3 of 4

The value of I is,

$$I_{C1} = I_{C2}$$
$$= I$$
$$= 9.5\ \mu A$$

Determine the current I_{C10}.

$$I_{C10} = 4I$$
$$= 4\times9.5$$
$$= 38\ \mu A$$

Determine the current I_{REF}.

$$I_{REF} = \frac{V_{CC} - V_{EB12} - V_{BE11} - (-V_{EE})}{R_5}$$
$$= \frac{15 - 0.7 - 0.7 - (-15)}{(39\times10^{3})}$$
$$= 0.73\ mA$$

Thus, the reference current I_{REF} of the Wildar current source is $\boxed{0.73\ mA}$.

Step 4 of 4

Refer to equation 12.75 in the text book.

Determine the series emitter resistance of the Wildar source.

$$R_4 = \frac{V_T}{I_{C10}} \ln\left(\frac{I_{REF}}{I_{C10}}\right)$$
$$= \left(\frac{25\times10^{-3}}{38\times10^{-6}}\right) \ln\left(\frac{0.73\times10^{-3}}{38\times10^{-6}}\right)$$
$$= 1.944\ k\Omega$$

Thus, the series emitter resistance R_4 of the Wildar source is $\boxed{1.944\ k\Omega}$.

13.040P

We don't have the solution to this problem yet.

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13.041P

Refer to Figure 12.16 in the text book.

Refer to equation 12.79 in the text book.

Write the expression for the collector current in the transistor Q_7 .

$$I_{C7} = \frac{2I}{\beta_N} + \frac{V_{BE6} + IR_2}{R_3}$$

Substitute 0 for R_2 .

$$I_{C7} = \frac{2I}{\beta_N} + \frac{V_{BE6}}{R_3}$$

Substitute $10.5\ \mu\text{A}$ for I_{C7} , 200 for β_N , $50\ \text{k}\Omega$ for R_3 and 517 mV for V_{BE6} .

$$10.5\times10^{-6} = \frac{2I}{200} + \frac{517\times10^{-3}}{50\times10^{3}}$$

$$10.5\times10^{-6} = \frac{2I}{200} + 10.53\times10^{-6}$$

$$\frac{2I}{200} = 0.03\times10^{-6}$$

$$I = 16\ \mu\text{A}$$

Thus, the current in transistor, Q_6 if R_2 is shorted is, $\boxed{16\ \mu\text{A}}$.

13.042P

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13.043P

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13.045P

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Step 1 of 2

Determine the current I_{B17}.

$$I_{B17} = \frac{I_{C17}}{\beta}$$
$$= \frac{550\ \mu A}{200}$$
$$= 2.75\ \mu A$$

Determine the current I_{E17}.

$$I_{E17} = \left(\frac{\beta}{\beta+1}\right) I_{C17}$$
$$= \left(\frac{200}{200+1}\right)(550\ \mu A)$$
$$= \left(\frac{200}{201}\right)(550\ \mu A)$$
$$= 547.26\ \mu A$$

Step 2 of 2

Write the expression for the current I_{C16}.

$$I_{C16} = I_{B17} + \frac{I_{E17} R_8 + V_{BE17}}{R_9}$$

Substitute $9.5\ \mu A$ for I_{C16}, $547.26\ \mu A$ for I_{E17}, $2.75\ \mu A$ for I_{B17}, $618\ mV$ for V_{BE17} and $100\ \Omega$ for R_8.

$$9.5\times10^{-6} = \left(2.75\times10^{-6}\right) + \frac{\left(547.26\times10^{-6}\right)(100) + \left(618\times10^{-3}\right)}{R_9}$$
$$\frac{\left(547.26\times10^{-6}\right)(100) + \left(618\times10^{-3}\right)}{R_9} = 6.75\times10^{-6}$$
$$\frac{\left(547.26\times10^{-6}\right)(100) + \left(618\times10^{-3}\right)}{6.75\times10^{-6}} = R_9$$
$$R_9 = 99.663\ k\Omega$$

Thus, the value of the resistance R_9 is $\boxed{99.663\ k\Omega}$.

Step 1 of 3

Refer to Figure 12.17 in the text book.

The current I_{C23} is,

$$\begin{aligned} I_{C23} &= I_{E23} \\ &= 0.25 I_{REF} \\ &= 180\ \mu A \end{aligned}$$

Neglect the base currents of transistors Q_{18} and Q_{19}.

The currents I_{C19} and I_{C18} are;

$$\begin{aligned} I_{C19} &= I_{C18} \\ &= \frac{180\times10^{-6}}{2} \\ &= 90\ \mu A \end{aligned}$$

Step 2 of 3

Determine the base-emitter voltage of transistor Q_{18}.

$$\begin{aligned} V_{BE18} &= V_T \ln\left(\frac{I_{C18}}{I_S}\right) \\ &= 25\times10^{-3} \ln\left(\frac{90\times10^{-6}}{10^{-14}}\right) \\ &= 0.573\ V \end{aligned}$$

Determine the value of the resistor R_{18}.

$$\begin{aligned} R_{18} &= \frac{V_{BE18}}{I_{C18}} \\ &= \frac{0.573}{90\times10^{-6}} \\ &= 6.37\ k\Omega \end{aligned}$$

Thus, the value of resistor to make is .

Step 3 of 3

The collector currents of transistors and are equal.

Determine the values for I_{C14} and I_{C20}.

$$\begin{aligned} I_{C14} &= I_{S14} e^{\frac{V_{BE18}}{V_T}} \\ &= 3\times10^{-14} \times e^{\frac{0.573}{25\times10^{-3}}} \\ &= 270\ \mu A \end{aligned}$$

$I_{C20} = 270\ \mu A$

Thus, the value of I_{C14} and I_{C20} is $\boxed{270\ \mu A}$.

Step 1 of 6

Refer to Figure P12.39 in the text book.

Consider the terminal voltage V.

Draw the modified circuit.

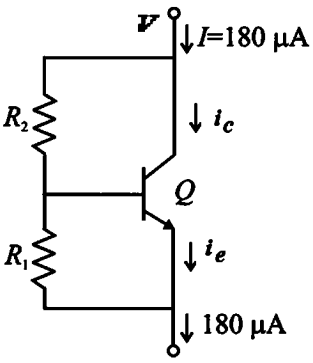


Figure 1

Step 2 of 6

Apply voltage division rule to write the expression for the voltage across the resistor R_1.

$$V_{R1}=\left(\frac{R_1}{R_1+R_2}\right)V \quad \text{..... (1)}$$

The current passing through the resistor is one half of $180\ \mu A$.

The voltage across R_1 is,

$$V_{R1}=\left(\frac{180\ \mu A}{2}\right)R_1 \quad \text{..... (2)}$$

Equate the equations (1) and (2).

$$\left(\frac{180\times10^{-6}}{2}\right)R_1=\left(\frac{R_1}{R_1+R_2}\right)V$$

$$R_1+R_2=\frac{V}{90\times10^{-6}}$$

$$=\frac{1.118}{90\times10^{-6}}$$

$$=12.42\ k\Omega$$

Step 3 of 6

Apply Kirchhoff's current law at the emitter node of the transistor Q as shown in Figure 1.

$$\frac{180}{2}\ \mu A+i_e=180\ \mu A$$

$$i_e=90\ \mu A$$

Determine the collector current i_c.

$$i_c=\left(\frac{\beta}{1+\beta}\right)i_e$$

$$=\left(\frac{200}{1+200}\right)\left(90\times10^{-6}\right)$$

$$=(0.995)\left(90\times10^{-6}\right)$$

$$=89.55\ \mu A$$

Step 4 of 6

Write the expression for V_{BE} of the transistor Q.

$$V_{BE}=V_T\ln\frac{i_C}{I_S}.$$

Substitute $V_T=25\ mV$, $I_S=10^{-14}\ A$ and $i_c=89.55\ \mu A$.

$$V_{BE}=\left(25\times10^{-3}\right)\ln\left(\frac{89.55\times10^{-6}}{10^{-14}}\right)$$

$$=\left(25\times10^{-3}\right)\ln\left(89.55\times10^{8}\right)$$

$$=\left(25\times10^{-3}\right)(22.9)$$

$$=0.573\ V$$

Thus, the value of V_{BE} is $\boxed{0.573\ V}$.

Step 5 of 6

The voltage V_{BE} is,

$$V_{BE}=V_{R1}$$

Substitute $0.573\ V$ for V_{BE} and $(90\ \mu A)R_1$ for V_{R1}.

$$90\times10^{-6}R_1=0.573$$

$$R_1=\frac{0.573}{90\times10^{-6}}$$

$$R_1=6.37\ k\Omega$$

Thus, the value of R_1 is $\boxed{6.37\ k\Omega}$.

Step 6 of 6

Substitute $6.37\ k\Omega$ for R_1 in $R_1+R_2=12.42\ k\Omega$.

$$R_2=12.42\ k\Omega-R_1$$

$$=12.42-6.37$$

$$=6.05\ k\Omega$$

Thus the value of R_2 is $\boxed{6.05\ k\Omega}$.

13.050P

We don't have the solution to this problem yet.

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13.051P

We don't have the solution to this problem yet.

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Step 1 of 5

(a)

Refer to Figure 12.19 in the textbook.

Consider that the resistances are,

$$\begin{aligned} R_1 &= R_2 \\ &= 2\ \text{k}\Omega \end{aligned}$$

Refer to Table 12.1 in the textbook for the values of DC collector currents of 741 circuit.

Note that the collector current for the transistors, Q_5 and Q_6 is $9.5\ \mu\text{A}$.

Consider that the collector current of transistors, Q_5 and Q_6 , and the current through the transistor, Q_7 is equal.

$$\begin{aligned} I_{C5} &= I_{C6} \\ &\simeq I_{C7} \end{aligned}$$

Calculate the small-signal emitter resistance for the transistors,

$$\begin{aligned} r_{e5} &= r_{e6} \\ &= r_{e7} \\ &= \frac{V_T}{I_C} \end{aligned}$$

Substitute $9.5\ \mu\text{A}$ for I_C , and 25 mV for thermal equivalent voltage, V_T .

$$\begin{aligned} r_e &= \frac{25\ \text{mV}}{9.5\ \mu\text{A}} \\ &= 2.63\ \text{k}\Omega \end{aligned}$$

The signal voltage at the base of transistor, Q_6 is,

$$\begin{aligned} V_{B6} &= (r_{e6} + R_2) i_e \\ &= (2.63\ \text{k}\Omega + 2\ \text{k}\Omega) i_e \\ &= (4.63\ \text{k}\Omega) i_e \end{aligned}$$

Thus, the signal voltage at the base of transistor, Q_6 is, $\boxed{(4.63\ \text{k}\Omega) i_e}$.

Step 2 of 5

(b)

The signal current in the emitter of transistor, Q_7 is,

$$I_{E7} = \frac{V_{B6}}{R_B}$$

Find the value of base resistance of transistor, Q_6 .

$$R_B = R_3 \| (r_{e5} \| r_{\pi 6})$$

Find the small-signal resistance, r_π .

$$\begin{aligned} r_\pi &= r_{\pi 5} \\ &= r_{\pi 6} \\ &= \frac{V_T}{I_B} \\ &= \frac{\beta V_T}{I_C} \end{aligned}$$

Substitute 200 for current gain, β , $9.5\ \mu\text{A}$ for I_C , and 25 mV for V_T .

$$\begin{aligned} r_\pi &= \frac{(200)(25\ \text{mV})}{9.5\ \mu\text{A}} \\ &= 526.3\ \text{k}\Omega \end{aligned}$$

Therefore,

$$\begin{aligned} R_B &= 50\ \text{k}\Omega \| (526.3\ \text{k}\Omega \| 526.3\ \text{k}\Omega) \\ &= 50\ \text{k}\Omega \| 263.15\ \text{k}\Omega \\ &= \frac{(50\ \text{k}\Omega)(263.15\ \text{k}\Omega)}{50\ \text{k}\Omega + 263.15\ \text{k}\Omega} \\ &= 42\ \text{k}\Omega \end{aligned}$$

Therefore,

$$\begin{aligned} I_{E7} &= \frac{(4.63\ \text{k}\Omega) i_e}{42\ \text{k}\Omega} \\ &= 0.11 i_e \end{aligned}$$

Thus, the signal current in the emitter of transistor, Q_7 is, $\boxed{0.11 i_e}$.

Step 3 of 5

(c)

The signal current in the base of transistor, Q_7 is,

$$I_{B7} = \frac{I_{E7}}{\beta + 1}$$

Substitute 200 for β .

$$\begin{aligned} I_{B7} &= \frac{0.11 i_e}{200 + 1} \\ &= 0.00055 i_e \end{aligned}$$

Thus, the signal current in the base of transistor, Q_7 is, $\boxed{0.00055 i_e}$.

Step 4 of 5

(d)

The signal voltage at the base of transistor, Q_7 is,

$$\begin{aligned} V_{B7} &= r_{e7} I_{C7} + V_{B6} \\ &= r_{e7} I_{E7} + V_{B6} \quad (\text{since}, I_{C7} \simeq I_{E7}) \end{aligned}$$

Substitute $2.63\ \text{k}\Omega$ for r_{e7} , $0.11 i_e$ for I_{E7} , and $(4.63\ \text{k}\Omega) i_e$ for V_{B6} .

$$\begin{aligned} V_{B7} &= (2.63\ \text{k}\Omega)(0.11 i_e) + (4.63\ \text{k}\Omega) i_e \\ &= 289.3 i_e + 4630 i_e \\ &= (4.91\ \text{k}\Omega) i_e \end{aligned}$$

Thus, the signal voltage at the base of transistor, Q_7 is, $\boxed{(4.91\ \text{k}\Omega) i_e}$.

Step 5 of 5

(e)

The input resistance seen by the left-hand-side signal current source, αi_e is,

$$R_{in} = \frac{V_{B7}}{i_e}$$

Substitute $(4.91\ \text{k}\Omega) i_e$ for V_{B7} .

$$\begin{aligned} R_{in} &= \frac{(4.91\ \text{k}\Omega) i_e}{i_e} \\ &= 4.91\ \text{k}\Omega \end{aligned}$$

The input resistance, R_{in} seen by the left-hand-side signal current source, αi_e is, $\boxed{4.91\ \text{k}\Omega}$.

13.053P

We don't have the solution to this problem yet.

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13.055P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

13.056P

We don't have the solution to this problem yet.

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13.057P

We don't have the solution to this problem yet.

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Step 1 of 3

13.058P

Refer to Figure 12.19 in the text book for the load circuit of the input stage of the 741 operational amplifier fed by the two complementary current signals.

The current gain for a resistor degenerated mirror is,

$$\frac{i_o}{i_e} = 2\alpha$$

Substitute 1 for α .

Thus the current gain of the 741 operational amplifier is,

$$\frac{i_o}{i_e} = 2$$

When the resistor R_1 is short-circuited, the current gain is,

$$\frac{i_o}{i_i} = \frac{r_{e1}}{r_{e2} + R_2}$$

Substitute $2.63\ \text{k}\Omega$ for r_e and $1\ \text{k}\Omega$ for R_2 .

$$\frac{i_o}{i_i} = \frac{2.63}{2.63+1}$$

$$i_o = 0.725 i_i$$

$$\frac{i_o}{i_e} = 1 + 0.725$$

$$= 1.725$$

Determine the reduction in the differential gain.

$$\left(\frac{2 - 1.725}{2}\right) \times 100 = 13.75\%$$

Thus, the differential gain is reduced when the resistor R_1 is short-circuited and the reduction percentage is $\boxed{13.75\%}$.

Step 2 of 3

When the resistor R_2 is short-circuited, the current gain is,

$$\frac{i_o}{i_i} = \frac{r_{e1} + R_1}{r_{e2}}$$

Substitute $2.63\ \text{k}\Omega$ for r_e and $1\ \text{k}\Omega$ for R_1 .

$$\frac{i_o}{i_i} = \frac{2.63+1}{2.63}$$

$$= 1.38$$

Thus,

$$\frac{i_o}{i_e} = 1 + 1.38$$

$$= 2.38$$

Determine the increase in the differential gain.

$$\left(\frac{2.38 - 2}{2}\right) \times 100 = 19\%$$

Thus, the differential gain is increased when the resistor R_2 is short-circuited and the increment percentage is $\boxed{19\%}$.

Step 3 of 3

When both R_1 and R_2 are short-circuited, the current gain is,

$$\frac{i_o}{i_i} = \frac{r_{e1}}{r_{e2}}$$

$$= 1$$

Thus,

$$\frac{i_o}{i_e} = 2$$

Thus, the differential gain is unaffected when both R_1 and R_2 are short-circuited.

Step 1 of 3

13.059P

Refer to Figure 12.23 in the text book for the 741 input stage.

Since the negative feedback increases the resistance at node Y by an amount of negative feedback, break the loop at Y and connect a resistance, $R_f = R_o(1 + A\beta)$ between the common base connection of $Q_3 - Q_4$ and ground.

Write the node equation at Y.

$$\frac{2i}{\beta_p} = \frac{v_{icm}}{R_o(1 + A\beta)}$$

$$i = \frac{v_{icm}\beta_p}{2R_o(1 + A\beta)}$$

Since the loop gain is approximately equal to β_p, substitute β_p for $(1 + A\beta)$.

$$i = \frac{v_{icm}\beta_p}{2R_o\beta_p}$$

$$= \frac{v_{icm}}{2R_o}$$

Thus, the current i is,

$$\boxed{i = \frac{v_{icm}}{2R_o}}.$$

Step 2 of 3

Determine the common-mode transconductance at the input stage, G_{mcm}.

$$G_{mcm} = \frac{i_o}{V_{icm}}$$

From the circuit, the current i_o is,

$$i_o = \varepsilon_m i$$

Here,

ε_m is the error due to mismatches in the current-mirror load

Therefore,

$$G_{mcm} = \frac{\varepsilon_m i}{V_{icm}}$$

Substituting $\frac{v_{icm}}{2R_o}$ for i.

$$G_{mcm} = \left(\frac{\varepsilon_m}{v_{icm}}\right)\left(\frac{v_{icm}}{2R_o}\right)$$

$$= \frac{\varepsilon_m}{2R_o}$$

Thus, the common-mode transconductance G_{mcm} of the input stage is $\boxed{\frac{\varepsilon_m}{2R_o}}$.

Step 3 of 3

Determine the $\mathbf{CMRR}$ as the ratio of the differential transconductance G_{m1} and the common-mode transconductance G_{mcm}.

$$\mathbf{CMRR} = \frac{G_{m1}}{G_{mcm}}$$

$$= \frac{2g_{m1}R_o}{\varepsilon_m}$$

$$= \frac{2g_{m1}R_o}{\varepsilon_m}$$

Here, g_{m1} is the transconductance of Q_1.

Substitute $R_{o9} \parallel R_{o10}$ for R_o.

$$\mathbf{CMRR} = \frac{2g_{m1}(R_{o9} \parallel R_{o10})}{\varepsilon_m}$$

Thus, the CMRR is $\boxed{\frac{2g_{m1}(R_{o9} \parallel R_{o10})}{\varepsilon_m}}$.

The result agrees with the equation 12.95 in the text book.

Step 1 of 6

Refer to Figure 12.13 in the text book.

The second or intermediate stage of the 741 operational amplifier is composed of Q_{16}, Q_{17} and Q_{13B}, and the two resistors R_8 and R_9.

Transistor Q_{16} acts as an emitter follower, thus giving the second stage a high input resistance. This minimizes the loading on the input stage and avoids loss of gain.

Transistor Q_{17} acts as a common-emitter amplifier with a $100\ \Omega$ resistor in the emitter. Its load is composed of high output resistance of the pnp current source Q_{13B} in parallel with the input resistance of the output stage.

The variation on the design of the 741 second stage is, $R_8 = 50\ \Omega$.

Find the ac emitter resistance r_{e16} of transistor Q_{16}.

$$r_{e16} = \frac{V_T}{I_{C16}}$$

Substitute $16.2\ \mu A$ for I_{C16} and 25 mV for V_T.

$$\begin{aligned} r_{e16} &= \frac{25\ \text{mV}}{16.2\ \mu\text{A}} \\ &= 1.54\ \text{k}\Omega \end{aligned}$$

Step 2 of 6

Find the ac emitter resistance, r_{e17}.

$$r_{e17} = \frac{V_T}{I_{C17}}$$

The collector current flowing through the transistor Q_{17} is,

$$I_{C17} = 550\ \mu A$$

Substitute $550\ \mu A$ for I_{C17} and 25 mV for V_T.

$$\begin{aligned} r_{e17} &= \frac{25\ \text{mV}}{550\ \mu\text{A}} \\ &= 45.45\ \Omega \end{aligned}$$

Determine the resistance R_{i17}.

$$R_{i17} = (\beta + 1)(r_{e17} + R_8)$$

Substitute $45.45\ \Omega$ for r_{e17}, 200 for β and $50\ \Omega$ for R_8.

$$\begin{aligned} R_{i17} &= (200+1)(45.45+50) \\ &= 19.2\ \text{k}\Omega \end{aligned}$$

Step 3 of 6

The input resistance of the second stage of the 741 operational amplifier is,

$$R_{i2} = (\beta + 1)\left[r_{e16} + R_9 \parallel R_{i17}\right]$$

Substitute $1.54\ \text{k}\Omega$ for r_{e16}, $50\times10^3\ \Omega$ for R_9, $19.2\times10^3\ \Omega$ for R_{i17} and 200 for β.

$$\begin{aligned} R_{i2} &= (200+1)\left[1.54\times10^3 + 50\times10^3 \parallel 19.2\times10^3\right] \\ &= (201)\left[1.54\times10^3 + \frac{(50\times10^3)(19.2\times10^3)}{50\times10^3 + 19.2\times10^3}\right] \\ &= (201)\left[1.54\times10^3 + 13.873\times10^3\right] \\ &= 3.1\ \text{M}\Omega \end{aligned}$$

Step 4 of 6

The resistance R_9 is $50\ \text{k}\Omega$.

Determine the resistance $R_{i17} \parallel R_9$.

$$\begin{aligned} R_{i17} \parallel R_9 &= \frac{R_{i17} \times R_9}{R_{i17} + R_9} \\ &= \frac{19.2\times10^3 \times 50\times10^3}{19.2\times10^3 + 50\times10^3} \\ &= 13.873\ \text{k}\Omega \end{aligned}$$

The base voltage at the transistor Q_{17} is,

$$\begin{aligned} v_{b17} &= \frac{R_{i17} \parallel R_9}{r_{e16} + R_{i17} \parallel R_9} v_{i2} \\ &= \left(\frac{13.873\times10^3}{1.54\times10^3 + 13.873\times10^3}\right) v_{i2} \\ &= 900.08 v_{i2}\ \text{V} \end{aligned}$$

Determine α.

$$\begin{aligned} \alpha &= \frac{\beta - 1}{\beta + 1} \\ &= \frac{200-1}{200+1} \\ &= 0.99 \end{aligned}$$

Step 5 of 6

The current through the transistor is.

$$\begin{aligned} i_{c17} &= \alpha\left(\frac{v_{b17}}{r_{e17} + R_8}\right) \\ &= 0.99\left(\frac{900.08 v_{i2}}{45.45+50}\right) v_{i2} \\ &= 9.335 v_{i2}\ \text{A} \end{aligned}$$

Step 6 of 6

Thus the transconductance is,

$$\begin{aligned} G_{m2} &= \frac{i_{c17}}{v_{i2}} \\ &= \frac{9.335 v_{i2}}{v_{i2}} \\ &= 9.335\ \text{A/V} \end{aligned}$$

Thus the parameters corresponding to the variation in the design of the 741 second stage are

$$\boxed{\begin{aligned} R_{i2} &= 3.1\ \text{M}\Omega \\ G_{m2} &= 9.335\ \text{A/V} \end{aligned}}$$

Step 1 of 5

In the analysis of the 741 second stage, the effect of placing appropriate resistors at the emitters of Q_{12}, Q_{13A} and Q_{13B} is considered.

The requirements are,

$$R_{o13B} = R_{o17}.$$

$$R_{o2} = \frac{R_{o17}}{2}.$$

For the 741 operational amplifier the parameter values are,

$$R_{o17} = 787\ \text{k}\Omega$$
$$I_{C13B} = 550\ \mu\text{A}$$
$$g_{m13B} = 22\ \text{mA/V}$$

Determine the small-signal input resistance between base and emitter looking into the base.

$$r_{\pi 13B} = \frac{(\beta+1)}{g_m}$$
$$= \frac{50+1}{22\times10^{-3}}$$
$$= 2.32\ \text{k}\Omega$$

Determine the output resistance.

$$r_o = \frac{\beta}{I_{C13B}}$$
$$= \frac{50}{550\times10^{-6}}$$
$$= 90.9\ \text{k}\Omega$$

Step 2 of 5

Determine the output resistance at the transistor, Q_{13B}.

$$R_{o13B} = r_o\left[1+g_m\left(R_E \| r_\pi\right)\right]$$
$$787\times10^3 = 90.9\times10^3\left[1+22\times10^{-3}\left(R_E \| 2.32\ \text{k}\right)\right]$$
$$1+22\times10^{-3}\left(R_E \| 2.32\ \text{k}\right) = \frac{787}{90.9}$$
$$1+22\times10^{-3}\left(R_E \| 2.32\ \text{k}\right) = 8.66$$

$$22\times10^{-3}\left(R_E \| 2.32\ \text{k}\right) = 8.66-1$$
$$22\times10^{-3}\left(R_E \| 2.32\ \text{k}\right) = 7.66$$
$$R_E \| 2.32\ \text{k} = \frac{7.66}{22\times10^{-3}}$$
$$R_E \| 2.32\ \text{k} = 0.348\ \text{k}$$

$$R_E \| 2.32 = \frac{2.32R_E}{R_E+2.32}$$

Step 3 of 5

Simplify further.

$$\frac{2.32R_E}{R_E+2.32} = 0.348$$
$$2.32R_E = 0.348R_E + 0.808$$
$$2.32R_E - 0.348R_E = 0.808$$
$$1.972R_E = 0.808$$

$$R_E = \frac{0.808}{1.972}$$
$$= 0.4097\ \text{k}\Omega$$
$$= 409.7\ \Omega$$

Step 4 of 5

Consider the ratio, $\frac{R_{E12}}{R_E}$.

$$\frac{R_{E12}}{R_E} = \frac{I_E}{I_{E12}}$$
$$= \frac{550}{730}$$
$$R_{E12} = \frac{550}{730}\times R_E$$

Substitute $409.7\ \Omega$ for R_E.

$$R_{E12} = \left(\frac{550}{730}\right)(409.7)$$
$$= 308.7\ \Omega$$

Step 5 of 5

Consider the ratio, $\frac{R_{E13A}}{R_E}$.

$$\frac{R_{E13A}}{R_E} = \frac{I_E}{I_{E13A}}$$
$$= \frac{550}{180}$$

$$R_{E13A} = \left(\frac{550}{180}\right)R_E$$

Substitute $409.7\ \Omega$ for R_E.

$$R_{E13A} = \left(\frac{550}{180}\right)(409.7\ \Omega)$$
$$= 1.25\ \text{k}\Omega$$

Thus, the resistance values in each of the emitters are,

$$\boxed{\begin{aligned} R_E &= 409.7\ \Omega \\ R_{E12} &= 308.9\ \Omega \\ R_{E13A} &= 1.25\ \text{k}\Omega \end{aligned}}$$

13.062P

The supply voltages of the 741 operational amplifier are,

$$V_{CC} = V_{EE} = \pm 5 \text{ V}.$$

The base to emitter voltage of the transistors is,

$$|V_{BE}| = 0.6 \text{ V}.$$

The saturation value of the collector to emitter voltage is,

$$|V_{CEsat}| = 0.2 \text{ V}.$$

Determine the minimum value of the output voltage of the 741 operational amplifier.

$$\begin{aligned} v_{O\min} &= -V_{EE} + V_{CEsat} + V_{EB23} + V_{EB20} \\ &= -5 + 0.2 + 0.6 + 0.6 \\ &= -3.6 \text{ V} \end{aligned}$$

Determine the maximum value of the output voltage of the 741 operational amplifier.

$$\begin{aligned} v_{O\max} &= V_{CC} - |V_{CEsat}| - V_{BE14} \\ &= 5 - 0.2 - 0.6 \\ &= 4.2 \text{ V} \end{aligned}$$

Thus, the output voltage limits are,

$$\boxed{\begin{aligned} v_{O\min} &= -3.6 \text{ V} \\ v_{O\max} &= 4.2 \text{ V} \end{aligned}}.$$

Step 1 of 4

With the transistor Q_{23} removed the current in the transistor Q_{17} increases to $I_{C17} = 730\ \mu A$.

The emitter resistance of the transistor Q_{17} is,

$$\begin{aligned} r_{e17} &= \frac{V_T}{I_{C17}} \\ &= \frac{25\times10^{-3}}{730\times10^{-6}} \\ &= 34.25\ \Omega \end{aligned}$$

Determine the resistance, R_{i17}.

$$\begin{aligned} R_{i17} &= (\beta+1)(r_{e17}+R_8) \\ &= (201)(34.25+100) \\ &= 27\ \text{k}\Omega \end{aligned}$$

Determine the resistance, $R_9 \parallel R_{i17}$

$$\begin{aligned} R_9 \parallel R_{i17} &= \frac{R_9 \times R_{i17}}{R_9 + R_{i17}} \\ &= \frac{50\times27}{50+27} \\ &= 17.5\ \text{k}\Omega \end{aligned}$$

Step 2 of 4

Determine α.

$$\begin{aligned} \alpha &= \frac{\beta-1}{\beta+1} \\ &= \frac{200-1}{200+1} \\ &= 0.99 \end{aligned}$$

The transconductance of the transistor is,

$$\begin{aligned} G_{m2} &= \frac{i_{C17}}{v_{i2}} \\ &= \frac{1}{v_{i2}}\left[\alpha v_{i2}\frac{(R_9 \parallel R_{i17})}{(R_9 \parallel R_{i17})+r_{e16}}\right] \\ &= \frac{\alpha(R_9 \parallel R_{i17})}{(R_9 \parallel R_{i17})+r_{e16}} \end{aligned}$$

The resistance, $r_{e16} = r_{e17}$.

Substitute 0.99 for α, $17.5\ \text{k}\Omega$ for $R_9 \parallel R_{i17}$ and $34.25\ \Omega$ for r_{e16}.

$$\begin{aligned} G_{m2} &= \frac{\alpha(R_9 \parallel R_{i17})}{(R_9 \parallel R_{i17})+r_{e16}} \\ &= \frac{(0.99)(17.5\times10^3)}{17.5\times10^3+34.25} \\ &= 0.988\ \text{A/V} \end{aligned}$$

Step 3 of 4

Determine the reflection of the load resistance to the collector of the transistor Q_{17}.

$$\begin{aligned} R_{in3} &= (\beta+1)R_L \parallel r_{O13,4} \\ &= (201)(2\times10^3) \parallel 90.9\times10^3 \\ &= \frac{(201)(2\times10^3)(90.9\times10^3)}{201\times2\times10^3+90.9\times10^3} \\ &= 74.1\ \text{k}\Omega \end{aligned}$$

Thus the reflection of the load resistance to the collector R_{in3} of the transistor Q_{17} is

$\boxed{74.1\ \text{k}\Omega}$.

Step 4 of 4

Determine the voltage gain of the second stage of the 741 operational amplifier.

$$A_2 = -G_{m2}R_{o2}\frac{R_{in3}}{R_{in3}+R_{o2}}.$$

The resistance, $R_{02} = 81\ \text{k}\Omega$.

$$\begin{aligned} A_2 &= (-0.988)(81\times10^3)\left(\frac{74.1\times10^3}{74.1\times10^3+81\times10^3}\right) \\ &= 38\times10^6\ \text{V/V} \end{aligned}$$

Thus, the voltage gain A_2 of the second stage of the 741 operational amplifier is,

$\boxed{38\times10^6\ \text{V/V}}$.

13.064P

We don't have the solution to this problem yet.

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Step 1 of 3

13.066P

Refer to Figure 12.13 in the textbook for the 741 op-amp circuit.

In the 741 sinking-current limit involving R_7, Q_{21}, Q_{24}, R_{11}, and Q_{22}, the current in the transistor Q_{22} is equal to the maximum current available from the output stage, that is, the current in the transistor Q_8. The reduced current limit is $10\ \text{mA}$.

The maximum current available from the output stage, that is, the current in the transistor Q_8 is $I_{C_{max}} = 19\ \mu\text{A}$.

The base to emitter voltage of the transistors is,

$$\begin{aligned} V_{BE22} &= V_{BE24} \\ &= 0.534\ \text{V} \end{aligned}$$

Thus the current through the resistor R_{11} is,

$$\begin{aligned} I_{R11} &= \frac{V_{BE22}}{R_{11}} \\ &= \frac{0.534}{50\times10^3} \\ &= 10.68\ \mu\text{A} \end{aligned}$$

Step 2 of 3

The collector current of the transistor Q_{21} is,

$$\begin{aligned} I_{C21} &= I_{C_{max}} + I_{R11} \\ &= 19\times10^{-6} + 10.68\times10^{-6} \\ &= 29.68\ \mu\text{A} \end{aligned}$$

The reverse saturation current of transistor is, $I_S = 10^{-14}\ \text{A}$

The base to emitter voltage of the transistor Q_{21} is,

$$\begin{aligned} V_{BE21} &= V_T \ln\left(\frac{I}{I_S}\right) \\ &= \left(25\times10^{-3}\right)\ln\left(\frac{29.68\times10^{-6}}{10^{-14}}\right) \\ &= 0.545\ \text{V} \end{aligned}$$

Step 3 of 3

Now, the current through the resistor R_7 is,

$$\begin{aligned} I &= \frac{V_{BE21}}{R_7} \\ &= \frac{0.545}{27} \\ &= 20.2\ \text{mA} \end{aligned}$$

Thus, the current through the resistor R_7 for maximum current at the input stage is, $\boxed{20.2\ \text{mA}}$.

If this current is reduced to $10\ \text{mA}$, then the simple change would be to double the resistance value.

$$10\ \text{mA} = \frac{0.545}{R_7}$$

$$R_7 \approx 54\ \Omega$$

That is,

$$\begin{aligned} R_{7new} &= 2(R_7) \\ &= 2(27) \\ &= 54\ \Omega \end{aligned}$$

Therefore, make the resistance R_7 double to reduce the current limit to 10 mA.

13.067P

We don't have the solution to this problem yet.

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13.068P

We don't have the solution to this problem yet.

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Ask an expert

13.069P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

13.070P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

13.071P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

13.072P

We don't have the solution to this problem yet.

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Step 1 of 2

13.073P

Write the expression for the slew rate for the 741-operational amplifier.

$$SR = 2\pi f_m V_0$$

Here,

The output of the amplifier, V_0

The slew rate of the amplifier, SR

The frequency, f_m

Rearrange the expression for f_m.

$$f_m = \frac{SR}{2\pi V_0}$$

Substitute $10\ \text{V}/\mu\text{s}$ for SR and 10 V for V_0

$$f_m = \frac{10\times10^6}{2\pi(10)}$$
$$= \frac{10^6}{2\pi}$$
$$= 159.2\ \text{kHz}$$

Thus, the full-power bandwidth is, $\boxed{159.2\ \text{kHz}}$.

Step 2 of 2

The slew rate of 741 operational amplifier is,

$$SR = 4V_T\omega_t$$

Rearrange the expression for the unity gain band width.

$$\omega_t = \frac{SR}{4V_T}$$
$$2\pi f_t = \frac{SR}{4V_T}$$
$$f_t = \frac{SR}{8\pi V_T}$$

Substitute $10\ \text{V}/\mu\text{s}$ for SR and 25 mV for V_T

$$f_t = \frac{10\times10^6}{8\pi\left(25\times10^{-3}\right)}$$
$$= 15.9\times10^6$$
$$= 15.9\ \text{MHz}$$

Thus, the unity gain band width is, $\boxed{15.9\ \text{MHz}}$.

13.074P

We don't have the solution to this problem yet.

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Step 1 of 8

(a)

Refer to Figure P12.63 in the textbook for the circuit diagram. Consider $\beta = 100$, and $V_{BE} = 0.7\ \text{V}$ and $r_o = \infty$ which are same for all transistors.

Neglect base currents. In other words, the base current of each transistor indicated with $I_{B1}, I_{B2}, I_{B3}, I_{B4}, I_{B5}, I_{B6}$, and I_{B7} is equal to zero.

The general expression for the emitter current (applicable to all the transistors) of a simple transistor is,

$$\begin{aligned} I_E &= I_B + I_C \quad \{\text{since } I_B = 0\} \\ &= 0 + I_C \\ &= I_C \end{aligned}$$

From the circuit, observed that the current of 0.1 mA is applied to the differential amplifier section in the circuit. Due to differential circuit, the current 0.1 mA divides equally to each half circuit. So, the current flows through the Q_1 and Q_3 section is $0.05\ \text{mA}$ and the current flows through the Q_2 and Q_4 section is $0.05\ \text{mA}$.

Step 2 of 8

The inputs are grounded and outputs are held at 0 V (by negative feedback). Also observe that the transistors Q_1 and Q_3 are connected in series (that is, in a single half circuit). So, the collector current of Q_2 and Q_4 is,

$$I_{C1} = I_{C3} = 50\ \mu\text{A}$$

The collector current of Q_1 and Q_3 is,

$$I_{C2} = I_{C4} = 50\ \mu\text{A}$$

Therefore, the collector current of the each transistor Q_1, Q_2, Q_3 and Q_4 is, $\boxed{50\ \mu\text{A}}$.

Since the base current of transistors Q_5, Q_6, and Q_7 is zero, then the current flows through the transistor Q_5 is 1 mA.

Therefore, the collector current of the transistor Q_5 is, $\boxed{1\ \text{mA}}$.

The base to emitter voltage of the each transistor is 0.7 V. the collector current of the transistor is,

$$I_C = \beta I_B$$

Since the base current is zero, so,

$$\begin{aligned} I_C &= 100(0) \\ &= 0\ \text{A} \end{aligned}$$

Therefore, the collector current of the transistor Q_6 and Q_7 is, $\boxed{0\ \text{A}}$.

Step 3 of 8

(b)

The collector current though the transistor Q_1 and Q_3 is $50\ \mu\text{A}$. That is, $I_C = 50\ \mu\text{A}$. From the BJT equivalent circuit, observe that the input resistance of each transistor Q_1 and Q_3 is r_e.

Write the expression for the trans-conductance.

$$g_m = \frac{I_C}{V_T}$$

Substitute $50\ \mu\text{A}$ for I_C and 25 mV for V_T.

$$\begin{aligned} g_m &= \frac{50 \times 10^{-6}}{25 \times 10^{-3}} \\ &= 2\ \text{mA/V} \end{aligned}$$

Write the expression for the r_π.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute 100 for β and 2 mA/V for g_m.

$$\begin{aligned} r_\pi &= \frac{100}{2 \times 10^{-3}} \\ &= 50\ \text{k}\Omega \end{aligned}$$

Step 4 of 8

The expression for the small signal emitter resistance is,

$$r_e = \frac{r_\pi}{\beta + 1}$$

Substitute 100 for β and $50\ \text{k}\Omega$ for r_π.

$$\begin{aligned} R_{in} &= \frac{50 \times 10^3}{100 + 1} \quad \{\text{since } R_{in} = r_e\} \\ &\simeq 500\ \Omega \end{aligned}$$

Therefore, the input resistance of Q_1, Q_2, Q_4 and Q_3 is, $\boxed{\simeq 500\ \Omega}$.

Step 5 of 8

The collector current though the transistor Q_5 is $1\ \text{mA}$. That is, $I_C = 1\ \text{mA}$. From the BJT equivalent circuit, observe that the input resistance of each transistor Q_5 is r_e

Write the expression for the trans-conductance.

$$g_m = \frac{I_C}{V_T}$$

Substitute $1\ \text{mA}$ for I_C and 25 mV for V_T.

$$\begin{aligned} g_m &= \frac{1 \times 10^{-3}}{25 \times 10^{-3}} \\ &= 0.04\ \text{A/V} \end{aligned}$$

Write the expression for the r_π.

$$r_\pi = \frac{\beta}{g_m}$$

Substitute 100 for β and 0.04 A/V for g_m.

$$\begin{aligned} r_\pi &= \frac{100}{0.04} \\ &= 2.5\ \text{k}\Omega \end{aligned}$$

Step 6 of 8

The expression for the small signal emitter resistance is,

$$r_e = \frac{r_\pi}{\beta + 1}$$

Substitute 100 for β and $2.5\ \text{k}\Omega$ for r_π.

$$\begin{aligned} R_{in} &= \frac{2.5 \times 10^3}{100 + 1} \quad \{\text{since } R_{in} = r_e\} \\ &\simeq 25\ \Omega \end{aligned}$$

Therefore, the input resistance of Q_5 is, $\boxed{\simeq 25\ \Omega}$.

The collector current though the transistor Q_6 and Q_7 is $0\ \text{A}$. That is, $I_C = 0\ \text{A}$. So, the input resistance is **infinite**.

Step 7 of 8

(c)

Consider the following expression.

$$G_m = \frac{2}{r_e}$$

Substitute the corresponding values.

$$\begin{aligned} G_m &= \frac{2}{500} \\ &= 4\ \text{mA/V} \end{aligned}$$

Find the value for R_o

$$\begin{aligned} R_O &= (\beta + 1)(r_e \| r_e \| R_L) \\ &= (100 + 1)\left(\frac{(25)(25)}{25 + 25} \| 5000\right) \\ &= (101)(12.46) \\ &= 1.25\ \text{k}\Omega \end{aligned}$$

Calculate the value of the gain.

$$\begin{aligned} A &= G_m R_O \\ &= \left(4 \times 10^{-3}\right)\left(1.25 \times 10^3\right) \\ &= 5\ \text{V/V} \end{aligned}$$

Thus, the gain of the amplifier is, $\boxed{5\ \text{V/V}}$.

Step 8 of 8

(d)

Write the expression for the 3 dB frequency.

$$f_H = \frac{1}{2\pi R_L C}$$

Substitute 100 Hz for f_H, $5\ \text{k}\Omega$ for R_L to calculate the value of capacitance.

$$\begin{aligned} C &= \frac{1}{2\pi(5000)(100)} \\ &= 7.5\ \mu\text{F} \end{aligned}$$

Therefore, the value of the capacitance is, $\boxed{7.5\ \mu\text{F}}$.

13.076P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

Step 1 of 6

13.077P

Refer to Figure 12.38 in the text book for a self-biased current-reference source utilizing a Widler circuit.

The circuit is augmented with emitters connected to ground and bases connected to V_{BIAS1} to generate constant currents of $10\ \mu A$ and $40\ \mu A$ respectively. Draw the circuit.

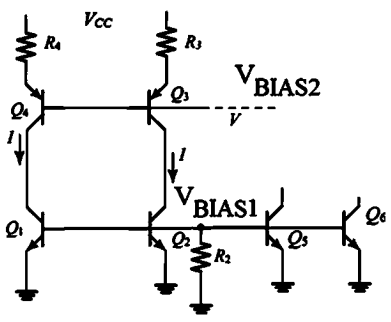


Figure 1

Step 2 of 6

The base-emitter voltage of the transistor Q_5 is equal to that of the transistor, Q_1.

$$V_{BE5} = V_{BE1}$$

$$V_T \ln\left(\frac{I_5}{I_{s5}}\right) = V_T \ln\left(\frac{I_1}{I_{s1}}\right)$$

$$\left(\frac{I_5}{I_{s5}}\right) = \left(\frac{I_1}{I_{s1}}\right)$$

As $I_1 = I_5 = 10\ \mu A$, $\frac{I_5}{I_{s5}} = 1$.

Thus, the transistors Q_1 and Q_5 have the same emitter area.

The base-emitter voltage of the transistor Q_6 is equal to that of the transistor Q_1.

$$V_{BE6} = V_{BE1}$$

$$V_T \ln\left(\frac{I_6}{I_{s6}}\right) = V_T \ln\left(\frac{I_1}{I_{s1}}\right)$$

$$\left(\frac{I_6}{I_{s6}}\right) = \left(\frac{I_1}{I_{s1}}\right)$$

As $I_1 = 10\ \mu A$ and $I_6 = 40\ \mu A$, $\frac{I_{S6}}{I_{S1}} = 4$.

Thus, the emitter area of the transistor Q_6 is four times the emitter area of the transistor Q_1.

Step 3 of 6

Draw the circuit by connecting a resistor R_6 to the emitter of Q_6 and by reducing the current I_6 to $10\ \mu A$.

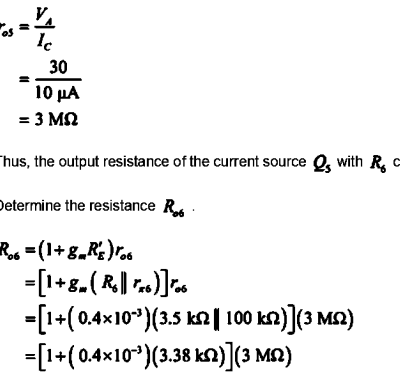


Figure 2

Step 4 of 6

Observe from the circuit that the base-emitter voltages,

$$V_{BE1} - V_{BE6} = R_6 I_6$$

$$V_T \ln\frac{I_1}{I_{S1}} - V_T \ln\frac{I_6}{I_{S6}} = I_6 R_6$$

$$V_T \ln\left[\frac{\left(\frac{I_1}{I_{S1}}\right)}{\left(\frac{I_6}{I_{S6}}\right)}\right] = I_6 R_6$$

$$\left(25\times10^{-3}\right)\ln(4) = R_6\left(10\times10^{-6}\right)$$

$$R_6 = \frac{\left(25\times10^{-3}\right)(1.386)}{10\times10^{-6}}$$

$$= 3.5\ \text{k}\Omega$$

Thus, the resistance, R_6 is $\boxed{3.5\ \text{k}\Omega}$.

Step 5 of 6

The output resistance of Q_5 is simply r_o.

Determine the resistance R_{o5}.

$$R_{o5} = r_{o5}$$

$$= \frac{V_A}{I_C}$$

$$= \frac{30}{10\ \mu\text{A}}$$

$$= 3\ \text{M}\Omega$$

Thus, the resistance R_{o5} is $\boxed{3\ \text{M}\Omega}$.

For Q_6, the output resistance is increased by a factor of $\left(1 + g_m R_E'\right)$ where $R_E' = R_6 \| r_{\pi 6}$.

Thus,

$$g_{m6} = \frac{I_{C6}}{V_T}$$

$$= \frac{10}{0.025}$$

$$= 0.4\ \text{mA/V}$$

The resistance r_π is,

$$r_\pi = \frac{\beta_n}{g_m}$$

$$= \frac{40}{0.4\times10^{-3}}$$

$$= 100\ \text{k}\Omega$$

Step 6 of 6

Determine the output resistance of the current source Q_5 with R_6 connected.

$$r_{o5} = \frac{V_A}{I_C}$$

$$= \frac{30}{10\ \mu\text{A}}$$

$$= 3\ \text{M}\Omega$$

Thus, the output resistance of the current source Q_5 with R_6 connected is $\boxed{3\ \text{M}\Omega}$.

Determine the resistance R_{o6}.

$$R_{o6} = (1 + g_m R_E')r_{o6}$$

$$= \left[1 + g_m\left(R_6 \| r_{\pi 6}\right)\right]r_{o6}$$

$$= \left[1 + \left(0.4\times10^{-3}\right)\left(3.5\ \text{k}\Omega \| 100\ \text{k}\Omega\right)\right](3\ \text{M}\Omega)$$

$$= \left[1 + \left(0.4\times10^{-3}\right)(3.38\ \text{k}\Omega)\right](3\ \text{M}\Omega)$$

$$= 7.06\ \text{M}\Omega$$

Thus, the output resistance R_{o6} of the current source Q_6 is $\boxed{7.06\ \text{M}\Omega}$.

13.078P

We don't have the solution to this problem yet.

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Step 1 of 5

(a)

Refer to Figure12.40 (a) in the textbook for the differential input stage with current mirror load. The input common mode range provides the range of values for which the input stage remains in the active state.

In order to determine the input common mode range of the differential input stage with current mirror load, determine the minimum and maximum value of the input common mode range.

Determine the minimum value or low end $V_{ICM_{min}}$ of the input common mode range. The minimum value or low end $V_{ICM_{min}}$ of the input common mode range is determined by the saturation of the transistors Q_1 and Q_2.

In the circuit the transistor Q_3 is collector feedback- biased. Since the base terminal and the collector terminal of the transistor are shorted, the voltage developed at the base of the transistor Q_3 appears at the collector terminal of the transistor. Hence, the voltage at the collector terminal of the transistor Q_3 is $0.7\ \text{V}$. Consequently, the collector of the transistor Q_1 is at the same voltage of $0.7\ \text{V}$.

Step 2 of 5

In order to keep the pnp transistor Q_1 in saturation, the collector voltage should be higher than the base voltage and to keep it in forward active state the voltage difference between the collector and the base should be $0.6\ \text{V}$.

$$V_{CB} = 0.6\ \text{V}$$
$$V_C - V_B = 0.6\ \text{V}$$
$$0.7\ \text{V} - V_B = 0.6\text{V}$$
$$V_B = 0.1\ \text{V}$$

The voltage at the collector of the transistor Q_1 cannot fall beyond $0.1\ \text{V}$, otherwise it drives the transistor into the cutoff mode.

Thus, the minimum value or low end value, $V_{ICM_{min}}$ of the input common mode range is,

$$V_{ICM_{min}} = 0.1\ \text{V}$$

Step 3 of 5

From the calculation of $V_{ICM_{min}}$, the input common mode range observe that the minimum voltage that can be applied to the base of the transistor Q_3 is $0.1\ \text{V}$.

Obtain the upper end $V_{ICM_{max}}$ of the input common mode range by considering the minimum voltage as well the voltage V_{EB1}.

Determine the maximum value or upper end $V_{ICM_{max}}$ of the input common mode range using the following relation:

$$V_{ICM_{max}} = V_{CC} - V_{ICM_{min}} - V_{EB1}$$

Substitute the value $3\ \text{V}$ for V_{CC}, $0.1\ \text{V}$ for $V_{ICM_{min}}$ and $0.7\ \text{V}$ for V_{EB1} in the expression for maximum value or upper end maximum value or upper end $V_{ICM_{max}}$ $V_{ICM_{max}}$.

$$V_{ICM_{max}} = 3 - 0.1 - 0.7$$
$$= 2.2\ \text{V}$$

Thus, the value of the minimum value of the input common mode range $V_{ICM_{min}}$ is $\boxed{0.1\ \text{V}}$ and the value of the maximum value of the input common mode range $V_{ICM_{max}}$ is $\boxed{2.2\ \text{V}}$.

Hence, the range of the input common mode range is $\boxed{0.1\ \text{V} \le V_{ICM} \le 2.2\ \text{V}}$.

Step 4 of 5

(b)

Draw the complementary version of the circuit in Figure 12.40(a) in the text book.

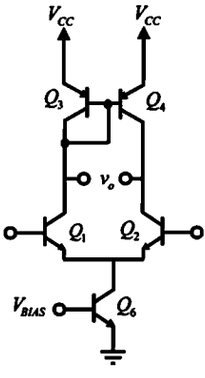


Figure 1

The base terminal and the collector terminal of the transistor in the circuit are short-circuited. The voltage developed at the base of the transistor Q_3 appears at the collector terminal of the transistor. Hence, the voltage at the collector terminal of the transistor Q_3 is,

$$V_{C3} = 3 - 0.7\ \text{V}$$
$$= 2.3\ \text{V}$$

Step 5 of 5

Consequently, the collector of the transistor Q_1 is at the same voltage $2.3\ \text{V}$. In order to keep the npn transistor Q_1 in saturation, the collector voltage should be higher than the base voltage and to keep it in forward active state the minimum voltage that has to be applied to the transistor Q_1 is $0.4\ \text{V}$.
Thus, the minimum value or low end value, $V_{ICM_{min}}$ of the input common mode range is

$0.4\ \text{V}$.

Determine the maximum value or upper end $V_{ICM_{max}}$ of the input common mode range using the following formula.

$$V_{ICM_{max}} = V_{CC} + V_{CEX(sat)}$$

Substitute $3\ \text{V}$ for V_{CC} and $0.2\ \text{V}$ for $V_{ICM_{min}}$.

$$V_{ICM_{max}} = 3 + 0.2$$
$$= 3.2\ \text{V}$$

Thus, the value of the minimum value of the input common mode range $V_{ICM_{min}}$ is $\boxed{0.4\ \text{V}}$ and the value of the maximum value of the input common mode range $V_{ICM_{max}}$ is $\boxed{3.2\ \text{V}}$.

Hence, the range of the input common mode range is $\boxed{0.4\ \text{V} \le V_{ICM} \le 3.2\ \text{V}}$.

13.080P

We don't have the solution to this problem yet.

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13.081P

We don't have the solution to this problem yet.

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13.082P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

13.083P

We don't have the solution to this problem yet.

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Step 1 of 3

13.084P

Refer to Figure 12.44 in the text book for the circuit.

(a)

The output resistance of Q_7 is R_{07} and that of Q_9 is R_{09}.

These two resistances are in parallel at the output node of Q_7 and Q_9 with respect to ground. That is, the input resistance to the feedback loop is $R_{07} \| R_{09}$. Input current at the feedback loop is $I_{C7} = I_{C9}$.

Therefore, the input voltage at the feedback loop is $I_{C7}(R_{07} \| R_{09})$.

The output voltage at the feedback loop is $V_B = I_{C7}(r_{e7} + R_7)$.

The feedback loop gain $A\beta$ is,

$$\begin{aligned} A\beta &= \frac{I_{C7}(R_{07} \| R_{09})}{I_{C7}(r_{e7} + R_7)} \\ &= \frac{(R_{07} \| R_{09})}{(r_{e7} + R_7)} \end{aligned}$$

Therefore, the feedback loop gain is $\boxed{A\beta = \frac{(R_{07} \| R_{09})}{(r_{e7} + R_7)}}$.

Step 2 of 3

(b)

Determine the value of the resistance, r_{e7}.

$$\begin{aligned} r_{e7} &= \frac{V_T}{I_C} \\ &= \frac{25\ \text{mV}}{10\ \mu\text{A}} \\ &= 2.5\ \text{k}\Omega \end{aligned}$$

Determine the loop gain $A\beta$.

$$\begin{aligned} A\beta &= \frac{23\ \text{M}\Omega \| 12.9\ \text{M}\Omega}{2.5\ \text{k}\Omega + 20\ \text{k}\Omega} \\ &= \frac{8.264\ \text{M}\Omega}{22.5\ \text{k}\Omega} \\ &= 367.3 \end{aligned}$$

Thus, the loop gain $A\beta$ is $\boxed{367.3}$.

Step 3 of 3

(c)

Without feedback, $\Delta V_{CM} = 2.5\ \text{V}$.

Determine the loop gain, $A\beta$.

$$\begin{aligned} A\beta &= \frac{R_{o9} \| R_{o7}}{r_{e7} + R_7} \\ &= \frac{23\ \text{M}\Omega \| 12.9\ \text{M}\Omega}{2.5\ \text{k}\Omega + 20\ \text{k}\Omega} \\ &= \frac{8.2646\ \text{M}\Omega}{22.5\ \text{k}\Omega} \\ &= 367.3 \end{aligned}$$

Therefore,

$1 + A\beta = 367.3$.

With negative feedback, the required change in V_B and V_{CM} gets reduced by a factor of $1 + A\beta$.

Therefore, the ΔV_{CM} with feedback is, $\Delta V'_{CM} = \frac{\Delta V_{CM}\ \textbf{without feedback}}{1 + A\beta}$.

Find $\Delta V'_{CM}$.

$$\begin{aligned} \Delta V'_{CM} &= \frac{2.5}{367.38} \\ &= 6.8\ \text{mV} \end{aligned}$$

Thus, the expected ΔV_{CM} is,

$\boxed{\Delta V'_{CM} = 6.8\ \text{mV}}$.

Hence the expected ΔV_{CM} found is approximately same as the value of ΔV_{CM} found by using different approach in Example 12.6 in the text book.

13.085P

We don't have the solution to this problem yet.

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13.086P

We don't have the solution to this problem yet.

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13.087P

Refer to Figure 12.48 in the text book for the complex version of the output stage of the amplifier circuit.

The emitter voltage V_E is,

$$v_E = v_{EB7} + v_{BEN}$$

Substitute $V_T \ln \frac{i_{C7}}{I_{S7}}$ for v_{BE} and $V_T \ln \frac{i_N}{I_{SN}}$ for v_{BEN}.

$$\begin{aligned} v_E &= V_T \ln \frac{i_{C7}}{I_{S7}} + V_T \ln \frac{i_N}{I_{SN}} \\ &= V_T \ln \left[\frac{i_{C7} i_N}{I_{S7} I_{SN}} \right] \end{aligned}$$

Substitute $I \frac{i_P}{i_P + i_N}$ for i_{C7}.

$$\begin{aligned} v_E &= V_T \ln \left[\frac{i_{C7} i_N}{I_{S7} I_{SN}} \right] \\ &= V_T \ln \left[\frac{i_N}{I_{S7} I_{SN}} \left(I \frac{i_P}{i_P + i_N} \right) \right] \\ &= V_T \ln \left[\frac{i_N i_P}{i_P + i_N} \frac{I}{I_{SN} I_{S7}} \right] \end{aligned}$$

Thus, the emitter voltage is,

$$\boxed{v_E = V_T \ln \left[\frac{i_N i_P}{i_N + i_P} \frac{I}{I_{SN} I_{S7}} \right]}.$$

13.088P

We don't have the solution to this problem yet.

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Step 1 of 5

Refer to Figure 14.2 (a) in the textbook.

Determine the value of small-signal resistance, r_{DS}.

$$r_{DS} = \frac{V_{OL}}{I_{DD}}$$

Substitute $90\ \text{mV}$ for V_{OL} and $30\ \mu\text{A}$ for I_{DD} in the equation.

$$\begin{aligned} r_{DS} &= \frac{90\times10^{-3}}{30\times10^{-6}} \\ &= \left(3\times10^{3}\right) \\ &= 3\ \text{k}\Omega \end{aligned}$$

Step 2 of 5

Determine the value of width-to-length ratio, W/L.

$$r_{DS} = \frac{1}{\mu_n C_{ox}\frac{W}{L}\left(V_{GS} - V_t\right)}$$

$$r_{DS} = \frac{1}{\mu_n C_{ox}\frac{W}{L}\left(V_{DD} - V_t\right)} \quad \left(\text{since } V_{GS} = V_{DD}\right)$$

Substitute $3\ \text{k}\Omega$ for r_{DS}, $1.8\ \text{V}$ for V_{DD}, $125\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$ and $0.4\ \text{V}$ for V_t in the equation.

$$\left(3\times10^{3}\right) = \frac{1}{\left(125\times10^{-6}\right)\left(\frac{W}{L}\right)(1.8-0.4)}$$

$$\left(3\times10^{3}\right) = \frac{1}{\left(175\times10^{-6}\right)\left(\frac{W}{L}\right)}$$

$$\begin{aligned} \frac{W}{L} &= \frac{1}{\left(250\times10^{-6}\right)\left(2\times10^{3}\right)} \\ &= 1.9 \end{aligned}$$

Therefore, the determined value of width-to-length ratio, W/L is $\boxed{1.9}$.

Step 3 of 5

Determine the value of resistor, R_D.

$$V_{OL} = V_{DD}\frac{r_{DS}}{R_D + r_{DS}}$$

Substitute $3\ \text{k}\Omega$ for r_{DS}, $1.8\ \text{V}$ for V_{DD} and $90\ \text{mV}$ for V_{OL} in the equation.

$$90\times10^{-3} = (1.8)\frac{\left(3\times10^{3}\right)}{R_D + \left(3\times10^{3}\right)}$$

$$90\times10^{-3}R_D + 270 = 5400$$

$$\begin{aligned} R_D &= \frac{5400-270}{90\times10^{-3}} \\ &= 57\times10^{3}\ \Omega \end{aligned}$$

Therefore, the value of resistor R_D is $\boxed{57\ \text{k}\Omega}$.

Step 4 of 5

Determine the value of power drawn from the supply when the switch is open.

When the switch is open, there is no current flow in the loop.

Thus, the value of power drawn from the supply when the switch is open, P_{open} is $\boxed{0\ \text{W}}$.

Step 5 of 5

Determine the value of power drawn from the supply when the switch is closed.

$$P_{\text{closed}} = V_{DD}I_{DD}$$

Substitute $1.8\ \text{V}$ for V_{DD} and $30\ \mu\text{A}$ for I_{DD} in the equation.

$$\begin{aligned} P_{DD} &= V_{DD}I_{DD} \\ &= 1.8\times30\times10^{-6} \\ &= 54\ \mu\text{W} \end{aligned}$$

Thus, the value of power drawn from the supply when the switch is closed, P_{closed} is $\boxed{54\ \mu\text{W}}$.

14.001P

We don't have the solution to this problem yet.

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Step 1 of 2

14.002E

Refer to Figure 13.9 for the current-steering circuit in the text book.

Determine the value of high-level output voltage.

$$V_{OH} = V_{CC}$$

Substitute $5\ \text{V}$ for V_{CC} in the equation.

$$V_{OH} = 5\ \text{V}$$

Thus, the value of high-level output voltage, V_{OH} is $\boxed{5\ \text{V}}$.

Step 2 of 2

Determine the value of low-level output voltage.

$$V_{OL} = V_{CC} - I_{EE}R_{C2}$$

Substitute $5\ \text{V}$ for V_{CC}, $1\ \text{mA}$ for I_{EE} and $2\ \text{k}\Omega$ for R_{C2} in the equation.

$$\begin{aligned} V_{OL} &= 5 - \left(1\times10^{-3}\right)\left(2\times10^{3}\right) \\ &= 5 - 2 \\ &= 3\ \text{V} \end{aligned}$$

Thus, the value of low-level output voltage, V_{OL} is $\boxed{3\ \text{V}}$.

14.002P

We don't have the solution to this problem yet.

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Step 1 of 6

Refer to Figure 13.10 in the text book.

Determine the value of voltage, V_x at $V_M = \frac{V_{DD}}{2}$.

$$V_x\Big|_{V_M=\frac{V_{DD}}{2}} = \frac{\left(\frac{V_{DD}}{2} - V_t\right)^2}{V_{DD}}$$

Substitute $1.8\ \text{V}$ for V_{DD} and $0.5\ \text{V}$ for V_t in the equation.

$$\begin{aligned} V_x\Big|_{V_M=\frac{V_{DD}}{2}} &= \frac{\left(\frac{1.8}{2} - 0.5\right)^2}{1.8} \\ &= \frac{0.16}{1.8} \\ &= 0.0889\ \text{V} \end{aligned}$$

Step 2 of 6

Determine the new value of width-to-length ratio.

$$k_n R_D = \frac{1}{V_x}$$

$$k_n'\left(\frac{W}{L}\right)R_D = \frac{1}{V_x} \quad \left(\text{since } k_n = k_n'\left(\frac{W}{L}\right)\right)$$

$$\frac{W}{L} = \frac{1}{k_n' R_D V_x}$$

Substitute $300\ \mu\text{A/V}^2$ for k_n', $0.0889\ \text{V}$ for V_x and $10\ \text{k}\Omega$ for R_D in the equation.

$$\begin{aligned} \frac{W}{L} &= \frac{1}{\left(300\times10^{-6}\right)\left(10\times10^{3}\right)\left(0.0889\right)} \\ &\approx 3.75 \end{aligned}$$

Thus, the new value of width-to-length ratio, $\frac{W}{L}$ is $\boxed{3.75}$.

Step 3 of 6

The expression for maximum value of input interpreted by the inverter as a logic-0 is,

$$V_{IL} = V_t + V_x$$

The expression for minimum value of input interpreted by the inverter as a logic-1 is,

$$V_{IH} = V_t + \left(1.63\sqrt{V_{DD}V_x}\right) - V_x$$

The expression for high-level output voltage is,

$$V_{OH} = V_{DD}$$

The expression for low-level output voltage is,

$$V_{OL} = \frac{V_{DD}}{1 + \left(\frac{V_{DD} - V_t}{V_x}\right)}$$

The expression for low-input noise margin is,

$$NM_L = V_{IL} - V_{OL}$$

The expression for high-input noise margin is,

$$NM_H = V_{OH} - V_{IH}$$

From the expressions, the noise margins depends on the voltage, V_x and independent of resistance, R_D and width-to-length ratio, $\left(\frac{W}{L}\right)$.

Thus, the noise margins are unchanged while reducing the value of resistance, R_D and increasing the value of width-to-length ratio, $\left(\frac{W}{L}\right)$.

Step 4 of 6

Determine the value of current drawn from the supply.

$$I_{DD} = \frac{V_{DD} - V_{OL}}{R_D}$$

Substitute $1.8\ \text{V}$ for V_{DD}, $0.12\ \text{V}$ for V_{OL} and $10\ \text{k}\Omega$ for R_D in the equation.

$$\begin{aligned} I_{DD} &= \frac{1.8 - 0.12}{10\times10^{3}} \\ &= 168\times10^{-6} \\ &= 168\ \mu\text{A} \end{aligned}$$

Step 5 of 6

Determine the value of power drawn from the supply during the low-output interval.

$$P_D = V_{DD} I_{DD}$$

Substitute $1.8\ \text{V}$ for V_{DD} and $168\ \mu\text{A}$ for I_{DD} in the equation.

$$\begin{aligned} P_D &= (1.8)\left(168\times10^{-6}\right) \\ &= 302.4\times10^{-6} \\ &= 302.4\ \mu\text{W} \end{aligned}$$

Step 6 of 6

Determine the value of average dissipation power.

$$P_{D\text{average}} = \frac{P_D}{2}$$

Substitute $302.4\ \mu\text{W}$ for P_D in the equation.

$$\begin{aligned} P_{D\text{average}} &= \frac{302.4\times10^{-6}}{2} \\ &= 151.2\times10^{-6} \\ &\approx 151\ \mu\text{W} \end{aligned}$$

Thus, the value of average dissipation power, $P_{D\text{average}}$ is $\boxed{151\ \mu\text{W}}$.

14.003P

We don't have the solution to this problem yet.

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Step 1 of 9

Refer to Figure 13.10 in the text book.

Determine the new value of voltage, V_x.

$$k_n R_D = \frac{1}{V_x}$$

$$V_x = \frac{1}{k_n R_D}$$

$$V_x = \frac{1}{k_n'\left(\frac{W}{L}\right)R_D} \quad \left(\text{since } k_n = k_n'\left(\frac{W}{L}\right)\right)$$

Substitute $300\ \mu A/V^2$ for k_n', 1.5 for $\left(\frac{W}{L}\right)$ and $10\ k\Omega$ for R_D in the equation.

$$V_x = \frac{1}{(300\times10^{-6})(1.5)(10\times10^{3})}$$
$$= 0.22\ V$$

Thus, the new value of voltage, V_x is $\boxed{0.22\ V}$.

Step 2 of 9

Determine the maximum value of input interpreted by the inverter as a logic-0.

$$V_{IL} = V_t + V_x$$

Substitute $0.5\ V$ for V_t and $0.22\ V$ for V_x in the equation.

$$V_{IL} = 0.5 + 0.22$$
$$= 0.72\ V$$

Determine the value of high-level output voltage.

$$V_{OH} = V_{DD}$$

Substitute $1.8\ V$ for V_{DD} in the equation.

$$V_{OH} = 1.8\ V$$

Step 3 of 9

Determine the value of low-level output voltage.

$$V_{OL} = \frac{V_{DD}}{1 + \left(\frac{V_{DD} - V_t}{V_x}\right)}$$

Substitute $1.8\ V$ for V_{DD}, $0.5\ V$ for V_t and $0.22\ V$ for V_x in the equation.

$$V_{OL} = \frac{1.8}{1 + \left(\frac{1.8 - 0.5}{0.22}\right)}$$
$$= \frac{1.8}{6.91}$$
$$= 0.26\ V$$

Step 4 of 9

Determine the minimum value of input interpreted by the inverter as a logic-1.

$$V_{IH} = V_t + \left(1.63\sqrt{V_{DD}V_x}\right) - V_x$$

Substitute $1.8\ V$ for V_{DD}, $0.5\ V$ for V_t and $0.22\ V$ for V_x in the equation.

$$V_{IH} = 0.5 + \left(1.63\sqrt{(1.8)(0.22)}\right) - 0.22$$
$$= 1.306\ V$$

Step 5 of 9

Determine the value of low-input noise margin.

$$NM_L = V_{IL} - V_{OL}$$

Substitute $0.72\ V$ for V_{IL} and $0.26\ V$ for V_{OL} in the equation.

$$NM_L = 0.72 - 0.26$$
$$= 0.46\ V$$

Thus, the value of low-input noise margin, NM_L is $\boxed{0.46\ V}$.

Step 6 of 9

Determine the value of high-input noise margin.

$$NM_H = V_{OH} - V_{IH}$$

Substitute $1.8\ V$ for V_{OH} and $1.306\ V$ for V_{IH} in the equation.

$$NM_H = 1.8 - 1.306$$
$$= 0.494$$
$$\approx 0.49\ V$$

Thus, the value of high-input noise margin, NM_H is $\boxed{0.49\ V}$.

Step 7 of 9

Determine the value of current drawn from the supply.

$$I_{DD} = \frac{V_{DD} - V_{OL}}{R_D}$$

Substitute $1.8\ V$ for V_{DD}, $0.26\ V$ for V_{OL} and $10\ k\Omega$ for R_D in the equation.

$$I_{DD} = \frac{1.8 - 0.26}{10\times10^{3}}$$
$$= 154\times10^{-6}$$
$$= 154\ \mu A$$

Step 8 of 9

Determine the value of power drawn from the supply during the low-output interval.

$$P_D = V_{DD}I_{DD}$$

Substitute $1.8\ V$ for V_{DD} and $154\ \mu A$ for I_{DD} in the equation.

$$P_D = (1.8)(154\times10^{-6})$$
$$= 277.2\times10^{-6}$$
$$= 277.2\ \mu W$$

Step 9 of 9

Determine the value of average dissipation power.

$$P_{Daverage} = \frac{P_D}{2}$$

Substitute $277.2\ \mu W$ for P_D in the equation.

$$P_{Daverage} = \frac{277.2\times10^{-6}}{2}$$
$$= 138.6\times10^{-6}$$
$$\approx 139\ \mu W$$

Thus, the value of average dissipation power, $P_{Daverage}$ is $\boxed{139\ \mu W}$.

14.004P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

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14.005E

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

14.005P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

Step 1 of 13

(a)

Determine the value of switching parameter, r .

$$V_M = \frac{r\left(V_{DD} - |V_{tp}|\right) + V_{tn}}{r+1}$$

Substitute 0.6 V for V_M , 1.2 V for V_{DD} , 0.4 V for V_{tn} and -0.4 V for V_{tp} in the equation.

$$0.6 = \frac{r\left(1.2 - |-0.4|\right) + 0.4}{r+1}$$
$$0.6r + 0.6 = 0.8r + 0.4$$
$$0.2r = 0.2$$
$$r = 1$$

Step 2 of 13

Determine the value of width for NMOS transistor.

$$\left(\frac{W}{L}\right)_n = 1.0$$
$$W_n = 1.0L_n$$

Substitute $0.13\ \mu m$ for L_n in the equation.

$$W_n = (1.0)(0.13\ \mu m)$$
$$= 0.13\ \mu m$$

Step 3 of 13

Determine the value of width for PMOS transistor.

$$r = \sqrt{\frac{W_p \mu_p}{W_n \mu_n}}$$

Substitute 1 for r and 4 for $\frac{\mu_n}{\mu_p}$ in the equation.

$$1 = \sqrt{\frac{W_p}{W_n}\left(\frac{1}{4}\right)}$$
$$1 = \frac{W_p}{W_n}\left(\frac{1}{4}\right)$$
$$W_p = 4W_n$$

Substitute $0.13\ \mu m$ for W_n in the equation.

$$W_p = 4W_n$$
$$= 4\left(0.13 \times 10^{-6}\right)$$
$$= 0.52\ \mu m$$

Thus, the value of width for PMOS transistor, W_p is $\boxed{0.52\ \mu m}$.

Step 4 of 13

(b)

Determine the value of high level output voltage for matched case.

$$V_{OH} = V_{DD}$$

Substitute 1.2 V for V_{DD} in the equation.

$$V_{OH} = 1.2\ V$$

Thus, the value of high level output voltage, V_{OH} is $\boxed{1.2\ V}$.

Step 5 of 13

Determine the value of low level output voltage for matched case.

$$V_{OL} = 0.0\ V$$

Thus, the value of low level output voltage, V_{OL} is $\boxed{0.0\ V}$.

Step 6 of 13

Determine the minimum value of input interpreted by the inverter as a logic-1.

$$V_{IH} = \frac{1}{8}\left(5V_{DD} - 2V_t\right)$$

Substitute 1.2 V for V_{DD} and 0.4 V for V_t in the equation.

$$V_{IH} = \frac{1}{8}\left(5(1.2) - 2(0.4)\right)$$
$$= 0.65\ V$$

Thus, the minimum value of input interpreted by the inverter as a logic-1, V_{IH} is $\boxed{0.65\ V}$.

Step 7 of 13

Determine the maximum value of input interpreted by the inverter as a logic-0.

$$V_{IL} = \frac{1}{8}\left(3V_{DD} + 2V_t\right)$$

Substitute 1.2 V for V_{DD} and 0.4 V for V_t in the equation.

$$V_{IL} = \frac{1}{8}\left[3(1.2) + 2(0.4)\right]$$
$$= 0.55\ V$$

Thus, the maximum value of input interpreted by the inverter as a logic-0, V_{IL} is $\boxed{0.55\ V}$.

Step 8 of 13

Determine the value of high-input noise margin.

$$NM_H = V_{OH} - V_{IH}$$

Substitute 1.2 V for V_{OH} and 0.65 V for V_{IH} in the equation.

$$NM_H = 1.2 - 0.65$$
$$= 0.55\ V$$

Thus, the value of high-input noise margin, NM_H is $\boxed{0.55\ V}$.

Step 9 of 13

Determine the value of low-input noise margin.

$$NM_L = V_{IL} - V_{OL}$$

Substitute 0.55 V for V_{IL} and 0.0 V for V_{OL} in the equation.

$$NM_L = 0.55 - 0.0$$
$$= 0.55\ V$$

Thus, the value of low-input noise margin, NM_L is $\boxed{0.55\ V}$.

Step 10 of 13

(c)

Determine the output resistance of the inverter in the low output state.

$$r_{DSN} = \frac{1}{\mu_n C_{ox}\left(\frac{W}{L}\right)_n \left(V_{DD} - V_{tn}\right)}$$

Substitute $430\ \mu A/V^2$ for $\mu_n C_{ox}$, 1.0 for $\left(\frac{W}{L}\right)_n$, 1.2 V for V_{DD} , 0.4 V for V_{tn} in the equation.

$$r_{DSN} = \frac{1}{\left(430 \times 10^{-6}\right)(1)(1.2 - 0.4)}$$
$$= 2.9 \times 10^3$$
$$= 2.9\ k\Omega$$

Thus, the output resistance of the inverter in the low output state, r_{DSN} is $\boxed{2.9\ k\Omega}$.

Step 11 of 13

Determine the output resistance of the inverter in the high output state.

$$r_{DSP} = \frac{1}{\mu_p C_{ox}\left(\frac{W}{L}\right)_p \left(V_{DD} - |V_{tp}|\right)}$$

$$r_{DSP} = \frac{1}{\left(\frac{\mu_n C_{ox}}{4}\right)\left(4 \times \left(\frac{W}{L}\right)_n\right)\left(V_{DD} - |V_{tp}|\right)} \quad \begin{pmatrix} \text{since } W_p = 4W_n \text{ and} \\ \mu_n = 4\mu_p \end{pmatrix}$$

Substitute $430\ \mu A/V^2$ for $\mu_n C_{ox}$, 1.0 for $\left(\frac{W}{L}\right)_n$, 1.2 V for V_{DD} , -0.4 V for V_{tp} in the equation.

$$r_{DSP} = \frac{1}{\left(\frac{430 \times 10^{-6}}{4}\right)(4 \times 1)\left(1.2 - |-0.4|\right)}$$
$$= 2.9 \times 10^3$$
$$= 2.9\ k\Omega$$

Thus, the output resistance of the inverter in the high output state, r_{DSP} is $\boxed{2.9\ k\Omega}$.

Step 12 of 13

(d)

Determine the value of switching parameter, r .

$$r = \sqrt{\frac{W_p \mu_p}{W_n \mu_n}}$$
$$= \sqrt{(1.0)\frac{\mu_p}{\mu_n}} \quad \left(\text{since } \left(\frac{W}{L}\right)_p = \left(\frac{W}{L}\right)_n = 1.0\right)$$
$$= \sqrt{\frac{\mu_p}{\mu_n}}$$

Substitute 4 for μ_n / μ_p in the equation.

$$r = \sqrt{\frac{1}{4}}$$
$$= 0.5$$

Step 13 of 13

Determine the value of switching voltage.

$$V_M = \frac{r\left(V_{DD} - |V_{tp}|\right) + V_{tn}}{r+1}$$

Substitute 0.5 for r , 1.2 V for V_{DD} , 0.4 V for V_{tn} and -0.4 V for V_{tp} in the equation.

$$V_M = \frac{(0.5)\left(1.2 - |-0.4|\right) + 0.4}{0.5+1}$$
$$= 0.53\ V$$

Thus, the value of switching voltage, V_M is $\boxed{0.53\ V}$.

14.006P

We don't have the solution to this problem yet.

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Step 1 of 3

Determine the value of width-to-length ratio of NMOS transistor.

$$i_{DN} = k_n'\left(\frac{W}{L}\right)_n\left[(v_I - V_{tn})v_O - \frac{1}{2}v_O^2\right]$$

$$i_{DN} = \mu_n C_{ox}\left(\frac{W}{L}\right)_n\left[(V_{DD} - V_{tn})v_O - \frac{1}{2}v_O^2\right] \qquad (\text{since } v_I = V_{DD})$$

Substitute $50\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$, $1\ \text{V}$ for V_{tn}, $0.2\ \text{mA}$ for i_{DN}, $0.2\ \text{V}$ for v_O and $5\ \text{V}$ for V_{DD} in the equation.

$$(0.2\times10^{-3}) = (50\times10^{-6})\left(\frac{W}{L}\right)_n\left[(5-1)(0.2) - \frac{1}{2}(0.2^2)\right]$$

$$\left(\frac{W}{L}\right)_n = \frac{0.2\times10^{-3}}{(50\times10^{-6})\left[(5-1)(0.2) - \frac{1}{2}(0.2^2)\right]}$$

$$= 5.12$$

$$= 5$$

Thus, the value of width-to-length ratio of NMOS transistor, $\left(\frac{W}{L}\right)_n$ is $\boxed{5}$.

Step 2 of 3

Determine the value of switching parameter, r.

$$V_M = \frac{r\left(V_{DD} - |V_{tp}|\right) + V_{tn}}{r+1}$$

Substitute $-1\ \text{V}$ for V_{tp}, $1\ \text{V}$ for V_{tn}, $2.5\ \text{V}$ for V_M and $5\ \text{V}$ for V_{DD} in the equation.

$$2.5 = \frac{r\left(5 - |-1|\right) + 1}{r+1}$$

$$2.5(r+1) = 4r + 1$$

$$1.5r = 1.5$$

$$r = 1$$

Step 3 of 3

Determine the value of width-to-length ratio of PMOS transistor.

$$r = \sqrt{\frac{W_p\mu_p}{W_n\mu_n}}$$

$$= \sqrt{\frac{\frac{W_p}{L}\mu_p}{\frac{W_n}{L}\mu_n}}$$

$$r = \sqrt{\frac{\left(\frac{W}{L}\right)_p\left(\frac{\mu_p}{\mu_n}\right)}{\left(\frac{W}{L}\right)_n}}$$

Substitute 1 for r and 2 for $\frac{\mu_p}{\mu_n}$ in the equation.

$$1 = \sqrt{\frac{\left(\frac{W}{L}\right)_p\left(\frac{1}{2}\right)}{\left(\frac{W}{L}\right)_n}}$$

$$1 = \frac{\left(\frac{W}{L}\right)_p\left(\frac{1}{2}\right)}{\left(\frac{W}{L}\right)_n}$$

$$\left(\frac{W}{L}\right)_p = 2\left(\frac{W}{L}\right)_n$$

Substitute 5 for $\left(\frac{W}{L}\right)_n$ in the equation.

$$\left(\frac{W}{L}\right)_p = 2(5)$$

$$= 10$$

Thus, the value of width-to-length ratio of PMOS transistor, $\left(\frac{W}{L}\right)_p$ is $\boxed{10}$.

14.007P

We don't have the solution to this problem yet.

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Step 1 of 2

14.008E

Determine the time t_{PLH} at which the capacitor voltage reaches $V_{DD}/2$.

Write the fundamental relationship in analyzing the dynamic operation of a circuit.

$$I\Delta t = C\Delta V$$

$$It_{PLH} = C\left(\frac{V_{DD}}{2} - 0\right)$$

$$It_{PLH} = C\frac{V_{DD}}{2}$$

$$t_{PLH} = \frac{CV_{DD}}{2I}$$

Thus, the low-to-high propagation delay at which the capacitor voltage reaches $V_{DD}/2$ is,

$$\boxed{t_{PLH} = \frac{CV_{DD}}{2I}}.$$

Step 2 of 2

Determine the value of current, I .

$$t_{PLH} = \frac{CV_{DD}}{2I}$$

$$I = \frac{CV_{DD}}{2t_{PLH}}$$

Substitute $10\ \text{ps}$ for t_{PLH} , $10\ \text{fF}$ for C and $1.8\ \text{V}$ for V_{DD} in the equation.

$$I = \frac{\left(10\times10^{-15}\right)(1.8)}{2\left(10\times10^{-12}\right)}$$

$$= 0.9\times10^{-3}$$

$$= 0.9\ \text{mA}$$

Thus, the value of current, I is $\boxed{0.9\ \text{mA}}$.

Step 1 of 4

Consider the following logic function:

$$Y = \overline{A + B(C + D)}$$

The CMOS logic gate is combination of pull up network (PUN) and pull down network (PDN). The PUN comprises PMOS transistors, and the PDN comprises NMOS transistors.

The PMOS transistor conducts when the input signal at its gate is low, and the NMOS transistor conducts when the input signal at its gate is high.

In the CMOS gate circuit, the PDN and the PUN are dual networks where series branch exist in one, a parallel branch exists in other.

Step 2 of 4

Draw the pull up network for the logic function $Y = \overline{A + B(C + D)}$, which comprises PMOS transistors only.

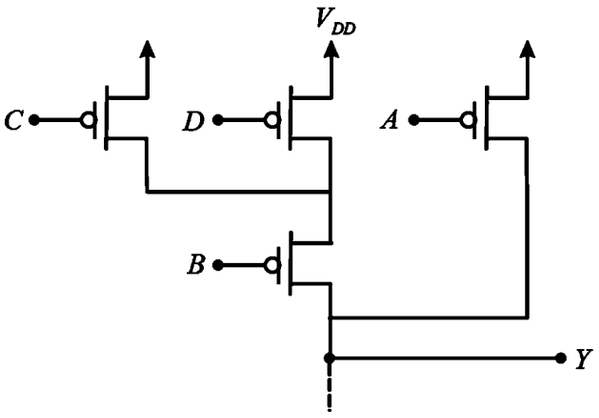


Figure 1

Step 3 of 4

Draw the pull down network for the logic function $Y = \overline{A + B(C + D)}$, which comprises NMOS transistors only.

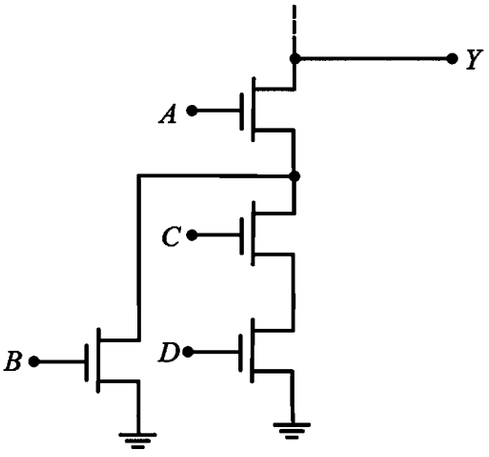


Figure 2

Step 4 of 4

Sketch the CMOS realization for the logic function $Y = \overline{A + B(C + D)}$.

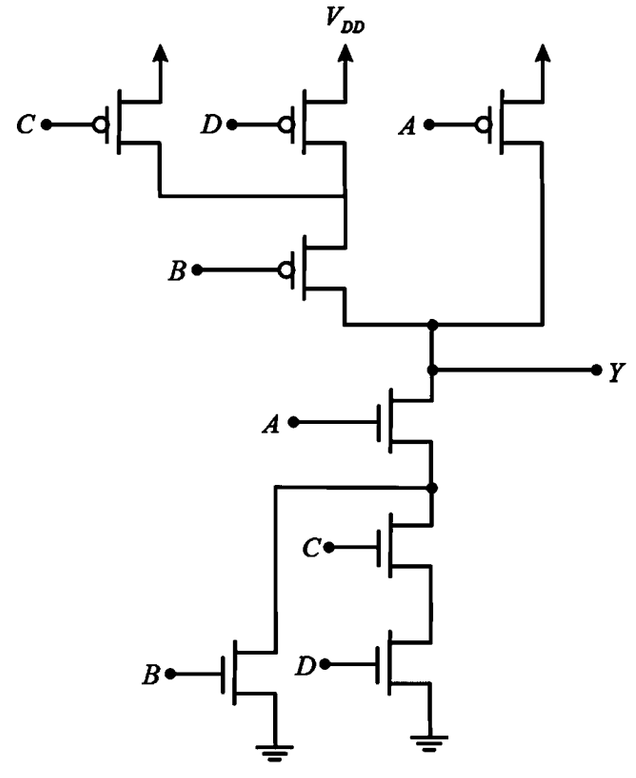


Figure 3

Thus, the CMOS realization for the logic function $Y = \overline{A + B(C + D)}$ is sketched as shown in Figure 3.

Step 1 of 3

14.009E

Refer to Figure 13.8 (a) in the text book.

Determine the value of propagation delay for the output going from low to high.

$$t_{PLH} = 0.69 R_{PU} C$$

Substitute $20\ \text{k}\Omega$ for R_{PU} and $10\ \text{fF}$ for C in the equation.

$$\begin{aligned} t_{PLH} &= 0.69\left(20\times10^{3}\right)\left(10\times10^{-15}\right) \\ &= 138\times10^{-12} \\ &= 138\ \text{ps} \end{aligned}$$

Thus, the value of low-to-high propagation delay, t_{PLH} is $\boxed{138\ \text{ps}}$.

Step 2 of 3

Determine the value of propagation delay for the output going from high to low.

$$t_{PHL} = 0.69 R_{PD} C$$

Substitute $10\ \text{k}\Omega$ for R_{PD} and $10\ \text{fF}$ for C in the equation.

$$\begin{aligned} t_{PHL} &= 0.69\left(10\times10^{3}\right)\left(10\times10^{-15}\right) \\ &= 69\times10^{-12} \\ &= 69\ \text{ps} \end{aligned}$$

Thus, the value of high-to-low propagation delay, t_{PHL} is $\boxed{69\ \text{ps}}$.

Step 3 of 3

Determine the value of propagation delay.

$$t_P = \frac{1}{2}\left(t_{PHL} + t_{PLH}\right)$$

Substitute $138\ \text{ps}$ for t_{PLH} and $69\ \text{ps}$ for t_{PHL} in the equation.

$$\begin{aligned} t_P &= \frac{\left(69\times10^{-12}\right)+\left(138\times10^{-12}\right)}{2} \\ &= 103.5\times10^{-12} \\ &= 104\ \text{ps} \end{aligned}$$

Thus, the value of propagation delay, t_P is $\boxed{104\ \text{ps}}$.

14.009P

We don't have the solution to this problem yet.

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Determine the fall time of the capacitor voltage.

$$t_f = 2.2RC$$

Substitute $2\ \text{k}\Omega$ for R and $100\ \text{fF}$ for C in the equation.

$$\begin{aligned} t_f &= (2.2)\left(2\times10^{3}\right)\left(100\times10^{-15}\right) \\ &= 4.4\times10^{-10} \\ &= 0.44\ \text{ns} \end{aligned}$$

Thus, the fall time of the capacitor voltage, t_f is $\boxed{0.44\ \text{ns}}$.

14.010P

We don't have the solution to this problem yet.

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Step 1 of 5

Determine the value of factor, α_n .

$$\alpha_n = \frac{2}{\left[\frac{7}{4} - \frac{3V_{tn}}{V_{DD}} + \left(\frac{V_{tn}}{V_{DD}}\right)^2\right]}$$

Substitute $0.5\ \text{V}$ for V_{tn} and $1.8\ \text{V}$ for V_{DD} in the equation.

$$\begin{aligned}\alpha_n &= \frac{2}{\left[\frac{7}{4} - \frac{3(0.5)}{1.8} + \left(\frac{0.5}{1.8}\right)^2\right]} \\ &= \frac{2}{1.75 - 0.8333 + 0.0772} \\ &= 2.01 \\ &\approx 2\end{aligned}$$

Step 2 of 5

Determine the value of low-to-high propagation delay.

$$t_{PLH} = \frac{\alpha_n C}{k_n' \left(\frac{W}{L}\right)_n V_{DD}}$$

Substitute $300\ \mu\text{A/V}^2$ for k_n', 1.5 for $\left(\frac{W}{L}\right)_n$, $1.8\ \text{V}$ for V_{DD}, $10\ \text{fF}$ for C and 2 for α_n in the equation.

$$\begin{aligned}t_{PLH} &= \frac{2\left(10\times10^{-15}\right)}{\left(300\times10^{-6}\right)(1.5)(1.8)} \\ &= 24.69\times10^{-12} \\ &\approx 24.7\ \text{ps}\end{aligned}$$

Thus, the value of low-to-high propagation delay, t_{PLH} is $\boxed{24.7\ \text{ps}}$.

Step 3 of 5

Determine the value of factor, α_p .

$$\alpha_p = \frac{2}{\left[\frac{7}{4} - \frac{3|V_{tp}|}{V_{DD}} + \left|\frac{V_{tp}}{V_{DD}}\right|^2\right]}$$

Substitute $-0.5\ \text{V}$ for V_{tp} and $1.8\ \text{V}$ for V_{DD} in the equation.

$$\begin{aligned}\alpha_p &= \frac{2}{\left[\frac{7}{4} - \frac{3|-0.5|}{1.8} + \left|\frac{-0.5}{1.8}\right|^2\right]} \\ &= \frac{2}{1.75 - 0.8333 + 0.0772} \\ &= 2.01 \\ &\approx 2\end{aligned}$$

Step 4 of 5

Determine the value of high-to-low propagation delay.

$$\begin{aligned}t_{PHL} &= \frac{\alpha_p C}{k_p' \left(\frac{W}{L}\right)_p V_{DD}} \\ &= \frac{\alpha_p C}{\frac{k_n'}{4}\left(\frac{W}{L}\right)_p V_{DD}} \quad \left(\text{since } k_n' = 4k_p'\right)\end{aligned}$$

Substitute $300\ \mu\text{A/V}^2$ for k_n', 3 for $\left(\frac{W}{L}\right)_p$, $1.8\ \text{V}$ for V_{DD}, $10\ \text{fF}$ for C and 2 for α_p in the equation.

$$\begin{aligned}t_{PHL} &= \frac{2\left(10\times10^{-15}\right)}{\left(\frac{300\times10^{-6}}{4}\right)(3)(1.8)} \\ &= 49.38\times10^{-12} \\ &\approx 49.4\ \text{ps}\end{aligned}$$

Thus, the value of high-to-low propagation delay, t_{PHL} is $\boxed{49.4\ \text{ps}}$.

Step 5 of 5

Determine the value of propagation delay.

$$t_p = \frac{t_{PLH} + t_{PHL}}{2}$$

Substitute $24.7\ \text{ps}$ for t_{PLH} and $49.4\ \text{ps}$ for t_{PHL} in the equation.

$$\begin{aligned}t_p &= \frac{\left(24.7\times10^{-12}\right) + \left(49.4\times10^{-12}\right)}{2} \\ &= 37.05\times10^{-12} \\ &\approx 37\ \text{ps}\end{aligned}$$

Thus, the value of propagation delay, t_p is $\boxed{37\ \text{ps}}$.

14.011P

We don't have the solution to this problem yet.

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Step 1 of 2

14.012E

Consider a CMOS inverter fabricated in $0.13\ \mu m$ process.

Determine the value of width to length ratio of NMOS device.

The empirical resistance value is,

$$R_N = \frac{12.5}{\left(\frac{W}{L}\right)_n}\ \text{k}\Omega$$

Consider that the propagation delay for the output going from high to low, t_{PHL} is 50 ps.

The general expression for t_{PHL} is,

$$\begin{aligned} t_{PHL} &= 0.69 R_N C \\ &= 0.69 \frac{12500}{\left(\frac{W}{L}\right)_n} C \end{aligned}$$

Simplify the expression for $\left(\frac{W}{L}\right)_n$.

$$\begin{aligned} \left(\frac{W}{L}\right)_n &= 0.69 \frac{12500}{t_{PHL}} C \\ &= 8625\left(\frac{C}{t_{PHL}}\right) \end{aligned}$$

Consider that the effective load capacitance, C is 20 fF.

Substitute 50 ps for t_{PHL}, and 20 fF for C.

$$\begin{aligned} \left(\frac{W}{L}\right)_n &= 8625\left(\frac{20\ \text{fF}}{50\ \text{ps}}\right) \\ &= 8625\left(0.4\times10^{-3}\right) \\ &= 3.45 \end{aligned}$$

Thus, the width to length ratio of NMOS device, $\left(\frac{W}{L}\right)_n$ is, $\boxed{3.45}$.

Step 2 of 2

Determine the value of width to length ratio of PMOS device.

The empirical resistance value is,

$$R_P = \frac{30}{\left(\frac{W}{L}\right)_p}\ \text{k}\Omega$$

Consider that the propagation delay for the output going from low to high, t_{PLH} is 50 ps.

The general expression for t_{PLH} is,

$$\begin{aligned} t_{PLH} &= 0.69 R_P C \\ &= 0.69 \frac{30000}{\left(\frac{W}{L}\right)_p} C \end{aligned}$$

Simplify the expression for $\left(\frac{W}{L}\right)_p$.

$$\begin{aligned} \left(\frac{W}{L}\right)_p &= 0.69 \frac{30000}{t_{PLH}} C \\ &= 20700\left(\frac{C}{t_{PLH}}\right) \end{aligned}$$

Substitute 50 ps for t_{PLH}, and 20 fF for C.

$$\begin{aligned} \left(\frac{W}{L}\right)_p &= 20700\left(\frac{20\ \text{fF}}{50\ \text{ps}}\right) \\ &= 20700\left(0.4\times10^{-3}\right) \\ &= 8.28 \end{aligned}$$

Thus, the width to length ratio of PMOS device, $\left(\frac{W}{L}\right)_p$ is, $\boxed{8.28}$.

Step 1 of 4

Consider the following logic function:

$$Y = ABC + \overline{A}\,\overline{B}\,\overline{C}$$

$$= \overline{\overline{ABC + \overline{A}\,\overline{B}\,\overline{C}}} \qquad \left(\text{since } M = \overline{\overline{M}}\right)$$

$$= \overline{\left(\overline{ABC}\right)\left(\overline{\overline{A}\,\overline{B}\,\overline{C}}\right)} \qquad \left(\text{since } \overline{M+N} = \overline{M}.\overline{N}\right)$$

$$= \overline{\left(\overline{A}+\overline{B}+\overline{C}\right)\left(A+B+C\right)} \qquad \left(\text{since } \overline{MNO} = \overline{M}+\overline{N}+\overline{O}\right)$$

The CMOS logic gate is combination of pull up network (PUN) and pull down network (PDN). The PUN comprises PMOS transistors, and the PDN comprises NMOS transistors.

The PMOS transistor conducts when the input signal at its gate is low, and the NMOS transistor conducts when the input signal at its gate is high.

In the CMOS gate circuit, the PDN and the PUN are dual networks where series branch exist in one, a parallel branch exists in other.

Step 2 of 4

Draw the pull up network for the logic function $Y = \overline{\left(\overline{A}+\overline{B}+\overline{C}\right)\left(A+B+C\right)}$, which comprises PMOS transistors only.

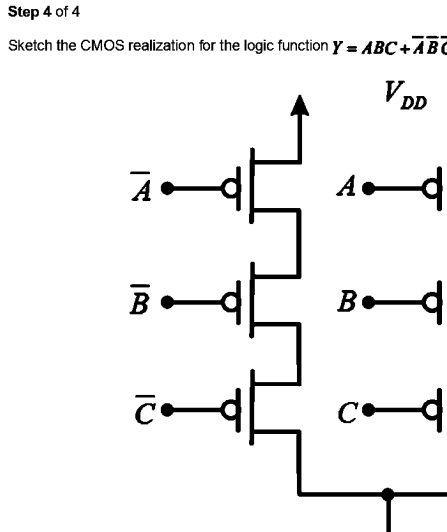


Figure 1

Step 3 of 4

Draw the pull down network for the logic function $Y = \overline{\left(\overline{A}+\overline{B}+\overline{C}\right)\left(A+B+C\right)}$, which comprises NMOS transistors only.

Figure 2

Step 4 of 4

Sketch the CMOS realization for the logic function $Y = ABC + \overline{A}\,\overline{B}\,\overline{C}$.

Figure 3

Thus, the CMOS realization for the logic function $Y = ABC + \overline{A}\,\overline{B}\,\overline{C}$ is sketched as shown in Figure 3.

Step 1 of 3

Consider a CMOS inverter.

Calculate the equivalent capacitance, C .

$$C = 2C_{gd1} + 2C_{gd2} + C_{db1} + C_{db2} + C_{g3} + C_{g4} + C_w$$

The gate-source and the gate-drain overlap capacitances are $0.3\ \frac{\text{fF}}{\mu\text{m}}$ of the gate width.

The gate-drain capacitance of Q_1 is,

$$\begin{aligned} C_{gd1} &= \left(0.3\ \frac{\text{fF}}{\mu\text{m}}\right)(W_n) \\ &= (0.3)(0.375) \\ &= 0.1125\ \text{fF} \end{aligned}$$

The gate-drain capacitance of Q_2 is,

$$\begin{aligned} C_{gd2} &= \left(0.3\ \frac{\text{fF}}{\mu\text{m}}\right)(W_p) \\ &= (0.3)(1.125) \\ &= 0.3375\ \text{fF} \end{aligned}$$

The capacitance per square meter, $C_{ox} = 6\ \text{fF/m}^2$.

The input capacitance of the load inverter is,

$$\begin{aligned} C_{g3} + C_{g4} &= (WL)_3 C_{ox} + C_{gsov3} + C_{gdov3} + (WL)_4 C_{ox} + C_{gsov4} + C_{gdov4} \\ &= (0.375)(0.25)(6) + 2(0.3)(0.375) + (1.125)(0.25)(6) + (2)(0.3)(1.125) \\ &= 0.7875\ \text{fF} + 2.3625\ \text{fF} \\ &= 3.15\ \text{fF} \end{aligned}$$

The wiring capacitance, $C_w = 0.2\ \text{fF}$.

The drain body capacitances are,

$$\begin{aligned} C_{dbn} &= C_{dbp} \\ &= 1\ \text{fF} \end{aligned}$$

Therefore, the initial equivalent capacitance is,

$$\begin{aligned} C &= 2C_{gd1} + 2C_{gd2} + C_{db1} + C_{db2} + C_{g3} + C_{g4} + C_w \\ &= 2(0.1125\ \text{fF}) + 2(0.3375\ \text{fF}) + 1\ \text{fF} + 1\ \text{fF} + 3.15\ \text{fF} + 0.2\ \text{fF} \\ &= 6.25\ \text{fF} \end{aligned}$$

Consider that an additional load capacitance of 0.1 pF is added to this capacitance.

The new value of the capacitance is,

$$\begin{aligned} C_{new} &= C + 0.1\ \text{pF} \\ &= 6.25\ \text{fF} + 0.1\ \text{pF} \\ &= 6.25\ \text{fF} + 100\ \text{fF} \\ &= 106.25\ \text{fF} \end{aligned}$$

Step 2 of 3

Calculate the propagation delay from high-to-low transition, t_{PHL} .

$$t_{PHL} = \frac{\alpha_n C}{k_n' \left(\frac{W}{L}\right)_n V_{DD}}$$

Here,

$$\alpha_n = \frac{2}{\frac{7}{4} - \frac{3V_{tn}}{V_{DD}} + \left(\frac{V_{tn}}{V_{DD}}\right)^2}$$

Here, V_{tn} is the threshold voltage, and V_{DD} is the supply voltage. Therefore,

$$\begin{aligned} \alpha_n &= \frac{2}{\frac{7}{4} - \frac{3(0.5)}{2.5} + \left(\frac{0.5}{2.5}\right)^2} \\ &= \frac{2}{1.19} \\ &\cong 1.7 \end{aligned}$$

Substitute 1.7 for α_n , 106.25 fF for C , $115\ \mu\text{A/V}^2$ for k_n' , 2.5 V for V_{DD} , and $\frac{0.375\ \mu\text{m}}{0.25\ \mu\text{m}}$ for $\left(\frac{W}{L}\right)_n$ in the expression for t_{PHL} .

$$\begin{aligned} t_{PHL} &= \frac{(1.7)(106.25\ \text{fF})}{(115\ \mu\text{A/V}^2)\left(\frac{0.375\ \mu\text{m}}{0.25\ \mu\text{m}}\right)(2.5\ \text{V})} \\ &= \frac{180.625 \times 10^{-15}}{431.25 \times 10^{-6}} \\ &= 418.84 \times 10^{-12} \\ &= 418.84\ \text{ps} \end{aligned}$$

Step 3 of 3

Calculate the propagation delay from low-to-high transition, t_{PLH} .

$$t_{PHL} = \frac{\alpha_p C}{k_p' \left(\frac{W}{L}\right)_p V_{DD}}$$

Here,

$$\alpha_n = \frac{2}{\frac{7}{4} - \frac{3V_{tp}}{V_{DD}} + \left(\frac{V_{tp}}{V_{DD}}\right)^2}$$

Since, the threshold voltages are same for PMOS and NMOS devices, $\alpha_n = \alpha_p$.

Substitute 1.7 for α_n , 106.25 fF for C , $30\ \mu\text{A/V}^2$ for k_p' , 2.5 V for V_{DD} , and $\frac{1.125\ \mu\text{m}}{0.25\ \mu\text{m}}$ for $\left(\frac{W}{L}\right)_p$ in the expression for t_{PLH} .

$$\begin{aligned} t_{PLH} &= \frac{(1.7)(106.25\ \text{fF})}{(30\ \mu\text{A/V}^2)\left(\frac{1.125\ \mu\text{m}}{0.25\ \mu\text{m}}\right)(2.5\ \text{V})} \\ &= \frac{180.625 \times 10^{-15}}{337.5 \times 10^{-6}} \\ &= 53.5 \times 10^{-12} \\ &= 53.5\ \text{ps} \end{aligned}$$

The average value of the propagation delay is,

$$\begin{aligned} t_p &= \frac{1}{2}(t_{PLH} + t_{PHL}) \\ &= \frac{1}{2}(418.84\ \text{ps} + 53.5\ \text{ps}) \\ &\cong 477\ \text{ps} \end{aligned}$$

Thus, the average value of the propagation delay, t_p is, $\boxed{477\ \text{ps}}$.

Step 1 of 5

(a)

The output is low when an even number of one's exists in inputs.

Draw the truth table for three-input even-parity checker.

Table 1

A	B	C	Y (even-parity generator)
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

From table 1, write the Boolean expression for $\overline{Y}$.

$$\overline{Y} = \overline{A}\overline{B}\overline{C} + \overline{A}BC + A\overline{B}C + AB\overline{C}$$

Thus, the Boolean expression for three-input even-parity checker is,

$$\boxed{\overline{Y} = \overline{A}\overline{B}\overline{C} + \overline{A}BC + A\overline{B}C + AB\overline{C}}$$

Step 2 of 5

(b)

The pull-down network can be most directly synthesized by expressing $\overline{Y}$ as a function of un-complemented variables. If any complemented variables are appear in the expression, additional inverters are required to generate the network.

The number of NMOS transistors required for pull-down network is,

$$3 \times 4 = 12 \text{ transistors}$$

Draw the pull-down network directly from the expression for $\overline{Y}$.

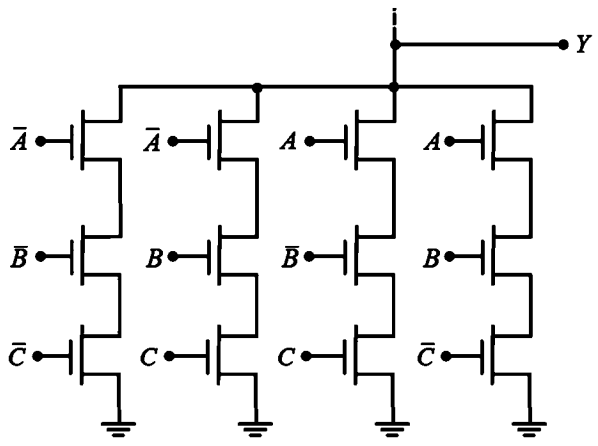


Figure 1

Step 3 of 5

(c)

The Boolean expression can be simplified as follows.

$$\begin{aligned}\overline{Y} &= \overline{A}\overline{B}\overline{C} + \overline{A}BC + A\overline{B}C + AB\overline{C} \\ &= \overline{A}\left(\overline{B}\overline{C} + BC\right) + A\left(\overline{B}C + B\overline{C}\right)\end{aligned}$$

The number of NMOS transistors required for pull-down network is,

$$1 + (2 \times 4) + 1 = 10 \text{ transistors}$$

Redraw the pull-down network directly from the expression for $\overline{Y}$.

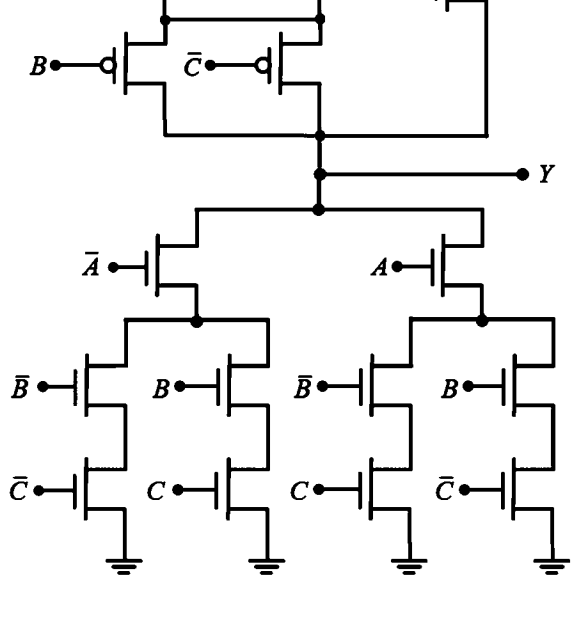


Figure 2

Step 4 of 5

(d)

The pull up network is dual of pull down network. The pull-up network can be most directly synthesized by expressing Y as a function of un-complemented variables and then applying the un-complemented variables to the gates of the PMOS transistors. If any complemented variables are appear in the expression, additional inverters are required to generate the network.

Draw the pull-up network for active-low output of even-parity checker.

Figure 3

Step 5 of 5

Sketch the complete CMOS realization for active-low output of even-parity checker.

Figure 4

Thus, the complete CMOS realization for active-low output of even-parity checker is sketched as shown in Figure 4.

Step 1 of 4

Consider a CMOS inverter.

The initial value of width to length ratio of PMOS device is,

$$\left(\frac{W}{L}\right)_{p1}=\frac{1.125\ \mu m}{0.25\ \mu m}$$

Consider that the width to length ratio of PMOS device is made equal to width to length ratio of NMOS device in order to reduce the area of the inverter. Therefore,

$$\left(\frac{W}{L}\right)_{p2}=\left(\frac{W}{L}\right)_{n}$$
$$=\frac{0.375\ \mu m}{0.25\ \mu m}$$

Observe that the width is changed from $1.125\ \mu m$ to $0.375\ \mu m$ in order to make the width to length ratio of PMOS device equal to width to length ratio of NMOS device.

Calculate the percentage reduction of width, W_p .

$$\%reduction=\left(\frac{0.375\ \mu m}{1.125\ \mu m}\right)(100)$$
$$=33.34\%$$

Therefore, the percentage reduction in the area is, $\boxed{33\%}$.

Step 2 of 4

Consider that the capacitance, C_{dbp} does not change.

The drain body capacitances are,

$$C_{dbn}=C_{dbp}$$
$$=1\ fF$$

Calculate the equivalent capacitance, C_{in} .

$$C=2C_{gd1}+2C_{gd2}+C_{db1}+C_{db2}+C_{g3}+C_{g4}+C_{w}$$

The gate-source and the gate-drain overlap capacitances are $0.3\ \frac{fF}{\mu m}$ of the gate width.

The gate-drain capacitance of Q_1 and Q_2 are,

$$C_{gd1}=C_{gd2}$$
$$=\left(0.3\ \frac{fF}{\mu m}\right)(W_n)\qquad (\text{since, } W_n=W_p)$$
$$=(0.3)(0.375)$$
$$=0.1125\ fF$$

The capacitance per square meter, $C_{ox}=6\ fF/m^2$.

The input capacitance of the load inverter is,

$$C_{g3}+C_{g4}=(WL)_3 C_{ox}+C_{gov3}+C_{gdov3}+(WL)_4 C_{ox}+C_{gov4}+C_{gdov4}$$
$$=(0.375)(0.25)(6)+2(0.3)(0.375)+(0.375)(0.25)(6)+(2)(0.3)(0.375)$$
$$=0.7875\ fF+0.7875\ fF$$
$$=1.575\ fF$$

The wiring capacitance, $C_w=0.2\ fF$.

Therefore, the initial equivalent capacitance is,

$$C=2C_{gd1}+2C_{gd2}+C_{db1}+C_{db2}+C_{g3}+C_{g4}+C_w$$
$$=2(0.1125\ fF)+2(0.1125\ fF)+1\ fF+1\ fF+1.575\ fF+0.2\ fF$$
$$=4.225\ fF$$

Thus, the new value of capacitance, C is, $\boxed{4.225\ fF}$.

Step 3 of 4

Calculate the propagation delay from high-to-low transition, t_{PHL} .

$$t_{PHL}=\frac{\alpha_n C}{k_n'\left(\frac{W}{L}\right)_n V_{DD}}$$

Here,

$$\alpha_n=\frac{2}{\frac{7}{4}-\frac{3V_{tn}}{V_{DD}}+\left(\frac{V_{tn}}{V_{DD}}\right)^2}$$

Here, V_{tn} is the threshold voltage, and V_{DD} is the supply voltage. Therefore,

$$\alpha_n=\frac{2}{\frac{7}{4}-\frac{3(0.5)}{2.5}+\left(\frac{0.5}{2.5}\right)^2}$$
$$=\frac{2}{1.19}$$
$$\cong 1.7$$

Substitute 1.7 for α_n , 4.225 fF for C , $115\ \mu A/V^2$ for k_n' , and $\frac{0.375\ \mu m}{0.25\ \mu m}$ for $\left(\frac{W}{L}\right)_n$ in the expression for t_{PHL} .

$$t_{PHL}=\frac{(1.7)(4.225\ fF)}{(115\ \mu A/V^2)\left(\frac{0.375\ \mu m}{0.25\ \mu m}\right)(2.5\ V)}$$
$$=\frac{7.1\times10^{-15}}{431.25\times10^{-6}}$$
$$=16.65\times10^{-12}$$
$$=16.65\ ps$$

Thus, the new value of the propagation delay from high-to-low transition, t_{PHL} is, $\boxed{16.65\ ps}$.

Step 4 of 4

Calculate the propagation delay from low-to-high transition, t_{PLH} .

$$t_{PHL}=\frac{\alpha_p C}{k_p'\left(\frac{W}{L}\right)_p V_{DD}}$$

Here,

$$\alpha_n=\frac{2}{\frac{7}{4}-\frac{3V_{tp}}{V_{DD}}+\left(\frac{V_{tp}}{V_{DD}}\right)^2}$$

Since, the threshold voltages are same for PMOS and NMOS devices, $\alpha_n=\alpha_p$.

Substitute 1.7 for α_n , 4.225 fF for C , $30\ \mu A/V^2$ for k_p' , and $\frac{0.375\ \mu m}{0.25\ \mu m}$ for $\left(\frac{W}{L}\right)_p$ in the expression for t_{PLH} .

$$t_{PLH}=\frac{(1.7)(4.225\ fF)}{(30\ \mu A/V^2)\left(\frac{0.375\ \mu m}{0.25\ \mu m}\right)(2.5\ V)}$$
$$=\frac{7.1\times10^{-15}}{112.5\times10^{-6}}$$
$$=63.12\times10^{-12}$$
$$=63.12\ ps$$

Thus, the new value of the propagation delay from low-to-high transition, t_{PLH} is, $\boxed{63.12\ ps}$.

The average value of the propagation delay is,

$$t_p=\frac{1}{2}(t_{PLH}+t_{PHL})$$
$$=\frac{1}{2}(63.12\ ps+16.6\ ps)$$
$$\cong 40\ ps$$

Thus, the average value of the propagation delay, t_p is, $\boxed{40\ ps}$.

Step 1 of 6

The output is high when an odd number of one's exists in inputs.

Draw the truth table for three-input odd-parity checker.

Table 1

A	B	C	Y (odd-parity generator)
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Step 2 of 6

From Table 1, write the Boolean expression for Y.

$$\begin{aligned} Y &= \bar{A}B\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC \\ &= \overline{\left(\bar{A}B\bar{C} + \bar{A}\bar{B}C + A\bar{B}\bar{C} + ABC\right)} \qquad \left(\text{since } M = \overline{\bar{M}}\right) \\ &= \overline{\left(\left(\bar{A}B\bar{C}\right)\left(\bar{A}\bar{B}C\right)\left(A\bar{B}\bar{C}\right)\left(ABC\right)\right)} \qquad \left(\text{since } \overline{M+N} = \bar{M}.\bar{N}\right) \\ &= \overline{\left(A+\bar{B}+C\right)\left(A+B+\bar{C}\right)\left(\bar{A}+\bar{B}+C\right)\left(\bar{A}+B+\bar{C}\right)} \quad \left(\text{since } \overline{MNO} = \bar{M}+\bar{N}+\bar{O}\right) \end{aligned}$$

Thus, the Boolean expression for three-input odd-parity checker is,

$$\boxed{Y = \overline{\left(A+B+\bar{C}\right)\left(A+\bar{B}+C\right)\left(\bar{A}+B+C\right)\left(\bar{A}+\bar{B}+\bar{C}\right)}}.$$

Step 3 of 6

Draw the pull-up network for active-high output of odd-parity checker.

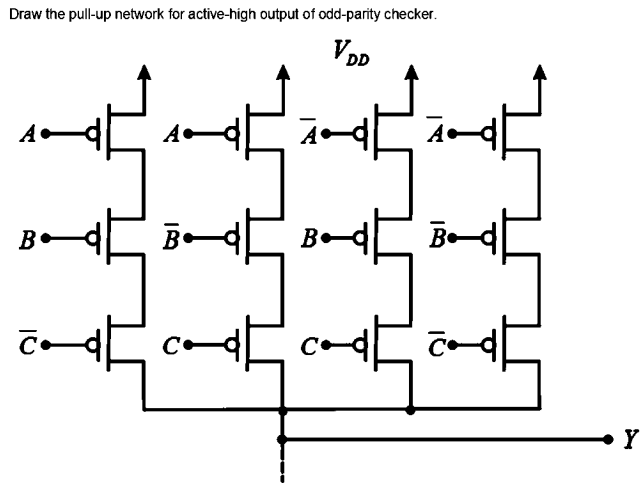


Figure 1

Step 4 of 6

The Boolean expression can be simplified as follows.

$$\begin{aligned} Y &= \overline{\left(A+B+\bar{C}\right)\left(A+\bar{B}+C\right)\left(\bar{A}+B+C\right)\left(\bar{A}+\bar{B}+\bar{C}\right)} \\ &= \overline{\left[A+\left(B+\bar{C}\right)\left(\bar{B}+C\right)\right]\left[\bar{A}+\left(B+C\right)\left(\bar{B}+\bar{C}\right)\right]} \end{aligned}$$

The number of PMOS transistors required for pull-up network is,

$$1+(2)(4)+1 = 10 \text{ transistors}$$

Redraw the pull-up network for active-high output of odd-parity checker.

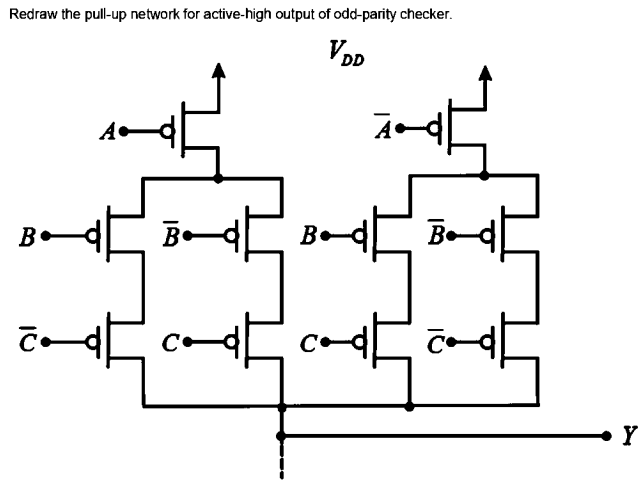


Figure 2

Step 5 of 6

The pull-down network is dual of pull-up network.
Draw the pull-down network for active-high output of odd-parity checker.

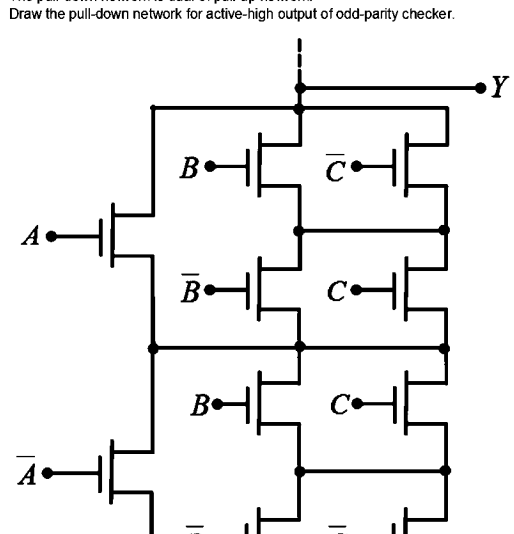


Figure 3

Step 6 of 6

Sketch the complete CMOS realization for active-high output of odd-parity checker.

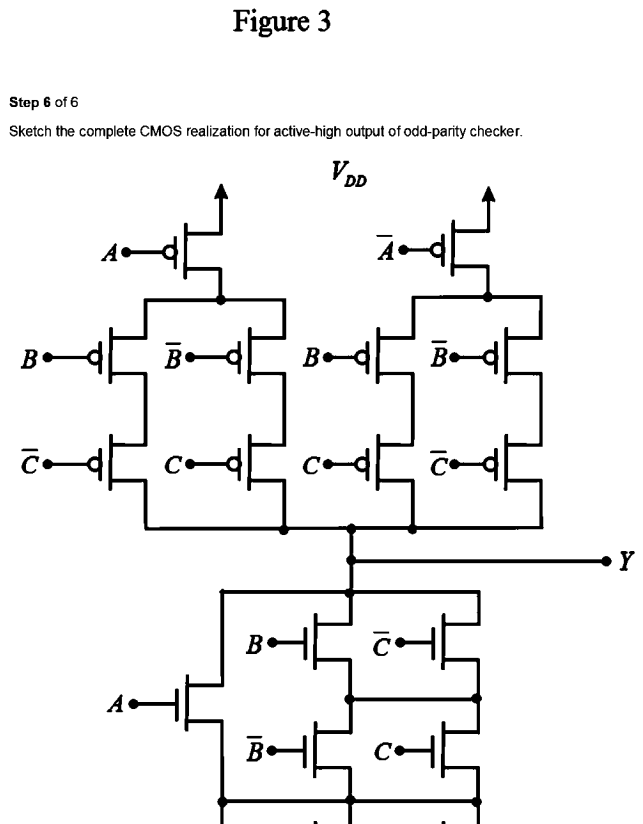


Figure 4

Thus, the complete CMOS realization for active-high output of odd-parity checker is sketched as shown in Figure 4.

14.015E

Consider a CMOS inverter with propagation delay, t_p as 28 ps.

The maximum frequency at which the inverter can be operated is,

$$f = \frac{1}{2t_p}$$

Substitute 28 ps for t_p.

$$\begin{aligned} f &= \frac{1}{2(28\text{ ps})} \\ &= \frac{1}{56\times10^{-12}} \\ &= 17.85\times10^{9} \\ &= 17.85\text{ GHz} \end{aligned}$$

Thus, the maximum operation, f for the inverter is, $\boxed{17.85\text{ GHz}}$.

Step 1 of 9

The CMOS full adder output sum is 1, if one or three inputs are 1 and the carry output is 1 , if two or more inputs are 1.

Draw the truth table for full adder circuit.

Table 1

A	B	C	S	C_0
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Step 2 of 9

From Table 1, write the Boolean expression for sum output.

$$S = \overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC$$
$$= \overline{\overline{A}\overline{B}C + \overline{A}B\overline{C} + A\overline{B}\overline{C} + ABC} \qquad (\text{since } M = \overline{\overline{M}})$$
$$= \overline{\left(\overline{A}\overline{B}C\right)\left(\overline{A}B\overline{C}\right)\left(A\overline{B}\overline{C}\right)\left(ABC\right)} \qquad (\text{since } \overline{M+N} = \overline{M}.\overline{N})$$
$$= \overline{\left(A+B+\overline{C}\right)\left(A+\overline{B}+C\right)\left(\overline{A}+B+C\right)\left(\overline{A}+\overline{B}+\overline{C}\right)} \quad (\text{since } \overline{MNO} = \overline{M}+\overline{N}+\overline{O})$$
$$= \overline{\left[\left(A+B\right)\left(\overline{A}+\overline{B}\right)+\overline{C}\right]\left[\left(A+\overline{B}\right)\left(\overline{A}+B\right)+C\right]}$$

Step 3 of 9

Draw the pull-up network for sum output of full adder.

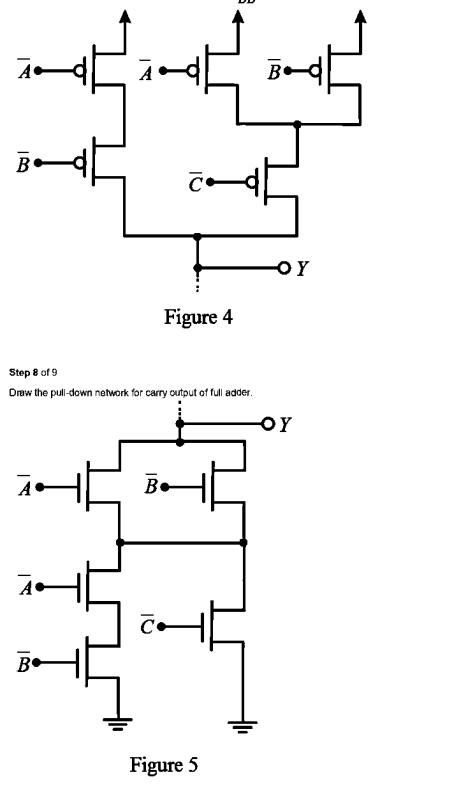


Figure 1

Step 4 of 9

The pull-down network is dual of pull-up network.

Draw the pull-down network for sum output of full adder.

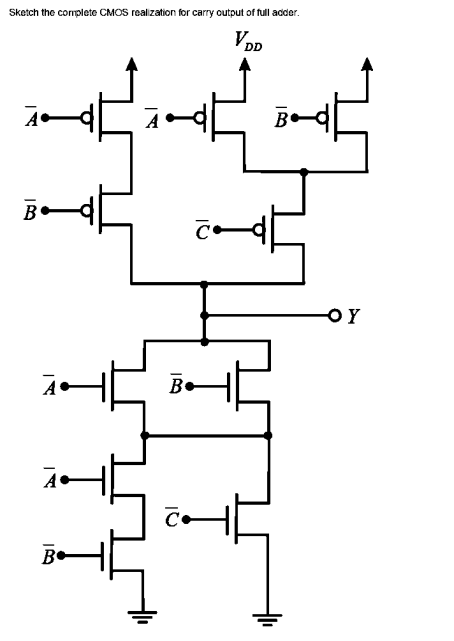


Figure 2

Step 5 of 9

Sketch the complete CMOS realization for sum output of full adder.

Figure 3

Thus, the complete CMOS realization for sum output of full adder is sketched as shown in Figure 3.

Step 6 of 9

From Table 1, write the Boolean expression for carry output.

$$C_0 = \overline{A}BC + A\overline{B}C + AB\overline{C} + ABC$$
$$= \left(\overline{A}BC + ABC\right) + \left(A\overline{B}C + ABC\right) + \left(AB\overline{C} + ABC\right) \qquad (\text{since } M + M = M)$$
$$= \left(\overline{A} + A\right)BC + \left(\overline{B} + B\right)AC + AB\left(\overline{C} + C\right)$$
$$= BC + AC + AB \qquad (\text{since } M + \overline{M} = 1)$$
$$C_0 = \left(A + B\right)C + AB$$

Further simplification as follows,

$$C_0 = \overline{\left(A+B\right)C + AB} \qquad (\text{since } M = \overline{\overline{M}})$$
$$= \overline{\left(\left(A+B\right)C\right)\left(AB\right)} \qquad (\text{since } \overline{M+N} = \overline{M}.\overline{N})$$
$$= \overline{\left(\left(A+B\right)C\right)\left(AB\right)} \qquad (\text{since } \overline{MN} = \overline{M} + \overline{N})$$
$$= \overline{\left(\overline{A+B}\right) + \overline{C}}\left(\overline{A+B}\right)$$
$$= \overline{\left(\overline{AB} + \overline{C}\right)\left(\overline{A+B}\right)}$$
$$= \overline{\left(\overline{A+B}\right)\left(\overline{AB} + \overline{C}\right)}$$

Step 7 of 9

Draw the pull-up network for carry output of full adder.

Figure 4

Step 8 of 9

Draw the pull-down network for carry output of full adder.

Figure 5

Step 9 of 9

Sketch the complete CMOS realization for carry output of full adder.

Figure 6

Thus, the complete CMOS realization for carry output of full adder is sketched as shown in Figure 6.

Step 1 of 7

(a)

Consider a CMOS inverter.

Calculate the intrinsic capacitance of the inverter.

$$C_{int} = 2C_{gd1} + 2C_{gd2} + C_{db1} + C_{db2}$$

The gate-source and the gate-drain overlap capacitance are $0.3\ \frac{\text{fF}}{\mu\text{m}}$ of the gate width.

The gate-drain capacitance of Q_1 is,

$$\begin{aligned} C_{gd1} &= \left(0.3\ \frac{\text{fF}}{\mu\text{m}}\right)(W_n) \\ &= (0.3)(0.375) \\ &= 0.1125\ \text{fF} \end{aligned}$$

The gate-drain capacitance of Q_2 is,

$$\begin{aligned} C_{gd2} &= \left(0.3\ \frac{\text{fF}}{\mu\text{m}}\right)(W_p) \\ &= \left(0.3\ \frac{\text{fF}}{\mu\text{m}}\right)(1.125) \\ &= 0.3375\ \text{fF} \end{aligned}$$

The drain body capacitances are,

$$\begin{aligned} C_{db1} &= C_{db2} \\ &= 1\ \text{fF} \end{aligned}$$

Substitute 0.1125 fF for C_{gd1} , 0.3375 fF for C_{gd2} , 1 fF for C_{db1} and C_{db2} to calculate the internal capacitance.

$$\begin{aligned} C_{int} &= 2C_{gd1} + 2C_{gd2} + C_{db1} + C_{db2} \\ &= (2)(0.1125\ \text{fF}) + (2)(0.3375\ \text{fF}) + 1\ \text{fF} + 1\ \text{fF} \\ &= 2.9\ \text{fF} \end{aligned}$$

Therefore, the internal capacitance of the inverter, C_{int} is, $\boxed{2.9\ \text{fF}}$.

Step 2 of 7

Calculate the external capacitance.

$$C_{ext} = C_{g3} + C_{g4} + C_w$$

The input capacitance of the load inverter is,

$$\begin{aligned} C_{g3} + C_{g4} &= (WL)_3 C_{ox} + C_{gdov3} + C_{gdov3} + (WL)_4 C_{ox} + C_{gdov4} + C_{gdov4} \\ &= (0.375)(0.25)(6) + 2(0.3)(0.375) + (1.125)(0.25)(6) + (2)(0.3)(1.125) \\ &= 0.7875\ \text{fF} + 2.3625\ \text{fF} \\ &= 3.15\ \text{fF} \end{aligned}$$

The wiring capacitance, $C_w = 0.2\ \text{fF}$.

Substitute 3.15 fF for $C_{g3} + C_{g4}$, and 0.2 fF for C_w to find the external capacitance.

$$\begin{aligned} C_{ext} &= C_{g3} + C_{g4} + C_w \\ &= 3.15\ \text{fF} + 0.2\ \text{fF} \\ &= 3.35\ \text{fF} \end{aligned}$$

Therefore, the external capacitance of the inverter, C_{ext} is, $\boxed{3.35\ \text{fF}}$.

Step 3 of 7

(b)

Consider that the extrinsic part of propagation delay is reduced by a factor of 2.

The general expression for the propagation delay, t_p is,

$$t_p = 0.69\left(R_{eq0}C_{int0} + \frac{1}{S}R_{eq0}C_{ext}\right)$$

The extrinsic part of propagation delay is,

$$t_{pext} = 0.69\frac{1}{S}R_{eq0}C_{ext}$$

Here, the equivalent resistance for the minimum-size inverter is, R_{eq0} , and the factor, S is,

$$S = \frac{R_{eq0}}{R_{eq}}$$

In order to reduce the extrinsic part of propagation delay by a factor of 2, S has to be increased by a factor of 2, in other hand, the equivalent resistance, R_{eq} has to be reduce by a factor of 2.

The expression for the equivalent resistance, R_{eq} is,

$$R_{eq} = \frac{1}{2}(R_N + R_P)$$

Here, the resistances, R_N and R_P are,

$$R_N = \frac{12.5}{\left(\frac{W}{L}\right)_n}\ \text{k}\Omega$$

$$R_P = \frac{30}{\left(\frac{W}{L}\right)_p}\ \text{k}\Omega$$

In order to reduce the equivalent resistance, R_{eq} by a factor of 2, the resistances, R_N and R_P has to be reduced by a factor of 2, in other hand, the width to length ratios of transistors, $\left(\frac{W}{L}\right)_n$ and $\left(\frac{W}{L}\right)_p$ has to be increased by a factor of 2.

Thus, the width to length ratios of transistors, $\left(\frac{W}{L}\right)_n$ and $\left(\frac{W}{L}\right)_p$ must be increased by a factor of $\boxed{2}$, in order to reduce the extrinsic part of propagation delay by a factor of 2.

Step 4 of 7

(c)

Calculate the propagation delay for the circuit without scaling of width to length ratios.

Calculate the propagation delay from high-to-low transition, t_{PHL} .

$$t_{PHL} = \frac{\alpha_n C}{k_n'\left(\frac{W}{L}\right)_n V_{DD}}$$

Here,

$$\alpha_n = \frac{2}{\frac{7}{4} - \frac{3V_{tn}}{V_{DD}} + \left(\frac{V_{tn}}{V_{DD}}\right)^3}$$

Here, V_{tn} is the threshold voltage, and V_{DD} is the supply voltage. Therefore,

$$\begin{aligned} \alpha_n &= \frac{2}{\frac{7}{4} - \frac{3(0.5)}{2.5} + \left(\frac{0.5}{2.5}\right)^3} \\ &= \frac{2}{1.19} \\ &\cong 1.7 \end{aligned}$$

The equivalent capacitance, C is.

$$\begin{aligned} C &= C_{int} + C_{ext} \\ &= 2.9 + 3.35 \\ &= 6.25\ \text{pF} \end{aligned}$$

Substitute 1.7 for α_n , 6.25 fF for C , $115\ \mu\text{A/V}^2$ for k_n' , 2.5 V for V_{DD} , and $\frac{0.375\ \mu\text{m}}{0.25\ \mu\text{m}}$ for $\left(\frac{W}{L}\right)_n$ in the expression for t_{PHL} .

$$\begin{aligned} t_{PHL} &= \frac{(1.7)(6.25\ \text{fF})}{\left(115\ \mu\text{A/V}^2\right)\left(\frac{0.375\ \mu\text{m}}{0.25\ \mu\text{m}}\right)(2.5\ \text{V})} \\ &= \frac{10.625\times10^{-15}}{431.25\times10^{-6}} \\ &= 24.6\ \text{ps} \end{aligned}$$

Step 5 of 7

Calculate the propagation delay from low-to-high transition, t_{PLH} .

$$t_{PHL} = \frac{\alpha_p C}{k_p'\left(\frac{W}{L}\right)_p V_{DD}}$$

Here,

$$\alpha_n = \frac{2}{\frac{7}{4} - \frac{3V_{tp}}{V_{DD}} + \left(\frac{V_{tp}}{V_{DD}}\right)^2}$$

Since, the threshold voltages are same for PMOS and NMOS devices, $\alpha_n = \alpha_p$.

Substitute 1.7 for α_n , 6.25 fF for C , $30\ \mu\text{A/V}^2$ for k_p' , 2.5 V for V_{DD} , and $\frac{1.125\ \mu\text{m}}{0.25\ \mu\text{m}}$ for $\left(\frac{W}{L}\right)_p$ in the expression for t_{PLH} .

$$\begin{aligned} t_{PLH} &= \frac{(1.7)(6.25\ \text{fF})}{\left(30\ \mu\text{A/V}^2\right)\left(\frac{1.125\ \mu\text{m}}{0.25\ \mu\text{m}}\right)(2.5\ \text{V})} \\ &= \frac{10.625\times10^{-15}}{337.5\times10^{-6}} \\ &= 31.5\ \text{ps} \end{aligned}$$

The average value of the propagation delay is,

$$\begin{aligned} t_p &= \frac{1}{2}(t_{PLH} + t_{PHL}) \\ &= \frac{1}{2}(24.6\ \text{ps} + 31.5\ \text{ps}) \\ &\cong 28\ \text{ps} \end{aligned}$$

Step 6 of 7

Consider that the width to length ratios of transistors, $\left(\frac{W}{L}\right)_n$ and $\left(\frac{W}{L}\right)_p$ are increased by a factor, $S = 2$.

Calculate the propagation delay, t_p .

$$t_p = 0.69\left(R_{eq0}C_{int0} + \frac{1}{S}R_{eq0}C_{ext}\right)$$

The ratio of propagation delay for new and old devices is,

$$\begin{aligned} \frac{t_p(\text{new})}{t_p(\text{old})} &= \frac{0.69\left(R_{eq0}C_{int0} + \frac{1}{S}R_{eq0}C_{ext}\right)}{0.69\left(R_{eq0}C_{int} + R_{eq0}C_{ext}\right)} \\ &= \frac{0.69R_{eq0}\left(C_{int0} + \frac{1}{S}C_{ext}\right)}{0.69R_{eq0}\left(C_{int} + C_{ext}\right)} \\ &= \frac{C_{int0} + \frac{1}{S}C_{ext}}{C_{int} + C_{ext}} \end{aligned}$$

$$t_p(\text{new}) = t_p(\text{old})\left[\frac{C_{int0} + \frac{1}{S}C_{ext}}{C_{int} + C_{ext}}\right]$$

Substitute 2 for S , 28 ps for $t_p(\text{old})$, 2.9 fF for C_{int0} , and 3.35 fF for C_{ext} .

$$\begin{aligned} t_p(\text{new}) &= (28\ \text{ps})\left(\frac{2.9\ \text{pF} + \frac{3.35\ \text{pF}}{2}}{2.9\ \text{pF} + 3.35\ \text{pF}}\right) \\ &= (28\ \text{ps})\left(\frac{4.575\ \text{pF}}{6.25\ \text{pF}}\right) \\ &= (28\ \text{ps})(0.732) \\ &= 20.5\ \text{ps} \end{aligned}$$

Thus, the new value of propagation delay, t_p is, $\boxed{20.5\ \text{ps}}$.

Step 7 of 7

(d)

Consider the following expression for the area of the inverter.

$$A = WL$$

Modify the equation to get the expression in terms of width to length ratio.

$$A = \left(\frac{W}{L}\right)L^2$$

Consider that the width to length ratios of transistors, $\left(\frac{W}{L}\right)_n$ and $\left(\frac{W}{L}\right)_p$ are increased by a factor, $S = 2$.

Since, there is no change in length individually, the area of the inverter increases by a factor of 2.

Thus, the are of the inverter is increased by a factor of $\boxed{2}$, with the increase in the width to length ratios of transistors, $\left(\frac{W}{L}\right)_n$ and $\left(\frac{W}{L}\right)_p$ by a factor of 2.

14.016P

We don't have the solution to this problem yet.

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Step 1 of 3

(a)

Consider a four-input NOR gate.

The width to length ratio of NMOS device is,

$$\frac{W}{L}=n$$

Here, W is the width, L is the length, n is the width to length ratio of NMOS device.

Substitute $0.18\ \mu m$ for L, 1.5 for n to find the value of W.

$$\frac{W}{0.18\ \mu m}=1.5$$

$$W=1.5(0.18\ \mu m)$$
$$=0.27\ \mu m$$

Thus, the width to length ratio, $\frac{W}{L}$ for NMOS transistor is, $\boxed{\frac{0.27}{0.18}}$.

The width to length ratio of PMOS device is,

$$\frac{W}{L}=4p$$

Here, p is the width to length ratio of PMOS device.

Substitute $0.18\ \mu m$ for L, 3 for p to find the value of W.

$$\frac{W}{0.18\ \mu m}=4(3)$$

$$W=12(0.18\ \mu m)$$
$$=2.16\ \mu m$$

Thus, the width to length ratio, $\frac{W}{L}$ for PMOS transistor is, $\boxed{\frac{2.16}{0.18}}$.

Step 2 of 3

(b)

Consider a four-input NAND gate.

The width to length ratio of NMOS device is,

$$\frac{W}{L}=4n$$

Substitute $0.18\ \mu m$ for L, 1.5 for n to find the value of W.

$$\frac{W}{0.18\ \mu m}=4(1.5)$$

$$W=6(0.18\ \mu m)$$
$$=1.08\ \mu m$$

Thus, the width to length ratio, $\frac{W}{L}$ for NMOS transistor is, $\boxed{\frac{1.08}{0.18}}$.

The width to length ratio of PMOS device is,

$$\frac{W}{L}=p$$

Substitute $0.18\ \mu m$ for L, 3 for p to find the value of W.

$$\frac{W}{0.18\ \mu m}=3$$

$$W=3(0.18\ \mu m)$$
$$=0.54\ \mu m$$

Thus, the width to length ratio, $\frac{W}{L}$ for PMOS transistor is, $\boxed{\frac{0.54}{0.18}}$.

Step 3 of 3

The relative areas of the two gates is,

$$\text{Relative area}=\frac{\text{NOR area}}{\text{NAND area}}$$
$$=\frac{4(0.27\ \mu m)(0.18\ \mu m)+4(2.16\ \mu m)(0.18\ \mu m)}{4(1.08\ \mu m)(0.18\ \mu m)+4(0.54\ \mu m)(0.18\ \mu m)}$$
$$=\frac{0.27\ \mu m+2.16\ \mu m}{1.08\ \mu m+0.54\ \mu m}$$
$$=1.5$$

Thus, the relative area of the two gates is, $\boxed{1.5}$.

14.017P

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Step 1 of 3

14.018E

(a)

Refer four-input NAND gate shown in Figure 10.17 in the text book.

The PMOS transistors are connected in parallel and the NMOS transistors are connected in series.

The load capacitor is charged by the supply voltage V_{CC} through the PMOS transistors.

The maximum current available to load capacitor when the four PMOS transistors in ON condition.

Thus, **four** PMOS transistors provide maximum current $(4I_P)$ to charge a load capacitor.

Step 2 of 3

The load capacitor starts charging when at least one PMOS transistor is ON.

Thus, **one** PMOS transistor provides minimum current (I_P) to charge a load capacitor.

Determine the ratio of the maximum to minimum current available to available to charge a load capacitance.

$$\text{Current ratio} = \frac{4I_P}{I_P}$$
$$= 4$$

Thus, the ratio of the maximum to minimum current available to charge a load capacitance is $\boxed{4}$.

Step 3 of 3

(b)

Four NMOS transistors in series are connected between the load capacitor and ground.

The load capacitor discharges through the NMOS transistors.

To provide discharge path to the load capacitor, the four NMOS transistors must be in ON condition. Since, all four transistors are connected in series.

Only one path is available to the load capacitor for discharging.

Thus maximum discharge current and the minimum discharge current available to the load capacitor are same.

Hence, the ratio of the maximum to minimum current available to charge a load capacitance is $\boxed{1}$.

14.018P

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14.019E

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14.019P

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14.020E

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14.021E

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14.022E

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14.022P

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Step 1 of 9

Determine the ideal value of low-level output voltage.

For an ideal logic circuit family,

$$V_{OL} = 0.0 \text{ V}$$

Thus, the ideal value of low-level output voltage, V_{OL} is $\boxed{0.0 \text{ V}}$.

Step 2 of 9

Determine the ideal value of high-level output voltage.

$$V_{OH} = V_{DD}$$

Substitute 1.2 V for V_{DD} in the equation.

$$V_{OH} = 1.2 \text{ V}$$

Thus, the ideal value of high-level output voltage, V_{OH} is $\boxed{1.2 \text{ V}}$.

Step 3 of 9

The gain of one points are separated by $\frac{1}{3}V_{DD}$.

$$(V_{IH} - V_{IL}) \le \frac{1}{3}V_{DD}$$

Substitute 1.2 V for V_{DD} in the equation.

$$(V_{IH} - V_{IL}) \le \frac{1.2}{3}$$

$$V_{IH} - V_{IL} \le 0.4 \text{ V} \quad \ldots\ldots (1)$$

Step 4 of 9

The noise margins are within 30% of one other.

The value of high input noise margin is,

$$\begin{aligned} NM_H &= \left(1 \pm \frac{30}{100}\right) NM_L \\ &= (1 \pm 0.3) NM_L \\ &= (1+03) NM_L \text{ and } (1-0.3) NM_L \\ &= 1.3NM_L \text{ and } 0.7NM_L \end{aligned}$$

The value of low input noise margin is,

$$\begin{aligned} NM_L &= \left(1 \pm \frac{30}{100}\right) NM_H \\ &= (1 \pm 0.3) NM_H \\ &= (1+03) NM_H \text{ and } (1-0.3) NM_H \\ &= 1.3NM_H \text{ and } 0.7NM_H \end{aligned}$$

Step 5 of 9

Determine the value of voltages V_{IL} and V_{IH} when $NM_L = 0.7NM_H$.

$$NM_L = 0.7NM_H$$

$$V_{IL} - V_{OL} = 0.7(V_{OH} - V_{IH})$$

Substitute 0.0 V for V_{OL} and 1.2 V for V_{OH} in the equation.

$$1.2 - V_{IH} = 0.7(V_{IL} - 0)$$

$$V_{IH} = 1.2 - 0.7V_{IL}$$

Substitute $(1.2 - 0.7V_{IL})$ for V_{IH} in equation (1).

$$(1.2 - 0.7V_{IL}) - V_{IL} = 0.4$$

$$-1.7V_{IL} = 0.4 - 1.2$$

$$V_{IL} = \frac{1.2 - 0.4}{1.7}$$

$$V_{IL} = 0.47 \text{ V}$$

Step 7 of 9

Substitute 0.47 V for V_{IL} in equation (1), to calculate the value of V_{IH}.

$$V_{IH} - 0.47 = 0.4$$

$$V_{IH} = 0.4 + 0.47$$

$$V_{IH} = 0.87 \text{ V}$$

Thus, the ranges of values for V_{IL} and V_{IH} are, $\boxed{0.329 \text{ V} \le V_{IL} \le 0.470 \text{ V}}$ and $\boxed{0.729 \text{ V} \le V_{IH} \le 0.870 \text{ V}}$ respectively.

Step 8 of 9

Determine the minimum value of low-input noise margin.

$$NM_{L(\min)} = V_{IL(\min)} - V_{OL}$$

Substitute 0.329 V for $V_{IL(\min)}$ and 0.0 V for V_{OL} in the equation.

$$\begin{aligned} NM_{L(\min)} &= 0.329 - 0 \\ &= 0.329 \text{ V} \end{aligned}$$

Determine the minimum value of high-input noise margin.

$$NM_{H(\min)} = V_{OH} - V_{IH(\max)}$$

Substitute 1.2 V for V_{OH} and 0.870 V for $V_{IH(\max)}$ in the equation.

$$\begin{aligned} NM_{H(\min)} &= 1.2 - 0.87 \\ &= 0.33 \text{ V} \end{aligned}$$

Step 9 of 9

Determine the maximum value of low-input noise margin.

$$NM_{L(\max)} = V_{IL(\max)} - V_{OL}$$

Substitute 0.470 V for $V_{IL(\min)}$ and 0.0 V for V_{OL} in the equation.

$$\begin{aligned} NM_{L(\max)} &= 0.470 - 0 \\ &= 0.470 \text{ V} \end{aligned}$$

Determine the maximum value of high-input noise margin.

$$NM_{H(\max)} = V_{OH} - V_{IH(\min)}$$

Substitute 1.2 V for V_{OH} and 0.729 V for $V_{IH(\max)}$ in the equation.

$$\begin{aligned} NM_{H(\max)} &= 1.2 - 0.729 \\ &= 0.471 \text{ V} \end{aligned}$$

Thus, the ranges of values for noise margins are, $\boxed{0.329 \text{ V} \le NM_L \le 0.470 \text{ V}}$ and $\boxed{0.330 \text{ V} \le NM_H \le 0.471 \text{ V}}$ respectively.

14.024P

We don't have the solution to this problem yet.

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14.025P

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14.026P

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14.027P

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14.043P

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14.044P

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Step 1 of 3

Determine the factor α_n using the average current approach.

$$\alpha_n = \frac{2}{\left[\frac{7}{4} - \frac{3V_{tn}}{V_{DD}} + \left(\frac{V_{tn}}{V_{DD}}\right)^2\right]}$$

Substitute 0.4 V for threshold voltage, V_{tn} and 1.2 V for supply voltage, V_{DD}.

$$\alpha_n = \frac{2}{\left[\frac{7}{4} - \frac{3(0.4)}{1.2} + \left(\frac{0.4}{1.2}\right)^2\right]}$$
$$= 2.3226$$

$$\alpha_n = \alpha_p = 2.3226$$

Determine the inverter delay for high-to-low transition.

$$t_{PHL} = \frac{\alpha_n C}{k_n'\left(\frac{W}{L}\right)_n V_{DD}}$$

Substitute 2.3226 for α_n, 10×10^{-15} F for C, 430×10^{-6} A/V^2 for k_n', 1.5 for $\left(\frac{W}{L}\right)_n$ and 1.2 V for V_{DD}.

$$t_{PHL} = \frac{(2.3226)(10\times10^{-15})}{(430\times10^{-6})(1.5)(1.2)}$$
$$= 30 \text{ ps}$$

Thus, the inverter delay t_{PHL} is $\boxed{30 \text{ ps}}$.

Step 2 of 3

Determine the inverter delay low-to-high transition.

$$t_{PLH} = \frac{\alpha_p C}{k_p'\left(\frac{W}{L}\right)_p V_{DD}}$$

Substitute 2.3226 for α_p, 10×10^{-15} F for C, $\frac{430}{4}\times10^{-6}$ A/V^2 for k_p', 3 for $\left(\frac{W}{L}\right)_p$ and 1.2 V for V_{DD}.

$$t_{PLH} = \frac{(2.3226)(10\times10^{-15})}{\left(\frac{430}{4}\times10^{-6}\right)(3)(1.2)}$$
$$= 60 \text{ ps}$$

Thus, the inverter delay t_{PLH} is $\boxed{60 \text{ ps}}$.

Step 3 of 3

Determine the propagation delay t_p.

$$t_p = \frac{1}{2}(t_{PHL} + t_{PLH})$$

Substitute 60 ps for t_{PLH} and 30 ps for t_{PHL}.

$$t_p = \frac{1}{2}(30\times10^{-12} + 60\times10^{-12})$$
$$= 45 \text{ ps}$$

Thus, the propagation delay t_p is $\boxed{45 \text{ ps}}$.

14.046P

We don't have the solution to this problem yet.

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Step 1 of 4

Determine the equivalent resistance of the transistor, Q_N.

$$R_N = \frac{12.5}{\left(\frac{W}{L}\right)_n}\ \text{k}\Omega$$

Substitute 1.5 for $\left(\frac{W}{L}\right)_n$.

$$R_N = \frac{12.5}{1.5}\ \text{k}\Omega$$
$$= 8.33\ \text{k}\Omega$$

Determine the equivalent resistance of the transistor, Q_P.

$$R_P = \frac{30}{\left(\frac{W}{L}\right)_p}\ \text{k}\Omega$$

Substitute 3 for $\left(\frac{W}{L}\right)_p$.

$$R_P = \frac{30}{3}\ \text{k}\Omega$$
$$= 10\ \text{k}\Omega$$

Step 2 of 4

Determine the inverter delay for high-to-low transition.

$$t_{PHL} = 0.69 R_N C$$

Substitute $8.33\times10^3\ \Omega$ for R_N and $10\times10^{-15}\ \text{F}$ for C.

$$t_{PHL} = (0.69)(8.33\times10^3)(10\times10^{-15})$$
$$= 57.5\times10^{-12}\ \text{s}$$
$$= 57.5\ \text{ps}$$

Thus, the inverter delay t_{PHL} is $\boxed{57.5\ \text{ps}}$.

Step 3 of 4

Determine the inverter delay for low-to-high transition.

$$t_{PLH} = 0.69 R_P C$$

Substitute $10\times10^3\ \Omega$ for R_P and $10\times10^{-15}\ \text{F}$ for C.

$$t_{PLH} = (0.69)(10\times10^3)(10\times10^{-15})$$
$$= 69\times10^{-12}\ \text{s}$$
$$= 69\ \text{ps}$$

Thus, the inverter delay t_{PLH} is $\boxed{69\ \text{ps}}$.

Step 4 of 4

Determine the propagation delay t_P.

$$t_P = \frac{1}{2}(t_{PHL} + t_{PLH})$$

Substitute $57.5\ \text{ps}$ for t_{PHL} and $69\ \text{ps}$ for t_{PLH}.

$$t_P = \frac{1}{2}(57.5\times10^{-12} + 69\times10^{-12})$$
$$= 63.25\ \text{ps}$$

Thus, the propagation delay t_P is $\boxed{63.25\ \text{ps}}$.

14.048P

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14.049P

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Step 1 of 2

14.050P

Write the formula for inverter delay for high-to-low transition.

$$t_{PHL} = \frac{CV_{DD}}{2I_{av}}$$

$$I_{av} = \frac{CV_{DD}}{2t_{PHL}}$$

Substitute 10×10^{-15} F for C, 2.5 V for V_{DD} and 41.2×10^{-12} s for t_{PHL}.

$$I_{av} = \frac{(10 \times 10^{-15})(2.5)}{2(41.2 \times 10^{-12})}$$

$$= 303.4\ \mu A$$

Thus, the average current without taking velocity saturation into consideration is $303.4\ \mu A$.

Step 2 of 2

Determine the average current considering velocity saturation.

The value of t_{PHL} when velocity saturation is considered is 57.5 pF

$$t_{PHL} = \frac{CV_{DD}}{2I_{av}}$$

$$I_{av} = \frac{CV_{DD}}{2t_{PHL}}$$

Substitute 10×10^{-15} F for C, 2.5 V for V_{DD} and 57.5×10^{-12} s for t_{PHL}.

$$I_{av} = \frac{(10 \times 10^{-15})(2.5)}{2(57.5 \times 10^{-12})}$$

$$= 217.4\ \mu A$$

Thus, the average current by taking velocity saturation into consideration is $217.4\ \mu A$.

Determine the current reduction factor.

$$\frac{303.4}{217.4} = 1.396$$

Thus, due to velocity saturation, the average current is reduced by a factor of $\boxed{1.396}$.

As the inverter delay for low-to-high transition does not change for a PMOS transistor, the average current remains constant.

Thus, the PMOS transistor is not affected by velocity saturation.

14.051P

We don't have the solution to this problem yet.

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Ask an expert

14.052P

We don't have the solution to this problem yet.

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Step 1 of 8

The two devices are designed to have equal lengths.

The ratio of the widths of the PMOS and NMOS transistors are,

$$\frac{W_p}{W_n}=\frac{\mu_n}{\mu_p}$$

$$\frac{W_p}{0.75}=\frac{\left(\frac{180}{3.7}\right)}{\left(\frac{45}{3.7}\right)}$$

$$W_p=4(0.75)$$

$$W_p=3\ \mu\text{m}$$

Determine the value of equivalent capacitance C. $C=2C_{gd1}+2C_{gd2}+C_{db1}+C_{db2}+C_{g3}+C_{g4}+C_w$

The gate drain overlap capacitance per micrometer of gate width is $0.4\ \text{fF}$.

$$C_{gd1}=(0.4\ \text{fF/}\mu\text{m})W_n$$
$$=(0.4)(0.75)$$
$$=0.3\ \text{fF}$$

$$C_{gd2}=(0.4\ \text{fF/}\mu\text{m})W_p$$
$$=(0.4)(3)$$
$$=1.2\ \text{fF}$$

Step 2 of 8

The effective drain body capacitance per micrometer of gate width is 1.0fF

$$C_{db1}=(1.0\text{fF/}\mu\text{m})W_n$$
$$=(1.0)(0.75)$$
$$=0.75\ \text{fF}$$

$$C_{db2}=(1.0\text{fF/}\mu\text{m})W_p$$
$$=1.0(3)$$
$$=3\ \text{fF}$$

Determine the input capacitance of the load inverter.

$$C_{g3}+C_{g4}=(WL)_3C_{ox}+(WL)_4C_{ox}+C_{gsov3}+C_{gdov3}+C_{gsov4}+C_{gdov4}$$
$$=\left[\begin{array}{l}(0.75\times0.5)(3.7)+(3\times0.5)(3.7)+\\(0.4\times0.75)+(0.4\times0.75)+(0.4\times3)+(0.4\times3)\end{array}\right]$$
$$=1.3875+5.55++0.3+0.3+1.2+1.2$$
$$=9.9375\ \text{fF}$$

Step 3 of 8

Determine the equivalent capacitance.

$$C=2C_{gd1}+2C_{gd2}+C_{db1}+C_{db2}+C_{g3}+C_{g4}+C_w$$
$$=2(0.3)+2(1.2)+0.75+3+9.9375+2$$
$$=18.6875\ \text{fF}$$

Determine the factor α_p.

$$\alpha_p=\frac{2}{\left[\frac{7}{4}-\frac{3|V_{tp}|}{V_{DD}}+\left|\frac{V_{tp}}{V_{DD}}\right|^2\right]}$$
$$=\frac{2}{\left[\frac{7}{4}-\frac{3|-0.7|}{3.3}+\left|\frac{-0.7}{3.3}\right|^2\right]}$$
$$=\frac{2}{1.75-0.6364+0.045}$$
$$=1.726$$

Step 4 of 8

Determine the factor α_n.

$$\alpha_n=\frac{2}{\left[\frac{7}{4}-\frac{3V_{tn}}{V_{DD}}+\left(\frac{V_{tp}}{V_{DD}}\right)^2\right]}$$
$$=\frac{2}{\left[\frac{7}{4}-\frac{3(0.7)}{3.3}+\left(\frac{0.7}{3.3}\right)^2\right]}$$
$$=\frac{2}{1.75-0.6364+0.045}$$
$$=1.726$$

Step 5 of 8

Determine the low to high propagation delay.

$$t_{PLH}=\frac{\alpha_pC}{k'_p\left(\frac{W}{L}\right)_pV_{DD}}$$
$$=\frac{1.726\times18.6875\times10^{-15}}{\left(180\times10^{-6}\right)\left(\frac{0.75}{0.5}\right)(3.3)}$$
$$=36.2\ \text{ps}$$

Thus, the low to high propagation delay t_{PLH} is $\boxed{36.2\ \text{ps}}$.

Step 6 of 8

Determine the high to low propagation delay.

$$t_{PHL}=\frac{\alpha_nC}{k'_n\left(\frac{W}{L}\right)_nV_{DD}}$$
$$=\frac{1.726\times18.6875\times10^{-15}}{\left(45\times10^{-6}\right)\left(\frac{3}{0.5}\right)(3.3)}$$
$$=36.2\ \text{ps}$$

Thus, the high to low propagation delay t_{PHL} is $\boxed{36.2\ \text{ps}}$.

Step 7 of 8

Determine the propagation delay, t_p.

$$t_p=\frac{1}{2}(t_{PLH}+t_{PHL})$$

Substitute $36.2\ \text{ps}$ for t_{PHL} and $36.2\ \text{ps}$ for t_{PLH}.

$$t_p=\frac{1}{2}(36.2\ \text{ps}+36.2\ \text{ps})$$
$$=36.2\ \text{ps}$$

Thus, the propagation delay t_p is $\boxed{36.2\ \text{ps}}$.

Step 8 of 8

The value of capacitance is proportional to the propagation delay.

If the propagation delay increases by 50% then the capacitance also increases by 50%.

Determine the additional load capacitance.

$$C_L=(18.6875\text{fF})\left(\frac{50}{100}\right)$$
$$=9.344\ \text{fF}$$

Thus, the additional capacitance load is $\boxed{9.344\text{fF}}$.

14.054P

We don't have the solution to this problem yet.

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14.055P

We don't have the solution to this problem yet.

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14.057P

We don't have the solution to this problem yet.

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14.058P

Refer to Figure 13.36 for a four-input NAND gate in the text book.

For Four-input NAND gate, the W/L ratio of each PMOS transistor to be equal to that of the PMOS transistor of the basic inverter. That is, equals to p. The W/L ratio of each NMOS transistor to be equals to four times of Q_N transistor of the basic inverter. That is, equals to $4n$.

Thus, the low-to-high propagation delay for the non-matched four-input CMOS NAND gate is, $\boxed{t_{PLH} = t_P}$.

Thus, the high-to-low propagation delay for the non-matched four-input CMOS NAND gate is, $\boxed{t_{PHL} = \frac{t_P}{4}}$.

For the matched four-input CMOS NAND gate, the low-to-high and high-to-low propagation delays are equal.

$$\boxed{\begin{aligned} t_{PLH} &= t_{PHL} \\ &= t_P \end{aligned}}$$

Hence, the low-to-high propagation delays are same in both matched and non-matched cases. The high-to-low propagation delay of non-matched four-input NAND gate is four times lesser with matched four-input NAND gate.

14.059P

We don't have the solution to this problem yet.

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Step 1 of 9

Refer to Figure 13.31 for a two-input CMOS NOR-gate in the text book.

For a two-input NOR gate, the W/L ratio of each NMOS transistor to be equal to that of the NMOS transistor of the basic inverter. That is, equals to n.

$$n_A = n_B$$
$$= n$$

The W/L ratio of each PMOS transistor to be equals to two times of Q_p transistor of the basic inverter. That is, equals to $2p$.

$$p_A = p_B$$
$$= 2p$$

Thus, the relationship between k_p **and** k_n for two-input CMOS NOR-gate is,

$$k_p = 2k_n$$

Step 2 of 9

(a)

The input terminal A is connected to ground. The modified circuit diagram is shown in Figure 1.

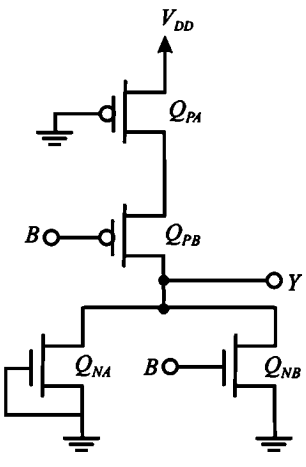


Figure 1

Step 3 of 9

Consider the threshold voltage, V_{th} is $\frac{V_{DD}}{2}$. The transistors Q_{PB} **and** Q_{NB} are in saturation-region and the transistor Q_{PA} in triode-region.

Write the general expression for the drain current in triode region.

$$i_D = k_p v_{DS}\left[\left(V_{GS} - |V_t|\right) - \frac{1}{2}v_{DS}\right]$$

The expression for drain current of transistor Q_{PA} is,

$$i_D = k_p V\left[\left(V_{DD} - |V_t|\right) - \frac{1}{2}V\right] \quad (\text{since } V_{GS} = V_{DD})$$

Here, the voltage across the transistor Q_{PA} is, $v_{DS} = V$.

$$i_D = k_p\left[\left(V_{DD} - |V_t|\right)V - \frac{1}{2}V^2\right]$$

Substitute 5 V for V_{DD}, 1 V for $|V_t|$ and $2k_n$ for k_p in the equation.

$$i_D = 2k_n\left[(5-1)V - \frac{V^2}{2}\right]$$
$$= 2k_n\left[4V - \frac{V^2}{2}\right]$$
$$= 2k_n\left[\frac{8V - V^2}{2}\right]$$
$$i_D = k_n\left(8V - V^2\right) \quad \ldots\ldots (1)$$

Step 4 of 9

Write the general expression for the drain current in saturation region.

$$i_D = \frac{1}{2}k_p\left(V_{GS} - |V_t|\right)^2$$

The expression for drain current of transistor Q_{PB} is,

$$i_D = \frac{1}{2}k_p\left[\left(V_{DD} - V - V_{th}\right) - |V_t|\right]^2$$

Substitute 5 V for V_{DD}, 1 V for $|V_t|$ and $2k_n$ for k_p in the equation.

$$i_D = \frac{1}{2}(2k_n)(5 - V - V_{th} - 1)^2$$

$$i_D = k_n\left(4 - V - V_{th}\right)^2 \quad \ldots\ldots (2)$$

Step 5 of 9

Write the general expression for the drain current in saturation region.

$$i_D = \frac{1}{2}k_n\left(V_{GS} - V_t\right)^2$$

The expression for drain current of transistor Q_{NB} is,

$$i_D = \frac{1}{2}k_n\left(V_{th} - V_t\right)^2$$

Substitute 1 V for V_t in the equation.

$$i_D = \frac{1}{2}k_n\left(V_{th} - 1\right)^2 \quad \ldots\ldots (3)$$

Substitute $\frac{1}{2}k_n\left(V_{th} - 1\right)^2$ for i_D in equation (2).

$$\frac{1}{2}k_n\left(V_{th} - 1\right)^2 = k_n\left(4 - V - V_{th}\right)^2$$
$$\frac{1}{2}\left(V_{th} - 1\right)^2 = \left(4 - V - V_{th}\right)^2$$
$$\frac{1}{\sqrt{2}}\left(V_{th} - 1\right) = \left(4 - V - V_{th}\right)$$
$$0.707V_{th} - 0.707 = 4 - V - V_{th}$$
$$V = 4.707 - 1.707V_{th}$$

Step 6 of 9

Substitute $\frac{1}{2}k_n\left(V_{th} - 1\right)^2$ for i_D in equation (1).

$$\frac{1}{2}k_n\left(V_{th} - 1\right)^2 = k_n\left(8V - V^2\right)$$
$$\left(V_{th} - 1\right)^2 = 2\left(8V - V^2\right)$$
$$V_{th}^2 - 2V_{th} + 1 = 16V - 2V^2$$

Substitute $\left(4.707 - 1.707V_{th}\right)$ for V in the equation.

$$V_{th}^2 - 2V_{th} + 1 = 16\left(4.707 - 1.707V_{th}\right) - 2\left(4.707 - 1.707V_{th}\right)^2$$
$$V_{th}^2 - 2V_{th} + 1 = 75.312 - 27.312V_{th} - 44.312 + 32.1394V_{th} - 5.828V_{th}^2$$
$$6.828V_{th}^2 - 6.8274V_{th} - 30 = 0$$
$$V_{th} = 2.65 \text{ V and } -1.65 \text{ V}$$

Since $2.65 > \frac{V_{DD}}{2}$, one PMOS is full ON.

Thus, the value of gate threshold voltage, V_{th} is $\boxed{2.65 \text{ V}}$.

Step 7 of 9

(b)

The two inputs are connected together in Figure 13.31. The modified circuit diagram is shown in Figure 2.

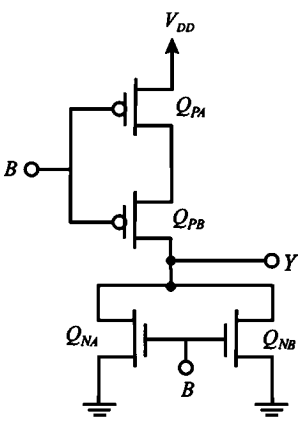


Figure 2

Step 8 of 9

The relationship between k_p **and** k_n for two-input matched CMOS NOR-gate is,

$$k_p = k_n = k.$$

Write the general expression for the drain current in saturation region.

$$i_D = \frac{1}{2}k_p\left(V_{GS} - |V_t|\right)^2$$

The expression for drain current of PMOS transistor is,

$$i_{DP} = \frac{1}{2}k_p\left[\left(V_{DD} - V_{th}\right) - |V_t|\right]^2$$

Substitute 5 V for V_{DD} and 1 V for $|V_t|$ in the equation.

$$i_{DP} = \frac{1}{2}(k_n)(5 - V_{th} - 1)^2$$

$$i_{DP} = \frac{1}{2}k_p\left(4 - V_{th}\right)^2$$

Step 9 of 9

Write the general expression for the drain current in saturation region.

$$i_D = \frac{1}{2}k_n\left(V_{GS} - V_t\right)^2$$

The expression for drain current of transistor NMOS transistor is,

$$i_{DN} = \frac{1}{2}k_n\left(V_{th} - V_t\right)^2$$

Substitute 1 V for V_t in the equation.

$$i_{DN} = \frac{1}{2}k_n\left(V_{th} - 1\right)^2$$

In Figure 2, the NMOS transistors connected in parallel. So, the two drain currents are added. In case of PMOS transistors, the drain current is same due to series connection.

$$2i_{DN} = i_{DP}$$

Substitute $\frac{1}{2}k_n\left(V_{th} - 1\right)^2$ for i_{DN} and $\frac{1}{2}k_p\left(4 - V_{th}\right)^2$ for i_{DP} in the equation.

$$2\left[\frac{1}{2}k_n\left(V_{th} - 1\right)^2\right] = \frac{1}{2}k_p\left(4 - V_{th}\right)^2$$
$$k\left(V_{th} - 1\right)^2 = \frac{1}{2}k\left(4 - V_{th}\right)^2 \quad \left(\text{since } k_p = k_n = k\right)$$
$$2\left(V_{th} - 1\right)^2 = \left(4 - V_{th}\right)^2$$
$$\sqrt{2}\left(V_{th} - 1\right) = 4 - V_{th}$$
$$1.414V_{th} - 1.414 = 4 - V_{th}$$
$$2.414V_{th} = 5.414$$
$$V_{th} = \frac{5.414}{2.414}$$
$$V_{th} = 2.24 \text{ V}$$

Thus, the value of gate threshold voltage, V_{th} is $\boxed{2.24 \text{ V}}$.

14.061P

We don't have the solution to this problem yet.

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14.062P

We don't have the solution to this problem yet.

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14.063P

We don't have the solution to this problem yet.

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14.064P

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Ask an expert

14.065P

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Step 1 of 4

Write the expression for dissipated power.

$$P_D = f_C V_{DD}^2$$
$$P_D \propto V_{DD}^2$$

The supply voltage is reduced from $5\ \text{V}$ to $3.3\ \text{V}$.

Thus, the power reduces by a factor is,

$$\left(\frac{3.3}{5}\right)^2 = \frac{10.89}{25}$$
$$= 0.4356$$

Determine the dissipated power when supply voltage is reduced.

$$P_{D(3.3\text{ V})} = P_{D(5\text{ V})}(0.4356)$$

Substitute $10\ \text{mW}$ for $P_{D(5\text{ V})}$ in the equation.

$$P_{D(3.3\text{ V})} = (10\times10^{-3})(0.4356)$$
$$= 4.356\times10^{-3}$$
$$= 4.356\ \text{mW}$$

Thus, the new value of dissipated power, $P_{D(3.3\text{ V})}$ is $\boxed{4.356\ \text{mW}}$.

Step 2 of 4

The frequency of operation is reduced by a same factor as the supply voltage.

$$P_D = f_C V_{DD}^2$$
$$P_D \propto f_C$$

Thus, the frequency reduces by a factor is,

$$\left(\frac{3.3}{5}\right) = 0.66$$

Step 3 of 4

Determine the value of dissipated power when the frequency is reduced.

$$P_D = (0.66)\left(P_{D(3.3\text{ V})}\right)$$

Substitute $4.356\ \text{mW}$ for $P_{D(3.3\text{ V})}$ in the equation.

$$P_D = (0.66)\left(4.356\times10^{-3}\right)$$
$$= 2.875\times10^{-3}$$
$$= 2.875\ \text{mW}$$

Thus, the new value of dissipated power, P_D is $\boxed{2.875\ \text{mW}}$.

Step 4 of 4

Determine the value of additional saved power.

$$P_{dyn} = \left[\frac{(5-3.3)}{5}\right] P_{D(3.3\text{ V})}$$

Substitute $4.356\ \text{mW}$ for $P_{D(3.3\text{ V})}$ in the equation.

$$P_{dyn} = \left(\frac{1.7}{5}\right)\left(4.356\times10^{-3}\right)$$
$$\approx 1.48\times10^{-3}$$
$$= 1.48\ \text{mW}$$

Thus, the value of additional saved power, P_{dyn} is $\boxed{1.48\ \text{mW}}$.

14.067P

We don't have the solution to this problem yet.

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14.068P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

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14.069P

We don't have the solution to this problem yet.

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Step 1 of 3

15.001E

Consider an inverter circuit is fabricated with 0.25-μm technology.

Consider the propagation delay of the component of the inverter circuit is t_{pd} and the dynamic power dissipation of the component of the inverter circuit is P_{dyn}.

The power –delay product (PDP) of the inverter circuit is defined as the product of the propagation delay (t_{pd}) of the component of the inverter circuit and the power dissipation (P_{dyn}) of the component of the inverter circuit.

$$PDP = (t_{pd})(P_{dyn})$$

Step 2 of 3

Consider a scaling is provided in device dimensions to 0.13-μm.

Consider an inverter circuit is fabricated with 0.13-μm technology.

Refer to Table 15.1 in the textbook.

Consider S is the scaling factor.

The propagation delay of the inverter circuit is scaled by $\frac{1}{S}$.

The propagation delay of the component of the inverter circuit designed with 0.13-μm technology is becomes $\left(\frac{t_{pd}}{S}\right)$.

Step 3 of 3

The dynamic power dissipation of the component of the inverter circuit is scaled by $\frac{1}{S^2}$.

The dynamic power dissipation of the component of the inverter circuit designed with 0.13-μm technology is becomes $\left(\frac{P_{dyn}}{S^2}\right)$.

Determine the power-delay product of the inverter circuit designed with 0.13-μm technology.

$$\begin{aligned} PDP_{0.13-\mu m} &= (t_{pd})(P_{dyn}) \\ &= \left(\frac{t_{pd}}{S}\right)\left(\frac{P_{dyn}}{S^2}\right) \\ &= \left(\frac{t_{pd}P_{dyn}}{S^3}\right) \\ &= \left(\frac{PDP_{0.25-\mu m}}{S^3}\right) \end{aligned}$$

Consider scaling factor $S = 2$.

$$\begin{aligned} PDP_{0.13-\mu m} &= \left(\frac{PDP_{0.25-\mu m}}{S^3}\right) \\ &= \left(\frac{PDP_{0.25-\mu m}}{2^3}\right) \\ &= \left(\frac{PDP_{0.25-\mu m}}{8}\right) \end{aligned}$$

Therefore, the power-delay product of the inverter circuit fabricated with 0.13-μm technology is decreases by a factor of $\boxed{8}$.

Step 1 of 2

15.001P

Consider a chip is designed using 5-μm technology in 1970s with 20,000 transistors.

According to definition of Moore's law the number of transistors per square inch is doubled for every 18 months.

Moore's law is also concentrated on speed of the chip and it is more accurate to apply the law to number of transistors than to speed.

Step 2 of 2

Determine the number of transistors on the chip fabricated with 32-nm technology in 2013.

$$2013-1970$$

$$=43\text{ years}$$

Determine the total number of months.

$$43*12$$

$$=516\text{ months}$$

Consider the number of transistors are doubled for every 18 months.

$$\frac{516}{18}=29$$

$$\text{total number of transistor in 2013}=\left(2^{29}\right)*(20000)$$

$$=\left(10.7*10^{12}\right)$$

Therefore, the number of transistors on the chip fabricated with 32-nm technology in 2013 is $\boxed{\left(10.7*10^{12}\right)}$

Step 1 of 5

Refer to table 15.1 in the textbook.

Consider V_{DD} and V_t are constant.

The width (W) of the component and the length of the component (L) does not depends upon V_{DD} and V_t.

The scaling factor of the width of the component and length of the component remains same.

The scaling factor of the Area of the component is also remains same because it depends upon the width and length of the component.

Step 2 of 5

The scaling factor of the oxide capacitance (C_{ox}), gate capacitance (C_{gate}), and the transconductance parameters k_n' and k_p' also remains same (S) because they are independent of V_{DD} and V_t.

Consider $t_p(\text{intrinsic}) = \dfrac{\alpha C}{k' V_{DD}}$.

Determine the propagation delay of the component after scaling.

$$\begin{aligned} t_p(\text{intrinsic})_{\text{After scaling}} &= \frac{\alpha\left(\frac{C}{S}\right)}{(Sk')V_{DD}} \\ &= \frac{1}{S^2}\left[\frac{\alpha C}{k' V_{DD}}\right] \\ &= \frac{1}{S^2}\left[t_p(\text{intrinsic})\right] \end{aligned}$$

Therefore, the propagation delay of the component after scaling is reduced by $\boxed{\dfrac{1}{S^2}}$.

Step 3 of 5

Consider energy/switching cycle $= CV^2{}_{DD}$.

Determine the energy/switching cycle of the component after scaling.

$$\begin{aligned} (\text{energy/switching cycle})_{\text{After scaling}} &= CV^2{}_{DD} \\ &= \left(\frac{C}{S}\right)V^2{}_{DD} \\ &= \left(\frac{CV^2{}_{DD}}{S}\right) \\ &= \frac{1}{S}\left[\text{energy/switching cycle}\right] \end{aligned}$$

Therefore, the energy/switching cycle of the component after scaling is reduced by $\boxed{\dfrac{1}{S}}$.

Step 4 of 5

Consider dynamic power dissipation $P_{dyn} = \dfrac{CV^2{}_{DD}}{2t_p}$.

Determine the dynamic power dissipation of the component after scaling.

$$\begin{aligned} \left(P_{dyn}\right)_{\text{After scaling}} &= \frac{CV^2{}_{DD}}{2t_p} \\ &= \frac{\left(\frac{C}{S}\right)V^2{}_{DD}}{2\left(\frac{t_p}{1/S^2}\right)} \\ &= S\left[P_{dyn}\right] \end{aligned}$$

Therefore, the dynamic power dissipation of the component after scaling is increased by $\boxed{S}$.

Step 5 of 5

Consider the power density $= \dfrac{P_{dyn}}{\text{device area}}$.

Determine the power density of the component after scaling.

$$\begin{aligned} (\text{power density})_{\text{After scaling}} &= \frac{P_{dyn}}{\text{device area}} \\ &= \frac{(S)(P_{dyn})}{\left(\frac{\text{Area}}{S^2}\right)} \\ &= S^3\left[\text{power density}\right] \end{aligned}$$

Therefore, the power density of the component after scaling is increased by $\boxed{S^3}$.

15.002P

We don't have the solution to this problem yet.

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Step 1 of 3

15.003E

Consider a MOSFET is fabricated in deep-submicron process.

Velocity saturation is one of the most important effects due to short channel used in sub-micron process.

Consider v is the drift velocity of the electrons.

$v = \mu_n E$ ..... (1)

Consider E is the longitudinal electric field established by V_{DS} .

$E = \frac{V_{DS}}{L}$ ..... (2)

Step 2 of 3

In a short channel MOSFET the electric field is greater than the critical electric field (E_{cr}).

Consider $V_{DS\,sat}$ is the drain to source voltage at which velocity saturation occurs.

$E_{cr} = \frac{V_{DS\,sat}}{L}$ ..... (3)

Replace $E_{cr} = \frac{V_{DS\,sat}}{L}$ for E in equation 1.

$v_{sat} = \mu_n \left(\frac{V_{DS\,sat}}{L} \right)$

$V_{DS\,sat} = \left(\frac{L}{\mu_n} \right) v_{sat}$ ..... (4)

Step 3 of 3

Replace $400\ \text{cm}^2/\text{V.s}$ for μ_n , $0.25\ \mu\text{m}$ for L , and $10^7\ \text{cm/s}$ for v_{sat} in equation 4.

$$\begin{aligned} V_{DS\,sat} &= \left(\frac{L}{\mu_n} \right) v_{sat} \\ &= \left(\frac{(0.25 * 10^{-6})}{(400 * 10^{-4})} \right) (10^7 * 10^{-2}) \\ &= 0.625\ \text{V} \end{aligned}$$

Therefore, the value of V_{DS} at which velocity saturation occurs is $\boxed{0.625\ \text{V}}$.

15.003P

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Step 1 of 3

(a)

Refer to Figure 15.6 in the textbook.

Consider a straight line representing subthreshold conduction.

Determine the slope of the straight line.

Consider a straight line equation with m as the slope of the straight line.

$$y = mx$$

$$m = \frac{y}{x} = \frac{\text{change in drain current in logscale}}{\text{change in gate to source votage}}$$

$$m = \frac{\Delta \log i_D}{\Delta v_{GS}}$$

Consider change in drain current in log scale is decade and change in gate to source voltage is $2.3nV_T$.

$$m = \left(\frac{1}{2.3nV_T}\right) \text{decade}/V$$

Determine the inverse of the slope of the straight line.

$$\text{Inverse of the slope of the straight line} = \frac{1}{m}$$

$$= \frac{1}{\left(\frac{1}{2.3nV_T}\right) \text{decade}/V}$$

$$= 2.3nV_T \frac{V}{\text{decade}}$$

Therefore, the inverse of the slope of the straight line representing subthreshold conduction is $\boxed{(2.3nV_T)\ V/\text{decade}}$.

Step 2 of 3

(b)

Consider i_D is the small amount of drain current that flows for $V_{GS} < V_t$.

$$i_D = I_S e^{\frac{v_{GS}}{nV_T}} \quad \text{..... (1)}$$

Consider $V_T = \frac{kT}{q}$ is the thermal voltage, n is a constant, and I_s is the saturation current.

Replace 1.22 for n, $100\ \text{nA}$ for i_D, $25\ \text{mV}$ for V_T, and $0.21\ \text{V}$ for v_{GS} in equation 1 and determine saturation current (I_S).

$$i_D = I_S e^{\frac{v_{GS}}{nV_T}}$$

$$\left(100 * 10^{-9}\right) = I_S e^{\left(\frac{0.21}{(1.22)(25*10^{-3})}\right)}$$

$$I_S = \frac{\left(100 * 10^{-9}\right)}{e^{\left(\frac{0.21}{(1.22)(25*10^{-3})}\right)}}$$

$$I_S = \left(0.1 * 10^{-9}\right) = 0.1\ \text{nA}$$

Determine the drain current (i_D) for $v_{GS} = 0$.

Replace 1.22 for n, $25\ \text{mV}$ for V_T, $0.1\ \text{nA}$ for I_S, and $0\ \text{V}$ for v_{GS} in equation 1.

$$i_D = I_S e^{\frac{v_{GS}}{nV_T}}$$

$$= \left(0.1 * 10^{-9}\right) e^{\left(\frac{0}{(1.22)(25*10^{-3})}\right)}$$

$$= 0.1\ \text{nA}$$

Therefore, the drain current for $v_{GS} = 0$ is $\boxed{0.1\ \text{nA}}$.

Step 3 of 3

(c)

Refer to sub part (b).

As a result of subthreshold conduction, a small amount of drain current flows in transistor for $v_{GS} = 0$.

Consider the drain current of a single transistor for $v_{GS} = 0$ is $0.1\ \text{nA}$.

Determine the drain current for 500 million transistors in sub threshold conduction.

$$i_D = \left(500 * 10^6\right)(0.1\ \text{nA})$$

$$= 0.05\ \text{A}$$

$$= 50\ \text{mA}$$

Determine the power dissipation.

$$P_d = (i_D)(V_{DD})$$

$$= \left(50 * 10^{-3}\right)(1.2)$$

$$= 60\ \text{mW}$$

Therefore, the current drawn for 500 million transistor is $\boxed{50\ \text{mA}}$ .and the amount of power dissipated is $\boxed{60\ \text{mW}}$.

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Step 1 of 9

Refer to the Example 15.2 in the textbook.

Consider a pseudo-NMOS inverter is fabricated in a 0.25-μm technology.

Consider static power dissipation is reduced to $75\ \mu W$.

Determine static current (I_{stat}).

$$P_D = I_{stat} V_{DD}$$
$$75\ \mu W = I_{stat}(2.5)$$
$$I_{stat} = \frac{75\ \mu W}{2.5}$$
$$I_{stat} = 30\ \mu A$$

Step 2 of 9

Determine the value of k_p.

$$I_{stat} = \frac{1}{2} k_p (V_{DD} - V_t)^2$$
$$(30*10^{-6}) = \frac{1}{2} k_p' \left(\frac{W}{L}\right)_p (V_{DD} - V_t)^2$$
$$(30*10^{-6}) = \frac{1}{2}\left((30*10^{-6})\left(\frac{W}{L}\right)_p\right)(2.5-0.5)^2$$
$$\left(\frac{W}{L}\right)_p = \frac{2}{4}$$
$$= 0.5$$

Therefore, the value of aspect ratio $\left(\frac{W}{L}\right)_p$ is $\boxed{0.5}$.

Step 3 of 9

Determine the value of k_n.

Consider r is the ratio of k_n and k_p.

$$r = \frac{k_n}{k_p} \quad \text{..... (1)}$$

Replace 9 for r in equation 1.

$$r = \frac{k_n}{k_p}$$
$$9 = \frac{k_n}{k_p' \left(\frac{W}{L}\right)_p}$$
$$9 = \frac{k_n}{\mu_p C_{ox} \left(\frac{W}{L}\right)_p}$$
$$9 = \frac{k_n}{(30*10^{-6})(0.5)}$$
$$k_n = (30*10^{-6})(0.5)(9)$$
$$= 135\ \mu A$$

Step 4 of 9

Consider the value of $k_n = k_n'(W/L)_n$.

$$k_n = k_n'(W/L)_n$$
$$k_n = \mu_n C_{ox}(W/L)_n$$
$$(135*10^{-6}) = (115*10^{-6})(W/L)_n$$
$$(W/L)_n = \frac{(135*10^{-6})}{(115*10^{-6})}$$
$$(W/L)_n = \frac{(135*10^{-6})}{(115*10^{-6})}$$
$$= 1.1739$$

Therefore, the aspect ratio $\left(\frac{W}{L}\right)_n$ is $\boxed{1.173}$.

Step 5 of 9

Determine the low-to-high propagation delay.

$$t_{PLH} = \frac{\alpha_p C}{k_p V_{DD}} \quad \text{..... (2)}$$

Determine the value of α_p.

$$\alpha_p = \frac{2}{\left[\frac{7}{4} - 3\left(\frac{V_t}{V_{DD}}\right) + \left(\frac{V_t}{V_{DD}}\right)^2\right]}$$
$$= \frac{2}{\left[\frac{7}{4} - 3\left(\frac{0.5}{2.5}\right) + \left(\frac{0.5}{2.5}\right)^2\right]}$$
$$= 1.68$$

Step 6 of 9

Replace 1.68 for α_p, 7 fF for C, $(15*10^{-6})$ for k_p, and 2.5 for V_{DD} in equation 2.

$$t_{PLH} = \frac{\alpha_p C}{k_p V_{DD}}$$
$$= \frac{(1.68)(7*10^{-15})}{(15*10^{-6})(2.5)}$$
$$= 0.3136\ ns$$

Therefore, the low-to-high propagation delay (t_{PLH}) is $\boxed{0.3136\ ns}$.

Step 7 of 9

Determine the high-to-low propagation delay (t_{PHL}).

$$t_{PHL} = \frac{\alpha_n C}{k_n V_{DD}} \quad \text{..... (3)}$$

Determine the value of α_n.

Replace 9 for r in equation 3.

$$\alpha_n = \frac{2}{\left[1 + \frac{3}{4}\left(1 - \frac{1}{r}\right) - \left(3 - \frac{1}{r}\right)\left(\frac{V_t}{V_{DD}}\right) + \left(\frac{V_t}{V_{DD}}\right)^2\right]}$$
$$= \frac{2}{\left[1 + \frac{3}{4}\left(1 - \frac{1}{9}\right) - \left(3 - \frac{1}{9}\right)\left(\frac{0.5}{2.5}\right) + \left(\frac{0.5}{2.5}\right)^2\right]}$$
$$= 1.77$$

Determine the value of k_n.

$$k_n = k_n'\left(\frac{W}{L}\right)_n$$
$$= (115*10^{-6})(1.173)$$
$$= 134.8\ \mu A$$

Replace 1.77 for α_n, 7 fF for C, $(134.8*10^{-6})$ for k_n, and 2.5 for V_{DD} in equation 3.

$$t_{PHL} = \frac{\alpha_n C}{k_n V_{DD}}$$
$$= \frac{(1.77)(7*10^{-15})}{(134.8*10^{-6})(2.5)}$$
$$= 0.037\ ns$$

Therefore, the high-to-low propagation delay (t_{PHL}) is $\boxed{0.037\ ns}$.

Step 8 of 9

Determine the propagation delay (t_p).

$$t_p = \frac{1}{2}(t_{PHL} + t_{PLH})$$
$$= \frac{1}{2}(0.037 + 0.3136)$$
$$= 0.175\ ns$$

Therefore, the propagation delay (t_p) is $\boxed{0.175\ ns}$.

Step 9 of 9

Determine the noise margins (NM_L) and (NM_H).

Replace $r = 9$, 2.5 for V_{DD}, and 0.5 for V_t in following equation.

Determine the low value of the noise margin (NM_L).

$$NM_L = V_t - (V_{DD} - V_t)\left[1 - \sqrt{1 - \frac{1}{r}} - \frac{1}{\sqrt{r(r+1)}}\right]$$
$$= 0.5 - (2.5 - 0.5)\left[1 - \sqrt{1 - \frac{1}{9}} - \frac{1}{\sqrt{9(9+1)}}\right]$$
$$= 0.5 - 2\left[1 - \sqrt{\frac{8}{9}} - \frac{1}{\sqrt{90}}\right]$$
$$= 0.59\ V$$

Determine the high value of the noise margin (NM_H).

$$NM_H = (V_{DD} - V_t)\left(1 - \frac{2}{\sqrt{3r}}\right)$$
$$= (2.5 - 0.5)\left(1 - \frac{2}{\sqrt{3(9)}}\right)$$
$$= 2\left(1 - \frac{2}{\sqrt{27}}\right)$$
$$= 1.23\ V$$

Therefore, the noise margins (NM_L) and (NM_H) are same as the noise margins of the Example 15.2 in textbook.

Step 1 of 4

Refer to Figure 13.22 in the text book.

(a)

Just prior to the leading edge of the input pulse (i.e., at $t = 0^-$), the output voltage is equal to V_{DD} . At time $t = 0$, v_I raises to V_{DD} , causing Q_P turn OFF and Q_N to turn ON. Thus, the transistor Q_N operates in saturation at $t = 0^+$ and supplies relatively large current to begin the process of discharging capacitance. The output voltage, v_o reduces from V_{DD} to $\frac{V_{DD}}{2}$ during the interval t_{PHL} .

$$I_{av} t_{PHL} = C\Delta V$$

$$I_{Dsat} t_{PHL} = C\Delta V \quad (\text{since } I_{av} = I_{Dsat})$$

$$t_{PHL} = \frac{C\Delta V}{I_{Dsat}}$$

$$= \frac{C}{I_{Dsat}}\left[V_{DD} - \frac{V_{DD}}{2}\right]$$

$$= \frac{CV_{DD}}{2I_{Dsat}}$$

Thus, the expression for high to low propagation delay is, $\boxed{t_{PHL} = \frac{CV_{DD}}{2I_{Dsat}}}$.

Step 2 of 4

(b)

Write the expression for equivalent resistance of Q_N .

$$R_N = \frac{12.5}{\left(\frac{W}{L}\right)_n}\ \text{k}\Omega$$

Derive the expression for high to low propagation delay.

$$t_{PHL} = 0.69 R_N C$$

Substitute $\frac{12.5}{\left(\frac{W}{L}\right)_n}\ \text{k}\Omega$ for R_N in the equation.

$$t_{PHL} = 0.69C\frac{12.5\times10^3}{\left(\frac{W}{L}\right)_n}$$

Thus, the expression for high to low propagation delay is, $\boxed{t_{PHL} = 0.69C\frac{12.5\times10^3}{(W/L)_n}}$.

Step 3 of 4

(c)

Determine the value of voltage, V_{DSsat} for NMOS transistor.

$$I_{Dsat} = \frac{CV_{DD}}{2t_{PHL}}$$

Substitute $0.69C\frac{12.5\times10^3}{(W/L)_n}$ for t_{PHL} in the equation.

$$I_{Dsat} = \frac{CV_{DD}}{2(0.69)C\frac{12.5\times10^3}{(W/L)_n}}$$

$$I_{Dsat} = \frac{V_{DD}(W/L)_n}{2(0.69)(12.5\times10^3)}$$

Substitute $\mu_n C_{ox}\left(\frac{W}{L}\right)_n V_{DSsat}\left(V_{DD} - V_t - \frac{1}{2}V_{DSsat}\right)$ for I_{Dsat} in the equation.

$$\mu_n C_{ox}\left(\frac{W}{L}\right)_n V_{DSsat}\left(V_{DD} - V_t - \frac{1}{2}V_{DSsat}\right) = \frac{V_{DD}\left(\frac{W}{L}\right)_n}{2(0.69)(12.5\times10^3)}$$

$$\mu_n C_{ox} V_{DSsat}\left(V_{DD} - V_t - \frac{1}{2}V_{DSsat}\right) = \frac{V_{DD}}{2(0.69)(12.5\times10^3)} \quad \ldots\ldots (1)$$

Step 4 of 4

Substitute $1.2\ \text{V}$ for V_{DD}, $0.4\ \text{V}$ for V_t, $325\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$ in equation (1).

$$(325\times10^{-6})V_{DSsat}\left(1.2 - 0.4 - \frac{1}{2}V_{DSsat}\right) = \frac{1.2}{2(0.69)(12.5\times10^3)}$$

$$0.8V_{DSsat} - 0.5(V_{DSsat})^2 = 0.214$$

$$0.5(V_{DSsat})^2 - 0.8V_{DSsat} + 0.214 = 0$$

$$V_{DSsat} = 1.26\ \text{V} \text{ and } 0.339\ \text{V}$$

The value of $V_{DSsat} = 1.26\ \text{V}$ is greater than the supply voltage. So, this value is not acceptable.

Thus, the value of voltage, V_{DSsat} for NMOS transistor is $\boxed{0.339\ \text{V}}$.

Step 1 of 11

Refer to Example 15.2 in the textbook.

Determine the value of V_{OL}.

$$V_{OL} = (V_{DD} - V_t)\left(1 - \sqrt{1 - \frac{1}{r}}\right)$$

Substitute 2.5V for the drain voltage V_{DD}, 0.5V for the threshold voltage V_t, and 4 for the ratio r.

$$\begin{aligned} V_{OL} &= (2.5 - 0.5)\left(1 - \sqrt{1 - \frac{1}{4}}\right) \\ &= (2)(1 - 0.866) \\ &= 0.268 \text{ V} \end{aligned}$$

Hence, the value of V_{OL} is $\boxed{0.268 \text{ V}}$.

Step 2 of 11

Determine the value of noise margin for logic LOW level.

$$NM_L = V_{IL} - V_{OL}$$

The expression for NM_L in terms of V_t, V_{DD} and r is,

$$\begin{aligned} NM_L &= V_t + \frac{V_{DD} - V_t}{\sqrt{r(r+1)}} - 0.268 \\ &= 0.5 + \frac{2.5 - 0.5}{\sqrt{4(4+1)}} - 0.268 \\ &= 0.232 + \frac{2}{2\sqrt{5}} \\ &= 0.68 \text{ V} \end{aligned}$$

Therefore, the value of noise margin, NM_L, is $\boxed{0.68 \text{ V}}$.

Step 3 of 11

Determine the value of noise margin for logic HIGH level.

$$NM_H = V_{OH} - V_{IH}$$

The expression for NM_H in terms of V_t, V_{DD} and r is,

$$\begin{aligned} NM_H &= V_{OH} - \left[V_t + \frac{2}{\sqrt{3r}}(V_{DD} - V_t)\right] \\ &= 2.5 - \left[0.5 + \frac{2}{\sqrt{3 \times 4}}(2.5 - 0.5)\right] \\ &= 2.5 - \left[0.5 + \frac{4}{3.464}\right] \\ &= 0.85 \text{ V} \end{aligned}$$

Therefore, the value of noise margin, NM_H, is $\boxed{0.85 \text{ V}}$.

Step 4 of 11

Determine the value of width-to-length ratio of PMOS transistor.

$$r = \frac{\mu_n C_{ox}\left(\frac{W}{L}\right)_n}{\mu_p C_{ox}\left(\frac{W}{L}\right)_p}$$

$$4 = \frac{115 \times \frac{0.375}{0.25}}{30 \times \left(\frac{W}{L}\right)_p}$$

The value of $\left(\frac{W}{L}\right)_p$ is,

$$\begin{aligned} \left(\frac{W}{L}\right)_p &= \frac{115 \times 1.5}{30 \times 4} \\ &= \frac{172.5}{120} \\ &= 1.44 \end{aligned}$$

Therefore, the value of width-to-length ratio of PMOS transistor, $\left(\frac{W}{L}\right)_p$, is $\boxed{1.44}$.

Step 5 of 11

Determine the value of dc current when the output is low, I_{stat}.

$$\begin{aligned} I_{stat} &= \frac{1}{2}k_p(V_{DD} - V_t)^2 \\ &= \frac{1}{2}\left(\mu_p C_{ox}\left(\frac{W}{L}\right)_p\right)(V_{DD} - V_t)^2 \end{aligned}$$

Substitute 30 μA/V^2 for $\mu_p C_{ox}$, 1.44 for $\left(\frac{W}{L}\right)_p$, 2.5V for the drain voltage V_{DD}, and 0.5V for the threshold voltage V_t. (Refer to Example 14.1 in the textbook).

$$\begin{aligned} I_{stat} &= \frac{1}{2}\left(30 \times 10^{-6}(1.44)\right)(2.5 - .05)^2 \\ &= \frac{1}{2}\left(43.2 \times 10^{-6}\right)(4) \\ &= 86.3 \text{ μA/V} \end{aligned}$$

Hence, the value of drain saturation current, I_{stat}, is $\boxed{86.3 \text{ μA}}$.

Step 6 of 11

Determine the value of power dissipation.

$$\begin{aligned} P_D &= I_{stat} V_{DD} \\ &= \left(86.3 \times 10^{-6}\right)(2.5) \\ &= 0.21 \text{mW} \end{aligned}$$

Hence, the power dissipated, P_D, is $\boxed{0.21 \text{mW}}$.

Step 7 of 11

Determine the value of α_p.

$$\begin{aligned} \alpha_p &= \frac{2}{\frac{7}{4} - 3\left(\frac{V_t}{V_{DD}}\right) + \left(\frac{V_t}{V_{DD}}\right)^2} \\ &= \frac{2}{\frac{7}{4} - 3\left(\frac{0.5}{2.5}\right) + \left(\frac{0.5}{2.5}\right)^2} \\ &= \frac{2}{1.75 - 0.6 + 0.04} \\ \alpha_p &= 1.68 \end{aligned}$$

Step 8 of 11

Determine the value of low-to-high propagation delay, t_{PLH}.

$$t_{PLH} = \frac{\alpha_p C}{k_p V_{DD}}$$

Substitute 7 fF for C, 1.68 for α_p, and 2.5V for V_{DD}.

Substitute $\mu_p C_{ox}\left(\frac{W}{L}\right)_p$ in place of k_p.

$$\begin{aligned} t_{PLH} &= \frac{\alpha_p C}{\mu_p C_{ox}\left(\frac{W}{L}\right)_p V_{DD}} \\ &= \frac{1.68 \times 7 \times 10^{-15}}{30 \times 10^{-6} \times 1.44 \times 2.5} \\ &= 0.11 \text{ ns} \end{aligned}$$

Thus, the value of low-to-high propagation delay, t_{PLH}, is $\boxed{0.11 \text{ ns}}$.

Step 9 of 11

Determine the value of α_n.

$$\begin{aligned} \alpha_n &= \frac{2}{1 + \frac{3}{4}\left(1 - \frac{1}{r}\right) - \left(3 - \frac{1}{r}\right)\left(\frac{V_t}{V_{DD}}\right) + \left(\frac{V_t}{V_{DD}}\right)^2} \\ &= \frac{2}{1 + \frac{3}{4}\left(1 - \frac{1}{4}\right) - \left(3 - \frac{1}{4}\right)\left(\frac{0.5}{2.5}\right) + \left(\frac{0.5}{2.5}\right)^2} \\ &= \frac{2}{1 + 0.5625 - (2.75)(0.2) + (0.04)} \\ \alpha_n &= 1.9 \end{aligned}$$

Step 10 of 11

The value of high-to-low propagation delay, t_{PHL}, is,

$$t_{PHL} = \frac{\alpha_n C}{k_n V_{DD}}$$

Substitute 7 fF for C, 1.68 for α_p, and 2.5V for V_{DD}.

Substitute $\mu_n C_{ox}\left(\frac{W}{L}\right)_n$ in place of k_n.

The value of $\mu_n C_{ox}$ is 115 μA/V and $\left(\frac{W}{L}\right)_n$ is $\frac{0.375 \text{ μm}}{0.25 \text{ μm}}$.

(Refer to Example 15.2 in the textbook).

$$\begin{aligned} t_{PHL} &= \frac{\alpha_n C}{\mu_n C_{ox}\left(\frac{W}{L}\right)_n V_{DD}} \\ &= \frac{1.9 \times 7 \times 10^{-15}}{115 \times 10^{-6} \times 1.5 \times 2.5} \\ &= 0.03 \text{ ns} \end{aligned}$$

Thus, the value of high-to-low propagation delay, t_{PHL}, is $\boxed{0.03 \text{ ns}}$.

Step 11 of 11

Determine the value of propagation delay.

$$\begin{aligned} t_p &= \frac{1}{2}(t_{PLH} + t_{PHL}) \\ &= \frac{1}{2}(0.03 + 0.11) \\ t_p &= 0.07 \text{ ns} \end{aligned}$$

Therefore, the propagation delay, t_p, is $\boxed{0.07 \text{ ns}}$.

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The high output voltage, V_{OH} , is,

$$V_{OH} = V_{DD} - V_t$$
$$= 1.8 - 0.65$$
$$V_{OH} = 1.15\,\text{V}$$

Therefore, the value of V_{OH} is $\boxed{1.15\,\text{V}}$.

Step 1 of 8

Consider I_S is the small amount of current that flows in sub threshold condition.

Consider i_D is the small amount of drain current that flows for $V_{GS} < V_t$.

$$i_D = I_S e^{\frac{v_{GS}}{nV_T}} \quad \text{..... (1)}$$

Consider $V_T = \frac{kT}{q}$ is the thermal voltage, n is a constant, and I_S is the saturation current.

Write the equation for I_S .

$$I_S = i_D e^{\frac{-V_t}{nV_T}}$$

Consider I_S is proportional to $\frac{-V_t}{e^{nV_T}}$ . ..... (2)

Step 2 of 8

Consider the threshold voltage of the NMOS transistor is reduced by 0.1 V .

Determine I_S after reducing V_t by 0.1.

Replace 0.025 for V_T , and 2 for n in equation 2.

$$\begin{aligned} I_S\Big|_{\substack{\text{after reducing} \\ \text{in } V_t}} &= e^{\frac{-(V_t - 0.1)}{nV_T}} \\ &= \left(\frac{-V_t}{e^{nV_t}} \times e^{\frac{0.1}{nV_t}} \right) \\ &= I_S \left(e^{\frac{(0.1)}{2(0.025)}} \right) \\ &= 7.39 I_S \text{ A} \end{aligned}$$

The current is increased by 7.39 times.

Step 3 of 8

Determine the drain current i_D .

Replace 0 for V_{GS} in equation 1.

$$\begin{aligned} i_D &= I_S e^{\frac{v_{GS}}{nV_T}} \\ &= 7.39 I_S e^{\frac{0}{2(0.025)}} \\ &= 7.39 I_S \end{aligned}$$

The drain current is also increased 7.39 times.

Step 4 of 8

Determine the static power dissipation.

$$\begin{aligned} P_{disp}\Big|_{\substack{\text{after reduction} \\ \text{in } V_t}} &= (i_D)(V_{DD}) \\ &= (7.39 i_D)(V_{DD}) \\ &= 7.39 (i_D)(V_{DD}) \\ &= 7.39 (P_{disp}) \end{aligned}$$

Therefore, the static power dissipation is also increased by 7.39 times.

Step 5 of 8

Consider the threshold voltage of the NMOS transistor is reduced by 0.2 V .

Determine I_S after reducing V_t by 0.2.

Replace 0.025 for V_T , and 2 for n in equation 2.

$$\begin{aligned} I_S\Big|_{\substack{\text{after reducing} \\ \text{in } V_t}} &= e^{\frac{-(V_t - 0.2)}{nV_T}} \\ &= \left(\frac{-V_t}{e^{nV_t}} \times e^{\frac{0.2}{nV_t}} \right) \\ &= I_S \left(e^{\frac{(0.2)}{2(0.025)}} \right) \\ &= 54.5 I_S \text{ A} \end{aligned}$$

The current is increased by 54.5 times.

Step 6 of 8

Determine the drain current i_D .

Replace 0 for V_{GS} in equation 1.

$$\begin{aligned} i_D &= I_S e^{\frac{v_{GS}}{nV_T}} \\ &= 54.5 I_S e^{\frac{0}{2(0.025)}} \\ &= 54.5 I_S \end{aligned}$$

The drain current is also increased 54.5 times.

Step 7 of 8

Determine the static power dissipation .

$$\begin{aligned} P_{disp}\Big|_{\substack{\text{after reduction} \\ \text{in } V_t}} &= (i_D)(V_{DD}) \\ &= (54.5 i_D)(V_{DD}) \\ &= 54.5 (i_D)(V_{DD}) \\ &= 54.5 (P_{disp}) \end{aligned}$$

Therefore, the static power dissipation is also increased by 54.5 times.

Step 8 of 8

Consider if the threshold voltage is reduced by 0.1 $\mathbf{V}$ then the amount of power dissipated by the inverter is increase by 7.39 times.
If the threshold voltage is reduced by 0.2 $\mathbf{V}$ then the amount of power dissipated by inverter is increased by 54.5 times.

Therefore, to reduce the static power dissipation, the current that flows for $v_{GS} = 0$ must be reduced.

Therefore, the current that flows for $v_{GS} = 0$ can be reduced by selecting **high threshold voltage.**.

Step 1 of 15

Refer to Figure 14.12(a) and Figure 14.12(b) in the textbook which shows the transmission gate with high and low inputs respectively.

(a)

In the absence of loading, the output voltage when the input is high, the transistor Q_p will be in triode region and the capacitor will charge to V_{DD} voltage.

Hence, the value of V_{OH} is $\boxed{5\text{ V}}$.

Similarly, the output voltage when input is low, the transistor Q_n conducts until the capacitor fully discharges to 0V.

Hence, the value of V_{OL} is $\boxed{0\text{ V}}$.

Step 2 of 15

(b)

The current in NMOS at time $t = 0$ is,

$$\begin{aligned} i_{DN}(0) &= \frac{1}{2}k_n'\left(\frac{W}{L}\right)_n (V_{DD} - V_{t0})^2 \\ &= \frac{1}{2}\left(50\ \frac{\mu\text{A}}{\text{V}^2}\right)\left(\frac{4\ \mu\text{m}}{2\ \mu\text{m}}\right)(5\text{ V} - 1\text{ V})^2 \\ &= \frac{1}{2}(50)(16)\ \mu\text{A} \\ &= 800\ \mu\text{A} \end{aligned}$$

Therefore, the current in NMOS at time $t = 0$ is: $\boxed{i_{DN}(0) = 800\ \mu\text{A}}$.

Step 3 of 15

The current in PMOS at time $t = 0$ is,

$$\begin{aligned} i_{DP}(0) &= \frac{1}{2}k_p'\left(\frac{W}{L}\right)_p (V_{DD} - V_{t0})^2 \\ &= \frac{1}{2}\left(20\ \frac{\mu\text{A}}{\text{V}^2}\right)\left(\frac{4\ \mu\text{m}}{2\ \mu\text{m}}\right)(5\text{ V} - 1\text{ V})^2 \\ &= \frac{1}{2}(20)(2)(16)\ \mu\text{A} \\ &= 320\ \mu\text{A} \end{aligned}$$

Therefore, the current in NMOS at time $t = 0$ is: $\boxed{i_{DP}(0) = 320\ \mu\text{A}}$.

The current in capacitor at time $t = 0$ is,

$$\begin{aligned} i_C(0) &= i_{DP}(0) + i_{DN}(0) \\ &= 320\ \mu\text{A} + 800\ \mu\text{A} \end{aligned}$$

$$i_C(0) = 1120\ \mu\text{A} \quad \ldots\ldots (1)$$

Step 4 of 15

The value of V_m is,

$$V_m = V_{t0} + \gamma\left(\sqrt{v_0 + 2\phi_f} - \sqrt{2\phi_f}\right)$$

$$\begin{aligned} V_m &= 1 + 0.5\left(\sqrt{\left(\frac{5}{2}\right) + 0.6} - \sqrt{0.6}\right) \\ &= 1.493\text{ V} \end{aligned}$$

The value of k_n is,

$$\begin{aligned} k_n &= k_n'\left(\frac{W}{L}\right)_n \\ &= 50 \times \left(\frac{4\ \mu\text{m}}{2\ \mu\text{m}}\right) \\ &= 100\ \mu\text{A/V}^2 \end{aligned}$$

The value of $i_{DN}(t_{PLH})$ as the transistor Q_n is operating in saturation is,

$$\begin{aligned} i_{DN}(t_{PLH}) &= \frac{1}{2}k_n(V_{DD} - v_0 - V_m)^2 \\ &= \frac{1}{2}(100 \times 10^{-6})\left(5 - \frac{5}{2} - 1.493\right)^2 \\ &= 50.70\ \mu\text{A} \end{aligned}$$

Hence, the value of $i_{DN}(t_{PLH})$ is, $\boxed{50.70\ \mu\text{A}}$.

Step 5 of 15

The value of V_{tp} is,

$$\begin{aligned} |V_{tp}| &= V_m \\ &= 1.493\text{ V} \end{aligned}$$

The value of k_p is,

$$\begin{aligned} k_p &= k_p'\left(\frac{W}{L}\right)_p \\ &= 20\left(\frac{4\ \mu\text{m}}{2\ \mu\text{m}}\right) \\ &= 40\ \mu\text{A/V}^2 \end{aligned}$$

Step 6 of 15

The value of $i_{DP}(t_{PLH})$ when the transistor Q_p operates in triode region is,

$$i_{DP}(t_{PLH}) = k_p\left[(V_{DD} - V_t)(V_{DD} - v_0) - \frac{1}{2}(V_{DD} - v_0)^2\right]$$

$$\begin{aligned} i_{DP}(t_{PLH}) &= \left(20\ \frac{\mu\text{A}}{\text{V}^2}\right)\left(\frac{4}{2}\right)\left[(5\text{ V} - 1\text{ V})\frac{(5\text{ V})}{2} - \frac{1}{2}\left(\frac{5\text{ V}}{2}\right)^2\right] \\ &= (40)(6.825)\ \mu\text{A} \\ &= 275\ \mu\text{A} \end{aligned}$$

The value of the current, $i_{DP}(t_{PLH})$, is $\boxed{275\ \mu\text{A}}$.

Step 7 of 15

The current through the capacitor at t_{PLH} is,

$$\begin{aligned} i_C(t_{PLH}) &= i_{DP}(t_{PLH}) + i_{DN}(t_{PLH}) \\ &= 50\ \mu\text{A} + 275\ \mu\text{A} \end{aligned}$$

$$i_C(t_{PLH}) = 325\ \mu\text{A} \quad \ldots\ldots (2)$$

The average current through the capacitor is,

$$\begin{aligned} i_{C(avg)} &= \frac{1}{2}(1120\ \mu\text{A} + 325\ \mu\text{A}) \\ &= 722.5\ \mu\text{A} \end{aligned}$$

The low-to-high propagation delay is,

$$\begin{aligned} t_{PLH} &= \frac{C\left(\frac{V_{DD}}{2}\right)}{i_{C(avg)}} \\ &= \frac{(70\text{ fF})\left(\frac{5\text{ V}}{2}\right)}{722.5\ \mu\text{A}} \\ &= \frac{(70 \times 10^{-15}\text{ F})(2.5\text{ V})}{722.5 \times 10^{-6}\text{ A}} \\ &= 0.24\text{ ns} \end{aligned}$$

The value of t_{PLH} is $\boxed{0.24\text{ ns}}$.

Step 8 of 15

(c)

The value of $i_{DN}(0)$ is,

$$\begin{aligned} i_{DN}(0) &= \frac{1}{2}k_n'\left(\frac{W}{L}\right)_n (V_{DD} - V_{t0})^2 \\ &= \frac{1}{2}\left(50\ \frac{\mu\text{A}}{\text{V}^2}\right)\left(\frac{4\ \mu\text{m}}{2\ \mu\text{m}}\right)(5\text{ V} - 1\text{ V})^2 \\ &= \frac{1}{2}(50)(32)\ \mu\text{A} \\ &= 800\ \mu\text{A} \end{aligned}$$

Therefore, the current in NMOS at time $t = 0$, $i_{DN}(0)$, is $\boxed{800\ \mu\text{A}}$.

Step 9 of 15

The value of $i_{DP}(0)$ is,

$$\begin{aligned} i_{DP}(0) &= \frac{1}{2}k_p'\left(\frac{W}{L}\right)_p (V_{DD} - V_{t0})^2 \\ &= \frac{1}{2}\left(20\ \frac{\mu\text{A}}{\text{V}^2}\right)\left(\frac{4\ \mu\text{m}}{2\ \mu\text{m}}\right)(5\text{ V} - 1\text{ V})^2 \\ &= \frac{1}{2}(20)(2)(16)\ \mu\text{A} \\ &= 320\ \mu\text{A} \end{aligned}$$

Therefore, the current in NMOS at time $t = 0$, $i_{DP}(0)$, is $\boxed{320\ \mu\text{A}}$.

The current in capacitor at time $t = 0$ is,

$$\begin{aligned} i_C(0) &= i_{DP}(0) + i_{DN}(0) \\ &= 320\ \mu\text{A} + 800\ \mu\text{A} \\ &= 1120\ \mu\text{A} \end{aligned}$$

Step 10 of 15

The drain current for the NMOS at $t = t_{PHL}$ is,

$$\begin{aligned} i_{DP}(t_{PHL}) &= k_n'\left(\frac{W}{L}\right)_n \left[(V_{DD} - V_{t0})\frac{V_{DD}}{2} - \frac{1}{2}\left(\frac{V_{DD}}{2}\right)^2\right] \\ &= \left(50\ \frac{\mu\text{A}}{\text{V}^2}\right)\left(\frac{4}{2}\right)\left[(5\text{ V} - 1\text{ V})\frac{(5\text{ V})}{2} - \frac{1}{2}\left(\frac{5\text{ V}}{2}\right)^2\right] \\ &= (100)(6.882)\ \mu\text{A} \\ &= 688.2\ \mu\text{A} \end{aligned}$$

Therefore, the drain current in the PMOS at $t = t_{PHL}$, $i_{DD}(t_{PHL})$ is $\boxed{688.2\ \mu\text{A}}$.

Step 11 of 15

The PMOS will stop conduction when the output voltage falls to V_{tp}.

The value of the voltage V_{tp} is,

$$\begin{aligned} |V_{tp}| &= V_{t0} + \gamma\left[\sqrt{V_{DD} - v_O + 2\phi_f} - \sqrt{2\phi_f}\right] \\ &= (1\text{ V}) + \left(\frac{1}{2}\right)\left[\sqrt{5\text{ V} - 0 + 0.6} - \sqrt{0.6\text{ V}}\right]\text{ V} \\ &= 1.49\text{ V} \end{aligned}$$

Step 12 of 15

The drain current in the PMOS is,

$$i_{DP}(t_{PHL}) = \frac{1}{2}k_p'\left(\frac{W}{L}\right)_p \left(V_{SG} - |V_{tp}|\right)^2$$

At the time instant t_{PHL}, the value of V_{SG} will be $\frac{V_{DD}}{2}$.

$$\begin{aligned} i_{DP}(t_{PHL}) &= \frac{1}{2}\left(20\ \frac{\mu\text{A}}{\text{V}^2}\right)\left(\frac{4\ \mu\text{m}}{2\ \mu\text{m}}\right)\left(\frac{5}{2}\text{V} - 1.49\text{ V}\right)^2 \\ &= 20\ \mu\text{A} \end{aligned}$$

Therefore, the value of $i_{DP}(t_{PHL})$ is $\boxed{20\ \mu\text{A}}$.

Step 13 of 15

The average current in the capacitor is,

$$\begin{aligned} i_{C(avg)} &= \frac{1}{2}(1120\ \mu\text{A} + 708\ \mu\text{A}) \\ &= 914\ \mu\text{A} \end{aligned}$$

The propagation delay from high to low is,

$$\begin{aligned} t_{PHL} &= \frac{C\left(\frac{V_{DD}}{2}\right)}{i_{C(avg)}} \\ &= \frac{(70\text{ fF})\left(\frac{5\text{ V}}{2}\right)}{914\ \mu\text{A}} \\ &= \frac{(70 \times 10^{-15}\text{ F})(2.5\text{ V})}{914 \times 10^{-6}\text{ A}} \\ &= 0.192\text{ ns} \end{aligned}$$

Therefore, the value of propagation delay, t_{PHL}, is $\boxed{0.192\text{ ns}}$.

Step 14 of 15

The transistor Q_p will be turned off when the output voltage is $v_O = |V_{tp}|$.

$$|V_{tp}| = V_{t0} + \gamma\left[\sqrt{V_{DD} - |V_{tp}| + 2\phi_f} - \sqrt{2\phi_f}\right]$$

$$\left(|V_{tp}| - 1\text{ V} + \frac{1}{2}\sqrt{0.6}\right)^2 = \frac{1}{2}\left(5\text{ V} - |V_{tp}| + 0.6\text{ V}\right)$$

$$|V_{tp}|^2 - 0.976|V_{tp}| - 1.024 = 0$$

Therefore, the voltage when the transistor, Q_p, will be turned off is $\boxed{|V_{tp}| = 1.6\text{ V}}$.

Step 15 of 15

(d)

The total propagation delay of the transmission gate is the average of the low-to-high propagation delay and the high-to-low propagation delay.

$$\begin{aligned} t_p &= \frac{1}{2}(t_{PLH} + t_{PHL}) \\ &= \frac{1}{2}(0.24\text{ ns} + 0.19\text{ ns}) \\ &= \frac{1}{2}(0.43\text{ ns}) \\ &= 0.22\text{ ns} \end{aligned}$$

Therefore, the propagation delay, t_p, is $\boxed{0.22\text{ ns}}$.

15.009P

Step 1 of 5

(a)

Consider a MOSFET is operating in sub-threshold conduction region.

Consider the current changes by a factor of 10 for every 80- $\mathbf{mV}$ change in v_{GS} .

Determine the slope of the $i_D - v_{GS}$ characteristics of the MOSFET.

$$\text{Slope} = \frac{\text{change in drain current}}{\text{change in } v_{GS} \text{ voltage}}$$

$$\text{Slope} = \frac{10}{80\,\text{mV}}$$

Step 2 of 5

Consider the inverse of the slope of the characteristics of the MOSFET is $(2.3nV_T)\ V/\text{decade}$.

$$(2.3nV_T)\ V/\text{decade} = \frac{1}{\left(\frac{10}{80\,\text{mV}}\right)}$$

$$(2.3nV_T)\ V/\text{decade} = \frac{80\,\text{mV}}{10} \quad \text{..... (1)}$$

Replace 0.025 V for V_T in equation 1.

$$2.3nV_T = 80\ \text{mV}$$

$$n = \frac{80\ \text{mV}}{2.3V_T}$$

$$= \frac{80\ \text{mV}}{2.3(0.025)}$$

$$= 1.4$$

Step 3 of 5

Consider i_D drain current.

$$i_D = I_S e^{\frac{v_{GS}}{nV_T}} \quad \text{..... (2)}$$

Replace 0.16 V for v_{GS} , 20 $\mathbf{nA}$ for i_D , and $n = 1.4$ in equation 2.

$$i_D = I_S e^{\frac{v_{GS}}{nV_T}}$$

$$(20\times10^{-9}) = I_S e^{\frac{0.16}{(1.4)(25\times10^{-3})}}$$

$$I_S = \frac{(20\times10^{-9})}{\left(e^{\frac{0.16}{(1.4)(25\times10^{-3})}}\right)}$$

$$= 0.207\ \text{nA}$$

Step 4 of 5

Determine the drain current in sub-threshold conduction.

Write the equation for drain current in sub threshold conduction $(v_{GS} = 0)$.

$$i_D = I_S e^{\frac{v_{GS}}{nV_T}}$$

Replace 0 for v_{GS} , and 0.207 $\mathbf{nA}$ for I_S , and 0.025 V for V_T in equation 1.

$$i_D = I_S e^{\frac{v_{GS}}{nV_T}}$$

$$= (0.207\times10^{-9}) e^{\frac{0}{nV_T}}$$

$$= 0.207\ \text{nA}$$

Therefore, the drain current in sub-threshold conduction is $\boxed{0.207\ \text{nA}}$.

Step 5 of 5

(b)

Refer to sub part (a).

The drain current in sub-threshold conduction is 0.207 $\mathbf{nA}$.

Determine the drain current for 1 billion transistors in sub-threshold conduction.

$$i_D = (1\times10^9)(0.207\ \text{nA})$$

$$= 0.207\ \text{A}$$

Determine the static power dissipation.

Consider $V_{DD} = 1\ \text{V}$.

$$P_d = (i_D)(V_{DD})$$

$$= (0.207)(1)$$

$$= 0.207\ \text{W}$$

Therefore, the drain current that flows for 1 billion transistors is $\boxed{0.207\ \text{A}}$ and the static power dissipated through MOSFET in sub-threshold conduction is $\boxed{0.207\ \text{W}}$.

Step 1 of 2

15.010E

The average resistance can be expressed as,

$$R_{avg} = \frac{R_{TG(0\ V)} + R_{TG(2.5\ V)}}{2}$$

Substitute $4.5\ \text{k}\Omega$ for the resistance at $v_O = 0\ \text{V}$, that is $R_{TG(0\ V)}$ and $6.5\ \text{k}\Omega$ for the resistance $v_O = 2.5\ \text{V}$, that is $R_{TG(2.5\ V)}$.

$$\begin{aligned} R_{avg} &= \frac{4.5\ \text{k}\Omega + 6.5\ \text{k}\Omega}{2} \\ &= \frac{11\ \text{k}\Omega}{2} \\ &= 5.5\ \text{k}\Omega \end{aligned}$$

Step 2 of 2

The charging delay of the transmission gate, t_{PLH}, is,

$$t_{PLH} = 0.69 R_{avg} C$$

Substitute 70 fF for *C* and $5.5\ \text{k}\Omega$ for R_{avg}.

$$\begin{aligned} t_{PLH} &= 0.69\left(5.5\times10^{3}\right)\left(70\times10^{-15}\right) \\ &= 265.65\times10^{-12} \\ &= 0.265\times10^{-9}\ \text{s} \end{aligned}$$

Therefore, the charging delay of transmission gate, t_{PLH}, is $\boxed{0.265\ \text{ns}}$.

Step 1 of 5

(a)

Consider a NMOS transistor is operating in saturation region.

Consider $0.2\ \text{mA}$ is the drain current I_D in saturation region.

Determine the gate to source voltage V_{GS}.

consider the equation for drain current in saturation region.

$$I_{Dsat}=\frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)(V_{GS}-V_t)^2$$

$$I_{Dsat}=\frac{1}{2}k_n(V_{GS}-V_t)^2 \quad \text{..... (1)}$$

Step 2 of 5

Replace $0.4\ \text{mA/V}^2$ for k_n, 0.2 mA for I_{Dsat}, and 0.4 for V_t in equation 1.

$$(0.2\times10^{-3})=\frac{1}{2}(0.4\times10^{-3})(V_{GS}-0.4)^2$$

$$(V_{GS}-0.4)^2=\frac{2(0.2\times10^{-3})}{(0.4\times10^{-3})}$$

$$(V_{GS}-0.4)^2=1$$

$$V_{GS}=0.6\ \text{V}$$

Step 3 of 5

Consider threshold voltage (V_t) is varied as $\pm10\%$.

V_t is ranged from $0.3\ \text{V}-0.5\ \text{V}$.

Determine the range of I_D.

Consider $V_t=0.3$.

$$I_{Dsat}=\frac{1}{2}k_n(V_{GS}-V_t)^2 \quad \text{..... (2)}$$

Replace $0.4\ \text{mA/V}^2$ for k_n, 0.6 V for V_{GS}, and 0.3 for V_t in equation 2.

$$I_{Dsat}=\frac{1}{2}(0.4\times10^{-3})(0.6-0.3)^2$$

$$=18\ \mu\text{A}$$

Step 4 of 5

Consider $V_t=0.5$.

$$I_{Dsat}=\frac{1}{2}k_n(V_{GS}-V_t)^2 \quad \text{..... (3)}$$

Replace $0.4\ \text{mA/V}^2$ for k_n, 0.6 V for V_{GS}, and 0.5 for V_t in equation 3.

$$I_{Dsat}=\frac{1}{2}(0.4\times10^{-3})(0.6-0.5)^2$$

$$=2\ \mu\text{A}$$

Therefore, the drain current varies from $\boxed{2\ \mu\text{A to}18\ \mu\text{A}}$.

Step 5 of 5

(b)

Consider the drain current in NMOS transistor is discharged by a $100\ \text{fF}$ capacitor.

Write the equation for the current through load capacitor.

$$t_{Pe}=\frac{C(\Delta v_0)}{2I_{Dsat}} \quad \text{..... (4)}$$

Substitute $100\ \text{fF}$ for C, 0.1 for dV_0, and 0.2◊ mA for I_{sat} in equation 4.

$$t_{Pe}=\frac{C(\Delta v_0)}{2I_{Dsat}}$$

$$=\frac{(100\times10^{-15})(0.1)}{2(0.2\times10^{-3})}$$

$$=25\ \text{ns}$$

Therefore, the variation in delay time for the change in output voltage is $\boxed{25\ \text{ns}}$.

The value of the equivalent resistance of the transmission gate, R_{TG} , is,

$$R_{TG} = \frac{12.5}{\left(\frac{W}{L}\right)_n} \text{ k}\Omega$$

Substitute, 1.5 for the width-to-length ratio of the NMOS transistor, $\left(\frac{W}{L}\right)_n$.

$$R_{TG} = \frac{12.5}{1.5} \text{ k}\Omega$$
$$= 8.333 \text{ k}\Omega$$

Therefore, the value of the equivalent resistance of the transmission gate, R_{TG} , is $\boxed{8.333 \text{ k}\Omega}$.

Step 1 of 4

(a)

Refer to Figure 15.11 in the textbook.

Consider the resistance of the aluminum wire is $R = \rho\frac{L}{A}$.

Consider $\frac{\rho}{T}$ is the sheet resistance of the wire.

Determine the resistance of aluminum wire.

$$R = \rho\frac{L}{A}$$

$$R = \frac{\rho L}{WT} \quad \text{..... (1)}$$

Step 2 of 4

Replace 5 $\mathbf{mm}$ for L, 0.5- $\mu\mathbf{m}$ for W, and 27 $\Omega/\mathbf{square}$ for $\frac{\rho}{T}$ in equation 1.

$$\begin{aligned} R &= \frac{\rho L}{WT} \\ &= \left(\frac{\rho}{T}\right)\left(\frac{L}{W}\right) \\ &= \left(27\times10^{-3}\right)\left(\frac{5\times10^{-3}}{0.5\times10^{-6}}\right) \\ &= 270\ \Omega \end{aligned}$$

Therefore, the resistance of the aluminum wire is $\boxed{270\ \Omega}$.

Step 3 of 4

(b)

Consider C_g is the wire capacitance to ground and L is the length of the wire.

Determine the total wire capacitance C.

$$C = C_g L \quad \text{..... (2)}$$

Replace 0.1- $fF/\mu\mathbf{m}$ for C_g, and 5 $\mathbf{mm}$ for L in equation 2.

$$\begin{aligned} C &= C_g L \\ &= \left(\frac{0.1\times10^{-15}}{10^{-6}}\right)\left(5\times10^{-3}\right) \\ &= 0.5\ \mathbf{pF} \end{aligned}$$

Therefore, the total capacitance of the wire is $\boxed{0.5\ \mathbf{pF}}$.

Step 4 of 4

(c)

Refer to Figure 15.11(b) in textbook.

Determine the delay introduced by the wire.

Consider R is the resistance of the wire and C is the capacitance of the wire.

$$t_{delay} = 0.69RC \quad \text{..... (3)}$$

Replace 270 Ω for R, and $0.5-\mathbf{pF}$ for C in equation 3.

$$\begin{aligned} t_{delay} &= 0.69RC \\ &= 0.69(270)\left(0.5\times10^{-12}\right) \\ &= 0.1863\ \mathbf{ns} \end{aligned}$$

Therefore, the delay introduced by the wire is $\boxed{0.1863\ \mathbf{ns}}$.

Step 1 of 3

Refer to Figure 15.23 in textbook.

Calculate the value of R_{P1} using the empirical relation. The value of the resistance is,

$$R_{P1} = \frac{30}{\left(\frac{W}{L}\right)_p}\ \text{k}\Omega$$

Substitute 2 for width-to-length ratio of PMOS transistor, $\left(\frac{W}{L}\right)_p$.

$$\begin{aligned} R_{P1} &= \frac{30}{2}\ \text{k}\Omega \\ &= 15\ \text{k}\Omega \end{aligned}$$

Therefore, the value resistance, R_{P1}, is $\boxed{15\ \text{k}\Omega}$.

Step 2 of 3

Calculate the value of R_{TG} using the empirical relation. The value of the resistance is,

$$R_{TG} = \frac{12.5}{\left(\frac{W}{L}\right)_n}\ \text{k}\Omega$$

Substitute, 1 for the width-to-length of NMOS transistor, $\left(\frac{W}{L}\right)_n$.

$$\begin{aligned} R_{TG} &= \frac{12.5}{1}\ \text{k}\Omega \\ &= 12.5\ \text{k}\Omega \end{aligned}$$

Therefore, the value resistance, R_{TG} is $\boxed{12.5\ \text{k}\Omega}$.

Step 3 of 3

The propagation delay for the transmission gate is,

$$t_p = 0.69\left[\left(C_{out1} + C_{TG1}\right)R_{P1} + \left(C_{in2} + C_{TG2}\right)\left(R_{P1} + R_{TG}\right)\right]$$

Substitute 10 fF for C_{out1}, 5 fF for C_{TG1}, 5 fF for C_{TG2}, $15\ \text{k}\Omega$ for R_{P1} and $12.5\ \text{k}\Omega$ for R_{TG}.

$$\begin{aligned} t_p &= 0.69\left[\left(10\ \text{fF} + 5\ \text{fF}\right)\left(15\ \text{k}\Omega\right) + \left(10\ \text{fF} + 5\ \text{fF}\right)\left(15\ \text{k}\Omega + 12.5\ \text{k}\Omega\right)\right] \\ &= 0.64\ \text{ns} \end{aligned}$$

Therefore, the propagation delay, t_p, is $\boxed{0.64\ \text{ns}}$.

Step 1 of 3

15.012P

Refer to Figure P14.1 (a) in the textbook for a pseudo-NMOS logic circuit with resistive load.

The propagation delay from low to high transition, t_{PLH} is the time at which the output reaches half of the supply voltage.

The expression for the output voltage for resistive load is,

$$v_O = V_{DD}\left(1 - e^{-\frac{t}{R_D C}}\right)$$

Find the time at which the output reaches half of supply voltage.

Observe that the following condition for the output to reach $\frac{V_{DD}}{2}$:

$$1 - e^{-\frac{t}{R_D C}} = 0.5$$

Simplify the expression to get the time at which the output reaches half of supply voltage.

$$e^{-\frac{t_{PLH}}{R_D C}} = 1 - 0.5$$

$$-\frac{t_{PLH}}{R_D C} = \ln(0.5)$$

$$t_{PLH} = 0.693 R_D C$$

Thus, the value of propagation delay from low to high transition, t_{PLH} for a pseudo-NMOS logic circuit with resistive load is, $\boxed{0.693 R_D C}$.

Step 2 of 3

Refer to Figure P14.1 (b) in the textbook for the pseudo-NMOS logic circuit with current-source load.

Find the time at which the output reaches half of supply voltage.

Consider the following condition for the output to reach $\frac{V_{DD}}{2}$:

$$t_{PLH} = \frac{C\left(\frac{V_{DD}}{2}\right)}{I}$$

Consider that the current source is,

$$I = \frac{V_{DD}}{R_D}$$

Substitute the expression of I in the expression of t_{PLH} .

$$\begin{aligned} t_{PLH} &= \frac{C\left(\frac{V_{DD}}{2}\right)}{\left(\frac{V_{DD}}{R_D}\right)} \\ &= C\left(\frac{V_{DD}}{2}\right)\left(\frac{R_D}{V_{DD}}\right) \\ &= C\left(\frac{R_D}{2}\right) \\ &= 0.5 R_D C \end{aligned}$$

Thus, the value of propagation delay from low to high transition, t_{PLH} for a pseudo-NMOS logic circuit with current-source load is, $\boxed{0.5 R_D C}$.

Step 3 of 3

Find the percentage reduction of propagation from low to high transition, t_{PLH} when the current source-load is used from the expressions of t_{PLH} for the both circuits.

$$\begin{aligned} \%of_{PLH} &= \frac{0.693 R_D C}{0.5 R_D C} \times 100 \\ &= \frac{0.693}{0.5} \times 100 \\ &= 1.38 \times 100 \\ &= 138\% \end{aligned}$$

Thus, the percentage reduction of propagation from low to high transition, t_{PLH} when the current source-load is used is, $\boxed{138\%}$.

Step 1 of 6

(a)

Refer to Figure 15.27 in the textbook.

Draw the following modified circuit diagram:

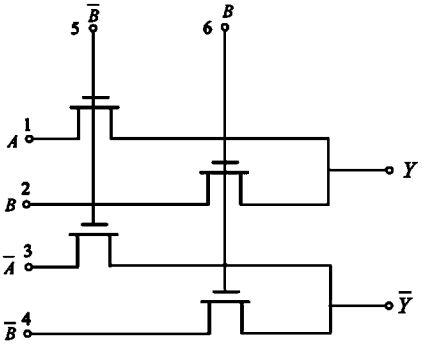


Figure 1

Step 2 of 6

From Figure 1, write the expression of output *Y*.

$$\begin{aligned} Y &= A\overline{B} + BB \\ &= A\overline{B} + B \\ &= (A+B)(\overline{B}+B) \\ &= A+B \end{aligned}$$

Therefore, the expression of *Y* is $\boxed{A+B}$.

Step 3 of 6

From Figure 1, write the expression of output $\overline{Y}$.

$$\begin{aligned} Y &= \overline{A\overline{B} + \overline{B}B} \\ &= \overline{A\overline{B} + 0} \\ &= \overline{A\overline{B}} \\ &= \overline{A+\overline{B}} \end{aligned}$$

Therefore, the expression of $\overline{Y}$ is $\boxed{\overline{A+B}}$.

Step 4 of 6

(b)

Draw the following modified circuit diagram:

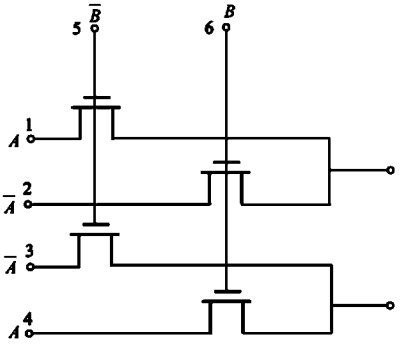


Figure 2

Step 5 of 6

From Figure 2, write the expression of output *Y*.

$$\begin{aligned} Y &= A\overline{B} + \overline{A}B \\ &= A \oplus B \end{aligned}$$

Therefore, the expression of *Y* is $\boxed{A \oplus B}$.

Step 6 of 6

From Figure 2, write the expression of output $\overline{Y}$.

$$\begin{aligned} Y &= \overline{A\overline{B} + AB} \\ &= A \odot B \end{aligned}$$

Therefore, the expression of $\overline{Y}$ is $\boxed{A \odot B}$.

Step 1 of 5

Consider a pseudo-NMOS transistor is fabricated with 0.13-μm technology.

Determine the output high voltage V_{OH}.

The maximum output voltage of the NMOS transistor is V_{DD}.

$$v_0 = V_{OH} = V_{DD}$$

Consider $V_{DD} = 1.3\ V$.

$$v_0 = V_{OH} = V_{DD}$$
$$V_{OH} = 1.3\ V$$

Therefore, the output voltage of the NMOS transistor is $\boxed{1.3\ V}$.

Step 2 of 5

Consider V_{tn} is the threshold voltage of NMOS transistor.

Determine the low value of the output voltage V_{OL}.

$$V_{OL} = (V_{DD} - V_{tn})\left[1 - \sqrt{1 - \frac{1}{r}}\right] \quad \text{..... (1)}$$

Consider $r = \frac{k_n}{k_p}$.

$$r = \frac{k_n}{k_p}$$
$$= \frac{500}{100}$$
$$= 5$$

Step 3 of 5

Replace 1.3 V for V_{DD}, 0.4 V for V_{tn}, and 5 for r in equation 1.

$$V_{OL} = (V_{DD} - V_{tn})\left[1 - \sqrt{1 - \frac{1}{r}}\right]$$
$$= (1.3 - 0.4)\left[1 - \sqrt{1 - \frac{1}{5}}\right]$$
$$= 0.9\left[1 - \sqrt{\frac{4}{5}}\right]$$
$$= 0.09\ V$$

Therefore, the low value of the output is $\boxed{0.09\ V}$.

Step 4 of 5

Determine the static current I_{stat} in the low output state.

$$I_{stat} = \frac{1}{2} k_p (V_{DD} - V_{tn})^2 \quad \text{..... (2)}$$

Replace 100-$\mu A/V^2$ for k_p, 1.3 for V_{DD}, and 0.4 V for V_{tn} in equation 2.

$$I_{stat} = \frac{1}{2} k_p (V_{DD} - V_{tn})^2$$
$$= \frac{1}{2}\left(100 \times 10^{-6}\right)(1.3 - 0.4)^2$$
$$= 40.5\ \mu A$$

Therefore, the static current of the NMOS transistor in the low output state is $\boxed{40.5\ \mu A}$.

Step 5 of 5

Determine the static power dissipation in the low output state.

$$P_{D\ stat} = I_{stat} V_{DD} \quad \text{..... (3)}$$

Replace 40.5-μA for I_{stat}, and 1.3 for V_{DD} in equation 3.

$$P_{D\ stat} = I_{stat} V_{DD}$$
$$= \left(40.5 \times 10^{-6}\right)(1.3)$$
$$= 52.65\ \mu W$$

Therefore, the static power dissipation in low output state is $\boxed{52.65\ \mu W}$.

Step 1 of 2

Refer to Figure 14.20 in the textbook.

The drain current, i_D , is,

$$i_D(V_{DD}) = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_{eqv}(V_{DD} - V_t)^2$$

Substitute, 1.8V for V_{DD} , 0.5V for V_t and 300 for $\mu_n C_{ox}$.

$$i_D(V_{DD}) = 300 \times 0.3(1.8 - 0.5)^2$$
$$= 152.1\ \mu A$$

When the value v_y is $\frac{V_{DD}}{2}$. the transistor Q_{eq} operates in triode region.

The drain current, i_D , is,

$$i_D\left(\frac{V_{DD}}{2}\right) = \mu_n C_{ox}\left(\frac{W}{L}\right)_{eqv}\left[(V_{DD} - V_t)\frac{V_{DD}}{2} - \frac{1}{2}\left(\frac{V_{DD}}{2}\right)^2\right]$$
$$= 2 \times 300 \times 0.3\left[(1.8 - 0.5)\frac{1.8}{2} - \frac{1}{2}\left(\frac{1.8}{2}\right)^2\right]$$
$$i_D\left(\frac{V_{DD}}{2}\right) = 137.8\ \mu A$$

The average current flowing through the capacitor is,

$$I_{av} = \frac{137.8 + 152.1}{2}$$
$$I_{av} = 144.95\ \mu A$$

Step 2 of 2

The propagation delay of the circuit, t_{PLH} , is,

$$t_{PLH} = \frac{C\left(V_{DD} - \frac{V_{DD}}{2}\right)}{I_{av}}$$

Substitute, 30fF for C , 1.8V for V_{DD} , and $144.95\ \mu A$ for I_{av} .

$$t_{PLH} = \frac{30 \times 10^{-15}(1.8 - 0.9)}{144.95 \times 10^{-6}}$$
$$= 0.19\ ns$$

The value of transmission delay, t_{PLH} , is $\boxed{0.19\ ns}$.

Step 1 of 9

Consider a pseudo-NMOS inverter is fabricated in 0.13-μm process.

Determine the low value of the output (V_{OL}).

$$V_{OL}=(V_{DD}-V_t)\left[1-\sqrt{1-\frac{1}{r}}\right] \quad \text{..... (1)}$$

Consider $r=\frac{k_n}{k_p}$

Consider $k_n=500\ \mu A/V^2$ and $k_p=100\ \mu A/V^2$.

$$\begin{aligned} r&=\frac{k_n}{k_p}\\ &=\frac{(500\times10^{-6})}{(100\times10^{-6})}\\ &=5 \end{aligned}$$

Step 2 of 9

Replace 1.3 V for V_{DD}, 0.4 V for V_t, and 5 for r in equation 1.

$$\begin{aligned} V_{OL}&=(V_{DD}-V_t)\left[1-\sqrt{1-\frac{1}{r}}\right]\\ &=0.095(1.3-0.4)\left[1-\sqrt{1-\frac{1}{5}}\right]\\ &=0.095 \end{aligned}$$

Therefore, the low value of the output (V_{OL}) is $\boxed{0.095\ \text{V}}$.

Step 3 of 9

Determine the value of V_{IL}.

$$V_{IL}=V_t+\frac{V_{DD}-V_t}{\sqrt{r(r+1)}} \quad \text{..... (2)}$$

Replace 1.3 V for V_{DD}, 0.4 V for V_t, and 5 for r in equation 2.

$$\begin{aligned} V_{IL}&=V_t+\frac{V_{DD}-V_t}{\sqrt{r(r+1)}}\\ &=0.4+\frac{1.3-0.4}{\sqrt{5(5+1)}}\\ &=0.56\ \text{V} \end{aligned}$$

Therefore, the value of V_{IL} is $\boxed{0.56\ \text{V}}$.

Step 4 of 9

Determine the threshold voltage V_M.

$$V_M=V_t+\frac{V_{DD}-V_t}{\sqrt{(r+1)}} \quad \text{..... (3)}$$

Replace 1.3 V for V_{DD}, 0.4 V for V_t, and 5 for r in equation 3.

$$\begin{aligned} V_M&=V_t+\frac{V_{DD}-V_t}{\sqrt{(r+1)}}\\ &=0.4+\frac{1.3-0.4}{\sqrt{(5+1)}}\\ &=0.76\ \text{V} \end{aligned}$$

Therefore, the threshold voltage V_M is $\boxed{0.76\ \text{V}}$.

Step 5 of 9

Determine the value of V_{IH}.

$$V_{IH}=V_t+\frac{2}{\sqrt{3r}}(V_{DD}-V_t) \quad \text{..... (4)}$$

Replace 1.3 V for V_{DD}, 0.4 V for V_t, and 5 for r in equation 4.

$$\begin{aligned} V_{IH}&=V_t+\frac{2}{\sqrt{3r}}(V_{DD}-V_t)\\ &=0.4+\frac{2}{\sqrt{3(5)}}(1.3-0.4)\\ &=0.864\ \text{V} \end{aligned}$$

Therefore, the value of the V_{IH} is $\boxed{0.866\ \text{V}}$.

Step 6 of 9

Determine the high value of the output V_{OH}.

The high value reached by the NMOS inverter is V_{DD}.

$$\begin{aligned} V_{OH}&=V_{DD}\\ &=1.3\ \text{V} \end{aligned}$$

Therefore, the high value of the output is $\boxed{1.3\ \text{V}}$.

Step 7 of 9

Determine the low value of noise margin NM_L.

$$NM_L=V_t-(V_{DD}-V_t)\left[1-\sqrt{1-\frac{1}{r}}-\frac{1}{\sqrt{r(r+1)}}\right] \quad \text{..... (5)}$$

Step 8 of 9

Replace 1.3 V for V_{DD}, 0.4 V for V_t, and 5 for r in equation 5.

$$\begin{aligned} NM_L&=V_t-(V_{DD}-V_t)\left[1-\sqrt{1-\frac{1}{r}}-\frac{1}{\sqrt{r(r+1)}}\right]\\ &=0.4-(1.3-0.4)\left[1-\sqrt{1-\frac{1}{5}}-\frac{1}{\sqrt{5(5+1)}}\right]\\ &=0.4-(1.3-0.4)[0.1055-0.1825]\\ &=0.4693\ \text{V} \end{aligned}$$

Therefore, the low value of the noise margin is $\boxed{0.4693\ \text{V}}$.

Step 9 of 9

Determine the high value of noise margin NM_H.

$$NM_H=(V_{DD}-V_t)\left(1-\frac{2}{\sqrt{3r}}\right) \quad \text{..... (6)}$$

Replace 1.3 V for V_{DD}, 0.4 V for V_t, and 5 for r in equation 6.

$$\begin{aligned} NM_H&=(V_{DD}-V_T)\left(1-\frac{2}{\sqrt{3r}}\right)\\ &=(1.3-0.4)\left(1-\frac{2}{\sqrt{3(5)}}\right)\\ &=0.4352\ \text{V} \end{aligned}$$

Therefore, the high value of the noise margin NM_H is $\boxed{0.4352\ \text{V}}$.

Step 1 of 5

(a)

Refer to Figure 14.22 in the textbook for a two-single input dynamic logic gates connected in cascade.

Refer to Figure E14.10 in the textbook for the equivalent circuit with a series combination of the transistors.

Consider that the equivalent transistor, Q_{eq1} is the series equivalent of the two transistors, Q_1 **and** Q_{n1} in the circuit. The width to length ratio of equivalent transistor, Q_{eq1} is,

$$\left(\frac{W}{L}\right)_{eq1} = \frac{1}{2}\left(\frac{W}{L}\right)$$

Consider that the width to length ratio for the NMOS transistors is,

$$\frac{W}{L} = \frac{4\ \mu m}{2\ \mu m}$$

Therefore,

$$\left(\frac{W}{L}\right)_{eq1} = \frac{1}{2}\left(\frac{4\ \mu m}{2\ \mu m}\right)$$
$$= 1$$

Consider that the equivalent transistor, Q_{eq2} is the series equivalent of the two transistors, Q_2 **and** Q_{n2} in the circuit. The width to length ratio of equivalent transistor, Q_{eq2} is,

$$\left(\frac{W}{L}\right)_{eq2} = \frac{1}{2}\left(\frac{4\ \mu m}{2\ \mu m}\right)$$
$$= 1$$

Therefore, the width to length ratio of the equivalent transistors, Q_{eq1} **and** Q_{eq2} is, $\boxed{1}$.

Step 2 of 5

(b)

Consider that the output of first logic gate, $v_{Y1} = V_{DD}$.

The transistor, Q_1 works in saturation region. Therefore, the expression for the drain current in saturation region of operation is,

$$i_{D1} = \frac{1}{2}k_n\left(\frac{W}{L}\right)_{eq1}(V_{DD} - V_t)^2$$

Here, k_n is the process transconductance parameter, V_{DD} is the supply voltage, and V_t is the threshold voltage.

Substitute $50\ \mu A/V^2$ for k_n , 1 for $\left(\frac{W}{L}\right)_{eq1}$, 5 V for V_{DD} , and 1 V for V_t .

$$i_{D1} = \frac{1}{2}(50\ \mu A/V^2)(1)(5\ V - 1\ V)^2$$
$$= 400\ \mu A$$

Consider that the output of first logic gate, $v_{Y1} = V_t$.

The transistor, Q_1 works in triode region. Therefore, the expression for the drain current in triode region of operation is,

$$i_{D1} = k_n\left(\frac{W}{L}\right)_{eq1}\left[(V_{DD} - V_t)V_t - \frac{1}{2}V_t^2\right]$$

Substitute $50\ \mu A/V^2$ for k_n , 1 for $\left(\frac{W}{L}\right)_{eq1}$, 5 V for V_{DD} , and 1 V for V_t .

$$i_{D1} = (50\ \mu A/V^2)(1)\left[(5\ V - 1\ V)(1\ V) - \frac{1}{2}(1\ V)^2\right]$$
$$= (50\ \mu A/V^2)(3.5\ V^2)$$
$$= 175\ \mu A$$

Thus, the values of the drain current i_{D1} are, $\boxed{400\ \mu A}$ at $v_{Y1} = V_{DD}$, $\boxed{175\ \mu A}$ at $v_{Y1} = V_t$.

The average value of the drain current is,

$$i_{D1(avg)} = \frac{400\ \mu A + 175\ \mu A}{2}$$
$$= 287.5\ \mu A$$

Thus, the average value of drain current, $i_{D1(avg)}$ is, $\boxed{287.5\ \mu A}$.

Step 3 of 5

(c)

Consider that the values of capacitances are,

$$C_{L1} = C_{L2}$$
$$= 40\ fF$$

Consider that the average current over the time interval is the time the capacitor, C_{L1} has a voltage change of Δv_{Y1} .

Thus, the time delay, Δt is,

$$\Delta t = \frac{C_{L1}(V_{DD} - V_t)}{i_{D1(avg)}}$$

Substitute $40\ fF$ for C_{L1} , $287.5\ \mu A$ for $i_{D1(avg)}$, 5 V for V_{DD} , and 1 V for V_t .

$$\Delta t = \frac{(40\ fF)(5\ V - 1\ V)}{287.5\ \mu A}$$
$$= \frac{160}{287.5}$$
$$= 0.56\ ns$$

Thus, the time delay, Δt is, $\boxed{0.56\ ns}$.

Step 4 of 5

(d)

Consider that the output of first logic gate, $v_{Y1} = 3\ V$, and the transistor, Q_{eq2} works in saturation region during the time interval, Δt .

The average current, $i_{D2(avg)}$ is,

$$i_{D2(avg)} = \frac{1}{2}k_n\left(\frac{W}{L}\right)_{eq2}(v_{Y1} - V_t)^2$$

Substitute $50\ \mu A/V^2$ for k_n , 1 for $\left(\frac{W}{L}\right)_{eq2}$, 5 V for V_{DD} , and 1 V for V_t .

$$i_{D2(avg)} = \frac{1}{2}(50\ \mu A/V^2)(1)(3\ V - 1\ V)^2$$
$$= 100\ \mu A$$

Thus, the average value of drain current, $i_{D2(avg)}$ is, $\boxed{100\ \mu A}$.

Step 5 of 5

(e)

The change in voltage in the capacitor, C_{L2} is,

$$\Delta v_{Y2} = -\frac{(i_{D2(avg)})(\Delta t)}{C_{L2}}$$

Substitute $40\ fF$ for C_{L2} , $100\ \mu A$ for $i_{D2(avg)}$, and 0.56 ns for Δt .

$$\Delta v_{Y2} = -\frac{(100\ \mu A)(0.56\ ns)}{40\ fF}$$
$$= -1.4\ V$$

Therefore, the voltage, v_{Y2} decreased by a value of 1.4 V from V_{DD} .

Hence, the value of output voltage of second logic gate, v_{Y2} is, $\boxed{3.6\ V}$.

Step 1 of 5

Consider a pseudo-NMOS inverter is fabricated using 0.13-μm process.

Determine the low-high propagation delay t_{PLH}.

$$t_{PLH} = \frac{\alpha_p C}{k_p V_{DD}} \quad \text{..... (1)}$$

Determine α_p.

$$\alpha_p = 2\Big/\left[\frac{7}{4} - 3\left(\frac{V_t}{V_{DD}}\right) + \left(\frac{V_t}{V_{DD}}\right)^2\right] \quad \text{..... (2)}$$

Replace 1.3 V for V_{DD}, 0.4 V for V_t, in equation 2.

$$\begin{aligned}\alpha_p &= 2\Big/\left[\frac{7}{4} - 3\left(\frac{V_t}{V_{DD}}\right) + \left(\frac{V_t}{V_{DD}}\right)^2\right] \\ &= 2\Big/\left[\frac{7}{4} - 3\left(\frac{0.4}{1.3}\right) + \left(\frac{0.4}{1.3}\right)^2\right] \\ &= 2/[1.75 - 0.923 + 0.094] \\ &= 2.171\end{aligned}$$

Step 2 of 5

Consider $C = 10\ fF$.

Replace 100-$\mu A/V^2$ for k_p, 2.171 for α_p, and 1.3 V for V_{DD} in equation 1.

$$\begin{aligned}t_{PLH} &= \frac{\alpha_p C}{k_p V_{DD}} \\ &= \frac{(2.171)(10\times10^{-15})}{(100\times10^{-6})(1.3)} \\ &= 167\ ps\end{aligned}$$

Therefore, the low-high propagation delay t_{PLH} is $\boxed{167\ ps}$.

Step 3 of 5

Determine the high-low propagation delay t_{PHL}.

$$t_{PHL} = \frac{\alpha_n C}{k_n V_{DD}} \quad \text{..... (3)}$$

Determine α_n.

$$\alpha_n = 2\Big/\left[1 + \frac{3}{4}\left(1 - \frac{1}{r}\right) - \left(3 - \frac{1}{r}\right)\left(\frac{V_t}{V_{DD}}\right) + \left(\frac{V_t}{V_{DD}}\right)^2\right] \quad \text{..... (4)}$$

Replace 1.3 V for V_{DD}, 0.4 V for V_t, and 5 for r in equation 4.

$$\begin{aligned}\alpha_n &= 2\Big/\left[1 + \frac{3}{4}\left(1 - \frac{1}{r}\right) - \left(3 - \frac{1}{r}\right)\left(\frac{V_t}{V_{DD}}\right) + \left(\frac{V_t}{V_{DD}}\right)^2\right] \\ &= 2\Big/\left[1 + \frac{3}{4}\left(1 - \frac{1}{5}\right) - \left(3 - \frac{1}{5}\right)\left(\frac{0.4}{1.3}\right) + \left(\frac{0.4}{1.3}\right)^2\right] \\ &= 2/[1 + 0.6 - 0.86 + 0.094] \\ &= 2.398\end{aligned}$$

Step 4 of 5

Replace 500-$\mu A/V^2$ for k_n, 2.398 for α_n, and 1.3 V for V_{DD} in equation 3.

Consider $C = 10\ fF$.

$$\begin{aligned}t_{PHL} &= \frac{\alpha_n C}{k_n V_{DD}} \\ &= \frac{(2.398)(10\times10^{-15})}{(500\times10^{-6})1.3} \\ &= 36.89\ ps\end{aligned}$$

Therefore, the high-low propagation delay t_{PHL} is $\boxed{36.89\ ps}$.

Step 5 of 5

Determine the propagation delay of the inverter t_p.

$$\begin{aligned}t_p &= \frac{1}{2}(t_{PLH} + t_{PHL}) \\ &= \frac{1}{2}(167 + 36.89) \\ &= 101.945\ ps\end{aligned}$$

Therefore, the propagation delay of the inverter is $\boxed{101.945\ ps}$.

15.016E

We don't have the solution to this problem yet.

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Step 1 of 7

The NMOS device is in cutoff and the PMOS device is in triode region for the rising output.

The expression for the drain current of transistor, Q_p is,

$$i_{DP} = k_p\left[(V_{DD} - V_t)(V_{DD} - v_o) - \frac{1}{2}(V_{DD} - v_o)^2\right]$$

Here, k_p is the transconductance parameter, V_{DD} is the supply voltage, V_t is the threshold voltage, v_o is the output voltage.

Consider that the output, v_o is,

$$v_o = \frac{V_{DD}}{4}$$

Substitute 2.5 V for V_{DD}.

$$\begin{aligned} v_o &= \frac{2.5\text{ V}}{4} \\ &= 0.625\text{ V} \end{aligned}$$

Substitute 2.5 V for V_{DD}, 0.5 V for V_t, and 0.625 V for v_o.

$$\begin{aligned} i_{DP} &= k_p\left[(2.5\text{ V} - 0.5\text{ V})(2.5\text{ V} - 0.625\text{ V}) - \frac{1}{2}(2.5\text{ V} - 0.625\text{ V})^2\right] \\ &= k_p\left[(2\text{ V})(1.875\text{ V}) - \frac{1}{2}(1.875\text{ V})^2\right] \\ &= k_p\left[(2\text{ V})(1.875\text{ V}) - \frac{1}{2}(1.875\text{ V})^2\right] \\ &= k_p(3.75\text{ V} - 1.757\text{ V}) \\ &= 1.99k_p \end{aligned}$$

The net current extracted from the load is, $(i_{on} - i_{op})$ which should be i_D. Therefore,

$$\begin{aligned} i_D &= 2i_{DP} \\ &= 2(1.99k_p) \\ &= 3.98k_p \end{aligned}$$

Step 2 of 7

Find the drain current, i_{DN} of NMOS device.

$$i_{DN} = k_n\left[(V_{DD} - V_t)v_o - \frac{1}{2}v_o^2\right]$$

Substitute 2.5 V for V_{DD}, 0.5 V for V_t, and 0.625 V for v_o.

$$\begin{aligned} i_{DN} &= k_n\left[(2.5\text{ V} - 0.5\text{ V})(0.625\text{ V}) - \frac{1}{2}(0.625\text{ V})^2\right] \\ &= k_n\left[(2\text{ V})(0.625\text{ V}) - \frac{1}{2}(0.39\text{ V}^2)\right] \\ &= k_n(1.25\text{ V} - 0.195\text{ V}) \\ &= 1.055k_n \end{aligned}$$

Equate the expressions of i_{DN} **and** i_{DP}, since there is equal current flow through the NMOS and PMOS devices.

$$\begin{aligned} i_{DN} &= i_{DP} \\ 1.055k_n &= 3.98k_p \end{aligned}$$

The value of ratio of transconductances, r is,

$$\begin{aligned} r &= \frac{k_n}{k_p} \\ &= \frac{3.98}{1.055} \\ &= 3.78 \end{aligned}$$

Step 3 of 7

Write the general expression for r to find the value of width to length ratio of PMOS device.

$$\begin{aligned} r &= \frac{k_n}{k_p} \\ &= \frac{k_n'\left(\frac{W}{L}\right)_n}{k_p'\left(\frac{W}{L}\right)_p} \end{aligned}$$

Here, the process transconductances are, $k_n' = 115\ \mu\text{A/V}^2$, $k_p' = 30\ \mu\text{A/V}^2$, the width to length ratio of NMOS device, $\left(\frac{W}{L}\right)_n$ is, 1.5.

Substitute $115\ \mu\text{A/V}^2$ for k_n', $30\ \mu\text{A/V}^2$ for k_p', 3.78 for r, and 1.5 for $\left(\frac{W}{L}\right)_n$ to find the value of $\left(\frac{W}{L}\right)_p$.

$$\begin{aligned} 3.78 &= \frac{(115\ \mu\text{A/V}^2)(1.5)}{(30\ \mu\text{A/V}^2)\left(\frac{W}{L}\right)_p} \\ \left(\frac{W}{L}\right)_p &= \frac{172.5}{113.4} \\ &= 1.5 \end{aligned}$$

Step 4 of 7

Obtain the value of low level voltage of input, V_{IL}.

$$V_{IL} = V_t + \frac{V_{DD} - V_t}{\sqrt{r(r+1)}}$$

Substitute 0.5 V for V_t, 2.5 V for V_{DD}, and 3.78 for r.

$$\begin{aligned} V_{IL} &= 0.5\text{ V} + \frac{2.5\text{ V} - 0.5\text{ V}}{\sqrt{3.78(3.78+1)}} \\ &= 0.5\text{ V} + \frac{2\text{ V}}{\sqrt{18}} \\ &= 0.5\text{ V} + 0.47\text{ V} \\ &= 0.97\text{ V} \end{aligned}$$

Thus, the value of low level voltage of input, V_{IL} is, $\boxed{0.97\text{ V}}$.

Obtain the value of high level voltage of input, V_{IH}.

$$V_{IH} = V_t + \frac{2}{\sqrt{3r}}(V_{DD} - V_t)$$

Substitute 0.5 V for V_t, 2.5 V for V_{DD}, and 3.78 for r.

$$\begin{aligned} V_{IH} &= V_t + \frac{2}{\sqrt{3r}}(V_{DD} - V_t) \\ &= 0.5\text{ V} + \frac{2}{\sqrt{3(3.78)}}(2.5\text{ V} - 0.5\text{ V}) \\ &= 0.5\text{ V} + (0.59)(2\text{ V}) \\ &= 1.68\text{ V} \end{aligned}$$

Thus, the value of high level voltage of input, V_{IH} is, $\boxed{1.68\text{ V}}$.

Step 5 of 7

Calculate the value of V_M.

$$V_M = V_t + \frac{V_{DD} - V_t}{\sqrt{r+1}}$$

Substitute 0.5 V for V_t, 2.5 V for V_{DD}, and 3.78 for r.

$$\begin{aligned} V_M &= 0.5\text{ V} + \frac{2.5\text{ V} - 0.5\text{ V}}{\sqrt{3.78+1}} \\ &= 0.5\text{ V} + \frac{2\text{ V}}{\sqrt{4.78}} \\ &= 0.5\text{ V} + 0.914\text{ V} \\ &= 1.414\text{ V} \end{aligned}$$

Thus, the value of V_M is, $\boxed{1.414\text{ V}}$.

Step 6 of 7

Obtain the value of high level voltage of output, V_{ON}.

$$\begin{aligned} V_{OH} &= V_{DD} \\ &= 2.5\text{ V} \end{aligned}$$

Thus, the value of high level voltage of output, V_{ON} is, $\boxed{2.5\text{ V}}$.

Obtain the value of low level voltage of output, V_{OL}.

$$V_{OL} = (V_{DD} - V_t)\left[1 - \sqrt{1 - \frac{1}{r}}\right]$$

Substitute 0.5 V for V_t, 2.5 V for V_{DD}, and 3.78 for r.

$$\begin{aligned} V_{OL} &= (2.5\text{ V} - 0.5\text{ V})\left[1 - \sqrt{1 - \frac{1}{3.78}}\right] \\ &= (2\text{ V})\left(1 - \sqrt{0.735}\right) \\ &= (2\text{ V})(0.142) \\ &= 0.285\text{ V} \end{aligned}$$

Thus, the value of low level voltage of output, V_{OL} is, $\boxed{0.285\text{ V}}$.

Step 7 of 7

Determine the high level noise margin, NM_H.

$$NM_H = V_{OH} - V_{IH}$$

Substitute 2.5 V for V_{OH}, and 1.68 V for V_{IH}.

$$\begin{aligned} NM_H &= 2.5\text{ V} - 1.68\text{ V} \\ &= 0.82\text{ V} \end{aligned}$$

Thus, the high level noise margin, NM_H is, $\boxed{0.82\text{ V}}$.

Determine the low level noise margin, NM_L.

$$NM_L = V_{IL} - V_{OL}$$

Substitute 0.97 V for V_{IL}, and 0.285 V for V_{OH}.

$$\begin{aligned} NM_H &= 0.97\text{ V} - 0.285\text{ V} \\ &= 0.685\text{ V} \end{aligned}$$

Thus, the low level noise margin, NM_L is, $\boxed{0.685\text{ V}}$.

15.017E

Refer to Figure E14.12 in the textbook.

Consider that the base-emitter junction voltage of transistor, Q_1 and the voltage drop across the diodes, D_1 **and** D_2 is, 0.75 V.

Neglect the base current in the transistor, Q_1 to find the current through the diodes.

$$\begin{aligned} I &= \frac{0 - 0.75\ \text{V} - 0.75\ \text{V} - (-5.2\ \text{V})}{R_1 + R_2} \\ &= \frac{5.2\ \text{V} - 1.5\ \text{V}}{907\ \Omega + 4.98\ \text{k}\Omega} \\ &= \frac{3.7\ \text{V}}{5.887\ \text{k}\Omega} \\ &= 0.6285\ \text{mA} \end{aligned}$$

The base voltage at the transistor, Q_1 is,

$$\begin{aligned} V_B &= 0 - IR_1 \\ &= -(0.6285\ \text{mA})(907\ \Omega) \\ &= -0.570\ \text{V} \end{aligned}$$

Therefore, the emitter voltage, V_R is,

$$\begin{aligned} V_R &= V_B - V_{BE} \\ &= -0.57\ \text{V} - 0.75\ \text{V} \\ &= -1.32\ \text{V} \end{aligned}$$

Therefore, the voltage, V_R in the circuit is, $\boxed{-1.32\ \text{V}}$.

Step 1 of 7

Consider the equation for low value of noise margin NM_L .

$$NM_L = V_t - (V_{DD} - V_t)\left[1 - \sqrt{1 - \frac{1}{r}} - \frac{1}{\sqrt{r(r+1)}}\right] \quad \text{..... (1)}$$

Determine the value of r for maximum value of noise margin NM_L .

Differentiate equation 1 with respect to r .

$$\frac{d(NM_L)}{dr} = \left[V_t - (V_{DD} - V_t)\right]\frac{d}{dr}\left[1 - \sqrt{1 - \frac{1}{r}} - \frac{1}{\sqrt{r(r+1)}}\right]$$

$$= \left[V_t - (V_{DD} - V_t)\right]\left[0 - \frac{d}{dr}\left[\sqrt{1 - \frac{1}{r}}\right] - \frac{d}{dr}\left[\frac{1}{\sqrt{r(r+1)}}\right]\right]$$

$$= \left[V_t - (V_{DD} - V_t)\right]\left[0 - \frac{1}{2\sqrt{1 - \frac{1}{r}}}\left(-\frac{1}{r^2}\right) - \frac{1}{2\sqrt{r(r+1)}}(2r^2 + 1)\right]$$

Step 2 of 7

For maximum value of noise margin NM_L , $\frac{d(NM_L)}{dr} = 0$.

$$\left[V_t - (V_{DD} - V_t)\right]\left[0 - \frac{1}{2\sqrt{1 - \frac{1}{r}}}\left(-\frac{1}{r^2}\right) - \frac{1}{2\sqrt{r(r+1)}}(2r^2 + 1)\right] = 0$$

$$\left[0 - \frac{1}{2\sqrt{1 - \frac{1}{r}}}\left(-\frac{1}{r^2}\right) - \frac{1}{2\sqrt{r(r+1)}}(2r^2 + 1)\right] = 0$$

$$\frac{1}{2\sqrt{1 - \frac{1}{r}}}\left(-\frac{1}{r^2}\right) = \frac{1}{2\sqrt{r(r+1)}}(2r^2 + 1)$$

By squaring on both sides.

$$\frac{1}{r^4\left(1 - \frac{1}{r}\right)} = \frac{(2r^2 + 1)}{r(r+1)}$$

$$\frac{4r^2 + 1 + 4r}{r^2 + r} = \frac{1}{r^4\left(1 - \frac{1}{r}\right)}$$

$$(r^4 - r^3)(4r^2 + 1 + 4r) = r^2 + r$$

$$4r^6 - 3r^4 - r^3 - r^2 - r = 0 \quad \text{..... (2)}$$

Step 3 of 7

Solve equation 2 for roots. and consider positive root for r .

The roots are 0 , 1.149 .

consider positive root for r .

Therefore, the value of r for maximum NM_L is $\boxed{1.149}$.

Step 4 of 7

Consider $V_{DD} = 3.3$ and $V_t = 0.4$.

Determine the value of NM_L .

Replace 3.3 for V_{DD} , 0.4 for V_t , and 1.149 for r in equation 1.

$$NM_L = V_t - (V_{DD} - V_t)\left[1 - \sqrt{1 - \frac{1}{r}} - \frac{1}{\sqrt{r(r+1)}}\right]$$

$$= 0.4 - (1.3 - 0.4)\left[1 - \sqrt{1 - \frac{1}{1.149}} - \frac{1}{\sqrt{1.149(1.149+1)}}\right]$$

$$= 0.4 - (1.3 - 0.4)[1 - 0.36 - 0.63]$$

$$= 0.391 \text{ V}$$

Therefore, the value of noise margin NM_L for $r = 1.149$ is $\boxed{0.391 \text{ V}}$.

Step 5 of 7

Consider $r = 2$.

Consider $V_{DD} = 3.3$ and $V_t = 0.4$.

Determine the value of NM_L .

Replace 3.3 for V_{DD} , 0.4 for V_t , and 2 for r in equation 1.

$$NM_L = V_t - (V_{DD} - V_t)\left[1 - \sqrt{1 - \frac{1}{r}} - \frac{1}{\sqrt{r(r+1)}}\right]$$

$$= 0.4 - (1.3 - 0.4)\left[1 - \sqrt{1 - \frac{1}{2}} - \frac{1}{\sqrt{2(2+1)}}\right]$$

$$= 0.4 - (1.3 - 0.4)[1 - 0.707 - 0.408]$$

$$= 0.5 \text{ V}$$

Therefore, the value of noise margin NM_L for $r = 2$ is $\boxed{0.5 \text{ V}}$.

Step 6 of 7

Consider $r = 5$.

Consider $V_{DD} = 3.3$ and $V_t = 0.4$.

Determine the value of NM_L .

Replace 3.3 for V_{DD} , 0.4 for V_t , and 5 for r in equation 1.

$$NM_L = V_t - (V_{DD} - V_t)\left[1 - \sqrt{1 - \frac{1}{r}} - \frac{1}{\sqrt{r(r+1)}}\right]$$

$$= 0.4 - (1.3 - 0.4)\left[1 - \sqrt{1 - \frac{1}{5}} - \frac{1}{\sqrt{5(5+1)}}\right]$$

$$= 0.4 - (1.3 - 0.4)[1 - 0.894 - 0.18]$$

$$= 0.466 \text{ V}$$

Therefore, the value of noise margin NM_L for $r = 5$ is $\boxed{0.466 \text{ V}}$.

Step 7 of 7

Consider $r = 10$.

Consider $V_{DD} = 3.3$ and $V_t = 0.4$.

Determine the value of NM_L .

Replace 3.3 for V_{DD} , 0.4 for V_t , and 10 for r in equation 1.

$$NM_L = V_t - (V_{DD} - V_t)\left[1 - \sqrt{1 - \frac{1}{r}} - \frac{1}{\sqrt{r(r+1)}}\right]$$

$$= 0.4 - (1.3 - 0.4)\left[1 - \sqrt{1 - \frac{1}{10}} - \frac{1}{\sqrt{10(10+1)}}\right]$$

$$= 0.4 - (1.3 - 0.4)[1 - 0.948 - 0.09]$$

$$= 0.4342 \text{ V}$$

Therefore, the value of noise margin NM_L for $r = 10$ is $\boxed{0.4342 \text{ V}}$.

Therefore, the value of noise margin does not change very much for $r = 2$, 5 , and 10 .

Step 1 of 2

15.018E

Refer to Figure 14.26 in the textbook for the basic circuit of the emitter-coupled logic (ECL) 10K logic-gate family.

Consider that the input terminals are left open.

The base emitter voltage of Q_R is, 0.75 V, and the voltage V_R is, $-1.32\ \text{V}$.

Neglect the base currents in all the transistors.

Apply Ohm's law to obtain the emitter current in the common emitter.

$$\begin{aligned} I_E &= \frac{V_R - V_{BE}\big|_{Q_R} - (-V_{EE})}{R_E} \\ &= \frac{-1.32\ \text{V} - 0.75\ \text{V} + 5.2\ \text{V}}{779\ \Omega} \\ &= \frac{3.13\ \text{V}}{0.779\ \text{k}\Omega} \\ &= 4\ \text{mA} \end{aligned}$$

Therefore, the emitter current, I_E is, $\boxed{4\ \text{mA}}$.

Step 2 of 2

The emitter current of transistors, Q_A and Q_B is same as the emitter current of transistor, Q_R .

Find the voltage at the collector of transistor, Q_R .

$$\begin{aligned} V_C &= -I_C R_C \\ &= -(4\ \text{mA})(245\ \Omega) \quad (\text{since, } \beta \text{ is high, } I_E \approx I_C) \\ &\simeq -1\ \text{V} \end{aligned}$$

Thus, the voltage at the collector of transistor, Q_R is, $\boxed{-1\ \text{V}}$.

Since, the input terminals are left open, the voltage at the base of transistors, Q_A and Q_B is $-V_{EE} = -5.2\ \text{V}$. As the emitter voltage cannot drop below $-V_{EE}$, both the transistors cannot conduct any current.

Hence, the current in the resistance, R_{C1} is zero, in other words the collector voltage for the transistors, Q_A and Q_B is zero.

Thus, the collector voltage for the transistors, Q_A and Q_B is, $\boxed{0\ \text{V}}$.

Step 1 of 4

15.018P

Consider a pseudo-NMOS inverter is fabricated using process technology.

Determine the value of r for which the value of NM_H of a pseudo-NMOS inverter is 0.

Consider the equation for high value of the noise margin NM_H.

$$NM_H = (V_{DD} - V_t)\left(1 - \frac{2}{\sqrt{3r}}\right)$$

Step 2 of 4

Consider $NM_H = 0$.

$$0 = (V_{DD} - V_t)\left(1 - \frac{2}{\sqrt{3r}}\right)$$

$$\left(1 - \frac{2}{\sqrt{3r}}\right) = 0$$

$$1 = \frac{2}{\sqrt{3r}}$$

$$3r = 4$$

$$r = \frac{4}{3}$$

$$= 1.33$$

Therefore, the value of r for which the value of NM_H of a pseudo-NMOS inverter is 0 is $\boxed{1.33}$.

Step 3 of 4

Consider $V_{DD} = 1.3\ \text{V}$ and $V_t = 0.4\ \text{V}$ and draw the table by varying $r = 2$ to 10.

r	$NM_H = (V_{DD} - V_t)\left[1 - \frac{2}{\sqrt{3r}}\right]$	$NM_L = V_t - (V_{DD} - V_t)\left[1 - \sqrt{1 - \frac{1}{r}} - \frac{1}{\sqrt{r(r+1)}}\right]$
2	$NM_H = (1.3 - 0.4)\left[1 - \frac{2}{\sqrt{3(2)}}\right]$ $= 0.165$	$NM_L = 0.4 - (1.3 - 0.4)\left[1 - \sqrt{1 - \frac{1}{2}} - \frac{1}{\sqrt{2(2+1)}}\right]$ $= 0.50$
3	$NM_H = (1.3 - 0.4)\left[1 - \frac{2}{\sqrt{3(3)}}\right]$ $= 0.3$	$NM_L = 0.4 - (1.3 - 0.4)\left[1 - \sqrt{1 - \frac{1}{3}} - \frac{1}{\sqrt{3(3+1)}}\right]$ $= 0.49$
4	$NM_H = (1.3 - 0.4)\left[1 - \frac{2}{\sqrt{3(4)}}\right]$ $= 0.38$	$NM_L = 0.4 - (1.3 - 0.4)\left[1 - \sqrt{1 - \frac{1}{4}} - \frac{1}{\sqrt{4(4+1)}}\right]$ $= 0.48$
5	$NM_H = (1.3 - 0.4)\left[1 - \frac{2}{\sqrt{3(5)}}\right]$ $= 0.435$	$NM_L = 0.4 - (1.3 - 0.4)\left[1 - \sqrt{1 - \frac{1}{5}} - \frac{1}{\sqrt{5(5+1)}}\right]$ $= 0.469$
6	$NM_H = (1.3 - 0.4)\left[1 - \frac{2}{\sqrt{3(6)}}\right]$ $= 0.47$	$NM_L = 0.4 - (1.3 - 0.4)\left[1 - \sqrt{1 - \frac{1}{6}} - \frac{1}{\sqrt{6(6+1)}}\right]$ $= 0.46$
7	$NM_H = (1.3 - 0.4)\left[1 - \frac{2}{\sqrt{3(7)}}\right]$ $= 0.50$	$NM_L = 0.4 - (1.3 - 0.4)\left[1 - \sqrt{1 - \frac{1}{7}} - \frac{1}{\sqrt{7(7+1)}}\right]$ $= 0.45$
8	$NM_H = (1.3 - 0.4)\left[1 - \frac{2}{\sqrt{3(8)}}\right]$ $= 0.53$	$NM_L = 0.4 - (1.3 - 0.4)\left[1 - \sqrt{1 - \frac{1}{8}} - \frac{1}{\sqrt{8(8+1)}}\right]$ $= 0.44$
9	$NM_H = (1.3 - 0.4)\left[1 - \frac{2}{\sqrt{3(9)}}\right]$ $= 0.55$	$NM_L = 0.4 - (1.3 - 0.4)\left[1 - \sqrt{1 - \frac{1}{9}} - \frac{1}{\sqrt{9(9+1)}}\right]$ $= 0.443$
10	$NM_H = (1.3 - 0.4)\left[1 - \frac{2}{\sqrt{3(10)}}\right]$ $= 0.57$	$NM_L = 0.4 - (1.3 - 0.4)\left[1 - \sqrt{1 - \frac{1}{10}} - \frac{1}{\sqrt{10(10+1)}}\right]$ $= 0.43$

Step 4 of 4

Observe the values for $r = 6$.

For $r = 6$, the high value of the noise margin NM_H is approximately equal to low value of the noise margin NM_L.

Therefore, the value of r for which the high value of the noise margin NM_H is equal to low value of the noise margin NM_L is $\boxed{r = 6}$.

Therefore, the resulting noise margin for which the high value of the noise margin NM_H is equal to low value of the noise margin NM_L is $\boxed{0.46\ \text{V}}$.

Step 1 of 3

15.019E

Refer to Figure 14.37 (e) in the textbook for the bipolar complementary metal oxide semiconductor (BiCMOS) inverter.

Consider the following expression for the current, I of MOS transistor Q_N at $v_I = V_M - 0.7\ \text{V}$.

$$I_n = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_n (V_M - 0.7\ \text{V} - V_{tn})^2$$

Here, the process trans-conductance parameter is $k_n' = \mu_n C_{ox}$.

The mobility of electrons in the channel is μ_n, and

The capacitance of the oxide layer is C_{ox}.

The aspect ratio of the NMOS device is $\left(\frac{W}{L}\right)_n$ and

The threshold voltage is V_{tn}.

Substitute $\frac{V_{DD}}{2}$ for V_M.

$$I_n = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_n \left(\frac{V_{DD}}{2} - 0.7 - V_{tn}\right)^2$$

Substitute 5 V for V_{DD}, and 0.6 V for V_{tn} to find the peak current.

$$I_n = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_n \left(\frac{5\ \text{V}}{2} - 0.7\ \text{V} - 0.6\ \text{V}\right)^2$$

$$I_n = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_n (1.2)^2$$

Step 2 of 3

Consider the following expression for the current, I for Q_P at $v_I = V_{DD} - V_M$.

$$I_p = \frac{1}{2}k_p'\left(\frac{W}{L}\right)_p (V_{DD} - V_M - V_{tp})^2$$

Here, the process trans-conductance parameter is $k_p' = \mu_p C_{ox}$.

The mobility of holes in the channel is μ_p, and

The capacitance of the oxide layer is C_{ox}.

The aspect ratio of the device is $\left(\frac{W}{L}\right)_p$ and

The threshold voltage is V_{tp}.

Substitute $\frac{V_{DD}}{2}$ for V_M.

$$I_p = \frac{1}{2}k_p'\left(\frac{W}{L}\right)_p \left(V_{DD} - \frac{V_{DD}}{2} - V_{tp}\right)^2$$

Substitute 5 V for V_{DD}, and 0.6 V for V_{tp} to find the peak current.

$$I_p = \frac{1}{2}k_p'\left(\frac{W}{L}\right)_p \left(5\ \text{V} - \frac{5\ \text{V}}{2} - 0.6\ \text{V}\right)^2$$

$$I_p = \frac{1}{2}\mu_p C_{ox}\left(\frac{W}{L}\right)_p (1.9)^2$$

Step 3 of 3

Consider that the both transistors, Q_N **and** Q_P are conducting equal currents and operating in saturation. Equate the current equations for these transistors.

$$I_n = I_p$$

$$\frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_n (1.2)^2 = \frac{1}{2}\mu_p C_{ox}\left(\frac{W}{L}\right)_p (1.9)^2$$

$$1.44\mu_n \frac{W_n}{L_n} = 3.61\mu_p \frac{W_p}{L_p}$$

Consider that the transistors, Q_N **and** Q_P have same channel lengths and $\mu_n = 2.5\mu_p$. Therefore,

$$1.44(2.5\mu_p)\frac{W_n}{L_n} = 3.61\mu_p \frac{W_p}{L_p}$$

$$1.44(2.5)W_n = 3.61W_p$$

Solve for $\frac{W_p}{W_n}$.

$$\frac{W_p}{W_n} = \frac{1.44(2.5)}{3.61}$$

$$= 1$$

Thus, the required ratio of widths, $\frac{W_p}{W_n}$ for the design of the inverter is, $\boxed{1}$.

Step 1 of 7

15.019P

consider a pseudo-NMOS inverter is fabricated design process technology .

To design a NMOS inverter V_{DD} and V_t are selected from process technology.

Since r is being the design parameter to control the values of noise margin and V_{OL} select a proper value for r .

Consider the equation for V_{OL} .

$$V_{OL}=(V_{DD}-V_t)\left[1-\sqrt{1-\frac{1}{r}}\right] \quad \text{..... (1)}$$

Step 2 of 7

Determine the value of r for $V_{OL}=0.1\ \text{V}$.

Replace $1.8\ \text{V}$ for V_{DD} , 0.5 for V_t in equation 1.

$$V_{OL}=(V_{DD}-V_t)\left[1-\sqrt{1-\frac{1}{r}}\right]$$

$$0.1=(1.8-0.5)\left[1-\sqrt{1-\frac{1}{r}}\right]$$

$$1-\sqrt{1-\frac{1}{r}}=\frac{0.1}{1.3}$$

$$1-\sqrt{1-\frac{1}{r}}=0.076$$

$$\sqrt{1-\frac{1}{r}}=1-0.076$$

$$\sqrt{1-\frac{1}{r}}=0.924$$

$$r=6.835$$

Therefore, to design a pseudo-NMOS inverter with $V_{OL}=0.1\ \text{V}$ select $\boxed{r=6.835}$.

Step 3 of 7

Determine the value of $\left(\frac{W}{L}\right)_n$.

Consider $r=\frac{k_n}{k_p}$.

$$r=\frac{k_n'\left(\frac{W}{L}\right)_n}{k_p'\left(\frac{W}{L}\right)_p} \quad \text{..... (2)}$$

Replace 400- $\mu\text{A/V}^2$ for k_n , 100- $\mu\text{A/V}^2$ for k_p , 6.835 for r and 1 for $\left(\frac{W}{L}\right)_p$ in equation 2.

$$r=\frac{k_n'\left(\frac{W}{L}\right)_n}{k_p'\left(\frac{W}{L}\right)_p}$$

$$6.835=\frac{(400\times10^{-6})\left(\left(\frac{W}{L}\right)_n\right)}{(100\times10^{-6})(1)}$$

$$\left(\frac{W}{L}\right)_n=\frac{(6.835)(100\times10^{-6})}{(400\times10^{-6})}$$

$$\left(\frac{W}{L}\right)_n=1.70$$

Therefore, the value of $\left(\frac{W}{L}\right)_n$ is $\boxed{1.70}$.

Step 4 of 7

Determine the low value of the noise margin NM_L .

$$NM_L=V_t-(V_{DD}-V_t)\left[1-\sqrt{1-\frac{1}{r}}-\frac{1}{\sqrt{r(r+1)}}\right] \quad \text{..... (3)}$$

Replace $1.8\ \text{V}$ for V_{DD} , 0.5 for V_t , and 6.835 for r in equation 3.

$$NM_L=0.5-(1.8-0.5)\left[1-\sqrt{1-\frac{1}{6.835}}-\frac{1}{\sqrt{6.835(6.835+1)}}\right]$$

$$=0.5-(1.8-0.5)[1-0.923-0.136]$$

$$=0.576\ \text{V}$$

Therefore, the low value of the noise margin NM_L is $\boxed{0.576}$.

Step 5 of 7

Determine the dc current in the low-output state I_{stat} .

$$I_{stat}=\frac{1}{2}k_p(V_{DD}-V_t)^2$$

$$I_{stat}=\frac{1}{2}k_p'\left(\frac{W}{L}\right)_p(V_{DD}-V_t)^2 \quad \text{..... (4)}$$

Step 6 of 7

Replace $1.8\ \text{V}$ for V_{DD} , 0.5 for V_t , 100- $\mu\text{A/V}^2$ for k_p , and 1 for $\left(\frac{W}{L}\right)_p$ in equation 4.

$$I_{stat}=\frac{1}{2}k_p'\left(\frac{W}{L}\right)_p(V_{DD}-V_t)^2$$

$$=\frac{1}{2}(100\times10^{-6})(1)(1.8-0.5)^2$$

$$=\frac{1}{2}(100\times10^{-6})(1.69)$$

$$=84.5\ \mu\text{A}$$

Step 7 of 7

Determine the static power P_D .

$$P_D=(I_{stat})(V_{DD}) \quad \text{..... (5)}$$

Replace $1.8\ \text{V}$ for V_{DD} , and 84.5- μA for I_{stat} in equation 5.

$$P_D=(I_{stat})(V_{DD})$$

$$=(84.5\times10^{-6})(1.8)$$

$$=152.1\ \mu\text{w}$$

Therefore, the static power dissipation of the pseudo-NMOS inverter is $\boxed{152.1\ \mu\text{w}}$.

Step 1 of 16

Consider a minimum area pseudo-NMOS inverter is fabricated in process technology.

Consider $r = 5.7$.

Determine the high value of the noise margin NM_H .

Consider the equation for the high value of the noise margin NM_H .

$$NM_H = (V_{DD} - V_t)\left[1 - \frac{2}{\sqrt{3r}}\right] \quad \text{..... (1)}$$

Step 2 of 16

Replace 1.3 for V_{DD} , 0.4 for V_t , and 5.7 for r in equation 1.

$$\begin{aligned} NM_H &= (V_{DD} - V_t)\left[1 - \frac{2}{\sqrt{3r}}\right] \\ &= (1.3 - 0.4)\left[1 - \frac{2}{\sqrt{3(5.7)}}\right] \\ &= (1.3 - 0.4)[1 - 0.483] \\ &= 0.46 \text{ V} \end{aligned}$$

Therefore, the high value of the noise margin for $r = 5.7$ is $\boxed{0.46 \text{ V}}$.

Step 3 of 16

Determine the low value of the noise margin NM_L .

Consider the equation for the low value of the noise margin NM_L .

$$NM_L = V_t - (V_{DD} - V_t)\left[1 - \sqrt{1 - \frac{1}{r}} - \frac{1}{\sqrt{r(r+1)}}\right] \quad \text{..... (2)}$$

Replace 1.3 for V_{DD} , 0.4 for V_t , and 5.7 for r in equation 2.

$$\begin{aligned} NM_L &= V_t - (V_{DD} - V_t)\left[1 - \sqrt{1 - \frac{1}{r}} - \frac{1}{\sqrt{r(r+1)}}\right] \\ &= 0.4 - (1.3 - 0.4)\left[1 - \sqrt{1 - \frac{1}{5.7}} - \frac{1}{\sqrt{5.7(5.7+1)}}\right] \\ &= 0.4 - (1.3 - 0.4)[1 - 0.908 - 0.161] \\ &= 0.462 \text{ V} \end{aligned}$$

Therefore, the low value of the noise margin of the NMOS inverter for $r = 5.7$ is $\boxed{0.462 \text{ V}}$.

Therefore, the for $r = 5.7$ the high and low values of noise margin is approximately equal.

Step 4 of 16

To get minimum area NMOS inverter select a small value of $\left(\frac{W}{L}\right)_n$.

Consider the aspect ratio of the minimum size device $\left(\frac{W}{L}\right) = 1$.

Consider $r = 5.7$ and $\left(\frac{W}{L}\right)_n = 1$.

$$r = \frac{k_n}{k_p}$$

$$r = \frac{k_n' \left(\frac{W}{L}\right)_n}{k_p' \left(\frac{W}{L}\right)_p} \quad \text{..... (3)}$$

Step 5 of 16

Replace 500- $\mu A/V^2$ for k_n , 125- $\mu A/V^2$ for k_p , and 1 for $\left(\frac{W}{L}\right)_n$ in equation 3.

$$r = \frac{k_n' \left(\frac{W}{L}\right)_n}{k_p' \left(\frac{W}{L}\right)_p}$$

$$5.7 = \frac{(500 \times 10^{-6})1}{(125 \times 10^{-6})\left(\frac{W}{L}\right)_p}$$

$$\left(\frac{W}{L}\right)_p = \frac{(500 \times 10^{-6})1}{(125 \times 10^{-6})5.7}$$

$$\left(\frac{W}{L}\right)_p = 0.701$$

Therefore, the aspect ratio of $\left(\frac{W}{L}\right)_n$ is $\boxed{1}$. And the aspect ratio of the $\left(\frac{W}{L}\right)_p$ is $\boxed{0.701}$.

Step 6 of 16

Determine the static current I_{stat} .

$$I_{stat} = \frac{1}{2} k_p (V_{DD} - V_t)^2$$

$$I_{stat} = \frac{1}{2} k_p' \left(\frac{W}{L}\right)_p (V_{DD} - V_t)^2 \quad \text{..... (4)}$$

Step 7 of 16

Replace 1.3 for V_{DD} , 0.4 for V_t , 125- $\mu A/V^2$ for k_p , and 0.701 for $\left(\frac{W}{L}\right)_p$ in equation 4.

$$\begin{aligned} I_{stat} &= \frac{1}{2}(125 \times 10^{-6})(0.701)(1.3 - 0.4)^2 \\ &= \frac{1}{2}(125 \times 10^{-6})(0.701)(1.3 - 0.4)^2 \\ &= 35.48 \text{ } \mu\text{A} \end{aligned}$$

Determine the static power dissipation P_D .

$$P_D = (I_{stat})(V_{DD})$$

Consider $I_{stat} = 35.48 \text{ } \mu\text{A}$ and 1.3 V for V_{DD} .

$$\begin{aligned} P_D &= (35.48 \times 10^{-6})(1.3) \\ &= 46.12 \text{ } \mu\text{w} \end{aligned}$$

Therefore, the static power dissipation is $\boxed{46.12 \text{ } \mu\text{w}}$.

Step 8 of 16

Consider the equation for low-high propagation delay t_{PLH} .

$$t_{PLH} = \frac{\alpha_p C}{k_p V_{DD}}$$

Consider the equation for high-to propagation delay t_{PHL} .

$$t_{PHL} = \frac{\alpha_n C}{k_n V_{DD}}$$

Determine the ratio of the propagation delay for low-high and high-low transitions.

$$\frac{t_{PLH}}{t_{PHL}} = \frac{\frac{\alpha_p C}{k_p V_{DD}}}{\frac{\alpha_n C}{k_n V_{DD}}}$$

$$\frac{t_{PLH}}{t_{PHL}} = \left(\frac{\alpha_p}{\alpha_n}\right)\left(\frac{k_n}{k_p}\right) \quad \text{..... (5)}$$

Step 9 of 16

Consider the equation for α_p .

$$\alpha_p = 2 \Big/ \left[\frac{7}{4} - 3\left(\frac{V_t}{V_{DD}}\right) + \left(\frac{V_t}{V_{DD}}\right)^2\right] \quad \text{..... (6)}$$

Replace 1.3 for V_{DD} , 0.4 for V_t in equation 6.

$$\begin{aligned} \alpha_p &= 2 \Big/ \left[\frac{7}{4} - 3\left(\frac{V_t}{V_{DD}}\right) + \left(\frac{V_t}{V_{DD}}\right)^2\right] \\ &= 2 \Big/ \left[\frac{7}{4} - 3\left(\frac{0.4}{1.3}\right) + \left(\frac{0.4}{1.3}\right)^2\right] \\ &= 2.17 \end{aligned}$$

Step 18 of 16

Consider the equation for α_n .

$$\alpha_n = 2 \Big/ \left[1 + \frac{3}{4}\left(1 - \frac{1}{r}\right) - \left(3 - \frac{1}{r}\right)\left(\frac{V_t}{V_{DD}}\right) + \left(\frac{V_t}{V_{DD}}\right)^2\right] \quad \text{..... (7)}$$

Replace 1.3 for V_{DD} , 0.4 for V_t , and 5.7 for r in equation 7.

$$\begin{aligned} \alpha_n &= 2 \Big/ \left[1 + \frac{3}{4}\left(1 - \frac{1}{r}\right) - \left(3 - \frac{1}{r}\right)\left(\frac{V_t}{V_{DD}}\right) + \left(\frac{V_t}{V_{DD}}\right)^2\right] \\ &= 2 \Big/ \left[1 + \frac{3}{4}\left(1 - \frac{1}{5.7}\right) - \left(3 - \frac{1}{5.7}\right)\left(\frac{0.4}{1.3}\right) + \left(\frac{0.4}{1.3}\right)^2\right] \\ &= 2 \Big/ [1 + 0.618 - 0.869 + 0.094] \\ &= 2.37 \end{aligned}$$

Step 11 of 16

Replace 2.17 for α_p , 2.37 for α_n , 125- $\mu A/V^2$ for k_p , and 500- $\mu A/V^2$ for k_n in equation 5.

$$\begin{aligned} \frac{t_{PLH}}{t_{PHL}} &= \left(\frac{\alpha_p}{\alpha_n}\right)\left(\frac{k_n}{k_p}\right) \\ &= \left(\frac{2.17}{2.37}\right)\left(\frac{125 \times 10^{-6}}{500 \times 10^{-6}}\right) \\ &= 0.22 \end{aligned}$$

Therefore, the ratio of the propagation delay for low-high and high-low transition is $\boxed{0.22}$.

Step 12 of 16

Consider load capacitance $C = 10 \text{ fF}$.

Consider $\alpha_p = 2.17$ and 125- $\mu A/V^2$ for k_p .

Determine the low-high propagation delay t_{PLH} .

$$\begin{aligned} t_{PLH} &= \frac{\alpha_p C}{k_p V_{DD}} \\ &= \frac{(2.17)(10 \times 10^{-15})}{(125 \times 10^{-6})1.3} \\ &= 133 \text{ ps} \end{aligned}$$

Therefore, the low-high propagation delay t_{PLH} is $\boxed{133 \text{ ps}}$.

Step 13 of 16

Consider $\alpha_n = 2.37$ and 500- $\mu A/V^2$ for k_n .

Determine the high-low propagation delay t_{PHL} .

$$\begin{aligned} t_{PHL} &= \frac{\alpha_n C}{k_n V_{DD}} \\ &= \frac{(2.37)(10 \times 10^{-15})}{(500 \times 10^{-6})1.3} \\ &= 36.46 \text{ ps} \end{aligned}$$

Therefore, the low-high propagation delay t_{PHL} is $\boxed{36.46 \text{ ps}}$.

Step 14 of 16

Determine the propagation delay t_p .

$$t_p = \frac{t_{PLH} + t_{PHL}}{2}$$

Consider $t_{PLH} = 133 \text{ ps}$ and $t_{PHL} = 36.46 \text{ ps}$.

$$\begin{aligned} t_p &= \frac{133 + 36.46}{2} \\ &= 84.73 \text{ ps} \end{aligned}$$

Therefore, the propagation delay of the NMOS inverter is $\boxed{84.73 \text{ ps}}$.

Step 15 of 16

Consider the static power dissipation is equal to static power dissipation.

Consider load capacitance $C = 10 \text{ fF}$.

Consider static power dissipation $P_{Dstatic} = 46.12 \text{ } \mu\text{w}$.

Determine the frequency for which the static and dynamic powers are equal.

Consider the equation for dynamic power dissipation $P_{dynamic} = Cf(V_{DD})^2$.

$$P_{Dstatic} = P_{dynamic}$$

$$(46.12 \times 10^{-6}) = CfV_{DD}^2$$

$$f = \frac{(46.12 \times 10^{-6})}{CV_{DD}^2}$$

$$f = \frac{(46.12 \times 10^{-6})}{(10 \times 10^{-15})(1.3)^2}$$

$$f = \frac{(46.12 \times 10^{-6})}{(1.69 \times 10^{-14})}$$

$$= 2.72 \text{ GHz}$$

Therefore, the frequency for which the static power dissipation is equals to dynamic power dissipation is $\boxed{2.72 \text{ GHz}}$.

Step 16 of 16

consider the minimum time delay of the inverter is two times of the propagation delay $T_{min} = 2t_p$.

Determine the minimum propagation delay.

$$T = \frac{1}{f}$$

$$= \frac{1}{2.72 \text{ GHz}}$$

$$= 0.367 \text{ ns}$$

Determine $2t_p$.

$$2t_p = 2(84.73 \text{ ps})$$

$$= 0.169 \text{ ns}$$

Compare T with $2t_p$.

Therefore, for the inverter circuit $T > 2t_p$.

Therefore, the circuit can be operated at 0.3 ns .

15.021P

Realize the function, $Y = \overline{A + B(C + D)}$ using pseudo-NMOS transistors.

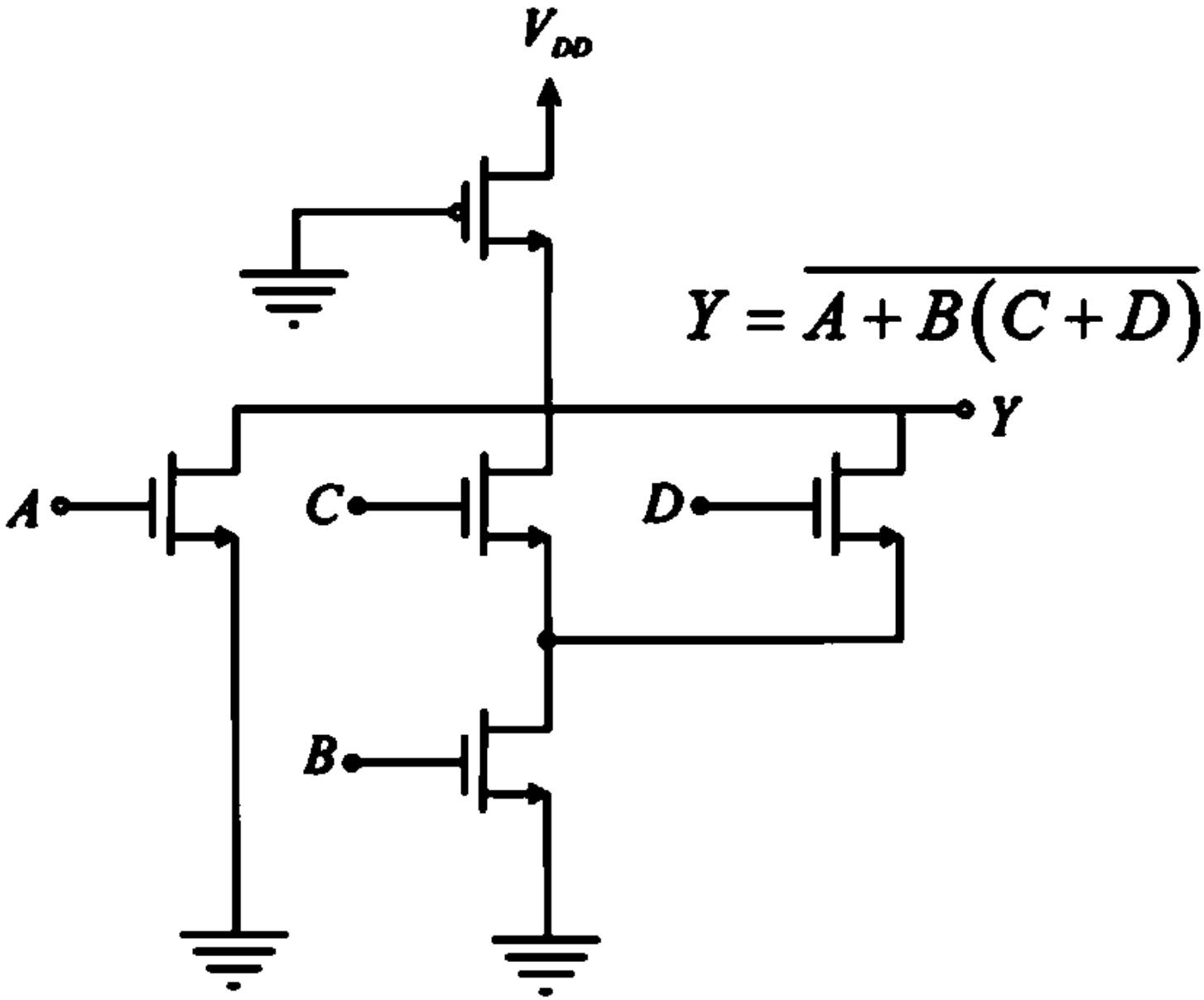


Figure 1

Thus, the pseudo-NMOS realization of the function is shown in Figure 1.

Step 1 of 2

15.022P

Sketch a pseudo-NMOS realization of the following exclusive-OR function:

$$Y = A\overline{B} + \overline{A}B$$

The pseudo-NMOS technique always results the complemented outputs. Therefore, modify the function, Y :

$$\begin{aligned}\overline{Y} &= \overline{A\overline{B} + \overline{A}B} \\ &= \left(\overline{A\overline{B}}\right)\left(\overline{\overline{A}B}\right) \\ &= \left(\overline{A} + B\right)\left(A + \overline{B}\right) \\ &= \overline{A}A + AB + \overline{A}\overline{B} + B\overline{B}\end{aligned}$$

$$\overline{Y} = \overline{A}\overline{B} + AB$$

Step 2 of 2

Realize the function.

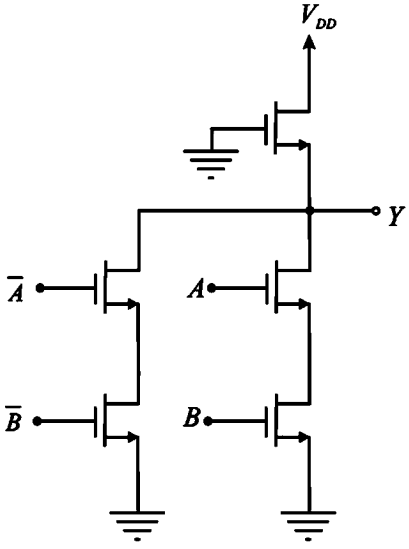


Figure 1

Thus, the pseudo-NMOS realization is shown in Figure 1.

Step 1 of 2

15.023P

Consider a four–input pseudo-NMOS NOR gate is fabricated using process technology.

Consider the equation for low value of output voltage.

$$V_{OL} = (V_{DD} - V_t)\left[1 - \sqrt{1 - \frac{1}{r}}\right] \quad \text{..... (1)}$$

Determine the value of r .

Replace 1.3 for V_{DD} , 0.4 for V_t , and 0.1 for V_{OL} in equation 1.

$$0.1 = (1.3 - 0.4)\left[1 - \sqrt{1 - \frac{1}{r}}\right]$$

$$\left[1 - \sqrt{1 - \frac{1}{r}}\right] = 0.11$$

$$\sqrt{1 - \frac{1}{r}} = 0.89$$

$$1 - \frac{1}{r} = 0.79$$

$$\frac{1}{r} = 0.21$$

$$r = 4.76$$

Therefore, the value of the r is $\boxed{4.76}$.

Step 2 of 2

Determine the aspect ratio of PMOS device $\left(\frac{W}{L}\right)_p$.

Determine the value for k_p' .

$$k_n' = 4k_p'$$

$$k_p' = \frac{k_n'}{4}$$

$$= \frac{500\ \mu A/V^2}{4}$$

$$= 250\ \mu A/V^2$$

Write the expression of r .

$$r = \frac{k_n}{k_p}$$

$$r = \frac{k_n'\left(\frac{W}{L}\right)_n}{k_p'\left(\frac{W}{L}\right)_p} \quad \text{..... (2)}$$

Replace 500-$\mu A/V^2$ for k_n' , 125-$\mu A/V^2$ for k_p' , 4.34 for r , and 1.5 for $\left(\frac{W}{L}\right)_n$ in equation 3.

$$r = \frac{k_n'\left(\frac{W}{L}\right)_n}{k_p'\left(\frac{W}{L}\right)_p}$$

$$4.76 = \frac{(500\times10^{-6})(1.5)}{(125\times10^{-6})\left(\frac{W}{L}\right)_p}$$

$$\left(\frac{W}{L}\right)_p = \frac{(500\times10^{-6})(1.5)}{(125\times10^{-6})4.76}$$

$$\left(\frac{W}{L}\right)_p = 1.26$$

Therefore, the aspect ratio of PMOS device is $\boxed{1.26}$.

15.024P

We don't have the solution to this problem yet.

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15.025P

We don't have the solution to this problem yet.

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Ask an expert

15.026P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

15.027P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

15.028P

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Ask an expert

Step 1 of 4

15.029P

Refer to Figure 15.17 in the textbook.

Consider NMOS transistor switch is fabricated with 0.18-μm CMOS process and the output of the NMOS transistor is connected to the input of the CMOS inverter.

Consider the output voltage of the NMOS switch is 1.15 V.

For input of 1.15 V to the inverter, PMOS transistor is operating in saturation region.

Determine the static current of the inverter (I_{stat}).

$$I_{stat} = \frac{1}{2}k_p(V_{DD} - v_0 - V_t)^2$$

$$I_{stat} = \frac{1}{2}\mu_p C_{ox}\left(\frac{W}{L}\right)_p (V_{DD} - v_0 - V_t)^2 \quad \text{..... (1)}$$

Step 2 of 4

Write the equation for threshold voltage (V_t).

$$V_t = V_{t0} + \gamma\left(\sqrt{v_o + 2\phi_f} - \sqrt{2\phi_f}\right)$$

Consider $v_0 = V_{DD} - V_t$.

$$V_t = V_{t0} + \gamma\left(\sqrt{V_{DD} - V_t + 2\phi_f} - \sqrt{2\phi_f}\right) \quad \text{..... (2)}$$

Replace 0.5 for V_{t0}, 0.3 for γ, and 0.85 for $2\phi_f$, 1.8 for V_{DD} in equation 2.

$$V_t = V_{t0} + \gamma\left(\sqrt{V_{DD} - V_t + 2\phi_f} - \sqrt{2\phi_f}\right)$$
$$V_t = 0.5 + 0.3\left(\sqrt{1.8 - V_t + 0.85} - \sqrt{0.85}\right)$$
$$V_t = 0.5 + 0.3\left(\sqrt{2.65 - V_t} - \sqrt{0.85}\right)$$
$$(V_t - 0.223)^2 = \left(0.3\sqrt{2.65 - V_t}\right)^2$$

$$V_t^2 + 0.0497 - 2V_t(0.223) = 0.09(2.65 - V_t)$$
$$V_t^2 + 0.0497 - 0.446V_t = 0.2385 - 0.09V_t$$

$$V_t^2 - 0.356V_t - 0.188 = 0$$
$$V_t = 0.64, -0.290$$

Consider the positive root as the threshold voltage $V_t = 0.64$ V.

Step 3 of 4

Replace 75 μA for $\mu_p C_{ox}$, and 3 for $\left(\frac{W}{L}\right)_p$, 1.15 for v_0, and 0.64 for V_t in equation 1.

$$I_{stat} = \frac{1}{2}\mu_p C_{ox}\left(\frac{W}{L}\right)_p (V_{DD} - v_0 - V_t)^2$$
$$= \frac{1}{2}\left(75\times10^{-6}\right)(3)(1.8 - 1.15 - 0.64)^2$$
$$= 0.112\ \mu A$$

Therefore the static current I_{stat} is $\boxed{0.112\ \mu A}$.

Step 4 of 4

Determine the power dissipation of the inverter (P_D).

$$P_D = V_{DD} I_{stat}$$
$$= (1.8)\left(0.112\times10^{-6}\right)$$
$$= 2\ \mu W$$

Therefore, the power dissipation of the inverter is $\boxed{2\ \mu W}$.

15.030P

We don't have the solution to this problem yet.

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Step 1 of 5

(a)

Refer to Figure 15.31 in the textbook.

Consider initially the output voltage is zero and at $t = 0$, the input voltage is raised to V_{DD} The PMOS transistor acts as ON switch and the output voltage reaches to V_{DD} .

Therefore the final value of the output V_{OH} is $\boxed{V_{DD}}$.

Step 2 of 5

(b)

Consider initially the output voltage is V_{DD} and at $t = 0$ v_i is lowered to $0\ \text{V}$.

The PMOS transistor acts as OFF switch and the output voltage reaches minimum value that is equal to V_t .

Therefore the final value of the output V_{OL} is $\boxed{V_t}$.

Step 3 of 5

(c)

Determine the average discharge current $\left(i_D\big|_{avg}\right)$.

Average discharge current is the average of the i_D at $t = 0$ and the i_D at $t = t_{PLH}$.

Determine i_D at $t = 0 \left(v_0 = 0, V_t = V_{t0}\right)$ ..... (1)

Replace 125 $\mu\text{A/V}^2$ for k_p , 1.2 for V_{DD} , 0 for v_0 , and 0.4 for V_t in equation 1.

$$\begin{aligned} i_D(t=0) &= \frac{1}{2}k_p\left(V_{DD} - v_0 - V_t\right)^2 \\ &= \frac{1}{2}\left(125\times10^{-6}\right)\left(1.2-0-0.4\right)^2 \\ &= 40\ \mu\text{A} \end{aligned}$$

Step 4 of 5

Determine i_D at $t = t_{PLH}\left(v_0 = \frac{V_{DD}}{2}, V_t = V_{to}\right)$.

$$i_D\left(t = t_{PLH}\right) = \frac{1}{2}k_p\left(V_{DD} - v_0 - V_t\right)^2 \quad \text{..... (2)}$$

Replace 125 $\mu\text{A/V}^2$ for k_p , 1.2 for V_{DD} , 0.6 for v_0 , and 0.4 for V_t in equation 2.

$$\begin{aligned} i_D\left(t = t_{PLH}\right) &= \frac{1}{2}k_p\left(V_{DD} - v_0 - V_t\right)^2 \\ &= \frac{1}{2}\left(125\times10^{-6}\right)\left(1.2-0.6-0.4\right)^2 \\ &= 2.5\ \mu\text{A} \end{aligned}$$

Step 5 of 5

Consider $i_D\big|_{avg}$.

$$\begin{aligned} i_D\big|_{avg} &= \frac{i_D(t=0) + i_D\left(t = t_{PLH}\right)}{2} \\ &= \frac{40\ \mu\text{A} + 2.5\ \mu\text{A}}{2} \\ &= 21.25\ \mu\text{A} \end{aligned}$$

Consider t_{PLH} .

$$\begin{aligned} t_{PLH} &= \frac{C\left(\frac{V_{DD}}{2}\right)}{i_D\big|_{avg}} \\ &= \frac{C(0.6)}{(21.25\ \mu\text{A})} \\ &= 28235.2C\ \text{s} \end{aligned}$$

Therefore, the low-to-high propagation delay of PMOS transistor is $\boxed{28235.2C\ \text{s}}$.

Step 1 of 3

15.032P

(a)

Refer to Figure P14.14 (a) in the textbook for the circuit.

From the circuit, functions X and Y are,

$$X = 1 \cdot (\overline{A})$$
$$= \overline{A}$$

And

$$Y = X \cdot \overline{B}$$
$$= \overline{A} \cdot \overline{B}$$

(b)

Refer to Figure P14.14 (b) in the textbook for the circuit.

From the circuit, functions X and Y are,

$$X = 1$$
$$Y = \overline{A} \cdot \overline{B}$$

Step 2 of 3

For each circuit, node *Y* nominally satisfies both conditions. That is node, *Y* is the function of *A* and *B*.

General function of the circuits:

In Figure (a), with *A* high and *B* low, *Y* is not pulled down completely to ground but remains at V_{tp}, due to PMOS threshold voltage.

Circuit (b) does not have any problem. But node *X* is floating foe *A*, *B* both high. However, *X* is not an output node. The body effect makes this worse. Notice that here both the circuits are complementary (CMOS) NOR gate function.

$$Y = \overline{A} \cdot \overline{B}$$
$$= \overline{A + B}$$

Step 3 of 3

If in each case, the terminal connected to V_{DD} is instead connected to a signal *C*, then the node *Y* is,

$$Y = C(\overline{A} \cdot \overline{B})$$
$$= \overline{A} \cdot \overline{B} \cdot \overline{C}$$
$$= \overline{A + B + C}$$

Thus, it gives a three input NOR gate function for both the circuits (a) and (b). Here, there is a practical problem in the circuit that the series PMOS do not operate well with a low input. The output *Y* is pulled down only one threshold drop below ground voltage, when *C* is high.

Step 1 of 1

15.033P

Refer to Figure P14.14 in the textbook for the circuits.

Replace all the PMOS transistors in the circuits with NMOS, and all NMOS transistors by PMOS, and interchange ground and V_{DD} connections.

Due to interchange of the transistors, the output is high only when B low or A is low.

For both the circuits the output is,

$$Y = \overline{A} + \overline{B}$$
$$= \overline{A \cdot B}$$

Therefore, if we interchange the transistors the output becomes a **NAND** function.

The circuit (b) acts as fully complementary CMOS gate. However circuit (b) gives a low value of high output for *A* low with *B* is high. In this case, the output becomes $V_{DD} - V_{tn}$, when V_{tn} is raised from V_{ton} by the body effect.

Step 1 of 7

Refer to Figure 15.19 in the textbook.

Consider Q_1 is a pass transistor controlled by input B and the output of pass transistor is connected to CMOS inverter.

Consider Q_R is the PMOS transistor and its gate is controlled by the output voltage of the inverter.

Consider v_A is the input voltage applied to pass transistor.

Step 2 of 7

The output voltage of the pass transistor (v_{01}) is depends on the input voltage applied at v_A. If v_A starts raising to V_{DD}, the transistor operates in saturation mode and capacitor starts charging and the output voltage of the pass transistor starts raising to V_{DD}.

Consider if no input is applied at v_A then the voltage at the output of the pass transistor (v_{01}) is low.

The output at the inverter v_{02} is high for low voltage at v_{01} and the transistor Q_R is OFF.

Step 3 of 7

If $v_{01} = \frac{V_{DD}}{2}$, then the output of the inverter v_{02} is low and the transistor Q_R is ON.

Consider v_{02} is the output voltage of the inverter.

$$\begin{aligned} v_{02} &= V_{DD} - V_{t0} \\ &= 3.3 - 0.8 \\ &= 2.5 \text{ V} \end{aligned}$$

Consider the drain current of the NMOS transistor in saturation region.

$$i_{DN} = \frac{1}{2} k_n (v_1 - V_t)^2$$

$$i_{DN} = \frac{1}{2} k_n' \left(\frac{W}{L}\right)_n (v_{01} - V_t)^2 \quad \text{..... (1)}$$

Replace 75 for k_n, 1.5 for $\left(\frac{W}{L}\right)_n$, and 0.8 for V_{t0} in the equation 1.

$$\begin{aligned} i_{DN} &= \frac{1}{2} k_n' \left(\frac{W}{L}\right)_n (v_{01} - V_t)^2 \\ &= \frac{1}{2} (75 \times 10^{-6})(1.5)(v_{01} - 0.8)^2 \end{aligned}$$

$$i_{DN} = (56.25 \times 10^{-6})(v_{01} - 0.8)^2 \quad \text{..... (2)}$$

Step 4 of 7

PMOS transistor of the inverter operates in triode region for the output voltage greater then threshold voltage.

Consider the current equation of the PMOS transistor in triode region.

$$i_{Dp} = k_p \left[(V_{DD} - V_t)(V_{DD} - v_0) - \frac{1}{2}(V_{DD} - v_0)^2 \right]$$

$$i_{Dp} = k_p \left(\frac{W}{L}\right)_p \left[(V_{DD} - V_t)(V_{DD} - v_0) - \frac{1}{2}(V_{DD} - v_0)^2 \right] \quad \text{..... (3)}$$

Replace 25 for k_p, 4.5 for $\left(\frac{W}{L}\right)_p$, 3.3 for V_{DD}, and 0.8 for V_{t0} in the equation 3.

$$\begin{aligned} i_{Dp} &= k_p \left(\frac{W}{L}\right)_p \left[(V_{DD} - V_t)(V_{DD} - v_0) - \frac{1}{2}(V_{DD} - v_0)^2 \right] \\ &= k_p \left(\frac{W}{L}\right)_p \left[(V_{DD} - V_t) V_t - \frac{1}{2} V_t^2 \right] \\ &= (25 \times 10^{-6})(4.5) \left[(3.3 - v_{01} - 0.8)0.8 - \frac{1}{2}(0.8)^2 \right] \\ &= (112.5 \times 10^{-6}) \left[(2.5 - 0.8 v_{01}) - \frac{1}{2}(0.8)^2 \right] \end{aligned}$$

$$i_{Dp} = (112.5 \times 10^{-6}) [2.18 - 0.8 v_{01}] \quad \text{..... (4)}$$

Step 5 of 7

Equate equation 3 and equation 4 to determine the output voltage v_{01}.

$$(56.25 \times 10^{-6})(v_{01} - 0.8)^2 = (112.5 \times 10^{-6})[2.18 - 0.8 v_{01}]$$

$$(v_{01} - 0.8)^2 = 2[2.18 - 0.8 v_{01}]$$

$$v_{01}^2 + 0.64 - 1.6 v_{01} = 4.36 - 1.6 v_{01}$$

$$v_{01}^2 = 3.72$$

$$\begin{aligned} v_{01} &= \sqrt{3.72} \\ &= 1.92 \text{ V} \end{aligned}$$

Therefore, the value of the output voltage of the NMOS transistor is $\boxed{1.92 \text{ V}}$.

Step 6 of 7

Determine the threshold voltage of the NMOS transistor.

$$\begin{aligned} V_{tn} &= V_{t0} + \gamma \left(\sqrt{v_{01} - 2\phi_f} - \sqrt{2\phi_f} \right) \\ &= 0.8 + 0.5 \left(\sqrt{1.92 - 0.6} - \sqrt{0.6} \right) \\ &= 0.98 \text{ V} \end{aligned}$$

Therefore, the threshold voltage of the NMOS transistor is $\boxed{0.98 \text{ V}}$.

Step 7 of 7

Determine the capacitor charging current.

$$\begin{aligned} i_{DN} &= \frac{1}{2} k_n' \left(\frac{W}{L}\right)_n (v_{01} - V_t)^2 \\ &= \frac{1}{2} (75 \times 10^{-6})(1.5)(1.92 - 0.8)^2 \\ &= 70.56 \ \mu\text{A} \end{aligned}$$

Therefore, the capacitor charging current is $\boxed{70.56 \ \mu\text{A}}$.

Step 1 of 7

Refer to Figure 15.19 in the textbook.

Consider the voltage at v_A is decreasing towards zero, the transistor Q_1 starts conducting and discharges the capacitor.

Determine the voltage at v_{01} .

$$V_{IH} = \frac{1}{8}(5V_{DD} - 2V_{t0}) \quad \text{..... (1)}$$

Replace 3.3 for V_{DD} , and 0.8 for V_{t0} in the equation 1.

$$\begin{aligned} V_{IH} &= \frac{1}{8}(5V_{DD} - 2V_{t0}) \\ &= \frac{1}{8}(5(3.3) - 2(0.8)) \\ &= 1.9875 \text{ V} \end{aligned}$$

Therefore, the voltage at which inverter begins to switch is $\boxed{1.9875 \text{ V}}$.

Step 2 of 7

Determine the current for which Q_1 starts conducting at 1.9875 V .

$$i_{D|_{Q1}} = k_n' \left(\frac{W}{L}\right)_n \left((V_{DD} - V_{t0})v_{01} - \frac{1}{2}(v_{01})^2\right) \quad \text{..... (2)}$$

Replace 75 $\mu\text{A/V}^2$ for k_n' , 1.5 for $\left(\frac{W}{L}\right)_n$, 3.3 for V_{DD} , 0.8 for V_{t0} , and 1.9875 V for v_{01} in equation 2.

$$\begin{aligned} i_{D|_{Q1}} &= (75\times10^{-6})(1.5)\left((3.3-0.8)1.9875 - \frac{1}{2}(1.9875)^2\right) \\ &= (75\times10^{-6})(1.5)((2.5)(1.9875) - 1.975) \\ &= 0.336 \text{ mA} \end{aligned}$$

Therefore, the current for which Q_1 starts conducting at v_{01} is $\boxed{0.336 \text{ mA}}$.

Step 3 of 7

Consider the maximum current through Q_R is limited to $\left(\frac{i_{D|_{Q1}}}{2}\right)$.

Determine the $\left(\frac{W}{L}\right)$ of Q_R .

$$i_{D|_{QR}} = k_p' \left(\frac{W}{L}\right)_R (V_{DD} - v_{t0})^2$$

$$\frac{i_{D|_{Q1}}}{2} = k_p' \left(\frac{W}{L}\right)_R (V_{DD} - v_{t0})^2 \quad \text{..... (3)}$$

Step 4 of 7

Replace 25 $\mu\text{A/V}^2$ for k_p' , 3.3 for V_{DD} , 0.8 for V_{t0} , and 0.336 mA for $i_{D|_{Q1}}$ in equation 3.

$$\frac{(0.336\times10^{-3})}{2} = (25\times10^{-6})\left(\frac{W}{L}\right)_R (3.3-0.8)^2$$

$$1.68\times10^{-4} = (1.5625\times10^{-4})\left(\frac{W}{L}\right)_R$$

$$\left(\frac{W}{L}\right)_R = 1.0752$$

Therefore, the value of $\left(\frac{W}{L}\right)$ for which the maximum current through Q_R is limited to one-half of the current through Q_1 is $\boxed{1.072}$.

Step 5 of 7

Consider at $t = 0$ the voltage at v_{01} is V_{DD} .

$$i_{D|_{V_{DD}}} = \frac{1}{2}(k_n')\left(\frac{W}{L}\right)_n (V_{DD} - V_t)^2 \quad \text{..... (4)}$$

Replace 75 $\mu\text{A/V}^2$ for k_n' , 1.5 for $\left(\frac{W}{L}\right)_n$, 3.3 for V_{DD} , 0.8 for V_{t0} in equation 4.

$$\begin{aligned} i_{D|_{V_{DD}}} &= \frac{1}{2}(75\times10^{-6})(1.5)(3.3-0.8)^2 \\ &= 0.351 \text{ mA} \end{aligned}$$

Step 6 of 7

Consider the voltage at v_{01} is $\left(\frac{V_{DD}}{2}\right)$.

$$i_{D|_{\frac{V_{DD}}{2}}} = \frac{1}{2}(k_n')\left(\frac{W}{L}\right)_n \left((V_{DD} - V_t)\frac{V_{DD}}{2} - \frac{1}{2}\left(\frac{V_{DD}}{2}\right)^2\right) \quad \text{..... (5)}$$

Replace 75 $\mu\text{A/V}^2$ for k_n' , 1.5 for $\left(\frac{W}{L}\right)_n$, 3.3 for V_{DD} , 0.8 for V_{t0} in equation 5.

$$\begin{aligned} i_{D|_{\frac{V_{DD}}{2}}} &= \frac{1}{2}(75\times10^{-6})(1.5)\left((3.3-0.8)\frac{3.3}{2} - \frac{1}{2}\left(\frac{3.3}{2}\right)^2\right) \\ &= \frac{1}{2}(75\times10^{-6})(1.5)(4.125-1.361) \\ &= 0.155 \text{ mA} \end{aligned}$$

Determine the current in PMOS transistor Q_R at v_{01} .

$$i_{DR} = k_p'\left(\frac{W}{L}\right)_R \left((V_{DD} - V_{t0})(V_{DD} - v_{01}) - \frac{1}{2}(V_{DD} - v_{01})\right) \quad \text{..... (6)}$$

Replace 25 $\mu\text{A/V}^2$ for k_p' , 3.3 for V_{DD} , 0.8 for V_{t0} , and 0.336 mA for $i_{D|_{Q1}}$ in equation 6.

$$\begin{aligned} i_{DR} &= (25\times10^{-6})(1.072)\left((3.3-0.8)(3.3-1.98) - \frac{1}{2}(3.3-1.98)\right) \\ &= (25\times10^{-6})(1.072)(3.3-0.66) \\ &= 70.752 \text{ µA} \end{aligned}$$

Step 7 of 7

Determine the average current I_{av} .

$$\begin{aligned} I_{av} &= \frac{i_{D|_{V_{DD}}} + i_{D|_{\frac{V_{DD}}{2}}} + i_{DR}}{2} \\ &= \frac{(0.351\times10^{-3}) + (0.155\times10^{-3}) + (70.752\times10^{-6})}{2} \\ &= 0.28 \text{ mA} \end{aligned}$$

Determine the propagation delay t_{PHL} .

$$t_{PHL} = \frac{C(V_{DD} - v_{01})}{I_{av}} \quad \text{..... (7)}$$

Replace 20 fF for C, 3.3 for , and 1.98 for v_{01} in equation 7.

$$\begin{aligned} t_{PHL} &= \frac{C(V_{DD} - v_{01})}{I_{av}} \\ &= \frac{(20\times10^{-15})(3.3-1.98)}{(0.28\times10^{-3})} \\ &= 94.28 \text{ ns} \end{aligned}$$

Therefore, the propagation delay is $\boxed{94.28 \text{ ns}}$.

Step 1 of 15

(a)

Refer to Figure 15.21 (a) and 15.21 (b) in the textbook.

Determine the high value reached by output V_{OH} and low value V_{OL} reached by output.

If the capacitor is fully charged then the maximum output voltage V_{OH} reached by the transmission gate is $1.2\ \text{V}$.

If the capacitor is fully discharged then the minimum output voltage reached by the transmission gate is $0\ \text{V}$.

Step 2 of 15

(b)

Refer to Figure 15.21 (a) in the textbook.

Consider the input is rising towards V_{DD} at $t = 0$.

Determine the current through NMOS transistor at $t = 0$ $i_{OH}(0)$.

$$i_{OH}(0) = \frac{1}{2}k_n(V_{DD} - V_t)^2$$

$$i_{OH}(0) = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_n(V_{DD} - V_t)^2 \quad \text{..... (1)}$$

Replace 500- $\mu A/V^2$ for k_n', 1.5 for $\left(\frac{W}{L}\right)_n$, 1.2 for V_{DD}, and 0.4 for V_t in equation 1.

$$\begin{aligned} i_{OH}(0) &= \frac{1}{2}k_n'\left(\frac{W}{L}\right)_n(V_{DD} - V_t)^2 \\ &= \frac{1}{2}\left(500\times10^{-6}\right)(1.5)(1.2-0.4)^2 \\ &= 2.4\times10^{-4}\ \text{A} \end{aligned}$$

Therefore, the current through NMOS transistor at $t = 0$ is $\boxed{\left(2.4\times10^{-4}\right)\text{A}}$.

Step 3 of 15

Determine the current through PMOS transistor at $t = 0$.

$$i_{DP}(0) = \frac{1}{2}k_p\left(V_{DD} - |V_\phi|\right)^2$$

$$i_{DP}(0) = \frac{1}{2}k_p'\left(\frac{W}{L}\right)_p\left(V_{DD} - |V_\phi|\right)^2 \quad \text{..... (2)}$$

Replace 125- $\mu A/V^2$ for k_p', 1.5 for $\left(\frac{W}{L}\right)_p$, 1.2 for V_{DD}, and 0.4 for V_ϕ in equation 2.

$$\begin{aligned} i_{DP}(0) &= \frac{1}{2}k_p'\left(\frac{W}{L}\right)_p\left(V_{DD} - |V_\phi|\right)^2 \\ &= \frac{1}{2}\left(125\times10^{-6}\right)(1.5)(1.2-0.4)^2 \\ &= 6\times10^{-5}\ \text{A} \end{aligned}$$

Therefore, the current through PMOS transistor at $t = 0$ is $\boxed{\left(6\times10^{-5}\right)\text{A}}$.

Step 4 of 15

Determine the threshold voltage (V_t).

$$V_t = V_{t0} + \gamma\left(\sqrt{v_0 + 2\phi_f} - \sqrt{2\phi_f}\right)$$

$$V_t = V_{t0} + \gamma\left(\sqrt{(V_{DD}/2) + 2\phi_f} - \sqrt{2\phi_f}\right) \quad \text{..... (3)}$$

Replace 0.4 for V_{t0}, 1.2 for V_{DD}, 0.2 for γ, and 0.88 for $2\phi_f$ in equation 3.

$$\begin{aligned} V_t &= 0.4 + 0.2\left(\sqrt{(1.2/2) + 0.88} - \sqrt{0.88}\right) \\ &= 0.4 + 0.2(1.21 - 0.93) \\ &= 0.456\ \text{V} \end{aligned}$$

Determine the current at low-high propagation delay of NMOS transistor $i_{OH}(t_{PLH})$.

$$i_{OH}(t_{PLH}) = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_n\left(V_{DD} - \frac{V_{DD}}{2} - V_t\right)^2 \quad \text{..... (4)}$$

Step 5 of 15

Replace 500- $\mu A/V^2$ for k_n', 1.5 for $\left(\frac{W}{L}\right)_n$, 1.2 for V_{DD}, and 0.4 for V_t in equation 4.

$$\begin{aligned} i_{OH}(t_{PLH}) &= \frac{1}{2}k_n'\left(\frac{W}{L}\right)_n\left(V_{DD} - \frac{V_{DD}}{2} - V_t\right)^2 \\ &= \frac{1}{2}\left(500\times10^{-6}\right)(1.5)(1.2-0.6-0.4)^2 \\ &= 1.5\times10^{-5}\ \text{A} \end{aligned}$$

Therefore, the current at low-high propagation delay is $\boxed{\left(1.5\times10^{-5}\right)\text{A}}$.

Determine the current at low-high propagation delay of PMOS transistor $i_{DP}(t_{PLH})$.

$$i_{DP}(t_{PLH}) = \frac{1}{2}k_p'\left(\frac{W}{L}\right)_p\left(\left(V_{DD} - V_{t0}\right)\left(\frac{V_{DD}}{2}\right) - \frac{1}{2}\left(\frac{V_{DD}}{2}\right)^2\right) \quad \text{..... (5)}$$

Step 6 of 15

Replace 125- $\mu A/V^2$ for k_p', 1.5 for $\left(\frac{W}{L}\right)_p$, 1.2 for V_{DD}, and 0.4 for V_{t0} in equation 5.

$$\begin{aligned} i_{DP}(t_{PLH}) &= \frac{1}{2}k_p'\left(\frac{W}{L}\right)_p\left(\left(V_{DD} - V_{t0}\right)\left(\frac{V_{DD}}{2}\right) - \frac{1}{2}\left(\frac{V_{DD}}{2}\right)^2\right) \\ &= \frac{1}{2}\left(125\times10^{-6}\right)(1.5)\left((1.2-0.4)\left(\frac{1.2}{2}\right) - \frac{1}{2}\left(\frac{1.2}{2}\right)^2\right) \\ &= 1.125\times10^{-5}\ \text{A} \end{aligned}$$

Therefore, the current at low-high propagation delay of PMOS transistor $i_{DP}(t_{PLH})$ is $\boxed{\left(1.125\times10^{-5}\right)\text{A}}$.

Step 7 of 15

Determine the low-high propagation delay t_{PLH}.

$$t_{PLH} = \frac{C\left(\frac{V_{DD}}{2}\right)}{i_{D_{avg}}} \quad \text{..... (6)}$$

$$\begin{aligned} i_{D_{avg}} &= \frac{\left(i_{OH}(0) + i_{DP}(0) + i_{DP}(t_{PLH}) + i_{ON}(t_{PLH})\right)}{2} \\ &= \frac{1}{2}\left[\left(2.4\times10^{-4}\right) + \left(6\times10^{-5}\right) + \left(1.5\times10^{-5}\right) + \left(1.125\times10^{-5}\right)\right] \\ &= 1.63\times10^{-4}\ \text{A} \end{aligned}$$

Replace 15 fF for C, 1.2 for V_{DD}, and $\left(1.63\times10^{-4}\right)$ for $i_{D_{avg}}$ in equation 6.

$$\begin{aligned} t_{PLH} &= \frac{\left(1.5\times10^{-15}\right)\left(\frac{1.2}{2}\right)}{\left(1.63\times10^{-4}\right)} \\ &= 5.52\ \text{ns} \end{aligned}$$

Therefore, the low-high propagation delay is $\boxed{5.52\ \text{ns}}$.

Step 8 of 15

(c)

Refer to Figure 15.21 (b) in the textbook.

Consider the input is rising towards V_{DD} at $t = 0$.

Determine the current through NMOS transistor at $t = 0$ $i_{ON}(0)$.

$$i_{ON}(0) = \frac{1}{2}k_n(V_{DD} - V_t)^2$$

$$i_{OH}(0) = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_n(V_{DD} - V_t)^2 \quad \text{..... (7)}$$

Replace 500- $\mu A/V^2$ for k_n', 1.5 for $\left(\frac{W}{L}\right)_n$, 1.2 for V_{DD}, and 0.4 for V_t in equation 7.

$$\begin{aligned} i_{OH}(0) &= \frac{1}{2}k_n'\left(\frac{W}{L}\right)_n(V_{DD} - V_t)^2 \\ &= \frac{1}{2}\left(500\times10^{-6}\right)(1.5)(1.2-0.4)^2 \\ &= 2.4\times10^{-4}\ \text{A} \end{aligned}$$

Therefore, the current through NMOS transistor at $t = 0$ is $\boxed{\left(2.4\times10^{-4}\right)\text{A}}$.

Step 9 of 15

Determine the current through PMOS transistor at $t = 0$.

$$i_{DP}(0) = \frac{1}{2}k_p\left(V_{DD} - |V_\phi|\right)^2$$

$$i_{DP}(0) = \frac{1}{2}k_p'\left(\frac{W}{L}\right)_p\left(V_{DD} - |V_\phi|\right)^2 \quad \text{..... (8)}$$

Replace 125- $\mu A/V^2$ for k_p', 1.5 for $\left(\frac{W}{L}\right)_p$, 1.2 for V_{DD}, and 0.4 for V_ϕ in equation 2.

$$\begin{aligned} i_{DP}(0) &= \frac{1}{2}k_p'\left(\frac{W}{L}\right)_p\left(V_{DD} - |V_\phi|\right)^2 \\ &= \frac{1}{2}\left(125\times10^{-6}\right)(1.5)(1.2-0.4)^2 \\ &= 6\times10^{-5}\ \text{A} \end{aligned}$$

Therefore, the current through PMOS transistor at $t = 0$ is $\boxed{\left(6\times10^{-5}\right)\text{A}}$.

Step 10 of 15

Consider $v_0 = \frac{V_{DD}}{2}$.

Determine the drain current of NMOS transistor at high-low propagation delay $i_{ON}(t_{PHL})$.

$$i_{OH}(t_{PHL}) = k_n'\left(\frac{W}{L}\right)_n\left(\left(V_{DD} - V_{t0}\right)v_0 - \frac{1}{2}(v_0)^2\right) \quad \text{..... (9)}$$

Replace 500- $\mu A/V^2$ for k_n', 1.5 for $\left(\frac{W}{L}\right)_n$, 1.2 for V_{DD}, 0.6 for v_0, and 0.4 for V_{t0} in equation 9.

$$\begin{aligned} i_{ON}(t_{PHL}) &= k_n'\left(\frac{W}{L}\right)_n\left(\left(V_{DD} - V_{t0}\right)v_0 - \frac{1}{2}(v_0)^2\right) \\ &= \left(500\times10^{-6}\right)(1.5)\left(1.2-0.4\right)0.6 - \frac{1}{2}(0.6)^2 \\ &= 2.25\times10^{-4}\ \text{A} \end{aligned}$$

Step 11 of 15

Therefore, the current of NMOS transistor at high-low propagation delay $i_{ON}(t_{PHL})$ is $\boxed{\left(2.25\times10^{-4}\right)\text{A}}$.

Step 12 of 15

Determine the threshold voltage (V_t).

$$V_t = V_{t0} + \gamma\left(\sqrt{v_0 + 2\phi_f} - \sqrt{2\phi_f}\right)$$

$$V_t = V_{t0} + \gamma\left(\sqrt{V_{DD} - v_0 + 2\phi_f} - \sqrt{2\phi_f}\right) \quad \text{..... (10)}$$

Replace 0.4 for V_{t0}, 1.2 for V_{DD}, 0.2 for γ, 0.6 for v_0, and 0.88 for $2\phi_f$ in equation 10.

$$\begin{aligned} V_t &= 0.4 + 0.2\left(\sqrt{1.2 - 0.6 + 0.88} - \sqrt{0.88}\right) \\ &= 0.4 + 0.2(1.21 - 0.93) \\ &= 0.456\ \text{V} \end{aligned}$$

Step 13 of 15

Determine the drain current of PMOS transistor at high-low propagation delay $i_{DP}(t_{PHL})$.

$$i_{DP}(t_{PHL}) = \frac{1}{2}k_p'\left(\frac{W}{L}\right)_p\left(V_{DD} - v_0 - V_\phi\right)^2 \quad \text{..... (11)}$$

Replace 125- $\mu A/V^2$ for k_p', 1.5 for $\left(\frac{W}{L}\right)_p$, 1.2 for V_{DD}, 0.6 for v_0, and 0.456 for V_ϕ in equation 11.

$$\begin{aligned} i_{DP}(t_{PHL}) &= \frac{1}{2}\left(125\times10^{-6}\right)(1.5)(1.2-0.6-0.456)^2 \\ &= 1.944\ \mu\text{A} \end{aligned}$$

Therefore, the drain current of PMOS transistor at high-low propagation delay $i_{DP}(t_{PHL})$ is $\boxed{1.944\ \mu\text{A}}$.

Step 14 of 15

Determine the high-low propagation delay t_{PHL}.

$$t_{PHL} = \frac{C\left(\frac{V_{DD}}{2}\right)}{i_{D_{avg}}} \quad \text{..... (12)}$$

$$\begin{aligned} i_{D_{avg}} &= \frac{\left(i_{OH}(0) + i_{DP}(0) + i_{DP}(t_{PLH}) + i_{ON}(t_{PLH})\right)}{2} \\ &= \frac{1}{2}\left[\left(2.4\times10^{-4}\right) + \left(6\times10^{-5}\right) + \left(2.25\times10^{-4}\right) + \left(1.944\times10^{-4}\right)\right] \\ &= 2.63\times10^{-4}\ \text{A} \end{aligned}$$

Replace 15 fF for C, 1.2 for V_{DD}, and $\left(2.63\times10^{-4}\right)$ for $i_{D_{avg}}$ in equation 12.

$$\begin{aligned} t_{PLH} &= \frac{\left(1.5\times10^{-15}\right)\left(\frac{1.2}{2}\right)}{\left(2.63\times10^{-4}\right)} \\ &= 3.42\ \text{ns} \end{aligned}$$

Therefore, the low-high propagation delay is $\boxed{3.42\ \text{ns}}$ and Q_p turn off at $0.5\ \text{V}$.

Step 15 of 15

(d)

Determine the propagation delay (t_p).

$$t_p = \frac{1}{2}\left(t_{PLH} + t_{PHL}\right) \quad \text{..... (13)}$$

Replace 5.52 ns for t_{PLH}, and 3.42 for t_{PHL} in equation 13.

$$\begin{aligned} t_p &= \frac{1}{2}\left(t_{PLH} + t_{PHL}\right) \\ &= \frac{1}{2}(5.52 + 3.42) \\ &= 4.47\ \text{ns} \end{aligned}$$

Therefore, the propagation delay of the CMOS inverter is $\boxed{4.47\ \text{ns}}$.

Step 1 of 10

Refer to Figure 15.21 in the textbook.

Consider the output voltage $v_0 = 0\ \text{V}$.

Determine the equivalent resistance of NMOS transistor.

$$R_{N_{eq}} = \frac{V_{DD} - v_0}{\frac{1}{2}k_n\left(V_{DD} - v_0 - V_{t0}\right)^2}$$

$$R_{N_{eq}} = \frac{V_{DD} - v_0}{\frac{1}{2}k_n'\left(\frac{W}{L}\right)_n\left(V_{DD} - v_0 - V_{t0}\right)^2} \quad \text{..... (1)}$$

Step 2 of 10

Replace 1.2 V for V_{DD} , 0 for v_0 , 500- $\mu\text{A/V}^2$ for k_n' , 1.5 for $\left(\frac{W}{L}\right)_n$, and 0.4 for V_{t0} in equation 1.

$$\begin{aligned} R_{N_{eq}} &= \frac{1.2 - 0}{\frac{1}{2}\left(500\times10^{-6}\right)(1.5)(1.2 - 0 - 0.4)^2} \\ &= \frac{1.2}{\frac{1}{2}\left(500\times10^{-6}\right)(1.5)(0.64)} \\ &= 5\ \text{k}\Omega \end{aligned}$$

Therefore, the equivalent resistance of NMOS transistor at $v_0 = 0\ \text{V}$ is $\boxed{5\ \text{k}\Omega}$.

Step 3 of 10

Determine the equivalent resistance of PMOS transistor.

$$R_{P_{eq}} = \frac{V_{DD} - v_0}{\frac{1}{2}k_p\left(V_{DD} - V_{t0}\right)^2}$$

$$R_{P_{eq}} = \frac{V_{DD} - v_0}{\frac{1}{2}k_p'\left(\frac{W}{L}\right)_p\left(V_{DD} - V_{t0}\right)^2} \quad \text{..... (2)}$$

Replace 1.2 V for V_{DD} , 125- $\mu\text{A/V}^2$ for k_p' , 1.5 for $\left(\frac{W}{L}\right)_p$, and 0.4 for V_{t0} in equation 2.

$$\begin{aligned} R_{P_{eq}} &= \frac{1.2 - 0}{\frac{1}{2}\left(125\times10^{-6}\right)(1.5)(1.2 - 0.4)^2} \\ &= \frac{1.2}{\frac{1}{2}\left(125\times10^{-6}\right)(1.5)(0.64)} \\ &= 20\ \text{k}\Omega \end{aligned}$$

Therefore, the equivalent resistance of PMOS transistor at $v_0 = 0\ \text{V}$ is $\boxed{20\ \text{k}\Omega}$.

Step 4 of 10

Determine the equivalent resistance R_{TG} .

$$R_{TG} = R_{N_{eq}} \| R_{P_{eq}} \quad \text{..... (3)}$$

Replace 5 $\text{k}\Omega$ for $R_{N_{eq}}$, and 20 $\text{k}\Omega$ for $R_{P_{eq}}$ in equation 3.

$$\begin{aligned} R_{TG} &= \frac{\left(R_{N_{eq}}\right)\left(R_{P_{eq}}\right)}{R_{N_{eq}} + R_{P_{eq}}} \\ &= \frac{(5\ \text{k}\Omega)(20\ \text{k}\Omega)}{5\ \text{k}\Omega + 20\ \text{k}\Omega} \\ &= \frac{100}{25} \\ &= 4\ \text{k} \end{aligned}$$

Therefore, the equivalent resistance R_{TG} at $v_0 = 0\ \text{V}$ is $\boxed{4\ \text{k}\Omega}$.

Step 5 of 10

Consider the output voltage $v_0 = 0.6\ \text{V}$.

Determine the threshold voltage V_t .

$$V_t = V_{t0} + \gamma\left(\sqrt{v_0 + 2\phi_f} - \sqrt{2\phi_f}\right) \quad \text{..... (4)}$$

Replace 0.6 for v_0 , 0.2 for γ , 0.88 for $2\phi_f$, and 0.4 for V_{t0} in equation 4.

$$\begin{aligned} V_t &= V_{t0} + \gamma\left(\sqrt{v_0 + 2\phi_f} - \sqrt{2\phi_f}\right) \\ &= (0.4) + (0.2)\left(\sqrt{0.6 + 0.88} - \sqrt{0.88}\right) \\ &= 0.456\ \text{V} \end{aligned}$$

Determine the equivalent resistance of NMOS transistor.

$$R_{N_{eq}} = \frac{V_{DD} - v_0}{\frac{1}{2}k_n\left(V_{DD} - v_0 - V_{t0}\right)^2}$$

$$R_{N_{eq}} = \frac{V_{DD} - v_0}{\frac{1}{2}k_n'\left(\frac{W}{L}\right)_n\left(V_{DD} - v_0 - V_t\right)^2} \quad \text{..... (5)}$$

Step 6 of 10

Replace 1.2 V for V_{DD} , 0.6 for v_0 , 500- $\mu\text{A/V}^2$ for k_n' , 1.5 for $\left(\frac{W}{L}\right)_n$, and 0.456 for V_{tn} in equation 5.

$$\begin{aligned} R_{N_{eq}} &= \frac{1.2 - 0.6}{\frac{1}{2}\left(500\times10^{-6}\right)(1.5)(1.2 - 0.6 - 0.456)^2} \\ &= \frac{0.6}{\frac{1}{2}\left(500\times10^{-6}\right)(1.5)(0.0207)} \\ &= 77\ \text{k}\Omega \end{aligned}$$

Therefore, the equivalent resistance of NMOS transistor at $v_0 = 0.6\ \text{V}$ is $\boxed{77\ \text{k}\Omega}$.

Step 7 of 10

Determine the equivalent resistance of PMOS transistor.

$$R_{P_{eq}} = \frac{1}{k_p\left[V_{DD} - |V_{tp}| - \frac{1}{2}(V_{DD} - v_0)\right]}$$

$$R_{P_{eq}} = \frac{1}{k_p'\left(\frac{W}{L}\right)_p\left[V_{DD} - |V_{tp}| - \frac{1}{2}(V_{DD} - v_0)\right]} \quad \text{..... (6)}$$

Replace 1.2 V for V_{DD} , 0.6 for v_0 , 125- $\mu\text{A/V}^2$ for k_p' , 1.5 for $\left(\frac{W}{L}\right)_p$, and 0.456 for V_{tp} in equation 6.

$$\begin{aligned} R_{P_{eq}} &= \frac{1}{\left(125\times10^{-6}\right)(1.5)\left[1.2 - 0.456 - \frac{1}{2}(1.2 - 0.6)\right]} \\ &= \frac{1}{\left(125\times10^{-6}\right)(1.5)\left[1.2 - 0.456 - \frac{1}{2}(1.2 - 0.6)\right]} \\ &= 12\ \text{k}\Omega \end{aligned}$$

Therefore, the equivalent resistance of PMOS transistor at $v_0 = 0.6\ \text{V}$ is $\boxed{12\ \text{k}\Omega}$.

Step 8 of 10

Determine the equivalent resistance R_{TG} .

$$R_{TG} = R_{N_{eq}} \| R_{P_{eq}} \quad \text{..... (7)}$$

Replace 77 $\text{k}\Omega$ for $R_{N_{eq}}$, and 12 $\text{k}\Omega$ for $R_{P_{eq}}$ in equation 7.

$$\begin{aligned} R_{TG} &= \frac{\left(R_{N_{eq}}\right)\left(R_{P_{eq}}\right)}{R_{N_{eq}} + R_{P_{eq}}} \\ &= \frac{(75\ \text{k}\Omega)(12\ \text{k}\Omega)}{75\ \text{k}\Omega + 12\ \text{k}\Omega} \\ &= 10.34\ \text{k}\Omega \end{aligned}$$

Therefore, the equivalent resistance R_{TG} at $v_0 = 0.6\ \text{V}$ is $\boxed{10.34\ \text{k}\Omega}$.

Step 9 of 10

Determine the average of R_{TG} at $v_0 = 0\ \text{V}$ and R_{TG} at $v_0 = 0.6\ \text{V}$.

$$R_{TG}\Big|_{avg} = \frac{R_{TG}(v_0 = 0\ \text{V}) + R_{TG}(v_0 = 0.6\ \text{V})}{2} \quad \text{..... (8)}$$

Replace $4\ \text{k}$ for $R_{TG}(v_0 = 0\ \text{V})$, and 10.34 $\text{k}\Omega$ for $R_{TG}(v_0 = 0.6\ \text{V})$ in equation 8.

$$\begin{aligned} R_{TG}\Big|_{avg} &= \frac{R_{TG}(v_0 = 0\ \text{V}) + R_{TG}(v_0 = 0.6\ \text{V})}{2} \\ &= \frac{4\ \text{k} + 10.34\ \text{k}}{2} \\ &= 7.17\ \text{k}\Omega \end{aligned}$$

Step 10 of 10

Determine the propagation delay t_{PLH} .

Consider $C = 15\ \text{fF}$.

$$\begin{aligned} t_{PLH} &= 0.693 R_{TG|avg} C \\ &= 0.693(7.17\ \text{k})(15\text{fF}) \\ &= 74.5\ \text{ns} \end{aligned}$$

Therefore, the propagation delay of the transmission gate is $\boxed{74.5\ \text{ns}}$.

Step 1 of 8

Refer to Figure 14.12 (b) in the textbook for the transmission gate with low input.

The value of electron mobility constant is,

$$k_n' = 300\ \frac{\mu A}{V^2}$$

The value of hole mobility constant is,

$$k_p' = 75\ \frac{\mu A}{V^2}$$

The supply voltage is,

$$V_{DD} = 5\ V$$

The value of γ is,

$$\gamma = 0.3\ V^{\frac{1}{2}}$$

The width to length ratio of MOS transistor is,

$$\left(\frac{W}{L}\right)_n = \left(\frac{W}{L}\right)_p = 1.5$$

The total capacitance at the output node is,

$$C = 15\ fF$$

Step 2 of 8

When the input is $0\ V$, the transistor Q_N will be in saturation region.

It will have a drain current which is given by,

$$i_{DN} = \frac{1}{2}k_n(V_{DD} - V_{t0})^2$$

Until the drain source voltage is,

$$\begin{aligned} V_{DSN} &= V_{GS} - V_{tn} \\ &= V_{DD} - V_{tn} \end{aligned}$$

The expression for equivalent resistance of Q_N is,

$$R_{Neq} = \frac{v_O}{\frac{1}{2}k_n'(V_{DD} - V_{t0})^2}$$

Step 3 of 8

If the output voltage is less than the one V_{tn}, the expression for equivalent resistance of Q_N is,

$$R_{Neq} = \frac{v_O}{k_n\left[(V_{DD} - V_{t0})v_O - \frac{v_O^2}{2}\right]^2} \quad (\text{For } v_O \le V_{DD} - V_{t0})$$

Step 4 of 8

When the output voltage is,

$$v_O = V_{DD}$$

The equivalent resistance of Q_N is,

$$\begin{aligned} R_{Neq} &= \frac{V_{DD} - v_O}{\frac{1}{2}k_n'\left(\frac{W}{L}\right)_n(V_{DD} - V_{t0} - v_O)^2} \\ &= \frac{1.8\ V - 0\ V}{\frac{1}{2}\left(300\ \frac{\mu A}{V^2}\right)(1.5)(1.8\ V - 0.5\ V - 0\ V)^2} \\ &= 4.7\ k\Omega \end{aligned}$$

Therefore, the resistance R_{Neq} is, $\boxed{4.7\ k\Omega}$.

Step 5 of 8

The equivalent resistance of the transistor Q_P is given by,

$$\begin{aligned} R_{Peq} &= \frac{V_{DD} - 0}{\frac{1}{2}k_P'\left(\frac{W}{L}\right)_p(V_{DD} - V_{t0} - v_O)^2} \\ &= \frac{1.8\ V - 0\ V}{\frac{1}{2}\left(75\ \frac{\mu A}{V^2}\right)(1.5)(1.8\ V - 1.5\ V)^2} \\ &= 18.9\ k\Omega \end{aligned}$$

Therefore, the resistance R_{Peq} is, $\boxed{18.9\ k\Omega}$.

Therefore, the transmission gate equivalent resistance is,

$$\begin{aligned} R_{TG(v_O = V_{DD})} &= R_{Neq} \parallel R_{Peq} \\ &= \frac{(4.7\ k\Omega)(18.9\ k\Omega)}{(4.7\ k\Omega) + (18.9\ k\Omega)} \\ &= 3.763\ k\Omega \end{aligned}$$

Therefore, the transmission gate equivalent resistance, $R_{TG(v_O = V_{DD})}$ is, $\boxed{3.763\ k\Omega}$.

Step 6 of 8

If the output voltage is,

$$\begin{aligned} v_O &= \frac{V_{DD}}{2} \\ &= 0.9\ V \end{aligned}$$

The value of $V_{DD} - V_{tn}$ is,

$$\begin{aligned} V_{DD} - V_{tn} &= 1.8\ V - 0.5\ V \\ &= 1.3\ V \end{aligned}$$

As the output voltage is less than the $V_{DD} - V_{tn}$, the transistor Q_N is in the triode region.

The equivalent resistance of Q_N is,

$$\begin{aligned} R_{Neq} &= \frac{1}{k_n'\left(\frac{W}{L}\right)_n(V_{DD} - V_{tn} - v_O)} \\ &= \frac{1}{\frac{1}{2}\left(300\ \frac{\mu A}{V^2}\right)(1.5)(1.8\ V - 0.5\ V - 0.9\ V)} \\ &= 2.6\ k\Omega \end{aligned}$$

Therefore, the resistance R_{Neq} is, $\boxed{2.6\ k\Omega}$.

Step 7 of 8

The value of the voltage V_{tn} is,

$$\begin{aligned} V_{tn} &= V_{t0} + \gamma\left(\sqrt{v_O + 2\phi_f} - \sqrt{2\phi_f}\right) \\ &= 0.5\ V + (0.3)\left(\sqrt{0.9\ V + 0.85\ V} - \sqrt{0.85\ V}\right) \\ &= 0.62\ V \end{aligned}$$

The equivalent resistance of the transistor Q_P is,

$$\begin{aligned} R_{Peq} &= \frac{2v_O}{k_P'\left(\frac{W}{L}\right)_p\left[v_O - |V_{tp}|\right]^2} \\ &= \frac{2(0.9\ V)}{\left(75\ \frac{\mu A}{V^2}\right)(1.5)(0.9\ V - 0.62\ V)^2} \\ &= \frac{12}{(75 \times 10^{-4})(0.28)^2} \\ &= 204\ k\Omega \end{aligned}$$

Therefore, the resistance R_{Peq} is, $\boxed{204\ k\Omega}$.

Step 8 of 8

Therefore, the transmission gate equivalent resistance is,

$$\begin{aligned} R_{TG\left(v_O = \frac{V_{DD}}{2}\right)} &= R_{Neq} \parallel R_{Peq} \\ &= \frac{(2.6\ k\Omega)(204\ k\Omega)}{(2.6\ k\Omega) + (204\ k\Omega)} \\ &= 2.57\ k\Omega \end{aligned}$$

Therefore, the transmission gate equivalent resistance, $R_{TG\left(v_O = \frac{V_{DD}}{2}\right)}$ is, $\boxed{2.57\ k\Omega}$.

The average transmission gate resistance is,

$$\begin{aligned} R_{TG(avg)} &= \frac{R_{TG}(v_O = 0\ V) + R_{TG}(v_O = 0.9\ V)}{2} \\ &= \frac{1}{2}(3.76\ k\Omega + 2.57\ k\Omega) \\ &= 3.17\ k\Omega \end{aligned}$$

Therefore, the average transmission gate resistance, $R_{TG(avg)}$ is, $\boxed{3.17\ k\Omega}$.

The time taken t_{PLH} for the transmission is,

$$\begin{aligned} t_{PLH} &= 0.69R_{TG(avg)}C \\ &= 0.69(3.17\ k\Omega)(15\ fF) \\ &= 32.8\ ps \end{aligned}$$

Therefore, the time taken for transmission, t_{PLH} is, $\boxed{32.8\ ps}$.

Step 1 of 3

15.039P

Consider a transmission gate is fabricated with 0.13-μm technology.

Consider $\left(\frac{W}{L}\right)_n = \left(\frac{W}{L}\right)_p = 1.5$.

Determine the equivalent resistance of transmission gate R_{TG} .

$$R_{TG} = \frac{12.5}{\left(\frac{W}{L}\right)_n} \quad \text{..... (1)}$$

Step 2 of 3

Substitute 1.5 for $\left(\frac{W}{L}\right)_n$ in equation (1).

$$\begin{aligned} R_{TG} &= \frac{12.5}{\left(\frac{W}{L}\right)_n}\ \text{k}\Omega \\ &= \frac{12.5}{1.5}\ \text{k}\Omega \\ &= 8.33\ \text{k}\Omega \end{aligned}$$

Therefore, the equivalent resistance of the transmission gate is $\boxed{8.33\ \text{k}\Omega}$.

Step 3 of 3

Determine the propagation delay of the transmission gate (t_p) .

$$\begin{aligned} t_p &= 0.693 R_{TG} C \\ &= 0.693\left(8.33\times10^{3}\right)\left(10\times10^{-15}\right) \\ &= 57.7\ \text{ns} \end{aligned}$$

Therefore, the propagation delay of the transmission gate is $\boxed{57.7\ \text{ns}}$.

Step 1 of 4

Refer to Figure 15.40 in textbook.

Consider the propagation delay t_p from Elmore delay.

$$t_p = 0.69\sum_{k=0}^{n} kCR_{TG}$$

Consider if n is the number of transmission gates that are connected in a chain form.

Consider the propagation delay of the series of the transmission gate.

$$t_p = 0.69CR_{TG}\frac{n(n+1)}{2} \quad \text{..... (1)}$$

Step 2 of 4

Replace 16 for n, $10\ \text{k}\Omega$ for R_{TG}, and $10\ \text{fF}$ for C in equation 1.

$$\begin{aligned} t_p &= 0.69\left(10\times10^{-15}\right)\left(10\times10^{3}\right)\left(\frac{16(16+1)}{2}\right) \\ &= \left(0.69\times10^{-10}\right)(136) \\ &= 9.384\ \text{ns} \end{aligned}$$

Therefore, the propagation delay of the series of transmission gate is $\boxed{9.384\ \text{ns}}$.

Step 3 of 4

Consider ramp signal is applied as input to the transmission gate.

Consider the propagation delay t_p from Elmore delay.

$$t_p = \sum_{k=0}^{n} kCR_{TG}$$

Determine the propagation delay.

$$t_p = CR_{TG}\frac{n(n+1)}{2} \quad \text{..... (2)}$$

Step 4 of 4

Replace 16 for n, $10\ \text{k}\Omega$ for R_{TG}, and $10\ \text{fF}$ for C in equation 2.

$$\begin{aligned} t_p &= \left(10\times10^{-15}\right)\left(10\times10^{3}\right)\left(\frac{16(16+1)}{2}\right) \\ &= 13.6\ \text{ns} \end{aligned}$$

Therefore, the propagation delay of the series of the transmission gate with ramp input is $\boxed{13.6\ \text{ns}}$.

Step 1 of 2

Refer to Figure 14.17 in the textbook.

(a)

The expression $\bar{Y} = AB + \bar{A}\bar{B}$ can be realized by the following circuit:

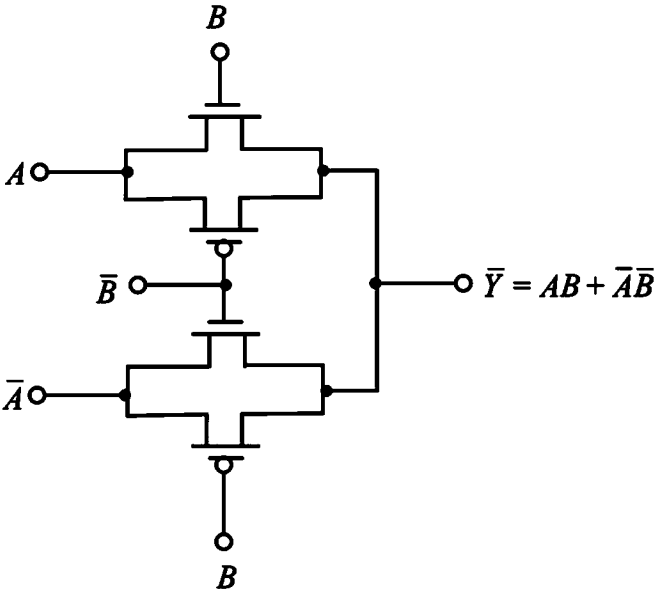


Figure 1

Thus, the expression for $\bar{y}$ is realized with two transmission gates in Figure 1.

Step 2 of 2

(b)

The expression $Z = \bar{Y}C + Y\bar{C}$ can be realized by the following circuit:

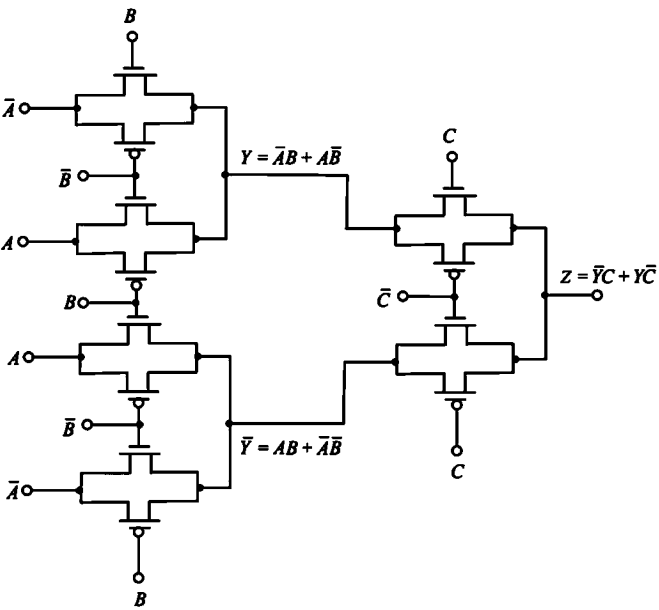


Figure 2

In Figure 2, three transmission gates are used and a total of 12 transistors are used.

Thus, the expression for $\bar{Z}$ is realized with two transmission gates in Figure 1.

15.042P

Draw the CPL (complementary pass transistor logic) for the expression $Y = A\overline{B} + \overline{A}B$ and $Y = A\overline{B} + \overline{A}B$

.

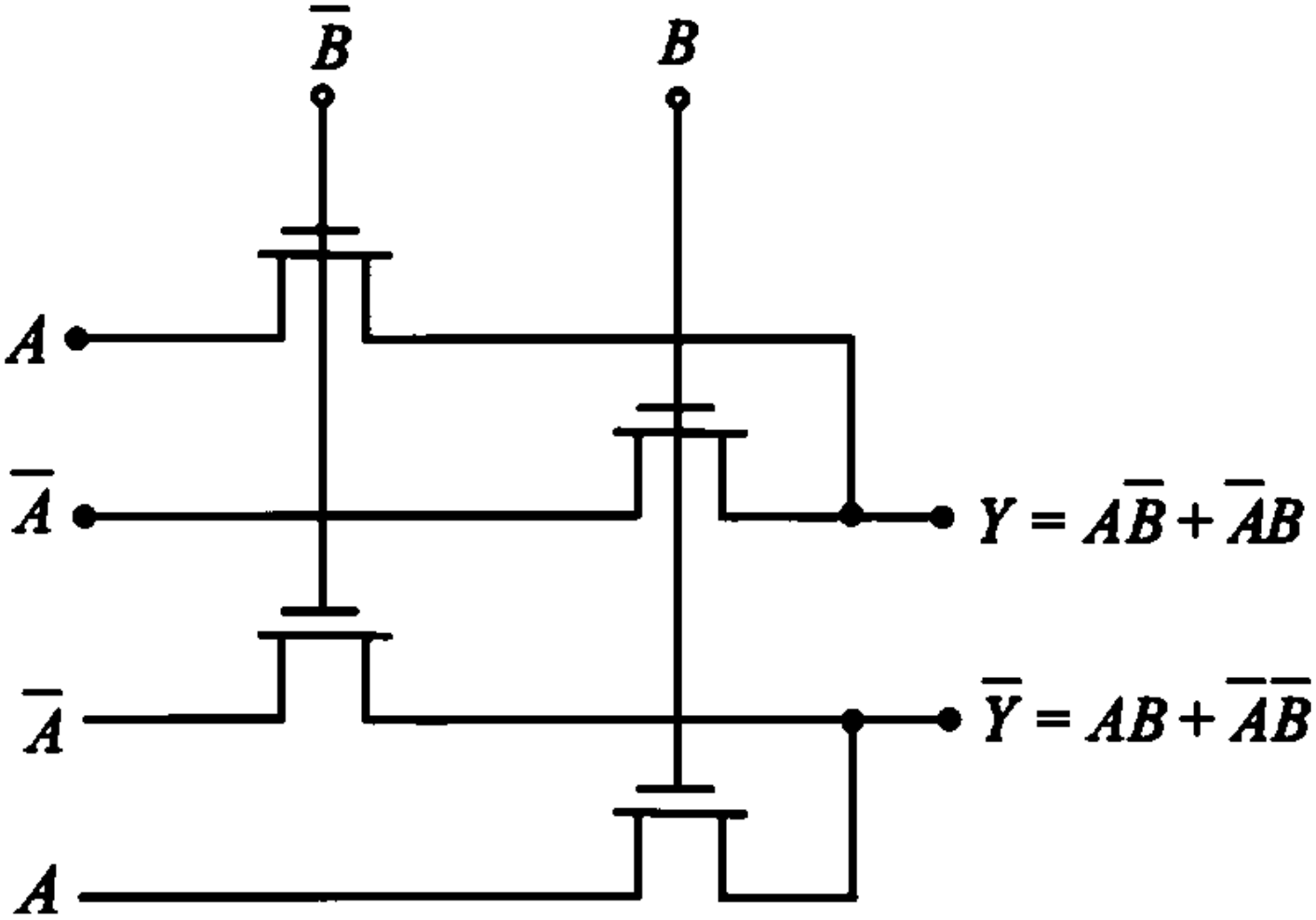


Figure 1

Thus, the CPL's for the expressions $Y = A\overline{B} + \overline{A}B$ and $Y = A\overline{B} + \overline{A}B$ is shown in Figure 1.

Step 1 of 2

15.043P

The CPL (complementary pass transistor logic) for the expression $Z = ABC$ and $\overline{Z} = \overline{A+B+C} = \overline{A}+\overline{B}+\overline{C}$ as follows:

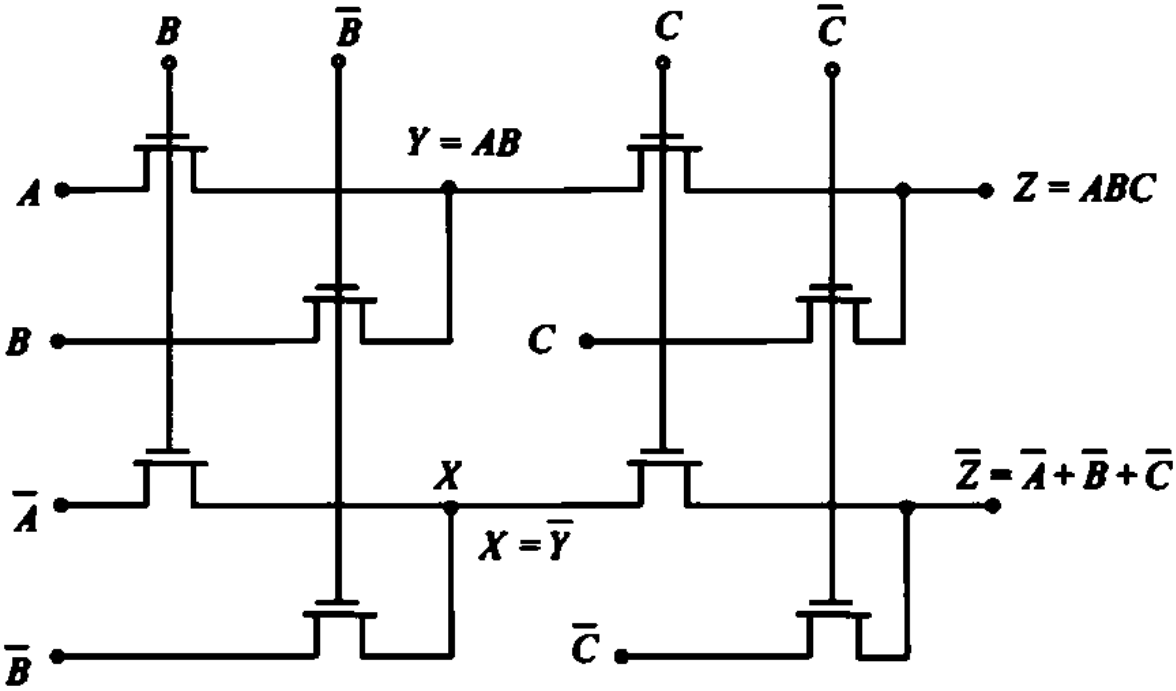


Figure 1

Step 2 of 2

From Figure 1, the expression at node *Y* is the logical AND operation of *A* and *B*. the output *Z* is the logical AND operation of *Y* and the input *C*.

$$Z = ABC$$

In Figure 1, the expression at the node *X* is,

$$X = \overline{A}B + \overline{B}\overline{B}$$
$$X = \overline{A}+\overline{B}$$
$$X = \overline{Y}$$

The output $\overline{Z}$ is,

$$\begin{aligned}\overline{Z} &= \overline{Y}C + \overline{C}\overline{C} \\ &= \left(\overline{A}+\overline{B}\right)C + \overline{C} \\ &= \overline{A}+\overline{B}+\overline{C} \qquad \left(\text{since } X + \overline{X}Y = X + Y\right)\end{aligned}$$

Thus, the CPL's for the expression $Z = ABC$ and $\overline{Z} = \overline{A}+\overline{B}+\overline{C}$ is shown in Figure 1.

Step 1 of 4

Refer to Figure 14.19 in the textbook.

Based on the dynamic logic circuit, the NOT gate logic diagram is as follows:

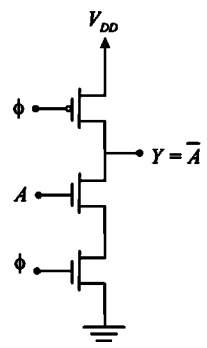


Figure 1

Figure 1 shows the implementation of NOT gate.

Step 2 of 4

Based on the dynamic logic circuit, the NAND gate logic diagram is as follows:

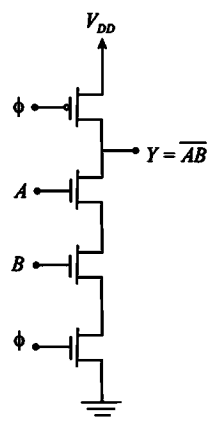


Figure 2

Figure 2 shows the implementation of NAND gate.

Step 3 of 4

Based on the dynamic logic circuit, the NOR gate logic diagram is as follows:

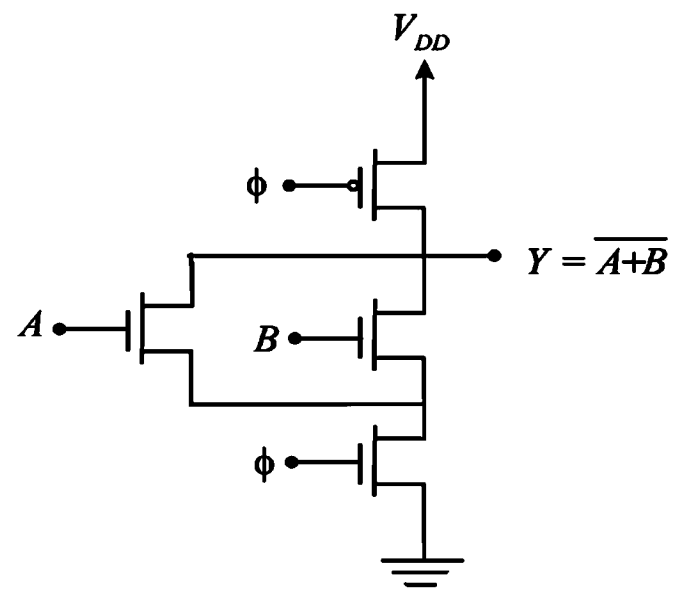


Figure 3

Figure 3 shows the implementation of NOR gate.

Step 4 of 4

The complete circuit for the expression $Y = \overline{AB+CD}$ is,

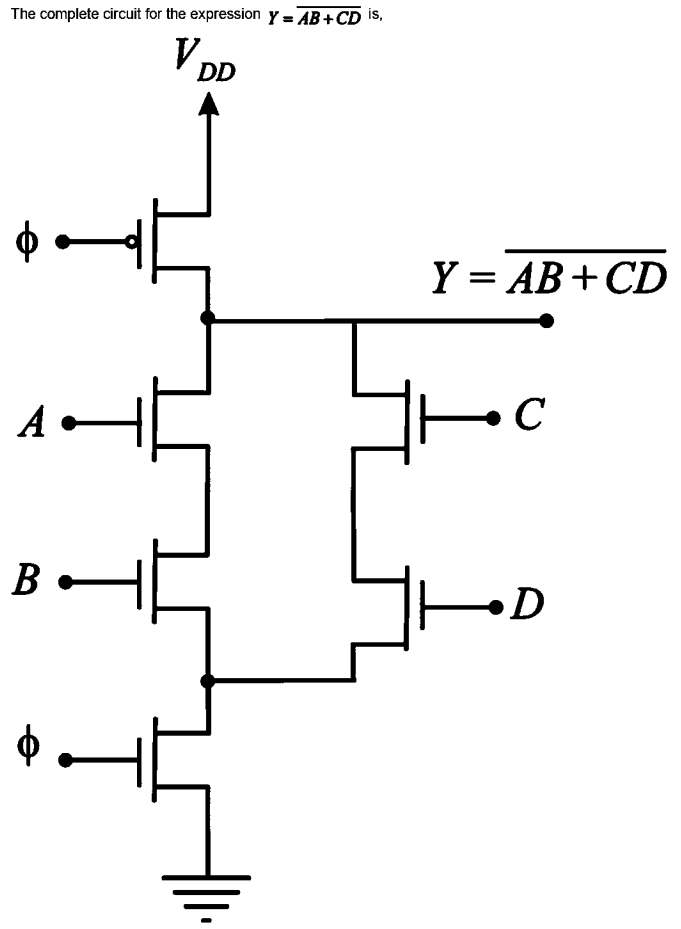


Figure 4

Figure 4 shows the implementation of the expression $Y = \overline{AB+CD}$.

Step 1 of 4

Refer to Figure 15.29 (b) in textbook.

Consider the gate is in precharge operation with Q_p at $0\ V$, and at $t = 0$.

Consider C_L is fully discharged.

For $v_Y = 0.1V_{DD} = 0.12$ the transistor Q_p is operating in the saturation region.

Determine the drain current at $0.1V_{DD}$.

$$i_D(0.1\,V_{DD}) = \frac{1}{2}k_p\left(\frac{W}{L}\right)_p\left(V_{DD} - |V_{tp}|\right)^2 \quad \text{..... (1)}$$

Step 2 of 4

Replace 1.2 V for V_{DD}, 125-$\mu A/V^2$ for k_p', 1.5 for $\left(\frac{W}{L}\right)_p$, and 0.4 for V_{t0} in equation 1.

$$\begin{aligned} i_D(0.1\,V_{DD}) &= \frac{1}{2}\left(125\times10^{-6}\right)(1.5)(1.2-0.4)^2 \\ &= \frac{1}{2}\left(125\times10^{-6}\right)(1.5)(0.64) \\ &= 60\ \mu A \end{aligned}$$

Determine the drain current at $0.9V_{DD}$.

$$i_D(0.9\,V_{DD}) = k_p\left(\frac{W}{L}\right)_p\left[\left(V_{DD} - |V_{tp}|\right)(V_{DD} - 0.9V_{DD}) - \frac{1}{2}(V_{DD} - 0.9V_{DD})^2\right] \quad \text{..... (2)}$$

Step 3 of 4

Replace 1.2 V for V_{DD}, 125-$\mu A/V^2$ for k_p', 1.5 for $\left(\frac{W}{L}\right)_p$, and 0.4 for V_{t0} in equation 2.

$$\begin{aligned} i_D(0.9\,V_{DD}) &= \left(125\times10^{-6}\right)(1.5)\left[(1.2-0.4)(1.2-0.9(1.2)) - \frac{1}{2}(1.2-0.9(1.2))^2\right] \\ &= \left(125\times10^{-6}\right)(1.5)\left[(0.8)(0.12) + 0.048\right] \\ &= 27\ \mu A \end{aligned}$$

Determine the average capacitor discharge current I_{av}.

$$\begin{aligned} I_{av} &= \frac{\left(i_D(0.1V_{DD}) + i_D(0.9V_{DD})\right)}{2} \\ &= \frac{(60\ \mu A + 27\ \mu A)}{2} \\ &= 43.5\ \mu A \end{aligned}$$

Step 4 of 4

Determine the rise time t_r.

$$t_r = \frac{C\Delta v_Y}{I_{av}} \quad \text{..... (3)}$$

Consider $C = 15\ fF$.

Replace $0.9V_{DD} - 0.1V_{DD}$ for v_Y, and 43.5 μA for I_{av} in equation 3.

$$\begin{aligned} t_r &= \frac{\left(15\times10^{-6}\right)(0.9V_{DD} - 0.1V_{DD})}{\left(43.5\times10^{-6}\right)} \\ &= \frac{\left(15\times10^{-6}\right)(0.9(1.2) - 0.1(1.2))}{\left(43.5\times10^{-6}\right)} \\ &= \frac{\left(15\times10^{-6}\right)(1.08 - 0.12)}{\left(43.5\times10^{-6}\right)} \\ &= 0.331\ ns \end{aligned}$$

Therefore, the rise time t_r is $\boxed{0.331\ ns}$.

Step 1 of 5

Refer to Figure 15.29 (c) in the textbook.

Consider a 2 input NAND gate is realized in dynamic logic form and fabricated in CMOS process.

Determine the drain current at $v_y = V_{DD}$.

$$i_D(V_{DD}) = \frac{1}{2}k_n\left(\frac{W}{L}\right)_n (V_{DD} - V_t)^2$$

$$i_D(V_{DD}) = \frac{1}{2}(k_n')\left(\frac{W}{L}\right)_{eq} (V_{DD} - V_t)^2 \quad \text{..... (1)}$$

Step 2 of 5

Determine the aspect ratio of equivalent device (Q_{eq}).

$$\left(\frac{W}{L}\right)_{eq} = \frac{1}{2}\left(\frac{W}{L}\right)$$
$$= \frac{1}{2}(1.5)$$
$$= 0.75$$

Replace $500\ \mu A/V^2$ for k_n', 0.75 for $\left(\frac{W}{L}\right)_{eq}$, 1.2 for V_{DD}, and 0.4 for V_t in equation 1.

$$i_D(V_{DD}) = \frac{1}{2}(500\times10^{-6})(0.75)(1.2-0.4)^2$$
$$= 1.2\times10^{-4}\ A$$

Step 3 of 5

Determine the drain current at $v_y = V_{DD}$.

$$i_D(V_{DD}/2) = \frac{1}{2}k_n\left(\frac{W}{L}\right)_{eq}\left[(V_{DD} - V_t)\frac{V_{DD}}{2} - \frac{1}{2}\left(\frac{V_{DD}}{2}\right)^2\right] \quad \text{..... (2)}$$

Replace $125\ \mu A/V^2$ for k_n', 0.75 for $\left(\frac{W}{L}\right)_{eq}$, 1.2 for V_{DD}, and 0.4 for V_t in equation 2.

$$i_D(V_{DD}/2) = \frac{1}{2}(125\times10^{-6})(0.75)\left[(1.2-0.4)\frac{1.2}{2} - \frac{1}{2}\left(\frac{1.2}{2}\right)^2\right]$$
$$= \frac{1}{2}(125\times10^{-6})(0.75)\left[(0.8)(0.6) - \frac{1}{2}(0.36)\right]$$
$$= 1.40\times10^{-5}\ A$$

Step 4 of 5

Determine the average capacitor discharge current I_{av}.

$$I_{av} = \frac{i_D(V_{DD}) + i_D(V_{DD}/2)}{2}$$
$$= \frac{(1.2\times10^{-4}) + (1.40\times10^{-5})}{2}$$
$$= (6.7\times10^{-5})\ A$$

Determine the high-low propagation delay t_{PHL}.

$$t_{PHL} = \frac{C(V_{DD} - V_{DD}/2)}{I_{av}} \quad \text{..... (3)}$$

Step 5 of 5

Replace 15 fF for C, 1.2 for V_{DD}, and 6.7×10^{-5} for I_{av} in equation 3.

$$t_{PHL} = \frac{(15\times10^{-15})(1.2-0.6)}{(6.7\times10^{-5})}$$
$$= 0.134\ ns$$

Therefore, the propagation delay t_{PHL} is $\boxed{0.134\ ns}$.

15.047P

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15.048P

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15.049P

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Step 1 of 6

Refer to Figure E 14.10 in the text book.

The source of Q_{eq1} is earthed and its gate is at $+V_{DD}$. The transistor conducts in the saturation mode. Hence discharge current from C_{L1} is I_{Dsat} .

The drain current is,

$$I_{Dsat} = \frac{1}{2} K_n \left(\frac{W}{L}\right)_n V_{OV_n}^2$$

Since the capacitor C_{L1} is discharging, the voltage v_{Y1} drops below V_m

The change in the voltage is ΔV .

$$\Delta V = V_{DD} - V_m$$
$$\Delta V = V_{OV_n}$$

Step 2 of 6

The expression for charge is,

$$Q = I \cdot t \text{ and } Q = C \cdot V$$

Therefore, $CV = It$

Here, the capacitor C_{L1} discharges and the discharge current, I_{Dsat} resulting in the change in the voltage of ΔV , therefore,

$$C_{L1} \cdot \Delta V = I_{Dsat} \cdot t \quad \text{..... (1)}$$

$$C_{L1} \cdot V_{OV_n} = I_{Dsat} \cdot t \left(\because \nabla V = V_{OV_n}\right)$$

$$C_{L1} V_{OV_n} = \frac{1}{2} K_n (W/L)\left(V_{OV_n}^2\right)(t)$$

$$t = \frac{C_{L1} V_{OV_n}}{\frac{1}{2} K_n (W/L)\left(V_{OV_n}^2\right)}$$

$$t = \frac{C_{L1}}{\frac{1}{2} K_n (W/L)\left(V_{OV_n}\right)}$$

Step 3 of 6

C_{L1} discharges at a constant rate, but C_{L2} discharges at a decreasing rate, as the overdrive voltage of Q_{eq2} is $\left(V_{C_{L1}} - V_t\right)$

From equation (1) , it is clear that is,

$$C_{L1} \cdot \Delta V = I_{Dsat} \cdot t$$
$$\Delta V = \frac{(I_{Dsat})(t)}{C_{L1}}$$

So, in a time Δt , V_{CL1} decreases by $\frac{(\Delta t)(I_{Deq1})}{C_{L1}}$

Where, $I_{Deq1} = I_{Dsat} = \frac{1}{2} K_n \left(\frac{W}{L}\right)_n V_{OV_n}$

Therefore,

$$\Delta V_{CL} = \frac{(\Delta t)(I_{Deq1})}{C_{L1}} = \left(\frac{\Delta t}{C_{L1}}\right)\left[\frac{1}{2} K_n \left(\frac{W}{L}\right)\left(V_{OV_n}^2\right)\right] \quad \text{..... (2)}$$

Step 4 of 6

Now, the overdrive voltage, V_{OV_n} of Q_{eq2} becomes $\left(V_{OV_n} - \Delta V_{CL}\right)$.

In a time Δt , I_D of Q_{eq2} changes from I_{Dsat} to the new value.

I_D of Q_{eq2} is $\frac{1}{2} K_n \left(\frac{W}{L}\right)_n \left(V_{OV_n} - \Delta V_{CL}\right)^2$

Therefore, the average value of the current is,

$$I_{av} = \frac{I_D \text{ of } Q_{eq1} + I_D \text{ of } Q_{eq2}}{2}$$
$$= \frac{\left[\frac{1}{2} K_n \left(\frac{W}{L}\right)_n \left(V_{OV_n}^2\right) + \frac{1}{2} K_n \left(\frac{W}{L}\right)_n \left(V_{OV_n} - \Delta V_{CL}\right)^2\right]}{2}$$
$$= \frac{\frac{1}{2} K_n (W/L)\left[V_{OV_n}^2 + \left(V_{OV_n} - \Delta V_{CL}\right)^2\right]}{2}$$

Step 5 of 6

Now, the change in voltage of C_{L2} is,

$$\Delta V_{C_{L2}} = \frac{(\text{Average current})(\Delta t)}{C_{L2}}$$
$$= \frac{\frac{1}{2} K_n (W/L)\left[V_{OV_n}^2 + \left(V_{OV_n} - \Delta V_{CL}\right)^2\right]}{2}\left(\frac{\Delta t}{C_{L2}}\right)$$

From equation (2), it is clear that is,

$$\Delta V_{CL} = \left(\frac{\Delta t}{C_{L1}}\right)\left[\frac{1}{2} K_n \left(\frac{W}{L}\right)\left(V_{OV_n}^2\right)\right]$$

Therefore,

$$\Delta V_{C_{L2}} = \frac{\frac{1}{2} K_n (W/L)\left[V_{OV_n}^2 + \left(V_{OV_n} - \left(\frac{\Delta t}{C_{L1}}\right)\left[\frac{1}{2} K_n \left(\frac{W}{L}\right)\left(V_{OV_n}^2\right)\right]\right)^2\right]}{2}\left(\frac{\Delta t}{C_{L2}}\right) \quad \text{..... (3)}$$

Step 6 of 6

The voltage V_{OV_n} is,

$$V_{OV_n} = V_{DD} - V_m$$
$$= V_{DD} - 0.2V_{DD}$$
$$= 0.8V_{DD}$$

Substitute $0.8V_{DD}$ for V_{OV_n} in equation (3) .

$$\Delta V_{C_{L2}} = \frac{\frac{1}{2} K_n (W/L)\left[0.8^2 V_{DD}^2 + \left(0.8V_{DD} - \left(\frac{\Delta t}{C_{L1}}\right)\left[\frac{1}{2} K_n \left(\frac{W}{L}\right)\left(0.8^2 V_{DD}^2\right)\right]\right)^2\right]}{2}\left(\frac{\Delta t}{C_{L2}}\right)$$

From the circuit, it is clear that, the change in the voltage of C_{L2} is nothing but the change in the output voltage v_{Y2} .

$$\Delta V_{C_{L2}} = \Delta v_{Y2}$$

Therefore Δv_{Y2} is,

$$\boxed{\frac{\frac{1}{2} K_n (W/L)\left[0.8^2 V_{DD}^2 + \left(0.8V_{DD} - \left(\frac{\Delta t}{C_{L1}}\right)\left[\frac{1}{2} K_n \left(\frac{W}{L}\right)\left(0.8^2 V_{DD}^2\right)\right]\right)^2\right]}{2}\left(\frac{\Delta t}{C_{L2}}\right)}$$

Step 1 of 8

Refer to Example 14.3 in the text book.

During rise in voltage, current in I_{Dsat} change from v_y to $\frac{V_{DD}}{2}$ and this change will be in the triode region.

Finding the $i_D(0.1V_{DD})$:

$$i_D(0.1V_{DD}) = \frac{1}{2}\mu_p C_{ox}\left(\frac{W}{L}\right)_p \left(V_{DD} - |V_{tp}|\right)^2$$

Substitute $0.54\ \mu m/0.18\ \mu m$ for $\left(\frac{W}{L}\right)_p$, $75\ \mu A/V^2$ for $\mu_p C_{ox}$, $1.8\ V$ for V_{DD} and $0.5\ V$ for V_t in equation.

$$\begin{aligned} i_D(0.1V_{DD}) &= \frac{1}{2} \times 75 \times \frac{0.54}{0.18}(1.8 - 0.5)^2 \\ &= 190.1\ \mu A \end{aligned}$$

Therefore, the value of I_{Dsat} for Q_p is $\boxed{190.1\ \mu A}$.

Step 2 of 8

At $v_y = 0.9V_{DD} = 1.62\ V$, Q_p will be operating in the triode region.

Thus, finding the $i_D(0.9V_{DD})$:

$$\begin{aligned} i_D(0.9V_{DD}) &= \mu_p C_{ox}\left(\frac{W}{L}\right)_p \left[\left(V_{DD} - |V_{tp}|\right)(V_{DD} - 0.9V_{DD}) - \frac{1}{2}(V_{DD} - 0.9V_{DD})^2\right] \\ &= 75 \times \frac{0.54}{0.18}\left[(1.8 - 0.5)(1.8 - 1.62) - \frac{1}{2}(1.8 - 1.62)^2\right] \\ i_D(0.9V_{DD}) &= 49\ \mu A \end{aligned}$$

Step 3 of 8

I_{DD} in triode region starts from $190.1\ \mu A$ and when v_y is $\frac{V_{DD}}{2}$ current is computed as follows.

Finding the average capacitor charging current:

$$\begin{aligned} I_{av} &= \frac{1}{2}(190.1 + 49)\mu A \\ I_{av} &= 119.6\ \mu A \end{aligned}$$

Finding the rise time t_r:

$$\begin{aligned} t_r &= \frac{C\Delta v_y}{I_{av}} \\ &= \frac{(20 \times 10^{-15})(0.9V_{DD} - 0.1V_{DD})}{119.6 \times 10^{-6}} \\ t_r &= 0.19\ ns \end{aligned}$$

Step 4 of 8

During, current change of v_y from v_{dd} to v_y is $V_{DD} - V_t$

Finding the $i_D(V_{DD})$:

$$\begin{aligned} i_D(V_{DD}) &= \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_{eq}\left(V_{DD} - |V_t|\right)^2 \\ &= \frac{1}{2} \times 300 \times 0.3(1.8 - 0.5)^2 \\ i_D(V_{DD}) &= 76.1\ \mu A \end{aligned}$$

Step 5 of 8

At $v_y = \frac{V_{DD}}{2}$, Q_{eq} will be operating in triode region.

Finding the $i_D\left(\frac{V_{DD}}{2}\right)$:

$$\begin{aligned} i_D\left(\frac{V_{DD}}{2}\right) &= \mu_n C_{ox}\left(\frac{W}{L}\right)_{eq}\left[\left(V_{DD} - |V_t|\right)\left(\frac{V_{DD}}{2}\right) - \frac{1}{2}\left(\frac{V_{DD}}{2}\right)^2\right] \\ &= 300 \times 0.3\left[(1.8 - 0.5)\left(\frac{1.8}{2}\right) - \frac{1}{2}\left(\frac{1.8}{2}\right)^2\right] \\ i_D\left(\frac{V_{DD}}{2}\right) &= 68.9\ \mu A \end{aligned}$$

Step 6 of 8

Find the average capacitor discharge current:

$$\begin{aligned} I_{av} &= \frac{1}{2}(76.1 + 68.9) \\ &= 72.5\ \mu A \end{aligned}$$

Step 7 of 8

Calculate the fall time, t_f.

$$t_{PHL} = \frac{C\left(V_{DD} - \frac{V_{DD}}{2}\right)}{I_{av}}$$

Substitute $1.8\ V$ for V_{DD} and $72.5\ \mu A$ for I_{av} in equation.

$$\begin{aligned} t_{PHL} &= \frac{(20 \times 10^{-15})(1.8 - 0.9)}{72.5 \times 10^{-6}} \\ &= 0.25\ ns \end{aligned}$$

Therefore, the value of t_{PHL} is $\boxed{0.25\ ns}$.

Step 8 of 8

Finding the maximum frequency:

$$\begin{aligned} f_{max} &= \frac{1}{t_r + t_{PHL}} \\ &= \frac{1}{(0.19 + 0.25) \times 10^{-9}} \end{aligned}$$

Therefore, the maximum frequency is $\boxed{2.27\ GHz}$.

15.052P

We don't have the solution to this problem yet.

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15.053P

We don't have the solution to this problem yet.

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15.054P

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15.055P

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15.056P

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15.057P

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15.058P

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15.059P

We don't have the solution to this problem yet.

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Ask an expert

15.060P

We don't have the solution to this problem yet.

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15.061P

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Step 1 of 2

16.001E

Refer to Figure 15.5 in the text book for the relevant portion of the SR-flip-flop circuit.

When the flip-flop inputs, *S* and ϕ are at $\frac{V_{DD}}{2}$ then the equivalent transistor, Q_{eq} gets saturated. In this case, the output voltage for state 1 is, $v_Q = \frac{V_{DD}}{2}$ and the transistor Q_2 is operating in the triode region.

Write the relation of the width-to-length ratio of the two transistors.

$$\left(\frac{W}{L}\right)_{Q_{eq}} = \frac{1}{2}\left(\frac{W}{L}\right)_{5,6}$$

From the circuit, the current flowing through the drain terminal of NMOS transistor is equal to that of PMOS transistor.

$$I_{Deq} = I_{D2}$$

$$\frac{1}{2}k_n'\left(\frac{W}{L}\right)_{eq}\left(\frac{V_{DD}}{2} - V_t\right)^2 = \frac{1}{2}k_p'\left(\frac{W}{L}\right)_2\left[(V_{DD} - V_t)\frac{V_{DD}}{2} - \frac{1}{2}\left(\frac{V_{DD}}{2}\right)^2\right]$$

Here,

k_n' and k_p' are the design parameters of NPN and PNP transistors respectively

V_t is the transition voltage.

Step 2 of 2

Substitute 300×10^{-6} A/V^2 for k_n', $\frac{300}{4}\times10^{-6}$ A/V^2 for k_p', 0.5 V for V_t, 1.8 V for V_{DD}, $\frac{1}{2}\left(\frac{W}{L}\right)_5$ for $\left(\frac{W}{L}\right)_{Q_{eq}}$ and $\frac{1.08}{0.18}$ for $\left(\frac{W}{L}\right)_2$ in the equation.

$$\frac{1}{2}(300\times10^{-6})\frac{1}{2}\left(\frac{W}{L}\right)_5\left(\frac{1.8}{2} - 0.5\right)^2 = \frac{1}{2}\left(\frac{300}{4}\times10^{-6}\right)\left(\frac{1.08}{0.18}\right)\left[(1.8-0.5)\frac{1.8}{2} - \frac{1}{2}\left(\frac{1.8}{2}\right)^2\right]$$

$$12\times10^{-6}\left(\frac{W}{L}\right)_5 = 172.125\times10^{-6}$$

$$\left(\frac{W}{L}\right)_5 = 14.34$$

Consider *L* as $0.18\ \mu m$ and determine *W*.

$$W = (14.34)L$$
$$= (14.34)(0.18)$$
$$= 2.58\ \mu m$$

Thus, the minimum $\frac{W}{L}$ that can be achieved when inputs are at $\frac{V_{DD}}{2}$ is $\boxed{\frac{2.58\ \mu m}{0.18\ \mu m} \text{ or } 14.34}$.

16.001P

We don't have the solution to this problem yet.

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Step 1 of 3

16.002E

The number of rows in each block is, 1024

Determine the number of bits required for row address.

$$1024 = 2^{10}$$

Therefore, the number of bits required for row address is $\boxed{10}$.

Step 2 of 3

The number of columns in each block is, 128

Determine the number of bits required for column address.

$$128 = 2^{7}$$

Therefore, the number of bits required for row address is $\boxed{7}$.

Step 3 of 3

The number of blocks in the 1M-bit memory chip is 32-blocks.

Determine the number of bits required for block address.

$$32 = 2^{5}$$

Therefore, the number of bits required for row address is $\boxed{5}$.

16.002P

Refer to Figure 15.3 in the textbook for the set/reset SR flip-flop, and its truth table.

The CMOS circuit implementation of the SR flip-flop is,

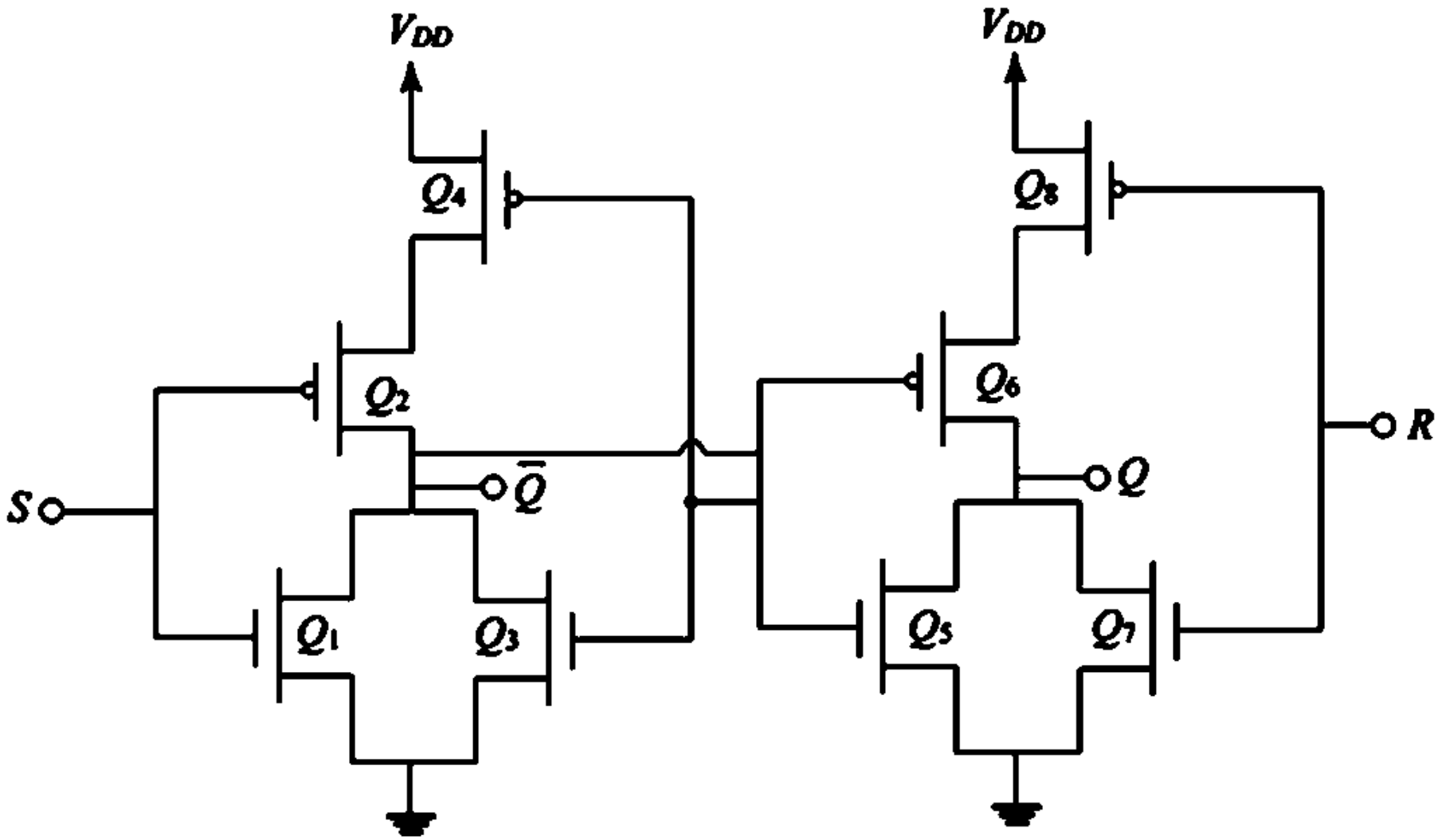


Figure 1

Step 1 of 3

Consider the RC circuit shown in Figure 1.

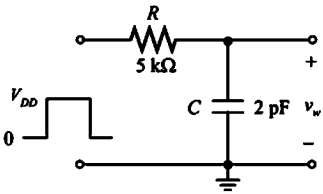


Figure 1

Step 2 of 3

Calculate the time constant, τ using the following formula:

$$\tau = RC$$

Substitute $2\ \text{pF}$ for C and $5\ \text{k}\Omega$ for R.

$$\begin{aligned}\tau &= \left(5\times10^{3}\right)\left(2\times10^{-12}\right)\\ &= 10\times10^{-9}\\ &= 10\ \text{ns}\end{aligned}$$

The voltage on the word line to be reached is, $v_w = \frac{V_{DD}}{2}$.

Step 3 of 3

Consider the expression for the voltage on the word line to calculate the time required to reach $\frac{V_{DD}}{2}$.

$$v_w = V_{DD}\left(1-e^{-\frac{t}{\tau}}\right)$$

Substitute $\frac{V_{DD}}{2}$ for v_w and $10\ \text{ns}$ for τ.

$$\frac{V_{DD}}{2} = V_{DD}\left(1-e^{-\frac{t}{10\times10^{-9}}}\right)$$

$$1-e^{-\frac{t}{10\times10^{-9}}} = \frac{1}{2}$$

$$e^{-\frac{t}{10\times10^{-9}}} = \frac{1}{2}$$

$$-\frac{t}{10\times10^{-9}} = \ln\left(\frac{1}{2}\right)$$

$$-\frac{t}{10\times10^{-9}} = -0.693$$

$$t = 0.693\left(10\times10^{-9}\right)$$

$$t = 6.93\times10^{-9}\ \text{s}$$

$$t = 6.93\ \text{ns}$$

Thus, the time required for the voltage on the word line to reach $\frac{V_{DD}}{2}$ is $\boxed{6.93\ \text{ns}}$.

Step 1 of 5

The logic gate implementation of SR flip-flop using NAND gate is,

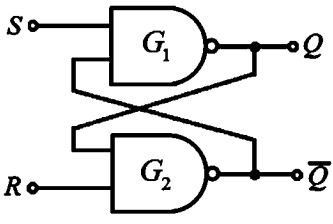


Figure 1

Step 2 of 5

The following is the truth table for the flip-flop:

Table 1

R	S	Q_{n+1}
0	0	Not used
0	1	0
1	0	1
1	1	Q_n

Step 3 of 5

Operation:

The set and reset functions are active when low for the flip-flop implemented by cross-coupling of NAND gates.

Similar to the $S=1$, $R=1$ state of the SR flip-flop using cross-coupled NOR gate, in active low SR flip-flop the $S=0$, $R=0$ state is an undefined state and it cannot be used.

When $S=1$, $R=0$ and if $Q_n=0$, then the inputs to G_2 are, $R=0$, and $Q_n=0$.

Therefore, $\overline{Q}_{n+1}=1$.

The inputs to G_1 are, $S=1$, and $\overline{Q}_{n+1}=1$.

Therefore, $Q_{n+1}=0$.

Alternatively if $Q_n=1$, then inputs to G_2 are, $R=0$, and $Q_n=1$.

Therefore, $\overline{Q}_{n+1}=0$.

The inputs to G_1 are, $S=1$, and $\overline{Q}_{n+1}=1$.

Therefore, $Q_{n+1}=0$.

Thus, for $S=1$ and $R=0$, irrespective of the present state the output, Q_{n+1} goes to reset state, $(Q_{n+1}=0)$.

Step 4 of 5

When $S=0$, $R=1$ and if $Q_n=1$, then the inputs to G_1 are, $S=0$, and $\overline{Q}_n=0$.

Therefore, $Q_{n+1}=1$.

The inputs to G_2 are, $R=1$, and $Q_{n+1}=1$.

Therefore, $\overline{Q}_{n+1}=0$.

Alternatively if $Q_n=0$, then inputs to G_1 are, $S=0$, and $\overline{Q}_n=1$.

Therefore, $Q_{n+1}=1$.

The inputs to G_2 are, $R=1$, and $Q_{n+1}=1$.

Therefore, $\overline{Q}_{n+1}=0$.

Thus, for $S=0$ and $R=1$, irrespective of the present state the output, Q_{n+1} goes to set state, $(Q_{n+1}=1)$.

Step 5 of 5

If $S=1$, $R=1$, and if $Q_n=1$, then the inputs to G_2 are $R=1$, and $Q_n=1$.

Therefore, $\overline{Q}_{n+1}=0$.

The inputs to G_1 are, $S=1$, and $\overline{Q}_{n+1}=0$.

Therefore, $Q_{n+1}=1$.

Alternatively if $Q_n=0$, then inputs to G_2 are, $R=1$, and $Q_n=0$.

Therefore, $\overline{Q}_{n+1}=1$.

The inputs to G_1 are, $S=1$, and $\overline{Q}_{n+1}=1$.

Therefore, $Q_{n+1}=0$.

Thus, for $S=1$ and $R=1$, irrespective of the present state the output, Q_{n+1} remains in same state, $(Q_{n+1}=Q_n)$.

Step 1 of 2

16.004E

Refer to Figure 15.12 in the textbook for the CMOS SRAM memory cell.

The condition for the ratio of $\left(\frac{W}{L}\right)_a$ to $\left(\frac{W}{L}\right)_n$ is,

$$\frac{\left(\frac{W}{L}\right)_a}{\left(\frac{W}{L}\right)_n} \le \frac{1}{\left(1-\frac{V_{tn}}{V_{DD}-V_{tn}}\right)^2} - 1$$

$$\left(\frac{W}{L}\right)_a \le \left(\frac{W}{L}\right)_n \left[\frac{1}{\left(1-\frac{V_{tn}}{V_{DD}-V_{tn}}\right)^2} - 1\right]$$

Step 2 of 2

Substitute 1.5 for $\left(\frac{W}{L}\right)_n$, $0.5\ \text{V}$ for V_{tn}, and $1.8\ \text{V}$ for V_{DD} in the equation.

$$\left(\frac{W}{L}\right)_a \le (1.5)\left(\frac{1}{\left(1-\frac{0.5}{1.8-0.5}\right)^2} - 1\right)$$

$$\le (1.5)\left(\frac{1}{(1-0.385)^2} - 1\right)$$

$$\le (1.5)\left(\frac{1}{0.378} - 1\right)$$

$$\le 2.5$$

Therefore, the maximum allowable W/L for the access transistors of SRAM cell is $\boxed{\left(\frac{W}{L}\right)_a \le 2.5}$.

Step 1 of 4

Refer to Figure 15.4 in the textbook for the CMOS implementation of clocked SR flip-flop.

The two inverters use matched transistors; that is,

$$\left(\frac{W}{L}\right)_p = \left(\frac{\mu_n}{\mu_p}\right)\left(\frac{W}{L}\right)_n$$

The fabrication of the transistors of SR-flip is done in $0.13\ \mu m$ process. Thus,

$$L = 0.13\ \mu m$$

$$\mu_n = 4\mu_p$$

Equate the current supplied by the transistors, Q_5 **and** Q_6 to the current supplied by Q_2 at $v_{\bar{Q}} = \frac{V_{DD}}{2}$, to obtain the minimum required $\left(\frac{W}{L}\right)$ for Q_5 **and** Q_6.

Step 2 of 4

Observe from Figure 15.4 that, current flows through the transistors, Q_5 **and** Q_6 when both the transistors are ON and since the transistors, Q_5 **and** Q_6 are in series, so the current flowing through both the transistors is same, so $\frac{W}{L}$ is half of the $\frac{W}{L}$ of each of the transistors, Q_5 **and** Q_6.

Consider the current flowing through the transistors, Q_5 **and** Q_6 as I_{Deq}.

I_{Deq} is the current flowing through the NMOS transistors, Q_5 **and** Q_6, so consider $\frac{W}{L}$ as $\frac{1}{2}\left(\frac{W}{L}\right)_5$.

I_{D2} is the current flowing through the PMOS transistor, Q_2.

Equate $I_{Deq} = I_{D2}$ and calculate $\left(\frac{W}{L}\right)_5$.

$$\mu_n C_{ox}\frac{1}{2}\left(\frac{W}{L}\right)_5\left[(V_{DD} - V_m)v_{\bar{Q}} - \frac{(v_{\bar{Q}})^2}{2}\right] = \mu_p C_{ox}\left(\frac{W}{L}\right)_p\left[(V_{DD} - V_m)v_{\bar{Q}} - \frac{(v_{\bar{Q}})^2}{2}\right]$$

$$\left(\frac{W}{L}\right)_5 = 2\frac{\mu_p}{\mu_n}\left(\frac{W}{L}\right)_p$$

Substitute $\left(\frac{\mu_n}{\mu_p}\right)\left(\frac{W}{L}\right)_n$ for $\left(\frac{W}{L}\right)_p$.

$$\left(\frac{W}{L}\right)_5 = 2\frac{\mu_p}{\mu_n}\left(\frac{\mu_n}{\mu_p}\right)\left(\frac{W}{L}\right)_n$$
$$= 2\left(\frac{W}{L}\right)_n$$

Hence, shown that minimum $\frac{W}{L}$ for each of $Q_5 - Q_8$ for switching to occur is, $\boxed{2\left(\frac{W}{L}\right)_n}$.

Step 3 of 4

Since, $\frac{W}{L} = 1$ for the NMOS transistors, Q_1 **and** Q_3, so, $\left(\frac{W}{L}\right)_n = 1$. Thus,

$$\left(\frac{W}{L}\right)_1 = \left(\frac{W}{L}\right)_n$$
$$= 1$$

Substitute $0.13\ \mu m$ for L.

$W_1 = 0.13\ \mu m$

W_3 is also $0.13\ \mu m$.

Observe from Figure 15.4 that the transistors, Q_2 and Q_4 are PMOS. Thus,

$$\left(\frac{W}{L}\right)_2 = \left(\frac{W}{L}\right)_p$$
$$= \frac{\mu_n}{\mu_p}\left(\frac{W}{L}\right)_n$$

Substitute $\mu_n = 4\mu_p$.

$$\left(\frac{W}{L}\right)_2 = \left(\frac{W}{L}\right)_p$$
$$= \frac{4\mu_p}{\mu_p}\left(\frac{W}{L}\right)_n$$

$$W_2 = 4\left(\frac{W}{L}\right)_n L_p$$

Substitute 1 for $\left(\frac{W}{L}\right)_n$ and $0.13\ \mu m$ for L.

$$W_2 = 4(1)(0.13\ \mu m)$$
$$= 0.52\ \mu m$$

Since, $\frac{W}{L}$ for both Q_2 and Q_4 transistors are same, thus, W_4 is $0.52\ \mu m$.

Step 4 of 4

Observe that the transistors, Q_5, Q_6, Q_7 **and** Q_8 are NMOS and $\frac{W}{L}$ ratio is same for all these transistors. Thus,

$$\left(\frac{W}{L}\right)_5 = 2\left(\frac{W}{L}\right)_n$$
$$W_5 = 2\left(\frac{W}{L}\right)_n L_5$$

Substitute 1 for $\left(\frac{W}{L}\right)_n$ and $0.13\ \mu m$ for L_5.

$$W_5 = 2(1)(0.13\ \mu m)$$
$$= 0.26\ \mu m$$

Thus, the values of W_6, W_7 **and** W_8 are $0.26\ \mu m$.

Thus, the sizes of eight transistors when fabricated in $0.13\ \mu m$ process are,

$W_1 = 0.13\ \mu m$
$W_2 = 0.52\ \mu m$
$W_3 = 0.13\ \mu m$
$W_4 = 0.52\ \mu m$
$W_5 = 0.26\ \mu m$
$W_6 = 0.26\ \mu m$
$W_7 = 0.26\ \mu m$
$W_8 = 0.26\ \mu m$

Step 1 of 4

Refer to Figure 15.12 in the textbook for the CMOS SRAM memory cell.

(a)

Calculate the value of current I_s of the $\overline{B}$ bit line.

$$I_s = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_a \left(V_{DD} - V_{in} - V_{tl}\right)^2$$

Substitute $300\times10^{-6}\ \text{A/V}^2$ for $\mu_n C_{ox}$, 2.5 for $\left(\frac{W}{L}\right)_a$, $1.8\ \text{V}$ for V_{DD}, 0.5 for V_{in}, and 0.5 for V_{tl}.

$$\begin{aligned} I_s &= \frac{1}{2}\left(300\times10^{-6}\right)(2.5)(1.8-0.5-0.5)^2 \\ &= \frac{1}{2}\left(300\times10^{-6}\right)(2.5)(0.8)^2 \\ &= \frac{1}{2}\left(480\times10^{-6}\right) \\ &= 240\,\mu\text{A} \end{aligned}$$

Step 2 of 4

Determine the read delay Δt.

$$\Delta t = \frac{\overline{C}_B \Delta V}{I_s}$$

Substitute $2\ \text{pF}$ for $C_{\overline{B}}$, $0.2\ \text{V}$ for ΔV, and $240\,\mu\text{A}$ for I_s in the equation.

$$\begin{aligned} \Delta t &= \frac{\left(2\times10^{-12}\right)(0.2)}{240\times10^{-6}} \\ &= \frac{0.4\times10^{-12}}{240\times10^{-6}} \\ &= 1.66\times10^{-9}\ \text{s} \\ &= 1.66\ \text{ns} \end{aligned}$$

Therefore, the value of read delay is, $\boxed{\Delta t = 1.7\ \text{ns}}$.

Step 3 of 4

(b)

Calculate the value of current I_s of the $\overline{B}$ bit line.

$$I_s = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_a \left(V_{DD} - V_{in} - V_{tl}\right)^2$$

Substitute $300\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$, 1.5 for $\left(\frac{W}{L}\right)_a$, $1.8\ \text{V}$ for V_{DD}, 0.5 for V_{in}, and 0.5 for V_{tl}.

$$\begin{aligned} I_s &= \frac{1}{2}\left(300\times10^{-6}\right)(1.5)(1.8-0.5-0.5)^2 \\ &= \frac{1}{2}\left(300\times10^{-6}\right)(1.5)(0.8)^2 \\ &= \frac{1}{2}\left(288\times10^{-6}\right) \\ &= 144\,\mu\text{A} \end{aligned}$$

Step 4 of 4

Determine the read delay Δt.

$$\Delta t = \frac{\overline{C}_B \Delta V}{I_s}$$

Substitute $2\ \text{pF}$ for $C_{\overline{B}}$, $0.2\ \text{V}$ for ΔV, and $240\,\mu\text{A}$ for I_s in the equation.

$$\begin{aligned} \Delta t &= \frac{\left(2\times10^{-12}\right)(0.2)}{144\times10^{-6}} \\ &= \frac{0.4\times10^{-12}}{144\times10^{-6}} \\ &= 2.8\times10^{-9}\ \text{s} \\ &= 2.8\ \text{ns} \end{aligned}$$

Therefore, the value of read delay is, $\boxed{\Delta t = 2.8\ \text{ns}}$.

Step 1 of 4

Refer to Fig. 15.4 in the text book for the CMOS implementation of clocked SR flip-flop.

The two inverters are unmatched transistors.

So, the values of $\left(\frac{W}{L}\right)_p$ and $\left(\frac{W}{L}\right)_n$ are $\frac{0.27\ \mu m}{0.18\ \mu m}$.

The fabrication of the transistors of SR-flip is done in $0.13\ \mu m$ process. Thus,

$L = 0.13\ \mu m$

Equate the current supplied by the transistors, Q_5 and Q_6 to the current supplied by Q_2 at $v_{\bar{Q}} = \frac{V_{DD}}{2}$.

Step 2 of 4

Observe from Figure 15.4 that, current flows through the transistors, Q_5 and Q_6 when both the transistors are ON and since the transistors, Q_5 and Q_6 are in series, so the current flowing through both the transistors is same, so $\left(\frac{W}{L}\right)_{eq}$ is half of the width to length ratio of each of the transistors, Q_5 and Q_6.

Consider the current flowing through the transistors, Q_5 and Q_6 as I_{Deq}.

I_{Deq} is the current flowing through the NMOS transistors, Q_5 and Q_6, so consider width to length ratio as $\frac{1}{2}\left(\frac{W}{L}\right)_5$.

Since Q_5 is an NMOS transistor, value of $\left(\frac{W}{L}\right)_{eq}$ is,

$$\left(\frac{W}{L}\right)_{eq} = \frac{1}{2}\left(\frac{W}{L}\right)_n = \frac{1}{2}\left(\frac{0.27\ \mu m}{0.18\ \mu m}\right)$$

I_{D2} is the current flowing through the PMOS transistor, Q_2.

Equate $I_{Deq} = I_{D2}$.

$$\mu_n C_{ox}\left(\frac{W}{L}\right)_{eq}\left[(V_{DD} - V_{tn})v_{\bar{Q}} - \frac{(v_{\bar{Q}})^2}{2}\right] = \mu_p C_{ox}\left(\frac{W}{L}\right)_p\left[(V_{DD} - V_{tp})v_{\bar{Q}} - \frac{(v_{\bar{Q}})^2}{2}\right]$$

Step 3 of 4

The formula for the threshold voltage is,

$$V_{th} = \frac{V_{tn} + (V_{DD} + V_{tp})\sqrt{\frac{\mu_p\left(\frac{W}{L}\right)_p}{\mu_n\left(\frac{W}{L}\right)_{eq}}}}{1 + \sqrt{\frac{\mu_p\left(\frac{W}{L}\right)_p}{\mu_n\left(\frac{W}{L}\right)_{eq}}}}$$

Substitute $0.5\ V$ for V_{tn}, V_{tp}, 1.8 V for V_{DD}, $4\mu_p$ for μ_n, $\frac{0.27\ \mu m}{0.18\ \mu m}$ for $\left(\frac{W}{L}\right)_p$ and $\frac{1}{2}\left(\frac{0.27\ \mu m}{0.18\ \mu m}\right)$ for $\left(\frac{W}{L}\right)_{eq}$ in the expression of the threshold voltage.

$$V_{th} = \frac{0.5 + (1.8 + 0.5)\sqrt{\frac{\mu_p\left(\frac{0.27\ \mu m}{0.18\ \mu m}\right)}{4\mu_p \cdot \frac{1}{2}\left(\frac{0.27\ \mu m}{0.18\ \mu m}\right)}}}{1 + \sqrt{\frac{\mu_p\left(\frac{0.27\ \mu m}{0.18\ \mu m}\right)}{4\mu_p \cdot \frac{1}{2}\left(\frac{0.27\ \mu m}{0.18\ \mu m}\right)}}}$$

$$= \frac{0.5 + (1.8 + 0.5)\sqrt{\frac{1}{2}}}{1 + \sqrt{\frac{1}{2}}}$$

$$= 1.25\ V$$

Thus, the threshold voltage of each inverter is, $\boxed{1.25\ V}$.

Step 4 of 4

Calculate the value of width to length ratio of transistor, Q_5, so that the flip-flop switches.

Substitute $0.5\ V$ for V_{tn}, V_{tp}, 1.8 V for V_{DD}, $4\mu_p$ for μ_n, $\frac{0.27\ \mu m}{0.18\ \mu m}$ for $\left(\frac{W}{L}\right)_p$ and $\frac{1}{2}\left(\frac{W}{L}\right)_5$ for $\left(\frac{W}{L}\right)_{eq}$ in the equation of $I_{Deq} = I_{D2}$.

$$\mu_n C_{ox}\left(\frac{W}{L}\right)_{eq}\left[(V_{DD} - V_{tn})v_{\bar{Q}} - \frac{(v_{\bar{Q}})^2}{2}\right] = \mu_p C_{ox}\left(\frac{W}{L}\right)_p\left[(V_{DD} - V_{tp})v_{\bar{Q}} - \frac{(v_{\bar{Q}})^2}{2}\right]$$

$$4\mu_p C_{ox}\frac{1}{2}\left(\frac{W}{L}\right)_5\left[(1.8 - 0.5)v_{\bar{Q}} - \frac{(v_{\bar{Q}})^2}{2}\right] = \mu_p C_{ox}\left(\frac{0.27\ \mu m}{0.18\ \mu m}\right)\left[(1.8 - 0.5)v_{\bar{Q}} - \frac{(v_{\bar{Q}})^2}{2}\right]$$

$$2\left(\frac{W}{L}\right)_5 = \left(\frac{0.27\ \mu m}{0.18\ \mu m}\right)$$

$$\left(\frac{W}{L}\right)_5 = \frac{1}{2}\left(\frac{0.27\ \mu m}{0.18\ \mu m}\right)$$

Observe from Figure 15.4 that the all the transistors, Q_5, Q_6, Q_7 and Q_8 are NMOS transistors and have same functionality as that of transistor, Q_5.

Thus, the required value of $\left(\frac{W}{L}\right)$ for each of the transistor, Q_5 to Q_8 is, $\boxed{\frac{0.135\ \mu m}{0.18\ \mu m}}$.

Step 1 of 2

16.006E

Refer to Figure 15.12 in the textbook for the CMOS SRAM memory cell.

For $0.18\,\mu m$ technology, $\frac{\mu_n}{\mu_p} \simeq 4$.

The upper bound on $\left(\frac{W}{L}\right)_p$ in terms of $\left(\frac{W}{L}\right)_a$ is,

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_a} \le \frac{\mu_n}{\mu_p}\left[1-\left(1-\frac{V_{tn}}{V_{DD}-V_{tn}}\right)^2\right]$$

Substitute 4 for $\frac{\mu_n}{\mu_p}$, $0.5\ V$ for V_{tn}, and $1.8\ V$ for V_{DD}.

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_a} \le 4\left[1-\left(1-\frac{0.5}{1.8-0.5}\right)^2\right]$$

$$\le 4\left[1-(1-0.385)^2\right]$$

$$\le 4[1-0.3787]$$

$$\le 2.5$$

Thus, the upper bound on $\left(\frac{W}{L}\right)_p$ is, $\boxed{\left(\frac{W}{L}\right)_p \le 2.5\left(\frac{W}{L}\right)_a}$

Step 2 of 2

Substitute 2.5 for $\left(\frac{W}{L}\right)_a$ in the equation.

$$\left(\frac{W}{L}\right)_p \le 2.5(2.5)$$

$$\le 6.25$$

Therefore, the maximum allowable value of $\left(\frac{W}{L}\right)_p$ is, $\boxed{\left(\frac{W}{L}\right)_p \le 6.25}$.

For minimum area, the width should be minimum. Here, the minimum allowable width is $0.18\ \mu m$. Thus, for minimum area select, $\boxed{W_n = W_p = W_a = 0.18\ \mu m}$.

Step 1 of 4

The expression for the drain current of an NMOS transistor operating in velocity saturation is,

$$i_D = \mu_n C_{ox}\left(\frac{W}{L}\right)V_{DSsat}\left(v_{GS} - V_t - \frac{1}{2}V_{DSsat}\right)(1+\lambda v_{DS})$$

Consider that the saturation drain voltage, V_{DSsat} for n- channel device is 0.6 V and for p- channel device is 1 V.

Assume the channel length parameters,

$$\lambda_n = |\lambda_p|$$
$$= 0.1\ \text{V}^{-1}$$

Refer to example 15.1 in the text book.

Refer to answer of part (a) in the text book.

Equate the current supplied by the transistors, Q_5 and Q_6 to the current supplied by Q_2 at $v_Q = \frac{V_{DD}}{2}$, to obtain the minimum required $\left(\frac{W}{L}\right)$ for Q_5 and Q_6.

Observe from Figure 15.4 that, current flows through the transistors, Q_5 and Q_6 when both the transistors are ON and since the transistors, Q_5 and Q_6 are in series, so the current flowing through both the transistors is same, so $\frac{W}{L}$ is half of the $\frac{W}{L}$ of each of the transistors, Q_5 and Q_6.

Step 2 of 4

Consider the current flowing through the transistors, Q_5 and Q_6 as I_{Deq}.

I_{Deq} is the current flowing through the NMOS transistors, Q_5 and Q_6 so consider $\frac{W}{L}$ as $\frac{1}{2}\left(\frac{W}{L}\right)_5$.

$$I_{eq} = \mu_n C_{ox}\frac{1}{2}\left(\frac{W}{L}\right)_5 V_{DSsat}\left(V_{GS} - V_t - \frac{1}{2}V_{Dsat}\right)(1+\lambda v_{DS})$$

Consider v_{GS} equal to V_{DD}, then $v_{GS} - V_t$ is,

$$v_{GS} - V_t = 1.8 - 0.5$$
$$= 1.3$$

V_{DSsat} of NMOS transistor is 0.6 V, and observe that $v_{Gs} - V_t > V_{DSsat}$, thus, required conditions are satisfied and the NMOS transistor operates in velocity saturation region, so v_{DS} value in the equation of I_{eq} becomes equal to V_{DSsat}.

$$I_{eq} = \mu_n C_{ox}\frac{1}{2}\left(\frac{W}{L}\right)_5 V_{DSsat}\left(V_{GS} - V_t - \frac{1}{2}V_{Dsat}\right)(1+\lambda V_{DSsat})$$

Substitute $300\ \mu\text{A/V}^2$ for $\mu_n C_{ox}$, 0.6 V for V_{DSsat}, 0.5 V for V_t and $0.1\ \text{V}^{-1}$ for λ in the equation of I_{eq}.

$$I_{eq} = (300\ \mu\text{A/V}^2)\frac{1}{2}\left(\frac{W}{L}\right)_5 0.6\left(1.8 - 0.5 - \frac{0.6}{2}\right)[1 + (0.1)(0.6)]$$
$$= (300\ \mu\text{A/V}^2)\frac{1}{2}\left(\frac{W}{L}\right)_5 (1.06)$$

Step 3 of 4

I_{D2} is the current flowing through the P-MOS transistor Q_2.

$$I_{D2} = \mu_p C_{ox}\left(\frac{W}{L}\right)_p V_{Dsatp}\left(V_{GS} - V_t - \frac{1}{2}V_{Dsat}\right)(1+\lambda v_{DS})$$

Consider v_{GS} equal to V_{DD}, then $v_{GS} - V_t$ is,

$$v_{GS} - V_t = 1.8 - 0.5$$
$$= 1.3$$

V_{DSsat} of PMOS transistor is 1 V, and observe that $v_{Gs} - V_t > V_{DSsat}$, thus, required conditions are satisfied and the PMOS transistor operates in velocity saturation region, so v_{DS} value in the equation of I_{D2} becomes equal to V_{DSsat}.

$$I_{D2} = \mu_p C_{ox}\left(\frac{W}{L}\right)_p V_{Dsatp}\left(V_{GS} - V_t - \frac{1}{2}V_{Dsat}\right)(1+\lambda v_{DSsat})$$

Substitute $\frac{300}{4}\ \mu\text{A/V}^2$ for $\mu_p C_{ox}$, $4\left(\frac{0.27\ \mu\text{m}}{0.18\ \mu\text{m}}\right)$ for $\left(\frac{W}{L}\right)_p$, 1 V for V_{DSsat}, 0.5 V for V_t and $0.1\ \text{V}^{-1}$ for λ in the equation of I_{D2}.

$$I_{D2} = \frac{300}{4}\ \mu\text{A/V}^2\left[4\left(\frac{0.27\ \mu\text{m}}{0.18\ \mu\text{m}}\right)\right]1\left(1.8 - 0.5 - \frac{1}{2}\right)[1 + (0.1)(1)]$$
$$= \frac{300}{4}\ \mu\text{A/V}^2\left(\frac{1.08\ \mu\text{m}}{0.18\ \mu\text{m}}\right)0.88$$

Step 4 of 4

Equate $I_{Deq} = I_{D2}$ and calculate $\left(\frac{W}{L}\right)_5$ when transistors are operating in velocity saturation.

$$(300\ \mu\text{A/V}^2)\frac{1}{2}\left(\frac{W}{L}\right)_5 (1.06) = \frac{300}{4}\ \mu\text{A/V}^2\left(\frac{1.08\ \mu\text{m}}{0.18\ \mu\text{m}}\right)0.88$$

$$\left(\frac{W}{L}\right)_5 = \left(\frac{2}{4}\right)\left(\frac{0.88}{1.06}\right)4\left(\frac{0.27\ \mu\text{m}}{0.18\ \mu\text{m}}\right)$$
$$= 1.66\left(\frac{0.27\ \mu\text{m}}{0.18\ \mu\text{m}}\right)$$
$$= \left(\frac{0.448\ \mu\text{m}}{0.18\ \mu\text{m}}\right)$$

Thus, the minimum required value of $\left(\frac{W}{L}\right)_5$ and $\left(\frac{W}{L}\right)_6$ is, $\boxed{\frac{0.448\ \mu\text{m}}{0.18\ \mu\text{m}}}$.

Observe that in example 15.1, the value of $\left(\frac{W}{L}\right)_5$ and $\left(\frac{W}{L}\right)_6$ is $\frac{0.54\ \mu\text{m}}{0.18\ \mu\text{m}}$.

Thus,

$$\frac{\left(\frac{0.448\ \mu\text{m}}{0.18\ \mu\text{m}}\right)}{\left(\frac{W}{L}\right)_5\ \text{of example 15.1}} = \frac{\left(\frac{0.448\ \mu\text{m}}{0.18\ \mu\text{m}}\right)}{\left(\frac{0.54\ \mu\text{m}}{0.18\ \mu\text{m}}\right)}$$
$$= 0.83$$

Hence, the value of $\left(\frac{W}{L}\right)_5$ and $\left(\frac{W}{L}\right)_6$ obtained is relatively 0.83 times of the value $\left(\frac{W}{L}\right)_5$ and $\left(\frac{W}{L}\right)_6$ in example 15.1.

Step 1 of 2

16.007E

Calculate the output readout voltage for a stored 1.

$$\Delta V(1) = \frac{C_s}{C_B}\left(\frac{V_{DD}}{2}\right)$$

Substitute $30\ \text{fF}$ for C_S , $0.3\ \text{pF}$ for C_B , and $1.2\ \text{V}$ for V_{DD} in the equation.

$$\Delta V(1) = \frac{30 \times 10^{-15}}{0.3 \times 10^{-12}}\left(\frac{1.2}{2}\right)$$
$$= 60\ \text{mV}$$

Thus, the output readout voltage for a stored 1 is, $\boxed{\Delta V(1) = 60\ \text{mV}}$.

Step 2 of 2

Calculate the output readout voltage for a stored 0.

$$\Delta V(0) = -\frac{C_s}{C_B}\left(\frac{V_{DD}}{2}\right)$$

Substitute $30\ \text{fF}$ for C_S , $0.3\ \text{pF}$ for C_B , and $1.2\ \text{V}$ for V_{DD} in the equation.

$$\Delta V(0) = -\frac{30 \times 10^{-15}}{0.3 \times 10^{-12}}\left(\frac{1.2}{2}\right)$$
$$= -60\ \text{mV}$$

Thus, the output readout voltage for a stored 0 is, $\boxed{\Delta V(0) = -60\ \text{mV}}$.

16.007P

We don't have the solution to this problem yet.

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Step 1 of 4

The area of the storage array,

$$A = nA_0$$

Here,

n is the n -bit DRAM chip

A_0 is the area per each cell

Substitute 64-Mbit for n and $2\ \mu m^2$ for A_0 .

$$\begin{aligned} A &= (64\text{-Mbit})(2\ \mu m^2) \\ &= (64 \times 2^{20})(2)\ \mu m^2 \\ &= 128 \times 2^{20}\ \mu m^2 \end{aligned}$$

The storage array is a square.

Step 2 of 4

Calculate the side of the storage array of a square.

$$A = s^2$$
$$s = \sqrt{A}$$

Substitute $128 \times 2^{20}\ \mu m^2$ for A .

$$\begin{aligned} s &= \sqrt{128 \times 2^{20}\ \mu m^2} \\ &= 11.6 \times 10^3\ \mu m \\ &= 11.6\ mm \end{aligned}$$

Calculate the dimensions of the storage array.

$$\text{Dimensions} = s \times s$$

Substitute $11.6\ mm$ for s .

$$\text{Dimensions} = 11.6\ mm \times 11.6\ mm$$

Thus, the dimensions of the storage array is $\boxed{11.6\ mm \times 11.6\ mm}$.

Step 3 of 4

The peripheral circuitry adds about 30% to the chip area.

Calculate the new area of the storage array, A_{new} .

$$\begin{aligned} A_{new} &= A + 30\%(A) \\ &= A + \frac{30A}{100} \\ &= A + 0.3A \\ &= 1.3A \end{aligned}$$

Substitute $128 \times 2^{20}\ \mu m^2$ for A .

$$\begin{aligned} A_{new} &= 1.3(128 \times 2^{20}\ \mu m^2) \\ &= 166.4 \times 2^{20}\ \mu m^2 \end{aligned}$$

Step 4 of 4

Calculate the new side of the storage array of a square, s_{new} .

$$A_{new} = s_{new}^2$$
$$s_{new} = \sqrt{A_{new}}$$

Substitute $166.4 \times 2^{20}\ \mu m^2$ for A_{new} .

$$\begin{aligned} s_{new} &= \sqrt{166.4 \times 2^{20}\ \mu m^2} \\ &= 13.2 \times 10^3\ \mu m \\ &= 13.2\ mm \end{aligned}$$

Calculate the new dimensions of the storage array.

$$\text{Dimensions} = s_{new} \times s_{new}$$

Substitute $13.2\ mm$ for s_{new} .

$$\text{Dimensions} = 13.2\ mm \times 13.2\ mm$$

Thus, the new dimensions of the storage array is $\boxed{13.2\ mm \times 13.2\ mm}$.

Step 1 of 2

16.008P

Refer to Figure 15.4 in the textbook for a CMOS implementation of clocked SR flip-flop.

It is not a fully complementary CMOS circuit.

Draw the fully complementary version of the SR flip-flop. The following figure is the modified version the CMOS implementation of clocked SR flip-flop. This is a fully complementary circuit that uses 12 transistors. However, a 10-T version exists in which Q_{10} **and** Q_{12} transistors omitted.

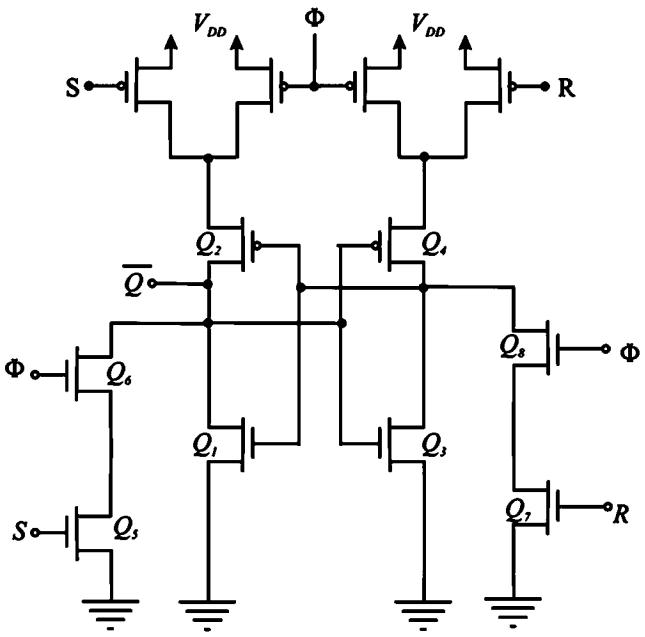


Figure 1: Fully complementary circuit of the SR-flip-flop.

Step 2 of 2

Note further that, an effective version exists in which the transistors, Q_6 **and** Q_8 are moved below and transistors, Q_5 **and** Q_7 are merged together and form a single transistor. The following is the later design with less number of the transistors:

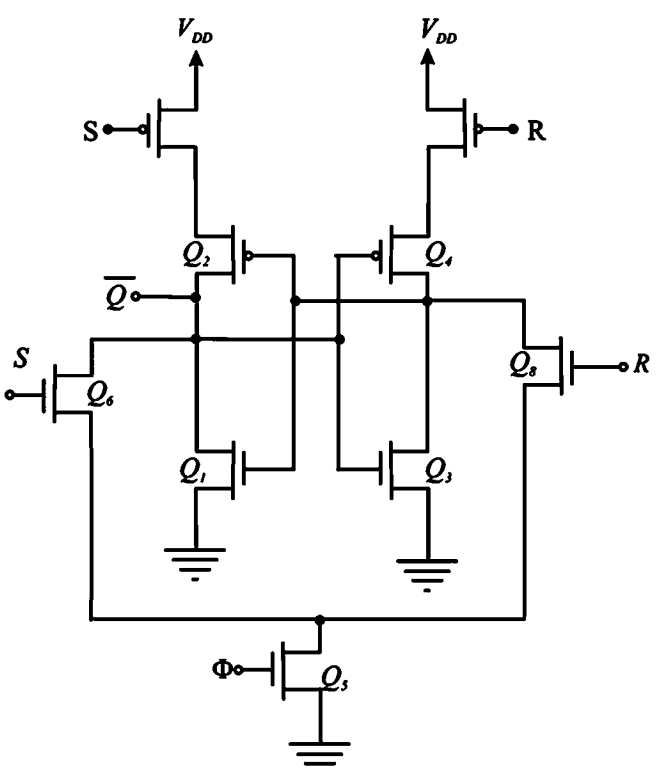


Figure 2

The main problem with the SR-flip flop is that, it suffers only from the fact that unclocked changes in *S* and *R* secondary impact on the output, Q **and** $\overline{Q}$. Since raising *S* or *R* disconnects Q, $\overline{Q}$ form V_{DD} . In some cases this problem may lead to system noise sensitivity in which case or other or both Q_{10} **and** Q_{12} may be added.

Step 1 of 4

Refer to Figure 15.20 in the textbook for the sense amplifier.

The voltage v_B is,

$$v_B = \frac{V_{DD}}{2} - \Delta V e^{\frac{\Delta t}{\tau}}$$

The time τ, for v_B to reach $0.9\ V_{DD}$ is,

$$0.9V_{DD} = \frac{V_{DD}}{2} - \Delta V(1) e^{\frac{\Delta t}{\tau}}$$

Substitute $1.8\ \text{V}$ for V_{DD}, $0.1\ \text{V}$ for $\Delta V(1)$, and $1.4\ \text{ns}$ for Δt in the equation.

$$0.9(1.8) = 0.9 + (0.1) e^{\frac{1.4 \times 10^{-9}}{\tau}}$$

$$e^{\frac{1.4 \times 10^{-9}}{\tau}} = \frac{0.9(1.8) - 0.9}{0.1}$$

$$\frac{1.4 \times 10^{-9}}{\tau} = \log(7.2)$$

$$\tau = \frac{1.4 \times 10^{-9}}{\log(7.2)}$$

$$\tau = 7.09 \times 10^{-10}\ \text{s}$$

Step 2 of 4

Calculate the value of G_m.

$$\tau = \frac{C_g}{G_m}$$

$$G_m = \frac{C_g}{\tau}$$

$$= \frac{1 \times 10^{-12}}{7.09 \times 10^{-10}}$$

$$= 1.41 \times 10^{-3}\ \text{A/V}$$

It is known that,

$$g_{mn} + g_{mp} = G_m$$

$$g_{mn} + g_{mn} = G_m \qquad (\text{Since}, g_{mn} = g_{mp})$$

$$g_{mn} = G_m$$

$$g_{mn} = \frac{G_m}{2}$$

$$g_{mn} = \frac{1.41 \times 10^{-3}}{2}$$

$$= 7.05 \times 10^{-4}\ \text{A/V}$$

Now, the value of g_{mp} is,

$$g_{mp} = g_{mn}$$

$$= 7.05 \times 10^{-4}\ \text{A/V}$$

Step 3 of 4

The transconductance g_{mn} is,

$$\mu_n C_{ox} \left(\frac{W}{L}\right)_n (V_{GS} - V_t) = g_{mn}$$

$$\left(\frac{W}{L}\right)_n = \frac{7.05 \times 10^{-4}}{(300 \times 10^{-6})(0.9 - 0.5)}$$

$$\left(\frac{W}{L}\right)_n \simeq 6$$

Thus, the value of $\left(\frac{W}{L}\right)_n$ is $\boxed{6}$.

Step 4 of 4

Calculate the value of $\left(\frac{W}{L}\right)_p$.

$$\left(\frac{W}{L}\right)_p \simeq 4\left(\frac{W}{L}\right)_n$$

$$\simeq 4(6)$$

$$= 24$$

Thus, the value of $\left(\frac{W}{L}\right)_p$ is $\boxed{24}$.

16.009P

We don't have the solution to this problem yet.

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Step 1 of 3

Refer to Figure 15.20 in the textbook for the sense amplifier.

Calculate the value of τ .

$$\tau = \frac{C_B}{G_m}$$

$$= \frac{C_B}{g_{mp} + g_{mn}} \qquad (\text{Since, } G_m = g_{mp} + g_{mn})$$

Substitute $1\ \text{pF}$ for C_B , $0.36\ \text{mA/V}$ for g_{mn} , and $0.36\ \text{mA/V}$ for g_{mp} in the equation.

$$\tau = \frac{1\times10^{-12}}{0.36\times10^{-3} + 0.36\times10^{-3}}$$

$$= 1.4\ \text{ns}$$

Step 2 of 3

The voltage v_B is,

$$v_B = \frac{V_{DD}}{2} - \Delta V e^{\frac{\Delta t}{\tau}}$$

The time τ , for v_B to reach $0.9\ V_{DD}$ is,

$$0.9V_{DD} = \frac{V_{DD}}{2} - \Delta V(1) e^{\frac{\Delta t}{\tau}}$$

Substitute $1.8\ \text{V}$ for V_{DD} , $0.05\ \text{V}$ for $\Delta V(1)$, and $1.4\ \text{ns}$ for τ in the equation.

$$0.9(1.8) = 0.9 + 0.05e^{\frac{\Delta t}{1.4\times10^{-9}}}$$

$$\frac{0.9(1.8) - 0.9}{0.05} = e^{\frac{\Delta t}{1.4\times10^{-9}}}$$

$$\Delta t = \left(1.4\times10^{-9}\right)\log\left(\frac{0.9(1.8) - 0.9}{0.05}\right)$$

$$\Delta t = 3.7\ \text{ns}$$

Thus, the value of Δt is $\boxed{3.7\ \text{ns}}$.

Step 3 of 3

The increase in Δt is,

$$\%\ \text{increase} = \frac{3.7 - 2.8}{2.8}\times100$$

$$= 32\%$$

Thus, the increase in Δt is $\boxed{32\%}$.

Step 1 of 4

16.010P

Refer to Figure P15.11 in the textbook for a commonly used circuit of a D flip-flop that is triggered by the negative-going edge of the clock ϕ .

(a)

Consider that the clock, ϕ is high. The transistors, Q_5 **and** Q_6 are ON because high signal is applied to its gate. Since, ϕ is high, Q_2 is off because it is a P-type device. The input, D is the input to G_2 since Q_5 **and** Q_6 are ON. So, $\bar{Q} = \bar{D}$, and $Q = D$.

If D is high, Q_1 become ON and Q_4 become OFF. If D is low, Q_1 become OFF and Q_4 become ON.

Step 2 of 4

(b)

Consider that the input, D is high and the clock, ϕ goes low. The transistors, Q_2 and Q_3 conduct and transistors, Q_5 **and** Q_6 move to cutoff. So, input to G_2 become high.

Therefore, $\bar{Q}$ is low and Q is high. Since $\bar{Q}$ is low Q_1 conducts and Q_4 does not conduct.

Step 3 of 4

(c)

Consider that the input, D is low and the clock, ϕ goes low. The transistors, Q_5 **and** Q_6 move to cutoff, and transistors, Q_3 and Q_2 conduct. Since, the input, D is low the output of G_2 is low.

Therefore, $\bar{Q}$ is high and Q is low. Thus, Q_1 is OFF and Q_4 is ON.

Step 4 of 4

(d)

If the clock, ϕ is high, the node capacitance of G_1 **and** G_2 are charged to the appropriate voltages corresponding to the value of input, D just before ϕ went down.

This output value remains constant for one clock period. So, the operation of the circuit relay on the node capacitance to maintain most of their charge.

Step 1 of 2

16.011E

Refer to Figure 15.24 in the textbook for the sense amplifier.

Calculate the value of current I .

$$I = \frac{CV_{DD}}{\Delta t}$$

Substitute $50\ \text{fF}$ for C , $1.8\ \text{V}$ for V_{DD} , and $0.5\ \text{ns}$ for Δt .

$$I = \frac{\left(50\times10^{-15}\right)(1.8)}{\left(0.5\times10^{-9}\right)}$$
$$= \frac{90\times10^{-15}}{0.5\times10^{-9}}$$
$$= 180\ \mu\text{A}$$

Therefore, the value of current I is $\boxed{180\ \mu\text{A}}$.

Step 2 of 2

The power dissipation is,

$$P = V_{DD}I$$

Substitute $1.8\ \text{V}$ for V_{DD} and $180\times10^{-6}\ \text{A}$ for I .

$$P = (1.8)\left(180\times10^{-6}\right)$$
$$= 324\ \mu\text{W}$$

Therefore, the dissipated power P is $\boxed{324\ \mu\text{W}}$.

16.011P

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16.012E

We don't have the solution to this problem yet.

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16.012P

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Step 1 of 2

16.013E

Refer to the circuit diagram of a tree column decoder in FIGURE 11.28 in the textbook.

Number of transistors required for 8 bit lines is 14. Among 14 transistors 8 transistors for selecting the LSB bit A_0 similarly 4 transistors for selecting the bit A_1 and 2 transistors for selecting the MSB bit A_2 . So, for 8 bit lines of the 3×8 decoder number of transistors required is 8+4+2 = 14.

Step 2 of 2

The number of transistors required for 3×8 is,

$$\underbrace{8+4+2+1-1}$$

$$2(8-1)$$

Here 8 is bit lines for 3×8 decoder.

Similarly for 2^N bit lines the number of transistor is,

$$2\left(2^N-1\right)$$

Here 2^N is a bit line representation and N is the data bit of a $N\times2^N$ decoder.

So number of transistor required for the 2^N bit lines is $\boxed{2\left(2^N-1\right)}$.

16.013P

Consider that a particular 1 M-bit square memory array has its peripheral circuits recognized to allow for readout of a 16-bit word.

Find the number of address bits required for the new design.

Observe that 1 M-bit array actually holds the following number of cells:

$$\begin{aligned} 2^{20} &= \left(2^{10}\right)^2 \\ &= (1024)^2 \\ &= 10,48,576 \text{ cells} \end{aligned}$$

The number of address bits required for 1 M-bits is,

$$2^n = 10^6$$

$$\log_{10}\left(2^n\right) = \log_{10}\left(10^6\right)$$

$$n \log_{10} 2 = 6$$

$$\begin{aligned} n &= \frac{6}{\log_{10} 2} \\ &= 19.93 \end{aligned}$$

Therefore, approximately 20-bits are needed to represent address of every cell.

For 16-bit words, $2^4 = 16$. That is four bits are not needed.

Therefore, $\boxed{16}$ bits of address are sufficient for the new design to access the 1 M-bit square memory array.

Step 1 of 2

Draw the ring oscillator formed by five inverters in cascade.

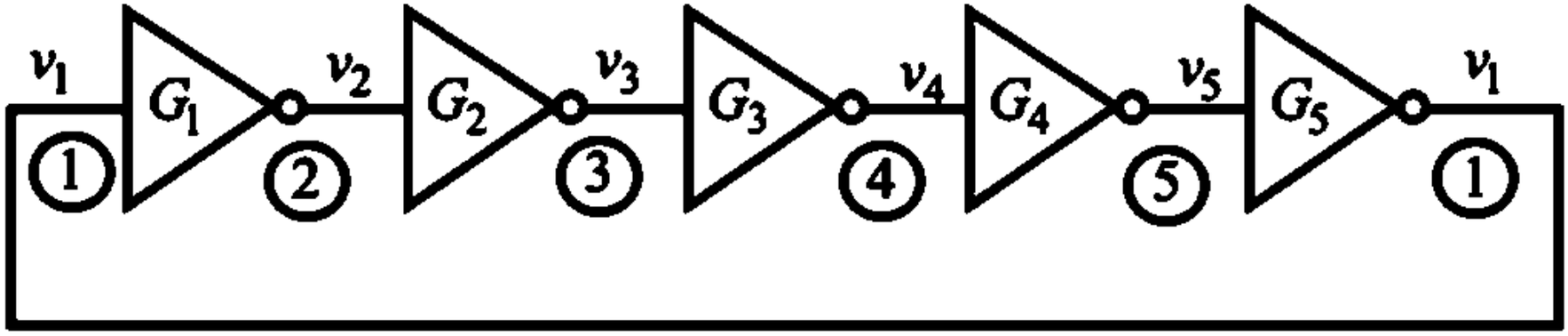


Figure 1

Step 2 of 2

Calculate the frequency of oscillation of ring, f .

$$f = \frac{1}{2Nt_p}$$

Substitute $1\ \text{ns}$ for t_p and 5 for N .

$$f = \frac{1}{2(5)\left(1\times10^{-9}\right)}$$
$$= \frac{10^9}{10}$$
$$= 100\times10^6$$
$$= 100\ \text{MHz}$$

Therefore, the frequency of oscillation of ring of five inverters is $\boxed{100\ \text{MHz}}$.

16.014P

We don't have the solution to this problem yet.

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Step 1 of 9

Refer to Figure 15.30 in the textbook for the organization of MOS ROM into 8 words and 4 bits.

(a)

Draw the PMOS transistor circuit to calculate the time required to charge the bit line.

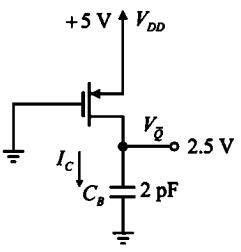


Figure 1

Step 2 of 9

Calculate the current through the capacitor, I_C.

$$I_C = (\mu_p C_{ox})\left(\frac{W}{L}\right)_p \left[(V_{DD} - V_{tp})V_{\bar{Q}} - \frac{1}{2}V_{\bar{Q}}^2\right]$$

Here,

$V_{\bar{Q}}$ is the pre-charge voltage

V_{tp} is the threshold voltage of the PMOS transistor

Substitute $20\ \mu A/V^2$ for $\mu_p C_{ox}$, $\frac{24\ \mu m}{2\ \mu m}$ for $\left(\frac{W}{L}\right)_p$, $5\ V$ for V_{DD}, $1\ V$ for V_{tp} and $5\ V$ for $V_{\bar{Q}}$.

$$\begin{aligned} I_C &= (20\times10^{-4})\left(\frac{24\ \mu m}{2\ \mu m}\right)\left[(5-1)(2.5)-\frac{1}{2}(2.5)^2\right] \\ &= (20\times10^{-4})(12)(10-3.125) \\ &= 1.65\times10^{-3} \\ &= 1.65\ mA \end{aligned}$$

Step 3 of 9

Calculate the time required to charge a bit line from $0\ V$ to $5\ V$.

$$\begin{aligned} \Delta t &= \frac{C_B \Delta V}{I_C} \\ &= \frac{C_B (V_2 - V_1)}{I_C} \end{aligned}$$

Substitute $2\ pF$ for C_B, $1.65\ mA$ for I_C, $0\ V$ for V_1 and $5\ V$ for V_2.

$$\begin{aligned} \Delta t &= \frac{(2\times10^{-12})(5-0)}{1.65\times10^{-3}} \\ &= \frac{10\times10^{-12}}{1.65\times10^{-3}} \\ &= 6.06\times10^{-9} \\ &\approx 6.1\ ns \end{aligned}$$

Thus, the time required to charge a bit line from $0\ V$ to $5\ V$ is $\boxed{6.1\ ns}$.

Step 4 of 9

(b)

Draw the following *RC* circuit representing the word line.

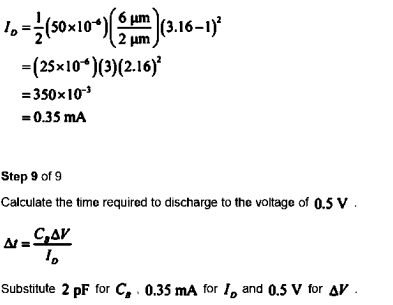


Figure 2

Step 5 of 9

Calculate the **10% to 90%** rise time of the word-line voltage, t_r.

$$t_r = 2.2RC$$

Substitute $3\ pF$ for C and $3\ k\Omega$ for R.

$$\begin{aligned} t_r &= (2.2)(3\times10^3)(3\times10^{-12}) \\ &= (2.2)(9\times10^{-9}) \\ &= 19.8\times10^{-9} \\ &= 19.8\ ns \end{aligned}$$

Thus, the **10% to 90%** rise time of the word-line voltage, t_r is $\boxed{19.8\ ns}$.

Step 6 of 9

Calculate the voltage at the end of one time constant, v_w.

$$v_w = V_{DD}\left(1 - e^{-\frac{t}{\tau}}\right)$$

Here t is one time constant.

Substitute $5\ V$ for V_{DD} and τ for t.

$$\begin{aligned} v_w &= 5\left(1 - e^{-\frac{\tau}{\tau}}\right) \\ &= 5\left(1 - e^{-1}\right) \\ &= 5(1 - 0.368) \\ &= 3.16\ V \end{aligned}$$

Thus, voltage reached at the end of one time constant, v_w is $\boxed{3.16\ V}$.

Step 7 of 9

(c)

Draw the NMOS transistor circuit to calculate the time required to discharge.

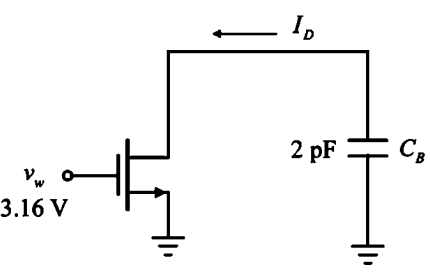


Figure 3

Step 8 of 9

Calculate the current through the capacitor, I_D.

$$I_D = \frac{1}{2}(\mu_n C_{ox})\left(\frac{W}{L}\right)_n (v_w - V_{tn})^2$$

Here,

v_w is the voltage reached at the end of one time constant

V_{tn} is the threshold voltage of the NMOS transistor

Substitute $50\ \mu A/V^2$ for $\mu_n C_{ox}$, $\frac{6\ \mu m}{2\ \mu m}$ for $\left(\frac{W}{L}\right)_n$, $1\ V$ for V_{tn} and $3.16\ V$ for v_w.

$$\begin{aligned} I_D &= \frac{1}{2}(50\times10^{-4})\left(\frac{6\ \mu m}{2\ \mu m}\right)(3.16-1)^2 \\ &= (25\times10^{-4})(3)(2.16)^2 \\ &= 350\times10^{-3} \\ &= 0.35\ mA \end{aligned}$$

Step 9 of 9

Calculate the time required to discharge to the voltage of $0.5\ V$.

$$\Delta t = \frac{C_B \Delta V}{I_D}$$

Substitute $2\ pF$ for C_B, $0.35\ mA$ for I_D and $0.5\ V$ for ΔV.

$$\begin{aligned} \Delta t &= \frac{(2\times10^{-12})(0.5)}{0.35\times10^{-3}} \\ &= \frac{1\times10^{-12}}{0.35\times10^{-3}} \\ &= 2.857\times10^{-9}\ s \\ &\approx 2.9\ ns \end{aligned}$$

Thus, the time required to discharge to the voltage of $0.5\ V$ is $\boxed{2.9\ ns}$.

Step 1 of 2

16.015P

Consider a 1-Gbit dynamic RAM (DRAM) with a cell size of $0.38\times0.76\ \mu m^2$.

Calculate the cell area.

$$\textbf{The cell area} = (\textbf{RAM capacity})(\textbf{cell size})$$

Substitute 1 G-bit for RAM capacity, and $0.38\times0.76\ \mu m^2$ for cell size.

$$\begin{aligned}\textbf{The cell area} &= \left(1\times10^{9}\right)\left[\left(0.38\times10^{-6}\right)\times\left(0.76\times10^{-6}\right)\right]\\ &= 0.289\times10^{-3}\ m^2\\ &= 0.289\ mm^2\end{aligned}$$

Consider that the chip size is, $19\times38\ mm^2$.

The total chip area in m^2 is,

$$\begin{aligned}\textbf{The total chip area} &= \left(19\times10^{-3}\right)\times\left(38\times10^{-3}\right)\\ &= 0.722\times10^{-3}\ m^2\\ &= 0.722\ mm^2\end{aligned}$$

Step 2 of 2

The space occupied by I/O connections, peripheral circuits on the chip is,

$$\begin{aligned}\textbf{Total chip area} - \textbf{Total cell area} &= 0.722\ mm^2 - 0.289\ mm^2\\ &= 0.433\times10^{-3}\ m^2\\ &= 0.433\ mm^2\end{aligned}$$

The percentage of space occupied by the peripherals in total chip area is,

$$\begin{aligned}\textbf{The fraction of chip area occupied by peripherals} &= \frac{0.433\ mm^2}{0.722\ mm^2}\times100\\ &= \frac{433}{722}\times100\\ &= 60\%\end{aligned}$$

Therefore, in total chip area 40% of space occupied by the cell area and remaining 60% space occupied by the I/O connections, peripheral circuits, and interconnect.

16.016P

Refer to Figure 15.12 in the textbook for a CMOS static RAM memory cell.

Consider that the voltages at Q and $\bar{Q}$ do not change by more than threshold voltage, V_t in a read operation.

Find the maximum allowable width to length ratio of transistor, Q_s .

$$\frac{\left(\frac{W}{L}\right)_s}{\left(\frac{W}{L}\right)_1} \le \frac{1}{\left(1-\frac{V_m}{V_{DD}-V_m}\right)^2} - 1$$

$$\left(\frac{W}{L}\right)_s \le \left(\frac{W}{L}\right)_1 \frac{1}{\left(1-\frac{V_m}{V_{DD}-V_m}\right)^2} - 1$$

Here, the width to length ratio of transistor, Q_1 is 1.5.

Substitute 0.5 V for V_t , and 2.5 V for the supply voltage, V_{DD} .

$$\left(\frac{W}{L}\right)_s \le (1.5)\frac{1}{\left(1-\frac{0.5}{2.5-0.5}\right)^2} - 1$$

$$\le (1.5)\frac{1}{(0.75)^2} - 1$$

$$\le 2.67 - 1$$

$$\le 1.67$$

Thus, the maximum allowable width to length ratio of transistor, Q_s , $\left(\frac{W}{L}\right)_s$ is $\boxed{1.67}$.

16.017P

Refer to Figure 15.12 in the textbook for a CMOS static RAM memory cell.

Consider that the voltages at Q **and** $\bar{Q}$ do not change by more than threshold voltage, V_t in a read operation.

Find the maximum allowable width to length ratio of transistor, Q_5 .

$$\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1} \le \frac{1}{\left(1-\frac{V_m}{V_{DD}-V_m}\right)^2}-1$$

$$\left(\frac{W}{L}\right)_5 \le \left(\frac{W}{L}\right)_1 \frac{1}{\left(1-\frac{V_m}{V_{DD}-V_m}\right)^2}-1$$

Here, the width to length ratio of transistor, Q_1 is 1.5.

Substitute 0.4 V for V_t , and 1.2 V for the supply voltage, V_{DD} .

$$\left(\frac{W}{L}\right)_5 \le (1.5)\frac{1}{\left(1-\frac{0.4}{1.2-0.4}\right)^2}-1$$

$$\le (1.5)\frac{1}{(0.5)^2}-1$$

$$\le 6-1$$

$$\le 5$$

Thus, the maximum allowable width to length ratio of transistor, Q_5 , $\left(\frac{W}{L}\right)_5$ is $\boxed{5}$.

Step 1 of 5

Refer to Figure 15.14 in the textbook for the graph between the normalized value of output, $V_{\bar{Q}}$ versus the ratio of width to length ratios of transistors, Q_5 and Q_1.

The relation between the normalized value of output, $V_{\bar{Q}}$ and the ratio of width to length ratios of transistors, Q_5 and Q_1 is,

$$\frac{V_{\bar{Q}}}{V_{DD}-V_{in}}=1-\frac{1}{\sqrt{1+\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}}}$$

Simplify the equation to find the ratio of width to length ratios of transistors, Q_5 and Q_1.

$$\frac{1}{\sqrt{1+\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}}}=1-\frac{V_{\bar{Q}}}{V_{DD}-V_{in}}$$

Apply square on both sides.

$$\frac{1}{\left(\sqrt{1+\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}}\right)^2}=\left(1-\frac{V_{\bar{Q}}}{V_{DD}-V_{in}}\right)^2$$

$$\frac{1}{1+\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}}=\left(1-\frac{V_{\bar{Q}}}{V_{DD}-V_{in}}\right)^2$$

$$\frac{1}{\left(1-\frac{V_{\bar{Q}}}{V_{DD}-V_{in}}\right)^2}=1+\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}$$

$$\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}=\frac{1}{\left(1-\frac{V_{\bar{Q}}}{V_{DD}-V_{in}}\right)^2}-1$$

Impose the condition, $V_{\bar{Q}}=V_{in}$ in the expression.

$$\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}=\frac{1}{\left(1-\frac{V_{in}}{V_{DD}-V_{in}}\right)^2}-1$$

Step 2 of 5

(a)

Consider that the supply voltage, V_{DD} is 2.5 V, and the threshold voltage, V_t is 0.5 V.

Substitute the values in the expression for the ratio of width to length ratios of transistors, Q_5 and Q_1.

$$\begin{aligned}\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}&=\frac{1}{\left(1-\frac{0.5}{2.5-0.5}\right)^2}-1\\&=\frac{1}{(1-0.25)^2}-1 \qquad \left(\text{since, } \frac{V_{\bar{Q}}}{V_{DD}-V_{\bar{Q}}}=0.25\right)\\&=\frac{1}{0.5625}-1\\&=0.78\end{aligned}$$

Thus, the ratio of width to length ratios of transistors, Q_5 and Q_1, $\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}$ is $\boxed{0.778}$.

Step 3 of 5

(b)

Consider that the supply voltage, V_{DD} is 1.8 V, and the threshold voltage, V_t is 0.5 V.

Substitute the values in the expression for the ratio of width to length ratios of transistors, Q_5 and Q_1.

$$\begin{aligned}\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}&=\frac{1}{\left(1-\frac{0.5}{1.8-0.5}\right)^2}-1\\&=\frac{1}{(1-0.386)^2}-1\\&=\frac{1}{0.378}-1\\&=1.64\end{aligned}$$

Thus, the ratio of width to length ratios of transistors, Q_5 and Q_1, $\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}$ is, $\boxed{1.64}$.

Step 4 of 5

(c)

Consider that the supply voltage, V_{DD} is 1.2 V, and the threshold voltage, V_t is 0.4 V.

Substitute the values in the expression for the ratio of width to length ratios of transistors, Q_5 and Q_1.

$$\begin{aligned}\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}&=\frac{1}{\left(1-\frac{0.4}{1.2-0.4}\right)^2}-1\\&=\frac{1}{(1-0.5)^2}-1 \left(\text{since, } \frac{V_{\bar{Q}}}{V_{DD}-V_{\bar{Q}}}=0.25\right)\\&=\frac{1}{0.25}-1\\&=3\end{aligned}$$

Thus, the ratio of width to length ratios of transistors, Q_5 and Q_1, $\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}$ is, $\boxed{3}$.

Step 5 of 5

Locate the obtained values on the plot.

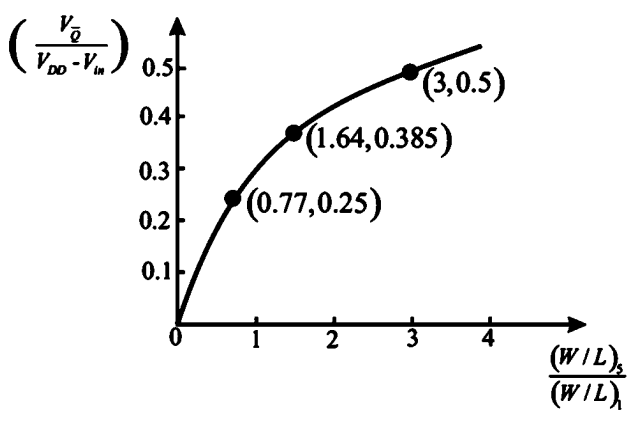


Figure 1

16.019P

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Step 1 of 3

16.020P

Refer to Figure 15.14 in the textbook for the graph between the normalized value of output, $V_{\bar{Q}}$ versus the ratio of width to length ratios of transistors, Q_5 and Q_1.

The relation between the normalized value of output, $V_{\bar{Q}}$ and the ratio of width to length ratios of transistors, Q_5 and Q_1 is,

$$\frac{V_{\bar{Q}}}{V_{DD}-V_{in}}=1-\frac{1}{\sqrt{1+\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}}}$$

Simplify the equation to find the ratio of width to length ratios of transistors, Q_5 and Q_1.

$$\frac{1}{\sqrt{1+\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}}}=1-\frac{V_{\bar{Q}}}{V_{DD}-V_{in}}$$

Apply square on both sides.

$$\frac{1}{1+\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}}=\left(1-\frac{V_{\bar{Q}}}{V_{DD}-V_{in}}\right)^2$$

$$\frac{1}{\left(1-\frac{V_{\bar{Q}}}{V_{DD}-V_{in}}\right)^2}=1+\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}$$

$$\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}=\frac{1}{\left(1-\frac{V_{\bar{Q}}}{V_{DD}-V_{in}}\right)^2}-1$$

Impose the condition, $V_{\bar{Q}}=V_{in}$ in the expression.

$$\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}=\frac{1}{\left(1-\frac{V_{in}}{V_{DD}-V_{in}}\right)^2}-1$$

Step 2 of 3

Consider that the supply voltage, V_{DD} is 1.8 V, and the threshold voltage, V_t is 0.5 V.

Substitute the values in the expression for the ratio of width to length ratios of transistors, Q_5 and Q_1.

$$\begin{aligned}\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1}&=\frac{1}{\left(1-\frac{0.5}{1.8-0.5}\right)^2}-1\\&=\frac{1}{(1-0.386)^2}-1\\&=\frac{1}{0.378}-1\\&=1.64\end{aligned}$$

Thus, the maximum allowable ratio of width to length ratios of transistors, Q_5 and Q_1 is $\boxed{1.64}$.

Step 3 of 3

Consider that the length of the transistors is,

$$\begin{aligned}L_1&=L_5\\&=0.18\ \mu m\end{aligned}$$

The minimum area is obtained when $\left(\frac{W}{L}\right)_5=1$, therefore, $W_5=0.18\ \mu m$. The transistor, Q_5 is saturated and Q_1 is in triode region. Then the currents are equal. Therefore,

$$I_5=I_1$$

$$\frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_5\left(V_{DD}-V_{in}-V_{\bar{Q}}\right)^2=\mu_n C_{ox}\left(\frac{W}{L}\right)_1\left[(V_{DD}-V_{in})V_{\bar{Q}}-\frac{1}{2}V_{\bar{Q}}^2\right]$$

$$\frac{1}{2}(1)(1.8-0.5-0.2)^2=\left(\frac{W}{0.18\ \mu m}\right)\left[\left[(1.8-0.5)(0.2)-\frac{1}{2}(0.2)^2\right]\right]$$

$$0.605=\left(\frac{W}{0.18\ \mu m}\right)(0.26-0.02)$$

$$W_1=0.45\ \mu m$$

Thus, the values of W_1 and W_5 that minimize the combined areas of Q_1 and Q_5 are $\boxed{0.45\ \mu m}$ and $\boxed{0.18\ \mu m}$ respectively.

Step 1 of 2

16.021P

Refer to Figure 15.12 in the textbook for the CMOS static RAM memory cell.

Consider the read operation of the 6T SRAM cell when it is storing a 0, that is, $V_Q = 0\ V$, and $V_{\bar{Q}} = V_{DD}$. Consider also that the bit lines are pre-charged to V_{DD} before the word-line voltage is raised to V_{DD} .

The relevant part of the circuit to describe the operation is,

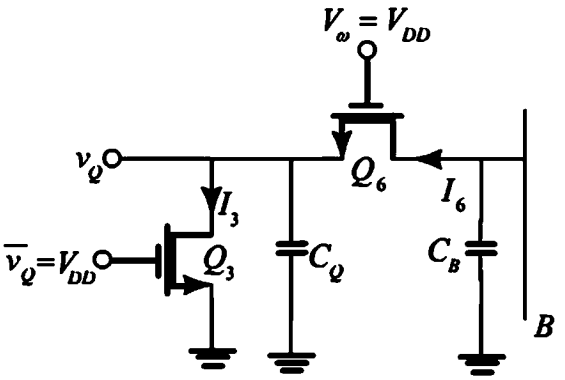


Figure 1

Step 2 of 2

Initially output voltage, V_Q is at 0 V. Before the read operation begins, the B and $\bar{B}$ lines are raised to a voltage in the range of $\frac{V_{DD}}{2}$ to V_{DD} .

When the word line is selected and the access transistors, Q_5 and Q_6 are turned on, examination of the circuit reveals that the only portion that is conducting is shown in circuit. As the initial value of V_Q is 0 V, current flows from the B line.

This current charges C_Q thus V_Q raises and Q_3 starts conducting and the equilibrium is reached when C_Q is charged to a voltage, V_Q at which $I_6 = I_3$.

Here, V_Q must not exceed the threshold voltage of $Q_1 - Q_2$.

Thus, this is same as the operation presented in the text but with Q_5 replaced by Q_6 , and Q_1 by Q_3 .

Step 1 of 2

16.022P

Refer to Fig.15.13 in the text book.

Consider the current of transistor, Q_5 operating in saturation to determine the ratio of width to length ratios of the transistors, Q_5 **and** Q_1 with $V_{\bar{Q}} \le V_{tn}$.

$$I_5 = \frac{1}{2}(\mu_n C_{ox})\left(\frac{W}{L}\right)_5 \left(V_{DD} - V_t - V_{\bar{Q}}\right)^2$$

Consider the current of transistor Q_1 operating in triode region.

$$I_1 = (\mu_n C_{ox})\left(\frac{W}{L}\right)_1 \left[(V_{DD} - V_t)V_{\bar{Q}} - \frac{1}{2}V_{\bar{Q}}^2\right]$$

Thus, equate the currents I_5 **and** I_1 considering the velocity saturation effects.

$$\frac{1}{2}(\mu_n C_{ox})\left(\frac{W}{L}\right)_5 \begin{bmatrix} V_{DD} - V_t \\ -V_{\bar{Q}} \end{bmatrix}^2 (1 + \lambda V_{DSsat}) = (\mu_n C_{ox})\left(\frac{W}{L}\right)_1 \begin{bmatrix} (V_{DD} - V_t)V_{\bar{Q}} \\ -\frac{1}{2}V_{\bar{Q}}^2 \end{bmatrix} (1 + \lambda V_{DSsat})$$

$$\frac{1}{2}\left(\frac{W}{L}\right)_5 \left(V_{DD} - V_t - V_{\bar{Q}}\right)^2 = \left(\frac{W}{L}\right)_1 \left[(V_{DD} - V_t)V_{\bar{Q}} - \frac{1}{2}V_{\bar{Q}}^2\right]$$

At $V_{\bar{Q}} = V_{tn}$, maximum value of the ratio of width to length ratios of the transistors, Q_5 **and** Q_1 is obtained.

Substitute $1.8\,\text{V}$ for V_{DD}, 0.5 V for V_t and $0.5\,\text{V}$ for $V_{\bar{Q}}$.

$$\frac{1}{2}\left(\frac{W}{L}\right)_5 (1.8 - 0.5 - 0.5)^2 = \left(\frac{W}{L}\right)_1 \left[(1.8 - 0.5)0.5 - \frac{1}{2}0.5^2\right]$$

$$\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1} = \frac{\left[(1.8 - 0.5)0.5 - \frac{1}{2}0.5^2\right]}{\frac{1}{2}(1.8 - 0.5 - 0.5)^2}$$

$$= \frac{0.65 - 0.125}{0.32}$$

$$= 1.641$$

Thus, maximum value of the ratio of width to length ratios of the transistors, Q_5 **and** Q_1, for $V_{\bar{Q}} \le V_t$, considering the velocity saturation effect is $\boxed{1.641}$.

Step 2 of 2

Calculate the value of the ratio of width to length ratios of the transistors, Q_5 **and** Q_1 considering the situation that only the transistor Q_5 is operating in velocity saturation.

Thus, equate the currents I_5 **and** I_1 considering the velocity saturation effect for only Q_5.

$$\frac{1}{2}(\mu_n C_{ox})\left(\frac{W}{L}\right)_5 \begin{bmatrix} V_{DD} - V_t \\ -V_{\bar{Q}} \end{bmatrix}^2 (1 + \lambda V_{DSsat}) = (\mu_n C_{ox})\left(\frac{W}{L}\right)_1 \left[(V_{DD} - V_t)V_{\bar{Q}} - \frac{1}{2}V_{\bar{Q}}^2\right]$$

$$\frac{1}{2}\left(\frac{W}{L}\right)_5 \left(V_{DD} - V_t - V_{\bar{Q}}\right)^2 (1 + \lambda V_{DSsat}) = \left(\frac{W}{L}\right)_1 \left[(V_{DD} - V_t)V_{\bar{Q}} - \frac{1}{2}V_{\bar{Q}}^2\right]$$

At $V_{\bar{Q}} = V_{tn}$, maximum value of the ratio of width to length ratios of the transistors, Q_5 **and** Q_1 is obtained.

Substitute $1.8\,\text{V}$ for V_{DD}, 0.5 V for V_t and $0.5\,\text{V}$ for $V_{\bar{Q}}$ and 0.6 V for V_{DSsat}.

$$\frac{1}{2}\left(\frac{W}{L}\right)_5 (1.8 - 0.5 - 0.5)^2 (1 + 0.6\lambda) = \left(\frac{W}{L}\right)_1 \left[(1.8 - 0.5)0.5 - \frac{1}{2}0.5^2\right]$$

$$\frac{\left(\frac{W}{L}\right)_5}{\left(\frac{W}{L}\right)_1} = \frac{\left[(1.8 - 0.5)0.5 - \frac{1}{2}0.5^2\right]}{\frac{1}{2}(1.8 - 0.5 - 0.5)^2 (1 + 0.6\lambda)}$$

$$= \frac{0.65 - 0.125}{0.32(1 + 0.6\lambda)}$$

$$= \frac{1.641}{(1 + 0.6\lambda)}$$

Thus, the maximum ratio of width to length ratios of the transistors, Q_5 **and** Q_1 taking velocity saturation into account is $\boxed{1 + 0.6\lambda}$ times the ratio of width to length ratios of the transistors, Q_5 **and** Q_1 obtained without considering the velocity saturation.

16.023P

We don't have the solution to this problem yet.

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16.024P

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In the case of writing a 1 into a 6T SRAM cell that is originally storing a 0, v_Q is lowered to 0 and then raised to V_{DD} .

Draw the relevant parts of the 6T SRAM circuit to explain this operation.

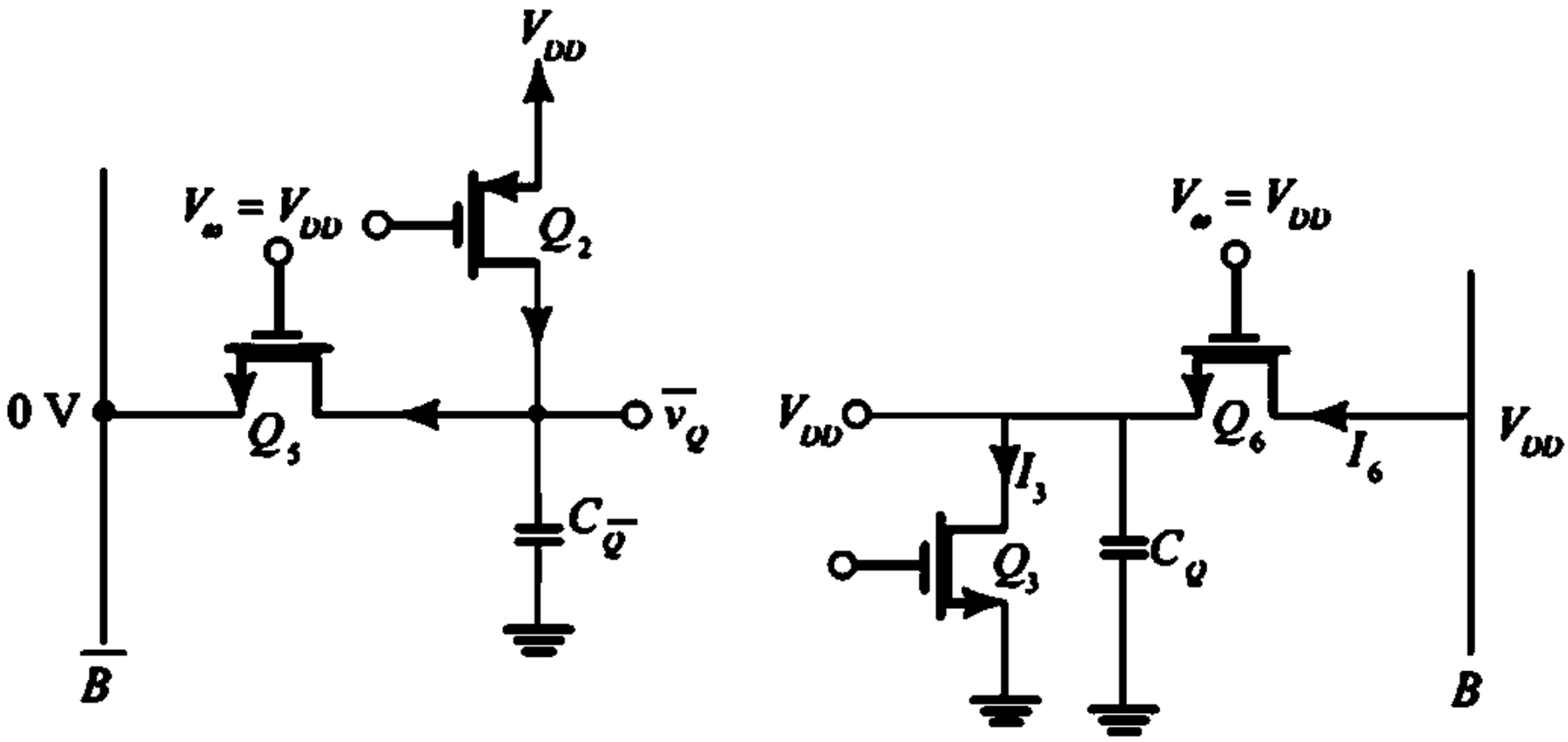


Figure 1

To write 1, the B line is raised to V_{DD} and $\overline{B}$ is lowered to 0. When $\overline{B}$ line voltage is lowered, Q_5 conducts and discharges $C_{\overline{Q}}$ to 0 V. Similarly, when B voltage is increased to V_{DD} , Q_6 conducts and charges C_Q to V_{DD} . The results are the same as that for writing a 0 because transistors Q_3 and Q_4 are active in this process.

Thus, the sketch for the relevant parts of the 6T SRAM circuit to write a 1 is shown in Figure 1.

16.026P

Consider the relationship that provides an upper bound on $\left(\frac{W}{L}\right)_p$ in terms of $\left(\frac{W}{L}\right)_a$.

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_a} \le \left(\frac{\mu_n}{\mu_p}\right)\left[1-\left(1-\frac{V_{tn}}{V_{DD}-V_{tn}}\right)^2\right]$$

Substitute $4\mu_p$ for μ_n, $1.2\ \text{V}$ for V_{DD} and $0.4\ \text{V}$ for V_{tn}.

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_a} \le \left(\frac{4\mu_p}{\mu_p}\right)\left[1-\left(1-\frac{0.4}{1.2-0.4}\right)^2\right]$$

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_a} \le (4)\left[1-\left(1-\frac{1}{2}\right)^2\right]$$

$$\left(\frac{W}{L}\right)_p \le (4)\left(1-\frac{1}{4}\right)\left(\frac{W}{L}\right)_a$$

$$\left(\frac{W}{L}\right)_p \le 3\left(\frac{W}{L}\right)_a$$

Therefore, the maximum permitted value of $\left(\frac{W}{L}\right)_p$ in terms of $\left(\frac{W}{L}\right)_a$ is $\boxed{3\left(\frac{W}{L}\right)_a}$.

16.027P

Consider the relationship that provides an upper bound on $\left(\frac{W}{L}\right)_p$ in terms of $\left(\frac{W}{L}\right)_a$.

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_a} \le \left(\frac{\mu_n}{\mu_p}\right)\left[1-\left(1-\frac{V_{tn}}{V_{DD}-V_{tn}}\right)^2\right]$$

Substitute $4\mu_p$ for μ_n, 2.5 V for V_{DD} and 0.5 V for V_{tn}.

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_a} \le \left(\frac{4\mu_p}{\mu_p}\right)\left[1-\left(1-\frac{0.5}{2.5-0.5}\right)^2\right]$$

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_a} \le (4)\left[1-\left(1-\frac{1}{4}\right)^2\right]$$

$$\left(\frac{W}{L}\right)_p \le (4)\left(1-\frac{9}{16}\right)\left(\frac{W}{L}\right)_a$$

$$\left(\frac{W}{L}\right)_p \le 1.75\left(\frac{W}{L}\right)_a$$

Therefore, the maximum permitted value of $\left(\frac{W}{L}\right)_p$ in terms of $\left(\frac{W}{L}\right)_a$ is $\boxed{1.75\left(\frac{W}{L}\right)_a}$.

Step 1 of 7

Refer to Figure 15.17 in the textbook.

(a)

Consider the relationship that provides an upper bound on $\left(\frac{W}{L}\right)_p$ in terms of $\left(\frac{W}{L}\right)_n$.

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_n} \le \left(\frac{\mu_n}{\mu_p}\right)\left[1-\left(1-\frac{V_{in}}{V_{DD}-V_{in}}\right)^2\right]$$

Substitute $4\mu_p$ for μ_n, $2.5\ \text{V}$ for V_{DD} and $0.5\ \text{V}$ for V_{in}.

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_n} \le \left(\frac{4\mu_p}{\mu_p}\right)\left[1-\left(1-\frac{0.5}{2.5-0.5}\right)^2\right]$$

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_n} \le (4)\left[1-\left(1-\frac{1}{4}\right)^2\right]$$

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_n} \le (4)\left(1-\frac{9}{16}\right)$$

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_n} \le 1.75$$

Step 2 of 7

Calculate the value of $\frac{V_{\bar{Q}}}{V_{DD}-V_{in}}$.

$$\frac{V_{\bar{Q}}}{V_{DD}-V_{in}} = \frac{V_{in}}{V_{DD}-V_{in}}$$

Substitute $2.5\ \text{V}$ for V_{DD} and $0.5\ \text{V}$ for V_{in}.

$$\frac{V_{\bar{Q}}}{V_{DD}-V_{in}} = \frac{0.5}{2.5-0.5} = 0.25$$

Thus, the point formed for $0.25\ \mu\text{m}$ fabrication process is $\boxed{A=(1.75,0.25)}$.

Step 3 of 7

(b)

Consider the relationship that provides an upper bound on $\left(\frac{W}{L}\right)_p$ in terms of $\left(\frac{W}{L}\right)_n$.

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_n} \le \left(\frac{\mu_n}{\mu_p}\right)\left[1-\left(1-\frac{V_{in}}{V_{DD}-V_{in}}\right)^2\right]$$

Substitute $4\mu_p$ for μ_n, $1.8\ \text{V}$ for V_{DD} and $0.5\ \text{V}$ for V_{in}.

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_n} \le \left(\frac{4\mu_p}{\mu_p}\right)\left[1-\left(1-\frac{0.5}{1.8-0.5}\right)^2\right]$$

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_n} \le (4)\left[1-\left(1-\frac{5}{13}\right)^2\right]$$

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_n} \le (4)\left(1-\frac{64}{169}\right)$$

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_n} \le 2.485$$

Step 4 of 7

Calculate the value of $\frac{V_{\bar{Q}}}{V_{DD}-V_{in}}$.

$$\frac{V_{\bar{Q}}}{V_{DD}-V_{in}} = \frac{V_{in}}{V_{DD}-V_{in}}$$

Substitute $1.8\ \text{V}$ for V_{DD} and $0.5\ \text{V}$ for V_{in}.

$$\frac{V_{\bar{Q}}}{V_{DD}-V_{in}} = \frac{0.5}{1.8-0.5} = 0.384$$

Thus, the point formed for $0.18\ \mu\text{m}$ fabrication process is $\boxed{B=(2.485,0.384)}$.

Step 5 of 7

(c)

Consider the relationship that provides an upper bound on $\left(\frac{W}{L}\right)_p$ in terms of $\left(\frac{W}{L}\right)_n$.

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_n} \le \left(\frac{\mu_n}{\mu_p}\right)\left[1-\left(1-\frac{V_{in}}{V_{DD}-V_{in}}\right)^2\right]$$

Substitute $4\mu_p$ for μ_n, $1.2\ \text{V}$ for V_{DD} and 0.4 for V_{in}.

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_n} \le \left(\frac{4\mu_p}{\mu_p}\right)\left[1-\left(1-\frac{0.4}{1.2-0.4}\right)^2\right]$$

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_n} \le (4)\left[1-\left(1-\frac{1}{2}\right)^2\right]$$

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_n} \le (4)\left(1-\frac{1}{4}\right)$$

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_n} \le 3$$

Step 6 of 7

Calculate the value of $\frac{V_{\bar{Q}}}{V_{DD}-V_{in}}$.

$$\frac{V_{\bar{Q}}}{V_{DD}-V_{in}} = \frac{V_{in}}{V_{DD}-V_{in}}$$

Substitute $1.2\ \text{V}$ for V_{DD} and 0.4 for V_{in}.

$$\frac{V_{\bar{Q}}}{V_{DD}-V_{in}} = \frac{0.4}{1.2-0.4} = 0.5$$

Thus, the point formed for $0.13\ \mu\text{m}$ fabrication process is $\boxed{C=(3,0.5)}$.

Step 7 of 7

Draw the plot to show the points A, B and C corresponding to the CMOS fabrication processes.

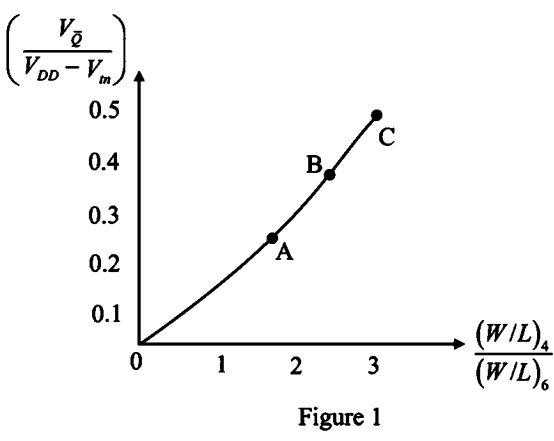


Figure 1

Thus, the plot to show the points A, B and C corresponding to the CMOS fabrication processes is shown in Figure 1.

Step 1 of 4

Refer to Figure 15.12 in the textbook for 6TSRAM.

Calculate the value of $\left(\frac{W}{L}\right)_a$.

$$\left(\frac{W}{L}\right)_{Q_3,Q_4} = \left(\frac{W}{L}\right)_a$$
$$= \frac{0.13\ \mu m}{0.13\ \mu m}$$
$$= 1$$

Consider the relationship between $\left(\frac{W}{L}\right)_n$ in terms of $\left(\frac{W}{L}\right)_a$ to calculate $\left(\frac{W}{L}\right)_{Q_1,Q_2}$.

$$\frac{\left(\frac{W}{L}\right)_a}{\left(\frac{W}{L}\right)_{Q_1,Q_2}} = \frac{1}{\left(1-\frac{V_t}{V_{DD}-V_t}\right)^2} - 1$$

Substitute 1 for $\left(\frac{W}{L}\right)_a$, $1.2\ V$ for V_{DD} and 0.4 for V_t.

$$\frac{1}{\left(\frac{W}{L}\right)_{Q_1,Q_2}} = \frac{1}{\left(1-\frac{0.4}{1.2-0.4}\right)^2} - 1$$
$$\frac{1}{\left(\frac{W}{L}\right)_{Q_1,Q_2}} = \frac{1}{\left(1-\frac{1}{2}\right)^2} - 1$$
$$\frac{1}{\left(\frac{W}{L}\right)_{Q_1,Q_2}} = 4-1$$
$$\left(\frac{W}{L}\right)_{Q_1,Q_2} = \frac{1}{3}$$

Thus, the value of $\left(\frac{W}{L}\right)_{Q_1,Q_2}$ is $\boxed{\frac{1}{3}}$.

Step 2 of 4

Consider the relationship between $\left(\frac{W}{L}\right)_p$ in terms of $\left(\frac{W}{L}\right)_a$ to calculate $\left(\frac{W}{L}\right)_{Q_3,Q_4}$.

$$\frac{\left(\frac{W}{L}\right)_{Q_3,Q_4}}{\left(\frac{W}{L}\right)_a} = \frac{\mu_n}{\mu_p}\left[1-\left(1-\frac{V_t}{V_{DD}-V_t}\right)^2\right]$$

Substitute 1 for $\left(\frac{W}{L}\right)_a$, $4\mu_p$ for μ_n, $1.2\ V$ for V_{DD} and 0.4 for V_t.

$$\frac{\left(\frac{W}{L}\right)_{Q_3,Q_4}}{1} = \left(\frac{4\mu_p}{\mu_p}\right)\left[1-\left(1-\frac{0.4}{1.2-0.4}\right)^2\right]$$
$$\left(\frac{W}{L}\right)_{Q_3,Q_4} = (4)\left[1-\left(1-\frac{1}{2}\right)^2\right]$$
$$\left(\frac{W}{L}\right)_{Q_3,Q_4} = (4)\left(1-\frac{1}{4}\right)$$
$$\left(\frac{W}{L}\right)_{Q_3,Q_4} = 3$$

Thus, the value of $\left(\frac{W}{L}\right)_{Q_3,Q_4}$ is $\boxed{3}$.

Step 3 of 4

Consider the relationship that provides an upper bound on $\left(\frac{W}{L}\right)_p$ in terms of $\left(\frac{W}{L}\right)_a$.

$$\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_a} \le \left(\frac{\mu_n}{\mu_p}\right)\left[1-\left(1-\frac{V_m}{V_{DD}-V_m}\right)^2\right]$$

Substitute 3 for $\left(\frac{W}{L}\right)_p$, 1 for $\left(\frac{W}{L}\right)_a$, $1.2\ V$ for V_{DD} and 0.4 for V_t.

$$\frac{3}{1} \le \left(\frac{4\mu_p}{\mu_p}\right)\left[1-\left(1-\frac{0.4}{1.2-0.4}\right)^2\right]$$
$$3 \le (4)\left[1-\left(1-\frac{1}{2}\right)^2\right]$$
$$3 \le 3$$

Thus, the expression, $\frac{\left(\frac{W}{L}\right)_p}{\left(\frac{W}{L}\right)_a} \le \left(\frac{\mu_n}{\mu_p}\right)\left[1-\left(1-\frac{V_m}{V_{DD}-V_m}\right)^2\right]$ is verified.

Step 4 of 4

Consider the relationship that provides an upper bound on $\left(\frac{W}{L}\right)_n$ in terms of $\left(\frac{W}{L}\right)_a$.

$$\frac{\left(\frac{W}{L}\right)_a}{\left(\frac{W}{L}\right)_n} = \frac{1}{\left(1-\frac{V_t}{V_{DD}-V_t}\right)^2} - 1$$

Substitute $\frac{1}{3}$ for $\left(\frac{W}{L}\right)_n$, 1 for $\left(\frac{W}{L}\right)_a$, $1.2\ V$ for V_{DD} and 0.4 for V_t.

$$\frac{1}{\frac{1}{3}} = \frac{1}{\left(1-\frac{0.4}{1.2-0.4}\right)^2} - 1$$
$$3 = \frac{1}{\left(1-\frac{1}{2}\right)^2} - 1$$
$$3 = 3$$

Thus, the expression, $\frac{\left(\frac{W}{L}\right)_a}{\left(\frac{W}{L}\right)_n} = \frac{1}{\left(1-\frac{V_t}{V_{DD}-V_t}\right)^2} - 1$ is verified.

16.030P

We don't have the solution to this problem yet.

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16.031P

We don't have the solution to this problem yet.

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16.032P

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16.033P

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16.034P

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16.035P

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16.036P

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16.037P

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16.038P

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16.039P

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16.040P

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16.041P

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16.042P

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16.043P

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16.044P

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16.045P

Refer to Figure 15.29 in the textbook.

Calculate the number of inverters required for the design.

$$T = Nt_p$$

$$N = \frac{T}{t_p}$$

Substitute $\mathbf{10\ ns}$ for T and $\mathbf{2.5\ ns}$ for t_p .

$$N = \frac{\mathbf{10\ ns}}{\mathbf{2.5\ ns}}$$
$$= \mathbf{4}$$

Thus, the number of inverters required for the design is $\boxed{4}$.

Draw the designed one shot circuit.

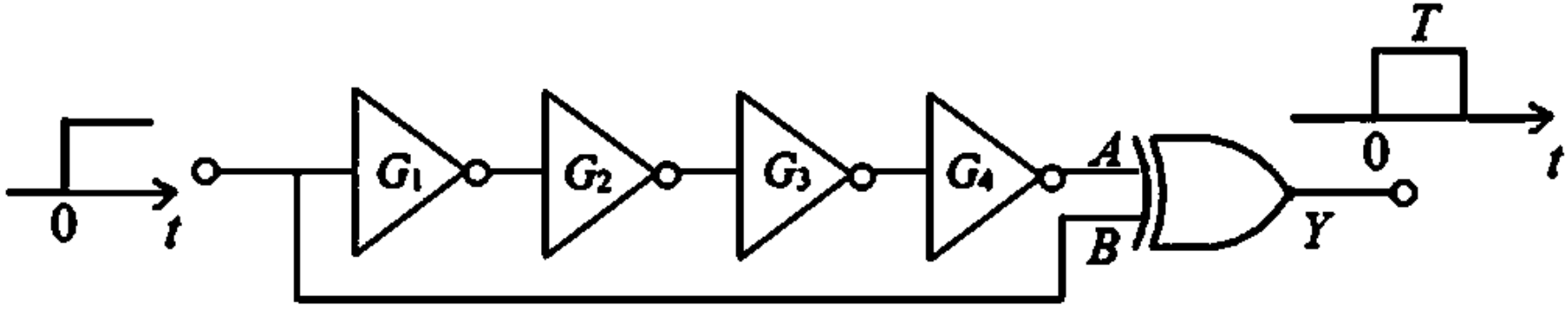


Figure 1

Thus, the one shot circuit is designed and shown in Figure 1.

16.046P

Refer to Figure 15.30 in the textbook for a simple MOS ROM organized as 8 words × 4 bits.

From the circuit diagram, the output is high if no word is selected. Thus, logically high must be corresponding to logic '0'.

No transistor represents logic '0' otherwise logic '1'.

The eight words stored in the ROM are,

$$W_1 = 0100$$
$$W_2 = 0000$$
$$W_3 = 1000$$
$$W_4 = 1001$$

$$W_5 = 0101$$
$$W_6 = 0001$$
$$W_7 = 0110$$
$$W_8 = 0010$$

Thus, the eight words stored in the ROM are,

$$\boxed{0100, 0000, 1000, 1001, 0101, 0001, 0110 \text{ and } 0010}.$$

Step 1 of 2

Refer to Figure 15.30 in the textbook for a simple MOS ROM organized as 8 words × 4 bits.

From the circuit diagram, the output is high if no word is selected. Thus, logically high must be corresponding to logic '0'.

No transistor represents logic '0' otherwise logic '1'.

Write the truth table such that, the four bit output, Z is the multiplication of two bit, X and Y.

Table 1

X	Y	Z
00	00	0000
00	01	0000
00	10	0000
00	11	0000
01	00	0000
01	01	0001
01	10	0010
01	11	0011
10	00	0000
10	01	0010
10	10	0100
10	11	0110
11	00	0000
11	01	0011
11	10	0110
11	11	1001

Step 2 of 2

Draw the MOS ROM organized by 16 words × 4 bits.

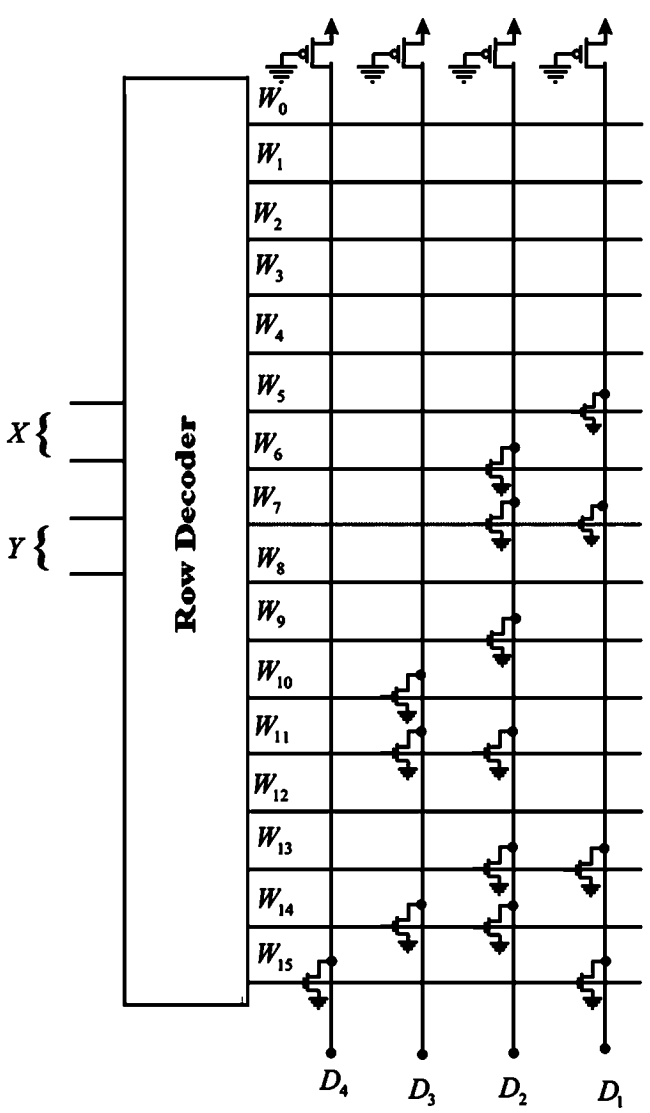


Figure 1

Thus, the designed circuit is shown in Figure 1.

Step 1 of 7

Refer to Figure 15.30 in the textbook for a simple MOS ROM organized as 8 words × 4 bits.

(a)

Calculate the average charging current of PMOS.

$$i_C = k_p'\left(\frac{W}{L}\right)_p\left[(V_{DD}-v_w)v_{\overline{Q}}-\frac{1}{2}v_{\overline{Q}}^2\right]$$

Substitute $\frac{90\ \mu A/V^2}{3}$ for k_p', $\frac{12\ \mu m}{1.2\ \mu m}$ for $\left(\frac{W}{L}\right)_p$, $5\ V$ for V_{DD}, $1\ V$ for v_w and $2.5\ V$ for $v_{\overline{Q}}$.

$$\begin{aligned} i_C &= \left(\frac{90\times10^{-6}}{3}\right)\left(\frac{12\times10^{-6}}{1.2\times10^{-6}}\right)\left[(5-1)2.5-\frac{(2.5)^2}{2}\right] \\ &= (30\times10^{-6})(10)(10-3.125) \\ &= 2.0625\times10^{-3} \\ &= 2.0625\ mA \end{aligned}$$

Step 2 of 7

Calculate the time required for charging a bit line from 0 V to 5 V.

$$\begin{aligned} Q &= C_B\Delta V \\ i_C t &= C_B(V_2-V_1) \\ t &= \frac{C_B(V_2-V_1)}{i_C} \end{aligned}$$

Substitute $5\ V$ for V_2, $0\ V$ for V_1, $1\ pF$ for C_B and $2.0625\ mA$ for i_C.

$$\begin{aligned} t &= \frac{(1\times10^{-12})(5-0)}{2.0625\times10^{-3}} \\ &= 2.42\times10^{-12} \\ &= 2.42\ ns \end{aligned}$$

Thus, the time for pre-charge is $\boxed{2.42\ ns}$.

Step 3 of 7

(b)

Calculate the value of time constant, τ.

$$\tau = RC$$

Substitute $5\ k\Omega$ for R and $2\ pF$ for C.

$$\begin{aligned} \tau &= (5\times10^3)\times(2\times10^{-12}) \\ &= 10\times10^{-9} \\ &= 10\ ns \end{aligned}$$

Calculate the word line voltage is at any time.

$$\begin{aligned} v_w &= v(\infty)+\left[v(0)-v(\infty)\right]e^{-\frac{t}{\tau}} \\ &= 5+(0-5)e^{\frac{t}{10\times10^{-9}}} \\ &= 5\left(1-e^{-10^8 t}\right)\ V \end{aligned}$$

Step 4 of 7

The 10% to 90 % rise time of the word-line, calculate the rise time.

$$\begin{aligned} 0.9V_{DD} &= v_w \\ 0.9(5) &= 5\left(1-e^{-10^8 t}\right) \\ 1-e^{-10^8 t} &= 0.9 \\ e^{-10^8 t} &= 0.1 \\ -10^8 t &= \ln(0.1) \\ -10^8 t &= -2.3 \\ t &= 23\times10^{-9} \\ t &= 23\ ns \end{aligned}$$

Thus, the rise time of the word-line is $\boxed{23\ ns}$.

Step 5 of 7

Calculate the word line voltage at the end of one time constant.

$$\begin{aligned} v_w &= 5\left(1-e^{\frac{-t}{\tau}}\right) \\ &= 5\left(1-e^{\frac{-t}{\tau}}\right) \\ &= 5\left(1-e^{-1}\right) \\ &= 3.16\ V \end{aligned}$$

Thus, the word line voltage at the end of one time constant is $\boxed{3.16\ V}$.

Step 6 of 7

(c)
For discharge, the current through the NMOS is,

$$i_{ds} = \frac{1}{2}k_n'\left(\frac{W}{L}\right)_n\left(V_{gs}-V_t\right)^2$$

Substitute $90\ \mu A/V^2$ for k_n', $\frac{3\ \mu m}{1.2\ \mu m}$ for $\left(\frac{W}{L}\right)_n$, $3.16\ V$ for V_{GS}, $1\ V$ for v_w.

$$\begin{aligned} i_{ds} &= \frac{1}{2}(90\times10^{-6})\left(\frac{3\times10^{-6}}{1.2\times10^{-6}}\right)(3.16-1)^2 \\ &= (45\times10^{-6})\left(\frac{3}{1.2}\right)(32.16)^2 \\ &= 525\times10^{-6} \\ &= 525\ \mu A \end{aligned}$$

Step 7 of 7

Calculate the required time interval Δt to discharge lower to one volt.

$$\Delta t = \frac{C\Delta V}{i_{ds}}$$

Substitute $1\ V$ for ΔV, $1\ pF$ for C_B and $2.0625\ mA$ for i_{ds}.

$$\begin{aligned} \Delta t &= \frac{(1\times10^{-12})(1)}{525\times10^{-6}} \\ &= 1.90\times10^{-9} \\ &= 1.9\ ns \end{aligned}$$

Thus, the required time interval Δt to discharge lower to one volt is $\boxed{1.9\ ns}$.

16.049P

We don't have the solution to this problem yet.

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Step 1 of 8

Determine the attenuation function in dB.

$$A(\omega) = -20\log|T(j\omega)|$$

Substitute 1 for $T(j\omega)$.

$$A(\omega) = -20\log(1)$$
$$= 0$$

Therefore, the value of attenuation in dB is **0 dB**.

Step 2 of 8

Determine the attenuation function in dB.

$$A(\omega) = -20\log|T(j\omega)|$$

Substitute 0.99 for $T(j\omega)$.

$$A(\omega) = -20\log(0.99)$$
$$= 0.0873\,\text{dB}$$

Therefore, the value of attenuation in dB is **0.0873 dB**.

Step 3 of 8

Determine the attenuation function in dB.

$$A(\omega) = -20\log|T(j\omega)|$$

Substitute 0.9 for $T(j\omega)$.

$$A(\omega) = -20\log(0.9)$$
$$= 0.915\,\text{dB}$$

Therefore, the value of attenuation in dB is **0.915 dB**.

Step 4 of 8

Determine the attenuation function in dB.

$$A(\omega) = -20\log|T(j\omega)|$$

Substitute 0.8 for $T(j\omega)$.

$$A(\omega) = -20\log(0.8)$$
$$= 1.938\,\text{dB}$$

Therefore, the value of attenuation in dB is **1.938 dB**.

Step 5 of 8

Determine the attenuation function in dB.

$$A(\omega) = -20\log|T(j\omega)|$$

Substitute 0.7 for $T(j\omega)$.

$$A(\omega) = -20\log(0.7)$$
$$= 3.098\,\text{dB}$$

Therefore, the value of attenuation in dB is **3.098 dB**.

Step 6 of 8

Determine the attenuation function in dB.

$$A(\omega) = -20\log|T(j\omega)|$$

Substitute 0.5 for $T(j\omega)$.

$$A(\omega) = -20\log(0.5)$$
$$= 6.02\,\text{dB}$$

Therefore, the value of attenuation in dB is **6.02 dB**.

Step 7 of 8

Determine the attenuation function in dB.

$$A(\omega) = -20\log|T(j\omega)|$$

Substitute 0.1 for $T(j\omega)$.

$$A(\omega) = -20\log(0.1)$$
$$= 20\,\text{dB}$$

Therefore, the value of attenuation in dB is **20 dB**.

Step 8 of 8

Determine the attenuation function in dB.

$$A(\omega) = -20\log|T(j\omega)|$$

Substitute 0 for $T(j\omega)$.

$$A(\omega) = -20\log(0)$$
$$= \infty\,\text{dB}$$

Therefore, the value of attenuation in dB is **∞ dB**.

Step 1 of 6

The transfer function of the first-order low pass filter is,

$$T(s)=\frac{\omega_0}{(s+\omega_0)}$$

Here, ω_0 = **3-db frequency of the filter.**

Step 2 of 6

The magnitude of the transfer function is,

$$T(s)=\frac{\omega_0}{(s+\omega_0)}$$

$$T(j\omega)=\frac{\omega_0}{(j\omega+\omega_0)}$$

$$|T(j\omega)|=\frac{\omega_0}{\sqrt{\omega^2+\omega^2_0}}$$

Thus, the magnitude of the transfer function is, $|T(j\omega)|=\frac{\omega_0}{\sqrt{\omega^2+\omega^2_0}}$

Step 3 of 6

The phase angle of the transfer function is,

$$\phi(\omega)=\frac{\tan^{-1}\left(\frac{0}{\omega_0}\right)}{\tan^{-1}\left(\frac{\omega}{\omega_0}\right)}$$

$$=\tan^{-1}\left(\frac{0}{\omega_0}\right)-\tan^{-1}\left(\frac{\omega}{\omega_0}\right)$$

$$=-\tan^{-1}\left(\frac{\omega}{\omega_0}\right)$$

Thus, the phase angle of the transfer function is, $\phi(\omega)=-\tan^{-1}\left(\frac{\omega}{\omega_0}\right)$.

Step 4 of 6

The gain G is,

$$G=20\log_{10}|T(j\omega)|$$

The attenuation is,

$$A=-20\log_{10}|T(j\omega)|.$$

Step 5 of 6

Calculate $|T|$ for $\omega=0$.

$$|T|=\frac{\omega_0}{\sqrt{0^2+\omega^2_0}}$$

$$=\frac{\omega_0}{\omega_0}$$

$$=1$$

Calculate ϕ for $\omega=0$.

$$\phi=-\tan^{-1}\left(\frac{0}{\omega_0}\right)$$

$$=0$$

Calculate G for $\omega=0$.

$$G=20\log_{10}|T|$$

$$=20\log_{10}|1|$$

$$=0$$

Calculate A for $\omega=0$.

$$A=-20\log_{10}|T|$$

$$=-20\log_{10}|1|$$

$$=0$$

Similarly, calculate the values of $|T|,\phi,G$ and A for $\omega=0.5\omega_0,\omega_0,2\omega_0,5\omega_0,10\omega_0,$ and $100\omega_0$.

Step 6 of 6

For different values of ω, the values of $|T|,\phi,G$ and A are shown in Table 1:

Table 1

ω	$\lvert T(j\omega)\rvert[V/V]$	$G(\mathrm{dB})$	$A[\mathrm{dB}]$	ϕ_o
0	1	0	0	0
$0.5\omega_o$	0.8944	-0.97	0.97	-26.57
ω_o	0.7071	-3.01	3.01	-45.0
$2\omega_o$	0.4472	-6.99	6.99	-63.43
$5\omega_o$	0.1961	-14.1	14.1	-78.69
$10\omega_o$	0.0995	-20	20	-84.29
$100\omega_o$	0.010	-40	40	-89.43

Step 1 of 4

The maximum amplitude in dB is 0.

Calculate the value of p.

$$p(\text{dB}) = 0$$
$$20\log p = 0$$
$$p = 10^0$$
$$= 1$$

The passband transmission is to be constant within $\pm 5\%$.

Therefore, the maximum amplitude is to be constant for $-5\% \text{ to } +5\% = 10\%$

Step 2 of 4

Calculate the value of q.

$$q = p - 10\%$$
$$= 1 - \left(\frac{10}{100}\right)$$
$$= 1 - 0.1$$
$$= 0.9$$

Calculate the value of r.

The stop band transmission is to be no greater than 1% of the pass band transmission.

$$r = 1\%$$
$$= \frac{1}{100}$$
$$= 0.01$$

Step 3 of 4

Calculate the value of A_{max}.

$$A_{max} = 20\log p - 20\log q$$

Substitute 1 for p and 0.9 for q.

$$A_{max} = 20\log(1) - 20\log(0.9)$$
$$= 0 - (-0.915)$$
$$= 0.915 \text{ dB}$$

Therefore, the value of A_{max} is $\boxed{0.915 \text{ dB}}$.

Step 4 of 4

Calculate the value of A_{min}.

$$A_{min} = 20\log p - 20\log r$$

Substitute 1 for p and 0.01 for r.

$$A_{min} = 20\log(1) - 20\log(0.01)$$
$$= 0 - (-40)$$
$$= 40 \text{ dB}$$

Therefore, the value of A_{min} is $\boxed{40 \text{ dB}}$.

17.002P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

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Step 1 of 2

17.003E

Write the general expression of transfer function of second-order filter.

$$T(s)=a_M\frac{(s-z_1)(s-z_2)}{(s-p_1)(s-p_2)}$$

Determine the expression of denominator using poles.

$$\begin{aligned}(s-p_1)(s-p_2)&=\left(s+\frac{1}{2}+j\frac{\sqrt{3}}{2}\right)\left(s+\frac{1}{2}-j\frac{\sqrt{3}}{2}\right)\\&=\left(s+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2\\&=s^2+s+\frac{1}{4}+\frac{3}{4}\\&=s^2+s+1\end{aligned}$$

Determine the expression of numerator using zeros.

$$\begin{aligned}(s-z_1)(s-z_2)&=(s+j2)(s-j2)\\&=s^2+4\end{aligned}$$

Step 2 of 2

Write the expression of transfer function.

$$T(s)=a_M\frac{s^2+4}{s^2+s+1}$$

The transmission is unity at dc. And the dc gain is obtained at $s=0$.

$$T(0)=1$$

$$a_M\left(\frac{0+4}{0+0+1}\right)=1$$

$$4a_M=1$$

$$a_M=\frac{1}{4}$$

Hence,

$$T(s)=\left(\frac{1}{4}\right)\frac{s^2+4}{s^2+s+1}$$

Therefore, the transfer function of the second-order filter is $\boxed{\left(\frac{1}{4}\right)\frac{s^2+4}{s^2+s+1}}$.

Step 1 of 12

Consider the transfer function $T(s)$.

$$T(s)=\frac{1}{(s+1)(s^2+s+1)}$$

Substitute $j\omega$ for s.

$$T(j\omega)=\frac{1}{(j\omega+1)((j\omega)^2+j\omega+1)}$$
$$=\frac{1}{(j\omega+1)(-\omega^2+j\omega+1)}$$
$$=\frac{1}{(j\omega)^3+(j\omega)^2+j\omega+(j\omega)^2+j\omega+1}$$
$$=\frac{1}{-j\omega^2-\omega^2+j\omega-\omega^2+j\omega+1}$$
$$=\frac{1}{-j\omega^3-2\omega^2+2j\omega+1}$$

$$T(j\omega)=\frac{1}{(1-2\omega^2)+j(-\omega^3+2\omega)} \quad \text{..... (1)}$$

Step 2 of 12

Consider equation (1). Assume that the magnitude of transfer function $T(j\omega)$ be $|T|$ and calculate the same.

$$|T|=\frac{1}{\sqrt{(1-2\omega^2)^2+(-\omega^3+2\omega)^2}}$$
$$=\frac{1}{\sqrt{(1+4\omega^4-4\omega^2)+(\omega^4+4\omega^2-4\omega^4)}}$$

$$|T|=\frac{1}{\sqrt{1+\omega^6}} \quad \text{..... (2)}$$

Hence, the magnitude $|T|$ is $\boxed{\frac{1}{\sqrt{1+\omega^6}}}$.

Step 3 of 12

Recall equation (1).

$$T(j\omega)=\frac{1}{(1-2\omega^2)+j(-\omega^3+2\omega)}$$

Assume that the phase response of $T(j\omega)$ be $\phi(\omega)$ and calculate the same.

$$\phi(\omega)=\tan^{-1}\left(\frac{0}{1}\right)-\tan^{-1}\left(\frac{-\omega^3+2\omega}{1-2\omega^2}\right)$$

$$\phi(\omega)=-\tan^{-1}\left(\frac{\omega(2-\omega^2)}{1-2\omega^2}\right) \quad \text{..... (3)}$$

Therefore, the expression for the phase response $\phi(\omega)$ is $\boxed{-\tan^{-1}\left(\frac{\omega(2-\omega^2)}{1-2\omega^2}\right)}$.

Step 4 of 12

Recall equabon (2).

$$|T|=\frac{1}{\sqrt{1+\omega^6}}$$

Substitute 0.1 for ω.

$$|T|=\frac{1}{\sqrt{1+(0.1)^6}}$$
$$=\frac{1}{\sqrt{1+10^{-6}}}$$

$$|T|=0.99 \quad \text{..... (4)}$$

Therefore, the value of $|T|$ for ω = 0.1 rad/s is $\boxed{0.99}$.

Step 5 of 12

Recall equation (2).

$$|T|=\frac{1}{\sqrt{1+\omega^6}}$$

Substitute 1 for ω.

$$|T|=\frac{1}{\sqrt{1+(1)^6}}$$
$$=\frac{1}{\sqrt{2}}$$

$$|T|=0.70 \quad \text{..... (5)}$$

Therefore, the value of $|T|$ for ω = 1 rad/s is $\boxed{0.7}$.

Step 6 of 12

Recall equabon (2).

$$|T|=\frac{1}{\sqrt{1+\omega^6}}$$

Substitute 10 for ω.

$$|T|=\frac{1}{\sqrt{1+10^6}}$$

$$|T|=999\times10^{-6} \quad \text{..... (6)}$$

Therefore, the value of $|T|$ for ω = 10 rad/s is $\boxed{999\times10^{-6}}$.

Step 7 of 12

Recall equabon (3).

$$\phi(\omega)=-\tan^{-1}\left(\frac{\omega(2-\omega^2)}{1-2\omega^2}\right)$$

Substitute 0.1 for ω.

$$\phi(\omega)=-\tan^{-1}\left(\frac{0.1\times(2-(0.1)^2)}{1-2(0.1)^2}\right)$$
$$=-\tan^{-1}\left(\frac{0.199}{0.98}\right)$$
$$=-\tan^{-1}(0.2)$$

$$\phi(\omega)=-11.48°$$

$$\phi(\omega)=-0.19 \text{ rad} \quad \text{..... (7)}$$

Therefore, the value of $\phi(\omega)$ for ω = 0.1 rad/s is $\boxed{-0.19 \text{ rad}}$.

Step 8 of 12

Recall equation (3).

$$\phi(\omega)=-\tan^{-1}\left(\frac{\omega(2-\omega^2)}{1-2\omega^2}\right)$$

Substitute 1 for ω.

$$\phi(\omega)=-\tan^{-1}\left(\frac{1\times(2-(1)^2)}{1-2(1)^2}\right)$$
$$=-\tan^{-1}\left(\frac{2-1}{1-2}\right)$$
$$=-\tan^{-1}(-1)$$
$$=-(135°)$$

$$\phi(\omega)=-2.35 \text{ rad} \quad \text{..... (8)}$$

Therefore, the value of $\phi(\omega)$ for ω = 1 rad/s is $\boxed{-2.35 \text{ rad}}$.

Step 9 of 12

Recall equabon (3).

$$\phi(\omega)=-\tan^{-1}\left(\frac{\omega(2-\omega^2)}{1-2\omega^2}\right)$$

Substitute 10 for ω.

$$\phi(\omega)=-\tan^{-1}\left(\frac{10\times(2-(10)^2)}{1-2(10)^2}\right)$$
$$=-\tan^{-1}\left(\frac{10\times(2-100)}{1-200}\right)$$
$$=-\tan^{-1}\left(\frac{-980}{-199}\right)$$
$$=-\tan^{-1}(4.92)$$

$$\phi(\omega)=-258.52°$$

$$\phi(\omega)=-4.51 \text{ rad} \quad \text{..... (9)}$$

Therefore, the value of $\phi(\omega)$ for ω = 10 rad/s is $\boxed{-4.51 \text{ rad}}$.

Step 10 of 12

Assume that the output of the filter be $v_o(t)$ and the input of the filter be $v_i(t)$.

(a)

Consider the following input signal:

$$v_i(t)=10\sin(0.1t) \text{ (volts)}$$

The output $v_o(t)$ is,

$$v_o(t)=|T|\times10\sin(0.1t+\phi(\omega)) \quad \text{..... (10)}$$

Here, the value of ω is 0.1 rad/s.

Recall equations (4) and (7).

$$|T|=0.99$$

$$\phi(\omega)=-0.19 \text{ rad}$$

Substitute 0.99 for $|T|$ and ◊◊◊◊ for $\phi(\omega)$ in equation (10).

$$v_o(t)=(0.99)\times10\sin(0.1t-0.19)$$
$$=9.9\sin(0.1t-0.19) \text{ V}$$

Therefore, the output of the filter $v_o(t)$ is $\boxed{9.9\sin(0.1t-0.19) \text{ V}}$.

Step 11 of 12

(b)
Consider the following input signal:

$$v_i(t)=10\sin(t) \text{ (volts)}$$

The output $v_o(t)$ is,

$$v_o(t)=|T|\times10\sin(t+\phi(\omega)) \quad \text{..... (11)}$$

Here, the value of ω is 1 rad/s.

Recall equations (5) and (8).

$$|T|=0.70$$

$$\phi(\omega)=-2.35 \text{ rad}$$

Substitute 0.7 for $|T|$ and ◊◊◊◊ for $\phi(\omega)$ in equation (11).

$$v_o(t)=0.7\times10\sin(t-2.35)$$
$$=7\sin(t-2.35) \text{ V}$$

Therefore, the output of the filter $v_o(t)$ is $\boxed{7\sin(t-2.35) \text{ V}}$.

Step 12 of 12

(c)

Consider the following input signal:

$$v_i(t)=10\sin(10t) \text{ (volts)}$$

The output $v_o(t)$ is,

$$v_o(t)=|T|\times10\sin(10t+\phi(\omega)) \quad \text{..... (12)}$$

Here, the value of ω is 10 rad/s.

Recall equations (6) and (9).

$$|T|=999\times10^{-6}$$

$$\phi(\omega)=-4.51 \text{ rad}$$

Substitute (999×10^{-6}) for $|T|$ and ◊◊◊◊ for $\phi(\omega)$ in equation (12).

$$v_o(t)=(999\times10^{-6})\times10\sin(10t-4.51)$$
$$=9.99\sin(10t-4.51) \text{ mV}$$

Therefore, the output of the filter $v_o(t)$ is $\boxed{9.99\sin(10t-4.51) \text{ mV}}$.

Step 1 of 2

17.004E

Write the transfer function of the fourth-order filter with zero at $\omega = 0$.

$$T(s) = a_M \frac{s(s-z_1)(s-z_2)}{(s-p_1)(s-p_2)(s-p_3)(s-p_4)}$$

Determine the expressions in denominator using poles.

$$\begin{aligned}(s-p_1)(s-p_2) &= (s+0.1+j0.8)(s+0.1-j0.8)\\ &= (s+0.1)^2+(0.8)^2\\ &= s^2+0.2s+0.01+0.64\\ &= s^2+0.2s+0.65\end{aligned}$$

$$\begin{aligned}(s-p_3)(s-p_4) &= (s+0.1+j1.2)(s+0.1-j1.2)\\ &= (s+0.1)^2+(1.2)^2\\ &= s^2+0.2s+0.01+1.44\\ &= s^2+0.2s+1.45\end{aligned}$$

Determine the expression of numerator using zeros.

$$\begin{aligned}s(s-z_1)(s-z_2) &= s(s+j2)(s-j2)\\ &= s(s^2+4)\end{aligned}$$

Step 2 of 2

Determine the value of M.

$$\begin{aligned}M &= N-1\\ &= 4-1\\ &= 3\end{aligned}$$

Write the expression of transfer function.

$$T(s) = a_3 \frac{s(s^2+4)}{(s^2+0.2s+0.65)(s^2+0.2s+1.45)}$$

Therefore, the expression of transfer function of fourth-order filter is

$$\boxed{a_3 \frac{s(s^2+4)}{(s^2+0.2s+0.65)(s^2+0.2s+1.45)}}.$$

Step 1 of 4

17.004P

Refer to figure 17.3 given in the textbook.

The maximum variation allowed in passband is,

$$A_{max} = 0.2 \text{ dB}$$

The minimum variation in stopband is,

$$A_{min} = 60 \text{ dB}$$

Step 2 of 4

The magnitude $|T|$ at $\omega = 0$ is zero as shown in figure 17.3 given in the textbook.

Therefore, the value of $|T|$ at $\omega = 0$ is $\boxed{0}$.

Step 3 of 4

The magnitude $|T|$ at $\omega = \omega_p$ is,

$$|T| = 0 - A_{max}$$

Substitute 0.2 dB for A_{max} .

$$\begin{aligned} |T| &= 0 - 0.2 \text{ dB} \\ &= -0.2 \text{ dB} \end{aligned}$$

Therefore, the value of $|T|$ at $\omega = \omega_p$ is $\boxed{-0.2 \text{ dB}}$.

Step 4 of 4

The magnitude $|T|$ at $\omega = \omega_s$ is,

$$|T| = 0 - A_{min}$$

Substitute 60 dB for A_{min} .

$$\begin{aligned} |T| &= 0 - 60 \text{ dB} \\ &= -60 \text{ dB} \end{aligned}$$

Therefore, the value of $|T|$ at $\omega = \omega_s$ is $\boxed{-60 \text{ dB}}$.

Step 1 of 6

Draw the poles representation.

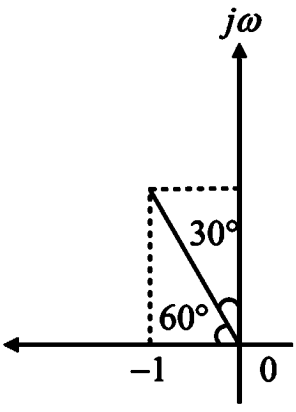


Figure 1

Step 2 of 6

The complex poles are at $30°$ angles from the $j\omega$ axis.

Calculate the poles of the transfer function.

$$p_1 = -(\cos 60° + j \sin 60°)$$
$$= -\left(\frac{1}{2} + j\frac{\sqrt{3}}{2}\right)$$
$$= -\frac{1}{2} - j\frac{\sqrt{3}}{2}$$

$$p_2 = -(\cos 60° - j \sin 60°)$$
$$= -\left(\frac{1}{2} - j\frac{\sqrt{3}}{2}\right)$$
$$= -\frac{1}{2} + j\frac{\sqrt{3}}{2}$$

$$p_3 = -1$$

Step 3 of 6

Calculate the transfer function, $T(s)$.

$$T(s) = \frac{K}{(s-p_1)(s-p_2)(s-p_3)}$$
$$= \frac{1}{\left[s-(-1)\right]\left[s-\left(-\frac{1}{2}-j\frac{\sqrt{3}}{2}\right)\right]\left[s-\left(-\frac{1}{2}+j\frac{\sqrt{3}}{2}\right)\right]}$$
$$= \frac{1}{(s+1)\left[\left(s+\frac{1}{2}\right)+\left(j\frac{\sqrt{3}}{2}\right)\right]\left[\left(s+\frac{1}{2}\right)-\left(j\frac{\sqrt{3}}{2}\right)\right]}$$
$$= \frac{1}{(s+1)\left(\left(s+\frac{1}{2}\right)^2-\left(j\frac{\sqrt{3}}{2}\right)^2\right)}$$

$$= \frac{1}{(s+1)\left(s^2+s+\frac{1}{4}+\frac{3}{4}\right)}$$
$$= \frac{1}{(s+1)(s^2+s+1)}$$

Therefore, the transfer function, $T(s)$ is $\boxed{\frac{1}{(s+1)(s^2+s+1)}}$.

Step 4 of 6

Consider the expression for the transfer function.

$$T(s) = \frac{1}{(s+1)(s^2+s+1)}$$

Substitute $j\omega$ for s.

$$T(j\omega) = \frac{1}{(j\omega+1)\left[(j\omega)^2+j\omega+1\right]}$$
$$= \frac{1}{(1+j\omega)(1-\omega^2+j\omega)}$$

Calculate the magnitude $|T(j\omega)|$.

$$|T(j\omega)| = \frac{1}{\left(\sqrt{1+\omega^2}\right)\left(\sqrt{(1-\omega^2)^2+(\omega)^2}\right)}$$
$$= \frac{1}{\sqrt{(1+\omega^2)\left[(\omega^4-2\omega^2+1)+(\omega)^2\right]}}$$
$$= \frac{1}{\sqrt{\omega^4-\omega^2+1+\omega^6-\omega^4+\omega^2}}$$
$$= \frac{1}{\sqrt{1+\omega^6}}$$

Therefore, the magnitude $|T(j\omega)| = \frac{1}{\sqrt{1+\omega^6}}$ is proved.

Step 5 of 6

Calculate the value of 3 dB frequency, ω_{3dB}.

$$-20\log|T(j\omega)| = 3$$
$$\log\frac{1}{\sqrt{1+\omega^6}} = \frac{-3}{20}$$
$$\frac{1}{\sqrt{1+\omega^6}} = 10^{\frac{-3}{20}}$$
$$\frac{1}{\sqrt{1+\omega^6}} = 0.707$$

$$\sqrt{1+\omega^6} = \sqrt{2}$$
$$1+\omega^6 = 2$$
$$\omega^6 = 1$$
$$\omega = 1 \text{ rad/s}$$

Therefore, the value of 3 dB frequency, ω_{3dB} is $\boxed{1 \text{ rad/s}}$.

Step 6 of 6

Calculate the attenuation at $\omega = 3 \text{ rad/s}$.

$$A(\omega) = -20\log|T(j\omega)|$$
$$= -20\log\frac{1}{\sqrt{1+\omega^6}}$$

Substitute 3 rad/s for ω in $A(\omega)$.

$$A(\omega) = -20\log\frac{1}{\sqrt{1+(3)^6}}$$
$$= -20\log(0.037)$$
$$= -20(-1.43)$$
$$= 28.63 \text{ dB}$$

Therefore, the attenuation, $A(\omega)$ at $\omega = 3 \text{ rad/s}$ is $\boxed{28.63 \text{ dB}}$.

Step 1 of 5

17.005P

Consider the transmission characteristics of a low-pass filter as shown in figure 1.

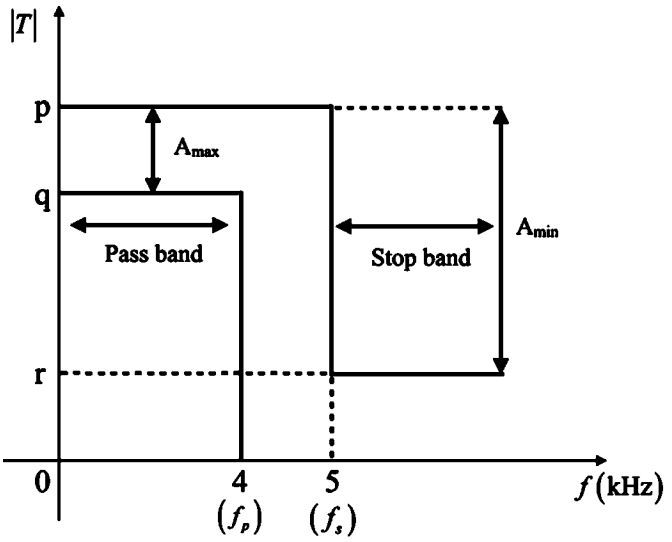


Figure 1

Step 2 of 5

Here, the magnitude $|T|$ is shown in linear scale, but the attenuation constants are in decibels (dB).

The range of pass-band is 0 to 4 kHz and that of stop-band is 5 kHz to ∞.

The transmission variation in the pass-band is at most 5%.

$$\frac{p}{q} = 1.05 \quad \text{..... (1)}$$

The transmission variation in the stop-band should not exceed 0.05% of the maximum pass-band transmission.

$$r = 0.05\% \times (p)$$
$$= \frac{0.05}{100} \times p$$

Rewrite as,

$$\frac{p}{r} = \frac{100}{0.05}$$

$$\frac{p}{r} = 2000 \quad \text{..... (2)}$$

Step 3 of 5

The maximum attenuation allowed in the pass-band is,

$$A_{max} = 20\log_{10}(p) - 20\log_{10}(q)$$
$$= 20\log_{10}\left(\frac{p}{q}\right)$$

Consider equation (1) and substitute 1.05 for $\frac{p}{q}$.

$$A_{max} = 20\log_{10}(1.05)$$
$$= 0.42 \text{ dB}$$

Therefore, the value of A_{max} is $\boxed{0.42 \text{ dB}}$.

Step 4 of 5

The minimum attenuation in the stop-band is,

$$A_{min} = 20\log_{10}(p) - 20\log_{10}(r)$$
$$= 20\log_{10}\left(\frac{p}{r}\right)$$

Consider equation (2) and substitute 2000 for $\left(\frac{p}{r}\right)$.

$$A_{min} = 20\log_{10}(2000)$$
$$= 66.02 \text{ dB}$$

Therefore, the value of A_{min} is $\boxed{66.02 \text{ dB}}$.

Step 5 of 5

The expression for the selectivity factor is,

$$\text{Selectivity} = \frac{\omega_s}{\omega_p}$$
$$= \frac{2\pi f_s}{2\pi f_p}$$
$$= \frac{f_s}{f_p}$$

Consider figure (1) and substitute 5 for f_s and 4 for f_p.

$$\text{Selectivity} = \frac{5}{4}$$
$$= 1.25$$

Therefore, the selectivity factor for this low-pass filter is $\boxed{1.25}$.

Step 1 of 5

Calculate the value of the parameter, ε which determines the maximum variation in the passband transmission.

$$\varepsilon = \sqrt{10^{\frac{A_{max}}{10}} - 1}$$

Substitute 1 for A_{max}.

$$\begin{aligned}\varepsilon &= \sqrt{10^{\frac{1}{10}} - 1} \\ &= \sqrt{1.2589 - 1} \\ &= \sqrt{0.2589} \\ &= 0.5089\end{aligned}$$

Step 2 of 5

Consider the expression for A_{min} to calculate the value of N.

$$A_{min} = 10\log\left[1 + \varepsilon^2\left(\frac{\omega_s}{\omega_p}\right)^{2N}\right]$$

Substitute 1.5 for $\frac{\omega_s}{\omega_p}$, 0.5089 for ε.

$$30 = 10\log\left[1 + 0.5089^2(1.5)^{2N}\right]$$

$$3 = \log\left[1 + 0.259(1.5)^{2N}\right]$$

$$1 + 0.259(1.5)^{2N} = 10^3$$

$$0.259(1.5)^{2N} = 1000 - 1$$

$$0.259(1.5)^{2N} = 999$$

$$(1.5)^{2N} = \frac{999}{0.259}$$

$$(1.5)^{2N} = 3857.15$$

Take logarithms on both sides.

$$2N\ln(1.5) = \ln(3857.15)$$

$$2N \times 0.405 = 8.258$$

$$N = \frac{8.258}{2 \times 0.405}$$

$$N = 10.2$$

Thus, the order of the Butterworth filter is $\boxed{N = 11}$.

Step 3 of 5

Calculate the minimum attenuation realized by the Butterworth filter.

$$A_{min} = 10\log\left[1 + \varepsilon^2\left(\frac{\omega_s}{\omega_p}\right)^{2N}\right]$$

Substitute 1.5 for $\frac{\omega_s}{\omega_p}$, 0.5089 for ε and 11 for N.

$$\begin{aligned}A_{min} &= 10\log\left[1 + 0.5089^2(1.5)^{2\times11}\right] \\ &= 10\log\left[1 + 0.5089^2(1.5)^{2\times11}\right] \\ &= 10 \times 3.287 \\ &= 32.87\ \text{dB}\end{aligned}$$

Thus, the minimum attenuation of the Butterworth filter is $\boxed{32.87\ \text{dB}}$.

Step 4 of 5

Calculate the value of the parameter, ε for minimum attenuation.

$$A_{min} = 10\log\left[1 + \varepsilon^2\left(\frac{\omega_s}{\omega_p}\right)^{2N}\right]$$

Substitute 1.5 for $\frac{\omega_s}{\omega_p}$, $30\ \text{dB}$ for A_{min} and 11 for N.

$$30 = 10\log\left[1 + \varepsilon^2(1.5)^{2\times11}\right]$$

$$10^3 = 1 + \varepsilon^2(1.5)^{2\times11}$$

$$\varepsilon^2(1.5)^{2\times11} = 10^3 - 1$$

$$\varepsilon^2 = \frac{10^3 - 1}{(1.5)^{2\times11}}$$

$$\varepsilon^2 = 0.1335$$

$$\varepsilon = 0.365$$

Step 5 of 5

Calculate the reduced maximum attenuation.

$$A_{max} = 10\log\left(1 + \varepsilon^2\right)$$

Substitute 0.365 for ε.

$$\begin{aligned}A_{max} &= 10\log\left(1 + 0.365^2\right) \\ &= 10\log\left(1.1332\right) \\ &= 0.543\end{aligned}$$

Thus, the maximum attenuation for A_{min} to be exactly at 30 dB is $\boxed{0.543\ \text{dB}}$.

17.006P

Step 1 of 11

The pass-band edge for a low-pass filter is,

$f_p = 5\ \text{kHz}$

The selectivity factor for the low-pass filter is,

$\text{Selectivity} = 10$ ..... (1)

Step 2 of 11

The expression for the selectivity factor is,

$$\text{Selectivity} = \frac{\omega_s}{\omega_p} = \frac{2\pi f_s}{2\pi f_p} \qquad \text{Selectivity} = \frac{f_s}{f_p} \quad \text{..... (2)}$$

Step 3 of 11

Consider equations (1) and (2); equate them.

$$10 = \frac{f_s}{f_p}$$

Substitute 5 for f_p .

$$10 = \frac{f_s}{5}$$
$$f_s = 50\ \text{kHz}$$

Therefore, the stop-band edge for the low-pass filter f_s is 50 kHz.

Step 4 of 11

Consider the transfer function $T(s)$.

$$T(s) = \frac{2\pi \times 10^4}{s + 2\pi \times 10^4}$$

Substitute $j\omega$ for *s*.

$$T(j\omega) = \frac{2\pi \times 10^4}{j\omega + 2\pi \times 10^4}$$

The expression for the magnitude $|T(j\omega)|$ is,

$$|T(j\omega)| = \frac{2\pi \times 10^4}{\sqrt{\omega^2 + (2\pi \times 10^4)^2}}$$

Substitute $2\pi f$ for ω .

$$|T(j\omega)| = \frac{2\pi \times 10^4}{\sqrt{(2\pi f)^2 + (2\pi \times 10^4)^2}}$$
$$= \frac{2\pi \times 10^4}{\sqrt{(2\pi)^2 \left(f^2 + (10^4)^2\right)}}$$
$$= \frac{2\pi \times 10^4}{2\pi\sqrt{f^2 + 10^8}}$$

$$|T(j\omega)| = \frac{10^4}{\sqrt{f^2 + 10^8}} \quad \text{..... (3)}$$

Therefore, the magnitude $|T(j\omega)|$ is $\dfrac{10^4}{\sqrt{f^2 + 10^8}}$.

Step 5 of 11

Consider equation (3) and substitute f_p for *f*.

$$|T(j\omega_p)| = \frac{10^4}{\sqrt{(f_p)^2 + 10^8}}$$

Substitute 5×10^3 ◊for◊ f_p .

$$|T(j\omega_p)| = \frac{10^4}{\sqrt{(5 \times 10^3)^2 + 10^8}}$$
$$= \frac{10^4}{\sqrt{25 \times 10^6 + 10^8}}$$
$$= \frac{10^4}{10^3\sqrt{25 + 100}}$$
$$= \frac{10}{\sqrt{125}}$$

Therefore, the magnitude $|T(j\omega_p)|$ is $\dfrac{10}{\sqrt{125}}$.

Step 6 of 11

Convert the magnitude $|T(j\omega_p)|$ into decibels.

$$|T(j\omega_p)| = 20\log_{10}\left(\frac{10}{\sqrt{125}}\right)$$
$$= -0.97\ \text{dB}$$

Therefore, the magnitude $|T(j\omega_p)|$ is ◊◊◊◊◊ dB.

Step 7 of 11

Recall equation (3).

$$|T(j\omega)| = \frac{10^4}{\sqrt{f^2 + 10^8}}$$

Substitute f_s for *f*.

$$|T(j\omega_s)| = \frac{10^4}{\sqrt{f_s^2 + 10^8}}$$

Substitute 50×10^3 ◊for◊ f_s .

$$|T(j\omega_s)| = \frac{10^4}{\sqrt{(50 \times 10^3)^2 + 10^8}}$$
$$= \frac{10^4}{\sqrt{(5 \times 10^4)^2 + 10^8}}$$
$$= \frac{10^4}{\sqrt{10^8(25 + 1)}}$$
$$= \frac{10^4}{10^4\sqrt{26}}$$
$$= \frac{1}{\sqrt{26}}$$

Therefore, the magnitude $|T(j\omega_s)|$ is $\dfrac{1}{\sqrt{26}}$.

Step 8 of 11

Convert the magnitude $|T(j\omega_s)|$ into decibels.

$$|T(j\omega_s)| = 20\log_{10}\left(\frac{1}{\sqrt{26}}\right)$$
$$= -14.15\ \text{dB}$$

Therefore, the magnitude $|T(j\omega_s)|$ is ◊◊◊◊◊ dB.

Step 9 of 11

Sketch the transmission specifications for the low-pass filter as shown in figure 1.

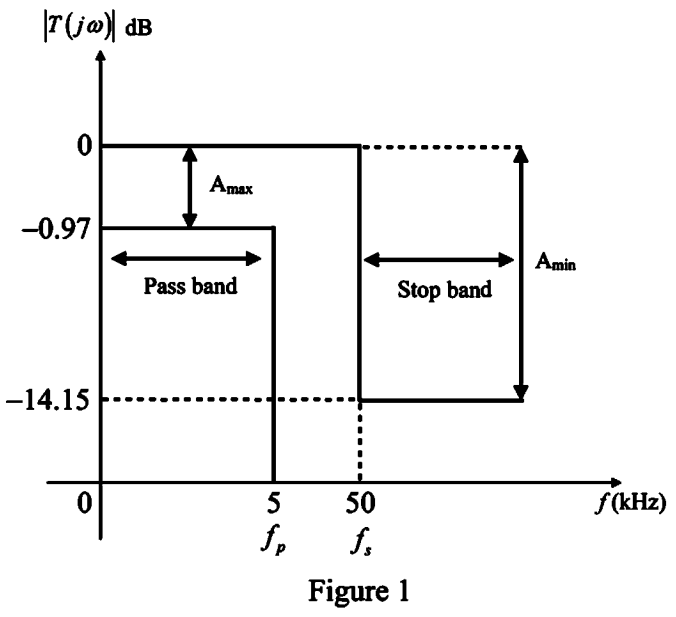


Figure 1

Step 10 of 11

Consider the figure 1 and calculate A_{max} .

$$A_{max} = 0 - (-0.97)$$
$$= 0.97\ \text{dB}$$

Therefore, the value of A_{max} is $\boxed{0.97\ \text{dB}}$.

Step 11 of 11

Consider the figure 1 and calculate A_{min} .

$$A_{min} = 0 - (-14.15)$$
$$= 14.15\ \text{dB}$$

Therefore, the value of A_{min} is $\boxed{14.15\ \text{dB}}$.

Step 1 of 2

17.007E

Calculate the value of resonant frequency, ω_0.

$$\omega_0 = \omega_p \left(\frac{1}{\varepsilon}\right)^{\frac{1}{N}}$$

Substitute $1\ \text{rad/s}$ for ω_p, 1 for ε and 3 for N.

$$\omega_0 = 1\left(\frac{1}{1}\right)^{\frac{1}{3}}$$
$$= 1\ \text{rad/s}$$

The natural modes or poles of the filter are,

$$p_{1,2} = \omega_0(\cos 60° \pm j \sin 60°)$$

Substitute $1\ \text{rad/s}$ for ω_0.

$$p_{1,2} = 1(\cos 60° \pm j \sin 60°)$$
$$= \frac{1}{2} \pm j\frac{\sqrt{3}}{2}$$

The other pole is at $\omega_p = 1\ \text{rad/s}$.

$$p_3 = -1$$

Thus, the natural modes of the 3rd order Butterworth filter are,

$$\boxed{\begin{array}{l} p_{1,2} = \frac{1}{2} \pm j\frac{\sqrt{3}}{2} \\ p_3 = -1 \end{array}}.$$

Step 2 of 2

Calculate the transfer function, $T(s)$.

$$T(s) = \frac{K\omega_0^N}{(s-p_1)(s-p_2)(s-p_3)} \qquad (K = 1 \text{ for unity gain})$$
$$= \frac{(1)(1)^3}{\left[s-(-1)\right]\left[s-\left(-\frac{1}{2}-j\frac{\sqrt{3}}{2}\right)\right]\left[s-\left(-\frac{1}{2}+j\frac{\sqrt{3}}{2}\right)\right]}$$
$$= \frac{1}{(s+1)\left[\left(s+\frac{1}{2}\right)+\left(j\frac{\sqrt{3}}{2}\right)\right]\left[\left(s+\frac{1}{2}\right)-\left(j\frac{\sqrt{3}}{2}\right)\right]}$$
$$= \frac{1}{(s+1)\left[\left(s+\frac{1}{2}\right)^2-\left(j\frac{\sqrt{3}}{2}\right)^2\right]}$$
$$= \frac{1}{(s+1)\left(s^2+s+\frac{1}{4}+\frac{3}{4}\right)}$$
$$= \frac{1}{(s+1)\left(s^2+s+1\right)}$$

Therefore, the transfer function of the 3rd order Butterworth filter, $T(s)$ is,

$$\boxed{\frac{1}{(s+1)\left(s^2+s+1\right)}}.$$

Step 1 of 6

17.007P

Consider the transmission characteristics of a low-pass filter shown in Figure 1.

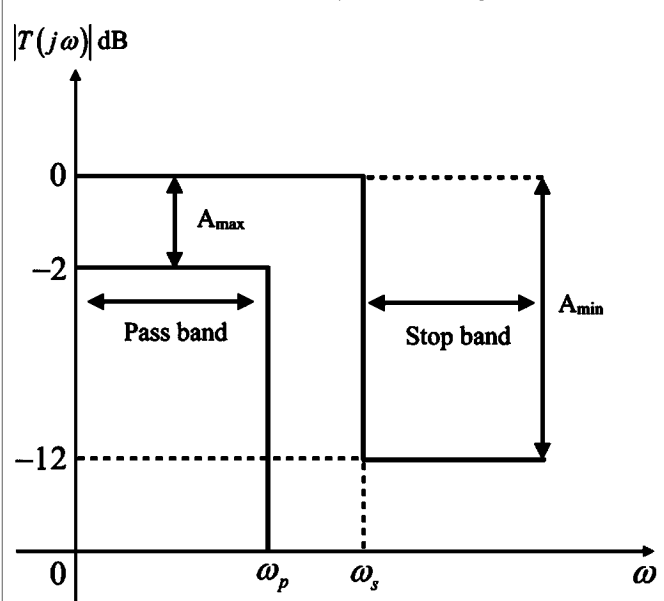


Figure 1

Step 2 of 6

The maximum attenuation in pass-band is,

$A_{max} = 1$ dB

The minimum attenuation in stop-band is,

$A_{min} = 12$ dB

The time constant τ of a RC low pass filter is 1 sec and the DC transmission *k* is unity.

The expression for the transfer function of a RC low-pass filter is,

$$T(s) = \frac{k}{1+s\tau}$$

Substitute 1 for τ and 1 for *k*.

$$T(s) = \frac{1}{1+s(1)}$$
$$= \frac{1}{1+s}$$

Substitute $j\omega$ for *s*.

$$T(j\omega) = \frac{1}{1+j\omega}$$

Calculate the magnitude $|T(j\omega)|$.

$$|T(j\omega)| = \frac{1}{\sqrt{1+\omega^2}} \quad \text{..... (1)}$$

Step 3 of 6

The expression for the gain function $G(\omega)$ is,

$$G(\omega) = 20\log|T(j\omega)| \text{ dB}$$

The expression for the magnitude $|T(j\omega)|$ is,

$$\log|T(j\omega)| = \frac{G(\omega)}{20}$$

$$|T(j\omega)| = 10^{\frac{G(\omega)}{20}} \quad \text{..... (2)}$$

Consider equations (1) and (2); equate them.

$$\frac{1}{\sqrt{1+\omega^2}} = 10^{\frac{G(\omega)}{20}}$$
$$\frac{1}{1+\omega^2} = 10^{\frac{G(\omega)}{10}}$$
$$1+\omega^2 = 10^{\frac{-G(\omega)}{10}}$$
$$\omega = \sqrt{10^{\frac{-G(\omega)}{10}} - 1} \quad \text{..... (3)}$$

Step 4 of 6

Consider equation (3). Substitute ω_p for ω.

$$\omega_p = \sqrt{10^{\frac{-G(\omega_p)}{10}} - 1}$$

Refer to Figure 1 and substitute ◊◊◊for $G(\omega_p)$.

$$\omega_p = \sqrt{10^{\frac{-(-2)}{10}} - 1}$$
$$= \sqrt{10^{\frac{1}{5}} - 1}$$
$$= 0.765 \text{ rad/s}$$

Therefore, the value of ω_p is **0.765 rad/s**.

Step 5 of 6

Recall equation (3).

$$\omega = \sqrt{10^{\frac{-G(\omega)}{10}} - 1}$$

Substitute ω_s for ω.

$$\omega_s = \sqrt{10^{\frac{-G(\omega_s)}{10}} - 1}$$

Refer to figure 1 and substitute ◊◊◊for $G(\omega_s)$.

$$\omega_s = \sqrt{10^{\frac{-(-12)}{10}} - 1}$$
$$= \sqrt{10^{\frac{12}{10}} - 1}$$
$$= 3.853 \text{ rad/s}$$

Therefore, the value of ω_s is **3.853 rad/s**.

Step 6 of 6

The expression for the selectivity factor is,

$$\text{Selectivity} = \frac{\omega_s}{\omega_p}$$

Substitute 3.853 for ω_s and 0.765 for ω_p.

$$\text{Selectivity} = \frac{3.853}{0.765}$$
$$= 5.04$$

Therefore, the selectivity factor is **5.04**.

Step 1 of 8

17.008E

Consider the expression for the magnitude of the transfer function of an Nth order Chebyshev filter with a passband edge (ripple bandwidth) ω_p.

$$|T(j\omega)| = \frac{1}{\sqrt{1+\varepsilon^2 \cos^2\left[N\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}}$$

The peaks are reached when $\cos^2\left[N\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]$ term is zero.

$$\cos^2\left[N\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 0$$

Substitute 5 for N.

$$\cos^2\left[5\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 0$$

$\cos^2\theta = 0$ when $\theta = (2k+1)\frac{\pi}{2}$ **where,** $k = 0,1,2,\cdots$.

$$5\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = (2k+1)\frac{\pi}{2} \text{ where, } k = 0,1,2,\cdots$$

Step 2 of 8

Calculate the frequency at peaks for $k = 0$.

$$5\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = (2(0)+1)\frac{\pi}{2}$$

$$5\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = \frac{\pi}{2}$$

$$\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = \frac{\pi}{2(5)}$$

$$\frac{\omega}{\omega_p} = \cos\left(\frac{\pi}{10}\right)$$

$$\frac{\omega}{\omega_p} = 0.95$$

$$\omega = 0.95\omega_p$$

Step 3 of 8

Calculate the frequency at peaks for $k = 1$.

$$5\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = (2(1)+1)\frac{\pi}{2}$$

$$5\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = \frac{3\pi}{2}$$

$$\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = \frac{3\pi}{2(5)}$$

$$\frac{\omega}{\omega_p} = \cos\left(\frac{3\pi}{10}\right)$$

$$\frac{\omega}{\omega_p} = 0.59$$

$$\omega = 0.59\omega_p$$

Step 4 of 8

Calculate the frequency at peaks for $k = 2$.

$$5\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = (2(2)+1)\frac{\pi}{2}$$

$$5\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = \frac{5\pi}{2}$$

$$\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = \frac{\pi}{2}$$

$$\frac{\omega}{\omega_p} = \cos\left(\frac{\pi}{2}\right)$$

$$\frac{\omega}{\omega_p} = 0$$

$$\omega = 0$$

Therefore, the peaks for a fifth order filter are at $\boxed{\omega = 0, 0.59\omega_p, \text{ and } 0.95\omega_p}$.

Step 5 of 8

The valleys are reached when $\cos^2\left[N\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]$ term is unity.

$$\cos^2\left[N\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 1$$

Substitute 5 for N.

$$\cos^2\left[5\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 1$$

$\cos^2\theta = 1$ when $\theta = k\pi$ **where,** $k = 0,1,2,\cdots$

$$5\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = k\pi \text{ where, } k = 0,1,2,\cdots$$

Step 6 of 8

Calculate the frequency at valleys for $k = 0$.

$$5\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = (0)\pi$$

$$5\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = 0$$

$$\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = 0$$

$$\frac{\omega}{\omega_p} = \cos(0)$$

$$\frac{\omega}{\omega_p} = 1$$

$$\omega = \omega_p$$

Step 7 of 8

Calculate the frequency at valleys for $k = 1$.

$$5\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = (1)\pi$$

$$5\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = \pi$$

$$\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = \frac{\pi}{5}$$

$$\frac{\omega}{\omega_p} = \cos\left(\frac{\pi}{5}\right)$$

$$\frac{\omega}{\omega_p} = 0.81$$

$$\omega = 0.81\omega_p$$

Step 8 of 8

Calculate the frequency at valleys for $k = 2$.

$$5\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = (2)\pi$$

$$5\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = 2\pi$$

$$\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = \frac{2\pi}{5}$$

$$\frac{\omega}{\omega_p} = \cos\left(\frac{2\pi}{5}\right)$$

$$\frac{\omega}{\omega_p} = 0.31$$

$$\omega = 0.31\omega_p$$

Therefore, the valleys for a fifth order filter are at $\boxed{\omega = 0.31\omega_p, 0.81\omega_p, \text{ and } \omega_p}$.

Step 1 of 5

The pass-band of a high pass filter is defined by $f \geq 3\ \text{kHz}$.

Therefore, the frequency f_p is,

$$f_p = 3\ \text{kHz}$$

The stop-band of the same is defined by $f \leq 2\ \text{kHz}$.

Therefore, the frequency f_s is,

$$f_s = 2\ \text{kHz}$$

Step 2 of 5

The expression for the radial frequency ω_p is,

$$\omega_p = 2\pi f_p$$

Substitute 3×10^3 for f_p .

$$\omega_p = 2\pi\left(3\times10^3\right)$$
$$= 6000\pi\ \text{rad/s}$$

Therefore, the value of ω_p is 6000π rad/s.

Step 3 of 5

The expression for the radial frequency ω_s is,

$$\omega_s = 2\pi f_s$$

Substitute 2×10^3 for f_s .

$$\omega_p = 2\pi\left(2\times10^3\right)$$
$$= 4000\pi\ \text{rad/s}$$

Therefore, the value of ω_s is 4000π rad/s.

Step 4 of 5

The maximum attenuation in pass-band is,

$$A_{max} = 0.4\ \text{dB}$$

The minimum attenuation in stop-band is,

$$A_{min} = 60\ \text{dB}$$

Step 5 of 5

Draw transmission specifications for the high pass filter as shown in Figure 1.

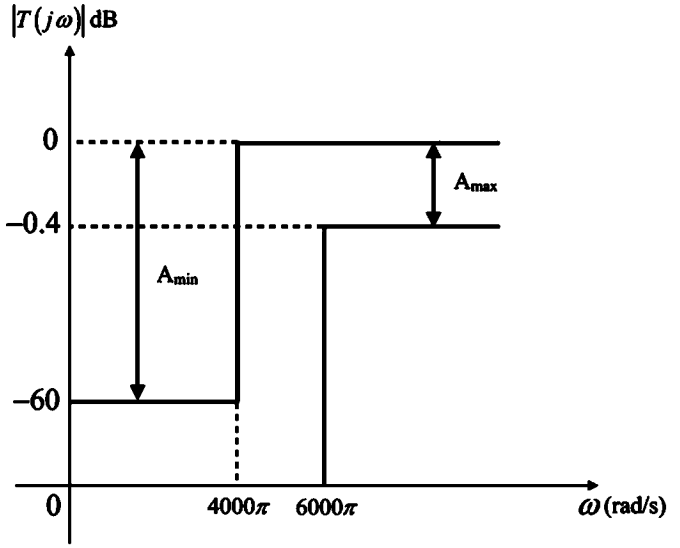


Figure 1

Step 1 of 2

Determine the attenuation with a pass band ripple of $0.5\ \text{dB}$.

$$\varepsilon = \sqrt{10^{\frac{A_{max}}{10}} - 1}$$

Substitute 0.5 for A_{max} to determine ε.

$$\begin{aligned}\varepsilon &= \sqrt{10^{\frac{0.5}{10}} - 1} \\ &= 0.349\end{aligned}$$

Calculate the value of attenuation provided by the Chebyshev filter.

$$A(\omega) = 10\log\left[1 + \varepsilon^2 \cosh^2\left(N\cosh^{-1}\left(\frac{\omega}{\omega_p}\right)\right)\right]$$

Substitute 0.349 for ε, 7 for N, $2\omega_p$ for ω.

$$\begin{aligned}A(\omega) &= 10\log\left[1 + 0.349^2 \cosh^2\left(7\cosh^{-1}\left(\frac{2\omega_p}{\omega_p}\right)\right)\right] \\ &= 10\log\left[1 + 0.349^2 \cosh^2\left(7\cosh^{-1}(2)\right)\right] \\ &= 10\log\left[1 + 0.349^2 \cosh^2(9.2187)\right] \\ &= 10\log(3096364.589) \\ &= 64.9\ \text{dB}\end{aligned}$$

Thus, the attenuation provided by the filter is $\boxed{64.9\ \text{dB}}$.

Step 2 of 2

Determine the attenuation with a pass band ripple of $1\ \text{dB}$.

$$\varepsilon = \sqrt{10^{\frac{A_{max}}{10}} - 1}$$

Substitute 1 for A_{max} to determine ε.

$$\begin{aligned}\varepsilon &= \sqrt{10^{\frac{1}{10}} - 1} \\ &= 0.509\end{aligned}$$

Calculate the value of attenuation provided by the Chebyshev filter.

$$A(\omega) = 10\log\left[1 + \varepsilon^2 \cosh^2\left(N\cosh^{-1}\left(\frac{\omega}{\omega_p}\right)\right)\right]$$

Substitute 0.509 for ε, 7 for N, $2\omega_p$ for ω.

$$\begin{aligned}A(\omega) &= 10\log\left[1 + 0.509^2 \cosh^2\left(7\cosh^{-1}\left(\frac{2\omega_p}{\omega_p}\right)\right)\right] \\ &= 10\log\left[1 + 0.509^2 \cosh^2\left(7\cosh^{-1}(2)\right)\right] \\ &= 10\log\left[1 + 0.509^2 \cosh^2(9.2187)\right] \\ &= 10\log(6586297.04) \\ &= 68.2\ \text{dB}\end{aligned}$$

Calculate the increase in stop band attenuation.

$$\begin{aligned}\text{Increase in } A(\omega) &= 68.2 - 64.9 \\ &= 3.3\ \text{dB}\end{aligned}$$

Thus, the increase in attenuation provided by the filter is $\boxed{3.3\ \text{dB}}$.

Step 1 of 2

17.009P

The frequency band over which a band-stop filter is required to pass signals is $0 \leq f \leq 10\ \text{kHz}$ and $20\ \text{kHz} \leq f \leq \infty$ with A_{max} of 0.5 dB.

The stop-band extends from f =12 kHz to f =18 kHz with minimum attenuation A_{min} of 50 dB.

Step 2 of 2

Draw transmission specifications for the band-stop filter as shown in figure 1.

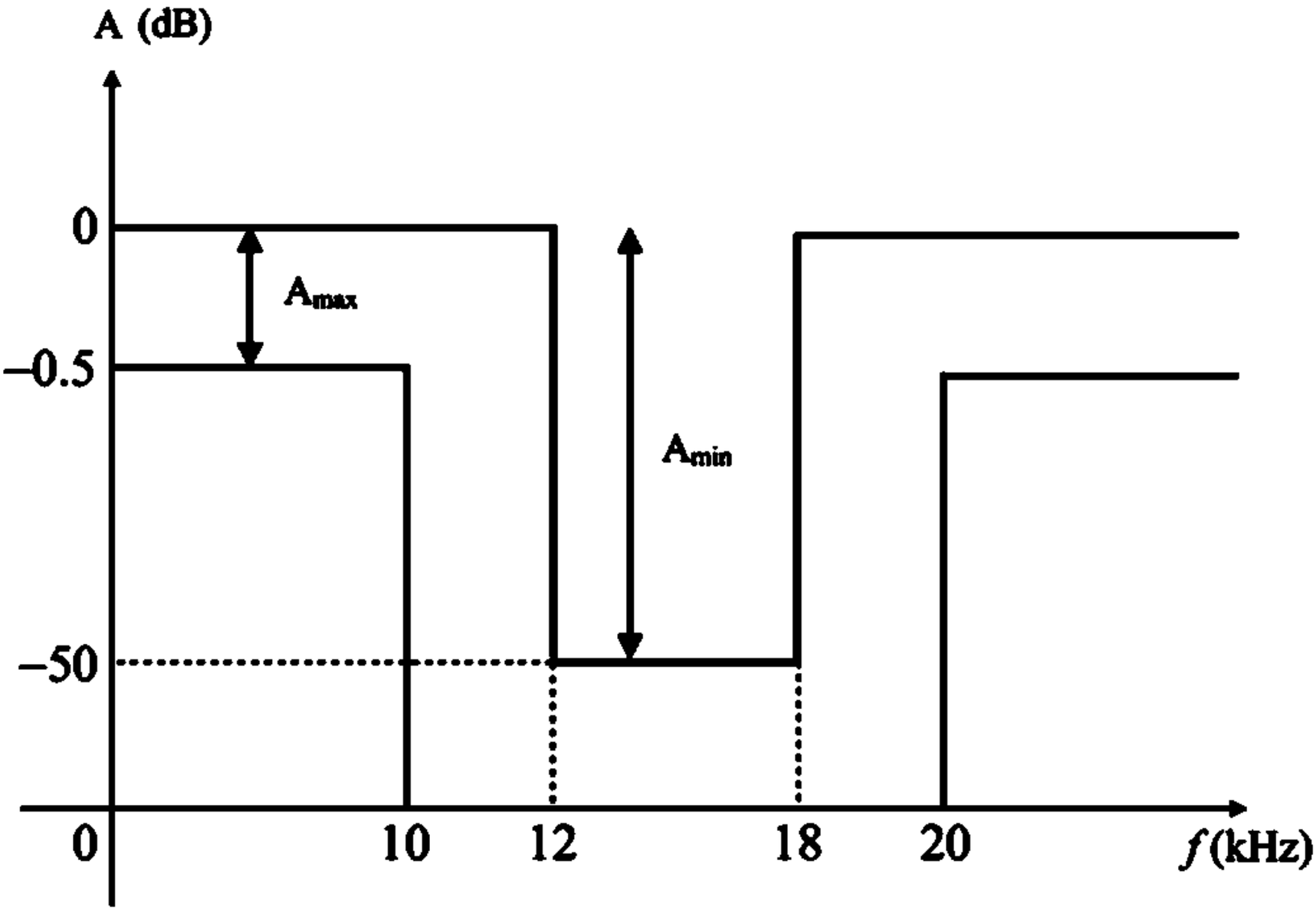


Figure 1

Step 1 of 6

Convert the maximum gain dB in Voltage to voltage ratio.

$$20\log A_{max} = 1$$

$$A_{max} = 10^{\frac{1}{20}}$$

$$A_{max} = 1.12\ \text{V}$$

Write the formula for passband ripple.

$$\epsilon = \sqrt{10^{\frac{A_{max}}{10}} - 1}$$

Substitute corresponding values.

$$\epsilon = \sqrt{10^{\frac{1}{10}} - 1}$$
$$= 0.5088$$

Step 2 of 6

Write the formula for attenuation of the Chebyshev filter.

$$A(\omega_s) = 10\log\left[1 + \epsilon^2 \cosh^2\left(N\cosh^{-1}\left(\frac{\omega_s}{\omega_p}\right)\right)\right]$$

Substitute different values of N, which is the lowest integer value of N that satisfies the condition $A(\omega_s) \geq A_{min}$

Substitute 8 for N in the equation.

$$A(\omega_s) = 10\log\left[1 + (0.5088)^2 \cosh^2\left(8\cosh^{-1}\left(\frac{1500}{1000}\right)\right)\right]$$
$$= 54.98\ \text{dB}$$

So, $A(\omega_s)$ is greater than A_{min}.

So, the order of the Chebyshev filter is $\boxed{8}$.

Step 3 of 6

Write the expression for transfer function at ω_s.

$$|T(j\omega_s)| = \frac{1}{\sqrt{1 + \epsilon^2 \cosh^2\left(N\cosh^{-1}\left(\frac{\omega_s}{\omega_p}\right)\right)}}$$

Substitute corresponding values.

$$|T(j\omega_s)| = \frac{1}{\sqrt{1 + (0.5088)^2 \cosh^2\left(8\cosh^{-1}\left(\frac{1.5}{1}\right)\right)}}$$
$$= 1.78\times10^{-3}$$

Convert into dB.

$$|T(j\omega_s)|_{dB} = 10(10\log(0.89))$$
$$= -54.98$$
$$= 54.98$$

So, the difference between the $|T(j\omega_s)|_{dB}$ and A_{min} is,

Excess stopband attenuation is,

$$|A_{min} - |T(j\omega_s)|_{dB}| = 50 - (54.98)$$
$$= 5\ \text{dB}$$

So, stop band attenuation is $\boxed{5\ \text{dB}}$.

Step 4 of 6

(b)

Write the formula for passband ripple.

$$\epsilon = \sqrt{10^{\frac{A_{max}}{10}} - 1}$$

Substitute corresponding values.

$$\epsilon = \sqrt{10^{\frac{1}{10}} - 1}$$
$$= 0.5088$$

Write the formula for attenuation of the Butterworth filter.

$$A(\omega_s) = 10\log\left[1 + \epsilon^2\left(\frac{\omega_s}{\omega_p}\right)^{2N}\right]$$

Substitute different values of N, which is the lowest integer value of N that satisfies the condition $A(\omega_s) \geq A_{min}$

Substitute 4 for N in the equation.

$$A(\omega_s) = 10\log\left[1 + (0.5088)^2\left(\frac{1500}{1000}\right)^{2(4)}\right]$$
$$= 8.827\ \text{dB}$$

So, $A(\omega_s)$ is not greater than A_{min}.

Step 5 of 6

Now for $N = 16$, the attenuation of the butter worth filter is,

$$A(\omega_s) = 10\log\left[1 + (0.5088)^2\left(\frac{1500}{1000}\right)^{2(16)}\right]$$
$$= 50.48\ \text{dB}$$

So, $A(\omega_s)$ is greater than $A_{min} = 50\ \text{dB}$.

So the order of butter worth filter is $\boxed{16}$.

Step 6 of 6

Write the expression for excess stop band attenuation.

$$|T(j\omega_s)| = \frac{1}{\sqrt{1 + \epsilon^2\left(\frac{\omega_s}{\omega_p}\right)^{2N}}}$$

Substitute corresponding values.

$$|T(j\omega_p)| = \frac{1}{\sqrt{1 + 0.5088^2\left(\frac{1500}{1000}\right)^{2(16)}}}$$
$$= 2.99\times10^{-3}$$

Convert into dB.

$$|T(j\omega_p)|_{dB} = 20\log\left(2.99\times10^{-3}\right)$$
$$= 50.5\ \text{dB}$$

So, the difference between the $|T(j\omega_s)|_{dB}$ and A_{min} is,

Excess stopband attenuation is,

$$|A_{min} - |T(j\omega_s)|_{dB}| = 50 - (50.5)$$
$$= 0.5\ \text{dB}$$

So, stop band attenuation is $\boxed{0.5\ \text{dB}}$.

Step 1 of 9

The pole-zero pattern of the filter is shown in figure 1.

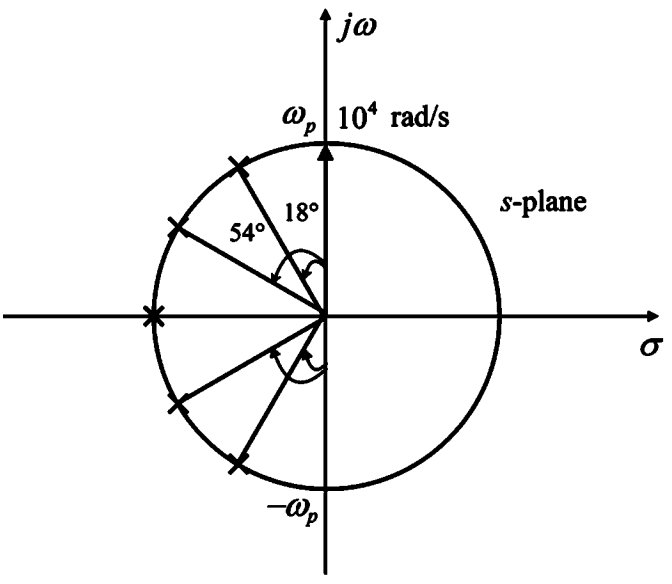


Figure 1

Step 2 of 9

As it is a fifth order filter, five poles exist as shown in figure 1.

The pair of complex conjugate poles at $18°$ from $j\omega$ axis is,

$$\begin{aligned} p_1 &= \omega_o\left(-\cos(90°-18°) \pm j\sin(90°-18°)\right) \\ &= \omega_o\left(-\cos(72°) \pm j\sin(72°)\right) \\ &= \omega_o\left(-0.309 \pm j0.951\right) \end{aligned}$$

Substitute 10^4 for ω_o.

$$p_1 = 10^4\left(-0.309 \pm j0.951\right)$$

$$p_1 = \left(-3090\right) \pm j\left(9510\right)$$

Step 3 of 9

The pair of complex conjugate poles at $52°$ from $j\omega$ axis is,

$$\begin{aligned} p_2 &= \omega_o\left(-\cos(90°-54°) \pm j\sin(90°-54°)\right) \\ &= \omega_o\left(-\cos(36°) \pm j\sin(36°)\right) \\ &= \omega_o\left(-0.809 \pm j0.588\right) \end{aligned}$$

Substitute 10^4 for ω_o.

$$p_2 = 10^4 \times \left(-0.809 \pm j0.588\right)$$

$$p_2 = \left(-8090\right) \pm j\left(5880\right)$$

Step 4 of 9

One pole on real axis is,

$$p_3 = -\omega_o$$

Substitute 10^4 for ω_o.

$$p_3 = -10^4$$

Step 5 of 9

The denominator term for the complex pair of poles at $18°$ from $j\omega$ axis is,

$$\begin{aligned} \left[s-\left(-3090+j9510\right)\right]\left[s-\left(-3090-j9510\right)\right] &= \left(s+3090-j9510\right)\left(s+3090+j9510\right) \\ &= \left(s+3090\right)^2+\left(9510\right)^2 \\ &= s^2+6180s+9548100+90440100 \\ &= s^2+6180s+99988200 \end{aligned}$$

Step 6 of 9

The denominator term for the complex pair of poles at $54°$ from $j\omega$ axis is,

$$\begin{aligned} \left[s-\left(-8090+j5880\right)\right]\left[s-\left(-8090-j5880\right)\right] &= \left(s+8090-j5880\right)\left(s+8090+j5880\right) \\ &= \left(s+8090\right)^2+\left(5880\right)^2 \\ &= s^2+16180s+65448100+34574400 \\ &= s^2+16180s+100022500 \end{aligned}$$

The denominator term for the pole on real axis is $\left(s+10^4\right)$.

Step 7 of 9

(a)

The transfer function of the fifth order filter with transmission zeros at $s=\infty$ and unity DC gain k is,

$$T(s) = \frac{a_o}{\left(s+10^4\right)\left(s^2+6180s+99988200\right)\left(s^2+16180s+100022500\right)} \quad \text{..... (1)}$$

For unity DC gain, the value of a_o is,

$$a_o = \left(\omega_o\right)^5$$

Substitute 10^4 for ω_o.

$$\begin{aligned} a_o &= \left(10^4\right)^5 \\ &= 10^{20} \end{aligned}$$

Substitute 10^{20} for a_o in equation (1).

$$T(s) = \frac{10^{20}}{\left(s+10^4\right)\left(s^2+6180s+99988200\right)\left(s^2+16180s+100022500\right)}$$

Therefore, the transfer function of the filter with all transmission zeros at $s=\infty$ and unity DC gain is

$$\boxed{\frac{10^{20}}{\left(s+10^4\right)\left(s^2+6180s+99988200\right)\left(s^2+16180s+100022500\right)}}.$$

The type of filter is **all-pole filter**.

Step 8 of 9

(b)

The number of zeros is,

$$Z = N-1$$

Here, N is the order of the filter. Substitute 4 for N.

$$Z = 4$$

Therefore, the number of zeros is 4.

Step 9 of 9

The numerator polynomial for all the transmission zeros at $s=0$ and unity high frequency gain a_4 is,

$$M = a_4 s^4$$

Substitute 1 for a_4.

$$M = s^4$$

Therefore, the numerator term is s^4.

Therefore, the transfer function of the filter with all transmission zeros at and unity high frequency gain is

$$\boxed{\frac{s^4}{\left(s+10^4\right)\left(s^2+6180s+99988200\right)\left(s^2+16180s+100022500\right)}}.$$

The type of filter is **all-pole filter**.

Step 1 of 3

17.011E

Refer to Figure 16.13 (b) in the textbook.

Draw the high pass first order filter circuit is

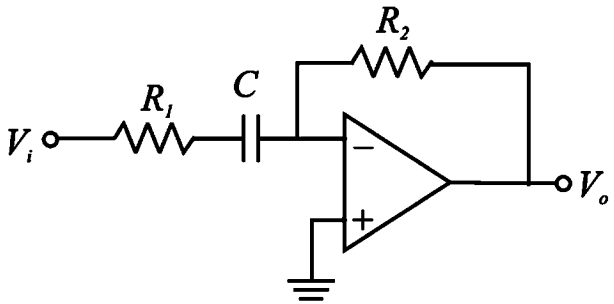


Figure 1

Step 2 of 3

Consider the expression for corner frequency to calculate the value of capacitor, C.

$$\omega_0 = \frac{1}{CR_1}$$

$$C = \frac{1}{\omega_0 R_1}$$

Substitute $10\ \text{k}\Omega$ for R_1 and $10^4\ \text{rad/s}$ for ω_0.

$$C = \frac{1}{(10^4)(10\times10^3)}$$
$$= 0.01\times10^{-6}$$
$$= 0.01\ \mu\text{F}$$

Thus, the value of the capacitor is $\boxed{0.01\ \mu\text{F}}$.

Step 3 of 3

At high frequencies, the capacitor acts as short circuit.

Consider the expression for high frequency gain to calculate the value of the resistor, R_2.

$$|A_v| = \frac{R_2}{R_1}$$

$$R_2 = |A_v| R_1$$

Substitute $10\ \text{k}\Omega$ for R_1 and $10\ \text{V/V}$ for $|A_v|$.

$$R_2 = (10)(10\times10^3)$$
$$= 100\times10^3$$
$$= 100\ \text{k}\Omega$$

Thus, the value of the resistor, R_2 is $\boxed{100\ \text{k}\Omega}$.

Step 1 of 3

The transfer function of the filter is,

$$T(s)=\frac{a_M(s-z_1)(s-z_2)}{(s-p_1)(s-p_2)(s-p_3)}$$

The order of the filter is,

$N=3$.

The transmission zeros of the filter are,

$\omega=2$ rad/s

$\omega=\infty$

M is the number of transfer function zeros or transmission zeros.

In this case, $M=2$.

The natural modes of the filter are,

$s=-1$

$s=-0.5\pm j0.8$

The dc gain is,

$\text{DCgain}=1$

From the given transmission modes of the filter, the denominator is,

$p_1=-1$

$$\begin{aligned}(s-p_2)(s-p_3)&=\left[s-(-0.5+j0.8)\right]\left[s-(-0.5-j0.8)\right]\\&=(s+0.5-j0.8)(s+0.5+j0.8)\\&=(s+0.5)^2+0.8^2\\&=s^2+s+0.25+0.64\end{aligned}\qquad =s^2+s+0.89$$

Step 2 of 3

Zeros at $\omega=\infty$ gives the numerator as follows:

$$\begin{aligned}(s-z_1)(s-z_2)&=(s+j2)(s-j2)\\&=s^2+4\end{aligned}$$

Now the transfer function is,

$$T(s)=\frac{a_M(s^2+4)}{(s+1)(s^2+s+0.89)} \quad \text{..... (1)}$$

$$\begin{aligned}T(s)&=\frac{4a_M\left(1+\frac{s^2}{4}\right)}{0.89(1+s)\left(1+\frac{s}{0.89}+\frac{s^2}{0.89}\right)}\\&=\left(\frac{4a_M}{0.89}\right)\frac{\left(1+\frac{s^2}{4}\right)}{(1+s)\left(1+\frac{s}{0.89}+\frac{s^2}{0.89}\right)}\end{aligned}$$

Step 3 of 3

The dc gain is,

$$\begin{aligned}\text{DC gain}&=T(0)\\&=\left(\frac{4a_M}{0.89}\right)\frac{\left(1+\frac{0^2}{4}\right)}{(1+0)\left(1+\frac{0}{0.89}+\frac{0^2}{0.89}\right)}\\&=\frac{4a_M}{0.89}\end{aligned}$$

Substitute 1 for dc gain in the equation.

$$\begin{aligned}1&=\frac{4a_M}{0.89}\\a_M&=\frac{0.89}{4}\\a_M&=0.2225\end{aligned}$$

Substitute 0.2225 for a_M in equation (1).

$$T(s)=\frac{0.225(s^2+4)}{(s+1)(s^2+s+0.89)}$$

Thus, the transfer function of the third-order low-pass filter is,

$$\boxed{T(s)=\frac{0.225(s^2+4)}{(s+1)(s^2+s+0.89)}}$$

Step 1 of 4

Refer to Figure 16.14 in the textbook for first order all pass filter.

Draw the circuit in s-domain representation.

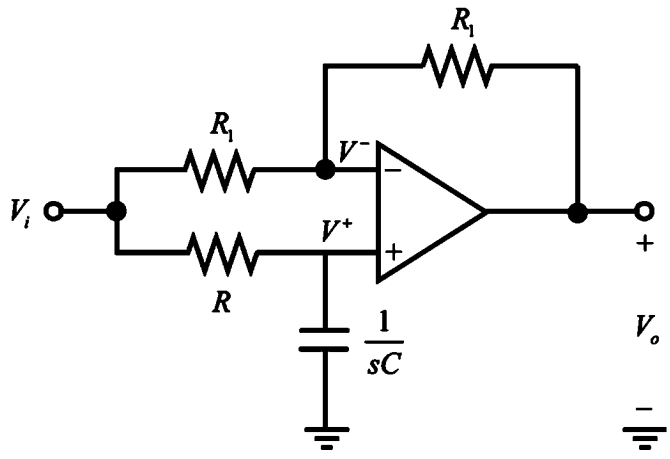


Figure 1

Step 2 of 4

Apply voltage division rule to calculate the voltage at the non-inverting terminal.

$$V^+ = \left(\frac{\frac{1}{sC}}{R+\frac{1}{sC}}\right)V_i$$
$$= \left(\frac{1}{sRC+1}\right)V_i$$

As per virtual ground concept, the voltage at the inverting terminal is equal to the voltage at the non-inverting terminal.

$$V^- = V^+$$
$$= \left(\frac{1}{sRC+1}\right)V_i$$

Step 3 of 4

Apply nodal analysis at the inverting terminal.

$$\frac{V^- - V_i}{R_1} + \frac{V^- - V_o}{R_1} = 0$$
$$V_o = 2V^- - V_i$$

Substitute $\left(\frac{1}{sRC+1}\right)V_i$ for V^-.

$$\frac{V^- - V_i}{R_1} + \frac{V^- - V_o}{R_1} = 0$$
$$V_o = \left(\frac{2}{sRC+1}\right)V_i - V_i$$
$$V_o = \left(\frac{2-sRC-1}{sRC+1}\right)V_i$$
$$\frac{V_o}{V_i} = \frac{1-sRC}{1+sRC}$$

Replace $j\omega$ for s.

$$\frac{V_o}{V_i} = \frac{1-j\omega RC}{1+j\omega RC}$$

Step 4 of 4

Calculate the phase shift from the transfer function.

$$\angle\frac{V_o}{V_i} = \tan^{-1}(-\omega RC) - \tan^{-1}(\omega RC)$$
$$= -\tan^{-1}(\omega RC) - \tan^{-1}(\omega RC)$$
$$= -2\tan^{-1}(\omega RC)$$

The phase of the all pass filter is $90°$ at $\omega = 10^3$ rad/s.

$$-2\tan^{-1}(\omega RC)\big|_{\omega=10^3} = 90°$$
$$\tan^{-1}(\omega RC)\big|_{\omega=10^3} = -45°$$
$$\tan^{-1}(10^3 RC) = -45°$$
$$10^3 RC = -1 \qquad (RC \text{ is always positive value})$$
$$RC = 10^{-3}$$

Assume the value of the resistors, $R_1 = R = 10\ \text{k}\Omega$ for one possible value of capacitor.

$$(10\times10^3)C = 10^{-3}$$
$$C = \frac{10^{-3}}{10^4}$$
$$= 10^{-7}$$
$$= 0.1\ \mu\text{F}$$

Thus, the values of the resistors are $\boxed{R_1 = R = 10\ \text{k}\Omega}$ and the value of the capacitor is $\boxed{0.1\ \mu\text{F}}$.

Step 1 of 5

The transmission zero z_1 is,

$$z_1 = 2$$

The poles of the transfer functions are,

$$p_1 = -0.25 + j$$
$$p_2 = -0.25 - j$$

Step 2 of 5

The expression for the transfer function of a second-order low-pass filter is,

$$T(s) = \frac{k(s - z_1)}{(s - p_1)(s - p_2)}$$

Substitute 2 for z_1, $-0.25 + j$ for p_1, and $-0.25 - j$ for p_2.

$$\begin{aligned} T(s) &= \frac{k \times (s-2)}{[s-(-0.25+j)][s-(-0.25-j)]} \\ &= \frac{k(s-2)}{(s+0.25-j)(s+0.25+j)} \\ &= \frac{k(s-2)}{(s+0.25)^2 - (j)^2} \\ &= \frac{k(s-2)}{s^2 + 2 \times (0.25)s + (0.25)^2 - (-1)} \end{aligned}$$

Simplify further.

$$\begin{aligned} T(s) &= \frac{k(s-2)}{s^2 + 0.5s + 0.0625 + 1} \\ &= \frac{k(s-2)}{s^2 + 0.5s + 1.0625} \end{aligned}$$

Step 3 of 5

Consider the expression $T(s)$.

$$T(s) = \frac{k(s-2)}{s^2 + 0.5s + 1.0625} \quad \text{..... (1)}$$

Substitute 0 for s and calculate DC gain $T(0)$.

$$\begin{aligned} T(0) &= \frac{k(0-2)}{0^2 + 0.5 \times 0 + 1.0625} \\ &= \frac{-2k}{1.0625} \end{aligned}$$

Substitute 1 for $T(0)$.

$$\begin{aligned} 1 &= \frac{-2k}{1.0625} \\ k &= \frac{-1.0625}{2} \\ k &= -0.53125 \end{aligned}$$

Step 4 of 5

Recall equation (1).

$$T(s) = \frac{k(s-2)}{s^2 + 0.5s + 1.0625}$$

Substitute ◊◊◊◊◊◊◊◊◊for k.

$$\begin{aligned} T(s) &= \frac{-0.53125(s-2)}{s^2 + 0.5s + 1.0625} \\ &= \frac{\frac{-17}{32}(s-2)}{s^2 + 0.5s + 1.0625} \\ &= \frac{-17(s-2)}{32\left(s^2 + 0.5s + 1.0625\right)} \\ &= \frac{-17(s-2)}{32s^2 + 16s + 34} \end{aligned}$$

Therefore, the transfer function $T(s)$ is $\boxed{\frac{-17(s-2)}{32s^2 + 16s + 34}}$.

Step 5 of 5

The expression for the gain as ω approaches to infinity is,

$$\text{DC gain}\big|_{\omega \to \infty} = \lim_{s \to \infty}\left[T(s)\right]$$

Substitute $\frac{-17(s-2)}{32s^2 + 16s + 34}$ for $T(s)$.

$$\begin{aligned} \text{DC gain}\big|_{\omega \to \infty} &= \lim_{s \to \infty}\left[\frac{-17(s-2)}{32s^2 + 16s + 34}\right] \\ &= \lim_{s \to \infty} \frac{-17s\left(1 - \frac{2}{s}\right)}{s\left(32s + 16 + \frac{34}{s}\right)} \\ &= \lim_{s \to \infty} \frac{-17\left(1 - \frac{2}{s}\right)}{\left(32s + 16 + \frac{34}{s}\right)} \\ &= 0 \end{aligned}$$

Therefore, the DC gain as ω approaches to infinity is $\boxed{0}$.

Step 1 of 2

17.013E

Consider the expression for transfer function of the second order low pass filter.

$$T(s)=\frac{\omega_0^2}{s^2+s\left(\frac{\omega_0}{Q}\right)+\omega_0^2}$$

Substitute $j\omega$ for s.

$$\begin{aligned}T(j\omega)&=\frac{\omega_0^2}{(j\omega)^2+j\omega\left(\frac{\omega_0}{Q}\right)+\omega_0^2}\\&=\frac{\omega_0^2}{\left(\omega_0^2-\omega^2\right)+j\omega\left(\frac{\omega_0}{Q}\right)}\end{aligned}$$

Substitute $\frac{1}{\sqrt{2}}$ for Q and ω_0 for ω.

$$\begin{aligned}T(j\omega)&=\frac{\omega_0^2}{\left(\omega_0^2-\omega_0^2\right)+j\omega_0\left(\sqrt{2}\omega_0\right)}\\&=\frac{\omega_0^2}{j\left(\sqrt{2}\omega_0^2\right)}\\&=\frac{1}{j\sqrt{2}}\end{aligned}$$

Step 2 of 2

Calculate the magnitude of the transfer function.

$$\left|T(j\omega)\right|=\frac{1}{\sqrt{2}}$$

Calculate the magnitude in dB.

$$\begin{aligned}\left|T(j\omega)\right|_{\text{dB}}&=20\log\left|T(j\omega)\right|\\&=20\log\frac{1}{\sqrt{2}}\\&=20\log\frac{1}{\sqrt{2}}\\&=-3\text{ dB}\end{aligned}$$

Thus, the magnitude response is 3 dB below the value of dc at $\omega=\omega_0$.

Step 1 of 4

The transmission zeros of a bandpass filter are at $\omega = 0, \omega_1 = 10^3 \text{ rad/s}, \omega_2 = 3\times10^3 \text{ rad/s}, \omega_3 = 6\times10^3 \text{ rad/s}$ and the other at $s = \infty$.

The filter has a monotonically decreasing passband transmission with a peak at the center frequency of $\omega_c = 2\times10^3 \text{ rad/s}$.

The filter has equiripple response in the stop bands.

Step 2 of 4

For the band pass filter has transmission zeros at $s = 0$, $s = \pm j\omega_1$, $s = \pm j\omega_2$ and $s = \pm j\omega_3$.

And one zero exists at each of $s = 0$ and $s = \infty$.

As there are seven transmission zeros and only one zero exists at each of $s = 0$ and $s = \infty$, the degree of the numerator is

$$M = 7.$$

Since, there is one zero at $s = \infty$, the number of poles are,

$$\begin{aligned} N &= 7+1 \\ &= 8 \end{aligned}$$

So, the degree of the denominator is 8.

Thus, the order of the band pass filter is, $\boxed{N = 8}$.

Step 3 of 4

The transfer function of the filter is of the following form:

$$\begin{aligned} T(s) &= \frac{a_7 s\left(s^2+\omega_1^2\right)\left(s^2+\omega_2^2\right)\left(s^2+\omega_3^2\right)}{s^8+b_7s^7+\ldots+b_0} \\ &= \frac{a_7 s\left(s^2+\left(10^3\right)^2\right)\left(s^2+\left(3\times10^3\right)^2\right)\left(s^2+\left(6\times10^3\right)^2\right)}{s^8+b_7s^7+\ldots+b_0} \\ &= \frac{a_7 s\left(s^2+10^6\right)\left(s^2+9\times10^6\right)\left(s^2+36\times10^3\right)}{s^8+b_7s^7+\ldots+b_0} \end{aligned}$$

Thus, the transfer function of the filter is,

$$\boxed{T(s) = \frac{a_7 s\left(s^2+10^6\right)\left(s^2+9\times10^6\right)\left(s^2+36\times10^3\right)}{s^8+b_7s^7+\ldots+b_0}}.$$

Step 4 of 4

The shape of $|T|$ is shown in Figure 1.

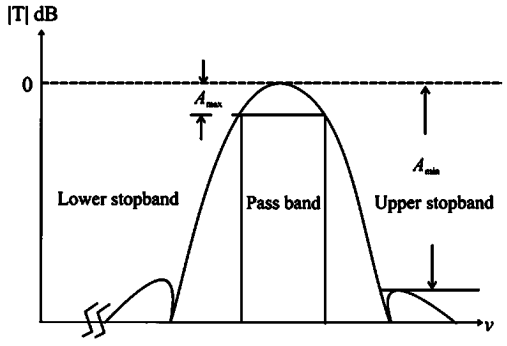


Figure 1

Step 1 of 3

Consider the expression for bandwidth to calculate the value of Quality factor.

$$BW = \frac{\omega_0}{Q}$$

$$Q = \frac{\omega_0}{BW}$$

Substitute 10^3 rad/s for BW and 10^5 rad/s for ω_0.

$$Q = \frac{10^5}{10^3}$$
$$= 100$$

Step 2 of 3

Consider the expression for center frequency gain to calculate the value of a_1.

$$\frac{a_1 Q}{\omega_0} = A$$

$$a_1 = \frac{A\omega_0}{Q}$$

Substitute 10 for A, 100 for Q and 10^5 rad/s for ω_0.

$$a_1 = \frac{(10)(10^5)}{100}$$
$$= 10^4$$

Step 3 of 3

Consider the expression for the transfer function of a second order band pass filter.

$$T(s) = \frac{a_1 s}{s^2 + s\frac{\omega_0}{Q} + \omega_0^2}$$

Substitute 10^4 for a_1, 100 for Q and 10^5 rad/s for ω_0.

$$T(s) = \frac{10^4 s}{s^2 + s\left(\frac{10^5}{100}\right) + (10^5)^2}$$
$$= \frac{10^4 s}{s^2 + 10^3 s + 10^{10}}$$

Thus, the transfer function of a second order band pass filter is $\boxed{\frac{10^4 s}{s^2 + 10^3 s + 10^{10}}}$.

Step 1 of 5

17.014P

Refer to Figure P16.11 in the textbook for the RLC network.

Redraw the circuit by representing nodal voltages as shown in Figure 1.

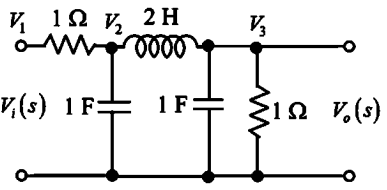


Figure 1

Step 2 of 5

From the circuit, write the following voltage equations:

$$V_1 = V_i$$
$$V_3 = V_0$$

The nodal equations at node (2) is,

$$\frac{V_2 - V_1}{1} + \frac{V_2}{\frac{1}{s}} + \frac{V_2 - V_0}{2s} = 0$$
$$V_2 - V_1 + sV_2 + \frac{V_2 - V_0}{2s} = 0$$
$$2s^2V_2 + 2sV_2 + V_2 = 2sV_i + V_0$$

Rearrange the terms.

$$(2s^2 + 2s + 1)V_2 = 2sV_i + V_0 \quad \text{..... (1)}$$

Step 3 of 5

The nodal equation at node (3) is,

$$\frac{V_0 - V_2}{2s} + \frac{V_0}{\frac{1}{s}} + \frac{V_0}{1} = 0$$
$$\frac{V_0 - V_2}{2s} + sV_0 + V_0 = 0$$
$$V_0 - V_2 + 2s^2V_0 + 2sV_0 = 0$$

Rearrange the terms.

$$V_2 = V_0(2s^2 + 2s + 1)$$

Step 4 of 5

Substitute $V_0(2s^2 + 2s + 1)$ for V_2 in equation (1).

$$(2s^2 + 2s + 1)(2s^2 + 2s + 1)V_0 = 2sV_i + V_0$$
$$(2s^2 + 2s + 1)(2s^2 + 2s + 1)V_0 - V_0 = 2sV_i$$
$$(4s^4 + 8s^3 + 8s^2 + 4s)V_0 = 2sV_i$$

Simplify further.

$$\frac{V_0}{V_i} = \frac{2s}{4s^4 + 8s^3 + 8s^2 + 4s}$$
$$= \frac{2}{4s^3 + 8s^2 + 8s + 4}$$
$$= \frac{2}{4(s^3 + 2s^2 + 2s + 1)}$$
$$= \frac{0.5}{s^3 + 2s^2 + 2s + 1}$$

Thus, the transfer function of the RLC network is, $\boxed{\frac{V_0}{V_i} = \frac{0.5}{s^3 + 2s^2 + 2s + 1}}$.

Step 5 of 5

There are no zeros as the numerator is a constant.

The poles are the roots of the equation,

$$s^3 + 2s^2 + 2s + 1 = 0$$
$$(s + 1)(s^2 + s + 1) = 0$$

The roots of the equation are,

$$s = -1, \frac{-1 \pm \sqrt{1 - 4}}{2}$$
$$= -1, \frac{-1 \pm j\sqrt{3}}{2}$$
$$= -1, -\frac{1}{2} \pm j\frac{\sqrt{3}}{2}$$

Thus, the poles of the transfer function are, $\boxed{s = -1, -\frac{1}{2} \pm j\frac{\sqrt{3}}{2}}$.

Step 1 of 7

(a)

Consider a second order filter with a center frequency, $\omega_n = \omega_0$.

Write the transfer function of a second order notch filter.

$$T(s) = \frac{s^2 + \omega_0^2}{s^2 + s\frac{\omega_0}{Q} + \omega_0^2}$$

Substitute $j\omega$ for s.

$$T(j\omega) = \frac{(j\omega)^2 + \omega_0^2}{(j\omega)^2 + j\omega\frac{\omega_0}{Q} + \omega_0^2}$$
$$= \frac{\omega_0^2 - \omega^2}{\omega_0^2 - \omega^2 + j\omega\frac{\omega_0}{Q}}$$

Step 2 of 7

Calculate the magnitude of the transfer function.

$$|T(j\omega)| = \frac{\omega_0^2 - \omega^2}{\sqrt{\left(\omega_0^2 - \omega^2\right)^2 + \frac{\omega^2\omega_0^2}{Q^2}}}$$
$$= \frac{1}{\sqrt{\frac{\left(\omega_0^2 - \omega^2\right)^2}{\left(\omega_0^2 - \omega^2\right)^2} + \frac{\omega^2\omega_0^2}{\left(\omega_0^2 - \omega^2\right)^2 Q^2}}}$$
$$= \frac{1}{\sqrt{1 + \frac{\omega^2\omega_0^2}{\left(\omega_0^2 - \omega^2\right)^2 Q^2}}}$$

Step 3 of 7

Assume two frequencies ω_1 and ω_2 at which $|T|$ is same.

$$|T(j\omega_1)| = |T(j\omega_1)|$$
$$\frac{1}{\sqrt{1 + \frac{\omega_1^2\omega_0^2}{\left(\omega_0^2 - \omega_1^2\right)^2 Q^2}}} = \frac{1}{\sqrt{1 + \frac{\omega_2^2\omega_0^2}{\left(\omega_0^2 - \omega_2^2\right)^2 Q^2}}}$$
$$\sqrt{1 + \frac{\omega_1^2\omega_0^2}{\left(\omega_0^2 - \omega_1^2\right)^2 Q^2}} = \sqrt{1 + \frac{\omega_2^2\omega_0^2}{\left(\omega_0^2 - \omega_2^2\right)^2 Q^2}}$$
$$1 + \frac{\omega_1^2\omega_0^2}{\left(\omega_0^2 - \omega_1^2\right)^2 Q^2} = 1 + \frac{\omega_2^2\omega_0^2}{\left(\omega_0^2 - \omega_2^2\right)^2 Q^2}$$

$$\frac{\omega_1^2\omega_0^2}{\left(\omega_0^2 - \omega_1^2\right)^2 Q^2} = \frac{\omega_2^2\omega_0^2}{\left(\omega_0^2 - \omega_2^2\right)^2 Q^2}$$
$$\frac{\omega_1^2}{\left(\omega_0^2 - \omega_1^2\right)^2} = \frac{\omega_2^2}{\left(\omega_0^2 - \omega_2^2\right)^2}$$
$$\frac{\omega_1}{\left(\omega_0^2 - \omega_1^2\right)} = \frac{\omega_2}{\left(\omega_0^2 - \omega_2^2\right)}$$
$$\omega_1\left(\omega_0^2 - \omega_2^2\right) = \omega_2\left(\omega_0^2 - \omega_1^2\right)$$

$$\omega_1\omega_0^2 - \omega_1\omega_2^2 = \omega_2\omega_0^2 - \omega_2\omega_1^2$$
$$\omega_1\omega_0^2 - \omega_2\omega_0^2 = \omega_1\omega_2^2 - \omega_2\omega_1^2$$
$$(\omega_1 - \omega_2)\omega_0^2 = \omega_1\omega_2(\omega_2 - \omega_1)$$
$$(\omega_1 - \omega_2)\omega_0^2 = -\omega_1\omega_2(\omega_1 - \omega_2)$$

$$\omega_0^2 = -\omega_1\omega_2$$

$$|\omega_0^2| = \omega_1\omega_2$$

Step 4 of 7

Obtain attenuation $\geq A$ **dB** at ω_1 and ω_2.

$$20\log\left(\left|\frac{1}{T(j\omega_1)}\right|\right) \geq A$$

$$20\log\left(\sqrt{1 + \frac{\omega_1^2\omega_0^2}{\left(\omega_0^2 - \omega_1^2\right)^2 Q^2}}\right) \geq A$$

$$\frac{20}{2}\log\left(1 + \frac{\omega_1^2\omega_0^2}{\left(\omega_0^2 - \omega_1^2\right)^2 Q^2}\right) \geq A$$

Step 5 of 7

Simplify further:

$$10\log\left(1 + \frac{\omega_1^2\omega_0^2}{\left(\omega_0^2 - \omega_1^2\right)^2 Q^2}\right) \geq A$$

$$\log\left(1 + \frac{\omega_1^2\omega_0^2}{\left(\omega_0^2 - \omega_1^2\right)^2 Q^2}\right) \geq \frac{A}{10}$$

$$1 + \frac{\omega^2\omega_0^2}{\left(\omega_0^2 - \omega_1^2\right)^2 Q^2} \geq 10^{\frac{A}{10}}$$

$$\frac{\omega^2\omega_0^2}{\left(\omega_0^2 - \omega_1^2\right)^2 Q^2} \geq 10^{\frac{A}{10}} - 1$$

Apply square root on both sides.

$$\sqrt{\frac{\omega_1^2\omega_0^2}{\left(\omega_0^2 - \omega_1^2\right)^2 Q^2}} \geq \sqrt{10^{\frac{A}{10}} - 1}$$

$$\frac{\omega_1\omega_0}{\left(\omega_0^2 - \omega_1^2\right)Q} \geq \sqrt{10^{\frac{A}{10}} - 1}$$

Step 6 of 7

Substitute $\omega_1\omega_2$ for ω_0^2 in the equation.

$$\frac{\omega_1\omega_0}{\left(\omega_1\omega_2 - \omega_1^2\right)Q} \geq \sqrt{10^{\frac{A}{10}} - 1}$$

$$\frac{\omega_1\omega_0}{\omega_1(\omega_2 - \omega_1)Q} \geq \sqrt{10^{\frac{A}{10}} - 1}$$

$$\frac{\omega_0}{(BW_a)Q} \geq \sqrt{10^{\frac{A}{10}} - 1} \qquad (\text{since } BW_a = \omega_2 - \omega_1)$$

$$\frac{1}{Q} \geq \frac{BW_a\sqrt{10^{\frac{A}{10}} - 1}}{\omega_0}$$

$$Q \leq \frac{\omega_0}{BW_a\sqrt{10^{\frac{A}{10}} - 1}}$$

Thus, it is proved that the quality factor is $\boxed{Q \leq \frac{\omega_0}{BW_a\sqrt{10^{\frac{A}{10}} - 1}}}$.

Step 7 of 7

(b)

Consider the expression for the Quality factor to calculate the bandwidth.

$$Q = \frac{\omega_0}{BW_a\sqrt{10^{\frac{A}{10}} - 1}}$$

$$BW_a = \frac{\omega_0}{Q\sqrt{10^{\frac{A}{10}} - 1}}$$

Substitute **3 dB** for A.

$$BW_3 = \frac{\omega_0}{Q\sqrt{10^{\frac{3}{10}} - 1}}$$
$$= \frac{\omega_0}{Q\sqrt{10^{0.3} - 1}}$$
$$= \frac{\omega_0}{Q\sqrt{2 - 1}}$$
$$= \frac{\omega_0}{Q}$$

Thus, the expression for 3-dB bandwidth is $\boxed{BW_3 = \frac{\omega_0}{Q}}$.

Step 1 of 7

The maximum variation in pass-band transmission is,

$$A_{max} = 0.5 \text{ dB}$$

The selectivity ratio is,

$$\frac{\omega_s}{\omega_p} = 1.7$$

The minimum variation in pass-band transmission is,

$$A_{min} = 20 \text{ dB}$$

Step 2 of 7

The attenuation of the Butterworth filter is,

$$A(\omega_s) = 10\log\left[1+\varepsilon^2\left(\frac{\omega_s}{\omega_p}\right)^{2N}\right]$$

Substitute 20 dB for $A(\omega_s)$ and 1.7 for $\frac{\omega_s}{\omega_p}$.

$$20 = 10\log\left[1+\varepsilon^2(1.7)^{2N}\right]$$
$$\log\left[1+\varepsilon^2(1.7)^{2N}\right] = 2$$
$$1+\varepsilon^2(1.7)^{2N} = 10^2$$
$$\varepsilon^2(1.7)^{2N} = 10^2 - 1 \quad \text{..... (1)}$$

Step 3 of 7

The expression for the parameter ε is,

$$\varepsilon = \sqrt{10^{\frac{A_{max}}{10}} - 1}$$

Substitute 0.5 dB for A_{max}.

$$\varepsilon = \sqrt{10^{\frac{0.5}{10}} - 1}$$
$$= 0.3493$$

Step 4 of 7

Substitute 0.3493 for ε in equation (1).

$$(0.3493)^2(1.7)^{2N} = 10^2 - 1$$
$$(1.7)^{2N} = \frac{100-1}{(0.3493)^2}$$
$$(1.7)^{2N} = \frac{99}{0.1220}$$
$$(1.7)^{2N} = 811.475$$

Apply natural logarithm.

$$2N\ln(1.7) = \ln(811.475)$$
$$2N \times 0.5306 = 6.6988$$
$$N = 6.312$$
$$N \approx 6$$

So, $N = 6$.

Therefore, the order of the Butterworth filter is $\boxed{6}$.

Step 5 of 7

The expression for the actual value of the minimum stop-band attenuation is,

$$A(\omega_s) = 10\log\left[1+\varepsilon^2\left(\frac{\omega_s}{\omega_p}\right)^{2N}\right]$$

Substitute 0.3493 for ε, 1.7 for $\frac{\omega_s}{\omega_p}$, and 6 for *N*.

$$A(\omega_s) = 10\log\left[1+(0.3493)^2(1.7)^{2(6)}\right]$$
$$= 10\log\left[1+(0.3493)^2(1.7)^{12}\right]$$
$$= 18.57 \text{ dB}$$

Therefore, the actual value of minimum stop-band attenuation is $\boxed{18.57 \text{ dB}}$.

Step 6 of 7

Consider the following expression for A_{min} to be exactly 20 dB:

$$A_{min} = 10\log\left[1+\varepsilon^2\left(\frac{\omega_s}{\omega_p}\right)^{2N}\right]$$

Substitute 20 for A_{min}, 1.7 for $\frac{\omega_s}{\omega_p}$, and 6 for *N*.

$$20 = 10\log\left[1+\varepsilon^2(1.7)^{2(6)}\right]$$
$$2 = \log\left[1+\varepsilon^2(1.7)^{12}\right]$$
$$1+\varepsilon^2(1.7)^{12} = 10^2$$
$$\varepsilon^2 = \frac{99}{(1.7)^{12}}$$
$$\varepsilon^2 = 0.1699$$

Step 7 of 7

If A_{min} is to be exactly 20 dB, the value of A_{max} is,

$$A_{max} = 20\log\sqrt{1+\varepsilon^2}$$

Substitute 0.1699 for ε^2.

$$A_{max} = 20\log\sqrt{1+0.1699}$$
$$= 20\log\sqrt{1.1699}$$
$$= 0.68 \text{ dB}$$

Therefore, if A_{min} is to be exactly 20 dB, the value of A_{max} is reduced to $\boxed{0.68 \text{ dB}}$.

Step 1 of 3

The transfer function of a second order low pass notch filter is given by

$$T(s)=a_2\frac{s^2+\omega_n^2}{s^2+s\frac{\omega_0}{Q}+\omega_0^2}$$

The dc gain of the filter is given by

$$a_2\frac{\omega_n^2}{\omega_0^2}=1.$$

Substitute **1.2 rad/s for ω_n , 1 rad/s for ω_0** in equation.

Thus the high frequency gain is given by

$$\begin{aligned}a_2&=\frac{1}{1.2^2}\\&=0.694\end{aligned}$$

Therefore, the high-frequency transmission of second order low pass notch filter is **0.694** .

Step 2 of 3

The frequency of the transmission peak is ,

$$\begin{aligned}\omega_{max}&=1\sqrt{\frac{1.2^2\left(1-\frac{1}{2\times10^2}\right)-1}{1.2^2+\frac{1}{2\times10^2}-1}}\\&=\sqrt{\frac{1.2^2\times0.995-1}{1.2^2+0.005-1}}\\&=\frac{0.4328}{0.445}\\&=0.972\text{ rad/s}\end{aligned}$$

Thus the frequency of the transmission peak of a second order low pass notch filter is **0.972 rad/s** .

Step 3 of 3

The magnitude of the transmission peak is given by

$$\begin{aligned}T_{max}&=\frac{a_2\left(\omega_n^2-\omega_{max}^2\right)}{\sqrt{\left(\omega_0^2-\omega_{max}^2\right)^2+\left(\frac{\omega_o}{Q}\right)^2\omega_{max}^2}}\\&=\frac{0.694\left(1.2^2-0.972^2\right)}{\sqrt{\left(1-0.972^2\right)^2+\left(\frac{1}{10}\right)^2 0.972^2}}\\&=\frac{0.694\times0.4952}{\sqrt{3.05\times10^{-3}+9.45\times10^{-3}}}\\&=3.07\end{aligned}$$

Therefore, the magnitude of the transmission peak of a second order low pass notch filter is **3.07** .

Step 1 of 2

17.016P

Refer to equation 17.15 given in the textbook.

Simplify further.

$$A(\omega_s) = 10\log\left[1+\varepsilon^2\left(\frac{\omega_s}{\omega_p}\right)^{2N}\right]$$

Neglect the unity term and rewrite as,

$$\begin{aligned} A(\omega_s) &= 10\log\left[\varepsilon^2\left(\frac{\omega_s}{\omega_p}\right)^{2N}\right] \\ &= 10\log\left[\varepsilon\left(\frac{\omega_s}{\omega_p}\right)^{N}\right]^2 \\ &= 20\log\left[\varepsilon\left(\frac{\omega_s}{\omega_p}\right)^{N}\right] \end{aligned}$$

Here, *N* is the order of the filter.

Step 2 of 2

The condition to determine the order of the filter *N* is:

$$A(\omega_s) \geq A_{min}$$

Substitute $20\log\left[\varepsilon\left(\frac{\omega_s}{\omega_p}\right)^{N}\right]$ for $A(\omega_s)$.

$$20\log\left[\varepsilon\left(\frac{\omega_s}{\omega_p}\right)^{N}\right] \geq A_{min}$$

$$20\left[\log\varepsilon + \log\left(\frac{\omega_s}{\omega_p}\right)^{N}\right] \geq A_{min}$$

$$20\log\left(\frac{\omega_s}{\omega_p}\right)^{N} \geq A_{min} - 20\log\varepsilon$$

$$\log\left(\frac{\omega_s}{\omega_p}\right)^{N} \geq \frac{A_{min} - 20\log\varepsilon}{20}$$

Simplify further.

$$N\log\left(\frac{\omega_s}{\omega_p}\right) \geq \frac{A_{min} - 20\log\varepsilon}{20}$$

$$N \geq \frac{A_{min} - 20\log\varepsilon}{20\log\left(\frac{\omega_s}{\omega_p}\right)}$$

Therefore, the order *N* of the Butterworth filter can be obtained from the approximate expression,

$$\boxed{N \geq \frac{A_{min} - 20\log\varepsilon}{20\log\left(\frac{\omega_s}{\omega_p}\right)}}$$

. Hence, it is proved.

Step 1 of 3

Refer to Figure 16.18(b) in the text book.

It is given that a second-order low-pass function of the maximally flat type with a 3-dB frequency of $f_o = 100\ \text{kHz}$.

Since the low-pass function of the maximally flat type, the quality factor is,

$$Q = \frac{1}{\sqrt{2}}$$

The angular frequency is,

$$\begin{aligned}\omega_o &= 2\pi f_o \\ &= 2\pi\left(100\times10^3\right) \\ &= 200\pi\times10^3\ \text{rad/s}\end{aligned}$$

Step 2 of 3

The quality factor is,

$$Q = \omega_o CR$$

$$C = \frac{Q}{\omega_o R}$$

$$C = \frac{\left(\frac{1}{\sqrt{2}}\right)}{\left(200\pi\times10^3\right)R}$$

$$C = \frac{1}{\left(200\sqrt{2}\pi\times10^3\right)R}$$

Consider the value of resistance R is $1\ \text{k}\Omega$.

$$\begin{aligned}C &= \frac{1}{\left(200\sqrt{2}\pi\times10^3\right)\left(1\times10^3\right)} \\ &= 1125\times10^{-12} \\ &= 1125\ \text{pF}\end{aligned}$$

Hence the value of capacitor is $\boxed{1125\ \text{pF}}$.

Step 3 of 3

It is also known that,

$$Q = \frac{R}{\omega_o L}$$

$$L = \frac{R}{\omega_o Q}$$

$$\begin{aligned}&= \frac{1\times10^3}{\left(200\pi\times10^3\right)\left(\frac{1}{\sqrt{2}}\right)} \\ &= 2.25\ \text{mH}\end{aligned}$$

Hence the value of inductor is $\boxed{2.25\ \text{mH}}$.

Step 1 of 4

The order N of the Butterworth filter is,

$$N = 7$$

The maximum variation in the pass-band transmission is,

$$A_{max} = 3 \text{ dB}$$

The frequency is 1.8 times the 3-dB frequency.

$$\omega = 1.8(\omega_p)$$

$$\frac{\omega}{\omega_p} = 1.8 \quad \text{..... (1)}$$

Step 2 of 4

The expression for the attenuation of the Butterworth filter is,

$$A(\omega) = 10\log\left[1 + \varepsilon^2\left(\frac{\omega}{\omega_p}\right)^{2N}\right]$$

Substitute 1.8 for $\left(\frac{\omega}{\omega_p}\right)$ and 7 for N.

$$A(\omega) = 10\log\left[1 + \varepsilon^2(1.8)^{2(7)}\right]$$

$$A(\omega) = 10\log\left[1 + \varepsilon^2(1.8)^{14}\right] \quad \text{..... (2)}$$

Step 3 of 4

The expression for the parameter ε is,

$$\varepsilon = \sqrt{10^{\frac{A_{max}}{10}} - 1}$$

Substitute 3 for A_{max}.

$$\begin{aligned}\varepsilon &= \sqrt{10^{\frac{3}{10}} - 1} \\ &= \sqrt{1.9953 - 1} \\ &= \sqrt{0.9953} \\ &= 0.9976\end{aligned}$$

Therefore, the value of ε is 0.9976.

Step 4 of 4

Recall equation (2).

$$A(\omega) = 10\log\left[1 + \varepsilon^2(1.8)^{14}\right]$$

Substitute 0.9976 for ε.

$$\begin{aligned}A(\omega) &= 10\log\left[1 + (0.9976)^2(1.8)^{14}\right] \\ &= 10\log\left[1 + 0.9953 \times (1.8)^{14}\right] \\ &= 10\log[3731.517] \\ &= 35.72 \text{ dB}\end{aligned}$$

Therefore, the value of attenuation obtained at a frequency 1.8 times the 3-dB frequency of a seventh order Butterworth filter is **35.72 dB**.

17.018E

Step 1 of 3

Refer to figure 16.18(e) in text book.

The 3-dB bandwidth in radians per second is,

$$BW_3 = 2\pi(10)$$
$$= 20\pi \text{ rad/s}$$

The 3-dB bandwidth of a notch filter is,

$$BW_3 = \frac{\omega_o}{Q}$$
$$BW_3 = \frac{2\pi f_o}{Q}$$

Substitute 10 Hz for BW_3 , 60 Hz for f_o and $10 \text{ k}\Omega$ for R in equation.

$$20\pi = \frac{2\pi(60)}{Q}$$
$$Q = \frac{2\pi(60)}{20\pi}$$
$$= 6$$

Step 2 of 3

It is known that,

$$Q = \omega_o CR$$
$$C = \frac{Q}{\omega_o R}$$
$$C = \frac{Q}{2\pi f_o R}$$

Substitute 60 Hz for f_o , $10 \text{ k}\Omega$ for R and 6 for Q in equation.

$$C = \frac{(6)}{2\pi(60)(10\times10^3)}$$
$$= 1.6 \text{ } \mu\text{F}$$

Therefore, the value of capacitor is $\boxed{1.6 \text{ } \mu\text{F}}$.

Step 3 of 3

Calculate the value of inductance.

$$Q = \frac{R}{\omega_o L}$$
$$L = \frac{R}{\omega_o Q}$$
$$L = \frac{R}{2\pi f_o Q}$$

Substitute $10 \text{ k}\Omega$ for R , 60 Hz for f_o and 6 for Q in equation.

$$L = \frac{10\times10^3}{2\pi(60)(6)}$$
$$= 4.42 \text{ H}$$

Therefore, the value of inductor is $\boxed{4.42 \text{ H}}$.

Step 1 of 4

The order N of the Butterworth filter is,

$$N = 5$$

The expression for the natural modes lie on a circle of radius ω_o is,

$$\omega_o = \omega_p \left(\frac{1}{\varepsilon}\right)^{\frac{1}{N}} \quad \text{..... (1)}$$

The expression for the pass-band deviation parameter ε is,

$$\varepsilon = \sqrt{10^{\frac{A_{max}}{10}} - 1}$$

Substitute 0.5 for A_{max}.

$$\begin{aligned}\varepsilon &= \sqrt{10^{\frac{0.5}{10}} - 1} \\ &= \sqrt{1.122 - 1} \\ &= \sqrt{0.122} \\ &= 0.3493\end{aligned}$$

Therefore, the pass-band deviation parameter ε is 0.3493.

Step 2 of 4

Recall equation (1).

$$\omega_o = \omega_p \left(\frac{1}{\varepsilon}\right)^{\frac{1}{N}}$$

Substitute 0.3493 for ε, 10^3 rad/s for ω_p, and 5 for N in equation (1).

$$\begin{aligned}\omega_o &= \omega_p \left(\frac{1}{\varepsilon}\right)^{\frac{1}{N}} \\ &= 10^3 \left(\frac{1}{0.3493}\right)^{\frac{1}{5}} \\ &= 10^3 (2.8627)^{\frac{1}{5}}\end{aligned}$$

$$\omega_o = 1234.12 \text{ rad/s}$$

Therefore, the radius ω_o is 1234.12 rad/s.

Step 3 of 4

The natural modes of the filter are spaced by equal angles of $\frac{\pi}{N}$.

$$\begin{aligned}\frac{\pi}{N} &= \frac{\pi}{5} \\ &= 36^\circ\end{aligned}$$

The first mode angle from $j\omega$ axis,

$$\begin{aligned}\frac{\pi}{2N} &= \frac{\pi}{2 \times 5} \\ &= 18^\circ\end{aligned}$$

Step 4 of 4

Sketch the natural modes of the fifth order Butterworth filter as shown in figure 1.

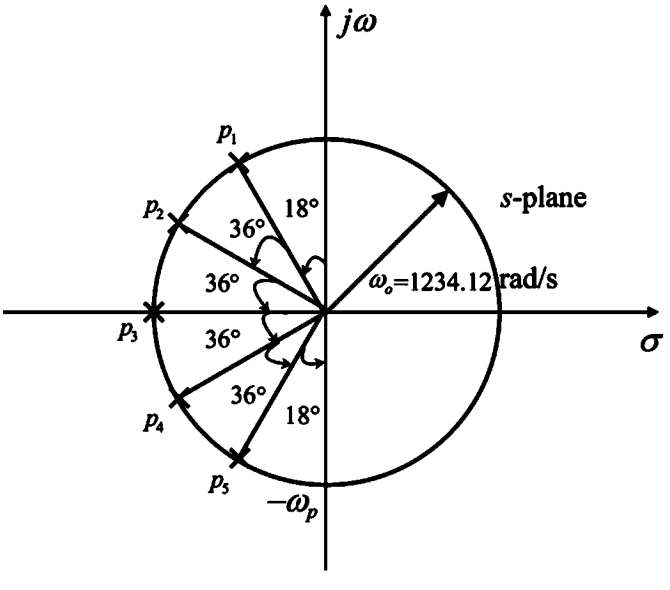


Figure 1

Step 1 of 6

Refer to Figure 16.22(c) in the text book.

The quality factor is,

$$Q = \frac{f_o}{\Delta f_{3dB}}$$

Substitute $500\ \text{Hz}$ for $\Delta f_{3\,dB}$, $10\ \text{kHz}$ for f_o in equation.

$$\begin{aligned} Q &= \frac{10\times10^3}{500} \\ &= 20 \end{aligned}$$

Step 2 of 6

The values of capacitances are,

$$\begin{aligned} C_4 &= C_6 \\ &= C \\ &= 1.2\ \text{nF} \end{aligned}$$

Therefore, the value of capacitances C_4 and C_6 are $\boxed{1.2\ \text{nF}}$.

Step 3 of 6

The values of the resistances are,

$$\begin{aligned} R_1 &= R_2 \\ &= R_3 \\ &= R_4 \\ &= \frac{1}{\omega_0 C} \\ &= \frac{1}{2\pi f_0 C} \end{aligned}$$

Substitute $1.2\ \text{nF}$ for C , $10\ \text{kHz}$ for f_o in equation.

$$\begin{aligned} R_1 &= \frac{1}{2\pi\left(10\times10^3\right)\left(1.2\times10^{-9}\right)} \\ &= 13.26\ \text{k}\Omega \end{aligned}$$

Thus, the value of resistances R_1, R_2 , R_3 , and R_4 are $\boxed{13.26\ \text{k}\Omega}$.

Step 4 of 6

Calculate the value of resistance, R_6 .

$$\begin{aligned} R_6 &= \frac{Q}{\omega_0 C} \\ &= \frac{Q}{2\pi f_0 C} \end{aligned}$$

Substitute $10\ \text{kHz}$ for f_o , $1.2\ \text{nF}$ for C in equation.

$$\begin{aligned} R_6 &= \frac{20}{2\pi\left(10\times10^3\right)\left(1.2\times10^{-9}\right)} \\ &= 265\ \text{k}\Omega \end{aligned}$$

Thus, the value of resistance R_6 is $\boxed{265\ \text{k}\Omega}$.

Step 5 of 6

The gain expression is,

$$1+\frac{r_2}{r_1} = \text{K}$$

Substitute 10 for K in equation.

$$\begin{aligned} 1+\frac{r_2}{r_1} &= 10 \\ \frac{r_2}{r_1} &= 10-1 \\ \frac{r_2}{r_1} &= 9 \end{aligned}$$

Step 6 of 6

Consider the value of r_1 is $10\ \text{k}\Omega$.

$$\begin{aligned} \frac{r_2}{10\times10^3} &= 9 \\ r_2 &= 9\left(10\times10^3\right) \\ r_2 &= 90\ \text{k}\Omega \end{aligned}$$

Therefore, the values of r_1 and r_2 are $\boxed{10\ \text{k}\Omega \text{ and } 90\ \text{k}\Omega}$.

Step 1 of 8

The expression for the attenuation of the Butterworth filter is,

$$A(\omega_s) = 10\log\left[1+\varepsilon^2\left(\frac{\omega_s}{\omega_p}\right)^{2N}\right]$$
$$= 10\log\left[1+\varepsilon^2\left(\frac{2\pi f_s}{2\pi f_p}\right)^{2N}\right]$$
$$A(\omega_s) = 10\log\left[1+\varepsilon^2\left(\frac{f_s}{f_p}\right)^{2N}\right] \quad \text{..... (1)}$$

Step 2 of 8

The expression for the passband deviation parameter ε is,

$$\varepsilon = \sqrt{10^{\frac{A_{max}}{10}} - 1}$$

Substitute 3 for A_{max}.

$$\varepsilon = \sqrt{10^{\frac{3}{10}} - 1}$$
$$= \sqrt{1.9952 - 1}$$
$$= \sqrt{0.9952}$$
$$= 1.4125$$

Step 3 of 8

Recall equation (1).

$$A(\omega_s) = 10\log\left[1+\varepsilon^2\left(\frac{f_s}{f_p}\right)^{2N}\right]$$

Substitute 20×10^3 for f_s, 10×10^3 for f_p, 1.4125 for ε, and 20 for $A(\omega_s)$.

$$20 = 10\log\left[1+(1.4125)^2\left(\frac{20\times10^3}{10\times10^3}\right)^{2N}\right]$$
$$2 = \log\left[1+(0.9952)(2)^{2N}\right]$$
$$1+(0.9952)(2)^{2N} = 10^2$$
$$(2)^{2N} = \frac{10^2-1}{0.9952}$$

Simplify further.

$$(2)^{2N} = \frac{99}{0.9952}$$

Apply natural logarithm.

$$2N\ln(2) = \ln\left(\frac{99}{0.9952}\right)$$
$$2N\times(0.6931) = \ln(99.4775)$$
$$N = 3.31$$
$$N \approx 3$$

Therefore, the order of the Butterworth filter is $\boxed{3}$.

Step 4 of 8

The natural modes of the filter are spaced by equal angles of $\frac{\pi}{N}$.

$$\frac{\pi}{N} = \frac{\pi}{3}$$
$$= 60°$$

The first mode angle from $j\omega$ axis is,

$$\frac{\pi}{2N} = \frac{\pi}{12}$$
$$= 15°$$

Since, the natural modes all have equal radial distance from the origin, their frequencies are same.

$$\omega_o = \omega_p\left(\frac{1}{\varepsilon}\right)^{\frac{1}{N}}$$
$$= 2\pi f_p\left(\frac{1}{\varepsilon}\right)^{\frac{1}{N}}$$

Substitute 10×10^3 for f_p, 1.4125 for ε, and 3 for *N*.

$$\omega_o = 2\pi\times10\times10^3\left(\frac{1}{1.4125}\right)^{\frac{1}{3}}$$
$$= 5.8638\times10^4 \text{ rad/s}$$

Therefore, the radius ω_o is 5.8638×10^4 rad/s.

Sketch the natural modes of the third order Butterworth filter as shown in figure 1.

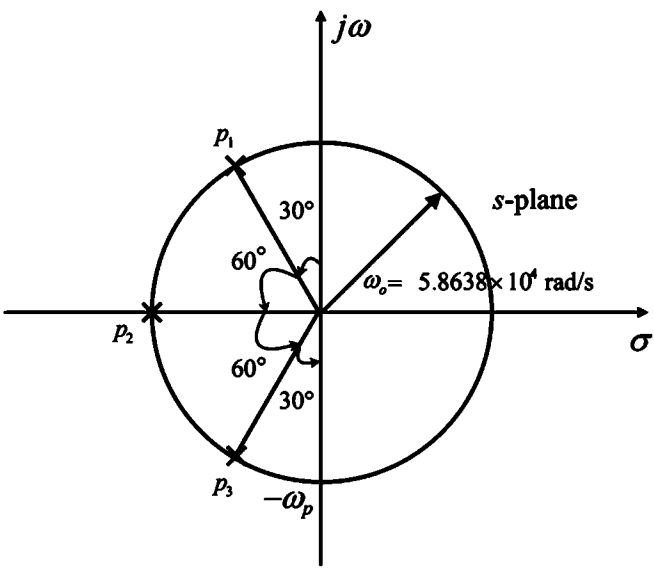


Figure 1

Step 5 of 8

Here, p_1, p_2, p_3 are the poles of the third-order Butterworth filter.

The expression for the pair of complex conjugate poles p_1 and p_3 are,

$$p_{1,3} = \omega_o\angle\left(\frac{\pi}{2}+\frac{\pi}{2N}\right)$$

Substitute 5.8638×10^4 for ω_o, and 3 for *N*.

$$p_{1,3} = 5.8638\times10^4\angle\left(\frac{\pi}{2}+\frac{\pi}{2(3)}\right)$$
$$= 5.8638\times10^4\angle\left(\frac{\pi}{2}+\frac{\pi}{6}\right)$$
$$= 5.8638\times10^4\angle(90°+30°)$$
$$= 5.8638\times10^4\angle(120°)$$

Simplify further.

$$p_{1,3} = 5.8638\times10^4[\cos120°\pm j\sin120°]$$
$$= 5.8638\times10^4[-0.5\pm0.866]$$
$$= [-2.9319\pm j5.0782]\times10^4$$

The pole p_2 on real axis is,

$$p_2 = -5.8638\times10^4 + j0$$

Therefore, the poles of the Butterworth filter are,

$p_1 = [-2.9319 + j5.0782]\times10^4$,

$p_2 = -5.8638\times10^4$, and

$p_3 = [-2.9319 - j5.0782]\times10^4$

Step 6 of 8

The expression for the transfer function $T(s)$ is,

$$T(s) = \frac{k}{(s-p_1)(s-p_2)(s-p_3)}$$

Substitute $[-2.9319+j5.0782]\times10^4$ for p_1, -5.8638×10^4 for p_2, and $[-2.9319-j5.0782]\times10^4$ for p_3.

$$T(s) = \frac{k}{\left(s-\left(\left[\begin{matrix}-2.9319+\\ j5.0782\end{matrix}\right]\times10^4\right)\right)\left(s-(-5.8638\times10^4)\right)\left(s-\left(\left[\begin{matrix}-2.9319-\\ j5.0782\end{matrix}\right]\times10^4\right)\right)}$$
$$= \frac{k}{\left[\left(s+2.9319\times10^4\right)^2+5.0782^2\times10^8\right]\left[s+5.8638\times10^4\right]}$$
$$= \frac{k}{\left[\left(s^2+5.8638s+8.5960\times10^8\right)+25.7881\times10^8\right]\left[s+5.8638\times10^4\right]}$$
$$T(s) = \frac{k}{\left(s^2+5.8638s+34.384\times10^8\right)\left(s+5.8638\times10^4\right)} \quad \text{..... (1)}$$

Substitute 0 for *s*.

$$T(0) = \frac{k}{\left(0^2+5.8638(0)+34.384\times10^8\right)\left(0+5.8638\times10^4\right)}$$

Substitute 1 for $T(0)$.

$$1 = \frac{k}{\left(0^2+5.8638(0)+34.384\times10^8\right)\left(0+5.8638\times10^4\right)}$$
$$k = \left(34.384\times10^8\right)\times\left(5.8638\times10^4\right)$$
$$= 201.6216\times10^{12}$$

Step 7 of 8

Substitute 201.6216×10^{12} for *k* in equation (1).

$$T(s) = \frac{201.6216\times10^{12}}{\left(s^2+5.8638s+34.384\times10^8\right)\left(s+5.8638\times10^4\right)}$$

Therefore, the transfer function $T(s)$ is $\boxed{\frac{201.6216\times10^{12}}{\left(s^2+5.8638s+34.384\times10^8\right)\left(s+5.8638\times10^4\right)}}$.

Step 8 of 8

The expression for the attenuation is,

$$A(\omega_s) = 10\log\left[1+\varepsilon^2\left(\frac{f_s}{f_p}\right)^{2N}\right]$$

Substitute 20×10^3 for f_s, 10×10^3 for f_p, 1.4125 for ε, and 3 for *N*.

$$A(\omega_s) = 10\log\left[1+(1.4125)^2\left(\frac{20\times10^3}{10\times10^3}\right)^{2(3)}\right]$$
$$= 10\log\left[1+(1.4125)^2(2)^6\right]$$
$$= 10\log[1+127.69]$$
$$= 21.095 \text{ dB}$$

Therefore, the attenuation at $f_s = 20$ kHz is $\boxed{21.095 \text{ dB}}$.

Step 1 of 7

Design a Chebyshev filter whose transfer function is shown in equation 16.25 in the textbook.

$$T(s)=\left\{\left[\frac{\omega_p^5}{8.1408\left(s+0.2895\omega_p\right)\left(s^2+s0.4684\omega_p+0.4293\omega_p^2\right)}\right]\left(\frac{1}{s^2+s0.1789\omega_p+0.9883\omega_p^2}\right)\right\}$$

Here, $\omega_p=2\pi\times10^4$ rad/s .

Divide the function into three parts to design the circuit as a cascade connection of three circuits.

$$T(s)=\left\{\left[\frac{\omega_p^5}{8.1408\left(s+0.2895\omega_p\right)}\right]\left(\frac{1}{s^2+s0.4684\omega_p+0.4293\omega_p^2}\right)\left(\frac{1}{s^2+s0.1789\omega_p+0.9883\omega_p^2}\right)\right\}$$

$$=\left\{\left[\frac{0.2895\omega_p}{s+0.2895\omega_p}\right]\left(\frac{0.4293\omega_p^2}{s^2+s0.4684\omega_p+0.4293\omega_p^2}\right)\left(\frac{0.9883\omega_p^2}{s^2+s0.1789\omega_p+0.9883\omega_p^2}\right)\right\}$$

Step 2 of 7

Refer to Figure 16.13 (a) in the textbook for the op amp-RC realization of a first-order low pass filter. The following is the transfer function of the circuit:

$$T(s)=\frac{a_0}{s+\omega_0}$$

Compare the transfer function with $\frac{0.2895\omega_p}{s+0.2895\omega_p}$.

Make the dc gain of the circuit equal to unity.

The expression for the dc gain of the circuit is,

$$\text{DC gain}=-\frac{R_2}{R_1}$$

Consider that the resistance, $R_1=10\ \text{k}\Omega$. Therefore,

$$\frac{R_2}{R_1}=1$$
$$R_2=R_1$$
$$R_2=10\ \text{k}\Omega$$

Step 3 of 7

The resonant frequency of the first-order low pass filter is,

$$\omega_0=\frac{1}{CR_2}$$

Therefore,

$$0.2895\omega_p=\frac{1}{CR_2}$$

Substitute $2\pi\times10^4$ rad/s for ω_p , and $10\ \text{k}\Omega$ for R_2 to find the value of C .

$$0.2895\left(2\pi\times10^4\right)=\frac{1}{C\left(10\times10^3\right)}$$

$$C=\frac{1}{0.2895\left(2\pi\times10^4\right)\left(10\times10^3\right)}$$
$$=\frac{1}{181.8\times10^6}$$
$$=5.5\times10^{-9}$$
$$=5.5\ \text{nF}$$

Thus, the first-stage circuit is an op amp-RC realization of a first-order low pass filter as shown in Figure 16.13 (a) in the textbook with R_1 and R_2 equal to $\boxed{10\ \text{k}\Omega}$ and C equal to $\boxed{5.5\ \text{nF}}$.

Step 4 of 7

Refer to Figure 16.22 (a) in the textbook for the op amp-RC realization of a second-order low pass filter. The following is the transfer function of the circuit:

$$T(s)=\frac{\frac{KR_2}{C_4C_6R_1R_3R_5}}{s^2+s\frac{1}{C_6R_6}+\frac{R_2}{C_4C_6R_1R_3R_6}}$$

Compare the transfer function with $\frac{0.4293\omega_p^2}{s^2+s0.4684\omega_p+0.4293\omega_p^2}$.

Make the dc gain, K of the circuit equal to unity.

Assume that the resistances, R_1, R_2, R_3, and R_5 equal to $10\ \text{k}\Omega$.

The resonant frequency of the second-order low pass filter is,

$$\omega_0=\frac{1}{\sqrt{\frac{C_4C_6R_1R_3R_5}{R_2}}}$$

Therefore,

$$\sqrt{0.4293}\omega_p=\sqrt{\frac{R_2}{C_4C_6R_1R_3R_5}}$$

Step 5 of 7

Substitute $2\pi\times10^4$ rad/s for ω_p , and $10\ \text{k}\Omega$ for R_1, R_2, R_3, and R_5 to find the value of C .

$$\sqrt{0.4293}\left(2\pi\times10^4\right)=\frac{\sqrt{10\times10^3}}{C\sqrt{\left(10\times10^3\right)\left(10\times10^3\right)\left(10\times10^3\right)}}$$

$$C=\frac{10^2}{\sqrt{0.4293}\left(2\pi\times10^4\right)\left(10^6\right)}$$
$$=\frac{100}{41.168\times10^9}$$
$$=2.43\times10^{-9}$$
$$=2.43\ \text{nF}$$

The quality factor of the second-order low pass filter is,

$$Q=R_6\sqrt{\frac{C_6}{C_4}\frac{R_2}{R_1R_3R_5}}$$

Find the value of quality factor from the transfer function of the second-order low pass filter.

$$Q=\frac{\sqrt{0.4293}\omega_p}{0.4684\omega_p}$$
$$=\frac{0.6552}{0.4684}$$
$$=1.398$$

Substitute 1.398 for Q , 2.43 nF for C_4 and C_6 , and $10\ \text{k}\Omega$ for R_1, R_2, R_3, and R_5 to find the value of R_6 .

$$1.398=R_6\sqrt{\left(\frac{2.43\ \text{nF}}{2.43\ \text{nF}}\right)\left[\frac{10\ \text{k}\Omega}{(10\ \text{k}\Omega)(10\ \text{k}\Omega)(10\ \text{k}\Omega)}\right]}$$

$$R_6=1.398\times10^4$$
$$\approx14\ \text{k}\Omega$$

Thus, the second stage is an op amp-RC realization of a second-order low pass filter as shown in Figure 16.22 (a) in the textbook with R_1, R_2, R_3, and R_5 equal to $\boxed{10\ \text{k}\Omega}$, R_6 equal to $\boxed{14\ \text{k}\Omega}$ and C_4 and C_6 equal to $\boxed{2.43\ \text{nF}}$.

Step 6 of 7

Refer to Figure 16.22 (a) in the textbook for the op amp-RC realization of a second-order low pass filter.

Compare the transfer function of the circuit with $\frac{0.9883\omega_p^2}{s^2+s0.1789\omega_p+0.9883\omega_p^2}$.

Make the dc gain, K of the circuit equal to unity.

Assume that the resistances, R_1, R_2, R_3, and R_5 equal to $10\ \text{k}\Omega$.

The resonant frequency of the second-order low pass filter is,

$$\omega_0=\frac{1}{\sqrt{\frac{C_4C_6R_1R_3R_5}{R_2}}}$$

Therefore,

$$\sqrt{0.9883}\omega_p=\sqrt{\frac{R_2}{C_4C_6R_1R_3R_5}}$$

Substitute $2\pi\times10^4$ rad/s for ω_p , and $10\ \text{k}\Omega$ for R_1, R_2, R_3, and R_5 to find the value of C .

$$\sqrt{0.9883}\left(2\pi\times10^4\right)=\frac{\sqrt{10\times10^3}}{C\sqrt{\left(10\times10^3\right)\left(10\times10^3\right)\left(10\times10^3\right)}}$$

$$C=\frac{10^2}{\sqrt{0.9883}\left(2\pi\times10^4\right)\left(10^6\right)}$$
$$=\frac{100}{62.46\times10^9}$$
$$=1.6\times10^{-9}$$
$$=1.6\ \text{nF}$$

Step 7 of 7

The quality factor of the second-order low pass filter is,

$$Q=R_6\sqrt{\frac{C_6}{C_4}\frac{R_2}{R_1R_3R_5}}$$

Find the value of quality factor from the transfer function of the second-order low pass filter.

$$Q=\frac{\sqrt{0.9883}\omega_p}{0.1789\omega_p}$$
$$=\frac{0.9941}{0.1789}$$
$$=5.556$$

Substitute 5.556 for Q , 1.6 nF for C_4 and C_6 , and $10\ \text{k}\Omega$ for R_1, R_2, R_3, and R_5 to find the value of R_6 .

$$5.556=R_6\sqrt{\left(\frac{1.6\ \text{nF}}{1.6\ \text{nF}}\right)\left[\frac{10\ \text{k}\Omega}{(10\ \text{k}\Omega)(10\ \text{k}\Omega)(10\ \text{k}\Omega)}\right]}$$

$$R_6=5.556\times10^4$$
$$\approx55.6\ \text{k}\Omega$$

Thus, the third stage is an op amp-RC realization of a second-order low pass filter as shown in Figure 16.22 (a) in the textbook with R_1, R_2, R_3, and R_5 equal to $\boxed{10\ \text{k}\Omega}$, R_6 equal to $\boxed{55.6\ \text{k}\Omega}$ and C_4 and C_6 equal to $\boxed{1.6\ \text{nF}}$.

Step 1 of 14

The expression for the magnitude of the transfer function of an Nth-order Chebyshev filter is

$$\left|T(j\omega)\right| = \frac{1}{\sqrt{1+\varepsilon^2\cos^2\left[N\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}} \quad \ldots (1)$$

(a)

Consider the magnitude $\left|T(j\omega)\right|$.

$$\left|T(j\omega)\right| = \frac{1}{\sqrt{1+\varepsilon^2\cos^2\left[N\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}}$$

Substitute 6 for N

$$\left|T(j\omega)\right| = \frac{1}{\sqrt{1+\varepsilon^2\cos^2\left[6\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}} \quad \ldots\ldots (2)$$

Substitute 1 for $\left|T(j\omega)\right|$.

$$1 = \frac{1}{\sqrt{1+\varepsilon^2\cos^2\left[6\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}}$$

$$1+\varepsilon^2\cos^2\left[6\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 1$$

$$\cos^2\left[6\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 0$$

$$6\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = \cos^{-1}(0)$$

Simplify further.

$$\frac{\omega_i}{\omega_p} = \cos\left(\frac{(2i+1)\pi}{2\times 6}\right)$$

$$\omega_i = \omega_p \cos\left(\frac{(2i+1)\pi}{12}\right)$$

Here, $i = 0,1,\ldots,\frac{N-1}{2}$.

Substitute 6 for N.

$$\omega_i = \omega_p \cos\left(\frac{(2i+1)\pi}{12}\right) \quad \ldots\ldots (3)$$

Here, $i = 0,1,2$

Step 2 of 14

Recall equation (3).

$$\omega_i = \omega_p \cos\left(\frac{(2i+1)\pi}{12}\right)$$

Calculate frequency ω_0.

Substitute 0 for i.

$$\omega_0 = \omega_p \cos\left(\frac{(2\times 0+1)\pi}{12}\right)$$

$$= \omega_p \cos\left(\frac{\pi}{12}\right)$$

$$= (0.966)\omega_p \text{ rad/s}$$

Recall equation (3).

$$\omega_i = \omega_p \cos\left(\frac{(2i+1)\pi}{12}\right)$$

Calculate frequency ω_1.

Substitute 1 for i.

$$\omega_1 = \omega_p \cos\left(\frac{(2\times 1+1)\pi}{12}\right)$$

$$= \omega_p \cos\left(\frac{3\pi}{12}\right)$$

$$= (0.707)\omega_p \text{ rad/s}$$

Step 3 of 14

Recall equation (3).

$$\omega_i = \omega_p \cos\left(\frac{(2i+1)\pi}{12}\right)$$

Calculate frequency ω_2.

Substitute 2 for i.

$$\omega_2 = \omega_p \cos\left(\frac{(2\times 2+1)\pi}{12}\right)$$

$$= \omega_p \cos\left(\frac{5\pi}{12}\right)$$

$$= (0.259)\omega_p \text{ rad/s}$$

Therefore, the magnitude $\left|T(j\omega)\right|$ is 1 for the frequencies $(0.966)\omega_p$ rad/s, $(0.707)\omega_p$ rad/s, and $(0.259)\omega_p$ rad/s.

Step 4 of 14

Recall equation (2).

$$\left|T(j\omega)\right| = \frac{1}{\sqrt{1+\varepsilon^2\cos^2\left[6\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}}$$

Substitute $\frac{1}{\sqrt{1+\varepsilon^2}}$ for $\left|T(j\omega)\right|$.

$$\frac{1}{\sqrt{1+\varepsilon^2}} = \frac{1}{\sqrt{1+\varepsilon^2\cos^2\left[6\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}}$$

$$1+\varepsilon^2\cos^2\left[6\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 1+\varepsilon^2$$

$$\cos^2\left[6\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 1$$

$$6\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = \cos^{-1}(1)$$

Simplify further.

$$6\cos^{-1}\left(\frac{\omega_i}{\omega_p}\right) = i\pi$$

$$\frac{\omega_i}{\omega_p} = \cos\left(\frac{i\pi}{6}\right)$$

$$\omega_i = \omega_p \cos\left(\frac{i\pi}{6}\right)$$

Here, $i = 0,1,\ldots,\frac{N}{2}$.

Substitute 6 for N.

$$\omega_i = \omega_p \cos\left(\frac{i\pi}{6}\right) \quad (4)$$

Here, $i = 0,1,2,3$

Step 5 of 14

Recall equation (4).

$$\omega_i = \omega_p \cos\left(\frac{i\pi}{6}\right)$$

Calculate frequency ω_0.

Substitute 0 for i.

$$\omega_0 = \omega_p \cos\left(\frac{(0)\pi}{6}\right)$$

$$= \omega_p \cos(0^\circ)$$

$$= \omega_p \text{ rad/s}$$

Recall equation (4).

$$\omega_i = \omega_p \cos\left(\frac{i\pi}{6}\right)$$

Calculate frequency ω_1.

Substitute 1 for i.

$$\omega_1 = \omega_p \cos\left(\frac{1\times\pi}{6}\right)$$

$$= \omega_p \cos(30^\circ)$$

$$= (0.866)\omega_p \text{ rad/s}$$

Recall equation (4).

$$\omega_i = \omega_p \cos\left(\frac{i\pi}{6}\right)$$

Calculate frequency ω_2.

Substitute 2 for i.

$$\omega_2 = \omega_p \cos\left(\frac{2\times\pi}{6}\right)$$

$$= \omega_p \cos(60^\circ)$$

$$= (0.5)\omega_p \text{ rad/s}$$

Recall equation (4).

$$\omega_i = \omega_p \cos\left(\frac{i\pi}{6}\right)$$

Step 6 of 14

Calculate frequency ω_3.

Substitute 3 for i.

$$\omega_3 = \omega_p \cos\left(\frac{3\pi}{6}\right)$$

$$= 0 \text{ rad/s}$$

Therefore, the magnitude $\left|T(j\omega)\right|$ is $\frac{1}{\sqrt{1+\varepsilon^2}}$ for the frequencies ω_p rad/s, $(0.866)\omega_p$ rads, $(0.5)\omega_p$ rads, and 0 rad/s

Step 7 of 14

Sketch the magnitude $\left|T(j\omega)\right|$ of the transfer function for low-pass Chebyshev filter of sixth-order as shown in figure 1

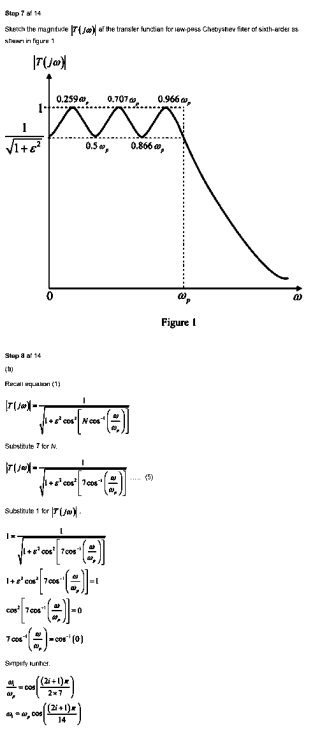


Figure 1

Step 8 of 14

(b)

Recall equation (1)

$$\left|T(j\omega)\right| = \frac{1}{\sqrt{1+\varepsilon^2\cos^2\left[N\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}}$$

Substitute 7 for N.

$$\left|T(j\omega)\right| = \frac{1}{\sqrt{1+\varepsilon^2\cos^2\left[7\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}} \quad \ldots\ldots (5)$$

Substitute 1 for $\left|T(j\omega)\right|$.

$$1 = \frac{1}{\sqrt{1+\varepsilon^2\cos^2\left[7\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}}$$

$$1+\varepsilon^2\cos^2\left[7\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 1$$

$$\cos^2\left[7\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 0$$

$$7\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = \cos^{-1}(0)$$

Simplify further.

$$\frac{\omega_i}{\omega_p} = \cos\left(\frac{(2i+1)\pi}{2\times 7}\right)$$

$$\omega_i = \omega_p \cos\left(\frac{(2i+1)\pi}{14}\right)$$

Here, $i = 0,1,\ldots,\frac{N-1}{2}$.

Substitute 7 for N.

$$\omega_i = \omega_p \cos\left(\frac{(2i+1)\pi}{14}\right) \quad \ldots (6)$$

Here, $i = 0,1,2,3$

Step 9 of 14

Recall equation (6).

$$\omega_i = \omega_p \cos\left(\frac{(2i+1)\pi}{14}\right)$$

Calculate frequency ω_0.

Substitute 0 for i.

$$\omega_0 = \omega_p \cos\left(\frac{(2\times 0+1)\pi}{14}\right)$$

$$= \omega_p \cos\left(\frac{\pi}{14}\right)$$

$$= (0.975)\omega_p \text{ rad/s}$$

Step 10 of 14

Recall equation (6).

$$\omega_i = \omega_p \cos\left(\frac{(2i+1)\pi}{14}\right)$$

Calculate frequency ω_1.

Substitute 1 for i.

$$\omega_1 = \omega_p \cos\left(\frac{(2\times 1+1)\pi}{14}\right)$$

$$= \omega_p \cos\left(\frac{3\pi}{14}\right)$$

$$= (0.781)\omega_p \text{ rad/s}$$

Recall equation (6).

$$\omega_i = \omega_p \cos\left(\frac{(2i+1)\pi}{14}\right)$$

Calculate frequency ω_2.

Substitute 2 for i.

$$\omega_2 = \omega_p \cos\left(\frac{(2\times 2+1)\pi}{14}\right)$$

$$= \omega_p \cos\left(\frac{5\pi}{14}\right)$$

$$= (0.434)\omega_p \text{ rad/s}$$

Recall equation (6).

$$\omega_i = \omega_p \cos\left(\frac{(2i+1)\pi}{14}\right)$$

Calculate frequency ω_3.

Substitute 3 for i.

$$\omega_3 = \omega_p \cos\left(\frac{(2\times 3+1)\pi}{14}\right)$$

$$= \omega_p \cos\left(\frac{7\pi}{14}\right)$$

$$= 0 \text{ rad/s}$$

Therefore, the magnitude $\left|T(j\omega)\right|$ is 1 for the frequencies $(0.975)\omega_p$ rads, $(0.781)\omega_p$ rads, $(0.434)\omega_p$ rads, and 0 rad/s

Step 11 of 14

Recall equation (5).

$$\left|T(j\omega)\right| = \frac{1}{\sqrt{1+\varepsilon^2\cos^2\left[7\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}}$$

Substitute $\frac{1}{\sqrt{1+\varepsilon^2}}$ for $\left|T(j\omega)\right|$.

$$\frac{1}{\sqrt{1+\varepsilon^2}} = \frac{1}{\sqrt{1+\varepsilon^2\cos^2\left[7\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}}$$

$$1+\varepsilon^2\cos^2\left[7\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 1+\varepsilon^2$$

$$\cos^2\left[7\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 1$$

$$7\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = \cos^{-1}(1)$$

Simplify further.

$$7\cos^{-1}\left(\frac{\omega_i}{\omega_p}\right) = i\pi$$

$$\frac{\omega_i}{\omega_p} = \cos\left(\frac{i\pi}{7}\right)$$

$$\omega_i = \omega_p \cos\left(\frac{i\pi}{7}\right)$$

Here, $i = 0,1,\ldots,\frac{N}{2}$.

Substitute 7 for N.

$$\omega_i = \omega_p \cos\left(\frac{i\pi}{7}\right) \quad (7)$$

Here, $i = 8,1,2,3$.

Step 12 of 14

Recall equation (7).

$$\omega_i = \omega_p \cos\left(\frac{i\pi}{7}\right)$$

Calculate frequency ω_0.

Substitute 0 for i.

$$\omega_0 = \omega_p \cos\left(\frac{8\times\pi}{7}\right)$$

$$= \omega_p \text{ rad/s}$$

Step 13 of 14

Recall equation (7).

$$\omega_i = \omega_p \cos\left(\frac{i\pi}{7}\right)$$

Calculate frequency ω_1.

Substitute 1 for i.

$$\omega_1 = \omega_p \cos\left(\frac{1\times\pi}{7}\right)$$

$$= \omega_p \cos\left(\frac{\pi}{7}\right)$$

$$= (0.901)\omega_p \text{ rad/s}$$

Recall equation (7).

$$\omega_i = \omega_p \cos\left(\frac{i\pi}{7}\right)$$

Calculate frequency ω_2.

Substitute 2 for i.

$$\omega_2 = \omega_p \cos\left(\frac{2\times\pi}{7}\right)$$

$$= \omega_p \cos\left(\frac{2\pi}{7}\right)$$

$$= (0.623)\omega_p \text{ rad/s}$$

Recall equation (7).

$$\omega_i = \omega_p \cos\left(\frac{i\pi}{7}\right)$$

Calculate frequency ω_3.

Substitute 3 for i.

$$\omega_3 = \omega_p \cos\left(\frac{3\times\pi}{7}\right)$$

$$= \omega_p \cos\left(\frac{3\pi}{7}\right)$$

$$= (0.222)\omega_p \text{ rad/s}$$

Therefore, the magnitude $\left|T(j\omega)\right|$ is $\frac{1}{\sqrt{1+\varepsilon^2}}$ for the frequencies ω_p rad/s, $(0.901)\omega_p$ rad/s, $(0.623)\omega_p$ rads, and $(0.222)\omega_p$ rad/s

Step 14 of 14

Sketch the magnitude $\left|T(j\omega)\right|$ of the transfer function for low-pass Chebyshev filter of seventh-order as shown in figure 2

Figure 2

Step 1 of 5

17.021E

The Kerwin-Huelsman-Newcomb (KHN) biquad circuit is shown in Figure 1.

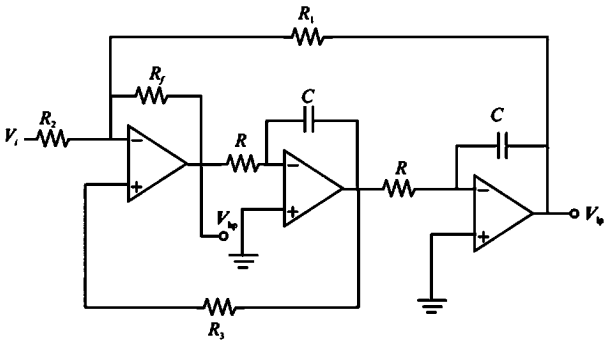


Figure 1

Step 2 of 5

The center frequency of the KHN biquad filter is given by

$$\omega_0 = \frac{1}{RC}$$
$$R = \frac{1}{2\pi f_0 C}$$

Substitute 1 nF for C, 10 kHz for f_0 in equation.

$$R = \frac{1}{2\pi \times 10 \times 10^3 \times 10^{-9}}$$
$$= 15.915\text{ k}\Omega$$

Thus, the value of R is, $\boxed{15.915\text{ k}\Omega}$.

Step 3 of 5

Consider the value of R_1, R_f and R_2 as $10\text{ k}\Omega$.

$$\frac{R_3}{R_2} = 2Q - 1$$
$$\frac{R_3}{10 \times 10^3} = 2 \times 2 - 1$$
$$\frac{R_3}{10 \times 10^3} = 3$$
$$R_3 = 30\text{ k}\Omega$$

Therefore, the parameter values are R is $15.915\text{ k}\Omega$, R_1, R_f and R_2 are $\boxed{10\text{ k}\Omega}$ and R_3 is $\boxed{30\text{ k}\Omega}$.

Step 4 of 5

The center frequency gain is given by

$$KQ = 2Q - 1$$

Substitute 2 for Q in equation.

$$KQ = 2 \times 2 - 1$$
$$= 3$$

Therefore, the center frequency gain of the KHN biquad filter is $\boxed{3}$.

Step 5 of 5

The high frequency gain is,

$$K = 2 - \frac{1}{Q}$$

Substitute 2 for Q in equation.

$$K = 2 - \frac{1}{2}$$
$$= 1.5$$

Therefore, the high frequency gain of the KHN biquad filter is $\boxed{1.5}$.

Step 1 of 10

The expression for the magnitude of the transfer function of a low-pass Butterworth filter is,

$$|T(j\omega)| = \frac{1}{\sqrt{1+\varepsilon^2\left(\frac{\omega}{\omega_p}\right)^{2N}}}$$

Substitute 5 for N.

$$|T(j\omega)| = \frac{1}{\sqrt{1+\varepsilon^2\left(\frac{\omega}{\omega_p}\right)^{2(5)}}}$$

$$|T(j\omega)| = \frac{1}{\sqrt{1+\varepsilon^2\left(\frac{\omega}{\omega_p}\right)^{10}}} \quad \ldots\ldots (1)$$

Substitute 1 for $|T(j\omega)|$.

$$1 = \frac{1}{\sqrt{1+\varepsilon^2\left(\frac{\omega}{\omega_p}\right)^{10}}}$$

$$1+\varepsilon^2\left(\frac{\omega}{\omega_p}\right)^{10} = 1$$

$$\frac{\omega}{\omega_p} = 0$$

$$\omega = 0 \text{ rad/s}$$

Therefore, the magnitude $|T(j\omega)|$ is 1 at $\omega = 0$ rad/s .

Step 2 of 10

Recall equation (1).

$$|T(j\omega)| = \frac{1}{\sqrt{1+\varepsilon^2\left(\frac{\omega}{\omega_p}\right)^{10}}}$$

Substitute $\frac{1}{\sqrt{1+\varepsilon^2}}$ for $|T(j\omega)|$.

$$\frac{1}{\sqrt{1+\varepsilon^2}} = \frac{1}{\sqrt{1+\varepsilon^2\left(\frac{\omega}{\omega_p}\right)^{10}}}$$

$$1+\varepsilon^2\left(\frac{\omega}{\omega_p}\right)^{10} = 1+\varepsilon^2$$

$$\left(\frac{\omega}{\omega_p}\right)^{10} = 1$$

$$\omega = \omega_p \text{ rad/s}$$

Therefore, the magnitude $|T(j\omega)|$ is $\frac{1}{\sqrt{1+\varepsilon^2}}$ at $\omega = \omega_p$ rad/s .

Step 3 of 10

The expression for the magnitude of the transfer function of a low-pass Chebyshev filter is,

$$|T(j\omega)| = \frac{1}{\sqrt{1+\varepsilon^2\cos^2\left[N\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}}$$

Substitute 5 for N.

$$|T(j\omega)| = \frac{1}{\sqrt{1+\varepsilon^2\cos^2\left[5\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}} \quad \ldots\ldots (2)$$

Substitute 1 for $|T(j\omega)|$.

$$1 = \frac{1}{\sqrt{1+\varepsilon^2\cos^2\left[5\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}}$$

$$1+\varepsilon^2\cos^2\left[5\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 1$$

$$\cos^2\left[5\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 0$$

$$5\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = \cos^{-1}(0)$$

Step 4 of 10

Simplify further.

$$\left(\frac{\omega_i}{\omega_p}\right) = \cos\left(\frac{(2i+1)\pi}{2\times 5}\right)$$

$$\omega_i = \omega_p \cos\left(\frac{(2i+1)\pi}{10}\right)$$

Here, $i = 0, 1, \ldots, \frac{N-1}{2}$.

Substitute 5 for N.

$$\omega_i = \omega_p \cos\left(\frac{(2i+1)\pi}{10}\right) \quad \ldots\ldots (3)$$

Here, $i = 0, 1, 2$.

Step 5 of 10

Recall equation (3).

$$\omega_i = \omega_p \cos\left(\frac{(2i+1)\pi}{10}\right)$$

Calculate frequency ω_0 .

Substitute 0 for i.

$$\omega_0 = \omega_p \cos\left(\frac{(2\times 0+1)\pi}{10}\right)$$

$$= \omega_p \cos\left(\frac{\pi}{10}\right)$$

$$= (0.951)\omega_p \text{ rad/s}$$

Recall equation (3).

$$\omega_i = \omega_p \cos\left(\frac{(2i+1)\pi}{10}\right)$$

Calculate frequency ω_1 .

Substitute 1 for i.

$$\omega_1 = \omega_p \cos\left(\frac{(2\times 1+1)\pi}{10}\right)$$

$$= \omega_p \cos\left(\frac{3\pi}{10}\right)$$

$$= (0.588)\omega_p \text{ rad/s}$$

Recall equation (3).

$$\omega_i = \omega_p \cos\left(\frac{(2i+1)\pi}{10}\right)$$

Calculate frequency ω_1 .

Substitute 2 for i.

$$\omega_2 = \omega_p \cos\left(\frac{(2\times 2+1)\pi}{10}\right)$$

$$= \omega_p \cos\left(\frac{5\pi}{10}\right)$$

$$= 0 \text{ rad/s}$$

Therefore, the magnitude $|T(j\omega)|$ is 1 for the frequencies $(0.951)\omega_p$ rad/s , $(0.588)\omega_p$ rad/s , and 0 rad/s.

Step 6 of 10

Recall equation (2).

$$|T(j\omega)| = \frac{1}{\sqrt{1+\varepsilon^2\cos^2\left[5\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}}$$

Substitute $\frac{1}{\sqrt{1+\varepsilon^2}}$ for $|T(j\omega)|$.

$$\frac{1}{\sqrt{1+\varepsilon^2}} = \frac{1}{\sqrt{1+\varepsilon^2\cos^2\left[5\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}}$$

$$1+\varepsilon^2\cos^2\left[5\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 1+\varepsilon^2$$

$$\cos^2\left[5\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 1$$

$$5\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = \cos^{-1}(1)$$

Simplify further.

$$\cos^{-1}\left(\frac{\omega_i}{\omega_p}\right) = \frac{i\pi}{5}$$

$$\omega_i = \omega_p \cos\left(\frac{i\pi}{5}\right)$$

Here, $i = 0, 1, \ldots, \frac{N}{2}$.

Substitute 5 for N.

$$\omega_i = \omega_p \cos\left(\frac{i\pi}{5}\right) \quad \ldots\ldots (4)$$

Here, $i = 0, 1, 2$.

Step 7 of 10

Recall equation (4).

$$\omega_i = \omega_p \cos\left(\frac{i\pi}{5}\right)$$

Calculate frequency ω_0 .

Substitute 0 for i.

$$\omega_0 = \omega_p \cos\left(\frac{0\times\pi}{5}\right)$$

$$= \omega_p \cos(0^\circ)$$

$$= \omega_p \text{ rad/s}$$

Recall equation (4).

$$\omega_i = \omega_p \cos\left(\frac{i\pi}{5}\right)$$

Calculate frequency ω_1 .

Substitute 1 for i.

$$\omega_1 = \omega_p \cos\left(\frac{1\times\pi}{5}\right)$$

$$= \omega_p \cos\left(\frac{\pi}{5}\right)$$

$$= (0.809)\omega_p \text{ rad/s}$$

Recall equation (4).

$$\omega_i = \omega_p \cos\left(\frac{i\pi}{5}\right)$$

Calculate frequency ω_2 .

Substitute 2 for i.

$$\omega_2 = \omega_p \cos\left(\frac{2\times\pi}{5}\right)$$

$$= \omega_p \cos\left(\frac{2\pi}{5}\right)$$

$$= (0.309)\omega_p \text{ rad/s}$$

Therefore, the magnitude $|T(j\omega)|$ is $\frac{1}{\sqrt{1+\varepsilon^2}}$ for the frequencies ω_p rad/s , $(0.809)\omega_p$ rad/s , and $(0.309)\omega_p$ rad/s .

Step 8 of 10

Sketch the magnitude of the transfer function of the Butterworth and the Chebyshev low-pass filters of fifth-order as shown in figure 1.

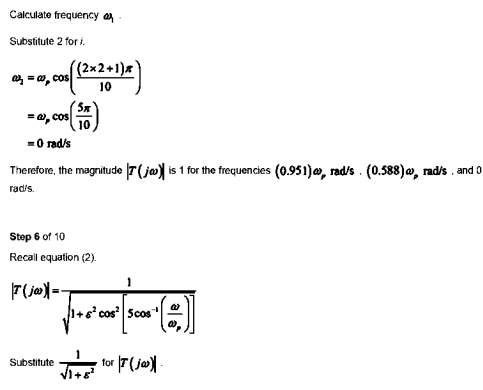


Figure 1

Step 9 of 10

As the values of A_{max} are same, assume the value of parameter ε as 1 for both Butterworth and Chebyshev filters. Assume that the edge of stop-band is at $\omega_s = 2\omega_p$.

The expression for the attenuation of the Butterworth filter at the edge of stop-band $\omega = \omega_s$ is,

$$A(\omega_s) = 10\log\left[1+\varepsilon^2\left(\frac{\omega_s}{\omega_p}\right)^{2N}\right]$$

Substitute $2\omega_p$ for ω_s , 1 for ε and 5 for N.

$$A(\omega_s) = 10\log\left[1+(1)^2\left(\frac{2\omega_p}{\omega_p}\right)^{2(5)}\right]$$

$$= 10\log\left[1+(2)^{10}\right]$$

$$= 30.107 \text{ dB}$$

Therefore, the attenuation of the Butterworth filter at the edge of stop-band $\omega = \omega_s$ is 30.107 dB.

Step 10 of 10

The expression for the attenuation of the Chebyshev filter at the edge of stop-band $\omega = \omega_s$ and $\omega \geq \omega_p$ is,

$$A(\omega_s) = 10\log\left[1+\varepsilon^2\cosh^2\left\{N\cosh^{-1}\left(\frac{\omega_s}{\omega_p}\right)\right\}\right]$$

Substitute $2\omega_p$ for ω_s , 1 for ε and 5 for N.

$$A(\omega_s) = 10\log\left[1+(1)^2\cosh^2\left\{5\cosh^{-1}\left(\frac{2\omega_p}{\omega_p}\right)\right\}\right]$$

$$= 10\log\left[1+\cosh^2\left\{5\cosh^{-1}(2)\right\}\right]$$

$$= 10\log[131045]$$

$$= 51.174 \text{ dB}$$

Therefore, the attenuation of the Chebyshev filter at the edge of stop-band $\omega = \omega_s$ is 51.174 dB.

Hence, the filter that gives greater attenuation at the stopband edge ω_s is **Chebyshev filter** .

Step 1 of 5

17.022E

The Kerwin-Huelsman-Newcomb (KHN) biquad circuit used as summing amplifier is shown in Figure 1.

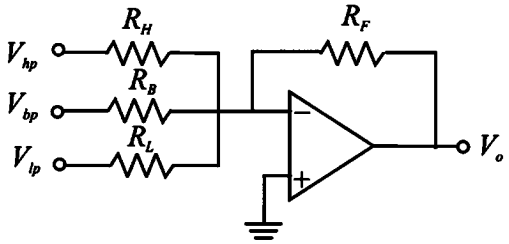


Figure 1

Step 2 of 5

The center frequency of the KHN biquad filter is given by

$$\omega_0 = \frac{1}{RC}$$

$$R = \frac{1}{2\pi f_0 C}$$

Substitute $5\ \text{kHz}$ for f_0 , $1\ \text{nF}$ for C in equation.

$$R = \frac{1}{2\pi \times 5\times 10^3 \times 10^{-9}}$$

$$= 31.83\ \text{k}\Omega$$

Step 3 of 5

Choose the values of R_1 , R_f and R_2 as $10\ \text{k}\Omega$.

$$\frac{R_3}{R_2} = 2Q - 1$$

$$\frac{R_3}{10\times 10^3} = 2\times 5 - 1$$

$$\frac{R_3}{10\times 10^3} = 9$$

$$R_3 = 90\ \text{k}\Omega$$

Step 4 of 5

Consider the expression.

$$R_L = R_H \left(\frac{\omega_0}{\omega_n}\right)^2$$

$$= R_H \left(\frac{f_0}{f_n}\right)^2$$

Substitute $8\ \text{kHz}$ for f_n , $5\ \text{kHz}$ for f_0 in equation.

$$10\times 10^3 = R_H \left(\frac{5}{8}\right)^2$$

$$R_H = 10\times 10^3 \times \left(\frac{8}{5}\right)^2$$

$$R_H = 25.6\ \text{k}\Omega$$

Step 5 of 5

For low pass notch filter, the dc gain is given by

$$K = 1 + \frac{R_F}{R_H \parallel R_L}.$$

$R_H \parallel R_L$ is given by

$$R_H \parallel R_L = \frac{R_H \times R_L}{R_H + R_L}$$

$$= \frac{25.6\times 10}{25.6+10}$$

$$= 7.19\ \text{k}\Omega$$

Therefore, K is ,

$$3 = 1 + \frac{R_F}{7.19}$$

$$\frac{R_F}{7.19} = 2$$

$$R_F = 14.38\ \text{k}\Omega$$

Therefore, the parameter values are R is $\boxed{31.83\ \text{k}\Omega}$, R_1 , R_f and R_2 are $\boxed{10\ \text{k}\Omega}$, R_3 is $\boxed{90\ \text{k}\Omega}$, R_H is $\boxed{25.6\ \text{k}\Omega}$, R_F is $\boxed{14.38\ \text{k}\Omega}$ and R_B is $\boxed{\infty}$.

Step 1 of 10

The expression for the magnitude of the transfer function of a Nth order Chebyshev filter is,

$$|T| = \frac{1}{\sqrt{1+\varepsilon^2 \cos^2\left[N\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}} \quad \text{..... (1)}$$

The maximum variation in passband transmission is,

$$A_{max} = 10\log\left(1+\varepsilon^2\right)$$

Substitute 0.5 for A_{max}.

$$0.5 = 10\log\left(1+\varepsilon^2\right)$$
$$\log\left(1+\varepsilon^2\right) = 0.05$$
$$1+\varepsilon^2 = 10^{0.05}$$
$$1+\varepsilon^2 = 1.122$$

Simplify further.

$$\varepsilon^2 = 1.122 - 1$$
$$\varepsilon^2 = 0.122$$
$$\varepsilon = 0.3493$$

Step 2 of 10

Recall equation (1).

$$|T| = \frac{1}{\sqrt{1+\varepsilon^2 \cos^2\left[N\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}}$$

Substitute 1 for $|T|$.

$$1 = \frac{1}{\sqrt{1+\varepsilon^2 \cos^2\left[N\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}}$$
$$1+\varepsilon^2 \cos^2\left[N\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 1$$
$$\cos^2\left[N\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 0$$
$$N\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = \cos^{-1}(0)$$

Substitute 1 for ω_p.

$$N\cos^{-1}\left(\frac{\omega}{1}\right) = \cos^{-1}(0)$$
$$N\cos^{-1}(\omega) = \frac{(2i+1)\pi}{2}$$
$$\omega_i = \cos\left(\frac{(2i+1)\pi}{2N}\right)$$

Here, $i = 0, 1, \ldots, \frac{N-1}{2}$.

Substitute 7 for N.

$$\omega_i = \cos\left(\frac{(2i+1)\pi}{2(7)}\right)$$
$$\omega_i = \cos\left(\frac{(2i+1)\pi}{14}\right) \quad \text{..... (2)}$$

Here, $i = 0, 1, 2, 3$.

Step 3 of 10

Recall equation (2).

$$\omega_i = \cos\left(\frac{(2i+1)\pi}{14}\right)$$

Calculate frequency ω_0.

Substitute 0 for i.

$$\omega_0 = \cos\left(\frac{(2(0)+1)\pi}{14}\right)$$
$$= \cos\left(\frac{\pi}{14}\right)$$
$$= 0.975 \text{ rad/s}$$

Recall equation (2).

$$\omega_i = \cos\left(\frac{(2i+1)\pi}{14}\right)$$

Calculate frequency ω_1.

Substitute 1 for i.

$$\omega_1 = \cos\left(\frac{(2(1)+1)\pi}{14}\right)$$
$$= \cos\left(\frac{3\pi}{14}\right)$$
$$= 0.782 \text{ rad/s}$$

Step 4 of 10

Recall equation (2).

$$\omega_i = \cos\left(\frac{(2i+1)\pi}{14}\right)$$

Calculate frequency ω_2.

Substitute 2 for i.

$$\omega_2 = \cos\left(\frac{(2(2)+1)\pi}{14}\right)$$
$$= \cos\left(\frac{5\pi}{14}\right)$$
$$= 0.433 \text{ rad/s}$$

Recall equation (2).

$$\omega_i = \cos\left(\frac{(2i+1)\pi}{14}\right)$$

Calculate frequency ω_3.

Substitute 3 for i.

$$\omega_3 = \cos\left(\frac{(2(3)+1)\pi}{14}\right)$$
$$= \cos\left(\frac{7\pi}{14}\right)$$
$$= 0 \text{ rad/s}$$

Therefore, the values of ω at $|T| = 1$ are,

$$\boxed{\begin{array}{l}\omega_0 = 0.975 \text{ rad/s}, \\ \omega_1 = 0.782 \text{ rad/s}, \\ \omega_2 = 0.433 \text{ rad/s, and} \\ \omega_3 = 0 \text{ rad/s}\end{array}}$$

Step 5 of 10

Recall equation (1).

$$|T| = \frac{1}{\sqrt{1+\varepsilon^2 \cos^2\left[N\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}}$$

Substitute $\frac{1}{\sqrt{1+\varepsilon^2}}$ for $|T|$.

$$\frac{1}{\sqrt{1+\varepsilon^2}} = \frac{1}{\sqrt{1+\varepsilon^2 \cos^2\left[N\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}}$$
$$1+\varepsilon^2 \cos^2\left[N\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 1+\varepsilon^2$$
$$\cos^2\left[N\cos^{-1}\left(\frac{\omega}{\omega_p}\right)\right] = 1$$
$$N\cos^{-1}\left(\frac{\omega}{\omega_p}\right) = \cos^{-1}(1)$$

Substitute 1 for ω_p.

$$N\cos^{-1}\left(\frac{\omega}{1}\right) = \cos^{-1}(1)$$
$$N\cos^{-1}(\omega) = i\pi$$
$$\omega_i = \cos\left(\frac{i\pi}{N}\right)$$

Here, $i = 0, 1, 2, \ldots, \frac{N}{2}$.

Substitute 7 for N.

$$\omega_i = \cos\left(\frac{i\pi}{7}\right) \quad \text{..... (3)}$$

Here, $i = 0, 1, 2, 3$.

Step 6 of 10

Recall equation (3).

$$\omega_i = \cos\left(\frac{i\pi}{7}\right)$$

Calculate frequency ω_0.

Substitute 0 for i.

$$\omega_0 = \cos\left(\frac{(0)\pi}{7}\right)$$
$$= \cos(0)$$
$$= 1 \text{ rad/s}$$

Step 7 of 10

Recall equation (3).

$$\omega_i = \cos\left(\frac{i\pi}{7}\right)$$

Calculate frequency ω_1.

Substitute 1 for i.

$$\omega_1 = \cos\left(\frac{(1)\pi}{7}\right)$$
$$= \cos\left(\frac{\pi}{7}\right)$$
$$= 0.9 \text{ rad/s}$$

Recall equation (3).

$$\omega_i = \cos\left(\frac{i\pi}{7}\right)$$

Calculate frequency ω_2.

Substitute 2 for i.

$$\omega_2 = \cos\left(\frac{(2)\pi}{7}\right)$$
$$= \cos\left(\frac{2\pi}{7}\right)$$
$$= 0.623 \text{ rad/s}$$

Recall equation (3).

$$\omega_i = \cos\left(\frac{i\pi}{7}\right)$$

Calculate frequency ω_3.

Substitute 3 for i.

$$\omega_3 = \cos\left(\frac{(3)\pi}{7}\right)$$
$$= \cos\left(\frac{3\pi}{7}\right)$$
$$= 0.222 \text{ rad/s}$$

Therefore, the values of ω for $|T| = \frac{1}{\sqrt{1+\varepsilon^2}}$ are,

$$\boxed{\begin{array}{l}\omega_0 = 1 \text{ rad/s}, \\ \omega_1 = 0.9 \text{ rad/s} \\ \omega_2 = 0.623 \text{ rad/s, and} \\ \omega_3 = 0.222 \text{ rad/s}\end{array}}$$

Step 8 of 10

Plot the magnitude $|T|$ of the seventh-order Chebyshev filter as shown in figure 1.

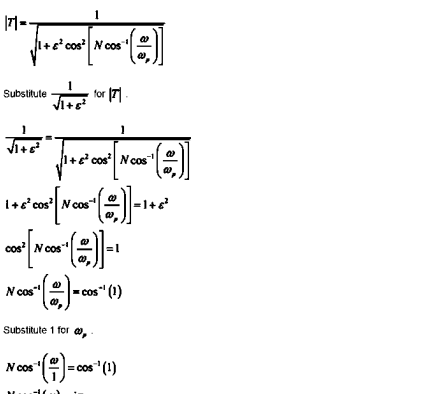


Figure 1

Step 9 of 10

The expression for the magnitude $|T|$ of the Nth - order Chebyshev filter for $\omega \geq \omega_p$ is,

$$|T| = \frac{1}{\sqrt{1+\varepsilon^2 \cosh^2\left[N\cosh^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}}$$

Substitute 2 for ω, 1 for ω_p, 0.3493 for ε and 7 for N.

$$|T| = \frac{1}{\sqrt{1+(0.3493)^2 \cosh^2\left[7\cosh^{-1}\left(\frac{2}{1}\right)\right]}}$$
$$= \frac{1}{\sqrt{1+0.122\cosh^2\left[7\cosh^{-1}(2)\right]}}$$
$$= \frac{1}{\sqrt{1+0.122\cosh^2(9.218)}}$$
$$= \frac{1}{1759.85}$$
$$= 5.68\times10^{-4}$$

Therefore, the magnitude $|T|$ of the seventh-order Chebyshev filter for $\omega = 2$ rad/s is $\boxed{5.68\times10^{-4}}$.

indicate this value as shown in figure 1.

Step 10 of 10

The expression for the rate at which the transmission rate decreases is,

$$\text{Rate} = 20\log\left(\frac{1}{2^N}\right)$$

Substitute 7 for N.

$$\text{Rate} = 20\log\left(\frac{1}{2^7}\right)$$
$$= 20\times(-2.1072)$$
$$= -42.14 \text{ dB/octave}$$

Therefore, the rate at which the transmission rate decreases is $\boxed{-42.14 \text{ dB/octave}}$.

Step 1 of 4

Refer to Figure 16.25(b) in the text book.

The center frequency of the KHN biquad filter is ,

$$\omega_0 = \frac{1}{RC}$$

$$C = \frac{1}{2\pi f_0 R}$$

Substitute **10 kHz for** f_0 , **20 for** Q in equation.

$$C = \frac{1}{2\pi \times 10 \times 10^3 \times 10 \times 10^3}$$
$$= 1.59 \text{ nF}$$

Thus the capacitor value is **1.59 nF** .

Step 2 of 4

The Q determining resistance R_d is given by

$$R_d = QR$$
$$= 20 \times 10$$
$$= 200 \text{ k}\Omega$$

Thus the resistance R_d is **200 kΩ** .

Step 3 of 4

The center frequency gain is given by

$KQ = 1$.

$$K = \frac{1}{Q}$$
$$= \frac{1}{20}$$

Step 4 of 4

The resistance R_g is given by

$$R_g = \frac{R}{K}$$
$$= \frac{10}{\frac{1}{20}}$$
$$= 200 \text{ k}\Omega$$

Therefore, the resistance R_g is **200 kΩ** .

Step 1 of 3

The expression for the parameter ε that determines the maximum variation in passband transmission is,

$$\varepsilon = \sqrt{10^{\frac{A_{max}}{10}} - 1}$$

Substitute 1 for A_{max}.

$$\begin{aligned}\varepsilon &= \sqrt{10^{\frac{1}{10}} - 1} \\ &= \sqrt{0.2589} \\ &= 0.5088\end{aligned}$$

The expression for the attenuation of the Butterworth filter is,

$$A_B(\omega_s) = 10\log\left[1+\varepsilon^2\left(\frac{\omega_s}{\omega_p}\right)^{2N}\right]$$

Substitute 0.5088 for ε, $2\omega_p$ for ω_s, and 6 for N.

$$\begin{aligned}A_B(\omega_s) &= 10\log\left[1+(0.5088)^2\left(\frac{2\omega_p}{\omega_p}\right)^{2(6)}\right] \\ &= 10\log\left[1+0.2589(2)^{12}\right] \\ &= 10\log(1061.45) \\ &= 30.26\ \text{dB}\end{aligned}$$

Therefore, the attenuation provided by the sixth-order Butterworth filter at $\omega_s = 2\omega_p$ is **30.26 dB**.

Step 2 of 3

The expression for the attenuation of the Chebyshev filter is,

$$A_C(\omega_s) = 10\log\left[1+\varepsilon^2\cosh^2\left\{N\cosh^{-1}\left(\frac{\omega_s}{\omega_p}\right)\right\}\right]$$

Substitute 0.5088 for ε, $2\omega_p$ for ω_s, and 6 for N.

$$\begin{aligned}A_C(\omega_s) &= 10\log\left[1+(0.5088)^2\cosh^2\left\{6\cosh^{-1}\left(\frac{2\omega_p}{\omega_p}\right)\right\}\right] \\ &= 10\log\left[1+(0.2588)\cosh^2\left\{6\cosh^{-1}(2)\right\}\right] \\ &= 10\log\left[1+(0.2588)\cosh^2\left\{7.901\right\}\right] \\ &= 10\log\left[472363.0188\right] \\ &= 56.742\ \text{dB}\end{aligned}$$

Therefore, the attenuation provided by the sixth-order Chebyshev filter at $\omega_s = 2\omega_p$ is **56.742 dB**.

Step 3 of 3

Plot the magnitude $|T|$ for both Butterworth and Chebyshev filters on same axes as shown in figure 1.

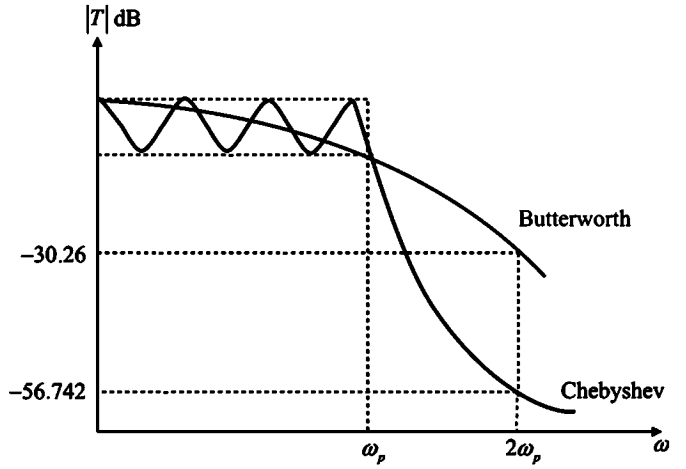


Figure 1

Step 1 of 4

17.024E

Refer to Figure 16.26 in the text book.

The center frequency of the KHN biquad filter is ,

$$\omega_0 = \frac{1}{RC}$$

$$R = \frac{1}{\omega_0 C}$$

Substitute 10^4 **rad/s** for ω_0 , **10 nF** for C in equation.

$$R = \frac{1}{10^4 \times 10 \times 10^{-9}}$$
$$= 10\ \text{k}\Omega$$

Step 2 of 4

Consider the equation.

$$R_d = QR$$

Substitute **5** for Q , **10 kΩ** for R in equation.

$$R_d = 5 \times 10$$
$$= 50\ \text{k}\Omega$$

The capacitance C_1 is ,

$$C_1 = C \times \textbf{flat gain}$$

Substitute 1 for flat gain in equation.

$$C_1 = 10 \times 1\ \text{nF}$$
$$= 10\ \text{nF}$$

Step 3 of 4

The resistance R_2 is,

$$R_2 = \frac{R}{\textbf{gain}}$$

Substitute 1 for gain, **10 kΩ** for R in equation.

$$R_2 = \frac{10\ \text{k}}{1}$$
$$= 10\ \text{k}\Omega$$

Step 4 of 4

The resistance R_3 is given by

$$R_3 = \frac{Qr}{\textbf{gain}}$$

Substitute **5** for Q , **10 kΩ** for r and 1 for gain.

$$R_3 = \frac{5 \times 10}{1}$$
$$= 50\ \text{k}\Omega$$

Therefore, the design parameters C_1 is $\boxed{\textbf{10 nF}}$, R_d is $\boxed{\textbf{50 kΩ}}$, R is $\boxed{\textbf{10 kΩ}}$, R_1 is $\boxed{\infty}$ R_2 is $\boxed{\textbf{10 kΩ}}$ and R_3 is $\boxed{\textbf{50 kΩ}}$.

Step 1 of 5

(a)

The parameter ε which determines the maximum variation in passband transmission is,

$$\begin{aligned}\varepsilon &= \sqrt{10^{\frac{A_{min}}{10}} - 1}\\ &= \sqrt{10^{\frac{1}{10}} - 1}\\ &= \sqrt{0.2589}\\ &= 0.5088\end{aligned}$$

The magnitude of the transfer function of an N[th] order Chebyshev filter for $\omega \geq \omega_p$ is,

$$|T(j\omega)| = \frac{1}{\sqrt{1+\varepsilon^2 \cosh^2\left[N\cosh^{-1}\left(\frac{\omega}{\omega_p}\right)\right]}}.$$

The attenuation achieved at the stopband edge is,

$$\begin{aligned}A(\omega_s) &= 10\log\left[1+\varepsilon^2\cosh^2\left[N\cosh^{-1}\left(\frac{\omega_s}{\omega_p}\right)\right]\right]\\ &= 10\log\left[1+(0.5088)^2\cosh^2\left[N\cosh^{-1}\left(\frac{2\pi f_s}{2\pi f_p}\right)\right]\right] \qquad \text{(Since, } \varepsilon = 0.5088\text{)}\\ &= 10\log\left[1+(0.5088)^2\cosh^2\left[N\cosh^{-1}\left(\frac{2\pi(4\times10^3)}{2\pi(3.4\times10^3)}\right)\right]\right]\\ &= 10\log\left[1+(0.5088)^2\cosh^2(0.5857N)\right]\end{aligned}$$

Step 2 of 5

For $N = 8$, the stop band attenuation is,

$$\begin{aligned}A(\omega_s) &= 10\log\left[1+(0.5088)^2\cosh^2(0.5857(8))\right]\\ &= 28.8\text{ dB}\end{aligned}$$

For $N = 9$, the stop band attenuation is,

$$\begin{aligned}A(\omega_s) &= 10\log\left[1+(0.5088)^2\cosh^2(0.5857(9))\right]\\ &= 33.9\text{ dB}\end{aligned}$$

For $N = 10$, the stop band attenuation is,

$$\begin{aligned}A(\omega_s) &= 10\log\left[1+(0.5088)^2\cosh^2(0.5857(10))\right]\\ &= 39\text{ dB}\end{aligned}$$

Thus the order of the Chebyshev filter above 35 dB is, $\boxed{N = 10}$.

The excess attenuation obtained is,

$$\begin{aligned}\text{Excess attenuation} &= 39 - 35\\ &= 4\text{ dB}\end{aligned}$$

Thus, the excess attenuation obtained is $\boxed{4\text{ dB}}$.

Step 3 of 5

(b)

The poles of the transfer function are,

$$\begin{aligned}p_k &= -\omega_p \sin\left(\frac{2k-1}{N}\frac{\pi}{2}\right)\sinh\left(\frac{1}{N}\sinh^{-1}\frac{1}{\varepsilon}\right) + j\omega_p\cos\left(\frac{2k-1}{N}\frac{\pi}{2}\right)\cosh\left(\frac{1}{N}\sinh^{-1}\frac{1}{\varepsilon}\right)\\ &= -2\pi f_p \sin\left(\frac{2k-1}{N}\frac{\pi}{2}\right)\sinh\left(\frac{1}{N}\sinh^{-1}\frac{1}{\varepsilon}\right) + j2\pi f_p\cos\left(\frac{2k-1}{N}\frac{\pi}{2}\right)\cosh\left(\frac{1}{N}\sinh^{-1}\frac{1}{\varepsilon}\right).\end{aligned}$$

Substitute 0.5088 for ε, 3.4 kHz for f_p and 10 for N in the equation.

$$\begin{aligned}P_k &= \begin{bmatrix}-2\pi(3.4\times10^3)\sin\left(\frac{2k-1}{10}\frac{\pi}{2}\right)\sinh\left(\frac{1}{10}\sinh^{-1}\frac{1}{0.5088}\right) +\\ j2\pi(3.4\times10^3)\cos\left(\frac{2k-1}{10}\frac{\pi}{2}\right)\cosh\left(\frac{1}{10}\sinh^{-1}\frac{1}{0.5088}\right)\end{bmatrix}\\ &= 21.36\times10^3\sin\left(\frac{2k-1}{10}\frac{\pi}{2}\right)\sinh(0.1426) + j21.36\times10^3\cos\left(\frac{2k-1}{10}\frac{\pi}{2}\right)\cosh(0.1426)\\ &= 21.36\times10^3\sin\left(\frac{2k-1}{10}\frac{\pi}{2}\right)(0.143) + j21.36\times10^3\cos\left(\frac{2k-1}{10}\frac{\pi}{2}\right)(1.0102)\\ &= 3.06\times10^3\sin\left(\frac{2k-1}{10}\frac{\pi}{2}\right) + j21.5775\times10^3\cos\left(\frac{2k-1}{10}\frac{\pi}{2}\right)(1.0102)\end{aligned}$$

The pole P_1 is,

$$\begin{aligned}P_1 &= 3.06\times10^3\sin\left(\frac{2(1)-1}{10}\frac{\pi}{2}\right) + j21.5775\times10^3\cos\left(\frac{2(1)-1}{10}\frac{\pi}{2}\right)(1.0102)\\ &= 478.7 + j3389.4\end{aligned}$$

The pole P_2 is,

$$\begin{aligned}P_2 &= 3.06\times10^3\sin\left(\frac{2(2)-1}{10}\frac{\pi}{2}\right) + j21.5775\times10^3\cos\left(\frac{2(2)-1}{10}\frac{\pi}{2}\right)(1.0102)\\ &= 1.3892\times10^3 + j10.168\times10^3\end{aligned}$$

Step 4 of 5

The pole P_3 is,

$$\begin{aligned}P_3 &= 3.06\times10^3\sin\left(\frac{2(3)-1}{10}\frac{\pi}{2}\right) + j21.5775\times10^3\cos\left(\frac{2(3)-1}{10}\frac{\pi}{2}\right)(1.0102)\\ &= 2.1637\times10^3 + j16.947\times10^3\end{aligned}$$

The pole P_4 is,

$$\begin{aligned}P_4 &= 3.06\times10^3\sin\left(\frac{2(4)-1}{10}\frac{\pi}{2}\right) + j21.5775\times10^3\cos\left(\frac{2(4)-1}{10}\frac{\pi}{2}\right)(1.0102)\\ &= 2.7265\times10^3 + j23.726\times10^3\end{aligned}$$

The pole P_5 is,

$$\begin{aligned}P_5 &= 3.06\times10^3\sin\left(\frac{2(5)-1}{10}\frac{\pi}{2}\right) + j21.5775\times10^3\cos\left(\frac{2(5)-1}{10}\frac{\pi}{2}\right)(1.0102)\\ &= 3.0223\times10^3 + j30.504\times10^3\end{aligned}$$

The remaining poles are the complex conjugates of the poles P_1 to P_5

$$\begin{aligned}P_1^* &= 478.7 - j3389.4\\ P_2^* &= 1.3892\times10^3 - j10.168\times10^3\\ P_3^* &= 2.1637\times10^3 - j16.947\times10^3\\ P_4^* &= 2.7265\times10^3 - j23.726\times10^3\\ P_5^* &= 3.0223\times10^3 - j30.504\times10^3\end{aligned}$$

Thus, the poles are,

$$\boxed{\begin{aligned}&P_1^* = 478.7 - j3389.4, P_2^* = 1.3892\times10^3 - j10.168\times10^3, P_3^* = 2.1637\times10^3 - j16.947\times10^3,\\ &P_4^* = 2.7265\times10^3 - j23.726\times10^3, P_5^* = 3.0223\times10^3 - j30.504\times10^3, P_1 = 478.7 + j3389.4\\ &P_2 = 1.3892\times10^3 + j10.168\times10^3, P_3 = 2.1637\times10^3 + j16.947\times10^3,\\ &P_4 = 2.7265\times10^3 + j23.726\times10^3, P_5 = 3.0223\times10^3 + j30.504\times10^3\end{aligned}}$$

Step 5 of 5

The transfer function of the chebyshev filter is,

$$\begin{aligned}T(s) &= \frac{K\omega_p^N}{\varepsilon 2^{N-1}(s-P_1)(s-P_2)\ldots(s-P_N)}\\ &= \frac{K(2\pi f_p)}{\left[\begin{matrix}(0.5088)2^{10-1}(s-P_1)(s-P_1^*)(s-P_2)(s-P_2^*)\\ (s-P_3)(s-P_3^*)(s-P_4)(s-P_4^*)(s-P_5)(s-P_5^*)\end{matrix}\right]}\\ &= \frac{K(2\pi(3.4\times10^3))^{10}}{\left[\begin{matrix}(0.5088)2^9(s-P_1)(s-P_1^*)(s-P_2)(s-P_2^*)\\ (s-P_3)(s-P_3^*)(s-P_4)(s-P_4^*)(s-P_5)(s-P_5^*)\end{matrix}\right]}\end{aligned}$$

The dc gain is,

$$\begin{aligned}K &= \frac{1}{1+\varepsilon^2}\\ &= \frac{1}{\sqrt{1+(0.5088)^2}}\\ &= 0.8913\end{aligned}$$

Thus, the transfer function of the chebyshev filter is,

$$\boxed{T(s) = \frac{0.8913(2\pi(3.4\times10^3))^{10}}{\left[\begin{matrix}(0.5088)2^9(s-P_1)(s-P_1^*)(s-P_2)(s-P_2^*)\\ (s-P_3)(s-P_3^*)(s-P_4)(s-P_4^*)(s-P_5)(s-P_5^*)\end{matrix}\right]}}$$

Here,

$$\begin{aligned}&P_1^* = 478.7 - j3389.4, P_2^* = 1.3892\times10^3 - j10.168\times10^3, P_3^* = 2.1637\times10^3 - j16.947\times10^3,\\ &P_4^* = 2.7265\times10^3 - j23.726\times10^3, P_5^* = 3.0223\times10^3 - j30.504\times10^3, P_1 = 478.7 + j3389.4\\ &P_2 = 1.3892\times10^3 + j10.168\times10^3, P_3 = 2.1637\times10^3 + j16.947\times10^3,\\ &P_4 = 2.7265\times10^3 + j23.726\times10^3, P_5 = 3.0223\times10^3 + j30.504\times10^3\end{aligned}$$

Step 1 of 3

17.025E

Refer to Figure 16.29 in the text book.

The resonant frequency of the active-filter feedback loop generated using the bridged-T network is given by

$$\omega_0 = \frac{1}{\sqrt{C_1 C_2 R_3 R_4}}$$

Substitute $1\ \text{nF}$ for C_2 , $1\ \text{nF}$ for C_1 , $10^4\ \text{rad/s}$ for ω_0 in equation.

$$10^4 = \frac{1}{\sqrt{10^{-9} \times 10^{-9} R_3 R_4}}$$

$$10^4 = \frac{1}{10^{-9}\sqrt{R_3 R_4}}$$

$$\sqrt{R_3 R_4} = \frac{1}{10^{-9} 10^4}$$

$$\sqrt{R_3 R_4} = 10^5$$

$$R_3 R_4 = 10^{10} \quad \text{..... (1)}$$

Step 2 of 3

The quality factor is given by

$$Q = \left[\frac{1}{\omega_0 R_3}\left(\frac{1}{C_1} + \frac{1}{C_2}\right)\right]^{-1}.$$

Substituting the given values we get

$$1 = \left[\frac{1}{10^4 R_3}\left(\frac{1}{1\times10^{-9}} + \frac{1}{1\times10^{-9}}\right)\right]^{-1}$$

$$1 = \left[\frac{1}{10^4 R_3} 2\times10^9\right]^{-1}$$

$$1 = \frac{1}{10^4 R_3} 2\times10^9$$

Thus the resistance value is

$$R_3 = \frac{1}{10^4} 2\times10^9$$
$$= 200\ \text{k}\Omega$$

Step 3 of 3

Substitute $200\ \text{k}\Omega$ for R_3 in equation (1).

$$R_4 = \frac{10^{10}}{R_3}$$
$$= \frac{10^{10}}{200\times10^3}$$
$$= 50\ \text{k}\Omega$$

Therefore, the design parameters are R_3 is $\boxed{200\ \text{k}\Omega}$, and R_4 is $\boxed{50\ \text{k}\Omega}$.

Step 1 of 4

Refer to Figure 17.13 for the first-order RC low pass filter in the textbook.

The input resistance is,

$R_{in} = 12\ \text{k}\Omega$.

Consider the circuit from Figure 17.13 and its input resistance is,

$$\begin{aligned} R_1 &= R_{in} \\ &= 12\ \text{k}\Omega \end{aligned}$$

Therefore, the value of resistance R_1 is $\boxed{12\ \text{k}\Omega}$.

Step 2 of 4

The expression for the DC gain magnitude is,

$$|\text{DC gain}| = \frac{R_2}{R_1}$$

Substitute 10 for DC gain, 12×10^3 for R_1.

$$\begin{aligned} 10 &= \frac{R_2}{12\times10^3} \\ R_2 &= 120\ \text{k}\Omega \end{aligned}$$

Therefore, the value of resistance R_2 is $\boxed{120\ \text{k}\Omega}$.

Step 3 of 4

The expression for 3-dB frequency is,

$$f_o = \frac{1}{2\pi R_2 C}$$

$$C = \frac{1}{2\pi R_2 f_o}$$

Substitute 5×10^3 for f_o, 120×10^3 for R_2.

$$\begin{aligned} C &= \frac{1}{2\pi\times120\times10^3\times5\times10^3} \\ &= \frac{10^{-6}}{3769.911} \\ &= 265.26\ \text{pF} \end{aligned}$$

Therefore, the value of capacitance C is $\boxed{265.26\ \text{pF}}$.

Step 4 of 4

Design the first-order op amp-RC low pass filter as shown in Figure 1.

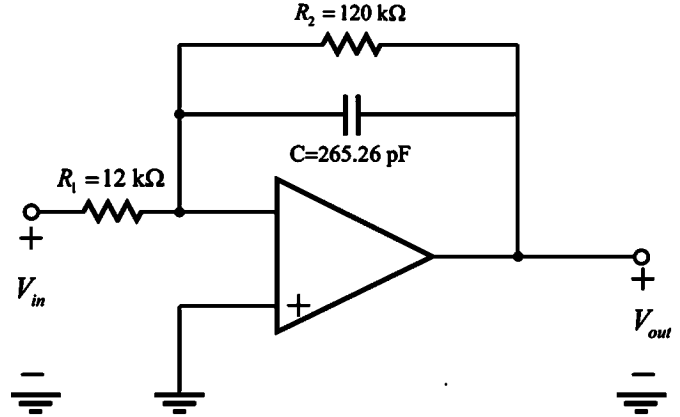


Figure 1

Hence, the required first-order operation amplifier-RC low-pass filter is designed as shown in Figure 1.

17.026E

Refer to Figure 16.29 in the text book.

The poles of the RC network are given by the equation

$$s^2+s\left(\frac{1}{C_1R_3}+\frac{1}{C_2R_3}+\frac{1}{C_1R_4}\right)+\frac{1}{C_1C_2R_3R_4}=0$$

Substitute $50\ \text{k}\Omega$ for R_4 , $200\ \text{k}\Omega$ for R_3 , $1\ \text{nF}$ for C_1 and $1\ \text{nF}$ for C_2 in equation.

$$s^2+s\left(\frac{1}{10^{-9}\times200\times10^3}+\frac{1}{10^{-9}\times200\times10^3}+\frac{1}{10^{-9}\times50\times10^3}\right)+\frac{1}{10^{-9}\times10^{-9}\times200\times10^3\times50\times10^3}=0$$

$$s^2+s\left(5000+5000+20000\right)+10^8=0$$

$$s^2+30000s+10^8=0$$

The roots of the equation are

$$s_1=-0.382\times10^4$$

$$s_2=-2.618\times10^4$$

Thus the location of the poles of the RC network in the feedback loop is

$$\boxed{-0.382\times10^4\ \text{rad/s and}\ -2.618\times10^4\ \text{rad/s}}$$

Step 1 of 4

17.026P

Refer to figure 17.13 for the first-order op amp-RC high-pass filter given in the textbook.

The input resistance of high-pass RC filter is,

$R_{in} = 120\ \text{k}\Omega$.

Consider the circuit from figure 17.13 and its input resistance is,

$R_{in} = R_1$.

The value of resistance R_1 is $\boxed{120\ \text{k}\Omega}$.

Step 2 of 4

The expression for the magnitude of high frequency gain is,

$$|\text{High frequency gain}| = \frac{R_2}{R_1}$$

Substitute 120×10^3 for R_1, and 1 for high frequency gain.

$$|1| = \frac{R_2}{120\times10^3}$$
$$R_2 = 120\ \text{k}\Omega$$

Therefore, the value of resistance R_2 is $\boxed{120\ \text{k}\Omega}$.

Step 3 of 4

The expression for 3-dB frequency,

$$f_o = \frac{1}{2\pi R_1 C}$$
$$C = \frac{1}{2\pi f_o R_1}$$

Substitute 120×10^3 for R_1, and 200 for f_o.

$$C = \frac{1}{2\pi\times200\times120\times10^3}$$
$$= 6.63\ \text{nF}$$

Therefore, the value of capacitance C is $\boxed{6.63\ \text{nF}}$.

Step 4 of 4

Design the first-order op amp-RC high-pass filter as shown in figure 1.

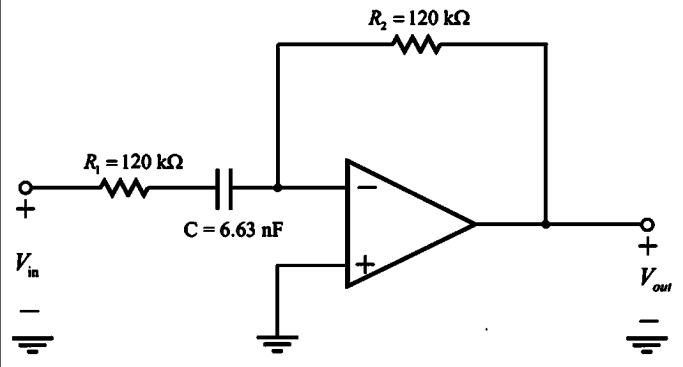


Figure 1

Step 1 of 4

Refer to Figure 16.30(a) in the text book.

The center frequency gain is,

$$KQ = 1$$

Substitute 1 for Q in equation.

$$K(1) = 1$$
$$K = 1$$

Step 2 of 4

Consider the expression.

$$\frac{\alpha}{C_1 R_4} = K\omega_0$$
$$\frac{\alpha}{R_4} = C_1 K\omega_0$$

Substitute $10^4\ \text{rad/s}$ for ω_0, 1 for K, $1\ \text{nF}$ for C_1 in equation.

$$\frac{\alpha}{R_4} = 10^{-9} \times 1 \times 10^4$$
$$= 10^{-5}$$

$$\frac{R_4}{\alpha} = 10^5$$
$$= 100\ \text{k}\Omega$$

Step 3 of 4

Rearrange the equation.

$$\alpha = R_4 \times 10^{-5}$$
$$= 50 \times 10^3 \times 10^{-5}$$
$$= 0.5$$

Step 4 of 4

Consider the expression.

$$\frac{R_4}{1-\alpha} = \frac{50 \times 10^3}{1-0.5}$$
$$= 100\ \text{k}\Omega$$

Therefore, the required values are $\frac{R_4}{\alpha}$, $\frac{R_4}{1-\alpha}$ are $\boxed{100\ \text{k}\Omega}$ and $\boxed{100\ \text{k}\Omega}$ respectively.

Step 1 of 6

Consider the following circuit:

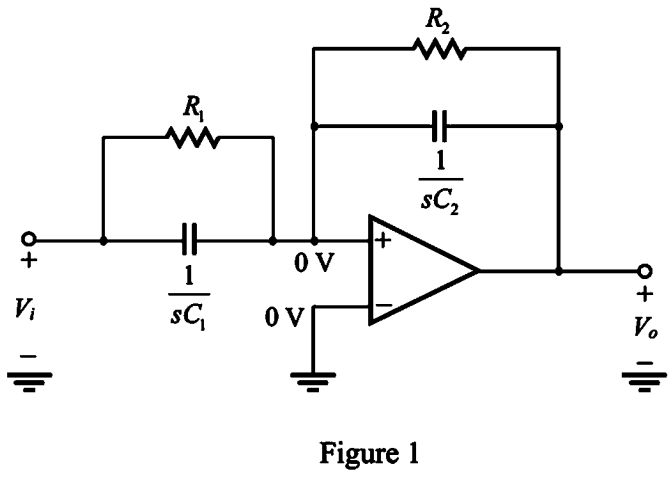


Figure 1

Step 2 of 6

Apply Nodal analysis at the non-inverting terminal of voltage 0 V.

$$\frac{0-V_i}{R_1}+\frac{0-V_i}{\left(\frac{1}{sC_1}\right)}+\frac{0-V_o}{R_2}+\frac{0-V_o}{\left(\frac{1}{sC_2}\right)}=0$$

$$-\frac{V_i}{R_1}-sC_1V_i-\frac{V_o}{R_2}-sC_2V_o=0$$

$$V_o\left(\frac{1}{R_2}+sC_2\right)=-V_i\left(\frac{1}{R_1}+sC_1\right)$$

$$\frac{V_o}{V_i}=\frac{-\left(\frac{1}{R_1}+sC_1\right)}{\left(\frac{1}{R_2}+sC_2\right)}$$

Simplify further.

$$T(s)=\frac{-C_1\left(s+\frac{1}{R_1C_1}\right)}{C_2\left(s+\frac{1}{R_2C_2}\right)} \quad \text{..... (1)}$$

Therefore, the transfer function $T(s)$ is $\boxed{\frac{-C_1\left(s+\frac{1}{R_1C_1}\right)}{C_2\left(s+\frac{1}{R_2C_2}\right)}}$.

Step 3 of 6

Consider the transfer function $T(s)$.

$$T(s)=\frac{-C_1\left(s+\frac{1}{R_1C_1}\right)}{C_2\left(s+\frac{1}{R_2C_2}\right)}$$

Equate the numerator to zero and determine the frequency of transmission zero.

$$s+\frac{1}{R_1C_1}=0$$

$$s=\frac{-1}{R_1C_1}$$

$$|s|=\left|\frac{-1}{R_1C_1}\right|$$

$$\omega_z=\frac{1}{R_1C_1}\ \text{rad/s}$$

Therefore, the frequency of transmission zero ω_z is $\boxed{\frac{1}{R_1C_1}\ \text{rad/s}}$.

Step 4 of 6

Consider the transfer function $T(s)$.

$$T(s)=\frac{-C_1\left(s+\frac{1}{R_1C_1}\right)}{C_2\left(s+\frac{1}{R_2C_2}\right)}$$

Equate the denominator to zero and determine the frequency of transmission pole.

$$s+\frac{1}{R_2C_2}=0$$

$$s=\frac{-1}{R_2C_2}$$

$$|s|=\left|\frac{-1}{R_2C_2}\right|$$

$$\omega_p=\frac{1}{R_2C_2}\ \text{rad/s}$$

Therefore, the frequency of transmission pole ω_p is $\boxed{\frac{1}{R_2C_2}\ \text{rad/s}}$.

Step 5 of 6

Recall equation (1).

$$T(s)=\frac{-C_1\left(s+\frac{1}{R_1C_1}\right)}{C_2\left(s+\frac{1}{R_2C_2}\right)}$$

Calculate DC gain.

Substitute 0 for *s*.

$$T(0)=\frac{-C_1\left(0+\frac{1}{R_1C_1}\right)}{C_2\left(0+\frac{1}{R_2C_2}\right)}$$

$$=\frac{\left(\frac{-C_1}{R_1C_1}\right)}{\left(\frac{C_2}{R_2C_2}\right)}$$

$$=\frac{-R_2}{R_1}$$

Therefore, the DC gain is $\boxed{\frac{-R_2}{R_1}}$.

Step 6 of 6

Recall equation (1).

$$T(s)=\frac{-C_1\left(s+\frac{1}{R_1C_1}\right)}{C_2\left(s+\frac{1}{R_2C_2}\right)}$$

$$=\frac{-sC_1\left(1+\frac{1}{sR_1C_1}\right)}{sC_2\left(1+\frac{1}{sR_2C_2}\right)}$$

$$=\frac{-C_1\left(1+\frac{1}{sR_1C_1}\right)}{C_2\left(1+\frac{1}{sR_2C_2}\right)}$$

Calculate high frequency gain.

Substitute ∞ for *s*.

$$T(\infty)=\frac{-C_1\left(1+\frac{1}{\infty(R_1C_1)}\right)}{C_2\left(1+\frac{1}{\infty(R_2C_2)}\right)}$$

$$=\frac{-C_1(1+0)}{C_2(1+0)}$$

$$=\frac{-C_1}{C_2}$$

Therefore, the high frequency gain is $\boxed{\frac{-C_1}{C_2}}$.

Step 1 of 4

Refer to Figure 16.34 (c) in the text book.

Redraw the circuit diagram as shown in Figure 1.

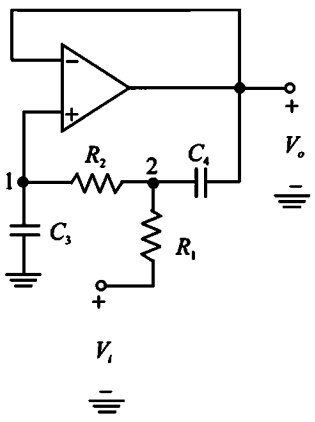


Figure 1

From Figure 1, it is clear that is,

The out put voltage is equal to voltage at node 1.

$$V_o = V_1$$

Apply Kirchhoff's current law at node 1.

$$\frac{V_1}{\frac{1}{sC_3}} + \frac{V_1 - V_2}{R_2} = 0$$

$$\frac{V_o}{\frac{1}{sC_3}} + \frac{V_o - V_2}{R_2} = 0$$

$$sC_3V_o + \frac{V_o - V_2}{R_2} = 0$$

$$\left(sC_3 + \frac{1}{R_2}\right)V_o - \frac{V_2}{R_2} = 0$$

$$(sC_3R_2 + 1)V_o - V_2 = 0 \quad \text{..... (1)}$$

Step 2 of 4

Apply Kirchhoff's current law at node 2.

$$\frac{V_2 - V_o}{R_2} + \frac{V_2 - V_i}{R_1} + \frac{V_2 - V_0}{\frac{1}{sC_4}} = 0$$

$$\frac{V_2 - V_o}{R_2} + \frac{V_2 - V_i}{R_1} + sC_4(V_2 - V_0) = 0$$

$$V_2\left(\frac{1}{R_2} + \frac{1}{R_1} + sC_4\right) - V_0\left(sC_4 + \frac{1}{R_2}\right) - \frac{V_i}{R_1} = 0$$

$$V_2\left(\frac{1}{R_2} + \frac{1}{R_1} + sC_4\right) - \frac{V_i}{R_1} = V_0\left(sC_4 + \frac{1}{R_2}\right)$$

$$V_2\frac{(R_1 + R_2 + sC_4R_1R_2)}{R_1R_2} - \frac{V_i}{R_1} = V_0\left(\frac{1 + R_2sC_4}{R_2}\right)$$

$$V_2(R_1 + R_2 + sC_4R_1R_2) - R_2V_i = V_0R_1(1 + R_2sC_4)$$

$$V_2 = \frac{R_2V_i + V_0R_1(1 + R_2sC_4)}{(R_1 + R_2 + sC_4R_1R_2)}$$

Substitute $\frac{R_2V_i + V_0R_1(1 + R_2sC_4)}{(R_1 + R_2 + sC_4R_1R_2)}$ for V_2 in equation (1).

$$(sC_3R_2 + 1)V_o - \frac{R_2V_i + V_0R_1(1 + R_2sC_4)}{(R_1 + R_2 + sC_4R_1R_2)} = 0$$

$$(R_1 + R_2 + sC_4R_1R_2)(sC_3R_2 + 1)V_o - (R_2V_i + V_0R_1(1 + R_2sC_4)) = 0$$

$$V_o(s)\left((R_1 + R_2 + sC_4R_1R_2)(sC_3R_2 + 1) - R_1(1 + R_2sC_4)\right) = R_2V_i(s)$$

$$V_o(s)\left(s^2(C_4R_1R_2C_3R_2) + s(C_4R_1R_2 + C_3R_2(R_1 + R_2)) + R_1 + R_2 - R_1 - sR_1R_2C_4\right) = R_2V_i(s)$$

$$V_o(s)\left(s^2(C_4R_1R_2C_3R_2) + s(C_3R_2(R_1 + R_2)) + R_2\right) = R_2V_i(s)$$

$$V_o(s)\left(s^2(C_4R_1R_2C_3) + s(C_3(R_1 + R_2)) + 1\right) = V_i(s)$$

$$\frac{V_o(s)}{V_i(s)} = \frac{1}{\left(s^2(C_4R_1R_2C_3) + s(R_1C_3 + R_2C_3) + 1\right)}$$

$$\frac{V_o(s)}{V_i(s)} = \frac{\frac{1}{(C_4R_1R_2C_3)}}{\left(s^2 + s\frac{(R_1C_3 + R_2C_3)}{(C_4R_1R_2C_3)} + \frac{1}{(C_4R_1R_2C_3)}\right)}$$

Therefore , the transfer function $\frac{V_o(s)}{V_i(s)}$ is $\boxed{\frac{\frac{1}{(C_4R_1R_2C_3)}}{\left(s^2 + s\frac{(R_1C_3 + R_2C_3)}{(C_4R_1R_2C_3)} + \frac{1}{(C_4R_1R_2C_3)}\right)}}$

Compare the denominator of the transfer function with standard form.

$$s^2 + \frac{\omega_0}{Q}s + \omega_0^2 = 0 \quad \text{..... (2)}$$

The angular frequency is,

$$\omega_0^2 = \frac{1}{(C_4R_1R_2C_3)}$$

$$\omega_0 = \frac{1}{\sqrt{(C_3C_4R_1R_2)}}$$

Therefore, the angular frequency is $\boxed{\frac{1}{\sqrt{(C_3C_4R_1R_2)}}}$.

Step 3 of 4

From equation (2), it is clear that is the quality factor is,

$$\frac{\omega_0}{Q} = \frac{(R_1C_3 + R_2C_3)}{(C_4R_1R_2C_3)}$$

$$Q = \frac{(C_4R_1R_2C_3)\omega_0}{(R_1C_3 + R_2C_3)}$$

$$= \frac{(C_4R_1R_2C_3)\frac{1}{\sqrt{(C_4R_1R_2C_3)}}}{(R_1C_3 + R_2C_3)}$$

$$= \frac{C_4}{\sqrt{(C_4R_1R_2C_3)}}\frac{(R_1R_2)}{(R_1 + R_2)}$$

$$Q = \frac{C_4}{\sqrt{(C_4R_1R_2C_3)}}\frac{1}{\frac{(R_1 + R_2)}{(R_1R_2)}}$$

$$= \frac{1}{\frac{\sqrt{(C_4R_1R_2C_3)}}{C_4}\left(\frac{1}{R_1} + \frac{1}{R_2}\right)}$$

$$= \left(\frac{\sqrt{(C_4R_1R_2C_3)}}{C_4}\left(\frac{1}{R_1} + \frac{1}{R_2}\right)\right)^{-1}$$

Therefore , the quality factor is $\boxed{\left(\frac{\sqrt{(C_4R_1R_2C_3)}}{C_4}\left(\frac{1}{R_1} + \frac{1}{R_2}\right)\right)^{-1}}$.

Step 4 of 4

The dc gain is,

Substitute **0 for s** in transfer function.

$$\frac{V_o(0)}{V_i(0)} = \frac{\frac{1}{(C_4R_1R_2C_3)}}{\left((0)^2 + (0)\frac{(R_1C_3 + R_2C_3)}{(C_4R_1R_2C_3)} + \frac{1}{(C_4R_1R_2C_3)}\right)}$$

$$= \frac{\frac{1}{(C_4R_1R_2C_3)}}{\frac{1}{(C_4R_1R_2C_3)}}$$

$$= 1$$

Therefore, the dc gain is $\boxed{\text{unity}}$.

Step 1 of 13

Refer to figure 17.13 (c) op amp-RC spectrum-shaping network given in the textbook.

The expression for the transmission zero frequency ω_z is,

$$\omega_z = 2\pi f_z$$

Substitute 100 for f_z.

$$\omega_z = 2\pi \times 100$$
$$= 200\pi \text{ rad/s}$$

Therefore, the transmission zero frequency ω_z is 200π rad/s.

Step 2 of 13

The expression for the pole frequency ω_b is,

$$\omega_b = 2\pi f_p$$

Substitute 10×10^3 for f_p.

$$\omega_b = 2\pi \times \left(10\times10^3\right)$$
$$= 2\times10^4\pi \text{ rad/s}$$

Therefore, the pole frequency ω_b is $2\times10^4\pi$ rad/s.

Step 3 of 13

Consider the transfer function $T(s)$.

$$T(s) = \frac{a_1 s + a_0}{s + \omega_b}$$

$$T(s) = \frac{a_1\left(s + \frac{a_0}{a_1}\right)}{s + \omega_b}$$

$$T(s) = \frac{a_1(s + \omega_z)}{s + \omega_b}$$

Substitute 200π for ω_z, and $2\times10^4\pi$ for ω_b.

$$T(s) = \frac{a_1(s + 200\pi)}{s + 2\times10^4\pi}$$

Step 4 of 13

Recall equation (1).

$$T(s) = \frac{a_1(s + 200\pi)}{s + 2\times10^4\pi}$$

Calculate DC gain magnitude.

Substitute 0 for *s*.

$$|\text{DC gain}| = \left|\frac{a_1(0 + 200\pi)}{0 + 2\times10^4\pi}\right|$$

Substitute 1 for DC gain.

$$1 = \frac{a_1 \times 200\pi}{2\times10^4\pi}$$
$$a_1 = 100$$

Substitute 100 for a_1 in equation (1).

$$T(s) = \frac{100(s + 200\pi)}{s + 2\times10^4\pi} \quad \ldots\ldots (2)$$

Step 5 of 13

Consider the op-amp circuit in figure 17.13 (c) given in the textbook and its input resistance is,

$$R_{in} = R_1$$

Substitute $10\ \text{k}\Omega$ for R_{in}.

$$R_1 = 10\ \text{k}\Omega$$

Therefore, the value of resistance R_1 is $10\ \text{k}\Omega$.

Step 6 of 13

The expression for the DC gain magnitude is,

$$|\text{DC gain}| = \left|\frac{-R_2}{R_1}\right|$$

$$|\text{DC gain}| = \frac{R_2}{R_1}$$

Substitute 1 for $|\text{DC gain}|$.

$$1 = \frac{R_2}{R_1}$$
$$R_1 = R_2$$

Substitute $10\ \text{k}\Omega$ for R_1.

$$R_2 = 10\ \text{k}\Omega$$

Therefore, the value of resistance R_2 is $10\ \text{k}\Omega$.

Step 7 of 13

Consider the following expression:

$$\frac{a_1}{a_0} = C_1 R_1$$

$$\frac{a_0}{a_1} = \frac{1}{C_1 R_1}$$

$$\omega_z = \frac{1}{C_1 R_1}$$

Substitute 200π for ω_z, and 10×10^3 for R_1.

$$200\pi = \frac{1}{C_1 \times 10\times10^3}$$

$$C_1 = \frac{1}{200\pi \times 10\times10^3}$$
$$= 159.15\text{ nF}$$

Therefore, the value of capacitance C_1 is159.15 nF.

Step 8 of 13

Consider the following expression:

$$C_2 R_2 = \frac{1}{\omega_b}$$

Substitute $2\times10^4\pi$ for ω_b, and 10×10^3 for R_2.

$$C_2 \times \left(10\times10^3\right) = \frac{1}{2\times10^4\pi}$$

$$C_2 = \frac{1}{2\times10^4\pi \times 10\times10^3}$$

$$C_2 = \frac{1}{2\times10^8\pi}$$

$$C_2 = 1.59\text{ nF}$$

Therefore, the value of capacitance C_2 is 1.591 nF.

Step 9 of 13

The expression for the high frequency gain is,

$$\text{HF gain} = \frac{-C_1}{C_2}$$

Substitute 1.591×10^{-9} for C_2, and 159.15×10^{-9} for C_1.

$$\text{HF gain} = \frac{-159.15\times10^{-9}}{1.591\times10^{-9}}$$
$$= 100$$

Therefore, the high frequency gain is $\boxed{100}$.

Step 10 of 13

Recall equation (2).

$$T(s) = \frac{100(s + 200\pi)}{s + 2\times10^4\pi}$$

Substitute $s = j\omega$.

$$T(j\omega) = \frac{100(j\omega + 200\pi)}{j\omega + 2\times10^4\pi}$$

Calculate the magnitude of $T(j\omega)$.

$$|T| = \frac{100\sqrt{(200\pi)^2 + \omega^2}}{\sqrt{\omega^2 + \left(2\times10^4\pi\right)^2}} \quad \ldots\ldots (3)$$

The value of $|T|$ at $\omega = 0$ rad/s is,

$$|T| = \frac{100\sqrt{(200\pi)^2 + 0}}{\sqrt{0 + \left(2\times10^4\pi\right)^2}}$$
$$= \frac{100\sqrt{(200\pi)^2}}{\sqrt{\left(2\times10^4\pi\right)^2}}$$
$$= 1$$

Convert $|T|$ into decibels (dB).

$$|T|_{dB} = 20\log|T|$$

Substitute 1 for $|T|$.

$$|T|_{dB} = 20\log(1)$$
$$= 0\text{ dB}$$

Therefore, the value of $|T|$ at $\omega = 0$ rad/s is 0 dB.

Step 11 of 13

Recall equation (3).

$$|T| = \frac{100\sqrt{(200\pi)^2 + \omega^2}}{\sqrt{\omega^2 + \left(2\times10^4\pi\right)^2}}$$

The value of $|T|$ at $\omega = \omega_z$ rad/s is,

$$|T| = \frac{100\sqrt{(200\pi)^2 + \omega_z^2}}{\sqrt{\omega_z^2 + \left(2\times10^4\pi\right)^2}}$$

Substitute 200π for ω_z.

$$|T| = \frac{100\sqrt{(200\pi)^2 + (200\pi)^2}}{\sqrt{(200\pi)^2 + \left(2\times10^4\pi\right)^2}}$$
$$= \frac{100\times200\pi\sqrt{(1)^2 + (1)^2}}{200\pi\times\sqrt{(1)^2 + \left(10^2\right)^2}}$$
$$= \frac{100\times\sqrt{2}}{\sqrt{(1)^2 + \left(10^4\right)}}$$
$$= 1.414$$

Convert $|T|$ into decibels (dB).

$$|T|_{dB} = 20\log|T|$$

Substitute 1.414 for $|T|$.

$$|T|_{dB} = 20\log(1.414)$$
$$= 3\text{ dB}$$

Therefore, the value of $|T|$ at $\omega = 200\pi$ rad/s is 3 dB.

Step 12 of 13

Recall equation (3).

$$|T| = \frac{100\sqrt{(200\pi)^2 + \omega^2}}{\sqrt{\omega^2 + \left(2\times10^4\pi\right)^2}}$$

The value of $|T|$ at $\omega = \omega_b$ rad/s is,

$$|T| = \frac{100\sqrt{(200\pi)^2 + \omega_b^2}}{\sqrt{\omega_b^2 + \left(2\times10^4\pi\right)^2}}$$

Substitute $2\times10^4\pi$ for ω_b.

$$|T| = \frac{100\sqrt{(200\pi)^2 + \left(2\times10^4\pi\right)^2}}{\sqrt{\left(2\times10^4\pi\right)^2 + \left(2\times10^8\pi\right)^2}}$$
$$= \frac{100\times200\pi\times\sqrt{(1)^2 + \left(10^4\right)^2}}{2\times10^4\pi\times\sqrt{(1)^2 + (1)^2}}$$
$$= \frac{\sqrt{(1)^2 + \left(10^4\right)}}{\sqrt{2}}$$
$$= 70.714$$

Convert $|T|$ into decibels (dB).

$$|T|_{dB} = 20\log|T|$$

Substitute 70.714 for $|T|$.

$$|T|_{dB} = 20\log(70.714)$$
$$= 37\text{ dB}$$

Therefore, the value of $|T|$ at $\omega = 2\times10^4\pi$ rad/s is 37 dB.

Step 13 of 13

Plot the magnitude of the transfer function versus frequency as shown in figure 1.

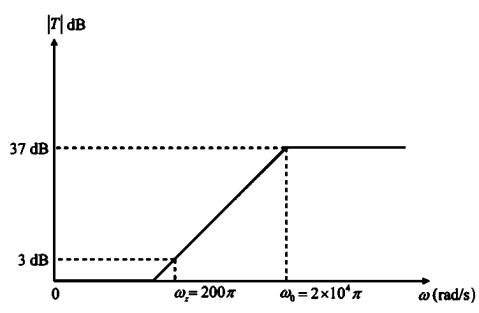


Figure 1

Step 1 of 2

17.029E

Refer to Figure 16.34(c) in the text book.

Consider the expression.

$$C_4 R = \frac{2Q}{\omega_0}$$

$$C_4 = \frac{2Q}{2\pi f_0 R}$$

Substitute $\frac{1}{\sqrt{2}}$ for Q , $4\ \text{kHz}$ for f_0 and $10\ \text{k}\Omega$ for R in equation.

$$C_4 = \frac{2 \times \frac{1}{\sqrt{2}}}{2\pi \times 4 \times 10^3 \times 10 \times 10^3}$$
$$= 5.626\ \text{nF}$$

Step 2 of 2

Consider the expression.

$$C_3 = \frac{C_4}{m}$$

$$m = 4Q^2$$

Substitute $\frac{1}{\sqrt{2}}$ for Q in equation.

$$m = 4 \times \frac{1}{2}$$
$$= 2$$

Thus the capacitance value is

$$C_3 = \frac{5.626}{2}$$
$$= 2.813\ \text{nF}$$

Therefore, the design parameters are C_3 is $\boxed{2.813\ \text{nF}}$, R_1 , R_2 are $\boxed{10\ \text{k}\Omega}$ and C_4 is $\boxed{5.626\ \text{nF}}$.

Step 1 of 7

The cascading of first-order op amp-RC low-pass filter and first-order op amp-RC high-pass filter is shown in figure 1.

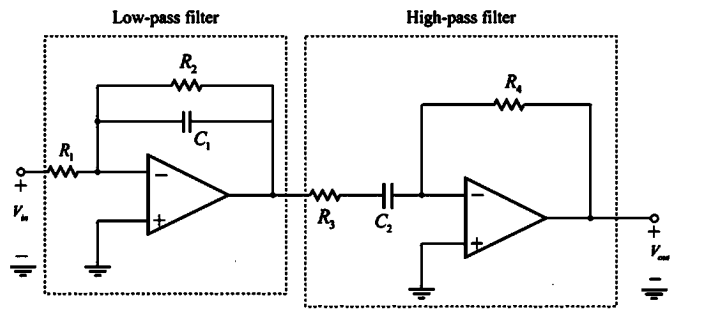


Figure 1

Step 2 of 7

The expression for the DC gain of the band-pass filter is,

$$|A_{BP}| = 10^{\frac{G}{20}}$$

Substitute 12 for mid-band gain *G*.

$$|A_{BP}| = 10^{\frac{12}{20}}$$
$$= 4$$

Therefore, the DC gain of the band-pass filter is 4.

Step 3 of 7

The input resistance R_{in} must be as high as possible. Ensure that the value of any resistor is not higher than $100\ \text{k}\Omega$.

Refer to figure 1 and the input resistance is,

$$R_{in} = R_1$$

Substitute $100\ \text{k}\Omega$ for R_{in}.

$$R_1 = 100\ \text{k}\Omega$$

Therefore, the resistance value of R_1 is $\boxed{100\ \text{k}\Omega}$.

Step 4 of 7

The expression for 3-dB frequency of the RC low-pass filter shown in figure 1 is,

$$f_o = \frac{1}{2\pi R_1 C_1}$$

Calculate the value of capacitance C_1.

$$C_1 = \frac{1}{2\pi R_1 f_o}$$

Substitute 100×10^3 for R_1, and 50 for f_o.

$$C_1 = \frac{1}{2\pi\times100\times10^3\times50}$$
$$= 31.83\ \text{nF}$$

Therefore, the capacitance value of C_1 is $\boxed{31.83\ \text{nF}}$.

Step 5 of 7

The gain of wide-band band-pass filter is the multiple of individual gains of low-pass and high-pass filters.

The expression for the gain of wide-band band-pass filter is,

$$|A_{BP}| = |A_{LP}|\times|A_{HP}|$$

Here, $|A_{LP}|$ and $|A_{HP}|$ are the individual gains of low-pass and high-pass filters; $|A_{BP}|$ is the gain of wide-band of band-pass filter.

Substitute 4 for $|A_{BP}|$.

$$|A_{LP}|\times|A_{HP}| = 4$$

The expression for DC gain of low-pass filter shown in figure 1 is,

$$|A_{LP}| = \left|\frac{-R_2}{R_1}\right|$$
$$R_2 = |A_{LP}|\times R_1$$
$$R_2 = |A_{LP}|\times(100\ \text{k}\Omega)$$

No resistance should exceed the resistance value of $100\ \text{k}\Omega$.

Assume the value of $|A_{LP}|$ as 1 and substitute for the same.

$$R_2 = 1\times(100\ \text{k}\Omega)$$
$$= 100\ \text{k}\Omega$$

Therefore, the value of resistance R_2 is $\boxed{100\ \text{k}\Omega}$.

Step 6 of 7

Consider the following expression:

$$|A_{LP}|\times|A_{HP}| = 4$$

Substitute 1 for $|A_{LP}|$.

$$1\times|A_{HP}| = 4$$
$$|A_{HP}| = 4$$

Therefore, the DC gain of high-pass filter is 4.

The expression for the DC gain of high-pass filter shown in figure 1 is,

$$|A_{HP}| = \left|\frac{-R_4}{R_3}\right|$$

Substitute 4 for $|A_{HP}|$.

$$4 = \left|\frac{R_4}{R_3}\right|$$
$$R_4 = 4R_3$$

No resistance should exceed the resistance value of $100\ \text{k}\Omega$.

Assume the value of R_4 as $100\ \text{k}\Omega$.

$$100\ \text{k}\Omega = 4R_3$$
$$R_3 = 25\ \text{k}\Omega$$

Therefore, the value of R_4 is $\boxed{100\ \text{k}\Omega}$ and that of R_3 is $\boxed{25\ \text{k}\Omega}$.

Step 7 of 7

The expression for 3-dB frequency of the RC high-pass filter shown in figure 1 is,

$$f_o = \frac{1}{2\pi R_3 C_2}$$

Calculate the value of capacitance C_2.

$$C_2 = \frac{1}{2\pi R_3 f_o}$$

Substitute 25×10^3 for R_3 and 50×10^3 for f_o.

$$C_2 = \frac{1}{2\pi\times25\times10^3\times50\times10^3}$$
$$= 127.32\ \text{pF}$$

Therefore, the value of capacitance C_2 is $\boxed{127.32\ \text{pF}}$.

Step 1 of 10

Refer to Figure 16.29 in the text book.

The resonant frequency is,

$$\omega_0 = \frac{1}{\sqrt{C_1 C_2 R_3 R_4}}.$$

The quality factor is

$$Q = \left[\frac{\sqrt{C_1 C_2 R_3 R_4}}{R_3}\left(\frac{1}{C_1}+\frac{1}{C_2}\right)\right]^{-1}.$$

With $C_1 = C_2$, the resonant frequency and quality factor are,

$$\omega_0 = \frac{1}{C\sqrt{R_3 R_4}}$$

And

$$\begin{aligned} Q &= \frac{C\sqrt{R_3 R_4}}{R_3}\left(\frac{1}{C}+\frac{1}{C}\right) \\ &= \left[\frac{2\sqrt{R_3 R_4}}{R_3}\right]^{-1} \\ &= \frac{R_3}{2\sqrt{R_3 R_4}} \end{aligned}$$

Step 2 of 10

(a)

If R_3 is 2% high, the resistance is,

$$R_3 = 1.02R_3,$$

The resonant frequency is,

$$\begin{aligned} \omega_0' &= \frac{1}{C\sqrt{1.02R_3 R_4}} \\ &= \frac{1}{1.01C\sqrt{R_3 R_4}} \\ &= \frac{0.99}{C\sqrt{R_3 R_4}} \\ &= 0.99\omega_0 \end{aligned}$$

The decrease in the resonant frequency is,

$$\begin{aligned} \%\ \text{decrease} &= \left(\frac{\omega_0 - 0.99\omega_0}{\omega_0}\right)\times 100 \\ &= 1\% \end{aligned}$$

Thus, the percentage change in the resonant frequency is $\boxed{-1\%}$.

Step 3 of 10

The quality factor is,

$$\begin{aligned} Q' &= \frac{1.02R_3}{2\sqrt{1.02R_3 R_4}} \\ &= 1.01Q \end{aligned}$$

The decrease in the quality factor is,

$$\begin{aligned} \%\ \text{increase} &= \left(\frac{1.01Q - Q}{Q}\right)\times 100 \\ &= 1\% \end{aligned}$$

Thus, the percentage change in the quality factor is $\boxed{+1\%}$.

Step 4 of 10

(b)

If R_4 is 2% high, the resistance is,

$$R_4 = 1.02R_4,$$

The resonant frequency is,

$$\begin{aligned} \omega_0' &= \frac{1}{C\sqrt{R_3 \times 1.02R_4}} \\ &= \frac{1}{1.01C\sqrt{R_3 R_4}} \\ &= \frac{0.99}{C\sqrt{R_3 R_4}} \\ &= 0.99\omega_0 \end{aligned}$$

The decrease in the resonant frequency is,

$$\begin{aligned} \%\ \text{decrease} &= \left(\frac{\omega_0 - 0.99\omega_0}{\omega_0}\right)\times 100 \\ &= 1\% \end{aligned}$$

Thus the percentage change in the resonant frequency is $\boxed{-1\%}$.

Step 5 of 10

The quality factor is,

$$\begin{aligned} Q' &= \frac{R_3}{2\sqrt{1.02R_3 R_4}} \\ &= 0.99Q \end{aligned}$$

Step 6 of 10

The decrease in the quality factor is,

$$\begin{aligned} \%\ \text{decrease} &= \left(\frac{Q - 0.99Q}{Q}\right)\times 100 \\ &= 1\% \end{aligned}$$

Thus the percentage change in the quality factor is $\boxed{-1\%}$

Step 7 of 10

(c)

If both R_3 and R_4 are 2% high, then the resistance are,

$$R_4 = 1.02R_4$$

$$R_3 = 1.02R_3.$$

The resonant frequency is,

$$\begin{aligned} \omega_0' &= \frac{1}{C\sqrt{1.02R_3 \times 1.02R_4}} \\ &= \frac{1}{1.02C\sqrt{R_3 R_4}} \\ &= \frac{0.98}{C\sqrt{R_3 R_4}} \\ &= 0.98\omega_0 \end{aligned}$$

The decrease in the resonant frequency is,

$$\begin{aligned} \%\ \text{decrease} &= \left(\frac{\omega_0 - 0.98\omega_0}{\omega_0}\right)\times 100 \\ &= 2\% \end{aligned}$$

Thus, the percentage change in the resonant frequency is $\boxed{-2\%}$.

Step 8 of 10

The quality factor is,

$$\begin{aligned} Q' &= \frac{1.02R_3}{2\sqrt{1.02R_3 \times 1.02R_4}} \\ &\frac{1.02R_3}{2\times 1.02\sqrt{R_3 R_4}} \\ &= Q \end{aligned}$$

Thus, the change in the quality factor is $\boxed{0\%}$.

Step 9 of 10

(d)

If both R_3 and R_4 are 2% high, then the resistance are,

$$R_4 = 1.02R_4$$

$$R_3 = 1.02R_3.$$

If both capacitors are 2% low, then the capacitors value is,

$$C = 0.98C.$$

The resonant frequency is,

$$\begin{aligned} \omega_0' &= \frac{1}{0.98C\sqrt{1.02R_3 \times 1.02R_4}} \\ &= \frac{1}{0.98\times 1.02C\sqrt{R_3 R_4}} \\ &= \frac{1}{C\sqrt{R_3 R_4}} \\ &= \omega_0 \end{aligned}$$

Thus, the change in the resonant frequency is $\boxed{0\%}$.

Step 10 of 10

The quality factor is,

$$\begin{aligned} Q' &= \frac{1.02R_3}{2\sqrt{1.02R_3 \times 1.02R_4}} \\ &\frac{1.02R_3}{2\times 1.02\sqrt{R_3 R_4}} \\ &= Q \end{aligned}$$

Thus, the change in the quality factor is $\boxed{0\%}$.

Step 1 of 11

17.030P

Refer to the Op Amp circuit diagram of Figure 17.14 in the textbook .

The 3-db frequency of the circuit is,

$$CR = \frac{1}{\omega_o}$$

The flat gain is, $(a_1) = 1$

The voltage at non-inverting terminal is,

$$V^+ = V_i \frac{\frac{1}{sC}}{\frac{1}{sC} + R_1}$$
$$= V_i \frac{1}{1 + s\tau}$$

Here, $\tau = RC$

Step 2 of 11

Due to virtual ground concept, the voltage at $V^+ = V^-$.

$$V^- = V_i \frac{1}{1 + s\tau} \quad \text{..... (1)}$$

Then the current flowing through the resistor R_1 is,

$$I_1 = \frac{V_i - V^-}{R_1} \quad \text{..... (2)}$$

Substitute equation (2) in equation (1).

$$I_1 = \frac{V_i - V_i \frac{1}{1 + s\tau}}{R_1}$$
$$= \frac{V_i \left(1 - \frac{1}{1 + s\tau}\right)}{R_1}$$

Step 3 of 11

The output voltage V_O is,

$$V_O = V^- - I_1 R_1$$

$$V_O = \frac{V_i}{1 + s\tau} - \frac{V_i \left(1 - \frac{1}{1 + s\tau}\right) R_1}{R_1}$$

$$\frac{V_O}{V_i} = \frac{1 - (1 + s\tau) + 1}{1 + s\tau}$$

$$\frac{V_O}{V_i} = \frac{1 - s\tau}{1 + s\tau}$$
$$= \frac{\frac{1}{\tau} - s}{\frac{1}{\tau} + s}$$

Simplify further.

$$T(s) = \frac{\omega_o - s}{\omega_o + s}$$

Here, $\omega_o = \frac{1}{\tau}$

$$T(s) = -\frac{s - \omega_o}{s + \omega_o}$$
$$= -\frac{j\omega - \omega_o}{j\omega + \omega_o}$$

Thus, the transfer function is, $\boxed{T(j\omega) = -\frac{j\omega - \omega_o}{j\omega + \omega_o}}$

Step 4 of 11

The phase angle is,

$$\phi(\omega) = 180° + \tan^{-1}\left(\frac{\omega}{-\omega_o}\right) - \tan^{-1}\left(\frac{\omega}{\omega_o}\right)$$
$$= 360° - 2\tan^{-1}\left(\frac{\omega}{\omega_o}\right)$$
$$= -2\tan^{-1}\left(\frac{\omega}{\omega_o}\right)$$

Therefore, the phase angle ϕ is, $\boxed{-2\tan^{-1}\left(\frac{\omega}{\omega_o}\right)}$.

Step 5 of 11

Further simplification is as follows:

$$\tan\left(-\frac{\phi(\omega)}{2}\right) = \frac{\omega}{\omega_o}$$
$$\tan\left(-\frac{\phi(\omega)}{2}\right) = \frac{\omega}{\frac{1}{CR}}$$
$$\tan\left(-\frac{\phi(\omega)}{2}\right) = \omega CR$$

$$R = \frac{\tan\left(-\frac{\phi(\omega)}{2}\right)}{\omega C} \quad \text{..... (3)}$$

Step 6 of 11

Determine the resistance, R .
Substitute $5\times10^3\ \text{rad/s}$ for ω , $10\ \text{nF}$ for C in equation (3).

$$R = \frac{\tan\left(-\frac{\phi(\omega)}{2}\right)}{(5\times10^3)(10\times10^{-9})}$$
$$= 2\times10^4\left(\tan\left(-\frac{\phi(\omega)}{2}\right)\right)$$

Step 7 of 11

For $\phi = -30°$, the resistance R is,

$$R = 2\times10^4\left(\tan\left(-\frac{-30°}{2}\right)\right)$$
$$= 5.36\ \text{k}\Omega$$

Hence, the value of R for is, $\boxed{5.36\ \text{k}\Omega}$.

Step 8 of 11

For $\phi = -60°$, the resistance R is,

$$R = 2\times10^4\left(\tan\left(-\frac{-60°}{2}\right)\right)$$
$$= 11.54\ \text{k}\Omega$$

Hence, the value of R for is, $\boxed{11.54\ \text{k}\Omega}$.

Step 9 of 11

For $\phi = -90°$, the resistance R is,

$$R = 2\times10^4\left(\tan\left(-\frac{-90°}{2}\right)\right)$$
$$= 20\ \text{k}\Omega$$

Hence, the value of R is, $\boxed{20\ \text{k}\Omega}$.

Step 10 of 11

For $\phi = -120°$, the resistance R is,

$$R = 2\times10^4\left(\tan\left(-\frac{-120°}{2}\right)\right)$$
$$= 34.64\ \text{k}\Omega$$

Hence, the value of R is, $\boxed{34.64\ \text{k}\Omega}$.

Step 11 of 11

For $\phi = -150°$, the resistance R is,

$$R = 2\times10^4\left(\tan\left(-\frac{-150°}{2}\right)\right)$$
$$= 74.64\ \text{k}\Omega$$

Hence, the value of R is $\boxed{74.64\ \text{k}\Omega}$.

17.031E

We don't have the solution to this problem yet.

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Step 1 of 5

Refer to Figure 16.14 in the textbook for the op amp-RC circuit.

Redraw the circuit by interchanging resistor R and capacitor C as shown in Figure 1.

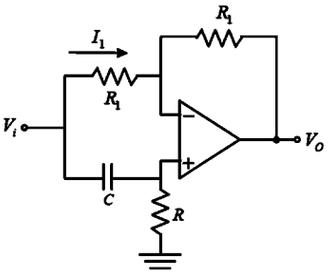


Figure 1

Step 2 of 5

For the circuit, the voltage at non-inverting terminal is,

$$\begin{aligned} V^+ &= V_i \frac{R}{R+\frac{1}{sC}} \\ &= V_i \frac{sCR}{sCR+1} \\ &= V_i \frac{CRs}{CR\left(s+\frac{1}{CR}\right)} \\ &= V_i \frac{s}{s+\omega_O} \qquad \left(\text{Where, } \omega_o = \frac{1}{RC}\right) \end{aligned}$$

The current I_1 is,

$$I_1 = \frac{V_i - V^-}{R_1}$$

By virtual ground concept, $V^+ = V^-$.

$$\begin{aligned} I_1 &= \frac{V_i - V^+}{R_1} \\ &= \frac{V_i - V_i \frac{s}{s+\omega_O}}{R_1} \qquad \left(\text{Since, } V^+ = V_i \frac{s}{s+\omega_O}\right) \\ &= \frac{V_i\left(1-\frac{s}{s+\omega_O}\right)}{R_1} \end{aligned}$$

Step 3 of 5

The output voltage is,

$$\begin{aligned} \frac{V^- - V_O}{R_1} &= I_1 \\ V_O &= V^- - I_1 R_1 \\ V_O &= V^+ - I_1 R_1 \qquad \left(\text{Since, } V^+ = V^-\right) \end{aligned}$$

Substitute $\frac{V_i\left(1-\frac{s}{s+\omega_O}\right)}{R_1}$ for I_1 and $V_i \frac{s}{s+\omega_O}$ for V^+ in the equation.

$$V_O = V_i \frac{s}{s+\omega_O} - \left(\frac{V_i\left(1-\frac{s}{s+\omega_O}\right)}{R_1}\right) R_1$$

$$\frac{V_O}{V_i} = \frac{s}{s+\omega_O} - \left(\frac{\left(1-\frac{s}{s+\omega_O}\right)}{R_1}\right) R_1$$

Simplify further.

$$\begin{aligned} \frac{V_O}{V_i} &= \left[\frac{s}{s+\omega_O} - 1 + \frac{s}{s+\omega_O}\right] \\ &= \left[\frac{2s-s-\omega_O}{s+\omega_O}\right] \\ &= \frac{s-\omega_O}{s+\omega_O} \\ &= \frac{j\omega-\omega_O}{j\omega+\omega_O} \end{aligned}$$

Step 4 of 5

The phase angle is,

$$\begin{aligned} \phi(\omega) &= \tan^{-1}\left(\frac{\omega}{-\omega_O}\right) - \tan^{-1}\left(\frac{\omega}{\omega_O}\right) \\ &= 180° - \tan^{-1}\left(\frac{\omega}{\omega_o}\right) - \tan^{-1}\left(\frac{\omega}{\omega_O}\right) \qquad \left(\text{Since, } \tan^{-1}\left(\frac{\omega}{-\omega_o}\right) = 180 - \tan^{-1}\left(\frac{\omega}{\omega_o}\right)\right) \\ &= 180° - 2\tan^{-1}\left(\frac{\omega}{\omega_O}\right) \end{aligned}$$

Step 5 of 5

At low frequencies, the phase shift is,

$$\begin{aligned} \phi(0) &= 180° - 2\tan^{-1}\left(\frac{0}{\omega_O}\right) \\ &= 180° \end{aligned}$$

At high frequencies, the phase shift is,

$$\begin{aligned} \phi(\infty) &= 180° - 2\tan^{-1}\left(\frac{\infty}{\omega_O}\right) \\ &= 180° - 2(90°) \\ &= 0° \end{aligned}$$

Therefore, it is proved that, for the circuit, the resulting phase shift covers **0° to 180°** over zero to infinite frequencies.

Step 1 of 4

For switched capacitor active-RC biquad filter, the relation between capacitances is,

$$C_3 = C_4 = KC$$
$$= 20K \text{ pF}$$

The capacitance C_5 is,

$$C_5 = 2\pi f_0 T_C \frac{C}{Q}$$
$$= 2\pi f_0 \frac{C}{f_C Q}$$

Substitute 20 pF for C, 10 kHz for f_0, 200 kHz for f_C and 20 for Q in equation.

$$C_5 = 2\pi \times 10 \times 10^3 \times \frac{1}{200 \times 10^3} \frac{20 \times 10^{-12}}{20}$$
$$= 0.314 \text{ pF}$$

Step 2 of 4

The constant K is,

$$K = \frac{C_5 Q}{C}$$
$$= \frac{0.314 \times 10^{-12} \times 20}{20 \times 10^{-12}}$$
$$= 0.314$$

Step 3 of 4

The value of the capacitances is,

$$C_3 = C_4$$
$$= 20K \text{ pF}$$
$$= 20 \times 0.314$$
$$= 6.28 \text{ pF}$$

Step 4 of 4

The center frequency gain is,

$$\text{Center-frequency gain} = \frac{C_6}{C_5}$$
$$C_6 = \text{Center-frequency gain} \times C_5$$

The capacitance C_6 is,

$$C_6 = \text{Center-frequency gain} \times C_5$$
$$= 1 \times 0.314 \text{ pF}$$
$$= 0.314 \text{ pF}$$

Therefore, the capacitance values are C_3, C_4 are $\boxed{6.28 \text{ pF}}$, C_5 and C_6 are $\boxed{0.314 \text{ pF}}$.

Step 1 of 3

17.032P

Refer to Figure P16.27 in the textbook for the phasor diagram.

From the phasor diagram, write the following results:

The voltage V_2 lags V_1 by $120°$,

The voltage V_3 lags V_2 by $120°$.

The cascaded connection of two first order op amp-RC all-pass circuits is shown in Figure 1

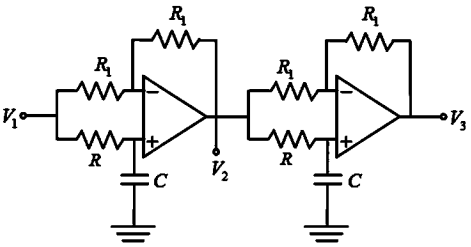


Figure 1

Step 2 of 3

The transfer function of the above circuit is,

$$T(s)=\frac{s-\omega_O}{s+\omega_O}$$

Here, $\omega_O=\frac{1}{RC}$

The phase angle is,

$$\phi(\omega)=180°+\tan^{-1}\left(\frac{\omega}{-\omega_O}\right)-\tan^{-1}\left(\frac{\omega}{\omega_O}\right)$$

$$=180°-2\tan^{-1}\left(\frac{\omega}{\omega_O}\right)$$

Step 3 of 3

Substitute $-120°$ for ϕ in the equation.

$$-120°=-2\tan^{-1}\left(\frac{\omega}{\omega_O}\right)$$

$$60°=\tan^{-1}(\omega RC)$$

$$\tan 60°=\omega RC$$

$$\tan 60°=2\pi fRC$$

Substitute $6\ \text{Hz}$ for f and $1\ \mu\text{F}$ for C in the equation.

$$\tan 60°=(2\pi\times 60)R(1\times 10^{-6})$$

$$1.732=(376.99\times 10^{-6})R$$

$$R=\frac{1.732}{376.99\times 10^{-6}}$$

$$R=4.59\ \text{k}\Omega$$

Thus, value of R is, $\boxed{R=4.59\ \text{k}\Omega}$.

And the value of R_1 arbitrarily choosen as $\boxed{R_1=10\ \text{k}\Omega}$.

17.033E

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Step 1 of 5

Refer to Figure 17.16(a) in the textbook.

Consider the second order low pass filter of the transfer function.

$$T(s)=\frac{a_0}{s^2+s\frac{\omega_0}{Q}+\omega_0^2} \quad \text{..... (1)}$$

Here,

ω_o is angular frequency,

Q is quality factor.

Step 2 of 5

Determine the coefficient a_0 from the transfer function $T(s)$.

Substitute $s=0$ in the transfer function to get dc gain.

$$T(0)=\frac{a_0}{(0)^2+(0)\frac{\omega_0}{Q}+\omega_0^2}$$

$$T(0)=\frac{a_0}{\omega_0^2} \quad \text{..... (2)}$$

Substitute 10^4 rad/sec for ω_0, 1 for $T(0)$ in the equation (2).

$$1=\frac{a_0}{(10^4)^2}$$
$$a_0=(10^4)^2$$
$$=10^8$$

Thus, the coefficient a_0 is 10^8.

Step 3 of 5

Determine the transfer function $T(s)$.

Substitute 10^4 rad/sec for ω_0, 10^8 for a_0, 2 for Q in the equation (1).

$$T(s)=\frac{10^8}{s^2+s\frac{10^4}{2}+(10^4)^2}$$
$$=\frac{10^8}{s^2+s(5\times10^3)+10^8}$$

Therefore, the transfer function $T(s)$ of the filter is, $\boxed{\frac{10^8}{s^2+s(5\times10^3)+10^8}}$.

Step 4 of 5

The maximum peak of $|T|$ occurs at frequency of ω_{max}.

Hence, the maximum frequency is,

$$\omega_{max}=\omega_0\sqrt{1-\frac{1}{2Q^2}}$$

Substitute 10^4 rad/s for ω_0, 2 for Q in the equation.

$$\omega_{max}=10^4\left(\sqrt{1-\frac{1}{2(2)^2}}\right)$$
$$=10^4\left(\sqrt{1-\frac{1}{8}}\right)$$
$$=10^4\left(\sqrt{\frac{7}{8}}\right)$$
$$=935.41 \text{ rad/sec}$$

Thus, the magnitude is maximum at $\boxed{\omega_{max}=935.41 \text{ rad/sec}}$.

Step 5 of 5

The peak transmission is,

$$|T_{max}|=\frac{|a_o|Q}{\omega_0^2\sqrt{1-\frac{1}{4Q^2}}}$$

Substitute 10^8 for a_0, 2 for Q and 10^4 rad/s for ω_0 in the equation.

$$|T_{max}|=\frac{10^8(2)}{(10^4)^2\sqrt{1-\frac{1}{4(2)^2}}}$$
$$=\frac{10^8(2)}{(10^8)\sqrt{1-\frac{1}{16}}}$$
$$=\frac{2}{\sqrt{\frac{15}{16}}}$$
$$=2.06 \text{ V/V}$$

The peak value in terms of decibels is,

$$|T_{max}|=20\log(2.06)$$
$$=6.27 \text{ dB}$$

Therefore, the peak transmission $|T_{max}|$ is, $\boxed{6.27 \text{ dB}}$.

Step 1 of 3

Refer to Figure 16.42(a) in the text book.

The quality factor is,

$$Q = \frac{(R_1 \| R_{in})}{\omega_0 L}$$
$$= \frac{(R_1 \| R_{in})}{2\pi f_0 L}$$

Substitute $455\ \text{kHz}$ for f_0, $1\ \text{k}\Omega$ for R_1 and $1\ \text{k}\Omega$ for R_{in} in equation.

$$Q = \frac{(1\times10^3) \| (1\times10^3)}{2\pi(455\times10^3)(5\times10^{-6})}$$
$$= \frac{500}{2\pi(2.275)}$$
$$= 35$$

Step 2 of 3

Consider the expression.

$$BW = \frac{f_0}{Q}$$

Substitute $455\ \text{kHz}$ for f_0, 35 for Q in equation.

$$BW = \frac{455\times10^3}{35}$$
$$= 13\ \text{kHz}$$

Therefore, the actual bandwidth is $\boxed{13\ \text{kHz}}$.

Step 3 of 3

Consider the expression.

$$C_1 + C_{in} = \frac{1}{\omega_0^2 L}$$
$$C_1 = \frac{1}{(2\pi f_0)^2 L} - C_{in}$$

Substitute $200\ \text{pF}$ for C_{in}, $455\ \text{kHz}$ for f_0 and $5\ \mu\text{H}$ for L in equation.

$$C_1 = \frac{1}{\left(2\pi(455\times10^3)\right)^2 (5\times10^{-6})} - 200\times10^{-12}$$
$$= 24.47\times10^{-9} - 0.2\times10^{-9}$$
$$= 24.27\ \text{nF}$$

Therefore, the value of capacitance C_1 is $\boxed{24.27\ \text{nF}}$.

Step 1 of 14

Refer to Figure 16.16(a) in the textbook for the |T| characteristics and S-plane singularities of second order Low pass filter.

For second order low pass filter, the transfer function is,

$$T(s)=\frac{a_0}{s^2+s\frac{\omega_0}{Q}+\omega^2_0} \text{ and}$$

$$\text{DC gain}=\frac{a_0}{\omega^2_0}$$

The value of $\omega_p = 1$ rad/sec .

The maximum attenuation allowed in the pass band is, $A_{max(dB)} = 3$ dB .

The value of $\omega_s = 2$ rad/sec .

Step 2 of 14

The value of A_{max} is,

$$A_{max(dB)}=-20\log\left(A_{max}\right)$$
$$3=-20\log\left(A_{max}\right)$$
$$A_{max}=10^{-\frac{3}{20}}$$
$$=0.708$$
$$=\frac{1}{\sqrt{2}}$$

Step 3 of 14

There are many Q-values which may be used. Consider the second order low pass filter characteristics as shown in Figure 1.

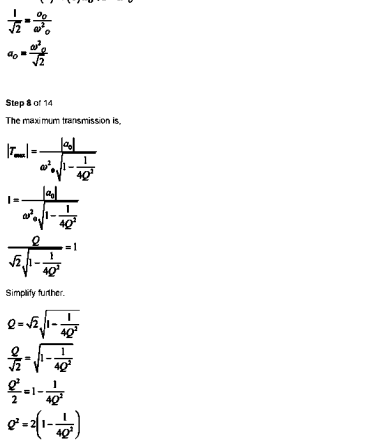


Figure 1

Step 4 of 14

From Figure 1, for $Q \le \frac{1}{\sqrt{2}}$, there is no peaking and for $Q > \frac{1}{\sqrt{2}}$ there is a peak in the above characteristics. For $Q = \frac{1}{\sqrt{2}}$, the response is maximally flat. So, use $Q = \frac{1}{\sqrt{2}}$ which is desirable.

Case1: $\left(Q \le \frac{1}{\sqrt{2}}\right)$

For $Q = \frac{1}{\sqrt{2}}$, the response is maximally flat.

The transfer function of the second order filter is,

$$T(s)=\frac{a_0}{s^2+s\omega_0\sqrt{2}+\omega^2_0}$$
$$T(0)=\frac{a_0}{(0)^2+(0)\omega_0\sqrt{2}+\omega^2_0}$$
$$1=\frac{a_0}{\omega^2_0}$$
$$\omega^2_0=a_0$$

Step 5 of 14

Now the transfer function is,

$$T(s)=\frac{\omega^2_0}{s^2+s\omega_0\sqrt{2}+\omega^2_0}$$
$$T(j\omega)=\frac{\omega^2_0}{-\omega^2+j\omega\omega_0\sqrt{2}+\omega^2_0}$$
$$|T(j\omega)|=\frac{\omega^2_0}{\sqrt{\left(\omega^2_0-\omega^2\right)^2+2\omega^2\omega^2_0}}$$

Step 6 of 14

Simplify further.

$$|T(j1)|=\frac{\omega^2_0}{\sqrt{\left(\omega^2_0-1^2\right)^2+2\left(1^2\right)\omega^2_0}}$$
$$\frac{1}{\sqrt{2}}=\frac{\omega^2_0}{\sqrt{\left(\omega^2_0-1^2\right)^2+2\left(1^2\right)\omega^2_0}} \qquad \left(\text{Since } |T(j1)|=\frac{1}{\sqrt{2}}\right)$$
$$\frac{1}{2}=\frac{\omega^4_0}{\left(\omega^2_0-1^2\right)^2+2\omega^2_0}$$
$$\frac{1}{2}=\frac{\omega^4_0}{\omega^4_0+1}$$
$$\omega^4_0+1=2\omega^4_0$$
$$\omega^4_0=1$$
$$\omega_0=1 \text{ rad/sec}$$

Thus, the values of Q and ω_0 are, $\boxed{Q=\frac{1}{\sqrt{2}} \text{ and } \omega_0=1 \text{ rad/sec}}$

For $\omega_0 = 1$ rad/sec , the transfer function is,

$$T(s)=\frac{(1)^2}{s^2+s(1)\sqrt{2}+(1)^2}$$

Hence, the transfer function of the second order low pass filter for $Q \le \frac{1}{\sqrt{2}}$ is,

$$T(s)=\frac{1}{s^2+s\sqrt{2}+1}$$

Step 7 of 14

Case2: $\left(Q > \frac{1}{\sqrt{2}}\right)$

The dc transfer function is,

$$T(0)=\frac{a_0}{(0)^2+(0)\omega_0\sqrt{2}+\omega^2_0}$$
$$\frac{1}{\sqrt{2}}=\frac{a_0}{\omega^2_0}$$
$$a_0=\frac{\omega^2_0}{\sqrt{2}}$$

Step 8 of 14

The maximum transmission is,

$$|T_{max}|=\frac{|a_0|}{\omega^2_0\sqrt{1-\frac{1}{4Q^2}}}$$
$$1=\frac{|a_0|}{\omega^2_0\sqrt{1-\frac{1}{4Q^2}}}$$
$$\frac{Q}{\sqrt{2}\sqrt{1-\frac{1}{4Q^2}}}=1$$

Simplify further.

$$Q=\sqrt{2}\sqrt{1-\frac{1}{4Q^2}}$$
$$\frac{Q}{\sqrt{2}}=\sqrt{1-\frac{1}{4Q^2}}$$
$$\frac{Q^2}{2}=1-\frac{1}{4Q^2}$$
$$Q^2=2\left(1-\frac{1}{4Q^2}\right)$$
$$Q^2=2-\frac{1}{2Q^2}$$
$$Q^4-2Q^2+\frac{1}{2}=0$$

Solve the equation to obtain the value of Q^2 as follows:

$$Q^2=\frac{2\pm\sqrt{(-2)^2-4(1)\left(\frac{1}{2}\right)}}{2}$$
$$=\frac{2\pm\sqrt{4-2}}{2}$$
$$=1\pm\frac{1}{\sqrt{2}}$$
$$=1.7071,\ 0.2929$$

Now, calculate Q .

$$Q=\sqrt{1.7071},\ \sqrt{0.2929}$$
$$=1.3066,\ 0.5412$$

Since, Q should obey the condition $Q > \frac{1}{\sqrt{2}}$, the value of Q is,

$$\boxed{Q=1.3066}$$

Step 9 of 14

Now at the passband edge, the transfer gain is,

$$T(j1)=\frac{1}{\sqrt{2}}$$

Consider the transfer function.

$$T(s)=\frac{a_0}{s^2+s\frac{\omega_0}{Q}+\omega^2_0}$$
$$T(s)=\frac{\left(\frac{\omega^2_0}{\sqrt{2}}\right)}{s^2+s\frac{\omega_0}{Q}+\omega^2_0} \qquad \left(\text{Since, } a_0=\frac{\omega^2_0}{\sqrt{2}}\right)$$
$$T(j\omega)=\frac{\left(\frac{\omega^2_0}{\sqrt{2}}\right)}{-\omega^2+j\frac{\omega_0}{Q}+\omega^2_0}$$
$$|T(j\omega)|=\frac{\left(\frac{\omega^2_0}{\sqrt{2}}\right)}{\sqrt{\left(\omega^2_0-\omega^2\right)^2+\frac{\omega^2_0}{Q^2}}}$$

Step 10 of 14

Substitute 1 for ω in the equation.

$$|T(j1)|^2=\frac{\left(\frac{\omega^2_0}{\sqrt{2}}\right)^2}{\left(\omega^2_0-1\right)^2+\frac{\omega^2_0}{Q^2}}$$
$$\frac{1}{2}=\frac{\left(\frac{\omega^2_0}{\sqrt{2}}\right)^2}{\left(\omega^2_0-1\right)^2+\frac{\omega^2_0}{Q^2}} \qquad \left(\text{Since, } T(j1)=\frac{1}{\sqrt{2}}\right)$$

Simplify further.

$$\frac{\omega^4_0}{2}=\frac{1}{2}\left[\omega^4_0-2\omega^2_0+1+\frac{\omega^2_0}{Q^2}\right]$$
$$\omega^4_0=\omega^4_0-2\omega^2_0+1+\frac{\omega^2_0}{Q^2}$$
$$2\omega^2_0-\frac{\omega^2_0}{Q^2}=1$$

Simplify further.

$$2\omega^2_0-\frac{\omega^2_0}{Q^2}=1$$
$$\omega^2_0\left(2-\frac{1}{Q^2}\right)=1$$
$$\omega_0=\sqrt{\frac{1}{2-\frac{1}{Q^2}}}$$
$$\omega_0=\sqrt{\frac{1}{2-\frac{1}{1.3066^2}}}$$
$$\omega_0=0.841$$

Thus, the value of ω_0 is, $\boxed{\omega_0=0.841}$

Step 11 of 14

Therefore, the transfer function is,

$$T_1(s)=\frac{\frac{\omega^2_0}{\sqrt{2}}}{s^2+\frac{\omega_0}{Q}s+\omega^2_0}$$
$$=\frac{\frac{(0.841)^2}{\sqrt{2}}}{s^2+\frac{0.841}{Q}s+(0.841)^2}$$
$$=\frac{0.5}{s^2+0.644s+0.707}$$

Step 12 of 14

Consider the transfer function $T_1(s)$.

$$T_1(s)=\frac{1}{s^2+\sqrt{2}s+1}$$
$$T_1(j\omega)=\frac{1}{-\omega^2+j\sqrt{2}\omega+1}$$
$$|T_1(j\omega)|=\frac{1}{\sqrt{\left(1-\omega^2\right)^2+\left(\sqrt{2}\omega\right)^2}}$$
$$=\frac{1}{\sqrt{\left(1-\omega^2\right)^2+2\omega^2}}$$

Step 13 of 14

If $\omega_s = 2$ rad/sec , the transfer function is,

$$|T_1(j2)|=\frac{1}{\sqrt{\left(1-2^2\right)^2+2\left(2^2\right)}}$$
$$=0.242$$

Calculate the value in dB.

$$20\log\left(|T_1(j2)|\right)=20\log(0.242)$$
$$=-12.3 \text{ dB}$$

Thus, the value of A_{min} in case 1 is $\boxed{-12.3 \text{ dB}}$.

Step 14 of 14

Consider the transfer function $T_2(s)$.

$$T_2(s)=\frac{0.5}{s^2+0.644s+0.707}$$
$$T_1(j\omega)=\frac{0.5}{-\omega^2+j0.644\omega+0.707}$$
$$|T_2(j\omega)|=\frac{0.5}{\sqrt{\left(0.707-\omega^2\right)^2+\left(0.644\omega\right)^2}}$$

If $\omega_s = 2$ rad/sec , the transfer function is,

$$|T_2(j2)|=\frac{0.5}{\sqrt{\left(0.707-2^2\right)^2+\left(0.644(2)\right)^2}}$$
$$=0.1414$$

Calculate the value in dB.

$$20\log\left(|T_1(j2)|\right)=20\log(0.1414)$$
$$=-17 \text{ dB}$$

Thus, the value of A_{min} in case 2 is $\boxed{-17 \text{ dB}}$.

Step 1 of 4

Refer to Figure 16.42(a) in the text book.

Consider the expression.

$$Q = \frac{f_0}{BW}$$

Substitute $10\ \text{kHz}$ for BW , in equation.

$$Q = \frac{455\times10^3}{10\times10^3}$$
$$= 45.5$$

According to specifications, it is required that,

$$\frac{R_1 \| n^2 R_{in}}{\omega_0 L} = Q$$
$$\frac{R_1 \| n^2 R_{in}}{2\pi f_0 L} = 45.5$$

Substitute $1\ \text{k}\Omega$ for R_{in} , $1\ \text{k}\Omega$ for R_1 , $455\ \text{kHz}$ for f_0 and $5\ \mu\text{H}$ for L in equation.

$$\frac{\left(1\times10^3\right) \| n^2 \left(1\times10^3\right)}{2\pi\left(455\times10^3\right)\left(5\times10^{-6}\right)} = 45.5$$
$$\frac{1 \| n^2}{2\pi(455)\left(5\times10^{-6}\right)} = 45.5$$
$$1 \| n^2 = 45.5(2\pi)(455)\left(5\times10^{-6}\right)$$
$$\frac{n^2}{1+n^2} = 0.65$$
$$n^2 = 0.65 + 0.65n^2$$
$$0.35n^2 = 0.65$$
$$n^2 = \frac{0.65}{0.35}$$
$$n = \sqrt{\frac{0.65}{0.35}}$$
$$= 1.36$$

Therefore, the value of n is $\boxed{1.36}$.

Step 2 of 4

Calculate the value of C_1 .

$$C_1 + \frac{C_{in}}{n^2} = \frac{1}{\omega_0^2 L}$$
$$C_1 + \frac{C_{in}}{n^2} = \frac{1}{\left(2\pi f_0\right)^2 L}$$

Substitute $200\ \text{pF}$ for C_{in} , $455\ \text{kHz}$ for f_0 , 1.36 for n and $5\ \mu\text{H}$ for L in equation.

$$C_1 + \frac{200\times10^{-12}}{(1.36)^2} = \frac{1}{\left(2\pi\left(455\times10^3\right)\right)^2\left(5\times10^{-6}\right)}$$
$$C_1 + 0.11\times10^{-9} = 24.47\times10^{-9}$$
$$C_1 = \left(24.47\times10^{-9}\right) - \left(0.11\times10^{-9}\right)$$
$$= 24.36\ \text{nF}$$

Therefore, the value of C_1 is $\boxed{24.36\ \text{nF}}$.

Step 3 of 4

The voltage developed across resistor R_1 at resonance is

$$V_{R1} = I\left(R_1 \| n^2 R_{in}\right)$$

Substitute $1\ \text{k}\Omega$ for R_{in} , $1\ \text{k}\Omega$ for R_1 and 1.36 for n in equation.

$$V_{R1} = I\left(\left(1\times10^3\right) \| (1.36)^2\left(1\times10^3\right)\right)$$
$$= I(1 \| 1.86)\times10^3$$
$$= I\left(\frac{1.86}{1+1.86}\right)\times10^3$$
$$= 650I$$

Step 4 of 4

Thus the voltage developed V_{De} is

$$V_{De} = \frac{V_{R1}}{n}$$
$$= \frac{650I}{n} \qquad \left(\text{Since } V_{R1} = 650I\right)$$

$$V_{De} = 478I \quad \text{..... (1)}$$

The current I_C is

$$I_C = g_m V_{De} \quad \text{..... (2)}$$

Substitute equation (1) in equation (2).

$$I_C = g_m(478I)$$
$$I_C = \left(40\times10^{-3}\right)(478I)$$
$$\frac{I_C}{I} = \left(40\times10^{-3}\right)(478)$$
$$= 19.1\ \text{A/A}$$

Therefore, the current gain at resonance $\frac{I_C}{I}$ is $\boxed{19.1\ \text{A/A}}$.

Step 1 of 4

Refer to the Figure 17.16(b) in the textbook.

The transfer function $T(s)$ of second order high pass filter is,

$$T(s)=\frac{a_2 s^2}{s^2+s\left(\frac{\omega_0}{Q}\right)+\omega_0^2} \quad \text{..... (1)}$$

Consider quality factor Q is $\frac{1}{\sqrt{2}}$. Since, it is maximally flat response in passband.

Determine the center frequency ω_0.

Consider the 3-dB bandwidth ω is as,

$$BW=\frac{\omega_0}{Q}$$

Substitute $1\ \text{rad/s}$ for BW and $\frac{1}{\sqrt{2}}$ for Q in the equation.

$$1=\frac{\omega_0}{\left(\frac{1}{\sqrt{2}}\right)}$$

$$\omega_0=\frac{1}{\sqrt{2}}\ \text{rad/s}$$

Therefore, the center frequency ω_0 is, $\boxed{\frac{1}{\sqrt{2}}\ \text{rad/s}}$.

Step 2 of 4

Determine the coefficient a_2.

Consider the high frequency gain $|T(s)|_{s=\infty}$.

$$|T(s)|_{s=\infty}=a_2$$

Substitute 1 for $|T(s)|_{s=\infty}$ in the above expression.

$$1=a_2$$

Therefore, the coefficient a_2 is, $\boxed{1}$.

Step 3 of 4

Determine the transfer function of second order high pass filter.

Substitute 1 for a_2, $\frac{1}{\sqrt{2}}\ \text{rad/s}$ for ω_0, $\frac{1}{\sqrt{2}}$ for Q in equation (1).

$$T(s)=\frac{(1)s^2}{s^2+s\left(\frac{\frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}}\right)+\left(\frac{1}{\sqrt{2}}\right)^2}$$

$$=\frac{s^2}{s^2+s+0.5}$$

Therefore, the transfer function of $T(s)$ is, $\boxed{\frac{s^2}{s^2+s+0.5}}$.

Step 4 of 4

The poles of the function is,

$$P_1,P_2=\frac{-1\pm\sqrt{1^2-4(0.5)}}{2}$$

$$=\frac{-1\pm\sqrt{1-2}}{2}$$

$$=-0.5\pm j0.5$$

Therefore, the system's pair of complex conjugate poles are $\boxed{-0.5\pm j0.5}$. And the pair of zeros are placed at $\boxed{\text{origin}}$.

Step 1 of 3

Consider the design of an intermediate frequency (IF) amplifier for an FM radio receiver.

Consider the following expressions for the overall bandwidth:

$$B = \frac{\omega_0}{Q}\sqrt{2^{\frac{1}{N}} - 1} \qquad (B \text{ in rad/s})$$

$$B = \frac{f_0}{Q}\sqrt{2^{\frac{1}{N}} - 1} \qquad (B \text{ in Hz})$$

Here, N is the number of synchronously tuned stages.

Consider that the overall bandwidth, $B = 200 \text{ kHz}$.

The 3-dB bandwidth required by the individual stages is,

$$\frac{f_0}{Q} = \frac{B}{\sqrt{2^{\frac{1}{N}} - 1}}$$

Substitute 200 kHz for B, and 2 for N.

$$\begin{aligned}\frac{f_0}{Q} &= \frac{200 \text{ kHz}}{\sqrt{2^{\frac{1}{2}} - 1}} \\ &= \frac{200 \times 10^3}{\sqrt{1.414 - 1}} \\ &= \frac{200 \times 10^3}{0.6435} \\ &= 310.75 \text{ kHz}\end{aligned}$$

Thus, the 3-dB bandwidth of the each stage is, $\boxed{310.75 \text{ kHz}}$.

Step 2 of 3

Find the values of the components in each stage of the tuned amplifier.

Consider that the center frequency of the tuned stages, $f_0 = 10.7 \text{ MHz}$.

Calculate the angular frequency.

$$\begin{aligned}\omega_0 &= 2\pi f_0 \\ &= 2\pi(10.7 \text{ MHz}) \\ &= 67.23 \text{ Mrad/s}\end{aligned}$$

Consider the following expression for the center frequency:

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Simplify the expression to get an expression for the capacitance.

$$\sqrt{LC} = \frac{1}{\omega_0}$$

$$LC = \frac{1}{\omega_0^2}$$

$$C = \frac{1}{L\omega_0^2}$$

Substitute $3 \ \mu\text{H}$ for L and 67.23 Mrad/s for ω_0.

$$\begin{aligned}C &= \frac{1}{(3 \ \mu\text{H})(67.23 \text{ Mrad/s})^2} \\ &= \frac{1}{(3 \times 10^{-6})(4.52 \times 10^{15})} \\ &= \frac{1}{13.56 \times 10^9} \\ &= 73.7 \text{ pF}\end{aligned}$$

Thus, the value of capacitance, C is $\boxed{73.7 \text{ pF}}$.

Step 3 of 3

Consider the following expression for the bandwidth:

$$\frac{f_0}{Q} = \frac{1}{2\pi RC}$$

Simplify the equation to get an expression for the resistance.

$$RC = \frac{1}{2\pi \frac{f_0}{Q}}$$

$$R = \frac{1}{2\pi C \frac{f_0}{Q}}$$

Substitute 73.7 pF for C and 310.75 kHz for $\frac{f_0}{Q}$.

$$\begin{aligned}R &= \frac{1}{2\pi(73.7 \text{ pF})(310.75 \text{ kHz})} \\ &= \frac{1}{2\pi(73.7 \times 10^{-12})(310.75 \times 10^3)} \\ &= \frac{1}{1.43 \times 10^{-4}} \\ &= 6.95 \text{ k}\Omega\end{aligned}$$

Thus, the value of resistance, R is $\boxed{6.95 \text{ k}\Omega}$.

Step 1 of 5

17.036P

Refer to Figure 17.16(6) in the textbook.

Consider the transfer function $T(s)$ of high pass filter.

$$T(s)=\frac{a_2 s^2}{s^2+s\frac{\omega_0}{Q}+\omega_0^2} \quad \text{..... (1)}$$

Here,

ω_0 is angular frequency,

Q is quality factor.

Step 2 of 5

Consider the s-Plane singularities of second order High pass filter, which is shown in Figure 1.

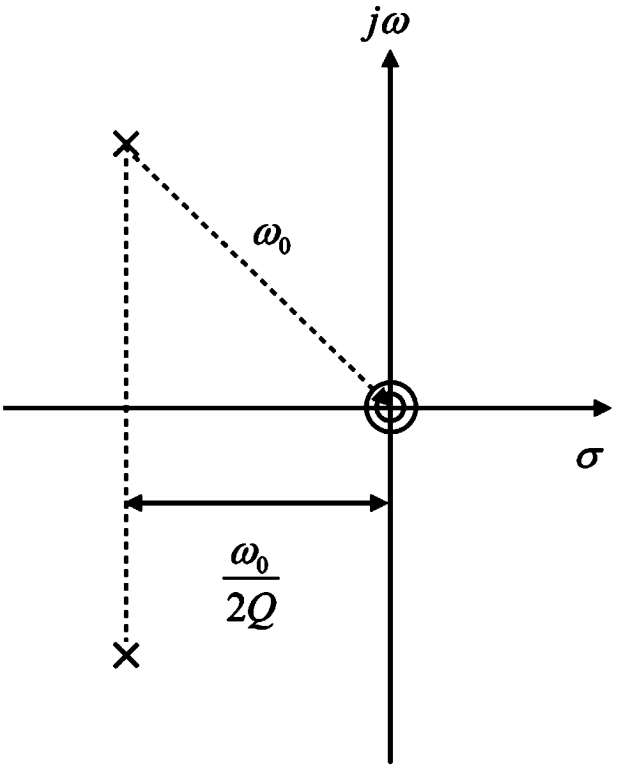


Figure1

Consider the natural modes or poles of the high pass filter are $-0.5 \pm j\frac{\sqrt{3}}{2}$.

From the s-plane of Figure 1, the value of ω_0 is,

$$\omega_0=\sqrt{(0.5)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$$
$$=\sqrt{0.25+0.75}$$
$$=1 \text{ rad/sec}$$

Therefore, the angular frequency ω_0 is, $\boxed{1 \text{ rad/s}}$.

Step 3 of 5

Determine the quality factor Q.

From the Figure 1, the real component is $\frac{\omega_0}{2Q}$.

$$\frac{\omega_0}{2Q}=0.5$$

Substitute 1 rad/sec for ω_0 in the equation.

$$\frac{1}{2Q}=\frac{1}{2}$$
$$Q=1$$

Thus, the quality factor Q is, $\boxed{1}$.

Step 4 of 5

Consider the high frequency gain $|T(j\infty)|$ of high-pass filter is unity.

$$T(s)=\frac{a_2 s^2}{s^2+s\frac{\omega_0}{Q}+\omega_0^2}$$

Substitute $j\omega$ for s in the transfer function.

$$T(j\omega)=\frac{-a_2\omega^2}{-\omega^2+j\omega\frac{\omega_0}{Q}+\omega_0^2}$$
$$T(j\omega)=\frac{-a_2}{-1+j\frac{\omega_0}{Q\omega}+\frac{\omega_0^2}{\omega^2}}$$
$$|T(j\omega)|=\frac{a_2}{\sqrt{\left(\frac{\omega_0^2}{\omega^2}-1\right)^2+\left(\frac{\omega_0}{Q\omega}\right)^2}}$$

The high-frequency gain is,

$$|T(j\infty)|=\frac{a_2}{\sqrt{\left(\frac{\omega_0^2}{\infty^2}-1\right)^2+\left(\frac{\omega_0}{Q(\infty)}\right)^2}}$$
$$|T(j\infty)|=\frac{a_2}{\sqrt{(0-1)^2+(0)^2}}$$
$$|T(j\infty)|=a_2$$

Substitute 1 for $|T(j\infty)|$ in the equation.

$$a_2=1$$

Thus, the coefficient a_2 is high frequency gain of transfer function.

Step 5 of 5

Determine the transfer function of the second order high-pass filter.
Substitute 1 rad/sec for ω_0, 1 for Q, 1 for a_2 in the equation (1).

$$T(s)=\frac{(1)s^2}{s^2+s\left(\frac{1}{1}\right)+(1)^2}$$
$$=\frac{s^2}{s^2+s+1}$$

Therefore, the transfer function $T(s)$ of second order high-pass filter is $\boxed{\frac{s^2}{s^2+s+1}}$.

Step 1 of 4

Refer to the Figure 17.16(c) in the textbook.

Consider the transfer function $T(s)$ of band pass filter as,

$$T(s)=\frac{a_1 s}{s^2+s\left(\frac{\omega_0}{Q}\right)+\omega_0^2}$$

Determine the quality factor Q.

3-dB frequency of bandpass filter is,

$$BW=\omega_2-\omega_1$$

$$BW=\frac{\omega_0}{Q}$$

$$=\frac{f_0}{Q}$$

Substitute $500\,\text{Hz}$ for BW and $10\,\text{kHz}$ for f_0 in the equation.

$$500=\frac{10\times10^3}{Q}$$

$$Q=\frac{10\times10^3}{500}$$

$$=20$$

Therefore, the quality factor Q is, $\boxed{20}$.

Step 2 of 4

Consider the central frequency gain a_1 as,

$$\left|T(s)\right|_{s=\omega_0}=\frac{a_1 Q}{\omega_0}$$

$$\left|T(s)\right|_{s=f_0}=\frac{a_1 Q}{2\pi f_0} \quad \text{..... (1)}$$

Determine the coefficient a_1.

Substitute 20 for Q, $10\,\text{kHz}$ for f_0 and 10 for $\left|T(s)\right|_{s=f_0}$ in the equation (1).

$$10=\frac{a_1(20)}{2\pi\left(10\times10^3\right)}$$

$$a_1=\frac{628\times10^3}{20}$$

$$=31400$$

Therefore, the coefficient a_1 is, $\boxed{31400}$.

Step 3 of 4

The transfer function $T(s)$ of band pass filter is,

$$T(s)=\frac{a_1 s}{s^2+s\left(\frac{\omega_0}{Q}\right)+\omega_0^2}$$

Substitute 20 for Q, $10\,\text{kHz}$ for f_0 and 31400 for a_1 in the equation.

$$T(s)=\frac{(31400)s}{s^2+s\left(\frac{2\pi\left(10\times10^3\right)}{20}\right)+\left(2\pi\left(10\times10^3\right)\right)}$$

$$=\frac{(31400)s}{s^2+s(3140)+(62800)^2}$$

$$=\frac{31400s}{s^2+3140s+3943.8\times10^6}$$

Therefore, the transfer function of second order bandpass filter $T(s)$ is,

$$\boxed{\frac{31400s}{s^2+3140s+3943.8\times10^6}}.$$

Step 4 of 4

The poles P_1, P_2 are,

$$P_1,P_2=\frac{-3140\pm\sqrt{3140^2-4\left(3943.8\times10^6\right)}}{2}$$

$$=-1570\pm j62780.1$$

Therefore, the location of pair of complex conjugate poles are $\boxed{-1570\pm j62780.1}$.

And one zero placed at $\boxed{\textit{origin}}$ and another zero placed at $\boxed{\infty}$.

Step 1 of 10

(a)

The transfer function of a second order band pass filter is,

$$T(s)=\frac{a_1 s}{s^2+s\frac{\omega_0}{Q}+\omega_0^2} \quad \ldots\ldots (1)$$

Substitute $j\omega$ for s in the equation (1).

$$\begin{aligned}T(j\omega)&=\frac{a_1(j\omega)}{(j\omega)^2+(j\omega)\frac{\omega_0}{Q}+\omega_0^2}\\&=\frac{ja_1\omega}{\omega_0^2-\omega^2+j\frac{\omega\omega_0}{Q}}\end{aligned}$$

The magnitude of the transfer function is,

$$|T(j\omega)|=\frac{a_1\omega}{\sqrt{\left(\omega_0^2-\omega^2\right)^2+\left(\frac{\omega\omega_0}{Q}\right)^2}}$$

Step 2 of 10

Determine the magnitude of transfer function at $\omega=\omega_1$.

$$|T(j\omega_1)|=\frac{a_1\omega_1}{\sqrt{\left(\omega_0^2-\omega_1^2\right)^2+\left(\frac{\omega_1\omega_0}{Q}\right)^2}} \quad \ldots\ldots (2)$$

Determine the magnitude of transfer function at $\omega=\omega_2$.

$$|T(j\omega_2)|=\frac{a_1\omega_2}{\sqrt{\left(\omega_0^2-\omega_2^2\right)^2+\left(\frac{\omega_2\omega_0}{Q}\right)^2}} \quad \ldots\ldots (3)$$

Equate the equation (2) to the equation (3).

$$|T(j\omega_1)|=|T(j\omega_2)|$$

$$\begin{aligned}\frac{a_1\omega_1}{\sqrt{\left(\omega_0^2-\omega_1^2\right)^2+\left(\frac{\omega_1\omega_0}{Q}\right)^2}}&=\frac{a_1\omega_2}{\sqrt{\left(\omega_0^2-\omega_2^2\right)^2+\left(\frac{\omega_2\omega_0}{Q}\right)^2}}\\\frac{\omega_1}{\sqrt{\left(\omega_0^2-\omega_1^2\right)^2+\left(\frac{\omega_1\omega_0}{Q}\right)^2}}&=\frac{\omega_2}{\sqrt{\left(\omega_0^2-\omega_2^2\right)^2+\left(\frac{\omega_2\omega_0}{Q}\right)^2}}\end{aligned}$$

Step 3 of 10

Square on both sides of the equation.

$$\begin{aligned}\frac{\omega_1^2}{\left[\left(\omega_0^2-\omega_1^2\right)^2+\left(\frac{\omega_1\omega_0}{Q}\right)^2\right]}&=\frac{\omega_2^2}{\left[\left(\omega_0^2-\omega_2^2\right)^2+\left(\frac{\omega_2\omega_0}{Q}\right)^2\right]}\\\omega_1^2\left[\left(\omega_0^2-\omega_2^2\right)^2+\left(\frac{\omega_2\omega_0}{Q}\right)^2\right]&=\omega_2^2\left[\left(\omega_0^2-\omega_1^2\right)^2+\left(\frac{\omega_1\omega_0}{Q}\right)^2\right]\\\omega_1^2\left(\omega_0^2-\omega_2^2\right)^2+\left(\frac{\omega_1\omega_2\omega_0}{Q}\right)^2&=\omega_2^2\left(\omega_0^2-\omega_1^2\right)^2+\left(\frac{\omega_1\omega_2\omega_0}{Q}\right)^2\\\omega_1^2\left(\omega_0^2-\omega_2^2\right)^2&=\omega_2^2\left(\omega_0^2-\omega_1^2\right)^2\end{aligned}$$

Expand the terms in the equation.

$$\begin{aligned}\omega_1^2\left(\omega_0^4+\omega_2^4-2\omega_0^2\omega_2^2\right)&=\omega_2^2\left(\omega_0^4+\omega_1^4-2\omega_0^2\omega_1^2\right)\\\omega_1^2\omega_0^4+\omega_1^2\omega_2^4-2\omega_0^2\omega_1^2\omega_2^2&=\omega_2^2\omega_0^4+\omega_2^2\omega_1^4-2\omega_0^2\omega_1^2\omega_2^2\\\omega_1^2\omega_0^4+\omega_1^2\omega_2^4&=\omega_2^2\omega_0^4+\omega_2^2\omega_1^4\\\omega_1^2\omega_0^4-\omega_2^2\omega_0^4&=\omega_2^2\omega_1^4-\omega_1^2\omega_2^4\\\omega_0^4\left(\omega_1^2-\omega_2^2\right)&=\omega_1^2\omega_2^2\left(\omega_1^2-\omega_2^2\right)\end{aligned}$$

$$\omega_0^4=\left(\omega_1\omega_2\right)^2$$

$$\omega_0^2=\left(\omega_1\omega_2\right)$$

Therefore, the members of each pair of frequencies are ω_1 and ω_2 for which $|T(j\omega_1)|=|T(j\omega_2)|$ are related by $\boxed{\omega_0^2=\omega_1\omega_2}$.

Step 4 of 10

(b)

Determine the center frequency ω_0.

The relation between the center frequency and ω_{p1},ω_{p2} is,

$$\omega_0^2=\omega_{p1}\omega_{p2}$$

Substitute 8100 rad/s for ω_{p1} , 10000 rad/s for ω_{p2} in the equation.

$$\begin{aligned}\omega_0^2&=8100\times10000\\\omega_0&=\sqrt{8100\times10000}\\&=9000\text{ rad/s}\end{aligned}$$

Step 5 of 10

Calculate the quality factor Q .

Consider the magnitude at $\omega=\omega_0$ is.

$$|T(j\omega)|=\frac{a_1\omega}{\sqrt{\left(\omega_0^2-\omega^2\right)^2+\left(\frac{\omega\omega_0}{Q}\right)^2}}$$

$$|T(j\omega_0)|=\frac{a_1\omega_0}{\left[\frac{\omega\omega_0}{Q}\right]}$$

$$|T(j\omega_0)|=\frac{a_1Q}{\omega_0}$$

Substitute $\mathbf{1}$ for $|T(j\omega)|$, 9000 rad/s for ω_0 in the equation.

$$\begin{aligned}1&=\frac{a_1Q}{9000}\\Q&=\frac{9000}{a_1}\end{aligned}$$

Step 6 of 10

The magnitudes of the filter at $\omega=\omega_{p1}$ and $\omega=\omega_{p2}$ are equal.

Consider the $A_{\max}$ is 3 dB .

$$|T(j\omega_{p1})|=|T(j\omega_{p2})|$$

$$A_{\max}=-20\log\left(|T(j\omega)|\right)_{\omega=\omega_{p1}=\omega_{p2}}$$

$$3\text{ dB}=-20\log\left(|T(j\omega)|\right)_{\omega=\omega_{p1}=\omega_{p2}}$$

$$\begin{aligned}T(j\omega)&=10^{\frac{-3}{20}}\\&=0.7079\end{aligned}$$

Step 7 of 10

The magnitude at $\omega=\omega_{p1}$ is,

$$|T(j\omega_{p1})|^2=\frac{a_1^2\omega_{p1}^2}{\left[\left(\omega_0^2-\omega_{p1}^2\right)^2+\left(\frac{\omega_{p1}\omega_0}{Q}\right)^2\right]}$$

Substitute 8100 rad/s for ω_{p1} , 9000 rad/s for ω_0 and 0.7079 for $|T(j\omega_{p1})|$ in the equation.

$$(0.7079)^2=\frac{a_1^2(8100)^2}{\left[\left((9000)^2-(8100)^2\right)^2+\left(\frac{8000\times9000}{Q}\right)^2\right]}$$

Substitute $\frac{9000}{a_1}$ for Q in the equation.

$$\begin{aligned}(0.7079)^2&=\frac{a_1^2(8100)^2}{\left[(9000)^2-8100^2\right)^2+\left(\frac{8000\times9000}{\frac{9000}{a_1}}\right)^2\Bigg]}\\(0.7079)^2&=\frac{a_1^2(8100)^2}{\left[\left((9000)^2-(8100)^2\right)^2+(8000a_1)^2\right]}\\(0.7079)^2&=\frac{a_1^2(65.61\times10^6)}{\left[236.8521\times10^{12}+(64\times10^6)a_1^2\right]}\\118.69\times10^{12}+(32.071\times10^6)a_1^2&=a_1^2(65.61\times10^6)\\118.69\times10^{12}+(32.071\times10^6)a_1^2-a_1^2(65.61\times10^6)&=0\\(33.539\times10^6)a_1^2&=118.69\times10^{12}\\a_1^2&=\frac{118.69\times10^{12}}{33.539\times10^6}\\a_1&=\sqrt{3.538\times10^6}\\&=1.881\times10^3\end{aligned}$$

Step 8 of 10

The quality factor is,

$$Q=\frac{9000}{a_1}$$

Substitute 1.881×10^3 for a_1 in equation.

$$\begin{aligned}Q&=\frac{9000}{1.881\times10^3}\\&=4.78\end{aligned}$$

Step 9 of 10

Determine the transfer function of a second order band pass filter.

$$T(s)=\frac{a_1 s}{s^2+s\frac{\omega_0}{Q}+\omega_0^2}$$

Substitute 1.88×10^3 for a_1 , 4.78 for Q , 9000 for ω_0 in the equation.

$$\begin{aligned}T(s)&=\frac{\left(1.881\times10^3\right)s}{s^2+s\frac{9000}{4.78}+9000^2}\\&=\frac{1881s}{s^2+1881s+9000^2}\end{aligned}$$

Therefore, the transfer function of a second order band pass filter is,

$$\boxed{T(s)=\frac{1881s}{s^2+1881s+9000^2}}$$

Step 10 of 10

The lower stop band frequency is,

$$\omega_{s1}=3000\text{ rad/s}$$

The minimum required stopband attenuation is,

$$A_{\min}=-20\log\left(|T(j\omega_{s1})|\right).$$

The magnitude at $\omega_{s1}=3000$ rad/s is.

$$\begin{aligned}|T(j\omega_{s1})|&=\frac{a_1\omega_{s1}}{\left[\left(\omega_0^2-\omega_{s1}^2\right)^2+\left(\frac{\omega_{s1}\omega_0}{Q}\right)^2\right]^{\frac{1}{2}}}\\&=\frac{1881(3000)}{\left[\left((9000)^2-(3000)^2\right)^2+\left(\frac{3000\times9000}{1881}\right)^2\right]^{\frac{1}{2}}}\\&=0.0781\end{aligned}$$

The minimum required stopband attenuation is,

$$\begin{aligned}A_{\min}&=-20\log(0.0781)\\&=22.14\text{ dB}\end{aligned}$$

Therefore, the minimum required stopband attenuation $A_{\min}$ is. $\boxed{22.14\text{ dB}}$.

Step 1 of 3

17.039P

The quality factor of a second order notch filter with notch frequency, $\omega_n = \omega_0$ is,

$$Q = \frac{\omega_0}{BW_n \sqrt{10^{\frac{A}{10}} - 1}}$$

The interference frequency is, $f_0 = 60\ \text{Hz}$.

The DC gain is, $a_2 = 1$.

The attenuation is, $A = 20\ \text{db}$.

The Band width is,

$$\begin{aligned} BW_n &= 2\pi \times 6 \\ &= 12\pi\ \text{Hz} \end{aligned}$$

The interference frequency is,

$$\begin{aligned} \omega_0 &= 2\pi f_0 \\ &= 2\pi \times 60 \\ &= 120\pi\ \text{rad/s} \end{aligned}$$

Step 2 of 3

The quality factor is,

$$\begin{aligned} Q &= \frac{120\pi}{12\pi\sqrt{10^{\frac{20}{10}} - 1}} \\ &= \frac{120\pi}{12\pi\sqrt{99}} \\ &= 1.005 \end{aligned}$$

Step 3 of 3

The transfer function of a notch filter is,

$$T(s) = a_2 \frac{s^2 + \omega_0^2}{s^2 + s\frac{\omega_0}{Q} + \omega_0^2}$$

Substitute 1 for a_2, $120\pi\ \text{rad/s}$ for ω_0, and 1.005 for Q in the equation.

$$\begin{aligned} T(s) &= (1)\frac{s^2 + (120\pi)^2}{s^2 + s\frac{120\pi}{1.005} + (120\pi)^2} \\ &= \frac{s^2 + 141978.24}{s^2 + 374.93s + 141978.24} \\ &= \frac{s^2 + 1.42 \times 10^5}{s^2 + 374.93s + 1.42 \times 10^5} \end{aligned}$$

Thus, the transfer function of the notch filter is, $\boxed{T(s) = \frac{s^2 + 1.42 \times 10^5}{s^2 + 374.93s + 1.42 \times 10^5}}$.

Step 1 of 3

Case 1:

The frequency of the zeros of second order all pass filter is slightly lower than that of the poles.

The transfer function of second order all-pass filter is,

$$T(s)=a_2\left(\frac{s^2-s\frac{\omega_n}{Q}+\omega_0^2}{s^2+s\frac{\omega_0}{Q}+\omega_0^2}\right).$$

The transfer function at $s=j\omega_n$ is,

$$T(j\omega_n)=a_2\left(\frac{(j\omega_n)^2-(j\omega_n)\frac{\omega_n}{Q}+\omega_0^2}{(j\omega_n)^2+(j\omega_n)\frac{\omega_0}{Q}+\omega_0^2}\right)$$

$$=a_2\left(\frac{\omega_0^2-\omega_n^2-(j\omega_n)\frac{\omega_n}{Q}}{\omega_0^2-\omega_n^2+(j\omega_n)\frac{\omega_0}{Q}}\right)$$

At DC, the transfer function is,

$|T|=a_2\frac{\omega_n^2}{\omega_0^2}$ where $\omega_n^2<\omega_0^2$.

The magnitude of the transfer function at $s=j\omega_n$, for frequency of the zeros lower than that of the poles is,

$\boxed{|T|=a_2\frac{\omega_n^2}{\omega_0^2}}$ where $\omega_n^2<\omega_0^2$.

Step 2 of 3

The sketch of the magnitude, $|T|$, for frequency of the zeros lower than that of the poles is shown in Figure 1.

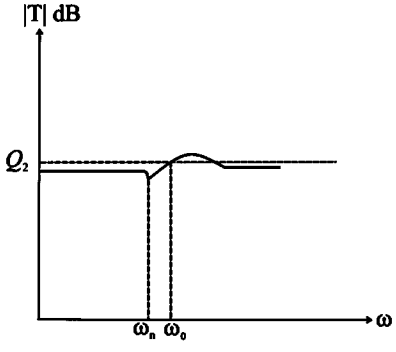


Figure 1

Step 3 of 3

Case 2:

The frequency of the zeros of second order all pass filter is slightly highly than that of the poles.

At DC, the transfer function is,

$|T|=a_2\frac{\omega_n^2}{\omega_0^2}$ where $\omega_n^2>\omega_0^2$.

The magnitude of the transfer function at $s=j\omega_n$, for frequency of the zeros higher than that of the poles is given by

$\boxed{|T|=a_2\frac{\omega_n^2}{\omega_0^2}}$ where $\omega_n^2>\omega_0^2$.

The sketch of the magnitude, $|T|$, for frequency of the zeros higher than that of the poles is shown in Figure 2.

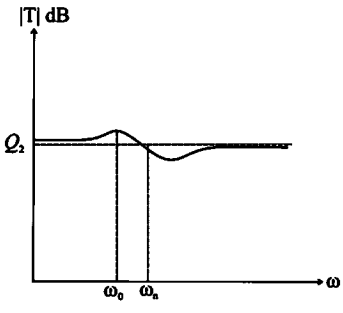


Figure 2

Step 1 of 3

The transfer function of second order all-pass filter is,

$$T(s)=a_2\left(\frac{s^2-s\frac{\omega_0}{Q'}+\omega_0^2}{s^2+s\frac{\omega_0}{Q_0}+\omega_0^2}\right).$$

The transfer function at $s=j\omega_0$ is,

$$\begin{aligned}T(j\omega_0)&=a_2\left(\frac{(j\omega_0)^2-(j\omega_0)\frac{\omega_0}{Q'}+\omega_0^2}{(j\omega_0)^2+(j\omega_0)\frac{\omega_0}{Q_0}+\omega_0^2}\right)\\&=a_2\left(\frac{-(j\omega_0)\frac{\omega_0}{Q'}}{+(j\omega_0)\frac{\omega_0}{Q_0}}\right)\end{aligned}$$

Step 2 of 3

The magnitude of the transfer function at $s=j\omega_0$ is,

$$|T(j\omega_0)|=\frac{a_2Q_0}{Q'}$$

Case 1:

The Q factor of the zeros of a second-order all-pass filter is greater than the Q factor of the poles.

For $Q'>Q_0$, the transfer function is,

$$\boxed{|T(j\omega_0)|=\frac{a_2Q_0}{Q'}<a_2}.$$

Case 2:

The Q factor of the zeros of a second-order all-pass filter is lower than the Q factor of the poles.

For $Q'<Q_0$, the transfer function is,

$$\boxed{|T(j\omega_0)|=\frac{a_2Q_0}{Q'}>a_2}.$$

Step 3 of 3

The sketch of the magnitude $|T|$ for both the cases is shown in Figure 1.

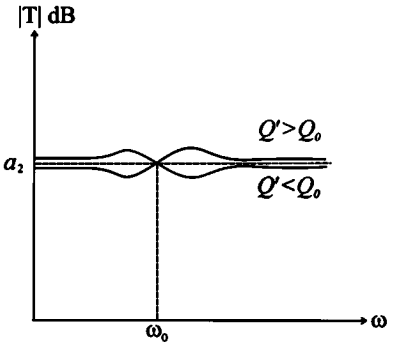


Figure 1

Step 1 of 3

Refer to the Figure 17.17(c) in the textbook.

Determine the transfer function $\frac{V_0}{V_i}$.

Consider the impedances $Z_1(s)$ and $Z_2(s)$ as follows:

The resistance R and capacitor C are in parallel.

Impedance $Z_2(s)$ is,

$$\begin{aligned} Z_2(s) &= R \left\| \left(\frac{1}{Cs}\right) \right. \\ &= \frac{R\left(\frac{1}{Cs}\right)}{R+\left(\frac{1}{Cs}\right)} \\ &= \frac{R}{1+RCs} \end{aligned}$$

Impedance $Z_1(s)$ is,

$$Z_1(s) = sL.$$

Step 2 of 3

Apply voltage division rule across impedance $Z_2(s)$.

$$V_0 = V_i\left(\frac{Z_2(s)}{Z_2(s)+Z_1(s)}\right)$$

Substitute sL for $Z_1(s)$ and $\frac{R}{1+RCs}$ for $Z_2(s)$ in equation.

$$V_0 = V_i\left(\frac{\frac{R}{1+RCs}}{\left(\frac{R}{1+RCs}\right)+Ls}\right)$$

$$\begin{aligned} \frac{V_0}{V_i} &= \left(\frac{R}{R+Ls(1+RCs)}\right) \\ &= \left(\frac{R}{R+Ls+RLCs^2}\right) \\ &= \frac{\left(\frac{1}{LC}\right)}{s^2+\left(\frac{1}{RC}\right)s+\frac{1}{LC}} \end{aligned}$$

Therefore, the transfer function $\frac{V_0}{V_i}$ is $\boxed{\frac{\left(\frac{1}{LC}\right)}{s^2+\left(\frac{1}{RC}\right)s+\frac{1}{LC}}}$.

Step 3 of 3

Equate the denominator term of the transfer function $\frac{V_0}{V_i}$ to the standard form $\left[s^2+\left(\frac{\omega_0}{Q}\right)s+\omega_0^2\right]$.

$$\omega_0^2 = \frac{1}{LC}$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\frac{\omega_0}{Q} = \frac{1}{CR}$$

$$Q = \omega_0 CR$$

Thus, the frequency ω_0 is $\boxed{\frac{1}{\sqrt{LC}}}$, and the quality factor Q is $\boxed{\omega_0 CR}$.

Step 1 of 4

Consider the LCR resonator circuit diagram of Figure 17.17(a) in the textbook.

The pair of complex conjugate poles characterized with frequency ω_0 is,

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Substitute $10^5\ \text{rad/s}$ for ω_0 in equation.

$$10^5 = \frac{1}{\sqrt{LC}} \quad \text{..... (1)}$$

Step 2 of 4

Determine the capacitor C.

Consider the pole quality factor Q is,

$$Q = \omega_0 CR$$

Substitute $10^5\ \text{rad/s}$ for ω_0 , 5 for Q and $10\ \text{k}\Omega$ for R in equation.

$$5 = 10^5 \left(10 \times 10^3\right) C$$
$$C = 5 \times 10^{-9}$$
$$= 5\ \text{nF}$$

Therefore, the value of capacitor C is, $\boxed{5\ \text{nF}}$.

Step 3 of 4

Determine the inductor L.

Substitute $5\ \text{nF}$ for C in the equation (1).

$$10^5 = \frac{1}{\sqrt{L\left(5 \times 10^{-9}\right)}}$$
$$10^{10} = \frac{1}{L\left(5 \times 10^{-9}\right)}$$
$$L = \frac{1}{10^{10}\left(5 \times 10^{-9}\right)}$$
$$= \frac{1}{50}$$
$$= 0.02$$
$$= 20\ \text{mH}$$

Therefore, the value of inductor L is, $\boxed{20\ \text{mH}}$.

Step 4 of 4

The circuit design of the second order parallel LCR resonator is shown in Figure 1.

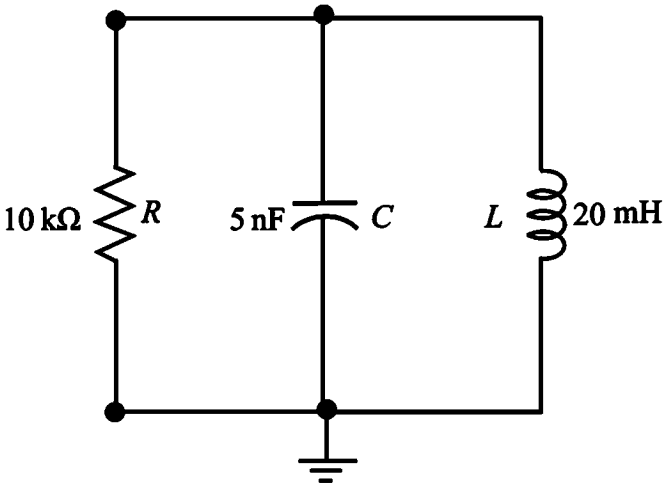


Figure 1

Thus, the design of LCR resonator is done.

Step 1 of 5

Refer to Figure 17.17(a) in the textbook.

Consider the resonant frequency ω_0 of second-order parallel LCR resonator.

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad \text{..... (1)}$$

(a)

Consider inductance L is increased by 1% . Then, the new value of L is, $1.01L$.

Determine resonant frequency $\omega_{0(new)}$.

Substitute $1.01L$ for L in equation (1).

$$\begin{aligned}\omega_{0(new)} &= \frac{1}{\sqrt{1.01LC}} \\ &= \frac{0.995}{\sqrt{LC}} \\ &= 0.995\omega_0\end{aligned}$$

Step 2 of 5

The decrease in the resonant frequency is,

$$\begin{aligned}\Delta\omega_0 &= \left(\frac{\omega_0 - \omega_{0(new)}}{\omega_0}\right)\times 100 \\ &= \left(\frac{\omega_0 - 0.995\omega_0}{\omega_0}\right)\times 100 \\ &= \left(\frac{1-0.995}{1}\right)\times 100 \\ &= 0.5\%\end{aligned}$$

Therefore, the decrease in the resonant frequency $\Delta\omega_0$ is, $\boxed{0.5\%}$.

Step 3 of 5

(b)

Consider capacitance C is increased by 1% . Then, the new value of C is, $1.01C$.

Determine resonant frequency $\omega_{0(new)}$.

Substitute $1.01C$ for C in equation (1).

$$\begin{aligned}\omega_{0(new)} &= \frac{1}{\sqrt{1.01LC}} \\ &= \frac{0.995}{\sqrt{LC}} \\ &= 0.995\omega_0\end{aligned}$$

Step 4 of 5

The decrease in the resonant frequency is,

$$\begin{aligned}\Delta\omega_0 &= \left(\frac{\omega_0 - \omega_{01}}{\omega_0}\right)\times 100 \\ &= \left(\frac{\omega_0 - 0.995\omega_0}{\omega_0}\right)\times 100 \\ &= \left(\frac{1-0.995}{1}\right)\times 100 \\ &= 0.5\%\end{aligned}$$

Therefore, the decrease in the resonant frequency $\Delta\omega_0$ is, $\boxed{0.5\%}$.

Step 5 of 5

(c)

The resistance, R of the parallel LCR resonator is increased by 1% .

From the equation (1), the resonant frequency is independent of the resistance. Hence, the change in the resistance has no effect on the resonant frequency.

That is, the resonant frequency is, $\omega_0 = \frac{1}{\sqrt{LC}}$.

Therefore, the change in the resonant $\Delta\omega_0$ is, $\boxed{0\%}$.

Step 1 of 10

(a)

Refer to the Figure P17.45 (a) in the textbook.

Consider the impedances $Z_1(s), Z_2(s)$ as follows:

Impedance $Z_1(s)$ is,

$$Z_1(s) = \left(\frac{1}{C_1 s}\right)$$

Impedance $Z_2(s)$ is,

$$Z_2(s) = \left(\frac{1}{C_2 s}\right)$$

Apply the Voltage division law.

$$V_0 = V_i\left(\frac{Z_2(s)}{Z_1(s)+Z_2(s)}\right)$$

Substitute $\frac{1}{C_1 s}$ for $Z_1(s)$ and $\frac{1}{C_2 s}$ for $Z_2(s)$ in equation.

$$V_0 = V_i\left(\frac{\frac{1}{C_2 s}}{\frac{1}{C_1 s}+\frac{1}{C_2 s}}\right)$$

Step 2 of 10

As $s = 0$,

$$\begin{aligned} Z_1(s) &= \frac{1}{C_1(0)} \\ &= \infty \end{aligned}$$

Similarly, $Z_2(s) = \infty$.

Simultaneously, $Z_1(s)$ and $Z_2(s)$ are can't be infinite. Hence, no transmission zeros at $s = 0$.

As $s = \infty$,

$$\begin{aligned} Z_1(s) &= \frac{1}{C_1(\infty)} \\ &= 0 \end{aligned}$$

Similarly, $Z_2(s) = 0$.

Simultaneously, $Z_1(s)$ and $Z_2(s)$ are can't be zero. Hence, no transmission zeros at $s = \infty$.

Thus, the circuit has **no transmission zeros** .

Step 3 of 10

(b)

Refer to the Figure P17.45 (b) in the textbook.

Consider the impedances $Z_1(s), Z_2(s)$ as follows:

Impedance $Z_1(s)$ is,

$$Z_1(s) = \left(\frac{1}{C_1 s}\right)$$

Impedance $Z_2(s)$ is,

$$\begin{aligned} Z_2(s) &= \left(\frac{1}{C_2 s} \| R\right) \\ &= \frac{R}{1+RC_2 s} \end{aligned}$$

Apply the Voltage division law.

$$V_0 = V_i\left(\frac{Z_2(s)}{Z_1(s)+Z_2(s)}\right)$$

Substitute $\frac{1}{C_1 s}$ for $Z_1(s)$ and $\frac{R}{1+RC_2 s}$ for $Z_2(s)$ in equation.

$$V_0 = \left(\frac{\frac{R}{1+RC_2 s}}{\frac{R}{1+RC_2 s}+\frac{1}{C_1 s}}\right)$$

$$\frac{V_0}{V_i} = \left(\frac{RC_1 s}{RC_1 s+1+RC_2 s}\right)$$

Step 4 of 10

As $s = 0$,

$$\begin{aligned} Z_1(s) &= \frac{1}{C_1(0)} \\ &= \infty \end{aligned}$$

Similarly,

$$\begin{aligned} Z_2(s) &= \frac{R}{1+RC_2 s} \\ &= \frac{R}{1+RC_2(0)} \\ &= R \end{aligned}$$

The impedance $Z_1(s)$ is infinite and $Z_2(s)$ is 4. Hence, it has transmission zeros at $s = 0$.

Step 5 of 10

As $s = \infty$,

$$\begin{aligned} Z_1(s) &= \frac{1}{C_1(\infty)} \\ &= 0 \end{aligned}$$

$$\begin{aligned} Z_2(s) &= \frac{R}{1+RC_2 s} \\ &= \frac{R}{1+RC_2(\infty)} \\ &= 0 \end{aligned}$$

The impedance $Z_1(s)$ is zero and $Z_2(s)$ is also zero. Hence, it has no transmission zero at $s = \infty$.

Thus, the circuit has **one transmission zero** .

Step 6 of 10

(c)

Refer to the Figure P17.45(c) in the textbook.

Consider the impedances $Z_1(s), Z_2(s)$ as follows:

Impedance $Z_1(s)$ is,

$$Z_1(s) = L_1 s$$

Impedance $Z_2(s)$ is,

$$Z_2(s) = L_2 s$$

Apply the Voltage division law.

$$V_0 = V_i\left(\frac{Z_2(s)}{Z_1(s)+Z_2(s)}\right)$$

Substitute $L_1 s$ for $Z_1(s)$ and $L_2 s$ for $Z_2(s)$ in equation.

$$V_0 = V_i\left(\frac{L_2 s}{L_1 s+L_2 s}\right)$$

Step 7 of 10

As $s = 0$.

$$\begin{aligned} Z_1(s) &= L_1 s \\ &= L_1(0) \\ &= 0 \end{aligned}$$

Similarly, $Z_2(s) = 0$.

Simultaneously, $Z_1(s)$ and $Z_2(s)$ are can't be zero. Hence, no transmission zeros at $s = 0$.

As $s = \infty$,

$$\begin{aligned} Z_1(s) &= L_1 s \\ &= L_1(\infty) \\ &= \infty \end{aligned}$$

Similarly, $Z_2(s) = \infty$.

Simultaneously, $Z_1(s)$ and $Z_2(s)$ are can't be infinite. Hence, no transmission zeros at $s = \infty$.

Thus, the circuit has **no transmission zeros** .

Step 8 of 10

(d)

Refer to the Figure P17.45 (d) in the textbook.

Consider the impedances $Z_1(s), Z_2(s)$ as follows:

Impedance $Z_1(s)$ is,

$$Z_1(s) = (R + L_1 s)$$

Impedance $Z_2(s)$ is,

$$Z_2(s) = L_2 s$$

Apply the Voltage division law.

$$V_0 = V_i\left(\frac{Z_2(s)}{Z_1(s)+Z_2(s)}\right)$$

Substitute $R + L_1 s$ for $Z_1(s)$ and $L_2 s$ for $Z_2(s)$ in equation.

$$\begin{aligned} V_0 &= V_i\left(\frac{L_2 s}{(R+L_1 s)+L_2 s}\right) \\ &= V_i\left(\frac{L_2 s}{R+(L_1+L_2)s}\right) \end{aligned}$$

Step 9 of 10

As $s = 0$,

$$\begin{aligned} Z_1(s) &= L_1 s + R \\ &= L_1(0) + R \\ &= R \end{aligned}$$

Similarly,

$$\begin{aligned} Z_2(s) &= L_2 s \\ &= L_2(0) \\ &= 0 \end{aligned}$$

The impedances $Z_1(s)$ is R and $Z_2(s)$ is zero. Hence, it has one transmission zero at $s = 0$.

Step 10 of 10

As $s = \infty$,

$$\begin{aligned} Z_1(s) &= L_1 s + R \\ &= L_1(\infty) + R \\ &= \infty \end{aligned}$$

Similarly,

$$\begin{aligned} Z_2(s) &= L_2 s \\ &= L_2(\infty) \\ &= \infty \end{aligned}$$

Simultaneously, $Z_1(s)$ and $Z_2(s)$ are can't be infinite. Hence, no transmission zeros at $s = \infty$.

Thus, the circuit has **one transmission zero** .

Step 1 of 2

17.046P

Refer to Figure 16.18(c) in the textbook for the high-pass circuit.

The impedance of the parallel combination of the resistor and inductor is,

$$\begin{aligned} Z &= R \| sL \\ &= \frac{R \times sL}{R + sL} \\ &= \frac{sLR}{R + sL} \end{aligned}$$

Redraw the circuit as shown in Figure 1.

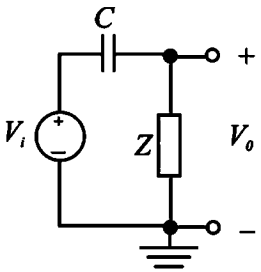


Figure 1

Step 2 of 2

Apply voltage divider rule to the circuit to calculate the output voltage.

$$\begin{aligned} V_0 &= \frac{Z}{Z + \frac{1}{sC}} V_s \\ &= \frac{\frac{sLR}{R + sL}}{\frac{sLR}{R + sL} + \frac{1}{sC}} V_s \\ &= \frac{s^2 LRC}{LRCs^2 + sL + R} V_s \end{aligned}$$

Simplify further.

$$\begin{aligned} V_0 &= \frac{s^2 LRC}{LRC\left(s^2 + \frac{1}{RC}s + \frac{1}{LC}\right)} V_s \\ \frac{V_0}{V_s} &= \frac{s^2}{\left(s^2 + \frac{1}{RC}s + \frac{1}{LC}\right)} \end{aligned}$$

Thus, the transfer function of the high-pass filter is,

$$\boxed{T(s) = \frac{V_0}{V_s} = \frac{s^2}{\left(s^2 + \frac{1}{RC}s + \frac{1}{LC}\right)}}$$

Step 1 of 3

17.047P

Refer to circuit diagram of Figure 17.18(b) in the textbook.

Determine the inductor L.

The resonant frequency ω_0 is,

$$\omega_0 = \frac{1}{\sqrt{LC}}$$
$$\omega_0^2 = \frac{1}{LC}$$
$$L = \frac{1}{C\omega_0^2}$$

Substitute $10^6\ \text{rad/s}$ for ω_0 and $1\ \text{nF}$ for C in the equation.

$$L = \frac{1}{C\omega_0^2}$$
$$= \frac{1}{\left(1\times10^{-9}\right)\left(10^6\right)^2}$$
$$= \frac{1}{10^3}$$
$$= 1\times10^{-3}\ \text{H}$$

Therefore, the value of inductor is, $\boxed{L = 1\ \text{mH}}$.

Step 2 of 3

Consider the quality factor Q is as follows:

$$Q = \omega_0 CR$$

Determine the resistance R.

$$R = \frac{Q}{\omega_0 C}$$

Substitute $10^6\ \text{rad/s}$ for ω_0, $\frac{1}{\sqrt{2}}$ for Q, and $1\ \text{nF}$ for C in the equation.

$$R = \frac{\left(\frac{1}{\sqrt{2}}\right)}{\left(10^6\right)\left(1\times10^{-9}\right)}$$
$$= \frac{1}{\left(\sqrt{2}\right)(0.001)}$$
$$= 0.7071\times10^3$$
$$= 707.1\ \Omega$$

Therefore, the value of resistor is, $\boxed{R = 707.1\ \Omega}$.

Step 3 of 3

The designed Low-pass filter circuit is shown in Figure 1.

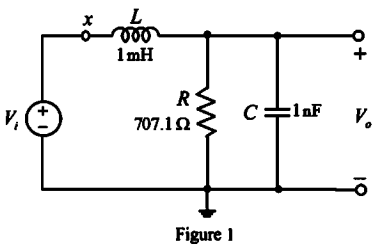


Figure 1

Therefore, the design of low-pass filter is done.

Step 1 of 2

17.048P

Refer to Figure 16.18(d) in the textbook for the band-pass circuit.

For the LCR resonator in the figure, the resonant frequency is,

$$\omega_0 = \frac{1}{\sqrt{LC}}.$$

The quality factor is,

$$Q = \omega_0 CR$$

The center-frequency gain is,

$$A_{mid} = 1$$

With out changing the center frequency or quality factor, the center-frequency gain can be changed to 0.5 by modifying the circuit as shown in Figure 1.

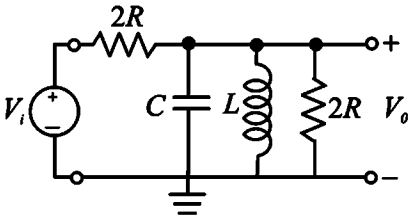


Figure 1

Step 2 of 2

Since the resonant frequency is independent of R, the resonant frequency doesn't change. Hence, the resonant frequency is,

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

The quality factor of the circuit shown in Figure 1 is,

$$\begin{aligned} Q &= \omega_0 C(2R \| 2R) \\ &= \omega_0 C\left(\frac{2R \times 2R}{2R + 2R}\right) \\ &= \omega_0 C\left(\frac{2R \times 2R}{4R}\right) \\ &= \omega_0 CR \end{aligned}$$

The center-frequency of the circuit shown in Figure 1 is,

$$\begin{aligned} A_{mid} &= \frac{2R}{2R + 2R} \\ &= \frac{2R}{4R} \\ &= \frac{1}{2} \\ &= 0.5 \end{aligned}$$

Thus, the resonant frequency and quality factor are unaltered and the center-frequency gain is changed to 0.5.

Step 1 of 4

17.049P

Refer to Figure 16.17 in the textbook for the second order parallel LCR resonator circuit.

Redraw the circuit indicating the voltages as shown in Figure 1.

The transfer function of the Band-pass structure is,

$$T_{BP}(s)=\left.\frac{V_0}{V_z}\right|_{V_x=0,V_y=0}$$

The transfer function of the High-pass structure is,

$$T_{HP}(s)=\left.\frac{V_0}{V_y}\right|_{V_x=0,V_z=0}$$

The transfer function of the Low-pass structure is,

$$T_{LP}(s)=\left.\frac{V_0}{V_x}\right|_{V_y=0,V_z=0}$$

Step 2 of 4

The voltage that develops across the resonator using super position theorem is,

$$V_0=\frac{V_0}{V_x}V_x+\frac{V_0}{V_y}V_y+\frac{V_0}{V_z}V_z$$

$$V_0=T_{LP}V_x+T_{HP}V_y+T_{BP}V_z \quad \text{..... (1)}$$

Apply Kirchhoff's current law to the circuit.

$$\frac{V_0-V_z}{R}+\frac{V_0-V_y}{\frac{1}{sC}}+\frac{V_0-V_x}{sL}=0$$

$$V_0\left(\frac{1}{R}+sC+\frac{1}{sL}\right)=\frac{V_z}{R}+sCV_y+\frac{V_x}{sL}$$

$$V_0=\frac{\frac{V_z}{R}+sCV_y+\frac{V_x}{sL}}{\frac{1}{R}+sC+\frac{1}{sL}}$$

$$V_0=\frac{\left(\frac{sLV_z+s^2LCRV_y+RV_x}{sLR}\right)}{\left(\frac{sL+s^2LCR+R}{sLR}\right)}$$

Simplify further.

$$V_0=\frac{sLV_z+s^2LCRV_y+RV_x}{sL+s^2LCR+R}$$

$$=\frac{RV_x+s^2LCRV_y+sLV_z}{s^2LCR+sL+R}$$

$$=\frac{\left(\frac{1}{LC}\right)V_x+s^2V_y+\left(\frac{1}{RC}s\right)V_z}{s^2+\frac{1}{RC}s+\frac{1}{LC}}$$

$$V_0=\left(\frac{\frac{1}{LC}}{s^2+\frac{1}{RC}s+\frac{1}{LC}}\right)V_x+\left(\frac{s^2}{s^2+\frac{1}{RC}s+\frac{1}{LC}}\right)V_y+\left(\frac{\frac{1}{RC}s}{s^2+\frac{1}{RC}s+\frac{1}{LC}}\right)V_z \quad \text{..... (2)}$$

Step 3 of 4

Compare equations (1) and (2)

$$T_{LP}=\frac{\frac{1}{LC}}{s^2+\frac{1}{RC}s+\frac{1}{LC}}$$

$$T_{HP}=\frac{s^2}{s^2+\frac{1}{RC}s+\frac{1}{LC}}$$

$$T_{BP}=\frac{\frac{1}{RC}s}{s^2+\frac{1}{RC}s+\frac{1}{LC}}$$

Step 4 of 4

Thus, the voltage that develops across the resonator is,

$$\boxed{V_0=\left(\frac{\frac{1}{LC}}{s^2+\frac{1}{RC}s+\frac{1}{LC}}\right)V_x+\left(\frac{s^2}{s^2+\frac{1}{RC}s+\frac{1}{LC}}\right)V_y+\left(\frac{\frac{1}{RC}s}{s^2+\frac{1}{RC}s+\frac{1}{LC}}\right)V_z}.$$

Step 1 of 4

17.050P

Refer to the low-pass notch (LPN) second-order filter circuit of Figure 17.18(g) in the textbook.

Consider the transfer function of HPN filter is,

$$T(s)=a_2\frac{s^2+\frac{1}{L(C_1+C_2)}}{s^2+\frac{1}{(C_1+C_2)R}s+\frac{1}{L(C_1+C_2)}}$$

Here, a_2 is high frequency gain.

$$a_2=\frac{C_1}{C_1+C_2}$$

Step 2 of 4

The resonant frequency is,

$$\omega_0^2=\frac{1}{L(C_2+C_1)} \quad \text{..... (1)}$$

The notch frequency is,

$$\omega_n=\frac{1}{\sqrt{LC_1}}$$

$$L=\frac{1}{\omega_n^2C_1}$$

Substitute $\frac{1}{\omega_n^2C_1}$ for L in equation (1).

$$\omega_0^2=\frac{1}{\left(\frac{1}{\omega_n^2C_1}\right)(C_2+C_1)}$$

$$=\frac{C_1}{(C_2+C_1)}\omega_n^2$$

Substitute $1.1\omega_0$ for ω_n in the equation.

$$\omega_0^2=\frac{C_1}{C_1+C_2}(1.1\omega_0)^2$$

$$\omega_0^2(C_1+C_2)=C_1\left(1.21\omega_0^2\right)$$

$$\frac{C_1+C_2}{C_1}=1.21$$

$$1+\frac{C_2}{C_1}=1.21$$

$$\frac{C_2}{C_1}=0.21$$

Thus, the ratio of C_2 to C_1 is, $\boxed{\frac{C_2}{C_1}=0.21}$.

Step 3 of 4

Consider the transfer function of LPN filter.

$$T(s)=\left(\frac{C_1}{C_1+C_2}\right)\left(\frac{s^2+\frac{1}{L(C_1+C_2)}}{s^2+\frac{1}{(C_1+C_2)R}s+\frac{1}{L(C_1+C_2)}}\right)$$

Substitute $j\omega$ for s in the equation.

$$T(j\omega)=\left(\frac{C_1}{C_1+C_2}\right)\left(\frac{-\omega^2+\frac{1}{L(C_1+C_2)}}{-\omega^2+\frac{1}{(C_1+C_2)R}(j\omega)+\frac{1}{L(C_1+C_2)}}\right)$$

For $\omega<<\omega_0$, the magnitude of the transfer function is,

$$|T|=\left(\frac{C_1}{C_1+C_2}\right)\left(\frac{\frac{1}{L(C_1+C_2)}}{\frac{1}{L(C_1+C_2)}}\right)$$

$$=\left(\frac{C_1}{C_1+C_2}\right)$$

Thus, the magnitude of the transfer function for $\omega<<\omega_0$ is, $\boxed{|T|=\frac{C_1}{C_1+C_2}}$.

Step 4 of 4

For $\omega>>\omega_0$, the inductors L is open and capacitors C_1 and C_2 are short circuited. Therefore, the magnitude of the transfer function for $\omega>>\omega_0$ is, $\boxed{|T|=0}$.

Step 1 of 3

Refer to the Antoniou inductance-simulation circuit of Figure 17.20 in the textbook.

Consider the equivalent inductance of the circuit is as,

$$L = \frac{C_4 R_1 R_2 R_3}{R_2} \quad \text{..... (1)}$$

Choose resistors as $R_1 = R_2 = R_3 = R_3 = 10\ \text{k}\Omega$.

(a)

Substitute $15\ \text{H}$ for L and $10\ \text{k}\Omega$ for resistors $R_1, R_2, R_3,$ and R_3 in the equation (1).

$$15 = \frac{C_4 (10\times10^3)(10\times10^3)(10\times10^3)}{(10\times10^3)}$$

$$15 = C_4 (10\times10^3)(10\times10^3)$$

$$15 = C_4 (10^8)$$

Simplify further.

$$C_4 = \frac{15}{10^8}$$

$$= 15\times10^{-8}$$

$$= 150\times10^{-9}$$

$$= 150\ \text{nF}$$

Thus, the component values of the Antoniou Inductance-simulation circuit are,

$$\boxed{\begin{array}{l} R_1 = R_2 = R_3 = R_3 = 10\ \text{k}\Omega \\ C_4 = 150\ \text{nF} \end{array}}$$

Step 2 of 3

(b)

Substitute $1.5\ \text{H}$ for L and $10\ \text{k}\Omega$ for resistors $R_1, R_2, R_3,$ and R_3 in the equation (1).

$$1.5 = \frac{C_4 (10\times10^3)(10\times10^3)(10\times10^3)}{(10\times10^3)}$$

$$1.5 = C_4 (10\times10^3)(10\times10^3)$$

$$1.5 = C_4 (10^8)$$

Simplify further.

$$C_4 = \frac{1.5}{10^8}$$

$$= 1.5\times10^{-8}$$

$$= 15\times10^{-9}$$

$$= 15\ \text{nF}$$

Thus, the component values of the Antoniou Inductance-simulation circuit are,

$$\boxed{\begin{array}{l} R_1 = R_2 = R_3 = R_3 = 10\ \text{k}\Omega \\ C_4 = 15\ \text{nF} \end{array}}$$

Step 3 of 3

(c)

Substitute $0.15\ \text{H}$ for L and $10\ \text{k}\Omega$ for resistors $R_1, R_2, R_3,$ and R_3 in the equation (1).

$$0.15 = \frac{C_4 (10\times10^3)(10\times10^3)(10\times10^3)}{(10\times10^3)}$$

$$0.15 = C_4 (10\times10^3)(10\times10^3)$$

$$0.15 = C_4 (10^8)$$

Simplify further.

$$C_4 = \frac{0.15}{10^8}$$

$$= 0.15\times10^{-8}$$

$$= 1.5\times10^{-9}$$

$$= 1.5\ \text{nF}$$

Thus, the component values of the Antoniou Inductance-simulation circuit are,

$$\boxed{\begin{array}{l} R_1 = R_2 = R_3 = R_3 = 10\ \text{k}\Omega \\ C_4 = 1.5\ \text{nF} \end{array}}$$

Step 1 of 6

Refer to the Figure P17.52 in the textbook.

(a)

The modified circuit diagram is shown in Figure 1.

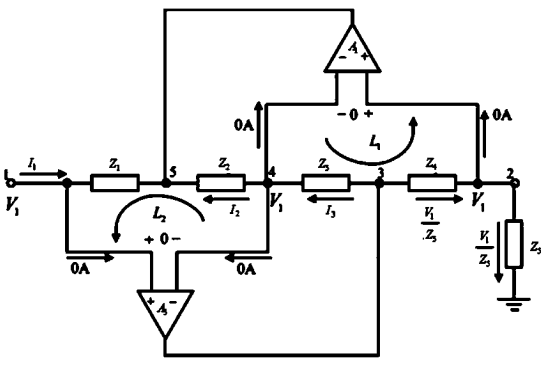


Figure1

The node voltages at nodes 1,4 and 2 are equal to V_1.

Consider the current in the impedances Z_1, Z_2, Z_3, Z_4 and Z_5 are I_1, I_2, I_3, I_4 and I_5 respectively.

From Figure 1, the current in impedance Z_5 is,

$$I_5 = \frac{V_1}{Z_5}$$

Current in impedance Z_4 is,

$$I_4 = I_5$$

Voltage across the impedance Z_4 is,

$$V_4 = I_4 Z_4$$
$$= \frac{V_1 Z_4}{Z_5}$$

Step 2 of 6

Voltage across the impedance Z_3 is,

Apply Kirchhoff's voltage law in Loop L_1.

$$-V_1 - Z_3 I_3 + V_4 + V_1 = 0$$

Substitute $\frac{V_1 Z_4}{Z_5}$ for V_4 in equation.

$$Z_3 I_3 = \frac{V_1 Z_4}{Z_5}$$
$$I_3 = \frac{V_1 Z_4}{Z_5 Z_3}$$

Current in impedance Z_2 is,

$$I_2 = I_3$$

Step 3 of 6

Apply Kirchhoff's voltage law in Loop L_2.

$$-V_1 + Z_2 I_2 - I_1 Z_1 + V_1 = 0$$
$$Z_2 I_2 = I_1 Z_1$$
$$Z_2\left(\frac{V_1 Z_4}{Z_5 Z_3}\right) = I_1 Z_1 \qquad \begin{pmatrix}\text{since } I_2 = I_3 \\ I_3 = \frac{V_1 Z_4}{Z_5 Z_3}\end{pmatrix}$$
$$I_1 = \frac{Z_2\left(\frac{V_1 Z_4}{Z_5 Z_3}\right)}{Z_1}$$

Thus, the current I_1 is,

$$I_1 = Z_2\left(\frac{V_1 Z_4}{Z_5 Z_3 Z_1}\right)$$

Determine the impedance Z_{11} at port 1.

$$Z_{11} = \frac{V_1}{I_1}$$
$$= \left(\frac{Z_3 Z_1}{Z_4 Z_2}\right) Z_5$$

Therefore, the impedance Z_{11} at port 1 is, $\boxed{\left(\frac{Z_1 Z_3}{Z_2 Z_4}\right) Z_5}$.

(b)

Consider the modified circuit diagram with impedance Z_6 is connected at port 1 is shown in Figure 2.

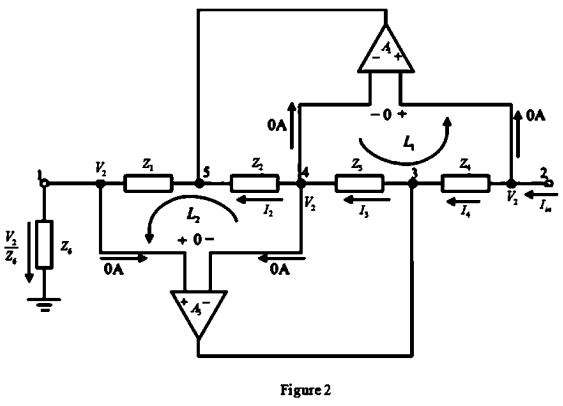


Figure 2

Step 4 of 6

The node voltages at nodes 1,4 and 2 are equal to V_2.

Consider the current in the impedances Z_1, Z_2, Z_3, Z_4 and Z_6 are I_1, I_2, I_3, I_4 and I_6 respectively.

Current leaving from port 2 is I_{in}.

From Figure 2, $I_{in} = I_6$.

The current in impedance Z_6 is,

$$I_6 = \frac{V_2}{Z_6}$$

Current in impedance Z_1 is,

$$I_1 = -I_6$$

Step 5 of 6

Apply Kirchhoff's voltage law in Loop L_2.

$$-V_2 + Z_2 I_2 + I_1 Z_1 + V_2 = 0$$
$$Z_2 I_2 + I_1 Z_1 = 0$$
$$Z_2 I_2 - I_6 Z_1 = 0$$
$$Z_2 I_2 = \frac{V_2 Z_1}{Z_6} \qquad \begin{pmatrix}\text{since } I_1 = -I_6 \\ I_6 = \frac{V_2}{Z_6}\end{pmatrix}$$
$$I_2 = \frac{V_2 Z_1}{Z_6 Z_2}$$

From Figure 2, the currents $I_2 = I_3$.

Step 6 of 6

Apply Kirchhoff's voltage law in Loop L_1.

$$-V_2 - Z_3 I_3 - V_4 + V_2 = 0$$
$$-Z_3 I_3 - V_4 = 0$$
$$Z_3 I_2 = V_4 \qquad (\text{since } I_2 = I_3)$$

Substitute $\frac{V_2 Z_1}{Z_6 Z_2}$ for I_2 in equation.

$$Z_3\left(\frac{V_2 Z_1}{Z_6 Z_2}\right) = I_4 Z_4$$
$$Z_3\left(\frac{Z_1}{Z_6 Z_2 Z_4}\right) =$$
$$\frac{V_2}{I_4} = \left(\frac{Z_6 Z_2 Z_4}{Z_1 Z_3}\right)$$

Determine the impedance Z_{22} is at port 2.

$$Z_{22} = \frac{V_2}{I_4}$$
$$= \frac{V_2}{I_{in}}$$
$$= \left(\frac{Z_2 Z_4}{Z_1 Z_3}\right) Z_6$$

Therefore, the impedance Z_{22} at port 2 is, $\boxed{\left(\frac{Z_2 Z_4}{Z_1 Z_3}\right) Z_6}$.

(c)

From part (a) and part (b), the results are as follows:

$$Z_{11} = \left(\frac{Z_3 Z_1}{Z_4 Z_2}\right) Z_5$$

$$Z_{22} = \left(\frac{Z_2 Z_4}{Z_1 Z_3}\right) Z_6$$

Choose the impedances $Z_4 = Z_3 = Z_2 = Z_1 = Z$. Then, the impedance at port 1 and port 2 are,

$$Z_{11} = Z_5$$

$$Z_{22} = Z_6$$

Thus, the load impedance of Z_5 is appeared at impedance at port1. Similarly, the load impedance of Z_6 is appeared at impedance at port2. Therefore, the circuit works as complete **impedance transformer**.

Step 1 of 4

Refer to Figure 16.20(a) in the textbook for the Antoniou Inductance-simulation circuit.

Redraw the circuit by eliminating resistor R_5, connecting capacitor C_6 between the node 1 and ground and connecting voltage source V_2 at node 2 as shown in Figure 1.

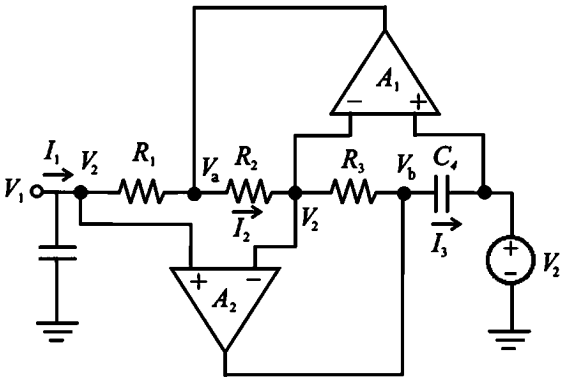


Figure 1

Step 2 of 4

Because of the virtual short at the input terminals of the operational amplifier, all nodes are at voltage V_2.

The current I_1 is,

$$\begin{aligned} I_1 &= \frac{-V_2}{\left(\frac{1}{sC_6}\right)} \\ &= -sC_6V_2 \end{aligned}$$

Since no current enters into the operational amplifier input terminals, the voltage V_a is,

$$V_a = V_2 - I_1R_1$$

Substitute $-sC_6V_2$ for I_1 in the equation.

$$\begin{aligned} V_a &= V_2 - sC_6V_2R_1 \\ &= V_2(1 - sC_6R_1) \end{aligned}$$

Step 3 of 4

From the circuit, the current through the resistor R_2 is,

$$\begin{aligned} I_2 &= \left(\frac{V_a - V_2}{R_2}\right) \\ &= \frac{V_2(1 - sC_6R_1) - V_2}{R_2} \quad (\text{Since, } V_a = V_2(1 - sC_6R_1)) \\ &= -\frac{sC_6R_1V_2}{R_2} \end{aligned}$$

From the circuit, the voltage at the node b is,

$$\begin{aligned} V_b &= V_2 - I_2R_3 \\ &= V_2 + \frac{sC_6R_1R_3V_2}{R_2} \quad \left(\text{Since } I_2 = -\frac{sC_6R_1V_2}{R_2}\right) \end{aligned}$$

From the circuit, the current I_3 is,

$$\begin{aligned} I_3 &= (V_b - V_2)sC_4 \\ &= \left(V_2 + \frac{sC_6R_1R_3V_2}{R_2} - V_2\right)sC_4 \\ &= \frac{s^2C_4C_6R_1R_3V_2}{R_2} \end{aligned}$$

Step 4 of 4

The input impedance seen by the voltage source V_2 is,

$$\begin{aligned} Z_{in} &= \frac{V_2}{I_3} \\ &= \frac{V_2}{\frac{s^2C_4C_6R_1R_3V_2}{R_2}} \quad \left(\text{Since, } I_3 = \frac{s^2C_4C_6R_1R_3V_2}{R_2}\right) \\ &= \frac{R_2}{s^2C_4C_6R_1R_3} \end{aligned}$$

Thus the input impedance seen by the voltage V_2 is, $\boxed{Z_{in} = \frac{R_2}{s^2C_4C_6R_1R_3}}$.

Substitute $j\omega$ for s in the equation.

$$\begin{aligned} Z_{in} &= \frac{R_2}{(j\omega)^2C_4C_6R_1R_3} \\ &= -\frac{R_2}{\omega^2C_4C_6R_1R_3} \end{aligned}$$

Thus, the impedance behaves as a frequency-dependent pure negative resistance.

Step 1 of 6

17.054P

Refer to Figure 17.22(a) for the second-order filter circuit using the operational amplifier-RC resonator circuit.

Redraw the circuit by representing the nodes as shown in Figure 1.

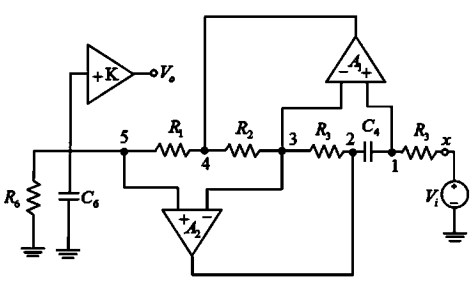


Figure 1

Step 2 of 6

From the Figure 1, the voltage at nodes 1, 3 and 5 are equal.

Thus, the voltages are,

$$V_1 = V_3 = V_5 = \frac{V_o}{K}$$

The current from the source V_i is,

$$I_1 = \frac{V_i - V_1}{R_3}$$

Substitute $\frac{V_o}{K}$ for V_1 in the equation.

$$I_1 = \frac{V_i - \frac{V_o}{K}}{R_3}$$

The voltage at node 2 is,

$$V_2 = \frac{V_0}{K} - \frac{I_1}{sC_4}$$

Substitute $\frac{V_i - \frac{V_o}{K}}{R_3}$ for I_1 in equation.

$$V_2 = \frac{V_0}{K} - \frac{\left(\frac{V_i - \frac{V_0}{K}}{R_3}\right)}{sC_4}$$

$$= \frac{V_0}{K} - \left(\frac{V_i - \frac{V_0}{K}}{sC_4R_3}\right)$$

$$= \frac{sC_4R_3V_0 - (KV_i - V_0)}{sC_4R_3K}$$

$$V_2 = \frac{(sC_4R_3 + 1)V_0 - KV_i}{sC_4R_3K}$$

Step 3 of 6

The current through the resistor R_3 is,

$$I_2 = \frac{V_2 - \frac{V_0}{K}}{R_3}$$

Substitute $\frac{(sC_4R_3 + 1)V_0 - KV_i}{sC_4R_3K}$ for V_2 in the equation.

$$I_2 = \frac{\frac{(sC_4R_3 + 1)V_0 - KV_i}{sC_4R_3K} - \frac{V_0}{K}}{R_3}$$

$$= \frac{(sC_4R_3 + 1)V_0 - KV_i}{sC_4R_3R_3K} - \frac{V_0}{KR_3}$$

$$= -\frac{V_i}{sC_4R_3R_3} + \frac{V_0}{sC_4R_3R_3K}$$

Step 4 of 6

The voltage at node 4 is,

$$V_4 = \frac{V_0}{K} - I_2R_2$$

Substitute $-\frac{V_i}{sC_4R_3R_3} + \frac{V_0}{sC_4R_3R_3K}$ for I_2 in the equation.

$$V_4 = \frac{V_0}{K} - \left(-\frac{V_i}{sC_4R_3R_3} + \frac{V_0}{sC_4R_3R_3K}\right)R_2$$

$$= \frac{V_0}{K} - \frac{\left(\frac{V_0}{K} - V_i\right)R_2}{sC_4R_3R_3}$$

Step 5 of 6

The current through the resistor R_1 is,

$$I_1 = \frac{V_4 - \frac{V_0}{K}}{R_1}$$

$$= \frac{V_4 - \frac{V_0}{K}}{R_1}$$

Substitute $\frac{V_0}{K} - \frac{\left(\frac{V_0}{K} - V_i\right)R_2}{sC_4R_3R_3}$ for V_4 in the equation.

$$I_1 = \frac{V_0}{KR_1} - \frac{\left(\frac{V_0}{K} - V_i\right)R_2}{sC_4R_3R_3R_1} - \frac{V_0}{KR_1}$$

$$= -\frac{\left(\frac{V_0}{K} - V_i\right)R_2}{sC_4R_3R_2R_1}$$

The current I_1 flows into the resistor R_6 and capacitor C_6.

For ideal operational amplifier, the input impedance is $R_{in} = \infty$. Thus, the voltage at node 5 is,

$$\frac{V_0}{K} = I_1\left(R_6 \parallel \frac{1}{sC_6}\right)$$

$$= I_1\left(\frac{R_6 \times \frac{1}{sC_6}}{R_6 + \frac{1}{sC_6}}\right)$$

$$= I_1\left(\frac{R_6}{1 + sC_6R_6}\right)$$

Substitute $-\frac{\left(\frac{V_0}{K} - V_i\right)R_2}{sC_4R_3R_3R_1}$ for I_1 in the equation.

$$\frac{V_0}{K} = -\frac{\left(\frac{V_0}{K} - V_i\right)R_2}{sC_4R_3R_3R_1}\left(\frac{R_6}{1 + sC_6R_6}\right)$$

$$\frac{V_0}{K} + \frac{V_0R_2R_6}{(KsC_4R_3R_3R_1)(1 + sC_6R_6)} = \frac{R_6R_2V_i}{(1 + sC_6R_6)sC_4R_2R_3R_1}$$

$$V_0\left(\frac{1}{K} + \frac{R_2R_6}{(KsC_4R_3R_3R_1)(1 + sC_6R_6)}\right) = \frac{R_6R_2V_i}{(1 + sC_6R_6)sC_4R_2R_3R_1}$$

$$V_0\left(\frac{sC_4R_3R_3R_1(1 + sC_6R_6) + R_2R_6}{KsC_4R_3R_3R_1(1 + sC_6R_6)}\right) = \frac{R_6R_2V_i}{(1 + sC_6R_6)sC_4R_3R_3R_1}$$

Step 6 of 6

Now, the transfer function is,

$$\frac{V_0}{V_i} = \frac{KR_6R_2}{sC_4R_3R_3R_1(1 + sC_6R_6) + R_2R_6}$$

$$= \frac{KR_6R_2}{s^2C_4R_3R_3R_1C_6R_6 + sC_4R_3R_3R_1 + R_2R_6}$$

$$= \frac{\frac{KR_6R_2}{C_4R_3R_3R_1C_6R_6}}{s^2 + s\left(\frac{C_4R_3R_3R_1}{C_4R_3R_3R_1C_6R_6}\right) + \frac{R_2R_6}{C_4R_3R_3R_1C_6R_6}}$$

$$= \frac{\frac{KR_2}{C_4R_2R_3R_1C_6}}{s^2 + s\left(\frac{1}{C_6R_6}\right) + \frac{R_2}{C_4R_3R_3R_1C_6}}$$

Thus, the transfer function is, $\boxed{\frac{V_0}{V_i} = \frac{\frac{KR_2}{C_4R_2R_3R_1C_6}}{s^2 + \frac{s}{C_6R_6} + \frac{R_2}{C_4R_3R_3R_1C_6}}}$.

Step 1 of 5

17.055P

Refer to Figure 17.22(a) for the second-order filter circuit and also refer to figure 17.13(a) for the first-order op amp-RC circuit given in the textbook.

Cascade both the second-order filter circuit and the first-order op amp-RC circuit and design the required Butterworth filter.

The 3-dB bandwidth of a fifth-order (*N*=5) Butterworth filter is,

$$BW = 10^5 \text{ rad/s}$$

The expression for the maximum gain A_{max} is,

$$A_{max} = 20\log\sqrt{1+\varepsilon^2}$$

Substitute 3 for A_{max}.

$$3 = 10\log\left(1+\varepsilon^2\right)$$
$$\log\left(1+\varepsilon^2\right) = 0.3$$
$$1+\varepsilon^2 = 10^{0.3}$$
$$\varepsilon^2 = 10^{0.3} - 1$$

Simplify further.

$$\varepsilon^2 = 0.9953$$
$$\varepsilon = 0.9976$$
$$\varepsilon \approx 1$$

Step 2 of 5

The resonant frequency is equal to the pass-band edge frequency.

$$\omega_0 = \omega_p = 10^5 \text{ rad/s}$$

The expression for the DC gain magnitude of the first-order op amp-RC filter circuit is,

$$\left|\text{DC gain}\right| = \frac{R_2}{R_1}$$

Substitute 1 for $\left|\text{DC gain}\right|$.

$$1 = \frac{R_2}{R_1}$$
$$R_1 = R_2$$

Choose $10\ \text{k}\Omega$ for R_1 and R_2.

The expression for resonant frequency is,

$$\omega_0 = \frac{1}{CR_2}$$

Calculate the capacitance *C*.

$$C = \frac{1}{\omega_0 R_2}$$

Substitute 10^5 for ω_0, and 10×10^3 for R_2.

$$C = \frac{1}{10^5 \times 10\times10^3}$$
$$= 1 \text{ nF}$$

Therefore, the value of capacitance *C* is 1 nF.

Step 3 of 5

Consider the transfer function of the second-order filter circuit.

$$T(s) = \frac{\left(\frac{KR_2}{C_4C_6R_1R_3R_3}\right)}{\left(s^2 + s\frac{1}{C_6R_6} + \frac{R_2}{C_4C_6R_1R_3R_3}\right)}$$

Here, *K* is the DC gain.

The resonant frequency is,

$$\omega_0^2 = \frac{R_2}{C_4C_6R_1R_3R_3}$$

Substitute *R* for R_1, R_3, R_2, and R_3; and *C* for C_4, and C_6.

$$\omega_0^2 = \frac{R}{C^2R^3}$$
$$\omega_0^2 = \frac{R}{C^2R^2}$$
$$R = \frac{1}{\omega_0 C}$$

Consider the value of capacitance *C* is 100 nF.

Substitute 100×10^{-9} for *C* and 10^3 for ω_0.

$$R = \frac{1}{10^5 \times 100\times10^{-9}}$$
$$= 100\ \Omega$$

Therefore, the value of resistances R_1, R_3, R_2, and R_3 is $100\ \Omega$ and the value of capacitances C_4, and C_6 is 100 nF.

Step 4 of 5

The expression for the quality factor of the filter is,

$$\frac{\omega_0}{2Q} = \omega_0 \cos\left(\frac{\pi}{N}\right)$$

Substitute 5 for *N*.

$$\frac{1}{2Q} = \cos\left(\frac{\pi}{5}\right)$$
$$\frac{1}{2Q} = 0.809$$
$$Q = 0.618$$

Therefore, the quality factor Q is 0.618.

Step 5 of 5

Consider the relation between resonant frequency ω_0 and quality factor Q.

$$\frac{\omega_0}{Q} = \frac{1}{C_6R_6}$$

Substitute 10^3 for ω_0, 0.618 for Q, and 100×10^{-9} for C_6.

$$\frac{10^5}{0.618} = \frac{1}{100\times10^{-9}\times R_6}$$
$$R_6 = \frac{0.618}{100\times10^{-9}\times10^5}$$
$$R_6 = 61.8\ \Omega$$

Therefore, the value of resistance R_6 is $61.8\ \Omega$.

Therefore, the component values of the fifth-order Butterworth filter circuit are,

$$\boxed{\begin{array}{l} R_1 = R_2 = 10\ \text{k}\Omega \\ C = 1\ \text{nF} \end{array}}$$ for the first-order op amp-RC filter, and

$$\boxed{\begin{array}{l} R_1 = R_2 = R_3 = R_3 = 100\ \Omega \\ R_6 = 61.8\ \Omega \\ C_4 = C_6 = 100\ \text{nF} \end{array}}$$ for the second-order filter circuit.

Step 1 of 3

Refer to Figure 17.22(e) for the Low-Pass Notch (LPN) filter in the textbook.

The expression for the resonant frequency is,

$$\omega_0 = \frac{1}{\sqrt{\frac{C_4(C_{61}+C_{62})R_1R_3R_5}{R_2}}}$$

$$2\pi f_0 = \frac{1}{\sqrt{\frac{C_4(C_{61}+C_{62})R_1R_3R_5}{R_2}}}$$

Substitute C_6 for $C_{61}+C_{62}$.

$$2\pi f_0 = \frac{1}{\sqrt{\frac{C_4(C_6)R_1R_3R_5}{R_2}}}$$

Select 10 nF for capacitors C_4, and C_6; and R for R_1, R_3, and R_5.

Substitute 10×10^3 for f_0, R for R_1, R_2, R_3 and R_5; and 10×10^{-9} for C_4 and C_6.

$$2\pi(10\times10^3) = \frac{1}{\sqrt{\frac{10\times10^{-9}\times10\times10^{-9}\times R^3}{R}}}$$

$$\sqrt{10^{-16}\times R^2} = \frac{1}{2\pi(10\times10^3)}$$

$$R = \frac{1}{20\pi\times10^3\times10^{-8}}$$

$$R = 1.6\ \text{k}\Omega$$

The value of resistances R_1, R_2, R_3 and R_5 is $1.6\ \text{k}\Omega$, and that of capacitances C_4 and C_6 is 10 nF.

Step 2 of 3

The expression for the notch frequency is,

$$\omega_n = \frac{1}{\sqrt{\frac{C_4C_{61}R_1R_3R_5}{R_2}}}$$

$$2\pi f_n = \frac{1}{\sqrt{\frac{C_4C_{61}R_1R_3R_5}{R_2}}}$$

Substitute 10×10^{-9} for C_4, 12×10^3 for f_n, and R for R_1, R_3, and R_5.

$$2\pi(12\times10^3) = \frac{1}{\sqrt{\frac{10\times10^{-9}\times C_{61}\times R^3}{R}}}$$

$$\sqrt{10\times10^{-9}\times C_{61}\times R^2} = \frac{1}{24\times10^3\pi}$$

$$10^{-4}R\sqrt{C_{61}} = \frac{1}{24\times10^3\pi}$$

$$\sqrt{C_{61}} = \frac{1}{(24\times10^{-1}\pi)R}$$

Substitute 1.6×10^3 for R.

$$C_{61} = \left(\frac{1}{(24\times10^{-1}\pi)\times1.6\times10^3}\right)^2$$

$$= 6.87\ \text{nF}$$

Therefore, the value of capacitance C_{61} is 6.87 nF.

Calculate the value of capacitance C_{62}.

Consider the following expression:

$$C_{62} = 10\ \text{nF} - C_{61}$$

Substitute 6.87 nF for C_{61}.

$$C_{62} = 10\ \text{nF} - 6.87\ \text{nF}$$

$$= 3.12\ \text{nF}$$

Therefore, the value of capacitance C_{62} is 3.12 nF.

Step 3 of 3

The expression for the quality factor is,

$$Q = R_6\sqrt{\frac{(C_{61}+C_{62})}{C_4}\times\frac{R_2}{R_1R_3R_5}}$$

Substitute C_6 for $(C_{61}+C_{62})$.

$$Q = R_6\sqrt{\frac{C_6}{C_4}\times\frac{R_2}{R_1R_3R_5}}$$

Substitute 10 for Q, R for R_1, R_3, and R_5; 10×10^{-9} for C_4 and C_6.

$$10 = R_6\sqrt{\frac{10\times10^{-9}}{10\times10^{-9}}\times\frac{R}{R^3}}$$

$$10 = R_6\sqrt{\frac{1}{R^2}}$$

$$R_6 = 10R$$

Substitute 1.6×10^3 for R.

$$R_6 = 10\times(1.6\times10^3)$$

$$= 16\ \text{k}\Omega$$

Therefore, the value of resistance R_6 is $16\ \text{k}\Omega$.

Therefore, the design parameters are,

$$\boxed{\begin{array}{l} C_6 = C_4 = 10\ \text{nF} \\ R_1 = R_2 = R_3 = R_5 = 1.6\ \text{k}\Omega \\ C_{61} = 6.87\ \text{nF} \\ C_{62} = 3.12\ \text{nF} \\ R_6 = 16\ \text{k}\Omega \end{array}}$$

Step 1 of 3

17.057P

Refer to Figure 17.22(g) for the all-pass filter circuit in the textbook.

The expression for the resonant frequency of all-pass filter circuit is,

$$\omega_0 = \sqrt{\frac{R_2}{C_4 C_6 R_1 R_3 R_5}}$$

$$2\pi f_0 = \sqrt{\frac{R_2}{C_4 C_6 R_1 R_3 R_5}}$$

Select R for the resistances R_1, R_2, R_3, and R_5; and 1 nF for C_4 and C_6.

Substitute 2×10^3 for f_0, R for R_1, R_2, R_3, and R_5; and 10^{-9} for C_4 and C_6.

$$2\pi\left(2\times10^3\right) = \sqrt{\frac{R}{10^{-9}\times10^{-9}\times R^3}}$$

$$10^{-9}R = \frac{1}{4\pi\times10^3}$$

$$R = \frac{1}{4\pi\times10^3\times10^{-9}}$$

$$R = 79.58\ \text{k}\Omega$$

Therefore, the value of resistances R_1, R_2, R_3, and R_5 is $79.58\ \text{k}\Omega$ and that of capacitances C_4 and C_6 is 1 nF.

Step 2 of 3

Consider the following relation:

$$\frac{\omega_0}{Q} = \frac{1}{R_6 C_6}$$

$$\frac{2\pi f_0}{Q} = \frac{1}{R_6 C_6}$$

Substitute 2×10^3 for f_0, 10^{-9} for C_6, 2 for Q.

$$\frac{2\pi\left(2\times10^3\right)}{2} = \frac{1}{R_6\times\left(10^{-9}\right)}$$

$$R_6 = \frac{1}{2\pi\times10^3\times10^{-9}}$$

$$R_6 = 159.2\ \text{k}\Omega$$

Therefore, the value of resistance R_6 is $159.2\ \text{k}\Omega$.

Step 3 of 3

Select the values of r_1 and r_2 arbitrarily such that the following condition is satisfied.

$$r_1 = r_2$$

Therefore, the design parameters are,

$$C_4 = C_6 = 1\ \text{nF}$$
$$R_1 = R_2 = R_3 = R_5 = 79.58\ \text{k}\Omega$$
$$R_6 = 159.2\ \text{k}\Omega$$
$$r_1 = r_2 = 1\ \text{k}\Omega$$

Step 1 of 5

Refer to Figure 16.22 (e) in the textbook for the Low pass Notch (LPN) filter circuit.

The transfer function of the low pass notch filter is,

$$T(s)=\frac{KC_{61}}{C_{61}+C_{62}}\times\frac{s^2+\dfrac{R_2}{C_4C_{61}R_5R_3R_1}}{s^2+\dfrac{s}{(C_{61}+C_{62})R_6}+\dfrac{R_2}{C_4(C_{61}+C_{62})R_3R_3R_1}}$$

Substitute 0 for s in the equation for the DC gain.

$$\begin{aligned}T(0)&=\frac{KC_{61}}{C_{61}+C_{62}}\left(\frac{\dfrac{R_2}{C_4C_{61}R_5R_3R_1}}{\dfrac{R_2}{C_4(C_{61}+C_{62})R_3R_3R_1}}\right)\\&=\frac{KC_{61}}{C_{61}+C_{62}}\left(\frac{R_2C_4(C_{61}+C_{62})R_3R_3R_1}{R_2C_4C_{61}R_3R_3R_1}\right)\\&=K\end{aligned}$$

Thus, the DC gain is, $\boxed{\text{DCgain}=K}$.

Step 2 of 5

The transfer function of the low pass notch filter is,

$$T(s)=\frac{KC_{61}}{C_{61}+C_{62}}\frac{s^2+\dfrac{R_2}{C_4C_{61}R_5R_3R_1}}{s^2+\dfrac{s}{(C_{61}+C_{62})R_6}+\dfrac{R_2}{C_4(C_{61}+C_{62})R_3R_3R_1}}$$

Compare the transfer function with $T(s)=a_2\dfrac{s^2+\omega_n^2}{s^2+s\left(\dfrac{\omega_0}{Q}\right)+\omega_0^2}$.

$$\omega_n^2=\frac{R_2}{C_4C_{61}R_3R_3R_1}$$

$$\omega_n=\sqrt{\frac{R_2}{C_4C_{61}R_5R_3R_1}}$$

$$\omega_n=\frac{1}{\sqrt{\dfrac{C_4C_{61}R_5R_3R_1}{R_2}}}$$

Thus, the notch frequency is, $\boxed{\omega_n=\dfrac{1}{\sqrt{\dfrac{C_4C_{61}R_5R_3R_1}{R_2}}}}$.

Step 3 of 5

Derive the expression for resonant frequency of the low pass notch filter.

$$\omega_0^2=\frac{R_2}{C_4(C_{61}+C_{62})R_3R_3R_1}$$

$$\begin{aligned}\omega_0&=\sqrt{\frac{R_2}{C_4(C_{61}+C_{62})R_3R_3R_1}}\\&=\frac{1}{\sqrt{\dfrac{C_4(C_{61}+C_{62})R_3R_3R_1}{R_2}}}\end{aligned}$$

Thus, resonant frequency of the low pass notch filter is, $\boxed{\omega_0=\dfrac{1}{\sqrt{\dfrac{C_4(C_{61}+C_{62})R_3R_3R_1}{R_2}}}}$

Step 4 of 5

Derive the expression for quality factor of the low pass notch filter.

$$\frac{\omega_0}{Q}=\frac{1}{(C_{61}+C_{62})R_6}$$

$$Q=\omega_0(C_{61}+C_{62})R_6$$

$$=\sqrt{\frac{R_2}{C_4(C_{61}+C_{62})R_3R_3R_1}}(C_{61}+C_{62})R_6\quad\left(\text{Since, }\omega_0=\sqrt{\frac{R_2}{C_4(C_{61}+C_{62})R_3R_3R_1}}\right)$$

$$=R_6\sqrt{\frac{R_2(C_{61}+C_{62})}{C_4R_3R_3R_1}}$$

Thus, the quality factor of the low pass notch filter is, $\boxed{Q=R_6\sqrt{\dfrac{R_2(C_{61}+C_{62})}{C_4R_3R_3R_1}}}$.

Step 5 of 5

The ratio $\left(\dfrac{\omega_0}{\omega_n}\right)^2$ is,

$$\left(\frac{\omega_0}{\omega_n}\right)^2=\left(\frac{\dfrac{1}{C_4(C_{61}+C_{62})R_3R_3R_1}}{\dfrac{1}{\dfrac{C_4C_{61}R_3R_3R_1}{R_3}}}\right)$$

$$=\left(\frac{\dfrac{R_2}{C_4(C_{61}+C_{62})R_3R_3R_1}}{\dfrac{R_2}{C_4C_{61}R_3R_3R_1}}\right)$$

$$=\frac{C_{61}}{(C_{61}+C_{62})}$$

Choose $C_{61}+C_{62}=C_6=C$.

$$\left(\frac{\omega_0}{\omega_n}\right)^2=\frac{C_{61}}{C}$$

$$C_{61}=C\left(\frac{\omega_0}{\omega_n}\right)^2$$

Thus the design equations are,

$$\boxed{\begin{aligned}&C_{61}=C\left(\frac{\omega_0}{\omega_n}\right)^2\\&C_{62}=C-C_{61}\\&C_{61}+C_{62}=C_6=C\end{aligned}}$$.

Step 1 of 4

17.059P

Refer to Figure 16.22 (f) for the HPN filter.

The transfer function of the high pass notch filter is,

$$T(s)=K\frac{s^2+\frac{R_2}{C_4C_6R_{51}R_3R_1}}{s^2+\frac{s}{C_6R_6}+\frac{R_2}{C_4C_6R_1R_3}\left(\frac{1}{R_{51}}+\frac{1}{R_{52}}\right)}$$

The high frequency gain is calculated at $s=\infty$.

$$T(s)=K\frac{1+\frac{R_2}{C_4C_6R_{51}R_3R_1s^2}}{1+\frac{1}{C_6R_6s}+\frac{R_2}{s^2C_4C_6R_1R_3}\left(\frac{1}{R_{51}}+\frac{1}{R_{52}}\right)}$$

$$T(\infty)=K$$

Thus the high frequency gain is $\boxed{K}$.

Step 2 of 4

The transfer function of the low pass notch filter is,

$$T(s)=K\frac{s^2+\frac{R_2}{C_4C_6R_{51}R_3R_1}}{s^2+\frac{s}{C_6R_6}+\frac{R_2}{C_4C_6R_1R_3}\left(\frac{1}{R_{51}}+\frac{1}{R_{52}}\right)}$$

Compare the transfer function with $T(s)=a_2\frac{s^2+\omega_n^2}{s^2+s\left(\frac{\omega_0}{Q}\right)+\omega_0^2}$.

$$\omega_n^2=\frac{R_2}{C_4C_6R_{51}R_3R_1}$$

$$\omega_n=\sqrt{\frac{R_2}{C_4C_6R_{51}R_3R_1}}$$

$$\omega_n=\frac{1}{\sqrt{\frac{C_4C_6R_{31}R_3R_1}{R_2}}}$$

Thus, the notch frequency is, $\boxed{\omega_n=\frac{1}{\sqrt{\frac{C_4C_6R_{51}R_3R_1}{R_2}}}}$.

$$\omega_0^2=\frac{R_2}{C_4C_6R_1R_3}\left(\frac{1}{R_{51}}+\frac{1}{R_{52}}\right)$$

$$\omega_0=\sqrt{\frac{R_2}{C_4C_6R_1R_3}\left(\frac{1}{R_{51}}+\frac{1}{R_{52}}\right)}$$

Thus, resonant frequency of the low pass notch filter is, $\boxed{\omega_0=\sqrt{\frac{R_2}{C_4C_6R_1R_3}\left(\frac{1}{R_{51}}+\frac{1}{R_{52}}\right)}}$

And

$$\frac{\omega_0}{Q}=\frac{1}{C_6R_6}$$

$$Q=\omega_0C_6R_6$$

$$=\sqrt{\frac{R_2}{C_4C_6R_1R_3}\left(\frac{1}{R_{51}}+\frac{1}{R_{52}}\right)}C_6R_6 \quad \left(\text{Since, } \omega_0=\sqrt{\frac{R_2}{C_4C_6R_1R_3}\left(\frac{1}{R_{51}}+\frac{1}{R_{52}}\right)}\right)$$

$$=R_6\sqrt{\frac{R_2C_6}{C_4R_1R_3}\left(\frac{1}{R_{51}}+\frac{1}{R_{52}}\right)}$$

Thus, the quality factor of the low pass notch filter is, $\boxed{Q=R_6\sqrt{\frac{R_2C_6}{C_4R_1R_3}\left(\frac{1}{R_{51}}+\frac{1}{R_{52}}\right)}}$.

The ratio $\left(\frac{\omega_0}{\omega_n}\right)^2$ is,

$$\left(\frac{\omega_0}{\omega_n}\right)^2=\left(\frac{\frac{R_2}{C_4C_6R_1R_3\left(\frac{1}{R_{51}}+\frac{1}{R_{52}}\right)}}{\frac{1}{\frac{C_4C_6R_{51}R_3R_1}{R_2}}}\right)$$

$$=\left(\frac{\frac{R_2}{C_4C_6R_1R_3\left(\frac{1}{R_{51}}+\frac{1}{R_{52}}\right)}}{\frac{R_2}{C_4C_6R_{51}R_3R_1}}\right)$$

$$=\frac{R_{51}}{\left(\frac{1}{R_{51}}+\frac{1}{R_{52}}\right)}$$

Step 3 of 4

Consider the following relation.

$$\boxed{\frac{1}{R_{51}}+\frac{1}{R_{52}}=\frac{1}{R_5}} \quad \text{..... (1)}$$

Now the ratio is,

$$\left(\frac{\omega_0}{\omega_n}\right)^2=R_5$$

$$R_{51}=\frac{1}{R_5}\left(\frac{\omega_0}{\omega_n}\right)^2$$

Thus, the design equation is,

$$\boxed{R_{51}=\frac{1}{R_5}\left(\frac{\omega_0}{\omega_n}\right)^2}$$

Step 4 of 4

From equation (1) we have

$$\frac{1}{R_{52}}=\frac{1}{R_5}-\frac{1}{R_{51}}$$

$$=\frac{1}{R_5}-\frac{1}{R_5\left(\frac{\omega_0}{\omega_n}\right)^2}$$

$$=\frac{1}{R_5}\left[1-\left(\frac{\omega_n}{\omega_0}\right)^2\right]$$

Thus, the design equation is,

$$\boxed{R_{52}=\frac{R_5}{\left[1-\left(\frac{\omega_n}{\omega_0}\right)^2\right]}}$$

Step 1 of 11

Consider the transfer function of the third-order low-pass filter.

$$T(s)=\frac{0.4508\left(s^2+1.6996\right)}{\left(s+0.7294\right)\left(s^2+0.2786s+1.0504\right)} \quad \ldots\ldots (1)$$

The passband edge frequency of the actual filter is,

$$\omega_p = 10^5 \text{ rad/s}$$

Step 2 of 11

(a)

Recall equation (1).

$$T(s)=\frac{0.4508\left(s^2+1.6996\right)}{\left(s+0.7294\right)\left(s^2+0.2786s+1.0504\right)}$$

Substitute $\frac{s}{10^5}$ for s and determine the transfer function of the actual filter.

$$\begin{aligned}T(s)&=\frac{0.4508\left[\left(\frac{s}{10^5}\right)^2+1.6996\right]}{\left(\frac{s}{10^5}+0.7294\right)\left[\left(\frac{s}{10^5}\right)^2+0.2786\left(\frac{s}{10^5}\right)+1.0504\right]}\\&=\frac{\frac{0.4508}{10^{10}}\left[s^2+1.6996\times10^{10}\right]}{\frac{1}{10^5}\left(s+0.7294\times10^5\right)\frac{1}{10^{10}}\left[s^2+0.2786\times10^5 s+1.0504\times10^{10}\right]}\\&=\frac{0.4508\times10^5\left[s^2+1.6996\times10^{10}\right]}{\left(s+72940\right)\left[s^2+27860s+1.0504\times10^{10}\right]}\\&=\frac{45080\left[s^2+1.6996\times10^{10}\right]}{\left(s+72940\right)\left[s^2+27860s+1.0504\times10^{10}\right]}\end{aligned}$$

Therefore, the actual transfer function of the filter is

$$\boxed{\frac{45080\left[s^2+1.6996\times10^{10}\right]}{\left(s+72940\right)\left[s^2+27860s+1.0504\times10^{10}\right]}}.$$

Step 3 of 11

(b)

Consider the transfer function of the cascaded connection of the first-order low-pass filter and the second-order low-pass notch filter.

$$T(s)=T_1(s)\times T_2(s)$$

Here, $T_1(s)$ is the transfer function of the first-order low-pass filter and $T_2(s)$ is the transfer function of the second-order low-pass notch filter.

$$T(s)=\frac{K_1}{\left(s+72940\right)}\times\frac{K_2\left(s^2+1.6996\times10^{10}\right)}{\left(s^2+27860s+1.0504\times10^{10}\right)}$$

Here,

$$T_1(s)=\frac{K_1}{\left(s+72940\right)}$$

$$T_2(s)=\frac{K_2\left(s^2+1.6996\times10^{10}\right)}{\left(s^2+27860s+1.0504\times10^{10}\right)}$$

Calculate the DC gain of $T_1(s)$.

Substitute 0 for s.

$$T_1(0)=\frac{K_1}{\left(0+72940\right)}$$

Substitute 1 for $T_1(0)$.

$$\begin{aligned}1&=\frac{K_1}{72940}\\K_1&=72940\end{aligned}$$

Therefore, the transfer function of the first-order low-pass filter $T_1(s)$ is $\boxed{\frac{72940}{\left(s+72940\right)}}$.

Step 4 of 11

Consider the following expression:

$$K_1\times K_2 = 45080$$

Substitute 72940 for K_1.

$$\begin{aligned}72940\times K_2&=45080\\K_2&=\frac{45080}{72940}\\K_2&=0.618\end{aligned}$$

Therefore, the transfer function of the second-order low-pass filter $T_2(s)$ is

$$\boxed{\frac{0.618\left(s^2+1.6996\times10^{10}\right)}{\left(s^2+27860s+1.0504\times10^{10}\right)}}$$

Step 5 of 11

Consider the transfer function $T(s)$.

$$T(s)=\frac{45080\left[s^2+1.6996\times10^{10}\right]}{\left(s+72940\right)\left[s^2+27860s+1.0504\times10^{10}\right]}$$

Substitute 0 for s.

$$\begin{aligned}T(0)&=\frac{45080\left[0^2+1.6996\times10^{10}\right]}{\left(0+72940\right)\left[0^2+27860\times0+1.0504\times10^{10}\right]}\\&=\frac{45080\times1.6996\times10^{10}}{72940\times1.0504\times10^{10}}\\&=1\end{aligned}$$

Hence, the DC gain is unity.

Step 6 of 11

Refer to figure 17.13(a) for the first-order op amp-RC low-pass filter.

The expression for the transfer function of the first-order RC low-pass filter is,

$$T_1(s)=\frac{\omega_0}{s+\omega_0}$$

Compare the above transfer function with $T_1(s)=\frac{72940}{\left(s+72940\right)}$.

Therefore, the value of ω_0 is 72940 rad/s.

Step 7 of 11

Consider the following expression:

$$\omega_0=\frac{1}{R_2C}$$

Substitute 72940 for ω_0.

$$72940=\frac{1}{R_2C}$$

Select 10 nF for the capacitance C.

Substitute 10×10^{-9} for C.

$$\begin{aligned}72940&=\frac{1}{R_2\times10\times10^{-9}}\\R_2&=\frac{1}{72940\times10\times10^{-9}}\\R_2&=1.37\ \text{k}\Omega\end{aligned}$$

Step 8 of 11

Consider the expression for the magnitude of DC gain.

$$\begin{aligned}\left|\text{DC gain}\right|&=\left|\frac{-R_2}{R_1}\right|\\\left|\text{DC gain}\right|&=\frac{R_2}{R_1}\end{aligned}$$

Substitute 1 for $\left|\text{DC gain}\right|$.

$$\begin{aligned}1&=\frac{R_2}{R_1}\\R_1&=R_2\end{aligned}$$

Substitute $1.37\ \text{k}\Omega$ for R_2.

$$R_1 = 1.37\ \text{k}\Omega$$

Therefore, the design parameters of the first-order op amp-RC low-pass filter is,

$$\boxed{\begin{aligned}R_1&=R_2=1.37\ \text{k}\Omega\\C&=10\ \text{nF}\end{aligned}}$$

Step 9 of 11

Refer to figure 17.22(e) for the second-order op amp-RC Low-Pass Notch filter.

Consider the transfer function of the Low-Pass Notch filter $T_2(s)$.

$$T_2(s)=\frac{0.618\left(s^2+1.6996\times10^{10}\right)}{\left(s^2+27860s+1.0504\times10^{10}\right)}$$

The value of resonant frequency is,

$$\begin{aligned}\omega_0^2&=1.0504\times10^{10}\\\omega_0&=\sqrt{1.0504\times10^{10}}\\&=1.025\times10^5\ \text{rad/s}\end{aligned}$$

The expression for resonant frequency is,

$$\omega_0=\frac{1}{\sqrt{\frac{C_4\left(C_{41}+C_{42}\right)R_1R_2R_3}{R_2}}}$$

Substitute C_4 for $C_{41}+C_{42}$.

$$\omega_0=\frac{1}{\sqrt{\frac{C_4\times C_4\times R_1R_2R_3}{R_2}}}$$

Choose 1 μF for C_4, and C_4 ; and R for R_1,R_2,R_3, and R_3.

Substitute 10^{-6} for C_4, and C_4 ; and 1.025×10^5 for ω_0.

$$\begin{aligned}1.025\times10^5&=\frac{1}{\sqrt{\frac{\left(10^{-6}\right)\times\left(10^{-6}\right)\times R^3}{R}}}\\10^{-6}\times R&=\frac{1}{1.025\times10^5}\\R&=\frac{1}{1.025\times10^5\times10^{-6}}\\R&=9.756\ \Omega\end{aligned}$$

Step 10 of 11

Consider the transfer function $T_2(s)$.

$$T_2(s)=\frac{0.618\left(s^2+1.6996\times10^{10}\right)}{\left(s^2+27860s+1.0504\times10^{10}\right)}$$

The value of notch frequency from the transfer function is,

$$\omega_n = 1.6996\times10^{10}\ \text{rad/s}$$

The expression for the notch frequency ω_n is,

$$\omega_n=\frac{1}{\sqrt{\frac{C_4C_{41}R_1R_2R_3}{R_2}}}$$

Substitute 1.6996×10^{10} for ω_n , 10^{-6} for C_4 , and R for R_1,R_2,R_3, and R_3.

$$\begin{aligned}1.6996\times10^{10}&=\frac{1}{\sqrt{\frac{\left(10^{-6}\right)C_{41}R^3}{R}}}\\R\sqrt{\left(10\times10^{-6}\right)C_{41}}&=\frac{1}{1.6996\times10^{05}}\end{aligned}$$

Substitute $9.756\ \Omega$ for R.

$$\begin{aligned}9.756\times\sqrt{\left(10^{-6}\right)C_{41}}&=\frac{1}{1.6996\times10^{10}}\\\sqrt{C_{41}}&=\frac{1}{1.6996\times10^{10}\times9.756\times10^{-2}}\\C_{41}&=0.036\times10^{-15}\ \text{F}\\C_{41}&=0.036\ \text{fF}\end{aligned}$$

Calculate the value of C_{42}.

$$\begin{aligned}C_{42}&=10^{-6}-C_{41}\\&=10^{-6}-0.036\times10^{-15}\\&=1\ \mu\text{F}\end{aligned}$$

Step 11 of 11

Consider the transfer function $T_2(s)$.

$$T_2(s)=\frac{0.618\left(s^2+1.6996\times10^{10}\right)}{\left(s^2+27860s+1.0504\times10^{10}\right)}$$

From the transfer function,

$$\frac{\omega_0}{Q}=27860$$

Substitute 1.025×10^5 for ω_0.

$$\begin{aligned}\frac{1.025\times10^5}{Q}&=27860\\Q&=\frac{1.025\times10^5}{27860}\\Q&=3.68\end{aligned}$$

The expression for the quality factor is,

$$Q=R_4\sqrt{\frac{\left(C_{41}+C_{42}\right)}{C_4}\times\frac{R_2}{R_1R_2R_3}}$$

Substitute C_4 for $C_{41}+C_{42}$.

$$Q=R_4\sqrt{\frac{C_4}{C_4}\times\frac{R_2}{R_1R_2R_3}}$$

Substitute 3.68 for Q. 10^{-6} for C_4 , and C_4 ; and R for R_1,R_2,R_3, and R_2.

$$\begin{aligned}3.68&=R_4\sqrt{1\times\frac{R}{R^3}}\\R_4&=3.68R\end{aligned}$$

Substitute $9.756\ \Omega$ for R.

$$\begin{aligned}R_4&=3.68\times9.756\\&=35.9\ \Omega\end{aligned}$$

Therefore, the design parameters of the second-order Low-Pass Notch filter are,

$$\boxed{\begin{aligned}R_1&=R_2=R_3=9.756\ \Omega\\R_4&=35.9\ \Omega\\C_4&=1\ \mu\text{F}\\C_{41}&=0.03\ \text{fF}\\C_{42}&=1\ \mu\text{F}\end{aligned}}$$

Step 1 of 6

Refer to figure 17.24(a) for the KHN biquad circuit given in the textbook.

The expression for the 3-dB bandwidth of KHN biquad circuit is,

$$B = \frac{\omega_0}{Q}$$

$$Q = \frac{\omega_0}{B}$$

$$Q = \frac{2\pi f_0}{B}$$

Substitute 2×10^3 for f_0, and $2\pi\times50$ for *B*.

$$Q = \frac{2\pi\times2\times10^3}{2\pi\times50}$$
$$= 40$$

Therefore, the quality factor *Q* is 40.

Step 2 of 6

The expression for the center frequency of the KHN biquad circuit is,

$$\omega_0 = \frac{1}{RC}$$

$$R = \frac{1}{2\pi f_0 C}$$

Substitute 2×10^3 for f_0, and 10×10^{-9} for *C*.

$$R = \frac{1}{2\pi\times2\times10^3\times10\times10^{-9}}$$
$$= 7.95\ \text{k}\Omega$$

Therefore, the value of resistance *R* is $7.95\ \text{k}\Omega$, and that of capacitance *C* is 10 nF.

Step 3 of 6

Determine the value of feedback resistance R_f.

Consider the following expression:

$$\frac{R_f}{R_1} = 1$$

$$R_f = R_1$$

Assume that the value of R_1 as $10\ \text{k}\Omega$ and substitute the same for R_1.

$$R_f = 10\ \text{k}\Omega$$

Therefore, the value of feedback resistance R_f is $10\ \text{k}\Omega$ and that of R_1 is $10\ \text{k}\Omega$.

Step 4 of 6

Determine the value of resistance R_2.

Consider the following expression:

$$\frac{R_2}{R_2} = 2Q - 1$$

Substitute 40 for *Q*.

Assume that the value of R_2 as $10\ \text{k}\Omega$ and substitute 10×10^3 for R_2.

$$\frac{R_2}{10\times10^3} = 2\times40 - 1$$

$$R_2 = 790\ \text{k}\Omega$$

Therefore, the value of resistance R_2 is $790\ \text{k}\Omega$ and that of R_2 is $10\ \text{k}\Omega$.

Therefore, the component values of the KHN biquad circuit is,

$$\boxed{\begin{aligned} R &= 7.95\ \text{k}\Omega \\ R_f &= 10\ \text{k}\Omega \\ R_1 &= 10\ \text{k}\Omega \\ R_2 &= 10\ \text{k}\Omega \\ R_2 &= 790\ \text{k}\Omega \\ C &= 10\ \text{nF} \end{aligned}}$$

Step 5 of 6

Design the KHN biquad circuit as shown in figure 1.

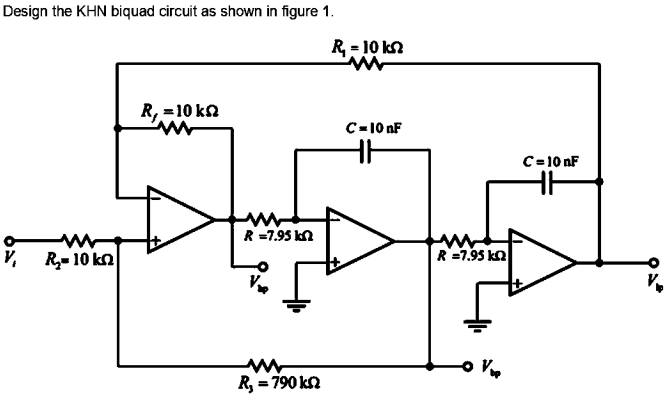


Figure 1

Step 6 of 6

The expression for the center frequency gain is,

$$KQ = 2Q - 1$$

Substitute 40 for *Q*.

$$KQ = 2(40) - 1$$
$$= 79$$

Therefore, the center frequency gain of the KHN biquad circuit is $\boxed{79}$.

Step 1 of 13

Refer to figure 17.24(b) for the output summing amplifier given in the textbook.

Consider the following condition:

$$R_L = R_H$$
$$= \frac{R_g}{Q}$$

$$\frac{R_H}{R_L} = 1 \quad \text{..... (1)}$$

$$\frac{R_H}{R_g} = \frac{1}{Q} \quad \text{..... (2)}$$

Step 2 of 13

The expression for the transfer function of the KHN biquad filter is,

$$\frac{V_o}{V_i} = -K\left[\frac{\left(\frac{R_F}{R_H}\right)s^2 - s\left(\frac{R_F}{R_g}\right)\omega_0 + \left(\frac{R_F}{R_L}\right)\omega_0^2}{s^2 + s\left(\frac{\omega_0}{Q}\right) + \omega_0^2}\right]$$
$$= -K\left(\frac{R_F}{R_H}\right)\left[\frac{s^2 - s\left(\frac{R_H}{R_g}\right)\omega_0 + \left(\frac{R_H}{R_L}\right)\omega_0^2}{s^2 + s\left(\frac{\omega_0}{Q}\right) + \omega_0^2}\right]$$

Consider equations (1) and (2).

Substitute 1 for $\left(\frac{R_H}{R_L}\right)$, and $\left(\frac{1}{Q}\right)$ for $\left(\frac{R_H}{R_g}\right)$.

$$\frac{V_o}{V_i} = -K\left(\frac{R_F}{R_H}\right)\left[\frac{s^2 - s\left(\frac{1}{Q}\right)\omega_0 + (1)\omega_0^2}{s^2 + s\left(\frac{\omega_0}{Q}\right) + \omega_0^2}\right]$$

$$\frac{V_o}{V_i} = -K\left(\frac{R_F}{R_H}\right)\left[\frac{s^2 - s\left(\frac{\omega_0}{Q}\right) + \omega_0^2}{s^2 + s\left(\frac{\omega_0}{Q}\right) + \omega_0^2}\right] \quad \text{..... (3)}$$

Step 3 of 13

Consider the general transfer function of the all-pass filter.

$$T(s) = K\left[\frac{s^2 - \left(\frac{\omega_0}{Q}\right)s + \omega_0^2}{s^2 + \left(\frac{\omega_0}{Q}\right)s + \omega_0^2}\right] \quad \text{..... (4)}$$

Here, K is the gain of the filter.

Compare the equations (3) and (4).

Therefore, the transfer function of the KHN biquad filter is an all-pass filter.

Hence, the all-pass filter function is realized by selecting $\boxed{R_L = R_H = \frac{R_g}{Q}}$.

Step 4 of 13

Recall equation (3).

$$\frac{V_o}{V_i} = -K\left(\frac{R_F}{R_H}\right)\left[\frac{s^2 - s\left(\frac{\omega_0}{Q}\right) + \omega_0^2}{s^2 + s\left(\frac{\omega_0}{Q}\right) + \omega_0^2}\right]$$

Therefore, the flat band gain is $\boxed{K\left(\frac{R_F}{R_H}\right)}$

Step 5 of 13

(b)

Refer to Figure 17.24(a) for the KHN biquad circuit in the textbook.

Assume that the value of capacitance C is 10 nF.

Consider the following relation:

$$\omega_0 = \frac{1}{RC}$$
$$R = \frac{1}{\omega_0 C}$$

Substitute 10^3 for ω_0, and 10×10^{-9} for C.

$$R = \frac{1}{10^3 \times 10\times10^{-9}}$$
$$= 1\ \text{k}\Omega$$

Therefore, the value of resistance R is $1\ \text{k}\Omega$.

Step 6 of 13

Assume that the value of resistances R_1 and R_2 is $10\ \text{k}\Omega$.

Determine the feedback resistance R_f.

Consider the following relation:

$$\frac{R_f}{R_1} = 1$$
$$R_f = R_1$$

Substitute $10\ \text{k}\Omega$ for R_1.

$$R_f = 10\ \text{k}\Omega$$

Therefore, the value of feedback resistance R_f is $10\ \text{k}\Omega$.

Step 7 of 13

Determine the value of resistance R_3.

Consider the following relation:

$$\frac{R_3}{R_2} = 2Q - 1$$

Substitute 4 for Q, and 10×10^3 for R_3.

$$\frac{R_3}{10\times10^3} = 2(4) - 1$$
$$R_3 = 70\ \text{k}\Omega$$

Therefore, the value of resistance R_3 is $70\ \text{k}\Omega$.

Therefore, the design parameters of the KHN biquad circuit are,

$$\boxed{R = 1\ \text{k}\Omega,\ C = 10\ \text{nF},\ R_1 = 10\ \text{k}\Omega,\ R_2 = 10\ \text{k}\Omega,\ R_f = 10\ \text{k}\Omega \text{ and } R_3 = 70\ \text{k}\Omega}$$.

Step 8 of 13

Refer to figure 17.24(b) for the output summing amplifier given in the textbook.

Determine the value of K.

Consider the following relation:

$$K = 2 - \left(\frac{1}{Q}\right)$$

Substitute 4 for Q.

$$K = 2 - \left(\frac{1}{4}\right)$$
$$= 1.75$$

Step 9 of 13

Consider the following expression:

$$\text{Flat gain} = K\left(\frac{R_F}{R_H}\right)$$

Substitute 10 for flat gain, and 1.75 for K.

$$10 = (1.75)\left(\frac{R_F}{R_H}\right)$$
$$\frac{R_H}{R_F} = 0.175$$

Step 10 of 13

Assume that the value of feedback resistance R_F as $100\ \text{k}\Omega$.

Determine the value of resistance R_H.

$$R_H = 0.175 \times R_F$$

Substitute $100\ \text{k}\Omega$ for R_F.

$$R_H = 0.175 \times (100\ \text{k}\Omega)$$
$$= 17.5\ \text{k}\Omega$$

Therefore, the value of feedback resistance R_F is $100\ \text{k}\Omega$ and that of R_H is $17.5\ \text{k}\Omega$.

Step 11 of 13

Determine the value of resistance R_g.

Consider the following relation:

$$R_g = QR_H$$

Substitute 4 for Q, and 17.5×10^3 for R_H.

$$R_g = 4\times17.5\times10^3$$
$$= 70\ \text{k}\Omega$$

Therefore, the value of resistance R_g is $70\ \text{k}\Omega$.

Step 12 of 13

Determine the value of resistance R_L.

Consider the following relation:

$$R_L = R_H$$

Substitute $17.5\ \text{k}\Omega$ for R_H.

$$R_L = 17.5\ \text{k}\Omega$$

Therefore, the design parameters of summing amplifier are,

$$\boxed{\begin{aligned} R_F &= 100\ \text{k}\Omega \\ R_H &= 17.5\ \text{k}\Omega \\ R_g &= 70\ \text{k}\Omega \\ R_L &= 17.5\ \text{k}\Omega \end{aligned}}$$.

Step 13 of 13

Design the all-pass filter by cascading the KHN biquad filter and the summing amplifier as shown in figure 1.

Figure 1

Step 1 of 5

Refer to equation (17.56) for the high-pass filter transfer function given in the textbook.

$$T_{hp} = \frac{V_{hp}}{V_i}$$

$$T_{hp} = \frac{Ks^2}{s^2 + s\left(\frac{\omega_0}{Q}\right) + \omega_0^2}$$

Replace high frequency gain *K* by *G*.

$$T_{hp} = \frac{Gs^2}{s^2 + s\left(\frac{\omega_0}{Q}\right) + \omega_0^2} \quad \text{..... (1)}$$

Step 2 of 5

Refer to equation (17.59) for the band-pass filter transfer function given in the transfer function.

$$T_{bp} = \frac{V_{bp}}{V_i}$$

$$T_{bp} = \frac{K\omega_0 s}{s^2 + s\left(\frac{\omega_0}{Q}\right) + \omega_0^2}$$

Replace high frequency gain *K* by *G*.

$$T_{bp} = \frac{G\omega_0 s}{s^2 + s\left(\frac{\omega_0}{Q}\right) + \omega_0^2} \quad \text{..... (2)}$$

Step 3 of 5

Refer to equation (17.60) for the low-pass filter transfer function given in the transfer function.

$$T_{lp} = \frac{V_{lp}}{V_i}$$

$$T_{lp} = \frac{K\omega_0^2}{s^2 + s\left(\frac{\omega_0}{Q}\right) + \omega_0^2}$$

Replace high frequency gain *K* by *G*.

$$T_{lp} = \frac{G\omega_0^2}{s^2 + s\left(\frac{\omega_0}{Q}\right) + \omega_0^2} \quad \text{..... (3)}$$

Step 4 of 5

Refer to figure 17.24(b) for the summing amplifier given in the textbook.

Apply Nodal analysis at the non-inverting terminal of summing amplifier.

$$\frac{0 - V_{hp}}{R_H} + \frac{0 - V_{bp}}{R_B} + \frac{0 - V_{lp}}{R_L} + \frac{0 - V_o}{R_F} = 0$$

$$\frac{V_o}{R_F} = -\left(\frac{V_{hp}}{R_H} + \frac{V_{bp}}{R_B} + \frac{V_{lp}}{R_L}\right)$$

$$V_o = -R_F\left(\frac{V_{hp}}{R_H} + \frac{V_{bp}}{R_B} + \frac{V_{lp}}{R_L}\right)$$

$$V_o = -R_F\left(\frac{V_i T_{hp}}{R_H} + \frac{V_i T_{bp}}{R_B} + \frac{V_i T_{lp}}{R_L}\right) \quad \text{..... (4)}$$

Consider equations (1), (2), and (3). Substitute the expressions for T_{hp}, T_{bp}, and T_{lp}.

Recall equation (1).

$$T_{hp} = \frac{Gs^2}{s^2 + s\left(\frac{\omega_0}{Q}\right) + \omega_0^2}$$

Recall equation (2).

$$T_{bp} = \frac{G\omega_0 s}{s^2 + s\left(\frac{\omega_0}{Q}\right) + \omega_0^2}$$

Recall equation (3).

$$T_{lp} = \frac{G\omega_0^2}{s^2 + s\left(\frac{\omega_0}{Q}\right) + \omega_0^2}$$

Substitute the expressions for T_{hp}, T_{bp}, and T_{lp} in equation (4).

Refer to equation (17.66) for the overall-transfer function of cascaded filter circuit.

$$\frac{V_o}{V_i} = -G\frac{\left(\frac{R_F}{R_H}\right)s^2 - \left(\frac{R_F}{R_B}\right)\omega_0 s + \left(\frac{R_F}{R_L}\right)\omega_0^2}{s^2 + \left(\frac{\omega_0}{Q}\right)s + \omega_0^2}$$

$$T(s) = -G\frac{\left(\frac{R_F}{R_H}\right)s^2 - \left(\frac{R_F}{R_B}\right)\omega_0 s + \left(\frac{R_F}{R_L}\right)\omega_0^2}{s^2 + \left(\frac{\omega_0}{Q}\right)s + \omega_0^2}$$

$$T(s) = -G\frac{R_F}{R_H} \times \frac{s^2 - \left(\frac{1}{R_H R_B}\right)\omega_0 s + \left(\frac{R_H}{R_L}\right)\omega_0^2}{s^2 + \left(\frac{\omega_0}{Q}\right)s + \omega_0^2} \quad \text{..... (5)}$$

Step 5 of 5

Consider the general transfer function of a notch filter.

$$T(s) = a_2\frac{s^2 + \omega_n^2}{s^2 + \left(\frac{\omega_0}{Q}\right)s + \omega_0^2} \quad \text{..... (6)}$$

Compare the equations (5) and (6).

$$\left(\frac{R_H}{R_L}\right)\omega_0^2 = \omega_n^2$$

$$\frac{R_H}{R_L} = \left(\frac{\omega_n}{\omega_0}\right)^2$$

Compare the equations (5) and (6).

The coefficient of *s* is 0.

Select the value of resistance R_B as ∞.

Therefore, the expressions for the values required of the resistances associated with the summing amplifier are,

$$\boxed{\begin{array}{l}\frac{R_H}{R_L} = \left(\frac{\omega_n}{\omega_0}\right)^2 \\ R_B = \infty\end{array}}.$$

Step 1 of 3

17.064P

Refer to Figure 16.24 (b) in the textbook for the op-amp summing circuit.

The Notch filter realized using KHN biquad circuit with the notch frequency equal to the resonant frequency, that is, $\omega_n = \omega_0$

The tolerance of the summing resistors used is 1% .

The notch filter is obtained by the both low pass and high pass filters. From the op-amp summing circuit, the notch filter is obtained by selecting or eliminating the resistor R_B (that is $R_B = \infty$). Therefore,

$$\frac{R_H}{R_L} = \left(\frac{\omega_n}{\omega_0}\right)^2$$

Here,

$$R_H = R_L \pm 1\%$$

Step 2 of 3

For $R_H = R_L + 1\%$, the worst case deviation between ω_n and ω_0 is,

$$\frac{1.01}{0.99} = \left(\frac{\omega_n}{\omega_0}\right)^2$$

Simplify further.

$$\frac{\omega_n}{\omega_0} = \sqrt{\frac{1.01}{0.99}}$$

$$\omega_n = \omega_0\sqrt{\frac{1.01}{0.99}}$$

$$\omega_n = 1.01\omega_0$$

$$\frac{\omega_n}{\omega_0} = 1.01$$

The worst case percentage deviation between ω_n and ω_0 is,

$$\begin{aligned}\left(\frac{\omega_n}{\omega_0} - 1\right)100 &= (1.01 - 1)100 \\ &= (0.01)100 \\ &= +1\%\end{aligned}$$

Step 3 of 3

For $R_H = R_L - 1\%$, the worst case deviation between ω_n and ω_0 is,

$$\frac{0.99}{1.01} = \left(\frac{\omega_n}{\omega_0}\right)^2$$

Simplify further.

$$\frac{\omega_n}{\omega_0} = \sqrt{\frac{0.99}{1.01}}$$

$$\omega_n = \omega_0\sqrt{\frac{0.99}{1.01}}$$

$$\omega_n = 0.99\omega_0$$

$$\frac{\omega_n}{\omega_0} = 0.99$$

The worst case percentage deviation between ω_n and ω_0 is,

$$\begin{aligned}\left(\frac{\omega_n}{\omega_0} - 1\right)100 &= (0.99 - 1)100 \\ &= (-0.01)100 \\ &= -1\%\end{aligned}$$

Thus, the worst case percentage deviation between ω_n and ω_0 is, $\boxed{\pm 1\%}$.

Step 1 of 3

Refer to Figure 17.26 for the low-pass notch filter in the textbook.

Refer to Table 16.2 for the design data in the textbook.

The expression for the resistance *R* of the filter is,

$$R = \frac{1}{\omega_0 C}$$

Substitute 10×10^{-9} for the capacitance *C*, and 10^5 for the resonant frequency ω_0.

$$\begin{aligned} R &= \frac{1}{10^5 \times 10\times10^{-9}} \\ &= 1\ \text{k}\Omega \end{aligned}$$

Therefore, the value of resistance *R* is $1\ \text{k}\Omega$.

Step 2 of 3

The expression for the DC gain is,

$$\text{DC gain} = a_2\left(\frac{\omega_n}{\omega_0}\right)^2$$

Calculate high frequency gain a_2.

Substitute 1 for DC gain, 1.3×10^5 for ω_n, and 10^5 for ω_0.

$$1 = a_2\left(\frac{1.3\times10^5}{10^5}\right)^2$$

$$a_2 = \frac{1}{(1.3)^2}$$

$$a_2 = 0.591$$

Calculate the value of capacitance *C*.

$$C_1 = Ca_2$$

Substitute 0.591 for a_2, and 10×10^{-9} for *C*.

$$\begin{aligned} C_1 &= \left(10\times10^{-9}\right)(0.591) \\ &= 5.91\ \text{nF} \end{aligned}$$

Therefore, the value of capacitance C_1 is 5.91 nF.

Step 3 of 3

Calculate resistance R_2.

Consider the following expression:

$$\begin{aligned} R_2 &= \frac{R\left(\frac{\omega_0}{\omega_n}\right)^2}{a_2} \\ &= \frac{R\left(\frac{\omega_0}{\omega_n}\right)^2}{\left(\frac{\omega_0}{\omega_n}\right)^2} \\ &= R \end{aligned}$$

Substitute $1\ \text{k}\Omega$ for *R*.

Therefore, the value of resistance R_2 is $1\ \text{k}\Omega$.

Consider the table 17.2.

The value of resistances R_1 and R_3 is ∞.

Therefore, the design parameters of low-pass notch filter is,

$$\boxed{R = 1\ \text{k}\Omega,\ C = 1\ \text{nF},\ C_1 = 5.91\ \text{nF},\ R_2 = 1\ \text{k}\Omega,\ R_1 = R_3 = \infty \text{ and } r = 20\ \text{k}\Omega}$$

Step 1 of 2

17.066P

(a)

Refer to Figure 16.26 in the textbook for the all-pass realization of the Tow-Thomas biquad filter circuit with feedforward path.

Write the transfer function of the filter.

$$\frac{V_o}{V_i}=-\frac{s^2\left(\frac{C_1}{C}\right)+s\frac{1}{C}\left(\frac{1}{R_1}-\frac{r}{RR_3}\right)+\frac{1}{C^2RR_2}}{s^2+s\left(\frac{1}{QCR}\right)+\frac{1}{C^2R^2}}$$

Consider the numerator expression of the transfer function.

$$s^2\left(\frac{C_1}{C}\right)+s\frac{1}{C}\left(\frac{1}{R_1}-\frac{r}{RR_3}\right)+\frac{1}{C^2RR_2}=0$$

$$s^2+s\frac{\frac{1}{C}\left(\frac{1}{R_1}-\frac{r}{RR_3}\right)}{\left(\frac{C_1}{C}\right)}+\frac{\frac{1}{C^2RR_2}}{\left(\frac{C_1}{C}\right)}=0$$

$$s^2+sC_1\left(\frac{1}{R_1}-\frac{r}{RR_3}\right)+\frac{1}{C^2RR_3}\frac{C}{C_1}=0$$

Compare the expression with the standard second order system, $s^2+2\zeta\omega_z s+\omega_z^2=0$

The all pass frequency is,

$$\omega_z^2=\frac{1}{C^2RR_2}\frac{C}{C_1}$$

$$\omega_z=\frac{1}{C\sqrt{RR_2}}\sqrt{\frac{C}{C_1}}$$

Here, observe that the expression ω_z is independent of r or R_3

Thus, the components one needs to trim to adjust ω_z are, $\boxed{R_2 \text{ and } (r \text{ or } R_3)}$.

Step 2 of 2

(b)

The quality factor of the all pass Tow-Thomas biquad filter is,

$$Q_z=\frac{\omega_z}{\frac{1}{C}\left(\frac{1}{R_1}-\frac{r}{RR_3}\right)\frac{C}{C_1}}$$

Substitute $\frac{1}{C\sqrt{RR_2}}\sqrt{\frac{C}{C_1}}$ for ω_z.

$$Q_z=\frac{\frac{1}{C\sqrt{RR_2}}\sqrt{\frac{C}{C_1}}}{\frac{1}{C}\left(\frac{1}{R_1}-\frac{r}{RR_3}\right)\frac{C}{C_1}}$$

$$=\frac{1}{\sqrt{RR_2}\left(\frac{1}{R_1}-\frac{r}{RR_3}\right)\sqrt{\frac{C}{C_1}}}$$

Refer to Table 16.2 in the textbook. For all pass filter, the value of the resistor, $R_1=\infty$.

Thus, the quality factor is,

$$Q_z=\frac{1}{\sqrt{RR_3}\left(0-\frac{r}{RR_3}\right)\sqrt{\frac{C}{C_1}}} \qquad \left\{\text{since } R_1\to\infty, \frac{1}{R_1}\to 0\right\}$$

$$=\frac{1}{\sqrt{RR_2}\left(-\frac{r}{RR_3}\right)\sqrt{\frac{C}{C_1}}}$$

Thus, the components one needs to trim to adjust Q_z are, $\boxed{C_1 \text{ and } (r \text{ or } R_3)}$.

Step 1 of 7

(a)

Refer to Figure 17.26 for the Tow-Thomas biquad with feedforward circuit in the textbook.

Consider the transfer function of the third-order low-pass filter.

$$T(s)=\frac{0.4508\left(s^2+1.6996\right)}{(s+0.7294)\left(s^2+0.2786s+1.0504\right)}$$

The passband edge frequency of the actual filter is, $\omega_p=10^5$ rad/s .

Replace *s* by $\frac{s}{\omega_p}$ and obtain the transfer function of Tow-Thomas biquad filter circuit.

$$\begin{aligned}T(s)&=\frac{0.4508\left(\left(\frac{s}{\omega_p}\right)^2+1.6996\right)}{\left(\frac{s}{\omega_p}+0.7294\right)\left(\left(\frac{s}{\omega_p}\right)^2+0.2786\left(\frac{s}{\omega_p}\right)+1.0504\right)}\\&=\frac{0.4508\left(\frac{s^2+1.6996\omega_p^2}{\omega_p^2}\right)}{\left(\frac{s+0.7294\omega_p}{\omega_p}\right)\left(\frac{s^2+\left(0.2786\omega_p\right)s+1.0504\omega_p^2}{\omega_p^2}\right)}\\&=\frac{0.4508\omega_p\left(s^2+1.6996\omega_p^2\right)}{\left(s+0.7294\omega_p\right)\left(s^2+\left(0.2786\omega_p\right)s+1.0504\omega_p^2\right)}\end{aligned}$$

Step 2 of 7

Substitute 10^5 for ω_p .

$$T(s)=\frac{0.4508\omega_p\left(s^2+1.6996\omega_p^2\right)}{\left(s+0.7294\omega_p\right)\left(s^2+\left(0.2786\omega_p\right)s+1.0504\omega_p^2\right)}$$

Substitute 10^5 for ω_p .

$$\begin{aligned}T(s)&=\frac{0.4508\times10^5\left(s^2+1.6996\left(10^5\right)^2\right)}{\left(s+0.7294\times10^5\right)\left[s^2+\left(0.2786\times10^5\right)s+1.0504\left(10^5\right)^2\right]}\\&=\frac{45080\left(s^2+1.6996\times10^{10}\right)}{(s+72940)\left(s^2+27860s+1.0504\times10^{10}\right)}\end{aligned}$$

Therefore, the transfer function of the actual filter is,

$$\boxed{\frac{45080\left(s^2+1.6996\times10^{10}\right)}{(s+72940)\left(s^2+27860s+1.0504\times10^{10}\right)}}$$

Step 3 of 7

Refer to figure 17.13 (a) for the first-order op amp-RC low-pass filter given in the textbook.

The resonant frequency is, $\omega_0=72940$ rad/s and the DC gain magnitude is 1.

Select the value of capacitance *C* as 10 nF.

Consider the following expression:

$$R_1=R_2=\frac{1}{\omega_0 C}$$

Substitute 10×10^{-9} for *C*, and 72940 for ω_0 .

$$\begin{aligned}R_1&=\frac{1}{72940\times10\times10^{-9}}\\&=1.371\ \text{k}\Omega\end{aligned}$$

Step 4 of 7

Recall the transfer function of the actual filter.

$$T(s)=\frac{45080\left(s^2+1.6996\times10^{10}\right)}{(s+72940)\left(s^2+27860s+1.0504\times10^{10}\right)}$$

Compare the denominator term $\left(s^2+27860s+1.0504\times10^{10}\right)$ with $s^2+\frac{\omega_0}{Q}s+\omega_0^2$.

Determine the resonant frequency.

$$\omega_0^2=1.0504\times10^{10}$$

$$\omega_0=1.025\times10^5\ \text{rad/s}$$

Determine the quality factor *Q*.

$$\frac{\omega_0}{Q}=27860$$

$$Q=\frac{\omega_0}{27860}$$

Substitute 1.025×10^5 for ω_0 .

$$\begin{aligned}Q&=\frac{1.025\times10^5}{27860}\\&=3.68\end{aligned}$$

Step 5 of 7

Recall the transfer function of the actual filter.

$$T(s)=\frac{45080\left(s^2+1.6996\times10^{10}\right)}{(s+72940)\left(s^2+27860s+1.0504\times10^{10}\right)}$$

Compare the numerator term $\left(s^2+1.6996\times10^{10}\right)$ with $s^2+\omega_n^2$.

Determine the notch frequency.

$$\omega_n^2=1.6996\times10^{10}$$

$$\omega_n=1.304\times10^5\ \text{rad/s}$$

Choose the capacitance value *C* as 10 nF.

Calculate the resistance value *R*.

$$R=\frac{1}{\omega_0 C}$$

Substitute 1.025×10^5 for ω_0 , and 10×10^{-9} for *C*.

$$\begin{aligned}R&=\frac{1}{1.025\left(10^5\right)\left(10\times10^{-9}\right)}\\&=0.976\ \text{k}\Omega\end{aligned}$$

Step 6 of 7

Refer to Figure 17.16 (e) for low-pass notch filter in the textbook.

Choose the value of resistance *r* as $20\ \text{k}\Omega$.

The expression for the DC gain of low-pass notch filter is,

$$\text{DC gain}=a_2\frac{\omega_n^2}{\omega_0^2}$$

Here, a_2 is the high frequency gain.

Substitute 1 for DC gain, 1.025×10^5 for ω_0^2 , and 1.6996×10^{10} for ω_n^2 .

$$1=a_2\frac{1.6996\times10^{10}}{1.0504\times10^{10}}$$

$$a_2=0.62$$

Therefore, the high frequency gain is 0.62.

Step 7 of 7

Refer to Table 16.2 given in the textbook.

Determine the value of capacitance C_1 .

$$C_1=Ca_2$$

Substitute 0.62 for a_2 , and 10×10^{-9} for *C*.

$$\begin{aligned}C_1&=\left(10\times10^{-9}\right)(0.62)\\&=6.2\ \text{nF}\end{aligned}$$

Determine the value of resistance R_2 .

$$\begin{aligned}R_2&=R\left(\frac{\omega_0}{\omega_n}\right)^2\\&=(0.976\ \text{k}\Omega)(0.62)\\&=0.605\ \text{k}\Omega\end{aligned}$$

And, the resistances $R_1=R_3=\infty$.

Therefore, the design parameters of Tow-Thomas biquad filter of low-pass filter are,

$$\boxed{C=10\ \text{nF},\ r=20\ \text{k}\Omega,\ R=0.976\ \text{k}\Omega,\ C_1=6.2\ \text{nF},\ R_2=0.605\ \text{k}\Omega,\ \text{and}\ R_1=R_3=\infty}$$

Step 1 of 6

17.068P

Refer to Figure 17.29 for the active-filter feedback loop generated using the bridged-T network in the textbook.

Consider the feedback loop using the bridged-T network in *s*-domain as shown in Figure 1.

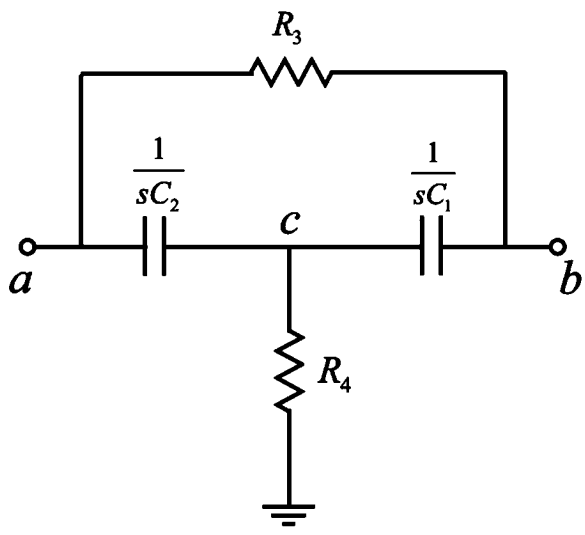


Figure 1

Step 2 of 6

Assume that the voltages of nodes *a*, *b*, and *c* as $V_a(s)$, $V_b(s)$, and $V_c(s)$.

Apply Nodal analysis at the node-*c*.

$$\frac{V_c(s)-V_a(s)}{\frac{1}{sC_2}}+\frac{V_c(s)-V_b(s)}{\frac{1}{sC_1}}+\frac{V_c(s)}{R_4}=0$$

$$\left(sC_1+sC_2+\frac{1}{R_4}\right)V_c(s)=sC_1V_b(s)+sC_2V_a(s)$$

$$V_c(s)=\frac{sC_1V_b(s)+sC_2V_a(s)}{\left(sC_1+sC_2+\frac{1}{R_4}\right)} \quad \text{..... (1)}$$

Apply Nodal analysis at the node-*b*.

$$\frac{V_b(s)-V_c(s)}{\frac{1}{sC_1}}+\frac{V_b(s)-V_a(s)}{R_3}=0$$

$$\left(sC_1+\frac{1}{R_3}\right)V_b(s)=sC_1V_c(s)+\frac{V_a(s)}{R_3} \quad \text{..... (2)}$$

Step 3 of 6

Consider equation (1) and substitute the expression for $V_c(s)$ in equation (2).

$$\left(sC_1+\frac{1}{R_3}\right)V_b(s)=sC_1\left[\frac{sC_1V_b(s)+sC_2V_a(s)}{\left(sC_1+sC_2+\frac{1}{R_4}\right)}\right]+\frac{V_a(s)}{R_3}$$

$$\left[sC_1+\frac{1}{R_3}-\frac{(sC_1)^2}{\left(sC_1+sC_2+\frac{1}{R_4}\right)}\right]V_b(s)=\left[\frac{s^2C_1C_2}{\left(sC_1+sC_2+\frac{1}{R_4}\right)}+\frac{1}{R_3}\right]V_a(s)$$

$$\frac{V_a(s)}{V_b(s)}=\frac{\left[sC_1+\frac{1}{R_3}-\frac{(sC_1)^2}{\left(sC_1+sC_2+\frac{1}{R_4}\right)}\right]}{\left[\frac{s^2C_1C_2}{\left(sC_1+sC_2+\frac{1}{R_4}\right)}+\frac{1}{R_3}\right]}$$

$$\frac{V_a(s)}{V_b(s)}=\frac{s^2+s\left(\frac{1}{R_4C_2}+\frac{1}{R_3C_2}+\frac{1}{R_3C_1}\right)+\frac{1}{R_3R_4C_1C_2}}{s^2+s\left(\frac{1}{C_2}+\frac{1}{C_1}\right)\frac{1}{R_3}+\frac{1}{R_3R_4C_1C_2}}$$

$$\frac{V_b(s)}{V_a(s)}=\frac{s^2+s\left(\frac{1}{C_2}+\frac{1}{C_1}\right)\frac{1}{R_3}+\frac{1}{R_3R_4C_1C_2}}{s^2+s\left(\frac{1}{R_4C_2}+\frac{1}{R_3C_2}+\frac{1}{R_3C_1}\right)+\frac{1}{R_3R_4C_1C_2}}$$

Therefore, the transfer function seen from *b* to *a* is,

$$t(s)=\frac{s^2+s\left(\frac{1}{C_2}+\frac{1}{C_1}\right)\frac{1}{R_3}+\frac{1}{R_3R_4C_1C_2}}{s^2+s\left(\frac{1}{R_4C_2}+\frac{1}{R_3C_2}+\frac{1}{R_3C_1}\right)+\frac{1}{R_3R_4C_1C_2}}$$

Step 4 of 6

Consider the transfer function $t(s)$.

$$t(s)=\frac{s^2+s\left(\frac{1}{C_2}+\frac{1}{C_1}\right)\frac{1}{R_3}+\frac{1}{R_3R_4C_1C_2}}{s^2+s\left(\frac{1}{R_4C_2}+\frac{1}{R_3C_2}+\frac{1}{R_3C_1}\right)+\frac{1}{R_3R_4C_1C_2}}$$

Compare the numerator expression with the standard second-order expression.

$$s^2+\frac{\omega_0}{Q}s+\omega_0^2$$

Here, ω_0 is the center frequency and Q is the quality factor.

$$\omega_0^2=\frac{1}{R_3R_4C_1C_2}$$

$$\omega_0=\frac{1}{\sqrt{R_3R_4C_1C_2}} \quad \text{..... (3)}$$

Also,

$$\frac{\omega_0}{Q}=\frac{1}{R_3}\left(\frac{1}{C_1}+\frac{1}{C_2}\right)$$

$$Q=\frac{1}{\omega_0R_3}\left(\frac{1}{C_1}+\frac{1}{C_2}\right)$$

Substitute $\frac{1}{\sqrt{R_3R_4C_1C_2}}$ for ω_0 .

$$Q=\frac{1}{\frac{1}{\sqrt{R_3R_4C_1C_2}}R_3}\left(\frac{1}{C_1}+\frac{1}{C_2}\right)$$

$$Q=\frac{\sqrt{R_3R_4C_1C_2}}{R_3}\left(\frac{1}{C_1}+\frac{1}{C_2}\right) \quad \text{..... (4)}$$

Step 5 of 6

Consider equation (3) for the expression of resonant frequency ω_0 of the active-filter feedback loop generated using the bridged-T network.

$$\omega_0=\frac{1}{\sqrt{R_3R_4C_1C_2}}$$

Substitute 10^{-9} for C_1 and C_2 ; and 10^3 for ω_0 .

$$10^3=\frac{1}{\sqrt{R_3R_4\left(10^{-9}\right)\left(10^{-9}\right)}}$$

$$\sqrt{R_3R_4}=\frac{1}{10^3\times10^{-9}}$$

$$R_3R_4=10^6$$

Step 6 of 6

Consider equation (4) for the expression of the quality factor Q.

$$Q=\frac{\sqrt{R_3R_4C_1C_2}}{R_3}\left(\frac{1}{C_1}+\frac{1}{C_2}\right)$$

Substitute 10^{-9} for C_1 and C_2 ; and 10^6 for R_3R_4 , and $\frac{1}{\sqrt{2}}$ for Q.

$$\frac{1}{\sqrt{2}}=\frac{\sqrt{10^6\left(10^{-9}\right)\left(10^{-9}\right)}}{R_3}\left(\frac{1}{10^{-9}}+\frac{1}{10^{-9}}\right)$$

$$R_3=\sqrt{2}\left(10^3\right)\left(10^{-9}\right)\left(\frac{2}{10^{-9}}\right)$$

$$R_3=2\sqrt{2}\left(10^3\right)$$

$$R_3=14.142\ \text{k}\Omega$$

Determine the resistance R_4 .

Consider the following expression.

$$R_3R_4=10^6$$

Substitute 14.142×10^3 for R_3 .

$$14.142\left(10^3\right)R_4=10^6$$

$$R_4=\frac{10^6}{14.142\times10^3}$$

$$R_4=\frac{10^6}{14.142\times10^3}$$

$$R_4=7.07\ \text{k}\Omega$$

Therefore, the design parameters of the circuit are $\boxed{R_3=14.142\ \text{k}\Omega \text{ and } R_4=7.07\ \text{k}\Omega}$.

Step 1 of 5

Refer to Figure 17.28 in the textbook.

Redraw the circuit as shown in Figure 1.

Consider V is the voltage across the resistor R_4.

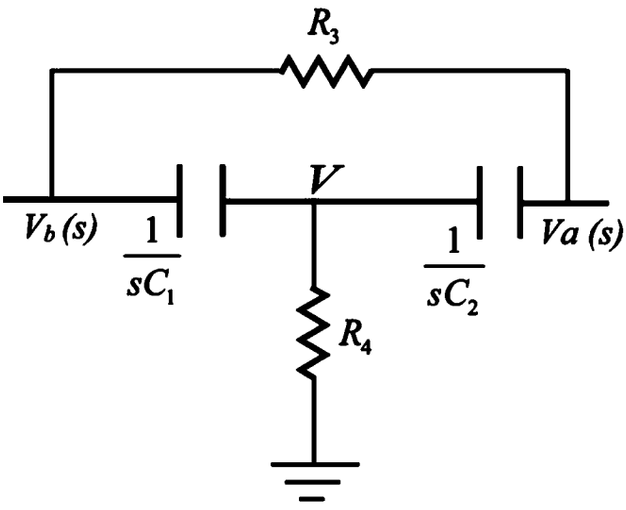


Figure 1

Step 2 of 5

Determine the transfer function.

Apply nodal analysis at node V_b.

$$\frac{(V_b(s)-V)}{1/sC_1}+\frac{(V_b(s)-V_a(s))}{R_3}=0$$

$$sC_1(V_b(s)-V)+\frac{(V_b(s)-V_a(s))}{R_3}=0$$

$$V_b(s)\left[sC_1+\frac{1}{R_3}\right]=\frac{V_a(s)}{R_3}+sC_1V$$

$$\left(\frac{V_b(s)}{sC_1}\right)\left[sC_1+\frac{1}{R_3}\right]-\frac{V_a(s)}{R_3}=V \quad \text{..... (1)}$$

Step 3 of 5

Apply nodal analysis at node V.

$$\frac{V-V_b(s)}{1/sC_1}+\frac{V}{R_4}+\frac{(V-V_a(s))}{sC_2}=0$$

$$sC_1(V-V_b(s))+\frac{V}{R_4}+sC_2(V-V_a(s))=0$$

$$sC_1V-sC_1V_b(s)+\frac{V}{R_4}+sC_2V-sC_2V_a(s)=0$$

$$\left[sC_1+\frac{1}{R_4}+sC_2\right]V-sC_2V_a(s)=sC_1V_b(s)$$

$$\left[1+\frac{1}{sC_1R_4}+\frac{C_2}{C_1}\right]V-\frac{C_2}{C_1}V_a(s)=V_b(s) \quad \text{..... (2)}$$

Step 4 of 5

Substitute equation 1 in equation 2.

$$\left[1+\frac{1}{sC_1R_4}+\frac{C_2}{C_1}\right]\left(\left(\frac{V_b(s)}{sC_1}\right)\left[sC_1+\frac{1}{R_3}\right]-\frac{V_a(s)}{R_3}\right)-\frac{C_2}{C_1}V_a(s)=V_b(s)$$

$$\left[1+\frac{1}{sC_1R_4}+\frac{C_2}{C_1}\right]\left(V_b(s)+\frac{V_b(s)}{sC_1R_3}-\frac{V_a(s)}{sC_1R_3}\right)-\frac{C_2}{C_1}V_a(s)=V_b(s)$$

$$V_b(s)+\frac{V_b(s)C_2}{C_1}+\frac{V_b(s)}{sR_4C_1}+\frac{V_b(s)}{sR_3C_1}+\frac{V_b(s)}{sR_3C_1^2}+\frac{V_b(s)C_2}{sR_3C_1^2}-\frac{V_a(s)C_2}{sR_3C_1^2}$$

$$-\frac{V_a(s)}{sR_4C_1}+\frac{V_b(s)}{s^2R_4R_3C_1^2}-\frac{V_a(s)}{s^2R_4R_3C_1^2}-\frac{C_2}{C_1}V_a(s)=V_b(s)$$

$$V_b(s)\left[1+\frac{V_b(s)C_2}{C_1}+\frac{V_b(s)}{sR_4C_1}+\frac{V_b(s)}{sR_3C_1}+\frac{V_b(s)C_2}{sR_3C_1^2}+\frac{V_b(s)}{s^2R_4R_3C_1^2}-1\right]$$

$$=V_a(s)\left[\frac{V_a(s)C_2}{sR_3C_1^2}+\frac{V_a(s)}{sR_4C_1}+\frac{V_a(s)}{s^2R_4R_3C_1^2}+\frac{C_2}{C_1}\right]$$

$$\left\{V_b(s)\left[\frac{s^2R_4R_3C_1C_2+s[R_3C_1+R_4C_1+R_4C_2]+1}{S^2R_4R_3C_1^2}\right]\right.$$

$$\left.=V_a(s)\left[\frac{sR_4C_1+sR_4C_2+1+s^2R_4R_3C_1C_1}{S^2R_4R_3C_1^2}\right]\right\}$$

$$\frac{V_b(s)}{V_a(s)}=\frac{\left[sR_4C_1+sR_4C_2+1+s^2R_4R_3C_1C_1\right]}{s^2R_4R_3C_1C_2+s[R_3C_1+R_4C_1+R_4C_2]+1}$$

Step 5 of 5

Further simplification as follows,

$$\frac{V_b(s)}{V_a(s)}=\frac{R_4R_3C_2C_1\left[s^2+s\left(\frac{R_4(C_1+C_2)}{R_4R_3C_1C_2}\right)+\frac{1}{R_4R_3C_2C_1}\right]}{\left[s^2+s\left(\frac{[R_3C_1+R_4C_1+R_4C_2]}{R_4R_3C_1C_2}\right)+\frac{1}{R_4R_3C_1C_2}\right]R_4R_3C_1C_2}$$

$$\frac{V_b(s)}{V_a(s)}=\frac{\left[s^2+s\left(\frac{1}{C_1}+\frac{1}{C_2}\right)\frac{1}{R_3}+\frac{1}{R_4R_3C_2C_1}\right]}{s^2+s\left(\frac{1}{C_1R_3}+\frac{1}{C_2R_3}+\frac{1}{C_1R_4}\right)+\frac{1}{R_4R_3C_1C_2}}$$

Therefore, the transfer function of the bridged-T network is,

$$\boxed{\frac{V_b(s)}{V_a(s)}=\frac{\left[s^2+s\left(\frac{1}{C_1}+\frac{1}{C_2}\right)\frac{1}{R_3}+\frac{1}{R_4R_3C_2C_1}\right]}{s^2+s\left(\frac{1}{C_1R_3}+\frac{1}{C_2R_3}+\frac{1}{C_1R_4}\right)+\frac{1}{R_4R_3C_1C_2}}}$$

Step 1 of 4

17.070P

Refer to Figure 16.28 (a) in the textbook for the RC network (the bridged-T network).

Write the transfer function of the bridged-T network.

$$T(s)=\frac{s^2+\frac{1}{R_3}\left(\frac{1}{C_1}+\frac{1}{C_2}\right)s+\frac{1}{C_1C_2R_3R_4}}{s^2+\left(\frac{1}{R_3C_1}+\frac{1}{C_2R_3}+\frac{1}{C_1R_4}\right)s+\frac{1}{C_1C_2R_3R_4}}$$

The element values are, $R_3=R_4=R$ and $C_1=C_2=C$ and $CR=\tau$.

Substitute the corresponding expressions.

$$T(s)=\frac{s^2+s\left(\frac{1}{C}+\frac{1}{C}\right)\frac{1}{R}+\frac{1}{C^2R^2}}{s^2+s\left(\frac{1}{CR}+\frac{1}{CR}+\frac{1}{CR}\right)+\frac{1}{C^2R^2}}$$
$$=\frac{s^2+s\frac{2}{CR}+\frac{1}{C^2R^2}}{s^2+s\frac{3}{CR}+\frac{1}{C^2R^2}}$$

Substitute CR for τ.

$$T(s)=\frac{s^2+s\frac{2}{\tau}+\frac{1}{\tau^2}}{s^2+s\frac{3}{\tau}+\frac{1}{\tau^2}}$$

Step 2 of 4

The numerator term is used to find the zeros and denominator is used to find poles.

The numerator is,

$$s^2+s\frac{2}{\tau}+\frac{1}{\tau^2}=0$$
$$\left(s+\frac{1}{\tau}\right)^2=0$$
$$s_{1,2}=-\frac{1}{\tau}$$

Therefore, the zeros of the bridged T network are, $\boxed{-\frac{1}{\tau}\text{ and }-\frac{1}{\tau}}$.

Step 3 of 4

The denominator is,

$$s^2+s\frac{3}{\tau}+\frac{1}{\tau^2}=0$$

Calculate the poles.

$$s_{1,2}=\frac{-\frac{3}{\tau}\pm\sqrt{\frac{9}{\tau^2}-4(1)\left(\frac{1}{\tau^2}\right)}}{2(1)}$$
$$=\frac{-\frac{3}{\tau}\pm\sqrt{\frac{9}{\tau^2}-\frac{4}{\tau^2}}}{2}$$
$$=\frac{-\frac{3}{\tau}\pm\frac{\sqrt{5}}{\tau}}{2}$$
$$=-\frac{3}{2\tau}\pm\frac{\sqrt{5}}{2\tau}$$

Thus, the poles of the bridged T-network are, $\boxed{\frac{-3+\sqrt{5}}{2\tau}\text{ and }\frac{-3-\sqrt{5}}{2\tau}}$.

Step 4 of 4

If the network is placed in the negative feedback path of an ideal infinite-gain operational amplifier, then the poles are given by the zeros of the bridged T-network.

$$s^2+s\frac{2}{\tau}+\frac{1}{\tau^2}=0$$
$$\left(s+\frac{1}{\tau}\right)^2=0$$
$$s_{1,2}=-\frac{1}{\tau}$$

Therefore, the zeros of the bridged T network are, $\boxed{-\frac{1}{\tau}\text{ and }-\frac{1}{\tau}}$.

Step 1 of 11

Refer to Figure 17-28 (b) in the textbook.

Draw the circuit as shown in Figure 1.

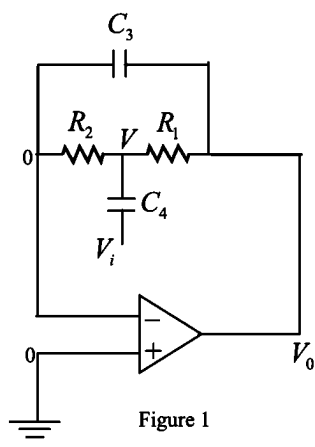


Figure 1

Step 2 of 11

Consider $R_1 = R_2 = R$ and $C_4 = C$. $C_3 = \frac{C}{36}$.

Determine the transfer function $\frac{V_0(s)}{V_i(s)}$.

Apply Laplace transform and draw the Figure 2.

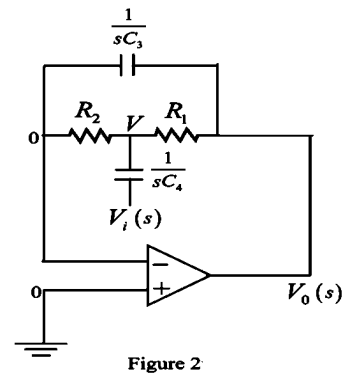


Figure 2

Step 3 of 11

Apply Kirchhoff current law at node V .

$$\frac{V-0}{R_2}+\frac{V-V_i(s)}{\left(\frac{1}{sC_4}\right)}+\frac{V-V_0(s)}{R_1}=0 \quad \text{..... (1)}$$

Apply Kirchhoff current law at node 0.

$$\frac{0-V}{R_2}+\frac{0-V_0(s)}{\left(\frac{1}{sC_3}\right)}+0$$

$$\frac{-V}{R_2}-\frac{V_0(s)}{\left(\frac{1}{sC_3}\right)}=0$$

$$sC_3V_0(s)=\frac{-V}{R}$$

$$V=-sR_2C_3V_0(s) \quad \text{..... (2)}$$

Step 4 of 11

Recall the equation (1) as follows,

$$\frac{V-0}{R_2}+\frac{V-V_i(s)}{\left(\frac{1}{sC_4}\right)}+\frac{V-V_0(s)}{R_1}=0$$

Substitute $-sR_2C_3V_0(s)$ and V in the equation..

$$\frac{\left(-sR_2C_3V_0(s)\right)-0}{R_2}+\frac{\left(-sR_2C_3V_0(s)\right)-V_i(s)}{\left(\frac{1}{sC_4}\right)}+\frac{\left(-sR_2C_3V_0(s)\right)-V_0(s)}{R_1}=0$$

$$-sC_3V_0(s)-s^2R_2C_3C_4V_0(s)-sC_4V_i(s)-\frac{sR_2C_3V_0(s)}{R_1}-\frac{V_0(s)}{R_1}=0$$

$$sC_3V_0(s)+s^2R_2C_3C_4V_0(s)+\frac{sR_2C_3V_0(s)}{R_1}+\frac{V_0(s)}{R_1}=-sC_4V_i(s)$$

$$sC_3V_0(s)+s^2R_2C_3C_4V_0(s)+\frac{sR_2C_3V_0(s)}{R_1}+\frac{V_0(s)}{R_1}=-sC_4V_i(s)$$

$$V_0(s)\left[sC_3+s^2R_2C_3C_4+\frac{sR_2C_3}{R_1}+\frac{1}{R_1}\right]=-sC_4V_i(s)$$

$$\frac{V_0(s)}{V_i(s)}=\frac{sC_4}{\left[sC_3+s^2R_2C_3C_4+\frac{sR_2C_3}{R_1}+\frac{1}{R_1}\right]}$$

$$\frac{V_0(s)}{V_i(s)}=-\frac{sC_4R_1^2}{sR_1C_3+s^2R_1R_2C_3C_4+sR_2C_3+1}$$

Step 5 of 11

Consider $R_1 = R_2 = R$ and $C_4 = C$. $C_3 = \frac{C}{36}$.

$$\frac{V_0(s)}{V_i(s)}=-\frac{sCR^2}{sR(C/36)+s^2R^2(C/36)C+sR(C/36)+1}$$

$$=-\frac{36sCR^2}{sRC+s^2R^2C^2+sRC+36}$$

$$=-\frac{36sCR^2}{s^2R^2C^2+2RCs+36}$$

$$=-\frac{\left(\frac{36}{C}\right)s}{s^2+\left(\frac{2}{RC}\right)s+\frac{36}{R^2C^2}}$$

Therefore, the transfer function is $\boxed{\frac{V_0(s)}{V_i(s)}=-\frac{\left(\frac{36}{C}\right)s}{s^2+\left(\frac{2}{RC}\right)s+\frac{36}{R^2C^2}}}$.

Step 6 of 11

Determine the gain at $s=0$ in the transfer function.

$$\left|\frac{V_0(s)}{V_i(s)}\right|=\left(\frac{\left(\frac{36}{C}\right)s}{s^2+\left(\frac{2}{RC}\right)s+\frac{36}{R^2C^2}}\right)$$

$$=\left(\frac{\left(\frac{36}{C}\right)(0)}{(0)^2+\left(\frac{2}{RC}\right)(0)+\frac{36}{R^2C^2}}\right)$$

$$=0$$

Therefore, the gain at $s=0$ is $\boxed{0}$.

Step 7 of 11

Determine the gain at $s=\infty$ in the transfer function.

$$\left|\frac{V_0(s)}{V_i(s)}\right|=\frac{\left(\frac{36}{C}\right)s}{s^2+\left(\frac{2}{RC}\right)s+\frac{36}{R^2C^2}}$$

$$=\frac{\left(\frac{36}{C}\right)s}{s^2\left[1+\frac{\left(\frac{2}{RC}\right)}{s}+\frac{\frac{36}{R^2C^2}}{s^2}\right]}$$

$$=\frac{\left(\frac{36}{C}\right)}{s\left[1+\frac{\left(\frac{2}{RC}\right)}{s}+\frac{\frac{36}{R^2C^2}}{s^2}\right]}$$

$$=\frac{\left(\frac{36}{C}\right)}{(\infty)\left[1+\frac{\left(\frac{2}{RC}\right)}{(\infty)}+\frac{\frac{36}{R^2C^2}}{(\infty)}\right]}$$

$$\left|\frac{V_0(s)}{V_i(s)}\right|=\frac{\left(\frac{36}{C}\right)}{(\infty)[1+0+0]}$$

$$=0$$

Therefore, the gain at $s=\infty$ is $\boxed{0}$.

Therefore, the gain at $s=0$ is zero and the gain at $s=\infty$ is zero.

Therefore, the given circuit is band pass filter.

Step 8 of 11

Consider the transfer function.

$$\left|\frac{V_0(s)}{V_i(s)}\right|=\frac{\left(\frac{36}{C}\right)s}{s^2+\left(\frac{2}{RC}\right)s+\frac{36}{R^2C^2}} \quad \text{..... (3)}$$

Compare equation (3) with standard transfer function of band pass filter.

$$\frac{a_1s}{s^2+s\left(\frac{\omega_b}{Q}\right)+\omega_b^2}=\frac{\left(\frac{36}{C}\right)s}{s^2+\left(\frac{2}{RC}\right)s+\frac{36}{R^2C^2}}$$

Therefore,

$$\omega_b^2=\frac{36}{R^2C^2}$$

$$a_1=\frac{36}{C}$$

$$\frac{\omega_b}{Q}=\frac{2}{RC}$$

Step 9 of 11

From the transfer function, the resonant frequency ω_b is,

$$\omega_b^2=\frac{36}{R^2C^2}$$

$$\omega_b=\sqrt{\frac{36}{R^2C^2}}$$

$$\omega_b=\frac{6}{RC}$$

Therefore, the value of resonant frequency is, $\boxed{\omega_b=\frac{6}{RC}}$.

Step 10 of 11

Determine the quality factor.

$$\frac{\omega_b}{Q}=\frac{2}{RC}$$

$$\frac{(6/RC)}{Q}=\frac{2}{RC}$$

$$\frac{(6/RC)}{\left(\frac{2}{RC}\right)}=Q$$

$$Q=3$$

Thus, the quality factor of the band pass filter is, $\boxed{3}$

Step 11 of 11

Determine the center frequency gain of the bridged-T network.

$$|T(s)|=\left(\frac{\left(\frac{36}{C}\right)s}{s^2+\left(\frac{2}{RC}\right)s+\frac{36}{R^2C^2}}\right)$$

$$=-\frac{6\frac{6}{RC}s}{s^2+\frac{1}{3}\frac{6}{RC}s+\left(\frac{6}{RC}\right)^2}$$

$$=-\frac{6\omega_0 s}{s^2+\frac{1}{3}\omega_0 s+(\omega_0)^2}$$

Therefore, the center frequency gain is,

$$|T(j\omega_0)|=\frac{6\omega_0|j\omega|}{\left|-(\omega_0)^2+j\frac{\omega_0^2}{3}+\omega_0^2\right|}$$

$$=\frac{6\omega_0^2}{\frac{\omega_0^2}{3}}$$

$$=18$$

Thus, the center frequency gain of the bridged-T network is, $\boxed{18\ V/V}$.

Step 1 of 6

Refer to Figure 17.30 (b) in the textbook.

Consider the transfer function.

$$\frac{V_0}{V_i}=\frac{-s\left(\alpha/C_1R_4\right)}{s^2+s\left(\frac{1}{C_1}+\frac{1}{C_1}\right)\frac{1}{R_3}+\frac{1}{C_1C_2R_3R_4}} \quad \text{..... (1)}$$

Compare the equation 1 with the standard second order equation.

$$s^2+s\left(\frac{\omega_0}{Q}\right)+\omega_0^2$$

$$\omega_0^2=\frac{1}{C_1C_2R_3R_4}$$

$$\left(\frac{\omega_0}{Q}\right)=\left(\frac{1}{C_1}+\frac{1}{C_1}\right)\frac{1}{R_3}$$

Step 2 of 6

Consider $\alpha=1$.

Consider the center frequency of the bandpass filter is equal to the pole frequency ω_0 .

Consider the center frequency of the band pass filter $\omega_0=10\ \text{KHz}$.

Consider the bandwidth of the band pass filter is $2\ \text{KHz}$.

$$BW=\frac{\omega_0}{Q}$$

$$\left(\frac{1}{C_1}+\frac{1}{C_1}\right)\frac{1}{R_3}=\left(2\times10^3\right)$$

Consider $C_1=C_2=10\ \text{nF}$.

$$\left(\frac{1}{10}+\frac{1}{10}\right)\frac{1}{R_3}=\left(2\times10^3\right)$$

$$\frac{1}{\left(5\times10^{-9}\right)R_3}=\left(2\times10^3\right)$$

$$R_3=\frac{1}{\left(5\times10^{-9}\right)\left(2\times10^3\right)}$$

$$R_3=\frac{1}{\left(5\times10^{-9}\right)\left(2\times10^3\right)}$$

$$R_3=100\ \text{k}\Omega$$

Therefore, the value of R_3 is $\boxed{100\ \text{k}\Omega}$.

Step 3 of 6

Consider $\omega_0^2=\frac{1}{C_1C_2R_3R_4}$.

$$\omega_0^2=\frac{1}{C_1C_2R_3R_4}$$

$$\left(10\times10^3\right)^2=\frac{1}{C_1C_2R_3R_4}$$

$$\frac{1}{C_1C_2R_3R_4}=10^8$$

Consider $C_1=C_2=10\ \text{nF}$.

$$\frac{1}{\left(10\times10^{-9}\right)\left(10\times10^{-9}\right)R_3R_4}=10^8$$

$$\frac{1}{\left(10\times10^{-9}\right)\left(10\times10^{-9}\right)10^8}=R_3R_4$$

$$R_3R_4=10^8$$

Step 4 of 6

Consider $R_3=100\ \text{k}\Omega$.

$$R_3R_4=10^8$$

$$\left(10^5\right)R_4=10^8$$

$$R_4=\frac{10^8}{10^5}$$

$$R_4=10^3$$

Therefore, the value of the resistance R_4 is $\boxed{1\ \text{k}\Omega}$.

Step 5 of 6

Consider $R_2=10^5\,\Omega$, $R_4=10^3\,\Omega$.

Calculate the value of m .

$$m=\frac{R_2}{R_4}$$

$$=\frac{10^5}{10^3}$$

$$=100$$

Determine the quality factor Q .

$$m=4Q^2$$

$$Q=\sqrt{\frac{m}{4}}$$

$$Q=\sqrt{\frac{100}{4}}$$

$$Q=5$$

Step 6 of 6

Determine the center frequency gain.

$$\text{Center-frequency gain}=\frac{a_1Q}{\omega_0}$$

$$a_1=-\frac{\alpha}{C_1R_4}$$

Consider $R_4=10^3\,\Omega$, $C_1=10\ \text{nF}$, and $\alpha=1$.

$$a_1=-\frac{\alpha}{C_1R_4}$$

$$=-\frac{1}{\left(10\times10^{-9}\right)\left(10^3\right)}$$

$$=-10^5$$

Therefore, the values of design parameters is $\boxed{C_1=C_2=10\ \text{nF}}$, $\boxed{R_3=10^5\ \text{and}\ R_4=10^3}$ and the center –frequency gain is $\boxed{-10^5}$.

17.073P

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Step 1 of 9

Refer to Figure 17.33(b) in the textbook.

Redraw the circuit as shown in Figure .

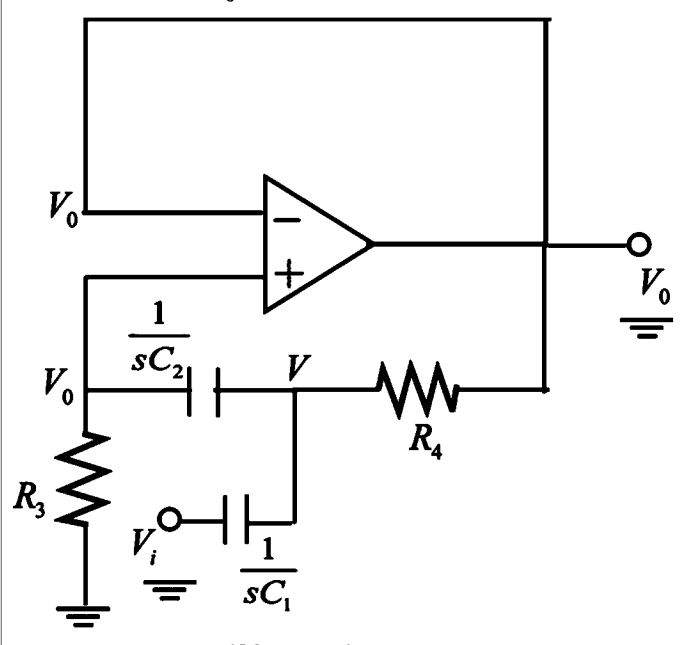


Figure 1

Step 2 of 9

Apply Kirchhoff current law at node V_0 .

$$\frac{0-V_0(s)}{R_3}+\frac{V_0(s)-V}{(1/sC_2)}=0$$

$$\frac{V_0(s)}{R_3}+sC_2\left(V_0(s)-V\right)=0$$

$$\frac{V_0(s)}{R_3}+sC_2V_0(s)=sC_2V$$

$$V_0(s)\left[1+\frac{1}{sC_2R_3}\right]=V$$

Step 3 of 9

Apply Kirchhoff current law at node V .

$$\frac{V-V_0(s)}{(1/sC_2)}+\frac{V-V_i(s)}{(1/sC_1)}+\frac{V-V_0(s)}{R_4}=0$$

$$sC_2V-sC_2V_0(s)+sC_1V-sC_1V_i(s)+\frac{V}{R_4}=\frac{V_0(s)}{R_4}$$

$$V\left[sC_2+sC_1+\frac{1}{R_4}\right]-\left[sC_2+\frac{1}{R_4}\right]V_0(s)=sC_1V_i(s) \quad \text{..... (1)}$$

Substitute $V=V_0(s)\left[1+\frac{1}{sC_2R_3}\right]$ in equation 1.

$$V_0(s)\left[1+\frac{1}{sC_2R_3}\right]\left[sC_2+sC_1+\frac{1}{R_4}\right]-\left[sC_2+\frac{1}{R_4}\right]V_0(s)=sC_1V_i(s)$$

$$V_0(s)\left[sC_2+sC_1+\frac{1}{R_4}+\frac{1}{R_3}+\frac{C_1}{C_2R_3}+\frac{1}{sC_2R_3R_4}-\left(sC_2+\frac{1}{R_4}\right)\right]=sC_1V_i(s)$$

$$\frac{V_0(s)}{V_i(s)}=\frac{sC_1}{\left[sC_2+sC+\frac{1}{R_4}+\frac{1}{R_3}+\frac{C_1}{C_2R_3}+\frac{1}{sC_2R_3R_4}-sC_2-\frac{1}{R_4}\right]}$$

Step 4 of 9

$$\frac{V_0(s)}{V_i(s)}=\frac{sC_1}{\left[sC_2+sC_1+\frac{1}{R_4}+\frac{1}{R_3}+\frac{C_1}{C_2R_3}+\frac{1}{sC_2R_3R_4}-sC_2-\frac{1}{R_4}\right]}$$

$$=\frac{S^2\left(R_3R_4C_1C_2\right)}{S^2R_3R_4C_2\left(C_1+C_2\right)+sR_3C_2+sR_4C_2+sR_4C_1+1-S^2R_3R_4C_2-sR_3C_2}$$

$$=\frac{S^2\left(R_3R_4C_1C_2\right)}{S^2R_3R_4C_1C_2+sR_4\left(C_1+C_2\right)+1}$$

$$=\frac{S^2}{S^2+s\left(\frac{R_4\left(C_1+C_2\right)}{\left(R_3R_4C_1C_2\right)}\right)+\frac{1}{\left(R_3R_4C_1C_2\right)}}$$

Therefore, the transfer function is $\boxed{\frac{S^2}{S^2+s\left(\frac{R_4\left(C_1+C_2\right)}{\left(R_3R_4C_1C_2\right)}\right)+\frac{1}{\left(R_3R_4C_1C_2\right)}}}$.

Step 5 of 9

Consider the transfer function.

$$\frac{V_0(s)}{V_i(s)}=\frac{S^2}{S^2+s\left(\frac{R_4\left(C_1+C_2\right)}{\left(R_3R_4C_1C_2\right)}\right)+\frac{1}{\left(R_3R_4C_1C_2\right)}}$$

Substitute $s=0$ in the transfer function.

$$\frac{V_0(s)}{V_i(s)}=\frac{S^2}{S^2+s\left(\frac{R_4\left(C_1+C_2\right)}{\left(R_3R_4C_1C_2\right)}\right)+\frac{1}{\left(R_3R_4C_1C_2\right)}}$$

$$=\frac{(0)^2}{(0)^2+(0)\left(\frac{R_4\left(C_1+C_2\right)}{\left(R_3R_4C_1C_2\right)}\right)+\frac{1}{\left(R_3R_4C_1C_2\right)}}$$

$$=0$$

Step 6 of 9

Substitute $s=\infty$ in the transfer function.

$$\frac{V_0(s)}{V_i(s)}=\frac{S^2}{S^2+s\left(\frac{R_4\left(C_1+C_2\right)}{\left(R_3R_4C_1C_2\right)}\right)+\frac{1}{\left(R_3R_4C_1C_2\right)}}$$

$$=\frac{1}{1+\frac{\left(\frac{R_4\left(C_1+C_2\right)}{\left(R_3R_4C_1C_2\right)}\right)}{s}+\frac{\frac{1}{\left(R_3R_4C_1C_2\right)}}{s^2}}$$

$$=\frac{1}{1+\frac{\left(\frac{R_4\left(C_1+C_2\right)}{\left(R_3R_4C_1C_2\right)}\right)}{\infty}+\frac{\frac{1}{\left(R_3R_4C_1C_2\right)}}{\infty}}$$

$$=1$$

For $s=0$, the gain is 0 and for $s=\infty$, the gain is 1.

Therefore, the circuit is $\boxed{\text{high pass filter}}$.

Therefore, the high frequency gain is $\boxed{1}$.

Step 7 of 9

To design the circuit for flat response, determine the values of R_3 and R_4 .

Consider the transfer function.

$$\frac{V_0(s)}{V_i(s)}=\frac{S^2}{S^2+s\left(\frac{R_4\left(C_1+C_2\right)}{\left(R_3R_4C_1C_2\right)}\right)+\frac{1}{\left(R_3R_4C_1C_2\right)}}$$

Compare the transfer function with the standard second order equation.

$$S^2+2\xi\omega_0s+\omega_0{}^2=S^2+s\left(\frac{R_4\left(C_1+C_2\right)}{\left(R_3R_4C_1C_2\right)}\right)+\frac{1}{\left(R_3R_4C_1C_2\right)}$$

$$S^2+\left(\frac{\omega_0}{Q}\right)s+\omega_0{}^2=S^2+s\left(\frac{R_4\left(C_1+C_2\right)}{\left(R_3R_4C_1C_2\right)}\right)+\frac{1}{\left(R_3R_4C_1C_2\right)}$$

Step 8 of 9

Compare the equations.

$$S^2+\left(\frac{\omega_0}{Q}\right)s+\omega_0{}^2=S^2+s\left(\frac{R_4\left(C_1+C_2\right)}{\left(R_3R_4C_1C_2\right)}\right)+\frac{1}{\left(R_3R_4C_1C_2\right)}$$

$$\left(\frac{\omega_0}{Q}\right)=\left(\frac{\left(C_1+C_2\right)}{\left(R_3C_1C_2\right)}\right) \quad \text{..... (2)}$$

$$\omega_0=\frac{1}{\sqrt{\left(R_3R_4C_1C_2\right)}} \quad \text{..... (3)}$$

Substitute $10^4\ \text{rad/s}$ for ω_0 , $10\ \text{nF}$ for C_1, C_2 , and $\frac{1}{\sqrt{2}}$ for Q in equation 2.

$$\left(\frac{\omega_0}{Q}\right)=\left(\frac{\left(C_1+C_2\right)}{\left(R_3C_1C_2\right)}\right)$$

$$\frac{10^4}{\left(1/\sqrt{2}\right)}=\frac{\left(10\times10^{-9}\right)+\left(10\times10^{-9}\right)}{R_3\left(10\times10^{-9}\right)\left(10\times10^{-9}\right)}$$

$$14142.13=\frac{0.2}{R_3}$$

$$R_3=14.142\ \text{k}\Omega$$

Step 9 of 9

Substitute $10^4\ \text{rad/s}$ for ω_0 , $10\ \text{nF}$ for C_1, C_2 , and $\frac{1}{\sqrt{2}}$ for Q in equation 2.

$$\omega_0{}^2=\frac{1}{\left(R_3R_4C_1C_2\right)}$$

$$10^4=\frac{1}{\left(R_3R_4C_1C_2\right)}$$

$$10^8=\frac{1}{\left(R_3R_4C_1C_2\right)}$$

$$R_3=\frac{1}{\left(10^8R_4C_1C_2\right)}$$

$$R_4=\frac{1}{\left(10^8\left(14.142\times10^3\right)\left(10\times10^{-9}\right)\left(10\times10^{-9}\right)\right)}$$

$$=7.07\ \text{k}\Omega$$

Therefore, the design parameters are $\boxed{R_3=14.142\ \text{k}\Omega}$ and $\boxed{R_4=7.07\ \text{k}\Omega}$.

Step 1 of 5

Refer to Figure 17.34(c) in the textbook.

The filter is designed by cascading Sallen-and-Key filter with the first-order section.

The value of ε is,

$$\begin{aligned}\varepsilon &= \left(10^{\frac{A_{max}}{10}} - 1\right)^{-\frac{1}{2}} \\ &= \left(10^{\frac{3}{10}} - 1\right)^{-\frac{1}{2}} \quad (\text{since } A_{max} = 3 \text{ dB}) \\ &= 1\end{aligned}$$

Determine the resonant frequency ω_0.

$$\omega_0 = \omega_p \left(\frac{1}{\varepsilon}\right)^{\frac{1}{N}}$$

Replace 10 kHz for f_p, 1 for ε, and 5 for N in the equation.

$$\begin{aligned}\omega_0 &= 2\pi f_p \left(\frac{1}{\varepsilon}\right)^{\frac{1}{N}} \\ &= 2\pi\left(10\times10^3\right)\left(\frac{1}{1}\right)^{\frac{1}{5}} \\ &= \left(20\times10^3\right)\pi \text{ rad/sec}\end{aligned}$$

Step 2 of 5

For the first order section, the resonant frequency is, $\omega_0 = \left(20\times10^3\right)\pi \text{ rad/sec}$ and the dc gain is unity.

Refer to Figure 17.13 (a) in the textbook for the low pass first order circuit.

Write the expression for the resonant frequency.

$$\omega_0 = \frac{1}{CR_2}$$

Substitute $\left(20\times10^3\right)\pi \text{ rad/sec}$ for ω_0, $10 \text{ k}\Omega$ for R_2 in the equation.

$$\begin{aligned}\left(20\times10^3\right)\pi &= \frac{1}{C\left(10\times10^3\right)} \\ C &= \frac{1}{\left(\left(20\times10^3\right)\pi\right)\left(10\times10^3\right)} \\ C &= 1.59 \text{ nF}\end{aligned}$$

Thus, the component values first order low pass filter are, $\boxed{R_2 = 10 \text{ k}\Omega}$ and $\boxed{C = 1.59 \text{ nF}}$.

Step 3 of 5

Refer to Figure 17.10 (a) for graphical construction for determining the poles of butter-worth filter of order *N*.

The quality factor of the filter is,

$$\begin{aligned}\frac{\omega_0}{2Q} &= \omega_0 \cos\left(\frac{\pi}{N}\right) \\ \frac{1}{2Q} &= \cos\left(\frac{180^\circ}{5}\right) \quad (\text{since } N = 5) \\ Q &= \frac{1}{2(\cos 36^\circ)} \\ Q &= 0.618\end{aligned}$$

Step 4 of 5

Calculate the value of m for the first order section.

$$m = 4Q^2$$

Substitute 0.618 for Q in the equation.

$$\begin{aligned}m &= 4(0.618)^2 \\ &= 1.528\end{aligned}$$

The resistor values are $R_1 = R_2 = R = 10 \text{ k}\Omega$.

Determine the value of the capacitance C.

$$C = \frac{2Q}{\omega_0 R}$$

Substitute $\left(20\times10^3\right)\pi \text{ rad/sec}$, $10 \text{ k}\Omega$ for R, and 0.618 for Q

$$\begin{aligned}C &= \frac{2(0.618)}{\left(\left(20\times10^3\right)\pi\right)\left(10\times10^3\right)} \\ &= 1.97 \text{ nF}\end{aligned}$$

From the circuit,

$C_4 = C = 1.97 \text{ nF}$.

Step 5 of 5

Calculate the capacitance, C_3.

$$\begin{aligned}C_3 &= \frac{C}{m} \\ &= \frac{1.97 \text{ nF}}{1.528} \\ &= 1.29 \text{ nF}\end{aligned}$$

Thus, the component values of a cascade connection of two Sallen-and-Key circuits are,

$$\boxed{\begin{array}{l} R_1 = R_2 = R = 10 \text{ k}\Omega, \\ C_4 = C = 1.97 \text{ nF and} \\ C_3 = 1.29 \text{ nF} \end{array}}$$

Step 1 of 2

17.076P

Refer to Figure 16.31 in the textbook for the linear network. The circuit in Figure 16.31 reveals the interchanging input and ground results in a compliment of the transfer function.

From the Figure 16.31 (a), the transfer function from b to a with c grounded is,

$$\frac{V_a}{V_b} = t$$

From the Figure 16.31 (b), the transfer function from c to a with b grounded is,

$$\frac{V_a}{V_c} = 1 - t$$

Here, t is the transfer function of the network.

Interchanging input and ground results in the complement of the transfer function. The linear network with input and ground interchanged is shown in Figure 16.31 (b) in the textbook.

Refer to Figure 16.18 (d) for the band-pass filter with RLC components.

If the linear network in the Figure 16.31 (a) is band-pass filter, then the transfer function of a band pass filter is,

$$T(s) = \frac{\frac{1}{RC}s}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}$$

$$T(s) = \frac{\frac{\omega_0}{Q}s}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2} \quad \text{..... (1)}$$

Here, $\omega_0 = \frac{1}{\sqrt{LC}}$ and $\frac{\omega_0}{Q} = \frac{1}{RC}$

Step 2 of 2

The center frequency gain is 1. The complementary function is obtained by substituting

$t' = 1 - t$.

Therefore,

$$t' = 1 - \frac{\frac{\omega_0}{Q}s}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

$$= \frac{s^2 + \frac{\omega_0}{Q}s + \omega_0^2 - \frac{\omega_0}{Q}s}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

$$t' = \frac{s^2 + \omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2} \quad \text{..... (2)}$$

The obtained complemented transfer function t' is same as the transfer function of the notch filter in the Figure 16.18 (e) in the textbook.

Interchanging of the input voltage and ground in the band pass filter results the notch filter as shown in Figure 16.18 (e).

The complement of the realization of second-order Band pass filter circuit is shown in Figure 16.18 (d) results a notch filter as shown in Figure 16.18 (e). Therefore, from equations (1) and (2), it is clear that complement of a band pass filter is a notch filter.

Thus, the complement of a band pass filter is a notch filter.

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Step 1 of 5

(a)

Consider the function, $y = uv$.

The sensitivity of y to x is,

$$\begin{aligned} S_x^y &= \frac{\partial y}{\partial x}\frac{x}{y} \\ &= \frac{\partial (uv)}{\partial x}\frac{x}{y} \\ &= \left(u\frac{\partial v}{\partial x} + v\frac{\partial u}{\partial x}\right)\frac{x}{uv} \\ &= \frac{x}{v}\frac{\partial v}{\partial x} + \frac{x}{u}\frac{\partial u}{\partial x} \end{aligned}$$

Consider the following identities.

$$S_x^v = \frac{x}{v}\frac{\partial v}{\partial x}$$

$$S_x^u = \frac{x}{u}\frac{\partial u}{\partial x}$$

Therefore, the sensitivity of y to x is,

$$\begin{aligned} S_x^y &= \frac{x}{v}\frac{\partial v}{\partial x} + \frac{x}{u}\frac{\partial u}{\partial x} \\ &= S_x^v + S_x^u \end{aligned}$$

Thus, the identity $\boxed{S_x^y = S_x^v + S_x^u}$ is verified.

Step 2 of 5

(b)

The function is, $y = \frac{u}{v}$.

The sensitivity of y to x is,

$$\begin{aligned} S_x^y &= \frac{\partial y}{\partial x}\frac{x}{y} \\ &= \frac{\partial\left(\frac{u}{v}\right)}{\partial x}\frac{x}{y} \\ &= \left(-\frac{u}{v^2}\frac{\partial v}{\partial x} + \frac{1}{v}\frac{\partial u}{\partial x}\right)\frac{x}{\frac{u}{v}} \\ &= -\frac{x}{v}\frac{\partial v}{\partial x} + \frac{x}{u}\frac{\partial u}{\partial x} \end{aligned}$$

Consider the following identities.

$$S_x^v = \frac{x}{v}\frac{\partial v}{\partial x}$$

$$S_x^u = \frac{x}{u}\frac{\partial u}{\partial x}$$

Therefore, the sensitivity of y to x is,

$$S_x^y = S_x^u - S_x^v$$

Thus, the identity $\boxed{S_x^y = S_x^u - S_x^v}$ is verified.

Step 3 of 5

(c)

The function is, $y = ku$.

Here k is a constant.

The sensitivity of y to x is,

$$\begin{aligned} S_x^y &= \frac{\partial y}{\partial x}\frac{x}{y} \\ &= \left(k\frac{\partial u}{\partial x}\right)\frac{x}{y} \\ &= \left(k\frac{\partial u}{\partial x}\right)\frac{x}{ku} \\ &= \frac{x}{u}\frac{\partial u}{\partial x} \end{aligned}$$

Consider the following identity.

$$S_x^u = \frac{x}{u}\frac{\partial u}{\partial x}$$

Therefore, the sensitivity of y to x is,

$$S_x^y = S_x^u$$

Thus, the identity $\boxed{S_x^y = S_x^u}$ is verified.

Step 4 of 5

(d)

The function is, $y = u^n$.

Here n is a constant.

The sensitivity of y to x is,

$$\begin{aligned} S_x^y &= \frac{\partial y}{\partial x}\frac{x}{y} \\ &= \left(nu^{n-1}\frac{\partial u}{\partial x}\right)\frac{x}{y} \\ &= \left(nu^{n-1}\frac{\partial u}{\partial x}\right)\frac{x}{u^n} \\ &= n\frac{x}{u}\frac{\partial u}{\partial x} \end{aligned}$$

Consider the following identity.

$$S_x^u = \frac{x}{u}\frac{\partial u}{\partial x}$$

Therefore, the sensitivity of y to x is,

$$S_x^y = nS_x^u$$

Thus, the identity $\boxed{S_x^y = nS_x^u}$ is verified.

Step 5 of 5

(e)

The function is,

$$y = f_1(u)$$
$$u = f_2(x)$$

The sensitivity of y to x is,

$$\begin{aligned} S_x^y &= \frac{\partial y}{\partial x}\frac{x}{y} \\ &= \left(\frac{\partial f_1}{\partial x} \times \frac{\partial u}{\partial x}\right)\frac{x}{y} \\ &= \left(\frac{\partial f_1}{\partial x} \times \frac{\partial f_2}{\partial x}\right)\frac{x}{f_1}\frac{u}{u} \\ &= \left(\frac{\partial f_1}{\partial x} \times \frac{\partial f_2}{\partial x}\right)\frac{x}{f_1}\frac{u}{f_2} \end{aligned}$$

Rearrange the terms.

$$\begin{aligned} S_x^y &= \left(\frac{x}{f_2}\frac{\partial f_1}{\partial x}\right)\left(\frac{u}{f_1}\frac{\partial f_2}{\partial x}\right) \\ &= S_u^{f_1} S_x^{f_2} \\ &= S_u^y S_x^u \end{aligned}$$

Thus, the identity $\boxed{S_x^y = S_u^y S_x^u}$ is verified.

Step 1 of 3

Refer to Figure 16.33 (b) in the textbook for the high pass filter op-amp circuit.

The characteristic equation of Figure 16.33 (b) and Figure 16.29 are same.

Assume the op-amp has finite gain, A, then the characteristic equation for the loop is,

$$1+AT(s)=0$$

Write the transfer function of the high pass filter op-amp circuit.

$$T(s)=\frac{s^2+s\frac{\omega_0}{Q}+\omega_0^2}{s^2+s\frac{\omega_0}{Q}\left(2Q^2+1\right)+\omega_0^2}$$

Substitute the transfer function in the characteristic equation.

$$1+A\left\{\frac{s^2+s\frac{\omega_0}{Q}+\omega_0^2}{s^2+s\frac{\omega_0}{Q}\left(2Q^2+1\right)+\omega_0^2}\right\}=0$$

$$s^2+s\frac{\omega_0}{Q}\left(2Q^2+1\right)+\omega_0^2+As^2+As\frac{\omega_0}{Q}+A\omega_0^2=0$$

Divide with $A+1$ on both sides.

$$s^2\frac{1}{A+1}+s\frac{\omega_0}{Q}\left(2Q^2+1\right)\frac{1}{A+1}+\omega_0^2\frac{1}{A+1}+\frac{A}{A+1}\left(s^2+s\frac{\omega_0}{Q}+\omega_0^2\right)=0$$

$$s^2+s\frac{\omega_0}{Q}\left\{\left(2Q^2+1\right)\frac{1}{A+1}+\frac{A}{A+1}\right\}+\omega_0^2\frac{1}{A+1}+\frac{A}{A+1}\omega_0^2=0$$

$$s^2+s\frac{\omega_0}{Q}\left\{2Q^2\frac{1}{A+1}+\frac{1}{A+1}+\frac{A}{A+1}\right\}+\omega_0^2\left(\frac{1}{A+1}+\frac{A}{A+1}\right)=0$$

$$s^2+s\frac{\omega_0}{Q}\left\{1+\frac{2Q^2}{A+1}\right\}+\omega_0^2=0$$

Step 2 of 3

Now, the characteristic equation is,

$$s^2+s\left(\frac{1}{C_1}+\frac{1}{C_2}\right)\frac{1}{R_3}+\frac{1}{R_1R_2C_1C_2}=0$$

Compare the characteristic equation with the standard second order system $s^2+s\frac{\omega_0}{Q}+\omega_0^2$.

The resonant frequency of high-pass filter is,

$$\omega_0=\frac{1}{\sqrt{C_3C_4R_1R_2}}$$

$$\omega_{0a}=\omega_0$$

The quality factor is,

$$Q=\left[\frac{\sqrt{C_3C_4R_1R_2}}{R_4}\left(\frac{1}{C_1}+\frac{1}{C_2}\right)\right]^{-1}$$

$$Q_a=\frac{Q}{1+\frac{2Q^2}{A+1}}$$

Step 3 of 3

The sensitivity of ω_0 to the capacitance A is,

$$S_A^{\omega_{0a}}=\frac{\partial\omega_{0a}}{\partial A}\frac{A}{\omega_{0a}}$$

As the resonant frequency is independent of the gain, so, its sensitivity to gain is zero.

Thus, the sensitivity of resonant frequency to gain is, $\boxed{0}$.

The sensitivity of quality factor to gain is,

$$S_A^{Q_a}=\frac{\partial Q_a}{\partial A}\frac{A}{Q_a}$$

Differentiate the quality factor with respect to A

$$S_A^{Q_a}=\frac{\partial}{\partial A}\left\{\frac{Q}{1+\frac{2Q^2}{A+1}}\right\}$$

$$=\left(\frac{A}{A+1}\right)\frac{\left(\frac{2Q^2}{A+1}\right)}{\left(1+\frac{2Q^2}{A+1}\right)}$$

For $A>>2Q^2$ and $A>>1$, the $\frac{2Q^2}{A}<<1$, thus

$$S_A^{Q_a}=\left(\frac{A}{A}\right)\frac{\left(\frac{2Q^2}{A}\right)}{\left(1+\frac{2Q^2}{A}\right)}$$

$$=\frac{2Q^2}{A}$$

Thus, the sensitivity of quality factor to gain is, $\boxed{\frac{2Q^2}{A}}$.

Step 1 of 8

Refer to Figure 16.34 (a) in the textbook for the op-amp circuit.

Write the expression for the transfer function.

$$T(s)=\frac{s^2+s\left(\frac{1}{R_1}+\frac{1}{R_2}\right)\frac{1}{C_4}+\frac{1}{R_1R_2C_3C_4}}{s^2+s\left(\frac{1}{C_4R_1}+\frac{1}{C_4R_2}+\frac{1}{C_3R_2}\right)+\frac{1}{R_1R_2C_3C_4}}$$

Compare the denominator equation with the standard second order system $s^2+s\frac{\omega_0}{Q}+\omega_0^2$.

The resonant frequency of active filter poles of a feedback loop is,

$$\omega_0=\frac{1}{\sqrt{C_3C_4R_1R_2}}$$

The quality factor of active filter poles of a feedback loop is,

$$Q=\left[\frac{\sqrt{C_3C_4R_1R_2}}{C_4}\left(\frac{1}{R_1}+\frac{1}{R_2}\right)\right]^{-1}$$

Since $R_1=R_2=R$, the resonant frequency is,

$$\omega_0=\frac{1}{R\sqrt{C_3C_4}}$$

The quality factor is,

$$\begin{aligned}Q&=\left[\frac{R\sqrt{C_3C_4}}{C_4}\left(\frac{2}{R}\right)\right]^{-1}\\&=\left[\frac{2\sqrt{C_3C_4}}{C_4}\right]^{-1}\\&=\left[2\sqrt{\frac{C_3}{C_4}}\right]^{-1}\\&=\frac{1}{2}\sqrt{\frac{C_4}{C_3}}\end{aligned}$$

Step 2 of 8

The sensitivity of the resonant frequency to capacitance C_4 is,

$$S_{C_4}^{\omega}=\frac{\partial\omega_0}{\partial C_4}\frac{C_4}{\omega_0}$$

Differentiate the resonant frequency with respect to C_4

$$\begin{aligned}\frac{\partial\omega_0}{\partial C_4}&=\frac{\partial}{\partial C_4}\left\{\frac{1}{R\sqrt{C_3C_4}}\right\}\\&=\frac{1}{R\sqrt{C_3}}\frac{\partial}{\partial C_4}(C_4)^{-\frac{1}{2}}\\&=\frac{1}{R\sqrt{C_3}}\left(-\frac{1}{2}(C_4)^{-\frac{3}{2}}\right)\\&=-\frac{1}{2}\frac{1}{R\sqrt{C_3C_4}}\frac{1}{C_4}\end{aligned}$$

The sensitivity of the resonant frequency to capacitance C_4 is,

$$\begin{aligned}S_{C_4}^{\omega}&=-\frac{1}{2}\frac{1}{R\sqrt{C_3C_4}}\frac{1}{C_4}\frac{C_4}{\frac{1}{R\sqrt{C_3C_4}}}\\&=-\frac{1}{2}\frac{1}{R\sqrt{C_3C_4}}\frac{1}{\frac{1}{R\sqrt{C_3C_4}}}\\&=-\frac{1}{2}\end{aligned}$$

Thus, the sensitivity of the resonant frequency to the element C_4 is, $\boxed{-\frac{1}{2}}$.

Step 3 of 8

The sensitivity of the resonant frequency to capacitance C_3 is,

$$S_{C_3}^{\omega}=\frac{\partial\omega_0}{\partial C_3}\frac{C_3}{\omega_0}$$

Differentiate the resonant frequency with respect to C_3

$$\begin{aligned}\frac{\partial\omega_0}{\partial C_3}&=\frac{\partial}{\partial C_3}\left\{\frac{1}{R\sqrt{C_3C_4}}\right\}\\&=\frac{1}{R\sqrt{C_4}}\frac{\partial}{\partial C_3}(C_3)^{-\frac{1}{2}}\\&=\frac{1}{R\sqrt{C_4}}\left(-\frac{1}{2}(C_3)^{-\frac{3}{2}}\right)\\&=-\frac{1}{2}\frac{1}{R\sqrt{C_4C_3}}\frac{1}{C_3}\end{aligned}$$

Step 4 of 8

The sensitivity of the resonant frequency to capacitance C_3 is,

$$\begin{aligned}S_{C_3}^{\omega}&=-\frac{1}{2}\frac{1}{R\sqrt{C_4C_3}}\frac{1}{C_3}\frac{C_3}{\frac{1}{R\sqrt{C_3C_4}}}\\&=-\frac{1}{2}\frac{1}{R\sqrt{C_3C_4}}\frac{1}{\frac{1}{R\sqrt{C_3C_4}}}\\&=-\frac{1}{2}\end{aligned}$$

Thus, the sensitivity of the resonant frequency to the element C_3 is, $\boxed{-\frac{1}{2}}$.

Step 5 of 8

The sensitivity of the resonant frequency to capacitance R is,

$$S_R^{\omega}=\frac{\partial\omega_0}{\partial R}\frac{R}{\omega_0}$$

Differentiate the resonant frequency with respect to R

$$\begin{aligned}\frac{\partial\omega_0}{\partial R}&=\frac{\partial}{\partial R}\left\{\frac{1}{R\sqrt{C_3C_4}}\right\}\\&=\frac{1}{\sqrt{C_3C_4}}\frac{\partial}{\partial R}R^{-1}\\&=-\frac{1}{\sqrt{C_3C_4}}\frac{1}{R^2}\end{aligned}$$

The sensitivity of the resonant frequency to capacitance R is,

$$\begin{aligned}S_R^{\omega}&=-\frac{1}{\sqrt{C_3C_4}}\frac{1}{R^2}\frac{R}{\frac{1}{R\sqrt{C_3C_4}}}\quad\{\text{since } R_1=R_2=R\}\\&=-1\end{aligned}$$

Thus, the sensitivity of the resonant frequency to the element $R(R_1 \text{ or } R_2)$ is, $\boxed{-1}$.

Step 6 of 8

The sensitivity of quality factor to capacitance C_4 is,

$$S_{C_4}^{Q}=\frac{\partial Q}{\partial C_4}\frac{C_4}{Q}$$

Differentiate the quality factor with respect to C_4

$$\begin{aligned}\frac{\partial Q}{\partial C_4}&=\frac{\partial}{\partial C_4}\frac{1}{2}\sqrt{\frac{C_4}{C_3}}\\&=\frac{1}{2}\sqrt{\frac{1}{C_3}}\frac{\partial}{\partial C_4}\sqrt{C_4}\\&=\frac{1}{4}\sqrt{\frac{1}{C_4C_3}}\end{aligned}$$

The sensitivity of quality factor to capacitance C_4 is,

$$\begin{aligned}S_{C_4}^{Q}&=\frac{1}{4}\sqrt{\frac{1}{C_4C_3}}\frac{C_4}{\frac{1}{2}\sqrt{\frac{C_4}{C_3}}}\\&=\frac{1}{2}\sqrt{\frac{1}{C_4}}\sqrt{\frac{1}{C_4}}C_4\\&=\frac{1}{2}\end{aligned}$$

Thus, the sensitivity of quality factor to capacitance C_4 is, $\boxed{\frac{1}{2}}$.

Step 7 of 8

The sensitivity of quality factor to capacitance C_3 is,

$$S_{C_3}^{Q}=\frac{\partial Q}{\partial C_3}\frac{C_3}{Q}$$

Differentiate the quality factor with respect to C_3

$$\begin{aligned}\frac{\partial Q}{\partial C_3}&=\frac{\partial}{\partial C_3}\frac{1}{2}\sqrt{\frac{C_4}{C_3}}\\&=\frac{1}{2}\sqrt{C_4}\frac{\partial}{\partial C_3}(C_3)^{-\frac{1}{2}}\\&=-\frac{1}{4}\sqrt{C_4}(C_3)^{-\frac{1}{2}-1}\\&=-\frac{1}{4}\frac{\sqrt{C_4}}{C_3\sqrt{C_3}}\end{aligned}$$

The sensitivity of quality factor to capacitance C_3 is,

$$\begin{aligned}S_{C_3}^{Q}&=-\frac{1}{4}\frac{\sqrt{C_4}}{C_3\sqrt{C_3}}\frac{C_3}{\frac{1}{2}\sqrt{\frac{C_4}{C_3}}}\\&=-\frac{1}{2}\end{aligned}$$

Thus, the sensitivity of quality factor to capacitance C_3 is, $\boxed{-\frac{1}{2}}$.

Step 8 of 8

As the quality factor is independent of resistances, its sensitivity to resistances is zero.

Thus, the sensitivity of the quality factor to the element $R(R_1 \text{ or } R_2)$ is, $\boxed{0}$.

Step 1 of 5

17.081P

Refer to Figure 16.21 (b) in the textbook for the op-amp second order RC resonator circuit.

The resonant frequency of operational amplifier –RC resonator is,

$$\omega_0 = \frac{1}{\sqrt{\frac{C_4 C_6 R_1 R_3 R_5}{R_2}}}$$

The quality factor of operational amplifier –RC resonator is,

$$Q = R_6\sqrt{\frac{C_6 R_2}{C_4 R_1 R_3 R_5}}$$

The sensitivity of the resonant frequency to capacitance C_4 is,

$$S_{C_4}^{\omega} = \frac{\partial \omega_0}{\partial C_4}\frac{C_4}{\omega_0}$$

Differentiate the resonant frequency with respect to C_4

$$\begin{aligned}\frac{\partial \omega_0}{\partial C_4} &= \frac{\partial}{\partial C_4}\left[\frac{1}{\sqrt{\frac{C_4 C_6 R_1 R_3 R_5}{R_2}}}\right]\\ &= \sqrt{\frac{R_2}{C_6 R_1 R_3 R_5}}\frac{\partial}{\partial C_4}(C_4)^{\frac{1}{2}}\\ &= \sqrt{\frac{R_2}{C_6 R_1 R_3 R_5}}\left(-\frac{1}{2}(C_4)^{\frac{1}{2}-1}\right)\\ &= -\frac{1}{2C_4}\sqrt{\frac{R_2}{C_6 C_4 R_1 R_3 R_5}}\end{aligned}$$

Now the sensitivity is,

$$\begin{aligned}S_{C_4}^{\omega} &= -\frac{1}{2C_4}\sqrt{\frac{R_2}{C_6 C_4 R_1 R_3 R_5}}\left\{\frac{C_4}{\sqrt{\frac{R_2}{C_6 C_4 R_1 R_3 R_5}}}\right\}\\ &= -\frac{1}{2}\end{aligned}$$

Since the capacitance $C_4 = C_6$ then the sensitivity is also same.

Thus, the sensitivity of the resonant frequency to capacitance C_4 **or** C_6 is, $\boxed{-\frac{1}{2}}$.

Step 2 of 5

The sensitivity of the resonant frequency to capacitance R_1 is,

$$S_{R_1}^{\omega} = \frac{\partial \omega_0}{\partial R_1}\frac{R_1}{\omega_0}$$

Differentiate the resonant frequency with respect to R_1

$$\begin{aligned}\frac{\partial \omega_0}{\partial R_1} &= \frac{\partial}{\partial R_1}\left[\frac{1}{\sqrt{\frac{C_4 C_6 R_1 R_3 R_5}{R_2}}}\right]\\ &= \sqrt{\frac{R_2}{C_6 C_4 R_3 R_5}}\frac{\partial}{\partial R_1}(R_1)^{\frac{1}{2}}\\ &= \sqrt{\frac{R_2}{C_6 C_4 R_3 R_5}}\left(-\frac{1}{2}(R_1)^{\frac{1}{2}-1}\right)\\ &= -\frac{1}{2R_1}\sqrt{\frac{R_2}{C_6 C_4 R_1 R_3 R_5}}\end{aligned}$$

Now the sensitivity is,

$$\begin{aligned}S_{C_4}^{\omega} &= -\frac{1}{2R_1}\sqrt{\frac{R_2}{C_6 C_4 R_1 R_3 R_5}}\left\{\frac{R_1}{\sqrt{\frac{R_2}{C_6 C_4 R_1 R_3 R_5}}}\right\}\\ &= -\frac{1}{2}\end{aligned}$$

Since the capacitance $R_1 = R_3 = R_5$ then the sensitivity is also same.

Thus, the sensitivity of the resonant frequency to resistance R_1 **or** R_3 **or** R_5 is, $\boxed{-\frac{1}{2}}$.

Step 3 of 5

The sensitivity of the resonant frequency to capacitance R_2 is,

$$S_{R_2}^{\omega} = \frac{\partial \omega_0}{\partial R_2}\frac{R_2}{\omega_0}$$

Differentiate the resonant frequency with respect to R_2

$$\begin{aligned}\frac{\partial \omega_0}{\partial R_2} &= \frac{\partial}{\partial R_2}\left[\frac{1}{\sqrt{\frac{C_4 C_6 R_1 R_3 R_5}{R_2}}}\right]\\ &= \sqrt{\frac{1}{C_4 C_6 R_1 R_3 R_5}}\frac{\partial}{\partial R_2}(R_2)^{\frac{1}{2}}\\ &= \sqrt{\frac{1}{C_4 C_6 R_1 R_3 R_5}}\left(\frac{1}{2\sqrt{R_2}}\right)\end{aligned}$$

Now the sensitivity is,

$$\begin{aligned}S_{R_2}^{\omega} &= \sqrt{\frac{1}{C_4 C_6 R_1 R_3 R_5}}\left(\frac{1}{2\sqrt{R_2}}\right)\left\{\frac{R_2}{\sqrt{\frac{R_2}{C_6 C_4 R_1 R_3 R_5}}}\right\}\\ &= \frac{1}{2}\end{aligned}$$

Thus, the sensitivity of the resonant frequency to resistance R_2 is, $\boxed{\frac{1}{2}}$.

Step 4 of 5

The sensitivity of quality factor to resistance R_6 is,

$$S_{R_6}^{Q} = \frac{\partial Q}{\partial R_6}\frac{R_6}{Q}$$

Differentiate the quality factor with respect to R_6

$$\frac{\partial Q}{\partial R_6} = \sqrt{\frac{C_6 R_2}{C_4 R_1 R_3 R_5}}$$

The sensitivity of quality factor to resistance R_6 is,

$$\begin{aligned}S_{R_6}^{Q} &= \sqrt{\frac{C_6 R_2}{C_4 R_1 R_3 R_5}}\frac{R_6}{R_6\sqrt{\frac{C_6 R_2}{C_4 R_1 R_3 R_5}}}\\ &= 1\end{aligned}$$

Thus, the sensitivity of quality factor to resistance R_6 is, $\boxed{1}$.

Step 5 of 5

The sensitivity of quality factor to capacitance C_6 is,

$$S_{C_6}^{Q} = \frac{\partial Q}{\partial C_6}\frac{C_6}{Q}$$

The terms C_6 **and** R_2 are in the numerator term of the quality factor expression.

Thus, the sensitivity of quality factor to capacitance C_6 is, $\boxed{\frac{1}{2}}$.

The sensitivity of quality factor to capacitance C_4 is,

$$S_{C_4}^{Q} = \frac{\partial Q}{\partial C_4}\frac{C_4}{Q}$$

The terms $C_4, R_1, R_3,$ **and** R_5 are in the denominator term of the quality factor expression.

Thus, the sensitivity of quality factor to resistance C_4, R_1, R_3, R_5 is, $\boxed{-\frac{1}{2}}$.

17.082P

We don't have the solution to this problem yet.

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17.083P

We don't have the solution to this problem yet.

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17.084P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

17.085P

We don't have the solution to this problem yet.

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Step 1 of 4

17.086P

Refer to Figure 17.36(b) for the integrator in the textbook.

Redraw the internal circuit with the output resistance R_o and capacitance C_o in s-domain as shown in figure 1.

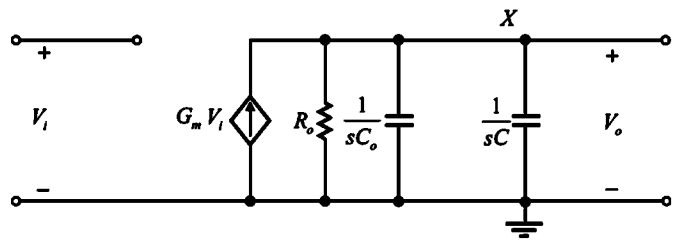


Figure 1

Step 2 of 4

Apply Nodal analysis at node-*X*.

$$-G_m V_i + \frac{V_o}{R_o} + \frac{V_o}{\left(\frac{1}{sC_o}\right)} + \frac{V_o}{\left(\frac{1}{sC}\right)} = 0$$

$$V_o\left(sC_o + sC + \frac{1}{R_o}\right) = G_m V_i$$

$$\frac{V_o}{V_i} = \frac{\frac{G_m}{C_o + C}}{s + \frac{1}{(C_o + C)R_o}}$$

Therefore the transfer function is $\frac{V_o}{V_i}$ is $\boxed{\frac{\frac{G_m}{C_o + C}}{s + \frac{1}{(C_o + C)R_o}}}$.

Step 3 of 4

Consider the transfer function $\frac{V_o}{V_i}$.

$$\frac{V_o}{V_i} = \frac{\frac{G_m}{C_o + C}}{s + \frac{1}{(C_o + C)R_o}}$$

Rewrite in time-constant form.

$$\frac{V_o}{V_i} = \frac{G_m R_o}{\left[(C_o + C)R_o\right]s + 1}$$

Therefore, the integrator time-constant is,

$$\tau = (C_o + C)R_o$$

Here, τ is the time-constant.

Differentiate partially with respect to *C*.

$$\frac{\partial \tau}{\partial C} = R_o$$

$$\partial \tau = R_o \partial C$$

$$\left(\frac{\partial \tau}{\tau}\right)100 = \left(\frac{R_o}{\tau}\right)(\partial C) \times 100$$

Substitute 1 for $\left(\frac{\partial \tau}{\tau}\right)100$.

$$1 = \left(\frac{R_o}{\tau}\right)(\partial C)100$$

$$\partial C = \frac{0.01\tau}{R_o}$$

Therefore, the smallest value of *C* that can be used is $\boxed{\frac{0.01\tau}{R_o}}$.

Step 4 of 4

Consider the transfer function $\frac{V_o}{V_i}$.

$$\frac{V_o}{V_i} = \frac{\frac{G_m}{C_o + C}}{s + \frac{1}{(C_o + C)R_o}}$$

Determine the unity-gain frequency ω_0.

$$\left|\frac{V_o}{V_i}\right| = \frac{\frac{G_m}{C_o + C}}{\sqrt{\omega_0^2 + \left[\frac{1}{(C_o + C)R_o}\right]^2}}$$

Substitute 1 for $\left|\frac{V_o}{V_i}\right|$.

$$1 = \frac{\frac{G_m}{C_o + C}}{\sqrt{\omega_0^2 + \left[\frac{1}{(C_o + C)R_o}\right]^2}}$$

$$\left(\frac{G_m}{C_o + C}\right)^2 = \omega_0^2 + \left[\frac{1}{(C_o + C)R_o}\right]^2$$

$$\omega_0^2 = \frac{R_o^2 G_m^2 - 1}{R_o^2 (C + C_o)^2}$$

$$\omega_0 = \frac{\sqrt{R_o^2 G_m^2 - 1}}{R_o (C + C_o)}$$

Consider that the low frequency pole is 2 decades lower than ω_0.

$$\frac{\sqrt{R_o^2 G_m^2 - 1}}{R_o (C + C_o)} - \frac{1}{R_o (C + C_o)} = 20$$

$$\sqrt{R_o^2 G_m^2 - 1} - 1 = 20 R_o (C + C_o)$$

$$G_m = \frac{1 + \left[1 + 20R_o (C + C_o)\right]^2}{\sqrt{R_o}}$$

Therefore, the smallest G_m that the transconductor must have is,

$$\boxed{\frac{1 + \left[1 + 20R_o (C + C_o)\right]^2}{\sqrt{R_o}}}$$

Step 1 of 3

17.087P

Refer to Figure 17.36(c) for the first-order low-pass filter in the textbook.

Redraw the same in *s*-domain as shown in Figure 1.

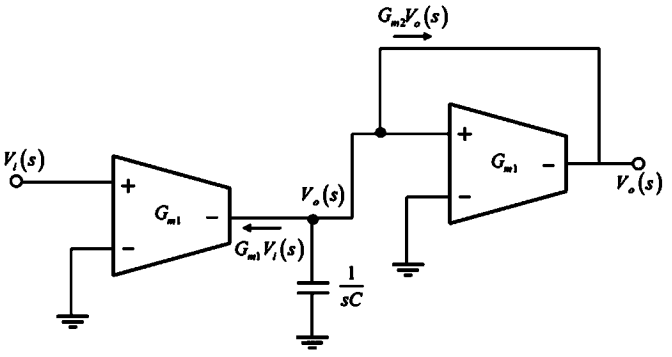


Figure 1

Step 2 of 3

Determine the transfer function $\frac{V_o(s)}{V_i(s)}$.

Apply Nodal analysis at node-*X*.

$$G_{m1}V_i(s)+\frac{V_o(s)}{\left(\frac{1}{sC}\right)}+G_{m2}V_o(s)=0$$

$$V_o(s)(sC+G_{m2})=-G_{m1}V_i(s)$$

$$\frac{V_o(s)}{V_i(s)}=\frac{-G_{m1}}{sC+G_{m2}}$$

$$\frac{V_o(s)}{V_i(s)}=\frac{-\frac{G_{m1}}{C}}{s+\frac{G_{m2}}{C}}$$

Therefore, the transfer function is,

$$T(s)=\frac{-\left(\frac{G_{m1}}{C}\right)}{s+\frac{G_{m2}}{C}}$$

The pole frequency ω_0 is,

$$\omega_0=\frac{G_{m2}}{C}$$

Substitute 20×10^6 for ω_0, and 2×10^{-12} for *C*.

$$\frac{G_{m2}}{2\times10^{-12}}=20\times10^6$$

$$G_{m2}=\left(20\times10^6\right)\left(2\times10^{-12}\right)$$

$$G_{m2}=40\ \mu\text{A/V}$$

Step 3 of 3

Consider the transfer function $T(s)$.

$$T(s)=\frac{-\left(\frac{G_{m1}}{C}\right)}{s+\frac{G_{m2}}{C}}$$

$$T(j\omega)=\frac{-\left(\frac{G_{m1}}{C}\right)}{j\omega+\frac{G_{m2}}{C}}$$

The DC gain is,

$$\left|T(j\omega)\right|=\frac{\left(\frac{G_{m1}}{C}\right)}{\sqrt{\omega_0^2+\left(\frac{G_{m2}}{C}\right)^2}}$$

Substitute 10 for $\left|T(j\omega)\right|$, 2×10^{-12} for *C*, 20×10^6 for ω_0, and 40×10^{-6} for G_{m2}.

$$10=\frac{\left(\frac{G_{m1}}{2\times10^{-12}}\right)}{\sqrt{\left(20\times10^6\right)^2+\left(\frac{40\times10^{-6}}{2\times10^{-12}}\right)^2}}$$

$$\frac{G_{m1}}{2\times10^{-12}}=10\sqrt{\left(40\times10^{12}\right)+\left(20\times10^{12}\right)}$$

$$G_{m1}=10\left(2\times10^{-12}\right)\left(10^6\sqrt{60}\right)$$

$$G_{m1}=0.155\ \text{mA/V}$$

Therefore, the design parameters are,

$$\boxed{\begin{aligned}G_{m1}&=0.155\ \text{mA/V}\\G_{m2}&=40\ \mu\text{A/V}\\C&=2\ \text{pF}\end{aligned}}$$

Step 1 of 2

17.088P

Refer to Figure 17.36(c) in the textbook.

Redraw the circuit by connecting a capacitor C_1 between the input node and node-X in

s-domain as shown in figure 1.

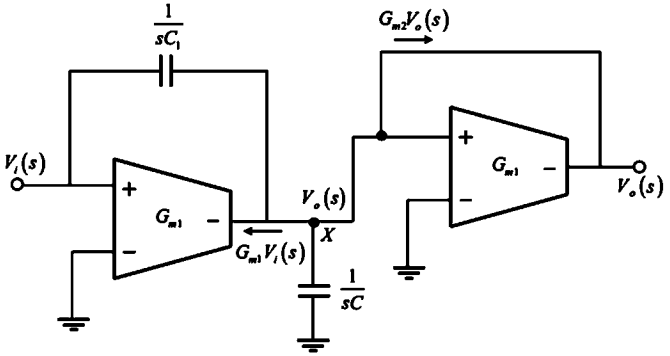


Figure 1

Step 2 of 2

Apply nodal analysis at node-X.

$$\frac{V_o(s)-V_i(s)}{\left(\frac{1}{sC_1}\right)}+G_{m1}V_i(s)+\frac{V_o(s)}{\left(\frac{1}{sC}\right)}+G_{m2}V_o(s)=0$$

$$sC_1\left(V_o(s)-V_i(s)\right)+G_{m1}V_i(s)+sCV_o(s)+G_{m2}V_o(s)=0$$

$$V_i(s)\left(G_{m1}-sC_1\right)+V_o(s)\left(sC_1+sC+G_{m2}\right)=0$$

$$\frac{V_o(s)}{V_i(s)}=\frac{sC_1-G_{m1}}{sC_1+sC+G_{m2}}$$

Simplify further.

$$\frac{V_o(s)}{V_i(s)}=\frac{C_1\left(s-\frac{G_{m1}}{C_1}\right)}{(C_1+C)\left(s+\frac{G_{m2}}{C_1+C}\right)}$$

Therefore, the transfer function $\frac{V_o}{V_i}$ is $\boxed{\frac{\left(\frac{C_1}{C_1+C}\right)\left(s-\frac{G_{m1}}{C_1}\right)}{\left(s+\frac{G_{m2}}{C_1+C}\right)}}$.

Step 1 of 7

17.089P

Refer to Figure P17.89 in the textbook.

Redraw the circuit as shown in Figure 1.

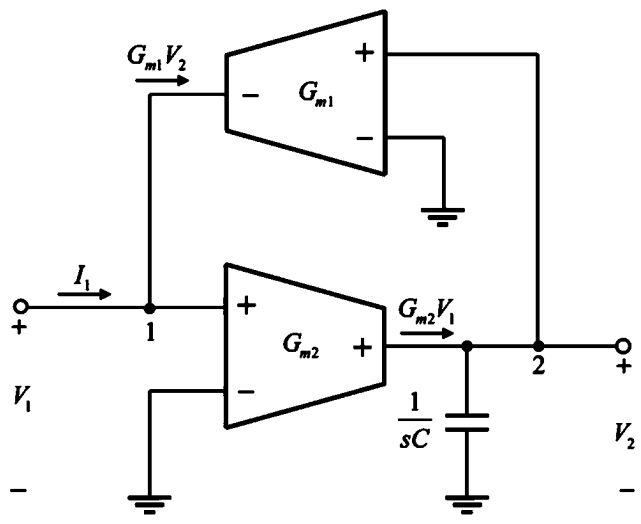


Figure 1

Step 2 of 7

Assume that the voltages of the nodes 1 and 2 are V_1 and V_2.

Apply Kirchhoff's current law at node-1.

$$I_1 = G_{m1}V_2$$

$$V_2 = \frac{I_1}{G_{m1}} \quad \text{..... (1)}$$

Apply voltage division principle at node-2.

$$V_2 = \frac{1}{sC}G_{m2}V_1 \quad \text{..... (2)}$$

Consider equation (1). Substitute $\frac{I_1}{G_{m1}}$ for V_2 in equation (2).

$$\frac{I_1}{G_{m1}} = \frac{1}{sC}G_{m2}V_1$$

$$\frac{V_1}{I_1} = s\left(\frac{C}{G_{m1}G_{m2}}\right)$$

Therefore, the input impedance is,

$$Z_{in} = s\left(\frac{C}{G_{m1}G_{m2}}\right) \quad \text{..... (3)}$$

Consider the expression for the impedance of an inductor.

$$Z_L = sL \quad \text{..... (4)}$$

Compare the equations (3) and (4).

Hence, the input impedance Z_{in} is that of an inductance L. And the expression for L is $\boxed{L = \frac{C}{G_{m1}G_{m2}}}$.

Step 3 of 7

(b)

Refer to Figure 17.17(b) in the textbook for the second -order parallel LCR resonator.

Realize the resistance R using a third transconductor G_{m3}, and use the inductor obtained in part (a) to form an LCR resonator as shown in Figure 2.

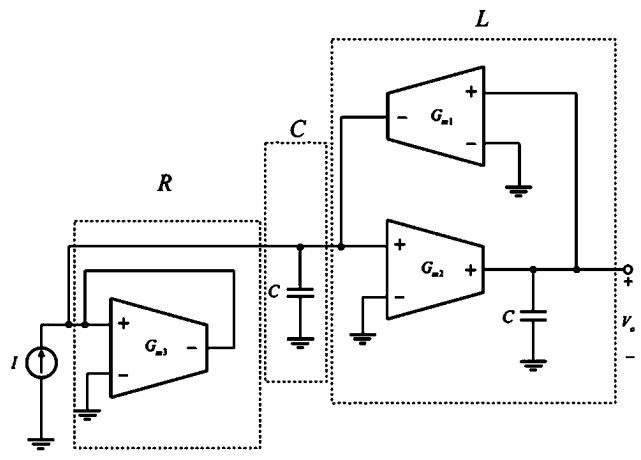


Figure 2

Step 4 of 7

(c)

Redraw the circuit shown in Figure P17.89 in the textbook using a fourth transconductor G_{m4} with its input connected to an input voltage V_i and its output connected to node 1.

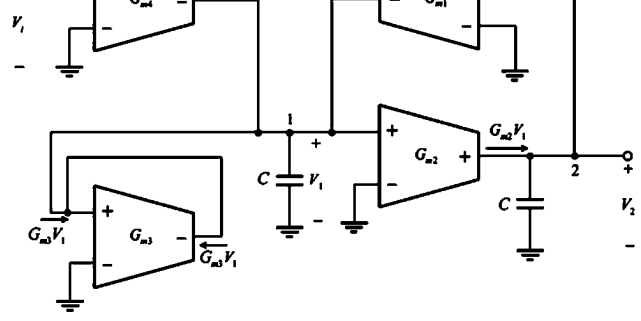


Figure 3

Compare the circuit shown in Figure 3 with the circuit shown in Figure 17.37(b) in the textbook.

Hence, both the circuits represent the same $G_m - C$ implementation of the block diagram of the two-integrator-loop biquad, but the capacitances C_1 and C_2 are equal to C.

Step 5 of 7

(d) (i)

Consider Figure 3.

Consider that the impedance of capacitor C in s-domain is $\frac{1}{sC}$.

Apply Nodal analysis at node-2.

$$-G_{m2}V_1 + \frac{V_2}{\left(\frac{1}{sC}\right)} = 0$$

$$G_{m2}V_1 = sCV_2$$

$$V_2 = \frac{G_{m2}}{sC}V_1 \quad \text{..... (5)}$$

Apply Nodal analysis at node-1.

$$G_{m4}V_i + G_{m3}V_1 + \frac{V_1}{sC} + G_{m1}V_2 = 0 \quad \text{..... (6)}$$

Step 6 of 7

Consider equation (5). Substitute the expression for V_2 in equation (6).

$$G_{m4}V_i + G_{m3}V_1 + \frac{V_1}{\left(\frac{1}{sC}\right)} + G_{m1}\left(\frac{G_{m2}}{sC}V_1\right) = 0$$

$$V_1\left(\frac{G_{m1}G_{m2}}{sC} + sC + G_{m3}\right) = -G_{m4}V_i$$

$$\frac{V_1}{V_i} = \frac{-G_{m4}}{\frac{G_{m1}G_{m2}}{sC} + sC + G_{m3}}$$

$$\frac{V_1}{V_i} = \frac{-sCG_{m4}}{G_{m1}G_{m2} + s^2C^2 + sCG_{m3}}$$

Simplify further.

$$\frac{V_1}{V_i} = \frac{-sCG_{m4}}{C^2\left(s^2 + \frac{sG_{m3}}{C} + \frac{G_{m1}G_{m2}}{C^2}\right)}$$

$$\frac{V_1}{V_i} = \frac{-\left(s\frac{G_{m4}}{C}\right)}{s^2 + s\left(\frac{G_{m3}}{C}\right) + \frac{G_{m1}G_{m2}}{C^2}}$$

Therefore, the transfer function $\frac{V_1}{V_i}$ is $\boxed{\frac{-\left(s\frac{G_{m4}}{C}\right)}{s^2 + s\left(\frac{G_{m3}}{C}\right) + \frac{G_{m1}G_{m2}}{C^2}}}$.

Step 7 of 7

(ii)

Consider the following transfer function:

$$\frac{V_1}{V_i} = \frac{-\left(s\frac{G_{m4}}{C}\right)}{s^2 + s\left(\frac{G_{m3}}{C}\right) + \frac{G_{m1}G_{m2}}{C^2}}$$

$$V_1 = \left(\frac{-\left(s\frac{G_{m4}}{C}\right)}{s^2 + s\left(\frac{G_{m3}}{C}\right) + \frac{G_{m1}G_{m2}}{C^2}}\right)V_i \quad \text{..... (7)}$$

Recall equation (5).

$$V_2 = \frac{G_{m2}}{sC}V_1$$

Consider equation (7). Substitute the expression for V_1.

$$V_2 = \frac{G_{m2}}{sC}\left(\frac{-\left(s\frac{G_{m4}}{C}\right)}{s^2 + s\left(\frac{G_{m3}}{C}\right) + \frac{G_{m1}G_{m2}}{C^2}}\right)(V_i)$$

$$\frac{V_2}{V_i} = \frac{-\left(\frac{G_{m2}G_{m4}}{C^2}\right)}{s^2 + s\left(\frac{G_{m3}}{C}\right) + \frac{G_{m1}G_{m2}}{C^2}}$$

Therefore, the filter function $\frac{V_2}{V_i}$ is $\boxed{\frac{-\left(\frac{G_{m2}G_{m4}}{C^2}\right)}{s^2 + s\left(\frac{G_{m3}}{C}\right) + \frac{G_{m1}G_{m2}}{C^2}}}$.

Step 1 of 2

17.090P

Refer to Figure 17.37(b) for the block diagram of $G_m - C$ second-order bandpass filter in the textbook.

Consider the following relations:

$$G_{m1} = G_{m2} = G_{m3} = G_m$$

$$C_2 = C$$

The output from the transconductors G_{m2} and G_{m3} are equal to G_m

Refer to equation 17.96 given in the textbook.

$$Q = \frac{\sqrt{G_{m1}G_{m2}}}{G_{m3}}\sqrt{\frac{C_1}{C_2}}$$

Substitute G_m for G_{m1}, G_{m2}, and G_{m3}; and C for C_2.

$$Q = \frac{\sqrt{G_m G_m}}{G_m}\sqrt{\frac{C_1}{C}}$$

$$Q = \frac{G_m}{G_m}\sqrt{\frac{C_1}{C}}$$

$$Q = \sqrt{\frac{C_1}{C}}$$

$$C_1 = Q^2 C$$

Therefore, the expression for C_1 is, $\boxed{C_1 = Q^2 C}$.

Step 2 of 2

Refer to equation 17.95 given in the textbook.

$$\omega_0 = \sqrt{\frac{G_{m1}G_{m2}}{C_1 C_2}}$$

Substitute G_m for G_{m1}, and G_{m2}; and C for C_2.

$$\omega_0 = \sqrt{\frac{G_m G_m}{C_1 C}}$$

$$\omega_0 = \frac{G_m}{\sqrt{C_1 C}}$$

$$\sqrt{C_1 C} = \left(\frac{G_m}{\omega_0}\right)$$

$$C_1 = \frac{G_m^2}{\omega_0^2 C}$$

Substitute Q^2C for C_1.

$$Q^2 C = \frac{G_m^2}{\omega_0^2 C}$$

$$G_m^2 = \omega_0^2 Q^2 C^2$$

$$G_m = \omega_0 Q C$$

Therefore, the expression for G_m is, $\boxed{G_m = \omega_0 Q C}$.

Step 1 of 4

(a)

Refer to Figure 17.37(b) for the $G_m - C$ circuit in the textbook. Connect a capacitor C_3 between the positive input terminal and the output terminal of transconductance G_{m4} and add a fifth negative transconductor G_{m5} as shown in Figure 1.

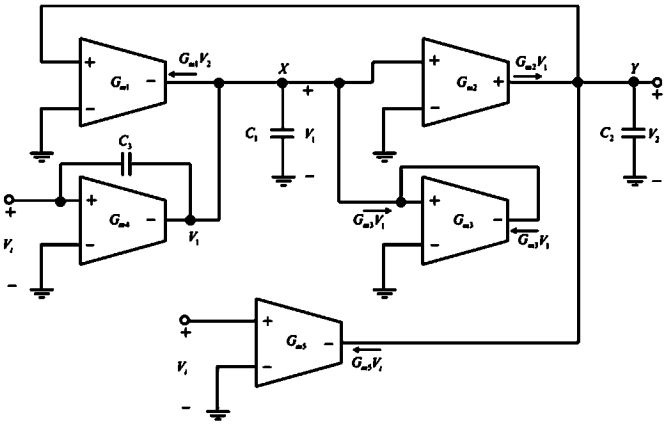


Figure 1

Step 2 of 4

Determine the transfer function $\frac{V_1}{V_i}$.

Apply Nodal analysis at node-Y.

$$-G_{m2}V_1 + G_{m5}V_i + \frac{V_2}{\left(\frac{1}{sC_2}\right)} = 0$$

$$G_{m2}V_1 = G_{m5}V_i + sC_2V_2$$

$$V_2 = \frac{G_{m2}V_1 - G_{m5}V_i}{sC_2} \quad \text{..... (1)}$$

Step 3 of 4

Apply Nodal analysis at node-X.

$$G_{m1}V_2 + \frac{V_1}{\left(\frac{1}{sC_1}\right)} + G_{m3}V_1 + \frac{V_1 - V_i}{\left(\frac{1}{sC_3}\right)} = 0$$

$$G_{m1}V_2 + sC_1V_1 + G_{m3}V_1 + (V_1 - V_i)sC_3 = 0 \quad \text{..... (2)}$$

Recall equation (1).

$$V_2 = \frac{G_{m2}V_1 - G_{m5}V_i}{sC_2}$$

Substitute the above expression for V_2 in equation (2).

$$G_{m1}\left(\frac{G_{m2}V_1 - G_{m5}V_i}{sC_2}\right) + sC_1V_1 + G_{m3}V_1 + (V_1 - V_i)sC_3 = 0$$

$$G_{m1}G_{m2}V_1 - G_{m1}G_{m5}V_i + s^2C_1C_2V_1 + sC_2G_{m3}V_1 + s^2C_2C_3(V_1 - V_i) = 0$$

$$V_1\left(G_{m1}G_{m2} + s^2C_1C_2 + s^2C_2C_3 + sC_2G_{m3}\right) = V_i\left(s^2C_2C_3 + G_{m1}G_{m5}\right)$$

$$\frac{V_1}{V_i} = \frac{s^2C_2C_3 + G_{m1}G_{m5}}{G_{m1}G_{m2} + s^2C_2(C_1 + C_3) + sC_2G_{m3}}$$

$$= \frac{C_2C_3\left(s^2 + \frac{G_{m1}G_{m5}}{C_2C_3}\right)}{C_2(C_1 + C_3)\left(s^2 + \frac{sC_2G_{m3}}{C_2(C_1 + C_3)} + \frac{G_{m1}G_{m2}}{C_2(C_1 + C_3)}\right)}$$

Step 4 of 4

Simplify further.

$$\frac{V_1}{V_i} = \frac{C_3}{C_1 + C_3}\left[\frac{s^2 + \frac{G_{m1}G_{m5}}{C_2C_3}}{s^2 + s\frac{G_{m3}}{(C_1 + C_3)} + \frac{G_{m1}G_{m2}}{C_2(C_1 + C_3)}}\right]$$

Therefore, the transfer function $\frac{V_1}{V_i}$ is,

$$\boxed{\frac{C_3}{C_1 + C_3}\left[\frac{s^2 + \frac{G_{m1}G_{m5}}{C_2C_3}}{s^2 + s\frac{G_{m3}}{(C_1 + C_3)} + \frac{G_{m1}G_{m2}}{C_2(C_1 + C_3)}}\right]}.$$

Step 1 of 3

17.092P

Refer to Figure 17.37(c) for the bandpass filter in the textbook.

Consider the following expression:

$$G_{m1} = G_{m2} = G_m$$

Consider the following expression and determine the value G_m.

$$\frac{G_m}{C} = \omega_0$$
$$G_m = \omega_0 C$$
$$G_m = (2\pi f_0) C$$

Substitute 5×10^{-12} for the capacitance C, 25×10^{6} for the center frequency f_0.

$$G_m = (2\pi)(25\times10^{6})(5\times10^{-12})$$
$$= 785.4 \text{ mA/V}$$

Step 2 of 3

Consider the expression of quality factor Q.

$$Q = \frac{\sqrt{G_{m1}G_{m2}}}{G_{m3}}\sqrt{\frac{C_1}{C_2}}$$
$$= \frac{\sqrt{G_m G_m}}{G_{m3}}\sqrt{\frac{C_1}{C_2}}$$
$$= \frac{G_m}{G_{m3}}\sqrt{\frac{C_1}{C_2}}$$

Substitute 5 for Q, 5×10^{-12} for C_1 and C_2; 785.4×10^{-3} for G_m.

$$5 = \frac{785.4\times10^{-3}}{G_{m3}}\sqrt{\frac{5\times10^{-12}}{5\times10^{-12}}}$$
$$G_{m3} = \frac{785.4\times10^{-3}}{5}$$
$$G_{m3} = 157.08 \text{ mA/V}$$

Step 3 of 3

The expression for G_{m4} is,

$$G_{m4} = \frac{G_m}{Q}\left|\text{Center-frequency gain}\right|$$

Substitute 1 for Center-frequency gain, 785.4×10^{-3} for G_m, and 5 for Q.

$$G_{m4} = \frac{785.4\times10^{-3}}{5}|1|$$
$$= 157.08 \text{ mA/V}$$

Therefore, the design parameters are,

$$G_{m3} = 157.08 \text{ mA/V}$$
$$G_{m4} = 157.08 \text{ mA/V}$$
$$G_{m1} = G_{m2} = G_m = 785.4 \text{ mA/V}$$
$$C_1 = C_2 = 5 \text{ pF}$$

Step 1 of 4

17.093P

Refer to Figure 17.38 (b) for the switched-capacitor input circuit in the textbook.

The expression for the time-constant of the switched-capacitor integrator circuit is,

$$T_C = C_1 R_{in}$$

The expression for clock frequency f_C is,

$$f_C = \frac{1}{R_{in} C_1}$$

Obtain the expression for R_{in}.

$$R_{in} = \frac{1}{f_C C_1}$$

Substitute 200×10^3 for f_C.

$$R_{in} = \frac{1}{200 \times 10^3 \times C_1} \quad \text{..... (1)}$$

Step 2 of 4

Calculate input resistance R_{in} for the capacitance 1 pF.

Recall equation (1).

$$R_{in} = \frac{1}{200 \times 10^3 \times C_1}$$

Substitute 10^{-12} for C_1.

$$\begin{aligned} R_{in} &= \frac{1}{200 \times 10^3 \times 10^{-12}} \\ &= 5\ \text{M}\Omega \end{aligned}$$

Therefore, the input resistance correspond to capacitance 1 pF is $\boxed{5\ \text{M}\Omega}$.

Step 3 of 4

Calculate input resistance R_{in} for the capacitance 5 pF.

Recall equation (1).

$$R_{in} = \frac{1}{200 \times 10^3 \times C_1}$$

Substitute 5×10^{-12} for C_1.

$$\begin{aligned} R_{in} &= \frac{1}{200 \times 10^3 \times 5 \times 10^{-12}} \\ &= 1\ \text{M}\Omega \end{aligned}$$

Therefore, the input resistance correspond to capacitance 5 pF is $\boxed{1\ \text{M}\Omega}$.

Step 4 of 4

Calculate input resistance R_{in} for the capacitance 10 pF.

Recall equation (1).

$$R_{in} = \frac{1}{200 \times 10^3 \times C_1}$$

Substitute 10×10^{-12} for C_1.

$$\begin{aligned} R_{in} &= \frac{1}{200 \times 10^3 \times 10 \times 10^{-12}} \\ &= 500\ \text{k}\Omega \end{aligned}$$

Therefore, the input resistance correspond to capacitance 10 pF is $\boxed{500\ \text{k}\Omega}$.

Step 1 of 5

Refer to Figure 16.35 (b) in the textbook for the switched-capacitor integrator circuit.

A dc voltage of 1 V is applied to the input of the circuit. Therefore, $v_i = 1\ \text{V}$.

The capacitance value is, $C_1 = 1\ \text{pF}$.

The charge transferred for each cycle of the two-phase clock is,

$$Q = CV$$

Substitute 1 pF for C and 1 V for V

$$\begin{aligned} Q &= \left(10^{-12}\right)(1) \\ &= 1\ \text{pC} \end{aligned}$$

Thus, the charge transferred for each cycle of the two-phase clock is, $\boxed{1\ \text{pC}}$.

Step 2 of 5

The clock frequency is, $f_c = 100\ \text{kHz}$.

The average current drawn from the input source is,

$$\begin{aligned} I &= \frac{Q}{T} \\ &= Qf \end{aligned}$$

Substitute 1 pC for Q and 100 kHz for f_c

$$\begin{aligned} I &= \left(10^{-12}\right)\left(100\times10^{3}\right) \\ &= 10^{-7} \\ &= 0.1\ \mu\text{A} \end{aligned}$$

Thus, the average current drawn from the input source is, $\boxed{0.1\ \mu\text{A}}$.

Step 3 of 5

The feedback capacitance is, $C_2 = 10\ \text{pF}$.

For each clock cycle, the output is changed by the same amount as the change in voltage across capacitor C_2

The change in output voltage is,

$$\Delta V = \frac{Q}{C_2}$$

Substitute 1 pC for Q and 10 pF for C_2

$$\begin{aligned} \Delta V &= \frac{10^{-12}}{10\times10^{-12}} \\ &= 0.1\ \text{V} \end{aligned}$$

Thus, the change in the output for each cycle of the clock is, $\boxed{0.1\ \text{V}}$.

Step 4 of 5

The saturation voltage of the amplifier is, $V = \pm10\ \text{V}$.

The number of cycles to take to saturate the amplifier is,

$$N = \frac{|V|}{\Delta V}$$

Substitute $\pm10\ \text{V}$ for V and 0.1 V for ΔV .

$$\begin{aligned} N &= \frac{10}{0.1} \\ &= 100\ \text{cycles} \end{aligned}$$

Thus, the number of cycles it would take to saturate the amplifier is, $\boxed{100\ \text{cycles}}$.

Step 5 of 5

The slope of the staircase output voltage produced is,

$$\begin{aligned} m &= \frac{\Delta V}{\Delta t} \\ &= \frac{\Delta V}{N\dfrac{1}{f_c}} \end{aligned}$$

Substitute 0.1 V for ΔV , 100 for N , and 100 kHz for f_c

$$\begin{aligned} m &= \frac{0.1}{100\left(\dfrac{1}{100\times10^{3}}\right)} \\ &= 100\ \text{V/s} \end{aligned}$$

Thus, the slope of the staircase output voltage produced is, $\boxed{100\ \text{V/s}}$.

Step 1 of 3

17.095P

Refer to figure 17.40 (b) for the switched capacitor active-RC biquad filter.

Assume that the values of two integrating capacitors are equal.

$$C_1 = C_2$$
$$= C$$

$$C = 20 \text{ pF}$$

The relation between the capacitances C_3 and C_4 for switched capacitor active-RC biquad filter is,

$$C_3 = C_4$$
$$= KC$$
$$= \omega_0 T_c C$$
$$= 2\pi f_0 \left(\frac{1}{f_c}\right) C$$

Here, f_0 is the resonant frequency and f_c is the clock frequency.

Substitute 500×10^3 for f_c, 10×10^3 for f_0, and 20×10^{-12} for C.

$$C_3 = 2\pi\left(10\times10^3\right)\left(\frac{1}{500\times10^3}\right)\left(20\times10^{-12}\right)$$
$$= 2.51 \text{ pF}$$

Therefore, the value of capacitances C_3 and C_4 is $\boxed{2.51 \text{ pF}}$.

Step 2 of 3

The expression for the capacitance C_5 is,

$$C_5 = \omega_0 T_c \frac{C}{Q}$$
$$= 2\pi f_0 \frac{1}{f_c} \frac{C}{Q}$$

Substitute 500×10^3 for f_c, 10×10^3 for f_0, 20×10^{-12} for C, and 20 for Q.

$$C_5 = 2\pi\left(10\times10^3\right)\left(\frac{1}{500\times10^3}\right)\left(\frac{10\times10^{-12}}{20}\right)$$
$$= 0.063 \text{ pF}$$

Step 3 of 3

The expression for the center frequency gain is,

$$\text{Centre frequency gain} = \frac{C_6}{C_5}$$

Rewrite as,

$$C_6 = \left[\text{Centre frequency gain}\right] C_5$$

Substitute 1 for Center frequency gain, and 0.063 pF for C_5.

$$C_6 = 0.063 \text{ pF}$$

Therefore, the value of capacitances C_5 and C_6 is 0.063 pF.

Step 1 of 3

17.096P

Refer to figure 17.40 (b) for the switched capacitor active-RC biquad filter.

Assume that the values of two integrating capacitors are equal.

$$C_1 = C_2 = C$$
$$C = 20\ \text{pF}$$

The relation between the capacitances C_3 and C_4 for switched capacitor active-RC biquad filter is,

$$\begin{aligned} C_3 &= C_4 \\ &= KC \\ &= \omega_0 T_c C \\ &= 2\pi f_0 \left(\frac{1}{f_c}\right) C \end{aligned}$$

Here, f_0 is the resonant frequency and f_c is the clock frequency.

Substitute 200×10^3 for f_c, 10×10^3 for f_0, and 20×10^{-12} for C.

Therefore, the value of capacitances C_3 and C_4 is $\boxed{6.28\ \text{pF}}$.

Step 2 of 3

The expression for the capacitance C_5 is,

$$C_5 = \omega_0 T_c \frac{C}{\quad}$$

Substitute for , for , for C, and 50 for Q.

Step 3 of 3

The expression for the center frequency gain is,

Rewrite as,

Substitute **1** for Center frequency gain, and 0.063 pF for .

Therefore, the value of capacitances and is .

Step 1 of 3

17.097P

Refer to figure 17.40(b) for the switched capacitor active-RC biquad filter given in the textbook.

The relation between the values of capacitances C_1 and C_2 is,

$$\begin{aligned} C_1 &= C_2 \\ &= C \end{aligned}$$

$$C = 5\ \text{pF}$$

The relation between the values of capacitances C_3 and C_4 is,

$$\begin{aligned} C_3 &= C_4 \\ &= KC \\ &= \omega_0\left(\frac{1}{f_c}\right)C \end{aligned}$$

Here, f_c is the clock frequency and ω_0 is the 3-dB frequency.

Substitute 100×10^3 for f_c, 5×10^{-12} for C, and 10^3 for ω_0.

$$\begin{aligned} C_3 &= 10^3\left(\frac{1}{100\times10^3}\right)\left(5\times10^{-12}\right) \\ &= 0.05\ \text{pF} \end{aligned}$$

Therefore, the value of capacitances C_3 and C_4 is $\boxed{0.05\ \text{pF}}$.

Step 2 of 3

The expression for the capacitance C_5 is,

$$C_5 = \omega_0\frac{1}{f_c}\frac{C}{Q}$$

Substitute 10^3 for ω_0, 100×10^3 for f_c, 5×10^{-12} for C, and $\frac{1}{\sqrt{2}}$ for Q.

$$\begin{aligned} C_5 &= 10^3\left(\frac{1}{100\times10^3}\right)\left(\frac{5\times10^{-12}}{\frac{1}{\sqrt{2}}}\right) \\ &= 0.353\ \text{pF} \end{aligned}$$

Step 3 of 3

The expression for the Center frequency gain is,

$$\text{Center frequency gain} = \frac{C_6}{C_5}$$

Rewrite as,

$$C_6 = C_5\left[\text{Center frequency gain}\right]$$

Substitute 1 for Center frequency gain, and $0.353\ \text{pF}$ for C_5.

$$\begin{aligned} C_6 &= (0.353\ \text{pF})(1) \\ &= 0.353\ \text{pF} \end{aligned}$$

Therefore, the value of capacitances C_5 and C_6 is $\boxed{0.353\ \text{pF}}$.

Step 1 of 7

Consider the following circuit:

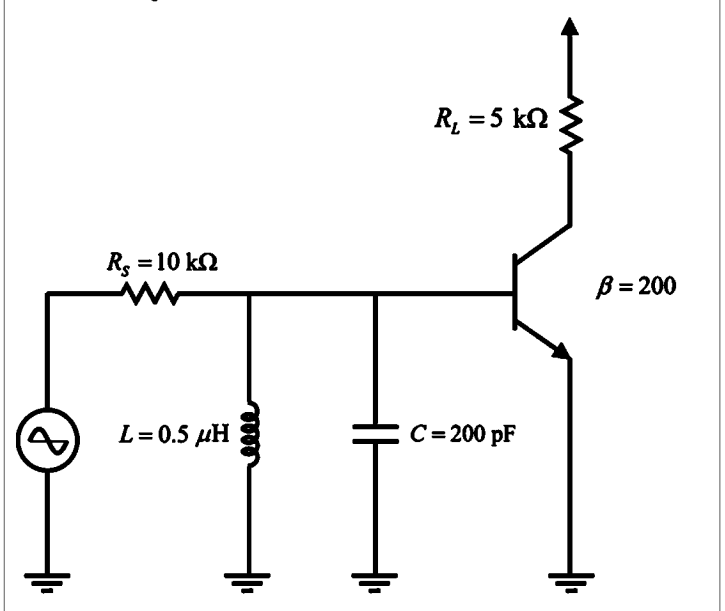


Figure 1

Step 2 of 7

The value of collector current is,

$$I_C = 1\ \text{mA}$$

The expression for the emitter current I_E is,

$$I_E = \left(1 + \frac{1}{\beta}\right) I_C$$

Substitute 200 for β, and $10^{-3}\ \text{A}$ for I_C.

$$\begin{aligned} I_E &= \left(1 + \frac{1}{200}\right) 10^{-3}\ \text{A} \\ &= 1.005\ \text{mA} \end{aligned}$$

The expressions for emitter resistance r_e is,

$$r_e = \frac{V_T}{I_E}$$

Substitute 25 mV for V_T, and $1.005\ \text{mA}$ for I_E.

$$\begin{aligned} r_e &= \frac{25\ \text{mV}}{1.005\ \text{mA}} \\ &= 24.88\ \Omega \end{aligned}$$

Step 3 of 7

The expression for the open-loop gain of the amplifier is,

$$\begin{aligned} A_v &= \frac{-V_o}{V_{in}} \\ &= \frac{-I_C R_L}{I_E r_e} \\ &= \frac{-(\beta I_b) R_L}{(1+\beta) I_b r_e} \\ &= \frac{-\beta R_L}{(1+\beta) r_e} \end{aligned}$$

Substitute 200 for β, $5\ \text{k}\Omega$ for R_L, and $24.88\ \Omega$ for r_e.

$$\begin{aligned} A_v &= \frac{-200(5\ \text{k}\Omega)}{(1+200)(24.88\ \Omega)} \\ &= -200\ \text{V/V} \end{aligned}$$

The expression for the value of resistance r_π is,

$$r_\pi = (1+\beta) r_e$$

Substitute 200 for β, and $24.88\ \Omega$ for r_e.

$$\begin{aligned} r_\pi &= (1+200) \times 24.88 \\ &= 5\ \text{k}\Omega \end{aligned}$$

Step 4 of 7

Determine the value of total capacitance C_t.

$$C_t = C_\pi + C + (1 - A_v) C_\mu$$

Substitute 0.5 pF for C_π, 200 pF for C, $-200\ \text{V/V}$ for A_v, and 0.5 pF for C_μ.

$$\begin{aligned} C_t &= 0.5\ \text{pF} + 200\ \text{pF} + (1 - (-200)) 0.5\ \text{pF} \\ &= 0.5\ pF + 200\ pF + (1+200) 0.5\ \text{pF} \\ &= 301\ \text{pF} \end{aligned}$$

Determine the value of resonant frequency ω_0.

$$\omega_0 = \frac{1}{\sqrt{LC_t}}$$

Substitute $0.5\ \mu\text{H}$ for L, and 301 pF for C_t.

$$\begin{aligned} \omega_0 &= \frac{1}{\sqrt{(0.5\ \mu\text{H})(301\ \text{pF})}} \\ &= \frac{1}{\sqrt{(0.5 \times 10^{-6}\ \text{H})(301 \times 10^{-12}\ \text{F})}} \\ &= 81.51 \times 10^6\ \text{rad/s} \end{aligned}$$

Therefore, the value of resonant frequency ω_0 is $\boxed{81.51 \times 10^6\ \text{rad/s}}$.

Step 5 of 7

Determine the value of center frequency gain A.

$$A = \left(\frac{r_\pi}{R_s + r_\pi}\right) A_v$$

Substitute $-200\ \text{V/V}$ for A_v, $5\ \text{k}\Omega$ for r_π, and $10\ \text{k}\Omega$ for R_s.

$$\begin{aligned} A &= \left(\frac{5\ \text{k}\Omega}{10\ \text{k}\Omega + 5\ \text{k}\Omega}\right)(-200\ \text{V/V}) \\ &= -66.66\ \text{V/V} \end{aligned}$$

Therefore, the value of center frequency gain is $\boxed{-66.66\ \text{V/V}}$.

Step 6 of 7

Determine the value of 3-dB bandwidth BW.

$$\begin{aligned} BW &= \frac{1}{RC_t} \\ &= \frac{1}{(R_s \| r_\pi) C_t} \end{aligned}$$

Substitute $10\ \text{k}\Omega$ for R_s, $5\ \text{k}\Omega$ for r_π, and 301 pF for C_t.

$$\begin{aligned} BW &= \frac{1}{(10\ \text{k}\Omega \| 5\ \text{k}\Omega)(301\ \text{pF})} \\ &= 996.68\ \text{rad/s} \end{aligned}$$

Therefore, the value of 3-dB bandwidth is $\boxed{996.68 \times 10^3\ \text{rad/s}}$.

Step 7 of 7

Determine the quality factory Q.

$$Q = \frac{\omega_0}{BW}$$

Substitute $81.51 \times 10^6\ \text{rad/s}$ for ω_0, and $996.68 \times 10^3\ \text{rad/s}$ for BW.

$$\begin{aligned} Q &= \frac{81.51 \times 10^6\ \text{rad/s}}{996.68 \times 10^3\ \text{rad/s}} \\ &= 81.78 \end{aligned}$$

Therefore, the value of quality factor is $\boxed{81.78}$.

Step 1 of 4

Consider the expression of quality factor Q.

$$Q = \frac{R_P}{\omega_0 L}$$

Therefore, the expression for the equivalent parallel resistance R_P is,

$$R_P = Q\omega_0 L$$
$$R_P = Q(2\pi f_0)L$$

Substitute 250 for Q, 10×10^{-6} for L, and 10^6 for f_0.

$$R_P = 250\times(2\pi\times10^6)(10\times10^{-6})$$
$$= 15.708\ \text{k}\Omega$$

Therefore, the equivalent parallel resistance R_P is $\boxed{15.708\ \text{k}\Omega}$.

Step 2 of 4

Consider the expression of resonant frequency ω_0.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$
$$LC = \frac{1}{\omega_0^2}$$
$$C = \frac{1}{(2\pi f_0)^2 L}$$

Substitute 10^6 for f_0, 10×10^{-6} for L.

$$C = \frac{1}{(2\pi\times10^6)^2(10\times10^{-6})}$$
$$= 2.533\ \text{nF}$$

Therefore, the value of capacitor required to produce resonance at 1 MHz is $\boxed{2.533\ \text{nF}}$.

Step 3 of 4

Consider the following expression of 3-dB bandwidth:

$$B = \frac{1}{R_r C}$$
$$R_r = \frac{1}{BC}$$
$$R_r = \frac{1}{(2\pi f)C}$$

Substitute 12×10^3 for f, and 2.533×10^{-9} for C.

$$R_r = \frac{1}{2\pi(12\times10^3)(2.533\times10^{-9})}$$
$$= 5.236\ \text{k}\Omega$$

Step 4 of 4

Determine the value of additional parallel resistance R.

Consider the following expression:

$$\frac{1}{R} + \frac{1}{R_P} = \frac{1}{R_r}$$

Substitute 32.898×10^3 for R_r, 15.708×10^3 for R_P.

$$\frac{1}{R} + \frac{1}{15.708\times10^3} = \frac{1}{5.236\times10^3}$$
$$\frac{1}{R} = \frac{1}{5.236\times10^3} - \frac{1}{15.708\times10^3}$$
$$R = 7.854\ \text{k}\Omega$$

Therefore, the value of additional parallel resistance R required to produce a 3-dB bandwidth of 12 kHz is $\boxed{7.854\ \text{k}\Omega}$.

Step 1 of 3

17.100P

Draw the following resonant tuned circuit.

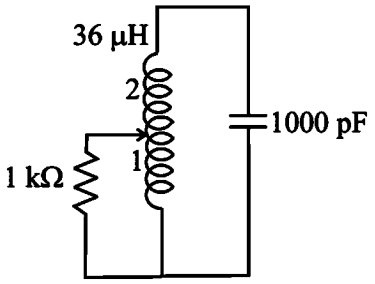


Figure 1

Step 2 of 3

Determine the value of resonant frequency.

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

Substitute $36\ \mu\text{H}$ for L and $1000\ \text{pF}$ for C.

$$\begin{aligned} f_0 &= \frac{1}{2\pi\sqrt{LC}} \\ &= \frac{1}{2\pi\sqrt{(36\times10^{-6})(1000\times10^{-12})}} \\ &= \frac{1}{2\pi\sqrt{36\times10^{-15}}} \\ &= 838.8\,\text{kHz} \end{aligned}$$

Therefore, the resonant frequency of the LC resonator tuned circuit is **838.8 kHz**.

Step 3 of 3

Determine the value of resistance R_p.

$$R_p = n^2 R$$

Substitute $\frac{1}{3}$ for n and $1\ \text{k}\Omega$ for R.

$$\begin{aligned} R_p &= 3^2(1\,\text{k}\Omega) \\ &= 9\,\text{k}\Omega \end{aligned}$$

Determine the value of quality factor of the tuned circuit.

$$\begin{aligned} Q &= \frac{R_p}{\omega_0 L} \\ &= \frac{R_p}{2\pi f_0 L} \end{aligned}$$

Substitute $36\ \mu\text{H}$ for L and $9\ \text{k}\Omega$ for R_p.

$$\begin{aligned} Q &= \frac{9\times10^3}{2\pi(838.8\times10^3)(36\times10^{-6})} \\ &= \frac{9\times10^3}{2\pi(30196.8\times10^{-3})} \\ &= 47.44 \end{aligned}$$

Therefore, the value of quality factor of the tuned LC circuit is **47.44**.

Step 1 of 5

17.101P

Draw the following small signal model for common emitter transistor amplifier loaded with an inductance *L*.

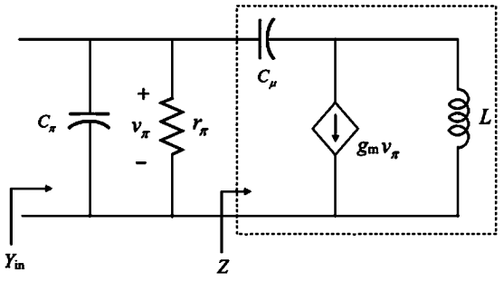


Figure 1

Step 2 of 5

Determine the input admittance of circuit using Figure 1.

$$Y_{in} = j\omega C_\pi + \frac{1}{r_\pi} + \frac{1}{Z} \quad \text{..... (1)}$$

Apply test source of 1 V to determine the impedance *Z* in the circuit.

Draw the following circuit diagram to determine the impedance *Z*.

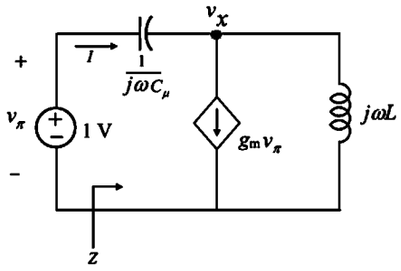


Figure 2

Step 3 of 5

Apply Kirchhoff's current law at node v_x.

$$\frac{v_x - 1}{\left(\frac{1}{j\omega C_\mu}\right)} + \frac{v_x}{j\omega L} + g_m v_\pi = 0$$

$$\left(j\omega C_\mu\right)(v_x - 1) + \frac{v_x}{j\omega L} + g_m v_\pi = 0$$

$$\left(j\omega C_\mu + \frac{1}{j\omega L}\right) v_x = j\omega C_\mu - g_m v_\pi$$

Substitute 1 V for v_π.

$$\left(j\omega C_\mu + \frac{1}{j\omega L}\right) v_x = j\omega C_\mu - g_m (1)$$

$$\left(-\omega^2 LC_\mu + 1\right) v_x = j\omega L\left(j\omega C_\mu - g_m\right)$$

$$\left(-\omega^2 LC_\mu + 1\right) v_x = \left(-\omega^2 LC_\mu - j\omega L g_m\right)$$

$$v_x = -\frac{\left(\omega^2 LC_\mu + j\omega L g_m\right)}{\left(1 - \omega^2 LC_\mu\right)}$$

Step 4 of 5

Determine the expression of current *I*.

$$I = \frac{1 - v_x}{\left(\frac{1}{j\omega C_\mu}\right)}$$

$$= j\omega C_\mu \left(1 - v_x\right)$$

$$= j\omega C_\mu \left(1 - \frac{-\left(\omega^2 LC_\mu + j\omega L g_m\right)}{\left(1 - \omega^2 LC_\mu\right)}\right)$$

$$= j\omega C_\mu \left(\frac{\left(1 - \omega^2 LC_\mu\right) + \left(\omega^2 LC_\mu + j\omega L g_m\right)}{\left(1 - \omega^2 LC_\mu\right)}\right)$$

$$= j\omega C_\mu \left(\frac{1 + j\omega L g_m}{\left(1 - \omega^2 LC_\mu\right)}\right)$$

$$= \frac{j\omega C_\mu - \omega^2 C_\mu L g_m}{1 - \omega^2 LC_\mu}$$

$$= \frac{j\omega C_\mu - \omega^2 C_\mu L g_m}{1 - 0} \qquad \left(\text{Since } \omega C_\mu \ll \frac{1}{\omega L},\ \omega^2 LC_\mu \ll 1\right)$$

$$= j\omega C_\mu - \omega^2 C_\mu L g_m$$

Step 5 of 5

Determine the expression of impedance *Z* using Ohms law.

$$V = IZ$$

$$(1) = IZ$$

$$Z = \frac{1}{I}$$

$$Z = \frac{1}{j\omega C_\mu - \omega^2 C_\mu L g_m} \quad \text{..... (2)}$$

Substitute equation (2) in equation (1).

$$Y_{in} = j\omega C_\pi + \frac{1}{r_\pi} + j\omega C_\mu - \omega^2 C_\mu L g_m$$

$$= j\omega\left(C_\pi + C_\mu\right) + \frac{1}{r_\pi} - \omega^2 C_\mu L g_m$$

$$= \frac{1}{r_\pi} - \omega^2 C_\mu L g_m + j\omega\left(C_\pi + C_\mu\right)$$

Therefore, the input admittance of the amplifier $\boxed{\frac{1}{r_\pi} - \omega^2 C_\mu L g_m + j\omega\left(C_\pi + C_\mu\right)}$ is proved.

Step 1 of 5

(a)

Write the expression of transfer function of second-order band pass filter.

$$T(s)=\frac{a_1 s}{s^2+\frac{\omega_0}{Q}s+\omega_0^2}$$

Substitute $j\omega$ for s.

$$T(j\omega)=\frac{a_1 j\omega}{(j\omega)^2+\left(\frac{\omega_0}{Q}\right)(j\omega)+\omega_0^2}$$
$$=\frac{ja_1\omega}{\left(\omega_0^2-\omega^2\right)+j\left(\frac{\omega_0\omega}{Q}\right)}$$

The magnitude of the transfer function is,

$$|T(j\omega)|=\left|\frac{ja_1\omega}{\left(\omega_0^2-\omega^2\right)+j\frac{\omega_0\omega}{Q}}\right|$$
$$=\frac{|ja_1\omega|}{\left|\left(\omega_0^2-\omega^2\right)+j\frac{\omega_0\omega}{Q}\right|}$$
$$=\frac{a_1\omega}{\sqrt{\left(\omega_0^2-\omega^2\right)^2+\left(\frac{\omega_0\omega}{Q}\right)^2}}$$
$$=\frac{a_1\omega Q}{\sqrt{\left(\omega_0^2-\omega^2\right)^2Q^2+(\omega_0\omega)^2}}$$

Further simplification yields

$$|T(j\omega)|=\frac{a_1\omega Q}{\omega_0\omega\sqrt{1+\left(\frac{\omega_0^2-\omega^2}{\omega_0\omega}\right)^2Q^2}}$$

$$|T(j\omega)|=\frac{\left(\frac{a_1Q}{\omega_0}\right)}{\sqrt{1+\left(\frac{\omega_0^2-\omega^2}{\omega_0\omega}\right)^2Q^2}} \quad \text{..... (1)}$$

Step 2 of 5

For ω in the vicinity of ω_0,

$$\omega=\omega_0+\delta\omega$$
$$=\omega_0\left(1+\frac{\delta\omega}{\omega_0}\right)$$

Squaring on both sides.

$$\omega^2=\omega_0^2\left(1+\frac{\delta\omega}{\omega_0}\right)^2$$
$$=\omega_0^2\left(1+2\left(\frac{\delta\omega}{\omega_0}\right)+\left(\frac{\delta\omega}{\omega_0}\right)^2\right)$$

As $\frac{\delta\omega}{\omega_0}<<1$, $\left(\frac{\delta\omega}{\omega_0}\right)^2$ can be neglected.

$$\omega^2=\omega_0^2\left(1+2\frac{\delta\omega}{\omega_0}\right)$$

$$\omega_0^2-\omega^2=-2\omega_0^2\frac{\delta\omega}{\omega_0} \quad \text{..... (2)}$$

Step 3 of 5

Substitute equation (2) in equation (1).

$$|T(j\omega)|=\frac{\left(\frac{a_1Q}{\omega_0}\right)}{\sqrt{1+\left(\frac{-2\omega_0^2\frac{\delta\omega}{\omega_0}}{\omega_0\omega}\right)^2Q^2}}$$
$$=\frac{\left(\frac{a_1Q}{\omega_0}\right)}{\sqrt{1+\left(-2\frac{\delta\omega}{\omega}\right)^2Q^2}}$$

$$|T(j\omega)|=\frac{\left(\frac{a_1Q}{\omega_0}\right)}{\sqrt{1+4Q^2\left(\frac{\delta\omega}{\omega}\right)^2}}$$

Step 4 of 5

For $Q>>1$, and since $\omega=\omega_0$

$$Q^2\left(\frac{\delta\omega}{\omega}\right)^2=Q^2\left(\frac{\delta\omega}{\omega_0}\right)^2$$

Now the magnitude is

$$|T(j\omega)|=\frac{\frac{a_1Q}{\omega_0}}{\sqrt{1+4Q^2\left(\frac{\delta\omega}{\omega_0}\right)^2}}$$
$$=\frac{|T(j\omega_0)|}{\sqrt{1+4Q^2\left(\frac{\delta\omega}{\omega_0}\right)^2}}$$

Where,

$$|T(j\omega_0)|=\frac{a_1Q}{\omega_0}$$

Therefore, the magnitude of the transfer function of second order band pass filter is $\boxed{\frac{|T(j\omega_0)|}{\sqrt{1+4Q^2\left(\frac{\delta\omega}{\omega_0}\right)^2}}}$.

Step 5 of 5

(b)

The 3 dB band width is,

$$\left(\frac{1}{\sqrt{1+4Q^2\left(\frac{\delta\omega}{\omega_0}\right)^2}}\right)^N=\frac{1}{\sqrt{2}}$$
$$\left(1+4Q^2\left(\frac{\delta\omega}{\omega_0}\right)^2\right)^N=2$$
$$1+4Q^2\left(\frac{\delta\omega}{\omega_0}\right)^2=2^{\frac{1}{N}}$$
$$4Q^2\left(\frac{\delta\omega}{\omega_0}\right)^2=2^{\frac{1}{N}}-1$$

Further simplification yields

$$4Q^2\left(\frac{\delta\omega}{\omega_0}\right)^2=2^{\frac{1}{N}}-1$$
$$Q^2=\frac{1}{4}\left(\frac{\omega_0}{\delta\omega}\right)^2\left(2^{\frac{1}{N}}-1\right)$$
$$\delta\omega^2=\frac{1}{4}\left(\frac{\omega_0}{Q}\right)^2\left(2^{\frac{1}{N}}-1\right)$$
$$\delta\omega=\frac{1}{2}\left(\frac{\omega_0}{Q}\right)\sqrt{\left(2^{\frac{1}{N}}-1\right)}$$

Therefore, the expression of band width is $\boxed{\frac{1}{2}\left(\frac{\omega_0}{Q}\right)\sqrt{\left(2^{\frac{1}{N}}-1\right)}}$.

17.103P

We don't have the solution to this problem yet.

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Step 1 of 2

18.001E

A sinusoidal oscillator consisting of an amplifier gain is $A = 2$.

Frequency of sustained oscillations is $f = 1\,\text{kHz}$.

The conditions for oscillation are

Frequency of oscillation is $f_0 = f$.

The condition for sustainable oscillation is $A\beta = 1$.

Where, β is positive center frequency gain and f_o is pole frequency of second order band pass filter.

Therefore frequency of oscillation is equal to the band pass filter pole frequency.

Thus, the Pole frequency is $\boxed{1\,\text{kHz}}$.

Step 2 of 2

The condition for sustainable oscillation is $A\beta = 1$.

Therefore the centre frequency gain is $\beta = \frac{1}{A}$.

$$\beta = \frac{1}{2}$$
$$= 0.5$$

Thus, the center frequency gain β is $\boxed{0.5}$.

A sinusoidal oscillator consisting of an amplifier gain is A.

Pole frequency of second order band pass filter is ω_0.

Positive centre frequency gain is k.

Gain of the sinusoidal oscillator is greater than zero $A>0$.

Frequency selective network is $\beta(s)=\dfrac{k\frac{\omega_0}{Q}s}{s^2+s\frac{\omega_0}{Q}+\omega_0^2}$.

Where k is positive centre frequency gain,

ω_0 is pole frequency of second order band pass filter,

Q is a pole.

Condition for sinusoidal oscillations : $1-A\beta(s)=0$.

$$A\beta(s)=1$$

$$\frac{Ak\frac{\omega_0}{Q}s}{s^2+s\frac{\omega_0}{Q}+\omega_0^2}=1$$

$$Ak\frac{\omega_0}{Q}s=s^2+s\frac{\omega_0}{Q}+\omega_0^2$$

$$Ak\frac{\omega_0}{Q}j\omega=(j\omega)^2+j\omega\frac{\omega_0}{Q}+\omega_0^2 \qquad (\text{Since } s=j\omega)$$

$$Ak\frac{\omega_0}{Q}j\omega=-\omega^2+j\omega\frac{\omega_0}{Q}+\omega_0^2$$

Separate real and imaginary parts in the above equation.

$$j\left(Ak\frac{\omega\omega_0}{Q}-\frac{\omega\omega_0}{Q}\right)=-\omega^2+\omega_0^2$$

$$j\frac{\omega\omega_0}{Q}(Ak-1)-(\omega_0^2-\omega^2)=0$$

At the frequency of oscillation both real and imaginary parts are zero.

$$\omega_0^2-\omega^2=0$$

$$\omega_0^2=\omega^2$$

$$\omega_0=\omega$$

Therefore frequency of oscillation is equal to the band pass filter pole frequency.

Frequency of oscillation is $\boxed{\omega_0=\omega}$.

Now equate imaginary part to zero to find the condition for sustainable oscillation.

$$Ak-1=0$$

$$\boxed{Ak=1}$$

The condition for sustainable oscillation is $\boxed{Ak=1}$.

We don't have the solution to this problem yet.

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18.003E

We don't have the solution to this problem yet.

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Ask an expert

Step 1 of 4

A sinusoidal oscillator consisting of an amplifier gain is A.

Pole frequency of second order band pass filter is ω_0.

Positive centre frequency gain is k.

Gain of the sinusoidal oscillator is greater than zero $A>0$.

Frequency selective network is $\beta(s)=\dfrac{k\dfrac{\omega_0}{Q}s}{s^2+s\dfrac{\omega_0}{Q}+\omega_0^2}$.

Where k is positive centre frequency gain,

ω_0 is pole frequency of second order band pass filter,

Q is a pole.

Condition for feedback loop to provide sinusoidal oscillations of frequency ω_0 : $1-A\beta(s)=0$.

$$A\beta(s)=1$$

$$\frac{Ak\dfrac{\omega_0}{Q}s}{s^2+s\dfrac{\omega_0}{Q}+\omega_0^2}=1$$

$$Ak\frac{\omega_0}{Q}s=s^2+s\frac{\omega_0}{Q}+\omega_0^2$$

$$Ak\frac{\omega_0}{Q}s=s^2+s\frac{\omega_0}{Q}+\omega_0^2$$

$$s^2+s\frac{\omega_0}{Q}(1-Ak)+\omega_0^2=0$$

Poles of the quadratic equation are at : $s=\dfrac{-\dfrac{\omega_0}{Q}(1-Ak)\pm\sqrt{\left(\dfrac{\omega_0}{Q}\right)^2(1-Ak)^2-4\omega_0^2}}{2}$.

$$s=\frac{-\dfrac{\omega_0}{Q}(1-Ak)\pm 2\omega_0\sqrt{\left(\dfrac{1}{2Q}\right)^2(1-Ak)^2-1}}{2}$$

$$s=-\omega_0\left[\frac{(1-Ak)}{2Q}\pm\sqrt{\left(\frac{1-Ak}{2Q}\right)^2-1}\right]$$

$$s=-\omega_0\left(\frac{1-Ak}{2Q}\right)\left[1\pm\sqrt{1-\left(\frac{2Q}{1-Ak}\right)^2}\right]$$

$$s=-\omega_0\left(\frac{1-Ak}{2Q}\right)\left[1\pm j\sqrt{\left(\frac{2Q}{1-Ak}\right)^2-1}\right]$$

Radial distance of ω_0 is $|s^2|$.

$$\begin{aligned}|s^2|&=\left(-\omega_0\left(\frac{1-Ak}{2Q}\right)\right)^2\left[1^2+\left(\pm j\sqrt{\left(\frac{2Q}{1-Ak}\right)^2-1}\right)^2\right]\\&=\omega_0^2\left(\frac{1-Ak}{2Q}\right)^2\left[1+\left(\frac{2Q}{1-Ak}\right)^2-1\right]\\&=\omega_0^2\left(\frac{1-Ak}{2Q}\right)^2\left[\left(\frac{2Q}{1-Ak}\right)^2\right]\\&=\omega_0^2\end{aligned}$$

Therefore the radial distance is independent of the values of A and k.

Step 2 of 4

(a)

For the poles on the $j\omega$ -axis real part should be zero.

Poles are $s=-\omega_0\left(\dfrac{1-Ak}{2Q}\right)\left[1\pm j\sqrt{\left(\dfrac{2Q}{1-Ak}\right)^2-1}\right]$.

Equate real parts to zero.

$$-\omega_0\left(\frac{1-Ak}{2Q}\right)=0$$

$$1-Ak=0$$

$$Ak=1$$

Therefore the value of Ak for the poles on the $j\omega$ -axis is $\boxed{Ak=1}$.

Step 3 of 4

(b)

For the right half of the s plane, at horizontal distance from the $j\omega$ -axis real part should be greater than zero.

Poles are $s=-\omega_0\left(\dfrac{1-Ak}{2Q}\right)\left[1\pm j\sqrt{\left(\dfrac{2Q}{1-Ak}\right)^2-1}\right]$.

$$-\omega_0\left(\frac{1-Ak}{2Q}\right)\geq 0$$

$$1-Ak\geq 0$$

$$Ak\geq 1$$

Step 4 of 4

Therefore the value of Ak for the poles on the right half of the s plane is $\boxed{Ak\geq 1}$.

Step 1 of 3

Refer to Figure 17.3 (a) in the textbook for a popular limiter circuit.

The negative limiting level is,

$$L_- = -V\frac{R_3}{R_2} - V_D\left(1+\frac{R_3}{R_2}\right)$$

Here, V is the supply voltage, V_D is the diode voltage.

Substitute 15 V for V, 0.7 V for V_D, $3\ \text{k}\Omega$ for R_3, and $9\ \text{k}\Omega$ for R_2.

$$\begin{aligned} L_- &= -(15\ \text{V})\left(\frac{3\ \text{k}\Omega}{9\ \text{k}\Omega}\right) - (0.7\ \text{V})\left(1+\frac{3\ \text{k}\Omega}{9\ \text{k}\Omega}\right) \\ &= -5\ \text{V} - (0.7\ \text{V})(1.34) \\ &= -5\ \text{V} - 0.934\ \text{V} \\ &= -5.934\ \text{V} \end{aligned}$$

The positive limiting level is,

$$L_+ = V\frac{R_4}{R_5} + V_D\left(1+\frac{R_4}{R_5}\right)$$

Substitute 15 V for V, 0.7 V for V_D, $3\ \text{k}\Omega$ for R_4, and $9\ \text{k}\Omega$ for R_3.

$$\begin{aligned} L_+ &= (15\ \text{V})\left(\frac{3\ \text{k}\Omega}{9\ \text{k}\Omega}\right) + (0.7\ \text{V})\left(1+\frac{3\ \text{k}\Omega}{9\ \text{k}\Omega}\right) \\ &= 5\ \text{V} + (0.7\ \text{V})(1.34) \\ &= 5\ \text{V} + 0.934\ \text{V} \\ &= 5.934\ \text{V} \end{aligned}$$

Therefore, the limiting levels are, $\boxed{L_\pm = \pm 5.934\ \text{V}}$.

Step 2 of 3

All of the input current, $\frac{v_I}{R_1}$ flows through the feedback resistance, R_f.

The output voltage is,

$$v_o = -\frac{R_f}{R_1}v_I$$

Thus, the limiter gain is,

$$\begin{aligned} \frac{v_o}{v_I} &= -\frac{R_f}{R_1} \\ &= \frac{-60\ \text{k}\Omega}{30\ \text{k}\Omega} \\ &= -2\ \text{V/V} \end{aligned}$$

Therefore, the gain of the limiter is $\boxed{-2\ \text{V/V}}$.

The limiting levels occurs at,

$$\begin{aligned} v_I &= \frac{\pm 5.934}{2} \\ &= \pm 2.967\ \text{V} \end{aligned}$$

Therefore, the value of v_I at which the limiting levels occur is $\boxed{\pm 2.967\ \text{V}}$.

Step 3 of 3

The slope of the transfer characteristic in the negative limiting region is,

$$\begin{aligned} \text{slope} &= \frac{-(R_f \parallel R_3)}{R_1} \\ &= -\left(\frac{R_f R_3}{R_f + R_3}\right)\frac{1}{R_1} \end{aligned}$$

Substitute $60\ \text{k}\Omega$ for R_f, $3\ \text{k}\Omega$ for R_3, and $30\ \text{k}\Omega$ for R_1.

$$\begin{aligned} \text{slope} &= -\left[\frac{(60\ \text{k}\Omega)(3\ \text{k}\Omega)}{60\ \text{k}\Omega + 3\ \text{k}\Omega}\right]\left(\frac{1}{30\ \text{k}\Omega}\right) \\ &= \frac{-2.857\ \text{k}\Omega}{30\ \text{k}\Omega} \\ &= -0.095 \end{aligned}$$

Therefore, the slope of the transfer characteristic in the negative limiting region is $\boxed{-0.095}$.

The slope of the transfer characteristic in the positive limiting region is,

$$\begin{aligned} \text{slope} &= \frac{-(R_f \parallel R_4)}{R_1} \\ &= -\left(\frac{R_f R_4}{R_f + R_4}\right)\frac{1}{R_1} \end{aligned}$$

Substitute $60\ \text{k}\Omega$ for R_f, $3\ \text{k}\Omega$ for R_4, and $30\ \text{k}\Omega$ for R_1.

$$\begin{aligned} \text{slope} &= -\left[\frac{(60\ \text{k}\Omega)(3\ \text{k}\Omega)}{60\ \text{k}\Omega + 3\ \text{k}\Omega}\right]\left(\frac{1}{30\ \text{k}\Omega}\right) \\ &= \frac{-2.857\ \text{k}\Omega}{30\ \text{k}\Omega} \\ &= -0.095 \end{aligned}$$

Therefore, the slope of the transfer characteristic in the positive limiting region is $\boxed{-0.095}$.

Step 1 of 6

(a)

For the oscillator circuit in the Fig. 18.3.(a), the oscillating frequency at resonance is $\omega_0 = \frac{1}{\sqrt{LC}}$.

$+1\%$ change in the value of L is

$$\begin{aligned} L' &= L + \frac{1}{100}L \\ &= L + 0.01L \\ &= 1.01L \end{aligned}$$

Step 2 of 6

For $+1\%$ change in the value of L, oscillating frequency is $\omega_0' = \frac{1}{\sqrt{L'C}}$.

$$\begin{aligned} \omega_0' &= \frac{1}{\sqrt{1.01LC}} \\ &= \frac{1}{\sqrt{1.01}\sqrt{LC}} \\ &= \frac{\omega_0}{\sqrt{1.01}} \\ &= \frac{\omega_0}{\sqrt{1.01}} \end{aligned}$$

Simplify further.

$$\begin{aligned} \omega_0' &= \frac{\omega_0}{1.00498} \\ &= 0.995\omega_0 \end{aligned}$$

Step 3 of 6

Determine the percentage change in the oscillation frequency.

$$\begin{aligned} \%\Delta\omega_0 &= \frac{\omega_0 - \omega_0'}{\omega_0} \times 100 \\ &= \frac{\omega_0 - 0.995\omega_0}{\omega_0} \times 100 \\ &= 0.5\% \end{aligned}$$

Therefore, for $+1\%$ change in the value of L, the percentage change in the oscillation frequency is $\boxed{0.5\%}$.

Step 4 of 6

(b)

The oscillating frequency at resonance is $\omega_0 = \frac{1}{\sqrt{LC}}$.

$+1\%$ change in the value of C is

$$\begin{aligned} C' &= C + \frac{1}{100}C \\ &= C + 0.01C \\ &= 1.01C \end{aligned}$$

For $+1\%$ change in the value of C, oscillating frequency is $\omega_0' = \frac{1}{\sqrt{LC'}}$.

$$\begin{aligned} \omega_0' &= \frac{1}{\sqrt{L(1.01C)}} \\ &= \frac{1}{\sqrt{1.01}\sqrt{LC}} \\ &= \frac{\omega_0}{\sqrt{1.01}} \\ &= \frac{\omega_0}{\sqrt{1.01}} \end{aligned}$$

Simplify further.

$$\begin{aligned} \omega_0' &= \frac{\omega_0}{1.00498} \\ &= 0.995\omega_0 \end{aligned}$$

Step 5 of 6

Determine the percentage change in the oscillation frequency.

$$\begin{aligned} \%\Delta\omega_0 &= \frac{\omega_0 - \omega_0'}{\omega_0} \times 100 \\ &= \frac{\omega_0 - 0.995\omega_0}{\omega_0} \times 100 \\ &= 0.5\% \end{aligned}$$

Therefore, for $+1\%$ change in the value of C, the percentage change in the oscillation frequency is $\boxed{0.5\%}$.

Step 6 of 6

(c)

The oscillating frequency at resonance is $\omega_0 = \frac{1}{\sqrt{LC}}$.

The oscillating frequency at resonance is independent of the value of R.

So there is no change in oscillating frequency.

Therefore, for $+1\%$ change in the value of R, the oscillating frequency is **unchanged**.

Thus, the for $+1\%$ change in the value of R, the percentage change in the oscillation frequency is $\boxed{0\%}$

Step 1 of 5

(a)

Refer to Figure P17.5 in the textbook for a Wien-bridge oscillator circuit with a limiter used for amplitude control.

The loop gain of the circuit is,

$$L(s)=\left[1+\frac{R_2}{R_1}\right]\left(\frac{Z_p}{Z_p+Z_s}\right)$$

Simplify the expression further.

$$L(s)=\left[1+\frac{R_2}{R_1}\right]\left(\frac{1}{1+Z_sY_p}\right)$$

$$=\left[1+\frac{R_2}{R_1}\right]\left[\frac{1}{1+\left(R_s+\frac{1}{sC_s}\right)\left(\frac{1}{R_p}+sC_p\right)}\right]$$

$$=\left[1+\frac{R_2}{R_1}\right]\left[\frac{1}{1+\frac{(1+sR_sC_s)(1+sR_pC_p)}{sR_pC_s}}\right]$$

Step 2 of 5

Put $R_p=R_s$ and $C_p=C_s$ to simplify the function further.

$$L(s)=\left[1+\frac{R_2}{R_1}\right]\left[\frac{1}{1+\frac{(1+sR_sC_s)^2}{sR_sC_s}}\right]$$

$$=\left[1+\frac{R_2}{R_1}\right]\left[\frac{sR_sC_s}{sR_sC_s+1+(sR_sC_s)^2+2sR_sC_s}\right]$$

$$=\left[1+\frac{R_2}{R_1}\right]\left[\frac{sR_sC_s}{1+3sR_sC_s+(sR_sC_s)^2}\right]$$

$$=\left[1+\frac{R_2}{R_1}\right]\left[\frac{1}{\frac{1}{sR_sC_s}+3+sR_sC_s}\right]$$

Substitute $20.3\ \text{k}\Omega$ for R_2, $10\ \text{k}\Omega$ for R_1, $10\ \text{k}\Omega$ for R_s, and 16 nF for C_s.

$$L(s)=\left[1+\frac{20.3\ \text{k}\Omega}{10\ \text{k}\Omega}\right]\left[\frac{1}{\frac{1}{s(10\ \text{k}\Omega)(16\ \text{nF})}+3+s(10\ \text{k}\Omega)(16\ \text{nF})}\right]$$

$$=[1+2.03]\left[\frac{1}{\frac{1}{0.00016s}+3+0.00016s}\right]$$

$$=3.03\left[\frac{0.00016s}{1+0.00048s+0.0000000256s^2}\right]$$

Step 3 of 5

Set the loop gain function equal to 1, to find the location of the closed-loop poles of the system.

$$1=3.03\left[\frac{0.00016s}{1+0.00048s+0.0000000256s^2}\right]$$

$$\frac{1+0.00048s+0.0000000256s^2}{0.00016s}=3.03$$

$$1+0.00048s+0.0000000256s^2=0.0004848s$$

$$1-0.0000048s+0.0000000256s^2=0$$

Simplify and solve the equation to find the values for s which are the closed-loop poles.

$$s^2-187.5s+6250^2=0$$

The solutions are,

$$s=93.75\pm j6250$$

$$=\frac{10^5}{16}(0.015\pm j)$$

Thus, the closed loop poles of the Wien-bridge oscillator disregarding the limiter circuit are,

$$\boxed{\frac{10^5}{16}(0.015\pm j)}.$$

Step 4 of 5

(b)
Find the angular frequency of the Wien-bridge oscillator.

$$\omega_0=\frac{1}{CR}$$

Substitute $10\ \text{k}\Omega$ for R, and 16 nF for C.

$$\omega_0=\frac{1}{(16\ \text{nF})(10\ \text{k}\Omega)}$$

$$=\frac{1}{0.00016}$$

$$=6250\ \text{rad/s}$$

Find the frequency of oscillation of the Wien-bridge oscillator.

$$f_0=\frac{\omega_0}{2\pi}$$

Substitute 6250 rad/s for ω_0.

$$f_0=\frac{6250}{2\pi}$$

$$\approx 1000\ \text{Hz}$$

$$=1\ \text{kHz}$$

Thus, the frequency of oscillation, f_0 of the Wien-bridge oscillator is, $\boxed{1\ \text{kHz}}$.

Step 5 of 5

(c)

Refer to Figure 17.5 in the textbook for the Wien-bridge oscillator with output voltage limiter arrangement.

At the maximum output voltage, v_o, the voltage at node b is v_1+V_{D2}.

Consider that the voltage drop across the diode is 0.7 V. Therefore,

$$v_b=v_1+0.7\ \text{V}$$

The voltage, v_1 is about $\frac{1}{3}v_o$. Therefore,

$$v_b=\frac{v_o}{3}+0.7\ \text{V}$$

Neglect the current through the diode, D_2 to apply Kirchhoff's current law at node b.

$$\frac{v_o-v_b}{R_5}=\frac{v_b-(-15\ \text{V})}{R_6}$$

$$(v_o-v_b)R_6=(v_b+15)R_5$$

Substitute the expression for v_b.

$$\left(v_o-\frac{v_o}{3}-0.7\ \text{V}\right)R_6=\left(\frac{v_o}{3}+0.7\ \text{V}+15\right)R_5$$

$$\left(\frac{2v_o}{3}-0.7\ \text{V}\right)R_6=\left(\frac{v_o}{3}+15.7\right)R_5$$

Substitute $3\ \text{k}\Omega$ for R_6, and $1\ \text{k}\Omega$ for R_5 to find the value of v_o.

$$\left(\frac{2v_o}{3}-0.7\ \text{V}\right)(3\ \text{k}\Omega)=\left(\frac{v_o}{3}+15.7\right)(1\ \text{k}\Omega)$$

$$2000v_o-2100=333.4v_o+15700$$

$$1666.67v_o=17800$$

$$v_o=\frac{17800}{1667.67}$$

$$=10.68\ \text{V}$$

Find the peak-to-peak value of the output sine wave.

$$v_{o(p\text{-}p)}=2(10.68\ \text{V})$$

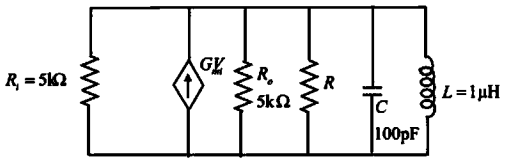
$$=21.36\ \text{V}$$

Thus, the amplitude of the output sine wave, v_o with the limiter in place is, $\boxed{21.36\ \text{V (peak-to-peak)}}$.

Step 1 of 2

Find the frequency of RLC oscillator circuit.

Draw the equivalent RLC circuit with specification.



The oscillating frequency at resonance is $\omega_0 = \frac{1}{\sqrt{LC}}$.

Substitute $L = 1\mu\text{H}$ and $C = 100\text{pF}$ in $\omega_0 = \frac{1}{\sqrt{LC}}$.

$$\omega_0 = \frac{1}{\sqrt{\left(1\times10^{-6}\right)\times\left(100\times10^{-12}\right)}}$$

$$\omega_0 = \frac{1}{\sqrt{10^{-6}\times10^{-10}}}$$

$$\omega_0 = \frac{1}{10^{-8}}$$

$$\omega_0 = 10^8 \text{ rad/s}$$

The oscillating frequency at resonance is $\boxed{\omega_0 = 10^8 \text{ rad/s}}$.

Step 2 of 2

The relation between the Q - factor and oscillating frequency is $\frac{\omega_0}{Q} = \frac{1}{RC}$.

$$R = \frac{Q}{\omega_0 C}$$

Find the value of R using above relation.

Substitute the values of $Q = 50, C = 100\text{pF}$ and $\omega_0 = 10^8 \text{ rad/s}$.

$$R = \frac{50}{\left(10^8\right)\times\left(100\times10^{-12}\right)}$$

$$R = \frac{50}{10^{-2}}$$

$$R = 5000$$

$$R = 5\text{k}\Omega$$

Oscillation will occur at $\omega_0 = 10^8 \text{ rad/s}$ when gain is unity.

Derive an expression unity gain using the equation. **(18.24)**.

$$G_m\left(R_i \| R \| R_o\right) = 1$$

$$G_m\left(5\text{k}\Omega \| 5\text{k}\Omega \| 5\text{k}\Omega\right) = 1$$

$$G_m\left(\frac{5000}{3}\right) = 1$$

$$G_m = \frac{3}{5000}$$

$$G_m = 0.6 \text{ mA/V}$$

Therefore the value of G_m is $\boxed{G_m = 0.6 \text{ mA/V}}$.

Step 1 of 2

(a)

Refer to Figure 17.6 in the textbook a Wien-bridge oscillator circuit diagram.

Represent the feedback resistor by R_2 and potentiometer resistance by R_1 in the circuit.

The condition to start oscillations is,

$$\frac{R_2}{R_1} = 2$$

Here, the value of the feedback resistance, R_2 is $10\ \text{k}\Omega$.

Substitute $10\ \text{k}\Omega$ for R_2 .

$$\frac{10\ \text{k}\Omega}{R_1} = 2$$

$$R_1 = \frac{10\ \text{k}\Omega}{2}$$
$$= 5\ \text{k}\Omega$$

Thus, the potentiometer adjusted such that, $\boxed{5\ \text{k}\Omega \text{ to ground}}$ for oscillations to just start.

Step 2 of 2

(b)

The frequency of oscillation is,

$$f = \frac{1}{2\pi RC}$$

Substitute $10\ \text{k}\Omega$ for R and 16 nF for C .

$$f = \frac{1}{2\pi(10\ \text{k}\Omega)(16\ \text{nF})}$$
$$= \frac{1}{0.001}$$
$$= 1000$$
$$= 1\ \text{kHz}$$

Thus, the frequency of oscillation, f is, $\boxed{1\ \text{kHz}}$.

Step 1 of 2

18.006P

Gain of the frequency selective network is $-12\,\mathrm{dB}$,

where negative sign indicates the loss in the gain of network.

Find the gain and phase shift of amplifier.

Conditions for oscillations are

1. Phase shift around the loop must be $0°$ or $360°$.

2. The oscillations start from $\left|A(j\omega_0)\beta(j\omega_0)\right| \geq 1$.

Where $\beta(j\omega_0)$ is gain of frequency selective network,

$A(j\omega_0)$ is gain of amplifier.

$$\left|A(j\omega_0)(-12)\right| \geq 1$$
$$12\left|A(j\omega_0)\right| \geq 1$$
$$\left|A(j\omega_0)\right| \geq \frac{1}{12}$$
$$\frac{-1}{12} < A(j\omega_0) < \frac{1}{12}$$

Therefore the minimum gain that the amplifier must have the oscillation to begin is $\boxed{\frac{-1}{12} < A(j\omega_0) < \frac{1}{12}}$.

Step 2 of 2

Phase shift of the frequency selective network is $180°$.

Phase shift around the loop must be $0°$ or $360°$.

Sum of the phases need to be $0°$ or $360°$.

Consider $0°$, then the phase shift of the amplifier is $-180°$.

Consider $360°$, then the phase shift of the amplifier is $180°$.

Hence the phase shift the amplifier is $\boxed{\pm 180°}$.

Step 1 of 6

Refer to Figure 17.8 in the textbook for a practical phase-shift oscillator with a limiter for amplitude stabilization. Break the feedback loop at X and omit the limiter circuit to draw the phase-shift oscillator circuit without the limiter.

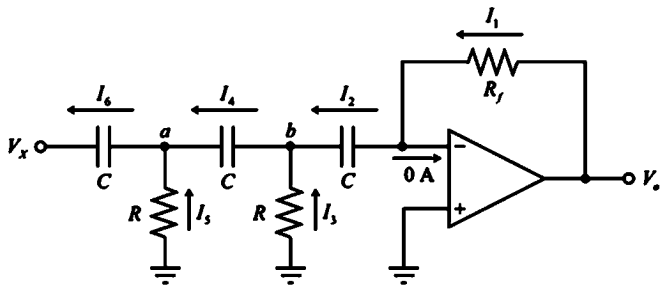


Figure 1

Step 2 of 6

Start at the output and work backward, to find the various currents and voltages in the circuit. Consider that the current flow in to the negative terminal of the op amp is zero.

Consider that the voltage at the negative terminal of the op amp as zero, since the positive terminal is grounded. Find the current, I_1.

$$I_1 = \frac{V_o}{R_f}$$

Find the voltage at node b.

$$\begin{aligned} v_b &= 0 - \frac{I_2}{sC} \\ &= -\frac{I_1}{sC} \left(\text{since, } I_2 = I_1\right) \\ &= \frac{-1}{sC}\left(\frac{V_o}{R_f}\right) \\ &= -\frac{V_o}{sCR_f} \end{aligned}$$

Find the current, I_3.

$$\begin{aligned} I_3 &= \frac{0 - v_b}{R} \\ &= \frac{1}{R}\left(\frac{V_o}{sCR_f}\right) \\ &= \frac{V_o}{sCR_fR} \end{aligned}$$

Step 3 of 6

Find the voltage at node a.

$$\begin{aligned} v_a &= v_b - \frac{I_4}{sC} \\ &= -\frac{V_o}{sCR_f} - \frac{1}{sC}(I_3 + I_2) \\ &= -\frac{V_o}{sCR_f} - \frac{1}{sC}\left(\frac{V_o}{sCR_fR} + \frac{V_o}{R_f}\right) \end{aligned}$$

Find the current, I_5.

$$\begin{aligned} I_5 &= \frac{0 - v_a}{R} \\ &= \frac{1}{R}\left[\frac{V_o}{sCR_f} + \frac{1}{sC}\left(\frac{V_o}{sCR_fR} + \frac{V_o}{R_f}\right)\right] \\ &= \frac{V_o}{sCR_fR} + \frac{1}{sCR}\left(\frac{V_o}{sCR_fR} + \frac{V_o}{R_f}\right) \end{aligned}$$

Step 4 of 6

Use the Kirchhoff's current law to find the current in the circuit.

$$\begin{aligned} I &= I_6 \\ &= I_5 + I_4 \\ &= I_5 + I_3 + I_2 \\ &= I_5 + I_3 + I_1 \end{aligned}$$

Substitute the expressions for I_5, I_3, **and** I_1 in the expression of current, I.

$$\begin{aligned} I &= \left[\frac{V_o}{sCR_fR} + \frac{1}{sCR}\left(\frac{V_o}{R_f} + \frac{V_o}{sCR_fR}\right)\right] + \frac{V_o}{sCR_fR} + \frac{V_o}{R_f} \\ &= \frac{V_o}{sCR_fR} + \left(\frac{1}{sCR}\right)\frac{V_o}{R_f} + \left(\frac{1}{sCR}\right)\frac{V_o}{sCR_fR} + \frac{V_o}{sCR_fR} + \frac{V_o}{R_f} \\ &= \frac{V_o}{sCR_fR}\left(1 + \frac{1}{sCR} + 1\right) + \frac{V_o}{R_f}\left(1 + \frac{1}{sCR}\right) \\ &= \frac{V_o}{sCR_fR}\left(2 + \frac{1}{sCR}\right) + \frac{V_o}{R_f}\left(1 + \frac{1}{sCR}\right) \end{aligned}$$

Step 5 of 6

Find the voltage, V_x.

$$\begin{aligned} V_x &= v_a - \frac{I_6}{sC} \\ &= v_a - \frac{I}{sC} \end{aligned}$$

Substitute the expressions for v_a, **and** I.

$$\begin{aligned} V_x &= \left\{\begin{array}{l} -\frac{V_o}{sCR_f} - \frac{1}{sC}\left(\frac{V_o}{sCR_fR} + \frac{V_o}{R_f}\right) \\ -\frac{1}{sC}\left[\frac{V_o}{sCR_fR}\left(2 + \frac{1}{sCR}\right) + \frac{V_o}{R_f}\left(1 + \frac{1}{sCR}\right)\right] \end{array}\right\} \\ &= \left\{\begin{array}{l} -\frac{V_o}{sCR_f} - \frac{V_o}{sCR_f}\left(\frac{1}{sCR}\right) - \frac{V_o}{sCR_f} \\ -\left(\frac{1}{sCR}\right)\left(\frac{V_o}{sCR_f}\right)\left(2 + \frac{1}{sCR}\right) - \frac{V_o}{sCR_f}\left(1 + \frac{1}{sCR}\right) \end{array}\right\} \\ &= -\frac{V_o}{sCR_f}\left[1 + \frac{1}{sCR} + 1 + \frac{1}{sCR}\left(2 + \frac{1}{sCR}\right) + \left(1 + \frac{1}{sCR}\right)\right] \\ &= -\frac{V_o}{sCR_f}\left[3 + \frac{4}{sCR} + \frac{1}{s^2C^2R^2}\right] \end{aligned}$$

Step 6 of 6

Modify the equation into $j\omega$ term.

$$V_x(j\omega) = -\frac{V_o(j\omega)}{j\omega CR_f}\left[3 + \frac{4}{j\omega CR} - \frac{1}{\omega^2C^2R^2}\right]$$

Simplify the expression further to find the loop gain.

$$\begin{aligned} A\beta &= \frac{V_o(j\omega)}{V_x(j\omega)} \\ &= -\frac{j\omega CR_f}{\left[3 + \frac{4}{j\omega CR} - \frac{1}{\omega^2C^2R^2}\right]} \end{aligned}$$

Simplify the expression further.

$$\begin{aligned} A\beta &= -\frac{j\omega CR_f}{\frac{1}{j\omega CR}\left[3j\omega CR + 4 - \frac{j}{\omega CR}\right]} \\ &= \frac{\omega^2C^2RR_f}{4 + j\left(3\omega CR - \frac{1}{\omega CR}\right)} \end{aligned}$$

Thus, the loop gain, $A\beta$ of the phase-shift oscillator without the limiter is, $\boxed{\frac{\omega^2C^2RR_f}{4 + j\left(3\omega CR - \frac{1}{\omega CR}\right)}}$.

18.007P

We don't have the solution to this problem yet.

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Step 1 of 5

Refer to Figure 17.8 in the textbook for a practical phase-shift oscillator.

Consider the following expression for the loop gain.

$$A\beta \equiv \frac{\omega^2 C^2 R R_f}{4 + j\left(3\omega CR - \frac{1}{\omega CR}\right)}$$

This oscillator circuit produces steady-state oscillations when the following Barkhausen stability conditions are satisfied.

1) The magnitude of the loop gain is unity.

$$|A\beta| = 1$$

2) The phase of the loop gain equals to $0°$ or $360°$.

$$\angle\beta = 0° \text{ or } 360°$$

Step 2 of 5

Use the second condition of Barkhausen criterion to find the frequency of oscillation.

$$\angle A\beta = 0°$$

$$\angle \frac{\omega_0^2 C^2 R R_f}{4 + j\left(3\omega_0 CR - \frac{1}{\omega_0 CR}\right)} = 0°$$

$$0° - \tan^{-1}\left(\frac{3\omega_0 CR - \frac{1}{\omega_0 CR}}{4}\right) = 0°$$

$$\frac{3\omega_0 CR - \frac{1}{\omega_0 CR}}{4} = \tan(0°)$$

Simply the expression to get an expression for ω_0.

$$3\omega_0 CR - \frac{1}{\omega_0 CR} = 0$$

$$3\omega_0 CR = \frac{1}{\omega_0 CR}$$

$$3(\omega_0 CR)^2 = 1$$

$$(\omega_0 CR)^2 = \frac{1}{3}$$

$$\omega_0 CR = \frac{1}{\sqrt{3}}$$

$$\omega_0 = \frac{1}{CR\sqrt{3}}$$

Step 3 of 5

Substitute $10\ \text{k}\Omega$ for R, 16 nF for C.

$$\omega_0 = \frac{1}{(16\ \text{nF})(10\ \text{k}\Omega)\sqrt{3}}$$
$$= \frac{1}{2.77 \times 10^{-4}}$$
$$= 3.6 \times 10^3\ \text{rad/s}$$
$$= 3.6\ \text{krad/s}$$

Calculate the frequency of oscillation, f_0.

$$f_0 = \frac{\omega_0}{2\pi}$$
$$= \frac{3.6\ \text{krad/s}}{2\pi}$$
$$= 574.3\ \text{Hz}$$

Thus, the frequency of oscillation of the circuit f_0 is $\boxed{574.3\ \text{Hz}}$.

Step 4 of 5

Use the first condition of Barkhausen criterion to find minimum required value for R_f.

$$|A\beta| \geq 1$$

$$\left|\frac{\omega_0^2 C^2 R R_f}{4 + j\left(3\omega_0 CR - \frac{1}{\omega_0 CR}\right)}\right| \geq 1$$

Substitute the expression for ω_0.

$$\left|\frac{\left(\frac{1}{CR\sqrt{3}}\right)^2 C^2 R R_f}{4 + j\left(3\left(\frac{1}{CR\sqrt{3}}\right)CR - \frac{1}{\left(\frac{1}{CR\sqrt{3}}\right)CR}\right)}\right| \geq 1$$

Step 5 of 5

Simplify the expression to get an expression for R_f.

$$\left|\frac{\frac{1}{3}\left(\frac{R_f}{R}\right)}{4 + j\left(\sqrt{3} - \sqrt{3}\right)}\right| \geq 1$$

$$\frac{1}{3}\left(\frac{R_f}{R}\right) \geq 4$$

$$R_f \geq 12R$$

Substitute $10\ \text{k}\Omega$ for R to find R_f.

$$R_f = 12R$$
$$= 12(10\ \text{k}\Omega)$$
$$= 120\ \text{k}\Omega$$

Thus, the minimum required value of R_f for oscillations to start in the circuit is $\boxed{120\ \text{k}\Omega}$.

Step 1 of 3

Negative limiting level when R_f is removed is $L_- = -V\frac{R_3}{R_2} - V_D\left(1+\frac{R_3}{R_2}\right)$.

Where L_- is negative limiting level,

V is supply voltage,

V_D is voltage drop of the conducting diode.

Substitute $L_- = 3, V_D = -0.7$ and $V = -5$ in $L_- = -V\frac{R_3}{R_2} - V_D\left(1+\frac{R_3}{R_2}\right)$.

$$3 = -(-5)\frac{R_3}{R_2} - (-0.7)\left(1+\frac{R_3}{R_2}\right)$$

$$3 = 5\frac{R_3}{R_2} + 0.7\left(1+\frac{R_3}{R_2}\right)$$

$$3 = 5.7\frac{R_3}{R_2} + 0.7$$

$$5.7\frac{R_3}{R_2} = 2.3$$

$$\frac{R_3}{R_2} = \frac{2.3}{5.7}$$

$$\frac{R_3}{R_2} = 0.403$$

Step 2 of 3

From the property of symmetry $\frac{R_3}{R_2} = \frac{R_4}{R_3}$.

Assume $R_2 = R_3 = 10\,\text{k}\Omega$.

$$\begin{aligned} R_3 &= 0.403R_2 \\ &= 0.403(10000) \\ &= 4030\,\Omega \\ &\approx 4\,\text{k}\Omega \end{aligned}$$

Thus from the property of symmetry $R_4 = 4\,\text{k}\Omega$.

Step 3 of 3

From the Fig. $18.4(c)$, slope of limiting characteristics when R_f removed is $-\frac{R_4}{R_1}$.

$$-\frac{R_4}{R_1} = 0.05$$

Here negative sign indicates increasing or decreasing of a slope.

$$\frac{R_4}{R_1} = 0.05$$

$$R_1 = \frac{R_4}{0.05}$$

$$R_1 = \frac{4000}{0.05}$$

$$R_1 = 80\,\text{k}\Omega$$

Therefore the resistor values are $\boxed{R_1 = 80\,\text{k}\Omega, R_2 = 10\,\text{k}\Omega, R_3 = 4\,\text{k}\Omega, R_4 = 4\,\text{k}\Omega, R_3 = 10\,\text{k}\Omega}$.

Step 1 of 2

18.009E

Refer to Figure 13.11 in the textbook for a practical implementation of the active-filter-tuned oscillator.

Consider that the circuit produces 1 kHz sinewaves.

The expression for the frequency of oscillation is,

$$f_0 = \frac{1}{2\pi RC}$$

Find the value of resistance.

$$R = \frac{1}{2\pi f_0 C}$$

Substitute 1 kHz for f_0 and 16 nF for C.

$$\begin{aligned} R &= \frac{1}{2\pi(1000)\left(16\times10^{-9}\right)} \\ &= \frac{1}{1\times10^{-4}} \\ &= 10\ \text{k}\Omega \end{aligned}$$

Thus, the required value of resistance R is $\boxed{10\ \text{k}\Omega}$.

Step 2 of 2

The output of the resonator is nothing but the diode drop.

$$V_r = 0.7\ \text{V}$$

Note that the output of the circuit directly proportional to (twice as large as) the voltage across the resonator. Therefore,

$$\begin{aligned} V_o &= 2V_r \\ &= 2(0.7\ \text{V}) \\ &= 1.4\ \text{V} \end{aligned}$$

The peak-to-peak output voltage is,

$$\begin{aligned} V_{o(p\text{-}p)} &= 2V_o \\ &= 2(1.4\ \text{V}) \\ &= 2.8\ \text{V} \end{aligned}$$

The fundamental component of peak-to-peak output voltage is,

$$\begin{aligned} v_o &= \frac{4V_{o(p\text{-}p)}}{\pi} \\ &= \frac{4(2.8\ \text{V})}{\pi} \\ &= 3.6\ \text{V} \end{aligned}$$

Thus, the peak-to-peak amplitude of output voltage v_o is $\boxed{3.6\ \text{V}}$.

Step 1 of 4

The characteristics curve when R_f is removed is shown in the Fig. $18.4(c)$.

A dc source v_B is connected to the virtual ground through the resistor R_B.

Current flows into the node is $\frac{v_B}{R_B}$.

To compensate, v_I must be moved by small amount in the direction opposite to v_B to produce a current $\frac{v_I}{R_1}=\frac{-v_B}{R_B}$.

Thus, $v_I=\frac{-R_1}{R_B}v_B$.

Therefore the characteristics curve moved to left with an amount $v_I=\frac{-R_1}{R_B}v_B$.

The figure 18.9 gives the characteristic curve for the given condition.

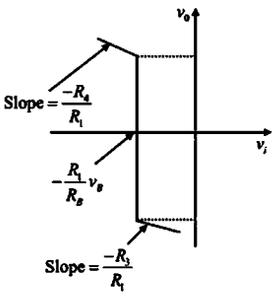


Figure 18.9

Step 2 of 4

Input resistance of the comparator is $R_1=100\ \text{k}\Omega$

Slope of limiting region is $\frac{R_4}{R_1}\le 0.05$.

$R_4\le 0.05R_1$

$R_4\le 0.05(100000)$

$R_4\le 5000$

$R_4\le 5k\Omega$

From the appendix J, the value of resistor is $R_4=4.7\ \text{k}\Omega$

From the property of symmetry $\frac{R_3}{R_2}=\frac{R_4}{R_5}$.

Therefore $R_5=4.7\ \text{k}\Omega$.

Step 3 of 4

Consider the equation (18.8) when R_f is removed is $L_-=-V\frac{R_3}{R_2}-V_D\left(1+\frac{R_3}{R_2}\right)$.

Assume voltage drop of the conducting diode $V_D=0$.

$L_-=-V\frac{R_3}{R_2}$.

Substitute $L_-=3$ and $V=5$ in $L_-=-V\frac{R_3}{R_2}$.

$3=-5\frac{R_3}{R_2}$

Here negative sign indicates increasing or decreasing of a slope.

$3=5\frac{R_3}{R_2}$

$\frac{R_3}{R_2}=\frac{3}{5}$

$\frac{R_3}{R_2}=0.6$

Simplify further.

$R_2=\frac{R_3}{0.6}$

$R_2=\frac{4.7\text{k}\Omega}{0.6}$

$R_2=\frac{4700}{0.6}$

$R_2=7833.33\Omega$

Step 4 of 4

From the appendix J, the value of resistor is $R_2=7.5\ \text{k}\Omega$.

From the property of symmetry $R_5=7.5\text{k}\Omega$.

The Threshold of comparator is $v_I=2\ \text{V}$.

Characteristics curve moved to left with an amount $v_I=\frac{-R_1}{R_B}v_B$.

$2=\frac{-R_1}{R_B}(-5)$

$R_B=\frac{5R_1}{2}$

$R_B=2.5R_1$

$R_B=2.5(100\text{k})$

$R_B=250k$

From the appendix J, the value of resistor is $R_B=240\ \text{k}\Omega$.

Therefore the value of resistors are,

$\boxed{R_1=100\text{k}\Omega, R_2=7.5\text{k}\Omega, R_3=4.7\text{k}\Omega, R_4=4.7\text{k}\Omega, R_5=7.5\text{k}\Omega, R_B=240\text{k}\Omega}$.

18.010E

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Step 1 of 11

(a)

Refer to Figure P17.8 in the text book.

Consider the cases $v_I \geq V_R$ and $v_I < V_R$.

Determine the output voltage expression for $v_I \geq V_R$.

The diode Z_1 is in forward bias and the diode Z_2 is in reverse bias.

Replace the diode Z_1 with 0.7 V and replace the diode Z_2 with the zener voltage V_{Z2}.

Draw the equivalent circuit.

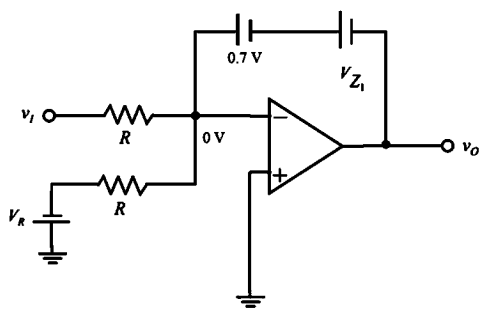


Figure 1

Step 2 of 11

Apply Kirchhoff's voltage law from the output terminal of op amp to the virtual ground.

$$-v_o - V_{Z1} - 0.7 = 0$$
$$v_o = -(V_{Z1} + 0.7)$$

Thus, the output voltage v_o expression for $v_I \geq V_R$ is $-(V_{Z1} + 0.7)$.

Step 3 of 11

Determine the output voltage expression for $v_I < V_R$.

The diode Z_2 is in forward bias and the diode Z_1 is in reverse bias.

Replace the diode Z_2 with 0.7 V and replace the diode Z_1 with the zener voltage V_{Z1}.

Draw the equivalent circuit.

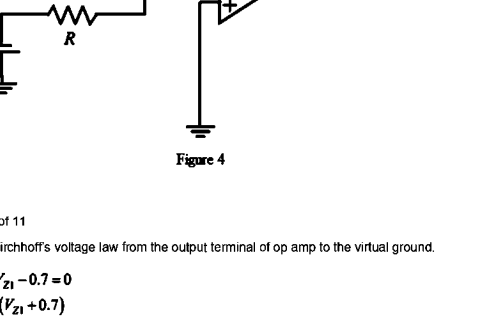


Figure 2

Step 4 of 11

Apply Kirchhoff's voltage law from the output terminal of op amp to the virtual ground.

$$-v_o + 0.7 + V_{Z2} = 0$$
$$v_o = V_{Z2} + 0.7$$

Thus, the output voltage v_o expression for $v_I < V_R$ is $V_{Z2} + 0.7$.

Step 5 of 11

Draw the transfer characteristics.

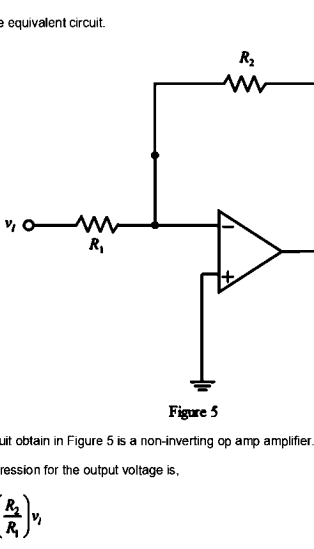


Figure 3

Step 6 of 11

(b)
Refer to Figure P17.8 (b) in the text book.

Consider the cases $v_I < -V_{Z2}$, $-V_{Z2} \leq v_I < V_{Z1}$ and $v_I \geq V_{Z1}$.

Determine the output voltage expression for $v_I < -V_{Z2}$.

The diode Z_1 is in forward bias and the diode Z_2 is in reverse bias.

Replace the diode Z_1 with 0.7 V and replace the diode Z_2 with the zener voltage V_{Z2}.

Draw the equivalent circuit.

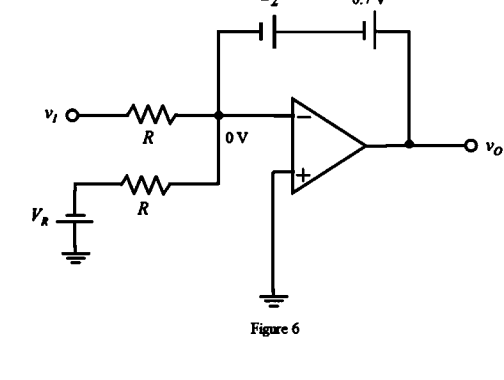


Figure 4

Step 7 of 11

Apply Kirchhoff's voltage law from the output terminal of op amp to the virtual ground.

$$-v_o - V_{Z1} - 0.7 = 0$$
$$v_o = -(V_{Z1} + 0.7)$$

Thus, the output voltage v_o expression for $v_I \geq V_R$ is $-(V_{Z1} + 0.7)$.

Step 8 of 11

Determine the output voltage for $-V_{Z2} \leq v_I < V_{Z1}$.

The diode Z_1 is in reverse bias, but not in break down condition and the diode.

Replace the diode Z_1 with open circuit.

Draw the equivalent circuit.

Figure 5

The circuit obtain in Figure 5 is a non-inverting op amp amplifier.

The expression for the output voltage is,

$$v_o = -\left(\frac{R_2}{R_1}\right) v_I$$

Step 9 of 11

Determine the output voltage expression for $v_I \geq V_{Z1}$.

The diode Z_2 is in forward bias and the diode Z_1 is in reverse bias.

Replace the diode Z_2 with 0.7 V and replace the diode Z_1 with the zener voltage V_{Z1}.

Draw the equivalent circuit.

Figure 6

Step 10 of 11

Apply Kirchhoff's voltage law from the output terminal of op amp to the virtual ground.

$$-v_o + 0.7 + V_{Z2} = 0$$
$$v_o = V_{Z2} + 0.7$$

Thus, the output voltage v_o expression for $v_I < V_R$ is $V_{Z2} + 0.7$.

Step 11 of 11

Draw the transfer characteristics.

Figure 7

18.011E

We don't have the solution to this problem yet.

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Step 1 of 4

Refer to Figure 17.4 in the text book.

Write the expression for the transfer function of the feedback network.

$$\frac{V_o(s)}{V_o(s)} = \frac{Z_P}{Z_P + Z_S}$$

Substitute $\frac{1}{SC} \| R$ for Z_P and $\left(\frac{1}{SC} + R\right)$ for Z_S.

$$\begin{aligned}\frac{V_o(s)}{V_o(s)} &= \frac{\frac{1}{SC} \| R}{\left(\frac{1}{SC} \| R\right) + \frac{1}{SC} + R} \\ &= \frac{\frac{R}{SC} \Big/ \frac{1}{SC} + R}{\left(\frac{R}{SC} \Big/ \frac{1}{SC} + R\right) + \left(\frac{1}{SC} + R\right)} \\ &= \frac{\frac{R}{SC}}{\frac{R}{SC} + \left(\frac{1}{SC} + R\right)^2}\end{aligned}$$

Simplify further.

$$\begin{aligned}\frac{V_o(s)}{V_o(s)} &= \frac{\frac{R}{SC}}{\frac{R}{SC} + \frac{1}{S^2C^2} + \frac{2R}{SC} + R^2} \\ &= \frac{\frac{R}{SC}}{\frac{RSC + 1 + 2RSC + R^2S^2C^2}{S^2C^2}} \\ &= \frac{SRC}{S^2R^2C^2 + 3SRC + 1}\end{aligned}$$

Step 2 of 4

Divide the numerator and denominator with R^2C^2.

$$\frac{V_o(S)}{V_o(S)} = \frac{\frac{1}{RC}S}{S^2 + 3S\frac{1}{RC} + \frac{1}{R^2C^2}}$$

Compare the transfer function obtain with the expression for the standard transfer function

$$\frac{V_o(S)}{V_o(S)} = \frac{K\frac{\omega_o}{Q}S}{S^2 + S\frac{\omega_o}{Q} + \omega_o^2}.$$

$$\omega_o^2 = \frac{1}{R^2C^2}$$

$$\omega_o = \frac{1}{RC}$$

The expression for the center frequency ω_o is $\boxed{\frac{1}{RC}}$.

Step 3 of 4

Determine the value for the quality factor Q.

$$\frac{\omega_o}{Q} = \frac{3}{RC}$$

$$Q = \frac{\omega_o}{\frac{3}{RC}}$$

$$Q = \frac{\frac{1}{RC}}{\frac{3}{RC}}$$

$$Q = \frac{1}{3}$$

Thus, the value of Q is $\boxed{\frac{1}{3}}$.

Step 4 of 4

Determine the gain at the center frequency.

Substitute $\omega_o = \frac{1}{RC}$ in the expression of $\frac{V_o(S)}{V_o(S)}$.

$$\begin{aligned}\frac{V_o(S)}{V_o(S)} &= \frac{\frac{1}{RC}S}{S^2 + 3S\frac{1}{RC} + \frac{1}{R^2C^2}} \\ &= \frac{\frac{1}{RC}(j\omega_o)}{(j\omega_o)^2 + 3(j\omega_o)\frac{1}{RC} + \frac{1}{R^2C^2}} \\ &= \frac{\frac{1}{RC}\left(\frac{j}{RC}\right)}{\left(\frac{j}{RC}\right)^2 + 3\left(\frac{j}{RC}\right)\frac{1}{RC} + \frac{1}{R^2C^2}} \\ &= \frac{\frac{j}{R^2C^2}}{\frac{j^2}{R^2C^2} + 3\frac{j}{R^2C^2} + \frac{1}{R^2C^2}}\end{aligned}$$

Simplify further.

$$\begin{aligned}\frac{V_o(S)}{V_o(S)} &= \frac{\frac{j}{R^2C^2}}{\frac{-1}{R^2C^2} + 3\frac{j}{R^2C^2} + \frac{1}{R^2C^2}} \qquad \left(\text{Since, } j^2 = -1\right) \\ &= \frac{\frac{j}{R^2C^2}}{3\frac{j}{R^2C^2}} \\ &= \frac{1}{3}\end{aligned}$$

Therefore, the center frequency gain is $\boxed{\frac{1}{3}}$.

Step 1 of 3

Find the value of C and minimum value of g_m in cross coupled oscillator circuit.

The oscillating frequency at resonance is $\omega_0 = \frac{1}{\sqrt{LC}}$.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$
$$C = \frac{1}{L(\omega_0)^2}$$

Substitute $L = 10\,\text{nH}$ and $\omega_0 = 10\,\text{G rad/s}$ in $C = \frac{1}{L(\omega_0)^2}$.

$$C = \frac{1}{10\times10^{-9}\left(10\times10^{9}\right)^2}$$
$$= \frac{1}{10^{-8}\left(10^{10}\right)^2}$$
$$= \frac{1}{10^{-8}\times10^{20}}$$
$$= 1\,\text{pF}$$

The value of C is $\boxed{1\,\text{pF}}$.

Step 2 of 3

The relation between the Q factor and oscillating frequency is $\frac{\omega_o}{Q} = \frac{1}{RC}$.

$$R = \frac{Q}{\omega_0 C}$$

Find the value of R using above relation.

Substitute the values of $Q = 10, C = 1\,\text{pF}$ and $\omega_0 = 10^{10}\ \text{rad/sec}$.

$$R = \frac{10}{\left(10^{10}\right)\times\left(1\times10^{-12}\right)}$$
$$= \frac{10}{10^{-2}}$$
$$= 1000$$
$$= 1\,\text{k}\Omega$$

Step 3 of 3

Oscillation will occur at $\omega_o = 10^{10}\ \text{rad/sec}$ when gain is unity.

Use the equation (18.24)

$$G_m\left(R \| r_o\right) = 1$$
$$G_m\left(10\,\text{k}\Omega \| 1\,\text{k}\Omega\right) = 1$$
$$G_m\left(\frac{\left(10\times10^{3}\right)\times\left(1\times10^{3}\right)}{\left(10\times10^{3}\right)+\left(1\times10^{3}\right)}\right) = 1$$
$$G_m\left(\frac{10000}{11}\right) = 1$$
$$G_m = \frac{11}{10000}$$
$$= 1.1\,\text{mA/V}$$

Therefore the value of G_m is $\boxed{1.1\,\text{mA/V}}$.

Step 1 of 3

The loop gain of the wien-bridge oscillator from the equation (18.11) is, $L(j\omega)=\dfrac{1+\dfrac{R_2}{R_1}}{3+j\left(\omega CR-\dfrac{1}{\omega CR}\right)}$

Find the phase shift of the loop.

$$\phi(\omega)=-\tan^{-1}\left(\frac{\text{Im}}{\text{Re}}\right)$$
$$=-\tan^{-1}\left(\frac{\omega CR-\dfrac{1}{\omega CR}}{3}\right)$$

Differentiate on each side with respect to ω.

$$\frac{d\phi(\omega)}{d\omega}=\frac{-1}{1+\left(\dfrac{\omega CR-\dfrac{1}{\omega CR}}{3}\right)^2}\frac{d}{d\omega}\left(\frac{\omega CR-\frac{1}{\omega CR}}{3}\right)\quad\left(\text{Since }\frac{d}{dx}\left(\tan^{-1}t\right)=\frac{1}{1+t^2}\frac{dt}{dx}\right)$$

$$\frac{d\phi(\omega)}{d\omega}=\frac{-1}{1+\left(\dfrac{\omega CR-\dfrac{1}{\omega CR}}{3}\right)^2}\frac{1}{3}\left(CR+\frac{1}{\omega^2 CR}\right)$$

Step 2 of 3

Evaluate the $\dfrac{d\phi(\omega)}{d\omega}$ when $\omega=\dfrac{1}{CR}$.

$$\left(\frac{d\phi(\omega)}{d\omega}\right)_{\omega=\frac{1}{CR}}=\frac{-1}{1+\left(\dfrac{1-\frac{1}{1}}{3}\right)^2}\frac{1}{3}\left(CR+\frac{1}{\left(\frac{1}{CR}\right)^2 CR}\right)$$
$$=\frac{-1}{1+0}\frac{1}{3}\left(CR+\frac{1}{\left(\frac{1}{CR}\right)}\right)$$
$$=\frac{-1}{3}(CR+CR)$$
$$=\frac{-2}{3}CR$$
$$=\frac{-2}{3\omega}\qquad\left(\text{Since }\omega=\frac{1}{CR}\right)$$

Step 3 of 3

Change in oscillator frequency with respect to phase shift is $\Delta\omega=\dfrac{\Delta\phi}{\dfrac{d\phi}{d\omega}}$.

$$\Delta\omega=\frac{-0.0523}{-\frac{2}{3\omega}}\qquad(\text{Since }1^\circ=0.01745\,\text{rad})$$
$$=0.0785\omega$$

Therefore the frequency of oscillation is $\omega'=\omega+\Delta\omega$.

$$\omega'=\omega+0.0785\omega$$
$$=1.0785\omega$$
$$=\frac{1.0785}{RC}$$

The frequency of oscillation is $\boxed{\omega'=\dfrac{1.0785}{RC}}$.

Step 1 of 3

Consider a 2 MHz quartz crystal.

Find the series resonant frequency.

$$\omega_s = \frac{1}{\sqrt{LC_s}}$$

Here, L is the crystal inductance, C_s is the series capacitance of crystal.

Substitute 0.52 H for L, and 0.012 pF for C_s.

$$\omega_s = \frac{1}{\sqrt{(0.52\text{ H})(0.012\text{ pF})}}$$
$$= \frac{1}{\sqrt{0.00624\times10^{-12}}}$$
$$= 12.66\text{ Mrad/s}$$

Find the frequency.

$$\omega_s = 2\pi f_s$$
$$12.66\text{ Mrad/s} = 2\pi f_s$$

$$f_s = \frac{12.66\times10^6}{2\pi}$$
$$= 2.015\text{ MHz}$$

Thus, the series resonant frequency, f_s is, $\boxed{2.015\text{ MHz}}$.

Step 2 of 3

Find the parallel resonant frequency.

$$\omega_p = \frac{1}{\sqrt{L\left(\frac{C_sC_p}{C_s+C_p}\right)}}$$

Here, the parallel capacitance is C_p.

Substitute 0.52 H for L, 0.012 pF for C_s and 4 pF for C_p.

$$\omega_p = \frac{1}{\sqrt{(0.52)\left(\frac{(0.012\text{ pF})(4\text{ pF})}{0.012\text{ pF}+4\text{ pF}}\right)}}$$
$$= \frac{1}{\sqrt{(0.52\text{ H})\left(\frac{0.048\times10^{-24}}{4.012\times10^{-12}}\right)}}$$
$$= \frac{1}{\sqrt{(6.22\times10^{-15})}}$$
$$= 12.68\text{ Mrad/s}$$

Find the frequency.

$$\omega_p = 2\pi f_p$$
$$12.68\text{ Mrad/s} = 2\pi f_p$$

$$f_p = \frac{12.68\text{ Mrad/s}}{2\pi}$$
$$= 2.018\text{ MHz}$$

Thus, the parallel resonant frequency, f_p is, $\boxed{2.018\text{ MHz}}$.

Step 3 of 3

Find the Q factor of quartz crystal.

$$Q = \frac{\omega_o L}{r}$$
$$\frac{\omega_o L}{r} \approx \frac{\omega_s L}{r}$$
$$Q = \frac{\omega_s L}{r}$$

Here, r is the resistance of crystal.

Substitute 12.66 Mrad/s for ω_s, 0.52 H for L, and $120\ \Omega$ for r.

$$Q = \frac{(12.66\text{ Mrad/s})(0.52\text{ H})}{120\ \Omega}$$
$$= \frac{6.5832\times10^6}{120}$$
$$= 54860$$

Therefore, the Q factor of quartz crystal is, $\boxed{54860}$.

Step 1 of 2

Refer to Figure 17.4 in the textbook.

Refer to equation 17.10 in the text book.

The expression for the loop gain is,

$$L(s)=\frac{1+\frac{R_2}{R_1}}{3+sRC+\frac{1}{sRC}}$$

The poles of the closed loop system are the roots of the characteristic equation $1-L(s)=0$.

$$1-\left(\frac{1+\frac{R_2}{R_1}}{3+sRC+\frac{1}{sRC}}\right)=0$$

$$1=\frac{1+\frac{R_2}{R_1}}{3+sRC+\frac{1}{sRC}}$$

$$3+sRC+\frac{1}{sRC}=1+\frac{R_2}{R_1}$$

$$3-1+sRC+\frac{1}{sRC}-\frac{R_2}{R_1}=0$$

Simplify further.

$$2s+s^2RC+\frac{1}{RC}-s\frac{R_2}{R_1}=0$$

$$s^2RC+s\left(2-\frac{R_2}{R_1}\right)+\frac{1}{RC}=0$$

$$s^2+\frac{s}{RC}\left(2-\frac{R_2}{R_1}\right)+\frac{1}{R^2C^2}=0$$

$$s^2+s\omega_o\left(2-\frac{R_2}{R_1}\right)+\frac{1}{R^2C^2}=0 \quad \text{..... (1)}$$

Step 2 of 2

Write the expression of the characteristic equation of the standard characteristic equation.

$$s^2+s\frac{\omega_o}{Q}+\omega_o^2=0 \quad \text{..... (2)}$$

Compare the equations (1) and (2).

$$Q=\frac{1}{2-\frac{R_2}{R_1}}$$

$$\frac{1}{Q}=2-\frac{R_2}{R_1}$$

Thus, the expression for the quality factor is, $\boxed{Q=\frac{1}{2-\frac{R_2}{R_1}}}$.

The poles lie on the imaginary axis for $Q=\infty$.

$$\frac{1}{\infty}=2-\frac{R_2}{R_1}$$

$$2-\frac{R_2}{R_1}=0$$

$$\frac{R_2}{R_1}=2$$

Thus, the condition for the poles to locate in the right half of the *s* plane is $\boxed{\frac{R_2}{R_1}>2}$.

Refer to Figure 17.19 (a) in the textbook.

Consider that the op-amp has the output saturation voltages of $L_{\pm} = \pm 13\ \text{V}$.

If the circuit is in L_+ state, it can be switched to L_- state by applying an input, v_I greater than, $V_{TH} = \beta L_+$. Such an input causes a net negative voltage to appear between the input terminals of the op amp, which initiates the negative cycle that culminates in the circuit switching to L_- stable state. Here it is important to note that the input, v_I merely initiates or triggers regeneration. Thus, it is possible to remove v_I with no effect on the regeneration process. In other words, v_I can be simply a pulse of short duration. The input signal v_I is thus referred to as a trigger signal, or simply a trigger. Therefore,

$$\begin{aligned} V_{TH} &= \beta L_+ \\ &= \frac{R_1}{R_1 + R_2} L_+ \end{aligned}$$

Substitute $10\ \text{k}\Omega$ for R_1 , 5 V for V_{TH} , and 13 V for L_+ to find the value of R_2 .

$$5\ \text{V} = \frac{10\ \text{k}\Omega}{10\ \text{k}\Omega + R_2}(13\ \text{V})$$

$$\frac{10\ \text{k}\Omega}{10\ \text{k}\Omega + R_2} = \frac{5\ \text{V}}{13\ \text{V}}$$

$$10\ \text{k}\Omega + R_2 = \frac{10\ \text{k}\Omega}{0.384}$$

$$\begin{aligned} R_2 &= 26\ \text{k}\Omega - 10\ \text{k}\Omega \\ &= 16\ \text{k}\Omega \end{aligned}$$

Therefore, the value of resistance, R_2 is, $\boxed{16\ \text{k}\Omega}$.

Step 1 of 3

At positive peak of the Fig. **18.6**.

$$v_0 = \left(1 + \frac{R_2}{R_s}\right) v_i \quad \text{..... (1)}$$

$$v_0 - \left[\frac{R_5}{R_5 + R_6}(v_0 + 15)\right] - 0.7 = v_i \quad \text{..... (2)}$$

Consider the values of resistors from the Fig.18.6.

$R_2 = 20.3\,\text{k}\Omega, R_s = 10\,\text{k}\Omega, R_5 = 1\,\text{k}\Omega$

Find the values of the resistors R_5 and R_6 when peak to peak output voltage is $8\,\text{V}$.

$$v_0 = \frac{\text{peak to peak voltage}}{2}$$
$$= \frac{8}{2}$$
$$= 4\,\text{V}$$

Step 2 of 3

Substitute the values $R_2 = 20.3\,\text{k}\Omega, R_s = 10\,\text{k}\Omega$ and $v_0 = 4\,\text{V}$ in equation (1).

$$4 = \left(1 + \frac{20.3}{10}\right) v_i$$
$$4 = (1 + 2.03) v_i$$
$$4 = 3.03 v_i$$
$$v_i = \frac{4}{3.03}$$
$$v_i = 1.32\,\text{V}$$

Substitute the values $R_5 = 1\,\text{k}\Omega$, $v_i = 1.32\,\text{V}$ and $v_0 = 4\,\text{V}$ in equation (2).

$$4 - \left[\frac{1}{1 + R_6}(4 + 15)\right] - 0.7 = 1.32$$
$$\frac{1}{1 + R_6}(19) = 1.98$$
$$\frac{1}{1 + R_6} = 0.104$$
$$1 + R_6 = 9.59$$
$$R_6 = 8.59\,\text{k}\Omega$$

From the Fig.18.6 $R_6 = R_5$.

Therefore the values of the resistors $\boxed{R_5 = 8.59\,\text{k}\Omega, R_6 = 8.59\,\text{k}\Omega}$.

Step 3 of 3

Find the peak to peak voltage if the resistors R_5 and R_6 are open circuited.

$$v_0 = \left(1 + \frac{20.3}{10}\right) v_i$$
$$= (1 + 2.03) v_i$$
$$= 3.03 v_i$$
$$v_i = \frac{v_0}{3.03}$$

Substitute the values $R_5 = 1\,\text{k}\Omega$, $R_6 = \infty, R_5 = \infty$ and $v_i = \frac{v_0}{3.03}$ in equation (2).

$$v_0 - \left[\frac{1}{1 + \infty}(v_0 + 15)\right] - 0.7 = \frac{v_0}{3.03}$$
$$v_0 - \frac{v_0}{3.03} = 0.7$$
$$\frac{(3.03 - 1) v_0}{3.03} = 0.7$$
$$v_0 = \frac{0.7 \times 3.03}{2.03}$$
$$v_0 = 1.04\,\text{V}$$

Output peak to peak voltage = $2 \times v_0$.

Therefore peak to peak voltage when the resistors R_5 and R_6 are open circuited is $\boxed{2.08\,\text{V}}$.

Refer to Figure 17.20 (a) in the textbook.

Consider that the output saturation voltages of op amp are $L_{\pm} = \pm 10\ \text{V}$.

If the circuit is in the positive stable state, $v_o = L_+$, positive value for v_I have no effect. To trigger the circuit into L_- state, v_I must be made negative and of such a value as to make v_+ decrease below zero.

Find the lower threshold value, V_{TL} .

$$V_{TL} = -L_+\left(\frac{R_1}{R_2}\right)$$

Substitute $-5\ \text{V}$ for V_{TL} , and 10 V for L_+ .

$$-5\ \text{V} = -(10\ \text{V})\left(\frac{R_1}{R_2}\right)$$

$$\frac{R_1}{R_2} = \frac{5\ \text{V}}{10\ \text{V}}$$

$$\frac{R_1}{R_2} = \frac{1}{2}$$

$$R_2 = 2R_1$$

Take $R_1 = 10\,\text{k}\Omega$.

Therefore, the feedback resistance, R_2 is,

$$R_2 = 2(10\ \text{k}\Omega)$$
$$= 20\ \text{k}\Omega$$

Therefore, the suitable values of components are, $\boxed{R_1 = 10\ \text{k}\Omega}$ and $\boxed{R_2 = 20\ \text{k}\Omega}$.

Step 1 of 9

18.015P

Draw the feedback network of the circuit shown in Figure P17.13 in the text book.

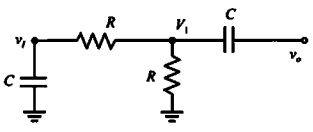


Figure 1

Step 2 of 9

Apply Kirchhoff's voltage law at node voltage v_I.

$$\frac{v_I}{\left(\frac{1}{sC}\right)}+\frac{v_I-V_1}{R}=0$$

$$v_I\left(sC+\frac{1}{R}\right)=\frac{V_1}{R}$$

$$v_I\left(\frac{sC+1}{R}\right)=\frac{V_1}{R}$$

$$V_1=v_I\left(1+sRC\right) \quad \text{..... (1)}$$

Step 3 of 9

Apply Kirchhoff's voltage law at node voltage V_1.

$$\frac{V_1-v_I}{R}+\frac{V_1}{R}+\frac{V_1-v_o}{\left(\frac{1}{sC}\right)}=0$$

$$V_1\left(\frac{1}{R}+\frac{1}{R}+sC\right)-\frac{v_I}{R}=sCv_o$$

$$V_1\left(2+sRC\right)-v_I=sRCv_o$$

Substitute $v_I\left(1+sRC\right)$ for V_1.

$$v_I\left(1+sRC\right)\left(2+sRC\right)-v_I=sRCv_o$$

$$v_I\left(2+sRC+2sRC+\left(sRC\right)^2-1\right)=sRCv_o$$

$$v_I\left(s^2R^2C^2+3sRC+1\right)=sRCv_o$$

$$\frac{v_o}{v_I}=\frac{s^2R^2C^2+3sRC+1}{sRC}$$

Step 4 of 9

Determine the expression for the feedback factor β.

$$\beta=\frac{v_I}{v_o}$$

$$\beta(s)=\frac{sRC}{s^2R^2C^2+3sRC+1}$$

$$\beta(s)=\frac{1}{sRC+3+\frac{1}{sRC}}$$

$$\beta(s)=\frac{1}{3+sRC+\frac{1}{sRC}}$$

Substitute $s=j\omega$.

$$\beta(j\omega)=\frac{1}{3+j\omega RC+\frac{1}{j\omega RC}}$$

$$\beta(j\omega)=\frac{1}{3+j\left(\omega RC-\frac{1}{\omega RC}\right)}$$

Step 5 of 9

Find the frequency for zero loop phase.

Equate the imaginary part of $\beta(j\omega)$ to zero for zero loop phase.

$$\omega RC-\frac{1}{\omega RC}=0$$

$$\omega RC=\frac{1}{\omega RC}$$

$$\omega^2=\frac{1}{R^2C^2}$$

$$\omega=\frac{1}{RC}$$

Thus, the frequency for zero loop phase is $\boxed{\frac{1}{RC}}$.

Step 6 of 9

Determine the feedback factor at zero phase frequency.

$$\beta_{\left(\omega=\frac{1}{RC}\right)}=\frac{1}{3+j\left(\left(\frac{1}{RC}\right)RC-\frac{1}{\omega RC}\right)}$$

$$=\frac{1}{3+j(0)}$$

$$=\frac{1}{3}$$

The gain of the non-inverting op-amp is,

$$A_v=1+\frac{R_2}{R_1}$$

Step 7 of 9

Write the expression for the condition for oscillations.

$$A_v\beta\geq 1$$

Substitute $1+\frac{R_2}{R_1}$ for A_v and $\frac{1}{3}$ for β.

$$\left(1+\frac{R_2}{R_1}\right)\left(\frac{1}{3}\right)\geq 1$$

$$1+\frac{R_2}{R_1}\geq 3$$

$$\frac{R_2}{R_1}\geq 2$$

Thus, $\frac{R_2}{R_1}$ for oscillations is $\boxed{\geq 2}$.

Step 8 of 9

Write the expression for the loop gain $L(s)$.

$$L(s)=A_v\beta$$

Substitute $1+\frac{R_2}{R_1}$ for A_v and $\frac{1}{3+sRC+\frac{1}{sRC}}$ for $\beta(s)$.

$$L(s)=\left(1+\frac{R_2}{R_1}\right)\left(\frac{1}{3+sRC+\frac{1}{sRC}}\right)$$

$$=\frac{1+R_2/R_1}{3+sRC+\frac{1}{sRC}}$$

Thus, the expression of $L(s)$ is $\boxed{\frac{1+R_2/R_1}{3+sRC+\frac{1}{sRC}}}$.

Step 9 of 9

Substitute $j\omega$ for s in the expression of $L(s)$.

$$L(j\omega)=\frac{1+R_2/R_1}{3+(j\omega)RC+\frac{1}{(j\omega)RC}}$$

$$=\frac{1+R_2/R_1}{3+j\left(\omega RC-\frac{1}{\omega RC}\right)}$$

Thus, the expression of $L(j\omega)$ is $\boxed{\frac{1+R_2/R_1}{3+j\left(\omega RC-\frac{1}{\omega RC}\right)}}$.

18.016E

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

Ask an expert

Step 1 of 8

Apply Kirchhoff's voltage law at node voltage v_t.

$$\frac{v_t}{R}+\frac{v_t-V_1}{\left(\frac{1}{cs}\right)}=0$$

$$v_t\left(sC+\frac{1}{R}\right)=sCV_1$$

$$v_t\left(\frac{sC+\frac{1}{R}}{sC}\right)=V_1$$

$$V_1=v_t\left(1+\frac{1}{sRC}\right) \quad \text{..... (1)}$$

Step 2 of 8

Apply Kirchhoff's voltage law at node voltage V_1.

$$\frac{V_1-v_t}{\left(\frac{1}{sC}\right)}+\frac{V_1}{\left(\frac{1}{sC}\right)}+\frac{V_1-v_o}{R}=0$$

$$V_1\left(sC+sC+\frac{1}{R}\right)-sCv_t=\frac{v_o}{R}$$

$$V_1(2sRC+1)-sRCv_t=v_o$$

Substitute $v_t\left(1+\frac{1}{sRC}\right)$ for V_1.

$$V_1(2sRC+1)-sRCv_t=v_o$$

$$v_t\left(1+\frac{1}{sRC}\right)(2sRC+1)-sRCv_t=v_o$$

$$v_t\left(2sRC+1+\frac{2sRC}{sRC}+\frac{1}{sRC}-sRC\right)=v_o$$

$$v_t\left(sRC+1+2+\frac{1}{sRC}\right)=v_o$$

Simplify further.

$$v_t\left(\frac{s^2R^2C^2+3sRC+1}{sRC}\right)=v_o$$

$$\frac{v_o}{v_t}=\frac{s^2R^2C^2+3sRC+1}{sRC}$$

Step 3 of 8

Determine the expression for the feedback factor β.

$$\beta=\frac{v_t}{v_o}$$

$$\beta(s)=\frac{sRC}{s^2R^2C^2+3sRC+1}$$

$$\beta(s)=\frac{1}{sRC+3+\frac{1}{sRC}}$$

$$\beta(s)=\frac{1}{3+sRC+\frac{1}{sRC}}$$

Substitute $s=j\omega$.

$$\beta(j\omega)=\frac{1}{3+j\omega RC+\frac{1}{j\omega RC}}$$

$$\beta(j\omega)=\frac{1}{3+j\left(\omega RC-\frac{1}{\omega RC}\right)}$$

Step 4 of 8

Find the frequency for zero loop phase.

Equate the imaginary part of $\beta(j\omega)$ to zero for zero loop phase.

$$\omega RC-\frac{1}{\omega RC}=0$$

$$\omega RC=\frac{1}{\omega RC}$$

$$\omega=\frac{1}{RC}$$

Thus, the frequency for zero loop phase is $\boxed{\frac{1}{RC}}$.

Step 5 of 8

Determine the feedback factor at zero phase frequency.

$$\beta_{\left(\omega=\frac{1}{RC}\right)}=\frac{1}{3+j\left(\left(\frac{1}{RC}\right)RC-\frac{1}{\omega RC}\right)}$$

$$=\frac{1}{3+j(0)}$$

$$=\frac{1}{3}$$

The gain of the non-inverting op-amp is,

$$A_v=1+\frac{R_2}{R_1}$$

Step 6 of 8

Write the expression for the condition for oscillations.

$$A_v\beta\geq 1$$

Substitute $1+\frac{R_2}{R_1}$ for A_v and $\frac{1}{3}$ for β.

$$\left(1+\frac{R_2}{R_1}\right)\left(\frac{1}{3}\right)\geq 1$$

$$1+\frac{R_2}{R_1}\geq 3$$

$$\frac{R_2}{R_1}\geq 2$$

Thus, $\frac{R_2}{R_1}$ for oscillations is $\boxed{\geq 2}$.

Step 7 of 8

Write the expression for the loop gain $L(s)$.

$$L(s)=A_v\beta.$$

Substitute $1+\frac{R_2}{R_1}$ for A_v and $\frac{1}{3+sRC+\frac{1}{sRC}}$ for $\beta(s)$.

$$L(s)=\left(1+\frac{R_2}{R_1}\right)\left(\frac{1}{3+sRC+\frac{1}{sRC}}\right)$$

$$=\frac{1+R_2/R_1}{3+sRC+\frac{1}{sRC}}$$

Thus, the expression of $L(s)$ is $\boxed{\frac{1+R_2/R_1}{3+sRC+\frac{1}{sRC}}}$.

Step 8 of 8

Substitute $j\omega$ for s in the expression of $L(s)$.

$$L(j\omega)=\frac{1+R_2/R_1}{3+(j\omega)RC+\frac{1}{(j\omega)RC}}$$

$$=\frac{1+R_2/R_1}{3+j\left(\omega RC-\frac{1}{\omega RC}\right)}$$

Thus, the expression of $L(j\omega)$ is $\boxed{\frac{1+R_2/R_1}{3+j\left(\omega RC-\frac{1}{\omega RC}\right)}}$.

Step 1 of 2

18.017E

Refer to Figure 17.21 (a) in the textbook for the block diagram representation and transfer characteristic for a comparator.

Consider that the op amp saturation levels are, $\pm 12\ \text{V}$.

Note that the op amp inverting terminal connected to $+3\ \text{V}$, and non-inverting terminal connected to input, v_I .

In the comparator circuit, the reference voltage is connected to inverting terminal.

Therefore, the reference voltage, V_R is, $\boxed{+3\ \text{V}}$.

The $L_\pm$ are the saturation levels of comparator.

Therefore, the saturation levels, $L_\pm$ are, $\boxed{\pm 12\ \text{V}}$.

Step 2 of 2

The characteristics of comparator is,

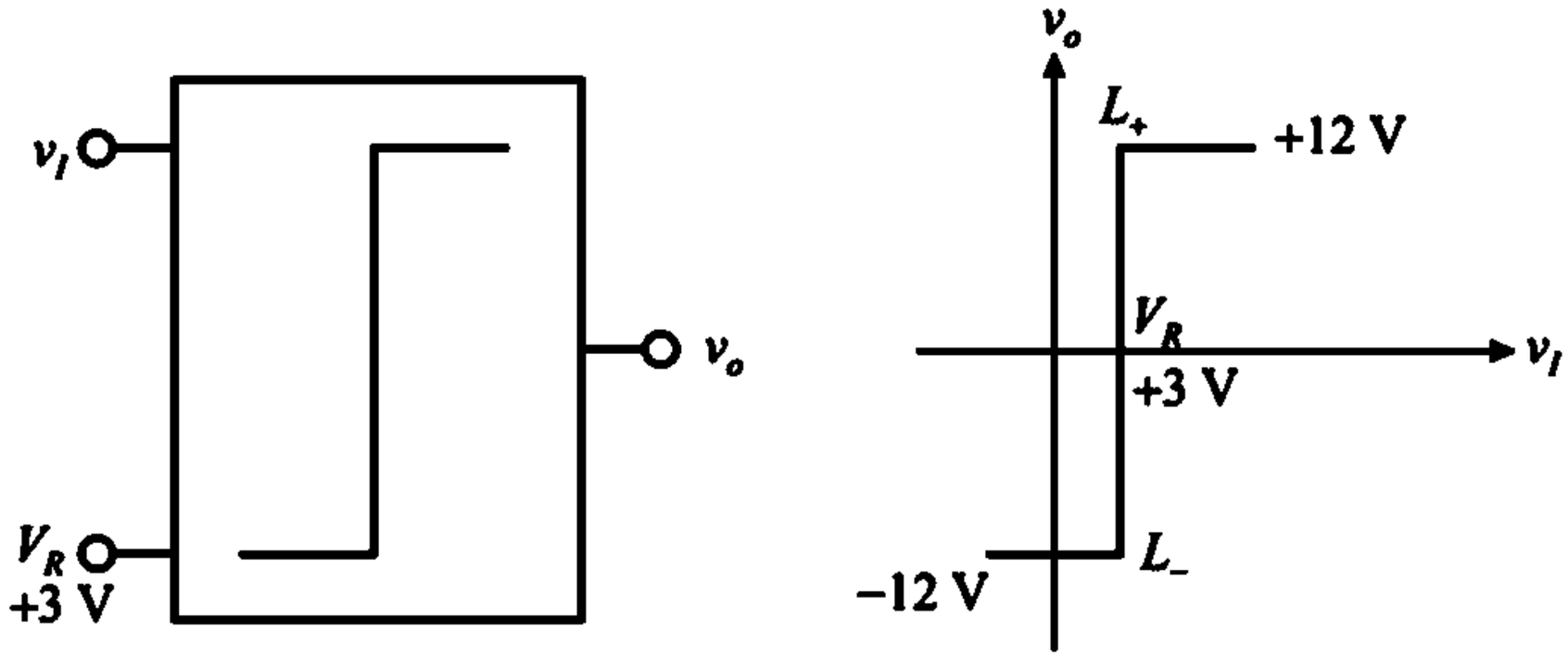
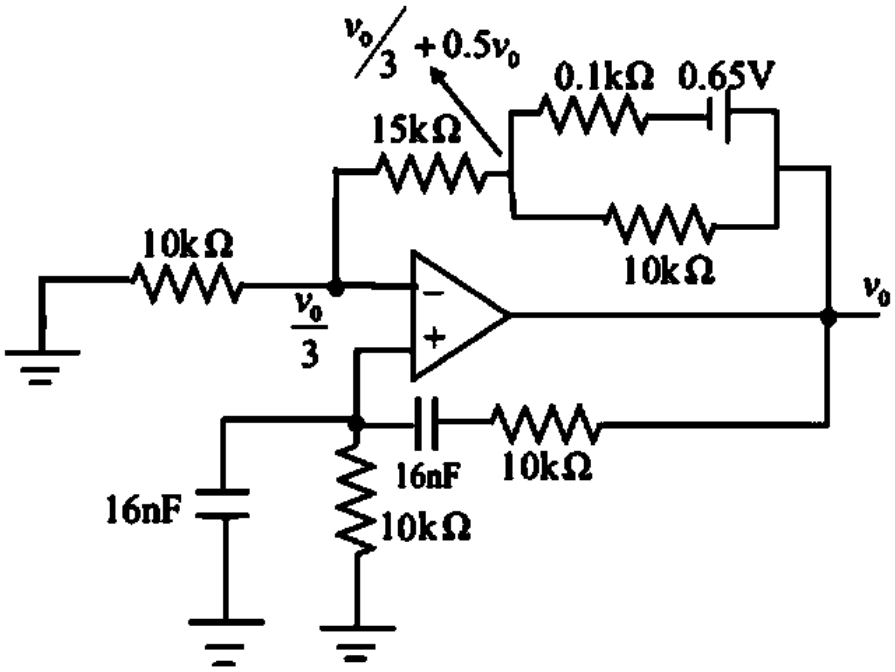


Figure 1

Step 1 of 2

18.017P

Redraw the circuit after replacing the potentiometer with two resistors.



Step 2 of 2

Apply the Kirchhoff's Current Law at positive node of op amp.

$$\frac{\frac{v_0}{3}}{10} = \frac{v_0 - \frac{v_0}{3} - 0.5v_0}{10} + \frac{v_0 - 0.65 - \frac{v_0}{3} - 0.5v_0}{0.1}$$

$$\frac{v_0}{30} = \frac{0.1667v_0}{10} + \frac{0.1667v_0 - 0.65}{0.1}$$

$$\frac{v_0}{30} = 0.01667v_0 + 1.667v_0 - 6.5$$

$$0.033v_0 = 1.65v_0 - 6.5$$

Simplify further.

$$1.65v_0 - 0.033v_0 = 6.5$$

$$1.616v_0 = 6.5$$

$$v_0 = 4.02\,\text{V}$$

$$v_{p-p} = 2v_0$$

$$v_{p-p} = 8.04\,\text{V}$$

Therefore maximum peak to peak voltage is $\boxed{8.04\,\text{V}}$.

Step 1 of 2

Refer to Figure 17.20 (a) in the textbook.

Consider that the width of hysteresis is 100 mV.

The hysteresis width is difference between threshold limits.

$$V_{TH} - (-V_{TL}) = 100 \text{ mV}$$
$$V_{TH} + V_{TL} = 100 \text{ mV}$$
$$|V_T| = \frac{100 \text{ mV}}{2}$$
$$|V_T| = 50 \text{ mV}$$

Therefore, the threshold voltages are, $V_{TH} = +0.05 \text{ V}$ and $V_{TL} = -0.05 \text{ V}$.

Step 2 of 2

If the circuit is in the positive stable state, $v_o = L_+$, positive value for v_I have no effect. To trigger the circuit into L_- state, v_I must be made negative and of such a value as to make v_+ decrease below zero.

Find the value of lower threshold, V_{TL} .

$$V_{TL} = -L_+\left(\frac{R_1}{R_2}\right)$$

Consider that the output saturation voltages are 10 V.

Substitute -0.05 V for V_{TL} , and 10 V for L_+ .

$$-0.05 \text{ V} = -(10 \text{ V})\left(\frac{R_1}{R_2}\right)$$
$$\frac{R_1}{R_2} = \frac{0.05 \text{ V}}{10 \text{ V}}$$
$$\frac{R_1}{R_2} = \frac{1}{200}$$
$$R_2 = 200R_1$$

Substitute $1 \text{ k}\Omega$ for resistance, R_1 .

$$R_2 = 200(1 \text{ k}\Omega)$$
$$= 200 \text{ k}\Omega$$

Therefore, the value of resistance, R_2 is, $\boxed{200 \text{ k}\Omega}$.

Step 1 of 3

The frequency of oscillations is $\omega = 10\,\text{kHz}$.

Find the frequency of oscillations when the excess phase shift is $5.7°$.

The loop gain of the Wien-bridge oscillator from the equation (18.11) is $L(j\omega) = \dfrac{1 + {R_2}/{R_1}}{3 + j\left(\omega CR - {1}/{\omega CR}\right)}$.

Find the phase shift of the loop.

$$\phi(\omega) = -\tan^{-1}\left(\frac{\text{Im}}{\text{Re}}\right)$$

$$\phi(\omega) = -\tan^{-1}\left(\frac{\omega CR - {1}/{\omega CR}}{3}\right)$$

Differentiate on each side with respect to ω.

$$\frac{d\phi(\omega)}{d\omega} = \frac{-1}{1+\left(\frac{\omega CR - {1}/{\omega CR}}{3}\right)^2}\frac{d}{d\omega}\left(\frac{\omega CR - {1}/{\omega CR}}{3}\right) \qquad \left(\text{Since } \frac{d}{dx}\left(\tan^{-1} t\right) = \frac{1}{1+t^2}\frac{dt}{dx}\right)$$

$$\frac{d\phi(\omega)}{d\omega} = \frac{-1}{1+\left(\frac{\omega CR - {1}/{\omega CR}}{3}\right)^2}\frac{1}{3}\left(CR + \frac{1}{\omega^2 CR}\right)$$

Evaluate the $\dfrac{d\phi(\omega)}{d\omega}$ when $\omega = \dfrac{1}{CR}$.

$$\left(\frac{d\phi(\omega)}{d\omega}\right)_{\omega=\frac{1}{CR}} = \frac{-1}{1+\left(\frac{1-1/1}{3}\right)^2}\frac{1}{3}\left(CR + \frac{1}{\left(\frac{1}{CR}\right)^2 CR}\right)$$

$$= \frac{-1}{1+0}\frac{1}{3}\left(CR + \frac{1}{\left(\frac{1}{CR}\right)}\right)$$

$$= \frac{-1}{3}(CR + CR)$$

$$= \frac{-2}{3}CR$$

$$= \frac{-2}{3\omega} \qquad \left(\text{Since } \omega = \frac{1}{CR}\right)$$

Change in oscillator frequency with respect to phase shift is $\Delta\omega = \dfrac{\Delta\phi}{{d\phi}/{d\omega}}$.

$$\Delta\omega = \frac{-0.1}{-2/3\omega} \qquad (\text{Since } 5.7° = 0.1\,\text{rad})$$

$$\Delta\omega = 0.15\omega$$

Therefore the frequency of oscillation is $\omega' = \omega - \Delta\omega$.

$$\omega' = 10 - 0.15(10)$$

$$\omega' = 10 - 1.5$$

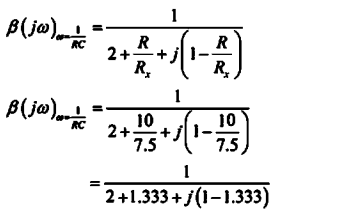
$$\omega' = 8.5\,\text{kHz}$$

The change in frequency of oscillation is $\boxed{\omega' = 8.5\,\text{kHz}}$.

Step 2 of 3

To restore the frequency of oscillation we need to change the parameters of feedback loop.

Consider the feedback loop.



Feedback loop gain is $\beta(s) = \dfrac{R_x \parallel {1}/{sC}}{\left(R_x \parallel {1}/{sC}\right) + R + {1}/{sC}}$.

$$\beta(s) = \frac{\frac{R_x/sC}{R_x + \frac{1}{sC}}}{\frac{R_x/sC}{R_x + \frac{1}{sC}} + R + 1/sC}$$

$$\beta(s) = \frac{R_x/sC}{R_x/sC + RR_x + R_x/sC + 1/s^2C^2 + R/sC}$$

$$\beta(s) = \frac{1}{1 + sCR + 1 + \frac{1}{sCR_x} + \frac{R}{R_x}}$$

$$\beta(s) = \frac{1}{2 + \frac{R}{R_x} + sCR + \frac{1}{sCR_x}}$$

$$\beta(j\omega) = \frac{1}{2 + \frac{R}{R_x} + j\omega CR + \frac{1}{j\omega CR_x}}$$

$$\beta(j\omega) = \frac{1}{2 + \frac{R}{R_x} + j\left(\omega CR - \frac{1}{\omega CR_x}\right)}$$

Find the phase shift of the loop.

$$\phi(\omega) = -\tan^{-1}\left(\frac{\text{Im}}{\text{Re}}\right)$$

$$\phi(\omega) = -\tan^{-1}\left(\frac{\omega CR - \frac{1}{\omega CR_x}}{2 + \frac{R}{R_x}}\right)$$

We require the phase shift is $5.7°$.

$$5.7 = -\tan^{-1}\left(\frac{\omega CR - \frac{1}{\omega CR_x}}{2 + \frac{R}{R_x}}\right)$$

$$\omega CR - \frac{1}{\omega CR_x} = \left(2 + \frac{R}{R_x}\right)\tan(-5.7)$$

$$\omega CR - \frac{1}{\omega CR_x} = \left(2 + \frac{R}{R_x}\right)\tan(-0.1)$$

Find the value of R_x when $\omega = \dfrac{1}{RC}$.

$$1 - \frac{1}{\left(\frac{1}{RC}\right)CR_x} = \left(2 + \frac{R}{R_x}\right)(-0.1)$$

$$1 - \frac{1}{\frac{R_x}{R}} = -0.2 - 0.1\frac{R}{R_x}$$

$$1 - \frac{R}{R_x} = -0.2 - 0.1\frac{R}{R_x}$$

$$1 + 0.2 = \frac{R}{R_x} - 0.1\frac{R}{R_x}$$

$$1.2 = 0.9\frac{R}{R_x}$$

$$R_x = 0.75R$$

Substitute $R = 10\,\text{k}\Omega$.

$$R_x = 0.75(10)$$

$$R_x = 7.5\,\text{k}\Omega$$

To restore the operation we need shunt resistance of $\boxed{7.5\,\text{k}\Omega}$.

Step 3 of 3

Now find the $\dfrac{R_2}{R_1}$ value.

Magnitude of the feedback network loop gain.

$$\beta(j\omega) = \frac{1}{2 + \frac{R}{R_x} + j\left(\omega CR - \frac{1}{\omega CR_x}\right)}$$

$$\beta(j\omega)_{\omega=\frac{1}{RC}} = \frac{1}{2 + \frac{R}{R_x} + j\left(1 - \frac{R}{R_x}\right)}$$

$$\beta(j\omega)_{\omega=\frac{1}{RC}} = \frac{1}{2 + \frac{10}{7.5} + j\left(1 - \frac{10}{7.5}\right)}$$

$$= \frac{1}{2 + 1.333 + j(1 - 1.333)}$$

$$= \frac{1}{3.333 - j(0.333)}$$

$$\left|\beta(j\omega)_{\omega=\frac{1}{RC}}\right| = \frac{1}{\sqrt{3.333^2 + (0.333)^2}}$$

$$\left|\beta(j\omega)_{\omega=\frac{1}{RC}}\right| = \frac{1}{3.35}$$

The condition for the oscillation is $1 + \dfrac{R_2}{R_1} = 3.35$.

$$\frac{R_2}{R_1} = 3.35 - 1$$

$$\frac{R_2}{R_1} = 2.35$$

The value of $\dfrac{R_2}{R_1}$ is $\boxed{\dfrac{R_2}{R_1} = 2.35}$.

Step 1 of 2

18.019E

Refer to Figure 17.24 (b) in the textbook.

Consider that the saturation voltages of op amp are, $L_{\pm} = \pm 10\ \text{V}$.

The time period, T of output wave is,

$$T = 2\tau \ln\left(\frac{1+\beta}{1-\beta}\right)$$

Here, the time constant, τ is,

$$\tau = RC$$

Substitute $1\ \text{M}\Omega$ for R , and $0.01\ \mu\text{F}$ for C .

$$\begin{aligned}\tau &= (1\ \text{M}\Omega)(0.01\ \mu\text{F}) \\ &= 0.01\ \text{s}\end{aligned}$$

The feedback gain, β is,

$$\beta = \frac{R_1}{R_1 + R_2}$$

Substitute $100\ \text{k}\Omega$ for R_1 , and $1\ \text{M}\Omega$ for R_2 .

$$\begin{aligned}\beta &= \frac{100\ \text{k}\Omega}{100\ \text{k}\Omega + 1\ \text{M}\Omega} \\ &= \frac{100\ \text{k}\Omega}{1100\ \text{k}\Omega} \\ &= 0.09\end{aligned}$$

Step 2 of 2

The time period is,

$$\begin{aligned}T &= 2\tau \ln\left(\frac{1+\beta}{1-\beta}\right) \\ &= 2(0.01)\ln\left(\frac{1+0.09}{1-0.09}\right) \\ &= (0.02)(0.18) \\ &= 3.6\ \text{ms}\end{aligned}$$

Therefore, the frequency of oscillation is,

$$f = \frac{1}{T}$$

Substitute 3.6 ms for T .

$$\begin{aligned}f &= \frac{1}{3.6\ \text{ms}} \\ &= 274.24\ \text{Hz}\end{aligned}$$

Thus, the frequency, f is, $\boxed{274.24\ \text{Hz}}$.

Step 1 of 10

Refer to Figure 17.8 in the text book.

Break circuit in Figure 17.8 in the text book at node X and connect a $10\ \text{k}\Omega$ resistor to the right most capacitor.

Draw the modified circuit.

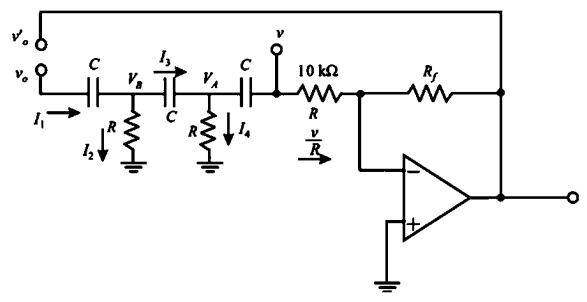


Figure 1

Step 2 of 10

Determine the expression for the voltage V_A.

$$V_A = \left(\frac{1}{SC}\right)\left(\frac{v}{R}\right) + v$$

$$V_A = v\left(1 + \frac{1}{SCR}\right) \quad \text{..... (1)}$$

Apply Kirchhoff's current law at the node voltage V_A.

$$I_3 = I_4 + \frac{v}{R}$$
$$= \frac{V_A}{R} + \frac{v}{R}$$

Substitute $v\left(1 + \frac{1}{SCR}\right)$ for V_A from Equation (1).

$$I_3 = \frac{v\left(1 + \frac{1}{SCR}\right)}{R} + \frac{v}{R}$$
$$= \frac{v}{R} + \frac{v}{SCR^2} + \frac{v}{R}$$

$$I_3 = \frac{2v}{R} + \frac{v}{SCR^2} \quad \text{..... (2)}$$

Step 3 of 10

Determine the expression for the current I_2.

$$I_2 = \frac{V_B}{R}$$
$$= \frac{V_A + \frac{I_3}{SC}}{R}$$

Substitute the expressions for V_A and I_3 from Equations (1) and (2).

$$I_2 = \frac{v\left(1 + \frac{1}{SCR}\right) + \frac{\frac{2v}{R} + \frac{v}{SCR^2}}{SC}}{R}$$
$$= \frac{v + \frac{v}{SCR} + \frac{2v}{SRC} + \frac{v}{S^2C^2R^2}}{R}$$
$$= \frac{v}{R} + \frac{3v}{SCR^2} + \frac{v}{S^2C^2R^3}$$

$$I_2 = \frac{v}{R} + \frac{3v}{SCR^2} + \frac{v}{S^2C^2R^3} \quad \text{..... (3)}$$

Step 4 of 10

Apply Kirchhoff's current law at node voltage V_B.

$$I_1 = I_2 + I_3$$

Substitute the expressions for the currents I_2 and I_3 from Equations (2) and (3).

$$I_1 = \frac{v}{R} + \frac{3v}{SCR^2} + \frac{v}{S^2C^2R^3} + \frac{2v}{R} + \frac{v}{SCR^2}$$
$$= \frac{3v}{R} + \frac{4v}{SCR^2} + \frac{v}{S^2C^2R^3}$$

$$I_1 = \frac{3v}{R} + \frac{4v}{SCR^2} + \frac{v}{S^2C^2R^3} \quad \text{..... (4)}$$

Step 5 of 10

Determine the expression for the voltage V_B.

$$V_B = V_A + \frac{I_3}{SC}$$

Substitute the expressions for V_A and I_3 from Equations (1) and (2).

$$V_B = v\left(1 + \frac{1}{SCR}\right) + \frac{\frac{2v}{R} + \frac{v}{SCR^2}}{SC}$$
$$= v + \frac{v}{SCR} + \frac{2v}{SRC} + \frac{v}{S^2C^2R^2}$$
$$= v + \frac{3v}{SRC} + \frac{v}{S^2C^2R^2}$$

$$V_B = v + \frac{3v}{SRC} + \frac{v}{S^2C^2R^2} \quad \text{..... (5)}$$

Step 6 of 10

The expression for the voltage v_o is,

$$v_o = V_B + \frac{I_1}{SC}$$

Substitute the expression for the current I_1 from equation (4) and the expression for the voltage V_B from Equation (5).

$$v_o = v + \frac{3v}{SRC} + \frac{v}{S^2C^2R^2} + \frac{\frac{3v}{R} + \frac{4v}{SCR^2} + \frac{v}{S^2C^2R^3}}{SC}$$
$$= v + \frac{3v}{SRC} + \frac{v}{S^2C^2R^2} + \frac{3v}{SCR} + \frac{4v}{S^2C^2R^2} + \frac{v}{S^3C^3R^3}$$
$$= v + \frac{6v}{SRC} + \frac{5v}{S^2C^2R^2} + \frac{v}{S^3C^3R^3}$$

$$v_o = v + \frac{6v}{SRC} + \frac{5v}{S^2C^2R^2} + \frac{v}{S^3C^3R^3} \quad \text{..... (6)}$$

Step 7 of 10

Determine expression for the loop gain $L(S)$.

$$L(S) = \frac{-v_o'}{v_o}$$
$$= \frac{-\frac{R_f}{R}v}{v + \frac{6v}{SRC} + \frac{5v}{S^2C^2R^2} + \frac{v}{S^3C^3R^3}}$$
$$= \frac{-\frac{R_f}{R}}{1 + \frac{6}{SRC} + \frac{5}{S^2C^2R^2} + \frac{1}{S^3C^3R^3}}$$
$$= \frac{S^3\left(\frac{R_f}{R}\right)}{S^3 + \frac{6S^2}{RC} + \frac{5S}{C^2R^2} + \frac{1}{C^3R^3}}$$

Step 8 of 10

Replace $S = j\omega$

$$L(j\omega) = \frac{(j\omega)^3\left(\frac{R_f}{R}\right)}{(j\omega)^3 + \frac{6(j\omega)^2}{RC} + \frac{5(j\omega)}{C^2R^2} + \frac{1}{C^3R^3}}$$
$$= \frac{-j\omega^3\left(\frac{R_f}{R}\right)}{\frac{1}{C^3R^3} - \frac{6\omega^2}{RC} + j\frac{5\omega}{C^2R^2} - j\omega^3}$$
$$= \frac{-j\omega^3\left(\frac{R_f}{R}\right)}{\frac{1}{C^3R^3} - \frac{6\omega^2}{RC} + j\left(\frac{5\omega}{C^2R^2} - \omega^3\right)}$$

Equate the imaginary part of $L(j\omega)$ to zero.

$$\frac{1}{C^3R^3} - \frac{6\omega_o^2}{RC} = 0$$
$$6\omega_o^2 = \frac{1}{R^2C^2}$$
$$\omega_o^2 = \frac{1}{6R^2C^2}$$
$$f_o = \frac{1}{2\pi\sqrt{6}RC}$$

Therefore, the frequency f_o to start oscillations is $\boxed{\frac{1}{2\pi\sqrt{6}RC}}$.

Step 9 of 10

Simplify the expression of the loop gain at $\omega = \omega_o$.

$$L(j\omega_o) = \frac{-j\omega_o^3\left(\frac{R_f}{R}\right)}{0 + j\left(\frac{5\omega_o}{C^2R^2} - \omega_o^3\right)}$$
$$= \frac{-\omega_o^3\left(\frac{R_f}{R}\right)}{\left(\frac{5\omega_o}{C^2R^2} - \omega_o^3\right)}$$
$$= \frac{\omega_o^2\left(\frac{R_f}{R}\right)}{-\frac{5}{C^2R^2} + \omega_o^2}$$

Step 10 of 10

Substitute $\frac{1}{RC}$ for ω_o.

$$L(j\omega_o) = \frac{\omega_o^2\left(\frac{R_f}{R}\right)}{-5\left(6\omega_0^2\right) + \omega_o^2}$$
$$= \frac{-\frac{R_f}{R}}{29}$$

Substitute 1 for $L(j\omega_o)$.

$$1 = \frac{R_f}{R}$$
$$R_f = 29R$$

Thus, the minimum value of R_f is $\boxed{29R}$.

Step 1 of 7

Refer to Figure 17.24 (b) in the textbook for a bistable multivibrator with inverting transfer characteristics in a feedback loop with an RC circuit.

Replace the resistor R_1 by a pair of diodes D_1 and D_2 connected in parallel in opposite directions.

The following are the resistance values.

$R_2 = 10\ \text{k}\Omega$

$R = 10\ \text{k}\Omega$

The capacitance value, $C = 0.1\ \mu\text{F}$.

The diode voltage is V_D .

Draw the modified circuit.

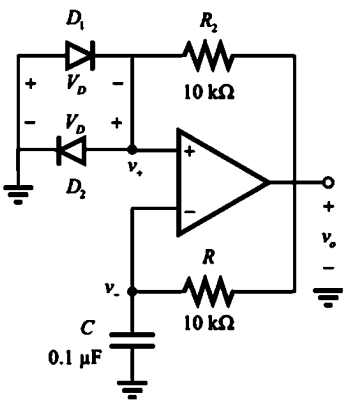


Figure 1

Step 2 of 7

Consider that the following are the two possible levels for the output of the circuit.

$L_- = -12\ \text{V}$

$L_+ = 12\ \text{V}$

Assume that the output of the circuit is at one of the possible level $L_+ = 12\ \text{V}$.

The following is the voltage at positive terminal of the operational amplifier when the output of the circuit is at level L_+ .

$v^+ = \beta L_+$

From the circuit in Figure 2, the following is the voltage at the positive terminal of the operational amplifier for the positive output $L_+ = 12\ \text{V}$.

$v^+ = V_D$

Compare the expressions for the voltage at the positive terminal of the operational amplifier to get the following.

$\beta L_+ = V_D$

$\beta = \dfrac{V_D}{L_+}$

Substitute 12 V for L_+ .

$\beta = \dfrac{V_D}{12}$ ..... (1)

Step 3 of 7

The following is the expression for the time period of the circuit.

$T = 2\tau \ln\left(\dfrac{1+\beta}{1-\beta}\right)$ ..... (2)

Calculate the time constant τ using the following expression.

$\tau = CR$

Substitute 0.1 μF for C and $10\ \text{k}\Omega$ for R .

$\tau = (0.1\ \mu\text{F})(10\ \text{k}\Omega)$

$= 1\times10^{-3}\ \text{s}$

$= 1\ \text{ms}$

Substitute 1 ms for τ and the expression for β from equation (1) in equation (2).

$T = 2(1\ \text{ms})\ln\left(\dfrac{1+\frac{V_D}{12}}{1-\frac{V_D}{12}}\right)$

$T = 0.002\ln\left(\dfrac{12+V_D}{12-V_D}\right)$ ..... (3)

Find the expression for frequency using the expression for time period.

$f = \dfrac{1}{T}$

$= \dfrac{1}{0.002\ln\left(\frac{12+V_D}{12-V_D}\right)}$

$f = \dfrac{500}{\ln\left(\frac{12+V_D}{12-V_D}\right)}$ ..... (4)

Thus, the following is the expression for the frequency as a function of diode voltage.

$$\boxed{f = \frac{500}{\ln\left(\frac{12+V_D}{12-V_D}\right)}\ \text{Hz}}$$

Step 4 of 7

Consider the following data for the diodes in the circuit of Figure 2.
The diode voltage at $25°\text{C}$, $V_D = 0.70\ \text{V}$ and

The diode voltage reduces by 2 mV for $1°\text{C}$ increase in temperature.

Consider that the temperature, $T = 0°\text{C}$.

Calculate the diode voltage corresponding to $0°\text{C}$ temperature.

$V_D = 0.70\ \text{V} + (-2\ \text{mV})(0-25)$

$= 0.70\ \text{V} + 0.05$

$= 0.75\ \text{V}$

Substitute 0.75 V for V_D in equation (4) to calculate frequency at $0°\text{C}$.

$f = \dfrac{500}{\ln\left(\frac{12+0.75}{12-0.75}\right)}$

$= \dfrac{500}{\ln(1.134)}$

$= \dfrac{500}{0.125}$

$\approx 3995\ \text{Hz}$

Thus, the frequency f at $0°\text{C}$ is $\boxed{3995\ \text{Hz}}$.

Step 5 of 7

Consider that the temperature, $T = 25°\text{C}$.

Calculate the diode voltage corresponding to $25°\text{C}$ temperature.

$V_D = 0.70\ \text{V} + (-2\ \text{mV})(25-25)$

$= 0.70\ \text{V} + 0$

$= 0.70\ \text{V}$

Substitute 0.70 V for V_D in equation (4) to calculate frequency at $25°\text{C}$.

$f = \dfrac{500}{\ln\left(\frac{12+0.70}{12-0.70}\right)}$

$= \dfrac{500}{\ln(1.123)}$

$= \dfrac{500}{0.116}$

$\approx 4281\ \text{Hz}$

Thus, the frequency f at $25°\text{C}$ is $\boxed{4281\ \text{Hz}}$.

Step 6 of 7

Consider that the temperature, $T = 50°\text{C}$.

Calculate the diode voltage corresponding to $50°\text{C}$ temperature.

$V_D = 0.70\ \text{V} + (-2\ \text{mV})(50-25)$

$= 0.70\ \text{V} - 0.05$

$= 0.65\ \text{V}$

Substitute 0.65 V for V_D in equation (4) to calculate frequency at $50°\text{C}$.

$f = \dfrac{500}{\ln\left(\frac{12+0.65}{12-0.65}\right)}$

$= \dfrac{500}{\ln(1.114)}$

$= \dfrac{500}{0.108}$

$= 4611\ \text{Hz}$

Thus, the frequency f at $50°\text{C}$ is $\boxed{4611\ \text{Hz}}$.

Step 7 of 7

Consider that the temperature, $T = 100°\text{C}$.

Calculate the diode voltage corresponding to $100°\text{C}$ temperature.

$V_D = 0.70\ \text{V} + (-2\ \text{mV})(100-25)$

$= 0.70\ \text{V} - 0.15$

$= 0.55\ \text{V}$

Substitute 0.55 V for V_D in equation (4) to calculate frequency at $100°\text{C}$.

$f = \dfrac{500}{\ln\left(\frac{12+0.55}{12-0.55}\right)}$

$= \dfrac{500}{\ln(1.096)}$

$= \dfrac{500}{0.09}$

$= 5451\ \text{Hz}$

Thus, the frequency f at $100°\text{C}$ is $\boxed{5451\ \text{Hz}}$.

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Step 1 of 2

Refer to Figure 17.25 (a) in the textbook for a general scheme of generating triangular and square waveforms.

Refer to Figure 17.20 (a) in the textbook to realize the bistable circuit.

The frequency of oscillation, $f = 1\ \text{kHz}$.

Determine the time period.

$$\begin{aligned} T &= \frac{1}{f} \\ &= \frac{1}{1\ \text{kHz}} \\ &= 1\ \text{ms} \end{aligned}$$

The general expression for the time period, T is,

$$\begin{aligned} T &= T_1 + T_2 \\ &= CR\left(\frac{V_{TH} - V_{TL}}{L_+}\right) + CR\left(\frac{V_{TH} - V_{TL}}{-L_-}\right) \end{aligned}$$

The output saturation voltage levels, $L_+ = -L_-$. Therefore,

$$\begin{aligned} T &= CR\left(\frac{V_{TH} - V_{TL}}{L_+}\right) + CR\left(\frac{V_{TH} - V_{TL}}{L_+}\right) \\ &= 2CR\left(\frac{V_{TH} - V_{TL}}{L_+}\right) \end{aligned}$$

Here, V_{TH} **and** V_{TL} are the threshold voltages.

Peak-to-peak value of the triangular waveform is 10 V.

Thus the upper and lower threshold values are 5 V.

Substitute 0.001 s for T, $0.01\times10^{-6}\ \text{F}$ for C, 10 H for L_+ , $+5\ \text{V}$ for V_{TH} and $-5\ \text{V}$ for V_{TL} to find the value of resistance, R .

$$0.001 = 2\left(0.01\times10^{-6}\right)R\left(\frac{5+5}{10}\right)$$

$$0.001 = (2)\left(0.01\times10^{-6}\right)R(1)$$

$$R = \frac{0.001}{(2)\left(0.01\times10^{-6}\right)}$$

$$R = 50\ \text{k}\Omega$$

Thus, the resistance, R in the circuit is, $\boxed{50\ \text{k}\Omega}$.

Step 2 of 2

Write the formula for peak-to-peak voltage of the bistable circuit.

$$V_{TH} = -L_-\left(\frac{R_1}{R_2}\right)$$

Substitute $-10\ \text{V}$ for L_- , 5 V for V_{TH} , and 10×10^3 for R_1 .

$$5\ \text{V} = 10\ \text{V}\left(\frac{10\times10^3}{R_2}\right)$$

$$R_2 = \frac{\left(10\times10^3\right)10}{5}$$

$$R_2 = 20\ \text{k}\Omega$$

Thus, the value of resistance, R_2 in the circuit is, $\boxed{20\ \text{k}\Omega}$.

Step 1 of 9

Refer to Figure 17.9 in the text book.

The quadrature oscillator is based on the second integrator.

As an active filter, the loop is damped to locate the poles in the left half of the s-plane. Here in the quadrature oscillator amplifier 1 is connected as an inverting miller integrator with a limiter in the feedback for amplitude control. Amplifier 2 is connected as a non- inverting integrator.

Step 2 of 9

Draw the circuit without the limiter.

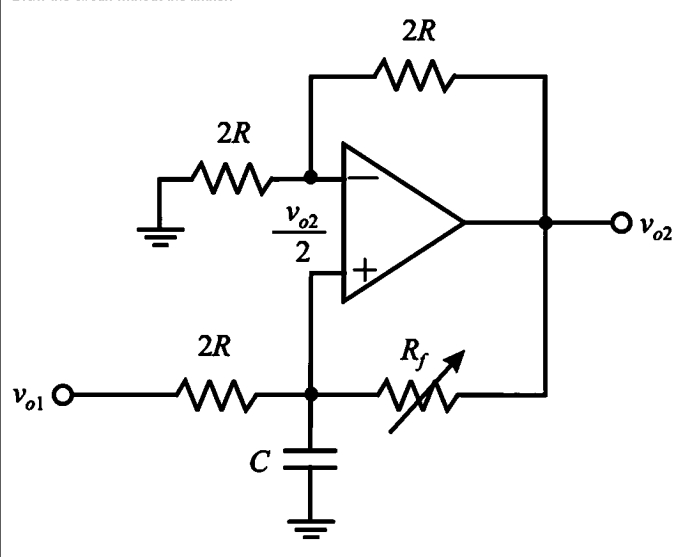


Figure 1

Step 3 of 9

Use voltage division principle to write the expression of the fraction of its input voltage.

$$v^+ = v^- = \left(\frac{v_{o2}}{2R+2R}\right)(2R)$$

$$v^+ = v^- = \frac{1}{2}v_{o2}$$

Step 4 of 9

Apply Kirchhoff's current law at the non-inverting terminal of the op-amp.

$$\frac{\frac{v_{o2}}{2}-v_{o1}}{2R}+\frac{\frac{v_{o2}}{2}}{\frac{1}{sC}}+\frac{\frac{v_{o2}}{2}-v_{o2}}{R_f}=0$$

$$\frac{v_{o2}-2v_{o1}}{4R}+sC\frac{v_{o2}}{2}+\frac{v_{o2}-2v_{o2}}{2R_f}=0$$

$$\frac{1}{2}\left[\frac{v_{o2}-2v_{o1}}{2R}+sCv_{o2}-\frac{v_{o2}}{R_f}\right]=0$$

$$\frac{v_{o2}-2v_{o1}}{2R}+sC\,v_{o2}-\frac{v_{o2}}{R_f}=0$$

Step 5 of 9

Substitute $\frac{2R}{1+\Delta}$ for R_f .

$$\frac{v_{o2}-2v_{o1}}{2R}+sC\,v_{o2}-\frac{v_{o2}}{2R}(1+\Delta)=0$$

$$v_{o2}\left(\frac{1}{2R}+sC-\frac{1+\Delta}{2R}\right)=\frac{v_{o1}}{R}$$

$$v_{o2}(1+2sRC-1-\Delta)=2v_{o1}$$

$$\frac{v_{o2}}{v_{o1}}=\frac{2}{2sRC-\Delta}$$

Simplify further.

$$\frac{v_{o2}}{v_{o1}}=\frac{1}{sRC-\frac{\Delta}{2}}$$

Step 6 of 9

If we disregard the limiter and,

Break the loop at X .

The expression for the loop gain is,

$$\frac{v_{o1}}{v_x}=\frac{-1}{sCR}$$

According to definition of loop gain in the circuit is ,

$$\begin{aligned}L(S)&=\frac{v_{o1}}{v_x}\times\frac{v_{o2}}{v_{o1}}\\&=\frac{-1}{sCR}\times\frac{1}{sCR-\frac{\Delta}{2}}\\&=\frac{-1}{s^2C^2R^2-\frac{sCR\Delta}{2}}\\&=\frac{1}{-s^2C^2R^2+\frac{sCR\Delta}{2}}\end{aligned}$$

Step 7 of 9

The characteristic equation of the system is,

$$1-L(s)=0$$

$$L(s)=1$$

$$-s^2C^2R^2+\frac{sCR\Delta}{2}=1$$

$$s^2C^2R^2-\frac{sCR\Delta}{2}+1=0$$

Step 8 of 9

Write the expression for the roots of the quadratic equation.

$$s_p=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

Substitute $a=C^2R^2$, $b=\frac{-CR\Delta}{2}$ and $c=1$.

$$\begin{aligned}s_p&=\frac{-\left(-\frac{CR\Delta}{2}\right)\pm\sqrt{\left(-\frac{CR\Delta}{2}\right)^2-4\left(C^2R^2\right)(1)}}{2C^2R^2}\\&=\frac{(RC)\left(\frac{\Delta}{2}\pm\sqrt{\left(\frac{\Delta}{2}\right)^2-4}\right)}{2C^2R^2}\\&=\frac{\frac{\Delta}{2}\pm\sqrt{\left(\frac{\Delta}{2}\right)^2-4}}{2RC}\\&=\frac{\frac{\Delta}{2}\pm j2\sqrt{1-\left(\frac{\Delta}{4}\right)^2}}{2RC}\end{aligned}$$

For $\Delta\ll 1$;

$$\left(1-\left(\frac{\Delta}{4}\right)^2\right)^{\frac{1}{2}}=1-\frac{1}{2}\left(\frac{\Delta}{4}\right)^2$$

Neglect the higher order terms.

$$\begin{aligned}s_p&=\frac{\frac{\Delta}{2}\pm j2\left(1-\frac{1}{2}\left(\frac{\Delta}{4}\right)^2\right)}{2RC}\\&=\frac{\frac{\Delta}{2}\pm j\left(2-\left(\frac{\Delta}{4}\right)^2\right)}{2RC}\end{aligned}$$

Step 9 of 9

The poles of the characteristic equation are in the right half of s-plane.

$$\mathrm{Re}(s_p)>0$$

Simplify further the expression of the poles.

$$\begin{aligned}s_p&=\frac{\frac{\Delta}{2}\pm j\left(2-\left(\frac{\Delta}{4}\right)^2\right)}{2RC}\qquad(\text{for }\Delta\ll 1)\\&=\frac{\frac{\Delta}{2}\pm 2j}{2RC}\\&=\frac{2\left(\frac{\Delta}{4}\pm j\right)}{2RC}\\&=\frac{\frac{\Delta}{4}\pm j}{RC}\end{aligned}$$

Therefore, the poles of the characteristic equation are in the right half of s-plane and the poles location is $\boxed{\frac{\frac{\Delta}{4}\pm j}{RC}}$.

Refer to Figure 17.26 in the textbook for an op amp monostable circuit.

A negative pulse is generated at the output during the quasi-stable state.

The expression for the duration of the output pulse is,

$$T = C_1 R_3 \ln\left(\frac{V_{D1} - L_-}{\beta L_- - L_-}\right)$$

Here, L_- is the output saturation level, the diode voltage is V_{D1}, and feedback gain is β.

Substitute $100\ \mu s$ for T, $0.1\ \mu F$ for C_1, 0.7 V for V_{D1}, 0.1 for β, and 12 V for L_- to find the value of resistance, R_3.

$$100\times10^{-6} = \left(0.1\times10^{-6}\right)R_3 \ln\left(\frac{0.7-(-12)}{(0.1)(-12)-(-12)}\right)$$

$$100 = (0.1)R_3 \ln\left(\frac{0.7+12}{-1.2+12}\right)$$

$$\frac{100}{0.1} = R_3 \ln\left(\frac{12.7}{10.8}\right)$$

$$1000 = R_3 \ln(1.175)$$

$$1000 = 0.162R_3$$

$$R_3 = \frac{1000}{0.162}$$
$$= 6.17\ k\Omega$$

Therefore, the value of resistance, R_3 is, $\boxed{6.17\ k\Omega}$.

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Refer to Figure 17.28 (a) in the textbook for the monostable circuit.

Consider that the output pulse duration is $100\ \mu s$.

The monostable multivibrator produces an output pulse, v_o .

The width of the pulse is,

$$T = CR\ln 3$$

Substitute $10\times10^{-9}\ \text{F}$ for C , $100\times10^{-6}\ \text{s}$ for T to find the value of resistor, R .

$$100\times10^{-6} = \left(10\times10^{-9}\right)R(1.0986)$$

$$\begin{aligned} R &= \frac{100\times10^{-6}}{\left(10\times10^{-9}\right)(1.0986)} \\ &= \frac{100}{10.986\times10^{-3}} \\ &= 9.1\times10^{3}\ \Omega \\ &= 9.1\ \text{k}\Omega \end{aligned}$$

Therefore, the value of resistance, R is $\boxed{9.1\ \text{k}\Omega}$.

Step 1 of 7

Refer to Figure 17.11 in the text book.

The transfer function of the filter normalised to the centre frequency ω_o is,

$$\begin{aligned}|T(j\omega)| &= \frac{\frac{\omega\omega_o}{Q}}{\sqrt{\left(\omega_o^2-\omega^2\right)^2+\frac{\omega^2\omega_o^2}{Q^2}}}\\ &= \frac{\frac{\omega\omega_o}{Q}}{\sqrt{\omega^4\left(\frac{\omega_o^2}{\omega^2}-1\right)^2+\frac{\omega^2\omega_o^2}{Q^2}}}\\ &= \frac{\frac{\omega\omega_o}{Q}}{\omega^2\sqrt{\left(\frac{\omega_o^2}{\omega^2}-1\right)^2+\frac{\omega_o^2}{Q^2\omega^2}}}\\ &= \frac{\frac{1}{Q}\left(\frac{\omega_o}{\omega}\right)}{\sqrt{\left(\frac{\omega_o^2}{\omega^2}-1\right)^2+\frac{1}{Q^2}\left(\frac{\omega_o^2}{\omega^2}\right)}}\end{aligned}$$

Substitute $Q=20$.

$$|T(j\omega)| = \frac{\frac{1}{Q}\left(\frac{\omega_o}{\omega}\right)}{\sqrt{\left(\frac{\omega_o^2}{\omega^2}-1\right)^2+\frac{1}{Q^2}\left(\frac{\omega_o^2}{\omega^2}\right)}}$$

Step 2 of 7

(a)

Represent the square wave of amplitude V and frequency ω in series.

$$\frac{4V}{\pi}\left(\cos\omega t-\frac{1}{3}\cos 3\omega t+\frac{1}{5}\cos 5\omega t-\frac{1}{7}\cos 7\omega t+\ldots\right)$$

The series contains no even harmonics.

The even harmonics relative to the fundamental amplitude is 0 .

Thus, the second harmonic relative to the fundamental amplitude is $\boxed{0}$.

Step 3 of 7

(b)

Determine the magnitude of third harmonic.

$$\begin{aligned}|T(3\omega_o)| &= \frac{1}{3}\left(\frac{\frac{1}{20}\left(\frac{1}{3}\right)}{\sqrt{\left(\frac{1}{3^2}-1\right)^2+\frac{1}{20^2}\left(\frac{1}{3}\right)^2}}\right)\\ &= \frac{1}{3}\left(\frac{\frac{1}{60}}{\sqrt{\left(\frac{-8}{9}\right)^2+\frac{1}{3600}}}\right)\\ &= \frac{1}{3}\left(\frac{0.016667}{\sqrt{0.79+2.78\times10^{-4}}}\right)\\ &= 6.25\times10^{-3}\end{aligned}$$

Thus, the magnitude of third harmonic is $\boxed{6.25\times10^{-3}}$.

Step 4 of 7

(c)

Determine the magnitude of fifth harmonic.

$$\begin{aligned}|T(5\omega_o)| &= \frac{1}{5}\left(\frac{\frac{1}{20}\left(\frac{1}{5}\right)}{\sqrt{\left(\frac{1}{5^2}-1\right)^2+\frac{1}{20^2}\left(\frac{1}{5}\right)^2}}\right)\\ &= \frac{1}{5}\left(\frac{\frac{1}{100}}{\sqrt{\left(\frac{-24}{25}\right)^2+\frac{1}{10000}}}\right)\\ &= \frac{1}{5}\left(\frac{0.01}{\sqrt{0.9216+0.0001}}\right)\\ &= 2.08\times10^{-3}\end{aligned}$$

Thus, the magnitude of fifth harmonic is $\boxed{2.08\times10^{-3}}$.

Step 5 of 7

(d)

The magnitude relative to 4th, 6th, 8th and 10th harmonics is 0 (zero).

Determine the magnitude of seventh harmonic.

$$\begin{aligned}|T(7\omega_o)| &= \frac{1}{7}\left(\frac{\frac{1}{20}\left(\frac{1}{7}\right)}{\sqrt{\left(\frac{1}{7^2}-1\right)^2+\frac{1}{20^2}\left(\frac{1}{7}\right)^2}}\right)\\ &= \frac{1}{7}\left(\frac{\frac{1}{140}}{\sqrt{\left(\frac{-48}{49}\right)^2+\frac{1}{19600}}}\right)\\ &= \frac{1}{7}\left(\frac{0.0071}{\sqrt{0.9596+0.000051}}\right)\\ &= 1.035\times10^{-3}\end{aligned}$$

Step 6 of 7

Determine the magnitude of ninth harmonic.

$$\begin{aligned}|T(9\omega_o)| &= \frac{1}{9}\left(\frac{\frac{1}{20}\left(\frac{1}{9}\right)}{\sqrt{\left(\frac{1}{9^2}-1\right)^2+\frac{1}{20^2}\left(\frac{1}{9}\right)^2}}\right)\\ &= \frac{1}{9}\left(\frac{\frac{1}{180}}{\sqrt{\left(\frac{-80}{81}\right)^2+\frac{1}{32400}}}\right)\\ &= \frac{1}{9}\left(\frac{0.00556}{\sqrt{0.9755+0.00003086}}\right)\\ &= 0.6255\times10^{-3}\end{aligned}$$

Step 7 of 7

Determine the rms of harmonics to the tenth.

$$\begin{aligned}X_{rms} &= \sqrt{\left(6.25\times10^{-3}\right)^2+\left(2.08\times10^{-3}\right)^2+\left(1.035\times10^{-3}\right)^2+\left(0.6255\times10^{-3}\right)^2}\\ &= \sqrt{6.25^2+2.08^2+1.035^2+0.6255^2}\times10^{-3}\\ &= \sqrt{39.0625+4.3264+1.071225+0.39125025}\times10^{-3}\\ &= 6.697\times10^{-3}\end{aligned}$$

Thus, the rms of harmonics to the tenth is $\boxed{6.697\times10^{-3}}$.

Step 1 of 2

Refer to Figure 17.29 (a) in the textbook for an astable multivibrator circuit.

The frequency of the oscillation is, $f = 100\ \text{kHz}$.

Calculate the time period.

$$T = \frac{1}{f}$$
$$= \frac{1}{100\ \text{kHz}}$$
$$= 10\ \mu\text{s}$$

The expression for the time period, T of the output wave for the astable multivibrator is,

$$T = 0.69C(R_A + 2R_B)$$

Substitute $10\ \mu\text{s}$ for T , $10^{-9}\ \text{F}$ for C .

$$10\times10^{-6} = (0.69)(1\times10^{-9})(R_A + 2R_B)$$

$$R_A + 2R_B = \frac{10\times10^{-6}}{(0.69)(10^{-9})}$$
$$= 14492.75$$

The duty cycle of output wave is,

$$\text{Duty cycle} = \frac{T_H}{T_H + T_L}$$
$$= \frac{R_A + R_B}{R_A + 2R_B}$$

Substitute 75% for duty cycle.

$$\frac{75}{100} = \frac{R_A + R_B}{R_A + 2R_B}$$
$$R_A + 2R_B = \frac{100}{75}(R_A + R_B)$$

Substitute 14492.75 for $R_A + 2R_B$.

$$14492.75 = 1.34(R_A + R_B)$$

$$R_A + R_B = \frac{14492.75}{1.34}$$
$$= 10869.565$$

Step 2 of 2

Subtract expression of $R_A + R_B$ from $R_A + 2R_B$.

$$(R_A + 2R_B) - (R_A + R_B) = 14492.75 - 10869.565$$

$$R_B = 3623.189$$
$$= 3.6\ \text{k}\Omega$$

Substitute this value into the equation of .

$$R_A + 3.6\ \text{k}\Omega = 10869.565$$

$$R_A = 10869.565 - 3623.189$$
$$= 7.246\ \text{k}\Omega$$

Therefore, the values of R_A and respectively are $\boxed{7.246\ \text{k}\Omega}$ and $\boxed{3.6\ \text{k}\Omega}$.

Step 1 of 4

To determine the oscillation condition for the Colpitts oscillator consider its equivalent circuit.

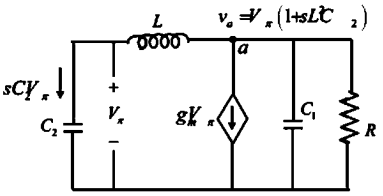


Figure24

Where v_π is the transistor input voltage.

To simplify the analysis , we have neglected the transistor capacitance C_μ and input resistance r_π .

Step 2 of 4

Assume the resistance R includes r_0 of the transistor.

Find the voltage at node a .

$$v_a = V_\pi + Ls(sC_2V_\pi)$$
$$v_a = V_\pi(1+s^2LC_2)$$

The node equation at node a can be found out using alternate analysis approach.

Apply Kirchhoff's current law at node a .

$$sC_2V_\pi + g_mV_\pi + sC_1V_\pi(1+s^2LC_2) + \frac{V_\pi(1+s^2LC_2)}{R} = 0$$

Since $V_\pi \neq 0$, assume $V_\pi = 1$ to eliminate the term.

$$sC_2 + g_m + sC_1(1+s^2LC_2) + \frac{(1+s^2LC_2)}{R} = 0$$
$$g_m + \frac{1}{R} + sC_2 + sC_1(1+s^2LC_2) + \frac{s^2LC_2}{R} = 0$$

Substitute $s = j\omega$.

$$g_m + \frac{1}{R} + j\omega C_2 + j\omega C_1(1-\omega^2LC_2) + \frac{-\omega^2LC_2}{R} = 0$$
$$g_m + \frac{1-\omega^2LC_2}{R} + j\left[\omega(C_2+C_1) - \omega^3LC_1C_2\right] = 0$$

Step 3 of 4

Find the frequency of oscillation by equating the imaginary part to zero.

$$\left[\omega(C_2+C_1) - \omega^3LC_1C_2\right] = 0$$
$$(C_2+C_1) = \omega^2LC_1C_2$$
$$\omega^2 = \frac{C_2+C_1}{LC_1C_2}$$
$$\omega = \sqrt{\frac{C_2+C_1}{LC_1C_2}}$$

Therefore the frequency of oscillation is $\boxed{\omega = \sqrt{\frac{C_2+C_1}{LC_1C_2}}}$.

Step 4 of 4

Find the condition on the gain by equating the real part to zero.

$$g_m + \frac{1-\omega^2LC_2}{R} = 0$$
$$g_mR = \omega^2LC_2 - 1$$
$$g_mR = \left(\frac{C_2+C_1}{LC_1C_2}\right)LC_2 - 1$$
$$g_mR = \left(\frac{C_2+C_1}{C_1}\right) - 1$$
$$g_mR = \frac{C_2}{C_1} + 1 - 1$$
$$g_mR = \frac{C_2}{C_1}$$

Therefore the condition on the gain is $\boxed{g_mR = \frac{C_2}{C_1}}$.

Step 1 of 8

Refer to Figure E18.25 in the text book.

Consider the following relation between the voltage and current in the circuit.

$i = 0.1v^2$

Here,

v is the voltage,

i is current in milliamps.

Determine the value of current at voltage $v = 2\,\text{V}$.

$$\begin{aligned} i &= 0.1(2)^2\ \text{mA} \\ &= 0.4\,\text{mA} \end{aligned}$$

The two diodes in the circuit does not conduct for $2\,\text{V}$. Therefore, the only active resistance in the circuit is R_1.

Determine the value of R_1.

$$R_1 = \frac{v}{i}$$

Substitute $2\,\text{V}$ for v and $0.4\,\text{mA}$ for i in the equation.

$$\begin{aligned} R_1 &= \frac{2}{0.4\,\text{mA}} \\ &= 5\,\text{k}\Omega \end{aligned}$$

Thus, the value of resistance, R_1 is $\boxed{5\,\text{k}\Omega}$.

Step 2 of 8

Determine the value of current at voltage $v = 4\,\text{V}$.

$$\begin{aligned} i &= 0.1(4)^2\ \text{mA} \\ &= 1.6\,\text{mA} \end{aligned}$$

The diode in the second branch of the circuit conducts for $4\,\text{V}$. Therefore, the active resistances in the circuit are R_1 and R_2.

Apply Kirchhoff's current law to the circuit, to determine the value of R_2.

$$i = \frac{v}{R_1} + \frac{v-3}{R_2}$$

Substitute $5\,\text{k}\Omega$ for R_1, $4\,\text{V}$ for v and $1.6\,\text{mA}$ for i in the equation.

$$\begin{aligned} 1.6\times10^{-3} &= \frac{4}{5\times10^3} + \frac{4-3}{R_2} \\ 1.6\times10^{-3} &= 0.8\times10^{-3} + \frac{1}{R_2} \\ R_2 &= \frac{1}{1.6\times10^{-3} - 0.8\times10^{-3}} \\ R_2 &= 1.25\,\text{k}\Omega \end{aligned}$$

Thus, the value of resistance, R_2 is $\boxed{1.25\,\text{k}\Omega}$.

Step 3 of 8

Determine the value of current at voltage $v = 8\,\text{V}$.

$$\begin{aligned} i &= 0.1(8)^2\ \text{mA} \\ &= 6.4\,\text{mA} \end{aligned}$$

The diode in the second branch of the circuit conducts for $8\,\text{V}$. Therefore, the active resistances in the circuit are R_1, R_2 and R_3.

Apply Kirchhoff's current law to the circuit, to determine the value of R_3.

$$i = \frac{v}{R_1} + \frac{v-3}{R_2} + \frac{v-7}{R_3}$$

Substitute $5\,\text{k}\Omega$ for R_1, $1.25\,\text{k}\Omega$ for R_2, $8\,\text{V}$ for v and $6.4\,\text{mA}$ for i in the equation.

$$\begin{aligned} 6.4\times10^{-3} &= \frac{8}{5\times10^3} + \frac{8-3}{1.25\times10^3} + \frac{8-7}{R_3} \\ 6.4\times10^{-3} &= \left(1.6\times10^{-3}\right) + \left(4\times10^{-3}\right) + \frac{1}{R_3} \\ R_3 &= \frac{1}{6.4\times10^{-3} - 1.6\times10^{-3} - 4\times10^{-3}} \\ R_3 &= 1.25\,\text{k}\Omega \end{aligned}$$

Thus, the value of resistance, R_3 is $\boxed{1.25\,\text{k}\Omega}$.

Step 4 of 8

Determine the ideal value of current at voltage $v = 3\,\text{V}$.

$$\begin{aligned} i &= 0.1(3)^2\ \text{mA} \\ &= 0.9\,\text{mA} \end{aligned}$$

The two diodes in the circuit does not conduct for $3\,\text{V}$. Therefore, the only active resistance in the circuit is R_1.

Determine the value of current i.

$$i = \frac{v}{R_1}$$

Substitute $3\,\text{V}$ for v and $5\,\text{k}\Omega$ for R_1 in the equation.

$$\begin{aligned} i &= \frac{3}{5\times10^3} \\ &= 0.6\,\text{mA} \end{aligned}$$

Step 5 of 8

Determine the error in the value of current.

$$\begin{aligned} \varepsilon &= 0.6\,\text{mA} - 0.9\,\text{mA} \\ &= -0.3\,\text{mA} \end{aligned}$$

Thus, the error in the current value is $\boxed{-0.3\,\text{mA}}$.

Step 6 of 8

Determine the ideal value of current at voltage $v = 5\,\text{V}$.

$$\begin{aligned} i &= 0.1(5)^2\ \text{mA} \\ &= 2.5\,\text{mA} \end{aligned}$$

The diode in the second branch of the circuit conducts for $5\,\text{V}$. Therefore, the active resistances in the circuit are R_1 and R_2.

Apply Kirchhoff's current law to the circuit, to determine the value of current.

$$i = \frac{v}{R_1} + \frac{v-3}{R_2}$$

Substitute $5\,\text{k}\Omega$ for R_1, $5\,\text{V}$ for v and $1.25\,\text{k}\Omega$ for R_2 in the equation.

$$\begin{aligned} i &= \frac{5}{5\times10^3} + \frac{5-3}{1.25\times10^3} \\ &= \left(1\times10^{-3}\right) + \left(1.6\times10^{-3}\right) \\ &= 2.6\times10^{-3} \\ &= 2.6\,\text{mA} \end{aligned}$$

Determine the error in the value of current.

$$\begin{aligned} \varepsilon &= 2.6\,\text{mA} - 2.5\,\text{mA} \\ &= 0.1\,\text{mA} \end{aligned}$$

Thus, the error in the current value is $\boxed{0.1\,\text{mA}}$.

Step 7 of 8

Determine the value of current at voltage $v = 7\,\text{V}$.

$$\begin{aligned} i &= 0.1(7)^2\ \text{mA} \\ &= 4.9\,\text{mA} \end{aligned}$$

The diode in the second branch of the circuit conducts for $7\,\text{V}$. Therefore, the active resistances in the circuit are R_1 and R_2.

Apply Kirchhoff's current law to the circuit, to determine the value of current.

$$i = \frac{v}{R_1} + \frac{v-3}{R_2}$$

Substitute $5\,\text{k}\Omega$ for R_1, $7\,\text{V}$ for v and $1.25\,\text{k}\Omega$ for R_2 in the equation.

$$\begin{aligned} i &= \frac{7}{5\times10^3} + \frac{7-3}{1.25\times10^3} \\ &= \left(1.4\times10^{-3}\right) + \left(3.2\times10^{-3}\right) \\ &= 4.6\times10^{-3} \\ &= 4.6\,\text{mA} \end{aligned}$$

Determine the error in the value of current.

$$\begin{aligned} \varepsilon &= 4.6\,\text{mA} - 4.9\,\text{mA} \\ &= -0.3\,\text{mA} \end{aligned}$$

Thus, the error in the current value is $\boxed{-0.3\,\text{mA}}$.

Step 8 of 8

Determine the value of current at voltage $v = 10\,\text{V}$.

$$\begin{aligned} i &= 0.1(10)^2\ \text{mA} \\ &= 10\,\text{mA} \end{aligned}$$

The diode in the second branch of the circuit conducts for $10\,\text{V}$. Therefore, the active resistances in the circuit are R_1, R_2 and R_3.

Apply Kirchhoff's current law to the circuit, to determine the value of R_3.

$$i = \frac{v}{R_1} + \frac{v-3}{R_2} + \frac{v-7}{R_3}$$

Substitute $5\,\text{k}\Omega$ for R_1, $1.25\,\text{k}\Omega$ for R_2, $10\,\text{V}$ for v and $1.25\,\text{k}\Omega$ for R_3 in the equation.

$$\begin{aligned} i &= \frac{10}{5\times10^3} + \frac{10-3}{1.25\times10^3} + \frac{10-7}{1.25\times10^3} \\ &= \left(2\times10^{-3}\right) + \left(5.6\times10^{-3}\right) + \left(2.4\times10^{-3}\right) \\ &= 10\times10^{-3} \\ &= 10\,\text{mA} \end{aligned}$$

Determine the error in the value of current.

$$\begin{aligned} \varepsilon &= 10\,\text{mA} - 10\,\text{mA} \\ &= 0\,\text{mA} \end{aligned}$$

Thus, the error in the current value is $\boxed{0\,\text{mA}}$.

Step 1 of 4

To determine the oscillation condition for the Colpitts oscillator consider its equivalent circuit.

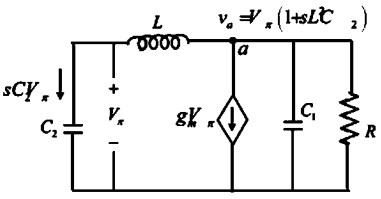


Figure25

Where v_π is the MOSFET input voltage.

To simplify the analysis , we have neglected the MOSFET capacitance C_μ and input resistance r_π .

Step 2 of 4

Assume the resistance R includes r_0 of the MOSFET.

Find the voltage at node a .

$$v_a = V_\pi + Ls(sC_2V_\pi)$$
$$v_a = V_\pi(1+s^2LC_2)$$

The node equation at node a can be found out using alternate analysis approach.

Apply Kirchhoff's current law at node a .

$$sC_2V_\pi + g_mV_\pi + sC_1V_\pi(1+s^2LC_2) + \frac{V_\pi(1+s^2LC_2)}{R} = 0$$

Since $V_\pi \neq 0$, assume $V_\pi = 1$ to eliminate the term.

$$sC_2 + g_m + sC_1(1+s^2LC_2) + \frac{(1+s^2LC_2)}{R} = 0$$
$$g_m + \frac{1}{R} + sC_2 + sC_1(1+s^2LC_2) + \frac{s^2LC_2}{R} = 0$$

Step 3 of 4

Substitute $s = j\omega$.

$$g_m + \frac{1}{R} + j\omega C_2 + j\omega C_1(1-\omega^2LC_2) + \frac{-\omega^2LC_2}{R} = 0$$
$$g_m + \frac{1-\omega^2LC_2}{R} + j\left[\omega(C_2+C_1) - \omega^3LC_1C_2\right] = 0$$

Find the frequency of oscillation by equating the imaginary part to zero.

$$\left[\omega(C_2+C_1) - \omega^3LC_1C_2\right] = 0$$
$$(C_2+C_1) = \omega^2LC_1C_2$$
$$\omega^2 = \frac{C_2+C_1}{LC_1C_2}$$
$$\omega = \sqrt{\frac{C_2+C_1}{LC_1C_2}}$$

Therefore the frequency of oscillation is $\boxed{\omega = \sqrt{\frac{C_2+C_1}{LC_1C_2}}}$.

Step 4 of 4

Find the condition on the gain by equating the real part to zero.

$$g_m + \frac{1-\omega^2LC_2}{R} = 0$$
$$g_mR = \omega^2LC_2 - 1$$
$$g_mR = \left(\frac{C_2+C_1}{LC_1C_2}\right)LC_2 - 1$$
$$g_mR = \left(\frac{C_2+C_1}{C_1}\right) - 1$$
$$g_mR = \frac{C_2}{C_1} + 1 - 1$$
$$g_mR = \frac{C_2}{C_1}$$

Therefore the condition on the gain is $\boxed{g_mR = \frac{C_2}{C_1}}$.

Step 1 of 5

Refer to Figure 18.34 in the text book.

The optimum performance relation between I and R is,

$RI = 2.5V_T$.

Here,

V_T is thermal voltage.

Determine the value of R at room temperature.

$$RI = 2.5V_T$$
$$R = \frac{2.5V_T}{I}$$

Substitute $25\ \text{mV}$ for V_T and $0.25\ \text{mA}$ for I in the equation.

$$R = \frac{2.5\left(25\times10^{-3}\right)}{0.25\times10^{-3}}$$
$$= 250\ \Omega$$

Step 2 of 5

The expression for emitter current of the first transistor is,

$$I_{E1} = I + \frac{2.42V_T}{R}$$
$$= I\left(1 + \frac{2.42V_T}{RI}\right)$$

Substitute $2.5V_T$ for RI in the equation.

$$I_{E1} = I\left(1 + \frac{2.42V_T}{2.5V_T}\right)$$
$$= I\left(1 + \frac{2.42}{2.5}\right)$$

Step 3 of 5

The expression for collector current of the first transistor is,

$$I_{C1} \approx I\left(1 + \frac{2.42}{2.5}\right)$$
$$= I(1 + 0.968)$$
$$I_{C1} = 1.968I$$

Substitute $0.25\ \text{mA}$ for I in the equation.

$$I_{C1} = 1.968\left(0.25\times10^{-3}\right)$$
$$= 492\times10^{-6}$$
$$= 492\ \mu\text{A}$$

Step 4 of 5

The expression for collector current of the second transistor is,

$$I_{C2} \approx I\left(1 - \frac{2.42}{2.5}\right)$$
$$= I(1 - 0.968)$$
$$I_{C2} = 0.032I$$

Substitute $0.25\ \text{mA}$ for I in the equation.

$$I_{C2} = 0.032\left(0.25\times10^{-3}\right)$$
$$= 8\times10^{-6}$$
$$= 8\ \mu\text{A}$$

Step 5 of 5

Determine the peak amplitude of the sine-wave output.

$$v_o = \left(V_{CC} - I_{C2}R_C\right) - \left(V_{CC} - I_{C1}R_C\right)$$
$$v_o = \left(I_{C1} - I_{C2}\right)R_C$$

Substitute $492\ \mu\text{A}$ for I_{C1} , $8\ \mu\text{A}$ for I_{C2} and $10\ \text{k}\Omega$ for R_C in the equation.

$$v_o = \left(492\times10^{-6} - 8\times10^{-6}\right)\left(10\times10^{3}\right)$$
$$= \left(484\times10^{-6}\right)\left(10\times10^{3}\right)$$
$$= 4.84\ \text{V}$$

Thus, the peak amplitude of the sine wave output, v_o is $\boxed{4.84\ \text{V}}$.

Step 1 of 5

Refer to Figure P18.26 in the text book for the Colpitts oscillator circuit.

Draw the small signal model of the circuit.

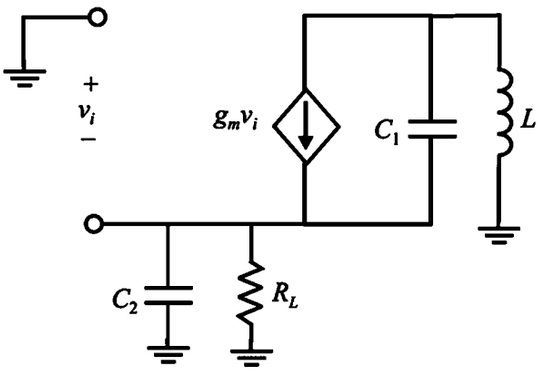


Figure 1

Step 2 of 5

Redraw the circuit shown in Figure 1 is as follows.

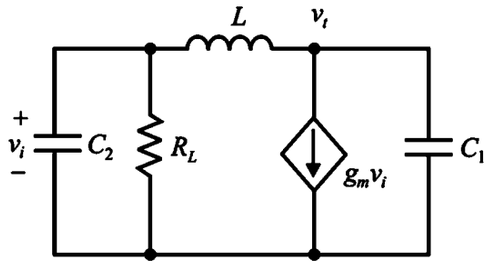


Figure 2

Step 3 of 5

From Figure 2, the voltage v_t is,

$$v_t = 1 + \left(sC_2 + \frac{1}{R}\right)(sL)$$

Consider the input voltage, $v_i = 1$.

Apply Kirchhoff's current law to the node v_t .

$$sC_1 v_t + g_m v_i + sC_2 + \frac{1}{R} = 0$$

$$sC_1 v_t + g_m + sC_2 + \frac{1}{R} = 0 \quad (\text{since } v_i = 1)$$

Substitute $1 + \left(sC_2 + \frac{1}{R}\right)(sL)$ for v_t in the equation.

$$sC_1\left[1 + \left(sC_2 + \frac{1}{R}\right)(sL)\right] + g_m + sC_2 + \frac{1}{R} = 0$$

$$sC_1 + s^2 LC_1\left(sC_2 + \frac{1}{R}\right) + g_m + sC_2 + \frac{1}{R} = 0$$

$$g_m + \frac{1}{R} + s(C_1 + C_2) + \frac{s^2 LC_1}{R} + s^3 LC_1 C_2 = 0$$

Substitute $j\omega$ for s in the equation.

$$g_m + \frac{1}{R} + \frac{(j\omega)^2 LC_1}{R} + j\omega(C_1 + C_2) + (j\omega)^3 LC_1 C_2 = 0$$

$$g_m + \frac{1}{R} - \frac{\omega^2 LC_1}{R} + j\omega(C_1 + C_2) - j\omega^3 LC_1 C_2 = 0$$

$$g_m + \frac{1}{R} - \frac{\omega^2 LC_1}{R} + j\left[\omega(C_1 + C_2) - \omega^3 LC_1 C_2\right] = 0 \quad \ldots\ldots (1)$$

Step 4 of 5

Equate the imaginary part of equation (1) to zero to get the frequency of oscillations, ω_0 .

$$\omega_0(C_1 + C_2) - \omega_0^3 LC_1 C_2 = 0$$

$$C_1 + C_2 = \omega_0^2 LC_1 C_2$$

$$\omega_0^2 = \frac{C_1 + C_2}{LC_1 C_2}$$

$$\omega_0 = \sqrt{\frac{C_1 + C_2}{LC_1 C_2}}$$

Thus, the frequency of oscillation, ω_0 is $\boxed{\sqrt{\frac{C_1 + C_2}{LC_1 C_2}}}$.

Step 5 of 5

Equate the real part of equation (1) to zero to get the gain condition at oscillation frequency, ω_0 .

$$g_m + \frac{1}{R} - \frac{\omega_0^2 LC_1}{R} = 0$$

$$g_m R = \omega_0^2 LC_1 - 1$$

Substitute $\sqrt{\frac{C_1 + C_2}{LC_1 C_2}}$ for ω_0 in the equation.

$$\begin{aligned} g_m R &= \left(\sqrt{\frac{C_1 + C_2}{LC_1 C_2}}\right)^2 (LC_1) - 1 \\ &= \left(\frac{C_1 + C_2}{LC_1 C_2}\right)(LC_1) - 1 \\ &= \left(\frac{C_1 + C_2}{C_2}\right) - 1 \\ &= \frac{C_1}{C_2} \end{aligned}$$

Thus, the gain condition that ensures the start of oscillations is $\boxed{g_m R = \frac{C_1}{C_2}}$.

Step 1 of 5

Refer to Figure P18.27 in the text book for the Colpitts oscillator circuit.

Draw the small signal model of the circuit.

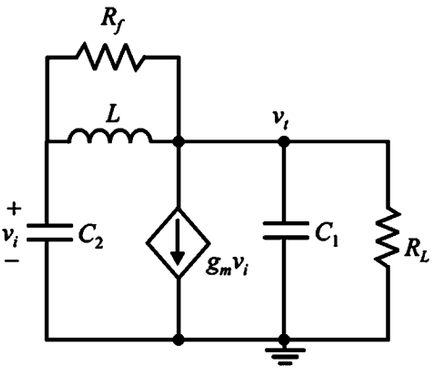


Figure 1

Step 2 of 5

Redraw the circuit shown in Figure 1 by assuming that $R_f \gg \omega_0 L$.

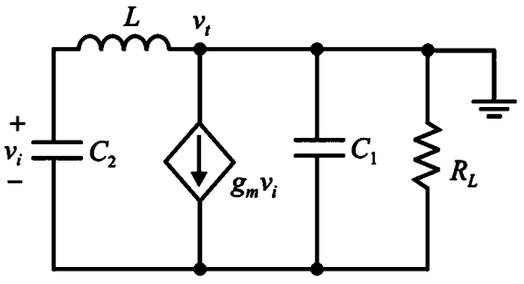


Figure 2

Step 3 of 5

From Figure 2, the expression for voltage v_t is,

$$v_t = 1 + (sC_2)(sL)$$
$$v_t = 1 + s^2 LC_2$$

Consider the input voltage, $v_i = 1$

Apply Kirchhoff's current law to the node v_t .

$$sC_1 v_t + g_m v_t + sC_2 + \frac{v_t}{R} = 0$$

$$sC_1 v_t + g_m + sC_2 + \frac{v_t}{R} = 0 \qquad (\text{since } v_t = 1)$$

Substitute $1 + s^2 LC_2$ for v_t in the equation.

$$sC_1\left[1 + s^2 LC_2\right] + g_m + sC_2 + \frac{1 + s^2 LC_2}{R} = 0$$

$$g_m + sC_1 + sC_2 + s^3 LC_1 C_2 + \frac{1}{R} + \frac{s^2 LC_2}{R} = 0$$

$$g_m + \frac{1}{R} + \frac{s^2 LC_2}{R} + s(C_1 + C_2) + s^3 LC_1 C_2 = 0$$

Substitute $j\omega$ for s in the equation.

$$g_m + \frac{1}{R} + \frac{(j\omega)^2 LC_2}{R} + j\omega(C_1 + C_2) + (j\omega)^3 LC_1 C_2 = 0$$

$$g_m + \frac{1}{R} - \frac{\omega^2 LC_2}{R} + j\omega(C_1 + C_2) - j\omega^3 LC_1 C_2 = 0$$

$$g_m + \frac{1}{R} - \frac{\omega^2 LC_2}{R} + j\left[\omega(C_1 + C_2) - \omega^3 LC_1 C_2\right] = 0 \quad \ldots\ldots (1)$$

Step 4 of 5

Equate the imaginary part of equation (1) to zero to get the frequency of oscillations, ω_0 .

$$\omega_0(C_1 + C_2) - \omega_0^3 LC_1 C_2 = 0$$
$$C_1 + C_2 = \omega_0^2 LC_1 C_2$$
$$\omega_0^2 = \frac{C_1 + C_2}{LC_1 C_2}$$
$$\omega_0 = \sqrt{\frac{C_1 + C_2}{LC_1 C_2}}$$

Thus, the frequency of oscillation, ω_0 is $\boxed{\sqrt{\frac{C_1 + C_2}{LC_1 C_2}}}$.

Step 5 of 5

Equate the real part of equation (1) to zero to get the gain condition at oscillation frequency, ω_0 .

$$g_m + \frac{1}{R} - \frac{\omega_0^2 LC_2}{R} = 0$$
$$g_m + \frac{1 - \omega_0^2 LC_2}{R} = 0$$
$$g_m R = \omega_0^2 LC_2 - 1$$

Substitute $\sqrt{\frac{C_1 + C_2}{LC_1 C_2}}$ for ω_0 in the equation.

$$g_m R = \left(\sqrt{\frac{C_1 + C_2}{LC_1 C_2}}\right)^2 (LC_2) - 1$$
$$= \left(\frac{C_1 + C_2}{LC_1 C_2}\right)(LC_2) - 1$$
$$= \left(\frac{C_1 + C_2}{C_1}\right) - 1$$
$$= \frac{C_2}{C_1}$$

Thus, the gain condition that ensures the start of oscillations is $\boxed{g_m R = \frac{C_2}{C_1}}$.

18.028P

We don't have the solution to this problem yet.

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Step 1 of 3

18.030P

Refer to Figure 17.16 in the text book for the Pierce crystal oscillator circuit utilizing a CMOS inverter.

The resulting circuit oscillates at the resonance frequency of the crystal inductance L with the series equivalence of C_s and $C_p + \frac{C_1 C_2}{C_1 + C_2}$.

Since C_s is much smaller than the three other capacitances, it dominates the other capacitances.

$$\omega_0 \approx \frac{1}{\sqrt{LC_s}} = \omega_s$$

Determine the equivalent capacitance.

$$C_{eq} = \frac{C_s\left(C_p + \frac{C_1 C_2}{C_1 + C_2}\right)}{C_s + C_p + \frac{C_1 C_2}{C_1 + C_2}}$$

Substitute 0.012 pF for C_s, 4 pF for C_p, 1 pF for C_1 and 10 pF for C_2 to determine the lowest equivalent capacitance.

$$\begin{aligned} C_L &= \frac{(0.012)\left(4 + \frac{10 \times 1}{10 + 1}\right)}{\left(0.012 + 4 + \frac{10 \times 1}{10 \times 1}\right)} \\ &= \frac{(0.012)(4 + 0.9091)}{(0.012 + 4 + 0.909)} \\ &= \frac{0.0589092}{4.9211} \\ &= 0.0119707\,\text{pF} \end{aligned}$$

Step 2 of 3

Substitute 0.012 pF for C_s, 4 pF for C_p, 1 pF for C_1 and 10 pF for C_2 to determine the highest equivalent capacitance.

$$\begin{aligned} C_H &= \frac{(0.012)\left(4 + \frac{10 \times 10}{10 + 10}\right)}{\left(0.012 + 4 + \frac{10 \times 10}{10 \times 10}\right)} \\ &= \frac{(0.012)(4 + 5)}{(0.012 + 4 + 5)} \\ &= 0.011984\,\text{pF} \end{aligned}$$

Step 3 of 3

Determine the upper oscillation frequency.

$$\begin{aligned} f_H &= \frac{1}{2\pi\sqrt{LC_L}} \\ &= \frac{1}{2\pi\sqrt{0.52 \times 0.0119707 \times 10^{-12}}} \\ &= 2.017 \times 10^6\ \text{Hz} \\ &= 2.017\ \text{MHz} \end{aligned}$$

Determine the lower oscillation frequency.

$$\begin{aligned} f_L &= \frac{1}{2\pi\sqrt{LC_H}} \\ &= \frac{1}{2\pi\sqrt{0.52 \times 0.011984 \times 10^{-12}}} \\ &= 2.016 \times 10^6\ \text{Hz} \\ &= 2.016\ \text{MHz} \end{aligned}$$

Thus, the range of the oscillation frequency is $\boxed{2.016\ \text{MHz}}$ to $\boxed{2.017\ \text{MHz}}$.

Step 1 of 2

18.031P

Refer to Figure 18.21 (a) in the textbook for the bi-stable circuit.

The op-amp operates at $\pm 5\ \mathrm{V}$. Observe that the characteristic is that of a comparator with a threshold voltage denoted as V_{TH} .

At positive saturation, $V_{TH} = \beta L_+$.

Similarly at negative saturation, $V_{TL} = \beta L_-$.

From the transfer characteristics of the circuit, the width of the hysteresis width is, $V_{TH} - V_{TL}$. Therefore,

$$\begin{aligned} V_{TH} - V_{TL} &= \beta L_+ - \beta L_- \\ &= \beta (L_+ - L_-) \end{aligned}$$

Here L_+ is positive saturation level, L_- is negative saturation level, and β is a parameter $\beta = \dfrac{R_1}{R_1 + R_2}$

Step 2 of 2

Substitute 2 V for $(V_{TH} - V_{TL})$, 5 V for L_+ , $-5\ \mathrm{V}$ for L_- , and $10\ \mathrm{k\Omega}$ for R_1 .

$$V_{TH} - V_{TL} = \frac{R_1}{R_1 + R_2}(L_+ - L_-)$$

$$2 = \frac{10 \times 10^3}{10 \times 10^3 + R_2}(5 - (-5))$$

$$1 = \frac{5 \times 10^3}{10 \times 10^3 + R_2}(10)$$

$$10 \times 10^3 + R_2 = 50 \times 10^3$$

$$R_2 = 40\ \mathrm{k\Omega}$$

Thus, the value of resistor R_2 is $\boxed{40\ \mathrm{k\Omega}}$.

Step 1 of 7

(a)

Refer to Figure 18.21 (a) in the textbook for the bi-stable circuit. The positive terminal of the op-amp in the circuit is connected to the voltage V through the resistor R_3. The modified circuit is shown in Figure 1.

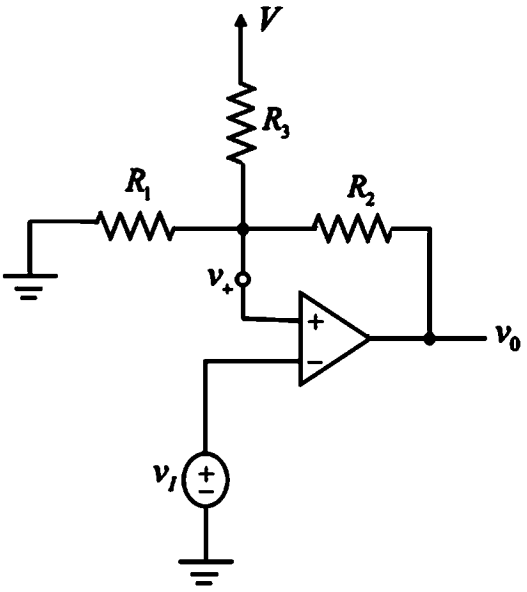


Figure 1

Step 2 of 7

Apply Kirchhoff Current Law at positive node of the op amp.

$$\frac{0-v_+}{R_1}+\frac{V-v_+}{R_3}+\frac{v_o-v_+}{R_2}=0$$

$$\frac{V}{R_3}+\frac{v_o}{R_2}=\frac{v_+}{R_1}+\frac{v_+}{R_3}+\frac{v_+}{R_2}$$

$$\frac{V}{R_3}+\frac{v_o}{R_2}=v_+\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}\right)$$

$$v_+\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}\right)=\frac{VR_2+v_oR_3}{R_3R_2}$$

Rearrange the expression.

$$v_+\left(\frac{R_3R_2+R_3R_1+R_1R_2}{R_1R_3R_2}\right)=\frac{VR_2+v_oR_3}{R_3R_2}$$

$$v_+=\frac{R_1\left(VR_2+v_oR_3\right)}{R_3R_2+R_3R_1+R_1R_2}$$

Step 3 of 7

When v_+ reaches lower threshold voltage V_{TL}, the output voltage v_o becomes L_-. Thus,

$$V_{TL}=\frac{R_1\left(VR_2+L_-R_3\right)}{R_3R_2+R_3R_1+R_1R_2}$$

Thus, the expression for threshold voltage is, $\boxed{V_{TL}=\frac{R_1\left(VR_2+L_-R_3\right)}{R_3R_2+R_3R_1+R_1R_2}}$.

When v_+ reaches higher threshold voltage V_{TH}, the output voltage v_o becomes L_+. Thus,

$$V_{TH}=\frac{R_1\left(VR_2+L_+R_3\right)}{R_3R_2+R_3R_1+R_1R_2}$$

Thus, the expression for threshold voltage is, $\boxed{V_{TH}=\frac{R_1\left(VR_2+L_+R_3\right)}{R_3R_2+R_3R_1+R_1R_2}}$.

Step 4 of 7

(b)

Substitute -10 V for L_-, 15 V for V, $10\text{ k}\Omega$ for R_1, and 4.9 V for V_{TL}.

$$\frac{V}{R_3}+\frac{L_-}{R_2}=V_{TL}\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}\right)$$

$$\frac{15}{R_3}+\frac{-10}{R_2}=4.9\left(\frac{1}{10000}+\frac{1}{R_2}+\frac{1}{R_3}\right)$$

$$\frac{15}{R_3}-\frac{10}{R_2}=0.00049+\frac{4.9}{R_2}+\frac{4.9}{R_3}$$

$$\frac{10.1}{R_3}-\frac{14.9}{R_2}=0.00049 \quad \text{..... (1)}$$

Step 5 of 7

Substitute -10 V for L_-, 15 V for V, $10\text{ k}\Omega$ for R_1, and 5.1 V for V_{TH}.

$$\frac{V}{R_3}+\frac{L_+}{R_2}=V_{TH}\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}\right)$$

$$\frac{15}{R_3}+\frac{10}{R_2}=5.1\left(\frac{1}{10000}+\frac{1}{R_2}+\frac{1}{R_3}\right)$$

$$\frac{15}{R_3}+\frac{10}{R_2}=0.00051+\frac{5.1}{R_2}+\frac{5.1}{R_3}$$

$$\frac{9.9}{R_3}+\frac{4.9}{R_2}=0.00051$$

Step 6 of 7

Multiply both sides with $\frac{10.1}{9.9}$.

$$\frac{10.1}{9.9}\left(\frac{9.9}{R_3}+\frac{4.9}{R_2}\right)=0.00051\left(\frac{10.1}{9.9}\right)$$

$$\frac{10.1}{R_3}+\frac{4.9989}{R_2}=0.0005203 \quad \text{..... (2)}$$

Subtract equation (1) from equation (2).

$$\frac{4.9989+14.9}{R_2}=0.0005203-0.00049$$

$$\frac{19.89}{R_2}=0.0000303$$

$$R_2=656.43\text{ k}\Omega$$

Step 7 of 7

Substitute $656.43\text{ k}\Omega$ for R_2 in equation (1).

$$\frac{9.9}{a}+\frac{4.9}{656.43\times10^3}=0.00051$$

$$\frac{9.9}{R_3}=0.00051-\frac{4.9}{656.43\times10^3}$$

$$R_3=19.7\text{ k}\Omega$$

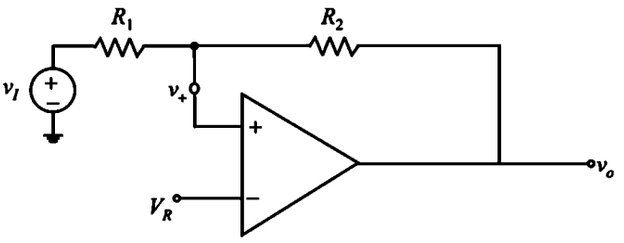
Thus, the resistor values are $\boxed{R_2=656.43\text{ k}\Omega}$ and $\boxed{R_3=19.7\text{ k}\Omega}$.

Step 1 of 5

Refer to Figure 17.20 (a) in the text book for the bistable multivibrator circuit with positive feedback loop.

Disconnect the negative input terminal from ground and connect it to a reference voltage

V_R.



Step 2 of 5

(a)

The op-amp's saturation levels are L_+ and L_-

If the circuit is in the positive stable state with $v_o = L_+$, positive values for v_I have no effect. To trigger the circuit into the L_- state, v_I must be made negative and of such a value as to make v_+ decrease below zero. Thus, the low threshold voltage V_{TL} is found by substituting $v_o = L_+, v_+ = 0$ and $v_I = V_{TL}$.

Apply Kirchhoff's current law to the positive input of the op-amp.

$$\frac{L_+ - V_R}{R_2} = \frac{V_R - V_{TL}}{R_1}$$

$$\frac{V_{TL}}{R_1} = \frac{V_R}{R_1} + \frac{V_R}{R_2} - \frac{L_+}{R_2}$$

$$V_{TL} = V_R\left(\frac{1}{R_1} + \frac{1}{R_2}\right)R_1 - \frac{L_+}{R_2}R_1$$

$$= V_R\left(1 + \frac{R_1}{R_2}\right) - L_+\frac{R_1}{R_2}$$

Thus, the threshold voltage V_{TL} is $\boxed{V_R\left(1 + \frac{R_1}{R_2}\right) - L_+\frac{R_1}{R_2}}$.

Step 3 of 5

Similarly, when the circuit is in the negative output state with $v_o = L_1$, negative values of v_I will make v_+ more negative with no effect on operation. To initiate the regeneration process that causes the circuit to switch to the positive state, v_+ must be made to go slightly positive. The value of v_I that causes this to happen is the high threshold voltage V_{TH}, which can be found by substituting $v_o = L_-, v_+ = 0$ and $v_I = V_{TH}$.

Apply Kirchhoff's current law to the positive input of the op-amp.

$$\frac{L_- - V_R}{R_2} = \frac{V_R - V_{TH}}{R_1}$$

$$\frac{V_{TH}}{R_1} = \frac{V_R}{R_1} + \frac{V_R}{R_2} - \frac{L_-}{R_2}$$

$$V_{TH} = V_R\left(\frac{1}{R_1} + \frac{1}{R_2}\right)R_1 - \frac{L_-}{R_2}R_1$$

$$= V_R\left(1 + \frac{R_1}{R_2}\right) - L_-\frac{R_1}{R_2}$$

Thus, the threshold voltage V_{TH} is $\boxed{V_R\left(1 + \frac{R_1}{R_2}\right) - L_-\frac{R_1}{R_2}}$.

Step 4 of 5

(b)

Consider the equation for threshold voltage, V_{TH}.

$$V_{TH} = V_R\left(1 + \frac{R_1}{R_2}\right) - L_-\frac{R_1}{R_2}$$

Substitute V for L_-, $10 \times 10^3\ \Omega$ for R_1 and $\frac{V}{10}$ for V_{TH}.

$$\frac{V}{10} = V_R\left(1 + \frac{10 \times 10^3}{R_2}\right) + V\left(\frac{10 \times 10^3}{R_2}\right) \quad \text{..... (1)}$$

Consider the equation for threshold voltage, V_{TL}.

$$V_{TL} = V_R\left(1 + \frac{R_1}{R_2}\right) - L_+\frac{R_1}{R_2}$$

Substitute V for L_+, $10 \times 10^3\ \Omega$ for R_1 and 0 for V_{TL}.

$$0 = V_R\left(1 + \frac{10 \times 10^3}{R_2}\right) - V\left(\frac{10 \times 10^3}{R_2}\right) \quad \text{..... (2)}$$

Step 5 of 5

Subtract equation (1) from (2).

$$\frac{V}{10} - 0 = V_R\left(1 + \frac{10000}{R_2}\right) + V\frac{10000}{R_2} - V_R\left(1 + \frac{10000}{R_2}\right) + V\left(\frac{10000}{R_2}\right)$$

$$\frac{V}{10} = 2V\left(\frac{10000}{R_2}\right)$$

$$R_2 = 200000\ \Omega$$

$$= 200\ \text{k}\Omega$$

Thus, the resistance R_2 is $\boxed{200\ \text{k}\Omega}$.

Substitute $200000\ \Omega$ for R_2 in equation (2).

$$0 = V_R\left(1 + \frac{10 \times 10^3}{200000}\right) - V\left(\frac{10 \times 10^3}{200000}\right)$$

$$V_R\left(1 + \frac{1}{20}\right) = V\frac{1}{20}$$

$$V_R\left(\frac{21}{20}\right) = \frac{V}{20}$$

$$V_R = \frac{V}{21}$$

$$= 0.0476V$$

Thus, the voltage V_R is $\boxed{0.0476V}$.

Step 1 of 3

18.034P

Refer to Figure P17.26 in the text book for the op-amp circuit.

The output of the circuit is the voltage across the diodes as the diodes are connected in parallel across the output.

The forward voltage drop of the diodes is $+0.7\ \text{V}$

When the voltage across the diodes is greater than $+0.7\ \text{V}$, one diode conducts and is directly connected to ground. Therefore, the maximum positive output voltage is $+0.7\ \text{V}$.

Similarly, when the voltage across the diodes less than $-0.7\ \text{V}$, the other diode conducts and is directly connected to ground. Therefore, the maximum negative output voltage is $-0.7\ \text{V}$.

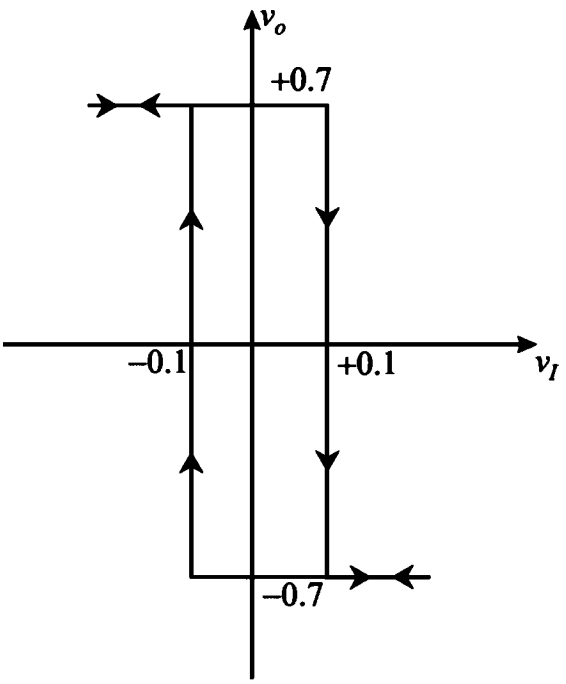
Therefore, the output levels are $\pm 0.7\ \text{V}$.

Step 2 of 3

Apply voltage division rule to find the threshold levels.

$$\begin{aligned} V_{TH} &= \left[\frac{10\times10^3}{(10\times10^3)+(60\times10^3)}\right](\pm 0.7) \\ &= \pm\left(\frac{10\times10^3}{70\times10^3}\right)(0.7) \\ &= \pm(0.14286)(0.7) \\ &= \pm 0.1 \end{aligned}$$

Draw the transfer characteristics, v_o versus v_I .



Step 3 of 3

Determine the maximum diode current.

$$i_{D\max} = \frac{V_{sat} - V_D}{10\times10^3} - \frac{V_D}{(10\times10^3)+(60\times10^3)}$$

Substitute 12 V for saturation voltage, V_{sat} and 0.7 V for the diode voltage, V_D .

$$\begin{aligned} i_{D\max} &= \frac{12-0.7}{10\times10^3} - \frac{0.7}{70\times10^3} \\ &= \frac{11.3}{10\times10^3} - \frac{0.7}{70\times10^3} \\ &= (11.3-0.1)\times10^{-4} \end{aligned}$$

$$\begin{aligned} i_{D\max} &= 1.12\times10^{-3}\ \text{A} \\ &= 1.12\ \text{mA} \end{aligned}$$

Thus, the maximum diode current, $i_{D\max}$ is $\boxed{1.12\ \text{mA}}$.

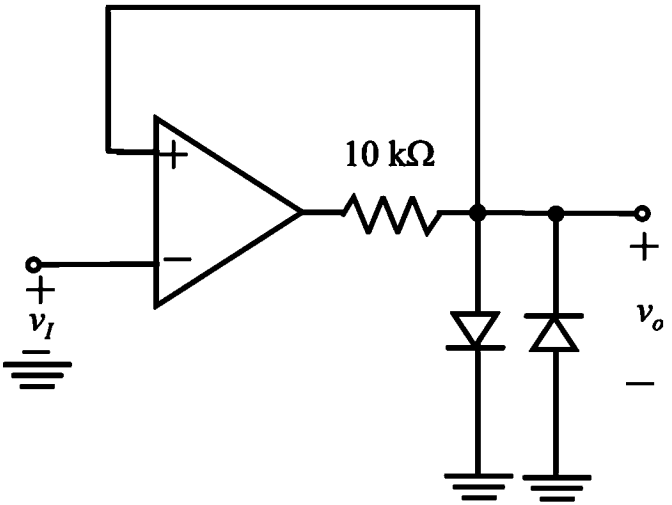
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Step 1 of 3

18.035P

Refer to Figure P17.26 in the text book for the op-amp circuit.

Remove the resistance, R_1 and short-circuit the resistance, R_2. Redraw the circuit.



Step 2 of 3

The output of the circuit is the voltage across the diodes as the diodes are connected in parallel across the output.

The forward voltage drop of the diodes is **+0.7 V**

When the voltage across the diodes is greater than **+0.7 V** , one diode conducts and is directly connected to ground. Therefore, the maximum positive output voltage is **+0.7 V** .

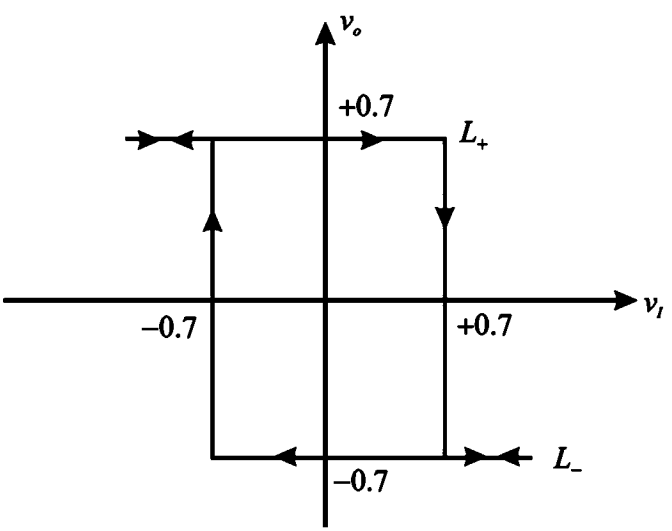
Similarly, when the voltage across the diodes less than **–0.7 V** , the other diode conducts and is directly connected to ground. Therefore, the maximum negative output voltage is **–0.7 V** .

Therefore, the output levels are **±0.7 V** .

Step 3 of 3

The threshold levels also are **±0.7 V** as the output appears at the non-inverting terminal.

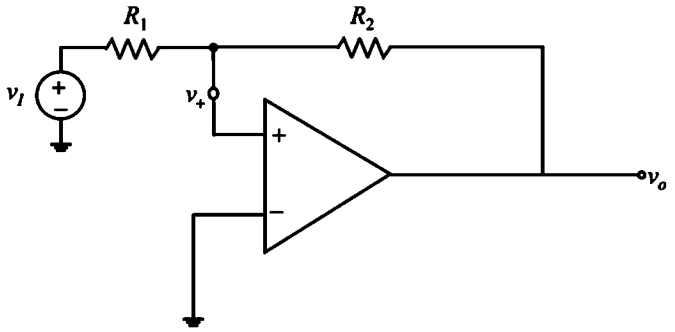
Draw the transfer characteristics, v_o versus v_I .



Thus, the transfer characteristics are drawn.

Step 1 of 3

Draw the bistable non-inverting op-amp circuit.



Step 2 of 3

(a)

The input is 0.5 V amplitude sine wave

When the circuit is in the positive stable state with $v_o = L_+$, the positive values for v_I have no effect.

When the circuit is in the negative output state with $v_o = L_-$, the negative values of v_I make v_+ more negative with no effect on the operation.

The amplitude of the input is very less, so, it is unable to change the state of the circuit, Hence, the output is at saturation levels.

Thus, the output voltage is $\boxed{\pm 12\,\mathrm{V}}$.

Step 3 of 3

(b)

The amplitude of 1.1 V is applied at the input.

There is change the state under the following condition:

$$1.1\sin\phi = 1$$

$$\sin\phi = \frac{1}{1.1}$$

$$\sin\phi = 0.9091$$

$$\phi = \sin^{-1}(0.9091)$$

$$\phi = 65.38^\circ$$

Hence, the output is a symmetric square wave with saturation voltages, $L_+ = -L_- = 12\,\mathrm{V}$ at a frequency, f and lagging with the sine wave by an angle of 65.38° .

Since, $V_{TH} = -V_{TL} = 1\,\mathrm{V}$, there is no change in the state when the average shift, either positive or negative swing is less than 1 V.

Thus, the output is a constant DC voltage if the shift is 0.1 V.

18.037P

We don't have the solution to this problem yet.

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Step 1 of 2

18.038P

Refer to Figure 18.26 (b) in the textbook for the circuit.

Write the expression for the time period of symmetrical square wave.

$$T = 2\tau \ln\left(\frac{1+\beta}{1-\beta}\right)$$

Here τ is a time constant $\tau = RC$ and β is a parameter $\beta = \frac{R_1}{R_1 + R_2}$

Calculate the value of β .

$$\beta = \frac{10\text{ k}}{10\text{ k} + 16\text{ k}}$$
$$= \frac{10}{26}$$

Step 2 of 2

Substitute $62\,\text{k}\Omega$ for R , 5 nF for C and $\frac{10}{26}$ for β .

$$T = 2RC \ln\left(\frac{1+\beta}{1-\beta}\right)$$
$$= 2\left(62\times10^{3}\right)\left(5\times10^{-9}\right)\ln\left(\frac{1+\frac{10}{26}}{1-\frac{10}{26}}\right)$$
$$= \left(62\times10^{-5}\right)\ln\left(\frac{1.3846}{0.6153}\right)$$
$$= 50.2776\times10^{-5}$$

Frequency is the inverse of time period. That is, $f = \frac{1}{T}$

Calculate the value of frequency of oscillations.

$$f = \frac{1}{50.2776\times10^{-5}}$$
$$= 1988.95$$
$$= 1.98\text{ kHz}$$

Therefore, the frequency of oscillation is $\boxed{1.98\text{ kHz}}$.

18.039P

We don't have the solution to this problem yet.

Get help from a Chegg subject expert.

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18.040P

We don't have the solution to this problem yet.

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Step 1 of 5

Refer to Figure P17.33 in the text book for the inverting bistable multivibrator circuit.

The resistances, $R_1 = R_2 = R_3 = R_4 = R_5 = R_6 = R$

Determine the Zener voltage V_Z.

$$V_Z = v_o - v_D$$

Substitute 7.5 V for v_o and 0.7 V for v_D.

$$\begin{aligned} V_Z &= 7.5 - 0.7 \\ &= 6.8\ \text{V} \end{aligned}$$

Thus, the Zener voltage V_Z is $\boxed{6.8\ \text{V}}$.

Step 2 of 5

Determine the period T by substituting $10\times10^3\ \text{Hz}$ for f.

$$\begin{aligned} T &= \frac{1}{f} \\ &= \frac{1}{10\times10^3} \\ &= 10^{-4}\ \text{s} \end{aligned}$$

The pulse width of the square wave is,

$$\begin{aligned} t_1 - t_o &= \frac{T}{2} \\ &= \frac{10^{-4}}{2}\ \text{s} \end{aligned}$$

Step 3 of 5

When a square wave is applied to an integrator, the output is a triangular wave.

The capacitor charges and discharges to the upper and lower threshold values respectively.

The voltage across capacitor is,

$$V_C(t) = \frac{1}{RC}\int_{t_o}^{t_1} V dt + v(t_o)$$

The capacitor charges from the square wave whose peak amplitudes are $\pm7.5\,\text{V}$

$$\frac{7.5}{2} = \frac{1}{RC}V\left[t\right]_{t_o}^{t_1} + \frac{-7.5}{2}$$

$$\frac{7.5}{2} + \frac{7.5}{2} = \frac{1}{R\left(0.5\times10^{-9}\right)}(15)(t_1 - t_o)$$

$$7.5 = \frac{1}{R\left(0.5\times10^{-9}\right)}(15)\left(\frac{10^{-4}}{2}\right)$$

$$1 = \frac{1}{R\left(0.5\times10^{-5}\right)}$$

$$\begin{aligned} R &= \frac{1}{0.5\times10^{-5}} \\ &= 200\ \text{k}\Omega \end{aligned}$$

Therefore, the resistance values are,

$$\boxed{R_1 = R_2 = R_3 = R_4 = R_5 = R_6 = 200\ \text{k}\Omega}.$$

Step 4 of 5

Apply Kirchhoff's current law to the output node of the bistable multivibrator circuit.

$$\frac{L_+ - V_{peak}}{R_7} = I_{min} + \frac{V_{peak}}{R_1 + R_2} + \frac{V_{peak} - V_C}{R_5}$$

$$\frac{13 - 7.5}{R_7} = 1 + \frac{7.5}{200 + 200} + \frac{7.5 - \frac{-7.5}{2}}{200}$$

$$\frac{5.5}{R_7} = 1 + 0.01875 + 0.05625$$

$$R_7 = \frac{5.5}{1.075}$$

$$R_7 = 5.116\ \text{k}\Omega$$

Therefore, the resistance R_7 is $\boxed{5.116\ \text{k}\Omega}$.

Step 5 of 5

The integrator output is a triangular wave with a period of $10^{-4}\,\text{s}$ and peak amplitude values of $\pm7.5\,\text{V}$.

Draw the integrator output voltage waveform.

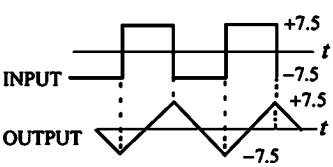


Figure 1

Thus, the output voltage waveform is drawn.

Step 1 of 4

Refer to Figure 17.26 (a) in the text book for the monostable op-amp circuit.

A negative pulse is generated at the output during the quasi stable state.

Determine the duration of the output pulse.

$$T = C_1 R_3 \ln\left(\frac{V_{D1} - L_-}{\beta L_- - L_-}\right)$$

Substitute 0.1×10^{-6} F for C_1, 0.7 V for V_{D1} and -12 V for L_-.

$$100\times10^{-6} = (0.1\times10^{-6})R_3 \ln\left[\frac{0.7-(-12)}{(0.1)(-12)-(-12)}\right]$$

$$100 = (0.1)R_3 \ln\left(\frac{0.7+12}{-1.2+12}\right)$$

$$\frac{100}{0.1} = R_3 \ln\left(\frac{12.7}{10.8}\right)$$

$$1000 = R_3 \ln(1.1759)$$

$$1000 = 0.0162R_3$$

$$R_3 = \frac{1000}{0.0162}$$

$$= 6170.7\ \Omega$$

Step 2 of 4

Draw the signal waveform for the circuit.

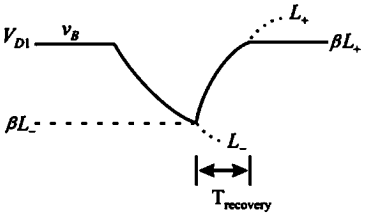


Figure 1

Step 3 of 4

The duration T of the output pulse is determined from the exponential waveform of v_B.

$$v_B(t) = \beta L_- + (L_+ - \beta L_-)\left(1 - e^{-\frac{t}{C_1R_3}}\right)$$

$$= (0.1)(-12) + (12 - (0.1)(-12))\left(1 - e^{-\frac{t}{C_1R_3}}\right)$$

$$= -1.2 + (12 + 1.2)\left(1 - e^{-\frac{t}{C_1R_3}}\right)$$

$$= 12 - 13.2e^{-t/C_1R_3}$$

Step 4 of 4

At $t = T_r$ (recovery time), $v_B(t) = V_{D1}$.

$$V_{D1} = 12 - 13.2e^{-\frac{T_r}{C_1R_3}}$$

$$V_{D1} - 12 = -13.2e^{-\frac{T_r}{C_1R_3}}$$

$$\frac{12 - V_{D1}}{13.2} = e^{-\frac{T_r}{C_1R_3}}$$

$$-\frac{T_r}{C_1R_3} = \ln\left(\frac{12 - V_{D1}}{13.2}\right)$$

$$T_r = -C_1R_3 \ln\left(\frac{12 - V_{D1}}{13.2}\right)$$

Substitute $6170.7\ \Omega$ for R_3, 0.1×10^{-6} F for C_1 and 0.7 V for V_{D1}.

$$Tr = -C_1R_3 \ln\left(\frac{12 - V_{D1}}{13.2}\right)$$

$$= -(0.1\times10^{-6})(6170.7)\ln\left(\frac{12-0.7}{13.2}\right)$$

$$= -617.07\times10^{-6}\ln\left(\frac{11.3}{13.2}\right)$$

$$= -(617.07\times10^{-6})(-0.1554)$$

$$= 95.9\times10^{-6}\ \text{s}$$

$$= 95.9\ \mu\text{s}$$

Thus, the recovery time is $\boxed{95.9\ \mu\text{s}}$.

Step 1 of 5

Refer to Figure P17.34 in the text book for the monostable multivibrator circuit.

Draw the trigger pulse.

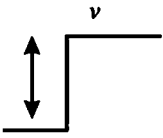


Figure 1

Step 2 of 5

Draw the voltage, v_B waveform at the inverting terminal.

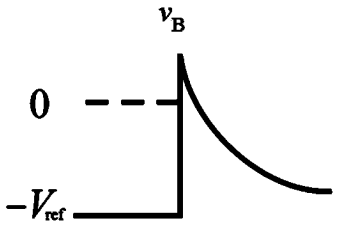


Figure 2

Step 3 of 5

The non-inverting terminal is fed back from the output.

Draw the voltage, v_A waveform at the non- inverting terminal.

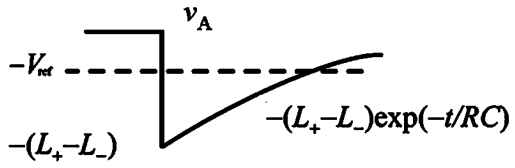


Figure 3

Step 4 of 5

The output of the circuit is positive saturation level when $v_A > v_B$ and negative saturation level when $v_A < v_B$.

Draw the output voltage waveform.

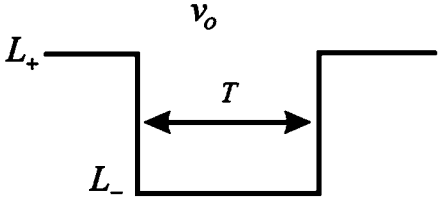


Figure 4

Step 5 of 5

At time, $t = T$, the voltage V_{ref} is,

$$-V_{ref} = -(L_+ - L_-)e^{-T/RC}$$

$$e^{-T/RC} = \frac{V_{ref}}{L_+ - L_-}$$

$$-\frac{T}{RC} = \ln\left(\frac{V_{ref}}{L_+ - L_-}\right)$$

$$-\frac{T}{CR} = -\ln\left(\frac{L_+ - L_-}{V_{ref}}\right)$$

$$\frac{T}{CR} = \ln\left(\frac{L_+ - L_-}{V_{ref}}\right)$$

$$T = CR\ln\left(\frac{L_+ - L_-}{V_{ref}}\right)$$

Therefore, the output pulse width T is $\boxed{CR\ln\left(\frac{L_+ - L_-}{V_{ref}}\right)}$.

Step 1 of 5

A negative pulse is generated at the output during the quasi stable state.

Determine the duration of the output pulse.

Substitute for , 0.7 V for and for .

Step 2 of 5

The resistance, is very much greater than the resistor .

Choose the resistor as

Determine the minimum trigger voltage.

Thus, the minimum trigger voltage is .

Step 3 of 5

Draw the signal waveform of the circuit.

Step 4 of 5

The duration of the output pulse is determined from the exponential waveform of .

Step 5 of 5

At $t = T_r$ (recovery time), $v_B(t) = V_{D1}$.

$$T_r = -C_1 R_3 \ln\left(\frac{13 - V_{D1}}{19.5}\right)$$

Substitute for R_3, 1×10^{-9} F for C_1 and 0.7 V for .

Thus, the recovery time is $\boxed{61.805\ \mu s}$.

Step 1 of 2

(a)

Refer to Figure 18.30 (a) in the textbook for a mono-stable multi-vibrator circuit.

Write the expression for the pulse width of the mono-stable multi-vibrator.

$$T \approx 1.1CR$$

Substitute for , 0.5 nF for .

Therefore, the value of resistance is .

Step 2 of 2

(b)

Exponential wave form of capacitor voltage is,

Step 1 of 2

Refer to Figure 18.31 (a) in the textbook for the astable circuit.

Write the expression for the Duty cycle of the astable circuit.

$$\text{Duty cycle} = \frac{R_A + R_B}{R_A + 2R_B}$$

Here, the Duty cycle is 80% . Find the relation between R_A and R_B .

$$\frac{R_A + R_B}{R_A + 2R_B} = \frac{80}{100}$$
$$\frac{R_A + R_B}{R_A + 2R_B} = \frac{4}{5}$$
$$5(R_A + R_B) = 4(R_A + 2R_B)$$
$$R_A = 3R_B$$

Step 2 of 2

The frequency of the circuit is $f = 20\ \text{kHz}$. Calculate the time period.

$$T = \frac{1}{20 \times 10^3}$$
$$= 50 \times 10^{-6}$$
$$= 50\ \mu\text{s}$$

Write the expression for the time period of the astable circuit.

$$T = 0.69C(R_A + 2R_B)$$

Substitute the corresponding values.

$$50 \times 10^{-6} = 0.69\left(680 \times 10^{-12}\right)(3R_B + 2R_B)$$
$$50 = \left(469.2 \times 10^{-6}\right)(5R_B)$$
$$R_B = \frac{50}{5\left(469.2 \times 10^{-6}\right)}$$
$$R_B = 21.31\ \text{k}\Omega$$

Calculate the value of R_A .

$$R_A = 3(21.31\ \text{k}\Omega)$$
$$= 63.93\ \text{k}\Omega$$

Therefore, the resistor values are $\boxed{R_A = 63.93\ \text{k}\Omega \text{ and } R_B = 21.31\ \text{k}\Omega}$.

Step 1 of 10

Refer to Figure 17.29 in the text book for the 555 timer connected to implement an astable multivibrator circuit.

(a)

The voltage across capacitor C is,

$$v_C = V_{CC} - (V_{CC} - V_{TL})\left(e^{-\frac{t}{(R_A+R_B)C}}\right)$$

The maximum charge on the capacitor is,

$V_C = V_{TH}$ at $t = T_H$.

$$V_{TH} = V_{CC} - (V_{CC} - V_{TL})\left(e^{-\frac{T_H}{(R_A+R_B)C}}\right)$$

$$V_{TH} - V_{CC} = -(V_{CC} - V_{TL})\left(e^{-\frac{T_H}{(R_A+R_B)C}}\right)$$

$$\frac{V_{CC} - V_{TH}}{V_{CC} - V_{TL}} = \left(e^{-\frac{T_H}{(R_A+R_B)C}}\right)$$

$$-\frac{T_H}{(R_A+R_B)C} = \ln\left(\frac{V_{CC} - V_{TH}}{V_{CC} - V_{TL}}\right)$$

$$\frac{T_H}{C(R_A+R_B)} = \ln\left(\frac{V_{CC} - V_{TL}}{V_{CC} - V_{TH}}\right)$$

$$T_H = C(R_A+R_B)\ln\left(\frac{V_{CC} - V_{TL}}{V_{CC} - V_{TH}}\right)$$

Thus, the expression for the time T_H during which the output is high is $\boxed{T_H = C(R_A+R_B)\ln\left(\frac{V_{CC} - V_{TL}}{V_{CC} - V_{TH}}\right)}$.

Step 2 of 10

The discharge voltage of the capacitor is,

$$V_C = V_{TH}\, e^{-t/CR_B}$$

The charge on the capacitor at $t = T_L$ is $V_C = V_{TH}$.

$$V_{TL} = V_{TH}\, e^{-\frac{T_L}{CR_B}}$$

$$\frac{V_{TL}}{V_{TH}} = e^{-\frac{T_L}{CR_B}}$$

$$-\frac{T_L}{CR_B} = \ln\left(\frac{V_{TL}}{V_{TH}}\right)$$

$$\frac{T_L}{CR_B} = \ln\left(\frac{V_{TH}}{V_{TL}}\right)$$

$$T_L = CR_B \ln\left(\frac{V_{TH}}{V_{TL}}\right)$$

Thus, the expression for the time T_L during which the output is low is $\boxed{T_L = CR_B \ln\left(\frac{V_{TH}}{V_{TL}}\right)}$.

Step 3 of 10

(b)

The period T of the output wave for an astable multivibrator is,

$$T = 0.69C(R_A + 2R_B)$$

Substitute 10^{-9} F for C, $7.2\times10^3\ \Omega$ for R_A and $3.6\times10^3\ \Omega$ for R_B.

$$\begin{aligned} T &= (0.69)\left(10^{-9}\right)\left(7.2\times10^3 + 2\times3.6\times10^3\right) \\ &= (0.69)\left(10^{-9}\right)\left(14.4\times10^3\right) \\ &= 9.936\times10^{-6}\ \text{s} \\ &= 9.936\ \mu\text{s} \end{aligned}$$

Determine the oscillation frequency.

$$\begin{aligned} f &= \frac{1}{T} \\ &= \frac{1}{9.936\times10^{-6}} \\ &= 100.6\ \text{kHz} \end{aligned}$$

Thus, the frequency of oscillation f is $\boxed{100.6\ \text{kHz}}$.

Step 4 of 10

Determine the duty cycle of the output waveform.

$$\begin{aligned} \text{Duty cycle} &= \frac{T_H}{T_H + T_L} \\ &= \frac{R_A + R_B}{R_A + 2R_B} \end{aligned}$$

Substitute $7.2\times10^3\ \Omega$ for R_A and $3.6\times10^3\ \Omega$ for R_B.

$$\begin{aligned} \text{Duty cycle} &= \frac{7.2\times10^3 + 3.6\times10^3}{7.2\times10^3 + 2\times3.6\times10^3} \\ &= \frac{10.8}{14.4} \\ &= 0.75 \\ &= 75\% \end{aligned}$$

Thus, the duty cycle of the square wave is $\boxed{75\%}$.

Step 5 of 10

(c)

Determine the voltage V_{TH}.

$$\begin{aligned} V_{TH} &= \frac{2}{3}V_{CC} \\ &= \frac{2}{3}(5\ \text{V}) \\ &= 3.333\ \text{V} \end{aligned}$$

Determine the two extreme values of V_{TH} for 1 V input.

$$\begin{aligned} V_{TH} &= 3.333 \pm 1 \\ &= 4.333\ \text{V and } 2.333\ \text{V} \end{aligned}$$

Thus, the two extreme values of the High-Threshold voltage are $\boxed{4.333\ \text{V}}$ and $\boxed{2.333\ \text{V}}$.

Determine the two extreme values of the voltage V_{TL}.

$$\begin{aligned} V_{TL} &= \frac{1}{2}V_{TH} \\ &= \frac{1}{2}(4.333\ \text{V}) \\ &= 2.167\ \text{V} \end{aligned}$$

$$\begin{aligned} V_{TL} &= \frac{1}{2}V_{TH} \\ &= \frac{1}{2}(2.333\ \text{V}) \\ &= 1.167\ \text{V} \end{aligned}$$

Step 6 of 10

Determine T_H using the following formula:

$$T_H = C(R_A + R_B)\ln\left(\frac{V_{CC} - V_{TL}}{V_{CC} - V_{TH}}\right)$$

Substitute 10^{-9} F for C, $7.2\times10^3\ \Omega$ for R_A, $3.6\times10^3\ \Omega$ for R_B, 5 V for V_{CC}, 4.333 V for V_{TH} and 2.167 V for V_{TL}.

$$\begin{aligned} T_H &= \left(10^{-9}\right)\left(7.2\times10^3 + 3.6\times10^3\right)\ln\left(\frac{5 - 2.167}{5 - 4.333}\right) \\ &= \left(10.8\times10^{-6}\right)\ln(4.2474) \\ &= 15.62\times10^{-6}\ \text{s} \\ &= 15.62\ \mu\text{s} \end{aligned}$$

Thus, the time duration T_H is $\boxed{15.62\ \mu\text{s}}$.

Step 7 of 10

Determine T_L using the following formula:

$$T_L = CR_B \ln\left(\frac{V_{TH}}{V_{TL}}\right)$$

Substitute 10^{-9} F for C, $3.6\times10^3\ \Omega$ for R_B, 4.333 V for V_{TH} and 2.167 V for V_{TL}.

$$\begin{aligned} T_L &= \left(10^{-9}\right)\left(3.6\times10^3\right)\ln\left(\frac{4.333}{2.17}\right) \\ &= 2.489\times10^{-6}\ \text{s} \\ &= 2.489\ \mu\text{s} \end{aligned}$$

Determine the frequency of oscillation.

$$\begin{aligned} f &= \frac{1}{T_L + T_H} \\ &= \frac{1}{2.489\times10^{-6} + 15.62\times10^{-6}} \\ &= 55.2\ \text{kHz} \end{aligned}$$

Thus, the frequency of oscillation f is $\boxed{55.2\ \text{kHz}}$.

Step 8 of 10

Determine T_H using the following formula:

$$T_H = C(R_A + R_B)\ln\left(\frac{V_{CC} - V_{TL}}{V_{CC} - V_{TH}}\right)$$

Substitute 10^{-9} F for C, $7.2\times10^3\ \Omega$ for R_A, $3.6\times10^3\ \Omega$ for R_B, 5 V for V_{CC}, 2.333 V for V_{TH} and 1.167 V for V_{TL}.

$$\begin{aligned} T_H &= \left(10^{-9}\right)\left(7.2\times10^3 + 3.6\times10^3\right)\ln\left(\frac{5 - 1.167}{5 - 2.333}\right) \\ &= \left(10.8\times10^{-6}\right)\ln(1.4372) \\ &= 3.917\times10^{-6}\ \text{s} \\ &= 3.917\ \mu\text{s} \end{aligned}$$

Thus, the time duration T_H is $\boxed{3.917\ \mu\text{s}}$.

Step 9 of 10

Determine T_L using the following formula:

$$T_L = CR_B \ln\left(\frac{V_{TH}}{V_{TL}}\right)$$

Substitute 10^{-9} F for C, $3.6\times10^3\ \Omega$ for R_B, 2.333 V for V_{TH} and 1.167 V for V_{TL}.

$$\begin{aligned} T_L &= \left(10^{-9}\right)\left(3.6\times10^3\right)\ln\left(\frac{2.333}{1.167}\right) \\ &= 2.49\times10^{-6}\ \text{s} \\ &= 2.49\ \mu\text{s} \end{aligned}$$

Determine the frequency of oscillation.

$$\begin{aligned} f &= \frac{1}{T_L + T_H} \\ &= \frac{1}{2.49\times10^{-6} + 15.62\times10^{-6}} \\ &= 55.2\ \text{kHz} \end{aligned}$$

Thus, the frequency of oscillation f is $\boxed{55.2\ \text{kHz}}$.

Step 10 of 10

Determine the duty cycle of the output waveform.

$$\begin{aligned} \text{Duty cycle} &= \frac{T_H}{T_H + T_L} \end{aligned}$$

Substitute $15.62\ \mu\text{s}$ for T_H and $2.489\ \mu\text{s}$ for T_L.

$$\begin{aligned} \text{Duty cycle} &= \frac{15.62}{15.62 + 2.489} \\ &= 0.863 \\ &= 86.3\% \end{aligned}$$

Substitute $3.917\ \mu\text{s}$ for T_H and $2.49\ \mu\text{s}$ for T_L.

$$\begin{aligned} \text{Duty cycle} &= \frac{3.917}{3.917 + 2.49} \\ &= 0.611 \\ &= 61.1\% \end{aligned}$$

Thus, the duty cycle at the two extreme values of T_H are $\boxed{86.3\%}$ and $\boxed{61.1\%}$.

Step 1 of 6

Refer to Figure P17.41 in the text book for the two-diode circuit.

The expression for the output sine wave is,

$$\begin{aligned} v_o &= 0.7\sin\omega t \\ &= 0.7\sin\omega t \\ &= 0.7\sin 2\pi ft \end{aligned}$$

Here,

T is time period of the waveform

Determine the slope of the sine wave at $t = 0$.

The slope of the sine wave is,

$$\begin{aligned} \left.\frac{dv_o}{dt}\right|_{t=0} &= \left.\frac{d}{dt}\left[0.7\sin 2\pi ft\right]\right|_{t=0} \\ &= 0.7(2\pi f)\left(\cos\omega t\right)\Big|_{t=0} \\ &= 0.7(2\pi f) \end{aligned}$$

Step 2 of 6

The peak voltage of the triangular wave is *V*.

The slope of the triangular wave is,

$$\begin{aligned} \text{Slope} &= \frac{V-(-V)}{\frac{T}{2}} \\ &= \frac{2V}{\frac{1}{2f}} \\ &= 4Vf \end{aligned}$$

The zero-crossing slope of the sine wave should match with the slope of the triangular wave.

$$0.7(2\pi f) = 4Vf$$

$$V = \frac{0.7\times 2\pi}{4}$$

$$V = 1.0996\ \text{V}$$

Thus, the value the peak voltage V of the triangular waveform is $\boxed{1.0996\ \text{V}}$.

Step 3 of 6

Determine the resistance *R*.

$$\begin{aligned} R &= \frac{V - V_D}{I_D} \\ &= \frac{1.0996 - 0.7}{1\times 10^{-3}} \\ &= 0.3996\times 10^{3} \\ &\approx 400\ \Omega \end{aligned}$$

Thus, the resistance *R* is $\boxed{400\ \Omega}$.

Step 4 of 6

The voltage drops of the diodes change at the rate of 0.1 V per decade change of the current, I_D.

$$v_o = 0.7 - 0.1\log\left(\frac{1\times 10^{-3}}{i}\right)$$

$$v_o - 0.7 = -0.1\log\left(\frac{1\times 10^{-3}}{i}\right)$$

$$10(v_o - 0.7) = \log\left(\frac{i}{1\times 10^{-3}}\right)$$

$$\frac{i}{1\times 10^{-3}} = 10^{10(v_o-0.7)}$$

$$i = 10^{10(v_o-0.7)}\ \text{mA}$$

The input voltage is,

$$\begin{aligned} v_i &= v_o + iR \\ &= v_o + 400i \end{aligned}$$

$$\begin{aligned} \theta &= \left(\frac{v_i}{V}\right)\times 90° \\ &= \left(\frac{v_i}{1.0996}\right)\times 90° \end{aligned}$$

Step 5 of 6

Tabulate the current and θ values for different voltage values.

v_o (V)	i (mA)	v_i (V)	$\theta = \left(\frac{v_i}{1.0996}\right)\times 90°$
0.7	1	1.1	**90°**
0.65	0.316	0.7764	**63.5°**
0.6	0.1	0.64	**52.4°**
0.55	0.0316	0.563	**46.1°**
0.5	0.01	0.504	**41.25°**
0.4	10^{-3}	0.4004	**32.8°**
0.3	10^{-4}	0.30004	**24.6°**
0.2	10^{-5}	0.2	**16.4°**
0.1	10^{-6}	0.1	**8.2°**
0	10^{-7}	0	**0°**

Step 6 of 6

Determine the output voltages using the determined θ values and compare the values with the original values to determine the percentage error.
Tabulate the values.

v_o (V)	θ	$v_{o\text{exact}} = 0.7\sin\theta$	$\%e = \frac{v_o - v_{o\text{exact}}}{v_{o\text{exact}}}\times 100$
0.7	**90°**	0.7	0
0.65	**63.5°**	0.626	3.83
0.6	**52.4°**	0.55	9.09
0.55	**46.1°**	0.504	9.12
0.5	**41.25°**	0.4615	8.34
0.4	**32.8°**	0.38	5.26
0.3	**24.6°**	0.29	3.45
0.2	**16.4°**	0.198	1
0.1	**8.2°**	0.0998	0.2
0	**0°**	0	0

Thus, the percentage error is determined.

18.050P

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Step 1 of 2

18.051P

Refer to Figure P17.43 in the text book for the logarithmic amplifier circuit.

Consider that the op-amp is ideal. So, the current passing through it is zero.

The total input current flows through the feedback path (through diode) only $\left(I_R = i_D\right)$.

The output voltage is the negative of the voltage across the diode $\left(v_o = -V_D\right)$.

By the virtual ground concept, as the non-inverting terminal connected to ground, the voltage across inverting terminal is also zero.

Therefore, the current passing through resistor R for $v_I > 0$ is,

$$I_R = \frac{v_I}{R}$$

The current passing through diode is,

$$i_D = \frac{v_I}{R}$$

Step 2 of 2

The general expression for the current passing through diode is,

$$i_D = I_S e^{\frac{v_D}{\eta V_T}}$$

Here,

I_S is the saturation current

V_T is thermal voltage

V_D is diode drop in the forward condition.

Substitute $\frac{v_I}{R}$ for i_D.

$$\frac{v_I}{R} = I_S e^{v_D/\eta V_T}$$

$$\frac{v_I}{I_S R} = e^{v_D/\eta V_T}$$

$$\frac{V_D}{\eta V_T} = \ln\left(\frac{v_I}{RI_S}\right)$$

The output voltage, $v_o = -V_D$

Substitute $-v_o$ for V_D.

$$-\frac{v_0}{\eta V_T} = \ln\left(\frac{v_I}{RI_S}\right)$$

$$v_0 = -\eta V_T \ln\left(\frac{v_I}{RI_S}\right)$$

Therefore, the output voltage of the circuit is,

$$\boxed{v_o = -\eta V_T \ln\left(\frac{v_I}{RI_S}\right), \quad v_I > 0}.$$

Step 1 of 14

Refer to Figure P17.44 in the textbook.

Refer to Figure P17.43 in the textbook for the part of the circuit in Figure P17.44.

The expression for the output of the circuit in Figure P17.43 is,

$$v_o = -\eta V_T \ln\left(\frac{v_I}{RI_S}\right)$$

Therefore, the voltages at nodes A, B, **and** C in the overall circuit are,

$$V_A = -\eta V_T \ln\left(\frac{v_1}{RI_S}\right)$$

$$V_B = -\eta V_T \ln\left(\frac{v_2}{RI_S}\right)$$

$$V_C = \eta V_T \ln\left(\frac{1}{RI_S}\right)$$

The voltage at the node D (summing amplifier output) is,

$$\begin{aligned} V_D &= -\frac{1\ \text{k}\Omega}{1\ \text{k}\Omega}(V_A + V_B + V_C) \\ &= -\left[-\eta V_T \ln\left(\frac{v_1}{RI_S}\right) - \eta V_T \ln\left(\frac{v_2}{RI_S}\right) + \eta V_T \ln\left(\frac{1}{RI_S}\right)\right] \\ &= \eta V_T \left[\ln\left(\frac{v_1}{RI_S}\right) + \ln\left(\frac{v_2}{RI_S}\right) - \ln\left(\frac{1}{RI_S}\right)\right] \\ &= \eta V_T \ln\left[\left(\frac{v_1}{RI_S}\right)\left(\frac{v_2}{RI_S}\right)\left(\frac{RI_S}{1}\right)\right] \end{aligned}$$

$$\left(\textbf{since, } \ln(ab) = \ln a + \ln b \textbf{ and } \ln\left(\frac{a}{b}\right) = \ln a - \ln b\right)$$

Step 2 of 14

Simplify the expression further.

$$V_D = \eta V_T \ln\left(\frac{v_1 v_2}{RI_S}\right)$$

$$\frac{V_D}{\eta V_T} = \ln\left(\frac{v_1 v_2}{RI_S}\right)$$

$$\frac{v_1 v_2}{RI_S} = e^{\frac{V_D}{\eta V_T}}$$

The current through the diode, D_4 is,

$$i_{D4} = I_S e^{\frac{V_D}{\eta V_T}}$$

Substitute the expression for $e^{\frac{V_D}{\eta V_T}}$.

$$\begin{aligned} i_{D4} &= I_S\left(\frac{v_1 v_2}{RI_S}\right) \\ &= \frac{v_1 v_2}{R} \end{aligned}$$

The output voltage for the circuit is,

$$\begin{aligned} v_o &= -i_{D4} R \\ &= -\left(\frac{v_1 v_2}{R}\right) R \\ &= -v_1 v_2 \end{aligned}$$

Thus, the expression for the output voltage is, $\boxed{v_o = -v_1 v_2}$, and hence, the circuit is an analog multiplier.

Step 3 of 14

Consider that all the diode are identical with the voltage drop, $v_D = 700\ \text{mV}$ at the current, $I_S = 1\ \text{mA}$.

Observe that the resistance, $R = 1\ \text{k}\Omega$.

Consider the following input combination for the circuit.

$$v_1 = 0.5\ \text{V}$$
$$v_2 = 2\ \text{V}$$

The current through the diode, D_1 is,

$$\begin{aligned} i_{D1} &= \frac{v_1}{1\ \text{k}\Omega} \\ &= \frac{0.5\ \text{V}}{1\ \text{k}\Omega} \\ &= 0.5\ \text{mA} \end{aligned}$$

The voltage at node A is,

$$V_A = -v_{D1} - \eta V_T \ln\left(\frac{v_1}{RI_S}\right)$$

Substitute 700 mV for v_{D1}, 2 for diode parameter, η, 25 mV for thermal equivalent voltage, V_T, 0.5 V for v_1, 1 mA for I_S, and $1\ \text{k}\Omega$ for R.

$$\begin{aligned} V_A &= -700\ \text{mV} - (2)(25\ \text{mV})\ln\left[\frac{0.5\ \text{V}}{(1\ \text{k}\Omega)(1\ \text{mA})}\right] \\ &= -0.7 - (0.05)(-0.693) \\ &= -0.665\ \text{V} \end{aligned}$$

Step 4 of 14

The current through the diode, D_2 is,

$$\begin{aligned} i_{D2} &= \frac{v_2}{1\ \text{k}\Omega} \\ &= \frac{2\ \text{V}}{1\ \text{k}\Omega} \\ &= 2\ \text{mA} \end{aligned}$$

The voltage at node B is,

$$V_B = -v_{D2} - \eta V_T \ln\left(\frac{v_1}{RI_S}\right)$$

Substitute 700 mV for v_{D2}, 2 for η, 25 mV for V_T, 2 V for v_2, 1 mA for I_S, and $1\ \text{k}\Omega$ for R.

$$\begin{aligned} V_B &= -700\ \text{mV} - (2)(25\ \text{mV})\ln\left[\frac{2\ \text{V}}{(1\ \text{k}\Omega)(1\ \text{mA})}\right] \\ &= -0.7 - (0.05)(0.693) \\ &= -0.734\ \text{V} \end{aligned}$$

Step 5 of 14

The current through the diode D_3 is,

$$\begin{aligned} i_{D3} &= \frac{-1\ \text{V}}{1\ \text{k}\Omega} \\ &= -1\ \text{mA} \end{aligned}$$

The voltage at node C is,

$$V_C = v_{D3} + \eta V_T \ln\left(\frac{1}{RI_S}\right)$$

Substitute 700 mV for v_{D3}, 2 for η, 25 mV for V_T, 1 mA for I_S, and $1\ \text{k}\Omega$ for R.

$$\begin{aligned} V_C &= 700\ \text{mV} + (2)(25\ \text{mV})\ln\left[\frac{1\ \text{V}}{(1\ \text{k}\Omega)(1\ \text{mA})}\right] \\ &= 0.7 + (0.05)(0) \\ &= 0.7\ \text{V} \end{aligned}$$

The voltage at the node D (summing amplifier output) is,

$$\begin{aligned} V_D &= -\frac{1\ \text{k}\Omega}{1\ \text{k}\Omega}(V_A + V_B + V_C) \\ &= -(-0.665 - 0.734 + 0.7) \\ &= 0.7\ \text{V} \end{aligned}$$

Step 6 of 14

The current through the diode, D_4 is,

$$\begin{aligned} \frac{i_{D4}}{1\ \text{mA}} &= \frac{I_S e^{\frac{V_D}{\eta V_T}}}{I_S e^{\frac{V_{D3}}{\eta V_T}}} \\ &= e^{\frac{V_D - V_{D3}}{\eta V_T}} \qquad \left(\textbf{since, } \frac{a^m}{a^n} = a^{m-n}\right) \end{aligned}$$

$$i_{D4} = (1\ \text{mA})e^{\frac{V_D - V_{D3}}{\eta V_T}}$$

Substitute 0.7 V for V_D **and** V_{D3}, 2 for η, and 25 mV for V_T.

$$\begin{aligned} i_{D4} &= (1\ \text{mA})e^{\frac{0.7 - 0.7}{(2)(0.025)}} \\ &= (1\ \text{mA})(1) \\ &= 1\ \text{mA} \end{aligned}$$

The output voltage is,

$$\begin{aligned} v_o &= -i_{D4} R \\ &= -(1\ \text{mA})(1\ \text{k}\Omega) \\ &= -1\ \text{V} \end{aligned}$$

The output of the analog multiplier is,

$$\begin{aligned} v_o &= -v_1 v_2 \\ &= -(0.5)(2) \\ &= -1\ \text{V} \end{aligned}$$

Hence, the circuit performs an analog multiplier operation.

Step 7 of 14

Consider the following input combination for the circuit.

$$v_1 = 3\ \text{V}$$
$$v_2 = 1\ \text{V}$$

The current through the diode, D_1 is,

$$\begin{aligned} i_{D1} &= \frac{v_1}{1\ \text{k}\Omega} \\ &= \frac{3\ \text{V}}{1\ \text{k}\Omega} \\ &= 3\ \text{mA} \end{aligned}$$

The voltage at node A is,

$$V_A = -v_{D1} - \eta V_T \ln\left(\frac{v_1}{RI_S}\right)$$

Substitute 700 mV for v_{D1}, 2 for η, 25 mV for V_T, 3 V for v_1, 1 mA for I_S, and $1\ \text{k}\Omega$ for R.

$$\begin{aligned} V_A &= -700\ \text{mV} - (2)(25\ \text{mV})\ln\left[\frac{3\ \text{V}}{(1\ \text{k}\Omega)(1\ \text{mA})}\right] \\ &= -0.7 - (0.05)(1.098) \\ &= -0.754\ \text{V} \end{aligned}$$

Step 8 of 14

The current through the diode, D_2 is,

$$\begin{aligned} i_{D2} &= \frac{v_2}{1\ \text{k}\Omega} \\ &= \frac{1\ \text{V}}{1000\ \Omega} \\ &= 1\ \text{mA} \end{aligned}$$

The voltage at node B is,

$$V_B = -v_{D2} - \eta V_T \ln\left(\frac{v_1}{RI_S}\right)$$

Substitute 700 mV for v_{D2}, 2 for η, 25 mV for V_T, 1 V for v_2, 1 mA for I_S, and $1\ \text{k}\Omega$ for R.

$$\begin{aligned} V_B &= -700\ \text{mV} - (2)(25\ \text{mV})\ln\left[\frac{1\ \text{V}}{(1\ \text{k}\Omega)(1\ \text{mA})}\right] \\ &= -0.7 - (0.05)(0) \\ &= -0.7\ \text{V} \end{aligned}$$

Step 9 of 14

The current through the diode D_3 is,

$$\begin{aligned} i_{D3} &= \frac{-1\ \text{V}}{1\ \text{k}\Omega} \\ &= -1\ \text{mA} \end{aligned}$$

The voltage at node C is,

$$V_C = v_{D3} + \eta V_T \ln\left(\frac{1}{RI_S}\right)$$

Substitute 700 mV for v_{D3}, 2 for η, 25 mV for V_T, 1 mA for I_S, and $1\ \text{k}\Omega$ for R.

$$\begin{aligned} V_C &= 700\ \text{mV} + (2)(25\ \text{mV})\ln\left[\frac{1\ \text{V}}{(1\ \text{k}\Omega)(1\ \text{mA})}\right] \\ &= 0.7 + (0.05)(0) \\ &= 0.7\ \text{V} \end{aligned}$$

The voltage at the node D (summing amplifier output) is,

$$\begin{aligned} V_D &= -\frac{1\ \text{k}\Omega}{1\ \text{k}\Omega}(V_A + V_B + V_C) \\ &= -(-0.754 - 0.7 + 0.7) \\ &= 0.754\ \text{V} \end{aligned}$$

Step 10 of 14

The current through the diode, D_4 is,

$$\begin{aligned} \frac{i_{D4}}{1\ \text{mA}} &= \frac{I_S e^{\frac{V_D}{\eta V_T}}}{I_S e^{\frac{V_{D3}}{\eta V_T}}} \\ &= e^{\frac{V_D - V_{D3}}{\eta V_T}} \qquad \left(\textbf{since, } \frac{a^m}{a^n} = a^{m-n}\right) \end{aligned}$$

$$i_{D4} = (1\ \text{mA})e^{\frac{V_D - V_{D3}}{\eta V_T}}$$

Substitute 0.754 V for V_D, 0.7 V for V_{D4}, 2 for η, and 25 mV for V_T.

$$\begin{aligned} i_{D4} &= (1\ \text{mA})e^{\frac{0.754 - 0.7}{(2)(0.025)}} \\ &= (1\ \text{mA})(3) \\ &= 3\ \text{mA} \end{aligned}$$

The output voltage is,

$$\begin{aligned} v_o &= -i_{D4} R \\ &= -(3\ \text{mA})(1\ \text{k}\Omega) \\ &= -3\ \text{V} \end{aligned}$$

The output of the analog multiplier is,

$$\begin{aligned} v_o &= -v_1 v_2 \\ &= -(3)(1) \\ &= -3\ \text{V} \end{aligned}$$

Hence, the circuit performs an analog multiplier operation.

Step 11 of 14

Consider the following single input for the circuit.

$$v_1 = 2\ \text{V}$$

Consider that the resistance value, $R = 0.5\ \text{k}\Omega$.

The current through the diode, D_1 is,

$$\begin{aligned} i_{D1} &= \frac{v_1}{0.5\ \text{k}\Omega} \\ &= \frac{2\ \text{V}}{0.5\ \text{k}\Omega} \\ &= 4\ \text{mA} \end{aligned}$$

The voltage at node A is,

$$V_A = -v_{D1} - \eta V_T \ln\left(\frac{v_1}{RI_S}\right)$$

Substitute 700 mV for v_{D1}, 2 for η, 25 mV for V_T, 2 V for v_1, 1 mA for I_S, and $0.5\ \text{k}\Omega$ for R.

$$\begin{aligned} V_A &= -700\ \text{mV} - (2)(25\ \text{mV})\ln\left[\frac{2\ \text{V}}{(0.5\ \text{k}\Omega)(1\ \text{mA})}\right] \\ &= -0.7 - (0.05)(1.386) \\ &= -0.769\ \text{V} \end{aligned}$$

Step 12 of 14

The current through the diode, D_2 is,

$$\begin{aligned} i_{D2} &= \frac{v_1}{0.5\ \text{k}\Omega} \\ &= \frac{1\ \text{V}}{0.5\ \text{k}\Omega} \\ &= 4\ \text{mA} \end{aligned}$$

The voltage at node B is,

$$V_B = -v_{D2} - \eta V_T \ln\left(\frac{v_1}{RI_S}\right)$$

Substitute 700 mV for v_{D2}, 2 for η, 25 mV for V_T, 2 V for v_1, 1 mA for I_S, and $0.5\ \text{k}\Omega$ for R

$$\begin{aligned} V_B &= -700\ \text{mV} - (2)(25\ \text{mv})\ln\left[\frac{2\ \text{V}}{(0.5\ \text{k}\Omega)(1\ \text{mA})}\right] \\ &= -0.7 - (0.05)(1.386) \\ &= -0.769\ \text{V} \end{aligned}$$

Step 13 of 14

The current through the diode, D_3 is,

$$\begin{aligned} i_{D3} &= \frac{-1\ \text{V}}{0.5\ \text{k}\Omega} \\ &= -2\ \text{mA} \end{aligned}$$

The voltage at node C is,

$$V_C = v_{D3} + \eta V_T \ln\left(\frac{1}{RI_S}\right)$$

Substitute 700 mV for v_{D3}, 2 for η, 25 mV for V_T, 1 mA for I_S, and $0.5\ \text{k}\Omega$ for R.

$$\begin{aligned} V_C &= 700\ \text{mV} + (2)(25\ \text{mV})\ln\left[\frac{1\ \text{V}}{(0.5\ \text{k}\Omega)(1\ \text{mA})}\right] \\ &= 0.7 + (0.05)(0.693) \\ &= 0.734\ \text{V} \end{aligned}$$

The voltage at the node D (summing amplifier output) is,

$$\begin{aligned} V_D &= -\frac{1\ \text{k}\Omega}{1\ \text{k}\Omega}(V_A + V_B + V_C) \\ &= -(-0.769 - 0.769 + 0.734) \\ &= 0.8\ \text{V} \end{aligned}$$

Step 14 of 14

The current through the diode, D_4 is,

$$\begin{aligned} \frac{i_{D4}}{1\ \text{mA}} &= \frac{I_S e^{\frac{V_D}{\eta V_T}}}{I_S e^{\frac{V_{D3}}{\eta V_T}}} \\ &= e^{\frac{V_D - V_{D3}}{\eta V_T}} \qquad \left(\textbf{since, } \frac{a^m}{a^n} = a^{m-n}\right) \end{aligned}$$

$$i_{D4} = (1\ \text{mA})e^{\frac{V_D - V_{D3}}{\eta V_T}}$$

Substitute 0.8 V for V_D, 0.7 V for V_{D4}, 2 for η, and 25 mV for V_T.

$$\begin{aligned} i_{D4} &= (1\ \text{mA})e^{\frac{0.8 - 0.7}{(2)(0.025)}} \\ &= (1\ \text{mA})(8) \\ &= 8\ \text{mA} \end{aligned}$$

The output voltage is,

$$\begin{aligned} v_o &= -i_{D4} R \\ &= -(8\ \text{mA})(0.5\ \text{k}\Omega) \\ &= -4\ \text{V} \end{aligned}$$

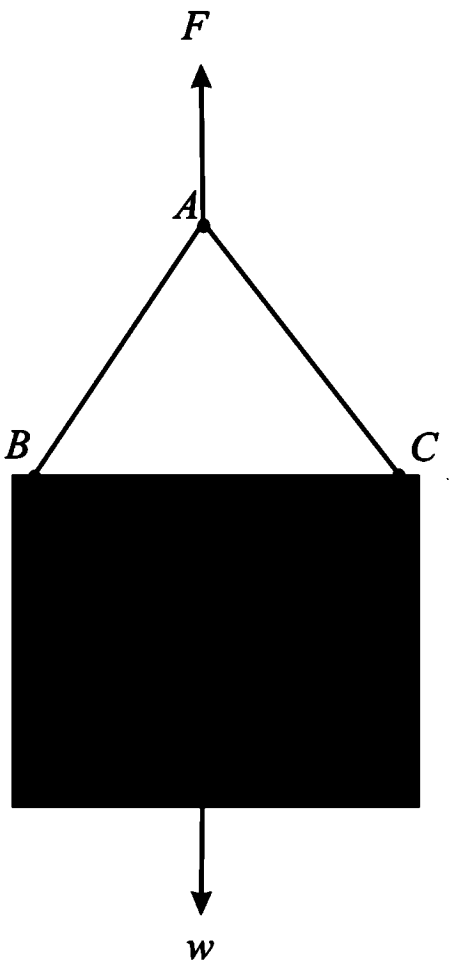
The output of the analog multiplier is,

$$\begin{aligned} v_o &= -v_1 v_1 \\ &= -(2)(2) \\ &= -4\ \text{V} \end{aligned}$$

Hence, the circuit performs a squarer operation for single input with $0.5\ \text{k}\Omega$ resistance.

Step 1 of 7

Draw the free body diagram of the concrete panel and pulley.



Step 2 of 7

Consider the entire system is in equilibrium.

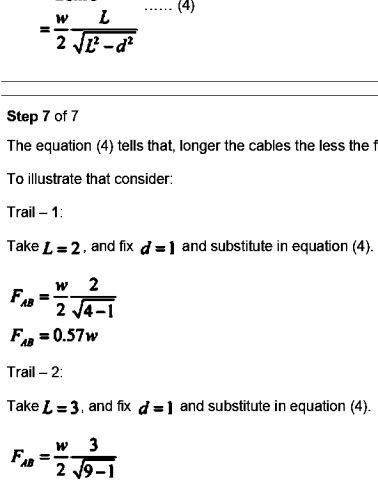
Find the force exerted on the pulley from the vertical equilibrium equations:

$$\sum F_y = 0$$
$$F - w = 0 \quad \text{..... (1)}$$
$$F = w$$

Here, force exerted on the pulley is F , and the weight of the concrete panel is w.

Step 3 of 7

Consider free body diagram of joint A,



Step 4 of 7

Equate the sum of forces in x direction.

$$\sum F_x = 0$$
$$F_{AB}\cos\theta = F_{AC}\cos\theta$$
$$F_{AB} = F_{AC}$$

Equate the sum of forces in y direction.

$$\sum F_y = 0$$
$$F - F_{AB}\sin\theta - F_{AC}\sin\theta = 0$$

Substitute F_{AB} for F_{AC} and w for F.

$$F - F_{AB}\sin\theta - F_{AC}\sin\theta = 0$$
$$w - F_{AB}\sin\theta - F_{AB}\sin\theta = 0$$
$$w - 2F_{AB}\sin\theta = 0 \quad \text{..... (2)}$$
$$F_{AB} = \frac{w}{2\sin\theta}$$

Step 5 of 7

Consider the triangle ABC

Take length of the cable's AB and AC as 'L' and the distance between OB and OC as'd'

Step 6 of 7

Calculate the angle made at B and C from the triangle ABC.

$$\sin\theta = \frac{AO}{AC} = \frac{\sqrt{L^2 - d^2}}{L} \quad \text{..... (3)}$$

Substitute $\frac{\sqrt{L^2 - d^2}}{L}$ for $\sin\theta$ in equation (2).

$$F_{AB} = \frac{w}{2\sin\theta} = \frac{w}{2}\frac{L}{\sqrt{L^2 - d^2}} \quad \text{..... (4)}$$

Step 7 of 7

The equation (4) tells that, longer the cables the less the force in each cable.

To illustrate that consider:

Trail – 1:

Take $L = 2$, and fix $d = 1$ and substitute in equation (4).

$$F_{AB} = \frac{w}{2}\frac{2}{\sqrt{4-1}}$$
$$F_{AB} = 0.57w$$

Trail – 2:

Take $L = 3$, and fix $d = 1$ and substitute in equation (4).

$$F_{AB} = \frac{w}{2}\frac{3}{\sqrt{9-1}}$$
$$F_{AB} = 0.53w$$

Hence, it is shown that longer the cables the less the force in each cable.